

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

JOSNELLY C. APONTE

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Projected 12-Month Period Ending Dec 31, 2021
Version 1
4CP 75/0/25 Production and 12CP Transmission
(thousands of dollars)

Summary RETURN (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional		
1	Rate Base	11,893,424	11,845,434	6,625,291	3,033,254	2,038,373	139,095	9,421	47,990		
	Revenue										
2	Total Rate Revenue	4,017,792	3,993,182	1,949,148	1,003,525	997,036	41,750	1,722	24,610		
3	Total Revenue Credits	379,354	378,396	171,280	87,702	113,061	2,119	4,235	958		
4	Total Revenue	4,397,146	4,371,578	2,120,428	1,091,227	1,110,096	43,869	5,957	25,568		
5	Expenses:										
6	Fuel and P&I Expense	1,683,201	1,664,958	744,413	389,454	519,499	8,637	2,956	18,243		
7	Transmission Expense	477,556	472,579	225,419	106,588	137,459	2,362	750	4,976		
8	Other O & M Expense	684,695	681,844	397,323	166,772	109,853	7,356	539	2,850		
9	Depreciation & Amortization Expense	705,582	702,002	391,217	177,500	121,587	11,054	644	3,580		
10	Other Taxes	263,853	263,093	138,806	68,962	50,878	4,198	250	760		
11	Federal Income Taxes	53,928	54,376	20,677	16,852	15,821	950	76	(448)		
12	Total Expenses	3,868,814	3,838,853	1,917,855	926,128	955,097	34,557	5,215	29,961		
13	Net Operating Income	528,332	532,725	202,573	165,099	155,000	9,312	742	(4,394)		
14	Other Income Adjustments	6,234	6,203	3,347	1,557	1,213	80	6	30		
15	Adjusted Net Operating Income	534,565	538,928	205,920	166,655	156,213	9,392	749	(4,363)		
16	Rate of Return on Rate Base	4.49%	4.55%	3.11%	5.49%	7.66%	6.75%	7.94%	-9.09%		
17	Index of Return (Jurisdictional)		100	68	121	168	148	175			
18	Return on Rate Base @ 6.09%	724,604	721,680	403,644	184,800	124,187	8,474	574	2,924		
19	Income Deficiency (Sufficiency)	190,039	182,752	197,724	18,145	(32,025)	(918)	(175)	7,287		
20	Revenue Deficiency (Sufficiency)	254,475	244,718	264,767	24,298	(42,884)	(1,229)	(234)	9,758		
21	Revenue Requirement/Total Cost of Service	4,651,621	4,616,296	2,385,195	1,115,525	1,067,212	42,641	5,723	35,326		
22	Less: Revenue Credits	379,354	378,396	171,280	87,702	113,061	2,119	4,235	958		
23	Proposed Rate Design Revenue	4,272,267	4,237,899	2,213,915	1,027,823	954,151	40,522	1,489	34,368		
24	Production: Net Capacity Cost	1,182,798	1,171,269	593,086	264,400	309,886	2,272	1,626	11,530		
25	Production: Capacity Related Cost Offset	505,456	497,714	223,525	113,845	156,266	3,008	1,071	7,742		
26	Production: Non-Capacity Related Cost	1,138,083	1,122,237	489,029	264,988	362,402	7,325	(1,507)	15,847		
27	Distribution: Demand Related Cost	1,213,565	1,214,459	734,013	340,302	113,666	26,210	268	(894)		
28	Distribution: Customer Related Cost	232,365	232,222	174,263	44,287	11,932	1,708	32	143		
29	Full Service MWH Sales	30,813,122	30,433,455	12,447,668	7,086,449	10,627,497	208,358	63,483	379,667		
30	ROA MWH Sales	3,697,427	3,697,427	-	205,339	3,492,088	-	-	-		
31	MWH Sales	34,510,549	34,130,882	12,447,668	7,291,788	14,119,585	208,358	63,483	379,667		
32	Customers	1,841,963	1,841,961	1,620,698	216,313	4,116	822	12	2		

Projected 12-Month Period Ending Dec 31, 2021
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(thousands of dollars)

Residential/Secondary RETURN (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Total Commercial Secondary
1	Rate Base		6,625,291	-	6,625,291	1,775,204	1,123,602	52,239	82,210	3,033,254
	Revenue									
2	Total Rate Revenue		1,949,148	-	1,949,148	564,822	400,954	13,055	24,694	1,003,525
3	Total Revenue Credits		171,280	-	171,280	47,524	37,035	1,174	1,968	87,702
4	Total Revenue		2,120,428	-	2,120,428	612,347	437,989	14,229	26,662	1,091,227
5	Expenses:									
6	Fuel and P&I Expense		744,413	-	744,413	207,788	168,269	5,014	8,383	389,454
7	Transmission Expense		225,419	-	225,419	57,391	45,429	1,482	2,286	106,588
8	Other O & M Expense		397,323	-	397,323	100,429	59,407	2,704	4,231	166,772
9	Depreciation & Amortization Expense		391,217	-	391,217	104,571	65,674	2,869	4,387	177,500
10	Other Taxes		138,806	-	138,806	40,081	25,928	1,066	1,886	68,962
11	Federal Income Taxes		20,677	-	20,677	9,455	6,787	101	508	16,852
12	Total Expenses		1,917,855	-	1,917,855	519,716	371,495	13,236	21,681	926,128
13	Net Operating Income		202,573	-	202,573	92,631	66,494	993	4,981	165,099
14	Other Income Adjustments		3,347	-	3,347	900	590	26	40	1,557
15	Adjusted Net Operating Income		205,920	-	205,920	93,531	67,084	1,019	5,021	166,655
16	Rate of Return on Rate Base		3.11%	0.00%	3.11%	5.27%	5.97%	1.95%	6.11%	5.49%
17	Index of Return (Jurisdictional)		68	-	68	116	131	43	134	121
18	Return on Rate Base @ 6.09%		403,644	-	403,644	108,154	68,455	3,183	5,009	184,800
19	Income Deficiency (Sufficiency)		197,724	-	197,724	14,623	1,371	2,164	(13)	18,145
20	Revenue Deficiency (Sufficiency)		264,767	-	264,767	19,581	1,836	2,898	(17)	24,298
21	Revenue Requirement/Total Cost of Service		2,385,195	-	2,385,195	631,928	439,825	17,127	26,645	1,115,525
22	Less: Revenue Credits		171,280	-	171,280	47,524	37,035	1,174	1,968	87,702
23	Proposed Rate Design Revenue		2,213,915	-	2,213,915	584,403	402,790	15,952	24,677	1,027,823
24	Production: Net Capacity Cost		593,086	-	593,086	143,993	111,929	3,663	4,815	264,400
25	Production: Capacity Related Cost Offset		223,525	-	223,525	60,537	49,683	1,498	2,127	113,845
26	Production: Non-Capacity Related Cost		489,029	-	489,029	140,658	114,854	3,414	6,061	264,988
27	Distribution: Demand Related Cost		734,013	-	734,013	201,199	120,664	7,036	11,403	340,302
28	Distribution: Customer Related Cost		174,263	-	174,263	38,017	5,659	341	270	44,287
29	Full Service MWH Sales		12,447,668	-	12,447,668	3,742,026	3,093,372	87,692	163,359	7,086,449
30	ROA MWH Sales		-	-	-	9,785	116,816	14,695	64,043	205,339
31	MWH Sales		12,447,668	-	12,447,668	3,751,811	3,210,188	102,387	227,402	7,291,788
32	Customers		1,620,698	-	1,620,698	194,349	19,604	1,580	780	216,313

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Primary & Lighting RETURN (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description		Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Total Lighting & Unmetered
1	Rate Base		212,424	-	175,123	340,332	1,193,422	37,710	23,475	59	9,175	46,654	2,038,373	4,725	76,172	36,741	21,457	139,095
	Revenue																	
2	Total Rate Revenue		101,921	-	102,260	173,239	562,996	14,966	22,612	14	3,256	15,771	997,036	1,551	23,398	7,927	8,874	41,750
3	Total Revenue Credits		9,357	-	16,662	21,215	60,174	1,388	2,455	1	296	1,514	113,061	119	783	254	964	2,119
4	Total Revenue		111,278	-	118,922	194,454	623,169	16,354	25,067	15	3,552	17,285	1,110,096	1,670	24,181	8,181	9,838	43,869
5	Expenses:																	
6	Fuel and P&I Expense		43,765	-	68,599	99,272	281,183	6,371	12,425	-	1,252	6,633	519,499	482	2,824	735	4,596	8,637
7	Transmission Expense		12,279	-	17,679	26,346	77,030	1,886	-	-	359	1,879	137,459	149	876	228	1,109	2,362
8	Other O & M Expense		10,520	-	10,917	19,168	63,030	1,838	1,667	2	431	2,280	109,853	268	5,299	580	1,209	7,356
9	Depreciation & Amortization Expense		12,022	-	12,730	22,375	68,571	1,886	1,335	2	510	2,155	121,587	275	5,923	3,561	1,295	11,054
10	Other Taxes		5,478	-	4,241	8,719	29,248	895	961	2	250	1,084	50,878	121	2,340	1,243	494	4,198
11	Federal Income Taxes		2,520	-	440	1,720	9,642	322	804	1	69	301	15,821	35	641	170	105	950
12	Total Expenses		86,585	-	114,607	177,600	528,705	13,198	17,192	7	2,871	14,332	955,097	1,330	17,903	6,517	8,808	34,557
13	Net Operating Income		24,693	-	4,315	16,854	94,465	3,156	7,876	8	681	2,953	155,000	340	6,278	1,664	1,030	9,312
14	Other Income Adjustments		119	-	105	225	698	21	13	0	6	25	1,213	2	42	24	12	80
15	Adjusted Net Operating Income		24,812	-	4,420	17,079	95,163	3,177	7,888	8	687	2,979	156,213	342	6,320	1,688	1,042	9,392
16	Rate of Return on Rate Base		11.68%	0.00%	2.52%	5.02%	7.97%	8.42%	33.60%	14.03%	7.49%	6.38%	7.66%	7.24%	8.30%	4.59%	4.86%	6.75%
17	Index of Return (Jurisdictional)		257	-	55	110	175	185	739	308	165	140	168	159	182	101	107	148
18	Return on Rate Base @ 6.09%		12,942	-	10,669	20,735	72,709	2,297	1,430	4	559	2,842	124,187	288	4,641	2,238	1,307	8,474
19	Income Deficiency (Sufficiency)		(11,870)	-	6,249	3,655	(22,454)	(879)	(6,458)	(5)	(128)	(136)	(32,025)	(54)	(1,679)	551	265	(918)
20	Revenue Deficiency (Sufficiency)		(15,895)	-	8,368	4,895	(30,067)	(1,177)	(8,648)	(6)	(171)	(183)	(42,884)	(73)	(2,249)	737	355	(1,229)
21	Revenue Requirement/Total Cost of Service		95,383	-	127,290	199,349	593,102	15,177	16,419	9	3,381	17,103	1,067,212	1,597	21,932	8,919	10,193	42,641
22	Less: Revenue Credits		9,357	-	16,662	21,215	60,174	1,388	2,455	1	296	1,514	113,061	119	783	254	964	2,119
23	Proposed Rate Design Revenue		86,026	-	110,628	178,134	532,928	13,789	13,964	8	3,085	15,589	954,151	1,478	21,150	8,665	9,229	40,522
24	Production: Net Capacity Cost		30,041	-	35,453	55,753	178,723	4,278	-	-	1,068	4,569	309,886	-	-	-	2,272	2,272
25	Production: Capacity Related Cost Offset		10,918	-	23,816	31,825	82,308	1,604	3,660	0	295	1,841	156,266	189	1,064	268	1,486	3,008
26	Production: Non-Capacity Related Cost		29,812	-	47,731	71,062	195,336	4,481	8,711	(0)	761	4,508	362,402	474	2,779	724	3,348	7,325
27	Distribution: Demand Related Cost		12,876	-	2,388	17,769	70,708	3,173	1,406	5	912	4,429	113,666	781	15,696	7,664	2,069	26,210
28	Distribution: Customer Related Cost		2,379	-	1,240	1,725	5,853	254	188	3	50	242	11,932	34	1,611	9	54	1,708
29	Full Service MWH Sales		830,685	-	1,514,473	2,094,112	5,560,720	121,808	358,537	-	21,349	125,813	10,627,497	13,509	79,208	20,629	95,012	208,358
30	ROA MWH Sales		46,609	-	1,089,479	1,247,199	854,891	27,722	-	2,690	70,229	153,269	3,492,088	-	-	-	-	-
31	MWH Sales		877,294	-	2,603,952	3,341,311	6,415,611	149,530	358,537	2,690	91,578	279,082	14,119,585	13,509	79,208	20,629	95,012	208,358
32	Customers		1,510	-	49	160	2,145	142	19	1	6	84	4,116	343	-	-	479	822

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Summary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description			Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Net Plant										
2	Production			3,675,605	3,639,776	1,761,999	832,316	1,028,226	11,642	5,594	35,829
3	Transmission			(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution			6,328,753	6,325,225	3,792,578	1,743,687	675,595	110,945	2,419	3,528
5	General/Common/Intangible			721,822	718,966	420,086	177,020	113,694	7,619	547	2,856
6	Plant Purchased/Sold			0	0	0	0	0	0	0	0
7	Total Net Plant			10,726,180	10,683,967	5,974,663	2,753,023	1,817,514	130,207	8,560	42,213
8	Working Capital										
9	Total Current Assets			2,236,176	2,226,380	1,255,789	554,101	389,033	25,791	1,665	9,796
10	Total Current Liabilities			1,009,093	1,005,104	569,465	256,986	162,616	15,256	782	3,989
11	Total Working Capital			1,227,083	1,221,275	686,324	297,115	226,417	10,535	884	5,807
	Adjustments to Rate Base										
12	Additions to Rate Base			0	0	0	0	0	0	0	0
13	Deductions from Rate Base			59,839	59,808	35,696	16,884	5,559	1,647	22	30
14	Total Adjustments to Rate Base			(59,839)	(59,808)	(35,696)	(16,884)	(5,559)	(1,647)	(22)	(30)
15	Total Rate Base			11,893,424	11,845,434	6,625,291	3,033,254	2,038,373	139,095	9,421	47,990

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Residential/Secondary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Net Plant									
2	Production		1,761,999	-	1,761,999	449,710	355,192	11,211	16,202	832,316
3	Transmission		(0)	-	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution		3,792,578	-	3,792,578	1,053,256	601,797	33,901	54,733	1,743,687
5	General/Common/Intangible		420,086	-	420,086	106,434	63,183	2,916	4,487	177,020
6	Plant Purchased/Sold		0	-	0	0	0	0	0	0
7	Total Net Plant		5,974,663	-	5,974,663	1,609,401	1,020,172	48,028	75,422	2,753,023
8	Working Capital									
9	Total Current Assets		1,255,789	-	1,255,789	328,411	202,683	8,897	14,110	554,101
10	Total Current Liabilities		569,465	-	569,465	152,315	93,509	4,361	6,800	256,986
11	Total Working Capital		686,324	-	686,324	176,095	109,175	4,536	7,310	297,115
	Adjustments to Rate Base									
12	Additions to Rate Base		0	-	0	0	0	0	0	0
13	Deductions from Rate Base		35,696	-	35,696	10,292	5,745	325	522	16,884
14	Total Adjustments to Rate Base		(35,696)	-	(35,696)	(10,292)	(5,745)	(325)	(522)	(16,884)
15	Total Rate Base		6,625,291	-	6,625,291	1,775,204	1,123,602	52,239	82,210	3,033,254

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Primary & Lighting RATE BASE (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(a)			-												-		Total
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Total Lighting & Unmetered
1	Net Plant																
2	Production	94,102	-	125,390	189,918	577,759	13,501	10,156	-	3,101	14,298	1,028,226	411	2,412	628	8,191	11,642
3	Transmission	(0)	-	(0)	(0)	(0)	(0)	(0)	0	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution	85,652	-	16,567	92,275	423,098	18,751	8,599	55	4,801	25,797	675,595	3,659	65,864	31,558	9,864	110,945
5	General/Common/Intangible	11,243	-	10,517	19,580	66,223	1,977	1,199	2	487	2,466	113,694	264	3,775	2,355	1,226	7,619
6	Plant Purchased/Sold	0	-	0	0	0	0	0	-	0	0	0	0	0	0	0	0
7	Total Net Plant	190,997	-	152,474	301,774	1,067,080	34,229	19,954	57	8,389	42,561	1,817,514	4,335	72,051	34,541	19,280	130,207
8	Working Capital																
9	Total Current Assets	38,477	-	37,295	67,664	224,175	6,523	5,433	7	1,580	7,880	389,033	861	13,266	7,606	4,057	25,791
10	Total Current Liabilities	16,381	-	14,512	28,234	94,431	2,891	1,835	4	749	3,579	162,616	430	8,194	4,845	1,787	15,256
11	Total Working Capital	22,096	-	22,783	39,430	129,744	3,632	3,598	3	831	4,301	226,417	431	5,072	2,762	2,270	10,535
Adjustments to Rate Base																	
12	Additions to Rate Base	0	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	Deductions from Rate Base	669	-	134	872	3,402	150	77	0	45	208	5,559	41	951	562	94	1,647
14	Total Adjustments to Rate Base	(669)	-	(134)	(872)	(3,402)	(150)	(77)	(0)	(45)	(208)	(5,559)	(41)	(951)	(562)	(94)	(1,647)
15	Total Rate Base	212,424	-	175,123	340,332	1,193,422	37,710	23,475	59	9,175	46,654	2,038,373	4,725	76,172	36,741	21,457	139,095

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Summary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description			Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Expense										
2	Fuel Expense			542,894	536,543	224,202	127,726	179,934	3,627	1,054	6,351
3	Purchased & Interchange Power Expense			1,140,307	1,128,416	520,211	261,728	339,565	5,010	1,902	11,892
4	Total Fuel and P&I Expense			1,683,201	1,664,958	744,413	389,454	519,499	8,637	2,956	18,243
5	Fossil O&M Exp			100,702	99,605	44,298	23,294	31,294	540	179	1,097
6	Nuclear O&M Exp			0	0	0	0	0	0	0	0
7	Hydro O&M Exp			22,159	21,932	10,234	5,066	6,504	92	36	227
8	Peaker O&M Exp			67,853	67,191	32,527	15,365	18,981	215	103	661
9	Other O&M			10,761	10,656	5,159	2,437	3,010	34	16	105
10	Total Production O&M Expense			201,475	199,384	92,217	46,161	59,790	881	335	2,091
11	Total Production (Inc. Fuel and P&I) O&M Expense			1,884,676	1,864,343	836,630	435,615	579,289	9,518	3,291	20,334
12	Transmission & Distribution Expense										
13	Trans O&M Exp			477,556	472,579	225,419	106,588	137,459	2,362	750	4,976
14	Other O&M Adjustments			0	0	0	0	0	0	0	0
15	Distr Oper Exp			89,212	89,198	57,042	23,144	5,522	3,474	16	14
16	Distr Maint Exp			175,919	175,880	105,801	51,727	16,992	1,301	59	39
17	Total Transmission & Distribution O&M Expense			742,687	737,658	388,262	181,459	159,974	7,138	826	5,029
18	Customer Related Expense										
19	Customer Accounts Exp			54,093	54,093	47,618	6,356	105	13	0	0
20	Customer Service Exp			16,912	16,770	8,173	3,199	5,295	80	24	142
21	Sales Expense			365	365	321	43	1	0	0	0
22	Total Customer O&M Expense			71,370	71,228	56,113	9,597	5,401	93	24	142
23	Admin & General Expense			146,719	146,154	86,151	36,143	22,147	1,607	105	565
24	Total O&M Expense			2,845,452	2,819,382	1,367,156	662,815	766,811	18,355	4,245	26,070

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Residential/Secondary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Production Expense									
2	Fuel Expense		224,202	-	224,202	67,490	55,701	1,587	2,949	127,726
3	Purchased & Interchange Power Expense		520,211	-	520,211	140,299	112,568	3,427	5,434	261,728
4	Total Fuel and P&I Expense		744,413	-	744,413	207,788	168,269	5,014	8,383	389,454
5	Fossil O&M Exp		44,298	-	44,298	12,415	10,076	299	503	23,294
6	Nuclear O&M Exp		0	-	0	0	0	0	0	0
7	Hydro O&M Exp		10,234	-	10,234	2,720	2,175	67	103	5,066
8	Peaker O&M Exp		32,527	-	32,527	8,302	6,557	207	299	15,365
9	Other O&M		5,159	-	5,159	1,317	1,040	33	47	2,437
10	Total Production O&M Expense		92,217	-	92,217	24,754	19,848	605	953	46,161
11	Total Production (Inc. Fuel and P&I) O&M Expense		836,630	-	836,630	232,542	188,117	5,619	9,336	435,615
12	Transmission & Distribution Expense									
13	Trans O&M Exp		225,419	-	225,419	57,391	45,429	1,482	2,286	106,588
14	Other O&M Adjustments		0	-	0	0	0	0	0	0
15	Distr Oper Exp		57,042	-	57,042	14,553	7,490	427	673	23,144
16	Distr Maint Exp		105,801	-	105,801	31,725	17,433	989	1,580	51,727
17	Total Transmission & Distribution O&M Expense		388,262	-	388,262	103,670	70,353	2,898	4,539	181,459
18	Customer Related Expense									
19	Customer Accounts Exp		47,618	-	47,618	5,710	576	46	23	6,356
20	Customer Service Exp		8,173	-	8,173	1,826	1,244	42	87	3,199
21	Sales Expense		321	-	321	39	4	0	0	43
22	Total Customer O&M Expense		56,113	-	56,113	7,575	1,824	88	110	9,597
23	Admin & General Expense		86,151	-	86,151	21,823	12,811	595	915	36,143
24	Total O&M Expense		1,367,156	-	1,367,156	365,609	273,106	9,200	14,900	662,815

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Primary & Lighting O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vit 1	Rate GPD GEI Vit 2	Rate GPD GEI Vit 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered	
1	Production Expense																	
2	Fuel Expense	14,287	-	24,970	35,149	94,917	2,103	5,986	-	361	2,161	179,934	230	1,348	351	1,697	3,627	
3	Purchased & Interchange Power Expense	29,478	-	43,629	64,123	186,266	4,268	6,439	-	891	4,472	339,565	252	1,476	384	2,898	5,010	
4	Total Fuel and P&I Expense	43,765	-	68,599	99,272	281,183	6,371	12,425	-	1,252	6,633	519,499	482	2,824	735	4,596	8,637	
5	Fossil O&M Exp	2,611	-	4,168	6,003	16,893	380	769	-	74	396	31,294	31	183	48	279	540	
6	Nuclear O&M Exp	0	-	0	0	0	0	0	-	0	0	0	0	0	0	0	0	
7	Hydro O&M Exp	571	-	827	1,223	3,587	82	109	-	18	87	6,504	4	26	7	55	92	
8	Peaker O&M Exp	1,737	-	2,315	3,506	10,666	249	187	-	57	264	18,981	8	45	12	151	215	
9	Other O&M	276	-	367	556	1,692	40	30	-	9	42	3,010	1	7	2	24	34	
10	Total Production O&M Expense	5,195	-	7,677	11,289	32,837	751	1,096	-	158	788	59,790	44	260	68	509	881	
11	Total Production (Inc. Fuel and P&I) O&M Expense	48,960	-	76,276	110,560	314,020	7,122	13,521	-	1,410	7,421	579,289	526	3,084	803	5,104	9,518	
12	Transmission & Distribution Expense																	
13	Trans O&M Exp	12,279	-	17,679	26,346	77,030	1,886	-	-	359	1,879	137,459	149	876	228	1,109	2,362	
14	Other O&M Adjustments	0	-	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
15	Distr Oper Exp	722	-	36	606	3,694	162	44	0	31	226	5,522	70	1,897	1,384	124	3,474	
16	Distr Maint Exp	2,031	-	144	2,294	11,080	480	164	0	119	681	16,992	87	661	267	285	1,301	
17	Total Transmission & Distribution O&M Expense	15,032	-	17,859	29,246	91,804	2,528	208	1	509	2,786	159,974	306	3,433	1,879	1,519	7,138	
18	Customer Related Expense																	
19	Customer Accounts Exp	39	-	1	4	55	4	0	0	0	2	105	6	-	-	7	13	
20	Customer Service Exp	332	-	975	1,251	2,406	56	134	1	34	105	5,295	6	30	8	37	80	
21	Sales Expense	0	-	0	0	0	0	0	0	0	0	1	0	-	-	0	0	
22	Total Customer O&M Expense	371	-	976	1,255	2,462	60	135	1	34	107	5,401	12	30	8	44	93	
23	Admin & General Expense	2,202	-	2,084	3,724	12,957	384	228	0	88	478	22,147	55	801	504	248	1,607	
24	Total O&M Expense	66,565	-	97,195	144,786	421,243	10,095	14,092	2	2,042	10,792	766,811	899	7,348	3,194	6,914	18,355	

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Summary ALLOCATORS 1										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	100.000	98.830	41.228	23.471	33.244	0.690	0.196	1.170
2	Energy On-Peak @ Generation	101	100.000	98.830	39.479	24.827	33.893	0.454	0.177	1.170
3	Energy Off-Peak @ Generation	102	100.000	98.830	43.220	21.927	32.506	0.959	0.218	1.170
4	Energy On-Peak @ Generation Summer	103	100.000	98.830	40.357	24.808	33.142	0.352	0.171	1.170
5	Energy Off-Peak @ Generation Summer	104	100.000	98.830	42.773	22.113	32.864	0.861	0.218	1.170
6	Energy On-Peak @ Generation Non-Summer	105	100.000	98.830	38.977	24.838	34.323	0.512	0.181	1.170
7	Energy Off-Peak @ Generation Non-Summer	106	100.000	98.830	43.463	21.826	32.311	1.012	0.217	1.170
8	Energy Critical On-Peak @ Gen	107	100.000	98.830	43.091	24.281	31.037	0.262	0.159	1.170
9	Energy Summer Mid-Peak @ Gen	108	100.000	98.830	39.455	24.982	33.837	0.382	0.175	1.170
10	12CP Dmd @ Generation	120	100.000	98.958	47.203	22.320	28.784	0.495	0.157	1.042
11	4CP Dmd @ Generation	121	100.000	99.025	50.143	22.354	26.199	0.192	0.137	0.975
12	Class Peak @ Subtransmission	122	100.000	99.942	48.167	24.049	27.015	0.543	0.167	0.058
13	Classpeak @ Transmission	127	100.000	99.117	44.989	22.462	31.002	0.507	0.156	0.883
14	Total Rate Revenue	143	100.000	99.407	48.269	24.924	25.145	1.028	0.041	0.593
15	Billed Sales	150	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
16	Billed Sales Excluding Rate E1	151	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
17	Number Of Customers	160	100.000	100.000	87.988	11.744	0.223	0.045	0.001	0.000
18	Weighted Customer	170	100.000	99.998	77.379	16.151	6.429	0.022	0.018	0.002
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	100.000	99.022	49.823	22.129	26.522	0.413	0.135	0.978
20	4CP 75/0/25	220	100.000	99.025	47.938	22.644	27.974	0.317	0.152	0.975
21	4CP 75/0/25 Exc WFR	222	100.000	99.963	48.389	22.859	28.241	0.320	0.154	0.037
22	4CP Dmd @ Gen Jurisdictional	224	100.000	100.000	50.636	22.574	26.457	0.194	0.139	-
23	12CP Demand @ Subtrans	226	100.000	100.000	49.608	23.457	26.360	0.520	0.056	-
24	Classpeak @ Primary	230	100.000	100.000	52.189	26.057	21.162	0.588	0.004	-
25	Classpeak @ Secondary	231	100.000	100.000	66.201	33.053	-	0.746	-	-
26	Classpeak for Streetlighting	233	100.000	99.188	-	-	-	99.188	-	0.812
27	Classpeak @ Single Phase	235	100.000	100.000	66.201	33.053	-	0.746	-	-
28	Billed Sales ROA	252	100.000	100.000	-	5.554	94.446	-	-	-
29	Billed Sales - Primary	253	100.000	100.000	44.960	26.337	27.945	0.753	0.005	-
30	Customers - Residential	260	100.000	100.000	100.000	-	-	-	-	-
31	Customers - Drops	261	100.000	100.000	-	100.000	-	-	-	-
32	Customers - NonPID	263	100.000	100.000	88.208	11.773	-	0.019	-	-
33	Customers - NonMunicipal	264	100.000	100.000	88.027	11.749	0.224	-	0.001	0.000
34	PIS - 138kV Distribution	301	100.000	99.117	44.989	22.462	31.002	0.507	0.156	0.883
35	PIS - 46kV Distribution	302	100.000	99.942	48.167	24.049	27.015	0.543	0.167	0.058
36	PIS - 138kV Dist Subs S&E	303	100.000	99.117	44.989	22.462	31.002	0.507	0.156	0.883
37	PIS - 46kV Dist Subs S&E	304	100.000	99.942	48.167	24.049	27.015	0.543	0.167	0.058

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Residential/Secondary ALLOCATORS 1										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	
Line No.	Description	Alloc	Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	41.228	-	41.228	12.394	10.246	0.290	0.541	23.471
2	Energy On-Peak @ Generation	101	39.479	-	39.479	13.321	10.566	0.343	0.597	24.827
3	Energy Off-Peak @ Generation	102	43.220	-	43.220	11.339	9.881	0.231	0.477	21.927
4	Energy On-Peak @ Generation Summer	103	40.357	-	40.357	13.304	10.706	0.309	0.489	24.808
5	Energy Off-Peak @ Generation Summer	104	42.773	-	42.773	11.251	10.205	0.211	0.446	22.113
6	Energy On-Peak @ Generation Non-Summer	105	38.977	-	38.977	13.331	10.486	0.362	0.660	24.838
7	Energy Off-Peak @ Generation Non-Summer	106	43.463	-	43.463	11.386	9.705	0.242	0.494	21.826
8	Energy Critical On-Peak @ Gen	107	43.091	-	43.091	12.929	10.603	0.284	0.465	24.281
9	Energy Summer Mid-Peak @ Gen	108	39.455	-	39.455	13.428	10.740	0.318	0.497	24.982
10	12CP Dmd @ Generation	120	47.203	-	47.203	12.018	9.513	0.310	0.479	22.320
11	4CP Dmd @ Generation	121	50.143	-	50.143	12.174	9.463	0.310	0.407	22.354
12	Class Peak @ Subtransmission	122	48.167	-	48.167	13.440	9.232	0.516	0.861	24.049
13	Classpeak @ Transmission	127	44.989	-	44.989	12.554	8.623	0.482	0.804	22.462
14	Total Rate Revenue	143	48.269	-	48.269	14.002	9.986	0.324	0.612	24.924
15	Billed Sales	150	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
16	Billed Sales Excluding Rate E1	151	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
17	Number Of Customers	160	87.988	-	87.988	10.551	1.064	0.086	0.042	11.744
18	Weighted Customer	170	77.379	-	77.379	13.902	2.054	0.113	0.082	16.151
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	49.823	-	49.823	12.058	9.363	0.307	0.401	22.129
20	4CP 75/0/25	220	47.938	-	47.938	12.235	9.663	0.305	0.441	22.644
21	4CP 75/0/25 Exc WFR	222	48.389	-	48.389	12.351	9.755	0.308	0.445	22.859
22	4CP Dmd @ Gen Jurisdictional	224	50.636	-	50.636	12.294	9.556	0.313	0.411	22.574
23	12CP Demand @ Subtrans	226	49.608	-	49.608	12.630	9.998	0.326	0.503	23.457
24	Classpeak @ Primary	230	52.189	-	52.189	14.563	10.003	0.559	0.933	26.057
25	Classpeak @ Secondary	231	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-
27	Classpeak @ Single Phase	235	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
28	Billed Sales ROA	252	-	-	-	0.265	3.159	0.397	1.732	5.554
29	Billed Sales - Primary	253	44.960	-	44.960	13.551	11.595	0.370	0.821	26.337
30	Customers - Residential	260	100.000	-	100.000	-	-	-	-	-
31	Customers - Drops	261	-	-	-	89.846	9.063	0.730	0.361	100.000
32	Customers - NonPID	263	88.208	-	88.208	10.578	1.067	0.086	0.042	11.773
33	Customers - NonMunicipal	264	88.027	-	88.027	10.556	1.065	0.086	0.042	11.749
34	PIS - 138kV Distribution	301	44.989	-	44.989	12.554	8.623	0.482	0.804	22.462
35	PIS - 46kV Distribution	302	48.167	-	48.167	13.440	9.232	0.516	0.861	24.049
36	PIS - 138kV Dist Subs S&E	303	44.989	-	44.989	12.554	8.623	0.482	0.804	22.462
37	PIS - 46kV Dist Subs S&E	304	48.167	-	48.167	13.440	9.232	0.516	0.861	24.049

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Primary & Lighting ALLOCATORS 1		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Alloc	Rate GP	Rate GPTU	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vit 1	Rate GPD GEI Vit 2	Rate GPD GEI Vit 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered
<u>Input Allocation Schedules</u>																		
1	Energy @ Generation	100	2.616	-	4.647	6.517	17.513	0.384	1.105	-	0.066	0.396	33.244	0.045	0.262	0.068	0.315	0.690
2	Energy On-Peak @ Generation	101	3.026	-	4.464	6.356	17.885	0.467	1.200	-	0.068	0.426	33.893	0.020	0.120	0.031	0.282	0.454
3	Energy Off-Peak @ Generation	102	2.150	-	4.855	6.700	17.090	0.288	0.996	-	0.065	0.362	32.506	0.072	0.425	0.111	0.352	0.959
4	Energy On-Peak @ Generation Summer	103	3.018	-	4.220	6.062	17.816	0.413	1.121	-	0.077	0.415	33.142	0.011	0.064	0.017	0.260	0.352
5	Energy Off-Peak @ Generation Summer	104	2.200	-	4.780	6.628	17.589	0.262	0.965	-	0.080	0.360	32.864	0.063	0.370	0.096	0.332	0.861
6	Energy On-Peak @ Generation Non-Summer	105	3.030	-	4.603	6.525	17.925	0.498	1.246	-	0.062	0.433	34.323	0.026	0.152	0.040	0.295	0.512
7	Energy Off-Peak @ Generation Non-Summer	106	2.123	-	4.895	6.739	16.819	0.303	1.012	-	0.057	0.363	32.311	0.077	0.454	0.118	0.363	1.012
8	Energy Critical On-Peak @ Gen	107	2.839	-	3.711	5.600	17.020	0.385	0.995	-	0.076	0.412	31.037	-	-	-	0.262	0.262
9	Energy Summer Mid-Peak @ Gen	108	3.078	-	4.388	6.214	18.078	0.423	1.163	-	0.077	0.416	33.837	0.015	0.086	0.022	0.259	0.382
10	12CP Dmd @ Generation	120	2.571	-	3.702	5.517	16.130	0.395	-	-	0.075	0.394	28.784	0.031	0.183	0.048	0.232	0.495
11	4CP Dmd @ Generation	121	2.540	-	2.997	4.714	15.110	0.362	-	-	0.090	0.386	26.199	-	-	-	0.192	0.192
12	Class Peak @ Subtransmission	122	2.676	-	-	6.738	15.231	0.659	0.411	-	0.352	0.948	27.015	0.046	0.269	0.070	0.158	0.543
13	Classpeak @ Transmission	127	2.500	-	4.943	6.293	14.226	0.615	1.197	0.014	0.329	0.885	31.002	0.043	0.251	0.065	0.148	0.507
14	Total Rate Revenue	143	2.543	-	2.624	4.395	14.152	0.373	0.583	0.000	0.081	0.393	25.145	0.039	0.572	0.193	0.224	1.028
15	Billed Sales	150	2.542	-	7.545	9.682	18.590	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
16	Billed Sales Excluding Rate E1	151	2.542	-	7.545	9.682	18.590	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
17	Number Of Customers	160	0.082	-	0.003	0.009	0.116	0.008	0.001	0.000	0.000	0.005	0.223	0.019	-	-	0.026	0.045
18	Weighted Customer	170	2.285	-	0.078	0.255	3.420	0.215	0.030	0.002	0.010	0.134	6.429	0.022	-	-	-	0.022
<u>Calculated Allocation Schedules</u>																		
19	4CP Average & Excess	219	2.515	-	2.943	4.641	14.932	0.358	0.660	-	0.090	0.383	26.522	0.027	0.157	0.041	0.188	0.413
20	4CP 75/0/25	220	2.560	-	3.411	5.167	15.719	0.367	0.276	-	0.084	0.389	27.974	0.011	0.066	0.017	0.223	0.317
21	4CP 75/0/25 Exc WFR	222	2.584	-	3.444	5.216	15.868	0.371	0.279	-	0.085	0.393	28.241	0.011	0.066	0.017	0.225	0.320
22	4CP Dmd @ Gen Jurisdictional	224	2.565	-	3.027	4.760	15.259	0.365	-	-	0.091	0.390	26.457	-	-	-	0.194	0.194
23	12CP Demand @ Subtrans	226	2.702	-	-	5.798	16.952	0.415	-	-	0.079	0.414	26.360	0.033	0.193	0.050	0.244	0.520
24	Classpeak @ Primary	230	2.900	-	-	-	16.502	0.714	0.019	-	-	1.027	21.162	0.050	0.291	0.076	0.171	0.588
25	Classpeak @ Secondary	231	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-	-	-	-	11.822	69.314	18.052	-	99.188
27	Classpeak @ Single Phase	235	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
28	Billed Sales ROA	252	1.261	-	29.466	33.732	23.121	0.750	-	0.073	1.899	4.145	94.446	-	-	-	-	-
29	Billed Sales - Primary	253	3.169	-	-	-	23.173	0.540	0.055	-	-	1.008	27.945	0.049	0.286	0.075	0.343	0.753
30	Customers - Residential	260	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
31	Customers - Drops	261	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	Customers - NonPID	263	-	-	-	-	-	-	-	-	-	-	-	0.019	-	-	-	0.019
33	Customers - NonMunicipal	264	0.082	-	0.003	0.009	0.117	0.008	0.001	0.000	0.000	0.005	0.224	-	-	-	-	-
34	PIS - 138kV Distribution	301	2.500	-	4.943	6.293	14.226	0.615	1.197	0.014	0.329	0.885	31.002	0.043	0.251	0.065	0.148	0.507
35	PIS - 46kV Distribution	302	2.676	-	-	6.738	15.231	0.659	0.411	-	0.352	0.948	27.015	0.046	0.269	0.070	0.158	0.543
36	PIS - 138kV Dist Subs S&E	303	2.500	-	4.943	6.293	14.226	0.615	1.197	0.014	0.329	0.885	31.002	0.043	0.251	0.065	0.148	0.507
37	PIS - 46kV Dist Subs S&E	304	2.676	-	-	6.738	15.231	0.659	0.411	-	0.352	0.948	27.015	0.046	0.269	0.070	0.158	0.543

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Summary ALLOCATORS 2										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
No.										
	<u>Calculated Allocation Schedules</u>									
1	PIS - Overhead Primary System	305	100.000	100.000	66.201	33.053	-	0.746	-	(0.000)
2	PIS - Distribution Distribution	306	100.000	99.998	63.000	29.455	4.177	3.364	0.002	0.002
3	Overhead Distribution	307	100.000	99.986	61.383	30.696	7.324	0.561	0.022	0.014
4	Underground Distribution	308	100.000	100.000	65.070	32.488	1.709	0.734	0.000	(0.000)
5	Total Dist PIS	309	100.000	99.949	59.654	28.216	9.289	2.753	0.038	0.051
6	Distribution Services	310	100.000	100.000	66.512	33.488	-	-	-	-
7	Streetlighting Equipment	311	100.000	99.928	-	-	-	99.928	-	0.072
8	Line Equipment	312	100.000	100.000	64.412	32.160	2.702	0.726	0.001	-
9	Meters	313	100.000	99.998	77.379	16.151	6.429	0.022	0.018	0.002
10	PIS - System Power Control	314	100.000	199.059	93.156	46.511	58.018	1.050	0.323	(99.059)
11	PIS - General	315	100.000	99.604	58.198	24.524	15.751	1.056	0.076	0.396
12	Total PIS	316	100.000	99.584	55.294	25.861	16.631	1.715	0.082	0.416
13	Distribution Depreciation	317	100.000	99.961	59.005	29.525	6.640	4.754	0.037	0.039
14	CWIP	330	100.000	99.512	53.695	24.970	19.463	1.283	0.102	0.488
15	Working Capital	340	100.000	99.527	55.931	24.213	18.452	0.859	0.072	0.473
16	Rate Base	390	100.000	99.596	55.705	25.504	17.139	1.170	0.079	0.404
17	Operations - Distribution excl Sup & Eng	400	100.000	99.988	64.397	26.001	5.583	3.993	0.014	0.012
18	Maintenance - Distribution excl Sup & Eng	401	100.000	99.988	61.510	30.031	7.665	0.764	0.019	0.012
19	Operations - 138kV Distribution excl Sup & Eng	402	100.000	99.025	50.143	22.354	26.199	0.192	0.137	0.975
20	Maintenance - 138kV Distribution excl Sup & Eng	403	100.000	99.456	47.155	22.505	29.259	0.387	0.150	0.544
21	Operations - 46kV Distribution excl Sup & Eng	404	100.000	99.942	48.167	24.049	27.015	0.543	0.167	0.058
22	Maintenance - 46kV Distribution excl Sup & Eng	405	100.000	99.942	48.167	24.049	27.015	0.543	0.167	0.058
23	HV Distribution O&M exp.	406	100.000	99.884	48.077	23.875	27.242	0.524	0.165	0.116
24	Distribution O&M, excl. HV Dist	407	100.000	99.989	62.558	28.607	6.907	1.900	0.017	0.011
25	Customer Accounting	408	100.000	100.000	88.031	11.749	0.195	0.024	0.001	0.000
26	Customer Accounts & Service	409	100.000	99.800	78.574	13.456	7.605	0.131	0.034	0.200
27	Distribution O&M	410	100.000	99.980	61.420	28.239	8.492	1.801	0.028	0.020
28	Customer & Sales O&M	411	100.000	99.801	78.622	13.447	7.567	0.130	0.034	0.199
29	Administrative and General O&M	412	100.000	99.615	58.719	24.635	15.095	1.095	0.071	0.385
30	Jurisdictional Distribution O&M	414	100.000	100.000	61.432	28.245	8.494	1.801	0.028	-
31	O&M Excluding Adjustments	438	100.000	99.084	48.047	23.294	26.949	0.645	0.149	0.916
32	Pre Tax NOI	439	100.000	100.832	38.342	31.249	29.338	1.763	0.140	(0.832)
33	R&PP Tax	440	100.000	99.602	55.222	25.838	16.771	1.685	0.084	0.398
34	Depreciation & Amortization Expense	442	100.000	99.493	55.446	25.157	17.232	1.567	0.091	0.507
35	Non PSCR O&M Expense	443	100.000	99.584	58.029	24.357	16.044	1.074	0.079	0.416
36	Distribution Depreciation Expense	444	100.000	99.959	62.546	28.152	6.095	3.135	0.031	0.041
37	Gen/Comm/Int Depreciation Expense	445	100.000	99.604	58.198	24.524	15.751	1.056	0.076	0.396
38	Production Labor	500	100.000	99.025	47.938	22.644	27.974	0.317	0.152	0.975
39	Total Labor	502	100.000	99.604	58.198	24.524	15.751	1.056	0.076	0.396
40	50% O&M, 50% Net Plant	600	100.000	99.497	54.097	25.169	19.042	1.095	0.094	0.503
41	50/50 PIS & Labor	601	100.000	99.594	56.746	25.193	16.191	1.385	0.079	0.406

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Residential/Secondary ALLOCATORS 2										
	(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Alloc	Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Calculated Allocation Schedules</u>										
1	PIS - Overhead Primary System	305	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
2	PIS - Distribution Distribution	306	63.000	-	63.000	18.287	9.737	0.553	0.878	29.455
3	Overhead Distribution	307	61.383	-	61.383	19.142	10.068	0.575	0.911	30.696
4	Underground Distribution	308	65.070	-	65.070	18.157	12.471	0.696	1.163	32.488
5	Total Dist PIS	309	59.654	-	59.654	17.200	9.601	0.543	0.872	28.216
6	Distribution Services	310	66.512	-	66.512	30.087	3.035	0.245	0.121	33.488
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-
8	Line Equipment	312	64.412	-	64.412	17.973	12.345	0.689	1.151	32.160
9	Meters	313	77.379	-	77.379	13.902	2.054	0.113	0.082	16.151
10	PIS - System Power Control	314	93.156	-	93.156	25.994	17.855	0.997	1.665	46.511
11	PIS - General	315	58.198	-	58.198	14.745	8.753	0.404	0.622	24.524
12	Total PIS	316	55.294	-	55.294	15.180	9.544	0.444	0.693	25.861
13	Distribution Depreciation	317	59.005	-	59.005	18.295	9.784	0.558	0.887	29.525
14	CWIP	330	53.695	-	53.695	14.437	9.468	0.418	0.648	24.970
15	Working Capital	340	55.931	-	55.931	14.351	8.897	0.370	0.596	24.213
16	Rate Base	390	55.705	-	55.705	14.926	9.447	0.439	0.691	25.504
17	Operations - Distribution excl Sup & Eng	400	64.397	-	64.397	16.399	8.371	0.478	0.752	26.001
18	Maintenance - Distribution excl Sup & Eng	401	61.510	-	61.510	18.567	9.990	0.569	0.905	30.031
19	Operations - 138kV Distribution excl Sup & Eng	402	50.143	-	50.143	12.174	9.463	0.310	0.407	22.354
20	Maintenance - 138kV Distribution excl Sup & Eng	403	47.155	-	47.155	12.454	8.981	0.417	0.653	22.505
21	Operations - 46kV Distribution excl Sup & Eng	404	48.167	-	48.167	13.440	9.232	0.516	0.861	24.049
22	Maintenance - 46kV Distribution excl Sup & Eng	405	48.167	-	48.167	13.440	9.232	0.516	0.861	24.049
23	HV Distribution O&M exp.	406	48.077	-	48.077	13.328	9.208	0.504	0.836	23.875
24	Distribution O&M, excl. HV Dist	407	62.558	-	62.558	17.804	9.415	0.537	0.851	28.607
25	Customer Accounting	408	88.031	-	88.031	10.556	1.065	0.086	0.042	11.749
26	Customer Accounts & Service	409	78.574	-	78.574	10.613	2.564	0.124	0.155	13.456
27	Distribution O&M	410	61.420	-	61.420	17.455	9.400	0.534	0.850	28.239
28	Customer & Sales O&M	411	78.622	-	78.622	10.613	2.556	0.124	0.154	13.447
29	Administrative and General O&M	412	58.719	-	58.719	14.874	8.732	0.405	0.623	24.635
30	Jurisdictional Distribution O&M	414	61.432	-	61.432	17.458	9.402	0.534	0.850	28.245
31	O&M Excluding Adjustments	438	48.047	-	48.047	12.849	9.598	0.323	0.524	23.294
32	Pre Tax NOI	439	38.342	-	38.342	17.533	12.586	0.188	0.943	31.249
33	R&PP Tax	440	55.222	-	55.222	15.148	9.553	0.444	0.693	25.838
34	Depreciation & Amortization Expense	442	55.446	-	55.446	14.821	9.308	0.407	0.622	25.157
35	Non PSCR O&M Expense	443	58.029	-	58.029	14.668	8.676	0.395	0.618	24.357
36	Distribution Depreciation Expense	444	62.546	-	62.546	17.673	9.141	0.519	0.819	28.152
37	Gen/Comm/Int Depreciation Expense	445	58.198	-	58.198	14.745	8.753	0.404	0.622	24.524
38	Production Labor	500	47.938	-	47.938	12.235	9.663	0.305	0.441	22.644
39	Total Labor	502	58.198	-	58.198	14.745	8.753	0.404	0.622	24.524
40	50% O&M, 50% Net Plant	600	54.097	-	54.097	14.552	9.529	0.422	0.666	25.169
41	50/50 PIS & Labor	601	56.746	-	56.746	14.963	9.149	0.424	0.657	25.193

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Primary & Lighting ALLOCATORS 2		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Alloc	Rate GP	Rate GPTU	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vit 1	Rate GPD GEI Vit 2	Rate GPD GEI Vit 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered
<u>Calculated Allocation Schedules</u>																		
1	PIS - Overhead Primary System	305	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
2	PIS - Distribution Distribution	306	0.699	-	0.007	0.027	3.110	0.141	0.006	0.000	0.001	0.186	4.177	0.074	1.954	1.178	0.157	3.364
3	Overhead Distribution	307	0.868	-	0.039	0.848	4.941	0.214	0.062	0.000	0.044	0.307	7.324	0.047	0.278	0.072	0.163	0.561
4	Underground Distribution	308	0.234	-	-	-	1.332	0.058	0.002	-	-	0.083	1.709	0.062	0.363	0.095	0.214	0.734
5	Total Dist PIS	309	1.118	-	0.224	1.458	5.686	0.251	0.128	0.001	0.076	0.348	9.289	0.068	1.589	0.939	0.157	2.753
6	Distribution Services	310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-	-	-	-	1.046	55.182	43.700	-	99.928
8	Line Equipment	312	0.370	-	-	-	2.107	0.091	0.002	-	-	0.131	2.702	0.061	0.360	0.094	0.212	0.726
9	Meters	313	2.285	-	0.078	0.255	3.420	0.215	0.030	0.002	0.010	0.134	6.429	0.022	-	-	-	0.022
10	PIS - System Power Control	314	5.176	-	4.943	13.031	29.457	1.274	1.609	0.014	0.682	1.833	58.018	0.089	0.520	0.135	0.306	1.050
11	PIS - General	315	1.558	-	1.457	2.713	9.174	0.274	0.166	0.000	0.067	0.342	15.751	0.037	0.523	0.326	0.170	1.056
12	Total PIS	316	1.679	-	1.488	2.913	9.630	0.295	0.185	0.000	0.078	0.362	16.631	0.045	0.940	0.549	0.182	1.715
13	Distribution Depreciation	317	0.657	-	0.140	1.504	3.749	0.162	0.115	0.000	0.079	0.233	6.640	0.088	2.686	1.821	0.159	4.754
14	CWIP	330	1.914	-	1.691	3.614	11.201	0.333	0.205	0.000	0.098	0.407	19.463	0.037	0.673	0.385	0.187	1.283
15	Working Capital	340	1.801	-	1.857	3.213	10.573	0.296	0.293	0.000	0.068	0.351	18.452	0.035	0.413	0.225	0.185	0.859
16	Rate Base	390	1.786	-	1.472	2.862	10.034	0.317	0.197	0.000	0.077	0.392	17.139	0.040	0.640	0.309	0.180	1.170
17	Operations - Distribution excl Sup & Eng	400	0.755	-	0.036	0.506	3.817	0.169	0.040	0.000	0.026	0.234	5.583	0.080	2.181	1.594	0.138	3.993
18	Maintenance - Distribution excl Sup & Eng	401	0.984	-	0.034	0.703	5.296	0.231	0.053	0.000	0.037	0.327	7.665	0.050	0.390	0.162	0.163	0.764
19	Operations - 138kV Distribution excl Sup & Eng	402	2.540	-	2.997	4.714	15.110	0.362	0.000	0.000	0.090	0.386	26.199	0.000	0.000	0.000	0.192	0.192
20	Maintenance - 138kV Distribution excl Sup & Eng	403	2.525	-	4.208	5.705	14.622	0.519	0.738	0.008	0.238	0.695	29.259	0.026	0.155	0.040	0.165	0.387
21	Operations - 46kV Distribution excl Sup & Eng	404	2.676	-	-	6.738	15.231	0.659	0.411	-	0.352	0.948	27.015	0.046	0.269	0.070	0.158	0.543
22	Maintenance - 46kV Distribution excl Sup & Eng	405	2.676	-	-	6.738	15.231	0.659	0.411	-	0.352	0.948	27.015	0.046	0.269	0.070	0.158	0.543
23	HV Distribution O&M exp.	406	2.659	-	0.461	6.615	15.167	0.642	0.442	0.001	0.338	0.917	27.242	0.043	0.255	0.066	0.159	0.524
24	Distribution O&M, excl. HV Dist	407	0.901	-	0.033	0.626	4.764	0.209	0.048	0.000	0.033	0.293	6.907	0.060	1.019	0.667	0.154	1.900
25	Customer Accounting	408	0.071	-	0.002	0.008	0.101	0.007	0.001	0.000	0.000	0.004	0.195	0.011	-	-	0.013	0.024
26	Customer Accounts & Service	409	0.522	-	1.375	1.768	3.466	0.084	0.190	0.001	0.049	0.150	7.605	0.017	0.042	0.011	0.061	0.131
27	Distribution O&M	410	1.038	-	0.068	1.094	5.572	0.242	0.079	0.000	0.056	0.342	8.492	0.059	0.965	0.623	0.154	1.801
28	Customer & Sales O&M	411	0.519	-	1.368	1.759	3.449	0.084	0.189	0.001	0.048	0.150	7.567	0.017	0.042	0.011	0.061	0.130
29	Administrative and General O&M	412	1.501	-	1.421	2.538	8.832	0.262	0.155	0.000	0.060	0.326	15.095	0.037	0.546	0.344	0.169	1.095
30	Jurisdictional Distribution O&M	414	1.039	-	0.068	1.094	5.573	0.242	0.079	0.000	0.057	0.342	8.494	0.059	0.965	0.623	0.154	1.801
31	O&M Excluding Adjustments	438	2.339	-	3.416	5.088	14.804	0.355	0.495	0.000	0.072	0.379	26.949	0.032	0.258	0.112	0.243	0.645
32	Pre Tax NOI	439	4.674	-	0.817	3.190	17.880	0.597	1.491	0.002	0.129	0.559	29.338	0.064	1.188	0.315	0.195	1.763
33	R&PP Tax	440	1.704	-	1.376	2.989	9.779	0.301	0.170	0.000	0.082	0.371	16.771	0.044	0.922	0.537	0.182	1.685
34	Depreciation & Amortization Expense	442	1.704	-	1.804	3.171	9.718	0.267	0.189	0.000	0.072	0.305	17.232	0.039	0.839	0.505	0.183	1.567
35	Non PSCR O&M Expense	443	1.537	-	1.594	2.799	9.206	0.268	0.243	0.000	0.063	0.333	16.044	0.039	0.533	0.326	0.177	1.074
36	Distribution Depreciation Expense	444	0.827	-	0.188	1.175	3.384	0.156	0.103	0.001	0.061	0.200	6.095	0.070	1.811	1.108	0.146	3.135
37	Gen/Comm/Int Depreciation Expense	445	1.558	-	1.457	2.713	9.174	0.274	0.166	0.000	0.067	0.342	15.751	0.037	0.523	0.326	0.170	1.056
38	Production Labor	500	2.560	-	3.411	5.167	15.719	0.367	0.276	-	0.084	0.389	27.974	0.011	0.066	0.017	0.223	0.317
39	Total Labor	502	1.558	-	1.457	2.713	9.174	0.274	0.166	0.000	0.067	0.342	15.751	0.037	0.523	0.326	0.170	1.056
40	50% O&M, 50% Net Plant	600	1.898	-	1.840	3.290	10.966	0.327	0.251	0.000	0.077	0.393	19.042	0.039	0.585	0.278	0.193	1.095
41	50/50 PIS & Labor	601	1.618	-	1.473	2.813	9.402	0.284	0.176	0.000	0.073	0.352	16.191	0.041	0.731	0.438	0.176	1.385

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NET PLANT (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Plant in Service									
2	Production		6,150,396	6,090,443	2,948,356	1,392,715	1,720,531	19,481	9,360	59,953
3	Transmission		0	0	0	0	0	0	0	0
4	Distribution		9,310,561	9,305,855	5,554,089	2,627,099	864,889	256,280	3,499	4,706
5	General/Common/Intangible		1,603,974	1,597,628	933,481	393,360	252,641	16,931	1,216	6,346
6	Plant Purchased/Sold		0	0	0	0	0	0	0	0
7	Total Plant in Service		17,064,931	16,993,926	9,435,926	4,413,174	2,838,061	292,691	14,075	71,005
8	Depreciation Reserve									
9	Production		2,626,131	2,600,532	1,258,906	594,669	734,642	8,318	3,997	25,599
10	Transmission		0	0	0	0	0	0	0	0
11	Distribution		3,130,436	3,129,204	1,847,104	924,248	207,862	148,819	1,171	1,232
12	General/Common/Intangible		942,168	938,441	548,323	231,058	148,400	9,945	714	3,727
13	Total Depreciation Reserve		6,698,735	6,668,176	3,654,333	1,749,975	1,090,904	167,082	5,882	30,558
14	Construction Work In Progress (CWIP)									
15	Production		151,340	149,865	72,549	34,270	42,336	479	230	1,475
16	Transmission		(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
17	Distribution		146,083	146,051	84,448	40,264	17,779	3,472	88	32
18	General/Common/Intangible		60,016	59,778	34,928	14,718	9,453	634	46	237
19	Total CWIP		357,439	355,694	191,925	89,253	69,568	4,585	364	1,744
20	Future Use									
21	Production	220	0	0	0	0	0	0	0	0
22	Distribution	127	2,545	2,523	1,145	572	789	13	4	22
23	Common/General	231	0	0	0	0	-	0	-	-
24	PHFFU Depreciation Reserve	127	0	0	0	0	0	0	0	0
25	Total Future Use		2,545	2,523	1,145	572	789	13	4	22
26	Total Net Plant		10,726,180	10,683,967	5,974,663	2,753,023	1,817,514	130,207	8,560	42,213

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PLANT IN SERVICE (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Plant in Service									
2	Production Plant in Service		6,150,396	6,090,443	2,948,356	1,392,715	1,720,531	19,481	9,360	59,953
3	Generation Step Ups		0	0	0	0	0	0	0	0
4	Total Production		6,150,396	6,090,443	2,948,356	1,392,715	1,720,531	19,481	9,360	59,953
5	Transmission Plant in Service									
6	Bulk Power Transm		0	0	0	0	0	0	0	0
7	Transm; Subtrans		0	0	0	0	0	0	0	0
8	Subtransmission		0	0	0	0	0	0	0	0
9	Total Transmission		0	0	0	0	0	0	0	0
10	Distribution Plant in Service									
11	Stations and Equipment		2,517,327	2,513,478	1,376,457	687,254	432,052	15,477	2,237	3,850
12	Overhead System		4,169,223	4,168,522	2,513,927	1,255,155	369,993	28,344	1,103	700
13	Underground System		875,240	875,228	565,877	282,531	20,401	6,380	39	13
14	Meters and Svc Drops		1,563,604	1,563,593	1,097,829	402,157	42,442	21,045	120	11
15	St Lgts and OPL		185,167	185,034	-	-	-	185,034	-	133
16	Total Distribution		9,310,561	9,305,855	5,554,089	2,627,099	864,889	256,280	3,499	4,706
17	General/Common/Intangible Plant in Service									
18	Total Gen/Comm/Int Plant		1,603,974	1,597,628	933,481	393,360	252,641	16,931	1,216	6,346
19	Plant Purchased/Sold		0	0	0	0	0	0	0	0
20	Total Plant in Service		17,064,931	16,993,926	9,435,926	4,413,174	2,838,061	292,691	14,075	71,005

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PLANT IN SERVICE (PRODUCTION & TRANSMISSION)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Production Plant in Service										
2	Fossil	220	4,655,905	4,610,520	2,231,932	1,054,298	1,302,458	14,747	7,086	45,385	
3	Nuclear	220	0	0	0	0	0	0	0	0	
4	Total Hydro	220	707,453	700,557	339,136	160,198	197,905	2,241	1,077	6,896	
5	Other Production/Combustion Turbine	220	787,038	779,366	377,288	178,219	220,169	2,493	1,198	7,672	
6	7 Classics	220	0	0	0	0	0	0	0	0	
7	Jackson Gas Plant	220	0	0	0	0	0	0	0	0	
6	Distribution GSUs	220	0	0	0	0	0	0	0	0	
7	Total Production Plant in Service		6,150,396	6,090,443	2,948,356	1,392,715	1,720,531	19,481	9,360	59,953	
8	Transmission Plant in Service										
9	Transmission Direct	228	0	0	0	0	0	0	0	0	
10	Transmission	228	0	0	0	0	0	0	0	0	
11	XYZ	127	0	0	0	0	0	0	0	0	
12	Total Transmission Plant in Service		0	0	0	0	0	0	0	0	

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PLANT IN SERVICE (DISTRIBUTION)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
Distribution Plant in Service										
1	Distribution Land & ROW (360)									
2	345/138kV Substations/Overheads (METC) Land & ROW	127	0	0	0	0	0	0	0	0
3	345/138kV Substations/Overheads Land & ROW	127	23,955	23,743	10,777	5,381	7,427	122	37	211
4	46/23kV Substations/Overheads Land & ROW	124	60,975	60,939	29,370	14,664	16,472	331	102	36
5	Distribution Substation Land & ROW	230	2,027	2,027	1,058	528	429	12	0	-
6	138kV Substations/Overheads (Assignable) Land & ROW	DIR	0	-	-	-	-	-	-	-
7	46kV Substations/Overheads (Assignable) Land & ROW	DIR	0	-	-	-	-	-	-	-
8	Overhead Lines Land & ROW	307	32,331	32,326	19,845	9,924	2,368	181	7	4
9	Total Distribution Land & ROW		119,287	119,036	61,050	30,497	26,696	646	147	252
10	Distribution Substations & Equipment (361/362/363)									
11	138kV Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
12	46kV Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
13	138kV HV Subtran/Dist Substations	127	346,679	343,619	155,968	77,872	107,479	1,759	542	3,060
14	138kV HV Subtran/Dist Substations-Proposed Trans Reclass	127	0	0	0	0	0	0	0	0
15	46kV Subtran/Dist Substations	124	920,673	920,135	443,462	221,412	248,722	5,000	1,540	538
16	Distribution Substations	230	100,810	100,810	52,611	26,268	21,333	593	4	-
17	Total Distribution Substations & Equipment		1,368,162	1,364,564	652,042	325,552	377,534	7,352	2,085	3,598
18	Distribution Overhead System (364/365)									
19	138kV HV Subtran/Dist Overhead Lines	127	39,694	39,344	17,858	8,916	12,306	201	62	350
20	138kV HV Subtran/Dist Overhead Lines-Proposed Trans Reclass	121	0	0	0	0	0	0	0	0
21	46kV Subtran Overheads & Transformer Platforms	122	599,020	598,670	288,531	144,058	161,826	3,253	1,002	350
22	Transformer Platforms	231	4,223	4,223	2,796	1,396	-	32	-	-
23	Three Phase Primary	230	925,538	925,538	483,027	241,166	195,861	5,446	39	-
24	Single Phase Primary	235	757,259	757,259	501,312	250,295	-	5,652	-	-
25	Single Phase Secondary	231	1,843,488	1,843,488	1,220,404	609,324	-	13,760	-	-
26	Total Distribution Overhead System		4,169,223	4,168,522	2,513,927	1,255,155	369,993	28,344	1,103	700
27	Distribution Underground System (366/367)									
28	Three Phase Primary	230	68,925	68,925	35,971	17,960	14,586	406	3	-
29	Single Phase Primary	235	628,860	628,860	416,311	207,856	-	4,694	-	-
30	Single Phase Secondary	231	155,930	155,930	103,227	51,539	-	1,164	-	-
31	46kV Subtran/Distribution Underground Lines	122	21,525	21,512	10,368	5,176	5,815	117	36	13
32	Total Distribution Underground System		875,240	875,228	565,877	282,531	20,401	6,380	39	13
33	Distribution Line Equipment (368)									
34	Primary	230	131,476	131,476	68,616	34,259	27,823	774	6	-
35	Secondary	231	898,402	898,402	594,749	296,947	-	6,706	-	-
36	Total Distribution Line Equipment		1,029,878	1,029,878	663,365	331,205	27,823	7,479	6	-
37	Distribution Services (369)									
38	Residential Overhead & Burial Services	260	586,964	586,964	586,964	-	-	-	-	-
39	C&I Overhead & Burial Services	261	295,526	295,526	-	295,526	-	-	-	-
40	Total Distribution Services		882,491	882,491	586,964	295,526	-	-	-	-

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PLANT IN SERVICE (DISTRIBUTION & GENERAL)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Distribution Metering Equipment (370)										
2	Metering Equipment (Mass)	170	660,213	660,203	510,864	106,631	42,442	145	120	11	
3	Total Distribution Metering Equipment		660,213	660,203	510,864	106,631	42,442	145	120	11	
4	Distribution Installations on Customer Premises (371)										
5	Installations on Premises L4 (Assignable)	DIR	20,900	20,900	-	-	-	20,900	-	-	
6	Total Distribution Installations on Customer Premises		20,900	20,900	-	-	-	20,900	-	-	
7	Distribution Streetlighting Equipment (373)										
8	Luminaires/Suspensions/Poles/Transformers	DIR	168,779	168,779	-	-	-	168,779	-	-	
9	Underground Cable & Conduits	233	9,376	9,299	-	-	-	9,299	-	76	
10	Photoelectric Switches	233	7,012	6,955	-	-	-	6,955	-	57	
11	Total Distribution Streetlighting Equipment		185,167	185,034	-	-	-	185,034	-	133	
12	Total Distribution Plant in Service		9,310,561	9,305,855	5,554,089	2,627,099	864,889	256,280	3,499	4,706	
13	Total Distribution Plant in Service	309	9,310,561	9,305,855	5,554,089	2,627,099	864,889	256,280	3,499	4,706	
14	Electric Plant Purchased & Sold	220	0	0	0	0	0	0	0	0	
15	General, Common & Intangible Plant in Service										
16	General: Production Related	220	0	0	0	0	0	0	0	0	
17	General: Merchant Control	226	0	0	0	0	0	0	0	-	
18	General: Power Control Center 138kV	301	0	0	0	0	0	0	0	0	
19	General: Power Control Center 46kV	302	0	0	0	0	0	0	0	0	
20	General: Functionalized	502	378,463	376,966	220,258	92,815	59,611	3,995	287	1,497	
21	General: Reallocated from/(to) Gas	DIR	0	-	-	-	-	-	-	-	
22	Common: Functionalized (Elec)	502	410,409	408,785	238,850	100,649	64,643	4,332	311	1,624	
23	Franchises & Consents - Generation	220	0	0	0	0	0	0	0	0	
24	Intangible PIS	502	815,102	811,877	474,373	199,896	128,386	8,604	618	3,225	
25	Total General, Common & Intangible Plant in Service		1,603,974	1,597,628	933,481	393,360	252,641	16,931	1,216	6,346	

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DEPRECIATION RESERVE (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Reserve									
2	Production Depreciation Reserve		2,626,131	2,600,532	1,258,906	594,669	734,642	8,318	3,997	25,599
3	Generation Step Ups		0	0	0	0	0	0	0	0
4	Total Production Depreciation Reserve		2,626,131	2,600,532	1,258,906	594,669	734,642	8,318	3,997	25,599
5	Transmission Depreciation Reserve									
6	Bulk Power Transm	225	-	-	-	-	-	-	-	-
7	Transm; Subtrans	122	0	0	0	0	0	0	0	0
8	Subtransmission	230	0	0	0	0	0	0	0	0
9	Total Transmission Depreciation Reserve		0	0	0	0	0	0	0	0
10	Distribution Depreciation Reserve									
11	Stations and Equipment		706,226	705,391	402,826	201,129	96,426	4,525	484	835
12	Overhead System		1,478,203	1,478,083	961,867	480,242	24,977	10,845	152	120
13	Underground System		324,031	323,844	156,675	78,225	86,641	1,766	536	187
14	Meters and Svc Drops		496,619	496,619	325,736	164,651	(183)	6,414	(1)	(0)
15	St Lgts and OPL		125,358	125,268	-	-	-	125,268	-	90
16	Total Distribution Depreciation Reserve		3,130,436	3,129,204	1,847,104	924,248	207,862	148,819	1,171	1,232
17		0								
18	Total General, Common & Intangible Depreciation Reserve		942,168	938,441	548,323	231,058	148,400	9,945	714	3,727
19	Total Depreciation Reserve	####	6,698,735	6,668,176	3,654,333	1,749,975	1,090,904	167,082	5,882	30,558

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DEPRECIATION RESERVE (PRODUCTION & TRANSMISSION)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Reserve									
2	Fossil	220	1,834,050	1,816,172	879,200	415,308	513,063	5,809	2,791	17,878
3	Nuclear	220	0	0	0	0	0	0	0	0
4	Hydro	220	405,182	401,233	194,235	91,751	113,347	1,283	617	3,950
5	Other Production/Combustion Turbine	220	386,899	383,127	185,470	87,611	108,232	1,225	589	3,771
6	Jackson Gas Plant	220	0	0	0	0	0	0	0	0
7	7 Classics	220	0	0	0	0	0	0	0	0
6	Distribution GSUs	220	0	0	0	0	0	0	0	0
7	Total Production Depreciation Reserve		2,626,131	2,600,532	1,258,906	594,669	734,642	8,318	3,997	25,599
8	Transmission Depreciation Reserve									
9	Total Transmission Direct	DIR	0	-	-	-	-	-	-	-
10	Total Subtransmission	228	0	0	0	0	0	0	0	0
11	XYZ	123	0	0	0	0	0	0	0	0
12	Total Transmission Depreciation Reserve		0	0	0	0	0	0	0	0

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DEPRECIATION RESERVE (DISTRIBUTION)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Depreciation Reserve									
2	Distribution Land & Right of Way (360)		-	-	-	-	-	-	-	-
3	345/138kV Substations/Overheads (METC)	127	0	0	0	0	0	0	0	0
4	345/138kV Substations/Overheads	127	2,023	2,006	910	455	627	10	3	18
5	46/23kV Substations/Overheads	124	10,175	10,169	4,901	2,447	2,749	55	17	6
6	Substations/Overheads (Assignnable)	DIR	0	-	-	-	-	-	-	-
7	Overhead Lines Land & ROW	307	12,415	12,414	7,621	3,811	909	70	3	2
8	Total Distribution Land & ROW Depreciation Reserve		24,614	24,588	13,432	6,713	4,285	135	23	26
9	Distribution Substations & Equipment (361/362/363)									
10	Customer Substations (Assignnable)	DIR	0	-	-	-	-	-	-	-
11	138kV HV Subtran/Dist Substations	127	78,367	77,675	35,257	17,603	24,296	398	122	692
12	46kV Subtran /Dist Substations	122	200,951	200,833	96,792	48,327	54,287	1,091	336	117
13	Distribution Substations	230	14,570	14,570	7,604	3,796	3,083	86	1	-
14	Total Distribution, Substations & Equipment Depreciation Reserve		293,888	293,078	139,653	69,726	81,666	1,575	459	809
15	Distribution Overhead System (364/365)									
16	138kV HV Subtran/Dist Overhead Lines	127	8,091	8,020	3,640	1,817	2,508	41	13	71
17	46kV Subtran Overheads & Transformer Platforms	122	83,171	83,122	40,061	20,002	22,469	452	139	49
18	Single Phase Primary & Secondary	305	1,386,940	1,386,940	918,166	458,423	-	10,352	-	-
19	Total Distribution Overhead System Depreciation Reserve		1,478,203	1,478,083	961,867	480,242	24,977	10,845	152	120
20	Distribution Underground System (366/367)									
21	Distribution UG System	308	3,542	3,542	2,305	1,151	61	26	0	-
22	46kV Subtran/Distribution UG Lines	122	320,489	320,302	154,370	77,074	86,581	1,740	536	187
23	Total Distribution Underground System Depreciation Reserve		324,031	323,844	156,675	78,225	86,641	1,766	536	187
24	Distribution Line Equipment (368)									
25	Capacitors/Regulators/Transformers	312	387,725	387,725	249,741	124,691	10,475	2,816	2	-
26	Total Distribution Line Equipment Depreciation Reserve		387,725	387,725	249,741	124,691	10,475	2,816	2	-
27	Distribution Services (369)									
28	C&I and Residential Services	310	493,050	493,050	327,939	165,111	-	-	-	-
29	Total Distribution Services Depreciation Reserve		493,050	493,050	327,939	165,111	-	-	-	-

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DEPRECIATION RESERVE (DISTRIBUTION & GENERAL)										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Metering Equipment (370)									
2	Metering Equipment (Mass)	170	(2,846)	(2,846)	(2,202)	(460)	(183)	(1)	(1)	(0)
3	Total Distribution Metering Equipment Depreciation Reserve		(2,846)	(2,846)	(2,202)	(460)	(183)	(1)	(1)	(0)
4	Distribution Installations on Customer Premises (371)									
5	Installations on Premises L4 (Assignable)	DIR	6,415	6,415	-	-	-	6,415	-	-
6	Total Distribution Installations on Customer Premises Depreciation Reserve		6,415	6,415	-	-	-	6,415	-	-
7	Distribution Streetlighting Equipment (373)									
8	Streetlighting Equipment Depreciation Reserve	311	125,358	125,268	-	-	-	125,268	-	90
9	Total Distribution Streetlighting Depreciation Reserve		125,358	125,268	-	-	-	125,268	-	90
10	Total Distribution Depreciation Reserve		3,130,436	3,129,204	1,847,104	924,248	207,862	148,819	1,171	1,232
11	Total Distribution Depreciation Reserve	317	3,130,436	3,129,204	1,847,104	924,248	207,862	148,819	1,171	1,232
12	General, Common & Intangible Depreciation Reserve									
13	General: Power Control Center	314	0	0	0	0	0	0	0	0
14	General: Functionalized	502	124,100	123,609	72,224	30,434	19,547	1,310	94	491
15	General: Reallocated to Gas	DIR	0	-	-	-	-	-	-	-
16	Common: Functionalized	502	203,546	202,740	118,459	49,918	32,060	2,149	154	805
17	Intangible Amortization Reserve	502	614,523	612,091	357,640	150,706	96,793	6,487	466	2,431
18	Total General, Common & Intangible Depreciation Reserve		942,168	938,441	548,323	231,058	148,400	9,945	714	3,727

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CWIP										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Production CWIP									
2	Production	220	151,340	149,865	72,549	34,270	42,336	479	230	1,475
3	Production: Gas Plant	220	0	0	0	0	0	0	0	0
3	Production: 7 Classics	220	0	0	0	0	0	0	0	0
4	Total Production CWIP		151,340	149,865	72,549	34,270	42,336	479	230	1,475
5	Transmission CWIP									
6	Transmission	228	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
7	Subtransmission	122	0	0	0	0	0	0	0	0
8	Total Transmission CWIP		(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
####										
9	Distribution CWIP									
10	HV Distribution	122	51,129	51,099	24,627	12,296	13,813	278	86	30
11	Distribution	306	94,954	94,952	59,821	27,968	3,966	3,194	2	2
12	Other	220	0	0	0	0	0	0	0	0
13	Total Distribution CWIP		146,083	146,051	84,448	40,264	17,779	3,472	88	32
####										
14	General/Common/Intangible CWIP									
15	General	502	16,401	16,336	9,545	4,022	2,583	173	12	65
16	Intangible	502	23,893	23,799	13,905	5,860	3,763	252	18	95
17	Common	502	19,721	19,643	11,477	4,836	3,106	208	15	78
18	Plant Held for Future Use	502	0	0	0	0	0	0	0	0
19	Other	502	0	0	0	0	0	0	0	0
20	Total General, Common & Intangible CWIP		60,016	59,778	34,928	14,718	9,453	634	46	237

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WORKING CAPITAL										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Current Assets									
2	Cash & Cash Equivalents	316	47,915	47,728	26,658	12,463	7,717	853	38	187
3	Accts Receivable	143	297,842	296,065	144,226	74,346	74,249	3,117	127	1,778
4	Material and Supplies	316	102,014	101,616	56,756	26,534	16,429	1,816	81	398
5	Fuel Stock	100	47,517	46,961	19,591	11,153	15,797	328	93	556
6	Real & Personal Property Taxes	316	179,599	178,898	99,920	46,714	28,924	3,197	142	701
7	Other Cur Assets	502	472,857	470,986	275,193	115,964	74,479	4,991	358	1,871
8	Deferred Debits	502	1,088,431	1,084,125	633,445	266,928	171,438	11,489	825	4,306
9	Total Current Assets		2,236,176	2,226,380	1,255,789	554,101	389,033	25,791	1,665	9,796
10	Current Liabilities									
11	Accounts Payable	316	417,607	415,978	232,337	108,620	67,256	7,433	331	1,629
12	Customer Deposits	143	15,511	15,418	7,511	3,872	3,867	162	7	93
13	Dividends Declared	316	29,338	29,223	16,322	7,631	4,725	522	23	114
14	Accrued Interest	316	44,231	44,059	24,608	11,505	7,123	787	35	173
15	Accrued Taxes - Federal	502	6,176	6,152	3,594	1,515	973	65	5	24
16	Accrued Taxes - State	601	1,508	1,502	858	381	240	21	1	6
17	Accrued Taxes - R&PP & Other	316	143,778	143,217	79,991	37,397	23,155	2,559	114	561
18	Other Current Liabilities	502	40,608	40,447	23,633	9,959	6,396	429	31	161
19	Deferred CR	502	310,336	309,109	180,610	76,107	48,881	3,276	235	1,228
20	Total Current Liabilities		1,009,093	1,005,104	569,465	256,986	162,616	15,256	782	3,989
21	Total Working Capital		1,227,083	1,221,275	686,324	297,115	226,417	10,535	884	5,807

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ADJUSTMENTS TO RATE BASE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Additions to Rate Base									
2	Sales and Use Tax Adjustment	309	0	0	0	0	0	0	0	0
3		0 100	0	0	0	0	0	0	0	0
4		0 220	0	0	0	0	0	0	0	0
5		0 263	0	0	0	0	-	0	-	-
6	Total Additions		0	0	0	0	0	0	0	0
	Deductions to Rate Base									
7	Construction Funds Retained from Contractors	330	0	0	0	0	0	0	0	0
8	Customer Advances	309	59,839	59,808	35,696	16,884	5,559	1,647	22	30
9	0	263	0	0	0	0	-	0	-	-
10	0	100	0	0	0	0	0	0	0	0
11	0	100	0	0	0	0	0	0	0	0
12	Total Deductions		59,839	59,808	35,696	16,884	5,559	1,647	22	30
13	Total Adjustments to Rate Base		(59,839)	(59,808)	(35,696)	(16,884)	(5,559)	(1,647)	(22)	(30)

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REVENUE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
Rate Revenue										
1	Non PSCR Rate Revenue									
2	Revenue From Electric Sales	DIR	2,119,720	2,108,954	1,176,864	563,864	337,680	28,823	1,722	10,766
3	Provision for Rate Refund	DIR	0	-	-	-	-	-	-	-
4	Unbilled Revenue	DIR	0	-	-	-	-	-	-	-
5	Total Non PSCR Rate Revenue		2,119,720	2,108,954	1,176,864	563,864	337,680	28,823	1,722	10,766
6	PSCR Base Revenue	DIR	1,898,071	1,884,228	772,284	439,661	659,356	12,927	-	13,844
7	Unbilled PSCR Base Revenue	DIR	0	-	-	-	-	-	-	-
8	0	DIR	0	-	-	-	-	-	-	-
9	Total PSCR Rate Revenue		1,898,071	1,884,228	772,284	439,661	659,356	12,927	-	13,844
10	Total Rate Revenue		4,017,792	3,993,182	1,949,148	1,003,525	997,036	41,750	1,722	24,610
Non-PSCR Rate Revenue Credits										
12	450 Late Payment Charge Revenue	DIR	9,448	9,448	5,345	2,561	1,534	-	8	-
13	451 Miscellaneous Service & Reconnect Fees	253	1,061	1,061	477	279	296	8	0	-
14	454 Pole Rental Rev	307	11,341	11,340	6,962	3,481	831	64	2	2
15	454 Other Rents	316	0	0	0	0	0	0	0	0
16	456 Purchased Power Administrative Fees	100	934	923	385	219	311	6	2	11
17	Other Revenues	150	8,329	8,237	3,004	1,760	3,408	50	15	92
18	Job Work Revenue	414	12,822	12,822	7,877	3,621	1,089	231	4	-
19	LTILRR: Production	DIR	75,176	74,367	34,027	16,974	22,791	441	134	809
20	LTILRR: Delivery	390	929	929	564	253	96	16	0	1
21	0		0	-	-	-	-	-	-	-
22	0		0	-	-	-	-	-	-	-
23	0		0	-	-	-	-	-	-	-
24	0		0	-	-	-	-	-	-	-
25	0		0	-	-	-	-	-	-	-
26	Non PSCR Revenue Credits		120,040	119,126	58,641	29,149	30,355	816	165	914
27	PSCR Factor Revenue	DIR	134,502	134,502	55,128	31,384	47,067	923	-	-
28	Unbilled PSCR Factor Revenue	DIR	0	-	-	-	-	-	-	-
29	Intersystem Sales	222	118,852	118,808	57,511	27,168	33,565	380	183	44
30	GSG and GI Market Price Revenue	DIR	5,960	5,960	-	-	2,073	-	3,887	-
31	PSCR Revenue Credits		259,314	259,270	112,639	58,553	82,705	1,303	4,070	44
32	Total Revenue Credits		379,354	378,396	171,280	87,702	113,061	2,119	4,235	958
33	Total Revenue		4,397,146	4,371,578	2,120,428	1,091,227	1,110,096	43,869	5,957	25,568

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PRODUCTION O&M (1)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Fuel and Purchased Power									
2	Mid-Peak Summer Fuel for Gen	103	75,006	74,129	30,270	18,607	24,859	264	128	877
3	On-Peak Winter Fuel for Gen	105	194,331	192,057	75,744	48,268	66,699	995	351	2,273
4	Off-Peak Summer Fuel for Gen	104	82,601	81,635	35,331	18,266	27,146	712	180	966
5	Off-Peak Winter Fuel for Gen	106	154,052	152,250	66,955	33,624	49,776	1,560	335	1,802
6	Critical Summer Peak Energy	107	36,904	36,473	15,902	8,961	11,454	97	59	432
7	Total Fuel Expense		542,894	536,543	224,202	127,726	179,934	3,627	1,054	6,351
8	Mid-Peak Summer Purchased Power	103	54,984	54,341	22,190	13,640	18,223	194	94	643
9	On-Peak Winter Purchased Power	105	142,457	140,791	55,525	35,384	48,895	729	257	1,667
10	Off-Peak Summer Purchased Power	104	60,552	59,844	25,900	13,390	19,900	522	132	708
11	Off-Peak Winter Purchased Power	106	112,930	111,609	49,083	24,649	36,489	1,143	246	1,321
12	Critical Peak Summer Purchased Power	107	27,053	26,737	11,658	6,569	8,396	71	43	316
13	Purchased Power Capacity	220	742,330	735,094	355,856	168,096	207,662	2,351	1,130	7,236
14	Total Purchased & Interchange Power Expense		1,140,307	1,128,416	520,211	261,728	339,565	5,010	1,902	11,892
15	Total Fuel and Purchased & Interchange Power Expense		1,683,201	1,664,958	744,413	389,454	519,499	8,637	2,956	18,243

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PRODUCTION O&M (2)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Fossil Plant O&M Expense									
2	Capacity Related Operations	220	36,951	36,590	17,713	8,367	10,337	117	56	360
3	Capacity Related Maintenance	220	4,478	4,435	2,147	1,014	1,253	14	7	44
4	Energy Related Operations	100	8,226	8,130	3,392	1,931	2,735	57	16	96
5	Energy Related Maintenance	100	38,509	38,058	15,877	9,039	12,802	266	76	451
6	Capacity Related Fuel Handling	220	0	0	0	0	0	0	0	0
7	Energy Related Fuel Handling	100	12,538	12,391	5,169	2,943	4,168	87	25	147
8	Total Fossil O&M Expense		100,702	99,605	44,298	23,294	31,294	540	179	1,097
9	Nuclear Plant O&M Expense									
10	Capacity Related Operations	220	0	0	0	0	0	0	0	0
11	Capacity Related Maintenance	220	0	0	0	0	0	0	0	0
12	Energy Related Maintenance	100	0	0	0	0	0	0	0	0
13	XYZ	220	0	0	0	0	0	0	0	0
14	XYZ	220	0	0	0	0	0	0	0	0
15	Total Nuclear Plant O&M Expense		0	0	0	0	0	0	0	0
16	Hydro Plant O&M Expense									
17	Capacity Related Operations	220	9,619	9,525	4,611	2,178	2,691	30	15	94
18	Capacity Related Maintenance	220	6,746	6,680	3,234	1,528	1,887	21	10	66
19	Energy Related Operations	100	725	717	299	170	241	5	1	8
20	Energy Related Maintenance	100	5,069	5,010	2,090	1,190	1,685	35	10	59
21	XYZ	220	0	0	0	0	0	0	0	0
22	Total Hydro O&M Total		22,159	21,932	10,234	5,066	6,504	92	36	227
23	Peaker O&M Expense									
24	Capacity Related Operations & Maintenance	220	67,853	67,191	32,527	15,365	18,981	215	103	661
25	Energy Related Operations & Maintenance	100	0	0	0	0	0	0	0	0
26	XYZ	100	0	0	0	0	0	0	0	0
27	Total Peaker O&M Expense		67,853	67,191	32,527	15,365	18,981	215	103	661
28	Other O&M Expense									
29	Capacity Related Sys Cntl Load Disp	220	10,761	10,656	5,159	2,437	3,010	34	16	105
30	Energy Related Sys Cntl Load Disp	100	0	0	0	0	0	0	0	0
31	Total Other O&M Expense		10,761	10,656	5,159	2,437	3,010	34	16	105
32	Disposition of Allowances	220	0	0	0	0	0	0	0	0
33	Total Production O&M (excluding Fuel and P&I)		201,475	199,384	92,217	46,161	59,790	881	335	2,091
34	Total Production O&M Expense		1,884,676	1,864,343	836,630	435,615	579,289	9,518	3,291	20,334

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TRANSMISSION O&M										
Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Transmission O&M Expense									
2	Transmission	120	477,556	472,579	225,419	106,588	137,459	2,362	750	4,976
3	Reclassified Transmission	127	0	0	0	0	0	0	0	0
4	Other	120	0	0	0	0	0	0	0	0
5	Total Transmission O&M Expense		477,556	472,579	225,419	106,588	137,459	2,362	750	4,976
6	Other O&M Adjustments									
7	Tax Benefit of Proforma Interest & Interest Synchronization Adjustment	150	0	0	0	0	0	0	0	0
8	Other Advertising Programs - Disallowance	412	0	0	0	0	0	0	0	0
9	Income Tax Effect of Interest	390	0	0	0	0	0	0	0	0
10	Charitable, Civic, Dues & Donations	412	0	0	0	0	0	0	0	0
11	Transmission reclass (indirect costs)	DIR	0	-	-	-	-	-	-	-
12	Streetlighting O&M	DIR	0	-	-	-	-	-	-	-
13	Customer O&M	411	0	0	0	0	0	0	0	0
14	Administrative and General O&M	412	0	0	0	0	0	0	0	0
15	Other O&M Inflation	443	0	0	0	0	0	0	0	0
16	Other O&M Adjmts	438	0	0	0	0	0	0	0	0
17	Total Other O&M Adjustments		0	0	0	0	0	0	0	0

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DISTRIBUTION O&M										
Line No.	Description	(a) Alloc	(b) Total Electric	(c) Total Jurisdictional Electric	(d) Total Residential	(e) Total Commercial Secondary	(f) Total Primary	(g) Total Lighting & Unmetered	(h) Rate GSG	(i) Total Non Jurisdictional
1	Distribution Operation Expense									
2	580 Supv & Engineering - Distribution	400	31,106	31,102	20,031	8,088	1,737	1,242	4	4
3	580 Supv & Engineering - 138kV	402	100	99	50	22	26	0	0	1
4	580 Supv & Engineering - 46kV	404	1,513	1,512	729	364	409	8	3	1
5	581 Load Dispatch - Distribution	301	0	0	0	0	0	0	0	0
6	582 Station Expense - Distribution	230	1,485	1,485	775	387	314	9	0	-
7	582 Station Expense - 138kV	303	0	0	0	0	0	0	0	0
8	582 Station Expense - 46kV	304	0	0	0	0	0	0	0	0
9	583 Overhead Expense - Distribution	307	16,231	16,228	9,963	4,982	1,189	91	4	2
10	583 Overhead Expense - 138kV	121	57	57	29	13	15	0	0	1
11	583 Overhead Expense - 46kV	122	862	861	415	207	233	5	1	1
12	584 Underground Exp	308	6,338	6,338	4,124	2,059	108	46	0	-
13	585 St Lt	311	1,160	1,159	-	-	-	1,159	-	1
14	586 - Metering Expense	313	1,364	1,364	1,055	220	88	0	0	0
15	587 Cust Instl Expense	160	5,514	5,514	4,852	648	12	2	0	0
16	588 Miscellaneous	400	21,312	21,309	13,724	5,541	1,190	851	3	3
17	589 Rents	309	2,170	2,169	1,295	612	202	60	1	1
18	Total Distribution Operation Expense		89,212	89,198	57,042	23,144	5,522	3,474	16	14
19	Distribution Maintenance Expense									
20	590 Supv & Engineering - Distribution	401	5,865	5,864	3,607	1,761	450	45	1	1
21	590 Supv & Engineering - 138kV	403	113	112	53	25	33	0	0	1
22	590 Supv & Engineering - 46kV	405	837	836	403	201	226	5	1	0
23	591 Structures - Distribution	230	492	492	257	128	104	3	0	-
24	591 Structures - 138kV	303	30	30	13	7	9	0	0	0
25	591 Structures - 46kV	304	80	79	38	19	21	0	0	0
26	592 Station Equipment - Distribution	230	8,436	8,436	4,403	2,198	1,785	50	0	-
27	592 Station Equipment - 138kV	303	1,209	1,198	544	272	375	6	2	11
28	592 Station Equipment - 46kV	304	3,211	3,209	1,546	772	867	17	5	2
29	593 Overhead Lines - Distribution	307	124,531	124,514	76,441	38,226	9,121	699	27	17
30	593 Overhead Lines - 138kV	224	771	771	390	174	204	1	1	-
31	593 Overhead Lines - 46kV	122	11,632	11,625	5,603	2,797	3,142	63	19	7
32	594 Underground Lines - Distribution	308	3,195	3,195	2,079	1,038	55	23	0	-
33	594 Underground Lines - 138kV	121	0	0	0	0	0	0	0	0
34	594 Underground Lines - 46kV	122	0	0	0	0	0	0	0	0
35	595 Line Xfmrs	312	10,185	10,185	6,560	3,275	275	74	0	-
36	596 St Lts & OPL	311	312	311	-	-	-	311	-	0
37	597 Meters	313	4,870	4,870	3,768	787	313	1	1	0
38	598 Miscellaneous	401	153	153	94	46	12	1	0	0
39	Total Distribution Maintenance Expense		175,919	175,880	105,801	51,727	16,992	1,301	59	39
40	Total Distribution O&M Expense		265,131	265,078	162,843	74,871	22,515	4,775	75	53

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CUSTOMER RELATED AND ADMINISTRATIVE & GENERAL EXPENSE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Customer Accounts Expense									
2	901 Supervision	408	7,119	7,119	6,267	836	14	2	0	0
3	902 Meter Reading	263	6,090	6,090	5,372	717	-	1	-	-
4	903 Rcrds & Collection	160	22,806	22,806	20,066	2,678	51	10	0	0
5	904 Uncollectibles	264	18,077	18,077	15,913	2,124	40	-	0	0
6	905 Misc Expenses	408	0	0	0	0	0	0	0	0
7	Total Customer Accounts		54,093	54,093	47,618	6,356	105	13	0	0
8	Customer Services Expense									
9	907 Supervision	160	3,436	3,436	3,024	404	8	2	0	0
10	908 Customer Assist	603	12,920	12,778	4,660	2,730	5,286	78	24	142
11	909 Info & Inst	160	556	556	489	65	1	0	0	0
12	910 Miscellaneous	160	0	0	0	0	0	0	0	0
13	Total Customer Services		16,912	16,770	8,173	3,199	5,295	80	24	142
14	Sales Expense									
15	911 Supervision	160	0	0	0	0	0	0	0	0
16	912 Demo & Selling	160	365	365	321	43	1	0	0	0
17	913 Advertising	160	0	0	0	0	0	0	0	0
18	916 Miscellaneous	160	0	0	0	0	0	0	0	0
19	Total Sales Expense		365	365	321	43	1	0	0	0
20	Administrative & General Expense									
21	Production	500	53,303	52,784	25,552	12,070	14,911	169	81	520
22	HV Distribution	406	3,290	3,286	1,582	786	896	17	5	4
23	Distribution	407	73,664	73,655	46,083	21,073	5,088	1,400	12	8
24	Customer	409	16,461	16,428	12,934	2,215	1,252	22	6	33
25	Total Administrative & General Expense		146,719	146,154	86,151	36,143	22,147	1,607	105	565
26	Total O&M Expense (excluding PSCR Expense)		684,695	681,844	397,323	166,772	109,853	7,356	539	2,850
27	Total O&M Expense		2,845,452	2,819,382	1,367,156	662,815	766,811	18,355	4,245	26,070

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DEPRECIATION EXPENSE (SUMMARY)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Production Depreciation Expense									
2	Production Depreciation Expense		309,278	306,263	148,261	70,034	86,518	980	471	3,015
3	GSU Depreciation Expense		0	0	0	0	0	0	0	0
4	Test Year Production Change	220	0	0	0	0	0	0	0	0
5	Total Production		309,278	306,263	148,261	70,034	86,518	980	471	3,015
6	Transmission									
7	Bulk Power Transm		0	-	-	-	-	-	-	-
8	Transm; Subtrans		0	0	0	0	0	0	0	0
9	Subtransmission		0	0	0	0	0	0	0	0
10	Total Transmission		0	0	0	0	0	0	0	0
11	Distribution									
12	Stations and Equipment		66,468	66,374	36,846	18,397	10,658	415	58	94
13	Overhead System		117,722	117,704	75,440	37,666	3,725	851	23	18
14	Underground System		20,930	20,930	13,619	6,800	358	154	0	-
15	Meters and Svc Drops		71,255	71,255	51,185	16,846	2,515	701	7	1
16	St Lgts and OPL		6,762	6,757	-	-	-	6,757	-	5
17	Sales and Use Tax Adjustment	444	129	129	81	36	8	4	0	0
18	Total Distribution		283,267	283,149	177,171	79,745	17,264	8,881	87	118
19	General/Common/Intangible									
20	Total Gen/Comm/Int		113,037	112,590	65,785	27,721	17,804	1,193	86	447
21	Test Year Gen/Comm/Int Change	445	0	0	0	0	0	0	0	0
22	Total General/Common/Intangible Dep Expense		113,037	112,590	65,785	27,721	17,804	1,193	86	447
23	Other Amortization Expense		0	0	0	0	0	0	0	0
24	Total Depreciation & Amortization Expense		705,582	702,002	391,217	177,500	121,587	11,054	644	3,580

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DEPRECIATION EXPENSE (PRODUCTION & TRANSMISSION)										
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Expense									
2	Direct		-	-	-	-	-	-	-	-
3	Fossil	220	224,639	222,449	107,687	50,868	62,841	712	342	2,190
4	Nuclear	220	0	0	0	0	0	0	0	0
5	Hydro	220	58,105	57,539	27,854	13,158	16,255	184	88	566
6	Other Production	220	26,534	26,275	12,720	6,008	7,423	84	40	259
7	Distribution GSUs	220	0	0	0	0	0	0	0	0
8	Jackson Gas Plant	220	0	0	0	0	0	0	0	0
9	7 Classics	220	0	-	-	-	-	-	-	-
10	Total Production Depreciation Expense		309,278	306,263	148,261	70,034	86,518	980	471	3,015
11	Transmission Depreciation Expense									
12	Direct		0	-	-	-	-	-	-	-
13	Transmission	127	0	0	0	0	0	0	0	0
14	Subtransmission	123	0	0	0	0	0	0	0	0
15	Total Transmission Depreciation Expense		0	0	0	0	0	0	0	0

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DEPRECIATION EXPENSE (DISTRIBUTION & GENERAL)										
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Depreciation Expense									
2	Distribution Land & Right of Way (360)		-	-	-	-	-	-	-	-
3	138kV Substations/Overheads (METC)	127	0	0	0	0	0	0	0	0
4	138kV Substations/Overheads	127	150	149	67	34	46	1	0	1
5	46kV Substations/Overheads	124	622	621	299	149	168	3	1	0
6	Substations/Overheads (Assignable)	DIR	0	-	-	-	-	-	-	-
7	Overhead Lines Land & ROW	307	503	503	309	154	37	3	0	0
8	Total Distribution Land & ROW Depreciation Expense		1,275	1,273	676	338	251	7	1	2
9	Distribution Substations & Equipment (361/362/363)									
10	Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
11	138kV HV Subtran/Dist Substations	127	8,811	8,733	3,964	1,979	2,732	45	14	78
12	Distribution Substations	124	25,302	25,288	12,187	6,085	6,836	137	42	15
13	Total Distribution Substations & Equipment Depreciation Expense		34,114	34,021	16,152	8,064	9,567	182	56	93
14	Distribution Overhead System (364/365)									
15	138kV HV Subtran/Dist Overhead Lines	127	1,183	1,172	532	266	367	6	2	10
16	46kV Subtran Overheads & Transformer Platforms	122	12,432	12,424	5,988	2,990	3,358	68	21	7
17	Overhead System	305	104,108	104,108	68,920	34,411	-	777	-	-
19	Total Distribution Overhead Depreciation Expense		117,722	117,704	75,440	37,666	3,725	851	23	18
20	Distribution Underground System (366/367)									
21	Underground System	308	20,930	20,930	13,619	6,800	358	154	0	-
22	Total Distribution Underground Depreciation Expense		20,930	20,930	13,619	6,800	358	154	0	-
23	Distribution Line Equipment (368)									
24	Line Equipment	312	31,080	31,080	20,019	9,995	840	226	0	-
25	Total Distribution Line Equipment Depreciation Expense		31,080	31,080	20,019	9,995	840	226	0	-
26	Distribution Services (369)									
27	Overhead & Underground Services	310	31,435	31,435	20,908	10,527	-	-	-	-
28	Total Distribution Services Depreciation Expense		31,435	31,435	20,908	10,527	-	-	-	-

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DEPRECIATION EXPENSE (DISTRIBUTION & GENERAL)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Distribution (cont.)									
2	Distribution Metering Equipment (370)		-	-	-	-	-	-	-	-
3	Metering Equipment	170	39,128	39,127	30,277	6,320	2,515	9	7	1
4	Total Distribution Metering Equipment Depreciation Expense		39,128	39,127	30,277	6,320	2,515	9	7	1
5	Distribution Installations on Customer Premises Depreciation Expense (371)	DIR	692	692	-	-	-	692	-	-
6	Distribution Streetlighting Equipment Depreciation Expense (373)	311	6,762	6,757	-	-	-	6,757	-	5
7	Total Distribution Depreciation Expense		283,138	283,020	177,091	79,709	17,256	8,877	87	118
8	General/Common/Intangible									
9	General	502	9,750	9,711	5,674	2,391	1,536	103	7	39
10	Common	502	22,116	22,029	12,871	5,424	3,484	233	17	87
11	Intangible Amortization	502	81,171	80,850	47,240	19,906	12,785	857	62	321
12	Total Gen/Comm/Int Depreciation Expense		113,037	112,590	65,785	27,721	17,804	1,193	86	447
13	Other Amortization									
14	Amort of 7 Classics Inventory	220	0	0	0	0	0	0	0	0
15	AFUDC in Excess of FERC Rate	330	0	0	0	0	0	0	0	0
16	Securitized Regulatory Assets	150	0	0	0	0	0	0	0	0
17	ARO Accretion/Transition Expense	220	0	0	0	0	0	0	0	0
18	Total Other Amortization Expense		0	0	0	0	0	0	0	0
19	Total Depreciation & Amortization Expense		705,453	701,873	391,136	177,464	121,579	11,050	644	3,579

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TAX										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	City Income Tax	150	1,077	1,065	388	227	440	6	2	12
2	Michigan Single Business Tax	601	0	0	0	0	0	0	0	0
3	Michigan Business Tax	439	30,672	30,927	11,760	9,585	8,998	541	43	(255)
RP&P Tax										
3	R&PP Taxes - Prod	220	71,896	71,195	34,465	16,280	20,112	228	109	701
4	R&PP Taxes - High Voltage Dist	302	25,086	25,072	12,083	6,033	6,777	136	42	15
5	R&PP Taxes - Low Voltage Dist	306	82,804	82,803	52,167	24,390	3,459	2,785	2	2
6	R&PP Taxes - General	315	4,505	4,487	2,626	1,106	705	48	3	18
7	R&PP Taxes - Common/Intangible	502	14,479	14,422	8,427	3,551	2,281	153	11	57
8	R&PP Taxes - PHFFU	226	29	29	14	7	8	0	0	-
9	R&PP Taxes - CWIP	330	0	0	0	0	0	0	0	0
10	Total R&PP Taxes		198,800	198,008	109,782	51,366	33,342	3,350	168	792
Payroll and Miscellaneous Tax										
11	Payroll Related Taxes	502	21,973	21,886	12,788	5,389	3,461	232	17	87
12	Miscellaneous General Taxes	150	0	0	0	0	0	0	0	0
13	Total Payroll/Miscellaneous Taxes		21,973	21,886	12,788	5,389	3,461	232	17	87
14	MPSC Assessment Fee	150	11,332	11,207	4,087	2,394	4,636	68	21	125
15	Total Other Taxes		263,853	263,093	138,806	68,962	50,878	4,198	250	760
16	Federal Income Tax Provision	439	53,928	54,376	20,677	16,852	15,821	950	76	(448)
17	Total Taxes Other Than Income		233,181	232,166	127,045	59,377	41,879	3,657	207	1,015
18	Total Income Taxes		84,600	85,304	32,437	26,437	24,820	1,491	119	(704)
19	Total Taxes		317,781	317,469	159,483	85,813	66,699	5,148	326	312

INCOME STATEMENT ADJUSTMENTS

Wages And Salaries	502	0	0	0	0	0	0	0
Pension Expense	502	0	0	0	0	0	0	0
Employee Life,Med Ins	502	0	0	0	0	0	0	0
Employee Savings Plan	502	0	0	0	0	0	0	0
Payroll Taxes	502	0	0	0	0	0	0	0
Prop,Inj,Dmg Insurance	600	0	0	0	0	0	0	0
Tot Increase Beyond Year		0	0	0	0	0	0	0

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Summary RETURN (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description		Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Rate Base		11,893,424	11,844,757	6,681,066	3,026,102	1,975,347	135,908	26,334	48,667
	Revenue									
2	Total Rate Revenue		4,017,792	3,993,182	1,949,148	1,003,525	997,036	41,750	1,722	24,610
3	Total Revenue Credits		379,354	378,394	174,158	87,956	109,987	2,023	4,270	961
4	Total Revenue		4,397,146	4,371,576	2,123,306	1,091,481	1,107,023	43,773	5,992	25,571
5	Expenses:									
6	Fuel and P&I Expense		1,683,201	1,664,639	759,860	390,952	502,750	8,163	2,913	18,563
7	Transmission Expense		477,556	472,640	222,685	105,296	141,584	2,334	741	4,916
8	Other O & M Expense		684,695	681,800	400,294	166,739	106,398	7,223	1,145	2,895
9	Depreciation & Amortization Expense		705,582	701,945	397,787	177,909	114,142	10,815	1,292	3,637
10	Other Taxes		263,853	263,097	139,036	68,868	50,512	4,176	505	756
11	Federal Income Taxes		53,928	54,409	18,861	16,830	17,749	1,025	(56)	(481)
12	Total Expenses		3,868,814	3,838,529	1,938,523	926,594	933,136	33,735	6,540	30,285
13	Net Operating Income		528,332	533,047	184,783	164,887	173,887	10,038	(548)	(4,715)
14	Other Income Adjustments		6,234	6,203	3,395	1,556	1,157	78	18	31
15	Adjusted Net Operating Income		534,565	539,249	188,178	166,442	175,044	10,116	(531)	(4,684)
16	Rate of Return on Rate Base		4.49%	4.55%	2.82%	5.50%	8.86%	7.44%	-2.02%	-9.62%
17	Index of Return (Jurisdictional)			100	62	121	195	163	(44)	
18	Return on Rate Base @ 6.09%		724,604	721,639	407,042	184,365	120,348	8,280	1,604	2,965
19	Income Deficiency (Sufficiency)		190,039	182,390	218,864	17,922	(54,696)	(1,836)	2,135	7,649
20	Revenue Deficiency (Sufficiency)		254,475	244,233	293,075	23,999	(73,242)	(2,458)	2,859	10,242
21	Revenue Requirement/Total Cost of Service		4,651,621	4,615,808	2,416,381	1,115,480	1,033,781	41,315	8,851	35,813
22	Less: Revenue Credits		379,354	378,394	174,158	87,956	109,987	2,023	4,270	961
23	Proposed Rate Design Revenue		4,272,267	4,237,415	2,242,223	1,027,524	923,794	39,292	4,581	34,852
24	Production: Net Capacity Cost		1,182,798	1,171,068	603,368	268,984	294,751	2,311	1,654	11,730
25	Production: Capacity Related Cost Offset		505,456	497,565	252,732	113,443	128,754	1,719	917	7,891
26	Production: Non-Capacity Related Cost		1,138,080	1,122,089	483,709	263,457	369,017	7,411	(1,506)	15,991
27	Distribution: Demand Related Cost		1,191,789	1,192,763	710,685	334,323	118,175	26,115	3,465	(974)
28	Distribution: Customer Related Cost		254,144	253,930	191,729	47,317	13,097	1,735	52	214
29	Full Service MWH Sales		30,813,122	30,433,455	12,447,668	7,086,449	10,627,497	208,358	63,483	379,667
30	ROA MWH Sales		3,697,427	3,697,427	-	205,339	3,492,088	-	-	-
31	MWH Sales		34,510,549	34,130,882	12,447,668	7,291,788	14,119,585	208,358	63,483	379,667
32	Customers		1,841,963	1,841,961	1,620,698	216,313	4,116	822	12	2

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Residential/Secondary RETURN (SUMMARY)									
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Total Commercial Secondary
1	Rate Base	6,681,066	-	6,681,066	1,773,386	1,119,398	52,213	81,105	3,026,102
	Revenue								
2	Total Rate Revenue	1,949,148	-	1,949,148	564,822	400,954	13,055	24,694	1,003,525
3	Total Revenue Credits	174,158	-	174,158	47,739	37,083	1,184	1,949	87,956
4	Total Revenue	2,123,306	-	2,123,306	612,561	438,038	14,239	26,643	1,091,481
5	Expenses:								
6	Fuel and P&I Expense	759,860	-	759,860	208,991	168,603	5,068	8,290	390,952
7	Transmission Expense	222,685	-	222,685	56,695	44,878	1,464	2,259	105,296
8	Other O & M Expense	400,294	-	400,294	100,499	59,345	2,707	4,189	166,739
9	Depreciation & Amortization Expense	397,787	-	397,787	104,978	65,711	2,888	4,332	177,909
10	Other Taxes	139,036	-	139,036	40,041	25,885	1,065	1,877	68,868
11	Federal Income Taxes	18,861	-	18,861	9,388	6,818	97	528	16,830
12	Total Expenses	1,938,523	-	1,938,523	520,592	371,240	13,289	21,474	926,594
13	Net Operating Income	184,783	-	184,783	91,970	66,798	950	5,169	164,887
14	Other Income Adjustments	3,395	-	3,395	901	589	26	40	1,556
15	Adjusted Net Operating Income	188,178	-	188,178	92,871	67,387	976	5,209	166,442
16	Rate of Return on Rate Base	2.82%	0.00%	2.82%	5.24%	6.02%	1.87%	6.42%	5.50%
17	Index of Return (Jurisdictional)	62	-	62	115	132	41	141	121
18	Return on Rate Base @ 6.09%	407,042	-	407,042	108,043	68,199	3,181	4,941	184,365
19	Income Deficiency (Sufficiency)	218,864	-	218,864	15,172	812	2,205	(267)	17,922
20	Revenue Deficiency (Sufficiency)	293,075	-	293,075	20,317	1,088	2,952	(358)	23,999
21	Revenue Requirement/Total Cost of Service	2,416,381	-	2,416,381	632,878	439,125	17,191	26,285	1,115,480
22	Less: Revenue Credits	174,158	-	174,158	47,739	37,083	1,184	1,949	87,956
23	Proposed Rate Design Revenue	2,242,223	-	2,242,223	585,139	402,042	16,007	24,336	1,027,524
24	Production: Net Capacity Cost	603,368	-	603,368	146,489	113,870	3,726	4,899	268,984
25	Production: Capacity Related Cost Offset	252,732	-	252,732	61,339	48,701	1,584	1,820	113,443
26	Production: Non-Capacity Related Cost	483,709	-	483,709	139,739	114,284	3,384	6,050	263,457
27	Distribution: Demand Related Cost	710,685	-	710,685	197,150	118,965	6,945	11,264	334,323
28	Distribution: Customer Related Cost	191,729	-	191,729	40,422	6,222	369	303	47,317
29	Full Service MWH Sales	12,447,668	-	12,447,668	3,742,026	3,093,372	87,692	163,359	7,086,449
30	ROA MWH Sales	-	-	-	9,785	116,816	14,695	64,043	205,339
31	MWH Sales	12,447,668	-	12,447,668	3,751,811	3,210,188	102,387	227,402	7,291,788
32	Customers	1,620,698	-	1,620,698	194,349	19,604	1,580	780	216,313

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Primary & Lighting RETURN (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate - GU-XL	Rate GU	Total Lighting & Unmetered	
1	Rate Base	211,953	841,758	89,652	181,808	540,466	37,378	16,787	58	9,283	46,204	1,975,347	4,429	74,430	36,287	20,762	135,908	
	Revenue																	
2	Total Rate Revenue	101,921	400,343	84,697	98,218	255,237	14,966	22,612	14	3,256	15,771	997,036	1,551	23,398	7,927	8,874	41,750	
3	Total Revenue Credits	9,395	45,469	12,404	11,083	26,029	1,390	2,396	1	302	1,519	109,987	110	730	240	944	2,023	
4	Total Revenue	111,316	445,811	97,101	109,301	281,266	16,357	25,008	15	3,559	17,290	1,107,023	1,661	24,128	8,167	9,818	43,773	
5	Expenses:																	
6	Fuel and P&I Expense	43,971	214,705	47,604	50,665	120,183	6,387	11,280	-	1,287	6,668	502,750	437	2,563	668	4,495	8,163	
7	Transmission Expense	12,130	59,541	14,484	14,321	32,534	1,863	4,500	-	355	1,857	141,584	148	865	225	1,096	2,334	
8	Other O & M Expense	10,531	45,560	5,783	10,258	28,349	1,828	1,385	2	437	2,266	106,398	255	5,224	562	1,182	7,223	
9	Depreciation & Amortization Expense	12,094	49,366	5,992	11,202	30,125	1,887	793	2	523	2,158	114,142	253	5,791	3,527	1,244	10,815	
10	Other Taxes	5,471	19,791	3,295	5,260	13,772	891	703	2	250	1,078	50,512	119	2,328	1,240	489	4,176	
11	Federal Income Taxes	2,512	5,265	1,847	1,630	5,215	324	588	1	65	302	17,749	42	681	180	121	1,025	
12	Total Expenses	86,708	394,229	79,005	93,337	230,179	13,180	19,248	7	2,917	14,327	933,136	1,253	17,453	6,402	8,628	33,735	
13	Net Operating Income	24,608	51,583	18,096	15,964	51,087	3,177	5,760	8	641	2,963	173,887	407	6,675	1,765	1,190	10,038	
14	Other Income Adjustments	119	498	48	119	312	21	8	0	6	25	1,157	2	41	24	11	78	
15	Adjusted Net Operating Income	24,727	52,081	18,144	16,084	51,399	3,197	5,768	8	647	2,988	175,044	410	6,716	1,789	1,201	10,116	
16	Rate of Return on Rate Base	11.67%	6.19%	20.24%	8.85%	9.51%	8.55%	34.36%	14.23%	6.97%	6.47%	8.86%	9.25%	9.02%	4.93%	5.79%	7.44%	
17	Index of Return (Jurisdictional)	256	136	445	194	209	188	755	312	153	142	195	203	198	108	127	163	
18	Return on Rate Base @ 6.09%	12,913	51,284	5,462	11,077	32,928	2,277	1,023	4	566	2,815	120,348	270	4,535	2,211	1,265	8,280	
19	Income Deficiency (Sufficiency)	(11,814)	(797)	(12,682)	(5,007)	(18,471)	(920)	(4,745)	(5)	(82)	(173)	(54,696)	(140)	(2,181)	422	64	(1,836)	
20	Revenue Deficiency (Sufficiency)	(15,820)	(1,067)	(16,982)	(6,705)	(24,734)	(1,232)	(6,354)	(6)	(109)	(232)	(73,242)	(187)	(2,921)	565	85	(2,458)	
21	Revenue Requirement/Total Cost of Service	95,496	444,744	80,119	102,596	256,532	15,124	18,654	9	3,449	17,058	1,033,781	1,473	21,207	8,732	9,903	41,315	
22	Less: Revenue Credits	9,395	45,469	12,404	11,083	26,029	1,390	2,396	1	302	1,519	109,987	110	730	240	944	2,023	
23	Proposed Rate Design Revenue	86,101	399,275	67,715	91,513	230,503	13,734	16,258	8	3,147	15,540	923,794	1,363	20,477	8,492	8,959	39,292	
24	Production: Net Capacity Cost	30,562	133,644	16,508	26,263	77,687	4,352	-	-	1,087	4,648	294,751	-	-	-	2,311	2,311	
25	Production: Capacity Related Cost Offset	11,002	61,100	7,577	12,323	31,515	1,586	1,422	0	368	1,859	128,754	71	374	88	1,186	1,719	
26	Production: Non-Capacity Related Cost	29,609	153,510	40,420	39,221	83,308	4,451	13,271	(0)	750	4,479	369,017	483	2,832	738	3,359	7,411	
27	Distribution: Demand Related Cost	12,407	47,629	1,840	12,249	34,476	3,071	1,338	5	883	4,279	118,175	771	15,652	7,655	2,037	26,115	
28	Distribution: Customer Related Cost	2,520	3,392	1,370	1,458	3,518	274	228	3	60	275	13,097	38	1,619	12	67	1,735	
29	Full Service MWH Sales	830,685	4,365,549	1,272,911	1,142,370	2,388,475	121,808	358,537	-	21,349	125,813	10,627,497	13,509	79,208	20,629	95,012	208,358	
30	ROA MWH Sales	46,609	-	1,089,479	1,247,199	854,891	27,722	-	2,690	70,229	153,269	3,492,088	-	-	-	-	-	
31	MWH Sales	877,294	4,365,549	2,362,390	2,389,569	3,243,366	149,530	358,537	2,690	91,578	279,082	14,119,585	13,509	79,208	20,629	95,012	208,358	
32	Customers	1,510	1,032	33	85	1,204	142	19	1	6	84	4,116	343	-	-	479	822	

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Summary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description			Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Net Plant										
2	Production			3,675,605	3,639,155	1,838,790	840,694	945,056	9,234	5,382	36,450
3	Transmission			(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution			6,328,753	6,325,280	3,761,932	1,728,259	706,260	110,595	18,234	3,473
5	General/Common/Intangible			721,822	718,922	424,067	176,813	109,351	7,425	1,266	2,900
6	Plant Purchased/Sold			0	0	0	0	0	0	0	0
7	Total Net Plant			10,726,180	10,683,357	6,024,789	2,745,766	1,760,666	127,253	24,883	42,823
8	Working Capital										
9	Total Current Assets			2,236,176	2,226,255	1,266,441	553,706	377,103	25,286	3,718	9,921
10	Total Current Liabilities			1,009,093	1,005,046	574,709	256,607	156,647	14,988	2,095	4,047
11	Total Working Capital			1,227,083	1,221,209	691,732	297,099	220,457	10,299	1,623	5,873
	Adjustments to Rate Base										
12	Additions to Rate Base			0	0	0	0	0	0	0	0
13	Deductions from Rate Base			59,839	59,809	35,454	16,763	5,776	1,644	172	30
14	Total Adjustments to Rate Base			(59,839)	(59,809)	(35,454)	(16,763)	(5,776)	(1,644)	(172)	(30)
15	Total Rate Base			11,893,424	11,844,757	6,681,066	3,026,102	1,975,347	135,908	26,334	48,667

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Residential/Secondary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Net Plant									
2	Production		1,838,790	-	1,838,790	456,246	357,167	11,503	15,778	840,694
3	Transmission		(0)	-	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution		3,761,932	-	3,761,932	1,044,661	595,851	33,569	54,177	1,728,259
5	General/Common/Intangible		424,067	-	424,067	106,454	63,012	2,920	4,426	176,813
6	Plant Purchased/Sold		0	-	0	0	0	0	0	0
7	Total Net Plant		6,024,789	-	6,024,789	1,607,361	1,016,031	47,992	74,382	2,745,766
8	Working Capital									
9	Total Current Assets		1,266,441	-	1,266,441	328,539	202,304	8,909	13,955	553,706
10	Total Current Liabilities		574,709	-	574,709	152,290	93,238	4,364	6,714	256,607
11	Total Working Capital		691,732	-	691,732	176,249	109,066	4,544	7,240	297,099
	Adjustments to Rate Base									
12	Additions to Rate Base		0	-	0	0	0	0	0	0
13	Deductions from Rate Base		35,454	-	35,454	10,224	5,698	322	518	16,763
14	Total Adjustments to Rate Base		(35,454)	-	(35,454)	(10,224)	(5,698)	(322)	(518)	(16,763)
15	Total Rate Base		6,681,066	-	6,681,066	1,773,386	1,119,398	52,213	81,105	3,026,102

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Primary & Lighting RATE BASE (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(a)			-												-		Total
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Total Lighting & Unmetered
1	Net Plant																
2	Production	95,311	426,064	55,809	86,324	245,615	13,617	4,546	-	3,279	14,490	945,056	184	1,080	281	7,689	9,234
3	Transmission	(0)	(0)	(0)	(0)	(0)	(0)	(0)	0	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution	83,938	277,666	14,730	64,640	208,564	18,327	8,446	54	4,709	25,186	706,260	3,630	65,691	31,513	9,762	110,595
5	General/Common/Intangible	11,240	47,099	5,276	10,306	29,741	1,962	782	2	495	2,448	109,351	246	3,668	2,327	1,184	7,425
6	Plant Purchased/Sold	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0
7	Total Net Plant	190,489	750,829	75,815	161,270	483,919	33,906	13,774	56	8,484	42,124	1,760,666	4,060	70,438	34,121	18,634	127,253
8	Working Capital																
9	Total Current Assets	38,486	160,133	21,278	36,184	100,742	6,488	4,351	7	1,602	7,833	377,103	814	12,989	7,534	3,949	25,286
10	Total Current Liabilities	16,367	66,961	7,322	15,034	42,519	2,869	1,263	4	759	3,550	156,647	405	8,048	4,806	1,729	14,988
11	Total Working Capital	22,119	93,171	13,956	21,150	58,223	3,619	3,089	3	844	4,283	220,457	409	4,941	2,728	2,221	10,299
Adjustments to Rate Base																	
12	Additions to Rate Base	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	Deductions from Rate Base	656	2,243	119	612	1,676	147	75	0	45	204	5,776	40	949	561	93	1,644
14	Total Adjustments to Rate Base	(656)	(2,243)	(119)	(612)	(1,676)	(147)	(75)	(0)	(45)	(204)	(5,776)	(40)	(949)	(561)	(93)	(1,644)
15	Total Rate Base	211,953	841,758	89,652	181,808	540,466	37,378	16,787	58	9,283	46,204	1,975,347	4,429	74,430	36,287	20,762	135,908

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Summary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description			Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Expense										
2	Fuel Expense			542,894	536,431	224,167	127,615	179,962	3,634	1,054	6,463
3	Purchased & Interchange Power Expense			1,140,307	1,128,208	535,693	263,338	322,789	4,529	1,859	12,099
4	Total Fuel and P&I Expense			1,683,201	1,664,639	759,860	390,952	502,750	8,163	2,913	18,563
5	Fossil O&M Exp			100,702	99,586	45,594	23,633	29,659	520	179	1,116
6	Nuclear O&M Exp			0	0	0	0	0	0	0	0
7	Hydro O&M Exp			22,159	21,928	10,618	5,127	6,066	82	36	231
8	Peaker O&M Exp			67,853	67,180	33,945	15,519	17,446	170	99	673
9	Other O&M			10,761	10,654	5,383	2,461	2,767	27	16	107
10	Total Production O&M Expense			201,475	199,348	95,540	46,741	55,937	800	330	2,127
11	Total Production (Inc. Fuel and P&I) O&M Expense			1,884,676	1,863,986	855,400	437,693	558,688	8,962	3,243	20,690
12	Transmission & Distribution Expense										
13	Trans O&M Exp			477,556	472,640	222,685	105,296	141,584	2,334	741	4,916
14	Other O&M Adjustments			0	0	0	0	0	0	0	0
15	Distr Oper Exp			89,212	89,198	56,752	22,998	5,857	3,471	120	13
16	Distr Maint Exp			175,919	175,881	104,944	51,298	17,896	1,291	452	38
17	Total Transmission & Distribution O&M Expense			742,687	737,719	384,381	179,592	165,337	7,096	1,313	4,968
18	Customer Related Expense										
19	Customer Accounts Exp			54,093	54,093	47,618	6,356	105	13	0	0
20	Customer Service Exp			16,912	16,770	8,173	3,199	5,295	80	24	142
21	Sales Expense			365	365	321	43	1	0	0	0
22	Total Customer O&M Expense			71,370	71,228	56,113	9,597	5,401	93	24	142
23	Admin & General Expense			146,719	146,145	86,945	36,105	21,307	1,569	219	574
24	Total O&M Expense			2,845,452	2,819,078	1,382,839	662,987	750,733	17,720	4,799	26,374

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Residential/Secondary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Production Expense									
2	Fuel Expense		224,167	-	224,167	67,422	55,664	1,584	2,945	127,615
3	Purchased & Interchange Power Expense		535,693	-	535,693	141,569	112,939	3,484	5,345	263,338
4	Total Fuel and P&I Expense		759,860	-	759,860	208,991	168,603	5,068	8,290	390,952
5	Fossil O&M Exp		45,594	-	45,594	12,619	10,206	305	504	23,633
6	Nuclear O&M Exp		0	-	0	0	0	0	0	0
7	Hydro O&M Exp		10,618	-	10,618	2,762	2,194	68	102	5,127
8	Peaker O&M Exp		33,945	-	33,945	8,422	6,593	212	291	15,519
9	Other O&M		5,383	-	5,383	1,336	1,046	34	46	2,461
10	Total Production O&M Expense		95,540	-	95,540	25,139	20,039	619	944	46,741
11	Total Production (Inc. Fuel and P&I) O&M Expense		855,400	-	855,400	234,130	188,642	5,687	9,234	437,693
12	Transmission & Distribution Expense									
13	Trans O&M Exp		222,685	-	222,685	56,695	44,878	1,464	2,259	105,296
14	Other O&M Adjustments		0	-	0	0	0	0	0	0
15	Distr Oper Exp		56,752	-	56,752	14,472	7,434	424	668	22,998
16	Distr Maint Exp		104,944	-	104,944	31,486	17,269	979	1,564	51,298
17	Total Transmission & Distribution O&M Expense		384,381	-	384,381	102,653	69,581	2,867	4,491	179,592
18	Customer Related Expense									
19	Customer Accounts Exp		47,618	-	47,618	5,710	576	46	23	6,356
20	Customer Service Exp		8,173	-	8,173	1,826	1,244	42	87	3,199
21	Sales Expense		321	-	321	39	4	0	0	43
22	Total Customer O&M Expense		56,113	-	56,113	7,575	1,824	88	110	9,597
23	Admin & General Expense		86,945	-	86,945	21,828	12,778	596	903	36,105
24	Total O&M Expense		1,382,839	-	1,382,839	366,185	272,826	9,238	14,737	662,987

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Primary & Lighting O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered	
1	Production Expense																	
2	Fuel Expense	14,265	74,236	20,965	19,175	40,725	2,098	5,979	-	361	2,159	179,962	231	1,353	352	1,698	3,634	
3	Purchased & Interchange Power Expense	29,706	140,469	26,640	31,491	79,459	4,288	5,301	-	927	4,509	322,789	206	1,210	315	2,797	4,529	
4	Total Fuel and P&I Expense	43,971	214,705	47,604	50,665	120,183	6,387	11,280	-	1,287	6,668	502,750	437	2,563	668	4,495	8,163	
5	Fossil O&M Exp	2,652	13,063	2,116	2,977	7,269	385	718	-	77	402	29,659	29	170	44	276	520	
6	Nuclear O&M Exp	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	
7	Hydro O&M Exp	579	2,704	394	580	1,533	83	85	-	19	88	6,066	3	20	5	53	82	
8	Peaker O&M Exp	1,759	7,865	1,030	1,594	4,534	251	84	-	61	267	17,446	3	20	5	142	170	
9	Other O&M	279	1,247	163	253	719	40	13	-	10	42	2,767	1	3	1	23	27	
10	Total Production O&M Expense	5,269	24,880	3,703	5,403	14,056	759	900	-	166	800	55,937	36	214	56	494	800	
11	Total Production (Inc. Fuel and P&I) O&M Expense	49,240	239,585	51,307	56,069	134,239	7,146	12,180	-	1,453	7,468	558,688	474	2,777	723	4,989	8,962	
12	Transmission & Distribution Expense																	
13	Trans O&M Exp	12,130	59,541	14,484	14,321	32,534	1,863	4,500	-	355	1,857	141,584	148	865	225	1,096	2,334	
14	Other O&M Adjustments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
15	Distr Oper Exp	706	2,434	29	422	1,813	158	43	0	31	220	5,857	70	1,895	1,383	123	3,471	
16	Distr Maint Exp	1,983	7,365	117	1,602	5,419	468	161	0	116	664	17,896	86	657	266	283	1,291	
17	Total Transmission & Distribution O&M Expense	14,819	69,340	14,630	16,345	39,766	2,490	4,704	1	502	2,740	165,337	303	3,416	1,874	1,502	7,096	
18	Customer Related Expense																	
19	Customer Accounts Exp	39	26	1	2	31	4	0	0	0	2	105	6	-	-	7	13	
20	Customer Service Exp	332	1,637	884	895	1,217	56	134	1	34	105	5,295	6	30	8	37	80	
21	Sales Expense	0	0	0	0	0	0	0	0	0	0	1	0	-	-	0	0	
22	Total Customer O&M Expense	371	1,663	885	897	1,248	60	135	1	34	107	5,401	12	30	8	44	93	
23	Admin & General Expense	2,202	9,218	1,048	1,934	5,813	382	146	0	90	475	21,307	51	780	499	239	1,569	
24	Total O&M Expense	66,632	319,806	67,870	75,245	181,067	10,078	17,165	2	2,079	10,790	750,733	840	7,003	3,104	6,773	17,720	

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Summary ALLOCATORS 1										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	100.000	98.809	41.956	23.886	32.066	0.702	0.200	1.191
2	Energy On-Peak @ Generation	101	100.000	98.809	39.214	24.661	34.308	0.451	0.176	1.191
3	Energy Off-Peak @ Generation	102	100.000	98.809	43.535	22.087	32.002	0.966	0.219	1.191
4	Energy On-Peak @ Generation Summer	103	100.000	98.809	40.151	24.681	33.457	0.350	0.170	1.191
5	Energy Off-Peak @ Generation Summer	104	100.000	98.809	43.129	22.298	32.294	0.869	0.220	1.191
6	Energy On-Peak @ Generation Non-Summer	105	100.000	98.809	38.679	24.649	34.794	0.508	0.179	1.191
7	Energy Off-Peak @ Generation Non-Summer	106	100.000	98.809	43.755	21.973	31.844	1.019	0.219	1.191
8	Energy Critical On-Peak @ Gen	107	100.000	98.809	42.963	24.209	31.218	0.261	0.159	1.191
9	Energy Summer Mid-Peak @ Gen	108	100.000	98.809	39.226	24.837	34.193	0.380	0.174	1.191
10	12CP Dmd @ Generation	120	100.000	98.971	46.630	22.049	29.648	0.489	0.155	1.029
11	4CP Dmd @ Generation	121	100.000	99.008	51.012	22.741	24.920	0.195	0.140	0.992
12	Class Peak @ Subtransmission	122	100.000	99.943	47.174	23.553	27.367	0.532	1.317	0.057
13	Classpeak @ Transmission	127	100.000	99.131	44.315	22.126	30.952	0.500	1.237	0.869
14	Total Rate Revenue	143	100.000	99.407	48.269	24.924	25.145	1.028	0.041	0.593
15	Billed Sales	150	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
16	Billed Sales Excluding Rate E1	151	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
17	Number Of Customers	160	100.000	100.000	87.988	11.744	0.223	0.045	0.001	0.000
18	Weighted Customer	170	100.000	99.998	77.444	16.165	6.350	0.022	0.018	0.002
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	100.000	99.013	50.838	22.423	25.206	0.410	0.135	0.987
20	4CP 89/0/11	220	100.000	99.008	50.027	22.872	25.712	0.251	0.146	0.992
21	4CP 89/0/11 Exc WFR	222	100.000	99.968	50.510	23.094	25.962	0.254	0.148	0.032
22	4CP Dmd @ Gen Jurisdictional	224	100.000	100.000	51.523	22.969	25.169	0.197	0.141	-
23	12CP Demand @ Subtrans	226	100.000	100.000	49.013	23.175	27.243	0.514	0.055	-
24	Classpeak @ Primary	230	100.000	100.000	50.621	25.274	23.501	0.571	0.033	-
25	Classpeak @ Secondary	231	100.000	100.000	66.201	33.053	-	0.746	-	-
26	Classpeak for Streetlighting	233	100.000	99.188	-	-	-	99.188	-	0.812
27	Classpeak @ Single Phase	235	100.000	100.000	66.201	33.053	-	0.746	-	-
28	Billed Sales ROA	252	100.000	100.000	-	5.554	94.446	-	-	-
29	Billed Sales - Primary	253	100.000	100.000	43.102	25.249	30.922	0.721	0.005	-
30	Customers - Residential	260	100.000	100.000	100.000	-	-	-	-	-
31	Customers - Drops	261	100.000	100.000	-	100.000	-	-	-	-
32	Customers - NonPID	263	100.000	100.000	88.208	11.773	-	0.019	-	-
33	Customers - NonMunicipal	264	100.000	100.000	88.027	11.749	0.224	-	0.001	0.000
34	PIS - 138kV Distribution	301	100.000	99.131	44.315	22.126	30.952	0.500	1.237	0.869
35	PIS - 46kV Distribution	302	100.000	99.943	47.174	23.553	27.367	0.532	1.317	0.057
36	PIS - 138kV Dist Subs S&E	303	100.000	99.131	44.315	22.126	30.952	0.500	1.237	0.869
37	PIS - 46kV Dist Subs S&E	304	100.000	99.943	47.174	23.553	27.367	0.532	1.317	0.057

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Residential/Secondary ALLOCATORS 1										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	
Line No.	Description	Alloc	Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	41.956	-	41.956	12.613	10.426	0.296	0.551	23.886
2	Energy On-Peak @ Generation	101	39.214	-	39.214	13.231	10.495	0.340	0.593	24.661
3	Energy Off-Peak @ Generation	102	43.535	-	43.535	11.421	9.953	0.233	0.480	22.087
4	Energy On-Peak @ Generation Summer	103	40.151	-	40.151	13.236	10.651	0.308	0.486	24.681
5	Energy Off-Peak @ Generation Summer	104	43.129	-	43.129	11.345	10.290	0.213	0.449	22.298
6	Energy On-Peak @ Generation Non-Summer	105	38.679	-	38.679	13.229	10.406	0.359	0.655	24.649
7	Energy Off-Peak @ Generation Non-Summer	106	43.755	-	43.755	11.463	9.770	0.243	0.497	21.973
8	Energy Critical On-Peak @ Gen	107	42.963	-	42.963	12.891	10.572	0.283	0.464	24.209
9	Energy Summer Mid-Peak @ Gen	108	39.226	-	39.226	13.350	10.677	0.316	0.494	24.837
10	12CP Dmd @ Generation	120	46.630	-	46.630	11.872	9.397	0.306	0.473	22.049
11	4CP Dmd @ Generation	121	51.012	-	51.012	12.385	9.627	0.315	0.414	22.741
12	Class Peak @ Subtransmission	122	47.174	-	47.174	13.163	9.042	0.505	0.843	23.553
13	Classpeak @ Transmission	127	44.315	-	44.315	12.366	8.494	0.474	0.792	22.126
14	Total Rate Revenue	143	48.269	-	48.269	14.002	9.986	0.324	0.612	24.924
15	Billed Sales	150	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
16	Billed Sales Excluding Rate E1	151	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
17	Number Of Customers	160	87.988	-	87.988	10.551	1.064	0.086	0.042	11.744
18	Weighted Customer	170	77.444	-	77.444	13.914	2.056	0.113	0.082	16.165
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	50.838	-	50.838	12.230	9.478	0.312	0.403	22.423
20	4CP 89/0/11	220	50.027	-	50.027	12.413	9.717	0.313	0.429	22.872
21	4CP 89/0/11 Exc WFR	222	50.510	-	50.510	12.533	9.812	0.316	0.433	23.094
22	4CP Dmd @ Gen Jurisdictional	224	51.523	-	51.523	12.509	9.724	0.318	0.418	22.969
23	12CP Demand @ Subtrans	226	49.013	-	49.013	12.479	9.878	0.322	0.497	23.175
24	Classpeak @ Primary	230	50.621	-	50.621	14.125	9.702	0.542	0.905	25.274
25	Classpeak @ Secondary	231	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-
27	Classpeak @ Single Phase	235	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
28	Billed Sales ROA	252	-	-	-	0.265	3.159	0.397	1.732	5.554
29	Billed Sales - Primary	253	43.102	-	43.102	12.991	11.116	0.355	0.787	25.249
30	Customers - Residential	260	100.000	-	100.000	-	-	-	-	-
31	Customers - Drops	261	-	-	-	89.846	9.063	0.730	0.361	100.000
32	Customers - NonPID	263	88.208	-	88.208	10.578	1.067	0.086	0.042	11.773
33	Customers - NonMunicipal	264	88.027	-	88.027	10.556	1.065	0.086	0.042	11.749
34	PIS - 138kV Distribution	301	44.315	-	44.315	12.366	8.494	0.474	0.792	22.126
35	PIS - 46kV Distribution	302	47.174	-	47.174	13.163	9.042	0.505	0.843	23.553
36	PIS - 138kV Dist Subs S&E	303	44.315	-	44.315	12.366	8.494	0.474	0.792	22.126
37	PIS - 46kV Dist Subs S&E	304	47.174	-	47.174	13.163	9.042	0.505	0.843	23.553

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Primary & Lighting ALLOCATORS 1																		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	
Line No.	Description	Alloc	Rate GP	Rate GPTU	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vit 1	Rate GPD GEI Vit 2	Rate GPD GEI Vit 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered
<u>Input Allocation Schedules</u>																		
1	Energy @ Generation	100	2.662	13.936	2.508	3.381	7.593	0.390	1.124	-	0.068	0.403	32.066	0.046	0.267	0.070	0.320	0.702
2	Energy On-Peak @ Generation	101	3.005	14.480	3.688	3.459	7.528	0.464	1.192	-	0.067	0.423	34.308	0.020	0.119	0.031	0.280	0.451
3	Energy Off-Peak @ Generation	102	2.166	12.781	4.155	3.664	7.513	0.290	1.003	-	0.066	0.365	32.002	0.073	0.428	0.111	0.354	0.966
4	Energy On-Peak @ Generation Summer	103	3.003	14.099	3.489	3.316	7.533	0.411	1.116	-	0.076	0.413	33.457	0.011	0.064	0.017	0.259	0.350
5	Energy Off-Peak @ Generation Summer	104	2.218	12.902	4.099	3.640	7.754	0.264	0.973	-	0.080	0.363	32.294	0.064	0.373	0.097	0.334	0.869
6	Energy On-Peak @ Generation Non-Summer	105	3.007	14.698	3.801	3.541	7.525	0.494	1.236	-	0.062	0.430	34.794	0.026	0.151	0.039	0.292	0.508
7	Energy Off-Peak @ Generation Non-Summer	106	2.137	12.716	4.185	3.677	7.382	0.305	1.019	-	0.057	0.366	31.844	0.078	0.457	0.119	0.365	1.019
8	Energy Critical On-Peak @ Gen	107	2.831	13.150	3.054	3.076	7.245	0.384	0.992	-	0.076	0.410	31.218	-	-	-	0.261	0.261
9	Energy Summer Mid-Peak @ Gen	108	3.060	14.412	3.633	3.396	7.627	0.420	1.156	-	0.077	0.413	34.193	0.015	0.085	0.022	0.258	0.380
10	12CP Dmd @ Generation	120	2.540	12.468	3.033	2.999	6.813	0.390	0.942	-	0.074	0.389	29.648	0.031	0.181	0.047	0.230	0.489
11	4CP Dmd @ Generation	121	2.584	11.299	1.396	2.220	6.568	0.368	-	-	0.092	0.393	24.920	-	-	-	0.195	0.195
12	Class Peak @ Subtransmission	122	2.621	10.211	-	4.735	7.478	0.645	0.403	-	0.345	0.928	27.367	0.045	0.263	0.069	0.155	0.532
13	Classpeak @ Transmission	127	2.462	9.592	4.430	4.448	7.025	0.606	1.179	0.013	0.324	0.872	30.952	0.042	0.247	0.064	0.146	0.500
14	Total Rate Revenue	143	2.543	10.107	2.176	2.487	6.402	0.373	0.583	0.000	0.081	0.393	25.145	0.039	0.572	0.193	0.224	1.028
15	Billed Sales	150	2.542	12.650	6.845	6.924	9.398	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
16	Billed Sales Excluding Rate E1	151	2.542	12.650	6.845	6.924	9.398	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
17	Number Of Customers	160	0.082	0.056	0.002	0.005	0.065	0.008	0.001	0.000	0.000	0.005	0.223	0.019	-	-	0.026	0.045
18	Weighted Customer	170	2.287	1.563	0.053	0.136	1.921	0.215	0.030	0.002	0.010	0.134	6.350	0.022	-	-	-	0.022
<u>Calculated Allocation Schedules</u>																		
19	4CP Average & Excess	219	2.550	11.045	1.465	2.193	6.454	0.363	0.657	-	0.092	0.388	25.206	0.027	0.156	0.041	0.187	0.410
20	4CP 89/0/11	220	2.593	11.592	1.518	2.349	6.682	0.370	0.124	-	0.089	0.394	25.712	0.005	0.029	0.008	0.209	0.251
21	4CP 89/0/11 Exc WFR	222	2.618	11.704	1.533	2.372	6.747	0.374	0.125	-	0.090	0.398	25.962	0.005	0.030	0.008	0.211	0.254
22	4CP Dmd @ Gen Jurisdictional	224	2.610	11.412	1.410	2.243	6.634	0.372	-	-	0.093	0.397	25.169	-	-	-	0.197	0.197
23	12CP Demand @ Subtrans	226	2.670	13.105	-	3.152	7.161	0.410	0.258	-	0.078	0.409	27.243	0.032	0.190	0.050	0.241	0.514
24	Classpeak @ Primary	230	2.813	10.957	-	-	8.024	0.692	0.018	-	-	0.996	23.501	0.048	0.283	0.074	0.166	0.571
25	Classpeak @ Secondary	231	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-	-	-	-	11.822	69.314	18.052	-	99.188
27	Classpeak @ Single Phase	235	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
28	Billed Sales ROA	252	1.261	-	29.466	33.732	23.121	0.750	-	0.073	1.899	4.145	94.446	-	-	-	-	-
29	Billed Sales - Primary	253	3.038	15.116	-	-	11.231	0.518	0.053	-	-	0.966	30.922	0.047	0.274	0.071	0.329	0.721
30	Customers - Residential	260	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
31	Customers - Drops	261	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	Customers - NonPID	263	-	-	-	-	-	-	-	-	-	-	-	0.019	-	-	-	0.019
33	Customers - NonMunicipal	264	0.082	0.056	0.002	0.005	0.065	0.008	0.001	0.000	0.000	0.005	0.224	-	-	-	-	-
34	PIS - 138kV Distribution	301	2.462	9.592	4.430	4.448	7.025	0.606	1.179	0.013	0.324	0.872	30.952	0.042	0.247	0.064	0.146	0.500
35	PIS - 46kV Distribution	302	2.621	10.211	-	4.735	7.478	0.645	0.403	-	0.345	0.928	27.367	0.045	0.263	0.069	0.155	0.532
36	PIS - 138kV Dist Subs S&E	303	2.462	9.592	4.430	4.448	7.025	0.606	1.179	0.013	0.324	0.872	30.952	0.042	0.247	0.064	0.146	0.500
37	PIS - 46kV Dist Subs S&E	304	2.621	10.211	-	4.735	7.478	0.645	0.403	-	0.345	0.928	27.367	0.045	0.263	0.069	0.155	0.532

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Summary ALLOCATORS 2										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
No.										
	<u>Calculated Allocation Schedules</u>									
1	PIS - Overhead Primary System	305	100.000	100.000	66.201	33.053	-	0.746	-	(0.000)
2	PIS - Distribution Distribution	306	100.000	99.998	62.740	29.323	4.566	3.361	0.008	0.002
3	Overhead Distribution	307	100.000	99.986	60.973	30.491	7.794	0.556	0.172	0.014
4	Underground Distribution	308	100.000	100.000	64.943	32.425	1.897	0.732	0.003	(0.000)
5	Total Dist PIS	309	100.000	99.950	59.249	28.013	9.652	2.748	0.287	0.050
6	Distribution Services	310	100.000	100.000	66.512	33.488	-	-	-	-
7	Streetlighting Equipment	311	100.000	99.928	-	-	-	99.928	-	0.072
8	Line Equipment	312	100.000	100.000	64.212	32.060	3.000	0.724	0.004	-
9	Meters	313	100.000	99.998	77.444	16.165	6.350	0.022	0.018	0.002
10	PIS - System Power Control	314	100.000	199.073	91.490	45.679	58.319	1.032	2.554	(99.073)
11	PIS - General	315	100.000	99.598	58.750	24.495	15.149	1.029	0.175	0.402
12	Total PIS	316	100.000	99.578	55.879	25.830	15.957	1.686	0.226	0.422
13	Distribution Depreciation	317	100.000	99.961	58.757	29.401	6.758	4.751	0.295	0.039
14	CWIP	330	100.000	99.504	54.461	24.956	18.557	1.248	0.282	0.496
15	Working Capital	340	100.000	99.521	56.372	24.212	17.966	0.839	0.132	0.479
16	Rate Base	390	100.000	99.591	56.174	25.443	16.609	1.143	0.221	0.409
17	Operations - Distribution excl Sup & Eng	400	100.000	99.988	64.089	25.846	5.962	3.989	0.102	0.012
18	Maintenance - Distribution excl Sup & Eng	401	100.000	99.989	61.068	29.809	8.209	0.759	0.144	0.011
19	Operations - 138kV Distribution excl Sup & Eng	402	100.000	99.008	51.012	22.741	24.920	0.195	0.140	0.992
20	Maintenance - 138kV Distribution excl Sup & Eng	403	100.000	99.464	47.080	22.449	28.734	0.384	0.817	0.536
21	Operations - 46kV Distribution excl Sup & Eng	404	100.000	99.943	47.174	23.553	27.367	0.532	1.317	0.057
22	Maintenance - 46kV Distribution excl Sup & Eng	405	100.000	99.943	47.174	23.553	27.367	0.532	1.317	0.057
23	HV Distribution O&M exp.	406	100.000	99.886	47.194	23.432	27.490	0.514	1.256	0.114
24	Distribution O&M, excl. HV Dist	407	100.000	99.989	62.164	28.409	7.393	1.896	0.128	0.011
25	Customer Accounting	408	100.000	100.000	88.031	11.749	0.195	0.024	0.001	0.000
26	Customer Accounts & Service	409	100.000	99.800	78.574	13.456	7.605	0.131	0.034	0.200
27	Distribution O&M	410	100.000	99.980	60.987	28.022	8.959	1.796	0.216	0.020
28	Customer & Sales O&M	411	100.000	99.801	78.622	13.447	7.567	0.130	0.034	0.199
29	Administrative and General O&M	412	100.000	99.609	59.260	24.608	14.523	1.069	0.149	0.391
30	Jurisdictional Distribution O&M	414	100.000	100.000	60.999	28.028	8.961	1.796	0.216	-
31	O&M Excluding Adjustments	438	100.000	99.073	48.598	23.300	26.384	0.623	0.169	0.927
32	Pre Tax NOI	439	100.000	100.892	34.975	31.209	32.912	1.900	(0.104)	(0.892)
33	R&PP Tax	440	100.000	99.595	55.797	25.801	16.102	1.656	0.239	0.405
34	Depreciation & Amortization Expense	442	100.000	99.485	56.377	25.214	16.177	1.533	0.183	0.515
35	Non PSCR O&M Expense	443	100.000	99.577	58.463	24.352	15.540	1.055	0.167	0.423
36	Distribution Depreciation Expense	444	100.000	99.959	62.364	28.059	6.177	3.133	0.226	0.041
37	Gen/Comm/Int Depreciation Expense	445	100.000	99.598	58.750	24.495	15.149	1.029	0.175	0.402
38	Production Labor	500	100.000	99.008	50.027	22.872	25.712	0.251	0.146	0.992
39	Total Labor	502	100.000	99.598	58.750	24.495	15.149	1.029	0.175	0.402
40	50% O&M, 50% Net Plant	600	100.000	99.490	54.582	25.117	18.505	1.068	0.219	0.510
41	50/50 PIS & Labor	601	100.000	99.588	57.314	25.163	15.553	1.358	0.201	0.412

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Residential/Secondary ALLOCATORS 2										
	(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Alloc	Rate Residential	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Calculated Allocation Schedules</u>										
1	PIS - Overhead Primary System	305	66.201	-	66.201	18.473	12.688	0.709	1.183	33.053
2	PIS - Distribution Distribution	306	62.740	-	62.740	18.214	9.686	0.550	0.874	29.323
3	Overhead Distribution	307	60.973	-	60.973	19.028	9.989	0.571	0.903	30.491
4	Underground Distribution	308	64.943	-	64.943	18.122	12.447	0.695	1.161	32.425
5	Total Dist PIS	309	59.249	-	59.249	17.087	9.523	0.539	0.865	28.013
6	Distribution Services	310	66.512	-	66.512	30.087	3.035	0.245	0.121	33.488
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-
8	Line Equipment	312	64.212	-	64.212	17.918	12.307	0.687	1.148	32.060
9	Meters	313	77.444	-	77.444	13.914	2.056	0.113	0.082	16.165
10	PIS - System Power Control	314	91.490	-	91.490	25.529	17.535	0.979	1.635	45.679
11	PIS - General	315	58.750	-	58.750	14.748	8.730	0.404	0.613	24.495
12	Total PIS	316	55.879	-	55.879	15.182	9.518	0.445	0.684	25.830
13	Distribution Depreciation	317	58.757	-	58.757	18.226	9.737	0.556	0.883	29.401
14	CWIP	330	54.461	-	54.461	14.453	9.446	0.419	0.637	24.956
15	Working Capital	340	56.372	-	56.372	14.363	8.888	0.370	0.590	24.212
16	Rate Base	390	56.174	-	56.174	14.911	9.412	0.439	0.682	25.443
17	Operations - Distribution excl Sup & Eng	400	64.089	-	64.089	16.312	8.312	0.475	0.747	25.846
18	Maintenance - Distribution excl Sup & Eng	401	61.068	-	61.068	18.443	9.905	0.564	0.897	29.809
19	Operations - 138kV Distribution excl Sup & Eng	402	51.012	-	51.012	12.385	9.627	0.315	0.414	22.741
20	Maintenance - 138kV Distribution excl Sup & Eng	403	47.080	-	47.080	12.421	8.965	0.414	0.649	22.449
21	Operations - 46kV Distribution excl Sup & Eng	404	47.174	-	47.174	13.163	9.042	0.505	0.843	23.553
22	Maintenance - 46kV Distribution excl Sup & Eng	405	47.174	-	47.174	13.163	9.042	0.505	0.843	23.553
23	HV Distribution O&M exp.	406	47.194	-	47.194	13.080	9.038	0.494	0.820	23.432
24	Distribution O&M, excl. HV Dist	407	62.164	-	62.164	17.694	9.339	0.532	0.844	28.409
25	Customer Accounting	408	88.031	-	88.031	10.556	1.065	0.086	0.042	11.749
26	Customer Accounts & Service	409	78.574	-	78.574	10.613	2.564	0.124	0.155	13.456
27	Distribution O&M	410	60.987	-	60.987	17.334	9.317	0.529	0.842	28.022
28	Customer & Sales O&M	411	78.622	-	78.622	10.613	2.556	0.124	0.154	13.447
29	Administrative and General O&M	412	59.260	-	59.260	14.877	8.709	0.406	0.615	24.608
30	Jurisdictional Distribution O&M	414	60.999	-	60.999	17.337	9.319	0.530	0.842	28.028
31	O&M Excluding Adjustments	438	48.598	-	48.598	12.869	9.588	0.325	0.518	23.300
32	Pre Tax NOI	439	34.975	-	34.975	17.408	12.643	0.180	0.978	31.209
33	R&PP Tax	440	55.797	-	55.797	15.147	9.525	0.445	0.684	25.801
34	Depreciation & Amortization Expense	442	56.377	-	56.377	14.878	9.313	0.409	0.614	25.214
35	Non PSCR O&M Expense	443	58.463	-	58.463	14.678	8.667	0.395	0.612	24.352
36	Distribution Depreciation Expense	444	62.364	-	62.364	17.622	9.104	0.517	0.816	28.059
37	Gen/Comm/Int Depreciation Expense	445	58.750	-	58.750	14.748	8.730	0.404	0.613	24.495
38	Production Labor	500	50.027	-	50.027	12.413	9.717	0.313	0.429	22.872
39	Total Labor	502	58.750	-	58.750	14.748	8.730	0.404	0.613	24.495
40	50% O&M, 50% Net Plant	600	54.582	-	54.582	14.542	9.497	0.422	0.657	25.117
41	50/50 PIS & Labor	601	57.314	-	57.314	14.965	9.124	0.425	0.649	25.163

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Primary & Lighting ALLOCATORS 2		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Alloc	Rate GP	Rate GPTU	Rate GPD Vit 1	Rate GPD Vit 2	Rate GPD Vit 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vit 1	Rate GPD GEI Vit 2	Rate GPD GEI Vit 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered
<u>Calculated Allocation Schedules</u>																		
1	PIS - Overhead Primary System	305	-	-	-	-	-	-	-	-	-	-	-	0.063	0.370	0.096	0.217	0.746
2	PIS - Distribution Distribution	306	0.684	2.001	0.005	0.015	1.536	0.137	0.006	0.000	0.001	0.181	4.566	0.074	1.952	1.178	0.156	3.361
3	Overhead Distribution	307	0.845	3.294	0.035	0.596	2.412	0.208	0.060	0.000	0.043	0.299	7.794	0.047	0.276	0.072	0.162	0.556
4	Underground Distribution	308	0.227	0.885	-	-	0.648	0.056	0.001	-	-	0.080	1.897	0.062	0.363	0.094	0.213	0.732
5	Total Dist PIS	309	1.096	3.748	0.199	1.023	2.800	0.245	0.126	0.001	0.075	0.340	9.652	0.068	1.587	0.938	0.156	2.748
6	Distribution Services	310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-	-	-	-	1.046	55.182	43.700	-	99.928
8	Line Equipment	312	0.359	1.399	-	-	1.024	0.088	0.002	-	-	0.127	3.000	0.061	0.359	0.093	0.211	0.724
9	Meters	313	2.287	1.563	0.053	0.136	1.921	0.215	0.030	0.002	0.010	0.134	6.350	0.022	-	-	-	0.022
10	PIS - System Power Control	314	5.083	19.803	4.430	9.184	14.503	1.251	1.582	0.013	0.669	1.800	58.319	0.087	0.511	0.133	0.300	1.032
11	PIS - General	315	1.557	6.525	0.731	1.428	4.120	0.272	0.108	0.000	0.069	0.339	15.149	0.034	0.508	0.322	0.164	1.029
12	Total PIS	316	1.679	6.836	0.725	1.539	4.323	0.293	0.124	0.000	0.079	0.360	15.957	0.042	0.924	0.545	0.176	1.686
13	Distribution Depreciation	317	0.643	2.513	0.125	1.058	1.840	0.159	0.113	0.000	0.077	0.228	6.758	0.088	2.684	1.820	0.159	4.751
14	CWIP	330	1.916	7.996	0.767	1.915	4.999	0.331	0.130	0.000	0.099	0.405	18.557	0.034	0.654	0.380	0.180	1.248
15	Working Capital	340	1.803	7.593	1.137	1.724	4.745	0.295	0.252	0.000	0.069	0.349	17.966	0.033	0.403	0.222	0.181	0.839
16	Rate Base	390	1.782	7.078	0.754	1.529	4.544	0.314	0.141	0.000	0.078	0.388	16.609	0.037	0.626	0.305	0.175	1.143
17	Operations - Distribution excl Sup & Eng	400	0.738	2.507	0.031	0.353	1.875	0.164	0.039	0.000	0.026	0.227	5.962	0.079	2.179	1.594	0.137	3.989
18	Maintenance - Distribution excl Sup & Eng	401	0.960	3.504	0.030	0.493	2.591	0.225	0.052	0.000	0.036	0.318	8.209	0.050	0.387	0.161	0.161	0.759
19	Operations - 138kV Distribution excl Sup & Eng	402	2.584	11.299	1.396	2.220	6.568	0.368	0.000	0.000	0.092	0.393	24.920	0.000	0.000	0.000	0.195	0.195
20	Maintenance - 138kV Distribution excl Sup & Eng	403	2.519	10.290	3.271	3.602	6.875	0.516	0.727	0.008	0.235	0.690	28.734	0.026	0.153	0.040	0.165	0.384
21	Operations - 46kV Distribution excl Sup & Eng	404	2.621	10.211	-	4.735	7.478	0.645	0.403	-	0.345	0.928	27.367	0.045	0.263	0.069	0.155	0.532
22	Maintenance - 46kV Distribution excl Sup & Eng	405	2.621	10.211	-	4.735	7.478	0.645	0.403	-	0.345	0.928	27.367	0.045	0.263	0.069	0.155	0.532
23	HV Distribution O&M exp.	406	2.610	10.228	0.351	4.598	7.408	0.630	0.434	0.001	0.332	0.899	27.490	0.043	0.250	0.065	0.156	0.514
24	Distribution O&M, excl. HV Dist	407	0.879	3.146	0.029	0.438	2.333	0.203	0.047	0.000	0.032	0.286	7.393	0.060	1.017	0.666	0.153	1.896
25	Customer Accounting	408	0.071	0.049	0.002	0.004	0.057	0.007	0.001	0.000	0.000	0.004	0.195	0.011	-	-	0.013	0.024
26	Customer Accounts & Service	409	0.522	2.342	1.247	1.263	1.757	0.084	0.190	0.001	0.049	0.150	7.605	0.017	0.042	0.011	0.061	0.131
27	Distribution O&M	410	1.014	3.696	0.055	0.763	2.728	0.236	0.077	0.000	0.055	0.333	8.959	0.059	0.962	0.622	0.153	1.796
28	Customer & Sales O&M	411	0.519	2.330	1.240	1.257	1.748	0.084	0.189	0.001	0.048	0.150	7.567	0.017	0.042	0.011	0.061	0.130
29	Administrative and General O&M	412	1.501	6.283	0.714	1.318	3.962	0.260	0.099	0.000	0.061	0.324	14.523	0.035	0.531	0.340	0.163	1.069
30	Jurisdictional Distribution O&M	414	1.015	3.697	0.055	0.764	2.728	0.236	0.077	0.000	0.055	0.333	8.961	0.059	0.963	0.622	0.153	1.796
31	O&M Excluding Adjustments	438	2.342	11.239	2.385	2.644	6.363	0.354	0.603	0.000	0.073	0.379	26.384	0.030	0.246	0.109	0.238	0.623
32	Pre Tax NOI	439	4.658	9.763	3.425	3.022	9.670	0.601	1.090	0.002	0.121	0.561	32.912	0.077	1.263	0.334	0.225	1.900
33	R&PP Tax	440	1.702	6.938	0.621	1.590	4.394	0.298	0.109	0.000	0.083	0.368	16.102	0.042	0.906	0.533	0.176	1.656
34	Depreciation & Amortization Expense	442	1.714	6.997	0.849	1.588	4.270	0.267	0.112	0.000	0.074	0.306	16.177	0.036	0.821	0.500	0.176	1.533
35	Non PSCR O&M Expense	443	1.538	6.654	0.845	1.498	4.140	0.267	0.202	0.000	0.064	0.331	15.540	0.037	0.522	0.323	0.173	1.055
36	Distribution Depreciation Expense	444	0.817	2.168	0.166	0.821	1.695	0.153	0.102	0.001	0.060	0.196	6.177	0.070	1.810	1.108	0.145	3.133
37	Gen/Comm/Int Depreciation Expense	445	1.557	6.525	0.731	1.428	4.120	0.272	0.108	0.000	0.069	0.339	15.149	0.034	0.508	0.322	0.164	1.029
38	Production Labor	500	2.593	11.592	1.518	2.349	6.682	0.370	0.124	-	0.089	0.394	25.712	0.005	0.029	0.008	0.209	0.251
39	Total Labor	502	1.557	6.525	0.731	1.428	4.120	0.272	0.108	0.000	0.069	0.339	15.149	0.034	0.508	0.322	0.164	1.029
40	50% O&M, 50% Net Plant	600	1.895	7.889	1.059	1.743	4.900	0.324	0.228	0.000	0.078	0.390	18.505	0.036	0.571	0.274	0.187	1.068
41	50/50 PIS & Labor	601	1.618	6.681	0.728	1.483	4.222	0.282	0.116	0.000	0.074	0.349	15.553	0.038	0.716	0.434	0.170	1.358

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NET PLANT (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Plant in Service									
2	Production		6,150,396	6,089,404	3,076,850	1,406,735	1,581,363	15,451	9,006	60,992
3	Transmission		0	0	0	0	0	0	0	0
4	Distribution		9,310,561	9,305,929	5,516,452	2,608,182	898,688	255,850	26,756	4,633
5	General/Common/Intangible		1,603,974	1,597,530	942,327	392,899	242,992	16,498	2,814	6,444
6	Plant Purchased/Sold		0	0	0	0	0	0	0	0
7	Total Plant in Service		17,064,931	16,992,862	9,535,629	4,407,816	2,723,043	287,799	38,575	72,069
8	Depreciation Reserve									
9	Production		2,626,131	2,600,088	1,313,771	600,656	675,219	6,597	3,845	26,043
10	Transmission		0	0	0	0	0	0	0	0
11	Distribution		3,130,436	3,129,223	1,839,342	920,372	211,544	148,731	9,234	1,213
12	General/Common/Intangible		942,168	938,383	553,519	230,787	142,732	9,691	1,653	3,785
13	Total Depreciation Reserve		6,698,735	6,667,694	3,706,632	1,751,815	1,029,495	165,020	14,732	31,041
14	Construction Work In Progress (CWIP)									
15	Production		151,340	149,839	75,711	34,615	38,912	380	222	1,501
16	Transmission		(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
17	Distribution		146,083	146,052	83,694	39,886	18,328	3,463	681	31
18	General/Common/Intangible		60,016	59,775	35,259	14,701	9,092	617	105	241
19	Total CWIP		357,439	355,665	194,663	89,202	66,332	4,461	1,008	1,773
20	Future Use									
21	Production	220	0	0	0	0	0	0	0	0
22	Distribution	127	2,545	2,523	1,128	563	788	13	31	22
23	Common/General	231	0	0	0	0	-	0	-	-
24	PHFFU Depreciation Reserve	127	0	0	0	0	0	0	0	0
25	Total Future Use		2,545	2,523	1,128	563	788	13	31	22
26	Total Net Plant		10,726,180	10,683,357	6,024,789	2,745,766	1,760,666	127,253	24,883	42,823

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PLANT IN SERVICE (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Plant in Service									
2	Production Plant in Service		6,150,396	6,089,404	3,076,850	1,406,735	1,581,363	15,451	9,006	60,992
3	Generation Step Ups		0	0	0	0	0	0	0	0
4	Total Production		6,150,396	6,089,404	3,076,850	1,406,735	1,581,363	15,451	9,006	60,992
5	Transmission Plant in Service									
6	Bulk Power Transm		0	0	0	0	0	0	0	0
7	Transm; Subtrans		0	0	0	0	0	0	0	0
8	Subtransmission		0	0	0	0	0	0	0	0
9	Total Transmission		0	0	0	0	0	0	0	0
10	Distribution Plant in Service									
11	Stations and Equipment		2,517,327	2,513,539	1,360,406	679,240	440,949	15,296	17,647	3,789
12	Overhead System		4,169,223	4,168,535	2,493,205	1,244,809	393,728	28,110	8,683	688
13	Underground System		875,240	875,228	564,583	281,885	22,089	6,366	306	12
14	Meters and Svc Drops		1,563,604	1,563,593	1,098,258	402,247	41,923	21,045	120	11
15	St Lgts and OPL		185,167	185,034	-	-	-	185,034	-	133
16	Total Distribution		9,310,561	9,305,929	5,516,452	2,608,182	898,688	255,850	26,756	4,633
17	General/Common/Intangible Plant in Service									
18	Total Gen/Comm/Int Plant		1,603,974	1,597,530	942,327	392,899	242,992	16,498	2,814	6,444
19	Plant Purchased/Sold		0	0	0	0	0	0	0	0
20	Total Plant in Service		17,064,931	16,992,862	9,535,629	4,407,816	2,723,043	287,799	38,575	72,069

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PLANT IN SERVICE (PRODUCTION & TRANSMISSION)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Production Plant in Service										
2	Fossil	220	4,655,905	4,609,733	2,329,203	1,064,911	1,197,106	11,696	6,817	46,172	
3	Nuclear	220	0	0	0	0	0	0	0	0	
4	Total Hydro	220	707,453	700,437	353,916	161,811	181,897	1,777	1,036	7,016	
5	Other Production/Combustion Turbine	220	787,038	779,233	393,731	180,013	202,360	1,977	1,152	7,805	
6	7 Classics	220	0	0	0	0	0	0	0	0	
7	Jackson Gas Plant	220	0	0	0	0	0	0	0	0	
6	Distribution GSUs	220	0	0	0	0	0	0	0	0	
7	Total Production Plant in Service		6,150,396	6,089,404	3,076,850	1,406,735	1,581,363	15,451	9,006	60,992	
8	Transmission Plant in Service										
9	Transmission Direct	228	0	0	0	0	0	0	0	0	
10	Transmission	228	0	0	0	0	0	0	0	0	
11	XYZ	127	0	0	0	0	0	0	0	0	
12	Total Transmission Plant in Service		0	0	0	0	0	0	0	0	

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PLANT IN SERVICE (DISTRIBUTION)										
Line No.	Description	(a) Alloc	(b) Total Electric	(c) Total Jurisdictional Electric	(d) Total Residential	(e) Total Commercial Secondary	(f) Total Primary	(g) Total Lighting & Unmetered	(h) Rate GSG	(i) Total Non Jurisdictional
Distribution Plant in Service										
1	Distribution Land & ROW (360)									
2	345/138kV Substations/Overheads (METC) Land & ROW	127	0	0	0	0	0	0	0	0
3	345/138kV Substations/Overheads Land & ROW	127	23,955	23,747	10,616	5,300	7,415	120	296	208
4	46/23kV Substations/Overheads Land & ROW	124	60,975	60,940	28,764	14,361	16,687	324	803	35
5	Distribution Substation Land & ROW	230	2,027	2,027	1,026	512	476	12	1	-
6	138kV Substations/Overheads (Assignable) Land & ROW	DIR	0	-	-	-	-	-	-	-
7	46kV Substations/Overheads (Assignable) Land & ROW	DIR	0	-	-	-	-	-	-	-
8	Overhead Lines Land & ROW	307	32,331	32,326	19,713	9,858	2,520	180	56	4
9	Total Distribution Land & ROW		119,287	119,040	60,119	30,032	27,098	635	1,156	248
10	Distribution Substations & Equipment (361/362/363)									
11	138kV Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
12	46kV Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
13	138kV HV Subtran/Dist Substations	127	346,679	343,665	153,632	76,706	107,305	1,732	4,289	3,014
14	138kV HV Subtran/Dist Substations-Proposed Trans Reclass	127	0	0	0	0	0	0	0	0
15	46kV Subtran/Dist Substations	124	920,673	920,146	434,319	216,847	251,957	4,897	12,126	527
16	Distribution Substations	230	100,810	100,810	51,031	25,479	23,691	575	33	-
17	Total Distribution Substations & Equipment		1,368,162	1,364,621	638,983	319,032	382,954	7,204	16,448	3,541
18	Distribution Overhead System (364/365)									
19	138kV HV Subtran/Dist Overhead Lines	127	39,694	39,349	17,591	8,783	12,286	198	491	345
20	138kV HV Subtran/Dist Overhead Lines-Proposed Trans Reclass	121	0	0	0	0	0	0	0	0
21	46kV Subtran Overheads & Transformer Platforms	122	599,020	598,677	282,582	141,088	163,932	3,186	7,890	343
22	Transformer Platforms	231	4,223	4,223	2,796	1,396	-	32	-	-
23	Three Phase Primary	230	925,538	925,538	468,521	233,923	217,510	5,282	302	-
24	Single Phase Primary	235	757,259	757,259	501,312	250,295	-	5,652	-	-
25	Single Phase Secondary	231	1,843,488	1,843,488	1,220,404	609,324	-	13,760	-	-
26	Total Distribution Overhead System		4,169,223	4,168,535	2,493,205	1,244,809	393,728	28,110	8,683	688
27	Distribution Underground System (366/367)									
28	Three Phase Primary	230	68,925	68,925	34,891	17,420	16,198	393	23	-
29	Single Phase Primary	235	628,860	628,860	416,311	207,856	-	4,694	-	-
30	Single Phase Secondary	231	155,930	155,930	103,227	51,539	-	1,164	-	-
31	46kV Subtran/Distribution Underground Lines	122	21,525	21,512	10,154	5,070	5,891	114	283	12
32	Total Distribution Underground System		875,240	875,228	564,583	281,885	22,089	6,366	306	12
33	Distribution Line Equipment (368)									
34	Primary	230	131,476	131,476	66,555	33,230	30,898	750	43	-
35	Secondary	231	898,402	898,402	594,749	296,947	-	6,706	-	-
36	Total Distribution Line Equipment		1,029,878	1,029,878	661,304	330,177	30,898	7,456	43	-
37	Distribution Services (369)									
38	Residential Overhead & Burial Services	260	586,964	586,964	586,964	-	-	-	-	-
39	C&I Overhead & Burial Services	261	295,526	295,526	-	295,526	-	-	-	-
40	Total Distribution Services		882,491	882,491	586,964	295,526	-	-	-	-

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PLANT IN SERVICE (DISTRIBUTION & GENERAL)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional	
1	Distribution Metering Equipment (370)										
2	Metering Equipment (Mass)	170	660,213	660,203	511,294	106,721	41,923	145	120	11	
3	Total Distribution Metering Equipment		660,213	660,203	511,294	106,721	41,923	145	120	11	
4	Distribution Installations on Customer Premises (371)										
5	Installations on Premises L4 (Assignable)	DIR	20,900	20,900	-	-	-	20,900	-	-	
6	Total Distribution Installations on Customer Premises		20,900	20,900	-	-	-	20,900	-	-	
7	Distribution Streetlighting Equipment (373)										
8	Luminaires/Suspensions/Poles/Transformers	DIR	168,779	168,779	-	-	-	168,779	-	-	
9	Underground Cable & Conduits	233	9,376	9,299	-	-	-	9,299	-	76	
10	Photoelectric Switches	233	7,012	6,955	-	-	-	6,955	-	57	
11	Total Distribution Streetlighting Equipment		185,167	185,034	-	-	-	185,034	-	133	
12	Total Distribution Plant in Service		9,310,561	9,305,929	5,516,452	2,608,182	898,688	255,850	26,756	4,633	
13	Total Distribution Plant in Service	309	9,310,561	9,305,929	5,516,452	2,608,182	898,688	255,850	26,756	4,633	
14	Electric Plant Purchased & Sold	220	0	0	0	0	0	0	0	0	
15	General, Common & Intangible Plant in Service										
16	General: Production Related	220	0	0	0	0	0	0	0	0	
17	General: Merchant Control	226	0	0	0	0	0	0	0	-	
18	General: Power Control Center 138kV	301	0	0	0	0	0	0	0	0	
19	General: Power Control Center 46kV	302	0	0	0	0	0	0	0	0	
20	General: Functionalized	502	378,463	376,943	222,345	92,706	57,335	3,893	664	1,520	
21	General: Reallocated from/(to) Gas	DIR	0	-	-	-	-	-	-	-	
22	Common: Functionalized (Elec)	502	410,409	408,760	241,113	100,531	62,174	4,221	720	1,649	
23	Franchises & Consents - Generation	220	0	0	0	0	0	0	0	0	
24	Intangible PIS	502	815,102	811,827	478,869	199,662	123,483	8,384	1,430	3,275	
25	Total General, Common & Intangible Plant in Service		1,603,974	1,597,530	942,327	392,899	242,992	16,498	2,814	6,444	

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DEPRECIATION RESERVE (SUMMARY)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Production Depreciation Reserve									
2	Production Depreciation Reserve		2,626,131	2,600,088	1,313,771	600,656	675,219	6,597	3,845	26,043
3	Generation Step Ups		0	0	0	0	0	0	0	0
4	Total Production Depreciation Reserve		2,626,131	2,600,088	1,313,771	600,656	675,219	6,597	3,845	26,043
5	Transmission Depreciation Reserve									
6	Bulk Power Transm	225	-	-	-	-	-	-	-	-
7	Transm; Subtrans	122	0	0	0	0	0	0	0	0
8	Subtransmission	230	0	0	0	0	0	0	0	0
9	Total Transmission Depreciation Reserve		0	0	0	0	0	0	0	0
10	Distribution Depreciation Reserve									
11	Stations and Equipment		706,226	705,405	399,133	199,285	98,685	4,484	3,818	821
12	Overhead System		1,478,203	1,478,085	960,987	479,802	25,266	10,835	1,196	118
13	Underground System		324,031	323,847	153,488	76,634	87,774	1,731	4,221	183
14	Meters and Svc Drops		496,619	496,619	325,734	164,651	(181)	6,414	(1)	(0)
15	St Lgts and OPL		125,358	125,268	-	-	-	125,268	-	90
16	Total Distribution Depreciation Reserve		3,130,436	3,129,223	1,839,342	920,372	211,544	148,731	9,234	1,213
17		0								
18	Total General, Common & Intangible Depreciation Reserve		942,168	938,383	553,519	230,787	142,732	9,691	1,653	3,785
19	Total Depreciation Reserve	####	6,698,735	6,667,694	3,706,632	1,751,815	1,029,495	165,020	14,732	31,041

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DEPRECIATION RESERVE (PRODUCTION & TRANSMISSION)										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Reserve									
2	Fossil	220	1,834,050	1,815,862	917,517	419,489	471,563	4,607	2,685	18,188
3	Nuclear	220	0	0	0	0	0	0	0	0
4	Hydro	220	405,182	401,164	202,700	92,674	104,179	1,018	593	4,018
5	Other Production/Combustion Turbine	220	386,899	383,062	193,553	88,492	99,478	972	567	3,837
6	Jackson Gas Plant	220	0	0	0	0	0	0	0	0
7	7 Classics	220	0	0	0	0	0	0	0	0
6	Distribution GSUs	220	0	0	0	0	0	0	0	0
7	Total Production Depreciation Reserve		2,626,131	2,600,088	1,313,771	600,656	675,219	6,597	3,845	26,043
8	Transmission Depreciation Reserve									
9	Total Transmission Direct	DIR	0	-	-	-	-	-	-	-
10	Total Subtransmission	228	0	0	0	0	0	0	0	0
11	XYZ	123	0	0	0	0	0	0	0	0
12	Total Transmission Depreciation Reserve		0	0	0	0	0	0	0	0

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DEPRECIATION RESERVE (DISTRIBUTION)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Depreciation Reserve									
2	Distribution Land & Right of Way (360)		-	-	-	-	-	-	-	-
3	345/138kV Substations/Overheads (METC)	127	0	0	0	0	0	0	0	0
4	345/138kV Substations/Overheads	127	2,023	2,006	897	448	626	10	25	18
5	46/23kV Substations/Overheads	124	10,175	10,169	4,800	2,397	2,785	54	134	6
6	Substations/Overheads (Assignnable)	DIR	0	-	-	-	-	-	-	-
7	Overhead Lines Land & ROW	307	12,415	12,414	7,570	3,786	968	69	21	2
8	Total Distribution Land & ROW Depreciation Reserve		24,614	24,589	13,267	6,630	4,378	133	180	25
9	Distribution Substations & Equipment (361/362/363)									
10	Customer Substations (Assignnable)	DIR	0	-	-	-	-	-	-	-
11	138kV HV Subtran/Dist Substations	127	78,367	77,686	34,729	17,339	24,256	392	970	681
12	46kV Subtran /Dist Substations	122	200,951	200,836	94,797	47,330	54,994	1,069	2,647	115
13	Distribution Substations	230	14,570	14,570	7,375	3,682	3,424	83	5	-
14	Total Distribution, Substations & Equipment Depreciation Reserve		293,888	293,091	136,901	68,352	82,674	1,544	3,621	796
15	Distribution Overhead System (364/365)									
16	138kV HV Subtran/Dist Overhead Lines	127	8,091	8,021	3,586	1,790	2,504	40	100	70
17	46kV Subtran Overheads & Transformer Platforms	122	83,171	83,123	39,235	19,589	22,761	442	1,095	48
18	Single Phase Primary & Secondary	305	1,386,940	1,386,940	918,166	458,423	-	10,352	-	-
19	Total Distribution Overhead System Depreciation Reserve		1,478,203	1,478,085	960,987	479,802	25,266	10,835	1,196	118
20	Distribution Underground System (366/367)									
21	Distribution UG System	308	3,542	3,542	2,300	1,148	67	26	0	-
22	46kV Subtran/Distribution UG Lines	122	320,489	320,306	151,188	75,485	87,707	1,705	4,221	183
23	Total Distribution Underground System Depreciation Reserve		324,031	323,847	153,488	76,634	87,774	1,731	4,221	183
24	Distribution Line Equipment (368)									
25	Capacitors/Regulators/Transformers	312	387,725	387,725	248,965	124,304	11,632	2,807	16	-
26	Total Distribution Line Equipment Depreciation Reserve		387,725	387,725	248,965	124,304	11,632	2,807	16	-
27	Distribution Services (369)									
28	C&I and Residential Services	310	493,050	493,050	327,939	165,111	-	-	-	-
29	Total Distribution Services Depreciation Reserve		493,050	493,050	327,939	165,111	-	-	-	-

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DEPRECIATION RESERVE (DISTRIBUTION & GENERAL)										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Metering Equipment (370)									
2	Metering Equipment (Mass)	170	(2,846)	(2,846)	(2,204)	(460)	(181)	(1)	(1)	(0)
3	Total Distribution Metering Equipment Depreciation Reserve		(2,846)	(2,846)	(2,204)	(460)	(181)	(1)	(1)	(0)
4	Distribution Installations on Customer Premises (371)									
5	Installations on Premises L4 (Assignable)	DIR	6,415	6,415	-	-	-	6,415	-	-
6	Total Distribution Installations on Customer Premises Depreciation Reserve		6,415	6,415	-	-	-	6,415	-	-
7	Distribution Streetlighting Equipment (373)									
8	Streetlighting Equipment Depreciation Reserve	311	125,358	125,268	-	-	-	125,268	-	90
9	Total Distribution Streetlighting Depreciation Reserve		125,358	125,268	-	-	-	125,268	-	90
10	Total Distribution Depreciation Reserve		3,130,436	3,129,223	1,839,342	920,372	211,544	148,731	9,234	1,213
11	Total Distribution Depreciation Reserve	317	3,130,436	3,129,223	1,839,342	920,372	211,544	148,731	9,234	1,213
12	General, Common & Intangible Depreciation Reserve									
13	General: Power Control Center	314	0	0	0	0	0	0	0	0
14	General: Functionalized	502	124,100	123,601	72,908	30,399	18,800	1,276	218	499
15	General: Reallocated to Gas	DIR	0	-	-	-	-	-	-	-
16	Common: Functionalized	502	203,546	202,728	119,582	49,859	30,836	2,094	357	818
17	Intangible Amortization Reserve	502	614,523	612,054	361,029	150,529	93,096	6,321	1,078	2,469
18	Total General, Common & Intangible Depreciation Reserve		942,168	938,383	553,519	230,787	142,732	9,691	1,653	3,785

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CWIP										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Production CWIP									
2	Production	220	151,340	149,839	75,711	34,615	38,912	380	222	1,501
3	Production: Gas Plant	220	0	0	0	0	0	0	0	0
3	Production: 7 Classics	220	0	0	0	0	0	0	0	0
4	Total Production CWIP		151,340	149,839	75,711	34,615	38,912	380	222	1,501
5	Transmission CWIP									
6	Transmission	228	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
7	Subtransmission	122	0	0	0	0	0	0	0	0
8	Total Transmission CWIP		(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
####										
9	Distribution CWIP									
10	HV Distribution	122	51,129	51,100	24,120	12,042	13,992	272	673	29
11	Distribution	306	94,954	94,952	59,574	27,844	4,335	3,191	8	2
12	Other	220	0	0	0	0	0	0	0	0
13	Total Distribution CWIP		146,083	146,052	83,694	39,886	18,328	3,463	681	31
####										
14	General/Common/Intangible CWIP									
15	General	502	16,401	16,335	9,636	4,018	2,485	169	29	66
16	Intangible	502	23,893	23,797	14,037	5,853	3,620	246	42	96
17	Common	502	19,721	19,642	11,586	4,831	2,988	203	35	79
18	Plant Held for Future Use	502	0	0	0	0	0	0	0	0
19	Other	502	0	0	0	0	0	0	0	0
20	Total General, Common & Intangible CWIP		60,016	59,775	35,259	14,701	9,092	617	105	241

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WORKING CAPITAL										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Current Assets									
2	Cash & Cash Equivalents	316	47,915	47,726	26,904	12,442	7,429	840	110	190
3	Accts Receivable	143	297,842	296,065	144,226	74,346	74,249	3,117	127	1,778
4	Material and Supplies	316	102,014	101,610	57,281	26,490	15,817	1,788	234	404
5	Fuel Stock	100	47,517	46,952	19,936	11,350	15,237	334	95	566
6	Real & Personal Property Taxes	316	179,599	178,888	100,844	46,636	27,846	3,148	413	711
7	Other Cur Assets	502	472,857	470,957	277,801	115,828	71,635	4,864	830	1,900
8	Deferred Debits	502	1,088,431	1,084,059	639,449	266,615	164,890	11,195	1,909	4,373
9	Total Current Assets		2,236,176	2,226,255	1,266,441	553,706	377,103	25,286	3,718	9,921
10	Current Liabilities									
11	Accounts Payable	316	417,607	415,954	234,486	108,439	64,749	7,321	959	1,653
12	Customer Deposits	143	15,511	15,418	7,511	3,872	3,867	162	7	93
13	Dividends Declared	316	29,338	29,222	16,473	7,618	4,549	514	67	116
14	Accrued Interest	316	44,231	44,056	24,836	11,485	6,858	775	102	175
15	Accrued Taxes - Federal	502	6,176	6,151	3,628	1,513	936	64	11	25
16	Accrued Taxes - State	601	1,508	1,502	866	380	231	21	3	6
17	Accrued Taxes - R&PP & Other	316	143,778	143,208	80,731	37,334	22,292	2,520	330	569
18	Other Current Liabilities	502	40,608	40,445	23,857	9,947	6,152	418	71	163
19	Deferred CR	502	310,336	309,090	182,321	76,018	47,014	3,192	544	1,247
20	Total Current Liabilities		1,009,093	1,005,046	574,709	256,607	156,647	14,988	2,095	4,047
21	Total Working Capital		1,227,083	1,221,209	691,732	297,099	220,457	10,299	1,623	5,873

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ADJUSTMENTS TO RATE BASE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Additions to Rate Base									
2	Sales and Use Tax Adjustment	309	0	0	0	0	0	0	0	0
3		0 100	0	0	0	0	0	0	0	0
4		0 220	0	0	0	0	0	0	0	0
5		0 263	0	0	0	0	-	0	-	-
6	Total Additions		0	0	0	0	0	0	0	0
	Deductions to Rate Base									
7	Construction Funds Retained from Contractors	330	0	0	0	0	0	0	0	0
8	Customer Advances	309	59,839	59,809	35,454	16,763	5,776	1,644	172	30
9	0	263	0	0	0	0	-	0	-	-
10	0	100	0	0	0	0	0	0	0	0
11	0	100	0	0	0	0	0	0	0	0
12	Total Deductions		59,839	59,809	35,454	16,763	5,776	1,644	172	30
13	Total Adjustments to Rate Base		(59,839)	(59,809)	(35,454)	(16,763)	(5,776)	(1,644)	(172)	(30)

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REVENUE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
Rate Revenue										
1	Non PSCR Rate Revenue									
2	Revenue From Electric Sales	DIR	2,119,986	2,109,220	1,176,973	563,926	337,773	28,825	1,722	10,766
3	Provision for Rate Refund	DIR	0	-	-	-	-	-	-	-
4	Unbilled Revenue	DIR	0	-	-	-	-	-	-	-
5	Total Non PSCR Rate Revenue		2,119,986	2,109,220	1,176,973	563,926	337,773	28,825	1,722	10,766
6	PSCR Base Revenue	DIR	1,897,806	1,883,962	772,175	439,599	659,263	12,925	-	13,844
7	Unbilled PSCR Base Revenue	DIR	0	-	-	-	-	-	-	-
8	0	DIR	0	-	-	-	-	-	-	-
9	Total PSCR Rate Revenue		1,897,806	1,883,962	772,175	439,599	659,263	12,925	-	13,844
10	Total Rate Revenue		4,017,792	3,993,182	1,949,148	1,003,525	997,036	41,750	1,722	24,610
Non-PSCR Rate Revenue Credits										
12	450 Late Payment Charge Revenue	DIR	9,448	9,448	5,345	2,561	1,534	-	8	-
13	451 Miscellaneous Service & Reconnect Fees	253	1,061	1,061	457	268	328	8	0	-
14	454 Pole Rental Rev	307	11,341	11,340	6,915	3,458	884	63	19	2
15	454 Other Rents	316	0	0	0	0	0	0	0	0
16	456 Purchased Power Administrative Fees	100	934	923	392	223	300	7	2	11
17	Other Revenues	150	8,329	8,237	3,004	1,760	3,408	50	15	92
18	Job Work Revenue	414	12,822	12,822	7,821	3,594	1,149	230	28	-
19	LTILRR: Production	DIR	75,176	74,358	34,503	17,009	22,288	425	132	818
20	LTILRR: Delivery	390	929	929	559	251	101	16	3	1
21	0		0	-	-	-	-	-	-	-
22	0		0	-	-	-	-	-	-	-
23	0		0	-	-	-	-	-	-	-
24	0		0	-	-	-	-	-	-	-
25	0		0	-	-	-	-	-	-	-
26	Non PSCR Revenue Credits		120,040	119,117	58,997	29,124	29,991	799	207	923
27	PSCR Factor Revenue	DIR	134,502	134,502	55,128	31,384	47,067	923	-	-
28	Unbilled PSCR Factor Revenue	DIR	0	-	-	-	-	-	-	-
29	Intersystem Sales	222	118,852	118,814	60,033	27,448	30,856	302	176	38
30	GSG and GI Market Price Revenue	DIR	5,960	5,960	-	-	2,073	-	3,887	-
31	PSCR Revenue Credits		259,314	259,277	115,161	58,832	79,996	1,224	4,063	38
32	Total Revenue Credits		379,354	378,394	174,158	87,956	109,987	2,023	4,270	961
33	Total Revenue		4,397,146	4,371,576	2,123,306	1,091,481	1,107,023	43,773	5,992	25,571

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PRODUCTION O&M (1)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Fuel and Purchased Power									
2	Mid-Peak Summer Fuel for Gen	103	75,006	74,113	30,116	18,512	25,095	263	128	893
3	On-Peak Winter Fuel for Gen	105	194,331	192,017	75,166	47,900	67,615	987	349	2,314
4	Off-Peak Summer Fuel for Gen	104	82,601	81,618	35,625	18,418	26,675	717	182	983
5	Off-Peak Winter Fuel for Gen	106	154,052	152,218	67,405	33,850	49,056	1,570	337	1,834
6	Critical Summer Peak Energy	107	36,904	36,465	15,855	8,934	11,521	96	59	439
7	Total Fuel Expense		542,894	536,431	224,167	127,615	179,962	3,634	1,054	6,463
8	Mid-Peak Summer Purchased Power	103	54,984	54,330	22,077	13,571	18,396	193	94	655
9	On-Peak Winter Purchased Power	105	142,457	140,761	55,102	35,114	49,566	724	256	1,696
10	Off-Peak Summer Purchased Power	104	60,552	59,831	26,115	13,502	19,555	526	133	721
11	Off-Peak Winter Purchased Power	106	112,930	111,586	49,412	24,814	35,961	1,151	247	1,344
12	Critical Peak Summer Purchased Power	107	27,053	26,731	11,623	6,549	8,445	71	43	322
13	Purchased Power Capacity	220	742,330	734,969	371,364	169,788	190,865	1,865	1,087	7,362
14	Total Purchased & Interchange Power Expense		1,140,307	1,128,208	535,693	263,338	322,789	4,529	1,859	12,099
15	Total Fuel and Purchased & Interchange Power Expense		1,683,201	1,664,639	759,860	390,952	502,750	8,163	2,913	18,563

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PRODUCTION O&M (2)										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Fossil Plant O&M Expense									
2	Capacity Related Operations	220	36,951	36,584	18,485	8,451	9,501	93	54	366
3	Capacity Related Maintenance	220	4,478	4,434	2,240	1,024	1,151	11	7	44
4	Energy Related Operations	100	8,226	8,128	3,451	1,965	2,638	58	16	98
5	Energy Related Maintenance	100	38,509	38,050	16,157	9,198	12,348	270	77	458
6	Capacity Related Fuel Handling	220	0	0	0	0	0	0	0	0
7	Energy Related Fuel Handling	100	12,538	12,389	5,260	2,995	4,020	88	25	149
8	Total Fossil O&M Expense		100,702	99,586	45,594	23,633	29,659	520	179	1,116
9	Nuclear Plant O&M Expense									
10	Capacity Related Operations	220	0	0	0	0	0	0	0	0
11	Capacity Related Maintenance	220	0	0	0	0	0	0	0	0
12	Energy Related Maintenance	100	0	0	0	0	0	0	0	0
13	XYZ	220	0	0	0	0	0	0	0	0
14	XYZ	220	0	0	0	0	0	0	0	0
15	Total Nuclear Plant O&M Expense		0	0	0	0	0	0	0	0
16	Hydro Plant O&M Expense									
17	Capacity Related Operations	220	9,619	9,523	4,812	2,200	2,473	24	14	95
18	Capacity Related Maintenance	220	6,746	6,679	3,375	1,543	1,734	17	10	67
19	Energy Related Operations	100	725	717	304	173	233	5	1	9
20	Energy Related Maintenance	100	5,069	5,009	2,127	1,211	1,625	36	10	60
21	XYZ	220	0	0	0	0	0	0	0	0
22	Total Hydro O&M Total		22,159	21,928	10,618	5,127	6,066	82	36	231
23	Peaker O&M Expense									
24	Capacity Related Operations & Maintenance	220	67,853	67,180	33,945	15,519	17,446	170	99	673
25	Energy Related Operations & Maintenance	100	0	0	0	0	0	0	0	0
26	XYZ	100	0	0	0	0	0	0	0	0
27	Total Peaker O&M Expense		67,853	67,180	33,945	15,519	17,446	170	99	673
28	Other O&M Expense									
29	Capacity Related Sys Cntl Load Disp	220	10,761	10,654	5,383	2,461	2,767	27	16	107
30	Energy Related Sys Cntl Load Disp	100	0	0	0	0	0	0	0	0
31	Total Other O&M Expense		10,761	10,654	5,383	2,461	2,767	27	16	107
32	Disposition of Allowances	220	0	0	0	0	0	0	0	0
33	Total Production O&M (excluding Fuel and P&I)		201,475	199,348	95,540	46,741	55,937	800	330	2,127
34	Total Production O&M Expense		1,884,676	1,863,986	855,400	437,693	558,688	8,962	3,243	20,690

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TRANSMISSION O&M										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Transmission O&M Expense									
2	Transmission	120	477,556	472,640	222,685	105,296	141,584	2,334	741	4,916
3	Reclassified Transmission	127	0	0	0	0	0	0	0	0
4	Other	120	0	0	0	0	0	0	0	0
5	Total Transmission O&M Expense		477,556	472,640	222,685	105,296	141,584	2,334	741	4,916
6	Other O&M Adjustments									
7	Tax Benefit of Proforma Interest & Interest Synchronization Adjustment	150	0	0	0	0	0	0	0	0
8	Other Advertising Programs - Disallowance	412	0	0	0	0	0	0	0	0
9	Income Tax Effect of Interest	390	0	0	0	0	0	0	0	0
10	Charitable, Civic, Dues & Donations	412	0	0	0	0	0	0	0	0
11	Transmission reclass (indirect costs)	DIR	0	-	-	-	-	-	-	-
12	Streetlighting O&M	DIR	0	-	-	-	-	-	-	-
13	Customer O&M	411	0	0	0	0	0	0	0	0
14	Administrative and General O&M	412	0	0	0	0	0	0	0	0
15	Other O&M Inflation	443	0	0	0	0	0	0	0	0
16	Other O&M Adjmts	438	0	0	0	0	0	0	0	0
17	Total Other O&M Adjustments		0	0	0	0	0	0	0	0

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DISTRIBUTION O&M										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Distribution Operation Expense									
2	580 Supv & Engineering - Distribution	400	31,106	31,102	19,935	8,039	1,854	1,241	32	4
3	580 Supv & Engineering - 138kV	402	100	99	51	23	25	0	0	1
4	580 Supv & Engineering - 46kV	404	1,513	1,512	714	356	414	8	20	1
5	581 Load Dispatch - Distribution	301	0	0	0	0	0	0	0	0
6	582 Station Expense - Distribution	230	1,485	1,485	752	375	349	8	0	-
7	582 Station Expense - 138kV	303	0	0	0	0	0	0	0	0
8	582 Station Expense - 46kV	304	0	0	0	0	0	0	0	0
9	583 Overhead Expense - Distribution	307	16,231	16,228	9,896	4,949	1,265	90	28	2
10	583 Overhead Expense - 138kV	121	57	57	29	13	14	0	0	1
11	583 Overhead Expense - 46kV	122	862	861	407	203	236	5	11	0
12	584 Underground Exp	308	6,338	6,338	4,116	2,055	120	46	0	-
13	585 St Lt	311	1,160	1,159	-	-	-	1,159	-	1
14	586 - Metering Expense	313	1,364	1,364	1,056	220	87	0	0	0
15	587 Cust Instl Expense	160	5,514	5,514	4,852	648	12	2	0	0
16	588 Miscellaneous	400	21,312	21,309	13,658	5,508	1,271	850	22	3
17	589 Rents	309	2,170	2,169	1,286	608	209	60	6	1
18	Total Distribution Operation Expense		89,212	89,198	56,752	22,998	5,857	3,471	120	13
19	Distribution Maintenance Expense									
20	590 Supv & Engineering - Distribution	401	5,865	5,864	3,581	1,748	481	44	8	1
21	590 Supv & Engineering - 138kV	403	113	112	53	25	32	0	1	1
22	590 Supv & Engineering - 46kV	405	837	836	395	197	229	4	11	0
23	591 Structures - Distribution	230	492	492	249	124	116	3	0	-
24	591 Structures - 138kV	303	30	30	13	7	9	0	0	0
25	591 Structures - 46kV	304	80	79	38	19	22	0	1	0
26	592 Station Equipment - Distribution	230	8,436	8,436	4,270	2,132	1,983	48	3	-
27	592 Station Equipment - 138kV	303	1,209	1,198	536	267	374	6	15	11
28	592 Station Equipment - 46kV	304	3,211	3,209	1,515	756	879	17	42	2
29	593 Overhead Lines - Distribution	307	124,531	124,514	75,930	37,971	9,706	693	214	17
30	593 Overhead Lines - 138kV	224	771	771	397	177	194	2	1	-
31	593 Overhead Lines - 46kV	122	11,632	11,625	5,487	2,740	3,183	62	153	7
32	594 Underground Lines - Distribution	308	3,195	3,195	2,075	1,036	61	23	0	-
33	594 Underground Lines - 138kV	121	0	0	0	0	0	0	0	0
34	594 Underground Lines - 46kV	122	0	0	0	0	0	0	0	0
35	595 Line Xfmrs	312	10,185	10,185	6,540	3,265	306	74	0	-
36	596 St Lts & OPL	311	312	311	-	-	-	311	-	0
37	597 Meters	313	4,870	4,870	3,772	787	309	1	1	0
38	598 Miscellaneous	401	153	153	93	46	13	1	0	0
39	Total Distribution Maintenance Expense		175,919	175,881	104,944	51,298	17,896	1,291	452	38
40	Total Distribution O&M Expense		265,131	265,079	161,696	74,296	23,753	4,762	572	52

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CUSTOMER RELATED AND ADMINISTRATIVE & GENERAL EXPENSE										
Line No.	(a) Description	(b) Alloc	(c) Total Electric	(d) Total Jurisdictional Electric	(e) Total Residential	(f) Total Commercial Secondary	(g) Total Primary	(h) Total Lighting & Unmetered	(i) Rate GSG	(j) Total Non Jurisdictional
1	Customer Accounts Expense									
2	901 Supervision	408	7,119	7,119	6,267	836	14	2	0	0
3	902 Meter Reading	263	6,090	6,090	5,372	717	-	1	-	-
4	903 Rcrds & Collection	160	22,806	22,806	20,066	2,678	51	10	0	0
5	904 Uncollectibles	264	18,077	18,077	15,913	2,124	40	-	0	0
6	905 Misc Expenses	408	0	0	0	0	0	0	0	0
7	Total Customer Accounts		54,093	54,093	47,618	6,356	105	13	0	0
8	Customer Services Expense									
9	907 Supervision	160	3,436	3,436	3,024	404	8	2	0	0
10	908 Customer Assist	603	12,920	12,778	4,660	2,730	5,286	78	24	142
11	909 Info & Inst	160	556	556	489	65	1	0	0	0
12	910 Miscellaneous	160	0	0	0	0	0	0	0	0
13	Total Customer Services		16,912	16,770	8,173	3,199	5,295	80	24	142
14	Sales Expense									
15	911 Supervision	160	0	0	0	0	0	0	0	0
16	912 Demo & Selling	160	365	365	321	43	1	0	0	0
17	913 Advertising	160	0	0	0	0	0	0	0	0
18	916 Miscellaneous	160	0	0	0	0	0	0	0	0
19	Total Sales Expense		365	365	321	43	1	0	0	0
20	Administrative & General Expense									
21	Production	500	53,303	52,775	26,666	12,192	13,705	134	78	529
22	HV Distribution	406	3,290	3,286	1,553	771	904	17	41	4
23	Distribution	407	73,664	73,656	45,792	20,927	5,446	1,396	94	8
24	Customer	409	16,461	16,428	12,934	2,215	1,252	22	6	33
25	Total Administrative & General Expense		146,719	146,145	86,945	36,105	21,307	1,569	219	574
26	Total O&M Expense (excluding PSCR Expense)		684,695	681,800	400,294	166,739	106,398	7,223	1,145	2,895
27	Total O&M Expense		2,845,452	2,819,078	1,382,839	662,987	750,733	17,720	4,799	26,374

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DEPRECIATION EXPENSE (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Expense									
2	Production Depreciation Expense		309,278	306,211	154,722	70,739	79,520	777	453	3,067
3	GSU Depreciation Expense		0	0	0	0	0	0	0	0
4	Test Year Production Change	220	0	0	0	0	0	0	0	0
5	Total Production		309,278	306,211	154,722	70,739	79,520	777	453	3,067
6	Transmission									
7	Bulk Power Transm		0	-	-	-	-	-	-	-
8	Transm; Subtrans		0	0	0	0	0	0	0	0
9	Subtransmission		0	0	0	0	0	0	0	0
10	Total Transmission		0	0	0	0	0	0	0	0
11	Distribution									
12	Stations and Equipment		66,468	66,375	36,464	18,206	10,840	410	454	93
13	Overhead System		117,722	117,705	75,309	37,600	3,768	849	178	17
14	Underground System		20,930	20,930	13,593	6,786	397	153	1	-
15	Meters and Svc Drops		71,255	71,255	51,210	16,852	2,485	701	7	1
16	St Lgts and OPL		6,762	6,757	-	-	-	6,757	-	5
17	Sales and Use Tax Adjustment	444	129	129	80	36	8	4	0	0
18	Total Distribution		283,267	283,151	176,656	79,481	17,498	8,875	641	116
19	General/Common/Intangible									
20	Total Gen/Comm/Int		113,037	112,583	66,409	27,689	17,124	1,163	198	454
21	Test Year Gen/Comm/Int Change	445	0	0	0	0	0	0	0	0
22	Total General/Common/Intangible Dep Expense		113,037	112,583	66,409	27,689	17,124	1,163	198	454
23	Other Amortization Expense		0	0	0	0	0	0	0	0
24	Total Depreciation & Amortization Expense		705,582	701,945	397,787	177,909	114,142	10,815	1,292	3,637

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DEPRECIATION EXPENSE (PRODUCTION & TRANSMISSION)										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Production Depreciation Expense									
2	Direct		-	-	-	-	-	-	-	-
3	Fossil	220	224,639	222,411	112,380	51,380	57,758	564	329	2,228
4	Nuclear	220	0	0	0	0	0	0	0	0
5	Hydro	220	58,105	57,529	29,068	13,290	14,940	146	85	576
6	Other Production	220	26,534	26,270	13,274	6,069	6,822	67	39	263
7	Distribution GSUs	220	0	0	0	0	0	0	0	0
8	Jackson Gas Plant	220	0	0	0	0	0	0	0	0
9	7 Classics	220	0	-	-	-	-	-	-	-
10	Total Production Depreciation Expense		309,278	306,211	154,722	70,739	79,520	777	453	3,067
11	Transmission Depreciation Expense									
12	Direct		0	-	-	-	-	-	-	-
13	Transmission	127	0	0	0	0	0	0	0	0
14	Subtransmission	123	0	0	0	0	0	0	0	0
15	Total Transmission Depreciation Expense		0	0	0	0	0	0	0	0

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DEPRECIATION EXPENSE (DISTRIBUTION & GENERAL)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution Depreciation Expense									
2	Distribution Land & Right of Way (360)		-	-	-	-	-	-	-	-
3	138kV Substations/Overheads (METC)	127	0	0	0	0	0	0	0	0
4	138kV Substations/Overheads	127	150	149	66	33	46	1	2	1
5	46kV Substations/Overheads	124	622	621	293	146	170	3	8	0
6	Substations/Overheads (Assignable)	DIR	0	-	-	-	-	-	-	-
7	Overhead Lines Land & ROW	307	503	503	307	153	39	3	1	0
8	Total Distribution Land & ROW Depreciation Expense		1,275	1,273	667	333	256	7	11	2
9	Distribution Substations & Equipment (361/362/363)									
10	Customer Substations (Assignable)	DIR	0	-	-	-	-	-	-	-
11	138kV HV Subtran/Dist Substations	127	8,811	8,735	3,905	1,950	2,727	44	109	77
12	Distribution Substations	124	25,302	25,288	11,936	5,960	6,924	135	333	14
13	Total Distribution Substations & Equipment Depreciation Expense		34,114	34,023	15,841	7,909	9,652	179	442	91
14	Distribution Overhead System (364/365)									
15	138kV HV Subtran/Dist Overhead Lines	127	1,183	1,172	524	262	366	6	15	10
16	46kV Subtran Overheads & Transformer Platforms	122	12,432	12,425	5,865	2,928	3,402	66	164	7
17	Overhead System	305	104,108	104,108	68,920	34,411	-	777	-	-
19	Total Distribution Overhead Depreciation Expense		117,722	117,705	75,309	37,600	3,768	849	178	17
20	Distribution Underground System (366/367)									
21	Underground System	308	20,930	20,930	13,593	6,786	397	153	1	-
22	Total Distribution Underground Depreciation Expense		20,930	20,930	13,593	6,786	397	153	1	-
23	Distribution Line Equipment (368)									
24	Line Equipment	312	31,080	31,080	19,957	9,964	932	225	1	-
25	Total Distribution Line Equipment Depreciation Expense		31,080	31,080	19,957	9,964	932	225	1	-
26	Distribution Services (369)									
27	Overhead & Underground Services	310	31,435	31,435	20,908	10,527	-	-	-	-
28	Total Distribution Services Depreciation Expense		31,435	31,435	20,908	10,527	-	-	-	-

Projected 12-Month Period Ending Dec 31, 2021
Version 2
4CP 89/0/11 Production and 12CP Transmission
(thousands of dollars)

DEPRECIATION EXPENSE (DISTRIBUTION & GENERAL)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Distribution (cont.)									
2	Distribution Metering Equipment (370)		-	-	-	-	-	-	-	-
3	Metering Equipment	170	39,128	39,127	30,302	6,325	2,485	9	7	1
4	Total Distribution Metering Equipment Depreciation Expense		39,128	39,127	30,302	6,325	2,485	9	7	1
5	Distribution Installations on Customer Premises Depreciation Expense (371)	DIR	692	692	-	-	-	692	-	-
6	Distribution Streetlighting Equipment Depreciation Expense (373)	311	6,762	6,757	-	-	-	6,757	-	5
7	Total Distribution Depreciation Expense		283,138	283,022	176,576	79,445	17,490	8,871	641	116
8	General/Common/Intangible									
9	General	502	9,750	9,711	5,728	2,388	1,477	100	17	39
10	Common	502	22,116	22,027	12,993	5,417	3,350	227	39	89
11	Intangible Amortization	502	81,171	80,845	47,688	19,883	12,297	835	142	326
12	Total Gen/Comm/Int Depreciation Expense		113,037	112,583	66,409	27,689	17,124	1,163	198	454
13	Other Amortization									
14	Amort of 7 Classics Inventory	220	0	0	0	0	0	0	0	0
15	AFUDC in Excess of FERC Rate	330	0	0	0	0	0	0	0	0
16	Securitized Regulatory Assets	150	0	0	0	0	0	0	0	0
17	ARO Accretion/Transition Expense	220	0	0	0	0	0	0	0	0
18	Total Other Amortization Expense		0	0	0	0	0	0	0	0
19	Total Depreciation & Amortization Expense		705,453	701,816	397,707	177,872	114,134	10,811	1,292	3,637

Projected 12-Month Period Ending Dec 31, 2021
Version 2
4CP 89/0/11 Production and 12CP Transmission
(thousands of dollars)

TAX										
(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	City Income Tax	150	1,077	1,065	388	227	440	6	2	12
2	Michigan Single Business Tax	601	0	0	0	0	0	0	0	0
3	Michigan Business Tax	439	30,672	30,946	10,728	9,572	10,095	583	(32)	(274)
RP&P Tax										
3	R&PP Taxes - Prod	220	71,896	71,183	35,967	16,444	18,486	181	105	713
4	R&PP Taxes - High Voltage Dist	302	25,086	25,072	11,834	5,909	6,865	133	330	14
5	R&PP Taxes - Low Voltage Dist	306	82,804	82,803	51,951	24,281	3,781	2,783	7	2
6	R&PP Taxes - General	315	4,505	4,487	2,650	1,104	679	47	8	18
7	R&PP Taxes - Common/Intangible	502	14,479	14,421	8,507	3,547	2,194	149	25	58
8	R&PP Taxes - PHFFU	226	29	29	14	7	8	0	0	-
9	R&PP Taxes - CWIP	330	0	0	0	0	0	0	0	0
10	Total R&PP Taxes		198,800	197,995	110,924	51,291	32,012	3,293	476	805
Payroll and Miscellaneous Tax										
11	Payroll Related Taxes	502	21,973	21,885	12,909	5,382	3,329	226	39	88
12	Miscellaneous General Taxes	150	0	0	0	0	0	0	0	0
13	Total Payroll/Miscellaneous Taxes		21,973	21,885	12,909	5,382	3,329	226	39	88
14	MPSC Assessment Fee	150	11,332	11,207	4,087	2,394	4,636	68	21	125
15	Total Other Taxes		263,853	263,097	139,036	68,868	50,512	4,176	505	756
16	Federal Income Tax Provision	439	53,928	54,409	18,861	16,830	17,749	1,025	(56)	(481)
17	Total Taxes Other Than Income		233,181	232,151	128,308	59,296	40,417	3,594	537	1,030
18	Total Income Taxes		84,600	85,355	29,589	26,403	27,844	1,607	(88)	(755)
19	Total Taxes		317,781	317,506	157,897	85,698	68,261	5,201	449	275

INCOME STATEMENT ADJUSTMENTS

Wages And Salaries	502	0	0	0	0	0	0	0
Pension Expense	502	0	0	0	0	0	0	0
Employee Life,Med Ins	502	0	0	0	0	0	0	0
Employee Savings Plan	502	0	0	0	0	0	0	0
Payroll Taxes	502	0	0	0	0	0	0	0
Prop,Inj,Dmg Insurance	600	0	0	0	0	0	0	0
Tot Increase Beyond Year		0	0	0	0	0	0	0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Electric Cost-of-Service Study
Capacity Related Cost and Charge Calculation
Test Year 2021

Case No.: U-20697
Exhibit No.: A-17 (JCA-3)
Page: 1 of 1
Witness: JCAponte
Date: February 2020

Line No.	(a) <u>Description</u>	(b) <u>Total Electric</u> (thousands of dollars)	(c) <u>Capacity Charge</u>	(d) <u>Formulae</u>
1	Total Production Related Cost	\$ 2,826,337		
	<u>Non-Capacity Related Cost:</u>			
2	Fuel Expense	\$ 542,894		
3	Purchased & Interchanged	397,977		
4	Energy Related Other O&M Expense	65,068		
5	PSCR Revenue Credits	(259,314)		
6	Non-PSCR Revenue Credits	(86,097)		
7	Transmission Expense	477,556		
8	Total Non-Capacity Related Cost	\$ 1,138,083		Σ Lines 2:7
9	Total Capacity Related Cost	\$ 1,688,254		Line 1 - Line 8
	<u>Offsets:</u>			
10	Energy Market Sales	\$ 962,313		
11	Off-System Energy Sales	11,475		
12	Ancillary Service Sales	12,128		
13	Bilateral Energy Sales	-		
14	Total Revenue	\$ 985,916		Σ Lines 10:13
15	Related Fuel Cost	480,460		
16	Total Revenue Less Fuel Cost	\$ 505,456		Line 14 - Line 15
17	Net Capacity Cost	\$ 1,182,798		Line 9 - Line 16
18	Capacity Charge Demand (MW)		7,238	
19	Capacity Charge (\$/MW-Day)		\$447.71	$[(\text{Line 17} \times 1,000) \div \text{Line 18}] \div 365$

Source:
Lines 1-7: Electric Cost-of-Service Study model, "Prod" tab
Lines 10,15,18: Testimony of Company Witness KGTroyer
Lines 11-13: 2018 Actuals from accounting records

Line No.	(a) Generating Plants Cost (in thousands)	(b) Status	(c) Plant Type	(d) 2016	(e) 2017	(f) 2018
1	Campbell 1&2		Coal	\$ 1,004,790	\$ 1,018,672	\$ 1,052,381
2	BC Cobb 4&5	RETIRED	Coal	821	-	-
3	Whiting	RETIRED	Coal	508	-	-
4	Karn 1&2		Coal	1,176,869	1,170,630	1,183,311
5	Karn 3&4		Gas	323,197	341,733	348,060
6	Weadock 7&8	RETIRED	Coal	3,956	-	-
7	Campbell 3		Coal	1,647,655	1,654,427	1,687,701
8	Zeeland		Gas	338,463	339,551	378,577
9	Weadock	RETIRED	Gas	1,622	-	-
10	Thetford		Gas	20,795	20,698	20,743
11	Whiting	RETIRED	Gas	1,738	-	-
12	Morrow	RETIRED	Gas	233	224	-
13	Gaylord		Gas	5,077	5,195	318
14	Straits		Gas	2,199	2,199	52
15	Campbell	RETIRED	Gas	1,788	1,782	-
16	Jackson		Gas	364,000	374,120	384,907
17	Hardy		Storage	14,913	17,574	20,353
18	Hodenpyl		Run-of-River Hydro	9,725	10,228	10,852
19	Tippy		Run-of-River Hydro	9,651	9,746	10,989
20	Ludington-51%		Pumped Storage	290,224	353,898	411,124
21	Foote - FPC #2436		Hydro	7,254	7,730	8,579
22	Cooke - FPC #2450		Hydro	3,873	3,929	6,152
23	Five Channels - FPC #2453		Hydro	5,483	5,918	6,686
24	Loud - FPC #2449		Hydro	3,929	3,979	4,589
25	Alcona - FPC #2447		Hydro	5,502	5,583	6,560
26	Mio - FPC #2448		Hydro	6,081	6,204	8,794
27	Croton - FPC #2468		Hydro	12,089	16,989	17,623
28	Rogers - FPC #2451		Hydro	8,617	11,679	13,259
29	Webber - FPC #2566		Hydro	11,084	12,887	13,139
30	Calkins Bridge (Allegan) - FPC #785		Hydro	5,563	7,026	7,902
31	Lake Winds Energy Park		Wind	225,512	226,953	227,158
32	Cross Winds Energy Park		Wind	241,624	240,767	326,035
33	GVSU Solar Garden		Solar	7,605	7,960	7,960
34	WMU Solar Garden		Solar	4,125	3,817	3,824
35	Circuit West Solar		Solar			2,885
36	Total Plants	Σ(1:35)		\$ 5,766,567	\$ 5,882,096	\$ 6,170,512
	Base Load (Coal Plants)	Min Load		2016	2017	2018
37	BC Cobb 4&5	56%		\$ 462	\$ -	\$ -
38	Whiting	52%		265	-	-
39	Weadock 7&8	69%		2,744	-	-
40	Campbell 1&2	34%		338,884	343,565	354,935
41	Karn 1&2	61%		718,168	714,361	722,099
42	Campbell 3	48%		783,665	786,886	802,711
43	Total Base Load (Coal Plants)	Σ(37:42)		\$ 1,844,187	\$ 1,844,812	\$ 1,879,745
44	Total Base Load over Total Plants	Ln43÷Ln36		32.0%	31.4%	30.5%
	Load at generation (kw)					
45	Minimum Hour			2,656,380	2,674,215	2,748,047
46	Maximum Hour or System Peak			7,635,347	7,057,054	7,567,978
47	Min hr as % of Max hr	Ln45÷Ln46		34.8%	37.9%	36.3%
48	Percent to be allocated on energy	Ln44xLn47		11.1%	11.9%	11.1%

Source:
MPSC Form P-521, pages 402-410
Part III, Attachment 57

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy
 Substation Ownership Credit
 General Service Primary Demand
 (thousands of dollars)

Case No.: U-20697
 Exhibit No.: A-19 (JCA-5)
 Page: 1 of 1
 Witness: JCAponte
 Date: February 2020

<u>Line</u>	<u>(a)</u> <u>Description</u>	<u>(b)</u> <u>Rate GPD</u> <u>Voltage 1</u>	<u>(c)</u> <u>Rate GPD</u> <u>Voltage 2</u>
1	Plant in Service	\$ 15,900	\$ 46,484
2	Depreciation Reserve	<u>(3,565)</u>	<u>(10,087)</u>
3	Net Plant	12,335	36,396
4	Working Capital	1,650	2,623
5	CWIP (HV Distribution)	-	1,182
6	Substation Rate Base	\$ 13,985	\$ 40,202
7	Pre-Tax ROE	7.10%	7.10%
8	Pre-Tax Return	\$ 992	\$ 2,853
9	Depreciation Expense	397	1,231
10	O&M Expense	55	156
11	Other Taxes	582	1,387
12	Revenue Credits	<u>(212)</u>	<u>(217)</u>
13	Total Revenue Requirement	\$ 1,814	\$ 5,409
14	Max Demand (MW)	<u>5,322</u>	<u>5,618</u>
15	Substation Ownership Credit (kW)	\$ 0.34	\$ 0.96

Source: Exhibit A-16 (JCA-2) Test Year COSS – Version 2
 WP-HWM-3
 WP-JCA-193-194

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Allocation of Surcharges

(thousands of dollars)

Case No.: U-20697

Exhibit No.: A-20 (JCA-6)

Page: 1 of 3

Witness: JCAponte

Date: February 2020

Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
Financial Compensation Mechanism (FCM)										
1	FCM (Energy) ¹		2,354	2,326	993	542	768	18	5	28
2	Mid-Peak Summer Fuel for Gen	103	325	321	131	80	109	1	1	4
3	On-Peak Winter Fuel for Gen	105	843	832	363	188	272	7	2	10
4	Off-Peak Summer Fuel for Gen	104	358	354	139	88	125	2	1	4
5	Off-Peak Winter Fuel for Gen	106	668	660	292	147	213	7	1	8
6	Critical Summer Peak Energy	107	160	158	69	39	50	0	0	2
7	FCM (Capacity) ¹	220	677	670	345	154	169	1	1	7
8	Total FCM		\$ 3,031	\$ 2,996	\$ 1,339	\$ 696	\$ 937	\$ 19	\$ 6	\$ 35
FCM Allocators²										
9	Energy On-Peak @ Generation Summer	103	100.0000	98.8095	40.1509	24.6812	33.4568	0.3504	0.1702	1.1905
10	Energy On-Peak @ Generation Non-Summer	105	100.0000	98.8095	43.1289	22.2976	32.2943	0.8686	0.2201	1.1905
11	Energy Off-Peak @ Generation Summer	104	100.0000	98.8095	38.6795	24.6489	34.7938	0.5079	0.1794	1.1905
12	Energy Off-Peak @ Generation Non-Summer	106	100.0000	98.8095	43.7547	21.9730	31.8438	1.0192	0.2189	1.1905
13	Energy Critical On-Peak @ Gen	107	100.0000	98.8095	42.9628	24.2088	31.2177	0.2614	0.1588	1.1905
14	4CP 89/0/11	220	100.0000	99.0083	51.0120	22.7414	24.9198	0.1954	0.1398	0.9917
15	Demand Response Refund³		\$ (2,125)	\$ (2,101)	\$ (1,014)	\$ (470)	\$ (608)	\$ (7)	\$ (3)	\$ (24)
16	Electric Rate Case Deferral³		\$ 12,600	\$ 12,606	\$ 7,858	\$ 3,323	\$ 1,151	\$ 243	\$ 31	\$ (7)

(1) Total FCM energy and capacity source: Exhibit No. A-110 (KGT-2). Summer and winter peak split source: WP-JCA-89

(2) Exhibit No. A-16 (JCA-2) Pages 10-12

(3) WP-JCA-201_203

Mid-Peak Summer Fuel for Gen	13.82%
On-Peak Winter Fuel for Gen	35.80%
Off-Peak Summer Fuel for Gen	15.21%
Off-Peak Winter Fuel for Gen	28.38%
Critical Summer Peak Energy	6.80%

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Allocation of Surcharges

(thousands of dollars)

Case No.: U-20697

Exhibit No.: A-20 (JCA-6)

Page: 2 of 3

Witness: JCAponte

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
										Total
Line	<u>Description</u>	<u>Alloc</u>	<u>Rate</u>	<u>Rate</u>	<u>Total</u>	<u>Rate</u>	<u>Rate</u>	<u>Rate</u>	<u>Rate</u>	<u>Commercial</u>
No.			<u>RS</u>	<u>RT</u>	<u>Residential</u>	<u>GS</u>	<u>GSD</u>	<u>GS</u>	<u>GSD</u>	<u>Secondary</u>
								<u>GEI</u>	<u>GEI</u>	
1	FCM (Energy) ¹		993	-	993	283	241	6	12	542
2	Mid-Peak Summer Fuel for Gen	103	131	-	131	43	35	1	2	80
3	On-Peak Winter Fuel for Gen	105	363	-	363	96	87	2	4	188
4	Off-Peak Summer Fuel for Gen	104	139	-	139	47	37	1	2	88
5	Off-Peak Winter Fuel for Gen	106	292	-	292	77	65	2	3	147
6	Critical Summer Peak Energy	107	69	-	69	21	17	0	1	39
7	FCM (Capacity) ¹	220	345	-	345	84	65	2	3	154
8	Total		\$ 1,339	\$ -	\$ 1,339	\$ 367	\$ 306	\$ 8	\$ 15	\$ 696
Allocators										
9	Energy On-Peak @ Generation Summer	103	40.1509	-	40.1509	13.2360	10.6512	0.3076	0.4864	24.6812
10	Energy On-Peak @ Generation Non-Summer	105	43.1289	-	43.1289	11.3450	10.2904	0.2129	0.4493	22.2976
11	Energy Off-Peak @ Generation Summer	104	38.6795	-	38.6795	13.2289	10.4062	0.3593	0.6545	24.6489
12	Energy Off-Peak @ Generation Non-Summer	106	43.7547	-	43.7547	11.4626	9.7701	0.2431	0.4972	21.9730
13	Energy Critical On-Peak @ Gen	107	42.9628	-	42.9628	12.8908	10.5717	0.2827	0.4636	24.2088
14	4CP 89/0/11	220	51.0120	-	51.0120	12.3850	9.6272	0.3150	0.4142	22.7414
15	Demand Response Refund³		\$ (1,014)	\$ -	\$ (1,014)	\$ (254)	\$ (201)	\$ (6)	\$ (9)	\$ (470)
16	Electric Rate Case Deferral³		\$ 7,858	\$ -	\$ 7,858	\$ 2,069	\$ 1,090	\$ 64	\$ 101	\$ 3,323

(1) Total FCM energy and capacity source: Exhibit No. A-110 (KGT-2). Summer and winter peak split source: WP-JCA-89

(2) Exhibit No. A-16 (JCA-2) Pages 10-12

(3) WP-JCA-201_203

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Allocation of Surcharges

(thousands of dollars)

Case No.: U-20697

Exhibit No.: A-20 (JCA-6)

Page: 3 of 3

Witness: JCAponte

Date: February 2020

Line No.	(a) Description	(b) Alloc	(c) Rate GP	(d) Rate GPTU	(e) Rate GPD Vlt 1	(f) Rate GPD Vlt 2	(g) Rate GPD Vlt 3	(h) Rate GP GEI	(i) Rate EIP	(j) Rate GPD GEI Vlt 1	(k) Rate GPD GEI Vlt 2	(l) Rate GPD GEI Vlt 3	(m) Total Primary	(n) Rate GML	(o) Rate GUL	(p) Rate GU-XL	(q) Rate GU	(r) Total Lighting & Unmetered
1	FCM (Energy) ¹		58	313	92	84	178	8	25	-	2	9	768	1	7	2	8	18
2	Mid-Peak Summer Fuel for Gen	103	10	46	11	11	24	1	4	-	0	1	109	0	0	0	1	1
3	On-Peak Winter Fuel for Gen	105	19	109	35	31	65	2	8	-	1	3	272	1	3	1	3	7
4	Off-Peak Summer Fuel for Gen	104	11	53	14	13	27	2	4	-	0	2	125	0	1	0	1	2
5	Off-Peak Winter Fuel for Gen	106	14	85	28	25	49	2	7	-	0	2	213	1	3	1	2	7
6	Critical Summer Peak Energy	107	5	21	5	5	12	1	2	-	0	1	50	-	-	-	0	0
7	FCM (Capacity) ¹	220	17	76	9	15	44	2	-	-	1	3	169	-	-	-	1	1
8	Total		\$ 76	\$ 390	\$ 102	\$ 99	\$ 222	\$ 10	\$ 25	\$ -	\$ 2	\$ 12	\$ 937	\$ 1	\$ 7	\$ 2	\$ 9	\$ 19
Allocators																		
9	Energy On-Peak @ Generation Summer	103	3.0030	14.0994	3.4895	3.3164	7.5327	0.4113	1.1155	-	0.0764	0.4127	33.4568	0.0109	0.0640	0.0167	0.2588	0.35041
10	Energy On-Peak @ Generation Non-Summer	105	2.2180	12.9019	4.0991	3.6405	7.7541	0.2639	0.9733	-	0.0804	0.3631	32.2943	0.0637	0.3734	0.0973	0.3343	0.86862
11	Energy Off-Peak @ Generation Summer	104	3.0065	14.6977	3.8012	3.5409	7.5253	0.4944	1.2363	-	0.0620	0.4295	34.7938	0.0257	0.1506	0.0392	0.2924	0.50792
12	Energy Off-Peak @ Generation Non-Summer	106	2.1374	12.7158	4.1851	3.6768	7.3819	0.3048	1.0190	-	0.0575	0.3656	31.8438	0.0779	0.4569	0.1190	0.3653	1.01916
13	Energy Critical On-Peak @ Gen	107	2.8307	13.1497	3.0543	3.0760	7.2452	0.3838	0.9918	-	0.0759	0.4104	31.2177	-	-	-	0.2614	0.26136
14	4CP 89/0/11	220	2.5839	11.2990	1.3957	2.2204	6.5681	0.3679	-	-	0.0919	0.3930	24.9198	-	-	-	0.1954	0.19540
15	Demand Response Refund³		\$ (51)	\$ (248)	\$ (92)	\$ (59)	\$ (136)	\$ (7)	\$ (5)	\$ (0)	\$ (2)	\$ (8)	\$ (608)	\$ (0)	\$ (1)	\$ (0)	\$ (5)	\$ (7)
16	Electric Rate Case Deferral³		\$ 130	\$ 444	\$ 36	\$ 119	\$ 331	\$ 29	\$ 14	\$ 0	\$ 8	\$ 40	\$ 1,151	\$ 7	\$ 150	\$ 67	\$ 18	\$ 243

(1) Total FCM energy and capacity source: Exhibit No. A-110 (KGT-2). Summer and winter peak split source: WP-JCA-89

(2) Exhibit No. A-16 (JCA-2) Pages 10-12

(3) WP-JCA-201_203

Analysis of Consumers' Standby Customers

PRESENTED TO
Consumers Energy

PRESENTED BY
Agustin J. Ros
Philip Q. Hanser
Lynn Zhang

February 3, 2020

THE **Brattle** GROUP

Overview

The Brattle Group is assisting Consumers Energy with developing a study analyzing the issue of the cost to serve customers who take standby service.

- Electric Rate Case No. U-20134 issued on January 9, 2019 requires a study focused on customers with behind-the-meter generation capacity that exceeds 550 kW.
- The study will review both, the actual demands that standby customer place on the system, as well as, the cost of the investments that are in place to provide standby service.

The analysis presented here provides an overview of contracted capacity and consumption patterns for each Standby customer. The analysis also compares Standby customers' total delivery service charges based on the tariff to the Total Demand-Related Distribution Cost Allocated to GSG-2 Customers from 2018 COS as well other allocation approaches using actual 2018 Standby NCP, contracted demand, and adjusted contracted demand.

General Service Self Generation Rate GSG-2

Consumers Energy Company Rate Book For Electric Service

- Standby rate is available to Full Service customers with self generation of 550 kW or greater.
- The contracted capacity of a Standby customer is the amount of capacity Consumers is expected to occasionally provide the customer during outages. Consumers is not required to supply more than the contracted amount to the customer.

The monthly rate of a Standby customer has the following components:

- Power supply charge, based on MISO LMPs and Consumers' capacity costs.
- Delivery charge, based on customer's voltage level and maximum demand (in current or previous 11 months):
 - Voltage Level 3: \$3.60 per kW Maximum Demand
 - Voltage Level 2: \$1.86 per kW Maximum Demand
 - Voltage Level 1: \$0.91 per kW Maximum Demand

We understand these charges are the same as for Primary Customers.

- Credit for: substation ownership, transmission interconnection, energy sales to Consumers.

Standby Customer Overview

There are currently fifteen standby customers on Consumers' system, under the rate class GSG-2. In 2018, the total contracted capacity for these customers was 219 MW.

Consumers Standby Customers

Contract No	Industry	Voltage	Contract Capacity (kW)			
			2015	2016	2017	2018
		2	1,426	1,426	1,426	1,426
		1	1,857	2,855	2,855	3,049
		1	1,397	1,397	1,397	1,397
		3	n/a	478	478	478
		1	6,320	10,062	10,062	15,717
		1	1,940	3,520	3,520	3,520
		2	3,381	3,381	3,381	3,381
		2	2,707	2,707	2,707	2,765
		1	2,980	2,980	2,980	2,980
		2	1,749	1,749	1,872	2,285
		1	59,019	59,019	59,019	59,019
		1	26,356	24,000	70,000	70,000
		1	25,000	25,000	25,000	25,000
		2	n/a	n/a	18,250	19,261
		2	n/a	n/a	8,456	8,732
Total			134,132	138,574	211,403	219,010

Sources: GSG-2 Contract accounts with Industry+Contracted capacity_2018.xlsx ("2018 Standby Customer List") and GSG2_contractdemand.xlsx ("2015-17 Standby Customer List").

Standby Customer Overview

Load and Energy Summary

As a class, Standby customer energy, coincident peak (CP), and non-coincident peak (NCP) have increased significantly from 2015 to 2018.

Annual Load and Energy Summary for Standby Customers, 2015 - 2018

Contract No	Industry	2018 Contract Capacity (kW)	Energy (MWh)				CP (kW)				NCP (kW)			
			2015	2016	2017	2018	2015	2016	2017	2018	2015	2016	2017	2018
		59,019	6	3	66	59	0	0	0	0	3,038	2,700	28,506	34,026
		25,000	n/a	0	2	16	n/a	0	0	0	n/a	0	1,644	8,724
		15,717	87	205	406	278	0	0	0	0	1,253	3,410	4,116	9,210
		1,426	39	175	237	114	0	0	0	0	289	1,262	1,247	1,140
		3,049	30	85	89	136	0	0	0	0	296	1,178	1,130	1,268
		3,381	42	226	488	230	0	0	0	0	517	2,295	2,309	2,552
		478	n/a	40	13	40	n/a	1	0	0	n/a	468	408	464
		3,520	87	361	277	394	0	0	0	0	463	1,940	1,927	1,892
		1,397	n/a	1,421	1,407	1,564	n/a	0	1,148	0	n/a	2,295	1,347	1,347
		2,765	360	410	280	1,340	0	0	0	547	637	2,376	2,318	2,693
		70,000	6,725	373	14,877	46,984	0	n/a	0	25	4,025	4,255	34,133	22,819
		2,980	1,222	3,682	4,581	3,623	0	0	0	0	354	1,433	1,138	1,203
		2,285	n/a	2,590	3,607	4,821	n/a	323	290	2,279	n/a	1,749	1,872	2,285
		19,261	n/a	n/a	5,845	8,591	n/a	n/a	924	154	n/a	n/a	8,919	17,877
		8,732	n/a	n/a	6,544	13,559	n/a	n/a	1,394	2,552	n/a	n/a	8,732	8,732
Total		219,010	8,598	9,571	38,720	81,748	0	324	3,755	5,557	4,166	5,147	42,446	52,316

Sources and Notes: Calculation based on GSG-2 (Stand by) Interval Data.xlsx ("2016-18 Standby Customer Load Data") and GSG-2 (Stand by) Interval Data_2015.xlsx ("2015 Standby Customer Load Data"). 2015 load data is missing for customers 100000231090, 103015037676, and 100000231611. Customers are sorted from lowest to highest % non-zero 2018 consumption.

Rate and Cost Comparison

Consumers Cost of Service Model

Consumers provided Brattle with two version of its Cost of Service (COS) Model:

- **U-20134 COSS Settlement_Unofficial.xlsx (“U-20134 COSS”)**, *received 7/10/2019*
 - Consumers’ existing COS model, which calculates allocators using average 2015-2017 load shapes applied to projected test year sales.
 - 2015-2017 load shapes are based on 2015-2017 load study estimates of class load.
- **U-20134 COSS Settlement_Unofficial_2018 Load Profile.xlsx (“U-20134 COSS – 2018 Load Study”)**, *received 8/8/2019*
 - Alternate COS model, which only uses 2018 load shape to allocate costs.
 - Brattle adjusted this model such that allocators reflect load for 2018, rather than that of the projected test year.

The U-20134 COSS includes ten GSG-2 customers as a rate class (the “GSG” class) and allocates costs to these customers using the same methodology as it uses for other customers.

- Customer-related distribution costs are primarily allocated using meter and customer count.
- Demand-related distribution costs are primarily allocated using class peak, which is the average 2015-2017 non-coincident peak demand of each class.

Rate and Cost Comparison

Rate Design and COSS Target

The U-20134 COSS implies that Consumers is under-collecting from Standby customers for production and transmission, but over-collecting for distribution.

- The allocation methodology of distribution costs in U-20134 COSS is discussed further in the next few slides.

Rate Design and COSS Target Comparison

	Rate Design Target	U-20134 COSS Target	(Under)/Over-Collection
	[1]	[2]	[3] = [1] - [2]
Production	\$1,268,196	\$2,220,994	(\$952,798)
Transmission	\$280,114	\$494,000	(\$213,886)
Distribution	\$1,260,526	\$35,037	\$1,225,489
Total	\$2,808,835	\$2,750,030	\$58,805

Sources and Notes:

[1] : Consumers' analysis from Standby Analysis_GSG-2_U-20134_hwm.xlsx ("Consumers' Standby Target Analysis"). Rate design target is calculated based on current rates for Standby customers applied to determinants from the U-20134 COSS.

[2]: Consumers' Standby Target Analysis. COSS targets are based on revenue requirement allocated to Standby customers from the U-20134 COSS (allocators based on average 2015-17 determinants).

[3]: Represents how much Consumers is under/over-collecting in rate revenues from the Standby customers in comparison to the costs allocated to these customers under U-20134 COSS.

Rate and Cost Comparison

Alternative Distribution Cost Allocation Methodology

Contract demand, rather than observed demand, may be a more appropriate allocator of demand-related distribution costs for Standby customers because Consumers must plan for Standby customers' contract demand.

- However, total contracted demand may overestimate Consumers' capacity obligation towards these customers because the individual Standby customers don't necessarily peak at the same time.
- Instead, we can adjust total contracted demand by a peak coincidence factor for the Standby class, which can be estimated by the Peak of Total Customer Demand divided by Sum of Individual Customer Peaks:

2016-2018 Standby Class Coincidence Factor

		2015	2016	2017	2015-2017 Average
Peak of Total Customer Demand (kW)	[A]	4,166	5,147	42,446	
Sum of Individual Customer Peaks (kW)	[B]	10,871	25,360	99,746	
Coincidence Factor (%)	[C] = [A] / [B]	38%	20%	43%	
Contract Capacity (kW)	[D]	134,132	138,574	211,403	161,370
Contract Capacity Adj. by Coincidence Factor (kW)	[E] = [D] x [C]	51,400	28,124	89,960	56,495

Sources:

[A] - [B] : Calculated from 2016-18 Standby Customer Load Data and 2015 Standby Customer Load Data.

[D]: 2015-17 Standby Customer List.

Rate and Cost Comparison

Alternative Distribution Cost Allocation Demand Measures

The GSG-2 class 2015-2017 load study class peak is much lower than the actual consumption or contracted demand of these customers.

GSG-2 Class Demand Measure Comparison

2015-17 Average Load Study Class Peak <i>Used in U-20134 COSS</i>	5,953 kW
2018 Actual Class Peak	52,316 kW
2015-17 Average Contract Capacity <i>Adjusted by Coincidence Factor</i>	56,495 kW
2015-17 Average Contract Capacity	161,370 kW

We provide alternative demand-related distribution cost allocations for GSG-2 customers using the following demand measures:

- 2018 actual class peak
- 2015-17 average contract capacity adjusted by coincidence factor of customer peaks (calculated on previous slide)
- 2015-17 average contract capacity (calculated on previous slide)

Rate and Cost Comparison

Comparison of Total Demand-Related Costs

We re-allocated distribution demand-related costs to Standby customers using alternate measures of Standby customer demand.

- The table below compares total distribution revenues collected from Standby customers with total distribution costs allocated to Standby customers under various demand measures.

GSG Class Allocated Distribution Costs

		Total \$	\$/kW-mo Charge
		<i>Revenues Collected</i>	<i>Actual by Voltage</i>
Distribution Rate Design Target	[A]	\$1,260,526	\$0.91 (Vlt 1)
			\$1.86 (Vlt 2)
			\$3.60 (Vlt 3)
		<i>Costs Allocated</i>	<i>Implied</i>
COSS Target (from U-20134)	[B]	\$35,037	\$0.49
<i>Allocated Using 2018 Actual Class Peak</i>	<i>[C]</i>	<i>\$1,123,274</i>	<i>\$1.79</i>
<i>Allocated Using 2015-17 Avg Contract Capacity Adj. by CF</i>	<i>[D]</i>	<i>\$1,275,181</i>	<i>\$1.88</i>
<i>Allocated Using 2015-17 Avg Contract Capacity</i>	<i>[E]</i>	<i>\$3,799,241</i>	<i>\$1.96</i>

Sources and Notes:

[A]: Consumers' Standby Target Analysis and Consumers Rate Book.

[B]: Consumers' Standby Target Analysis. Distribution demand-related costs are in the U-20134 COSS are allocated based on 2015-17 average load study NCP.

[C]: Calculated by replacing GSG-2 class load data in U-20134 COSS – 2018 Load Study with 2018 actual load (class peak of 52,316 kW).

[D]: Calculated by replacing GSG-2 class peak in U-20134 COSS with 2015-17 average contracted capacity adjusted by coincidence factor of customer peaks (56,495 kW).

[E]: Calculated by replacing GSG-2 class peak in U-20134 COSS with 2015-17 average contracted capacity (161,370 kW).

Rate and Cost Comparison

Rate Design and COSS Target

If demand-related distribution costs were allocated to Standby customers using 2015-17 average CF-adjusted contract capacity, the costs allocated to these customers are comparable to the distribution revenues collected from these customers.

Rate Design and COSS Target Comparison *Alternative Allocation for Demand-Related Distribution Costs*

	U-20134 COSS Target		
	Rate Design Target	Adj. Distribution Allocator	(Under)/Over-Collection
	[1]	[2]	[3] = [1] - [2]
Production	\$1,268,196	\$2,220,994	(\$952,798)
Transmission	\$280,114	\$494,000	(\$213,886)
Distribution	\$1,260,526	\$1,275,181	(\$14,656)
Total	\$2,808,835	\$3,990,175	(\$1,181,340)

Sources and Notes:

[1] : Consumers' Standby Target Analysis.

[2]: Consumers' Standby Target Analysis, with demand-related distribution costs allocated using 2015-17 average contracted capacity adjusted by coincidence factor of customer peaks (56,495 kW).

[3]: Represents how much Consumers is under/over-collecting in rate revenues from the Standby customers in comparison to the costs allocated to these customers.

Appendix

Appendix

Consumers' Engineering Approach for Standby Customers

1. When a Standby customer is added to the system, Consumers planners determine whether a network upgrade is required to accommodate the new customer based on the customer's contracted demand.
 - If an upgrade is needed, the customer generally is not responsible for the costs, unless:
 - the upgrade is determined to be dedicated to the customer, or
 - an upgrade is needed to allow the customer to sell energy back to the company.
2. A key engineering consideration in determining Standby Costs is whether:
 - the Consumers distribution system is planned to be able to *simultaneously* provide the contracted demand of *all* Standby customers, *holding constant all other classes demand*, or
 - Consumers takes into account the likely demand diversity among the Standby customers when planning its system.
3. After the customer is connected to the system, the utility doesn't generally monitor the customers (*e.g.*, doesn't track if the customer's consumption is exceeding its contracted demand).

Appendix

Public Utilities Regulatory Policy Act of 1978

The Public Utilities Regulatory Policy Act of 1978 (PURPA, amended in 2005 by the Energy Policy Act) mandates that rates for sales of backup and maintenance power:

- Shall not be based upon an assumption (unless supported by factual data) that forced outages or other reductions in electric output by all qualifying facilities on an electric utility's system will occur simultaneously, or during the system peak, or both.
- Shall take into account the extent to which scheduled outages of the qualifying facilities can be usefully coordinated with the scheduled outages of the utility's facilities.

Appendix

Customer Consumption Regularity

Consumption is primarily driven by a few customers—the majority of Standby customers consumed in less than 25% of hours per year in 2016-2018.

- Four customers (highlighted in grey below) consumed regularly across the years.
- Two customers (highlighted in yellow below) changed consumption behavior across the years, becoming more like regular consumers by 2018.

Annual Consumption Summary for Standby Customers, 2016 - 2018

Contract No	Industry	% Non-Zero Consumption Hours			Avg Non-Zero Consumption (kW)		
		2016	2017	2018	2016	2017	2018
		0%	0%	0%	1,575	16,577	29,730
		0%	0%	0%	n/a	1,644	8,070
		1%	3%	1%	1,768	1,527	2,336
		4%	5%	3%	483	569	508
		3%	2%	3%	388	428	536
		7%	9%	5%	386	630	546
		4%	2%	6%	112	64	76
		13%	10%	16%	323	323	280
		20%	20%	22%	801	788	816
		6%	5%	49%	764	680	314
		18%	36%	74%	1,371	4,708	7,257
		69%	94%	78%	604	558	533
		91%	100%	82%	324	412	671
		n/a	100%	100%	n/a	994	981
		n/a	100%	100%	n/a	1,116	1,548

Source: Brattle analysis of Hourly Standby Customer Data. Customers are sorted from lowest to highest % non-zero 2018 consumption.

Appendix

Monthly Customer-Level Summaries

Brattle summarized the 2018 hourly consumption data for Standby customers at the monthly level.

- About half of Standby customers do not consume regularly across the months, with some months in which they don't consume at all.
- The majority of Standby customers have no consumption during the system peak.

2018 monthly energy and load summaries are provided on the next couple of slides. Zero consumption months are highlighted in blue.

Appendix: Monthly Summary

2018 Monthly Energy and Load Factor

2018 Monthly Energy by Customer (kWh)

Contract No	Industry	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		59,460	0	0	0	0	0	0	0	0	0	0	0
		16,140	0	0	0	0	0	0	0	0	0	0	0
		0	0	114,220	3,264	0	0	0	0	79,854	68,794	3,867	7,942
		0	0	0	0	77,253	0	0	0	0	0	10,621	26,481
		0	0	19,861	78,371	0	0	0	0	8,101	0	0	29,238
		33,331	0	0	53,888	0	13,985	4,434	0	0	124,156	0	0
		1,783	1,395	14,107	496	3,714	7,234	3,650	1,056	765	1,467	2,954	1,579
		17,914	0	37,110	20,936	0	20,769	49,342	22,765	16,590	151,961	0	56,715
		72,399	62,570	91,160	86,370	149,588	145,846	214,353	182,370	181,522	73,696	113,613	190,453
		185,875	176,630	182,938	163,454	127,699	198,058	254,016	49,176	0	0	0	1,958
		2,721,902	3,608,010	3,764,767	3,738,974	4,571,141	5,198,483	6,275,027	4,191,516	3,182,017	3,768,647	3,653,887	2,309,933
		330,345	466,600	374,973	132,924	220,738	312,532	316,201	337,830	282,560	265,336	280,007	302,826
		466,940	677,879	57,583	22,511	366,703	577,303	504,821	432,827	283,266	196,717	591,664	642,295
		2,056,922	1,838,045	1,394,382	1,126,675	596,750	256,676	121,986	123,629	111,402	790,045	80,604	93,491
		663,554	637,916	710,803	1,126,336	1,268,976	1,752,186	1,541,303	1,609,450	1,422,864	1,407,974	645,576	771,658

2018 Monthly Load Factor by Customer (%)

Contract No	Industry	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		0%	-	-	-	-	-	-	-	-	-	-	-
		0%	-	-	-	-	-	-	-	-	-	-	-
		-	-	2%	0%	-	-	-	-	3%	3%	0%	0%
		-	-	-	-	9%	-	-	-	-	-	1%	3%
		-	-	2%	9%	-	-	-	-	1%	-	-	3%
		2%	-	-	3%	-	1%	0%	-	-	7%	-	-
		1%	1%	8%	1%	2%	2%	1%	1%	1%	1%	2%	1%
		1%	-	3%	2%	-	2%	4%	2%	4%	11%	-	4%
		8%	7%	9%	10%	16%	16%	23%	20%	19%	8%	12%	19%
		11%	79%	81%	75%	19%	13%	15%	2%	-	-	-	0%
		23%	34%	33%	23%	33%	32%	38%	27%	26%	27%	28%	24%
		41%	83%	62%	17%	30%	46%	37%	38%	34%	48%	51%	56%
		34%	54%	5%	2%	22%	35%	30%	26%	20%	16%	36%	38%
		15%	21%	57%	19%	11%	12%	21%	13%	39%	13%	44%	3%
		53%	54%	60%	18%	65%	28%	24%	80%	32%	22%	10%	12%

Source: Brattle analysis of Hourly Standby Customer Data. Customers are sorted from lowest to highest % non-zero 2018 consumption. Zero consumption months are highlighted in blue

Appendix: Monthly Summary

2018 Monthly Peaks

2018 Monthly CP by Customer (kW)

Contract No	Industry	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		0	0	0	0	0	0	0	0	0	0	0	0
		0	0	0	0	0	0	0	0	0	0	0	0
		0	0	0	0	0	0	0	0	0	0	0	0
		0	0	0	0	0	0	0	0	0	0	0	0
		0	0	0	0	0	0	0	0	0	0	0	0
		0	0	0	884	0	1,098	0	0	0	0	0	0
		0	0	0	0	12	0	0	0	0	0	0	0
		0	0	520	0	0	0	0	0	0	166	0	360
		0	0	0	0	0	0	0	0	0	0	0	1,148
		0	331	259	230	144	202	547	0	0	0	0	0
		189	4,637	7,925	12,386	0	9,551	25	0	0	4,511	1,386	0
		0	794	741	0	0	0	0	0	0	0	454	480
		409	399	270	43	2,208	923	2,279	841	844	468	826	819
		10,008	1,950	1,830	787	650	513	154	171	154	1,796	86	103
		714	720	743	1,267	2,540	2,598	2,552	2,569	2,701	5,777	605	795

2018 Monthly NCP by Customer (kW)

Contract No	Industry	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		34,026	0	0	0	0	0	0	0	0	0	0	0
		8,724	0	0	0	0	0	0	0	0	0	0	0
		0	0	9,210	2,827	0	0	0	0	4,220	3,077	2,017	3,805
		0	0	0	0	1,098	0	0	0	0	0	1,140	1,062
		0	0	1,268	1,150	0	0	0	0	929	0	0	1,261
		2,096	0	0	2,295	0	1,497	1,868	0	0	2,552	0	0
		200	233	241	107	256	464	425	145	80	215	240	208
		1,823	0	1,739	1,857	0	1,490	1,684	1,892	513	1,809	0	1,712
		1,247	1,297	1,297	1,247	1,247	1,247	1,247	1,247	1,297	1,297	1,347	1,347
		2,318	331	302	302	907	2,117	2,347	2,693	0	0	0	1,872
		16,052	15,637	15,309	22,819	18,446	22,327	22,390	21,118	17,312	18,522	17,816	13,079
		1,090	835	820	1,117	988	943	1,152	1,203	1,161	736	763	733
		1,872	1,872	1,426	1,285	2,279	2,279	2,279	2,279	1,966	1,686	2,285	2,285
		17,877	13,207	3,280	8,280	7,253	2,874	787	1,283	393	8,331	257	4,225
		1,670	1,768	1,590	8,732	2,644	8,732	8,732	2,719	6,249	8,732	8,732	8,732

Source: Brattle analysis of Hourly Standby Customer Data. Customers are sorted from lowest to highest % non-zero 2018 consumption.

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Projected 12-Month Period Ending Dec 31, 2021
Version 2 - Separating RS and RT with Projected Load Reductions
4CP 89/0/11 Production and 12CP Transmission
(thousands of dollars)

Summary RETURN (SUMMARY)										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description		Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Rate Base		11,893,424	11,843,224	6,691,753	3,018,826	1,970,008	136,130	26,507	50,200
	Revenue									
2	Total Rate Revenue		4,017,792	3,993,182	1,949,148	1,003,525	997,036	41,750	1,722	24,610
3	Total Revenue Credits		379,354	378,393	174,409	87,812	109,867	2,030	4,275	962
4	Total Revenue		4,397,146	4,371,575	2,123,557	1,091,336	1,106,903	43,781	5,997	25,572
5	Expenses:									
6	Fuel and P&I Expense		1,683,201	1,664,372	761,226	390,042	501,948	8,210	2,947	18,829
7	Transmission Expense		477,556	472,642	222,790	105,252	141,526	2,333	741	4,914
8	Other O & M Expense		684,695	681,731	400,775	166,407	106,163	7,233	1,153	2,963
9	Depreciation & Amortization Expense		705,582	701,819	398,552	177,381	113,745	10,835	1,307	3,763
10	Other Taxes		263,853	263,090	139,113	68,814	50,480	4,177	506	763
11	Federal Income Taxes		53,928	54,452	18,626	16,990	17,879	1,018	(61)	(524)
12	Total Expenses		3,868,814	3,838,107	1,941,081	924,886	931,741	33,806	6,592	30,708
13	Net Operating Income		528,332	533,468	182,476	166,450	175,161	9,975	(595)	(5,136)
14	Other Income Adjustments		6,234	6,202	3,402	1,551	1,153	78	18	32
15	Adjusted Net Operating Income		534,565	539,669	185,877	168,001	176,315	10,053	(577)	(5,104)
16	Rate of Return on Rate Base		4.49%	4.56%	2.78%	5.57%	8.95%	7.39%	-2.18%	-10.17%
17	Index of Return (Jurisdictional)			100	61	122	196	162	(48)	
18	Return on Rate Base @ 6.09%		724,604	721,546	407,693	183,921	120,022	8,294	1,615	3,058
19	Income Deficiency (Sufficiency)		190,039	181,876	221,816	15,920	(56,292)	(1,760)	2,192	8,162
20	Revenue Deficiency (Sufficiency)		254,475	243,545	297,027	21,318	(75,379)	(2,356)	2,936	10,930
21	Revenue Requirement/Total Cost of Service		4,651,621	4,615,120	2,420,584	1,112,655	1,031,523	41,425	8,933	36,502
22	Less: Revenue Credits		379,354	378,393	174,409	87,812	109,867	2,030	4,275	962
23	Proposed Rate Design Revenue		4,272,267	4,236,727	2,246,175	1,024,843	921,656	39,394	4,658	35,540
24	Production: Net Capacity Cost		1,182,798	1,170,644	605,821	267,377	293,338	2,395	1,714	12,154
25	Production: Capacity Related Cost Offset		505,456	497,298	253,777	112,717	128,107	1,756	943	8,158
26	Production: Non-Capacity Related Cost		1,138,080	1,122,092	483,584	263,545	369,071	7,403	(1,511)	15,988
27	Distribution: Demand Related Cost		1,191,789	1,192,763	711,264	333,888	118,043	26,106	3,462	(974)
28	Distribution: Customer Related Cost		254,144	253,930	191,729	47,317	13,097	1,735	52	214
29	Full Service MWH Sales		30,813,122	30,433,455	12,447,668	7,086,449	10,627,497	208,358	63,483	379,667
30	ROA MWH Sales		3,697,427	3,697,427	-	205,339	3,492,088	-	-	-
31	MWH Sales		34,510,549	34,130,882	12,447,668	7,291,788	14,119,585	208,358	63,483	379,667
32	Customers		1,841,963	1,841,961	1,620,698	216,313	4,116	822	12	2

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Residential/Secondary RETURN (SUMMARY)									
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Rate RS	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Total Commercial Secondary
1	Rate Base	6,691,753	-	6,691,753	1,769,394	1,116,399	52,097	80,936	3,018,826
	Revenue								
2	Total Rate Revenue	1,949,148	-	1,949,148	564,822	400,954	13,055	24,694	1,003,525
3	Total Revenue Credits	174,409	-	174,409	47,661	37,022	1,182	1,946	87,812
4	Total Revenue	2,123,557	-	2,123,557	612,483	437,977	14,237	26,640	1,091,336
5	Expenses:								
6	Fuel and P&I Expense	761,226	-	761,226	208,495	168,218	5,055	8,274	390,042
7	Transmission Expense	222,790	-	222,790	56,672	44,860	1,463	2,258	105,252
8	Other O & M Expense	400,775	-	400,775	100,317	59,208	2,702	4,181	166,407
9	Depreciation & Amortization Expense	398,552	-	398,552	104,689	65,491	2,880	4,321	177,381
10	Other Taxes	139,113	-	139,113	40,011	25,863	1,064	1,875	68,814
11	Federal Income Taxes	18,626	-	18,626	9,475	6,885	99	531	16,990
12	Total Expenses	1,941,081	-	1,941,081	519,659	370,525	13,263	21,439	924,886
13	Net Operating Income	182,476	-	182,476	92,824	67,452	973	5,201	166,450
14	Other Income Adjustments	3,402	-	3,402	898	587	26	40	1,551
15	Adjusted Net Operating Income	185,877	-	185,877	93,722	68,039	1,000	5,241	168,001
16	Rate of Return on Rate Base	2.78%	0.00%	2.78%	5.30%	6.09%	1.92%	6.48%	5.57%
17	Index of Return (Jurisdictional)	61	-	61	116	134	42	142	122
18	Return on Rate Base @ 6.09%	407,693	-	407,693	107,800	68,016	3,174	4,931	183,921
19	Income Deficiency (Sufficiency)	221,816	-	221,816	14,078	(23)	2,174	(310)	15,920
20	Revenue Deficiency (Sufficiency)	297,027	-	297,027	18,851	(30)	2,912	(415)	21,318
21	Revenue Requirement/Total Cost of Service	2,420,584	-	2,420,584	631,334	437,946	17,149	26,225	1,112,655
22	Less: Revenue Credits	174,409	-	174,409	47,661	37,022	1,182	1,946	87,812
23	Proposed Rate Design Revenue	2,246,175	-	2,246,175	583,673	400,924	15,966	24,279	1,024,843
24	Production: Net Capacity Cost	605,821	-	605,821	145,614	113,189	3,704	4,869	267,377
25	Production: Capacity Related Cost Offset	253,777	-	253,777	60,943	48,393	1,574	1,807	112,717
26	Production: Non-Capacity Related Cost	483,584	-	483,584	139,787	114,321	3,385	6,051	263,545
27	Distribution: Demand Related Cost	711,264	-	711,264	196,907	118,798	6,935	11,248	333,888
28	Distribution: Customer Related Cost	191,729	-	191,729	40,422	6,222	369	303	47,317
29	Full Service MWH Sales	12,447,668	-	12,447,668	3,742,026	3,093,372	87,692	163,359	7,086,449
30	ROA MWH Sales	-	-	-	9,785	116,816	14,695	64,043	205,339
31	MWH Sales	12,447,668	-	12,447,668	3,751,811	3,210,188	102,387	227,402	7,291,788
32	Customers	1,620,698	-	1,620,698	194,349	19,604	1,580	780	216,313

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Primary & Lighting RETURN (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description		Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Total Lighting & Unmetered
1	Rate Base		211,282	838,861	89,318	182,355	538,737	37,275	16,779	58	9,257	46,086	1,970,008	4,425	74,407	36,281	21,018	136,130
	Revenue																	
2	Total Rate Revenue		101,921	400,343	84,697	98,218	255,237	14,966	22,612	14	3,256	15,771	997,036	1,551	23,398	7,927	8,874	41,750
3	Total Revenue Credits		9,379	45,400	12,396	11,101	25,989	1,388	2,396	1	302	1,516	109,867	110	730	240	951	2,030
4	Total Revenue		111,300	445,743	97,092	109,319	281,226	16,354	25,008	15	3,558	17,288	1,106,903	1,661	24,128	8,167	9,825	43,781
5	Expenses:																	
6	Fuel and P&I Expense		43,867	214,253	47,548	50,772	119,920	6,372	11,280	-	1,284	6,652	501,948	437	2,563	668	4,542	8,210
7	Transmission Expense		12,125	59,517	14,478	14,315	32,521	1,862	4,498	-	355	1,856	141,526	147	865	225	1,096	2,333
8	Other O & M Expense		10,502	45,432	5,768	10,283	28,273	1,823	1,385	2	436	2,260	106,163	255	5,222	562	1,194	7,233
9	Depreciation & Amortization Expense		12,044	49,146	5,966	11,250	29,996	1,879	792	2	521	2,149	113,745	253	5,790	3,526	1,265	10,835
10	Other Taxes		5,467	19,774	3,293	5,263	13,762	891	703	2	250	1,077	50,480	119	2,328	1,240	491	4,177
11	Federal Income Taxes		2,528	5,337	1,856	1,615	5,256	327	588	1	66	305	17,879	42	682	180	115	1,018
12	Total Expenses		86,532	393,458	78,909	93,498	229,728	13,154	19,246	7	2,911	14,299	931,741	1,253	17,450	6,402	8,701	33,806
13	Net Operating Income		24,769	52,285	18,183	15,821	51,498	3,200	5,762	8	647	2,988	175,161	408	6,678	1,766	1,124	9,975
14	Other Income Adjustments		119	496	48	120	310	21	8	0	6	25	1,153	2	41	24	11	78
15	Adjusted Net Operating Income		24,888	52,781	18,231	15,941	51,808	3,221	5,770	8	653	3,013	176,315	410	6,719	1,789	1,135	10,053
16	Rate of Return on Rate Base		11.78%	6.29%	20.41%	8.74%	9.62%	8.64%	34.39%	14.24%	7.06%	6.54%	8.95%	9.27%	9.03%	4.93%	5.40%	7.39%
17	Index of Return (Jurisdictional)		259	138	448	192	211	190	755	312	155	143	196	203	198	108	119	162
18	Return on Rate Base @ 6.09%		12,872	51,107	5,442	11,110	32,822	2,271	1,022	4	564	2,808	120,022	270	4,533	2,210	1,280	8,294
19	Income Deficiency (Sufficiency)		(12,015)	(1,674)	(12,789)	(4,831)	(18,986)	(950)	(4,748)	(5)	(89)	(206)	(56,292)	(140)	(2,185)	421	145	(1,760)
20	Revenue Deficiency (Sufficiency)		(16,089)	(2,242)	(17,125)	(6,469)	(25,423)	(1,272)	(6,358)	(6)	(119)	(275)	(75,379)	(188)	(2,926)	564	194	(2,356)
21	Revenue Requirement/Total Cost of Service		95,211	443,501	79,967	102,850	255,803	15,082	18,650	9	3,439	17,012	1,031,523	1,472	21,201	8,731	10,020	41,425
22	Less: Revenue Credits		9,379	45,400	12,396	11,101	25,989	1,388	2,396	1	302	1,516	109,867	110	730	240	951	2,030
23	Proposed Rate Design Revenue		85,831	398,101	67,571	91,750	229,814	13,694	16,254	8	3,137	15,496	921,656	1,363	20,472	8,491	9,069	39,394
24	Production: Net Capacity Cost		30,379	132,846	16,409	26,455	77,223	4,326	-	-	1,080	4,621	293,338	-	-	-	2,395	2,395
25	Production: Capacity Related Cost Offset		10,920	60,738	7,532	12,404	31,305	1,575	1,422	0	365	1,846	128,107	71	374	88	1,222	1,756
26	Production: Non-Capacity Related Cost		29,619	153,551	40,422	39,196	83,333	4,452	13,269	(0)	750	4,480	369,071	483	2,832	737	3,351	7,403
27	Distribution: Demand Related Cost		12,394	47,575	1,838	12,236	34,436	3,067	1,336	5	882	4,274	118,043	770	15,647	7,654	2,034	26,106
28	Distribution: Customer Related Cost		2,520	3,392	1,370	1,458	3,518	274	228	3	60	275	13,097	38	1,619	12	67	1,735
29	Full Service MWH Sales		830,685	4,365,549	1,272,911	1,142,370	2,388,475	121,808	358,537	-	21,349	125,813	10,627,497	13,509	79,208	20,629	95,012	208,358
30	ROA MWH Sales		46,609	-	1,089,479	1,247,199	854,891	27,722	-	2,690	70,229	153,269	3,492,088	-	-	-	-	-
31	MWH Sales		877,294	4,365,549	2,362,390	2,389,569	3,243,366	149,530	358,537	2,690	91,578	279,082	14,119,585	13,509	79,208	20,629	95,012	208,358
32	Customers		1,510	1,032	33	85	1,204	142	19	1	6	84	4,116	343	-	-	479	822

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Summary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description			Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
1	Net Plant										
2	Production			3,675,605	3,637,836	1,845,500	836,214	941,110	9,465	5,547	37,769
3	Transmission			(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution			6,328,753	6,325,283	3,764,506	1,726,420	705,586	110,553	18,217	3,470
5	General/Common/Intangible			721,822	718,825	424,707	176,375	109,027	7,439	1,278	2,996
6	Plant Purchased/Sold			0	0	0	0	0	0	0	0
7	Total Net Plant			10,726,180	10,681,944	6,034,714	2,739,008	1,755,724	127,457	25,042	44,235
8	Working Capital										
9	Total Current Assets			2,236,176	2,226,003	1,268,128	552,550	376,253	25,324	3,747	10,173
10	Total Current Liabilities			1,009,093	1,004,914	575,611	255,988	156,199	15,007	2,110	4,179
11	Total Working Capital			1,227,083	1,221,089	692,517	296,563	220,054	10,317	1,638	5,994
	Adjustments to Rate Base										
12	Additions to Rate Base			0	0	0	0	0	0	0	0
13	Deductions from Rate Base			59,839	59,809	35,478	16,745	5,770	1,644	172	30
14	Total Adjustments to Rate Base			(59,839)	(59,809)	(35,478)	(16,745)	(5,770)	(1,644)	(172)	(30)
15	Total Rate Base			11,893,424	11,843,224	6,691,753	3,018,826	1,970,008	136,130	26,507	50,200

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Residential/Secondary RATE BASE (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate RS	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Net Plant									
2	Production		1,845,500	-	1,845,500	453,805	355,270	11,441	15,697	836,214
3	Transmission		(0)	-	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution		3,764,506	-	3,764,506	1,043,634	595,146	33,529	54,112	1,726,420
5	General/Common/Intangible		424,707	-	424,707	106,214	62,831	2,913	4,417	176,375
6	Plant Purchased/Sold		0	-	0	0	0	0	0	0
7	Total Net Plant		6,034,714	-	6,034,714	1,603,653	1,013,247	47,883	74,225	2,739,008
8	Working Capital									
9	Total Current Assets		1,268,128	-	1,268,128	327,906	201,826	8,890	13,928	552,550
10	Total Current Liabilities		575,611	-	575,611	151,950	92,983	4,355	6,700	255,988
11	Total Working Capital		692,517	-	692,517	175,956	108,843	4,536	7,228	296,563
	Adjustments to Rate Base									
12	Additions to Rate Base		0	-	0	0	0	0	0	0
13	Deductions from Rate Base		35,478	-	35,478	10,215	5,691	322	517	16,745
14	Total Adjustments to Rate Base		(35,478)	-	(35,478)	(10,215)	(5,691)	(322)	(517)	(16,745)
15	Total Rate Base		6,691,753	-	6,691,753	1,769,394	1,116,399	52,097	80,936	3,018,826

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Primary & Lighting RATE BASE (SUMMARY)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Total Lighting & Unmetered
1	Net Plant																
2	Production	94,802	423,838	55,534	86,852	244,321	13,545	4,546	-	3,261	14,413	941,110	184	1,080	281	7,920	9,465
3	Transmission	(0)	(0)	(0)	(0)	(0)	(0)	(0)	0	(0)	(0)	(0)	(0)	(0)	(0)	(0)	(0)
4	Distribution	83,868	277,393	14,717	64,577	208,364	18,309	8,438	54	4,705	25,161	705,586	3,626	65,670	31,507	9,750	110,553
5	General/Common/Intangible	11,199	46,921	5,256	10,342	29,635	1,956	782	2	494	2,441	109,027	246	3,667	2,326	1,200	7,439
6	Plant Purchased/Sold	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0
7	Total Net Plant	189,869	748,152	75,506	161,771	482,319	33,810	13,766	56	8,460	42,015	1,755,724	4,056	70,417	34,115	18,869	127,457
8	Working Capital																
9	Total Current Assets	38,379	159,667	21,224	36,277	100,465	6,471	4,350	7	1,598	7,815	376,253	813	12,985	7,533	3,992	25,324
10	Total Current Liabilities	16,310	66,717	7,293	15,081	42,374	2,860	1,262	4	757	3,540	156,199	405	8,046	4,806	1,751	15,007
11	Total Working Capital	22,068	92,950	13,931	21,196	58,091	3,611	3,088	3	842	4,274	220,054	409	4,940	2,727	2,241	10,317
Adjustments to Rate Base																	
12	Additions to Rate Base	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	Deductions from Rate Base	655	2,241	119	611	1,674	147	75	0	45	203	5,770	40	949	561	93	1,644
14	Total Adjustments to Rate Base	(655)	(2,241)	(119)	(611)	(1,674)	(147)	(75)	(0)	(45)	(203)	(5,770)	(40)	(949)	(561)	(93)	(1,644)
15	Total Rate Base	211,282	838,861	89,318	182,355	538,737	37,275	16,779	58	9,257	46,086	1,970,008	4,425	74,407	36,281	21,018	136,130

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Summary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional		
1	Production Expense										
2	Fuel Expense	542,894	536,431	224,173	127,611	179,959	3,634	1,054	6,463		
3	Purchased & Interchange Power Expense	1,140,307	1,127,942	537,053	262,430	321,989	4,575	1,893	12,366		
4	Total Fuel and P&I Expense	1,683,201	1,664,372	761,226	390,042	501,948	8,210	2,947	18,829		
5	Fossil O&M Exp	100,702	99,571	45,670	23,583	29,614	523	181	1,131		
6	Nuclear O&M Exp	0	0	0	0	0	0	0	0		
7	Hydro O&M Exp	22,159	21,922	10,648	5,107	6,048	83	36	237		
8	Peaker O&M Exp	67,853	67,156	34,068	15,437	17,373	175	102	697		
9	Other O&M	10,761	10,651	5,403	2,448	2,755	28	16	111		
10	Total Production O&M Expense	201,475	199,299	95,789	46,575	55,791	808	336	2,176		
11	Total Production (Inc. Fuel and P&I) O&M Expense	1,884,676	1,863,671	857,015	436,617	557,739	9,018	3,282	21,005		
12	Transmission & Distribution Expense										
13	Trans O&M Exp	477,556	472,642	222,790	105,252	141,526	2,333	741	4,914		
14	Other O&M Adjustments	0	0	0	0	0	0	0	0		
15	Distr Oper Exp	89,212	89,198	56,782	22,975	5,851	3,470	120	13		
16	Distr Maint Exp	175,919	175,881	105,019	51,243	17,877	1,290	451	38		
17	Total Transmission & Distribution O&M Expense	742,687	737,721	384,591	179,470	165,254	7,093	1,312	4,966		
18	Customer Related Expense										
19	Customer Accounts Exp	54,093	54,093	47,618	6,356	105	13	0	0		
20	Customer Service Exp	16,912	16,770	8,173	3,199	5,295	80	24	142		
21	Sales Expense	365	365	321	43	1	0	0	0		
22	Total Customer O&M Expense	71,370	71,228	56,113	9,597	5,401	93	24	142		
23	Admin & General Expense	146,719	146,126	87,072	36,017	21,244	1,572	221	593		
24	Total O&M Expense	2,845,452	2,818,746	1,384,791	661,701	749,637	17,775	4,840	26,706		

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Residential/Secondary O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description		Rate RS	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
1	Production Expense									
2	Fuel Expense		224,173	-	224,173	67,421	55,662	1,584	2,945	127,611
3	Purchased & Interchange Power Expense		537,053	-	537,053	141,075	112,555	3,472	5,329	262,430
4	Total Fuel and P&I Expense		761,226	-	761,226	208,495	168,218	5,055	8,274	390,042
5	Fossil O&M Exp		45,670	-	45,670	12,591	10,184	304	503	23,583
6	Nuclear O&M Exp		0	-	0	0	0	0	0	0
7	Hydro O&M Exp		10,648	-	10,648	2,751	2,186	68	102	5,107
8	Peaker O&M Exp		34,068	-	34,068	8,377	6,558	211	290	15,437
9	Other O&M		5,403	-	5,403	1,329	1,040	33	46	2,448
10	Total Production O&M Expense		95,789	-	95,789	25,048	19,969	617	941	46,575
11	Total Production (Inc. Fuel and P&I) O&M Expense		857,015	-	857,015	233,544	188,186	5,672	9,215	436,617
12	Transmission & Distribution Expense									
13	Trans O&M Exp		222,790	-	222,790	56,672	44,860	1,463	2,258	105,252
14	Other O&M Adjustments		0	-	0	0	0	0	0	0
15	Distr Oper Exp		56,782	-	56,782	14,459	7,425	424	667	22,975
16	Distr Maint Exp		105,019	-	105,019	31,455	17,248	978	1,562	51,243
17	Total Transmission & Distribution O&M Expense		384,591	-	384,591	102,586	69,533	2,865	4,487	179,470
18	Customer Related Expense									
19	Customer Accounts Exp		47,618	-	47,618	5,710	576	46	23	6,356
20	Customer Service Exp		8,173	-	8,173	1,826	1,244	42	87	3,199
21	Sales Expense		321	-	321	39	4	0	0	43
22	Total Customer O&M Expense		56,113	-	56,113	7,575	1,824	88	110	9,597
23	Admin & General Expense		87,072	-	87,072	21,780	12,742	594	901	36,017
24	Total O&M Expense		1,384,791	-	1,384,791	365,484	272,285	9,220	14,712	661,701

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Primary & Lighting O&M (SUMMARY)		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered	
1	Production Expense																	
2	Fuel Expense	14,264	74,235	20,964	19,175	40,724	2,098	5,979	-	361	2,159	179,959	231	1,353	352	1,698	3,634	
3	Purchased & Interchange Power Expense	29,603	140,018	26,584	31,597	79,197	4,274	5,301	-	923	4,493	321,989	206	1,210	315	2,844	4,575	
4	Total Fuel and P&I Expense	43,867	214,253	47,548	50,772	119,920	6,372	11,280	-	1,284	6,652	501,948	437	2,563	668	4,542	8,210	
5	Fossil O&M Exp	2,647	13,038	2,112	2,983	7,255	384	718	-	77	401	29,614	29	170	44	279	523	
6	Nuclear O&M Exp	0	0	0	0	0	0	0	-	0	0	0	0	0	0	0	0	
7	Hydro O&M Exp	576	2,695	393	583	1,528	83	85	-	18	88	6,048	3	20	5	54	83	
8	Peaker O&M Exp	1,750	7,824	1,025	1,603	4,510	250	84	-	60	266	17,373	3	20	5	146	175	
9	Other O&M	278	1,241	163	254	715	40	13	-	10	42	2,755	1	3	1	23	28	
10	Total Production O&M Expense	5,251	24,797	3,693	5,423	14,008	757	900	-	165	797	55,791	36	214	56	502	808	
11	Total Production (Inc. Fuel and P&I) O&M Expense	49,118	239,050	51,241	56,195	133,928	7,129	12,180	-	1,449	7,449	557,739	474	2,777	723	5,044	9,018	
12	Transmission & Distribution Expense																	
13	Trans O&M Exp	12,125	59,517	14,478	14,315	32,521	1,862	4,498	-	355	1,856	141,526	147	865	225	1,096	2,333	
14	Other O&M Adjustments	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
15	Distr Oper Exp	705	2,431	29	422	1,811	158	43	0	31	220	5,851	70	1,895	1,383	123	3,470	
16	Distr Maint Exp	1,981	7,357	117	1,600	5,413	468	161	0	116	663	17,877	86	656	266	282	1,290	
17	Total Transmission & Distribution O&M Expense	14,812	69,305	14,624	16,337	39,745	2,488	4,702	1	502	2,739	165,254	303	3,415	1,874	1,501	7,093	
18	Customer Related Expense																	
19	Customer Accounts Exp	39	26	1	2	31	4	0	0	0	2	105	6	-	-	7	13	
20	Customer Service Exp	332	1,637	884	895	1,217	56	134	1	34	105	5,295	6	30	8	37	80	
21	Sales Expense	0	0	0	0	0	0	0	0	0	0	1	0	-	-	0	0	
22	Total Customer O&M Expense	371	1,663	885	897	1,248	60	135	1	34	107	5,401	12	30	8	44	93	
23	Admin & General Expense	2,194	9,183	1,044	1,941	5,793	381	146	0	90	474	21,244	51	779	499	242	1,572	
24	Total O&M Expense	66,493	319,201	67,794	75,370	180,714	10,058	17,162	2	2,074	10,768	749,637	840	7,001	3,104	6,831	17,775	

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Summary ALLOCATORS 1										
Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	100.000	98.809	41.956	23.886	32.066	0.702	0.200	1.191
2	Energy On-Peak @ Generation	101	100.000	98.809	39.220	24.658	34.305	0.451	0.176	1.191
3	Energy Off-Peak @ Generation	102	100.000	98.809	43.529	22.089	32.005	0.966	0.219	1.191
4	Energy On-Peak @ Generation Summer	103	100.000	98.809	40.170	24.673	33.446	0.350	0.170	1.191
5	Energy Off-Peak @ Generation Summer	104	100.000	98.809	43.134	22.295	32.291	0.869	0.220	1.191
6	Energy On-Peak @ Generation Non-Summer	105	100.000	98.809	38.677	24.650	34.795	0.508	0.179	1.191
7	Energy Off-Peak @ Generation Non-Summer	106	100.000	98.809	43.743	21.978	31.851	1.019	0.219	1.191
8	Energy Critical On-Peak @ Gen	107	100.000	98.809	42.991	24.196	31.202	0.261	0.159	1.191
9	Energy Summer Mid-Peak @ Gen	108	100.000	98.809	39.241	24.830	34.185	0.380	0.174	1.191
10	12CP Dmd @ Generation	120	100.000	98.971	46.652	22.040	29.636	0.488	0.155	1.029
11	4CP Dmd @ Generation	121	100.000	98.972	51.219	22.605	24.800	0.202	0.145	1.028
12	Class Peak @ Subtransmission	122	100.000	99.943	47.227	23.530	27.339	0.531	1.316	0.057
13	Classpeak @ Transmission	127	100.000	99.131	44.367	22.105	30.923	0.499	1.236	0.869
14	Total Rate Revenue	143	100.000	99.407	48.269	24.924	25.145	1.028	0.041	0.593
15	Billed Sales	150	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
16	Billed Sales Excluding Rate E1	151	100.000	98.900	36.069	21.129	40.914	0.604	0.184	1.100
17	Number Of Customers	160	100.000	100.000	87.988	11.744	0.223	0.045	0.001	0.000
18	Weighted Customer	170	100.000	99.998	77.444	16.165	6.350	0.022	0.018	0.002
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	100.000	98.986	51.521	22.166	24.752	0.410	0.137	1.014
20	4CP 89/0/11	220	100.000	98.972	50.209	22.750	25.604	0.258	0.151	1.028
21	4CP 89/0/11 Exc WFR	222	100.000	99.967	50.713	22.979	25.862	0.260	0.152	0.033
22	4CP Dmd @ Gen Jurisdictional	224	100.000	100.000	51.751	22.840	25.058	0.205	0.146	-
23	12CP Demand @ Subtrans	226	100.000	100.000	49.035	23.165	27.231	0.513	0.055	-
24	Classpeak @ Primary	230	100.000	100.000	50.674	25.247	23.476	0.570	0.033	-
25	Classpeak @ Secondary	231	100.000	100.000	66.248	33.007	-	0.745	-	-
26	Classpeak for Streetlighting	233	100.000	99.188	-	-	-	99.188	-	0.812
27	Classpeak @ Single Phase	235	100.000	100.000	66.248	33.007	-	0.745	-	-
28	Billed Sales ROA	252	100.000	100.000	-	5.554	94.446	-	-	-
29	Billed Sales - Primary	253	100.000	100.000	43.102	25.249	30.922	0.721	0.005	-
30	Customers - Residential	260	100.000	100.000	100.000	-	-	-	-	-
31	Customers - Drops	261	100.000	100.000	-	100.000	-	-	-	-
32	Customers - NonPID	263	100.000	100.000	88.208	11.773	-	0.019	-	-
33	Customers - NonMunicipal	264	100.000	100.000	88.027	11.749	0.224	-	0.001	0.000
34	PIS - 138kV Distribution	301	100.000	99.131	44.367	22.105	30.923	0.499	1.236	0.869
35	PIS - 46kV Distribution	302	100.000	99.943	47.227	23.530	27.339	0.531	1.316	0.057
36	PIS - 138kV Dist Subs S&E	303	100.000	99.131	44.367	22.105	30.923	0.499	1.236	0.869
37	PIS - 46kV Dist Subs S&E	304	100.000	99.943	47.227	23.530	27.339	0.531	1.316	0.057

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Residential/Secondary ALLOCATORS 1										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	
Line No.	Description	Alloc	Rate RS	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Input Allocation Schedules</u>										
1	Energy @ Generation	100	41.956	-	41.956	12.613	10.426	0.296	0.551	23.886
2	Energy On-Peak @ Generation	101	39.220	-	39.220	13.230	10.494	0.340	0.593	24.658
3	Energy Off-Peak @ Generation	102	43.529	-	43.529	11.423	9.954	0.233	0.480	22.089
4	Energy On-Peak @ Generation Summer	103	40.170	-	40.170	13.232	10.648	0.308	0.486	24.673
5	Energy Off-Peak @ Generation Summer	104	43.134	-	43.134	11.344	10.289	0.213	0.449	22.295
6	Energy On-Peak @ Generation Non-Summer	105	38.677	-	38.677	13.229	10.407	0.359	0.655	24.650
7	Energy Off-Peak @ Generation Non-Summer	106	43.743	-	43.743	11.465	9.772	0.243	0.497	21.978
8	Energy Critical On-Peak @ Gen	107	42.991	-	42.991	12.884	10.566	0.283	0.463	24.196
9	Energy Summer Mid-Peak @ Gen	108	39.241	-	39.241	13.346	10.675	0.316	0.494	24.830
10	12CP Dmd @ Generation	120	46.652	-	46.652	11.867	9.394	0.306	0.473	22.040
11	4CP Dmd @ Generation	121	51.219	-	51.219	12.311	9.570	0.313	0.412	22.605
12	Class Peak @ Subtransmission	122	47.227	-	47.227	13.150	9.033	0.504	0.842	23.530
13	Classpeak @ Transmission	127	44.367	-	44.367	12.354	8.486	0.474	0.791	22.105
14	Total Rate Revenue	143	48.269	-	48.269	14.002	9.986	0.324	0.612	24.924
15	Billed Sales	150	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
16	Billed Sales Excluding Rate E1	151	36.069	-	36.069	10.871	9.302	0.297	0.659	21.129
17	Number Of Customers	160	87.988	-	87.988	10.551	1.064	0.086	0.042	11.744
18	Weighted Customer	170	77.444	-	77.444	13.914	2.056	0.113	0.082	16.165
<u>Calculated Allocation Schedules</u>										
19	4CP Average & Excess	219	51.521	-	51.521	12.114	9.350	0.311	0.391	22.166
20	4CP 89/0/11	220	50.209	-	50.209	12.346	9.666	0.311	0.427	22.750
21	4CP 89/0/11 Exc WFR	222	50.713	-	50.713	12.471	9.763	0.314	0.431	22.979
22	4CP Dmd @ Gen Jurisdictional	224	51.751	-	51.751	12.439	9.669	0.316	0.416	22.840
23	12CP Demand @ Subtrans	226	49.035	-	49.035	12.473	9.873	0.322	0.497	23.165
24	Classpeak @ Primary	230	50.674	-	50.674	14.110	9.692	0.541	0.904	25.247
25	Classpeak @ Secondary	231	66.248	-	66.248	18.447	12.671	0.708	1.182	33.007
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-
27	Classpeak @ Single Phase	235	66.248	-	66.248	18.447	12.671	0.708	1.182	33.007
28	Billed Sales ROA	252	-	-	-	0.265	3.159	0.397	1.732	5.554
29	Billed Sales - Primary	253	43.102	-	43.102	12.991	11.116	0.355	0.787	25.249
30	Customers - Residential	260	100.000	-	100.000	-	-	-	-	-
31	Customers - Drops	261	-	-	-	89.846	9.063	0.730	0.361	100.000
32	Customers - NonPID	263	88.208	-	88.208	10.578	1.067	0.086	0.042	11.773
33	Customers - NonMunicipal	264	88.027	-	88.027	10.556	1.065	0.086	0.042	11.749
34	PIS - 138kV Distribution	301	44.367	-	44.367	12.354	8.486	0.474	0.791	22.105
35	PIS - 46kV Distribution	302	47.227	-	47.227	13.150	9.033	0.504	0.842	23.530
36	PIS - 138kV Dist Subs S&E	303	44.367	-	44.367	12.354	8.486	0.474	0.791	22.105
37	PIS - 46kV Dist Subs S&E	304	47.227	-	47.227	13.150	9.033	0.504	0.842	23.530

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Primary & Lighting ALLOCATORS 1																		
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	
Line		Rate	-	Rate	Rate	Rate	Rate		Rate GPD	Rate GPD	Rate GPD		Rate	Rate	Rate	Rate		
No.	Description	Alloc	GP	GPTU	GPD Vit 1	GPD Vit 2	GP GEI	EIP	GEI Vit 1	GEI Vit 2	GEI Vit 3	Primary	GML	GUL	GU-XL	GU	Lighting & Unmetered	
<u>Input Allocation Schedules</u>																		
1	Energy @ Generation	100	2.662	13.936	2.508	3.381	7.593	0.390	1.124	-	0.068	0.403	32.066	0.046	0.267	0.070	0.320	0.702
2	Energy On-Peak @ Generation	101	3.005	14.479	3.688	3.459	7.527	0.464	1.192	-	0.067	0.423	34.305	0.020	0.119	0.031	0.280	0.451
3	Energy Off-Peak @ Generation	102	2.166	12.782	4.155	3.664	7.513	0.290	1.003	-	0.066	0.365	32.005	0.073	0.428	0.111	0.354	0.966
4	Energy On-Peak @ Generation Summer	103	3.002	14.095	3.488	3.315	7.530	0.411	1.115	-	0.076	0.413	33.446	0.011	0.064	0.017	0.259	0.350
5	Energy Off-Peak @ Generation Summer	104	2.218	12.901	4.099	3.640	7.753	0.264	0.973	-	0.080	0.363	32.291	0.064	0.373	0.097	0.334	0.869
6	Energy On-Peak @ Generation Non-Summer	105	3.007	14.698	3.801	3.541	7.526	0.494	1.236	-	0.062	0.430	34.795	0.026	0.151	0.039	0.292	0.508
7	Energy Off-Peak @ Generation Non-Summer	106	2.138	12.718	4.186	3.678	7.383	0.305	1.019	-	0.057	0.366	31.851	0.078	0.457	0.119	0.365	1.019
8	Energy Critical On-Peak @ Gen	107	2.829	13.143	3.053	3.074	7.241	0.384	0.991	-	0.076	0.410	31.202	-	-	-	0.261	0.261
9	Energy Summer Mid-Peak @ Gen	108	3.059	14.408	3.632	3.395	7.625	0.420	1.156	-	0.077	0.413	34.185	0.015	0.085	0.022	0.258	0.380
10	12CP Dmd @ Generation	120	2.539	12.463	3.032	2.998	6.810	0.390	0.942	-	0.074	0.389	29.636	0.031	0.181	0.047	0.229	0.488
11	4CP Dmd @ Generation	121	2.568	11.231	1.387	2.237	6.529	0.366	-	-	0.091	0.391	24.800	-	-	-	0.202	0.202
12	Class Peak @ Subtransmission	122	2.618	10.201	-	4.731	7.470	0.645	0.403	-	0.345	0.927	27.339	0.045	0.263	0.069	0.155	0.531
13	Classpeak @ Transmission	127	2.460	9.583	4.426	4.444	7.018	0.606	1.178	0.013	0.324	0.871	30.923	0.042	0.247	0.064	0.145	0.499
14	Total Rate Revenue	143	2.543	10.107	2.176	2.487	6.402	0.373	0.583	0.000	0.081	0.393	25.145	0.039	0.572	0.193	0.224	1.028
15	Billed Sales	150	2.542	12.650	6.845	6.924	9.398	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
16	Billed Sales Excluding Rate E1	151	2.542	12.650	6.845	6.924	9.398	0.433	1.039	0.008	0.265	0.809	40.914	0.039	0.230	0.060	0.275	0.604
17	Number Of Customers	160	0.082	0.056	0.002	0.005	0.065	0.008	0.001	0.000	0.000	0.005	0.223	0.019	-	-	0.026	0.045
18	Weighted Customer	170	2.287	1.563	0.053	0.136	1.921	0.215	0.030	0.002	0.010	0.134	6.350	0.022	-	-	-	0.022
<u>Calculated Allocation Schedules</u>																		
19	4CP Average & Excess	219	2.524	10.793	1.463	2.142	6.339	0.358	0.656	-	0.093	0.384	24.752	0.027	0.156	0.041	0.187	0.410
20	4CP 89/0/11	220	2.579	11.531	1.511	2.363	6.647	0.369	0.124	-	0.089	0.392	25.604	0.005	0.029	0.008	0.215	0.258
21	4CP 89/0/11 Exc WFR	222	2.605	11.647	1.526	2.387	6.714	0.372	0.125	-	0.090	0.396	25.862	0.005	0.030	0.008	0.218	0.260
22	4CP Dmd @ Gen Jurisdictional	224	2.595	11.348	1.402	2.260	6.597	0.370	-	-	0.092	0.395	25.058	-	-	-	0.205	0.205
23	12CP Demand @ Subtrans	226	2.669	13.099	-	3.151	7.158	0.410	0.258	-	0.078	0.408	27.231	0.032	0.190	0.050	0.241	0.513
24	Classpeak @ Primary	230	2.810	10.945	-	-	8.016	0.692	0.018	-	-	0.995	23.476	0.048	0.282	0.074	0.166	0.570
25	Classpeak @ Secondary	231	-	-	-	-	-	-	-	-	-	-	-	0.063	0.369	0.096	0.217	0.745
26	Classpeak for Streetlighting	233	-	-	-	-	-	-	-	-	-	-	-	11.822	69.314	18.052	-	99.188
27	Classpeak @ Single Phase	235	-	-	-	-	-	-	-	-	-	-	-	0.063	0.369	0.096	0.217	0.745
28	Billed Sales ROA	252	1.261	-	29.466	33.732	23.121	0.750	-	0.073	1.899	4.145	94.446	-	-	-	-	-
29	Billed Sales - Primary	253	3.038	15.116	-	-	11.231	0.518	0.053	-	-	0.966	30.922	0.047	0.274	0.071	0.329	0.721
30	Customers - Residential	260	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
31	Customers - Drops	261	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
32	Customers - NonPID	263	-	-	-	-	-	-	-	-	-	-	-	0.019	-	-	-	0.019
33	Customers - NonMunicipal	264	0.082	0.056	0.002	0.005	0.065	0.008	0.001	0.000	0.000	0.005	0.224	-	-	-	-	-
34	PIS - 138kV Distribution	301	2.460	9.583	4.426	4.444	7.018	0.606	1.178	0.013	0.324	0.871	30.923	0.042	0.247	0.064	0.145	0.499
35	PIS - 46kV Distribution	302	2.618	10.201	-	4.731	7.470	0.645	0.403	-	0.345	0.927	27.339	0.045	0.263	0.069	0.155	0.531
36	PIS - 138kV Dist Subs S&E	303	2.460	9.583	4.426	4.444	7.018	0.606	1.178	0.013	0.324	0.871	30.923	0.042	0.247	0.064	0.145	0.499
37	PIS - 46kV Dist Subs S&E	304	2.618	10.201	-	4.731	7.470	0.645	0.403	-	0.345	0.927	27.339	0.045	0.263	0.069	0.155	0.531

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Summary ALLOCATORS 2										
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
Line No.	Description	Alloc	Total Electric	Total Jurisdictional Electric	Total Residential	Total Commercial Secondary	Total Primary	Total Lighting & Unmetered	Rate GSG	Total Non Jurisdictional
<u>Calculated Allocation Schedules</u>										
1	PIS - Overhead Primary System	305	100.000	100.000	66.248	33.007	-	0.745	-	(0.000)
2	PIS - Distribution Distribution	306	100.000	99.998	62.777	29.292	4.562	3.360	0.008	0.002
3	Overhead Distribution	307	100.000	99.986	61.013	30.460	7.786	0.556	0.172	0.014
4	Underground Distribution	308	100.000	100.000	64.991	32.380	1.895	0.731	0.003	(0.000)
5	Total Dist PIS	309	100.000	99.950	59.289	27.983	9.643	2.747	0.287	0.050
6	Distribution Services	310	100.000	100.000	66.512	33.488	-	-	-	-
7	Streetlighting Equipment	311	100.000	99.928	-	-	-	99.928	-	0.072
8	Line Equipment	312	100.000	100.000	64.260	32.016	2.997	0.723	0.004	-
9	Meters	313	100.000	99.998	77.444	16.165	6.350	0.022	0.018	0.002
10	PIS - System Power Control	314	100.000	199.074	91.594	45.635	58.263	1.031	2.552	(99.074)
11	PIS - General	315	100.000	99.585	58.838	24.435	15.104	1.031	0.177	0.415
12	Total PIS	316	100.000	99.564	55.975	25.764	15.909	1.689	0.228	0.436
13	Distribution Depreciation	317	100.000	99.961	58.796	29.369	6.751	4.750	0.295	0.039
14	CWIP	330	100.000	99.487	54.570	24.882	18.499	1.251	0.284	0.513
15	Working Capital	340	100.000	99.512	56.436	24.168	17.933	0.841	0.133	0.488
16	Rate Base	390	100.000	99.578	56.264	25.382	16.564	1.145	0.223	0.422
17	Operations - Distribution excl Sup & Eng	400	100.000	99.988	64.121	25.820	5.956	3.989	0.102	0.012
18	Maintenance - Distribution excl Sup & Eng	401	100.000	99.989	61.109	29.778	8.201	0.758	0.143	0.011
19	Operations - 138kV Distribution excl Sup & Eng	402	100.000	98.972	51.219	22.605	24.800	0.202	0.145	1.028
20	Maintenance - 138kV Distribution excl Sup & Eng	403	100.000	99.465	47.199	22.387	28.674	0.386	0.818	0.535
21	Operations - 46kV Distribution excl Sup & Eng	404	100.000	99.943	47.227	23.530	27.339	0.531	1.316	0.057
22	Maintenance - 46kV Distribution excl Sup & Eng	405	100.000	99.943	47.227	23.530	27.339	0.531	1.316	0.057
23	HV Distribution O&M exp.	406	100.000	99.886	47.254	23.404	27.459	0.514	1.255	0.114
24	Distribution O&M, excl. HV Dist	407	100.000	99.989	62.201	28.379	7.386	1.895	0.127	0.011
25	Customer Accounting	408	100.000	100.000	88.031	11.749	0.195	0.024	0.001	0.000
26	Customer Accounts & Service	409	100.000	99.800	78.574	13.456	7.605	0.131	0.034	0.200
27	Distribution O&M	410	100.000	99.980	61.027	27.993	8.950	1.796	0.216	0.020
28	Customer & Sales O&M	411	100.000	99.801	78.622	13.447	7.567	0.130	0.034	0.199
29	Administrative and General O&M	412	100.000	99.596	59.346	24.548	14.479	1.071	0.151	0.404
30	Jurisdictional Distribution O&M	414	100.000	100.000	61.039	27.998	8.951	1.796	0.216	-
31	O&M Excluding Adjustments	438	100.000	99.061	48.667	23.255	26.345	0.625	0.170	0.939
32	Pre Tax NOI	439	100.000	100.972	34.538	31.505	33.154	1.888	(0.113)	(0.972)
33	R&PP Tax	440	100.000	99.581	55.893	25.734	16.054	1.658	0.241	0.419
34	Depreciation & Amortization Expense	442	100.000	99.467	56.486	25.140	16.121	1.536	0.185	0.533
35	Non PSCR O&M Expense	443	100.000	99.567	58.533	24.304	15.505	1.056	0.168	0.433
36	Distribution Depreciation Expense	444	100.000	99.959	62.399	28.030	6.172	3.132	0.226	0.041
37	Gen/Comm/Int Depreciation Expense	445	100.000	99.585	58.838	24.435	15.104	1.031	0.177	0.415
38	Production Labor	500	100.000	98.972	50.209	22.750	25.604	0.258	0.151	1.028
39	Total Labor	502	100.000	99.585	58.838	24.435	15.104	1.031	0.177	0.415
40	50% O&M, 50% Net Plant	600	100.000	99.477	54.669	25.057	18.460	1.070	0.220	0.523
41	50/50 PIS & Labor	601	100.000	99.574	57.406	25.099	15.507	1.360	0.202	0.426

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Residential/Secondary ALLOCATORS 2										
	(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	Alloc	Rate RS	Rate RT	Total Residential	Rate GS	Rate GSD	Rate GS GEI	Rate GSD GEI	Commercial Secondary
<u>Calculated Allocation Schedules</u>										
1	PIS - Overhead Primary System	305	66.248	-	66.248	18.447	12.671	0.708	1.182	33.007
2	PIS - Distribution Distribution	306	62.777	-	62.777	18.196	9.674	0.549	0.873	29.292
3	Overhead Distribution	307	61.013	-	61.013	19.010	9.977	0.570	0.902	30.460
4	Underground Distribution	308	64.991	-	64.991	18.097	12.430	0.694	1.159	32.380
5	Total Dist PIS	309	59.289	-	59.289	17.070	9.511	0.538	0.864	27.983
6	Distribution Services	310	66.512	-	66.512	30.087	3.035	0.245	0.121	33.488
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-
8	Line Equipment	312	64.260	-	64.260	17.893	12.290	0.686	1.146	32.016
9	Meters	313	77.444	-	77.444	13.914	2.056	0.113	0.082	16.165
10	PIS - System Power Control	314	91.594	-	91.594	25.504	17.518	0.978	1.634	45.635
11	PIS - General	315	58.838	-	58.838	14.715	8.705	0.404	0.612	24.435
12	Total PIS	316	55.975	-	55.975	15.146	9.491	0.444	0.683	25.764
13	Distribution Depreciation	317	58.796	-	58.796	18.209	9.725	0.555	0.881	29.369
14	CWIP	330	54.570	-	54.570	14.413	9.416	0.418	0.636	24.882
15	Working Capital	340	56.436	-	56.436	14.339	8.870	0.370	0.589	24.168
16	Rate Base	390	56.264	-	56.264	14.877	9.387	0.438	0.681	25.382
17	Operations - Distribution excl Sup & Eng	400	64.121	-	64.121	16.298	8.301	0.474	0.746	25.820
18	Maintenance - Distribution excl Sup & Eng	401	61.109	-	61.109	18.426	9.893	0.563	0.896	29.778
19	Operations - 138kV Distribution excl Sup & Eng	402	51.219	-	51.219	12.311	9.570	0.313	0.412	22.605
20	Maintenance - 138kV Distribution excl Sup & Eng	403	47.199	-	47.199	12.387	8.940	0.413	0.647	22.387
21	Operations - 46kV Distribution excl Sup & Eng	404	47.227	-	47.227	13.150	9.033	0.504	0.842	23.530
22	Maintenance - 46kV Distribution excl Sup & Eng	405	47.227	-	47.227	13.150	9.033	0.504	0.842	23.530
23	HV Distribution O&M exp.	406	47.254	-	47.254	13.064	9.027	0.494	0.819	23.404
24	Distribution O&M, excl. HV Dist	407	62.201	-	62.201	17.678	9.328	0.532	0.843	28.379
25	Customer Accounting	408	88.031	-	88.031	10.556	1.065	0.086	0.042	11.749
26	Customer Accounts & Service	409	78.574	-	78.574	10.613	2.564	0.124	0.155	13.456
27	Distribution O&M	410	61.027	-	61.027	17.317	9.306	0.529	0.841	27.993
28	Customer & Sales O&M	411	78.622	-	78.622	10.613	2.556	0.124	0.154	13.447
29	Administrative and General O&M	412	59.346	-	59.346	14.845	8.685	0.405	0.614	24.548
30	Jurisdictional Distribution O&M	414	61.039	-	61.039	17.321	9.308	0.529	0.841	27.998
31	O&M Excluding Adjustments	438	48.667	-	48.667	12.844	9.569	0.324	0.517	23.255
32	Pre Tax NOI	439	34.538	-	34.538	17.569	12.767	0.184	0.984	31.505
33	R&PP Tax	440	55.893	-	55.893	15.111	9.497	0.444	0.683	25.734
34	Depreciation & Amortization Expense	442	56.486	-	56.486	14.837	9.282	0.408	0.612	25.140
35	Non PSCR O&M Expense	443	58.533	-	58.533	14.651	8.647	0.395	0.611	24.304
36	Distribution Depreciation Expense	444	62.399	-	62.399	17.606	9.093	0.516	0.815	28.030
37	Gen/Comm/Int Depreciation Expense	445	58.838	-	58.838	14.715	8.705	0.404	0.612	24.435
38	Production Labor	500	50.209	-	50.209	12.346	9.666	0.311	0.427	22.750
39	Total Labor	502	58.838	-	58.838	14.715	8.705	0.404	0.612	24.435
40	50% O&M, 50% Net Plant	600	54.669	-	54.669	14.509	9.472	0.421	0.655	25.057
41	50/50 PIS & Labor	601	57.406	-	57.406	14.931	9.098	0.424	0.647	25.099

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(thousands of dollars)

Primary & Lighting ALLOCATORS 2		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
Line No.	Description	Alloc	Rate GP	Rate GPTU	Rate GPD Vlt 1	Rate GPD Vlt 2	Rate GPD Vlt 3	Rate GP GEI	Rate EIP	Rate GPD GEI Vlt 1	Rate GPD GEI Vlt 2	Rate GPD GEI Vlt 3	Total Primary	Rate GML	Rate GUL	Rate GU-XL	Rate GU	Lighting & Unmetered
<u>Calculated Allocation Schedules</u>																		
1	PIS - Overhead Primary System	305	-	-	-	-	-	-	-	-	-	-	-	0.063	0.369	0.096	0.217	0.745
2	PIS - Distribution Distribution	306	0.684	1.999	0.005	0.015	1.534	0.137	0.006	0.000	0.001	0.181	4.562	0.074	1.952	1.178	0.156	3.360
3	Overhead Distribution	307	0.845	3.290	0.035	0.596	2.410	0.208	0.060	0.000	0.043	0.299	7.786	0.047	0.275	0.072	0.162	0.556
4	Underground Distribution	308	0.227	0.884	-	-	0.647	0.056	0.001	-	-	0.080	1.895	0.062	0.362	0.094	0.213	0.731
5	Total Dist PIS	309	1.095	3.745	0.199	1.022	2.797	0.245	0.126	0.001	0.074	0.340	9.643	0.067	1.586	0.938	0.155	2.747
6	Distribution Services	310	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
7	Streetlighting Equipment	311	-	-	-	-	-	-	-	-	-	-	-	1.046	55.182	43.700	-	99.928
8	Line Equipment	312	0.359	1.397	-	-	1.023	0.088	0.002	-	-	0.127	2.997	0.061	0.358	0.093	0.211	0.723
9	Meters	313	2.287	1.563	0.053	0.136	1.921	0.215	0.030	0.002	0.010	0.134	6.350	0.022	-	-	-	0.022
10	PIS - System Power Control	314	5.078	19.784	4.426	9.175	14.489	1.250	1.581	0.013	0.669	1.798	58.263	0.087	0.510	0.133	0.300	1.031
11	PIS - General	315	1.551	6.500	0.728	1.433	4.106	0.271	0.108	0.000	0.068	0.338	15.104	0.034	0.508	0.322	0.166	1.031
12	Total PIS	316	1.673	6.810	0.721	1.544	4.308	0.292	0.124	0.000	0.079	0.359	15.909	0.042	0.924	0.545	0.178	1.689
13	Distribution Depreciation	317	0.643	2.511	0.125	1.057	1.838	0.159	0.113	0.000	0.077	0.228	6.751	0.088	2.684	1.820	0.158	4.750
14	CWIP	330	1.909	7.964	0.763	1.922	4.980	0.330	0.130	0.000	0.099	0.404	18.499	0.034	0.654	0.380	0.183	1.251
15	Working Capital	340	1.798	7.575	1.135	1.727	4.734	0.294	0.252	0.000	0.069	0.348	17.933	0.033	0.403	0.222	0.183	0.841
16	Rate Base	390	1.776	7.053	0.751	1.533	4.530	0.313	0.141	0.000	0.078	0.387	16.564	0.037	0.626	0.305	0.177	1.145
17	Operations - Distribution excl Sup & Eng	400	0.737	2.505	0.031	0.353	1.873	0.164	0.039	0.000	0.026	0.227	5.956	0.079	2.179	1.594	0.137	3.989
18	Maintenance - Distribution excl Sup & Eng	401	0.959	3.500	0.030	0.492	2.588	0.225	0.052	0.000	0.036	0.318	8.201	0.050	0.387	0.161	0.161	0.758
19	Operations - 138kV Distribution excl Sup & Eng	402	2.568	11.231	1.387	2.237	6.529	0.366	0.000	0.000	0.091	0.391	24.800	0.000	0.000	0.000	0.202	0.202
20	Maintenance - 138kV Distribution excl Sup & Eng	403	2.512	10.260	3.266	3.606	6.856	0.515	0.726	0.008	0.235	0.688	28.674	0.026	0.152	0.040	0.168	0.386
21	Operations - 46kV Distribution excl Sup & Eng	404	2.618	10.201	-	4.731	7.470	0.645	0.403	-	0.345	0.927	27.339	0.045	0.263	0.069	0.155	0.531
22	Maintenance - 46kV Distribution excl Sup & Eng	405	2.618	10.201	-	4.731	7.470	0.645	0.403	-	0.345	0.927	27.339	0.045	0.263	0.069	0.155	0.531
23	HV Distribution O&M exp.	406	2.607	10.215	0.350	4.594	7.399	0.629	0.433	0.001	0.331	0.898	27.459	0.043	0.250	0.065	0.157	0.514
24	Distribution O&M, excl. HV Dist	407	0.879	3.142	0.029	0.438	2.331	0.203	0.047	0.000	0.032	0.285	7.386	0.060	1.016	0.666	0.153	1.895
25	Customer Accounting	408	0.071	0.049	0.002	0.004	0.057	0.007	0.001	0.000	0.000	0.004	0.195	0.011	-	-	0.013	0.024
26	Customer Accounts & Service	409	0.522	2.342	1.247	1.263	1.757	0.084	0.190	0.001	0.049	0.150	7.605	0.017	0.042	0.011	0.061	0.131
27	Distribution O&M	410	1.013	3.692	0.055	0.763	2.725	0.236	0.077	0.000	0.055	0.333	8.950	0.059	0.962	0.622	0.153	1.796
28	Customer & Sales O&M	411	0.519	2.330	1.240	1.257	1.748	0.084	0.189	0.001	0.048	0.150	7.567	0.017	0.042	0.011	0.061	0.130
29	Administrative and General O&M	412	1.495	6.259	0.711	1.323	3.948	0.259	0.099	0.000	0.061	0.323	14.479	0.035	0.531	0.340	0.165	1.071
30	Jurisdictional Distribution O&M	414	1.014	3.693	0.055	0.763	2.725	0.236	0.077	0.000	0.055	0.333	8.951	0.059	0.962	0.622	0.153	1.796
31	O&M Excluding Adjustments	438	2.337	11.218	2.383	2.649	6.351	0.353	0.603	0.000	0.073	0.378	26.345	0.030	0.246	0.109	0.240	0.625
32	Pre Tax NOI	439	4.688	9.896	3.442	2.994	9.747	0.606	1.091	0.002	0.122	0.566	33.154	0.077	1.264	0.334	0.213	1.888
33	R&PP Tax	440	1.696	6.912	0.618	1.595	4.378	0.298	0.108	0.000	0.083	0.367	16.054	0.041	0.906	0.533	0.178	1.658
34	Depreciation & Amortization Expense	442	1.707	6.965	0.846	1.594	4.251	0.266	0.112	0.000	0.074	0.305	16.121	0.036	0.821	0.500	0.179	1.536
35	Non PSCR O&M Expense	443	1.534	6.635	0.842	1.502	4.129	0.266	0.202	0.000	0.064	0.330	15.505	0.037	0.522	0.323	0.174	1.056
36	Distribution Depreciation Expense	444	0.816	2.166	0.166	0.820	1.693	0.153	0.101	0.001	0.060	0.196	6.172	0.070	1.809	1.108	0.145	3.132
37	Gen/Comm/Int Depreciation Expense	445	1.551	6.500	0.728	1.433	4.106	0.271	0.108	0.000	0.068	0.338	15.104	0.034	0.508	0.322	0.166	1.031
38	Production Labor	500	2.579	11.531	1.511	2.363	6.647	0.369	0.124	-	0.089	0.392	25.604	0.005	0.029	0.008	0.215	0.258
39	Total Labor	502	1.551	6.500	0.728	1.433	4.106	0.271	0.108	0.000	0.068	0.338	15.104	0.034	0.508	0.322	0.166	1.031
40	50% O&M, 50% Net Plant	600	1.889	7.865	1.056	1.747	4.885	0.323	0.228	0.000	0.078	0.389	18.460	0.036	0.570	0.274	0.189	1.070
41	50/50 PIS & Labor	601	1.612	6.655	0.725	1.488	4.207	0.281	0.116	0.000	0.074	0.348	15.507	0.038	0.716	0.434	0.172	1.360

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

RACHEL L. BARNES

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

1. Tariff Sheet Nos. A-4.00 through A-6.00 – **Index** – Updated index to reflect Residential Rate Restructuring, removal of Load Profiling, and the addition of Distributed Generation, Long Term Industrial Load Retention Rate LTILRR, and General Service Unmetered Light Emitting Diode Rate GU-LED.
2. Tariff Sheet No. C-4.00 – Rule C1 – **Characteristics of Service** – Updated Contribution in Aid of Construction Allowance Sheet.
3. Tariff Sheet No. C-15.00 – Rule C4.4 – **Resale** – Updated all-inclusive rates for campgrounds and boat harbor slips, removed a reference to Rate RT, and clarified RSM as the standard rate for resale residential use.
4. Tariff Sheet No. C-19.00 – Rule C5.2 – **Bills and Payments** – Updated interval data process within Meter Reads and Estimated Bills.
5. Tariff Sheet No. C-26.00 – Rule C5.6 – **Customer-Selected Due Date Program** – Updated availability to reflect current residential rate options.
6. Tariff Sheet No. C-28.00 – Rule C6.2 – **Underground Policy** – Updated to reflect adjustments to contribution in aid of construction when Underground service is required by state or local law or regulation.
7. Tariff Sheet Nos. C-36.00, C-54.00, D-67.00, D-82.00, E-22.00, E-24.00, and E-26.00 – **Line Loss Factors** – Updated line loss factors.
8. Tariff Sheet No. C-52.00 – Rule C10.5 – **Pilot Solar Program** – Updated Customer Eligibility to reflect current residential rate options.
9. Tariff Sheet Nos. C-58.00 through C-64.80, D-17.00, D-38.00, D-39.00, D-43.00, D-44.00, D-46.00, D-47.00, D-49.00, D-50.00, D-53.00, D-54.00, D-57.00, D-58.00, D-64.00, D-68.00, D-73.00, D-77.00, D-78.00 and D-86.00 – Rule C11 – **Self-Generation, Net Metering Program and Distributed Generation Program** – Combined the tariffs for Self-Generation, the Net Metering Program, and the Distributed Generation Program into Rule C11 and reflected all associated changes on the individual rate schedules.
10. Tariff Sheet Nos. D-4.00 and D-4.10 – **Surcharges** – Added the Electric Rate Case Deferral Surcharge, the Financial Compensation Mechanism Surcharge, the Demand Response Reconciliation Surcharge, and the Conservation Voltage Reduction Incentive Surcharge.
11. Tariff Sheet Nos. D-8.00, D-9.00, and D-10.00 – **Rate Categories and Provisions** – Updated pages to reflect proposed new and deleted Rate Categories and Provisions.
12. Tariff Sheet Nos. D-14.00, D-15.00, D-16.00, D-36.00, D-37.00, D-38.00, D-40.00, D-41.00, D-42.00, D-43.00, D-44.10, D-44.20, D-44.30, D-45.00, D-46.00, D-48.00, D-49.00, D-51.00, D-53.00, D-55.00, D-56.00, D-57.00, D-59.00 through D-63.00, D-66.00, D-71.00, D-72.00, D-75.00, D-76.00, D-77.00, D-82.00, D-83.00, D-84.00, D-85.00, D-86.00, D-90.00, D-94.00, D-94.10, and D-96.00 – **Rate Schedules** – Revised prices.
13. Tariff Sheet No. D-14.00 – **Residential Summer On-Peak Basic Rate RSP** – Removed references to the Pilot which terminated on December 31, 2019.
14. Tariff Sheet Nos. D-15.00, D-37.00, D-41.00, and D-44.20 – **Low Income Assistance Credit (LIAC)** – Added LIAC language to Residential Summer On-Peak Basic, Residential Smart Hours, Residential Nighttime Savers, and Residential Service Secondary Non-Transmitting Meter Rates.

15. Tariff Sheet Nos. D-15.10, D-16.00, D-37.00, D-38.00, D-41.00 and D-42.00 – **Peak Power Savers Programs** – Added language allowing test events between October 1 and May 31. The Air Conditioning Peak Cycling Program is being modified to allow auto-enrollment of customers into the program who move into a premises with existing Load Management equipment. The Peak Reward Program is being modified to allow up to five critical peak events between October 1 and May 31.
16. Tariff Sheet Nos. D-19.00 through D-35.00 – **Residential Rates** – Canceling tariff sheets for Residential Service Secondary Rate RS, Residential Dynamic Pricing Program, Experimental Residential Plug-In Electric Vehicle Charging Program, and Residential Service Time-of-Day Secondary Rate RT.
17. Tariff Sheet No. D-40.00 – **Residential Nighttime Savers Rate RPM** – Updated the Availability clause to remove erroneous language referencing critical peak events.
18. Tariff Sheet No. D-44.10 – **Residential Service Secondary Non-Transmitting Meter Rate RSM** – Added language to include non-communicating meters into the RSM Rate.
19. Tariff Sheet No. D-45.00 – **General Service Secondary Rate GS** – Closed rate to new business as of January 1, 2021.
20. Tariff Sheet Nos. D-48.00 and D-49.00 – **General Service Secondary Time-of-Use Rate GSTU** – Added standby service for solar installations, a definition for Peak Demand, and the Interruptible Service Provision.
21. Tariff Sheet Nos. D-52.00 and D-53.00 – **General Service Secondary Demand Rate GSD** – Modified Adjustment for Power Factor to align with primary rates and added Interruptible Service Provision.
22. Tariff Sheet No. D-55.00 – **General Service Primary Rate GP** – Closed rate to new business as of January 1, 2021.
23. Tariff Sheet No. D-62.00, D-64.00 through D-66.00 – **Large General Service Primary Demand Rate GPD** – Added language to clarify the on-peak billing demand is determined using the four preceding summer months, not including the current month. Updated maximum interruptible amounts for the Interruptible Service Provision (GI) and notification dates to renew the GI Provision. Added information to clarify the Interruptible Service Provision – Market-Price Option (GI2).
24. Tariff Sheet No. D-70.00 – **General Service Primary Time-of-Use Rate GPTU** – Modified the availability to include standby service for solar installations equal to or greater than 150 kW.
25. Tariff Sheet Nos. D-74.00 and D-75.00 – **Energy Intensive Primary Rate EIP** – Modified the availability of the rate to include former metal melting customers and revised Critical Peak Hours to include all hours during a Critical Peak Event to align with existing Critical Peak Event Determination language which states a System Integrity Event shall occur at any time for any duration.
26. Tariff Sheet No. D-79.00 – **Experimental Advanced Renewable Program AR** – Updated reference to residential rate schedule from Residential Service Rate RS to Residential Service Secondary Non-Transmitting Meter Rate RSM.
27. Tariff Sheet Nos. D-84.10 through D-84.60 – **Long Term Industrial Load Retention Rate LTILRR** – Added Rate LTILRR.
28. Tariff Sheet Nos. D-93.00 through D-95.00 – **General Service Unmetered Light Emitting Diode Rate GU-LED** – Replaced “experimental” with “Light Emitting Diode” or “LED” throughout the rate, including in the rate name. Tariff language has been updated to offer additional options and to align with current practices. Term of contract has been increased from two years to five years to align with Rate GUL.
29. Tariff Sheet Nos. E-7.00 and E-20.00 – **Load Profiling** – Eliminated references to load profiling for retail open access customers.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Proposed Electric Tariff Sheets (M.P.S.C No. 14 – Redlined Version)

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. A-4.00

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M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. C-4.00

(Continued From Sheet No. C-3.00)

C1. CHARACTERISTICS OF SERVICE (Contd)

C1.4 Extraordinary Facility Requirements and Charges (Contd)

Contribution In Aid of Construction Allowance Schedule							
Schedule	Customer Voltage Level(CVL)	With a Full Service Contract, by Contract Duration					Without Full Service Contract
		1 Year	2 Year	3 Year	4 Year	5 Year	
General Service Primary Rate GP	1	\$0.020 \$0.029/kWh	\$0.038 \$0.057/kWh	\$0.055 \$0.082/kWh	\$0.070 \$0.106/kWh	\$0.085 \$0.128/kWh	\$0.020 \$0.018/kWh
	2	\$0.034/kWh	\$0.045 \$0.066/kWh	\$0.064 \$0.095/kWh	\$0.083 \$0.122/kWh	\$0.100 \$0.148/kWh	\$0.034 \$0.024/kWh
	3	\$0.052 \$0.042/kWh	\$0.053 \$0.079/kWh	\$0.076 \$0.114/kWh	\$0.098 \$0.147/kWh	\$0.199 \$0.178/kWh	\$0.052 \$0.042/kWh
Large General Service Primary Demand Rate GPD	1	\$60.425 /kW	\$115.240 /kW	\$165.350 /kW	\$215.450 /kW	\$260.545 /kW	\$35.40 /kW
	2	\$105.455 /kW	\$160.295 /kW	\$230.425 /kW	\$295.550 /kW	\$355.665 /kW	\$105.75 /kW
	3	\$175.190 /kW	\$200.365 /kW	\$290.525 /kW	\$375.680 /kW	\$455.820 /kW	\$175.140 /kW
General Service Primary Time-of-Use Rate GPTU	1	\$0.015 \$0.022/kWh	\$0.029 \$0.042/kWh	\$0.042 \$0.061/kWh	\$0.054 \$0.079/kWh	\$0.066 \$0.095/kWh	NA
	2	\$0.018 \$0.027/kWh	\$0.035 \$0.052/kWh	\$0.050 \$0.075/kWh	\$0.065 \$0.096/kWh	\$0.078 \$0.117/kWh	NA
	3	\$0.021 \$0.033/kWh	\$0.041 \$0.064/kWh	\$0.060 \$0.092/kWh	\$0.077 \$0.119/kWh	\$0.093 \$0.144/kWh	NA
Energy Intensive Primary Rate EIP	1	\$0.006/kWh	\$0.011/kWh	\$0.016/kWh	\$0.020/kWh	\$0.024/kWh	NA
	2	\$0.008 \$0.010/kWh	\$0.016 \$0.019/kWh	\$0.024 \$0.027/kWh	\$0.031 \$0.035/kWh	\$0.037 \$0.042/kWh	NA
	3	\$0.011 \$0.012/kWh	\$0.022/kWh	\$0.032/kWh	\$0.041 \$0.042/kWh	\$0.049 \$0.050/kWh	NA

The Company reserves the right to make special contractual arrangements as to the provision of necessary Service Facilities, duration of contract, minimum bills, require upfront deposit and other service conditions, including, but not limited to, when the customer's load requirements are of a short-term duration, temporary or a transient nature, or if in the opinion of the Company, the customer does not have acceptable credit history or represents an unacceptable credit risk or other reasons within the sound discretion of the Company.

Contributions in Aid of Construction otherwise required by the Company may be suspended for publicly available AC Level 2 or DC Fast Charge sites participating in the PowerMIDrive pilot. Suspension is at the Company's sole discretion, for a term of three years from the date of Commission approval of the PowerMIDrive pilot.

C1.5 Invalidity of Oral Agreements or Representations

When a written contract is required, no employee or agent of the Company is authorized to modify or supplement the Rules and Regulations and Rate Schedules of the Electric Rate Book by oral agreement or representation, and no such oral agreement or representation shall be binding upon the Company.

(Continued on Sheet No. C-5.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-15.00

(Continued From Sheet No. C-14.00)

C4. APPLICATION OF RATES (Contd)

C4.3 Application of Residential Usage and Non-Residential Usage (Contd)

D. Rate Application for Seasonal Condominium Campgrounds (Contd)

- (6) The customer must notify individuals and co-owners utilizing the customer's property that requests and concerns regarding electric service will be addressed between the single legal entity and ownership and primary operating authority, not with individuals.
- (7) The customer shall be responsible for ensuring that the electrical facilities are adequate to meet the needs of the units placed within the Seasonal Condominium Campground in their entirety and shall pay the Company for any charges incurred for modifications necessary to accommodate load according to other portions of this Electric Rate Book.

C4.4 Resale

This provision is closed to resale for general unmetered service, unmetered or metered lighting service and new or expanded service for resale for residential use.

No customer shall resell electric service to others except when the customer is served under a Company rate expressly made available for resale purposes, and then only as permitted under such rate and under this rule.

Where, in the Company's opinion, the temporary or transient nature of the proposed ultimate use, physical limitation upon extensions, or other circumstances, make it impractical for the Company to extend or render service directly to the ultimate user, the Company may allow a customer to resell electric service to others.

For the purposes of this tariff, the provision of electric vehicle charging service for which there is no direct per kWh charge shall not be considered resale of service.

A resale customer is required to take service under the resale provision of one of the following rates for which they qualify: General Service Secondary Rate GS, General Service Secondary Time-of-Use Rate GSTU, General Service Secondary Demand Rate GSD, General Service Primary Rate GP, Large General Service Primary Demand Rate GPD, or General Service Primary Time-of-Use Rate GPTU. Resale Service is provided pursuant to a service contract providing for such resale privilege. Service to each ultimate user shall be separately metered.

- A. If the resale customer elects to take service under a Company Full Service resale rate, the ultimate user shall be served and charged for such service under standard Rate ~~RS~~ RSM for residential use or under the appropriate standard General Service Rate applicable in the Company's Electric Rate Book available for similar service under like conditions. Reselling customers are not required to offer or administer any additional service provisions or nonstandard rates contained in the Electric Rate Book, such as the Income Assistance Service Provision, ~~Residential Service Time-of-Day Secondary Rate RT~~ or the Educational Institution Service Provision.
- B. If the resale customer elects to take service under a Company Retail Open Access Service rate, the ultimate user shall be served and charged for such service under Rate ROA-R for residential use or under Rate ROA-S or ROA-P applicable in the Company's Electric Rate Book available for similar service under like conditions.
- C. If the ultimate user is a campground lot or boat harbor slip, the resale customer has the option to charge a maximum of the following all inclusive rate per kWh in place of billing the ultimate customer on the appropriate standard Company tariff rate:

\$0.153111 ~~0.449825~~ per kWh for all kWh during the months of June-September

\$0.152162 ~~0.448848~~ per kWh for all kWh during the months of October-May

The Company shall be under no obligation to furnish or maintain meters or other facilities for the resale of service by the reselling customer to the ultimate user.

The service contract shall provide that the reselling customer's billings to the ultimate user shall be audited each year by February's month end, for the previous calendar year. The audit shall be conducted either by the Company, if the Company elects to conduct such audit, or by an independent auditing firm approved by the Company. The reselling customer shall be assessed a reasonable fee for an audit conducted by the Company. If the audit is conducted by an independent auditing firm, the customer shall submit a copy of the results of such audit to the Company in a form approved by the Company.

(Continued on Sheet No. C-16.00)

(Continued From Sheet No. C-18.00)

C4. APPLICATION OF RATES (Contd)

C4.5 Mobile Home Park - Individually Served (Contd)

Notwithstanding other provisions of this rule, an overhead distribution system shall be incorporated into the Company's electric distribution system originally as an overhead distribution system. Any subsequent conversion to underground distribution facilities shall be in accordance with the provisions of this rule. The mobile home park owner shall be responsible to provide a recordable easement granting rights-of-way suitable for installation and maintenance of the electric facilities; to provide, own, install and maintain suitable meter supports; to remove any unused existing electrical equipment not transferred to the utility; to make any necessary wiring changes to separate the electrical responsibilities of the park owner from those of the tenant; and to move mobile homes or other equipment as required to provide access to easements to facilitate maintenance or required upgrading of the existing system.

F. Extension Policy

Service to mobile home parks shall be subject to the provisions of Rule C6., Distribution Systems, Line Extensions and Service Connections.

G. Any charges, contributions or deposits may be required In Advance of commencement of construction.

C5. CUSTOMER RESPONSIBILITIES

C5.1 Access to Customer's Premises

The Company's authorized agents shall have access to the customer's premises at all reasonable hours, to install, inspect, read, repair or remove its meters; to install, operate, maintain or remove other Company property, and to inspect and determine the connected electrical load on the customer's premises. Neglect or refusal on the part of the customer to provide reasonable access shall be sufficient cause for shutoff of service by the Company, and assurance of access may be required before service is restored.

C5.2 Bills and Payments

A. Billing Frequency

Bills for electric service shall be rendered on approximately a monthly basis, and shall be due and payable on or before the due date shown on each bill.

B. Meter Reads and Estimated Bills

The Company shall schedule meters to be read on approximately a monthly basis and will attempt to read meters in accordance with such schedule.

When the Company is unable to obtain an actual meter reading, the bill shall be estimated on the basis of past service records, adjusted, as may be appropriate. Where past service records are not available or suitable for use, such billing shall be based upon whatever other service data are available. Each such account shall be adjusted as necessary each time an actual meter reading is obtained.

(1) Interval Data Estimation

The Company requests usage data (including index and interval data) from smart meters daily. The usage data is stored in the meter data management system. The billing system requests the ending index read and time of use interval billing determinants (i.e., consumption of usage in each tier) for the billing cycle.

The billing system performs an industry standard sum check calculation comparing the sum of the interval billing determinant tiers to the difference of the starting and ending index.

When index data is missing, the billing system estimates the missing index read.

When interval data is missing, the delta is placed in the lower-tiered rate.

In the case where there is no missing data and the sum of the interval tiers exceeds difference of the index reads, the bill will be adjusted by a billing agent such that the excess interval tiered usage is removed from the higher priced tier.

~~When data is requested from a smart meter for a specific timeframe, the Company shall check the interval consumption data provided by the meter using industry standard validation rules. If the validation process fails because of missing or invalid interval data, that interval data is estimated by the meter data management system. The system estimates using linear interpolation for gaps of 4 hours or less, followed by the California historical estimation algorithm, both of which are industry standard. In the case where actual register reads are not available in the meter data management system the Company will manually insert the register reads extracted from the billing system into the meter data management system. The estimation algorithms will be executed again in the meter data management system to ensure the estimated intervals match the register reads from the billing system.~~

(Continued on Sheet No. C-20.00)

See Barnes Testimony, Page 3, Lines 1-13; Exhibit A-23 (RLB-1) Item #4
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(Continued From Sheet No. C-25.00)

C5. CUSTOMER RESPONSIBILITIES (Contd)

C5.6 Customer-Selected Due Date Program

Notwithstanding other provisions in this tariff book, the Company, at its discretion, may provide its electric service customers and combination electric and gas service customers the option to select the day of the month on which their bill is due, regardless of the meter read date. Participating customers must have an electric AMI transmitting technology meter.

Participation in the Customer-Selected Due Date Program is available to customers, as determined by the Company, when technically feasible based on the customer's selected rate and billing options. Customers not eligible to participate include, but not limited to, customers billed on a calendar-month basis, customers participating in Retail Open Access and customers participating in the Net Metering Program.

The Customer-Selected Due Date Program is only available for the following rate categories: Residential Summer On-Peak Basic (RSP), Residential Smart Hours (RSH), Residential Nighttime Savers (RPM), ~~Residential (RS)~~, ~~Residential Service Time of Day Secondary (RT)~~, ~~Plug-In Electric Vehicle Charging (PEV)~~, General Service Secondary (GS), General Service Secondary Demand (GSD), General Service Primary (GP) and General Service Metered Lighting (GML).

C6. DISTRIBUTION SYSTEMS, LINE EXTENSIONS AND SERVICE CONNECTIONS

C6.1 Overhead Extension Policy

Applications for electric service which require the construction of an overhead distribution system shall be granted under the following conditions:

A. Residential Customers

The Company shall construct single-phase distribution line extensions at its own cost a distance of 600 feet, for each residential dwelling.

The length of the distribution line extension shall be measured from the nearest point of connection to the Company's facilities from which the extension can be made to the point from which the service line to the customer shall be run.

Distribution line extensions in excess of the above 600 feet shall require a deposit for the estimated cost of such excess footage. The required deposit for such excess footage shall be \$3.50 per lineal foot less 25%.

(Continued on Sheet No. C-27.00)

**M.P.S.C. No. 14 – Electric
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Sheet No. C-28.00

(Continued From Sheet No. C-27.00)

C6. DISTRIBUTION SYSTEMS, LINE EXTENSIONS AND SERVICE CONNECTIONS (Contd)

C6.1 Overhead Extension Policy (Contd)

C. General (Contd)

- (6) The Company reserves the right to make special contractual arrangements as to the provision of necessary Service Facilities, duration of contract, amount of deposit and refunds thereon, minimum bills or other service conditions with respect to the customers or prospective customers whose load requirements exceed the capacity of the available distribution system in the area, or whose load characteristics or special service needs require unusual investments by the Company in Service Facilities or where there is not sufficient assurance of the permanence of the use of the service. The Company shall construct overhead electric distribution facilities and extensions only in the event it is able to obtain or use the necessary materials, equipment and supplies. The Company, subject only to review by the Commission, reserves the right, in its discretion, to allocate the use of such materials, equipment and supplies it may have on hand from time to time among the various classes of customers and prospective customers and among various customers and prospective customers of the same class.
- (7) Contributions in Aid of Construction otherwise required by the Company may be suspended for publicly available AC Level 2 or DC Fast Charge sites participating in the PowerMIDrive pilot. Suspension is at the Company's sole discretion, for a term of three years from the date of Commission approval of the PowerMIDrive pilot.
- (8) All service rendered shall be subject to the Company's Standard Contract forms and to its Electric Rate Book.
- (9) Any charges, deposits or contributions may be required In Advance of commencement of construction.

C6.2 Underground Policy

A. General

This rule sets forth the conditions under which the Company shall install direct burial underground electric distribution systems and underground service connections for residential and General Service customers. For the purpose of this rule, such underground distribution facilities are defined as those facilities operated at 15,000 Volts or less phase to ground wye connected or 20,000 Volts or less phase to phase delta connected.

The general policy of the Company is that real estate developers, property owners or other applicants for underground service shall make a contribution in aid of construction to the Company in an amount equal to the estimated difference in cost between underground and equivalent overhead facilities. Methods for determining this cost differential for specific classifications of service are provided herein. In cases, where the nature of service or the construction conditions are such that these conditions are not applicable, the general policy stated above shall apply.

In cases where the Company does not require underground electric distribution systems and/or underground service connections, but is required to underground such facilities by state or local law or regulation, the Company may adjust the contribution in aid of construction to account for such requirement.

It shall be mandatory that all original electric distribution systems installed in new residential subdivisions and in existing residential subdivisions in which overhead electric distribution facilities have not already been constructed be placed underground, except that a lot within a subdivision facing a previously existing street or county road and having an existing overhead distribution line on its side of the street or county road shall be served with an underground service from these facilities and shall be considered a part of the underground service area. It shall also be mandatory that all original service connections installed to serve one-family or two-family dwellings from an underground distribution system be placed underground.

Except as otherwise provided in the following paragraph, it shall be mandatory that all new General Service distribution systems and service connections installed in the vicinity of or on the customer's premises to be served, and constructed solely to serve the customer or a group of adjacent customers, be placed underground.

(Continued on Sheet No. C-29.00)

(Continued From Sheet No. C-35.00)

C8. POWER SUPPLY COST RECOVERY (PSCR) CLAUSE (Contd)

A. Applicability of Clause (Contd)

"Power Supply Costs" means those elements of the costs of fuel and purchased and net interchanged power as determined by the Commission to be included in the calculation of the Power Supply Cost Recovery Factor. The Commission determined in its Order in Case No. U-10335 dated May 10, 1994 that the fossil plant emissions permit fees over or under the amount included in base rates charged the Company are an element of fuel costs for the purpose of the clause.

B. Billing

- (1) The Power Supply Cost Recovery Factor shall consist of an adjustment factor of ~~1.07933~~ 1.08378 applied to projected average booked cost of fuel burned for electric generation and purchased and net interchange power incurred above or below a cost base of \$0.05570 per kWh (excluding line losses). Average booked costs of fuel burned and purchased and net interchange power shall be equal to the booked costs in that period divided by that period's net system kWh requirements. The average booked costs so determined shall be truncated to the full \$0.00001 cost per Kilowatt-hour. Net system kWh requirements shall be the sum of the net kWh generation and net kWh purchased and interchange power.
- (2) Each month the Company shall include in its rates a Power Supply Cost Recovery Factor up to the maximum authorized by the Commission as shown on Sheet No. D-6.00.

Should the Company apply lesser factors than those shown on Sheet No. D-6.00, or if the factors are later revised pursuant to Commission Orders or Michigan Compiled Laws, Annotated, 460.6 et seq., the Company shall notify the Commission if necessary and file a revised Sheet No. D-6.00.

C. General Conditions

- (1) The power supply and cost review shall be conducted not less than once a year for the purpose of evaluating the Power Supply Cost Recovery Plan filed by the Company and to authorize appropriate Power Supply Cost Recovery Factors. Contemporaneously with its Power Supply Cost Recovery Plan, the Company shall file a 5-year forecast of the power supply requirements of its customers, its anticipated sources of supply and projections of Power Supply Costs.
- (2) Not more than 45 days following the last day of each billing month in which a Power Supply Cost Recovery Factor has been applied to customers' bills, the Company shall file with the Commission a detailed statement for that month of the revenues recorded pursuant to the Power Supply Cost Recovery Factor and the allowance for cost of power included in the base rates established in the latest Commission order for the Company, and the cost of power supply.
- (3) All revenues collected pursuant to the Power Supply Cost Recovery Factors and the allowance for power included in the base rates are subject to annual reconciliation proceedings.

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Sheet No. C-52.00

(Continued from Sheet No. C-51.00)

C10. RENEWABLE ENERGY PLAN (REP) (Contd)

C10.5 Pilot Solar Program

The purpose of this rule is to develop and test programs to enable the development of Michigan's renewable energy resources. The Pilot Solar Program (Solar Program) is a voluntary program intended to further the deployment of solar energy in Michigan and meet customer demand. The Solar Program will consist of up to 10 MW of large scale solar facilities. The Company will extend the pilot program to follow the PA 342 2016 Amendatory Act for a period of three years following Commission approval of this tariff in docket No. U-18231. The Company will own and maintain all facilities under this pilot and/or contract with Independent Power Producers for the solar energy output of facilities located within Consumers Energy's electric distribution service area.

Eligible customers will have an opportunity to subscribe to the Solar Program. A subscription is equal to 0.5 kW of solar energy. Customers may subscribe to more than one subscription; however, a customer's total subscriptions shall not exceed the customer's Annual Net Usage. A subscribed customer will receive a Solar Energy Credit for the subscription's percentage of the solar energy generated in the Solar Program. This Solar Energy Credit includes the energy and capacity value of the program production as defined herein, and avoided line losses. The Company will retire the Renewable Energy Credits (REC), as defined in Public Act 295 of 2008 and in compliance with MCL 460.1011.

A. Definitions

Annual Net Usage - the average annual kWh usage or the annual Imputed Customer Usage in kWhs if enrolled in Net Metering.

Long Term Program Capacity Value - the product of the Zonal Resource Credits for the facilities, as determined by Mid-Continent Independent System Operator (MISO), and 75% of the applicable MISO published Cost of New Entry for the resource zone in the lower peninsula of Michigan, adjusted annually.

Long Term Program Energy Value - the kWh production of the Solar Program at each hourly interval, multiplied by the hourly day ahead Locational Marginal Price (LMP) at the CONS.CETR pricing node, adjusted for applicable line losses.

Short Term Program Energy and Capacity Value - the monthly kWh production of the Solar Program multiplied by the fixed rate of \$0.075/kWh.

Solar Energy Credit - the monthly bill credit provided to the enrolled customer based on enrollment level, program solar energy production and the value of the energy credit and capacity credit described below.

Subscription Payment - a payment to participate in the Solar Program, equal to the cost of 0.5 kW of solar capacity.

B. Customer Eligibility

Subject to any restrictions, the Solar Program is available to any Full Service customer served on Rate RSP, RSH, RPM, RS, RDP, RDPR, RT, RSM, GS, GSTU, GSD, GP, GPD, EIP, and GPTU. Customers will not be eligible for the Solar Program if they have received a shutoff notice within nine months preceding their application.

(Continued on Sheet No. C-53.00)

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Sheet No. C-54.00

(Continued from Sheet No. C-53.00)

C10. RENEWABLE ENERGY PLAN (REP) (Contd)

C10.5 Pilot Solar Program (Contd)

E. Solar Energy Credits

Solar Energy Credits applied to the customer's monthly bill are based on the customer's subscription level, the energy credit and the capacity credit.

The Solar Energy Credits in years one through five will be based on the Short Term Program Energy and Capacity Value and in years six through twenty-five on the sum of the Long Term Program Energy Value and the Long Term Program Capacity Value.

The Long Term Program Energy Value includes a factor to account for avoided line losses attributable to the distributed resource location on the distribution system. The avoided line loss factor is 2.38 ~~2.32~~%. This value will be revised when line losses are updated in general electric rate cases, as approved by the Commission.

Customers that chose to have the REC sold when this option was initially available will be credited quarterly. The REC credit is based on a Michigan Renewable Portfolio Standard REC value published quarterly in the Midwest Market Notes by Clear Energy Brokerage and Consulting, LLC, or successor publication, multiplied by the RECs generated. Alternatively, the REC value may be based on the actual sale of the RECs.

If the monthly Solar Energy Credit is greater than the customer's bill, the excess credit will be rolled over and applied to the next month's bill. If a Solar Energy Credit accumulates to an amount greater than \$100, the Company shall pay the balance to the customer.

F. Reporting

Solar Program production data will be available on the Company's website. Each participating customer's monthly energy bill will include the Subscription Payment and Solar Energy Credit.

The Company will provide quarterly reports to the MPSC detailing the enrollment status and Solar Program production.

G. Cost Recovery

Costs will be recovered as set forth in the Commission Order in Case No. U-17752.

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Consumers Energy Company

Sheet No. C-58.00

(Continued from Sheet No. C-57.00)

C10. RENEWABLE ENERGY PLAN (REP) (Contd)

C10.6 Voluntary Large Customer Renewable Energy Pilot (LC-REP) Program (Contd)

B. Monthly Rate (Contd)

(1) Billing and Credits (Contd)

- (c) Administrative Charge: The customer or the customer's third party renewable energy provider is responsible for delivery and sale of renewable energy to MISO. As mutually agreed, the Company may act as the administrator of the customer's renewable energy and the customer will compensate the Company through a negotiated service contract. If the Company acts as the customer's administrator for renewable energy deliveries, then the Company will bid the customer's renewable energy into the MISO energy market at the generator node and bid the generator capacity into the MISO annual capacity auction on the customer's behalf.

C. Term and Form of Contract

The LC-REP Program shall require a written negotiated service contract. Except for the initial year in which this tariff is approved, the enrollment period is open from June through September 30th each year and the program year runs from January to December.

D. Early Termination of Contract

Customers who choose to terminate their service under Option B of the LC-REP Program early will be required to take service under the existing rate schedule for the remainder of their contract year. Customers who opted for the Company to manage their renewable energy will also be responsible for any costs to the Company not yet recovered under their negotiated service contract.

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION

C11.1 Self-Generation

A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility, but elects not to participate in the Company's Standard Offer under Rule C18, Distributed Generation Program, or Net Metering Program, may elect to sell energy to the Company under an energy-only contract. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.

A. Distribution Requirements for Sellers Connected to Company System

- (1) All facilities operated in parallel with the Company's system must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall install, own, operate, and maintain all metering and auxiliary devices (including any telecommunication links, if applicable) connected to the Company System. Meters furnished, installed, and maintained by the Company shall meter generation equipment.
- (2) Energy delivered to the Company shall be alternating current, 60-hertz, single-phase or three-phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) service. The Company will determine the particular nature of the voltage in each case.
- (3) If the seller's QF is connected to a distribution line serving other Company customers, then the point of delivery for energy measurement purposes shall be at the high voltage side of the generating facility's isolation transformer connecting the seller's generating facility to the Company's distribution system. If the seller's generating facility is not connected to a distribution line serving other Company customers, then the point of delivery for energy measurement purposes shall be at the point at which the radial line connecting the seller's generating facility to the Company's distribution system terminates at the first substation beyond the generating facility's isolation transformer.

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Sheet No. C-58.10

(Continued from Sheet No. C-58.00)

- (4) Interval Data Meters are required for each generating unit served under this rate. For a seller in which the measurement of energy delivered to the Company is not located at the point of delivery, then electric losses as determined by the Company for losses between the energy measurement location and the point of delivery shall be deducted for billing purposes from the energy measurements thus made.
- (5) The seller must meet the requirements contained in Rule B8., Electric Interconnection and Net Metering Standards, R 460.615 - R 460.628, for the category of generator installed. Per these standards, testing and utility approval of the interconnection and execution of a parallel operating agreement must be completed prior to the equipment operating in parallel with the distribution system of the utility. Additionally, the Company will confirm and ensure that an electric generator installation at the seller's site meets the IEEE 1547 anti-islanding requirements.
- (6) The seller is required to obtain the characteristics of service from the Company prior to the installation of equipment. The Company shall provide the characteristics in writing upon request. In the event that the equipment proposed for connection is not compatible with these characteristics, the Company shall have no obligation to modify its distribution system or provide any monetary compensation to the seller.
- Any service facilities shall be dedicated to the generator and shall not be shared with those providing service to any seller. The Company shall determine the characteristics of service. Should the installation of new Company distribution facilities be necessary for the equipment, all costs for the distribution facilities installed may be charged to the applicant in advance of construction as a nonrefundable contribution. If the applicant desires underground service facilities, the difference in cost between overhead and underground service facilities shall be charged to the applicant in advance of construction as a nonrefundable contribution.
- (7) If, in the sole judgment of the Company, it appears that connection of the equipment and subsequent service through the Company's facilities may cause a safety hazard, endanger the Company facilities or the seller's equipment or to disturb the Company's service to customers and other sellers, the Company may refuse or delay connection of the equipment to its facilities.
- A seller who contracts to sell energy to the Company on an energy-only contract is not eligible to participate in the Company's Standard Offer, Net Metering Program or its Distributed Generation Program during the term the energy-only contract is in effect. Sellers with unsatisfactory payment history on their delivery account are not eligible to participate.
- (8) The Company may discontinue purchases during system emergencies, maintenance, and other operational circumstances.

Administrative Cost Charge: \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less

Energy Purchase:
An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance, and other operational circumstances.

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(Continued from Sheet No. C-58.10)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program

- A. The Net Metering Program is offered as authorized by 2008 PA 295 and the Commission in Case Nos. U-15787, U-15803 and U-15919.
- B. Net Metering Definitions
 - (1) Category 1 – eligible electric generator(s) with aggregate generation of 20 kW or less that use equipment certified by a nationally recognized testing laboratory to IEEE 1547.1 testing standards and in compliance with UL 1741 scope 1.1A.
 - (2) Category 2 – eligible electric generator(s) with aggregate generation greater than 20 kW and not more than 150 kW.
 - (3) Category 3 – methane digester(s) with aggregate generation greater than 150 kW but not more than 550 kW.
 - (4) Eligible Electric Generator – a renewable energy system or a methane digester with a generation capacity limited to the customer's electric need and that does not exceed the following:
 - (i) For a renewable energy system, 150 kW of aggregate generation at a single site
 - (ii) For a methane digester, 550 kW of aggregate generation at a single site
 - (5) Full Retail Rate – the power supply and distribution components of the cost of electric service. Full Retail Rate does not include surcharges, the system access charge or other charges that are assessed on a per meter basis.
 - (6) Imputed Customer Usage – calculated as the sum of the metered on-site generation and the net of the bidirectional flow of power across the customer interconnection during the billing period.
 - (7) Modified Net Metering – a utility billing method that applies the power supply energy component of the customer's otherwise applicable tariff rate to the net of the bidirectional flow of kWh across the customer interconnection with the utility distribution system during a billing period or time-of-use period. Category 2 and Category 3 customers qualify for Modified Net Metering.
 - (8) Net Customer Consumption – when a positive value is the result of subtracting metered outflow from the customer from metered inflow supplied by the Company. The customer has consumed electricity in excess of what is generated on premises and returned to the Company's system.
 - (9) Net Excess Generation – when a negative value is the result of subtracting metered outflow from the customer from metered inflow supplied by the Company. The customer has generated and returned more electricity to the Company's system than the amount of electricity supplied by the Company to the customer's premises.
 - (10) Program Capacity – maximum program limit of 1% of the Company's average Peak Demand for Full-Service Customers during the previous five calendar years. Within the Program Capacity, 0.5% is reserved for Category 1 Net Metering Customers, 0.25% is reserved for Category 2 Net Metering Customers and 0.25% is reserved for Category 3 Net Metering Customers.

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C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program (Contd)

B. Net Metering Definitions (Contd)

- (11 10) Renewable Energy Resource – a resource that naturally replenishes over a human, not a geological, timeframe and that is ultimately derived from solar power, water power or wind power. Renewable energy resource does not include petroleum, nuclear, natural gas, or coal. A renewable energy resource comes from the sun or from thermal inertia of the earth and minimizes the output of toxic material in the conversion of the energy and includes, but is not limited to, all of the following:
- (i) Biomass
 - (ii) Solar and solar thermal energy
 - (iii) Wind energy
 - (iv) Kinetic energy of moving water, including the following:
 - (a) waves, tides or currents
 - (b) water released through a dam
 - (v) Geothermal energy
 - (vi) Municipal solid waste
 - (vii) Landfill gas produced by municipal solid waste
- (12 11) True Net Metering – a utility billing method that applies the full retail rate to the net of the bidirectional flow of kWh across the customer interconnection with the utility distribution system during a billing period or time-of-use period. Category 1 customers with a system capable of generating 20 kW or less qualify for True Net Metering.

C. Net Metering Program Availability

The Net Metering Program is available for eligible Net Metering customers beginning with the first day of the August 2009 Bill Month. Net Metering Program participation is contingent upon available Program Capacity under the authorized cap for each Category. Electronic Interconnection Applications for the Net Metering Program will be accepted until 4:59 P.M. Eastern Standard Time on January 2, 2021. Corresponding payments for electronic applications and hard copy applications must be postmarked no later than January 15, 2021. Applications received after the Category specific Program Capacity has been met, and before 4:59 P.M. Eastern Standard Time on January 2, 2021, will be added to the Net Metering Program waitlist.

The Net Metering Program is voluntary and is available on a first come, first served basis until the nameplate capacity of all participating generators is equal to the maximum program limit of 1.0% of the Company's average peak demand for Full-Service customers during the previous five calendar years. Within the Program Capacity, 0.5% is reserved for Category 1 Net Metering customers, 0.25% is reserved for Category 2 Net Metering customers and 0.25% is reserved for Category 3 Net Metering customers.

D. Customer Eligibility

In order to be eligible to participate in the Net Metering Program, customers must generate a portion or all of their own retail electricity requirements with an Eligible Electric Generator which utilizes a Renewable Energy Resource, as defined in Rule C11.2.B, Net Metering Definitions.

A customer's eligibility to participate in the Net Metering Program is conditioned on the full satisfaction of any payment term or condition imposed on the customer by pre-existing contracts or tariffs with the Company, including those imposed by participation in the Net Metering Program, or those required by the interconnection of the customer's Eligible Electric Generator to the Company's distribution system.

A customer eligible to participate in the Net Metering Program will be placed into the appropriate Net Metering Category based on the aggregate nameplate capacity of the Eligible Electric Generator(s) located on the customer's premises.

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C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program (Contd)

D. Customer Eligibility (Contd)

- (1) A Category 1 Net Metering customer has one or more Eligible Electric Generators with an aggregate nameplate capacity of 20 kW or less that use equipment certified by a nationally recognized testing laboratory to IEEE 1547.1 testing standards and is in compliance with UL 1741 scope 1.1A located on the customer's premises and metered at a single point of contact.
- (2) A Category 2 Net Metering customer has one or more Eligible Electric Generators with an aggregate nameplate capacity greater than 20 kW but not more than 150 kW located on the customer's premises and metered at a single point of contact.
- (3) A Category 3 Net Metering customer has one or more methane digesters with an aggregate nameplate capacity greater than 150 kW but not more than 550 kW located on the customer's premises and metered at a single point of contact.

E. Customer Billing and Net Excess Generation Credit

(1) Category 1 Customers

(a) Full Service Customers

- (i) The customer will be billed at the Full Retail Rate, plus surcharges, Power Plant Securitization Charges and Power Supply Cost Recovery (PSCR) Factor on Net Customer Consumption for the billing month.
- (ii) The customer will be credited at the Full Retail Rate on Net Excess Generation for the billing month. The credit shall appear on the bill for the following billing period and shall be used to offset total utility charges on that bill. Any excess credit not used to offset total utility charges will be carried forward to subsequent billing periods. Net Excess Generation Credit is non-transferrable. In months when the customer has zero Net Customer Consumption or Net Excess Generation, all applicable surcharges will be billed on the metered inflow supplied by the Company to the customer.

(b) Retail Open Access Customers

- (i) The customer will be billed for the distribution components, including applicable surcharges, and Power Plant Securitization Charges, if applicable, as stated on the customer's Retail Open Access Rate Schedule on Net Customer Consumption for the billing month.
- (ii) The Retail Open Access customer will be credited for distribution components as stated on the ROA customer's otherwise applicable Company Full Service Rate Schedule on Net Excess Generation for the billing month. The credit shall appear on the bill for the following billing period and shall be used to offset utility distribution charges on that bill. Any excess credit not used to offset utility distribution charges will be carried forward to subsequent billing periods. Net Excess Generation Credit is non-transferrable. In months when the customer has zero Net Customer Consumption or Net Excess Generation, all applicable surcharges will be billed on the metered inflow delivered by the Company to the customer.

(2) Category 2 Customers

(a) Full Service Customers

- (i) The customer will be billed for power supply energy components, including Power Supply Cost Recovery (PSCR) Factor, on Net Customer Consumption. The customer will be billed for distribution components, surcharges, and Power Plant Securitization Charges on metered inflow supplied by the Company to the customer. General Service Secondary Demand Rate GSD and Large General Service Primary Demand Rate GPD customers will be billed for demand based capacity charges as stated on the applicable Rate Schedule.

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C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program: (Contd)

E. Customer Billing and Net Excess Generation Credit (Contd)

(2) Category 2 Customers (Contd)

(a) Full Service Customers (Contd)

- (ii) The customer will be credited for power supply energy components on Net Excess Generation. The credit shall appear on the bill for the following billing period and shall be used to offset total power supply charges on that bill. Net Excess Generation Credit exceeding total power supply charges shall be carried forward and applied to power supply charges in subsequent billing periods. Net Excess Generation Credit is non-transferrable.

(b) Retail Open Access Customers

- (i) The customer will be billed for the distribution components, including applicable surcharges, and Power Plant Securitization Charges, if applicable, as stated on the ROA customer's otherwise applicable Company Full Service Rate Schedule on metered inflow supplied by the Company to the customer. The customer will be billed for demand based capacity charges in accordance with the ROA customer's otherwise applicable Company Full Service Rate Schedule.

- (ii) Retail Open Access customers will not receive distribution credit on Net Excess Generation.

(3) Category 3 Customers

(a) Full Service Customers on General Service Secondary Rate GS or General Service Primary Rate GP

- (i) The customer will be billed for power supply energy components, including Power Supply Cost Recovery (PSCR) Factor, on Net Customer Consumption. The customer will be billed for surcharges, and Power Plant Securitization Charges on the metered inflow supplied by the Company to the customer. The customer will be billed for distribution components on Imputed Customer Usage.
- (ii) The customer will be credited for power supply energy components on Net Excess Generation. The credit shall appear on the bill for the following billing period and shall be used to offset total power supply charges on that bill. Net Excess Generation Credit exceeding total power supply charges will be carried forward and applied to power supply charges in subsequent billing periods. Net Excess Generation Credit is non-transferrable.

(b) Full Service Customers on General Service Secondary Demand Rate GSD or Large General Service Primary Demand Rate GPD

- (i) The customer will be billed for power supply components, including Power Supply Cost Recovery (PSCR) Factor, on Net Customer Consumption. The customer will be billed for surcharges, and Power Plant Securitization Charges on the metered inflow supplied by the Company to the customer. The customer will be billed for distribution components on Imputed Customer Usage. General Service Secondary Demand Rate GSD and Large General Service Primary Demand Rate GPD customers will be billed for demand based capacity charges as stated on the applicable Rate Schedule.
- (ii) The customer will be credited for power supply energy components on Net Excess Generation. The credit shall appear on the bill for the following billing period and shall be used to offset total power supply charges on that bill. Net Excess Generation Credit exceeding total power supply charges will be carried forward and applied to power supply charges in subsequent billing periods. Net Excess Generation Credit is non-transferrable.

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Sheet No. C-62.00

(Continued From Sheet No. C-61.00)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program: (Contd)

E. Customer Billing and Net Excess Generation Credit (Contd)

(3) Category 3 Customers (Contd)

(c) Retail Open Access Customers

(i) The customer will be billed for distribution components as stated on the ROA customer's otherwise applicable Company Full Service Rate Schedule on Imputed Customer Usage. The customer will be billed for surcharges, and Power Plant Securitization Charges, if applicable, on the metered inflow supplied by the Company to the customer. The customer will be billed for demand based capacity charges as stated on the ROA customer's otherwise applicable Company Full Service Rate Schedule.

(ii) Retail Open Access customers will not receive a distribution credit on Net Excess Generation.

No refunds shall be made for any customer contribution required under Paragraphs H, I or J of this tariff or for any other costs incurred by the customer in connection with participation in the Net Metering Program.

F. Application for Service

In order to participate in the Net Metering Program, a customer shall submit a completed Interconnection Application, including application fee of \$75 and a completed Net Metering Program Application, including application fee of \$25 to the Company. The Net Metering Program application fee is refundable if the customer withdraws the application prior to commencing service under the Net Metering Program.

G. Generator Requirements

The Eligible Electric Generator(s) must be located on the customer's premises, serving only the customer's premises and must be intended primarily to offset a portion or all of the customer's requirement for electricity.

The customer's requirement for electricity shall be determined by one of the following methods:

- (1) The customer's annual energy usage, measured in kWh, during the previous 12-month period
- (2) When metered demand is available, the maximum integrated hourly demand measured in kW during the previous 12-month period
- (3) In instances where complete and correct data is not available or where the customer is making changes on-site that will affect total usage, the Company and the customer shall mutually agree on a method to determine the customer's electric requirement for electricity.

The aggregate capacity of Eligible Electric Generators shall be determined by one of the following methods:

- (1) Aggregate nameplate capacity of the generator(s)
- (2) Aggregate projected annual kWh output of the generator(s)

The customer is required to provide the Company with a capacity rating in kW of the generating unit and a projected monthly and annual Kilowatt-hour output of the generating unit when completing the Company's Net Metering Application.

The customer need not be the owner or operator of the eligible generation equipment, but is ultimately responsible for ensuring compliance with all technical, engineering and operational requirements suitable for the Company's distribution system.

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Consumers Energy Company

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(Continued From Sheet No. C-62.00)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program: (Contd)

H. Generator Interconnection Requirements

The requirements for interconnecting a generator with the Company's facilities are contained in Rule B8., Electric Interconnection and Net Metering Standards, the Michigan Electric Utility Generator Interconnection Requirements and the Company's Generator Interconnection Supplement to Michigan Electric Utility Generator Interconnection Requirements. All such interconnection requirements must be met prior to the effective date of a customer's participation in the Net Metering Program. The customer must sign an Interconnection and Operating Agreement with the Company and fulfill all requirements as specified in the Agreement. A customer with a system capable of generating more than 20 kW shall pay actual interconnection costs associated with participating in the Net Metering Program, subject to limits established by the Michigan Public Service Commission.

I. Metering Requirements

Metering requirements shall be specified by the Company, as detailed below. All metering, including the generator meter where applicable, must be capable of recording all parameters metered on the customer's otherwise applicable tariff rate, for both Full Service and Retail Open Access customers.

(1) Category 1 Metering Requirements

The Company will utilize a meter capable of measuring the flow of energy in both directions. At the Company's option, either the customer's existing meter will be used or a single meter with separate registers capable of measuring power flow in each direction will be installed. If the existing meter is used, the Company shall test and calibrate the meter to ensure accuracy in both directions. If a meter is installed, the Company shall provide the metering equipment without cost to the customer. The customer may purchase a meter from the Company to measure generator output. The customer shall be responsible for installation and maintenance of the generator meter if purchased. The Company has no obligation to read the generator meter.

(2) Category 2 Metering Requirements

The Company will utilize a meter or meters capable of measuring the flow of energy in both directions and the generator output. If meter upgrades are necessary to provide such functionality, the Company shall provide metering equipment without cost to the customer.

(3) Category 3 Metering Requirements

The Company will utilize a meter or meters capable of measuring the flow of energy in both directions and the generator output. If meter upgrades are necessary to provide such functionality, the customer shall pay the costs incurred. Metering costs must be paid in full prior to participation in the Net Metering Program.

(Continued on Sheet No. C-64.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-64.00

(Continued From Sheet No. C-63.00)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.2 Net Metering Program: (Contd)

J. Distribution Line Extension and/or Extraordinary Facilities

The Company reserves the right to make special contractual arrangements with Net Metering Program customers whose utility service requires investment in electric facilities, as authorized by the Company's Rule C1.4, Extraordinary Facility Requirements and Charges, Rule C1.6, General Provisions of Service, and Rule C6., Distribution Systems, Line Extensions and Service Connections, as set out in the Company's Electric Rate Book. The Company further reserves the right to condition a customer's participation in the Net Metering Program on a satisfactory completion of any such contractual requirements. Category 1 Net Metering customers are not responsible for incremental costs associated with participation in the Net Metering Program.

K. Customer Termination from the Net Metering Program

A participating customer may terminate participation in the Company's Net Metering Program at any time for any reason on sixty days' notice. In the event that a customer who terminates participation in the Net Metering Program wishes to re-enroll, that customer must reapply as a new program participant, subject to program size limitations, application queue and application fees.

The Company may terminate a customer from the Net Metering Program if the customer fails to maintain the eligibility requirements, fails to comply with the terms of the operating agreement, or if the customer's facilities are determined not to be in compliance with technical, engineering, or operational requirements suitable for the Company's distribution system. The Company will provide sixty days' notice to the customer prior to termination from the Net Metering Program, except in situations the Company deems dangerous or hazardous. Such notice will include the reason(s) for termination.

Upon customer termination from the Net Metering Program, any existing credit on the customer's account will either be applied to the customer's final bill or refunded to the customer. The Company will refund to the customer any remaining credit in excess of the final bill amount. Net Excess Generation Credit is non-transferrable.

L. Company Termination of the Net Metering Program

Company termination of the Net Metering Program may occur upon receipt of Commission approval. Upon Company termination of the Net Metering Program, any existing credit on the customer's account will either be applied to the customer's final bill or refunded to the customer. The Company will refund to the customer any remaining credit in excess of the final bill amount. Net Excess Generation Credit is non-transferrable.

M. Net Metering Program Status and Evaluation Reports

The Company will submit an annual status report to the Commission Staff by March 31 of each year including Net Metering Program data for the previous 12 months, ending December 31. The Company's status report shall maintain customer confidentiality.

N. Renewable Energy Credits

Renewable Energy Credits (RECs) are owned by the customer. The Company may purchase Renewable Energy Credits from participating Net Metering customers who are willing to sell RECs generated if the customer has a generator meter in place to accurately measure and verify generator output. REC certification costs are the responsibility of the customer. The Company will enter into a separate agreement with the customer for the purchase of any RECs.

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(Continued from Sheet No. C-64.00)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM

- A. The Distributed Generation Program is offered as authorized by 2008 PA 295 as amended, 1939 PA 3, as amended by 2016 PA 341, Section (6)(a)(14), and the Commission in Case No. U-20697.
- B. Distributed Generation Definitions
1. A Category 1 distributed generation customer has one or more Eligible Electric Generators with an aggregate nameplate capacity of 20 kW or less that use equipment certified by a nationally recognized testing laboratory to IEEE 1547.1 testing standards and is in compliance with UL 1741 scope 1.1A located on the customer's premises and metered at a single point of contact.
 2. A Category 2 distributed generation customer has one or more Eligible Electric Generators with an aggregate capacity greater than 20 kW but not more than 150 kW located on the customer's premises and metered at a single point of contact.
 3. A Category 3 distributed generation customer has one or more methane digesters with an aggregate nameplate capacity greater than 150 kW but not more than 550 kW located on the customer's premises and metered at a single point of contact.
 4. Eligible Electric Generator – a renewable energy system or a methane digester with a generation capacity limited to no more than 100% of the customer's electricity consumption for the previous 12 months and does not exceed the following:
 - a. For a renewable energy system, 150 kW of aggregate generation at a single site
 - b. For a methane digester, 550 kW of aggregate generation at a single site
 5. Inflow – the metered inflow delivered by the Company to the customer during the billing month or time-based pricing period.
 6. Outflow – the metered quantity of the customer's generation not used on site and exported to the utility during the billing month or time-based pricing period.
 7. Program Capacity – maximum program limit of 1% of the Company's average Peak Demand for Full-Service Customers during the previous five calendar years. Within the Program Capacity, 0.5% is reserved for Category 1 Net Metering Customers, 0.25% is reserved for Category 2 Net Metering Customers and 0.25% is reserved for Category 3 Net Metering Customers.
 8. Renewable Energy Resource – a resource that naturally replenishes over a human, not geological, timeframe and that is ultimately derived from solar power, water power, or wind power. Renewable energy resource does not include petroleum, nuclear, natural gas, or coal. A renewable energy resource comes from the sun or from thermal inertia of the earth and minimizes the output of toxic material in the conversion of the energy and includes, but is not limited to, all of the following:
 - a. Biomass
 - b. Solar and solar thermal energy
 - c. Wind energy
 - d. Kinetic energy of moving water, including the following:
 - i. Waves, tides or currents
 - ii. Water released through a dam
 - e. Geothermal energy
 - f. Thermal energy produced from a geothermal heat pump
 - g. Any of the following cleaner energy resources:
 - i. Municipal solid waste, including the biogenic and anthropogenic fractions
 - ii. Landfill gas produced by municipal solid waste
 - iii. Fuel that has been manufactured in whole or significant part from waste, including, but not limited to, municipal solid waste. Fuel that meets the requirements of this subparagraph includes, but is not limited to, material that is listed under 40 CFR 241.3(b) or 241.4(a) for which a nonwasted determination is made by the United States Environmental Protection Agency pursuant to 40 CFR 241.3(c). Pet coke, hazardous waste, or scrap tires are not fuel that meeting the requirements of this subparagraph.

(Continued on Sheet No. C-64.20)

(Continued from Sheet No. C-64.10)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM (Contd)

C. Distributed Generation Program Availability

The Distributed Generation Program is available for eligible Distributed Generation customers for service rendered on and after (insert effective date of rate case order approving this tariff).

A customer participating in a net metering program approved by the Commission before (insert effective date of the rate case order approving this tariff) shall have the option to take service under this tariff at the time service under the terms and conditions of the previous net metering program terminates in accordance with MCL 460.1183.

The Distributed Generation Program is voluntary and available on a first come, first served basis for new customer participants or existing customer participants increasing their aggregate generation. The combined legacy Net Metering and DG program size is equal to 1.0% of the Company's average instate peak load for Full-Service customers during the previous 5 calendar years. Within the Program Capacity, 0.5% is reserved for Category 1 Distributed Generation customers, 0.25% is reserved for Category 2 Distributed Generation customers and 0.25% is reserved for Category 3 Distributed Generation customers. The Company shall notify the Commission upon the Program reaching capacity in any Category.

If an existing Net Metering customer increases the aggregate generation following the effective date of the Distributed Generation Program, all onsite generation will be subject to the terms and conditions of the Distributed Generation Program.

D. Customer Eligibility

In order to be eligible to participate in the Distributed Generation Program, customers must generate a portion or all of their own retail electricity requirements with an Eligible Electric Generator which utilizes a Renewable Energy Resource, as defined in C11.3 B, Distributed Generation Definitions.

A customer's eligibility to participate in the Distributed Generation Program is conditioned on the full satisfaction of any payment term or condition imposed on the customer by pre-existing contracts or tariffs with the Company, including those imposed by participation in the Distributed Generation Program, or those required by the interconnection of the customer's Eligible Electric Generator to the Company's distribution system.

E. Customer Billing – Category 1, 2 and 3 Customers

1. Inflow

a. Full Service Customers

The customer will be billed according to their retail rate schedule, plus surcharges, and Power Supply Cost Recovery (PSCR) Factor on metered Inflow for the billing period or time-based pricing period.

b. Retail Open Access Customers

The customer will be billed as stated on the customer's Retail Open Access Rate Schedule on metered Inflow for the billing period or time-based pricing period.

2. Customer Billing – Outflow Credit

The customer will be credited on Outflow for the billing period or time-based pricing period. The credit shall be applied to the current billing month and shall be used to offset power supply charges on that bill. Any excess credit not used will be carried forward to subsequent billing periods. Unused Outflow Credit from previous months will be applied to power supply charges in the current billing month, if applicable. Outflow credit is non-transferrable.

(Continued on Sheet No. C-64.30)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-64.30

(Continued from Sheet No. C-64.20)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM (Contd)

E. Customer Billing – Category 1, 2 and 3 Customers (Cont)

a. Full Service Customers Outflow Credit

Customers will be credited per kWh or per kW of Outflow based on the power supply rates (which exclude transmission costs) of their Full Service Rate Schedule as shown below, plus the PSCR factor as shown on Tariff Sheet No. D-6.00.

<u>Residential Rates</u>		
<u>Summer</u>	<u>\$(0.125355)</u>	<u>per kWh of On-Peak Outflow between June 1 and September 30</u>
<u>On-Peak Basic</u>	<u>\$(0.084319)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
<u>Rate RSP</u>	<u>\$(0.088869)</u>	<u>per kWh of all Outflow kWh between October 1 and May 31</u>
	<u>\$(0.125355)</u>	<u>per kWh of On-Peak Outflow between June 1 and September 30</u>
<u>Smart Hours</u>	<u>\$(0.084319)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
<u>Rate RSH</u>	<u>\$(0.095128)</u>	<u>per kWh of On-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.086523)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.125355)</u>	<u>per kWh of On-Peak Outflow between June 1 and September 30</u>
<u>Nighttime Savers</u>	<u>\$(0.097334)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
<u>Rate RPM</u>	<u>\$(0.062420)</u>	<u>per kWh of Super Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.095128)</u>	<u>per kWh of On-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.094440)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.070323)</u>	<u>per kWh of Super Off-Peak Outflow between October 1 and May 31</u>
<u>Secondary Rates</u>		
<u>Rate GS</u>	<u>\$(0.078927)</u>	<u>per kWh of Outflow during the billing months of June through September</u>
	<u>\$(0.077383)</u>	<u>per kWh of Outflow during the billing months of October through May</u>
<u>Rate GSTU⁽¹⁾</u>	<u>\$(0.109499)</u>	<u>per kWh of On-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.087158)</u>	<u>per kWh of Mid-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.057855)</u>	<u>per kWh of Off-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.089069)</u>	<u>per kWh of On-Peak Outflow during the billing months of October through May</u>
	<u>\$(0.069143)</u>	<u>per kWh of Off-Peak Outflow during the billing months of October through May</u>
<u>Rate GSD⁽¹⁾</u>	<u>\$(0.033511)</u>	<u>per kWh of Outflow during the billing months of June through September</u>
	<u>\$(0.030961)</u>	<u>per kWh of Outflow during the billing months of October through May</u>
	<u>\$17.42</u>	<u>per kW of Outflow Demand during the billing months of June through September</u>
	<u>\$14.34</u>	<u>per kW of Outflow Demand during the billing months of October through May</u>
<u>⁽¹⁾ Outflow credit will be reduced by \$0.017518 per kWh for GSTU and GSD customers participating on GIS Provision.</u>		

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M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-64.40

(Continued from Sheet No. C-64.30)

<u>Primary Rates</u>		
<u>Rate GP</u>		
<u>Customer Voltage Level 1</u>	<u>\$(0.068555)</u>	<u>per kWh of outflow during the billing months of June through September</u>
	<u>\$(0.067281)</u>	<u>per kWh of outflow during the billing months of October through May</u>
<u>Customer Voltage Level 2</u>	<u>\$(0.069524)</u>	<u>per kWh of outflow during the billing months of June through September</u>
	<u>\$(0.068240)</u>	<u>per kWh of outflow during the billing months of October through May</u>
<u>Customer Voltage Level 3</u>	<u>\$(0.071038)</u>	<u>per kWh of outflow during the billing months of June through September</u>
	<u>\$(0.069734)</u>	<u>per kWh of outflow during the billing months of October through May</u>
<u>Rate GPD⁽²⁾</u>		
<u>Customer Voltage Level 1</u>	<u>\$(0.027643)</u>	<u>per kWh of On-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.017803)</u>	<u>per kWh of Off-Peak Outflow during the billing months of June through September</u>
	<u>\$(20.60)</u>	<u>per kW of Outflow Demand during the billing months of June through September</u>
	<u>\$(0.022639)</u>	<u>per kWh of On-Peak Outflow during the billing months of October through May</u>
	<u>\$(0.021052)</u>	<u>per kWh of Off-Peak Outflow during the billing months of October through May</u>
	<u>\$(18.60)</u>	<u>per kW of Outflow Demand during the billing months of October through May</u>
<u>Customer Voltage Level 2</u>	<u>\$(0.027983)</u>	<u>per kWh of On-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.018022)</u>	<u>per kWh of Off-Peak Outflow during the billing months of June through September</u>
	<u>\$(20.91)</u>	<u>per kW of Outflow Demand during the billing months of June through September</u>
	<u>\$(0.022917)</u>	<u>per kWh of On-Peak Outflow during the billing months of October through May</u>
	<u>\$(0.021311)</u>	<u>per kWh of Off-Peak Outflow during the billing months of October through May</u>
	<u>\$(18.88)</u>	<u>per kW of Outflow Demand during the billing months of October through May</u>
<u>Customer Voltage Level 3</u>	<u>\$(0.028533)</u>	<u>per kWh of On-Peak Outflow during the billing months of June through September</u>
	<u>\$(0.018376)</u>	<u>per kWh of Off-Peak Outflow during the billing months of June through September</u>
	<u>\$(21.38)</u>	<u>per kW of Outflow Demand during the billing months of June through September</u>
	<u>\$(0.023368)</u>	<u>per kWh of On-Peak Outflow during the billing months of October through May</u>
	<u>\$(0.021730)</u>	<u>per kWh of Off-Peak Outflow during the billing months of October through May</u>
	<u>\$(19.30)</u>	<u>per kW of Outflow Demand during the billing months of October through May</u>
<u>(²) For customers on Rate GPD GI Provision, On-Peak kW Outflow Credit shall be reduced by \$7.00 per kW during the billing months of June through September and \$6.00 per kW during the billing months of October through May.</u>		

(Continued on Sheet No. C-64.50)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-64.50

(Continued from Sheet No. C-64.40)

<u>Rate GPTU</u>		
<u>Customer Voltage Level 1</u>	<u>\$(0.103597)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.094602)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.074855)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.052393)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.068334)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.066157)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.059486)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
<u>Customer Voltage Level 2</u>	<u>\$(0.105068)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.095954)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.075927)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.053139)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.069281)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.067077)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.060309)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
<u>Customer Voltage Level 3</u>	<u>\$(0.107363)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.098060)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.077596)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.054304)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.070767)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.068520)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.061602)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
<u>Rate EIP</u>		
<u>Customer Voltage Level 1</u>	<u>\$(0.087482)</u>	<u>per kWh of Critical Peak Outflow between June 1 and September 30</u>
	<u>\$(0.058321)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.053084)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.042728)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.028623)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.062912)</u>	<u>per kWh of Critical Peak Outflow between October 1 and May 31</u>
	<u>\$(0.041941)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.040216)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.035524)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
<u>Customer Voltage Level 2</u>	<u>\$(0.088648)</u>	<u>per kWh of Critical Peak Outflow between June 1 and September 30</u>
	<u>\$(0.059098)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.053795)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.043301)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.029006)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.063731)</u>	<u>per kWh of Critical Peak Outflow between October 1 and May 31</u>
	<u>\$(0.042486)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.040741)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.035987)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>
<u>Customer Voltage Level 1</u>	<u>\$(0.090495)</u>	<u>per kWh of Critical Peak Outflow between June 1 and September 30</u>
	<u>\$(0.060330)</u>	<u>per kWh of High-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.054921)</u>	<u>per kWh of Mid-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.044208)</u>	<u>per kWh of Low-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.029612)</u>	<u>per kWh of Off-Peak Outflow between June 1 and September 30</u>
	<u>\$(0.065036)</u>	<u>per kWh of Critical Peak Outflow between October 1 and May 31</u>
	<u>\$(0.043357)</u>	<u>per kWh of High-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.041575)</u>	<u>per kWh of Mid-Peak Outflow between October 1 and May 31</u>
	<u>\$(0.036725)</u>	<u>per kWh of Off-Peak Outflow between October 1 and May 31</u>

b. Retail Open Access Customers

The Outflow Credit will be determined by the Retail Service Supplier

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(Continued from Sheet No. C-64.50)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM (Contd)

F. Application for Service

In order to participate in the Distributed Generation Program, a customer shall submit completed Interconnection and Distributed Generation Program Applications, including the application fee of \$50 to the Company. The application fees are non-refundable.

The Distributed Generation Program application fee is waived if the customer is transitioning from the Net Metering Program to the DG Program, without a system modification.

If a customer does not act or provide proper response to the Company on an application, when some action is required by the customer, the application will be voided by the Company after 6 months.

G. Generator Requirements

The customer's requirement for electricity shall be determined by one of the following methods:

1. The customer's annual energy usage, measured in kWh, during the previous 12-month period
2. In instances where complete and correct data is not available or where the customer is making changes on-site that will affect total usage, the Company and the customer shall mutually agree on a method to determine the customer's annual electric requirement.

The aggregate capacity of Eligible Electric Generators shall be determined by the aggregate projected annual kWh output of the generator(s).

The customer is required to provide the Company with a capacity rating in kW_{AC} and kW_{DC} of the generating unit and a projected monthly and annual Kilowatt-hour output of the generating unit when completing the Company's Distributed Generation Program Application.

The customer need not be the owner or operator of the eligible generation equipment, but is ultimately responsible for ensuring compliance with all technical, engineering, and operational requirements suitable for the Company's distribution system.

H. Generator Interconnection Requirements

The requirements for interconnecting a generator with the Company's facilities are contained in Rule B8., Electric Interconnection and Distributed Generation Standards, the Michigan Electric Utility Generator Interconnection Requirements, and the Company's Generation Interconnection Supplement to Michigan Electric Utility Generator Interconnection Requirements. All such interconnection requirements must be met prior to the effective date of a customer's participation in the Distributed Generation Program. The customer must sign an Interconnection and Operating Agreement with the Company and fulfill all requirements as specified in the Agreement. The customer shall pay actual interconnection costs associated with participating in the Distributed Generation Program, subject to limits established by the Michigan Public Service Commission.

(Continued on Sheet No. C-64.70)

(Continued from Sheet No. C-64.60)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM (Contd)

I. Metering Requirements

Metering requirements shall be specified by the Company, as detailed below.
All metering must be capable of recording inflow and outflow, and all parameters metered on the customer's otherwise applicable retail rate schedule, for both Full Service and Retail Open Access customers.

J. Distribution Line Extension and/or Extraordinary Facilities

The Company reserves the right to make special contractual arrangements with Distributed Generation Program customers whose utility service requires investment in electric facilities, as authorized by the Company's Rule C1.4, Extraordinary Facility Requirements and Charges, Rule C1.6, General Provisions of Service, and Rule C6., Distribution Systems, Line Extensions and Service Connections, as set out in the Company's Electric Rate Book. The Company further reserves the right to condition a customer's participation in the Distributed Generation Program on a satisfactory completion of any such contractual requirements.

K. Customer Termination from the Distributed Generation Program

A participating customer may terminate participation in the Company's Distributed Generation Program at any time for any reason on sixty days' notice. In the event that a customer who terminates participation in the Distributed Generation Program wishes to re-enroll, that customer must reapply as a new program participant, subject to program size limitations, application queue, and application fee.

The Company may terminate a customer from the Distributed Generation Program if the customer fails to maintain eligibility requirements, fails to comply with the terms of the operating agreement, or if the customer's facilities are determined not to be in compliance with technical, engineering, or operational requirements suitable for the Company's distribution system. The Company will provide sixty days' notice to the customer prior to termination from the Distributed Generation Program, except in situations the Company deems dangerous or hazardous. Such notice will include reason(s) for termination.

Upon customer termination from the Distributed Generation Program, any existing credit on the customer's account will either be applied to the customer's final bill or refunded to the customer. The Company will refund to the customer any remaining credit in excess of the final bill amount. Distributed Generation Program credit is non-transferrable.

L. Company Termination of the Distributed Generation Program

Company termination of the Distributed Generation Program may occur upon receipt of Commission approval.

Upon Company termination of the Distributed Generation Program, any existing credit on the customer's account will either be applied to the customer's final bill or refunded to the customer. The Company will refund to the customer any remaining credit in excess of the final bill amount. Distributed Generation Program credit is non-transferrable.

(Continued on Sheet No. C-64.80)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. C-64.80

(Continued from Sheet No. C-64.70)

C11. SELF-GENERATION, NET METERING AND DISTRIBUTED GENERATION (Contd)

C11.3 DISTRIBUTED GENERATION PROGRAM (Contd)

M. Distributed Generation Program Status and Evaluation Reports

The Company will submit an annual status report to the Commission Staff by March 31 of each year including Distributed Generation Program data for the 12 months ending December 31. The Company's status report shall maintain customer confidentiality.

N. Renewable Energy Credits

Renewable Energy Credits (RECs) are owned by the customer.

The Company may, but is not obligated to, purchase Renewable Energy Credits from participating Distributed Generation Program customers who are willing to sell RECs generated if the customer has a generator meter in place to accurately measure and verify generator output. REC certification costs are the responsibility of the customer.

The Company will enter into a separate agreement with the customer for the purchase of any RECs.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-4.00

SURCHARGES

<u>Rate Schedule</u>	<u>Electric Rate Case Deferral Surcharge (Case No. U-20697) Effective for service rendered January 1, 2021 through December 31, 2021</u>	<u>Financial Compensation Mechanism Surcharge (Case No. U-20697) Effective for service rendered On and after January 1, 2021</u>
<u>Rate RSP</u>	<u>\$0.000631/kWh</u>	<u>\$0.000108/kWh</u>
<u>Rate RDP</u>	<u>0.000631/kWh</u>	<u>0.000108/kWh</u>
<u>Rate RDPR</u>	<u>0.000631/kWh</u>	<u>0.000108/kWh</u>
<u>Rate RSM</u>	<u>0.000631/kWh</u>	<u>0.000108/kWh</u>
<u>Rate GS</u>	<u>0.000553/kWh</u>	<u>0.000098/kWh</u>
<u>Rate GSTU</u>	<u>0.000553/kWh</u>	<u>0.000098/kWh</u>
<u>Rate GSD</u>	<u>0.000346/kWh</u>	<u>0.000098/kWh</u>
<u>Rate GP</u>		
<u>Customer Voltage Level 1</u>	<u>0.000013/kWh</u>	<u>0.000078/kWh</u>
<u>Customer Voltage Level 2</u>	<u>0.000078/kWh</u>	<u>0.000085/kWh</u>
<u>Customer Voltage Level 3</u>	<u>0.000160/kWh</u>	<u>0.000091/kWh</u>
<u>Rate GPD</u>		
<u>Customer Voltage Level 1</u>	<u>0.000009/kWh</u>	<u>0.000080/kWh</u>
<u>Customer Voltage Level 2</u>	<u>0.000051/kWh</u>	<u>0.000087/kWh</u>
<u>Customer Voltage Level 3</u>	<u>0.000105/kWh</u>	<u>0.000093/kWh</u>
<u>Rate GPTU</u>		
<u>Customer Voltage Level 1</u>	<u>0.000010/kWh</u>	<u>0.000079/kWh</u>
<u>Customer Voltage Level 2</u>	<u>0.000059/kWh</u>	<u>0.000085/kWh</u>
<u>Customer Voltage Level 3</u>	<u>0.000121/kWh</u>	<u>0.000091/kWh</u>
<u>Rate EIP</u>		
<u>Customer Voltage Level 1</u>	<u>0.000015/kWh</u>	<u>0.000067/kWh</u>
<u>Customer Voltage Level 2</u>	<u>0.000088/kWh</u>	<u>0.000073/kWh</u>
<u>Customer Voltage Level 3</u>	<u>0.000181/kWh</u>	<u>0.000078/kWh</u>
<u>Rate LTILRR</u>	<u>0.000009/kWh</u>	<u>NA</u>
<u>Rate GSG-2</u>		
<u>Customer Voltage Level 1</u>	<u>0.000179/kWh</u>	<u>0.000087/kWh</u>
<u>Customer Voltage Level 2</u>	<u>0.001058/kWh</u>	<u>0.000095/kWh</u>
<u>Customer Voltage Level 3</u>	<u>0.002165/kWh</u>	<u>0.000102/kWh</u>
<u>Rate GML</u>	<u>0.001978/kWh</u>	<u>0.000088/kWh</u>
<u>Rate GUL</u>	<u>0.001978/kWh</u>	<u>0.000088/kWh</u>
<u>Rate GU-LED</u>	<u>0.001978/kWh</u>	<u>0.000088/kWh</u>
<u>Rate GU</u>	<u>0.000193/kWh</u>	<u>0.000094/kWh</u>
<u>Rate PA</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-R</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-S</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-P</u>	<u>NA</u>	<u>NA</u>

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-4.10

SURCHARGES

<u>Rate Schedule</u>	<u>Demand Response (DR)</u> <u>Reconciliation Surcharge</u> <u>(Case No. U-20697)</u> <u>Effective for service rendered</u> <u>On and after January 1, 2021</u>	<u>Conservation Voltage Reduction (CVR)</u> <u>Incentive Surcharge</u> <u>(Case No. U-20697)</u> <u>Effective for service rendered</u> <u>On and after January 1, 2021</u>
<u>Rate RSP</u>	<u>\$(0.000081)/kWh</u>	<u>\$0.000034/kWh</u>
<u>Rate RDP</u>	<u>(0.000081)/kWh</u>	<u>0.000034/kWh</u>
<u>Rate RDPR</u>	<u>(0.000081)/kWh</u>	<u>0.000034/kWh</u>
<u>Rate RSM</u>	<u>(0.000081)/kWh</u>	<u>0.000034/kWh</u>
<u>Rate GS</u>	<u>(0.000068)/kWh</u>	<u>0.000032/kWh</u>
<u>Rate GSTU</u>	<u>(0.000068)/kWh</u>	<u>0.000032/kWh</u>
<u>Rate GSD</u>	<u>(0.000061)/kWh</u>	<u>0.000026/kWh</u>
<u>Rate GP</u>		
<u>Customer Voltage Level 1</u>	<u>(0.000031)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 2</u>	<u>(0.000035)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 3</u>	<u>(0.000058)/kWh</u>	<u>0.000030/kWh</u>
<u>Rate GPD</u>		
<u>Customer Voltage Level 1</u>	<u>(0.000022)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 2</u>	<u>(0.000025)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 3</u>	<u>(0.000041)/kWh</u>	<u>0.000021/kWh</u>
<u>Rate GPTU</u>		
<u>Customer Voltage Level 1</u>	<u>(0.000035)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 2</u>	<u>(0.000038)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 3</u>	<u>(0.000064)/kWh</u>	<u>0.000029/kWh</u>
<u>Rate EIP</u>		
<u>Customer Voltage Level 1</u>	<u>(0.000012)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 2</u>	<u>(0.000013)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 3</u>	<u>(0.000022)/kWh</u>	<u>0.000010/kWh</u>
<u>Rate LTILRR</u>	<u>(0.000022)/kWh</u>	<u>NA</u>
<u>Rate GSG-2</u>		
<u>Customer Voltage Level 1</u>	<u>(0.000051)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 2</u>	<u>(0.000056)/kWh</u>	<u>NA</u>
<u>Customer Voltage Level 3</u>	<u>(0.000094)/kWh</u>	<u>0.002596/kWh</u>
<u>Rate GML</u>	<u>(0.000017)/kWh</u>	<u>0.000030/kWh</u>
<u>Rate GUL</u>	<u>(0.000017)/kWh</u>	<u>0.000030/kWh</u>
<u>Rate GU-LED</u>	<u>(0.000017)/kWh</u>	<u>0.000030/kWh</u>
<u>Rate GU</u>	<u>(0.000049)/kWh</u>	<u>0.000014/kWh</u>
<u>Rate PA</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-R</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-S</u>	<u>NA</u>	<u>NA</u>
<u>Rate ROA-P</u>	<u>NA</u>	<u>NA</u>

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-8.00

RATE CATEGORIES AND PROVISIONS

Description	Full Service	Retail Open Access
RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP		
Residential Provisions	1001	Not Applicable
Residential Summer On-Peak Basic With Income Assistance (RIA) *	Applicable	Not Applicable
Residential Summer On-Peak Basic With Senior Citizen (RSC) *	Applicable	Not Applicable
Peak Power Savers – Air Conditioner Peak Cycling Program	Applicable	Not Applicable
Peak Power Savers – Peak Reward ***	Applicable	Not Applicable
Peak Power Savers – Critical Peak Pricing ***	Applicable	Not Applicable
Residential Summer On-Peak Basic With Self-Generation (SG) **	1701	Not Applicable
Net Metering Program	Applicable	Not Applicable
Green Generation Program ****	Applicable	Not Applicable
RESIDENTIAL SERVICE SECONDARY RATE RS		
Residential Provisions	1000	2000
Residential With Income Assistance (RIA) *	Applicable	Applicable
Residential With Senior Citizen (RSC) *	Applicable	Applicable
Peak Power Savers – Air Conditioner Peak Cycling Program	1005	Not Applicable
Residential With Self-Generation (SG) **	1700	Not Applicable
Net Metering Program	Applicable	Applicable
Green Generation Program ****	Applicable	Not Applicable
Non-Transmitting Meter Provision	Applicable	Applicable
RESIDENTIAL SERVICE DYNAMIC PROGRAM		
Peak Power Savers – Critical Peak Time Of Use (RDP)	1007	Not Applicable
Peak Power Savers – Peak Rewards Time Of Use (RDPR)	1008	Not Applicable
Provisions		
Residential Dynamic Pricing With Income Assistance (RIA) *	Applicable	Not Applicable
Residential Dynamic Pricing With Senior Citizen (RSC) *	Applicable	Not Applicable
Residential Dynamic Pricing With Self-Generation (SG) **	1700	Not Applicable
Green Generation Program ****	Applicable	Not Applicable
RESIDENTIAL SERVICE TIME OF DAY SECONDARY RATE RT		
Residential Time-of-Day Provisions	1010	2010
Residential Time-of-Day With Income Assistance (RIA) *	Applicable	Applicable
Residential Time-of-Day With Senior Citizen (RSC) *	Applicable	Applicable
Residential Time-of-Day With Self-Generation (SG) **	1705	Not Applicable
Net Metering Program	Applicable	Applicable
Green Generation Program ****	Applicable	Not Applicable

* Provisions shall not be taken in conjunction with each other.

** Provisions shall not be taken in conjunction with the Net Metering Program.

*** Peak Reward and Critical Peak Pricing shall not be taken in conjunction with each other.

**** Closed to new customers, effective April 5, 2019.

(Continued on Sheet No. D-9.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-9.00

RATE CATEGORIES AND PROVISIONS
(Continued From Sheet No. D-8.00)

<u>Description</u>	<u>Full Service</u>	<u>Retail Open Access</u>
EXPERIMENTAL RESIDENTIAL PLUG-IN ELECTRIC VEHICLE CHARGING PROGRAM		
Residential Electric Vehicle Service (REV-1)	1020	Not Applicable
Residential Electric Vehicle Service (REV-1) With Self-Generation (SG) **	1710	Not Applicable
Residential Electric Vehicle Service (REV-2)	1030	Not Applicable
Green Generation Program ****	Applicable	Not Applicable
RESIDENTIAL SMART HOURS RATE RSH		
Residential	1040	Not Applicable
<u>Provisions</u>		
Residential Smart Hours With Income Assistance (RIA) *	Applicable	Not Applicable
Residential Smart Hours With Senior Citizen (RSC) *	Applicable	Not Applicable
Peak Power Savers – Air Conditioner Peak Cycling Program	Applicable	Not Applicable
Peak Power Savers – Peak Reward ***	Applicable	Not Applicable
Peak Power Savers – Critical Peak Pricing ***	Applicable	Not Applicable
Residential Smart Hours With Self-Generation (SG) **	1702	Not Applicable
Net Metering Program	Applicable	Not Applicable
Green Generation Program ****	Applicable	Not Applicable
RESIDENTIAL NIGHTTIME SAVERS RATE RPM		
Residential	1050	Not Applicable
<u>Provisions</u>		
Residential Nighttime Savers With Income Assistance (RIA) *	Applicable	Not Applicable
Residential Nighttime Savers With Senior Citizen (RSC) *	Applicable	Not Applicable
Residential Nighttime Savers – Plug-In Electric Vehicle Only Credit	Applicable	Not Applicable
Peak Power Savers – Air Conditioner Peak Cycling Program	Applicable	Not Applicable
Peak Power Savers – Peak Reward ***	Applicable	Not Applicable
Peak Power Savers – Critical Peak Pricing ***	Applicable	Not Applicable
Residential Nighttime Savers With Self-Generation (SG) **	1703	Not Applicable
Net Metering Program	Applicable	Not Applicable
Green Generation Program ****	Applicable	Not Applicable
RESIDENTIAL SECONDARY SERVICE NON-TRANSMITTING METER RATE RSM		
Residential	1000	Not Applicable
<u>Provisions</u>		
Residential Non-Transmitting Meter With Income Assistance (RIA) *	Applicable	Applicable
Residential Non-Transmitting Meter With Senior Citizen (RSC) *	Applicable	Applicable
Green Generation Program ****	Applicable	Not Applicable

- * Provisions shall not be taken in conjunction with each other.
** Provisions shall not be taken in conjunction with the Net Metering Program.
*** Peak Reward and Critical Peak Pricing shall not be taken in conjunction with each other.
**** Closed to new customers, effective April 5, 2019.

(Continued on Sheet No. D-10.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-10.00

RATE CATEGORIES AND PROVISIONS
(Continued From Sheet No. D-9.00)

<u>Description</u>	<u>Full Service</u>	<u>Retail Open Access</u>
GENERAL SERVICE SECONDARY RATE GS		
Commercial	1100	2100
Commercial – Temporary Construction Service	1999	Not Applicable
<u>Provisions</u>		
Commercial Billboards/Outdoor Advertising Signs – Dusk to Dawn	Applicable	Not Applicable
Commercial Billboards/Outdoor Advertising Signs – Fixed Hours of Operation	Applicable	Not Applicable
Commercial Miscellaneous	Applicable	Not Applicable
Commercial Resale	Applicable	Applicable
Commercial With Educational Institution (GEI)	Applicable	Applicable
Commercial With Self-Generation (SG) *	1715	Not Applicable
Net Metering Program	Applicable	Applicable
Green Generation Program **	Applicable	Not Applicable
Non-Transmitting Meter Provision	Applicable	Applicable
GENERAL SERVICE SECONDARY TIME-OF-USE RATE GSTU		
Commercial	1121	Not Applicable
<u>Provisions</u>		
Commercial With Educational Institution (GEI)	Applicable	Not Applicable
<u>Commercial With Interruptible (GSI)</u>	<u>Applicable</u>	<u>Not Applicable</u>
Commercial With Self-Generation (SG) *	1716	Not Applicable
Commercial Resale	Applicable	Not Applicable
Green Generation Program **	Applicable	Not Applicable
GENERAL SERVICE SECONDARY DEMAND RATE GSD		
Commercial	1120	2120
Commercial (100 kW Billing Demand Guarantee)	1140	2140
<u>Provisions</u>		
Commercial Resale	Applicable	Applicable
Commercial With Educational Institution (GEI)	Applicable	Applicable
<u>Commercial With Interruptible (GSI)</u>	<u>Applicable</u>	<u>Not Applicable</u>
Commercial With Self-Generation (SG) *	1725	Not Applicable
Commercial (100 kW Billing Demand Guarantee) With Self-Generation (SG) *	1735	Not Applicable
Net Metering Program	Applicable	Applicable
Green Generation Program **	Applicable	Not Applicable

*Provisions shall not be taken in conjunction with the Net Metering Program.

** Closed to new customers, effective April 5, 2019.

(Continued on Sheet No. D-11.00)

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-14.00

RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP

Availability:

Subject to any restrictions, this rate is available to any Full Service Customer desiring electric service for any usual residential use in: (i) private family dwellings; (ii) tourist homes, rooming houses, dormitories, nursing homes and other similarly occupied buildings containing sleeping accommodations for up to six persons; or (iii) existing multifamily dwellings containing up to four households served through a single meter. Service for single-phase or three-phase equipment may be included under this rate, provided the individual capacity of such equipment does not exceed 3 hp or 3 kW, nor does the total connected load of the home exceed 10 kW, without the specific consent of the Company.

~~The Residential Summer On-Peak Basic Rate Pilot will commence by June 1, 2019. Customer eligibility to participate in the Pilot is determined solely by the Company. Selected customers must remain in the pilot through December 31, 2019.~~

This rate is not available for: (i) resale purposes; (ii) multifamily dwellings containing more than four living units served through a single meter; (iii) tourist homes, rooming houses, dormitories, nursing homes and similarly occupied buildings containing sleeping accommodations for more than six persons; (iv) any other Non-Residential usage; or (v) Rule C5.5 – Non-Transmitting Meter Provision participants.

Residences in conjunction with commercial or industrial enterprises and mobile home parks may take service on this rate only under the Rules and Regulations contained in the Company's Electric Rate Book.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service Customers.

Energy Charge:

Non-Capacity	Capacity	Total	
\$0.056766	\$0.047710	\$0.104476	per kWh for Off-Peak kWh between June 1 and September 30
0.061121	0.035660	0.096781	
\$0.084361	\$0.070983	\$0.155344	per kWh for On-Peak kWh between June 1 and September 30
0.090785	0.052967	0.143752	
\$0.057734	\$0.046840	\$0.104574	per kWh for all kWh between October 1 and May 31
0.061121	0.035660	0.096781	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service Customers.

System Access Charge:	\$8.50 7.50	per customer per month
Distribution Charge:	\$0.060209 0.047054	per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-15.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-15.00

RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP
(Continued From Sheet No. D-14.00)

Monthly Rate: (Contd)

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2, Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102, Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service Customers.

Income Assistance Credit: \$(~~8.50~~ ~~7.50~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

Low Income Assistance Credit (LIAC):

Company selected Residential customers may receive LIAC for up to 12 consecutive months. The number of customers enrolled may be adjusted, at the Company's discretion, in order to dispense Commission-approved LIAC funding on an annual basis. Any shortfall in the dispensing of annual LIAC funds to qualified customers shall be carried over into the subsequent LIAC program year. LIAC customer selection will be based on highest need chosen from one or more of the following eligibility criteria:

1. Customers with an approved critical care certification where the total household income does not exceed 150% of the Federal Poverty level within the last 12 months, as verified by an authorized State, Federal or community agency.
2. Customers who are enrolled in the Company's Consumers Affordable Resource for Energy (CARE) program.
3. Customers who have received a Home Heating Credit in the previous 12 months.
4. Customers whose total household income does not exceed 150% of the Federal Poverty level as verified by an authorized State, Federal or community agency.

The monthly credit for LIAC shall be applied as follows:

Low Income Assistance Credit: \$(30.00) per meter per month

If a credit balance occurs, the credit shall apply to the customer's future electric utility charges. Re-enrollment, if applicable, and confirmation of qualification is required for each annual period of participation.

Customers selected for LIAC will not be eligible for the RIA Provision while enrolled in LIAC.

Senior Citizen Service Provision (RSC):

When service is supplied to the Principal Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.

The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.

Senior Citizen Credit: \$(~~4.25~~ ~~3.75~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).

(Continued on Sheet D-15.1000)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-15.10

RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP

(Continued From Sheet No. D-15.00)

Peak Power Savers:

Customers can elect to participate in the Air Conditioning Peak Cycling Program and the Peak Reward Program as described in this tariff. When a customer participated in both programs, the customer's incremental energy savings earned under the Peak Reward is compared to the Peak Power Savers – Air Conditioning Peak Cycling Program Credit. The greater of the two credits will be applied to the customer's invoice for that billing month. Both credits will not apply in a single billing month. Customers participating in the Peak Reward Program cannot participate in the Critical Peak Price Program. The Company reserves the right to call test events between October 1 and May 31 for customers participating in Peak Power Savers Programs.

Air Conditioner Peak Cycling Program – ~~(Available on a Date to be Announced by the Company):~~

A customer in a single family residence who is taking service from the Company may be eligible to participate in the Company's voluntary Peak Power Savers – Air Conditioner Peak Cycling Program for load management of eligible electric central air conditioning, central heat pump, or other qualifying electric equipment. Customer eligibility to participate is determined solely by the Company. The Company will accept a customer's central air conditioning, central heat pump, and other qualifying electric equipment under this program only if it has the capability to be controlled by the Company. Load Management of a customer's swimming pool pump is permitted under this program only if the customer is allowing Load Management of their air conditioner or heat pump unit. The Company will install the required equipment at the customer's premises which will allow Load Management upon signal from the Company. When Load Management equipment is installed at a premises, future customers will be auto-enrolled into the Peak Power Savers-Air Conditioner Peak Cycling Program. Upon move in, the customer will be notified confirming participation in the Peak Power Savers-Air Conditioner Peak Cycling Program and will have 30 days to opt out. Such equipment shall be furnished, installed, maintained and owned by the Company at the Company's expense. Equipment installations must conform to the Company's specifications.

The Company reserves the right to specify the term or duration of the program. The customer's enrollment shall be terminated if the voluntary program ceases, if the customer tampers with the control switch or the Company's equipment or any reasons as provided for in Rule C1.3, Use of Service.

Load Management may occur any day of the week including weekends between the hours of 7:00 AM and 8:00 PM for no more than an eight hour period in any one day. Load management may be implemented for, but not limited to, maintaining system integrity, making an emergency purchase, economic reasons, or when there is insufficient system generation available to meet anticipated system load. Load Management may only occur outside of the hours of 7:00 AM and 8:00 PM during a declared emergency event as directed by MISO.

The customer may contact the Company to request to override a Load Management event for one Load Management event during the June through September months in any one calendar year for the balance of the hours left in that Load Management event with no penalty. The request shall be granted at the discretion of the Company. If the override request was granted by the Company and the customer requests and is granted any additional overrides in the same calendar year, the Peak Power Savers – Air Conditioner Peak Cycling Credit may be forfeited for that billing month.

Rule C1.1 Character of Service, Rule C3 Emergency Electrical Procedures and other rules and regulations contained in the Company's Electric Rate Book apply to customers taking service under this Peak Power Savers – Air Conditioner Peak Cycling Program.

The monthly credit for the Peak Power Savers Program shall be applied as follows:

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Power Savers – Air Conditioner Peak Cycling Credit: \$(8.00) per customer per month during the billing months of June – September

(Continued on Sheet No. D-16.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-16.00

RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP

(Continued From Sheet No. D-15.10~~00~~)

Monthly Rate: (Contd)

Peak Power Savers: (Contd)

Peak Reward ~~–(Available on a Date to be Announced by the Company):~~

Participating customers are able to manage electric costs by reducing load during critical peak events. The Company may call up to fourteen critical peak events between June 1 and September 30 and up to five critical peak events between October 1 and May 31. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers will be credited the Peak Reward per kWh of incremental energy reductions.

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Reward: \$(~~0.95~~ 1.00) per kWh of incremental energy reduction during a critical peak event ~~between June 1 and September 30~~

Critical Peak Price ~~–(Available on a Date to be Announced by the Company):~~

Participating customers are able to manage electric costs by shifting load during critical peak events to a lower cost pricing period. The Company may call up to fourteen critical peak events between June 1 and September 30. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers will be charged the Critical Peak Price per kWh consumed during the critical peak event.

Power Supply Charges: These charges are applicable to Full Service Customers.

Critical Peak Price: \$1.00 ~~0.95~~ per kWh of energy consumed during a critical peak event between June 1 and September 30

Off-Peak Discount: \$(0.015122 ~~0.032260~~) per kWh of Off-Peak kWh between June 1 and September 30

(Continued on Sheet No. D-17.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-17.00

RESIDENTIAL SUMMER ON-PEAK BASIC RATE RSP
(Continued From Sheet No. D-16.00)

Monthly Rate: (Contd)

Self-Generation ~~Provision~~ (SG):

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8, Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge:~~ \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C 11.2., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C 11.2., Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

(Continued on Sheet No. D-18.00)

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-19.00

~~RESIDENTIAL SERVICE SECONDARY RATE RS~~

~~Availability:~~

~~Subject to any restrictions, this rate is available to any customer desiring electric service for any usual residential use in: (i) private family dwellings; (ii) tourist homes, rooming houses, dormitories, nursing homes and other similarly occupied buildings containing sleeping accommodations for up to six persons; or (iii) existing multifamily dwellings containing up to four households served through a single meter. Service for single phase or three phase equipment may be included under this rate, provided the individual capacity of such equipment does not exceed 3 hp or 3 kW, nor does the total connected load of the home exceed 10 kW, without the specific consent of the Company.~~

~~As of January 1, 2020 this rate is closed to new business. After January 1, 2020 this rate is only available to customers electing a Non-Transmitting Meter in accordance with Rule C5.5, Non-Transmitting Meter Provision or customers determined to be eligible at the Company's sole discretion.~~

~~This rate is not available for: (i) resale purposes; (ii) multifamily dwellings containing more than four living units served through a single meter; (iii) tourist homes, rooming houses, dormitories, nursing homes and similarly occupied buildings containing sleeping accommodations for more than six persons; or (iv) any other Non-Residential usage.~~

~~Residences in conjunction with commercial or industrial enterprises and mobile home parks may take service on this rate only under the Rules and Regulations contained in the Company's Electric Rate Book.~~

~~Nature of Service:~~

~~Service under this rate shall be alternating current, 60 Hertz, single phase or three phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~The Company will schedule meter readings on a monthly basis and attempt to obtain an actual meter reading for all tourist and/or occasional residence customers at intervals of not more than six months.~~

~~Monthly Rate:~~

~~Power Supply Charges: These charges are applicable to Full Service customers.~~

~~Energy Charge:~~

Non-Capacity	Capacity	Total	
\$ 0.060483	\$ 0.035102	\$0.095585	per kWh for the first 600 kWh per month during the billing months of June–September
\$ 0.079987	\$ 0.046421	\$0.126408	per kWh for all kWh over 600 kWh per month during the billing months of June–September
\$ 0.060483	\$ 0.035102	\$0.095585	per kWh for all kWh during the billing months of October–May

~~This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.~~

~~Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.~~

System Access Charge:	\$7.50	per customer per month
Distribution Charge:	\$0.047054	per kWh for all kWh

~~This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.~~

~~(Continued on Sheet No. D-20.00)~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-20.00

~~RESIDENTIAL SERVICE SECONDARY RATE RS~~
~~(Continued From Sheet No. D-19.00)~~

~~Monthly Rate: (Contd)~~

~~Income Assistance Service Provision (RIA):~~

~~When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2, Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102, Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.~~

~~The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:~~

~~Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.~~

~~Income Assistance Credit: \$(7.50) per customer per month~~

~~This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).~~

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. D-21.00

~~RESIDENTIAL SERVICE SECONDARY RATE RS~~

~~(Continued From Sheet No. D-20.00)~~

~~Monthly Rate: (Contd)~~

~~Senior Citizen Service Provision (RSC):~~

~~When service is supplied to the Principal Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.~~

~~The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:~~

~~Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.~~

~~Senior Citizen Credit: \$(3.75) per customer per month~~

~~This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).~~

~~Self-Generation Provision (SG):~~

~~As of June 8, 2012, this provision may be required for any Full Service Customer with a generating installation less than 550 kW operating in parallel with the Company's system, which may employ cogeneration or small power production technology.~~

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge:~~ ~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

~~(Continued on Sheet No. D-22.00)~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-22.00

~~RESIDENTIAL SERVICE SECONDARY RATE RS~~

~~(Continued From Sheet No. D-21.00)~~

~~Monthly Rate: (Contd)~~

~~Peak Power Savers—Air Conditioner Peak Cycling Program:~~

~~A customer in a single family residence who is taking service from the Company may be eligible to participate in the Company's voluntary Peak Power Savers—Air Conditioner Peak Cycling Program for load management of eligible electric central air conditioning, central heat pump, or other qualifying electric equipment. Customer eligibility to participate in this program is determined solely by the Company. The customer must be located within an area in which Advanced Metering Infrastructure (AMI) is deployed and have a fully operational AMI meter for purposes of this program. The Company will accept a customer's central air conditioning, central heat pump, and other qualifying electric equipment under this program only if it has the capability to be controlled by the Company. Load Management of a customer's swimming pool pump is permitted under this program only if the customer is allowing Load Management of their air conditioner or heat pump unit. The Company will install the required equipment at the customer's premises which will allow Load Management upon signal from the Company. Such equipment shall be furnished, installed, maintained and owned by the Company at the Company's expense. Equipment installations must conform to the Company's specifications.~~

~~The customer's enrollment shall be terminated if the voluntary program ceases, if the customer tampers with the control switch or the Company's equipment or any reasons as provided for in Rule C1.3, Use of Service.~~

~~Load Management may occur any day of the week including weekends between the hours of 7:00 AM and 8:00 PM for no more than an eight hour period in any one day. Load Management may be implemented for, but not limited to, maintaining system integrity, making an emergency purchase, economic reasons, or when there is insufficient system generation available to meet anticipated system load. Load Management may only occur outside of the hours of 7:00 AM and 8:00 PM during a declared emergency event as directed by MISO.~~

~~The Customer may contact the Company to request to override a Load Management event for one Load Management event during the June through September months in any one calendar year for the balance of the hours left in that Load Management event with no penalty. The request shall be granted at the discretion of the Company. If the override request was granted by the Company and the customer requests and is granted any additional overrides in the same calendar year, the Peak Power Savers—Air Conditioner Peak Cycling Credit may be forfeited for that billing month.~~

~~Rule C1.1 Character of Service, Rule C3 Emergency Electrical Procedures and other rules and regulations contained in the Company's Electric Rate Book apply to customers taking service under this Peak Power Savers—Air Conditioner Peak Cycling Program.~~

~~The monthly credit for the Peak Power Savers—Air Conditioner Peak Cycling Program shall be applied as follows:~~

~~Power Supply Charges: These charges are applicable to Full Service Customers:~~

Peak Power Savers—Air Conditioner Peak Cycling Credit:	\$(8.00) per customer per month during the billing months of June-September
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~~(Continued on Sheet No. D-23.00)~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-23.00

~~RESIDENTIAL SERVICE SECONDARY RATE RS~~
~~(Continued From Sheet No. D-22.00)~~

~~Monthly Rate: (Contd)~~

~~Net Metering Program:~~

~~The Net Metering Program is available to any eligible customer as described in Rule C11., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C11.B, Net Metering Definitions.~~

~~A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C11., Net Metering Program.~~

~~Green Generation Program:~~

~~Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.~~

~~A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.~~

~~Non-Transmitting Meter Provision:~~

~~A customer who chooses a non-transmitting meter is subject to the provisions contained in Rule C5.5, Non-Transmitting Meter Provision.~~

~~General Terms:~~

~~This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.~~

~~Minimum Charge:~~

~~The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non-consumption-based surcharges.~~

~~Due Date and Late Payment Charge:~~

~~The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.~~

~~Term and Form of Contract:~~

~~Service under this rate shall not require a written contract except for the Green Generation Program participants.~~

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. D-24.00

~~RESIDENTIAL DYNAMIC PRICING PROGRAM~~

~~Availability:~~

~~The Residential Dynamic Pricing Program is voluntary and available to Full Service residential customers who have, or are selected to have, the required metering equipment and infrastructure installed. Customer eligibility to participate in this program is determined solely by the Company. The Company will furnish, install, maintain and own the required equipment at the customers' premises at the Company's expense. At the sole discretion of the Company, customers may be allowed to furnish, install and maintain the equipment at the customer's expense. Equipment and installations must conform to the Company's specifications. By enrolling in the program, the customer agrees to provide an email address, to participate in surveys and understands that the metering data will be used for evaluation purposes.~~

~~The customer's enrollment shall be terminated if the program ceases or for any reasons as provided for in Rule C1.3, Use of Service.~~

~~Deployment of the Residential Dynamic Pricing Program is at the sole discretion of the Company and is dependent upon installation of advanced metering infrastructure and supporting critical systems.~~

~~Customers participating in the Residential Dynamic Pricing Program shall not participate in any other Demand Response Program or Net Metering.~~

~~The program is not available for: (i) resale purposes; (ii) multifamily dwellings containing more than four living units served through a single meter; (iii) tourist homes, rooming houses, dormitories, nursing homes and similarly occupied buildings containing sleeping accommodations for more than six persons; or (iv) any other Non-Residential usage.~~

~~Residences in conjunction with commercial or industrial enterprises and mobile home parks may take service on this program only under the Rules and Regulations contained in the Company's Electric Rate Book.~~

~~Nature of Service:~~

~~Service under this program shall be alternating current, 60 Hertz, single phase or three phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Rate Options:~~

~~Customers are able to manage electric costs by either reducing load during high cost pricing periods or shifting load from high cost pricing periods to lower cost pricing periods. Upon enrollment of the customer in the Residential Dynamic Pricing Program, the customer shall take service under one of the following rate options:~~

~~**Option 1 – Peak Power Savers – Critical Peak Time of Use (RDP)** During a critical peak event, customers on Rate RDP will be charged the Critical Peak Event charge for all kWh consumed during the critical peak event.~~

~~**Option 2 – Peak Power Savers – Peak Rewards Time of Use (RDPR)** During a critical peak event, customers on Rate RDPR will be credited the Critical Peak Rebate for incremental energy reductions.~~

~~The Company may call up to fourteen critical peak events between June 1 and September 30. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer.~~

(Continued on Sheet No. D-25.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-25.00

RESIDENTIAL DYNAMIC PRICING PROGRAM
(Continued From Sheet No. D-24.00)

Monthly Rate:

Option 1 – Peak Power Savers – Critical Peak Time of Use Rate (RDP):

Power Supply Charges: These charges are applicable to Full Service customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak – Summer	\$0.048820	\$0.028333	\$0.077153	per kWh for all Off-Peak kWh between June 1 and September 30
Mid-Peak – Summer	\$0.069461	\$0.040312	\$0.109773	per kWh for all Mid-Peak kWh between June 1 and September 30
On-Peak – Summer	\$0.088400	\$0.051304	\$0.139704	per kWh for all On-Peak kWh between June 1 and September 30
Off-Peak – Winter	\$0.048820	\$0.028333	\$0.077153	per kWh for all Off-Peak kWh between October 1 and May 31
On-Peak – Winter	\$0.061087	\$0.035452	\$0.096539	per kWh for all On-Peak kWh between October 1 and May 31
Critical Peak Event	\$0.601129	\$0.348871	\$0.950000	per kWh during a critical peak event between June 1 and September 30

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service Customers.

System Access Charge: \$7.50 per customer per month

Distribution Charge: \$0.047054 per kWh for all kWh for a Full Service Customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Option 2 – Peak Power Savers – Peak Rewards Time of Use Rate (RDPR):

Power Supply Charges: These charges are applicable to Full Service Customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak – Summer	\$0.057112	\$0.033146	\$0.090258	per kWh for all Off-Peak kWh between June 1 and September 30
Mid-Peak – Summer	\$0.081003	\$0.047012	\$0.128015	per kWh for all Mid-Peak kWh between June 1 and September 30
On-Peak – Summer	\$0.102910	\$0.059726	\$0.162636	per kWh for all On-Peak kWh between June 1 and September 30
Off-Peak – Winter	\$0.048820	\$0.028333	\$0.077153	per kWh for all Off-Peak kWh between October 1 and May 31
On-Peak – Winter	\$0.061087	\$0.035452	\$0.096539	per kWh for all On-Peak kWh between October 1 and May 31
Critical Peak Reward	\$(0.601129)	\$(0.348871)	\$(0.950000)	per kWh during a critical peak event between June 1 and September 30

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

(Continued on Sheet No. D-26.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-26.00

RESIDENTIAL DYNAMIC PRICING PROGRAM

(Continued From Sheet No. D-25.00)

Monthly Rate: (Contd)

Option 2—Peak Power Savers—Peak Rewards Time-of-Use Rate (RDPR): (Contd)

Delivery Charges: These charges are applicable to Full Service Customers:

System Access Charge: \$7.50 per customer per month

Distribution Charge: \$0.047054 per kWh for all kWh for a Full Service Customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2, Consumer Standards and Billing Practices for Electric and Gas Residential Customers, R 460.102, Definitions. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers:

Income Assistance Credit: \$(7.50) per customer per month

This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

Senior Citizen Service Provision (RSC):

When service is supplied to the Principal Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.

The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers:

Senior Citizen Credit: \$(3.75) per customer per month

This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).

(Continued on Sheet No. D-27.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-27.00

RESIDENTIAL DYNAMIC PRICING PROGRAM
(Continued From Sheet No. D-26.00)

Monthly Rate: (Contd)

Self-Generation Provision (SG):

~~As of June 8, 2012, this provision may be required for any Full Service Customer with a generating installation less than 550 kW operating in parallel with the Company's system, which may employ cogeneration or small power production technology.~~

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

Sales of Self-Generated Energy to the Company:

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~**Administrative Cost Charge:** \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

Energy Purchase:

~~An energy purchase by the Company shall be bought at the Midecontinent Independent System Operator's, Inc. (MISO) real time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Green Generation Program:

~~Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.~~

~~A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.~~

General Terms:

~~This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.~~

Minimum Charge:

~~The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non-consumption based surcharges.~~

(Continued on Sheet No. D-28.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-28.00

RESIDENTIAL DYNAMIC PRICING PROGRAM
(Continued From Sheet No. D-27.00)

Monthly Rate: (Contd)

Due Date and Late Payment Charge:

~~The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.~~

Schedule of Hours:

~~The following schedule shall apply Monday through Friday (except holidays designated by the Company). Weekends and holidays are off peak. Holidays designated by the Company include: New Year's Day—January 1, Memorial Day—Last Monday in May, Independence Day—July 4, Labor Day—First Monday in September, Thanksgiving Day—Fourth Thursday in November, and Christmas Day—December 25. Whenever January 1, July 4, or December 25 falls on Sunday, extended holiday periods such as Monday, January 2, Monday, July 5 and Monday, December 26 shall not be considered as holidays for application of off peak hours.~~

~~Summer Billing Months of June through September:~~

- ~~(1) Off Peak Hours: 12:00 AM to 7:00 AM and 11:00 PM to 12:00 AM~~
- ~~(2) Mid Peak Hours: 7:00 AM to 2:00 PM and 6:00 PM to 11:00 PM~~
- ~~(3) On Peak Hours: 2:00 PM to 6:00 PM~~

~~Winter Billing Months of January through May and October through December:~~

- ~~(1) Off Peak Hours: 11:00 PM to 7:00 AM~~
- ~~(2) On Peak Hours: 7:00 AM to 11:00 PM~~

Term and Form of Contract:

~~Service under this rate shall not require a written contract except for the Green Generation Program participants.~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-29.00

EXPERIMENTAL RESIDENTIAL PLUG-IN ELECTRIC VEHICLE CHARGING PROGRAM

Availability:

The Experimental Residential Plug-In Electric Vehicle Charging Program is a voluntary pilot available to Full Service residential customers. Upon enrollment of the customer in the program, the customer may take service under one of the following options as applicable:

Option 1 – Residential Home and Plug-in Electric Vehicle Time-of-Day Rate (REV-1)—Level 1 or Level 2 Charging of an electric vehicle combined with household electric usage such as space conditioning, cooking, water heating, refrigeration, clothes drying, incineration or lighting based upon on-peak, mid-peak and off-peak periods and through a single meter.

Option 2 – Residential Plug-In Electric Vehicle Only Time-of-Day Rate (REV-2)—Level 2 Charging of the electric vehicle based upon on-peak, mid-peak and off-peak periods through a separate meter. Electric usage for the household will be billed under the RS or RT Rate Schedule.

“Level 1 Charging” is defined as voltage connection of 120 volts and a maximum load of 12 amperes or 1.4 kVA.

“Level 2 Charging” is defined as voltage connection of either 240 volts or 208 volts and a maximum load of 32 amperes or 7.7 kVA at 240 volts or 6.7 kVA at 208 volts.

“Electric Vehicle Supply Equipment (EVSE)” is defined as the conductors, including the ungrounded, grounded and equipment grounding conductors, the electric vehicle connectors, attachment plugs, and all other fittings, devices, power outlets, or apparatus installed specifically for the purpose of delivering energy from the premise wiring to the electric vehicle.

Vehicles shall be registered and operable on public highways in the State of Michigan to qualify for this rate. Low-speed electric vehicles including golf carts are not eligible to take service under this rate even if licensed to operate on public streets. The customer may be required to provide proof of registration of the electric vehicle to qualify for program.

The total connected load of the home including the electric vehicle charging shall not exceed 10 kW, without the specific consent of the Company.

Customers shall not back-feed or transmit stored energy from the electric vehicle’s battery to the Company’s distribution system.

Nature of Service:

Service under this rate shall be alternating current, 60 Hertz, single phase or three phase (at the Company's option) Secondary Voltage service.

Monthly Rate:

Option 1 – REV-1:

Power Supply Charges:

These charges are applicable to Full Service customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak—Summer	\$0.049127	\$0.028511	\$0.077638	per kWh for all Off-Peak kWh between June 1 and September 30
Mid-Peak—Summer	\$0.073691	\$0.042767	\$0.116458	per kWh for all Mid-Peak kWh between June 1 and September 30
On-Peak—Summer	\$0.117906	\$0.068427	\$0.186333	per kWh for all On-Peak kWh between June 1 and September 30
Off-Peak—Winter	\$0.049127	\$0.028511	\$0.077638	per kWh for all Off-Peak kWh between October 1 and May 31
On-Peak—Winter	\$0.073691	\$0.042767	\$0.116458	per kWh for all On-Peak kWh between October 1 and May 31

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

(Continued on Sheet No. D-30.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-30.00

EXPERIMENTAL RESIDENTIAL PLUG-IN ELECTRIC VEHICLE CHARGING PROGRAM

(Continued From Sheet No. D-29.00)

Monthly Rate (Contd)

Option 1 – REV – 1 (Contd)

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.

System Access Charge: \$7.50 per customer per month

Distribution Charge: \$0.047054 per kWh for all kWh for a Full Service customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

General Terms:

These rates are subject to all general terms and conditions shown on Sheet No. D-1.00.

Self-Generation Provision (SG):

As of June 8, 2012, this provision may be required for any Full Service Customer with a generating installation less than 550 kW operating in parallel with the Company's system, which may employ cogeneration or small power production technology.

All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.

Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.

Sales of Self-Generated Energy to the Company:

A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.

Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.

Administrative Cost Charge:

\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.

Energy Purchase:

An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstance.

(Continued on Sheet No. D-31.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-31.00

~~EXPERIMENTAL RESIDENTIAL PLUG-IN ELECTRIC VEHICLE CHARGING PROGRAM~~

~~(Continued From Sheet No. D-30.00)~~

~~Monthly Rate (Contd)~~

~~Option 2 – REV 2:~~

~~Power Supply Charges:~~

~~These charges are applicable to Full Service customers.~~

~~Energy Charge:~~

	Non-Capacity	Capacity	Total	
Off Peak – Summer	\$0.049127	\$0.028511	\$0.077638	per kWh for all Off Peak kWh between June 1 and September 30
Mid Peak – Summer	\$0.073691	\$0.042767	\$0.116458	per kWh for all Mid Peak kWh between June 1 and September 30
On Peak – Summer	\$0.117906	\$0.068427	\$0.186333	per kWh for all On Peak kWh between June 1 and September 30
Off Peak – Winter	\$0.049127	\$0.028511	\$0.077638	per kWh for all Off Peak kWh between October 1 and May 31
On Peak – Winter	\$0.073691	\$0.042767	\$0.116458	per kWh for all On Peak kWh between October 1 and May 31

~~This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.~~

~~Delivery Charges:~~

~~These charges are applicable to Full Service and Retail Open Access customers~~

~~Distribution Charge:~~

~~\$0.047054~~

~~for all kWh~~

~~This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00. The REP Surcharge shown on Sheet No. D-2.00 shall not apply.~~

~~General Terms:~~

~~These rates are subject to all general terms and conditions shown on Sheet No. D-1.00.~~

~~(Continued on Sheet No. D-32.00)~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-32.00

~~EXPERIMENTAL RESIDENTIAL PLUG-IN ELECTRIC VEHICLE CHARGING PROGRAM~~
~~(Continued From Sheet No. D-31.00)~~

~~Schedule of On-Peak Hours:~~

~~The following schedule shall apply Monday through Friday (except holidays designated by the Company). Weekends and holidays are off-peak. Holidays designated by the Company include: New Year's Day—January 1, Memorial Day—Last Monday in May, Independence Day—July 4, Labor Day—First Monday in September, Thanksgiving Day—Fourth Thursday in November, and Christmas Day—December 25. Whenever January 1, July 4, or December 25 falls on Sunday, extended holiday periods such as Monday, January 2, Monday, July 5 and Monday, December 26 shall not be considered as holidays for application of off-peak hours.~~

~~Summer:~~

- | | | |
|----------------|----------------------------|---|
| (1) | Off-Peak Hours: | 11:00 PM to 7:00 AM |
| (2) | Mid-Peak Hours: | 7:00 AM to 2:00 PM and 6:00 PM to 11:00 PM |
| (3) | On-Peak Hours: | 2:00 PM to 6:00 PM |

~~Winter:~~

- | | | |
|----------------|----------------------------|--------------------------------|
| (1) | Off-Peak Hours: | 11:00 PM to 7:00 AM |
| (2) | On-Peak Hours: | 7:00 AM to 11:00 PM |

~~Minimum Charge:~~

~~REV-1—The System Access Charge included in the rate adjusted for qualified service provision credit and any applicable non-consumption based surcharges.~~

~~REV-2—Any applicable non-consumption based surcharges.~~

~~Due Date and Late Payment Charge:~~

~~The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.~~

~~Green Generation Program:~~

~~Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2., Green Generation Program.~~

~~A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.~~

~~Term and Form of Contract:~~

~~Service under this rate shall not require a written contract, except for a customer who receives a contribution for purchase or installation of the EVSE, installation of the separately metered service from the Company, or who participates in the Green Generation Program. The Company reserves the right to specify the term of duration of the experimental rates. The participating customer may elect to terminate service for any reason giving the Company ten business days notice prior to the customer's next billing cycle. The customer's enrollment shall be terminated if the voluntary pilot ceases or for any reason as provided for in Rule C1.3, Use of Service.~~

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-33.00

RESIDENTIAL SERVICE TIME-OF-DAY SECONDARY RATE RT

Availability:

Subject to any restrictions, this rate is available to any residential customer desiring electric service who chooses to have their electric consumption metered based upon on-peak and off-peak periods. In addition, this rate is available to customers desiring electric service for electric vehicle battery charging where such service is in addition to all other household requirements. Battery charging service is limited to four-wheel vehicles licensed for operation on public streets and highways. Service for single-phase or three-phase equipment may be included under this rate, provided the individual capacity of such equipment does not exceed 3 hp or 3 kW, nor does the total connected load of the home exceed 10 kW, without the specific consent of the Company.

Service under this rate is limited to 10,000 customers.

This rate is not available for resale purposes or for any Non-Residential usage.

Nature of Service:

Service under this rate shall be alternating current, 60 Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service Customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
On-Peak Summer	\$0.078143	\$0.045351	\$0.123494	per kWh for all On-Peak kWh between June 1 and September 30
Off-Peak Summer	\$0.053674	\$0.031150	\$0.084824	per kWh for all Off-Peak kWh between June 1 and September 30
On-Peak Winter	\$0.065300	\$0.037898	\$0.103198	per kWh for all On-Peak kWh between October 1 and May 31
Off-Peak Winter	\$0.057190	\$0.033191	\$0.090381	per kWh for all Off-Peak kWh between October 1 and May 31

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

System Access Charge:	\$7.50	per customer per month
Distribution Charge:	\$0.047054	per kWh for all kWh for a Full Service Customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit, the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102 Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Income Assistance Credit:	\$(7.50)	per customer per month
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This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

(Continued on Sheet No. D-34.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-34.00

RESIDENTIAL SERVICE TIME-OF-DAY SECONDARY RATE RT

(Continued From Sheet No. D-33.00)

Monthly Rate: (Contd)

Senior Citizen Service Provision (RSC):

~~When service is supplied to the Principle Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.~~

~~The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:~~

~~**Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.**~~

~~Senior Citizen Credit: \$(3.75) per customer per month~~

~~This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).~~

Self-Generation Provision (SG):

~~As of June 8, 2012, this provision may be required for any Full Service Customer with a generating installation less than 550 kW operating in parallel with the Company's system, which may employ cogeneration or small power production technology.~~

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

Sales of Self-Generated Energy to the Company:

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

Administrative Cost Charge:

~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

Energy Purchase:

~~An energy purchase by the Company shall be bought at the Midecontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstance.~~

(Continued on Sheet No. D-35.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-35.00

~~RESIDENTIAL SERVICE TIME-OF-DAY SECONDARY RATE RT~~
~~(Continued From Sheet No. D-34.00)~~

~~Monthly Rate: (Contd)~~

~~Net Metering Program:~~

~~The Net Metering Program is available to any eligible customer as described in Rule C11., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C11.B., Net Metering Definitions.~~

~~A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C11., Net Metering Program.~~

~~Green Generation Program:~~

~~Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2., Green Generation Program.~~

~~A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2., Green Generation Program.~~

~~General Terms:~~

~~This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.~~

~~Minimum Charge:~~

~~The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non-consumption based surcharges.~~

~~Due Date and Late Payment Charge:~~

~~The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.~~

~~Schedule of On Peak and Off Peak Hours:~~

~~The following schedule shall apply Monday through Friday, including holidays when applicable:~~

~~(1) — On Peak Hours: 11:00 AM to 7:00 PM~~

~~(2) — Off Peak Hours: 7:00 PM to 11:00 AM~~

~~Saturday and Sunday are Off Peak.~~

~~Term and Form of Contract:~~

~~Service under this rate shall not require a written contract.~~

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-36.00

RESIDENTIAL SMART HOURS RATE RSH

Availability:

The Residential Smart Hour Rate will be available on a date to be announced by the Company.

Subject to any restrictions, this rate is available to any Full Service residential customers who have the required metering equipment and infrastructure installed. The Company will furnish, maintain and own the required equipment at the customers' premises at the Company's request. By selecting this rate schedule, the customer agrees to provide an email address. Electric consumption is billed using on-peak and off-peak periods year-round on the Residential Smart Hours Rate.

This rate is not available for resale purposes or for any Non-Residential usage.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service Customers.

	Non-Capacity	Capacity	Total	
Off-Peak – Summer	\$0.056766 0.059280	\$0.047710 0.034404	\$0.104476 0.093684	per kWh for all Off-Peak kWh between June 1 and September 30
On-Peak – Summer	\$0.084361 0.088051	\$0.070983 0.051101	\$0.155344 0.139152	per kWh for all On-Peak kWh between June 1 and September 30
Off-Peak – Winter	\$0.056878 0.059280	\$0.045217 0.034404	\$0.102095 0.093684	per kWh for all Off-Peak kWh between October 1 and May 31
On-Peak – Winter	\$0.061405 0.066561	\$0.051437 0.038629	\$0.112842 0.105190	per kWh for all On-Peak kWh between October 1 and May 31

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service Customers.

System Access Charge: ~~\$8.50~~ ~~7.50~~ per customer per month

Distribution Charge: ~~\$0.060209~~
~~0.047054~~ per kWh for all kWh for a Full Service customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2, Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102, Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service Customers.

Income Assistance Credit: ~~\$(8.50)~~ ~~7.50~~ per customer per month

This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

(Continued on Sheet No. D-37.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-37.00

RESIDENTIAL SMART HOURS RATE RSH

(Continued From Sheet No. D-36.00)

Monthly Rate: (Contd)

Low Income Assistance Credit (LIAC):

Company selected Residential customers may receive LIAC for up to 12 consecutive months. The number of customers enrolled may be adjusted, at the Company's discretion, in order to dispense Commission-approved LIAC funding on an annual basis. Any shortfall in the dispensing of annual LIAC funds to qualified customers shall be carried over into the subsequent LIAC program year. LIAC customer selection will be based on highest need chosen from one or more of the following eligibility criteria:

1. Customers with an approved critical care certification where the total household income does not exceed 150% of the Federal Poverty level within the last 12 months, as verified by an authorized State, Federal or community agency.
2. Customers who are enrolled in the Company's Consumers Affordable Resource for Energy (CARE) program.
3. Customers who have received a Home Heating Credit in the previous 12 months.
4. Customers whose total household income does not exceed 150% of the Federal Poverty level as verified by an authorized State, Federal or community agency.

The monthly credit for LIAC shall be applied as follows:

Low Income Assistance Credit: \$(30.00) per meter per month

If a credit balance occurs, the credit shall apply to the customer's future electric utility charges. Re-enrollment, if applicable, and confirmation of qualification is required for each annual period of participation.

Customers selected for LIAC will not be eligible for the RIA Provision while enrolled in LIAC.

Senior Citizen Service Provision (RSC):

When service is supplied to the Principle Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.

The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service Customers.

Senior Citizen Credit: \$(4.25 ~~3.75~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).

Peak Power Savers:

Customers can elect to participate in the Air Conditioning Peak Cycling Program and the Peak Reward Program as described in this tariff. When a customer participates in both programs, the customer's incremental energy savings earned under the Peak Reward is compared to the Peak Power Savers – Air Conditioner Peak Cycling Program Credit. The greater of the two credits will be applied to the customer's invoice for that billing month. Both credits will not apply in a single billing month. Customers participating in the Peak Reward Program cannot participate in the Critical Peak Price Program. The Company reserves the right to call test events between October 1 and May 31 for customers participating in Peak Power Savers Programs.

Air Conditioner Peak Cycling Program—(Available on a Date to be Announced by the Company):

A customer in a single family residence who is taking service from the Company may be eligible to participate in the Company's voluntary Peak Power Savers – Air Conditioner Peak Cycling Program for load management of eligible electric central air conditioning, central heat pump, or other qualifying electric equipment. Customer eligibility to participate is determined solely by the Company. The Company will accept a customer's central air conditioning, central heat pump, and other qualifying electric equipment under this program only if it has the capability to be controlled by the Company. Load Management of a customer's swimming pool pump is permitted under this program only if the customer is allowing Load Management of their air conditioner or heat pump unit. The Company will install the required equipment at the customer's premises which will allow Load Management upon signal from the Company. When Load Management equipment is installed at a premises, future customers will be auto-enrolled into the Peak Power Savers-Air Conditioner Peak Cycling Program. Upon move in, the customer will be notified confirming participation in the Peak Power Savers-Air Conditioner Peak Cycling Program and will have 30 days to opt out. Such equipment shall be furnished, installed, maintained and owned by the Company at the Company's expense. Equipment installations must conform to the Company's specifications.

The Company reserves the right to specify the term or duration of the program. The customer's enrollment shall be terminated if the voluntary program ceases, if the customer tampers with the control switch or the Company's equipment or any reasons as provided for in Rule C1.3, Use of Service.

Load Management may occur any day of the week including weekends between the hours of 7:00 AM and 8:00 PM for no more than an eight hour period in any one day. Load Management may be implemented for, but not limited to, maintaining system integrity, making an emergency purchase, economic reasons, or when there is insufficient system generation available to meet anticipated system load. Load Management may only occur outside of the hours of 7:00 AM and 8:00 PM during a declared emergency event as directed by MISO.

The Customer may contact the Company to request to override a Load Management event for one Load Management event during the June through September months in any one calendar year for the balance of the hours left in that Load Management event with no penalty. The request shall be granted at the discretion of the Company. If the override request was granted by the Company and the customer requests and is granted any additional overrides in the same calendar year, the Peak Power Savers – Air Conditioner Peak Cycling Credit may be forfeited for that billing month.

Rule C1.1 Character of Service, Rule C3 Emergency Electrical Procedures and other rules and regulations contained in the Company's Electric Rate Book apply to customers taking service under this Peak Power Savers – Air Conditioner Peak Cycling Program.

The monthly credit for the Peak Power Savers Program shall be applied as follows:

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Power Savers – Air Conditioner Peak Cycling Credit: \$(8.00) per customer per month during the billing months of June-September

(Continued on Sheet No. D-38.00)

See Barnes Testimony, Page 5, Lines 13-20; McLean Testimony, Page 50, Line 13 through Page 54, Line 5, Page 73, Lines 3-25 and Page 74, Lines 1-7; Miller Testimony Page 17, Lines 1-7; Exhibit A-23 (RLB-1) Items #12, 14 and 15; Exhibit A-16 (HWM-3), Page 2; Exhibit A-16 (HWM-2) Sch F-2.1, line 21

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-38.00

RESIDENTIAL SMART HOURS RATE RSH
(Continued From Sheet No. D-37.00)

Monthly Rate: (Contd)

Peak Power Savers: (Contd)

Peak Reward—~~(Available on a Date to be Announced by the Company):~~

Participating customers are able to manage electric costs by reducing load during critical peak events. The Company may call up to fourteen critical peak events between June 1 and September 30 and up to five critical peak events between October 1 and May 31. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers on will be credited the Peak Reward per kWh of incremental energy reductions.

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Reward \$(1.00 ~~0.95~~) per kWh of incremental energy reduction during a critical peak event ~~between June 1 and September 30~~

Critical Peak Price—~~(Available on a Date to be Announced by the Company)~~

Participating customers are able to manage electric costs by shifting load during critical peak events to a lower cost pricing period. The Company may call up to fourteen critical peak events ~~between June 1 and September 30~~. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers on will be charged the Critical Peak Price per kWh consumed during the critical peak event.

Power Supply Charges: These charges are applicable to Full Service Customers.

Critical Peak Price \$(1.00 ~~0.95~~) per kWh of energy consumed during a critical peak event between June 1 and September 30

Off-Peak Discount \$(0.015122 ~~0.032260~~) per kWh for Off-Peak kWh between June 1 and September 30

Self-Generation ~~Provision~~ (SG):

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data/billing determinants necessary for billing purposes.~~

(Continued on Sheet No. D-39.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-39.00

RESIDENTIAL SMART HOURS RATE RSH

(Continued From Sheet No. D-38.00)

Monthly Rate: (Contd)

~~Self-Generation Provision (SG): (Contd)~~

~~Administrative Cost Charge:-~~

~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:-~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstance.~~

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C11.2., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C11.2., Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2., Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2., Green Generation Program.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non consumption based surcharges.

Due Date and Late Payment Charge:

The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.

Schedule of On-Peak and Off-Peak Hours:

The following schedule shall apply Monday through Friday, including weekday holidays when applicable:

Summer: June 1 through September 30

Winter: October 1 through May 31

(1) On-Peak Hours: 2:00 PM to 7:00 PM

(2) Off-Peak Hours: 7:00 PM to 2:00 PM

Saturday and Sunday are Off-Peak.

Term and Form of Contract:

Service under this rate shall not require a written contract.

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-40.00

RESIDENTIAL NIGHTTIME SAVERS RATE RPM

Availability:

The Residential Nighttime Savers Rate will be available on a date to be announced by the Company.

The Residential Nighttime Savers Rate is voluntary and available to Full Service residential customers who have the required metering equipment and infrastructure installed. The Company will furnish, install, maintain and own the required equipment at the customers' premises at the Company's expense. ~~By selecting this rate schedule, the customer agrees to provide an email address.~~

~~Customers taking service on the Residential Nighttime Savers Rate are able to manage electric costs by reducing load during high cost pricing periods and shifting load from high cost pricing periods to lower cost pricing periods. During a critical peak event, customers on the Residential Nighttime Savers Rate will be credited the Peak Reward per kWh of incremental energy reductions.~~

~~The Company may call up to fourteen critical peak events between June 1 and September 30. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer.~~

This rate is not available for: (i) resale purposes; (ii) multifamily dwellings containing more than four living units served through a single meter; (iii) tourist homes, rooming houses, dormitories, nursing homes and similarly occupied buildings containing sleeping accommodations for more than six persons; (iv) any other Non-Residential usage or (v) customers being served under Rule C5.5 Non-Transmitting Meter Provision.

Residences in conjunction with commercial or industrial enterprises and mobile home parks may take service on this program only under the Rules and Regulations contained in the Company's Electric Rate Book.

Nature of Service:

Service under this program shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service Customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
Super Off-Peak - Summer	\$0.043748 0.047573	\$0.031810 0.027609	\$0.075558 0.075182	per kWh for all Off-Peak kWh between June 1 and September 30
Off-Peak - Summer	\$0.066664 0.080874	\$0.052252 0.046936	\$0.118916 0.127810	per kWh for all Mid-Peak kWh between June 1 and September 30
On-Peak - Summer	\$0.084361 0.095146	\$0.070983 0.055219	\$0.155344 0.150365	per kWh for all On-Peak kWh between June 1 and September 30
Super Off-Peak - Winter	\$0.048724 0.047573	\$0.032981 0.027609	\$0.081705 0.075182	per kWh for all Off-Peak kWh between June 1 and September 30
Off-Peak - Winter	\$0.064325 0.061845	\$0.045985 0.035892	\$0.110310 0.097737	per kWh for all Off-Peak kWh between October 1 and May 31
On-Peak - Winter	\$0.061405 0.066602	\$0.051437 0.038653	\$0.112842 0.105255	per kWh for all On-Peak kWh between October 1 and May 31

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service Customers.

System Access Charge: ~~\$8.50~~ ~~7.50~~ per customer per month

Distribution Charge: ~~\$0.060209~~ ~~0.047054~~ per kWh for all kWh for a Full Service Customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-41.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-41.00

RESIDENTIAL NIGHTTIME SAVERS RATE RPM

(Continued From Sheet No. D-40.00)

Monthly Rate: (Contd)

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102, Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service Customers.

Income Assistance Credit: \$(~~8.50~~ ~~7.50~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

Low Income Assistance Credit (LIAC):

Company selected Residential customers may receive LIAC for up to 12 consecutive months. The number of customers enrolled may be adjusted, at the Company's discretion, in order to dispense Commission-approved LIAC funding on an annual basis. Any shortfall in the dispensing of annual LIAC funds to qualified customers shall be carried over into the subsequent LIAC program year. LIAC customer selection will be based on highest need chosen from one or more of the following eligibility criteria:

1. Customers with an approved critical care certification where the total household income does not exceed 150% of the Federal Poverty level within the last 12 months, as verified by an authorized State, Federal or community agency.
2. Customers who are enrolled in the Company's Consumers Affordable Resource for Energy (CARE) program.
3. Customers who have received a Home Heating Credit in the previous 12 months.
4. Customers whose total household income does not exceed 150% of the Federal Poverty level as verified by an authorized State, Federal or community agency.

The monthly credit for LIAC shall be applied as follows:

Low Income Assistance Credit: \$(30.00) per meter per month

If a credit balance occurs, the credit shall apply to the customer's future electric utility charges. Re-enrollment, if applicable, and confirmation of qualification is required for each annual period of participation.

Customers selected for LIAC will not be eligible for the RIA Provision while enrolled in LIAC.

Senior Citizen Service Provision (RSC):

When service is supplied to the Principal Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.

The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service Customers.

Senior Citizen Credit: \$(~~4.25~~ ~~3.75~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).

Residential Plug-In Electric Vehicle Only Credit (REV):

When service is supplied for Level 2 Charging of a separately metered electric vehicle, a credit shall be applied during all billing months. Electric usage for the household will be billed under the Residential Summer On-Peak Basic Rate or the Residential Smart Hours Rate.

"Level 2 Charging" is defined as voltage connection of either 240 volts or 208 volts and a maximum load of 32 amperes or 7.7 kVA at 240 volts or 6.7 kVA at 208 volts.

Vehicles shall be registered and operable on public highways in the State of Michigan to qualify for this credit. Low-speed electric vehicles including golf carts are not eligible for this credit even if licensed to operate on public streets. The customer may be required to provide proof of registration of the electric vehicle to qualify for this credit.

Delivery Charges: These charges are applicable to Full Service Customers.

Residential Plug-In Electric Vehicle Only Credit: \$(8.50 ~~7.50~~) per customer per month

Peak Power Savers:

Customers can elect to participate in the Air Conditioner Peak Cycling Program and the Peak Reward Program as described in this tariff. When a customer participates in both programs, the customer's incremental energy savings earned under the Peak Reward is compared to the Peak Power Savers – Air Conditioner Peak Cycling Program Credit. The greater of the two credits will be applied to the customer's invoice for that billing month. Both credits will not apply in a single billing month. Customers participating in the Peak Reward Program cannot participate in the Critical Peak Price Program. [The Company reserves the right to call test events between October 1 and May 31 for customers participating in Peak Power Savers Programs.](#)

(Continued on Sheet No. D-42.00)

See Barnes Testimony, Page 5, Lines 13-20; McLean Testimony, Page 50, Line 13 through Page 54, Line 5, Page 73, Lines 3-25 and Page 74, Lines 1-7; Miller Testimony Page 17, Lines 1-7; Exhibit A-23 (RLB-1) Item #12, 14 and 15; Exhibit A-16 (HWM-3), Page 3; Exhibit A-16 (HWM-2) Sch F-2.1, line 21

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-42.00

RESIDENTIAL NIGHTTIME SAVERS RATE RPM

(Continued From Sheet No. D-41.00)

Monthly Rate: (Contd)

Peak Power Savers: (Contd)

Air Conditioner Peak Cycling Program—~~(Available on a Date to be Announced by the Company):~~

A customer in a single family residence who is taking service from the Company may be eligible to participate in the Company's voluntary Peak Power Savers – Air Conditioner Peak Cycling Program for load management of eligible electric central air conditioning, central heat pump, or other qualifying electric equipment. Customer eligibility to participate is determined solely by the Company. The Company will accept a customer's central air conditioning, central heat pump, and other qualifying electric equipment under this program only if it has the capability to be controlled by the Company. Load Management of a customer's swimming pool pump is permitted under this program only if the customer is allowing Load Management of their air conditioner or heat pump unit. The Company will install the required equipment at the customer's premises which will allow Load Management upon signal from the Company. When Load Management equipment is installed at a premises, future customers will be auto-enrolled into the Peak Power Savers-Air Conditioner Peak Cycling Program. Upon move in, the customer will be notified confirming participation in the Peak Power Savers-Air Conditioner Peak Cycling Program and will have 30 days to opt out. Such equipment shall be furnished, installed, maintained and owned by the Company at the Company's expense. Equipment installations must conform to the Company's specifications.

The Company reserves the right to specify the term or duration of the program. The customer's enrollment shall be terminated if the voluntary program ceases, if the customer tampers with the control switch or the Company's equipment or any reasons as provided for in Rule C1.3, Use of Service.

Load Management may occur any day of the week including weekends between the hours of 7:00 AM and 8:00 PM for no more than an eight hour period in any one day. Load Management may be implemented for, but not limited to, maintaining system integrity, making an emergency purchase, economic reasons, or when there is insufficient system generation available to meet anticipated system load. Load Management may only occur outside of the hours of 7:00 AM and 8:00 PM during a declared emergency event as directed by MISO.

The Customer may contact the Company to request to override a Load Management event for one Load Management event during the June through September months in any one calendar year for the balance of the hours left in that Load Management event with no penalty. The request shall be granted at the discretion of the Company. If the override request was granted by the Company and the customer requests and is granted any additional overrides in the same calendar year, the Peak Power Savers – Air Conditioner Peak Cycling Credit may be forfeited for that billing month.

Rule C1.1 Character of Service, Rule C3 Emergency Electrical Procedures and other rules and regulations contained in the Company's Electric Rate Book apply to customers taking service under this Peak Power Savers – Air Conditioner Peak Cycling Program.

The monthly credit for the Peak Power Savers Program shall be applied as follows:

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Power Savers – Air Conditioner Peak Cycling Credit: \$(8.00) per customer per month during the billing months of June-September

Peak Reward—~~(Available on a Date to be Announced by the Company):~~

Participating customers are able to manage electric costs by reducing load during critical peak events. The Company may call up to fourteen critical peak events between June 1 and September 30 and up to five critical peak events between October 1 and May 31. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers on will be credited the Peak Reward per kWh of incremental energy reductions.

Power Supply Charges: These charges are applicable to Full Service Customers.

Peak Reward ~~\$(0.95~~ 1.00) per kWh of incremental energy reduction during a critical peak event ~~between~~
June 1 and September 30

(Continued on Sheet No. D-43.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-43.00

RESIDENTIAL NIGHTTIME SAVERS RATE RPM (Continued From Sheet No. D-42.00)

Monthly Rate: (Contd)

Peak Power Savers: (Contd)

Critical Peak Price—~~(Available on a Date to be Announced by the Company)~~

Participating customers are able to manage electric costs by shifting load during critical peak events to a lower cost pricing period. The Company may call up to fourteen critical peak events between June 1 and September 30. Customers will be notified by 11:59 PM the day before a critical peak event is expected to occur. Receipt of such notice is the responsibility of the participating customer. Customers must have a transmitting meter to participate in Peak Power Savers.

During a critical peak event, customers on will be charged the Critical Peak Price per kWh consumed during the critical peak event.

Power Supply Charges: These charges are applicable to Full Service Customers.

Critical Peak Price ~~\$1.00~~ 0.95 per kWh of energy consumed during a critical peak event between June 1 and September 30

Off-Peak Discount ~~\$(0.015122)~~ 0.032260 per kWh for Off-Peak kWh between June 1 and September 30

Self-Generation ~~Provision~~ (SG):

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C 1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge: \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

(Continued on Sheet No. D-44.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-44.00

RESIDENTIAL NIGHTTIME SAVERS RATE RPM
(Continued From Sheet No. D-43.00)

Monthly Rate: (Contd)

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C 11.2., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C 11.2., Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge:

The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.

Schedule of Hours:

The following schedule shall apply Monday through Friday including weekday holidays.

Summer: June 1 through September 30

Winter: October 1 through May 31

- | | | |
|-----|-----------------------|--|
| (1) | Super Off-Peak Hours: | 11:00 PM to 6:00 AM |
| (2) | Off-Peak Hours: | 6:00 AM to 2:00 PM and 7:00 PM to 11:00 PM |
| (3) | On-Peak Hours: | 2:00 PM to 7:00 PM |

Saturday and Sunday are Super Off-Peak.

Term and Form of Contract:

Service under this rate shall not require a written contract except for the Green Generation Program participants.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-44.10

RESIDENTIAL SERVICE SECONDARY NON-TRANSMITTING METER RATE RSM

Availability:

Subject to any restrictions, this rate is available to any customer desiring electric service for any usual residential use in: (i) private family dwellings; (ii) tourist homes, rooming houses, dormitories, nursing homes and other similarly occupied buildings containing sleeping accommodations for up to six persons; or (iii) existing multifamily dwellings containing up to four households served through a single meter. Service for single-phase or three-phase equipment may be included under this rate, provided the individual capacity of such equipment does not exceed 3 hp or 3 kW, nor does the total connected load of the home exceed 10 kW, without the specific consent of the Company.

This rate is only available to customers electing a Non-Transmitting Meter in accordance with Rule C5.5, Non-Transmitting Meter Provision, customers with a Non-Communicating Advanced Metering Infrastructure (AMI) Meter, or customers determined to be eligible at the Company's sole discretion.

A Non-Communicating AMI meter is unable to consistently transmit interval data to the Company's billing system. Non-Communicating Meters are determined at the Company's sole discretion and are subject to a minimum of one communication review per calendar year. When the meter has been determined to successfully communicate interval data, the customer will be notified and transferred to Residential Service Secondary On-Peak Summer Basic Rate RSP. The transfer to Rate RSP shall not occur between June 1 and September 30.

This rate is not available for: (i) resale purposes; (ii) multifamily dwellings containing more than four living units served through a single meter; (iii) tourist homes, rooming houses, dormitories, nursing homes and similarly occupied buildings containing sleeping accommodations for more than six persons; or (iv) any other Non-Residential usage.

Residences in conjunction with commercial or industrial enterprises and mobile home parks may take service on this rate only under the Rules and Regulations contained in the Company's Electric Rate Book.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

The Company will schedule meter readings on a monthly basis and attempt to obtain an actual meter reading for all tourist and/or occasional residence customers at intervals of not more than six months.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service customers.

Energy Charge:

Non-Capacity	Capacity	Total	
\$0.062414	\$0.046840	\$0.109254	per kWh for the first 600 kWh per month during the billing months of June – September
0.060483	0.035102	0.095585	
\$0.078673	\$0.055837	\$0.134510	per kWh for all kWh over 600 kWh per month during the billing months of June - September
0.079987	0.046421	0.126408	
\$0.057734	\$0.046840	\$0.104574	per kWh for all kWh during the billing months of October-May
0.060483	0.035102	0.095585	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.

System Access Charge: ~~\$8.50~~ 7.50 per customer per month

Distribution Charge: ~~\$0.060209~~ 0.047054 per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-44.20)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-44.20

RESIDENTIAL SERVICE SECONDARY NON-TRANSMITTING METER RATE RSM
(Continued From Sheet No. D-44.10)

Monthly Rate: (Contd)

Income Assistance Service Provision (RIA):

When service is supplied to a Principal Residence Customer, where the household receives a Home Heating Credit (HHC) in the State of Michigan, a credit shall be applied during all billing months. For an income assistance customer to qualify for this credit the Company shall require annual evidence of the HHC energy draft or warrant. The customer may also qualify for this credit by meeting the requirements under Rule B2, Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.102, Definitions; A to F. Confirmation shall be required by an authorized State or Federal agency to verify that the customer's total household income does not exceed 150% of the Federal poverty level.

The monthly credit for the residential Income Assistance Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Income Assistance Credit: \$(8.50 ~~7.50~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Senior Citizen Service Provision (RSC).

Low Income Assistance Credit (LIAC):

Company selected Residential customers may receive LIAC for up to 12 consecutive months. The number of customers enrolled may be adjusted, at the Company's discretion, in order to dispense Commission-approved LIAC funding on an annual basis. Any shortfall in the dispensing of annual LIAC funds to qualified customers shall be carried over into the subsequent LIAC program year. LIAC customer selection will be based on highest need chosen from one or more of the following eligibility criteria:

1. Customers with an approved critical care certification where the total household income does not exceed 150% of the Federal Poverty level within the last 12 months, as verified by an authorized State, Federal or community agency.
2. Customers who are enrolled in the Company's Consumers Affordable Resource for Energy (CARE) program.
3. Customers who have received a Home Heating Credit in the previous 12 months.
4. Customers whose total household income does not exceed 150% of the Federal Poverty level as verified by an authorized State, Federal or community agency.

The monthly credit for LIAC shall be applied as follows:

Low Income Assistance Credit: \$(30.00) per meter per month

If a credit balance occurs, the credit shall apply to the customer's future electric utility charges. Re-enrollment, if applicable, and confirmation of qualification is required for each annual period of participation.

Customers selected for LIAC will not be eligible for the RIA Provision while enrolled in LIAC.

(Continued on Sheet No. D-44.30)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-44.30

RESIDENTIAL SERVICE SECONDARY NON-TRANSMITTING METER RATE RSM
(Continued From Sheet No. D-44.20)

Monthly Rate: (Contd)

Senior Citizen Service Provision (RSC):

When service is supplied to the Principal Residence Customer who is 65 years of age or older and head of household, a credit shall be applied during all billing months.

The monthly credit for the residential Senior Citizen Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.

Senior Citizen Credit: \$(~~4.25~~ ~~3.75~~) per customer per month

This credit shall not be taken in conjunction with a credit for the Income Assistance Service Provision (RIA).

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

Non-Transmitting Meter Provision:

A customer who chooses a non-transmitting meter is subject to the provisions contained in Rule C5.5, Non-Transmitting Meter Provision.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate, adjusted for qualified service provision credit and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge:

The due date of the customer's bill shall be 21 days from the date of transmittal. A late payment charge of 2%, not compounded, of the portion of the bill, net of taxes, shall be assessed to any bill that is delinquent. A customer who participates in the Winter Protection Plan or who is 65 years of age or older and who has notified the Company the customer is 65 years of age or older, shall be exempt from a late payment charge as described in Rule B2., Consumer Standards and Billing Practices for Electric and Natural Gas Service, R 460.125, Late payment charges.

Term and Form of Contract:

Service under this rate shall not require a written contract except for the Green Generation Program participants.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-45.00

GENERAL SERVICE SECONDARY RATE GS

Availability:

As of January 1, 2021, this rate is closed to new business. Subject to any restrictions, this rate is available to any general use customer, political subdivision or agency of the State of Michigan, either acting separately or in combinations permitted under the laws of this state, desiring Secondary Voltage service for any of the following: (i) standard secondary service, (ii) public potable water pumping and/or waste water system(s), or (iii) resale purposes. This rate is also available for service to any Primary Rate Customer where the Company elects to provide one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer.

This rate is not available for: (i) private family dwellings, (ii) lighting service except for private streets, mobile home parks or service to temporary lighting installations, (iii) heating water for industrial processing, (iv) resale for lighting service, or (v) new or expanded service for resale to residential customers. Unmetered Billboard Service is not available to Retail Open Access service.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Three-phase, 3-wire service requires that the customer furnishes all transformation facilities required for single-phase load and so arranges the load as to avoid excessive unbalance of the three-phase load. When the service is single-phase, or 4-wire, three-phase, the single-phase individual motor capacity shall not exceed 3 hp, nor the total single-phase motor capacity of 10 hp, without the specific consent of the Company.

Where the Company elects to measure the service on the Primary side of the transformers, 3% shall be deducted for billing purposes from the energy measurements thus made. Where the Company elected to provide a Primary Rate Customer one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer, 3% shall not be deducted for billing purposes from the energy measurements thus made.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service customers.

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.055373</u>	<u>\$0.038384</u>	<u>\$0.093757</u>	per kWh for all kWh during the billing months of June-September
0.062210	0.034294	0.096504	
<u>\$0.052883</u>	<u>\$0.039925</u>	<u>\$0.092808</u>	per kWh for all kWh during the billing months of October-May
0.061580	0.033947	0.095527	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service and Retail Open Access customers.

System Access Charge: \$20.00 per customer per month

Distribution Charge: \$0.052253
~~0.042472~~ per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Billboard Service Provision:

Monthly kWh shall be determined by multiplying the total connected load in kW (including the lamps, ballasts, transformers, amplifiers, and control devices) times 730 hours. The kWh for cyclical devices shall be adjusted for the average number of hours used.

(Continued on Sheet No. D-46.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-46.00

GENERAL SERVICE SECONDARY RATE GS
(Continued From Sheet No. D-45.00)

Monthly Rate: (Contd)

Resale Service Provision:

Subject to any restrictions, this provision is available to customers desiring Secondary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

Educational Institution Service Provision (GEI):

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, “school” shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. “College” or “University” shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Education Institution Credit: \$(~~0.000829~~ 0.000748) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

Self-Generation ~~Provision~~ (SG):

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company’s system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission’s (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data/billing determinants necessary for billing purposes.~~

(Continued on Sheet No. D-47.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-47.00

GENERAL SERVICE SECONDARY RATE GS
(Continued From Sheet No. D-46.00)

Monthly Rate: (Contd)

~~**Administrative Cost Charge:**~~

~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~**Energy Purchase:**~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C 11.2., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C 11.2., Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C 10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provision contained in Rule C 10.2, Green Generation Program.

Non-Transmitting Meter Provision:

A customer who chooses a non-transmitting meter is subject to the provisions contained in Rule C 5.5, Non-Transmitting Meter Provision.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate and any applicable non-consumption based surcharges. Special Minimum Charges shall be billed in accordance with Rule C 15., Special Minimum Charges.

Due Date and Late Payment Charge:

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract:

Service under this rate shall not require a written contract except for: (i) resale service, (ii) service under the Green Generation Program, (iii) for Special Minimum Charges, (iv) service for lighting or where mobile home parks are involved, (v) service under the Educational Institution Service Provision, (vi) service under the Net Metering Program, or (vii) at the option of the Company. If a contract is deemed necessary by the Company, the appropriate contract form shall be used and the contract shall require a minimum term of one year.

**M.P.S.C. No. 14 - Electric
Consumers Energy Company**

Sheet No. D-48.00

GENERAL SERVICE SECONDARY TIME-OF-USE RATE GSTU

Availability

Subject to any restrictions, General Service Secondary Time-of-Use Rate GSTU is available to any Full Service Customer taking service at the Company's Secondary Voltage level with advanced metering infrastructure and supporting critical systems. Standby service shall be provided on this rate for secondary customers with solar installations equal to or greater than 150 kW.

This rate is not available for: (i) private family dwellings, (ii) lighting service except for private streets, mobile home parks or service to temporary lighting installations, (iii) heating water for industrial processing, (iv) resale for lighting service, or (v) new or expanded service for resale to residential customers.

This rate shall not be taken in conjunction with any other Demand Response Program or Net Metering.

Nature of Service

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Three-phase, 3-wire service requires that the customer furnishes all transformation facilities required for single-phase load and so arranges the load as to avoid excessive unbalance of the three-phase load. When the service is single-phase, or 4-wire, three-phase, the single-phase individual motor capacity shall not exceed 3 hp, nor the total single-phase motor capacity of 10 hp, without the specific consent of the Company.

Where the Company elects to measure the service on the Primary side of the transformers, 3% shall be deducted for billing purposes from the energy measurements thus made. Where the Company elected to provide a Primary Rate Customer one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer, 3% shall not be deducted for billing purposes from the energy measurements thus made.

Monthly Rate

Power Supply Charges: These charges are applicable to Full Service Customers.

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	\$0.040002 0.055495	\$0.029094 0.030592	\$0.069096 0.086087	per kWh for all Off-Peak kWh during the billing months of June-September
Mid-Peak-Summer	\$0.059494 0.086391	\$0.045082 0.047624	\$0.104576 0.134015	per kWh for all Mid-Peak kWh during the billing months of June-September
On-Peak-Summer	\$0.076203 0.109946	\$0.054260 0.060609	\$0.130463 0.170555	per kWh for all On-Peak kWh during the billing months of June-September
Off-Peak-Winter	\$0.048284 0.050720	\$0.033991 0.027960	\$0.082275 0.078680	per kWh for all Off-Peak kWh during the billing months of October-May
On-Peak -Winter	\$0.061149 0.057913	\$0.045497 0.031925	\$0.106646 0.089838	per kWh for all On-Peak kWh during the billing months of October-May

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges: These charges are applicable to Full Service Customers.

System Access Charge:	\$30.00 20.00	per customer per month
<u>Capacity Charge:</u>	<u>\$1.34</u>	<u>per kW for all kW of Peak Demand</u>
Distribution Charge:	\$0.034680 0.042472	per kWh for all kWh for a Full Service Customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-49.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-49.00

GENERAL SERVICE SECONDARY TIME-OF-USE RATE GSTU
(Continued From Sheet No. D-48.00)

Monthly Rate (Contd)

Schedule of Hours

The following schedule shall apply Monday through Friday (except holidays designated by the Company). Weekends and holidays are off-peak. Holidays designated by the Company include: New Year's Day – January 1, Memorial Day – Last Monday in May, Independence Day – July 4, Labor Day – First Monday in September, Thanksgiving Day – Fourth Thursday in November and Christmas Day – December 25. Whenever January 1, July 4, or December 25 falls on Sunday, extended holiday periods such as Monday, January 2, Monday, July 5 and Monday, December 26 shall not be considered as holidays for application of off-peak hours.

Summer Billing Months of June through September:

- | | |
|--------------------|--|
| (1) Off-Peak Hours | 12:00 AM to 7:00 AM and 11:00 PM to 12:00 AM |
| (2) Mid-Peak Hours | 7:00 AM to 2:00 PM and 6:00 PM to 11:00 PM |
| (3) On-Peak Hours | 2:00 PM to 6:00 PM |

Winter Billing Months of January through May and October through December:

- | | |
|--------------------|---------------------|
| (1) Off-Peak Hours | 11:00 PM to 7:00 AM |
| (2) On-Peak Hours | 7:00 AM to 11:00 PM |

Peak Demand

The Peak Demand shall be the Kilowatts (kW) supplied during the period of highest use in the billing month but not less than 60% of the highest Peak Demand created during the preceding billing months of June through September.

The Company reserves the right to make special determination of the Peak Demand and/or the Minimum Charge should the equipment which creates momentary high demands be included in the customer's installation.

Resale Service Provision

Subject to any restrictions, the provision is available to customers desiring Secondary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

Educational Institution Service Provision (GEI)

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, "school" shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. "College" or "University" shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service Customers.

Education Institution Credit: \$(0.000829 ~~0.000748~~) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

General Service Secondary Interruptible (GSI) Provision:

This provision is available to no more than 200 Full Service Customers desiring interruptible service in conjunction with service taken under General Service Secondary Demand Rate GSD or General Service Secondary Time-of-Use Rate GSTU. Service to interruptible load shall be taken through separately metered circuits and permanently wired. The design and method of installation for application of this rate shall be subject to the approval of the Company.

Any load designated as interruptible by the customer is subject to Midcontinent Independent System Operator's, Inc. (MISO) requirements for Load Modifying Resources and the Company shall inform the Customer of such MISO requirements. Interruption under this provision may occur if MISO issues a Maximum Generation Emergency Event Step 2b order or NERC Emergency Event Alert 2 notice indicating that MISO is experiencing or expects to experience

a shortage of economic resources and the Company has declared Emergency Status.

Under this provision, the customer shall be interrupted at any time the Company deems it necessary to maintain system integrity. Service to interruptible load shall not be transferred to firm service circuits to avoid interruption. The Company shall provide the Customer at least 30 minutes notice in advance of a required interruption. Failure to acknowledge receipt of such notice shall not relieve the Customer of the obligation for interruption under the GSI provision. Failure by a customer to comply with a system integrity interruption order of the Company shall be considered unauthorized used and billed at (1) the higher of the actual damages incurred by the Company or (ii) the rate of \$25.00 per kW for the highest 15-minute kW of demand created during the interruption period in addition to the prescribed monthly rate.

This rate is not available for loads that are primarily off-peak, for example parking lot lighting. Participation requires a minimum term of one year.

The monthly credit for the Interruptible Service Provision shall be applied as follows:

Power Supply Charges - These charges are applicable to Full Service Customers.

Capacity Credit: These charges are applicable to Full Service Customers.
Interruptible Credit: \$(0.017518) per kWh for all kWh

Self-Generation ~~Provision~~ (SG)

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C 1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data/ billing determinants necessary for billing purposes.~~

(Continued on Sheet No. D-50.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-50.00

GENERAL SERVICE SECONDARY TIME-OF-USE RATE GSTU
(Continued From Sheet No. D-49.00)

Monthly Rate: (Contd)

~~Sales of Self-Generated Energy to the Company (Contd):~~

~~Administrative Cost Charge:~~

~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provision contained in Rule C 10.2, Green Generation Program.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate and any applicable non-consumption based surcharges. Special Minimum Charges shall be billed in accordance with Rule C15., Special Minimum Charges.

Due Date and Late Payment Charge:

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract:

Service under this rate shall not require a written contract except for: (i) resale service, (ii) service under the Green Generation Program, (iii) for Special Minimum Charges, (iv) service for lighting or where mobile home parks are involved, (v) service under the Educational Institution Service Provision, or (vi) at the option of the Company. If a contract is deemed necessary by the Company, the appropriate contract form shall be used and the contract shall require a minimum term of one year.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-51.00

GENERAL SERVICE SECONDARY DEMAND RATE GSD

Availability:

Subject to any restrictions, this rate is available to any customer desiring Secondary Voltage service, either for general use or resale purposes, where the Peak Demand is 5 kW or more. This rate is also available for service to any Primary Rate Customer where the Company elects to provide one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer.

This rate is not available for: (i) private family dwellings, (ii) lighting service, (iii) resale for lighting service, or (iv) new or expanded service for resale to residential customers.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Secondary Voltage service. The Company will determine the particular nature of the voltage in each case.

Three-phase, 3-wire service requires that the customer furnishes all transformation facilities required for single-phase load and so arranges the load as to avoid excessive unbalance of the three-phase load. When the service is single-phase, or 4-wire, three-phase, the single-phase individual motor capacity shall not exceed 3 hp, nor the total single-phase motor capacity of 10 hp, without the specific consent of the Company.

Where the Company elects to measure the service on the Primary side of the transformers, 3% shall be deducted for billing purposes from the demand and energy measurements thus made. Where the Company elected to provide a Primary Rate Customer one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer, 3% shall not be deducted for billing purposes from the energy measurements thus made.

Monthly Rate:

Power Supply Charges: These Charges are applicable to Full Service customers.

Peak Demand Charge:

Non-Capacity	Capacity	Total	
\$8.10	\$13.04	\$21.14	per kW for all kW of Peak Demand during the
<u>\$8.83</u>	<u>\$14.23</u>	<u>\$23.06</u>	billing months of June-September
\$6.10	\$11.04	\$17.14	per kW for all kW of Peak Demand during the
<u>\$6.67</u>	<u>\$12.69</u>	<u>\$19.36</u>	billing months of October-May

Energy Charge:

Non-Capacity	
\$0.043298	per kWh for all kWh during the billing months of June-September
<u>\$0.033511</u>	
\$0.040994	per kWh for all kWh during the billing months of October-May
<u>\$0.030961</u>	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factors shown on Sheet No. D-6.00.

Delivery Charges: These Charges are applicable to Full Service and Retail Open Access (ROA) customers.

System Access Charge:	\$30.00	per customer per month
Capacity Charge:	\$1.15	per kW for all kW of Peak Demand
	<u>\$1.34</u>	
Distribution Charge:	\$0.029722	per kWh for all kWh
	<u>\$0.034680</u>	

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-52.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-52.00

GENERAL SERVICE SECONDARY DEMAND RATE GSD

(Continued From Sheet No. D-51.00)

Monthly Rate: (Contd)

Adjustment for Power Factor:

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

<u>Power Factor</u>	<u>Penalty</u>
<u>0.800 to 0.849</u>	<u>0.50%</u>
<u>0.750 to 0.799</u>	<u>1.00%</u>
<u>0.700 to 0.749</u>	<u>2.00%</u>
<u>Below 0.700</u>	<u>3% first 2 months</u>

Adjustment for Power Factor shall not be applied when the Peak Billing Demand is based on 60% of the highest On-Peak Billing Demand created during the preceding bill months of June through September or on a Minimum Peak Billing Demand.

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

~~When the average Power Factor during the billing month is less than 80% lagging, the Company reserves the right to increase the capacity charges for such billing month in the ratio that 80% bears to such Power Factor. The Company may, at its option, determine the Power Factor by test or by permanently installed measuring equipment. The capacity charges shall not be increased when the Peak Demand is based on 60% of the highest Peak Demand created during the preceding billing months of June through September or on a minimum Peak Demand.~~

Peak Demand:

The Peak Demand shall be the Kilowatts (kW) supplied during the period of highest use in the billing month but not less than 60% of the highest Peak Demand created during the preceding billing months of June through September, nor less than 5 kW.

The Company reserves the right to make special determination of the Peak Demand and/or the Minimum Charge should the equipment which creates momentary high demands be included in the customer's installation.

When a customer guarantees a Peak Demand of 100 kW, the current month Peak Demand shall be the greatest of (1) the highest actual Peak Demand created during the on-peak hours in the current billing month, (2) 1/3 of the highest Peak Demand created during the off-peak hours in the current billing month, (3) 100 kW, or (4) 60% of the highest Peak Demand created during the previous billing months of June through September. For the purpose of applying the 60% provision, only the Peak Demands created after a customer guarantees 100 kW minimum shall be considered. On-peak and off-peak hours are contained in Rule C14., Provisions Governing the Application of On-Peak and Off-Peak Rates.

Resale Service Provision:

Subject to any restrictions, this provision is available to customers desiring Secondary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

(Continued on Sheet No. D-53.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-53.00

GENERAL SERVICE SECONDARY DEMAND RATE GSD
(Continued From Sheet No. D-52.00)

Monthly Rate: (Contd)

Educational Institution Service Provision (GEI):

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, “school” shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. “College” or “University” shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Education Institution Credit: \$ (0.000659 ~~0.000616~~) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

General Service Secondary Interruptible (GSI) Provision:

This provision is available to no more than 200 Full Service Customers desiring interruptible service in conjunction with service taken under General Service Secondary Demand Rate GSD or General Service Secondary Time-of-Use Rate GSTU. Service to interruptible load shall be taken through separately metered circuits and permanently wired. The design and method of installation for application of this rate shall be subject to the approval of the Company.

Any load designated as interruptible by the customer is subject to Midcontinent Independent System Operator’s, Inc. (MISO) requirements for Load Modifying Resources and the Company shall inform the Customer of such MISO requirements. Interruption under this provision may occur if MISO issues a Maximum Generation Emergency Event Step 2b order or NERC Emergency Event Alert 2 notice indicating that MISO is experiencing or expects to experience a shortage of economic resources and the Company has declared Emergency Status.

Under this provision, the customer shall be interrupted at any time the Company deems it necessary to maintain system integrity. Service to interruptible load shall not be transferred to firm service circuits to avoid interruption. The Company shall provide the Customer at least 30 minutes notice in advance of a required interruption. Failure to acknowledge receipt of such notice shall not relieve the Customer of the obligation for interruption under the GSI provision. Failure by a customer to comply with a system integrity interruption order of the Company shall be considered unauthorized use and billed at (i) the higher of the actual damages incurred by the Company or (ii) the rate of \$25.00 per kW for the highest 15-minute kW of demand created during the interruption period in addition to the prescribed monthly rate.

This rate is not available for loads that are primarily off-peak, for example parking lot lighting. Participation requires a minimum term of one year.

The monthly credit for the Interruptible Service Provision shall be applied as follows:

Power Supply Charges - These charges are applicable to Full Service Customers.

Capacity Credit: These charges are applicable to Full Service Customers.
Interruptible Credit: \$(0.017518) per kWh for all kWh

Self-Generation ~~Provision~~ (SG):

~~Subject to any restrictions, as of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities operated in parallel with the Company's system must meet the Parallel Operation Requirements set forth in Rule C1.6 B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~There shall be no double billing of demand under the base rate and the Self-Generation Provision.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~**Administrative Cost Charge:** \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less~~

(Continued on Sheet No. D-54.00)

See Barnes Testimony, Page 4, Lines 3-18 and Page 9, Lines 2-4; McLean Testimony, Page 70, Lines 19-21 and Page 71, Lines 1-14; Miller Testimony, Page 19, Lines 1-10; Exhibit A-23 (RLB-1) Items #9, 12 and 21; Exhibit A-16 (HWM-3), Page 7

M.P.S.C. No. 14 – Electric
Consumers Energy Company

First Revised Sheet No. D-54.00

GENERAL SERVICE SECONDARY DEMAND RATE GSD
(Continued From Sheet No. D-24.00)

Self-Generation Provision (SG) (Contd)

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C 11.2, Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.2.B, Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C 11.2, Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3, Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B, Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3, Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge:

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract:

Service under this rate shall not require a written contract except for: (i) resale service, (ii) service under the Green Generation Program, (iii) service under the Educational Institution Service Provision, (iv) service under the Net Metering program, or (v) at the option of the Company. If a contract is deemed necessary by the Company, the appropriate contract form shall be used and the contract shall require a minimum term of one year.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-55.00

GENERAL SERVICE PRIMARY RATE GP

Availability:

As of January 1, 2021, this rate is closed to new business. Subject to any restrictions, this rate is available to any customer, political subdivision or agency of the State of Michigan, either acting separately or in combinations permitted under the laws of this state, desiring Primary Voltage service for general use or for public potable water pumping and/or waste water system(s).

This rate is available to existing Full Service Customers with an electric generating facility interconnected at a primary voltage level utilizing General Service Primary Rate GP for standby service on or before June 7, 2012. The amount of retail usage shall be determined on an hourly basis. Customers with a generating installation are required to have an Interval Data Meter.

This rate is not available to a Primary Rate Customer where the Company elects to provide one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer.

This rate is not available for lighting service, except for temporary service for lighting installations.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

Where service is supplied at a nominal voltage of 25,000 Volts or less, the customer shall furnish, install and maintain all necessary transforming, controlling and protective equipment.

Where the Company elects to measure the service at a nominal voltage above 25,000 Volts, 1% shall be deducted for billing purposes, from the energy measurements thus made.

Where the Company elects to measure the service at a nominal voltage of less than 2,400 Volts, 3% shall be added for billing purposes, to the energy measurements thus made.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service customers.

Charges for Customer Voltage Level 3 (CVL3)

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.049961</u>	<u>\$0.035443</u>	<u>\$0.085404</u>	per kWh for all kWh during the billing months of June-September
0.058898	0.047432	0.106330	
<u>\$0.047823</u>	<u>\$0.036847</u>	<u>\$0.084670</u>	per kWh for all kWh during the billing months of October-May
0.058404	0.047054	0.105458	

Charges for Customer Voltage Level 2 (CVL2)

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.048939</u>	<u>\$0.034615</u>	<u>\$0.083554</u>	per kWh for all kWh during the billing months of June-September
0.053957	0.042491	0.096448	
<u>\$0.046840</u>	<u>\$0.035987</u>	<u>\$0.082827</u>	per kWh for all kWh during the billing months of October-May
0.053463	0.042113	0.095576	

Charges for Customer Voltage Level 1 (CVL1)

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.048294</u>	<u>\$0.034070</u>	<u>\$0.082364</u>	per kWh for all kWh during the billing months of June-September
0.048787	0.037321	0.086108	
<u>\$0.046218</u>	<u>\$0.035420</u>	<u>\$0.081638</u>	per kWh for all kWh during the billing months of October-May
0.048293	0.036943	0.085236	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

(Continued on Sheet No. D-56.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-56.00

GENERAL SERVICE PRIMARY RATE GP
(Continued From Sheet No. D-55.00)

Monthly Rate (Contd)

Delivery Charges: These charges are applicable to Full Service and Retail Open Access (ROA) customers.

System Access Charge: \$100.00 per customer per month

Charges for Customer Voltage Level 3 (CVL3)

\$0.016643

Distribution Charge: ~~0.013386~~ per kWh for all kWh

Charges for Customer Voltage Level 2 (CVL2)

\$0.010934

Distribution Charge: ~~0.007723~~ per kWh for all kWh

Charges for Customer Voltage Level 1 (CVL1)

\$0.006421

Distribution Charge: ~~0.005733~~ per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Adjustment for Power Factor

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

Power Factor	Penalty
0.800 to 0.849	0.50%
0.750 to 0.799	1.00%
0.700 to 0.749	2.00%
Below 0.700	3% first 2 months

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

Resale Service Provision

Subject to any restrictions, this provision is available to customers desiring Primary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

(Continued on Sheet No. D-57.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-57.00

GENERAL SERVICE PRIMARY RATE GP
(Continued From Sheet No. D-27.10)

Monthly Rate (Contd)

Substation Ownership Credit

Where service is supplied at a nominal voltage of more than 25,000 volts, and the customer provides all of the necessary transforming, controlling and protective equipment for all of the service there shall be deducted from the bill a monthly credit.

The monthly credit for the substation ownership shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service and Retail Open Access customers.

Charges for Customer Voltage Level 2 (CVL 2)

Substation Ownership Credit: \$ (0.002180) per kWh for all kWh

Charges for Customer Voltage Level 1 (CVL 1)

Substation Ownership Credit: \$ (0.000767 0.000287) per kWh for all kWh

For those customers served by more than one substation where one or more of the substations is owned by the customer, the credit will be applied to the customer's coincident Maximum Demand for those substations owned by the customer. This credit shall not operate to reduce the customer's billing below the prescribed minimum charges included in the rate. The credit shall be based on the kW after the 1% deduction or 3% addition has been applied to the metered kWh.

Educational Institution Service Provision (GEI)

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, "school" shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. "College" or "University" shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service and Retail Open Access Customers.

Educational Institution Credit: \$(0.000516 0.000571) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

Self-Generation ~~Provision~~ (SG):

~~As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities operated in parallel with the Company's system must meet the Parallel Operation Requirements set forth in Rule C1.6B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

(Continued on Sheet No. D-58.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-58.00

GENERAL SERVICE PRIMARY RATE GP (Continued From Sheet No. D-57.00)

Monthly Rate (Contd)

~~Self-Generation Provision (SG) (Contd):~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge: \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C 11.2., Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C 11.2., Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

General Terms:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access charge included in the rate and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract

For customers with monthly demands of 300 kW or more, all service under this rate shall require a written contract with a minimum term of one year.

For customers with monthly demands of less than 300 kW, service under this rate shall not require a written contract except for: (i) service under the Green Generation Program, (ii) service under the Educational Institution provision, (iii) service under the Resale Service Provision, (iv) service under the Net Metering Program, or (v) at the option of the

Company. If a contract is deemed necessary by the Company, the appropriate contract form shall be used and the contract shall require a minimum term of one year.

A new contract will not be required for existing customers who increase their demand requirements after initiating service, unless new or additional facilities are required or service provisions deem it necessary.

See Barnes Testimony, Page 4, Lines 3-18; Exhibit A-23 (RLB-1) Item #9

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-59.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

Availability

Subject to any restrictions, this rate is available to any customer desiring Primary Voltage service, either for general use or resale purposes, where the On-Peak Billing Demand is 25 kW or more. This rate is also available to any political subdivision or agency of the State of Michigan, either acting separately or in combinations permitted under the laws of this state, for Primary Voltage service for potable water pumping and/or waste water system(s).

This rate is not available to a Primary Rate Customer where the Company elects to provide one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer.

This rate is also not available for lighting service, for resale for lighting service, or for new or expanded service for resale to residential customers.

Nature of Service

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

Where service is supplied at a nominal voltage of 25,000 Volts or less, the customer shall furnish, install and maintain all necessary transforming, controlling and protective equipment.

Where the Company elects to measure the service at a nominal voltage above 25,000 Volts, 1% shall be deducted for billing purposes, from the demand and energy measurements thus made.

Where the Company elects to measure the service at a nominal voltage of less than 2,400 Volts, 3% shall be added for billing purposes, to the demand and energy measurements thus made.

Interval Data Meters are required for service under this rate. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.

Monthly Rate:

Power Supply Charges: These charges are applicable to Full Service customers.

Charges for Customer Voltage Level 3 (CVL3)

Demand Charge:

Capacity	Non-Capacity	Total	
\$14.85	\$6.53	\$21.38	per kW of On-Peak Billing Demand during the billing months of June-September
14.82	10.34	25.16	
\$13.80	\$5.50	\$19.30	per kW of On-Peak Billing Demand during the billing months of October-May
13.82	9.34	23.16	

Transmission Charge:

Capacity	
\$7.31	per kW of On-Peak Billing Demand during the billing months of June-September
6.98	
\$6.81	per kW of On-Peak Billing Demand during the billing months of October-May
6.98	

Energy Charge:

Non-Capacity	
\$0.028533	per kWh for all On-Peak kWh during the billing months of June-September
0.042156	
\$0.018376	per kWh for all Off-Peak kWh during the billing months of June-September
0.027322	
\$0.023368	per kWh for all On-Peak kWh during the billing months of October-May
0.034413	
\$0.021730	per kWh for all Off-Peak kWh during the billing months of October-May
0.030139	

(Continued on Sheet No. D-60.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-60.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-59.00)

Monthly Rate: (Contd)

Power Supply Charges: These charges are applicable to Full Service customers. (Contd)

Charges for Customer Voltage Level 2 (CVL2)

Demand Charge:

Capacity	Non-Capacity	Total	
\$14.50	\$6.41	\$20.91	per kW of On-Peak Billing Demand during the billing months of June-September
14.32	9.84	24.16	
\$13.48	\$5.40	\$18.88	per kW of On-Peak Billing Demand during the billing months of October-May
13.32	8.84	22.16	

Transmission Charge:

Capacity	
\$7.14	per kW of On-Peak Billing Demand during the billing months of June-September
6.71	
\$6.65	per kW of On-Peak Billing Demand during the billing months of October-May
6.71	

Energy Charge:

Non-Capacity	
\$0.027983	per kWh for all On-Peak kWh during the billing months of June-September
0.034239	
\$0.018022	per kWh for all Off-Peak kWh during the billing months of June-September
0.022191	
\$0.022917	per kWh for all On-Peak kWh during the billing months of October-May
0.027950	
\$0.021311	per kWh for all Off-Peak kWh during the billing months of October-May
0.024479	

Charges for Customer Voltage Level 1 (CVL1)

Demand Charge:

Capacity	Non-Capacity	Total	
\$14.27	\$6.33	\$20.60	per kW of On-Peak Billing Demand during the billing months of June-September
13.82	9.34	23.16	
\$13.27	\$5.33	\$18.60	per kW of On-Peak Billing Demand during the billing months of October-May
12.82	8.34	21.16	

Transmission Charge:

Capacity	
\$7.03	per kW of On-Peak Billing Demand during the billing months of June-September
6.58	
\$6.55	per kW of On-Peak Billing Demand during the billing months of October-May
6.58	

Energy Charge:

Non-Capacity	
\$0.027643	per kWh for all On-Peak kWh during the billing months of June-September
0.026510	
\$0.017803	per kWh for all Off-Peak kWh during the billing months of June-September
0.017182	
\$0.022639	per kWh for all On-Peak kWh during the billing months of October-May
0.021641	
\$0.021052	per kWh for all Off-Peak kWh during the billing months of October-May
0.018953	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

(Continued on Sheet No. D-61.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-61.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-60.00)

Monthly Rate: (Contd)

Delivery Charges: These charges are applicable to Full Service and Retail Open Access (ROA) customers.

System Access Charge: \$200.00 per customer per month

Charges for Customer Voltage Level 3 (CVL3)

Capacity Charge: ~~3.60~~ ^{\$4.58} per kW of Maximum Demand

Charges for Customer Voltage Level 2 (CVL2)

Capacity Charge: ~~1.86~~ ^{\$2.69} per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL1)

Capacity Charge: ~~0.91~~ ^{\$0.70} per kW of Maximum Demand

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Adjustment for Power Factor:

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

Power Factor	Penalty
0.800 to 0.849	0.50%
0.750 to 0.799	1.00%
0.700 to 0.749	2.00%
Below 0.700	3% first 2 months

Adjustment for Power Factor shall not be applied when the On-Peak Billing Demand is based on 60% of the highest On-Peak Billing Demand created during the preceding bill months of June through September or on a Minimum On-Peak Billing Demand.

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

(Continued on Sheet No. D-62.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-62.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-31.10)

Monthly Rate: (Contd)

Maximum Demand:

The Maximum Demand shall be the highest 15-minute demand created during the current month or previous 11 months.

On-Peak Billing Demand:

The On-Peak Billing Demand shall be based on the highest on-peak demand created during the current billing month, but never less than 60% of the highest on-peak billing demand of the four preceding summer billing months ~~of~~ (June through September), nor less than 25 kW.

The On-Peak Billing Demand shall be the Kilowatts (kW) supplied during the 15-minute period of maximum use during on-peak hours, as described in Rule C14., Provisions Governing the Application of On-Peak and Off-Peak Rates.

The Company reserves the right to make special determination of the On-Peak Billing Demand, and/or the Minimum Charge, should the equipment which creates momentary high demands be included in the customer's installation.

Transmission On-Peak Billing Demand:

The Transmission On-Peak Billing Demand for each billing month shall be the Kilowatts (kW) supplied during the 15-minute period of maximum use during on-peak hours, as described in Rule C14., Provisions Governing the Application of On-Peak and Off-Peak Rates.

Resale Service Provision:

Subject to any restrictions, this provision is available to customers desiring Primary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

Substation Ownership Credit:

Where service is supplied at a nominal voltage of more than 25,000 Volts, energy is measured through an Interval Data Meter, and the customer provides all of the necessary transforming, controlling and protective equipment for all of the service there shall be deducted from the bill a monthly credit. For those customers, part of whose load is served through customer-owned equipment, the credit shall be based on the Maximum Demand.

The monthly credit for the substation ownership shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Charges for Customer Voltage Level 2 (CVL 2)

Substation Ownership Credit: \$(0.96 ~~0.97~~) per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL 1)

Substation Ownership Credit: \$(0.34 ~~0.45~~) per kW of Maximum Demand

For those customers served by more than one substation where one or more of the substations is owned by the customer, the credit will be applied to the customer's coincident Maximum Demand for those substations owned by the customer. This credit shall not operate to reduce the customer's billing below the prescribed minimum charges included in the rate. The credit shall be based on the kW after the 1% deduction or 3% addition has been applied to the metered kW.

(Continued on Sheet No. D-63.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-63.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-62.00)

Monthly Rate: (Contd)

Aggregate Peak Demand Service Provision (GAP):

This provision is available to any customer with 7 accounts or more who desire to aggregate their On-Peak Billing Demands for power supply billing purposes. To be eligible, each account must have a minimum average On-Peak Billing Demand of 250 kW and be located within the same billing district. The customer's aggregated accounts shall be billed under the same rate schedule and service provisions. The aggregate maximum capacity of all customers served under this provision shall be limited to 200,000 kW.

This provision commences with service rendered on and after June 20, 2008 and remains in effect until terminated by a Commission Order.

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

Interval Data Meters are required for service under this provision.

The aggregated accounts shall be summarized for each interval time period registered and a comparison shall be performed to determine the on-peak time at which the summarized value of the aggregated accounts reached a maximum for the billing month. The individual aggregated accounts shall be billed for their corresponding On-Peak Billing Demand occurring at that point in time.

Educational Institution Service Provision (GEI):

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, "school" shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. "College" or "University" shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges: These charges are applicable to Full Service and Retail Open Access Customers.

Educational Institution Credit: \$(0.000259 ~~0.000314~~) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

(Continued on Sheet No. D-64.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-64.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-63.00)

Monthly Rate: (Contd)

Self-Generation ~~Provision~~ (SG):

~~Subject to any restrictions, as of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities operated in parallel with the Company's system must meet the Parallel Operation Requirements set forth in Rule C1.6B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~There shall be no double billing of demand under the base rate and the Self-Generation Provision.~~

~~Sales of Self-Generated Energy to the Company:~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge: \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase:~~

~~An energy purchase by the Company shall be bought at the Midecontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Interruptible Service Provision (GI):

This provision is available to any customer account willing to contract for at least 500 kW of On-Peak Billing Demand as interruptible. The Company reserves the right to limit the amount of load contracted as interruptible, but in no case shall it exceed ~~100,000~~ 300,000 kW per customer. Customers served under Rate GPD shall have no more than 50% of their annual On-Peak Billing Demand contracted as interruptible when contracting for more than 50,000 kW of interruptible load. The aggregate amount of monthly On-Peak Billing Demand subscribed under this provision shall be limited to ~~300,000~~ 400,000 kW.

Consumers Energy may require the Customer to monitor and provide real-time, Internet-enabled power monitoring. If such monitoring is required, Consumers Energy will provide the metering or monitoring devices necessary, which shall be owned by Consumers Energy and provided to the Customer at the Company's expense. The Customer may be required to provide suitable space for such monitoring equipment and either a static or non-static, as applicable, Internet Protocol (IP) address and Local Area Network (LAN) access that allows for Internet-based communication of the Customer's site electricity consumption and interruption event performance.

(Continued on Sheet No. D-65.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-65.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-64.00)

Monthly Rate: (Contd)

Interruptible Service Provision (GI): (Contd)

For billing purposes, the monthly interruptible On-Peak Billing Demand shall be billed first and discounted under this interruptible service provision. The actual On-Peak Billing Demand for the interruptible load supplied shall be credited by the amount specified under the Power Supply Charges - Interruptible Credit listed below. Subsequently all firm service used during the billing period in excess of the contracted interruptible shall be billed at the appropriate firm rate. All contracts under this provision shall be negotiated on an annual basis. ~~The Customer must notify the Company by December 31st of each year of their desire to renew the GI provision and the amount of interruptible kW for the following capacity planning year (June 1 through May 31)~~ and the Customer must notify the Company by December 10th of each year of their desire to renew the GI provision, unless the Customer chooses to lengthen the term of their commitment (up to five years). Annual changes to the amount of interruptible kW for long term contracts are open to adjustment through December 10th of each year. Within 30 minutes of receiving an interruption notice, the customer shall reduce their total load level by the amount of contracted interruptible capacity.

The minimum On-Peak Billing Demand that shall be billed for the interruptible portion of a customer's bill is the contracted interruptible amount. At the Company's discretion, the customer may reduce the contracted amount one time within the annual contract period.

Any load designated as interruptible by the customer is also subject to Midcontinent Independent System Operator's Inc. (MISO) requirements for Load Modifying Resources and the Company shall inform the Customer of such MISO requirements. Interruption under this provision may occur if MISO issues a Maximum Generation Emergency Event Step 2b order or NERC Emergency Event Alert 2 notice indicating that MISO is experiencing or expects to experience a shortage of economic resources and the Company has declared Emergency Status. Participation in the GI provision does not limit the Company's ability to implement emergency electrical procedures as described in the Company's Electric Rate Book including interruption of service as required to maintain system integrity.

Conditions of Interruption

Under this provision, the customer shall be interrupted at any time, on-peak or off-peak, the Company deems it necessary to maintain system integrity. The Company shall provide the Customer at least thirty minutes advance notice of a required interruption, and if possible, a second notice. The notice will be communicated by telephone to the contact numbers provided by the Customer. The Customer shall confirm the receipt of such notice through the automated response process. Failure to acknowledge receipt of such notice shall not relieve the customer of the obligation for interruption under the GI Provision. The customer shall be informed, when possible, of the estimated duration of the interruption at the time of interruption.

The Company shall not be liable for any loss or damage caused by or resulting from any interruption of service under this provision.

Interruptions beyond the Company's control, described in Rules C1.1, Character of Service, and C3., Emergency Electrical Procedures, of the Company's Electric Rate Book, shall not be considered as interruptions for purposes of this provision.

Should the Company be ordered by Governmental authority during a national emergency to supply firm instead of interruptible service, billing shall be made on an applicable firm power schedule.

Cost of Customer Non-Interruption

Failure by a customer to comply with a system integrity interruption order of the Company shall be considered as unauthorized use and billed at (i) the higher of the actual damages incurred by the Company or (ii) the rate of \$25.00 per kW for the highest 15-minute kW of Interruptible On-Peak Billing demand created during the interruption period, in addition to the prescribed monthly rate. In addition, the interruptible contract capacity of a customer who does not interrupt within one hour following notice shall be immediately reduced by the amount which the customer failed to interrupt, unless the customer demonstrates that failure to interrupt was beyond its control.

The monthly credit for the Interruptible Service Provision shall be applied as follows:

Power Supply Charges: These charges are applicable to Full Service Customers.

Interruptible Credit:	\$(7.00)	per kW of On-Peak Billing Demand during the billing months of June-September
	\$(6.00)	per kW of On-Peak Billing Demand during the billing months of October-May

(Continued on Sheet No. D-66.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-66.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-65.00)

Interruptible Service Provision – Market-Price Option (GI2)

Availability:

This provision is available to any Full Service GPD customer account willing to ~~contract for~~ designate at least 3,000 kW of On-Peak Billing Demand as ~~interruptible~~ Defined Interruptible Capacity. The Company reserves the right to limit the amount of designated interruptible load available to any single customer ~~contracted as interruptible~~, but in no case shall it exceed 100,000 kW. The combined aggregate amount of monthly On-Peak Billing Demand subscribed under the GI and GI2 provisions shall be limited to 400,000 kW.

In the event the combined aggregate amount of monthly On-Peak Demand subscribed is less than the approved limit specified above, the Company may offer the remaining capacity, to otherwise eligible customers willing to ~~contract for~~ designate less than the minimum ~~contract capacity~~ amounts specified above.

The customer may choose to have the interruptible load separately metered. The customer shall bear any expense incurred by the Company in providing a separate service for the interruptible portion of an existing customer load. The customer must provide space suitable for the separate metering. Consumers Energy may require the Customer to monitor and provide real-time, Internet-enabled power monitoring. If such monitoring is required, Consumers Energy will provide the metering or monitoring devices necessary, which shall be owned by Consumers Energy and provided to the Customer at the Company's expense. The Customer may be required to provide suitable space for such monitoring equipment and either a static or non-static, as applicable, Internet Protocol (IP) address and Local Area Network (LAN) access that allows for Internet-based communication of the Customer's site electricity consumption and interruption event performance.

Contracted Firm Capacity and Defined Interruptible Capacity

Defined Interruptible Capacity shall be the amount of the customer's On-Peak Billing Demand at the time of the most recent annual MISO peak hour that exceeds the Customer's Firm Contract Capacity.

The minimum difference between the Customer's Contracted Firm Capacity and the Customer's On-Peak Billing Demand required to participate in the GI2 Provision is 3,000 kW and is subject to Company verification.

Customers shall contract for a specified capacity in kilowatts sufficient to meet the customers' maximum interruptible requirements, but not less than the minimum contract capacity amounts, specified above. The contract capacity shall not be decreased during the term of the contract and subsequent renewal periods as long as service is required unless there is a verified reduction in connected load. Capacity disconnected from service under this provision shall not be subsequently served under any other tariff during the term of this contract and subsequent renewal periods. The Customer must notify and contract with the Company by December 10th ~~31st~~ of each year of their desire to renew the GI2 provision and the amount of interruptible kW for the following capacity planning year (June 1 through May 31).

Monthly Billing

For billing purposes, the ~~monthly firm service~~ Contracted Firm Capacity will be billed first on Rate GPD, with the load in excess of contracted firm being billed on the GI2 charges specified in this rate schedule.

Power Supply Charges - These charges are applicable to contracted interruptible capacity.

The customer shall be responsible for the MISO Real-Time Locational Market Price (LMP) for the Company's load node (designated as "CONS.CETR" as the date of this Rate Schedule), multiplied by the customer's consumption (kWh), plus the Market Settlement Fee of \$0.002/kWh.

Charges for Customer Voltage Level 3 (CVL 3)

LMP Energy Charge:

MISO Real-Time LMP per kWh for all kWh

Capacity & Transmission Charge:

\$0.030487 per kWh for all kWh during the billing months of June-September

~~0.049807~~

\$0.030403 per kWh for all kWh during the billing months of October-May

~~0.046476~~

Charges for Customer Voltage Level 2 (CVL 2)

LMP Energy Charge: MISO Real-Time LMP per kWh for all kWh

Capacity & Transmission Charge: \$~~0.039453~~ 0.026704 per kWh for all kWh during the billing months of June-September

~~0.036122~~ 0.025668 per kWh for all kWh during the billing months of October-May

Charges for Customer Voltage Level 1 (CVL 1)

LMP Energy Charge: MISO Real-Time LMP per kWh for all kWh

Capacity & Transmission Charge: \$~~0.028890~~ 0.024857 per kWh for all kWh during the billing months of June-September

~~0.025559~~ 0.023762 per kWh for all kWh during the billing months of October-May

(Continued on Sheet No. D-67.00)

See Barnes Testimony, Page 2, Lines 20-22 and Page 9, Lines 4-7; Exhibit A-23 (RLB-1) Item #12 and 23; Exhibit A-16 (HWM-3), Pages

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-67.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD
(Continued From Sheet No. D-66.00)

Interruptible Service Provision – Market-Price Option (GI2) (Cont)

The MISO Real-Time LMP per kWh shall be adjusted for losses based on the customer's point of metering as shown below:

	Meter Point	
	High Side	Low Side
Customer Voltage Level 1	0.000%	<u>0.728</u> 0.705 %
Customer Voltage Level 2	<u>1.325</u> 1.274 %	<u>2.189</u> 2.366 %
Customer Voltage Level 3	<u>3.329</u> 3.221 %	<u>8.082</u> 7.643 %

Delivery Charges – These charges are applicable to contract capacity

Rate GPD Delivery Charges will apply to all Delivery service, including contracted capacity designated as GI2 interruptible service.

System Access Charge:

If contracted capacity is separately metered: \$100.00 per additional meter installation per month

This provision is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00. as well as the System Access Charge, Delivery Charges, General Terms, Adjustment for Power Factor, Substation Ownership Credit, Minimum Charge and the Due Date and Late Payment Charge applicable to Rate GPD.

Conditions of Interruption

The Company will notify the customer as to the amount of total load on this rider to be curtailed. Load identified as monthly firm service and billed on Rate GPD is not considered as interruptible and does not need to be curtailed under the terms of GI2. Although actual load at time of interruption may vary from contract capacity, the total measured load on this provision shall be subject to curtailment by the Company.

The Company shall provide the Customer at least thirty minutes advance notice of a required interruption, and if possible, a second notice. The notice will be communicated by telephone to the contact numbers provided by the Customer. The Customer shall confirm the receipt of such notice through the automated response process. Failure to acknowledge receipt of such notice shall not relieve the customer of the obligation for interruption under the GI Provision. The customer shall be informed, when possible, of the estimated duration of the interruption at the time of interruption. Within 30 minutes of receiving an interruption notice, the customer shall reduce their total load level by the amount of contracted interruptible capacity or have the total facility subject to interruption.

Any load designated as interruptible by the customer may require the installation and maintenance of equipment that allow the Company to remotely interrupt the customer's load. If the company determines it is required to install and maintain equipment at the customer's site to comply with any requirements associated with the GI service provision then it shall do so at the customer's expense. In addition, the customer shall also adhere to any advance notification requirements the Company deems are necessary to comply with its obligations to MISO under this provision.

Any load designated as interruptible by the customer is also subject to Midcontinent Independent System Operator's Inc. (MISO) requirements for Load Modifying Resources and the Company shall inform the Customer of such MISO requirements. Interruption under this provision may occur if MISO issues a Maximum Generation Emergency Event Step 2b order or NERC Emergency Event Alert 2 notice indicating that MISO is experiencing or expects to experience a shortage of economic resources and the Company has declared Emergency Status. Participation in the GI provision does not limit the Company's ability to implement emergency electrical procedures as described in the Company's Electric Rate Book including interruption of service as required to maintain system integrity.

(Continued on Sheet No. D-68.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-68.00

LARGE GENERAL SERVICE PRIMARY DEMAND RATE GPD

(Continued From Sheet No. D-67.00)

Interruptible Service Provision – Market-Price Option (GI2) (Cont)

Conditions of Interruption (Cont)

Under this provision, the customer shall be interrupted at any time, on-peak or off-peak, the Company deems it necessary to maintain system integrity. The Company shall provide notice in advance of probable interruption, and if possible, a second notice of positive interruption. The notice will be communicated by telephone to the contact numbers provided by the Customer. The Customer shall confirm the receipt of such notice through the automated response process. Failure to acknowledge receipt of such notice shall not relieve the Customer of the obligation for interruption under the GI2 provision. The customer shall be informed, when possible, of the estimated duration of the interruption at the time of interruption.

The Company shall not be liable for any loss or damage caused by or resulting from any interruption of service under this provision.

Interruptions beyond the Company's control, described in Rules C1.1, Character of Service, and C3., Emergency Electrical Procedures, of the Company's Electric Rate Book, shall not be considered as interruptions for purposes of this provision.

Should the Company be ordered by Governmental authority during a national emergency to supply firm instead of interruptible service, billing shall be made on an applicable firm power schedule.

Cost of Customer Non-Interruption

Failure by a customer to comply with a system integrity interruption order of the Company shall be considered as unauthorized use and billed at (i) the higher of the actual damages incurred by the Company or (ii) the rate of \$25.00 per kW for the highest 15-minute kW of Interruptible On-Peak Billing demand created during the interruption period, in addition to the prescribed monthly rate. In addition, the interruptible contract capacity of a customer who does not interrupt within one hour following notice shall be immediately reduced by the amount which the customer failed to interrupt, unless the customer demonstrates that failure to interrupt was beyond its control.

Net Metering Program:

The Net Metering Program is available to any eligible customer as described in Rule C11.2, Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C11.2.B., Net Metering Definitions.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C11.2, Net Metering Program.

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

(Continued on Sheet No. D-69.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Original Sheet No. D-70.00

GENERAL SERVICE PRIMARY TIME-OF-USE RATE GPTU

Availability:

Subject to any restrictions, this General Service Primary Time-Of-Use (GPTU) Rate is available to any Full Service Customer taking service at the Company's Primary Voltage level. Standby service shall be provided on this rate for primary customers with solar installations equal to or greater than 150 kW.

This rate is not available for Standby service with generators that exceed 550 kW, except for solar installations, nor available for lighting service, except for temporary service for lighting installations.

Nature of Service:

Service under the rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

Where service is supplied at a normal voltage of 25,000 Volts or less, the customer shall furnish, install and maintain all necessary transforming, controlling, and protective equipment.

Where the Company elects to measure the service at a nominal voltage above 25,000 Volts, 1% shall be deducted for billing purposes, from the demand and energy measurements thus made.

Where the Company elects to measure the service at a nominal voltage of less than 2,400 Volts, 3% shall be added for billing purposes, to the demand and energy measurements thus made.

Interval Data Meters are required for service under this rate. Meter reading will be accomplished electronically through telecommunication links or other electronic measuring equipment available to provide the Company with the metering data necessary for billing purposes.

Schedule of Hours:

The following schedule shall apply Monday through Friday (except holidays designated by the Company):

Summer:

Off-Peak Hours:	12:00 AM to 6:00 AM and 11:00 PM to 12:00 AM
Low-Peak Hours:	6:00 AM to 12:00 PM and 7:00 PM to 11:00 PM
Mid-Peak Hours:	12:00 PM to 2:00 PM and 5:00 PM to 7:00 PM
High-Peak Hours:	2:00 PM to 5:00 PM

Winter:

Off-Peak Hours:	12:00 AM to 2:00 PM and 9:00 PM to 12:00 AM
Mid-Peak Hours:	2:00 PM to 4:00 PM and 7:00 PM to 9:00 PM
High-Peak Hours:	4:00 PM to 7:00 PM

Weekends and holidays are off-peak. Designated Company holidays are: New Year's Day - January 1; Memorial Day - Last Monday in May; Independence Day - July 4; Labor Day - First Monday in September; Thanksgiving Day - Fourth Thursday in November; and Christmas Day - December 25. Whenever January 1, July 4 or December 25 fall on a Sunday, extended holiday periods such as Monday, January 2, Monday, July 5 and Monday, December 26 shall not be considered as holidays for application of off-peak hours.

(Continued on Sheet No. D-71.00)

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-71.00

GENERAL SERVICE PRIMARY TIME-OF-USE RATE GPTU

(Continued from Sheet No. D-70.00)

Monthly Rate:

Power Supply Charges:

Charges for Customer Voltage Level 3 (CVL3)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	<u>\$0.038396</u>	<u>\$0.028667</u>	<u>\$0.067063</u>	per kWh during the calendar months of June-September
	\$0.056286	\$0.020681	\$0.076967	
Low-Peak-Summer	<u>\$0.054047</u>	<u>\$0.042439</u>	<u>\$0.096486</u>	per kWh during the calendar months of June-September
	\$0.072924	\$0.025299	\$0.098223	
Mid-Peak-Summer	<u>\$0.068738</u>	<u>\$0.052842</u>	<u>\$0.121580</u>	per kWh during the calendar months of June-September
	\$0.088694	\$0.029676	\$0.118370	
High-Peak-Summer	<u>\$0.076646</u>	<u>\$0.055354</u>	<u>\$0.132000</u>	per kWh during the calendar months of June-September
	\$0.100060	\$0.032831	\$0.132891	
Off-Peak-Winter	<u>\$0.047317</u>	<u>\$0.025743</u>	<u>\$0.073060</u>	per kWh during the calendar months of October-May
	\$0.056918	\$0.020856	\$0.077774	
Mid-Peak-Winter	<u>\$0.051922</u>	<u>\$0.029911</u>	<u>\$0.081833</u>	per kWh during the calendar months of October-May
	\$0.063823	\$0.022773	\$0.086596	
High-Peak-Winter	<u>\$0.054162</u>	<u>\$0.029924</u>	<u>\$0.084086</u>	per kWh during the calendar months of October-May
	\$0.065501	\$0.023239	\$0.088740	

Charges for Customer Voltage Level 2 (CVL2)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	<u>\$0.037603</u>	<u>\$0.027997</u>	<u>\$0.065600</u>	per kWh during the calendar months of June-September
	\$0.051286	\$0.015681	\$0.066967	
Low-Peak-Summer	<u>\$0.052928</u>	<u>\$0.041448</u>	<u>\$0.094376</u>	per kWh during the calendar months of June-September
	\$0.067924	\$0.020299	\$0.088223	
Mid-Peak-Summer	<u>\$0.067317</u>	<u>\$0.051608</u>	<u>\$0.118925</u>	per kWh during the calendar months of June-September
	\$0.083694	\$0.024676	\$0.108370	
High-Peak-Summer	<u>\$0.075069</u>	<u>\$0.054061</u>	<u>\$0.129130</u>	per kWh during the calendar months of June-September
	\$0.095060	\$0.027831	\$0.122891	
Off-Peak - Winter	<u>\$0.046357</u>	<u>\$0.025142</u>	<u>\$0.071499</u>	per kWh during the calendar months of October-May
	\$0.054918	\$0.015856	\$0.067774	
Mid-Peak - Winter	<u>\$0.050867</u>	<u>\$0.029212</u>	<u>\$0.080079</u>	per kWh during the calendar months of October-May
	\$0.058823	\$0.017773	\$0.076596	
High-Peak - Winter	<u>\$0.053064</u>	<u>\$0.029225</u>	<u>\$0.082289</u>	per kWh during the calendar months of October-May
	\$0.060501	\$0.018239	\$0.078740	

Charges for Customer Voltage Level 1 (CVL1)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	<u>\$0.037102</u>	<u>\$0.027556</u>	<u>\$0.064658</u>	per kWh during the calendar months of June-September
	\$0.049286	\$0.013681	\$0.062967	
Low-Peak-Summer	<u>\$0.052218</u>	<u>\$0.040795</u>	<u>\$0.093013</u>	per kWh during the calendar months of June-September
	\$0.065924	\$0.018299	\$0.084223	
Mid-Peak-Summer	<u>\$0.066416</u>	<u>\$0.050795</u>	<u>\$0.117211</u>	per kWh during the calendar months of June-September
	\$0.081694	\$0.022676	\$0.104370	
High-Peak-Summer	<u>\$0.074070</u>	<u>\$0.053210</u>	<u>\$0.127280</u>	per kWh during the calendar months of June-September
	\$0.093060	\$0.025831	\$0.118891	
Off-Peak - Winter	<u>\$0.045754</u>	<u>\$0.024746</u>	<u>\$0.070500</u>	per kWh during the calendar months of October-May
	\$0.049918	\$0.013856	\$0.063774	
Mid-Peak - Winter	<u>\$0.050202</u>	<u>\$0.028752</u>	<u>\$0.078954</u>	per kWh during the calendar months of October-May
	\$0.056823	\$0.015773	\$0.072596	
High-Peak - Winter	<u>\$0.052372</u>	<u>\$0.028765</u>	<u>\$0.081137</u>	per kWh during the calendar months of October-May
	\$0.058501	\$0.016239	\$0.074740	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges:

System Access Charge:	\$200.00	per customer per month
<u>Charges for Customer Voltage Level 3 (CVL3)</u>		
Capacity Charge:	\$3.60 <u>\$4.58</u>	per kW of Maximum Demand
<u>Charges for Customer Voltage Level 2 (CVL2)</u>		
Capacity Charge:	\$1.86 <u>\$2.69</u>	per kW of Maximum Demand
<u>Charges for Customer Voltage Level 1 (CVL1)</u>		
Capacity Charge:	\$0.94 <u>\$0.70</u>	per kW of Maximum Demand

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Adjustment for Power Factor

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

(Continued on Sheet No. D-72.00)

See Barnes Testimony, Page 2, Lines 20-22; Exhibit A-23 (RLB-1) Item #12; Exhibit A-16 (HWM-3), Pages 14-16

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-72.00

GENERAL SERVICE PRIMARY TIME-OF-USE RATE GPTU

(Continued from Sheet No. D-71.00)

Monthly Rate:

Adjustment for Power Factor (Contd)

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

Power Factor	Penalty
0.800 to 0.849	0.50%
0.750 to 0.799	1.00%
0.700 to 0.749	2.00%
Below 0.700	3% first 2 months

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

Maximum Demand

The Maximum Demand shall be the highest 15-minute demand created during the current month or previous 11 months.

Resale Service Provision

Subject to any restrictions, this provision is available to customers desiring Primary Voltage service for resale purposes in accordance with Rule C4.4, Resale.

Substation Ownership Credit

Where service is supplied at a nominal voltage of more than 25,000 volts, energy is measured through an Interval Data Meter, and the customer provides all the necessary transforming, controlling and protective equipment for all the service there shall be deducted from the bill a monthly credit. For those customers, part of whose load is served through customer-owned equipment, the credit shall be based on the Maximum Demand.

The monthly substation ownership credit shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service Customers.

Charges for Customer Voltage Level 2 (CVL 2)

Substation Ownership Credit: \$(0.96 ~~0.97~~) per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL 1)

Substation Ownership Credit: \$(0.34 ~~0.45~~) per kW of Maximum Demand

For those customers served by more than one substation where one or more of the substations is owned by the customer, the credit will be applied to the customer's coincident Maximum Demand for those substations owned by the customer. This credit shall not operate to reduce the customer's billing below the prescribed minimum charges included in the rate. The credit shall be based on the kW after the 1% deduction or 3% addition has been applied to the metered kW.

Educational Institution Service Provision (GEI)

When service is supplied to a school, college or university, a credit shall be applied during all billing months. As used in this provision, "school" shall mean buildings, facilities, playing fields, or property directly or indirectly used for school purposes for children in grades kindergarten through twelve, when provided by a public or nonpublic school. School does not include instruction provided in a private residence or proprietary trade, vocational, training, or occupational school. "College" or "University" shall mean buildings located on the same campus and used to impart instruction, including all adjacent and appurtenant buildings owned by the same customer which are located on the same campus and which constitute an integral part of such college or university facilities.

The monthly credit for the Educational Institution Service Provision shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service Customers.

Educational Institution Credit: \$ (0.000259 ~~0.000314~~) per kWh for all kWh

Customers on this provision shall require a written contract, with a minimum term of one year, and shall be evaluated annually to determine whether or not the accounts shall remain on the service provision.

(Continued on Sheet No. D-73.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-73.00

GENERAL SERVICE PRIMARY TIME-OF-USE RATE GPTU

(Continued from Sheet No. D-71.10)

Monthly Rate (Contd)

Self-Generation ~~Provision~~ (SG)

~~Subject to any restrictions, as of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities operated in parallel with the Company's system must meet the Parallel Operation Requirements set forth in Rule C1.6B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~There shall be no double billing of demand under the base rate and the Self Generation Provision.~~

~~Sales of Self-Generated Energy to the Company~~

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data / billing determinants necessary for billing purposes.~~

~~Administrative Cost Charge~~

~~\$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

~~Energy Purchase~~

~~An energy purchase by the Company shall be bought at the Midecontinent Independent System Operator's, Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Program

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

General Terms

The rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge

The System Access Charge included in the rate, and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract

Service under this rate shall require a written contract with a minimum term of one year.

(Continued on Sheet No. D-74.00)

See Barnes Testimony, Page 4, Lines 3-18; Exhibit A-23 (RLB-1) Item #9
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**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-74.00

ENERGY INTENSIVE PRIMARY RATE EIP

Availability

Subject to any restrictions, the Energy Intensive Primary Rate EIP is available to any Full Service electric metal melting customer taking service at the Company's Primary Voltage levels, where the electric load on this rate is utilized for industrial metal melting processes such as electric arc or induction furnaces or to any Full Service electric industrial customer who qualified as energy intensive as defined herein. For metal melting customers, only electric load that directly supports the process of melting metal using electricity as the main melting source qualifies as load to be served under this rate. Ancillary equipment required for the metal melting process is not intended to be served on this rate.

Existing or former metal melting customers taking service under the Company's ~~former~~ Metal Melting Primary Pilot as of November 30, 2015 are eligible for service on Rate EIP. An additional 200 MW of Maximum Demand capacity will be available on a first-come, first-served basis to Full Service customers with new electric metal melting or energy intensive industrial load not previously served by the Company. To qualify as energy intensive load, the customer must demonstrate viable options to site the production outside of the state and the customer's incremental load must exceed 2 MW at a single site with an annual load factor that exceeds 70% or the customer's incremental load must exceed 15 MW with a minimum of 75% of their total consumption occurring during Off-Peak Hours. New electric metal melting load must be separately metered. The customer must provide a special circuit or circuits in order for the Company to install separate metering.

Nature of Service

Service under the rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

Where service is supplied at a nominal voltage of 25,000 Volts or less, the customer shall furnish, install and maintain all necessary transforming, controlling and protective equipment.

Where the Company elects to measure the service at a nominal voltage above 25,000 Volts, 1% shall be deducted for billing purposes, from the demand and energy measurements thus made.

Where the Company elects to measure the service at a nominal voltage of less than 2,400 Volts, 3% shall be added for billing purposes, to the demand and energy measurements thus made.

Interval Data Meters are required for service under this rate. Meter reading will be accomplished electronically through telecommunication links or other electronic measuring equipment available to provide the Company with the metering data necessary for billing purposes.

For purposes of this rate, the appropriate measure of market price is the Real-Time LMP for the Company's retail aggregating node CONS.CETR established by the Midcontinent Independent System Operator Inc. (MISO).

Critical Peak Event Determination

The Company shall call a Critical Peak Event to signal either the market price has exceeded an Economic Trigger Price or a system integrity event is enacted.

A System Integrity Event is enacted when MISO declares that a Maximum Generation Emergency Event has occurred and MISO has instructed the Company to implement Load Management Measures using Load Modifying Resources and Load Management Measures - Stage 1. A System Integrity Event shall occur at any time for any duration. A Critical Peak Event caused by a System Integrity Event shall be billed at the greater of 150% of the High Peak Energy Charge or the average market price during the duration of the event.

The Summer Economic Trigger Price is the greater of 150% of the High Peak Energy Charge, Customer Voltage Level 1 or the average market price during the hours of 3:00 PM to 5:00 PM for the period of June 1 through September 30 of the previous year. The Summer Economic Trigger Price will be set on January 30 of each year by the Company.

The Winter Economic Trigger Price is the greater of 150% of the High Peak Energy Charge, Customer Voltage Level 1 or the average market price during the hours of 5:00 PM to 7:00 PM for the period of October 1 through May 31 of the previous year. The Winter Economic Trigger Price will be set on July 31 of each year by the Company.

Energy Intensive Primary Rate customers will be notified after the Summer and Winter Economic Trigger Prices are set. The Company shall endeavor to provide notice in advance of a probable System Integrity Event.

(Continued on Sheet No. D-75.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-75.00

ENERGY INTENSIVE PRIMARY RATE EIP
(Continued from Sheet No. D-74.00)

Schedule of Hours:

The following schedule shall apply Monday through Friday (except holidays designated by the Company):

Summer:

Off-Peak Hours: 12:00 AM to 6:00 AM and 11:00 PM to 12:00 AM
Low-Peak Hours: 6:00 AM to 2:00 PM and 6:00 PM to 11:00 PM
Mid-Peak Hours: 2:00 PM to 3:00 PM and 5:00 PM to 6:00 PM
High-Peak Hours: 3:00 PM to 5:00 PM
Critical Peak Hours: ~~3:00 PM to 5:00 PM~~ All hours during a Critical Peak Event

Winter:

Off-Peak Hours: 12:00 AM to 4:00 PM and 8:00 PM to 12:00 AM
Mid-Peak Hours: 4:00 PM to 5:00 PM and 7:00 PM to 8:00 PM
High-Peak Hours: 5:00 PM to 7:00 PM
Critical Peak Hours: ~~5:00 PM to 7:00 PM~~ All hours during a Critical Peak Event

Weekends and holidays are off-peak. Designated Company holidays are: New Year's Day - January 1; Memorial Day - Last Monday in May; Independence Day - July 4; Labor Day - First Monday in September; Thanksgiving Day - Fourth Thursday in November; and Christmas Day - December 25. Whenever January 1, July 4, or December 25 fall on Sunday, extended holiday periods such as Monday, January 2, Monday, July 5 and Monday, December 26 shall not be considered as holidays for application of off-peak hours.

Monthly Rate:

Power Supply Charges:

Charges for Customer Voltage Level 3 (CVL3)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	<u>\$0.033493</u>	<u>\$0.008628</u>	<u>\$0.042121</u>	per kWh during the calendar months of June-September
	\$0.035312	\$0.017089	\$0.052401	
Low-Peak-Summer	<u>\$0.050271</u>	<u>\$0.013483</u>	<u>\$0.063754</u>	per kWh during the calendar months of June-September
	\$0.049172	\$0.023367	\$0.072539	
Mid-Peak-Summer	<u>\$0.062294</u>	<u>\$0.016393</u>	<u>\$0.078687</u>	per kWh during the calendar months of June-September
	\$0.060664	\$0.028572	\$0.089236	
High-Peak-Summer	<u>\$0.067872</u>	<u>\$0.016771</u>	<u>\$0.084643</u>	per kWh during the calendar months of June-September
	\$0.065164	\$0.030610	\$0.095774	
Critical Peak-Summer				the greater of either 150% of the High-Peak - Summer Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of June - September
Off-Peak - Winter	<u>\$0.039994</u>	<u>\$0.007271</u>	<u>\$0.047265</u>	per kWh during the calendar months of October-May
	\$0.035492	\$0.017170	\$0.052662	
Mid-Peak - Winter	<u>\$0.045307</u>	<u>\$0.008296</u>	<u>\$0.053603</u>	per kWh during the calendar months of October-May
	\$0.049652	\$0.023584	\$0.073236	
High-Peak - Winter	<u>\$0.047139</u>	<u>\$0.008410</u>	<u>\$0.055549</u>	per kWh during the calendar months of October-May
	\$0.062539	\$0.029421	\$0.091960	
Critical Peak-Winter				the greater of either 150% of the High-Peak Winter Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of October - May

(Continued on Sheet No. D-76.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-76.00

ENERGY INTENSIVE PRIMARY RATE EIP
(Continued from Sheet No. D-75.00)

Monthly Rate (Contd):

Power Supply Charges: (Contd)

Charges for Customer Voltage Level 2 (CVL2)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	\$0.032795 \$0.038312	\$0.008427 \$0.020089	\$0.041222 \$0.058401	per kWh during the calendar months of June-September
Low-Peak-Summer	\$0.049223 \$0.052172	\$0.013168 \$0.026367	\$0.062391 \$0.078539	per kWh during the calendar months of June-September
Mid-Peak-Summer	\$0.060996 \$0.063664	\$0.016010 \$0.031572	\$0.077006 \$0.095236	per kWh during the calendar months of June-September
High-Peak-Summer	\$0.066464 \$0.068164	\$0.016379 \$0.033610	\$0.082843 \$0.101774	per kWh during the calendar months of June-September
Critical Peak-Summer				the greater of either 150% of the High-Peak-Summer Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of June-September
Off-Peak - Winter	\$0.039180 \$0.038492	\$0.007101 \$0.020170	\$0.046281 \$0.058662	per kWh during the calendar months of October - May
Mid-Peak - Winter	\$0.044385 \$0.052652	\$0.008103 \$0.026584	\$0.052488 \$0.079236	per kWh during the calendar months of October - May
High-Peak - Winter	\$0.046181 \$0.065539	\$0.008213 \$0.032421	\$0.054394 \$0.097960	per kWh during the calendar months of October - May
Critical Peak-Winter				the greater of either 150% of the High-Peak Winter Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of October - May

Charges for Customer Voltage Level 1(CVL1)

Energy Charge:

	Non-Capacity	Capacity	Total	
Off-Peak-Summer	\$0.032353 \$0.033312	\$0.008294 \$0.015089	\$0.040647 \$0.048401	per kWh during the calendar months of June-September
Low-Peak-Summer	\$0.048556 \$0.047172	\$0.012961 \$0.021367	\$0.061517 \$0.068539	per kWh during the calendar months of June-September
Mid-Peak-Summer	\$0.060171 \$0.058664	\$0.015758 \$0.026572	\$0.075929 \$0.085236	per kWh during the calendar months of June-September
High-Peak-Summer	\$0.065571 \$0.063164	\$0.016121 \$0.028610	\$0.081692 \$0.091774	per kWh during the calendar months of June-September
Critical Peak-Summer				the greater of either 150% of the High-Peak-Summer Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of June-September
Off-Peak - Winter	\$0.038667 \$0.033492	\$0.006989 \$0.015170	\$0.045656 \$0.048662	per kWh during the calendar months of October - May
Mid-Peak - Winter	\$0.043803 \$0.047652	\$0.007975 \$0.021584	\$0.051778 \$0.069236	per kWh during the calendar months of October - May
High-Peak - Winter	\$0.045577 \$0.060539	\$0.008084 \$0.027421	\$0.053661 \$0.087960	per kWh during the calendar months of October - May
Critical Peak-Winter				the greater of either 150% of the High-Peak Winter Energy Charge or the average Market price per kWh for a Critical Peak Event during the calendar months of October - May

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges:

System Access Charge: \$200.00 per customer per month

Charges for Customer Voltage Level 3 (CVL3)

Capacity Charge: \$~~4.58~~ ~~3.60~~ per kW of Maximum Demand

Charges for Customer Voltage Level 2 (CVL2)

Capacity Charge: \$~~2.69~~ ~~1.86~~ per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL1)

Capacity Charge: \$~~0.70~~ ~~0.91~~ per kW of Maximum Demand

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-77.00)

See Barnes Testimony, Page 2, Lines 20-22; Exhibit A-23 (RLB-1) Item #12; Exhibit A-16 (HWM-3), Pages 17-19

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-77.00

ENERGY INTENSIVE PRIMARY RATE EIP
(Continued from Sheet No. D-76.00)

Adjustment for Power Factor:

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

Power Factor	Penalty
0.800 to 0.849	0.50%
0.750 to 0.799	1.00%
0.700 to 0.749	2.00%
Below 0.700	3% first 2 months

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

Maximum Demand:

The Maximum Demand shall be the highest 15-minute demand created during the current month or previous 11 months.

Substation Ownership Credit:

Where service is supplied at a nominal voltage of more than 25,000 volts, energy is measured through an Interval Data Meter, and the customer provides all the necessary transforming, controlling and protective equipment for all the service there shall be deducted from the bill a monthly credit. For those customers, part of whose load is served through customer-owned equipment, the credit shall be based on the Maximum Demand.

The monthly substation ownership credit shall be applied as follows:

Delivery Charges - These charges are applicable to Full Service and Retail Open Access Customers.

Charges for Customer Voltage Level 2 (CVL 2)

Substation Ownership Credit: \$(~~0.96~~ ~~0.97~~) per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL 1)

Substation Ownership Credit: \$(~~0.34~~ ~~0.45~~) per kW of Maximum Demand

For those customers served by more than one substation where one or more of the substations is owned by the customer, the credit will be applied to the customer's coincident Maximum Demand for those substations owned by the customer. This credit shall not operate to reduce the customer's billing below the prescribed minimum charges included in the rate. The credit shall be based on the kW after the 1% deduction or 3% addition has been applied to the metered kW.

Self-Generation ~~Provision~~ (SG):

~~Subject to any restrictions, As of June 8, 2012, this provision may be required for any Full Service~~ To be eligible for Self-Generation, a Customer with a generating installation ~~less than 550 kW~~ operating in parallel with the Company's system, ~~which may employ cogeneration or small power production technology~~ must meet the requirements described in Rule C 11.1., Self-Generation.

~~All facilities must meet the Parallel Operation Requirements set forth in Rule C1.6B. The Company shall own, operate and maintain all metering and auxiliary devices (including telecommunication links) at the customer's expense. Meters furnished, installed and maintained by the Company shall meter generation equipment for customers that sell energy to the Company. No refund shall be made for any customer contribution required.~~

(Continued on Sheet No. D-78.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-78.00

ENERGY INTENSIVE PRIMARY RATE EIP
(Continued from Sheet No. D-77.00)

Self-Generation Provision (SG) (Contd)

~~Energy delivered to the Company shall be alternating current, 60 hertz, single phase or three phase (as governed by Rule B8., Electric Interconnection and Net Metering Standards) Secondary Voltage or Primary Voltage service. The Company will determine the particular nature of the voltage in each case.~~

~~Self-generation customers requiring Company delivery service for any portion of the load that has been self-generated will be charged as described in the Delivery Charges section of this Rate Schedule.~~

~~There shall be no double billing of demand under the base rate and the Self-Generation Provision.~~

Sales of Self-Generated Energy to the Company:

~~A customer who meets the Federal Energy Regulatory Commission's (FERC) criteria for a Qualifying Facility may elect to sell energy to the Company. The Company has the right to refuse to contract for the purchase of energy. Sales of energy to the Company under this provision shall require a written contract with a minimum term of one year.~~

~~Where the customer elects to sell energy to the Company, an Interval Data Meter (IDM) or other applicable meter is required for their generator. Meter reading will be accomplished electronically through telecommunication links or other electronic data methods able to provide the Company with the metering data /billing determinants necessary for billing purposes.~~

~~**Administrative Cost Charge:** \$0.0010 per kWh purchased for generation installations with a capacity of 550 kW or less.~~

Energy Purchase:

~~An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule). The Company may discontinue purchases during system emergencies, maintenance and other operational circumstances.~~

Distributed Generation Program:

The Distributed Generation Program is available to any eligible customer as described in Rule C 11.3., Distributed Generation Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C 11.3.B., Distributed Generation Definitions.

A customer who participates in the Distributed Generation Program is subject to the provisions contained in Rule C 11.3., Distributed Generation Program.

Green Generation Programs:

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

General Terms:

The rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Minimum Charge:

The System Access Charge included in the rate and any applicable non-consumption based surcharges.

Due Date and Late Payment Charge:

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract:

Service under this rate shall require a written contract with a minimum term of one year.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-79.00

EXPERIMENTAL ADVANCED RENEWABLE PROGRAM AR

Availability:

Subject to any restrictions and requirements of Rule C10.3, an individual or entity who is a delivery customer of the Company that generates electricity from a solar energy system owned by the customer and constructed using Michigan workforce labor, or using equipment made in the state of Michigan is eligible to sell power to the Company under the terms set forth in this schedule.

Monthly Rate:

System Access Charge: Equal to the System Access Charge of the Customer's Delivery Account but not in excess of \$50, assessed per generator meter, to be paid to the Company by the customer or to be deducted from the payment to the customer by the Company

Sales of Energy to the Company that begin service no later than December 31, 2009:

\$0.650 per kWh purchased by the Company payable to a Residential customer
\$0.450 per kWh purchased by the Company, payable to a Non-Residential customer

Sales of Energy to the Company that begin service after December 31, 2009 but no later than October 1, 2011:

\$0.525 per kWh purchased by the Company, payable to a Residential customer
\$0.375 per kWh purchased by the Company, payable to a Non-Residential customer

Sales of Energy to the Company that begin service after October 1, 2011:

Price set contractually, in accordance with conditions specified in Rule C10.3.

Purchases of Energy from the Company for generator station power:

For all energy supplied by the Company, the charges shall be as provided for under the Residential Service Secondary Non-Transmitting Meter Rate ~~RSM~~ ~~RS~~ Rate Schedule for residential customers or the General Service Secondary Rate GS Rate Schedule, for all per kWh charges only, including additional charges such as, but not limited to, applicable surcharges, Power Plant Securitization Charges and Power Supply Cost Recovery (PSCR) Factor.

General Terms:

This program is subject to all general terms and conditions shown on Sheet No. D-1.00.

Payment of Energy Purchases:

The Company reserves the right to transfer amounts due to the Company or the customer under this schedule to an active account for energy purchases from the Company.

Term and Form of Contract:

Sales of energy to the Company under this schedule shall require a written contract with a minimum term of one year and a maximum term of 15 years; however, no contract term may extend beyond August 31, 2029.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-82.00

GENERAL SERVICE SELF GENERATION RATE GSG-2
(Continued From Sheet No. D-81.00)

Nature of Service (Contd)

Where service is supplied at a nominal voltage equal to or greater than 2,400 volts and the Company elects to measure the service at a nominal voltage above 25,000 volts, 1% shall be deducted for billing purposes, from the demand and energy measurements thus made.

Where service is supplied at a nominal voltage equal to or greater than 2,400 volts and the Company elects to measure the service at a nominal voltage of less than 2,400 volts, 3% shall be added for billing purposes, to the demand and energy measurements thus made.

Where service is supplied at a nominal voltage less than 2,400 volts and the Company elects to measure the service at a nominal voltage equal to or greater than 2,400 volts, 3% shall be deducted for billing purposes from the energy measurements thus made.

There shall be no double billing of demand under the base rate and Rate GSG-2.

Monthly Rate

Standby Charges

Power Supply Standby Charges

For all standby energy supplied by the Company, the customer shall be responsible for the MISO Real-Time Locational Market Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule), multiplied by the customer's consumption (kWh), plus the Market Settlement Fee of \$0.002/kWh. In addition capacity charges will be assessed monthly, calculated using the highest 15 minute kW demand associated with Standby Service occurring during the Company's On-Peak billing hours will be multiplied by the highest contracted capacity purchased by the Company in that month, plus allocated transmission and ancillaries. The capacity charges will be prorated based on the number of On-Peak days that Standby Service was used during the billing month.

A customer with a generator(s) nameplate rating more than 550 kW must provide written notice to the Company by December 1 if they desire standby service in the succeeding calendar months of June through September. Written notice shall be submitted on Company Form 500. If the customer fails to meet this written notice requirement, the LMP shall be increased by applying a 10% adder.

Real Power Losses

Real Power Losses shall be measured based on the transmission loss factor of 2.10 ~~2.20~~% plus the associated meter point as listed below:

	Meter Point	
	High Side	Low Side
Customer Voltage Level 1	0.000%	<u>0.728</u> 0.705 %
Customer Voltage Level 2	<u>1.325</u> 1.274 %	<u>2.189</u> 2.366 %
Customer Voltage Level 3	<u>3.329</u> 3.224 %	<u>8.082</u> 7.643 %

Delivery Standby Charges

System Access Charge:

Generator that does not meet or exceed load:	\$100.00	per generator installation per month
Generator that meets or exceeds load:	\$200.00	per generator installation per month

Charges for Customer Voltage Level 3 (CVL 3)

Capacity Charge: \$4.58 ~~3.60~~ per kW of Maximum Demand

Charges for Customer Voltage Level 2 (CVL 2)

Capacity Charge: \$2.69 ~~1.86~~ per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL 1)

Capacity Charge: \$0.70 ~~0.91~~ per kW of Maximum Demand

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued On Sheet No. D-83.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-83.00

GENERAL SERVICE SELF GENERATION RATE GSG-2
(Continued From Sheet No. D-82.00)

Monthly Rate (Contd)

Standby Charges (Contd)

Adjustment for Power Factor

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar -hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor. Whenever the average Power Factor during the billing period is above .899 or below .850, the customer bill shall be adjusted as follows:

- (a) If the average Power Factor during the billing period is .900 or higher, a 0.50% credit will be applied to all metered-based charges, excluding surcharges. This credit shall not in any case be used to reduce the prescribed Minimum Charge.
- (b) If the average Power Factor during the billing period is less than .850, a penalty will be applied to all metered-based charges, excluding surcharges, in accordance with the following table:

Power Factor Penalty

0.800 to 0.849	0.50%
0.750 to 0.799	1.00%
0.700 to 0.749	2.00%
Below 0.700	3% first 2 months

- (c) A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, it is necessary to complete two consecutive months below 0.700 before the 15% penalty applies again.

Substation Ownership Credit

Where service is supplied at a nominal voltage of more than 25,000 volts, energy is measured through an Interval Data Meter, and the customer provides all of the necessary transforming, controlling and protective equipment for all of the service there shall be deducted from the bill a monthly credit. For those customers, part of whose load is served through customer-owned equipment, the credit shall be based on the billed Standby Demand.

The monthly credit for the substation ownership shall be applied as follows:

Delivery Charges

Charges for Customer Voltage Level 2 (CVL 2)

Substation Ownership Credit: \$(~~0.96~~ ~~0.97~~) per kW of Maximum Demand

Charges for Customer Voltage Level 1 (CVL 1)

Substation Ownership Credit: \$(~~0.34~~ ~~0.45~~) per kW of Maximum Demand

For those customers served by more than one substation where one or more of the substations is owned by the customer, the credit will be applied to the customer's coincident Maximum Demand for those substations owned by the customer. This credit shall not operate to reduce the customer's billing below the prescribed minimum charges included in the rate. The credit shall be based on the kW after the 1% deduction or 3% addition has been applied to the metered kW.

(Continued on Sheet No. D-84.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-84.00

GENERAL SERVICE SELF GENERATION RATE GSG-2
(Continued From Sheet No. D-83.00)

Monthly Rate (Contd)

Standby Charges (Contd)

Transmission Interconnect Credit

Where standby service is provided to a non-utility electric generator located within the Company's service territory and taking power through its transmission interconnect, where the Company has no owned infrastructure other than metering, including billing grade current transformers and potential transformers, telemetry facilities and associated wiring, the following monthly credit shall be applied to the bill:

Delivery Charges

Transmission Interconnect Credit: \$ ~~(0.70 0.91)~~ per kW of Maximum Demand

This credit shall be based on the kW after the 1% deduction has been applied to the metered kW. The credit supersedes any applicable substation ownership credit.

Sales of Energy to the Company

Administrative Cost Charge

Generation installation with a capacity of over 550 kW but less than or equal to 2,000 kW
As negotiated or \$0.0010 per kWh purchased, at the option of the customer

Generation installation with a capacity of over 2,000 kW
As negotiated

Energy Purchase:

An energy purchase by the Company shall be bought at the Midcontinent Independent System Operator's Inc. (MISO) real-time Locational Marginal Price (LMP) for the Company's load node (designated as "CONS.CETR" as of the date of this Rate Schedule).

General Terms

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Green Generation Program

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

Minimum Charge

The System Access Charge included in this Rate Schedule in addition to the customer's contracted Standby Capacity multiplied by the net of any Substation Ownership Credit and Delivery Capacity Charges of this Rate Schedule.

Due Date and Late Payment Charge

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract

Standby service and/or sales of energy to the Company under this rate shall require a written contract with a minimum term of one year.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-84.10

LONG TERM INDUSTRIAL LOAD RETENTION RATE - LTILRR

Availability:

Subject to any restrictions, the Long Term Industrial Load Retention Rate (“LTILRR”) is available to any industrial Full Service Customer taking electric service at the Company's Primary Voltage levels that, at the time the rate contract is executed 1) has an Average Demand of at least 200,000 kW at a single site, and 2) has a minimum Annual Load Factor of 75%. Customers must execute a long-term rate contract under this Rate Schedule for a minimum of 100,000 kW of Firm Contracted Capacity, and for service at a site where the Average Demand is at least 200,000 kW at the time the rate contract is executed. Customers must enter into a contract for a term, equal to: i) the term of the designated power purchase agreement or agreements, which in no case shall be for less than 15 years for one or more designated power supply resource if the resource is a power purchase agreement or agreements, or ii) the expected remaining life of one or more designated utility-owned power supply resources.

A corporate officer of the customer taking service under this rate must submit a sworn affidavit stating that the customer would no longer purchase standard tariff service from the electric utility absent the customer being able to purchase power supply under the LTILRR.

Service under this rate is not available for intrastate facility consolidation or relocation of the customer's existing facilities, for standby service, for new or expanded service for resale or new customers or for expanded service for the benefit of parties other than the customer. Electric service provided under this Rate Schedule may not be transported off the customer's Site. A single customer shall not aggregate load from multiple sites to meet the requirements under this rate, and multiple customers shall not aggregate load to meet the requirements under this rate.

A customer shall be considered an industrial customer if the customer's operation meets the qualifications as determined by the NAICS as defined by the Energy Information Administration.

The rate contract shall require a written agreement approved by the Michigan Public Service Commission (“Commission”), specifying the terms of the electric service and shall include creditworthiness requirements to the Company's satisfaction.

Contracted Capacity and Annual Nominations:

The Maximum Contracted Capacity available to any customer under this Rate Schedule shall be specified in a written agreement approved by the Commission. The customer must nominate annually, at the time the agreement is executed, and subsequently at least eight months before the start of the subsequent Midcontinent Independent System Operator, Inc. (“MISO”) Planning Year, the amount of Annual Forecast Capacity, which shall be based on the customer's highest expected Maximum Monthly Demand adjusted for known and verifiable changes. The Annual Forecast Capacity shall not exceed the Maximum Contracted Capacity. If the customer's Maximum Monthly Demand in any month exceeds the Annual Forecast Capacity for the current Planning Year, the Annual Forecast Capacity shall be increased to the Maximum Monthly Demand, up to the Maximum Contracted Capacity, and customer shall be billed for the increase in Annual Forecast Capacity for the entire current MISO Planning Year.

The difference between the Annual Forecast Capacity and the Maximum Contracted Capacity shall be the Reserved Capacity. The Reserved Capacity shall be made available to the customer for load growth as specified in the customer's written agreement for electric service.

At the time the agreement is executed, and no later than eight months prior to the start of each subsequent MISO Planning Year, the customer must specify the level of Firm Contracted Capacity, which shall not exceed the Annual Forecast Capacity. The difference between the Annual Forecast Capacity and the Firm Contracted Capacity shall be Interruptible Service Capacity, which shall be subject to the Interruptible Service Provision as specified in this Rate Schedule.

(Continued on Sheet No. D-84.20)

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Original Sheet No. D-84.20

LONG TERM INDUSTRIAL LOAD RETENTION RATE - LTILRR

(Continued From Sheet No. D-84.10)

Nature of Service:

Service under the rate shall be alternating current, 60-Hertz, three-phase Primary Voltage service. The particular nature of the voltage service provided to the customer shall be specified in a written agreement.

Interval Data Meters are required for service under this rate. Meter reading will be accomplished electronically through telecommunication links or other electronic measuring equipment available to provide the Company with the metering data necessary for billing purposes.

Line losses shall be applied to the customer's monthly metered energy and capacity values to reflect the energy consumed in moving electric power through the Transmission system and the Company's distribution system to the customer's point of delivery as determined by the Company and approved by the Commission.

Monthly Rate:

System Access Charge:

Fixed charge per billing month as specified in the customer's written agreement for electric service

Power Supply Charges:

Capacity Charge:

\$ per kW per month for contracted Annual Forecast Capacity as specified in the customer's written agreement for electric service

Reserved Capacity Charge:

\$ per kW per month for the difference between the Maximum Contract Capacity and the Annual Forecast Capacity as specified in the customer's written agreement for electric service

Excess Capacity Charge:

\$ per kW per month for Maximum Monthly Demand in excess of the Maximum Contract Capacity based on the Power Supply Demand Charges (for Capacity and Non-Capacity) per the Large General Service Primary Demand Rate GPD Rate Schedule at the customer's applicable Customer Voltage Level

Interruptible Credit:

Equivalent to the Commission-approved \$ per kW per month Rate GPD Interruptible Service Provision (GI) Interruptible Credit, applied to Interruptible Service Capacity, not to exceed the Capacity Charge

Energy Charge:

The monthly energy charges shall be based on the designated power supply resource's actual variable fuel and variable operations and maintenance expense, or the displacement costs of such expense, as applicable, associated with the customer's actual energy consumption as specified in the customer's written agreement for electric service

(Continued on Sheet No. D-84.30)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-84.30

LONG TERM INDUSTRIAL LOAD RETENTION RATE - LTILRR

(Continued From Sheet No. D-84.20)

Power Supply Charges: (Contd)

Excess Energy Charge:

\$ per kWh for energy used in excess of the Maximum Contracted Capacity based on the Power Supply Energy Charges per the Rate GPD Rate Schedule at the customer's applicable Customer Voltage Level, including the applicable non-transmission PSCR Factor charges

Transmission Charges:

Transmission Charge:

Monthly charge per billing month based on the Company's costs to acquire transmission service to serve the customer's load as specified in the customer's written agreement for electric service

Delivery Charges:

Distribution Charges:

Monthly charge per billing month based on the dedicated distribution facilities in place to serve the customer

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and Securitization Charges shown on Sheet No. D-7.00. This rate is not subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00

Interruptible Service Provision

The monthly credit under this Interruptible Service Provision shall be set by the Commission and shall be equivalent to the credit provided to customers receiving an Interruptible Credit under the Large General Service Primary Demand Rate GPD, Interruptible Service Provision (GI). The monthly credit available to the customer under this Interruptible Service Provision shall not exceed the Monthly Capacity Charge specified in the customer's written agreement for electric service.

The Company reserves the right to limit the amount of load contracted as Interruptible Service Capacity under this rate schedule, but in no case shall it exceed 300,000 kW.

Customers contracting for interruptible service under this rate schedule shall be required to monitor and provide real-time, Internet-enabled power monitoring. The Company will provide the metering or monitoring devices necessary, which shall be owned by the Company and provided to the customer at the Company's expense. The customer may be required to provide suitable space for such monitoring equipment and either a static or non-static, as applicable, Internet Protocol (IP) address and Local Area Network (LAN) access that allows for Internet-based communication of the customer's site electricity consumption and interruption event performance.

The interruptible load is subject to the MISO Load Modifying Resource requirements. Within 30 minutes of receiving an interruption notice from the Company, the customer shall reduce its total load level down to the Firm Contracted Capacity level or as required by the MISO partial curtailment request.

Any load designated as interruptible is subject to MISO requirements for Load Modifying Resources and Company shall inform customer of such MISO requirements. Interruption under this Interruptible Service Provision may occur if MISO issues a Maximum Generation Emergency Event Step 2b order or North American Electric Reliability Corporation Emergency Event Alert 2 notice indicating that MISO is experiencing or expects to experience a shortage of economic resources and the Company has declared emergency status. Participation in the Interruptible Service Provision does not limit the Company's ability to implement emergency electrical procedures as described in the Company's Electric Rate Book including interruption of service as required to maintain system integrity.

(Continued on Sheet No. D-84.40)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-84.40

LONG TERM INDUSTRIAL LOAD RETENTION RATE – LTILRR

(Continued From Sheet No. D-84.30)

Interruptible Service Provision (Contd)

Conditions of Interruption

Under this Interruptible Service Provision, the customer shall be interrupted at any time MISO deems it necessary to maintain system integrity. The Company shall endeavor to provide notice to the customer in advance of probable interruption by MISO. The Company shall provide the customer at least thirty minutes advance notice of a required interruption, and if possible, a second notice. Notices will be communicated by telephone to the contact numbers provided by the customer. The customer shall confirm the receipt of such notice through the automated response process. Failure to acknowledge receipt of such notice shall not relieve the customer of the obligation for interruption. The customer shall be informed, when possible, of the estimated duration of the interruption at the time of interruption.

The Company shall not be liable for any loss or damage caused by or resulting from any interruption of service under this Interruptible Service Provision.

Interruptions beyond the Company's control, described in Rules C1.1, Character of Service, and C3, Emergency Electrical Procedures, of the Company's Electric Rate Book, shall not be considered as interruptions for purposes of this Interruptible Service Provision.

Should the Company be ordered by Governmental authority during a national emergency to supply firm instead of interruptible service, billing shall reflect firm service capacity as provided under this rate schedule.

Cost of Non-Compliance with Interruption

Failure by customer to comply with an interruption order under this Interruptible Service Provision shall be considered as unauthorized use and billed at (i) the higher of the customer's pro rata share of any actual MISO penalties incurred by the Company or (ii) the rate of \$25.00 per kW for the highest 15-minute kW of Interruptible Peak Billing Demand created during the interruption period in excess of the Firm Contracted Capacity or the partial curtailment requested amount, in addition to the prescribed monthly rate.

Adjustment for Power Factor

This rate requires a determination of the average Power Factor maintained by the customer during the billing period. Such average Power Factor shall be determined through metering of lagging Kilovar-hours and Kilowatt-hours during the billing period. The calculated ratio of lagging Kilovar-hours to Kilowatt-hours shall then be converted to the average Power Factor for the billing period by using the appropriate conversion factor.

A Power Factor less than 0.700 is not permitted and necessary corrective equipment must be installed by the customer. A 15% penalty will be applied to any metered-based charges, excluding surcharges, after two consecutive months below 0.700 Power Factor and will continue as long as the Power Factor remains below 0.700. Once the customer's Power Factor exceeds 0.700, the 15% penalty shall apply again if the Power Factor falls below 0.700 for two consecutive months.

General Terms:

The rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

(Continued on Sheet No. D-84.50)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Original Sheet No. D-84.50

LONG TERM INDUSTRIAL LOAD RETENTION RATE – LTLRR

(Continued From Sheet No. D-84.40)

Monthly Minimum Charge:

The Monthly Minimum Charge shall be the lower of the total amount due on the invoice or the sum of (i) the System Access Charge, (ii) the Distribution Charge, (iii) the monthly Capacity Charge, (iv) the monthly Reserved Capacity Charge, (v) any applicable non-consumption-based Surcharges, plus (vi) the monthly Interruptible Credit.

Due Date and Late Payment Charge:

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Term and Form of Contract:

Service under this rate shall require a written agreement, approved by the Commission. Customers served under this Rate Schedule must contract for a minimum of 100,000 kW of Firm Contracted Capacity.

Definitions Applicable to the Long Term Industrial Load Retention Rate:

Annual Forecast Capacity

Annual Forecast Capacity is the higher of the customer's maximum forecasted amount of electric capacity nominated, or actual Maximum Monthly Demand used, by the customer during the MISO Planning Year beginning June 1 and ending May 31 of the following calendar year, subject to the limitations and adjustments as specified in the customer's written agreement.

Annual Load Factor

Annual Load Factor shall be calculated as an average of the prior 12 monthly load factors. Each monthly load factor shall be determined by dividing the customer's actual monthly kWh sales by the product of the customer's Maximum Monthly Demand times the number of hours in the month.

Average Demand

Shall mean the average of the most recent 12 monthly site Maximum Monthly Demands.

Capacity Charge

The Capacity Charge shall be the Company's levelized cost of capacity, including fixed operation and maintenance expense, associated with the designated power supply resource at the time the customer's agreement for electric service is executed, or the Company's cost of capacity, including fixed operation and maintenance expense, associated with a designated power purchase agreement or agreements.

Energy Charge

The Energy Charge shall be the Company's actual variable fuel and actual variable operation and maintenance expense based on the customer's actual energy consumption and associated with the designated power supply resource, or the Company's actual energy and capacity purchases, if any, based on the customer's actual consumption, as applicable.

Excess Capacity

The Excess Capacity is the customer's actual Maximum Monthly Demand in excess of the Maximum Contracted Capacity in any billing month.

(Continued on Sheet No. D-84.60)

LONG TERM INDUSTRIAL LOAD RETENTION RATE - LTILRR

(Continued From Sheet No. D-84.50)

Definitions Applicable to the Long Term Industrial Load Retention Rate: (Contd)

Firm Contracted Capacity

The amount of electric capacity of at least 100,000 kW and not more than the Annual Forecast Capacity that the Company will supply to qualifying customers as specified in a written agreement that is not subject to the Interruptible Service Provision.

Interruptible Peak Billing Demand

The highest measured 15-minute interval demand in excess of the Firm Contracted Capacity that is consumed by the customer during an interruption event.

Interruptible Service Capacity

Interruptible Service Capacity is the difference between the Annual Forecast Capacity and the Firm Contracted Capacity which shall be subject to interruption per the Long Term Industrial Load Retention Rate Interruptible Service Provision.

Interval Data Meters

Interval Data Meters are meters that register customer kilowatt-hour use, peak demand, on-peak demand, and Maximum Monthly Demand.

Maximum Contracted Capacity

The maximum amount of electric capacity eligible for purchase by eligible customer under this Rate Schedule for the term of a written agreement.

Maximum Monthly Demand

The Maximum Monthly Demand shall be the highest 15-minute demand created by customer during the billing month.

MISO Planning Year

MISO Planning Year means a period extending from June 1st of a calendar year to May 31st of the following calendar year.

Reserved Capacity

The difference between the Maximum Contracted Capacity and the Annual Forecast Capacity held in reserve for future customer growth during the term of the customer's written agreement for electric service under the LTILRR

Site

An industrial site or contiguous industrial site or single commercial establishment as specified in the written agreement for electric service pursuant to the LTILRR. A site that is divided by an inland body of water or by a public highway, road, or street but that otherwise meets this definition meets the contiguous requirements.

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-85.00

GENERAL SERVICE METERED LIGHTING RATE GML

Availability

Subject to any restrictions, this rate is available to any political subdivision or agency of the State of Michigan having jurisdiction over public streets or roadways, for Primary or Secondary Voltage energy-only metered lighting service where the Company has existing distribution lines available for supplying energy for such service. Luminaires which are served under the Company's unmetered lighting rates shall not be intermixed with luminaires served under this metered lighting rate. Luminaire types in addition to those served on Rate Schedule GUL, such as light-emitting diode (LED) streetlights, may receive service under this Rate Schedule.

This rate is not available for resale purposes or for Retail Open Access Service.

Nature of Service

Secondary Voltage

Service under this rate shall be alternating current, 60-hertz, single-phase or three-phase (at the Company's option), 120/240 nominal Volt service for a minimum of ten luminaires located within a clearly defined area. Control equipment shall be furnished, owned and maintained by the Company. The customer shall furnish, install, own and maintain the rest of the equipment comprising the metered lighting system including, but not limited to, the overhead wires or underground cables between the luminaires, protective equipment, and the supply circuits extending to the point of attachment with the Company's distribution system. The Company shall connect the customer's equipment to the Company's lines and supply the energy for its operation. All of the customer's equipment shall be subject to the Company's approval. The customer shall not change the capacity requirements of the equipment owned by it without first notifying the Company in writing of such changes and the date that they shall be made.

Dusk to Midnight Service

Dusk to midnight service shall be the same as Secondary service except:

The customer shall pay the difference between the cost of the control equipment necessary for dusk to midnight service and control equipment normally installed for Secondary service. Circuits shall be arranged approximating minimum loads of 3 kW.

Primary Voltage

Service under this rate shall be alternating current, 60-hertz, single-phase or three-phase (at the Company's option), Primary Voltage service for actual kW demands of not less than 100 kW for each point of delivery and where the customer guarantees a minimum of 4,000 annual hours' use of the actual demand. The Company will determine the particular nature of the voltage in each case. The customer shall furnish, install, own and maintain all equipment comprising the metered lighting system including, but not limited to, controls, protective equipment, transformers and overhead or underground metered lighting circuits extending to the point of attachment with the Company's distribution system. The Company shall furnish, install, own and maintain the metering equipment and connect the customer's metered lighting circuit to its distribution system and supply the energy for operation of the customer's metered lighting system.

Monthly Rate

Secondary Power Supply Charge

Energy Charge:

Non-Capacity	Capacity	Total	
\$0.041960	\$0.000000	\$0.041960	per kWh for all kWh
0.051457		0.051457	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

(Continued on Sheet No. D-86.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-86.00

GENERAL SERVICE METERED LIGHTING RATE GML
(Continued From Sheet No. D-85.00)

Monthly Rate (Contd)

Secondary Delivery Charge

System Access Charge: \$10.00 per customer per month
Distribution Charge: \$0.058006 ~~0.062025~~ per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Primary Power Supply Charge

Energy Charge:

Non-Capacity	Capacity	Total	
\$ <u>0.020592</u>	\$0.000000	\$ <u>0.020592</u>	per kWh for all kWh
0.025253		0.025253	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Primary Delivery Charge

System Access Charge: \$20.00 per customer per month
Distribution Charge: \$0.044205 ~~0.047268~~ per kWh for all kWh

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

Net Metering Program

The Net Metering Program is available to any eligible customer as described in Rule C11.2, Net Metering Program, who desires to generate a portion or all of their own retail electricity requirements using a Renewable Energy Resource as defined in Rule C11.2.B., Net Metering Program.

A customer who participates in the Net Metering Program is subject to the provisions contained in Rule C11.2, Net Metering Program.

Green Generation Program

Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.

A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.

(Continued on Sheet No. D-87.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-90.00

GENERAL SERVICE UNMETERED LIGHTING RATE GUL
(Continued From Sheet No. D-89.00)

Monthly Rate

The charge per luminaire per month shall be:

<u>Type of Luminaire</u>	<u>Nominal Rating of Lamps (One Lamp per Luminaire) (1)</u>			<u>Service Charge per Luminaire (4)</u>		<u>Total</u>	<u>Fixture Charge per Luminaire (4)</u>
	<u>Watts</u>	<u>Watts Including Ballast (2)</u>	<u>Lumens</u>	<u>Non-Capacity</u>	<u>Capacity</u>		
Mercury Vapor (3)	100	128	3,500	\$ <u>8.01</u> 8.31	\$0.00	\$ <u>8.01</u> 8.31	\$ <u>5.00</u> 6.00
Mercury Vapor (3)	175	209	7,500	<u>13.07</u> 13.58	0.00	<u>13.07</u> 13.58	\$ <u>5.00</u> 6.00
Mercury Vapor (3)	250	281	10,000	<u>17.58</u> 18.25	0.00	<u>17.58</u> 18.25	\$ <u>5.00</u> 6.00
Mercury Vapor (3)	400	458	20,000	<u>28.65</u> 29.75	0.00	<u>28.65</u> 29.75	\$ <u>5.00</u> 6.00
Mercury Vapor (3)	700	770	35,000	<u>48.16</u> 50.02	0.00	<u>48.16</u> 50.02	\$ <u>5.00</u> 6.00
Mercury Vapor (3)	1,000	1,080	50,000	<u>67.55</u> 70.15	0.00	<u>67.55</u> 70.15	\$ <u>5.00</u> 6.00
High-Pressure Sodium (3)	70	83	5,000	<u>5.19</u> 5.39	0.00	<u>5.19</u> 5.39	\$ <u>5.00</u> 6.00
High-Pressure Sodium	100	117	8,500	<u>7.32</u> 7.60	0.00	<u>7.32</u> 7.60	\$ <u>5.00</u> 6.00
High-Pressure Sodium	150	171	14,000	<u>10.70</u> 11.11	0.00	<u>10.70</u> 11.11	\$ <u>5.00</u> 6.00
High-Pressure Sodium (3)	200	247	20,000	<u>15.45</u> 16.04	0.00	<u>15.45</u> 16.04	\$ <u>5.00</u> 6.00
High-Pressure Sodium	250	318	24,000	<u>19.89</u> 20.66	0.00	<u>19.89</u> 20.66	\$ <u>5.00</u> 6.00
High-Pressure Sodium	400	480	45,000	<u>30.02</u> 31.18	0.00	<u>30.02</u> 31.18	\$ <u>5.00</u> 6.00
Fluorescent (3)	380	470	20,000	<u>29.40</u> 30.53	0.00	<u>29.40</u> 30.53	\$ <u>5.00</u> 6.00
Incandescent (3)	202	202	2,500	<u>12.63</u> 13.12	0.00	<u>12.63</u> 13.12	\$ <u>5.00</u> 6.00
Incandescent (3)	305	305	4,000	<u>19.08</u> 19.81	0.00	<u>19.08</u> 19.81	\$ <u>5.00</u> 6.00
Incandescent (3)	405	405	6,000	<u>25.33</u> 26.31	0.00	<u>25.33</u> 26.31	\$ <u>5.00</u> 6.00
Incandescent (3)	690	690	10,000	<u>43.16</u> 44.82	0.00	<u>43.16</u> 44.82	\$ <u>5.00</u> 6.00
Metal Halide (3)	150	170	9,750	<u>10.63</u> 11.04	0.00	<u>10.63</u> 11.04	\$ <u>5.00</u> 6.00
Metal Halide (3)	175	210	10,500	<u>13.14</u> 13.64	0.00	<u>13.14</u> 13.64	\$ <u>5.00</u> 6.00
Metal Halide (3)	250	290	15,500	<u>18.14</u> 18.84	0.00	<u>18.14</u> 18.84	\$ <u>5.00</u> 6.00
Metal Halide (3)	400	460	24,000	<u>28.77</u> 29.88	0.00	<u>28.77</u> 29.88	\$ <u>5.00</u> 6.00

(1) Ratings for fluorescent lighting apply to all lamps in one luminaire.

- (2) Watts including ballast used for monthly billing of the Power Supply Cost Recovery (PSCR) Factor, the Power Plant Securitization Charges and surcharges.
- (3) Rates apply to existing luminaires only and are not open to new business.
- (4) For Customer-Owned lighting fixtures that are assessed a Service Charge (but not a Fixture Charge), the charge per luminaire represents a 22.6 ~~26.9~~% Power Supply Charge and a 77.4 ~~73.1~~% Distribution Charge.

For Company-Owned lighting fixtures that are assessed both a Service Charge and a Fixture Charge, the charge per luminaire represents a 15.0 ~~17.2~~% Power Supply Charge and a 85.0 ~~82.8~~% Distribution Charge.

For energy conservation purposes, customers may, at their option, elect to have any or all luminaires served under this rate disconnected for a period of six months or more. The charge per luminaire per month, for each disconnected luminaire, shall be 40% of the monthly rate set forth above. However, should any such disconnected luminaire be reconnected at the customer's request after having been disconnected for less than six months, the monthly rate set forth above shall apply to the period of disconnection. An \$8.00 per luminaire disconnect/reconnect charge shall be made at the time of disconnection except that when the estimated disconnect/reconnect cost is significantly higher than \$8.00, the estimated cost per luminaire shall be charged.

For 24-hour mercury-vapor service, the charge per luminaire shall be 125% of the foregoing rates.

(Continued on Sheet No. D-91.00)

See Barnes Testimony, Page 2, Lines 20-22; Exhibit A-23 (RLB-1) Item #12; Exhibit A-16 (HWM-3), Page 21

M.P.S.C. No. 14 – Electric

Sheet No. D-93.00

Consumers Energy Company

**GENERAL UNMETERED ~~EXPERIMENTAL~~ LIGHT EMITTING DIODE
LIGHTING RATE GU-~~XL~~ LED**

Availability:

Subject to any restrictions, this rate is available to any political subdivision or agency of the State of Michigan having jurisdiction over public streets or roadways for unmetered streetlighting service where the Company has existing distribution lines available for supplying energy for unmetered ~~experimental lighting technology including~~ light-emitting diode (LED) lighting or for any Company-owned LED streetlighting system consisting of one or more luminaires. This rate is not available for resale purposes or for Retail Open Access Service. Installations under this rate shall require ~~advanced approval by the Company and~~ a written agreement.

Nature of Service:

Company-Owned Option

In Company-owned systems, the Company shall select, furnish, install and own all equipment for any new unmetered ~~experimental~~ LED lighting or for any modifications to existing Company-owned equipment. The Company shall supply the energy and maintain all equipment. In areas where the Company's facilities are underground or required to be placed underground or the customer requests underground facilities, the unmetered lighting system shall be served from underground cables pursuant to the provisions contained in this Rate Schedule. In all other areas, the unmetered lighting system shall normally be served from overhead lines pursuant to the provisions contained in this Rate Schedule.

Customer-Owned Option

The capacity requirements of the customer-owned Unmetered ~~Experimental~~ LED Lighting served under this rate shall be determined by the Company based on verifiable documentation supplied by the customer. The Company shall have the right to test such capacity requirements. In the event that said tests show capacity requirements different from those indicated by the documentation supplied by the customer, the Company's test capacity value shall be used for billing purposes.

In customer-owned systems, control equipment shall be furnished and owned by the Company. The customer shall furnish, install and maintain the equipment comprising the unmetered ~~experimental~~ LED lighting system including, but not limited to, poles, the overhead wires or underground cables between the luminaires and the supply circuits extending to the point of attachment with the Company's lines. The customer's ~~experimental~~ LED lighting fixtures and equipment must be approved in advance by the Company before purchase and installation for service under this rate. The Company shall connect the customer's equipment to the Company's lines in a manner consistent with the Company's engineering standards, supply the energy and control the burning hours of the experimental lighting. Maintenance and replacement of the customer-owned equipment shall be the responsibility of the customer.

Existing unmetered installations with customer-owned fixtures on Company-owned distribution equipment must be converted to the customer-owned system described above or the Company-owned system described below to receive service under this Rate Schedule. Such installations may also be converted to a customer-owned metered system and receive service under Rate Schedule GML. Conversion costs shall be the responsibility of the customer.

Facilities Policy:

Company-Owned Option

~~At the customer's request and if~~ Following execution of a written agreement, the Company shall install ~~experimental~~ LED lighting and associated facilities ~~it will make~~ available under this rate under the following guidelines:

- A. The installation of all new, standard unmetered lights shall require a customer contribution of \$100 per luminaire. This policy includes the extension of up to 350 feet of distribution facilities to serve any individual light. Any extension beyond this amount shall require a contribution based on the Company's general service line extension policy. For unmetered lighting systems ~~installed~~ fed by underground electric lines, the customer shall be required to contribute the estimated difference in cost between the equivalent standard overhead construction and required underground construction.
- B. The conversion of existing unmetered lights to LED shall require a customer contribution per luminaire equal to the incremental additional cost to be incurred by the Company, ~~less a discount~~ A credit of \$200 per light shall be applied to the incremental cost for the conversion of existing luminaires that are closed to new business ~~if when~~ converted to the ~~nearest equivalent fixture size available and approved~~ luminaire recommended by the Company.
- C. For light upgrades, such as the replacement of fixtures to a size greater or less than the next equivalent value, Company expenditures for additional facilities beyond those described above shall be calculated in accordance with the Company's general service line extension policy.

(Continued on Sheet No. D-94.00)

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. D-94.00

GENERAL UNMETERED ~~EXPERIMENTAL~~ LIGHT EMITTING DIODE
LIGHTING RATE GU-~~XL~~ LED
(Continued From Sheet No. D-93.00)

Facilities Policy (Contd)

Company-Owned Option (Contd)

- D. The Company will determine ~~the type and size of all experimental~~ LED lighting fixtures to be offered under this rate. The list of approved fixtures is subject to modification at the sole discretion of the Company to accommodate new product development and advances in technology. Upon customer request, the Company shall provide a list of ~~experimental~~ LED lighting available under this rate.
- E. For customer requested material requiring special order, an additional per luminaire per month charge may apply for procurement and material handling. The Company and the Customer shall mutually agree to the monthly charge prior to procurement and installation of the special order material.
- F. The Company shall determine all associated equipment necessary to provide service under the Company-Owned Unmetered ~~Experimental~~ LED Lighting option.
- ~~FG.~~ Any charges, deposits or contributions may be required in advance of commencement of construction.
- ~~GH.~~ At the Company's discretion, any ~~failed lighting~~ fixtures may be converted to ~~an equivalent~~ LED at no cost to the customer ~~if the customer agrees to the conversion~~. The replaced fixture will ~~then~~ be moved to General Unmetered Light Emitting Diode ~~Experimental~~ Lighting Rate LED GU-XL upon completion of the installation and reconciliation of the community's streetlighting inventory for billing accuracy.

Customer-Owned Option

If it is necessary for the Company to install distribution facilities to serve a customer-owned system, contributions and/or deposits for such additional facilities shall be calculated in accordance with the Company's general service line extension policy. Any charges, deposits or contributions may be required in advance of commencement of construction.

Monthly Rate

Transitional Power Supply Charges, effective January 1, 2021 through June 30, 2021:

Power Supply Charges

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.029059</u>	\$0.000000	<u>\$0.029059</u>	per kWh for all kWh
0.048281		0.048281	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-4.00.

Delivery Charges Customer-Owned Option

Distribution Charge: \$0.098495 ~~0.042193~~ per kWh for all kWh

Delivery Charges Company-Owned Option

Distribution Charge: \$0.120810 ~~0.051752~~ per kWh for all kWh

Fixture Charge per Luminaire: \$5.00 ~~6.00~~ per month

Company-Owned Conversion Credit:

A conversion credit may be available to Customers who converted to LED municipal streetlighting.

Customers who converted to LED streetlighting before February 28, 2017 are eligible for the following Conversion Credit per billing month beginning with the January 2021 billing month through the December 2024 billing month:

Fixture Credit per Luminaire: \$(3.52) per month

(Continued on Sheet No. D-94.10)

M.P.S.C. No. 14 - Electric
Consumers Energy Company

Sheet No. D-94.10

**GENERAL UNMETERED ~~EXPERIMENTAL~~ LIGHT EMITTING DIODE
LIGHTING RATE GU-~~XL~~ LED**
(Continued From Sheet No. D-94.00)

Monthly Rate (Contd)

Final Rates, effective for service rendered on and after July 1, 2021, following implementation period:

<u>Company-Owned Equipment</u>		<u>Energy Charges</u>			<u>Delivery</u>	<u>Monthly Cost</u>
		<u>Non-Capacity</u>	<u>Capacity</u>	<u>Total</u>		
<u>15-24 W</u>	<u>Per Light</u>	<u>\$0.28</u>	<u>\$0.00</u>	<u>\$0.28</u>	<u>\$5.28</u>	<u>\$5.56</u>
<u>25-34 W</u>	<u>Per Light</u>	<u>\$0.42</u>	<u>\$0.00</u>	<u>\$0.42</u>	<u>\$5.84</u>	<u>\$6.26</u>
<u>35-44 W</u>	<u>Per Light</u>	<u>\$0.56</u>	<u>\$0.00</u>	<u>\$0.56</u>	<u>\$6.40</u>	<u>\$6.96</u>
<u>45-54 W</u>	<u>Per Light</u>	<u>\$0.70</u>	<u>\$0.00</u>	<u>\$0.70</u>	<u>\$6.97</u>	<u>\$7.67</u>
<u>55-64 W</u>	<u>Per Light</u>	<u>\$0.84</u>	<u>\$0.00</u>	<u>\$0.84</u>	<u>\$7.53</u>	<u>\$8.37</u>
<u>65-74 W</u>	<u>Per Light</u>	<u>\$0.98</u>	<u>\$0.00</u>	<u>\$0.98</u>	<u>\$8.09</u>	<u>\$9.07</u>
<u>75-84 W</u>	<u>Per Light</u>	<u>\$1.12</u>	<u>\$0.00</u>	<u>\$1.12</u>	<u>\$8.66</u>	<u>\$9.78</u>
<u>85-94 W</u>	<u>Per Light</u>	<u>\$1.26</u>	<u>\$0.00</u>	<u>\$1.26</u>	<u>\$9.22</u>	<u>\$10.48</u>
<u>95-104 W</u>	<u>Per Light</u>	<u>\$1.40</u>	<u>\$0.00</u>	<u>\$1.40</u>	<u>\$9.78</u>	<u>\$11.18</u>
<u>105-114 W</u>	<u>Per Light</u>	<u>\$1.54</u>	<u>\$0.00</u>	<u>\$1.54</u>	<u>\$10.35</u>	<u>\$11.89</u>
<u>115-124 W</u>	<u>Per Light</u>	<u>\$1.68</u>	<u>\$0.00</u>	<u>\$1.68</u>	<u>\$10.91</u>	<u>\$12.59</u>
<u>125-134 W</u>	<u>Per Light</u>	<u>\$1.82</u>	<u>\$0.00</u>	<u>\$1.82</u>	<u>\$11.47</u>	<u>\$13.29</u>
<u>135-144 W</u>	<u>Per Light</u>	<u>\$1.96</u>	<u>\$0.00</u>	<u>\$1.96</u>	<u>\$12.04</u>	<u>\$14.00</u>
<u>145-154 W</u>	<u>Per Light</u>	<u>\$2.10</u>	<u>\$0.00</u>	<u>\$2.10</u>	<u>\$12.60</u>	<u>\$14.70</u>
<u>155-164 W</u>	<u>Per Light</u>	<u>\$2.24</u>	<u>\$0.00</u>	<u>\$2.24</u>	<u>\$13.16</u>	<u>\$15.40</u>
<u>165-174 W</u>	<u>Per Light</u>	<u>\$2.38</u>	<u>\$0.00</u>	<u>\$2.38</u>	<u>\$13.73</u>	<u>\$16.11</u>
<u>175-184 W</u>	<u>Per Light</u>	<u>\$2.52</u>	<u>\$0.00</u>	<u>\$2.52</u>	<u>\$14.29</u>	<u>\$16.81</u>
<u>185-194 W</u>	<u>Per Light</u>	<u>\$2.66</u>	<u>\$0.00</u>	<u>\$2.66</u>	<u>\$14.85</u>	<u>\$17.51</u>
<u>195-204 W</u>	<u>Per Light</u>	<u>\$2.80</u>	<u>\$0.00</u>	<u>\$2.80</u>	<u>\$15.41</u>	<u>\$18.21</u>
<u>205-214 W</u>	<u>Per Light</u>	<u>\$2.94</u>	<u>\$0.00</u>	<u>\$2.94</u>	<u>\$15.97</u>	<u>\$18.91</u>
<u>Customer-Owned Equipment</u>		<u>Energy Charges</u>			<u>Delivery</u>	<u>Monthly Cost Per Light</u>
		<u>Non-Capacity</u>	<u>Capacity</u>	<u>Total</u>		
<u>15-24 W</u>	<u>Per Light</u>	<u>\$0.28</u>	<u>\$0.00</u>	<u>\$0.28</u>	<u>\$3.37</u>	<u>\$3.65</u>
<u>25-34 W</u>	<u>Per Light</u>	<u>\$0.42</u>	<u>\$0.00</u>	<u>\$0.42</u>	<u>\$3.75</u>	<u>\$4.17</u>
<u>35-44 W</u>	<u>Per Light</u>	<u>\$0.56</u>	<u>\$0.00</u>	<u>\$0.56</u>	<u>\$4.14</u>	<u>\$4.70</u>
<u>45-54 W</u>	<u>Per Light</u>	<u>\$0.70</u>	<u>\$0.00</u>	<u>\$0.70</u>	<u>\$4.53</u>	<u>\$5.23</u>
<u>55-64 W</u>	<u>Per Light</u>	<u>\$0.84</u>	<u>\$0.00</u>	<u>\$0.84</u>	<u>\$4.92</u>	<u>\$5.76</u>
<u>65-74 W</u>	<u>Per Light</u>	<u>\$0.98</u>	<u>\$0.00</u>	<u>\$0.98</u>	<u>\$5.30</u>	<u>\$6.28</u>
<u>75-84 W</u>	<u>Per Light</u>	<u>\$1.12</u>	<u>\$0.00</u>	<u>\$1.12</u>	<u>\$5.70</u>	<u>\$6.82</u>
<u>85-94 W</u>	<u>Per Light</u>	<u>\$1.26</u>	<u>\$0.00</u>	<u>\$1.26</u>	<u>\$6.08</u>	<u>\$7.34</u>
<u>95-104 W</u>	<u>Per Light</u>	<u>\$1.40</u>	<u>\$0.00</u>	<u>\$1.40</u>	<u>\$6.47</u>	<u>\$7.87</u>
<u>105-114 W</u>	<u>Per Light</u>	<u>\$1.54</u>	<u>\$0.00</u>	<u>\$1.54</u>	<u>\$6.86</u>	<u>\$8.40</u>
<u>115-124 W</u>	<u>Per Light</u>	<u>\$1.68</u>	<u>\$0.00</u>	<u>\$1.68</u>	<u>\$7.25</u>	<u>\$8.93</u>
<u>125-134 W</u>	<u>Per Light</u>	<u>\$1.82</u>	<u>\$0.00</u>	<u>\$1.82</u>	<u>\$7.63</u>	<u>\$9.45</u>
<u>135-144 W</u>	<u>Per Light</u>	<u>\$1.96</u>	<u>\$0.00</u>	<u>\$1.96</u>	<u>\$8.03</u>	<u>\$9.99</u>
<u>145-154 W</u>	<u>Per Light</u>	<u>\$2.10</u>	<u>\$0.00</u>	<u>\$2.10</u>	<u>\$8.41</u>	<u>\$10.51</u>
<u>155-164 W</u>	<u>Per Light</u>	<u>\$2.24</u>	<u>\$0.00</u>	<u>\$2.24</u>	<u>\$8.80</u>	<u>\$11.04</u>
<u>165-174 W</u>	<u>Per Light</u>	<u>\$2.38</u>	<u>\$0.00</u>	<u>\$2.38</u>	<u>\$9.19</u>	<u>\$11.57</u>
<u>175-184 W</u>	<u>Per Light</u>	<u>\$2.52</u>	<u>\$0.00</u>	<u>\$2.52</u>	<u>\$9.58</u>	<u>\$12.10</u>
<u>185-194 W</u>	<u>Per Light</u>	<u>\$2.66</u>	<u>\$0.00</u>	<u>\$2.66</u>	<u>\$9.96</u>	<u>\$12.62</u>
<u>195-204 W</u>	<u>Per Light</u>	<u>\$2.80</u>	<u>\$0.00</u>	<u>\$2.80</u>	<u>\$10.36</u>	<u>\$13.16</u>
<u>205-214 W</u>	<u>Per Light</u>	<u>\$2.94</u>	<u>\$0.00</u>	<u>\$2.94</u>	<u>\$10.74</u>	<u>\$13.68</u>

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00 ~~Previous Rate GUL Company or Customer Owned Energy Efficiency Program customers shall pay monthly the Rate GUL Company Owned Energy Efficiency Program Surcharge shown on Sheet No. D-2.00 once they have converted to Rate GU-XL.~~

(Continued on Sheet No. D-95.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. D-95.00

GENERAL UNMETERED ~~EXPERIMENTAL~~ LIGHT EMITTING DIODE
LIGHTING RATE GU-~~XL~~ LED
(Continued From Sheet No. D-~~94.00~~ 94.10)

General Terms

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Due Date and Late Payment Charge

The due date of the customer bill shall be 21 days from the date of mailing. A late payment charge of 2% of the unpaid balance, net of taxes, shall be assessed to any bill which is not paid on or before the due date shown thereon.

Determination of Monthly Kilowatt-Hours and Burning Hours per Month Based on 4,200 Burning Hours per Year

The monthly kilowatt-hours shall be determined by multiplying the total capacity requirements in watts (including the lamps, ballasts, drivers, and control devices) times the monthly Burning Hours as defined below divided by 1,000. The customer shall not change the capacity requirements of the equipment owned by it without first notifying the Company in writing of such changes and the date that they shall be made, and modifying the lighting contract with the Company accordingly.

Jan	Feb	Mar	April	May	June	July	Aug	Sept	Oct	Nov	Dec	Total
457.8	382.2	369.6	306.6	264.6	226.8	252.0	298.2	336.0	399.0	432.6	474.6	4,200

Hours of Lighting

Unmetered ~~Experimental~~ LED Lighting shall be burning at all times when the natural general level of illumination is lower than about 3/4 footcandle, and under normal conditions this is approximately one-half hour after sunset until approximately one-half hour before sunrise. Lighting service will be supplied from dusk to dawn every night and all night on an operating schedule of approximately 4,200 hours per year.

Maintenance of Lighting:

The Company shall replace or repair, at its own cost, Company-Owned Unmetered ~~Experimental~~ LED Lighting equipment that is out of service. If, for some reason, the Company is not able to make such restoration within one full billing month from the date the outage is first reported to the Company, the Company shall provide a credit to the customer's bill for unmetered lighting service. The credit shall be applied to the customer's bill beginning with the second full billing month after the outage is reported.

Outages caused by factors beyond the Company's reasonable control as provided for in Rules C1.1, Character of Service, and C3., Emergency Electrical Procedures, of the Company's Electric Rate Schedule are not covered by this policy. Such outages would be handled consistent with the particular circumstances and no credit would be made for such outages.

~~Green Generation Program:~~

~~Customer contracts for participation in the Green Generation Program shall be available to any eligible customer as described in Rule C10.2, Green Generation Program.~~

~~A customer who participates in the Green Generation Program is subject to the provisions contained in Rule C10.2, Green Generation Program.~~

Term and Form of Contract:

All service under this rate shall require a written contract with an initial term of ~~two~~ five years or more.

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. D-96.00

GENERAL SERVICE UNMETERED RATE GU

Availability:

Subject to any restrictions, this rate is available to the US Government, any political subdivision or agency of the State of Michigan, and any public or private school district for filament and/or gaseous discharge lamp installations maintained for traffic regulation or guidance, as distinguished from street illumination and police signal systems. Lighting for traffic regulation may use experimental lighting technology including light-emitting diode (LED). This rate is also available to Community Antenna Television Service Companies (CATV), Wireless Access Companies or Security Camera Companies for unmetered Power Supply Units. Where the Company's total investment to serve an individual location exceeds three times the annual revenue to be derived from such location, a contribution to the Company shall be required for the excess.

This rate is not available for resale purposes, new roadway lighting or for Retail Open Access Service.

Nature of Service:

Customer furnishes and installs all fixtures, lamps, ballasts, controls, amplifiers and other equipment, including wiring to point of connection with Company's overhead or underground system, as directed by the Company. Company furnishes and installs, where required for center suspended overhead traffic light signals, messenger cable and supporting wood poles and also makes final connections to its lines. If, in the Company's opinion, the installation of wood poles for traffic lights is not practical, the customer shall furnish, install and maintain suitable supports other than wood poles. The customer shall maintain the equipment, including lamp renewals, and the Company shall supply the energy for the operation of the equipment. Conversion and/or relocation costs of existing facilities shall be paid for by the customer except when initiated by the Company.

The capacity requirements of the lamp(s), associated ballast(s) and control equipment for each luminaire shall be determined by the Company from the specifications furnished by the manufacturers of such equipment, provided that the Company shall have the right to test such capacity requirements from time to time. In the event that said tests shall show capacity requirements different from those indicated by the manufacturers' specifications, the capacity requirements shown by said tests shall control. The customer shall not change the capacity requirements of the equipment owned by it without first notifying the Company in writing of such changes and the date that they shall be made.

Monthly Rate:

Power Supply Charges:

Energy Charge:

Non-Capacity	Capacity	Total	
<u>\$0.047810</u>	<u>\$0.024416</u>	<u>\$0.072226</u>	per kWh for all kWh
0.055846	0.020186	0.076032	

This rate is subject to the Power Supply Cost Recovery (PSCR) Factor shown on Sheet No. D-6.00.

Delivery Charges:

System Access Charge: \$2.00 per customer per month

Distribution Charge: \$0.022542 per kWh for all kWh
~~0.017235~~

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00.

(Continued on Sheet No. D-97.00)

**M.P.S.C. No. 14 – Electric
Consumers Energy Company**

Sheet No. E-7.00

(Continued From Sheet No. E-6.00)

E2. ROA CUSTOMER SECTION (Contd)

E2.2 Metering (Contd)

Metering equipment for a ROA Customer shall be furnished, installed, read, maintained and owned by the Company.

For a ROA Customer with an Interval Data Meter that is not a Wireless Under Glass Meter, meter reading will be accomplished electronically through a ROA Customer-provided telephone line or other communication links that allow access to the meter data by the Company and are compatible with the Company's metering and billing systems. The communication link must be installed and operating prior to the ROA Customer receiving ROA Service.

A ROA ~~load-profiled~~ customer with maximum demand of 20 kW or less may receive meter reads by conventional means. If the ~~load-profiled~~ account exceeds a maximum demand of 20 kW and the customer does not have a Wireless Under Glass Meter, the customer will be required to install a communication line to access the Interval Data Meter electronically in order to continue ROA service if the customer is located in an area where electric Advanced Metering Infrastructure (AMI) transmitting technology meters are not available.

The ROA Customer, not being metered with a Wireless Under Glass Meter shall obtain a separate telephone line for such purposes paying all charges in connection therewith. The ROA Customer is responsible for assuring the performance of the telephone line or other communication links at the time of meter interrogation for billing purposes. If the Company is unable to access meter data electronically, the Company will retrieve the data manually. If the Company is unable to access meter data electronically for two or more billing months within a 12 month period, the Company will assess a \$45 charge for the second and all subsequent manual meter reads unless the inability to access the meter data electronically is the fault of the Company. The ROA Customer will be notified of the \$45 manual meter read policy following the first incident requiring a manual meter read within the 12 month period. In the event that the Company is unable to access meter data electronically for three consecutive months, the ROA Customer's ROA Service shall be terminated and the ROA Customer shall be transferred to Company Full Service and be subject to the "Return to Company Full Service" provision unless telephonic access failure is due to non-performance of the telecommunications service provider or the Company. The 60-day notice requirement to terminate the ROA Customer's service does not apply in the event the Company is unable to access the ROA Customer's meter data electronically for three consecutive months and is subsequently returned to Company Full Service. In the event the Company is unable to access the meter data electronically for 12 consecutive months due to non-performance of the telecommunications service provider, the customer will be returned to full service. It is the customer's responsibility to notify the Company the status of any known telephonic communication issues that may inhibit the Company's ability to access meter data electronically.

A hardship exception may be made for cases where installation of both land-line and cellular telephone service is impractical and a Wireless Under Glass Meter is not an option. The burden of proving hardship rests on the customer. If the hardship exception is granted, the customer's meter will be manually read once a month, on a date the Company selects, for an additional charge of \$45 month.

For a Wireless Under Glass Meter, an Energy-Only Registering or Energy and Maximum Demand Registering metered ROA Customer, the meter will be read by conventional means and the ROA Customer will not be required to provide a telephone service or other communication link.

E2.3 Character of Service

- A. Refer to the "Nature of Service" provision of the applicable ROA Rate Schedule.
- B. The ROA Customer with a monthly-Maximum Demand greater than or equal to 1,000 kW is not required to utilize an Aggregator.

(Continued on Sheet No. E-8.00)

(Continued From Sheet No. E-19.00)

E3. RETAILER SECTION (Contd)

~~E3.7 Load Profiling~~

~~Retailers with ROA Customers who do not have an Interval Data Meter or a Wireless Under Glass Meter shall comply with the following provisions:~~

~~A. The Company will provide the Retailer with the rate class profile and applicable loss factor for the Retailer's customers as a basis for scheduling energy with MISO and reporting energy to MISO. The rate class profile will be the most recent profile approved for the Company by the MPSC.~~

~~B. Hourly Energy Reporting:~~

~~The Retailer or entity serving as the MDMA for the Retailer will report the hourly energy usage determined in (1) below to the MISO as the actual usage for the Retailer in the MISO energy market.~~

~~Hourly energy usage for MISO settlement shall be determined as follows:~~

~~(1) The Power consumed by the Retailer's ROA Customers shall be determined as the total of (a) and (b) as follows:~~

~~(a) For customers with Interval Data Meters or Wireless Under Glass Meters, by actual hourly energy usage, adjusted for losses.~~

~~(b) For customers with Energy Only Registering Meters or Energy and Demand Registering Meters, hourly usage data for these customers will be determined by the use of the profile for the customer class to distribute the total weather adjusted usage (actual or estimated) in the billing period across all the hours in that billing period, adjusted for losses.~~

E3.8 Customer Protections

The maximum early termination fee for residential contracts of one year or less shall not exceed \$50. The maximum early termination fee for residential contracts of longer than one year shall not exceed \$100. It is the Retailer's responsibility to have a current valid contract with the customer at all times. Any contract that is not signed by the customer or Legally Authorized Person shall be considered null and void. Only the customer account holder or Legally Authorized Person shall be permitted to sign a contract. A Retailer and its agent shall make reasonable inquiries to confirm that the individual signing the contract is a Legally Authorized Person. For each customer, a Retailer must be able to demonstrate that a customer has made a knowing selection of the Retailer by at least one of the following verification records:

- (1) An original signature from the customer account holder or Legally Authorized Person.
- (2) Independent third party verification with an audio recording of the entire verification call.
- (3) An e-mail address if signed up through the Internet.

The Commission or its Staff may request a reasonable number of records from a Retailer to verify compliance with this customer verification provision, and in addition, may request records for any customer due to a dispute.

A Retailer must distribute a confirmation letter to residential customers by U.S. mail. The confirmation letter must be postmarked within seven (7) days of the customer or Legally Authorized Person signing a contract with the Retailer. The confirmation letter must include the date the letter was sent, the date the contract was signed, the term of the contract with end date, the fixed or variable rate charged, the unconditional cancellation period, any early termination fee, the Retailer's phone number, the Commission's toll-free number and the Company's emergency contact information.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. E-22.00

RETAIL OPEN ACCESS RESIDENTIAL SECONDARY RATE ROA-R

(Continued From Sheet No. E-21.00)

RETAILER

Monthly Rate - Retailer:

Transmission Service:

Subject to Rule E1.5, Transmission Service must be obtained from the appropriate transmission service providers and the charges for such service shall be as specified in the Applicable FERC Open Access Tariff.

Real Power Losses:

The Retailer is responsible for replacing Real Power Losses of 8.082 ~~7.643~~% on the Company's Distribution System associated with the movement of Power and for compensation for losses.

General Terms and Conditions:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Term and Form of Contract - Retailer:

All service under this rate shall require a written ROA Service Contract between the Company and a Retailer.

ROA CUSTOMER

Monthly Rate – ROA Customer:

ROA System Access Charge, Distribution Charge, General Terms, Minimum Charge and Due Date and Late Payment Charge:

The System Access Charge, Distribution Charge, General Terms, Minimum Charge and the Due Date and Late Payment Charge shall be as provided for under the ROA Customer's otherwise applicable Company Full Service rate.

This rate is subject to the Surcharges shown on Sheet Nos. D-2.00 through D-5.00 and the Power Plant Securitization Charges shown on Sheet No. D-7.00. Customers taking ROA service on December 6, 2013 are excluded from the Power Plant Securitization Charges. This exclusion does not apply to customers first taking ROA service after December 6, 2013 or to customers taking service on December 6, 2013 who discontinue taking ROA service any time after December 6, 2013. Customers who discontinue taking ROA service any time after December 6, 2013 and who return to ROA service shall pay the Power Plant Securitization Charges applicable to the customer's otherwise applicable Company Full Service Rate Schedule.

State Reliability Mechanism for ROA:

Beginning June 1, 2018 all ROA customers may be subject to a State Reliability Mechanism Capacity Charge. This charge shall not apply to ROA customers for any planning year in which their Alternative Electric Supplier can demonstrate to the Commission that it can meet its capacity obligations by the seventh business day of February each year starting in 2018.

If a capacity charge is required to be paid in the planning year beginning June 1, 2018, or any of the three subsequent planning years, due to the Alternative Electric Supplier not meeting its capacity obligations, then the capacity charge is applicable for each of those planning years. Any capacity charged required to be paid any time after the first initial four-year period shall be applicable for a single year. The planning year is defined as being June 1 through the following May 31 of each year. The capacity charge paid by ROA customers will be the same amount as a Full Service Customer on the otherwise applicable Rate Schedule. Non-capacity charges shall not apply.

ROA Customer Switching Service Charge:

A \$5.00 switching fee shall be charged the ROA Customer each time a ROA Customer switches (i) from one Retailer to another or (ii) from ROA to a Company Full Service rate. The ROA Customer may switch Retailers at the end of any billing month by having their new Retailer give the Company at least 30 days' written notice. The Company will notify the ROA Customer's previous Retailer and new Retailer electronically of the effective date of the switch. The ROA Customer may choose to return to Company Full Service at the end of any billing month in compliance with Rule E2.5 D., Return to Company Full Service - Residential ROA Customers. The ROA Customer Switching Service Charge shall not be applied (i) for the initial switch to ROA Service or (ii) at the time the ROA Customer returns to Company Full Service or another Retailer because the ROA Customer was Slammed by the Retailer.

Term and Form of Contract - ROA Customer:

Service under this rate shall not require a ROA Service Contract between the Company and a ROA Customer.

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Sheet No. E-24.00

RETAIL OPEN ACCESS SECONDARY RATE ROA-S
(Continued From Sheet No. E-23.00)

Metering Requirements:

The ROA Customer with a Maximum Demand of less than 20 kW shall be separately metered by a Wireless Under Glass Meter or an Energy Registering Meter, with or without maximum demand registers, of billing quality. Such metering equipment shall be furnished, installed, maintained and owned by the Company.

The ROA Customer with a Maximum Demand of less than 20 kW may elect to install an Interval Data Meter. Such metering equipment shall be furnished, installed, maintained and owned by the Company. The requesting ROA Customer shall be required to pay the System Access Charge, as provided for under the ROA Customer's otherwise applicable Company Full Service rate, for all such metering equipment.

The ROA Customer with a Maximum Demand of 20 kW or more shall be separately metered by a Wireless Under Glass Meter or an Interval Data Meter of billing quality. Such metering equipment shall be furnished, installed, maintained and owned by the Company. The ROA Customer shall be required to pay the System Access Charge, as provided for under the ROA Customer's otherwise applicable Company Full Service rate, for all such metering equipment.

The ROA Customer with an Interval Data Meter shall be responsible for (i) the communication links that allow access to the meter data by the Company and are compatible with the Company's metering and billing systems, and (ii) all associated costs relating to the communication links including other accompanying equipment and monthly fees.

RETAILER:

Monthly Rate - Retailer:

Transmission Service:

Subject to Rule E1.5, Transmission Service must be obtained from the appropriate transmission service providers and the charges for such service shall be as specified in the Applicable FERC Open Access Tariff.

Real Power Losses:

The Retailer is responsible for replacing Real Power Losses of ~~7.643~~ 8.082% on the Company's Distribution System associated with the movement of Power and for compensation for losses.

General Terms and Conditions:

This rate is subject to all general terms and conditions shown on Sheet No. D-1.00.

Term and Form of Contract - Retailer:

All service under this rate shall require a written ROA Service Contract between the Company and a Retailer.

(Continued on Sheet No. E-25.00)

M.P.S.C. No. 14 – Electric
Consumers Energy Company

Original Sheet No. E-26.00

RETAIL OPEN ACCESS PRIMARY RATE ROA-P

Availability:

Subject to any restrictions, this rate is available to any customer receiving service at a Primary Voltage for the delivery of Power from the Point of Receipt to the Point of Delivery and for resale service in accordance with Rule C4.4, Resale.

This rate is not available to a ROA-P Customer where the Company elects to provide one transformation from the available Primary Voltage to another available Primary Voltage desired by the customer. This ROA Customer must take service under Retail Open Access Secondary Rate ROA-S.

This rate is not available for unmetered general service or for any unmetered or metered lighting service.

Service under this rate shall be separately metered. The Retailer shall deliver a flat, fixed amount of power every hour of every day.

Any ROA Customer whose monthly minimum Maximum Demand is less than 1,000 kW must utilize an Aggregator.

Nature of Service:

Service under this rate shall be alternating current, 60-Hertz, single-phase or three-phase (at the Company's option) Primary Voltage service. The Company will determine the particular nature of the voltage in each case.

The Company shall not be required to, but may expand its existing facilities to make deliveries under this tariff. The ROA Customer and/or Retailer shall be liable for any and all costs incurred as a result of an expansion of facilities made to make deliveries under this tariff.

Metering Requirements:

The load under this tariff shall be separately metered by a Wireless Under Glass Meter or an Interval Data Meter of billing quality. Such metering equipment shall be furnished, installed, maintained and owned by the Company. The ROA customer shall be required to pay the System Access Charge, as provided for under the ROA customer's otherwise applicable Company Full Service rate, for all such metering equipment.

The ROA Customer with an Interval Data Meter shall be responsible for (i) the communication links that allow access to the meter data by the Company and are compatible with the Company's metering and billing systems, and (ii) all associated costs relating to the communication links including other accompanying equipment and monthly fees.

RETAILER

Monthly Rate - Retailer:

Transmission Service:

Subject to Rule E1.5, Transmission Service must be obtained from the appropriate transmission service providers and the charges for such service shall be as specified in the Applicable FERC Open Access Tariff.

Real Power Losses:

The Retailer is responsible for replacing Real Power Losses as shown below on the Company's Distribution System associated with the movement of Power and for compensation for losses.

	Meter Point	
	High Side	Low Side
Customer Voltage Level 1	0.000%	<u>0.728</u> 0.705 %
Customer Voltage Level 2	<u>1.325</u> 1.271 %	<u>2.189</u> 2.366 %
Customer Voltage Level 3	<u>3.329</u> 3.221 %	<u>8.082</u> 7.643 %

(Continued on Sheet No. E-27.00)

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

MARC R. BLECKMAN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule D-1

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Overall Rate of Return Summary
for the Projected Year Ending December 31, 2021

Case No.: U-20697
Exhibit No.: A-14 (MRB-1)
Schedule: D-1
Page: 1 of 1
Witness: MRBleckman
Date: February 2020

Line No.	(a) Description	(b) Amount (\$000,000) (1)	(c) Capital Structure		(e) Cost Rate %	(g) Weighted Cost			
			(d) Percent Permanent Capital (2)	(d) Percent of Total Capital		(f) Permanent Capital (7)	(g) Total Cost % (8)	(h) Conversion Factor	(i) Pre-Tax Return (9)
1									
2	Long-Term Debt	\$ 8,178	47.29%	38.55%	3.95% (3)	1.87%	1.52%	1.0000	1.52%
3									
4	Preferred Stock	37	0.22%	0.18%	4.50% (4)	0.01%	0.01%	1.3391	0.01%
5									
6	Common Shareholder's Equity	<u>9,080</u>	<u>52.50%</u>	42.80%	10.50% (5)	5.51%	4.49%	1.3391	6.02%
7									
8	Total Permanent Capital	\$ 17,295	<u>100.00%</u>						
9									
10	Short-Term Debt	139		0.65%	3.46% (6)		0.02%	1.0000	0.02%
11									
12	Deferred Income Taxes	3,655		17.23%	0.00%		0.00%	1.0000	0.00%
13									
14	<u>Investment Tax Credit</u>								
15	Long-Term Debt	59		0.28%	3.95%		0.01%	1.0000	0.01%
16	Preferred Stock	0		0.00%	4.50%		0.00%	1.3391	0.00%
17	Common Equity	65		0.31%	10.50%		0.03%	1.3391	0.04%
18									
19	Total	<u>\$ 21,214</u>		<u>100.00%</u>			<u>6.09%</u>		<u>7.63%</u>

(1) See Exhibit A-14 (MRB-2), Schedule D-1a, Page 1.

(2) Excludes Short-term Debt, Deferred Income Taxes, and Investment Tax Credit to calculate the rate of return for Investment Tax Credit purposes in accordance with Internal Revenue Service Income Tax Regulation Section 1.46-6.

(3) See Exhibit A-14 (MRB-4), Schedule D-2.

(4) See Exhibit A-14 (MRB-6), Schedule D-4.

(5) See Exhibit A-14 (TAW-1), Schedule D-5.

(6) See Exhibit A-14 (MRB-5), Schedule D-3, Page 1.

(7) Column (c) x column (e).

(8) Column (d) x column (e).

(9) Column (g) x column (h).

Schedule D-1a

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Capital Structure Development
 for the Projected Year Ending December 31, 2021

Case No.: U-20697
 Exhibit No.: A-14 (MRB-2)
 Schedule: D-1a
 Page: 1 of 4
 Witness: MRBleckman
 Date: February 2020

Line No.	(a) Description	(b) Historical Capital Structure 13-Month Avg. For Yr. Ended December 31, 2018			(e) Balances as of Dec. 31, 2018 (000,000)	(f) Test Year Adjustments (000,000)	(g) Recommended Capital Structure		
		Amount Outstanding (000,000)	% of Permanent Capital	% of Total Capital			Amount Outstanding (000,000)	% of Permanent Capital	% of Total Capital
1									
2	Long-Term Debt	\$ 5,846	46.24%	36.48%	\$ 6,312	\$ 1,867	\$ 8,178	47.29%	38.55%
3									
4	Preferred Stock	37	0.30%	0.23%	37	-	37	0.22%	0.18%
5									
6	Common Equity	<u>6,758</u>	<u>53.46%</u>	42.18%	<u>6,905</u>	<u>2,175</u>	<u>9,080</u>	<u>52.50%</u>	42.80%
7									
8	Total Permanent Capital	\$ 12,641	<u>100.00%</u>		\$ 13,254	\$ 4,041	\$ 17,295	<u>100.00%</u>	
9									
10	Short-Term Debt, Incl Renewable Liability	125		0.78%	354	(215)	139		0.65%
11									
12	Deferred Income Taxes	3,163		19.74%	3,329	326	3,655		17.23%
13									
14	<u>Investment Tax Credit</u>								
15	Long-Term Debt	44		0.27%	47	12	59		0.28%
16	Preferred Stock	0		0.00%	0	0	0		0.00%
17	Common Equity	<u>50</u>		<u>0.31%</u>	<u>52</u>	<u>14</u>	<u>65</u>		<u>0.31%</u>
18									
19	Total Investment Tax Credit	<u>94</u>		<u>0.59%</u>	<u>99</u>	<u>25</u>	<u>125</u>		<u>0.59%</u>
20									
21	Total Capitalization	<u>\$ 16,023</u>		<u>100.00%</u>	<u>\$ 17,037</u>		<u>\$ 21,214</u>		<u>100.00%</u>

Sources and Support, by Column:

- (b) Consumers Energy General Ledger 13-month average balances as of December 31, 2018
- (c) Each line 2, 4, and 6 in column (b) is divided by line 8, column (b)
- (d) Each line 2, 4, 6, 10, 12, 15, 16, 17, and 19 in column (b) is divided by line 21, column (b)
- (e) Consumers Energy General Ledger balances as of December 31, 2018
- (f) Line 2 Debt maturities and debt issues, line 6 Adjustment for retained earnings and equity contributions, line 10 Adjustment to project short-term debt and renewable liability balance, line 12 Adjustment to project Deferred Income Taxes balance, lines 15-17 Adjustment to project Investment Tax Credit balance.
- (g) Column (e) + column (f). Represents 13-month averages.
- (h) Each line 2, 4, and 6 in column (g) is divided by line 8, column (g)
- (i) Each line 2, 4, 6, 10, 12, 15, 16, 17, and 19 in column (g) is divided by line 21, column (g)

Schedule D-1a

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Capital Structure Development - Projected Long-Term Debt Balance
for the Projected Year Ending December 31, 2021
(in millions)

Case No.: U-20697
Exhibit No.: A-14 (MRB-2)
Schedule: D-1a
Page: 2 of 4
Witness: MRBleckman
Date: February 2020

Line No.	(a) Long-Term Debt	(b) Actual Jan-18	(c) Actual Feb-18	(d) Actual Mar-18	(e) Actual Apr-18	(f) Actual May-18	(g) Actual Jun-18	(h) Actual Jul-18	(i) Actual Aug-18	(j) Actual Sep-18	(k) Actual Oct-18	(l) Actual Nov-18	(m) Actual Dec-18
1													
2	Beginning Balance	\$ 5,638	\$ 5,638	\$ 5,638	\$ 5,638	\$ 5,570	\$ 5,870	\$ 5,870	\$ 5,870	\$ 5,870	\$ 5,870	\$ 6,370	\$ 6,370
3	Add: Issuances	-	-	-	-	550	-	-	-	-	500	850	-
4	Less: Retirements	-	-	-	(68)	(250)	-	-	-	-	-	(850)	-
5	Subtotal	\$ 5,638	\$ 5,638	\$ 5,638	\$ 5,570	\$ 5,870	\$ 5,870	\$ 5,870	\$ 5,870	\$ 5,870	\$ 6,370	\$ 6,370	\$ 6,370
6													
7	Less: Unamortized Fees	(38)	(37)	(37)	(37)	(47)	(47)	(47)	(47)	(47)	(49)	(58)	(58)
8													
9	Ending Balance	\$ 5,600	\$ 5,601	\$ 5,601	\$ 5,533	\$ 5,823	\$ 5,823	\$ 5,823	\$ 5,823	\$ 5,823	\$ 6,321	\$ 6,312	\$ 6,312
10													
11													
12													
13	Long-Term Debt	Jan-19	Feb-19	Mar-19	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19
14													
15	Beginning Balance	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,996	\$ 7,071	\$ 7,071
16	Add: Issuances	-	-	-	-	300	-	-	-	626	75	-	-
17	Less: Retirements	-	-	-	-	(300)	-	-	-	-	-	-	-
18	Subtotal	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,370	\$ 6,996	\$ 7,071	\$ 7,071	\$ 7,071
19													
20	Less: Unamortized Fees	(56)	(56)	(56)	(56)	(62)	(63)	(63)	(63)	(73)	(73)	(74)	(72)
21													
22	Ending Balance	\$ 6,314	\$ 6,314	\$ 6,314	\$ 6,314	\$ 6,308	\$ 6,307	\$ 6,307	\$ 6,307	\$ 6,923	\$ 6,998	\$ 6,997	\$ 6,999
23													
24													
25													
26	Long-Term Debt	Jan-20	Feb-20	Mar-20	Apr-20	May-20	Jun-20	Jul-20	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20
27													
28	Beginning Balance	\$ 7,071	\$ 7,071	\$ 7,071	\$ 7,071	\$ 7,071	\$ 7,371	\$ 7,371	\$ 7,371	\$ 8,121	\$ 8,121	\$ 8,021	\$ 8,021
29	Add: Issuances	-	-	-	-	300	-	-	750	-	-	-	-
30	Less: Retirements	-	-	-	-	-	-	-	-	-	(100)	-	-
31	Subtotal	\$ 7,071	\$ 7,071	\$ 7,071	\$ 7,071	\$ 7,371	\$ 7,371	\$ 7,371	\$ 8,121	\$ 8,121	\$ 8,021	\$ 8,021	\$ 8,021
32													
33	Less: Unamortized Fees	(72)	(72)	(71)	(71)	(74)	(74)	(74)	(82)	(82)	(82)	(82)	(81)
34													
35	Ending Balance	\$ 6,999	\$ 6,999	\$ 7,000	\$ 7,000	\$ 7,297	\$ 7,297	\$ 7,297	\$ 8,039	\$ 8,039	\$ 7,939	\$ 7,939	\$ 7,940
36													
37													
38													
39	Long-Term Debt	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21
40													
41	Beginning Balance	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,646	\$ 8,646	\$ 8,646	\$ 8,646
42	Add: Issuances	-	-	-	-	-	-	-	625	-	-	-	-
43	Less: Retirements	-	-	-	-	-	-	-	-	-	-	-	-
44	Subtotal	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,021	\$ 8,646	\$ 8,646	\$ 8,646	\$ 8,646	\$ 8,646
45													
46	Less: Unamortized Fees	(81)	(81)	(81)	(80)	(80)	(80)	(80)	(86)	(86)	(86)	(86)	(85)
47													
48	Ending Balance	\$ 7,940	\$ 7,940	\$ 7,940	\$ 7,941	\$ 7,941	\$ 7,941	\$ 7,941	\$ 8,560	\$ 8,560	\$ 8,560	\$ 8,560	\$ 8,561
49													
50	13-Month Average:												
51	Subtotal												\$ 8,261
52	Less: Unamortized Fees												(83)
53	Ending Balance												<u>\$ 8,178</u>

13-Month Average

Schedule D-1a

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Capital Structure Development - Projected Common Equity Balance
for the Projected Year Ending December 31, 2021

(in millions)

Case No.: U-20697
Exhibit No.: A-14 (MRB-2)
Schedule: D-1a
Page: 3 of 4
Witness: MRBleckman
Date: February 2020

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	Common Equity	Actual Jan-18	Actual Feb-18	Actual Mar-18	Actual Apr-18	Actual May-18	Actual Jun-18	Actual Jul-18	Actual Aug-18	Actual Sep-18	Actual Oct-18	Actual Nov-18	Actual Dec-18
1													
2	Beginning Balance	\$ 6,464	\$ 6,672	\$ 6,612	\$ 6,705	\$ 6,639	\$ 6,680	\$ 6,880	\$ 6,828	\$ 6,898	\$ 6,914	\$ 6,804	\$ 6,851
3	Ret. Earnings	108	(60)	93	(66)	41	50	(52)	70	16	(110)	47	54
4	Equity Infusion	100	-	-	-	-	150	-	-	-	-	-	-
5	Ending Balance	<u>\$ 6,672</u>	<u>\$ 6,612</u>	<u>\$ 6,705</u>	<u>\$ 6,639</u>	<u>\$ 6,680</u>	<u>\$ 6,880</u>	<u>\$ 6,828</u>	<u>\$ 6,898</u>	<u>\$ 6,914</u>	<u>\$ 6,804</u>	<u>\$ 6,851</u>	<u>\$ 6,905</u>
6													
7													
8													
9	Common Equity	Jan-19	Feb-19	Mar-19	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19
10													
11	Beginning Balance	\$ 6,905	\$ 7,188	\$ 7,228	\$ 7,308	\$ 7,236	\$ 7,263	\$ 7,631	\$ 7,596	\$ 7,658	\$ 7,720	\$ 7,563	\$ 7,659
12	Ret. Earnings	(67)	40	80	(72)	27	43	(35)	63	62	(157)	96	54
13	Equity Infusion	350	-	-	-	-	325	-	-	-	-	-	-
14	Ending Balance	<u>\$ 7,188</u>	<u>\$ 7,228</u>	<u>\$ 7,308</u>	<u>\$ 7,236</u>	<u>\$ 7,263</u>	<u>\$ 7,631</u>	<u>\$ 7,596</u>	<u>\$ 7,658</u>	<u>\$ 7,720</u>	<u>\$ 7,563</u>	<u>\$ 7,659</u>	<u>\$ 7,713</u>
15													
16													
17													
18	Common Equity	Jan-20	Feb-20	Mar-20	Apr-20	May-20	Jun-20	Jul-20	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20
19													
20	Beginning Balance	\$ 7,713	\$ 7,725	\$ 8,086	\$ 8,098	\$ 8,110	\$ 8,122	\$ 8,433	\$ 8,445	\$ 8,457	\$ 8,469	\$ 8,480	\$ 8,492
21	Ret. Earnings	12	12	12	12	12	12	12	12	12	12	12	12
22	Equity Infusion	-	350	-	-	-	300	-	-	-	-	-	-
23	Ending Balance	<u>\$ 7,725</u>	<u>\$ 8,086</u>	<u>\$ 8,098</u>	<u>\$ 8,110</u>	<u>\$ 8,122</u>	<u>\$ 8,433</u>	<u>\$ 8,445</u>	<u>\$ 8,457</u>	<u>\$ 8,469</u>	<u>\$ 8,480</u>	<u>\$ 8,492</u>	<u>\$ 8,504</u>
24													
25													
26													
27	Common Equity	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21
28													
29	Beginning Balance	\$ 8,504	\$ 8,516	\$ 8,927	\$ 8,939	\$ 8,951	\$ 8,963	\$ 9,284	\$ 9,296	\$ 9,308	\$ 9,320	\$ 9,331	\$ 9,343
30	Ret. Earnings	12	12	12	12	12	12	12	12	12	12	12	12
31	Equity Infusion	-	400	-	-	-	310	-	-	-	-	-	-
32	Ending Balance	<u>\$ 8,516</u>	<u>\$ 8,927</u>	<u>\$ 8,939</u>	<u>\$ 8,951</u>	<u>\$ 8,963</u>	<u>\$ 9,284</u>	<u>\$ 9,296</u>	<u>\$ 9,308</u>	<u>\$ 9,320</u>	<u>\$ 9,331</u>	<u>\$ 9,343</u>	<u>\$ 9,355</u>

13-Month Average \$ 9,080

	Test Year Impact of Retained Earnings			Test Year Impact of Equity Infusions		
	2019	2020	2021	2019	2020	2021
Jan	\$ (67)	\$ 145	\$ 286	\$ 350	\$ 675	\$ 1,325
Feb	\$ (27)	\$ 156	\$ 297	\$ 350	\$ 1,025	\$ 1,725
Mar	\$ 53	\$ 168	\$ 309	\$ 350	\$ 1,025	\$ 1,725
Apr	\$ (19)	\$ 180	\$ 321	\$ 350	\$ 1,025	\$ 1,725
May	\$ 8	\$ 192	\$ 333	\$ 350	\$ 1,025	\$ 1,725
Jun	\$ 51	\$ 203	\$ 344	\$ 675	\$ 1,325	\$ 2,035
Jul	\$ 16	\$ 215	\$ 356	\$ 675	\$ 1,325	\$ 2,035
Aug	\$ 79	\$ 227	\$ 368	\$ 675	\$ 1,325	\$ 2,035
Sep	\$ 140	\$ 239	\$ 380	\$ 675	\$ 1,325	\$ 2,035
Oct	\$ (17)	\$ 250	\$ 391	\$ 675	\$ 1,325	\$ 2,035
Nov	\$ 79	\$ 262	\$ 403	\$ 675	\$ 1,325	\$ 2,035
Dec	\$ 133	\$ 274	\$ 415	\$ 675	\$ 1,325	\$ 2,035
13-Month Avg.		\$ 344				\$ 1,831

MICHIGAN PUBLIC SERVICE COMMISSION

Capital Structure Development - Projected Deferred Income Tax Balance

(in millions)

Date: February 2020

Line No.	(a) Actual Jan-18	(b) Actual Feb-18	(c) Actual Mar-18	(d) Actual Apr-18	(e) Actual May-18	(f) Actual Jun-18	(g) Actual Jul-18	(h) Actual Aug-18	(i) Actual Sep-18	(j) Actual Oct-18	(k) Actual Nov-18	(l) Actual Dec-18
1	\$ 3,140	\$ 3,138	\$ 3,126	\$ 3,122	\$ 3,118	\$ 3,152	\$ 3,148	\$ 3,144	\$ 3,150	\$ 3,146	\$ 3,265	\$ 3,329
2												
3												
4												
5	Jan-19	Feb-19	Mar-19	Apr-19	May-19	Jun-19	Jul-19	Aug-19	Sep-19	Oct-19	Nov-19	Dec-19
6	\$ 3,341	\$ 3,353	\$ 3,365	\$ 3,377	\$ 3,389	\$ 3,401	\$ 3,412	\$ 3,424	\$ 3,436	\$ 3,448	\$ 3,460	\$ 3,472
7												
8												
9												
10	Jan-20	Feb-20	Mar-20	Apr-20	May-20	Jun-20	Jul-20	Aug-20	Sep-20	Oct-20	Nov-20	Dec-20
11	\$ 3,483	\$ 3,494	\$ 3,505	\$ 3,516	\$ 3,527	\$ 3,539	\$ 3,550	\$ 3,561	\$ 3,572	\$ 3,583	\$ 3,594	\$ 3,605
12												
13												
14												
15	Jan-21	Feb-21	Mar-21	Apr-21	May-21	Jun-21	Jul-21	Aug-21	Sep-21	Oct-21	Nov-21	Dec-21
16	\$ 3,614	\$ 3,622	\$ 3,630	\$ 3,639	\$ 3,647	\$ 3,655	\$ 3,664	\$ 3,672	\$ 3,680	\$ 3,689	\$ 3,697	\$ 3,705
17												
18												
19										13-Month Average		\$ 3,655

Schedule D-1b

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Development of Capital Structure
for the Projected Year Ending December 31, 2021

Case No.: U-20697
Exhibit No.: A-14 (MRB-3)
Schedule: D-1b
Page: 1 of 1
Witness: MRBleckman
Date: February 2020

Line No.	(a) Description	(b) Financial Basis	(c) MPSC Ratemaking Basis
I.	LONG-TERM DEBT:		
a.	First Mortgage Bonds	Include	Include
b.	Trust Preferred Securities	Include	Include
c.	Other Subordinated LTD	Include	Include
d.	Unamortized Debt Premium	Include	Include
e.	Unamortized Debt Discount	Include	Include
f.	Unamortized Debt Expense	Include	Include
g.	Current Maturities	Exclude	Include
h.	Capitalized Leases	Include	Exclude
II.	SHORT-TERM DEBT	Include	Include
III.	PREFERRED STOCK EXPENSE		
a.	Preferred Stock	Include	Include
b.	Trust Preferred Securities	Exclude	Exclude
c.	Preferred Stock Expense	Include	Include
IV.	COMMON EQUITY		
a.	Common Stock Issued	Include	Include
b.	Premium on Common Stock	Include	Include
c.	Donations Received From Stockholders	Include	Include
d.	Common Stock Expense	Include	Include
e.	Gain on Reacquired Stock	Include	Include
f.	Miscellaneous Paid In Capital	Include	Include
g.	Mark-to-Market Accounting	Include	Exclude
h.	Appropriated Retained Earnings	Include	Include
i.	Unappropriated Retained Earnings	Include	Include
j.	FAS 90 (Abandoned Plant)	Include	Include
V.	DEFERRED ITC	Exclude	Include
VI.	DEFERRED TAXES	Exclude	Include
VII.	DEFERRED JDITC	Exclude	Include

Schedule D-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Short-Term Debt - Revolver / Commercial Paper
for the Projected Year Ending December 31, 2021
(in millions)

Case No.: U-20697
Exhibit No.: A-14 (MRB-5)
Schedule: D-3
Page: 1 of 2
Witness: MRBleckman
Date: February 2020

Line No.	Description	(a) Average Borrowings	(b) Cost of Borrowings	(c) Cost Rate
1	Short-Term Debt - Revolver / Commercial Paper	\$ 137.3	\$ 4.8	
2				
3	Short-Term Debt - Renewable Liability	1.5	-	
4				
5	Total Short-Term Debt	<u>\$ 138.8</u>	<u>\$ 4.8</u>	<u>3.46%</u>

Sources:

Column (a): Average borrowings per Exhibit A-14 (MRB-7), Schedule D-6.

Column (b): Short-Term Debt - Revolver cost per Exhibit A-14 (MRB-5), Schedule D-3, page 2.

Short-Term Debt - Renewable Liability cost equal to the average
borrowings (column (a)) times the projected interest on borrowings
rate of 2.65%, per Exhibit A-14 (MRB-5), Schedule D-3, page 2.

Column (c) = column (b)/column (a).

Schedule D-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Short-Term Debt - Revolver / Commercial Paper

for the Projected Year Ending December 31, 2021

(in millions)

Case No.: U-20697

Exhibit No.: A-14 (MRB-5)

Schedule: D-3

Page: 2 of 2

Witness: MRBleckman

Date: February 2020

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
	Summary				Test Year Assumption			
	Facility	Agreement Date	Expiration	Facility Amount	Less: Avg. Borrowings	Less: Letters of Credit	Amount Unused^a	Upfront Fee Amort
					(A)	(B)	(C)	(D)
1	1. JPMorgan Revolver	Jun. 2018	Jun. 2023	\$ 850.0	\$ -	\$ 7.1	\$ 842.9	\$ 0.2
2	2. Commercial Paper Facility	Sep. 2014	N/A	\$ 500.0	\$ 137.3	\$ -	\$ -	\$ -
3	3. Scotiabank Revolver	Nov. 2018	Nov. 2021	\$ 250.0	\$ -	\$ 10.0	\$ 240.0	\$ 0.3
4	4. JPMorgan Letter of Credit Facility 1	Sep. 2011	Apr. 2022	\$ 30.0	\$ -	\$ 30.0	\$ -	\$ -
5	5. JPMorgan Letter of Credit Facility 2	Aug. 2012	Aug. 2020	\$ 35.5	\$ -	\$ 35.5	\$ -	\$ -
	Cost of Short-Term Debt - Revolver / Commercial Paper							
	(A) Interest on Borrowings - Calculated on the projected drawn balance at LIBOR plus the spread on borrowings							
	Facility	LIBOR^b	Plus: Spread	Projected Rate	Avg. Borrowings	Cost		
	1. JPMorgan Revolver	2.10%	0.875%	2.98%	\$ -	\$ -		
	2. Commercial Paper Facility	2.10%	0.000%	2.10%	\$ 137.3	\$ 2.9		
	3. Scotiabank Revolver	2.10%	0.750%	2.85%	\$ -	\$ -		
	4. JPMorgan Letter of Credit Facility 1	N/A - Letter of Credit Facility			\$ -	\$ -		
	5. JPMorgan Letter of Credit Facility 2	N/A - Letter of Credit Facility			\$ -	\$ -		
						\$ 2.9		
	(B) Letter of Credit Fees - Calculated on the projected letters of credit outstanding at a rate equal to the spread.							
	Facility	Letter of Credit Type	Projected Rate	Letters of Credit	Cost			
	1. JPMorgan Revolver	Regular Operating	0.875%	\$ 7.1	\$ 0.1			
	2. Commercial Paper Facility	N/A	N/A	\$ -	\$ -			
	3. Scotiabank Revolver	MISO Margin	0.750%	\$ 10.0	\$ 0.1			
	4. JPMorgan Letter of Credit Facility 1	Palisades PPA	0.550%	\$ 30.0	\$ 0.2			
	5. JPMorgan Letter of Credit Facility 2	Tax Exempt Bonds	0.550%	\$ 35.5	\$ 0.2			
					\$ 0.6			
	(C) Unused (Commitment) Fees - Calculated on the unused portion of the revolver at a rate stated in the facility agreement.							
			Projected Rate	Amount Unused^a	Cost			
	1. JPMorgan Revolver		0.075%	\$ 842.9	\$ 0.6			
	2. Commercial Paper Facility		N/A	\$ -	\$ -			
	3. Scotiabank Revolver		0.075%	\$ 240.0	\$ 0.2			
	4. JPMorgan Letter of Credit Facility 1		0.400%	\$ -	\$ -			
	5. JPMorgan Letter of Credit Facility 2		N/A	\$ -	\$ -			
					\$ 0.8			
	(D) Amortization / Expense of Facility Fees - Fees paid upfront at the inception or amendment to the facility, amortized over the life of the facility.							
					Annual Cost			
	1. JPMorgan Revolver				\$ 0.2			
	2. Commercial Paper Facility				\$ -			
	3. Scotiabank Revolver				\$ 0.3			
	4. JPMorgan Letter of Credit Facility 1				\$ -			
	5. JPMorgan Letter of Credit Facility 2				\$ -			
					\$ 0.5			
	Total Cost of Short-Term Debt - Revolver / Commercial Paper					\$ 4.8		

^a Commercial Paper drawn balances go against, or are "backstopped," by the Company's JPMorgan Revolver.

To the extent amounts are borrowed under the Commercial Paper Facility, the availability of the JPMorgan Revolver are reduced.

These borrowings do not, however, reduce the "unused" portion of the revolver in calculating the unused (commitment) fees.

^b Projected LIBOR rate per Exhibit A-14 (MRB-4), Schedule D-2. Forecasted LIBOR assumed to closely approximate commercial paper rate.

Schedule D-4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Preferred Stock

for the Projected Year Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (MRB-6)

Schedule: D-4

Page: 1 of 1

Witness: MRBleckman

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
						Net Proceeds Received By Company	Number Of Shares Outstanding	Total Value Of Net Proceeds (000)	Cost Rates	Annual Cost (000)
Line No.	Description	Annual Dividend	Par Value	Finance Expense	(Premium) Discount					
	PREFERRED STOCK:									
1	\$4.500 Series	\$4.500	\$100.00	\$0.00	\$0.00	\$100.00	373,148	<u>\$37,315</u>	4.50%	<u>\$1,679</u>

Calculations:

Column (i) = (column (h)*column (b)/column (f))/column (h).

Column (j) = column (h)*column (i).

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Short-Term Debt Utilization
for the Projected Year Ending December 31, 2021
(in millions)

Ending Short-Term Debt - Revolver / Commercial Paper Balance

Ending Short-Term Debt - Renewable Liability Balance ^a

^aProjected year-end Renewable Liability balances are consistent with the Company's Renewable Energy Plan reflected in Case No. U-18231.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Current and Historical Credit Ratings
for the Projected Year Ending December 31, 2021

Case No.: U-20697
Exhibit No.: A-24 (MRB-8)
Page: 1 of 1
Witness: MRBleckman
Date: February 2020

Line No.	Credit Ratings					
	(a)	(b)	(c)	(d)	(e)	(f)
	Standard & Poors Ratings at Year End					
	Current	2018	2017	2016	2015	2014
1						
2						
3	Consumers Energy Company					
4	Senior Secured Debt	A	A	A	A	A
5	Commercial Paper	A-2	A-2	A-2	A-2	A-2
6	Senior Unsecured Debt	N/A	N/A	N/A	N/A	N/A
7	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
8	Preferred Stock	N/A	N/A	N/A	BB+	BB+
9	Outlook	Stable	Stable	Stable	Stable	Stable
10						
11	CMS Energy Corporation					
12	Senior Secured Debt	N/A	N/A	N/A	N/A	N/A
13	Senior Unsecured Debt	BBB	BBB	BBB	BBB	BBB
14	Junior Subordinated Debt	BBB-	BBB-	N/A	N/A	N/A
15	Hybrid Preferred Securities	N/A	N/A	N/A	BB+	BB+
16	Preferred Stock	N/A	N/A	N/A	N/A	N/A
17	Outlook	Stable	Stable	Stable	Stable	Stable
18						
19						
20		Moody's Ratings at Year End				
21	Current	2018	2017	2016	2015	2014
22	Consumers Energy Company					
23	Senior Secured Debt	Aa3	Aa3	A1	A1	A1
24	Commercial Paper	P-1	P-1	P-2	P-2	P-2
25	Senior Unsecured Debt	N/A	N/A	N/A	A3	A3
26	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
27	Preferred Stock	A3	A3	Baa1	Baa1	Baa1
28	Outlook	Stable	Stable	Positive	Stable	Stable
29						
30	CMS Energy Corporation					
31	Senior Secured Debt	A3	A3	Baa1	Baa1	Baa1
32	Senior Unsecured Debt	Baa1	Baa1	Baa2	Baa2	Baa2
33	Junior Subordinated Debt	Baa2	Baa2	N/A	N/A	N/A
34	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
35	Preferred Stock	N/A	N/A	N/A	N/A	N/A
36	Outlook	Stable	Stable	Positive	Stable	Stable
37						
38						
39		Fitch Ratings at Year End				
40	Current	2018	2017	2016	2015	2014
41	Consumers Energy Company					
42	Senior Secured Debt	A+	A+	A+	A-	A-
43	Commercial Paper	F-2	F-2	F-2	F-3	N/A
44	Senior Unsecured Debt	A	A	A	BBB+	BBB+
45	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
46	Preferred Stock	BBB+	BBB+	BBB+	BBB-	BBB-
47	Outlook	Stable	Stable	Stable	Stable	Stable
48						
49	CMS Energy Corporation					
50	Senior Secured Debt	BBB+	BBB+	BBB+	BBB-	BBB-
51	Senior Unsecured Debt	BBB	BBB	BBB	BBB-	BBB-
52	Junior Subordinated Debt	BB+	BB+	N/A	N/A	N/A
53	Hybrid Preferred Securities	N/A	N/A	N/A	N/A	N/A
54	Preferred Stock	N/A	N/A	N/A	N/A	N/A
55	Outlook	Stable	Stable	Stable	Stable	Stable

Source: S&P Global Market Intelligence (SNL Energy)								
(a) Line No.	(b) Company Name	(d) Dec 31, 2018 (amounts in \$000s)				(g) Long-Term Debt	(h) % of Total Preferred Stock	(i) Common Stock
		(c) Long-Term Debt	(d) Preferred Stock	(e) Common Stock	(f) Total			
1	<u>Alliant Energy Corporation</u>							
2	Interstate Power and Light Company	2,635,380	200,000	3,034,635	5,870,015			
3	Wisconsin Power and Light Company	1,953,011	0	2,175,425	4,128,436			
4		4,588,391	200,000	5,210,060	9,998,451	45.9%	2.0%	52.1%
5								
6	<u>Ameren Corporation</u>							
7	Ameren Illinois Company	3,310,035	61,632	3,711,714	7,083,381			
8	Ameren Transmission Company of Illinois	525,000	0	659,982	1,184,982			
9	Union Electric Company	3,751,502	80,760	4,148,988	7,981,250			
10		7,586,537	142,392	8,520,684	16,249,613	46.7%	0.9%	52.4%
11								
12	<u>American Electric Power Company</u>							
13	AEP Appalachian Transmission Company, Inc.	14,997	0	25,569	40,566			
14	AEP Generating Company	195,000	0	221,401	416,401			
15	AEP Generation Resources Inc.	NA	NA	NA	NA			
16	AEP Indiana Michigan Transmission Company, Inc.	775,600	0	973,069	1,748,669			
17	AEP Kentucky Transmission Company, Inc.	42,808	0	55,155	97,963			
18	AEP Ohio Transmission Company, Inc.	1,175,099	0	1,496,674	2,671,773			
19	AEP Oklahoma Transmission Company, Inc.	379,933	0	460,218	840,151			
20	AEP Southwestern Transmission Company, Inc.	0	0	-1,270	(1,270)			
21	AEP Texas Central Company	NA	NA	NA	NA			
22	AEP Texas Inc.	3,106,567	0	2,580,521	5,687,088			
23	AEP Texas North Company	NA	NA	NA	NA			
24	AEP West Virginia Transmission Company, Inc.	454,642	0	561,660	1,016,302			
25	Appalachian Power Company	4,085,297	0	4,006,253	8,091,550			
26	Indiana Michigan Power Company	2,919,741	0	2,352,848	5,272,589			
27	Kentucky Power Company	870,000	0	732,879	1,602,879			
28	Kingsport Power Company	50,000	0	51,596	101,596			
29	Ohio Power Company	1,677,282	0	2,297,416	3,974,698			
30	Public Service Company of Oklahoma	1,290,562	0	1,248,093	2,538,655			
31	Southwestern Electric Power Company	2,613,646	0	2,315,251	4,928,897			
32	Transource Maryland, LLC	3,533	0	2,538	6,071			
33	Transource Missouri	123,144	0	150,872	274,016			
34	Transource Pennsylvania, LLC	8,407	0	6,068	14,475			
35	Transource West Virginia, LLC	16,956	0	28,483	45,439			
36	Wheeling Power Company	350,000	0	421,269	771,269			
37		20,153,214	0	19,986,563	40,139,777	50.2%	0.0%	49.8%
38								
39	<u>Dominion Resources, Inc.</u>							
40	Dominion Energy Carolina Gas Transmission, LLC	0	0	446,844	446,844			
41	Dominion Energy Cove Point LNG, LP	3,000,000	0	5,056,663	8,056,663			
42	Dominion Energy Overthrust Pipeline, LLC	55,000	0	158,201	213,201			
43	Dominion Energy Questar Pipeline, LLC	428,513	0	533,856	962,369			
44	Dominion Energy South Carolina, Inc.	5,078,121	100	4,135,362	9,213,583			
45	Dominion Energy Transmission, Inc.	1,895,000	0	1,850,014	3,745,014			
46	Hope Gas, Inc.	105,297	0	122,432	227,729			
47	Public Service Company of North Carolina, Incorporated	700,000	0	860,689	1,560,689			
48	Questar Gas Company	750,000	0	1,011,808	1,761,808			
49	Questar Southern Trails Pipeline Company	64,120	0	-17,127	46,993			
50	SCANA Corporation	5,151,386	100	4,308,698	9,460,184			
51	South Carolina Generating Company, Inc.	73,265	0	173,336	246,601			
52	The East Ohio Gas Company	1,300,000	0	1,728,096	3,028,096			
53	Virginia Electric and Power Company	11,748,117	0	13,046,045	24,794,162			
54		30,348,819	200	33,414,917	63,763,936	47.6%	0.0%	52.4%
55								
56	<u>DTE Energy Company</u>							
57	Citizens Gas Fuel Company	6,541,391	0	6,797,686	13,339,077			
58	DTE Electric Company	1,549,633	0	1,668,209	3,217,842			
59	DTE Gas Company	0	0	1,709	1,709			
60		8,091,024	0	8,467,604	16,558,628	48.9%	0.0%	51.1%
61								
62	<u>Evergy, Inc.</u>							
63	Evergy Kansas Central, Inc.	3,716,640	0	6,972,875	10,689,515			
64	Evergy Kansas South, Inc.	970,877	0	2,907,333	3,878,210			
65	Evergy Metro, Inc.	2,544,607	0	2,493,301	5,037,908			
66	Evergy Missouri West, Inc.	975,114	0	1,178,137	2,153,251			
67	Great Plains Energy Incorporated	3,519,721	0	3,671,438	7,191,159			
68	Westar Energy (KPL)	2,745,763	0	3,964,902	6,710,665			
69	Westar Generating, Inc.	0	0	100,640	100,640			
70		14,472,722	0	21,288,626	35,761,348	40.5%	0.0%	59.5%
71								
72	<u>NISource, Inc.</u>							
73	Bay State Gas Company	428,646	0	-4,343	424,303			
74	Central Kentucky Transmission Company	0	0	644	644			
75	Columbia Gas of Kentucky, Incorporated	127,375	0	153,333	280,708			
76	Columbia Gas of Maryland, Incorporated	49,355	0	64,624	113,979			
77	Columbia Gas of Ohio, Inc.	1,332,926	0	1,589,969	2,922,895			
78	Columbia Gas of Pennsylvania, Inc.	705,515	0	886,397	1,591,912			
79	Columbia Gas of Virginia, Incorporated	371,275	0	276,735	648,010			
80	Northern Indiana Public Service Company	2,144,367	0	2,770,972	4,915,339			
81		5,159,459	0	5,738,331	10,897,790	47.3%	0.0%	52.7%
82								
83	<u>OGE Energy Corp.</u>							
84	Oklahoma Gas and Electric Company	3,169,826	0	3,603,266	6,773,092	46.8%	0.0%	53.2%
85								
86	<u>Pinnacle West Capital Corp.</u>							
87	Arizona Public Service Company	4,752,874	0	5,661,007	10,413,881	45.6%	0.0%	54.4%
88								
89	<u>Portland General Electric Company</u>							
90	Portland General Electric Company	2,487,382	0	2,506,442	4,993,824	49.8%	0.0%	50.2%
91								
92	<u>WEC Energy Group, Inc.</u>							
93	Integrus Holding, Inc.	NA	NA	NA	NA			
94	Michigan Gas Utilities Corporation	90,000	0	163,121	253,121			
95	Minnesota Energy Resources Corporation	NA	NA	NA	NA			
96	North Shore Gas Company	132,000	0	142,651	274,651			
97	Peoples Gas Light and Coke Company	1,195,000	0	1,459,276	2,654,276			
98	Upper Michigan Energy Resources Corporation	150,000	0	133,088	283,088			
99	Wisconsin Electric Power Company	2,715,643	30,450	3,460,847	6,206,940			
100	Wisconsin Gas LLC	489,434	0	844,358	1,333,792			
101	Wisconsin Public Service Corporation	1,324,322	0	1,779,477	3,103,799			
102		6,096,399	30,450	7,982,818	14,109,667	43.2%	0.2%	56.6%
103								
104	<u>Xcel Energy Inc.</u>							
105	Northern States Power Company - MN	4,979,364	0	5,573,159	10,552,523			
106	Northern States Power Company - WI	815,642	0	942,375	1,758,017			
107	Public Service Company of Colorado	4,885,966	0	6,298,264	11,184,230			
108	Southwestern Public Service Company	2,146,459	0	2,536,626	4,683,085			
109	WestGas InterState, Inc.	0	0	706	706			
110		12,827,431	0	15,351,130	28,178,561	45.5%	0.0%	54.5%
111								
112	Average Proxy Group					46.5%	0.0%	53.2%

Column (b) = Company ROE witness Wehner's proxy group from Exhibit A- (TAW-1), Schedule D-5, page 1.
Columns (c), (d) & (e): Balances at December 31, 2018 per S&P Global Market Intelligence (formerly SNL Energy).
Data for each company is equal to the sum of the regulated subsidiaries of each proxy group company (where applicable).
Column (f) = sum (c), (d), (e).
Column (g) = (c)/(f).
Column (h) = (d)/(f).
Column (i) = (e)/(f).

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Rating Agency Adjusted FFO Analysis
for the Projected Year Ending December 31, 2021

Case No.: U-20697
Exhibit No.: A-27 (MRB-11)
Page: 1 of 1
Witness: MRBleckman
Date: February 2020

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	
	(000's)						
	Consumers Energy						
	<u>S&P</u>	<u>2018 - Actual</u>	<u>Adjustment: Tax Reform</u>	<u>2018 - Adjusted Tax Reform</u>	<u>Adjustment: 52.05% Equity Ratio</u>	<u>Adjustment: 9.90% ROE</u>	<u>2018 - Adjusted Tax Reform / Equity Ratio / ROE</u>
1	FFO (adjusted)	\$ 1,665	(138)	\$ 1,527	(18)	(19)	\$ 1,490
2	Debt (adjusted)	<u>\$ 7,774</u>	<u>69</u>	<u>\$ 7,843</u>	<u>178</u>	<u>-</u>	<u>\$ 8,021</u>
3	FFO / Debt (Line 1 / Line 2)	21.4%		<u>19.5%</u>			<u>18.6%</u>

	(000's)						
	Consumers Energy						
	Moody's	2018 - Actual	Adjustment: Tax Reform	2018 - Adjusted Tax Reform	Adjustment: 52.05% Equity Ratio	Adjustment: 9.90% ROE	2018 - Adjusted Tax Reform / Equity Ratio / ROE
4	CFO Pre-W/C (adjusted)	\$ 1,659	(138)	\$ 1,521	(18)	(19)	\$ 1,484
5	Debt (adjusted)	\$ 7,505	69	\$ 7,574	178	-	\$ 7,752
6	(CFO Pre-W/C) / Debt	22.1%		20.1%			19.1%
	(Line 4 / Line 5)						

Column (a): For S&P, data from S&P's January 29, 2020 credit opinion for Consumers Energy. For Moody's, data from Moody's June 19, 2019 credit opinion for Consumers Energy.

Column (b): FFO reduction \$(161) for Credit A and \$(46) for Calc C, offset by \$69 Credit A already applied in 2018 for a total of \$(138). Debt and equity assumed to increase equally.

Column (c): Lines 1, 2, 4, and 5, equal to Column (a) + Column (b)

Column (d): Equity ratio reduced from 53.46% in 2018 to 52.05%.

Column (e): Total Company earned ROE in 2018 from 10.5% to 9.9%.

Column (f): Lines 1, 2, 4, and 5, equal to Column (c) + Column (d) + Column (e)



Rating Action **Moody's downgrades DTE Gas to A3 from A2; outlook stable**

22 Jul 2019

Approximately \$1.3 billion of debt downgraded

New York, July 22, 2019 -- Moody's Investors Service ("Moody's") downgraded ratings of DTE Gas Company (DTE Gas) including its senior unsecured rating to A3 from A2, senior secured rating to A1 from A2, and short-term rating for commercial paper to Prime-2 from Prime-1. The outlook was revised to stable from negative. See the debt list below for a complete list of rating actions.

Downgrades:

..Issuer: DTE Gas Company

....Commercial Paper, Downgraded P-2 from P-1

....Senior Secured First Mortgage Bonds Downgraded to A1 from Aa3

....Senior Secured Regular Bond/Debenture Downgraded to A1 from Aa3

....Underlying Senior Secured Regular Bond/Debenture Downgraded to A1 from Aa3

....Senior Unsecured Bank Credit Facility Downgraded to A3 from A2

Outlook Actions:

..Issuer: DTE Gas Company

....Outlook, Changed To Stable From Negative

RATINGS RATIONALE

"The robust investment program of DTE Gas, combined with the negative cash flow effect of federal tax reform, continue to put pressure on its financial metrics, weakening its overall profile," stated Jairo Chung, Moody's analyst. "The downgrade reflects our expectation that these lower financial metrics will continue on a sustained basis," added Chung.

DTE Gas plans to invest approximately \$2.5 billion over the next five years with approximately \$1.3 billion of the investments related to its gas main renewal and accelerated pipeline integrity. These investments are expected to be financed mostly with debt, increasing the company's leverage and pressuring its financial metrics. In addition, the negative impact of federal tax reform resulted in a reduction in its operating cash flow as a result of lower retail rates and a customer refund related to excess deferred taxes. Moody's expects DTE Gas' cash flow from operations before changes in working capital (CFO pre-WC) to decline around 17% on a sustained basis, which is more characteristic of an A3 rating.

As a low risk natural gas local distribution company (LDC), DTE Gas operates in a credit supportive regulatory environment in Michigan. DTE Gas benefits from limited regulatory lag for its investment cost recovery and a suite of rate adjustment mechanisms such as an infrastructure recovery mechanism and a revenue decoupling mechanism. Moody's expects the constructive regulatory environment in Michigan to continue.

Rating Outlook

The stable outlook reflects Moody's expectation that the Michigan regulatory environment will remain supportive. Also, the stable outlook incorporates our expectation that DTE Gas will produce stable and predictable financial metrics that are supportive of its A3 rating.

Liquidity

DTE Gas's liquidity is expected to be adequate over the next 12 months. DTE Gas has access to a \$300

million bank credit facility expiring in April 2024, which was extended in April 2019, primarily used to backstop its \$300 million commercial paper program. Over the next 12 months, DTE Gas's capital expenditures and its dividend to the parent, DTE Energy Company (Baa1 stable), will be higher than its internally generated cash flow, resulting in negative free cash flow. Moody's expects DTE Gas to fund the shortfall with a combination of long-term debt as well as some equity infusions from its parent.

Factors that could lead to an upgrade

A rating upgrade could be considered if there are legislative actions that reduce regulatory lag in the recovery of costs, and the regulatory environment in Michigan become even more formulaic and transparent. Also, a rating upgrade could be possible if DTE Gas's financial metrics improve such that its CFO pre-W debt is in excess of 18% on a sustained basis, for example.

Factors that could lead to a downgrade

A downgrade could be considered if the company's financial profile continues to deteriorate. For example, if CFO pre-W debt falls to 15% or below, a downgrade could be possible. Also, if there is an adverse change in Michigan's regulatory environment, increasing regulatory lag or contentiousness, the company experiences insufficient cost recovery and returns, a rating downgrade could be triggered.

DTE Gas Company is a local distribution company (LDC) for natural gas in Michigan and is a subsidiary of DTE Energy Company (DTE, Baa1 stable). DTE Gas is regulated by the Michigan Public Service Commission (MPSC), serving approximately 1.3 million customers with approximately \$4.2 billion in rate base.

The principal methodology used in these ratings was Regulated Electric Gas Utilities published in June 2017. Please see the Rating Methodologies page on www.moodys.com for a copy of the methodology.

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STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

RICHARD T. BLUMENSTOCK

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule B-5.1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Electric Distribution

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (RTB-1)

Schedule: B-5.1

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	New Business Program	105,147	131,799	131,695	263,495	145,162
	Contractor	12,148	19,814	20,784	40,598	24,081
	Labor	31,008	32,100	31,483	63,584	34,316
	Materials	26,087	29,589	29,403	58,992	31,743
	Business Expenses	41	445	457	902	521
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	35,863	49,850	49,568	99,419	54,501
2	Reliability Program	179,640	230,207	201,165	431,372	331,234
	Contractor	58,064	90,912	67,255	158,167	102,171
	Labor	19,140	20,620	19,893	40,512	28,099
	Materials	27,977	39,625	44,462	84,088	68,030
	Business Expenses	2,064	1,660	1,619	3,279	1,735
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	72,395	77,390	67,935	145,325	131,199
3	Capacity Program	58,211	56,220	56,640	112,860	66,323
	Contractor	13,322	17,009	15,886	32,895	18,115
	Labor	7,046	5,632	6,290	11,922	7,444
	Materials	13,590	12,259	13,014	25,273	15,913
	Business Expenses	159	74	72	146	70
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	24,093	21,246	21,379	42,624	24,781
4	Demand Failures Program	144,758	158,586	111,154	269,740	122,568
	Contractor	24,799	28,399	16,622	45,021	18,559
	Labor	28,542	30,649	23,538	54,186	28,172
	Materials	30,967	32,288	22,598	54,886	22,777
	Business Expenses	339	508	387	895	445
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	60,111	66,743	48,008	114,751	52,615
5	Asset Relocation Program	40,505	44,512	41,675	86,187	45,976
	Contractor	10,952	16,226	15,644	31,871	17,027
	Labor	9,396	10,220	9,333	19,553	10,414
	Materials	5,266	5,510	5,367	10,877	5,757
	Business Expenses	17	83	76	159	85
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	14,874	12,473	11,255	23,727	12,692
6	Electric Other Program	4,907	7,540	9,813	17,353	11,412
	Contractor	366	2,153	2,876	5,029	2,953
	Labor	67	305	625	930	580
	Materials	3,974	3,474	4,058	7,532	4,349
	Business Expenses	15	3	3	5	3
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	484	1,605	2,252	3,857	3,527
7	Total Capital	533,167	628,865	552,142	1,181,007	722,675

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Projected Capital Expenditures
Electric Distribution
Summary of 5yr Historical Electric Capital Expenditures
(\$000)

Case No.: U-20697
Exhibit No.: A-29 (RTB-2)
Page: 1 of 1
Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Program / Sub-Program	(b) Witness	(c) 2014 Actuals	(d) 2015 Actuals	(e) 2016 Actuals	(f) 2017 Actuals	(g) 2018 Actuals	(h) Syr Average	(i) 2019 Projected Bridge Year	(j) 2020 Projected Bridge Year	(k) 2021 Projected Test Year	(l) Test Year vs Syr Ave Variance (k - h)
1	Lines New Business - LVD	RTBlumenstock	40,058	45,644	43,039	65,878	87,057	56,335	85,578	84,661	94,788	38,453
2	Large New Business Projects	RTBlumenstock	680	29	-37	0	-	134	-	-	-	(134)
3	Lines Strategic Customers - HVD	RTBlumenstock	8,213	7,035	27,864	7,113	(6,179)	8,809	5,586	12,114	17,281	8,472
4	Metering New Business - LVD	RTBlumenstock	2,745	5,171	5,266	8,075	9,396	6,131	13,546	10,562	10,868	4,738
5	Transformers New Business - LVD	RTBlumenstock	8,306	12,136	8,852	13,063	11,421	10,756	19,034	19,858	18,724	7,969
6	Metro New Business	RTBlumenstock	2,700	2,760	3,243	2,411	3,451	2,913	8,055	4,500	3,500	587
7	New Business		62,702	72,775	88,227	96,540	105,146	85,078	131,799	131,695	145,162	60,084
8	Lines Reliability - LVD	RTBlumenstock	48,215	25,092	48,617	37,877	36,866	39,333	37,272	37,884	44,929	5,595
9	Lines Reliability - HVD	RTBlumenstock	26,702	14,640	37,825	17,325	42,708	27,840	48,110	16,324	78,129	50,289
10	Substations Reliability - LVD	RTBlumenstock	6,780	8,936	11,135	14,112	10,198	10,232	14,344	11,500	15,502	5,270
11	Substations Reliability - HVD	RTBlumenstock	2,021	3,458	3,850	4,342	2,848	3,304	4,981	5,364	5,223	1,919
12	System Protection	RTBlumenstock	2,372	1,899	1,569	4,244	3,000	2,617	2,989	2,323	2,344	(273)
13	Repetitive Outages - LVD	RTBlumenstock	16,086	10,322	8,353	6,270	4,367	9,080	6,988	7,348	9,710	630
14	Metro Reliability	RTBlumenstock	1,755	4,209	2,518	949	969	2,080	3,694	1,750	1,647	(433)
15	Substations Comm Upgrades	RTBlumenstock	10	508	1,324	11903	24,114	7,572	31,649	2,100	-	(7,572)
16	Grid Capabilities: Automation	RTBlumenstock	9,216	13,758	17,601	13924	22,620	15,424	42,255	44,456	60,421	44,997
17	Grid Capabilities: Advanced Technologies	RTBlumenstock	0	0	0	0	-	-	17,452	19,857	7,833	7,833
18	Lines and Subs Rehabilitation HVD	RTBlumenstock	-	-	-	-	-	-	-	14,222	38,921	38,921
19	Substations Rehabilitation LVD	RTBlumenstock	-	-	-	-	-	-	-	11,500	14,500	14,500
20	Lines Rehabilitation - LVD	RTBlumenstock	-	-	-	-	31,949	31,949	20,473	20,597	37,723	5,773
21	Lines Rehabilitation Metro	RTBlumenstock	-	-	-	-	-	-	-	955	4,353	4,353
22	Grid Storage	RTBlumenstock	-	-	-	-	-	-	-	4,985	10,000	10,000
23	Reliability		113,157	82,822	132,792	110,946	179,639	123,871	230,207	201,165	331,234	207,363
24	Lines Capacity - LVD	RTBlumenstock	13,608	16,871	14,517	18,332	12,295	15,125	4,715	8,038	11,321	(3,803)
25	Lines & Subs Capacity - HVD	RTBlumenstock	13,596	15,612	20,965	16,823	16,545	16,708	22,874	21,501	20,203	3,495
26	Substations Capacity - LVD	RTBlumenstock	10927	7209	18,044	13,696	15,890	13,153	10,720	10,000	14,002	849
27	Transformers Capacity - LVD	RTBlumenstock	3692	3944	3,219	4,610	3,221	3,737	2,514	2,755	2,473	(1,264)
28	New Business Cap - LVD	RTBlumenstock	0	0	0	0	10,260	2,052	15,397	11,559	11,777	9,725
29	Conservative Voltage Reduction CVR	RTBlumenstock	-	-	-	-	-	-	-	2,737	2,851	2,851
30	Interconnections - LVD Lines	RTBlumenstock	-	-	-	-	-	-	-	50	-	-
31	Interconnections - HVD Lines	RTBlumenstock	-	-	-	-	-	-	-	-	2,062	2,062
32	Interconnections - LVD Substations	RTBlumenstock	-	-	-	-	-	-	-	-	-	-
33	Interconnections - HVD Substations	RTBlumenstock	-	-	-	-	-	-	-	-	1,634	1,634
34	Capacity		41,823	43,636	56,745	53,461	58,211	50,775	56,220	56,640	66,323	15,548
35	Lines Failures - LVD	RTBlumenstock	47,674	76,151	66,860	84,508	66,302	68,299	88,388	67,960	78,538	10,239
36	Lines & Subs Failures - HVD	RTBlumenstock	11,688	14,877	13,206	17,623	27,756	17,030	20,944	3,780	4,180	(12,850)
37	Substations Failures - LVD	RTBlumenstock	11,125	7,613	9,399	15,451	20,039	12,725	18,657	6,997	7,001	(5,724)
38	Metering Failures - LVD	RTBlumenstock	3,021	5,719	7,272	11,805	11,632	7,890	11,375	10,318	10,617	2,727
39	Transformers Failures - LVD	RTBlumenstock	18,766	14,260	14,754	20,747	14,642	16,634	14,366	15,742	14,132	(2,502)
40	Streetlight-Mercury Vapor / LED	RTBlumenstock	2,877	2,701	2,193	2,080	1,840	2,338	2,847	3,357	-	(2,338)
41	Streetlighting Center Suspension	RTBlumenstock	-	-	-	-	-	-	-	-	5,000	5,000
42	Metro Failures	RTBlumenstock	1,970	1517	5,047	3,643	2,547	2,945	2,009	3,000	3,100	155
43	Demand Failures		97,121	122,838	118,731	155,857	144,758	127,861	158,586	111,154	122,568	-5,293
44	Lines Relocations - LVD	RTBlumenstock	16,712	19,368	14,362	23,154	34,092	21,538	40,563	36,585	41,226	19,688
45	Lines Relocations - HVD	RTBlumenstock	364	1056	288	168	1,986	772	248	900	900	128
46	Metro Relocations	RTBlumenstock	2,338	7,325	4,854	4,791	4,428	4,747	3,701	4,190	3,850	(897)
47	Asset Relocations		19,414	27,749	19,504	28,113	40,505	27,057	44,512	41,675	45,976	18,919
48	Computer & Equipment	RTBlumenstock	393	113	76	430	77	218	24	100	100	(118)
49	Tools	RTBlumenstock	2,305	2,178	3,377	1,903	3,822	2,717	5,521	5,691	5,792	3,075
50	System Control Projects	RTBlumenstock	174	88	2	619	1,002	377	1,996	4,022	4,170	3,793
51	Facilities Security	RTBlumenstock	0	0	1	-1	-	-	-	-	-	-
52	NERC/NESC Compliance	RTBlumenstock	0	0	0	0	-	-	-	-	-	-
53	Substation Fall Protection	RTBlumenstock	251	196	80	6	5	108	-	-	-	(108)
54	Grid Technologies	RTBlumenstock	-	-	-	-	-	-	-	-	1,350	1,350
55	Electric Other		3,123	2,575	3,536	2,957	4,906	3,419	7,540	9,813	11,412	7,993
56	Total Capital Electric D&T		337,340	352,395	419,535	447,874	533,167	418,062	628,865	552,142	722,675	304,613

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Projected Capital Expenditures

New Business Program

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-30 (RTB-3)

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Lines New Business - LVD	87,058	85,578	84,661	170,239	94,788
	Contractor	10,000	15,036	14,875	29,912	16,655
	Labor	25,797	24,329	24,068	48,396	26,947
	Materials	13,393	12,445	12,311	24,756	13,784
	Business Expenses	26	429	424	854	475
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	37,842	33,339	32,982	66,321	36,927
2	Large New Business Projects	-	-	-	-	-
	Contractor	-	-	-	-	-
	Labor	-	-	-	-	-
	Materials	-	-	-	-	-
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
3	Lines Strategic Customers - HVD	(6,178)	5,586	12,114	17,700	17,281
	Contractor	1,477	2,012	4,363	6,375	6,225
	Labor	362	267	578	845	825
	Materials	2,981	1,508	3,270	4,778	4,665
	Business Expenses	15	15	32	46	45
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	(11,013)	1,785	3,871	5,655	5,521
4	Metering New Business - LVD	9,396	13,546	10,562	24,108	10,868
	Contractor	-	-	-	-	-
	Labor	1,831	2,440	1,903	4,343	1,958
	Materials	4,110	6,278	4,895	11,173	5,037
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	3,455	4,828	3,764	8,592	3,874
5	Transformers New Business - LVD	11,420	19,034	19,858	38,893	18,724
	Contractor	-	-	-	-	-
	Labor	2,445	4,344	4,532	8,876	4,273
	Materials	4,719	7,631	7,961	15,591	7,506
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	4,256	7,060	7,366	14,425	6,945
6	Metro New Business	3,451	8,055	4,500	12,556	3,500
	Contractor	671	2,766	1,545	4,311	1,202
	Labor	573	721	403	1,124	313
	Materials	884	1,728	966	2,694	751
	Business Expenses	-	1	1	2	1
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	1,323	2,839	1,586	4,424	1,233
					-	-
					-	-
					-	-
7	Total Capital	105,147	131,799	131,695	263,495	145,162

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Projected Capital Expenditures

Reliability Program

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-31 (RTB-4)

Page: 1 of 2

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Lines Reliability - LVD	36,867	37,272	37,884	75,156	44,929
	Contractor	8,114	11,970	12,167	24,136	14,429
	Labor	6,923	5,588	5,680	11,268	6,736
	Materials	3,000	2,915	2,963	5,878	3,514
	Business Expenses	1,580	1,281	1,302	2,584	1,545
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	17,250	15,518	15,772	31,290	18,705
2	Lines Reliability - HVD	42,708	48,110	16,324	64,434	78,129
	Contractor	20,393	23,768	8,065	31,833	38,599
	Labor	1,129	1,369	465	1,834	2,223
	Materials	6,750	7,387	2,506	9,893	11,996
	Business Expenses	35	32	11	42	52
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	14,401	15,554	5,278	20,832	25,259
3	Substations Reliability - LVD	10,198	14,344	11,500	25,844	15,502
	Contractor	2,273	4,218	3,381	7,599	4,558
	Labor	775	1,306	1,047	2,354	1,412
	Materials	3,509	4,035	3,235	7,270	4,361
	Business Expenses	20	6	5	12	7
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	3,621	4,779	3,831	8,610	5,165
4	Substations Reliability - HVD	2,848	4,981	5,364	10,345	5,223
	Contractor	233	713	768	1,481	748
	Labor	406	861	927	1,789	903
	Materials	1,101	1,603	1,727	3,330	1,681
	Business Expenses	1	0	0	1	0
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	1,107	1,803	1,941	3,744	1,890
5	Grid Capabilities: Automation	22,620	42,255	44,456	86,711	60,421
	Contractor	4,258	10,384	10,925	21,309	14,848
	Labor	1,844	2,736	2,878	5,614	3,912
	Materials	7,938	15,173	15,963	31,135	21,695
	Business Expenses	361	(42)	(45)	(87)	(61)
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	8,218	14,005	14,735	28,740	20,027
6	Grid Capabilities: Advanced Technologies	0	17,452	19,857	37,309	7,833
	Contractor	0	14,146	16,096	30,242	6,349
	Labor	-	1,025	1,167	2,192	460
	Materials	-	1,462	1,664	3,126	656
	Business Expenses	-	248	282	531	111
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	570	649	1,219	256
7	Substations Comm Upgrades	24,114	31,649	2,100	33,749	-
	Contractor	14,246	17,202	1,141	18,343	-
	Labor	476	1,252	83	1,335	-
	Materials	735	2,657	176	2,834	-
	Business Expenses	15	40	3	42	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	8,641	10,498	697	11,195	-
8	System Protection	3,000	2,989	2,323	5,312	2,344
	Contractor	885	588	457	1,045	461
	Labor	276	315	245	560	247
	Materials	743	1,092	849	1,941	857
	Business Expenses	0	1	1	1	1
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	1,095	993	771	1,764	778

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Projected Capital Expenditures
Reliability Program
Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-31 (RTB-4)

Page: 2 of 2

Witness: RTBlumenstock

Date: February 2020

9	Repetitive Outages - LVD	4,367	6,988	7,348	14,336	9,710
	Contractor	864	1,356	1,426	2,782	1,884
	Labor	915	1,594	1,676	3,269	2,214
	Materials	458	894	940	1,834	1,242
	Business Expenses	0	1	2	3	2
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	2,130	3,143	3,305	6,448	4,367
10	Metro Reliability	969	3,694	1,750	5,444	1,647
	Contractor	206	1,622	769	2,391	723
	Labor	72	355	168	523	158
	Materials	345	501	237	738	223
	Business Expenses	-	65	31	96	29
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	345	1,150	545	1,695	513
11	Lines and Subs Rehabilitation HVD	-	-	14,222	14,222	38,921
	Contractor	-	-	4,827	4,827	6,115
	Labor	-	-	466	466	577
	Materials	-	-	3,987	3,987	5,566
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	4,942	4,942	26,663
12	Substations Rehabilitation LVD	-	-	11,500	11,500	14,500
	Contractor	-	-	1,127	1,127	1,409
	Labor	-	-	751	751	939
	Materials	-	-	5,637	5,637	7,044
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	3,985	3,985	5,108
13	Lines Rehabilitation - LVD	31,949	20,473	20,597	41,070	37,723
	Contractor	6,593	4,944	4,974	9,918	9,109
	Labor	6,323	4,219	4,244	8,463	7,773
	Materials	3,397	1,906	1,917	3,823	3,512
	Business Expenses	51	27	27	54	49
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	15,585	9,378	9,434	18,812	17,279
14	Lines Rehabilitation Metro	-	-	955	955	4,353
	Contractor	-	-	299	299	1,267
	Labor	-	-	95	95	544
	Materials	-	-	161	161	683
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	401	401	1,859
15	Grid Storage	-	-	4,985	4,985	10,000
	Contractor	-	-	835	835	1,670
	Labor	-	-	-	-	-
	Materials	-	-	2,500	2,500	5,000
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	1,650	1,650	3,330
				-	-	-
				-	-	-
16	Total Capital	179,640	230,207	201,165	431,372	331,234

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Projected Capital Expenditures

Capacity Program

 Summary of Actual and Projected Electric Capital Expenditures
 (\$000)

Case No.: U-20697

Exhibit No.: A-32 (RTB-5)

Page: 1 of 2

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Lines Capacity - LVD	12,295	4,715	8,038	12,753	11,321
	Contractor	3,488	953	1,625	2,578	2,288
	Labor	1,592	936	1,596	2,533	2,249
	Materials	1,505	773	1,317	2,090	1,856
	Business Expenses	17	3	5	7	7
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	5,693	2,050	3,495	5,545	4,922
2	Lines & Subs Capacity - HVD	16,545	22,874	21,501	44,375	20,203
	Contractor	5,347	9,471	8,902	18,373	8,365
	Labor	1,322	1,135	1,067	2,202	1,003
	Materials	3,249	3,653	3,434	7,087	3,226
	Business Expenses	108	69	64	133	61
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	6,518	8,547	8,034	16,580	7,549
3	Substations Capacity - LVD	15,890	10,720	10,000	20,720	14,002
	Contractor	3,310	2,280	2,126	4,406	2,978
	Labor	830	777	725	1,503	1,016
	Materials	6,240	4,340	4,048	8,388	5,669
	Business Expenses	30	0	0	1	1
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	5,480	3,323	3,099	6,422	4,340
4	Transformers Capacity - LVD	3,221	2,514	2,755	5,269	2,473
	Contractor	-	-	-	-	-
	Labor	690	574	629	1,203	564
	Materials	1,331	1,008	1,104	2,112	991
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	1,201	932	1,022	1,954	917
5	New Business Cap - LVD	10,260	15,397	11,559	26,956	11,777
	Contractor	1,177	4,306	3,233	7,539	3,294
	Labor	2,612	2,209	1,658	3,867	1,690
	Materials	1,264	2,486	1,866	4,352	1,901
	Business Expenses	5	3	2	5	2
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	5,202	6,394	4,800	11,194	4,891
6	Conservative Voltage Reduction CVR	-	-	2,737	2,737	2,851
	Contractor	-	-	-	-	-
	Labor	-	-	598	598	619
	Materials	-	-	1,235	1,235	1,273
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	904	904	959
7	Interconnections - LVD Lines	-	-	50	50	-
	Contractor	-	-	-	-	-
	Labor	-	-	16	16	-
	Materials	-	-	9	9	-
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	25	25	-
8	Interconnections - HVD Lines	-	-	-	-	2,062
	Contractor	-	-	-	-	699
	Labor	-	-	-	-	140
	Materials	-	-	-	-	559
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	664

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Projected Capital Expenditures

Capacity Program

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-32 (RTB-5)

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Witness: RTBlumenstock

Date: February 2020

9	Interconnections - LVD Substations	-	-	-	-
	Contractor			-	
	Labor			-	
	Materials			-	
	Business Expenses			-	
	Contingency			-	
	Other (Loadings, Chargebacks)			-	
10	Interconnections - HVD Substations	-	-	-	1,634
	Contractor			-	492
	Labor			-	164
	Materials			-	438
	Business Expenses			-	
	Contingency			-	
	Other (Loadings, Chargebacks)			-	540
				-	-
				-	-
				-	-
11	Total Capital	<u>58,211</u>	<u>56,220</u>	<u>56,640</u>	<u>112,860</u>
					<u>66,323</u>

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Projected Capital Expenditures

Demand Failures Program

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-33 (RTB-6)

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

(a)		(b)	(c)	(d)	(e)	(f)
Line No.	Description	Capital Expenditures				Projected Test Year 12 Mos Ending 12/31/2021
		Historical 12 Mos Ended 12/31/2018	Projected Bridge Year			
			12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	
1	Lines Failures - LVD	66,302	88,388	67,960	156,348	78,538
	Contractor	10,925	16,394	12,605	28,999	14,567
	Labor	17,722	20,832	16,017	36,849	18,510
	Materials	5,688	8,463	6,507	14,970	7,520
	Business Expenses	308	498	383	881	443
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	31,659	42,202	32,448	74,649	37,498
2	Lines & Subs Failures - HVD	27,756	20,944	3,780	24,724	4,180
	Contractor	10,039	8,461	1,527	9,988	1,689
	Labor	2,153	1,672	302	1,974	334
	Materials	5,823	3,911	706	4,617	781
	Business Expenses	10	7	1	8	1
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	9,731	6,892	1,244	8,136	1,376
3	Substations Failures - LVD	20,039	18,657	6,997	25,654	7,001
	Contractor	2,751	2,451	919	3,370	920
	Labor	2,169	1,909	716	2,625	716
	Materials	7,768	7,896	2,961	10,858	2,963
	Business Expenses	21	1	0	2	0
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	7,328	6,400	2,400	8,800	2,401
4	Metering Failures - LVD	11,632	11,375	10,318	21,693	10,617
	Contractor	1	-	-	-	-
	Labor	2,740	2,308	2,094	4,402	2,154
	Materials	4,546	4,949	4,489	9,438	4,619
	Business Expenses	-	-	-	-	-
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	4,345	4,118	3,736	7,854	3,844
5	Transformers Failures - LVD	14,642	14,366	15,742	30,108	14,132
	Contractor	-	-	-	-	-
	Labor	3,135	3,279	3,593	6,872	3,225
	Materials	6,050	5,759	6,311	12,070	5,665
	Business Expenses	-	-	-	-	-
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	5,457	5,328	5,838	11,166	5,241
6	Streetlight-Mercury Vapor / LED	1,840	2,847	3,357	6,204	
	Contractor	92	197	232	428	
	Labor	372	484	571	1,055	
	Materials	667	1,052	1,240	2,292	
	Business Expenses		2	2	3	
	Contingency		-	-	-	
	Other (Loadings, Chargebacks)	708	1,113	1,312	2,425	
	Streetlighting Center Suspension					5,000
	Contractor					
	Labor					2,978
	Materials					832
	Business Expenses					
	Contingency					
	Other (Loadings, Chargebacks)					1,190
7	Metro Failures	2,547	2,009	3,000	5,009	3,100
	Contractor	991	897	1,339	2,236	1,384
	Labor	250	164	246	410	254
	Materials	425	257	384	642	397
	Business Expenses	-	0	1	1	1
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	882	690	1,030	1,720	1,065
					-	-
					-	-
					-	-
8	Total Capital	144,758	158,586	111,154	269,740	122,568

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Asset Relocation Program

Summary of Actual and Projected Electric Capital Expenditures
(\$000)

Case No.: U-20697

Exhibit No.: A-34 (RTB-7)

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Lines Relocations - LVD	34,092	40,563	36,585	77,148	41,226
	Contractor	7,872	13,845	12,487	26,332	14,071
	Labor	9,009	9,789	8,829	18,619	9,949
	Materials	3,890	4,325	3,901	8,226	4,396
	Business Expenses	8	81	73	154	83
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	13,314	12,523	11,294	23,817	12,727
2	Lines Relocations - HVD	1,986	248	900	1,148	900
	Contractor	1,479	185	670	855	670
	Labor	54	7	24	31	24
	Materials	398	50	181	230	181
	Business Expenses	1	0	0	1	0
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	54	7	24	31	24
3	Metro Relocations	4,428	3,701	4,190	7,891	3,850
	Contractor	1,602	2,197	2,487	4,684	2,285
	Labor	333	424	480	904	441
	Materials	978	1,135	1,285	2,421	1,181
	Business Expenses	8	2	2	4	2
	Contingency		-	-	-	-
	Other (Loadings, Chargebacks)	1,506	(57)	(64)	(121)	(59)
					-	-
					-	-
					-	-
4	Total Capital	40,505	44,512	41,675	86,187	45,976

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Projected Capital Expenditures

Electric Other Program

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-35 (RTB-8)

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Test Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Computer & Equipment	77	24	100	124	100
	Contractor	0	-	-	-	-
	Labor	-	-	100	100	-
	Materials	77	24	-	24	100
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	0	-	-	-	-
2	Tools	3,822	5,521	5,691	11,212	5,792
	Contractor	82	1,486	1,532	3,018	1,559
	Labor	21	91	94	186	96
	Materials	3,580	2,942	3,033	5,975	3,087
	Business Expenses	2	3	3	5	3
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	137	998	1,029	2,027	1,047
3	System Control Projects	1,002	1,996	4,022	6,018	4,170
	Contractor	284	667	1,344	2,011	1,393
	Labor	46	214	431	644	447
	Materials	315	508	1,025	1,533	1,062
	Business Expenses	14	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	344	607	1,223	1,830	1,268
4	Facilities Security	-	-	-	-	-
	Contractor	-	-	-	-	-
	Labor	-	-	-	-	-
	Materials	-	-	-	-	-
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
5	NERC/NESC Compliance	-	-	-	-	-
	Contractor	-	-	-	-	-
	Labor	-	-	-	-	-
	Materials	-	-	-	-	-
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	-
6	Substation Fall Protection	5	-	-	-	-
	Contractor	-	-	-	-	-
	Labor	0	-	-	-	-
	Materials	3	-	-	-	-
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	2	-	-	-	-
7	Grid Technologies	-	-	-	-	1,350
	Contractor	-	-	-	-	-
	Labor	-	-	-	-	38
	Materials	-	-	-	-	100
	Business Expenses	-	-	-	-	-
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	-	-	-	-	1,212
		-	-	-	-	-
		-	-	-	-	-
8	Total Capital	4,907	7,540	9,813	17,353	11,412

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual & Projected Electric & Common O&M Expenses
For the Year 2018 & Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-36 (RTB-9)
Page: 1 of 2
Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)
		2018 Actual	2019 Projected	2020 Projected	12 Months Ending December 31, 2021 Projected
1	Electric Division Electric & Common O&M Expenses	\$ 140,650	\$ 166,195	\$ 124,359	\$ 170,744

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual & Projected Electric & Common O&M Expenses
For the Year 2018 & Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-37 (RTB-10)
Page: 1 of 1
Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Description	(b) 2018 Actual	(c) 2019 Projected	(d) 2020 Projected	(e) 12 Months Ending December 31, 2021 Projected
1	Electric Operations	\$ 127,566	\$ 150,841	\$ 108,954	\$ 151,941
2	Electric Engineering & Support	\$ 13,084	\$ 15,353	\$ 15,405	\$ 18,803
3	Total Electric & Common O&M Expenses	\$ 140,650	\$ 166,195	\$ 124,359	\$ 170,744

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Summary of Actual & Projected Electric & Common O&M Expenses
For the Year 2018 & Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-38 (RTB-11)
Page: 1 of 2
Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Description	(b) 2018 Actual	(c) 2019 Projected	(d) 2020 Projected	(e) 12 Months Ending December 31, 2021 Projected
1	O&M Assoc w/Construction	8,121	7,791	9,000	7,716
2	Transformer Credits	(7,247)	(8,559)	(9,000)	(7,716)
3	O&M Assoc w/Construction	874	(768)	-	-
4	Lines Reliability - LVD	56	72	280	1,452
5	Lines Reliability - HVD	177	132	125	125
6	Substations Reliability - LVD	2,090	1,800	1,700	2,820
7	Substations Reliability - HVD	1,422	1,207	1,353	2,728
8	Non-Forestry Reliability	3,745	3,211	3,458	7,125
9	Lines Demand - HVD	1,681	325	735	940
10	Substations Demand - LVD	3,246	2,939	2,953	3,975
11	Substations Demand - HVD	2,496	2,280	2,160	3,835
12	Corrective Maintenance	5,007	4,897	4,549	5,549
13	Staking	3,466	2,876	2,929	3,017
14	Meter Services (and Credits)	1,020	57	2,418	2,832
15	Streetlighting	2,206	1,488	1,168	1,156
16	Service Calls	4,172	4,510	3,589	4,589
17	Alma Equipment Repair	1,242	1,002	1,159	1,169
18	Meter Reading	1,813	1,708	1,778	1,827
19	Meter Tech & Mgmt Sys Support	1,343	1,222	1,273	1,320
20	Smart Energy MTC - Elec	7,836	8,694	8,316	8,729
21	Ops, Mtc & Mtr w/o Svc Rest	35,529	31,998	33,027	38,937
22	Service Restoration - LVD	53,924	83,670	38,726	65,100
23	Service Restoration	53,924	83,670	38,726	65,100
24	Training	6,160	6,866	7,582	10,247
25	Tools	1,900	1,504	1,471	1,868
26	Field Operations Expenses	2,691	2,540	2,161	3,161
27	Indirect Labor/Labor Variations	155	0	-	-
28	Supervision / Admin-Staff	7,634	7,715	7,998	9,689
29	Smart Energy Operations Center	1,183	1,051	1,001	1,001
30	Grid Management - Distr	4,091	3,925	4,063	4,513
31	Field Operations	23,814	23,602	24,275	30,479
32	Compliance and Controls	-	444	529	540
33	Compliance and Controls	-	444	529	540
34	Resource Planning & Closeout	365	336	364	372
35	Scheduling & Dispatch	5,390	4,970	5,183	5,732
36	Contract Administration	345	285	325	333
37	Planning & Scheduling	6,100	5,591	5,871	6,437
38	OP Distribution & Generation	677	575	534	548
39	OP Business Services	1,229	1,026	1,140	1,168
40	Operations Performance	1,906	1,601	1,674	1,716
41	Operations Management	1,673	1,492	1,393	1,608
42	Total Electric Operations	127,566	150,841	108,954	151,941

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual & Projected Electric & Common O&M Expenses
For the Year 2018 & Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-38 (RTB-11)
Page: 2 of 2
Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Description	(b) 2018 Actual	(c) 2019 Projected	(d) 2020 Projected	(e) 12 Months Ending December 31, 2021 Projected
1	Rate Case	102	96	77	79
2	Regulatory & Compliance-Elec	151	92	149	154
3	CES	362	452	448	513
4	Engineering Support	616	640	675	746
5	Geospatial Mgmt & Data Quality - Elec	107		-	-
6	Planning - LVD System	3,103	3,849	2,635	2,769
7	Planning - HVD System	3,139	3,318	2,934	3,939
8	System Protection	1,951	1,360	1,554	1,749
9	Electric Planning	8,299	8,527	7,124	8,457
10	Design - LVD	-	1,031	1,107	1,153
11	Design - DER / I&C	-	338	368	387
12	Standards & Document Control	247	472	560	549
13	Design - HVD	1,544	1,135	678	1,155
14	Joint Pole Rental	1,857	1,976	2,102	2,239
15	Grid Technologies - Dist	-		605	623
16	LVD Conservation Voltage Reduction (CVR)	-		86	177
17	Electric Design	3,648	4,952	5,505	6,283
18	ERSI Analytics	521	1,235	881	1,379
19	Electric Grid Analytics	521	1,235	881	1,379
20	IT-Grid Mod	-	0	1,220	1,938
21	Electric Engineering & Support	13,084	15,353	15,405	18,803

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual & Projected Electric & Common O&M Expenses
For the Year 2018 & Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-39 (RTB-12)
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Witness: RTBlumenstock
Date: February 2020

Line No.	(a) Description	(b) 2014 Actual	(c) 2015 Actual	(d) 2016 Actual	(e) 2017 Actual	(f) 2018 Actual	(g) 5-Year Average	(h) 2019 Projected	(i) 2020 Projected	(j) 2021 Projected	(k) 2021 vs. 5-Year Average
1	O&M Assoc w/Construction	4,806	4,381	7,228	6,405	8,121	6,188	7,791	9,000	7,716	1,528
2	Transformer Credits	(6,788)	(6,146)	(6,134)	(8,925)	(7,247)	(7,048)	(8,559)	(9,000)	(7,716)	(668)
3	O&M Assoc w/Construction	(1,982)	(1,765)	1,094	(2,520)	874	(860)	(768)	-	-	860
4	Lines Reliability - LVD	1,054	235	56	157	56	312	72	280	1,452	1,140
5	Lines Reliability - HVD	628	236	317	147	177	301	132	125	125	(176)
6	Substations Reliability - LVD	1,999	1,549	1,794	1,697	2,090	1,826	1,800	1,700	2,820	994
7	Substations Reliability - HVD	839	1,069	957	1,145	1,422	1,086	1,207	1,353	2,728	1,642
8	Non-Forestry Reliability	4,520	3,089	3,124	3,146	3,745	3,525	3,211	3,458	7,125	3,600
9	Lines Demand - HVD	2,007	1,167	533	785	1,681	1,235	325	735	940	(295)
10	Substations Demand - LVD	3,748	3,393	3,321	2,728	3,246	3,287	2,939	2,953	3,975	688
11	Substations Demand - HVD	2,826	2,420	2,150	2,054	2,496	2,389	2,280	2,160	3,835	1,446
12	Corrective Maintenance	10,324	8,519	3,483	4,586	5,007	6,384	4,897	4,549	5,549	(835)
13	Staking	2,554	3,868	3,221	3,285	3,466	3,279	2,876	2,929	3,017	(262)
14	Meter Services (and Credits)	8,582	5,705	2,992	437	1,020	3,747	57	2,418	2,832	(916)
15	Streetlighting	1,809	2,153	1,933	1,637	2,206	1,948	1,488	1,168	1,156	(792)
16	Service Calls	2,549	2,451	2,108	2,839	4,172	2,824	4,510	3,589	4,589	1,765
17	Alma Equipment Repair	1,248	980	1,136	1,058	1,242	1,133	1,002	1,159	1,169	36
18	Meter Reading	12,038	10,697	11,582	4,982	1,813	8,222	1,708	1,778	1,827	(6,395)
19	Meter Tech & Mgmt Sys Support	1,317	1,343	1,133	965	1,343	1,220	1,222	1,273	1,320	99
20	Smart Energy MTC - Elec	-	-	-	7,476	7,836	3,062	8,694	8,316	8,729	5,666
21	Ops, Mtc & Mtr w/o Svc Rest	49,002	42,697	33,592	32,832	35,529	38,730	31,998	33,027	38,937	206
22	Service Restoration - LVD	47,005	38,167	35,504	50,172	53,924	44,954	83,670	38,726	65,100	20,146
23	Service Restoration	47,005	38,167	35,504	50,172	53,924	44,954	83,670	38,726	65,100	20,146
24	Training	5,279	6,047	4,174	6,075	6,160	5,547	6,866	7,582	10,247	4,700
25	Tools	1,330	1,920	1,811	1,461	1,900	1,684	1,504	1,471	1,868	183
26	Field Operations Expenses	1,565	2,346	2,360	2,604	2,691	2,313	2,540	2,161	3,161	848
27	Indirect Labor/Labor Variations	752	1,509	868	515	155	760	0	-	-	(760)
28	Supervision / Admin-Staff	6,586	6,556	6,063	6,725	7,634	6,713	7,715	7,998	9,689	2,977
29	Smart Energy Operations Center	-	-	-	1,166	1,183	470	1,051	1,001	1,001	531
30	Grid Management - Distr	2,628	2,807	2,778	4,073	4,091	3,275	3,925	4,063	4,513	1,237
31	Field Operations	18,140	21,185	18,054	22,619	23,814	20,762	23,602	24,275	30,479	9,716
32	Compliance and Controls	-	-	-	-	-	-	444	529	540	540
33	Compliance and Controls	-	-	-	-	-	-	444	529	540	540
34	Resource Planning & Closeout	96	-	39	495	365	199	336	364	372	173
35	Scheduling & Dispatch	5,189	3,249	3,605	5,273	5,390	4,541	4,970	5,183	5,732	1,191
36	Contract Administration	-	-	229	353	345	185	285	325	333	147
37	Planning & Scheduling	5,285	3,249	3,873	6,121	6,100	4,926	5,591	5,871	6,437	1,512
38	OP Distribution & Generation	208	361	722	1,552	677	704	575	534	548	(156)
39	OP Business Services	234	230	255	202	1,229	430	1,026	1,140	1,168	738
40	Operations Performance	442	591	977	1,754	1,906	1,134	1,601	1,674	1,716	582
41	Operations Management	1,750	3,415	2,640	1,167	1,673	2,129	1,492	1,393	1,608	(521)
42	Total Electric Operations	124,162	110,628	98,858	115,291	127,566	115,301	150,841	108,954	151,941	36,641
43	Rate Case	-	-	-	57	102	32	96	77	79	48
44	Regulatory & Compliance-Elec	168	168	170	140	151	159	92	149	154	(6)
45	CES	587	499	360	354	362	432	452	448	513	80
46	Engineering Support	755	667	529	550	616	623	640	675	746	122
47	Geospatial Mgmt & Data Quality - Elec	236	253	385	598	107	316	-	-	-	(316)
48	Planning - LVD System	3,373	3,190	1,797	2,183	3,103	2,729	3,849	2,635	2,769	40
49	Planning - HVD System	3,519	2,715	3,270	3,972	3,139	3,323	3,318	2,934	3,939	616
50	System Protection	2,118	2,828	2,549	1,425	1,951	2,174	1,360	1,554	1,749	(425)
51	Electric Planning	9,247	8,986	8,001	8,178	8,299	8,542	8,527	7,124	8,457	(85)
52	Design - LVD	-	-	-	-	-	-	1,031	1,107	1,153	1,153
53	Design - DER / I&C	-	-	-	-	-	-	338	368	387	387
54	Standards & Document Control	197	179	151	353	247	225	472	560	549	324
55	Design - HVD	2,458	1,522	1,209	1,287	1,544	1,604	1,135	678	1,155	(449)
56	Joint Pole Rental	1,733	1,791	1,789	1,805	1,857	1,795	1,976	2,102	2,239	444
57	Grid Technologies - Dist	-	-	-	-	-	-	-	605	623	623
58	LVD Conservation Voltage Reduction (CVR)	-	-	-	-	-	-	-	86	177	177
59	Electric Design	4,388	3,492	3,149	3,445	3,648	3,624	4,952	5,505	6,283	2,659
60	ERSI Analytics	-	-	-	420	521	188	1,235	881	1,379	1,191
61	Electric Grid Analytics	-	-	-	420	521	188	1,235	881	1,379	1,191
62	IT-Grid Mod	-	-	-	-	-	-	-	1,220	1,938	1,938
63	Electric Engineering & Support	14,389	13,144	11,679	12,593	13,084	12,978	15,353	15,405	18,803	5,825
64	Total O&M	138,552	123,772	110,537	127,884	140,650	128,279	166,195	124,359	170,744	42,466

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual Capital Expenditures per Case No. U-20134 Settlement
\$0

Case No.: U-20697

Exhibit No.: A-40 (RTB-13)

Page: 1 of 1

Witness: RTBlumenstock

Date: February 2020

2019 SPENDING (in thousands \$)	Actuals												U-20134 Settlement Amount	
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		2019 Total
Capital: New Business	\$ 13,685	\$ 9,829	\$ 11,684	\$ 7,442	\$ 8,564	\$ 11,862	\$ 10,721	\$ 9,736	\$ 10,982	\$ 11,896	\$ 12,553	\$ 13,600	\$ 132,554	\$ 94,000
Capital: Asset Relocations	\$ 5,240	\$ 1,306	\$ 5,176	\$ 2,256	\$ 3,828	\$ 5,155	\$ 3,402	\$ 3,716	\$ 4,061	\$ 4,079	\$ 2,883	\$ 2,579	\$ 43,681	\$ 24,000
Capital: "Reactive" Demand Failures	\$ 8,397	\$ 25,077	\$ 11,210	\$ 5,882	\$ 9,978	\$ 15,279	\$ 15,321	\$ 9,840	\$ 7,741	\$ 8,573	\$ 10,137	\$ 11,364	\$ 138,799	\$ 87,000

Line No	(a) Sub-Program	(b) Contents of Concept Approvals in Workpaper WP-5	(c) Status as of 1/17/2020
1	HVD Strategic Customers New Business	New substation and lines, relocated substations in south-west Michigan *includes information related to a new substation project in the LVD Substations Capacity sub-program	Approved
2		New substation in north-west Michigan	Approved
3	LVD Lines Reliability	10 LVD pole replacement projects	Approved
4		9 targeted circuit improvement projects	Approved
5	HVD Lines Reliability	10 line rebuild projects	Approved
6		12 line rebuild projects	In Progress
7	LVD Substations Reliability	4 substation rebuild projects	In Progress
8		1 new mobile substation project	Approved
9		6 transformer replacement projects *including 3 transformer replacement projects for LVD Substations Rehabilitation sub-program	In Progress
10	HVD Substations Reliability	2 circuit breaker/circuit switcher replacement projects	In Progress
11	Grid Modernization	ATR Loops project charter	In Progress
12	System Protection	5 line exit relay replacement projects	Approved
13	Metro Reliability	1 obsolete equipment replacement project	Approved
14	HVD Lines and Substations Rehabilitation	3 HVD substation failure projects	Approved
15		2 HVD substation failure projects *includes information related to a right-of-way procurement project in the HVD Lines and Substations Capacity sub-program	In Progress
16	LVD Substations Rehabilitation	16 Allis Chalmers transformer replacements	In Progress
17		1 equipment replacement and regulatory project *3 transformer replacement projects in LVD Substations Rehabilitation sub-program included in LVD Substations Reliability concept approvals	Approved
18	LVD Lines Rehabilitation	3 security assessment repair projects	Approved
19	Metro Rehabilitation	2 vault or manhole roof replacement projects	Approved
20		1 crushed duct replacement project	Approved
21	Grid Storage	Cadillac solar farm battery project	Approved
22		Standish portable battery project and Fort Custer islanded battery project	In Progress
23	LVD Lines Capacity	3 overloaded equipment upgrade projects	Approved
24		2 substation capacity-associated line work projects	Approved
25	HVD Lines and Substations Capacity	3 transmission coordination projects	Approved
26		1 new interconnection project	Approved
27		1 load carrying capability/voltage support project	Approved
28		1 improved functionality projects	Approved
29		3 improved functionality projects	In Progress
30		1 new interconnection project *includes information related to a new substation project in the LVD Substations Capacity sub-program	In Progress
31	LVD Substations Capacity	3 new substation projects	Approved
32		1 new substations project	In Progress
33		1 existing substation capacity increase project	In Progress
34	Substation Inspection O&M	Description of Initiative	Approved

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Distribution Projects

Summary Projected Electric Capital Expenditures

For the Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-42 (RTB-15)

Page: 1 of 30

Witness: RTBlumenstock

Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) (d) (e) (f) Projected 2021 Test Year Spending Units Unit Type Investment Category			
			Test Year Spending	Units	Unit Type	Investment Category
New Business						
1	HVD Strategic Customers New Business	New 1.3 mile 138 kV line #1 to connect to a new dedicated customer substation in south-west Michigan	1,800	1	Projects	
2		New 1.3 mile 138 kV line #2 to connect to a new dedicated customer substation in south-west Michigan	1,800	1	Projects	
3		New 138 kV dedicated customer substation in south-west Michigan	6,000	1	Projects	
4		Relocate Cooper 46kV Line to accomodate new 138 kV line to new dedicated customer substation in south-west Michigan	143	1	Projects	
5		Relocate Ampersee 46kV Line to accomodate new 138 kV line to new dedicated customer substation in south-west Michigan	98	1	Projects	
6		New 138 kV dedicated customer substation in north-west Michigan	5,550	1	Projects	
7		Additional projects to be identified	1,891			
8		HVD Strategic Customers New Business Total	17,281			
Metro New Business						
9	Metro New Business	FLINT MARKETPLACE	120	1	Project	
10		GRAND RAPIDS S-CURVE PARKING LOT	325	1	Project	
11		GRAND RAPIDS VAN ANDEL TRIANGLE LOT	325	1	Project	
12		GRAND RAPIDS VARIOUS MICROCELLS	200	1	Project	
13		Additional projects to be identified	2,530			
		Metro New Business Total	3,500			
Reliability						
14	LVD Lines Reliability	POLE ALABAMA/KING ROAD	268	28	Poles	Pole Replacements
15		POLE ALDER CREEK/LEE LAKE	513	53	Poles	Pole Replacements
16		POLE AUBIL LAKE/TOWERS	235	24	Poles	Pole Replacements
17		POLE BEECH-NUT/BEECH-NUT	353	52	Poles	Pole Replacements
18		POLE BRECKENRIDGE/VILLAGE	636	93	Poles	Pole Replacements
19		POLE BRIDGE STREET/HUPP AVENUE	78	8	Poles	Pole Replacements
20		POLE BURR OAK/DOUGLAS	51	6	Poles	Pole Replacements
21		POLE BURROWS/WHEELER	31	3	Poles	Pole Replacements
22		POLE CHEESMAN/WARWICK	79	8	Poles	Pole Replacements
23		POLE COLEMAN/COLEMAN	52	5	Poles	Pole Replacements
24		POLE COOLEY/NORTH STREET	1,354	137	Poles	Pole Replacements
25		POLE DAVENPORT/CONGRESS	51	4	Poles	Pole Replacements
26		POLE DELTON/DELTON	1,587	164	Poles	Pole Replacements
27		POLE DIXIE/HARVARD	297	31	Poles	Pole Replacements
28		POLE GOGUAC/LAKEVIEW	37	4	Poles	Pole Replacements
29		POLE GOODALE/ROOSEVELT	346	35	Poles	Pole Replacements
30		POLE GRANT/GRANT	205	23	Poles	Pole Replacements
31		POLE GROVER/TRIANGLE	816	104	Poles	Pole Replacements
32		POLE HOGSBACK/PINE TREE	805	83	Poles	Pole Replacements
33		POLE KILGORE/MOUNT EVEREST	156	16	Poles	Pole Replacements
34		POLE LEVEL PARK/COLLIER	556	59	Poles	Pole Replacements
35		POLE LIBERTY/WASHINGTON	266	28	Poles	Pole Replacements
36		POLE LOVEJOY/BRADEN	623	115	Poles	Pole Replacements
37		POLE MILL GROVE/BABYLON	181	13	Poles	Pole Replacements
38		POLE NIAGARA/HAMILTON	152	12	Poles	Pole Replacements
39		POLE NIAGARA/NIAGARA	220	23	Poles	Pole Replacements
40		POLE NORTH ALLEGAN/HUBBARD	96	10	Poles	Pole Replacements
41		POLE NORTH LANSING/VALLEY FARMS	50	6	Poles	Pole Replacements
42		POLE OAK STREET/WEST GANSON	52	5	Poles	Pole Replacements
43		POLE ORCHARD ROAD/ST ANDREWS	964	102	Poles	Pole Replacements
44		POLE OTSEGO/OTSEGO	104	11	Poles	Pole Replacements
45		LVD Lines Reliability subtotal	11,215			

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Distribution Projects

Summary Projected Electric Capital Expenditures

For the Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-42 (RTB-15)

Page: 2 of 30

Witness: RTBlumenstock

Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Reliability (cont.)	POLE OTTAWA BEACH/PORT SHELDON	474	50 Poles		Pole Replacements
2		POLE PARKWAY/BALCH	25	3 Poles		Pole Replacements
3		POLE PHILLIPS/FACTORY	102	10 Poles		Pole Replacements
4		POLE PHILLIPS/MILWOOD	16	2 Poles		Pole Replacements
5		POLE PLAINWELL/COMMERCIAL	90	11 Poles		Pole Replacements
6		POLE PRINCETON/WATTLES	55	7 Poles		Pole Replacements
7		POLE RANGER LAKE/LUPTON	77	9 Poles		Pole Replacements
8		POLE RAVINE/PATTERSON	73	8 Poles		Pole Replacements
9		POLE RED ARROW/MARKHAM	194	23 Poles		Pole Replacements
10		POLE SARANAC/KEENE	22	2 Poles		Pole Replacements
11		POLE SAUGATUCK/DOUGLAS	171	19 Poles		Pole Replacements
12		POLE SAUGATUCK/SAUGATUCK	290	30 Poles		Pole Replacements
13		POLE STADIUM/NORMAL	222	22 Poles		Pole Replacements
14		POLE STANDISH/STANDISH	167	17 Poles		Pole Replacements
15		POLE TWELFTH STREET/WESTFIELD	12	1 Poles		Pole Replacements
16		POLE WATKINS/HAMILTON	191	19 Poles		Pole Replacements
17		POLE FREEPORT/BOWNE CENTER	620	54 Poles		Pole Replacements
18		POLE ATHENS/SHERWOOD	141	12 Poles		Pole Replacements
19		POLE HOPKINS/MONTEREY	131	12 Poles		Pole Replacements
20		POLE TITUS LAKE/TENTH STREET	433	30 Poles		Pole Replacements
21		RLBY MT MORRIS/NEFF ROAD SUB	243	1 Project		Targeted Circuit Improvements
22		RLBY STANWOOD/RIVERSWAY 916	1,620	1 Project		Targeted Circuit Improvements
23		RLBY EDMORE/CEDAR LAKE 461	1,949	1 Project		Targeted Circuit Improvements
24		RLBY ALBER/TERRITORIAL 984	3	1 Project		Targeted Circuit Improvements
25		RLBY ALDER CREEK/EAST LEROY 846	3	1 Project		Targeted Circuit Improvements
26		RLBY ALDER CREEK/LEE LAKE 889	16	1 Project		Targeted Circuit Improvements
27		RLBY AMPERSEE/BORGESS 917	8	1 Project		Targeted Circuit Improvements
28		RLBY AUSTIN/WEST LAKE 473	174	1 Project		Targeted Circuit Improvements
29		RLBY BEADLE/SPAULDING 308	5	1 Project		Targeted Circuit Improvements
30		RLBY BEECH-NUT/HOLAGAN 227	15	1 Project		Targeted Circuit Improvements
31		RLBY BELL ROAD/ALBEE 856	24	1 Project		Targeted Circuit Improvements
32		RLBY BELLEVUE/ASSYRIA 308	5	1 Project		Targeted Circuit Improvements
33		RLBY BREEDSVILLE/GRAND JUNCTION SUB	3	1 Project		Targeted Circuit Improvements
34		RLBY BURR OAK/INDUSTRIAL 400	2	1 Project		Targeted Circuit Improvements
35		RLBY BURROWS/GRATIOT 725	76	1 Project		Targeted Circuit Improvements
36		RLBY CASCO/HAWKHEAD 508	26	1 Project		Targeted Circuit Improvements
37		RLBY COLEMAN/COLEMAN 671	97	1 Project		Targeted Circuit Improvements
38		RLBY COMSTOCK/TUNIER 464	54	1 Project		Targeted Circuit Improvements
39		RLBY DRAKE ROAD/WURZBURG SUB	6	1 Project		Targeted Circuit Improvements
40		RLBY ELM STREET/VERONA 162	47	1 Project		Targeted Circuit Improvements
41		RLBY EMERALD/BANBURY 604	55	1 Project		Targeted Circuit Improvements
42		RLBY FIFTEEN MILE ROAD/15 MILE ROAD 352	31	1 Project		Targeted Circuit Improvements
43		RLBY KILGORE/MILHAM 117	5	1 Project		Targeted Circuit Improvements
44		RLBY LITCHFIELD/QUAKER LAKE SUB	3	1 Project		Targeted Circuit Improvements
45		RLBY MILL GROVE/DUMONT SUB	50	1 Project		Targeted Circuit Improvements
46		RLBY MILLERS POINT/ARCADIA SUB	3	1 Project		Targeted Circuit Improvements
47		RLBY MILLERS POINT/CONCORD SUB	9	1 Project		Targeted Circuit Improvements
48		RLBY MORGAN/ORCHARD 108	6	1 Project		Targeted Circuit Improvements
49		RLBY MORGAN/ST MARYS 596	3	1 Project		Targeted Circuit Improvements
50		LVD Lines Reliability subtotal	8,044			

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Distribution Projects

Summary Projected Electric Capital Expenditures

For the Test Year 12 Months Ending December 31, 2021

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Case No.: U-20697

Exhibit No.: A-42 (RTB-15)

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Witness: RTBlumenstock

Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Reliability (cont.)	RLBY NORTH ALLEGAN/HUBBARD 361		2	1 Project	Targeted Circuit Improvements
2		RLBY NORTHERN FIBRE/NORTHERN FIBRE 400		147	1 Project	Targeted Circuit Improvements
3		RLBY OTTAWA BEACH/LAKEWOOD 868		16	1 Project	Targeted Circuit Improvements
4		RLBY PHILLIPS/ALCOTT 277		42	1 Project	Targeted Circuit Improvements
5		RLBY PRINCETON/WATTLES SUB		11	1 Project	Targeted Circuit Improvements
6		RLBY QUINCY/CHICAGO ROAD SUB		4	1 Project	Targeted Circuit Improvements
7		RLBY RAVINE/PATTERSON SUB		2	1 Project	Targeted Circuit Improvements
8		RLBY SAUGATUCK/SAUGATUCK 202		10	1 Project	Targeted Circuit Improvements
9		RLBY THAYER/LUFKIN RULE SUB		31	1 Project	Targeted Circuit Improvements
10		RLBY WARNER/MILO SUB		32	1 Project	Targeted Circuit Improvements
11		RLBY WILLIAMS/ELY SUB		83	1 Project	Targeted Circuit Improvements
12		RLBY ABBE/ABBE 428		123	1 Project	Targeted Circuit Improvements
13		RLBY ALAMO/PINE GROVE 646		95	1 Project	Targeted Circuit Improvements
14		RLBY ALDER CREEK/LEE LAKE 603		11	1 Project	Targeted Circuit Improvements
15		RLBY ALGER/SKIDWAY 5220		100	1 Project	Targeted Circuit Improvements
16		RLBY ALGER/FOREST LAKE 720		821	1 Project	Targeted Circuit Improvements
17		RLBY ALTO/MCCORDS 704		364	1 Project	Targeted Circuit Improvements
18		RLBY AU GRES/POINT LOOK-OUT 523		49	1 Project	Targeted Circuit Improvements
19		RLBY BACKUS/MAPLE VALLEY 531		160	1 Project	Targeted Circuit Improvements
20		RLBY BELDING/MALL 183		170	1 Project	Targeted Circuit Improvements
21		RLBY BENNETT/DOBBIE ROAD 174		292	1 Project	Targeted Circuit Improvements
22		RLBY BOSTON SQUARE/NELAND 51		146	1 Project	Targeted Circuit Improvements
23		RLBY BRICKER/BRICKER 228		162	1 Project	Targeted Circuit Improvements
24		RLBY BROADWAY/PHILLIPS 731		103	1 Project	Targeted Circuit Improvements
25		RLBY BROGAN/BROGAN 762		39	1 Project	Targeted Circuit Improvements
26		RLBY CAMELOT LAKE/COLEMAN 128		96	1 Project	Targeted Circuit Improvements
27		RLBY CASCADE/PEACE STREET 872		124	1 Project	Targeted Circuit Improvements
28		RLBY CENTER ROAD/EVERGREEN 304		52	1 Project	Targeted Circuit Improvements
29		RLBY CHAPIN/MARION SUB		194	1 Project	Targeted Circuit Improvements
30		RLBY CONKLIN PARK/CROTON 946		472	1 Project	Targeted Circuit Improvements
31		RLBY COURT/KENT SUB		42	1 Project	Targeted Circuit Improvements
32		RLBY EAST JORDAN/IRONTON 499		382	1 Project	Targeted Circuit Improvements
33		RLBY EASTWOOD/TEXEL SUB		122	1 Project	Targeted Circuit Improvements
34		RLBY FOREMAN/CUMBERLAND 621		162	1 Project	Targeted Circuit Improvements
35		RLBY FOREMAN/CUMBERLAND SUB		261	1 Project	Targeted Circuit Improvements
36		RLBY FRANKFORT/GATEWAY 5537		29	1 Project	Targeted Circuit Improvements
37		RLBY FROST/LONG LAKE 267		410	1 Project	Targeted Circuit Improvements
38		RLBY FULTON/ALTA DALE SUB		11	1 Project	Targeted Circuit Improvements
39		RLBY GLEN LAKE/BURDICKVILLE 693		242	1 Project	Targeted Circuit Improvements
40		RLBY GLEN LAKE/BURDICKVILLE 772		310	1 Project	Targeted Circuit Improvements
41		RLBY GLENDALE/KEYES 921		64	1 Project	Targeted Circuit Improvements
42		RLBY GREENWOOD/RAU ROAD SUB		305	1 Project	Targeted Circuit Improvements
43		RLBY HARRIETTA/BOON 411		127	1 Project	Targeted Circuit Improvements
44		RLBY HARRIETTA/BOON 744		308	1 Project	Targeted Circuit Improvements
45		RLBY HASTINGS/BOLTWOOD 238		3	1 Project	Targeted Circuit Improvements
46		RLBY HOGSBACK/PINE TREE 819		2	1 Project	Targeted Circuit Improvements
47		RLBY HONOR/INDIAN HILL 6418		392	1 Project	Targeted Circuit Improvements
48		RLBY HOUGHTON HEIGHTS/PRUDENVILLE 983		69	1 Project	Targeted Circuit Improvements
49		LVD Lines Reliability subtotal	7,192			

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Consumers Energy Company

Distribution Projects

Summary Projected Electric Capital Expenditures

For the Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

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Witness: RTBlumenstock

Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Reliability (cont.)	RLBY HULL STREET/CRANBERRY 604	168	1	Project	Targeted Circuit Improvements
2		RLBY LABARGE/ALASKA 208	176	1	Project	Targeted Circuit Improvements
3		RLBY LELAND/LELAND 234	250	1	Project	Targeted Circuit Improvements
4		RLBY LELAND/LELAND 294	205	1	Project	Targeted Circuit Improvements
5		RLBY LEONARD/IONIA 326	150	1	Project	Targeted Circuit Improvements
6		RLBY LEVEL PARK/COLLIER 279	38	1	Project	Targeted Circuit Improvements
7		RLBY LINCOLN/LOST LAKE 628	29	1	Project	Targeted Circuit Improvements
8		RLBY LINDEN/LOBDELL 519	6	1	Project	Targeted Circuit Improvements
9		RLBY MAPLE GROVE/SUMMIT AVENUE 382	181	1	Project	Targeted Circuit Improvements
10		RLBY MARION/GASCOM 14	222	1	Project	Targeted Circuit Improvements
11		RLBY MARTIN/SHELBYVILLE 97	57	1	Project	Targeted Circuit Improvements
12		RLBY MENDON/PINHOOK 645	102	1	Project	Targeted Circuit Improvements
13		RLBY MONA LAKE/AIRPORT 574	170	1	Project	Targeted Circuit Improvements
14		RLBY MORGAN/ST MARYS 240	100	1	Project	Targeted Circuit Improvements
15		RLBY NESTROM/SOUTH SHORE 550	17	1	Project	Targeted Circuit Improvements
16		RLBY NORTH MUSKEGON/DALTON 347	122	1	Project	Targeted Circuit Improvements
17		RLBY NORTH MUSKEGON/STATE PARK 53	102	1	Project	Targeted Circuit Improvements
18		RLBY ONEKAMA/CHIEF 608	54	1	Project	Targeted Circuit Improvements
19		RLBY ORLEANS/ORLEANS 95	5	1	Project	Targeted Circuit Improvements
20		RLBY OTSEGO/OTSEGO 135	204	1	Project	Targeted Circuit Improvements
21		RLBY OTTAWA BEACH/LAKEWOOD 764	270	1	Project	Targeted Circuit Improvements
22		RLBY OTTAWA BEACH/PORT SHELDON 538	78	1	Project	Targeted Circuit Improvements
23		RLBY PALO/CHARLES ROAD 314	22	1	Project	Targeted Circuit Improvements
24		RLBY PETTIS ROAD/PETTIS ROAD 959	356	1	Project	Targeted Circuit Improvements
25		RLBY PICKEREL/INDIAN LAKE 909	70	1	Project	Targeted Circuit Improvements
26		RLBY RED ARROW/OGEMA 273	81	1	Project	Targeted Circuit Improvements
27		RLBY ROGUE RIVER/ROGUE RIVER 318	151	1	Project	Targeted Circuit Improvements
28		RLBY ROSCOMMON/PIONEER 161	93	1	Project	Targeted Circuit Improvements
29		RLBY SAUGATUCK/DOUGLAS 140	85	1	Project	Targeted Circuit Improvements
30		RLBY SAUGATUCK/SILVER LAKE 913	44	1	Project	Targeted Circuit Improvements
31		RLBY SILVER LAKE/SECOR 324	16	1	Project	Targeted Circuit Improvements
32		RLBY SKYLARK/ROCKINGCHAIR 336	56	1	Project	Targeted Circuit Improvements
33		RLBY ST CHARLES/FERGUS 310	63	1	Project	Targeted Circuit Improvements
34		RLBY SURREY/MAIN STREET 386	59	1	Project	Targeted Circuit Improvements
35		RLBY SURREY/MAIN STREET 400	27	1	Project	Targeted Circuit Improvements
36		RLBY TALLMAN/WRIGHT ROAD SUB	61	1	Project	Targeted Circuit Improvements
37		RLBY VIRGINIA PARK/MACATAWA SUB	21	1	Project	Targeted Circuit Improvements
38		RLBY WHITEHALL/ALICE 874	322	1	Project	Targeted Circuit Improvements
39		RLBY WIRTZ ROAD/WILDWOOD 309	509	1	Project	Targeted Circuit Improvements
40		RLBY WITHEY LAKE/PETTIT 804	145	1	Project	Targeted Circuit Improvements
41		RLBY YORKVILLE/HIGHLAND PARK 182	9	1	Project	Targeted Circuit Improvements
42		RLBY READING/CAMBRIA 603	291	1	Project	Targeted Circuit Improvements
43		RLBY RAMONA/ROBINSON SUB	470	1	Project	Targeted Circuit Improvements
44		RLBY COCHRAN/SNOW SUB	111	1	Project	Targeted Circuit Improvements
45		RLBY KINDERHOOK/LAKE DRIVE 681	669	1	Project	Targeted Circuit Improvements
46		RLBY COWAN LAKE/GRATTAN 814	383	1	Project	Targeted Circuit Improvements
47		RLBY HOWARD CITY/CORAL SUB	300	1	Project	Targeted Circuit Improvements
48		RLBY LAKE MITCHELL/GOLF CLUB 6001	428	1	Project	Targeted Circuit Improvements
49		RLBY HOLTON/HOLTON 488	911	1	Project	Targeted Circuit Improvements
50		LVD Lines Reliability subtotal	8,459			

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
1	Reliability (cont.)					
1	LVD Lines Reliability (cont.)	RLBY NAPOLEON/STONEY LAKE SUB	142	1	Project	Targeted Circuit Improvements
2		RLBY GOLDEN/CONTINENTAL SUB	15	1	Project	Targeted Circuit Improvements
3		RLBY GOLDEN/ROCKWELL SUB	68	1	Project	Targeted Circuit Improvements
4		RLBY GOLDEN/SMITHS CROSSING SUB	19	1	Project	Targeted Circuit Improvements
5		RLBY MAYFAIR/HAMADY SUB	211	1	Project	Targeted Circuit Improvements
6		RLBY CHAPIN/MARION 936	106	1	Project	Targeted Circuit Improvements
7		RLBY GENESEEVILLE/GENESEE 30	243	1	Project	Targeted Circuit Improvements
8		RLBY LAINGSBURG/ROUND LAKE 217	711	1	Project	Targeted Circuit Improvements
9		RLBY FOURTEENTH STREET/LIBERTY STREET Secondary	450	1	Project	Targeted Circuit Improvements
10		RLBY GETTY/MARQUETTE Secondary	188	1	Project	Targeted Circuit Improvements
11		RLBY OWOSSO/GOULD 653	852	1	Project	Targeted Circuit Improvements
12		RLBY Underground Injection	1,658	1	Project	Targeted Circuit Improvements
13		RLBY GLEN LAKE/ARBOR 844	271	1	Project	Targeted Circuit Improvements
14		RLBY MARKEY/FOREST ESTATES 452	395	1	Project	Targeted Circuit Improvements
15		RLBY CONVIS/WALNUT POINT 461	18	1	Project	Targeted Circuit Improvements
16		RLBY BELDING/MALL SUB	45	1	Project	Targeted Circuit Improvements
17		RLBY COLLEGE PARK/RIVERSIDE 199	362	1	Project	Targeted Circuit Improvements
18		RLBY KRAFT AVENUE/CENTENNIAL 119	100	1	Project	Targeted Circuit Improvements
19		EXIT CASCADE/THORNCREST SUB	7	1	Project	Circuit Exit Projects
20		EXIT CORUNNA/COURTHOUSE SUB	21	1	Project	Circuit Exit Projects
21		EXIT DURAND/GAINES SUB	21	1	Project	Circuit Exit Projects
22		EXIT HUBBARD LAKE/HUBBARD LAKE SUB	21	1	Project	Circuit Exit Projects
23		EXIT HUDSONVILLE/HUDSONVILLE SUB	11	1	Project	Circuit Exit Projects
24		EXIT JOPPA/JOPPA SUB	4	1	Project	Circuit Exit Projects
25		EXIT JOPPA/BECKLEY SUB	4	1	Project	Circuit Exit Projects
26		EXIT LEFFINGWELL/MICHIGAN OAKS SUB	13	1	Project	Circuit Exit Projects
27		EXIT LEONARD/IONIA SUB	16	1	Project	Circuit Exit Projects
28		EXIT MANISTEE/PARKDALE SUB	33	1	Project	Circuit Exit Projects
29		EXIT MAYFAIR/SHERATON SUB	8	1	Project	Circuit Exit Projects
30		EXIT NORTHERN FIBRE/FIBRE SUB	27	1	Project	Circuit Exit Projects
31		EXIT PIGEON LAKE/PIGEON SUB	15	1	Project	Circuit Exit Projects
32		EXIT PRINCETON/BROWNLEE SUB	27	1	Project	Circuit Exit Projects
33		EXIT PRINCETON/WATTLES SUB	27	1	Project	Circuit Exit Projects
34		EXIT SUMMIT/FRANCIS STREET SUB	8	1	Project	Circuit Exit Projects
35		EXIT PULLMAN/CHICORA SUB	14	1	Project	Circuit Exit Projects
36		EXIT BYRON CENTER/RAILSIDE SUB	43	1	Project	Circuit Exit Projects
37		EXIT HASKELITE/RICHMOND SUB	19	1	Project	Circuit Exit Projects
38		EXIT WYOMING PARK/PORTER SUB	16	1	Project	Circuit Exit Projects
39		EXIT CHICAGO/CHICAGO SUB	54	1	Project	Circuit Exit Projects
40		EXIT THORNAPPLE/BUTTRICK SUB	22	1	Project	Circuit Exit Projects
41		EXIT CALVIN/WOODCLIFF SUB	5	1	Project	Circuit Exit Projects
42		EXIT PEARLINE/COPPER BEACH SUB	13	1	Project	Circuit Exit Projects
43		EXIT AGNEW SUB	28	1	Project	Circuit Exit Projects
44		EXIT ALLENDALE SUB	28	1	Project	Circuit Exit Projects
45		EXIT AUSTIN SUB	28	1	Project	Circuit Exit Projects
46		EXIT BABCOCK SUB	28	1	Project	Circuit Exit Projects
47		EXIT BALZER SUB	21	1	Project	Circuit Exit Projects
48		EXIT BARNUM CREEK SUB	28	1	Project	Circuit Exit Projects
49		EXIT BAY HARBOR SUB	28	1	Project	Circuit Exit Projects
50		EXIT BENNINGTON SUB	28	1	Project	Circuit Exit Projects
51		LVD Lines Reliability subtotal	6,519			

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	Reliability (cont.)					
1	LVD Lines Reliability (cont.)	EXIT BITTERSWEET SUB	28	1 Project	Circuit Exit Projects	
2		EXIT BLISSFIELD SUB	28	1 Project	Circuit Exit Projects	
3		EXIT BOSTON SQUARE SUB	21	1 Project	Circuit Exit Projects	
4		EXIT BOWEN SUB	21	1 Project	Circuit Exit Projects	
5		EXIT BOYNE MTN SUB	28	1 Project	Circuit Exit Projects	
6		EXIT BREEDSVILLE SUB	28	1 Project	Circuit Exit Projects	
7		EXIT BROADMOOR SUB	20	1 Project	Circuit Exit Projects	
8		EXIT BRYE ROAD SUB	28	1 Project	Circuit Exit Projects	
9		EXIT BURLINGAME SUB	20	1 Project	Circuit Exit Projects	
10		EXIT CENTREVILLE SUB	21	1 Project	Circuit Exit Projects	
11		EXIT CHAPIN SUB	21	1 Project	Circuit Exit Projects	
12		EXIT CLARKSVILLE SUB	28	1 Project	Circuit Exit Projects	
13		EXIT CLIMAX SUB	28	1 Project	Circuit Exit Projects	
14		EXIT CRANBROOK SUB	28	1 Project	Circuit Exit Projects	
15		EXIT CURTIS SUB	28	1 Project	Circuit Exit Projects	
16		EXIT DALE ROAD SUB	28	1 Project	Circuit Exit Projects	
17		EXIT DEER LAKE SUB	28	1 Project	Circuit Exit Projects	
18		EXIT DOEHLER JARVIS SUB	28	1 Project	Circuit Exit Projects	
19		EXIT DUFFIELD SUB	28	1 Project	Circuit Exit Projects	
20		EXIT EASTLAWN SUB	28	1 Project	Circuit Exit Projects	
21		EXIT EASTWOOD SUB	21	1 Project	Circuit Exit Projects	
22		EXIT ENGLISHVILLE SUB	28	1 Project	Circuit Exit Projects	
23		EXIT FENNVILLE SUB	28	1 Project	Circuit Exit Projects	
24		EXIT FLETCHER SUB	28	1 Project	Circuit Exit Projects	
25		EXIT FREMONT SUB	28	1 Project	Circuit Exit Projects	
26		EXIT GILSON SUB	28	1 Project	Circuit Exit Projects	
27		EXIT GRAND BLANC SUB	21	1 Project	Circuit Exit Projects	
28		EXIT GRAND LEDGE SUB	28	1 Project	Circuit Exit Projects	
29		EXIT GUN LAKE SUB	20	1 Project	Circuit Exit Projects	
30		EXIT HAGER PARK SUB	28	1 Project	Circuit Exit Projects	
31		EXIT HAMMOND RD SUB	28	1 Project	Circuit Exit Projects	
32		EXIT HARVEY STREET SUB	20	1 Project	Circuit Exit Projects	
33		EXIT HENDERSHOT SUB	28	1 Project	Circuit Exit Projects	
34		EXIT HUBBARDSTON ROAD SUB	21	1 Project	Circuit Exit Projects	
35		EXIT LAINGSBURG SUB	28	1 Project	Circuit Exit Projects	
36		EXIT LAKE LEANN SUB	28	1 Project	Circuit Exit Projects	
37		EXIT LEE STREET SUB	20	1 Project	Circuit Exit Projects	
38		EXIT LESLIE SUB	28	1 Project	Circuit Exit Projects	
39		EXIT LESLIE INDUSTRIAL SUB	21	1 Project	Circuit Exit Projects	
40		EXIT MACATAWA SUB	28	1 Project	Circuit Exit Projects	
41		EXIT MANNSIDING SUB	28	1 Project	Circuit Exit Projects	
42		EXIT MAYNARD SUB	28	1 Project	Circuit Exit Projects	
43		EXIT MCCracken SUB	21	1 Project	Circuit Exit Projects	
44		EXIT MIDLAND SUB	20	1 Project	Circuit Exit Projects	
45		EXIT MORENCI SUB	28	1 Project	Circuit Exit Projects	
46		EXIT TWELFTH STREET SUB	28	1 Project	Circuit Exit Projects	
47		Right of Way, Salary, and Outside Service Capital Costs for LVD Lines Reliability	2,321			Right of Way
48		LVD Lines Reliability subtotal	3,499			
49		LVD Lines Reliability Total	44,929			

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	Reliability (cont.)					
1	HVD Lines Reliability	St. Charles	1,035	1.6	Miles	HVD Line Rebuilds
2		Grand Valley - GVSU	556	0.8	Miles	HVD Line Rebuilds
3		Morrice North	2,101	4.6	Miles	HVD Line Rebuilds
4		Morrice - Morrice Sub Tap	3	0.0	Miles	HVD Line Rebuilds
5		Hughes Rd	64	0.1	Miles	HVD Line Rebuilds
6		Union City	2,439	5.3	Miles	HVD Line Rebuilds
7		Middleville	1,187	2.6	Miles	HVD Line Rebuilds
8		Rosebush	3,654	8.0	Miles	HVD Line Rebuilds
9		Remus	5,892	12.9	Miles	HVD Line Rebuilds
10		Goodale	1,324	2.9	Miles	HVD Line Rebuilds
11		Goodale - Brigden Spur	91	0.2	Miles	HVD Line Rebuilds
12		Hudsonville	365	0.8	Miles	HVD Line Rebuilds
13		Gun Lake - Martin Spur	2,466	5.4	Miles	HVD Line Rebuilds
14		Newaygo - Grant Spur	2,466	4.9	Miles	HVD Line Rebuilds
15		West Branch	685	1.5	Miles	HVD Line Rebuilds
16		Rankin	320	0.7	Miles	HVD Line Rebuilds
17		Ionia #2	594	1.3	Miles	HVD Line Rebuilds
18		Grand Blanc	137	0.3	Miles	HVD Line Rebuilds
19		Wirtz	502	1.1	Miles	HVD Line Rebuilds
20		Lake Shore	2,436	0.8	Miles	HVD Line Rebuilds
21		Galesburg	1,827	4.0	Miles	HVD Line Rebuilds
22		Hodenpyl	1,416	3.1	Miles	HVD Line Rebuilds
23		Grover	228	0.5	Miles	HVD Line Rebuilds
24		Grover - Grover Spur	46	0.1	Miles	HVD Line Rebuilds
25		Grand Valley	1,736	3.8	Miles	HVD Line Rebuilds
26		Van Slyke #1 - Reroute	1,005	2.2	Miles	HVD Line Rebuilds
27		Van Slyke #1/Atherton/Aldrich	3,806	2.5	Miles	HVD Line Rebuilds
28		Union City	311	0.7	Miles	HVD Line Rebuilds
29		Nashville	2,740	0.0*	Miles	HVD Line Rebuilds
30		Nashville - Casite Sub Tap	3	0.0*	Miles	HVD Line Rebuilds
31		Wayland	2,250	0.0*	Miles	HVD Line Rebuilds
32		Morrice - Center	1,827	0.0*	Miles	HVD Line Rebuilds
33		Morrice - Perry Sub Tap	3	0.0*	Miles	HVD Line Rebuilds
34		New Richmond	1,142	0.0*	Miles	HVD Line Rebuilds
35		Manitou Beach	228	2.5	Miles	Pole Top Rehabilitation
36		Waldron	228	2.5	Miles	Pole Top Rehabilitation
37		Diesel	192	2.1	Miles	Pole Top Rehabilitation
38		Agnew	1,078	11.8	Miles	Pole Top Rehabilitation
39		Lake Shore	987	10.8	Miles	Pole Top Rehabilitation
40		Bellevue	1,516	16.6	Miles	Pole Top Rehabilitation
41		Surrey	1,297	14.2	Miles	Pole Top Rehabilitation
42		Davison	174	1.9	Miles	Pole Top Rehabilitation
43		Galesburg	530	5.8	Miles	Pole Top Rehabilitation
44		Hodenpyl	265	2.9	Miles	Pole Top Rehabilitation
45		Kalarama	311	3.4	Miles	Pole Top Rehabilitation
46		Plainwell	512	5.6	Miles	Pole Top Rehabilitation
47		Wirtz #1	512	5.6	Miles	Pole Top Rehabilitation
48		Wirtz #2	210	2.3	Miles	Pole Top Rehabilitation
49		Wirtz #3	27	0.3	Miles	Pole Top Rehabilitation
50		Lake Odessa	667	7.3	Miles	Pole Top Rehabilitation
51		Davison #1	183	2.0	Miles	Pole Top Rehabilitation
52		Davison #2	347	3.8	Miles	Pole Top Rehabilitation
53		Belsay	146	1.6	Miles	Pole Top Rehabilitation
54		Stadium	256	2.8	Miles	Pole Top Rehabilitation
		HVD Lines Reliability subtotal	56,321			

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	Reliability (cont.)					
1	HVD Lines Reliability (cont.)	Pole Replacements to be identified	17,416	880	Poles	Pole Replacements
2		MOABS SCADA projects	3,752		Locations	Switches
3		ABS replacements	639	7	Switches	Switches
4		HVD Lines Reliability subtotal	21,807			
5		HVD Lines Reliability Total	78,128			
6	LVD Substations Reliability	MT MORRIS	1,500	1	Substation	Rebuild Substation
7		MAPLE CITY	640	1	Substation	Rebuild Substation
8		TAWAS	1,500	1	Substation	Rebuild Substation
9		THORNTON	1,500	1	Substation	Rebuild Substation
10		MOBILE #24	3,360	1	Substation	New Mobile Substation
11		BEAUGRAND	90	1	Projects	Animal Mitigation
12		DRAKE ROAD	120	1	Projects	Animal Mitigation
13		ELM STREET	120	1	Projects	Animal Mitigation
14		ENSLEY	90	1	Projects	Animal Mitigation
15		FOREMAN	90	1	Projects	Animal Mitigation
16		GOGUAC	90	1	Projects	Animal Mitigation
17		HASTINGS	120	1	Projects	Animal Mitigation
18		HOMER	90	1	Projects	Animal Mitigation
19		INDIAN RIVER	90	1	Projects	Animal Mitigation
20		JOPPA	90	1	Projects	Animal Mitigation
21		LAKE CITY	120	1	Projects	Animal Mitigation
22		PHILLIPS	150	1	Projects	Animal Mitigation
23		PLAINWELL	120	1	Projects	Animal Mitigation
24		PRINCETON	90	1	Projects	Animal Mitigation
25		SALEM	90	1	Projects	Animal Mitigation
26		THORNAPPLE	225	1	Projects	Animal Mitigation
27		WHITE CLOUD	90	1	Projects	Animal Mitigation
28		WILLIAMS	150	1	Projects	Animal Mitigation
29		YORKVILLE	90	1	Projects	Animal Mitigation
30		BALZER	90	1	Projects	Animal Mitigation
31		BELL ROAD	180	1	Projects	Animal Mitigation
32		BENNINGTON	180	1	Projects	Animal Mitigation
33		BROOKLYN	90	1	Projects	Animal Mitigation
34		COLEMAN	90	1	Projects	Animal Mitigation
35		DIMONDALE	90	1	Projects	Animal Mitigation
36		FERGUSON	90	1	Projects	Animal Mitigation
37		FROST	90	1	Projects	Animal Mitigation
38		GRASS LAKE	90	1	Projects	Animal Mitigation
39		GREGORY	90	1	Projects	Animal Mitigation
40		HOUGHTON HEIGHTS	90	1	Projects	Animal Mitigation
41		NEWARK	90	1	Projects	Animal Mitigation
42		NEWBURG	90	1	Projects	Animal Mitigation
43		ROSCOMMON	90	1	Projects	Animal Mitigation
44		SPRING ARBOR	90	1	Projects	Animal Mitigation
45		TALLMAN	180	1	Projects	Animal Mitigation
46		WIRTZ ROAD	90	1	Projects	Animal Mitigation
47		WISNER	90	1	Projects	Animal Mitigation
48		LVD Substations Reliability subtotal	12,505			

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	Reliability (cont.)					
1	LVD Substations Reliability (cont.)	DUTTON	90	3 Regulators	Regulator Replacement	
2		GOGUAC	165	9 Regulators	Regulator Replacement	
3		KILGORE	90	3 Regulators	Regulator Replacement	
4		WILMOTT	150	3 Regulators	Regulator Replacement	
5		CARROLTON	36	1 Regulators	Regulator Replacement	
6		FINDLEY	36	1 Regulators	Regulator Replacement	
7		HEMLOCK	90	3 Regulators	Regulator Replacement	
8		JANES	90	3 Regulators	Regulator Replacement	
9		LETTIS ROAD	90	3 Regulators	Regulator Replacement	
10		MCKEIGHAN	90	3 Regulators	Regulator Replacement	
11		SHATTUCK	35	1 Regulators	Regulator Replacement	
12		WIRTZ ROAD	35	1 Regulators	Regulator Replacement	
13		LELAND	500	1 Transformers	Transformer Replacements	
14		LAMBERTVILLE	750	1 Transformers	Transformer Replacements	
15		SCHUSS MOUNTAIN	750	1 Transformers	Transformer Replacements	
16		LVD Substations Reliability subtotal	2,997			
17		LVD Substations Reliability Total	15,502			
18	HVD Substations Reliability	Frankfort TB1 U	100	1 Projects	Transformer Bushing Replacements	
19		Reading TB1 U	100	1 Projects	Transformer Bushing Replacements	
20		Travis TB1 U	100	1 Projects	Transformer Bushing Replacements	
21		Scottville TB1 U	100	1 Projects	Transformer Bushing Replacements	
22		Twin Lake TB1 U	100	1 Projects	Transformer Bushing Replacements	
23		Morrell TB1 U	100	1 Projects	Transformer Bushing Replacements	
24		United Parcel TB1 U	100	1 Projects	Transformer Bushing Replacements	
25		Au Gres TB1-1 U	100	1 Projects	Transformer Bushing Replacements	
26		Au Gres TB1-2 S	100	1 Projects	Transformer Bushing Replacements	
27		Lewiston TB1-1 U	100	1 Projects	Transformer Bushing Replacements	
28		Lewiston TB1-2 U	100	1 Projects	Transformer Bushing Replacements	
29		Lewiston TB1-3 U	100	1 Projects	Transformer Bushing Replacements	
30		Moore Road 165, 1375, 1382, 1384	200	4 Switches	Switch Replacements	
31		Esco 199 HVH	50	1 Switches	Switch Replacements	
32		Hart 199 HVH	50	1 Switches	Switch Replacements	
33		Jonesville 199 GE TA1	50	1 Switches	Switch Replacements	
34		Portland 199 RVL	50	1 Switches	Switch Replacements	
35		Riga 199 UHT	50	1 Switches	Switch Replacements	
36		Shelby 199 HVH	50	1 Switches	Switch Replacements	
37		Thornapple 199 RVL	50	1 Switches	Switch Replacements	
38		Nashville 199 RBS with USCO live parts	50	1 Switches	Switch Replacements	
39		Moore Road - Replace 166, 1377, 1388, 1488	600	4 Breakers	Circuit Breaker/Switcher Replacements	
40		Muskegon Heights - Replace 499, 500	300	2 Breakers	Circuit Breaker/Switcher Replacements	
41		Raisin - Replace 100, 166, 1077, 1088, 1177	750	5 Breakers	Circuit Breaker/Switcher Replacements	
42		Stronach - Replace 166, 1188, 1388	450	3 Breakers	Circuit Breaker/Switcher Replacements	
43		Bingham - Replace 146, 166, 1188	450	3 Breakers	Circuit Breaker/Switcher Replacements	
44		Bullock - Replace 166, 366, 1077	450	3 Breakers	Circuit Breaker/Switcher Replacements	
45		Brown Paper SPT (1)	35	1 Transformers	Other	
46		Trowbridge Merson Line PTs (3)	105	3 Transformers	Other	
47		Oceana Alt SPT (1)	35	1 Transformers	Other	
48		Broadmoor G Bus, Q Bus, 11JQ (7)	248	7 Transformers	Other	
49		HVD Substations Reliability Total	5,223			

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	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_FAIRFIELD_JASPER_576	N/A	1 Line Reg Location	Automation - Regulator Controllers	
2		REG21_FAIRFIELD_JASPER_630	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_FRONTIER_RANSOM_430	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_NORTH ADAMS_JEROME_988	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_TRIPP ROAD_TRIPP ROAD_405	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_WALDRON_MUNSON_314	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_FRONTIER_RANSOM_418	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_FRONTIER_RANSOM_529	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_HENDERSHOT_CENTENNIAL_905	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_MANITOU BEACH_DEVILS LAKE_277	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_MANITOU BEACH_DEVILS LAKE_548	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_NORTH ADAMS_JEROME_800	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_NORTH ADAMS_JEROME_935	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_NORTH ADAMS_JEROME_985	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_PALMYRA_VICTORSVILLE_737	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		REG21_RUSSELL RD_RUSSELL ROAD_247	N/A	1 Line Reg Location	Automation - Regulator Controllers	
17		REG21_COLLEGE PARK_CARLETON_472	N/A	1 Line Reg Location	Automation - Regulator Controllers	
18		REG21_COLLEGE PARK_RIVERSIDE_213	N/A	1 Line Reg Location	Automation - Regulator Controllers	
19		REG21_FRONTIER_FRONTIER_552	N/A	1 Line Reg Location	Automation - Regulator Controllers	
20		REG21_FRONTIER_FRONTIER_635	N/A	1 Line Reg Location	Automation - Regulator Controllers	
21		REG21_FRONTIER_RANSOM_436	N/A	1 Line Reg Location	Automation - Regulator Controllers	
22		REG21_FRONTIER_RANSOM_530	N/A	1 Line Reg Location	Automation - Regulator Controllers	
23		REG21_FRONTIER_RANSOM_625	N/A	1 Line Reg Location	Automation - Regulator Controllers	
24		REG21_HENDERSHOT_CENTENNIAL_191	N/A	1 Line Reg Location	Automation - Regulator Controllers	
25		REG21_HENDERSHOT_MONROE ROAD_792	N/A	1 Line Reg Location	Automation - Regulator Controllers	
26		REG21_MANITOU BEACH_ADDISON_534	N/A	1 Line Reg Location	Automation - Regulator Controllers	
27		REG21_MANITOU BEACH_DEVILS LAKE_281	N/A	1 Line Reg Location	Automation - Regulator Controllers	
28		REG21_NORTH ADAMS_NORTH ADAMS_356	N/A	1 Line Reg Location	Automation - Regulator Controllers	
29		REG21_PALMYRA_VICTORSVILLE_732	N/A	1 Line Reg Location	Automation - Regulator Controllers	
30		REG21_RIGA_BIERMAN_607	N/A	1 Line Reg Location	Automation - Regulator Controllers	
31		REG21_RIGA_GOETZ_507	N/A	1 Line Reg Location	Automation - Regulator Controllers	
32		REG21_RIGA_GOETZ_599	N/A	1 Line Reg Location	Automation - Regulator Controllers	
33		REG21_ROUND LAKE ASPHALT_153	N/A	1 Line Reg Location	Automation - Regulator Controllers	
34		REG21_RUSSELL RD_RUSSELL ROAD_215	N/A	1 Line Reg Location	Automation - Regulator Controllers	
35		REG21_WAMPLERS_FRANKLIN_100	N/A	1 Line Reg Location	Automation - Regulator Controllers	
36		REG21_WAMPLERS_WAMPLERS_173	N/A	1 Line Reg Location	Automation - Regulator Controllers	
37		REG21_BEHNKE_RIVER RD_414	N/A	1 Line Reg Location	Automation - Regulator Controllers	
38		REG21_HOMER_INDUSTRIAL_719	N/A	1 Line Reg Location	Automation - Regulator Controllers	
39		REG21_JONESVILLE_JONESVILLE_593	N/A	1 Line Reg Location	Automation - Regulator Controllers	
40		REG21_LITCHFIELD_QUAKER LAKE_548	N/A	1 Line Reg Location	Automation - Regulator Controllers	
41		REG21_READING_CAMBRIA_511	N/A	1 Line Reg Location	Automation - Regulator Controllers	
42		REG21_READING_CAMBRIA_550	N/A	1 Line Reg Location	Automation - Regulator Controllers	
43		REG21_READING_CAMBRIA_576	N/A	1 Line Reg Location	Automation - Regulator Controllers	
44		REG21_READING_CAMBRIA_590	N/A	1 Line Reg Location	Automation - Regulator Controllers	
45		REG21_TEKONSHA_TEKONSHA_251	N/A	1 Line Reg Location	Automation - Regulator Controllers	
46		REG21_TEKONSHA_TEKONSHA_661	N/A	1 Line Reg Location	Automation - Regulator Controllers	
47		REG21_TEKONSHA_WAGNER_515	N/A	1 Line Reg Location	Automation - Regulator Controllers	
48		REG21_BEHNKE_ANGOLA ROAD_421	N/A	1 Line Reg Location	Automation - Regulator Controllers	
49		REG21_BEHNKE_RIVER RD_777	N/A	1 Line Reg Location	Automation - Regulator Controllers	
50		REG21_CAMDEN_CAMDEN_131	N/A	1 Line Reg Location	Automation - Regulator Controllers	
51		REG21_CAMDEN_MONTGOMERY_326	N/A	1 Line Reg Location	Automation - Regulator Controllers	
52		REG21_GIRARD_GIRARD_321	N/A	1 Line Reg Location	Automation - Regulator Controllers	
53		REG21_GIRARD_GIRARD_571	N/A	1 Line Reg Location	Automation - Regulator Controllers	
54		REG21_GIRARD_GIRARD_586	N/A	1 Line Reg Location	Automation - Regulator Controllers	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_JONESVILLE_JONESVILLE_681	N/A	1 Line Reg Location	Automation - Regulator Controllers	
2		REG21_KINDERHOOK_GILEAD_346	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_TEKONSHA_TEKONSHA_336	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_TEKONSHA_TEKONSHA_677	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_TEKONSHA_WAGNER_95	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_BEHNKE_RIVER RD_361	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_BEHNKE_RIVER RD_774	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_BRONSON_BRONSON_613	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_BRONSON_INDUSTRIAL_229	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_CAMDEN_CAMDEN_113	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_CAMDEN_CAMDEN_143	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_CAMDEN_CAMDEN_145	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_CAMDEN_CAMDEN_249	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_CARLETON ROAD_BECK ROAD_589	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_GIRARD_DAYBURG_164	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		REG21_GIRARD_DAYBURG_166	N/A	1 Line Reg Location	Automation - Regulator Controllers	
17		REG21_HOMER_HOMER_417	N/A	1 Line Reg Location	Automation - Regulator Controllers	
18		REG21_HOMER_HOMER_593	N/A	1 Line Reg Location	Automation - Regulator Controllers	
19		REG21_HOMER_INDUSTRIAL_757	N/A	1 Line Reg Location	Automation - Regulator Controllers	
20		REG21_KINDERHOOK_GILEAD_133	N/A	1 Line Reg Location	Automation - Regulator Controllers	
21		REG21_KINDERHOOK_GILEAD_134	N/A	1 Line Reg Location	Automation - Regulator Controllers	
22		REG21_KINDERHOOK_GILEAD_138	N/A	1 Line Reg Location	Automation - Regulator Controllers	
23		REG21_KINDERHOOK_GILEAD_355	N/A	1 Line Reg Location	Automation - Regulator Controllers	
24		REG21_KINDERHOOK_GILEAD_367	N/A	1 Line Reg Location	Automation - Regulator Controllers	
25		REG21_KINDERHOOK_GILEAD_398	N/A	1 Line Reg Location	Automation - Regulator Controllers	
26		REG21_KINDERHOOK_GILEAD_439	N/A	1 Line Reg Location	Automation - Regulator Controllers	
27		REG21_KINDERHOOK_LAKE DRIVE_114	N/A	1 Line Reg Location	Automation - Regulator Controllers	
28		REG21_KINDERHOOK_LAKE DRIVE_781	N/A	1 Line Reg Location	Automation - Regulator Controllers	
29		REG21_KINDERHOOK_LAKE DRIVE_833	N/A	1 Line Reg Location	Automation - Regulator Controllers	
30		REG21_KINDERHOOK_LAKE DRIVE_889	N/A	1 Line Reg Location	Automation - Regulator Controllers	
31		REG21_KINDERHOOK_LAKE DRIVE_891	N/A	1 Line Reg Location	Automation - Regulator Controllers	
32		REG21_KINDERHOOK_LAKE DRIVE_893	N/A	1 Line Reg Location	Automation - Regulator Controllers	
33		REG21_KINDERHOOK_LAKE DRIVE_900	N/A	1 Line Reg Location	Automation - Regulator Controllers	
34		REG21_LITCHFIELD_ADAMS RD_526	N/A	1 Line Reg Location	Automation - Regulator Controllers	
35		REG21_LITCHFIELD_ADAMS RD_566	N/A	1 Line Reg Location	Automation - Regulator Controllers	
36		REG21_LITCHFIELD_QUAKER LAKE_433	N/A	1 Line Reg Location	Automation - Regulator Controllers	
37		REG21_LITCHFIELD_QUAKER LAKE_482	N/A	1 Line Reg Location	Automation - Regulator Controllers	
38		REG21_LITCHFIELD_QUAKER LAKE_546	N/A	1 Line Reg Location	Automation - Regulator Controllers	
39		REG21_LITCHFIELD_QUAKER LAKE_604	N/A	1 Line Reg Location	Automation - Regulator Controllers	
40		REG21_READING_CAMBRIA_535	N/A	1 Line Reg Location	Automation - Regulator Controllers	
41		REG21_READING_CAMBRIA_773	N/A	1 Line Reg Location	Automation - Regulator Controllers	
42		REG21_TEKONSHA_TEKONSHA_333	N/A	1 Line Reg Location	Automation - Regulator Controllers	
43		REG21_TEKONSHA_WAGNER_240	N/A	1 Line Reg Location	Automation - Regulator Controllers	
44		REG21_TEKONSHA_WAGNER_425	N/A	1 Line Reg Location	Automation - Regulator Controllers	
45		REG21_SCIPIO_POPE ROAD_945	N/A	1 Line Reg Location	Automation - Regulator Controllers	
46		REG21_SPRINGPORT_SPRINGPORT_289	N/A	1 Line Reg Location	Automation - Regulator Controllers	
47		REG21_WINGATE_NORTH_53	N/A	1 Line Reg Location	Automation - Regulator Controllers	
48		REG21_WINGATE_NORTH_54	N/A	1 Line Reg Location	Automation - Regulator Controllers	
49		REG21_SCIPIO_MOSHERVILLE_354	N/A	1 Line Reg Location	Automation - Regulator Controllers	
50		REG21_SPRINGPORT_DEVEREAUX_430	N/A	1 Line Reg Location	Automation - Regulator Controllers	
51		REG21_SPRINGPORT_DEVEREAUX_477	N/A	1 Line Reg Location	Automation - Regulator Controllers	
52		REG21_VANDERCOOK LAKE_HAGUE ROAD_547	N/A	1 Line Reg Location	Automation - Regulator Controllers	
53		REG21_WINGATE_NORTH_616	N/A	1 Line Reg Location	Automation - Regulator Controllers	
54		REG21_BROOKLYN_BROOKLYN_725	N/A	1 Line Reg Location	Automation - Regulator Controllers	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_BROOKLYN_BROOKLYN_768	N/A	1 Line Reg Location	Automation - Regulator Controllers	
2		REG21_SCIPIO_MOSHERVILLE_389	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_SCIPIO_POPE ROAD_990	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_SYLVAN_CITY_304	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_SYLVAN_CITY_555	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_SYLVAN_CITY_633	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_SYLVAN_CITY_634	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_SYLVAN_CITY_684	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_SYLVAN_RURAL_290	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_SYLVAN_RURAL_312	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_SYLVAN_RURAL_479	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_VANDERCOOK LAKE_ACKERSON LAKE_45	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_VANDERCOOK LAKE_HAGUE ROAD_549	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_WINGATE_SOUTH_210	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_BECK ROAD_CONSEAR_37	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		REG21_STERNS ROAD_POINT PLACE_239	N/A	1 Line Reg Location	Automation - Regulator Controllers	
17		REG21_BELLEVUE_ASSYRIA_410	N/A	1 Line Reg Location	Automation - Regulator Controllers	
18		REG21_FINE LAKE_BRISTOL_252	N/A	1 Line Reg Location	Automation - Regulator Controllers	
19		REG21_FINE LAKE_DOWLING_594	N/A	1 Line Reg Location	Automation - Regulator Controllers	
20		REG21_MORGAN_ST MARYS_585	N/A	1 Line Reg Location	Automation - Regulator Controllers	
21		REG21_WAKESHMA_LEONIDAS_655	N/A	1 Line Reg Location	Automation - Regulator Controllers	
22		REG21_BELLEVUE_ASSYRIA_262	N/A	1 Line Reg Location	Automation - Regulator Controllers	
23		REG21_BELLEVUE_BELLEVUE_421	N/A	1 Line Reg Location	Automation - Regulator Controllers	
24		REG21_FINE LAKE_DOWLING_544	N/A	1 Line Reg Location	Automation - Regulator Controllers	
25		REG21_FINE LAKE_DOWLING_684	N/A	1 Line Reg Location	Automation - Regulator Controllers	
26		REG21_LOMBARD_SHERIDAN_571	N/A	1 Line Reg Location	Automation - Regulator Controllers	
27		REG21_OLIVET_AINGER_593	N/A	1 Line Reg Location	Automation - Regulator Controllers	
28		REG21_OLIVET_COLLEGE_144	N/A	1 Line Reg Location	Automation - Regulator Controllers	
29		REG21_WILDER_WILDER_206	N/A	1 Line Reg Location	Automation - Regulator Controllers	
30		REG21_BELLEVUE_ASSYRIA_731	N/A	1 Line Reg Location	Automation - Regulator Controllers	
31		REG21_BELLEVUE_BELLEVUE_235	N/A	1 Line Reg Location	Automation - Regulator Controllers	
32		REG21_BELLEVUE_BELLEVUE_416	N/A	1 Line Reg Location	Automation - Regulator Controllers	
33		REG21_BRIGDEN_VERONA PUMPING STATION BUS_1	N/A	1 Line Reg Location	Automation - Regulator Controllers	
34		REG21_CHAUNCEY_CASS_273	N/A	1 Line Reg Location	Automation - Regulator Controllers	
35		REG21_FINE LAKE_BRISTOL_217	N/A	1 Line Reg Location	Automation - Regulator Controllers	
36		REG21_FINE LAKE_BRISTOL_288	N/A	1 Line Reg Location	Automation - Regulator Controllers	
37		REG21_HALLS LAKE_HALLS LAKE_454	N/A	1 Line Reg Location	Automation - Regulator Controllers	
38		REG21_LEVEL PARK_COLLIER_202	N/A	1 Line Reg Location	Automation - Regulator Controllers	
39		REG21_LOMBARD_LOMBARD_863	N/A	1 Line Reg Location	Automation - Regulator Controllers	
40		REG21_LOMBARD_SHERIDAN_721	N/A	1 Line Reg Location	Automation - Regulator Controllers	
41		REG21_MORGAN_ORCHARD_418	N/A	1 Line Reg Location	Automation - Regulator Controllers	
42		REG21_MORGAN_ST MARYS_253	N/A	1 Line Reg Location	Automation - Regulator Controllers	
43		REG21_MORGAN_ST MARYS_289	N/A	1 Line Reg Location	Automation - Regulator Controllers	
44		REG21_MORGAN_ST MARYS_293	N/A	1 Line Reg Location	Automation - Regulator Controllers	
45		REG21_MORGAN_ST MARYS_489	N/A	1 Line Reg Location	Automation - Regulator Controllers	
46		REG21_OLIVET_COLLEGE_352	N/A	1 Line Reg Location	Automation - Regulator Controllers	
47		REG21_WAKESHMA_FULTON_612	N/A	1 Line Reg Location	Automation - Regulator Controllers	
48		REG21_WAKESHMA_LEONIDAS_571	N/A	1 Line Reg Location	Automation - Regulator Controllers	
49		REG21_OTTAWA BEACH_LAKEWOOD_121	N/A	1 Line Reg Location	Automation - Regulator Controllers	
50		REG21_OTTAWA BEACH_PORT SHELDON_260	N/A	1 Line Reg Location	Automation - Regulator Controllers	
51		REG21_PULLMAN_CHICORA_560	N/A	1 Line Reg Location	Automation - Regulator Controllers	
52		REG21_PULLMAN_PULLMAN_94	N/A	1 Line Reg Location	Automation - Regulator Controllers	
53		REG21_SALEM_NORTH DORR_438	N/A	1 Line Reg Location	Automation - Regulator Controllers	
54		REG21_SALEM_NORTH DORR_439	N/A	1 Line Reg Location	Automation - Regulator Controllers	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Spent	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_BILMAR_POLK_592	N/A	1	Line Reg Location	Automation - Regulator Controllers
2		REG21_PULLMAN_CHICORA_928	N/A	1	Line Reg Location	Automation - Regulator Controllers
3		REG21_BENTHEIM_BENTHEIM_163	N/A	1	Line Reg Location	Automation - Regulator Controllers
4		REG21_BENTHEIM_BENTHEIM_263	N/A	1	Line Reg Location	Automation - Regulator Controllers
5		REG21_BILMAR_PIERCE_230	N/A	1	Line Reg Location	Automation - Regulator Controllers
6		REG21_BILMAR_PIERCE_528	N/A	1	Line Reg Location	Automation - Regulator Controllers
7		REG21_BILMAR_POLK_623	N/A	1	Line Reg Location	Automation - Regulator Controllers
8		REG21_BLACK RIVER_ZEELAND_273	N/A	1	Line Reg Location	Automation - Regulator Controllers
9		REG21_BLACK RIVER_ZEELAND_460	N/A	1	Line Reg Location	Automation - Regulator Controllers
10		REG21_NORTHERN FIBRE_FIBRE_310	N/A	1	Line Reg Location	Automation - Regulator Controllers
11		REG21_NORTHERN FIBRE_FIBRE_459	N/A	1	Line Reg Location	Automation - Regulator Controllers
12		REG21_OTTAWA BEACH_LAKEWOOD_590	N/A	1	Line Reg Location	Automation - Regulator Controllers
13		REG21_OTTAWA BEACH_LAKEWOOD_687	N/A	1	Line Reg Location	Automation - Regulator Controllers
14		REG21_OTTAWA BEACH_PORT SHELDON_462	N/A	1	Line Reg Location	Automation - Regulator Controllers
15		REG21_SALEM_NORTH DORR_342	N/A	1	Line Reg Location	Automation - Regulator Controllers
16		REG21_SALEM_NORTH DORR_647	N/A	1	Line Reg Location	Automation - Regulator Controllers
17		REG21_BROGAN_BROGAN_515	N/A	1	Line Reg Location	Automation - Regulator Controllers
18		REG21_WOODLAND_BARNUM_923	N/A	1	Line Reg Location	Automation - Regulator Controllers
19		REG21_WOODLAND_WOODBURY_180	N/A	1	Line Reg Location	Automation - Regulator Controllers
20		REG21_BROGAN_BROGAN_828	N/A	1	Line Reg Location	Automation - Regulator Controllers
21		REG21_FREEPORT_BOWNE CENTER_509	N/A	1	Line Reg Location	Automation - Regulator Controllers
22		REG21_FREEPORT_BOWNE CENTER_510	N/A	1	Line Reg Location	Automation - Regulator Controllers
23		REG21_FREEPORT_CARLTON CENTER_482	N/A	1	Line Reg Location	Automation - Regulator Controllers
24		REG21_FREEPORT_CARLTON CENTER_494	N/A	1	Line Reg Location	Automation - Regulator Controllers
25		REG21_FREEPORT_CARLTON CENTER_495	N/A	1	Line Reg Location	Automation - Regulator Controllers
26		REG21_FREEPORT_BOWNE CENTER_328	N/A	1	Line Reg Location	Automation - Regulator Controllers
27		REG21_FREEPORT_BOWNE CENTER_635	N/A	1	Line Reg Location	Automation - Regulator Controllers
28		REG21_HASTINGS_BOLTWOOD_235	N/A	1	Line Reg Location	Automation - Regulator Controllers
29		REG21_HASTINGS_BOLTWOOD_295	N/A	1	Line Reg Location	Automation - Regulator Controllers
30		REG21_HASTINGS_BOLTWOOD_314	N/A	1	Line Reg Location	Automation - Regulator Controllers
31		REG21_HASTINGS_BOLTWOOD_825	N/A	1	Line Reg Location	Automation - Regulator Controllers
32		REG21_HASTINGS_BROADWAY_69	N/A	1	Line Reg Location	Automation - Regulator Controllers
33		REG21_HASTINGS_BROADWAY_417	N/A	1	Line Reg Location	Automation - Regulator Controllers
34		REG21_HASTINGS_VIKING_182	N/A	1	Line Reg Location	Automation - Regulator Controllers
35		REG21_HASTINGS_VIKING_325	N/A	1	Line Reg Location	Automation - Regulator Controllers
36		REG21_COOPER_NAGEL_422	N/A	1	Line Reg Location	Automation - Regulator Controllers
37		REG21_COMSTOCK_TUNIER_403	N/A	1	Line Reg Location	Automation - Regulator Controllers
38		REG21_COOPER_COOPER CENTER_123	N/A	1	Line Reg Location	Automation - Regulator Controllers
39		REG21_COMSTOCK_SHIELDS_477	N/A	1	Line Reg Location	Automation - Regulator Controllers
40		REG21_COMSTOCK_TUNIER_714	N/A	1	Line Reg Location	Automation - Regulator Controllers
41		REG21_OSHTEMO_ALMENA_422	N/A	1	Line Reg Location	Automation - Regulator Controllers
42		REG21_RAVINE_RAVINE_322	N/A	1	Line Reg Location	Automation - Regulator Controllers
43		REG21_Texas_EAGLE LAKE_872	N/A	1	Line Reg Location	Automation - Regulator Controllers
44		REG21_TWILIGHT_GULL ROAD_32	N/A	1	Line Reg Location	Automation - Regulator Controllers
45		REG21_TWILIGHT_GULL ROAD_103	N/A	1	Line Reg Location	Automation - Regulator Controllers
46		REG21_MULLINS_ROYAL VISTA_310	N/A	1	Line Reg Location	Automation - Regulator Controllers
47		REG21_CHAFFEE_BOULEVARD_411	N/A	1	Line Reg Location	Automation - Regulator Controllers
48		REG21_GRANDVILLE_KENOWA_592	N/A	1	Line Reg Location	Automation - Regulator Controllers
49		REG21_MARNE_MARNE_466	N/A	1	Line Reg Location	Automation - Regulator Controllers
50		REG21_FOREMAN_CUMBERLAND_659	N/A	1	Line Reg Location	Automation - Regulator Controllers
51		REG21_THORNAPPLE_BUTTRICK_657	N/A	1	Line Reg Location	Automation - Regulator Controllers
52		REG21_ALTO_ALTO_982	N/A	1	Line Reg Location	Automation - Regulator Controllers
53		REG21_ALTO_MCCORDS_433	N/A	1	Line Reg Location	Automation - Regulator Controllers
54		REG21_FOREMAN_CUMBERLAND_379	N/A	1	Line Reg Location	Automation - Regulator Controllers

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
1	Reliability (cont.)					
2	Grid Modernization (Automation)	REG21_FOREMAN_CUMBERLAND_618	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_FOREMAN_VERGENNES_262	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_FOREMAN_VERGENNES_561	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_ALTO_ALTO_745	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_ALTO_ALTO_746	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_ALTO_ALTO_957	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_ALTO_MCCORDS_417	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_ALTO_MCCORDS_451	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_ALTO_MCCORDS_591	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_FOREMAN_VERGENNES_179	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_FOREMAN_VERGENNES_594	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_KENTWOOD_PARIS PARK_209	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_LABARGE_ALASKA_895	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_LABARGE_ALASKA_917	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		REG21_LABARGE_BLODGETT LAKE_21	N/A	1 Line Reg Location	Automation - Regulator Controllers	
17		REG21_LABARGE_BLODGETT LAKE_96	N/A	1 Line Reg Location	Automation - Regulator Controllers	
18		REG21_LABARGE_BLODGETT LAKE_941	N/A	1 Line Reg Location	Automation - Regulator Controllers	
19		REG21_THORNAPPLE_BUTTRICK_321	N/A	1 Line Reg Location	Automation - Regulator Controllers	
20		REG21_THORNAPPLE_BUTTRICK_927	N/A	1 Line Reg Location	Automation - Regulator Controllers	
21		REG21_THORNAPPLE_HEADLEY_415	N/A	1 Line Reg Location	Automation - Regulator Controllers	
22		REG21_BELLA VISTA_BLAKEY_667	N/A	1 Line Reg Location	Automation - Regulator Controllers	
23		REG21_CEDAR SPRINGS_EDGERTON_465	N/A	1 Line Reg Location	Automation - Regulator Controllers	
24		REG21_COWAN LAKE_GRATTAN_60	N/A	1 Line Reg Location	Automation - Regulator Controllers	
25		REG21_COWAN LAKE_GRATTAN_994	N/A	1 Line Reg Location	Automation - Regulator Controllers	
26		REG21_KENT CITY_CASNOVIA_878	N/A	1 Line Reg Location	Automation - Regulator Controllers	
27		REG21_KENT CITY_CASNOVIA_919	N/A	1 Line Reg Location	Automation - Regulator Controllers	
28		REG21_PIERSON_PIERSON_266	N/A	1 Line Reg Location	Automation - Regulator Controllers	
29		REG21_PISTON RING_STEBBINS_442	N/A	1 Line Reg Location	Automation - Regulator Controllers	
30		REG21_PLAINFIELD_KUTTSHILL_607	N/A	1 Line Reg Location	Automation - Regulator Controllers	
31		REG21_CEDAR SPRINGS_NELSON_244	N/A	1 Line Reg Location	Automation - Regulator Controllers	
32		REG21_CEDAR SPRINGS_NELSON_732	N/A	1 Line Reg Location	Automation - Regulator Controllers	
33		REG21_PETTIS ROAD_HONEY CREEK_394	N/A	1 Line Reg Location	Automation - Regulator Controllers	
34		REG21_PLAINFIELD_WOOD_124	N/A	1 Line Reg Location	Automation - Regulator Controllers	
35		REG21_ROCKFORD_WOLVERINE_624	N/A	1 Line Reg Location	Automation - Regulator Controllers	
36		REG21_ALPINE_ALPINE_566	N/A	1 Line Reg Location	Automation - Regulator Controllers	
37		REG21_CANNONSBURG_CANNONSBURG_211	N/A	1 Line Reg Location	Automation - Regulator Controllers	
38		REG21_CANNONSBURG_CANNONSBURG_213	N/A	1 Line Reg Location	Automation - Regulator Controllers	
39		REG21_CEDAR SPRINGS_NELSON_168	N/A	1 Line Reg Location	Automation - Regulator Controllers	
40		REG21_FOUR MILE_BAUMHOFF_333	N/A	1 Line Reg Location	Automation - Regulator Controllers	
41		REG21_FOUR MILE_WALKENT_244	N/A	1 Line Reg Location	Automation - Regulator Controllers	
42		REG21_FOUR MILE_WALKENT_245	N/A	1 Line Reg Location	Automation - Regulator Controllers	
43		REG21_KENT CITY_CASNOVIA_467	N/A	1 Line Reg Location	Automation - Regulator Controllers	
44		REG21_KENT CITY_CASNOVIA_880	N/A	1 Line Reg Location	Automation - Regulator Controllers	
45		REG21_PETTIS ROAD_HONEY CREEK_191	N/A	1 Line Reg Location	Automation - Regulator Controllers	
46		REG21_PIERSON_PIERSON_107	N/A	1 Line Reg Location	Automation - Regulator Controllers	
47		REG21_PIERSON_PIERSON_876	N/A	1 Line Reg Location	Automation - Regulator Controllers	
48		REG21_PIERSON_WHITEFISH_118	N/A	1 Line Reg Location	Automation - Regulator Controllers	
49		REG21_PIERSON_WHITEFISH_965	N/A	1 Line Reg Location	Automation - Regulator Controllers	
50		REG21_PISTON RING_STEBBINS_207	N/A	1 Line Reg Location	Automation - Regulator Controllers	
51		REG21_PLAINFIELD_BELMONT_265	N/A	1 Line Reg Location	Automation - Regulator Controllers	
52		REG21_PLAINFIELD_WOOD_102	N/A	1 Line Reg Location	Automation - Regulator Controllers	
53		REG21_ROCKFORD_WOLVERINE_622	N/A	1 Line Reg Location	Automation - Regulator Controllers	
54		REG21_BELDING_COOKS CORNERS_134	N/A	1 Line Reg Location	Automation - Regulator Controllers	
		REG21_CARSON CITY_BUTTERNUT_231	N/A	1 Line Reg Location	Automation - Regulator Controllers	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Spent	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_CARSON CITY_BUTTERNUT_437	N/A	1	Line Reg Location	Automation - Regulator Controllers
2		REG21_CARSON CITY_BUTTERNUT_441	N/A	1	Line Reg Location	Automation - Regulator Controllers
3		REG21_CARSON CITY_BUTTERNUT_682	N/A	1	Line Reg Location	Automation - Regulator Controllers
4		REG21_CARSON CITY_BUTTERNUT_838	N/A	1	Line Reg Location	Automation - Regulator Controllers
5		REG21_CLYDE ROAD_GLENN ROAD_134	N/A	1	Line Reg Location	Automation - Regulator Controllers
6		REG21_MARKER LAKE_KYSER ROAD_444	N/A	1	Line Reg Location	Automation - Regulator Controllers
7		REG21_ORLEANS_LONG LAKE_28	N/A	1	Line Reg Location	Automation - Regulator Controllers
8		REG21_ORLEANS_LONG LAKE_45	N/A	1	Line Reg Location	Automation - Regulator Controllers
9		REG21_ORLEANS_LONG LAKE_310	N/A	1	Line Reg Location	Automation - Regulator Controllers
10		REG21_ORLEANS_ORLEANS_154	N/A	1	Line Reg Location	Automation - Regulator Controllers
11		REG21_ORLEANS_ORLEANS_191	N/A	1	Line Reg Location	Automation - Regulator Controllers
12		REG21_ORLEANS_ORLEANS_271	N/A	1	Line Reg Location	Automation - Regulator Controllers
13		REG21_SANDERSON_M-57_96	N/A	1	Line Reg Location	Automation - Regulator Controllers
14		REG21_SARANAC_KEENE_776	N/A	1	Line Reg Location	Automation - Regulator Controllers
15		REG21_SARANAC_KEENE_815	N/A	1	Line Reg Location	Automation - Regulator Controllers
16		REG21_SARANAC_KEENE_850	N/A	1	Line Reg Location	Automation - Regulator Controllers
17		REG21_SARANAC_KEENE_919	N/A	1	Line Reg Location	Automation - Regulator Controllers
18		REG21_SARANAC_RIVERSIDE_527	N/A	1	Line Reg Location	Automation - Regulator Controllers
19		REG21_SARANAC_RIVERSIDE_530	N/A	1	Line Reg Location	Automation - Regulator Controllers
20		REG21_SARANAC_RIVERSIDE_532	N/A	1	Line Reg Location	Automation - Regulator Controllers
21		REG21_SARANAC_SARANAC_963	N/A	1	Line Reg Location	Automation - Regulator Controllers
22		REG21_SHERIDAN_FENWICK_96	N/A	1	Line Reg Location	Automation - Regulator Controllers
23		REG21_SHERIDAN_FENWICK_179	N/A	1	Line Reg Location	Automation - Regulator Controllers
24		REG21_SHERIDAN_FENWICK_194	N/A	1	Line Reg Location	Automation - Regulator Controllers
25		REG21_SHERIDAN_FENWICK_196	N/A	1	Line Reg Location	Automation - Regulator Controllers
26		REG21_SHERIDAN_FENWICK_333	N/A	1	Line Reg Location	Automation - Regulator Controllers
27		REG21_SHERIDAN_FENWICK_801	N/A	1	Line Reg Location	Automation - Regulator Controllers
28		REG21_TAMARACK_LAKEVIEW_308	N/A	1	Line Reg Location	Automation - Regulator Controllers
29		REG21_TAMARACK_LAKEVIEW_362	N/A	1	Line Reg Location	Automation - Regulator Controllers
30		REG21_TRUFANT_HUNTER LAKE_651	N/A	1	Line Reg Location	Automation - Regulator Controllers
31		REG21_TRUFANT_HUNTER LAKE_831	N/A	1	Line Reg Location	Automation - Regulator Controllers
32		REG21_TRUFANT_TRUFANT_109	N/A	1	Line Reg Location	Automation - Regulator Controllers
33		REG21_BELDING_COOKS CORNERS_178	N/A	1	Line Reg Location	Automation - Regulator Controllers
34		REG21_BELDING_COOKS CORNERS_304	N/A	1	Line Reg Location	Automation - Regulator Controllers
35		REG21_CARSON CITY_BUTTERNUT_767	N/A	1	Line Reg Location	Automation - Regulator Controllers
36		REG21_CARSON CITY_BUTTERNUT_818	N/A	1	Line Reg Location	Automation - Regulator Controllers
37		REG21_DERBY_BROWN_341	N/A	1	Line Reg Location	Automation - Regulator Controllers
38		REG21_DERBY_BROWN_342	N/A	1	Line Reg Location	Automation - Regulator Controllers
39		REG21_DERBY_DERBY_237	N/A	1	Line Reg Location	Automation - Regulator Controllers
40		REG21_GODFREY_FLAT RIVER_561	N/A	1	Line Reg Location	Automation - Regulator Controllers
41		REG21_GODFREY_LOWELL_180	N/A	1	Line Reg Location	Automation - Regulator Controllers
42		REG21_GREENVILLE_WASHINGTON STREET_802	N/A	1	Line Reg Location	Automation - Regulator Controllers
43		REG21_MARKER LAKE_KYSER ROAD_636	N/A	1	Line Reg Location	Automation - Regulator Controllers
44		REG21_ORLEANS_ORLEANS_156	N/A	1	Line Reg Location	Automation - Regulator Controllers
45		REG21_ORLEANS_ORLEANS_666	N/A	1	Line Reg Location	Automation - Regulator Controllers
46		REG21_PALO_CHARLES ROAD_241	N/A	1	Line Reg Location	Automation - Regulator Controllers
47		REG21_PALO_CHARLES ROAD_315	N/A	1	Line Reg Location	Automation - Regulator Controllers
48		REG21_PALO_PALO_220	N/A	1	Line Reg Location	Automation - Regulator Controllers
49		REG21_SARANAC_CENTERLINE_702	N/A	1	Line Reg Location	Automation - Regulator Controllers
50		REG21_SARANAC_KEENE_701	N/A	1	Line Reg Location	Automation - Regulator Controllers
51		REG21_SHERIDAN_FENWICK_83	N/A	1	Line Reg Location	Automation - Regulator Controllers
52		REG21_SHERIDAN_FENWICK_666	N/A	1	Line Reg Location	Automation - Regulator Controllers
53		REG21_SHERIDAN_SIDNEY_177	N/A	1	Line Reg Location	Automation - Regulator Controllers
54		REG21_STANTON_STANTON_327	N/A	1	Line Reg Location	Automation - Regulator Controllers

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_TRUFANT_HUNTER LAKE_502	N/A	1 Line Reg Location	Automation - Regulator Controllers	
2		REG21_BELDING_CITY_888	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_BELDING_COOKS CORNERS_147	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_BELDING_COOKS CORNERS_151	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_BELDING_COOKS CORNERS_466	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_CARSON CITY_BUTTERNUT_439	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_CARSON CITY_BUTTERNUT_780	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_CARSON CITY_BUTTERNUT_822	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_CARSON CITY_BUTTERNUT_851	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_CARSON CITY_BUTTERNUT_866	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_CARSON CITY_HOSPITAL_783	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_CLYDE ROAD_GLENN ROAD_43	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_DERBY_DERBY_176	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_DERBY_DERBY_288	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_DERBY_DERBY_888	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		REG21_GODFREY_FLAT RIVER_536	N/A	1 Line Reg Location	Automation - Regulator Controllers	
17		REG21_GODFREY_LOWELL_216	N/A	1 Line Reg Location	Automation - Regulator Controllers	
18		REG21_GREENVILLE_WASHINGTON STREET_550	N/A	1 Line Reg Location	Automation - Regulator Controllers	
19		REG21_GREENVILLE_WASHINGTON STREET_666	N/A	1 Line Reg Location	Automation - Regulator Controllers	
20		REG21_GREENVILLE_WASHINGTON STREET_708	N/A	1 Line Reg Location	Automation - Regulator Controllers	
21		REG21_GREENVILLE_WILLIAMS STREET_640	N/A	1 Line Reg Location	Automation - Regulator Controllers	
22		REG21_GREENVILLE_WILLIAMS STREET_761	N/A	1 Line Reg Location	Automation - Regulator Controllers	
23		REG21_LYONS_LYONS-MUIR_271	N/A	1 Line Reg Location	Automation - Regulator Controllers	
24		REG21_MARKER LAKE_JACKSON ROAD_581	N/A	1 Line Reg Location	Automation - Regulator Controllers	
25		REG21_MARKER LAKE_KYSER ROAD_17	N/A	1 Line Reg Location	Automation - Regulator Controllers	
26		REG21_MARKER LAKE_KYSER ROAD_217	N/A	1 Line Reg Location	Automation - Regulator Controllers	
27		REG21_MARKER LAKE_KYSER ROAD_301	N/A	1 Line Reg Location	Automation - Regulator Controllers	
28		REG21_MARKER LAKE_KYSER ROAD_320	N/A	1 Line Reg Location	Automation - Regulator Controllers	
29		REG21_ORLEANS_LONG LAKE_36	N/A	1 Line Reg Location	Automation - Regulator Controllers	
30		REG21_ORLEANS_LONG LAKE_59	N/A	1 Line Reg Location	Automation - Regulator Controllers	
31		REG21_ORLEANS_LONG LAKE_116	N/A	1 Line Reg Location	Automation - Regulator Controllers	
32		REG21_ORLEANS_LONG LAKE_153	N/A	1 Line Reg Location	Automation - Regulator Controllers	
33		REG21_ORLEANS_LONG LAKE_308	N/A	1 Line Reg Location	Automation - Regulator Controllers	
34		REG21_PALO_CHARLES ROAD_156	N/A	1 Line Reg Location	Automation - Regulator Controllers	
35		REG21_PALO_CHARLES ROAD_247	N/A	1 Line Reg Location	Automation - Regulator Controllers	
36		REG21_PALO_PALO_245	N/A	1 Line Reg Location	Automation - Regulator Controllers	
37		REG21_SARANAC_CENTERLINE_508	N/A	1 Line Reg Location	Automation - Regulator Controllers	
38		REG21_SARANAC_CENTERLINE_511	N/A	1 Line Reg Location	Automation - Regulator Controllers	
39		REG21_SARANAC_KEENE_916	N/A	1 Line Reg Location	Automation - Regulator Controllers	
40		REG21_SARANAC_KEENE_921	N/A	1 Line Reg Location	Automation - Regulator Controllers	
41		REG21_SARANAC_KEENE_922	N/A	1 Line Reg Location	Automation - Regulator Controllers	
42		REG21_SHERIDAN_FENWICK_111	N/A	1 Line Reg Location	Automation - Regulator Controllers	
43		REG21_SHERIDAN_FENWICK_777	N/A	1 Line Reg Location	Automation - Regulator Controllers	
44		REG21_STANTON_DICKERSON LAKE_664	N/A	1 Line Reg Location	Automation - Regulator Controllers	
45		REG21_STANTON_DICKERSON LAKE_665	N/A	1 Line Reg Location	Automation - Regulator Controllers	
46		REG21_STANTON_DICKERSON LAKE_670	N/A	1 Line Reg Location	Automation - Regulator Controllers	
47		REG21_TAMARACK_LAKEVIEW_221	N/A	1 Line Reg Location	Automation - Regulator Controllers	
48		REG21_TAMARACK_LAKEVIEW_285	N/A	1 Line Reg Location	Automation - Regulator Controllers	
49		REG21_TAMARACK_LAKEVIEW_385	N/A	1 Line Reg Location	Automation - Regulator Controllers	
50		REG21_TAMARACK_LAKEVIEW_390	N/A	1 Line Reg Location	Automation - Regulator Controllers	
51		REG21_TRUFANT_HUNTER LAKE_104	N/A	1 Line Reg Location	Automation - Regulator Controllers	
52		REG21_TRUFANT_HUNTER LAKE_444	N/A	1 Line Reg Location	Automation - Regulator Controllers	
53		REG21_TRUFANT_HUNTER LAKE_449	N/A	1 Line Reg Location	Automation - Regulator Controllers	
54		REG21_TRUFANT_HUNTER LAKE_567	N/A	1 Line Reg Location	Automation - Regulator Controllers	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	REG21_TRUFANT_HUNTER LAKE_577	N/A	1 Line Reg Location	Automation - Regulator Controllers	
2		REG21_TRUFANT_HUNTER LAKE_590	N/A	1 Line Reg Location	Automation - Regulator Controllers	
3		REG21_TRUFANT_HUNTER LAKE_603	N/A	1 Line Reg Location	Automation - Regulator Controllers	
4		REG21_TRUFANT_HUNTER LAKE_777	N/A	1 Line Reg Location	Automation - Regulator Controllers	
5		REG21_TRUFANT_HUNTER LAKE_915	N/A	1 Line Reg Location	Automation - Regulator Controllers	
6		REG21_NORTH MUSKEGON_STATE PARK_9331	N/A	1 Line Reg Location	Automation - Regulator Controllers	
7		REG21_NESTROM_SCENIC DRIVE_355	N/A	1 Line Reg Location	Automation - Regulator Controllers	
8		REG21_DUPONT_OLD CHANNEL_3	N/A	1 Line Reg Location	Automation - Regulator Controllers	
9		REG21_DUPONT_OLD CHANNEL_456	N/A	1 Line Reg Location	Automation - Regulator Controllers	
10		REG21_HAYES STREET_BUCCANEER_807	N/A	1 Line Reg Location	Automation - Regulator Controllers	
11		REG21_JOHNSON_LINCOLN_250	N/A	1 Line Reg Location	Automation - Regulator Controllers	
12		REG21_JOHNSON_LINCOLN_329	N/A	1 Line Reg Location	Automation - Regulator Controllers	
13		REG21_JOHNSON_LINCOLN_479	N/A	1 Line Reg Location	Automation - Regulator Controllers	
14		REG21_NESTROM_SCENIC DRIVE_356	N/A	1 Line Reg Location	Automation - Regulator Controllers	
15		REG21_NORTH MUSKEGON_STATE PARK_76	N/A	1 Line Reg Location	Automation - Regulator Controllers	
16		LSEN21_BURR OAK_DOUGLAS	N/A	17 Circuit	Automation - Line Sensors	
17		LSEN21_FINE LAKE_BRISTOL	N/A	17 Circuit	Automation - Line Sensors	
18		LSEN21_HOLTON_MAPLE ISLAND	N/A	17 Circuit	Automation - Line Sensors	
19		LSEN21_CAMELOT LAKE_LOOMIS	N/A	17 Circuit	Automation - Line Sensors	
20		LSEN21_FINE LAKE_DOWLING	N/A	17 Circuit	Automation - Line Sensors	
21		LSEN21_HULL STREET_LIME LAKE	N/A	17 Circuit	Automation - Line Sensors	
22		LSEN21_NESTROM_SOUTH SHORE	N/A	17 Circuit	Automation - Line Sensors	
23		LSEN21_KINDERHOOK_GILEAD	N/A	17 Circuit	Automation - Line Sensors	
24		LSEN21_BURR OAK_INDUSTRIAL	N/A	17 Circuit	Automation - Line Sensors	
25		LSEN21_MORGAN_ST MARYS	N/A	17 Circuit	Automation - Line Sensors	
26		LSEN21_EIGHT POINT_WHITE BIRCH	N/A	17 Circuit	Automation - Line Sensors	
27		LSEN21_MENDON_PINHOOK	N/A	17 Circuit	Automation - Line Sensors	
28		LSEN21_LOGISTIC_FELCH	N/A	17 Circuit	Automation - Line Sensors	
29		LSEN21_SPRING DRIVE_FERRIS	N/A	17 Circuit	Automation - Line Sensors	
30		LSEN21_HASTINGS_BOLTWOOD	N/A	17 Circuit	Automation - Line Sensors	
31		LSEN21_SPRING DRIVE_BISHOP LAKE	N/A	17 Circuit	Automation - Line Sensors	
32		LSEN21_OTISVILLE_STATE ROAD	N/A	17 Circuit	Automation - Line Sensors	
33		LSEN21_KOLASSA_KOSMERICK	N/A	17 Circuit	Automation - Line Sensors	
34		LSEN21_GREENVILLE_WASHINGTON ST	N/A	17 Circuit	Automation - Line Sensors	
35		LSEN21_GREENWOOD_INDIAN LAKE	N/A	17 Circuit	Automation - Line Sensors	
36		LSEN21_EASTON_PARMETER	N/A	17 Circuit	Automation - Line Sensors	
37		LSEN21_FOREMAN_VERGENNES	N/A	17 Circuit	Automation - Line Sensors	
38		LSEN21_COCHRAN_KALAMO	N/A	17 Circuit	Automation - Line Sensors	
39		LSEN21_NASHVILLE_NASHVILLE	N/A	17 Circuit	Automation - Line Sensors	
40		LSEN21_CONKLIN PARK_HOLLY	N/A	17 Circuit	Automation - Line Sensors	
41		LSEN21_LABARGE_BLODGETT LAKE	N/A	17 Circuit	Automation - Line Sensors	
42		LSEN21_BUSCH ROAD_CURTIS	N/A	17 Circuit	Automation - Line Sensors	
43		LSEN21_COLEMAN_RURAL	N/A	17 Circuit	Automation - Line Sensors	
44		LSEN21_QUINCY_BLACKHAWK	N/A	17 Circuit	Automation - Line Sensors	
45		LSEN21_NEFF ROAD_LEWIS ROAD	N/A	17 Circuit	Automation - Line Sensors	
46		LSEN21_MONTAGUE_NORTH SHORE	N/A	17 Circuit	Automation - Line Sensors	
47		LSEN21_HOWARD CITY_MORLEY	N/A	17 Circuit	Automation - Line Sensors	
48		LSEN21_RAVENNA_MOORLAND	N/A	17 Circuit	Automation - Line Sensors	
49		LSEN21_CENTREVILLE_BUSINESS	N/A	17 Circuit	Automation - Line Sensors	
50		LSEN21_GUN LAKE_TRAILS END	N/A	17 Circuit	Automation - Line Sensors	
51		LSEN21_ROSE CITY_KLACKING CREEK	N/A	17 Circuit	Automation - Line Sensors	
52		LSEN21_ENSLEY_DISTRIBUTION	N/A	17 Circuit	Automation - Line Sensors	
53		LSEN21_ALAMO_FISH HATCHERY	N/A	17 Circuit	Automation - Line Sensors	
54		LSEN21_ALLENDALE_BLENDON	N/A	17 Circuit	Automation - Line Sensors	

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Spent	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	LSEN21_LEVEL PARK_COLLIER	N/A	17	Circuit	Automation - Line Sensors
2		LSEN21_MERSON_DUCK LAKE	N/A	17	Circuit	Automation - Line Sensors
3		LSEN21_HANNAH_HANNAH	N/A	17	Circuit	Automation - Line Sensors
4		LSEN21_CENTREVILLE_INDUSTRIAL	N/A	17	Circuit	Automation - Line Sensors
5		LSEN21_BEADLE_SPAULDING	N/A	17	Circuit	Automation - Line Sensors
6		LSEN21_DEAN ROAD_SHANNON LAKE	N/A	17	Circuit	Automation - Line Sensors
7		LSEN21_MARNE_WRIGHT	N/A	17	Circuit	Automation - Line Sensors
8		LSEN21_TEKONSHA_TEKONSHA	N/A	17	Circuit	Automation - Line Sensors
9		LSEN21_TEMPERANCE_WOOD ROAD	N/A	17	Circuit	Automation - Line Sensors
10		LSEN21_GULL LAKE_TURNER	N/A	17	Circuit	Automation - Line Sensors
11		LSEN21_ROCKFORD_SUMMIT	N/A	17	Circuit	Automation - Line Sensors
12		LSEN21_FRONTIER_RANSOM	N/A	17	Circuit	Automation - Line Sensors
13		LSEN21_GREGORY_UNADILLA	N/A	17	Circuit	Automation - Line Sensors
14		LSEN21_LABARGE_ALASKA	N/A	17	Circuit	Automation - Line Sensors
15		LSEN21_CASINO_LEATON	N/A	17	Circuit	Automation - Line Sensors
16		LSEN21_STARKS_LEE	N/A	17	Circuit	Automation - Line Sensors
17		LSEN21_CASCO_HAWKHEAD	N/A	17	Circuit	Automation - Line Sensors
18		LSEN21_READING_CAMBRIA	N/A	17	Circuit	Automation - Line Sensors
19		LSEN21_BLACK RIVER_FILLMORE	N/A	17	Circuit	Automation - Line Sensors
20		LSEN21_VIRGINIA PARK_CASTLE	N/A	17	Circuit	Automation - Line Sensors
21		LSEN21_HESPERIA_RURAL	N/A	17	Circuit	Automation - Line Sensors
22		LSEN21_CANNONSBURG_CANNONSBURG	N/A	17	Circuit	Automation - Line Sensors
23		LSEN21_ATHENS_ATHENS	N/A	17	Circuit	Automation - Line Sensors
24		LSEN21_COTTAGE GROVE_PREVO	N/A	17	Circuit	Automation - Line Sensors
25		LSEN21_BALDWIN_IDLEWILD	N/A	17	Circuit	Automation - Line Sensors
26		LSEN21_BEAVERTON_ROSS STREET	N/A	17	Circuit	Automation - Line Sensors
27		LSEN21_OTISVILLE_IRISH ROAD	N/A	17	Circuit	Automation - Line Sensors
28		LSEN21_RODNEY_RODNEY	N/A	17	Circuit	Automation - Line Sensors
29		LSEN21_SCOTTS_SCOTTS	N/A	17	Circuit	Automation - Line Sensors
30		LSEN21_WALKER_ROSALIE	N/A	17	Circuit	Automation - Line Sensors
31		LSEN21_ALTO_ALTO	N/A	17	Circuit	Automation - Line Sensors
32		LSEN21_TEXAS_EAGLE LAKE	N/A	17	Circuit	Automation - Line Sensors
33		LSEN21_LYON MANOR_TOWN HALL	N/A	17	Circuit	Automation - Line Sensors
34		LSEN21_ORIOLE_BRYANT ROAD	N/A	17	Circuit	Automation - Line Sensors
35		LSEN21_BLUE WATER_SCOTT ROAD	N/A	17	Circuit	Automation - Line Sensors
36		LSEN21_PLAINFIELD_WOOD	N/A	17	Circuit	Automation - Line Sensors
37		LSEN21_DEAN ROAD_KELSEY-HAYES	N/A	17	Circuit	Automation - Line Sensors
38		LSEN21_ALDEN_TORCH	N/A	17	Circuit	Automation - Line Sensors
39		LSEN21_COLON_COLON	N/A	17	Circuit	Automation - Line Sensors
40		LSEN21_COOPERSVILLE_EASTMANVILLE	N/A	17	Circuit	Automation - Line Sensors
41		LSEN21_PALO_PALO	N/A	17	Circuit	Automation - Line Sensors
42		LSEN21_BELLEVUE_ASSYRIA	N/A	17	Circuit	Automation - Line Sensors
43		LSEN21_ALDER CREEK_LEE LAKE	N/A	17	Circuit	Automation - Line Sensors
44		LSEN21_NUNICA_WILSON	N/A	17	Circuit	Automation - Line Sensors
45		LSEN21_LAKE LEANN_LAKE LEANN	N/A	17	Circuit	Automation - Line Sensors
46		LSEN21_HOMESTEAD_BEULAH	N/A	17	Circuit	Automation - Line Sensors
47		LSEN21_MENDON_M-60	N/A	17	Circuit	Automation - Line Sensors
48		LSEN21_HANOVER_HORTON	N/A	17	Circuit	Automation - Line Sensors
49		LSEN21_PENNFIELD_PENNFIELD	N/A	17	Circuit	Automation - Line Sensors
50		LSEN21_HOWARD CITY_CORAL	N/A	17	Circuit	Automation - Line Sensors
51		LSEN21_CAMDEN_CAMDEN	N/A	17	Circuit	Automation - Line Sensors
52		LSEN21_BEDFORD_MEACHEM	N/A	17	Circuit	Automation - Line Sensors
53		LSEN21_QUINCY_CHICAGO ROAD	N/A	17	Circuit	Automation - Line Sensors
54		LSEN21_BRIDGEPORT_DIXIE	N/A	17	Circuit	Automation - Line Sensors

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	<u>Reliability (cont.)</u>					
1	Grid Modernization (Automation)	LSEN21_GRANT_GRANT	N/A	17	Circuit	Automation - Line Sensors
2		LSEN21_APPLE_CHATTERSON ROAD	N/A	17	Circuit	Automation - Line Sensors
3		LSEN21_MILL GROVE_DUMONT	N/A	17	Circuit	Automation - Line Sensors
4		LSEN21_BROGAN_BROGAN	N/A	17	Circuit	Automation - Line Sensors
5		LSEN21_GILSON_WYMAN	N/A	17	Circuit	Automation - Line Sensors
6		LSEN21_GLEN LAKE_BURDICKVILLE	N/A	17	Circuit	Automation - Line Sensors
7		LSEN21_COOPER_NAGEL	N/A	17	Circuit	Automation - Line Sensors
8		BRIDGEPORT	N/A	1	Substation	Automation - DSCADA
9		SHATTUCK	N/A	1	Substation	Automation - DSCADA
10		STATE STREET	N/A	1	Substation	Automation - DSCADA
11		HILL ROAD	N/A	1	Substation	Automation - DSCADA
12		BEERS	N/A	1	Substation	Automation - DSCADA
13		SUTTONS BAY	N/A	1	Substation	Automation - DSCADA
14		ROEDEL ROAD	N/A	1	Substation	Automation - DSCADA
15		WEBSTER	N/A	1	Substation	Automation - DSCADA
16		SEIDEL	N/A	1	Substation	Automation - DSCADA
17		CHAUNCEY	N/A	1	Substation	Automation - DSCADA
18		WALNUT	N/A	1	Substation	Automation - DSCADA
19		NIAGARA	N/A	1	Substation	Automation - DSCADA
20		COMSTOCK	N/A	1	Substation	Automation - DSCADA
21		ALGER	N/A	1	Substation	Automation - DSCADA
22		HARRIETTA	N/A	1	Substation	Automation - DSCADA
23		PENTWATER	N/A	1	Substation	Automation - DSCADA
24		AU GRES	N/A	1	Substation	Automation - DSCADA
25		PRESCOTT	N/A	1	Substation	Automation - DSCADA
26		GERRISH	N/A	1	Substation	Automation - DSCADA
27		LAKE CITY	N/A	1	Substation	Automation - DSCADA
28		CALKINS	N/A	1	Substation	Automation - DSCADA
29		OTISVILLE	N/A	1	Substation	Automation - DSCADA
30		PENINSULA	N/A	1	Substation	Automation - DSCADA
31		SHERIDAN	N/A	1	Substation	Automation - DSCADA
32		MCBAIN	N/A	1	Substation	Automation - DSCADA
33		ERIE	N/A	1	Substation	Automation - DSCADA
34		BALDWIN	N/A	1	Substation	Automation - DSCADA
35		FRONTIER	N/A	1	Substation	Automation - DSCADA
36		RIVERDALE	N/A	1	Substation	Automation - DSCADA
37		STANWOOD	N/A	1	Substation	Automation - DSCADA
38		ALDEN	N/A	1	Substation	Automation - DSCADA
39		PLAINFIELD	N/A	1	Substation	Automation - DSCADA
40		BRICKER	N/A	1	Substation	Automation - DSCADA
41		SANFORD DAM	N/A	1	Substation	Automation - DSCADA
42		DEAN ROAD	N/A	1	Substation	Automation - DSCADA
43		TAMARACK	N/A	1	Substation	Automation - DSCADA
44		GLADWIN	N/A	1	Substation	Automation - DSCADA
45		KENT CITY	N/A	1	Substation	Automation - DSCADA
46		SUNFIELD	N/A	1	Substation	Automation - DSCADA
47		NORTH PARK	N/A	1	Substation	Automation - DSCADA
48		STARKS	N/A	1	Substation	Automation - DSCADA
49		CHESANING	N/A	1	Substation	Automation - DSCADA
50		ASHMAN CIRCLE	N/A	1	Substation	Automation - DSCADA
51		ROCKFORD	N/A	1	Substation	Automation - DSCADA
52		SKYLARK	N/A	1	Substation	Automation - DSCADA
53		PECK ROAD	N/A	1	Substation	Automation - DSCADA
54		ST CHARLES	N/A	1	Substation	Automation - DSCADA

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	<u>Reliability (cont.)</u>					
1	Grid Modernization (Automation)	GREENBUSH	N/A	1	Substation	Automation - DSCADA
2		LASALLE	N/A	1	Substation	Automation - DSCADA
3		LEHRING	N/A	1	Substation	Automation - DSCADA
4		PIERSON	N/A	1	Substation	Automation - DSCADA
5		STANTON	N/A	1	Substation	Automation - DSCADA
6		ORLEANS	N/A	1	Substation	Automation - DSCADA
7		LINCOLN	N/A	1	Substation	Automation - DSCADA
8		CALVIN	N/A	1	Substation	Automation - DSCADA
9		TRUFANT	N/A	1	Substation	Automation - DSCADA
10		GENESEEVILLE	N/A	1	Substation	Automation - DSCADA
11		FOURTEENTH STREET	N/A	1	Substation	Automation - DSCADA
12		SANDERSON	N/A	1	Substation	Automation - DSCADA
13		GREENWOOD	N/A	1	Substation	Automation - DSCADA
14		TEMPERANCE	N/A	1	Substation	Automation - DSCADA
15		NORTHPORT	N/A	1	Substation	Automation - DSCADA
16		ALDER CREEK	N/A	1	Substation	Automation - DSCADA
17		BELDING	N/A	1	Substation	Automation - DSCADA
18		MT FOREST	N/A	1	Substation	Automation - DSCADA
19		KINDERHOOK	N/A	1	Substation	Automation - DSCADA
20		PORTSMOUTH	N/A	1	Substation	Automation - DSCADA
21		BELLEVUE	N/A	1	Substation	Automation - DSCADA
22		PELLSTON	N/A	1	Substation	Automation - DSCADA
23		NEFF ROAD	N/A	1	Substation	Automation - DSCADA
24		EDMORE	N/A	1	Substation	Automation - DSCADA
25		REED CITY	N/A	1	Substation	Automation - DSCADA
26		SARANAC	N/A	1	Substation	Automation - DSCADA
27		HEMLOCK	N/A	1	Substation	Automation - DSCADA
28		SILVER LAKE	N/A	1	Substation	Automation - DSCADA
29		DIXIE	N/A	1	Substation	Automation - DSCADA
30		BALLENGER	N/A	1	Substation	Automation - DSCADA
31		BELLA VISTA	N/A	1	Substation	Automation - DSCADA
32		NORTH KENT	N/A	1	Substation	Automation - DSCADA
33		LABARGE	N/A	1	Substation	Automation - DSCADA
34		CAMDEN	N/A	1	Substation	Automation - DSCADA
35		EAST GENESEE AVENUE	N/A	1	Substation	Automation - DSCADA
36		OLIVET	N/A	1	Substation	Automation - DSCADA
37		STERN'S ROAD	N/A	1	Substation	Automation - DSCADA
38		DERBY	N/A	1	Substation	Automation - DSCADA
39		WAGER	N/A	1	Substation	Automation - DSCADA
40		NEWARK	N/A	1	Substation	Automation - DSCADA
41		CUMBERLAND	N/A	1	Substation	Automation - DSCADA
42		KEARSLEY	N/A	1	Substation	Automation - DSCADA
43		RED ARROW	N/A	1	Substation	Automation - DSCADA
44		ENSLEY	N/A	1	Substation	Automation - DSCADA
45		FINDLEY	N/A	1	Substation	Automation - DSCADA
46		WEST RIVER	N/A	1	Substation	Automation - DSCADA
47		HACKETT	N/A	1	Substation	Automation - DSCADA
48		CROFTON	N/A	1	Substation	Automation - DSCADA
49		RUSSELLVILLE	N/A	1	Substation	Automation - DSCADA
50		MARNE	N/A	1	Substation	Automation - DSCADA
51		TEFT ROAD	N/A	1	Substation	Automation - DSCADA
52		BASS LAKE	N/A	1	Substation	Automation - DSCADA
53		IRON STREET	N/A	1	Substation	Automation - DSCADA
54		HOWARD CITY	N/A	1	Substation	Automation - DSCADA

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Spent	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	Grid Modernization (Automation)	GODFREY		1	Substation	Automation - DSCADA
2		CARSON CITY		1	Substation	Automation - DSCADA
3		BELLAIRE		1	Substation	Automation - DSCADA
4		REMUS		1	Substation	Automation - DSCADA
5		WHITE CLOUD		1	Substation	Automation - DSCADA
6		RANKIN		1	Substation	Automation - DSCADA
7		ROSCOMMON		1	Substation	Automation - DSCADA
8		LAMOREAUX		1	Substation	Automation - DSCADA
9		READING		1	Substation	Automation - DSCADA
10		NORTH MUSKEGON		1	Substation	Automation - DSCADA
11		FREEPORT		1	Substation	Automation - DSCADA
12		RIGA		1	Substation	Automation - DSCADA
13		MANCELONA		1	Substation	Automation - DSCADA
14		SHIELDS		1	Substation	Automation - DSCADA
15		MIDDLETON		1	Substation	Automation - DSCADA
16		BURROWS		1	Substation	Automation - DSCADA
17		LAWRENCE		1	Substation	Automation - DSCADA
18		STOCKBRIDGE01_GREGORY02		1	Loop	Automation - ATR Loops
19		BOWEN03_KENTWOOD04		1	Loop	Automation - ATR Loops
20		Cowan Lake02_Harvard Lake02		1	Loop	Automation - ATR Loops
21		SURREY02_CLARE01		1	Loop	Automation - ATR Loops
22		HOMESTEAD02_FRANKFORT04		1	Loop	Automation - ATR Loops
23		MONA LAKE01_MAPLE GROVE02		1	Loop	Automation - ATR Loops
24		MAPLE GROVE04_MAPLE GROVE05		1	Loop	Automation - ATR Loops
25		EASTON01_CLYDE ROAD01		1	Loop	Automation - ATR Loops
26		MAPLE GROVE01_MAPLE GROVE02		1	Loop	Automation - ATR Loops
27		GETTY02_KEATING01		1	Loop	Automation - ATR Loops
28		WESTERN AVENUE01_WESTERN AVENUE04		1	Loop	Automation - ATR Loops
29		WESTERN AVENUE02_WESTERN AVENUE01		1	Loop	Automation - ATR Loops
30		VAN BUREN03_HAGER PARK01		1	Loop	Automation - ATR Loops
31		JAMESTOWN01_HUDSONVILLE02		1	Loop	Automation - ATR Loops
32		SUMMIT01_MICOR01		1	Loop	Automation - ATR Loops
33		LAUNDRA01_KOCHVILLE01		1	Loop	Automation - ATR Loops
34		Eastwood03_Twilight01		1	Loop	Automation - ATR Loops
35		CHAFFEE01_KELLOGGSVILLE01		1	Loop	Automation - ATR Loops
36		SUTTONS BAY02_LEELANAU02		1	Loop	Automation - ATR Loops
37		LEELANAU02_LEELANAU01		1	Loop	Automation - ATR Loops
38		MT PLEASANT01_CRAWFORD01		1	Loop	Automation - ATR Loops
39		EAST GENESEE AVE01_EAST GENESEE AVE02		1	Loop	Automation - ATR Loops
40		GREENSPIRE02_PORTAGE01		1	Loop	Automation - ATR Loops
41		KIESEL02_PATTERSON03		1	Loop	Automation - ATR Loops
42		Nineteen Mile Road02_Milton03		1	Loop	Automation - ATR Loops
43		KIESEL01_KIESEL03		1	Loop	Automation - ATR Loops
44		QUINCY02_QUINCY03		1	Loop	Automation - ATR Loops
45		Kolassa01_Burr Oak01		1	Loop	Automation - ATR Loops
46		HURON01_KAWKAWLIN02		1	Loop	Automation - ATR Loops
47		BURROWS02_SEIDEL01		1	Loop	Automation - ATR Loops
48		JAMES SAVAGE01_ASHMAN CIRCLE03		1	Loop	Automation - ATR Loops
49		Withey Lake01_Ranger Lake03		1	Loop	Automation - ATR Loops
50		STERN ROAD01_STERNS ROAD02		1	Loop	Automation - ATR Loops
51		KEARSLEY06_KEARSLEY05		1	Loop	Automation - ATR Loops
52		EAST GENESEE AVE01_EAST GENESEE AVE03		1	Loop	Automation - ATR Loops
53		Blue star01_Fennville01		1	Loop	Automation - ATR Loops
54		Elm Street05_Princeton01		1	Loop	Automation - ATR Loops
55		hamilton01_bentheim01		1	Loop	Automation - ATR Loops
56		Webster01_Mt Morris01		1	Loop	Automation - ATR Loops

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	System Protection	Bingham	350	5	Relay Packages	Line Exit Relay Replacements
2		Lake Shore	350	5	Relay Packages	Line Exit Relay Replacements
3		Dort	1080	16	Relay Packages	Line Exit Relay Replacements
4		Gleaner	200	3	Relay Packages	Line Exit Relay Replacements
5		Beveridge (6 of 13 units to be completed in 2021)	364	6	Relay Packages	Line Exit Relay Replacements
6		System Protection Total	2,344			
7	Metro Reliability	SAGINAW WATER STREET CIVIL & ELECTRICAL	1,147	1	Project	Obsolete Equipment Replacements
8		WOLVERINE VAULT REBUILD	300	1	Project	Dead Fronting Equipment for Safety
9		JC PENNEY VAULT DEADFRONT	200	1	Project	Dead Fronting Equipment for Safety
10		Metro Reliability Total	1,647			
11	HVD Lines and Substations Rehabilitation	MORROW REBUILD PART 1	6,100	1	Projects	HVD Substation Failure Projects
12		TWINING REBUILD	10,500	1	Projects	HVD Substation Failure Projects
13		HIGGINS REBUILD PART 1	4,800	1	Projects	HVD Substation Failure Projects
14		BASS CREEK #1 TRANSFORMER BANK REPLACEMENT	1,800	1	Projects	HVD Substation Failure Projects
15		BARRY #1 TRANSFORMER BANK REPLACEMENT	1,800	1	Projects	HVD Substation Failure Projects
16		Additional projects to be identified	13,921			
17		HVD Lines and Substations Rehabilitation Total	38,921			
18	LVD Substations Rehabilitation	FAIRFIELD	800	2	Transformers	Allis Chalmers Substation Transformers
19		DAVISON	1,200	2	Transformers	Allis Chalmers Substation Transformers
20		METRO	800	2	Transformers	Allis Chalmers Substation Transformers
21		NORTH KENT	600	1	Transformers	Allis Chalmers Substation Transformers
22		ROSEWOOD	600	1	Transformers	Allis Chalmers Substation Transformers
23		KINGSLEY	600	1	Transformers	Allis Chalmers Substation Transformers
24		LINDEN	600	1	Transformers	Allis Chalmers Substation Transformers
25		MERSON	600	1	Transformers	Allis Chalmers Substation Transformers
26		SUTTONS BAY	600	1	Transformers	Allis Chalmers Substation Transformers
27		FIFTEEN MILE ROAD	800	1	Transformers	Allis Chalmers Substation Transformers
28		HALLS LAKE	600	1	Transformers	Allis Chalmers Substation Transformers
29		INGHAM	600	1	Transformers	Allis Chalmers Substation Transformers
30		MORLEY	600	1	Transformers	Allis Chalmers Substation Transformers
31		BRICKER / Working Clearance Code	450	1	Projects	Equipment Replacement and Regulatory
32		GREENWOOD / Working Clearance Code	750	1	Projects	Equipment Replacement and Regulatory
33		HARING / Working Clearance Code	450	1	Projects	Equipment Replacement and Regulatory
34		ROEDEL ROAD / Working Clearance Code	450	1	Projects	Equipment Replacement and Regulatory
35		MILLERS POINT / Working Clearance Code	450	1	Projects	Equipment Replacement and Regulatory
36		STONEGATE / Working Clearance Code	450	1	Projects	Equipment Replacement and Regulatory
37		SARANAC / Obsolete Reclosers	75	6	Projects	Equipment Replacement and Regulatory
38		KENT CITY / Obsolete Reclosers	150	6	Projects	Equipment Replacement and Regulatory
39		LEELANAU / Obsolete Reclosers	100	6	Projects	Equipment Replacement and Regulatory
40		STEELCASE / Obsolete Reclosers-Switches-Fuses	375	16	Projects	Equipment Replacement and Regulatory
41		PELLSTON	600	1	Transformers	Transformer Replacements (Imminent)
42		TUNNEL PARK	600	1	Transformers	Transformer Replacements (Imminent)
43		ARCADIA	600	1	Transformers	Transformer Replacements (Imminent)
44		LVD Substations Rehabilitation Total	14,500			

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Distribution Projects

Summary Projected Electric Capital Expenditures

For the Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-42 (RTB-15)

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Witness: RTBlumenstock

Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Rehabilitation	INSP APPLE/CHATTERSON ROAD	66	1	Project	Security Assessment Repairs
2		INSP BARNARD/HOSPITAL	89	1	Project	Security Assessment Repairs
3		INSP BEAVERTON/ROSS STREET	58	1	Project	Security Assessment Repairs
4		INSP BECKER/GILES	29	1	Project	Security Assessment Repairs
5		INSP BELL ROAD/RATHBUN	286	1	Project	Security Assessment Repairs
6		INSP BELLEVUE/ASSYRIA	306	1	Project	Security Assessment Repairs
7		INSP BIL-MAR/POLK	35	1	Project	Security Assessment Repairs
8		INSP BIRCH RUN/BIRCH RUN	123	1	Project	Security Assessment Repairs
9		INSP BIRCH RUN/SAGINAW ROAD	113	1	Project	Security Assessment Repairs
10		INSP BOMAN/COUTANT	124	1	Project	Security Assessment Repairs
11		INSP BOMAN/DELAND	124	1	Project	Security Assessment Repairs
12		INSP BRADFORD/DISTRIBUTION	497	1	Project	Security Assessment Repairs
13		INSP BREEDSVILLE/BREEDSVILLE	308	1	Project	Security Assessment Repairs
14		INSP BULLOCK/POSEYVILLE	199	1	Project	Security Assessment Repairs
15		INSP BULLOCK/PRAIRIE	199	1	Project	Security Assessment Repairs
16		INSP BURTCH ROAD/BURTCH ROAD	87	1	Project	Security Assessment Repairs
17		INSP BUSCH ROAD/CURTIS	165	1	Project	Security Assessment Repairs
18		INSP CALKINS/BEECHER ROAD	29	1	Project	Security Assessment Repairs
19		INSP CALKINS/DYE ROAD	58	1	Project	Security Assessment Repairs
20		INSP CAMDEN/CAMDEN	58	1	Project	Security Assessment Repairs
21		INSP CHAUNCEY/AUSTIN	72	1	Project	Security Assessment Repairs
22		INSP CHAUNCEY/CASS	276	1	Project	Security Assessment Repairs
23		INSP CHEESMAN/MONROE	25	1	Project	Security Assessment Repairs
24		INSP CHEESMAN/WARWICK	6	1	Project	Security Assessment Repairs
25		INSP CHESANING/OAKLEY	375	1	Project	Security Assessment Repairs
26		INSP CHESANING/PEET	200	1	Project	Security Assessment Repairs
27		INSP CHEYENNE/ARBUTUS	46	1	Project	Security Assessment Repairs
28		INSP CHEYENNE/LAWNDALE	110	1	Project	Security Assessment Repairs
29		INSP CHEYENNE/MANNION	70	1	Project	Security Assessment Repairs
30		INSP CHEYENNE/MCDONALDS	206	1	Project	Security Assessment Repairs
31		INSP COOKE DAM/VILLAGE	245	1	Project	Security Assessment Repairs
32		INSP COTTAGE GROVE/HURON	87	1	Project	Security Assessment Repairs
33		INSP DAVISON/CLARK STREET	246	1	Project	Security Assessment Repairs
34		INSP DAVISON/CONTINENTAL	124	1	Project	Security Assessment Repairs
35		INSP DAVISON/DELVE	124	1	Project	Security Assessment Repairs
36		INSP DAVISON/POTTER LAKE	19	1	Project	Security Assessment Repairs
37		INSP DEER LAKE/BALL AVENUE	44	1	Project	Security Assessment Repairs
38		INSP DEERFIELD/BUCHOLTZ	32	1	Project	Security Assessment Repairs
39		INSP DEERFIELD/DEERFIELD	264	1	Project	Security Assessment Repairs
40		INSP DEWITT/EAST HERBISON	2	1	Project	Security Assessment Repairs
41		INSP DUCK LAKE/DUCK LAKE	102	1	Project	Security Assessment Repairs
42		INSP EAST JACKSON/DONNELLY	116	1	Project	Security Assessment Repairs
43		INSP EAST JACKSON/TROJAN	116	1	Project	Security Assessment Repairs
44		INSP EAST JORDAN/IRONTON	58	1	Project	Security Assessment Repairs
45		INSP EAST TAWAS/LINCOLN STREET	332	1	Project	Security Assessment Repairs
46		INSP EASTLAWN/FLAJOLE	150	1	Project	Security Assessment Repairs
47		INSP EMERALD/BANBURY	197	1	Project	Security Assessment Repairs
48		INSP EMERALD/GOLFVIEW	115	1	Project	Security Assessment Repairs
49		INSP FERGUSON/KIBBY ROAD	290	1	Project	Security Assessment Repairs
50		INSP FERRIS STREET/FORREST PARK	59	1	Project	Security Assessment Repairs
51		INSP FINDLEY/NODULAR	20	1	Project	Security Assessment Repairs
52		INSP FINE LAKE/DOWLING	8	1	Project	Security Assessment Repairs
53		INSP FLUSHING/SEYMOUR ROAD	124	1	Project	Security Assessment Repairs
54		INSP FORDYCE/LINCOLN	16	1	Project	Security Assessment Repairs
		LVD Lines Rehabilitation subtotal	7,231			

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Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Spent	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Rehabilitation	INSP FOUNTAIN/SOUTH	17	1	Project	Security Assessment Repairs
2		INSP FRANKFORT/ELBERTA	38	1	Project	Security Assessment Repairs
3		INSP FREMONT/HIGH SCHOOL	62	1	Project	Security Assessment Repairs
4		INSP FREMONT/REEMAN	12	1	Project	Security Assessment Repairs
5		INSP FROST/LEOTA	35	1	Project	Security Assessment Repairs
6		INSP FRUITPORT/FRUITPORT	48	1	Project	Security Assessment Repairs
7		INSP GENESEEVILLE/ROGERS	125	1	Project	Security Assessment Repairs
8		INSP GETTY/MARQUETTE	48	1	Project	Security Assessment Repairs
9		INSP GLENDALE/HERCULES	34	1	Project	Security Assessment Repairs
10		INSP GRAND BLANC/GRAND BLANC	38	1	Project	Security Assessment Repairs
11		INSP GRAND BLANC/REID	124	1	Project	Security Assessment Repairs
12		INSP GRAND BLANC/WHIGVILLE	124	1	Project	Security Assessment Repairs
13		INSP GRAND LEDGE/ACADEMY	134	1	Project	Security Assessment Repairs
14		INSP GRANT/GRANT	70	1	Project	Security Assessment Repairs
15		INSP GREENBUSH/HARRISVILLE	319	1	Project	Security Assessment Repairs
16		INSP GROVER/TRIANGLE	96	1	Project	Security Assessment Repairs
17		INSP GULL LAKE/TURNER	96	1	Project	Security Assessment Repairs
18		INSP HARLEM/BORCULO	32	1	Project	Security Assessment Repairs
19		INSP HARLEM/NEW HOLLAND	22	1	Project	Security Assessment Repairs
20		INSP HARRISON/LILLEY LAKE	249	1	Project	Security Assessment Repairs
21		INSP HASTINGS/VIKING	29	1	Project	Security Assessment Repairs
22		INSP HAYES STREET/HAYES STREET	35	1	Project	Security Assessment Repairs
23		INSP HEMLOCK/HEMLOCK	285	1	Project	Security Assessment Repairs
24		INSP HEMLOCK/NELSON	342	1	Project	Security Assessment Repairs
25		INSP HOGAN ROAD/HOISINGTON	174	1	Project	Security Assessment Repairs
26		INSP HOGAN ROAD/MCCASLIN LAKE	205	1	Project	Security Assessment Repairs
27		INSP HOGAN ROAD/ROLSTON	249	1	Project	Security Assessment Repairs
28		INSP HOLLY/GRANGE HALL	203	1	Project	Security Assessment Repairs
29		INSP HOLLY/MAPLE STREET	124	1	Project	Security Assessment Repairs
30		INSP HOMER/INDUSTRIAL	75	1	Project	Security Assessment Repairs
31		INSP HYDE PARK/DUCK LAKE	12	1	Project	Security Assessment Repairs
32		INSP INTERLOCHEN/COPEMISH	50	1	Project	Security Assessment Repairs
33		INSP ITHACA/FAIR GROUNDS	194	1	Project	Security Assessment Repairs
34		INSP JACKMAN/DEAN ROAD	132	1	Project	Security Assessment Repairs
35		INSP JAMES SAVAGE/PATRICK	58	1	Project	Security Assessment Repairs
36		INSP JAMES SAVAGE/WASHINGTON	58	1	Project	Security Assessment Repairs
37		INSP JANES/BELAIR	343	1	Project	Security Assessment Repairs
38		INSP JANES/JANES	99	1	Project	Security Assessment Repairs
39		INSP JANES/WALNUT	116	1	Project	Security Assessment Repairs
40		INSP JOPPA/JOPPA	64	1	Project	Security Assessment Repairs
41		INSP KAWKAWLIN/LAGOON BEACH	271	1	Project	Security Assessment Repairs
42		INSP KAWKAWLIN/RUSSELL	87	1	Project	Security Assessment Repairs
43		INSP KAWKAWLIN/WHEELER ROAD	33	1	Project	Security Assessment Repairs
44		INSP KEARSLEY/BRIAR LANE	35	1	Project	Security Assessment Repairs
45		INSP KEARSLEY/CRAIG	125	1	Project	Security Assessment Repairs
46		INSP KEARSLEY/CROSBY ROAD	125	1	Project	Security Assessment Repairs
47		INSP KEARSLEY/SHILLELAGH	125	1	Project	Security Assessment Repairs
48		INSP KIESEL/WILDER	17	1	Project	Security Assessment Repairs
49		INSP KOLASSA/MATTESON	91	1	Project	Security Assessment Repairs
50		INSP LEHRING/MYERS LAKE	124	1	Project	Security Assessment Repairs
51		INSP LEITH STREET/LEITH	58	1	Project	Security Assessment Repairs
52		INSP LEITH STREET/WESTERN ROAD	17	1	Project	Security Assessment Repairs
53		INSP LOOMIS/TAFT ROAD	586	1	Project	Security Assessment Repairs
54		INSP MACATAWA/BEE LINE	34	1	Project	Security Assessment Repairs
		LVD Lines Rehabilitation subtotal	6,301			

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Distribution Projects

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Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Sper	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Rehabilitation	INSP MANCELONA/ALBA	14	1	Project	Security Assessment Repairs
2		INSP MANISTEE/MANISTEE SALT	37	1	Project	Security Assessment Repairs
3		INSP MAPLE GROVE/WESTGATE	17	1	Project	Security Assessment Repairs
4		INSP MCCracken/MCCracken	15	1	Project	Security Assessment Repairs
5		INSP MERSON/DUCK LAKE	134	1	Project	Security Assessment Repairs
6		INSP MERSON/MERSON	274	1	Project	Security Assessment Repairs
7		INSP MERSON/PIKE LAKE	156	1	Project	Security Assessment Repairs
8		INSP MILL GROVE/BABYLON	7	1	Project	Security Assessment Repairs
9		INSP MILLERS POINT/ARCADIA	10	1	Project	Security Assessment Repairs
10		INSP MOLINE/MOLINE	45	1	Project	Security Assessment Repairs
11		INSP MORRICE/SOUTH MORRICE	4	1	Project	Security Assessment Repairs
12		INSP NORTH MUSKOGON/STATE PARK	107	1	Project	Security Assessment Repairs
13		INSP NUNICA/WILSON	56	1	Project	Security Assessment Repairs
14		INSP OKEMOS/WKAR	23	1	Project	Security Assessment Repairs
15		INSP OVID/OVID	169	1	Project	Security Assessment Repairs
16		INSP PATTERSON/BRIDGE	33	1	Project	Security Assessment Repairs
17		INSP PATTERSON/PATTERSON	33	1	Project	Security Assessment Repairs
18		INSP PATTERSON/STATE	33	1	Project	Security Assessment Repairs
19		INSP PEACOCK/COLEMAN ROAD	58	1	Project	Security Assessment Repairs
20		INSP PICKEREL/EAST LAKE	131	1	Project	Security Assessment Repairs
21		INSP PICKEREL/INDIAN LAKE	42	1	Project	Security Assessment Repairs
22		INSP PIGEON LAKE/OLIVE	55	1	Project	Security Assessment Repairs
23		INSP PIGEON LAKE/PIGEON	24	1	Project	Security Assessment Repairs
24		INSP PORTSMOUTH/BLUMFIELD	222	1	Project	Security Assessment Repairs
25		INSP PRINCETON/BROWNLEE	225	1	Project	Security Assessment Repairs
26		INSP PRINCETON/WATTLES	258	1	Project	Security Assessment Repairs
27		INSP RANGER LAKE/KOKOSING	830	1	Project	Security Assessment Repairs
28		INSP RAVENNA/RAVENNA	71	1	Project	Security Assessment Repairs
29		INSP RIGA/BIERMAN	122	1	Project	Security Assessment Repairs
30		INSP RIGA/GOETZ	198	1	Project	Security Assessment Repairs
31		INSP ROBERTS STREET/BENDER	15	1	Project	Security Assessment Repairs
32		INSP ROBERTS STREET/LEROY STREET	58	1	Project	Security Assessment Repairs
33		INSP SALZBURG/SALZBURG	87	1	Project	Security Assessment Repairs
34		INSP SCIPPIO/POPE ROAD	60	1	Project	Security Assessment Repairs
35		INSP SCOTTVILLE/US-31	33	1	Project	Security Assessment Repairs
36		INSP SHELBY/PINE STREET	40	1	Project	Security Assessment Repairs
37		INSP SHEPHERD/FOREST HILL	81	1	Project	Security Assessment Repairs
38		INSP SIMMONS/CABIN LAKE	12	1	Project	Security Assessment Repairs
39		INSP SKYLARK/SUN VALLEY	124	1	Project	Security Assessment Repairs
40		INSP SOUTH WASHINGTON/FORDNEY	296	1	Project	Security Assessment Repairs
41		INSP SOUTH WASHINGTON/GALLAGHER	50	1	Project	Security Assessment Repairs
42		INSP SOUTH WASHINGTON/HOYT STREET	215	1	Project	Security Assessment Repairs
43		INSP SOUTH WASHINGTON/WATER STREET	99	1	Project	Security Assessment Repairs
44		INSP ST JOHNS/CITY	16	1	Project	Security Assessment Repairs
45		INSP STADIUM/NORMAL	38	1	Project	Security Assessment Repairs
46		INSP STARKS/HOMER	302	1	Project	Security Assessment Repairs
47		INSP STERNS ROAD/LEWIS	290	1	Project	Security Assessment Repairs
48		INSP SUNFIELD/MULLIKEN	4	1	Project	Security Assessment Repairs
49		INSP SUNFIELD/SUNFIELD	4	1	Project	Security Assessment Repairs
50		INSP SWAN CREEK/COPPER	66	1	Project	Security Assessment Repairs
51		INSP SWAN CREEK/MINING	100	1	Project	Security Assessment Repairs
52		INSP SYLVAN/CITY	85	1	Project	Security Assessment Repairs
53		INSP TALLMAN/CLARK ROAD	11	1	Project	Security Assessment Repairs
54		INSP TAWAS/TAWAS	264	1	Project	Security Assessment Repairs
		LVD Lines Rehabilitation subtotal	5,754			

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Date: February 2020

Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Year Sper	(d) Units	(e) Unit Type	(f) Investment Category
	Reliability (cont.)					
1	LVD Lines Rehabilitation	INSP ULMER/BURT ROAD	219	1	Project	Security Assessment Repairs
2		INSP ULMER/MARKET PLACE	30	1	Project	Security Assessment Repairs
3		INSP VILLAGE GREEN/CONSTITUTION	42	1	Project	Security Assessment Repairs
4		INSP VILLAGE GREEN/PLAZA	38	1	Project	Security Assessment Repairs
5		INSP VIRGINIA PARK/CASTLE	53	1	Project	Security Assessment Repairs
6		INSP VIRGINIA PARK/MACATAWA	46	1	Project	Security Assessment Repairs
7		INSP WALDO/JEFFERSON	132	1	Project	Security Assessment Repairs
8		INSP WALDO/LABORATORY	298	1	Project	Security Assessment Repairs
9		INSP WARNER/MILO	335	1	Project	Security Assessment Repairs
10		INSP WAYLAND/WAYLAND	206	1	Project	Security Assessment Repairs
11		INSP WEST FENTON/PARKWAY	124	1	Project	Security Assessment Repairs
12		INSP WHITTUM/KINNEVILLE	20	1	Project	Security Assessment Repairs
13		INSP WIRTZ ROAD/BOWMANVILLE	341	1	Project	Security Assessment Repairs
14		INSP WISNER/MONROE	9	1	Project	Security Assessment Repairs
15		INSP ATLAS/ATLAS	114	1	Project	Security Assessment Repairs
16		INSP ATLAS/GALE	114	1	Project	Security Assessment Repairs
17		INSP AUBURN/AUBURN	220	1	Project	Security Assessment Repairs
18		INSP AUBURN/ELEVATOR	256	1	Project	Security Assessment Repairs
19		INSP BALLENGER/BRADLEY	35	1	Project	Security Assessment Repairs
20		INSP BALLENGER/SALISBURY	32	1	Project	Security Assessment Repairs
21		INSP BRADFORD/MAGRUDER	100	1	Project	Security Assessment Repairs
22		INSP BROOKLYN/FORD	131	1	Project	Security Assessment Repairs
23		INSP BROUGHWELL/MINARD	131	1	Project	Security Assessment Repairs
24		INSP BULLOCK/STEWART	58	1	Project	Security Assessment Repairs
25		INSP CALKINS/FLUSHING ROAD	49	1	Project	Security Assessment Repairs
26		INSP CHAPIN/MARION	525	1	Project	Security Assessment Repairs
27		INSP CHEYENNE/MCCARTY	97	1	Project	Security Assessment Repairs
28		INSP COTTAGE GROVE/PREVO	74	1	Project	Security Assessment Repairs
29		INSP COURT/WOODLAWN	122	1	Project	Security Assessment Repairs
30		INSP DUNBAR/DUNBAR	30	1	Project	Security Assessment Repairs
31		INSP GRASS LAKE/GRASS LAKE	182	1	Project	Security Assessment Repairs
32		INSP MONTROSE/SHERIDAN ROAD	123	1	Project	Security Assessment Repairs
33		INSP MT MORRIS/LEADLEY	123	1	Project	Security Assessment Repairs
34		INSP NEW HAVEN/HENDERSON	130	1	Project	Security Assessment Repairs
35		INSP NEW HAVEN/JUDDVILLE	154	1	Project	Security Assessment Repairs
36		INSP NIAGARA/HAMILTON	54	1	Project	Security Assessment Repairs
37		INSP NIAGARA/NIAGARA	33	1	Project	Security Assessment Repairs
38		INSP ORCHARD ROAD/SAGINAW ROAD	83	1	Project	Security Assessment Repairs
39		INSP PARMA/BALDWIN	131	1	Project	Security Assessment Repairs
40		INSP PARNALL/DISPOSAL	171	1	Project	Security Assessment Repairs
41		INSP PARNALL/PARNALL RD	188	1	Project	Security Assessment Repairs
42		INSP RED CEDAR/BLUE HAVEN	23	1	Project	Security Assessment Repairs
43		INSP SHATTUCK/COUNTY FARM	30	1	Project	Security Assessment Repairs
44		INSP SHATTUCK/FOX GLEN	58	1	Project	Security Assessment Repairs
45		INSP SHIELDS/SHIELDS	59	1	Project	Security Assessment Repairs
46		INSP SKYLARK/RIVERBEND	124	1	Project	Security Assessment Repairs
47		INSP SLOAN/LAVELLE ROAD	110	1	Project	Security Assessment Repairs
48		INSP TOWN LINE/MACKINAW	258	1	Project	Security Assessment Repairs
49		INSP WAGER/FLINT PARK	125	1	Project	Security Assessment Repairs
50		INSP WAGER/PARKLAND	123	1	Project	Security Assessment Repairs
51		INSP WAMPLERS/WAMPLERS	130	1	Project	Security Assessment Repairs
52		INSP WEBSTER/WEBSTER	123	1	Project	Security Assessment Repairs
53		INSP WEST ROAD/WOOD ROAD	28	1	Project	Security Assessment Repairs
54		Imminent Rehabilitation Projects (Demand)	11,893	4,115	Orders	Imminent Rehabilitation
55		LVD Lines Rehabilitation subtotal	18,436			
56		LVD Lines Rehabilitation Total	37,723			

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
<u>Reliability (cont.)</u>						
1	Metro Rehabilitation	ASSOCIATION OF COMMERCE VAULT REPLACEMENT	1,500	1 Project		Vault or Manhole Roof Replacements
2		LOUIS IONIA VAULT REPLACEMENT	1,500	1 Project		Vault or Manhole Roof Replacements
3		SAGINAW JANES AVENUE	1,353	1 Project		Crushed Duct Replacements
4		Metro Rehabilitation Total	4,353			
5	Grid Storage	Standish Portable Battery - 2021 work	8,131	1 Project		Portable Battery
6		Ft Custer Islanded Battery - 2021 work	1,669	1 Project		Islanded Battery
7		Cadillac Solar Farm Battery - 2021 work	200	1 Project		Solar Farm Battery
8		Grid Storage Total	10,000			
<u>Capacity</u>						
9	LVD Lines Capacity	DARE ABBE/CALDWELL 660	35	1 Project		Overloaded Equipment Upgrades
10		DARE ALAMO/FISH HATCHERY 393	15	1 Project		Overloaded Equipment Upgrades
11		DARE AU GRES/AU GRES 837	20	1 Project		Overloaded Equipment Upgrades
12		DARE BALZER/SANDERS 143	8	1 Project		Overloaded Equipment Upgrades
13		DARE BELLA VISTA/BELLA VISTA 114	17	1 Project		Overloaded Equipment Upgrades
14		DARE BIG PRAIRIE/OXBOW SUB	5	1 Project		Overloaded Equipment Upgrades
15		DARE BREEDSVILLE/GRAND JUNCTION 939	15	1 Project		Overloaded Equipment Upgrades
16		DARE BREEDSVILLE/GRAND JUNCTION 955	13	1 Project		Overloaded Equipment Upgrades
17		DARE BRICKER/ELLIS 298	15	1 Project		Overloaded Equipment Upgrades
18		DARE CAMELOT LAKE/COLEMAN 513	419	1 Project		Overloaded Equipment Upgrades
19		DARE CARY ROAD/JEFFERSON 40	1,424	1 Project		Overloaded Equipment Upgrades
20		DARE CASCO/BLUFF SUB	135	1 Project		Overloaded Equipment Upgrades
21		DARE CEDAR LAKE/KINGS CORNER 431	163	1 Project		Overloaded Equipment Upgrades
22		DARE CEDAR LAKE/VAN ET TEN 103	81	1 Project		Overloaded Equipment Upgrades
23		DARE CHEBOYGAN/ALVERNO 761	306	1 Project		Overloaded Equipment Upgrades
24		DARE COGGINS/NEWBERG 197	160	1 Project		Overloaded Equipment Upgrades
25		DARE CONKLIN PARK/CROTON 971	37	1 Project		Overloaded Equipment Upgrades
26		DARE CRANBROOK/11 MILE ROAD 581	192	1 Project		Overloaded Equipment Upgrades
27		DARE CRANBROOK/11 MILE ROAD 708	199	1 Project		Overloaded Equipment Upgrades
28		DARE CRANBROOK/11 MILE ROAD 957	12	1 Project		Overloaded Equipment Upgrades
29		DARE CRYSTAL/CRYSTAL ROAD 490	27	1 Project		Overloaded Equipment Upgrades
30		DARE EAST GRANT/MUD FLAT 120	6	1 Project		Overloaded Equipment Upgrades
31		DARE ENSLEY/BAPTIST LAKE 741	305	1 Project		Overloaded Equipment Upgrades
32		DARE ENSLEY/DISTRIBUTION 621	45	1 Project		Overloaded Equipment Upgrades
33		DARE FIFTEEN MILE ROAD/15 MILE ROAD 605	224	1 Project		Overloaded Equipment Upgrades
34		DARE FIFTEEN MILE ROAD/A DRIVE 103	100	1 Project		Overloaded Equipment Upgrades
35		DARE GREGORY/GREGORY 606	60	1 Project		Overloaded Equipment Upgrades
36		DARE HARPER ROAD/EIFERT 163	1	1 Project		Overloaded Equipment Upgrades
37		DARE HARRISON/STOCKWELL 310	244	1 Project		Overloaded Equipment Upgrades
38		DARE HONOR/PLATTE 073J	100	1 Project		Overloaded Equipment Upgrades
39		DARE LAKE LANSING/WHITE HILLS 838	8	1 Project		Overloaded Equipment Upgrades
40		DARE LAKE LEANN/LAKE LEANN 851	118	1 Project		Overloaded Equipment Upgrades
41		DARE MAGNUS/EAGLE CORNER 822	153	1 Project		Overloaded Equipment Upgrades
42		DARE MAGNUS/EAGLE CORNER 831	121	1 Project		Overloaded Equipment Upgrades
43		DARE MILL GROVE/ALLEGAN HYDRO 914	15	1 Project		Overloaded Equipment Upgrades
44		DARE ORLEANS/LONG LAKE 406	30	1 Project		Overloaded Equipment Upgrades
45		DARE PLAINFIELD/KUTTSHILL 524	95	1 Project		Overloaded Equipment Upgrades
46		DARE RATIGAN/GREELEY 288	103	1 Project		Overloaded Equipment Upgrades
47		DARE REMUS/MECOSTA 811	6	1 Project		Overloaded Equipment Upgrades
48		DARE RODNEY/RODNEY 702	6	1 Project		Overloaded Equipment Upgrades
49		DARE ROSEBUSH/DELWIN 785	242	1 Project		Overloaded Equipment Upgrades
50		DARE SANDERSON/COUNTY FARM 483	177	1 Project		Overloaded Equipment Upgrades
		LVD Lines Capacity subtotal	5,459			

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Capacity (cont.)					
1	LVD Lines Capacity (cont.)	DARE SARANAC/RIVERSIDE 784	1,036	1	Project	Overloaded Equipment Upgrades
2		DARE SAUGATUCK/DOUGLAS 404	588	1	Project	Overloaded Equipment Upgrades
3		DARE SAUGATUCK/SILVER LAKE 984	154	1	Project	Overloaded Equipment Upgrades
4		DARE SIMMONS/DAM ROAD 647	35	1	Project	Overloaded Equipment Upgrades
5		DARE WARNER/MILO 186	15	1	Project	Overloaded Equipment Upgrades
6		DARE HARLEM/HARLEM 967	59	1	Project	Overloaded Equipment Upgrades
7		DARE HAMILTON/HAMILTON 241	133	1	Project	Overloaded Equipment Upgrades
8		DARE GREENWOOD/RAU RD 865	357	1	Project	Overloaded Equipment Upgrades
9		DARE PARMA/ BALDWIN 16	22	1	Project	Overloaded Equipment Upgrades
10		DARE TRUFANT/MASTON LAKE 590	55	1	Project	Overloaded Equipment Upgrades
11		DARE NASHVILLE/VERMONTVILLE 926	27	1	Project	Overloaded Equipment Upgrades
12		DARE TEKONSHA/WAGNER 245	9	1	Project	Overloaded Equipment Upgrades
13		DARE BLUE STAR/PIER COVE 812	138	1	Project	Overloaded Equipment Upgrades
14		DARE DOBSON ROAD/HALF MOON 375	55	1	Project	Overloaded Equipment Upgrades
15		DARE LEVELY/ALLBRIGHT 482	41	1	Project	Overloaded Equipment Upgrades
16		DARE KOLASSA/KOSMERICK 395	163	1	Project	Overloaded Equipment Upgrades
17		DARE - SUB GREENBUSH/GREENBUSH	381	1	Project	Sub Capacity-associated Line Work
18		DARE - SUB LINCOLN/LOST LAKE	797	1	Project	Sub Capacity-associated Line Work
19		DARE - SUB LINCOLN/MIKADO	13	1	Project	Sub Capacity-associated Line Work
20		DARE - SUB HUCKLEBERRY	250	1	Project	Sub Capacity-associated Line Work
21		DARE - SUB PEACH RIDGE/BALLARD	489	1	Project	Sub Capacity-associated Line Work
22		DARE - SUB TAMARACK/LAKEVIEW	1,046	1	Project	Sub Capacity-associated Line Work
23		LVD Lines Capacity subtotal	5,862			
24		LVD Lines Capacity Total	11,321			
25	HVD Lines and	WD1249 ALGOMA INST DUAL PILOT RLY & CCVT	750	1	Orders	Transmission Coordination
26	Substations Capacity	WD1213 EUREKA INSTALL CKT SWR	218	1	Orders	Transmission Coordination
27		WD1248 FARR RD INST RLY PROTECTION METC	539	1	Orders	Transmission Coordination
28		Moore Rd - Add Ckt Swrs/REPL Relays	923	1	Orders	Transmission Coordination
29		WD1134 HARING - ACQR PROPERTY	10	1	Orders	Right of Way Procurement
30		WD1680 SKYLINE SUB ACQUIRE PROPERTY	75	1	Orders	Right of Way Procurement
31		Van Slyke #1 46 kV Line Rebuild/Relocate R/W	200	1	Orders	Right of Way Procurement
32		Bulk Power Substation Rebuild R/W (2/yr) #1	250	1	Orders	Right of Way Procurement
33		Bulk Power Substation Rebuild R/W (2/yr) #2	250	1	Orders	Right of Way Procurement
34		Morrow New Bulk Power Substation (R/W)	500	1	Orders	Right of Way Procurement
35		Houghton Heights 138 kV Line R/W	1200	1	Orders	Right of Way Procurement
36		Five Channels Dist Sub TAP	140	1	Orders	Right of Way Procurement
37		NEW MORENCI AREA 46 KV LINE 7.0 MI ACQ R/W	410	1	Orders	Right of Way Procurement
38		LN029J REA RD Tap-new 46 kV In tap	210	1	Orders	New Interconnections
39		Bilmar Sub - Inst 138kV Tap to TB#2	300	1	Orders	New Interconnections
40		Thornton 46kV Tap	150	1	Orders	New Interconnections
41		Fifteen Mile Road Sub - Inst 138kV Tap	300	1	Orders	New Interconnections
42		Hawthorne Sub - Inst 138kV Tap	300	1	Orders	New Interconnections
43		Maple City Sub - Modify 46kV Tap	150	1	Orders	New Interconnections
44		Lincoln 138kV Tap & Line	1065	1	Orders	New Interconnections
45		VIPOND CREEK SPUR - New 69kV Line	1080	1	Orders	New Interconnections
46		MOSEL SPUR - New 46kV Line	450	1	Orders	New Interconnections
47		Geronimo Bingham Solar (Avery Solar Sub)	15	1	Orders	New Interconnections
48		Contribution Geronimo Bingham Solar	-15	0	Orders	New Interconnections
49		Sand Creek 138kV line & tap	149	1	Orders	New Interconnections
50		Skyline 46kV line & tap	300	1	Orders	New Interconnections
51		HVD Lines and Substations Capacity subtotal	9,917			

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Line No.	(a) Sub-Program	(b) Project Description, Line, Substation, or Location	(c) Projected 2021 Test Year Spending	(d) Units	(e) Unit Type	(f) Investment Category
	Capacity (cont.)					
1	HVD Lines and	SPOWER MIC 1 NEW 46 KV LINE	2250	1 Orders		New Interconnections
2	Substations Capacity (cont.)	Contribution SPOWER MIC 1 NEW 46 KV LINE	-2250	0 Orders		New Interconnections
3		SPOWER MIC 1 PARR RD NEW 46 KV EXIT	225	1 Orders		New Interconnections
4		Contribution SPOWER MIC 1 PARR RD NEW 46 KV EXIT	-225	0 Orders		New Interconnections
5		SPOWER MIC 1 NEW 46 KV SUB	128	1 Orders		New Interconnections
6		Contribution SPOWER MIC 1 NEW 46 KV SUB	-128	0 Orders		New Interconnections
7		SPOWER MIC 2 NEW 46 KV LINE	225	1 Orders		New Interconnections
8		Contribution SPOWER MIC 2 NEW 46 KV LINE	-225	0 Orders		New Interconnections
9		SPOWER MIC 2 NEW 46 KV SUB	128	1 Orders		New Interconnections
10		Contribution SPOWER MIC 2 NEW 46 KV SUB	-128	0 Orders		New Interconnections
11		SPOWER MIC 2 REPL DISTRIBUTION LA'S	165	1 Orders		New Interconnections
12		Contribution SPOWER MIC 2 REPL DISTRIBUTION LA'S	-165	0 Orders		New Interconnections
13		SPOWER MIC 4 NEW 46 KV LINE	563	1 Orders		New Interconnections
14		Contribution SPOWER MIC 4 NEW 46 KV LINE	-563	0 Orders		New Interconnections
15		SPOWER MIC 4 CEMENT CITY NEW 46 KV EXIT	470	1 Orders		New Interconnections
16		Contribution SPOWER MIC 4 CEMENT CITY NEW 46 KV EXIT	-470	0 Orders		New Interconnections
17		SPOWER MIC 4 NEW 46 KV SUB	128	1 Orders		New Interconnections
18		Contribution SPOWER MIC 4 NEW 46 KV SUB	-128	0 Orders		New Interconnections
19		SPOWER MIC 6 NEW 46 KV LINE	75	1 Orders		New Interconnections
20		Contribution SPOWER MIC 6 NEW 46 KV LINE	-75	0 Orders		New Interconnections
21		SPOWER MIC 6 SCOTT LK NEW 46 KV EXIT	188	1 Orders		New Interconnections
22		Contribution SPOWER MIC 6 SCOTT LK NEW 46 KV EXIT	-188	0 Orders		New Interconnections
23		SPOWER MIC 6 NEW 46 KV SUB	128	1 Orders		New Interconnections
24		Contribution SPOWER MIC 6 NEW 46 KV SUB	-128	0 Orders		New Interconnections
25		WD0190 MORROW REPL 156 VARMASTER	383	1 Orders		Load Carrying Capability/Voltage Support
26		HSC - Replace 399 Circuit Swr	450	1 Orders		Load Carrying Capability/Voltage Support
27		DOWLING NEW 46 KV LINE EXIT	473	1 Orders		Load Carrying Capability/Voltage Support
28		LN086AA HARBISON RBLD 4.8MI & INST MOABS	1650	1 Orders		Load Carrying Capability/Voltage Support
29		LN008AE MILLERS POINT REM 138KV LINE %	90	1 Orders		Improved Functionality
30		Four Mile Working Space	1125	1 Orders		Improved Functionality
31		Gleaner Working Space	375	1 Orders		Improved Functionality
32		Post Cereal Working Space	675	1 Orders		Improved Functionality
33		MOABS/ABS Program 2021 MOAB #1	300	1 Orders		Improved Functionality
34		MOABS/ABS Program 2021 MOAB #2	300	1 Orders		Improved Functionality
35		MOABS/ABS Program 2021 MOAB #3	300	1 Orders		Improved Functionality
36		MOABS/ABS Program 2021 MOAB #4	300	1 Orders		Improved Functionality
37		MOABS/ABS Program 2021 ABS #1	100	1 Orders		Improved Functionality
38		MOABS/ABS Program 2021 ABS #2	100	1 Orders		Improved Functionality
39		MOABS/ABS Program 2021 ABS #3	100	1 Orders		Improved Functionality
40		Moore Rd Working Space	642	1 Orders		Improved Functionality
41		Moore Rd Differential Bus Protection	720	1 Orders		Improved Functionality
42		Batavia Working Space	120	1 Orders		Improved Functionality
43		Anticipated projects yet to be specifically identified	2084	Orders		
44		HVD Lines and Substations Capacity subtotal	10,286			
45		HVD Lines and Substations Capacity Total	20,203			
46	LVD Substations Capacity	CLOVERLEAF	1,500	1 Project		New Substations
47		HUCKLEBERRY	2,000	1 Project		New Substations
48		MOSEL	2,000	1 Project		New Substations
49		SKYLINE	1,500	1 Project		New Substations
50		VIPOND	1,500	1 Project		New Substations
51		BILMAR	1,500	1 Project		Increase Capacity at Existing Substation
52		BREEDSVILLE	327	1 Project		Increase Capacity at Existing Substation
53		ELEVENTH STREET	600	1 Project		Increase Capacity at Existing Substation
54		MILLERS POINT	450	1 Project		Increase Capacity at Existing Substation
55		TREMAINE	225	1 Project		Increase Capacity at Existing Substation
56		WEALTHY STREET	1,500	1 Project		Increase Capacity at Existing Substation
57		WHITE CLOUD	900	1 Project		Increase Capacity at Existing Substation
		LVD Substation Capacity Total	14,002			

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	Capacity (cont.)					
1	Conservation Voltage	CVR21_IVANREST_IVANREST	38	1	Circuit	Conservation Voltage Reduction
2	Reduction	CVR21_APPLETON_FULLER	38	1	Circuit	Conservation Voltage Reduction
3		CVR21_BLACKMAN_SANDSTONE	38	1	Circuit	Conservation Voltage Reduction
4		CVR21_LEE STREET_LEE	38	1	Circuit	Conservation Voltage Reduction
5		CVR21_UPTON_SIMMONS	38	1	Circuit	Conservation Voltage Reduction
6		CVR21_CRAHEN_GREENBRIER	38	1	Circuit	Conservation Voltage Reduction
7		CVR21_PACKARD_PACKARD	38	1	Circuit	Conservation Voltage Reduction
8		CVR21_MONA LAKE_AIRPORT	38	1	Circuit	Conservation Voltage Reduction
9		CVR21_LEFFINGWELL_RAPIDSTAN	38	1	Circuit	Conservation Voltage Reduction
10		CVR21_O-AT-KA_EAST BAY	38	1	Circuit	Conservation Voltage Reduction
11		CVR21_MACKINAW CITY_POND STREET	38	1	Circuit	Conservation Voltage Reduction
12		CVR21_STEVENS_ALBANY	38	1	Circuit	Conservation Voltage Reduction
13		CVR21_HYDE PARK_MCMILLIAN	38	1	Circuit	Conservation Voltage Reduction
14		CVR21_ROBERTS STREET_DETTMAN	38	1	Circuit	Conservation Voltage Reduction
15		CVR21_LEE STREET_KIRTLAND	38	1	Circuit	Conservation Voltage Reduction
16		CVR21_HARVEY STREET_FULLER	38	1	Circuit	Conservation Voltage Reduction
17		CVR21_ARTHUR_BERLIN	38	1	Circuit	Conservation Voltage Reduction
18		CVR21_ROBERTS STREET_BENDER	38	1	Circuit	Conservation Voltage Reduction
19		CVR21_BOWEN_BLOSSOM	38	1	Circuit	Conservation Voltage Reduction
20		CVR21_LEE STREET_CENTURY	38	1	Circuit	Conservation Voltage Reduction
21		CVR21_BOYNE CITY_BOYNE CITY	38	1	Circuit	Conservation Voltage Reduction
22		CVR21_BIRCHWOOD_LARAWAY LAKE	38	1	Circuit	Conservation Voltage Reduction
23		CVR21_HARVEY STREET_SUNSHINE	38	1	Circuit	Conservation Voltage Reduction
24		CVR21_STANDALE_STANDALE	38	1	Circuit	Conservation Voltage Reduction
25		CVR21_WEALTHY STREET_INDIANA	38	1	Circuit	Conservation Voltage Reduction
26		CVR21_SPRING LAKE_SPRING LAKE	38	1	Circuit	Conservation Voltage Reduction
27		CVR21_BEALS ROAD_ALGER	38	1	Circuit	Conservation Voltage Reduction
28		CVR21_EAST MUSKEGON_MILL IRON	38	1	Circuit	Conservation Voltage Reduction
29		CVR21_MANCHESTER_DUNCAN	38	1	Circuit	Conservation Voltage Reduction
30		CVR21_BEALS ROAD_DIVISION ROAD	38	1	Circuit	Conservation Voltage Reduction
31		CVR21_HICKORY_BLACK LAKE	38	1	Circuit	Conservation Voltage Reduction
32		CVR21_BYRON CENTER_RAILSIDE	38	1	Circuit	Conservation Voltage Reduction
33		CVR21_DUTTON_GLENWOOD	38	1	Circuit	Conservation Voltage Reduction
34		CVR21_STANDALE_CHESTERFIELD	38	1	Circuit	Conservation Voltage Reduction
35		CVR21_DUTTON_GATEWAY CROSSING	38	1	Circuit	Conservation Voltage Reduction
36		CVR21_CHICAGO_PINEBROOK	38	1	Circuit	Conservation Voltage Reduction
37		CVR21_NEWBURG_VERNON	38	1	Circuit	Conservation Voltage Reduction
38		CVR21_BROADWAY_PHILLIPS	38	1	Circuit	Conservation Voltage Reduction
39		CVR21_MACATAWA_BEE LINE	38	1	Circuit	Conservation Voltage Reduction
40		CVR21_LEONARD_IONIA	38	1	Circuit	Conservation Voltage Reduction
41		CVR21_HICKORY_HARVEY	38	1	Circuit	Conservation Voltage Reduction
42		CVR21_MEDICAL PARK_HEALTHCARE VILLAGE	38	1	Circuit	Conservation Voltage Reduction
43		CVR21_SAVIDGE_KELLY STREET	38	1	Circuit	Conservation Voltage Reduction
44		CVR21_STANDALE_PARKSIDE	38	1	Circuit	Conservation Voltage Reduction
45		CVR21_HARVARD LAKE_COURTLAND	38	1	Circuit	Conservation Voltage Reduction
46		CVR21_LEFFINGWELL_MICHIGAN OAKS	38	1	Circuit	Conservation Voltage Reduction
47		CVR21_MAPLE GROVE_SHAW BOX	38	1	Circuit	Conservation Voltage Reduction
48		CVR21_WALKER_REMEMBRANCE	38	1	Circuit	Conservation Voltage Reduction
49		CVR21_UPTON_MARKET PLACE	38	1	Circuit	Conservation Voltage Reduction
50		CVR21_CUTLERVILLE_68TH STREET	38	1	Circuit	Conservation Voltage Reduction
51		CVR21_AUSTIN_WEST LAKE	38	1	Circuit	Conservation Voltage Reduction
52		CVR21_BABCOCK_DIVISION	38	1	Circuit	Conservation Voltage Reduction
53		CVR21_BEECH-NUT_HOLAGAN	38	1	Circuit	Conservation Voltage Reduction
54		CVR21_LAMBERTVILLE_WHITEFORD	38	1	Circuit	Conservation Voltage Reduction
55		CVR21_TECUMSEH_BRITTON	38	1	Circuit	Conservation Voltage Reduction
56		CVR21_DRAKE ROAD_COUNTRY CLUB	38	1	Circuit	Conservation Voltage Reduction
57		CVR21_HURON_TWO MILE	38	1	Circuit	Conservation Voltage Reduction
58		CVR21_FAIRFIELD_JASPER	38	1	Circuit	Conservation Voltage Reduction
59		CVR21_DUFFIELD_COLE CREEK	38	1	Circuit	Conservation Voltage Reduction
60		CVR21_MISSION_THREE LEAVES	38	1	Circuit	Conservation Voltage Reduction
61		CVR21_SPRING DRIVE_BISHOP LAKE	38	1	Circuit	Conservation Voltage Reduction
62		CVR21_DUQUITE_PINE RIVER	38	1	Circuit	Conservation Voltage Reduction
63		CVR21_HARPER ROAD_AURELIUS	38	1	Circuit	Conservation Voltage Reduction
64		CVR21_MILLER ROAD_YALE STREET	38	1	Circuit	Conservation Voltage Reduction
65		CVR21_HUBBARD LAKE_HUBBARD LAKE	38	1	Circuit	Conservation Voltage Reduction
66		CVR21_BUSCH ROAD_CANADA	38	1	Circuit	Conservation Voltage Reduction
67		CVR21_NOBLE_WHITNEY	38	1	Circuit	Conservation Voltage Reduction
68		CVR21_MIDLAND_BUTTLES	38	1	Circuit	Conservation Voltage Reduction
69		CVR21_TERRACE_MALL	38	1	Circuit	Conservation Voltage Reduction
70		CVR21_BYRON CENTER_FALCON	38	1	Circuit	Conservation Voltage Reduction
71		CVR21_NEELEY_DOSTER	38	1	Circuit	Conservation Voltage Reduction
72		CVR21_TALLMAN_WACOSTA	38	1	Circuit	Conservation Voltage Reduction
73		CVR21_HARLEM_BORCULO	38	1	Circuit	Conservation Voltage Reduction
74		CVR21_CRANBROOK_11 MILE ROAD	38	1	Circuit	Conservation Voltage Reduction
75		CVR21_EAST MUSKEGON_QUARTERLINE ROAD	38	1	Circuit	Conservation Voltage Reduction
76		Conservation Voltage Reduction Total	2,851			

**CONSUMERS ENERGY
ELECTRIC ASSET MANAGEMENT**

High Voltage Distribution(HVD) Pole Inspection Specifications

1. GENERAL SPECIFICATIONS

- 1.1. Consumers Energy shall provide all maps necessary for the wood pole inspection work to be performed.
- 1.2. The Contractor shall furnish all tools, equipment, inspection tags, bore tags, red tags and labor as required for the completion of wood pole inspection as prescribed in these specifications.
 - 1.2.1. Each foreman employed by the Contractor working for Consumers Energy shall be a permanent, full-time employee of the Contractor. Each foreman shall have adequate training and not less than eight weeks experience as a Pole Inspection Foreman.
 - 1.2.2. Each supervisor employed by the Contractor working for Consumers Energy shall be a permanent, full-time employee of the Contractor having not less than two years experience in supervising pole inspection crews.
 - 1.2.3. The Contractor shall supply to a Consumers Energy representative written verification of the training and experience of each foreman and supervisor working for Consumers Energy before such work begins.
- 1.3. The Contractor shall inspect all poles with Consumers Energy facilities on them, which are connected electrically to the circuit or line section specified. All stub poles that are used for guying purposes should also be inspected. Poles within the protective fence of an electric substation shall also be inspected. Vacated or cut-off poles that had/have 3rd party attachments located adjacent to a Consumers Energy pole shall not be inspected, but the adjacent pole inspection record shall include yes or vacant value in the *PULL POLE* field.
 - 1.3.1. For safety purposes, at least one individual per pole inspection team must attend substation entry training provided by Consumers Energy prior to entering a substation perimeter fence.
- 1.4. A representative of the Contractor shall be involved in a periodic conference call/meeting with Consumers Energy representatives to discuss inspection status and issues that may arise.
- 1.5. The Contractor shall electronically report the following information to Consumers Energy's representative each Monday by 12:00 noon (the format will be agreed upon with the Consumers Energy representative in advance):
 - 1.5.1 Number of people who worked the previous week.

- 1.5.2 Specific days of week that each person worked during the previous week.
- 1.5.3 Specific circuit or line section that each crew worked during the previous week.
- 1.5.4 Planned circuit or line section of each crew for the current week.
- 1.6. The Contractor shall obtain, at their expense, all necessary permits required to complete the work prior to beginning the work requiring such permits.
- 1.7. Monthly the Contractor shall submit electronically via E-mail to a designated Consumers Energy's representative an invoice for the entire circuits or line sections that were completed. The Contractor shall include with each invoice a statement summarizing all work performed in each category of billing units.
- 1.8. The Contractor shall submit pole inspection data per individual circuit or line section, with the following conditions:
 - 1.8.1. The data will be in Microsoft excel and either Microsoft Access format (MDB file) or ESRI Shapefile format.
 - 1.8.2. CE Attachment B - Data Format lists the required data to capture for each pole inspection with a specific schema for Microsoft excel and either Microsoft Access format or ESRI Shapefile format.
 - 1.8.3. A template for Microsoft excel and either Microsoft Access MDB file or ESRI Shapefile format with the required specific schema will be provided.
 - 1.8.4. Transfer of circuit or line section data, including photos, Microsoft excel file and Microsoft Access database file or ESRI Shapefile will be packaged as one zip file named with the name and number of the Line Section or Feeder ID. The zip file will be transferred using FTP (File Transfer Protocol).
 - 1.8.5. A sample set of twenty-five (25) pole data records fulfilling all data requirements specified in CE Attachment B - Data Format are to be submitted two weeks before start of pole inspections for testing and validation purposes.
- 1.9. The Contractor will specify and follow their quality assurance and audit process. The Contractor will work with Consumers Energy on an agreed upon acceptance process to ensure inspection and data quality requirements are achieved.
- 1.10. All work, including obtaining, recording, and reporting of the inspection data shall be entirely satisfactory to Consumers Energy and shall be subject to inspection by and approval of Consumers Energy. Any work submitted needing correction may be rejected and/or may delay payment. Common data errors need to be identified and corrected before data is submitted.

- 1.11. Consumers Energy may request a sample data set of an incomplete inspected circuit or line section for auditing purposes.
- 1.12. Contractor shall return any items/materials received from Consumers Energy and not used for inspection purposes.

2. VISUAL INSPECTION & GPS LOCATION SPECIFICATIONS

- 2.1. The Contractor will perform a visual pole inspection of the applicable poles on each circuit or line section assigned and record data per 1.8 above.
- 2.2. The Contractor shall report any system threatening or safety concerns promptly to the designated Consumers Energy representative(s). This includes but is not limited to:
 - 1) A pole that the contractor determines to be an immediate replacement.
 - 2) Any hazard that the contractor determines is a threat to public safety.
 - 3) A floating phase or neutral wire.
 - 4) A broken or severely cracked crossarm.
 - 2.2.1 The Contractor could be requested by a Consumers Energy representative to stay on site until relieved by Consumers Energy representative. Consumers Energy will reimburse Contractor for time spent on site at agreed upon time and material rate.
 - 2.2.2 A follow up E-mail shall be sent to a designated Consumers Energy representative for said pole or hazard within 3 days of call-in. A report template will be provided.
- 2.3. The Contractor will install a Consumers Energy pole number tag placed immediately above the inspection tag (if present), if a legible existing Consumers Energy pole number tag is not already present.
 - 2.3.1. The Consumers Energy pole number tag is a numbered circular metal tag that will be provided by Consumers Energy.
 - 2.3.2. The existing or new pole tag number will be recorded at each pole location.
- 2.4. The Contractor will obtain GPS location data accurate to 2-5 meters or better for each inspected pole. GPS location data will be recorded using the WGS84 (UTM) Coordinate System as a standard. See CE Attachment B – Data Format for electronic data reporting format.

2.5. The Contractor will take one in focus picture of the inspected pole. Picture should contain in its frame all the attachments on that pole.

2.5.1. Photos shall be captured with a device capable of 5.0 megapixels or greater.

2.5.2. The image file name will contain the tag number of the pole, followed by the compass heading of the direction the camera was facing when the photo was taken. The compass heading should be specified as the four cardinal directions and the four ordinal directions (N,NE,E,SE,S,SW,W,NW)

2.5.3. Image file must be in JPEG (.jpg) format.

2.5.4. For poles that are part of a muti-pole structure, photo shall include all poles of that structure.

3. POLE TEST INSPECTION SPECIFICATIONS

3.1. Consumer Energy's preferred pole test inspection method is a device assisted test that helps identify poor pole conditions and voids.

3.2. The Contractor shall conduct a pole test inspection of all poles eleven (11) years old or older and which have not been inspected within the last five (5) years.

3.3. Designated poles shall be tested from ground line to 6 feet above ground line.

3.4. The Contractor shall install an agreed upon inspection tag at chest height on the road side of all tested poles.

3.5. The inspection tag shall have the name of the Contractor and the year of inspection stamped into the tag.

4. BORE TEST INSPECTION SPECIFICATIONS

4.1. The Contractor will perform a bore test on poles for the following reasons:

4.1.1. The pole test indicates the pole is decaying.

4.1.2. Visual decay is present.

4.1.3. Insects appear to be in the pole.

4.1.4. The pole has been backfilled or is discolored.

4.1.5. To satisfy a doubt about a pole's condition.

4.1.6. To determine the condition of a pole which was previously reinforced.

- 4.2. The Contractor shall only probe the pole surface if the pole has indication of defects from the visual or pole test inspections.
- 4.3. The Contractor shall follow the following bore test procedure:
 - 4.3.1. Bore where either the pole test or visual inspection indicates the weakest point. Where two points are equally weak at the same level, bore the side of the pole perpendicular to the direction of the line.
 - 4.3.2. If the pole test indicates decay or voids near ground, excavate to a depth of 6 to 8 inches and bore downward at a 45 degree angle. For poles installed in concrete or black top, bore downward at grade level at a 45 degree angle. (Read shell thickness on the 45 degree scale.) Poles with 1" shell thickness or less shall be indicated for replacement (Section 5.3).
 - 4.3.3. For externally damaged sections, bore both the damaged side and 180 degrees from the damaged side.
 - 4.3.4. For insect colonies, bore weakest point found during pole test.
 - 4.3.5. Probe bored holes with depth gauge and record shell thickness. On all poles that are bored a shell thickness recorded in the appropriate data field is required. For poles where no void is found record a NV (No Void) in the *Bore* field. Record approximant height from ground line of minimum shell thickness in *Minimum Shell Thickness Height Field*.
 - 4.3.6. Bore HVD candidate poles at 15" and 54" above the groundline to determine if it can be reinforced. (**Note:** "Candidate" poles are poles that qualify as replacement candidates per Section 5.1.2 Report the minimum shell thickness at 15" and 54" above the groundline. Report the pole as reinforceable if there is an average 2" of shell or greater at 15" above the groundline and an average 4" shell or greater at 54" above the groundline.
 - 4.3.7. Plug all holes with 3" treated wood plugs.
 - 4.3.8. Bore previously reinforced poles at the top of the steel truss. Identify previously reinforced pole as a replacement candidate if less than average 4" shell.
 - 4.3.9. When inspection is complete replace any excavated soil, tamping it firmly around the pole butt. Make a small mound of soil to compensate for settling.
- 4.4. Inspect the pole for surface decay.
- 4.5. The Contractor will install an acceptable bore tag at chest height on the road side of all poles bore tested.

5. EVALUATION CRITERIA

- 5.1. The Contractor shall use the following bore test criteria to identify wood pole replacement candidates:

5.1.1 HVD Line Section Criteria

<u>Original Circumference Just Above Groundline</u>	<u>Reduction of Circumference At or Below Groundline (Decayed Area)</u>	<u>Hollow Heart Minimum Allowable Thickness of Shell At Outer Decay Location</u>	<u>Pole Is A Replacement Candidate If Shell Thickness Is Less Than</u>
35"	0"	2.0"	2.0"
	2"	2.0"	2.0"
	4.4"	Solid	Solid
	Greater than 4.4"	Candidate	-
40"	0"	2.0"	2.0"
	2"	2.0"	2.0"
	4"	2.6"	2.6"
	5.1"	Solid	Solid
	Greater than 5.1"	Candidate	-
50"	0"	2.0"	2.0"
	2"	2.2"	2.2"
	4"	2.8"	2.8"
	6.3"	Solid	Solid
	Greater than 6.3"	Candidate	-
60"	0"	2.3"	2.3"
	2"	2.6"	2.6"
	4"	3.1"	3.1"
	7.6"	Solid	Solid
	Greater than 7.6"	Candidate	-
70"	0"	2.7"	2.7"
	2"	3.0"	3.0"
	4"	3.4"	3.4"
	6"	4.0"	4.0"
	8"	5.4"	5.4"
	8.9"	Solid	Solid
80"	Greater than 8.9"	Candidate	-
	0"	3.0"	3.0"
	2"	3.4"	3.4"
	4"	3.8"	3.8"
	6"	4.3"	4.3"
	8"	5.2"	5.2"
	10.1"	Solid	Solid
	Greater than 10.1"	Candidate	-

- 5.2. Visual inspection can be used to identify rejected poles where there is severe decay at the top of the pole or where the pole is split or has large voids above chest height or other similar conditions. On all poles rejected due to visual inspection record reason in *Comments* field.

- 5.3. The Contractor shall install a red tag (white arrow) on all rejected poles at chest height on the road side of the pole, and indicate proper replacement value in the *ActionRequired* field.
- 5.3.1. Place the red tag "arrow up" if the pole was rejected from the visual inspection.
- 5.3.2. Place the red tag "arrow down" if the pole was rejected from the bore test inspection.
- 5.4. Replacement HVD candidate poles shall not be treated with wood preserving products unless the pole is reinforceable according to the criteria in 4.3.6.

6.0 Record Keeping

- 6.1 If the brand date is unreadable, estimate the manufactured year, height, and class of the pole and record as such.
- 6.2 Any poles that were unable to be inspected shall be recorded as follows:
- No in the *Accessible* field.
 - Nearby GPS coordinate in GPS coordinate field.
 - Additional location information in the *LocationDescription* field (ex. 200' N of Pole# 1234567).
 - A valid reason of why the pole is inaccessible in the *Comments* field (ex. Surrounded by water).
- 6.3 When testing multiple pole HVD structures, indicate which pole is being tested in the *HVD Structure Number* field. Example: #556N, #556S, #556W or #556E.
- 6.4 If the existing HVD pole is stubbed, indicate which pole is being tested in the *HVD Structure Number* field. Example: #556, #556stub.
- 6.5 If there is a guy pole attached to a HVD pole, indicate which pole is being tested in the *HVD Structure Number* field. Example: #556 (T pole), #556guy.
- 6.6 If there is a swamp fixture attached to the pole, indicate "swamp fixture" in the foundation type field.
- 6.7 A HVD pole with an air-break switch in single circuit lines may not have pole numbers on them; in such cases, use the switch number on the switch handle. Switches in double circuit lines should have pole numbers on them. Indicate the air-break switch number followed by "ABS" in the *HVD Structure Number* field. (ex. 177ABS)
- 6.8 For poles in substations, record "sub" in *HVD Structure Number* field for un-numbered substation poles.

7. HVD WOOD PRESERVATIVE TREATMENT SPECIFICATIONS

- 7.1. The Contractor will use only those wood preservative products listed on Attachment C for treating HVD wood poles.
- 7.2. The Contractor shall treat all HVD poles with groundline surface decay that reduces the original circumference of the pole by 2" or more.
- 7.3. The following method will be used to treat HVD poles with surface decay.
 - 7.3.1. Excavate to a depth of 18".
 - 7.3.2. Remove all external decay by chipping and wire brushing (avoid removing any good wood).
 - 7.3.3. Remove all shavings and decayed material from the hole and surrounding area.
 - 7.3.4. When possible, remove all attachments and replace when bandaging or preservative application is complete (never bandage over wire molding).
 - 7.3.5. Apply bandage or preservative application so that it extends 3" above groundline. Cover the bandage or preservative with an approved moisture carrier.
 - 7.3.6. Replace excavated soil and tamp. Do not place solid objects such as large rocks against the pole. Backfills shall be mounded around the pole to allow for settling.
 - 7.3.7. No debris, loose soil, etc, is to be left in the pole area and shall be properly disposed of. Replace turf removed from excavation with care in its original location.
- 7.4. The contractor will treat all HVD poles with internal decay except replacement candidates which are not reinforceable. The contractor will treat all HVD poles infested with termite or ant colonies that are not replacement candidates or are reinforceable candidates. The contractor will internally treat all previously reinforced HVD poles with an average 4" shell or greater at 54" above groundline. The contractor shall treat all internally decayed HVD poles with an approved preservative, and in addition, decay voids/pockets shall be treated with pentachlorophenol or liquid copper naphthenate.
 - 7.4.1. The Contractor will follow the manufacturer's labeled directions for internal treatment of wood poles.
 - 7.4.2. All holes will be plugged with 3" treated wood plugs following internal treatment.

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Sample HVD Pole Replacement Costs

Case No.: U-20697

Exhibit No.: A-44 (RTB-17)

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Witness: RTBlumenstock

Date: February 2020

Order	Description	# of Structures	Completed On	Total Costs	Capital Costs	Cost of Removal
19974015	LN022K ALLEGAN REPL 31 STRS	31	01/03/2017	\$ 616,528	\$ 534,834	\$ 81,694
25767441	LN042G EATON RAPIDS REPL 21 STRS %	21	02/23/2017	\$ 651,142	\$ 469,342	\$ 181,800
25745864	LN026D STOCKBRIDGE REPL 21 STRS %	21	02/24/2017	\$ 509,769	\$ 432,506	\$ 77,263
25747368	LN026D STOCKBRIDGE REPL 30 STRS %	30	04/27/2017	\$ 685,629	\$ 621,994	\$ 63,635
25757130	LN026D STOCKBRIDGE REPL 5 STRS %	5	04/27/2017	\$ 160,982	\$ 144,867	\$ 16,115
25458883	LN110G IONIA #2 REPL 18 STRS %	18	05/05/2017	\$ 271,311	\$ 244,947	\$ 26,364
25745015	LN026B GRASS LAKE REPL 11 STRS %	11	05/19/2017	\$ 163,650	\$ 143,074	\$ 20,577
26131209	LN078V MONTROSE REPL 9 STRS %	9	05/23/2017	\$ 128,546	\$ 119,006	\$ 9,540
25737037	LN026B GRASS LAKE REPL 6 STRS %	6	05/30/2017	\$ 140,461	\$ 133,538	\$ 6,923
25729462	LN021S BALZER REPL STR 5	1	06/08/2017	\$ 29,375	\$ 28,096	\$ 1,279
25790320	LN030F PARNALL REPL 8 STRS %	8	07/10/2017	\$ 115,966	\$ 109,355	\$ 6,611
26130957	LN078L MONTROSE REPL 9 STRS %	9	07/24/2017	\$ 251,244	\$ 225,868	\$ 25,376
20851891	LN072M OTTAWA BEACH REPL 22 STRS	22	07/28/2017	\$ 560,962	\$ 501,755	\$ 59,207
25796539	LN036J HENRY REPL 4 STRS %	4	08/02/2017	\$ 77,373	\$ 70,467	\$ 6,906
25818530	LN026R CHRYSLER REPL 7 STRS %	7	09/14/2017	\$ 217,096	\$ 205,244	\$ 11,851
25657577	LN055E BAUM ST REPL 18 STRS % "RR"	18	10/06/2017	\$ 268,789	\$ 246,315	\$ 22,475
25818279	LN026R CHRYSLER REPL 3 STRS %	3	10/09/2017	\$ 32,070	\$ 30,964	\$ 1,106
25735762	LN026A GRASS LAKE REPL 5 STRS	5	11/13/2017	\$ 127,131	\$ 110,226	\$ 16,904
19662246	LN033HH ROSEBUSH REPL 3 STRS	3	12/04/2017	\$ 46,415	\$ 42,164	\$ 4,251
25769868	LN042J EATON RAPIDS REPL 8 STRS %	8	12/21/2017	\$ 109,598	\$ 99,659	\$ 9,939

2018 Electric System Loss Study Report

Electric Grid Integration
October 22, 2019



Oct 22, 2019

2018 Electric System Loss Study Report

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Introduction

The purpose of this study is to allocate system energy and demand losses among the various components of the electric system by calculating a percentage Loss Factor for each system component. This information will be used to update loss calculations used in electric rate design. Customer and Service Infrastructure - HVD calculated Loss Factors for the 345, 138 and 46 kV systems and the low side of the 138 and 46 kV industrial systems. The Loss Factor for the Distribution Primary system was calculated with input from Customer and Service Infrastructure - LVD. Finally, the Loss Factor for the Distribution Secondary system (including secondary transformers) was calculated from the amount of system loss remaining after all other system component losses were allocated.

Definitions

- **System Component Losses:** Generated and purchased ("input" or "delivery") Power or Energy (including imports) minus consumed or distributed ("output") Power or Energy.

Note: The total generated and purchased energy MegaWatt-hours (MWh) for 2018 and the total MWh delivered (sold) at Each System Component during 2018 were provided by the Accounting Department and Rates Department, respectively. The overall electric system loss percentage is derived from the generated and purchased energy data rather than from system models.

- **Loss Factor (%):**

$$\left(1 - \frac{\text{Component Output Power or Energy}}{\text{Component Input Power or Energy}}\right) \times 100\%$$

$$= \left(\frac{\text{Component Power or Energy Loss}}{\text{Component Input Power or Energy}}\right) \times 100\%$$

- **Efficiency Factor:** [100% - Loss Factor (%)] or [1 - Loss Factor (per-unit)]
- **Energy Loss Factor:** Total System Component MWh Loss divided by Total System Component MWh Input.
- **Demand Loss Factor:** Average of Monthly Peak System Component MW Losses divided by Monthly Peak System Component MW Inputs.

NOTE: Average based on twelve monthly peak hours as identified in FERC Form 1, Page 401b.

- **Cumulative Loss Factor (Energy or Demand):** One minus the product of one minus the per-unit Loss Factor for that system component and one minus the per-unit loss factor for each of the upstream system components. Alternatively, one minus the product of the per-unit efficiency factors for the system component and all upstream system components.

NOTE: Cumulative loss factors are used to estimate the generation requirements necessary to serve a particular amount of load at any system component.

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2018 Electric System Loss Study Report

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Energy Losses - Method

1. Loadflow model cases were created for each of the 8760 hours in the 2018 study year. From each loadflow case (*i.e.* from each hour of the study year), delivery and loss data at points throughout the electric Transmission and High Voltage Distribution (HVD) systems were programmatically collected, processed, and recorded.
2. Utilizing the MWh loss and delivery data, loss percentages for each Transmission or HVD system component were then calculated as:

$$\frac{\sum \text{Component MWh Loss}}{\sum \text{Component MWh Delivery}}$$

3. For Distribution Primary, average load and loss data from approximately 400 representative circuits at nine different system gross load levels for all-switched-capacitors-on and all-switched-capacitors-off scenarios were provided by Customer and Service Infrastructure - LVD Substation Planning. A line loss percentage was calculated from load and loss figures interpolated between the caps-on and caps-off data, based on the per system load level distribution capacitor schedule (*tone groups*). The line loss, along with Distribution Primary transformer losses calculated in the hourly loadflow cases, comprise the total losses for Distribution Primary.
4. For Distribution Secondary, sales (output) data per system component were provided by Rates Department. Generation requirements (input) per system component were estimated based on cumulative loss percentages previously calculated for each of the other system components. Total system generation requirements were estimated based on an overall system loss percentage derived from Generation and Purchased Power data provided by Accounting Department.

The difference between the total system estimated generation requirements and the sum of the estimated generation requirements for each system component (except for Distribution Secondary) gives the estimated generation requirement for Distribution Secondary. The difference between Distribution Secondary estimated generation requirements and Distribution Secondary sales gives a cumulative loss percentage, from which the Distribution Secondary loss percentage was derived.

Demand Losses - Method

1. The loss and delivery data from the loadflow cases for the 12 monthly peak hours, as identified in FERC Form 1, were selected and the loss percentages for each Transmission or HVD system component were calculated as the average of ratios of losses to deliveries:

$$\left(\frac{\text{MW Loss}_{PK}}{\text{MW Delivery}_{PK}} \right)$$

2. For Distribution Primary, demand line loss percentages for each of the 12 monthly peak hours were calculated similarly as for energy loss. Demand loss percentages were calculated as the average of the loss to delivery ratios, as with the Transmission and HVD system components.

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Demand Losses - Method, continued

3. For Distribution Secondary, the ratios of monthly peak MW deliveries to the hourly average MW deliveries were used to estimate an average monthly peak MW delivery per system component. Similar to Distribution Secondary energy loss calculations, an average of monthly peak generation was then estimated from the system component cumulative demand loss percentages. Distribution Secondary average peak generation and, subsequently, cumulative demand loss percentage were estimated, similarly as with energy losses. From that, the Distribution Secondary demand loss percentage was derived. See "Cumulative Demand Loss Percentages Applied to MWh Deliveries", page 5, below.
4. Monthly peak demand MW deliveries and losses are tabulated for each system component. The loss factors and cumulative loss factors for each of the 12 monthly peaks were calculated using the previously described methods. For Distribution Secondary, the annual Demand Loss Factor was applied to each of the monthly peak MW deliveries to estimate monthly peak losses and monthly peak Cumulative Loss Factors. Using the annual Distribution Secondary Loss Factor in this manner was necessary since the MWh sales figures used to estimate Distribution Secondary losses are available on an annual, not monthly, basis.

Losses Applied to MWh Deliveries

1. The cumulative Loss Factors at each system component (Energy and Demand, in per-unit) were used to estimate the generation necessary to provide the given delivery to that system component, per the following formula:

$$\text{Generation} = \frac{\text{Delivery}}{1 - \text{Loss Factor}_{p.u.}}$$

2. The difference between the total power generated and purchased (for the entire system) and the sum of the generation requirements for all system components except Distribution Secondary yielded the cumulative losses for Distribution Secondary. The Loss Factor for Distribution Secondary was then found by rearranging the definition for cumulative losses, as follows:

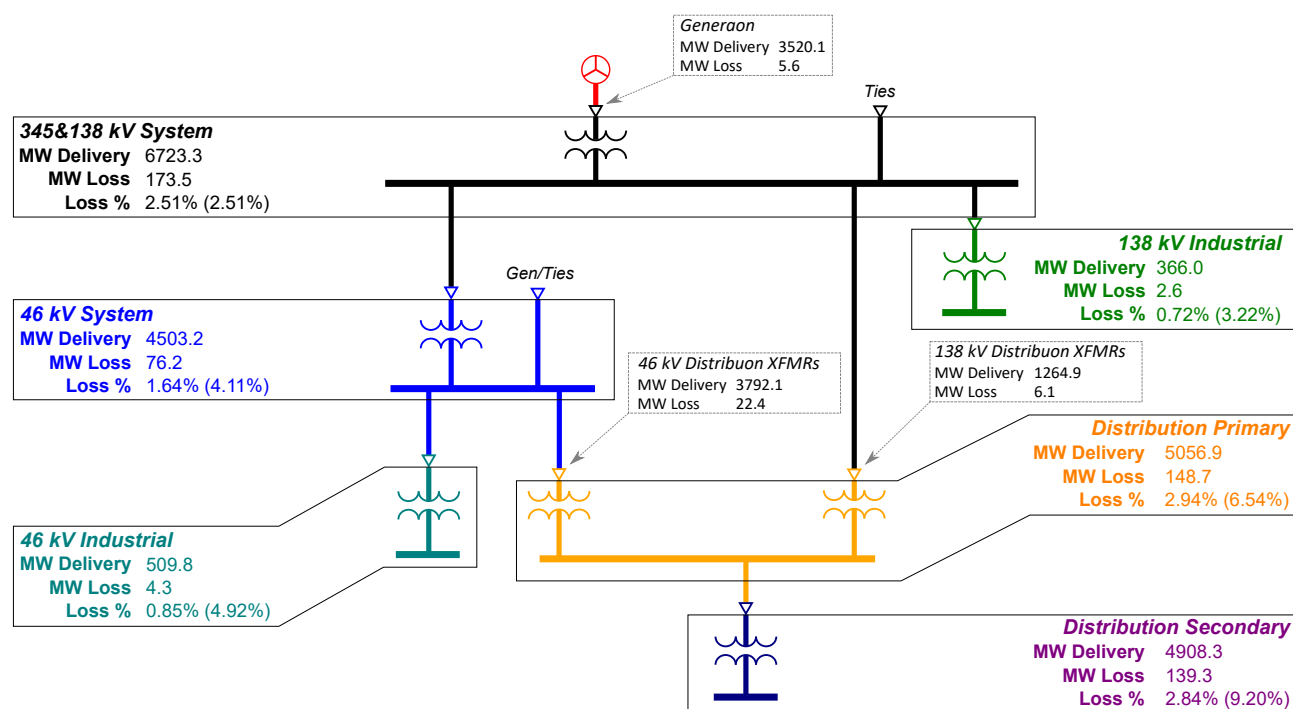
$$\text{Loss Factor}_{Sec.Dist.,p.u.} = 1 - \frac{1 - \text{Cum. Loss Factor}_{Sec.Dist.,p.u.}}{1 - \text{Cum. Loss Factor}_{Pri.Dist.,p.u.}}$$

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DEMAND LOSSES AND DELIVERIES (AVERAGE OF 12 MONTHLY PEAKS)
Combined Method (Each System Component Includes Transformation to that Component)



General Notes

- 138 kV Industrial Delivery and Loss does not include HSC volumes
- Monthly loss percentages are calculated as Loss % = (MW Loss/MW Delivery) x 100%
- Loss percentages presented are the average of the monthly loss percentages
- Loss percentages in parentheses are the Cumulative Loss percentages
- Cumulative Loss factors are calculated as one minus the product of one minus the per unit loss factor for that component and all upstream components.
- Loss % for Distribution Primary Lines (2.39%) provided by C&SI – LVD
- Generation Transformers were combined with the 345 & 138 kV system because all customers are connected at lower voltages (components).

Notes for Distribution Primary and Secondary

- MW Delivery is high-side sum of 138/DST and 46/DST XFMRs
- MW Loss includes 138 & 46 /DST XFMR AND primary line loss
- Cumulative Loss % is adjusted based on the weighted average amount of load served from 138 kV and 46 kV:
First, $[(3792.1) \times 4.11\% + (1264.9) \times 2.51\%] / (3792.1 + 1264.9) = 3.71\%$
Then, $[1 - (1 - 0.0294) \times (1 - 0.0371)] \times 100\% = 6.54\%$
- Distribution Secondary cumulative demand loss % estimated using the average monthly peak deliveries. Refer to estimating sheet (page 5).

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CUMULATIVE DEMAND LOSS PERCENTAGES APPLIED TO MWh DELIVERIES
Combined Method (Each System Component Includes Transformation to that Component)

Electrical System Voltage Level	MWh Deliveries (Total) ⁵	Average Hourly MW Delivery	Estimated Average Monthly Peak MW Delivery	Cumulative % Demand Loss ⁴	Estimated MW @ Generation	% Demand Losses	
138 kV System	778,409	88.9	117	2.51%	120	2.51%	Generation Transformers, 345 & 138 kV Systems Combined
V1: >120kV ^{1&6}	2,513,062	286.9	310	3.22%	321	0.72%	Cum. % = [1-(1-138IND%) x (1-138Sys_cum%)] x 100%
46 kV System ¹	404,955	46.2	65	4.11%	68	1.64%	Cum. % = [1-(1-46Sys%) x (1-138Sys_cum%)] x 100%
V2: 25kV - 120kV ¹	3,318,045	378.8	420	4.92%	442	0.85%	Cum. % = [1-(1-46IND%) x (1-46Sys_cum%)] x 100%
V3: <25kV, Primary Distribution ^{1&2}	8,253,184	942.1	1360	6.54%	1455	2.94%	Cum. % = [1-(1-DSTprimary%) x (1-46&138AVG_cum%)] x 100%
V4: Secondary Distribution ^{1,2&3}	21,079,315	2406.3	3473	9.20%	3824	2.84%	Cum. % = [1-(1-DSTprimary%) x (1-46&138AVG_cum%)] x 100%
	36,346,970	4149	5745		6230	7.78%	

NOTES:

1. The cumulative loss for any level is equal to one minus the product of one minus the per unit loss factor for that level and one minus the per unit cumulative loss factor one level higher.
2. The Cumulative Loss percentages for the Distribution Primary and Secondary were adjusted to account for load served from 138 kV distribution subs. This adjustment was based on a weighted average cumulative loss % from the total MWh delivered to either 46kV/Dist. or 138/Dist.
3. The Distribution Secondary Cumulative Loss % was calculated from the MWh Gen and MWh Del remaining. Then, the Demand Loss % was calculated in reverse from the cumulative.
4. All cumulative loss percentages are calculated assuming the 345 kV and 138 kV Systems are combined along with the Generation Transformers (GSUs)
5. MWh Delivery figures include ROA amounts.
6. V1: >120kV does not include HSC volumes.

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MONTHLY DEMAND LOSSES AND DELIVERIES

Combined Method (Each System Component Includes Transformation to that Component)

PEAKS		Jan 4, 2018 19:00	Feb 1, 2018 20:00	Mar 14, 2018 08:00	Apr 17, 2018 09:00	May 29, 2018 16:00	Jun 30, 2018 17:00	Jul 5, 2018 16:00	Aug 14, 2018 16:00	Sep 4, 2018 17:00	Oct 9, 2018 19:00	Nov 28, 2018 19:00	Dec 11, 2018 18:00
GENERATION	Deliveries (MW)	4067.0	3427.9	2361.9	1571.9	3278.6	4287.8	4691.6	4568.8	4494.0	3003.9	3176.1	3311.7
	Losses (MW)	6.3	5.1	3.4	2.0	6.4	7.2	7.9	7.2	7.8	4.2	4.6	5.1
345&138 KV SYSTEM	Deliveries (MW)	5910.7	6095.6	5422.3	5277.2	7639.3	7915.0	8238.5	7975.7	8314.8	6137.9	5707.1	6045.8
	Losses (MW)	148.2	117.6	104.6	100.3	214.6	189.5	277.8	227.4	269.9	183.6	116.0	132.9
	Loss Factor	2.51%	1.93%	1.93%	1.90%	2.81%	2.39%	3.37%	2.85%	3.25%	2.99%	2.03%	2.20%
	Deliveries (MW)	377.2	387.9	377.8	409.5	355.5	326.0	287.2	392.3	383.2	418.5	343.5	333.7
138 KV INDUSTRIAL ³	Losses (MW)	2.6	2.6	2.5	2.8	2.6	2.5	2.5	2.8	2.8	2.8	2.5	2.5
	Loss Factor	0.69%	0.68%	0.67%	0.67%	0.72%	0.78%	0.86%	0.71%	0.72%	0.68%	0.72%	0.75%
	Cumulative Loss Factor ²	3.18%	2.60%	2.58%	2.56%	3.51%	3.15%	4.20%	3.54%	3.95%	3.65%	2.74%	2.93%
	Deliveries (MW)	4019.4	3828.2	3537.5	3460.3	5231.0	5598.5	5803.7	5370.3	5566.1	4115.2	3741.5	3766.9
46 KV SYSTEM	Losses (MW)	62.8	55.2	47.9	46.3	94.4	112.7	119.3	100.4	106.8	63.4	52.3	52.8
	Loss Factor	1.56%	1.44%	1.35%	1.34%	1.81%	2.01%	2.06%	1.87%	1.92%	1.54%	1.40%	1.40%
	Cumulative Loss Factor ²	4.03%	3.35%	3.26%	3.21%	4.56%	4.36%	5.36%	4.67%	5.10%	4.49%	3.40%	3.57%
	Deliveries (MW)	500.5	508.4	516.5	495.6	547.1	471.4	481.3	557.2	565.7	507.5	478.5	488.2
46 KV INDUSTRIAL	Losses (MW)	4.2	4.2	4.2	4.2	4.5	4.2	4.2	4.6	4.6	4.3	4.2	4.2
	Loss Factor	0.85%	0.83%	0.81%	0.85%	0.82%	0.89%	0.87%	0.82%	0.81%	0.86%	0.87%	0.86%
	Cumulative Loss Factor ²	4.84%	4.15%	4.04%	4.04%	5.35%	5.21%	6.18%	5.45%	5.88%	5.30%	4.25%	4.40%
	Deliveries (MW)	1183.5	1048.6	1077.6	993.4	1602.5	1466.1	1552.1	1458.1	1574.2	1088.3	1051.6	1082.4
138 KV DISTRIBUTION	Losses (MW)	5.7	4.9	5.0	4.6	8.5	7.2	7.7	7.1	8.0	4.9	4.7	4.7
	Deliveries (MW)	3349.8	3151.2	2862.5	2808.5	4474.8	4856.9	5045.6	4558.0	4745.7	3430.8	3106.3	3114.8
46 KV DISTRIBUTION	Losses (MW)	19.4	17.8	16.0	16.0	27.2	30.8	32.2	27.5	29.1	19.1	17.0	16.6
	Deliveries (MW)	4533.3	4199.8	3940.1	3801.9	6077.3	6322.9	6597.7	6016.1	6319.9	4519.0	4157.9	4197.2
DISTRIBUTION PRIMARY	Losses (MW)	123.1	108.8	96.7	92.8	205.3	217.6	237.3	203.8	219.8	122.8	104.5	105.7
	Loss Factor	2.71%	2.59%	2.45%	2.44%	3.38%	3.44%	3.60%	3.39%	3.48%	2.72%	2.51%	2.52%
	Cumulative Loss Factor ²	6.25%	5.50%	5.28%	5.24%	7.34%	7.21%	8.31%	7.47%	7.96%	6.73%	5.49%	5.65%
	Deliveries (MW)	4410.2	4091.0	3843.5	3709.2	5872.0	6105.3	6360.4	5812.3	6100.1	4396.2	4053.5	4091.5
DISTRIBUTION SECONDARY	Losses (MW)	125.2	116.1	109.1	105.3	166.7	173.3	180.6	165.0	173.2	124.8	115.1	116.1
	Loss Factor ¹	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%	2.84%
	Cumulative Loss Factor ²	8.91%	8.19%	7.97%	7.93%	9.97%	9.84%	10.92%	10.10%	10.57%	9.38%	8.17%	8.33%

1. The Distribution Secondary Loss factor presented above represents an annual average loss percentage. This Loss factor is applied to each of the 12 monthly peaks to estimate monthly peak demand losses and cumulative demand Loss Factors.
2. See Page 5 for additional notes regarding Cumulative Losses.
3. 138 KV INDUSTRIAL Deliveries and Losses do not include HSC volumes.

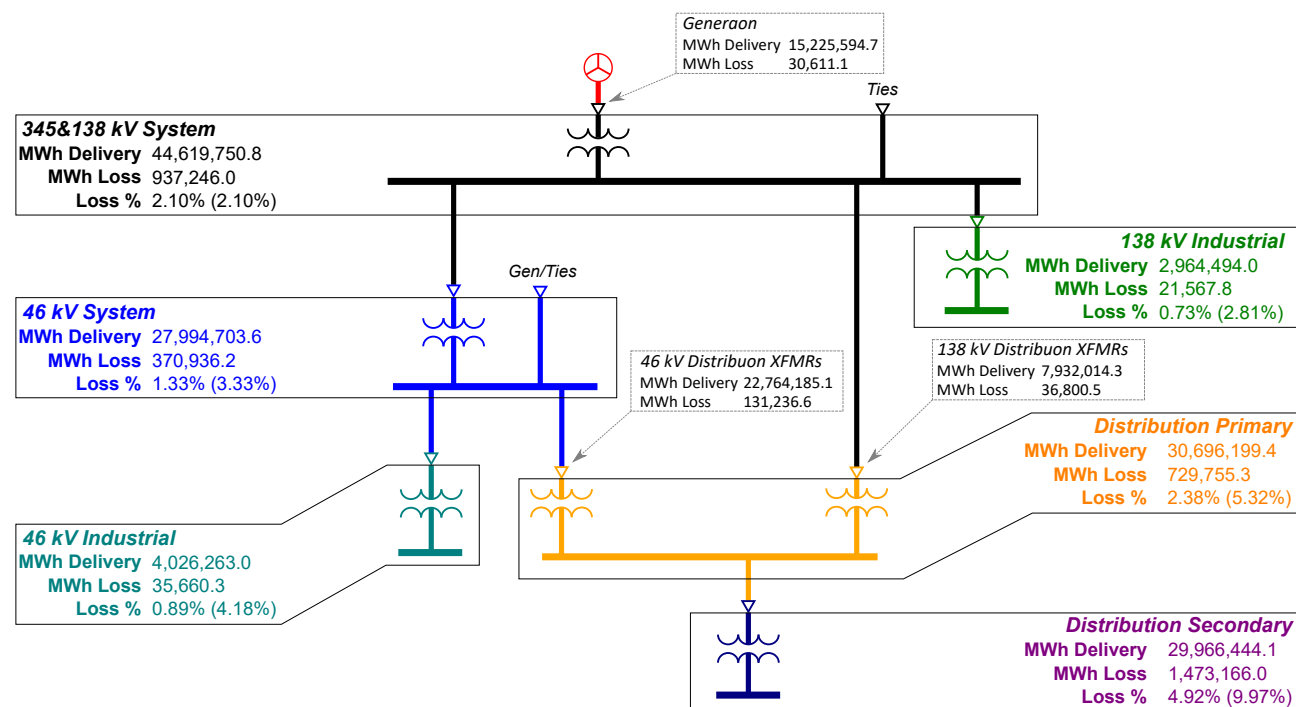
Oct 22, 2019

2018 Electric System Loss Study Report

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ENERGY LOSSES AND DELIVERIES (TOTAL YEARLY MWh)

Combined Method (Each System Component Includes Transformation to that Component)



General Notes

- 138 kV Industrial Delivery and Loss does not include HSC volumes
- Loss percentages are calculated as Loss % = (MWh Loss/MWh Delivery) x 100%
- Loss percentages in parentheses are the Cumulative Loss Percentages
- Cumulative Loss factors are calculated as one minus the product of one minus the per unit loss factor for that component and all upstream components.
- Loss % for Distribution Primary Lines (1.84%) provided by C&SI – LVD
- Generation Transformers were combined with the 345 & 138 kV system because all customers are connected at lower voltages (components).

Notes for Distribution Primary and Secondary

- MWh Delivery is high-side sum of 138/DST and 46/DST XFMRs
- MWh Loss includes 138 & 46 /DST XFMR AND primary line loss
- Cumulative Loss % is adjusted based on the weighted average amount of load served from 138 kV and 46 kV:
First, $[(22,764,185.1) \times 3.33\% + (7,932,014.3) \times 2.10\%] / (22,764,185.1 + 7,932,014.3) = 3.01\%$
Then, $[1 - (1 - 0.0238) \times (1 - 0.0301)] \times 100\% = 5.32\%$

Oct 22, 2019

2018 Electric System Loss Study Report

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CUMULATIVE ENERGY LOSS PERCENTAGES APPLIED TO MWh DELIVERIES

Combined Method (Each System Component Includes Transformation to that Component)

Electrical System Voltage Level	MWh Deliveries (Total) ⁵	Cumulative % Energy Loss ⁴	Estimated MWh @ Generation	% Energy Loss	Notes
138 kV System	778,409	2.10%	795,111	2.10%	
V1: >120kV ^{1&6}	2,513,062	2.81%	2,585,795	0.73%	
46 kV System ¹	404,955	3.33%	418,887	1.33%	
V2: 25kV - 120kV ¹	3,318,045	4.18%	3,462,868	0.89%	
V3: <25kV, Primary Distribution ^{1&2}	8,253,184	5.32%	8,716,471	2.38%	
V4: Secondary Distribution ^{1,2&3}	21,079,315	9.97%	23,413,615	4.92%	
TOTAL	36,346,970		39,392,746	7.73%	

NOTES:

1. The cumulative loss for any level is equal to one minus the product of one minus the per unit loss factor for that level and one minus the per unit cumulative loss factor one level higher.
2. The Cumulative Loss percentages for the Distribution Primary and Secondary were adjusted to account for load served from 138 kV distribution subs. This adjustment was based on a weighted average cumulative loss % from the total MWh delivered to either 46kV/Dist. or 138/Dist.
3. The Distribution Secondary Cumulative Loss % was calculated from the MWh Gen and MWh Del remaining. Then, the Demand Loss % was calculated in reverse from the cumulative.
4. All cumulative loss percentages are calculated assuming the 345 kV and 138 kV Systems are combined along with the Generation Transformers (GSUs)
5. MWh Delivery figures include ROA amounts.
6. V1: >120kV does not include HSC volumes.

Oct 22, 2019

2018 Electric System Loss Study Report

9

ENERGY AND DEMAND LOSS PERCENTAGES

4-Year Summary

ENERGY LOSS %	2015	2016	2017	2018	4 yr Avg.
345 & 138 kV System	2.04%	2.20%	2.24%	2.10%	2.15%
138 kV Industrial	0.69%	0.70%	0.69%	0.73%	0.70%
46 kV System	1.39%	1.27%	1.25%	1.33%	1.31%
46 kV Industrial	1.12%	1.12%	1.13%	0.89%	1.06%
Distribution Primary	2.71%	2.32%	2.43%	2.38%	2.46%
Distribution Secondary	4.40%	4.57%	3.13%	4.92%	4.25%

ENERGY LOSS CUMULATIVE%	2015	2016	2017	2018	4 yr Avg.
345 & 138 kV System	2.04%	2.20%	2.24%	2.10%	2.15%
138 kV Industrial	2.72%	2.89%	2.91%	2.81%	2.83%
46 kV System	3.41%	3.44%	3.37%	3.33%	3.39%
46 kV Industrial	4.49%	4.52%	4.46%	4.18%	4.41%
Distribution Primary	5.64%	5.35%	5.43%	5.32%	5.43%
Distribution Secondary	9.79%	9.68%	8.39%	9.97%	9.46%

DEMAND LOSS %	2015	2016	2017	2018	4 yr Avg.
345 & 138 kV System	2.30%	2.42%	2.38%	2.51%	2.40%
138 kV Industrial	0.68%	0.69%	0.68%	0.72%	0.69%
46 kV System	1.67%	1.52%	1.49%	1.64%	1.58%
46 kV Industrial	1.07%	1.06%	1.07%	0.85%	1.01%
Distribution Primary	3.20%	2.82%	2.95%	2.94%	2.98%
Distribution Secondary	3.75%	5.71%	4.43%	2.84%	4.18%

DEMAND LOSS CUMULATIVE%	2015	2016	2017	2018	4 yr Avg.
345 & 138 kV System	2.30%	2.42%	2.38%	2.51%	2.40%
138 kV Industrial	2.96%	3.10%	3.04%	3.22%	3.08%
46 kV System	3.92%	3.91%	3.83%	4.11%	3.94%
46 kV Industrial	4.95%	4.93%	4.85%	4.92%	4.91%
Distribution Primary	6.54%	6.22%	6.30%	6.54%	6.40%
Distribution Secondary	10.05%	11.58%	10.45%	9.20%	10.32%

General Notes

- Annual Loss percentages are calculated as Loss % = [MW(h) Loss/MW(h) Delivery] x 100%
- Annual Cumulative Loss percentages are calculated as one minus the product of one minus the per unit Loss Factor for that component and for all upstream components.
- 4-year Average Loss percentages and Cumulative Loss percentages are the averages of the annual loss percentages.
- 2018 138 kV Industrial loss percentages do not include HSC losses.

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

HEATHER A. BREINING

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Environmental - Air Quality Compliance

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-46 (HAB-1)

Page: 1 of 1

Witness: HABreining

Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				Projected Test Year	
		Historical	Projected Bridge Year				
Line No.	Description	12 Mos Ended 12/31/2018	12 Mos Ending 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 Mos Ending 12/31/2021	
col. (c)+(d)							
1	J.H. Campbell, Unit 1 - Dry Sorbent Injection	864	35	-	35	-	
2	Labor	219	32	-	32	-	
3	Contractor	466	2	-	2	-	
4	Materials	13	0	-	0	-	
5	Business Expenses	-	0	-	0	-	
6	Contingency	-	-	-	-	-	
7	Other (Loadings, Chargebacks)	166	1	-	1	-	
8	J.H. Campbell, Unit 2 - Dry Sorbent Injection	814	26	-	26	-	
9	Labor	227	23	-	23	-	
10	Contractor	407	2	-	2	-	
11	Materials	13	0	-	0	-	
12	Business Expenses	-	0	-	0	-	
13	Contingency	-	-	-	-	-	
14	Other (Loadings, Chargebacks)	166	1	-	1	-	
15	J.H. Campbell, Unit 3 - Spray Dry Absorber	(167)	-	-	-	-	
16	Labor	(227)	-	-	-	-	
17	Contractor	(181)	-	-	-	-	
18	Materials	344	-	-	-	-	
19	Business Expenses	0	-	-	-	-	
20	Contingency	-	-	-	-	-	
21	Other (Loadings, Chargebacks)	(103)	-	-	-	-	
22	Total Capital	1,510	61	-	61	-	

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Projected Capital Expenditures
 Environmental - Coal Combustion Residuals Compliance
 Summary of Actual and Projected Electric Capital Expenditures
 (\$000)

Case No.: U-20697
 Exhibit No.: A-47 (HAB-2)
 Page: 1 of 1
 Witness: HABreining
 Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				Projected Test Year	
		Historical	Projected Bridge Year				
Line		12 Mos Ended	12 Mos Ending	12 Mos Ending	24 Mos Ending	12 Mos Ending	
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12/31/2021	
col. (c)+(d)							
1	J.H. Campbell Site - RCRA	2,896	-	-	-	-	
2	Labor	179	-	-	-	-	
3	Contractor	2,759	-	-	-	-	
4	Materials	(317)	-	-	-	-	
5	Business Expenses	2	-	-	-	-	
6	Contingency	-	-	-	-	-	
7	Other (Loadings, Chargebacks)	274	-	-	-	-	
8	D.E. Karn/J.C. Weadock Site - RCRA	3,049	-	-	-	-	
9	Labor	90	-	-	-	-	
10	Contractor	2,409	-	-	-	-	
11	Materials	143	-	-	-	-	
12	Business Expenses	0	-	-	-	-	
13	Contingency	-	-	-	-	-	
14	Other (Loadings, Chargebacks)	407	-	-	-	-	
15	Total Capital	5,945	-	-	-	-	

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Environmental - 316(b) Compliance

Summary of Actual and Projected Electric Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-48 (HAB-3)

Page: 1 of 1

Witness: HABreining

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				Projected Test Year
Line No.	Description	Historical 12 Mos Ended 12/31/2018	Projected Bridge Year			12 Mos Ending 12/31/2021
			12 Mos Ending 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	
			<i>col. (c)+(d)</i>			
1	J.H. Campbell, Units 1&2 - 316(b)	16	-	-	-	500
2	Labor	-	-	-	-	-
3	Contractor	14	-	-	-	90
4	Materials	-	-	-	-	-
5	Business Expenses	-	-	-	-	-
6	Contingency	-	-	-	-	-
7	Other (Loadings, Chargebacks)	2	-	-	-	410
8	D.E. Karn, Units 1&2 - 316(b)	(8)	-	-	-	-
9	Labor	-	-	-	-	-
10	Contractor	(7)	-	-	-	-
11	Materials	-	-	-	-	-
12	Business Expenses	-	-	-	-	-
13	Contingency	-	-	-	-	-
14	Other (Loadings, Chargebacks)	(1)	-	-	-	-
15	Total Capital	8	-	-	-	500

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Environmental - SEEG Compliance

Summary of Actual and Projected Electric Capital Expenditures
(\$000)

Case No.: U-20697

Exhibit No.: A-49 (HAB-4)

Page: 1 of 1

Witness: HABreining

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
Line		Historical	Projected Bridge Year			Projected Test
No.	Description	12 Mos Ended	12 Mos Ending	12 Mos Ending	24 Mos Ending	Year
		12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
					col. (c)+(d)	
1	J.H. Campbell Site - SEEG	9	-	100	100	7,100
2	Labor	2	-	-	-	-
3	Contractor	5	-	97	97	6,234
4	Materials	-	-	-	-	-
5	Business Expenses	-	-	-	-	-
6	Contingency	-	-	3	3	710
7	Other (Loadings, Chargebacks)	1	-	-	-	156
8	D.E. Karn Site - SEEG	250	1	-	1	-
9	Labor	7	(1)	-	(1)	-
10	Contractor	201	-	-	-	-
11	Materials	7	2	-	2	-
12	Business Expenses	-	-	-	-	-
13	Contingency	-	-	-	-	-
14	Other (Loadings, Chargebacks)	36	1	-	1	-
					-	
15	Total Capital	259	1	100	101	7,100

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

EUGÈNE M.J.A. BREURING

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule E-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Annual Service Area Sales by Major Customer Classes and System Output

5-Year Historical

(GWh)

Case No.: U-20697
Exhibit No.: A-5 (EMB-1)
Schedule: E-1
Page: 1 of 1
Witness: EMBreuring
Date: February 2020

Line No.			(a)	(b)	(c)	(d)	(e)	(f)		(g)	(h)
			Total Company Electric Deliveries					Losses & Company Use		% of	System Output
	Year	Hist	Residential	Commercial	Industrial	Other	Total	GWh		Output	
1	2014	Hist	12,594	11,858	12,583	561	37,596	3,316	8.1%		40,912
2	2015	Hist	12,495	12,696	11,546	544	37,281	2,892	7.2%		40,173
3	2016	Hist	12,789	12,868	11,709	545	37,911	3,020	7.4%		40,931
4	2017	Hist	12,341	12,749	11,759	544	37,394	2,639	6.6%		40,032
5	2018	Hist	13,051	13,046	11,599	535	38,231	3,101	7.5%		41,331
Bundled Electric Deliveries											
	Year	Hist	Residential	Commercial	Industrial	Other	Total	GWh		% of Output	System Output
6	2014	Hist	12,594	10,848	9,614	561	33,617	3,214	8.7%		36,831
7	2015	Hist	12,495	11,699	8,605	544	33,342	2,771	7.7%		36,114
8	2016	Hist	12,789	11,843	8,839	545	34,016	2,964	8.0%		36,980
9	2017	Hist	12,341	11,763	8,966	544	33,614	2,522	7.0%		36,136
10	2018	Hist	13,051	12,034	8,833	535	34,453	2,997	8.0%		37,450
Choice Electric Deliveries											
	Year	Hist	Residential	Commercial	Industrial	Other	Total	GWh		% of Output	System Output
11	2014	Hist	-	1,010	2,969	-	3,979	102	2.5%		4,081
12	2015	Hist	-	997	2,941	-	3,938	121	3.0%		4,059
13	2016	Hist	-	1,025	2,870	-	3,895	56	1.4%		3,951
14	2017	Hist	-	986	2,793	-	3,780	117	3.0%		3,896
15	2018	Hist	-	1,012	2,766	-	3,778	104	2.7%		3,881

Schedule E-1

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Annual Service Area Sales by Major Customer Classes and System Output

5-Year Projected

(GWh)

Case No.: U-20697

Exhibit No.: A-15 (EMB-2)

Schedule: E-1

Page: 1 of 1

Witness: EMBreuring

Date: February 2020

Line No.			(a) <u>Total Company Electric Deliveries</u>	(b)	(c)	(d)	(e)	(f) <u>Losses & Company Use</u>	(g) <u>% of Output</u>	(h) <u>System Output</u>
	Year		Residential	Commercial	Industrial	Other	Total	GWh		
1	2019	Fcst	12,481	12,731	11,330	520	37,062	3,004	7.5%	40,066
2	2020	Fcst	12,364	12,584	11,005	526	36,479	2,852	7.3%	39,331
3	2021	Fcst	12,440	12,467	10,858	527	36,292	2,751	7.0%	39,043
4	2022	Fcst	12,707	12,242	10,773	529	36,251	2,672	6.9%	38,923
5	2023	Fcst	12,652	12,219	10,862	530	36,263	2,619	6.7%	38,882
<u>Bundled Electric Deliveries</u>								<u>Losses & Company Use</u>		
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	System Output
6	2019	Fcst	12,481	11,755	8,576	520	33,331	2,686	7.5%	36,018
7	2020	Fcst	12,364	11,614	8,252	526	32,755	2,540	7.2%	35,295
8	2021	Fcst	12,440	11,512	8,115	527	32,595	2,441	7.0%	35,036
9	2022	Fcst	12,707	11,295	8,057	529	32,588	2,365	6.8%	34,954
10	2023	Fcst	12,652	11,274	8,125	530	32,581	2,311	6.6%	34,892
<u>Choice Electric Deliveries</u>								<u>Losses & Company Use</u>		
	Year		Residential	Commercial	Industrial	Other	Total	GWh	% of Output	System Output
11	2019	Fcst	-	977	2,754	-	3,731	318	7.8%	4,048
12	2020	Fcst	-	970	2,754	-	3,724	312	7.7%	4,036
13	2021	Fcst	-	955	2,742	-	3,697	310	7.7%	4,007
14	2022	Fcst	-	946	2,716	-	3,662	307	7.7%	3,969
15	2023	Fcst	-	945	2,736	-	3,682	308	7.7%	3,990

Schedule E-2

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Test Year Total Company Electric Revenues & Deliveries
 January 2021 - December 2021

Case No.: U-20697
 Exhibit No.: A-15 (EMB-3)
 Schedule: E-2
 Page: 1 of 1
 Witness: EMBreuring
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	
		Less Non-Base Tariff Items							Test Year		Test Year		
Line		2018	PSCR	Other	L/T Industrial	Weather	Change In	U-18322	Test Year	Total	Non-		
No.	Description	Actual	Factor	Adjustments	Load Retention Rate	Normalization	Determinants	Pro Forma	Factor	Company	Jurisdictional	Jurisdictional	
	Adjusted Operating Revenues (\$000)												
	Cycle Billed Tariff Revenue												
1	Base Tariff	\$ 2,250,104	\$ -	\$ -	\$ -	\$ (142,305)	\$ (42,138)	\$ (34,774)	\$ 144,273	\$ -	\$ 2,175,161	\$ 10,675	\$ 2,164,486
2	GSG Power Supply (1)	4,903	(4,903)	-	-	-	-	-	-	5,960	5,960	-	5,960
3	Base PSCR	1,868,778	-	-	-	-	(37,847)	(27,764)	39,373	-	1,842,540	13,844	1,828,696
4	PSCR Factor	98,371	(98,371)	-	-	-	-	-	-	134,593	134,593	91	134,502
5	Total Cycle Billed Tariff Revenue	4,222,156	(103,274)	-	-	(142,305)	(79,985)	(62,538)	183,647	140,553	4,158,254	24,610	4,133,644
	Cycle Billed Surcharge Revenue												
6	Energy Optimization	123,448	-	(123,448)	-	-	-	-	-	-	-	-	-
7	PEM & OEM	-	-	-	-	-	-	-	-	-	-	-	-
8	Renewable Energy	5,170	-	(5,170)	-	-	-	-	-	-	-	-	-
9	Security Recovery Factor	-	-	-	-	-	-	-	-	-	-	-	-
10	Major Maintenance	(0)	-	0	-	-	-	-	-	-	-	-	-
11	Low-Income Assistance Fund	20,297	-	(20,297)	-	-	-	-	-	-	-	-	-
12	Stranded Cost Recovery	-	-	-	-	-	-	-	-	-	-	-	-
13	Securitization (Classic7)	34,963	-	(34,963)	-	-	-	-	-	-	-	-	-
14	Securitization	-	-	-	-	-	-	-	-	-	-	-	-
15	Securitization Tax	-	-	-	-	-	-	-	-	-	-	-	-
16	Regulatory Asset Recovery 10d(4)	-	-	-	-	-	-	-	-	-	-	-	-
17	ERIP	-	-	-	-	-	-	-	-	-	-	-	-
18	Other Provisions for Refund	(101,966)	-	101,966	-	-	-	-	-	-	-	-	-
19	Total Cycle Billed Surcharge Revenue	81,911	-	(81,911)	-	-	-	-	-	-	-	-	-
	Unbilled Revenue												
20	Base Tariff	(17,188)	-	-	17,188	-	-	-	-	-	-	-	-
21	Base PSCR	(4,574)	-	-	4,574	-	-	-	-	-	-	-	-
22	PSCR Factor	(2,451)	2,451	-	-	-	-	-	-	-	-	-	-
23	Total Unbilled Revenue	(24,214)	2,451	-	21,762	-	-	-	-	-	-	-	-
24	PSCR Over/Under Recovery	33,958	(33,958)	-	-	-	-	-	-	-	-	-	-
25	Total Calendar Revenue	4,313,811	(134,780)	(81,911)	21,762	(142,305)	(79,985)	(62,538)	183,647	140,553	4,158,254	24,610	4,133,644
26	Miscellaneous Revenue (2) (3)	84,099	(934)	-	(52,052)	76,105	-	-	-	-	107,218	923	106,295
27	Intersystem Sales Revenue	116,731	-	-	(116,731)	-	-	-	-	-	118,852	38	118,814
28	Total Operating Revenue	\$ 4,514,641	\$ (135,714)	\$ (81,911)	\$ (147,020)	\$ (66,200)	\$ (79,985)	\$ (62,538)	\$ 183,647	\$ 140,553	\$ 4,384,325	\$ 25,571	\$ 4,358,754
	Revenue Adjustments												
29	Jobwork Revenue	12,822	-	-	-	-	-	-	-	-	12,822	-	12,822
30	Jobwork Expense	(10,387)	-	-	-	-	-	-	-	-	(10,387)	-	(10,387)
31	Total Revenue Adjustments	2,435	-	-	-	-	-	-	-	-	2,435	-	2,435
32	Adjusted Total Operating Revenue	\$ 4,517,076	\$ (135,714)	\$ (81,911)	\$ (147,020)	\$ (66,200)	\$ (79,985)	\$ (62,538)	\$ 183,647	\$ 140,553	\$ 4,386,760	\$ 25,571	\$ 4,361,189
	Electric Deliveries (MWh)												
33	Cycle Billed Bundled Service	34,485,537	-	-	-	(2,257,495)	(966,053)	(512,350)	-	-	30,749,639	379,667	30,369,972
34	Cycle Billed Self-generation	92,606	-	-	-	-	-	(29,123)	-	-	63,483	-	63,483
35	Cycle Billed Electric Choice	3,776,151	-	-	-	-	(27,457)	(51,267)	-	-	3,697,428	-	3,697,428
36	Intersystem Sales	2,445,904	-	-	(2,445,904)	-	-	-	-	-	-	-	-
37	Unbilled Bundled Service	(125,183)	-	-	125,183	-	-	-	-	-	-	-	-
38	Unbilled Electric Choice	2,116	-	-	(2,116)	-	-	-	-	-	-	-	-
39	Total Electric Deliveries	40,677,131	-	-	(2,322,837)	(2,257,495)	(993,510)	(592,739)	-	-	34,510,550	379,667	34,130,883

Notes:

(1) GSG Power Supply (ln2, col. i), includes GI-2 PSCR revenue (EMB-6)

(2) Miscellaneous Revenues (ln26, col. B): PSCR Admin Fees (U-20202, A-17 (KGT-1)

(3) Miscellaneous Revenues (ln26, col.d): Authorized Return on Renewable Energy Assets (2018 P-521, pg 301.1) and Revenues from Transmission of Electricity of Others (2018 P-521, pg 300(M))

Schedule E-3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Electric Deliveries & Customer Counts by Rate Category

(Annual Deliveries in MWh)

Case No.: U-20697
Exhibit No.: A-15 (EMB-4)
Schedule: E-3
Page: 1 of 1
Witness: EMBreuring
Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)
		2018 Actual		Test Year			
Line		Avg. No. of	Annual	Avg. No. of	Annual	No. of	Annual
No.	Description	Customers	Deliveries	Customers	Deliveries	Customers	Deliveries
	<u>Bundled Residential Service</u>						
1	Standard Service RS	1,550,238	12,534,313	16,221	130,889	16,221	130,889
2	Peak Pwr Savers / Dynamic Pricing	49,706	494,902	-	-	-	-
3	Time-of-Day RT	2,461	62,837	-	-	-	-
4	Electric Vehicles REV-1	685	9,111	-	-	-	-
5	Electric Vehicles REV-2	35	61	-	-	-	-
6	Nighttime Savers (RPM)	-	-	676	8,045	676	8,045
7	Summer On-Peak (RSP)	-	-	1,601,135	12,247,759	1,601,135	12,247,759
8	Smart Hours (RSH)	-	-	2,666	60,975	2,666	60,975
9	Total Bundled Residential	1,603,125	13,101,224	1,620,698	12,447,669	1,620,698	12,447,669
	<u>Bundled Secondary Service</u>						
10	Secondary Energy-only GS	193,889	3,966,822	195,747	3,826,453	195,747	3,826,453
11	Secondary Demand GSD	19,934	3,489,883	19,907	3,256,731	19,907	3,256,731
12	Secondary GSTU	16	599	72	3,265	72	3,265
13	Total Bundled Secondary	213,839	7,457,304	215,726	7,086,449	215,726	7,086,449
	<u>Bundled Primary Service</u>						
14	Primary Energy-only GP	1,698	1,140,790	1,598	952,493	1,598	952,493
15	Primary Demand GPD	1,513	9,797,973	1,059	4,950,917	1,059	4,950,917
16	Primary Time-of-Use GPTU	463	1,921,250	1,032	4,365,549	1,032	4,365,549
17	General Service Primary (EIP)	19	470,014	19	358,537	19	358,537
18	Special Contract	-	-	-	-	-	-
19	Total Bundled Primary	3,693	13,330,026	3,708	10,627,496	3,708	10,627,496
	<u>Bundled Street Lighting Service</u>						
20	Unmetered Lighting GUL	4,203	110,660	3,735	79,208	3,735	79,208
21	Metered Lighting GML	366	6,108	343	13,509	343	13,509
22	Unmetered GU	480	94,523	479	95,012	479	95,012
23	Unmetered Lighting GU-XL	121	3,078	797	20,629	797	20,629
24	Total Bundled Street Lighting	5,170	214,369	5,354	208,357	5,354	208,357
	<u>Bundled Self-generation Service</u>						
25	Self-generation GSG-1	11	9,750	-	-	-	-
26	Self-generation GSG-2	11	82,856	11	63,483	11	63,483
27	Total Bundled Self-generation	22	92,606	11	63,483	11	63,483
	<u>Bundled Other Service</u>						
28	Wholesale	1	361,501	1	359,193	-	-
29	Grand Rapids	1	21,112	1	20,474	-	-
30	Total Bundled Other	2	382,613	2	379,667	-	-
25	Cycle Billed Bundled Service	1,825,851	34,578,143	1,845,500	30,813,122	1,845,498	30,433,455
31	Unbilled	-	(125,183)	-	-	-	-
32	Calendar Bundled Service	1,825,851	34,452,960	1,845,500	30,813,122	1,845,498	30,433,455
	<u>ROA Secondary Service</u>						
33	Secondary Energy-only GS	108	24,519	110	24,480	110	24,480
34	Secondary Demand GSD	479	191,179	477	180,859	477	180,859
35	Total ROA Secondary	587	215,698	587	205,339	587	205,339
	<u>ROA Primary Service</u>						
36	Primary Energy-only GP	47	74,250	54	74,331	54	74,331
37	Primary Demand GPD	359	3,486,203	354	3,417,757	354	3,417,757
38	Total ROA Primary	406	3,560,453	408	3,492,089	408	3,492,089
39	Cycle Billed ROA Service	993	3,776,151	995	3,697,428	995	3,697,428
40	Unbilled ROA	-	2,116	-	-	-	-
41	Calendar ROA Service	993	3,778,267	995	3,697,428	995	3,697,428
42	Cycle Billed Total Deliveries	1,826,844	38,354,294	1,846,495	34,510,550	1,846,493	34,130,883
43	Unbilled	-	(123,067)	-	-	-	-
44	Calendar Total Deliveries	1,826,844	38,231,227	1,846,495	34,510,550	1,846,493	34,130,883

Schedule E-4

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Calculation of Annual System Load Factor
2014 - 2019 Historical / 2019 - 2023 Forecast

Case No.: U-20697
Exhibit No.: A-15 (EMB-5)
Schedule: E-4
Page: 1 of 1
Witness: EMBreuring
Date: February 2020

Line No.			(a)	(b)	(c)
	Year		System Output GWh	System Peak Demand MW	Annual Load Factor
1	2014	Hist	40,912	7,498	62.3%
2	2015	Hist	40,173	7,812	58.7%
3	2016	Hist	40,931	8,227	56.6%
4	2017	Hist	40,032	7,634	59.9%
5	2018	Hist	41,331	8,084	58.4%
6	2019	Fcst	40,066	8,039	56.9%
7	2020	Fcst	39,331	7,654	58.5%
8	2021	Fcst	39,043	7,521	59.3%
9	2022	Fcst	38,923	7,464	59.5%
10	2023	Fcst	38,882	7,395	60.0%

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Estimated Electric Rate Case PSCR Factor
 January 2019 - December 2019

Case No.: U-20697
 Exhibit No.: A-50 (EMB-6)
 Page: 1 of 1
 Witness: EMBreuring
 Date: February 2020

Line No.	Description	(a) Total Company	(b) Jurisdictional Factors	(c) Jurisdictional	(d) Non-Jurisdictional Factors	(e) Non-Jurisdictional
<u>Expenses (\$000)</u>						
1	System Power Supply Costs	(1) \$ 1,683,201	98.90%	\$ 1,664,709	1.10%	\$ 18,492
2	Transmission & Market Administrative Expense	(2) 477,556	98.98%	472,662	1.02%	4,894
3	PSCR Expenses	<u>\$ 2,160,757</u>		<u>\$ 2,137,371</u>		<u>\$ 23,386</u>
<u>PSCR Revenue Contributions (\$000)</u>						
4	Intersystem Sales Revenue	(3) \$ 118,852	99.97%	\$ 118,814	0.03%	\$ 38
5	Self-Generation	(5) 3,887	100.00%	3,887	0.00%	-
6	GI-2	(9) 2,073	100.00%	2,073	0.00%	-
7	Base Jurisdictional PSCR x \$0.06037	1,828,696	100.00%	1,828,696	0.00%	-
8	Base Non-Jurisdictional PSCR x \$0.06037	1,236	0.00%	-	100.00%	1,236
9	Long-Term Industrial Load Retention Rate	(4) 49,168	100.00%	49,168	0.00%	-
10	Wholesale Fuel Revenue	(6) 12,608	0.00%	-	100.00%	12,608
11	PSCR Revenue Contributions	<u>\$ 2,016,520</u>		<u>\$ 2,002,638</u>		<u>\$ 13,882</u>
12	PSCR (Over)/Under Recovery	<u>\$ 144,237</u>		<u>\$ 134,733</u>		<u>\$ 9,504</u>
<u>Electric Deliveries (MWh)</u>						
13	Jurisdictional Bundled Deliveries	(7)		32,227,986		
14	Less: Self-Generation	(8)		(63,483)		
15	Less: GI-2			(76,651)		
16	Less: Long-Term Industrial Load Retention Rate			(1,794,531)		
17	Total Bundled PSCR Deliveries			<u>30,293,321</u>		
<u>Estimated Jurisdictional PSCR Factor</u>						
18	PSCR (Over)/Under Recovery (\$000)			\$ 134,733		
19	Total Bundled PSCR Deliveries (MWh)			<u>30,293,321</u>		
20	Total Jurisdictional PSCR Factor (\$ per kWh)			<u>\$ 0.00444</u>		
<u>Estimated PSCR Recovery</u>						
21	PSCR Revenue Contributions	\$ 2,016,520		\$ 2,002,638		\$ 13,882
22	Plus: PSCR (Over)/Under Recovery	134,502		134,502		-
23	Non-jurisdictional PSCR Factor Contribution	91		-		91
24	Total PSCR Revenues	<u>\$ 2,151,113</u>		<u>\$ 2,137,141</u>		<u>\$ 13,973</u>
25	Total PSCR (Over)/Under Recovery	<u>\$ 9,644</u>		<u>\$ 231</u>		<u>\$ 9,413</u>
Total Losses Less: Transmission System Losses System Efficiency PSCR Base @ Gen System Efficiency PSCR Base @ Del 0.0773 0.0000 0.0773 1.08378 0.05570 1.0838 0.06037						

Notes:

- (1) Exhibit A- (KGT-1), page 1, line 40; plus page 3, line 59; less page 1, line 34.
 (2) Exhibit A- (KGT-1), page 1, line 34.
 (3) Exhibit A- (KGT-1), page 3, line 59.
 (4) Exhibit A- (KGT-1), page 1, line 41.
 (5) Exhibit A- (EMB-3), Schedule E-2, line 2, column (i).
 (6) \$0.0351/kWh from U-20523, Exhibit A-1, Schedule 4, page 1 (Supplemental Power: Energy Charge rate).
 (7) Exhibit A-15 (EMB-4), Schedule E-3, line 27, column (f).
 (8) Exhibit A-15 (EMB-4), Schedule E-3, line 22, column (f).
 (9) GI-2 PSCR Revenue from HWM-1, WP-HWM-20

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

LORA B. CHRISTOPHER

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Benefits O&M Expenses

For the Years 2018, 2019, 2020 and 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-51 (LBC-1)

Page: 1 of 1

Witness: LBChristopher

Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)
		12 mos. ended 12/31/2018	12 mos. ended 12/31/2019	12 mos. ended 12/31/2020	12 mos. ended 12/31/2021
1	Pension Plans A/B	\$26,717	\$5,546	\$4,341	(\$4,394)
2	Defined Company Contribution Plan	7,628	8,567	9,741	11,202
3	401 (k) Employees' Savings Plan	7,982	8,273	8,672	9,189
4	Active Health Care/Life Insurance/LTD	24,194	25,353	25,948	27,473
5	Retiree Health Care and Life Insurance	(56,384)	(40,032)	(53,066)	(53,639)
6	Other Benefits	825	1,451	1,451	1,451
7	Total Expense	\$10,962	\$9,158	(\$2,913)	(\$8,718)

Consumers Energy Company

ASC 715 Pension Expense Estimates

(\$ millions)

Exhibit No.: A-52 (LBC-2)

Page: 1 of 3

Witness: LBChristopher

Date: February 2020

Prepared on January 21, 2020

Minimum Required Contributions - Baseline Scenario

[illegible]

Consumers Energy Company

ASC 715 Pension Expense Estimates

(\$ millions)

Case No.: U-20697

Exhibit No : A-52 (I BC-2)

Page: 2 of 3

Witness: LBChristopher

Date: February 2020

ASC 715 Pension Expense Estimates (\$ millions)

Prepared on January 21, 2020

[illegible]

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

CMS Energy - Pension Plans A and B -

ASC 715 Pension Expense Estimates

(\$ millions)

Case No.: U-20697

Exhibit No.: A-52 (LBC-2)

Page: 3 of 3

Witness: LBChristopher

Date: February 2020

2020-2027 projections reflect the following:

- January 1, 2019 census data.
- PBO discount rates of 3.37% (Plan A) and 3.17% (Plan B) in fiscal 2020, based on December 31, 2019 yield curve analysis.
- Service Cost effective interest rate of 3.44% (Plan A) for fiscal 2020, based on December 31, 2019 yield curve.
- Interest Cost effective interest rates of 2.92% (Plan A) and 2.74% (Plan B) in fiscal 2020, based on December 31, 2019 yield curve.
- Fiscal 2021+ effective interest rates determined based on December 31, 2019 yield curve.
- Expense is allocated to Utility/Non Utility based on PBO using the preliminary allocation percentages as will be provided for 2020 expense.
- Contributions are allocated to Utility/Non Utility based on the most recent allocation percentages provided by CMS for the January 2020 contribution.
- December 31, 2019 market assets provided by CMS, with expected returns thereafter.
- Expected and actual asset returns drop 25 basis points every other year, starting with a drop from 6.75% to 6.50% in 2022.
- The plans will waive Funding Balance when necessary, in order to maintain a Funded Percentage of at least 80%.
- Other provisions, assumptions and methods are the same as those used for December 31, 2019 ASC 715 disclosures.

CMS Energy
ASC 715 OPEB Expense Estimates (\$ millions)

Prepared on January 21, 2020

	2018	2019	2020	2021	2022	2023	2024	2025
Funded Status, January 1								
Accumulated Postretirement Benefit Obligation	\$ (1,095)	\$ (1,046)	\$ (1,165)	\$ (1,161)	\$ (1,153)	\$ (1,144)	\$ (1,133)	\$ (1,120)
Plan Assets at Fair Value	1,420	1,280	1,509	1,551	1,594	1,634	1,675	1,714
Funded Status	\$ 325	\$ 234	\$ 344	\$ 390	\$ 441	\$ 490	\$ 542	\$ 594
ASC 715 Accounting Expense								
Utility	\$ (90)	\$ (64)	\$ (86)	\$ (87)	\$ (88)	\$ (81)	\$ (72)	\$ (75)
Nonutility	(6)	(5)	(6)	(6)	(6)	(7)	(6)	(6)
Total	\$ (96)	\$ (69)	\$ (92)	\$ (93)	\$ (94)	\$ (88)	\$ (78)	\$ (81)
Components of Total Expense								
Service Cost	\$ 17	\$ 14	\$ 16	\$ 16	\$ 15	\$ 15	\$ 14	\$ 14
Interest Cost	36	41	33	33	32	32	32	31
Expected Return on Assets	(97)	(87)	(100)	(103)	(101)	(105)	(103)	(105)
Amortization of Net (Gain) or Loss	15	26	15	13	12	12	11	10
Amortization of Prior Service Cost	(67)	(63)	(56)	(52)	(52)	(42)	(32)	(31)
Total Expense	\$ (96)	\$ (69)	\$ (92)	\$ (93)	\$ (94)	\$ (88)	\$ (78)	\$ (81)
Assumptions								
APBO Discount Rate	3.74%	4.42%	3.32%	3.32%	3.32%	3.32%	3.32%	3.32%
Service Cost Effective Interest Rate	3.93%	4.63%	3.57%	3.57%	3.57%	3.57%	3.57%	3.57%
Interest Cost Effective Interest Rate	3.35%	4.03%	2.88%	2.88%	2.88%	2.88%	2.88%	2.88%
Expected Return on Assets	7.00%	7.00%	6.75%	6.75%	6.50%	6.50%	6.25%	6.25%
Trend Rate—Initial Pre-65	7.00%	6.75%	6.50%	6.25%	6.00%	5.75%	5.50%	5.25%
Trend Rate—Initial Post-65	7.75%	7.25%	7.00%	6.75%	6.50%	6.00%	5.75%	5.50%
Trend Rate—Ultimate	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%	4.75%
Trend Rate—Ultimate Year Pre-65	2027	2027	2027	2027	2027	2027	2027	2027
Trend Rate—Ultimate Year Post-65	2027	2027	2027	2027	2027	2027	2027	2027
Expected Contribution	\$ 0.4	\$ 0.4	\$ 0.4	\$ 0.4	\$ 0.4	\$ 0.4	\$ 0.4	\$ 0.4

2020-2025 expense projections reflect the following:

- January 1, 2019 census data.
- APBO discount rate of 3.32% in fiscal 2020+, based on December 31, 2019 yield curve.
- Service Cost effective interest rate of 3.57% in fiscal 2020+, based on December 31, 2019 yield curve.
- Interest Cost effective interest rate of 2.88% in fiscal 2020+, based on December 31, 2019 yield curve.
- December 31, 2019 market assets provided by CMS.
- Projected contributions provided by CMS:
 - \$0.4 million in all future years.
- Other provisions, assumptions and methods are the same as those used for December 31, 2019 ASC 715 disclosures.

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

AMY M. CONRAD

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

EICP Performance Measures

Case No.: U-20697

Exhibit No.: A-55 (AMC-1)

Page: 1 of 1

Witness: AMConrad

Date: February 2020

2019 Plan Year								
Operational				Target				
1. Customer Experience Index (CXi) <i>(Forester Index for Digital, Live Agent, and Interactive Voice Response)</i>				≥71				
2. Employee Safety (OSHA Recordable) <i>a) Incidents, and b) Incident Rate, and zero fatalities</i>				a) ≤54 b) ≤0.67				
3. Cyber Safety <i>(Phishing Click Rate (Level 2 Tests))</i>				≤5%				
4. Customer On-Time Delivery <i>(Orders Completed within Target Window)</i>				≥70%				
5. Service On-Time Commitments <i>a) Volume b) Adherence to Target Window</i>				a) ≥60% b) ≥95%				
6. Distribution Reliability - SAIDI <i>(System Average Interruption Duration Index - Customer Outage Minutes)</i>				≤165				
7. Generation Customer Value <i>(Fleet Availability at Least Cost Option and within Target Limits)</i>				≥72%				
8. Gas Flow Deliverability				≥90%				
9. Eliminate Vintage Services				≥9,250				
Number of targets achieved		0-3	4	5	6	7	8	9
Award Percentage		0%	50%	75%	100%	125%	150%	200%

Financial		Target
Earnings Per Share (EPS) <i>70% of financial award</i>		\$2.47
Incentive Operating Cash Flow (Billions) <i>30% of financial award</i>		\$1,700
Financial Award Percentage = %		

TOTAL EICP Award Payout Level = %

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Target Pay Level Market Analysis
Non-officer

Case No.: U-20697
Exhibit No.: A-56 (AMC-2)
Page: 1 of 1
Witness: AMConrad
Date: February 2020

Exempt Jobs:
2018 Market study

Job Family	# of Employees	Avg. Annual Salary	Market Data	Avg. Salary vs. Market
<u>ACCOUNTING ANALYST</u>	23	\$ 95,927	\$ 96,161	-0.5%
<u>IT BUSINESS</u>	14	\$ 106,220	\$ 95,283	10.0%
<u>IT PROJECT MGR</u>	4	\$ 159,069	\$ 156,758	1.5%
<u>BUSINESS SUPPORT</u>	193	\$ 84,655	\$ 82,267	2.8%
<u>CALL CNTR TEAM LDR</u>	11	\$ 79,146	\$ 85,832	-8.4%
<u>COMMUNICATIONS</u>	13	\$ 99,440	\$ 94,845	4.2%
<u>ELEC./GAS/FLEET FIELD LEADER</u>	244	\$ 118,615	\$ 122,647	-2.0%
<u>EMPLOYEE DEVELOPMENT</u>	21	\$ 103,022	\$ 102,949	0.5%
<u>ENGINEER</u>	267	\$ 126,862	\$ 130,427	12.3%
<u>ENGINEER TECH</u>	89	\$ 99,525	\$ 98,152	2.5%
<u>ENVIRONMENTAL PLANNER</u>	10	\$ 108,317	\$ 107,303	0.9%
<u>FORESTERY</u>	14	\$ 91,708	\$ 100,141	-9.2%
<u>GENERAL TECHICAL ANALYST</u>	114	\$ 85,316	\$ 85,430	-0.2%
<u>HUMAN RESOURCES</u>	22	\$ 120,377	\$ 113,208	7.0%
<u>SAFETY</u>	10	\$ 97,222	\$ 92,891	5.8%
<u>LABORATORY TECH</u>	3	\$ 85,972	\$ 83,358	3.0%
<u>RATE ANALYST</u>	13	\$ 98,092	\$ 98,487	1.9%
<u>EXCUTIVE ASSISTANT</u>	19	\$ 78,956	\$ 74,007	6.3%
<u>MAINTENANCE/PRODUCTION SUPV</u>	41	\$ 108,614	\$ 113,378	-1.8%
<u>SYSTEM CONTROLLER</u>	17	\$ 102,699	\$ 114,724	-10.4%
<u>PLANNER/SCHEDULER</u>	24	\$ 96,822	\$ 100,187	-2.9%
<u>CORPORATE ACCOUNT MANAGER</u>	22	\$ 126,884	\$ 133,858	-3.6%
<u>EPC PROJECT MANAGER</u>	23	\$ 138,408	\$ 138,760	1.9%
<u>GENERATION ASSET MANAGER</u>	10	\$ 93,723	\$ 100,378	-4.9%
<u>FINANCIAL ANALYST</u>	15	\$ 101,317	\$ 104,923	-2.5%
<u>TAX</u>	3	\$ 115,560	\$ 109,768	5.0%
<u>CLAIMS ADM</u>	11	\$ 72,612	\$ 77,756	-7.1%
<u>IT TECHNICAL</u>	161	\$ 96,202	\$ 92,213	3.7%
<u>IT SECURITY</u>	12	\$ 139,810	\$ 142,076	-1.6%
<u>IT ARCHITECTURE</u>	2	\$ 157,440	\$ 145,608	7.5%

Total exempt employees as a
% of market survey matches 45%

% of total base salaries Exempt & Non-Exempt 49%

Average	3.7%
Weighted Average	0.1%

Non-Exempt:
2018 Market study

Job Family	# of Employees	Avg. Annual Salary	Market Data	Avg. Salary vs. Market
Administrative Support Job Family	236	\$58,738	\$61,911	-5.6%
Customer Service Revenue Recovery Job Family	56	\$54,947	\$51,208	6.5%
Operations Support Job Family	12	\$59,418	\$61,294	-6.2%
Technical Support Job Family	220	\$59,318	\$62,262	-5.0%
Distribution Project Delivery Job Family	37	\$75,000	\$79,358	-6.0%
Electric System Owner Job Family	20	\$78,236	\$79,358	-1.2%
Technician Job Family	298	\$71,736	\$71,017	0.2%
Paralegal Job Family	4	\$77,777	\$90,541	-16.5%
Dispatcher Job Family	12	\$71,204	\$81,415	-14.3%

Total non-exempt employees as a
% of market survey matches 59%

Average	7.0%
Weighted Average	-1.2%

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Annual Incentive O&M Expenses

(\$000)

Case No.: U-20697

Exhibit No.: A-57 (AMC-3)

Page: 1 of 1

Witness: AMConrad

Date: February 2020

Line No.	(a) Description	(b) Historical		(c) 12 mos. ending		(d) Projected		(e) Test Year		(e) Source
		12 mos. ended 12/31/2018	12 mos. ending 12/31/2019	12 mos. ending 12/31/2020	12 mos. ending 12/31/2020	12 mos. ending 12/31/2020	12 mos. ending 12/31/2020	12/31/2021	12/31/2021	
1	Annual Incentive - Officer (1)	\$ 1,558	\$ 1,509	\$ 1,586	\$ 1,586	\$ 1,586	\$ 1,586	1,634		
2	Annual Incentive - Non Officer (EICP)	3,418	\$ 3,321	\$ 3,421	\$ 3,421	\$ 3,421	\$ 3,421	3,524		
3	Total Expense	\$ 4,976	\$ 4,830	\$ 5,007	\$ 5,007	\$ 5,007	\$ 5,007	5,157		

(2)

Footnotes

(1) Excludes named proxy officers

(2) Amounts represent 2018 EICP assuming payout at 100%

Amount of actual payout based on 2018 incentive program results were:

	(\$000)
Officer	\$ 2,072
Non-officer	4,427
Total	\$ 6,498

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

MICHAEL J. DELANEY

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Conversation Voltage Reduction Program Benefits and Costs (2021-2025)

Line	Item	2021	2022	2023	2024	2025
	(a)	(b)	(c)	(d)	(e)	(f)
<u>Operations</u>						
1	Number of Circuits	150	200	250	300	350
2	Capacity Reduction (MW)	33	44	56	67	78
3	Zonal Resource Credit (ZRC)	35	46	58	69	81
<u>Energy Reductions (MWh)</u>						
4	Voltage Reductions	75,520	101,054	126,683	152,513	178,251
5	Loss Reductions	2,644	3,538	4,435	5,339	6,240
6	Total Energy Reductions	78,164	104,592	131,118	157,852	184,491
<u>Total Expenses (\$M)</u>						
7	Capital Investments (CapEx)	\$ 2.9	\$ 2.9	\$ 2.9	\$ 3.0	\$ 3.0
8	Cost of Removal	-	-	-	-	-
9	Operating Expenses (O&M)	\$ 0.2	\$ 0.3	\$ 0.3	\$ 0.4	\$ 0.4
10	Total Expenses	\$ 3.1	\$ 3.2	\$ 3.3	\$ 3.3	\$ 3.4
<u>Capital Investments (\$M)</u>						
11	Beginning of Year	\$ 5.4	\$ 8.0	\$ 10.6	\$ 13.1	\$ 15.5
12	Additions	\$ 2.9	\$ 2.9	\$ 2.9	\$ 3.0	\$ 3.0
13	Depreciation	\$ (0.2)	\$ (0.3)	\$ (0.4)	\$ (0.5)	\$ (0.6)
14	End of Year	\$ 8.0	\$ 10.6	\$ 13.1	\$ 15.5	\$ 17.9
<u>Revenue Requirements (\$M)</u>						
15	CapEx Depreciation	\$ 0.2	\$ 0.3	\$ 0.4	\$ 0.5	\$ 0.6
16	CapEx Cost of Capital (Equity)	\$ 0.4	\$ 0.6	\$ 0.7	\$ 0.9	\$ 1.0
17	CapEx Cost of Capital (Debt)	\$ 0.1	\$ 0.1	\$ 0.2	\$ 0.2	\$ 0.3
18	Operating Expenses (O&M)	\$ 0.2	\$ 0.3	\$ 0.3	\$ 0.4	\$ 0.4
19	Total Revenue Requirement	\$ 1.0	\$ 1.3	\$ 1.7	\$ 2.0	\$ 2.3
<u>Avoided Costs (\$M)</u>						
20	Avoided Capacity Cost	\$ 3.2	\$ 4.7	\$ 6.2	\$ 7.7	\$ 9.2
21	Avoided Energy Cost	\$ 2.4	\$ 3.4	\$ 4.5	\$ 5.7	\$ 7.2
22	Avoided Transmission and Distribution Cost	\$ 0.9	\$ 1.2	\$ 1.5	\$ 1.9	\$ 2.2
23	Total Avoided Cost	\$ 6.5	\$ 9.3	\$ 12.3	\$ 15.2	\$ 18.6
<u>Conservation Voltage Reduction Incentive (\$M)</u>						
24	Incentive at 15% Shared Savings	\$ 0.8	\$ 1.2	\$ 1.6	\$ 2.0	\$ 2.4
25	Incentive Capped at 20% of O&M	\$ 0.05	\$ 0.06	\$ 0.07	\$ 0.07	\$ 0.08
Supply Side Alternatives						
26	Return on 50% Owned Solar / 50% PPA with FCM (IRP Supply Side Alternative)	\$ 2.1	\$ 3.1	\$ 3.9	\$ 4.6	\$ 5.3
27	Return on 100% Owned Solar	\$ 3.8	\$ 5.7	\$ 7.0	\$ 8.3	\$ 9.4
28	CVR Incentive and Capital Return Above/(Below) IRP Supply Side Alternative	\$ (0.9)	\$ (1.4)	\$ (1.6)	\$ (1.7)	\$ (1.8)
29	Net Customer Benefit (\$M)	\$ 4.7	\$ 6.8	\$ 9.0	\$ 11.2	\$ 13.8

SOURCES AND NOTES

Capacity reduction values and capital and O&M costs: U-20165 Exhibits A-67 through A-70 (MAO-1 through MAO-4)

Energy reduction values: RTBlumenstock DT, Figure 61

Net Customer Benefit Calculated as Total Avoided Cost (Line 23) less Total Revenue Requirement (Line 19) and Incentive at 15% Shared Savings (Line 24)

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Conservation Voltage Reduction Regulatory Return Comparison

Case No.: U-20697

Exhibit No.: A-59 (MJD-2)

Page: 1 of 1

Witness: MJDelaney

Date: February 2020

Line	Resource Type	Incentive Type	Incentive/Return Amount (\$M)	
1	Conservation Voltage Reduction	ROE Only	\$	8.1
2	Conservation Voltage Reduction	ROE + 20% O&M Spend	\$	9.2
3	Conservation Voltage Reduction	ROE + 15% Shared Savings	\$	32.6
4	IRP Solar (50% Company-Owned / 50% PPA with FCM)	ROE Only	\$	41.2
5	IRP Solar (100% Company-Owned)	ROE Only	\$	71.4

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBIT

OF

DOUG E. DETTERMAN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Electric Distribution

Summary of Actual and Projected Electric Field Maintenance and Construction Resources

(\$000) unless otherwise indicated

Case No.: U-20697

Exhibit No.: A-60 (DED-1)

Page: 1 of 1

Witness: DEDetterman

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
Line		Historical	Projected Bridge Year			Projected
No.	Description	12 Mos Ended 12/31/2018	12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	Test Year 12 Mos Ending 12/31/2021
Low Voltage Distribution Capital						
1	Expenditures - excluding metering, transformers, computers, etc.	299,584	329,412	309,381	638,793	371,741
2	Change vs Historical 12 Months Ended 12/31/2018		10%	3%	7%	24%
Low Voltage Distribution Field						
3	Maintenance and Construction Full Time Equivalents, FTEs	854	949	848	1797	1008
4	Change vs Historical 12 Months Ended 12/31/2018*		11%	-1%	5%	18%
High Voltage Distribution Capital						
5	Expenditures - excluding metering, transformers, computers, etc.	178,370	213,626	153,855	367,481	269,874
6	Change vs Historical 12 Months Ended 12/31/2018		20%	-14%	3%	51%
High Voltage Distribution Field						
7	Maintenance and Construction Full Time Equivalents, FTEs	270	308	222	530	392
8	Change vs Historical 12 Months Ended 12/31/2018*		14%	-18%	-2%	45%

* 2% net FTE cost escalation assumed for each 2020 and 2021.

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

KAREN M. GASTON

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Projected Electric & Common O&M Expenses

for the Years 2018, 2019, 2020; and the 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697

Exhibit No.: A-61 (KMG-1)

Page: 1 of 1

Witness: KMGaston

Date: February 2020

Corporate O&M

Line No.	(a) Program Description	(b) Source	(c) 2018 Actual	(d) 2019 Projected	(e) 2020 Projected	(f) 2021 Projected
1	Adjusted Corporate Services Expense	Exhibit A-62 (KMG-2)	\$ 45,482	\$ 47,950	\$ 49,278	\$ 56,787
2	Uncollectible Expense	Exhibit A-64 (KMG-4)	16,661	16,964	17,433	18,077
3	Injuries & Damages Expense	Exhibit A-65 (KMG-5)	3,975	4,050	4,115	4,531
4	TOTAL O&M EXPENSES		\$ 66,119	\$ 68,964	\$ 70,827	\$ 79,396

Consumers Energy Company

Case No.: U-20697
Exhibit No.: A-62 (KMG-2)
Page: 1 of 1
Witness: KMGaston
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
					1.032	1.02		1.032	1.015		1.032	1.023	
Line No.	Description	2018 Labor Actual	2018 Non-Labor Actual	Total 2018 Actual	2019 Labor Projected	2019 Non-Labor Projected	Total 2019 Projected	2020 Labor Projected	2020 Non-Labor Projected	Total 2020 Projected	2021 Labor Projected	2021 Non-Labor Projected	Total 2021 Projected
1	Governmental Regulatory and Public Affairs	\$ 5,613	\$ 1,250	\$ 6,863	\$ 5,792	\$ 1,275.34	\$ 7,068	\$ 5,977.66	\$ 1,294	\$ 7,272	\$ 6,169	\$ 1,324	\$ 7,493
2	General Counsel, Legal & Risk Management	6,209	9,755	15,964	6,407	9,950	16,357	6,612	10,099	16,712	6,824	16,316	23,140
3	Human Resources and Learning and Development	13,191	3,189	16,380	13,613	3,252	16,866	14,049	3,301	17,350	14,498	3,377	17,875
4	Transformation & Operations Support	3,927	1,422	5,349	4,052	1,450	5,503	4,182	1,472	5,654	4,316	1,506	5,822
5	Chief Financial Officer	13,765	7,353	21,117	14,205	7,500	21,705	14,660	7,612	22,272	15,129	7,787	22,916
6	Strategy	1,565	1,872	3,436	1,615	1,909	3,524	1,666	1,938	3,604	1,720	1,982	3,702
7	General Activities	(12,952)	(4,747)	(17,699)	(13,367)	(4,842)	(18,208)	(13,794)	(4,914)	(18,709)	(14,236)	(5,027)	(19,263)
13	Total	\$ 31,316	\$ 20,094	\$ 51,410	\$ 32,318	\$ 20,495	\$ 52,814	\$ 33,353	\$ 20,803	\$ 54,155	\$ 34,420	\$ 27,265	\$ 61,685
14	Less: EICP			4,147			3,127			3,174			3,247
15	Less: Transfers Out			-			-			-			-
16	Total Corporate Service Departments			\$ 47,263			\$ 49,687			\$ 50,982			\$ 58,438
17	Admin & Other-Electric Portion			2,191			2,234			2,268			2,320
18	Total Corporate Services			\$ 49,453			\$ 51,921			\$ 53,249			\$ 60,758
	Normalizations												
19	Storm Insurance			(3,346)			(3,346)			(3,346)			(3,346)
20	Total Normalizations			\$ (3,346)			\$ (3,346)			\$ (3,346)			\$ (3,346)
	Less Giving and Communications												
21	Corporate Giving, Communications, and Lobbying			(625)			(625)			(625)			(625)
22	Total Giving and Communications			\$ (625)			\$ (625)			\$ (625)			\$ (625)
23	TOTAL ADJUSTED CORPORATE SERVICES EXPENSE			\$ 45,482			\$ 47,950			\$ 49,278			\$ 56,787

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

S&P Global Market Intelligence ranking of Consumers Energy Electric A&G Costs for 2018

Ranked by A&G per Customer (less Pension and Benefits)

(Companies over 500K Customers)

Case No.: U-20697

Exhibit No.: A-63 (KMG-3)

Page: 1 of 1

Witness: KMGaston

Date: February 2020

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Rank	Company Name	Total Customers (actual)	Total Admin & Gen: O&M Expense (\$000)	Total Admin & Gen: Pension & Benefits (\$000)	Total Admin & Gen: Excluding Pension & Benefits (d) - (e) (\$000)	Total Admin & Gen: Excluding Pension & Benefits \$/Customer (f)/(c) x \$1,000	\$/Customer Ranking
1	Atmos Energy Corporation	205,101	111,212	3,126,009	93,889	\$30.03	1
2	Wisconsin Gas LLC	26,311	7,062	626,570	19,249	\$30.72	2
3	Public Service Electric and Gas Company	89,702	26,092	1,832,201	63,610	\$34.72	3
4	Spire Missouri Inc.	101,219	58,839	1,164,174	42,380	\$36.40	4
5	The East Ohio Gas Company	-19,674	-67,595	1,186,786	47,921	\$40.38	5
6	Consumers Energy Company	71,248	-973	1,763,743	72,221	\$40.95	6
7	Questar Gas Company	49,495	4,832	990,924	44,663	\$45.07	7
8	Public Service Company of North Carolina, Incorp	40,662	12,080	552,993	28,582	\$51.69	8
9	PECO Energy Co.	34,513	7,328	519,473	27,185	\$52.33	9
10	Puget Sound Energy, Inc.	58,756	15,307	820,624	43,449	\$52.95	10
11	New Mexico Gas Company, Inc.	39,524	11,536	524,655	27,988	\$53.35	11
12	Ameren Illinois Company	53,526	6,188	812,837	47,338	\$58.24	12
13	Southern California Gas Company	539,588	177,001	5,744,043	362,587	\$63.12	13
14	National Fuel Gas Distribution Corporation	91,663	41,253	744,885	50,410	\$67.67	14
15	Northern Illinois Gas Company	202,154	39,371	2,217,017	162,783	\$73.42	15
16	Niagara Mohawk Power Corporation	65,664	20,232	614,718	45,432	\$73.91	16
17	Consolidated Edison Company of New York, Inc.	141,177	61,541	1,077,394	79,636	\$73.92	17
18	Northwest Natural Gas Company	90,893	34,073	731,635	56,820	\$77.66	18
19	San Diego Gas & Electric Company	87,343	18,083	880,387	69,260	\$78.67	19
20	Atlanta Gas Light Company	150,262	23,420	1,577,838	126,842	\$80.39	20
21	Philadelphia Gas Works Co.	133,018	91,837	503,607	41,181	\$81.77	21
22	DTE Gas Company	134,105	30,030	1,253,989	104,075	\$83.00	22
23	Columbia Gas of Ohio, Inc.	142,466	22,473	1,439,462	119,993	\$83.36	23
24	Pacific Gas and Electric Company	511,046	131,966	4,494,485	379,080	\$84.34	24
25	Indiana Gas Company, Inc.	53,335	345	592,618	52,990	\$89.42	25
26	Northern Indiana Public Service Company	85,596	10,621	820,527	74,975	\$91.37	26
27	Kansas Gas Service Company, Inc.	84,087	25,508	638,049	58,579	\$91.81	27
28	Baltimore Gas and Electric Company	88,019	21,423	673,937	66,596	\$98.82	28
29	Boston Gas Company	130,199	53,389	700,052	76,810	\$109.72	29
30	New Jersey Natural Gas Company	88,383	25,591	534,425	62,792	\$117.49	30
31	Washington Gas Light Company	162,448	21,128	1,160,175	141,320	\$121.81	31
32	Brooklyn Union Gas Company	252,196	68,090	1,249,659	184,106	\$147.32	32
33	KeySpan Gas East Corporation	129,093	40,094	593,036	88,999	\$150.07	33
34	Peoples Gas Light and Coke Company	173,735	38,103	846,670	135,632	\$160.19	34

S&P Global Market Intelligence, 55 Water Street, New York, NY 10041

Columns c-e from S&P Global for regulated electric companies with more than 500,000 customers.

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Electric Uncollectible Accounts Expense

for the Years 2018, 2019, 2020; and the 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697
Exhibit No.: A-64 (KMG-4)
Page: 1 of 2
Witness: KMGaston
Date: February 2020

Line No.	(a) Description	(b) 2018 Actual	(c) 2019 Projected	(d) 2020 Projected	(e) 2021 Projected
1	Uncollectible Accounts Expense	\$ 16,661	\$ 16,964	\$ 17,433	\$ 18,077 ¹

¹ Exhibit A-64 (KMG-4) Page 2, row 10, column (e) (3-Year Average)

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Electric Uncollectible Accounts Expense
for 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-64 (KMG-4)
Page: 2 of 2
Witness: KMGaston
Date: February 2020

Line No.	(a) Year	(b) Gross Charge-Offs	(c) Less Recoveries	(d) Net Write-Offs	(e) Total Electric Service Revenue MPSC P-521 P. 304.1 col (c) + P. 305 col (c)	(f) BDLR col (d) / col (e)
1	2014	\$ 48,049	\$ 16,824	\$ 31,225	\$ 4,150,882	0.752%
2	2015	46,941	16,886	30,055	4,031,759	0.745%
3	2016	32,691	13,496	19,195	4,157,271	0.462%
4	2017	32,032	13,060	18,972	4,245,558	0.447%
5	2018	28,943	12,282	16,661	4,382,878	0.380%
6	3-Year Average	\$ 31,222	\$ 12,946	\$ 18,276	\$ 4,261,902	0.429%
7	5-Year Average	\$ 37,731	\$ 14,510	\$ 23,222	\$ 4,193,670	0.554%
8	Test Year Total Company Electric Revenues and Deliveries Exhibit A-15 (EMB-3), Schedule E-2, Page 1 of 1 Row 25, Column (I) - Row 25, Column (c)				\$ 4,215,555	
9	3-Year Average BDLR				0.429%	
10	Test Year Total Uncollectible Accounts Expense				<u><u>\$ 18,077</u></u>	

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Electric Injuries & Damages Expense

for the Years 2014 through 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-65 (KMG-5)

Page: 1 of 1

Witness: KMGaston

Date: February 2020

Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
No.	Program Description	2014	2015	2016	2017	2018	2019	2020	2021
		Actual	Actual	Actual	Actual	Actual	Projected @ 2.0%	Projected @ 1.5%	Projected 5-Yr Average
1	Electric Injuries & Damages	¹ \$ 4,382	\$ 1,290	\$ 3,111	\$ 2,933	\$ 2,958	\$ 3,017	\$ 3,062	\$ 2,935
2	Internal Legal Costs	² 563	563	562	617	559	570	579	573
3	Workers' Compensation	³ 1,737	1,115	870	937	458	467	474	1,023
4	Total Electric Injuries & Damages	<u>\$ 6,682</u>	<u>\$ 2,968</u>	<u>\$ 4,543</u>	<u>\$ 4,487</u>	<u>\$ 3,975</u>	<u>\$ 4,055</u>	<u>\$ 4,115</u>	<u>\$ 4,531</u>

¹ Electric Injuries & Damages costs are 2014 - 2018 (actual expense) and escalated using the noted inflation factors below for 2019. 2021 test year based on a five-year average (2014 - 2018).

² Legal costs are 2014 - 2018 (actual expense) and escalated using the noted inflation factors below for 2019. 2021 test year based on a five-year average (2014 - 2018).

³ Electric Workers' Compensation costs are 2014 - 2018 (actual expense) and escalated using the noted inflation factors below for 2019. 2021 test year based on a five-year average (2014 - 2018).

2019 Inflation Factor 2.0%

2020 Inflation Factor 1.5%

Source: November 2019 IHS Markit

Schedule B-5.4

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Electric Corporate Services Capital Expenditures

for the Years 2018, 2019, 2020; and the 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (KMG-6)

Schedule No.: B-5.4

Page: 1 of 1

Witness: KMGaston

Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures					
Line No.	Program Description	Historical	Projected Bridge Year			Projected Test	
		12 Mos Ended 12/31/2018	12 Mos Ending 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 mos. Ending 12/31/2021	
1	Labor	\$ 18	\$ 4	\$ 4	\$ 9	\$ 5	
2	Labor Overhead	-	-	-	-	-	
3	Materials	221	244	252	495	260	
4	Contracted	63	190	196	386	202	
5	Business Expenses	4	3	3	6	3	
6	Contingency	-	-	-	-	-	
7	Other (Loadings, Chargebacks)	8	2	2	4	2	
8	Total Corporate Capital	\$ 313	\$ 443	\$ 457	\$ 900	\$ 472	

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBIT

OF

DANIEL L. HARRY

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Consumers Energy Company
Amortization of Karn Plant Retention and Separation Costs
for the 12-Months Ended December 31, 2021
(\$000)

Exhibit No.: A-66 (DLH-1)
Page: 1 of 2
Witness: DLHarry
Date: February 2020

[illegible]

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy CompanyKarn Plant Retention and Separation Costs Unamortized Balance
(\$000)

Case No.: U-20697

Exhibit No.: A-66 (DLH-1)

Page: 2 of 2

Witness: DLHarry

Date: February 2020

(a)		(b)	(c)	(d)	(e)
Line		(Col. (c)+Col. (d))/2			
No.		Average Balance	12 Months Ending	12 Months Ending	Original Cost
		December 31, 2020	December 31, 2020	December 31, 2021	
1	Karn Expenditures Deferred 2020	\$ 12,626	12,967	12,285	\$ 12,967
2	Karn Expenditures Deferred 2021	<u>3,707</u>	-	7,413	\$ 7,413
3	Total Unamortized Balance to be included in rate base: (000)	<u><u>\$ 16,333</u></u>			

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

SCOTT A. HUGO

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Generating Unit Periodic Outages

January 1, 2021 Through December 31, 2021

Case No.: U-20697

Exhibit No.: A-67 (SAH-1)

Page: 1 of 1

Witness: SAHugo

Date: February 2020

MAJOR OUTAGES: FOSSIL GENERATION AND LUDINGTON

<u>Line</u> <u>No.</u>	(a) <u>Unit</u>	(b) <u>Planned Days in 2021</u>	(c) <u>Start</u>	(d) <u>Stop</u>
1	Campbell 2	60	3/19/2021	5/18/2021
2	Campbell 3	35	4/3/2021	5/8/2021
3	Ludington 1	42	5/2/2021	6/13/2021
4	Ludington 2	42	5/2/2021	6/13/2021
5	Ludington 3	42	5/2/2021	6/13/2021
6	Ludington 4	42	5/2/2021	6/13/2021
7	Ludington 5	42	5/2/2021	6/13/2021
8	Ludington 6	42	5/2/2021	6/13/2021
9	Zeeland 1	40	9/15/2021	10/25/2021
10	Zeeland 2	40	9/15/2021	10/25/2021
11	Campbell 1	45	9/24/2021	11/8/2021
12	Karn 3	69	10/1/2021	12/9/2021
13	Karn 1	31	10/8/2021	11/8/2021
14	Ludington 4	42	10/15/2021	11/26/2021
15	Karn 2	28	10/21/2021	11/18/2021
16	Karn 4	46	10/29/2021	12/14/2021

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Generating Unit Availability Projections
 January 1, 2021 Through December 31, 2021

Case No.: U-20697
 Exhibit No.: A-68 (SAH-2)
 Page: 1 of 1
 Witness: SAHugo
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Plant	Actual ROR 2014-2018	Projected ROR	Periodic Factor	Projected Availability	Actual NEV 2014-2018
1	Campbell 1	11.81%	23.00%	16.72%	64.13%	\$52,992,065
2	Campbell 2	13.75%	22.00%	35.86%	50.03%	\$58,456,638
3	Campbell 3	7.32%	5.00%	9.61%	85.87%	\$228,464,447
4	Karn 1	25.03%	42.00%	21.06%	45.79%	\$42,500,331
5	Karn 2	13.49%	27.00%	18.43%	59.55%	\$43,970,868
6	Karn 3	31.94%	32.00%	18.99%	55.09%	-\$5,767,010
7	Karn 4	33.62%	37.00%	12.63%	55.04%	-\$11,141,972
8	Ludington 1	8.74%	5.10%	14.00%	81.61%	\$27,510,874
9	Ludington 2	15.10%	5.10%	14.00%	81.61%	\$16,516,595
10	Ludington 3	8.98%	5.10%	14.00%	81.61%	\$22,302,623
11	Ludington 4	5.14%	5.10%	23.31%	72.78%	\$15,850,075
12	Ludington 5	30.92%	5.10%	14.00%	81.61%	\$12,290,523
13	Ludington 6	6.95%	5.10%	14.00%	81.61%	\$22,510,383
14	Hydros	6.73%	6.50%	2.94%	90.75%	\$46,972,451
15	Zeeland CC	3.78%	4.00%	8.91%	87.45%	\$102,948,822
16	Zeeland 1A	3.59%	4.00%	13.79%	82.76%	\$5,174,132
17	Zeeland 1B	3.16%	4.00%	13.78%	82.77%	\$5,204,696
18	Jackson ¹	3.23%	6.30%	2.19%	91.65%	\$40,171,131

¹Jackson acquired in December 2015.

MICHIGAN PUBLIC SERVICE COMMISSION
Schedule: B-5.2
Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

(\$000's)

Case No.: U-20697

Exhibit No.: A-12 (SAH-3)

Schedule: B-5.2

Page: 1 of 9

Witness: SAHugo

Date: February 2020

**Generation Capital Expenditures
(\$000)**

		(a)	(b)	(c)	(d)	(e)	(f)
		Historical		Projected Bridge Year		Projected Test Year	
Line		12 Months Ended	12 Months Ended	12 Months Ending	24 Months Ending	12 Months Ending	
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12/31/2021	
1	Steam Power Generation						
2	Environmental	\$ 36,792	\$ 5,987	\$ 6,648	\$ 12,635	\$ 19,058	
3	Routine and Small CapEx	\$ 79,304	\$ 93,544	\$ 72,847	\$ 166,391	\$ 85,198	
4	Total Steam Production	\$ 116,096	\$ 99,531	\$ 79,495	\$ 179,026	\$ 104,257	
5	Hydraulic Power Generation						
6	Routine and Small CapEx	\$ 19,209	\$ 31,586	\$ 15,073	\$ 46,659	\$ 31,542	
7	Total hydraulic production	\$ 19,209	\$ 31,586	\$ 15,073	\$ 46,659	\$ 31,542	
8	Pumped Storage Generation						
9	Ludington Overhaul	\$ 25,454	\$ 26,248	\$ 12,707	\$ 38,955	\$ 2,040	
10	Routine and Small CapEx	\$ 7,880	\$ 10,285	\$ 9,975	\$ 20,261	\$ 19,852	
11	Total Pumped Storage Generation	\$ 33,333	\$ 36,533	\$ 22,682	\$ 59,216	\$ 21,892	
12	Other Production Plant						
13	Routine and Small CapEx	\$ 7,065	\$ 6,393	\$ 2,395	\$ 8,789	\$ 3,384	
14	Total Other Production Plant	\$ 7,065	\$ 6,393	\$ 2,395	\$ 8,789	\$ 3,384	
15	Grand Total	<u>\$ 175,704</u>	<u>\$ 174,044</u>	<u>\$ 119,646</u>	<u>\$ 293,689</u>	<u>\$ 161,075</u>	

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Case No.: U-20697
 Exhibit No.: A-12 (SAH-3)
 Schedule: B-5.2
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Consumers Energy Company
 Summary of Actual and Projected Electric Capital Expenditures
 For the years 2018 through 2021
 (\$000's)

Generation Capital Expenditures

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
		Historical		12 Months Ended		Projected Bridge Year		24 Months Ending		Projected Test Year	
Line No.	Description	12/31/2018		12/31/2019		12/31/2020		12/31/2020		12/31/2021	
1	JHCampbell 1&2	\$ 6,584		\$ 7,272		\$ 15,161		\$ 22,434		\$ 12,869	
2	Contractor	\$ 3,166		\$ 3,822		\$ 10,001		\$ 13,823		\$ 9,983	
3	Labor	\$ 647		\$ 756		\$ 885		\$ 1,641		\$ 333	
4	Materials	\$ 1,686		\$ 2,041		\$ 2,019		\$ 4,060		\$ 893	
5	Business Expenses	\$ 5		\$ 3		\$ -		\$ 3		\$ -	
6	Contingency	\$ -		\$ -		\$ 717		\$ 717		\$ 738	
7	Other (Loadings, Chargebacks)	\$ 1,079		\$ 651		\$ 1,539		\$ 2,190		\$ 922	
8	JHCampbell 3	\$ 11,683		\$ 28,652		\$ 8,835		\$ 37,487		\$ 12,445	
9	Contractor	\$ 7,465		\$ 17,770		\$ 4,711		\$ 22,481		\$ 8,128	
10	Labor	\$ 974		\$ 2,171		\$ 438		\$ 2,609		\$ 520	
11	Materials	\$ 1,561		\$ 7,318		\$ 1,157		\$ 8,475		\$ 2,833	
12	Business Expenses	\$ 21		\$ 125		\$ -		\$ 125		\$ -	
13	Contingency	\$ -		\$ -		\$ 318		\$ 318		\$ 805	
14	Other (Loadings, Chargebacks)	\$ 1,662		\$ 1,268		\$ 2,211		\$ 3,479		\$ 158	
15	DEKarn 1&2	\$ 7,875		\$ 6,033		\$ 2,712		\$ 8,745		\$ 12,215	
16	Contractor	\$ 2,490		\$ 3,253		\$ 1,929		\$ 5,182		\$ 6,919	
17	Labor	\$ 1,030		\$ 490		\$ 126		\$ 616		\$ 22	
18	Materials	\$ 2,914		\$ 1,756		\$ 419		\$ 2,175		\$ 3,840	
19	Business Expenses	\$ 6		\$ (4)		\$ -		\$ (4)		\$ -	
20	Contingency	\$ -		\$ -		\$ 44		\$ 44		\$ 515	
21	Other (Loadings, Chargebacks)	\$ 1,436		\$ 537		\$ 194		\$ 731		\$ 919	
22	DEKarn 3&4	\$ 14,012		\$ 12,159		\$ 6,886		\$ 19,045		\$ 11,103	
23	Contractor	\$ 7,358		\$ 6,577		\$ 4,524		\$ 11,101		\$ 7,877	
24	Labor	\$ 816		\$ 1,381		\$ 267		\$ 1,648		\$ 293	
25	Materials	\$ 3,363		\$ 2,406		\$ 1,030		\$ 3,436		\$ 1,144	
26	Business Expenses	\$ 24		\$ 11		\$ -		\$ 11		\$ -	
27	Contingency	\$ -		\$ -		\$ 517		\$ 517		\$ 997	
28	Other (Loadings, Chargebacks)	\$ 2,451		\$ 1,783		\$ 548		\$ 2,331		\$ 793	
29	Zeeland	\$ 26,948		\$ 11,924		\$ 13,708		\$ 25,632		\$ 20,222	
30	Contractor	\$ 19,446		\$ 8,360		\$ 11,678		\$ 20,038		\$ 19,074	
31	Labor	\$ 692		\$ 439		\$ 132		\$ 571		\$ 40	
32	Materials	\$ 4,702		\$ 2,353		\$ 651		\$ 3,004		\$ 5,000	
33	Business Expenses	\$ 13		\$ 5		\$ -		\$ 5		\$ -	
34	Contingency	\$ -		\$ -		\$ 356		\$ 356		\$ 691	
35	Other (Loadings, Chargebacks)	\$ 2,095		\$ 768		\$ 891		\$ 1,659		\$ (4,583)	
36	Jackson Generating Station	\$ 11,869		\$ 26,904		\$ 24,644		\$ 51,548		\$ 16,345	
37	Contractor	\$ 12,426		\$ 20,962		\$ 21,876		\$ 42,839		\$ 15,549	
38	Labor	\$ 259		\$ 469		\$ -		\$ 469		\$ 14	
39	Materials	\$ (162)		\$ 1,600		\$ -		\$ 1,600		\$ 167	
40	Business Expenses	\$ 4		\$ 25		\$ -		\$ 25		\$ -	
41	Contingency	\$ -		\$ -		\$ 601		\$ 601		\$ 277	
42	Other (Loadings, Chargebacks)	\$ (659)		\$ 3,848		\$ 2,167		\$ 6,015		\$ 338	
43	CTs	\$ (84)		\$ -		\$ -		\$ -		\$ -	
44	Contractor	\$ (12)		\$ -		\$ -		\$ -		\$ -	
45	Labor	\$ (41)		\$ -		\$ -		\$ -		\$ -	
46	Materials	\$ 3		\$ -		\$ -		\$ -		\$ -	
47	Business Expenses	\$ -		\$ -		\$ -		\$ -		\$ -	
48	Contingency	\$ -		\$ -		\$ -		\$ -		\$ -	
49	Other (Loadings, Chargebacks)	\$ (34)		\$ -		\$ -		\$ -		\$ -	

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Case No.: U-20697
 Exhibit No.: A-12 (SAH-3)
 Schedule: B-5.2
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 Witness: SAHugo
 Date: February 2020

Consumers Energy Company
 Summary of Actual and Projected Electric Capital Expenditures
 For the years 2018 through 2021
 (\$000's)

Generation Capital Expenditures

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Description	Historical 12 Months Ended 12/31/2018	12 Months Ended 12/31/2019	Projected Bridge Year 12 Months Ending 12/31/2020	24 Months Ending 12/31/2020	Projected Test Year 12 Months Ending 12/31/2021					
50	Classic 7	\$ 416	\$ 601	\$ 900	\$ 1,501	\$ -					
51	Contractor	\$ 389	\$ 484	\$ 500	\$ 984	\$ -					
52	Labor	\$ (33)	\$ 50	\$ -	\$ 50	\$ -					
53	Materials	\$ (0)	\$ 4	\$ 333	\$ 337	\$ -					
54	Business Expenses	\$ 1	\$ -	\$ -	\$ -	\$ -					
55	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -					
56	Other (Loadings, Chargebacks)	\$ 58	\$ 63	\$ 67	\$ 130	\$ -					
57	Hydros	\$ 19,209	\$ 31,586	\$ 15,073	\$ 46,659	\$ 31,542					
58	Contractor	\$ 14,045	\$ 23,201	\$ 7,742	\$ 30,943	\$ 24,308					
59	Labor	\$ 1,404	\$ 2,851	\$ -	\$ 2,851	\$ 154					
60	Materials	\$ 573	\$ 2,246	\$ -	\$ 2,246	\$ -					
61	Business Expenses	\$ 87	\$ 195	\$ -	\$ 195	\$ -					
62	Contingency	\$ -	\$ -	\$ 758	\$ 758	\$ 2,585					
63	Other (Loadings, Chargebacks)	\$ 3,100	\$ 3,092	\$ 6,573	\$ 9,665	\$ 4,495					
64	Ludington	\$ 33,333	\$ 36,533	\$ 22,682	\$ 59,216	\$ 21,892					
65	Contractor	\$ 49,603	\$ 54,928	\$ 30,030	\$ 84,958	\$ 33,646					
66	Labor	\$ (246)	\$ 826	\$ 675	\$ 1,502	\$ 693					
67	Materials	\$ 2,463	\$ 3,405	\$ 335	\$ 3,740	\$ 1,195					
68	Business Expenses	\$ 306	\$ 251	\$ -	\$ 251	\$ -					
69	Contingency	\$ -	\$ -	\$ 3,793	\$ 3,793	\$ 2,143					
70	Other (Loadings, Chargebacks)	\$ (18,792)	\$ (22,877)	\$ (12,151)	\$ (35,028)	\$ (15,784)					
71	Admin and Other	\$ 7,065	\$ 6,393	\$ 2,395	\$ 8,789	\$ 3,384					
72	Contractor	\$ 4,803	\$ 5,067	\$ 485	\$ 5,552	\$ 485					
73	Labor	\$ 129	\$ 489	\$ -	\$ 489	\$ -					
74	Materials	\$ 425	\$ 636	\$ -	\$ 636	\$ -					
75	Business Expenses	\$ 1,714	\$ 7	\$ -	\$ 7	\$ -					
76	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -					
77	Other (Loadings, Chargebacks)	\$ (4)	\$ 194	\$ 1,910	\$ 2,105	\$ 2,899					
78	Air Quality	\$ 11,474	\$ 5,843	\$ 4,351	\$ 10,194	\$ 6,734					
79	Contractor	\$ 6,012	\$ 5,263	\$ 2,013	\$ 7,277	\$ 3,131					
80	Labor	\$ 882	\$ 362	\$ 592	\$ 954	\$ 376					
81	Materials	\$ 2,850	\$ 10	\$ 853	\$ 863	\$ 2,385					
82	Business Expenses	\$ 3	\$ 19	\$ -	\$ 19	\$ -					
83	Contingency	\$ -	\$ -	\$ 279	\$ 279	\$ 628					
84	Other (Loadings, Chargebacks)	\$ 1,727	\$ 189	\$ 614	\$ 803	\$ 213					
85	RCRA	\$ 5,945	\$ -	\$ -	\$ -	\$ -					
86	Contractor	\$ 5,168	\$ -	\$ -	\$ -	\$ -					
87	Labor	\$ 269	\$ -	\$ -	\$ -	\$ -					
88	Materials	\$ (174)	\$ -	\$ -	\$ -	\$ -					
89	Business Expenses	\$ 2	\$ -	\$ -	\$ -	\$ -					
90	Contingency	\$ -	\$ -	\$ -	\$ -	\$ -					
91	Other (Loadings, Chargebacks)	\$ 681	\$ -	\$ -	\$ -	\$ -					
92	316b	\$ 8	\$ -	\$ -	\$ -	\$ 500					
93	Contractor	\$ 7	\$ -	\$ -	\$ -	\$ 90					
94	Labor	\$ -	\$ -	\$ -	\$ -	\$ -					
95	Materials	\$ -	\$ -	\$ -	\$ -	\$ -					
96	Business Expenses	\$ -	\$ -	\$ -	\$ -	\$ -					
97	Contingency	\$ -	\$ -	\$ -	\$ -	\$ 50					
98	Other (Loadings, Chargebacks)	\$ 1	\$ -	\$ -	\$ -	\$ 360					
99	SEEG	\$ 259	\$ 1	\$ 100	\$ 101	\$ 7,100					
100	Contractor	\$ 206	\$ (1)	\$ 97	\$ 96	\$ 6,234					
101	Labor	\$ 9	\$ 0	\$ -	\$ 0	\$ -					
102	Materials	\$ 7	\$ 2	\$ -	\$ 2	\$ -					
103	Business Expenses	\$ -	\$ -	\$ -	\$ -	\$ -					
104	Contingency	\$ -	\$ -	\$ 3	\$ 3	\$ 710					
105	Other (Loadings, Chargebacks)	\$ 38	\$ 0	\$ -	\$ 0	\$ 156					
106	All Other Environmental	\$ 19,107	\$ 144	\$ 2,197	\$ 2,341	\$ 4,725					
107	Contractor	\$ 16,246	\$ 32	\$ 1,870	\$ 1,902	\$ 4,194					
108	Labor	\$ 354	\$ 72	\$ -	\$ 72	\$ -					
109	Materials	\$ 74	\$ 12	\$ -	\$ 12	\$ -					
110	Business Expenses	\$ 27	\$ 1	\$ -	\$ 1	\$ -					
111	Contingency	\$ -	\$ -	\$ 81	\$ 81	\$ 323					
112	Other (Loadings, Chargebacks)	\$ 2,405	\$ 27	\$ 246	\$ 273	\$ 209					
113	Total Capital	\$ 175,704	\$ 175,704	\$ 174,044	\$ 174,044	\$ 119,646	\$ 119,646	\$ 293,689	\$ 293,689	\$ 161,075	\$ 161,075

MICHIGAN PUBLIC SERVICE COMMISSION**Schedule: B-5.2**

Case No.: U-20697

Exhibit No.: A-12 (SAH-3)

Schedule: B-5.2

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Witness: SAHugo

Date: February 2020

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

(\$000's)

Generation Capital Expenditures

	(a)	(b)	(c)	(d)	(e)	(f)
	Historical		Projected Bridge Year			Projected Test Year
Line	12 Months Ended	12 Months Ended	12 Months Ending	24 Months Ending		12 Months Ending
No. Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020		12/31/2021
1 Contractor	\$ 148,819	\$ 149,718	\$ 97,456	\$ 247,174	\$	139,618
2 Labor	\$ 7,146	\$ 10,356	\$ 3,116	\$ 13,472	\$	2,443
3 Materials	\$ 20,286	\$ 23,788	\$ 6,798	\$ 30,586	\$	17,457
4 Business Expenses	\$ 2,211	\$ 638	\$ -	\$ 638	\$	-
5 Contingency	\$ -	\$ -	\$ 7,467	\$ 7,467	\$	10,461
6 Other (Loadings, Chargebacks)	\$ (2,757)	\$ (10,457)	\$ 4,809	\$ (5,648)	\$	(8,905)
Total	\$ 175,704	\$ 174,044	\$ 119,646	\$ 293,689	\$	161,075

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Case No.: U-20697
 Exhibit No.: A-12 (SAH-3)
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Consumers Energy Company
 Summary of Projected Electric Capital Expenditures
 For the year 2021
 (\$000's)

Generation Capital Expenditures

(a)		(b)	(c)	(d)	(e)	(f)
Line No.	Description	Projected Bridge Year 12 Months Ending 12/31/2020		Projected Test Year 12 Months Ending 12/31/2021		Reference
1	Campbell 1&2 Non-Environmental	\$	15,161	\$	12,869	A-12 (SAH-3) Page 2 line 1 Columns (f) and (j)
2	Campbell 1&2 "All Other Environmental"		\$ 2,390		\$ 11,333	
3	Campbell 3 Non-Environmental	\$	8,835	\$	12,445	A-12 (SAH-3) Page 2 line 2 Columns (f) and (j)
4	Campbell 3 "All Other Environmental"		\$ 1,119		\$ 7,025	
5	Karn 1&2 Non-Environmental	\$	2,712	\$	12,215	A-12 (SAH-3) Page 2 line 3 Columns (f) and (j)
6	Karn 1&2 "All Other Environmental"		\$ 3,140		\$ 700	
7	Total Other Environmental		<u>\$ 6,648</u>		<u>\$ 19,058</u>	A-12 (SAH-3) Page 2 lines 12-16 Columns (f) and (j)

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

Generation Capital Projects greater than \$1M

(\$000's)

Case No.: U-20697

Exhibit No.: A-12 (SAH-3)

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Date: February 2020

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Tier 3 Portfolio	(e) Work Item Description	(f) Projected ⁽¹⁾ Contingency	(g) Projected ⁽¹⁾ Amount	(h) Actual Amount
1	2018	Coal Generation	Campbell 1	Maintenance	Condenser Retube	\$ 161	\$ 1,982	\$ 1,769
2	2018	Coal Generation	Campbell 1&2 Site Commons	Environmental Maintenance	Fly Ash System	\$ 1,035	\$ 15,336	\$ 14,610
3	2018	Coal Generation	Campbell 1&2 Site Commons	Infrastructure	Pigeon Lake Channel South Jetty			\$ 1,182
4	2018	Coal Generation	Campbell 3	Maintenance	6A High Pressure Feed Water Heater	\$ 207	\$ 1,215	\$ 2,372
5	2018	Coal Generation	Campbell 3	Maintenance	No 4 Low Pressure Feed Water Heater	\$ 524	\$ 1,837	\$ 2,162
6	2018	Coal Generation	Campbell 3	Maintenance	7A High Pressure Feedwater Heater Replacement			\$ 1,075
7	2018	Coal Generation	Campbell 3	Environmental Maintenance	Spray Dry Absorber (PM2.5)			\$ 2,029
8	2018	Coal Generation	Campbell 3	Environmental Maintenance	Catalyst Layer 1 Replacement	\$ 314	\$ 2,367	\$ 1,690
9	2018	Coal Generation	Campbell 3	Safety	Install Foam Dust Suppression Sys	\$ 192	\$ 1,020	\$ 1,293
10	2018	Coal Generation	Campbell Site Commons	Environmental Maintenance	Bottom Ash Tanks	\$ 221	\$ 2,210	\$ 2,494
11	2018	Coal Generation	Campbell Site Commons	Environmental Maintenance	Design & Construct Ash Cell #5	\$ 57	\$ 2,502	\$ 4,544
12	2018	Coal Generation	Campbell Site Commons	Infrastructure	Dust Conveying Line to JHC3 Bunkers			\$ 1,213
13	2018	Coal Generation	Karn 1	Maintenance	Distributed Control System - Evergreen	\$ 114	\$ 1,142	\$ 1,144
14	2018	Coal Generation	Karn 1	Environmental Maintenance	Fabric Filter Bag Replacement	\$ 520	\$ 1,444	\$ 1,106
15	2018	Coal Generation	Karn 2	Environmental Maintenance	Selective Catalytic Reduction 2nd Layer Catalyst Replacement	\$ 82	\$ 1,540	\$ 1,770
16	2018	Coal Generation	Karn 2	Maintenance	C Mill Overhaul	\$ 105	\$ 1,049	\$ 1,048
17	2018	Coal Generation	Karn 1&2 Site Commons	Environmental Maintenance	Double Lined Pond	\$ -	\$ 5,098	\$ 3,049
18	2018	Coal Generation	Weadock	Environmental Maintenance	Slurry Wall	\$ -	\$ 2,376	\$ 1,188
19	2018	Gas/Oil Generation	Karn 3	Reliability	Distributed Control System Upgrade	\$ 23	\$ 3,500	\$ 6,132
20	2018	Gas/Oil Generation	Karn 4	Maintenance	Replace Cooling Tower Internal Structure	\$ 229	\$ 2,290	\$ 3,769
21	2018	Gas/Oil Generation	Karn 4	Maintenance	Breeching - Boiler to Stack			\$ 1,074
22	2018	Gas Generation	Zeeland Site Commons	Maintenance	Replace Auxiliary Boiler	\$ 5	\$ 1,993	\$ 3,009
23	2018	Gas Generation	Zeeland Site Commons	Maintenance	Milestone Outage Capital to GE	\$ 674	\$ 6,740	\$ 6,936
24	2018	Gas Generation	Zeeland Site Commons	Maintenance	Diaphragm Exchange Program			\$ 1,710
25	2018	Gas Generation	Zeeland Site Commons	Maintenance	Steam Turbine L-0 Bucket Replace			\$ 2,880
26	2018	Gas Generation	Zeeland Site Commons	Maintenance	Base Outage Capital			\$ 2,421
27	2018	Gas Generation	Zeeland Site Commons	Reliability	GE Long Term Service Agreement Payments -Combined Cycle	\$ 787	\$ 7,869	\$ 3,714
28	2018	Gas Generation	Jackson Site Commons	Reliability	Long Term Service Agreement spend with GE	\$ 670	\$ 6,700	\$ 1,937
29	2018	Gas Generation	Jackson Site Commons	Maintenance	7EA Replace Compressor Stator Vanes 1-8			\$ 1,000
30	2018	Gas Generation	Jackson Site Commons	Maintenance	7EA Upgrade Hot Gas Path	\$ 153	\$ 1,530	\$ 2,697
31	2018	Gas Generation	Jackson Site Commons	Maintenance	Cooling Tower Fill Overhaul			\$ 1,041
32	2018	Gas Generation	Jackson Site Commons	Reliability	GE Long Term Service Agreement Historical Extra Work	\$ 150	\$ 1,500	\$ 1,530
33	2018	Hydro Generation	Alcona	Regulatory	Emergency Spillway	\$ 160	\$ 1,795	\$ 1,885
34	2018	Hydro Generation	Cooke	Maintenance	Replace Wicket Gates & Bushings			\$ 1,523
35	2018	Hydro Generation	Hardy	Regulatory	Spill Tube Remediation Project			\$ 1,595
36	2018	Hydro Generation	Webber	Safety	Electrical Upgrades	\$ 84	\$ 1,400	\$ 1,631
37	2018	Hydro Generation	Webber	Reliability	2 Unit Overhaul	\$ 140	\$ 1,148	\$ 1,448
38	2018	Hydro Generation	Ludington Site Commons	Maintenance	Main Transformer Bank Project	\$ 114	\$ 5,849	\$ 5,818
39	2018	Hydro Generation	Ludington Site Commons	Infrastructure	Major Overhauls-All Six Units	\$ -	\$ 34,198	\$ 25,454
40	2018	Common Support	ERBS	Project Management	Muskegon waste water land - solar development			\$ 4,996
41	2018	Common Support	ERBS	Project Management	EPMO Transformation Proj Capital	\$ 646	\$ 6,463	\$ 1,778
42	Total 2018 Projects					\$ 7,367	\$ 124,093	\$ 131,717

Note:

(1) Projected amounts were taken from Part III filing requirement 115 for Case No. U-20134

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Case No.: U-20697
Exhibit No.: A-12 (SAH-3)
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Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

Generation Capital Projects greater than \$1M

(\$000's)

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Tier 3 Portfolio	(e) Work Item Description	(f) Projected ⁽¹⁾ Contingency	(g) Projected ⁽¹⁾ Amount	(h) Actual Amount
1	2019	Coal Generation	Campbell 1	Reliability	Distributed Control System and Simulator Upgrade	\$ 103	\$ 1,031	\$ 1,222
2	2019	Coal Generation	Campbell 3	Environmental Maintenance	Selective Catalytic Reduction Catalyst Management	\$ 552	\$ 5,524	\$ 2,151
3	2019	Coal Generation	Campbell 3	Maintenance	7A High Pressure Heater Replacement			\$ 3,007
4	2019	Coal Generation	Campbell 3	Maintenance	8 Bank 6 9kV Non-Seg Bus Insulators	\$ 115	\$ 1,150	\$ 2,103
5	2019	Coal Generation	Campbell 3	Maintenance	Control Room HVAC			\$ 2,037
6	2019	Coal Generation	Campbell 3	Maintenance	Install Cross-Tie Between 8-1 and 8-2 AQCS Start-Up Transformers			\$ 2,314
7	2019	Coal Generation	Campbell 3	Maintenance	Replace primary air economizer outlet and SCR bypass gas expansion joints	\$ 144	\$ 1,440	\$ 1,472
8	2019	Coal Generation	Campbell 3	Maintenance	Replace Burner Primary Air Tubes	\$ 124	\$ 1,237	\$ 3,389
9	2019	Coal Generation	Campbell 3	Maintenance	Replace Closed Loop Cooling Heat Exchangers	\$ 105	\$ 1,050	\$ 2,057
10	2019	Coal Generation	Campbell 3	Maintenance	Replace Forced Draft Fan Lube-Hydraulic Oil Skids			\$ 1,177
11	2019	Coal Generation	Campbell 3	Reliability	Distributed Control System and Simulator Upgrade	\$ 193	\$ 1,929	\$ 1,830
12	2019	Coal Generation	Karn 1	Environmental Maintenance	Selective Catalytic Reduction 2nd Layer Catalyst Replacement - Environmental			\$ 1,773
13	2019	Coal Generation	Karn 2	Regulatory	Distributed Control System Upgrade (Evergreen)			\$ 1,278
14	2019	Gas/Oil Generation	Karn 3	Reliability	Distributed Control System and Simulator Upgrade	\$ 400	\$ 4,000	\$ 2,352
15	2019	Gas/Oil Generation	Karn 4	Maintenance	Cooling Tower Internal Structure Replacement	\$ 265	\$ 2,650	\$ 3,675
16	2019	Gas/Oil Generation	Karn 4	Maintenance	Electro hydraulic Control System Retrofit	\$ 290	\$ 2,900	\$ 3,345
17	2019	Gas Generation	Jackson Site Commons	Maintenance	Reverse Osmosis Water Makeup Expansion			\$ 1,352
18	2019	Gas Generation	Jackson Site Commons	Regulatory	Turbine Distributed Control System Replacement	\$ 300	\$ 3,000	\$ 3,335
19	2019	Gas Generation	Jackson Site Commons	Reliability	GE Long Term Service Agreement Historical Extra Work Expected	\$ 100	\$ 1,000	\$ 5,077
20	2019	Gas Generation	Jackson Site Commons	Reliability	Long Term Service Agreement	\$ 780	\$ 7,800	\$ 12,388
21	2019	Gas Generation	Zeeland Site Commons	Reliability	Long Term Service Agreement / GE	\$ 803	\$ 8,026	\$ 5,901
22	2019	Hydro Generation	Alcona	Regulatory	Emergency Spillway	\$ 138	\$ 1,376	\$ 2,507
23	2019	Hydro Generation	Cooke	Maintenance	2 Wicket Gates			\$ 1,385
24	2019	Hydro Generation	Croton	Regulatory	Right Abutment Remediation			\$ 4,277
25	2019	Hydro Generation	Croton	Reliability	Rebuild Unit 4			\$ 1,557
26	2019	Hydro Generation	Hardy	Regulatory	Auxiliary Spillway			\$ 1,822
27	2019	Hydro Generation	Hardy	Regulatory	Spill Tube Remediation	\$ 198	\$ 1,978	\$ 1,543
28	2019	Hydro Generation	Rogers	Maintenance	Governor Replacement			\$ 1,594
29	2019	Hydro Generation	Rogers	Safety	Electrical Safety Project	\$ 170	\$ 1,699	\$ 3,238
30	2019	Hydro Generation	Ludington Site Commons	Maintenance	Ludington Generator Stepup Transformer Replacement	\$ 558	\$ 5,584	\$ 5,125
31	2019	Hydro Generation	Ludington Site Commons	Infrastructure	Ludington Overhaul	\$ -	\$ 32,658	\$ 26,248
32	2019	Common Support	ERBS	Project Management	EPMO Transformation - Enterprise Project Management Information System	\$ 160	\$ 1,600	\$ 1,404
33	Total 2019 Projects					\$ 5,497	\$ 87,633	\$ 113,935

Note:

(1) Projected amounts were taken from Part III filing requirement 115 for Case No. U-20134

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

Generation Capital Projects greater than \$1M

(\$000's)

Case No.: U-20697

Exhibit No.: A-12 (SAH-3)

Schedule: B-5.2

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Witness: SAHugo

Date: February 2020

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Tier 3 Portfolio	(e) Work Item Description	(f) Contingency Amount	(g) Total Amount
1	2020	Coal Generation	Campbell 1	General Maintenance	High Pressure Turbine Blading Replacement	\$ -	\$ 1,901
2	2020	Coal Generation	Campbell 1	General Maintenance	Low Pressure Turbine blade replacement, row L-0.	\$ 214	\$ 4,274
3	2020	Coal Generation	Campbell 2	Reliability	Distributed Control System and Simulator Upgrade	\$ -	\$ 1,333
4	2020	Coal Generation	Campbell Site Commons	Regulatory	Bottom Ash Tanks Chemical Treatment System	\$ 60	\$ 1,193
5	2020	Coal Generation	Karn 1 & 2 Site Commons	Environmental Maintenance	Landfill Remedial Action Plan	\$ 60	\$ 1,200
6	2020	Coal Generation	Karn 2	Environmental Maintenance	Selective Catalytic Reduction - Catalyst Replacement	\$ 50	\$ 1,000
7	2020	Gas/Oil Generation	Karn 4	General Maintenance	Replace Cooling Tower Internal Structure	\$ 200	\$ 2,000
8	2020	Gas/Oil Generation	Karn 4	General Maintenance	Electro-Hydraulic Control System Replacement	\$ 115	\$ 2,300
9	2020	Gas Generation	Zeeland Site Commons	Reliability	Zeeland Long Term Service Agreement - Running Capital Contract	\$ -	\$ 7,557
10	2020	Gas Generation	Jackson Site Commons	Regulatory	Increase Stack Height	\$ 75	\$ 1,500
11	2020	Gas Generation	Jackson Site Commons	Reliability	Jackson GE Long Term Service Agreement FFH	\$ -	\$ 10,600
12	2020	Gas Generation	Jackson Site Commons	Reliability	Jackson GE Long Term Service Agreement Historical Extra Work Expected	\$ -	\$ 1,200
13	2020	Gas Generation	Jackson Site Commons	Reliability	Jackson Turbine Control System Replacement	\$ 163	\$ 3,265
14	2020	Gas Generation	Jackson Site Commons	Infrastructure	Jackson Warehouse for Jackson Generating Station	\$ 220	\$ 4,400
15	2020	Hydro Generation	Alcona	Regulatory	Emergency Spillway	\$ 95	\$ 1,890
16	2020	Hydro Generation	Five Channels	Safety	Five Channels Electrical Safety Project	\$ 82	\$ 1,631
17	2020	Hydro Generation	Hardy	Regulatory	Auxiliary Spillway Remediation	\$ 100	\$ 1,000
18	2020	Hydro Generation	Webber	General Maintenance	Overhaul Webber Unit 1	\$ 85	\$ 1,690
19	2020	Hydro Generation	Ludington Site Commons	Infrastructure	Ludington 16-424 HVAC Replacement	\$ 219	\$ 4,371
20	2020	Hydro Generation	Ludington Site Commons	Infrastructure	Ludington Upgrade and Overhaul	\$ 3,177	\$ 12,707
21	2020	Common Support	ERBS	Project Management	EPMO Transformation - Enterprise Project Management Information System	\$ -	\$ 1,910
34	Total 2020 Projects					\$ 4,912	\$ 68,922

MICHIGAN PUBLIC SERVICE COMMISSION

Schedule: B-5.2

Case No.: U-20697

Exhibit No.: A-12 (SAH-3)

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Witness: SAHugo

Date: February 2020

Consumers Energy Company

Summary of Actual and Projected Electric Capital Expenditures

For the years 2018 through 2021

Generation Capital Projects greater than \$1M

(\$000's)

Generation Capital Expenditures

Line No.	(a) Calendar Year	(b) Tier 1 Portfolio	(c) Tier 2 Portfolio	(d) Tier 3 Portfolio	(e) Work Item Description	(f) Contingency Amount	(g) Total Amount
1	2021	Coal Generation	Campbell 1	General Maintenance	Re-align 4160V switchgear with AQCS implementation	\$ 100	\$ 1,000
2	2021	Coal Generation	Campbell 1&2 Site Commons	Environmental Maintenance	Steam Electric Effluent Guidelines - Compliance - Closed Loop W/ Recirc.	\$ 212	\$ 2,118
3	2021	Coal Generation	Campbell 2	Environmental Maintenance	Replace Secondary Air Heater baskets and seals	\$ 243	\$ 2,425
4	2021	Coal Generation	Campbell 2	Environmental Maintenance	Catalyst Management	\$ 150	\$ 1,500
5	2021	Coal Generation	Campbell 2	Environmental Maintenance	Pulse Jet Fabric Filter bag replacement	\$ 269	\$ 2,694
6	2021	Coal Generation	Campbell 2	Reliability	Low Pressure Turbine Component Replacement	\$ 330	\$ 3,300
7	2021	Coal Generation	Campbell 3	Environmental Maintenance	Steam Electric Effluent Guidelines - Waste Water Treatment	\$ 121	\$ 1,206
8	2021	Coal Generation	Campbell 3	Environmental Maintenance	Steam Electric Effluent Guidelines - Compliance - Closed Loop W/ Recirc.	\$ 289	\$ 2,893
9	2021	Coal Generation	Campbell 3	Environmental Maintenance	Selective Catalytic Reduction Catalyst Management	\$ 196	\$ 1,960
10	2021	Coal Generation	Campbell 3	General Maintenance	Replace CO-O2 monitors	\$ 104	\$ 1,045
11	2021	Coal Generation	Campbell 3	General Maintenance	Reheater Sootblower	\$ 125	\$ 1,250
12	2021	Coal Generation	Campbell 3	General Maintenance	Redundant Sootblowing Air Compressor	\$ 120	\$ 1,200
13	2021	Coal Generation	Campbell 3	Infrastructure	Replace Unit 3 Lake Michigan Intake Screens	\$ 127	\$ 1,270
14	2021	Coal Generation	Campbell 3	Reliability	Mill Complete Overhauls	\$ -	\$ 1,235
15	2021	Coal Generation	Campbell Site Commons	Other Environmental	Dry Ash Landfill Cell Construction & Permitting	\$ 274	\$ 5,483
16	2021	Coal Generation	Karn 1&2 Site Commons	Asset Retirement	Karn 3&4 decoupling and Karn 1&2 Decommissioning	\$ 515	\$ 10,296
17	2021	Gas/Oil Generation	Karn 3	Reliability	Cooling Tower Rebuild	\$ 208	\$ 2,500
18	2021	Gas/Oil Generation	Karn 3&4 Site Commons	General Maintenance	Startup Optimization	\$ 390	\$ 3,900
19	2021	Gas/Oil Generation	Karn 3&4 Site Commons	Infrastructure	Parking Lot Replacement	\$ 100	\$ 1,000
20	2021	Gas Generation	Zeeland Site Commons	Infrastructure	Build New Storage Building - Major Overhaul Work	\$ 225	\$ 4,500
21	2021	Gas Generation	Zeeland Site Commons	Reliability	Long Term Service Agreement - Running Capital Contract	\$ -	\$ 8,900
22	2021	Gas Generation	Jackson Site Commons	Reliability	Jackson GE Long Term Service Agreement FFH	\$ -	\$ 10,600
23	2021	Gas Generation	Jackson Site Commons	Reliability	Jackson GE LTSA Historical Extra Work Expected	\$ -	\$ 1,850
24	2021	Gas Generation	Jackson Site Commons	Reliability	Boiler Feed Pump Automatic Recirculation Valve Replacement	\$ 116	\$ 1,160
25	2021	Hydro Generation	Croton 3&4	General Maintenance	Croton 3 and 4 Governor Replacement	\$ 112	\$ 1,121
26	2021	Hydro Generation	Five Channels	Regulatory	Corewall Remediation	\$ 205	\$ 2,050
27	2021	Hydro Generation	Hardy	Regulatory	Auxiliary Spillway Remediation	\$ 800	\$ 8,000
28	2021	Hydro Generation	Hodenpyl	General Maintenance	Hodenpyl 1 Generator Rewind	\$ 158	\$ 1,579
29	2021	Hydro Generation	Hodenpyl	Safety	Spillway Hoist Replacement	\$ -	\$ 1,625
30	2021	Hydro Generation	Loud	Regulatory	Loud Training Wall Replacement Project	\$ -	\$ 2,200
31	2021	Hydro Generation	Mio	Safety	Mio Electrical Safety Project	\$ 230	\$ 2,300
32	2021	Hydro Generation	Webber	Regulatory	Downstream Training Wall Replacement	\$ 230	\$ 2,300
33	2021	Pumped Storage	Ludington Site Commons	Infrastructure	Ludington Upgrade and Overhaul	\$ -	\$ 2,040
34	2021	Pumped Storage	Ludington Site Commons	Infrastructure	Powerhouse Roof Wearing Surface and Weather Proofing Replacement	\$ 169	\$ 3,380
35	2021	Hydro Generation	Ludington Site Commons	Regulatory	Design & Install Net Barrier Net (AMP)	\$ 173	\$ 1,917
36	2021	Pumped Storage	Ludington Site Commons	Regulatory	Reservoir Liner Replacement	\$ 992	\$ 6,610
37	2021	Pumped Storage	Ludington Site Commons	Reliability	Replace Lower Penstock Expansion Joint Chamber Waterstop	\$ 248	\$ 1,656
38	2021	Common Support	ERBS	Project Management	EPMO Transformation - Enterprise Project Management Information System	\$ -	\$ 2,899
39	Total 2021 Projects					\$ 7,530	\$ 114,961

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Projected Electric Capital Expenditures
For the projected test year 2021

Case No.: U-20697

Exhibit No.: A-69 (SAH-4)

Page: 1 of 2

Witness: SAHugo

Date: February 2020

Generation Capital Expenditures
AVOIDABLE & INCREMENTAL UNDER CAMPBELL 1 & 2 EARLY RETIREMENT MAY 31, 2024
(\$000's)

Line No.	(a) Description	(b) Non-Environmental	(c) Environmental
Campbell 1 retirement scenario			
1	JHCampbell 1		
2	Unavoidable	\$ 5,707	\$ 1,789
3	Avoidable	\$ 200	\$ -
4	Incremental	\$ -	\$ -
Campbell 2 retirement scenario			
5	JHCampbell 2		
6	Unavoidable	\$ 4,876	\$ 10,098
7	Avoidable	\$ 982	\$ 550
8	Incremental	\$ -	\$ -
Campbell 1 and 2 retirement scenario			
9	JHCampbell 1		
10	Unavoidable	\$ 5,707	\$ 1,789
11	Avoidable	\$ 200	\$ -
12	Incremental	\$ -	\$ -
13	JHCampbell 2		
14	Unavoidable	\$ 4,876	\$ 10,098
15	Avoidable	\$ 982	\$ 550
16	Incremental	\$ -	\$ -
17	JHCampbell 3		
18	Unavoidable	\$ 12,445	\$ 7,025
19	Avoidable	\$ -	\$ -
20	Incremental	\$ 4,000	\$ -

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of Projected Electric Capital Expenditures
For the projected test year 2021

Case No.: U-20697
Exhibit No.: A-69 (SAH-4)
Page: 2 of 2
Witness: SAHugo
Date: February 2020

Generation Capital Expenditures
AVOIDABLE & INCREMENTAL UNDER CAMPBELL 1 & 2 EARLY RETIREMENT MAY 31, 2025
(\$000's)

Line No.	(a) Description	(b) Non-Environmental	(c) Environmental
Campbell 1 retirement scenario			
1	JHCampbell 1		
2	Unavoidable	\$ 5,707	\$ 1,789
3	Avoidable	\$ 200	\$ -
4	Incremental	\$ -	\$ -
Campbell 2 retirement scenario			
5	JHCampbell 2		
6	Unavoidable	\$ 5,858	\$ 10,098
7	Avoidable	\$ -	\$ 550
8	Incremental	\$ -	\$ -
Campbell 1 and 2 retirement scenario			
9	JHCampbell 1		
10	Unavoidable	\$ 5,707	\$ 1,789
11	Avoidable	\$ 200	\$ -
12	Incremental	\$ -	\$ -
13	JHCampbell 2		
14	Unavoidable	\$ 5,858	\$ 10,098
15	Avoidable	\$ -	\$ 550
16	Incremental	\$ -	\$ -
17	JHCampbell 3		
18	Unavoidable	\$ 12,445	\$ 7,025
19	Avoidable	\$ -	\$ -
20	Incremental	\$ 300	\$ -

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of the Generation O&M Expense
For the Years 2018 through 2021
(\$000's)

Case No.: U-20697
Exhibit No.: A-70 (SAH-5)
Page: 1 of 3
Witness: SAHugo
Date: February 2020

GENERATION OPERATION AND MAINTENANCE EXPENSES

Line No.	(a) Description	(b) Historical 12 Months Ended 12/31/2018		(c) Projected Bridge Year 12 Months Ended 12/31/2019		(d) 12 Months Ending 12/31/2020		(e) Projected Test Year 12 Months Ending 12/31/2021	
1	BASE O&M	\$	108,466	\$	111,683	\$	106,186	\$	113,621
2	ADJUSTED O&M								
3	Environmental Operations	\$	11,940	\$	11,850	\$	10,710	\$	11,002
4	Major Maintenance	\$	27,825	\$	18,817	\$	23,899	\$	34,756
5	Karn Retention & Separation	\$	-	\$	5,925	\$	12,967	\$	7,413
6	TOTAL O&M	\$	148,231	\$	148,274	\$	153,762	\$	166,793

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of the Generation O&M Expense

For the Years 2018 through 2021

(\$000's)

Case No.: U-20697

Exhibit No.: A-70 (SAH-5)

Page: 2 of 3

Witness: SAHugo

Date: February 2020

GENERATION OPERATION AND MAINTENANCE EXPENSES

Line No.	Description	(a)	(b)	(c)	(d)	(e)
		Historical 12 Months Ended 12/31/2018	Projected Bridge Year 12 Months Ended 12/31/2019	Projected Bridge Year 12 Months Ending 12/31/2020	Projected Test Year 12 Months Ending 12/31/2021	
1	Labor	\$ 85,600	\$ 98,518	\$ 104,733	\$ 111,313	
2	Material	\$ 10,038	\$ 9,160	\$ 8,985	\$ 10,162	
3	Contractor	\$ 24,921	\$ 16,013	\$ 15,725	\$ 17,787	
4	Non-Labor Other	\$ 27,672	\$ 24,583	\$ 24,320	\$ 27,531	
5	Total	<u>\$ 148,231</u>	<u>\$ 148,274</u>	<u>\$ 153,762</u>	<u>\$ 166,793</u>	

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of the Generation Major Maintenance O&M Expense

For the Years 2018 through 2021

(\$000's)

Case No.: U-20697

Exhibit No.: A-70 (SAH-5)

Page: 3 of 3

Witness: SAHugo

Date: February 2020

GENERATION MAJOR MAINTENANCE EXPENSES

Line No.	(a) Description	(b) (c) (d) (e)			
		Historical 12 Months Ended 12/31/2018	Projected Bridge Year 12 Months Ended 12/31/2019	12 Months Ending 12/31/2020	Projected Test Year 12 Months Ending 12/31/2021
1	Major Maintenance				
2	Campbell 1&2	\$ 2,261	\$ 1,157	\$ 8,262	\$ 10,647
3	Campbell 3	\$ 2,400	\$ 2,783	\$ 1,515	\$ 4,636
4	Karn 1&2	\$ 3,012	\$ 1,927	\$ 1,814	\$ 3,784
5	Karn 3&4	\$ 777	\$ 851	\$ 378	\$ 968
6	Classic 7	\$ 191	\$ 149	\$ 153	\$ 193
7	Zeeland Generating Station	\$ 6,762	\$ 2,872	\$ 3,351	\$ 4,110
8	Jackson Generating Station	\$ 4,061	\$ 2,492	\$ 2,510	\$ 3,045
9	Ludington	\$ 2,388	\$ 2,554	\$ 2,202	\$ 2,828
10	Hydros	\$ 5,973	\$ 4,032	\$ 3,715	\$ 4,546
11	Admin & Other	\$ -	\$ -	\$ -	\$ -
12	TOTAL Major Maintenance	\$ 27,825	\$ 18,817	\$ 23,899	\$ 34,756

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Projected Electric O&M Major Maintenance Expenses

For the projected test year 2021

(\$000's)

Case No.: U-20697

Exhibit No.: A-71 (SAH-6)

Page: 1 of 2

Witness: SAHugo

Date: February 2020

**Generation O&M Major Maintenance Expenses
AVOIDABLE UNDER CAMPBELL 1 & 2 EARLY RETIREMENT MAY 31, 2024**

Line No.	(a) Description	(b) Non-Environmental	(c) Environmental
Campbell 1 retirement scenario			
1	JHCampbell 1		
2	Unavoidable	\$ 1,819	\$ 90
3	Avoidable	\$ 129	\$ -
Campbell 2 retirement scenario			
4	JHCampbell 2		
5	Unavoidable	\$ 7,943	\$ 124
6	Avoidable	\$ 543	\$ -
Campbell 1 and 2 retirement scenario			
7	JHCampbell 1		
8	Unavoidable	\$ 1,819	\$ 90
9	Avoidable	\$ 129	\$ -
10	JHCampbell 2		
11	Unavoidable	\$ 7,943	\$ 124
12	Avoidable	\$ 543	\$ -
13	JHCampbell 3		
14	Unavoidable	\$ 3,904	\$ 732
15	Avoidable	\$ -	\$ -

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Summary of Projected Electric O&M Major Maintenance Expenses
For the projected test year 2021

Case No.: U-20697
Exhibit No.: A-71 (SAH-6)
Page: 2 of 2
Witness: SAHugo
Date: February 2020

**Generation O&M Major Maintenance Expenses
AVOIDABLE UNDER CAMPBELL 1 & 2 EARLY RETIREMENT MAY 31, 2025
(\$000's)**

Line No.	(a) Description	(b) Non-Environmental	(c) Environmental
Campbell 1 retirement scenario			
1	JHCampbell 1		
2	Unavoidable	\$ 1,819	\$ 90
3	Avoidable	\$ 129	\$ -
Campbell 2 retirement scenario			
4	JHCampbell 2		
5	Unavoidable	\$ 7,943	\$ 124
6	Avoidable	\$ 543	\$ -
Campbell 1 and 2 retirement scenario			
7	JHCampbell 1		
8	Unavoidable	\$ 1,819	\$ 90
9	Avoidable	\$ 129	\$ -
10	JHCampbell 2		
11	Unavoidable	\$ 7,943	\$ 124
12	Avoidable	\$ 543	\$ -
13	JHCampbell 3		
14	Unavoidable	\$ 3,904	\$ 732
15	Avoidable	\$ -	\$ -

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

KYLE P. JONES

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule B-5.7

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Fleet Services

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (KPJ-1)

Schedule: B-5.7

Page: 1 of 1

Witness: KPJones

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				Projected Test
		Historical	Projected Bridge Year			Year
Line		12 Mos Ended	12 Mos Ending	12 Mos Ending	24 Mos Ending	12 mos. Ending
No.	Program Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12/31/2021
1	Transportation Equipment	17,648	28,431	32,982	61,412	62,508
2	Contractor	3,263	5,257	6,098.00	11,355	11,557
3	Labor	-	-	-	-	-
4	Materials	14,168	22,824	26,478	49,302	50,182
5	Business Expenses	115	185	214	399	406
6	Other (Loadings, Chargebacks)	102	165	191	356	363
7	Other Equipment	319	243	240	483	240
8	Total Capital	17,967	28,674	33,222	61,896	62,749



2017 Replacement Report Consumers Energy

Presented by Utilimarc

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Summary

- Utilimarc's Vehicle Replacement Module (VRM) uses Consumers' historic ownership and operational data to develop a replacement strategy that is unique to the behavior and characteristics of the Consumers fleet.
- This report is based on the current state of fleet and does not take any fleet growth into account.
- This report has units in the class codes listed below removed from this analysis since Consumers does not plan to replace any units in these class codes.
 - Compact Sedan, Midsize Sedan, Fullsize Sedan, Hybrid Sedan, Electric Sedan, Van – Passenger, Van – Mini Cargo, Van – Cargo 150, Van – Cargo 350, Tankers, Heavy Trencher, and Brush Chipper.

Lifecycle Analysis

Lifecycles by Class

The table below shows the Consumers historic lifecycle compared to the Consumers recommended lifecycle.

Class	Unit Count	Consumers Historic Lifecycle		Consumers Recommended Lifecycle	
		Age	Mileage (000)	Age	Mileage (000)
Light Duty Pickup	551	10	168.5	10	168.5
Medium Duty Pickup	614	10	104.5	10	104.5
Heavy Duty Pickup	48	10	98.2	10	98.2
Compact SUV	10	6	109.0	10	181.7
Midsize SUV	19	10	196.4	10	196.4
Van - Cargo 250	137	10	110.9	10	110.9
Van - Cube/Step	6	10	24.6	10	24.6
Cube Van	5	10	117.2	10	117.2
Step Van	75	10	32.0	10	32.0
Dump Truck - Single	118	10	40.8	10	40.8
Dump Truck - Tandem	37	10	51.6	10	51.6
LD Service Truck	410	10	147.9	10	147.9
MD Service Truck	11	10	25.5	10	25.5
HD Service Truck	205	10	68.6	10	68.6
Stake Truck	87	10	67.4	10	67.4
Light Duty Aerial	105	10	266.7	6	160.0
Medium Duty Aerial	7	10	85.2	8	68.2
Heavy Duty Aerial	262	10	129.7	8	103.7
Super Heavy Duty Aerial	7	10	141.5	8	113.2
Diggers, Derricks	156	10	74.7	8	59.7
Super HD Diggers, Derricks	7	10	18.5	8	14.8

Class	Unit Count	Consumers Historic Lifecycle		Consumers Recommended Lifecycle	
		Age	Mileage (000)	Age	Mileage (000)
Semi-Tractor - Tandem	36	10	194.8	10	194.8
Track Unit - Mounted	16	12	0.0	12	0.0
Directional (Horizontal) Drill Unit	81	8	0.0	8	0.0
Frontend Loader & Backhoe	167	12	0.0	15	0.0
Light Skid-steer	29	12	0.0	15	0.0
Heavy Skid-steer	10	12	0.0	15	0.0
Dozer	11	12	0.0	20	0.0
Walk-Behind Trencher	71	12	0.0	15	0.0
Light Trencher	8	12	0.0	15	0.0
Medium Trencher	165	12	0.0	15	0.0
Light Loader	12	12	0.0	15	0.0
Mini Excavator	206	10	0.0	10	0.0
Light Excavator	31	10	0.0	10	0.0
Medium Excavator	8	10	0.0	10	0.0
Tensioner/Puller	50	12	0.0	20	0.0
Welder	5	10	0.0	20	0.0
Air Compressor	87	10	0.0	20	0.0
Utility Cart	66	8	0.0	15	0.0
Backyard Mobile Equipment	11	12	0.0	12	0.0
Mobile Aerial Platform	28	12	0.0	12	0.0
Forklift	159	15	0.0	20	0.0
Off Road Tractor with Equipment	26	15	0.0	15	0.0
Trailers	1667	20	0.0	20	0.0
Vacuum Truck	1	10	9.4	10	9.4
Total Count	5,828				

Units to Replace Annually

Utilimarc recommends replacing a consistent number of units every year. This will create an even distribution of older and younger units across your fleet, leading to predictable maintenance costs, staffing needs, and unit availability while avoiding large replacement bubbles. The table below shows the average number of units to replace each year while following the chosen lifecycles.

Class	Consumers Historic Lifecycle		Consumers Recommended Lifecycle	
	Average Units	Average Total Replacement Cost	Average Units	Average Total Replacement Cost
Light Duty Pickup	55.10	\$2,204,000	55.10	\$2,204,000
Medium Duty Pickup	61.40	\$3,070,000	61.40	\$3,070,000
Heavy Duty Pickup	4.80	\$240,000	4.80	\$240,000
Compact SUV	1.67	\$48,333	1.00	\$29,000
Midsize SUV	1.90	\$55,100	1.90	\$55,100
Van - Cargo 250	13.70	\$753,500	13.70	\$753,500
Van - Cube/Step	0.60	\$87,000	0.60	\$87,000
Cube Van	0.50	\$72,500	0.50	\$72,500
Step Van	7.50	\$1,087,500	7.50	\$1,087,500
Dump Truck - Single	11.80	\$1,032,500	11.80	\$1,032,500
Dump Truck - Tandem	3.70	\$555,000	3.70	\$555,000
LD Service Truck	41.00	\$3,280,000	41.00	\$3,280,000
MD Service Truck	1.10	\$126,500	1.10	\$126,500
HD Service Truck	20.50	\$3,772,000	20.50	\$3,772,000
Stake Truck	8.70	\$1,218,000	8.70	\$1,218,000
Light Duty Aerial	10.50	\$2,047,500	17.50	\$3,412,500
Medium Duty Aerial	0.70	\$136,500	0.88	\$170,625
Heavy Duty Aerial	26.20	\$8,777,000	32.75	\$10,971,250
Super Heavy Duty Aerial	0.70	\$280,000	0.88	\$350,000
Diggers, Derricks	15.60	\$3,884,400	19.50	\$4,855,500
Super HD Diggers, Derricks	0.70	\$212,100	0.88	\$265,125

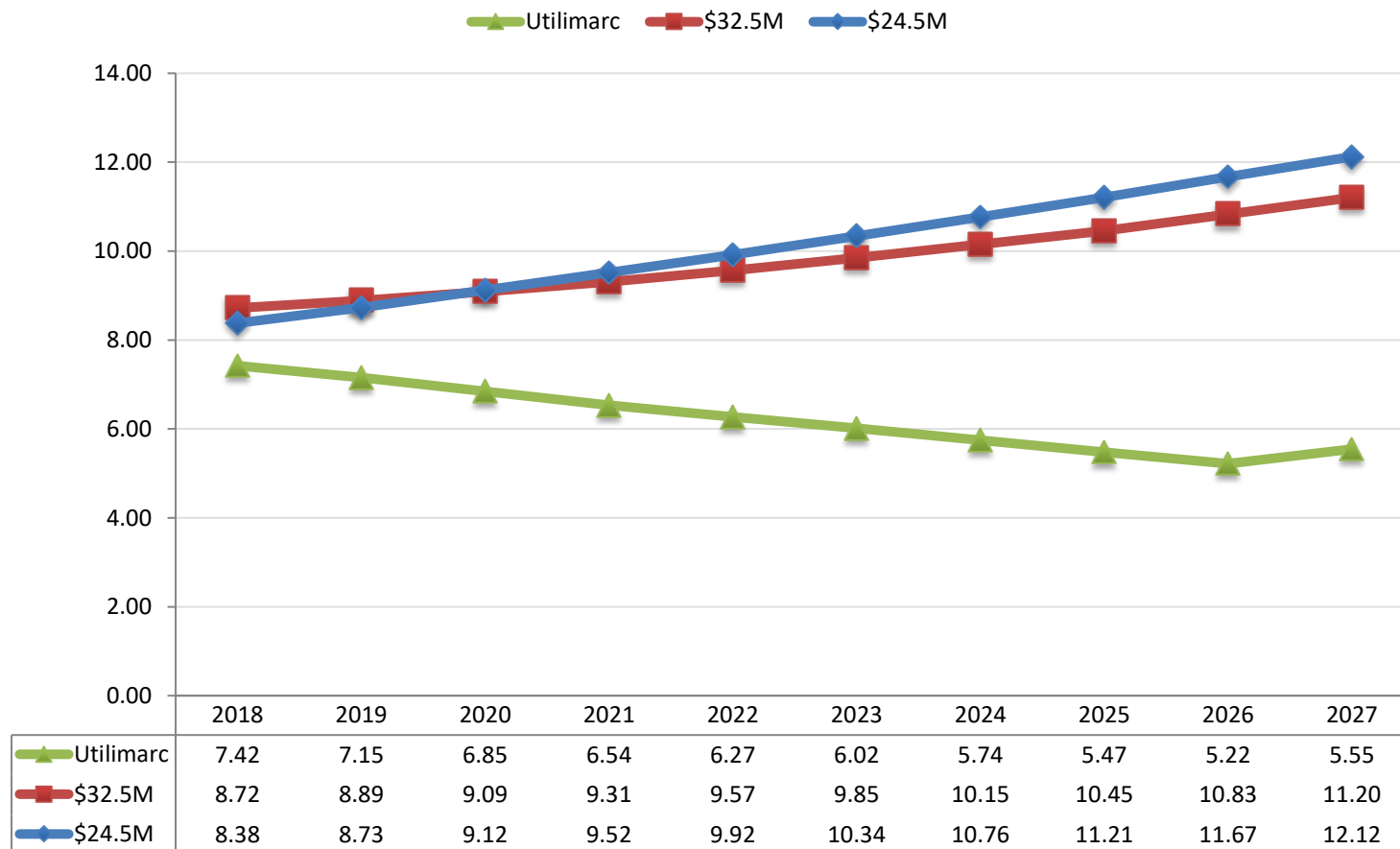
Class	Consumers Historic Lifecycle		Consumers Recommended Lifecycle	
	Average Units	Average Total Replacement Cost	Average Units	Average Total Replacement Cost
Semi-Tractor - Tandem	3.60	\$594,000	3.60	\$594,000
Track Unit - Mounted	1.33	\$466,667	1.33	\$466,667
Directional (Horizontal) Drill Unit	10.13	\$2,531,250	10.13	\$2,531,250
Frontend Loader & Backhoe	13.92	\$1,530,833	11.13	\$1,224,667
Light Skid-steer	2.42	\$263,417	1.93	\$210,733
Heavy Skid-steer	0.83	\$110,833	0.67	\$88,667
Dozer	0.92	\$487,392	0.55	\$292,435
Walk-Behind Trencher	5.92	\$473,333	4.73	\$378,667
Light Trencher	0.67	\$53,333	0.53	\$42,667
Medium Trencher	0.00	\$0	11.00	\$1,188,000
Light Loader	1.00	\$160,000	0.80	\$128,000
Mini Excavator	20.60	\$1,668,600	20.60	\$1,668,600
Light Excavator	3.10	\$368,900	3.10	\$368,900
Medium Excavator	0.80	\$174,400	0.80	\$174,400
Tensioner/Puller	4.17	\$250,000	2.50	\$150,000
Welder	0.50	\$9,000	0.25	\$4,500
Air Compressor	8.70	\$217,500	4.35	\$108,750
Utility Cart	8.25	\$1,320,000	4.40	\$704,000
Backyard Mobile Equipment	0.92	\$152,167	0.92	\$152,167
Mobile Aerial Platform	2.33	\$93,333	2.33	\$93,333
Forklift	10.60	\$848,000	7.95	\$636,000
Off Road Tractor with Equipment	1.73	\$1,040,000	1.73	\$1,040,000
Trailers	83.35	\$7,668,200	83.35	\$7,668,200
Vacuum Truck	0.10	\$3,500	0.10	\$3,500
Total	474.24	\$53,424,092	484.47	\$57,525,702

Projections

This section shows the effects of three different scenarios on a variety of fleet metrics. Each projection runs from 2018 to 2027. Each scenario uses the recommended lifecycles chosen by Consumers. The green line represents the replacement scenario based on Utilimarc's recommended budget (Utilimarc). The blue line represents the replacement scenario based on a \$24.5M budget (\$24.5M). The red line represents the replacement scenario based on a \$32.5M budget (\$32.5M). See the index for more details regarding each replacement scenario.

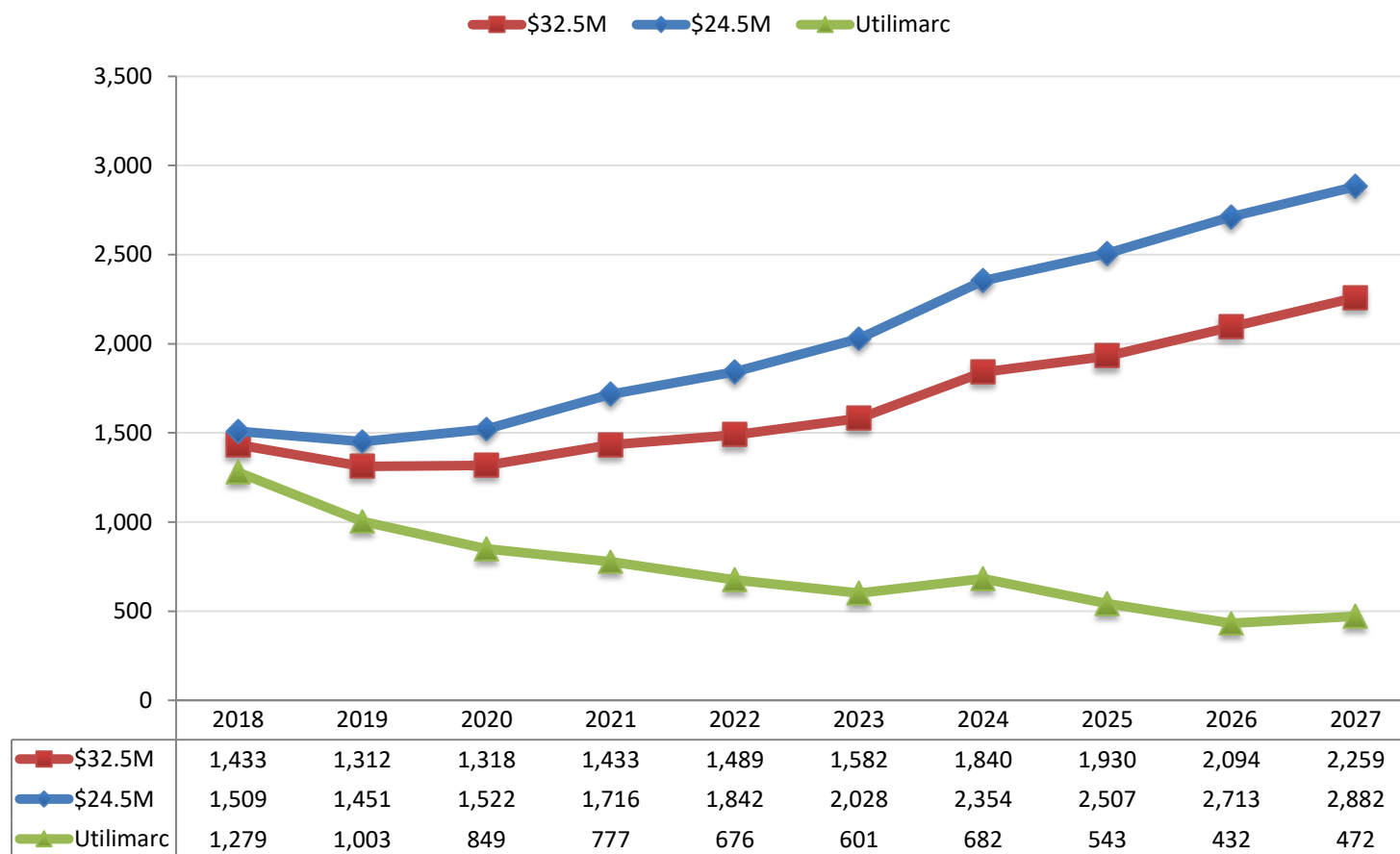
Unit Average Age

This graph shows average unit age of fleet over the next six years. Under the Utilimarc Budget scenario unit average age goes from 7.42 to 5.55 by 2027. Unit average age goes from 8.38 to 12.12 under the \$24.5M Budget scenario and from 8.72 to 11.20 under the \$32.5M Budget scenario.



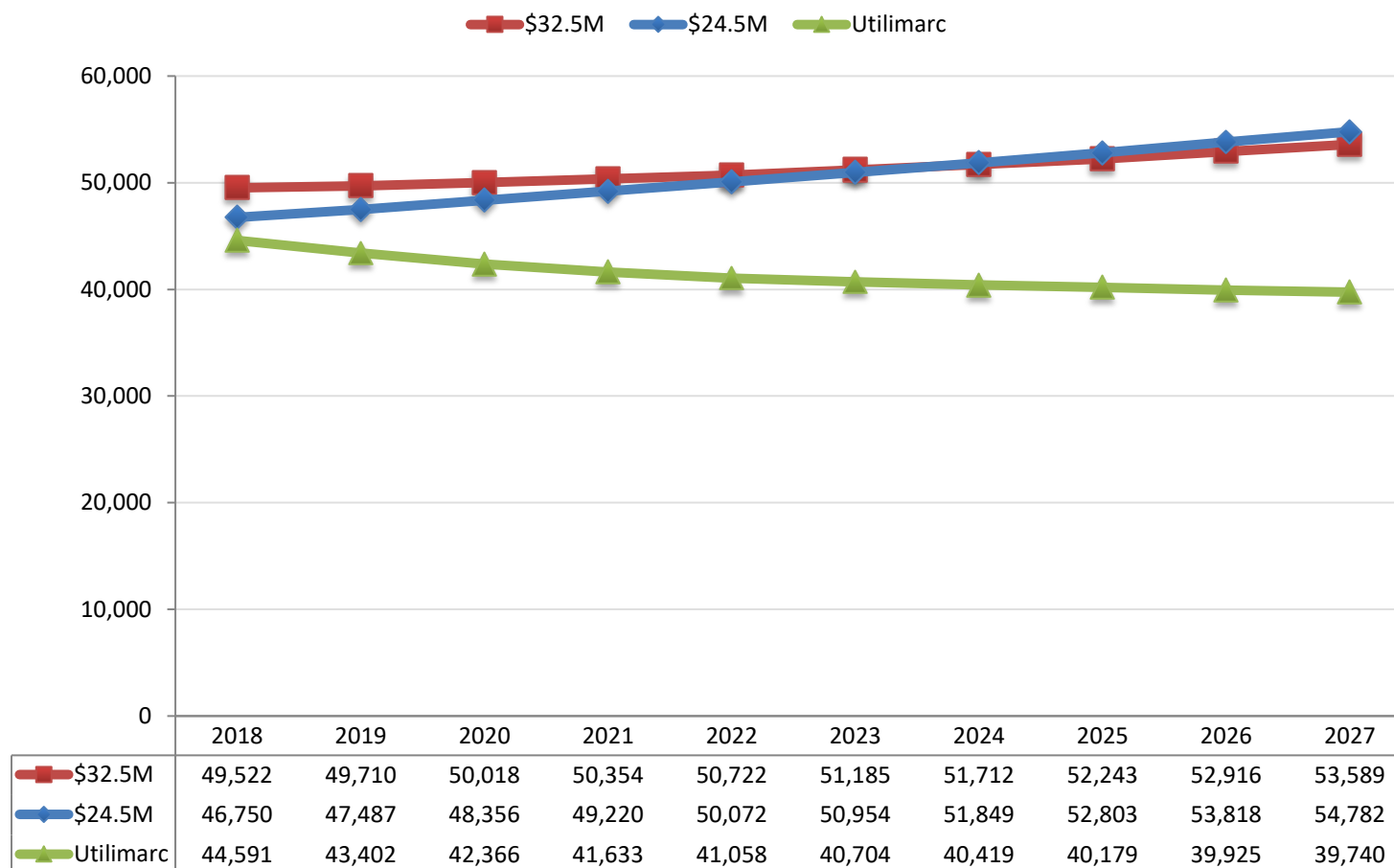
Units Out of Lifecycle

This graph shows the number of units outside the stated lifecycle for each scenario. The target value for this metric is close to zero. In the Utilimarc scenario the number of units out of lifecycle decrease from 1,279 to 472 by 2027. The number of units out of lifecycle increases from 1,509 to 2,882 in 2027 in the \$24.5M Budget scenario and increases from 1,433 to 2,259 in 2027 in the \$32.5M Budget Scenario.



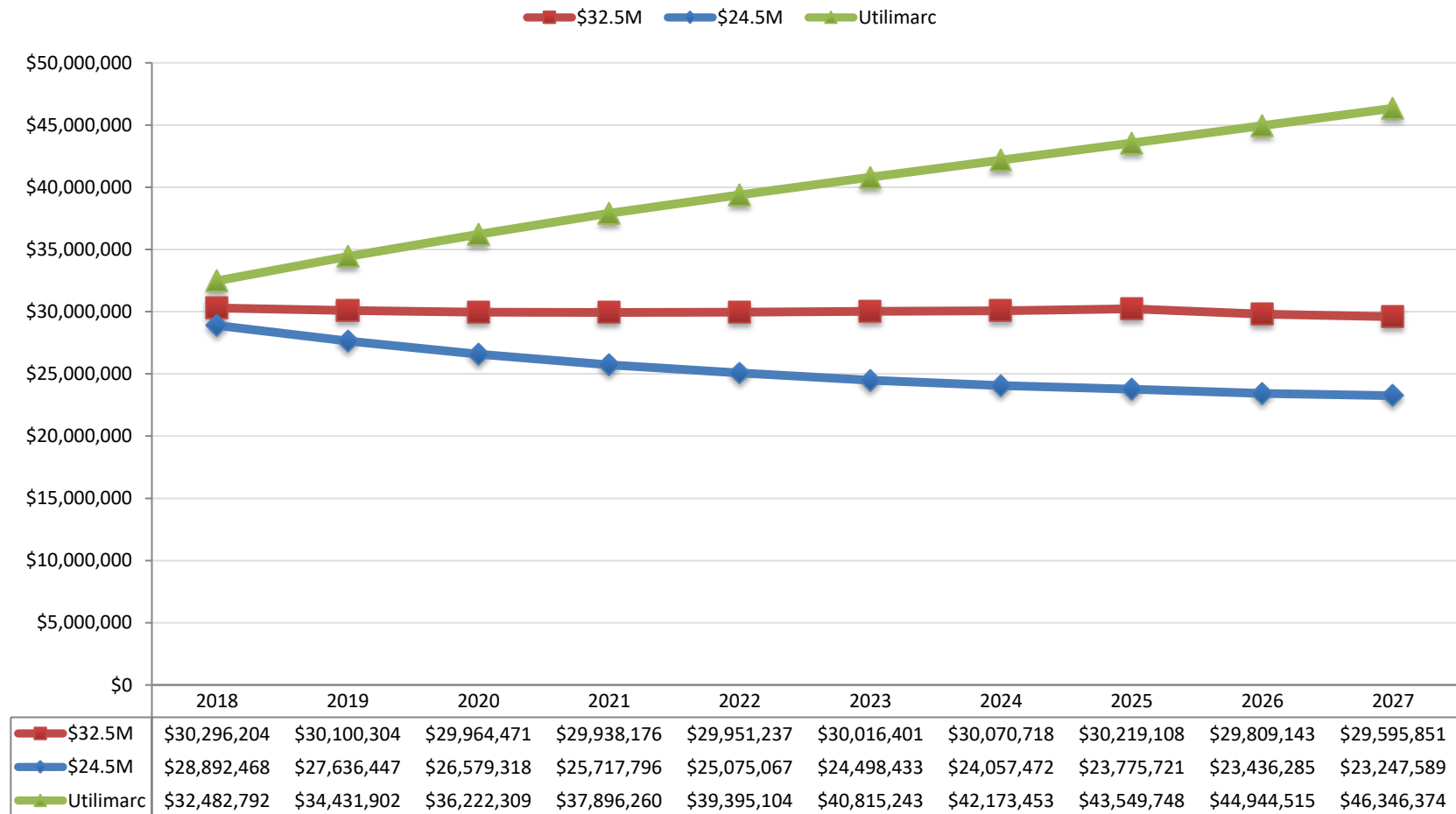
Demand Repair Work Orders

This graph shows the number of demand repair work orders required to support fleet under each scenario. Under the Utilimarc scenario, DR WO decreases from 44,594 work orders to 39,740 work orders by 2027. The number of DM WO increases from 46,750 to 54,782 in 2027 in the \$24.5M Budget scenario and increases from 49,522 to 53,589 in 2027 in the \$32.5M Budget Scenario.



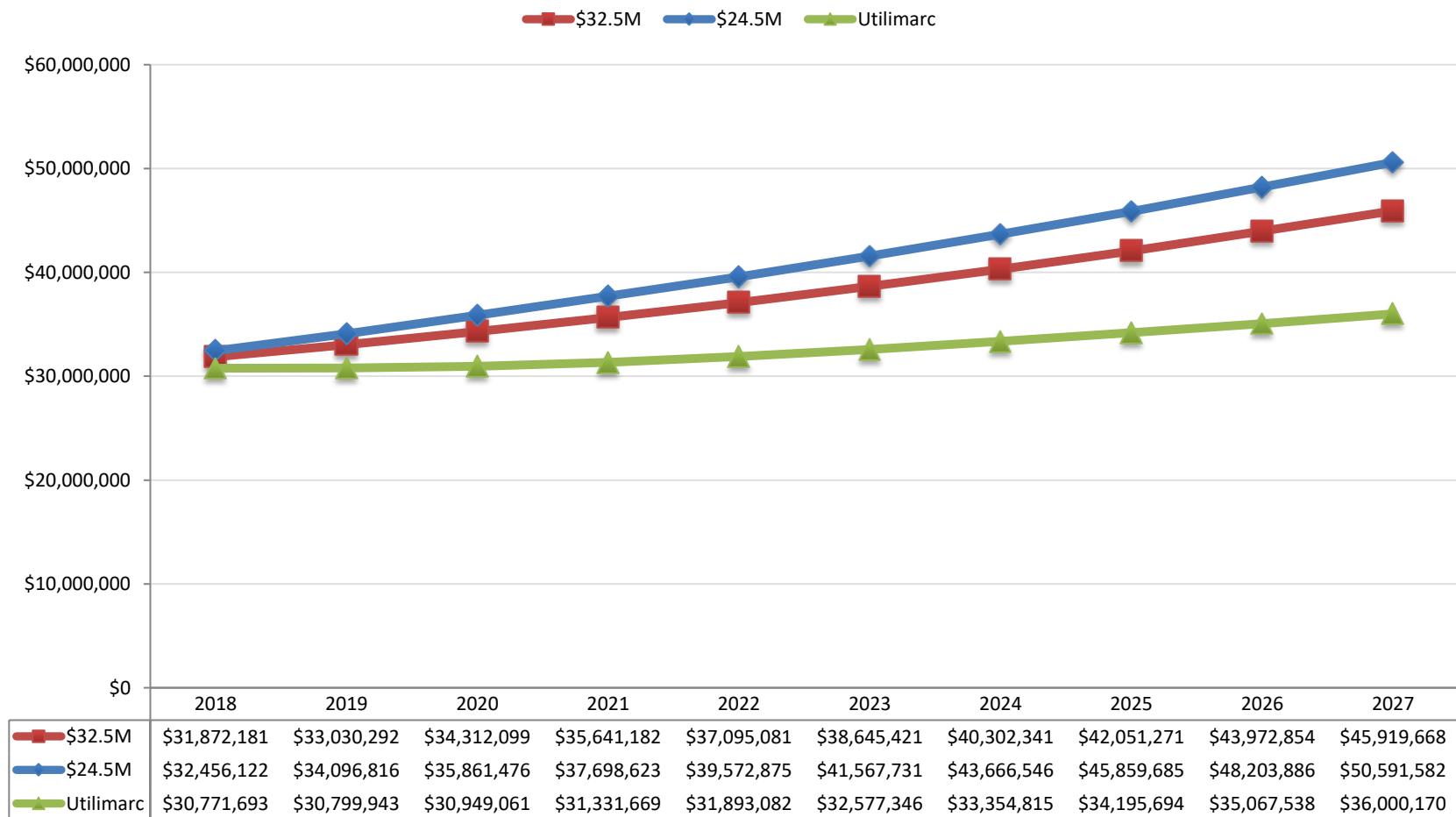
Annual Ownership Cost – Comparison

This graph shows annual total ownership cost under each scenario.



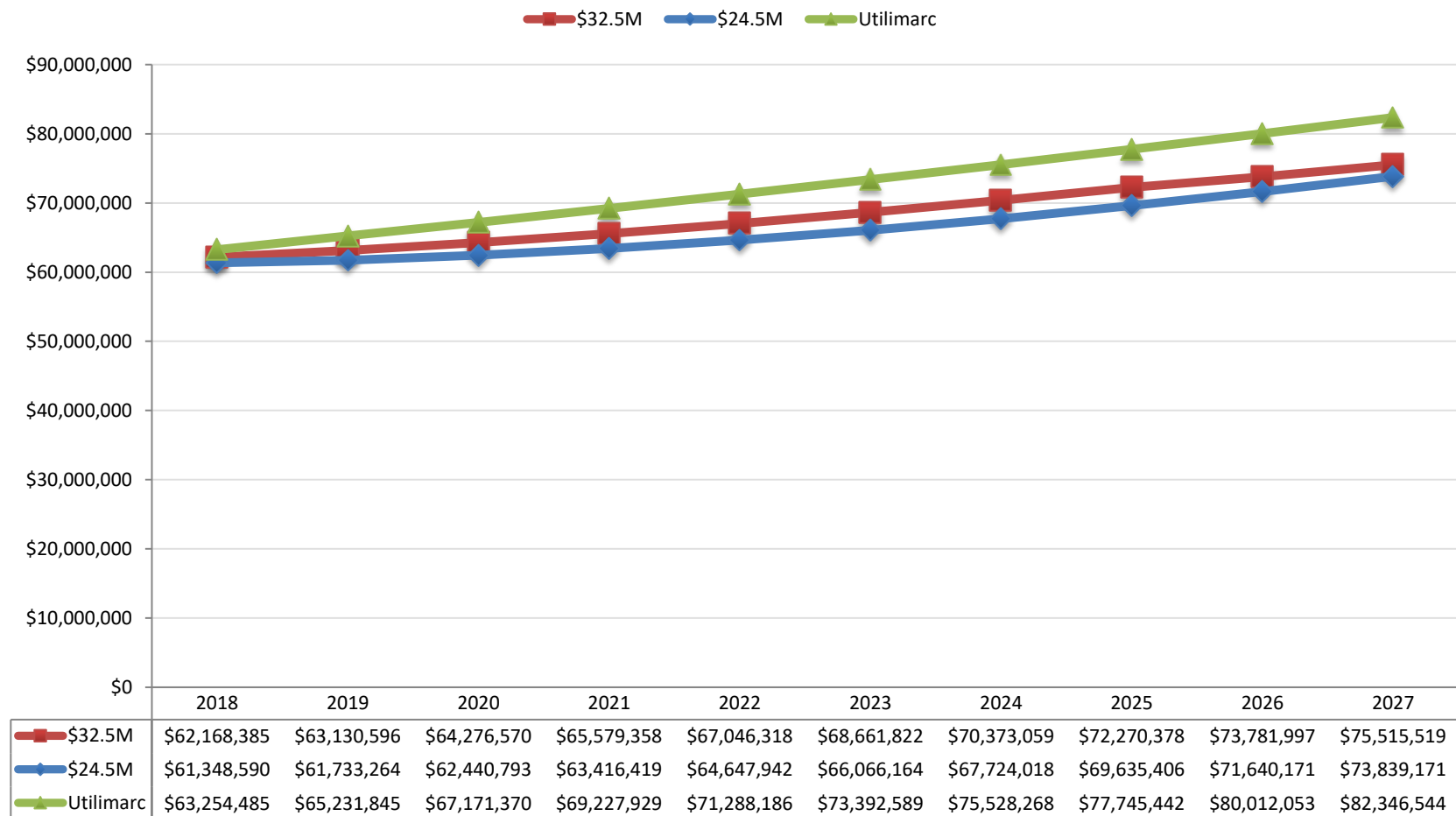
Annual Maintenance Cost – Comparison

This graph shows annual total maintenance cost under each scenario.



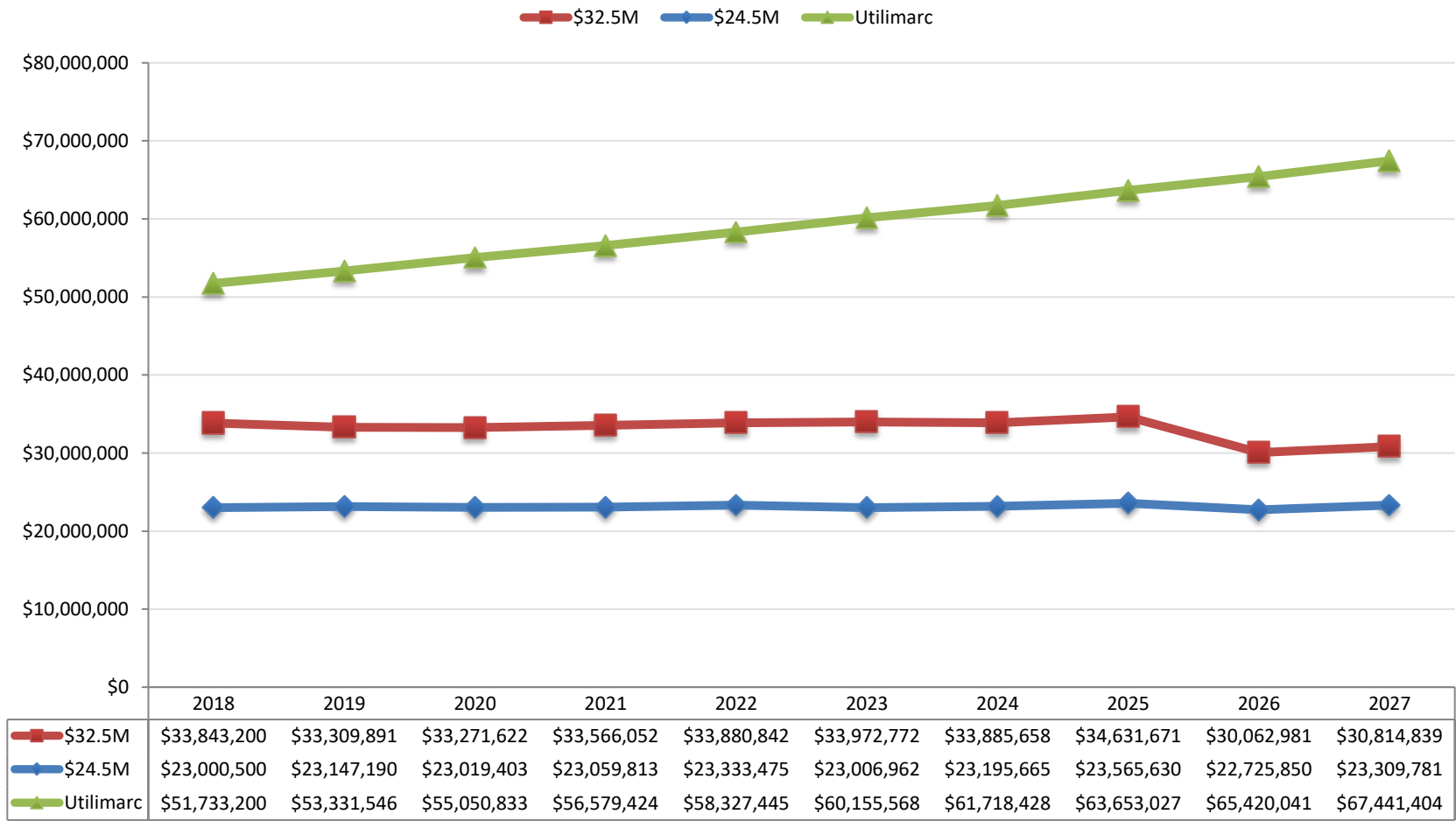
Annual Total Cost – Comparison

This graph shows annual total cost under each scenario.



Annual Capital Investment

This graph shows the amount spent on replacement each year under each scenario.



Methodology

Definitions

Lifecycle is the age at which a unit is prepared for replacement. This is typically stated as an age or limit on lifetime mileage (KM). This analysis relies primarily on age when determining whether or not to replace a unit.

Labor Hour Demand is the amount of labor hours needed to support a unit over a period of time. This includes labor hours handled internally or by an outside vendor.

Ownership Cost is the cost associated with owning a unit over time. This analysis interprets ownership cost as the difference between the sale value of a unit in the current year and the sale value of the same unit in the previous year.

Maintenance Cost is the cost to actively use a unit over time. This includes the cost of parts and repairs handled internally or by an outside vendor. For this analysis, fuel is not included as a maintenance cost.

Total Cost is the sum of maintenance and ownership cost.

Capital Investment is the amount spent on purchasing a new unit. This includes all costs to put the unit into service.

Residual Value is the amount received for the sale of a unit, expressed as a percent of the original purchase price.

Annualized Cost is the cost of the asset divided by the number of years the asset has been in service.

List of Assumptions

The following is a short list of important assumptions made by the model, provided for your reference and information:

- Inflation is included on all future costs, set to 3%.
- Annual mileage is assumed to be consistent among all vehicles of a given class. No adjustments in annual mileage are made based on the vintage of the unit.
- Replacement is assumed to be consistent year over year, in accordance with the most recent survey of active units in the fleet. No adjustments are made to anticipate future increases or decreases in funding or fleet size.
- Any units that are listed as “Removed from Fleet” were included in the 6 year projections, however will never be chosen for replacement since these units are not candidates for replacement.

Lifecycle Calculations

For each vehicle class, Utilimarc's Vehicle Replacement Module (VRM) determines what replacement age achieves the lowest total annualized cost. This guarantees the lowest total cost over the life of an average vehicle. In the example below, this principle is demonstrated in the calculation of the total annualized cost of an industry vehicle class over the lifetime of the vehicle.

Replacement Age	Total Annualized Cost	Deviation
1	\$7,811	34.4%
2	\$7,121	22.6%
3	\$6,613	13.8%
4	\$6,251	7.6%
5	\$6,010	3.5%
6	\$5,869	1.0%
7	\$5,810	0.0%
8	\$5,821	0.2%
9	\$5,892	1.4%
10	\$6,015	3.5%
11	\$6,183	6.4%
12	\$6,391	10.0%
13	\$6,636	14.2%
14	\$6,913	19.0%

The table on the previous page shows the annualized cost for the possible lifecycles of a typical light duty pickup truck. Consider the following three replacement scenarios over a 14-year financial period.

Scenario 1: A fleet manager plans to replace this vehicle every year. The annualized cost of this replacement strategy is \$7,811. Over the 14-year period, this replacement strategy will cost fleet $14 \times \$7,811 = \$109,354$.

Scenario 2: A fleet manager plans to replace this vehicle every seven years. The annualized cost of this replacement strategy is \$5,810. Over the 14-year period, this replacement strategy will cost fleet $14 \times \$5,810 = \$81,340$.

Scenario 3: A fleet manager plans to replace this vehicle every fourteen years. The annualized cost of this replacement strategy is \$6,913. Over the 14-year period, this strategy will cost fleet $14 \times \$6,913 = \$96,768$.

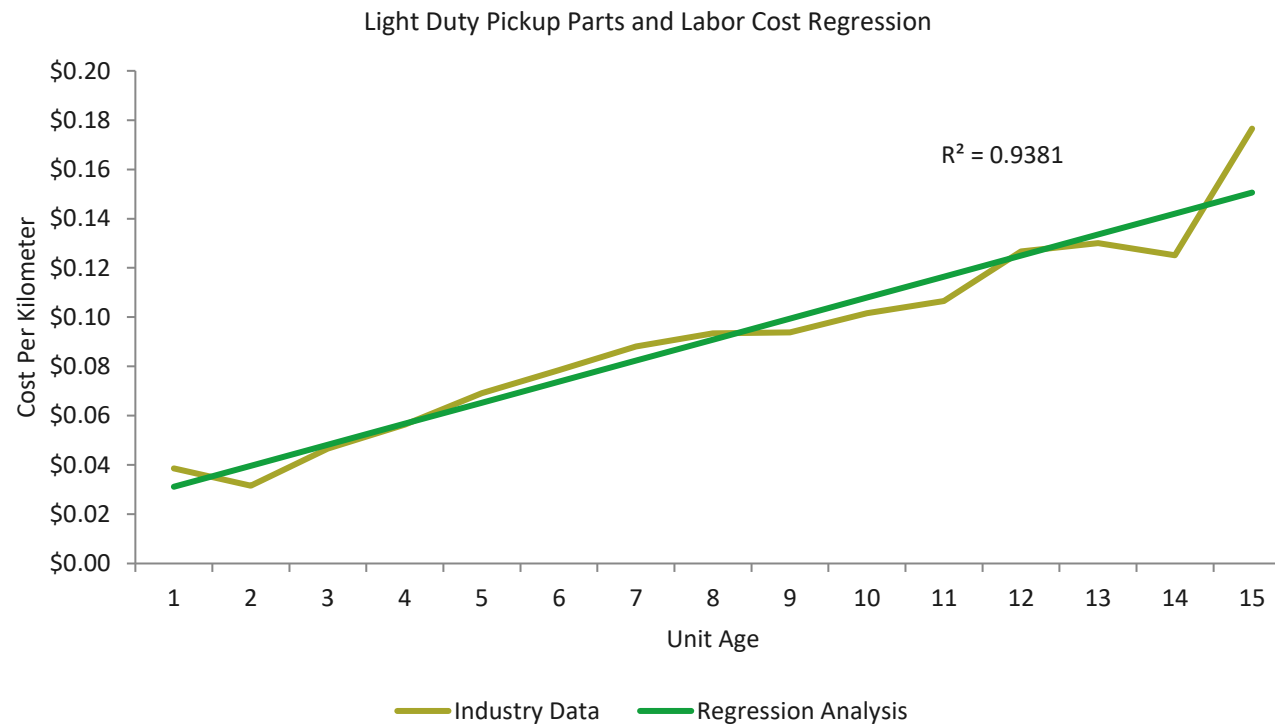
The table below summarizes the calculations in the previous example.

Scenario 1	1	14	\$7,811	\$109,354
Scenario 2	7	14	\$5,810	\$81,340
Scenario 3	14	14	\$6,913	\$96,768

This example illustrates that by minimizing total annualized cost achieves the lowest total cost of ownership over the life of the vehicle. Utilimarc recommends replacing units within 1.0% of the true lowest cost of ownership. This generally provides a three-year range for replacement, which allows for flexibility when planning replacement without dramatically effecting overall cost.

Regression Analysis

The VRM predicts future maintenance costs using vehicle class specific linear regressions. A regression model is built for each vehicle class based on historical, maintenance data from the Utilimarc database. The following example was built from historical industry data. It shows the total parts and labor regression for light duty pickup trucks. Note the R-squared value of .93. This means that 93% of the change in the vehicles cost per mile is a result of its increased age. For most vehicles in your fleet, the oldest units will cost you the most on a per kilometer basis. Utilimarc's replacement model prioritizes these units for replacement, followed by the units with the highest lifetime mileage (KM).



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Aging Table

The table below shows the number of units in your fleet by age and vehicle class. Your new replacement ages have been highlighted in red. High concentrations of vehicles have been highlighted in yellow. These "bubbles" can cause sudden increases in parts and labor cost, vehicle downtime, and technician requirements. Conversely, consistent replacement yields predictable parts and labor cost, vehicle downtime, and technician requirements.

Class	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	>15	Total
Light Duty Pickup	0	26	5	45	46	50	70	62	10	56	55	33	68	0	0	0	25	551
Medium Duty Pickup	0	24	14	126	14	57	48	36	15	0	86	4	5	8	0	13	164	614
Heavy Duty Pickup	0	8	5	2	2	1	1	1	1	0	5	5	3	2	0	0	12	48
Compact SUV	0	2	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	10
Midsize SUV	0	6	1	1	2	9	0	0	0	0	0	0	0	0	0	0	0	19
Van - Cargo 250	0	13	2	49	0	0	0	0	0	0	0	0	0	0	6	0	67	137
Van - Cube/Step	0	0	1	1	2	1	1	0	0	0	0	0	0	0	0	0	0	6
Cube Van	0	0	1	1	1	0	0	1	0	0	0	0	1	0	0	0	0	5
Step Van	0	0	0	2	68	0	0	2	0	0	0	0	0	0	0	1	2	75
Dump Truck - Single	0	12	17	20	39	12	10	0	0	0	0	0	0	0	1	0	7	118
Dump Truck - Tandem	0	0	6	2	11	6	8	0	0	2	0	0	0	0	0	1	1	37
LD Service Truck	0	5	36	52	37	34	62	30	15	67	12	10	9	8	5	1	27	410
MD Service Truck	2	0	2	0	1	1	0	0	0	0	0	0	0	1	0	0	4	11
HD Service Truck	0	7	16	29	25	25	0	0	4	30	18	0	25	0	0	0	26	205
Stake Truck	4	7	4	17	25	8	5	0	7	1	1	2	2	1	0	0	3	87
Light Duty Aerial	0	0	12	18	14	4	6	14	0	17	18	2	0	0	0	0	0	105
Medium Duty Aerial	0	0	0	0	0	0	6	0	0	1	0	0	0	0	0	0	0	7
Heavy Duty Aerial	0	47	0	35	28	19	1	4	42	2	0	8	47	26	0	0	3	262

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Inputs for Lifecycle Analysis

The following table includes some of the class specific inputs that were provided to Utilimarc to run the VRM. This includes class average annual mileage, purchase prices, historic lifecycles, and residual values. Any class code that did not have enough data to estimate a historic lifecycle or residual value used a default lifecycle of 10 years and a default residual value of 14%.

Class	Mileage	Average Purchase Price	Residual	Life
Light Duty Pickup	16,848	\$40,000	14%	10.70
Medium Duty Pickup	10,448	\$50,000	9%	13.93
Heavy Duty Pickup	9,823	\$50,000	12%	14.33
Compact SUV	18,166	\$29,000	14%	10.00
Midsized SUV	19,640	\$29,000	5%	12.64
Van - Cargo 250	11,085	\$55,000	4%	12.97
Van - Cube/Step	2,457	\$145,000	14%	10.00
Cube Van	11,719	\$145,000	14%	10.00
Step Van	3,201	\$145,000	14%	10.00
Dump Truck – Single < 5 Yard Dump	4,078	\$87,500	14%	10.00
Dump Truck – Single >= 5 Yard Dump	4,078	\$110,000	14%	10.00
Dump Truck - Tandem	5,156	\$150,000	14%	10.00
LD Service Truck	14,794	\$80,000	7%	13.91
MD Service Truck	2,552	\$115,000	4%	21.30
HD Service Truck	6,859	\$184,000	10%	15.14
Stake Truck	6,738	\$140,000	10%	17.92
Light Duty Aerial	26,665	\$195,000	11%	8.34
Medium Duty Aerial	8,523	\$195,000	5%	12.13
Heavy Duty Aerial	12,966	\$335,000	6%	12.85
Super Heavy Duty Aerial	14,154	\$400,000	14%	10.00
Diggers, Derricks	7,466	\$249,000	8%	15.40
Super HD Diggers, Derricks	1,846	\$303,000	14%	10.00
Semi-Tractor - Tandem	19,482	\$165,000	14%	10.00

Class	Mileage	Average Purchase Price	Residual	Life
Track Unit - Mounted	0	\$350,000	14%	10.00
Directional (Horizontal) Drill Unit	0	\$250,000	15%	6.87
Frontend Loader & Backhoe	0	\$110,000	28%	17.68
Light Skid-steer	0	\$109,000	14%	10.00
Heavy Skid-steer	0	\$133,000	14%	10.00
Dozer	0	\$531,700	14%	10.00
Walk-Behind Trencher	0	\$80,000	11%	18.12
Light Trencher	0	\$80,000	9%	13.12
Medium Trencher	0	\$108,000	9%	14.91
Light Loader	0	\$160,000	14%	10.00
Mini Excavator	0	\$81,000	14%	10.00
Light Excavator	0	\$119,000	14%	10.00
Medium Excavator	0	\$218,000	14%	10.00
Tensioner/Puller	0	\$60,000	14%	10.00
Welder	0	\$18,000	47%	3.65
Air Compressor	0	\$25,000	16%	20.99
Utility Cart	0	\$160,000	14%	10.00
Backyard Mobile Equipment	0	\$166,000	14%	10.00
Mobile Aerial Platform	0	\$40,000	14%	10.00
Forklift	0	\$80,000	14%	10.00
Off Road Tractor with Equipment	0	\$600,000	14%	10.00
Trailers	0	\$92,000	14%	10.00
Vacuum Truck	944	\$35,000	14%	20.00

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Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Directional (Horizontal) Drill Unit	5	5	4	4	4	4	4	4	4	4
Frontend Loader & Backhoe	5	5	4	4	5	4	4	4	3	4
Light Skid-steer	1	1	1	1	1	1	1	1	1	1
Heavy Skid-steer	0	0	0	0	0	0	0	0	0	0
Dozer	0	0	0	0	0	0	0	0	0	0
Walk-Behind Trencher	2	2	2	2	2	2	0	2	2	2
Light Trencher	0	0	0	0	0	0	0	0	0	0
Medium Trencher	5	5	5	5	5	4	4	4	4	4
Light Loader	0	0	0	0	0	0	0	0	0	0
Mini Excavator	8	8	8	8	8	9	8	8	7	8
Light Excavator	1	1	1	1	1	1	1	1	1	1
Medium Excavator	0	0	0	0	0	0	0	0	0	0
Tensioner/Puller	1	1	1	1	1	0	1	0	1	0
Welder	0	0	0	0	0	0	0	0	0	0
Air Compressor	2	2	2	2	2	2	2	2	1	1
Utility Cart	2	2	2	2	2	2	2	2	1	1
Backyard Mobile Equipment	0	0	0	0	0	0	0	0	0	0
Mobile Aerial Platform	1	1	1	1	1	1	1	1	1	1
Forklift	4	4	4	3	3	3	3	2	2	3
Off Road Tractor with Equipment	1	1	1	1	1	1	1	1	1	1
Trailers	36	33	36	34	30	27	27	26	24	26
Vacuum Truck	0	0	0	0	0	0	0	0	0	0
Total Vehicles	126	123	115	122	107	112	106	104	100	93
Total Equipment	74	71	72	69	66	61	59	58	53	57
Total Fleet	200	194	187	191	173	173	165	162	153	150

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Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Directional (Horizontal) Drill Unit	6	6	6	6	6	5	5	5	5	5
Frontend Loader & Backhoe	7	6	5	6	6	5	6	6	5	5
Light Skid-steer	1	1	1	1	1	1	1	1	1	1
Heavy Skid-steer	1	1	1	1	1	1	1	1	0	0
Dozer	1	1	1	1	1	1	1	1	0	0
Walk-Behind Trencher	3	3	3	3	1	3	3	3	2	2
Light Trencher	1	1	1	1	1	1	1	1	0	0
Medium Trencher	7	7	6	6	6	6	6	6	5	5
Light Loader	1	1	1	1	1	1	1	1	0	0
Mini Excavator	10	11	11	11	12	11	10	11	10	10
Light Excavator	2	2	2	2	2	2	2	0	1	1
Medium Excavator	1	1	1	1	1	1	1	1	0	0
Tensioner/Puller	2	2	1	1	1	2	1	1	1	0
Welder	0	0	0	0	0	0	0	0	0	0
Air Compressor	3	2	2	2	2	2	2	2	2	2
Utility Cart	3	2	2	2	2	2	2	2	2	2
Backyard Mobile Equipment	1	1	1	1	1	1	1	1	0	0
Mobile Aerial Platform	1	1	1	1	1	1	1	1	1	1
Forklift	5	5	5	5	4	3	3	4	4	4
Off Road Tractor with Equipment	1	1	1	1	1	1	1	1	1	1
Trailers	47	46	47	39	37	38	36	37	39	40
Vacuum Truck	0	0	0	0	0	0	0	0	0	0
Total Vehicles	175	165	168	152	156	154	141	149	141	140
Total Equipment	105	102	100	93	89	89	86	87	79	79
Total Fleet	280	267	268	245	245	243	227	236	220	219

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

MICHAEL P. KELLY

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

A-73 (MPK-1)
IS **CONFIDENTIAL** AND BEING FILED
UNDER SEAL WITH THE MPSC

A-74 (MPK-2)
IS **CONFIDENTIAL** AND BEING FILED
UNDER SEAL WITH THE MPSC

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

STEVEN Q. MCLEAN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Customer Experience and Operations

Summary of Actual and Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (SQM-1)

Schedule: B-5.5

Page: 1 of 5

Witness: SQMcLean

Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures					
Line No.	Description	Historical	Projected Bridge Year			Projected	
		12 Mos Ended 12/31/2018	12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 Mos Ending 12/31/2021	
1	Customer Experience and Operations	-	-	9,920	9,920	3,068	
	Contractor	0	0	2,458	2,458	1,083	
	Labor	0	0	3,201	3,201	328	
	Materials	0	0	0	0	0	
	Business Expenses	0	0	3,876	3,876	1,357	
	Contingency	0	0	0	0	0	
	Other (Loadings, Chargebacks)	0	0	385	385	300	
2	Demand Response	6,675	17,137	10,856	27,993	8,868	
	Contractor	2,082	7,816	820	8,636	215	
	Labor	12	2,861	592	3,453	615	
	Materials	4,500	5,730	492	6,223	273	
	Business Expenses	0	18	8,593	8,611	7,659	
	Contingency	-	-	-	-	-	
	Other (Loadings, Chargebacks)	82	711	359	1,070	108	
3	Total Capital	<u>6,675</u>	<u>17,137</u>	<u>20,776</u>	<u>37,913</u>	<u>11,937</u>	

Schedule B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Customer Experience & Operations

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (SQM-1)

Schedule: B-5.5

Page: 2 of 5

Witness: SQMcLean

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected Test Year
Line No.	Program Description	12 Mos Ended 12/31/2018	12 Mos Ending 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 mos. Ending 12/31/2021
	Analytics & Outreach					
1	Analytics & Outreach	0	0	8,706	8,706	3,065
	Contractor	0	0	2,367	2,367	1,083
	Labor	0	0	2,079	2,079	328
	Materials	0	0	0	0	0
	Business Expenses	0	0	3,876	3,876	1,353
	Contingency	0	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	385	385	300
	Billing & Payment					
2	Customer Payment Programs	0	0	1,199	1,199	0
	Contractor	0	0	77	77	0
	Labor	0	0	1,122	1,122	0
	Materials	0	0	0	0	0
	Business Expenses	0	0	0	0	0
	Contingency	0	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0	0
	Customer Interactions					
3	Digital Customer Experience	0	0	15	15	3
	Contractor	0	0	15	15	0
	Labor	0	0	0	0	0
	Materials	0	0	0	0	0
	Business Expenses	0	0	0	0	3
	Contingency	0	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0	0
4	Total CX&O Capital	-	-	9,920	9,920	3,068
	<i>Control Variance</i>			0	0	0
5	Demand Response					
	Commerical and Industrial	517	1,646	1,296	2,942	717
	Contractor	60	502	389	891	215
	Labor	11	268	207	476	115
	Materials	364	614	492	1,107	273
	Business Expenses	0	18	13	31	7
	Contingency	0	0	0	0	0
	Other (Loadings, Chargebacks)	81	242	194	437	108
6	Residential	6,158	15,491	9,560	25,051	8,151
	Contractor	2,022	7,313	431	7,745	-
	Labor	0	2,593	385	2,978	500
	Materials	4,135	5,116	-	5,116	-
	Business Expenses	-	-	8,580	8,580	7,651
	Contingency	-	-	-	-	-
	Other (Loadings, Chargebacks)	0	469	164	633	-
7	Total Demand Response	6,675	17,137	10,856	27,993	8,868

Schedule B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Customer Experience & Operations

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (SQM-1)

Schedule: B-5.5

Page: 3 of 5

Witness: SQMcLean

Date: February 2020

	(a)	(c)	(d)	(e)	(f)
Line No.	Program Description	Projected Bridge Year			Projected Test Year
		12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 Mos Ending 12/31/2021
	Analytics & Outreach				
	Analytics & Outreach				
1	Customer Relationship Management (CRM) Product Suite	0	2,608	2,608	2,310
	Contractor	0	442	442	328
	Labor	0	415	415	328
	Materials	0	0	0	0
	Business Expenses	0	1,367	1,367	1,353
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	385	385	300
2	Voice of the Customer Analytics	0	670	670	0
	Contractor	0	67	67	0
	Labor	0	134	134	0
	Materials	0	0	0	0
	Business Expenses	0	469	469	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
3	Customer Analytics Hub	0	1,195	1,195	755
	Contractor	0	1,195	1,195	755
	Labor	0	0	0	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
4	Customized Payment Alerts	0	153	153	0
	Contractor	0	153	153	0
	Labor	0	0	0	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
5	Service Tracker	0	2,040	2,040	0
	Contractor	0	510	510	0
	Labor	0	1,530	1,530	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
6	Online Work Scheduling	0	1,020	1,020	0
	Contractor	0	0	0	0
	Labor	0	0	0	0
	Materials	0	0	0	0
	Business Expenses	0	1,020	1,020	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
7	Streetlight Application	0	1,020	1,020	0
	Contractor	0	0	0	0
	Labor	0	0	0	0
	Materials	0	0	0	0
	Business Expenses	0	1,020	1,020	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
	Analytics & Outreach Total	0	8,706	8,706	3,065

Schedule B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Customer Experience & Operations

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (SQM-1)

Schedule: B-5.5

Page: 4 of 5

Witness: SQMcLean

Date: February 2020

	(a)	(c)	(d)	(e)	(f)
		Projected Bridge Year			Projected Test Year
Line No.	Program Description	12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 Mos Ending 12/31/2021
Billing & Payment					
Customer Payment Programs					
7	Flexible/Advanced Payment Options	0	1,020	1,020	0
	Contractor	0	0	0	0
	Labor	0	1,020	1,020	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
8	Payment Extension - Online/IVR	0	51	51	0
	Contractor	0	0	0	0
	Labor	0	51	51	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
9	In Person Payment Network Expansion	0	128	128	0
	Contractor	0	77	77	0
	Labor	0	51	51	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
	Customer Payment Programs Total	0	1,199	1,199	0
Customer Interactions					
Digital Customer Experience					
10	RPA - Single Address Change	0	15	15	3
	Contractor	0	15	15	0
	Labor	0	0	0	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	3
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
	Digital Customer Experience Total	0	15	15	3
Total CX&O Capital		-	9,920	9,920	3,068

Schedule B-5.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Customer Experience & Operations

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (SQM-1)

Schedule: B-5.5

Page: 5 of 5

Witness: SQMcLean

Date: February 2020

Line No.	(a) Program Description	(c) (d) (e)			(f)
		Projected Bridge Year			Projected Test Year
		12 Mos Ended 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 Mos Ending 12/31/2021
	Demand Response				
	Commercial and Industrial				
11	Demand Response Commercial and Industrial	1,014	1,296	2,310	717
	Contractor	6	389	395	215
	Labor	157	207	365	115
	Material	614	492	1,107	273
	Business Expenses	18	13	31	7
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	217	194	412	108
12	CIA	632	0	632	0
	Contractor	496	0	496	0
	Labor	111	0	111	0
	Material	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	25	0	25	0
	Residential				
13	A/C Peak Cycling	8,440	8,580	17,020	7,651
	Contractor	3,323	0	3,323	0
	Labor	1	0	1	0
	Materials	5,116	0	5,116	0
	Business Expenses	0	8,580	8,580	7,651
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	0	0	0
	CIA	3,951	0	3,951	0
	Contractor	1,281	0	1,281	0
	Labor	2,264	0	2,264	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	406	0	406	0
14	Residential DR Product Bundling	0	567	567	500
	Contractor	0	268	268	0
	Labor	0	235	235	500
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	0	64	64	0
14	Residential Demand Response Provisions	3,100	413	3,513	0
	Contractor	2,709	163	2,872	0
	Labor	329	150	479	0
	Materials	0	0	0	0
	Business Expenses	0	0	0	0
	Contingency	0	0	0	0
	Other (Loadings, Chargebacks)	63	100	163	0
15	Demand Response Total Capital	17,137	10,856	27,993	8,868

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

**Summary of Actual & Projected Customer Experience and Operations O&M Expenses
& Revenues Summary
(\$000)**

Case No.: U-20679

Exhibit No.: A-75 (SQM-2)

Page: 1 of 5

Witness: SQMcLean

Date: February 2020

	(a)	(b)	(c)	(d)	(e)
	O&M Expenses				
Line No.	Description	2018 Actual	2019 Projected	2020 Projected	12 Month Ending December 31, 2021 Projectd
	Analytics & Outreach				
1	Labor	\$ 1,714.9	\$ 2,155.1	\$ 3,385.2	\$ 2,616.6
2	Material	294.4	1.8	-	-
3	Contractor	3,351.6	1,047.2	2,700.8	2,024.5
4	Non Labor Overheads	-	-	-	-
5	Non Labor Other	191.4	189.5	1,217.1	892.3
6	Total	\$ 5,552.2	\$ 3,393.7	\$ 7,303.0	\$ 5,533.5
	Customer Interactions				
7	Labor	\$ 17,847.5	\$ 15,970.8	\$ 16,451.4	\$ 16,804.3
8	Material	674.7	11.1	47.9	47.9
9	Contractor	8,795.8	6,153.7	6,277.7	6,010.6
10	Non Labor Overheads	-	-	-	-
11	Non Labor Other	1,830.1	2,439.3	2,833.2	2,816.7
12	Total	\$ 29,148.1	\$ 24,575.0	\$ 25,610.2	\$ 25,679.5
	Billing & Payments				
13	Labor	\$ 2,765.4	\$ 4,603.3	\$ 6,003.7	\$ 6,248.5
14	Material	810.8	-	-	-
15	Contractor	9,702.0	6,960.8	7,361.9	7,804.5
16	Non Labor Overheads	-	-	-	-
17	Non Labor Other	5,979.0	6,741.7	7,549.3	8,752.8
18	Total	\$ 19,257.2	\$ 18,305.8	\$ 20,914.9	\$ 22,805.8
	Demand Response				
19	Labor	\$ 1,028.9	\$ 1,688.0	\$ 8,072.3	\$ 8,744.0
20	Material	934.5	63.7	-	-
21	Contractors	4,855.8	9,827.6	11,324.3	15,158.2
22	Non Labor Overheads	-	-	-	-
23	Non Labor Other	94.8	860.4	7,337.0	10,779.4
24	Total	\$ 6,913.9	\$ 12,439.6	\$ 26,733.6	\$ 34,681.6
25	Total CX&O Expenses	\$ 60,871.4	\$ 58,714.1	\$ 80,561.7	\$ 88,700.4

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Projected Analytics & Outreach O&M Expenses
Summary
(\$000)

Case No.: U-20679
Exhibit No.: A-75 (SQM-2)
Page: 2 of 5
Witness: SQMcLean
Date: February 2020

		(a)	(b)	(c)	(d)	(e)
		O&M Expenses				
Line	Description	2018	2019	2020	12 Months Ending	
No.		Actual	Projected	Projected	December 31, 2021	
					Projected	
	Analytics & Outreach					
1	Labor	\$ 1,714.9	\$ 2,155.1	\$ 3,385.2	\$	2,616.6
2	Material	294.4	1.8	-		-
3	Contractor	3,351.6	1,047.2	2,700.8		2,024.5
4	Non Labor Overheads	-	-	-		-
5	Non Labor Other	191.4	189.5	1,217.1		892.3
5	Total	\$ 5,552.2	\$ 3,393.7	\$ 7,303.0	\$	5,533.5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Actual & Projected Customer Interactions O&M Expenses
Summary

(\$000)

Case No.: U-20679

Exhibit No.: A-75 (SQM-2)

Page: 3 of 5

Witness: SQMcLean

Date: February 2020

		(a)	(b)	(c)	(d)	(e)
		O&M Expenses				
Line No.	Description	2018 Actual	2019 Projected	2020 Projected	12 Months Ending December 31, 2021 Projected	
	Digital Customer Experience					
1	Labor	\$ 1,768.4	\$ 1,193.1	\$ 1,306.8	\$	1,348.6
2	Material	74.8	1.8	-		-
3	Contractor	4,096.1	1,600.5	1,319.4		1,418.8
4	Non Labor Overhead	-	-	-		-
5	Non Labor Other	154.7	323.8	435.0		416.0
6	Total	\$ 6,093.9	\$ 3,119.2	\$ 3,061.2	\$	3,183.4
	Customer Contact Center					
7	Labor	\$ 11,076.8	\$ 9,887.7	\$ 10,178.9	\$	10,407.3
8	Material	464.1	-	-		-
9	Contractor	2,736.3	2,434.3	2,636.6		2,270.0
10	Non Labor Overhead	-	-	-		-
11	Non Labor Other	1,268.4	1,717.6	1,780.3		1,778.0
12	Total	\$ 15,545.6	\$ 14,039.7	\$ 14,595.8	\$	14,455.3
	Business Customer Care					
13	Labor	\$ 2,629.5	\$ 2,469.7	\$ 2,318.9	\$	2,318.9
14	Material	17.8	-	-		-
15	Contractor	256.9	181.3	274.1		274.1
16	Non Labor Overhead	-	-	-		-
	Non Labor Other	227.5	150.2	231.5		231.5
17	Total	\$ 3,131.5	\$ 2,801.1	\$ 2,824.5	\$	2,824.5
	Field Payment Channels and Claims					
18	Labor	\$ 1,668.2	\$ 1,393.5	\$ 1,501.1	\$	1,548.5
19	Material	18.5	-	-		-
20	Contractor	380.9	235.4	263.7		263.7
21	Non Labor Overhead	-	-	-		-
22	Non Labor Other	94.1	107.4	108.1		108.1
23	Total	\$ 2,161.7	\$ 1,736.3	\$ 1,872.9	\$	1,920.3
	Credit & Assistance					
24	Labor	\$ 704.7	\$ 1,026.9	\$ 1,145.8	\$	1,181.0
25	Material	99.5	9.3	47.9		47.9
26	Contractor	1,325.7	1,702.1	1,784.0		1,784.0
27	Non Labor Overhead	-	-	-		-
28	Non Labor Other	85.4	140.4	278.2		283.1
29	Total	\$ 2,215.3	\$ 2,878.7	\$ 3,255.9	\$	3,296.0
30	Total Customer Interactions	\$ 29,148.1	\$ 24,575.0	\$ 25,610.2	\$	25,679.5

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Projected Billing & Payment O&M Expenses
Summary
(\$000)

Case No.: U-20679
Exhibit No.: A-75 (SQM-2)
Page: 4 of 5
Witness: SQMcLean
Date: February 2020

		(a)	(b)	(c)	(d)	(e)
		O&M Expenses				
Line No.	Description	2018 Actual	2019 Projected	2020 Projected	12 Months Ending December 31, 2021 Projected	
	Customer Payment Programs					
1	Labor	\$ -	\$ -	\$ 737.3	\$	844.2
2	Contractor	7,538.0	6,712.7	7,134.4		7,577.1
3	Non Labor Other	(375.6)	49.1	728.6		1,929.6
4	Total	\$ 7,162.4	\$ 6,761.8	\$ 8,600.3	\$	10,350.9
	Customer Billing					
5	Labor	\$ 2,394.1	\$ 2,837.4	\$ 3,325.5	\$	3,423.0
6	Material	19.5	-	-		-
7	Contractor	371.7	101.4	-		-
8	Non Labor Overheads	-	-	-		-
9	Non Labor Other	234.8	126.7	206.7		209.2
10	Total	\$ 3,020.2	\$ 3,065.4	\$ 3,532.1	\$	3,632.2
	Business Support					
11	Labor	\$ 371.3	\$ 1,765.9	\$ 1,940.9	\$	1,981.3
12	Material	791.2	-	-		-
13	Contractor	1,792.3	146.7	227.5		227.5
14	Non Labor Overheads	-	-	-		-
15	Non Labor Other	6,119.8	6,565.9	6,614.0		6,614.0
16	Total	\$ 9,074.6	\$ 8,478.6	\$ 8,782.4	\$	8,822.8
17	Total Billing & Payment	\$ 19,257.2	\$ 18,305.8	\$ 20,914.9	\$	22,805.8

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Projected Demand Response O&M Expenses
Summary
(\$000)

Case No.: U-20679
Exhibit No.: A-75 (SQM2-2)
Page: 5 of 5
Witness: SQMcLean
Date: February 2020

	(a)	(b)	(c)	(d)	(e)
Line No.	Description	2018 Actual	2019 Projected	2020 Projected	12 Months Ending December 31, 2021 Projected
	Demand Response - Residential				
1	Labor	\$ 238.8	\$ -	\$ 5,419.9	\$ 6,074.0
2	Material	0.3	-	-	-
3	Contractor	3,616.8	7,220.6	6,726.1	8,677.1
4	Non Labor Overheads	-	-	-	-
5	Non Labor Other	32.3	(556.0)	2,562.5	4,181.8
6	Total	\$ 3,888.2	\$ 6,664.6	\$ 14,708.5	\$ 18,933.0
	Demand Response - C&I				
1	Labor	\$ 790.1	\$ 1,688.0	\$ 2,652.4	\$ 2,670.0
2	Material	934.2	63.7	-	-
3	Contractor	1,239.0	2,606.9	4,598.2	6,481.0
4	Non Labor Overheads	-	-	-	-
5	Non Labor Other	62.5	1,416.4	4,774.5	6,597.6
6	Total	\$ 3,025.8	\$ 5,775.0	\$ 12,025.1	\$ 15,748.6
	Total Demand Response	<u>\$ 6,913.9</u>	<u>\$ 12,439.6</u>	<u>\$ 26,733.6</u>	<u>\$ 34,681.6</u>

Case No.: U-20697
Exhibit No.: A-76 (SQM-3)
Page: 1 of 1
Witness: SQMcLean
Date: February 2020

Expenses	
Capital	O&M
5,209,551	926,970
1,105,813	46,215
-	165,000
1,336,508	138,270
-	116,000
1,462,828	175,973
\$ 5,114,707	\$ 1,668,428
-	79,200
3,184,331	434,445
2,528,027	164,670
401,911	36,135
\$ 6,114,268	\$ 714,450
166,272	7,920
2,094,646	149,110
\$ 2,240,918	\$ 157,030
-	115,500
\$ -	\$ 115,500
311,842	131,784
\$ 311,842	\$ 131,784
\$ 17,861,129	\$ 2,687,760

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

HUBERT W. MILLER III

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule

Total Revenues

Case No.: U-20697
Exhibit No.: A-16 (HWM-1)
Schedule: F-2.0
Page: 1 of 3
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)
Line				Revenue		Net Increase / (Decrease)	
No.	Description	Customers	Sales	Present	Proposed	Revenue	Percent
		Mthly	MWh	\$000	\$000	\$000	%
BUNDLED SERVICE							
Residential Class							
1	Summer On-peak RSP	1,601,135	12,247,759	\$ 1,972,286	\$ 2,247,732	\$ 275,446	14.0
2	Smart Hours RSH	2,666	60,975	9,320	10,725	1,405	15.1
3	Night Time Savers RPM	676	8,045	1,233	1,386	153	12.4
4	Non-Transmitting Meters RSM	16,221	130,889	21,438	24,670	3,231	15.1
5	Total Residential Class	1,620,698	12,447,668	\$ 2,004,276	\$ 2,284,513	\$ 280,236	14.0
Secondary Class							
6	Energy-only GS	195,747	3,826,453	\$ 593,299	\$ 620,276	\$ 26,977	4.5
7	Time-of-Use GSTU	72	3,265	484	479	(5)	(1.0)
8	Demand GSD	19,907	3,256,731	433,955	435,935	1,980	0.5
9	Total Secondary	215,726	7,086,449	\$ 1,027,738	\$ 1,056,690	\$ 28,952	2.8
Primary Class							
10	Energy-only GP	1,598	952,493	\$ 120,078	\$ 103,425	\$ (16,652)	(13.9)
11	Demand GPD	1,059	4,950,918	463,196	413,101	(50,096)	(10.8)
12	Time-of-Use GPTU	1,032	4,365,549	419,677	420,519	843	0.2
13	Energy Intensive EIP	19	358,537	24,200	21,404	(2,796)	(11.6)
14	Total Primary	3,708	10,627,497	\$ 1,027,150	\$ 958,450	\$ (68,701)	(6.7)
Lighting & Unmetered Class							
15	Metered Lighting GML	343	13,509	\$ 1,610	\$ 1,431	\$ (179)	(11.1)
16	Unmetered Lighting GUL	3,735	79,208	23,749	21,756	(1,993)	(8.4)
17	Unmetered GU-XL / LED	797	20,629	8,019	7,845	(174)	(2.2)
18	Unmetered GU	479	95,012	9,295	9,437	143	1.5
19	Total Lighting & Unmetered	5,354	208,358	\$ 42,673	\$ 40,470	\$ (2,203)	(5.2)
Self-generation Class							
20	Small Self-generation GSG-1	-	-	\$ -	\$ -	\$ -	NA
21	Large Self-generation GSG-2	12	63,483	5,609	5,648	39	0.7
22	Total Self-generation	12	63,483	\$ 5,609	\$ 5,648	\$ 39	0.7
23	Total Bundled Service	1,845,498	30,433,455	\$ 4,107,448	\$ 4,345,770	\$ 238,323	5.8
ROA SERVICE							
Secondary Class							
24	Energy-only GS	110	24,480	\$ 1,055	\$ 1,293	\$ 238	22.6
25	Demand GSD	477	180,859	6,116	7,111	994	16.3
26	Total Secondary	587	205,339	\$ 7,171	\$ 8,404	\$ 1,233	17.2
Primary Class							
27	Energy-only GP	54	74,331	\$ 1,028	\$ 1,272	\$ 244	23.7
28	Demand GPD	354	3,417,757	17,997	22,729	4,731	26.3
29	Total Primary	408	3,492,088	\$ 19,025	\$ 24,000	\$ 4,975	26.2
30	Total ROA Service	995	3,697,427	\$ 26,196	\$ 32,404	\$ 6,208	23.7
31	Total Jurisdictional Service	1,846,493	34,130,882	\$ 4,133,644	\$ 4,378,175	\$ 244,531	5.9
32	Less: PSCR Factor Revenues			134,502	134,502	-	
33	Less: GSG-2 and GI-2 PSCR Revenues			5,960	6,096	136	
34	Total Jurisdictional Base Revenues			\$ 3,993,181	\$ 4,237,576	\$ 244,395	
35	Rounding				(162)	(38)	
36	Total Jurisdictional Base Revenues				\$ 4,237,415	\$ 244,357	

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule
Production & Transmission Revenues

Case No.: U-20697
Exhibit No.: A-16 (HWM-1)
Schedule: F-2.0
Page: 2 of 3
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
Line			Revenue		Net Increase / (Decrease)	
No.	Description	Sales	Present	Proposed	Revenue	Percent
		MWh	\$000	\$000	\$000	%
BUNDLED SERVICE						
Residential Class						
1	Summer On-peak RSP	12,247,759	\$ 1,270,724	\$ 1,369,434	\$ 98,710	7.8
2	Smart Hours RSH	60,975	6,241	6,816	575	9.2
3	Night Time Savers RPM	8,045	793	833	39	4.9
4	Non-Transmitting Meters RNT	130,889	13,473	14,788	1,315	9.8
5	Total Residential Class	12,447,668	\$ 1,291,232	\$ 1,391,871	\$ 100,639	7.8
Secondary Class						
6	Energy-only GS	3,826,453	\$ 383,868	\$ 373,425	\$ (10,443)	(2.7)
7	Time-of-Use GSTU	3,265	328	319	(9)	(2.8)
8	Demand GSD	3,256,731	319,627	303,899	(15,728)	(4.9)
9	Total Secondary	7,086,449	\$ 703,823	\$ 677,643	\$ (26,180)	(3.7)
Primary Class						
10	Energy-only GP	952,493	\$ 105,669	\$ 85,905	\$ (19,764)	(18.7)
11	Demand GPD	4,950,918	430,127	372,137	(57,990)	(13.5)
12	Time-of-Use GPTU	4,365,549	378,499	368,044	(10,456)	(2.8)
13	Energy Intensive EIP	358,537	22,488	19,381	(3,107)	(13.8)
14	Total Primary	10,627,497	\$ 936,783	\$ 845,467	\$ (91,317)	(9.7)
Lighting & Unmetered Class						
15	Metered Lighting GML	13,509	\$ 740	\$ 614	\$ (125)	(17.0)
16	Unmetered Lighting GUL	79,208	4,381	3,559	(821)	(18.8)
17	Unmetered GU-XL / LED	20,629	1,088	917	(171)	(15.7)
18	Unmetered GU	95,012	7,646	7,284	(362)	(4.7)
19	Total Lighting & Unmetered	208,358	\$ 13,854	\$ 12,375	\$ (1,479)	(10.7)
Self-generation Class						
20	Small Self-generation GSG-1	-	\$ -	\$ -	\$ -	NA
21	Large Self-generation GSG-2	63,483	4,015	4,015	-	-
22	Total Self-generation	63,483	\$ 4,015	\$ 4,015	\$ -	-
23	Total Bundled Service	30,433,455	\$ 2,949,707	\$ 2,931,370	\$ (18,336)	(0.6)
ROA SERVICE						
Secondary Class						
24	Energy-only GS	-	\$ -	\$ -	\$ -	NA
25	Demand GSD	-	-	-	-	NA
26	Total Secondary	-	\$ -	\$ -	\$ -	NA
Primary Class						
27	Energy-only GP	-	\$ -	\$ -	\$ -	NA
28	Demand GPD	-	-	-	-	NA
29	Total Primary	-	\$ -	\$ -	\$ -	NA
30	Total ROA Service	-	\$ -	\$ -	\$ -	NA
31	Total Jurisdictional Service	30,433,455	\$ 2,949,707	\$ 2,931,370	\$ (18,336)	(0.6)
32	Less: PSCR Factor Revenues		134,502	134,502	-	
33	Less: GSG-2 and GI-2 PSCR Revenues		5,960	6,096	136	
34	Total Jurisdictional Base Revenues		\$ 2,809,244	\$ 2,790,772	\$ (18,472)	
35	Rounding			(50)		
36	Total Jurisdictional Base Revenues			\$ 2,790,722		

Schedule F-2.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Summary of Present and Proposed Pro Forma Revenues by Rate Schedule
Delivery Revenues

Case No.: U-20697
Exhibit No.: A-16 (HWM-1)
Schedule: F-2.0
Page: 3 of 3
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
Line			Revenue		Net Increase / (Decrease)	
No.	Description	Sales	Present	Proposed	Revenue	Percent
		MWh	\$000	\$000	\$000	%
BUNDLED SERVICE						
<u>Residential Class</u>						
1	Summer On-peak RSP	12,247,759	\$ 701,562	\$ 878,298	\$ 176,736	25.2
2	Smart Hours RSH	60,975	3,079	3,909	830	27.0
3	Night Time Savers RPM	8,045	439	553	114	25.9
4	Non-Transmitting Meters RNT	130,889	7,965	9,882	1,916	24.1
5	Total Residential Class	12,447,668	\$ 713,045	\$ 892,642	\$ 179,597	25.2
<u>Secondary Class</u>						
6	Energy-only GS	3,826,453	\$ 209,431	\$ 246,850	\$ 37,419	17.9
7	Time-of-Use GSTU	3,265	156	161	5	3.0
8	Demand GSD	3,256,731	114,329	132,036	17,707	15.5
9	Total Secondary	7,086,449	\$ 323,915	\$ 379,047	\$ 55,131	17.0
<u>Primary Class</u>						
10	Energy-only GP	952,493	\$ 14,409	\$ 17,521	\$ 3,112	21.6
11	Demand GPD	4,950,918	33,069	40,963	7,895	23.9
12	Time-of-Use GPTU	4,365,549	41,177	52,476	11,299	27.4
13	Energy Intensive EIP	358,537	1,712	2,023	311	18.2
14	Total Primary	10,627,497	\$ 90,367	\$ 112,983	\$ 22,616	25.0
<u>Lighting & Unmetered Class</u>						
15	Metered Lighting GML	13,509	\$ 871	\$ 817	\$ (54)	(6.2)
16	Unmetered Lighting GUL	79,208	19,368	18,197	(1,171)	(6.0)
17	Unmetered GU-XL / LED	20,629	6,931	6,928	(3)	(0.0)
18	Unmetered GU	95,012	1,649	2,153	504	30.6
19	Total Lighting & Unmetered	208,358	\$ 28,819	\$ 28,096	\$ (724)	(2.5)
<u>Self-generation Class</u>						
20	Small Self-generation GSG-1	-	\$ -	\$ -	\$ -	NA
21	Large Self-generation GSG-2	63,483	1,594	1,633	39	2.4
22	Total Self-generation	63,483	\$ 1,594	\$ 1,633	\$ 39	2.4
23	Total Bundled Service	30,433,455	\$ 1,157,741	\$ 1,414,400	\$ 256,659	22.2
ROA SERVICE						
<u>Secondary Class</u>						
24	Energy-only GS	24,480	\$ 1,055	\$ 1,293	\$ 238	22.6
25	Demand GSD	180,859	6,116	7,111	994	16.3
26	Total Secondary	205,339	\$ 7,171	\$ 8,404	\$ 1,233	17.2
<u>Primary Class</u>						
27	Energy-only GP	74,331	\$ 1,028	\$ 1,272	\$ 244	23.7
28	Demand GPD	3,417,757	17,997	22,729	4,731	26.3
29	Total Primary	3,492,088	\$ 19,025	\$ 24,000	\$ 4,975	26.2
30	Total ROA Service	3,697,427	\$ 26,196	\$ 32,404	\$ 6,208	23.7
31	Total Jurisdictional Service	34,130,882	\$ 1,183,937	\$ 1,446,804	\$ 262,867	22.2
32	Rounding			(112)		
33	Total Jurisdictional Base Revenues			\$ 1,446,693		

Schedule F-2.1

MICHIGAN PUBLIC SERVICE COMMISSION
 Consumers Energy Company
 Calculation of Rate Design Targets
 (\$000)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-2)
 Schedule: F-2.1
 Page: 1 of 1
 Witness: HWMiller
 Date: February 2020

Line	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
No.	Description	Jurisdictional	Residential	Secondary Class		GP	Primary Class		EIP	GML	GUL	GU-XL/LED	GU	Self Gen. Class		Source
				GS	GSD		GPD	GPTU						GSG-1	GSG-2	
Production																
1	Capacity (SRM)	\$ 1,171,068	\$ 603,368	\$ 150,216	\$ 118,769	\$ 34,914	\$ 126,193	\$ 133,644	\$ -	\$ -	\$ -	\$ -	\$ 2,311	\$ -	\$ 1,654	
	<u>Capacity Adjustments</u> ¹															
2	DR Credits	0	(2,220)	741	13	172	623	659	-	-	-	-	11	-	-	WP-HWM-9
3	Energy Intensive	(0)	(1,600)	(398)	(315)	(93)	(335)	(354)	3,101	-	-	-	(6)	-	-	Alloc. Line 1
4	Self Generation	-	853	212	168	49	178	189	-	-	-	-	3	-	(1,654)	Alloc. Line 1
5	Total Adjustments	(0)	(2,967)	555	(134)	129	466	494	3,101	-	-	-	9	-	(1,654)	
6	Capacity (SRM) Design Target	\$ 1,171,068	\$ 600,402	\$ 150,771	\$ 118,634	\$ 35,043	\$ 126,659	\$ 134,138	\$ 3,101	\$ -	\$ -	\$ -	\$ 2,320	\$ -	\$ -	
7	Energy ²	\$ 1,147,014	\$ 513,756	\$ 147,887	\$ 123,718	\$ 32,656	\$ 158,268	\$ 155,069	\$ 10,193	\$ 407	\$ 2,341	\$ 600	\$ 3,449	\$ -	\$ (1,330)	
	<u>Energy Adjustments</u> ¹															
8	Self Generation	\$ -	\$ (595)	\$ (171)	\$ (143)	\$ (38)	\$ (183)	\$ (180)	\$ (12)	\$ (0)	\$ (3)	\$ (1)	\$ (4)	\$ -	\$ 1,330	Alloc. Line 7
9	Total Adjustments	-	(595)	(171)	(143)	(38)	(183)	(180)	(12)	(0)	(3)	(1)	(4)	-	1,330	
10	Energy Design Target	\$ 1,147,014	\$ 513,161	\$ 147,715	\$ 123,575	\$ 32,618	\$ 158,085	\$ 154,889	\$ 10,182	\$ 406	\$ 2,338	\$ 599	\$ 3,445	\$ -	\$ -	
11	Total Production	\$ 2,318,082	\$ 1,113,562	\$ 298,486	\$ 242,209	\$ 67,661	\$ 284,744	\$ 289,028	\$ 13,283	\$ 406	\$ 2,338	\$ 599	\$ 5,764	\$ -	\$ -	
Transmission																
12	Transmission	472,640	222,685	58,159	47,137	13,993	63,551	59,541	4,500	148	865	225	1,096	-	741	
13	Self Generation	-	350	91	74	22	100	94	7	0	1	0	2	-	(741)	Alloc. Line 13
14	Transmission Design Target	\$ 472,640	\$ 223,035	\$ 58,250	\$ 47,211	\$ 14,015	\$ 63,650	\$ 59,635	\$ 4,507	\$ 148	\$ 866	\$ 226	\$ 1,098	\$ -	\$ -	
Delivery																
16	Distribution	\$ 1,192,763	\$ 710,685	\$ 204,094	\$ 130,229	\$ 15,478	\$ 53,731	\$ 47,629	\$ 1,338	\$ 771	\$ 15,652	\$ 7,655	\$ 2,037	\$ -	\$ 3,465	
17	Customer	253,930	191,729	40,791	6,526	2,794	6,683	3,392	228	38	1,619	12	67	-	52	
18	Subtotal Delivery	1,446,693	902,414	244,885	136,754	18,272	60,414	51,021	1,565	809	17,271	7,667	2,103	-	3,517	
	<u>Credit Adjustments</u> ³															
19	Substation Credits	\$ -	\$ 797	\$ 229	\$ 146	\$ (10)	\$ (475)	\$ (399)	\$ (117)	\$ 1	\$ 18	\$ 9	\$ 2	\$ -	\$ (200)	WP-HWM-16
20	Street Lighting Transition	-	-	-	-	-	-	-	-	-	800	(800)	-	-	-	
21	Life Line Credits	0	(10,571)	3,192	2,264	530	2,169	2,120	86	7	109	45	48	-	-	WP-HWM-10
22	Total Adjustments	0	(9,774)	3,421	2,410	520	1,694	1,722	(31)	8	926	(746)	50	-	(200)	
23	Total Delivery Design Target	\$ 1,446,693	\$ 892,640	\$ 248,306	\$ 139,165	\$ 18,792	\$ 62,108	\$ 52,742	\$ 1,534	\$ 817	\$ 18,197	\$ 6,921	\$ 2,153	\$ -	\$ 3,317	
24	Total Rate Design Target	\$ 4,237,415	\$ 2,229,237	\$ 605,042	\$ 428,585	\$ 100,468	\$ 410,503	\$ 401,405	\$ 19,324	\$ 1,371	\$ 21,402	\$ 7,746	\$ 9,016	\$ -	\$ 3,317	

Notes

1 Capacity and energy production costs adjusted to capture elements occurring outside the Cost-of-Service Study

2 Includes Non-SRM capacity related costs.

3 Credit adjustments approved in Case No. U-20134

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Residential Summer On-peak (RSP)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
 Page: 1 of 25
 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/(Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	On-peak kWh	773,421	MWh	@	0.090785	\$ 70,215	0.054372	\$ 42,052	\$ (28,163)
2	Off-peak kWh	3,618,847	MWh	@	0.061121	221,188	0.036609	132,482	(88,705)
Capacity									
3	On-peak kWh	773,421	MWh	@	0.052967	40,966	0.070983	54,900	13,934
4	Off-peak kWh	3,618,847	MWh	@	0.035660	129,048	0.047710	172,655	43,607
Winter (Oct. - May)									
5	Non-capacity All kWh	7,854,886	MWh	@	0.061121	480,099	0.042029	330,133	(149,965)
6	Capacity All kWh	7,854,886	MWh	@	0.035660	280,105	0.046840	367,923	87,818
Provisions									
7	Peak Savers (ACPC)	494,980	Bills	@	(8.00)	(3,960)	(8.00)	(3,960)	-
8	Peak Time Rewards	632	MWh	@	(0.95)	(600)	(1.00)	(632)	(32)
Critical Peak Pricing									
9	Critical-peak Charge	605	MWh	@	0.95	575	1.00	605	30
10	Non-critical Credits	40,008	MWh	@	(0.032260)	(1,291)	(0.015122)	(605)	686
11	Annual PSCR Factor	12,247,759	MWh	@	0.004440	54,380	0.004440	54,380	-
12	Total Production					\$ 1,270,724		\$ 1,149,934	\$ (120,790)
Transmission									
Summer (June - Sept.)									
13	On-peak kWh	773,421	MWh	@	-	\$ -	0.029989	\$ 23,194	\$ 23,194
14	Off-peak kWh	3,618,847	MWh	@	-	-	0.020157	72,945	72,945
Winter (Oct. - May)									
15	All kWh	7,854,886	MWh	@	-	-	0.015705	123,361	123,361
16	Total Transmission					\$ -		\$ 219,500	\$ 219,500
Delivery									
17	System Access	19,213,620	Bills	@	7.50	\$ 144,102	8.50	\$ 163,316	\$ 19,214
18	Distribution	12,247,759	MWh	@	0.047054	576,306	0.060209	737,425	161,119
Provisions									
19	Senior Citizen (RSC)	3,839,220	Bills	@	(3.75)	(14,397)	(4.25)	(16,317)	(1,920)
20	Low Income Credit (LIAC)	50,400	Bills	@	(7.50)	(378)	(30.00)	(1,512)	(1,134)
21	Income Assistance (RIA)	542,868	Bills	@	(7.50)	(4,072)	(8.50)	(4,614)	(543)
22	Total Delivery					\$ 701,562		\$ 878,298	\$ 176,736
23	Total Bundled Service					\$ 1,972,286		\$ 2,247,732	\$ 275,446
ROA SERVICE									
Delivery									
24	System Access	-	Bills	@	7.50	\$ -	8.50	\$ -	\$ -
25	Distribution	-	MWh	@	0.047054	-	0.060209	-	-
Provisions									
26	Senior Citizen (RSC)	-	Bills	@	(3.75)	-	(4.25)	-	-
27	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-	(30.00)	-	-
28	Income Assistance (RIA)	-	Bills	@	(7.50)	-	(8.50)	-	-
29	Total Delivery					\$ -		\$ -	\$ -
30	Total Bundled & ROA Service					\$ 1,972,286		\$ 2,247,732	\$ 275,446

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Residential Smart Hours (RSH)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
 Page: 2 of 25
 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kWh	3,958	MWh	@	0.088051	\$ 349	0.054372	\$ 215	\$ (133)
2	Off-peak kWh	15,482	MWh	@	0.059280	918	0.036609	567	(351)
	Capacity								
3	On-peak kWh	3,958	MWh	@	0.051101	202	0.070983	281	79
4	Off-peak kWh	15,482	MWh	@	0.034404	533	0.047710	739	206
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kWh	6,748	MWh	@	0.066561	449	0.043691	295	(154)
6	Off-peak kWh	34,787	MWh	@	0.059280	2,062	0.041306	1,437	(625)
	Capacity								
7	On-peak kWh	6,748	MWh	@	0.038629	261	0.051437	347	86
8	Off-peak kWh	34,787	MWh	@	0.034404	1,197	0.045217	1,573	376
	Provisions								
9	Peak Savers (ACPC)	-	Bills	@	(8.00)	-	(8.00)	-	-
10	Peak Time Rewards	-	MWh	@	(0.95)	-	(1.00)	-	-
	Critical Peak Pricing								
11	Critical-peak Charge	-	MWh	@	0.95	-	1.00	-	-
12	Non-critical Credits	-	MWh	@	(0.032260)	-	(0.015122)	-	-
13	Annual PSCR Factor	60,975	MWh	@	0.004440	271	0.004440	271	-
14	Total Production					\$ 6,241		\$ 5,724	\$ (517)
	Transmission								
	Summer (June - Sept.)								
15	On-peak kWh	3,958	MWh	@	-	\$ -	0.029989	\$ 119	\$ 119
16	Off-peak kWh	15,482	MWh	@	-	-	0.020157	312	312
	Winter (Oct. - May)								
17	On-peak kWh	6,748	MWh	@	-	-	0.017714	120	120
18	Off-peak kWh	34,787	MWh	@	-	-	0.015572	542	542
19	Total Transmission					\$ -		\$ 1,092	\$ 1,092
	Delivery								
20	System Access	31,992	Bills	@	7.50	\$ 240	8.50	\$ 272	\$ 32
21	Distribution	60,975	MWh	@	0.047054	2,869	0.060209	3,671	802
	Provisions								
22	Senior Citizen (RSC)	6,816	Bills	@	(3.75)	(26)	(4.25)	(29)	(3)
23	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-	(30.00)	-	-
24	Income Assistance (RIA)	624	Bills	@	(7.50)	(5)	(8.50)	(5)	(1)
25	Total Delivery					\$ 3,079		\$ 3,909	\$ 830
26	Total Bundled Service					\$ 9,320		\$ 10,725	\$ 1,405
	ROA SERVICE								
	Delivery								
27	System Access	-	Bills	@	7.50	\$ -	8.50	\$ -	\$ -
28	Distribution	-	MWh	@	0.047054	-	0.060209	-	-
	Provisions								
29	Senior Citizen (RSC)	-	Bills	@	(3.75)	-	(4.25)	-	-
30	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-	(30.00)	-	-
31	Income Assistance (RIA)	-	Bills	@	(7.50)	-	(8.50)	-	-
32	Total Delivery					\$ -		\$ -	\$ -
33	Total Bundled & ROA Service					\$ 9,320		\$ 10,725	\$ 1,405

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Present and Proposed Revenue Detail

Residential Nighttime Savers (RPM)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
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 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)	
				\$/unit	\$000	\$/unit	\$000	\$000	
	BUNDLED SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kWh	427	MWh	@	0.095146	\$ 41	0.054372	\$ 23	\$ (17)
2	Off-peak kWh	1,021	MWh	@	0.080874	83	0.045082	46	(37)
3	Super Off-peak kWh	1,394	MWh	@	0.047573	66	0.030610	43	(24)
	Capacity								
4	On-peak kWh	427	MWh	@	0.055219	24	0.070983	30	7
5	Off-peak kWh	1,021	MWh	@	0.046936	48	0.052252	53	5
6	Super Off-peak kWh	1,394	MWh	@	0.027609	38	0.031810	44	6
	Winter (Oct. - May)								
	Non-capacity								
7	On-peak kWh	892	MWh	@	0.066602	59	0.043691	39	(20)
8	Off-peak kWh	1,781	MWh	@	0.061845	110	0.048455	86	(24)
9	Super Off-peak kWh	2,530	MWh	@	0.047573	120	0.037342	94	(26)
	Capacity								
10	On-peak kWh	892	MWh	@	0.038653	34	0.051437	46	11
11	Off-peak kWh	1,781	MWh	@	0.035892	64	0.045985	82	18
12	Super Off-peak kWh	2,530	MWh	@	0.027609	70	0.032981	83	14
	Provisions								
13	Peak Savers (ACPC)	-	Bills	@	(8.00)	-	(8.00)	-	-
14	Peak Time Rewards	-	MWh	@	(0.95)	-	(1.00)	-	-
	Critical Peak Pricing								
15	Critical-peak Charge	-	MWh	@	0.95	-	1.00	-	-
16	Non-critical Credits	-	MWh	@	(0.032260)	-	(0.015122)	-	-
17	Annual PSQR Factor	8,045	MWh	@	0.004440	36	0.004440	36	-
18	Total Production					\$ 793		\$ 707	\$ (87)
	Transmission								
	Summer (June - Sept.)								
19	On-peak kWh	427	MWh	@	-	\$ -	0.029989	\$ 13	\$ 13
20	Off-peak kWh	1,021	MWh	@	-	-	0.021582	22	22
21	Super Off-peak kWh	1,394	MWh	@	-	-	0.013138	18	18
	Winter (Oct. - May)								
22	On-peak kWh	892	MWh	@	-	-	0.017714	16	16
23	Off-peak kWh	1,781	MWh	@	-	-	0.015870	28	28
24	Super Off-peak kWh	2,530	MWh	@	-	-	0.011382	29	29
25	Total Transmission					\$ -		\$ 126	\$ 126
	Delivery								
26	System Access	8,112	Bills	@	7.50	\$ 61	8.50	\$ 69	\$ 8
27	Distribution	8,045	MWh	@	0.047054	379	0.060209	484	106
	Provisions								
28	Senior Citizen (RSC)	-	Bills	@	(3.75)	-	(4.25)	-	-
29	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-	(30.00)	-	-
30	Income Assistance (RIA)	-	Bills	@	(7.50)	-	(8.50)	-	-
31	Total Delivery					\$ 439		\$ 553	\$ 114
32	Total Bundled Service					\$ 1,233		\$ 1,386	\$ 153
	ROA SERVICE								
	Delivery								
33	System Access	-	Bills	@	7.50	\$ -	8.50	\$ -	\$ -
34	Distribution	-	MWh	@	0.047054	-	0.060209	-	-
	Provisions								
35	Senior Citizen (RSC)	-	Bills	@	(3.75)	-	(4.25)	-	-
36	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-	(30.00)	-	-
37	Income Assistance (RIA)	-	Bills	@	(7.50)	-	(8.50)	-	-
38	Total Delivery					\$ -		\$ -	\$ -
39	Total Bundled & ROA Service					\$ 1,233		\$ 1,386	\$ 153

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Residential Non-Transmitting Meters (RSM)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Billing Determinants			Present				Proposed				Net Increase		
No.	Description	Quantity	Units		Rate	Revenue		Rate	Revenue		Rate	Revenue		/ (Decrease)	
					\$/unit	\$000		\$/unit	\$000		\$/unit	\$000		\$000	
BUNDLED SERVICE															
Production															
Summer (June - Sept.)															
Non-capacity															
1	First 600 kWh	33,877	MWh	@	0.060529	\$ 2,051		0.042029	\$ 1,424				\$	(627)	
2	Excess kWh	12,048	MWh	@	0.080047	964		0.054372	655					(309)	
Capacity															
3	First 600 kWh	33,877	MWh	@	0.035129	1,190		0.046840	1,587					397	
4	Excess kWh	12,048	MWh	@	0.046457	560		0.055837	673					113	
Winter (Oct. - May)															
5	Non-capacity All kWh	84,964	MWh	@	0.060529	5,143		0.042029	3,571					(1,572)	
6	Capacity All kWh	84,964	MWh	@	0.035129	2,985		0.046840	3,980					995	
7	Annual PSQR Factor	130,889	MWh	@	0.004440	581		0.004440	581					-	
8	Total Production					\$ 13,473			\$ 12,470				\$	(1,003)	
Transmission															
Summer (June - Sept.)															
9	First 600 kWh	33,877	MWh	@	-	\$ -		0.020385	\$ 691				\$	691	
10	Excess kWh	12,048	MWh	@	-	-		0.024301	293					293	
Winter (Oct. - May)															
11	All kWh	84,964	MWh	@	-	-		0.015705	1,334					1,334	
12	Total Transmission					\$ -			\$ 2,318				\$	2,318	
Delivery															
13	System Access	194,652	Bills	@	7.50	\$ 1,460		8.50	\$ 1,655				\$	195	
14	Opt-out Fee	115,452	Bills	@	3.00	346		3.00	346					-	
15	Distribution	130,889	MWh	@	0.047054	6,159		0.060209	7,881					1,722	
Provisions															
16	Senior Citizen (RSC)	-	Bills	@	(3.75)	-		(4.25)	-					-	
17	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-		(30.00)	-					-	
18	Income Assistance (RIA)	-	Bills	@	(7.50)	-		(8.50)	-					-	
19	Total Delivery					\$ 7,965			\$ 9,882				\$	1,916	
20	Total Bundled Service					\$ 21,438			\$ 24,670				\$	3,231	
ROA SERVICE															
Delivery															
21	System Access	-	Bills	@	7.50	\$ -		8.50	\$ -				\$	-	
22	Opt-out Fee	-	Bills	@	3.00	-		3.00	-					-	
23	Distribution	-	MWh	@	0.047054	-		0.060209	-					-	
Provisions															
24	Senior Citizen (RSC)	-	Bills	@	(3.75)	-		(4.25)	-					-	
25	Low Income Credit (LIAC)	-	Bills	@	(7.50)	-		(30.00)	-					-	
26	Income Assistance (RIA)	-	Bills	@	(7.50)	-		(8.50)	-					-	
27	Total Delivery					\$ -			\$ -				\$	-	
28	Total Bundled & ROA Service					\$ 21,438			\$ 24,670				\$	3,231	

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Present and Proposed Revenue Detail

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
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 Witness: HWMiller
 Date: February 2020

Secondary Energy-only (GS)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE							
	Production							
	Summer (June - Sept.)							
1	Non-capacity All kWh	1,380,836	MWh @	0.062210	\$ 85,902	0.040543	\$ 55,983	\$ (29,919)
2	Capacity All kWh	1,380,836	MWh @	0.034294	47,354	0.038384	53,002	5,648
	Winter (Oct. - May)							
3	Non-capacity All kWh	2,445,617	MWh @	0.061580	150,601	0.037458	91,608	(58,993)
4	Capacity All kWh	2,445,617	MWh @	0.033947	83,021	0.039925	97,641	14,620
5	Annual PSCR Factor	3,826,453	MWh @	0.004440	16,989	0.004440	16,989	-
6	Total Production				\$ 383,868		\$ 315,224	\$ (68,644)
	Transmission							
	Summer (June - Sept.)							
7	All kWh	1,380,836	MWh @	-	\$ -	0.014830	\$ 20,478	\$ 20,478
	Winter (Oct. - May)							
8	All kWh	2,445,617	MWh @	-	-	0.015425	37,724	37,724
9	Total Transmission				\$ -		\$ 58,201	\$ 58,201
	Delivery							
10	System Access	2,348,964	Bills @	20.00	\$ 46,979	20.00	\$ 46,979	\$ -
11	Distribution	3,826,453	MWh @	0.042472	162,517	0.052253	199,944	37,427
	Provisions							
12	Education GEI	87,692	MWh @	(0.000748)	(66)	(0.000829)	(73)	(7)
13	Total Delivery				\$ 209,431		\$ 246,850	\$ 37,419
14	Total Bundled Service				\$ 593,299		\$ 620,276	\$ 26,977
	ROA SERVICE							
	Delivery							
15	System Access	1,320	Bills @	20.00	\$ 26	20.00	\$ 26	\$ -
16	Distribution	24,480	MWh @	0.042472	1,040	0.052253	1,279	239
	Provisions							
17	Education GEI	14,695	MWh @	(0.000748)	(11)	(0.000829)	(12)	(1)
18	Total Delivery				\$ 1,055		\$ 1,293	\$ 238
19	Total Bundled & ROA Service				\$ 594,354		\$ 621,569	\$ 27,215

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Secondary Time-of-Use (GSTU)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

(a)		(b)		(c)	(d)		(e)	(f)		(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase			
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)			
					\$/unit	\$000	\$/unit	\$000	\$000			
	BUNDLED SERVICE											
	Production											
	<u>Summer (June - Sept.)</u>											
	Non-capacity											
1	On-peak kWh	154	MWh	@	0.109946	\$ 17	0.055239	\$ 9	\$	(8)		
2	Mid-peak kWh	474	MWh	@	0.086391	41	0.042076	20		(21)		
3	Off-peak kWh	605	MWh	@	0.055495	34	0.028761	17		(16)		
	Capacity											
4	On-peak kWh	154	MWh	@	0.060609	9	0.054260	8		(1)		
5	Mid-peak kWh	474	MWh	@	0.047624	23	0.045082	21		(1)		
6	Off-peak kWh	605	MWh	@	0.030592	19	0.029094	18		(1)		
	<u>Winter (Oct. - May)</u>											
	Non-capacity											
7	On-peak kWh	1,048	MWh	@	0.057913	61	0.043572	46		(15)		
8	Off-peak kWh	984	MWh	@	0.050720	50	0.035152	35		(15)		
	Capacity											
9	On-peak kWh	1,048	MWh	@	0.031925	33	0.045497	48		14		
10	Off-peak kWh	984	MWh	@	0.027960	28	0.033991	33		6		
	<u>Provisions</u>											
11	Interruptible (GSI)	-	MWh		-	-	(0.017518)	-		-		
12	Annual PSCR Factor	3,265	MWh	@	0.004440	14	0.004440	14		-		
13	Total Production					\$ 328		\$ 269	\$	(59)		
	Transmission											
	<u>Summer (June - Sept.)</u>											
14	On-peak kWh	154	MWh	@	-	\$ -	0.020964	\$ 3	\$	3		
15	Mid-peak kWh	474	MWh	@	-	-	0.017418	8		8		
16	Off-peak kWh	605	MWh	@	-	-	0.011241	7		7		
	<u>Winter (Oct. - May)</u>											
17	On-peak kWh	1,048	MWh	@	-	-	0.017577	18		18		
18	Off-peak kWh	984	MWh	@	-	-	0.013132	13		13		
19	Total Transmission					\$ -		\$ 50	\$	50		
	Delivery											
20	System Access	864	Bills	@	20.00	\$ 17	30.00	\$ 26	\$	9		
21	Peak kW	16	MW		-	-	1.34	21		21		
22	Distribution	3,265	MWh	@	0.042472	139	0.034680	113		(25)		
	<u>Provisions</u>											
23	Education GEI	-	MWh	@	(0.000748)	-	(0.000829)	-		-		
24	Total Delivery					\$ 156		\$ 161	\$	5		
25	Total Bundled Service					\$ 484		\$ 479	\$	(5)		

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Secondary Demand (GSD)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
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 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/(Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE								
Production								
<u>Summer (June - Sept.)</u>								
Non-capacity								
1	Peak kW	3,213	MW	@ 8.10	\$ 26,025	3.19	\$ 10,249	\$ (15,776)
2	All kWh	1,164,499	MWh	@ 0.043298	50,420	0.033511	39,024	(11,397)
Capacity								
3	Peak kW	3,213	MW	@ 13.04	41,898	14.23	45,721	3,823
4	All kWh	1,164,499	MWh	@ -	-	-	-	-
<u>Winter (Oct. - May)</u>								
Non-capacity								
5	Peak kW	5,795	MW	@ 6.10	35,350	1.65	9,562	(25,788)
6	All kWh	2,092,232	MWh	@ 0.040994	85,769	0.030961	64,778	(20,991)
Capacity								
7	Peak kW	5,795	MW	@ 11.04	63,977	12.69	73,539	9,562
8	All kWh	2,092,232	MWh	@ -	-	-	-	-
9	Power Factor Adjustment				1,728		(72)	(1,800)
<u>Provisions</u>								
10	Interruptible (GSI)	32,719	MWh	-	-	(0.017518)	(573)	(573)
11	Annual PSCR Factor	3,256,731	MWh	@ 0.004440	14,460	0.004440	14,460	-
12	Total Production				\$ 319,627		\$ 256,687	\$ (62,940)
Transmission								
<u>Summer (June - Sept.)</u>								
13	Peak kW	3,213	MW	@ -	\$ -	5.64	\$ 18,121	\$ 18,121
<u>Winter (Oct. - May)</u>								
14	Peak kW	5,795	MW	@ -	-	5.02	29,091	29,091
15	Total Transmission				\$ -		\$ 47,212	\$ 47,212
Delivery								
16	System Access	238,884	Bills	@ 30.00	\$ 7,167	30.00	\$ 7,167	-
17	Peak kW	9,008	MW	@ 1.15	10,359	1.34	12,071	1,712
18	Distribution	3,256,731	MWh	@ 0.029722	96,797	0.034680	112,943	16,147
19	Power Factor Adjustment				107		(37)	(144)
<u>Provisions</u>								
20	Education GEI	163,359	MWh	@ (0.000616)	(101)	(0.000659)	(108)	(7)
21	Total Delivery				\$ 114,329		\$ 132,036	\$ 17,707
22	Total Bundled Service				\$ 433,955		\$ 435,935	\$ 1,980
ROA SERVICE								
Delivery								
23	System Access	5,724	Bills	@ 30.00	\$ 172	30.00	\$ 172	-
24	Peak kW	529	MW	@ 1.15	608	1.34	709	101
25	Distribution	180,859	MWh	@ 0.029722	5,375	0.034680	6,272	897
26	Power Factor Adjustment				-		-	-
<u>Provisions</u>								
27	Education GEI	64,043	MWh	@ (0.000616)	(39)	(0.000659)	(42)	(3)
28	Total Delivery				\$ 6,116		\$ 7,111	\$ 994
29	Total Bundled & ROA Service				\$ 440,071		\$ 443,046	\$ 2,974

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 1 (GP VL 1)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Billing Determinants			Present			Proposed			Net Increase				
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	Rate	Revenue	Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE															
Production															
Summer (June - Sept.)															
1	Non-capacity All kWh	1,630	MWh	@	0.048787	\$ 80	0.034485	\$ 56	\$						(23)
2	Capacity All kWh	1,630	MWh	@	0.037321	61	0.034070	56							(5)
Winter (Oct. - May)															
3	Non-capacity All kWh	3,568	MWh	@	0.048293	172	0.031861	114							(59)
4	Capacity All kWh	3,568	MWh	@	0.036943	132	0.035420	126							(5)
5	Power Factor Adjustment					6		5							(1)
6	Annual PSCR Factor	5,198	MWh	@	0.004440	23	0.004440	23							-
7	Total Production					\$ 474		\$ 380							(94)
Transmission															
Summer (June - Sept.)															
8	All kWh	1,630	MWh	@	-	\$ -	0.013809	\$ 23	\$						23
Winter (Oct. - May)															
9	All kWh	3,568	MWh	@	-	-	0.014357	51							51
10	Total Transmission					\$ -		\$ 74							74
Delivery															
11	System Access	60	Bills	@	100.00	\$ 6	100.00	\$ 6							-
12	Distribution	5,198	MWh	@	0.005733	30	0.006421	33							4
13	Substation Ownership Credit	1,045	MWh	@	(0.000287)	(0)	(0.000767)	(1)							(1)
14	Power Factor Adjustment					0		0							0
Provisions															
15	Education GEI	-	MWh	@	(0.000571)	-	(0.000516)	-							-
16	Total Delivery					\$ 36		\$ 39							3
17	Total Bundled Service					\$ 509		\$ 492							(17)
ROA SERVICE															
Delivery															
18	System Access	-	Bills	@	100.00	\$ -	100.00	\$ -							-
19	Distribution	-	MWh	@	0.005733	-	0.006421	-							-
20	Substation Ownership Credit	-	MWh	@	(0.000287)	-	(0.000767)	-							-
21	Power Factor Adjustment					-		-							-
Provisions															
22	Education GEI	-	MWh	@	(0.000571)	-	(0.000516)	-							-
23	Total Delivery					\$ -		\$ -							-
24	Total Bundled & ROA Service					\$ 509		\$ 492							(17)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 2 (GP VL 2)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
	<u>Summer (June - Sept.)</u>								
1	Non-capacity All kWh	18,393	MWh	@	0.053957	\$ 992	0.034909	\$ 642	\$ (350)
2	Capacity All kWh	18,393	MWh	@	0.042491	782	0.034615	637	(145)
	<u>Winter (Oct. - May)</u>								
3	Non-capacity All kWh	36,264	MWh	@	0.053463	1,939	0.032253	1,170	(769)
4	Capacity All kWh	36,264	MWh	@	0.042113	1,527	0.035987	1,305	(222)
5	Power Factor Adjustment					117		84	(33)
6	Annual PSCR Factor	54,657	MWh	@	0.004440	243	0.004440	243	-
7	Total Production					\$ 5,600		\$ 4,080	\$ (1,520)
	Transmission								
	<u>Summer (June - Sept.)</u>								
8	All kWh	18,393	MWh	@	-	\$ -	0.014030	\$ 258	\$ 258
	<u>Winter (Oct. - May)</u>								
9	All kWh	36,264	MWh	@	-	-	0.014587	529	529
10	Total Transmission					\$ -		\$ 787	\$ 787
	Delivery								
11	System Access	564	Bills	@	100.00	\$ 56	100.00	\$ 56	\$ -
12	Distribution	54,657	MWh	@	0.007723	422	0.010934	598	176
13	Substation Ownership Credit	12,101	MWh	@	(0.000287)	(3)	(0.002180)	(26)	(23)
14	Power Factor Adjustment					9		13	3
	<u>Provisions</u>								
15	Education GEI	-	MWh	@	(0.000571)	-	(0.000516)	-	-
16	Total Delivery					\$ 484		\$ 640	\$ 156
17	Total Bundled Service					<u>\$ 6,084</u>		<u>\$ 5,507</u>	<u>\$ (577)</u>
	ROA SERVICE								
	Delivery								
18	System Access	24	Bills	@	100.00	\$ 2	100.00	\$ 2	\$ -
19	Distribution	3,058	MWh	@	0.007723	24	0.010934	33	10
20	Substation Ownership Credit	-	MWh	@	(0.000287)	-	(0.000287)	-	-
21	Power Factor Adjustment					-		-	-
	<u>Provisions</u>								
22	Education GEI	-	MWh	@	(0.000571)	-	(0.000516)	-	-
23	Total Delivery					\$ 26		\$ 36	\$ 10
24	Total Bundled & ROA Service					\$ 6,110		\$ 5,543	\$ (567)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy-only Voltage Level 3 (GP VL 3)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE							
	Production							
	Summer (June - Sept.)							
1	Non-capacity All kWh	312,395	MWh	@	0.058898	\$ 18,399	0.035595 \$ 11,120	\$ (7,280)
2	Capacity All kWh	312,395	MWh	@	0.047432	14,818	0.035443 11,072	(3,745)
	Winter (Oct. - May)							
3	Non-capacity All kWh	580,243	MWh	@	0.058404	33,889	0.032887 19,082	(14,806)
4	Capacity All kWh	580,243	MWh	@	0.047054	27,303	0.036847 21,380	(5,923)
5	Power Factor Adjustment					1,224	812	(412)
6	Annual PSCR Factor	892,638	MWh	@	0.004440	3,963	0.004440 3,963	-
7	Total Production					\$ 99,596	\$ 67,430	\$ (32,165)
	Transmission							
	Summer (June - Sept.)							
8	All kWh	312,395	MWh	@	-	\$ -	0.014366 \$ 4,488	\$ 4,488
	Winter (Oct. - May)							
9	All kWh	580,243	MWh	@	-	-	0.014936 8,667	8,667
10	Total Transmission					\$ -	\$ 13,154	\$ 13,154
	Delivery							
11	System Access	18,552	Bills	@	100.00	\$ 1,855	100.00 \$ 1,855	\$ -
12	Distribution	892,638	MWh	@	0.013386	11,949	0.016643 14,856	2,907
13	Power Factor Adjustment					154	193	39
	Provisions							
14	Education GEI	121,808	MWh	@	(0.000571)	(70)	(0.000516) (63)	7
15	Total Delivery					\$ 13,889	\$ 16,841	\$ 2,953
16	Total Bundled Service					\$ 113,484	\$ 97,426	\$ (16,058)
	ROA SERVICE							
	Delivery							
17	System Access	624	Bills	@	100.00	\$ 62	100.00 \$ 62	\$ -
18	Distribution	71,273	MWh	@	0.013386	954	0.016643 1,186	232
19	Power Factor Adjustment					1	2	0
	Provisions							
20	Education GEI	27,722	MWh	@	(0.000571)	(16)	(0.000516) (14)	2
21	Total Delivery					\$ 1,002	\$ 1,236	\$ 234
22	Total Bundled & ROA Service					\$ 114,486	\$ 98,662	\$ (15,824)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Demand Voltage Level 1 (GPD VL 1)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

(a)		(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line No.	Description	Billing Determinants		Present		Proposed		Net Increase
		Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE								
Production								
<u>Summer (June - Sept.)</u>								
Non-capacity								
1	On-peak kW	658	MW	@ 9.34	\$ 6,146	6.33	\$ 4,165	\$ (1,981)
2	On-peak kWh	86,071	MWh	@ 0.026510	2,282	0.027643	2,379	98
3	Off-peak kWh	292,468	MWh	@ 0.017182	5,025	0.017803	5,207	182
4	Capacity On-peak kW	658	MW	@ 13.82	9,094	14.27	9,390	296
<u>Winter (Oct. - May)</u>								
Non-capacity								
5	On-peak kW	1,406	MW	@ 8.34	11,726	5.33	7,494	(4,232)
6	On-peak kWh	183,943	MWh	@ 0.021641	3,981	0.022639	4,164	184
7	Off-peak kWh	633,778	MWh	@ 0.018953	12,012	0.021052	13,342	1,330
8	Capacity On-peak kW	1,406	MW	@ 12.82	18,025	13.27	18,658	633
<u>Provisions</u>								
Interruptible GI								
9	Summer On-peak kW	186	MW	@ (7.00)	(1,302)	(7.00)	(1,302)	-
10	Winter On-peak kW	371	MW	@ (6.00)	(2,226)	(6.00)	(2,226)	-
Interruptible GI-2								
11	Summer Capacity & Transmission	25,343	MWh	@ 0.028890	732	0.024857	630	(102)
12	Winter Capacity & Transmission	51,308	MWh	@ 0.025559	1,311	0.023762	1,219	(92)
13	LMP	76,651	MWh	@ 0.029146	2,234	0.029146	2,234	-
14	Power Factor Adjustment				(261)		(247)	14
15	Annual PSQR Factor	1,196,260	MWh	@ 0.004440	5,311	0.004440	5,311	-
16	Total Production				\$ 74,090		\$ 70,419	\$ (3,671)
Transmission								
<u>Summer (June - Sept.)</u>								
17	On-peak kW	658	MW	@ 6.58	\$ 4,330	7.03	\$ 4,626	\$ 296
<u>Winter (Oct. - May)</u>								
18	On-peak kW	1,406	MW	@ 6.58	9,251	6.55	9,209	(42)
19	Total Transmission				\$ 13,581		\$ 13,835	\$ 254
Delivery								
20	System Access	252	Bills	@ 200.00	\$ 50	200.00	\$ 50	-
21	Maximum kW	3,086	MW	@ 0.91	2,808	0.70	2,160	(648)
22	Substation Ownership	154	MW	@ (0.45)	(69)	(0.34)	(52)	17
23	Joint Substation Ownership	-	MW	@ (0.31)	-	(0.23)	-	-
24	Distribution	1,272,911	MWh	@ -	-	-	-	-
25	Power Factor Adjustment				(10)		(8)	2
<u>Provisions</u>								
26	Education GEI	-	MWh	@ (0.000314)	-	(0.000259)	-	-
27	Total Delivery				\$ 2,779		\$ 2,150	\$ (629)
28	Total Bundled Service				\$ 90,450		\$ 86,404	\$ (4,046)
ROA SERVICE								
Delivery								
29	System Access	156	Bills	@ 200.00	\$ 31	200.00	\$ 31	-
30	Maximum kW	2,236	MW	@ 0.91	2,035	0.70	1,565	(470)
31	Substation Ownership	197	MW	@ (0.45)	(89)	(0.34)	(67)	22
32	Distribution	1,092,169	MWh	@ -	-	-	-	-
33	Power Factor Adjustment				(5)		(4)	1
<u>Provisions</u>								
34	Education GEI	2,690	MWh	@ (0.000314)	(1)	(0.000259)	(1)	0
35	Total Delivery				\$ 1,971		\$ 1,525	\$ (447)
36	Total Bundled & ROA Service				\$ 92,421		\$ 87,928	\$ (4,493)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Demand Voltage Level 2 (GPD VL 2)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	On-peak kW	742	MW	@	9.84	\$ 7,301	6.41	\$ 4,755	\$ (2,547)
2	On-peak kWh	102,315	MWh	@	0.034239	3,503	0.027983	2,863	(640)
3	Off-peak kWh	304,472	MWh	@	0.022191	6,757	0.018022	5,487	(1,269)
4	Capacity On-peak kW	742	MW	@	14.32	10,625	14.50	10,759	134
	Winter (Oct. - May)								
	Non-capacity								
5	On-peak kW	1,375	MW	@	8.84	12,155	5.40	7,419	(4,736)
6	On-peak kWh	188,179	MWh	@	0.027950	5,260	0.022917	4,312	(947)
7	Off-peak kWh	568,753	MWh	@	0.024479	13,923	0.021311	12,121	(1,802)
8	Capacity On-peak kW	1,375	MW	@	13.32	18,315	13.48	18,535	220
	Provisions								
	Interruptible GI								
9	Summer On-peak kW	36	MW	@	(7.00)	(252)	(7.00)	(252)	-
10	Winter On-peak kW	71	MW	@	(6.00)	(426)	(6.00)	(426)	-
	Interruptible GI-2								
11	Summer Capacity & Transmission	-	MWh	@	0.039453	-	0.026704	-	-
12	Winter Capacity & Transmission	-	MWh	@	0.036122	-	0.025668	-	-
13	LMP	-	MWh	@	-	-	-	-	-
14	Power Factor Adjustment					(130)		(110)	20
15	Annual PSQR Factor	1,163,719	MWh	@	0.004440	5,167	0.004440	5,167	-
16	Total Production					\$ 82,197		\$ 70,629	\$ (11,568)
	Transmission								
	Summer (June - Sept.)								
17	On-peak kW	742	MW	@	6.71	\$ 4,979	7.14	\$ 5,298	\$ 319
	Winter (Oct. - May)								
18	On-peak kW	1,375	MW	@	6.71	9,226	6.65	9,144	(83)
19	Total Transmission					\$ 14,205		\$ 14,442	\$ 237
	Delivery								
20	System Access	612	Bills	@	200.00	\$ 122	200.00	\$ 122	\$ -
21	Maximum kW	2,717	MW	@	1.86	5,054	2.69	7,309	2,255
22	Substation Ownership	202	MW	@	(0.97)	(196)	(0.96)	(194)	2
23	Distribution	1,163,719	MWh	@	-	-	-	-	-
24	Power Factor Adjustment					(8)		(12)	(4)
	Provisions								
25	Education GEI	21,349	MWh	@	(0.000314)	(7)	(0.000259)	(6)	1
26	Total Delivery					\$ 4,965		\$ 7,219	\$ 2,254
27	Total Bundled Service					\$ 101,368		\$ 92,291	\$ (9,077)
	ROA SERVICE								
	Delivery								
28	System Access	480	Bills	@	200.00	\$ 96	200.00	\$ 96	\$ -
29	Maximum kW	2,901	MW	@	1.86	5,396	2.69	7,804	2,408
30	Substation Ownership	231	MW	@	(0.97)	(224)	(0.96)	(222)	2
31	Distribution	1,317,428	MWh	@	-	-	-	-	-
32	Power Factor Adjustment					15		22	7
	Provisions								
33	Education GEI	70,229	MWh	@	(0.000314)	(22)	(0.000259)	(18)	4
34	Total Delivery					\$ 5,261		\$ 7,682	\$ 2,421
35	Total Bundled & ROA Service					\$ 106,628		\$ 99,972	\$ (6,656)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Demand Voltage Level 3 (GPD VL 3)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
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 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	On-peak kW	1,885	MW	@	10.34	\$ 19,491	6.53	\$ 12,316	\$ (7,175)
2	On-peak kWh	251,101	MWh	@	0.042156	10,585	0.028533	7,165	(3,421)
3	Off-peak kWh	686,223	MWh	@	0.027322	18,749	0.018376	12,610	(6,139)
4	Capacity On-peak kW	1,885	MW	@	14.82	27,936	14.85	27,992	57
Winter (Oct. - May)									
Non-capacity									
5	On-peak kW	3,173	MW	@	9.34	29,636	5.50	17,457	(12,179)
6	On-peak kWh	416,607	MWh	@	0.034413	14,337	0.023368	9,735	(4,601)
7	Off-peak kWh	1,160,357	MWh	@	0.030139	34,972	0.021730	25,215	(9,757)
8	Capacity On-peak kW	3,173	MW	@	13.82	43,851	13.80	43,787	(63)
Provisions									
Interruptible GI									
9	Summer On-peak kW	9	MW	@	(7.00)	(63)	(7.00)	(63)	-
10	Winter On-peak kW	19	MW	@	(6.00)	(114)	(6.00)	(114)	-
Interruptible GI-2									
11	Summer Capacity & Transmission	-	MWh	@	0.049807	-	0.030487	-	-
12	Winter Capacity & Transmission	-	MWh	@	0.046476	-	0.030403	-	-
13	LMP	-	MWh	@	-	-	-	-	-
14	Power Factor Adjustment					206		161	(45)
15	Annual PSCR Factor	2,514,288	MWh	@	0.004440	11,163	0.004440	11,163	-
16	Total Production					\$ 210,749		\$ 167,425	\$ (43,324)
Transmission									
Summer (June - Sept.)									
17	On-peak kW	1,885	MW	@	6.98	\$ 13,157	7.31	\$ 13,779	\$ 622
Winter (Oct. - May)									
18	On-peak kW	3,173	MW	@	6.98	22,148	6.81	21,608	(539)
19	Total Transmission					\$ 35,305		\$ 35,387	\$ 83
Delivery									
20	System Access	11,844	Bills	@	200.00	\$ 2,369	200.00	\$ 2,369	-
21	Maximum kW	6,381	MW	@	3.60	22,972	4.58	29,225	6,253
22	Distribution	2,514,288	MWh	@	-	-	-	-	-
23	Power Factor Adjustment					24		33	9
Provisions									
24	Education GEI	125,813	MWh	@	(0.000314)	(40)	(0.000259)	(33)	7
25	Total Delivery					\$ 25,325		\$ 31,594	\$ 6,269
26	Total Bundled Service					\$ 271,378		\$ 234,406	\$ (36,972)
ROA SERVICE									
Delivery									
27	System Access	3,612	Bills	@	200.00	\$ 722	200.00	\$ 722	-
28	Maximum kW	2,799	MW	@	3.60	10,076	4.58	12,819	2,743
29	Distribution	1,008,160	MWh	@	-	-	-	-	-
30	Power Factor Adjustment					15		20	5
Provisions									
31	Education GEI	153,269	MWh	@	(0.000314)	(48)	(0.000259)	(40)	8
32	Total Delivery					\$ 10,766		\$ 13,522	\$ 2,757
33	Total Bundled & ROA Service					\$ 282,144		\$ 247,928	\$ (34,215)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 1 (GPTU VL 1)

Case No.: U-20697
Exhibit No.: A-16 (HWM-3)
Schedule: F-3.0
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Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	High-peak kWh	8,076	MWh	@	0.093060	\$ 752	0.050387	\$ 407	\$ (345)
2	Mid-peak kWh	10,913	MWh	@	0.081694	892	0.043807	478	(413)
3	Low-peak kWh	21,770	MWh	@	0.065924	1,435	0.034060	741	(694)
4	Off-peak kWh	40,946	MWh	@	0.049286	2,018	0.024837	1,017	(1,001)
Capacity									
5	High-peak kWh	8,076	MWh	@	0.025831	209	0.053210	430	221
6	Mid-peak kWh	10,913	MWh	@	0.022676	247	0.050795	554	307
7	Low-peak kWh	21,770	MWh	@	0.018299	398	0.040795	888	490
8	Off-peak kWh	40,946	MWh	@	0.013681	560	0.027556	1,128	568
Winter (Oct. - May)									
Non-capacity									
9	High-peak kWh	15,700	MWh	@	0.058501	918	0.039569	621	(297)
10	Mid-peak kWh	21,428	MWh	@	0.056823	1,218	0.037405	802	(416)
11	Off-peak kWh	122,729	MWh	@	0.049918	6,126	0.034740	4,264	(1,863)
Capacity									
12	High-peak kWh	15,700	MWh	@	0.016239	255	0.028765	452	197
13	Mid-peak kWh	21,428	MWh	@	0.015773	338	0.028752	616	278
14	Off-peak kWh	122,729	MWh	@	0.013856	1,701	0.024746	3,037	1,337
15	Power Factor Adjustment					69		62	(7)
15	Annual PSCR Factor	241,562	MWh	@	0.004440	1,073	0.004440	1,073	-
16	Total Production					\$ 18,208		\$ 16,570	\$ (1,638)
Transmission									
Summer (June - Sept.)									
17	High-peak kWh	8,076	MWh	@	-	\$ -	0.023683	\$ 191	\$ 191
18	Mid-peak kWh	10,913	MWh	@	-	-	0.022609	247	247
19	Low-peak kWh	21,770	MWh	@	-	-	0.018158	395	395
20	Off-peak kWh	40,946	MWh	@	-	-	0.012265	502	502
21	Winter (Oct. - May)								
22	High-peak kWh	15,700	MWh	@	-	-	0.012803	201	201
23	Mid-peak kWh	21,428	MWh	@	-	-	0.012797	274	274
24	Off-peak kWh	122,729	MWh	@	-	-	0.011014	1,352	1,352
25	Total Transmission					\$ -		\$ 3,162	\$ 3,162
Delivery									
26	System Access	192	Bills	@	200.00	\$ 38	200.00	\$ 38	\$ -
27	Maximum kW	423	MW	@	0.91	385	0.70	296	(89)
28	Substation Ownership	-	MW	@	(0.45)	-	(0.34)	-	-
29	Distribution	241,562	MWh	@	-	-	-	-	-
30	Power Factor Adjustment					2		1	(0)
Provisions									
31	Education GEI	-	MWh	@	(0.000314)	-	(0.000259)	-	-
32	Total Delivery					\$ 425		\$ 336	\$ (89)
33	Total Bundled Service					\$ 18,633		\$ 20,068	\$ 1,435

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 2 (GPTU VL 2)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	High-peak kWh	29,664	MWh	@	0.095060	\$ 2,820	0.051007	\$ 1,513	\$ (1,307)
2	Mid-peak kWh	39,567	MWh	@	0.083694	3,312	0.044346	1,755	(1,557)
3	Low-peak kWh	78,292	MWh	@	0.067924	5,318	0.034479	2,699	(2,618)
4	Off-peak kWh	186,338	MWh	@	0.051286	9,557	0.025142	4,685	(4,872)
	Capacity								
5	High-peak kWh	29,664	MWh	@	0.027831	826	0.054061	1,604	778
6	Mid-peak kWh	39,567	MWh	@	0.024676	976	0.051608	2,042	1,066
7	Low-peak kWh	78,292	MWh	@	0.020299	1,589	0.041448	3,245	1,656
8	Off-peak kWh	186,338	MWh	@	0.015681	2,922	0.027997	5,217	2,295
	Winter (Oct. - May)								
	Non-capacity								
9	High-peak kWh	60,498	MWh	@	0.060501	3,660	0.040056	2,423	(1,237)
10	Mid-peak kWh	82,058	MWh	@	0.058823	4,827	0.037865	3,107	(1,720)
11	Off-peak kWh	475,325	MWh	@	0.051918	24,678	0.035167	16,716	(7,962)
	Capacity								
12	High-peak kWh	60,498	MWh	@	0.018239	1,103	0.029225	1,768	665
13	Mid-peak kWh	82,058	MWh	@	0.017773	1,458	0.029212	2,397	939
14	Off-peak kWh	475,325	MWh	@	0.015856	7,537	0.025142	11,951	4,414
15	Power Factor Adjustment					(40)		(34)	5
16	Annual PSCR Factor	951,742	MWh	@	0.004440	4,226	0.004440	4,226	-
17	Total Production					\$ 74,769		\$ 65,313	\$ (9,456)
	Transmission								
	Summer (June - Sept.)								
18	High-peak kWh	29,664	MWh	@	-	\$ -	0.024062	\$ 714	\$ 714
19	Mid-peak kWh	39,567	MWh	@	-	-	0.022971	909	909
20	Low-peak kWh	78,292	MWh	@	-	-	0.018449	1,444	1,444
21	Off-peak kWh	186,338	MWh	@	-	-	0.012461	2,322	2,322
	Winter (Oct. - May)								
22	High-peak kWh	60,498	MWh	@	-	-	0.013008	787	787
23	Mid-peak kWh	82,058	MWh	@	-	-	0.013002	1,067	1,067
24	Off-peak kWh	475,325	MWh	@	-	-	0.011190	5,319	5,319
25	Total Transmission					\$ -		\$ 12,562	\$ 12,562
	Delivery								
26	System Access	900	Bills	@	200.00	\$ 180	200.00	\$ 180	\$ -
27	Maximum kW	2,552	MW	@	1.86	4,747	2.69	6,865	2,118
28	Substation Ownership	471	MW	@	(0.97)	(457)	(0.96)	(452)	5
29	Distribution	951,742	MWh	@	-	-	-	-	-
30	Power Factor Adjustment					(2)		(4)	(1)
	Provisions								
31	Education GEI	-	MWh	@	(0.000314)	-	(0.000259)	-	-
32	Total Delivery					\$ 4,467		\$ 6,589	\$ 2,122
33	Total Bundled Service					\$ 79,236		\$ 84,464	\$ 5,228

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Time-of-Use Voltage Level 3 (GPTU VL 3)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
	Summer (June - Sept.)								
	Non-capacity								
1	High-peak kWh	122,263	MWh	@	0.100060	\$ 12,234	0.052009	\$ 6,359	\$ (5,875)
2	Mid-peak kWh	161,365	MWh	@	0.088694	14,312	0.045218	7,297	(7,016)
3	Low-peak kWh	319,824	MWh	@	0.072924	23,323	0.035157	11,244	(12,079)
4	Off-peak kWh	528,442	MWh	@	0.056286	29,744	0.025637	13,548	(16,196)
	Capacity								
5	High-peak kWh	122,263	MWh	@	0.032831	4,014	0.055354	6,768	2,754
6	Mid-peak kWh	161,365	MWh	@	0.029676	4,789	0.052842	8,527	3,738
7	Low-peak kWh	319,824	MWh	@	0.025299	8,091	0.042439	13,573	5,482
8	Off-peak kWh	528,442	MWh	@	0.020681	10,929	0.028667	15,149	4,220
	Winter (Oct. - May)								
	Non-capacity								
9	High-peak kWh	213,477	MWh	@	0.065501	13,983	0.040843	8,719	(5,264)
10	Mid-peak kWh	292,339	MWh	@	0.063823	18,658	0.038609	11,287	(7,371)
11	Off-peak kWh	1,534,535	MWh	@	0.056918	87,343	0.035859	55,027	(32,316)
	Capacity								
12	High-peak kWh	213,477	MWh	@	0.023239	4,961	0.029924	6,388	1,427
13	Mid-peak kWh	292,339	MWh	@	0.022773	6,657	0.029911	8,744	2,087
14	Off-peak kWh	1,534,535	MWh	@	0.020856	32,004	0.025743	39,504	7,499
15	Power Factor Adjustment					397		310	(86)
16	Annual PSQR Factor	3,172,245	MWh	@	0.004440	14,085	0.004440	14,085	-
17	Total Production					\$ 285,523		\$ 226,527	\$ (58,995)
	Transmission								
	Summer (June - Sept.)								
18	High-peak kWh	122,263	MWh	@	-	\$ -	0.024637	\$ 3,012	\$ 3,012
19	Mid-peak kWh	161,365	MWh	@	-	-	0.023520	3,795	3,795
20	Low-peak kWh	319,824	MWh	@	-	-	0.018890	6,041	6,041
21	Off-peak kWh	528,442	MWh	@	-	-	0.012759	6,742	6,742
	Winter (Oct. - May)								
22	High-peak kWh	213,477	MWh	@	-	-	0.013319	2,843	2,843
23	Mid-peak kWh	292,339	MWh	@	-	-	0.013313	3,892	3,892
24	Off-peak kWh	1,534,535	MWh	@	-	-	0.011458	17,583	17,583
25	Total Transmission					\$ -		\$ 43,909	\$ 43,909
	Delivery								
26	System Access	11,292	Bills	@	200.00	\$ 2,258	200.00	\$ 2,258	\$ -
27	Maximum kW	9,438	MW	@	3.60	33,977	4.58	43,226	9,249
28	Distribution	3,172,245	MWh	@	-	-	-	-	-
29	Power Factor Adjustment					50		67	17
	Provisions								
30	Education GEI	-	MWh	@	(0.000314)	-	(0.000259)	-	-
31	Total Delivery					\$ 36,285		\$ 45,551	\$ 9,266
32	Total Bundled Service					\$ 321,808		\$ 315,988	\$ (5,820)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 1 (EIP VL 1)

Case No.: U-20697
Exhibit No.: A-16 (HWM-3)
Schedule: F-3.0
Page: 17 of 25
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)	
				\$/unit	\$000	\$/unit	\$000	\$000	
BUNDLED SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	Critical-peak kWh	269	MWh	@	0.094746	\$ 25	0.063300	\$ 17	\$ (8)
2	High-peak kWh	6,008	MWh	@	0.063164	379	0.042200	254	(126)
3	Mid-peak kWh	6,187	MWh	@	0.058664	363	0.037326	231	(132)
4	Low-peak kWh	43,799	MWh	@	0.047172	2,066	0.029767	1,304	(762)
5	Off-peak kWh	40,944	MWh	@	0.033312	1,364	0.020329	832	(532)
Capacity									
6	Critical-peak kWh	269	MWh	@	0.042915	12	0.024182	7	(5)
7	High-peak kWh	6,008	MWh	@	0.028610	172	0.016121	97	(75)
8	Mid-peak kWh	6,187	MWh	@	0.026572	164	0.015758	97	(67)
9	Low-peak kWh	43,799	MWh	@	0.021367	936	0.012961	568	(368)
10	Off-peak kWh	40,944	MWh	@	0.015089	618	0.008294	340	(278)
Winter (Oct. - May)									
Non-capacity									
11	Critical-peak kWh	106	MWh	@	0.090809	10	0.050786	5	(4)
12	High-peak kWh	11,528	MWh	@	0.060539	698	0.033857	390	(308)
13	Mid-peak kWh	11,757	MWh	@	0.047652	560	0.032241	379	(181)
14	Off-peak kWh	144,383	MWh	@	0.033492	4,836	0.028535	4,120	(716)
Capacity									
15	Critical-peak kWh	106	MWh	@	0.041132	4	0.012126	1	(3)
16	High-peak kWh	11,528	MWh	@	0.027421	316	0.008084	93	(223)
17	Mid-peak kWh	11,757	MWh	@	0.021584	254	0.007975	94	(160)
18	Off-peak kWh	144,383	MWh	@	0.015170	2,190	0.006989	1,009	(1,181)
19	Power Factor Adjustment					(27)	(17)		9
20	Annual PSCR Factor	264,981	MWh	@	0.004440	1,177	0.004440	1,177	-
21	Total Production				\$	16,117	\$	10,997	\$ (5,121)
Transmission									
Summer (June - Sept.)									
22	Critical-peak kWh	269	MWh	@	-	\$ -	0.035057	\$ 9	\$ 9
23	High-peak kWh	6,008	MWh	@	-	-	0.023371	140	140
24	Mid-peak kWh	6,187	MWh	@	-	-	0.022845	141	141
25	Low-peak kWh	43,799	MWh	@	-	-	0.018789	823	823
26	Off-peak kWh	40,944	MWh	@	-	-	0.012024	492	492
Winter (Oct. - May)									
27	Critical-peak kWh	106	MWh	@	-	-	0.017580	2	2
28	High-peak kWh	11,528	MWh	@	-	-	0.011720	135	135
29	Mid-peak kWh	11,757	MWh	@	-	-	0.011562	136	136
30	Off-peak kWh	144,383	MWh	@	-	-	0.010132	1,463	1,463
31	Total Transmission				\$	-	\$	3,342	\$ 3,342
Delivery									
32	System Access	72	Bills	@	200.00	\$ 14	200.00	\$ 14	\$ -
33	Maximum kW	695	MW	@	0.91	632	0.70	487	(146)
34	Substation Ownership	-	MW	@	(0.45)	-	(0.34)	-	-
35	Power Factor Adjustment					(1)	(1)		0
36	Distribution	264,981	MWh	@	-	-	-	-	-
37	Total Delivery				\$	646	\$	500	\$ (146)
38	Total Bundled Service				\$	16,763	\$	14,839	\$ (1,924)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 2 (EIP VL 2)

Case No.: U-20697
Exhibit No.: A-16 (HWM-3)
Schedule: F-3.0
Page: 18 of 25
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE								
Production								
<u>Summer (June - Sept.)</u>								
Non-capacity								
1	Critical-peak kWh	46	MWh @	0.099746	\$ 5	0.064079	\$ 3	\$ (2)
2	High-peak kWh	1,403	MWh @	0.068164	96	0.042719	60	(36)
3	Mid-peak kWh	1,473	MWh @	0.063664	94	0.037785	56	(38)
4	Low-peak kWh	10,345	MWh @	0.052172	540	0.030133	312	(228)
5	Off-peak kWh	12,755	MWh @	0.038312	489	0.020579	262	(226)
Capacity								
6	Critical-peak kWh	46	MWh @	0.047915	2	0.024569	1	(1)
7	High-peak kWh	1,403	MWh @	0.033610	47	0.016379	23	(24)
8	Mid-peak kWh	1,473	MWh @	0.031572	47	0.016010	24	(23)
9	Low-peak kWh	10,345	MWh @	0.026367	273	0.013168	136	(137)
10	Off-peak kWh	12,755	MWh @	0.020089	256	0.008427	107	(149)
<u>Winter (Oct. - May)</u>								
Non-capacity								
11	Critical-peak kWh	43	MWh @	0.095809	4	0.051411	2	(2)
12	High-peak kWh	3,031	MWh @	0.065539	199	0.034273	104	(95)
13	Mid-peak kWh	3,113	MWh @	0.052652	164	0.032638	102	(62)
14	Off-peak kWh	46,013	MWh @	0.038492	1,771	0.028886	1,329	(442)
Capacity								
15	Critical-peak kWh	43	MWh @	0.046132	2	0.012320	1	(1)
16	High-peak kWh	3,031	MWh @	0.032421	98	0.008213	25	(73)
17	Mid-peak kWh	3,113	MWh @	0.026584	83	0.008103	25	(58)
18	Off-peak kWh	46,013	MWh @	0.020170	928	0.007101	327	(601)
19	Power Factor Adjustment				(24)		(14)	10
20	Annual PSCR Factor	78,222	MWh @	0.004440	347	0.004440	347	-
21	Total Production				\$ 5,420		\$ 3,232	\$ (2,188)
Transmission								
<u>Summer (June - Sept.)</u>								
22	Critical-peak kWh	46	MWh @	-	\$ -	0.035618	\$ 2	\$ 2
23	High-peak kWh	1,403	MWh @	-	-	0.023745	33	33
24	Mid-peak kWh	1,473	MWh @	-	-	0.023211	34	34
25	Low-peak kWh	10,345	MWh @	-	-	0.019090	197	197
26	Off-peak kWh	12,755	MWh @	-	-	0.012216	156	156
<u>Winter (Oct. - May)</u>								
27	Critical-peak kWh	43	MWh @	-	-	0.017861	1	1
28	High-peak kWh	3,031	MWh @	-	-	0.011908	36	36
29	Mid-peak kWh	3,113	MWh @	-	-	0.011747	37	37
30	Off-peak kWh	46,013	MWh @	-	-	0.010294	474	474
31	Total Transmission				\$ -		\$ 970	\$ 970
Delivery								
32	System Access	84	Bills @	200.00	\$ 17	200.00	\$ 17	\$ -
33	Maximum kW	438	MW @	1.86	815	2.69	1,178	364
34	Substation Ownership	124	MW @	(0.97)	(120)	(0.96)	(119)	1
35	Power Factor Adjustment				(3)		(5)	(2)
36	Distribution	78,222	MWh @	-	-	-	-	-
37	Total Delivery				\$ 708		\$ 1,071	\$ 363
38	Total Bundled Service				\$ 6,128		\$ 5,273	\$ (855)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Primary Energy Intensive Level 3 (EIP VL 3)

Case No.: U-20697
Exhibit No.: A-16 (HWM-3)
Schedule: F-3.0
Page: 19 of 25
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
BUNDLED SERVICE									
Production									
Summer (June - Sept.)									
Non-capacity									
1	Critical-peak kWh	7	MWh	@	0.096746	\$ 1	0.065338	\$ 0	\$ (0)
2	High-peak kWh	212	MWh	@	0.065164	14	0.043559	9	(5)
3	Mid-peak kWh	232	MWh	@	0.060664	14	0.038528	9	(5)
4	Low-peak kWh	2,522	MWh	@	0.049172	124	0.030725	77	(47)
5	Off-peak kWh	1,967	MWh	@	0.035312	69	0.020984	41	(28)
Capacity									
6	Critical-peak kWh	7	MWh	@	0.044915	0	0.025157	0	(0)
7	High-peak kWh	212	MWh	@	0.030610	6	0.016771	4	(3)
8	Mid-peak kWh	232	MWh	@	0.028572	7	0.016393	4	(3)
9	Low-peak kWh	2,522	MWh	@	0.023367	59	0.013483	34	(25)
10	Off-peak kWh	1,967	MWh	@	0.017089	34	0.008628	17	(17)
Winter (Oct. - May)									
Non-capacity									
11	Critical-peak kWh	3	MWh	@	0.092809	0	0.052421	0	(0)
12	High-peak kWh	161	MWh	@	0.062539	10	0.034947	6	(4)
13	Mid-peak kWh	218	MWh	@	0.049652	11	0.033279	7	(4)
14	Off-peak kWh	10,012	MWh	@	0.035492	355	0.029454	295	(60)
Capacity									
15	Critical-peak kWh	3	MWh	@	0.043132	0	0.012615	0	(0)
16	High-peak kWh	161	MWh	@	0.029421	5	0.008410	1	(3)
17	Mid-peak kWh	218	MWh	@	0.023584	5	0.008296	2	(3)
18	Off-peak kWh	10,012	MWh	@	0.017170	172	0.007271	73	(99)
19	Power Factor Adjustment					(4)		(3)	1
20	Annual PSCR Factor	15,334	MWh	@	0.004440	68	0.004440	68	-
21	Total Production					\$ 951		\$ 645	\$ (305)
Transmission									
Summer (June - Sept.)									
22	Critical-peak kWh	7	MWh	@	-	\$ -	0.036470	\$ 0	\$ 0
23	High-peak kWh	212	MWh	@	-	-	0.024313	5	5
24	Mid-peak kWh	232	MWh	@	-	-	0.023766	6	6
25	Low-peak kWh	2,522	MWh	@	-	-	0.019546	49	49
26	Off-peak kWh	1,967	MWh	@	-	-	0.012509	25	25
Winter (Oct. - May)									
27	Critical-peak kWh	3	MWh	@	-	-	0.018288	0	0
28	High-peak kWh	161	MWh	@	-	-	0.012192	2	2
29	Mid-peak kWh	218	MWh	@	-	-	0.012028	3	3
30	Off-peak kWh	10,012	MWh	@	-	-	0.010540	106	106
31	Total Transmission					\$ -		\$ 195	\$ 195
Delivery									
32	System Access	72	Bills	@	200.00	\$ 14	200.00	\$ 14	\$ -
33	Maximum kW	96	MW	@	3.60	346	4.58	440	94
34	Power Factor Adjustment					(2)		(2)	(0)
35	Distribution	15,334	MWh	@	-	-	-	-	-
36	Total Delivery					\$ 358		\$ 452	\$ 94
37	Total Bundled Service					\$ 1,309		\$ 1,292	\$ (17)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Metered Lighting (GML)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)		(d)	(e)	(f)	(g)	(h)
Line		Billing Determinants			Present		Proposed		Net Increase
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	/ (Decrease)
					\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE								
	Production								
1	Secondary All kWh	12,916	MWh	@	0.051457	\$ 665	0.030773	\$ 397	\$ (267)
2	Primary All kWh	593	MWh	@	0.025253	15	0.015102	9	(6)
3	Annual PSCR Factor	13,509	MWh	@	0.004440	60	0.004440	60	-
4	Total Production					\$ 740		\$ 466	\$ (273)
	Transmission								
5	Secondary All kWh	12,916	MWh	@	-	-	0.011187	144	144
6	Primary All kWh	593	MWh	@	-	-	0.005490	3	3
7	Total Transmission	13,509				-		148	148
	Delivery								
	Secondary								
8	System Access	4,056	Bills	@	10.00	\$ 41	10.00	\$ 41	\$ -
9	Distribution	12,916	MWh	@	0.062025	801	0.058006	749	(52)
	Primary								
10	System Access	60	Bills	@	20.00	\$ 1	20.00	\$ 1	\$ -
11	Distribution	593	MWh	@	0.047268	28	0.044205	26	(2)
12	Total Delivery					\$ 871		\$ 817	\$ (54)
13	Total Bundled Service					\$ 1,610		\$ 1,431	\$ (179)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Present and Proposed Revenue Detail

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
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 Witness: HWMiller
 Date: February 2020

Unmetered Lighting (GUL)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Line	Billing Determinants						Present			Proposed			Net Increase
No.	Description	Customer	Company	Units	Watts	MWh	Rate		Revenue	Rate		Revenue	/ (Decrease)
							Service	Fixture	\$000	Service	Fixture	\$000	\$000
							\$/unit	\$/unit		\$/unit	\$/unit		
Mercury Vapor													
1	3,500 Lumens	-	-	Lights	128	- @	8.31	6.00	\$ -	8.01	5.00	\$ -	\$ -
2	7,500 Lumens	-	-	Lights	209	- @	13.58	6.00	-	13.07	5.00	-	-
3	10,000 Lumens	-	-	Lights	281	- @	18.25	6.00	-	17.58	5.00	-	-
4	20,000 Lumens	-	-	Lights	458	- @	29.75	6.00	-	28.65	5.00	-	-
5	35,000 Lumens	-	-	Lights	770	- @	50.02	6.00	-	48.16	5.00	-	-
6	50,000 Lumens	-	-	Lights	1,080	- @	70.15	6.00	-	67.55	5.00	-	-
		-	-			-			\$ -			\$ -	\$ -
High-Pressure Sodium													
7	5,000 Lumens	72	-	Lights	83	2 @	5.39	6.00	\$ 0	5.19	5.00	\$ 0	\$ (0)
8	8,500 Lumens	804	1,081,914	Lights	117	44,337 @	7.60	6.00	14,720	7.32	5.00	13,335	(1,385)
9	14,000 Lumens	1,404	161,698	Lights	171	9,762 @	11.11	6.00	2,782	10.70	5.00	2,554	(229)
10	20,000 Lumens	60	28,260	Lights	247	2,448 @	16.04	6.00	624	15.45	5.00	579	(45)
11	24,000 Lumens	540	118,944	Lights	318	13,299 @	20.66	6.00	3,182	19.89	5.00	2,971	(211)
12	45,000 Lumens	180	50,313	Lights	480	8,483 @	31.18	6.00	1,876	30.02	5.00	1,767	(109)
		3,060	1,441,129			78,331			\$ 23,185			\$ 21,207	\$ (1,978)
Incandescent													
13	2,500 Lumens	-	408	Lights	202	29 @	13.12	6.00	\$ 8	12.63	5.00	\$ 7	\$ (1)
14	4,000 Lumens	108	-	Lights	305	12 @	19.81	6.00	2	19.08	5.00	2	(0)
15	6,000 Lumens	24	1,968	Lights	405	282 @	26.31	6.00	64	25.33	5.00	60	(4)
16	10,000 Lumens	-	12	Lights	690	3 @	44.82	6.00	1	43.16	5.00	1	(0)
		132	2,388			326			\$ 75			\$ 70	\$ (5)
Fluorescent													
17	20,000 Lumens	-	36	Lights	470	6 @	30.53	6.00	\$ 1	29.40	5.00	\$ 1	\$ (0)
Metal Halide													
18	9,750 Lumens	-	672	Lights	170	40 @	11.04	6.00	\$ 11	10.63	5.00	\$ 11	\$ (1)
19	10,500 Lumens	468	3,312	Lights	210	278 @	13.64	6.00	71	13.14	5.00	66	(5)
20	15,500 Lumens	12	1,260	Lights	290	129 @	18.84	6.00	32	18.14	5.00	29	(2)
21	24,000 Lumens	24	588	Lights	460	99 @	29.88	6.00	22	28.77	5.00	21	(1)
		504	5,832			545			\$ 136			\$ 127	\$ (10)
22	Annual PSCR Factor					79,208	0.004440		\$ 352	0.004440		\$ 352	\$ -
23	Total Unmetered Lighting GUL	3,696	1,449,385			79,208			\$ 23,749			\$ 21,756	\$ (1,993)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Unmetered Experimental Lighting Service (GU-XL)

Transitional Rate Until Rate LED Implemented

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

(a)		(b)	(c)	(d)		(e)	(f)	(g)	(h)
Line		Billing Determinants		Present		Proposed		Net Increase	
No.	Description	Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)	
				\$/unit	\$000	\$/unit	\$000	\$000	
	BUNDLED SERVICE								
	Production								
1	Non-capacity All kWh	20,629	MWh	@	0.048281	\$ 996	0.029059	\$ 599	\$ (397)
2	Capacity All kWh	20,629	MWh	@	-	-	-	-	-
3	Annual PSCR Factor	20,629	MWh	@	0.004440	92	0.004440	92	-
4	Total Production					\$ 1,088		\$ 691	\$ (397)
	Transmission								
5	All kWh	20,629	MWh	@	-	-	0.010937	226	226
	Delivery								
	Company Owned Equipment								
6	Fixtures	977,388	Bills	@	6.00	\$ 5,864	5.00	\$ 4,887	\$ (977)
7	Conversion Credit	129,840	Bills		-	-	(3.52)	(457)	(457)
8	Distribution	20,561	MWh	@	0.051752	1,064	0.120810	2,484	1,420
9	Company Owned Delivery					\$ 6,928		\$ 6,914	\$ (15)
	Customer Owned Equipment								
10	Distribution	68	MWh	@	0.042193	3	0.098495	7	4
11	Total Delivery					\$ 6,931		\$ 6,921	\$ (11)
12	Total Bundled Service					\$ 8,019		\$ 7,837	\$ (182)

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Unmetered Light Emitting Diode Service (LED)

Case No.: U-20697
 Exhibit No.: A-16 (HWM-3)
 Schedule: F-3.0
 Page: 23 of 25
 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	
Line	Billing Determinants							Present					Proposed					Net Increase	
						Customer	Company	Customer Owned		Company Owned			Customer Owned		Company Owned				
No.	Description	Customer	Company	Units	Watts	MWh	MWh	Production	Delivery	Production	Delivery	Revenue	Production	Delivery	Production	Delivery	Revenue	/(Decrease)	
								\$/Light	\$/Light	\$/Light	\$/Light	\$000	\$/Light	\$/Light	\$/Light	\$/Light	\$000	\$000	
1	15 - 24 Watts	12	-	Lights	20	0	@	-	-	-	-	\$ -	0.28	3.37	0.28	5.28	\$ 0	\$ 0	
2	25 - 34 Watts	-	-	Lights	30	-	@	-	-	-	-	-	0.42	3.75	0.42	5.84	-	-	
3	35 - 44 Watts	1,524	-	Lights	40	21	@	-	-	-	-	-	0.56	4.14	0.56	6.40	7	7	
4	45 - 54 Watts	924	755,836	Lights	50	16	13,227	@	-	-	-	-	0.70	4.53	0.70	6.97	5,802	5,802	
5	55 - 64 Watts	1,092	-	Lights	60	23	@	-	-	-	-	-	0.84	4.92	0.84	7.53	6	6	
6	65 - 74 Watts	312	90,818	Lights	70	8	2,225	@	-	-	-	-	0.98	5.30	0.98	8.09	826	826	
7	75 - 84 Watts	-	2,004	Lights	80	-	56	@	-	-	-	-	1.12	5.70	1.12	8.66	20	20	
8	85 - 94 Watts	-	88,517	Lights	90	-	2,788	@	-	-	-	-	1.26	6.08	1.26	9.22	928	928	
9	95 - 104 Watts	-	132	Lights	100	-	5	@	-	-	-	-	1.40	6.47	1.40	9.78	1	1	
10	105 - 114 Watts	-	7,356	Lights	110	-	283	@	-	-	-	-	1.54	6.86	1.54	10.35	87	87	
11	115 - 124 Watts	-	24	Lights	120	-	1	@	-	-	-	-	1.68	7.25	1.68	10.91	0	0	
12	125 - 134 Watts	-	-	Lights	130	-	-	@	-	-	-	-	1.82	7.63	1.82	11.47	-	-	
13	135 - 144 Watts	-	-	Lights	140	-	-	@	-	-	-	-	1.96	8.03	1.96	12.04	-	-	
14	145 - 154 Watts	-	-	Lights	150	-	-	@	-	-	-	-	2.10	8.41	2.10	12.60	-	-	
15	155 - 164 Watts	-	-	Lights	160	-	-	@	-	-	-	-	2.24	8.80	2.24	13.16	-	-	
16	165 - 174 Watts	-	30,492	Lights	170	-	1,814	@	-	-	-	-	2.38	9.19	2.38	13.73	491	491	
17	175 - 184 Watts	-	-	Lights	180	-	-	@	-	-	-	-	2.52	9.58	2.52	14.29	-	-	
18	185 - 194 Watts	-	156	Lights	190	-	10	@	-	-	-	-	2.66	9.96	2.66	14.85	3	3	
19	195 - 204 Watts	-	48	Lights	200	-	3	@	-	-	-	-	2.80	10.36	2.80	15.41	1	1	
20	205 - 214 Watts	-	2,004	Lights	210	-	147	@	-	-	-	-	2.94	10.74	2.94	15.97	38	38	
21	Conversion Credit	-	129,840	Lights							-	\$ -				(3.52)	\$ (457)	\$ (457)	
22	Annual PSCR Factor	3,864	977,388	Lights		68	20,561	0.004440		0.004440		\$ -	0.004440		0.004440		\$ 92	\$ 92	
23	Total Unmetered LED	3,864	977,388	Lights		68	20,561					\$ -					\$ 7,845	\$ 7,845	

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Unmetered Service (GU)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Description	Billing Determinants		Present		Proposed		Net Increase
No.		Quantity	Units	Rate	Revenue	Rate	Revenue	/ (Decrease)
				\$/unit	\$000	\$/unit	\$000	\$000
	BUNDLED SERVICE							
	Production							
1	Non-capacity All kWh	95,012	MWh	@	0.055846 \$	5,306	0.036256 \$	3,445 \$ (1,861)
2	Capacity All kWh	95,012	MWh	@	0.020186	1,918	0.024416	2,320 402
3	Annual PSCR Factor	95,012	MWh	@	0.004440	422	0.004440	422 -
4	Total Production					\$ 7,646	\$ 6,186	\$ (1,459)
	Transmission							
5	All kWh	95,012	MWh	@	-	-	0.011554	1,098 1,098
	Delivery							
6	System Access	5,748	Bills	@	2.00 \$	11	2.00 \$	11 -
7	Distribution	95,012	MWh	@	0.017235	1,638	0.022542	2,142 504
8	Total Delivery					\$ 1,649	\$ 2,153	\$ 504
9	Total Bundled Service					\$ 9,295	\$ 9,437	\$ 143

Schedule F-3.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Present and Proposed Revenue Detail

Large Self-generation (GSG-2)

Case No.: U-20697

Exhibit No.: A-16 (HWM-3)

Schedule: F-3.0

Page: 25 of 25

Witness: HWMiller

Date: February 2020

(a)		(b)		(c)		(d)		(e)		(f)		(g)		(h)	
Line		Billing Determinants				Present		Proposed		Net Increase					
No.	Description	Quantity	Units		Rate	Revenue	Rate	Revenue	Rate	Revenue	/ (Decrease)				
					\$/unit	\$000	\$/unit	\$000	\$/unit	\$000	\$000				
	BUNDLED SERVICE														
	Production														
	Voltage Level 1														
1	On-peak kW-day	82	MW	@	11.56	\$ 948	11.56	\$ 948	\$	-					
2	All kWh	41,706	MWh	@	0.029692	1,238	0.029692	1,238		-					
	Voltage Level 2														
3	On-peak kW-day	63	MW	@	11.75	740	11.75	740		-					
4	All kWh	21,673	MWh	@	0.030033	651	0.030033	651		-					
	Voltage Level 3														
5	On-peak kW-day	3	MW	@	12.35	37	12.35	37		-					
6	All kWh	104	MWh	@	0.031345	3	0.031345	3		-					
7	Power Factor Adjustment					34		34		-					
8	Total Production					\$ 3,651		\$ 3,651	\$	-					
	Transmission														
	Voltage Level 1														
9	On-peak kW-day	82	MW	@	2.44	\$ 200	2.44	\$ 200	\$	-					
	Voltage Level 2														
10	On-peak kW-day	63	MW	@	2.48	156	2.48	156		-					
	Voltage Level 3														
11	On-peak kW-day	3	MW	@	2.61	8	2.61	8		-					
12	Total Transmission					\$ 364		\$ 364	\$	-					
	Delivery														
	System Access														
	Voltage Level 1														
13	Standby	60	Bills	@	200.00	\$ 12	200.00	\$ 12	\$	-					
14	Supplemental	-	Bills	@	100.00	-	100.00	-		-					
13	Maximum kW	1,291	MW	@	0.91	1,175	0.70	904		(271)					
14	Interconnect	81	MW	@	(0.91)	(74)	(0.70)	(57)		17					
15	Substation Ownership	-	MW	@	(0.45)	-	(0.34)	-		-					
	Voltage Level 2														
16	Standby	72	Bills	@	200.00	\$ 14	200.00	\$ 14	\$	-					
17	Supplemental	-	Bills	@	100.00	-	100.00	-		-					
16	Maximum kW	339	MW	@	1.86	631	2.69	912		281					
18	Substation Ownership	210	MW	@	(0.97)	(204)	(0.96)	(202)		2					
	Voltage Level 3														
19	Standby	12	Bills	@	200.00	\$ 2	200.00	\$ 2	\$	-					
20	Supplemental	-	Bills	@	100.00	-	100.00	-		-					
19	Maximum kW	7	MW	@	3.60	25	4.58	32		7					
21	Power Factor Adjustment					12		15		2					
22	Total Delivery					\$ 1,594		\$ 1,633	\$	39					
23	Total Standby Service					\$ 5,609		\$ 5,648	\$	39					

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 1 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.26	9.37	1.11	13.4%	187.4	8.22	9.32	1.10	13.4%	186.5
2	50	15.11	17.18	2.08	13.7%	34.4	14.69	16.74	2.05	13.9%	33.5
3	100	22.71	25.86	3.15	13.9%	25.9	21.88	24.98	3.09	14.1%	25.0
4	150	30.32	34.55	4.23	14.0%	23.0	29.08	33.22	4.14	14.2%	22.1
5	200	37.92	43.23	5.31	14.0%	21.6	36.27	41.46	5.19	14.3%	20.7
6	250	45.53	51.91	6.38	14.0%	20.8	43.46	49.70	6.24	14.4%	19.9
7	300	53.13	60.59	7.46	14.0%	20.2	50.65	57.93	7.28	14.4%	19.3
8	350	60.74	69.27	8.54	14.1%	19.8	57.84	66.17	8.33	14.4%	18.9
9	400	68.34	77.96	9.61	14.1%	19.5	65.03	74.41	9.38	14.4%	18.6
10	450	75.95	86.64	10.69	14.1%	19.3	72.23	82.65	10.43	14.4%	18.4
11	500	83.55	95.32	11.77	14.1%	19.1	79.42	90.89	11.47	14.4%	18.2
12	550	91.16	104.00	12.84	14.1%	18.9	86.61	99.13	12.52	14.5%	18.0
13	600	98.76	112.69	13.92	14.1%	18.8	93.80	107.37	13.57	14.5%	17.9
14	650	106.37	121.37	15.00	14.1%	18.7	100.99	115.61	14.62	14.5%	17.8
15	700	113.97	130.05	16.08	14.1%	18.6	108.18	123.85	15.66	14.5%	17.7
16	750	121.58	138.73	17.15	14.1%	18.5	115.38	132.09	16.71	14.5%	17.6
17	800	129.18	147.41	18.23	14.1%	18.4	122.57	140.33	17.76	14.5%	17.5
18	850	136.79	156.10	19.31	14.1%	18.4	129.76	148.57	18.81	14.5%	17.5
19	900	144.40	164.78	20.38	14.1%	18.3	136.95	156.80	19.85	14.5%	17.4
20	950	152.00	173.46	21.46	14.1%	18.3	144.14	165.04	20.90	14.5%	17.4
21	1,000	159.61	182.14	22.54	14.1%	18.2	151.34	173.28	21.95	14.5%	17.3
22	1,500	235.66	268.96	33.30	14.1%	17.9	223.25	255.67	32.42	14.5%	17.0
23	2,000	311.71	355.78	44.07	14.1%	17.8	295.17	338.07	42.90	14.5%	16.9
24	2,500	387.76	442.61	54.84	14.1%	17.7	367.09	420.46	53.37	14.5%	16.8
25	3,000	463.82	529.43	65.61	14.1%	17.6	439.01	502.85	63.84	14.5%	16.8
26	3,500	539.87	616.25	76.38	14.1%	17.6	510.92	585.24	74.32	14.5%	16.7
27	4,000	615.92	703.07	87.14	14.1%	17.6	582.84	667.63	84.79	14.5%	16.7
28	4,500	691.98	789.89	97.91	14.1%	17.6	654.76	750.02	95.27	14.5%	16.7
29	5,000	768.03	876.71	108.68	14.2%	17.5	726.68	832.42	105.74	14.6%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Senior Citizen Provision RSC

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.51	5.12	0.61	13.5%	102.4	4.47	5.07	0.60	13.5%	101.5
2	50	11.36	12.93	1.58	13.9%	25.9	10.94	12.49	1.55	14.1%	25.0
3	100	18.96	21.61	2.65	14.0%	21.6	18.13	20.73	2.59	14.3%	20.7
4	150	26.57	30.30	3.73	14.0%	20.2	25.33	28.97	3.64	14.4%	19.3
5	200	34.17	38.98	4.81	14.1%	19.5	32.52	37.21	4.69	14.4%	18.6
6	250	41.78	47.66	5.88	14.1%	19.1	39.71	45.45	5.74	14.4%	18.2
7	300	49.38	56.34	6.96	14.1%	18.8	46.90	53.68	6.78	14.5%	17.9
8	350	56.99	65.02	8.04	14.1%	18.6	54.09	61.92	7.83	14.5%	17.7
9	400	64.59	73.71	9.11	14.1%	18.4	61.28	70.16	8.88	14.5%	17.5
10	450	72.20	82.39	10.19	14.1%	18.3	68.48	78.40	9.93	14.5%	17.4
11	500	79.80	91.07	11.27	14.1%	18.2	75.67	86.64	10.97	14.5%	17.3
12	550	87.41	99.75	12.34	14.1%	18.1	82.86	94.88	12.02	14.5%	17.3
13	600	95.01	108.44	13.42	14.1%	18.1	90.05	103.12	13.07	14.5%	17.2
14	650	102.62	117.12	14.50	14.1%	18.0	97.24	111.36	14.12	14.5%	17.1
15	700	110.22	125.80	15.58	14.1%	18.0	104.43	119.60	15.16	14.5%	17.1
16	750	117.83	134.48	16.65	14.1%	17.9	111.63	127.84	16.21	14.5%	17.0
17	800	125.43	143.16	17.73	14.1%	17.9	118.82	136.08	17.26	14.5%	17.0
18	850	133.04	151.85	18.81	14.1%	17.9	126.01	144.32	18.31	14.5%	17.0
19	900	140.65	160.53	19.88	14.1%	17.8	133.20	152.55	19.35	14.5%	17.0
20	950	148.25	169.21	20.96	14.1%	17.8	140.39	160.79	20.40	14.5%	16.9
21	1,000	155.86	177.89	22.04	14.1%	17.8	147.59	169.03	21.45	14.5%	16.9
22	1,500	231.91	264.71	32.80	14.1%	17.6	219.50	251.42	31.92	14.5%	16.8
23	2,000	307.96	351.53	43.57	14.1%	17.6	291.42	333.82	42.40	14.5%	16.7
24	2,500	384.01	438.36	54.34	14.2%	17.5	363.34	416.21	52.87	14.6%	16.6
25	3,000	460.07	525.18	65.11	14.2%	17.5	435.26	498.60	63.34	14.6%	16.6
26	3,500	536.12	612.00	75.88	14.2%	17.5	507.17	580.99	73.82	14.6%	16.6
27	4,000	612.17	698.82	86.64	14.2%	17.5	579.09	663.38	84.29	14.6%	16.6
28	4,500	688.23	785.64	97.41	14.2%	17.5	651.01	745.77	94.77	14.6%	16.6
29	5,000	764.28	872.46	108.18	14.2%	17.4	722.93	828.17	105.24	14.6%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Summer On-peak RSP

Income Assistance Provision RIA

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.76	0.87	0.11	14.2%	17.4	0.72	0.82	0.10	14.6%	16.5
2	50	7.61	8.68	1.08	14.2%	17.4	7.19	8.24	1.05	14.6%	16.5
3	100	15.21	17.36	2.15	14.2%	17.4	14.38	16.48	2.09	14.6%	16.5
4	150	22.82	26.05	3.23	14.2%	17.4	21.58	24.72	3.14	14.6%	16.5
5	200	30.42	34.73	4.31	14.2%	17.4	28.77	32.96	4.19	14.6%	16.5
6	250	38.03	43.41	5.38	14.2%	17.4	35.96	41.20	5.24	14.6%	16.5
7	300	45.63	52.09	6.46	14.2%	17.4	43.15	49.43	6.28	14.6%	16.5
8	350	53.24	60.77	7.54	14.2%	17.4	50.34	57.67	7.33	14.6%	16.5
9	400	60.84	69.46	8.61	14.2%	17.4	57.53	65.91	8.38	14.6%	16.5
10	450	68.45	78.14	9.69	14.2%	17.4	64.73	74.15	9.43	14.6%	16.5
11	500	76.05	86.82	10.77	14.2%	17.4	71.92	82.39	10.47	14.6%	16.5
12	550	83.66	95.50	11.84	14.2%	17.4	79.11	90.63	11.52	14.6%	16.5
13	600	91.26	104.19	12.92	14.2%	17.4	86.30	98.87	12.57	14.6%	16.5
14	650	98.87	112.87	14.00	14.2%	17.4	93.49	107.11	13.62	14.6%	16.5
15	700	106.47	121.55	15.08	14.2%	17.4	100.68	115.35	14.66	14.6%	16.5
16	750	114.08	130.23	16.15	14.2%	17.4	107.88	123.59	15.71	14.6%	16.5
17	800	121.68	138.91	17.23	14.2%	17.4	115.07	131.83	16.76	14.6%	16.5
18	850	129.29	147.60	18.31	14.2%	17.4	122.26	140.07	17.81	14.6%	16.5
19	900	136.90	156.28	19.38	14.2%	17.4	129.45	148.30	18.85	14.6%	16.5
20	950	144.50	164.96	20.46	14.2%	17.4	136.64	156.54	19.90	14.6%	16.5
21	1,000	152.11	173.64	21.54	14.2%	17.4	143.84	164.78	20.95	14.6%	16.5
22	1,500	228.16	260.46	32.30	14.2%	17.4	215.75	247.17	31.42	14.6%	16.5
23	2,000	304.21	347.28	43.07	14.2%	17.4	287.67	329.57	41.90	14.6%	16.5
24	2,500	380.26	434.11	53.84	14.2%	17.4	359.59	411.96	52.37	14.6%	16.5
25	3,000	456.32	520.93	64.61	14.2%	17.4	431.51	494.35	62.84	14.6%	16.5
26	3,500	532.37	607.75	75.38	14.2%	17.4	503.42	576.74	73.32	14.6%	16.5
27	4,000	608.42	694.57	86.14	14.2%	17.4	575.34	659.13	83.79	14.6%	16.5
28	4,500	684.48	781.39	96.91	14.2%	17.4	647.26	741.52	94.27	14.6%	16.5
29	5,000	760.53	868.21	107.68	14.2%	17.4	719.18	823.92	104.74	14.6%	16.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 4 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.25	9.38	1.13	13.6%	187.5	8.21	9.32	1.11	13.5%	186.4
2	50	15.00	17.25	2.25	15.0%	34.5	14.63	16.70	2.07	14.2%	33.4
3	100	22.50	26.00	3.50	15.6%	26.0	21.76	24.91	3.14	14.4%	24.9
4	150	30.00	34.76	4.76	15.9%	23.2	28.89	33.11	4.22	14.6%	22.1
5	200	37.50	43.51	6.01	16.0%	21.8	36.02	41.31	5.29	14.7%	20.7
6	250	45.00	52.26	7.26	16.1%	20.9	43.15	49.51	6.36	14.7%	19.8
7	300	52.50	61.01	8.51	16.2%	20.3	50.28	57.72	7.43	14.8%	19.2
8	350	60.00	69.76	9.77	16.3%	19.9	57.41	65.92	8.50	14.8%	18.8
9	400	67.50	78.52	11.02	16.3%	19.6	64.54	74.12	9.58	14.8%	18.5
10	450	75.00	87.27	12.27	16.4%	19.4	71.67	82.32	10.65	14.9%	18.3
11	500	82.50	96.02	13.52	16.4%	19.2	78.80	90.53	11.72	14.9%	18.1
12	550	90.00	104.77	14.78	16.4%	19.0	85.93	98.73	12.79	14.9%	18.0
13	600	97.50	113.53	16.03	16.4%	18.9	93.06	106.93	13.87	14.9%	17.8
14	650	105.00	122.28	17.28	16.5%	18.8	100.19	115.13	14.94	14.9%	17.7
15	700	112.50	131.03	18.53	16.5%	18.7	107.33	123.34	16.01	14.9%	17.6
16	750	120.00	139.78	19.78	16.5%	18.6	114.46	131.54	17.08	14.9%	17.5
17	800	127.50	148.53	21.04	16.5%	18.6	121.59	139.74	18.15	14.9%	17.5
18	850	135.00	157.29	22.29	16.5%	18.5	128.72	147.94	19.23	14.9%	17.4
19	900	142.50	166.04	23.54	16.5%	18.4	135.85	156.15	20.30	14.9%	17.3
20	950	150.00	174.79	24.79	16.5%	18.4	142.98	164.35	21.37	14.9%	17.3
21	1,000	157.50	183.54	26.05	16.5%	18.4	150.11	172.55	22.44	15.0%	17.3
22	1,500	232.49	271.06	38.57	16.6%	18.1	221.41	254.58	33.16	15.0%	17.0
23	2,000	307.49	358.58	51.09	16.6%	17.9	292.71	336.60	43.89	15.0%	16.8
24	2,500	382.49	446.10	63.62	16.6%	17.8	364.02	418.63	54.61	15.0%	16.7
25	3,000	457.49	533.63	76.14	16.6%	17.8	435.32	500.65	65.33	15.0%	16.7
26	3,500	532.48	621.15	88.66	16.7%	17.7	506.63	582.68	76.05	15.0%	16.6
27	4,000	607.48	708.67	101.19	16.7%	17.7	577.93	664.70	86.77	15.0%	16.6
28	4,500	682.48	796.19	113.71	16.7%	17.7	649.23	746.73	97.49	15.0%	16.6
29	5,000	757.48	883.71	126.23	16.7%	17.7	720.54	828.75	108.21	15.0%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Senior Citizen Provision RSC

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.50	5.13	0.63	13.9%	102.5	4.46	5.07	0.61	13.6%	101.4
2	50	11.25	13.00	1.75	15.6%	26.0	10.88	12.45	1.57	14.4%	24.9
3	100	18.75	21.75	3.00	16.0%	21.8	18.01	20.66	2.64	14.7%	20.7
4	150	26.25	30.51	4.26	16.2%	20.3	25.14	28.86	3.72	14.8%	19.2
5	200	33.75	39.26	5.51	16.3%	19.6	32.27	37.06	4.79	14.8%	18.5
6	250	41.25	48.01	6.76	16.4%	19.2	39.40	45.26	5.86	14.9%	18.1
7	300	48.75	56.76	8.01	16.4%	18.9	46.53	53.47	6.93	14.9%	17.8
8	350	56.25	65.51	9.27	16.5%	18.7	53.66	61.67	8.00	14.9%	17.6
9	400	63.75	74.27	10.52	16.5%	18.6	60.79	69.87	9.08	14.9%	17.5
10	450	71.25	83.02	11.77	16.5%	18.4	67.92	78.07	10.15	14.9%	17.3
11	500	78.75	91.77	13.02	16.5%	18.4	75.05	86.28	11.22	15.0%	17.3
12	550	86.25	100.52	14.28	16.6%	18.3	82.18	94.48	12.29	15.0%	17.2
13	600	93.75	109.28	15.53	16.6%	18.2	89.31	102.68	13.37	15.0%	17.1
14	650	101.25	118.03	16.78	16.6%	18.2	96.44	110.88	14.44	15.0%	17.1
15	700	108.75	126.78	18.03	16.6%	18.1	103.58	119.09	15.51	15.0%	17.0
16	750	116.25	135.53	19.28	16.6%	18.1	110.71	127.29	16.58	15.0%	17.0
17	800	123.75	144.28	20.54	16.6%	18.0	117.84	135.49	17.65	15.0%	16.9
18	850	131.25	153.04	21.79	16.6%	18.0	124.97	143.69	18.73	15.0%	16.9
19	900	138.75	161.79	23.04	16.6%	18.0	132.10	151.90	19.80	15.0%	16.9
20	950	146.25	170.54	24.29	16.6%	18.0	139.23	160.10	20.87	15.0%	16.9
21	1,000	153.75	179.29	25.55	16.6%	17.9	146.36	168.30	21.94	15.0%	16.8
22	1,500	228.74	266.81	38.07	16.6%	17.8	217.66	250.33	32.66	15.0%	16.7
23	2,000	303.74	354.33	50.59	16.7%	17.7	288.96	332.35	43.39	15.0%	16.6
24	2,500	378.74	441.85	63.12	16.7%	17.7	360.27	414.38	54.11	15.0%	16.6
25	3,000	453.74	529.38	75.64	16.7%	17.6	431.57	496.40	64.83	15.0%	16.5
26	3,500	528.73	616.90	88.16	16.7%	17.6	502.88	578.43	75.55	15.0%	16.5
27	4,000	603.73	704.42	100.69	16.7%	17.6	574.18	660.45	86.27	15.0%	16.5
28	4,500	678.73	791.94	113.21	16.7%	17.6	645.48	742.48	96.99	15.0%	16.5
29	5,000	753.73	879.46	125.73	16.7%	17.6	716.79	824.50	107.71	15.0%	16.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Smart Hours RSH

Income Assistance Provision RIA

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.75	0.88	0.13	16.7%	17.5	0.71	0.82	0.11	15.0%	16.4
2	50	7.50	8.75	1.25	16.7%	17.5	7.13	8.20	1.07	15.0%	16.4
3	100	15.00	17.50	2.50	16.7%	17.5	14.26	16.41	2.14	15.0%	16.4
4	150	22.50	26.26	3.76	16.7%	17.5	21.39	24.61	3.22	15.0%	16.4
5	200	30.00	35.01	5.01	16.7%	17.5	28.52	32.81	4.29	15.0%	16.4
6	250	37.50	43.76	6.26	16.7%	17.5	35.65	41.01	5.36	15.0%	16.4
7	300	45.00	52.51	7.51	16.7%	17.5	42.78	49.22	6.43	15.0%	16.4
8	350	52.50	61.26	8.77	16.7%	17.5	49.91	57.42	7.50	15.0%	16.4
9	400	60.00	70.02	10.02	16.7%	17.5	57.04	65.62	8.58	15.0%	16.4
10	450	67.50	78.77	11.27	16.7%	17.5	64.17	73.82	9.65	15.0%	16.4
11	500	75.00	87.52	12.52	16.7%	17.5	71.30	82.03	10.72	15.0%	16.4
12	550	82.50	96.27	13.78	16.7%	17.5	78.43	90.23	11.79	15.0%	16.4
13	600	90.00	105.03	15.03	16.7%	17.5	85.56	98.43	12.87	15.0%	16.4
14	650	97.50	113.78	16.28	16.7%	17.5	92.69	106.63	13.94	15.0%	16.4
15	700	105.00	122.53	17.53	16.7%	17.5	99.83	114.84	15.01	15.0%	16.4
16	750	112.50	131.28	18.78	16.7%	17.5	106.96	123.04	16.08	15.0%	16.4
17	800	120.00	140.03	20.04	16.7%	17.5	114.09	131.24	17.15	15.0%	16.4
18	850	127.50	148.79	21.29	16.7%	17.5	121.22	139.44	18.23	15.0%	16.4
19	900	135.00	157.54	22.54	16.7%	17.5	128.35	147.65	19.30	15.0%	16.4
20	950	142.50	166.29	23.79	16.7%	17.5	135.48	155.85	20.37	15.0%	16.4
21	1,000	150.00	175.04	25.05	16.7%	17.5	142.61	164.05	21.44	15.0%	16.4
22	1,500	224.99	262.56	37.57	16.7%	17.5	213.91	246.08	32.16	15.0%	16.4
23	2,000	299.99	350.08	50.09	16.7%	17.5	285.21	328.10	42.89	15.0%	16.4
24	2,500	374.99	437.60	62.62	16.7%	17.5	356.52	410.13	53.61	15.0%	16.4
25	3,000	449.99	525.13	75.14	16.7%	17.5	427.82	492.15	64.33	15.0%	16.4
26	3,500	524.98	612.65	87.66	16.7%	17.5	499.13	574.18	75.05	15.0%	16.4
27	4,000	599.98	700.17	100.19	16.7%	17.5	570.43	656.20	85.77	15.0%	16.4
28	4,500	674.98	787.69	112.71	16.7%	17.5	641.73	738.23	96.49	15.0%	16.4
29	5,000	749.98	875.21	125.23	16.7%	17.5	713.04	820.25	107.21	15.0%	16.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighttime Savers RPM

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.26	9.32	1.05	12.8%	186.3	8.18	9.29	1.11	13.6%	185.7
2	50	15.12	16.67	1.54	10.2%	33.3	14.26	16.35	2.10	14.7%	32.7
3	100	22.74	24.83	2.09	9.2%	24.8	21.01	24.20	3.19	15.2%	24.2
4	150	30.37	33.00	2.63	8.7%	22.0	27.77	32.06	4.29	15.4%	21.4
5	200	37.99	41.17	3.18	8.4%	20.6	34.52	39.91	5.39	15.6%	20.0
6	250	45.61	49.33	3.72	8.2%	19.7	41.28	47.76	6.48	15.7%	19.1
7	300	53.23	57.50	4.27	8.0%	19.2	48.03	55.61	7.58	15.8%	18.5
8	350	60.85	65.67	4.81	7.9%	18.8	54.79	63.47	8.68	15.8%	18.1
9	400	68.48	73.83	5.36	7.8%	18.5	61.54	71.32	9.77	15.9%	17.8
10	450	76.10	82.00	5.90	7.8%	18.2	68.30	79.17	10.87	15.9%	17.6
11	500	83.72	90.17	6.45	7.7%	18.0	75.06	87.02	11.97	15.9%	17.4
12	550	91.34	98.33	6.99	7.7%	17.9	81.81	94.87	13.06	16.0%	17.2
13	600	98.96	106.50	7.54	7.6%	17.7	88.57	102.73	14.16	16.0%	17.1
14	650	106.59	114.67	8.08	7.6%	17.6	95.32	110.58	15.26	16.0%	17.0
15	700	114.21	122.83	8.62	7.6%	17.5	102.08	118.43	16.35	16.0%	16.9
16	750	121.83	131.00	9.17	7.5%	17.5	108.83	126.28	17.45	16.0%	16.8
17	800	129.45	139.16	9.71	7.5%	17.4	115.59	134.13	18.55	16.0%	16.8
18	850	137.07	147.33	10.26	7.5%	17.3	122.35	141.99	19.64	16.1%	16.7
19	900	144.69	155.50	10.80	7.5%	17.3	129.10	149.84	20.74	16.1%	16.6
20	950	152.32	163.66	11.35	7.5%	17.2	135.86	157.69	21.83	16.1%	16.6
21	1,000	159.94	171.83	11.89	7.4%	17.2	142.61	165.54	22.93	16.1%	16.6
22	1,500	236.16	253.50	17.34	7.3%	16.9	210.17	244.07	33.90	16.1%	16.3
23	2,000	312.38	335.16	22.78	7.3%	16.8	277.72	322.59	44.86	16.2%	16.1
24	2,500	388.60	416.83	28.23	7.3%	16.7	345.28	401.11	55.83	16.2%	16.0
25	3,000	464.82	498.49	33.68	7.2%	16.6	412.84	479.63	66.79	16.2%	16.0
26	3,500	541.04	580.16	39.12	7.2%	16.6	480.39	558.15	77.76	16.2%	15.9
27	4,000	617.26	661.82	44.57	7.2%	16.5	547.95	636.67	88.73	16.2%	15.9
28	4,500	693.47	743.49	50.02	7.2%	16.5	615.51	715.20	99.69	16.2%	15.9
29	5,000	769.69	825.16	55.46	7.2%	16.5	683.06	793.72	110.66	16.2%	15.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighttime Savers RPM

Senior Citizen Provision RSC

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

Page 8 of 46

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.51	5.07	0.55	12.3%	101.3	4.43	5.04	0.61	13.8%	100.7
2	50	11.37	12.42	1.04	9.2%	24.8	10.51	12.10	1.60	15.2%	24.2
3	100	18.99	20.58	1.59	8.4%	20.6	17.26	19.95	2.69	15.6%	20.0
4	150	26.62	28.75	2.13	8.0%	19.2	24.02	27.81	3.79	15.8%	18.5
5	200	34.24	36.92	2.68	7.8%	18.5	30.77	35.66	4.89	15.9%	17.8
6	250	41.86	45.08	3.22	7.7%	18.0	37.53	43.51	5.98	15.9%	17.4
7	300	49.48	53.25	3.77	7.6%	17.7	44.28	51.36	7.08	16.0%	17.1
8	350	57.10	61.42	4.31	7.6%	17.5	51.04	59.22	8.18	16.0%	16.9
9	400	64.73	69.58	4.86	7.5%	17.4	57.79	67.07	9.27	16.0%	16.8
10	450	72.35	77.75	5.40	7.5%	17.3	64.55	74.92	10.37	16.1%	16.6
11	500	79.97	85.92	5.95	7.4%	17.2	71.31	82.77	11.47	16.1%	16.6
12	550	87.59	94.08	6.49	7.4%	17.1	78.06	90.62	12.56	16.1%	16.5
13	600	95.21	102.25	7.04	7.4%	17.0	84.82	98.48	13.66	16.1%	16.4
14	650	102.84	110.42	7.58	7.4%	17.0	91.57	106.33	14.76	16.1%	16.4
15	700	110.46	118.58	8.12	7.4%	16.9	98.33	114.18	15.85	16.1%	16.3
16	750	118.08	126.75	8.67	7.3%	16.9	105.08	122.03	16.95	16.1%	16.3
17	800	125.70	134.91	9.21	7.3%	16.9	111.84	129.88	18.05	16.1%	16.2
18	850	133.32	143.08	9.76	7.3%	16.8	118.60	137.74	19.14	16.1%	16.2
19	900	140.94	151.25	10.30	7.3%	16.8	125.35	145.59	20.24	16.1%	16.2
20	950	148.57	159.41	10.85	7.3%	16.8	132.11	153.44	21.33	16.1%	16.2
21	1,000	156.19	167.58	11.39	7.3%	16.8	138.86	161.29	22.43	16.2%	16.1
22	1,500	232.41	249.25	16.84	7.2%	16.6	206.42	239.82	33.40	16.2%	16.0
23	2,000	308.63	330.91	22.28	7.2%	16.5	273.97	318.34	44.36	16.2%	15.9
24	2,500	384.85	412.58	27.73	7.2%	16.5	341.53	396.86	55.33	16.2%	15.9
25	3,000	461.07	494.24	33.18	7.2%	16.5	409.09	475.38	66.29	16.2%	15.8
26	3,500	537.29	575.91	38.62	7.2%	16.5	476.64	553.90	77.26	16.2%	15.8
27	4,000	613.51	657.57	44.07	7.2%	16.4	544.20	632.42	88.23	16.2%	15.8
28	4,500	689.72	739.24	49.52	7.2%	16.4	611.76	710.95	99.19	16.2%	15.8
29	5,000	765.94	820.91	54.96	7.2%	16.4	679.31	789.47	110.16	16.2%	15.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Nighttime Savers RPM

Income Assistance Provision RIA

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

Page 9 of 46

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	0.76	0.82	0.05	7.1%	16.3	0.68	0.79	0.11	16.2%	15.7
2	50	7.62	8.17	0.54	7.1%	16.3	6.76	7.85	1.10	16.2%	15.7
3	100	15.24	16.33	1.09	7.1%	16.3	13.51	15.70	2.19	16.2%	15.7
4	150	22.87	24.50	1.63	7.1%	16.3	20.27	23.56	3.29	16.2%	15.7
5	200	30.49	32.67	2.18	7.1%	16.3	27.02	31.41	4.39	16.2%	15.7
6	250	38.11	40.83	2.72	7.1%	16.3	33.78	39.26	5.48	16.2%	15.7
7	300	45.73	49.00	3.27	7.1%	16.3	40.53	47.11	6.58	16.2%	15.7
8	350	53.35	57.17	3.81	7.1%	16.3	47.29	54.97	7.68	16.2%	15.7
9	400	60.98	65.33	4.36	7.1%	16.3	54.04	62.82	8.77	16.2%	15.7
10	450	68.60	73.50	4.90	7.1%	16.3	60.80	70.67	9.87	16.2%	15.7
11	500	76.22	81.67	5.45	7.1%	16.3	67.56	78.52	10.97	16.2%	15.7
12	550	83.84	89.83	5.99	7.1%	16.3	74.31	86.37	12.06	16.2%	15.7
13	600	91.46	98.00	6.54	7.1%	16.3	81.07	94.23	13.16	16.2%	15.7
14	650	99.09	106.17	7.08	7.1%	16.3	87.82	102.08	14.26	16.2%	15.7
15	700	106.71	114.33	7.62	7.1%	16.3	94.58	109.93	15.35	16.2%	15.7
16	750	114.33	122.50	8.17	7.1%	16.3	101.33	117.78	16.45	16.2%	15.7
17	800	121.95	130.66	8.71	7.1%	16.3	108.09	125.63	17.55	16.2%	15.7
18	850	129.57	138.83	9.26	7.1%	16.3	114.85	133.49	18.64	16.2%	15.7
19	900	137.19	147.00	9.80	7.1%	16.3	121.60	141.34	19.74	16.2%	15.7
20	950	144.82	155.16	10.35	7.1%	16.3	128.36	149.19	20.83	16.2%	15.7
21	1,000	152.44	163.33	10.89	7.1%	16.3	135.11	157.04	21.93	16.2%	15.7
22	1,500	228.66	245.00	16.34	7.1%	16.3	202.67	235.57	32.90	16.2%	15.7
23	2,000	304.88	326.66	21.78	7.1%	16.3	270.22	314.09	43.86	16.2%	15.7
24	2,500	381.10	408.33	27.23	7.1%	16.3	337.78	392.61	54.83	16.2%	15.7
25	3,000	457.32	489.99	32.68	7.1%	16.3	405.34	471.13	65.79	16.2%	15.7
26	3,500	533.54	571.66	38.12	7.1%	16.3	472.89	549.65	76.76	16.2%	15.7
27	4,000	609.76	653.32	43.57	7.1%	16.3	540.45	628.17	87.73	16.2%	15.7
28	4,500	685.97	734.99	49.02	7.1%	16.3	608.01	706.70	98.69	16.2%	15.7
29	5,000	762.19	816.66	54.46	7.1%	16.3	675.56	785.22	109.66	16.2%	15.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 10 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	8.21	9.35	1.13	13.8%	186.9	8.21	9.32	1.11	13.5%	186.5
2	50	14.64	16.97	2.34	16.0%	33.9	14.64	16.74	2.10	14.4%	33.5
3	100	21.77	25.45	3.68	16.9%	25.4	21.77	24.98	3.21	14.7%	25.0
4	150	28.91	33.92	5.01	17.3%	22.6	28.91	33.22	4.31	14.9%	22.1
5	200	36.04	42.39	6.35	17.6%	21.2	36.04	41.46	5.41	15.0%	20.7
6	250	43.18	50.87	7.69	17.8%	20.3	43.18	49.70	6.52	15.1%	19.9
7	300	50.31	59.34	9.03	17.9%	19.8	50.31	57.93	7.62	15.1%	19.3
8	350	57.45	67.81	10.36	18.0%	19.4	57.45	66.17	8.72	15.2%	18.9
9	400	64.58	76.29	11.70	18.1%	19.1	64.58	74.41	9.83	15.2%	18.6
10	450	71.72	84.76	13.04	18.2%	18.8	71.72	82.65	10.93	15.2%	18.4
11	500	78.86	93.23	14.38	18.2%	18.6	78.86	90.89	12.04	15.3%	18.2
12	550	85.99	101.70	15.71	18.3%	18.5	85.99	99.13	13.14	15.3%	18.0
13	600	93.13	110.18	17.05	18.3%	18.4	93.13	107.37	14.24	15.3%	17.9
14	650	101.81	119.91	18.11	17.8%	18.4	100.26	115.61	15.35	15.3%	17.8
15	700	110.48	129.65	19.17	17.3%	18.5	107.40	123.85	16.45	15.3%	17.7
16	750	119.16	139.39	20.22	17.0%	18.6	114.53	132.09	17.55	15.3%	17.6
17	800	127.84	149.12	21.28	16.6%	18.6	121.67	140.33	18.66	15.3%	17.5
18	850	136.52	158.86	22.34	16.4%	18.7	128.81	148.57	19.76	15.3%	17.5
19	900	145.19	168.59	23.40	16.1%	18.7	135.94	156.80	20.86	15.3%	17.4
20	950	153.87	178.33	24.46	15.9%	18.8	143.08	165.04	21.97	15.4%	17.4
21	1,000	162.55	188.07	25.52	15.7%	18.8	150.21	173.28	23.07	15.4%	17.3
22	1,500	249.33	285.42	36.10	14.5%	19.0	221.57	255.67	34.11	15.4%	17.0
23	2,000	336.11	382.78	46.68	13.9%	19.1	292.92	338.07	45.14	15.4%	16.9
24	2,500	422.89	480.14	57.26	13.5%	19.2	364.28	420.46	56.18	15.4%	16.8
25	3,000	509.67	577.50	67.84	13.3%	19.3	435.64	502.85	67.21	15.4%	16.8
26	3,500	596.45	674.86	78.42	13.1%	19.3	506.99	585.24	78.25	15.4%	16.7
27	4,000	683.22	772.22	89.00	13.0%	19.3	578.35	667.63	89.28	15.4%	16.7
28	4,500	770.00	869.58	99.58	12.9%	19.3	649.70	750.02	100.32	15.4%	16.7
29	5,000	856.78	966.94	110.16	12.9%	19.3	721.06	832.42	111.36	15.4%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Senior Citizen Provision RSC

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.46	5.10	0.63	14.2%	101.9	4.46	5.07	0.61	13.7%	101.5
2	50	10.89	12.72	1.84	16.9%	25.4	10.89	12.49	1.60	14.7%	25.0
3	100	18.02	21.20	3.18	17.6%	21.2	18.02	20.73	2.71	15.0%	20.7
4	150	25.16	29.67	4.51	17.9%	19.8	25.16	28.97	3.81	15.1%	19.3
5	200	32.29	38.14	5.85	18.1%	19.1	32.29	37.21	4.91	15.2%	18.6
6	250	39.43	46.62	7.19	18.2%	18.6	39.43	45.45	6.02	15.3%	18.2
7	300	46.56	55.09	8.53	18.3%	18.4	46.56	53.68	7.12	15.3%	17.9
8	350	53.70	63.56	9.86	18.4%	18.2	53.70	61.92	8.22	15.3%	17.7
9	400	60.83	72.04	11.20	18.4%	18.0	60.83	70.16	9.33	15.3%	17.5
10	450	67.97	80.51	12.54	18.4%	17.9	67.97	78.40	10.43	15.3%	17.4
11	500	75.11	88.98	13.88	18.5%	17.8	75.11	86.64	11.54	15.4%	17.3
12	550	82.24	97.45	15.21	18.5%	17.7	82.24	94.88	12.64	15.4%	17.3
13	600	89.38	105.93	16.55	18.5%	17.7	89.38	103.12	13.74	15.4%	17.2
14	650	98.06	115.66	17.61	18.0%	17.8	96.51	111.36	14.85	15.4%	17.1
15	700	106.73	125.40	18.67	17.5%	17.9	103.65	119.60	15.95	15.4%	17.1
16	750	115.41	135.14	19.72	17.1%	18.0	110.78	127.84	17.05	15.4%	17.0
17	800	124.09	144.87	20.78	16.7%	18.1	117.92	136.08	18.16	15.4%	17.0
18	850	132.77	154.61	21.84	16.5%	18.2	125.06	144.32	19.26	15.4%	17.0
19	900	141.44	164.34	22.90	16.2%	18.3	132.19	152.55	20.36	15.4%	17.0
20	950	150.12	174.08	23.96	16.0%	18.3	139.33	160.79	21.47	15.4%	16.9
21	1,000	158.80	183.82	25.02	15.8%	18.4	146.46	169.03	22.57	15.4%	16.9
22	1,500	245.58	281.17	35.60	14.5%	18.7	217.82	251.42	33.61	15.4%	16.8
23	2,000	332.36	378.53	46.18	13.9%	18.9	289.17	333.82	44.64	15.4%	16.7
24	2,500	419.14	475.89	56.76	13.5%	19.0	360.53	416.21	55.68	15.4%	16.6
25	3,000	505.92	573.25	67.34	13.3%	19.1	431.89	498.60	66.71	15.4%	16.6
26	3,500	592.70	670.61	77.92	13.1%	19.2	503.24	580.99	77.75	15.4%	16.6
27	4,000	679.47	767.97	88.50	13.0%	19.2	574.60	663.38	88.78	15.5%	16.6
28	4,500	766.25	865.33	99.08	12.9%	19.2	645.95	745.77	99.82	15.5%	16.6
29	5,000	853.03	962.69	109.66	12.9%	19.3	717.31	828.17	110.86	15.5%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Residential Non-Transmitting Meters RSM

Income Assistance Provision RIA

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

Page 12 of 46

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	5	4.46	5.10	0.63	14.2%	101.9	4.46	5.07	0.61	13.7%	101.5
2	50	10.89	12.72	1.84	16.9%	25.4	10.89	12.49	1.60	14.7%	25.0
3	100	18.02	21.20	3.18	17.6%	21.2	18.02	20.73	2.71	15.0%	20.7
4	150	25.16	29.67	4.51	17.9%	19.8	25.16	28.97	3.81	15.1%	19.3
5	200	32.29	38.14	5.85	18.1%	19.1	32.29	37.21	4.91	15.2%	18.6
6	250	39.43	46.62	7.19	18.2%	18.6	39.43	45.45	6.02	15.3%	18.2
7	300	46.56	55.09	8.53	18.3%	18.4	46.56	53.68	7.12	15.3%	17.9
8	350	53.70	63.56	9.86	18.4%	18.2	53.70	61.92	8.22	15.3%	17.7
9	400	60.83	72.04	11.20	18.4%	18.0	60.83	70.16	9.33	15.3%	17.5
10	450	67.97	80.51	12.54	18.4%	17.9	67.97	78.40	10.43	15.3%	17.4
11	500	75.11	88.98	13.88	18.5%	17.8	75.11	86.64	11.54	15.4%	17.3
12	550	82.24	97.45	15.21	18.5%	17.7	82.24	94.88	12.64	15.4%	17.3
13	600	89.38	105.93	16.55	18.5%	17.7	89.38	103.12	13.74	15.4%	17.2
14	650	98.06	115.66	17.61	18.0%	17.8	96.51	111.36	14.85	15.4%	17.1
15	700	106.73	125.40	18.67	17.5%	17.9	103.65	119.60	15.95	15.4%	17.1
16	750	115.41	135.14	19.72	17.1%	18.0	110.78	127.84	17.05	15.4%	17.0
17	800	124.09	144.87	20.78	16.7%	18.1	117.92	136.08	18.16	15.4%	17.0
18	850	132.77	154.61	21.84	16.5%	18.2	125.06	144.32	19.26	15.4%	17.0
19	900	141.44	164.34	22.90	16.2%	18.3	132.19	152.55	20.36	15.4%	17.0
20	950	150.12	174.08	23.96	16.0%	18.3	139.33	160.79	21.47	15.4%	16.9
21	1,000	158.80	183.82	25.02	15.8%	18.4	146.46	169.03	22.57	15.4%	16.9
22	1,500	245.58	281.17	35.60	14.5%	18.7	217.82	251.42	33.61	15.4%	16.8
23	2,000	332.36	378.53	46.18	13.9%	18.9	289.17	333.82	44.64	15.4%	16.7
24	2,500	419.14	475.89	56.76	13.5%	19.0	360.53	416.21	55.68	15.4%	16.6
25	3,000	505.92	573.25	67.34	13.3%	19.1	431.89	498.60	66.71	15.4%	16.6
26	3,500	592.70	670.61	77.92	13.1%	19.2	503.24	580.99	77.75	15.4%	16.6
27	4,000	679.47	767.97	88.50	13.0%	19.2	574.60	663.38	88.78	15.5%	16.6
28	4,500	766.25	865.33	99.08	12.9%	19.2	645.95	745.77	99.82	15.5%	16.6
29	5,000	853.03	962.69	109.66	12.9%	19.3	717.31	828.17	110.86	15.5%	16.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

Page 13 of 46

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	54.74	56.50	1.76	3.2%	22.6	54.50	56.27	1.77	3.2%	22.5
2	500	89.49	93.01	3.52	3.9%	18.6	89.00	92.53	3.53	4.0%	18.5
3	750	124.23	129.51	5.28	4.2%	17.3	123.50	128.80	5.30	4.3%	17.2
4	1,000	158.98	166.01	7.03	4.4%	16.6	158.00	165.06	7.06	4.5%	16.5
5	1,500	228.46	239.02	10.55	4.6%	15.9	227.00	237.59	10.59	4.7%	15.8
6	2,000	297.95	312.02	14.07	4.7%	15.6	296.00	310.12	14.12	4.8%	15.5
7	2,500	367.44	385.03	17.58	4.8%	15.4	365.00	382.65	17.66	4.8%	15.3
8	3,000	436.93	458.03	21.10	4.8%	15.3	434.00	455.18	21.19	4.9%	15.2
9	3,500	506.42	531.04	24.62	4.9%	15.2	503.00	527.71	24.72	4.9%	15.1
10	4,000	575.90	604.04	28.14	4.9%	15.1	572.00	600.24	28.25	4.9%	15.0
11	4,500	645.39	677.05	31.65	4.9%	15.0	641.00	672.77	31.78	5.0%	15.0
12	5,000	714.88	750.05	35.17	4.9%	15.0	710.00	745.31	35.31	5.0%	14.9
13	6,000	853.86	896.06	42.20	4.9%	14.9	847.99	890.37	42.37	5.0%	14.8
14	7,000	992.83	1,042.07	49.24	5.0%	14.9	985.99	1,035.43	49.43	5.0%	14.8
15	8,000	1,131.81	1,188.08	56.27	5.0%	14.9	1,123.99	1,180.49	56.50	5.0%	14.8
16	9,000	1,270.78	1,334.09	63.31	5.0%	14.8	1,261.99	1,325.55	63.56	5.0%	14.7
17	10,000	1,409.76	1,480.10	70.34	5.0%	14.8	1,399.99	1,470.61	70.62	5.0%	14.7
18	15,000	2,104.64	2,210.15	105.51	5.0%	14.7	2,089.99	2,195.92	105.93	5.1%	14.6
19	20,000	2,799.52	2,940.20	140.68	5.0%	14.7	2,779.98	2,921.22	141.24	5.1%	14.6
20	25,000	3,494.40	3,670.25	175.85	5.0%	14.7	3,469.98	3,646.53	176.55	5.1%	14.6
21	30,000	4,189.28	4,400.30	211.02	5.0%	14.7	4,159.97	4,371.83	211.86	5.1%	14.6
22	35,000	4,884.16	5,130.35	246.19	5.0%	14.7	4,849.97	5,097.14	247.17	5.1%	14.6
23	40,000	5,579.04	5,860.40	281.36	5.0%	14.7	5,539.96	5,822.44	282.48	5.1%	14.6
24	45,000	6,273.92	6,590.45	316.53	5.0%	14.6	6,229.96	6,547.75	317.79	5.1%	14.6
25	50,000	6,968.80	7,320.50	351.70	5.0%	14.6	6,919.95	7,273.05	353.10	5.1%	14.5
26	55,000	7,663.68	8,050.55	386.87	5.0%	14.6	7,609.95	7,998.36	388.41	5.1%	14.5
27	60,000	8,358.56	8,780.60	422.04	5.0%	14.6	8,299.94	8,723.66	423.72	5.1%	14.5
28	65,000	9,053.44	9,510.65	457.21	5.1%	14.6	8,989.94	9,448.97	459.03	5.1%	14.5
29	70,000	9,748.32	10,240.70	492.38	5.1%	14.6	9,679.93	10,174.27	494.34	5.1%	14.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	54.56	56.30	1.74	3.2%	22.5	54.31	56.06	1.75	3.2%	22.4
2	500	89.11	92.59	3.48	3.9%	18.5	88.63	92.12	3.49	3.9%	18.4
3	750	123.67	128.89	5.21	4.2%	17.2	122.94	128.17	5.24	4.3%	17.1
4	1,000	158.23	165.18	6.95	4.4%	16.5	157.25	164.23	6.98	4.4%	16.4
5	1,500	227.34	237.77	10.43	4.6%	15.9	225.88	236.35	10.47	4.6%	15.8
6	2,000	296.46	310.36	13.91	4.7%	15.5	294.50	308.46	13.96	4.7%	15.4
7	2,500	365.57	382.95	17.38	4.8%	15.3	363.13	380.58	17.45	4.8%	15.2
8	3,000	434.68	455.54	20.86	4.8%	15.2	431.75	452.70	20.94	4.9%	15.1
9	3,500	503.80	528.13	24.34	4.8%	15.1	500.38	524.81	24.43	4.9%	15.0
10	4,000	572.91	600.72	27.81	4.9%	15.0	569.00	596.93	27.92	4.9%	14.9
11	4,500	642.03	673.31	31.29	4.9%	15.0	637.63	669.04	31.41	4.9%	14.9
12	5,000	711.14	745.91	34.76	4.9%	14.9	706.26	741.16	34.90	4.9%	14.8
13	6,000	849.37	891.09	41.72	4.9%	14.9	843.51	885.39	41.89	5.0%	14.8
14	7,000	987.60	1,036.27	48.67	4.9%	14.8	980.76	1,029.62	48.87	5.0%	14.7
15	8,000	1,125.82	1,181.45	55.62	4.9%	14.8	1,118.01	1,173.86	55.85	5.0%	14.7
16	9,000	1,264.05	1,326.63	62.58	5.0%	14.7	1,255.26	1,318.09	62.83	5.0%	14.6
17	10,000	1,402.28	1,471.81	69.53	5.0%	14.7	1,392.51	1,462.32	69.81	5.0%	14.6
18	15,000	2,093.42	2,197.72	104.30	5.0%	14.7	2,078.77	2,183.48	104.72	5.0%	14.6
19	20,000	2,784.56	2,923.62	139.06	5.0%	14.6	2,765.02	2,904.64	139.62	5.0%	14.5
20	25,000	3,475.70	3,649.53	173.83	5.0%	14.6	3,451.28	3,625.80	174.53	5.1%	14.5
21	30,000	4,166.84	4,375.43	208.59	5.0%	14.6	4,137.53	4,346.96	209.43	5.1%	14.5
22	35,000	4,857.98	5,101.34	243.36	5.0%	14.6	4,823.79	5,068.12	244.33	5.1%	14.5
23	40,000	5,549.12	5,827.24	278.12	5.0%	14.6	5,510.04	5,789.28	279.24	5.1%	14.5
24	45,000	6,240.26	6,553.15	312.88	5.0%	14.6	6,196.30	6,510.44	314.14	5.1%	14.5
25	50,000	6,931.40	7,279.05	347.65	5.0%	14.6	6,882.55	7,231.60	349.05	5.1%	14.5
26	55,000	7,622.54	8,004.96	382.41	5.0%	14.6	7,568.81	7,952.76	383.96	5.1%	14.5
27	60,000	8,313.68	8,730.86	417.18	5.0%	14.6	8,255.06	8,673.92	418.86	5.1%	14.5
28	65,000	9,004.82	9,456.77	451.94	5.0%	14.5	8,941.32	9,395.08	453.76	5.1%	14.5
29	70,000	9,695.96	10,182.67	486.71	5.0%	14.5	9,627.57	10,116.24	488.67	5.1%	14.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
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Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	97.04	97.52	0.48	0.5%	19.5	90.41	91.18	0.76	0.8%	18.2
2	1,000	164.09	165.04	0.95	0.6%	16.5	150.83	152.35	1.53	1.0%	15.2
3	1,500	231.13	232.56	1.43	0.6%	15.5	211.24	213.53	2.29	1.1%	14.2
4	2,000	298.18	300.08	1.90	0.6%	15.0	271.65	274.71	3.06	1.1%	13.7
5	2,500	365.22	367.60	2.38	0.7%	14.7	332.06	335.88	3.82	1.2%	13.4
6	3,000	432.27	435.12	2.86	0.7%	14.5	392.48	397.06	4.58	1.2%	13.2
7	3,500	499.31	502.64	3.33	0.7%	14.4	452.89	458.24	5.35	1.2%	13.1
8	4,000	566.35	570.16	3.81	0.7%	14.3	513.30	519.41	6.11	1.2%	13.0
9	4,500	633.40	637.68	4.28	0.7%	14.2	573.72	580.59	6.87	1.2%	12.9
10	5,000	700.44	705.20	4.76	0.7%	14.1	634.13	641.77	7.64	1.2%	12.8
11	6,000	834.53	840.24	5.71	0.7%	14.0	754.95	764.12	9.17	1.2%	12.7
12	7,000	968.62	975.28	6.66	0.7%	13.9	875.78	886.47	10.69	1.2%	12.7
13	8,000	1,102.71	1,110.32	7.61	0.7%	13.9	996.60	1,008.83	12.22	1.2%	12.6
14	9,000	1,236.80	1,245.36	8.57	0.7%	13.8	1,117.43	1,131.18	13.75	1.2%	12.6
15	10,000	1,370.88	1,380.40	9.52	0.7%	13.8	1,238.26	1,253.53	15.28	1.2%	12.5
16	11,000	1,504.97	1,515.44	10.47	0.7%	13.8	1,359.08	1,375.89	16.81	1.2%	12.5
17	12,000	1,639.06	1,650.48	11.42	0.7%	13.8	1,479.91	1,498.24	18.33	1.2%	12.5
18	13,000	1,773.15	1,785.52	12.37	0.7%	13.7	1,600.73	1,620.59	19.86	1.2%	12.5
19	14,000	1,907.24	1,920.56	13.33	0.7%	13.7	1,721.56	1,742.95	21.39	1.2%	12.4
20	15,000	2,041.33	2,055.60	14.28	0.7%	13.7	1,842.38	1,865.30	22.92	1.2%	12.4
21	16,000	2,175.42	2,190.65	15.23	0.7%	13.7	1,963.21	1,987.65	24.44	1.2%	12.4
22	17,000	2,309.50	2,325.69	16.18	0.7%	13.7	2,084.04	2,110.01	25.97	1.2%	12.4
23	18,000	2,443.59	2,460.73	17.13	0.7%	13.7	2,204.86	2,232.36	27.50	1.2%	12.4
24	19,000	2,577.68	2,595.77	18.08	0.7%	13.7	2,325.69	2,354.71	29.03	1.2%	12.4
25	20,000	2,711.77	2,730.81	19.04	0.7%	13.7	2,446.51	2,477.07	30.55	1.2%	12.4
26	21,000	2,845.86	2,865.85	19.99	0.7%	13.6	2,567.34	2,599.42	32.08	1.2%	12.4
27	22,000	2,979.95	3,000.89	20.94	0.7%	13.6	2,688.16	2,721.77	33.61	1.3%	12.4
28	23,000	3,114.04	3,135.93	21.89	0.7%	13.6	2,808.99	2,844.13	35.14	1.3%	12.4
29	24,000	3,248.12	3,270.97	22.84	0.7%	13.6	2,929.81	2,966.48	36.67	1.3%	12.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	96.74	97.19	0.45	0.5%	19.4	90.10	90.85	0.74	0.8%	18.2
2	1,000	163.47	164.38	0.91	0.6%	16.4	150.21	151.69	1.48	1.0%	15.2
3	1,500	230.21	231.57	1.36	0.6%	15.4	210.31	212.54	2.23	1.1%	14.2
4	2,000	296.94	298.76	1.82	0.6%	14.9	270.42	273.39	2.97	1.1%	13.7
5	2,500	363.68	365.95	2.27	0.6%	14.6	330.52	334.24	3.71	1.1%	13.4
6	3,000	430.42	433.14	2.73	0.6%	14.4	390.63	395.08	4.45	1.1%	13.2
7	3,500	497.15	500.33	3.18	0.6%	14.3	450.73	455.93	5.20	1.2%	13.0
8	4,000	563.89	567.53	3.64	0.6%	14.2	510.84	516.78	5.94	1.2%	12.9
9	4,500	630.63	634.72	4.09	0.6%	14.1	570.94	577.62	6.68	1.2%	12.8
10	5,000	697.36	701.91	4.54	0.7%	14.0	631.05	638.47	7.42	1.2%	12.8
11	6,000	830.83	836.29	5.45	0.7%	13.9	751.26	760.17	8.91	1.2%	12.7
12	7,000	964.31	970.67	6.36	0.7%	13.9	871.47	881.86	10.39	1.2%	12.6
13	8,000	1,097.78	1,105.05	7.27	0.7%	13.8	991.68	1,003.55	11.88	1.2%	12.5
14	9,000	1,231.25	1,239.43	8.18	0.7%	13.8	1,111.89	1,125.25	13.36	1.2%	12.5
15	10,000	1,364.72	1,373.81	9.09	0.7%	13.7	1,232.10	1,246.94	14.85	1.2%	12.5
16	11,000	1,498.20	1,508.19	10.00	0.7%	13.7	1,352.31	1,368.64	16.33	1.2%	12.4
17	12,000	1,631.67	1,642.58	10.91	0.7%	13.7	1,472.52	1,490.33	17.82	1.2%	12.4
18	13,000	1,765.14	1,776.96	11.81	0.7%	13.7	1,592.72	1,612.03	19.30	1.2%	12.4
19	14,000	1,898.61	1,911.34	12.72	0.7%	13.7	1,712.93	1,733.72	20.79	1.2%	12.4
20	15,000	2,032.09	2,045.72	13.63	0.7%	13.6	1,833.14	1,855.41	22.27	1.2%	12.4
21	16,000	2,165.56	2,180.10	14.54	0.7%	13.6	1,953.35	1,977.11	23.76	1.2%	12.4
22	17,000	2,299.03	2,314.48	15.45	0.7%	13.6	2,073.56	2,098.80	25.24	1.2%	12.3
23	18,000	2,432.50	2,448.86	16.36	0.7%	13.6	2,193.77	2,220.50	26.73	1.2%	12.3
24	19,000	2,565.98	2,583.24	17.27	0.7%	13.6	2,313.98	2,342.19	28.21	1.2%	12.3
25	20,000	2,699.45	2,717.63	18.18	0.7%	13.6	2,434.19	2,463.89	29.69	1.2%	12.3
26	21,000	2,832.92	2,852.01	19.09	0.7%	13.6	2,554.40	2,585.58	31.18	1.2%	12.3
27	22,000	2,966.39	2,986.39	19.99	0.7%	13.6	2,674.61	2,707.28	32.66	1.2%	12.3
28	23,000	3,099.87	3,120.77	20.90	0.7%	13.6	2,794.82	2,828.97	34.15	1.2%	12.3
29	24,000	3,233.34	3,255.15	21.81	0.7%	13.6	2,915.03	2,950.66	35.63	1.2%	12.3

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 17 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	98.77	95.60	(3.17)	-3.2%	19.1	83.45	97.82	14.37	17.2%	19.6
2	1,000	177.53	161.20	(16.33)	-9.2%	16.1	146.91	165.64	18.74	12.8%	16.6
3	1,500	256.30	226.80	(29.50)	-11.5%	15.1	210.36	233.46	23.10	11.0%	15.6
4	2,000	335.07	292.40	(42.67)	-12.7%	14.6	273.81	301.29	27.47	10.0%	15.1
5	2,500	413.83	358.00	(55.84)	-13.5%	14.3	337.27	369.11	31.84	9.4%	14.8
6	3,000	492.60	423.60	(69.00)	-14.0%	14.1	400.72	436.93	36.21	9.0%	14.6
7	3,500	571.37	489.20	(82.17)	-14.4%	14.0	464.17	504.75	40.58	8.7%	14.4
8	4,000	650.14	554.80	(95.34)	-14.7%	13.9	527.63	572.57	44.95	8.5%	14.3
9	4,500	728.90	620.39	(108.51)	-14.9%	13.8	591.08	640.39	49.31	8.3%	14.2
10	5,000	807.67	685.99	(121.67)	-15.1%	13.7	654.53	708.22	53.68	8.2%	14.2
11	6,000	965.20	817.19	(148.01)	-15.3%	13.6	781.44	843.86	62.42	8.0%	14.1
12	7,000	1,122.74	948.39	(174.34)	-15.5%	13.5	908.35	979.50	71.15	7.8%	14.0
13	8,000	1,280.27	1,079.59	(200.68)	-15.7%	13.5	1,035.25	1,115.14	79.89	7.7%	13.9
14	9,000	1,437.80	1,210.79	(227.01)	-15.8%	13.5	1,162.16	1,250.79	88.63	7.6%	13.9
15	10,000	1,595.34	1,341.99	(253.35)	-15.9%	13.4	1,289.07	1,386.43	97.36	7.6%	13.9
16	11,000	1,752.87	1,473.19	(279.68)	-16.0%	13.4	1,415.97	1,522.07	106.10	7.5%	13.8
17	12,000	1,910.41	1,604.39	(306.02)	-16.0%	13.4	1,542.88	1,657.72	114.84	7.4%	13.8
18	13,000	2,067.94	1,735.59	(332.35)	-16.1%	13.4	1,669.79	1,793.36	123.57	7.4%	13.8
19	14,000	2,225.47	1,866.78	(358.69)	-16.1%	13.3	1,796.69	1,929.00	132.31	7.4%	13.8
20	15,000	2,383.01	1,997.98	(385.02)	-16.2%	13.3	1,923.60	2,064.65	141.04	7.3%	13.8
21	16,000	2,540.54	2,129.18	(411.36)	-16.2%	13.3	2,050.51	2,200.29	149.78	7.3%	13.8
22	17,000	2,698.07	2,260.38	(437.69)	-16.2%	13.3	2,177.41	2,335.93	158.52	7.3%	13.7
23	18,000	2,855.61	2,391.58	(464.03)	-16.2%	13.3	2,304.32	2,471.57	167.25	7.3%	13.7
24	19,000	3,013.14	2,522.78	(490.36)	-16.3%	13.3	2,431.23	2,607.22	175.99	7.2%	13.7
25	20,000	3,170.68	2,653.98	(516.70)	-16.3%	13.3	2,558.13	2,742.86	184.73	7.2%	13.7
26	21,000	3,328.21	2,785.18	(543.03)	-16.3%	13.3	2,685.04	2,878.50	193.46	7.2%	13.7
27	22,000	3,485.74	2,916.38	(569.37)	-16.3%	13.3	2,811.95	3,014.15	202.20	7.2%	13.7
28	23,000	3,643.28	3,047.57	(595.70)	-16.4%	13.3	2,938.85	3,149.79	210.93	7.2%	13.7
29	24,000	3,800.81	3,178.77	(622.04)	-16.4%	13.2	3,065.76	3,285.43	219.67	7.2%	13.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	98.39	95.18	(3.21)	-3.3%	19.0	83.08	97.41	14.33	17.2%	19.5
2	1,000	176.79	160.37	(16.42)	-9.3%	16.0	146.16	164.81	18.66	12.8%	16.5
3	1,500	255.18	225.55	(29.62)	-11.6%	15.0	209.24	232.22	22.98	11.0%	15.5
4	2,000	333.57	290.74	(42.83)	-12.8%	14.5	272.32	299.63	27.31	10.0%	15.0
5	2,500	411.96	355.92	(56.04)	-13.6%	14.2	335.40	367.04	31.64	9.4%	14.7
6	3,000	490.36	421.11	(69.25)	-14.1%	14.0	398.48	434.44	35.97	9.0%	14.5
7	3,500	568.75	486.29	(82.46)	-14.5%	13.9	461.56	501.85	40.29	8.7%	14.3
8	4,000	647.14	551.48	(95.66)	-14.8%	13.8	524.63	569.26	44.62	8.5%	14.2
9	4,500	725.54	616.66	(108.87)	-15.0%	13.7	587.71	636.66	48.95	8.3%	14.1
10	5,000	803.93	681.85	(122.08)	-15.2%	13.6	650.79	704.07	53.28	8.2%	14.1
11	6,000	960.71	812.22	(148.50)	-15.5%	13.5	776.95	838.88	61.93	8.0%	14.0
12	7,000	1,117.50	942.59	(174.91)	-15.7%	13.5	903.11	973.70	70.59	7.8%	13.9
13	8,000	1,274.29	1,072.96	(201.33)	-15.8%	13.4	1,029.27	1,108.51	79.24	7.7%	13.9
14	9,000	1,431.07	1,203.33	(227.74)	-15.9%	13.4	1,155.43	1,243.33	87.90	7.6%	13.8
15	10,000	1,587.86	1,333.70	(254.16)	-16.0%	13.3	1,281.59	1,378.14	96.55	7.5%	13.8
16	11,000	1,744.64	1,464.07	(280.58)	-16.1%	13.3	1,407.75	1,512.95	105.21	7.5%	13.8
17	12,000	1,901.43	1,594.44	(306.99)	-16.1%	13.3	1,533.90	1,647.77	113.86	7.4%	13.7
18	13,000	2,058.22	1,724.81	(333.41)	-16.2%	13.3	1,660.06	1,782.58	122.52	7.4%	13.7
19	14,000	2,215.00	1,855.18	(359.82)	-16.2%	13.3	1,786.22	1,917.40	131.17	7.3%	13.7
20	15,000	2,371.79	1,985.55	(386.24)	-16.3%	13.2	1,912.38	2,052.21	139.83	7.3%	13.7
21	16,000	2,528.57	2,115.92	(412.65)	-16.3%	13.2	2,038.54	2,187.02	148.48	7.3%	13.7
22	17,000	2,685.36	2,246.29	(439.07)	-16.4%	13.2	2,164.70	2,321.84	157.14	7.3%	13.7
23	18,000	2,842.14	2,376.66	(465.49)	-16.4%	13.2	2,290.86	2,456.65	165.80	7.2%	13.6
24	19,000	2,998.93	2,507.03	(491.90)	-16.4%	13.2	2,417.02	2,591.47	174.45	7.2%	13.6
25	20,000	3,155.72	2,637.40	(518.32)	-16.4%	13.2	2,543.17	2,726.28	183.11	7.2%	13.6
26	21,000	3,312.50	2,767.77	(544.73)	-16.4%	13.2	2,669.33	2,861.09	191.76	7.2%	13.6
27	22,000	3,469.29	2,898.14	(571.15)	-16.5%	13.2	2,795.49	2,995.91	200.42	7.2%	13.6
28	23,000	3,626.07	3,028.51	(597.57)	-16.5%	13.2	2,921.65	3,130.72	209.07	7.2%	13.6
29	24,000	3,782.86	3,158.88	(623.98)	-16.5%	13.2	3,047.81	3,265.54	217.73	7.1%	13.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Secondary Interruptible Provision

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	98.77	86.84	(11.93)	-12.1%	17.4	83.45	89.06	5.61	6.7%	17.8
2	1,000	177.53	143.68	(33.85)	-19.1%	14.4	146.91	148.13	1.22	0.8%	14.8
3	1,500	256.30	200.52	(55.78)	-21.8%	13.4	210.36	207.19	(3.17)	-1.5%	13.8
4	2,000	335.07	257.36	(77.71)	-23.2%	12.9	273.81	266.25	(7.56)	-2.8%	13.3
5	2,500	413.83	314.20	(99.63)	-24.1%	12.6	337.27	325.31	(11.95)	-3.5%	13.0
6	3,000	492.60	371.04	(121.56)	-24.7%	12.4	400.72	384.38	(16.34)	-4.1%	12.8
7	3,500	571.37	427.88	(143.48)	-25.1%	12.2	464.17	443.44	(20.74)	-4.5%	12.7
8	4,000	650.14	484.72	(165.41)	-25.4%	12.1	527.63	502.50	(25.13)	-4.8%	12.6
9	4,500	728.90	541.57	(187.34)	-25.7%	12.0	591.08	561.56	(29.52)	-5.0%	12.5
10	5,000	807.67	598.41	(209.26)	-25.9%	12.0	654.53	620.63	(33.91)	-5.2%	12.4
11	6,000	965.20	712.09	(253.12)	-26.2%	11.9	781.44	738.75	(42.69)	-5.5%	12.3
12	7,000	1,122.74	825.77	(296.97)	-26.5%	11.8	908.35	856.88	(51.47)	-5.7%	12.2
13	8,000	1,280.27	939.45	(340.82)	-26.6%	11.7	1,035.25	975.00	(60.25)	-5.8%	12.2
14	9,000	1,437.80	1,053.13	(384.67)	-26.8%	11.7	1,162.16	1,093.13	(69.03)	-5.9%	12.1
15	10,000	1,595.34	1,166.81	(428.53)	-26.9%	11.7	1,289.07	1,211.25	(77.81)	-6.0%	12.1
16	11,000	1,752.87	1,280.49	(472.38)	-26.9%	11.6	1,415.97	1,329.38	(86.60)	-6.1%	12.1
17	12,000	1,910.41	1,394.17	(516.23)	-27.0%	11.6	1,542.88	1,447.50	(95.38)	-6.2%	12.1
18	13,000	2,067.94	1,507.85	(560.09)	-27.1%	11.6	1,669.79	1,565.63	(104.16)	-6.2%	12.0
19	14,000	2,225.47	1,621.54	(603.94)	-27.1%	11.6	1,796.69	1,683.75	(112.94)	-6.3%	12.0
20	15,000	2,383.01	1,735.22	(647.79)	-27.2%	11.6	1,923.60	1,801.88	(121.72)	-6.3%	12.0
21	16,000	2,540.54	1,848.90	(691.64)	-27.2%	11.6	2,050.51	1,920.00	(130.50)	-6.4%	12.0
22	17,000	2,698.07	1,962.58	(735.50)	-27.3%	11.5	2,177.41	2,038.13	(139.28)	-6.4%	12.0
23	18,000	2,855.61	2,076.26	(779.35)	-27.3%	11.5	2,304.32	2,156.25	(148.07)	-6.4%	12.0
24	19,000	3,013.14	2,189.94	(823.20)	-27.3%	11.5	2,431.23	2,274.38	(156.85)	-6.5%	12.0
25	20,000	3,170.68	2,303.62	(867.05)	-27.3%	11.5	2,558.13	2,392.50	(165.63)	-6.5%	12.0
26	21,000	3,328.21	2,417.30	(910.91)	-27.4%	11.5	2,685.04	2,510.63	(174.41)	-6.5%	12.0
27	22,000	3,485.74	2,530.98	(954.76)	-27.4%	11.5	2,811.95	2,628.76	(183.19)	-6.5%	11.9
28	23,000	3,643.28	2,644.67	(998.61)	-27.4%	11.5	2,938.85	2,746.88	(191.97)	-6.5%	11.9
29	24,000	3,800.81	2,758.35	(1,042.46)	-27.4%	11.5	3,065.76	2,865.01	(200.76)	-6.5%	11.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
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Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	145.92	144.39	(1.53)	-1.0%	28.9	145.48	144.03	(1.46)	-1.0%	28.8
2	1,000	191.84	188.79	(3.06)	-1.6%	18.9	190.97	188.06	(2.91)	-1.5%	18.8
3	1,500	237.76	233.18	(4.58)	-1.9%	15.5	236.45	232.09	(4.37)	-1.8%	15.5
4	2,000	283.68	277.57	(6.11)	-2.2%	13.9	281.94	276.12	(5.82)	-2.1%	13.8
5	2,500	329.60	321.96	(7.64)	-2.3%	12.9	327.42	320.15	(7.27)	-2.2%	12.8
6	3,000	375.52	366.36	(9.17)	-2.4%	12.2	372.91	364.18	(8.73)	-2.3%	12.1
7	4,000	467.36	455.14	(12.22)	-2.6%	11.4	463.88	452.24	(11.64)	-2.5%	11.3
8	5,000	559.21	543.93	(15.28)	-2.7%	10.9	554.85	540.30	(14.55)	-2.6%	10.8
9	6,000	651.05	632.71	(18.34)	-2.8%	10.5	645.81	628.35	(17.46)	-2.7%	10.5
10	7,000	742.89	721.50	(21.39)	-2.9%	10.3	736.78	716.41	(20.37)	-2.8%	10.2
11	8,000	834.73	810.28	(24.45)	-2.9%	10.1	827.75	804.47	(23.28)	-2.8%	10.1
12	9,000	926.57	899.07	(27.50)	-3.0%	10.0	918.72	892.53	(26.19)	-2.9%	9.9
13	10,000	1,018.41	987.85	(30.56)	-3.0%	9.9	1,009.69	980.59	(29.10)	-2.9%	9.8
14	15,000	1,477.62	1,431.78	(45.84)	-3.1%	9.5	1,464.54	1,420.89	(43.65)	-3.0%	9.5
15	20,000	1,936.82	1,875.70	(61.12)	-3.2%	9.4	1,919.38	1,861.18	(58.20)	-3.0%	9.3
16	25,000	2,396.03	2,319.63	(76.40)	-3.2%	9.3	2,374.23	2,301.48	(72.75)	-3.1%	9.2
17	30,000	2,855.23	2,763.55	(91.68)	-3.2%	9.2	2,829.07	2,741.77	(87.30)	-3.1%	9.1
18	35,000	3,314.44	3,207.48	(106.96)	-3.2%	9.2	3,283.92	3,182.07	(101.85)	-3.1%	9.1
19	40,000	3,773.64	3,651.40	(122.24)	-3.2%	9.1	3,738.76	3,622.36	(116.40)	-3.1%	9.1
20	45,000	4,232.85	4,095.33	(137.52)	-3.2%	9.1	4,193.61	4,062.66	(130.95)	-3.1%	9.0
21	50,000	4,692.05	4,539.25	(152.80)	-3.3%	9.1	4,648.45	4,502.95	(145.50)	-3.1%	9.0
22	60,000	5,610.46	5,427.10	(183.36)	-3.3%	9.0	5,558.14	5,383.54	(174.60)	-3.1%	9.0
23	70,000	6,528.87	6,314.95	(213.92)	-3.3%	9.0	6,467.83	6,264.13	(203.70)	-3.1%	8.9
24	80,000	7,447.28	7,202.80	(244.48)	-3.3%	9.0	7,377.52	7,144.72	(232.80)	-3.2%	8.9
25	90,000	8,365.69	8,090.65	(275.04)	-3.3%	9.0	8,287.21	8,025.31	(261.90)	-3.2%	8.9
26	100,000	9,284.10	8,978.50	(305.60)	-3.3%	9.0	9,196.90	8,905.90	(291.00)	-3.2%	8.9
27	110,000	10,202.51	9,866.35	(336.16)	-3.3%	9.0	10,106.59	9,786.49	(320.10)	-3.2%	8.9
28	120,000	11,120.92	10,754.20	(366.72)	-3.3%	9.0	11,016.28	10,667.08	(349.20)	-3.2%	8.9
29	130,000	12,039.33	11,642.05	(397.28)	-3.3%	9.0	11,925.97	11,547.67	(378.30)	-3.2%	8.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	145.64	144.13	(1.50)	-1.0%	28.8	145.20	143.77	(1.43)	-1.0%	28.8
2	1,000	191.27	188.27	(3.00)	-1.6%	18.8	190.40	187.54	(2.86)	-1.5%	18.8
3	1,500	236.91	232.40	(4.50)	-1.9%	15.5	235.60	231.31	(4.28)	-1.8%	15.4
4	2,000	282.54	276.54	(6.00)	-2.1%	13.8	280.80	275.09	(5.71)	-2.0%	13.8
5	2,500	328.18	320.67	(7.50)	-2.3%	12.8	326.00	318.86	(7.14)	-2.2%	12.8
6	3,000	373.81	364.81	(9.00)	-2.4%	12.2	371.19	362.63	(8.57)	-2.3%	12.1
7	4,000	465.08	453.08	(12.00)	-2.6%	11.3	461.59	450.17	(11.42)	-2.5%	11.3
8	5,000	556.35	541.35	(15.00)	-2.7%	10.8	551.99	537.72	(14.28)	-2.6%	10.8
9	6,000	647.62	629.61	(18.01)	-2.8%	10.5	642.39	625.26	(17.13)	-2.7%	10.4
10	7,000	738.89	717.88	(21.01)	-2.8%	10.3	732.79	712.80	(19.99)	-2.7%	10.2
11	8,000	830.16	806.15	(24.01)	-2.9%	10.1	823.18	800.34	(22.84)	-2.8%	10.0
12	9,000	921.43	894.42	(27.01)	-2.9%	9.9	913.58	887.89	(25.70)	-2.8%	9.9
13	10,000	1,012.70	982.69	(30.01)	-3.0%	9.8	1,003.98	975.43	(28.55)	-2.8%	9.8
14	15,000	1,469.05	1,424.04	(45.01)	-3.1%	9.5	1,455.97	1,413.15	(42.83)	-2.9%	9.4
15	20,000	1,925.40	1,865.38	(60.02)	-3.1%	9.3	1,907.96	1,850.86	(57.10)	-3.0%	9.3
16	25,000	2,381.75	2,306.73	(75.02)	-3.1%	9.2	2,359.95	2,288.58	(71.38)	-3.0%	9.2
17	30,000	2,838.10	2,748.07	(90.03)	-3.2%	9.2	2,811.94	2,726.29	(85.65)	-3.0%	9.1
18	35,000	3,294.45	3,189.42	(105.04)	-3.2%	9.1	3,263.93	3,164.01	(99.93)	-3.1%	9.0
19	40,000	3,750.80	3,630.76	(120.04)	-3.2%	9.1	3,715.92	3,601.72	(114.20)	-3.1%	9.0
20	45,000	4,207.15	4,072.11	(135.05)	-3.2%	9.0	4,167.91	4,039.44	(128.48)	-3.1%	9.0
21	50,000	4,663.50	4,513.45	(150.05)	-3.2%	9.0	4,619.90	4,477.15	(142.75)	-3.1%	9.0
22	60,000	5,576.20	5,396.14	(180.06)	-3.2%	9.0	5,523.88	5,352.58	(171.30)	-3.1%	8.9
23	70,000	6,488.90	6,278.83	(210.07)	-3.2%	9.0	6,427.86	6,228.01	(199.85)	-3.1%	8.9
24	80,000	7,401.60	7,161.52	(240.08)	-3.2%	9.0	7,331.84	7,103.44	(228.40)	-3.1%	8.9
25	90,000	8,314.30	8,044.21	(270.09)	-3.2%	8.9	8,235.82	7,978.87	(256.95)	-3.1%	8.9
26	100,000	9,227.00	8,926.90	(300.10)	-3.3%	8.9	9,139.80	8,854.30	(285.50)	-3.1%	8.9
27	110,000	10,139.70	9,809.59	(330.11)	-3.3%	8.9	10,043.78	9,729.73	(314.05)	-3.1%	8.8
28	120,000	11,052.40	10,692.28	(360.12)	-3.3%	8.9	10,947.76	10,605.16	(342.60)	-3.1%	8.8
29	130,000	11,965.10	11,574.97	(390.13)	-3.3%	8.9	11,851.74	11,480.59	(371.15)	-3.1%	8.8

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 22 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	152.09	147.24	(4.84)	-3.2%	29.4	151.65	146.88	(4.77)	-3.1%	29.4
2	1,000	204.17	194.49	(9.68)	-4.7%	19.4	203.30	193.76	(9.54)	-4.7%	19.4
3	1,500	256.26	241.73	(14.52)	-5.7%	16.1	254.95	240.64	(14.31)	-5.6%	16.0
4	2,000	308.34	288.98	(19.37)	-6.3%	14.4	306.60	287.52	(19.08)	-6.2%	14.4
5	2,500	360.43	336.22	(24.21)	-6.7%	13.4	358.25	334.40	(23.84)	-6.7%	13.4
6	3,000	412.51	383.46	(29.05)	-7.0%	12.8	409.90	381.28	(28.61)	-7.0%	12.7
7	4,000	516.68	477.95	(38.73)	-7.5%	11.9	513.20	475.04	(38.15)	-7.4%	11.9
8	5,000	620.86	572.44	(48.42)	-7.8%	11.4	616.50	568.81	(47.69)	-7.7%	11.4
9	6,000	725.03	666.93	(58.10)	-8.0%	11.1	719.79	662.57	(57.23)	-8.0%	11.0
10	7,000	829.20	761.42	(67.78)	-8.2%	10.9	823.09	756.33	(66.77)	-8.1%	10.8
11	8,000	933.37	855.90	(77.46)	-8.3%	10.7	926.39	850.09	(76.30)	-8.2%	10.6
12	9,000	1,037.54	950.39	(87.15)	-8.4%	10.6	1,029.69	943.85	(85.84)	-8.3%	10.5
13	10,000	1,141.71	1,044.88	(96.83)	-8.5%	10.4	1,132.99	1,037.61	(95.38)	-8.4%	10.4
14	15,000	1,662.57	1,517.32	(145.25)	-8.7%	10.1	1,649.49	1,506.42	(143.07)	-8.7%	10.0
15	20,000	2,183.42	1,989.76	(193.66)	-8.9%	9.9	2,165.98	1,975.22	(190.76)	-8.8%	9.9
16	25,000	2,704.28	2,462.20	(242.08)	-9.0%	9.8	2,682.48	2,444.03	(238.45)	-8.9%	9.8
17	30,000	3,225.13	2,934.64	(290.49)	-9.0%	9.8	3,198.97	2,912.83	(286.14)	-8.9%	9.7
18	35,000	3,745.99	3,407.08	(338.91)	-9.0%	9.7	3,715.47	3,381.64	(333.83)	-9.0%	9.7
19	40,000	4,266.84	3,879.52	(387.32)	-9.1%	9.7	4,231.96	3,850.44	(381.52)	-9.0%	9.6
20	45,000	4,787.70	4,351.96	(435.74)	-9.1%	9.7	4,748.46	4,319.25	(429.21)	-9.0%	9.6
21	50,000	5,308.55	4,824.40	(484.15)	-9.1%	9.6	5,264.95	4,788.05	(476.90)	-9.1%	9.6
22	60,000	6,350.26	5,769.28	(580.98)	-9.1%	9.6	6,297.94	5,725.66	(572.28)	-9.1%	9.5
23	70,000	7,391.97	6,714.16	(677.81)	-9.2%	9.6	7,330.93	6,663.27	(667.66)	-9.1%	9.5
24	80,000	8,433.68	7,659.04	(774.64)	-9.2%	9.6	8,363.92	7,600.88	(763.04)	-9.1%	9.5
25	90,000	9,475.39	8,603.92	(871.47)	-9.2%	9.6	9,396.91	8,538.49	(858.42)	-9.1%	9.5
26	100,000	10,517.10	9,548.80	(968.30)	-9.2%	9.5	10,429.90	9,476.10	(953.80)	-9.1%	9.5
27	110,000	11,558.81	10,493.68	(1,065.13)	-9.2%	9.5	11,462.89	10,413.71	(1,049.18)	-9.2%	9.5
28	120,000	12,600.52	11,438.56	(1,161.96)	-9.2%	9.5	12,495.88	11,351.32	(1,144.56)	-9.2%	9.5
29	130,000	13,642.23	12,383.44	(1,258.79)	-9.2%	9.5	13,528.87	12,288.93	(1,239.94)	-9.2%	9.5

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	151.80	146.99	(4.81)	-3.2%	29.4	151.36	146.62	(4.74)	-3.1%	29.3
2	1,000	203.60	193.97	(9.63)	-4.7%	19.4	202.73	193.25	(9.48)	-4.7%	19.3
3	1,500	255.40	240.96	(14.44)	-5.7%	16.1	254.09	239.87	(14.22)	-5.6%	16.0
4	2,000	307.20	287.94	(19.26)	-6.3%	14.4	305.46	286.49	(18.97)	-6.2%	14.3
5	2,500	359.00	334.93	(24.07)	-6.7%	13.4	356.82	333.11	(23.71)	-6.6%	13.3
6	3,000	410.80	381.92	(28.88)	-7.0%	12.7	408.18	379.74	(28.45)	-7.0%	12.7
7	4,000	514.40	475.89	(38.51)	-7.5%	11.9	510.91	472.98	(37.93)	-7.4%	11.8
8	5,000	618.00	569.86	(48.14)	-7.8%	11.4	613.64	566.23	(47.42)	-7.7%	11.3
9	6,000	721.60	663.83	(57.77)	-8.0%	11.1	716.37	659.47	(56.90)	-7.9%	11.0
10	7,000	825.20	757.80	(67.40)	-8.2%	10.8	819.10	752.72	(66.38)	-8.1%	10.8
11	8,000	928.80	851.78	(77.02)	-8.3%	10.6	921.82	845.96	(75.86)	-8.2%	10.6
12	9,000	1,032.40	945.75	(86.65)	-8.4%	10.5	1,024.55	939.21	(85.35)	-8.3%	10.4
13	10,000	1,136.00	1,039.72	(96.28)	-8.5%	10.4	1,127.28	1,032.45	(94.83)	-8.4%	10.3
14	15,000	1,654.00	1,509.58	(144.42)	-8.7%	10.1	1,640.92	1,498.68	(142.25)	-8.7%	10.0
15	20,000	2,172.00	1,979.44	(192.56)	-8.9%	9.9	2,154.56	1,964.90	(189.66)	-8.8%	9.8
16	25,000	2,690.00	2,449.30	(240.70)	-8.9%	9.8	2,668.20	2,431.13	(237.08)	-8.9%	9.7
17	30,000	3,208.00	2,919.16	(288.84)	-9.0%	9.7	3,181.84	2,897.35	(284.49)	-8.9%	9.7
18	35,000	3,726.00	3,389.02	(336.98)	-9.0%	9.7	3,695.48	3,363.58	(331.91)	-9.0%	9.6
19	40,000	4,244.00	3,858.88	(385.12)	-9.1%	9.6	4,209.12	3,829.80	(379.32)	-9.0%	9.6
20	45,000	4,762.00	4,328.74	(433.26)	-9.1%	9.6	4,722.76	4,296.03	(426.74)	-9.0%	9.5
21	50,000	5,280.00	4,798.60	(481.40)	-9.1%	9.6	5,236.40	4,762.25	(474.15)	-9.1%	9.5
22	60,000	6,316.00	5,738.32	(577.68)	-9.1%	9.6	6,263.68	5,694.70	(568.98)	-9.1%	9.5
23	70,000	7,352.00	6,678.04	(673.96)	-9.2%	9.5	7,290.96	6,627.15	(663.81)	-9.1%	9.5
24	80,000	8,388.00	7,617.76	(770.24)	-9.2%	9.5	8,318.24	7,559.60	(758.64)	-9.1%	9.4
25	90,000	9,424.00	8,557.48	(866.52)	-9.2%	9.5	9,345.52	8,492.05	(853.47)	-9.1%	9.4
26	100,000	10,460.00	9,497.20	(962.80)	-9.2%	9.5	10,372.80	9,424.50	(948.30)	-9.1%	9.4
27	110,000	11,496.00	10,436.92	(1,059.08)	-9.2%	9.5	11,400.08	10,356.95	(1,043.13)	-9.2%	9.4
28	120,000	12,532.00	11,376.64	(1,155.36)	-9.2%	9.5	12,427.36	11,289.40	(1,137.96)	-9.2%	9.4
29	130,000	13,568.00	12,316.36	(1,251.64)	-9.2%	9.5	13,454.64	12,221.85	(1,232.79)	-9.2%	9.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 24 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	159.86	151.02	(8.83)	-5.5%	30.2	159.42	150.66	(8.77)	-5.5%	30.1
2	1,000	219.72	202.05	(17.67)	-8.0%	20.2	218.84	201.31	(17.53)	-8.0%	20.1
3	1,500	279.57	253.07	(26.50)	-9.5%	16.9	278.27	251.97	(26.30)	-9.5%	16.8
4	2,000	339.43	304.09	(35.34)	-10.4%	15.2	337.69	302.63	(35.06)	-10.4%	15.1
5	2,500	399.29	355.12	(44.17)	-11.1%	14.2	397.11	353.28	(43.83)	-11.0%	14.1
6	3,000	459.15	406.14	(53.01)	-11.5%	13.5	456.53	403.94	(52.59)	-11.5%	13.5
7	4,000	578.86	508.19	(70.68)	-12.2%	12.7	575.38	505.25	(70.12)	-12.2%	12.6
8	5,000	698.58	610.24	(88.34)	-12.6%	12.2	694.22	606.57	(87.65)	-12.6%	12.1
9	6,000	818.30	712.28	(106.01)	-13.0%	11.9	813.06	707.88	(105.19)	-12.9%	11.8
10	7,000	938.01	814.33	(123.68)	-13.2%	11.6	931.91	809.19	(122.72)	-13.2%	11.6
11	8,000	1,057.73	916.38	(141.35)	-13.4%	11.5	1,050.75	910.50	(140.25)	-13.3%	11.4
12	9,000	1,177.44	1,018.42	(159.02)	-13.5%	11.3	1,169.60	1,011.82	(157.78)	-13.5%	11.2
13	10,000	1,297.16	1,120.47	(176.69)	-13.6%	11.2	1,288.44	1,113.13	(175.31)	-13.6%	11.1
14	15,000	1,895.74	1,630.71	(265.04)	-14.0%	10.9	1,882.66	1,619.70	(262.97)	-14.0%	10.8
15	20,000	2,494.32	2,140.94	(353.38)	-14.2%	10.7	2,476.88	2,126.26	(350.62)	-14.2%	10.6
16	25,000	3,092.90	2,651.18	(441.73)	-14.3%	10.6	3,071.10	2,632.83	(438.28)	-14.3%	10.5
17	30,000	3,691.48	3,161.41	(530.07)	-14.4%	10.5	3,665.32	3,139.39	(525.93)	-14.3%	10.5
18	35,000	4,290.06	3,671.65	(618.42)	-14.4%	10.5	4,259.54	3,645.96	(613.59)	-14.4%	10.4
19	40,000	4,888.64	4,181.88	(706.76)	-14.5%	10.5	4,853.76	4,152.52	(701.24)	-14.4%	10.4
20	45,000	5,487.22	4,692.12	(795.11)	-14.5%	10.4	5,447.98	4,659.09	(788.90)	-14.5%	10.4
21	50,000	6,085.80	5,202.35	(883.45)	-14.5%	10.4	6,042.20	5,165.65	(876.55)	-14.5%	10.3
22	60,000	7,282.96	6,222.82	(1,060.14)	-14.6%	10.4	7,230.64	6,178.78	(1,051.86)	-14.5%	10.3
23	70,000	8,480.12	7,243.29	(1,236.83)	-14.6%	10.3	8,419.08	7,191.91	(1,227.17)	-14.6%	10.3
24	80,000	9,677.28	8,263.76	(1,413.52)	-14.6%	10.3	9,607.52	8,205.04	(1,402.48)	-14.6%	10.3
25	90,000	10,874.44	9,284.23	(1,590.21)	-14.6%	10.3	10,795.96	9,218.17	(1,577.79)	-14.6%	10.2
26	100,000	12,071.60	10,304.70	(1,766.90)	-14.6%	10.3	11,984.40	10,231.30	(1,753.10)	-14.6%	10.2
27	110,000	13,268.76	11,325.17	(1,943.59)	-14.6%	10.3	13,172.84	11,244.43	(1,928.41)	-14.6%	10.2
28	120,000	14,465.92	12,345.64	(2,120.28)	-14.7%	10.3	14,361.28	12,257.56	(2,103.72)	-14.6%	10.2
29	130,000	15,663.08	13,366.11	(2,296.97)	-14.7%	10.3	15,549.72	13,270.69	(2,279.03)	-14.7%	10.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Education Provision GEI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	159.57	150.77	(8.81)	-5.5%	30.2	159.14	150.40	(8.74)	-5.5%	30.1
2	1,000	219.15	201.53	(17.61)	-8.0%	20.2	218.27	200.80	(17.48)	-8.0%	20.1
3	1,500	278.72	252.30	(26.42)	-9.5%	16.8	277.41	251.20	(26.21)	-9.4%	16.7
4	2,000	338.29	303.06	(35.23)	-10.4%	15.2	336.55	301.59	(34.95)	-10.4%	15.1
5	2,500	397.86	353.83	(44.04)	-11.1%	14.2	395.68	351.99	(43.69)	-11.0%	14.1
6	3,000	457.44	404.59	(52.84)	-11.6%	13.5	454.82	402.39	(52.43)	-11.5%	13.4
7	4,000	576.58	506.12	(70.46)	-12.2%	12.7	573.09	503.19	(69.90)	-12.2%	12.6
8	5,000	695.73	607.66	(88.07)	-12.7%	12.2	691.37	603.99	(87.38)	-12.6%	12.1
9	6,000	814.87	709.19	(105.68)	-13.0%	11.8	809.64	704.78	(104.86)	-13.0%	11.7
10	7,000	934.02	810.72	(123.30)	-13.2%	11.6	927.91	805.58	(122.33)	-13.2%	11.5
11	8,000	1,053.16	912.25	(140.91)	-13.4%	11.4	1,046.18	906.38	(139.81)	-13.4%	11.3
12	9,000	1,172.31	1,013.78	(158.53)	-13.5%	11.3	1,164.46	1,007.17	(157.28)	-13.5%	11.2
13	10,000	1,291.45	1,115.31	(176.14)	-13.6%	11.2	1,282.73	1,107.97	(174.76)	-13.6%	11.1
14	15,000	1,887.18	1,622.97	(264.21)	-14.0%	10.8	1,874.10	1,611.96	(262.14)	-14.0%	10.7
15	20,000	2,482.90	2,130.62	(352.28)	-14.2%	10.7	2,465.46	2,115.94	(349.52)	-14.2%	10.6
16	25,000	3,078.63	2,638.28	(440.35)	-14.3%	10.6	3,056.83	2,619.93	(436.90)	-14.3%	10.5
17	30,000	3,674.35	3,145.93	(528.42)	-14.4%	10.5	3,648.19	3,123.91	(524.28)	-14.4%	10.4
18	35,000	4,270.08	3,653.59	(616.49)	-14.4%	10.4	4,239.56	3,627.90	(611.66)	-14.4%	10.4
19	40,000	4,865.80	4,161.24	(704.56)	-14.5%	10.4	4,830.92	4,131.88	(699.04)	-14.5%	10.3
20	45,000	5,461.53	4,668.90	(792.63)	-14.5%	10.4	5,422.29	4,635.87	(786.42)	-14.5%	10.3
21	50,000	6,057.25	5,176.55	(880.70)	-14.5%	10.4	6,013.65	5,139.85	(873.80)	-14.5%	10.3
22	60,000	7,248.70	6,191.86	(1,056.84)	-14.6%	10.3	7,196.38	6,147.82	(1,048.56)	-14.6%	10.2
23	70,000	8,440.15	7,207.17	(1,232.98)	-14.6%	10.3	8,379.11	7,155.79	(1,223.32)	-14.6%	10.2
24	80,000	9,631.60	8,222.48	(1,409.12)	-14.6%	10.3	9,561.84	8,163.76	(1,398.08)	-14.6%	10.2
25	90,000	10,823.05	9,237.79	(1,585.26)	-14.6%	10.3	10,744.57	9,171.73	(1,572.84)	-14.6%	10.2
26	100,000	12,014.50	10,253.10	(1,761.40)	-14.7%	10.3	11,927.30	10,179.70	(1,747.60)	-14.7%	10.2
27	110,000	13,205.95	11,268.41	(1,937.54)	-14.7%	10.2	13,110.03	11,187.67	(1,922.36)	-14.7%	10.2
28	120,000	14,397.40	12,283.72	(2,113.68)	-14.7%	10.2	14,292.76	12,195.64	(2,097.12)	-14.7%	10.2
29	130,000	15,588.85	13,299.03	(2,289.82)	-14.7%	10.2	15,475.49	13,203.61	(2,271.88)	-14.7%	10.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	7.45	7.12	(0.33)	-4.4%	7.1	7.13	6.82	(0.30)	-4.2%	6.8
2	110	8.17	7.81	(0.36)	-4.4%	7.1	7.82	7.49	(0.33)	-4.2%	6.8
3	120	8.90	8.50	(0.39)	-4.4%	7.1	8.51	8.15	(0.36)	-4.2%	6.8
4	130	9.62	9.19	(0.43)	-4.4%	7.1	9.20	8.81	(0.39)	-4.3%	6.8
5	140	10.35	9.89	(0.46)	-4.4%	7.1	9.90	9.47	(0.42)	-4.3%	6.8
6	150	11.07	10.58	(0.49)	-4.4%	7.1	10.59	10.14	(0.45)	-4.3%	6.8
7	200	14.69	14.04	(0.66)	-4.5%	7.0	14.05	13.45	(0.60)	-4.3%	6.7
8	250	18.32	17.50	(0.82)	-4.5%	7.0	17.51	16.76	(0.75)	-4.3%	6.7
9	300	21.94	20.96	(0.99)	-4.5%	7.0	20.98	20.07	(0.90)	-4.3%	6.7
10	350	25.57	24.42	(1.15)	-4.5%	7.0	24.44	23.38	(1.05)	-4.3%	6.7
11	400	29.19	27.88	(1.31)	-4.5%	7.0	27.90	26.70	(1.20)	-4.3%	6.7
12	450	32.81	31.34	(1.48)	-4.5%	7.0	31.36	30.01	(1.35)	-4.3%	6.7
13	500	36.44	34.80	(1.64)	-4.5%	7.0	34.83	33.32	(1.50)	-4.3%	6.7
14	600	43.68	41.71	(1.97)	-4.5%	7.0	41.75	39.94	(1.81)	-4.3%	6.7
15	700	50.93	48.63	(2.30)	-4.5%	6.9	48.68	46.57	(2.11)	-4.3%	6.7
16	800	58.18	55.55	(2.63)	-4.5%	6.9	55.60	53.19	(2.41)	-4.3%	6.6
17	900	65.43	62.47	(2.96)	-4.5%	6.9	62.53	59.82	(2.71)	-4.3%	6.6
18	1,000	72.67	69.39	(3.28)	-4.5%	6.9	69.45	66.44	(3.01)	-4.3%	6.6
19	1,500	108.91	103.99	(4.93)	-4.5%	6.9	104.08	99.56	(4.51)	-4.3%	6.6
20	2,000	145.15	138.58	(6.57)	-4.5%	6.9	138.70	132.68	(6.02)	-4.3%	6.6
21	2,500	181.39	173.18	(8.21)	-4.5%	6.9	173.33	165.80	(7.52)	-4.3%	6.6
22	3,000	217.62	207.77	(9.85)	-4.5%	6.9	207.95	198.92	(9.03)	-4.3%	6.6
23	3,500	253.86	242.37	(11.49)	-4.5%	6.9	242.58	232.04	(10.53)	-4.3%	6.6
24	4,000	290.10	276.96	(13.14)	-4.5%	6.9	277.20	265.16	(12.04)	-4.3%	6.6
25	4,500	326.34	311.56	(14.78)	-4.5%	6.9	311.83	298.28	(13.54)	-4.3%	6.6
26	5,000	362.57	346.15	(16.42)	-4.5%	6.9	346.45	331.41	(15.05)	-4.3%	6.6
27	5,500	398.81	380.75	(18.06)	-4.5%	6.9	381.08	364.53	(16.55)	-4.3%	6.6
28	6,000	435.05	415.34	(19.70)	-4.5%	6.9	415.70	397.65	(18.06)	-4.3%	6.6
29	6,500	471.29	449.94	(21.35)	-4.5%	6.9	450.33	430.77	(19.56)	-4.3%	6.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Primary Interruptible Provision GI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	8.61	8.05	(0.56)	-6.5%	8.1	8.36	7.78	(0.58)	-7.0%	7.8
2	110	9.45	8.84	(0.62)	-6.5%	8.0	9.18	8.54	(0.64)	-7.0%	7.8
3	120	10.29	9.62	(0.67)	-6.5%	8.0	10.00	9.30	(0.70)	-7.0%	7.7
4	130	11.14	10.41	(0.73)	-6.6%	8.0	10.81	10.05	(0.76)	-7.0%	7.7
5	140	11.98	11.19	(0.79)	-6.6%	8.0	11.63	10.81	(0.82)	-7.0%	7.7
6	150	12.82	11.98	(0.84)	-6.6%	8.0	12.44	11.57	(0.87)	-7.0%	7.7
7	200	17.02	15.90	(1.12)	-6.6%	8.0	16.53	15.36	(1.17)	-7.1%	7.7
8	250	21.23	19.83	(1.40)	-6.6%	7.9	20.61	19.15	(1.46)	-7.1%	7.7
9	300	25.44	23.75	(1.69)	-6.6%	7.9	24.69	22.94	(1.75)	-7.1%	7.6
10	350	29.64	27.68	(1.97)	-6.6%	7.9	28.77	26.73	(2.04)	-7.1%	7.6
11	400	33.85	31.60	(2.25)	-6.6%	7.9	32.85	30.52	(2.33)	-7.1%	7.6
12	450	38.05	35.53	(2.53)	-6.6%	7.9	36.93	34.31	(2.62)	-7.1%	7.6
13	500	42.26	39.45	(2.81)	-6.6%	7.9	41.01	38.10	(2.91)	-7.1%	7.6
14	600	50.67	47.30	(3.37)	-6.7%	7.9	49.18	45.68	(3.50)	-7.1%	7.6
15	700	59.09	55.15	(3.93)	-6.7%	7.9	57.34	53.26	(4.08)	-7.1%	7.6
16	800	67.50	63.00	(4.49)	-6.7%	7.9	65.50	60.84	(4.66)	-7.1%	7.6
17	900	75.91	70.85	(5.06)	-6.7%	7.9	73.66	68.42	(5.24)	-7.1%	7.6
18	1,000	84.32	78.70	(5.62)	-6.7%	7.9	81.83	76.00	(5.83)	-7.1%	7.6
19	1,500	126.38	117.96	(8.43)	-6.7%	7.9	122.64	113.90	(8.74)	-7.1%	7.6
20	2,000	168.44	157.21	(11.23)	-6.7%	7.9	163.45	151.80	(11.65)	-7.1%	7.6
21	2,500	210.50	196.46	(14.04)	-6.7%	7.9	204.27	189.70	(14.56)	-7.1%	7.6
22	3,000	252.57	235.71	(16.85)	-6.7%	7.9	245.08	227.60	(17.48)	-7.1%	7.6
23	3,500	294.63	274.97	(19.66)	-6.7%	7.9	285.89	265.50	(20.39)	-7.1%	7.6
24	4,000	336.69	314.22	(22.47)	-6.7%	7.9	326.70	303.40	(23.30)	-7.1%	7.6
25	4,500	378.75	353.47	(25.28)	-6.7%	7.9	367.52	341.30	(26.21)	-7.1%	7.6
26	5,000	420.81	392.72	(28.09)	-6.7%	7.9	408.33	379.20	(29.13)	-7.1%	7.6
27	5,500	462.87	431.98	(30.90)	-6.7%	7.9	449.14	417.10	(32.04)	-7.1%	7.6
28	6,000	504.93	471.23	(33.70)	-6.7%	7.9	489.96	455.00	(34.95)	-7.1%	7.6
29	6,500	546.99	510.48	(36.51)	-6.7%	7.9	530.77	492.90	(37.87)	-7.1%	7.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
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Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	10.17	9.25	(0.92)	-9.0%	9.2	9.74	8.81	(0.93)	-9.6%	8.8
2	110	11.17	10.15	(1.01)	-9.1%	9.2	10.70	9.67	(1.03)	-9.6%	8.8
3	120	12.16	11.06	(1.10)	-9.1%	9.2	11.65	10.53	(1.12)	-9.6%	8.8
4	130	13.16	11.96	(1.20)	-9.1%	9.2	12.61	11.39	(1.21)	-9.6%	8.8
5	140	14.16	12.87	(1.29)	-9.1%	9.2	13.56	12.25	(1.31)	-9.6%	8.8
6	150	15.15	13.77	(1.38)	-9.1%	9.2	14.52	13.11	(1.40)	-9.6%	8.7
7	200	20.14	18.30	(1.84)	-9.1%	9.1	19.29	17.42	(1.87)	-9.7%	8.7
8	250	25.12	22.82	(2.30)	-9.2%	9.1	24.06	21.72	(2.33)	-9.7%	8.7
9	300	30.11	27.35	(2.76)	-9.2%	9.1	28.83	26.03	(2.80)	-9.7%	8.7
10	350	35.09	31.87	(3.22)	-9.2%	9.1	33.60	30.33	(3.27)	-9.7%	8.7
11	400	40.08	36.40	(3.68)	-9.2%	9.1	38.37	34.64	(3.74)	-9.7%	8.7
12	450	45.06	40.92	(4.14)	-9.2%	9.1	43.15	38.94	(4.20)	-9.7%	8.7
13	500	50.04	45.45	(4.60)	-9.2%	9.1	47.92	43.25	(4.67)	-9.7%	8.6
14	600	60.01	54.49	(5.52)	-9.2%	9.1	57.46	51.86	(5.60)	-9.8%	8.6
15	700	69.98	63.54	(6.44)	-9.2%	9.1	67.01	60.47	(6.54)	-9.8%	8.6
16	800	79.95	72.59	(7.36)	-9.2%	9.1	76.55	69.08	(7.47)	-9.8%	8.6
17	900	89.92	81.64	(8.28)	-9.2%	9.1	86.09	77.69	(8.40)	-9.8%	8.6
18	1,000	99.89	90.69	(9.20)	-9.2%	9.1	95.64	86.30	(9.34)	-9.8%	8.6
19	1,500	149.73	135.94	(13.80)	-9.2%	9.1	143.36	129.35	(14.01)	-9.8%	8.6
20	2,000	199.58	181.18	(18.39)	-9.2%	9.1	191.07	172.40	(18.68)	-9.8%	8.6
21	2,500	249.42	226.43	(22.99)	-9.2%	9.1	238.79	215.45	(23.34)	-9.8%	8.6
22	3,000	299.26	271.67	(27.59)	-9.2%	9.1	286.51	258.50	(28.01)	-9.8%	8.6
23	3,500	349.11	316.92	(32.19)	-9.2%	9.1	334.23	301.55	(32.68)	-9.8%	8.6
24	4,000	398.95	362.16	(36.79)	-9.2%	9.1	381.95	344.60	(37.35)	-9.8%	8.6
25	4,500	448.80	407.41	(41.39)	-9.2%	9.1	429.67	387.65	(42.02)	-9.8%	8.6
26	5,000	498.64	452.65	(45.99)	-9.2%	9.1	477.39	430.70	(46.69)	-9.8%	8.6
27	5,500	548.48	497.90	(50.59)	-9.2%	9.1	525.10	473.75	(51.36)	-9.8%	8.6
28	6,000	598.33	543.14	(55.18)	-9.2%	9.1	572.82	516.80	(56.03)	-9.8%	8.6
29	6,500	648.17	588.39	(59.78)	-9.2%	9.1	620.54	559.85	(60.69)	-9.8%	8.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Primary Interruptible Provision GI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	9.75	8.73	(1.01)	-10.4%	8.7	9.50	8.45	(1.05)	-11.0%	8.5
2	110	10.70	9.59	(1.11)	-10.4%	8.7	10.43	9.28	(1.15)	-11.1%	8.4
3	120	11.65	10.44	(1.21)	-10.4%	8.7	11.36	10.11	(1.26)	-11.1%	8.4
4	130	12.61	11.30	(1.31)	-10.4%	8.7	12.29	10.93	(1.36)	-11.1%	8.4
5	140	13.56	12.15	(1.41)	-10.4%	8.7	13.22	11.76	(1.47)	-11.1%	8.4
6	150	14.52	13.00	(1.52)	-10.4%	8.7	14.15	12.58	(1.57)	-11.1%	8.4
7	200	19.29	17.27	(2.02)	-10.5%	8.6	18.81	16.71	(2.10)	-11.2%	8.4
8	250	24.06	21.54	(2.53)	-10.5%	8.6	23.46	20.84	(2.62)	-11.2%	8.3
9	300	28.84	25.80	(3.03)	-10.5%	8.6	28.11	24.96	(3.15)	-11.2%	8.3
10	350	33.61	30.07	(3.54)	-10.5%	8.6	32.76	29.09	(3.67)	-11.2%	8.3
11	400	38.38	34.34	(4.04)	-10.5%	8.6	37.41	33.22	(4.19)	-11.2%	8.3
12	450	43.15	38.61	(4.55)	-10.5%	8.6	42.06	37.35	(4.72)	-11.2%	8.3
13	500	47.93	42.87	(5.05)	-10.5%	8.6	46.72	41.47	(5.24)	-11.2%	8.3
14	600	57.47	51.41	(6.06)	-10.6%	8.6	56.02	49.73	(6.29)	-11.2%	8.3
15	700	67.02	59.94	(7.07)	-10.6%	8.6	65.32	57.98	(7.34)	-11.2%	8.3
16	800	76.56	68.48	(8.09)	-10.6%	8.6	74.62	66.24	(8.39)	-11.2%	8.3
17	900	86.11	77.01	(9.10)	-10.6%	8.6	83.93	74.49	(9.44)	-11.2%	8.3
18	1,000	95.66	85.55	(10.11)	-10.6%	8.6	93.23	82.74	(10.49)	-11.2%	8.3
19	1,500	143.38	128.22	(15.16)	-10.6%	8.5	139.75	124.02	(15.73)	-11.3%	8.3
20	2,000	191.11	170.90	(20.21)	-10.6%	8.5	186.26	165.29	(20.97)	-11.3%	8.3
21	2,500	238.84	213.57	(25.27)	-10.6%	8.5	232.78	206.56	(26.21)	-11.3%	8.3
22	3,000	286.57	256.24	(30.32)	-10.6%	8.5	279.29	247.83	(31.46)	-11.3%	8.3
23	3,500	334.29	298.92	(35.37)	-10.6%	8.5	325.81	289.11	(36.70)	-11.3%	8.3
24	4,000	382.02	341.59	(40.43)	-10.6%	8.5	372.32	330.38	(41.94)	-11.3%	8.3
25	4,500	429.75	384.27	(45.48)	-10.6%	8.5	418.84	371.65	(47.19)	-11.3%	8.3
26	5,000	477.48	426.94	(50.53)	-10.6%	8.5	465.35	412.92	(52.43)	-11.3%	8.3
27	5,500	525.20	469.62	(55.59)	-10.6%	8.5	511.87	454.20	(57.67)	-11.3%	8.3
28	6,000	572.93	512.29	(60.64)	-10.6%	8.5	558.38	495.47	(62.91)	-11.3%	8.3
29	6,500	620.66	554.96	(65.70)	-10.6%	8.5	604.90	536.74	(68.16)	-11.3%	8.3

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 30 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	11.43	9.87	(1.57)	-13.7%	9.9	11.02	9.42	(1.60)	-14.5%	9.4
2	110	12.56	10.83	(1.72)	-13.7%	9.8	12.10	10.34	(1.76)	-14.5%	9.4
3	120	13.68	11.80	(1.88)	-13.7%	9.8	13.18	11.26	(1.92)	-14.5%	9.4
4	130	14.80	12.77	(2.03)	-13.7%	9.8	14.26	12.18	(2.08)	-14.6%	9.4
5	140	15.93	13.74	(2.19)	-13.8%	9.8	15.34	13.11	(2.24)	-14.6%	9.4
6	150	17.05	14.70	(2.35)	-13.8%	9.8	16.42	14.03	(2.40)	-14.6%	9.4
7	200	22.67	19.54	(3.13)	-13.8%	9.8	21.83	18.64	(3.20)	-14.6%	9.3
8	250	28.28	24.37	(3.91)	-13.8%	9.7	27.24	23.24	(3.99)	-14.7%	9.3
9	300	33.90	29.20	(4.70)	-13.9%	9.7	32.65	27.85	(4.79)	-14.7%	9.3
10	350	39.52	34.04	(5.48)	-13.9%	9.7	38.05	32.46	(5.59)	-14.7%	9.3
11	400	45.13	38.87	(6.26)	-13.9%	9.7	43.46	37.07	(6.39)	-14.7%	9.3
12	450	50.75	43.71	(7.04)	-13.9%	9.7	48.87	41.68	(7.19)	-14.7%	9.3
13	500	56.37	48.54	(7.83)	-13.9%	9.7	54.28	46.29	(7.99)	-14.7%	9.3
14	600	67.60	58.21	(9.39)	-13.9%	9.7	65.09	55.51	(9.59)	-14.7%	9.3
15	700	78.83	67.88	(10.96)	-13.9%	9.7	75.91	64.73	(11.18)	-14.7%	9.2
16	800	90.07	77.54	(12.52)	-13.9%	9.7	86.72	73.94	(12.78)	-14.7%	9.2
17	900	101.30	87.21	(14.09)	-13.9%	9.7	97.54	83.16	(14.38)	-14.7%	9.2
18	1,000	112.53	96.88	(15.65)	-13.9%	9.7	108.35	92.38	(15.98)	-14.7%	9.2
19	1,500	168.70	145.22	(23.48)	-13.9%	9.7	162.43	138.47	(23.96)	-14.8%	9.2
20	2,000	224.86	193.56	(31.30)	-13.9%	9.7	216.51	184.56	(31.95)	-14.8%	9.2
21	2,500	281.03	241.90	(39.13)	-13.9%	9.7	270.58	230.65	(39.94)	-14.8%	9.2
22	3,000	337.20	290.24	(46.96)	-13.9%	9.7	324.66	276.74	(47.93)	-14.8%	9.2
23	3,500	393.36	338.58	(54.78)	-13.9%	9.7	378.74	322.83	(55.91)	-14.8%	9.2
24	4,000	449.53	386.92	(62.61)	-13.9%	9.7	432.82	368.92	(63.90)	-14.8%	9.2
25	4,500	505.69	435.26	(70.44)	-13.9%	9.7	486.89	415.01	(71.89)	-14.8%	9.2
26	5,000	561.86	483.60	(78.26)	-13.9%	9.7	540.97	461.09	(79.88)	-14.8%	9.2
27	5,500	618.03	531.94	(86.09)	-13.9%	9.7	595.05	507.18	(87.86)	-14.8%	9.2
28	6,000	674.19	580.28	(93.91)	-13.9%	9.7	649.12	553.27	(95.85)	-14.8%	9.2
29	6,500	730.36	628.62	(101.74)	-13.9%	9.7	703.20	599.36	(103.84)	-14.8%	9.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Primary Interruptible Provision GI

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	11.15	9.47	(1.68)	-15.1%	9.5	10.91	9.18	(1.74)	-15.9%	9.2
2	110	12.24	10.40	(1.85)	-15.1%	9.5	11.98	10.07	(1.91)	-15.9%	9.2
3	120	13.34	11.32	(2.01)	-15.1%	9.4	13.05	10.97	(2.08)	-16.0%	9.1
4	130	14.43	12.25	(2.18)	-15.1%	9.4	14.13	11.87	(2.26)	-16.0%	9.1
5	140	15.53	13.18	(2.35)	-15.1%	9.4	15.20	12.77	(2.43)	-16.0%	9.1
6	150	16.62	14.10	(2.52)	-15.1%	9.4	16.27	13.66	(2.61)	-16.0%	9.1
7	200	22.09	18.74	(3.36)	-15.2%	9.4	21.62	18.15	(3.47)	-16.1%	9.1
8	250	27.57	23.37	(4.19)	-15.2%	9.3	26.98	22.64	(4.34)	-16.1%	9.1
9	300	33.04	28.01	(5.03)	-15.2%	9.3	32.34	27.13	(5.21)	-16.1%	9.0
10	350	38.52	32.64	(5.87)	-15.2%	9.3	37.69	31.61	(6.08)	-16.1%	9.0
11	400	43.99	37.28	(6.71)	-15.3%	9.3	43.05	36.10	(6.95)	-16.1%	9.0
12	450	49.46	41.91	(7.55)	-15.3%	9.3	48.40	40.59	(7.82)	-16.1%	9.0
13	500	54.94	46.55	(8.39)	-15.3%	9.3	53.76	45.08	(8.68)	-16.2%	9.0
14	600	65.88	55.82	(10.07)	-15.3%	9.3	64.47	54.05	(10.42)	-16.2%	9.0
15	700	76.83	65.09	(11.74)	-15.3%	9.3	75.19	63.03	(12.16)	-16.2%	9.0
16	800	87.78	74.36	(13.42)	-15.3%	9.3	85.90	72.00	(13.89)	-16.2%	9.0
17	900	98.73	83.63	(15.10)	-15.3%	9.3	96.61	80.98	(15.63)	-16.2%	9.0
18	1,000	109.67	92.90	(16.78)	-15.3%	9.3	107.32	89.96	(17.37)	-16.2%	9.0
19	1,500	164.41	139.24	(25.17)	-15.3%	9.3	160.88	134.83	(26.05)	-16.2%	9.0
20	2,000	219.15	185.59	(33.56)	-15.3%	9.3	214.44	179.71	(34.73)	-16.2%	9.0
21	2,500	273.88	231.94	(41.95)	-15.3%	9.3	268.01	224.59	(43.42)	-16.2%	9.0
22	3,000	328.62	278.29	(50.34)	-15.3%	9.3	321.57	269.47	(52.10)	-16.2%	9.0
23	3,500	383.36	324.63	(58.72)	-15.3%	9.3	375.13	314.34	(60.78)	-16.2%	9.0
24	4,000	438.09	370.98	(67.11)	-15.3%	9.3	428.69	359.22	(69.47)	-16.2%	9.0
25	4,500	492.83	417.33	(75.50)	-15.3%	9.3	482.25	404.10	(78.15)	-16.2%	9.0
26	5,000	547.57	463.68	(83.89)	-15.3%	9.3	535.81	448.98	(86.83)	-16.2%	9.0
27	5,500	602.30	510.02	(92.28)	-15.3%	9.3	589.37	493.85	(95.52)	-16.2%	9.0
28	6,000	657.04	556.37	(100.67)	-15.3%	9.3	642.93	538.73	(104.20)	-16.2%	9.0
29	6,500	711.78	602.72	(109.06)	-15.3%	9.3	696.49	583.61	(112.89)	-16.2%	9.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 1

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 32 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	8.45	9.13	0.69	8.1%	9.1	6.96	7.59	0.63	9.0%	7.6
2	110	9.27	10.02	0.75	8.1%	9.1	7.64	8.33	0.69	9.0%	7.6
3	120	10.09	10.92	0.82	8.2%	9.1	8.32	9.07	0.75	9.1%	7.6
4	130	10.92	11.81	0.89	8.2%	9.1	8.99	9.81	0.82	9.1%	7.5
5	140	11.74	12.70	0.96	8.2%	9.1	9.67	10.55	0.88	9.1%	7.5
6	150	12.57	13.60	1.03	8.2%	9.1	10.34	11.29	0.94	9.1%	7.5
7	200	16.69	18.06	1.37	8.2%	9.0	13.73	14.98	1.26	9.1%	7.5
8	250	20.81	22.53	1.72	8.2%	9.0	17.11	18.68	1.57	9.2%	7.5
9	300	24.94	26.99	2.06	8.3%	9.0	20.49	22.37	1.88	9.2%	7.5
10	350	29.06	31.46	2.40	8.3%	9.0	23.87	26.07	2.20	9.2%	7.4
11	400	33.18	35.93	2.75	8.3%	9.0	27.25	29.76	2.51	9.2%	7.4
12	450	37.30	40.39	3.09	8.3%	9.0	30.63	33.46	2.82	9.2%	7.4
13	500	41.43	44.86	3.43	8.3%	9.0	34.02	37.15	3.14	9.2%	7.4
14	600	49.67	53.79	4.12	8.3%	9.0	40.78	44.54	3.77	9.2%	7.4
15	700	57.92	62.72	4.80	8.3%	9.0	47.54	51.94	4.39	9.2%	7.4
16	800	66.16	71.65	5.49	8.3%	9.0	54.31	59.33	5.02	9.2%	7.4
17	900	74.41	80.58	6.18	8.3%	9.0	61.07	66.72	5.65	9.2%	7.4
18	1,000	82.65	89.52	6.86	8.3%	9.0	67.83	74.11	6.28	9.3%	7.4
19	1,500	123.88	134.17	10.29	8.3%	8.9	101.65	111.06	9.41	9.3%	7.4
20	2,000	165.11	178.83	13.73	8.3%	8.9	135.46	148.01	12.55	9.3%	7.4
21	2,500	206.33	223.49	17.16	8.3%	8.9	169.28	184.97	15.69	9.3%	7.4
22	3,000	247.56	268.15	20.59	8.3%	8.9	203.10	221.92	18.83	9.3%	7.4
23	3,500	288.78	312.81	24.02	8.3%	8.9	236.91	258.88	21.96	9.3%	7.4
24	4,000	330.01	357.46	27.45	8.3%	8.9	270.73	295.83	25.10	9.3%	7.4
25	4,500	371.24	402.12	30.88	8.3%	8.9	304.54	332.78	28.24	9.3%	7.4
26	5,000	412.46	446.78	34.32	8.3%	8.9	338.36	369.74	31.38	9.3%	7.4
27	5,500	453.69	491.44	37.75	8.3%	8.9	372.17	406.69	34.52	9.3%	7.4
28	6,000	494.92	536.10	41.18	8.3%	8.9	405.99	443.64	37.65	9.3%	7.4
29	6,500	536.14	580.75	44.61	8.3%	8.9	439.81	480.60	40.79	9.3%	7.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 2

Case No.: U-20697
Exhibit No.: A-16 (HWM-4)
Schedule F-4.0
Page 33 of 46
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	8.88	9.30	0.42	4.7%	9.3	7.70	8.29	0.59	7.7%	8.3
2	110	9.75	10.21	0.46	4.7%	9.3	8.45	9.10	0.65	7.7%	8.3
3	120	10.62	11.12	0.50	4.7%	9.3	9.20	9.91	0.71	7.7%	8.3
4	130	11.49	12.03	0.54	4.7%	9.3	9.95	10.72	0.77	7.7%	8.2
5	140	12.36	12.94	0.58	4.7%	9.2	10.70	11.53	0.83	7.7%	8.2
6	150	13.22	13.85	0.62	4.7%	9.2	11.45	12.34	0.89	7.7%	8.2
7	200	17.56	18.40	0.83	4.7%	9.2	15.20	16.38	1.18	7.8%	8.2
8	250	21.91	22.95	1.04	4.7%	9.2	18.95	20.43	1.48	7.8%	8.2
9	300	26.25	27.50	1.25	4.8%	9.2	22.70	24.48	1.77	7.8%	8.2
10	350	30.59	32.04	1.46	4.8%	9.2	26.46	28.52	2.07	7.8%	8.1
11	400	34.93	36.59	1.66	4.8%	9.1	30.21	32.57	2.36	7.8%	8.1
12	450	39.27	41.14	1.87	4.8%	9.1	33.96	36.61	2.66	7.8%	8.1
13	500	43.61	45.69	2.08	4.8%	9.1	37.71	40.66	2.95	7.8%	8.1
14	600	52.29	54.79	2.50	4.8%	9.1	45.21	48.75	3.54	7.8%	8.1
15	700	60.98	63.89	2.91	4.8%	9.1	52.71	56.84	4.13	7.8%	8.1
16	800	69.66	72.99	3.33	4.8%	9.1	60.21	64.94	4.72	7.8%	8.1
17	900	78.34	82.09	3.74	4.8%	9.1	67.71	73.03	5.31	7.8%	8.1
18	1,000	87.02	91.18	4.16	4.8%	9.1	75.22	81.12	5.91	7.9%	8.1
19	1,500	130.44	136.68	6.24	4.8%	9.1	112.72	121.58	8.86	7.9%	8.1
20	2,000	173.85	182.17	8.32	4.8%	9.1	150.23	162.04	11.81	7.9%	8.1
21	2,500	217.26	227.66	10.40	4.8%	9.1	187.74	202.50	14.76	7.9%	8.1
22	3,000	260.67	273.15	12.48	4.8%	9.1	225.25	242.96	17.72	7.9%	8.1
23	3,500	304.08	318.64	14.56	4.8%	9.1	262.75	283.42	20.67	7.9%	8.1
24	4,000	347.49	364.13	16.64	4.8%	9.1	300.26	323.88	23.62	7.9%	8.1
25	4,500	390.91	409.63	18.72	4.8%	9.1	337.77	364.34	26.57	7.9%	8.1
26	5,000	434.32	455.12	20.80	4.8%	9.1	375.28	404.80	29.53	7.9%	8.1
27	5,500	477.73	500.61	22.88	4.8%	9.1	412.78	445.26	32.48	7.9%	8.1
28	6,000	521.14	546.10	24.96	4.8%	9.1	450.29	485.72	35.43	7.9%	8.1
29	6,500	564.55	591.59	27.04	4.8%	9.1	487.80	526.18	38.38	7.9%	8.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Time-of-Use GPTU Voltage Level 3

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	10.76	10.58	(0.18)	-1.7%	10.6	9.29	9.11	(0.18)	-1.9%	9.1
2	110	11.82	11.62	(0.20)	-1.7%	10.6	10.20	10.00	(0.20)	-1.9%	9.1
3	120	12.88	12.65	(0.22)	-1.7%	10.5	11.11	10.89	(0.22)	-1.9%	9.1
4	130	13.93	13.69	(0.24)	-1.7%	10.5	12.02	11.78	(0.23)	-1.9%	9.1
5	140	14.99	14.73	(0.26)	-1.7%	10.5	12.93	12.67	(0.25)	-1.9%	9.1
6	150	16.04	15.77	(0.28)	-1.7%	10.5	13.83	13.56	(0.27)	-2.0%	9.0
7	200	21.33	20.96	(0.37)	-1.7%	10.5	18.38	18.02	(0.36)	-2.0%	9.0
8	250	26.61	26.15	(0.46)	-1.7%	10.5	22.92	22.47	(0.45)	-2.0%	9.0
9	300	31.89	31.34	(0.55)	-1.7%	10.4	27.47	26.93	(0.54)	-2.0%	9.0
10	350	37.17	36.53	(0.64)	-1.7%	10.4	32.01	31.38	(0.63)	-2.0%	9.0
11	400	42.45	41.72	(0.74)	-1.7%	10.4	36.56	35.84	(0.72)	-2.0%	9.0
12	450	47.73	46.91	(0.83)	-1.7%	10.4	41.10	40.29	(0.81)	-2.0%	9.0
13	500	53.01	52.09	(0.92)	-1.7%	10.4	45.65	44.75	(0.90)	-2.0%	8.9
14	600	63.58	62.47	(1.10)	-1.7%	10.4	54.74	53.66	(1.08)	-2.0%	8.9
15	700	74.14	72.85	(1.29)	-1.7%	10.4	63.83	62.57	(1.26)	-2.0%	8.9
16	800	84.70	83.23	(1.47)	-1.7%	10.4	72.92	71.48	(1.44)	-2.0%	8.9
17	900	95.26	93.61	(1.65)	-1.7%	10.4	82.01	80.39	(1.62)	-2.0%	8.9
18	1,000	105.83	103.99	(1.84)	-1.7%	10.4	91.10	89.30	(1.80)	-2.0%	8.9
19	1,500	158.64	155.88	(2.76)	-1.7%	10.4	136.54	133.85	(2.70)	-2.0%	8.9
20	2,000	211.45	207.78	(3.68)	-1.7%	10.4	181.99	178.39	(3.60)	-2.0%	8.9
21	2,500	264.27	259.67	(4.59)	-1.7%	10.4	227.44	222.94	(4.50)	-2.0%	8.9
22	3,000	317.08	311.57	(5.51)	-1.7%	10.4	272.89	267.49	(5.40)	-2.0%	8.9
23	3,500	369.89	363.46	(6.43)	-1.7%	10.4	318.34	312.04	(6.30)	-2.0%	8.9
24	4,000	422.71	415.36	(7.35)	-1.7%	10.4	363.78	356.59	(7.20)	-2.0%	8.9
25	4,500	475.52	467.25	(8.27)	-1.7%	10.4	409.23	401.14	(8.10)	-2.0%	8.9
26	5,000	528.33	519.15	(9.19)	-1.7%	10.4	454.68	445.68	(9.00)	-2.0%	8.9
27	5,500	581.15	571.04	(10.11)	-1.7%	10.4	500.13	490.23	(9.89)	-2.0%	8.9
28	6,000	633.96	622.94	(11.03)	-1.7%	10.4	545.58	534.78	(10.79)	-2.0%	8.9
29	6,500	686.77	674.83	(11.94)	-1.7%	10.4	591.02	579.33	(11.69)	-2.0%	8.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 1

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	6.73	5.89	(0.84)	-12.5%	5.9	5.82	5.11	(0.71)	-12.2%	5.1
2	110	7.38	6.46	(0.93)	-12.6%	5.9	6.38	5.60	(0.78)	-12.2%	5.1
3	120	8.04	7.02	(1.01)	-12.6%	5.9	6.94	6.09	(0.85)	-12.2%	5.1
4	130	8.69	7.59	(1.10)	-12.6%	5.8	7.51	6.58	(0.92)	-12.3%	5.1
5	140	9.34	8.16	(1.18)	-12.6%	5.8	8.07	7.08	(0.99)	-12.3%	5.1
6	150	10.00	8.73	(1.27)	-12.7%	5.8	8.63	7.57	(1.06)	-12.3%	5.0
7	200	13.26	11.57	(1.69)	-12.7%	5.8	11.44	10.02	(1.42)	-12.4%	5.0
8	250	16.53	14.42	(2.11)	-12.8%	5.8	14.25	12.48	(1.77)	-12.4%	5.0
9	300	19.79	17.26	(2.53)	-12.8%	5.8	17.06	14.93	(2.13)	-12.5%	5.0
10	350	23.06	20.10	(2.95)	-12.8%	5.7	19.87	17.39	(2.48)	-12.5%	5.0
11	400	26.32	22.95	(3.38)	-12.8%	5.7	22.68	19.84	(2.83)	-12.5%	5.0
12	450	29.59	25.79	(3.80)	-12.8%	5.7	25.49	22.30	(3.19)	-12.5%	5.0
13	500	32.85	28.63	(4.22)	-12.8%	5.7	28.30	24.75	(3.54)	-12.5%	5.0
14	600	39.38	34.32	(5.06)	-12.9%	5.7	33.92	29.66	(4.25)	-12.5%	4.9
15	700	45.91	40.01	(5.91)	-12.9%	5.7	39.54	34.58	(4.96)	-12.5%	4.9
16	800	52.44	45.69	(6.75)	-12.9%	5.7	45.16	39.49	(5.67)	-12.6%	4.9
17	900	58.97	51.38	(7.60)	-12.9%	5.7	50.78	44.40	(6.38)	-12.6%	4.9
18	1,000	65.50	57.06	(8.44)	-12.9%	5.7	56.39	49.31	(7.09)	-12.6%	4.9
19	1,500	98.16	85.50	(12.66)	-12.9%	5.7	84.49	73.86	(10.63)	-12.6%	4.9
20	2,000	130.81	113.93	(16.88)	-12.9%	5.7	112.59	98.42	(14.17)	-12.6%	4.9
21	2,500	163.46	142.36	(21.10)	-12.9%	5.7	140.69	122.97	(17.72)	-12.6%	4.9
22	3,000	196.11	170.79	(25.32)	-12.9%	5.7	168.78	147.52	(21.26)	-12.6%	4.9
23	3,500	228.76	199.23	(29.54)	-12.9%	5.7	196.88	172.08	(24.80)	-12.6%	4.9
24	4,000	261.42	227.66	(33.76)	-12.9%	5.7	224.98	196.63	(28.35)	-12.6%	4.9
25	4,500	294.07	256.09	(37.98)	-12.9%	5.7	253.08	221.19	(31.89)	-12.6%	4.9
26	5,000	326.72	284.52	(42.20)	-12.9%	5.7	281.17	245.74	(35.43)	-12.6%	4.9
27	5,500	359.37	312.96	(46.42)	-12.9%	5.7	309.27	270.30	(38.98)	-12.6%	4.9
28	6,000	392.02	341.39	(50.64)	-12.9%	5.7	337.37	294.85	(42.52)	-12.6%	4.9
29	6,500	424.68	369.82	(54.86)	-12.9%	5.7	365.47	319.40	(46.06)	-12.6%	4.9

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 2

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	8.06	6.65	(1.41)	-17.5%	6.6	7.15	5.86	(1.29)	-18.0%	5.9
2	110	8.84	7.29	(1.55)	-17.6%	6.6	7.84	6.42	(1.42)	-18.1%	5.8
3	120	9.63	7.93	(1.69)	-17.6%	6.6	8.53	6.99	(1.55)	-18.1%	5.8
4	130	10.41	8.58	(1.83)	-17.6%	6.6	9.23	7.55	(1.68)	-18.2%	5.8
5	140	11.20	9.22	(1.97)	-17.6%	6.6	9.92	8.12	(1.80)	-18.2%	5.8
6	150	11.98	9.87	(2.12)	-17.7%	6.6	10.62	8.69	(1.93)	-18.2%	5.8
7	200	15.91	13.09	(2.82)	-17.7%	6.5	14.09	11.51	(2.58)	-18.3%	5.8
8	250	19.84	16.31	(3.53)	-17.8%	6.5	17.56	14.34	(3.22)	-18.3%	5.7
9	300	23.77	19.54	(4.23)	-17.8%	6.5	21.04	17.17	(3.87)	-18.4%	5.7
10	350	27.70	22.76	(4.94)	-17.8%	6.5	24.51	20.00	(4.51)	-18.4%	5.7
11	400	31.63	25.98	(5.64)	-17.8%	6.5	27.98	22.83	(5.15)	-18.4%	5.7
12	450	35.55	29.21	(6.35)	-17.9%	6.5	31.45	25.66	(5.80)	-18.4%	5.7
13	500	39.48	32.43	(7.05)	-17.9%	6.5	34.93	28.48	(6.44)	-18.4%	5.7
14	600	47.34	38.87	(8.46)	-17.9%	6.5	41.87	34.14	(7.73)	-18.5%	5.7
15	700	55.19	45.32	(9.87)	-17.9%	6.5	48.82	39.80	(9.02)	-18.5%	5.7
16	800	63.05	51.76	(11.29)	-17.9%	6.5	55.76	45.45	(10.31)	-18.5%	5.7
17	900	70.91	58.21	(12.70)	-17.9%	6.5	62.71	51.11	(11.60)	-18.5%	5.7
18	1,000	78.76	64.66	(14.11)	-17.9%	6.5	69.65	56.77	(12.89)	-18.5%	5.7
19	1,500	118.04	96.88	(21.16)	-17.9%	6.5	104.38	85.05	(19.33)	-18.5%	5.7
20	2,000	157.33	129.11	(28.21)	-17.9%	6.5	139.11	113.33	(25.77)	-18.5%	5.7
21	2,500	196.61	161.34	(35.27)	-17.9%	6.5	173.84	141.62	(32.22)	-18.5%	5.7
22	3,000	235.89	193.57	(42.32)	-17.9%	6.5	208.56	169.90	(38.66)	-18.5%	5.7
23	3,500	275.17	225.80	(49.37)	-17.9%	6.5	243.29	198.19	(45.10)	-18.5%	5.7
24	4,000	314.45	258.02	(56.43)	-17.9%	6.5	278.02	226.47	(51.55)	-18.5%	5.7
25	4,500	353.73	290.25	(63.48)	-17.9%	6.5	312.74	254.75	(57.99)	-18.5%	5.7
26	5,000	393.02	322.48	(70.53)	-17.9%	6.4	347.47	283.04	(64.43)	-18.5%	5.7
27	5,500	432.30	354.71	(77.59)	-17.9%	6.4	382.20	311.32	(70.88)	-18.5%	5.7
28	6,000	471.58	386.94	(84.64)	-17.9%	6.4	416.92	339.60	(77.32)	-18.5%	5.7
29	6,500	510.86	419.16	(91.70)	-17.9%	6.4	451.65	367.89	(83.76)	-18.5%	5.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy Intensive EIP Voltage Level 3

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	8.05	7.08	(0.97)	-12.1%	7.1	7.14	6.60	(0.54)	-7.5%	6.6
2	110	8.83	7.76	(1.07)	-12.1%	7.1	7.83	7.24	(0.59)	-7.5%	6.6
3	120	9.62	8.45	(1.17)	-12.2%	7.0	8.53	7.88	(0.64)	-7.6%	6.6
4	130	10.40	9.14	(1.27)	-12.2%	7.0	9.22	8.52	(0.70)	-7.6%	6.6
5	140	11.19	9.83	(1.36)	-12.2%	7.0	9.91	9.16	(0.75)	-7.6%	6.5
6	150	11.97	10.51	(1.46)	-12.2%	7.0	10.61	9.80	(0.81)	-7.6%	6.5
7	200	15.90	13.95	(1.95)	-12.3%	7.0	14.08	13.00	(1.07)	-7.6%	6.5
8	250	19.82	17.39	(2.44)	-12.3%	7.0	17.55	16.20	(1.34)	-7.7%	6.5
9	300	23.75	20.83	(2.92)	-12.3%	6.9	21.02	19.41	(1.61)	-7.7%	6.5
10	350	27.67	24.26	(3.41)	-12.3%	6.9	24.49	22.61	(1.88)	-7.7%	6.5
11	400	31.60	27.70	(3.90)	-12.3%	6.9	27.96	25.81	(2.15)	-7.7%	6.5
12	450	35.52	31.14	(4.38)	-12.3%	6.9	31.43	29.01	(2.42)	-7.7%	6.4
13	500	39.45	34.58	(4.87)	-12.3%	6.9	34.90	32.21	(2.69)	-7.7%	6.4
14	600	47.30	41.45	(5.85)	-12.4%	6.9	41.83	38.61	(3.22)	-7.7%	6.4
15	700	55.15	48.33	(6.82)	-12.4%	6.9	48.77	45.01	(3.76)	-7.7%	6.4
16	800	63.00	55.20	(7.80)	-12.4%	6.9	55.71	51.41	(4.30)	-7.7%	6.4
17	900	70.85	62.08	(8.77)	-12.4%	6.9	62.65	57.82	(4.84)	-7.7%	6.4
18	1,000	78.70	68.96	(9.74)	-12.4%	6.9	69.59	64.22	(5.37)	-7.7%	6.4
19	1,500	117.95	103.33	(14.62)	-12.4%	6.9	104.29	96.23	(8.06)	-7.7%	6.4
20	2,000	157.20	137.71	(19.49)	-12.4%	6.9	138.98	128.24	(10.75)	-7.7%	6.4
21	2,500	196.45	172.09	(24.36)	-12.4%	6.9	173.68	160.25	(13.43)	-7.7%	6.4
22	3,000	235.70	206.47	(29.23)	-12.4%	6.9	208.37	192.25	(16.12)	-7.7%	6.4
23	3,500	274.95	240.84	(34.10)	-12.4%	6.9	243.07	224.26	(18.80)	-7.7%	6.4
24	4,000	314.20	275.22	(38.98)	-12.4%	6.9	277.76	256.27	(21.49)	-7.7%	6.4
25	4,500	353.45	309.60	(43.85)	-12.4%	6.9	312.46	288.28	(24.18)	-7.7%	6.4
26	5,000	392.70	343.98	(48.72)	-12.4%	6.9	347.15	320.29	(26.86)	-7.7%	6.4
27	5,500	431.95	378.36	(53.59)	-12.4%	6.9	381.85	352.30	(29.55)	-7.7%	6.4
28	6,000	471.20	412.73	(58.46)	-12.4%	6.9	416.54	384.31	(32.24)	-7.7%	6.4
29	6,500	510.45	447.11	(63.34)	-12.4%	6.9	451.24	416.32	(34.92)	-7.7%	6.4

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Energy-only GS

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

Page 38 of 46

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	250	30.62	33.06	2.45	8.0%	13.2	30.62	33.06	2.45	8.0%	13.2
2	500	41.24	46.13	4.89	11.9%	9.2	41.24	46.13	4.89	11.9%	9.2
3	750	51.85	59.19	7.34	14.1%	7.9	51.85	59.19	7.34	14.1%	7.9
4	1,000	62.47	72.25	9.78	15.7%	7.2	62.47	72.25	9.78	15.7%	7.2
5	1,500	83.71	98.38	14.67	17.5%	6.6	83.71	98.38	14.67	17.5%	6.6
6	2,000	104.94	124.51	19.56	18.6%	6.2	104.94	124.51	19.56	18.6%	6.2
7	2,500	126.18	150.63	24.45	19.4%	6.0	126.18	150.63	24.45	19.4%	6.0
8	3,000	147.42	176.76	29.34	19.9%	5.9	147.42	176.76	29.34	19.9%	5.9
9	3,500	168.65	202.89	34.23	20.3%	5.8	168.65	202.89	34.23	20.3%	5.8
10	4,000	189.89	229.01	39.12	20.6%	5.7	189.89	229.01	39.12	20.6%	5.7
11	4,500	211.12	255.14	44.01	20.8%	5.7	211.12	255.14	44.01	20.8%	5.7
12	5,000	232.36	281.27	48.91	21.0%	5.6	232.36	281.27	48.91	21.0%	5.6
13	6,000	274.83	333.52	58.69	21.4%	5.6	274.83	333.52	58.69	21.4%	5.6
14	7,000	317.30	385.77	68.47	21.6%	5.5	317.30	385.77	68.47	21.6%	5.5
15	8,000	359.78	438.02	78.25	21.7%	5.5	359.78	438.02	78.25	21.7%	5.5
16	9,000	402.25	490.28	88.03	21.9%	5.4	402.25	490.28	88.03	21.9%	5.4
17	10,000	444.72	542.53	97.81	22.0%	5.4	444.72	542.53	97.81	22.0%	5.4
18	15,000	657.08	803.80	146.72	22.3%	5.4	657.08	803.80	146.72	22.3%	5.4
19	20,000	869.44	1,065.06	195.62	22.5%	5.3	869.44	1,065.06	195.62	22.5%	5.3
20	25,000	1,081.80	1,326.33	244.53	22.6%	5.3	1,081.80	1,326.33	244.53	22.6%	5.3
21	30,000	1,294.16	1,587.59	293.43	22.7%	5.3	1,294.16	1,587.59	293.43	22.7%	5.3
22	35,000	1,506.52	1,848.86	342.34	22.7%	5.3	1,506.52	1,848.86	342.34	22.7%	5.3
23	40,000	1,718.88	2,110.12	391.24	22.8%	5.3	1,718.88	2,110.12	391.24	22.8%	5.3
24	45,000	1,931.24	2,371.39	440.15	22.8%	5.3	1,931.24	2,371.39	440.15	22.8%	5.3
25	50,000	2,143.60	2,632.65	489.05	22.8%	5.3	2,143.60	2,632.65	489.05	22.8%	5.3
26	55,000	2,355.96	2,893.92	537.96	22.8%	5.3	2,355.96	2,893.92	537.96	22.8%	5.3
27	60,000	2,568.32	3,155.18	586.86	22.8%	5.3	2,568.32	3,155.18	586.86	22.8%	5.3
28	65,000	2,780.68	3,416.45	635.77	22.9%	5.3	2,780.68	3,416.45	635.77	22.9%	5.3
29	70,000	2,993.04	3,677.71	684.67	22.9%	5.3	2,993.04	3,677.71	684.67	22.9%	5.3

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Demand GSD

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	46.44	49.18	2.74	5.9%	9.8	46.44	49.18	2.74	5.9%	9.8
2	1,000	62.87	68.35	5.48	8.7%	6.8	62.87	68.35	5.48	8.7%	6.8
3	1,500	79.31	87.53	8.22	10.4%	5.8	79.31	87.53	8.22	10.4%	5.8
4	2,000	95.75	106.70	10.96	11.4%	5.3	95.75	106.70	10.96	11.4%	5.3
5	2,500	112.18	125.88	13.70	12.2%	5.0	112.18	125.88	13.70	12.2%	5.0
6	3,000	128.62	145.05	16.44	12.8%	4.8	128.62	145.05	16.44	12.8%	4.8
7	3,500	145.05	164.23	19.17	13.2%	4.7	145.05	164.23	19.17	13.2%	4.7
8	4,000	161.49	183.40	21.91	13.6%	4.6	161.49	183.40	21.91	13.6%	4.6
9	4,500	177.93	202.58	24.65	13.9%	4.5	177.93	202.58	24.65	13.9%	4.5
10	5,000	194.36	221.76	27.39	14.1%	4.4	194.36	221.76	27.39	14.1%	4.4
11	6,000	227.24	260.11	32.87	14.5%	4.3	227.24	260.11	32.87	14.5%	4.3
12	7,000	260.11	298.46	38.35	14.7%	4.3	260.11	298.46	38.35	14.7%	4.3
13	8,000	292.98	336.81	43.83	15.0%	4.2	292.98	336.81	43.83	15.0%	4.2
14	9,000	325.85	375.16	49.31	15.1%	4.2	325.85	375.16	49.31	15.1%	4.2
15	10,000	358.73	413.51	54.79	15.3%	4.1	358.73	413.51	54.79	15.3%	4.1
16	11,000	391.60	451.86	60.26	15.4%	4.1	391.60	451.86	60.26	15.4%	4.1
17	12,000	424.47	490.21	65.74	15.5%	4.1	424.47	490.21	65.74	15.5%	4.1
18	13,000	457.34	528.57	71.22	15.6%	4.1	457.34	528.57	71.22	15.6%	4.1
19	14,000	490.22	566.92	76.70	15.6%	4.0	490.22	566.92	76.70	15.6%	4.0
20	15,000	523.09	605.27	82.18	15.7%	4.0	523.09	605.27	82.18	15.7%	4.0
21	16,000	555.96	643.62	87.66	15.8%	4.0	555.96	643.62	87.66	15.8%	4.0
22	17,000	588.84	681.97	93.14	15.8%	4.0	588.84	681.97	93.14	15.8%	4.0
23	18,000	621.71	720.32	98.61	15.9%	4.0	621.71	720.32	98.61	15.9%	4.0
24	19,000	654.58	758.67	104.09	15.9%	4.0	654.58	758.67	104.09	15.9%	4.0
25	20,000	687.45	797.02	109.57	15.9%	4.0	687.45	797.02	109.57	15.9%	4.0
26	21,000	720.33	835.38	115.05	16.0%	4.0	720.33	835.38	115.05	16.0%	4.0
27	22,000	753.20	873.73	120.53	16.0%	4.0	753.20	873.73	120.53	16.0%	4.0
28	23,000	786.07	912.08	126.01	16.0%	4.0	786.07	912.08	126.01	16.0%	4.0
29	24,000	818.94	950.43	131.49	16.1%	4.0	818.94	950.43	131.49	16.1%	4.0

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Secondary Time-of-Use GSTU

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	41.24	50.40	9.16	22.2%	10.1	41.24	50.40	9.16	22.2%	10.1
2	1,000	62.47	70.80	8.33	13.3%	7.1	62.47	70.80	8.33	13.3%	7.1
3	1,500	83.71	91.20	7.49	8.9%	6.1	83.71	91.20	7.49	8.9%	6.1
4	2,000	104.94	111.60	6.65	6.3%	5.6	104.94	111.60	6.65	6.3%	5.6
5	2,500	126.18	132.00	5.82	4.6%	5.3	126.18	132.00	5.82	4.6%	5.3
6	3,000	147.42	152.40	4.98	3.4%	5.1	147.42	152.40	4.98	3.4%	5.1
7	3,500	168.65	172.80	4.14	2.5%	4.9	168.65	172.80	4.14	2.5%	4.9
8	4,000	189.89	193.19	3.31	1.7%	4.8	189.89	193.19	3.31	1.7%	4.8
9	4,500	211.12	213.59	2.47	1.2%	4.7	211.12	213.59	2.47	1.2%	4.7
10	5,000	232.36	233.99	1.63	0.7%	4.7	232.36	233.99	1.63	0.7%	4.7
11	6,000	274.83	274.79	(0.04)	0.0%	4.6	274.83	274.79	(0.04)	0.0%	4.6
12	7,000	317.30	315.59	(1.71)	-0.5%	4.5	317.30	315.59	(1.71)	-0.5%	4.5
13	8,000	359.78	356.39	(3.39)	-0.9%	4.5	359.78	356.39	(3.39)	-0.9%	4.5
14	9,000	402.25	397.19	(5.06)	-1.3%	4.4	402.25	397.19	(5.06)	-1.3%	4.4
15	10,000	444.72	437.99	(6.73)	-1.5%	4.4	444.72	437.99	(6.73)	-1.5%	4.4
16	11,000	487.19	478.79	(8.41)	-1.7%	4.4	487.19	478.79	(8.41)	-1.7%	4.4
17	12,000	529.66	519.58	(10.08)	-1.9%	4.3	529.66	519.58	(10.08)	-1.9%	4.3
18	13,000	572.14	560.38	(11.75)	-2.1%	4.3	572.14	560.38	(11.75)	-2.1%	4.3
19	14,000	614.61	601.18	(13.43)	-2.2%	4.3	614.61	601.18	(13.43)	-2.2%	4.3
20	15,000	657.08	641.98	(15.10)	-2.3%	4.3	657.08	641.98	(15.10)	-2.3%	4.3
21	16,000	699.55	682.78	(16.77)	-2.4%	4.3	699.55	682.78	(16.77)	-2.4%	4.3
22	17,000	742.02	723.58	(18.45)	-2.5%	4.3	742.02	723.58	(18.45)	-2.5%	4.3
23	18,000	784.50	764.38	(20.12)	-2.6%	4.2	784.50	764.38	(20.12)	-2.6%	4.2
24	19,000	826.97	805.18	(21.79)	-2.6%	4.2	826.97	805.18	(21.79)	-2.6%	4.2
25	20,000	869.44	845.97	(23.47)	-2.7%	4.2	869.44	845.97	(23.47)	-2.7%	4.2
26	21,000	911.91	886.77	(25.14)	-2.8%	4.2	911.91	886.77	(25.14)	-2.8%	4.2
27	22,000	954.38	927.57	(26.81)	-2.8%	4.2	954.38	927.57	(26.81)	-2.8%	4.2
28	23,000	996.86	968.37	(28.49)	-2.9%	4.2	996.86	968.37	(28.49)	-2.9%	4.2
29	24,000	1,039.33	1,009.17	(30.16)	-2.9%	4.2	1,039.33	1,009.17	(30.16)	-2.9%	4.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 1

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	102.87	103.21	0.34	0.3%	20.6	102.87	103.21	0.34	0.3%	20.6
2	1,000	105.73	106.42	0.69	0.7%	10.6	105.73	106.42	0.69	0.7%	10.6
3	1,500	108.60	109.63	1.03	1.0%	7.3	108.60	109.63	1.03	1.0%	7.3
4	2,000	111.47	112.84	1.38	1.2%	5.6	111.47	112.84	1.38	1.2%	5.6
5	2,500	114.33	116.05	1.72	1.5%	4.6	114.33	116.05	1.72	1.5%	4.6
6	3,000	117.20	119.26	2.06	1.8%	4.0	117.20	119.26	2.06	1.8%	4.0
7	4,000	122.93	125.68	2.75	2.2%	3.1	122.93	125.68	2.75	2.2%	3.1
8	5,000	128.67	132.11	3.44	2.7%	2.6	128.67	132.11	3.44	2.7%	2.6
9	6,000	134.40	138.53	4.13	3.1%	2.3	134.40	138.53	4.13	3.1%	2.3
10	7,000	140.13	144.95	4.82	3.4%	2.1	140.13	144.95	4.82	3.4%	2.1
11	8,000	145.86	151.37	5.50	3.8%	1.9	145.86	151.37	5.50	3.8%	1.9
12	9,000	151.60	157.79	6.19	4.1%	1.8	151.60	157.79	6.19	4.1%	1.8
13	10,000	157.33	164.21	6.88	4.4%	1.6	157.33	164.21	6.88	4.4%	1.6
14	15,000	186.00	196.32	10.32	5.5%	1.3	186.00	196.32	10.32	5.5%	1.3
15	20,000	214.66	228.42	13.76	6.4%	1.1	214.66	228.42	13.76	6.4%	1.1
16	25,000	243.33	260.53	17.20	7.1%	1.0	243.33	260.53	17.20	7.1%	1.0
17	30,000	271.99	292.63	20.64	7.6%	1.0	271.99	292.63	20.64	7.6%	1.0
18	35,000	300.66	324.74	24.08	8.0%	0.9	300.66	324.74	24.08	8.0%	0.9
19	40,000	329.32	356.84	27.52	8.4%	0.9	329.32	356.84	27.52	8.4%	0.9
20	45,000	357.99	388.95	30.96	8.6%	0.9	357.99	388.95	30.96	8.6%	0.9
21	50,000	386.65	421.05	34.40	8.9%	0.8	386.65	421.05	34.40	8.9%	0.8
22	60,000	443.98	485.26	41.28	9.3%	0.8	443.98	485.26	41.28	9.3%	0.8
23	70,000	501.31	549.47	48.16	9.6%	0.8	501.31	549.47	48.16	9.6%	0.8
24	80,000	558.64	613.68	55.04	9.9%	0.8	558.64	613.68	55.04	9.9%	0.8
25	90,000	615.97	677.89	61.92	10.1%	0.8	615.97	677.89	61.92	10.1%	0.8
26	100,000	673.30	742.10	68.80	10.2%	0.7	673.30	742.10	68.80	10.2%	0.7
27	110,000	730.63	806.31	75.68	10.4%	0.7	730.63	806.31	75.68	10.4%	0.7
28	120,000	787.96	870.52	82.56	10.5%	0.7	787.96	870.52	82.56	10.5%	0.7
29	130,000	845.29	934.73	89.44	10.6%	0.7	845.29	934.73	89.44	10.6%	0.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 2

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	103.86	105.47	1.61	1.5%	21.1	103.86	105.47	1.61	1.5%	21.1
2	1,000	107.72	110.93	3.21	3.0%	11.1	107.72	110.93	3.21	3.0%	11.1
3	1,500	111.58	116.40	4.82	4.3%	7.8	111.58	116.40	4.82	4.3%	7.8
4	2,000	115.45	121.87	6.42	5.6%	6.1	115.45	121.87	6.42	5.6%	6.1
5	2,500	119.31	127.34	8.03	6.7%	5.1	119.31	127.34	8.03	6.7%	5.1
6	3,000	123.17	132.80	9.63	7.8%	4.4	123.17	132.80	9.63	7.8%	4.4
7	4,000	130.89	143.74	12.84	9.8%	3.6	130.89	143.74	12.84	9.8%	3.6
8	5,000	138.62	154.67	16.06	11.6%	3.1	138.62	154.67	16.06	11.6%	3.1
9	6,000	146.34	165.60	19.27	13.2%	2.8	146.34	165.60	19.27	13.2%	2.8
10	7,000	154.06	176.54	22.48	14.6%	2.5	154.06	176.54	22.48	14.6%	2.5
11	8,000	161.78	187.47	25.69	15.9%	2.3	161.78	187.47	25.69	15.9%	2.3
12	9,000	169.51	198.41	28.90	17.0%	2.2	169.51	198.41	28.90	17.0%	2.2
13	10,000	177.23	209.34	32.11	18.1%	2.1	177.23	209.34	32.11	18.1%	2.1
14	15,000	215.85	264.01	48.17	22.3%	1.8	215.85	264.01	48.17	22.3%	1.8
15	20,000	254.46	318.68	64.22	25.2%	1.6	254.46	318.68	64.22	25.2%	1.6
16	25,000	293.08	373.35	80.28	27.4%	1.5	293.08	373.35	80.28	27.4%	1.5
17	30,000	331.69	428.02	96.33	29.0%	1.4	331.69	428.02	96.33	29.0%	1.4
18	35,000	370.31	482.69	112.39	30.3%	1.4	370.31	482.69	112.39	30.3%	1.4
19	40,000	408.92	537.36	128.44	31.4%	1.3	408.92	537.36	128.44	31.4%	1.3
20	45,000	447.54	592.03	144.50	32.3%	1.3	447.54	592.03	144.50	32.3%	1.3
21	50,000	486.15	646.70	160.55	33.0%	1.3	486.15	646.70	160.55	33.0%	1.3
22	60,000	563.38	756.04	192.66	34.2%	1.3	563.38	756.04	192.66	34.2%	1.3
23	70,000	640.61	865.38	224.77	35.1%	1.2	640.61	865.38	224.77	35.1%	1.2
24	80,000	717.84	974.72	256.88	35.8%	1.2	717.84	974.72	256.88	35.8%	1.2
25	90,000	795.07	1,084.06	288.99	36.3%	1.2	795.07	1,084.06	288.99	36.3%	1.2
26	100,000	872.30	1,193.40	321.10	36.8%	1.2	872.30	1,193.40	321.10	36.8%	1.2
27	110,000	949.53	1,302.74	353.21	37.2%	1.2	949.53	1,302.74	353.21	37.2%	1.2
28	120,000	1,026.76	1,412.08	385.32	37.5%	1.2	1,026.76	1,412.08	385.32	37.5%	1.2
29	130,000	1,103.99	1,521.42	417.43	37.8%	1.2	1,103.99	1,521.42	417.43	37.8%	1.2

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Energy-only GP Voltage Level 3

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use kWh	Summer (June - Sept.)				Proposed Unit Cost ¢/kWh	Winter (Oct. - May)				Proposed Unit Cost ¢/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$	\$	\$	%		\$	\$	\$	%	
1	500	106.69	108.32	1.63	1.5%	21.7	106.69	108.32	1.63	1.5%	21.7
2	1,000	113.39	116.64	3.26	2.9%	11.7	113.39	116.64	3.26	2.9%	11.7
3	1,500	120.08	124.96	4.89	4.1%	8.3	120.08	124.96	4.89	4.1%	8.3
4	2,000	126.77	133.29	6.51	5.1%	6.7	126.77	133.29	6.51	5.1%	6.7
5	2,500	133.47	141.61	8.14	6.1%	5.7	133.47	141.61	8.14	6.1%	5.7
6	3,000	140.16	149.93	9.77	7.0%	5.0	140.16	149.93	9.77	7.0%	5.0
7	4,000	153.54	166.57	13.03	8.5%	4.2	153.54	166.57	13.03	8.5%	4.2
8	5,000	166.93	183.22	16.29	9.8%	3.7	166.93	183.22	16.29	9.8%	3.7
9	6,000	180.32	199.86	19.54	10.8%	3.3	180.32	199.86	19.54	10.8%	3.3
10	7,000	193.70	216.50	22.80	11.8%	3.1	193.70	216.50	22.80	11.8%	3.1
11	8,000	207.09	233.14	26.06	12.6%	2.9	207.09	233.14	26.06	12.6%	2.9
12	9,000	220.47	249.79	29.31	13.3%	2.8	220.47	249.79	29.31	13.3%	2.8
13	10,000	233.86	266.43	32.57	13.9%	2.7	233.86	266.43	32.57	13.9%	2.7
14	15,000	300.79	349.65	48.86	16.2%	2.3	300.79	349.65	48.86	16.2%	2.3
15	20,000	367.72	432.86	65.14	17.7%	2.2	367.72	432.86	65.14	17.7%	2.2
16	25,000	434.65	516.08	81.43	18.7%	2.1	434.65	516.08	81.43	18.7%	2.1
17	30,000	501.58	599.29	97.71	19.5%	2.0	501.58	599.29	97.71	19.5%	2.0
18	35,000	568.51	682.51	114.00	20.1%	2.0	568.51	682.51	114.00	20.1%	2.0
19	40,000	635.44	765.72	130.28	20.5%	1.9	635.44	765.72	130.28	20.5%	1.9
20	45,000	702.37	848.94	146.57	20.9%	1.9	702.37	848.94	146.57	20.9%	1.9
21	50,000	769.30	932.15	162.85	21.2%	1.9	769.30	932.15	162.85	21.2%	1.9
22	60,000	903.16	1,098.58	195.42	21.6%	1.8	903.16	1,098.58	195.42	21.6%	1.8
23	70,000	1,037.02	1,265.01	227.99	22.0%	1.8	1,037.02	1,265.01	227.99	22.0%	1.8
24	80,000	1,170.88	1,431.44	260.56	22.3%	1.8	1,170.88	1,431.44	260.56	22.3%	1.8
25	90,000	1,304.74	1,597.87	293.13	22.5%	1.8	1,304.74	1,597.87	293.13	22.5%	1.8
26	100,000	1,438.60	1,764.30	325.70	22.6%	1.8	1,438.60	1,764.30	325.70	22.6%	1.8
27	110,000	1,572.46	1,930.73	358.27	22.8%	1.8	1,572.46	1,930.73	358.27	22.8%	1.8
28	120,000	1,706.32	2,097.16	390.84	22.9%	1.7	1,706.32	2,097.16	390.84	22.9%	1.7
29	130,000	1,840.18	2,263.59	423.41	23.0%	1.7	1,840.18	2,263.59	423.41	23.0%	1.7

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 1

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present \$000	Proposed \$000	Amount \$000	Percent %		Present \$000	Proposed \$000	Amount \$000	Percent %	
1	100	0.36	0.32	(0.04)	-10.2%	0.3	0.36	0.32	(0.04)	-10.2%	0.3
2	110	0.37	0.33	(0.04)	-10.7%	0.3	0.37	0.33	(0.04)	-10.7%	0.3
3	120	0.39	0.35	(0.04)	-11.2%	0.3	0.39	0.35	(0.04)	-11.2%	0.3
4	130	0.41	0.36	(0.05)	-11.7%	0.3	0.41	0.36	(0.05)	-11.7%	0.3
5	140	0.42	0.37	(0.05)	-12.1%	0.3	0.42	0.37	(0.05)	-12.1%	0.3
6	150	0.44	0.38	(0.05)	-12.5%	0.3	0.44	0.38	(0.05)	-12.5%	0.3
7	200	0.52	0.44	(0.07)	-14.1%	0.2	0.52	0.44	(0.07)	-14.1%	0.2
8	250	0.59	0.50	(0.09)	-15.3%	0.2	0.59	0.50	(0.09)	-15.3%	0.2
9	300	0.67	0.56	(0.11)	-16.2%	0.2	0.67	0.56	(0.11)	-16.2%	0.2
10	350	0.75	0.62	(0.13)	-16.9%	0.2	0.75	0.62	(0.13)	-16.9%	0.2
11	400	0.83	0.69	(0.15)	-17.5%	0.2	0.83	0.69	(0.15)	-17.5%	0.2
12	450	0.91	0.75	(0.16)	-18.0%	0.2	0.91	0.75	(0.16)	-18.0%	0.2
13	500	0.99	0.81	(0.18)	-18.4%	0.2	0.99	0.81	(0.18)	-18.4%	0.2
14	600	1.15	0.93	(0.22)	-19.1%	0.2	1.15	0.93	(0.22)	-19.1%	0.2
15	700	1.30	1.05	(0.25)	-19.5%	0.1	1.30	1.05	(0.25)	-19.5%	0.1
16	800	1.46	1.17	(0.29)	-19.9%	0.1	1.46	1.17	(0.29)	-19.9%	0.1
17	900	1.62	1.29	(0.33)	-20.2%	0.1	1.62	1.29	(0.33)	-20.2%	0.1
18	1,000	1.78	1.41	(0.36)	-20.5%	0.1	1.78	1.41	(0.36)	-20.5%	0.1
19	1,500	2.57	2.02	(0.55)	-21.3%	0.1	2.57	2.02	(0.55)	-21.3%	0.1
20	2,000	3.36	2.63	(0.73)	-21.7%	0.1	3.36	2.63	(0.73)	-21.7%	0.1
21	2,500	4.14	3.23	(0.91)	-22.0%	0.1	4.14	3.23	(0.91)	-22.0%	0.1
22	3,000	4.93	3.84	(1.09)	-22.1%	0.1	4.93	3.84	(1.09)	-22.1%	0.1
23	3,500	5.72	4.45	(1.27)	-22.3%	0.1	5.72	4.45	(1.27)	-22.3%	0.1
24	4,000	6.51	5.06	(1.46)	-22.4%	0.1	6.51	5.06	(1.46)	-22.4%	0.1
25	4,500	7.30	5.66	(1.64)	-22.4%	0.1	7.30	5.66	(1.64)	-22.4%	0.1
26	5,000	8.09	6.27	(1.82)	-22.5%	0.1	8.09	6.27	(1.82)	-22.5%	0.1
27	5,500	8.88	6.88	(2.00)	-22.6%	0.1	8.88	6.88	(2.00)	-22.6%	0.1
28	6,000	9.67	7.48	(2.18)	-22.6%	0.1	9.67	7.48	(2.18)	-22.6%	0.1
29	6,500	10.46	8.09	(2.37)	-22.6%	0.1	10.46	8.09	(2.37)	-22.6%	0.1

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 2

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	0.62	0.81	0.19	30.3%	0.8	0.62	0.81	0.19	30.3%	0.8
2	110	0.67	0.88	0.21	31.2%	0.8	0.67	0.88	0.21	31.2%	0.8
3	120	0.71	0.94	0.23	32.0%	0.8	0.71	0.94	0.23	32.0%	0.8
4	130	0.75	1.00	0.25	32.8%	0.8	0.75	1.00	0.25	32.8%	0.8
5	140	0.79	1.06	0.27	33.4%	0.8	0.79	1.06	0.27	33.4%	0.8
6	150	0.84	1.12	0.28	34.0%	0.7	0.84	1.12	0.28	34.0%	0.7
7	200	1.05	1.43	0.38	36.1%	0.7	1.05	1.43	0.38	36.1%	0.7
8	250	1.26	1.74	0.47	37.5%	0.7	1.26	1.74	0.47	37.5%	0.7
9	300	1.47	2.04	0.57	38.6%	0.7	1.47	2.04	0.57	38.6%	0.7
10	350	1.69	2.35	0.66	39.3%	0.7	1.69	2.35	0.66	39.3%	0.7
11	400	1.90	2.66	0.76	39.9%	0.7	1.90	2.66	0.76	39.9%	0.7
12	450	2.11	2.96	0.85	40.4%	0.7	2.11	2.96	0.85	40.4%	0.7
13	500	2.32	3.27	0.95	40.8%	0.7	2.32	3.27	0.95	40.8%	0.7
14	600	2.75	3.88	1.14	41.4%	0.6	2.75	3.88	1.14	41.4%	0.6
15	700	3.17	4.50	1.33	41.8%	0.6	3.17	4.50	1.33	41.8%	0.6
16	800	3.60	5.11	1.52	42.1%	0.6	3.60	5.11	1.52	42.1%	0.6
17	900	4.02	5.73	1.71	42.4%	0.6	4.02	5.73	1.71	42.4%	0.6
18	1,000	4.45	6.34	1.89	42.6%	0.6	4.45	6.34	1.89	42.6%	0.6
19	1,500	6.57	9.41	2.84	43.3%	0.6	6.57	9.41	2.84	43.3%	0.6
20	2,000	8.69	12.48	3.79	43.6%	0.6	8.69	12.48	3.79	43.6%	0.6
21	2,500	10.82	15.55	4.74	43.8%	0.6	10.82	15.55	4.74	43.8%	0.6
22	3,000	12.94	18.62	5.68	43.9%	0.6	12.94	18.62	5.68	43.9%	0.6
23	3,500	15.06	21.70	6.63	44.0%	0.6	15.06	21.70	6.63	44.0%	0.6
24	4,000	17.19	24.77	7.58	44.1%	0.6	17.19	24.77	7.58	44.1%	0.6
25	4,500	19.31	27.84	8.53	44.2%	0.6	19.31	27.84	8.53	44.2%	0.6
26	5,000	21.43	30.91	9.47	44.2%	0.6	21.43	30.91	9.47	44.2%	0.6
27	5,500	23.56	33.98	10.42	44.2%	0.6	23.56	33.98	10.42	44.2%	0.6
28	6,000	25.68	37.05	11.37	44.3%	0.6	25.68	37.05	11.37	44.3%	0.6
29	6,500	27.80	40.12	12.32	44.3%	0.6	27.80	40.12	12.32	44.3%	0.6

Schedule F-4.0

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Comparison of Present and Proposed Monthly Bills

Primary Demand GPD Voltage Level 3

Retail Open Access

Case No.: U-20697

Exhibit No.: A-16 (HWM-4)

Schedule F-4.0

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Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line No.	Monthly Use MWh	Summer (June - Sept.)				Proposed Unit Cost c/kWh	Winter (Oct. - May)				Proposed Unit Cost c/kWh
		Monthly Bills		Difference			Monthly Bills		Difference		
		Present	Proposed	Amount	Percent		Present	Proposed	Amount	Percent	
		\$000	\$000	\$000	%		\$000	\$000	\$000	%	
1	100	1.02	1.25	0.22	21.9%	1.2	1.02	1.25	0.22	21.9%	1.2
2	110	1.10	1.35	0.25	22.3%	1.2	1.10	1.35	0.25	22.3%	1.2
3	120	1.19	1.45	0.27	22.6%	1.2	1.19	1.45	0.27	22.6%	1.2
4	130	1.27	1.56	0.29	22.9%	1.2	1.27	1.56	0.29	22.9%	1.2
5	140	1.35	1.66	0.31	23.2%	1.2	1.35	1.66	0.31	23.2%	1.2
6	150	1.43	1.77	0.34	23.4%	1.2	1.43	1.77	0.34	23.4%	1.2
7	200	1.84	2.29	0.45	24.3%	1.1	1.84	2.29	0.45	24.3%	1.1
8	250	2.25	2.81	0.56	24.8%	1.1	2.25	2.81	0.56	24.8%	1.1
9	300	2.67	3.34	0.67	25.2%	1.1	2.67	3.34	0.67	25.2%	1.1
10	350	3.08	3.86	0.78	25.5%	1.1	3.08	3.86	0.78	25.5%	1.1
11	400	3.49	4.38	0.89	25.7%	1.1	3.49	4.38	0.89	25.7%	1.1
12	450	3.90	4.91	1.01	25.8%	1.1	3.90	4.91	1.01	25.8%	1.1
13	500	4.31	5.43	1.12	26.0%	1.1	4.31	5.43	1.12	26.0%	1.1
14	600	5.13	6.47	1.34	26.2%	1.1	5.13	6.47	1.34	26.2%	1.1
15	700	5.95	7.52	1.57	26.3%	1.1	5.95	7.52	1.57	26.3%	1.1
16	800	6.78	8.57	1.79	26.4%	1.1	6.78	8.57	1.79	26.4%	1.1
17	900	7.60	9.61	2.01	26.5%	1.1	7.60	9.61	2.01	26.5%	1.1
18	1,000	8.42	10.66	2.24	26.6%	1.1	8.42	10.66	2.24	26.6%	1.1
19	1,500	12.53	15.88	3.36	26.8%	1.1	12.53	15.88	3.36	26.8%	1.1
20	2,000	16.64	21.11	4.47	26.9%	1.1	16.64	21.11	4.47	26.9%	1.1
21	2,500	20.75	26.34	5.59	27.0%	1.1	20.75	26.34	5.59	27.0%	1.1
22	3,000	24.86	31.57	6.71	27.0%	1.1	24.86	31.57	6.71	27.0%	1.1
23	3,500	28.97	36.80	7.83	27.0%	1.1	28.97	36.80	7.83	27.0%	1.1
24	4,000	33.08	42.03	8.95	27.1%	1.1	33.08	42.03	8.95	27.1%	1.1
25	4,500	37.19	47.25	10.07	27.1%	1.1	37.19	47.25	10.07	27.1%	1.1
26	5,000	41.30	52.48	11.19	27.1%	1.0	41.30	52.48	11.19	27.1%	1.0
27	5,500	45.41	57.71	12.31	27.1%	1.0	45.41	57.71	12.31	27.1%	1.0
28	6,000	49.52	62.94	13.42	27.1%	1.0	49.52	62.94	13.42	27.1%	1.0
29	6,500	53.62	68.17	14.54	27.1%	1.0	53.62	68.17	14.54	27.1%	1.0

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Long-term Industrial Load Retention Rate (LTILRR) Net Benefit Analysis
(\$000)

Case No.: U-20697
Exhibit No.: A-77 (HWM-5)
Page: 1 of 1
Witness: HWMiller
Date: February 2020

Description	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031
Transmission costs (including LTILRR) ⁽¹⁾	\$ 477,556	\$ 500,112	\$ 511,369	\$ 523,724	\$ 526,338	\$ 535,133	\$ 543,541	\$ 551,002	\$ 557,576	\$ 568,728	\$ 580,102
Transmission costs (excluding LTILRR) ⁽¹⁾	<u>472,128</u>	<u>495,103</u>	<u>506,533</u>	<u>518,735</u>	<u>521,308</u>	<u>530,137</u>	<u>538,572</u>	<u>546,071</u>	<u>552,659</u>	<u>563,712</u>	<u>574,987</u>
Incremental transmission cost of serving LTILRR load	\$ 5,428	\$ 5,010	\$ 4,836	\$ 4,989	\$ 5,031	\$ 4,997	\$ 4,969	\$ 4,931	\$ 4,917	\$ 5,015	\$ 5,116
Transmission cost contribution under LTILRR ⁽²⁾	\$ 17,072	\$ 18,004	\$ 18,409	\$ 18,854	\$ 18,948	\$ 19,265	\$ 19,567	\$ 19,836	\$ 20,073	\$ 20,474	\$ 20,884
Transmission cost contribution under standby service	<u>350</u>	<u>368</u>	<u>376</u>	<u>668</u>	<u>387</u>	<u>394</u>	<u>400</u>	<u>703</u>	<u>410</u>	<u>418</u>	<u>427</u>
Incremental transmission cost contribution	\$ 16,721	\$ 17,637	\$ 18,033	\$ 18,186	\$ 18,561	\$ 18,871	\$ 19,168	\$ 19,133	\$ 19,662	\$ 20,056	\$ 20,457
Incremental transmission cost contribution	\$ 16,721	\$ 17,637	\$ 18,033	\$ 18,186	\$ 18,561	\$ 18,871	\$ 19,168	\$ 19,133	\$ 19,662	\$ 20,056	\$ 20,457
Less: Incremental transmission cost of serving LTILRR load	<u>5,428</u>	<u>5,010</u>	<u>4,836</u>	<u>4,989</u>	<u>5,031</u>	<u>4,997</u>	<u>4,969</u>	<u>4,931</u>	<u>4,917</u>	<u>5,015</u>	<u>5,116</u>
Net transmission benefit/(cost) to other customers	<u>\$ 11,293</u>	<u>\$ 12,627</u>	<u>\$ 13,198</u>	<u>\$ 13,197</u>	<u>\$ 13,531</u>	<u>\$ 13,875</u>	<u>\$ 14,199</u>	<u>\$ 14,202</u>	<u>\$ 14,746</u>	<u>\$ 15,040</u>	<u>\$ 15,341</u>

Description	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	2042
Transmission costs (including LTILRR) ⁽¹⁾	\$ 591,704	\$ 603,538	\$ 615,609	\$ 627,921	\$ 640,480	\$ 653,289	\$ 666,355	\$ 679,682	\$ 693,276	\$ 707,141	\$ 721,284
Transmission costs (excluding LTILRR) ⁽¹⁾	<u>586,486</u>	<u>598,216</u>	<u>610,180</u>	<u>622,384</u>	<u>634,832</u>	<u>647,528</u>	<u>660,479</u>	<u>673,688</u>	<u>687,162</u>	<u>700,905</u>	<u>714,923</u>
Incremental transmission cost of serving LTILRR load	\$ 5,218	\$ 5,322	\$ 5,429	\$ 5,537	\$ 5,648	\$ 5,761	\$ 5,876	\$ 5,994	\$ 6,114	\$ 6,236	\$ 6,361
Transmission cost contribution under LTILRR ⁽²⁾	\$ 22,102	\$ 21,727	\$ 22,162	\$ 22,605	\$ 23,057	\$ 23,518	\$ 23,989	\$ 24,469	\$ 24,958	\$ 25,457	\$ 25,966
Transmission cost contribution under standby service	<u>755</u>	<u>444</u>	<u>453</u>	<u>462</u>	<u>817</u>	<u>481</u>	<u>490</u>	<u>500</u>	<u>884</u>	<u>520</u>	<u>531</u>
Incremental transmission cost contribution	\$ 21,347	\$ 21,283	\$ 21,709	\$ 22,143	\$ 22,240	\$ 23,038	\$ 23,499	\$ 23,968	\$ 24,074	\$ 24,937	\$ 25,436
Incremental transmission cost contribution	\$ 21,347	\$ 21,283	\$ 21,709	\$ 22,143	\$ 22,240	\$ 23,038	\$ 23,499	\$ 23,968	\$ 24,074	\$ 24,937	\$ 25,436
Less: Incremental transmission cost of serving LTILRR load	<u>5,218</u>	<u>5,322</u>	<u>5,429</u>	<u>5,537</u>	<u>5,648</u>	<u>5,761</u>	<u>5,876</u>	<u>5,994</u>	<u>6,114</u>	<u>6,236</u>	<u>6,361</u>
Net transmission benefit/(cost) to other customers	<u>\$ 16,129</u>	<u>\$ 15,961</u>	<u>\$ 16,280</u>	<u>\$ 16,606</u>	<u>\$ 16,592</u>	<u>\$ 17,277</u>	<u>\$ 17,622</u>	<u>\$ 17,975</u>	<u>\$ 17,960</u>	<u>\$ 18,701</u>	<u>\$ 19,075</u>
Net Benefit/(Cost) to other customers ⁽³⁾	<u>\$ 153,508</u>										

Sources

(1) Exhibit A-111 (KGT-3)

(2) Exhibit A-73 (MPK-1)

(3) Discount rate based on test-year weighted cost of capital. See Exhibit A-14 (MRB-1), line 19.

Notes

(a) Forecast includes a scheduled outage of 10 days in October plus 9 days of unplanned outages each year.

(b) Approximately 90% availability of the generator every four years (9 days of unplanned and 31 days of extended planned outages).

(c) Customer would reduce load to 102 MW during planned outages.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Calculation of the FCM Incentive Surcharge

Case No.: U-20697
 Exhibit No.: A-78 (HWM-6)
 Page: 1 of 1
 Witness: HWMiller
 Date: February 2020

	(a)	(b)	(c)	(d)
Line		Test-year	FCM	FCM
No.	Description	Sales	Incentive ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)
1	Residential Class	12,447,668	\$ 1,339	0.000108
	<u>Secondary Class</u>			
2	Energy-only GS	3,826,453	375	0.000098
3	Time-of-Use GSTU	3,265	0	0.000098
4	Demand GSD	3,256,731	321	0.000098
	<u>Primary Class</u>			
	Energy-only GP			
5	Voltage Level 1	5,198	0	0.000078
6	Voltage Level 2	54,657	5	0.000085
7	Voltage Level 3	892,638	81	0.000091
	Demand GPD			
8	Voltage Level 1	1,272,911	102	0.000080
9	Voltage Level 2	1,163,719	101	0.000087
10	Voltage Level 3	2,514,288	234	0.000093
	Time-of-Use GPTU			
11	Voltage Level 1	241,562	19	0.000079
12	Voltage Level 2	951,742	81	0.000085
13	Voltage Level 3	3,172,245	290	0.000091
	Energy Intensive EIP			
14	Voltage Level 1	264,981	18	0.000067
15	Voltage Level 2	78,222	6	0.000073
16	Voltage Level 3	15,334	1	0.000078
	<u>Lighting & Unmetered Class</u>			
17	Metered Lighting GML	13,509	1	0.000088
18	Unmetered Lighting GUL	79,208	7	0.000088
19	Unmetered GU-XL / LED	20,629	2	0.000088
20	Unmetered GU	95,012	9	0.000094
	<u>Self-generation Class</u>			
21	Large Self-generation GSG-2			
22	Voltage Level 1	41,706	4	0.000087
23	Voltage Level 2	21,673	2	0.000095
24	Voltage Level 3	104	0	0.000102
25	Total Jurisdictional Bundled Service	<u>30,433,455</u>	<u>\$ 2,996</u>	

Notes

(1) Exhibit A-20 (JCA-6), lines 1 and 7

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Conservation Voltage Reduction (CVR) Incentive Surcharge

Case No.: U-20697

Exhibit No.: A-79 (HWM-7)

Page: 1 of 1

Witness: HWMiller

Date: February 2020

	(a)	(b)	(c)	(d)	(e)
Line		Test-year		CVR	CVR
No.	Description	Sales	Allocator ^(2)	Incentive ^(3)	Surcharge
		(MWh)		(\$000)	(\$ / kWh)
1	Residential Class	12,447,668	0.506214	\$ 419	0.000034
	<u>Secondary Class</u>				
2	Energy-only GS	3,850,933	0.146671	121	0.000032
3	Time-of-Use GSTU ^(1)	3,265	---	---	0.000032
4	Demand GSD	3,437,590	0.106072	88	0.000026
	<u>Primary Class</u>				
	Energy-only GP				
5	Voltage Level 1	5,198	-	-	-
6	Voltage Level 2	57,715	-	-	-
7	Voltage Level 3	963,911	0.035050	29	0.000030
	Demand GPD				
8	Voltage Level 1	2,365,080	-	-	-
9	Voltage Level 2	2,481,147	-	-	-
10	Voltage Level 3	3,522,448	0.090203	75	0.000021
	Time-of-Use GPTU				
11	Voltage Level 1	241,562	-	-	-
12	Voltage Level 2	951,742	-	-	-
13	Voltage Level 3	3,172,245	0.109571	91	0.000029
	Energy Intensive EIP				
14	Voltage Level 1	264,981	-	-	-
15	Voltage Level 2	78,222	-	-	-
16	Voltage Level 3	15,334	0.000185	0	0.000010
	<u>Lighting & Unmetered Class</u>				
17	Metered Lighting GML	13,509	0.000482	0	0.000030
18	Unmetered Lighting GUL	79,208	0.002827	2	0.000030
19	Unmetered GU-XL / LED	20,629	0.000736	1	0.000030
20	Unmetered GU	95,012	0.001662	1	0.000014
	<u>Self-generation Class</u>				
21	Large Self-generation GSG-2				
22	Voltage Level 1	41,706	-	-	-
23	Voltage Level 2	21,673	-	-	-
24	Voltage Level 3	104	0.000326	0	0.002596
25	Total Jurisdictional Service	<u>34,130,882</u>		<u>\$ 828</u>	

Notes

(1) GSTU sales included in calculation of combined GS and GSTU surcharge.

(2) Allocator 230 from the COSS

(3) Exhibit A-58 (MJD-1), col (c), line 24.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Demand Response (DR) Reconciliation Surcharge

Case No.: U-20697
Exhibit No.: A-80 (HWM-8)
Page: 1 of 1
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)
Line		Test-year	2017-18 DR	DR
No.	Description	Sales	Refund ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)
1	Residential Class	12,447,668	\$ (1,014)	(0.000081)
	<u>Secondary Class</u>			
2	Energy-only GS	3,850,933	(260)	(0.000068)
3	Time-of-Use GSTU	3,265	(0)	(0.000068)
4	Demand GSD	3,437,590	(209)	(0.000061)
	<u>Primary Class</u>			
	Energy-only GP			
5	Voltage Level 1	5,198	(0)	(0.000031)
6	Voltage Level 2	57,715	(2)	(0.000035)
7	Voltage Level 3	963,911	(56)	(0.000058)
	Demand GPD			
8	Voltage Level 1	2,365,080	(52)	(0.000022)
9	Voltage Level 2	2,481,147	(61)	(0.000025)
10	Voltage Level 3	3,522,448	(144)	(0.000041)
	Time-of-Use GPTU			
11	Voltage Level 1	241,562	(8)	(0.000035)
12	Voltage Level 2	951,742	(37)	(0.000038)
13	Voltage Level 3	3,172,245	(203)	(0.000064)
	Energy Intensive EIP			
14	Voltage Level 1	264,981	(3)	(0.000012)
15	Voltage Level 2	78,222	(1)	(0.000013)
16	Voltage Level 3	15,334	(0)	(0.000022)
17	Load Retention Rate LTILRR	1,794,531	(40)	(0.000022)
	<u>Lighting & Unmetered Class</u>			
18	Metered Lighting GML	13,509	(0)	(0.000017)
19	Unmetered Lighting GUL	79,208	(1)	(0.000017)
20	Unmetered GU-XL / LED	20,629	(0)	(0.000017)
21	Unmetered GU	95,012	(5)	(0.000049)
	<u>Self-generation Class</u>			
22	Large Self-generation GSG-2			
23	Voltage Level 1	41,706	(2)	(0.000051)
24	Voltage Level 2	21,673	(1)	(0.000056)
25	Voltage Level 3	104	(0)	(0.000094)
26	Total Jurisdictional Service	<u>35,925,413</u>	<u>\$ (2,101)</u>	

Notes

(1) A-20 (JCA-6), line 15

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Electric Rate Case Deferral Surcharge

Case No.: U-20697
Exhibit No.: A-81 (HWM-9)
Page: 1 of 1
Witness: HWMiller
Date: February 2020

	(a)	(b)	(c)	(d)
Line	Test-year	ERC	ERC Deferral	
No.	Description	Sales	Deferral ⁽¹⁾	Surcharge
		(MWh)	(\$000)	(\$ / kWh)
1	Residential Class	12,447,668	\$ 7,858	0.000631
	<u>Secondary Class</u>			
2	Energy-only GS	3,850,933	2,131	0.000553
3	Time-of-Use GSTU	3,265	2	0.000553
4	Demand GSD	3,437,590	1,191	0.000346
	<u>Primary Class</u>			
	Energy-only GP			
5	Voltage Level 1	5,198	0	0.000013
6	Voltage Level 2	57,715	5	0.000078
7	Voltage Level 3	963,911	155	0.000160
	Demand GPD			
8	Voltage Level 1	2,365,080	21	0.000009
9	Voltage Level 2	2,481,147	128	0.000051
10	Voltage Level 3	3,522,448	371	0.000105
	Time-of-Use GPTU			
11	Voltage Level 1	241,562	2	0.000010
12	Voltage Level 2	951,742	57	0.000059
13	Voltage Level 3	3,172,245	385	0.000121
	Energy Intensive EIP			
14	Voltage Level 1	264,981	4	0.000015
15	Voltage Level 2	78,222	7	0.000088
16	Voltage Level 3	15,334	3	0.000181
17	Load Retention Rate LTILRR	1,794,531	16	0.000009
	<u>Lighting & Unmetered Class</u>			
18	Metered Lighting GML	13,509	27	0.001978
19	Unmetered Lighting GUL	79,208	157	0.001978
20	Unmetered GU-XL / LED	20,629	41	0.001978
21	Unmetered GU	95,012	18	0.000193
	<u>Self-generation Class</u>			
22	Large Self-generation GSG-2			
23	Voltage Level 1	41,706	7	0.000179
24	Voltage Level 2	21,673	23	0.001058
25	Voltage Level 3	104	0	0.002165
26	Total Jurisdictional Service	<u>35,925,413</u>	<u>\$ 12,606</u>	

Notes

(1) A-20 (JCA-6), line 16

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

HEIDI J. MYERS

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule A-1

Michigan Public Service Commission
 Consumers Energy Company
 Revenue Deficiency (Sufficiency)
 for the Historical Year Ended December 31, 2018

Case No: U-20697
 Exhibit No: A-1 (HJM-1)
 Schedule: A-1
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) (d)	
			Total (\$000)	Jurisdictional Amount (\$000)
1	Rate Base	Exhibit: A-2 (HJM-3)	\$ 10,252,937	\$ 10,210,952
2	Adjusted Net Operating Income	Exhibit: A-3 (HJM-9)	616,904	612,206
3	Overall Rate of Return	Line 2 / Line 1	6.02%	6.00%
4	Required Rate of Return	Exhibit: A-4 (HJM-34)	5.84%	5.84%
5	Income Required	Line 1 * Line 4	598,350	595,900
6	Income Deficiency/ (Sufficiency)	Line 5 - Line 2	(18,554)	(16,306)
7	Revenue Multiplier	Exhibit: A-3 (HJM-10)	1.3391	1.3391
8	Revenue Deficiency (Sufficiency)	Line 6 * Line 7	\$ (24,845)	\$ (21,835)

Schedule A-2

Michigan Public Service Commission
Consumers Energy Company
Financial Metrics - Financial Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule A-2
Page: 1 of 6
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		Calendar Year				
		2014	2015	2016	2017	2018
1	Operating Revenue	\$ 4,435,717	\$ 4,249,518	\$ 4,379,111	\$ 4,447,710	\$ 4,561,203
2	Operating Expenses	3,645,089	3,423,640	3,448,040	3,531,978	3,745,412
3	Pre-Tax Operating Income	790,628	825,878	931,071	915,731	815,791
4	Income Taxes	211,476	223,728	245,310	244,975	108,918
5	Net Operating Income	579,152	602,150	685,761	670,757	706,874
6	Other Income and Deductions	(17,658)	8,173	(34,820)	(16,239)	35,098
7	AFUDC	8,262	8,808	7,849	4,901	5,095
8	Interest Charges	(184,124)	(180,418)	(198,810)	(202,755)	(210,785)
9	Preferred Stock Dividends	(1,205)	(1,208)	(1,214)	(1,209)	(1,193)
10	Net Income Available for Common	384,427	437,505	458,767	455,455	535,088
11	Average Common Equity	3,588,651	3,838,084	4,116,231	4,488,159	4,808,892
12	Earned Rate of Return on Common Equity	10.71%	11.40%	11.15%	10.15%	11.13%
13	Authorized Rate of Return on Common Equity	10.30% ⁽¹⁾	10.30% ⁽²⁾	10.30% ⁽²⁾	10.10% ⁽³⁾	10.00% ⁽⁴⁾

Schedule A2

Michigan Public Service Commission
Consumers Energy Company
Financial Metrics - Financial Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule A-2
Page: 2 of 6
Witness: HJMyers

Line	(a)	(b)	(c)	(d)	(e)	(f)
No.	Description	2014	2015	2016	2017	2018
<u>EBIT Interest Coverage Ratio</u>						
14	Pre-Tax Operating Income	\$ 790,628	\$ 825,878	\$ 931,071	\$ 915,731	\$ 815,791
15	Other Income and Deductions	(17,658)	8,173	(34,820)	(16,239)	35,098
16	AFUDC	8,262	8,808	7,849	4,901	5,095
17	Total EBIT	781,232	842,859	904,101	904,394	855,985
18	Interest Charges	184,124	180,418	198,810	202,755	210,785
19	EBIT Interest Coverage Ratio	4.24	4.67	4.55	4.46	4.06
<u>EBITDA Interest Coverage Ratio</u>						
20	Total EBIT	\$ 781,232	\$ 842,859	\$ 904,101	\$ 904,394	\$ 855,985
21	Depreciation and Amortization	522,014	567,371	603,190	654,244	682,307
22	Total EBITDA	1,303,245	1,410,231	1,507,291	1,558,638	1,538,292
23	Interest Charges	184,124	180,418	198,810	202,755	210,785
24	EBITDA Interest Coverage Ratio	7.08	7.82	7.58	7.69	7.30
<u>Funds Flow from Operations (FFO) Interest Coverage Ratio</u>						
25	Net Operating Income	\$ 579,152	\$ 602,150	\$ 685,761	\$ 670,757	\$ 706,874
26	Depreciation and Amortization	522,014	567,371	603,190	654,244	682,307
27	Deferred Income Tax	192,660	143,401	341,939	(54,642)	124,791
28	AFUDC	8,262	8,808	7,849	4,901	5,095
29	Other Major Recurring Non-Cash Items	-	-	-	-	-
30	Interest Paid	180,423	165,485	190,245	193,634	196,495
31	Less:					
32	Operating Lease Adjustment to Depreciation	-	-	-	-	-
33	Subtotal	1,482,510	1,487,216	1,828,985	1,468,894	1,715,561
34	Interest Charges	184,124	180,418	198,810	202,755	210,785
35	FFO Interest Coverage Ratio	8.05	8.24	9.20	7.24	8.14

Schedule A2

Michigan Public Service Commission
Consumers Energy Company
Financial Metrics - Financial Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule A-2
Page: 3 of 6
Witness: HJMyers

Line	(a)	(b)	(c)	(d)	(e)	(f)
No.	Description	2014	2015	2016	2017	2018
<u>Overall Fixed Charge Coverage Ratio</u>						
36	Net Income Available for Common	\$ 384,427	\$ 437,505	\$ 458,767	\$ 455,455	\$ 535,088
37	Interest Charges	184,124	180,418	198,810	202,755	210,785
38	Subtotal Numerator	568,551	617,923	657,577	658,210	745,874
39	Interest Charges	184,124	180,418	198,810	202,755	210,785
40	Preferred Stock Dividends	1,205	1,208	1,214	1,209	1,193
41	Subtotal Denominator	185,329	181,626	200,024	203,964	211,978
42	Overall Fixed Charge Coverage Ratio	3.07	3.40	3.29	3.23	3.52
<u>Cash Flow Coverage of Dividends Ratio</u>						
43	Net Income Available for Common	\$ 384,427	\$ 437,505	\$ 458,767	\$ 455,455	\$ 535,088
44	Depreciation and Amortization	522,014	567,371	603,190	654,244	682,307
45	Deferred Taxes	192,660	143,401	341,939	(54,642)	124,791
46	Subtotal	1,099,100	1,148,277	1,403,896	1,055,057	1,342,186
47	Common Dividends ⁽⁴⁾	307,541	350,004	367,014	364,364	428,071
48	Cash Flow Coverage of Dividends Ratio	3.57	3.28	3.83	2.90	3.14
<u>Common Dividend Payout Ratio</u>						
49	Common Dividends ⁽⁴⁾	\$ 307,541	\$ 350,004	\$ 367,014	\$ 364,364	\$ 428,071
50	Net Income Available for Common	384,427	437,505	458,767	455,455	535,088
51	Common Dividend Payout Ratio	80%	80%	80%	80%	80%
<u>Permanent Capitalization</u>						
52	Long Term Debt	\$ 5,196,593	\$ 5,371,634	\$ 5,617,920	\$ 5,895,659	\$ 6,809,306
53	Preferred Stock	37,315	37,315	37,315	37,315	37,315
54	Common Equity	5,240,388	5,508,798	5,903,031	6,451,911	6,884,117
55	Total Permanent Capital	10,474,297	10,917,747	11,558,266	12,384,885	13,730,738
56	Long Term Debt	49.61%	49.20%	48.61%	47.60%	49.59%
57	Preferred Stock	0.36%	0.34%	0.32%	0.30%	0.27%
58	Common Equity	50.03%	50.46%	51.07%	52.10%	50.14%
59	Total Permanent Capital	100.00%	100.00%	100.00%	100.00%	100.00%

⁽¹⁾ Case No. U-17087 Settlement Agreement dated May 15, 2013

⁽²⁾ Case No. U-17735 Final Order dated November 19, 2015

⁽³⁾ Case No. U-17990 Final Order dated February 28, 2017

⁽⁴⁾ Case No. U-18322 Final Order dated July 24, 2018

⁽⁵⁾ Consumers Energy pays dividends to the parent as a total company. This number represents the Company's 80% dividend policy

Schedule A-2

Michigan Public Service Commission
Consumers Energy Company
Financial Metric - Ratemaking Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule: A-2
Page: 4 of 6
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)
		2014 ⁽⁷⁾	2015 ⁽⁷⁾	Calendar Year 2016 ⁽⁵⁾⁽⁷⁾	2017	2018
60	Operating Revenue	\$ 4,371,083	\$ 4,225,531	\$ 4,355,259	\$ 4,412,087	\$ 4,471,514
61	Operating Expense	3,672,570	3,453,224	3,493,232	3,551,832	3,792,887
62	Pre-Tax Operating Income	698,513	772,307	862,027	860,255	678,627
63	Income Taxes	171,563	180,456	239,709	220,597	58,632
64	Net Operating Income	526,951	591,852	622,319	639,659	619,995
65	Tax Impact of Pro-Forma Interest on NOI	(5,868)	(1,076)	(7,096)	(10,112)	(12,859)
66	AFUDC	8,262	8,808	7,849	4,901	5,070
67	Interest Charges	(168,371)	(178,276)	(181,322)	(177,389)	(159,794)
68	Preferred Stock Dividends	(1,205)	(1,230)	(1,215)	(1,175)	(1,070)
69	Net Income Available for Common and JDITC	359,768	420,078	440,535	455,884	451,342
70	Return Assignable to JDITC	(2,175)	(2,986)	(3,643)	(4,277)	(3,213)
71	Net Income Available for Common	357,593	417,092	436,892	451,607	448,130
72	Average Common Equity	3,671,144	3,973,729	4,174,355	4,400,725	4,306,516
73	Earned Rate of Return on Common Equity	9.74%	10.50%	10.47%	10.26%	10.41%
74	Authorized Return on Equity	10.30% ⁽¹⁾	10.30% ⁽²⁾	10.30% ⁽²⁾	10.10% ⁽³⁾	10.00% ⁽⁴⁾

Schedule A2

Michigan Public Service Commission
Consumers Energy Company
Financial Metric - Ratemaking Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule A-2
Page: 5 of 6
Witness: HJMyers

Line	(a)	(b)	(c)	(d)	(e)	(f)
No.	Description	Calendar Year				
		2014 ⁽⁷⁾	2015 ⁽⁷⁾	2016 ⁽⁵⁾	2017	2018
	<u>EBIT Interest Coverage Ratio</u>					
75	Pre-Tax Operating Income	\$ 698,513	\$ 772,307	\$ 862,027	\$ 860,255	\$ 678,627
76	AFUDC	8,262	8,808	7,849	4,901	5,070
77	Total EBIT	706,775	781,115	869,877	865,157	683,697
78	Interest Charges	162,503	177,200	174,225	167,277	146,935
79	EBIT Interest Coverage Ratio	4.35	4.41	4.99	5.17	4.65
	<u>EBITDA Interest Coverage Ratio</u>					
80	Total EBIT	\$ 706,775	\$ 781,115	\$ 869,877	\$ 865,157	\$ 683,697
81	Depreciation and Amortization	508,786	554,165	590,050	641,020	665,312
82	Total EBITDA	1,215,561	1,335,280	1,459,927	1,506,177	1,349,009
83	Interest Charges	162,503	177,200	174,225	167,277	146,935
84	EBITDA Interest Coverage Ratio	7.48	7.54	8.38	9.00	9.18
	<u>Funds Flow from Operations (FFO) Interest Coverage Ratio</u>					
85	Net Operating Income	\$ 526,951	\$ 591,852	\$ 622,319	\$ 639,659	\$ 619,995
86	Depreciation and Amortization	508,786	554,165	590,050	641,020	665,312
87	Deferred Income Tax	192,660	143,401	341,939	(54,642)	124,791
88	AFUDC	8,262	8,808	7,849	4,901	5,070
89	Other Major Recurring Non-Cash Items	-	-	-	-	-
90	Interest Paid	180,423	165,485	190,245	193,634	196,495
91	Less:					
92	Operating Lease Adjustment	-	-	-	-	-
93	Subtotal	1,417,081	1,463,711	1,752,402	1,424,572	1,611,662
94	Interest Charges	162,503	177,200	174,225	167,277	146,935
95	FFO Interest Coverage Ratio	8.72	8.26	10.06	8.52	10.97
	<u>Overall Fixed Charge Coverage Ratio</u>					
96	Net Income Available for Common	\$ 357,593	\$ 417,092	\$ 436,892	\$ 451,607	\$ 448,130
97	Interest Charges	162,503	177,200	174,225	167,277	146,935
98	Subtotal Numerator	520,096	594,292	611,117	618,884	595,065
99	Interest Charges	162,503	177,200	174,225	167,277	146,935
100	Preferred Stock Dividends	1,205	1,230	1,215	1,175	1,070
101	Subtotal Denominator	163,708	178,430	175,440	168,452	148,005
102	Overall Fixed Charge Coverage Ratio	3.18	3.33	3.48	3.67	4.02

Schedule A2

Michigan Public Service Commission
Consumers Energy Company
Financial Metric - Ratemaking Basis

Case No: U-20697
Exhibit No: A-1 (HJM-2)
Schedule A-2
Page: 6 of 6
Witness: HJMyers

Line	(a)	(b)	(c)	(d)	(e)	(f)
No.	Description	2014 ⁽⁷⁾	2015 ⁽⁷⁾	2016 ⁽⁵⁾	2017	2018
<u>Cash Flow Coverage of Dividends Ratio</u>						
103	Net Income Available for Common	\$ 357,593	\$ 417,092	\$ 436,892	\$ 451,607	\$ 448,130
104	Depreciation and Amortization	508,786	554,165	590,050	641,020	665,312
105	Deferred Taxes	192,660	143,401	341,939	(54,642)	124,791
106	Subtotal	1,059,039	1,114,658	1,368,881	1,037,985	1,238,232
107	Common Dividends ⁽⁶⁾	286,075	333,674	349,513	361,286	358,504
108	Cash Flow Coverage of Dividends Ratio	3.70	3.34	3.92	2.87	3.45
<u>Common Dividend Payout Ratio</u>						
109	Common Dividends ⁽⁶⁾	\$ 286,075	\$ 333,674	\$ 349,513	\$ 361,286	\$ 358,504
110	Net Income Available for Common	357,593	417,092	436,892	451,607	448,130
111	Common Dividend Payout Ratio	80%	80%	80%	80%	80%
<u>Permanent Capitalization</u>						
112	Long Term Debt (Excluding Securitization)	\$ 4,827,361	\$ 5,027,277	\$ 5,299,006	\$ 5,599,864	\$ 6,311,870
113	Preferred Stock	37,315	37,315	37,315	37,315	37,315
114	Common Equity	5,247,298	5,515,574	5,905,957	6,464,167	6,904,894
115	Total Permanent Capital	10,111,974	10,580,166	11,242,278	12,101,346	13,254,078
116	Long Term Debt (Excluding Securitization)	47.74%	47.52%	47.13%	46.27%	47.62%
117	Preferred Stock	0.37%	0.35%	0.33%	0.31%	0.28%
118	Common Equity	51.89%	52.13%	52.53%	53.42%	52.10%
119	Total Permanent Capital	100.00%	100.00%	100.00%	100.00%	100.00%

Footnotes

⁽¹⁾ Case No. U-17087 Settlement Agreement dated May 15, 2013

⁽²⁾ Case No. U-17735 Final Order dated November 19, 2015

⁽³⁾ Case No. U-17990 Final Order dated February 28, 2017

⁽⁴⁾ Case No. U-18322 Final Order dated July 24, 2018

⁽⁵⁾ 2016 Actuals do not reflect all ratemaking normalizations.

⁽⁶⁾ Consumers Energy pays dividends to the parent as a total company. This number represents the Company's 80% dividend policy applied to Net Income Available for Common.

⁽⁷⁾ Prior to 2017, all Transmission assets are reflected in Total Rate Base

⁽⁸⁾ Permanent capitalization is based on calendar year end balances.

Schedule B-1

Michigan Public Service Commission
Consumers Energy Company
Total Rate Base
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-3)
Schedule: B-1
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Rate Base (\$000)	(d) Jurisdictional Amount (\$000)
1	Plant in Service	Exhibit: A-2 (HJM-4)	\$ 14,671,638	\$ 14,609,677
2	Plant Held for Future Use	Exhibit: A-2 (HJM-4)	2,545	2,522
3	Construction Work in Progress	Exhibit: A-2 (HJM-4)	437,907	435,735
4	Total Utility Plant	Exhibit: A-2 (HJM-4)	15,112,090	15,047,934
5	Depreciation Reserve	Exhibit: A-2 (HJM-5)	5,500,305	5,475,065
6	Net Utility Plant	Line 4 - Line 5	9,611,785	9,572,869
7	Net Capital Lease Property		-	-
8	Total Utility Property and Plant	Line 6 + Line 7	9,611,785	9,572,869
9	Less: Capital Lease Obligations		-	-
10	Net Plant	Line 8 - Line 9	9,611,785	9,572,869
11	Working Capital	Exhibit: A-2 (HJM-6)	641,152	638,083
12	Total Historical Period Rate Base	Line 10 + line 11	\$ 10,252,937	\$ 10,210,952

Schedule B-2

Michigan Public Service Commission
Consumers Energy Company
Total Utility Plant
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-4)
Schedule B-2
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) MPSC Account Number	(c) Utility Plant (\$000)	(d) Jurisdictional Amount (\$000)
1	Plant in Service	101	\$ 14,671,638	\$ 14,609,677
2	Plant Purchased or Sold	102	-	-
3	Experimental Plant Unclassified	103	-	-
4	Plant Leased to Others	104	-	-
5	Completed Construction not Classified	106	-	-
6	Plant In Service		<u>\$ 14,671,638</u>	<u>\$ 14,609,677</u>
7	Property Held for Future Use	105	2,545	2,522
8	Deferred Debits		-	-
9	Construction Work in Progress	107	<u>437,907</u>	<u>435,735</u>
10	Total Historical Period Utility Plant	Sum Lines 6 - 9	<u><u>\$ 15,112,090</u></u>	<u><u>\$ 15,047,934</u></u>

Schedule B-3

Michigan Public Service Commission
Consumers Energy Company
Depreciation Reserve and Other Deductions
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-5)
Schedule B-3
Page: 1 of 1
Witness: HJMyers

(a)			(b)	(c)
			Historical Accumulated Provision for Depreciation (\$000)	Jurisdictional Amount (\$000)
Line No.	Description	Source		
1	Accumulated Depreciation, Amortization and Depletion of Plant in Service	WP-HJM-32	\$ 5,440,466	\$ 5,415,256
2	Electric Plant A/P - Amts Retained from Contractors	WP-HJM-33	-	-
3	Customer Advances for Construction	WP-HJM-33	59,839	59,809
4	Total Historical Period Accumulated Provision for Depreciation	Sum Lines 1 - 3	<u>\$ 5,500,305</u>	<u>\$ 5,475,065</u>

Schedule B-4

Michigan Public Service Commission
Consumers Energy Company
Working Capital
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-6)
Schedule B-4
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No	(a) Description	(b) Source	(c) Working Capital (\$000)
<u>Assets</u>			
1	Cash	WP-HJM-35	\$ 38,293
2	Accounts Receivable	WP-HJM-35	310,098
3	Sale of Receivables	WP-HJM-35	-
4	CE Receivable Funding	WP-HJM-35	-
5	Materials & Supplies	WP-HJM-35	97,880
6	Fuel Stock	WP-HJM-35	57,252
7	Clean Air Allowance	WP-HJM-35	14
8	Accrued Revenues	WP-HJM-35	247,189
9	Sale of Accrued Revenues	WP-HJM-35	-
10	Prepayments	WP-HJM-35	304,534
11	Real & Personal Property Taxes	WP-HJM-35	<u>182,905</u>
12	Total Current Assets	Sum Lines 1 - 11	1,238,166
13	Deferred Debits	WP-HJM-35	<u>713,538</u>
14	Total Assets	Line 12 + Line 13	<u><u>\$ 1,951,704</u></u>
<u>Liabilities</u>			
15	Accounts Payable	WP-HJM-35	\$ 423,205
16	Customer Deposits	WP-HJM-35	\$ 16,556
17	Dividends Payable	WP-HJM-35	22,614
18	Accrued Interest	WP-HJM-35	44,248
19	Accrued Taxes	WP-HJM-35	311,359
20	Other Current Liabilities	WP-HJM-35	<u>19,278</u>
21	Total Current Liabilities	Sum Lines 15 - 20	837,261
22	Deferred Credits & Operating Reserves	WP-HJM-35	<u>473,291</u>
23	Total Liabilities	Line 21 + Line 22	<u><u>\$ 1,310,552</u></u>
24	Net Working Capital	Line 14 - Line 23	<u><u>\$ 641,152</u></u>

Schedule B-5

Michigan Public Service Commission
Consumers Energy Company
13-Month Average Working Capital Balance Sheet Summary
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-7)
Schedule: B-5
Page: 1 of 2
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Total	(c) CECo Investor Supplied	(d) Subsidiary Investor Supplied	(e) Electric Plant	(f) Gas Plant	(g) MGP Plant	(h) Non-Utility Plant	(i) Total Working Capital	(j) Electric Working Capital	(k) Gas Working Capital	(l) Non-Utility Working Capital
ASSETS												
1	Utility Plant	\$ 22,331,831,996	\$ 377,809,732	\$ -	\$ 14,890,048,485	\$ 7,063,973,779	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Common Plant	1,507,890,854	6,249,510	-	963,483,733	538,157,610	-	-	-	-	-	-
3	Total Gross Plant	23,839,722,850	384,059,242	-	15,853,532,219	7,602,131,389	-	-	-	-	-	-
4	Utility Plant Depreciation	(5,945,127,706)	30,222,755	-	(4,115,507,074)	(1,859,843,388)	-	-	-	-	-	-
5	Common Plant Depreciation	(790,593,278)	(5,157,361)	-	(511,302,896)	(274,133,021)	-	-	-	-	-	-
6	Total Depreciation	(6,735,720,984)	25,065,394	-	(4,626,809,969)	(2,133,976,409)	-	-	-	-	-	-
7	Net Plant (Line 3 + Line 6)	17,104,001,865	409,124,636	-	11,226,722,249	5,468,154,980	-	-	-	-	-	-
8	Other Property	129,873,436	119,075,307	(5,329,285)	-	-	-	16,127,414	-	-	-	-
Current and Accrued Assets												
9	Cash	60,685,567	9,235	-	-	-	-	-	60,676,331	38,292,784	22,383,548	-
10	Cash Equivalents	2,991,999	2,991,999	-	-	-	-	-	-	-	-	-
11	Notes Receivable	14,434,861	14,434,861	-	-	-	-	-	-	-	-	-
12	Accounts Receivable	463,564,915	27,338,571	-	-	-	-	-	436,226,344	310,098,478	123,388,100	2,739,766
13	Sale of Receivables	-	-	-	-	-	-	-	-	-	-	-
14	CE Receivable Funding	9,909,671	-	-	-	-	-	-	9,909,671	-	-	9,909,671
15	Materials and Supplies	134,034,396	882,674	-	-	-	-	-	133,151,722	97,879,943	35,271,779	-
16	Fuel Stock	57,252,227	-	-	-	-	-	-	57,252,227	57,252,227	-	-
17	Clean Air Allowances	14,288	-	-	-	-	-	-	14,288	14,288	-	-
18	Gas Stored Underground	407,567,668	-	-	-	-	-	-	407,567,668	-	407,567,668	-
19	Accrued Revenues	356,511,792	6,805,219	-	-	-	-	-	349,706,573	247,188,690	102,517,883	-
20	Sale of Accrued Revenues	-	-	-	-	-	-	-	-	-	-	-
21	Prepayments	505,963,933	-	-	-	-	-	-	505,963,933	304,534,025	201,430,023	(115)
22	Real and Personal Property Tax	278,549,575	-	-	-	-	-	-	278,549,575	182,905,442	95,368,790	275,344
23	Total Current and Accrued Assets	2,291,480,891	52,462,559	-	-	-	-	-	2,239,018,332	1,238,165,876	987,927,790	12,924,666
24	Deferred Debits	4,478,066,214	3,279,960,565	-	-	-	137,976,493	-	1,060,129,155	713,537,869	346,459,044	132,242
25	Total Assets	\$ 24,003,422,406	\$ 3,860,623,068	\$ (5,329,285)	\$ 11,226,722,249	\$ 5,468,154,980	\$ 137,976,493	\$ 16,127,414	\$ 3,299,147,487	\$ 1,951,703,744	\$ 1,334,386,834	\$ 13,056,909

Source: WP-HJM-35

Schedule B-5

Michigan Public Service Commission
Consumers Energy Company
13-Month Average Working Capital Balance Sheet Summary
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-7)
Schedule: B-5
Page: 2 of 2
Witness: HJMyers

Line No.	(a) Description	(b) Total	(c) CECo Investor Supplied	(d) Subsidiary Investor Supplied	(e) Electric Plant	(f) Gas Plant	(g) MGP Plant	(h) Non-Utility Plant	(i) Total Working Capital	(j) Electric Working Capital	(k) Gas Working Capital	(l) Non-Utility Working Capital
<u>LIABILITIES AND EQUITY</u>												
26	Common Equity	(6,731,906,529)	(6,732,904,701)	998,173	-	-	-	-	-	-	-	-
27	Preferred Stock	(37,314,800)	(37,314,800)	-	-	-	-	-	-	-	-	-
28	Long Term Debt	(5,766,321,289)	(5,766,321,289)	-	-	-	-	-	-	-	-	-
29	Capital Leases	(102,324,299)	(102,324,299)	-	-	-	-	-	-	-	-	-
30	Total Capitalization	(12,637,866,917)	(12,638,865,090)	998,173	-	-	-	-	-	-	-	-
Current and Accrued Liabilities												
31	Notes Payable	(85,585,422)	(85,585,422)	-	-	-	-	-	-	-	-	-
32	Accounts Payable	(601,593,364)	(589,615)	-	-	-	-	-	(601,003,749)	(423,205,306)	(167,545,366)	(10,253,077)
33	Customer Deposits	(24,952,246)	-	-	-	-	-	-	(24,952,246)	(16,556,255)	(8,395,991)	-
34	Dividends Payable	(31,821,474)	-	-	-	-	-	-	(31,821,474)	(22,613,931)	(9,132,763)	(74,780)
35	Accrued Interest	(62,499,075)	(234,883)	-	-	-	-	-	(62,264,193)	(44,248,048)	(17,869,823)	(146,321)
36	Accrued Taxes	(443,406,038)	-	-	-	-	-	-	(443,406,038)	(311,359,316)	(146,140,874)	14,094,153
37	Current Maturities	(382,369,231)	(382,369,231)	-	-	-	-	-	-	-	-	-
38	Other Liabilities	(67,361,508)	(26,938,458)	-	-	-	(15,996,336)	-	(24,426,715)	(19,278,087)	(5,148,628)	-
39	Total Current and Accrued Liabilities	(1,699,588,358)	(495,717,608)	-	-	-	(15,996,336)	-	(1,187,874,414)	(837,260,944)	(354,233,445)	3,619,975
Deferred Credits												
40	Customer Advances	(64,636,906)	-	-	(59,838,506)	(4,798,400)	-	-	-	-	-	-
41	ITC/JDITC	(94,065,349)	(94,065,349)	-	-	-	-	-	-	-	-	-
42	Deferred Income Taxes	(5,877,356,522)	(5,877,356,522)	-	-	-	-	-	-	-	-	-
43	Deferred Credits/Oper. Res.	(3,629,908,354)	(870,916,616)	-	(918,341,907)	(989,447,318)	(66,588,717)	-	(784,613,796)	(473,291,013)	(309,173,384)	(2,149,399)
44	Total Deferred Credits	(9,665,967,131)	(6,842,338,487)	-	(978,180,413)	(994,245,718)	(66,588,717)	-	(784,613,796)	(473,291,013)	(309,173,384)	(2,149,399)
45	Total Liabilities & Equity	(24,003,422,406)	(19,976,921,185)	998,173	(978,180,413)	(994,245,718)	(82,585,053)	-	(1,972,488,210)	(1,310,551,957)	(663,406,829)	1,470,575
46	Net Assets	\$ -	\$ (16,116,298,117)	\$ (4,331,112)	\$ 10,248,541,836	\$ 4,473,909,262	\$ 55,391,441	\$ 16,127,414	\$ 1,326,659,277	\$ 641,151,788	\$ 670,980,005	\$ 14,527,484

Source: WP-HJM-35

Schedule B-6

Michigan Public Service Commission
Consumers Energy Company
Working Capital Balance Sheet Summary PIT
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-8)
Schedule: B-6
Page: 1 of 2
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Total	(c) CECo Investor Supplied	(d) Subsidiary Investor Supplied	(e) Electric Plant	(f) Gas Plant	(g) MGP Plant	(h) Non-Utility Plant	(i) Total Working Capital	(j) Electric Working Capital	(k) Gas Working Capital	(l) Non-Utility Working Capital
ASSETS												
1	Utility Plant	\$ 23,164,375,346	\$ 383,182,758	\$ -	\$ 15,239,546,885	\$ 7,541,645,703	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
2	Common Plant	1,552,471,545	5,875,074	-	992,327,861	554,268,610	-	-	-	-	-	-
3	Total Gross Plant	24,716,846,891	389,057,833	-	16,231,874,746	8,095,914,313	-	-	-	-	-	-
4	Utility Plant Depreciation	(6,130,163,970)	5,009,831	-	(4,240,698,397)	(1,894,475,404)	-	-	-	-	-	-
5	Common Plant Depreciation	(834,301,348)	(5,168,368)	-	(539,743,205)	(289,389,774)	-	-	-	-	-	-
6	Total Depreciation	(6,964,465,318)	(158,537)	-	(4,780,441,603)	(2,183,865,178)	-	-	-	-	-	-
7	Net Plant (Line 3 + Line 6)	17,752,381,574	388,899,295	-	11,451,433,143	5,912,049,135	-	-	-	-	-	-
8	Other Property	130,123,061	118,028,263	(5,454,013)	-	-	-	17,548,811	-	-	-	-
Current and Accrued Assets												
9	Cash	36,178,306	11,617	-	-	-	-	-	36,166,689	22,824,768	13,341,921	-
10	Cash Equivalents	2,570,452	2,570,452	-	-	-	-	-	-	-	-	-
11	Notes Receivable	758	758	-	-	-	-	-	-	-	-	-
12	Accounts Receivable	478,220,453	34,465,260	-	-	-	-	-	443,755,193	309,872,924	125,578,735	8,303,534
13	Sale of Receivables	-	-	-	-	-	-	-	-	-	-	-
14	CE Receivable Funding	9,909,671	-	-	-	-	-	-	9,909,671	-	-	9,909,671
15	Materials and Supplies	137,269,759	856,736	-	-	-	-	-	136,413,024	99,466,245	36,946,779	-
16	Fuel Stock	52,028,760	-	-	-	-	-	-	52,028,760	52,028,760	-	-
17	Clean Air Allowances	14,288	-	-	-	-	-	-	14,288	14,288	-	-
18	Gas Stored Underground	450,409,972	-	-	-	-	-	-	450,409,972	-	450,409,972	-
19	Accrued Revenues	430,125,260	19,715,283	-	-	-	-	-	410,409,977	252,660,222	157,749,755	-
20	Sale of Accrued Revenues	-	-	-	-	-	-	-	-	-	-	-
21	Prepayments	303,826,701	-	-	-	-	-	-	303,826,701	185,582,357	118,244,474	(130)
22	Real and Personal Property Tax	424,372,461	-	-	-	-	-	-	424,372,461	277,222,352	146,657,140	492,969
23	Total Current and Accrued Assets	2,324,926,841	57,620,105	-	-	-	-	-	2,267,306,736	1,199,671,916	1,048,928,776	18,706,044
24	Deferred Debits	4,395,062,191	3,158,889,387	-	-	-	133,297,889	-	1,102,874,914	737,735,112	364,975,962	163,841
25	Total Assets	\$ 24,602,493,666	\$ 3,723,437,051	\$ (5,454,013)	\$ 11,451,433,143	\$ 5,912,049,135	\$ 133,297,889	\$ 17,548,811	\$ 3,370,181,650	\$ 1,937,407,028	\$ 1,413,904,738	\$ 18,869,885

Source: WP-HJM-86

Schedule B-6

Michigan Public Service Commission
Consumers Energy Company
Working Capital Balance Sheet Summary PIT
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-2 (HJM-8)
Schedule: B-6
Page: 2 of 2
Witness: HJMyers

Line No.	(a) Description	(b) Total	(c) CECo Investor Supplied	(d) Subsidiary Investor Supplied	(e) Electric Plant	(f) Gas Plant	(g) MGP Plant	(h) Non-Utility Plant	(i) Total Working Capital	(j) Electric Working Capital	(k) Gas Working Capital	(l) Non-Utility Working Capital
<u>LIABILITIES AND EQUITY</u>												
26	Common Equity	(6,621,574,454)	(6,622,572,626)	998,173	-	-	-	-	-	-	-	-
27	Preferred Stock	(37,314,800)	(37,314,800)	-	-	-	-	-	-	-	-	-
28	Long Term Debt	(6,809,306,087)	(6,809,306,087)	-	-	-	-	-	-	-	-	-
29	Capital Leases	(91,328,811)	(91,328,811)	-	-	-	-	-	-	-	-	-
30	Total Capitalization	(13,559,524,151)	(13,560,522,324)	998,173	-	-	-	-	-	-	-	-
Current and Accrued Liabilities												
31	Notes Payable	(122,779,010)	(122,779,010)	-	-	-	-	-	-	-	-	-
32	Accounts Payable	(683,582,506)	-	-	-	-	-	-	(683,582,506)	(464,568,874)	(206,016,647)	(12,996,985)
33	Customer Deposits	(25,619,440)	-	-	-	-	-	-	(25,619,440)	(16,716,913)	(8,902,527)	-
34	Dividends Payable	-	-	-	-	-	-	-	-	-	-	-
35	Accrued Interest	(57,614,502)	(184,372)	-	-	-	-	-	(57,430,131)	(40,812,722)	(16,482,447)	(134,961)
36	Accrued Taxes	(485,775,191)	-	-	-	-	-	-	(485,775,191)	(326,271,561)	(163,970,364)	4,466,734
37	Current Maturities	-	-	-	-	-	-	-	-	-	-	-
38	Other Liabilities	(77,025,610)	(31,943,588)	-	-	-	(14,660,534)	-	(30,421,488)	(22,668,524)	(7,752,964)	-
39	Total Current and Accrued Liabilities	(1,452,396,259)	(154,906,970)	-	-	-	(14,660,534)	-	(1,282,828,756)	(871,038,595)	(403,124,949)	(8,665,212)
Deferred Credits												
40	Customer Advances	(69,402,803)	-	-	(50,061,447)	(19,341,356)	-	-	-	-	-	-
41	ITC/JDITC	(99,057,392)	(99,057,392)	-	-	-	-	-	-	-	-	-
42	Deferred Income Taxes	(5,893,119,264)	(5,893,119,264)	-	-	-	-	-	-	-	-	-
43	Deferred Credits/Oper. Res.	(3,266,441,196)	(845,969,117)	-	(957,510,982)	(1,007,996,996)	(58,398,698)	-	(396,565,404)	(246,467,321)	(146,873,729)	(3,224,354)
44	Total Deferred Credits	(9,328,020,654)	(6,838,145,772)	-	(1,007,572,429)	(1,027,338,352)	(58,398,698)	-	(396,565,404)	(246,467,321)	(146,873,729)	(3,224,354)
45	Total Liabilities & Equity	(24,339,941,065)	(20,553,575,066)	998,173	(1,007,572,429)	(1,027,338,352)	(73,059,232)	-	(1,679,394,159)	(1,117,505,916)	(549,998,678)	(11,889,565)
46	Net Assets	\$ 262,552,601	\$ (16,830,138,015)	\$ (4,455,840)	\$ 10,443,860,715	\$ 4,884,710,783	\$ 60,238,657	\$ 17,548,811	\$ 1,690,787,491	\$ 819,901,111	\$ 863,906,060	\$ 6,980,320

Source: WP-HJM-86

Schedule C-1

Michigan Public Service Commission
Consumers Energy Company
Adjusted Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-9)
Schedule C-1
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Net Operating Income (\$000)	(d) Jurisdictional Amount (\$000)
<u>Operating Revenues</u>				
1	Sales Revenue	Exhibit: A-3 (HJM-11)	\$ 4,254,127	\$ 4,221,419
2	Sales For Resale	Exhibit: A-3 (HJM-11)	140,140	\$ 139,063
3	Other Electric Revenue	Exhibit: A-3 (HJM-11)	111,892	\$ 111,032
4	Total Operating Revenue	Sum Lines 1 - 3	\$ 4,506,160	\$ 4,471,514
<u>Operating Expense</u>				
5	Fuel Expense	Exhibit: A-3 (HJM-12)	\$ 407,072	\$ 402,649
6	Purchased & Interchange Power	Exhibit: A-3 (HJM-12)	1,686,435	\$ 1,668,111
7	Other Operation and Maintenance	Exhibit: A-3 (HJM-13)	715,517	\$ 712,491
8	Depreciation and Amortization Expense	Exhibit: A-3 (HJM-14)	668,759	\$ 665,312
9	Property Taxes	Exhibit: A-3 (HJM-15)	158,224	\$ 157,583
10	General Taxes - Other	Exhibit: A-3 (HJM-15)	32,254	\$ 32,043
11	Other (or Local) Taxes	Exhibit: A-3 (HJM-18)	1,170	\$ 1,181
12	State Income Tax	Exhibit: A-3 (HJM-17)	46,992	\$ 47,412
13	Federal Income Tax	Exhibit: A-3 (HJM-16)	58,114	\$ 58,632
14	Total Operating Expense	Sum Lines 5 - 13	\$ 3,774,537	\$ 3,745,414
15	Net Operating Income	Line 4 - Line 14	731,622	\$ 726,101
16	Allowance for Funds Used During Construction	Exhibit: A-3 (HJM-19)	5,095	\$ 5,070
17	Subtotal	Line 15 + Line 16	\$ 736,717	\$ 731,170
<u>Net Operating Income Adjustments</u>				
18	Compensation Disallowances	Exhibit: A-3 (HJM-20)	(3,947)	\$ (3,928)
19	Dues and Donations Disallowances	Exhibit: A-3 (HJM-21)	458	\$ 455
20	Corporate Giving & Communications Disallowances	Exhibit: A-3 (HJM-23)	142	\$ 142
21	Weather Normalization	Exhibit: A-3 (HJM-24)	(59,731)	\$ (59,272)
22	PSCR Adjustments	Exhibit: A-3 (HJM-25)	(79)	\$ (79)
23	Customer Deposit Interest Expense	Exhibit: A-3 (HJM-26)	(324)	\$ (322)
24	Jobwork Revenue	Exhibit: A-3 (HJM-27)	9,575	\$ 9,528
25	Jobwork Expense	Exhibit: A-3 (HJM-28)	(7,757)	\$ (7,718)
26	Transmission Depreciation Expense	Exhibit: A-3 (HJM-30)	1,313	\$ 1,307
27	Property Tax Refund	Exhibit: A-3 (HJM-31)	(6,646)	\$ (6,619)
28	Surcharge Revenue, Expense, and Amortization	Exhibit: A-3 (HJM-29)	(39,906)	\$ (39,599)
29	Tax Benefit of Pro-Forma Interest	Exhibit: A-3 (HJM-32)	(12,742)	\$ (12,690)
30	Interest Synchronization	Exhibit: A-3 (HJM-33)	(169)	\$ (169)
31	Total Net Operating Income Adjustments	Sum Lines 18 - 30	(119,813)	\$ (118,964)
32	Adjusted Net Operating Income	Line 17 + Line 31	\$ 616,904	\$ 612,206

Schedule C-2

Michigan Public Service Commission
Consumers Energy Company
Computation of Revenue Multiplier
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-10)
Schedule C-2
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line	(a)	(b)	(c)
No.	Description	Calc. Logic/Source	Amount
1	Income Base - Before Taxes		100.0000
2	Michigan Corporate Income Tax	Line 1 * 5.31% MCIT Rate	5.3100
3	City Income Tax	Line 2 * 0.16% CIT Rate	<u>0.1600</u>
4	Federal Income Tax Base	Line 1 - Line 2 - Line 3	94.5300
5	Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>19.8513</u>
6	Income Base - After Taxes	Line 4 - Line 5	<u>74.6787</u>
7	Revenue Multiplier	Line 1 / Line 6	<u><u>1.3391</u></u>

Schedule C-3

Michigan Public Service Commission
Consumers Energy Company
Historical Operating Revenue
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-11)
Schedule C-3
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Sales Revenue (\$000)
1	Sales Revenue	WP-HJM-9	\$ 4,254,127
2	Sales For Resale	WP-HJM-9	140,140
3	Other Electric Revenue	WP-HJM-9	<u>111,892</u>
4	Total Historical Period Operating Revenue	Sum Lines 1 - 3	<u><u>\$ 4,506,160</u></u>

Schedule C-4

Michigan Public Service Commission
Consumers Energy Company
Historical Fuel and Purchased Power
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-12)
Schedule C-4
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)		(b)	(c)
			Fuel and Purchased Power
Line No.	Description	Source	(\$000)
1	Fuel	WP-HJM-2	\$ 407,072
2	Purchased and Net Interchange Power	WP-HJM-2	<u>1,686,435</u>
3	Total Historical Period Fuel and Purchased Power Expense	Line 1 + Line 2	<u>\$ 2,093,508</u>

Schedule C-5

Michigan Public Service Commission
Consumers Energy Company
Historical Operation and Maintenance Expenses
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-13)
Schedule C-5
Page: 1 of 4
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Operation and Maintenance Expenses (\$000)
1	Power Production Expense	WP-HJM-2	\$ 170,618
2	Transmission Expense	WP-HJM-16	1,680
3	Distribution Expense	WP-HJM-12	202,639
4	Customer Accounts Expense	WP-HJM-12	63,710
5	Customer Service & Information Expense	WP-HJM-13	160,662
6	Sales Expense	WP-HJM-13	365
7	Administrative & General Expense	WP-HJM-13	<u>116,264</u>
8	Total Other O&M Expense	Sum Lines 1 - 7	715,937
9	Gains From Disposition of Allowances	WP-HJM-6	<u>(421)</u>
10	Total Historical Period Other O&M	Line 8 + Line 9	<u><u>\$ 715,517</u></u>

Schedule C5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Other Operation and Maintenance Expense
for the Historical Year Ended December 31, 2018
O&M Expense By FERC Account and By Witness

Case No.: U-20697
Exhibit: A-3 (HJM-13)
Schedule C5
Witness: HJMyers
Filing Date: February 2020
Page: 2 of 4

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
Line	FERC Account	Description	Blumenstock/Houtz	Shellberg	Hugo	McLean	Tolonen	Ennis	Christopher	Conrad	Gaston	Total O&M	MPSC P-521 Pg 320-323	Variance	
11	94118000	Gain Sale NOX Allow	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	
12	94160000	EXP JOBBING & CTR	-	-	-	-	71,264	-	-	-	-	71,264	-	(71,264)	
13	94170020	EXP FROM NONUTI OP	-	-	-	-	-	-	-	-	-	-	-	-	
14	94264000	CIVIC/POLITICAL	-	-	-	-	-	-	-	-	-	-	-	-	
15	94265010	OTHER DEDUCTIONS	-	-	(1,322,140)	-	(78,780)	1,283,869	(0)	5,354,026	787	5,293,501	-	(5,293,501)	
16	95000010	OP SUPV & ENGIN	207,881	-	9,897,791	-	-	-	-	131,886	220,541	10,458,099	10,455,658	(2,441)	
17	95012010	FUEL - NONFUEL CLAUS	0	-	12,825,883	-	-	-	-	-	-	12,825,883	-	(12,825,883)	
18	95020010	STEAM EXPENSES	-	-	13,043,065	-	-	-	-	-	-	13,043,065	13,388,739	345,674	
19	95050010	ELECTRIC EXPENSES	-	-	8,481,598	-	-	-	-	-	-	8,481,598	8,481,391	(207)	
20	95060010	MISCELL STEAM PWR	-	-	3,354,576	-	8,489,750	285,224	-	-	1,758,031	13,887,582	13,886,968	(614)	
21	95090010	ALLOWANCES	-	-	-	-	-	-	-	-	-	-	-	-	
22	95100010	NTC SUPV & ENGIN	6,973	-	6,702,079	-	-	-	-	131,886	220,541	7,061,478	7,058,271	(3,207)	
23	95110010	NTC OF STRUCTURES	-	-	4,582,199	-	-	-	-	-	-	4,582,199	4,581,046	(1,153)	
24	95120010	NTC BOILER PLANT	-	-	27,441,408	-	-	-	-	-	-	27,441,408	27,108,623	(332,785)	
25	95130010	NTCOF ELEC PLANT	-	-	3,968,262	-	-	-	-	-	-	3,968,262	3,967,967	(295)	
26	95140010	MTC OF MISC STM PLT	-	-	1,257,319	-	-	-	-	-	-	1,257,319	1,256,976	(343)	
27	95350010	OP SUPV & ENGIN- HY	24,133	-	1,293,586	-	1,259,288	-	-	13,957	55,135	2,646,100	2,641,640	(4,460)	
28	95360010	WATER FOR PWR-HY	-	-	1,189,313	-	-	-	-	-	-	1,189,313	1,189,314	1	
29	95370010	HYDRAULIC EXP-HY	-	-	3,108,130	-	-	-	-	-	-	3,108,130	3,108,130	0	
30	95380010	ELECTRIC EXP-HY	-	-	2,194,051	-	-	-	-	-	-	2,194,051	2,194,051	(0)	
31	95390010	MISCELL HY PWR GEN	-	-	653,277	-	289,559	59,963	-	145	438,741	1,441,685	1,441,533	(152)	
32	95400010	RENTS-HYDRO PWR	-	-	6,342	-	-	-	-	-	-	6,342	6,342	-	
33	95410010	MTC SUPV & ENGIN	-	-	811,218	-	-	-	-	13,957	55,135	880,310	875,854	(4,456)	
34	95420010	MAINT OF STRUCT-HY	-	-	2,359,136	-	-	-	-	-	-	2,359,136	2,359,136	(0)	
35	95430010	MAINT OF RESV DAM	-	-	4,541,301	-	-	-	-	-	-	4,541,301	4,541,301	0	
36	95440010	MTC OF ELEC PLT-HY	-	-	1,942,673	-	-	-	-	-	-	1,942,673	1,942,672	(1)	
37	95450010	MTC OF HYDRA PROD	-	-	2,366,906	-	-	-	-	-	-	2,366,906	2,366,906	(0)	
38	95460010	SUPV & ENGG-OTH PWR	57,371	-	1,465,619	-	1,145,126	29,034	-	17,934	247,093	2,962,236	2,980,750	18,514	
39	95470010	FUEL - OTH PWR	(0)	-	448,216	-	-	-	-	-	-	448,216	-	(448,216)	
40	95480010	GEN EXPENSES	-	-	3,802,982	-	-	-	-	-	-	3,802,982	3,878,335	75,353	
41	95490010	MISC OTHER PWR GEN	-	-	3,860,143	-	-	-	-	-	-	3,860,143	4,532,044	671,901	
42	95500010	RENTS-OTHER PWR GEN	-	-	1,755	-	-	-	-	-	-	1,755	1,755	(0)	
43	95510010	NTC SUPV & ENGIN	15,108	-	1,214,611	-	-	-	-	21,869	66,162	1,317,749	1,667,404	349,655	
44	95520010	MAINT OF STRUCTURES	-	-	20,530	-	-	-	-	-	-	20,530	65,374	44,844	
45	95530010	MAINT OF ELECT PLT	-	-	16,256,293	-	-	-	-	-	-	16,256,293	20,805,661	4,549,368	
46	95540010	MTC OF MISC OTH GEN	-	-	55	-	-	-	-	-	-	55	55	(0)	
47	95550010	PURCHASED POWER	-	-	-	-	-	-	-	-	-	-	-	-	
48	95560010	SYS CONTROL AND DISP	925,799	-	9,011,640	-	1,070,490	-	-	-	-	11,007,930	11,007,929	(1)	
49	95600010	ETO Supervsn & Engg	461	-	-	-	-	-	-	-	-	461	-	(461)	
50	95614010	SCHED, SYS CONTROL & DISPATCH SERV	-	-	-	-	-	-	-	-	-	-	-	-	
51	95618010	RELIAB PLAN STND SVC	1,239,515	-	-	-	-	-	-	-	-	1,239,515	1,679,758	440,243	
52	95650010	TRANS OF ELECTRICITY BY OTHERS	-	-	-	-	-	-	-	-	-	-	-	-	

Schedule C5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
 Other Operation and Maintenance Expense
 for the Historical Year Ended December 31, 2018
 O&M Expense By FERC Account and By Witness

Case No.: U-20697
 Exhibit: A-3 (HJM-13)
 Schedule C5
 Witness: HJMyers
 Filing Date: February 2020
 Page: 3 of 4

Line	(a) FERC Account	(b) Description	(c) Blumenstock/Houtz	(d) Shellberg	(e) Hugo	(f) McLean	(g) Tolonen	(i) Saba	(j) Christopher	(k) Conrad	(l) Gaston	(m) Total O&M	(n) MPSC P-521 Pg 320-323	(o) Variance
53	95680010	ETM Supervsn & Engg	\$ 461	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 461	\$ -	(461)
54	95757010	MRK FACILITATION, MONITORING & COMPL	-	-	-	-	-	-	-	-	-	-	-	-
55	95800010	OP SUPV & ENGG-DISTR	10,827,427	2,485,699	307,570	1,452,454	12,495,031	48,284	-	112,904	571,056	28,300,425	28,298,408	(2,017)
56	95820010	STATION EXP-DISTR	1,284,169	-	-	-	-	-	-	-	-	1,284,169	1,284,169	(0)
57	95830010	OVERHEAD LINE EXP	14,829,094	-	-	-	-	-	-	-	-	14,829,094	14,832,411	3,317
58	95840010	UNDERGRD LINE EXP	5,481,567	-	-	-	-	-	-	-	-	5,481,567	5,481,694	127
59	95850010	STREET LIGHT SIGNAL	2,054,374	-	-	-	-	-	-	-	-	2,054,374	2,054,373	(1)
60	95860010	METER EXP-DISTR	1,179,571	-	-	-	-	-	-	-	-	1,179,571	1,179,571	(0)
61	95870010	CUST INSTALL EXP	4,769,280	-	-	-	-	-	-	-	-	4,769,280	4,769,325	45
62	95880010	MISCELL DISTR EXP	13,743,000	-	540,914	-	-	1,778,075	-	-	2,370,438	18,432,426	18,432,430	4
63	95890010	RENTS-DISTR	1,876,972	-	-	-	-	-	-	-	-	1,876,972	1,876,972	0
64	95900010	MTC SUPV AND ENGIN	4,039,413	1,800,051	-	-	-	-	-	53,825	-	5,893,289	5,893,291	2
65	95910010	MTC OF STRUCTURES	519,938	-	-	-	-	-	-	-	-	519,938	519,938	(0)
66	95920010	MTC OF STATION EQ	11,118,686	-	-	-	-	-	-	-	-	11,118,686	11,118,682	(4)
67	95930010	MTC OF OVH LINES	43,049,105	47,662,608	-	-	-	-	-	-	-	90,711,714	90,711,677	(37)
68	95940010	MTC OF LINES	2,763,294	-	-	-	-	-	-	-	-	2,763,294	2,763,290	(4)
69	95950010	MTC OF LINES TRANSF	8,808,519	-	-	-	-	-	-	-	-	8,808,519	8,808,519	0
70	95960010	MTC OF STREETLIGHT	269,497	-	-	-	-	-	-	-	-	269,497	269,497	(0)
71	95970010	MTC OF METERS	4,212,108	-	-	-	-	-	-	-	-	4,212,108	4,212,108	(0)
72	95980010	MTC MISC PLANT	132,342	-	-	-	-	-	-	-	-	132,342	132,342	(0)
73	99010010	SUPV CUST ACCT-E	641,197	-	-	5,706,767	-	63,746	-	13,770	693,911	7,119,392	7,119,392	0
74	99020010	MTR READING EXP-E	6,073,608	-	-	-	-	5,854	-	-	10,734	6,090,196	6,090,192	(4)
75	99030010	CUST REC COLL E	165,875	-	-	25,211,551	7,401,481	-	-	50,490	1,542,285	34,371,683	34,371,683	0
76	99041010	UNCOLL ACCTS-ELEC	-	-	-	-	-	-	-	-	16,661,090	16,661,090	16,113,351	(547,739)
77	99050010	MISC CUST ACCT E	-	-	15,101	-	-	-	-	-	-	15,101	15,100	(1)
78	99070010	SUPRV CUST SVC-ELEC	-	-	3,422,654	-	-	-	-	13,770	-	3,436,424	3,436,425	1
79	99080010	CUST ASSIST-ELEC	143,771	-	248,125	11,245,687	-	-	-	47,799	-	11,685,382	156,669,571	144,984,189
80	99090010	INFO AND INSTR-ELEC	-	-	-	-	-	-	-	-	93,538	93,538	556,107	462,569
81	99120010	DEMONSTR SELLING-E	-	-	-	365,144	-	-	-	-	-	365,144	365,144	0
82	99200010	A & G SAL-ELECT	-	-	-	730,945	6,972,595	2,850,884	4,477	829,214	31,536,597	42,924,713	43,115,738	191,025
83	99210010	OFFICE SUPP EXP-E	-	-	299	888,611	3,983,584	5,131,295	(4,477)	-	3,820,619	13,819,932	13,886,497	66,565
84	99220010	ADMIN TRANSF CR-E	-	-	-	-	-	-	-	(382,420)	(18,856,140)	(19,238,560)	(19,238,560)	-
85	99230010	OUTSIDE SVCS ELEC	125,885	-	-	181,805	17,647,303	749,401	557,328	4,074	11,999,291	31,265,088	31,539,992	274,904
86	99240010	PROP INSUR-ELEC	-	-	-	-	-	-	-	-	(2,673,533)	(2,673,533)	672,467	3,346,000
87	99250010	I&D ELECT	7,368	-	-	-	-	-	-	-	11,660,462	11,667,830	11,667,830	0
88	99260010	EMPL PEN & BENE-E	-	-	-	-	-	-	10,405,511	-	(2,260,802)	8,144,708	8,610,967	466,259
89	99280010	REGULATORY COM EXP-E	-	-	-	-	-	-	-	-	646,798	646,798	646,798	(0)
90	99302010	MISC GEN EXP-ELECT	-	-	352,492	11,650,673	1,261,309	-	-	-	5,597,471	18,861,944	19,512,040	650,096
91	99310010	RENTS-ELECTRIC	-	-	-	-	-	114,132	-	-	(151,565)	(37,434)	(37,435)	(1)
92	99350010	MTC OF GENL PLT-E	-	-	1,815	0	-	5,488,539	-	-	(205,708)	5,284,647	5,887,963	603,316
93		Total	\$ 140,649,501	\$ 51,948,358	\$ 148,231,095	\$ 60,871,391	\$ 62,008,000	\$ 17,888,301	\$ 10,962,839	\$ 6,498,327	\$ 66,118,708	\$ 565,176,521	\$ 703,111,542	\$ 137,935,021
<u>Adjustments to tie to P-521 Pages</u>														
94	Renewable wind/solar	-	-	7,453,639	-	-	-	-	-	-	-	7,453,639	-	-
95	Excess MDNR Fees	-	-	411,502	-	-	-	-	-	-	-	411,502	-	-
96	MISO Reliability	440,242	-	-	-	-	-	-	-	-	-	440,242	-	-
97	Non-Energy Expenses	219,325	-	-	-	-	-	-	-	-	(767,064)	(547,739)	-	-
98	Normalization for Insurance	-	-	-	-	-	-	-	-	-	3,346,000	3,346,000	-	-
99	Energy Efficiency Program	123,453,056	-	-	-	-	-	-	-	-	-	123,453,056	-	-
100	LIEAF	20,296,796	-	-	-	-	-	-	-	-	-	20,296,796	-	-
101	SERP & Other Benefits	-	-	-	-	-	-	-	870,492	-	-	870,492	-	-
102	BTL, true up, & proxy Incentive Compensation	-	-	-	-	-	-	-	-	(5,346,676)	-	(5,346,676)	-	-
103	Corporate Giving, Communications, & Lobbying	-	-	-	-	-	-	-	-	-	804,000	804,000	-	-
104	EXP JOBBING & CTR	-	-	-	-	(71,264)	-	-	-	-	-	(71,264)	-	-
105	Zeeland and Jackson plant non-Fuel	(448,216)	-	-	-	-	-	-	-	-	-	(448,216)	-	-
106	EXP JOBBING & CTR	-	-	75,429	-	-	-	-	-	-	-	75,429	-	-
107	Other Deductions	-	-	-	-	-	78,780	-	-	(69,240)	-	23,644	-	-
108		Total Adjustments	\$ 143,961,203	\$ -	\$ 7,940,570	\$ -	\$ 7,517	\$ -	\$ 870,492	\$ (5,415,916)	\$ 3,382,936	\$ 150,760,905	-	-

Schedule C5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
 Other Operation and Maintenance Expense
 for the Historical Year Ended December 31, 2018
 O&M Expense By FERC Account and By Witness

Case No.: U-20697
 Exhibit: A-3 (HJM-13)
 Schedule C5
 Witness: HJMyers
 Filing Date: February 2020
 Page: 4 of 4

Line	(a) Description	(b) Total	(c) Source
<u>Reconcile Witness Exhibits To MPSC P-521</u>			
109	Other O&M Expense Per BI Report	\$ 565,176,521	Line 93 Column (m)
110	Less:		
111	Fuel Handling	(12,825,883)	Line 17 Column (o)
112	Renewable wind/solar	7,453,639	
113	Excess MDNR Fees	411,502	
114	MISO Reliability	440,242	
115	Non-Energy Expenses	(547,739)	
116	Normalization for insurance	3,346,000	
117	Energy Efficiency Program	123,453,056	
118	LIEAF	20,296,796	
119	SERP & Other Benefits	870,492	
120	BTL, true up, & proxy Incentive Compensation	(5,346,676)	
121	Corporate Giving, Communications, & Lobbying	804,000	
122	EXP JOBBING & CTR	(71,264)	
123	Zeeland and Jackson plant non-Fuel	(448,216)	
124	Significant FR Entries net	75,429	
125	Other Deductions	23,644	
126	Other O&M Expense Per MPSC P-521	<u>\$ 703,111,542</u>	Line 93 Column (n)
<u>Reconcile MPSC P-521 To Schedule C5</u>			
127	Other O&M Expense Per MPSC P-521	\$ 703,111,542	Line 93 Column (n)
128	Fuel Handling	12,825,883	Line 17 Column (m)
129	Rounding		
130	Exhibit: A-3 (HJM-13) Schedule C5	<u>\$ 715,937,426</u>	Line 8

Schedule C-6

Michigan Public Service Commission
Consumers Energy Company
Depreciation and Amortization Expenses
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-14)
Schedule C-6
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Depreciation & Amortization Expense (\$000)
1	Power Production Expense	WP-HJM-14	\$ 288,051
2	Transmission Expense	WP-HJM-14	1,759
3	Distribution Expense	WP-HJM-14	234,991
4	General Plant Expense	WP-HJM-14	11,655
5	Allocated Portion of Common Plant	WP-HJM-14	18,230
6	Amortization of Limited Term Electric Plant	WP-HJM-14	522
7	Amortization of Other Electric Plant	WP-HJM-14	82,724
8	Amortization of Utility Plant Acq. Adj	WP-HJM-6	5,428
9	Regulatory Debits	WP-HJM-6	<u>25,398</u>
10	Total Historical Period Depreciation and Amortization Expense	Sum Lines 1 - 9	<u><u>\$ 668,759</u></u>

Schedule C-7

Michigan Public Service Commission
Consumers Energy Company
General Taxes
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-15)
Schedule C-7
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) General Taxes (\$000)
1	Real & Personal Property Tax	WP-HJM-8	\$ 158,224
2	Payroll Taxes	WP-HJM-3	20,645
3	Other General Taxes	WP-HJM-8	<u>11,609</u>
4	Total Historical Period General Taxes	Sum Lines 1 - 3	<u><u>\$ 190,478</u></u>

Schedule C-8

Michigan Public Service Commission
 Consumers Energy Company
 Federal Income Taxes
 for the Historical Year Ended December 31, 2018

Case No: U-20697
 Exhibit No: A-3 (HJM-16)
 Schedule C-8
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Federal Income Taxes (\$000)
1	Federal Income Tax	WP-HJM-6	\$ (34,607)
2	Provision for Deferred Income Tax	WP-HJM-6	345,627
3	Provision for Deferred Income Tax - Credit	WP-HJM-6	(233,920)
4	Investment Tax Credit Adjustment - Net	WP-HJM-6	13,084
4	Deferred State Income Tax	Tax Department	(30,944)
5	Deferred Local Income Tax	Tax Department	(1,127)
6	Total Federal Income Tax	Sum Lines 1 - 5	<u>\$ 58,114</u>

Schedule C-9

Michigan Public Service Commission
Consumers Energy Company
State Income Taxes
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-17)
Schedule C-9
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)		(b)	(c)
			State
			Income
			Taxes
Line	Description	Source	(\$000)
No.			
1	State Income Tax	WP-HJM-8	\$ 16,048
2	Deferred State Income Tax	Tax Department	30,944
3	Total State Income Tax	Line 1 + Line 2	\$ 46,992

Schedule C-10

Michigan Public Service Commission
Consumers Energy Company
Other (or Local) Taxes
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-18)
Schedule C-10
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Other (or Local) Taxes (\$000)
1	Other (or Local) Taxes	WP-HJM-8	\$ 44
2	Deferred Local Income Tax	Tax Department	<u>1,127</u>
3	Total Other (or Local) Taxes		<u><u>\$ 1,170</u></u>

Schedule C-11

Michigan Public Service Commission
Consumers Energy Company
Allowance for Funds Used During Construction
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-19)
Schedule C-11
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) AFUDC (\$000)
1	Total Historical Period Allowance for Funds Used During Construction	WP-HJM-17	\$ 5,095

Schedule C-12

Michigan Public Service Commission
 Consumers Energy Company
 Compensation Disallowances
 Impact on Net Operating Income
 for the Historical Year Ended December 31, 2018

Case No: U-20697
 Exhibit No: A-3 (HJM-20)
 Schedule C-12
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Executive Annual Physicals	WP-HJM-18	\$ 13
2	Request for Allowance - Incentive Compensation		<u>(5,299)</u>
3	Impact on Taxable Income	Line 1 + Line 2	(5,286)
4	Impact on State and Local Income Tax	Line 5 * State and Local Rate	<u>(289)</u>
5	Impact on Federal Taxable Income	Line 3 - Line 4	(4,997)
6	Impact on Federal Income Tax	Line 5 * 21.00% FIT Rate	<u>(1,049)</u>
7	Impact on Net Operating Income	Line 5 - Line 6	<u><u>\$ (3,947)</u></u>

Schedule C-13

Michigan Public Service Commission
 Consumers Energy Company
 Dues and Donations Disallowances
 Impact on Net Operating Income
 for the Historical Year Ended December 31, 2018

Case No: U-20697
 Exhibit No: A-3 (HJM-21)
 Schedule C-13
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Chamber of Commerce Expense	WP-HJM-19	\$ 105
2	Economic Development Expense	WP-HJM-19	5
3	Civic Dues / Other Expense	WP-HJM-19	14
4	Additional Corporate Dues & Lobbying	WP-HJM-19	<u>489</u>
5	Impact on Taxable Income	Sum Lines 1 - 4	613
6	Impact on State and Local Income Tax	Line 5 * State and Local Rate	<u>34</u>
7	Impact on Federal Taxable Income	Line 5 - Line 6	579
8	Impact on Federal Income Tax	Line 7 * 21.00% FIT Rate	<u>122</u>
9	Impact on Net Operating Income	Line 7 - Line 8	<u><u>\$ 458</u></u>

Schedule C-14

Michigan Public Service Commission
Consumers Energy Company
Advertising Classification and Disallowance
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-22)
Schedule C-14
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Gross Amount	(d) Net Amount
1	Customer Information	WP-HJM-25	\$ 1,491	\$ 1,113
2	Related to Employment Opportunities	WP-HJM-25	7	5
3	Required by Law	WP-HJM-25	100	75
4	Provision of Data to Educational Institutions	WP-HJM-25	-	-
5	Above the Line Advertising Expenses ¹	Sum Lines 1 - 4	1,598	1,193
6	Other Advertising Programs - Disallowance	WP-HJM-25	-	-
7	Total Historical Period Above the Line Advertising Expenses	Line 5 - Line 6	<u>\$ 1,598</u>	<u>\$ 1,193</u>

Footnotes

¹ Allowable categories of advertising expense are defined in Case No. U-6490 and U-8455, dated April 29, 1986.

Schedule C-15

Michigan Public Service Commission
Consumers Energy Company
Corporate Giving & Communications Disallowances
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-23)
Schedule C-15
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Corporate Giving	WP-HJM-26	\$ 10
2	Corporate Communications	WP-HJM-26	<u>\$ 180</u>
3	Impact on Taxable Income	Line 1 + Line 2	\$ 191
4	Impact on State and Local Income Tax	Line 3 * State and Local Rate	<u>\$ 10</u>
5	Impact on Federal Taxable Income	Line 3 - Line 4	\$ 180
6	Impact on Federal Income Tax	Line 5 * 21.00% FIT Rate	<u>\$ 38</u>
7	Impact on Net Operating Income	Line 5 - Line 6	<u><u>\$ 142</u></u>

Schedule C-16

Michigan Public Service Commission
Consumers Energy Company
Weather Normalization
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-24)
Schedule C-16
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Weather Normalization Impact on Revenue	Exhibit: A-15 (EMB-1)	<u>\$ (79,984)</u>
2	Impact on Taxable Income	Line 1	(79,984)
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>(4,375)</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	(75,609)
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>(15,878)</u>
6	Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ (59,731)</u></u>

Schedule C-17

Michigan Public Service Commission
 Consumers Energy Company
 PSCR Adjustments
 Impact on Net Operating Income
 for the Historical Year Ended December 31, 2018

Case No: U-20697
 Exhibit No: A-3 (HJM-25)
 Schedule C-17
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	MISO Reliability Schedule 10 Charge	WP-HJM-28	\$ 440
2	Excess MDNR Fees	WP-HJM-27	\$ 412
3	Purchased Power Administration Fee Revenue	WP-HJM-29	<u>\$ (958)</u>
4	Impact on Taxable Income	Sum Lines 1 - 3	\$ (106)
5	Impact on State and Local Income Tax	Line 4 * State and Local Rate	<u>\$ (6)</u>
6	Impact on Federal Taxable Income	Line 4 - Line 5	\$ (100)
7	Impact on Federal Income Tax	Line 6 * 21.00% FIT Rate	<u>\$ (21)</u>
8	Impact on Net Operating Income	Line 6 - Line 7	<u><u>\$ (79)</u></u>

Schedule C-18

Michigan Public Service Commission
Consumers Energy Company
Customer Deposit Interest Expense
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-26)
Schedule C-18
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Customer Deposit Interest Expense	WP-HJM-30	<u>\$ (434)</u>
2	Impact on Taxable Income	Line 1	(434)
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>(24)</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	(410)
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>(86)</u>
6	Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ (324)</u></u>

Schedule C-19

Michigan Public Service Commission
Consumers Energy Company
Jobwork Revenue
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-27)
Schedule C-19
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Jobwork Revenues	Exhibit: A-15 (EMB-1)	<u>\$ 12,822</u>
2	Impact on Taxable Income	Line 1	12,822
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>701</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	12,120
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>2,545</u>
6	Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ 9,575</u></u>

Schedule C-20

Michigan Public Service Commission
Consumers Energy Company
Jobwork Expense
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-28)
Schedule C-20
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Jobwork Expense	Exhibit: A-15 (EMB-1)	<u>\$ (10,387)</u>
2	Impact on Taxable Income	Line 1	(10,387)
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>(568)</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	(9,818)
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>(2,062)</u>
6	Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ (7,757)</u></u>

Schedule C-21

Michigan Public Service Commission
Consumers Energy Company
Surcharge Revenue, Expense, and Amortization
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-29)
Schedule C-21
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
<u>Surcharge Revenue</u>			
1	Energy Optimization	WP-HJM-31	\$ (123,448)
2	Green Generation	WP-HJM-31	(4,479)
3	Renewable Energy Plan	WP-HJM-31	(59,740)
4	Security Recovery Factor	WP-HJM-31	-
5	Stranded Cost Recovery	WP-HJM-31	-
6	ECC Implementation Revenue (ERIP)	WP-HJM-31	-
7	Coal Plant Securitization Credit	WP-HJM-31	-
8	Securitization	WP-HJM-31	-
9	Securitization (Classic 7)	WP-HJM-31	(34,963)
10	Securitization Tax Charge	WP-HJM-31	-
11	Major Maintenance	WP-HJM-31	-
12	Low Income Assistance Fund	WP-HJM-31	(20,297)
13	Total Surcharge Revenue	Sum Lines 1 - 12	(242,926)
<u>Surcharge Expense & Amortization</u>			
14	Energy Optimization Expense	WP-HJM-31	123,147
15	Green Generation	WP-HJM-31	4,479
16	Renewable Energy Plan	WP-HJM-31	16,169
17	Security Recovery Factor	WP-HJM-31	-
18	Stranded Cost Recovery	WP-HJM-31	-
19	ERIP	WP-HJM-31	-
20	Coal Plant Securitization Credit	WP-HJM-31	-
21	Securitization Amortization	WP-HJM-31	25,398
22	Major Maintenance	WP-HJM-31	-
23	Low Income Assistance Fund	WP-HJM-31	20,297
24	Total Surcharge Expenses & Amortization	Sum Lines 14 - 23	189,490
25	Impact on Taxable Income	Line 13 + Line 24	(53,436)
26	Impact on State and Local Income Tax	Line 26 * State and Local Rate	(2,923)
27	Impact on Federal Taxable Income	Line 25 - Line 26	(50,513)
28	Impact on Federal Income Tax	Line 27 * 21.00% FIT Rate	(10,608)
29	Impact on Net Operating Income	Line 27 - Line 28	\$ (39,906)

Schedule C-22

Michigan Public Service Commission
Consumers Energy Company
Transmission Depreciation Expense
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-30)
Schedule C-22
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Transmission Depreciation Expense	WP-HJM-14	<u>\$ 1,759</u>
2	Impact on Taxable Income	Line 1	1,759
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>96</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	1,663
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>349</u>
6	Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ 1,313</u></u>

Schedule C-23

Michigan Public Service Commission
Consumers Energy Company
Property Tax Refund
Impact on Net Operating Income
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-31)
Schedule C-23
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Property Tax Refund	WP-BJV-2	<u>\$ (8,900)</u>
2	Impact on Taxable Income	Line 1	(8,900)
3	Impact on State and Local Income Tax	Line 2 * State and Local Rate	<u>(487)</u>
4	Impact on Federal Taxable Income	Line 2 - Line 3	(8,413)
5	Impact on Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>(1,767)</u>
6	Property Tax Impact on Net Operating Income	Line 4 - Line 5	<u><u>\$ (6,646)</u></u>

Schedule C-24

Michigan Public Service Commission
Consumers Energy Company
Tax Effect of Pro-Forma Interest Adjustment
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-32)
Schedule C-24
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Rate Base	Exhibit: A-2 (HJM-3)	\$ 10,252,937
2	Weighted Cost of Debt ¹	Exhibit: A-4 (HJM-34)	1.56%
3	Allowable Interest Expense	Line 1 * Line 2	160,450
4	Actual Interest Expense	Exhibit: A-1 (HJM-2)	210,785
5	Increase/ (Decrease) in Allowable Interest Deduction	Line 3 - Line 4	(50,335)
6	Impact on Taxable Income	Line 5 * -1	50,335
7	CIT Rate	Tax Department	0.16%
8	Impact on City Income Tax	Line 6 * Line 7	81
9	Impact on State Taxable Income	Line 6 - Line 8	50,254
10	MCIT Rate	Tax Department	5.310%
11	Impact on State Income Tax	Line 9 * Line 10	2,668
12	Impact on Federal Taxable Income	Line 9 - Line 11	47,586
13	FIT Rate	Tax Department	21.00%
14	Impact on Federal Income Tax	Line 12 * Line 13	9,993
15	Impact on Net Operating Income	(Line 8 + Line 11 + Line 14) * -1	\$ (12,742)

Footnotes

¹ Excludes the Job Development Investment Tax Credit portion

Schedule C-25

Michigan Public Service Commission
Consumers Energy Company
Tax Effect of Interest Synchronization Adjustment
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-3 (HJM-33)
Schedule C-25
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)
1	Rate Base	Exhibit: A-2 (HJM-3)	\$ 10,252,937
2	Debt Related JDITC Portion of the Capital Structure ¹	Exhibit: A-4 (HJM-34)	<u>0.27%</u>
3	Portion of Rate Base Funded by JDITC	Line 1 * Line 2	27,834
4	Cost of JDITC - Debt Portion	Exhibit: A-4 (HJM-34)	<u>4.23%</u>
5	Allowable JDITC Interest Expense	Line 3 * Line 4	<u>1,177</u>
6	Impact on Taxable Income	Line 5 * -1	(1,177)
7	CIT Rate	Tax Department	<u>0.16%</u>
8	Impact on City Income Tax	Line 6 * Line 7	<u>(2)</u>
9	Impact on State Taxable Income	Line 6 - Line 8	(1,175)
10	MCIT Rate	Tax Department	<u>5.310%</u>
11	Impact on State Income Tax	Line 9 * Line 10	<u>(62)</u>
12	Impact on Federal Taxable Income	Line 9 - Line 11	(1,113)
13	FIT Rate	Tax Department	<u>21.00%</u>
14	Impact on Federal Income Tax	Line 12 * Line 13	<u>234</u>
15	Synchronization Adjustment Impact on Net Operating Income	(Line 8 + Line 11 + Line 14) * -1	<u><u>\$ (169)</u></u>

Footnotes

¹ Job Development Investment Tax Credit

Schedule D-1

Michigan Public Service Commission
Consumers Energy Company
Overall Rate of Return Summary
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-4 (HJM-34)
Schedule: D-1
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a) Line	(b) Description	(c) Source	Capital Structure		(e) Percent of Total Capital	(f) Cost Rate ¹ %	Weighted Cost			
			Amount (\$000)	Percent Permanent Capital			Permanent Capital	Total Cost %	Conversion Factor	Pre-Tax Return
1	Long Term Debt	WP-HJM-76	\$ 5,845,790	46.24%	36.48%	4.23%	1.96%	1.54%	1.54%	1.54%
2	Preferred Stock	WP-HJM-77	\$ 37,315	0.30%	0.23%	4.50%	0.01%	0.01%		0.01%
3	Common Equity	WP-HJM-77	\$ 6,757,947	53.46%	42.18%	10.00%	5.35%	4.22%		5.65%
4	Permanent Capital		\$ 12,641,052	100.00%			7.32%			
5	Total Short Term Debt	WP-HJM-76	\$ 125,130		0.78%	2.79%		0.02%	0.02%	0.02%
6	Deferred FIT	WP-HJM-78	\$ 3,163,165		19.74%	0.00%		0.00%		0.00%
<u>Deferred JDITC/ITC ²</u>										
7	Long Term Debt		\$ 43,500		0.27%	4.23%		0.01%	0.01%	0.01%
8	Preferred Stock		\$ 278		0.00%	4.50%		0.00%		0.00%
9	Common Equity		\$ 50,288		0.31%	10.00%		0.03%		0.04%
10	Total Deferred JDITC/ITC	WP-HJM-79	\$ 94,065		0.59%			0.04%	0.01%	0.05%
11	Total Capitalization		\$ 16,023,412		100.00%			5.84%	1.58%	7.28%

Footnotes

¹ Cost Rates: A-4 (HJM-35) to A-4 (HJM-38)

² Deferred JDITC & ITC is allocated on the basis of % of permanent capital in column (d)

Schedule D-2

Michigan Public Service Commission
Consumers Energy Company
Cost of Long-Term Debt
for the Historical Year Ended December 31, 2018
(Excluding Securitization)

Case No: U-20697
Exhibit No: A-4 (HJM-35)
Schedule: D-2
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Original Issue Date	(c) Stated Maturity Date	(d) Interest Rate (%)	(e) Amount of Offering (\$000)	(f) Price to Public %	(g) Underwriting & Financing Expenses %	(h) Net Proceeds to the Company %	(i) Cost Based on Net Proceeds %	(j) Amount Outstanding (\$000)	(k) Annual Cost (\$000)
<u>First Mortgage Bonds</u>											
1	FMB 5.15% - \$250M - Due 2/15/2017	08-17-2004	08-15-2016	5.5%	350,000	99.52%	0.76%	98.76%	5.64%	-	-
2	FMB 3.21% - \$100M - Due 10/15/2017	01-20-2005	02-15-2017	5.2%	250,000	99.56%	0.71%	98.85%	5.28%	-	-
3	FMB 5.65% - \$250M - Due 9/15/2018	10-15-2010	10-15-2017	3.2%	100,000	100.00%	0.54%	99.46%	3.30%	-	-
4	FMB 6.125% - \$350M - Due 3/15/2018	03-14-2008	09-15-2018	5.7%	250,000	99.62%	0.80%	98.81%	5.80%	-	-
5	FMB 6.70% - \$500M - Due 9/15/2018	09-12-2008	03-15-2019	6.1%	350,000	99.93%	0.72%	99.21%	6.23%	-	-
6	FMB 5.65% - \$300M - Due 4/15/2020	03-06-2009	09-15-2019	6.7%	500,000	99.95%	0.71%	99.24%	6.80%	-	-
7	FMB 3.77% - \$100M - Due 10/15/2020	03-24-2005	04-15-2020	5.7%	300,000	99.60%	0.91%	98.69%	5.78%	300,000	17,342
8	FMB 5.30% - \$250M - Due 9/1/2022	10-15-2010	10-15-2020	3.8%	100,000	100.00%	0.54%	99.46%	3.84%	100,000	3,835
9	FMB 2.85% - \$375M - Due 5/15/2022	09-01-2010	09-01-2022	5.3%	250,000	100.00%	0.39%	99.61%	5.34%	250,000	13,361
10	FMB 3.375% - \$325M - Due 8/15/2022	05-08-2012	05-15-2022	2.9%	375,000	99.99%	0.77%	99.22%	2.94%	375,000	11,025
11	FMB 3.19% - \$51.5M - Due 12/15/2024	08-09-2013	08-15-2023	3.4%	325,000	99.95%	0.84%	99.11%	3.48%	325,000	11,312
12	FMB 3.125% - \$250M - Due 8/31/2024	12-17-2012	12-15-2024	3.2%	51,500	100.00%	0.54%	99.46%	3.24%	51,500	1,671
13	FMB 3.68% - \$100M Due 10/15/2027	08-18-2014	08-31-2024	3.1%	250,000	99.90%	0.80%	99.10%	3.23%	250,000	8,077
14	FMB 3.39% - \$35.5M - Due 12/15/2027	10-01-2018	10-01-2027	3.7%	100,000	100.00%	0.16%	99.84%	3.70%	100,000	3,701
15	FMB 3.80% - \$300M - Due 11/15/2028	12-17-2012	12-15-2027	3.4%	35,500	100.00%	0.55%	99.45%	3.44%	35,500	1,220
16	FMB 3.18% - \$40M - Due 09/28/2032	11-13-2018	11-15-2028	3.8%	300,000	99.75%	0.69%	99.06%	3.91%	300,000	11,744
17	FMB 3.18% - \$60M - Due 11/15/2032	09-28-2017	09-28-2032	3.2%	40,000	100.00%	0.32%	99.68%	3.21%	40,000	1,283
18	FMB 3.18% - \$60M - Due 11/15/2032	11-15-2017	11-15-2032	3.2%	60,000	100.00%	0.32%	99.68%	3.21%	60,000	1,924
19	FMB 5.80% - \$175M - Due 9/15/2035	08-11-2005	09-15-2035	5.8%	175,000	99.81%	1.05%	98.76%	5.89%	175,000	10,304
20	FMB 3.52% - \$125M - Due 09/28/2037	09-28-2017	09-28-2037	3.5%	125,000	100.00%	0.42%	99.58%	3.55%	125,000	4,437
21	FMB 3.52% - \$210M - Due 11/15/2037	11-15-2017	11-15-2037	3.5%	210,000	100.00%	0.42%	99.58%	3.55%	210,000	7,454
22	FMB 6.17% - \$50M - Due 9/1/2040	10-01-2018	10-01-2038	4.0%	215,000	100.00%	0.35%	99.65%	4.04%	215,000	8,676
23	FMB 6.17% - \$50M - Due 9/1/2040	09-01-2010	09-01-2040	6.2%	50,000	100.00%	0.98%	99.02%	6.24%	50,000	3,121
24	FMB 4.97% - \$50M - Due 10/15/2040	10-15-2010	10-15-2040	5.0%	50,000	100.00%	0.55%	99.45%	5.01%	50,000	2,503
25	FMB 4.31% - \$263M - Due 12/15/2042	12-17-2012	12-15-2042	4.3%	263,000	100.00%	0.52%	99.48%	4.34%	263,000	11,418
26	FMB 3.95% - \$425M - Due 5/15/2042	05-17-2013	05-15-2043	4.0%	425,000	99.84%	1.01%	98.83%	4.02%	425,000	17,074
27	FMB 4.10% - \$250M - Due 11/15/2045	11-02-2015	11-15-2045	4.1%	250,000	99.91%	1.06%	98.85%	4.17%	250,000	10,418
28	FMB 3.25% - \$450M - Due 08/15/2046	08-10-2016	08-15-2046	3.3%	450,000	99.22%	1.06%	98.16%	3.35%	450,000	15,065
29	FMB 3.95% - \$350M - Due 07/15/2047	02-15-2017	07-15-2047	4.0%	350,000	99.58%	1.07%	98.51%	4.04%	350,000	14,124
30	FMB 4.05% - \$550M - Due 05/15/2048	05-14-2018	05-15-2048	4.1%	550,000	98.97%	1.07%	97.91%	4.17%	550,000	22,952
31	FMB 3.86% - \$20M - Due 09/28/2052	11-13-2018	04-15-2049	4.4%	550,000	99.62%	0.99%	98.63%	4.43%	550,000	24,379
32	FMB 3.86% - \$20M - Due 09/28/2052	09-28-2017	09-28-2052	3.9%	20,000	100.00%	0.74%	99.26%	3.90%	20,000	780
33	FMB 3.86% - \$30M - Due 11/15/2052	11-15-2017	11-15-2052	3.9%	30,000	100.00%	0.74%	99.26%	3.90%	30,000	1,170
34	FMB 4.35% - \$250M - Due 8/31/2064	10-01-2018	10-01-2057	4.3%	185,000	100.00%	0.67%	99.33%	4.32%	185,000	7,984
35	FMB 4.35% - \$250M - Due 8/31/2064	08-18-2014	08-31-2064	4.4%	250,000	99.14%	1.03%	98.10%	4.44%	250,000	11,112
36	Total First Mortgage Bonds				8,135,000				4.10%	6,335,000	259,466
<u>Other Long Term Debt</u>											
37	MSF \$35.0M VAR LORBs (2005) - Due 04/01/2035	04-01-2005	04-01-2035	1.7%	35,000	100.00%	4.28%	95.72%	1.92%	35,000	671
38	MSF \$67.7M VAR LORBs (2008) - Due 04/15/2018	03-01-2008	04-15-2018	0.0%	67,700	100.00%	0.65%	99.35%	0.06%	-	-
39	SR NOTE 6 7/8% - \$225M - Due 3/01/2018	03-01-1998	03-01-2018	6.9%	225,000	98.53%	0.97%	97.56%	7.10%	-	-
40	Total Other Long Term Debt				327,700				1.92%	35,000	671
41	Amortization of Loss on Recquired Debt									-	5,887
42	Revolver Fees									-	840
43	Revolver Upfront Fees Amortization 1									-	1,123
44	Letter of Credit Fees									-	758
45	Unamortized Debt Discount/(Premium)									(16,173)	-
46	Total Long Term Debt								4.23%	6,353,827	268,745
47	Unamortized Debt Costs									(41,957)	
48	Total Long Term Debt, Net of Unamortized Debt Costs									6,311,870	

1 The upfront issuance costs and ongoing activity fees of the revolving credit facilities are treated as interest expense instead of net proceeds. This is done in order to include the cost of financing as part of the cost of debt in months when the Company does not draw on its RCI

Source: Cost of Capital Study

Schedule D-3

Michigan Public Service Commission
Consumers Energy Company
Cost of Short-Term Debt
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-4 (HJM-36)
Schedule D-3
Page: 1 of 1
Witness: HJMyers
Date: February 2020

	(a)	(b)	(c)	(d)	(e)
			% of ²		Weighted Average
Line	Description	13-Month Average ¹	Total Capital	Cost Rate ³	Cost of Short Term Debt
1	Notes Payable	\$ 72,808	0.45%	2.79%	\$ 2,035
2	Advanced Renewable Regulatory Liability	\$ 52,321	0.33%	2.79%	\$ 1,462
3	Total Short Term Debt	\$ 125,130	0.78%	2.79%	<u>\$ 3,497</u>
4	Historical Period Weighted Average Cost of Short Term Debt				<u><u>2.79%</u></u>

Footnotes

¹ WP-HJM-76

² Exhibit: A-4 (HJM-34)

³ Treasury Department

Schedule D-4

Michigan Public Service Commission
Consumers Energy Company
Cost of Preferred Stock
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-4 (HJM-37)
Schedule D-4
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)
		Annual Dividend (\$000)	Par Value	Discount or Premium (\$000)	Finance Expenses (\$000)	Net Proceeds Received (\$000)	Number of Shares Outstanding (\$000)	Total Value of Outstanding Proceeds (\$000)	Cost Rate %	Annual Dollar Amount (\$000)
1	Preferred Stock \$4.50 Series	\$ 4.50	\$ 100.00	\$ -	\$ -	\$ 100.00	373,148	\$ 37,315	4.50%	\$ 1,679
2	Total Preferred Stock							<u>\$ 37,315</u>		<u>\$ 1,679</u>
3	Historical Period Weighted Average Cost of Preferred Stock									4.50%

Source: WP-HJM-77

Schedule D-5

Michigan Public Service Commission
Consumers Energy Company
Cost of Common Shareholder's Equity
for the Historical Year Ended December 31, 2018

Case No: U-20697
Exhibit No: A-4 (HJM-38)
Schedule D-5
Page: 1 of 1
Witness: HJMyers
Date: February 2020

The Cost of Common Equity is 10.00%. This rate was authorized by the
Michigan Public Service Commission in Electric Rate Case No. U-18322 issued on July 24, 2018

Schedule A-1

Michigan Public Service Commission
 Consumers Energy Company
 Revenue Deficiency (Sufficiency)
 Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
 Exhibit No: A-11 (HJM-39)
 Schedule A-1
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total (\$000)	(d) Jurisdictional Amount (\$000)
1	Rate Base	Exhibit: A-12 (HJM-42)	\$ 11,893,424	\$ 11,844,757
2	Adjusted Net Operating Income	Exhibit: A-13 (HJM-49)	534,565	539,156
3	Overall Rate of Return	Line 2 / Line 1	4.49%	4.55%
4	Required Rate of Return	Exhibit: A-14 (MRB-1)	6.09%	6.09%
5	Income Required	Line 1 * Line 4	724,604	721,639
6	Income Deficiency/ (Sufficiency)	Line 5 - Line 2	190,039	182,483
7	Revenue Multiplier	Exhibit: A-13 (HJM-50)	1.3391	1.3391
8	Revenue Deficiency (Sufficiency)	Line 6 * Line 7	254,475	244,357

Schedule A-2

Michigan Public Service Commission
 Consumers Energy Company
 Financial Metrics - Ratemaking Basis

Case No: U-20697
 Exhibit No: A-11 (HJM-40)
 Schedule: A-2
 Page: 1 of 3
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Fiscal Year (c)	
		No Rate Relief December 2021	Full Rate Relief December 2021
1	Operating Revenue	\$ 4,371,575	\$ 4,615,932
2	Operating Expense	3,784,212	3,797,188
3	Pre-Tax Operating Income	587,363	818,745
4	Income Taxes	54,409	102,999
5	Net Operating Income	532,954	715,745
6	Tax Impact of Pro-Forma Interest on NOI	-	-
7	AFUDC	6,203	6,203
8	Interest Charges	(184,564)	(184,564)
9	Preferred Stock Dividends	(938)	(938)
10	Net Income Available for Common and JDITC	353,654	536,446
11	Return Assignable to JDITC	(5,135)	(5,135)
12	Net Income Available for Common	348,519	531,311
13	Average Common Equity	\$ 5,060,104	\$ 5,060,104
14	Earned Rate of Return on Common Equity	6.89%	10.50%
15	Authorized Return on Common Equity ¹	10.50%	10.50%

Schedule A-2

Michigan Public Service Commission
Consumers Energy Company
Financial Metrics - Ratemaking Basis

Case No: U-20697
Exhibit No: A-11 (HJM-40)
Schedule: A-2
Page: 2 of 3
Witness: HJMyers

Line	(a) Description	(b) Fiscal Year		(c)	
		No Rate Relief		Full Rate Relief	
		December 2021		December 2021	
	<u>EBIT Interest Coverage Ratio</u>				
16	Pre-Tax Operating Income	\$	587,363	\$	818,744.62
17	AFUDC		6,203		6,203
18	Total EBIT		593,565		824,947
19	Interest Charges, Net of Tax Benefit		184,564		184,564
20	EBIT Interest Coverage Ratio		3.22		4.47
	<u>EBITDA Interest Coverage Ratio</u>				
21	Total EBIT		593,565		824,947
22	Depreciation and Amortization		701,945		701,945
23	Total EBITDA		1,295,510		1,526,892
24	Interest Charges, Net of Tax Benefit		184,564		184,564
25	EBITDA Interest Coverage Ratio		7.02		8.27
	<u>Funds Flow from Operations (FFO) Interest Coverage Ratio</u>				
26	Net Operating Income	\$	532,954	\$	715,745
27	Depreciation and Amortization		701,945		701,945
28	Deferred Income Tax		124,791		124,791
29	AFUDC		6,203		6,203
30	Other Major Recurring Non-Cash Items		-		-
31	Interest Paid		196,495		196,495
	Less:				
32	Operating Lease Adjustment to Depreciation		-		-
33	Subtotal		1,562,386		1,745,178
34	Interest Charges, Net of Tax Benefit		184,564		184,564
35	FFO Interest Coverage Ratio		8.47		9.46
	<u>Overall Fixed Charge Coverage Ratio</u>				
36	Net Income Available for Common	\$	348,519	\$	531,311
37	Interest Charges, Net of Tax Benefit		184,564		184,564
38	Subtotal Numerator		533,084		715,875
39	Interest Charges		184,564		184,564
40	Preferred Stock Dividends		938		938
41	Subtotal Denominator		185,502		185,502
42	Overall Fixed Charge Coverage Ratio		2.87		3.86

Schedule A-2

Michigan Public Service Commission
Consumers Energy Company
Financial Metrics - Ratemaking Basis

Case No: U-20697
Exhibit No: A-11 (HJM-40)
Schedule: A-2
Page: 3 of 3
Witness: HJMyers

	(a)	(b)	(c)
		Fiscal Year	
Line	Description	No Rate Relief December 2021	Full Rate Relief December 2021
	<u>Cash Flow Coverage of Dividend Ratio</u>		
43	Net Income Available for Common	\$ 348,519	\$ 531,311
44	Depreciation and Amortization	701,945	701,945
45	Deferred Taxes	124,791	124,791
46	Subtotal	1,175,255	1,358,046
47	Common Dividends ²	278,815	425,049
48	Cash Flow Coverage of Dividends Ratio	4.22	3.20
	<u>Common Dividend Payout Ratio</u>		
49	Common Dividends ²	\$ 278,815	\$ 425,049
50	Net Income Available for Common	348,519	531,311
51	Common Dividend Payout Ratio	80%	80%
	<u>Permanent Capitalization</u>		
52	Long Term Debt (Excluding Securitization)	\$ 8,178,495	\$ 8,178,495
53	Preferred Stock	37,315	37,315
54	Common Equity	9,079,687	9,079,687
55	Total Permanent Capital	17,295,497	17,295,497
56	Long Term Debt	47.29%	47.29%
57	Preferred Stock	0.22%	0.22%
58	Common Equity	52.50%	52.50%
59	Total Permanent Capital	100.00%	100.00%

Footnotes

¹ The Company is requesting an ROE of 10.50% in this case

² Consumers Energy pays dividends to the parent as a total company. This number represents the Company's 80% dividend policy applied to Net Income Available for Common

Schedule A-3

Michigan Public Service Commission
Consumers Energy Company
Comparison of the Electric Revenue Requirement
between the Historical Year and the Test Year

Case No: U-20697
Exhibit No: A-11 (HJM-41)
Schedule A-3
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source		(d)	(e) Jurisdictional		(f)
		Historical Year	Test Year	Historical Year	Change		Test Year
1	Net Utility Plant	Exhibit: A-2 (HJM-3)	Exhibit: A-12 (HJM-42)	\$ 9,572,869	\$ 1,050,678	\$	10,623,548
2	Working Capital	Exhibit: A-2 (HJM-3)	Exhibit: A-12 (HJM-42)	638,083	583,126		1,221,209
3	Rate Base	Line 1 + Line 2	Line 1 + Line 2	10,210,952	1,633,805		11,844,757
4	Rate of Return	Exhibit: A-4 (HJM-34)	Exhibit: A-14 (MRB-1)	5.84%			6.09%
5	Income Requirement	Line 3 * Line 4	Line 3 * Line 4	595,900	125,739		721,639
6	Adjusted Net Operating Income	Exhibit: A-3 (HJM-9)	Exhibit: A-13 (HJM-49)	612,206	(73,050)		539,156
7	Income Deficiency/ (Sufficiency)	Line 5 - Line 6	Line 5 - Line 6	(16,306)	198,789		182,483
8	Revenue Multiplier	Exhibit: A-3 (HJM-10)	Exhibit: A-13 (HJM-50)	1.3391			1.3391
9	Revenue Deficiency/(Sufficiency)	Line 7 * Line 8	Line 7 * Line 8	\$ (21,835)	\$ 266,192	\$	244,357

Schedule B-1

Michigan Public Service Commission
 Consumers Energy Company
 Projected Rate Base
 Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
 Exhibit No: A-12 (HJM-42)
 Schedule B-1
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line No.	(a) Description	(b) Source	(c) Rate Base (\$000)	(d) Jurisdictional Amount (\$000)
1	Total Utility Plant	Exhibit: A-12 (HJM-44)	\$ 17,424,914	\$ 17,351,050
2	Depreciation Reserve	Exhibit: A-12 (HJM-45)	6,698,735	6,667,694
3	Net Utility Plant	Line 1 - Line 2	\$ 10,726,180	\$ 10,683,357
4	Retainers & Customer Advances	Exhibit: A-2 (HJM-5)	(59,839)	(59,809)
5	Adjusted Net Utility Plant	Sum Lines 3 - 4	10,666,341	\$ 10,623,548
6	Working Capital	Exhibit: A-12 (HJM-46)	1,227,083	\$ 1,221,209
7	Total Projected Period Rate Base	Line 5 + Line 6	\$ 11,893,424	\$ 11,844,757

Schedule B-1a

Michigan Public Service Commission
Consumers Energy Company
Development of Projected Rate Base
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-12 (HJM-43)
Schedule B-1a
Page: 1 of 2
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c)	(d)
			Total Amount (\$000)	Jurisdictional Amount (\$000)
1	Net Utility Plant	Exhibit: A-2 (HJM-3)	\$ 9,611,785	\$ 9,572,869
2	Working Capital	Exhibit: A-2 (HJM-3)	641,152	638,083
3	Rate Base - Historical Year	Line 1 + Line 2	10,252,937	10,210,952
<u>Adjustments To Rate Base</u>				
4	Change In Net Utility Plant	Sch. B-1a, Page 2	1,054,556	1,050,325
5	Change in Net Plant Due to Change in Jurisdictional Factor	Sch. B-1a, Page 2		353
6	Change in Working Capital From Moving to Sept 2019 Working Capital	Sch. B-1a, Page 2	120,529	119,952
7	Change in Working Capital Due to Change in Pension	Sch. B-1a, Page 2	286,754	285,381
8	Change in Working Capital Due to Change In OPEB	Sch. B-1a, Page 2	72,007	71,662
9	Change in Working Capital Due to Change in Accrued Tax	Sch. B-1a, Page 2	105,275	104,771
10	Change in Working Capital Due to Cloud Prepayment	Sch. B-1a, Page 2	913	909
11	Change in Working Capital Due to PowerMIDrive	Sch. B-1a, Page 2	453	451
12	Change in Working Capital Due to Change in Jurisdictional Factor	Sch. B-1a, Page 2	-	-
13	Net Change In Rate Base	Sum Lines 4 - 12	1,640,487	1,633,805
14	Total Projected Period Rate Base	Line 3 + Line 13	\$ 11,893,424	\$ 11,844,757

Schedule B-1a

Michigan Public Service Commission
Consumers Energy Company
Development of Projected Rate Base
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-12 (HJM-43)
Schedule B-1a
Page: 2 of 2
Witness: HJMyers
Date: February 2020

(a)	(b)	(c)
Line		Total
No.	Description	Source
		Amount
		(\$000)
<u>Adjustments for Utility Plant</u>		
1	Adjusted Gross Plant In Service	Exhibit: A-12 (HJM-44) \$ 17,064,931
2	Plant Held For Future Use	Exhibit: A-12 (HJM-44) 2,545
3	Construction Work in Progress	Exhibit: A-12 (HJM-44) 357,439
4	Depreciation Reserve	Exhibit: A-12 (HJM-45) (6,698,735)
5	Retained from Contractors	Exhibit: A-2 (HJM-5) -
6	Customer Advances for Construction	Exhibit: A-2 (HJM-5) (59,839)
7	Net Utility Plant - Test Year	Sum Lines 1 - 6 10,666,341
8	Net Utility Plant - Historical Year	Exhibit: A-2 (HJM-3) 9,611,785
9	Change in Net Utility Plant	Line 7 - Line 8 \$ 1,054,556
<u>Adjustments to Working Capital</u>		
10	Adjust Dec 2017 Working Capital to Feb 2018 Working Capital	Exhibit: A-12 (HJM-46) 120,529
11	Pension Adjustment	Exhibit: A-12 (HJM-47) 286,754
12	OPEB Adjustment	Exhibit: A-12 (HJM-47) 72,007
13	Change in Accrued Taxes	Exhibit: A-12 (HJM-47) 105,275
14	Cloud Prepayment	Exhibit: A-12 (HJM-47) 913
15	PowerMIDrive	Exhibit: A-12 (HJM-47) 453
16	Change in Working Capital	Sum Lines 10 - 15 \$ 585,931
<u>Adjustments in Rate Base Due to Changes in Jurisdictional Factors</u>		
Net Utility Plant:		
17	Test Year Net Utility Plant - Jurisdictional	Exhibit: A-12 (HJM-42) \$ 10,623,548
18	Test Year Net Utility Plant - Total Electric	Exhibit: A-12 (HJM-42) 10,666,341
19	Jurisdictional Factor	Line 17 / Line 18 0.995988
20	Historical Year Net Utility Plant - Jurisdictional	Exhibit: A-2 (HJM-3) 9,572,869
21	Historical Year Net Utility Plant - Jurisdictional, Using New Jurisdictional Factor	Exhibit: A-2 (HJM-3) * Line 19 9,573,223
22	Change in Net Utility Plant Due to Change in Jurisdictional Factor	Line 21 - Line 20 \$ 353
Working Capital:		
23	Test Year Working Capital - Jurisdictional	Exhibit: A-12 (HJM-42) \$ 1,221,209
24	Test Year Working Capital - Total Electric	Exhibit: A-12 (HJM-42) 1,227,083
25	Jurisdictional Factor	Line 23 / Line 24 0.995214
26	Historical Working Capital - Jurisdictional	Exhibit: A-2 (HJM-3) 638,083
27	Historical Working Capital - Jurisdictional, Using New Jurisdictional Factor	Exhibit: A-2 (HJM-3) Line 23 638,083
28	Change in Working Capital Due to Change in Jurisdictional Factor	Line 27 - Line 26 \$ -

Schedule B-2

Michigan Public Service Commission
Consumers Energy Company
Total Utility Plant
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-12 (HJM-44)
Schedule B-2
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total Amount (\$000)	(d) Jurisdictional Amount (\$000)
1	Plant In Service	WP-HJM-63	\$ 17,064,931	\$ 16,992,862
2	Plant Held For Future Use	WP-HJM-66	2,545	2,522
3	Construction Work In Progress	WP-HJM-65	357,439	355,666
4	Total Projected Period Utility Plant	Sum Lines 1 - 3	<u>\$ 17,424,914</u>	<u>\$ 17,351,050</u>

Schedule B-3

Michigan Public Service Commission
 Consumers Energy Company
 Depreciation Reserve and Other Deductions
 Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
 Exhibit No: A-12 (HJM-45)
 Schedule B-3
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line	(a)	(b)	(c)
No.	Description	Source	Projected Accumulated Provision for Depreciation (\$000)
1	Distribution	WP-HJM-64	\$ 3,130,436
2	Production (Hydro)	WP-HJM-64	405,182
3	Production (Steam)	WP-HJM-64	1,761,826
4	Production (Other)	WP-HJM-64	459,122
5	Intangibles	WP-HJM-64	614,523
6	General Plant	WP-HJM-64	124,100
7	Common Plant	WP-HJM-64	203,546
8	Total Projected Period Accumulated Provision for Depreciation	Sum Lines 1 - 7	<u>\$ 6,698,735</u>

Schedule B-4

Michigan Public Service Commission
Consumers Energy Company
Working Capital Summary
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-12 (HJM-46)
Schedule B-4
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) December-18	(c) Adjustments	(d) September-19	(e) Test Year Adjustments	(f) Test Year Working Capital
<u>Assets</u>						
1	Cash	\$ 38,293	\$ 9,623	\$ 47,915	\$ -	\$ 47,915
2	Accounts Receivable	310,098	(12,256)	297,842	-	297,842
3	Sale of Receivables	-	-	-	-	-
4	CE Receivable Funding	-	-	-	-	-
5	Materials & Supplies	97,880	4,134	102,014	-	102,014
6	Fuel Stock	57,252	(9,735)	47,517	-	47,517
7	Clean Air Allowance	14	-	14	-	14
8	Accrued Revenues	247,189	(10,134)	237,055	-	237,055
9	Sale of Accrued Revenues	-	-	-	-	-
10	Prepayments	304,534	(69,660)	234,874	913	235,788
11	Real & Personal Property Taxes	182,905	(3,306)	179,599	-	179,599
12	Deferred Debits	713,538	15,680	729,217	359,214	1,088,431
13	Total Assets	1,951,704	(75,655)	1,876,048	360,127	2,236,176
<u>Liabilities</u>						
14	Accounts Payable	423,205	(5,598)	417,607	-	417,607
15	Customer Deposits	16,556	(1,046)	15,511	-	15,511
16	Dividends Payable	22,614	6,724	29,338	-	29,338
17	Accrued Interest	44,248	(17)	44,231	-	44,231
18	Accrued Taxes	311,359	(54,623)	256,737	(105,275)	151,462
19	Other Current Liabilities	19,278	21,330	40,608	-	40,608
20	Deferred Credits and Operating Reserves	473,291	(162,955)	310,336	-	310,336
21	Total Liabilities	1,310,552	(196,184)	1,114,368	(105,275)	1,009,093
22	Total Projected Working Capital	\$ 641,152	\$ 120,529	\$ 761,680	\$ 465,402	\$ 1,227,083

Source

Column (b): Exhibit: A-2 (HJM-6)
Column (c): Column (d) - Column (b)
Column (d): Exhibit: A-12 (HJM-47)
Column (e): Column (f) - Column (d)
Column (f): Exhibit: A-12 (HJM-47)

Schedule B-4a

Michigan Public Service Commission
Consumers Energy Company
Working Capital
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-12 (HJM-47)
Schedule: B-4a
Page: 1 of 1
Witness: HJMyers
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line No.	Description	13-Month Average Sep-2019	Pension Adjustment	OPEB Adjustment	Accrued Tax Adjustment	Cloud Prepayment	PowerMiDrive Adjustment	Total Adjustments	Test Year Working Capital
1	Cash	\$ 47,915	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 47,915
2	Accounts Receivable	297,842	-	-	-	-	-	-	297,842
3	Sale of Receivables	-	-	-	-	-	-	-	-
4	CEReivable Funding	-	-	-	-	-	-	-	-
5	Materials & Supplies	102,014	-	-	-	-	-	-	102,014
6	Fuel Stock	47,517	-	-	-	-	-	-	47,517
7	Clean Air Allowances	14	-	-	-	-	-	-	14
8	Accrued Revenues	237,055	-	-	-	-	-	-	237,055
9	Sale of Accrued Revenues	-	-	-	-	-	-	-	-
10	Prepayments	234,874	-	-	-	913	-	913	235,788
11	Real & Personal Property Tax	179,599	-	-	-	-	-	-	179,599
12	Deferred Debits	729,217	286,754	72,007	-	-	453	359,214	1,088,431
13	TOTAL ASSETS	1,876,048	286,754	72,007	-	913	453	360,127	2,236,176
14	Accounts Payable	417,607	-	-	-	-	-	-	417,607
15	Customer Deposits	15,511	-	-	-	-	-	-	15,511
16	Dividends Payable	29,338	-	-	-	-	-	-	29,338
17	Accrued Interest	44,231	-	-	-	-	-	-	44,231
18	Accrued Taxes	256,737	-	-	(105,275)	-	-	(105,275)	151,462
19	Other Liabilities	40,608	-	-	-	-	-	-	40,608
20	Deferred Credits	310,336	-	-	-	-	-	-	310,336
21	TOTAL LIABILITIES	1,114,368	-	-	(105,275)	-	-	(105,275)	1,009,093
22	Total Projected Working Capital	\$ 761,680	\$ 286,754	\$ 72,007	\$ 105,275	\$ 913	\$ 453	\$ 465,402	\$ 1,227,083

Source: WP-HJM-34

Schedule B-5

Michigan Public Service Commission
Consumers Energy Company
Projected Capital Expenditures
Summary
(\$000)

Case No: U-20697
Exhibit No: A-12 (HJM-48)
Schedule: B-5
Page: 1 of 1
Witness: HJMyers
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
		Capital Expenditures						Last Rate Case	
		Historical						Approved	Actual & Projected
		12 mos ended	12 mos. Ending	12 mos. Ending	24 mos. Ending	Test Year Ending		Spending Plan	Spending Test Year U-
		12/31/2018	12/31/2019	12/31/2020	12/31/2019	12/31/2021	Reference	U-20134	20134
Line	Description							12 mos. Ended	12 mos. Ended
No.								12/31/2019 ²	12/31/2019
					<i>col. (d)+(e)</i>				
1	Electric Distribution	533,167	628,865	552,142	1,181,007	722,675	Exh A-12 (RTB-1)		628,865
2	Generation	175,704	174,044	119,646	293,690	161,075	Exh A-12 (SAH-3)		174,044
3	Information Technology	74,245	56,370	55,586	111,956	73,758	Exh A-12 (JDT-3)		56,370
4	Electric Business Services	44,214	49,414	54,996	104,410	127,605	Footnote 1		49,414
5	Corporate	313	443	457	900	472	Exh A-12 (KMG-6)		443
6	Customer Experience	-	-	9,920	9,920	3,068	Exh A-12 (SQM-1)		-
7	Demand Response	6,675	17,137	10,856	27,993	8,868	Exh A-12 (SQM-1)		17,137
8	Total Capital Expenditures	<u>834,318</u>	<u>926,273</u>	<u>803,603</u>	<u>1,729,876</u>	<u>1,097,521</u>		<u>-</u>	<u>926,273</u>

Footnotes

¹ Electric Business Services by Witness

Operations Support	26,247	20,740	21,774	42,514	64,856	Exh A-12 (LDS-1)
Fleet	17,967	28,674	33,222	61,896	62,749	Exh. A-12 (KPJ-1)
	<u>44,214</u>	<u>49,414</u>	<u>54,996</u>	<u>104,410</u>	<u>127,605</u>	

² Case Number U-20134 resulted in a settlement agreement that did not state approved capital spending.

Schedule C-1

Michigan Public Service Commission
Consumers Energy Company
Adjusted Net Operating Income
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-49)
Schedule C-1
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Net Operating Income (\$000)	(d) Jurisdictional Amount (\$000)
<u>Operating Revenue</u>				
1	Sales Revenue	Exhibit: A-13 (HJM-51)	\$ 4,252,458	\$ 4,252,458
2	Wholesale Revenue	Exhibit: A-13 (HJM-51)	24,648	-
3	Other Electric Revenue	Exhibit: A-13 (HJM-51)	120,040	119,117
4	Total Operating Revenue	Sum Lines 1 - 3	\$ 4,397,146	\$ 4,371,575
<u>Operating Expense</u>				
5	Power Supply Costs	Exhibit: A-13 (HJM-52)	2,160,757	2,137,371
6	Other O&M Expense	Exhibit: A-13 (HJM-53)	684,695	681,800
7	Depreciation and Amortization	Exhibit: A-13 (HJM-54)	705,582	701,945
8	R&PP Tax	Exhibit: A-13 (HJM-55)	198,800	197,995
9	Other General Taxes	Exhibit: A-13 (HJM-55)	33,305	33,087
10	Other (or Local) Taxes ¹	Exhibit: A-13 (HJM-58)	1,077	1,070
11	State Income Tax ¹	Exhibit: A-13 (HJM-57)	30,672	30,946
12	Federal Income Tax ¹	Exhibit: A-13 (HJM-56)	53,928	54,409
13	Total Operating Expense	Sum Lines 5 - 12	3,868,814	3,838,621
14	Net Operating Income	Line 4 - Line 13	528,332	532,954
15	AFUDC	Exhibit: A-13 (HJM-59)	6,234	6,203
16	Subtotal	Line 14 + Line 15	534,565	539,156
<u>Operating Income Adjustments</u>				
17	Income Tax Effect of Interest ¹		Included in Lines 11 & 12	
18	Interest Synchronization Adjustment ¹		Included in Lines 11 & 12	
19	Total Net Operating Income Adjustments	Line 17 + Line 18	-	-
20	Adjusted Net Operating Income	Line 16 + Line 19	\$ 534,565	\$ 539,156

Footnotes

¹ Income Tax Effect of Interest and Interest Synchronization Adjustments are included in the calculation of CIT, MCIT and FIT. The separate calculations can be found on Exhibit: A-13 (HJM-61) & A-13 (HJM-62)

Schedule C-2

Michigan Public Service Commission
Consumers Energy Company
Projected Revenue Conversion Factor
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-50)
Schedule: C-2
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line	(a)	(b)	(c)
No.	Description	Source	Amount
1	Income Base - Before Taxes		100.0000
2	Michigan Corporate Income Tax	Line 1 * 5.310% MCIT Rate	5.3100
3	City Income Tax	Line 1 * 0.16% CIT Rate	<u>0.1600</u>
4	Federal Income Tax Base	Line 1 - Line 2 - Line 3	94.5300
5	Federal Income Tax	Line 4 * 21.00% FIT Rate	<u>19.8513</u>
6	Income Base - After Taxes	Line 4 - Line 5	<u>74.6787</u>
7	Revenue Multiplier	Line 1 / Line 6	<u><u>1.3391</u></u>

Schedule C-3

Michigan Public Service Commission
Consumers Energy Company
Projected Operating Revenue
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-51)
Schedule C-3
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)		(b)	(c)
Line			Projected
No.	Description	Source	Sales
			Revenue
			(\$000)
1	Sales Revenue	Exhibit: A-15 (EMB-3)	\$ 4,252,458
2	Wholesale Revenues	Exhibit: A-15 (EMB-3)	<u>24,648</u>
3	Subtotal	Line 1 + Line 2	4,277,106
4	Other Electric Revenue	Exhibit: A-15 (EMB-3)	<u>120,040</u>
5	Total Projected Period Operating Revenue	Line 3 + Line 4	<u><u>\$ 4,397,146</u></u>

Schedule C-4

Michigan Public Service Commission
Consumers Energy Company
Projected Fuel and Purchased Power/Cost of Gas Sold
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-52)
Schedule C-4
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line	(a)	(b)	(c)
No.	Description	Source	Projected Fuel and Purchased Power (\$000)
1	Total Projected Period Power Supply Costs	Exhibit: A-15 (EMB-6)	\$ <u>2,160,757</u>

Schedule C-5

Michigan Public Service Commission
Consumers Energy Company
Projected Operating and Maintenance Expenses
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-53)
Schedule C-5
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)		(b)	(c)	(d)	(e)
			Historical Operation and Maintenance Expenses (\$000)	Change in Operation and Maintenance Expenses (\$000)	Projected Operation and Maintenance Expenses (\$000)
Line No.	Description	Source			
1	Electric Distribution	Exhibit: A-36 (RTB-9)	\$ 140,650	\$ 30,094	\$ 170,744
2	Line Clearing	Exhibit: A-98 (CAS-1)	\$ 51,948	\$ 32,052	84,000
3	Fossil & Hydro Generation	Exhibit: A-70 (SAH-5)	148,231	18,562	166,793
4	Operations Support	Exhibit: A-93 (LDS-2)	17,888	(823)	17,065
5	Information Technology - Operations	Exhibit: A-104 (JDT-1)	45,905	3,382	49,287
6	Information Technology - Investment	Exhibit: A-105 (JDT-2)	16,103	5,781	21,884
7	Pension	Exhibit: A-51 (LBC-1)	26,717	(31,111)	(4,394)
8	Defined Company Contribution Plan	Exhibit: A-51 (LBC-1)	7,628	3,574	11,202
9	401 (K) Savings Plan	Exhibit: A-51 (LBC-1)	7,983	1,206	9,189
10	Active Health Care/ Insurance/ LTD	Exhibit: A-51 (LBC-1)	24,194	3,279	27,473
11	Retiree Health Care and Life Insurance	Exhibit: A-51 (LBC-1)	(56,385)	2,746	(53,639)
12	Other Benefits	Exhibit: A-51 (LBC-1)	826	625	1,451
13	Corporate	Exhibit: A-61 (KMG-1)	45,482	11,305	56,787
14	Uncollectibles	Exhibit: A-61 (KMG-1)	16,661	1,416	18,077
15	Injuries & Damages	Exhibit: A-61 (KMG-1)	3,975	556	4,531
16	Demand Response Other O&M Expense	Exhibit: A-75 (SQM-2)	6,914	27,768	34,682
17	Billing & Payments	Exhibit: A-75 (SQM-2)	19,257	3,549	22,806
18	Customer Experience	Exhibit: A-75 (SQM-2)	34,700	(3,488)	31,212
19	Jobwork Expense	Exhibit: A-15 (EMB-3)	10,387	-	10,387
20	Incentive Compensation	Exhibit: A-57 (AMC-3)	4,976	181	5,157
21	Total Projected Period Other O&M Expense	Sum Lines 1 - 20	\$ 574,040	\$ 110,654	\$ 684,695

Schedule C-6

Michigan Public Service Commission
Consumers Energy Company
Projected Depreciation and Amortization Expenses
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-54)
Schedule C-6
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)		(b)	(c)	
			Projected Depreciation & Amortization Expense (\$000)	
Line No.	Description	Source		
1	Distribution	WP-HJM-66	\$	283,138
2	Production (Hydro)	WP-HJM-66		58,105
3	Production (Steam)	WP-HJM-66		219,302
4	Production (Other)	WP-HJM-66		31,870
5	Intangibles	WP-HJM-66		81,171
6	General Plant	WP-HJM-66		9,750
7	Common Plant	WP-HJM-66		22,116
8	PowerMiDrive	Exh. A-84 (HJM-65)		129
9	Total Projected Period Depreciation and Amortization Expense		\$	<u>705,582</u>

Schedule C-7

Michigan Public Service Commission
Consumers Energy Company
Projected General Taxes
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-55)
Schedule C-7
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Projected General Taxes (\$000)
1	Projected R&PP Tax	Exhibit: A-112 (BJV-1)	\$ 198,800
2	Projected Payroll Taxes	WP-HJM-85	21,973
3	Projected Other General Taxes	WP-HJM-85	<u>11,332</u>
4	Total Projected Period General Taxes	Sum Lines 1 - 3	<u><u>\$ 232,105</u></u>

Schedule C-8

Michigan Public Service Commission
Consumers Energy Company
Projected Federal Income Taxes
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-56)
Schedule C-8
Page: 1 of 3
Witness: HJMyers
Date: February 2020

(a)	(b)	(c)
		Projected Federal Income Taxes (\$000)
Line	Description	Source
1	Operating Revenue	Exhibit: A-13 (HJM-51) \$ 4,397,146
2	Power Supply Costs	Exhibit: A-13 (HJM-52) (2,160,757)
3	Other O&M Expense	Exhibit: A-13 (HJM-53) (684,695)
4	Depreciation & Amortization	Exhibit: A-13 (HJM-54) (705,582)
5	General Taxes	Exhibit: A-13 (HJM-55) (232,105)
6	Other Taxes Based on Income	(Exhibit: A-13 (HJM-57), Line 16 + Exhibit: A-13 (HJM-58), Line 19) *-1) (31,749)
7	Net Operating Income	Sum Lines 1 - 6 582,260
8	Allowable Interest Expense	Exhibit: A-13 (HJM-60) (184,015)
9	JDITC Interest Expense	Exhibit: A-13 (HJM-61) (1,307)
10	Pre-Tax Operating Income	Sum Lines 7 - 9 396,937
<u>Permanent Differences</u>		
11	Meals and Entertainment	WP-HJM-72 988
12	Non-deductible Parking	WP-HJM-72 103
13	Total Permanent Differences	Sum Lines 11 - 12 1,091

Schedule C-8

Michigan Public Service Commission
Consumers Energy Company
Projected Federal Income Taxes
Projected 12 Month Period Ending December 31, 2019

Case No: U-20134
Exhibit No: A-13 (HJM-56)
Schedule C-8
Page: 2 of 3
Witness: HJMyers
Date: May 2018

(a)	(b)	(c)
Line	Description	Source
		Projected Federal Income Taxes (\$000)
	<u>Temporary Differences</u>	
14	Book Depreciation	WP-HJM-72 705,582
15	Tax Depreciation	WP-HJM-72 (579,080)
16	263A - Mixed Service Costs Deduction	WP-HJM-72 (43,800)
17	Tax Repairs	WP-HJM-72 (674,000)
18	Contributions In Aid of Construction	WP-HJM-72 24,800
19	Cost of Removal	WP-HJM-72 (130,379)
20	Gain/Loss on ACRS/MACRS Dispositions	WP-HJM-72 (10,303)
21	Software Expense	WP-HJM-72 (15,838)
22	Tax Interest During Construction	WP-HJM-72 17,566
23	Bad Debt Allowance	WP-HJM-72 -
24	OPEB - Book Expense	WP-HJM-72 (53,639)
25	OPEB - Payments	WP-HJM-72 -
26	Other Revenue Reserves	WP-HJM-72 -
27	Pension - Book Expense	WP-HJM-72 (4,394)
28	Pension - Payments	WP-HJM-72 -
29	Power Supply Cost Recovery	WP-HJM-72 -
30	Premium, Discount & Debt Expense Amortization	WP-HJM-72 3,196
31	Real and Personal Property Taxes	WP-HJM-72 (14,705)
32	Renewable Energy Reserve	WP-HJM-72 (4,310)
33	Deferred State Income Taxes	WP-HJM-72 76,224
34	Deferred City Income Taxes	WP-HJM-72 1,563
35	Total Temporary Differences	Sum Lines 14 - 32 (701,518)
36	Total Permanent & Temporary Differences	Line 13 + Line 35 (700,427)
37	Taxable Income	Line 10 + Line 36 (303,489)
38	FIT Rate	Tax Department 21%
39	Current Federal Income Tax	Line 37 * Line 38 (63,733)
	<u>Adjustments to Current Federal Income Tax</u>	
40	Research & Development Credit	WP-HJM-72 (1,615)
41	Total Current Federal Income Tax	Line 39 + Line 40 (65,348)
42	Deferred Federal Income Tax	Line 35 * Line 38 147,319

Schedule C-8

Michigan Public Service Commission
Consumers Energy Company
Projected Federal Income Taxes
Projected 12 Month Period Ending December 31, 2019

Case No: U-20134
Exhibit No: A-13 (HJM-56)
Schedule: C-8
Page: 3 of 3
Witness: HJMyers
Date: May 2018

(a)	(b)	(c)
Line	Description	Source
		Projected Federal Income Taxes (\$000)
43	<u>Adjustments to Deferred Federal Income Tax</u>	
44	FAS 109	WP-HJM-72 (145)
45	Investment Tax Credit Amortization	WP-HJM-72 (4,948)
46	TCJA Amortization - ARAM	WP-HJM-72 (22,510)
47	TCJA Amortization - NonARAM	WP-HJM-72 (4,148)
48	Reg Tax Asset Amort-Transmission Assets U-13224	WP-HJM-72 1,059
49	Repealed Medicare Subsidy Benefit	WP-HJM-72 2,649
50	State Deferred Amortization Offset Adjustment	WP-HJM-72 -
51	Total Deferred Federal Income Tax Expense	Sum Lines 42 - 50 119,276
52	Total Projected Federal Income Tax Expense	Line 41 + Line 51 53,928
53	Jurisdictional Factor	WP-HJM-83 1.008924
54	Total Projected Period Jurisdictional Federal Income Tax Expense	Line 52 * Line 53 \$ 54,409

Schedule C-9

Michigan Public Service Commission
Consumers Energy Company
Projected State Income Taxes
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-57)
Schedule C-9
Page: 1 of 1
Witness: HJMyers
Date: February 2020

(a)	(b)	(c)
Line	Description	Source
		Projected State Income Taxes (\$000)
1	Pre-Tax Operating Income	Exhibit: A-13 (HJM-56) \$ 396,937
2	Schedule M Permanent Differences	Exhibit: A-13 (HJM-56) 1,091
3	Schedule M Temporary Differences	Exhibit: A-13 (HJM-56) <u>(701,518)</u>
4	Federal Taxable Income	Sum Lines 1 - 3 (303,489)
5	Depreciation Adjustment	WP-HJM-73 (508,322)
6	Other Taxes Based on Income	Exhibit: A-13 (HJM-56) Line 6 - Line 34 - Line 35 <u>(46,038)</u>
7	State Taxable Income	Sum Lines 4 - 6 (857,850)
8	MCIT Rate	Tax Department <u>5.310%</u>
9	Current MCIT Expense	Line 7 * Line 8 (45,552)
10	Deferred MCIT Expense	WP-HJM-73 68,373
11	FAS 109 - AFUDC-Equity Flow-Through Reversal	WP-HJM-73 142
12	Deferred - Amortization of Regulatory Asset Per U-15986	WP-HJM-73 -
13	Deferred - Amortization of Medicare Subsidy Benefit - State	WP-HJM-73 454
14	Deferred - Amortization of MCIT Transition Regulatory Asset	WP-HJM-73 <u>7,255</u>
15	Total Deferred MCIT Expense	Sum Lines 10 - 14 76,224
16	Total Projected State Income Tax	Line 9 + Line 15 <u>30,672</u>
17	Jurisdictional Factor	WP-HJM-83 1.008924
18	Total Projected Period Jurisdictional State Income Tax	Line 16 * Line 17 <u>\$ 30,946</u>

Schedule C-10

Michigan Public Service Commission
Consumers Energy Company
Projected Other (or Local) Taxes
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-58)
Schedule C-10
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Projected Other (or Local) Taxes (\$000)
1	Operating Revenue	Exhibit: A-13 (HJM-49)	\$ 4,397,146
2	Power Supply Costs	Exhibit: A-13 (HJM-49)	(2,160,757)
3	Other O&M Expense	Exhibit: A-13 (HJM-49)	(684,695)
4	Depreciation and Amortization Expense	Exhibit: A-13 (HJM-49)	(705,582)
5	State Income Taxes	Exhibit: A-13 (HJM-49)	(30,672)
6	General Taxes	Exhibit: A-13 (HJM-49)	(232,105)
7	Net Operating Income	Sum Lines 1 - 6	583,336
8	Interest Expense	Exhibit: A-13 (HJM-60)	(184,015)
9	Interest Synchronization	Exhibit: A-13 (HJM-61)	(1,307)
10	Pre-Tax Income Excluding Other Taxes Based on Income	Sum Lines 7 - 9	398,014
11	Schedule M Permanent Differences	WP-HJM-74	1,091
12	Schedule M Temporary Differences Excluding DCIT	WP-HJM-74	(703,080)
13	City Taxable Income	Sum Lines 10 - 12	(303,976)
14	City Income Tax (CIT Rate)	Tax Department	0.16%
15	Current CIT Expense	Line 13 * Line 14	(486)
16	Deferred CIT Expense	(Line 12) * 0.16% CIT Rate	1,125
17	Deferred - Amortization of Regulatory Asset	WP-HJM-74	438
18	Total Deferred CIT Expense	Line 16 + Line 17	1,563
19	Total Projected CIT Expense	Line 15 + Line 18	1,077
20	Jurisdictional Factor	WP-HJM-83	0.993462
21	Total Projected Period Jurisdictional CIT Expense	Line 19 * Line 20	\$ 1,070

Schedule C-11

Michigan Public Service Commission
Consumers Energy Company
Projected Allowance for Funds Used During Construction
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-59)
Schedule C-11
Page: 1 of 1
Witness: HJMyers
Date: February 2020

	(a)	(b)	(c)
			Projected AFUDC (\$000)
Line	Description	Source	
1	Projected Period Allowance for Funds Used During Construction	WP-HJM-66	\$ 6,234

Schedule C-12

Michigan Public Service Commission
Consumers Energy Company
Tax Effect of Pro-Forma Interest Adjustment
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-60)
Schedule C-12
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total Amount (\$000)
1	Rate Base	Exhibit: A-12 (HJM-43)	\$ 11,893,424
2	Weighted Cost of Debt ¹	Exhibit: A-14 (MRB-1)	1.55%
3	Allowable Interest Expense	Line 1 * Line 2	184,015
4	Historical Year Pro-Forma Interest Expense	Exhibit: A-3 (HJM-32)	160,450
5	Increase/ (Decrease) in Allowable Interest Deduction	Line 3 - Line 4	23,565
6	Impact on Taxable Income	Line 5 * -1	(23,565)
7	CIT Rate	Tax Department	0.16%
8	Impact on City Income Tax	Line 6 * Line 7	(38)
9	Impact on State Taxable Income	Line 6 - Line 8	\$ (23,527)
10	MCIT Rate	Tax Department	5.310%
11	Impact on State Income Tax	Line 9 * Line 10	(1,249)
12	Impact on Federal Taxable Income	Line 9 - Line 11	(22,278)
13	FIT Rate	Tax Department	21%
14	Impact on Federal Income Tax	Line 12 * Line 13	(4,678)
15	Impact on Net Operating Income	(Line 8 + Line 11 + Line 14) * -1	\$ 5,965

Footnotes

¹ Excludes the Job Development Investment Tax Credit portion

Schedule C-13

Michigan Public Service Commission
Consumers Energy Company
Tax Effect of Interest Synchronization Adjustment
Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
Exhibit No: A-13 (HJM-61)
Schedule C-13
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line No.	(a) Description	(b) Source	(c) Total Amount (\$000)
1	Rate Base	Exhibit: A-12 (HJM-43)	\$ 11,893,424
2	Debt Related JDITC ¹ Portion of the Capital Structure	Exhibit: A-14 (MRB-1)	0.28%
3	Portion of Rate Base Funded by JDITC	Line 1 * Line 2	33,055
4	Cost of Debt - JDITC Portion	Exhibit: A-14 (MRB-1)	3.95%
5	JDITC Interest Expense	Line 3 * Line 4	1,307
6	Historical Year JDITC Interest Expense	Exhibit: A-3 (HJM-33)	1,177
7	Increase/ (Decrease) in Allowable JDITC Interest Expense	Line 5 - Line 6	130
8	Impact on Taxable Income	Line 7 * -1	(130)
9	CIT Rate	Tax Department	0.16%
10	Impact on Local Income Tax	Line 8 * Line 9	(0)
11	Impact on State Taxable Income	Line 8 - Line 10	(130)
12	MCIT Rate	Tax Department	5.310%
13	Impact on State Income Tax	Line 11 * Line 12	(7)
14	Impact on Federal Taxable Income	Line 11 - Line 13	(123)
15	FIT Rate	Tax Department	21%
16	Impact on Federal Income Tax	Line 14 * Line 15	(26)
17	Synchronization Adjustment to Net Operating Income	(Line 10 + Line 13 + Line 16) * -1	\$ 33

Footnotes

¹ Job Development Investment Tax Credit

Schedule C-14

Michigan Public Service Commission
 Consumers Energy Company
 Development of Net Operating Income
 Projected 12 Month Period Ending December 31, 2021

Case No: U-20697
 Exhibit No: A-13 (HJM-62)
 Schedule C-14
 Page: 1 of 1
 Witness: HJMyers
 Date: February 2020

Line	(a) Description	Revenue				Expense								(n) NOI	(o) AFUDC	(p) Adjusted NOI
		(b) Sales	(c) Wholesale	(d) Other Revenue	(e) Total	(f) Fuel P&I	(g) Other O&M	(h) Deprec.	(i) Property Tax	(j) Other Taxes	(k) CIT	(l) MCIT	(m) FIT			
1	Historical Year Net Operating Income	\$ 4,254,127	\$ 140,140	\$ 111,892	\$ 4,506,160	\$ 2,093,508	\$ 715,517	\$ 668,759	\$ 158,224	\$ 32,254	\$ 1,170	\$ 46,992	\$ 58,114	\$ 731,622	\$ 5,095	\$ 736,717
	<u>Historical Year Adjustments</u>															
2	Compensation Disallowances	-	-	-	-	-	5,286	-	-	-	(8)	(281)	(1,049)	(3,947)	-	(3,947)
3	Dues and Donations Disallowances	-	-	-	-	-	(613)	-	-	-	1	33	122	458	-	458
4	Corporate Giving & Communications Disallowances	-	-	-	-	-	(191)	-	-	-	0	10	38	142	-	142
5	Weather Normalization	(79,984)	-	-	(79,984)	-	-	-	-	-	(128)	(4,247)	(15,878)	(59,731)	-	(59,731)
6	PSCR Adjustments	(106)	-	-	(106)	-	-	-	-	-	(0)	(6)	(21)	(79)	-	(79)
7	Customer Deposit Interest Expense	-	-	-	-	-	434	-	-	-	(1)	(23)	(86)	(324)	-	(324)
8	Jobwork Revenue	-	-	12,822	12,822	-	-	-	-	-	21	681	2,545	9,575	-	9,575
9	Jobwork Expense	-	-	-	-	-	10,387	-	-	-	(17)	(552)	(2,062)	(7,757)	-	(7,757)
10	Transmission Depreciation Expense	-	-	-	-	-	(1,759)	-	-	-	3	93	349	1,313	-	1,313
11	Property Tax Refund	-	-	-	-	-	-	-	8,900	-	(14)	(473)	(1,767)	(6,646)	-	(6,646)
12	Surcharge Revenue, Expense, and Amortization	(242,926)	-	-	(242,926)	-	(164,092)	(25,398)	-	-	(85)	(2,837)	(10,608)	(39,906)	-	(39,906)
13	Tax Benefit of Pro-Forma Interest	-	-	-	-	-	-	-	-	-	81	2,668	9,993	(12,742)	-	(12,742)
14	Interest Synchronization	-	-	-	-	-	-	-	-	-	(2)	(62)	234	(169)	-	(169)
15	Total Historical Year Adjustments	(323,016)	-	12,822	(310,195)	-	(150,548)	(25,398)	8,900	-	(150)	(4,995)	(18,190)	(119,813)	-	(119,813)
16	Adjusted Historical Year Net Operating Income	3,931,110	140,140	124,714	4,195,965	2,093,508	564,969	643,361	167,124	32,254	1,020	41,997	39,924	611,809	5,095	616,904
	<u>Test Year Adjustments</u>															
17	Sales Revenue	321,348	-	-	321,348	-	-	-	-	-	514	17,064	63,792	239,978	-	239,978
18	Wholesale Revenues	-	(115,492)	-	(115,492)	-	-	-	-	-	(185)	(6,133)	(22,927)	(86,248)	-	(86,248)
19	Other Electric Revenue	-	-	(4,674)	(4,674)	-	-	-	-	-	(7)	(248)	(928)	(3,491)	-	(3,491)
20	Fuel and P&I	-	-	-	-	67,249	-	-	-	-	(108)	(3,571)	(13,350)	(50,221)	-	(50,221)
21	Other Operation & Maintenance	-	-	-	-	-	119,726	-	-	-	(192)	(6,357)	(23,767)	(89,409)	-	(89,409)
22	Depreciation & Amortization	-	-	-	-	-	-	62,221	-	-	(100)	(3,304)	(12,352)	(46,466)	-	(46,466)
23	Real and Personal Property Tax	-	-	-	-	-	-	-	31,676	-	(51)	(1,682)	(6,288)	(23,655)	-	(23,655)
24	Other General Taxes	-	-	-	-	-	-	-	-	1,051	(2)	(56)	(209)	(785)	-	(785)
25	Other (or Local) Taxes	-	-	-	-	-	-	-	-	-	186	-	(39)	(147)	-	(147)
26	State Income Tax Adjustments	-	-	-	-	-	-	-	-	-	-	(7,038)	1,478	5,560	-	5,560
27	AFUDC	-	-	-	-	-	-	-	-	-	-	-	-	-	1,139	1,139
28	FIT Adjustments	-	-	-	-	-	-	-	-	-	-	-	28,594	(28,594)	-	(28,594)
29	Total Projected Test Year Adjustments	321,348	(115,492)	(4,674)	201,181	67,249	119,726	62,221	31,676	1,051	57	(11,325)	14,004	(83,477)	1,139	(82,339)
30	Projected Test Year Net Operating Income	4,252,458	24,648	120,040	4,397,146	2,160,757	684,695	705,582	198,800	33,305	1,077	30,672	53,928	528,332	6,234	534,565
31	Jurisdictional Factor	1.000000	-	0.992311	0.994185	0.989177	0.995772	0.994845	0.995950	0.993462	0.993462	1.008924	1.008924	1.008748	0.995039	
32	Jurisdictional Amount - Test Year	\$ 4,252,458	\$ -	\$ 119,117	\$ 4,371,575	\$ 2,137,371	\$ 681,800	\$ 701,945	\$ 197,995	\$ 33,087	\$ 1,070	\$ 30,946	\$ 54,409	\$ 532,954	\$ 6,203	\$ 539,156

Source

Lines 1 through 14 Exhibit: A-3 (HJM-9) through A-3 (HJM-33)

Lines 17 through 28 Exhibit: A-13 (HJM-49) through A-13 (HJM-61)

Consumers Energy CompanyDemand Response Revenue Requirement
(000)Case No.: U-20697
Exhibit No.: A-82 (HJM-63)
Page: 1 of 2
Witness: HJMyers
Date: February 2020**Residential Demand Response**

Line	Description	2019	2020	2021	Total	
(a)	(b)	(c)	(d)	(e)	(f)	
1	Total Capital Spending	<u>\$ 15,491</u>	<u>\$ 9,560</u>	<u>\$ 8,151</u>	<u>\$ 33,202</u>	Exhibit A-12 (SQM-1)
<u>Revenue Requirement Calculation:</u>						
					2021 Average	
2	Beginning Plant	\$ 12,558	\$ 28,049	\$ 37,609		
3	Closings	15,491	9,560	8,151		
4	Ending Plant	<u>\$ 28,049</u>	<u>\$ 37,609</u>	<u>\$ 45,760</u>	<u>\$ 41,684</u>	
5	Beginning CWIP	\$ -	\$ -	\$ -		
6	Capital Spending	15,491	9,560	8,151		
7	Closings	15,491	9,560	8,151		
8	Ending CWIP	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	<u>\$ -</u>	
9	Beginning Depreciation Reserve	\$ 517	\$ 2,017	\$ 4,780		
10	Depreciation Expense	1,500	2,763	3,196		WP-HJM-87
11	End Depreciation Reserve	<u>\$ 2,017</u>	<u>\$ 4,780</u>	<u>\$ 7,976</u>	<u>\$ 6,378</u>	
12	Total Rate Base				<u>\$ 35,306</u>	Line 4 + Line 8 - Line 11
13	Pre-Tax Rate of Return				<u>7.63%</u>	Exhibit A-14 (MRB-1)
14	Return on Investment				\$2,694	Line 12 * Line 13
15	Depreciation Expense				3,196	Line 11
16	O&M				18,933	Exhibit A-75 (SQM-2)
17	R&PP Tax				<u>494</u>	Footnote ⁽¹⁾
18	Total Residential Revenue Requirement				<u>\$25,317</u>	

Footnotes(1) Average Plant Balance Dec 31, 2021
Property Tax Rate

\$	41,684
	0.011851675
<u>\$</u>	<u>494</u>

See Exh A-112 (BJV-1)

Consumers Energy CompanyDemand Response Revenue Requirement
(000)Case No.: U-20697
Exhibit No.: A-82 (HJM-63)
Page: 2 of 2
Witness: HJMyers
Date: February 2020**Commercial and Industrial Demand Response**

Line	Description	2019	2020	2021	Total	
(a)	(b)	(c)	(d)	(e)	(f)	
1	Total Capital Spending	<u>\$ 1,646</u>	<u>\$ 1,296</u>	<u>\$ 717</u>	<u>\$ 3,659</u>	Exhibit A-12 (SQM-1)
<u>Revenue Requirement Calculation:</u>						
					2021 Average	
2	Beginning Plant	\$ 1,617	\$ 3,583	\$ 4,879		
3	Closings	<u>1,966</u>	<u>1,296</u>	<u>717</u>		
4	Ending Plant	\$ 3,583	\$ 4,879	\$ 5,596	<u>\$ 5,237</u>	
5	Beginning CWIP	\$ 320	\$ 0	\$ 0		
6	Capital Spending	<u>1,646</u>	<u>1,296</u>	<u>717</u>		
7	Closings	<u>1,966</u>	<u>1,296</u>	<u>717</u>		
8	Ending CWIP	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 0</u>	<u>\$ 0</u>	
9	Beginning Depreciation Reserve	\$ 126	\$ 420	\$ 845		
10	Depreciation Expense	<u>293</u>	<u>425</u>	<u>456</u>		WP-HJM-87
11	End Depreciation Reserve	\$ 420	\$ 845	\$ 1,301	<u>\$ 1,073</u>	
12	Total Rate Base				<u>\$ 4,164</u>	Line 4 + Line 8 - Line 11
13	Pre-Tax Rate of Return				<u>7.63%</u>	Exhibit A-14 (MRB-1)
14	Return on Investment				\$318	Line 12 * Line 13
15	Depreciation Expense				456	Line 10
16	O&M				15,749	Exhibit 75 (SQM-2)
17	R&PP Tax				<u>62</u>	Footnote ⁽¹⁾
18	Total C&I Revenue Requirement				<u>\$16,584</u>	

Footnotes(1) Average Plant Balance Dec 31, 2021
Property Tax Rate

\$	5,237
	<u>0.011851675</u>
\$	<u>62</u>

See Exh A-112 (BJV-1)

Consumers Energy CompanyKarn Retention Revenue Requirement Comparison
(000)

Case No.: U-20697

Exhibit No.: A-83 (HJM-64)

Page: 1 of 1

Witness: HJMyers

Date: February 2020

Line	Description	Amount	Revenue Requirement	Source
(a)	(b)	(c)	(d)	(e)
1	Dec. 31, 2021 Regulatory Asset (13 Month Average)	\$ 16,333		Exh. A-66 (DLH-1)
2	Pre-tax Cost of Capital	<u>7.63%</u>		Exh. A-14 (MLB-1)
3	Return on Regulatory Asset		1,246	Line 1 X Line 2
4	2021 Amortization Expense		<u>682</u>	Exh. A-66 (DLH-1)
5	Regulatory Asset Revenue Requirement		1,928	Line 3 + Line 4
6	Karn Retention 2021 O&M Expense	7,413		
7	Revenue Requirement of O&M		<u>\$ 7,413</u>	Equals Line 6
8	Difference in Revenue Requirement		<u><u>\$ (5,485)</u></u>	Line 5 - Line 7

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Line No.	Item Description	2019												12-Months Ending 2019
		Actual Jan	Actual Feb	Actual Mar	Actual Apr	Actual May	Actual Jun	Actual Jul	Actual Aug	Actual Sep	Actual Oct	Forecast Nov	Forecast Dec	
1	Electric Vehicle Expenditures	\$ -	\$ 32	\$ 20	\$ 74	\$ 33	\$ 42	\$ 152	\$ 48	\$ 52	\$ 62	\$ 67	\$ 66	\$ 647
	Amortization Expense													
2	2020	129												
3	2021	129												

	Item Description	Avg Balance Dec 31, 2021 ⁽³⁾	12 Mos Ending Dec 31, 2020 ⁽¹⁾	12 Mos Ending Dec 31, 2021 ⁽²⁾
4	Electric Vehicle Deferred Balance	\$ 453	\$ 518	\$ 388

⁽¹⁾ Line 1 column (n) Less Line 2 column (b)

⁽²⁾ Line 1 column (n) Less Line 2 column (b) Less Line 3 column (b)

⁽³⁾ (Line 4 column (c) Plus Line 4 column (d))/2

Consumers Energy Company

Electric Rate Case Deferral
(000)

Case No.: U-20697
Exhibit No.: A-85 (HJM-66)
Page: 1 of 2
Witness: HJMyers
Date: February 2020

Line	Description	Actual 2019	U-20134 Settlement 2019	Difference 2019	Source
(a)	(b)	(c)	(d)	(e)	(f)
	<u>Capital Spending</u>				
1	New Business	132,554	94,000	38,554	Exh. A-XX(RTB-13)
2	Demand Failures	138,799	87,000	51,799	Exh. A-XX(RTB-13)
3	Asset Relocations	43,681	24,000	19,681	Exh. A-XX(RTB-13)
4	Total Capital Spending	\$ 315,034	\$ 205,000	\$ 110,034	

Consumers Energy CompanyElectric Rate Case Deferral
(000)Case No.: U-20697
Exhibit No.: A-85 (HJM-66)
Page: 2 of 2
Witness: HJMyers
Date: February 2020

Line	Description (a)	Amount (d)	Source (e)
<u>Revenue Requirement Calculation:</u>			
1	Beginning Plant in Service	\$ -	
2	Additional Capital Spending	110,034	Page 1, Line 4
3	Ending Plant	\$ 110,034	Line 1 + Line 2
4	2019 Average Plant in Service	\$ 55,017	Average (Line 1 & Line 3)
5	Beginning Depreciation Reserve	\$ -	
6	Depreciation Expense ⁽¹⁾	1,678	Line 4 * 3.05%
7	End Depreciation Reserve	\$ 1,678	Line 5 + Line 6
8	2019 Average Depreciation Reserve	\$ 839	Average (Line 5 & Line 7)
9	Total 2019 Average Rate Base	\$ 54,178	Line 3 Less Line 8
10	Return on Investment ⁽¹⁾	\$4,009	Line 9 X 7.40%
11	R&PP Tax ⁽¹⁾	613	Line 4 * 1.11%
12	Total 2019 Revenue Requirement	\$ 6,300	Line 6 + Line 10 + Line 11
13	Add 2020 Deferral	\$ 6,300	Line 12
14	Total Electric Rate Case Deferral	\$ 12,600	Line 12 + Line 13

⁽¹⁾ Depreciation rate of 3.05%, pretax cost of capital of 7.40%, and the property tax rate of 1.11% are from case No. U-20134.

Consumers Energy Company

Hemlock Contract

Levelized Revenue Requirement - Capacity

(000)

Case No.: U-20697

Exhibit No.: A-86 (HJM-67)

Page: 1 of 2

Witness: HJMyers

Date: February 2020

Line	Description	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	Source
	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)
1	Rate Base	295,496	293,207	292,660	290,835	284,876	279,636	280,501	279,725	294,392	306,685	293,898	280,398	271,310	260,398	242,063	222,003	208,085	180,510	130,684	78,415	24,461	Exh. A-86 (HJM-67) p2
2	Pretax Cost of Capital	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	WP-HJM-88
3	Return on Rate Base	21,865	21,696	21,656	21,520	21,080	20,692	20,756	20,698	21,784	22,693	21,747	20,748	20,076	19,268	17,912	16,427	15,397	13,357	9,670	5,802	1,810	Line 1 * Line 2
4	Depreciation Expense	10,105	10,763	11,374	12,368	13,000	13,700	14,589	16,361	17,334	22,215	23,464	24,896	26,554	29,852	32,168	34,986	38,549	47,484	48,629	49,514	50,425	WP-HJM-88
5	Amortization Expense	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	WP-HJM-88
6	Property Tax Expense	6,481	6,481	6,384	6,288	6,194	6,101	6,009	5,919	5,830	5,743	5,657	5,572	5,488	5,406	5,325	5,245	5,166	5,089	5,012	2,057	844	Tax Department
7	Property Ins Expense	889	913	937	962	988	1,015	1,042	1,070	1,099	1,128	1,159	1,190	1,222	1,255	1,289	1,323	1,359	1,395	1,433	1,472	630	WP-HJM-88
8	Gas Transportation Costs	1,013	1,031	989	1,009	1,029	1,050	1,072	1,094	1,116	1,140	1,164	1,188	1,220	1,253	1,287	1,321	1,357	1,393	1,431	1,469	629	WP-HJM-88
9	Overheads	1,600	1,632	1,665	1,709	1,755	1,803	1,851	1,901	1,952	2,004	2,058	2,114	2,171	2,229	2,289	2,351	2,414	2,479	2,545	2,614	1,089	Footnote ⁽¹⁾
10	Fixed O&M	6,315	6,441	6,570	6,747	6,928	7,115	7,306	7,503	7,705	7,912	8,125	8,343	8,568	8,798	9,035	9,278	9,527	9,784	10,047	10,317	4,299	Footnote ⁽²⁾
11	Revenue Requirement	52,996	53,686	54,303	55,332	55,703	56,203	57,354	59,274	61,549	67,564	68,102	68,780	70,027	72,790	74,032	75,659	78,498	85,709	83,496	77,974	64,454	Sum Lines 3-10
12	NPV Revenue Requirement		656,557																				
13	Levelized Revenue Requirement		62,552																				

Footnotes:⁽¹⁾ Allocation of Benefits, A&G, and Payroll Taxes⁽²⁾ Base O&M. Excludes major maintenance

Hemlock Contract
Levelized Revenue Requirement - Capacity
(000)

Exhibit No.: A-86 (HJM-67)
Page: 2 of 2
Witness: HJMyers
Date: February 2020

		2018 YE ⁽¹⁾		2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	Source (y)
		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)	(x)	(y)	
1	Beginning Balance PIS	277,793	277,793	289,239	298,405	311,573	323,182	341,067	351,817	363,013	376,355	401,156	413,813	472,378	486,120	500,441	515,366	541,747	557,962	574,866	592,684	628,421	631,857	633,627	634,538	Prior year Line 1 + Line 7	
2	Acquisition Add	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	147,294	Prior year line 2
3	Accum. Dep. less COR	(74,882.2)	(87,830)	(101,275)	(116,629)	(132,077)	(148,529)	(165,987)	(184,445)	(203,903)	(224,361)	(245,819)	(268,277)	(291,735)	(316,193)	(341,651)	(368,109)	(395,567)	(424,025)	(453,483)	(483,941)	(515,399)	(547,857)	(580,315)	(612,773)	(645,231)	Prior year line 3 - Line 14
4	Cumulative COR Corrected	(1,072)	(1,790)	(2,507)	(3,224)	(3,942)	(4,659)	(5,377)	(6,094)	(6,811)	(7,529)	(8,246)	(8,963)	(9,681)	(10,398)	(11,116)	(11,833)	(12,550)	(13,268)	(13,985)	(14,702)	(15,420)	(16,137)	(16,855)	(17,572)		
5	Cumulative COR Spent		666	1,251	1,664	2,220	2,670	3,135	3,630	4,275	4,963	5,651	6,228	6,954	7,728	8,525	9,152	9,970	10,841	11,738	12,444	12,904	13,156	13,286	13,342		
6	Accum. Amort. - Acq. Add	(40,428)	(44,310.35)	(47,993)	(52,721)	(57,450)	(62,179)	(66,907)	(71,636)	(76,365)	(81,093)	(85,822)	(90,550)	(95,279)	(100,008)	(104,736)	(109,465)	(114,194)	(118,922)	(123,651)	(128,379)	(133,108)	(137,837)	(142,565)	(147,294)		
7	Capital Additions		11,446	9,166	13,168	11,609	17,885	10,750	11,196	13,342	24,801	12,657	58,565	13,742	14,321	14,925	26,381	16,215	16,904	17,818	35,737	3,437	1,770	911	391		
8	Land		947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	947	
9	Rate Base	309,452	304,216	296,123	294,870	291,543	293,776	287,894	281,857	277,416	283,587	275,863	312,922	300,447	287,350	273,447	269,172	251,624	232,502	211,503	204,668	156,352	105,016	51,815	(2,892)		
10	Average Rate Base				295,496	293,207	292,660	290,835	284,876	279,636	280,501	279,725	294,392	306,685	293,898	280,398	273,310	260,398	242,003	222,003	208,085	180,510	130,684	78,415	24,461		
11	Depreciable Balance		283,516	293,822	304,989	317,377	332,124	346,442	357,415	369,684	388,755	407,484	443,095	479,249	493,280	507,903	528,556	549,854	566,414	583,775	610,552	630,139	632,742	633,627	634,538		
12	Dep'r Rate		4.86%	4.82%	3.08%	3.17%	3.21%	3.36%	3.44%	3.51%	3.57%	3.64%	3.75%	4.49%	4.61%	4.76%	4.89%	5.30%	5.55%	5.87%	6.20%	7.42%	7.57%	7.70%	7.83%		
13	Depreciation Expense		13,665	14,162	9,387	10,046	10,657	11,650	12,282	12,982	13,872	15,843	16,717	21,497	22,747	24,179	25,837	29,135	31,451	34,268	37,832	46,766	47,912	48,797	49,708		
14	Acc. Dep. Collected*		717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717	717		
15	Total Dep'r. Expense		13,665	14,162	10,105	10,763	11,374	12,368	13,000	13,700	14,589	16,361	17,334	22,215	23,464	24,896	26,559	29,852	32,168	34,986	38,549	47,484	48,629	49,514	50,425		
16	Amort. Expense Rate			2.5%	2.5%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%	3.2%		
17	Amort. Expense				4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729	4,729		

Consumers Energy Company

HSC Contract
Distribution Charge
(000)

Case No.: U-20697
Exhibit No.: A-87 (HJM-68)
Page: 1 of 1
Witness: HJMyers
Date: February 2020

Line		2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	2033	2034	2035	2036	2037	2038	2039	2040	2041	Source
1	Rate Base	4,351	4,255	4,158	4,061	3,964	3,867	3,771	3,674	3,577	3,480	3,384	3,287	3,190	3,093	2,996	2,900	2,803	2,706	2,609	2,513	2,416	WP-HJM-91
2	Pretax Cost of Capital	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	7.40%	WP-HJM-89
3	Return on Rate Base	322	315	308	300	293	286	279	272	265	258	250	243	236	229	222	215	207	200	193	186	179	Line 1 X Line 2
4	Depreciation Expense	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	97	WP-HJM-91
5	Property Tax Expense	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	108	WP-HJM-91
6	Property Ins Expense	1	1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	Footnote ¹
7	O&M & Overhead	271	278	286	293	301	309	318	326	335	344	353	363	373	383	393	404	414	426	437	449	461	Footnote ²
8	Revenue Requirement	799	799	800	800	801	802	803	805	806	808	810	813	815	818	821	825	829	833	837	842	847	Sum Lines 3-7
9	NPV Revenue Requirement		8,495																				
10	Levelized Revenue Requirement		809																				
11	Monthly Revenue Requirement		67.443																				

Footnotes

¹ 0.03% of Rate Base inflated through 2041

² 2019 O&M and Overheads inflated through 2041

Line	Date (a)	Description (b)	Details (c)
1	January 1, 2021	Implement FCM Surcharge	-Surcharge is approved in this rate case and covers the FCM for 2019 through 2021. -Surcharge is designed to collect the 2019 through 2021 FCM over 12 months.
2	March 31, 2022	File FCM Case	-Reconciles 2021 surcharge and calculates new surcharge to be implemented July 1, 2022. -New annual surcharge amount is calculated as follows: Actual FCM amount 2019 - 2021 Less Billed FCM surcharge during 2021 Plus Projected 2022 FCM Amount -The 2021 surcharge will remain in place until adjusted by the outcome of the March 31, 2022 filing.
3	July 1, 2022	FCM Surcharge Adjusted	-FCM Surcharge will be adjusted per the outcome of the March 31, 2022 filing.
4	March 31, 2023	File FCM Case	-Reconciles 2022 surcharges and calculates new surcharge to be implemented July 1, 2023. -New annual surcharge is calculated as follows: Actual FCM amount 2022 Less Billed FCM surcharge during 2022 Plus Projected 2023 FCM Amount -The July 1, 2022 surcharge will remain in place until adjusted by the outcome of the March 31, 2023 filing.
5	July 1, 2023	FCM Surcharge Adjusted	-FCM Surcharge will be adjusted per the outcome of the March 31, 2023 filing.

Smart Grid Program

Line No.	(a) Utility	(b) Description	(c) 2007	(d) 2008	(e) 2009	(f) 2010	(g) 2011	(h) 2012	(i) 2013	(j) 2014	(k) 2015	(l) 2016	(m) 2017	(n) 2018	(o) 2019	(p) 2020
1	Electric	Meters	\$ -	\$ 1,424	\$ 1,231	\$ 930	\$ 128	\$ 9,602	\$ 21,432	\$ 29,976	\$ 62,152	\$ 109,569	\$ 53,679	\$ -	\$ -	\$ -
2		Field Equipment/Facilities	\$ -	\$ 113	\$ 1,305	\$ 1,510	\$ 341	\$ 259	\$ 171	\$ 154	\$ 270	\$ 1,200	\$ 5,965	\$ 6,331	\$ 8,440	\$ 8,580
3		Software/Systems Development	\$ 7,889	\$ 8,214	\$ 12,259	\$ 15,909	\$ 19,654	\$ 20,770	\$ 32,100	\$ 31,990	\$ 45,038	\$ 24,918	\$ (1,734)	\$ -	\$ 15,181	\$ -
4		SG Infrastructure	\$ -	\$ -	\$ -	\$ 2,789	\$ 7,834	\$ 5,912	\$ 1,109	\$ 976	\$ 1,463	\$ 3,391	\$ 750	\$ 1,943	\$ 2,827	\$ 3,375
5		Pilot Prep & Project Management	\$ -	\$ 7,883	\$ 8,932	\$ 13,283	\$ 9,558	\$ 7,919	\$ 6,194	\$ 6,063	\$ 6,657	\$ 4,790	\$ 2,493	\$ -	\$ -	\$ -
6		Total Capital Costs before Avoidance - Electric	\$ 7,889	\$ 17,634	\$ 23,726	\$ 34,421	\$ 37,515	\$ 44,463	\$ 61,005	\$ 69,159	\$ 115,581	\$ 143,868	\$ 61,153	\$ 8,274	\$ 26,448	\$ 11,955
7		Avoided Capital Costs - Electric	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (306)	\$ (240)	\$ 410	\$ 352	\$ 263	\$ (2,221)
8		Total Capital Costs Net - Electric	\$ 7,889	\$ 17,634	\$ 23,726	\$ 34,421	\$ 37,515	\$ 44,463	\$ 61,005	\$ 69,159	\$ 115,274	\$ 143,628	\$ 61,563	\$ 8,625	\$ 26,711	\$ 9,735
9																
10	Gas	Modules	\$ -	\$ -	\$ -	\$ 5	\$ -	\$ -	\$ -	\$ -	\$ 6,663	\$ 25,404	\$ 15,845	\$ -	\$ -	\$ -
11		Field Equipment/Facilities	\$ -	\$ 15	\$ 162	\$ 95	\$ 41	\$ 37	\$ 23	\$ 21	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
12		Software/Systems Development	\$ 1,076	\$ 1,120	\$ 1,524	\$ 2,143	\$ 2,664	\$ 2,832	\$ 4,377	\$ 4,362	\$ 5,995	\$ 3,359	\$ (54)	\$ -	\$ -	\$ -
13		SG Infrastructure	\$ -	\$ -	\$ -	\$ 380	\$ 1,068	\$ 806	\$ 151	\$ 133	\$ 200	\$ 470	\$ 102	\$ 265	\$ 385	\$ 460
14		Pilot Prep & Project Management	\$ -	\$ 1,573	\$ 2,348	\$ 3,016	\$ 1,737	\$ 1,183	\$ 827	\$ 727	\$ 872	\$ 667	\$ 354	\$ -	\$ -	\$ -
15		Total Capital Costs before Avoidance - Gas	\$ 1,076	\$ 2,709	\$ 4,034	\$ 5,639	\$ 5,510	\$ 4,858	\$ 5,379	\$ 5,243	\$ 13,730	\$ 29,900	\$ 16,248	\$ 265	\$ 385	\$ 460
16		Avoided Capital Costs - Gas	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (10)	\$ (20)	\$ (34)	\$ (42)	\$ (43)	\$ (43)
17		Total Capital Costs Net - Gas	\$ 1,076	\$ 2,709	\$ 4,034	\$ 5,639	\$ 5,510	\$ 4,858	\$ 5,379	\$ 5,243	\$ 13,720	\$ 29,881	\$ 16,213	\$ 223	\$ 343	\$ 417
18																
19	Total	Meter/Module Purchases, including Set & Test	\$ -	\$ 1,424	\$ 1,231	\$ 935	\$ 128	\$ 9,602	\$ 21,432	\$ 29,976	\$ 68,816	\$ 134,973	\$ 69,524	\$ -	\$ -	\$ -
20		Field Equipment/Facilities	\$ -	\$ 128	\$ 1,466	\$ 1,605	\$ 382	\$ 296	\$ 194	\$ 174	\$ 270	\$ 1,200	\$ 5,965	\$ 6,331	\$ 8,440	\$ 8,580
21		Software/Systems Development	\$ 8,965	\$ 9,334	\$ 13,782	\$ 18,052	\$ 22,318	\$ 23,602	\$ 36,477	\$ 36,353	\$ 51,033	\$ 28,277	\$ (1,788)	\$ -	\$ 15,181	\$ -
22		SG Infrastructure	\$ -	\$ -	\$ -	\$ 3,170	\$ 8,903	\$ 6,719	\$ 1,260	\$ 1,109	\$ 1,663	\$ 3,861	\$ 853	\$ 2,208	\$ 3,212	\$ 3,835
23		Pilot Prep & Project Management	\$ -	\$ 9,456	\$ 11,280	\$ 16,299	\$ 11,294	\$ 9,102	\$ 7,021	\$ 6,790	\$ 7,529	\$ 5,457	\$ 2,848	\$ -	\$ -	\$ -
24		Total Capital Costs before Avoidance	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,320	\$ 66,384	\$ 74,402	\$ 129,311	\$ 173,768	\$ 77,401	\$ 8,539	\$ 26,833	\$ 12,416
25		Avoided Capital Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (316)	\$ (260)	\$ 375	\$ 309	\$ 220	\$ (2,264)
26		Total Capital Costs Net	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,320	\$ 66,384	\$ 74,402	\$ 128,994	\$ 173,508	\$ 77,776	\$ 8,848	\$ 27,054	\$ 10,152
27																
28																
29		Note:														
30		Cost of Removal	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 129	\$ 120	\$ 1	\$ -	\$ 59	\$ 23	\$ -	\$ -	\$ -
31		Total Capital Costs Incl COR before Avoidance	\$ 8,965	\$ 20,342	\$ 27,760	\$ 40,060	\$ 43,025	\$ 49,450	\$ 66,504	\$ 74,403	\$ 129,311	\$ 173,827	\$ 77,424	\$ 8,539	\$ 26,833	\$ 12,416
32		Cml Total Capital Costs Incl COR before Avoidance	\$ 8,965	\$ 29,307	\$ 57,067	\$ 97,127	\$ 140,152	\$ 189,602	\$ 256,106	\$ 330,509	\$ 459,820	\$ 633,647	\$ 711,071	\$ 719,610	\$ 746,443	\$ 758,858

Total NPV Benefit of \$160.5 million based on nominal values 2007 - 2020 and present values 2021 - 2032

Smart Grid Program

Line No.	(a) Utility	(b) Description	(c) 2021	(d) 2022	(e) 2023	(f) 2024	(g) 2025	(h) 2026	(i) 2027	(j) 2028	(k) 2029	(l) 2030	(m) 2031	(n) 2032	(o) Total 2007-2032
1	Electric	Meters	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 290,123
2		Field Equipment/Facilities	\$ 7,651	\$ 7,349	\$ 1,169	\$ 977	\$ 1,007	\$ 1,037	\$ 1,070	\$ 1,104	\$ 1,140	\$ 1,179	\$ 1,219	\$ 1,262	\$ 60,802
3		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 232,188
4		SG Infrastructure	\$ 2,916	\$ 2,342	\$ 2,569	\$ 3,064	\$ 2,985	\$ 2,617	\$ 2,577	\$ 2,850	\$ 2,943	\$ 2,756	\$ 2,648	\$ 2,760	\$ 65,395
5		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 73,771
6		Total Capital Costs before Avoidance - Electric	\$ 10,566	\$ 9,691	\$ 3,739	\$ 4,041	\$ 3,992	\$ 3,654	\$ 3,647	\$ 3,954	\$ 4,083	\$ 3,935	\$ 3,867	\$ 4,021	\$ 722,279
7		Avoided Capital Costs - Electric	\$ 359	\$ 377	\$ 388	\$ 388	\$ 345	\$ 323	\$ 323	\$ 344	\$ 544	\$ 544	\$ 544	\$ 544	\$ 3,281
8		Total Capital Costs Net - Electric	\$ 10,925	\$ 10,068	\$ 4,127	\$ 4,429	\$ 4,337	\$ 3,977	\$ 3,970	\$ 4,299	\$ 4,627	\$ 4,479	\$ 4,410	\$ 4,565	\$ 725,560
9															
10	Gas	Modules	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 47,918
11		Field Equipment/Facilities	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 394
12		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 29,398
13		SG Infrastructure	\$ 398	\$ 319	\$ 350	\$ 418	\$ 407	\$ 357	\$ 351	\$ 389	\$ 401	\$ 376	\$ 361	\$ 376	\$ 8,925
14		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 13,305
15		Total Capital Costs before Avoidance - Gas	\$ 398	\$ 319	\$ 350	\$ 418	\$ 407	\$ 357	\$ 351	\$ 389	\$ 401	\$ 376	\$ 361	\$ 376	\$ 99,939
16		Avoided Capital Costs - Gas	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (43)	\$ (707)
17		Total Capital Costs Net - Gas	\$ 355	\$ 276	\$ 307	\$ 375	\$ 364	\$ 314	\$ 309	\$ 346	\$ 358	\$ 333	\$ 318	\$ 333	\$ 99,232
18															
19	Total	Meter/Module Purchases, including Set & Test	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 338,040
20		Field Equipment/Facilities	\$ 7,651	\$ 7,349	\$ 1,169	\$ 977	\$ 1,007	\$ 1,037	\$ 1,070	\$ 1,104	\$ 1,140	\$ 1,179	\$ 1,219	\$ 1,262	\$ 61,196
21		Software/Systems Development	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 261,586
22		SG Infrastructure	\$ 3,313	\$ 2,661	\$ 2,920	\$ 3,482	\$ 3,392	\$ 2,974	\$ 2,929	\$ 3,239	\$ 3,344	\$ 3,132	\$ 3,009	\$ 3,136	\$ 74,320
23		Pilot Prep & Project Management	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 87,076
24		Total Capital Costs before Avoidance	\$ 10,964	\$ 10,010	\$ 4,089	\$ 4,459	\$ 4,399	\$ 4,011	\$ 3,998	\$ 4,343	\$ 4,484	\$ 4,311	\$ 4,228	\$ 4,398	\$ 822,218
25		Avoided Capital Costs	\$ 316	\$ 334	\$ 346	\$ 345	\$ 302	\$ 280	\$ 281	\$ 301	\$ 501	\$ 501	\$ 501	\$ 501	\$ 2,573
26		Total Capital Costs Net	\$ 11,280	\$ 10,344	\$ 4,435	\$ 4,804	\$ 4,701	\$ 4,291	\$ 4,279	\$ 4,644	\$ 4,985	\$ 4,812	\$ 4,728	\$ 4,898	\$ 824,792
27															
28															
29		Note:													
30		Cost of Removal	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 332
31		Total Capital Costs Incl COR before Avoidance	\$ 10,964	\$ 10,010	\$ 4,089	\$ 4,459	\$ 4,399	\$ 4,011	\$ 3,998	\$ 4,343	\$ 4,484	\$ 4,311	\$ 4,228	\$ 4,398	\$ 822,551
32		Cml Total Capital Costs Incl COR before Avoidance	\$ 769,822	\$ 779,832	\$ 783,921	\$ 788,380	\$ 792,778	\$ 796,789	\$ 800,788	\$ 805,131	\$ 809,615	\$ 813,926	\$ 818,153	\$ 822,551	

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
 Summary of Business Case Costs and Benefits
 2007 - 2032
 (000)

Case No.: U-20697
 Exhibit No.: A-89 (HJM-70)
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 Witness: HJMyers
 Date: February 2020

Smart Grid Program

Line No.	(a) Utility	(b) Description	(c) 2007	(d) 2008	(e) 2009	(f) 2010	(g) 2011	(h) 2012	(i) 2013	(j) 2014	(k) 2015	(l) 2016	(m) 2017	(n) 2018	(o) 2019	(p) 2020
33	Electric Meter Reading		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (116)	\$ (1,878)	\$ (5,151)	\$ (9,767)	\$ (16,577)	\$ (19,196)	\$ (19,926)	\$ (20,325)
34	Uncollectible Expense		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (128)	\$ (3,685)	\$ (6,227)	\$ (8,181)	\$ (8,344)	\$ (8,511)
35	Other O&M		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,271)	\$ (1,115)	\$ (2,917)	\$ (4,690)	\$ (7,375)	\$ (7,942)	\$ (8,183)	\$ (8,508)
36	AC Load Control Avoided Generation, Transmission		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (135)	\$ (960)	\$ (3,502)	\$ (4,674)
37	Demand Response Avoided Generation, Transmission		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (0)	\$ (47)	\$ (490)	\$ (954)	\$ (19,713)
38	Theft Reduction		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (93)	\$ (919)	\$ (1,837)	\$ (4,787)	\$ (6,943)	\$ (8,342)	\$ (9,810)	\$ (15,126)	\$ (22,561)
39	AMI Induced Conservation & Efficiency Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (2,906)	\$ (5,571)	\$ (7,991)	\$ (10,912)	\$ (12,405)
40	Load Management Conserved Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
41	Demand Response Conserved Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,417)
42	Property Tax (Savings) - Legacy Meters		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (33)	\$ (97)	\$ (235)	\$ (500)	\$ (835)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)
43	Terminal Value		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
44	Total Non Capital Benefits before Costs - Electric		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (126)	\$ (2,403)	\$ (5,064)	\$ (13,482)	\$ (28,827)	\$ (45,340)	\$ (55,636)	\$ (68,014)	\$ (99,180)
45	Common		\$ 382	\$ 67	\$ 545	\$ 1,165	\$ 1,094	\$ 2,586	\$ 5,977	\$ 4,853	\$ 6,466	\$ 6,821	\$ 8,606	\$ 9,183	\$ 9,399	\$ 9,643
46	Meters, Modules & Communications		\$ -	\$ -	\$ -	\$ -	\$ -	\$ 97	\$ 963	\$ 1,310	\$ 1,671	\$ 3,226	\$ 4,898	\$ 5,227	\$ 5,228	\$ 5,228
47	Customer Engagement		\$ -	\$ -	\$ -	\$ -	\$ -	\$ 1,538	\$ 2,444	\$ 1,956	\$ 1,833	\$ 4,179	\$ 5,275	\$ 7,938	\$ 21,493	\$ 11,199
48	Load Control Program		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 195	\$ 665	\$ 2,567	\$ 3,017	\$ 2,726	\$ 2,625
49	Demand Response Program		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 16	\$ 194	\$ 2,219	\$ 986	\$ 606	\$ 618
50	Property Taxes - AMI		\$ -	\$ 24	\$ 44	\$ 70	\$ 52	\$ 335	\$ 862	\$ 1,572	\$ 2,977	\$ 5,402	\$ 6,644	\$ 6,515	\$ 6,407	\$ 6,270
51	Total Non Capital Benefits Net - Electric		\$ 382	\$ 91	\$ 589	\$ 1,235	\$ 1,146	\$ 4,430	\$ 7,843	\$ 4,627	\$ (325)	\$ (8,340)	\$ (15,131)	\$ (22,771)	\$ (22,155)	\$ (63,597)
52																
53	Gas Meter Reading		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 0	\$ (591)	\$ (1,965)	\$ (2,507)	\$ (2,602)	\$ (2,654)
54	Uncollectible Expense		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (168)	\$ (1,335)	\$ (2,144)	\$ (2,187)	\$ (2,231)
55	Other O&M		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (68)	\$ 10	\$ (1)	\$ (214)	\$ (411)	\$ (444)	\$ (454)	\$ (468)
56	Theft Reduction		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (235)	\$ (462)	\$ (531)	\$ (667)	\$ (890)	\$ (3,460)
57	AMI Induced Conservation & Efficiency Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (199)	\$ (1,433)	\$ (2,247)	\$ (2,585)	\$ (2,705)
58	LAUF Gas Reduction Enabled		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (22)	\$ (173)	\$ (181)	\$ (190)
59	Terminal Value		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
60	Total Non Capital Benefits before Costs - Gas		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (68)	\$ 10	\$ (236)	\$ (1,634)	\$ (5,697)	\$ (8,182)	\$ (8,900)	\$ (11,709)
61	Common		\$ 367	\$ 64	\$ 87	\$ 143	\$ 73	\$ 127	\$ 36	\$ 11	\$ 216	\$ 746	\$ 1,173	\$ 1,252	\$ 1,282	\$ 1,315
62	Meters, Modules & Communications		\$ -	\$ -	\$ -	\$ -	\$ -	\$ 5	\$ 6	\$ -	\$ 42	\$ 466	\$ 1,237	\$ 1,371	\$ 1,371	\$ 1,371
63	Property Taxes - AMI		\$ -	\$ -	\$ -	\$ 5	\$ 5	\$ 8	\$ 12	\$ 11	\$ 141	\$ 692	\$ 891	\$ 831	\$ 778	\$ 728
64	Total Non Capital Benefits Net - Gas		\$ 367	\$ 64	\$ 87	\$ 148	\$ 78	\$ 140	\$ (14)	\$ 32	\$ 164	\$ 271	\$ (2,396)	\$ (4,728)	\$ (5,469)	\$ (8,294)
65																
66	Total Meter Reading		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (116)	\$ (1,878)	\$ (5,151)	\$ (10,358)	\$ (18,542)	\$ (21,704)	\$ (22,529)	\$ (22,979)
67	Uncollectible Expense		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (128)	\$ (3,653)	\$ (7,562)	\$ (10,325)	\$ (10,531)	\$ (10,742)
68	Other O&M		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,339)	\$ (1,105)	\$ (2,917)	\$ (4,905)	\$ (7,786)	\$ (8,386)	\$ (8,637)	\$ (8,976)
69	AC Load Control Avoided Generation, Transmission		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (135)	\$ (960)	\$ (3,502)	\$ (4,674)
70	Demand Response Avoided Generation, Transmission		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (0)	\$ (47)	\$ (490)	\$ (954)	\$ (19,713)
71	Theft Reduction		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (93)	\$ (919)	\$ (1,837)	\$ (5,023)	\$ (7,405)	\$ (8,873)	\$ (10,477)	\$ (16,016)	\$ (26,021)
72	AMI Induced Conservation & Efficiency Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (3,105)	\$ (7,004)	\$ (10,238)	\$ (13,497)	\$ (15,111)
73	Load Management Conserved Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
74	Demand Response Conserved Energy		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (1,417)
75	Property Tax (Savings) - Legacy Meters		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (33)	\$ (97)	\$ (235)	\$ (500)	\$ (835)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)
76	LAUF Gas Reduction Enabled		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (22)	\$ (173)	\$ (181)	\$ (190)
77	Terminal Value		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
78	Total Non Capital Benefits before Costs		\$ -	\$ -	\$ -	\$ -	\$ -	\$ (126)	\$ (2,471)	\$ (5,055)	\$ (13,718)	\$ (30,461)	\$ (51,037)	\$ (63,819)	\$ (76,914)	\$ (110,889)
79	Common		\$ 749	\$ 131	\$ 632	\$ 1,308	\$ 1,167	\$ 2,713	\$ 6,013	\$ 4,864	\$ 6,682	\$ 7,567	\$ 9,779	\$ 10,435	\$ 10,681	\$ 10,958
80	Meters, Modules & Communications		\$ -	\$ -	\$ -	\$ -	\$ -	\$ 102	\$ 969	\$ 1,310	\$ 1,713	\$ 3,692	\$ 6,135	\$ 6,598	\$ 6,599	\$ 6,599
81	Other O&M		\$ -	\$ -	\$ -	\$ -	\$ -	\$ 1,538	\$ 2,444	\$ 1,956	\$ 1,833	\$ 4,179	\$ 5,275	\$ 7,938	\$ 21,493	\$ 11,199
82	Load Control Program		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 195	\$ 665	\$ 2,567	\$ 3,017	\$ 2,726	\$ 2,625
83	Demand Response Program		\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ 16	\$ 194	\$ 2,219	\$ 986	\$ 606	\$ 618
84	Property Taxes - AMI		\$ -	\$ 24	\$ 44	\$ 75	\$ 57	\$ 343	\$ 874	\$ 1,583	\$ 3,118	\$ 6,094	\$ 7,535	\$ 7,346	\$ 7,185	\$ 6,999
85	Total Non Capital Benefits Net		\$ 749	\$ 155	\$ 676	\$ 1,383	\$ 1,224	\$ 4,569	\$ 7,829	\$ 4,659	\$ (161)	\$ (8,069)	\$ (17,526)	\$ (27,499)	\$ (27,624)	\$ (71,891)

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Summary of Business Case Costs and Benefits
2007 - 2032
(000)

Case No.: U-20697
Exhibit No.: A-89 (HJM-70)
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Witness: HJMyers
Date: February 2020

Smart Grid Program

Line No.	(a) Utility	(b) Description	(c) 2021	(d) 2022	(e) 2023	(f) 2024	(g) 2025	(h) 2026	(i) 2027	(j) 2028	(k) 2029	(l) 2030	(m) 2031	(n) 2032	(o) Total 2007-2032
33	Electric	Meter Reading	\$ (20,935)	\$ (21,563)	\$ (22,210)	\$ (22,876)	\$ (23,562)	\$ (24,269)	\$ (24,997)	\$ (25,747)	\$ (26,519)	\$ (27,315)	\$ (28,134)	\$ (28,978)	\$ (390,041)
34		Uncollectible Expense	\$ (8,681)	\$ (8,855)	\$ (9,032)	\$ (9,213)	\$ (9,397)	\$ (9,585)	\$ (9,776)	\$ (9,972)	\$ (10,171)	\$ (10,375)	\$ (10,582)	\$ (10,794)	\$ (151,509)
35		Other O&M	\$ (8,826)	\$ (9,104)	\$ (9,363)	\$ (9,613)	\$ (9,865)	\$ (10,124)	\$ (10,392)	\$ (10,669)	\$ (10,953)	\$ (11,247)	\$ (11,550)	\$ (11,863)	\$ (165,570)
36		AC Load Control Avoided Generation, Transmission	\$ (6,080)	\$ (7,375)	\$ (8,248)	\$ (8,493)	\$ (8,658)	\$ (8,834)	\$ (9,010)	\$ (9,190)	\$ (9,374)	\$ (9,561)	\$ (9,753)	\$ (9,948)	\$ (113,795)
37		Demand Response Avoided Generation, Transmission	\$ (20,173)	\$ (20,648)	\$ (21,519)	\$ (23,082)	\$ (25,126)	\$ (27,412)	\$ (29,780)	\$ (32,055)	\$ (32,696)	\$ (33,350)	\$ (34,017)	\$ (34,697)	\$ (355,758)
38		Theft Reduction	\$ (27,997)	\$ (35,010)	\$ (39,052)	\$ (39,833)	\$ (40,629)	\$ (41,442)	\$ (42,271)	\$ (43,116)	\$ (43,978)	\$ (44,858)	\$ (45,755)	\$ (46,670)	\$ (561,029)
39		AMI Induced Conservation & Efficiency Energy	\$ (13,339)	\$ (13,715)	\$ (13,890)	\$ (14,069)	\$ (14,251)	\$ (14,486)	\$ (14,776)	\$ (15,071)	\$ (15,373)	\$ (15,680)	\$ (15,994)	\$ (16,314)	\$ (216,740)
40		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
41		Demand Response Conserved Energy	\$ (1,577)	\$ (1,681)	\$ (1,744)	\$ (1,810)	\$ (1,865)	\$ (1,920)	\$ (1,978)	\$ (2,037)	\$ (2,097)	\$ (2,156)	\$ (2,215)	\$ (2,270)	\$ (24,767)
42		Property Tax (Savings) - Legacy Meters	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (18,767)
43		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (252,976)	\$ (252,976)
44		Total Non Capital Benefits before Costs - Electric	\$ (108,675)	\$ (119,016)	\$ (126,123)	\$ (130,055)	\$ (134,420)	\$ (139,139)	\$ (144,046)	\$ (148,923)	\$ (152,229)	\$ (155,609)	\$ (159,067)	\$ (162,576)	\$ (2,250,952)
45		Common	\$ 10,326	\$ 10,557	\$ 10,781	\$ 11,012	\$ 11,258	\$ 11,510	\$ 11,774	\$ 12,032	\$ 12,297	\$ 12,579	\$ 12,868	\$ 13,171	\$ 206,952
46		Meters, Modules & Communications	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 5,228	\$ 90,579
47		Customer Engagement	\$ 5,286	\$ 5,214	\$ 5,231	\$ 5,247	\$ 5,264	\$ 5,282	\$ 5,300	\$ 5,318	\$ 5,336	\$ 5,355	\$ 5,375	\$ 5,394	\$ 121,458
48		Load Control Program	\$ 2,530	\$ 2,425	\$ 631	\$ 582	\$ 599	\$ 617	\$ 636	\$ 656	\$ 677	\$ 698	\$ 721	\$ 745	\$ 23,311
49		Demand Response Program	\$ 874	\$ 879	\$ 1,226	\$ 669	\$ 683	\$ 696	\$ 710	\$ 724	\$ 739	\$ 754	\$ 769	\$ 784	\$ 14,147
50		Property Taxes - AMI	\$ 6,138	\$ 5,996	\$ 5,815	\$ 5,583	\$ 5,357	\$ 5,135	\$ 4,925	\$ 4,736	\$ 4,579	\$ 4,480	\$ 4,449	\$ 4,450	\$ 98,816
51		Total Non Capital Benefits Net - Electric	\$ (78,293)	\$ (88,716)	\$ (97,212)	\$ (101,734)	\$ (106,032)	\$ (110,672)	\$ (115,473)	\$ (120,230)	\$ (123,373)	\$ (126,516)	\$ (129,657)	\$ (132,805)	\$ (1,695,689)
52															
53	Gas	Meter Reading	\$ (2,734)	\$ (2,816)	\$ (2,901)	\$ (2,988)	\$ (3,077)	\$ (3,170)	\$ (3,265)	\$ (3,363)	\$ (3,464)	\$ (3,567)	\$ (3,674)	\$ (3,785)	\$ (49,122)
54		Uncollectible Expense	\$ (2,276)	\$ (2,321)	\$ (2,367)	\$ (2,415)	\$ (2,463)	\$ (2,512)	\$ (2,563)	\$ (2,614)	\$ (2,666)	\$ (2,719)	\$ (2,774)	\$ (2,829)	\$ (38,585)
55		Other O&M	\$ (484)	\$ (501)	\$ (519)	\$ (537)	\$ (556)	\$ (575)	\$ (595)	\$ (615)	\$ (636)	\$ (658)	\$ (680)	\$ (703)	\$ (9,111)
56		Theft Reduction	\$ (4,501)	\$ (5,570)	\$ (7,363)	\$ (7,601)	\$ (7,753)	\$ (7,908)	\$ (8,066)	\$ (8,227)	\$ (8,392)	\$ (8,560)	\$ (8,731)	\$ (8,906)	\$ (97,824)
57		AMI Induced Conservation & Efficiency Energy	\$ (2,846)	\$ (3,002)	\$ (3,174)	\$ (3,352)	\$ (3,419)	\$ (3,487)	\$ (3,557)	\$ (3,628)	\$ (3,701)	\$ (3,775)	\$ (3,850)	\$ (3,927)	\$ (50,888)
58		LAUF Gas Reduction Enabled	\$ (201)	\$ (213)	\$ (225)	\$ (238)	\$ (243)	\$ (248)	\$ (252)	\$ (256)	\$ (260)	\$ (263)	\$ (267)	\$ (270)	\$ (3,502)
59		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (34,497)	\$ (34,497)
60		Total Non Capital Benefits before Costs - Gas	\$ (13,042)	\$ (14,424)	\$ (16,550)	\$ (17,131)	\$ (17,511)	\$ (17,900)	\$ (18,297)	\$ (18,703)	\$ (19,119)	\$ (19,543)	\$ (19,977)	\$ (20,417)	\$ (283,529)
61		Common	\$ 1,408	\$ 1,440	\$ 1,470	\$ 1,502	\$ 1,535	\$ 1,570	\$ 1,606	\$ 1,641	\$ 1,677	\$ 1,715	\$ 1,755	\$ 1,796	\$ 26,006
62		Meters, Modules & Communications	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 1,371	\$ 22,324
63		Property Taxes - AMI	\$ 685	\$ 644	\$ 603	\$ 563	\$ 522	\$ 482	\$ 441	\$ 401	\$ 362	\$ 334	\$ 326	\$ 326	\$ 9,792
64		Total Non Capital Benefits Net - Gas	\$ (9,578)	\$ (10,969)	\$ (13,105)	\$ (13,695)	\$ (14,083)	\$ (14,477)	\$ (14,879)	\$ (15,290)	\$ (15,708)	\$ (16,123)	\$ (16,525)	\$ (16,924)	\$ (225,407)
65															
66	Total	Meter Reading	\$ (23,669)	\$ (24,379)	\$ (25,110)	\$ (25,864)	\$ (26,639)	\$ (27,439)	\$ (28,262)	\$ (29,110)	\$ (29,983)	\$ (30,882)	\$ (31,809)	\$ (32,763)	\$ (439,164)
67		Uncollectible Expense	\$ (10,957)	\$ (11,176)	\$ (11,399)	\$ (11,627)	\$ (11,860)	\$ (12,097)	\$ (12,339)	\$ (12,586)	\$ (12,838)	\$ (13,094)	\$ (13,356)	\$ (13,623)	\$ (190,094)
68		Other O&M	\$ (9,310)	\$ (9,605)	\$ (9,882)	\$ (10,150)	\$ (10,420)	\$ (10,699)	\$ (10,987)	\$ (11,284)	\$ (11,590)	\$ (11,905)	\$ (12,231)	\$ (12,566)	\$ (174,681)
69		AC Load Control Avoided Generation, Transmission	\$ (6,080)	\$ (7,375)	\$ (8,248)	\$ (8,493)	\$ (8,658)	\$ (8,834)	\$ (9,010)	\$ (9,190)	\$ (9,374)	\$ (9,561)	\$ (9,753)	\$ (9,948)	\$ (113,795)
70		Demand Response Avoided Generation, Transmission	\$ (20,173)	\$ (20,648)	\$ (21,519)	\$ (23,082)	\$ (25,126)	\$ (27,412)	\$ (29,780)	\$ (32,055)	\$ (32,696)	\$ (33,350)	\$ (34,017)	\$ (34,697)	\$ (355,758)
71		Theft Reduction	\$ (32,498)	\$ (40,580)	\$ (46,415)	\$ (47,434)	\$ (48,382)	\$ (49,350)	\$ (50,337)	\$ (51,344)	\$ (52,370)	\$ (53,418)	\$ (54,486)	\$ (55,576)	\$ (658,852)
72		AMI Induced Conservation & Efficiency Energy	\$ (16,185)	\$ (16,717)	\$ (17,064)	\$ (17,420)	\$ (17,670)	\$ (17,973)	\$ (18,333)	\$ (18,699)	\$ (19,073)	\$ (19,455)	\$ (19,844)	\$ (20,241)	\$ (267,628)
73		Load Management Conserved Energy	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
74		Demand Response Conserved Energy	\$ (1,577)	\$ (1,681)	\$ (1,744)	\$ (1,810)	\$ (1,865)	\$ (1,920)	\$ (1,978)	\$ (2,037)	\$ (2,097)	\$ (2,156)	\$ (2,215)	\$ (2,270)	\$ (24,767)
75		Property Tax (Savings) - Legacy Meters	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (1,067)	\$ (18,767)
76		LAUF Gas Reduction Enabled	\$ (201)	\$ (213)	\$ (225)	\$ (238)	\$ (243)	\$ (248)	\$ (252)	\$ (256)	\$ (260)	\$ (263)	\$ (267)	\$ (270)	\$ (3,502)
77		Terminal Value	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -	\$ (287,472)	\$ (287,472)
78		Total Non Capital Benefits before Costs	\$ (121,717)	\$ (133,440)	\$ (142,674)	\$ (147,185)	\$ (151,931)	\$ (157,039)	\$ (162,344)	\$ (167,827)	\$ (173,488)	\$ (179,315)	\$ (185,312)	\$ (191,489)	\$ (2,534,481)
79		Common	\$ 11,735	\$ 11,997	\$ 12,251	\$ 12,513	\$ 12,793	\$ 13,080	\$ 13,380	\$ 13,672	\$ 13,974	\$ 14,294	\$ 14,623	\$ 14,967	\$ 232,958
80		Meters, Modules & Communications	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 6,599	\$ 112,903
81		Other O&M	\$ 5,286	\$ 5,214	\$ 5,231	\$ 5,247	\$ 5,264	\$ 5,282	\$ 5,300	\$ 5,318	\$ 5,336	\$ 5,355	\$ 5,375	\$ 5,394	\$ 121,458
82		Load Control Program	\$ 2,530	\$ 2,425	\$ 631	\$ 582	\$ 599	\$ 617	\$ 636	\$ 656	\$ 677	\$ 698	\$ 721	\$ 745	\$ 23,311
83		Demand Response Program	\$ 874	\$ 879	\$ 1,226	\$ 669	\$ 683	\$ 696	\$ 710	\$ 724	\$ 739	\$ 754	\$ 769	\$ 784	\$ 14,147
84		Property Taxes - AMI	\$ 6,823	\$ 6,641	\$ 6,419	\$ 6,146	\$ 5,879	\$ 5,616	\$ 5,366	\$ 5,137	\$ 4,941	\$ 4,813	\$ 4,775	\$ 4,776	\$ 108,608
85		Total Non Capital Benefits Net	\$ (87,871)	\$ (99,685)	\$ (110,317)	\$ (115,429)	\$ (120,114)	\$ (125,149)	\$ (130,353)	\$ (135,520)	\$ (139,081)	\$ (142,638)	\$ (146,182)	\$ (149,718)	\$ (1,921,096)

Smart Grid Program

Line No.	(a) Year	(b) Description	(c) Pretax Return on AMI Investment		(d) AMI Depreciation		(e) Revenue Requirements - Avoided Capital Costs		(f) Total Non-Capital Benefits Net		(g) Net Revenue Requirements		(h) NPV Factor		(i) NPV of Net Revenue Requirements	
86	2009		\$3.3	+	\$0.1	+	\$0.0	+	\$0.7	=	\$4.1	x	1.000000	=	\$4.1	
87	2010		5.8	+	0.2	+	0.0	+	1.4	=	7.4	x	1.000000	=	7.4	
88	2011		8.8	+	4.8	+	0.0	+	1.2	=	14.8	x	1.000000	=	14.8	
89	2012	First Year of Electric AMI Meter Installations	11.7	+	10.9	+	0.0	+	4.6	=	27.2	x	1.000000	=	27.2	
90	2013		15.2	+	14.5	+	(0.1)	+	7.8	=	37.4	x	1.000000	=	37.4	
91	2014		19.2	+	18.8	+	(0.3)	+	4.7	=	42.4	x	1.000000	=	42.4	
92	2015	First Year of Gas Module Installations	25.4	+	24.7	+	(0.4)	+	(0.2)	=	49.6	x	1.000000	=	49.6	
93	2016		34.7	+	33.2	+	(0.4)	+	(8.1)	=	59.4	x	1.000000	=	59.4	
94	2017	Electric AMI Meter and Gas Module Installations complete	41.5	+	40.0	+	(0.4)	+	(17.5)	=	63.6	x	1.000000	=	63.6	
95	2018		41.7	+	42.5	+	(4.4)	+	(27.5)	=	52.3	x	1.000000	=	52.3	
96	2019		40.1	+	44.4	+	(4.2)	+	(27.6)	=	52.7	x	1.000000	=	52.7	
97	2020		38.5	+	46.5	+	(4.1)	+	(71.9)	=	9.0	x	1.000000	=	9.0	
98	2021		35.8	+	47.4	+	(4.0)	+	(87.9)	=	(8.7)	x	0.929109	=	(8.1)	
99	2022		33.0	+	48.2	+	(3.8)	+	(99.7)	=	(22.3)	x	0.863244	=	(19.3)	
100	2023		29.8	+	48.8	+	(3.5)	+	(110.3)	=	(35.3)	x	0.802047	=	(28.3)	
101	2024		26.4	+	49.1	+	(3.3)	+	(115.4)	=	(43.2)	x	0.745189	=	(32.2)	
102	2025		23.0	+	49.4	+	(3.0)	+	(120.1)	=	(50.8)	x	0.692362	=	(35.1)	
103	2026		19.7	+	45.2	+	(2.8)	+	(125.1)	=	(63.0)	x	0.643280	=	(40.5)	
104	2027		16.8	+	39.4	+	(2.6)	+	(130.4)	=	(76.7)	x	0.597677	=	(45.9)	
105	2028		14.2	+	36.2	+	(2.3)	+	(135.5)	=	(87.4)	x	0.555307	=	(48.5)	
106	2029		11.9	+	32.3	+	(2.1)	+	(139.1)	=	(96.9)	x	0.515941	=	(50.0)	
107	2030		10.0	+	27.7	+	(1.8)	+	(142.6)	=	(106.8)	x	0.479365	=	(51.2)	
108	2031		8.3	+	23.8	+	(1.6)	+	(146.2)	=	(115.6)	x	0.445383	=	(51.5)	
109	2032		6.9	+	21.5	+	(1.3)	+	(149.8)	=	(122.7)	x	0.413809	=	(50.8)	
110	2032	Model Terminal Value							(287.5)	=	(287.5)	x	0.413809	=	(119.0)	
111																
112		Total 2009-2032	\$521.7		\$749.5		(\$46.3)		(\$1,922.0)		(\$697.1)				(\$160.5)	

Smart Grid Program - Electric Portion

Line No.	(a) Year	(b) Description	(c)		(d)		(e)		(f)		(g)		(h)		(i)	
			Pretax Return on AMI Investment		AMI Depreciation		Revenue Requirements - Avoided Capital Costs		Total Non-Capital Benefits Net		Net Revenue Requirements		NPV Factor		NPV of Net Revenue Requirements	
113	2009		\$2.8	+	\$0.1	+	\$0.0	+	\$0.6	=	\$3.5	x	1.000000	=	\$3.5	
114	2010		5.1	+	0.2	+	0.0	+	1.2	=	6.4	x	1.000000	=	6.4	
115	2011		7.6	+	4.2	+	0.0	+	1.1	=	12.9	x	1.000000	=	12.9	
116	2012	First Year of Electric AMI Meter Installations	10.2	+	9.5	+	0.0	+	4.4	=	24.2	x	1.000000	=	24.2	
117	2013		13.4	+	12.8	+	(0.1)	+	7.8	=	33.9	x	1.000000	=	33.9	
118	2014		17.2	+	16.7	+	(0.3)	+	4.6	=	38.2	x	1.000000	=	38.2	
119	2015	First Year of Gas Module Installations	22.8	+	22.1	+	(0.4)	+	(0.3)	=	44.1	x	1.000000	=	44.1	
120	2016		30.7	+	29.3	+	(0.4)	+	(8.3)	=	51.3	x	1.000000	=	51.3	
121	2017	Electric AMI Meter and Gas Module Installations complete	36.1	+	34.9	+	(0.4)	+	(15.1)	=	55.5	x	1.000000	=	55.5	
122	2018		36.0	+	37.0	+	(4.4)	+	(22.8)	=	45.9	x	1.000000	=	45.9	
123	2019		34.8	+	38.9	+	(4.2)	+	(22.2)	=	47.4	x	1.000000	=	47.4	
124	2020		33.6	+	41.0	+	(4.1)	+	(63.6)	=	6.9	x	1.000000	=	6.9	
125	2021		31.3	+	41.8	+	(4.0)	+	(78.3)	=	(9.1)	x	0.929109	=	(8.5)	
126	2022		28.9	+	42.6	+	(3.8)	+	(88.7)	=	(21.0)	x	0.863244	=	(18.1)	
127	2023		26.1	+	43.2	+	(3.5)	+	(97.2)	=	(31.4)	x	0.802047	=	(25.2)	
128	2024		23.1	+	43.5	+	(3.3)	+	(101.7)	=	(38.4)	x	0.745189	=	(28.6)	
129	2025		20.1	+	43.8	+	(3.0)	+	(106.0)	=	(45.2)	x	0.692362	=	(31.3)	
130	2026		17.2	+	40.2	+	(2.8)	+	(110.7)	=	(56.1)	x	0.643280	=	(36.1)	
131	2027		14.6	+	35.1	+	(2.6)	+	(115.5)	=	(68.3)	x	0.597677	=	(40.8)	
132	2028		12.3	+	32.3	+	(2.3)	+	(120.2)	=	(78.0)	x	0.555307	=	(43.3)	
133	2029		10.3	+	28.6	+	(2.1)	+	(123.4)	=	(86.5)	x	0.515941	=	(44.6)	
134	2030		8.6	+	24.5	+	(1.8)	+	(126.5)	=	(95.3)	x	0.479365	=	(45.7)	
135	2031		7.2	+	20.9	+	(1.6)	+	(129.7)	=	(103.2)	x	0.445383	=	(46.0)	
136	2032		5.9	+	18.8	+	(1.3)	+	(132.8)	=	(109.5)	x	0.413809	=	(45.3)	
137	2032	Model Terminal Value							(253.0)	=	(253.0)	x	0.413809	=	(104.7)	
138																
139		Total 2009-2032	\$456.2		\$661.7		(\$46.3)		(\$1,696.2)		(\$624.5)				(\$147.8)	

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

SARAH R. NIELSEN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Case No.: U-20697
Exhibit No.: A-90 (SRN-1)
Page: 1 of 1
Witness: SRNielsen
Date: February 2020

Fleet EV Benefit Analysis for Consumers Energy Territory

SECONDARY CUSTOMERS						PRIMARY CUSTOMERS					
Source						Source					
Capacity Calculation											
Total Capacity	269,405	\$ in thousands	Exhibit A-1 (HWM-2), line 6			298,941	\$ in thousands	Exhibit A-1 (HWM-2), line 6			
Full Service Deliveries	7,086,449	MWh	Exhibit A-1 (HWM-1), line 9			10,627,497	MWh	Exhibit A-1 (HWM-1), line 14			
Capacity per kWh	0.0380	\$ / kWh				0.0281	\$ / kWh				
Distribution Calculation											
Total Distribution	334,323	\$ in thousands	Exhibit A-1 (HWM-2), line 16			118,175	\$ in thousands	Exhibit A-1 (HWM-2), line 16			
Total Deliveries	7,291,788	MWh	Exhibit A-1 (HWM-1), lines 9&26			14,119,585	MWh	Exhibit A-1 (HWM-1), lines 14&29			
Distribution per kWh	0.0458	\$ / kWh				0.0084	\$ / kWh				
Gross Margin Calculation - Additional Capacity and Distribution Revenue From Charging											
Capacity	18,717	kWh	0.0380	\$ / kWh	\$ 712	18,717	kWh	0.0281	\$ / kWh	\$ 526	
Distribution	18,717	kWh	0.0458	\$ / kWh	858	18,717	kWh	0.0084	\$ / kWh	157	
Total					\$ 1,570					\$ 683	
Lifetime Value Analysis											
Annual gross margin of additional EV on the grid				\$ 1,570					\$ 683		
Useful life (years)				10					10		
Discount rate (WACC from Rebuttal)				6.08%					6.08%		
NPV of an additional EV				\$ 11,510					\$ 5,009		

Assumptions

All charging is off-peak

Additional capacity or distribution required will be an incremental cost to this analysis

Annual EV energy use for a medium-duty delivery truck

Miles driven ¹	12,958	12,958
Avg mi/kWh ²	0.69	0.69
EV kWh	18,717	18,717

Benefit Summary for Select Electric Vehicle Types

Vehicle	Annual Mileage ¹	Avg mi/ kWh ²	Secondary Value	Primary Value
Medium-duty delivery truck	12,958	0.7	\$11,510	\$5,009
School bus	12,000	0.5	\$13,528	\$5,888
Transit bus	34,012	0.5	\$42,085	\$18,315

¹ Federal Highway Administration. (2016). Average Annual Vehicle Miles Traveled by Major Vehicle Categories. Retrieved by <https://afdc.energy.gov/data/10309>

² Vehicle manufacturer or vehicle manufacturer website

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

PowerMIDrive Costs

Summary of 2019 Actual and Projected O&M Costs
(\$000)

Case No.: U-20697
Exhibit No.: A-91 (SRN-2)
Page: 1 of 1
Witness: SRNielsen
Date: February 2020

		2019												
		Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Full Year
Cost Centers	Description	Actual	Actual	Actual	Actual	Actual	Actual	Actual	Actual	Actual	Actual	Projected	Projected	Projected
PLUG IN ELECTRIC VEHICLES														
14136 Labor		\$ -	\$ 21,168	\$ 13,684	\$ 17,229	\$ 17,419	\$ 17,229	\$ 17,469	\$ 17,469	\$ 17,469	\$ 17,469	\$ 17,469	\$ 17,469	\$ 191,546
14137 Non Labor		-	-	-	3,614	37	-	-	-	-	381	-	-	4,031
	Employee Training	-	-	-	-	-	-	-	-	-	-	-	-	-
	Employee Dues & Membership	-	-	-	-	-	-	-	-	-	-	-	-	-
	Misc Travel	-	-	-	-	-	1,939	636	-	-	-	1,288	351	4,214
	CGI	-	-	-	-	-	9,337	-	-	-	-	-	-	9,337
Internal Orders														
6509071	Other Outside Services (Labor Loadings)	-	10,372	6,705	8,442	8,535	8,442	8,560	8,560	8,560	8,560	8,522	8,522	93,782
6509071	Other Outside Services (Website & IT)	-	-	-	45,200	-	(4,884)	107	295	-	-	201	201	41,119
6509072	Other Outside Services (Educ & Outreach)	-	-	-	-	6,686	371	114,867	-	13,217	15,216	14,216	14,216	178,790
6509073	Residential L2 Rebates	-	-	-	-	-	10,000	10,000	6,400	12,400	15,700	15,000	15,000	84,500
6509074	Public/Workplace L2 Rebates	-	-	-	-	-	-	-	15,000	-	5,000	10,000	10,000	40,000
6509075	Public DCFC Rebates	-	-	-	-	-	-	-	-	-	-	-	-	-
Total		\$ -	\$ 31,541	\$ 20,389	\$ 74,485	\$ 32,677	\$ 42,435	\$ 151,639	\$ 47,725	\$ 51,646	\$ 62,326	\$ 66,696	\$ 65,760	\$ 647,319

References Summary

Testimony

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STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

LATINA D. SABA

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule B-5.6

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Operations Support

Summary of Actual & Projected Electric and Common Capital Expenditures

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (LDS-1)

Schedule: B-5.6

Page: 1 of 1

Witness: LDSaba

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
Line		Historical	Projected Bridge Year		Projected Test Year	
No.	Program Description	12 Mos Ended 12/31/2018	12 Mos Ending 12/31/2019	12 Mos Ending 12/31/2020	24 Mos Ending 12/31/2020	12 mos. Ending 12/31/2021
1	Asset Preservation	25,201	20,546	21,503	42,049	64,585
2	Contractor	21,039	17,153	17,952	35,105	53,919
3	Labor	562	459	480	939	1,442
4	Materials	2,482	2,023	2,118	4,141	6,360
5	Business Expenses	67	54	57	111	171
6	Other (Loadings, Chargebacks)	1,051	857	896	1,753	2,693
7	Computer & Other Equipment	1,046	194	271	465	271
8	Total Capital	26,247	20,740	21,774	42,514	64,856

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Operations Support O&M Expenses

For the Year 2018 and Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-93 (LDS-2)

Page: 1 of 1

Witness: LDSaba

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
Line No.	Description	Historical 12 mos. ended 12/31/2018	Projected 12 mos. ending 12/31/2019	Projected 12 mos. ending 12/31/2020	Projected 12 mos. ending 12/31/2021	Source
1	Facilities	\$14,312	\$ 13,315	\$ 13,144	\$ 13,345	
	Labor	\$3,353	\$ 3,119	\$ 3,509	\$ 3,538	
	Non Labor	\$10,959	\$ 10,196	\$ 9,635	\$ 9,807	
2	Real Estate	\$1,749	\$ 1,923	\$ 2,113	\$ 1,992	
	Labor	\$1,048	\$ 1,152	\$ 1,339	\$ 1,262	
	Non Labor	\$701	\$ 771	\$ 774	\$ 730	
3	Administrative Operations	\$ 1,827	\$ 1,608	\$ 1,699	\$ 1,728	
	Labor	\$ 744	\$ 655	\$ 673	\$ 684	
	Non Labor	\$ 1,083	\$ 953	\$ 1,026	\$ 1,044	

Michigan Public Service Commission

Consumers Energy Company

Detailed List of Projected Electric & Common Capital Expenditures

For the Years 12 months ending 12/31/2019, 12 months ending 12/31/2020 and 12 months ending 12/31/2021

Case No: U-20697

Exhibit No: A-94 (LDS-3)

Page: 1 of 1

Witness: LDSaba

Date: February 2020

Line No.	Program Description	12 months ending 12/31/2019 Projected	12 months ending 12/31/2020 Projected	12 months ending 12/31/2021 Projected
1	Wellness Equip	\$ 22	\$ 51	\$ 51
2	Computer Equipment	\$ 32	\$ 38	\$ 38
3	Print Equipment	\$ 28	\$ 28	\$ 28
4	Real Estate Survey Equip	\$ -	\$ 10	\$ 10
5	Supply Chain Equip	\$ 71	\$ 101	\$ 101
6	Facilities Tools	\$ 40	\$ 43	\$ 43
7	Asset Preservation - FIIB	\$ 1,844	\$ 1,103	\$ 1,097
8	Statewide Paving	\$ 2,165	\$ 1,865	\$ 1,933
9	Statewide Roofing	\$ 3,108	\$ 4,923	\$ 3,467
10	Statewide Mechanical/Electrical	\$ 2,547	\$ 619	\$ 890
11	Statewide Elevators	\$ 677	\$ 772	\$ 1,945
12	Furniture	\$ 544	\$ 81	\$ 624
13	Grand Rapids Circuit 501	\$ 1,570	\$ 2,805	\$ 26,484
14	Unified Control Center	\$ -	\$ -	\$ 1,000
15	Lansing Service Center	\$ 41	\$ 1,085	\$ 4,339
16	Hastings Service Center	\$ 38	\$ 542	\$ 5,695
17	OEP Parking Decks	\$ 153	\$ 163	\$ -
18	Tawas	\$ 4,800	\$ 4,346	\$ -
19	Kalamazoo Service Center	\$ 217	\$ 1,119	\$ 15,533
20	Parnall Renovation	\$ 1,098	\$ 1,866	\$ 1,389
21	EV Charging Stations	\$ -	\$ 213	\$ 190
22	OEP Fusion Center	\$ 557	\$ -	\$ -
23	Gratiot Wind Building	\$ 147	\$ -	\$ -
24	Crescent Wind Building	\$ 169	\$ -	\$ -
25	Zeeland Storage Building	\$ 318	\$ -	\$ -
26	Croton Storage Building	\$ 284	\$ -	\$ -
27	Coldwater Service Center	\$ 270	\$ -	\$ -
28	TOTAL CAPITAL EXPENDITURES	\$ 20,740	\$ 21,774	\$ 64,856
29	Exhibit Lines:			
30	Other Equipment			
31	Asset Preservation			

Facility Assessment - Lansing Service Center

	Lansing	Maximum Possible Points	Serviceable 45 +/- 15 Points	Poor < 30 Points	Good > 60 Points
Safety					
Compatible Use	1	5	3	2	4
Site Conditions	3	5	3	2	4
Traffic/Site Flow	1	5	3	2	4
Environmental Hazard	1	5	3	2	4
Quality					
<u>Employee/Customer Engagement</u>					
Branding	2	5	3	2	4
Employee Pride	2	5	3	2	4
Workplace Efficiency	1	5	3	2	4
Wellness	2	5	3	2	4
Retention	2	5	3	2	4
Cost					
Facility Operating Costs	4	5	3	2	4
Non Facility Cost Impact	2	5	3	2	4
Space Optimization	1	5	3	2	4
Delivery					
<u>Customer Service</u>					
Response Time	3	5	3	2	4
Customer Access	2	5	3	2	4
Sustainability	1	5	3	2	4
Total	28	75	45	30	60

Facilities Site Conditions - Lansing Service Center

Facilities Site Conditions

Lansing SC

*Former Manufactured Gas Plant site (EPA Part 201)
*Built outside of electric service territory
*Not centrally located within electric service territory
*Not centrally located within gas service territory
*Location far from highway
*PCB Contamination
*Lead paint contamination
*Asbestos insulation contamination (duct/pipe)
*Located within residential neighborhood
*Multiple law enforcement incidents on property and in neighborhood

Facilities Scoring Criteria - Lansing Service Center

Safety		Comments	Pro	Con
Compatible Use	Use of adjoining properties and surrounding area.	Industrial zoning and use	Residential use, retail use, criminal activity	
Site Conditions	Physical condition of site	Full utilities, good paving, fencing, drainage	No utilities, poor paving, fencing, drainage	
Traffic/Site Flow	Road network, site access drive, vehicle turning space	Close to highway, access from two roads, space for truck trailer to pull off road, one way gates	Secondary road, no truck pull off space at gate, two way gates, insufficient turning space	
Environmental Hazard	Environmental hazards	LEED certification, no environmental hazards	Former MGP site, soil contamination, abandoned underground fuel tanks, abandoned transformer storage, asbestos fire proofing, asbestos duct wrap, asbestos pipe wrap, lead paint,	
Quality				
<u>Employee/Customer Engagement</u>				
Branding	Projects positive image to community and potential employees	Well cared for facility, curb appeal, neat landscaping, coordinated site signage	Dilapidated facility, dated appearance, overgrown landscaping, mismatched site signage	
Workplace Efficiency	Space is fully utilized	Space provided for all operational functions, space size meets needs, proper inventory stored	Space not provided for all operational functions, space oversized, space not fully utilized, unused materials (junk) stored	
Wellness	Wellness facilities	Wellness room, space for trainer, drinking fountain, shower facilities nearby, space for wellness classes, well maintained exercise equipment, mirrors, lockers	Wellness room combined with non-compatible uses (ie: conference room), aged exercise equipment, poorly maintained exercise equipment, remote shower facilities	
Cost				
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	
Delivery				
<u>Customer Service</u>				
Response Time	Service area of proper size	Centrally located within service area(s)	Offset location within service area(s)	
Customer Access	Meets needs of customers	Direct payment office available, meter pick up available	No direct payment office available, no meter pick up available	
Sustainability	Site will serve business for foreseeable future	Area of growing service territory, area of stable territory	Customer base shrinking within service territory	
Morale				
<u>Employee Wellness</u>				
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	
Healthy Environment	Space environment supports healthy employees	Multiple types of work space, natural light, collaborative work areas	Uniform or static work areas, artificial light, no collaborative work areas	

Facility Assessment - Kalamazoo Service Center

	Kalamazoo	Maximum Possible Points	Servicable 45 +/- 15 Points	Poor < 30 Points	Good > 60 Points
Safety					
Compatible Use	3	5	3	2	4
Site Conditions	5	5	3	2	4
Traffic/Site Flow	2	5	3	2	4
Environmental Hazard	1	5	3	2	4
Quality					
<u>Employee/Customer Engagement</u>					
Branding	3	5	3	2	4
Employee Pride	3	5	3	2	4
Workplace Efficiency	1	5	3	2	4
Wellness	1	5	3	2	4
Retention	3	5	3	2	4
Cost					
Facility Operating Costs	5	5	3	2	4
Non Facility Cost Impact	4	5	3	2	4
Space Optimization	1	5	3	2	4
Delivery					
<u>Customer Service</u>					
Response Time	5	5	3	2	4
Customer Access	4	5	3	2	4
Sustainability	4	5	3	2	4
Total	45	75	45	30	60

Facilities Site Conditions - Kalamazoo Service Center

Facilities Site Conditions

Kalamazoo SC

*First floor areas have been vacated due to indoor air quality issues

*Unable to access ceiling plenum due to asbestos issues

*Asbestos contamination (fireproofing, duct/pipe insulation)

*Lead paint contamination

*Centrally located within service territory

*Location close to highway

Facilities Scoring Criteria - Kalamazoo Service Center

Safety		Comments	Pro	Con
Compatible Use	Use of adjoining properties and surrounding area.	Industrial zoning and use	Residential use, retail use, criminal activity	
Site Conditions	Physical condition of site	Full utilities, good paving, fencing, drainage	No utilities, poor paving, fencing, drainage	
Traffic/Site Flow	Road network, site access drive, vehicle turning space	Close to highway, access from two roads, space for truck trailer to pull off road, one way gates	Secondary road, no truck pull off space at gate, two way gates, insufficient turning space	
Environmental Hazard	Environmental hazards	LEED certification, no environmental hazards	Former MGP site, soil contamination, abandoned underground fuel tanks, abandoned transformer storage, asbestos fire proofing, asbestos duct wrap, asbestos pipe wrap, lead paint,	
Quality				
<u>Employee/Customer Engagement</u>				
Branding	Projects positive image to community and potential employees	Well cared for facility, curb appeal, neat landscaping, coordinated site signage	Dilapidated facility, dated appearance, overgrown landscaping, mismatched site signage	
Workplace Efficiency	Space is fully utilized	Space provided for all operational functions, space size meets needs, proper inventory stored	Space not provided for all operational functions, space oversized, space not fully utilized, unused materials (junk) stored	
Wellness	Wellness facilities	Wellness room, space for trainer, drinking fountain, shower facilities nearby, space for wellness classes, well maintained exercise equipment, mirrors, lockers	Wellness room combined with non-compatible uses (ie: conference room), aged exercise equipment, poorly maintained exercise equipment, remote shower facilities	
Cost				
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	
Delivery				
<u>Customer Service</u>				
Response Time	Service area of proper size	Centrally located within service area(s)	Offset location within service area(s)	
Customer Access	Meets needs of customers	Direct payment office available, meter pick up available	No direct payment office available, no meter pick up available	
Sustainability	Site will serve business for foreseeable future	Area of growing service territory, area of stable territory	Customer base shrinking within service territory	
Morale				
<u>Employee Wellness</u>				
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	
Healthy Environment	Space environment supports healthy employees	Multiple types of work space, natural light, collaborative work areas	Uniform or static work areas, artificial light, no collaborative work areas	

Facility Assessment - Hastings Service Center

	Hastings	Maximum Possible Points	Servicable 45 +/- 15 Points	Poor < 30 Points	Good > 60 Points
Safety					
Compatible Use	4	5	3	2	4
Site Conditions	1	5	3	2	4
Traffic/Site Flow	1	5	3	2	4
Environmental Hazard	4	5	3	2	4
Quality					
Employee/Customer Engagement					
Branding	2	5	3	2	4
Employee Pride	2	5	3	2	4
Workplace Efficiency	1	5	3	2	4
Wellness	1	5	3	2	4
Retention	2	5	3	2	4
Cost					
Facility Operating Costs	1	5	3	2	4
Non Facility Cost Impact	4	5	3	2	4
Space Optimization	3	5	3	2	4
Delivery					
Customer Service					
Response Time	4	5	3	2	4
Customer Access	3	5	3	2	4
Sustainability	1	5	3	2	4
Total	34	75	45	30	60

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Case No.: U-20697

Exhibit No.: A-97 (LDS-6)

Page: 2 of 3

Witness: LDSaba

Date: February 2020

Facilities Site Conditions - Hastings Service Center

*Centrally located within service territory

*Location close to State highway

Facilities Scoring Criteria - Hastings Service Center

Safety		Comments	Pro	Con
Compatible Use	Use of adjoining properties and surrounding area.	Industrial zoning and use	Residential use, retail use, criminal activity	
Site Conditions	Physical condition of site	Full utilities, good paving, fencing, drainage	No utilities, poor paving, fencing, drainage	
Traffic/Site Flow	Road network, site access drive, vehicle turning space	Close to highway, access from two roads, space for truck trailer to pull off road, one way gates	Secondary road, no truck pull off space at gate, two way gates, insufficient turning space	
Environmental Hazard	Environmental hazards	LEED certification, no environmental hazards	Former MGP site, soil contamination, abandoned underground fuel tanks, abandoned transformer storage, asbestos fire proofing, asbestos duct wrap, asbestos pipe wrap, lead paint,	
Quality				
Employee/Customer Engagement				
Branding	Projects positive image to community and potential employees	Well cared for facility, curb appeal, neat landscaping, coordinated site signage	Dilapidated facility, dated appearance, overgrown landscaping, mismatched site signage	
Workplace Efficiency	Space is fully utilized	Space provided for all operational functions, space size meets needs, proper inventory stored	Space not provided for all operational functions, space oversized, space not fully utilized, unused materials (junk) stored	
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Cost				
Facility Operating Costs	O&M costs to operate and maintain facility	Cost per square foot for facility less than statewide average cost per square foot (\$5.31)	Cost per square foot for facility more than statewide average cost per square foot (\$5.31)	
Non Facility Cost Impact	O&M cost impacts to operations	Building correct size, centrally located within customer base area, able to accommodate necessary materials, able to maneuver onsite	Building too large, eccentrically located within customer base area, unable to accommodate necessary materials, difficult to maneuver onsite	
Space Optimization	Reduce excess square footage	Space fully utilized, space utilized efficiently	Space under utilized, space utilized inefficiently, indoor parking area	
Delivery				
Customer Service				
Response Time	Service area of proper size	Centrally located within service area(s)	Offset location within service area(s)	
Customer Access	Meets needs of customers	Direct payment office available, meter pick up available	No direct payment office available, no meter pick up available	
Sustainability	Site will serve business for foreseeable future	Area of growing service territory, area of stable territory	Customer base shrinking within service territory	
Morale				
Employee Wellness				
Employee Pride	Employees are engaged	Employee engagement scores > company average	Employee engagement scores < company average	
Retention	Facilities located in a desirable area, space is attractive to Millennials and supports millennial work styles	Downtown location, local amenities available, space supports varied work styles	Remote location, few local amenities, fixed work space	
Healthy Environment	Space environment supports healthy employees	Multiple types of work space, natural light, collaborative work areas	Uniform or static work areas, artificial light, no collaborative work areas	

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

CHRIS A. SHELLBERG

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Line Clearing O&M Expense

2017 through 2019 Historical/2020 through 2025 Projected

Case No.: U-20697

Exhibit No.: A-98 (CAS-1)

Page: 1 of 1

Witness: CAShellberg

Date: February 2020

Line No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
				Historical			Projected					
				2017	2018	2019	2020	2021	2022	2023	2024	2025
1	Electric System	Expense Type										
2		Electric HVD		\$11,393,937	\$12,035,797	\$11,474,854	\$11,645,000	\$12,570,000	\$12,840,000	\$13,160,000	\$13,480,000	\$13,805,000
3		Contractor	Maintenance Clearing	\$7,658,022	\$8,438,081	\$7,390,660	\$6,755,000	\$6,925,000	\$7,100,000	\$7,270,000	\$7,450,000	\$7,635,000
4		Contractor	Brushing - Cut	\$2,388,351	\$2,682,986	\$3,296,287	\$2,340,000	\$2,400,000	\$2,460,000	\$2,520,000	\$2,585,000	\$2,650,000
5		Contractor	Brushing - Spray	N/A	N/A	N/A	\$1,750,000	\$2,310,000	\$2,320,000	\$2,390,000	\$2,450,000	\$2,505,000
6		Contractor	Demand Clearing	\$375,620	\$152,548	\$84,242	\$75,000	\$75,000	\$75,000	\$75,000	\$65,000	\$60,000
7		Contractor	Noxious Weed Control	\$20,546	\$70,093	\$24,009	\$35,000	\$40,000	\$45,000	\$45,000	\$50,000	\$50,000
8		Labor	HVD Salaries	\$709,395	\$666,847	\$653,501	\$660,000	\$760,000	\$775,000	\$790,000	\$810,000	\$830,000
9		Other	HVD Expenses	\$242,003	\$25,242	\$26,155	\$30,000	\$60,000	\$65,000	\$70,000	\$70,000	\$75,000
10			Miles	1,019	1,081	1,064	1,065	1,125	1,127	1,130	1,132	1,134
11		Electric LVD		\$38,359,929	\$39,912,561	\$41,525,147	\$41,355,000	\$71,430,000	\$81,485,000	\$86,840,000	\$104,100,000	\$106,520,000
12		Contractor	Maintenance Clearing	\$29,372,207	\$31,038,484	\$33,481,817	\$30,325,000	\$54,265,000	\$62,912,000	\$66,236,000	\$81,032,000	\$83,515,000
13		Contractor	Repetitive Outage Zone	\$934,465	\$2,054,288	\$795,749	\$1,500,000	\$2,000,000	\$1,900,000	\$1,700,000	\$1,500,000	\$1,350,000
14		Contractor	CEMI Clearing	N/A	N/A	N/A	\$1,700,000	\$3,125,000	\$4,125,000	\$5,125,000	\$6,125,000	\$6,125,000
15		Contractor	First Zone Clearing	\$399,793	\$102,600	\$1,172,518	\$900,000	\$1,875,000	\$1,975,000	\$2,175,000	\$2,375,000	\$2,525,000
16		Contractor	Demand Clearing	\$2,642,665	\$1,444,228	\$1,541,213	\$1,700,000	\$1,235,000	\$1,173,000	\$1,114,000	\$1,058,000	\$1,005,000
17		Contractor	Brushing - Spray	\$1,807,622	\$1,679,301	\$788,688	\$1,450,000	\$2,900,000	\$2,900,000	\$3,190,000.0	\$3,510,000.0	\$3,500,000
18		Labor	Distribution Salaries	\$2,388,399	\$2,532,425	\$2,671,931	\$2,700,000	\$4,305,000	\$4,640,000	\$5,210,000	\$6,070,000	\$6,070,000
19		Other	Distribution Expenses	\$814,778	\$1,061,235	\$1,073,231	\$1,080,000	\$1,725,000	\$1,860,000	\$2,090,000	\$2,430,000	\$2,430,000
20			Miles	3,503	3,218	3,106	3,166	5,223	5,986	6,346	7,654	7,914
21		Electric Total	O&M Expense	\$49,753,866	\$51,948,358	\$53,000,001	\$53,000,000	\$84,000,000	\$94,325,000	\$100,000,000	\$117,580,000	\$120,325,000
22		Contractor	Clearing	\$45,599,291	\$47,662,609	\$48,575,183	\$48,530,000	\$77,150,000	\$86,985,000	\$91,840,000	\$108,200,000	\$110,920,000
23		Labor	Salaries	\$3,097,794	\$3,199,272	\$3,325,432	\$3,360,000	\$5,065,000	\$5,415,000	\$6,000,000	\$6,880,000	\$6,900,000
24		Other	Expenses	\$1,056,781	\$1,086,477	\$1,099,386	\$1,110,000	\$1,785,000	\$1,925,000	\$2,160,000	\$2,500,000	\$2,505,000
25			O&M Miles	4,522	4,299	4,170	4,231	6,348	7,113	7,476	8,786	9,048
26			HVD Cost/Mile	\$11,181	\$11,134	\$10,785	\$10,934	\$11,173	\$11,391	\$11,651	\$11,911	\$12,173
27			LVD Cost/Mile	\$10,951	\$12,403	\$13,369	\$13,062	\$13,676	\$13,613	\$13,684	\$13,600	\$13,460
28			LVD Target Cycle Miles	8,362	8,362	8,362	8,362	8,362	8,362	8,362	8,362	8,362
29			LVD O&M Program Miles	3,503	3,218	3,106	3,166	5,223	5,986	6,346	7,654	7,914
30			Capital Clearing Miles	425	393	340	350	575	625	615	600	475
31			Percent of 7-year Cycle	47%	43%	41%	42%	69%	79%	83%	99%	100%

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Line Clearing Ramp-up Plan Estimated Service Restoration Reductions

Case No.: U-20697

Exhibit No.: A-99 (CAS-2)

Page: 1 of 1

Witness: CASHellberg

Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e) Projected	(f)	(g)	(h)
		2019	2020	2021	2022	2023	2024	2025
1	Line Clearing Program Funding Level (\$million)	\$53.0	\$53.0	\$84.0	\$94.3	\$100.0	\$117.6	\$120.3
2	Projected Tree-Related Primary Outage Incidents with MED	10,919	10,620	10,257	9,663	8,850	8,318	7,897
3	Outage Incidents Reduced with MED			363	957	1,770	2,302	2,723
4	Outage Incident Reduction Percent			3.42%	9.01%	16.67%	21.68%	25.64%
5	Service Restoration Savings Potential (\$million)			\$0.381	\$1.005	\$1.859	\$2.417	\$2.859

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
2021 HVD Line Clearing Work Plan

Case No.: U-20697
Exhibit No.: A-100 (CAS-3)
Page: 1 of 1
Witness: CASHellberg
Date: February 2020

Line No.	(a) Description	(b) Voltage	(c) Miles	(d) Last Year Cleared	(e) Line Name	(f) Voltage	(g) Miles	(h) Last Year Cleared
1	ACUGLASS	138 kV	20.46	2017	LAKE CITY	46 kV	15.77	2017
2	ALMA-BEGOLE	138 kV	0.98	2017	LAKE ODESSA	46 kV	15.75	2017
3	AMASTEEL	138 kV	1.42	2017	MACKINAW	46 kV	5.48	2017
4	AREGENTA-MILHAM #2	138 kV	1.18	2017	MAUMEE	46 kV	7.25	2017
5	BALDWIN	46 kV	9.29	2017	MCBAIN	46 kV	21.23	2017
6	BALLENGER	46 kV	3.51	2017	MENDON	46 kV	31.80	2017
7	BATH	46 kV	30.66	2017	MERSON	46 kV	13.78	2017
8	BEAVER CREEK - HIGGINS	138 kV	0.05	2017	MIO - AIRPORT	138 kV	0.07	2016
9	BELSAY	46 kV	7.28	2017	MIO - GAYLORD	138 kV	1.88	2017
10	BEVERIDGE - VAN SLYKE	46 kV	3.20	2017	MONA LAKE	46 kV	8.82	2017
11	BRISTOL	46 kV	13.66	2017	MT PLEASANT	46 kV	3.89	2017
12	BULLOCK - SAGINAW RIVER	138 kV	0.12	2017	MUSKEGON #2	138 kV	0.45	2017
13	CADILLAC	46 kV	2.73	2017	NEW RICHMOND	46 kV	27.00	2017
14	CAMPUS	46 kV	5.05	2017	NEWAYGO	46 kV	18.61	2017
15	CASINO	46 kV	9.72	2017	O-15 (CROTON - FOUR MILE)	138 kV	9.13	2017
16	CHEBOYGAN	46 kV	11.62	2017	PARR RD-WHITING	138 kV	0.47	2017
17	CHELSEA	46 kV	1.12	2017	PATTERSON	46 kV	9.19	2017
18	CHRYSLER	46 kV	11.91	2017	PEACH RIDGE	46 kV	22.13	2017
19	CLAREMONT - MANNING	138 kV	0.37	2017	PHILLIPS #1	46 kV	7.80	2017
20	COLLEGE PARK	46 kV	1.40	2017	PHILLIPS #2	46 kV	5.58	2017
21	EAST JORDAN	46 kV	19.99	2017	PORTAGE	46 kV	2.01	2017
22	EATON RAPIDS	46 kV	34.75	2017	RAISIN	46 kV	1.19	2017
23	ELK RAPIDS - BOARDMAN	46 kV	0.91	2016	RANGER LAKE	46 kV	24.63	2017
24	FRUITPORT	46 kV	19.65	2017	RAVENNA	46 kV	36.61	2017
25	GALLAGHER-TWINING	138 kV	3.15	2017	REED CITY	46 kV	56.74	2017
26	GETTY	46 kV	10.06	2017	RIGGSVILLE - LIVINGSTON	138 kV	2.67	2017
27	GLEN OAKS	46 kV	10.23	2017	ROSEBUSH	46 kV	23.12	2017
28	GOSS - BEVERIDGE	138 kV	0.06	2017	SARANAC	46 kV	33.57	2017
29	GRASS LAKE	46 kV	28.29	2017	SPARTAN (SPARTAN - DELHI)	46 kV	7.61	2017
30	GUN LAKE	46 kV	24.31	2017	ST CHARLES	46 kV	47.41	2017
31	HANOVER	46 kV	33.42	2017	STAUFFER	46 kV	11.96	2017
32	HARLEM	46 kV	6.13	2017	STOCKBRIDGE	46 kV	37.78	2017
33	HENRY	46 kV	6.11	2017	SUMMERTON	138 kV	0.74	2017
34	HICKORY	46 kV	3.66	2017	TECUMSEH	46 kV	0.06	2017
35	HILE ROAD	138 kV	0.03	2017	TECUMSEH PRODUCTS	46 kV	11.00	2017
36	HUDSONVILLE	46 kV	18.15	2017	TERRACE	46 kV	3.05	2017
37	HUNT ROAD	46 kV	11.60	2017	TUSTIN	46 kV	13.94	2017
38	INDIAN RIVER	46 kV	9.54	2017	VENICE	46 kV	47.92	2017
39	ITHACA	46 kV	12.88	2016	VIRGINIA PARK	46 kV	17.33	2017
40	KALARAMA	46 kV	2.43	2017	WEST BRANCH	46 kV	39.79	2017
41	KEATING	46 kV	8.73	2017	WESTERN AVENUE	46 kV	2.14	2017
42	KENT CITY (ALGOMA - FOUR MILE)	46 kV	30.01	2017	WESTPHALIA	46 kV	9.56	2017
43	KILGORE	46 kV	4.08	2017	WILLIAMS	46 kV	22.07	2017
44	L-12	138 kV	1.35	2017	WIRTZ ROAD	46 kV	28.57	2017
45	Total Plan Miles	1142.80						

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Line Clearing Reliability Results
Reliability Pre and Post Clearing for 2017 Cleared LVD Circuits

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f)	(g)	(h)
1	Circuits Cleared in 2017				Tree Related Outage Incidents			
2	FEEDER ID	WHQ	Substation	Circuit	Voltage	2016	2017	2018
3	403	Clare	COLEMAN	COLEMAN	4.8/8.32 kV Wye Grounded	6	9	4
4	4802	Jackson	PARMA	BALDWIN	4.8/8.32 kV Wye Grounded	8	4	5
5	7304	Big Rapids	MILTON	MILLPOND	7.2/12.47 kV Wye Grounded	4	0	3
6	9801	Flint	PORTER	PARTS	4.8/8.32 kV Wye Grounded	3	0	0
7	10001	West Kent	WEALTHY STREET	NORTHWEST	7.2/12.47 kV Wye Grounded	3	1	1
8	10403	Clare	GLADWIN	GLADWIN	4.8/8.32 kV Wye Grounded	13	8	1
9	13701	Midland	SANFORD DAM	AVERILL	4.8/8.32 kV Wye Grounded	5	5	6
10	13801	West Branch	SMALLWOOD DAM	WOODENSHOE	4.8/8.32 kV Wye Grounded	10	5	6
11	14501	Greenville	GREENVILLE	WILLIAMS	4.8/8.32 kV Wye Grounded	0	1	0
12	19602	Battle Creek	CONVIS	WALNUT PT	14.4/24.9 kV Wye Grounded	14	7	10
13	20502	West Branch	WEST BRANCH	REFINERY	4.8/8.32 kV Wye Grounded	6	1	0
14	21801	Jackson	WILDWOOD	YARDMAN	4.8/8.32 kV Wye Grounded	5	5	1
15	24102	West Kent	DOEHLER JARVIS	GRIGGS STREET	7.2/12.47 kV Wye Grounded	1	0	1
16	25103	Muskegon	NORTH MUSKEGON	STATE PARK	7.2/12.47 kV Wye Grounded	1	1	6
17	26003	Greenville	SARANAC	SARANAC	4.8/8.32 kV Wye Grounded	5	4	3
18	26004	Greenville	SARANAC	RIVERSIDE	4.8/8.32 kV Wye Grounded	6	3	2
19	29001	Alma	BRECKENRIDGE	WHEELER	4.8/8.32 kV Wye Grounded	2	1	6
20	29201	Jackson	MANCHESTER	LOGAN ROAD	4.8/8.32 kV Wye Grounded	6	1	0
21	29302	West Branch	MARKEY	CARRICK	4.8/8.32 kV Wye Grounded	3	4	0
22	29902	Muskegon	NORTON	HILE ROAD	7.2/12.47 kV Wye Grounded	12	1	1
23	30302	Battle Creek	JOPPA	JOPPA	4.8/8.32 kV Wye Grounded	10	2	5
24	32404	Big Rapids	HOWARD CITY	CORAL	14.4/24.9 kV Wye Grounded	7	2	11
25	32701	Kalamazoo	COMSTOCK	SHIELDS	4.8/8.32 kV Wye Grounded	3	3	2
26	33803	Clare	HARRISON	STOCKWELL	4.8/8.32 kV Wye Grounded	10	3	2
27	33901	Saginaw	BURROWS	WHEELER	4.8/8.32 kV Wye Grounded	0	0	1
28	34101	Saginaw	POTTER	KIRK	4.8/8.32 kV Wye Grounded	0	2	0
29	34203	Muskegon	HICKORY	DOGWOOD	7.2/12.47 kV Wye Grounded	1	1	0
30	35002	Muskegon	FRUITPORT	JUDSON	7.2/12.47 kV Wye Grounded	8	5	4
31	35802	Adrian	FRONTIER	FRONTIER	4.8/8.32 kV Wye Grounded	1	0	3
32	37002	Hamilton	OTTAWA BEACH	PORTSHELDN	4.8/8.32 kV Wye Grounded	4	5	6
33	37101	Hastings	DELTON	CLOVERDALE	4.8/8.32 kV Wye Grounded	13	7	6
34	37502	Coldwater	CAMDEN	MONTGOMERY	4.8/8.32 kV Wye Grounded	2	1	2
35	39302	Saginaw	NIAGARA	HAMILTON	4.8/8.32 kV Wye Grounded	3	0	1
36	39601	West Branch	HOUGHTON HEIGHTS	MERRITT	14.4/24.9 kV Wye Grounded	8	11	7
37	40702	Adrian	ONSTED	ONSTED	4.8/8.32 kV Wye Grounded	8	4	2
38	41601	Lansing	WHITTUM	KINNEVILLE	4.8/8.32 kV Wye Grounded	1	0	0
39	43901	Battle Creek	WAKESHMA	LEONIDAS	4.8/8.32 kV Wye Grounded	8	4	3
40	47302	Boyne City	DIETZ ROAD	BOYNE FALL	7.2/12.47 kV Wye Grounded	2	2	0
41	48001	Boyne City	PELLSTON	BURT LAKE	13.2 kV Delta	8	23	12
42	48801	Benzie	ARCADIA	STARKE	7.2/12.47 kV Wye Grounded	7	1	3
43	50301	North Kent	NORTH PARK	LAMBERTON	7.2/12.47 kV Wye Grounded	11	1	2
44	53302	Traverse City	GLEN LAKE	ARBOR	7.2/12.47 kV Wye Grounded	11	7	11
45	56602	Ludington	BALDWIN	BALDWIN	4.8/8.32 kV Wye Grounded	2	1	4
46	60602	Kalamazoo	BREEDSVILLE	GRAND JCT	14.4 kV Delta	21	16	6
47	63802	Kalamazoo	NEELEY	DOSTER	4.8/8.32 kV Wye Grounded	5	8	5
48	65401	Flint	BISHOP	RAINBOW	4.8/8.32 kV Wye Grounded	2	0	0
49	68604	Kalamazoo	PARKWAY	SO CENTRAL	4.8/8.32 kV Wye Grounded	0	1	0
50	70802	West Kent	MICHIGAN	LOOKOUT	7.2/12.47 kV Wye Grounded	1	0	0

51	71203	Adrian	COLLEGE PARK	RIVERSIDE	7.2/12.47 kV Wye Grounded	0	4	1
52	72103	Boyne City	PORT CALCITE	ROGERS CITY	7.2/12.47 kV Wye Grounded	2	1	0
53	75901	Jackson	REYNOLDS	REYNOLDS	14.4/24.9 kV Wye Grounded	17	10	8
54	76606	Flint	JUDD ROAD	MANDEVILLE	4.8/8.32 kV Wye Grounded	1	3	0
55	78802	Benzie	HONOR	PLATTE	7.2/12.47 kV Wye Grounded	26	17	11
56	79702	West Branch	GRAYLING	RIVER	4.8/8.32 kV Wye Grounded	2	4	3
57	81503	Lansing	HARPER ROAD	ONONDAGA	14.4/24.9 kV Wye Grounded	4	9	4
58	82501	Hamilton	NORTHERN FIBRE	FIBRE	4.8/8.32 kV Wye Grounded	3	0	2
59	84001	Traverse City	HOSPITAL	ELMWOOD	7.2/12.47 kV Wye Grounded	14	2	7
60	88501	Jackson	BURTCH ROAD	BURTCH ROAD	14.4/24.9 kV Wye Grounded	9	9	16
61	90801	Jackson	OAK STREET	COOPER STREET	4.8/8.32 kV Wye Grounded	2	0	2
62	91801	Battle Creek	LOMBARD	SHERIDAN	4.8/8.32 kV Wye Grounded	10	2	3
63	91901	West Branch	BESSINGER	QUARRY	11.0 kV Delta	15	5	6
64	95203	Greenville	PECK ROAD	WISE ROAD	14.4/24.9 kV Wye Grounded	16	6	4
65	97202	Midland	EASTLAWN	FLAJOLE	4.8/8.32 kV Wye Grounded	6	6	4
66	99001	Lansing	WATERTOWN	WATERTOWN	4.8/8.32 kV Wye Grounded	2	4	0
67	100501	Owosso	CHAPIN	CHAPIN	4.8/8.32 kV Wye Grounded	1	5	3
68	100502	Owosso	CHAPIN	MARION	4.8/8.32 kV Wye Grounded	17	18	8
69	100601	Lansing	RED CEDAR	BLUE HAVEN	4.8/8.32 kV Wye Grounded	1	0	0
70	100802	Cadillac	HARRIETTA	BOON	14.4 kV Delta	10	2	9
71	102702	Muskegon	SAVIDGE	KELLY ST	7.2/12.47 kV Wye Grounded	1	2	0
72	106503	Flint	FOURTEENTH STREET	LIBERTY STREET	4.8/8.32 kV Wye Grounded	1	2	0
73	110801	Hamilton	WILLIAMS	ELY	4.8/8.32 kV Wye Grounded	6	8	6
74	112101	West Branch	ABBE	ABBE	14.4/24.9 kV Wye Grounded	15	8	11
75	112502	Adrian	HUNT ROAD	HOSPITAL	7.2/12.47 kV Wye Grounded	0	0	2
76	112601	East Kent	CASCADE	THORNCREST	4.8/8.32 kV Wye Grounded	3	3	2
77	116801	Battle Creek	WILDER	WILDER	4.8/8.32 kV Wye Grounded	5	1	4
78	122704	Flint	STEEL DRIVE	VISTA	4.8/8.32 kV Wye Grounded	1	10	0
79	126302	Tawas	SPRUCE ROAD	BLACK RIVER	14.4/24.9 kV Wye Grounded	18	16	8
80	127002	Tawas	RANGER LAKE	GOODAR	14.4/24.9 kV Wye Grounded	15	8	7
81	127003	Tawas	RANGER LAKE	LUPTON	14.4/24.9 kV Wye Grounded	12	23	3
82	127301	Tawas	WHITTEMORE	M-65	14.4/24.9 kV Wye Grounded	22	15	13
83	129001	Lansing	PEACOCK	STOLL ROAD	4.8/8.32 kV Wye Grounded	8	9	1
84	129401	Jackson	BROUGHWELL	MINARD	14.4/24.9 kV Wye Grounded	40	27	23
85	133901	Battle Creek	WATKINS	CHRISTY	4.8/8.32 kV Wye Grounded	0	4	0
86	134002	Battle Creek	ALDER CREEK	EAST LEROY	14.4/24.9 kV Wye Grounded	15	9	6
87	145101	Tawas	NOBLE	DUBY	14.4/24.9 kV Wye Grounded	11	3	2
88	148301	Clare	MANNSIDING	CEDAR	14.4/24.9 kV Wye Grounded	8	7	4
89	148502	West Kent	CRAHEN	LOMOND	7.2/12.47 kV Wye Grounded	1	5	5
90	149601	Jackson	CLEAR LAKE	WATERLOO	14.4/24.9 kV Wye Grounded	14	6	8
91	150901	Jackson	WINGATE	SOUTH	4.8/8.32 kV Wye Grounded	4	1	4
92	150902	Jackson	WINGATE	NORTH	4.8/8.32 kV Wye Grounded	3	0	0
93	151602	Tawas	HUBBARD LAKE	MILLER RD	14.4/24.9 kV Wye Grounded	20	11	4
94	152703	North Kent	COIT AVENUE	RIFLE RANGE	7.2/12.47 kV Wye Grounded	2	4	1
95	154302	Greenville	HARVARD LAKE	COURTLAND	14.4/24.9 kV Wye Grounded	8	5	4
96	155501	Cadillac	WOODWARD	WOODWARD LAKE	14.4 kV Delta	3	8	2
97	157003	Flint	MILBOURNE	DUPONT	4.8/8.32 kV Wye Grounded	0	1	0
98	158201	West Branch	RYNO	MORENCI	14.4/24.9 kV Wye Grounded	14	26	5
99	159501	West Kent	PEARLINE	COPPER BEACH	7.2/12.47 kV Wye Grounded	1	1	1
100	159502	West Kent	PEARLINE	WINDFIELD	7.2/12.47 kV Wye Grounded	1	4	1
101	159802	Adrian	RUSSELL RD	RAISIN	7.2/12.47 kV Wye Grounded	1	2	0
102	161902	East Kent	EMERSON	KILMER	7.2/12.47 kV Wye Grounded	10	5	4
103					Total Outage Incidents	691	512	376
104					Percent Improvement		25.9%	45.6%

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company
Justification of 7-year Cycle VS Other Cycles

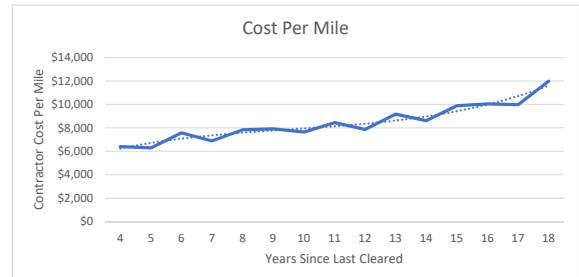
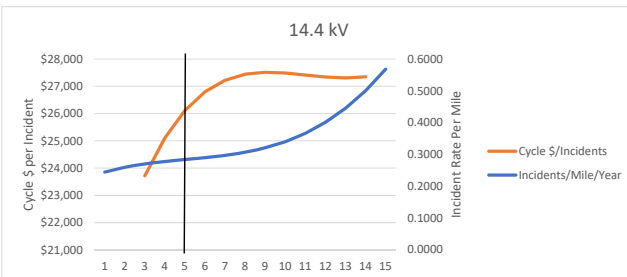
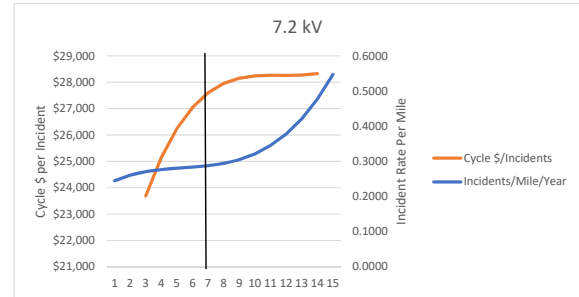
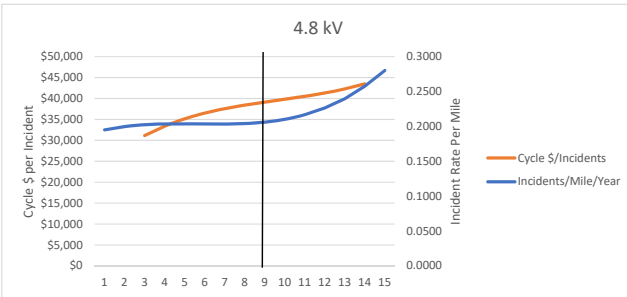
Case No.: U-20697
Exhibit No.: A-102 (CAS-5)
Page: 1 of 1
Witness: CASHellberg
Date: February 2020

Line No.	Description	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
1		4.8/8.32 kV System				7.2/12.47 kV System				14.4/24.9 kV System				
	Years Since Last Cleared/ Cycle Years	Incidents per Mile per Year	Incidents Per Year for Cycle Years	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost	Incidents per Mile per Year	Incidents Per Year for Cycle Years	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost	Incidents per Mile per Year	Incidents Per Year for Cycle Years	Contractor Cycle \$ Per Incident	Contractor \$ Annual LVD Cycle Cost	
2	1	0.1951				0.2449				0.2446				
3	2	0.1999	6197			0.2607	2501			0.2596	4364			
4	3	0.2026	6251			0.2709	2560			0.2703	4469			
5	4	0.2037	6286	\$31,153	\$48,956,333	0.2770	2605	\$23,693	\$15,431,074	0.2780	4554	\$23,718	\$27,005,549	
6	5	0.2038	6308	\$33,394	\$42,128,984	0.2808	2640	\$25,153	\$13,279,088	0.2841	4627	\$25,112	\$23,239,411	
7	6	0.2035	6321	\$35,150	\$37,029,747	0.2839	2668	\$26,252	\$11,671,805	0.2901	4693	\$26,117	\$20,426,543	
8	7	0.2034	6330	\$36,517	\$33,020,108	0.2879	2693	\$27,050	\$10,407,963	0.2973	4757	\$26,801	\$18,214,725	
9	8	0.2041	6339	\$37,584	\$29,780,809	0.2943	2721	\$27,603	\$9,386,934	0.3071	4827	\$27,225	\$16,427,846	
10	9	0.2061	6354	\$38,437	\$27,134,484	0.3050	2753	\$27,956	\$8,552,810	0.3210	4908	\$27,446	\$14,968,066	
11	10	0.2102	6378	\$39,160	\$24,974,714	0.3214	2796	\$28,155	\$7,872,049	0.3404	5007	\$27,516	\$13,776,683	
12	11	0.2167	6416	\$39,833	\$23,233,778	0.3452	2852	\$28,244	\$7,323,305	0.3666	5128	\$27,490	\$12,816,338	
13	12	0.2264	6473	\$40,535	\$21,866,527	0.3780	2926	\$28,266	\$6,892,346	0.4011	5280	\$27,415	\$12,062,128	
14	13	0.2398	6554	\$41,338	\$20,841,707	0.4215	3022	\$28,263	\$6,569,322	0.4453	5467	\$27,340	\$11,496,812	
15	14	0.2575	6663	\$42,308	\$20,136,991	0.4773	3143	\$28,272	\$6,347,195	0.5006	5695	\$27,306	\$11,108,072	
16	15	0.2802	6805	\$43,502	\$19,736,006	0.5470	3294	\$28,326	\$6,220,804	0.5684	5971	\$27,348	\$10,886,879	

	Cost per Incident Reduced	Expected Incidents	Incident Reduction from 2019 Total	Contractor Full Circuit Dollars	Cycle 14.4/7.2/4.8	Years Since Last Cleared	Miles Cleared	Total Contractor Dollars	\$ per Mile	Trendline \$ per Mile
18	\$11,412	13674	5326	\$60,781,858	5/7/9	4	171.1231272	\$1,095,666	\$6,403	\$6,240
19	\$12,462	13650	5350	\$66,667,483	5/7/7	5	853.7307575	\$5,371,519	\$6,292	\$6,713
20	\$14,496	13575	5425	\$78,647,482	5/5/5	6	2577.505368	\$19,533,047	\$7,578	\$7,080
21	\$12,869	13596	5404	\$69,538,607	5/5/7	7	3078.678373	\$21,201,586	\$6,887	\$7,366
22	\$11,832	13620	5380	\$63,652,983	5/5/9	8	4160.676719	\$32,609,605	\$7,838	\$7,592
23	\$11,594	13648	5352	\$62,045,700	5/6/9	9	4146.990126	\$32,802,966	\$7,910	\$7,782
24	\$12,637	13625	5375	\$67,931,324	5/6/7	10	4604.511367	\$35,166,933	\$7,637	\$7,959
25						11	4339.964728	\$36,632,176	\$8,441	\$8,144
26						12	3627.496385	\$28,521,785	\$7,863	\$8,362
27						13	3122.514371	\$28,638,438	\$9,172	\$8,634
28						14	2574.423293	\$22,158,274	\$8,607	\$8,984
29						15	2584.876247	\$25,524,567	\$9,875	\$9,434
30						16	1600.97892	\$16,078,818	\$10,043	\$10,007
31						17	1486.689814	\$14,825,354	\$9,972	\$10,726
32						18	981.2217905	\$11,757,733	\$11,983	\$11,614

14.4 kV needs 5-year cycle - too many outages/complaints when longer
7.2 kV vary between 5 and 7 years for analysis
4.8 kV vary between 7 and 9 years for analysis

kV System Miles
31380
9891
17310



STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBIT

OF

R. MICHAEL STUART

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

EICP Performance Measures

Case No.: U-20697

Exhibit No.: A-103 (RMS-1)

Page: 1 of 1

Witness: RMStuart

Date: February 2020

2019 Plan Year								
Operational				Target				
1. Customer Experience Index (CXi) <i>(Forester Index for Digital, Live Agent, and Interactive Voice Response)</i>				≥71				
2. Employee Safety (OSHA Recordable) <i>a) Incidents, and b) Incident Rate, and zero fatalities</i>				a) ≤54 b) ≤0.67				
3. Cyber Safety <i>(Phishing Click Rate (Level 2 Tests))</i>				≤5%				
4. Customer On-Time Delivery <i>(Orders Completed within Target Window)</i>				≥70%				
5. Service On-Time Commitments <i>a) Volume b) Adherence to Target Window</i>				a) ≥60% b) ≥95%				
6. Distribution Reliability - SAIDI <i>(System Average Interruption Duration Index - Customer Outage Minutes)</i>				≤165				
7. Generation Customer Value <i>(Fleet Availability at Least Cost Option and within Target Limits)</i>				≥72%				
8. Gas Flow Deliverability				≥90%				
9. Eliminate Vintage Services				≥9,250				
Number of targets achieved		0-3	4	5	6	7	8	9
Award Percentage		0%	50%	75%	100%	125%	150%	200%

Financial		Target
Earnings Per Share (EPS) <i>70% of financial award</i>		\$2.47
Incentive Operating Cash Flow (Billions) <i>30% of financial award</i>		\$1,700
Financial Award Percentage = %		

TOTAL EICP Award Payout Level = %

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

JEFFREY D. TOLONEN

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Information Technology Operations O&M Expense
For the Years 2018, 2019, 2020 and Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20697
Exhibit No.: A-104 (JDT-1)
Page: 1 of 1
Witness: JDTolonen
Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f) Source
		Historical		Projected		
		12 mos. ended	12 mos. ended	12 Months Ending	12 Months Ending	
		12/31/2018	12/31/2019	12/31/2020	12/31/2021	
1	Labor	\$ 10,649	\$ 10,541	\$ 11,493	\$ 12,014	
2	Contracts	31,684	31,141	34,943	35,269	
3	Business Expense	3,343	1,731	1,858	1,858	
4	Material	229	183	146	146	
5	Total Operations Expense	\$ 45,905	\$ 43,596	\$ 48,440	\$ 49,287	

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Summary of Actual and Projected Information Technology Investments O&M Expenses

For the Years 2018, 2019, 2020 and Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-105 (JDT-2)

Page: 1 of 1

Witness: JDTolonen

Date: February 2020

Line No.	(a) Description	(b)	(c)	(d)	(e)	(f) Source
		Historical 12 mos. ended 12/31/2018	12 mos. ended 12/31/2019	Projected 12 Months Ending 12/31/2020	12 Months Ending 12/31/2021	
1	Investments Planning	\$ 938	\$ 729	\$ 978	\$ 978	
2	Labor	431	471	809	809	
3	Contracts	470	256	24	24	
4	Business Expense	37	2	145	145	
5	Investments O&M	\$ 15,165	\$ 13,183	\$ 16,034	\$ 20,906	
6	Labor	2,659	3,034	8,718	10,775	
7	Software	1,821	620	1,238	2,267	
8	Hardware	844	314	483	619	
9	Prof & Contract Services	8,722	8,128	4,394	4,985	
10	Business Expense	254	241	239	1,276	
11	Overheads	865	846	962	984	
12	Total Investments Expense	<u>\$ 16,103</u>	<u>\$ 13,912</u>	<u>\$ 17,012</u>	<u>\$ 21,884</u>	

Schedule B-5.3

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected Capital Expenditures

Information Technology

Summary of Actual and Projected Electric Capital Expenditures

For Years 2018 Through 2020 and Test Year 2021

(\$000)

Case No.: U-20697

Exhibit No.: A-12 (JDT-3)

Schedule: B-5.3

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Witness: JDTolonen

Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)
		Capital Expenditures				
		Historical	Projected Bridge Year			Projected Test
Line		12 Mos Ended	12 Mos Ended	12 Mos Ending	24 Mos Ending	Year
No.	Description	12/31/2018	12/31/2019	12/31/2020	12/31/2020	12 Mos Ending
						12/31/2021
1	Upgrades & Replacements (Enterprise)	22,548	11,449	3,373	14,821	1,260
	Software	1,840	259	69	329	-
	Materials	9,909	4,316	1,178	5,494	797
	Labor	1,980	1,217	742	1,959	301
	Contractor Costs	7,103	4,897	990	5,887	-
	Engineering	-	-	-	-	-
	Overhead & Others	1,716	759	394	1,153	162
	Contingency	-	-	-	-	-
2	Upgrades & Replacements (Business Partner)	2,482	816	7,019	7,835	4,943
	Software	88	-	329	329	190
	Materials	46	211	2,213	2,424	127
	Labor	532	168	2,422	2,590	2,343
	Contractor Costs	1,535	344	627	971	788
	Engineering	-	-	-	-	-
	Overhead & Others	281	94	1,428	1,523	1,495
	Contingency	-	-	-	-	-
3	Security	6,278	4,764	6,160	10,925	5,985
	Software	518	884	1,420	2,305	1,489
	Materials	1,161	1,556	1,719	3,274	2,514
	Labor	1,072	490	1,867	2,357	1,086
	Contractor Costs	3,089	1,662	518	2,179	518
	Engineering	-	-	-	-	-
	Overhead & Others	439	172	637	809	378
	Contingency	-	-	-	-	-
4	IT Service Delivery	17,280	15,958	23,434	39,392	24,661
	Software	861	911	670	1,581	57
	Materials	12,973	10,762	19,119	29,880	20,877
	Labor	1,053	1,187	1,834	3,020	1,867
	Contractor Costs	1,837	2,609	910	3,519	931
	Engineering	-	-	-	-	-
	Overhead & Others	556	489	902	1,391	929
	Contingency	-	-	-	-	-
5	Enhancements	4,763	3,458	543	4,001	543
	Software	1,580	37	-	37	-
	Materials	63	110	-	110	-
	Labor	770	1,020	372	1,392	372
	Contractor Costs	1,795	1,664	-	1,664	-
	Engineering	-	-	-	-	-
	Overhead & Others	555	627	171	798	171
	Contingency	-	-	-	-	-
6	BP Functionality	20,894	19,926	15,056	34,982	36,366
	Software	5,857	656	1,998	2,654	4,090
	Materials	(1,395)	3,812	100	3,912	849
	Labor	3,733	2,392	4,675	7,066	12,298
	Contractor Costs	9,643	10,708	5,691	16,399	12,270
	Engineering	-	-	-	-	-
	Overhead & Others	3,056	2,358	2,593	4,951	6,859
	Contingency	-	-	-	-	-
7	Architecture	-	-	-	-	-
	Software	-	-	-	-	-
	Materials	-	-	-	-	-
	Labor	-	-	-	-	-
	Contractor Costs	-	-	-	-	-
	Engineering	-	-	-	-	-
	Overhead & Others	-	-	-	-	-
	Contingency	-	-	-	-	-
	Total Capital	74,245	56,370	55,586	111,956	73,758

Michigan Public Service Commission
Consumers Energy Company

Synopses Containing Descriptors, Scope, Benefits, Implementation Dates and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses
For the years 2018, 2019, 2020, and 2021

Information Technology Department

Case No.: U-20697
Exhibit No.: A-106 (JDT-4)
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No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)				
	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC				
1	2018	Application Performance Monitoring	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Application Performance Monitoring project will implement an end to end systems performance monitoring solution that provides the following capabilities: Application Performance Monitoring, Infrastructure and Network Performance Monitoring, Customer Experience and End User Monitoring. The value of completing the project is real-time contact center response time reporting vs. hourly reporting, improved user interface dashboard view vs. emailed report, visibility into the smart energy application network, targeted troubleshooting and faster root cause analysis, deep-dive analytics and drill-down capability, exception alerting capability, auto-discovery, low configuration solutions (i.e., less administrative work), and reducing waste.	23-Mar-18	(1.00)	121,025	113,154	0	20,488	(18,022)	15,405	(6,913)	0	0	10,712	(18,914)	1,288				
2	2018	ARP - Server	Upgrades & Replacements (Enterprise)	New Computers / Hardware	IT infrastructure generally becomes less reliable after 5 years, jeopardizing the stability of our business' critical applications running on top of our IT infrastructure. This Server ARP project will evaluate Computer Hardware with more than 5 years of continuous use and replace where appropriate. The project will intelligently and systematically replace critical infrastructure before a system failure that would disrupt business operations.	31-Dec-18	(1.07)	0	0	0	0	0	0	12,768	0	0	10,144	1,307	1,518				
3	2018	ATM Retirement	Upgrades & Replacements (Enterprise)	Network	To migrate Call Centers and 800 Mhz network off a legacy ATM network that is now obsolete and no longer supported to current industry standard and supported network infrastructure. To establish network connectivity through the IP network instead of the legacy TDM based ATM. This project will define alternatives for those two groups and then to finish the conversion and remove the 15 remaining ATM Nodes; move tower sites off the ATM network. This Project will mitigate the pending obsolescence. The ATM network is at the end of its design life; if ATM network stops working or more network outages occur, this will impact mission critical business processes.	01-Aug-15	(0.87)	3,929	0	0	2,860	0	1,069	0	0	0	0	0	0				
4	2018	BI ENH Bus Rptg & Enrtps Data Mgt	Upgrades & Replacements (Enterprise)	Software (Intangible)	Adjustment from prior year.	N/A	N/A	(1,782)	0	0	(1,286)	0	(496)	0	0	0	0	0	0				
5	2018	Data Center 2.0	Upgrades & Replacements (Enterprise)	New Computers / Hardware	The Data Center 2.0 project will mitigate the current significant risks (location risks, capacity risks, technology risks) with the Consumers Energy IT Disaster Recovery (DR) capabilities. The Company is co-locating to a Backup Recovery Center (BRC) at a vendor facility. A co-location facility is a data center facility in which a business can rent space for servers and other computing hardware. Typically, a co-location vendor provides the building, cooling, power, and physical security, while the business provides the servers, storage, and other computing and networking equipment. Project scope includes both Information Technology and Operational Technology data centers. The project will migrate the BRC to a vendor (Switch) colocation data center, expand DR system capacity to support 100% production load requirements, expand DR capabilities to all applications determined by the business as needing DR, and enable cloud capabilities to migrate non-production workloads to the cloud to allow for production disaster recovery on non-production infrastructure. At the end of the migration to Switch, the compute capacity at Switch will be greater than or equal to the compute capacity at Parallel. Some of the capacity will be on newly purchased hardware and some will be on hardware shifted from the Battle Creek Data Center. Currently, there are no plans to put compute for production DR in the cloud. The Company performed an analysis of two primary alternatives to expand DR capabilities and address constraints/risks: 1) at the current BRC location, and 2) at a third-party co-location facility. Based on the analysis, the Company decided to implement the third-party co-location alternative for the BRC. The value this brings the Company is that it mitigates risks to the corporation in the event of a disaster by significantly enhancing the disaster recovery capabilities, DR testing capabilities, and scalability constraints at the current data center locations.	01-Nov-20	(0.82)	11,283,492	887,621	7,936,031	569,385	1,539,043	351,632	1,615,571	489,726	297,256	78,815	720,871	28,904				
6	2018	Dense Wave Division Multiplexing - Fiber Backbone Refresh	Upgrades & Replacements (Enterprise)	Network	The Dense Wave Division Multiplexing - Fiber Backbone Refresh project will refresh the company's network hardware that is part of the Company's "Dark Fiber" network. Included in the implementation is to replace the existing Fiber Optic transport equipment at the ten Company locations that are part of the internal "Dark Fiber" network. This project provides value by insuring that the Company's internal Network connectivity for Call Centers, Computer Networks and the Radio Network is reliable and on modern, supportable hardware.	10-Nov-18	(0.99)	2,346,298	55,012	1,790,108	83,307	379,363	38,508	134,904	0	86,356	24,271	2,713	21,565				
7	2018	ESB Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Enterprise Service Bus (ESB) BWS 14 Upgrade project will upgrade and migrate the Business Works developer code from v5 to v5.14 and to upgrade all remaining applications and infrastructure as needed. BWS 14 provides the following value for Consumers Energy: accelerated productivity, continuous delivery and integration, open ecosystem for improved operational visibility, integration with web and mobile apps and APIs in real-time, and improved administrative and operational efficiencies. BWS 14 allows for easy integration with other cloud-based solutions such as Amazon Web Services and provides easy migration to cloud-based containers in the future. The risks of not upgrading to BWS 14 are slow deployments due to the increased number of physical servers to manually maintain and increased time to debug errors. BWS end of extended support is December 2019, after which the support costs will increase.	01-Nov-19	(0.93)	1,930,850	12,248	24,473	438,660	987,526	467,943	200,561	6,199	0	65,954	105,030	21,378				
8	2018	HR Support Pack and BSI Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will update the SAP system with HR Support Packs that are released annually by SAP to comply with HR and tax changes. This project creates value for the Company by: (1) ensuring that it is in compliance with new financial rules and regulations, and (2) ensuring that it can calculate and distribute payroll. The scope of this project is to add SAP HR corrections to ensure proper reporting of financial information by the Company. As this is an upgrade of an existing system, the alternative considered was to delay the upgrade. This alternative was not chosen due to the risk of not complying with financial rules and regulations.	01-Dec-20	(0.94)	0	0	0	0	0	0	601,033	36,256	0	27,483	520,984	16,310				
9	2018	Iron Enterprise Edition Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will upgrade IEE, which collects the reads from meters to ensure accurate and non-estimated bills are provided to our customers. This project creates value for the Company by: (1) ensuring the new features and functionality added to meet business requirements are available to business partners and IT; (2) meeting Information Security's requirement to keep applications patched to control cyber debt; and (3) allowing for validation, estimation, and editing functions for all data collected. The scope of this project includes: (1) an operating system refresh, (2) migrating from Windows 2012 to the next Windows version, and (3) migrating the database from SQL Server version 2012 to the next SQL Server version. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and cyber security of the system.	01-Sep-20	(1.00)	47,860	0	0	25,518	11,020	11,323	290,802	0	0	145,403	107,290	38,109				
10	2018	LandWorks Property Management System Upgrade V5.5	Upgrades & Replacements (Enterprise)	Software (Intangible)	The LandWorks Property Management System Upgrade V 5.5 project will upgrade the Land Property Management System (LMP by Landworks) to the next version, as well upgrade to arcGIS in order to support the version upgrade. The application is currently on an older version and at risk of no longer being supported by vendor.	01-Oct-21	(0.90)	42,978	0	0	9,833	26,530	6,215	6,826	0	0	3,722	0	3,104				
11	2018	Lotus Notes Application Migration & Retirement Wave 3	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Lotus Notes Application Migration & Retirement Wave 3 project will migrate Lotus Notes to SharePoint. Lotus Notes is an unsupported technology now at CE. Most of the 600+ LN applications can be moved to SharePoint, either from a direct move or customization. The applications are categorized into simple, medium, and complex. The migration is happening in 5 Phases or Waves and this is Wave 3. This next phase will further enable capabilities on our current collaboration platform standard (SharePoint), while reducing the risk footprint of using an unsupported standard (Lotus Notes). SharePoint gives many new enhancements to these applications including collaboration, versioning of documents, security, and automated auditing. With the use of the K2 the users can also modify their own sites once migrated to better tailor them to their business needs.	14-Sep-18	(1.00)	2,022,197	0	0	206,061	1,522,845	293,291	428,465	13,539	0	42,696	334,760	37,469				
12	2018	Oracle Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Oracle Server Database Upgrade project will upgrade Oracle server databases to the next version and supportability across business portfolios. This project will create value for the Company and its customers by (1) reducing the risk of system failure and the resulting impacts to business partners and customers, and (2) ensuring that systems are secure, supported, and have the latest features and functionality. The scope of this project includes upgrading to the next version of Oracle across all business systems. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and supportability of the system.	30-Apr-21	(1.00)	0	0	0	0	0	292,888	9,794	0	53,344	206,268	23,481					

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Consumers Energy Company

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Information Technology Department

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FISC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
13	2018	OWCE Nitrogen Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The OWCE Nitrogen Upgrade project will upgrade the OpenWay Collection Engine (OWCE1) to the latest version. This upgrade will be done in parallel with the hardware asset refresh to reduce overall risk and cost. OWCE is the head end software that communicates to smart meters. The upgrade to the Nitrogen version ensures that the Company sustains this critical business function on a current and supported version.	04-Jan-19	(0.98)				11,843	(44,504)	(4,207)	380,294	18,420	12,708	62,115	157,668	29,362
14	2018	Remote Application Delivery Strategy	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Remote Application Delivery Strategy project will upgrade VDI and bring Citrix into VDI to eliminate the Citrix platform. Along with this, there is the intent to bring down the cost of each VDI. There are new technologies being implemented via Thin-App to allow for streaming applications. Included in the implementation is eliminating a platform that has redundant functionality. We will be gaining functionality with VMware Thin-app giving us the capability to stream applications instead of having to push applications to the VDI's. This can reduce the size of the VDI footprint.	20-Dec-18	(0.98)	1,024,891	334,943	200,351	47,520	402,353	39,724	40,138	24,382	0	2,804	11,474	1,478
15	2018	TCOE HP ALM Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	Adjustment from prior year.	13-Jan-17	(0.86)	22	0	0	0	0	22	(253)	0	0	(0)	0	(253)
16	2018	TCOE Test Data & Environment Management	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Testing Center of Excellence (TCOE) Test Data & Environment Management project will provide the tools and techniques to be able to periodically (e.g. annually) refresh the data in our SAP development, QA and pre-production environments from production data. This will decrease the time and effort needed to test and improve the quality of testing, which will improve the quality of application changes. Included in the implementation is determining refresh requirements; using the Request for Information process to select tools based on the requirements; setting up environments for the tools; installing and configuring the tools; testing refresh processes in development and QA environments to ensure appropriate data is refreshed and that processes can be completed with minimal outages; testing refresh processes in development and QA environments to ensure sensitive, Personal Identifiable Information copied from production is obfuscated according to our Information Security data standards; implementing the tools; and putting the processes in place to ensure we refresh on an ongoing, periodic basis. The value of completing the project is that manual testing efforts and costs will decrease while speed to deliver and quality of testing will improve resulting in better quality application changes. This is true for all SAP applications and SAP-interfacing applications, including customer applications such as Contact Center applications, our Consumersenergy.com website, and customer billing. As part of our review process during the Plan phase, a Request for Information will be issued to explore solutions from a variety of vendors, including the following: SAP, Aptivity, EP-USSE, SNV, Delphix, Computer Associates, Informatica, and IBM.	01-Apr-19	(0.94)	784,135	206,546	184,062	133,868	183,258	76,401	85,089	0	0	48,885	27,511	8,693
17	2018	Time of Use Billing Rate Expansion	Upgrades & Replacements (Enterprise)	Software (Intangible)	Consumers Energy will be filing a rate order with the MPSC in May 2018 requesting that all residential customers who are currently being billed using register index reads be switched to a new Time-Of-Use (TOU) interval based billing rate. Expand the current TOU billing rate available to all electric residential customers. Provides for all electric customers to be able to take advantage of the Time of Use (TOU) rate during peak period times without having to sign up for the program, thereby allowing them to save on their energy bill during this period.	01-Mar-19	(0.94)	2,633,447	0	0	348,053	1,948,456	336,937	302,474	0	0	41,037	222,682	38,745
18	2018	TOA Phase II	Upgrades & Replacements (Enterprise)	Software (Intangible)	The overall of the scope of the project is to upgrade the Transmission Outage Application (TOA) based on the current business and technical requirements. The tasks include: the completion of the SOW, the testing of the upgraded application, the creation of new policies and procedures based on the upgraded TOA, and the change management associated with the education and training of the appropriate Grid Management Business Partners, and the implementation of the TOA II solution. IT and the Grid Management Business Partners will be working closely on this project. The Grid Management Business Partners will be responsible for certain tasks (scope development/formalization, testing and validation of functional scope through test results) and IT will be responsible for certain tasks (completion of the Solution Design Document, software installation, help with test case development and execution, and validation of technical scope through test results).	28-Mar-19	(1.00)	98,791	0	0	15,102	77,430	6,259	7,919	0	0	6,864	0	1,055
19	2018	Treasury Work Management	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Treasury Work Management project will determine and implement a Treasury solution for cash forecasting, bank account fee analysis, and treasury work management. The remaining phases of this project will be to 1) Develop cash forecasting process; 2) Automate bank fee reconciliation process; 3) Identify bank account management tool (i.e. bank account management including signatures, etc.); and 4) Other enhancement to the solution as needed. The scope includes replacing and automating Cash Forecasting process that is currently supported by an Access database that is unsupported and subject to maintenance risks, replacing and automating the current manual process over bank account fees. The current process is a manual process to compare bank statements to confirm that the bank service fees are correct. There are hundreds of bank service fees daily. The project impacts address: obsolete technology - current technology is obsolete and breaks every time there is an upgrade or service pack to SAP and the business support personnel is no longer available to maintain the Access database and financial impact: Savings in Consumers Energy's daily Liquidity/Cash Position can move as much as \$200 million per day (dependent on large transactions such as dividend payments, debt retirements, debt issuances, etc.). Daily changes in Liquidity/Cash Position can typically be as much as \$50 million.	17-Oct-18	(0.92)	0	0	0	0	0	0	(4,356)	0	0	0	(4,356)	0
20	2018	Tumbleweed Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Tumbleweed Upgrade project will upgrade the system with support, redundancy, disaster recovery, a new test environment and additional capacity to comply with corporate operational standards to support critical business functions. Included in the implementation is upgrading the application from 5.1 to version 5.3.6, creating redundancy in Prod and Non Prod environments, Disaster Recovery, increased capacity, training, support plan, and interfaces to other internal systems.	31-Oct-18	(1.06)	241,630	230,309	(230,309)	68,525	97,443	75,662	6,704	0	0	0	3,797	2,906
21	2018	vSphere Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project is to upgrade vSphere and avoid technology obsolescence by doing so. VMware vSphere is the most critical piece of software that runs our datacenter. CE is currently running vSphere 5.5, with v6 released in early 2015. Many systems are dependent on vSphere being up to date, including Nimbus, SRM (Our DR tool), and vRlock. By keeping vSphere on an older version, all of these dependent applications must also remain on older versions to stay compatible, which increasing risk of flaws in all systems. Upgrading vSphere to v6 will enable the support of big data analytic servers by increasing the maximum server size to 4TB of memory.	21-Jan-18	N/A	0	0	0	0	0	0	36,266	0	0	17,843	15,997	2,426
22	2018	WAN Transformation	Upgrades & Replacements (Enterprise)	Network	This project is to migrate the Wide Area Network connectivity at Company locations off of legacy T1 technology and on to newer Carrier Ethernet technology. All Company locations within the State of Michigan will be converted to new Wide Area Network technology. The new technology is more reliable and provides more bandwidth to enable more productivity at the Company's locations.	20-Nov-18	(1.02)	4,726	0	4,726	0	0	0	27,304	0	0	22,550	1,485	3,270
23	2018	Windows 10 Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Windows 10 Upgrade project will upgrade Consumers Energy corporate computing devices from Windows 7 to Windows 10. The project will convert 4,500 Windows 7 computing devices to Windows 10 with the assistance from a partnered vendor at various Consumers Energy locations for corporate, gas, and electric employees. Project tasks include assessing current tools and image process to ensure industry standards are being applied, communicating with employees on project and expectations, scheduling and completing device operating system upgrades, verifying that devices were upgraded with no impact to employees' data and performance, and providing on-site post support at Consumers Energy locations. The alternative is not to move forward with this project and pay additional support fees for Microsoft to continue supporting Windows 7 for up to four years. This will allow Consumers Energy time to replace the devices through our asset management refresh projects. Work station assets are currently replaced on a four-year cycle. This alternative was not pursued due to the additional support fees and risk to business processes.	01-Oct-19	(0.96)	0	0	0	0	0	158,279	0	0	30,849	118,278	9,151	
24	2018	SubTotal Upgrades & Replacements (Enterprise)	Upgrades & Replacements (Enterprise)					22,547,619	1,839,834	9,909,441	1,979,536	7,163,141	1,715,666	4,516,762	600,317	396,320	695,491	2,534,674	289,960

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
25	2018	Corp Portfolio Application Currency	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Corporate Portfolio Application Currency project is for specific application upgrades within the Corporate Services portfolio. Included in the implementation is upgrading application software, assessing any new functionality for value to the company, making necessary configuration changes, testing any integrations to or from the applications, testing upgrades, and updating documentation related to integration changes. The value of completing the project is lessening the number of incidents associated with out of date software, increasing application stability leading to fewer incidents due to out of date software, and allowing the company to leverage new functionality available in upgrades. As part of our review process the alternative considered was to stay at the current version of the applications. If this approach is chosen then the company risks applications not working, increased number of incidents, increased support costs and impact key business processes.	31-Dec-18	(0.90)	0	0	0	0	0	0	124,663	10,034	2,361	31,329	11,116	29,623
26	2018	Enhancements - APS	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Enhancements - APS is for small software enhancement work efforts performed for the analytics, Cloud, DevOps, and Architecture (ACDA) business area. Included in the implementation are enhancements or upgrades for Business Objects (BOBJ) Upgrade, Business Objects Data Services (BODS) Upgrade, Dev Ops, Business Planning and Consolidation (BPC), BW HANA, HANA DB, MEMSQL, Soap UI, R Studio, Tableau, and Attunity. The value of completing the project is compliance, hard cost savings, avoided costs, hitting corporate goals, and mitigating risk. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-18	(1.00)	0	0	0	0	0	0	311,991	57,723	0	12,113	215,873	26,282
27	2018	eSOMS - upgrade to Operations Management	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Electronic Shift Operations Management System Upgrade - Operations Mgmt project is an upgrade of the electronic Shift Operations Management System (eSOMS) application used by generation and gas compression stations. The upgrade includes the associated hardware such as servers.	18-Aug-18	(0.88)	299,207	6,678	7,849	138,055	61,002	85,623	34,148	1,386	3,975	24,880	0	3,908
28	2018	Government Reports	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Governments Report project is being implemented to address a technology obsolescence issue, ensure regulatory reports are filed on time, provide insight into when reports are due and remediate internal audit findings. This project includes implementation of a new tool that collects metadata for each regulatory report, automates notifications to users as due dates are approaching, automates workflows for approval and automates Email notifications with links to tasks. As part of our review process, the alternatives considered were a SaaS solution, leveraging existing technology or reverting to a manual process. The alternative selected was to leverage existing technology, as it was the most cost effective.	01-Dec-18	(0.98)	11,936	0	0	1,454	8,652	1,830	64,039	0	0	1,215	55,941	6,883
29	2018	Integrated Talent Mgmt and Learning Mgmt Upgrades	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Integrated Talent Management and Learning Management Upgrades project is for upgrades to the SAP SuccessFactors cloud solution for their talent management and learning management systems. All quarterly SuccessFactors enhancements are reviewed for impacts to the overall system and integrations, included in the implementation is making necessary configuration changes, testing the upgrade, validating integrations, and updating documentation related to integration changes. The value of completing the project is that quarterly SuccessFactors upgrades give us additional functionality to better serve both employees and potential candidates. Human Resources also benefits by being able to improve the employee lifecycle. As part of our review process no alternatives were considered because SAP SuccessFactors automatically pushes upgrades to our system every quarter, the company does not have an option to reject those upgrades. If nothing is done to test the system and integrations, it is possible that our Talent Management and Learning Management systems will not work and won't be accessible by applicants, managers, employees, HR and contractors.	01-Dec-19	(0.89)	0	0	0	0	0	0	249,706	0	0	37,879	197,063	14,765
30	2018	ITCP - Cold Water Service Center	Upgrades & Replacements (Business Partner)	Network	The ITCP - Cold Water Service Center - IT will be supporting the Facilities project to build a new service center in Coldwater. This work will require new connectivity to the site, security, data, video, voice as well as SAP changes. Included in the build of Coldwater is closing Branson Service Center and consolidating employees in Coldwater	14-Sep-18	(1.00)	293,869	0	37,946	44,875	190,731	20,317	0	0	0	0	0	0
31	2018	ITCP - Parnall East Renovation	Upgrades & Replacements (Business Partner)	Network	Adjustment from prior year.	18-Oct-18	(1.01)	208	0	0	139	0	69	373	0	0	337	0	36
32	2018	ITCP - Parnall Pt 1 Renovation	Upgrades & Replacements (Business Partner)	Network	Adjustment from prior year.	31-Dec-18	(1.00)	63	0	0	0	0	63	0	0	0	0	0	0
33	2018	Osisoft Pi Historian Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Osisoft Pi Historian Upgrade project will maintain application and hardware platform currency for the Osisoft Pi system. The project will create value for the Company and its customers by: (1) reducing security vulnerabilities, (2) improving efficiencies and increasing synergies between environments, and (3) enabling business partners to leverage new features that the vendor includes with major releases. The project scope includes the implementation of: (1) Osisoft Meter Operational Data Management, (2) Osisoft Electric Distribution Historian, (3) Osisoft Generation, (4) Osisoft Low Voltage Distribution/High Voltage Distribution, (5) Osisoft Gas Automated Meter Read, and (6) data archiving for analytics purposes. The alternative considered was to delay the upgrade until a future year. The hardware has not been refreshed since 2014. Continuing to delay the upgrade could lead to the inability to apply security and system patch sets. A timely upgrade will allow the Company to sustain system performance and supportability. Other vendors were not considered due to a longstanding Enterprise Agreement with Osisoft, Inc.	01-Oct-20	(1.00)	283,405	83,357	0	39,202	123,817	39,029	90,080	0	0	22,284	56,215	11,581
34	2018	PowerPlan 2018 Application Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The PowerPlan Upgrade 2018 project is to upgrade PowerPlan to current version, implement functionality to support the new Leasing Generally Accepted Accounting Principles ("GAAP") regulatory requirements effective January 1, 2019, and enable month-end close automation through the financial close cockpit. The Company's current version of PowerPlan is at end of life and PowerPlan will no longer offer extended support. The value of doing this project is to avoid risk of not being able to comply with new accounting standard and running an unsupported version of the software, as PowerPlan is critical to meeting our financial reporting requirements. As part of our review process, the alternative considered was to not upgrade PowerPlan. This alternative was not selected due to the risk on financial reporting.	18-Oct-18	(0.95)	1,501,363	0	0	291,444	1,083,270	126,649	234,816	1,945	0	77,374	143,078	12,420

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35	2018	Software Asset Mgmt Automation Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will add additional out of the box automation to the Software Asset Management program to remove repetitive, manual tasks and take advantage of continual system enhancements by keeping the system up to date with vendor release schedules and upgrade the supporting software to its latest versions. Project scope includes: - Implement additional out of the box software automation workbooks - Upgrade to the latest versions of Snow License Manager, Automation Platform and Inventory Manager (tools to manage software license compliance) This project would allow the IT Asset Management (ITAM) team to continuously monitor and mitigate corporate risk of ineffective ITAM. This project would: [1] add additional automation that would eliminate manual tasks that create a high risk of data errors. These manual tasks also add additional time to software requests; [2] increased ability to manage the company's increasing portfolio of cloud-based, subscription licenses; [3] decrease the time to completion for software requests. Alternatives considered are: - Stay at the same version and do not add automation and continue to struggle managing user and cloud-based licensing and processing tasks manually which would create the need for additional personnel (Full Time Equivalents "FTEs" or contractors) - Convert the Software License Management software to another system at an implementation cost of \$500k and annual recurring costs of \$250k These alternatives were not considered due to: [1] Option 1 does not address the risk of data errors and would cost between \$75k to \$200k to support the additional head count [2] Option 2 would have a large implementation fee and the net overall annual costs would be much higher than the current maintenance rates	20-Dec-18	(0.39)	56,098	0	0	3,901	49,497	2,700	21,468	0	0	4,461	13,990	1,017
36	2018	TSM-Enhancements	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Enhancements - Technology Strategy Management (TSM) initiative will apply O & M funding to make enhancements to existing software and to address requests generated by changing business requirements. The enhancement requests are fulfilled to provide capabilities across business areas. Requests for this funding are governed by a cross-functional board of representatives from infrastructure area which meets periodically to evaluate and prioritize the work. The board assesses requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing technology investments for infrastructure. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and rising failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-19	(1.00)	36,234	0	0	13,260	17,827	5,147	96,731	0	0	20,922	64,575	11,233
37	2018	SubTotal Upgrades & Replacements (Business Partner)	Upgrades & Replacements (Business Partner)					2,482,385	88,095	45,795	582,330	1,534,797	281,428	1,228,014	71,088	6,536	232,792	799,851	117,747
38	2018	AccessNOW	Security	Software (Intangible)	The Accessnow project is for the implementation of configurable identity and Access Management functionality and best practices with enforced compliance. This includes enterprise-level foundation architecture, technology, and end-to-end processes and controls. The project will deliver an integrated and synced enterprise system of record data. Additionally, processes will be fully automated for user self-service and access lifecycle management. Included in the implementation is the continuation of building on the One Identity Manager application against the identity and Access Management Program (IDAP) roadmap. As Phase 1 efforts concluded in mid-2017, 2018's main objective is to implement Phase 2, which includes designing and building a web portal with a request catalog for employees' access needs. The intent is to replace the legacy Computer Access and Authorization Request (CAAR) system. These requests will be driven by automated workflows and the tool will provide automatic Active Directory provisioning. Additional features aimed at making this process more user friendly are also included. The value of completing the project is automated workflows with escalations that will reduce wait times for end users to get the access they need for their jobs. As part of our review process the alternative considered was using manual processes but this strategy was deemed too costly and inefficient.	01-Aug-19	(0.96)	995,168	(147,863)	(4,602)	323,891	697,954	125,788	285,893	139,934	5,496	97,815	80,044	2,605
39	2018	ARP - Cyber Security	Security	Network	The ARP Cyber Security project will upgrade IT infrastructure to keep up with increasing user demands and applications, and prevent system failures and service interruptions. Refresh strategies are driven by business objectives including security, financial, and growth requirements. Continuing a refresh schedule based on historical performance and usage requirements will maintain existing levels of security. Included in the implementation is the replacement of end of life and obsolete systems. The main initiatives are Carbon Black Response server replacement, Ql radar server and licenses, and Blue Coat reporter upgrade. Completing the replacement strategy enables a consistent security platform, offers the flexibility to respond to change, and continues existing levels of security in support of protecting our customers' data and critical infrastructure. A deployment schedule is developed based on an analysis of the age and capabilities of the current security technology integrated into the existing infrastructure to ensure cyber security and the organization's ability to recover vital assets and information. As part of our review process the alternative considered was not to upgrade or replace assets as required. This approach will most likely introduce security risks, system vulnerabilities, and out-of-creativity repair costs.	31-Dec-18	(1.09)	230,773	865	148,813	40,901	5,705	14,489	36,344	0	3,944	21,021	9,835	1,544
40	2018	Cyber Security Enhancements	Security	Software (Intangible)	The Cyber Security Enhancements project is for emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape. It will address issues and bring value to the environment for initiatives that may not meet the criteria for a formal project. Included in the plan are the following initiatives: application whitelisting for Zone A, Cyber Ark upgrade, SharePoint enhancement, threat prevention subscription, multi-factor authentication for the payment card industry, system center endpoint protection, and centralized endpoints. The enhancements add new features to the software as requirements or process improvement initiatives dictate. The value in completing the enhancements is to allow software and hardware product performance scalability, to improve flexibility to respond to change in the threat landscape, and to promote continuous performance improvements. As part of our review process the alternative considered was not to upgrade or enhance software or hardware, which could reduce Consumers Energy's ability to respond to future threats and vulnerabilities.	31-Dec-18	(0.96)	0	0	0	0	0	0	69,178	24,168	364	20,499	20,123	4,023

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41	2018	Enterprise Incident Response Toolkit	Security	Software (Intangible)	The Enterprise Incident Response Toolkit project is for implementing an incident management platform that enables the Cyber Security Incident Response Team (CSIRT) to automate alert triage, which is currently a manual process through technical playbooks, to bring more capabilities to track and manage incidents, and to integrate threat intelligence with security orchestration. Included in the implementation is building five or six playbooks, providing basic training to the CSIRT, instituting firewall rules, setting up internet and security information and event management access, placing an F5 load balancer in front of the two Demisto servers, and deploying two Demisto servers that will operate in Disaster Recovery mode. The value of completing the project will be improving response time to events and limiting the risk to Consumers Energy from a cyber security incident. As part of our review process other automation tools (i.e. Swimlane, Phantom, Fire Eye) were vetted via on-site demonstrations or webinars. However, this was determined to be the best solution to align with the Consumers Energy requirements.	22-Jul-20	(0.93)	\$46,346	\$42,930	\$35,655	\$9,940	\$41,381	\$(1,160)	\$,220	0	0	4,228	923	69	
42	2018	Firewall Management Platform	Security	Software (Intangible)	The Firewall Management Platform project will implement a firewall management platform. Cyber security now manages nearly 100 firewalls and keeping configurations up to date, changes processed correctly and mistakes to a minimum is challenging. A firewall management platform will enable automation and workflow to provide greater security and efficiency for our team. Firewall security mistakes and failures are leading causes of data breaches.	30-Nov-19	(0.98)	\$0,700	\$6,760	\$(13,739)	\$19,071	\$,247	\$,461	\$68	0	0	\$68	0	0	
43	2018	FIS Release 4 - Appointment Booking & R9 Upgrade	Security	Software (Intangible)	The FIS Release 4 - Appointment Booking & R9 Upgrade project will add functionality to the SAP/Service Suite systems to allow the Call Center and Scheduling groups to provide 3 hour (ultimately 2-hour) appointment windows for customer booking of service appointments. This project will provide the ability to provide 3 hour (ultimately 2 hour) service appointment windows and provides a smaller wait time for customers for CE field employees to arrive and work.	28-Oct-18	4.32	\$71,009	\$3,545	0	\$29,748	\$44,372	\$163,344	\$38,473	0	0	\$26,106	\$,644	\$,724	
44	2018	Mass Notification	Security	Software (Intangible)	The Mass Notification project will interact with our current Siemens fire panels to take advantage of communications protocols in existence at Consumers Energy physical sites. Similar to current capabilities used to announce "fire" events, the mass notification tool will allow for other event notifications to building participants. In addition, mass notification has capabilities to involve personal device notification via texting, email alerts, and social media updates. Included in the project is implementing communications software to tap into current fire panels in physical locations, allowing for remote connectivity to sites for access to software, allowing for personal device communications, tapping into current building security software, allowing for emailing of key threats/events, and allowing for social media interfacing. The value of completing the project is directly tied to employee safety, which is increased by allowing better communication to employees during emergency situations. As part of our review process the alternative considered was not to implement the capability and to continue notifying individual Consumers Energy site managers of life safety situations through emails, phone calls, and/or text messages.	15-Aug-19	(1.06)	\$37,613	\$107,063	\$104,028	\$17,631	\$150,700	\$(5,809)	0	0	0	0	0	0	0
45	2018	NERC CIP v5	Security	New Computers / Hardware	Consumers Energy is required to be compliant with version 3 of the NERC Critical Infrastructure Protection (CIP) standards, including CIP V3 for High and Medium Impact BES Cyber Systems on 4/1/2014 and Low Impact BES Cyber Systems on 12/1/2016. Implementation of compliant processes and systems must be accomplished. This project will fund consulting, implementation service, and required tools in order for assets to meet the new requirements. Significant effort will be required related to network segmentation to minimize in-scope cyber assets and separate High, Medium, and Low Impact cyber assets while minimizing the risk to the reliability of the Bulk Electric System (BES). Version 5 is a significant change in the standards and requires additional assets to fall under the CIP requirements. The scope of the project encompasses: (1) network segmentation to minimize in-scope cyber assets and separate High, Medium, and Low Impact cyber assets while minimizing the risk to the reliability of the Bulk Electric System (BES); (2) moving additional assets under the Critical Infrastructure Protection requirements per the standards of Version 5; (3) implementation of service and required tools in order for assets to meet the new requirements. The value of completing the project lies in: (1) Identifying and classifying Bulk Electric System (BES) Cyber Assets and BES Cyber Systems, and (2) achieving compliance with NERC CIP Version 5 Standards. As part of the review process the alternative considered was to not be compliant with CIP Version 5. This option was not chosen because any noncompliance of the NERC CIP standards may result in a fine of up to \$1 million per day per violation.	04-Dec-18	(0.96)	\$18,332	0	\$(17,043)	\$2,283	\$17,333	\$5,759	\$10,757	0	\$75	\$9,576	\$607	0	
46	2018	OT Security Architecture	Security	Network	The Operational Technology (OT) Security Architecture project will implement consistent cyber security standards at generation plants to protect valuable Company assets from increasing cyber security threats and attack. The project will also transition sustainable cyber security operations to the Cyber Security Operation Center (CSOC). The project adds value to the Company by: (1) implementing cyber security standards to ensure Company policy and non-standard configurations are consistently applied, (2) mitigating cyber security risk to the most critical assets, (3) applying standardization for both policy and technical architecture, and (4) improves visibility to cyber security vulnerabilities, enabling proactive efforts to mitigate them. The scope of the project includes: (1) defining a cyber security standard for operational technology architecture at generating plants, (2) identifying controls that have a high impact on cyber security, (3) implementing a consistent cyber security across the generation fleet, and (4) conducting generation plant evaluations against the newly adopted standard to define maturity gaps. As part of the review process, the alternative considered was to not implement and build the standards. However, this alternative was not selected because it introduces a significant increase to the cyber security risk posture at the Company.	01-Nov-19	(1.00)	\$250,901	\$,486	\$12,368	\$172,253	\$,947	\$55,846	\$45,522	\$,498	\$407	\$39,526	0	91	
47	2018	Passive Vulnerability Assessment	Security	New Computers / Hardware	The Passive Vulnerability Assessment project will deploy a tool to passively provide vulnerability assessment services to Control Networks. Active vulnerability scanning can negatively impact the reliability of a network. To overcome that risk, passive vulnerability scanning acquires asset and vulnerability data from offline networks. A zero cost proof of concept will determine capabilities and outline an approach to use either an on-site or off-site deployment strategy. The value of completing the project is to obtain more accurate and timely lists of assets, improve the ability to make data-driven operational and security decisions, complete an assessment of existing vulnerabilities in the environment without the risk of negatively impacting the reliability of Consumers Energy electric generation and energy delivery systems, and increase incident response capability, speed, and effectiveness. As part of our review process the alternative considered was implementing a traditional vulnerability scanner to interrogate all systems on the network. However, this was determined to be the best solution to fit the Company's requirements.	31-Jan-20	(1.00)	\$171,629	0	0	\$659	\$170,724	\$247	0	0	0	0	0	0	0
48	2018	Physical Security Hardware Refresh	Security	Network	The Physical Security Hardware Refresh project will ensure continued compliance with Federal Energy Regulatory Commission (FERC) requirements as they pertain to physical security assets. Included in the project is implementation of assets including advanced door systems at Company buildings, security cameras for monitoring capabilities, and gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems. The value provided by completing the project is to maintain compliance, reduce redundancies and gaps in functionality, and optimize overall performance. An integrated solution is efficient and allows for centralized management, situational awareness, real-time monitoring, compliance with regulations and guidelines, and faster, more effective/consistent response to emergencies and incidents. As part of our review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	30-Nov-19	(1.07)	\$1,168,074	0	\$434,411	0	\$733,663	0	0	0	0	0	0	0	0

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49	2018	SAP IDM Integration	Security	Software (Intangible)	The SAP IDM Integration project will configure the Human Management Capital (HCM) component of SAP IDM (User Access Provisioning tool) to connect to the HCM modules of our applicable SAP environments (ECP, CTP, and SIP). This change will allow the SAP IDM tool to view and function with the HR organizational structure in SAP ECC, otherwise known as HCM. This will provide the tool the capability to automate access provisioning and de-provisioning to users in SAP via HR processes including onboarding, transfers, and termination. The workflows would be configured to execute based on HR triggers and actions. The value of completing the project is to automate access provisioning and deprovisioning based on HR actions and triggers including onboarding and transfers, the ability to view and manage access associated with HR components including Position IDs, org structures (CRM PRDMA), and business partner creation / management, enable the use of business roles in SAP IDM through configuration of HCM to respond to HR triggers, and business role mapping to applicable business units.	18-May-18	(0.71)	375,331	0	0	31,603	297,156	46,472	13,912	0	0	0	0	0
50	2018	SAP Security	Security	Software (Intangible)	The SAP Security project will provide vulnerability scanning of SAP-specific platforms. The product will enable the scanning of HANA, which current tools in the environment do not support. Included in the project is implementing a new SAP Cyber Security platform, tuning existing systems to include security monitoring, and integrating with Cyber Incident Response Team (CIRT) processes for security and user behavior monitoring. The value of completing the project is the ability to assess the security posture of SAP environments and continuously monitor those systems. This project will enable Information Security the ability to write rules to detect potentially malicious insider activity, and to perform routine vulnerability assessments against all SAP environments which identify missing security patches and poor configuration management practices. As part of our review process the alternative considered was to apply manual monitoring of Consumers Energy systems, which is prone to errors and inefficiencies.	01-Jan-19	(0.96)	586,570	275,124	261,525	13,708	27,107	9,106	0	0	0	0	0	0
51	2018	2018 Sub-Total Security	Security					6,278,345	517,510	1,161,418	1,071,688	3,089,187	438,543	905,568	169,600	10,785	182,181	130,945	12,057
52	2018	800MHz Tower Connectivity Optimization	IT Service Delivery	Network	The 800 MHz Tower Connectivity Optimization project will replace leased backhaul T1 circuits to all radio tower sites after telecommunication providers have announced the discontinuation of leased Time Division Multiplexing (TDM) services (i.e. T1s) by 2020. The project will avoid loss of tower site connectivity by migrating off legacy TDM copper circuits to carrier ethernet fiber services before the telecommunications providers discontinue this service. Alternatives considered were to do nothing and continue escalation of chronic circuit outages to the telecommunication providers, install Consumers Energy-owned fiber, or install point-to-point microwave. These alternatives were not selected due to option 1 being undesirable from a reliability perspective and options 2 and 3 being too costly.	01-Nov-19	(0.94)	1,154,998	0	86,650	24,830	1,031,065	12,453	0	0	0	0	0	0
53	2018	ARP - Collaboration Asset Refresh	IT Service Delivery	Network	This project will replace the Company's collaborative tools and equipment. This project creates value by: (1) ensuring that the Company's audio, visual, telephony and other communications systems are stable and reliable, and (2) beginning the foundational retirement of the legacy Avaya PBX systems that have reached end of mainstream manufacturer support. The project scope consists of: (1) annually replacing collaboration assets, and (2) installing new collaboration assets to account for organic growth requirements. The alternatives considered were to: (1) refresh visual assets and a portion of the audio assets, (2) refresh a portion of the audio assets only, and (3) refresh visual assets only. These alternatives were not chosen due to the risk inherent with a partial replacement of assets which includes: (1) a reduced supply of equivalent replacement Avaya parts that are no longer being produced, and (2) an erosion of the knowledge technicians possess on discounted systems in favor of education on the newest available technology.	31-Dec-18	(0.95)	1,203,259	117,449	689,579	159,503	161,851	74,876	43,231	0	2,728	12,945	22,232	5,326
54	2018	ARP - FS Refresh	IT Service Delivery	Network	The ARP -FS Refresh project will refresh the company's FS Load Balancing equipment. The FS hardware was purchased in 2011 and the industry average refresh cycle is 5 years. Included in the implementation is refresh obsolete FS equipment, determine required feature sets for new Load Balancer, and implement new FS according to industry best practices.	31-Dec-18	(0.96)	344,551	16,065	315,482	8,102	1,269	3,632	11,971	0	0	9,217	1,161	1,592
55	2018	ARP - Field Device Asset Management (FOAM)	IT Service Delivery	New Computers / Hardware	This project will replace field devices. This project creates value for the Company by: (1) mitigating potential costs for hardware repairs, and (2) allowing field workers to complete their job tasks. The project scope consists of replacing field device assets. The alternatives considered were to: (1) extend refresh cycles from four years to five years, and (2) run the assets to failure. The selection of a four-year cycle was deemed to be the best solution because replacement in less than four years would result in additional unnecessary expense for replacement of assets that are still in peak operating condition and replacement cycles that exceed four years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, which negatively impact employee productivity.	31-Dec-18	(1.09)	1,239,744	0	1,232,984	4,145	1,282	1,723	43	0	0	39	0	4
56	2018	ARP - IT Facilities	IT Service Delivery	New Computers / Hardware	Adjustment from prior year.	31-Dec-18	(1.08)	(664)	0	(1,014)	(90)	0	440	1,154	0	959	168	0	27
57	2018	ARP - Operational Technology Support	IT Service Delivery	Network	This project will replace dated and obsolete servers and workstations. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support the operation of the Company's critical electric infrastructure. The project scope consists of: (1) the annual replacement of compute hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. Extending maintenance is not a viable alternative as current systems do not provide sufficient capacity for new electric system support capabilities.	31-Dec-18	(1.04)	1,005,547	157,548	737,912	37,524	55,796	16,767	11,562	0	11,517	39	0	6
58	2018	ARP - Printer Asset Management (PAM)	IT Service Delivery	New Computers / Hardware	This project will replace printers, plotters, and multi-function printing devices. This project creates value for the Company by: (1) improving the dependability of these printer devices for employees, (2) averting increased costs due to hardware repairs, and (3) ensuring compatibility with enterprise print applications. The project scope consists of the annual replacement of printer assets under this program. The alternatives considered for the project included looking at refresh cycles from three to seven years and running the assets to failure. The selection of a five year cycle was deemed to be the best solution since anything less than five years would result in additional unneeded expense for replacement of assets that were still in peak operating condition, and anything greater than five years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, negatively impacting employee productivity.	31-Dec-18	(1.05)	646,648	0	591,481	28,547	12,626	13,994	0	0	0	0	0	0
59	2018	ARP - Server	IT Service Delivery	New Computers / Hardware	IT infrastructure generally becomes less reliable after 5 years, jeopardizing the stability of our business' critical applications running on top of our IT infrastructure. This Server ARP project will evaluate Computer Hardware with more than 5 years of continuous use and replace where appropriate. The project will intelligently and systematically replace critical infrastructure before a system failure that would disrupt business operations.	31-Dec-18	(1.07)	1,297,962	168,846	954,337	46,653	91,321	36,805	265,274	257,094	0	1,802	5,539	840
60	2018	ARP - Storage	IT Service Delivery	New Computers / Hardware	The ARP Storage project will evaluate, validate, replace, and increase capacity of storage infrastructure after five years of continuous use. The project objectives are to reduce hardware repairs, improve system performance, accommodate organic growth, and reduce potential downtime. The benefit of replacing data center storage on a proactive refresh cycle is that it improves the stability and availability of business critical applications by proactively replacing hardware prior to the chance of hardware failures increasing. The storage five year refresh cycle is based on industry best practice on the useful life of storage arrays. From 2013 through 2017, the Company experienced an average 49% annual growth in data center storage. The Company plans on 30% growth per year based off this growth based on assumed improvements in technology, no significant architectural changes and no changes to data retention policies. The alternative is to not maintain the proactive refresh cycle and purchase extended maintenance. However, this would add additional cost and therefore was not considered further.	31-Dec-18	(1.04)	2,804,355	386,636	2,253,017	54,712	75,952	34,038	26,715	0	146	3,630	2,302	20,637

Michigan Public Service Commission
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Synopses Containing Descriptions, Scope, Benefits, Implementation Dates and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses
For the years 2018, 2019, 2020, and 2021

Information Technology Department

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
61	2018	ARP - Wireless Network	IT Service Delivery	Network	This project will replace portions of the company's aging wireless systems. This project creates value for the Company by: (1) ensuring real-time communications between Company crews and dispatch locations, (2) ensuring efficient gas leak and electric outage response times for customers, and (3) maintaining critical infrastructure and regulatory compliance. The project scope consists of: (1) replacing wireless assets annually, and (2) installing additional new wireless assets to account for organic growth requirements. The alternatives considered for system replacement were: (1) running parallel systems while the new system is deployed and the old system is dismantled, and (2) leasing a system from a vendor. The first alternative was not chosen since it would be highly disruptive due to systems having to run independently of one another, and would have a high cost associated with the need to acquire additional radio frequency spectrum. The second alternative was not chosen since the Company would be dependent on the response times offered by a shared vendor system that offers lower system reliability. The Company chose to replace existing assets based on its refresh cycle to avoid extended support cost, and to provide a seamless transition that allows both the old and new systems to interact with little to no disruption to the end user.	31-Dec-18	(0.97)	1,128,436	0	946,376	28,975	139,514	13,572	37,406	0	177,861	1,561	(142,724)	649
62	2018	ARP - Workstation Asset Management (WAM)	IT Service Delivery	New Computers / Hardware	This project will replace and install new desktops, laptops, and tablets. This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations prior to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks. The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources. The alternatives considered were: to: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle only on desktops from four years to five years, and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ARP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	31-Dec-18	(1.06)	5,281,331	95	5,107,263	141,465	0	32,407	149,221	0	127,108	12,171	7,304	2,637
63	2018	Hydro Network Connectivity & Improvement	IT Service Delivery	Network	Provide reliable corporate network connectivity to three non-headquarter, Hydro-Electric Generation sites. Along with appropriate bandwidth necessary to deliver acceptable network performance for the Hydro Control and Monitoring System (operational technology), SAP, eSOMs, SharePoint, and other critical applications. Three Hydro/Dam Sites: Rogers Crook Footstie The benefits of this project include: 1. Provide reliable business and operational network connectivity to the ten non-headquarter, hydro-electric generating sites. Along with the bandwidth necessary to deliver commonly acceptable network performance for the Hydro Control and Monitoring System (operational technology), SAP, eSOMs, SharePoint, and other critical applications. 2. The networks must be available and reliable as they are an integral part of the Hydro Monitoring and Control System. For example, the Rogers, Hardy, and Croton sites (along the Muskegon river) plant control systems are each interdependent on inputs from the others to control within regulatory requirements. 3. To achieve expected productivity, Hydro operators must be able to use their critical applications like eSOMs, SharePoint, and SAP while physically at any of the hydro generation sites.	18-Nov-18	(1.06)	175,630	0	62,014	62,154	26,774	28,688	18,229	0	0	15,079	966	2,184
64	2018	GIS Upgrade	IT Service Delivery	Software (Intangible)	Adjustment from prior year.	18-Nov-17	(0.92)	3,032	0	1,011	0	2,021	0	0	0	0	0	0	0
65	2018	Nimbus Phase 4	IT Service Delivery	Software (Intangible)	In 2015, Consumers Energy implemented its private cloud. 2015's effort was focused around creating self-service and automated deployment for basic datacenter requests. In 2016 continued expanding the features of Consumers Energy's private cloud in phase 2 project. These additional features would include: • Self-Healing: Automated repairs to server and application errors. Instead of requiring manual intervention from an operation team when a problem ticket is created, the Private cloud with automatically resolve the issues and inform CE teams that the issue was resolved. • Server Application Packaging and Self Service Deployment allowing application teams to deploy and redeploy entire environments in a matter of minutes instead of weeks. • Storage/Network virtualization allowing the private cloud to provision storage/networks when needed without requiring manual intervention and labor. • Continue to reduce CE Build labor (Benchmarks), speed to market for IT infrastructure services. • Develop reactive healing countermeasures to reduce infrastructure/application unplanned outages. • Develop automatic benchmarking interfaces and reports. With successful implementation of this phase of the project effort, simple errors which cause critical systems outages will be resolved before outages occur. Critical business applications will increase in size before our customers and our business feels performance issues.	01-Jun-18	(1.03)	714,990	0	(6,093)	411,287	100,596	209,200	190,196	19,784	0	82,612	77,296	10,503
66	2018	Office 365	IT Service Delivery	Software (Intangible)	The Microsoft 365 project will implement in a standardized and secure manner the following Microsoft 365 services: Exchange, OneDrive, Skype for Business, Teams, and the enablement of SharePoint Online. This project will also implement a Cloud Access Security Broker service to be used for all cloud applications. This will allow for secure collaboration with internal and external parties. In addition, this project will enable mobile access from corporate devices, contain all aspects of conversion of mail, and activate e-discovery features. Two alternatives were considered. One option was to do nothing and run non-mainstream supported applications that are dependent on computing and storage that will no longer be supported. This was not selected due to the risk of running on an unsupported version for critical applications. The other alternative considered was to refresh on-premise instances for Exchange, File Share, SharePoint, and Skype. This option was not selected due to the Microsoft strategy and future support scenarios.	01-Dec-18	(0.94)	0	0	0	0	0	0	1,047,031	56,773	41	92,829	878,036	19,352
67	2018	Operations Application Currency	IT Service Delivery	Software (Intangible)	The Operations Application Currency project is for upgrades for the following applications: Primavera, Impres, Robosync, ArcPad, WECCO (meter test boards), MIMS (Mobile Information Mapping System), and OMS (Outage Management System). Included in the implementation is upgrading the application software, assessing any new functionality for value to the Company, making necessary configuration changes, testing any integrations to or from the application, testing the upgrade, and updating documentation related to integration changes. The value of completing the project is reducing the number of incidents associated with out-of-date software, increasing application stability, and allowing the Company to leverage new functionality available in upgrades. As part of our review process the alternative considered was to stay at the current version of the applications. If this approach is selected we risk applications not working, increased number of incidents, increased support costs, and employees will not be able to perform their tasks.	31-Dec-18	(0.94)	234,073	29,570	0	58,137	100,075	36,291	30,882	0	5,808	17,409	4,911	2,754
68	2018	Printer Document Management Platform	IT Service Delivery	Software (Intangible)	Adjustment from prior year.	28-Jul-17	(1.01)	(21,985)	0	0	(11,294)	(5,726)	(4,965)	18,217	0	0	10,688	5,419	2,110

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
69	2018	SAP Data Archiving	IT Service Delivery	Software (Intangible)	This project will move outdated data from an online database to offline storage. This project will create value for the Company by: (1) increasing system stability by reducing data growth, and (2) controlling maintenance costs associated with data storage. The project scope includes: (1) archiving data based on the fastest growing and largest objects in SAP; (2) building and archiving solutions that allow the business to retrieve archived data in the required form, and (3) setting up the solution so that the business areas meet compliance standards. Three alternatives were explored and determined non-viable for the project: (1) allow the database to grow in size, which would put system performance at risk and result in prohibitive storage costs; (2) decrease the overall scope and archive less projects, which would result in minimal positive impact to system growth and significant storage-related costs; and (3) increase the scope and archive more objects in a shorter timeframe, which would result in significant cost increase over shorter time period. After considering each of these options, it was determined that the current scope of the project was the best strategy to address the problem while balancing annual spending.	31-Dec-21	(1.00)	413,071	0	0	33,188	325,371	\$4,512	286,483	0	0	16,851	233,839	35,773
70	2018	SAP Data Encryption	IT Service Delivery	Software (Intangible)	The project will implement Cyber Security requirements for encryption of Personal Identifying Information (PII) data "at rest" and "data in transit" in Company SAP data bases.	01-May-21	(1.00)	0	0	0	0	0	0	178,347	0	0	35,549	128,331	14,466
71	2018	SAP Performance Tuning	IT Service Delivery	Software (Intangible)	The project aims to target two major work streams to help improve SAP Performance (1) Use Oracle compression on large tables to decelerate the data growth in ECC and improve performance (2) Analyze Custom Code that can be retired or remediated to improve performance. (3) Perform Custom Code Analysis to identify code to retire and remediate (2) Identify the top 30 tables in ECC for compression. (3) Identify SAP data purge targets and purge data to improve performance.	16-Dec-18	(0.97)	0	0	0	0	0	0	119,707	0	0	11,588	97,539	10,580
72	2018	SAP Platform Modernization	IT Service Delivery	New Computers / Hardware	The SAP Platform Modernization project will include the re-architecture and replacement of the 2007-2008 SAP infrastructure, which is well beyond its recommended useful life. The program also includes an upgrade of SAP applications to Enhancement Pack 8. This program is essential for the company to maintain support and stability of its core set of business applications, while also improving system availability, performance and resiliency.	04-Jun-18	(1.00)	(130,751)	(15,010)	2,609	(35,260)	(274,176)	(8,913)	186,702	15,375	0	61,740	99,575	10,012
73	2018	Service Now Quarterly Releases	IT Service Delivery	Software (Intangible)	The ServiceNow Bi-Annual Releases project will upgrade the cloud solution, ServiceNow, and validate/remediate the integrations. ServiceNow requires that customers move to the current version of the software as new versions are released. In addition, this project will increase the use of automated testing each year by 20%. Also included in 2019 is the upgrade of the ServiceNow Portal - it will reconfigure the portal and make it much more intuitive for end users to enter tickets, receive status, and avoid rework. Included in the implementation is performing regression testing on the new version to ensure it still functions as expected, validating and remediating integrations, analyzing new available functionality for features that may improve productivity, testing any new features that will be implemented, and implementing the new version. The value of completing the project is continuing to be on a vendor supported version, ensuring support in resolution of issues with the software, and implementing new functionality to improve productivity. As part of our review process, the alternative considered was waiting for ServiceNow to automatically upgrade us without us prepping or doing any testing. This alternative would potentially have significant impacts on operations and our ability to support the business. Without ServiceNow functioning, the Company cannot process, track, monitor, and execute break/fix and other operational incidents. This could lead to significant loss of services for the business and potentially customers.	01-Jul-19	(0.87)	0	0	0	0	0	0	160,726	0	0	19,868	121,175	19,682
74	2018	ServiceNow IT Asset Management	IT Service Delivery	Software (Intangible)	Adjustment from prior year.	25-Mar-17	5.00	(8,600)	0	0	0	(8,600)	0	8,138	0	0	0	8,138	0
75	2018	Software Platform Refresh 2019	IT Service Delivery	Software (Intangible)	The Software Platform Refresh project will upgrade server operating systems, hypervisors (virtual machine monitors), and databases to retain low cost, unlimited vendor support. The project scope includes: (1) upgrading operating systems and databases on servers that are within three years of end of support, and (2) maintaining hypervisors at the current version for stability and performance. The project will add value for the Company by: (1) avoiding costs for special maintenance agreements required at the end of normal manufacturer support, (2) ensuring reliability and compliance with Information Security requirements, (3) improving data center environment stability, and (4) avoiding the need for high risk upgrades that cross multiple versions. A funding options matrix was completed to review the alternatives. The options were: (1) fund the full project for \$1 million and pay no support liability in 2020, (2) fund \$750,000 of the project and pay \$1.4 million in support liability in 2020, (3) fund \$500,000 of the project and pay \$2.4 million in support liability in 2020, and (4) not funding the project and paying \$3.3 million for support liability in 2020. Alternatives 2-4 were not selected due to the high cost of support liability. Option 1 was chosen to avert these costs and to ensure system stability.	31-Dec-18	(1.00)	0	0	0	0	0	0	247,755	0	0	50,506	167,105	30,143
76	2018	The Clarity 15.4 Upgrade and Enhancements	IT Service Delivery	Software (Intangible)	The purpose of this project is the upgrading of the IT project and portfolio management application to the current version as scheduled in 2020. Included in the implementation is performing impact analysis on the new version to ensure it still functions as expected, analysis of new available functionality for features that may improve productivity of users, testing of the upgrade, and new features. The value of completing the project is continuing to be on a vendor supported version so that issues with the software can be resolved, and implementing new functionality to help improve the productivity of users. As part of our review process, the alternative considered was not upgrading and assuming risk of running software that is not supported by the vendor. If problems are encountered that require support (fixes) from the vendor, the Company would not be able to resolve them, resulting in potential business process impact. Therefore, the upgrade was selected over the alternative.	01-Nov-20	(0.89)	0	0	0	0	0	0	120,958	0	0	1,325	97,240	12,372
77	2018	SubTotal IT Service Delivery	IT Service Delivery					17,279,527	861,498	12,979,219	1,092,940	1,897,611	355,520	3,149,927	340,826	326,170	457,637	1,813,406	201,490
78	2018	e911 Compliance Project	Enhancements	Software (Intangible)	The e911 Compliance project will implement a corporate-wide enhanced 911 (e911) system to ensure fire, police, and paramedics have accurate location information and can find those in distress when a call is placed to a public safety answering point ("PSAP"). Included in the implementation is installing the on-premise application with standardized virtual compute and storage in a highly available configuration and configuring the application for interfacing with existing telephony systems and Local Area Networks. The system will be installed and configured by an integrator assisted by the publisher of the e911 application. The value of completing the project is increasing employee safety by increasing the accuracy of where a distress call has been placed within a building or facility, decreasing the response time of first responders as emergency calls will be routed directly to the PSAP instead of first to the Command Center, and ensuring compliance with the new MPSC Rule 484. As part of our review process the alternative considered was do nothing and continue to have 911 calls routed to the Command Center and become noncompliant with MPSC Rule 484.	01-Apr-19	(0.96)	400,213	83,517	7,976	8,355	15,316	5,552	221,822	1,595	0	10,119	187,593	3,116
79	2018	Enhancements - CEO	Enhancements	Software (Intangible)	The Enhancements - Cust Exp - Ops project is for software enhancements performed for Customer Experience and Operations business areas. The functionality includes mandatory regulatory changes, enhancements to billing, enhancements to dialing for low income customers, and changes to the Company website based on customer feedback. An alternative would be to not do these smaller system enhancements. However, that limits the company's ability to improve the applications that benefit employees who are serving customers and puts the company at risk of not meeting regulatory requirements. Customer benefit: System enhancements to comply with regulatory changes.	31-Dec-18	(0.98)	1,237,257	(7,811)	0	126,746	682,859	235,463	275,143	3,920	0	79,119	172,458	19,646

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80	2018	Enhancements - Corporate	Enhancements	Software (Intangible)	The Enhancements – Corp project is for small software enhancement work efforts performed for the Corporate Services business areas. Included in the implementation are enhancements for Finance, Human Resources (HR), Learning & Development, Legal, Governmental, Regulatory and Public Affairs, and Corporate Security. The value of completing the project is compliance, hard cost savings, avoided costs, hitting corporate goals, and mitigating risk. As part of our review process the alternative considered was to not to provide funding for enhancements. However, this then limits our ability to provide process improvements, make changes related to customer and employee safety, and meet legally required HR and Finance system changes.	31-Dec-18	(0.90)	642,719	41,720	23,433	175,780	276,914	124,872	290,910	269	2,017	87,638	176,875	24,111
81	2018	Enhancements - TEOS	Enhancements	Software (Intangible)	The Enhancements – Transformation, Engineering, & Operations Support ("TEOS") project is for small software enhancement work efforts performed for the Transmission, Engineering, and Operation Support business area. Throughout the course of a year small requests for system changes are requested. These small changes generally cannot be predicted in advance at a detailed level so we set aside a certain amount of funding each year to address them. Requests for this funding are assessed for value by representatives from each area using benefits that are categorized into hard cost savings, cost avoidance, compliance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are regulatory compliance mandatory changes, improvements to engineering and design applications, and minor functionality improvements across many production systems. As part of our review process the alternative considered was to not to fund enhancements, however, that limits the Company's ability to improve the applications that benefit employees who are serving customers. This will limit our ability to provide process improvements, retire applications, make changes related to customer and employee safety, and meet legally required changes.	31-Dec-18	(0.98)	223,233	0	0	98,194	73,235	51,804	161,332	0	0	35,152	114,124	12,056
82	2018	Enhancements - Operations	Enhancements	Software (Intangible)	The Enhancements – Operations project is for small software enhancement work efforts performed for the Operations business area. Throughout the course of a year small requests for system changes are requested. These small changes generally cannot be predicted in advance at a detailed level so we set aside a certain amount of funding each year to address them. Requests for this funding are assessed for value by representatives from each area using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small application changes and functionality improvements for Electric Operations, Gas Operations, Operations & Financial Planning, Generation Operations, and Distribution Performance. As part of our review process the alternative considered was to not fund enhancements, however, that limits the Company's ability to improve the applications that benefit employees who are serving utility operations. This will limit our ability to provide process improvements, retire applications, make changes related to customer and employee safety, and meet legally required changes.	31-Dec-18	(0.96)	182,952	0	31,746	45,391	73,704	32,111	130,075	0	0	28,060	90,529	11,486
83	2018	Enterprise Content Management	Enhancements	Software (Intangible)	The Enterprise Content Management project will provide an Enterprise Document Management solution including the strategies, tools, and processes to more easily manage, rapidly locate, and deliver CMS Energy content throughout its life cycle wherever that content exists into the form the business partner needs. Areas needing this include: Legal, Risk and Insurance, HR, Strategic Communications, Learning and Development, Distribution. Content management enables organizations to enforce policies and rules for the retention and disposition of content required for documenting business transactions, in addition to automating the management of their record-retention policies. These technologies, implemented with well-formulated and consistently enforced records retention policies, form an essential part of the life cycle management of information. As industry regulation and compliance requirements increase, along with the volume of digital content that must be retained and the demand for legal discovery.	01-Dec-18	(0.94)	2,076,724	1,183,345	0	115,036	673,231	105,112	402,789	0	0	45,043	343,926	13,820
84	2018	2018 SubTotal Enhancements	Enhancements					4,763,100	1,980,271	63,154	769,501	1,795,239	554,915	1,482,072	6,144	2,017	294,151	1,099,366	84,255
85	2018	5 Minute Settlements	BP Functionality	Software (Intangible)	PCI Back Office Suite, PCI Profit and Loss Analyzer, PCI Settlement Analyzer, PCI Energy Accounting, EA Data Mart, and GenTrader require a major redesign and upgrade to support the new MIS0 5-Minute Settlements solution (Requirement of FERC Order 825). Moving to 5-minute intervals for data will result in a significant increase in database size, requiring new hardware/OS for database servers. In order to continue settlements with MIS0, and to meet requirements of FERC Order 825, this upgrade is required by April 1, 2018. Not completing this work would have negative financial and compliance implications.	04-Sep-18	(0.95)	293,111	0	162,108	33,060	84,720	13,422	30,190	0	9,625	13,280	4,931	2,354
86	2018	API Management	BP Functionality	Software (Intangible)	Adjustment from prior year.	07-Dec-17	(0.95)	64	0	0	0	64	0	2,118	0	0	1,596	304	218
87	2018	Bill Redesign	BP Functionality	Software (Intangible)	The Bill Redesign project will update customer bill presentation, align and address changes driven from the Summer Peak Usage Rate project, and provide a more intuitive invoice of charges to avoid customer inquiries. Included in the implementation are paper and e-bill invoices for electric and gas-electric combo customers, comprehensive analysis of required bill data, and bill clarity analysis based on customer perception. The value of completing the project is reduced customer calls and improved customer experience. Customer benefit: Improved CX score of 5.0% by updating visual appearance of bills to make them easier to read. As part of the project review process the alternative considered was to continue with the current bill presentation and address anticipated increase in customer calls through staffing and other non-billing mechanisms.	01-Aug-19	(0.91)	0	0	0	0	0	81,297	0	0	2,045	70,496	8,756	
88	2018	BPC Automation and Reporting	BP Functionality	Software (Intangible)	The BPC Automation and Reporting project will provide a version upgrade that allows for functionality for automation of data loading and reporting. New functionality for data and hoc analysis for gas/electric price volume and general ledger validations. Included in the implementation is allowing for functionality for automation of data loading and reporting.	22-Apr-18	(1.02)	166,224	0	0	30,723	101,827	33,674	8,497	0	0	3,191	4,272	1,033
89	2018	Business Continuity Disaster Recovery Integration	BP Functionality	Software (Intangible)	The Business Continuity Disaster Recovery Integration Project will implement technology solution and supporting processes to integrate the Company's business continuity and disaster recovery programs to enhance program efficiency and effectiveness. The solution will improve program management and drive efficiency with the following: Plan management repository with workflow capabilities (in support of plan development/review, training and testing requirements; Maintain program schedules, monitor status and reporting capabilities; Risk analysis and interdependency mapping of critical business processes and IT applications; Flagging mechanism to ensure identification of restoration capability gaps to critical business processes and/or IT recovery capabilities; Business Impact Analysis (BIA) capabilities to quantify financial risks to critical process disruptions.	30-Apr-19	(0.74)	129,631	155,394	(107,400)	47,832	5,518	28,287	9,580	0	0	9,419	0	161
90	2018	Business Process Performance Monitoring	BP Functionality	Software (Intangible)	The Business Process Performance Monitoring project will deploy the AppDynamics tool across enterprise applications like SAP, the Outage Management System, and other business critical systems. The project scope includes (1) SAP, (2) Outage Management System (OMS), (3) Advanced Device Metering System (ADMS), and (4) other enterprise applications. This project will create value for Consumers Energy by providing for visibility into business process performance, artificial intelligence, and automation, leading to better system troubleshooting, root cause determination and a reduction in mean time to resolution. Consumers Energy evaluated multiple products for this project and AppDynamics fit the cost and functionality expectations. The others were cost prohibitive and lacked the functionality of the product selected. Please see the attachments Model Evaluations, Presentation 1 to Leadership, and Presentation 2 to Leadership for a detailed summary of alternatives.	01-Oct-23	(1.00)	0	0	0	0	0	274,022	105,383	0	27,626	134,389	6,624	

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
91	2018	Care Annual Updates	BP Functionality	Software (Intangible)	This project will: (1) implement software changes to improve the process for offering energy assistance to low-income customers, and (2) streamline the process for assistance agencies who utilize the application through improved user interface and updates to SAP to process various CARE requests. Upcoming modifications will be identified following an annual review of requests to prioritize the list of changes. The project will provide the following value: (1) complete modifications to the existing internal SAP application and Agency Portal to receive annual Low Income Home Energy Assistance Program ("LIHEAP") funding, which is used to provide customers bill credits and arrears forgiveness; and (2) improve the data within the assistance Agency Portal, thereby making it easier to assist customers in need of LIHEAP funding. The project scope includes: (1) updating the enrollment and status process, (2) allowing for flexible bill credits, (3) improving reporting, (4) updating the arrears forgiveness plan, and (5) satisfying additional regulatory requirements for the annual grant rule changes required by the Michigan Department of Health and Human Services and the Michigan Agency for Energy. The alternative considered was to continue with the current process. This alternative was not chosen since it would result in loss of grant funding, thus decreasing or eliminating energy assistance dollars for low income customers. Instead, making annual updates to the application will allow agencies to easily enroll customers on assistance programs, and allow placement of holds to stop or prolong credit activity until assistance decisions are granted. This option provides long-term proactive energy assistance to low income customers and prevents loss of grant funds. All changes are related to the existing internal SAP and Agency Portal, therefore a cloud alternative is not viable.	31-Dec-18	(0.90)	0	0	0	0	0	0	138,084	0	0	63,124	58,755	16,205	
92	2018	Complex Billing Automation	BP Functionality	Software (Intangible)	The Complex Billing Automation project will reduce manual intervention and decrease risk of error on large customer and specialty billing accounts. Complex billing encompasses, but is not limited to, all primary rates that require MV90 billing, large commercial and industrial rates, unique customer specific billing solutions (i.e., MSU, BioQuestor, some self-generators), and customer programs such as net metering. Included in the implementation is automating fragmented processes to support complex billing; updating/reviving complex account calculations based on rules while permitting customers to choose a billing method and delivery time, permitting multiple accounts to be summarized, addressing Net meter billing, Collective accounts, Gas Transportation accounts and unmetered billing accounts. The value of completing the project is improving the process of setting up and billing this customer segment by automating the process, improving Customer experience by providing this customer segment currently receiving multiple bills to have one consolidated bill, and allowing customers to pay multiple bills with one invoice. As part of our review process the alternative considered was to maintain the status quo and continue to leverage employees to manually set up, bill, and monitor the billing of these customers.	07-Mar-19	(0.94)	373,409	0	0	155,081	138,089	80,239	96,521	0	0	27,445	56,947	12,129	
93	2018	Contact Center Customer Experience Refresh	BP Functionality	Network	The Contact Center Customer Experience Refresh project will refresh of the Customer Call Center's IT infrastructure, including the three Automatic Call Distributor (ACD) systems, networking equipment, IVRs, Work Force Management, servers, and applications. The ACD Systems are 10 years old in 2015 and cannot readily adapt to best practice. Additionally, they are no longer vendor supported and hardware replacement parts are not available. Included in the implementation is speech enabled interactive voice response (IVR), Customer Service Representative Knowledge Management, eMail Management, Call Center Quality Monitoring, Optimize Skills Based routing, Customer Analytics Enhancements, Multi-Channel Inbound & Outbound Communications, Virtual Hold, and Click to Call.	24-May-18	(0.55)	1,537,236	246,822	(15,726)	241,112	809,315	255,713	324,421	51,827	30,934	71,701	25,823	144,137	
94	2018	Credit and Collections	BP Functionality	Software (Intangible)	The Credit and Collections project will make improvement to our outside agencies collections rate by enhancing our visibility and tracking of uncollectable expense via middleware. This project will implement the DebtNet cloud solution to support the lowering of uncollectable expenses. It will allow payments to be uploaded and credited to a customer's account automatically, provide reports that detail debt and aging of debt, and provide the ability to move accounts between various collection agencies. The value of completing the project is increased productivity due to reduction in time to audit and complete invoice reconciliation, new customer communication channels, the ability to easily manage third-party collections agencies and the accounts placed with them, the ability to quickly transfer accounts amongst agencies, decreased uncollectables, reduced costs, and improved operations. Failure to implement this project would result in unrealized savings listed above. Customer benefit: Reduced uncollectables expense by improved credit collection rate. Projected 5-to-1 payback.	01-Mar-19	5.00	2,661	0	0	480	1,756	426	0	0	0	0	0	0	0
95	2018	Crosswinds Energy Park II	BP Functionality	Network	CWEP II late 2017 project will add (19) GE 2.3-116 turbines (44MW total), requiring: (1) Adding OPC hardware, SCADA/PI reconfiguration; (2) Re-configure the RTU Skynet modem configuration (in the substation) from cellular antenna to POTS (fiber) connection for communication to HVD Fault Analysis group; (3) remote Access Upgrade (Jumpbox); (4) Bandwidth Evaluation; (5) Hardware / Rack Space (GE Server Rack); and (6) analyzing and adding Historian Storage / Archive Capacity. The project will additionally include: (1) integration of Phase II and Phase III assets into the PI Historian; (2) revise modem connection from cellular antenna to POTS (fiber) connection for improved reliability to HVD Fault Analysis group; and (3) reconfigure the SCADA server.	20-Jun-18	N/A	0	0	0	0	0	0	3,184	0	0	0	2,760	424	
96	2018	Customer Care Excellence (Interactions-SIP Based Implementation) (IVR Solutions)	BP Functionality	Software (Intangible)	Adjustment from prior year.	04-Apr-19	0.77	(7,543)	0	0	(4,651)	(1,505)	(1,387)	0	0	0	0	0	0	
97	2018	Customer Intelligence and Analytics	BP Functionality	Software (Intangible)	The Customer Intelligence and Analytics project will create a comprehensive and analytical view of customers across various classes encompassing: outages, billing history, payment history, customer usage and payment channel for the purpose of segmenting and providing improved customer experience. Included in the implementation is to include various business partner data, rates and billing data and history, various programs our customers are enrolled in, complaints filed by customer and customer preferences. This data will integrate with our Business Warehouse to provide a comprehensive view of our customer. The value of completing the project is providing a single source for all of the customer interaction data within the utility, providing analytics for modeling, and execution of one on one customer interactions, and these analytics will provide insights into the various customer journeys and experiences allowing Consumers Energy to determine opportunities to improve that experience across all channels. As part of our review process the alternative considered was to continue to operate as today, leaving us without a comprehensive view of our customers and their experiences with Consumers Energy.	01-Apr-20	(0.95)	1,488,368	321,396	191	252,634	741,087	173,080	143,421	41,913	0	30,256	62,180	9,073	

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98	2018	DCE R4 Hub and Spoke	BP Functionality	Software (Intangible)	The DCE Hub and Spoke project release will implement a payment workflow that creates a simplified user experience for customers looking to pay their bills online. Instead of traversing multiple steps to define: payment date, payment amount, and payment method, customers will be routed to a new "hub" experience where defaults will be supplied for all of the above elements and a customer can selectively choose to 'spoke' out to shortened user flows if they want to make any changes. This new experience will allow customers to make a payment in as few as two clicks. The release scope includes new reporting added to the hub & spoke experience that will allow business partners to gain insight into how many successful payments were made, how many payments failed, and how many customers go through unique payment flows such as installment plans, overdue or dunning, and auto-pay. In addition, the release adds value by re-purposing one of the existing preprod and production Sitecore servers, which will enable the Company to accelerate future releases (beginning with this one) with a new release strategy. This will allow future releases the ability to make isolated deployments to this server and throttle specific amounts of customers to the new version of the website, while the rest of the customer-base visits the standard version of the website. This creates the ability to effectively monitor the customer experience through new reporting that gauges whether the new experience is working as expected, and provides input on the decision to proceed with the deployment for all visitors.	28-Sep-18	(0.92)	617,739	0	(2,165)	70,517	440,737	88,650	3,829	0	0	0	881	2,567	381
99	2018	DCE Website Replacement_R3A 2017	BP Functionality	Software (Intangible)	This project release will implement Website search functionality. • Integrated search provides users the option of using search as a primary navigation source or for assistance in locating content of interest. Statistics vary on the percentage of customers who prefer search over traditional navigation. • Inability to find desired content is the top reason customers report having to call after first attempting to self-serve on the website • Search provides valuable diagnostics/analyses on what customers are looking for, which helps prioritize initiatives and optimize site content	17-Jul-17	(0.97)	(71,502)	0	251	(8,162)	(53,329)	(10,261)	0	0	0	0	0	0	
100	2018	DCE Website Replacement_R3B 2017	BP Functionality	Software (Intangible)	The DCE Website Replacement_R3B 2017 project will implement Sitecore Version 8.2. The value of completing the project is that version 8.2 is a significant upgrade to the current 7.5 version. It provides website analytics to help identify opportunities to improve site traffic and performance. This includes "value" assignment for advanced target marketing capabilities, includes launch of a "forms" module that will help accelerate development of online forms that allow customers to complete transactions on the new website, and will improve website performance	09-Mar-18	(0.97)	598,692	0	(2,098)	68,343	446,531	85,916	52,211	0	0	0	12,018	35,002	5,192
101	2018	DCE Website Replacement_R3C 2017	BP Functionality	Software (Intangible)	The DCE Website Replacement_R3C 2017 project will implement Profile Update/Profile Wizard. Included in the implementation is creating a new "create profile" and "profile update" user flow that introduces customers to Consumers Energy service options and will remind customers when their profiles are incomplete. This first version will include alerts, email and choose due date. Subsequent versions will include EE/RE options and rates. Additional options are expected to be added.	28-Jun-18	(0.84)	2,270,357	0	(7,956)	259,169	1,693,333	325,812	148,587	0	0	0	34,201	99,611	14,775
102	2018	EA - Capacitor Control Replacement	BP Functionality	Software (Intangible)	Adjustment from prior year.	01-Dec-16	(0.90)	(62)	0	0	(43)	0	(19)	0	0	0	0	0	0	
103	2018	ED - ARP OSI HVD and LVD SCADA Upgrades	BP Functionality	Software (Intangible)	The Electric Distribution (ED) Asset Refresh Project (ARP) upgrades OSI Supervisory Control and Data Acquisition (SCADA) application for High Voltage Distribution (HVD) and Low Voltage Distribution (LVD) systems. Biennial hardware refresh included. The scope of this project is a software upgrade to the High Voltage Distribution (HVD) supervisory control and data acquisition (SCADA) software. The HVD SCADA is used to control generation and high transmission and distribution voltage circuits. Upgrading the HVD SCADA software will ensure that when problems occur, they are promptly addressed. Additionally, upgrades include enhancements and fixes to the core product that support reliability of electric delivery, ensuring that customers receiving the energy they need when they need it. Because the reliability of the entire distribution system is dependent on availability of the HVD system, staying relatively current with the solution provider upgrades is critical. If upgrades are delayed, additional full time resources will be required for 1 year to address upgrade complexities.	19-Dec-18	(0.98)	141,176	0	10,657	76,018	7,037	47,463	21,460	0	1,137	16,541	664	3,118	
104	2018	ED - Cyme Upgrade	BP Functionality	Software (Intangible)	The CYME Distribution Analysis software is designed for planning studies and simulating the behavior of electrical distribution networks under different operating conditions and scenarios. It includes several built-in functions that are required for distribution network planning, operation and analysis. The Cyme solution currently runs on an unsupported version of software. The vendor has encouraged Consumers to upgrade, stating that they will not be able to support future problem resolution efforts. The software is currently loaded on individual workstations. Gateway will continue to reside on server. However, a network based solution to replace the PC based solution is not a feasible solution from Cyme. The scope of this project is to "replace in kind." Once upgraded, there will be the ability to evaluate other Cyme modules for their potential value for other departments. . The CYMDIST Distribution Analysis software is a suite of applications composed of a network editor, analysis modules and user-customizable model libraries. The program is designed for planning studies and simulating the behavior of electrical distribution networks under different operating conditions and scenarios. It includes several built-in functions that are required for distribution network planning, operation and analysis. This will also require an upgrade to Cyme Gateway (this will not require hardware/infrastructure changes. Consumers must utilize Cyme, or some other solution, in order to effectively perform load flow analysis that is critical to distribution planning, outage resolution. It's used on a daily basis for switching analysis. If the software and platform are not upgraded, the vendor will not support future problem resolution, putting the ability to perform required analysis at risk.	01-Aug-18	(0.95)	349,666	15,900	0	186,305	44,180	103,281	3,394	0	0	1,352	1,681	361	
105	2018	EHS Compliance	BP Functionality	Software (Intangible)	The EHS (Environmental Health & Safety) Compliance project will implement a cloud-based corporate-wide incident and risk management solution. The solution will ensure consistent accurate tracking, management, and reporting of environmental, health & safety compliance tasks using a centralized source of data. Modules to be implemented include Compliance, Health and Safety, Human Performance Management, Incident & Crisis, and Reporting & Analytics. The value of doing this project is to leverage best practice processes which are built into the new solution, including incident management and prevention to ensure a safe and productive workforce able to complete work for customers. The project also supports the Company's Planet goal through enhanced tracking and reporting for air quality, waste management, and sustainability. Other benefits include risk avoidance for environmental penalties, increased productivity, and increased quality. The addition of visual management and dashboards to enable transparency in tracking of goals; and create awareness of emerging environmental regulations and, to respond to them. One alternative is to continue "as is" through use of disparate Excel and SharePoint solutions. This requires much manual effort, and poses a risk of not being able to accurately and centrally track EHS data. Another alternative is to have separate projects; one for Environmental Compliance, and another for Safety and Health Compliance. This was the initial path for these two business areas. Investigation into potential solutions including a formal Request for Information (RFI) revealed that the standard for these market solutions is integrated functionality for Environmental and Safety/Health. Both on premise and cloud solutions were considered. The single solution was selected as the most efficient solution.	01-Dec-20	(0.86)	1,213,054	502,455	0	46,501	618,073	46,026	157,008	0	0	32,899	108,310	15,799	
106	2018	ECM - ProjectWise	BP Functionality	Software (Intangible)	This ECM (Engineering Content Management) project is to investigate and install drawing management software capable of bundling, managing and handling the versions of Gas records/documents related to design, proposals, contract resources and record-keeping. The solution will be capable of bundling, managing and handling the versions of documents related to design, proposals, contract resources and record-keeping. 1. It must be able to render, as one document, drawings that are composed of several files, as is the case with many CAD drawings such as those produced in Microstation and AutoCAD software. 2. It must be able to maintain spatial relationships within drawings and documents such that all documents pertinent to a geographical location can be quickly located and maintained. 3. It must be able to integrate with Microstation, Microsoft Office, AutoCAD, SharePoint, SAP and potentially other systems.	01-Nov-21	(1.00)	256,589	0	0	46,821	158,428	51,340	74,891	(94,438)	0	3,906	163,758	1,685	

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107	2018	Internal Customer Center Support	BP Functionality	Network	The External Customer Contact Support portal will implement access to Company applications for a third party contact center vendor, so that they may augment internal customer contact's day-to-day operations by supporting inbound customer calls. The project adds value by: (1) augmenting contact center staffing for day-to-day operations, which reduces customer wait times, increases Average Speed of Answer (ASA), and improves first contact resolution, especially during peak call volumes such as during storm restoration; (2) providing the ability to effectively respond to ever-changing business demands; (3) enhancing services provided to customers through optimization and applying best practices across all customer interactions; and (4) enables flexibility to manage call volume between multiple third party vendors. In addition, configuring the contact center environment to manage multiple third party vendors creates a dual vendor partnership and champion challenger model, which reduces operational costs. The scope of the project includes: (1) configuring third party vendor access and connectivity to Company environments; (2) configuring the Automatic Call Distributor (ACD) and Interactive Voice Response (IVR) to route calls to multiple third party vendors; and (3) enabling access to the Company contact center training environments. Alternatives considered include: (1) Continue to single source a third party vendor for the contact center. This alternative was not selected because it increases operational costs and offers no incentive to improve performance. (2) Implement multiple third party vendors. This alternative was selected because of the capability to improve performance by implementing a champion challenger dual vendor environment while optimizing ongoing operational spend.	22-Feb-19	N/A	0	0	0	0	0	0	449,247	177,928	0	33,866	226,474	30,979
108	2018	Field Connectivity	BP Functionality	Network	Adjustment from prior year.	N/A	N/A	802	0	802	0	0	0	0	0	0	0	0	0
109	2018	Field Service Solution Release 3	BP Functionality	Software (Intangible)	Adjustment from prior year.	11-Dec-17	(0.67)	(10,926)	0	0	(3,726)	(5,043)	(2,157)	5,487	0	0	2,861	2,062	564
110	2018	Financial Planning Transformation - R1 Basic Support	BP Functionality	Software (Intangible)	Adjustment from prior year.	18-Apr-19	0.61	(2,596)	0	0	0	0	(2,596)	0	0	0	0	0	0
111	2018	Generation and Compression & Storage Engineering Records Management	BP Functionality	Software (Intangible)	The Generation and Compression Engineering Records Management project is for implementation of the ProjectWise drawing/records management software to manage compression assets. The system will be capable of bundling, managing and handling the versions of drawings and documents related to design, project files, construction of As-Built and record-keeping. Included in the project is implementation of document sharing for collaboration with parties external to the company, and developing a data conversion solution to facilitate creation and importation of data. The value of completing the project is: creating a single "resource of truth" for projects and operations for all records with a regulated review process; meeting both compliance and engineering best practice standards; eliminating the possibility of work being executed in the field based on an out-of-date drawing or document; ensuring records needed for an audit are readily accessible and easily searchable; providing compliance and design control; managing and handling all versions of drawings and documents related to design, proposals, contract resources and record-keeping with real-time access; integrating with SAP, and replacing the use of multiple document repositories (e.g., SharePoint, network drives, Access Databases, and other systems). As part of our review process the alternative considered was the application OpenText, but the capability to work with engineering drawings and computer aided design (CAD) software was lacking.	01-Feb-19	3.02	750,445	550,004	0	39,301	141,755	19,385	66,665	5,913	0	6,539	47,463	6,750
112	2018	GIS-Integrated Design	BP Functionality	Software (Intangible)	This project replaces the CAD and Work Requirements and Design ("W&D") software with Bentley's Open Utilities Map ("BOMU") to leverage GIS asset data to generate engineering designs, implement workflows with SAP, and mitigate risk associated with the aging software application. The project provides value to the Company in three ways: i) the BOMU software is compatible with GIS integration and may be further developed after the Company implements the LUPDM as the basis of its GIS systems; ii) the project mitigates risk of manual asset record imports, asset record manipulation for design while space management, or manual asset record recreation for base design files, by enabling technology that creates a new design over existing electronic asset records; iii) it enables technology that is compatible with the asset system of record and the Project Wise engineering document management system. The project scope includes: i) replacing CAD engineering design software (Bentley Microstation (JVT) with BOMU; ii) leveraging GIS asset data for the purpose of generating engineering designs; iii) integrating workflows with SAP through Bentley Workflow Manager software; iv) implementing GIS replication databases to isolate data editing from data viewing; v) replacing the desktop View Graphics custom application with a web-based application for searching and viewing asset records; and vi) implementing a Primary Distribution Map ("PDM") generation service to generate electronic PDM files for printing during storm and other emergency events. See Richard T. Blumenthal's testimony for full project synopsis.	01-Mar-23	(0.95)	2,234,240	(15,569)	0	516,367	1,231,829	501,713	23,379	0	0	6,044	7,171	10,164
113	2018	DM - Distribution Management System - Release 1	BP Functionality	Software (Intangible)	The scope of this portion of the DMS is implementation of GIS's eMap solution. It includes integration with GIS and is a foundational component for the broader implementation of a comprehensive DMS/ADMS. The purpose of this project is to implement the first release of a Distribution Management System (DMS). DMS Modules to be implemented include: - Distribute on Network Operating Model (eMap) - GIS Interface Savings per customer for 30+ minute primary outages of and reduced outage time from looping implementation. Last 18 months 2.3% system coverage of loop schemes resulted in 2,959,564 minutes saved.10% crew time savings on each primary outage. This module will provide the foundation necessary for the broader implementation of a comprehensive DMS. A DMS system is an electric network management tool that provides advanced functionality in distribution grid analysis, operation, and restoration	01-Jan-19	(0.99)	538,813	444,960	(444,960)	36,962	402,599	99,252	857,743	0	38,754	104,889	675,192	38,908
114	2018	DM - Electric System Model Enhancement	BP Functionality	Software (Intangible)	Implement an Electric Grid System Model that will support the current and future needs of GIS, OMS, DMS (EMap), DMS and GIS (EMap). The Electric Grid System Model will be designed and implemented to serve as an extensible platform to efficiently and effectively manage and share the Electric GIS network model information with the rest of grid operational and planning systems. Establish a methodology for integration that will connect Electric GIS, SAP, Cascade and other Asset Management Systems in a common way to provide an integrated view of assets across asset management areas. The Electric Grid System Model will be designed and implemented to serve as an extensible platform to efficiently and effectively manage and share the Electric GIS network model information with the rest of grid operational and planning systems. The Project will achieve the following high level objectives in support of the Grid Modernization Program: (no specific order) 1. Capture all primary and secondary pole locations within one (1) meter accuracy and reconcile the existing EGIS connectivity model to these locations to support DMS and other advanced applications. 2. Evaluate and enhance EGIS integrations with SAP and OMS to provide a more flexible support model and data accuracy in preparation for DMS by following Consumers Energy's integration standards and where applicable leverage the IEEE 61850 CIM standards for integration and common networked model exchanges. 3. Consolidate disparate electric distribution GIS databases into a common shared electric GIS platform. 4. Implement change management associated with business processes for managing all electric distribution map records. 5. During execution of the project, all EGIS integrated processes will continue to operate without unplanned disruption.	01-Nov-18	(1.00)	747,213	(49)	0	270,910	235,404	240,949	27,656	49	0	20,557	3,943	3,107

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC			
115	2018	GM - GIS Connectivity Model Integration	BP Functionality	Software (Intangible)	The Grid Modernization (GM) Geographic Information System (GIS) Connectivity Model Integration project will design, develop, and implement a GIS extract using an industry standard to publish electric GIS data to the network, where it can be consumed by multiple applications. The extract will make the data available to the electric distribution historian as well as other applications in the future. The project provides value to the Company by: (1) leveraging the International Electrotechnical Commission (IEC) Common Information Model (CIM), which provides a basic ontology for data and data relationships and a standard repeatable approach that can be leveraged for sharing data about the electrical network across technical systems; (2) supporting an enterprise application integration framework, which keeps business process as simple as possible; (3) providing the ability to detect the physical location of metered assets, including circuit, substations and transformers; and (4) reducing the costs and time required to develop integrations in the future resulting from streamlined approaches and processes. Deliverables within the project scope include: (1) develop and implement the GIS extract, leveraging IEC CIM for data sharing; (2) leverage the enterprise service bus (ESB) as the high level communications protocol for sharing CIM data across the enterprise; (3) build the integration to the electric distribution historian; and (4) develop a data and integration framework that can be leveraged for future projects. The alternative considered was to continue to implement point-to-point application-specific integrations to meet the requirements specific to a single project. This alternative was not chosen because it continues to increase the technical debt associated with the maintenance of non-standard interfaces. The preferred approach as outlined in this business case will result in: (1) standard, repeatable approaches for future integrations, (2) lower ongoing maintenance costs and, (3) less complexity.	30-Nov-20	(1.00)	286,699	(29,460)	0	56,393	163,493	96,493	0	0	0	0	0	0	0	0	0
116	2018	GM - Grid Communication Modernization	BP Functionality	Network	Verizon has announced that they will no longer offer their analog, multi-drop phone service as of February 28, 2015 and their Frame Relay service after December 31, 2015. These services are an integral component of the SCADA communication infrastructure. The project scope includes: (1) a wired and wireless solution to replace Verizon's sunsetted services (analog multi-drop circuits and frame relay circuits); (2) Defined minimum and uptime requirements; (3) Sufficient site coverage. This IT project is to support the initial Energy Delivery line sensor roll-out as part of the Grid Modernization program. This IT portion of the broader scope includes communications design, modern configuration and purchase and installation of a head end/device management server and integration with SCADA systems (HVD and LVD) associated with the initial group ("300" of line sensing devices throughout Consumers Energy distribution system. Installation of up to 300 line sensors (with pre-installed modems) with larger roll-out funded separately. The implementation will be prioritized based upon most unreliable feeders. IT requirements include communications design, modern configuration and purchase and installation of a head end/device management server and integration with SCADA systems (HVD and LVD). To accurately and quickly pinpoint trouble location on feeders so immediate step restoration can proceed. To leverage the 10 year SCADA project to station information that is required can be obtained almost immediately. Communications costs will be dependent upon data volume, frequency and unsolicited/solicited capabilities of the devices.	15-Dec-18	(0.98)	646,721	0	27,600	317,205	100,852	201,065	33,507	1,138	12,700	13,981	1,019	4,670			
117	2018	GM - Line Sensor Initial Rollout	BP Functionality	Network	This IT project is to support the initial Energy Delivery line sensor roll-out as part of the Grid Modernization program. This IT portion of the broader scope includes communications design, modern configuration and purchase and installation of a head end/device management server and integration with SCADA systems (HVD and LVD) associated with the initial group ("300" of line sensing devices throughout Consumers Energy distribution system. Installation of up to 300 line sensors (with pre-installed modems) with larger roll-out funded separately. The implementation will be prioritized based upon most unreliable feeders. IT requirements include communications design, modern configuration and purchase and installation of a head end/device management server and integration with SCADA systems (HVD and LVD). To accurately and quickly pinpoint trouble location on feeders so immediate step restoration can proceed. To leverage the 10 year SCADA project to station information that is required can be obtained almost immediately. Communications costs will be dependent upon data volume, frequency and unsolicited/solicited capabilities of the devices.	18-Feb-18	(1.07)	3,957	0	2,234	557	1,166	1,392	0	0	1,245	0	147				
118	2018	ICS Resource Acquisition & Mgmt	BP Functionality	Software (Intangible)	The Incident Command System (ICS) Resource Acquisition & Management System project will implement a centralized, collaborative storm responder management system and restoration site that consolidates outage and restoration related documentation and streamlines storm response processes with ICS. The project adds value to the Company and its customers by: (1) aligning storm restoration standards and processes with standardized ICS framework; and (2) enhancing storm responder system navigation and communication resulting in overall enhanced experience. The scope of the project includes: (1) redesigning a streamlined storm restoration SharePoint site; and (2) upgrading the Catastrophic Crewing (CatCrew) application to incorporate resource management capabilities. Alternatives considered for this project include: (1) Utilize the Automated Resource Call Out System (ARCOS) solution for resource management. This alternative was not selected because of ongoing, incremental increase to O&M maintenance costs. Additionally, ARCOS does not provide on call schedule and storm responder profile functionality, which would have required software customization and increased project expenses. (2) Utilize the SharePoint solution for document management. This alternative was selected because it is the standard Company platform for document management.	21-Nov-18	(1.00)	85,269	0	0	45,944	19,460	19,866	21,151	0	0	5,407	12,478	3,267			
119	2018	Integrated Resource Planning Project	BP Functionality	Software (Intangible)	Energy Policy requires Integrated Resource Plan be filed. Solution selected will enable scenario driven, long term, economic capacity planning, production cost modeling, evaluating future years for as is & potential changes in future energy landscape. A robust modeling tool will enable the Company to have agility and flexibility in developing scenarios that consider fleet capability, demand outlook, financial and economic considerations, economic, and environmental factors. Utilization of industry leading models for evaluating energy portfolio optimization will inform Company leadership decision making for our energy fleet investments. Advanced capacity/cost modeling to meet energy policy RFP requirements. Replace/upgrade obsolete tools being used to ensure latest simulation data, modeling capability with capacity expansion considerations for portfolio optimization. An RFP enabled the team to consider other industry tools. Solutions for this niche functionality are limited. The team relied on requirements evaluation, vendor reference calls, and pricing to determine a solution. One option to address this outdated technology was to upgrade to the Strategist vendor's new platform. This new software platform is extremely new and is not yet proven as a viable solution. This would be a risk to replacing the existing tool.	01-Dec-18	(1.00)	879,616	784,420	0	23,503	62,496	9,197	36,444	0	0	5,381	30,241	822			
120	2018	Integrated Training (Dispatch Simulator)	BP Functionality	Software (Intangible)	The Integrated Training project will implement a training environment that integrates SAP, Outage Management System (OMS), and Service Suite to create a comprehensive, real-life training environment for the operations workforce, including dispatchers and storm support personnel. The project adds value by enabling start-to-finish training in an environment that mimics real-life processes, so that the operational workforce may complete training that complies with annual certification requirements and refresher training. The project scope includes: (1) Integrations between existing SAP, OMS, and Service Suite; (2) implementing a year-round integrated training environment available to practice work order process, including in a simulated storm sessions; (3) define a technology process to sustain cross-system periodic data refresh; and (4) determine additional storage and infrastructure components necessary to deliver the required solution.	31-Jul-18	N/A	145,771	0	0	36,180	69,370	40,221	0	0	0	0	0	0	0		
121	2018	Iron Performance Manager	BP Functionality	Software (Intangible)	The Iron Performance Manager project is for the enablement of the Smart Energy Operations Center with the functionality to handle daily exceptions, monitor the health of the system, and enhance procedures of smart energy devices to improve long-term performance. The ability to monitor, report, and track performance of electric smart meters, gas communication modules, and other smart CE devices in the field is instrumental to the success of smart energy. Included in the implementation is operational reporting, performance metrics, automated exception detection, intelligent prioritization of exceptions, historical trending of performance, and mapping and visualization. The value of completing the project is: Improvements to Time of Use billing by providing required meter data reporting; enables monitoring and near real-time data analysis of meter health and system integration; and provides process and financial benefits. As part of our review process the alternative considered was the "do nothing" approach and hire additional staff to handle the manual process which would be less efficient and more costly.	01-Jun-19	(0.18)	0	0	0	0	0	119,249	0	0	3,614	102,434	13,201				

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122	2018	IWP Replacement	BP Functionality	Software (Intangible)	The IWP Replacement project will develop a new Internet Web Portal ("IWP") platform that increases customer engagement and satisfaction among residential and business customers by allowing them to compare their current rate and bill amounts versus other rates and energy efficiency offerings. As part of the project review process, the alternative considered was continuing with the existing IWP. However, that tool lacks in rate comparison abilities and is no longer being enhanced by the vendor. If we do not replace the IWP, customers who move to the new Time of Use Billing rate will not have access to data that provides details on their peak time usage making it difficult to understand their billed charges. Additionally, it will not provide customers the opportunity to compare their usage and charges with other rates to determine the best rate for their needs. Customer benefit: Improved CII score of 5.0% and annual O&M savings of \$60,000.	01-Sep-19	(0.91)	0	0	0	0	0	0	167,097	0	0	7,780	146,315	13,082
123	2018	Large Customer Rate Tool	BP Functionality	Software (Intangible)	The Large Customer Rate Tool project is for automated rate analysis through a rate tool for our large customers. The project will automate currently manual rate evaluations and thereby reduce the time it takes to run individual rate analysis. Included in the implementation are secondary, primary and special rates for business customers, process automation, and accuracy of rate definition for large customers. The value of completing the project is the automation of a currently manual intensive process that account managers utilize in working with large businesses to ensure they are on the best possible rate. As part of the project review process, the alternative considered was to do nothing and maintain the manual process of extracting data from SAP and entering into spreadsheets to run rate comparisons. This alternative was not selected since it would not support the gains in customer satisfaction or the efficiency of providing the data. Customer benefit: improved functionality to better assist large business customers with evaluating different rate options.	01-Aug-20	(0.92)	0	0	0	0	0	0	27,849	0	0	128	24,401	3,320
124	2018	Legal Early Case Assessment & Legal Hold	BP Functionality	Software (Intangible)	Implement a solution to provide Early Case Assessment and Document Review tools for use by the CE Legal Department. Early Case Assessment features: • Reduce the volume of information being collected and reviewed • Improve productivity by reducing human effort required for case review • TAB - Technology Assisted Review Document Review features: • Keep case data internal to Consumers Energy • Reduce costs on external legal services and storage Solution should allow Legal to perform keyword searches and analysis of unstructured documents in the archive system(s) as well as perform bulk exports of the documents found. Legal hold management features: • The Legal department should be able to perform the following without IT Assistance: * perform global or custodian specific searches on unstructured documents, * manage and preserve searches related to a hold, * send and track notifications to custodians on a hold, * increase in Legal self-service	01-Sep-17	(0.99)	0	0	0	0	0	26,861	0	8,593	5,296	11,819	1,153	
125	2018	Manage Credits and Collections - Payment Plans	BP Functionality	Software (Intangible)	Streamline the standardization of the SPP (and WPP) payment plans, provide consistent eligibility criteria, enable enrollment consistently across channels and enable a multiple channel customer facing experience. The project scope and value includes: • Improve system processing to ensure that SPP enrollment reflects the tariff and guidelines: - o Restrict enrollment only to customers that qualify as low income - o Establish income verification process - requires document submission process (customer produces proof of income or proof of assistance) - o Improve tagging of low income customers (BP ID and Operands are not kept current) - o Improved communication and notification of down payments, monthly amounts etc. - o Expand enrollment capability to the Web • WPP (Winter Protection Program) should follow the same guidelines as SPP	18-Sep-18	(0.97)	0	0	0	0	0	2,714	0	0	2,424	0	290	
126	2018	Manage IVR	BP Functionality	Software (Intangible)	The Manage Interactive Voice Response (IVR) project is for continuous improvements to the Interactive Voice Response ("IVR") system which would include menu refinements around options and restructuring of the menus to drive more optimal option and queue selections. These improvements will enable the proper customer routing and appropriate call disposition. Included in the implementation is IVR menu selection/options refinement and appropriate customer choices for routing and queuing. The value of completing the project is enhanced customer experience and IVR containment. Failure to implement this project will result in longer wait times for customers and more manual intervention by Customer Service Representatives. Customer benefit: System updates to improve customer navigation of menu options.	01-Apr-19	(0.98)	1,079,150	543,670	0	85,443	377,648	72,389	50,396	0	0	9,232	34,788	6,375
127	2018	Manage Rev Act - Mg Rev Assurance (PID/FACT)	BP Functionality	Software (Intangible)	Adjustment from prior year.	N/A	N/A	0	0	0	0	0	0	1,793	0	0	1,330	255	208
128	2018	Rate Case Implementation	BP Functionality	Software (Intangible)	The Rate Case Implementation project will modify SAP billing in accordance with MPSC filings. These enhancements include, but are not limited to, overall price changes and structure changes to the rates. Project scope includes the large electric and gas rate changes that occur annually and/or monthly. The benefit of this project is to ensure that customers are billed accurately based on the timing of the rate structure and price changes. An alternative considered for this effort was an offshore development model. However, the risk of billing inaccuracies and customer complaints with that model were deemed too high based on the various complexities and timing of the development and testing.	30-Sep-19	(0.89)	0	0	0	0	0	0	28,871	0	0	3,706	24,566	600
129	2018	Reservoir Modeling Software	BP Functionality	Software (Intangible)	Adjustment from prior year.	27-Dec-16	(1.00)	(314)	(314)	(0)	0	0	0	0	0	0	0	0	0
130	2018	Residential Joint Invoicing	BP Functionality	Software (Intangible)	This project involves enablement of SAP's standard consolidated billing functionality in order to provide a comprehensive view of a customer's usage in a single bill. The "One Bill" approach. Supports increased convenience/satisfaction to the customer via delivery of a single bill rather than multiple. Generates cost savings to CE by decreasing printing, postage and payment processing costs. Estimated reduction of 2,400,000 bills annually	18-Oct-18	(1.03)	0	0	0	0	0	0	26,589	0	0	23,219	581	2,788
131	2018	Service Bench for VAPS	BP Functionality	Software (Intangible)	The Service Bench for VAPS project will enable better operational management, customer satisfaction and set the BU up for future expansion. It will improve the margin for the entire program which can be utilized at the corporate level to either offset utility customer rate increases or shareholder return. ServiceBench is a third party software platform specifically designed for service companies to utilize its modular software platform for effortless scheduling and dispatch, field work management, automated claims processing and real-time updates partnered with powerful analytics. This will be the replacement of SuperCOW as a new Service Mgmt System. This software performs scheduling, dispatching, field work management, automated claims processing, and data analytics.	28-Feb-19	2.93	1,071,409	0	0	233,022	743,116	95,271	0	0	0	0	0	0
132	2018	Supply Chain - Inventory Optimization	BP Functionality	Software (Intangible)	The Supply Chain - Inventory Optimization project will allow for accuracy and management of inventory (especially generation materials/parts) and how much inventory is on hand. Supply Chain Inventory Optimization will improve service levels. Tool assumes a 99.7% service level for critical materials; give us the ability to segment inventory by criticality; will improve, automate management by category segmentation; give Proactive Risk Identification; show notifications when service levels will be impacted; show inactive inventory management; give proper categorization of non-moving inventory for potential write-off. The tool calculates actual lead times and updates system and the tool updates SAP automatically once analyst reviews	22-Dec-17	(1.00)	383	0	0	276	0	107	(24,217)	(10,221)	0	(1,664)	(11,227)	(1,105)

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133	2018	Supply Chain - Sourcing Intelligence Tool	BP Functionality	Software (Intangible)	The Supply Chain Sourcing Intelligence Tool project will implement five modules of Power Advocates Energy Intelligence suite to provide a consolidated spend view and analysis for commodities and services across the Company. The project will add value to the Company and its customers by: (1) providing cost controls and visibility by enabling functionality to identify and control rogue spend; (2) increasing negotiation leverage and additional cost savings by capturing and aggregating spend for similar commodities and services across the Company; (3) identifying like commodities or services to ensure the Company is paying the same rates, leading to consistent spend and additional savings; and (4) replacing manual efforts to aggregate spend or control and identify rogue spend with an analysis tool, leading to process efficiencies and data quality improvements. The scope of the project includes: (1) purchasing and implementing the Power Advocates Energy Intelligence suite; (2) integrating the cloud-based solution with SAP data through automated data extractions, and (3) configuring secure single sign on (SSO) capabilities for users. Prior to implementing the project, alternatives considered include: (1) Continue efforts to manually investigate, identify, control, and aggregate cross-Company spend. This alternative was not selected because it does not provide the stated benefits of the sourcing intelligence tool and continues to perpetuate the problem of rogue spend. (2) Implement a spend intelligence tool from an alternate vendor. This alternative was not selected because the tool did not offer the insight to data and did not offer the same benefits to the Company. (3) Implement the cloud-based Energy Intelligence suite. This alternative was selected because it is part of a larger suite of tools for a well-rounded sourcing platform already in use by the Company. In addition, Spend Intelligence provides market and cost intelligence data through market analysis and cost comparisons to help the Company improve negotiations with vendors.	22-Aug-19	(1.00)	1,354,333	1,321,207	0	0	33,127	0	256,097	256,097	0	0	0	0	0	0
134	2018	Tax Work Management Tool	BP Functionality	Software (Intangible)	Adjustment from prior year.	04-Dec-17	(0.98)	(4,964)	0	0	0	0	(4,964)	0	4,697	0	0	0	4,697	0	
135	2018	Telecom Expense Mgmt (TEM) Strategic	BP Functionality	Software (Intangible)	The project will identify opportunities for cost savings in current Consumers Energy Telecommunications expenses, select and implement software to automate ongoing telecom expense management, and contract for a managed service provider to improve management of ongoing telecommunication expenses and payments. The project will utilize an optimization vendor to analyze current Consumers Energy telecommunication assets, expenses, and processes then implement telecommunication expense management software, which will provide invoice processing, invoice auditing, invoice dispute resolution, inventory management, contract validation, advanced reporting, mobile rate-plan optimization, and mobile device management. In addition the project will select and utilize a managed service provider to assist Consumers Energy in maximizing ongoing telecommunication expense management efficiency and cost-effectiveness. The value in completing this project is to ensure telecommunication assets, expenses, and processes are optimized to provide labor savings, reduce billing errors, and eliminate duplicate billings and unnecessary service charges, thereby reducing costs by an estimated \$154,000 annually. The savings will help offset the increases in IT operational O&M associated with new IT assets. Quality improvements will be implemented in the expense management process to improve efficiency, accuracy, and quality of future telecom expense results. The alternative approach considered was to develop a customized application in-house and internally analyze, develop, and implement process improvements by adding additional resources, which increases cost and does not effectively utilize the knowledge of industry experts.	01-Apr-19	0.76	0	0	0	0	0	0	61,275	0	0	6,192	49,089	5,994		
136	2018	TEOS Application Currency	BP Functionality	Software (Intangible)	The TEOS Application Currency project is for smaller application upgrades for the following applications: gReady, Inspection Manager, MAINS, Synergi Gas, Synergi Gas Customer Mgt Module, Synergi Gas ModelBuilder, Uptime TAMP, Align Fuel Management, Align Pipe Ops, Consumption Imports, Laboratory Information Mgmt System Chemistry, WinST, Modellogic, business center, Collector, CYME, DGN Out, FlowCal, Land Property Management, Maxload, MAOP, Next TLM. Included in the implementation is upgrading the application software, assessing any new functionality for value to the Company, making necessary configuration changes, testing any integrations to or from the application, testing the upgrade, and updating documentation related to integration changes. The value of completing the project is reducing the number of incidents associated with out of date software, increasing application stability, and allowing the Company to leverage new functionality available in upgrades. As part of our review process the alternative considered were stay at the current version of the applications. If this approach is selected, we risk applications not working after the application of operating system security patches, increased number of incidents, increased support costs, and inoperable systems for employees.	31-Dec-18	(0.87)	91,484	0	0	1,044	84,635	5,806	35,361	13,905	0	10,790	8,087	2,580		
137	2018	Theft Analytics Phase 2	BP Functionality	Software (Intangible)	The Theft Analytics Phase 2 project will use the Smart Meter data to identify customers who may be stealing electricity or gas from the company. Queries and algorithms created for several use cases will be automated and scheduled to run daily to produce theft cases where there is a high confidence level that theft is occurring. Included in the implementation is creating and fine tuning these algorithms to identify customers that are potentially stealing electricity or gas from the company, automatically creating theft cases and/or work orders in SAP for customer accounts identified, and creating executive summary screens for managing theft. The value of completing the project is ensuring the safety of customers and employees, as tampering with meters creates a potential safety risk. The company can also recover the losses incurred from the theft. The potential to turn customers back into bill paying status will also increase future revenues for the Company. Two vendor products failed to deliver on a Theft Analytics solution; therefore the Company is choosing to implement an in-house solution. Customer benefit: Reduced uncollectible expenses with a projected payback of 5-to-1.	01-Jun-19	5.00	655,524	0	484	187,460	339,302	128,278	128,403	0	0	50,129	62,185	16,089		
138	2018	Transformer Data Analytics	BP Functionality	Software (Intangible)	This project will provide the infrastructure and framework for data analytics across Consumers Energy. This integrated platform will be leveraged to support analysis of cross functional data that will result in actionable information. The implementation of intelligent field devices provides Consumers with an enormous amount of data. Turning that data into actionable information requires implementation of an enterprise wide framework.	18-May-18	(0.52)	54,727	1,016,460	(1,017,260)	16,704	26,803	12,021	99,856	0	0	81,948	4,676	13,232		
139	2018	Wholesale Contractual Settlements	BP Functionality	Software (Intangible)	Wholesale Contract Settlements currently uses a combination of Excel and Fortran programs to produce Remittance Statements to suppliers and NUGs (Non-Utility Generators). 70 contracts in total; 8 are Fortran which are 20+ years old and are for more complex contracts (more details, variables, scenarios and to the minute reads for the bill month). These contracts manage \$960M annually. * The project will: (1) Create and implement a contractual settlements solution that will leverage the PCI EA and data warehouse for MISO settlements, (2) Evaluate best platform for contracts which are currently in Excel and Fortran.	15-Sep-18	(0.70)	421,914	0	0	171,867	161,203	88,845	110,234	75,400	0	8,556	24,400	1,878		
140	2018	Workspace Desktop Edition Voice	BP Functionality	Software (Intangible)	The Workspace Desktop Edition Voice project will implement the Genesys Workspace Desktop Edition for Voice for the Customer Contact Centers. Included in the implementation is implementing Genesys Workspace Desktop Edition (WDE) for Voice and implementing SAP Workspace Plugin from Soft Phone. The value of completing the project is implementing Genesys WDE for Voice provides the framework for Contact Center CSR's to handle more than voice contacts such as Web Chat, SMS text or co-browsing. This allows customers the ability to engage with Consumers Energy in a manner of their choice.	01-Nov-18	(0.96)	505,070	0	0	194,492	203,621	106,957	63,839	0	0	32,040	20,640	11,159		
141	2018	2018 SubTotal BP Functionality	BP Functionality					20,893,912	5,857,075	(1,395,472)	3,733,297	9,643,122	3,055,890	4,282,350	624,892	101,743	796,970	2,346,097	412,647		
142	2018	2018 Total						74,344,888	10,743,923	22,757,555	9,138,932	25,002,516	6,601,961	15,164,693	1,821,066	843,570	2,659,223	8,722,478	1,118,555		

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
143	2019	BitTalk Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The BitTalk Upgrade project will upgrade the BitTalk software to meet the support and Information Security patching requirements. The scope of this project includes (1) procuring all new servers to replace the current hardware, (2) creating new databases to replace the current database version, (3) installing the new BitTalk version, (4) migrating the data from the old databases to new, (5) performing all testing required to meet the targeted Go-Live date, and (6) maintain the ability to create and support accurate financial transaction data for exchange between Consumers Energy and external partners. The value of completing the project is that upgrading to a current version will ensure vendor support for this critical application. The Information Security initiative to keep applications patched to a more current level will also be met by performing the upgrade. The alternative of not upgrading puts this financial transaction handling platform at risk because the project has already been delayed for two years.	15-Sep-21	(0.88)	301,670	4,853	0	116,017	129,424	\$1,376	25,783	13,592	0	16,318	(8,154)	2,038
144	2019	Data Center 2.0	Upgrades & Replacements (Enterprise)	New Computers / Hardware	The Data Center 2.0 project will mitigate the current significant risks (location risks, capacity risks, technology risks) with the Consumers Energy IT Disaster Recovery (DR) capabilities. The Company is co-locating to a Backup Recovery Center (BRC) at a vendor facility. A co-location facility is a data center facility in which a business can rent space for servers and other computing hardware. Typically, a co-location vendor provides the building, cooling, power, and physical security, while the business provides the servers, storage, and other computing and networking equipment. Project scope includes both Information Technology and Operational Technology data centers. The project will migrate the BRC to a vendor (Switch) colocation data center, expand DR system capacity to support 100% production load requirements, expand DR capabilities to all applications determined by the business as needing DR, and enable cloud capabilities to migrate non-production workloads to the cloud to allow for production disaster recovery on non-production infrastructure. At the end of the migration to Switch, the compute capacity at Switch will be greater than or equal to the compute capacity at Parnali. Some of the capacity will be on newly purchased hardware and some will be on hardware shifted from the Battle Creek Data Center. Currently, there are no plans to put compute for production DR in the cloud. The Company performed an analysis of two primary alternatives to expand DR capabilities and address constraints/risks: 1) at the current BRC location, and 2) at a third-party co-location facility. Based on the analysis, the Company decided to implement the third-party co-location alternative for the BRC. The value this brings the Company is that it mitigates risks to the corporation in the event of a disaster by significantly enhancing the disaster recovery capabilities, DR testing capabilities, and scalability constraints at the current data center locations.	01-Nov-20	(0.82)	5,005,213	175,088	3,963,574	306,549	389,929	170,072	2,129,717	4,083	4,637	480,108	1,514,421	126,468
145	2019	Dense Wave Division Multiplexing – Fiber Backbone Refresh	Upgrades & Replacements (Enterprise)	Network	The Dense Wave Division Multiplexing – Fiber Backbone Refresh project will refresh the company's Network hardware that is part of the Company's "Dark Fiber" network. Included in the implementation is to replace the existing Fiber Optic transport equipment at the ten Company locations that are part of the internal "Dark Fiber" network. This project provides value by insuring that the Company's internal Network connectivity for Call Centers, Computer Networks and the Radio Network is reliable and on modern, supportable hardware.	10-Nov-18	(0.99)	14,046	0	11,912	0	2,135	0	0	0	0	0	0	0
146	2019	EcoPrint Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The EcoPrint Upgrade project will issue ongoing updates to the EcoPrint project that was implemented in 2016. The manufacturer recommends Consumers Energy to be no more than two versions behind on printer applications so we do not lose support and are able to maintain our print environment for employees. Included in the project is the implementation of Equitrac Application Upgrade, the Global Scan (GSN) Application Upgrade, the Device Manager NX Enterprise (DMNX) Application Upgrade, and the Print Job Encryption Enhancement, which would remove a major vulnerability from our print environment. The value of completing the project is that Consumers Energy maintains manufacturer support, implements print job encryption resulting in a secure print environment, and meets Information Security requirements. In addition, the support provider could meet Consumers Energy resolution time requirements in the case of a major print outage, software and hardware compatibility would be maintained minimizing potential print outages, and any new upgrade features would be available to enhance our printing services allowing for potential efficiencies. As part of our review process the alternatives considered were to do nothing and risk a major outage of our print environment along with loss of vendor maintenance (which would also push the print upgrade cost to another year) or revert back to our old Microsoft print options. The cost of reverting back to Microsoft print environment would be significantly higher than maintaining the upgrades.	01-Jul-19	(0.90)	0	0	0	0	0	0	121,186	0	0	27,516	24,334	69,336
147	2019	ESB Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Enterprise Service Bus (ESB) BWS 14 Upgrade project will upgrade and migrate the Business Works developer code from v5 to v5.14 and to upgrade all remaining applications and infrastructure as needed. BWS 14 provides the following value for Consumers Energy: accelerated productivity, continuous delivery and integration, open ecosystem for improved operational visibility, integration with web and mobile apps and APIs in real-time, and improved administrative and operational efficiencies. BWS 14 allows for easy integration with other cloud-based solutions such as Amazon Web Services and provides easy migration to cloud-based containers in the future. The risks of not upgrading to BWS 14 are slow deployments due to the increased number of physical servers to manually maintain and increased time to debug errors. BWS end of extended support is December 2019, after which the support costs will increase.	01-Nov-19	(0.93)	0	0	0	0	0	0	194	0	0	194	0	0
148	2019	HR Support Pack and BSI Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will update the SAP system with HR Support Packs that are released annually by SAP to comply with HR and tax changes. This project creates value for the Company by: (1) ensuring that it is in compliance with new financial rules and regulations, and (2) ensuring that it can calculate and distribute payroll. The scope of this project is to add SAP HR corrections to ensure proper reporting of financial information by the Company. As this is an upgrade of an existing system, the alternative considered was to delay the upgrade. This alternative was not chosen due to the risk of not complying with financial rules and regulations.	01-Dec-20	(0.94)	0	0	0	0	0	0	1,301,466	75,787	0	134,054	1,047,437	44,188
149	2019	Iron Enterprise Edition Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will upgrade IEE, which collects the reads from meters to ensure accurate and non-estimated bills are provided to our customers. This project creates value for the Company by: (1) ensuring the new features and functionality added to meet business requirements are available to business partners and IT; (2) meeting Information Security's requirement to keep applications patched to control cyber debt; and (3) allowing for validation, estimation, and editing functions for all data collected. The scope of this project includes: (1) an operating system refresh, (2) migrating from Windows 2012 to the next Windows version, and (3) migrating the database from SQL Server version 2012 to the next SQL Server version. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and cyber security of the system.	01-Sep-20	(1.00)	44,934	0	0	532	42,112	2,289	(693)	0	0	0	(693)	0
150	2019	LandWorks Property Management System Upgrade V5.5	Upgrades & Replacements (Enterprise)	Software (Intangible)	The LandWorks Property Management System Upgrade V5.5 project will upgrade the Land Property Management System (LMP by Landworks) to the next version, as well upgrade to arcGIS in order to support the version upgrade. The application is currently on an older version and at risk of no longer being supported by vendor.	01-Oct-21	(0.90)	(128)	0	0	(248)	0	(79)	280	0	0	248	0	32
151	2019	Oracle Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Oracle Server Database Upgrade project will upgrade Oracle server databases to the next version and supportability across business portfolios. This project will create value for the Company and its customers by (1) reducing the risk of system failure and the resulting impacts to business partners and customers, and (2) ensuring that systems are secure, supported, and have the latest features and functionality. The scope of this project includes upgrading to the next version of Oracle across all business systems. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and supportability of the system.	30-Apr-21	(1.00)	(57,248)	0	0	(2,958)	(37,898)	(16,392)	54,862	0	0	2,958	37,898	14,005

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152	2020	OWCE Nitrogen Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The OWCE Nitrogen Upgrade project will upgrade the OpenWay Collection Engine (OWCE1) to the latest version. This upgrade will be done in parallel with the hardware asset refresh to reduce overall risk and cost. OWCE is the head end software that communicates to smart meters. The upgrade to the Nitrogen version ensures that the Company sustains this critical business function on a current and supported version.	04-Jan-19	(0.98)	0	0	0	0	0	0	3,250	0	0	0	2,709	480
153	2019	Remote Application Delivery Strategy	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Remote Application Delivery Strategy project will upgrade VDI and bring Citrix into VDI to eliminate the Citrix platform. Along with this, there is the intent to bring down the cost of each VDI. There are new technologies being implemented via Thin-App to allow for streaming applications. Included in the implementation is eliminating a platform that has redundant functionality. We will be gaining functionality with VMware Thin-App giving us the capability to stream applications instead of having to push applications to the VDI's. This can reduce the size of the VDI footprint.	20-Dec-18	(0.98)	49,751	0	0	0	49,751	0	1,362	0	0	0	1,362	0
154	2019	Secure Access Control System Refresh	Upgrades & Replacements (Enterprise)	New Computers / Hardware	The Secure Access Control System (ACS) Refresh project will replace the Cisco Secure Access Control System hardware and software with another selected authentication, authorization, and device access accounting solution. The scope includes replacing the existing hardware and software for the ACS, identifying new feature sets and determining which features complement existing security policies, and implementing the new access control system platform using best practices from the vendor. The value in doing this project is to have network automation device profiling capabilities, which enables business partners to move devices (e.g. laptops, printers, scanners) within offices without having to submit a Work Order request. Also, this project will provide remote access (Virtual Private Network, aka VPN) into the corporate network to allow users to work more efficiently as well as two-factor authentication. This allows contractors to securely access the corporate network. The only alternative would be to convert the physical servers to virtual servers; however, this would only be a short term (< 1 year) fix.	01-May-19	(1.06)	516,125	79,930	319,746	69,755	24,231	22,863	31,751	0	0	22,561	5,365	3,825
155	2019	SiteCore Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will refresh all components of the website hosting, delivery, search, and analytics applications to add new features and improve search capabilities. Sitecore is the content manager for the ConsumersEnergy.com website. The project will add value by providing these four benefits: (1) maintains currency with the web hosting application version; (2) allows business users to make use of new features and functionality; (3) neutralizes continually evolving cyber threats; and (4) continuously improves customer experience using ConsumersEnergy.com. The project scope includes: (1) upgrading the Sitecore content management software to include content hosting and delivery allowing the use of new features and functionality; (2) upgrading the Coveo software, which will allow for more intuitive search results and provide suggestions or recommendations based on the customers' search text; (3) upgrading the Mongo database, which provides the analytics functionality within Sitecore to better understand customer actions and errors on the web page; and (4) developing analytics across customers' web and contact center interactions, allowing the company to connect contact center call data with corresponding web data for improved customer experience. Alternatives considered but deemed not to be viable include: (1) implementing a two-year upgrade cycle, which would not position the Company to be able to keep up with constantly changing cyber threats, or gain value from the rapidly changing feature set being developed by the vendor; and (2) purchasing an existing cloud solution, which is cost prohibitive at this time. Annually upgrading the existing Sitecore platform provides the functionality and stability needed while meeting financial requirements and mitigating cyber security risks.	31-Oct-19	(0.91)	458,909	0	0	190,941	182,137	85,831	81,959	30,483	0	19,149	25,639	6,688
156	2019	TCOE Test Data & Environment Management	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Testing Center of Excellence (TCoE) Test Data & Environment Management project will provide the tools and techniques to be able to periodically (e.g. annually) refresh the data in our SAP development, QA and pre-production environments from production data. This will decrease the time and effort needed to test and improve the quality of testing, which will improve the quality of application changes. Included in the implementation is determining refresh requirements; using the Request for Information process to select tools based on the requirements; setting up environments for the tools; installing and configuring the tools; testing refresh processes in development and QA environments to ensure appropriate data is refreshed and that processes can be completed with minimal outages; testing refresh processes in development and QA environments to ensure sensitive, Personal Identifiable Information copied from production is obfuscated according to our Information Security data standards; implementing the tools; and putting the processes in place to ensure we refresh on an ongoing, periodic basis. The value of completing the project is that manual testing efforts and costs will decrease while speed to deliver and quality of testing will improve resulting in better quality application changes. This is true for all SAP applications and SAP-interfacing applications, including customer applications such as Contact Center applications, our ConsumersEnergy.com website, and customer billing. As part of our review process during the Plan phase, a Request for Information will be issued to explore solutions from a variety of vendors, including the following: SAP, Atunity, EPI-USE, SNP, Delphic, Computer Associates, Informatica, and IBM.	01-Apr-19	(0.94)	346,110	0	20,784	139,611	96,456	89,258	113,730	0	(20,784)	74,776	44,563	15,175
157	2019	Time of Use Billing Rate Expansion	Upgrades & Replacements (Enterprise)	Software (Intangible)	Consumers Energy will be filing a rate order with the MPSC in May 2018 requesting that all residential customers who are currently being billed using register index reads be switched to a new Time-Of-Use (TOU) interval based billing rate. Expand the current TOU billing rate available to all electric residential customers. Provides for all electric customers to be able to take advantage of the Time of Use (TOU) rate during peak period times without having to sign up for the program, thereby allowing them to save on their energy bill during this period.	01-Mar-19	(0.94)	4,735,471	0	0	388,230	3,999,932	347,309	444,405	0	(10,600)	76,852	332,577	45,576
158	2019	TOA Phase II	Upgrades & Replacements (Enterprise)	Software (Intangible)	The overall of the scope of the project is to upgrade the Transmission Outage Application (TOA) based on the current business and technical requirements. The tasks include: the completion of the SOW, the testing of the upgraded application, the creation of new policies and procedures based on the upgraded TOA, and the change management associated with the education and training of the appropriate Grid Management Business Partners, and the implementation of the TOA II solution. IT and the Grid Management Business Partners will be working closely on this project. The Grid Management Business Partners will be responsible for certain tasks (scope development/formalization, testing and validation of functional scope through test results) and it will be responsible for certain tasks (completion of the Solution Design Document, software installation, help with test case development and execution, and validation of technical scope through test results).	28-Mar-19	(1.00)	33,853	0	0	8,450	19,256	6,147	0	0	0	0	0	0
159	2019	Windows 10 Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Windows 10 Upgrade project will upgrade Consumers Energy corporate computing devices from Windows 7 to Windows 10. The project will convert 4,500 Windows 7 computing devices to Windows 10 with the assistance from a partnered vendor at various Consumers Energy locations for corporate, gas, and electric employees. Project tasks include assessing current tools and image process to ensure industry standards are being applied, communicating with employees on project and expectations, scheduling and completing device operating system upgrades, verifying that devices were upgraded with no impact to employees' data and performance, and providing on-site post support at Consumers Energy locations. The alternative is not to move forward with this project and pay additional support fees for Microsoft to continue supporting Windows 7 for up to four years. This will allow Consumers Energy time to replace the devices through our asset management refresh projects. Work station assets are currently replaced on a four-year cycle. This alternative was not pursued due to the additional support fees and risk to business processes.	01-Oct-19	(0.96)	0	0	0	0	0	730,489	0	9,599	130,990	544,908	44,991	
160	2019	SubTotal Upgrades & Replacements (Enterprise)	Upgrades & Replacements (Enterprise)					11,448,306	259,471	4,316,016	1,216,879	4,897,465	758,675	5,039,741	123,945	(17,147)	987,723	3,572,425	372,794
161	2019	Corp Portfolio Application Currency	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Corporate Portfolio Application Currency project is for specific application upgrades within the Corporate Services portfolio. Included in the implementation is upgrading application software, assessing any new functionality for value to the company, making necessary configuration changes, testing any integrations to or from the applications, testing upgrades, and updating documentation related to integration changes. The value of completing the project is lessening the number of incidents associated with out of date software, increasing application stability leading to fewer incidents due to out of date software, and allowing the company to leverage new functionality available in upgrades. As part of our review process the alternative considered was to stay at the current version of the applications. If this approach is chosen then the company risks applications not working, increased number of incidents, increased support costs and impact key business processes.	31-Dec-18	(0.90)	112,638	0	79,811	2,941	25,110	4,776	116,565	8,262	1,104	53,349	40,963	12,887

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
162	2019	Enhancements - APS	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Enhancements – APS is for small software enhancement work efforts performed for the analytics, Cloud, DevOps, and Architecture (ACDA) business area. Included in the implementation are enhancements or upgrades for Business Objects (BOBJ) Upgrade, Business Objects Data Services (BODS) Upgrade, Dev Ops, Business Planning and Consolidation (BPC), BW HANA, HANA DB, MEMQOL, Soap UI, R Studio, Tableau, and Attunity. The value of completing the project is compliance, hard cost savings, avoided costs, hitting corporate goals, and mitigating risk. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-18	(1.00)	0	0	0	0	0	0	70,837	0	4,157	914	64,336	1,410
163	2019	sXOMS - upgrade to Operations Management	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Electronic Shift Operations Management System (sXOMS) application used by generation and gas compression stations. The upgrade includes the associated hardware such as servers.	18-Aug-18	(0.88)	(5,260)	0	(5,260)	0	0	0	0	0	0	0	0	0
164	2019	Genesys Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Genesys Upgrade project is to upgrade and enhance Genesys Software features and functionality with improved call routing, reporting, and recording contacts. The scope will include all Genesys applications: Work Force Management, Speechminer recording application, interactive insights reporting, and Pulse contact monitoring. The upgrade ensures that the Company is on a Genesys version supported by the vendor and upgrade provides improved functionality across all applications. The improved functionality offers convenience to customers by letting them schedule interactions and callbacks with customer service over any channel or device. It supports transfer, supervise, and conference co-browse and chat sessions. Additionally, it will enhance social media and text functionality for customers. The alternative is to delay the upgrade to 2020 or beyond and risk experiencing system performance issues and support problems.	15-Apr-20	(0.94)	208,872	0	0	23,468	168,073	18,330	36,210	0	6,928	6,886	19,557	2,879
165	2019	Government Reports	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Governments Report project is being implemented to address a technology obsolescence issue, ensure regulatory reports are filed when reports are due and remediate internal audit findings. This project includes implementation of a new tool that collects metadata for each regulatory report, automates notifications to users as due dates are approaching, automates workflows for approval and automates Email notifications with links to tasks. As part of our review process, the alternatives considered were a SaaS solution, leveraging existing technology or reverting to a manual process. The alternative selected was to leverage existing technology, as it was the most cost effective.	01-Dec-18	(0.98)	334,658	0	50,886	107,914	120,746	55,112	106,184	0	0	15,727	82,876	7,380
166	2019	Integrated Talent Mgmt and Learning Mgmt Upgrades	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Integrated Talent Management and Learning Management Upgrades project is for upgrades to the SAP SuccessFactors cloud solution for their talent management and learning management systems. All quarterly SuccessFactors enhancements are reviewed for impacts to the overall system and integrations. Included in the implementation is making necessary configuration changes, testing the upgrade, validating integrations, and updating documentation related to integration changes. The value of completing the project is that quarterly SuccessFactors upgrades give us additional functionality to better serve both employees and potential candidates. Human Resources also benefits by being able to improve the employee lifecycle. As part of our review process no alternatives were considered because SAP SuccessFactors automatically pushes upgrades to our system every quarter, the company does not have an option to reject those upgrades. If nothing is done to test the system and integrations, it is possible that our Talent Management and Learning Management systems will not work and won't be accessible by applicants, managers, employees, HR and contractors.	01-Dec-19	(0.89)	0	0	0	0	0	0	9,393	0	0	513	8,794	86
167	2019	ITCP – Cold Water Service Center	Upgrades & Replacements (Business Partner)	Network	The ITCP – Cold Water Service Center - IT will be supporting the Facilities project to build a new service ctr in Coldwater. This work will require new connectivity to the site, security, data, video, voice as well as SAP changes. Included in the build of Coldwater is closing Bronson Service Center and consolidating employees in Coldwater	14-Sep-18	(1.00)	1,073	0	0	464	349	259	0	0	0	0	0	0
168	2019	OpenWay 3G to 4G Enablement	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will upgrade OpenWay Collection Engine (OWCE) software to the next version to enable the support of the 4G network for electric meters as the telecommunications carrier decommissions the 3G network. The project will add value by enabling the Company to maintain current capabilities for 680,000 electric meters and customers. The project scope includes: (1) implementation of the next version of OpenWay Collection Engine (OWCE), (2) equipping meters with communications boards, and (3) implementing active directory authentication for OWCE. The alternative considered was to defer the timing of the OpenWay Collection Engine (OWCE) upgrade. This alternative was not selected due to the retirement of the 3G Verizon network driving the requirement for Consumers Energy stay current with 4G smart meter technology. Failure to upgrade the network would result in the inability for smart meters to communicate over the network, and Consumers Energy would have to re-implement manual meter reading efforts. Upgrading to the next version of OpenWay Collection Engine was chosen because reverting to manual meter reading would increase costs to both the Company and its customers.	01-Jan-20	(1.00)	(7,090)	0	(7,090)	0	0	0	(1,270)	0	(1,270)	0	0	0
169	2019	Osloft Pi Historian Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Osloft Pi Historian Upgrade project will maintain application and hardware platform currency for the Osloft Pi system. The project will create value for the Company and its customers by (1) reducing security vulnerabilities, (2) improving efficiencies and increasing synergies between environments, and (3) enabling business partners to leverage new features that the vendor includes with major releases. The project scope includes the implementation of: (1) Osloft Meter Operational Data Management, (2) Osloft Electric Distribution Historian, (3) Osloft Generation, (4) Osloft Low Voltage Distribution/High Voltage Distribution, (5) Osloft Gas Automated Meter Read, and (6) data archiving for analytics purposes. The alternative considered was to delay the upgrade until a future year. The hardware has not been refreshed since 2014. Continuing to delay the upgrade could lead to the inability to apply security and system patch sets. A timely upgrade will allow the Company to sustain system performance and supportability. Other vendors were not considered due to a longstanding Enterprise Agreement with Osloft, Inc.	01-Oct-20	(1.00)	0	0	0	0	0	0	28,276	0	153	1,748	58,328	(31,953)
170	2019	PowerPlan 2018 Application Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The PowerPlan Upgrade 2018 project is to upgrade PowerPlan to current version, implement functionality to support the new Leasing Generally Accepted Accounting Principles ("GAAP") regulatory requirements effective January 1, 2019, and enable month-end close automation through the financial close cockpit. The Company's current version of PowerPlan is at end of life and PowerPlan will no longer offer extended support. The value of doing this project is to avoid risk of not being able to comply with new accounting standard and running an unsupported version of the software, as PowerPlan is critical to meeting our financial reporting requirements. As part of our review process, the alternative considered was to not upgrade PowerPlan. This alternative was not selected due to the risk on financial reporting.	18-Oct-18	(0.95)	42,323	0	9,563	28,762	3,998	114,590	0	0	31,966	75,913	6,711	
171	2019	Standish Field Office	Upgrades & Replacements (Business Partner)	Network	IT will be supporting the Facilities project to build a field office in Standish. This project will provide technology needed for the new facility, including new connectivity to the site, security, data, video, voice as well as SAP changes.	N/A	N/A	29,930	0	13,157	12,431	0	4,341	1,000	0	0	0	1,000	0

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172	2019	IS&M Enhancements	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Enhancements - Technology Strategy Management (TSM) initiative will apply O & M funding to make enhancements to existing software and to address requests generated by changing business requirements. The enhancement requests are fulfilled to provide capabilities across business areas. Requests for this funding are governed by a cross-functional board of representatives from infrastructure areas which meets periodically to evaluate and prioritize the work. The board assesses requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing technology investments for infrastructure. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-19	(1.00)	98,090	0	79,212	10,785	596	7,497	32,889	0	13,804	13,969	2,606	2,309
173	2019	SubTotal Upgrades & Replacements (Business Partner)	Upgrades & Replacements (Business Partner)					816,233	0	210,716	187,066	340,636	94,314	516,073	8,262	24,076	125,078	354,353	2,109
174	2019	AccessNOW	Security	Software (Intangible)	The AccessNow project is for the implementation of configurable identity and Access Management functionality and best practices with enforced compliance. This includes enterprise-level foundation architecture, technology, and end-to-end processes and controls. The project will deliver an integrated and synced enterprise system of record data. Additionally, processes will be fully automated for user self-service and access lifecycle management. Included in the implementation is the continuation of building on the One Identity Manager application against the Identity and Access Management Program (IDAP) roadmap. As Phase 1 efforts concluded in mid-2017, 2018's main objective is to implement Phase 2, which includes designing and building a web portal with a request catalog for employees' access needs. The intent is to replace the legacy Computer Access and Authorization Request (CAAR) system. These requests will be driven by automated workflows and the tool will provide automatic Active Directory provisioning. Additional features aimed at making this process more user friendly are also included. The value of completing the project is automated workflows with escalations that will reduce wait times for end users to get the access they need for their jobs. As part of our review process the alternative considered was using manual processes but this strategy was deemed too costly and inefficient.	01-Aug-19	(0.96)	155,382	0	0	55,371	78,791	20,918	118,079	0	0	125,078	146,865	5,658
175	2019	ARP - Cyber Security	Security	Network	The ARP Cyber Security project will upgrade its IT infrastructure to keep up with increasing user demands and applications, and prevent system failures and service interruptions. Refresh strategies are driven by business objectives including security, financial, and growth requirements. Continuing a refresh schedule based on historical performance and usage requirements will maintain existing levels of security. Included in the implementation is the replacement of end of life and obsolete systems. The main initiatives are Carbon Black Response server replacement, Qlador server and licenses, and Blue Coat reporter upgrade. Completing the replacement strategy enables a consistent security platform, offers the flexibility to respond to change, and continues existing levels of security in support of protecting our customers' data and critical infrastructure. A deployment schedule is developed based on an analysis of the age and capabilities of the current security technology integrated into the existing infrastructure to ensure cyber security and the organization's ability to recover vital assets and information. As part of our review process the alternative considered was not to upgrade or replace assets as required. This approach will most likely introduce security risks, system vulnerabilities, and out-of-warranty repair costs.	31-Dec-18	(1.09)	359,477	3,212	290,815	46,538	5,779	13,133	2,077	0	0	1,940	0	137
176	2019	Cyber Security Enhancements	Security	Software (Intangible)	The Cyber Security Enhancements project is for emerging or unpaired cyber security activities resulting from audits, incidents, or a changing threat landscape. It will address issues and bring value to the environment for initiatives that may not meet the criteria for a formal project. Included in the plan are the following initiatives: application whitelisting for Zone A, Cyber Ark upgrade, SharePoint enhancement, threat prevention subscription, multi-factor authentication for the payment card industry, system center endpoint protection, and centralized endpoints. The enhancements add new features to the software as requirements or process improvement initiatives dictate. The value in completing the enhancements is to allow software and hardware product performance scalability, to improve flexibility to respond to change in the threat landscape, and to promote continuous performance improvements. As part of our review process the alternative considered was not to upgrade or enhance software or hardware, which could reduce Consumers Energy's ability to respond to future threats and vulnerabilities.	31-Dec-18	(0.96)	0	0	0	0	0	0	176,622	6,927	1,457	121,212	38,580	8,445
177	2019	Enterprise Incident Response Toolset	Security	Software (Intangible)	The Enterprise Incident Response Toolset project is for implementing an incident management platform that enables the Cyber Security Incident Response Team (CSIRT) to automate alert triage, which is currently a manual process through technical playbooks, to bring more capabilities to track and manage incidents, and to integrate threat intelligence with security orchestration. Included in the implementation is building five or six playbooks, providing basic training to the CSIRT, instituting firewall rules, setting up internet and security information and event management access, placing an FS load balancer in front of the two Demisto servers, and deploying two Demisto servers that will operate in Disaster Recovery mode. The value of completing the project will be improving response time to events and limiting the risk to Consumers Energy from a cyber security incident. As part of our review process other automation tools (i.e. Swimlane, Phantom, Fire Eye) were vetted via on-site demonstrations or webinars. However, this was determined to be the best solution to align with the Consumers Energy requirements.	22-Jul-20	(0.93)	275,580	343	133,951	88,145	0	50,141	21,431	0	5,807	15,073	0	551
178	2019	Firewall Management Platform	Security	Software (Intangible)	The Firewall Management Platform project will implement a firewall management platform. Cyber security now manages nearly 100 firewalls and keeping configurations up to date, changes processed correctly and mistakes to a minimum is challenging. A firewall management platform will enable automation and workflow to provide greater security and efficiency for our team. Firewall security mistakes and failures are leading causes of data breaches.	30-Nov-19	(0.98)	41,757	0	0	0	41,797	0	0	0	0	0	0	0

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179	2019	Juion Center	Security	Network	This project will procure a location to convert the Command Center and the Cyber Security Incident Response Team ("CSIRT"). This project will focus on constructing a physical office space to facilitate these two teams working in a collaborative fashion. This project will add value by: (1) co-locating physical and cyber team members for process efficiencies and better integration of tools in the event of a security incident that requires both teams' collaboration; (2) having a dedicated emergency operations area for use in the event of a security incident (cyber or physical or both) providing the tools needed to keep stakeholders informed; (3) evaluating and potentially implementing a shared intelligence capability across both areas; and (4) further integrating existing processes and workflows. The project scope encompasses building an integrated Security Command Center across physical and cyber security domains that takes into consideration physical space, technology, process integration, analytics, and intelligence capabilities without changing existing organizational structures. As part of the review process, the alternative considered was to continue to run a separate Security Command Center for physical security incident response and a separate Cyber Security Incident Response Center. The two centers would continue to run separate, non-integrated systems and tools. This option was not chosen as it does not provide the integrated space and tools needed for physical and cyber security teams to collaborate to resolve critical incidents that pose both a physical and cyber threat.	01-Dec-21	(0.95)	\$33,239	0	\$9,731	\$4,470	\$59,745	\$4,293	\$11,400	0	\$8,491	\$9,817	\$92,058	1,034
180	2019	Mass Notification	Security	Software (Intangible)	The Mass Notification project will interact with our current Siemens fire panels to take advantage of communications protocols in existence at Consumers Energy physical sites. Similar to current capabilities used to announce "fire" events, the mass notification tool will allow for other event notifications to building participants. In addition, mass notification has capabilities to involve personal device notification via texting, email alerts, and social media updates. Included in the project is implementing communications software to tap into current fire panels in physical locations, allowing for remote connectivity to sites for access to software, allowing for personal device communications, tapping into current building security software, allowing for emailing of key threats/events, and allowing for social media interfacing. The value of completing the project is directly tied to employee safety, which is increased by allowing better communication to employees during emergency situations. As part of our review process the alternative considered was not to implement the capability and to continue notifying individual Consumers Energy site managers of life safety situations through emails, phone calls, and/or text messages.	15-Aug-19	(1.00)	\$65,117	0	\$127,259	\$21,727	\$104,146	\$11,984	\$10,273	\$4,088	0	0	0	6,235
181	2019	NERC CIP v5	Security	New Computers / Hardware	Consumers Energy is required to be in compliance with version 5 of the NERC Critical Infrastructure Protection (CIP) standards, including CIP V5 for High and Medium Impact BES Cyber Systems on 4/1/2018 and Low Impact BES Cyber Systems on 12/31/2019. Implementation of compliant processes and systems must be accomplished. This project will fund consulting, implementation service, and required tools in order for assets to meet the new requirements. Significant effort will be required related to network segmentation to minimize in-scope cyber assets and separate High, Medium, and Low Impact cyber assets while minimizing the risk to the reliability of the Bulk Electric System (BES). Version 5 is a significant change in the standards and requires additional assets to fall under the CIP requirements. The scope of the project encompasses: (1) network segmentation to minimize in-scope cyber assets and separate High, Medium, and Low Impact cyber assets while minimizing the risk to the reliability of the Bulk Electric System (BES); (2) moving additional assets under the Critical Infrastructure Protection requirements per the standards of Version 5; (3) implementation of service and required tools in order for assets to meet the new requirements. The value of completing the project lies in: (1) identifying and classifying Bulk Electric System (BES) Cyber Assets and BES Cyber Systems, and (2) achieving compliance with NERC CIP Version 5 Standards. As part of the review process the alternative considered was to not be compliant with CIP Version 5. This option was not chosen because any noncompliance of the NERC CIP standards may result in a fine of up to \$1 million per day per violation.	04-Dec-18	(0.96)	\$105,742	0	\$168,248	\$76,324	(\$64,347)	\$25,517	\$18,769	0	\$1,211	0	558	\$17,000
182	2019	Operational Security Assessment	Security	Software (Intangible)	The Operational Security Assessment project is for the implementation of an automate tracking tool to plan audits and inspections, monitor progress in real-time and track corrective actions in one central and secure location. The implementation of the identified product will help to decrease vulnerabilities that could threaten Company assets. This project will establish: (1) a centralized location to securely manage and store assessments; (2) implement an automated tracking system; (3) prioritize deficiencies; and (4) track completion with an interface to the SAP work order system. This will decrease the manual effort needed by the physical assessors to complete their day to day work. With the automation of tracking audits and inspections the physical security team can: (1) increase their audits by 20%; (2) reduce the likelihood of impactful security events to customers; and (3) increase efficiency, which in turn reduces potential operating costs for the Company. An additional feature of the identified solution is the ability for executive leadership to track the assessment scores in real-time and identify long trends enabling proactive prevention. As part of the review process the alternative considered was to continue using the manual process of tracking in an Excel spreadsheet and entering tickets in the Request Management Center (RMC) SharePoint site but this strategy was deemed too costly and inefficient.	13-Nov-19	(1.00)	\$2,161	\$11,431	0	\$16,196	0	\$4,534	\$18,413	\$1,732	0	\$3,370	\$11,443	\$1,868
183	2019	OT Security Architecture	Security	Network	The Operational Technology (OT) Security Architecture project will implement consistent cyber security standards at generation plants to protect valuable Company assets from increasing cyber security threats and attack. The project will also transition sustainable cyber security operations to the Cyber Security Operation Center (CSOC). The project adds value to the Company by: (1) implementing cyber security standards to ensure Company policy and non-standard configurations are consistently applied; (2) mitigating cyber security risk to the most critical assets; (3) applying standardization for both policy and technical architecture; and (4) improve visibility to cyber security vulnerabilities, enabling proactive efforts to mitigate them. The scope of the project includes: (1) defining a cyber security standard for operational technology architecture at generating plants; (2) identifying controls that have a high impact on cyber security; (3) implementing a consistent cyber security across the generation fleet; and (4) conducting generation plant evaluations against the newly adopted standard to define maturity gaps. As part of the review process, the alternative considered was to not implement and build the standards. However, this alternative was not selected because it introduces a significant increase to the cyber security risk posture at the Company.	01-Nov-19	(1.00)	\$1,533	0	0	\$1,262	(\$232)	\$504	\$692	0	0	\$603	0	\$9
184	2019	Passive Vulnerability Assessment	Security	New Computers / Hardware	The Passive Vulnerability Assessment project will deploy a tool to passively provide vulnerability assessment services to Control Networks. Active vulnerability scanning can negatively impact the reliability of a network. To overcome that risk, passive vulnerability scanning acquires asset and vulnerability data from offline networks. A zero cost proof of concept will determine capabilities and outline an approach to use either an on-site or off-site deployment strategy. The value of completing the project is to obtain more accurate and timely lists of assets, improve the ability to make data-driven operational and security decisions, complete an assessment of existing vulnerabilities in the environment without the risk of negatively impacting the reliability of Consumers Energy electric generation and energy delivery systems, and increase incident response capability, speed, and effectiveness. As part of our review process the alternative considered was implementing a traditional vulnerability scanner to interrogate all systems on the network. However, this was determined to be the best solution to fit the Company's requirements.	31-Jan-20	(1.00)	\$71,870	(\$0)	\$85,818	\$98,614	(\$39,114)	\$26,553	\$1,762	\$46,497	0	0	0	\$15,265

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185	2019	Physical Security Hardware Refresh	Security	Network	The Physical Security Hardware Refresh project will ensure continued compliance with Federal Energy Regulatory Commission (FERC) requirements as they pertain to physical security assets. Included in the project is implementation of assets including advanced door systems at Company buildings, security camera for monitoring capabilities, and gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems. The value provided by completing the project is to maintain compliance, reduce redundancies and gaps in functionality, and optimize overall performance. An integrated solution is efficient and allows for centralized management, situational awareness, real-time monitoring, compliance with regulations and guidelines, and faster, more effective/consistent response to emergencies and incidents. As part of our review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	30-Nov-19	(1.07)	1,694,773	0	449,767	48,226	1,183,400	13,379	6,021	0	767	5,097	(302)	359		
186	2019	PII Data Discovery Tool	Security	Software (Intangible)	The PII Data Discovery Tool project will implement software that enables data management decisions for unstructured data, reducing security risk and lowering costs associated with data. The file analysis software scans, maps, and manages unstructured data stores. The Company will gain value from this project through: (1) the added ability to monitor unstructured data for PII/privacy-related data, (2) closing the gap in security visibility, and (3) maturation of the Privacy Program. Additionally, the Company will benefit through discovery of the scope, risk, and opportunities associated with the ever-growing volume of unstructured data. The scope of this project encompasses evaluation, procurement and implementation of a software solution to analyze, index, search, track, and report on file metadata and content. The unstructured data types included in the project scope and analysis consist of: (1) file shares, (2) email databases, (3) content collaboration platforms (CCLs), (4) records management and enterprise content management (ECM) systems, (5) Microsoft SharePoint, (6) Log files, (7) Internet of Things (IoT) generated objects, and (8) data archives. No alternatives were considered beyond the data discovery tool due to the large amount of data that needs to be monitored. Manual or native monitoring techniques are not possible for effective real-time results. By not having a solution in place, the Company will not have visibility into PII data in unstructured data sources.	01-May-20	(0.96)	763,272	609,664	0	27,019	117,776	8,812	51,535	0	0	12,803	34,640	4,092		
187	2019	SAP IDM Integration	Security	Software (Intangible)	The SAP IDM Integration project will configure the Human Management Capital (HCM) component of SAP (SAP User Access Provisioning tool) to connect to the HCM modules of our applicable SAP environments (ECP, CTP, and S3P). This change will allow the SAP IDM tool to view and function with the HR organizational structure in SAP ECC, otherwise known as HCM. This will provide the tool the capability to automate access provisioning and de-provisioning to users in SAP via HR processes including onboarding, transfers, and termination. The workflows would be configured to execute based on HR triggers and actions. The value of completing the project is to automate access provisioning and deprovisioning based on HR actions and triggers including onboarding and transfers, the ability to view and manage access associated with HR components including Position IDs, org structures (CRM PHCMA), and business partner creation / management, enable the use of business roles in SAP IDM through configuration of HCM to respond to HR triggers, and business role mapping to applicable business units.	18-May-18	(0.71)	(7,408)	0	0	0	0	(7,408)	0	0	0	0	0	0	0	
188	2019	SAP Security	Security	Software (Intangible)	The SAP Security project will provide vulnerability scanning of SAP-specific platforms. The product will enable the scanning of HANA, which current tools in the environment do not support. Included in the project is implementing a new SAP Cyber Security platform, tuning existing systems to include security monitoring, and integrating with Cyber Incident Response Team (CIRT) processes for security and user behavior monitoring. The value of completing the project is the ability to assess the security posture of SAP environments and continuously monitor those systems. This project will enable Information Security the ability to write rules to detect potentially malicious insider activity, and to perform routine vulnerability assessments against all SAP environments which identify missing security patches and poor configuration management practices. As part of our review process the alternative considered was to apply manual monitoring of Consumers Energy systems, which is prone to errors and inefficiencies.	01-Jan-19	(0.96)	(25,980)	259,800	(259,800)	0	(25,980)	0	0	0	0	0	0	0	0	
189	2019	2019 SubTotal Security	Security					4,764,433	884,430	1,555,791	490,092	1,661,762	172,358	637,974	56,154	17,733	176,372	323,942	60,733		
190	2019	800 MHz Modernization Project	IT Service Delivery	Network	This project will replace antiquated head-end (the main audio routing switch for tower sites and dispatch consoles), tower site, dispatch, and radio infrastructure, as well as user subscribed equipment with infrastructure that meets current Project 25 ("P25") standards. P25 standards are for digital mobile radio communications that are used by North American public safety and dispatch organizations. The current equipment was implemented in 1994 and is no longer manufactured or sold, so the Company is unable to find replacement parts. This project creates value for the Company by: (1) moving the Company to current production equipment that can be replaced and is more stable, (2) enabling the organization to migrate from an unsupported to a supported platform, and (3) allowing for quicker response to electric outages, and (4) increasing customer and employee safety. The scope of the project includes: (1) the design; (2) configuration; and (3) implementation of P25 systems for headends, tower sites, dispatch consoles, and subscriber equipment. The alternatives considered were to: (1) remain with the current system, and (2) forklift the migration to other manufacturers and architecture. Option 1 was not selected because the equipment is not supported, and the organization is unable to find replacement parts. Option 2 was not chosen because there would be a larger learning curve, it would be more disruptive to business, and it would require a complete replacement of the existing system. The Company chose to go with the upgrade because it has been incrementally adopting the P25 standards through the wireless ARP, and this prudently utilizes current and ongoing investments. The upgrade is the easiest and least disruptive path to migrate to the P25 standards.	01-Jul-23	(1.00)	1,996,400	1,996,400	0	0	0	0	0	0	0	0	0	0	0	0
191	2019	800MHz Tower Connectivity Optimization	IT Service Delivery	Network	The 800 MHz Tower Connectivity Optimization project will replace leased backhaul T1 circuits to all radio tower sites after telecommunication providers have announced the discontinuation of leased Time Division Multiplexing (TDM) services (i.e. T1s) by 2020. The project will avoid loss of tower site connectivity by migrating off legacy TDM copper circuits to carrier ethernet fiber services before the telecommunications providers discontinue this service. Alternatives considered were to do nothing and continue escalation of chronic circuit outages to the telecommunication providers, install Consumers Energy-owned fiber, or install point-to-point microwave. These alternatives were not selected due to option 1 being undesirable from a reliability perspective and options 2 and 3 being too costly.	01-Nov-19	(0.94)	866,237	0	404,102	25,115	425,876	11,343	0	0	0	0	0	0		
192	2019	ARP - Collaboration Asset Refresh	IT Service Delivery	Network	This project will replace the Company's collaborative tools and equipment. This project creates value by: (1) ensuring that the Company's audio, visual, telephony, and other communications systems are stable and reliable; and (2) beginning the foundational retirement of the legacy Avaya PBX systems that have reached end of mainstream manufacturer support. The project scope consists of: (1) annually replacing collaboration assets, and (2) installing new collaboration assets to account for organic growth requirements. The alternatives considered were to: (1) refresh visual assets and a portion of the audio assets, (2) refresh a portion of the audio assets only, and (3) refresh visual assets only. These alternatives were not chosen due to the risk inherent with a partial replacement of assets which includes: (1) a reduced supply of equivalent replacement Avaya parts that are no longer being produced, and (2) an erosion of the knowledge technicians possess on discounted systems in favor of education on the newest available technology.	31-Dec-18	(0.95)	339,638	(2)	197,732	27,358	98,142	16,409	13,478	0	0	166	13,031	281		
193	2019	ARP - FS Refresh	IT Service Delivery	Network	The ARP -FS Refresh project will refresh the company's FS Load Balancing equipment. The FS hardware was purchased in 2013 and the industry average refresh cycle is 5 years. Included in the implementation is refresh obsolete FS equipment, determine required feature sets for new Load Balancer, and implement new FS according to industry best practices.	31-Dec-18	(0.96)	7,654	0	0	4,938	293	2,423	1,984	0	0	1,577	153	255		

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
194	2019	ARP - Field Device Asset Management (FDM)	IT Service Delivery	New Computers / Hardware	This project will replace and install new desktops, laptops, and tablets. This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations prior to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks. The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources. The alternatives considered were to: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle only on desktops from four years to five years; and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ARP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	31-Dec-18	(1.09)	1,358,167	0	0	83	1,358,010	74	0	0	0	0	0	0
195	2019	ARP - IT Facilities	IT Service Delivery	New Computers / Hardware	Adjustment from prior year.	31-Dec-18	(1.08)	(2,330)	0	(2,047)	(252)	0	(31)	2,333	0	2,047	252	0	34
196	2019	ARP - Operational Technology Support	IT Service Delivery	Network	This project will replace dated and obsolete servers and workstations. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support the operation of the Company's critical electric infrastructure. The project scope consists of: (1) the annual replacement of compute hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. Extending maintenance is not a viable alternative as current systems do not provide sufficient capacity for new electric system support capabilities.	31-Dec-18	(1.04)	385,140	0	241,745	95,833	11,469	36,093	6,811	0	2,281	1,242	2,934	314
197	2019	ARP - Printer Asset Management (PAM)	IT Service Delivery	New Computers / Hardware	This project will replace printers, plotters, and multi-function printing devices. This project creates value for the Company by: (1) improving the dependability of these printer devices for employees, (2) averting increased costs due to hardware repairs, and (3) ensuring compatibility with enterprise print applications. The project scope consists of the annual replacement of printer assets under this program. The alternatives considered for the project included looking at refresh cycles from three to seven years and running the assets to failure. The selection of a five-year cycle was deemed to be the best solution since anything less than five years would result in additional unneeded expense for replacement of assets that were still in peak operating condition, and anything greater than five years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, negatively impacting employee productivity.	31-Dec-18	(1.05)	640,625	0	615,200	23,224	6,773	(4,572)	1,477	0	0	1,203	80	194
198	2019	ARP - Server	IT Service Delivery	New Computers / Hardware	IT infrastructure generally becomes less reliable after 5 years, jeopardizing the stability of our business' critical applications running on top of our IT infrastructure. This Server ARP project will evaluate Computer Hardware with more than 5 years of continuous use and replace where appropriate. The project will intelligently and systematically replace critical infrastructure before a system failure that would disrupt business operations.	31-Dec-18	(1.07)	370,723	22,637	177,704	80,920	53,305	36,157	6,174	0	0	5,221	352	602
199	2019	ARP - Storage	IT Service Delivery	New Computers / Hardware	The ARP Storage project will evaluate, validate, replace, and increase capacity of storage infrastructure after five years of continuous use. The project objectives are to reduce hardware repairs, improve system performance, accommodate organic growth, and reduce potential downtime. The benefit of replacing data center storage on a proactive refresh cycle is that it improves the stability and availability of business critical applications by proactively replacing hardware prior to the chance of hardware failures increasing. The storage five year refresh cycle is based on industry best practice on the useful life of storage arrays. From 2013 through 2017, the Company experienced an average 49% annual growth in data center storage. The Company plans on 30% growth per year based off this growth based on assumed improvements in technology, no significant architectural changes and no changes to data retention policies. The alternative is to not maintain the proactive refresh cycle and purchase extended maintenance. However, this would add additional cost and therefore was not considered further.	31-Dec-18	(1.04)	1,296,105	616,191	451,209	147,045	25,383	56,277	205,856	200,981	0	4,452	102	320
200	2019	ARP - Wireless Network	IT Service Delivery	Network	This project will replace portions of the company's aging wireless systems. This project creates value for the Company by: (1) ensuring real-time communications between Company crews and dispatch locations, (2) ensuring efficient gas leak and electric outage response times for customers, and (3) maintaining critical infrastructure and regulatory compliance. The project scope consists of: (1) replacing wireless assets annually, and (2) installing additional new wireless assets to account for organic growth requirements. The alternatives considered for system replacement were: (1) running parallel systems while the new system is deployed and the old system is dismantled, and (2) leasing a system from a vendor. The first alternative was not chosen since it would be highly disruptive due to systems having to run independently of one another, and would have a high cost associated with the need to acquire additional radio frequency spectrum. The second alternative was not chosen since the Company would be dependent on the response times offered by a shared vendor system that offers lower system reliability. The Company chose to replace existing assets based on its refresh cycle to avoid extended support cost, and to provide a seamless transition that allows both the old and new systems to interact with little to no disruption to the end user.	31-Dec-18	(0.97)	1,443,837	0	1,363,544	26,990	47,568	5,735	85,526	0	70,010	187	14,755	575
201	2019	ARP - Workstation Asset Management (WAM)	IT Service Delivery	New Computers / Hardware	This project will replace and install new desktops, laptops, and tablets. This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations prior to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks. The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources. The alternatives considered were to: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle only on desktops from four years to five years; and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ARP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	31-Dec-18	(1.06)	2,793,991	0	2,562,906	183,731	0	47,355	45,253	0	12,723	13,310	15,308	3,913
202	2019	Native HANA 2019	IT Service Delivery	Software (Intangible)	The Native HANA 2019 project will replace two existing HP HANA appliances that are at end of support and replace them with new infrastructure. The Native HANA system is an analytics platform that contains SAP data, primarily around the customer domain. The project scope will include selecting the new infrastructure for Native HANA system, installing and confirming the new infrastructure, and retiring the existing appliances. The value in doing this is that Consumers Energy will maintain a stable data analytics environment and control costs related to fixing issues by staying current with the latest version. One alternative considered was accepting the risk of not staying current. This would introduce risk of system instability and unavailability in the event of an appliance failure. We currently have another analytics platform called MemSQL that contains meter data. The other alternative considered was to consolidate platforms by migrating the Native HANA models and data into MemSQL; however, this would be a large effort and equally or more expensive. Analytics platforms are newer technologies that are rapidly changing and it is safer to keep options open by having two platforms that do not overlap in business capability.	09-Oct-19	(1.06)	1,889,375	5,196	1,939,840	34,640	0	9,699	142,923	980	0	59,789	65,123	17,031

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203	2019	Nimbus Phase 4	IT Service Delivery	Software (Intangible)	In 2013, Consumers Energy implemented the 2013/2014 effort was focused around creating self-service and automated deployment for basic datacenter requests. In 2016 continued expanding the features of Consumers Energy's private cloud in phase 2 project. These additional features would include: • Self-Healing: Automated repairs to server and application errors. Instead of requiring manual intervention from an operation team when a problem ticket is created, the Private cloud with automatically resolve the issues and inform CE teams that the issue was resolved. • Server Application Packaging and Self-Service Deployment allowing application teams to deploy and redeploy entire environments in a matter of minutes instead of weeks. • Storage/Network virtualization allowing the private cloud to provision storage/networks when needed without requiring manual intervention and labor. • Continue to reduce CE Build labor (Benchmarks), speed to market for IT infrastructure services. • Develop reactive healing countermeasures to reduce infrastructure/application unplanned outages. • Develop automatic benchmarking interfaces and reports. With successful implementation of this phase of the project effort, simple errors which cause critical systems outages will be resolved before outages occur. Critical business applications will increase in size before our customers and our business feels performance issues.	01-Jun-18	(1.03)	\$51,432	7,403	0	276,460	124,800	142,788	86,652	44,307	0	27,435	9,598	5,312
204	2019	Nimbus Phase 5	IT Service Delivery	Software (Intangible)	The Nimbus Phase 5 project will add additional features and enhancements to Nimbus, Consumers Energy's IT cloud automation platform. This will continue to improve the efficiency, quality, and speed to market of Consumers Energy's customer facing and internal IT services. Included in the implementation is adding the following features to the Nimbus platform: network virtualization support, public cloud big data and analytics enablement, additional infrastructure as a Service (IaaS) automation for new platforms, support of deployment of higher tier (less critical) applications in the public cloud, support of automated horizontal and/or vertical scaling in the public cloud (IaaS boost) for certain applications, support of containers and container deployment, and additional detailed reporting. The value of completing the project is furthering the Company's ability to deploy, use, and decommission public and private cloud resources in an on-demand and secure fashion, increasing the agility of IT, lowering risk in running applications in the cloud, keeping Consumers Energy's systems and customer data available and safe, and increasing the ability of the Company to learn from its data to provide new and better services. As part of our review process the alternatives considered were to manually build the servers, to offer limited or no IT support to public cloud applications, and to use on-premise analytics exclusively. These were considered not to be viable due to the risk of increased costs, reduced speed to market, and associated waste.	01-Oct-19	(0.68)	273,408	79	0	145,488	68,370	59,471	126,160	0	0	80,637	33,967	11,556
205	2019	Office 365	IT Service Delivery	Software (Intangible)	The Microsoft 365 project will implement in a standardized and secure manner the following Microsoft 365 services: Exchange, OneDrive, Skype for Business, Teams, and the enablement of SharePoint Online. This project will also implement a Cloud Access Security Broker service to be used for all cloud applications. This will allow for secure collaboration with internal and external parties. In addition, this project will enable mobile access from corporate devices, contain all aspects of conversion of mail, and activate e discovery features. Two alternatives were considered. One option was to do nothing and run non-mainstream supported applications that are dependent on computing and storage that will no longer be supported. This was not selected due to the risk of running on an unsupported version for critical applications. The other alternative considered was to refresh on-premise instances for Exchange, File Share, SharePoint, and Skype. This option was not selected due to the Microsoft strategy and future support scenarios.	01-Dec-18	(0.94)	0	0	0	0	0	0	1,535,923	0	0	254,988	1,213,333	67,601
206	2019	Operations Application Currency	IT Service Delivery	Software (Intangible)	The Operations Application Currency project is for upgrades for the following applications: Primavera, Impress, Robosync, ArcPad, WECO (meter test boards), MMS (Mobile Information Mapping System), and OMS (Outage Management System). Included in the implementation is upgrading the application software, assessing any new functionality for value to the Company, making necessary configuration changes, testing any integrations to or from the application, testing the upgrade, and updating documentation related to integration changes. The value of completing the project is reducing the number of incidents associated with out-of-date software, increasing application stability, and allowing the Company to leverage new functionality available in upgrades. As part of our review process the alternative considered was to stay at the current version of the applications. If this approach is selected we risk applications not working, increased number of incidents, increased support costs, and employees will not be able to perform their tasks.	31-Dec-18	(0.94)	232,615	0	0	57,605	141,146	33,863	49,379	0	0	36,398	8,421	4,560
207	2019	SAP Data Archiving	IT Service Delivery	Software (Intangible)	This project will move outdated data from an online database to offline storage. This project will create value for the Company by: (1) increasing system stability by reducing data growth, and (2) controlling maintenance costs associated with data storage. The project scope includes: (1) archiving data based on the fastest growing and largest objects in SAP, (2) building and archiving solutions that allow the business to retrieve archived data in the required form, and (3) setting up the solution so that the business areas meet compliance standards. Three alternatives were explored and determined non-viable for the project: (1) allow the database to grow in size, which would put system performance at risk and result in prohibitive storage costs; (2) decrease the overall scope and archive less project, which would result in minimal positive impact to system growth and significant storage-related costs; and (3) increase the scope and archive more objects in a shorter timeframe, which would result in significant cost increase over shorter time period. After considering each of these options, it was determined that the current scope of the project was the best strategy to address the problem while balancing annual spending.	31-Dec-21	(1.00)	100,707	0	0	11,715	68,180	20,812	(3,146)	0	0	(9,212)	7,898	(1,832)
208	2019	SAP Data Encryption	IT Service Delivery	Software (Intangible)	The project will implement Cyber Security requirements for encryption of Personal Identifying Information (PII), data "at rest" and "data in transit" in Company SAP data bases.	01-May-21	(1.00)	887	0	0	693	0	194	13,137	0	0	8,593	3,055	1,488
209	2019	Service Now Quarterly Releases	IT Service Delivery	Software (Intangible)	The ServiceNow Annual Release project will upgrade the cloud solution ServiceNow, and validate/remediate the integrations. ServiceNow requires that customers move to the current version of the software as new versions are released. In addition, this project will increase the use of automated testing each year by 20%. Also included in 2020 is the upgrade of the ServiceNow Portal - it will reconfigure the portal and make it much more intuitive for end users to enter tickets, receive status, and avoid rework. Included in the implementation is performing regression testing on the new version to ensure it still functions as expected, validating and remediating integrations, analyzing new available functionality for features that may improve productivity, testing any new features that will be implemented, and implementing the new version. The value of completing the project is continuing to be on a vendor supported version, ensuring support in resolution of issues with the software, and implementing new functionality to improve productivity. As part of our review process, the alternative considered was waiting for ServiceNow to automatically upgrade us without us prepping or doing any testing. This alternative would potentially have significant impacts on operations and our ability to support the business. Without ServiceNow functioning, the Company cannot process, track, monitor, and execute break/fix and other operational incidents. This could lead to significant loss of services for the business and potentially customers.	01-Jul-19	(0.87)	343,121	259,965	0	45,186	22,807	15,362	374,858	2,218	0	73,167	250,539	48,914

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210	2019	Software Platform Refresh	IT Service Delivery	Software (Intangible)	The Software Platform Refresh project will upgrade server operating systems, hypervisors (virtual machine monitors), and databases to retain low-cost, unlimited vendor support. The project scope includes: (1) upgrading operating systems and databases on servers that are within three years of end of support, and (2) maintaining hypervisors at the current version for stability and performance. The project will add value for the Company by: (1) avoiding costs for special maintenance agreements required at the end of normal manufacturer support, (2) ensuring reliability and compliance with Information Security requirements, (3) improving data center environment stability, and (4) avoiding the need for high risk upgrades that cross multiple versions. A funding options matrix was completed to review the alternatives. The options were: (1) fund the project for \$1 million and pay no support liability in 2020, (2) fund \$750,000 of the project and pay \$2.5 million in support liability in 2020, (3) fund \$500,000 of the project and pay \$2.4 million in support liability in 2020, and (4) not funding the project and paying \$3.3 million for support liability in 2020. Alternatives 2-4 were not selected due to the high cost of support liability. Option 1 was chosen to avert these costs and to ensure system stability.	31-Dec-18	(1.00)	969,920	0	969,920	0	0	0	501,525	19,128	78,112	117,143	177,382	109,739
211	2019	SubTotal IT Service Delivery	IT Service Delivery					15,957,774	911,369	10,761,657	1,186,771	2,408,723	489,253	3,196,302	267,615	165,193	676,569	1,816,068	270,856
212	2019	#911 Compliance Project	Enhancements	Software (Intangible)	The #911 Compliance project will implement a corporate-wide enhanced 911 (e911) system to ensure fire, police, and paramedics have accurate location information and can find those in distress when a call is placed to a public safety answering point ("PSAP"). Included in the implementation is installing the on-premise application with standardized virtual compute and storage in a highly available configuration and configuring the application for interfacing with existing telephony systems and Local Area Networks. The system will be installed and configured by an integrator assisted by the publisher of the e911 application. The value of completing the project is increasing employee safety by increasing the accuracy of where a distress call has been placed within a building or facility, decreasing the response time of first responders as emergency calls will be routed directly to the PSAP instead of first to the Command Center, and ensuring compliance with the new MPSC Rule 484. As part of our review process the alternative considered was to do nothing and continue to have 911 calls routed to the Command Center and become noncompliant with MPSC Rule 484.	01-Apr-19	(0.96)	356,972	0	1,337	103,084	186,061	66,490	146,387	(2,052)	0	12,073	133,431	2,935
213	2019	Enhancements - CEO	Enhancements	Software (Intangible)	The Enhancements - Cust Exp - Ops project is for software enhancements performed for Customer Experience and Operations business areas. The functionality includes mandatory regulatory changes, enhancements to billing, enhancements to dialing for low income customers, and changes to the Company website based on customer feedback. An alternative would be to not do these smaller system enhancements. However, that limits the company's ability to improve the applications that benefit employees who are serving customers and puts the company at risk of not meeting regulatory requirements. Customer benefit: System enhancements to comply with regulatory changes.	31-Dec-18	(0.98)	1,115,847	(31,176)	51,465	210,610	757,651	127,297	317,854	0	0	51,332	236,091	30,432
214	2019	Enhancements - Corporate	Enhancements	Software (Intangible)	The Enhancements - Corp project is for small software enhancement work efforts performed for the Corporate Services business areas. Included in the implementation are enhancements for Finance, Human Resources (HR), Learning & Development, Legal, Governmental, Regulatory and Public Affairs, and Corporate Security. The value of completing the project is compliance, hard cost savings, avoided costs, hitting corporate goals, and mitigating risk. As part of our review process the alternative considered was to not provide funding for enhancements. However, this then limits our ability to provide process improvements, make changes related to customer and employee safety, and meet legally required HR and Finance system changes.	31-Dec-18	(0.90)	625,299	68,521	56,752	237,278	147,095	115,654	374,439	0	4,365	74,568	266,313	29,193
215	2019	Enhancements - TEOS	Enhancements	Software (Intangible)	The Enhancements - Transformation, Engineering, & Operations Support ("TEOS") project is for small software enhancement work efforts performed for the Transmission, Engineering, and Operation Support business area. Throughout the course of a year small requests for system changes are requested. These small changes generally cannot be predicted in advance at a detailed level so we set aside a certain amount of funding each year to address them. Requests for this funding are assessed for value by representatives from each area using benefits that are categorized into hard cost savings, cost avoidance, compliance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are regulatory compliance mandatory changes, improvements to engineering and design applications, and minor functionality improvements across many production systems. As part of our review process the alternative considered was to not do enhancements, however, that limits the Company's ability to improve the applications that benefit employees who are serving customers. This will limit our ability to provide process improvements, retire applications, make changes related to customer and employee safety, and meet legally required changes.	31-Dec-18	(0.98)	410,550	0	0	158,120	157,786	94,644	13,868	0	0	6,826	5,678	1,365
216	2019	Enhancements - Operations	Enhancements	Software (Intangible)	The Enhancements - Operations project is for small software enhancement work efforts performed for the Operations business area. Throughout the course of a year small requests for system changes are requested. These small changes generally cannot be predicted in advance at a detailed level so we set aside a certain amount of funding each year to address them. Requests for this funding are assessed for value by representatives from each area using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small application changes and functionality improvements for Electric Operations, Gas Operations, Operations & Financial Planning, Generation Operations, and Distribution Performance. As part of our review process the alternative considered was to not fund enhancements, however, that limits the Company's ability to improve the applications that benefit employees who are serving utility operations. This will limit our ability to provide process improvements, retire applications, make changes related to customer and employee safety, and meet legally required changes.	31-Dec-18	(0.96)	445,733	0	0	220,645	118,901	106,186	418,920	0	231	147,359	212,409	58,921
217	2019	Enterprise Content Management	Enhancements	Software (Intangible)	The Enterprise Content Management project will provide an Enterprise Document Management solution including the strategies, tools, and processes to more easily manage, rapidly locate, and deliver CMS Energy content throughout its life cycle wherever that content exists into the form the business partner needs. Areas needing this include: Legal, Risk and Insurance, HR, Strategic Communications, Learning and Development, Distribution. Content management enables organizations to enforce policies and rules for the retention and disposition of content required for documenting business transactions, in addition to automating the management of their record-retention policies. These technologies, implemented with well-formulated and consistently enforced records retention policies, form an essential part of the life cycle management of information. As industry regulation and compliance requirements increase, along with the volume of digital content that must be retained and the demand for legal discovery.	01-Dec-18	(0.94)	61,320	0	0	12,602	10,530	38,188	55,026	0	0	20,103	29,055	5,868

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
218	2019	New Move In/Move Out Customer Experience	Enhancements	Software (Intangible)	The New Move In/Move Out (NMMO) Customer Experience project is for enabling additional moves, transfers and moves-outs to be completed in the customer's channel of choice. This effort will expand baseline functionality to small and medium businesses and provide options for customers with outstanding bills that require security deposits to complete online move-in. Technologies to address current shortcoming will also be deployed along with reporting to assure customer experience is improved. The value of completing the project is an increase in digital customer experience and a reduction in calls to the contact center to complete MIMO transactions. Blocks will be removed that restrict on-line MIMO processing and new simplified options will be supported. As part of the project review process the alternative considered was maintaining the current MIMO functionality. However, this option would not yield anticipated customer experience objectives and would sustain current call levels. Customer benefit: Improved Live CX score of 1.0%	01-Jan-19	(0.99)	442,033	0	0	78,068	285,673	78,292	17,895	0	0	4,123	8,802	4,910
219	2019	2019 Sub/Total Enhancements	Enhancements					3,457,754	97,345	109,554	1,620,407	1,663,696	626,751	1,344,989	(2,052)	4,595	316,988	851,779	133,684
220	2019	4G SAP Implementation	BP Functionality	Software (Intangible)	There are approximately 680,000 smart meters in the Company's electric system that utilize the 3G wireless technology. The wireless carrier will be retiring the 3G network technology on December 31, 2022, requiring the Company to replace the 3G meters with 4G meters in order for them to continue functioning. This project will create the technology that enables firm to upgrade smart meter communication modules from 3G to 4G technology. This project will add value by continuing the ability to: (1) communicate with the smart meters, (2) provide accurate and timely bills to customers, (3) execute remote turn-ons and turn-offs, (4) receive outage information, and (5) administer demand response events. The project scope includes: (1) building technology interfaces to meter installation vendor technology, which include an interface to extract list of 3G meter information, an interface for posting work order completion data in SAP, and an interface for tracking exceptions and errors; and (2) end-to-end testing of the interfaces and back-end technology (meter-to-cash processes). The alternatives considered were: (1) migrating to a new meter platform, and (2) not implementing 4G technology in 3G meters. The first option would require a substantially larger modification to all supporting systems and infrastructure. The second option is not feasible since the deprecating communications technology will eventually make the meters non-communicative. The option of implementing 4G technology was chosen so the Company can continue to reap the benefits of the smart meter technology previously implemented. These benefits include: (1) improved timeliness and billing accuracy; (2) retaining ability for remote disconnect and reconnect of services; (3) administration of demand response programs; and (4) enabling automated meter reads, outage reporting, and faster restoration.	01-Jul-20	(0.93)	296,006	0	12,861	57,075	101,551	25,120	16,400	0	0	5,839	6,650	1,532
221	2019	Accounts Payable (AP) Automation	BP Functionality	Software (Intangible)	The Accounts Payable (AP) Automation project will provide an end-to-end AP solution to optimize invoice capture via Optical Character Recognition (OCR) software, leverage workflows to automate invoice approvals and processing, manage electronic document retention, provide insights via improved reporting, and minimize human intervention. The value of completing the project includes: (1) centralizing invoice processing resulting in reductions to invoice entry costs, number of paper invoices, duplicate vendor payments, and late vendor payments; (2) automating validation of an invoice against contract rates; (3) increasing transparency of invoice processing status; (4) providing more accurate accuracy; (5) improving the ability to capture discounts; (6) enabling cash flow benefits and reduced financing costs; and (7) improving internal controls. The scope of the project includes: (1) enabling OCR technology; (2) creating automated workflows for receiving, managing, routing, and monitoring invoices and related documentation; (3) automating posting of invoices; (4) creating new reports; (5) enabling electronic document retention; and (6) data cleanup. See Karen M. Gaston's testimony for full project synopsis.	17-Oct-20	0.69	277,120	0	0	0	0	277,120	56,952	0	0	36,990	0	19,962
222	2019	Bill Redesign	BP Functionality	Software (Intangible)	The Bill Redesign project will update customer bill presentation, align and address changes driven from the Summer Peak Usage Rate project, and provide a more intuitive invoice of charges to avoid customer inquiries. Included in the implementation are paper and e-bill invoices for electric and gas-electric combo customers, comprehensive analysis of required bill data, and bill clarity analysis based on customer perception. The value of completing the project is reduced customer calls and improved customer experience. Customer benefit: Improved CX score of 5.0% by updating visual appearance of bills to make them easier to read. As part of the project review process the alternative considered was to continue with the current bill presentation and address anticipated increase in customer calls through staffing and other non-billing mechanisms.	01-Aug-19	(0.91)	138,688	0	0	36,086	81,148	21,454	59,843	0	0	16,522	36,656	6,665
223	2019	BPS Enhancements	BP Functionality	Software (Intangible)	This is the second phase a technology roadmap to implement a company-wide suite of solutions to mature the company's operational planning activities. This phase supports monthly plan management while the first phase was limited to annual planning. The scope of release 2 is focused on supporting and facilitating monthly plan management. The visibility of the linkages between financials and performance will allow us to better optimize our portfolio of work. This will ensure we are investing the right amount in the right areas, functions and assets thus reducing risk. This will reduce ad hoc reporting, reduce report-to-report reconciliation and enable automation of utility wide plans down to the optimal level of detail.	17-Oct-19	(1.00)	236,245	0	0	33,705	157,714	44,826	57,988	356	0	10,112	40,245	7,275
224	2019	Business Continuity Disaster Recovery Integration	BP Functionality	Software (Intangible)	The Business Continuity Disaster Recovery Integration Project will implement technology solution and supporting processes to integrate the Company's business continuity and disaster recovery programs to enhance program efficiency and effectiveness. The solution will improve program management and drive efficiency with the following: Plan management repository with workflow capabilities in support of plan development/review, training and testing requirements; Maintain program schedules, monitor status and reporting capabilities; Risk analysis and interdependency mapping of critical business processes and IT applications; Flagging mechanism to ensure identification of restoration capability gaps to critical business processes and/or IT recovery capabilities; Business Impact Analysis (BIA) capabilities to quantify financial risks to critical process disruptions.	30-Apr-19	(0.74)	696	0	0	471	0	225	2,018	0	0	1,783	0	234
225	2019	Business Process Performance Monitoring	BP Functionality	Software (Intangible)	This project will deploy the AppDynamics tool across enterprise applications like SAP, OMS, and other business critical systems. This project will create value for the Company by providing faster system restoration through automation of system troubleshooting that pinpoints the software or hardware that is causing the problem. The project scope includes: (1) SAP, (2) OMS, (3) Advanced Device Metering System, and (4) other enterprise applications. The Company evaluated multiple products for this project and AppDynamics fit the cost and functionality expectations. The others were cost prohibitive and lacked the functionality of the product selected.	01-Oct-23	(1.00)	3,842,555	0	3,671,839	24,941	82,756	63,019	(7,668)	0	0	54,969	(69,038)	6,401

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226	2019	Care Annual Updates	BP Functionality	Software (Intangible)	This project will: (1) implement software changes to improve the process for offering energy assistance to low-income customers, and (2) streamline the process for assistance agencies who utilize the application through improved user interface and updates to SAP to process various CARE requests. Upcoming modifications will be identified following an annual review of requests to prioritize the list of changes. The project will provide the following value: (1) complete modifications to the existing internal SAP application and Agency Portal to receive annual Low Income Home Energy Assistance Program ("LIHEAP") funding, which is used to provide customers bill credits and arrears forgiveness; and (2) improve the data within the assistance Agency Portal, thereby making it easier to assist customers in need of LIHEAP funding. The project scope includes: (1) updating the enrollment and status process, (2) allowing for flexible bill credits, (3) improving reporting, (4) updating the arrears forgiveness plan, and (5) satisfying additional regulatory requirements for the annual grant rule changes required by the Michigan Department of Health and Human Services and the Michigan Agency for Energy. The alternative considered was to continue with the current process. This alternative was not chosen since it would result in loss of grant funding, thus decreasing or eliminating energy assistance dollars for low income customers. Instead, making annual updates to the application will allow agencies to easily enroll customers on assistance programs, and allow placement of holds to stop or prolog credit activity until assistance decisions are granted. This option provides long-term proactive energy assistance to low income customers and prevents loss of grant funds. All changes are related to the existing internal SAP and Agency Portal, therefore a cloud alternative is not viable.	31-Dec-18	(0.90)	0	0	0	0	0	0	376,563	0	0	200,000	130,602	45,911			
227	2019	Centralized Demand Response Management	BP Functionality	Software (Intangible)	The Centralized Demand Response Management project is intended to centralize triggering and reporting of all Demand Response (DR) events activities into one technical solution and will be informed by the outcomes of the Centralized Demand Response Management Assessment effort. The project will add value by providing functionality to ensure Demand Response (DR) events can be collectively managed from a central solution that includes all programs and potential load shed. The upgraded Demand Response Management System (DRMS) application will be able to support the various DR Programs including new Electric Charging Vehicle programs and Time of Use (TOU) rates implemented in 2019. Finally, this upgraded application will ensure accuracy of kilowatt hour (kWh) savings and peak load reduction when a DR event is called. The scope consists of: (1) centralization of the DR event management programs including residential DR, bring your own device, electric vehicle charging and large customer DR; (2) upgrade or replacement of the Demand Response Management System (DRMS) with a potential move to a cloud-based solution; (3) provision of new DRMS system functionality to support on-going DR programs; (4) addition of DRMS functionality to account for the new TOU rates. Alternatives considered during the assessment effort include: (1) continue to maintain the current hardware and software at the risk of continued inability to centralize information for all DR programs, (2) enhance the existing solution to address centralization needs, (3) transition the application to the vendor's hosted environment and configure to address requirements for additional DR programs, and (4) replace the current solution with a new cloud solution, or (5) replace the current solution with a new internally hosted solution. The selected alternative will be made as an outcome of the Centralized Demand Response Management Assessment effort.	31-Jul-22	(1.00)	480,481	0	17,997	0	462,484	0	0	0	0	0	0	0	0	0	0
228	2019	Complex Billing Automation	BP Functionality	Software (Intangible)	The Complex Billing Automation project will reduce manual intervention and decrease risk of error on large customer and specialty billing accounts. Complex billing encompasses, but is not limited to, all primary rates that require MVRD billing, large commercial and industrial rates, unique customer specific billing solutions (i.e., MSU, BioDigestor, some self-generators), and customer programs such as net metering. Included in the implementation is automating fragmented processes to support complex billings updating/revising complex account calculations based on rules while permitting customers to choose a billing method and delivery time, permitting multiple accounts to be summarized, addressing Net meter billing, Collective accounts, Gas Transportation accounts and unmetered billing accounts. The value of completing the project is improving the process of setting up and billing this customer segment by automating the process, improving Customer experience by providing this customer segment currently receiving multiple bills to have one consolidated bill, and allowing customers to pay multiple bills with one invoice. As part of our review process the alternative considered was to maintain the status quo and continue to leverage employees to manually set up, bill, and monitor the billing of these customers.	07-Mar-19	(0.94)	94,872	0	0	19,458	49,013	26,401	38,849	0	0	1,231	32,380	5,238			
229	2019	Contact Center Customer Experience Refresh	BP Functionality	Network	The Contact Center Customer Experience Refresh project will refresh of the Customer Call Center's IT infrastructure, including the three Automatic Call Distributor (ACD) systems, networking equipment, IVRS, Work Force Management, servers, and applications. The ACD Systems are 10 years old in 2015 and cannot readily adapt to best practice. Additionally, they are no longer vendor supported and hardware replacement parts are not available. Included in the implementation is speech enabled interactive voice response (IVR), Customer Service Representative Knowledge Management, eMail Management, Call Center Quality Monitoring, Optimize Skills Based routing, Customer Analytics Enhancements, Multi-Channel Inbound & Outbound Communications, Virtual Hold, and Click to Call.	24-May-18	(0.55)	(35,282)	(28,848)	0	0	(6,434)	0	36,464	28,848	1,181	0	6,434	0			
230	2019	Credit and Collections	BP Functionality	Software (Intangible)	The Credit and Collections project will make improvement to our outside agencies collections rate by enhancing our visibility and tracking of uncollectable expense via middleware. This project will implement the DelinText cloud solution to support the lowering of uncollectable expenses. It will allow payments to be uploaded and credited to a customer's account automatically, provide reports that detail debt and aging of debt, and provide the ability to move accounts between various collection agencies. The value of completing the project is increased productivity due to reduction in time to audit and complete invoice reconciliation, new customer communication channels, the ability to easily manage third-party collections agencies and the accounts placed with them, the ability to quickly transfer accounts amongst agencies, decreased uncollectables, reduced costs, and improved operations. Failure to implement this project would result in unrealized savings listed above. Customer benefit: Reduced uncollectables expense by improved credit collection rate. Projected 5-to-1 payback.	01-Mar-19	5.00	197,642	0	0	45,966	113,276	38,400	31,811	0	0	20,277	8,398	3,137			
231	2019	Customer Intelligence and Analytics	BP Functionality	Software (Intangible)	The Customer Intelligence and Analytics project will create a comprehensive and analytical view of customers across various classes encompassing outages, billing history, payment history, customer usage and payment channel for the purpose of segmenting and providing improved customer experience. Included in the implementation is to include various business partner data, rates and billing data and history; various programs our customers are enrolled in; complaints filed by customer and customer preferences. This data will integrate with our Business Warehouse to provide a comprehensive view of our customer. The value of completing the project is providing a single source for all of the customer interaction data within the utility, providing analytics for modeling, and execution of one on one customer interactions, and these analytics will provide insights into the various customer journeys and experiences allowing Consumers Energy to determine opportunities to improve that experience across all channels. As part of our review process the alternative considered was to continue to operate as today, leaving us without a comprehensive view of our customers and their experiences with Consumers Energy.	01-Apr-20	(0.95)	328,987	0	(190)	81,794	182,559	64,823	17,453	0	190	274	14,630	2,359			

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232	2019	DCE R4 Hub and Spoke	BP Functionality	Software (Intangible)	The DCE Hub and Spoke project release will implement a payment workflow that creates a simplified user experience for customers looking to pay their bills online. Instead of traversing multiple steps to define: payment date, payment amount, and payment method, customers will be routed to a new "hub" experience where defaults will be supplied for all of the above elements and a customer can selectively choose to "spoke" out to shortened user flows if they want to make any changes. This new experience will allow customers to make a payment in as few as two clicks. The release scope includes new reporting added to the hub & spoke experience that will allow business partners to gain insight into how many successful payments were made, how many payments failed, and how many customers go through unique payment flows such as installment plans, overdue or delinquent, and auto-pay. In addition, the release adds value by re-purposing one of the existing preprod and production Sitecore servers, which will enable the Company to accelerate future releases (beginning with this one) with a new release strategy. This will allow future releases the ability to make isolated deployments to this server and throttle specific amounts of customers to the new version of the website, while the rest of the customer-base visits the standard version of the website. This creates the ability to effectively monitor the customer experience through new reporting that gauges whether the new experience is working as expected, and provides input on the decision to proceed with the deployment for all visitors.	28-Sep-18	(0.92)	4,321,670	(157)	0	275,413	3,628,685	417,929	75,080	905	0	26,164	40,451	7,361	
233	2019	ECM Sharepoint Integration	BP Functionality	Software (Intangible)	The ECM - Sharepoint Integration project will connect specific SharePoint sites with the ECM system and apply taxonomy, metadata, and retention rules to that content. Included in the implementation is integrating content in specific SharePoint repositories into ECM, assigning taxonomy content values including classification to data in those repositories, and assigning retention rules to the content. The value of completing the project is to enable enterprise-wide search of content in SharePoint sites, better visibility to content in those SharePoint sites, the ability to purge content based on retention schedule to reduce risk, and a faster eDiscovery process. As part of our review process, the alternative considered was to continue what is being done today by not integrating SharePoint content into the ECM, which will not allow us to meet record retention standards set in the corporate policy without manual effort to find, classify, monitor and delete hundreds of thousands of records every year, which would be more costly over time.	19-Sep-19	(0.93)	0	0	0	0	0	0	342,881	0	0	99,291	199,563	44,028	
234	2019	ED - ARP OSI HVD and LVD SCADA Upgrades	BP Functionality	Software (Intangible)	The Electric Distribution (ED) Asset Refresh Project (ARP) upgrades OSI Supervisory Control and Data Acquisition (SCADA) application for High Voltage Distribution (HVD) and Low Voltage Distribution (LVD) systems. Biennial hardware refresh included. The scope of this project is a software upgrade to the High Voltage Distribution (HVD) supervisory control and data acquisition (SCADA) software. The HVD SCADA is used to control generation and high transmission and distribution voltage circuits. Upgrading the HVD SCADA software will ensure that when problems occur, they are promptly addressed. Additionally, upgrades include enhancements and fixes to the core product that support reliability of electric delivery, ensuring that customers receiving the energy they need when they need it. Because the reliability of the entire distribution system is dependent on availability of the HVD system, staying relatively current with the solution provider upgrades is critical. If upgrades are delayed, additional full time resources will be required for 1 year to address upgrade complexities.	19-Dec-18	(0.98)	0	0	0	0	0	0	103	0	103	0	0	0	0
235	2019	ED - Cyme Upgrade	BP Functionality	Software (Intangible)	The Cyme Distribution Analysis software is designed for planning studies and simulating the behavior of electrical distribution networks under different operating conditions and scenarios. It includes several built-in functions that are required for distribution network planning, operation and analysis. The Cyme solution currently runs on an unsupported version of software. The vendor has encouraged Consumers to upgrade, stating that they will not be able to support future problem resolution efforts. The software is currently loaded on individual workstations. Gateway will continue to reside on server. However, a network based solution to replace the PC based solution is not a feasible solution from Cyme. The scope of this project is to "replace in kind." Once upgraded, there will be the ability to evaluate other Cyme modules for their potential value for other departments. The CymDIST Distribution Analysis software is a suite of applications composed of a network editor, analysis modules and user-customizable model libraries. The program is designed for planning studies and simulating the behavior of electrical distribution networks under different operating conditions and scenarios. It includes several built-in functions that are required for distribution network planning, operation and analysis. This will also require an upgrade to Cyme Gateway (this will not require hardware /infrastructure changes. Consumers must utilize Cyme, or some other solution, in order to effectively perform load flow analysis that is critical to distribution planning, outage resolution. It's used on a daily basis for switching analysis. If the software and platform are not upgraded, the vendor will not support future problem resolution, putting the ability to perform required analysis at risk.	01-Aug-18	(0.95)	0	0	0	0	0	0	3,842	0	0	2,797	0	1,045	
236	2019	EHS Compliance	BP Functionality	Software (Intangible)	The EHS (Environmental Health & Safety) Compliance project will implement a cloud-based corporate-wide incident and risk management solution. The solution will ensure consistent accurate tracking, management, and reporting of environmental, health & safety compliance tasks using a centralized source of data. Modules to be implemented include Compliance, Health and Safety, Human Performance Management, Incident & Crisis, and Reporting & Analytics. The value of doing this project is to leverage best practice processes which are built into the new solution, including incident management and prevention to ensure a safe and productive workforce able to complete work for customers. The project also supports the Company's Planet goal through enhanced tracking and reporting for air quality, waste management, and sustainability. Other benefits include risk avoidance for environmental penalties, increased productivity, and increased quality. The addition of visual management and dashboards to enable transparency in tracking of goals; and create awareness of emerging environmental regulations and, to respond to them. One alternative is to continue "as is" through use of disparate Excel and SharePoint solutions. This requires much manual effort, and poses a risk of not being able to accurately and centrally track EHS data. Another alternative is to have separate projects; one for Environmental Compliance, and another for Safety and Health Compliance. This was the initial path for these two business areas. Investigation into potential solutions including a formal Request for Information (RFI) revealed that the standard for these market solutions is integrated functionality for Environmental and Safety/Health. Both on premise and cloud solutions were considered. The single solution was selected as the most efficient solution.	01-Dec-20	(0.86)	866,175	0	118	204,915	432,743	228,399	65,268	0	0	29,701	29,917	5,651	
237	2019	External Customer Contact Support	BP Functionality	Network	The External Customer Contact Support project will implement access to Company applications for a third party contact center vendor, so that they may augment internal customer contact's day-to-day operations by supporting inbound customer calls. The project adds value by: (1) augmenting contact center staffing for day-to-day operations, which reduces customer wait times, increases Average Speed of Answer (ASA), and improves first contact resolution, especially during peak call volumes such as during storm restoration; (2) providing the ability to effectively respond to ever-changing business demands; (3) enhancing services provided to customers through optimization and applying best practices across all customer interactions; and (4) enables flexibility to manage call volume between multiple third party vendors. In addition, configuring the contact center environment to manage multiple third party vendors creates a dual vendor partnership and champion challenger model, which reduces operational costs. The scope of the project includes: (1) configuring third party vendor access and connectivity to Company environments; (2) configuring the Automatic Call Distributor (ACD) and Interactive Voice Response (IVR) to route calls to multiple third party vendors; and (3) enabling access to the Company contact center training environments. Alternatives considered include: (1) Continue to single source a third party vendor for the contact center. This alternative was not selected because it increases operational costs and offers no incentive to improve performance. (2) Implement multiple third party vendors. This alternative was selected because of the capability to improve performance by implementing a champion challenger dual vendor environment while optimizing ongoing operational spend.	22-Feb-19	N/A	0	0	0	0	0	0	77,992	0	89	16,435	54,300	7,168	

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC		
238	2019	Field Service Solution Release 3	BP Functionality	Software (Intangible)	Adjustment from prior year.	11-Dec-17	(0.67)	(873)	0	0	0	0	(873)	0	873	0	0	0	873	0	
239	2019	Financial Planning Transformation - R1 Basic Support	BP Functionality	Software (Intangible)	Adjustment from prior year.	18-Apr-19	0.61	(69)	0	0	0	0	(69)	0	0	0	0	0	0	0	
240	2019	Generation and Compression & Storage Engineering Records Management	BP Functionality	Software (Intangible)	The Generation and Compression Engineering Records Management project is for implementation of the ProjectWise drawing/records management software to manage compression assets. The system will be capable of bundling, managing and handling the versions of drawings and documents related to design, project files, construction of As-Built and record-keeping. Included in the project is implementation of document sharing for collaboration with parties external to the company, and developing a data conversion solution to facilitate creation and importation of data. The value of completing the project is: creating a single "resource of truth" for projects and operations for all records with a regulated review process, meeting both compliance and engineering best practice standards; eliminating the possibility of work being executed in the field based on an out-of-date drawing or document; ensuring records needed for an audit are readily accessible and easily searchable; providing compliance and design control; managing and handling all versions of drawings and documents related to design, proposals, contract resources and record-keeping with real-time access; integrating with SAP; and replacing the use of multiple document repositories (e.g. SharePoint, network drives, Access Databases, and other systems). As part of our review process the alternative considered was the application OpenText, but the capability to work with engineering drawings and computer aided design (CAD) software was lacking.	01-Feb-19	3.02	18,602	0	0	(363)	(0)	18,965	25,455	0	0	409	23,357	1,690		
241	2019	Generation Unit Performance Optimization	BP Functionality	Software (Intangible)	Adjustment from prior year.	N/A	N/A	1,134	0	0	0	0	1,134	0	0	0	0	0	0	0	
242	2019	GIS-Integrated Design	BP Functionality	Software (Intangible)	This project replaces the CAD and Work Requirements and Design ("WinBd") software with Bentley's Open Utilities Map ("BOUM") to leverage GIS asset data to generate engineering designs, implement workflows with SAP, and mitigate risk associated with the aging software application. The project provides value to the Company in three ways: i) the BOUM software is compatible with GIS integration and may be further developed after the Company implements the UPM as the basis of its GIS systems; ii) the project mitigates risk of manual asset record imports, asset record manipulation for design white space management, or manual asset record recreation for base design files, by enabling technology that creates a new design over existing electronic asset records; iii) it enables technology that is compatible with the asset system of record and the Project Wise engineering document management system. The project scope includes: i) replacing CAD engineering design software (Bentley Microstation (iV7) with BOUM; ii) leveraging GIS asset data for the purpose of generating engineering designs; iii) integrating workflows with SAP through Bentley Workflow Manager software; iv) implementing GIS replication databases to isolate data editing from data viewing; v) replacing the desktop View Graphics custom application with a web-based application for searching and viewing asset records; and vi) implementing a Primary Distribution Map ("PDM") generation service to generate electric PDM files for printing during storm and other emergency events. See Richard T. Blumenstock's testimony for full project synopsis.	01-Mar-23	(0.95)	877,716	0	2,188	235,304	460,482	179,742	134,180	0	0	49,716	79,410	5,053		
243	2019	GIS - Distribution Management System - Release 1	BP Functionality	Software (Intangible)	The scope of this portion of the DMS is implementation of OSI's eMap solution. It includes integration with GIS and is a foundational component for the broader implementation of a comprehensive DMS/ADMS. The purpose of this project is to implement the first release of a Distribution Management System (DMS). OSI Modules to be implemented include: - Distribru on Network Operating Model (eMap) - GIS Interface Savings per customer for 30+ minute primary outages of and reduced outage time from looping implementation. Last 18 months 2.3% system coverage of loop schemes resulted in 2,959,564 minutes saved.10% crew time savings on each primary outage. This module will provide the foundation necessary for the broader implementation of a comprehensive DMS. A DMS system is an electric network management tool that provides advanced functionality in distribution grid analysis, oper. and restoration	01-Jan-19	(0.99)	706,177	0	0	110,153	465,116	130,908	28,538	0	1,340	(9,207)	30,087	6,318		
244	2019	GIS - Electric System Model Enhancement	BP Functionality	Software (Intangible)	Implement an Electric Grid System Model that will support the current and future needs of GIS, OMS, DPS (CYME), DMS and GIS Integrated Design Tool. The Electric Grid System Model will be designed and implemented to serve as an extensible platform to efficiently and effectively manage and share the Electric GIS network model information with the rest of grid operational and planning systems. Establish a methodology for integration that will connect Electric GIS, SAP, Cascade and other Asset Management Systems in a common way to provide an integrated view of assets across asset management areas. The Electric Grid System Model will be designed and implemented to serve as an extensible platform to efficiently and effectively manage and share the Electric GIS network model information with the rest of grid operational and planning systems. The Project will achieve the following high level objectives in support of the Grid Modernization Program: (no specific order) 1. Capture all primary and secondary pole locations within one (1) meter accuracy and reconcile the existing EGIS connectivity model to these locations to support DMS and other advanced applications. 2. Evaluate and enhance EGIS integrations with SAP and OMS to provide a more flexible support model and data accuracy in preparation for DMS by following Consumers Energy's integration standards and where applicable leverage the IEC 61968 CIM standards for integration and common networked model exchanges. 3. Consolidate disparate electric distribution GIS databases into a common shared electric GIS platform. 4. Implement change management associated with business processes for managing all electric distribru on map records. 5. During execution of the project, all EGIS integrated processes will continue to operate without unplanned disruption.	01-Nov-18	(1.00)	20,923	0	0	5,367	12,683	2,874	8,028	0	0	5,983	590	1,455		
245	2019	GIS - GIS Connectivity Model Integration	BP Functionality	Software (Intangible)	The Grid Modernization (GM) Geographic Information System (GIS) Connectivity Model Integration project will design, develop, and implement a GIS extract using an industry standard to publish electric GIS data to the network, where it can be consumed by multiple applications. The extract will make the data available to the electric distribution historian as well as other applications in the future. The project provides value to the Company by: (1) leveraging the International Electrotechnical Commission (IEC) Common Information Model (CIM), which provides a basic ontology for data and data relationships and a standard repeatable approach that can be leveraged for sharing data about the electrical network across technical systems; (2) supporting an enterprise application integration framework, which keeps business process as simple as possible; (3) providing the ability to detect the physical location of metered assets, including circuit, substations and transformers; and (4) reducing the costs and time required to develop integrations in the future resulting from streamlined approaches and processes. Deliverables within the project scope include: (1) develop and implement the GIS extract, leveraging IEC CIM for data sharing; (2) leverage the enterprise service bus (ESB) as the high-level communications protocol for sharing CIM data across the enterprise; (3) build the integration to the electric distribution historian; and (4) develop a data and integration framework that can be leveraged for future projects. The alternative considered was to continue to implement point-to-point application-specific integrations to meet the requirements specific to a single project. This alternative was not chosen because it continues to increase the technical debt associated with the maintenance of non-standard interfaces. The preferred approach as outlined in this business case will result in: (1) standard, repeatable approaches for future integrations, (2) lower ongoing maintenance costs and, (3) less complexity.	30-Nov-20	(1.00)	61,248	0	0	10,506	11,604	39,138	0	0	0	0	0	0	0	0

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC			
246	2019	GM - Grid Communication Modernization	BP Functionality	Network	Verizon has announced that they will no longer offer their analog, multi-drop phone service as of February 28, 2015 and their SCADA Relay service after December 31, 2015. These services are an integral component of the SCADA communication infrastructure. The project scope includes: (1) a wired and wireless solution to replace Verizon's submitted services (analog multi-drop circuits and frame relay circuits), (2) Defined minimum and uptime requirements, (3) Sufficient site coverage.	15-Dec-18	(0.98)	(266,597)	(5,650)		0	(151,126)	(43,995)	(65,826)	19,794	5,650	0	0	0	1,233		
247	2019	ICS Resource Acquisition & Mgmt	BP Functionality	Software (Intangible)	The Incident Command System (ICS) Resource Acquisition & Management System project will implement a centralized, collaborative storm responder management system and restoration site that consolidates outage and restoration related documentation and streamlines storm response processes with ICS. The project adds value to the Company and its customers by: (1) aligning storm restoration standards and processes with standardized ICS framework, and (2) enhancing storm responder system navigation and communication resulting in overall enhanced experience. The scope of the project includes: (1) redesigning a streamlined storm restoration SharePoint site, and (2) upgrading the Catastrophic Crewing (CatCrew) application to incorporate resource management capabilities. Alternatives considered for this project include: (1) Utilize the Automated Resource Call Out System (ARCOS) solution for resource management. This alternative was not selected because of ongoing, incremental increase to O&M maintenance costs. Additionally, ARCOS does not provide on call schedule and storm responder profile functionality, which would have required software customization and increased project expenses. (2) Utilize the SharePoint solution for document management. This alternative was selected because it is the standard Company platform for document management.	21-Nov-18	(1.00)	927	0	0	0	0	927	0	0	0	0	0	0	0	0	0
248	2019	Integrated Resource Planning Project	BP Functionality	Software (Intangible)	Energy Policy requires Integrated Resource Plan be filed. Solution selected will enable scenario driven, long term, economic capacity planning, production cost modeling, evaluating future years for as is & potential changes in future energy landscape. A robust modeling tool will enable the Company to have agility and flexibility in developing scenarios that consider fleet capability, demand outlook, financial and economic considerations, economic, and environmental factors. Utilization of industry leading models for evaluating energy portfolio optimization will inform Company leadership decision making for our energy fleet investments. Advanced capacity/cost modeling to meet energy policy IRP requirements. Replace/upgrade obsolete tools being used to ensure latest simulation data, modeling capability with capacity expansion considerations for portfolio optimization. An IRP enabled the team to consider other industry tools. Solutions for this niche functionality are limited. The team relied on requirements evaluation, vendor reference calls, and pricing to determine a solution. One option to address this outdated technology was to upgrade to the Strategist vendor's new platform. This new software platform is extremely new and is not yet proven as a viable solution. This would be a risk to replacing the existing tool.	01-Dec-18	(1.00)	0	0	0	0	0	0	1,728	0	0	1,472	(0)	255			
249	2019	ITCP-Tawas	BP Functionality	New Computers/ Hardware	To construct a new field office that will assist consumers energy in meeting customer needs in the Tawas area. In phase 1 a trailer needs to setup by the end of Q1, 2019. To construct a new field office in Tawas area that will allow consumers energy to meet customer SLAs in the area. The purpose of this Business Case document is to capture IT related cost to a corporate construction project. As such, any value for the project is captured at the larger project level. The purpose of this Business Case document is to capture IT related cost to a corporate construction project. As such, any alternatives for the project are captured at the larger project level.	01-Nov-20	(1.00)	30,365	0	173	8,314	19,204	2,674	182,029	0	1,386	9,303	169,520	1,820			
250	2019	Iron Performance Manager	BP Functionality	Software (Intangible)	The Iron Performance Manager project is for the enablement of the Smart Energy Operations Center with the functionality to handle daily exceptions, monitor the health of the system, and enhance procedures of smart energy devices to improve long-term performance. The ability to monitor, report, and track performance of electric smart meters, gas communication modules, and other smart CE devices in the field is instrumental to the success of smart energy, included in the implementation is operational reporting, performance metrics, automated exception detection, intelligent prioritization of exceptions, historical trending of performance, and mapping and visualization. The value of completing the project is: Improvements to Time of Use billing by providing required meter data reporting; enables monitoring and near real time data analysis of meter health and system integration; and provides process and financial benefits. As part of our review process the alternative considered was the 'do nothing' approach and hire additional staff to handle the manual process which would be less efficient and more costly.	01-Jun-19	(0.18)	2,333,647	(65,816)	103,453	231,903	1,936,005	128,102	81,944	63,628	0	9,442	5,262	3,612			
251	2019	IWP Replacement	BP Functionality	Software (Intangible)	The IWP Replacement project will develop a new internet Web Portal ("IWP") platform that increases customer engagement and satisfaction among residential and business customers by allowing them to compare their current rate and bill amounts versus other rates and energy efficiency offerings. As part of the project review process, the alternative considered was continuing with the existing IWP. However, that tool lacks in rate comparison abilities and is no longer being enhanced by the vendor. If we do not replace the IWP, customers who move to the new Time of Use Billing rate will not have access to data that provides details on their peak time usage making it difficult to understand their billed charges. Additionally, it will not provide customers the opportunity to compare their usage and charges with other rates to determine the best rate for their needs. Customer benefit: Improved CII score of 5.0% and annual O&M savings of \$66,000.	01-Sep-19	(0.91)	2,953,503	0	0	525,102	2,028,264	400,138	21,548	0	0	8,184	10,097	3,267			
252	2019	Large Customer Rate Tool	BP Functionality	Software (Intangible)	The Large Customer Rate Tool project is for automated rate analysis through a rate tool for our large customers. The project will automate currently manual rate evaluations and thereby reduce the time it takes to run individual rate analysis. Included in the implementation are secondary, primary and special rates for business customers, process automation, and accuracy of rate definition for large customers. The value of completing the project is the automation of a currently manual intensive process that account managers utilize in working with large businesses to ensure they are on the best possible rate. As part of the project review process, the alternative considered was to do nothing and maintain the manual process of extracting data from SAP and entering into spreadsheets to run rate comparisons. This alternative was not selected since it would not support the gains in customer satisfaction or the efficiency of providing the data. Customer benefit: Improved functionality to better assist large business customers with evaluating different rate options.	01-Aug-20	(0.92)	0	0	0	0	0	55,437	0	0	8,356	40,089	6,993				
253	2019	Manage IVR	BP Functionality	Software (Intangible)	The Manage Interactive Voice Response (IVR) project is for continuous improvements to the Interactive Voice Response ("IVR") system which would include menu refinements around options and restructuring of the menus to drive more optimal option and queue selections. These improvements will enable the proper customer routing and appropriate call disposition. Included in the implementation is IVR menu selections/options refinement and appropriate customer choices for routing and queuing. The value of completing the project is enhanced customer experience and IVR containment. Failure to implement this project will result in longer wait times for customers and more manual intervention by Customer Service Representatives. Customer benefit: System updates to improve customer navigation of menu options.	01-Apr-19	(0.98)	(234)	0	0	0	0	(234)	156	0	0	133	0	23			
254	2019	Rate Case Implementation	BP Functionality	Software (Intangible)	The Rate Case Implementation project will modify SAP billing in accordance with MPSC filings. These enhancements include, but are not limited to, overall price changes and structure changes to the rates. Project scope includes the large electric and gas rate changes that occur annually and/or monthly. The benefit of this project is to ensure that customers are billed accurately based on the timing of the rate structure and price changes. An alternative considered for this effort was an offshore development model. However, the risk of billing inaccuracies and customer complaints with that model were deemed too high based on the various complexities and timing of the development and testing.	30-Sep-19	(0.89)	0	0	0	0	0	0	204,114	0	110,893	12,200	69,819	11,203			
255	2019	Service Bench for VAPS	BP Functionality	Software (Intangible)	The Service Bench for VAPS project will enable better operational management, customer satisfaction and set the BU up for future expansion. It will improve the margin for the entire program which can be utilized at the corporate level to either offset utility customers rate increases or shareholder return. ServiceBench is a third party software platform specifically designed for service companies to utilize its modular software platform for effortless scheduling and dispatch, field work management, automated claims processing and real-time updates partnered with powerful analytics. This will be the replacement of SuperCOW as a new Service Mgmt System. This software performs scheduling, dispatching, field work management, automated claims processing, and data analytics.	28-Feb-19	2.93	142,550	0	31,676	98,049	12,825	0	0	0	0	0	0	0	0		

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256	2019	Service/In-house Customer Service Management	BP Functionality	Software (Intangible)	This project will implement the CRM model to enable the recently defined Supply Chain ("SC") service delivery model to support SC optimization. This project will provide value to the Company and its customers through: (1) waste elimination, (2) cost savings, (3) maturing processes, and (4) movement to a sustainable service delivery model to continually improve the customer experience. This project specifically will enable the newly designed processes and service delivery model as it will: (1) enhance the delivery of SC services to both internal customers and suppliers; (2) include an integrated knowledge base and case management system, improving the customer experience by having one place to go for information and help; and (3) provide standard workflow and funnel the previously disparate intake channels for SC support into one solution. This Technology solution is the backbone of an SC Support Center and will enable: (1) better management of work; (2) more efficient and accurate response to questions; and (3) improved satisfaction of internal customers and external suppliers. See Jeffrey D. Tolonen's testimony for full project synopsis.	28-Feb-20	(1.00)	924,021	647,168	3,118	87,001	148,098	41,037	176,202	41,222	3,118	29,916	83,867	16,081
257	2019	Supply Chain - Sourcing Intelligence Tool	BP Functionality	Software (Intangible)	The Supply Chain Sourcing Intelligence Tool project will implement five modules of Power Advocates Energy Intelligence suite to provide a consolidated spend view and analysis for commodities and services across the Company. The project will add value to the Company and its customers by: (1) providing cost controls and visibility by enabling functionality to identify and control rogue spend; (2) increasing negotiation leverage and additional cost savings by capturing and aggregating spend for similar commodities and services across the Company; (3) identifying like commodities or services to ensure the Company is paying the same rates, leading to consistent spend and additional savings; and (4) replacing manual efforts to aggregate spend or control and identify rogue spend with an analysis tool, leading to process efficiencies and data quality improvements. The scope of the project includes: (1) purchasing and implementing the Power Advocates Energy Intelligence suite, (2) integrating the cloud-based solution with SAP data through automated data extractions, and (3) configuring secure single sign on (SSO) capabilities for users. Prior to implementing the project, alternatives considered include: (1) Continue efforts to manually investigate, identify, control, and aggregate cross-Company spend. This alternative was not selected because it does not provide the stated benefits of the sourcing intelligence tool and continues to perpetuate the problem of rogue spend; (2) Implement a spend intelligence tool from an alternate vendor. This alternative was not selected because the tool did not offer the insight to data and did not offer the same benefits to the Company; (3) Implement the cloud-based Energy Intelligence suite. This alternative was selected because it is part of a larger suite of tools for a well-rounded sourcing platform already in use by the Company. In addition, Spend Intelligence provides market and cost intelligence data through market analysis and cost comparisons to help the Company improve negotiations with vendors.	22-Aug-19	(1.00)	0	0	0	0	0	0	34,395	0	0	26,506	3,942	3,947
258	2019	Telecom Expense Mgmt (TEM) Strategic	BP Functionality	Software (Intangible)	The project will identify opportunities for cost savings in current Consumers Energy Telecommunications expenses, select and implement software to automate ongoing telecom expense management, and contract for a managed service provider to improve management of ongoing telecommunication expenses and payments. The project will utilize an optimization vendor to analyze current Consumers Energy telecommunication assets, expenses, and processes then implement telecommunication expense management software, which will provide invoice processing, invoice auditing, invoice dispute resolution, inventory management, contract validation, advanced reporting, mobile rate-plan optimization, and mobile device management. In addition the project will select and utilize a managed service provider to assist Consumers Energy in maintaining ongoing telecommunication expense management efficiency and cost-effectiveness. The value in completing this project is to ensure telecommunication assets, expenses, and processes are optimized to provide labor savings, reduce billing errors, and eliminate duplicate billings and unnecessary service charges, thereby reducing costs by an estimated \$154,000 annually. The savings will help offset the increases in IT operational O&M associated with new IT assets. Quality improvements will be implemented in the expense management process to improve efficiency, accuracy, and quality of future telecom expense results. The alternative approach considered was to develop a customized application in-house and internally analyze, develop, and implement process improvements by adding additional resources, which increases cost and does not effectively utilize the knowledge of industry experts.	01-Apr-19	0.76	348,922	109,694	0	118,003	70,978	50,247	108,395	22,467	0	11,815	67,302	6,811
259	2019	TEOS Application Currency	BP Functionality	Software (Intangible)	The TEOS Application Currency project is for smaller application upgrades for the following applications: gReddy, Inspection Manager, MANIS, Synergi Gas, Synergi Gas Customer Mgt Module, Synergi Gas ModelBuilder, Liptime TMAP, Align Fuel Management, Align Pipe Ops, Consumption Imports, Laboratory Information Mgmt System Chemistry, WineST, Modellogix, business center, Collector, CYME, DGN Out, FlowCal, Land Property Management, Maaload, MAOP, Next TLM. Included in the implementation is upgrading the application software, assessing any new functionality for value to the Company, making necessary configuration changes, testing any integrations to or from the application, testing the upgrade, and updating documentation related to integration changes. The value of completing the project is reducing the number of incidents associated with out of date software, increasing application stability, and allowing the Company to leverage new functionality available in upgrades. As part of our review process the alternative considered were stay at the current version of the applications. If this approach is selected, we risk applications not working after the application of operating system security patches, increased number of incidents, increased support costs, and inoperable systems for employees.	31-Dec-18	(0.87)	56,980	0	0	33,249	10,185	13,546	37,069	0	0	24,461	8,423	4,185
260	2019	Theft Analytics Phase 2	BP Functionality	Software (Intangible)	The Theft Analytics Phase 2 project will use the Smart Meter data to identify customers who may be stealing electricity or gas from the company. Queries and algorithms created for several use cases will be automated and scheduled to run daily to produce theft cases where there is a high confidence level that theft is occurring. Included in the implementation is creating and fine tuning these algorithms to identify customers that are potentially stealing electricity or gas from the company, automatically creating theft cases and/or work orders in SAP for customer accounts identified, and creating executive summary screens for managing theft. The value of completing the project is ensuring the safety of customers and employees, as tampering with meters creates a potential safety risk. The company can also recover the losses incurred from the theft. The potential to turn customers back into bill paying status will also increase future revenues for the Company. Two vendor products failed to deliver on a Theft Analytics solution; therefore the Company is choosing to implement an in-house solution. Customer benefit: Reduced uncollectible expenses with a projected payback of 5-to-1.	01-Jun-19	5.00	773,573	162	(481)	366,013	218,364	189,515	66,369	0	481	33,010	25,323	7,555
261	2019	Wholesale Contractual Settlements	BP Functionality	Software (Intangible)	Wholesale Contractual Settlements currently uses a combination of Excel and Fortran programs to produce Remittance Statements to suppliers and NUGs (Non-Utility Generators). 70 contracts in total; 8 are Fortran which are 20+ years old and are for more complex contracts (more details, variables, scenarios) and to the minute reads for the bill month). These contracts manage \$960M annually. * The project will: (1) Create and implement a contractual settlements solution that will leverage the PCI EA and data warehouse for MISO settlements; (2) Evaluate best platform for contracts which are currently in Excel and Fortran. (3) Replace Fortran and Excel programs and process; robust analytic capability and reporting; improve remittance process; integration with Managed Meter Solution.	15-Sep-18	(0.70)	(8,620)	0	0	0	(8,620)	0	8,620	0	0	0	8,620	0

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
262	2019	Workspace Desktop Edition Voice	BP Functionality	Software (Intangible)	The Workspace Desktop Edition Voice project will implement the Genevys Workspace Desktop Edition for Voice for the Customer Contact Centers. Included in the implementation is implementing Genevys Workspace Desktop Edition (WDE) for Voice and implementing SAP Workplace Plugin from Soft Phone. The value of completing the project is implementing Genevys WDE for Voice provides the framework for Contact Center CSRs to handle more than voice contacts such as Web Chat, SMS text or co-browsing. This allows customers the ability to engage with Consumers Energy in a manner of their choice.	01-Nov-18	(0.96)	5,304	0	0	803	0	4,501	4,674	0	0	1,658	2,187	829
263	2019	2019 SubTotal BP Functionality	BP Functionality					19,925,656	656,154	3,811,877	2,391,730	10,708,037	2,357,858	2,449,387	163,076	118,781	751,436	1,169,200	246,894
264	2019	2019 Total						56,370,375	2,748,789	20,765,611	6,473,446	21,883,321	4,499,209	13,182,466	620,039	314,032	3,033,556	8,127,768	1,087,071
265	2020	Data Center 2.0	Upgrades & Replacements (Enterprise)	New Computers / Hardware	The Data Center 2.0 project will mitigate the current significant risks (location risks, capacity risks, technology risks) with the Consumers Energy IT Disaster Recovery (DR) capabilities. The Company is co-locating to a Backup Recovery Center (BRC) at a vendor facility. A co-location facility is a data center facility in which a business can rent space for servers and other computing hardware. Typically, a co-location vendor provides the building, cooling, power, and physical security, while the business provides the servers, storage, and other computing and networking equipment. Project scope includes both Information Technology and Operational Technology data centers. The project will migrate the BRC to a vendor (Switch) colocation data center, expand DR system capacity to support 100% production load requirements, expand DR capabilities to all applications determined by the business as needed DR, and enable cloud capabilities to migrate non-production workloads to the cloud to allow for production disaster recovery on non-production infrastructure. At the end of the migration to Switch, the compute capacity at Switch will be greater than or equal to the compute capacity at Paranal. Some of the capacity will be on newly purchased hardware and some will be on hardware shifted from the Battle Creek Data Center. Currently, there are no plans to put compute for production DR in the cloud. The Company performed an analysis of two primary alternatives to expand DR capabilities and address constraints/risks: 1) at the current BRC location, and 2) at a third-party co-location facility. Based on the analysis, the Company decided to implement the third-party co-location alternative for the BRC. The value this brings the Company is that it mitigates risks to the corporation in the event of a disaster by significantly enhancing the disaster recovery capabilities, DR testing capabilities, and scalability constraints at the current data center locations.	01-Nov-20	(1.00)	356,788	0	0	38,797	297,211	20,780	1,256,310	46,200	0	382,800	756,360	70,950
266	2020	HR Support Pack and BSI Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will update the SAP system with HR Support Packs that are released annually by SAP to comply with HR and tax changes. This project creates value for the Company by (1) ensuring that it is in compliance with new financial rules and regulations, and (2) ensuring that it can calculate and distribute payroll. The scope of this project is to add SAP HR corrections to ensure proper reporting of financial information by the Company. As this is an upgrade of an existing system, the alternative considered was to delay the upgrade. This alternative was not chosen due to the risk of not complying with financial rules and regulations.	01-Dec-20	(0.94)	0	0	0	0	0	0	680,942	57,737	0	99,000	514,800	9,405
267	2020	Oracle Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Oracle Server Database Upgrade project will upgrade Oracle server databases to the next version and supportability across business portfolios. This project will create value for the Company and its customers by (1) reducing the risk of system failure and the resulting impacts to business partners and customers, and (2) ensuring that systems are secure, supported, and have the latest features and functionality. The scope of this project includes upgrading to the next version of Oracle across all business systems. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and supportability of the system.	30-Apr-21	(1.00)	252,861	0	103,920	96,992	0	51,949	1,052,251	0	0	960,960	0	91,291
268	2020	Redwood Cronace Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Redwood Cronace Upgrade project will upgrade the Redwood Cronace batch job scheduling software. The scope of this project includes (1) upgrading Redwood Cronace and the associated database to the latest release of the software and (2) upgrading the operating system. This project will create value for Consumers Energy and its customers by providing a supported platform for billing, payment, payroll, and financial processing. The alternative considered was to not upgrade and agree to the vendor's terms. This alternative was not selected because those terms specify that the system would be unsupported. Consumers Energy selected the upgrade to resolve production issues that require vendor assistance.	01-Nov-20	(1.00)	814,629	69,280	207,840	259,800	138,560	139,149	128,783	0	0	115,500	0	13,283
269	2020	SharePoint Upgrade Project	Upgrades & Replacements (Enterprise)	Software (Intangible)	The SharePoint Upgrade project will implement Microsoft 365 Cloud-based hosting, which extends and enhances the existing SharePoint 2010 platform by providing additional functionalities and an enhanced user experience. This project will create value for the Company and its customers by: (1) maintaining system currency and security, (2) supporting mobile device browsing, (3) maintaining a vendor-supported platform, and (4) increasing audit capabilities. Current vendor support ends in October of 2020. The project scope includes: (1) creating a new SharePoint environment in the next current version, and (2) migrating all applications and data in the existing SharePoint 2010 environment to the new one. Alternatives considered include: (1) fund the full project, which would upgrade all 2010 SharePoint sites; (2) fund 50% of the project, which only would include business-critical SharePoint sites and non-complex user sites; and (3) fund 25% of the project, which only would include non-complex user sites. Alternatives 2 and 3 were not selected because of the increased cost to maintain two SharePoint environments, and the risk of cyber security, stability, and performance of the unsupported version. The selected alternative is to fund the full project to mitigate these risks and minimize application support costs.	01-Nov-20	(1.00)	1,316,320	0	762,080	0	564,240	0	180,675	0	0	165,000	0	15,675
270	2020	SiteCore Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will refresh all components of the website hosting, delivery, search, and analytics applications to add new features and improve search capabilities. Sitecore is the content manager for the ConsumersEnergy.com website. The project will add value by providing these four benefits: (1) maintains currency with the web hosting application version; (2) allows business users to make use of new features and functionality; (3) neutralizes continually evolving cyber threats; and (4) continuously improves customer experience using ConsumersEnergy.com. The project scope includes: (1) upgrading the Sitecore content management software to include content hosting and delivery allowing the use of new features and functionality; (2) upgrading the Coveo software, which will allow for more intuitive search results and provide suggestions or recommendations based on the customers' search text; (3) upgrading the Mongo database, which provides the analytics functionality within Sitecore to better understand customer actions and errors on the web page; and (4) developing analytics across customers' web and contact center interactions, allowing the company to connect contact center call data with corresponding web data for improved customer experience. Alternatives considered but deemed not to be viable include: (1) implementing a two-year upgrade cycle, which would not position the Company to be able to keep up with constantly changing cyber threats, or gain value from the rapidly changing feature set being developed by the vendor; and (2) purchasing an existing cloud solution, which is cost prohibitive at this time. Annually upgrading the existing Sitecore platform provides the functionality and stability needed while meeting financial requirements and mitigating cyber security risks.	31-Dec-19	(1.00)	0	0	0	0	0	505,398	0	0	330,132	145,200	30,066	

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC			
271	2020	SQL Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project, which supports critical applications such as the meter read collection systems and the customer contact center applications, will upgrade all SQL Server 2000-2014 instances to the latest version. This project will create value for the Company and its customers by: (1) reducing the risk of system failure and the resulting impacts to business partners and customers; and (2) ensuring that systems are secure, supported, and have the latest features and functionality. Project scope includes: (1) upgrades to all SQL Server 2000-2014 instances currently in use and not identified as part of another portfolio upgrade project, legal hold, or pending system retirement (approximately 400 instances); (2) installation and/or distribution of new SQL Server Client Tool software packages to affected workstations and application servers; (3) new Nimbus virtual machine templates for the new SQL Server release; and (4) technical database support to IT portfolio, business partners, and vendors during all project phases. The alternative considered was to migrate to Azure Cloud. As part of this option, the organization would obtain three years of extended support through Microsoft on SQL Server versions 2005-2008. This option was not selected because extended support would not be provided 2012 and 2014 versions, and the organization would not reap the benefits of new features offered through the upgrade. The Company decided in favor of the upgrade to avoid these issues and ensure system stability.	30-Apr-22	(1.00)	632,388	0	101,930	346,400	0	182,068	1,170,180	0	0	1,069,300	0	100,980			
272	2020	Windows 10 Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Windows 10 Upgrade project will upgrade Consumers Energy corporate computing devices from Windows 7 to Windows 10. The project will convert 4,500 Windows 7 computing devices to Windows 10 with the assistance from a partnered vendor at various Consumers Energy locations for corporate, gas, and electric employees. Project tasks include assessing current tools and image process to ensure industry standards are being applied, communicating with employees on project and expectations, scheduling and completing device operating system upgrades, verifying that devices were upgraded with no impact to employees' data and performance, and providing on-site post support at Consumers Energy locations. The alternative is not to move forward with this project and pay additional support fees for Microsoft to continue supporting Windows 7 for up to four years. This will allow Consumers Energy time to replace the devices through our asset management refresh projects. Work station assets are currently replaced on a four-year cycle. This alternative was not pursued due to the additional support fees and risk to business processes.	01-Oct-19	(0.96)	0	0	0	0	0	0	232,340	0	0	22,285	207,958	2,317			
273	2020	SubTotal Upgrades & Replacements (Enterprise)	Upgrades & Replacements (Enterprise)					3,372,985	65,280	1,177,760	741,989	990,011	393,945	5,206,879	103,957	0	3,144,877	1,624,298	333,767			
274	2020	Application Currency and Enhancements - ACDA - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency, and Enhancements Analytics, Cloud, DevOps, and Architecture ("ACDA") initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the ACDA Portfolio. The value of regular upgrades to applications in the ACDA Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the ACDA Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the ACDA Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	220,469	0	220,469	0	101,282	0	0	101,282	0	0	101,282	0	0		
275	2020	Application Currency and Enhancements - ACDA - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Analytics, Cloud, DevOps, and Architecture ("ACDA") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade the applications that support the ACDA Portfolio. The value of regular upgrades to applications in the ACDA Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for ACDA Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the ACDA Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	0	0	0	0	0	0	101,667	0	0	101,667	0	0	101,667	0	0

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
276	2020	Application Currency and Enhancements - CE&O - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Customer Experience and Operations ("CE&O") initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade the applications that support the CE&O portfolio. The value of regular upgrades to applications in the CE&O Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for CE&O. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Customer Experience and Operations Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	997,632	0	0	692,800	0	304,832	144,340	0	0	132,000	0	0	12,340
277	2020	Application Currency and Enhancements - CE&O - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Customer Experience and Operations ("CE&O") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Customer Experience and Operations Portfolio. The value of regular upgrades to applications in the Customer Experience and Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the CE&O Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the CE&O Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	0	0	0	0	0	119,988	0	0	102,168	0	0	17,820	
278	2020	Application Currency and Enhancements - Corporate Services - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade applications that support the Corporate Services Portfolio. The value of regular upgrades to applications in the Corporate Services Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Corporate Services. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as Finance, HR, Learning & Development, Legal, Governmental, Regulatory and Public Affairs, Corporate Security, and Strategy and IT Governance. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-20	(1.00)	248,022	0	10,392	0	0	237,630	81,736	0	0	75,900	0	5,836	
279	2020	Application Currency and Enhancements - Corporate Services - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Corporate Services Portfolio. The value of regular upgrades to applications in the Corporate Services Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Corporate Services. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as Finance, Human Resources ("HR"), Learning & Development, Legal, Governmental, Regulatory and Public Affairs, Corporate Security, Strategy and IT Governance. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	0	0	0	0	0	265,320	0	0	209,880	0	55,440		

Michigan Public Service Commission
Consumers Energy Company

Synopses Containing Descriptions, Scope, Benefits, Implementation Dates and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses
For the years 2018, 2019, 2020, and 2021

Information Technology Department

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
280	2020	Application Currency and Enhancements - Infrastructure Applications and Operations - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements - Infrastructure Applications and Operations ("IAO") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the IAO Portfolio. The value of regular upgrades to applications in the IAO Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the IAO Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the IAO Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-20	(1.00)	0	0	0	0	0	0	328,137	30,206	0	271,859	0	26,071
281	2020	Application Currency and Enhancements - Operational Technology - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade the following applications that support the OT Portfolio. The value of regular upgrades to applications in the OT Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for OT. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for OT. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-20	(1.00)	69,280	0	69,280	0	0	0	38,704	0	0	35,346	0	3,358
282	2020	Application Currency and Enhancements - Operations - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Operations Initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the Operations Portfolio. The value of regular upgrades to applications in the Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Operations area. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	556,180	0	6,928	381,040	0	168,212	43,362	0	0	39,600	0	3,762
283	2020	Application Currency and Enhancements - Operations - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements - Operations O&M initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Operations Portfolio. The value of regular upgrades to applications in the Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Operations Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	0	0	0	0	0	185,239	0	17,820	136,620	15,840	14,959	

Michigan Public Service Commission
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For the years 2018, 2019, 2020, and 2021

Information Technology Department

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC			
284	2020	Application Currency and Enhancements - TEOS - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Transmission, Engineering and Operations ("TEOS") Support initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the TEOS Portfolio. The value of regular upgrades to applications in the TEOS Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the TEOS Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the TEOS portfolio areas. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	304,832	0	17,320	187,056	0	100,456	4,417	3,673	0	0	0	0	0	0	744
285	2020	Application Currency and Enhancements - TEOS - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements for Transmission, Engineering, and Operational Support ("TEOS") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the TEOS Portfolio. The value of regular upgrades to applications in the TEOS Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the TEOS Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the TEOS Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-20	(1.00)	0	0	0	0	0	161,914	0	0	147,866	0	0	0	0	14,047	
286	2020	Electric Geographic Information System (GIS) Platform Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will upgrade the GIS platform applications, servers, application hosting servers, and databases to the next major versions to remediate the outdated platform technology. The project adds value by: (1) remediating GIS, Windows, and Citrix technology obsolescence, creating a stable and sustainable foundational platform for critical business functions; (2) improving platform resilience by remediating current single-point-of-failure areas; (3) sharing technology resources (servers, databases, etc.) across gas and electric platforms to optimize them, yet still ensuring the ability to meet diverse business area functional needs; and (4) adding new capabilities for spatial analytics, real-time visualization of data, and three-dimensional infrastructure network traceability. In addition, the GIS platform remediation work establishes the foundation necessary for the full implementation of the Electric Distribution Asset Management Program. Project scope includes: (1) building GIS platform servers, software, and data; (2) expanding GIS platform capabilities; (3) re-architecting current single-point-of-failure areas; (4) designing and developing new architecture; (5) upgrading database technology; (6) re-platforming the application hosting platform (Citrix); and (7) upgrading and reconfiguring applications that utilize the GIS platform. The alternatives considered for the project were to: (1) delay implementation, pay a premium for extended Windows support, and incur cyber security and business continuity risk for the application hosting platform and the GIS platform; (2) upgrade the platform to the Utility Network as a "big bang" approach without the incremental GIS version upgrade; and (3) perform an iterative platform upgrade to the next GIS version with application platform hosting upgrade, server replacements, and database upgrade, and shift the Utility Network upgrade to a future year. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-21	(1.00)	1,235,000	0	550,000	175,000	350,000	160,000	287,125	0	55,000	75,000	150,000	7,125			
287	2020	Genesys Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Genesys Upgrade project is to upgrade and enhance Genesys Software features and functionality with improved call routing, reporting, and recording contacts. The scope will include all Genesys applications: Work Force Management, Speechminer recording application, interactive insights reporting, and Pulse contact monitoring. The upgrade ensures that the Company is on a Genesys version supported by the vendor and upgrade provides improved functionality across all applications. The improved functionality offers convenience to customers by letting them schedule interactions and callbacks with customer service over any channel or device. It supports transfer, supervise, and conference co-browse and chat sessions. Additionally, it will enhance social media and text functionality for customers. The alternative is to delay the upgrade to 2020 or beyond and risk experiencing system performance issues and support problems.	15-Apr-20	(1.00)	74,822	0	0	24,248	27,712	22,862	16,124	0	0	10,063	4,741	1,320			
288	2020	MISO Market User Interface Changes	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Midcontinent Independent System Operator (MISO) Market User Interface (MUI) Changes project will update the Power Costs Inc. (PCI) software used for market interaction and settlements with MISO. These changes are required as a result of significant MISO redesign of the MUI to modernize technology and improve cyber security. The project will provide benefits to the Company by meeting required changes to the MUI initiated by MISO, modernizing technology and improving cyber security. This is a significant change that requires updates to the PCI software used for market interaction and settlements with MISO. These updates will be applied incrementally as MISO and PCI release the updates. Each update requires installation in the test environment, regression testing, and user acceptance testing prior to implementation in production. Alternatives considered in order to continue to interact with MISO include: (1) replacing PCI with a new product, or (2) updating the PCI solution. The alternative to replace PCI would require significant investment in research, product evaluation, people resources, and financial investment. Due to the schedule for updates provided by MISO, and the significant footprint of the PCI solution, the alternative to upgrade was selected.	01-May-21	(1.00)	275,000	185,000	0	25,000	50,000	15,000	16,000	0	0	10,000	5,000	1,000			

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
289	2020	OpenWay 3G to 4G Enablement	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will upgrade OpenWay Collection Engine (OWCE) software to the next version to enable the support of the 4G network for electric meters as the telecommunications carrier decommissions the 3G network. The project will add value by enabling the Company to maintain current capabilities for 680,000 electric meters and customers. The project scope includes: (1) implementation of the next version of OpenWay Collection Engine (OWCE), (2) equipping meters with communications boards, and (3) implementing active directory authentication for OWCE. The alternative considered was to defer the timing of the OpenWay Collection Engine (OWCE) upgrade. This alternative was not selected due to the retirement of the 3G Verizon network driving the requirement for Consumers Energy stay current with 4G smart meter technology. Failure to upgrade the network would result in the inability for smart meters to communicate over the network, and Consumers Energy would have to re-implement manual meter reading efforts. Upgrading to the next version of OpenWay Collection Engine was chosen because reverting to manual meter reading would increase costs to both the Company and its customers.	01-Jan-20	(1.00)	78,313	0	0	49,367	0	28,752	64,176	0	0	13,075	50,000	1,101	
290	2020	OSoft Pi Historian Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The OSOft Pi Historian Upgrade project will maintain application and hardware platform currency for the OSOft Pi system. The project will create value for the Company and its customers by: (1) reducing security vulnerabilities, (2) improving efficiencies and increasing synergies between environments, and (3) enabling business partners to leverage new features that the vendor includes with major releases. The project scope includes the implementation of: (1) OSOft Meter Operational Data Management, (2) OSOft Electric Distribution Historian, (3) OSOft Generation, (4) OSOft Low Voltage Distribution/High Voltage Distribution, (5) OSOft Gas Automated Meter Read, and (6) data archiving for analytics purposes. The alternative considered was to delay the upgrade until a future year. The hardware has not been refreshed since 2014. Continuing to delay the upgrade could lead to the inability to apply security and system patch sets. A timely upgrade will allow the Company to sustain system performance and supportability. Other vendors were not considered due to a longstanding Enterprise Agreement with OSOft, Inc.	01-Oct-20	(1.00)	2,485,431	0	1,262,282	666,820	199,180	357,149	131,531	0	0	120,120	0	11,411	
291	2020	Replacement of Testing Software for the Electric Meter Technology Center	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Replacement of Testing Software for the Electric Meter Technology Center (MTC) project will replace the existing Winboard software for MTC test boards and will implement four multi-position test boards, increasing the output of tested meters and ensuring future software stability. The project will add value by: (1) increasing meter testing output from approximately 150 to roughly 225 meters a day, enabling Company employees to work more efficiently, (2) implementing a multi-position test board that integrates meter qualification, meter demand, and meter accuracy, and (3) providing increasing Microsoft support costs from \$50.00 per computer to \$100.00 per computer in 2021. An additional benefit specific to the new software is the creation of a centralized database where all the boards access one database, Advance Metering Infrastructure (AMI) device support, power quality capture, and web reports. The project scope includes: (1) replacing the Winboard software with a version compatible with the Company's current Windows operation system version, (2) writing new interfaces for test boards to SAP, and (3) replacing four existing test boards with multi-position test boards. Alternatives considered include: (1) purchase new software, rebuild interfaces, and replace four single-position test boards with two three-position test boards; (2) purchase new software and rebuild interfaces, but do not replace the four single position test boards; and (3) do not complete the upgrade. The second and third options were not chosen because they do not allow us to reap the benefits of the new three-position test boards. The first option was chosen because it allows the Company to keep old test boards for additional capacity and creates more efficiencies, which result in a higher output of tested meters.	31-Dec-20	(1.00)	474,215	143,800	297,084	0	0	33,331	312,560	0	0	224,000	34,280	64,280	
292	2020	Time Entry and Expense Reports - Flash Remediation	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will replace Adobe Flash with HTML5 technology. Time entry and approvals, and expense report approvals currently leverage Adobe Flash technology. By the end of 2020, Microsoft will remove the ability to run Adobe Flash in Microsoft Internet Explorer. The value of the project includes: (1) reducing security vulnerabilities that exist in Adobe Flash, (2) ensuring employees and managers can enter and approve time so employees are paid on time, and (3) ensuring managers can approve expense requests for reimbursement. The scope of the project includes: (1) replacing Adobe Flash technology with HTML5, and (2) eliminating security risks associated with Adobe Flash. Alternatives considered for this project were to: (1) use SAP Fiori technology that provides mobile capabilities and reduces license and support costs with the external vendor, and (2) implement other software applications such as Concur. The first alternative would reduce functionality and impose a major training impact on more than 1,000 employees. The second alternative would require: (1) integration with SAP, (2) retraining of all employees, (3) additional licensing and support costs, and (4) custom development of rules and redevelopment of enhancements. The option to replace the current Adobe Flash solution with HTML5 technology was selected because it is the least costly option with little to no impact to the Company's employees. This solution does not require retraining or the development of new custom time entry rules or enhancements.	22-Oct-20	(1.00)	0	0	0	0	0	202,620	0	0	89,100	105,600	7,520		
293	2020	SubTotal Upgrades & Replacements (Business Partner)	Upgrades & Replacements (Business Partner)					7,015,203	328,800	2,213,286	2,422,000	626,892	1,428,225	2,606,442	33,879	72,820	1,895,547	355,461	248,735	
294	2020	AccessNOW	Security	Software (Intangible)	The AccessNow project is for the implementation of configurable Identity and Access Management functionality and best practices with enforced compliance. This includes enterprise-level foundation architecture, technology, and end-to-end processes and controls. The project will deliver an integrated and synced enterprise system of record data. Additionally, processes will be fully automated for user self-service and access lifecycle management. Included in the implementation is the continuation of building on the One Identity Manager application against the Identity and Access Management Program (IDAP) roadmap. As Phase 1 efforts concluded in mid-2017, 2018's main objective is to implement Phase 2, which includes designing and building a web portal with a request catalog for employees' access needs. The intent is to replace the legacy Computer Access and Authorization Request (CAAR) system. These requests will be driven by automated workflows and the tool will provide automatic Active Directory provisioning. Additional features aimed at making this process more user friendly are also included. The value of completing the project is automated workflows with escalations that will reduce wait times for end users to get the access they need for their jobs. As part of our review process the alternative considered was using manual processes but this strategy was deemed too costly and inefficient.	01-Aug-19	(0.96)	762,080	0	0	136,135	346,400	279,545	66,907	0	0	61,710	0	5,197	
295	2020	ARP - Cyber Security	Security	Network	This project will replace cyber security infrastructure to support increasing system and application demands, and to prevent system failures and service interruptions. This program includes projects that bring value to the Company by maintaining the currency of the security infrastructure and core enterprise software. These are used to ensure the stability of technology for business operations. This project will support continued systems stability. The scope encompasses: (1) evaluation, validation, and replacement of cyber security firewalls and servers, and (2) asset and application upgrades. As part of the review process, the alternative considered was not to upgrade or replace assets as required, which is likely to introduce security risks, system vulnerabilities, and out-of-warranty repair costs.	31-Dec-20	(1.00)	256,680	0	171,120	85,560	0	0	0	0	0	0	0	0	
296	2020	Cloud Access Security Broker Expansion	Security	Software (Intangible)	The Cloud Access Security Broker (CASB) Expansion project will integrate a security access solution to additional Software As-A-Service (SaaS) applications that the Company is using, which help address cloud service risks, enforce security policies, and assist in compliance to regulations. Completion of this project will provide value to the Company through: (1) increasing its security posture for cloud applications by mitigating cloud service risks, (2) enforcing security policies, and (3) assisting in compliance to regulations. The project will plan to move the top five high-risk applications to the Cloud Access Security Broker (CASB) solution. The scope of this project encompasses continued integration of the Cloud Access Security Broker (CASB) solution into the Company's technical network. The project will aim to integrate CASB with five additional high risk applications. Alternatives considered include: (1) remain at the current state and not continue to integrate the Cloud Access Security Broker (CASB) solution into the company's cloud environments, or (2) continue with additional integrations with Cloud Access Security Broker (CASB) solution the Company has invested in. The preferred option is to continue integrations to increase the Company's security posture.	31-Aug-20	(1.00)	83,136	0	0	83,136	0	3,960	0	0	3,960	0	0	0	0

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC		
297	2020	CRISP	Security	New Computers / Hardware	This project will implement the continuous readiness information security program, which is a cyber security tool coordinated by the Department of Homeland Security ("DHS"). The tool watches network traffic coming into and leaving our environment and attempts to identify attacks from highly skilled actors, typically associated with other nations. Details regarding identified attacks are then anonymized and sent to DHS for analysis. CRISP does not identify cyber crime attacks or commodity malware. CRISP focuses on attackers who are motivated to impact critical infrastructure. Nearly all major utilities in the United States have or are in the process of implementing CRISP. Completion of this project will provide value to the Company by: (1) Receiving alerts from DHS based upon classified intelligence regarding who is attacking the Company; (2) Analyzing alerts to determine if the Company was impacted; (3) Receiving alerts about attacks at other utilities, which will enable the Company to proactively mitigate potential future attacks; and (4) Enabling the company to share threat intelligence with other utilities for the benefit of our nation's critical infrastructure. The scope of this project encompasses: (1) forming an agreement with DHS; (2) procuring a network sensor; and (3) working through architecture designs, deployment, and testing of the network sensor. As part of the review process the alternative considered was not to implement CRISP, which limits the Company's ability to respond to future advanced threats and share information for the benefit of the nation.	31-Jul-20	(1.00)	116,390	0	0	0	86,600	0	29,790	115,500	115,500	0	0	0	0	0
298	2020	Cyber Security Enhancements	Security	Software (Intangible)	This project will: (1) fund emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape, and (2) support initiatives that may not meet the criteria for a formal project. Requests for this funding are governed by project management and the security governance board, which is comprised of representatives from each area of Security and meets monthly to evaluate and prioritize the work. The board assesses requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in scope are small changes and functionality improvements to existing software application investments for Security. The enhancement requests are fulfilled to provide functionality for areas such as security program management, cyber security incident response, corporate physical security, compliance, privacy and risk management, and cyber security engineering and standards. As part of the review process, the alternative considered was not to upgrade or enhance software or hardware, which could reduce the Company's ability to respond to future threats and vulnerabilities.	31-Dec-20	(1.00)	0	0	0	0	0	0	118,800	0	0	118,800	0	0	0	
299	2020	Email Protection	Security	Software (Intangible)	This project will implement an enterprise-wide email filtering toolset to ensure protection against current cyber security risks, including malware and non-malware threats, email fraud, and ransomware. Completion of this project will provide value to the Company by: (1) evaluating and procuring a leading email filtering toolset; (2) configuring the toolset to help eliminate human error; and (3) implementing the toolset enterprise-wide for company email. The scope of this project encompasses protection for both incoming and outbound emails. The alternative considered was to continue with focused cyber security and phishing training programs for employees. Security awareness training has reached peak effectiveness. The Company is choosing to implement a technical solution to further mitigate the risk of phishing attacks caused by human error.	12-Sep-20	(1.00)	0	0	0	0	0	0	108,504	92,664	0	15,840	0	0	0	
300	2020	Enterprise Incident Response Toolset	Security	Software (Intangible)	The Enterprise Incident Response Toolset project is for implementing an incident management platform that enables the Cyber Security Incident Response Team (CSIRT) to automate alert triage, which is currently a manual process through technical playbooks, to bring more capabilities to track and manage incidents, and to integrate threat intelligence with security orchestration. Included in the implementation is building five or six playbooks, providing basic training to the CSIRT, instituting firewall rules, setting up internet and security information and event management access, placing an FS load balancer in front of the two Demisto servers, and deploying two Demisto servers that will operate in Disaster Recovery mode. The value of completing the project will be improving response time to events and limiting the risk to Consumers Energy from a cyber security incident. As part of our review process other automation tools (i.e. Swimlane, Phantom, Fire Eye) were vetted via on-site demonstrations or webinars. However, this was determined to be the best solution to align with the Consumers Energy requirements.	22-Jul-20	(1.00)	138,560	0	0	138,560	0	0	6,600	0	0	6,600	0	0	0	
301	2020	Fusion Center	Security	Network	This project will procure a location to combine the Security Command Center and the Cyber Security Incident Response Team ("CSIRT"). This project will focus on constructing a physical office space to facilitate these two teams working in a collaborative fashion. This project will add value by: (1) co-locating physical and cyber team members for process efficiencies and better integration of tools in the event of a security incident that requires both teams' collaboration; (2) having a dedicated emergency operations area for use in the event of a security incident (cyber or physical or both) providing the tools needed to keep stakeholders informed; (3) evaluating and potentially implementing a shared intelligence capability across both areas; and (4) further integrating existing processes and workflows. The project scope encompasses building an integrated Security Command Center across physical and cyber security domains that takes into consideration physical space, technology, process integration, analytics, and intelligence capabilities without changing existing organizational structures. As part of the review process, the alternative considered was to continue to run a separate Security Command Center for physical security incident response and a separate Cyber Security Incident Response Center. The two centers would continue to run separate, non-integrated systems and tools. This option was not chosen as it does not provide the integrated space and tools needed for physical and cyber security teams to collaborate to resolve critical incidents that pose both a physical and cyber threat.	01-Dec-21	(1.00)	655,005	0	285,200	242,420	0	127,385	101,501	0	0	72,600	0	0	28,901	
302	2020	Fusion Center Technologies	Security	New Computers / Hardware	This project will further integrate the physical and cyber security teams through the implementation of new technologies. This project will evaluate, replace, and implement new capabilities for monitoring physical sites and technology assets, significantly improving the Company's quality and timeliness of detection and response to both physical and cyber security incidents. Completion of this project will provide value to the Company through: (1) implementing a new toolset to move MITRE Adversarial Tactics, Techniques, & Common Knowledge ("ATT&CK") coverage from 30% to 80% (MITRE ATT&CK framework is an industry-leading framework to detect malicious activity); (2) replacing the vulnerability management platform, Qualys, which has several capability gaps estimated to be remedied at a significant cost; and (3) evaluating and implementing a biometric toolset, such as facial recognition, to eliminate an unauthorized individual to follow an authorized individual into a facility and to proactively alert of unwanted or unknown personnel at Company facilities. The scope of this project encompasses: (1) all endpoints and servers, and all facilities that house employees; (2) defining system scalability requirements; (3) designing new platform architecture; and (4) purchasing and implementing hardware and software once alternatives are reviewed. The alternative considered was to remain on current toolsets and not implement additional physical security countermeasures, impacting the quality and timeliness of detection and response to both physical and cyber security incidents. The Company is choosing to replace and implement new capabilities to provide further efficiencies and capabilities.	31-Jul-21	(1.00)	1,039,200	692,800	0	242,480	0	103,920	72,600	0	0	72,600	0	0	0	0

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303	2020	Operational Security Assessment	Security	Software (Intangible)	The Operational Security Assessment project will be the implementation of an automate tracking tool to plan audits and inspections, monitor progress in real-time and track corrective actions in one central and secure location. The implementation of the identified product will help to decrease vulnerabilities that could threaten Company assets. This project will establish: (1) a centralized location to securely manage and store assessments; (2) implement an automated tracking system; (3) prioritize deficiencies; and (4) track completion with an interface to the SAP work order system. This will decrease the manual effort needed by the physical assessors to complete their day to day work. With the automation of tracking audits and inspections the physical security team can: (1) increase their audits by 20%; (2) reduce the likelihood of impactful security events to customers; and (3) increase efficiency, which in turn reduces potential operating costs for the Company. An additional feature of the identified solution is the ability for executive leadership to track the assessment scores in real-time and identify long trends enabling proactive prevention. As part of the review process the alternative considered was to continue using the manual process of tracking in an Excel spreadsheet and entering tickets in the Request Management Center (RMC) SharePoint site but this strategy was deemed too costly and inefficient.	13-Nov-19	(1.00)	207,840	0	0	207,840	0	0	16,500	0	0	0	0	0	0		
304	2020	Passive Vulnerability Assessment	Security	New Computers / Hardware	The Passive Vulnerability Assessment project will deploy a tool to passively provide vulnerability assessment services to Control Networks. Active vulnerability scanning can negatively impact the reliability of a network. To overcome that risk, passive vulnerability scanning acquires asset and vulnerability data from offline networks. A zero cost proof of concept will determine capabilities and outline an approach to use either an on-site or off-site deployment strategy. The value of completing the project is to obtain more accurate and timely lists of assets, improve the ability to make data-driven operational and security decisions, complete an assessment of existing vulnerabilities in the environment without the risk of negatively impacting the reliability of Consumers Energy electric generation and energy delivery systems, and increase incident response capability, speed, and effectiveness. As part of our review process the alternative considered was implementing a traditional vulnerability scanner to interrogate all systems on the network. However, this was determined to be the best solution to fit the Company's requirements.	31-Jan-20	(1.00)	103,920	0	0	103,920	0	0	6,600	0	0	0	6,600	0	0		
305	2020	Physical Security Hardware Refresh	Security	Network	This project will enhance or replace physical security assets to provide improved visibility into potential security concerns and resolve incidents. The Company has several thousand physical security asset devices currently in use, including security cameras, motion detectors, intrusion detection systems, and card access systems. The value provided by completing the project is to: (1) maintain compliance, (2) reduce redundancies and gaps in functionality, and (3) optimize overall performance. An integrated solution is efficient and allows for centralized management; situational awareness; real-time monitoring; compliance with regulations and guidelines; and faster, more effective and consistent responses to emergencies and incidents. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems. As part of the review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	30-Nov-19	(1.00)	1,197,840	0	1,026,720	0	171,120	0	0	0	0	0	0	0	0	0	
306	2020	PI Data Discovery Tool	Security	Software (Intangible)	The PI Data Discovery Tool project will implement software that enables data management decisions for unstructured data, reducing security risk and lowering costs associated with data. The file analysis software scans, maps, and manages unstructured data stores. The Company will gain value from this project through: (1) the added ability to monitor unstructured data for PI/privacy-related data; (2) closing the gap in security visibility; and (3) maturation of the Privacy Program. Additionally, the Company will benefit through discovery of the scope, risk, and opportunities associated with the ever-growing volume of unstructured data. The scope of this project encompasses evaluation, procurement and implementation of a software solution to analyze, index, search, track, and report on the file metadata and content. The unstructured data types included in the project scope and analysis consist of: (1) file shares, (2) email databases, (3) content collaboration platforms (CCPs), (4) records management and enterprise content management (ECM) systems, (5) Microsoft SharePoint, (6) log files, (7) Internet of Things (IoT) generated objects, and (8) data archives. No alternatives were considered beyond the data discovery tool due to the large amount of data that needs to be monitored. Manual or native monitoring techniques are not possible for effective real-time results. By not having a solution in place, the Company will not have visibility into PI data in unstructured data sources.	01-May-20	(1.00)	207,840	0	0	207,840	0	0	16,500	0	0	16,500	0	0	16,500	0	0
307	2020	Radar Intrusion Detection	Security	Software (Intangible)	This project will integrate and deploy new radar detection technology to enable the detection and tracking of individuals at Company sites. The radar would alert in real time and allow for the dispatch of Security or law enforcement to investigate. Completion of this project will provide value to the Company through evaluating and selecting a more reliable intrusion detection solution that reduces the number of false alarms and prevent resources from being sent to a site for unnecessary investigations. The scope of this project encompasses: (1) assessing radar intrusion detection solutions to obtain a solution that fits the needs of the Company; (2) establishing a priority list of sites where the solution will be rolled out; and (3) installing updated radar intrusion at physical sites in the Company's service territory with a focus on critical infrastructure sites, including but not limited to gas compressor stations, electric substations, and hydro-electric sites. Alternatives considered include: (1) keeping the current fence intrusion detection solution; or (2) evaluating, selecting, and implementing a new intrusion detection solution. The preferred option is to implement a new solution, given limitations with the current solution that result in multiple false alarms that must be investigated.	31-Aug-22	(1.00)	692,800	519,600	0	173,200	0	0	33,000	0	0	33,000	0	0	33,000	0	0
308	2020	Replace and Re-badge	Security	New Computers / Hardware	This project will enhance or replace physical security assets to provide improved visibility into potential security concerns and resolve incidents. The Company has several thousand physical security asset devices currently in use, including security cameras, motion detectors, intrusion detection systems, and card access systems. The value provided by completing the project is to: (1) maintain compliance, (2) reduce redundancies and gaps in functionality, and (3) optimize overall performance. An integrated solution is efficient and allows for centralized management; situational awareness; real-time monitoring; compliance with regulations and guidelines; and faster, more effective and consistent responses to emergencies and incidents. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems. As part of the review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	01-Dec-20	(1.00)	347,105	69,280	235,552	27,712	0	14,561	66,000	16,500	49,500	0	0	0	0	0	0
309	2020	Security Analytics	Security	Software (Intangible)	The Security Analytics project will implement the use of data collection, aggregation, and analysis tools for security monitoring and threat detection. These tools can incorporate large and diverse data sets into detection algorithms, which will assist in data analysis of security incidents, protection from unauthorized users, unintentional modification and compliance shortcomings. The project scope includes identifying and implementing a security analytics tool set that collects data from network traffic, endpoint and user behavior, cloud resources, business applications, non-IT contextual data, identity and access management, and external threat intelligence sources. Completing the project will provide the value of proactive security incident detection and response, regulatory compliance, and improved cyber forensics capabilities. As part of the review process the alternative considered was to not invest in a new tool. This choice was not selected as it would forfeit a proactive approach and enhanced capabilities for addressing potential cyber incidents. A final decision on tooling will be made after top solutions in this space are compared and vetted internally.	01-Sep-20	(1.00)	351,798	138,560	0	131,632	0	81,606	61,540	0	0	56,760	0	0	56,760	0	4,780
310	2020	2020 SubTotal Security	Security					6,180,194	1,420,240	1,718,592	1,967,025	517,520	636,807	794,511	224,664	49,500	481,470	0	38,877			

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311	2020	800 MHz Modernization Project	IT Service Delivery	Network	This project will replace antiquated head end and radio audio routing switch for tower sites and dispatch consoles), tower site, dispatch, and radio infrastructure, as well as user subscribed equipment with infrastructure that meets current Project 25 ("P25") standards. P25 standards are for digital mobile radio communications that are used by North American public safety and dispatch organizations. The current equipment was implemented in 1994 and is no longer manufactured or sold, so the Company is unable to find replacement parts. This project creates value for the Company by: (1) moving the Company to current production equipment that can be replaced and is more stable, (2) enabling the organization to migrate from an unsupported to a supported platform, and (3) allowing for quicker response to electric outages, and (4) increasing customer and employee safety. The scope of the project includes: (1) the design; (2) configuration; and (3) implementation of P25 systems for headends, tower sites, dispatch consoles, and subscriber equipment. The alternatives considered were to: (1) remain with the current system, and (2) forklift the migration to other manufacturers and architecture. Option 1 was not selected because the equipment is not supported, and the organization is unable to find replacement parts. Option 2 was not chosen because there would be a larger learning curve, it would be more disruptive to business, and it would require a complete replacement of the existing system. The Company chose to go with the upgrade because it has been incrementally adopting the P25 standards through the wireless ARP and this prudently utilizes current and ongoing investments. The upgrade is the easiest and least disruptive path to migrate to the P25 standards.	01-Jul-23	(1.00)	4,179,891	0	2,624,410	570,400	722,697	262,384	72,270	0	0	66,000	0	6,270
312	2020	ARP - Collaboration Asset Refresh	IT Service Delivery	Network	This project will replace the Company's collaborative tools and equipment. This project creates value by: (1) ensuring that the Company's audio, visual, telephony, and other communications systems are stable and reliable; and (2) beginning the foundational retirement of the legacy Avaya PBX systems that have reached end of mainstream manufacturer support. The project scope consists of: (1) annually replacing collaboration assets; and (2) installing new collaboration assets to account for organic growth requirements. The alternatives considered were to: (1) refresh visual assets and a portion of the audio assets, (2) refresh a portion of the audio assets only, and (3) refresh visual assets only. These alternatives were not chosen due to the risk inherent with a partial replacement of assets which includes: (1) a reduced supply of equivalent replacement Avaya parts that are no longer being produced, and (2) an erosion of the knowledge technicians possess on discounted systems in favor of education on the newest available technology.	31-Dec-20	(1.00)	479,342	57,040	153,614	96,968	114,080	57,640	246,637	36,300	131,992	25,740	46,200	6,405
313	2020	ARP - Field Device Asset Management (DAM)	IT Service Delivery	New Computers / Hardware	This project will replace field devices. This project creates value for the Company by: (1) mitigating potential costs for hardware repairs, and (2) allowing field workers to complete their job tasks. The project scope consists of replacing field device assets. The alternatives considered were to: (1) extend refresh cycles from four years to five years, and (2) run the assets to failure. The selection of a four-year cycle was deemed to be the best solution because replacement in less than four years would result in additional unnecessary expense for replacement of assets that are still in peak operating condition and replacement cycles that exceed four years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, which negatively impact employee productivity.	31-Dec-20	(1.00)	1,286,093	0	1,248,430	20,999	8,999	7,664	7,888	0	0	5,522	2,366	0
314	2020	ARP - Infoblox Refresh	IT Service Delivery	Network	The Asset Refresh Program (ARP) - Infoblox project will replace the Company's Infoblox system. The value of this program includes: (1) enabling the Company to efficiently manage and control their networks, and (2) providing Domain Name System, Dynamic Host Configuration Protocol, and Internet Protocol address management. The scope of this project includes the annual replacement of network assets under this program. The alternative considered was to continue operating on existing Infoblox equipment past the vendor's end-of-support date in February of 2021, and extended support is not an option. This alternative was not selected because it carries risks with not having vendor support, software bug fixes, security updates, and other software fixes. The alternative to replace the existing Infoblox equipment with the latest hardware and software provided by the vendor was selected to avoid these risks and continue a regular refresh cycle.	31-Dec-20	(1.00)	387,778	0	260,693	42,210	64,170	20,706	29,915	0	0	23,100	4,620	2,195
315	2020	ARP - Operational Technology Storage Area Network	IT Service Delivery	Network	This project will refresh aging SAN with Unity SANs. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support and enhance customer interactions, as well as ensure the stability of technology for business operations that are in service of the Company's customers. The project scope consists of: (1) annually replacing SAN hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. The alternative considered was to purchase extended maintenance. This alternative was not chosen due to the risk of increased downtime of critical infrastructure and maintenance costs. The cost of bringing personnel on site to make system corrections in the event of a serious interruption is higher than the cost of buying new.	31-Dec-20	(1.00)	409,547	114,080	253,828	28,520	0	13,119	37,587	30,360	0	6,600	0	627
316	2020	ARP - Operational Technology Support	IT Service Delivery	Network	This project will replace dated and obsolete servers and workstations. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support the operation of the Company's critical electric infrastructure. The project scope consists of: (1) the annual replacement of compute hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. Extending maintenance is not a viable alternative as current systems do not provide sufficient capacity for new electric system support capabilities.	31-Dec-20	(1.00)	358,154	0	262,384	65,596	0	30,174	34,568	16,500	0	16,500	0	1,568
317	2020	ARP - Printer Asset Management (PAM)	IT Service Delivery	New Computers / Hardware	This project will replace printers, plotters, and multi-function printing devices. This project creates value for the Company by: (1) improving the dependability of these printer devices for employees, (2) averting increased costs due to hardware repairs, and (3) ensuring compatibility with enterprise print applications. The project scope consists of the annual replacement of printer assets under this program. The alternatives considered for the project included looking at refresh cycles from three to seven years and running the assets to failure. The selection of a five-year cycle was deemed to be the best solution since anything less than five years would result in additional unneeded expense for replacement of assets that were still in peak operating condition, and anything greater than five years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, negatively impacting employee productivity.	31-Dec-20	(1.00)	396,894	0	322,044	51,267	0	23,583	6,721	0	0	6,138	0	583
318	2020	ARP - Server and Storage	IT Service Delivery	New Computers / Hardware	This project will replace or augment server and storage infrastructure for the Company. This project creates value for the Company through: (1) Improved stability and availability of business critical applications by proactively replacing server and storage hardware assets prior to the chance of hardware failures increasing, and (2) ensuring that adequate resources are available to support application demands after five to seven years of actual use. The scope of this project encompasses: (1) replacement of server and storage hardware assets, and (2) installation of additional new computers and storage capacity to account for organic growth requirements. The alternative considered was to purchase extended maintenance. This solution was not selected because full support would not be offered after seven years and maintenance costs would increase. The Company continues to refresh these critical technologies based on a five- to seven-year refresh cycle to mitigate the risk of failure.	31-Dec-20	(1.00)	6,422,548	0	5,624,843	546,647	0	251,458	411,177	330,000	0	13,860	66,000	1,317

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
319	2020	ARP - Wireless Network	IT Service Delivery	Network	This project will replace portions of the company's aging wireless systems. This project creates value for the Company by: (1) ensuring real-time communications between Company crews and dispatch locations, (2) ensuring efficient gas leak and electric outage response times for customers, and (3) maintaining critical infrastructure and regulatory compliance. The project scope consists of: (1) replacing wireless assets annually, and (2) installing additional new wireless assets to account for organic growth requirements. The alternatives considered for system replacement were: (1) running parallel systems while the new system is deployed and the old system is dismantled, (2) leasing a system from a vendor. The first alternative was not chosen since it would be highly disruptive due to systems having to run independently of one another, and would have a high cost associated with the need to acquire additional radio frequency spectrum. The second alternative was not chosen since the Company would be dependent on the response times offered by a shared vendor system that offers lower system reliability. The Company chose to replace existing assets based on its refresh cycle to avoid extended support cost, and to provide a seamless transition that allows both the old and new systems to interact with little to no disruption to the end user.	31-Dec-20	(1.00)	2,037,554	0	1,963,214	11,365	0	63,075	249,819	0	227,040	2,720	19,800	259
320	2020	ARP - Workstation Asset Management (WAM)	IT Service Delivery	New Computers / Hardware	This project will replace and install new desktops, laptops, and tablets. This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations prior to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks. The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources. The alternatives considered were: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle only on desktops from four years to five years, and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ARP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	31-Dec-20	(1.00)	6,710,795	0	6,405,224	209,295	0	96,376	86,734	0	0	79,200	0	7,534
321	2020	SAP Data Encryption	IT Service Delivery	Software (Intangible)	The project will implement Cyber Security requirements for encryption of Personal Identifying Information (PII) data "at rest" and "data in transit" in Company SAP data bases.	01-May-21	(1.00)	765,482	498,816	0	190,520	0	76,146	1,138,253	0	0	1,039,500	0	98,753
322	2020	Software Platform Refresh 2019	IT Service Delivery	Software (Intangible)	This project will upgrade server operating systems, hypervisors (virtual machine monitors), and databases to retain low-cost, unlimited vendor support. These platforms support all of the Company's applications including critical company and customer applications. The project will add value for the Company by: (1) avoiding costs for special maintenance agreements required at the end of normal manufacturer support; (2) ensuring reliability and compliance with information security requirements; (3) improving data center environment stability; and (4) avoiding the need for high risk upgrades that cross multiple versions. The project scope includes: (1) upgrading operating systems and databases on servers that are within three years of end of support, and (2) maintaining hypervisors at the current version for stability and performance. A funding options matrix was completed to review the alternatives. The options were: (1) fund the full project for \$1 million and pay no support liability in 2020; (2) fund \$750,000 of the project and pay \$1.6 million in support liability in 2020; (3) fund \$500,000 of the project and pay \$2.4 million in support liability in 2020; and (4) not funding the project and pay \$3.3 million for support liability in 2020. Alternative options 2-4 were not selected due to the high cost of support liability. Option 1 was chosen to avert these costs and to ensure system stability.	31-Dec-20	(1.00)	0	0	0	0	0	0	451,398	0	0	412,236	0	39,162
323	2020	SubTotal IT Service Delivery	IT Service Delivery					23,436,482	669,936	19,118,685	1,639,647	909,946	902,276	2,772,056	413,149	359,032	1,697,116	138,365	364,062
324	2020	Enhancements - Cloud Automation	Enhancements	Software (Intangible)	The Company's existing cloud automation platform consists of a number of scripts that automate the build of infrastructure like servers, operating systems, databases, and application installations in minutes or hours. These scripts include security hardening and application of the latest patches. This project will maintain and improve these scripts to ensure they meet the latest security policy and requirements, and any changes to the way the operating systems, databases, or applications are configured. The value of this project is inherent in the time savings and quality improvement, as it used to take weeks or even months to manually build infrastructure servers, which is prone to human error. The scope includes updating and improving the cloud automation platform to align with the Company's current security, database, operating system, network, and other application standards. The alternative would be to continue as-is with existing scripts; however, this introduces security risks and additional manual steps to build infrastructure, which will lead to increased costs and risk of manual errors.	31-Dec-20	(1.00)	542,769	0	0	371,760	0	171,009	117,326	0	0	91,778	17,490	8,009
325	2020	SubTotal Enhancements	Enhancements					542,769	0	0	371,760	0	171,009	117,326	0	0	91,778	17,490	8,009
326	2020	4G SAP Implementation	BP Functionality	Software (Intangible)	There are approximately 680,000 smart meters in the Company's electric system that utilize the 3G wireless technology. The wireless carrier will be retiring the 3G network technology on December 31, 2022, requiring the Company to replace the 3G meters with 4G meters in order for them to continue functioning. This project will create the technology that enables Itron to upgrade smart meter communication modules from 3G to 4G technology. This project will add value by continuing the ability to: (1) communicate with the smart meters, (2) provide accurate and timely bills to customers, (3) execute remote turn-ons and turn-offs, (4) receive outage information, and (5) administer demand response events. The project scope includes: (1) building technology interfaces to meter installation vendor technology, which include an interface to extract list of 3G meter information, an interface for posting work order completion data in SAP, and an interface for tracking exceptions and errors; and (2) end-to-end testing of the interfaces and back-end technology (meter-to-cash process). The alternatives considered were: (1) migrating to a new meter platform, and (2) not implementing 4G technology in 3G meters. The first option would require a substantially larger modification to all supporting systems and infrastructure. The second option is not feasible since the deprecating communications technology will eventually make the meters non-communicative. The option of implementing 4G technology was chosen so the Company can continue to reap the benefits of the smart meter technology previously implemented. These benefits include: (1) improved timeliness and billing accuracy; (2) retaining ability for remote disconnect and reconnect of services; (3) administration of demand response programs; and (4) enabling automated meter reads, outage reporting, and faster restoration.	01-Jul-20	(1.00)	749,412	0	0	620,056	0	129,356	60,839	0	0	47,520	0	13,319
327	2020	Accounts Payable (AP) Automation	BP Functionality	Software (Intangible)	The Accounts Payable (AP) Automation project will provide an end-to-end AP solution to optimize invoice capture via Optical Character Recognition (OCR) software; leverage workflows to automate invoice approvals and processing; manage electronic document retention; provide insights via improved reporting; and minimize human intervention. The value of completing the project includes: (1) centralizing invoice processing resulting in reductions to invoice entry costs, number of paper invoices, duplicate vendor payments, and late vendor payments; (2) automating validation of an invoice against contract rates; (3) increasing transparency of invoice processing status; (4) providing more accurate accruals; (5) improving the ability to capture discounts; (6) enabling cash flow benefits and reduced financing costs; and (7) improving internal controls. The scope of the project includes: (1) enabling OCR technology; (2) creating automated workflow for receiving, managing, routing, and monitoring invoices and related documentation; (3) automating posting of invoices; (4) creating new reports; (5) enabling electronic document retention; and (6) data cleanup. See Karen M. Gaston's testimony for full project synopsis.	17-Oct-20	0.69	547,312	0	0	225,160	190,520	131,632	109,890	13,200	1,650	16,500	27,060	51,480

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328	2020	Business Customer Internal Web Portal	BP Functionality	Software (Intangible)	The Business Customer Internal Web Portal project will develop a new Internal Web Platform (IWP) for Business Customers to provide energy usage insight to customers and support the Company's initiatives for reducing energy demand. This project will provide value to both the Company and its customers through: (1) increased customer engagement and satisfaction through self-service capability focused on energy waste reduction, and (2) reduced calls and contacts with the Company regarding customers' energy usage and energy waste reduction options. The project scope will include: (1) implementing an internal web portal to support business customers, (2) integrating with SAP and smart meter data to identify energy usage, (3) enabling reporting and analytics capabilities for deeper understanding of customers' energy usage, and (4) enabling features for energy efficiency and demand response within the internal web portal. Three alternatives were considered but not chosen for this project: (1) Build a custom IWP for business customers. This alternative was not chosen given the high cost and unfavorable return on investment because it requires complex logic for rate analysis, rate modeling, and energy waste management improvements as well as increased maintenance costs. (2) Defer to future years. This alternative was not selected because the Company is already lagging in the industry regarding availability of internal portals to customers, and this option would further limit offers to business customers for energy waste reduction and rate analysis based on time of use, which impacts customer savings resulting from energy waste reduction and limits the impact of the Company's plan for energy reduction. (3) Forego IWP for Business Customers. This option was not chosen as it limits customer offerings for energy management and still results in negative CX impacts. The option to implement a Business Customer IWP was chosen because it enables customer self-service for energy management, improves CX, and supports energy demand reduction.	14-Sep-20	(1.00)	1,887,164	1,039,200	0	242,480	119,600	85,884	322,363	82,500	0	0	132,000	107,863
329	2020	Business Process Performance Monitoring	BP Functionality	Software (Intangible)	This project will deploy the AppDynamics tool across enterprise applications like SAP, O&M, and other business critical systems. This project will create value for the Company by providing faster system restoration through automation of system troubleshooting that pinpoints the software or hardware that is causing the problem. The project scope includes: (1) SAP, (2) O&M, (3) Advanced Device Metering System, and (4) other enterprise applications. The Company evaluated multiple products for this project and AppDynamics fit the cost and functionality expectations. The others were cost prohibitive and lacked the functionality of the product selected.	01-Oct-23	(1.00)	473,313	0	0	137,174	235,815	100,323	172,541	18,390	0	7,920	139,075	7,156
330	2020	Care Annual Updates	BP Functionality	Software (Intangible)	This project will: (1) implement software changes to improve the process for offering energy assistance to low-income customers, and (2) streamline the process for assistance agencies who utilize the application through improved user interface and updates to SAP to process various CARE requests. Upcoming modifications will be identified following an annual review of requests to prioritize the list of changes. The project will provide the following value: (1) complete modifications to the existing internal SAP application and Agency Portal to receive annual Low Income Home Energy Assistance Program ("LIHEAP") funding, which is used to provide customers bill credits and arrears forgiveness; and (2) improve the data within the assistance Agency Portal, thereby making it easier to assist customers in need of LIHEAP funding. The project scope includes: (1) updating the enrollment and status process, (2) allowing for flexible bill credits, (3) improving reporting, (4) updating the arrears forgiveness plan, and (5) satisfying additional regulatory requirements for the annual grant rule changes required by the Michigan Department of Health and Human Services and the Michigan Agency for Energy. The alternative considered was to continue with the current process. This alternative was not chosen since it would result in loss of grant funding, thus decreasing or eliminating energy assistance dollars for low income customers. Instead, making annual updates to the application will allow agencies to easily enroll customers on assistance programs, and allow placement of holds to stop or prolong credit activity until assistance decisions are granted. This option provides long-term proactive energy assistance to low income customers and prevents loss of grant funds. All changes are related to the existing internal SAP and Agency Portal, therefore a cloud alternative is not viable.	31-Dec-20	(1.00)	0	0	0	0	0	198,508	0	0	167,232	17,194	14,083	
331	2020	Cloud Automation Phase 6	BP Functionality	Software (Intangible)	The Cloud Automation Phase 6 project will add additional features and enhancements to the Company's cloud automation platform. This project provides value to the Company by (1) extending the ability to deploy, use, and decommission public and private cloud services in an automated, on-demand, and secure fashion, (2) increasing the agility of IT, (3) lowering risk in running applications in the cloud, keeping systems and customer data available and safe, and (4) improving the efficiency, quality, and speed to market of customer-facing and internal IT services. The scope of this project includes adding between three and six features to the Company's cloud automation platform including: (1) support of deployment of lower tier (more critical) applications in the public cloud, (2) support of container deployment in the hybrid cloud, (3) data lake cloud storage automation, (4) automation of Machine Learning (ML) and Artificial Intelligence (AI) services, (5) additional lifecycle and governance tooling, and (6) Disaster Recovery as a Service (DRaaS) automation. Alternatives considered include (1) deploying all critical applications only in the on-premises and co-location data centers, (2) manually deploying containers, (3) avoiding container technology, (4) manually deploying data lake storage in the cloud, (5) deploying storage only in the on-premises or co-location data centers, (5) deploying services similar to the available public cloud ML and AI services in the on-premises and co-location data centers, (6) manually deploying ML and AI cloud services, (7) avoiding ML and AI services, (8) manually managing the lifecycle and governance of cloud services, (9) continuing with existing on-premises DR solutions, and (10) manually managing cloud DRaaS offerings. These options were not chosen as they would require significant investment in hardware and staff augmentation to perform the work. See Jeffrey D. Tolonen's testimony for full project synopsis.	01-May-20	(1.00)	681,077	76,741	0	336,092	65,816	202,428	119,824	3,848	0	86,536	20,790	8,650
332	2020	Cross-Channel Analytics	BP Functionality	Software (Intangible)	The Cross-Channel Analytics project will support the Operational Analytics team with furthering its technological ability through the enhancement and upgrades of server systems such as Speechminer, the speech analytics tool on the Genesys platform. Additionally, the team intends to purchase and implement the Tag Management System, Tealium iQ, enabling the use of website tags and providing a management system for those tags. This will provide the next layer to the team as they work to understand customer pain/break points that they may experience as they attempt to self serve. The Company will gain value from this project through: (1) improved ability to back test new phrases, topics, or categories; and (2) enable further analytics for the Core Team to understand and visually show what customers have called for historically and why they were not able to self serve on single web pages. The project scope includes: (1) implement hardware to retain 12 months of Speechminer data data, (2) archive and update every SMART Apply, (3) implement a tag management system for use in cross-channel analytics and self-service use cases, and (4) integrate with existing data stores for customer and Interactive Voice Response (IVR) data. Alternatives considered include: (1) Keeping Speech Miner retention at three months which results in a loss of considerable analytics ability, especially when it comes to marketing programs like Time of Use (TOU), (2) Currently no current Tag Management System so the alternative is to do nothing, but the Core Team will not have the ability to problem solve any further. Customer issues may not be identified and resolved resulting in continued calls in the contact center. (3) Utilize manual data stores/cross referencing including small databases and Excel. This option increased costs for manual processing for analytics, required complex processing to complete analysis on customer needs. (4) Increase data retention and implement a Tag Management System. This option was selected as the automation reduced the need for manual processes and increased data volume will improve our ability to identify customer needs.	01-Aug-20	(1.00)	0	0	0	0	0	79,200	79,200	0	0	0	0	0

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FISC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
333	2020	Digital Application Programming Interface Fabric	BP Functionality	Software (Intangible)	This project provides foundational capabilities around APIs, which are a set of technologies used to integrate applications within the Company and, more importantly, with external third-party applications. In short, APIs allow two applications to talk to each other. This is currently a gap and is a key technical capability needed to enable multiple IT projects related to the Company's strategy for clean energy like: Electric Interconnection Billing and Payments; MSGO Market User Interface Changes; Utility Analytics Mis-Phasing/Power Quality/Outage; Field Contractor Work Management Technology Enablement; Work Management Scheduling, Analytics and Reporting; DEMAS and Google X. Similarly, this API foundational capability will enable customer-focused projects like: Business Customer Interval Web Portal ("IWP"), Bring Your Own Thermostat ("BYOT") Pilot, Customer Relationship Management ("CRM") Product Suite, and Large Customer Rate Tool. The project will add value by: (1) implementing functionality to perform API services at-scale; (2) allowing partners to integrate with the Company's internal applications; (3) providing the ability to reuse integrations; (4) enabling monitoring of API traffic; (5) implementing functionality to perform API throttling (i.e., traffic management); (6) visualizing API traffic and analytics through key performance indicators ("KPI"); (7) enabling faster prototyping, testing, and deployment of integrations; and (8) providing operational tools for monitoring, incident management, and resolution. The project scope includes: (1) evaluating multiple API management tool vendors; (2) executing API onboarding for a number of external partners without scalability constraints; and (3) configuring and deploying API services with out-of-the-box operations tooling to achieve faster speed to market. Two alternatives were explored and determined to be non-viable (1) remain on current Tibco API product that is part of their ESB platform, which does not support the future needs of the Company including efforts to modernize the grid; and (2) scale back on the implementation of the API fabric and complete over a longer period, which would negatively impact other projects. These alternatives were not selected because the API fabric is foundational and needs to be completed to support other projects. The current scope and direction is the best fit to support current Company initiatives.	01-Nov-20	(1.00)	1,187,439	139,344	0	363,720	346,400	317,995	189,833	9,900	0	115,500	49,500	14,933
334	2020	EHS Compliance	BP Functionality	Software (Intangible)	The EHS (Environmental Health & Safety) Compliance project will implement a cloud-based corporate-wide incident and risk management solution. The solution will ensure consistent accurate tracking, management, and reporting of environmental, health & safety compliance tasks using a centralized source of data. Modules to be implemented include Compliance, Health and Safety, Human Performance Management, Incident & Crisis, and Reporting & Analytics. The value of doing this project is to leverage best practice processes which are built into the new solution, including incident management and prevention to ensure a safe and productive workforce able to complete work for customers. The project also supports the Company's Planet goal through enhanced tracking and reporting for air quality, waste management, and sustainability. Other benefits include risk avoidance for environmental penalties, increased productivity, and increased quality. The addition of visual management and dashboards to enable transparency in tracking of goals; and create awareness of emerging environmental regulations and, to respond to them. One alternative is to continue "as is" through use of disparate Excel and SharePoint solutions. This requires much manual effort, and poses a risk of not being able to accurately and centrally track EHS data. Another alternative is to have separate projects; one for Environmental Compliance, and another for Safety and Health Compliance. This was the initial path for these two business areas. Investigation into potential solutions including a formal Request for Information (RFI) revealed that the standard for these market solutions is integrated functionality for Environmental and Safety/Health. Both on premise and cloud solutions were considered. The single solution was selected as the most efficient solution.	01-Feb-20	(0.86)	775,936	0	0	277,120	346,400	152,416	118,381	32,465	0	42,900	37,796	5,280
335	2020	Electric Distribution Asset Management	BP Functionality	Software (Intangible)	The Electric Distribution Asset Management project will develop a Distribution Asset Management strategy with corresponding process and technology implementation plans and objectives. The Company will gain value from this project through: (1) creating a standardized process for designers and field resources to collect and manage asset data, reducing waste and rework associated with current processes; (2) optimizing planning and management of distribution assets, resulting in investments of highest value; (3) establishing a channel for the Company to develop and manage asset health conditions on the distribution system, optimizing the operation and maintenance investments of distribution assets; and (4) improving the quality of Company electric distribution asset data, optimizing asset performance and providing opportunities to leverage quality data to support other engineering and operations efforts. The project scope includes: (1) developing a Distribution Asset Management strategy with a corresponding process and technology implementation plans and objectives; (2) designing and implementing optimized processes and supporting technology solutions; (3) implementing an Asset Integrity Management program leveraging industry standards; (4) evaluating, designing and implementing tools to optimize investment planning and asset health management; (5) implementing a distribution asset system of record repository; and (6) designing and implementing the organizational change management necessary to ensure success. See Richard T. Blumentstock's testimony for full project synopsis.	31-Dec-24	(1.00)	992,000	200,000	100,000	200,000	400,000	92,000	0	0	0	0	0	0
336	2020	Enterprise Content Management - Managing Business Records	BP Functionality	Software (Intangible)	Using the Company's Enterprise Content Management ("ECM") system, this project will manage business records for high-focus areas contained in SharePoint and Shared Drives. The records will be classified, categorized, and placed under formal retention rules, via metadata assignment. This project will add value through: (1) continuation of the multi-year management action plan; (2) consistent practices for records management; (3) defensible process for validating completeness and accuracy of records produced and records deleted; (4) deletion of excess, irrelevant, or inappropriate information; (5) easy generation of electronic records in the event of litigation; (6) achieving Generally Accepted Recordkeeping Principle maturity of 3.0 or greater through functional capabilities of ECM coupled with business practices to align to ARMA (formerly the Association of Records Managers and Administrators) principles; and (7) mitigating the possibility of fines as a result of being unable to produce records. For high-focus areas, the scope includes: (1) integrating SharePoint and Shared Drive content into ECM; (2) assigning taxonomy and records classification values through metadata on content and records; (3) assigning standard retention rules; (4) mapping, building routines, and migrating existing content to new taxonomy and retention schedules; (5) ensuring records can be located; and (6) allowing legal holds and lineage audits to be conducted on content. See Karen M. Gaston's testimony for full project synopsis.	14-Nov-21	(1.00)	304,242	0	0	86,600	190,520	27,122	519,594	0	0	330,000	172,920	16,674
337	2020	GIS-Integrated Design	BP Functionality	Software (Intangible)	This project replaces the CAD and Work Requirements and Design ("WR&D") software with Bentley's Open Utilities Map ("BOM") to leverage GIS asset data to generate engineering designs, implement workflows with SAP, and mitigate risk associated with the aging software application. The project provides value to the Company in three ways: i) the BOM software is compatible with GIS integration and may be further developed after the Company implements the LUPDM as the basis of its GIS systems; ii) the project mitigates risk of manual asset record imports, asset record manipulation for design white space management, or manual asset record recreation for base design files, by enabling technology that creates a new design over existing electronic asset records; iii) it enables technology that is compatible with the asset system of record and the Project Wise engineering document management system. The project scope includes: i) replacing CAD engineering design software (Bentley Microstation Jv7) with BOM; ii) leveraging GIS asset data for the purpose of generating engineering designs; iii) integrating workflows with SAP through Bentley Workflow Manager software; iv) implementing GIS replication databases to isolate data editing from data viewing; v) replacing the desktop View Graphics custom application with a web-based application for searching and viewing asset records; and vi) implementing a Primary Distribution Map ("PDM") generation service to generate electronic PDF files for printing during storm and other emergency events. See Richard T. Blumentstock's testimony for full project synopsis.	01-Mar-23	(1.00)	2,911,732	412,048	0	992,090	512,672	994,922	417,071	175,560	0	36,300	144,540	60,671

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338	2020	GM - GIS Connectivity Model Integration	BP Functionality	Software (Intangible)	The Grid Modernization (GM) Geographic Information System (GIS) Connectivity Model Integration project will design, develop, and implement a GIS extract using an industry standard to publish electric GIS data to the network, where it can be consumed by multiple applications. The extract will make the data available to the electric distribution historian as well as other applications in the future. The project provides value to the Company by: (1) leveraging the International Electrotechnical Commission (IEC) Common Information Model (CIM), which provides a basic ontology for data and data relationships and a standard repeatable approach that can be leveraged for sharing data about the electrical network across technical systems; (2) supporting an enterprise application integration framework, which keeps business process as simple as possible; (3) providing the ability to detect the physical location of metered assets, including circuit, substations and transformers; and (4) reducing the costs and time required to develop integrations in the future resulting from streamlined approaches and processes. Deliverables within the project scope include: (1) develop and implement the GIS extract, leveraging IEC CIM for data sharing; (2) leverage the enterprise service bus (ESB) as the high level communications protocol for sharing CIM data across the enterprise; (3) build the integration to the electric distribution historian; and (4) develop a data and integration framework that can be leveraged for future projects. The alternative considered was to continue to implement point-to-point application-specific integrations to meet the requirements specific to a single project. This alternative was not chosen because it continues to increase the technical debt associated with the maintenance of non-standard interfaces. The preferred approach as outlined in this business case will result in: (1) standard, repeatable approaches for future integrations, (2) lower ongoing maintenance costs and, (3) less complexity.	31-Dec-20	(1.00)	301,332	0	0	132,210	149,022	20,000	8,000	0	0	0	0	0	0	8,000
339	2020	HR - 2020 Union Changes	BP Functionality	Software (Intangible)	The HR - 2020 Union Changes project will implement SAP and other system changes required as a result of three collective bargaining agreements which will be renegotiated and ratified. Collective bargaining agreements expire every five years for Operating Maintenance and Construction ("OM&C"), Virtual Contact Center ("VCC"), and Zealand employee groups. For OM&C, the current agreement ends June 1, 2020. The VCC agreement ends August 1, 2020. The Zealand agreement ends October 1, 2020. Completion of this project will provide value to the Company and its customers through: (1) waste elimination by making changes in the software for pay and benefits changes as required by the new agreements; (2) defect reduction by adding process automation to otherwise manual processes for tracking and recording work, premium and absence time; (3) improved employee engagement amongst the OM&C, VCC, and Zealand union employees. The scope of this project encompasses making system changes required to support the new working agreement language for the OM&C, VCC, and Zealand employees. Exact details will be finalized after the negotiation process is completed and contracts are approved. Three alternatives were considered for this project: (1) Make no system changes after the contracts are ratified. This option was not chosen because it exposes the Company to possible fines, disengaged employees, union grievances, significant manual processes leading to greater possibility of error, hiring additional staff to perform activities outlined in the agreements, and increased legal costs due to employee grievances. (2) Find other third party software to support the changes required by the union agreements. This option was not chosen because it would require SAP integration along with additional software licensing and maintenance costs. The option to make system changes to eliminate manual updating, comply with the working agreement language, support union employee engagement, and reduce grievances, leading to reduced legal costs, was chosen because it leverages current SAP technology, automates what would otherwise require manual processing, and is the least costly option.	18-Dec-20	(1.00)	0	0	0	0	0	386,412	0	0	142,560	213,840	30,012			
340	2020	Large Customer Rate Tool	BP Functionality	Software (Intangible)	The Large Customer Rate Tool project is for automated rate analysis through a rate tool for our large customers. The project will automate currently manual rate evaluations and thereby reduce the time it takes to run individual rate analysis. Included in the implementation are secondary, primary and special rates for business customers, process automation, and accuracy of rate definition for large customers. The value of completing the project is the automation of a currently manual intensive process that account managers utilize in working with large businesses to ensure they are on the best possible rate. As part of the project review process, the alternative considered was to do nothing and maintain the manual process of extracting data from SAP and entering into spreadsheets to run rate comparisons. This alternative was not selected since it would not support the gains in customer satisfaction or the efficiency of providing the data. Customer benefit: Improved functionality to better assist large business customers with evaluating different rate options.	01-Aug-20	(1.00)	1,257,393	110,848	0	347,747	736,446	62,352	102,960	0	0	35,310	60,390	7,260		
341	2020	Rate Case Implementation	BP Functionality	Software (Intangible)	The Rate Case Implementation project will modify SAP billing in accordance with MPSC filings. These enhancements include, but are not limited to, overall price changes and structure changes to the rates. Project scope includes the large electric and gas rate changes that occur annually and/or monthly. The benefit of this project is to ensure that customers are billed accurately based on the timing of the rate structure and price changes. An alternative considered for this effort was an offshore development model. However, the risk of billing inaccuracies and customer complaints with that model were deemed too high based on the various complexities and timing of the development and testing.	30-Sep-19	(1.00)	0	0	0	0	0	172,135	0	0	83,160	81,972	7,003			
342	2020	ServiceNow Customer Service Management	BP Functionality	Software (Intangible)	This project will implement the CSM module of ServiceNow to enable the recently defined Supply Chain ("SC") service delivery model to support SC optimization. This project will provide value to the Company and its customers through: (1) waste elimination, (2) cost savings, (3) maturing processes, and (4) movement to a sustainable service delivery model to continually improve the customer experience. This project specifically will enable the newly designed processes and service delivery model as it will: (1) enhance the delivery of SC services to both internal customers and suppliers; (2) include an integrated knowledge base and case management system, improving the customer experience by having one place to go for information and help; and (3) provide standard workflow and funnel the previously disparate intake channels for SC support into one solution. This technology solution is the backbone of an SC Support Center and will enable: (1) better management of work; (2) more efficient and accurate response to questions; and (3) improved satisfaction of internal customers and external suppliers. See Jeffrey D. Tolonen's testimony for full project synopsis.	28-Feb-20	(1.00)	198,141	0	0	114,312	17,320	66,509	88,011	47,065	0	29,040	9,900	2,006		

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343	2020	Summer Peak Use Rate (SPUR) Release 2	BP Functionality	Software (Intangible)	The Summer Peak Use Rate (SPUR) Release 2 project will continue the implementation of Time of Use (TOU) rates for Net Metered and PayGo customers that were not included in SPUR R1, allowing customers to lower their bills during peak summer usage. This project will add value by: (1) providing Net Metered, PayGo, and electric vehicle rate customers with a TOU rate that allows them to manage their consumption during peak summer hours; (2) converting all non-TOU customers to new TOU rates by June 2020; (3) implementing a rate structure that can lower a monthly bill for a customer, which improves customer experience and Net Promoter Scores; and (4) shifting off usage to off-peak and reducing on-peak demand during summer months. The scope of this project includes: (1) configuring new TOU rates for Net Meter, PayGo, and electric vehicle rate customers in SAP; and (2) modifying the conversion program and transitioning all new TOU customers to new TOU rates by June 2020. This project will also include modifications to the SAP Customer Relationship Management (CRM) application to ensure these customers are able to concurrently enroll in various customer programs (Budget Billing, Appliance Service Protection, Shut Off Protection, etc.). The alternatives considered include: (1) manually converting approximately 10,000 (Net Metered and PayGo) customers to the TOU rate; (2) restricting the scope of the project to exclude updating the CRM program, thereby preventing these customer from participating in customer programs; (3) requesting an exception from the Michigan Public Service Commission. None of these alternatives were selected due to the increased risk of errors associated with manual conversion, limiting customer participation in critical assistance or elective programs, and risk of non-compliance of MPSC billing rules.	01-Nov-20	(1.00)	2,790,000	0	0	0	600,000	1,980,000	230,000	1,421,026	0	0	0	225,000	1,150,500	45,526
344	2020	Workforce Connect - Talent Roadmap Enablement	BP Functionality	Software (Intangible)	The Workforce Connect - Talent Enablement project is part of an overarching talent enablement plan which will enable the company to move forward on its goal of creating an employee experience to better serve customers. Improving the employee experience is focused on two outcomes: (1) enabling employees to be more engaged and productive and (2) attracting and retaining new skill sets and talent, both of which provide for an improved customer experience and higher customer satisfaction. This particular project will provide the following value: (1) increased employee retention for continuity of knowledge and long-term customer relationships, as well as reduction in costs associated with recruiting and onboarding new employees; (2) positive direct customer interactions facilitated through strong employee engagement; (3) enhanced employee development to meet customer demands during a time where retirement eligibility is high, and risk of knowledge loss has the potential to negatively impact customer satisfaction; (4) enable the Company to operationalize a new career framework and competency model which will not only support training and development of employees to deliver strong customer service but will also enhance the ability to hire candidates capable of delivering on the Company's initiatives and goals; (5) advanced workforce analytics that provide data-driven insights to proactively optimize talent management strategies and programs in support of maintaining and enhancing the workforce to continue to deliver strong customer service and results; (6) reduce waste and defects leading to increased quality and simplifying manual processes; (7) improved insight and consistency in reporting (i.e. Ethics, Safety, Office of Federal Contract Compliance Programs); (8) provide transparency into key talent areas to identify retention risk within critical areas and develop succession strategies; and (9) ensure core systems are stable and operational interruptions are minimized. See Karen M. Gaston's testimony for full project synopsis.	15-Dec-22	(0.98)	0	0	0	0	0	0	49,500	0	0	0	42,075	0	7,425	
345	2020	2020 SubTotal BP Functionality	BP Functionality					15,056,412	1,398,181	100,000	4,074,761	5,690,512	2,592,839	4,536,087	462,127	1,650	1,407,552	2,257,417	497,341		
346	2020	2020 Total						55,586,044	4,486,437	24,328,322	11,911,231	8,794,901	6,125,152	16,034,202	1,237,767	483,002	8,718,340	4,393,653	1,201,440		
347	2021	Enterprise Service Bus Application Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Enterprise Service Bus (ESB) Application Upgrade project will upgrade and migrate the Business Works developer application to the next version. This value this project provides the Company includes (1) accelerated productivity, (2) continuous delivery and integration, (3) an open ecosystem for improved operational viability, (4) integration with web, mobile apps, and application programming interfaces in real-time, and (5) improved administrative and operational efficiencies. The upgrade will also provide for easy integration with other cloud-based solutions such as Amazon Web Services and provides for easy migration to cloud-based containers in the future. The project scope includes: (1) implementing new versions of all applications that are part of the Enterprise Service Bus (ESB) in two security zones, and (2) a server refresh. The new products will be implemented on the latest version of the Suse Enterprise Edition operating system. The alternative considered was to incur additional annual \$40,000 maintenance cost and lose supportability. Given the critical nature of this application, it is not recommended to lose mainstream support for any of the applications involved. Any sustained product deficiency would impact many areas of the company such as billing, revenue collection, and remote meters. This alternative was not chosen due to these reasons, and the additional expense.	01-Nov-21	(1.00)	497,780	0	415,680	51,960	0	30,140	975,645	0	0	0	89,000	0	84,645	
348	2021	HR Support Pack and BS Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will update the SAP system with HR Support Packs that are released annually by SAP to comply with HR and tax changes. This project creates value for the Company by: (1) ensuring that it is in compliance with new financial rules and regulations, and (2) ensuring that it can calculate and distribute payroll. The scope of this project is to add SAP HR corrections to ensure proper reporting of financial information by the Company. As this is an upgrade of an existing system, the alternative considered was to delay the upgrade. This alternative was not chosen due to the risk of not complying with financial rules and regulations.	01-Dec-21	(0.94)	0	0	0	0	0	0	680,942	57,737	0	99,000	514,800	9,405		
349	2021	Iron Enterprise Edition Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will upgrade IEI, which collects the reads from meters to ensure accurate and non-estimated bills are provided to our customers. This project creates value for the Company by: (1) ensuring the new features and functionality added to meet business requirements are available to business partners and IT; (2) meeting Information Security's requirement to keep applications patched to control cyber debt; and (3) allowing for validation, estimation, and editing functions for all data collected. The scope of this project includes: (1) an operating system refresh; (2) migrating from Windows 2012 to the next Windows version, and (3) migrating the database from SQL Server version 2012 to the next SQL Server version. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and cyber security of the system.	01-Nov-21	(1.00)	381,040	0	381,040	0	0	0	551,430	0	0	0	396,000	115,500	39,930	
350	2021	Oracle Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	The Oracle Server Database Upgrade project will upgrade Oracle server databases to the next version and supportability across business portfolios. This project will create value for the Company and its customers by (1) reducing the risk of system failure and the resulting impacts to business partners and customers, and (2) ensuring that systems are secure, supported, and have the latest features and functionality. The scope of this project includes upgrading to the next version of Oracle across all business systems. As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and supportability of the system.	30-Apr-21	(1.00)	63,832	0	0	41,568	0	22,264	450,965	0	0	0	411,840	0	39,125	

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351	2021	54 HANA Assessment	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will review options for moving to the new platform before the current SAP platform is no longer supported, which is currently projected to occur in 2024. The value of the project is to: (1) devise the best option to migrate to a new platform at the least cost, and (2) ensure that the Company is prepared to move to the new platform by projected 2024 end of SAP support, and (3) ensure that all alternatives have been explored so that the best option is implemented. The scope of the project includes: (1) reviewing SAP options for migrating to a new platform, (2) reviewing alternative platform options that the Company could utilize in place of S/4HANA, (3) reviewing support options for the existing SAP platform past 2024, and (4) providing cost and alternative options so the Company can develop a project for the best option to address the SAP platform. Three alternatives were explored and determined non-viable for the project: (1) completing the assessment as the initial phase of the implementation project would limit the amount of options considered and make assumptions about cost needed to stand up the project, (2) delaying the assessment until a later date would not give the Company enough time to prepare for the implementation, and (3) completing an assessment that had a scope limited only to migrating to the new SAP platform may result in the Company accepting a sub-optimal solution. The option selected gave the Company the best opportunity to look at all viable options and to have enough lead time for transition to the new database.	01-Oct-21	(1.00)	0	0	0	0	0	0	118,800	0	0	18,216	99,000	1,584	
352	2021	SiteCore Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project will refresh all components of the website hosting, delivery, search, and analytics applications to add new features and improve search capabilities. Sitecore is the content manager for the ConsumersEnergy.com website. The project will add value by providing these four benefits: (1) maintains currency with the web hosting application version; (2) allows business users to make use of new features and functionality; (3) neutralizes continually evolving cyber threats; and (4) continuously improves customer experience using ConsumersEnergy.com. The project scope includes: (1) upgrading the Sitecore content management software to include content hosting and delivery allowing the use of new features and functionality; (2) upgrading the Coveo software, which will allow for more intuitive search results and provide suggestions or recommendations based on the customers' search text; (3) upgrading the Mongo database, which provides the analytics functionality within Sitecore to better understand customer actions and errors on the web page; and (4) developing analytics across customers' web and contact center interactions, allowing the company to connect contact center call data with corresponding web data for improved customer experience. Alternatives considered but deemed not to be viable include: (1) implementing a two-year upgrade cycle, which would not position the Company to be able to keep up with constantly changing cyber threats, or gain value from the rapidly changing feature set being developed by the vendor; and (2) purchasing an existing cloud solution, which is cost prohibitive at this time. Annually upgrading the existing Sitecore platform provides the functionality and stability needed while meeting financial requirements and mitigating cyber security risks.	31-Dec-19	(1.00)	0	0	0	0	0	545,302	0	0	363,000	148,764	33,538		
353	2021	SQL Server Database Upgrade	Upgrades & Replacements (Enterprise)	Software (Intangible)	This project, which supports critical applications such as the meter read collection systems and the customer contact center applications, will upgrade all SQL Server 2008-2014 instances to the latest version. This project will create value for the Company and its customers by: (1) reducing the risk of system failure and the resulting impacts to business partners and customers; and (2) ensuring that systems are secure, supported, and have the latest features and functionality. Project scope includes: (1) upgrades to all SQL Server 2008-2014 instances currently in use and not identified as part of another portfolio upgrade project, legal hold, or pending system retirement (approximately 400 instances); (2) installation and/or distribution of new SQL Server Client Tool software packages to affected workstations and application servers; (3) new Nimbus virtual machine templates for the new SQL Server release; and (4) technical database support to IT portfolios, business partners, and vendors during all project phases. The alternative considered was to migrate to Azure Cloud. As part of this option, the organization would obtain three years of extended support through Microsoft on SQL Server versions 2008-2014. This option was not selected because extended support would not be provided 2012 and 2014 versions, and the organization would not reap the benefits of new features offered through the upgrade. The Company decided in favor of the upgrade to avoid these issues and ensure system stability.	30-Apr-22	(1.00)	317,081	0	0	207,840	0	109,241	701,019	0	0	640,200	0	60,819	
354	2021	2021 SubTotal Upgrades & Replacements (Enterprise)						1,239,733	0	796,720	301,368	0	161,645	20,058,304	1,239,504	483,002	11,537,596	5,271,717	1,470,486	
355	2021	Application Currency and Enhancements - ACDA - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency, and Enhancements Analytics, Cloud, DevOps, and Architecture ("ACDA") initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the ACDA Portfolio. The value of regular upgrades to applications in the ACDA Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the ACDA Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the ACDA Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	56,810	0	56,810	0	0	7,260	0	0	7,260	0	0	0	0

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Synopses Containing Descriptions, Scope, Benefits, Implementation Dates and Detailed Costs of Actual and Projected Electric & Common Capital Expenditures and O&M Expenses
For the years 2018, 2019, 2020, and 2021

Information Technology Department

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No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
	SPEND YEAR	PROJECT NAME	IT PROGRAM	FISC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
356	2021	Application Currency and Enhancements - ACDA - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Analytics, Cloud, DevOps, and Architecture ("ACDA") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade the applications that support the ACDA Portfolio. The value of regular upgrades to applications in the ACDA Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for ACDA Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the ACDA Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	0	0	113,067	0	0	113,067	0	0
357	2021	Application Currency and Enhancements - CE&O - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Customer Experience and Operations ("CE&O") initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade the applications that support the CE&O portfolio. The value of regular upgrades to applications in the CE&O Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for CE&O. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Customer Experience and Operations Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	374,112	0	0	69,280	0	304,832	144,540	0	0	132,000	0	12,540	
358	2021	Application Currency and Enhancements - CE&O - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Customer Experience and Operations ("CE&O") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Customer Experience and Operations Portfolio. The value of regular upgrades to applications in the Customer Experience and Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for CE&O Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Customer Experience and Operations Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	21,780	0	0	19,800	0	1,980		
359	2021	Application Currency and Enhancements - Corporate Services - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade applications that support the Corporate Services Portfolio. The value of regular upgrades to applications in the Corporate Services Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Corporate Services. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as Finance; HR; Learning & Development; Legal; Governmental, Regulatory and Public Affairs; Corporate Security; and Strategy and IT Governance. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-21	(1.00)	717,048	0	10,392	448,934	0	257,722	81,736	0	0	75,900	0	5,836	

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Information Technology Department

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No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
360	2021	Application Currency and Enhancements - Corporate Services - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Corporate Services Portfolio. The value of regular upgrades to applications in the Corporate Services Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Corporate Services. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as Finance; Human Resources ("HR"); Learning & Development; Legal; Governmental, Regulatory and Public Affairs; Corporate Security; Strategy and IT Governance. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	0	239,976	0	0	184,336	0	55,440
361	2021	Application Currency and Enhancements - Infrastructure Applications and Operations - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements - Infrastructure Applications and Operations ("IAO") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the IAO Portfolio. The value of regular upgrades to applications in the IAO Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the IAO Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the IAO Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	214,821	19,793	0	178,139	0	16,889	
362	2021	Application Currency and Enhancements - Operational Technology - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade the following applications that support the OT Portfolio. The value of regular upgrades to applications in the OT Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for OT. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for OT. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-21	(1.00)	17,320	0	17,320	0	0	19,436	0	0	17,750	0	1,686	
363	2021	Application Currency and Enhancements - Operations - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Operations initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the Operations Portfolio. The value of regular upgrades to applications in the Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Operations area. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	576,964	0	27,712	381,040	0	168,212	43,362	0	0	39,600	0	3,762

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
364	2021	Application Currency and Enhancements - Operations - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements - Operations O&M initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Operations Portfolio. The value of regular upgrades to applications in the Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Operations Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	0	170,438	9,900	19,800	126,720	0	14,018
365	2021	Application Currency and Enhancements - TEOS - Capital	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements Transmission, Engineering and Operations ("TEOS") Support initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the TEOS Portfolio. The value of regular upgrades to applications in the TEOS Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the TEOS Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the TEOS portfolio areas. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	304,832	0	17,320	187,056	0	100,456	4,417	3,673	0	0	0	744
366	2021	Application Currency and Enhancements - TEOS - O&M	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Application Currency and Enhancements for Transmission, Engineering, and Operational Support ("TEOS") initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the TEOS Portfolio. The value of regular upgrades to applications in the TEOS Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the TEOS Portfolio. The scope of upgrading these applications encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the TEOS Portfolio. Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	31-Dec-21	(1.00)	0	0	0	0	0	186,196	0	0	170,042	0	16,154	
367	2021	Electric High Voltage Distribution Monitoring System Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Electric High Voltage Distribution Monitoring System Upgrade project upgrades the software components of the High Voltage Distribution (HVD) Supervisory Control and Data Acquisition (SCADA) system, which is used to monitor and control generation, transmission, and high voltage distribution voltage circuits. The upgrade will provide new functionality and enable new capabilities. Upgrading the HVD SCADA software adds value to the Company by ensuring that problems are promptly addressed when they occur. Additionally, the upgrades include enhancements and fixes to the core product that support reliability of electric delivery, ensuring that customers are receiving the energy they need when they need it. Also, because the reliability of the re-distribution system is dependent on availability of the HVD system, staying current with the solution provider upgrades is critical to the reliability of the distribution system. The scope of this project is an upgrade to the HVD SCADA software which encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, (4) updating documentation related to the integration changes. The alternative considered was to defer the biennial upgrade of the SCADA application. The Company targets SCADA biennial application upgrades to stay within two versions of the current application state to maintain application stability and cyber security, given the critical nature of the SCADA system. This alternative was not chosen due to increased security risk and need for vendor support of the application.	30-Nov-22	(1.00)	142,000	0	0	30,000	100,000	12,000	5,000	0	0	0	5,000	

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Information Technology Department

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
368	2021	Electronic Geographic Information System (GIS) Platform Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will upgrade the GIS platform applications, servers, application hosting servers, and databases to the next major versions to remediate the outdated platform technology. The project adds value by: (1) remediating GIS, Windows, and Citrix technology obsolescence, creating a stable and sustainable foundational platform for critical business functions; (2) improving platform resilience by remediating current single-point-of-failure areas; (3) sharing technology resources (servers, databases, etc.) across gas and electric platforms to optimize them, yet still ensuring the ability to meet diverse business area functional needs; and (4) adding new capabilities for spatial analytics, real-time visualization of data, and three-dimensional infrastructure network traceability. In addition, the GIS platform remediation work establishes the foundation necessary for the full implementation of the Electric Distribution Asset Management Program. Project scope includes: (1) building GIS platform servers, software, and data; (2) expanding GIS platform capabilities; (3) re-architecting current single-point-of-failure areas; (4) designing and developing new architecture; (5) upgrading database technology; (6) re-platforming the application hosting platform (Citrix); and (7) upgrading and reconfiguring applications that utilize the GIS platform. The alternatives considered for the project were to: (1) delay implementation, pay a premium for extended Windows support, and incur cyber security and business continuity risk for the application hosting platform and the GIS platform; (2) upgrade the platform to the Utility Network as a "big bang" approach without the incremental GIS version upgrade; and (3) perform an iterative platform upgrade to the next GIS version with application platform hosting upgrade, server replacements, and database upgrade, and shift the Utility Network upgrade to a future year. See Jeffrey D. Tolonen's testimony for full project synopsis.	31-Dec-21	(1.00)	768,000	0	0	175,000	350,000	243,000	312,125	0	55,000	75,000	175,000	7,125
369	2021	Electronic Shift Operations Management System Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will perform a major upgrade of the eSOMS application, including servers. This project will completely rewrite the clearance and narrative logs, and the safety critical emailer functionality to enable compatibility with the new version. The project will add value by: (1) reducing the human struggle with manual workarounds and old applications; (2) empowering employees with proper electronic tools to meet customer expectations; (3) enabling process improvements to deliver outages more effectively; (4) reducing plant downtime; and (5) increasing reliability. The scope of this project includes: (1) assessing any new functionality for value to the Company; (2) replacing the servers and upgrading the application software; (3) making necessary configuration changes; (4) testing any integrations to or from the application; (5) testing the upgrade; and (6) updating documentation related to the integration changes. Alternatives considered include: (1) continuing to use the application without vendor support at the risk of a critical application issue that results in an employee safety incident, extended plant production outages, prolonged plant reliability issue, or regulatory or compliance violation; (2) instituting the manual business continuity process until the application upgrade is possible, which would increase the previously mentioned business risks; or (3) revisiting the decision to utilize eSOMS and replace the application by customizing SAP. The option to upgrade the existing eSOMS application was chosen to minimize cost and risk to the Company and its employees.	30-Nov-21	(1.00)	34,640	0	34,640	0	0	0	214,674	0	0	171,600	26,400	36,674
370	2021	GIS Payplus Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will upgrade the Payplus Objects suite of applications to the most recent version available per vendor recommendation. The GIS Payplus application is critical to creating electronic and paper correspondence for customers, including bills and dunning. The value of this project includes: (1) providing a more stable operational model by upgrading to the most recent version available; and (2) resolving tuning and stability issues with the vendor. The scope of the project is to upgrade the various licensed products that comprise the Payplus Objects suite of applications. As this is an upgrade of an existing system, the alternative considered was to delay the upgrade and continue operating with the current version. This alternative was not chosen due to the risk of application stability and the inability to maintain cyber security patching.	01-Sep-21	(1.00)	0	0	0	0	0	144,227	0	0	128,700	0	15,527	
371	2021	MISO Market System Replacement	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Midcontinent Independent System Operator (MISO) Market System Replacement project will implement software changes in response to the announcement that MISO has determined that its current market system will be obsolete in five to seven years given evolving cyber security standards and increasing market complexity. To address these concerns, MISO is proposing a \$134M, seven-year incremental (modular) replacement to transition to a new market platform. This project adds value by implementing software changes in response to MISO's system changes to ensure the company can participate in the MISO market. Project scope includes significant change requiring updates to the Power Costs Inc. (PCI) software used for market interaction and settlements with MISO. These updates will be applied incrementally as MISO and PCI release the updates. Alternatives considered in order to continue to interact with MISO include: (1) replacing PCI with a new product, or (2) updating the PCI solution. The alternative to replace PCI would require significant investment in research, product evaluation, people resources, and financial investment. Due to the schedule for updates provided by MISO, and the significant footprint of the PCI solution, the alternative to upgrade was selected.	16-Dec-25	(1.00)	440,000	190,000	20,000	75,000	90,000	65,000	66,000	35,000	0	20,000	10,000	1,000
372	2021	MISO Market User Interface Changes	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Midcontinent Independent System Operator (MISO) Market User Interface (MUI) Changes project will update the Power Costs Inc. (PCI) software used for market interaction and settlements with MISO. These changes are required as a result of significant MISO redesign of the MUI to modernize technology and improve cyber security. The project will provide benefits to the Company by meeting required changes to the MUI initiated by MISO, modernizing technology and improving cyber security. This is a significant change that requires updates to the PCI software used for market interaction and settlements with MISO. These updates will be applied incrementally as MISO and PCI release the updates. Each update requires installation in the test environment, regression testing, and user acceptance testing prior to implementation in production. Alternatives considered in order to continue to interact with MISO include: (1) replacing PCI with a new product, or (2) updating the PCI solution. The alternative to replace PCI would require significant investment in research, product evaluation, people resources, and financial investment. Due to the schedule for updates provided by MISO, and the significant footprint of the PCI solution, the alternative to upgrade was selected.	01-May-21	(1.00)	65,000	0	0	15,000	40,000	10,000	68,000	35,000	15,000	10,000	5,000	3,000
373	2021	Replacement of Testing Software for the Electric Meter Technology Center	Upgrades & Replacements (Business Partner)	Software (Intangible)	The Replacement of Testing Software for the Electric Meter Technology Center (MTC) project will replace the existing Winboard software for MTC test boards and will implement four multi-position test boards to increase the output of tested meters and ensure future software stability. The project will add value by: (1) increasing meter testing output from approximately 150 to roughly 225 meters a day, enabling Company employees to work more efficiently; (2) implementing a multi-position test board that integrates meter qualification, meter demand, and meter accuracy; and (3) avoiding increasing Microsoft support costs from \$50.00 per computer to \$100.00 per computer in 2021. An additional benefit specific to the new software is the creation of a centralized database where all the boards access one database, Advance Metering Infrastructure (AMI) device support, power quality capture, and web reports. The project scope includes: (1) replacing the Winboard software with a version compatible with the Company's current Windows operation system version; (2) writing new interfaces for test boards to SAP; and (3) replacing four existing test boards with multi-position test boards. Alternatives considered include: (1) purchase new software, rebuild interfaces, and replace four single-position test boards with two three-position test boards; (2) purchase new software and rebuild interfaces, but do not replace the four single position test boards; and (3) do not complete the upgrade. The second and third options were not chosen because they do not allow us to reap the benefits of the new three-position test boards. The first option was chosen because it allows the Company to keep old test boards for additional capacity and creates more efficiencies, which result in a higher output of tested meters.	31-Dec-20	(1.00)	0	0	0	0	0	24,280	0	0	0	34,280	0	

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
374	2021	Service Suite Upgrade	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will: (1) implement a new version of Service Suite Work Management that allows for easier distribution to the field, and (2) maintain a current version on a vendor supported platform for this critical enterprise application used across Operations and Engineering. The product enhancements in Service Suite and TC Technology's Mobile Information Management System ("MIMS") will be implemented to provide additional business value and benefits. The project will add value by: (1) implementing a new version that provides the highest level of vendor support for this 24x7 critical system that serves over 100,000 work orders weekly; (2) implementing Service Suite Fieldworker Mobile for a touch-based interface and migration to cloud-based configurations; (3) providing live traffic display on Dispatch Application; and (4) improving readability and usability of Dispatch Application schedule view. The scope of the project includes: (1) implementing the new version of Service Suite Work Management; and (2) implementing MIMS for mobile mapping on the field device to improve safety, response, usability, and supportability of the critical application. The alternatives considered and determined not to be viable included: (1) remaining on the current Service Suite version, which would cause continuation of additional manual steps in the emergency response and work assignment; or (2) creating a custom-developed solution, which would increase waste and inefficiency without the same level of Service Suite integration. Upgrading Service Suite was chosen because it allows the Company to stay on the current and supported version of the application already being used, and also adds functionality that will provide additional business value and benefits.	31-Dec-21	(0.63)	1,446,068	0	0	904,320	207,840	333,708	1,249,726	0	0	64,152	33,000	1,146,514	
375	2021	Time Entry and Expense Reports - Flash Remediation	Upgrades & Replacements (Business Partner)	Software (Intangible)	This project will replace Adobe Flash with HTML 5 technology. Time entry and approvals, and expense report approvals currently leverage Adobe Flash technology. By the end of 2020, Microsoft will remove the ability to run Adobe Flash in Microsoft Internet Explorer. The value of the project includes: (1) reducing security vulnerabilities that exist in Adobe Flash; (2) ensuring employees and managers can enter and approve time so employees are paid on time; and (3) ensuring managers can approve expense requests for reimbursement. The scope of the project includes: (1) replacing Adobe Flash technology with HTML 5, and (2) eliminating security risks associated with Adobe Flash. Alternatives considered for this project were to: (1) use SAP Fiori technology that provides mobile capabilities and reduces license and support costs with the external vendor, and (2) implement other software applications such as Concur. The first alternative would reduce functionality and impose a major training impact on more than 6,000 employees. The second alternative would require: (1) integration with SAP, (2) retraining of all employees, (3) additional licensing and support costs, and (4) custom development of rules and redevelopment of enhancements. The option to replace the current Adobe Flash solution with HTML 5 technology was selected because it is the least costly option with little to no impact to the Company's employees. This solution does not require retraining or the development of new custom time entry rules or enhancements.	22-Oct-20	(1.00)	0	0	0	0	0	19,800	0	0	0	19,800	0		
376	2021	SubTotal Upgrades & Replacements (Business Partner)	Upgrades & Replacements (Business Partner)					4,942,793	190,000	127,384	2,342,640	787,840	1,494,929	3,344,862	103,304	89,800	1,534,267	293,480	1,322,949	
377	2021	AccessNOW	Security	Software (Intangible)	This project will implement configurable identity access management functionality and best practices, and will enforce compliance. This project will add value by: (1) reducing waste and failure points through automation; (2) improving and standardizing the business partner experience; (3) centralizing access management; (4) ensuring regulatory compliance; and (5) ensuring system stability and continuous improvement. The project scope includes implementing additional integrations to Active Directory domains that are not currently connected to the AccessNOW application and integrating with the Company's SAP system to allow for automation of SAP role access provisioning. In addition, the project scope includes: (1) design, configuration, and testing of technical connections; (2) completion of application support pack upgrades to the next versions to maintain system stability and to stay current with vendor releases; and (3) review and implementation of enhancements to improve and standardize the business partner experience. As part of the review process the alternative considered was using manual processes. This option was not chosen because it was deemed too costly and inefficient.	01-Aug-19	(0.96)	346,400	0	0	0	346,400	0	33,000	0	0	0	33,000	0	0
378	2021	ARP - Cyber Security	Security	Network	This project will replace cyber security infrastructure to support increasing system and application demands, and to prevent system failures and service interruptions. This program includes projects that bring value to the Company by maintaining the currency of the security infrastructure and core enterprise software. These are used to ensure the stability of technology for business operations. This project will support continued systems stability. The scope encompasses: (1) evaluation, validation, and replacement of cyber security firewalls and servers; and (2) asset and application upgrades. As part of the review process, the alternative considered was not to upgrade or replace assets as required, which is likely to introduce security risks, system vulnerabilities, and out-of-warranty repair costs.	31-Dec-21	(1.00)	256,680	0	171,120	85,560	0	0	0	0	0	0	0	0	
379	2021	CRISP	Security	New Computers / Hardware	This project will implement the continuous readiness in information security program, which is a cyber security tool coordinated by the Department of Homeland Security ("DHS"). The tool watches network traffic coming into and leaving our environment and attempts to identify attacks from highly skilled actors, typically associated with other nations. Details regarding identified attacks are then anonymized and sent to DHS for analysis. CRISP does not identify cyber crime attacks or commodity malware. CRISP focuses on attackers who are motivated to impact critical infrastructure. Nearly all major utilities in the United States have or are in the process of implementing CRISP. Completion of this project will provide value to the Company by: (1) Receiving alerts from DHS based upon classified intelligence regarding who is attacking the Company; (2) Analyzing alerts to determine if the Company was impacted; (3) Receiving alerts about attacks at other utilities, which will enable the Company to proactively mitigate potential future attacks; and (4) Enabling the company to share threat intelligence with other utilities for the benefit of our nation's critical infrastructure. The scope of this project encompasses: (1) forming an agreement with DHS; (2) procuring a network sensor; and (3) working through architecture designs, deployment, and testing of the network sensor. As part of the review process the alternative considered was not to implement CRISP, which limits the Company's ability to respond to future advanced threats and share information for the benefit of the nation.	31-Jul-20	(1.00)	0	0	0	0	0	115,500	115,500	0	0	0	0	0	
380	2021	Cyber Security Enhancements	Security	Software (Intangible)	This project will: (1) fund emerging or unplanned cyber security activities resulting from audits, incidents, or a changing threat landscape; and (2) support initiatives that may not meet the criteria for a formal project. Requests for this funding are governed by project management and the security governance board, which is comprised of representatives from each area of Security and meets monthly to evaluate and prioritize the work. The board assesses requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in scope are small changes and functionality improvements to existing software application investments for Security. The enhancement requests are fulfilled to provide functionality for areas such as security program management, cyber security incident response, corporate physical security, compliance, privacy and risk management, and cyber security engineering and standards. As part of the review process, the alternative considered was not to upgrade or enhance software or hardware, which could reduce the Company's ability to respond to future threats and vulnerabilities.	31-Dec-21	(0.96)	0	0	0	0	0	0	118,800	0	0	118,800	0	0	

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381	2021	Data Governance	Security	Software (Intangible)	This project will develop an enterprise-wide data governance solution as an aspect of an Enterprise Data Privacy and Management Plan. Completion of this project will provide value to the Company through programmatic data governance resulting in increased data quality which will: (1) decrease operational costs; (2) increase employee satisfaction and productivity; (3) enhance system performance efficiencies; and (4) support better informed decision making. The scope of this project encompasses: (1) achieving consistency in collecting and reporting data across various organizational users and source systems; (2) achieving high quality in the collection, maintenance, analysis, and reporting of data; (3) responding to data issues; (4) promoting and ensuring consistent enterprise-wide data definitions, increasing controls regarding data creation, modification, and access; (5) proposing and implementing system controls for maintaining data quality (e.g., field-level validation, reporting, batch job cleaning); (6) facilitating the development of a data quality assurance process, including policy, process, and system (including system access) reviews against data; and (7) creating a process to include a holistic review of data management methodology for the effective collection, management, and access for data quality control. Alternatives considered include: (1) remain at the current state and not implement a data governance solution; (2) evaluate, select, and implement a technology solution to fulfill the Company's data governance needs; or (3) determine if any existing solutions the Company has built or purchased can meet the needs. The alternative selected is to move forward with implementing a data governance solution. A part of the project will be to evaluate both existing and new technology solutions.	30-Nov-21	(1.00)	1,108,480	692,800	0	270,192	0	145,488	33,000	0	0	0	33,000	0	0
382	2021	Email Protection	Security	Software (Intangible)	This project will implement an enterprise-wide email filtering toolset to ensure protection against current cyber security risks, including malware and non-malware threats, email fraud, and ransomware. Completion of this project will provide value to the Company by: (1) evaluating and procuring a leading email filtering toolset; (2) configuring the toolset to help eliminate human error; and (3) implementing the toolset enterprise-wide for company email. The scope of this project encompasses protection for both incoming and outbound emails. The alternative considered was to continue with focused cyber security and phishing training programs for employees. Security awareness training has reached peak effectiveness. The Company is choosing to implement a technical solution to further mitigate the risk of phishing attacks caused by human error.	12-Sep-20	(1.00)	0	0	0	0	0	0	99,000	99,000	0	0	0	0	0
383	2021	Fusion Center	Security	Network	This project will procure a location to combine the Security Command Center and the Cyber Security Incident Response Team ("CSIRT"). This project will focus on constructing a physical office space to facilitate these two teams working in a collaborative fashion. This project will add value by: (1) co-locating physical and cyber team members for process efficiencies and better integration of tools in the event of a security incident that requires both teams' collaboration; (2) having a dedicated emergency operations area for use in the event of a security incident (cyber or physical or both) providing the tools needed to keep stakeholders informed; (3) evaluating and potentially implementing a shared intelligence capability across both areas; and (4) further integrating existing processes and workflows. The project scope encompasses building an integrated Security Command Center across physical and cyber security domains that takes into consideration physical space, technology, process integration, analytics, and intelligence capabilities without changing existing organizational structures. As part of the review process, the alternative considered was to continue to run a separate Security Command Center for physical security incident response and a separate Cyber Security Incident Response Center. The two centers would continue to run separate, non-integrated systems and tools. This option was not chosen as it does not provide the integrated space and tools needed for physical and cyber security teams to collaborate to resolve critical incidents that pose both a physical and cyber threat.	01-Dec-21	(0.99)	304,545	0	0	199,640	0	104,905	77,116	0	0	66,000	0	11,116	
384	2021	Fusion Center Technologies	Security	New Computers / Hardware	This project will further integrate the physical and cyber security teams through the implementation of new technologies. This project will evaluate, replace, and implement new capabilities for monitoring physical sites and technology assets, significantly improving the Company's quality and timeliness of detection and response to both physical and cyber security incidents. Completion of this project will provide value to the Company through: (1) implementing a new toolset to move MITRE Adversarial Tactics, Techniques, & Common Knowledge ("ATT&CK") coverage from 30% to 80% (MITRE ATT&CK Framework is an industry-leading framework to detect malicious activity); (2) replacing the vulnerability management platform, Qualys, which has several capability gaps estimated to be remedied at a significant cost; and (3) evaluating and implementing a biometric toolset, such as facial recognition, to eliminate an unauthorized individual to follow an authorized individual into a facility and to proactively alert of unwanted or unknown personnel at Company facilities. The scope of this project encompasses: (1) all endpoints and servers, and all facilities that house employees; (2) defining system scalability requirements; (3) designing new platform architecture; and (4) purchasing and implementing hardware and software once alternatives are reviewed. The alternative considered was to remain on current toolsets and not implement additional physical security countermeasures, impacting the quality and timeliness of detection and response to both physical and cyber security incidents. The Company is choosing to replace and implement new capabilities to provide further efficiencies and capabilities.	31-Jul-21	(1.00)	692,800	346,400	0	242,480	0	103,920	66,000	0	0	66,000	0	0	
385	2021	Lock and Key Management System	Security	Software (Intangible)	This project will identify and implement a physical smart lock and key management system throughout the Company's service territories. Current estimates show there are approximately 10,000-14,000 locks throughout the state, and the Company has no system to properly manage ownership of specific physical keys and control who uses them to access sites. Completion of this project will provide value to the Company by: (1) assessing and taking inventory of current locks and keys that are used throughout the state; (2) determining core functionalities needed to ensure proper lock and key management state-wide; and (3) reviewing and implementing a solution to give the physical security team a lock and key management capability. The scope of this project includes: (1) assessing the type of locks and keys used, and at what sites they will be needed to properly plan this project; (2) determining the levels and functions of different smart lock systems available; and (3) purchasing and implementing a lock and key management system based on assessment findings. Alternatives considered include: (1) remain at the current state and forego implementing new lock and key management capabilities; or (2) evaluate, select, and implement a smart lock and key solution. The preferred option is to mitigate known and observed risks with the lock of lock and key management capabilities.	01-Oct-21	(1.00)	692,800	346,400	103,920	242,480	0	33,000	0	0	33,000	0	0		
386	2021	Physical Security Hardware Refresh	Security	Network	This project will enhance or replace physical security assets to provide improved visibility into potential security concerns and resolve incidents. The Company has several thousand physical security asset devices currently in use, including security cameras, motion detectors, intrusion detection systems, and card access systems. The value provided by completing the project is to: (1) maintain compliance; (2) reduce redundancies and gaps in functionality; and (3) optimize overall performance. An integrated solution is efficient and allows for centralized management; situational awareness; real-time monitoring; compliance with regulations and guidelines; and faster, more effective and consistent responses to emergencies and incidents. Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems. As part of the review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	30-Nov-19	(1.00)	1,197,840	0	1,026,720	0	171,120	0	0	0	0	0			
387	2021	Radar Intrusion Detection	Security	Software (Intangible)	This project will integrate and deploy new radar detection technology to enable the detection and tracking of individuals at Company sites. The radar would alert in real time and allow for the dispatch of Security or law enforcement to investigate. Completion of this project will provide value to the Company through evaluating and selecting a more reliable intrusion detection solution that reduces the number of false alarms and prevent resources from being sent to a site for unnecessary investigations. The scope of this project encompasses: (1) assessing radar intrusion detection solutions to obtain a solution that fits the needs of the Company; (2) establishing a priority list of sites where the solution will be rolled out; and (3) installing updated radar intrusion at physical sites in the Company's service territory with a focus on critical infrastructure sites, including but not limited to gas compressor stations, electric substations, and hydro-electric sites. Alternatives considered include: (1) keeping the current fence intrusion detection solution; or (2) evaluating, selecting, and implementing a new intrusion detection solution. The preferred option is to implement a new solution, given limitations with the current solution that result in multiple false alarms that must be investigated.	31-Aug-22	(1.00)	1,039,200	0	1,039,200	0	0	33,000	0	0	33,000	0	0		

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	
388	2021	Replace and Re-badge	Security	New Computers / Hardware	This project will: (1) replace and re-badge existing card readers with human interface device multiclass readers; and (2) re-badge employees and contractors with more secure badges, mitigating cloning vulnerabilities with the current legacy technology being used. The value of completing the project is avoidance of the security risk for one of the Company's key security controls. The scope of the implementation includes: (1) replacing existing card readers, (2) re-badging to i-class badges, and (3) encrypting the new badges. As part of the review process, the alternative considered was to continue with current badge readers, which have a potential to be cloned. This alternative was not chosen due to the potential to high likelihood of security vulnerabilities and risks.	01-Dec-20	(1.00)	346,525	103,514	173,200	49,725	0	24,027	33,000	16,500	16,500	0	0	0	
389	2021	2021 SubTotal Security	Security					5,985,271	1,489,174	2,514,540	1,086,977	517,540	278,340	641,416	231,000	16,500	382,800	0	11,116	
390	2021	800 MHz Modernization Project	IT Service Delivery	Network	This project will replace antiquated head-end (the main audio routing switch for tower sites and dispatch consoles), tower site, dispatch, and radio infrastructure, as well as user subscribed equipment with infrastructure that meets current Project 25 ("P25") standards. P25 standards are for digital mobile radio communications that are used by North American public safety and dispatch organizations. The current equipment was implemented in 1994 and is no longer manufactured or sold, so the Company is unable to find replacement parts. This project creates value for the Company by: (1) moving the Company to current production equipment that can be replaced and is more stable, (2) enabling the organization to migrate from an unsupported to a supported platform, and (3) allowing for quicker response to electric outages, and (4) increasing customer and employee safety. The scope of the project includes: (1) the design, (2) configuration, and (3) implementation of P25 systems for headends, tower sites, dispatch consoles, and subscriber equipment. The alternatives considered were to: (1) remain with the current system, and (2) forklift the migration to other manufacturers and architecture. Option 1 was not selected because the equipment is not supported, and the organization is unable to find replacement parts. Option 2 was not chosen because there would be a larger learning curve, it would be more disruptive to business, and it would require a complete replacement of the existing system. The Company chose to go with the upgrade because it has been incrementally adopting the P25 standards through the wireless ARP, and this prudently utilizes current and ongoing investments. The upgrade is the easiest and least disruptive path to migrate to the P25 standards.	01-Jul-23	(1.00)	5,091,390	0	3,608,546	427,800	808,257	196,788	0	0	0	0	0	0	0
391	2021	ARP - Collaboration Asset Refresh	IT Service Delivery	Network	This project will replace the Company's collaborative tools and equipment. This project creates value by: (1) ensuring that the Company's audio, visual, telephony, and other communications systems are stable and reliable, and (2) beginning the foundational retirement of the legacy Avaya P88 systems that have reached end of mainstream manufacturer support. The project scope consists of: (1) annually replacing collaboration assets, and (2) installing new collaboration assets to account for organic growth requirements. The alternatives considered were to: (1) refresh visual assets and a portion of the audio assets, (2) refresh a portion of the audio assets only, and (3) refresh visual assets only. These alternatives were not chosen due to the risk inherent with a partial replacement of assets which includes: (1) a reduced supply of equivalent replacement Avaya parts that are no longer being produced, and (2) an erosion of the knowledge technicians possess on discounted systems in favor of education on the newest available technology.	31-Dec-21	(1.00)	422,302	57,040	96,574	96,968	114,080	57,640	331,265	36,300	210,020	25,740	52,800	6,405	
392	2021	ARP - Field Device Asset Management (FDAM)	IT Service Delivery	New Computers / Hardware	This project will replace field devices. This project creates value for the Company by: (1) mitigating potential costs for hardware repairs, and (2) allowing field workers to complete their job tasks. The project scope consists of replacing field device assets. The alternatives considered were to: (1) extend refresh cycles from four years to five years, and (2) run the assets to failure. The selection of a four-year cycle was deemed to be the best solution because replacement in less than four years would result in additional unnecessary expense for replacement of assets that are still in peak operating condition and replacement cycles that exceed four years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, which negatively impact employee productivity.	31-Dec-21	(1.00)	1,575,480	0	1,537,817	20,999	8,999	7,664	7,888	0	0	5,522	2,366	0	
393	2021	ARP - Operational Technology Storage Area Network	IT Service Delivery	Network	This project will refresh aging SAN with Unity SANs. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support and enhance customer interactions, as well as ensure the stability of technology for business operations that are in service of the Company's customers. The project scope consists of: (1) annually replacing SAN hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. The alternative considered was to purchase extended maintenance. This alternative was not chosen due to the risk of increased downtime of critical infrastructure and maintenance costs. The cost of bringing personnel on site to make system corrections in the event of a serious interruption is higher than the cost of buying new.	31-Dec-21	(1.00)	278,355	0	236,716	28,520	0	13,119	30,360	30,360	0	0	0	0	
394	2021	ARP - Operational Technology Support	IT Service Delivery	Network	This project will replace dated and obsolete servers and workstations. This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support the operation of the Company's critical electric infrastructure. The project scope consists of: (1) the annual replacement of compute hardware under the program, and (2) installing additional new compute capacity to account for organic growth requirements. Extending maintenance is not a viable alternative as current systems do not provide sufficient capacity for new electric system support capabilities.	31-Dec-21	(1.00)	502,403	0	367,908	92,120	0	42,376	34,568	16,500	0	16,500	0	1,568	
395	2021	ARP - Printer Asset Management (PAM)	IT Service Delivery	New Computers / Hardware	This project will replace printers, plotters, and multi-function printing devices. This project creates value for the Company by: (1) improving the dependability of these printer devices for employees, (2) averting increased costs due to hardware repairs, and (3) ensuring compatibility with enterprise print applications. The project scope consists of the annual replacement of printer assets under this program. The alternatives considered for the project included looking at refresh cycles from three to seven years and running the assets to failure. The selection of a five-year cycle was deemed to be the best solution since anything less than five years would result in additional unneeded expense for replacement of assets that were still in peak operating condition, and anything greater than five years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, negatively impacting employee productivity.	31-Dec-21	(1.00)	715,356	0	640,506	51,267	0	23,583	11,997	0	0	10,956	0	1,041	
396	2021	ARP - Server and Storage	IT Service Delivery	New Computers / Hardware	This project will replace or augment server and storage infrastructure for the Company. This project creates value for the Company through: (1) improved stability and availability of business critical applications by proactively replacing server and storage hardware assets prior to the chance of hardware failures increasing, and (2) ensuring that adequate resources are available to support application demands after five to seven years of actual use. The scope of this project encompasses: (1) replacement of server and storage hardware assets, and (2) installation of additional new computers and storage capacity to account for organic growth requirements. The alternative considered was to purchase extended maintenance. This solution was not selected because full support would not be offered after seven years and maintenance costs would increase. The Company continues to refresh these critical technologies based on a five- to seven-year refresh cycle to mitigate the risk of failure.	31-Dec-21	(1.00)	8,550,238	0	7,472,194	738,586	0	339,658	411,177	330,000	0	13,860	66,000	1,317	
397	2021	ARP - Wireless Network	IT Service Delivery	Network	This project will replace portions of the company's aging wireless systems. This project creates value for the Company by: (1) ensuring real-time communications between Company crews and dispatch locations, (2) ensuring efficient gas leak and electric outage response times for customers, and (3) maintaining critical infrastructure and regulatory compliance. The project scope consists of: (1) replacing wireless assets annually, and (2) installing additional new wireless assets to account for organic growth requirements. The alternatives considered for system replacement were: (1) running parallel systems while the new system is deployed and the old system is dismantled, and (2) leasing a system from a vendor. The first alternative was not chosen since it would be highly disruptive due to systems having to run independently of one another, and would have a high cost associated with the need to acquire additional radio frequency spectrum. The second alternative was not chosen since the Company would be dependent on the response times offered by a shared vendor system that offers lower system reliability. The Company chose to replace existing assets based on its refresh cycle to avoid extended support cost, and to provide a seamless transition that allows both the old and new systems to interact with little to no disruption to the end user.	31-Dec-21	(1.00)	2,037,772	0	1,963,214	11,408	0	63,150	234,969	0	227,040	2,720	4,950	259	

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398	2021	SAP - Workstation Asset Management (WAM)	IT Service Delivery	New Computers / Hardware	This project will replace and install new desktops, laptops, and tablets. This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations prior to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks. The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources. The alternatives considered were to: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle only on desktops from four years to five years, and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ASP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	31-Dec-21	(1.00)	5,107,804	0	4,903,180	140,153	0	64,471	86,724	0	0	79,200	0	7,534
399	2021	SAP Data Archiving	IT Service Delivery	Software (Intangible)	This project will move outdated data from an online database to offline storage. This project will create value for the Company by: (1) increasing system stability by reducing data growth, and (2) controlling maintenance costs associated with data storage. The project scope includes: (1) archiving data based on the fastest growing and largest objects in SAP; (2) building and archiving solutions that allow the business to retrieve archived data in the required form, and (3) setting up the solution so that the business areas meet compliance standards. Three alternatives were explored and determined non-viable for the project: (1) allow the database to grow in size, which would put system performance at risk and result in prohibitive storage costs; (2) decrease the overall scope and archive less projects, which would result in minimal positive impact to system growth and significant storage-related costs; and (3) increase the scope and archive more objects in a shorter timeframe, which would result in significant cost increase over shorter time period. After considering each of these options, it was determined that the current scope of the project was the best strategy to address the problem while balancing annual spending.	31-Dec-21	(1.00)	380,001	0	0	259,800	0	120,201	171,270	0	33,000	66,000	66,000	6,270
400	2021	SAP Data Encryption	IT Service Delivery	Software (Intangible)	The project will implement Cyber Security requirements for encryption of Personal Identifying Information (PII) data "at rest" and "data in transit" in Company SAP data bases.	01-May-21	(1.00)	0	0	0	0	0	0	838,035	79,200	0	693,000	0	65,835
401	2021	SAP Optimization and Tuning	IT Service Delivery	Software (Intangible)	This project will maintain and improve the operation of the SAP system by addressing data issues within the system, optimizing database structures, and fixing sub-optimal code. The project creates value by improving the performance of the SAP system, which improves the customer's online self-service experience and allows employees serving the customer to complete timely transactions. The project scope includes: (1) normalizing multiple accounts assigned to a single business partner, (2) purging duplicate or unnecessary records, (3) purging unneeded technical data, (4) reviewing and optimizing custom code, and (5) implementing minor service pack updates provided by the vendor. The alternative considered was breaking the scope into individual work efforts to be individually completed. This alternative was not selected because the efforts are interrelated and completing them separately could lead to duplication of work effort. The selected project scope was determined to be the best balance of maintaining overall SAP performance and optimizing cost.	31-Dec-21	(1.00)	0	0	0	0	0	0	231,919	0	0	103,305	118,800	9,814
402	2021	Software Platform Refresh 2019	IT Service Delivery	Software (Intangible)	This project will upgrade server operating systems, hypervisors (virtual machine monitors), and databases to retain low-cost, unlimited server support. These platforms support all the Company's applications including critical company and customer applications. The project will add value for the Company by: (1) avoiding costs for special maintenance agreements required at the end of normal manufacturer support; (2) ensuring reliability and compliance with information security requirements; (3) improving data center environment stability; and (4) avoiding the need for high risk upgrades that cross multiple versions. The project scope includes: (1) upgrading operating systems and databases on servers that are within three years of end of support, and (2) maintaining hypervisors at the current version for stability and performance. A funding options matrix was completed to review the alternatives. The options were: (1) fund the full project for \$1 million and pay no support liability in 2020; (2) fund \$750,000 of the project and pay \$1.6 million in support liability in 2020; (3) fund \$500,000 of the project and pay \$2.4 million in support liability in 2020; and (4) not funding the project and pay \$3.3 million for support liability in 2020. Alternative options 2-4 were not selected due to the high cost of support liability. Option 1 was chosen to avert these costs and to ensure system stability.	31-Dec-21	(1.00)	0	0	0	0	0	451,398	0	0	412,236	0	39,162	
403	2021	SubTotal IT Service Delivery	IT Service Delivery					24,961,302	57,040	20,876,656	1,890,421	931,136	928,649	2,841,669	492,360	470,660	1,429,039	310,916	139,195
404	2021	Enhancements - Cloud Automation	Enhancements	Software (Intangible)	The Company's existing cloud automation platform consists of a number of scripts that automate the build of infrastructure like servers, operating systems, databases, and application installations in minutes or hours. These scripts include security hardening and application of the latest patches. This project will maintain and improve these scripts to ensure they meet the latest security policy and requirements, and any changes to the way the operating systems, databases, or applications are configured. The value of this project is inherent in the time savings and quality improvement, as it used to take weeks or even months to manually build infrastructure servers, which is prone to human error. The scope includes updating and improving the cloud automation platform to align with the Company's current security, database, operating system, network, and other application standards. The alternative would be to continue as-is with existing scripts; however, this introduces security risks and additional manual steps to build infrastructure, which will lead to increased costs and risk of manual errors.	31-Dec-20	(1.00)	542,769	0	0	371,760	0	171,009	117,986	0	0	81,778	17,490	8,719
405	2021	SubTotal Enhancements	Enhancements					542,769	0	0	371,760	0	171,009	117,986	0	0	91,778	17,490	8,719
406	2021	4G SAP Implementation	BP Functionality	Software (Intangible)	There are approximately 680,000 smart meters in the Company's electric system that utilize the 3G wireless technology. The wireless carrier will be retiring the 3G network technology on December 31, 2022, requiring the Company to replace the 3G meters with 4G meters in order for them to continue functioning. This project will create the technology that enables Itron to upgrade smart meter communication modules from 3G to 4G technology. This project will add value by continuing the ability to: (1) communicate with the smart meters, (2) provide accurate and timely bills to customers, (3) execute remote turn-ons and turn-offs, (4) receive outage information, and (5) administer demand response events. The project scope includes: (1) building technology interfaces to meter installation vendor technology, which include an interface to extract list of 3G meter information, an interface for posting work order completion data in SAP, and an interface for tracking exceptions and errors; and (2) end-to-end testing of the interfaces and back-end technology (meter-to-cash processes). The alternatives considered were: (1) migrating to a new meter platform, and (2) not implementing 4G technology in 3G meters. The first option would require a substantially larger modification to all supporting systems and infrastructure. The second option is not feasible since the depreciating communications technology will eventually make the meters non-communicative. The option of implementing 4G technology was chosen so the Company can continue to reap the benefits of the smart meter technology previously implemented. These benefits include: (1) improved timeliness and billing accuracy; (2) retaining ability for remote disconnect and reconnect of services; (3) administration of demand response programs; and (4) enabling automated meter reads, outage reporting, and faster restoration.	01-Jul-20	(1.00)	0	0	0	0	0	87,450	8,250	0	0	79,200	0	
407	2021	Accounts Payable (AP) Automation	BP Functionality	Software (Intangible)	The Accounts Payable (AP) Automation project will provide an end-to-end AP solution to optimize invoice capture via Optical Character Recognition (OCR) software; leverage workflows to automate invoice approvals and processing; manage electronic document retention; provide insights via improved reporting; and minimize human intervention. The value of completing the project includes: (1) centralizing invoice processing resulting in reductions to invoice entry costs, number of paper invoices, duplicate vendor payments, and late vendor payments; (2) automating validation of an invoice against contract rates; (3) increasing transparency of invoice processing status; (4) providing more accurate accruals; (5) improving the ability to capture discounts; (6) enabling cash flow benefits and reduced financing costs; and (7) improving internal controls. The scope of the project includes: (1) enabling OCR technology; (2) creating automated workflows for receiving, managing, routing, and monitoring invoices and related documentation; (3) automating posting of invoices; (4) creating new reports; (5) enabling electronic document retention; and (6) data cleanup. See Karen M. Gaston's testimony for full project synopsis.	17-Oct-20	0.69	0	0	0	0	0	104,280	66,000	6,600	0	31,680	0	

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408	2021	Bill Design and Delivery Transformation	BP Functionality	Software (Intangible)	The Bill Design and Delivery Transformation project will execute a bill redesign for all rates and program combinations, including necessary software replacements and a flexible print and delivery outsourcing initiative. Following this initiative all in-house services for bill print will be outsourced, reducing internal costs for the Company while providing more efficient means to update outbound printed materials. The Company and its customers will receive value from this project through: (1) reduced costs for bill print and delivery, (2) consistent and accurate delivery of bills and other printed materials, and (3) reduced calls to contact centers with improved bill format and content. The project scope includes: (1) redesigned customer billing documents with online availability, (2) an outsourced solution for bill printing and mailing, (3) an improved eBill product, (4) integrating customer billing information with third party printing source for bill print capability, and (5) an integrated digital and interactive voice response (IVR) channels for consistent customer experience. Three alternatives considered but not chosen for this project: (1) Keeping the bill delivery process the same. This option was not chosen because it would continue current state issues resulting in \$16M O&M annual expenses. Additionally, extensive time is required to modify bills which still generate 1.8M calls annually into call centers. (2) Modifying the bills for Summer Time of Use (TOU) only. This option was not selected because while modifying the bill to support the recent TOU rate is needed, this still would not address any issues with cumbersome processes, increased calls to call centers, or difficulty in modifying the bill. (3) Defer the project until a later date. This option was not chosen because the need to spend nearly \$2M to redesign the bill for TOU would still exist, along with continuing significant cost and issues with the current process. The option to transition bill print to a third party provider was selected as it provides O&M savings and reduces the complexity of the existing process. The Company will be able to more efficiently update the customer's bill to improve customer satisfaction and resolve billing issues.	14-Jun-21	(0.56)	5,205,551	0	0	2,386,240	1,732,000	1,191,111	926,970	0	0	856,000	0	0	0	0	68,970
409	2021	Business Customer Interval Web Portal	BP Functionality	Software (Intangible)	The Business Customer Interval Web Portal project will develop a new Interval Web Platform (IWP) for Business Customers to provide energy usage insight to customers and support the Company's initiatives for reducing energy demand. This project will provide value to both the Company and its customers through: (1) increased customer engagement and satisfaction through self-service capability focused on energy waste reduction, and (2) reduced calls and contacts with the Company regarding customers' energy usage and energy waste reduction options. The project scope will include: (1) implementing an interval web portal to support business customers, (2) integrating with SAP and smart meter data to identify energy usage, (3) enabling reporting and analytics capabilities for deeper understanding of customers' energy usage, and (4) enabling features for energy efficiency and demand response within the interval web portal. Three alternatives were considered but not chosen for this project: (1) Build a custom IWP for business customers. This alternative was not chosen given the high cost and unfavorable return on investment because it requires complex logic for rate analysis, rate modeling, and energy waste management improvements as well as increased maintenance costs. (2) Defer to future years. This alternative was not selected because the Company is already lagging in the industry regarding availability of interval portals to customers, and this option would further limit offers to business customers for energy waste reduction and rate analysis based on time of use, which impacts customer savings resulting from energy waste reduction and limits the impact of the Company's plan for energy reduction. (3) Forgo IWP for Business Customers. This option was not chosen as it limits customer offerings for energy management and still results in negative CXI impacts. The option to implement a Business Customer IWP was chosen because it enables customer self-service for energy management, improves CXI, and supports energy demand reduction.	14-Sep-20	(1.00)	0	0	0	0	0	0	0	165,000	165,000	0	0	0	0	0	0
410	2021	Business Process Performance Monitoring	BP Functionality	Software (Intangible)	This project will deploy the AppDynamics tool across enterprise applications like SAP, OMS, and other business critical systems. This project will create value for the Company by providing faster system restoration through automation of system troubleshooting that pinpoints the software or hardware that is causing the problem. The project scope includes: (1) SAP, (2) OMS, (3) Advanced Device Metering System, and (4) other enterprise applications. The Company evaluated multiple products for this project and AppDynamics fit the cost and functionality expectations. The others were cost prohibitive and lacked the functionality of the product selected.	01-Oct-23	(1.00)	0	0	0	0	0	0	18,390	18,390	0	0	0	0	0		
411	2021	Care Annual Updates	BP Functionality	Software (Intangible)	This project will: (1) implement software changes to improve the process for offering energy assistance to low-income customers, and (2) streamline the process for assistance agencies who utilize the application through improved user interface and updates to SAP to process various CARE requests. Upcoming modifications will be identified following an annual review of requests to prioritize the list of changes. The project will provide the following value: (1) complete modifications to the existing internal SAP application and Agency Portal to receive annual Low Income Home Energy Assistance Program ("LHEAP") funding, which is used to provide customers bill credits and arrears forgiveness; and (2) improve the data within the assistance Agency Portal, thereby making it easier to assist customers in need of LHEAP funding. The project scope includes: (1) updating the enrollment and status process, (2) allowing for flexible bill credits, (3) improving reporting, (4) updating the arrears forgiveness plan, and (5) satisfying additional regulatory requirements for the annual grant rule changes required by the Michigan Department of Health and Human Services and the Michigan Agency for Energy. The alternative considered was to continue with the current process. This alternative was not chosen since it would result in loss of grant funding, thus decreasing or eliminating energy assistance dollars for low income customers. Instead, making annual updates to the application will allow agencies to easily enroll customers on assistance programs, and allow placement of holds to stop or prolong credit activity until assistance decisions are granted. This option provides long-term proactive energy assistance to low income customers and prevents loss of grant funds. All changes are related to the existing internal SAP and Agency Portal, therefore a cloud alternative is not viable.	31-Dec-21	(1.00)	0	0	0	0	0	0	207,574	0	0	175,593	17,194	14,787			
412	2021	Centralized Demand Response Management	BP Functionality	Software (Intangible)	The Centralized Demand Response Management project is intended to centralize triggering and reporting of all Demand Response (DR) events activities into one technical solution and will be informed by the outcomes of the Centralized Demand Response Management Assessment effort. The project will add value by providing functionality to ensure Demand Response (DR) events can be collectively managed from a central solution that includes all programs and potential load shed. The upgraded Demand Response Management System (DRMS) application will be able to support the various DR Programs including new Electric Charging Vehicle programs and Time of Use (TOU) rates implemented in 2018. Finally, this upgraded application will ensure accuracy of kilowatt hour (kWh) savings and peak load reduction when a DR event is called. The scope consists of: (1) centralization of the DR event management programs including residential DR, bring your own device, electric vehicle charging and large customer DR, (2) upgrade or replacement of the Demand Response Management System (DRMS) with a potential move to a cloud-based solution; (3) provision of new DRMS system functionality to support on-going DR programs; (4) addition of DRMS functionality to account for the new TOU rates. Alternatives considered during the assessment effort include: (1) continue to maintain the current hardware and software at the risk of continued inability to centralize information for all DR programs, (2) enhance the existing solution to address centralization needs, (3) transition the application to the vendor's hosted environment and configure to address requirements for additional DR programs, and (4) replace the current solution with a new cloud solution, or (5) replace the current solution with a new internally hosted solution. The selected alternative will be made as an outcome of the Centralized Demand Response Management Assessment effort.	31-Jul-22	(1.00)	1,293,000	864,000	0	150,000	60,000	219,000	127,000	43,200	0	40,000	40,000	3,800			

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Information Technology Department

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413	2021	Centralized Demand Response Management Assessment	BP Functionality	Software (Intangible)	The Centralized Demand Response Management project is intended to explore feasibility and options to centralize control for triggering, managing, and reporting of all events for Demand Response (DR) programs and pilots. This project will provide clear understanding of the requirements needed to enable the Demand Response Management System (DRMS) to support the significant growth in Demand Response (DR) capacity as outlined in the Integrated Resource Plan (IRP). This effort will be completed prior to making a large investment in modifying or replacing the current DRMS technology. The scope for this assessment includes: (1) evaluating moving to a cloud solution; (2) fully vetting integration requirements for all proposed DR programs; (3) reviewing the backlog of requested enhancements to DRMS; (4) evaluating alternate DR solutions; (5) identifying opportunities to leverage additional functionality of current DRMS; (6) mapping DRMS features to requirements needed to support additional DR programs and the IRP; (7) documenting and reviewing "as-is" and "to-be" DR organization and processes to achieve centralized DR operated from one operations center; and (8) mapping future state for DRMS where it integrates with Distributed Energy Resource Management Solution (DERMS). Alternatives to be considered during this assessment include: (1) continuing to maintain the current hardware and software at the risk of continued inability to centralize information for all DR programs, (2) enhancing the existing solution to address centralization needs, (3) transitioning the application to the vendor's hosted environment and configure to address requirements for additional DR programs, (4) replacing the current solution with a new cloud solution, and (5) replacing the current solution with a new internally hosted solution. Due to the significant increase in DR capacity as outlined in the Integrated Resource Plan, further assessment of the technology requirements is needed.	01-Aug-21	(1.00)	0	0	0	0	0	0	305,700	0	0	60,000	240,000	5,700
414	2021	Contract Life Cycle Management	BP Functionality	Software (Intangible)	This project will implement a new contract management solution to manage the life cycle of contracts. This project will provide value to the Company and its customers through: (1) an improved user experience; (2) standardization of the supply chain platform for sourcing and contracts; (3) reduction of manual steps to select approvers; (4) integration with SAP; and (5) reduced annual subscription fee for the solution. The scope of this project includes: (1) implementing the new contract life cycle management solution; (2) integrating this solution with SAP supply chain; (3) transitioning active contract information from current solution; (4) discontinuing use of current solution for contract management. As part of the review process, alternatives considered included: (1) delay implementation of a new solution; or (2) remain on the current platform. The alternative to delay implementation was not selected because it defers a significant reduction in support costs and opportunities to reduce manual efforts. The alternative to remain on the current platform was not considered due to the intensive manual effort to route physical documents for approvals, inability to streamline source-to-pay ("S2P") workflows, and costly support model. The option to implement a new contract management life cycle management cloud-based solution was chosen after evaluation of leading vendor software applications for S2P solutions through a request for information and vendor demonstration process.	15-Feb-21	1.22	30,483	0	0	27,712	0	2,771	80,190	59,400	0	9,900	9,900	990
415	2021	Cross-Channel Analytics	BP Functionality	Software (Intangible)	The Cross Channel Analytics project will support the Operational Analytics team with furthering its technological ability through the enhancement and upgrades of several systems such as Speechminer, the speech analytics tool on the Genesys platform. Additionally, the team intends to purchase and implement the Tag Management System, Tealium IQ, enabling the use of website tags and providing a management system for those tags. This will provide the next layer to the team as they work to understand customer pain/break points that they may experience as they attempt to self serve. The Company will gain value from this project through: (1) improved ability to back test new phrases, topics, or categories; and (2) enable further analytics for the Core Team to understand and visually show what customers have called for historically and why they were not able to self serve on single web pages. The project scope includes: (1) implement hardware to retain 12 months of Speechminer data data, (2) archive and update every SMART Apply, (3) implement a tag management system for use in cross-channel analytics and self-service use cases, and (4) integrate with existing data stores for customer and Interactive Voice Response (IVR) data. Alternatives considered include: (1) Keeping Speech Miner retention at three months which results in a loss of considerable analytics ability, especially when it comes to marketing programs like Time of Use (TOU). (2) Currently no current Tag Management System so the alternative is to do nothing, but the Core Team will not have the ability to problem solve any further. Customer issues may not be identified and resolved resulting in continued calls in the contact center. (3) Utilize manual data stores/cross referencing including small databases and Excel. This option increased costs for manual processing for analytics, required complex processing to complete analysis on customer needs. (4) Increase data retention and implement a Tag Management System. This option was selected as the automation reduced the need for manual processes and increased data volume will improve our ability to identify customer needs.	01-Aug-20	(1.00)	0	0	0	0	0	0	79,200	79,200	0	0	0	0
416	2021	Customer Operations Commercial Theft	BP Functionality	Software (Intangible)	The Customer Operations Commercial Theft project will develop tools and complex reports to enable the Theft and Fraud teams to proactively identify incidents of commercial theft. This effort will provide high-confidence theft detection. The project will add value by: (1) assuring a high-probability of actual theft prior to on-site investigations; (2) increasing the safety of our field teams; and (3) eliminating unnecessary customer visits and the associated expense of dispatching a field crew. The benefits will result in reduced uncollectible expenses for our customers and the Company. The project scope includes: (1) defining high-value theft algorithms specific to the identification of commercial theft, (2) constructing and testing algorithms, and (3) creating and generating theft reports. An alternative considered was to identify a third party technical vendor that specializes in theft analytics to provide a solution rather than develop the necessary functionality using Company resources; however, the cost of a vendor's solution outweighed the implementation benefits.	04-May-20	(1.00)	311,842	0	0	106,345	138,560	66,937	131,784	0	0	67,980	58,080	5,724
417	2021	Dashboard Redesign	BP Functionality	Software (Intangible)	The Dashboard Redesign project will complete a refresh of the customer-facing website on mobile devices, including improved navigation capabilities for mobile users and progressive web application foundation for future implementation. A digital agency will be utilized to identify front end designs and page flows. This project will provide several benefits to the Company and its customers: (1) A new design of the customer dashboard on the customer-facing website that will include billing information, new widgets through an Internal Web Platform (IWP) solution, outage info, and easy accessibility to other experiences such as stop/start/transfer of service; (2) improved integration with SAP to address customer challenges with dunning messaging, filing payment arrangements, and allowing ability to engage payment options currently not available on the website. (3) Personalized and customized content in Sitecore based on the customer's profile and characteristics. (4) Improved performance testing of the customer-facing website. (5) Optimized views and navigation for mobile users to improve the Company's Customer Experience Index by two to four points. The project scope includes: (1) developing a new user interface design for the ConsumersEnergy.com website; (2) integrating widgets for usage information and energy efficiency capability; and (3) updating integrations for billing status and history, payment plans and options, account accessibility, and outage information. Three alternatives were considered and not chosen for this project: (1) Leave the customer dashboard in its current state. This option was not selected as it does not mitigate the negative Customer Experience Index (CXI) impacts resulting from customers struggling with payment confirmation, shut off indicator, past due indicator, editing payment methods, finding energy usage information, and overall navigation issues. (2) Adding IWP widgets into the bottom of the page to support the dashboard. This option was not selected because many customer issues would not be address or resolved, which negatively impacts CXI, and the return on investment would be minimal. (3) Deferring the effort to a future year. This alternative was not selected because it defers timely resolution to customer issues, it does not reduce inbound calls to the call center, and continues to negatively impact CXI. Redesigning the customer dashboard including support of mobile users was the selected alternative because it provides the necessary customer capabilities and results in a favorable return on investment.	30-Sep-21	(1.00)	2,528,027	0	0	692,800	1,385,600	449,627	164,670	0	0	66,000	92,400	6,270

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418	2021	Data Lake Entry	BP Functionality	Software (Intangible)	The Data Lake Entry project will enable the Customer Operations Data Analytics team to finalize the migration of all Customer Operations data from the Nimbus SQL Server to the MEMSQL Data Lake. The Company will gain value from this project by having all Customer Operations data in the data lake, reducing manual efforts to collect, consolidate, and analyze the data. The scope of the project includes: (1) extract, transformation, and loading of all Customer Operations data into the data lake automatically and in near real-time; and (2) implementing extracts from existing Customer Operations systems or tools, like Interactive Voice Response (IVR), into the data lake for further analysis. Three alternatives were considered but not chosen for this project: (1) Continue with segregated data which negatively impacts process efficiency for data gathering and analysis. This option was not chosen as it requires extensive manual processing to complete routine tasks. (2) Migrate a portion of the data. This option was not chosen because it provides minimal value compared to overall costs and processes would be more complicated with only part of the data migrated. (3) Defer this work to a future year. This option was not selected as it perpetuates an existing problem with no resolution. The option to migrate data into the existing MemSQL Data Lake was selected as the team already utilizes the tool for analytics with other data sets and adding this data provides the Company with access to key customer data without manual action.	01-Jul-21	(1.00)	401,911	0	0	294,440	0	107,471	36,135	0	0	0	0	33,000	0	3,135
419	2021	Digital - Application Programming Interface Fabric	BP Functionality	Software (Intangible)	This project provides foundational capabilities around APIs, which are a set of technologies used to integrate applications within the Company and, more importantly, with external third-party applications. In short, APIs allow two applications to talk to each other. This is currently a gap and is a key technical capability needed to enable multiple IT projects related to the Company's strategy for clean energy like: Electric Interconnection Billing and Payments, MISO Market User Interface Changes, Utility Analytics Mts-Phasing/Power Quality/Outage, Field Contractor Work Management Technology Enablement, Work Management Scheduling, Analytics and Reporting, DERMS, and Google X. Similarly, this API foundational capability will enable customer-focused projects like: Business Customer Interval Web Portal ("IWP"), Bring Your Own Thermostat ("BYOT") Pilot, Customer Relationship Management ("CRM") Product Suite, and Large Customer Rate Tool. The project will add value by: (1) implementing functionality to perform API services at-scale; (2) allowing partners to integrate with the Company's internal applications; (3) providing the ability to reuse integrations; (4) enabling monitoring of API traffic; (5) implementing functionality to perform API throttling (i.e., traffic management); (6) visualizing API traffic and analytics through key performance indicators ("KPI"); (7) enabling faster prototyping, testing, and deployment of integrations; and (8) providing operational tools for monitoring, incident management, and resolution. The project scope includes: (1) evaluating multiple API management tool vendors; (2) executing API onboarding for a number of external partners without scalability constraints; and (3) configuring and deploying API services with out-of-the-box operations tooling to achieve faster speed to market. Two alternatives were explored and determined to be non-viable: (1) remain on current Tibco API product that is part of their ESB platform, which does not support the future needs of the Company including efforts to modernize the grid; and (2) scale back on the implementation of the API fabric and complete over a longer period, which would negatively impact other projects. These alternatives were not selected because the API fabric is foundational and needs to be completed to support other projects. The current scope and direction is the best fit to support current Company initiatives.	01-Nov-20	(1.00)	602,736	0	0	259,800	152,416	190,520	175,692	0	0	132,000	29,700	13,992		
420	2021	Digital - Data and Analytics in the Cloud	BP Functionality	Software (Intangible)	This project will extend the Company's current data and analytics environment into a cloud environment, which will enable data and analytics at-scale and enable the delivery of outcomes for the Natural Gas Delivery Plan, Electric Grid Integration, customer programs, and other business needs. The project will add value by: (1) providing the ability to perform data analytics at-scale; (2) allowing the ability to leverage the leading machine learning ("ML") and artificial intelligence ("AI") tools to enable predictive and prescriptive analytics at-scale; (3) providing the ability to provision infrastructure at-scale rapidly; (4) enabling pay for use; (5) empowering faster prototyping, testing, and deployment of analytics solutions; (6) reducing total cost of ownership ("TCO"); and (7) providing operational tools for monitoring, incident management, and resolution. The project scope includes: (1) the execution of data analytics at-scale without scalability constraints; (2) flexible transitioning with technology platforms as they evolve; (3) the use of out-of-the-box ML and AI tools provided by cloud vendors; (4) new services and innovations on the cloud platform; (5) capacity pay per use; and (6) the foundation for future cloud migration and maintenance. The alternative considered to a cloud solution is to continue to expand the on-premise infrastructure and purchase multiple tools to solve individual capability gaps. The Company did not choose this alternative because it is more costly than cloud due to higher infrastructure costs and more manpower required for implementation.	31-Oct-21	(1.00)	1,048,400	242,480	277,120	173,200	207,840	147,760	132,254	0	0	120,780	0	11,474		
421	2021	Digital - Data Governance	BP Functionality	Software (Intangible)	This project will be used to establish data governance roles and responsibilities, processes, and the purchase of a tool to support best practices across the enterprise. The project will add value by: (1) increasing productivity of data analysts across the Company by reducing time spent cleaning data; (2) improving business planning; (3) maximizing the use of data to make decisions; and (4) discovering where data lives and the definition of data elements. The project scope includes: (1) initializing key data domains and ownership across the Company through the creation of an overarching data governance process, and establishing processes and cadence for introducing new data elements into their domain; (2) enabling people to become functional and technical data stewards through education; and (3) implementing technology through the selection of a data cleansing, data quality, data extract, and transformation tool, including enterprise-wide semantic definition. Alternatives considered include: (1) not implementing a data governance solution, potentially limiting the Company's productivity due to data quality and accessibility; (2) developing an internal tool to help manage the Company's data footprint; or (3) purchasing a third-party solution. Purchasing a third-party solution was selected because the skill sets required to internally develop such tools are not available and it would take a larger investment to upskill or hire individuals with this experience.	15-Nov-21	(1.00)	760,992	346,400	0	155,880	138,560	120,152	240,405	0	0	99,000	132,000	9,405		
422	2021	Digital - Foundation Enhancements	BP Functionality	Software (Intangible)	The Company's digital foundation consists of platforms that provide capabilities for analytics, cloud, integration, content management, and network communications. These capabilities are the underlying foundation for digital systems needed to help achieve the company's goals for clean energy, gas system reliability and safety, and the products, services, and experience our customers expect. This project ensures that we keep these foundational platforms updated with the latest software releases to keep the platforms secure while adding features like newer advanced analytics capabilities or new bundled and improved cloud service offerings. In the near future, it is expected that more smart devices will exist on the grid and pipelines, and in customer's homes. The ability to communicate securely and in a timely manner in order to manage the electric and gas systems effectively will require upgrading underlying platforms to conform with the latest security and communication requirements. The project will add value by: (1) enabling advanced and new functionality; (2) increasing the reliability and resiliency of Company applications; and (3) ensuring flexible, configurable platforms to scale and adapt with an evolving business landscape and customer expectations. As part of the review process, the alternative considered was not providing the funding for foundational enhancements. However, this limits the Company's ability to meet customer expectations, reduce security risks, and maintain system performance.	31-Dec-21	(1.00)	271,578	0	0	138,560	69,280	63,738	210,540	0	0	132,000	66,000	12,540		
423	2021	Digital - Work Automation	BP Functionality	Software (Intangible)	This project will implement and enhance Robotic Process Automation ("RPA"), ML, and AI tools. The project adds value to the Company by providing the foundation that will allow business areas to automate key processes based on individual use cases. Each use case will deliver benefits, reduce errors, and improve overall productivity to support Company customers and employees. The scope of the project will be to leverage the existing platforms to provide the foundation of RPA, ML, and AI to support use cases across all business units. The alternative considered was to continue with existing manual processes, which prevents the business areas from reducing the risk of manual errors or improving productivity through automation. Providing a foundational automation platform allows the Company to maximize the benefits of automation in support of customers and employees.	01-Nov-21	(1.00)	151,723	0	0	103,920	0	47,803	157,740	33,000	0	115,500	0	9,240		

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424	2021	ONS Compliance	BP Functionality	Software (Intangible)	The EHS (Environmental Health & Safety) Compliance project will implement a cloud based corporate-wide incident and risk management solution. The solution will ensure consistent accurate tracking, management, and reporting of environmental, health & safety compliance tasks using a centralized source of data. Modules to be implemented include Compliance, Health and Safety, Human Performance Management, Incident & Crisis, and Reporting & Analytics. The value of doing this project is to leverage best practice processes which are built into the new solution, including incident management and prevention to ensure a safe and productive workforce able to complete work for customers. The project also supports the Company's Planet goal through enhanced tracking and reporting for air quality, waste management, and sustainability. Other benefits include risk avoidance for environmental penalties, increased productivity, and increased quality. The addition of visual management and dashboards to enable transparency in tracking of goals; and create awareness of emerging environmental regulations and, to respond to them. One alternative is to continue "as is" through use of disparate Excel and SharePoint solutions. This requires much manual effort, and poses a risk of not being able to accurately and centrally track EHS data. Another alternative is to have separate projects; one for Environmental Compliance, and another for Safety and Health Compliance. This was the initial path for these two business areas. Investigation into potential solutions including a formal Request for Information (RFI) revealed that the standard for these market solutions is integrated functionality for Environmental and Safety/Health. Both on premise and cloud solutions were considered. The single solution was selected as the most efficient solution.	01-Feb-20	(0.89)	0	0	0	0	0	0	63,601	32,465	0	26,400	4,736	0
425	2021	Electric Grid Telecom Device Management	BP Functionality	Software (Intangible)	The Electric Grid Telecom Device Management project will implement a solution for managing the telecommunications device component of grid modernization that includes managing device metadata (version, software) and the ability to historically manage the real-time status of telecommunications devices. Completion of this project will benefit the Company by: (1) reducing hours spent combining data from several platforms; (2) providing visibility into devices on the electric distribution system, including a consolidated view of multiple device types; (3) providing highly reliable communications to field control devices in support of advanced grid functionality; and (4) improving the ability to troubleshoot and perform root cause analysis, leading to improved reliability and a reduction in dispatching field crews for troubleshooting devices. The implementation of a communications management system that can monitor and maintain multiple wide area network/field area network (WAN/FAN) device types also adds value by consolidating data from multiple disparate systems into a central repository which can then be used to manage and analyze the disposition and features of distribution grid telecommunications devices. Additionally, a platform that enables both GIS integration of asset location and telecommunication asset inventory for future substation communication needs will enable geo-based trouble analysis and proactive replacement of similar types of faulty communication equipment. The scope of the project consists of: (1) implementing a communications device management solution, (2) performing necessary data transformation, (3) developing analytic queries that leverage the cross-functional data, (4) defining an end-to-end business process that will support actionable findings, and (5) managing associated organizational changes. Several alternatives were considered for the execution of the project: (1) no change, which assumes the risk of failure of services for reliability and system optimization, (2) develop an in-house application consolidating device information from vendor-specific systems into a single view, (3) deploy a commercial solution (cloud based or on premise) that will consolidate and normalize vendor-specific data. Both alternatives two and three will be evaluated in the project planning phase to determine the most cost-effective solution, as they both support achievement of the investment objectives.	30-Nov-22	(1.00)	91,000	10,000	20,000	20,000	20,000	21,000	146,600	0	0	70,000	70,000	6,600
426	2021	Electric Agricultural Services Data Base	BP Functionality	Software (Intangible)	The Agricultural (Ag) Services Database project will update and stabilize the system with which data for the Agricultural Services Neutral Separation Program is managed. The Neutral Separation Program tracks and manages mandatory inspections for stray voltage at agricultural locations. Upgrading the Ag Services Database will provide value for both the Company and its customers by: (1) supporting consistent tracking of the agricultural services inspection program; (2) eliminating the need for processing paper forms which reduces rework and paper waste, and (3) maintaining up to date information on customers in the program, which ensures currency of participant data. The scope of this project will: (1) replace the existing database with one that is supported by IT; (2) integrate with SAP to provide access to customer information; and (3) integrate with GIS to capture customer geographic information, including details related to connectivity, voltage, service type, management headquarters, and circuit and location. Two alternatives were considered for execution of this project: (1) build a new database that uses current technology supported by the Company and includes functionality for keeping records up to date, or (2) restore the inoperative functionality and update current access database. Alternative two was not chosen because the current solution uses outdated and unsupported technology, and it would not be cost-effective to maintain it. Alternative one is the recommended solution because it will leverage current, supported database technology.	30-Nov-21	(1.00)	292,000	0	0	200,000	0	92,000	32,850	0	0	30,000	0	2,850
427	2021	Electric Distribution Asset Management	BP Functionality	Software (Intangible)	The Electric Distribution Asset Management project will develop a Distribution Asset Management strategy with corresponding process and technology implementation plans and objectives. The Company will gain value from this project through: (1) creating a standardized process for designers and field resources to collect and manage asset data, reducing waste and rework associated with current processes; (2) optimizing planning and management of distribution assets, resulting in investments of highest value; (3) establishing a channel for the Company to develop and manage asset health conditions on the distribution system, optimizing the operation and maintenance investments of distribution assets; and (4) improving the quality of Company electric distribution asset data, optimizing asset performance and providing opportunities to leverage quality data to support other engineering and operations efforts. The project scope includes: (1) developing a Distribution Asset Management strategy with a corresponding process and technology implementation plans and objectives, (2) designing and implementing optimized processes and supporting technology solutions, (3) implementing an Asset Integrity Management program leveraging industry standards, (4) evaluating, designing and implementing tools to optimize investment planning and asset health management, (5) implementing a distribution asset system of record repository, and (6) designing and implementing the organizational change management necessary to ensure success. One alternative considered is to continue to fund and implement electric distribution asset-based technology solutions as separate projects including: (1) upgrading the Geographic Information System (GIS) with the next major data model version and integrating affected applications, (2) integrating GIS with the work management system (SAP), (3) enhancing mobile data, and (4) designing asset data processes. This alternative was not selected as it does not consider technology, process, people and data inter-dependencies. The recommended alternative is that which is outlined in this business case because it creates the opportunity to identify, prioritize and address key gaps in current asset data collection processes, addresses the technology and process inter-dependencies and provides for distribution asset management planning and investment optimization.	31-Dec-24	(1.00)	1,984,000	400,000	200,000	400,000	800,000	184,000	751,000	0	0	275,000	450,000	26,000
428	2021	Electric Infrastructure Attachments	BP Functionality	Software (Intangible)	The Electric Infrastructure Attachment (IA) project will research, create a request for proposal (RFP), evaluate solution alternatives, design, develop and implement software to replace the existing functionality of the IA System Design and Standards (SDS) Microsoft Access database. Unlike the current solution, the new solution will be IT supported. Purchasing and implementing an IT-supported software platform that provides current functionality of the SDS Database, as well as new functionality will benefit both the Company and its customers by: (1) integrating data across several platforms including SAP, Geographic Information System (GIS), the National Joint Utility Notification System (NJUNS), and location-specific field design related materials as documentation of the assets inspected; (2) providing a more efficient approach to fulfill information requests; (3) affording a history of pole attachments at a specific location; and (4) reducing waste through improved processes. The scope of this project includes: (1) researching industry solutions, (2) creating a request for proposal (RFP), (3) evaluating options and selecting a solution, and (4) design and implement a solution. Alternatives considered include: (1) select and implement a fully integrated, IT-supported solution that replaces the existing functionality of the IA SDS database. (2) Develop an in-house replacement for the SDS database through updates to the existing database or SharePoint. Alternative two was not chosen because this option would not integrate with Corporate systems, exacerbating ongoing issues with data discrepancies and client dissatisfaction making it an inferior solution. (3) Continue to use the current SDS database. Alternative three is unacceptable due to the stated business risks. The first alternative was selected because it will provide for an integrated, IT-supported solution.	30-Dec-21	(1.00)	727,000	100,000	150,000	250,000	50,000	177,000	80,500	0	15,000	58,000	0	7,500

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Information Technology Department

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429	2021	Electric Interconnection Billing and Payment	BP Functionality	Software (Intangible)	Entities with a desire to interconnect to the Company's electric distribution network are required to submit applications and associated fees for processing. The Electric Interconnection Billing and Payment project will research, evaluate, and implement a solution that will provide for the electronic processing of applicant fees. Providing the capability for interconnection applicants to pay application fees electronically adds value through: (1) reduced manual internal processing of checks, (2) minimized risks associated with lost payments, and (3) increased likelihood in applicant satisfaction. The scope of this project includes: (1) researching industry solutions and defining requirements; (2) creating a request for proposal (RFP); (3) evaluating options and selecting a solution and; (4) designing and implementing the solution. The alternatives considered include: (1) implement an electronic billing and payment solution that the Company's current interconnection solution provider integrated into their solution for another client; (2) implement electronic billing and payment functionality that is currently used to process other types of payments to the Company; and (3) delay implementation of a solution until accepting electronic payments becomes a regulatory requirement. The third alternative was not selected because risks associated with handling physical checks will continue, while providing less time for solution evaluation and implementation once processing of electronic payments becomes a regulatory requirement. Alternatives one and two, both cloud-based solutions, will be evaluated once requirements are finalized.	30-Nov-21	(1.00)	1,095,000	0	0	750,000	0	345,000	148,000	0	0	0	135,000	0	13,000
430	2021	Electric Underground Conflation	BP Functionality	Software (Intangible)	The Electric Underground Conflation project will establish a single Geographic Information System (GIS) repository for electric distribution assets through conversion of Computer Aided Design (CAD) asset record files to assets in the electric GIS database and align the facility map records to aerial imagery and the GIS street and right-of-way maps ("land base"). This conversion and alignment process is referred to as conflation. Implementation will benefit the Company through: (1) increased accuracy of system planning and analysis models; (2) enhanced safety by providing a single, spatially accurate view for overhead and underground electric assets rather than multiple applications; (3) provides the framework for the conversion of underground electric services from Service Information Management System (SIMS); (3) improved GIS integrated design quality for electric underground assets through a consolidated view of electric assets that eliminates additional manual work-arounds to manipulate designs to align overhead and underground electric digital records; (4) improved productivity for electric designers, dispatchers, schedulers, and all office electric GIS users by eliminating additional manual work-arounds necessary to match the electric underground distribution records with the previously conflated land base; (5) improved safety for field operations by providing a spatially accurate representation of electric distribution assets over the previously conflated land base files, eliminating the misalignment and risk of misinterpretation of asset location between the conflated land base and an un-conflated electric distribution file; (6) improved damage prevention through better analysis for MISS DIG staking by providing a more larger view of the electric underground system and; (7) improved safety and reliability during storm restoration activities through enhancement of the outage management system mapping application via inclusion of a more detailed underground electric asset location view that provides dispatchers with more accurate data when locating faults and improves switching operations. See Richard T. Blumenstock's testimony for full project synopsis.	30-Sep-21	(1.00)	392,000	0	0	25,000	355,000	12,000	11,000	0	0	10,000	0	1,000	
431	2021	Enterprise Content Management - Managing Business Records	BP Functionality	Software (Intangible)	Using the Company's Enterprise Content Management ("ECM") system, this project will manage business records for high-focus areas contained in SharePoint and Shared Drives. The records will be classified, categorized, and placed under formal retention rules, via metadata assignment. This project will add value through: (1) continuation of the multi-year management action plan; (2) consistent practices for records management; (3) defensible process for validating completeness and accuracy of records produced and records deleted; (4) deletion of excess, irrelevant, or inappropriate information; (5) easy generation of electronic records in the event of litigation; (6) achieving Generally Accepted Recordkeeping Principle maturity of 3.0 or greater through functional capabilities of ECM coupled with business practices to align to ARMA (formerly the Association of Records Managers and Administrators) principles; and (7) mitigating the possibility of fines as a result of being unable to produce records. For high-focus areas, the scope includes: (1) integrating SharePoint and Shared Drive content into ECM; (2) assigning taxonomy and records classification values through metadata on content and records; (3) assigning standard retention rules; (4) mapping, building routines, and migrating existing content to new taxonomy and retention schedules; (5) ensuring records can be located; and (6) allowing legal holds and lineage audits to be conducted on content. Three alternatives were considered for this project: (1) Continue managing business records on shared network drives and in SharePoint without ECM integration. This option was not chosen as it would provide minimal visibility to the records from within the ECM system, retention rules will not be applied to the records, taxonomy or metadata will not be applied making it more difficult to find and manage records, the problem will continue to get worse over time as the amount of data grows, the audit finding is not remediated, and the Company could face substantial fines if it is unable to find records. (2) Integrate all SharePoint and shared drive content in the ECM. This option was not chosen as it will be costly and not all content is considered a business record. (3) Start by focusing on integrating business critical records with ECM for the high-focus areas. Alternative (3) was selected as it mitigates risk with the high-focus areas and requires less funding than managing all SharePoint and shared drive content in ECM.	14-Nov-21	(1.00)	251,486	0	0	86,600	138,560	26,326	486,420	0	0	330,000	139,920	16,500	
432	2021	Field Contractor Work Management Technology Enablement	BP Functionality	Software (Intangible)	This project will provide the ability to electronically manage contractor work, increasing accuracy and timeliness of information processing for field work deliverables. This project will create new opportunities to measure and optimize field work processes that support customer on-time delivery goals. The project will add value by: i) improving on-time delivery of customer work by providing electronic work order information to contractors; ii) improving customer satisfaction through efficiency in scheduling work and reporting on the progress electronically; iii) increasing safety by tracking work and contractor status; iv) improving material management; v) making it easier to move emergent work to contractors, which will better meet customer commitments and balance workload; and vi) enabling real time updates to work order information, increasing data accuracy and reducing invoice reconciliation time. The project scope includes: i) meeting with stakeholders to identify requirements for a Bring Your Own Device ("BYOD") field contractor work management technology solution and process; ii) developing, configuring, and testing interfaces, hardware, and software for the solution; iii) implementing the solution and process for the following work groups: ElectricRVD, Electric LVD, Mutual Assistance, Forestry, Gas Distribution, Gas Code Compliance, and Substation Operations Construction/Metro; iv) updating the following vendor contract types to support BYOD field contractor work management: zone, specific bid, ancillary, electric storm, and mutual assistance; and v) training field contractors on new technology and processes. See Richard T. Blumenstock's testimony for full project synopsis.	31-Aug-21	(0.61)	1,937,715	0	0	900,640	346,400	690,675	171,680	0	0	54,912	116,768	0	
433	2021	Field Mapping and Graphics	BP Functionality	Software (Intangible)	The Field Mapping and Graphics project will implement a replacement system for the mobile field mapping and data collection software ("ArcPad") that can search and view facility map data, view work order designs, and create work order as-built construction drawings in the field. This will make it possible to consolidate field graphics functionality into an efficient process while implementing a current and supported application and retiring the unsupported ArcPad solution. The project will add value by: (1) providing more accurate geospatial data, including facility map data, pre-construction designs, and as-built construction drawings; (2) consolidating daily tasks into a more simplified process; (3) eliminating the process waste from duplicating asset data in two systems resulting from current system limitations; (4) enabling the adoption of the Geographic Information System ("GIS") standard; and (5) allowing for growth of functionality and capabilities to make more mapping and graphics data available on field devices. The project scope includes: (1) installing a new mobile field mapping and graphics application; (2) creating the ability to search and view the facility map data in GIS format, and the ability to search by address; (3) producing the ability to view pre-construction work order designs in a new GIS format; and (4) enabling the creation of as-built construction drawings in GIS format for assigned work orders. See Richard T. Blumenstock's testimony for full project synopsis.	21-Feb-21	(0.91)	1,428,719	0	0	772,749	207,840	448,130	23,852	0	0	20,592	0	3,260	

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434	2021	Financial Planning Transformation - Intake and Monthly Plan Management	BP Functionality	Software (Intangible)	The Financial Planning Transformation - Intake and Monthly Plan Management project will improve portfolio management capabilities in financial planning processes (also referred to as integrated business planning). Benefits from this project include: (1) labor savings across the Company, (2) improved data quality for decision making, (3) improved data analysis and reporting, and (4) automation of manual processes and reconciliation. The project scope includes: (1) managing Work Intake (the process by which employees submit programs and projects for funding considerations within the financial planning process); (2) developing Intake Scenarios Prioritization and Workflow (the process through which all projects are reviewed then prioritized, creating different scenarios for management review and decision making, and implementing the workflow needed to take an idea from creation to inclusion in the financial plan); (3) connecting planning, prioritization, and decision making to strategic, objectives and world class goals; (4) storing plans and scenarios for future reference of historical data; (5) bringing actuals and plan/forecast together for monthly review and forecast updates; (6) creating the ability to track and store risks and opportunities; and (7) bringing both financial and work volume (units) together for forecasting and planning. The three alternatives considered for this project were: (1) Leverage and make changes to the home-grown Business Planning System ("BPS"). This alternative was not chosen due to technology limitations in the existing system, and use of a custom-developed program that does not reflect industry best practices. (2) Consider other third-party applications to provide the functionality. (3) Leverage the existing SAP system to provide the functionality. Advantages include existing integrations to forecasting and budgeting in SAP, planning system is directly connected to the financial management system, use of existing investments in SAP, and realizing benefits of updating processes and eliminating manual processes. The preferred options are (2) and (3) as they would integrate more seamlessly with the Company's existing technology and reflect industry best practices for a comprehensive planning solution. A final decision will be made after an RFP is issued and a vendor is selected.	21-Jun-22	0.56	3,407,190	692,800	0	207,840	2,078,400	428,150	287,100	0	0	66,000	165,000	56,100
435	2021	GIS-Integrated Design	BP Functionality	Software (Intangible)	This project replaces the CAD and Work Requirements and Design ("W&D") software with Bentley's Open Utilities Map ("BOUM") to leverage GIS asset data to generate engineering designs, implement workflows with SAP, and mitigate risk associated with the aging software application. The project provides value to the Company in three ways: i) the BOUM software is compatible with GIS integration and may be further developed after the Company implements the IJPM as the basis of its GIS systems; ii) the project mitigates risk of manual asset record imports, asset record manipulation for design white space management, or manual asset record recreation for base design files, by enabling technology that creates a new design over existing electronic asset records; iii) it enables technology that is compatible with the asset system of record and the Project Wise engineering document management system. The project scope includes: i) replacing CAD engineering design software (Bentley Microstation (V7i) with BOUM; ii) leveraging GIS asset data for the purpose of generating engineering designs; iii) integrating workflows with SAP through Bentley Workflow Manager software; iv) implementing GIS replication databases to isolate data editing from data viewing; v) replacing the desktop View Graphics custom application with a web-based application for searching and viewing asset records; and vi) implementing a Primary Distribution Map ("PDM") generation service to generate electric PDM files for printing during storm and other emergency events. See Richard T. Blumentack's testimony for full project synopsis.	01-Mar-23	(0.95)	0	0	0	0	0	0	349,206	225,060	0	0	134,146	0
436	2021	GM - GIS Connectivity Model Integration	BP Functionality	Software (Intangible)	The Grid Modernization (GM) Geographic Information System (GIS) Connectivity Model Integration project will design, develop, and implement a GIS extract using an industry standard to publish electric GIS data to the network, where it can be consumed by multiple applications. The extract will make the data available to the electric distribution historian as well as other applications in the future. The project provides value to the Company by: (1) leveraging the International Electrotechnical Commission (IEC) Common Information Model (CIM), which provides a basic ontology for data and data relationships and a standard repeatable approach that can be leveraged for sharing data about the electrical network across technical systems; (2) supporting an enterprise application integration framework, which keeps business process as simple as possible; (3) providing the ability to detect the physical location of metered assets, including circuit, substations and transformers; and (4) reducing the costs and time required to develop integrations in the future resulting from streamlined approaches and processes. Deliverables within the project scope include: (1) develop and implement the GIS extract, leveraging IEC CIM for data sharing; (2) leverage the enterprise service bus (ESB) as the high level communications protocol for sharing GM data across the enterprise; (3) build the integration to the electric distribution historian; and (4) develop a data and integration framework that can be leveraged for future projects. The alternative considered was to continue to implement point-to-point application-specific integrations to meet the requirements specific to a single project. This alternative was not chosen because it continues to increase the technical debt associated with the maintenance of non-standard interfaces. The preferred approach as outlined in this business case will result in: (1) standard, repeatable approaches for future integrations, (2) lower ongoing maintenance costs and, (3) less complexity.	31-Dec-20	(1.00)	0	0	0	0	0	25,000	0	0	25,000	0	0	0
437	2021	HR - 2020 Union Changes	BP Functionality	Software (Intangible)	The HR - 2020 Union Changes project will implement SAP and other system changes required as a result of three collective bargaining agreements which will be renegotiated and ratified. Collective bargaining agreements expire every five years for Operating Maintenance and Construction ("OM&C"), Virtual Contact Center ("VCC"), and Zeeland employee groups. For OM&C, the current agreement ends June 1, 2020. The VCC agreement ends August 1, 2020. The Zeeland agreement ends October 1, 2020. Completion of this project will provide value to the Company and its customers through: (1) waste elimination by making changes in the software for pay and benefits changes as required by the new agreements; (2) defect reduction by adding process automation to otherwise manual processes for tracking and recording work, premium and absence time; (3) improved employee engagement amongst the OM&C, VCC, and Zeeland union employees. The scope of this project encompasses making system changes required to support the new working agreement language for the OM&C, VCC, and Zeeland employees. Exact details will be finalized after the negotiation process is completed and contracts are approved. Three alternatives were considered for this project: (1) Make no system changes after the contracts are ratified. This option was not chosen because it exposes the Company to possible fines, disgruntled employees, union grievances, significant manual processes leading to greater possibility of error, hiring additional staff to perform activities outlined in the agreements, and increased legal costs due to employee grievances. (2) Find other third-party software to support the changes required by the union agreements. This option was not chosen because it would require SAP integration along with additional software licensing and maintenance costs. The option to make system changes to eliminate manual updating, comply with the working agreement language, support union employee engagement, and reduce grievances, leading to reduced legal costs, was chosen because it leverages current SAP technology, automates what would otherwise require manual processing, and is the least costly option.	18-Dec-20	(1.00)	0	0	0	0	0	177,679	0	0	63,360	100,980	13,339	

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438	2021	Landlord Small Business Portal	BP Functionality	Software (Intangible)	The Landlord Small Business Portal project will modernize the existing Landlord Portal to reflect current usability standards, style sheets and added features of the Company's latest website portal release. This will allow for increased adoption and utilization of digital services by landlords, multi-family dwelling owners, and management companies, resulting in a reduction of annual call volume to contact call centers. The project creates value to both the Company and its customers through: (1) improved customer satisfaction utilizing the Landlord Portal, (2) reduced deflection rates based on self-service capabilities, (3) increased adoption rates in Landlord Portal, and (4) reduced annual call volume to contact call centers. The scope of this project includes: (1) updating Landlord Portal style sheets and content templates, (2) implementing single sign-on capabilities, (3) updating password security policy to reflect current Company standards and incorporating two-factor authentication, (4) implementing mobile-friendly web pages, (5) adding record locks in SAP to prevent unwanted overwrites, (6) enabling multi-pane view within the Landlord Portal, (7) enabling self-service features for customers including ad-hoc bill payment, (8) enabling email and SMS notifications and verification, (9) enhancing exporting and reporting features, (10) implementing an administrative view for sub users, (11) adding heads-up display Section 9 export and formatting, and (12) integrating with smart meters to obtain interval consumption data. The alternatives considered include: (1) Submit ad-hoc requests to incrementally update Landlord Small Business Portal. This would require additional overhead for each request and extend the duration required to complete the anticipated business changes. (2) Continue to utilize the existing Landlord Portal. Continued use of the current portal does not mitigate the customer pain points or reduce inbound call volume to call centers, which will continue to adversely impact customer satisfaction scores and Digital CX scores. (3) Merge the Landlord Portal features into the core customer-facing portal. Providing these additional features in the existing portal would greatly increase the complexity of the solution while potentially impacting customer experiences. (4) Update the Landlord Portal to meet current standards and enable customer self-service features. This alternative was selected because it results in inbound call volume reduction and anticipated CX improvements.	30-May-22	(1.00)	2,094,646	0	0	866,000	692,800	535,846	149,111	0	0	75,900	66,000	7,211
439	2021	Large Customer Rate Tool	BP Functionality	Software (Intangible)	The Large Customer Rate Tool project is for automated rate analysis through a rate tool for our large customers. The project will automate currently manual rate evaluations and thereby reduce the time it takes to run individual rate analysis. Included in the implementation are secondary, primary and special rates for business customers, process automation, and accuracy of rate definition for large customers. The value of completing the project is the automation of a currently manual intensive process that account managers utilize in working with large businesses to ensure they are on the best possible rate. As part of the project review process, the alternative considered was to do nothing and maintain the manual process of extracting data from SAP and entering into spreadsheets to run rate comparisons. This alternative was not selected since it would not support the gains in customer satisfaction or the efficiency of providing the data. Customer benefit: Improved functionality to better assist large business customers with evaluating different rate options.	01-Aug-20	(0.92)	0	0	0	0	0	0	115,500	0	17,952	0	97,548	0
440	2021	Move In Move Out Version 3.0	BP Functionality	Software (Intangible)	The Move In Move Out (MIMO) Version 3.0 project will update and redesign the customer experience to remove barriers and increase ease of use for customers who desire a self-service MIMO experience. This will improve customer satisfaction and completion rates for MIMO and reduce inbound call center volume. The Company and its customers will gain value from this project through: (1) eliminating barriers and increasing the ease of use for customers who opt for self-service in the MIMO experience, and (2) utilizing the MIMO backlog and gantt charts to identify the highest value updates for customer self-service. The scope of the project includes: (1) implementing fixes currently identified within the MIMO backlog, (2) reducing complexity of customer facing sites including E.com where needed, (3) enhancing SAP back-end to MIMO functionality where needed, and (4) review and execute tasks identified in the MIMO backlog. Three alternatives were considered but not chosen for this project: (1) Increase resources in Call Centers. This option was not chosen because it would increase costs in the call center and avoid cost-reduction goals. (2) Defer to later years. This option was not chosen because it does not reduce call volume or positively impact customer satisfaction, and it does not achieve cost-reduction goals. (3) Implement a third-party tool to handle MIMO. This alternative was not chosen due to large cost associated with the implementation, including high complexity of integration with SAP. Additionally, any customer issues that could be experienced with a third-party tool would still result in calls to the call centers. The option to enhance the current implementation was chosen because it continues to expand customer self-service capabilities while reducing Company cost-reduction goals.	19-Oct-21	(1.00)	1,462,828	0	0	866,000	519,600	77,228	175,973	0	0	115,500	49,500	10,973
441	2021	Move In/Move Out Digital Redesign	BP Functionality	Software (Intangible)	The Move In-Move Out project will redesign the Move In and Move Out web pages to enable mobile device support, which improves the customer's ability to navigate on mobile devices and successfully complete move in and move out requests. The Company and its customers will gain value from this project through: (1) implementing mobile friendly pages for Move In-Move Out transactions, (2) replacing the modal format with a more appropriate technical framework to improve customer navigation capabilities, (3) increasing CXI Digital score for Move In-Move out to meet improvement targets, and (4) enabling tagging to track failure points within digital interactions with customers. The scope of the project includes: (1) redesigning the Move In-Move Out solution to utilize mobile-friendly architecture, (2) modifications to Sitecore and .Net to accommodate the new customer-facing web pages, and (3) testing the new pages for performance and security gaps. Three alternatives were considered but not chosen for this project: (1) Develop a mobile application instead of mobile pages. This option was not chosen because the cost is three to four times the current costs for a mobile redesign. (2) Defer to future years. This option was not selected because mobile users would continue to experience issues that result in inbound calls to the call centers for issue resolution, which reduces customer satisfaction. (3) Generate a submission form for mobile users. This option was not chosen because the customer would have two different workflows for the same process, which increases waste and support costs and adversely impacts customer satisfaction scores. The option to complete a Move In-Move Out Digital Redesign was chosen because it establishes a standard mobile presence to increase customer satisfaction and is the most cost-efficient solution.	30-Sep-21	(1.00)	1,105,813	0	0	346,400	554,240	205,173	46,215	0	0	16,500	26,400	8,315
442	2021	On-Bill Financing	BP Functionality	Software (Intangible)	The On-Bill Financing project will enable product purchases (both margin and Energy Efficiency, Demand Response) by customers utilizing on-bill financing. This process will allow the Company to increase sales and support customer investment in product offerings that drive better management of energy demand. This project provides value to the Company and customer through: (1) enabling the purchase Energy Efficiency, Demand Response, and other products via a monthly payment system; (2) utilizing third-party financing organization to support revenue recovery and setup of the financing practices; and (3) integrating with Customer Information and Customer Relationship systems to track products, financial terms, and impacts to customer bills. The scope of the project encompasses: (1) identification and partnering with a third-party financing vendor, (2) updating the bill presentation, customer facing website, and interactive voice response systems with information related to available on-bill financing, (3) creating marketing campaigns for on-bill financing and related products; and (4) connecting to internal systems to track customer information and products. Three alternatives were considered and not chosen for this project: (1) Generating invoices for month to month billing. This was not chosen as it creates more internal work to keep contracts updated, payments tracked, and complicates billing. (2) Adding the full amount of the product to next bill. This was not chosen as it requires extensive financial planning for customers to ensure they can pay the full amount and could deter customers from acquiring the Company's products and services which support energy waste reduction. (3) Partnering with a third party to provide products and financing to customers on behalf of the Company. This option was not chosen as it reduced revenue and margin on these transactions limiting future financial opportunities for the Company. The alternative to implement an On-Bill Financing solution was chosen as it supports our future retail strategy without a major up-front investment which would impact ROI.	02-Nov-21	(1.00)	1,336,508	692,800	0	346,400	103,920	193,388	138,270	0	0	66,000	66,000	6,270

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Information Technology Department

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443	2021	One Call Ticket Risk Analysis Model for Damage Prevention	BP Functionality	Software (Intangible)	The One Call Ticket Risk Assessment Model implements a risk analysis and data analytics program that identifies the riskiest excavation tickets, utilizing current one-call ticket data, damage history, and incorporating asset information from the Geographic Information System (GIS) to focus damage prevention resources and activities on locate requests with the highest risk. Completion of this project will provide value to both the Company and its customers. Each party will benefit from safety improvements and risk mitigation through: (1) implementation of automated screening daily one call tickets to enable smarter decisions that mitigate damages and support proactive communication; (2) creation of a risk analysis model to identify and prioritize the highest risk tickets; (3) creation of detailed data analytics for high risk tickets and actual damages to identify root cause or systemic issues; (4) mitigation of damage prevention risk and prevented damages, which results in fewer root cause investigations, facility repairs, lost service to customers, collection efforts, legal expenses, regulatory reporting and better public relations. Additionally, fewer damages result in reduced potential for serious injuries and property damage for customers. The customer directly and indirectly benefits from this implementation by reducing disruptions to service, reducing planned work interruptions for emergent work, and reducing public safety risk through proactive damage prevention measures. Together with improved business process, risk analysis, monitoring and proactive communication safe digging practices with highest risk excavations, damages can be reduced. The project scope includes: (1) vendor application hosting, maintenance, and support of one or more instances of the Cloud-based platform; (2) a solution to provide gas and electric asset information from the GIS databases for use within the vendor platform as either extracts (FTP or emailed files) on a regular interval (ie, quarterly, monthly) or through integration with the GIS platform; (3) ongoing and historical ticket locate data to vendor software through either an email format, web service, or integration; (4) copies of ongoing and historical excavation damage records that occurred on any of the locate requests provided above by either a spreadsheet or web service; and (5) a damage risk statistical model that calculates risk scores based on multiple factors to produce outputs of a daily summary of highest risk tickets, reports, and web access to the customer portal for detailed risk assessment data. See Richard T. Blumenstock's testimony for full project synopsis.	01-Dec-20	0.11	\$80,220	0	20,784	242,480	173,100	143,756	125,202	38,016	1,980	62,700	16,500	6,006
444	2021	Rate Case Implementation	BP Functionality	Software (Intangible)	The Rate Case Implementation project will modify SAP billing in accordance with MPSC filings. These enhancements include, but are not limited to, overall price changes and structure changes to the rates. Project scope includes the large electric and gas rate changes that occur annually and/or monthly. The benefit of this project is to ensure that customers are billed accurately based on the timing of the rate structure and price changes. An alternative considered for this effort was an offshore development model. However, the risk of billing inaccuracies and customer complaints with that model were deemed too high based on the various complexities and timing of the development and testing.	30-Sep-19	(1.00)	0	0	0	0	0	0	172,135	0	0	83,160	81,972	7,003
445	2021	Real Time Electric System Access in the Field	BP Functionality	Software (Intangible)	The Real Time Electric System Access in the Field project provides field crews with access to real-time network, asset, and outage information using an interface on mobile and hand-held devices that includes highly configurable map displays, substation one-lines, and real-time tabular displays. This project will implement and integrate these capabilities in the field, as well as provide for the organizational change management necessary for project success. This new technology will transform Corporate Grid and Field Operations processes by providing for: (1) a comprehensive, end-to-end integrated solution that will provide field personnel with the ability to monitor and update the real-time, as-operated grid status in the field, eliminating the need for continual dialog with System Operations teams; and (2) efficient and effective assignment of work to field technicians improving safety, communication, documentation, and productivity. The scope of this project includes: (1) assess the capabilities and value of each module of the OSI Compass solution to determine functionality to be included in scope; (2) evaluate work processes of field worker roles to determine those roles to be included in scope; (3) evaluate options and select a solution; (4) developing and executing organizational change management activities; and (5) designing and implementing the technology solution. Alternatives to be considered for this project include: (1) continue to follow existing field processes that requires back-and-forth radio communication to validate, update, and change system status; and (2) implement a solution to enable as-operated, real time electric system grid status changes from the field. The first alternative was not selected because although three-way communication procedures have been established, they are prone to human error and pose a safety risk. The second alternative was selected for the aforementioned benefits. A cloud-based application will be considered as an option for implementation.	31-Aug-23	(1.00)	0	0	0	0	0	252,000	0	0	230,000	0	22,000	
446	2021	Renewables Supervisory and Control Data Acquisition Overlay	BP Functionality	Software (Intangible)	The Renewables Supervisory Control and Data Acquisition (SCADA) Overlay project is the design, procurement, installation, and commissioning of a SCADA overlay software application for the Merchant Operations Center and Renewable Generation Operations to view and control Lake Wind, Cross Wind, Grand Valley Solar, Western Michigan Solar, and Circuit West Solar on a consolidated application pane. It will also provide the platform for monitoring and control of future wind and solar assets added to the generating fleet. The project will add value by: (1) bringing the renewables generating fleet operations and dispatch up to industry standards for its projected size by managing all the assets from a single screen, (2) allowing better monitoring of the renewables generating fleet, (3) automating the NERC required reporting and associated changes, and (4) improving quality while reducing time spent on reporting and data management for analysis. The project scope includes: (1) merging multiple different Renewable SCADA views (ie, command and control functions, alarms, warnings, and key performance indicators for wind turbines and solar inverters) into a single operational and dispatch view through off-the-shelf renewables SCADA overlay software, (2) installing on-premise servers and databases for daily operation and disaster recovery functions, and (3) integrating with the individual site native SCADA systems. Alternatives considered include: (1) an extensive manual build of screens into the existing SCADA screens; (2) building an overlay into the current historian system, though it would have limited capabilities for command and control; (3) implementing an additional SCADA vendor software and accept limited control capabilities; and (4) implementing vendor software that is more related to distributed energy resources than utility scale generation assets and accept limited functionality and usefulness. These alternatives were not selected due to limitations in standard functionality that decreases usefulness and increases the cost of customizing the existing SCADA solution. The selected off-the-shelf renewables SCADA overlay software is the preferred option due to preliminary fit with requirements and cost-effective implementation.	31-Oct-21	(1.00)	\$13,157	400,000	150,000	63,000	129,100	71,057	21,705	0	0	19,000	900	1,805
447	2021	Role Based Access Control	BP Functionality	Software (Intangible)	This project will develop business roles based on job functions for the SAP environment. The value of completing the project is: (1) more efficient access control policy maintenance and certification, (2) more efficient provisioning by network and systems administrators, (3) reduced new employee downtime from more efficient provisioning, (4) enhanced organizational productivity, and (5) enhanced system security and integrity. The scope of this project includes the following systems: (1) SAP Enterprise Resource Planning ("ERP") Central Component, (2) Customer Relationship Management, (3) Business Warehouse, (4) Governance Risk and Compliance, (5) Solution Manager, (6) Process Orchestration, and (7) NetWeaver Development Infrastructure. As part of the review process, the alternative considered was to continue with access provisioning based on selection of specific functions and processing multiple approvals. However, this alternative was not selected because it does not eliminate waste in the form of human struggle, rework, and extra processing. The alternative to implement role-based access control was selected because of the financial savings and process improvements.	01-Nov-21	0.58	0	0	0	0	0	200,017	0	0	47,048	148,500	4,470	

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448	2021	SAP Access Controls	BP Functionality	Software (Intangible)	This project replaces the current software tool that manages and monitors Sarbanes-Oxley Act ("SOX") compliance for SAP transactions. The project will add value by: (1) reducing risk and eliminating waste through new tools that provide higher levels of automation and process optimization; (2) decreasing maintenance and upgrade costs by implementing the out-of-the-box solution without customization; (3) increasing the ability to meet requirements changes from auditors and the Public Company Accounting Oversight Board; and (4) creating complete and accurate report capabilities. The project scope includes: (1) automating the SAP periodic user access review; (2) automating the emergency access management process; (3) displaying areas of risk and identifying how to mediate that risk; (4) tracking mitigations; and (5) converting the current rule set into the new tool. The alternatives that were considered and deemed to be non-viable are: (1) continue using the current application as-is, accepting the audit risks stemming from the known issues and the risk associated with standard SAP support ending on December 31, 2020, and continue following current error-prone manual processes; (2) use robotic process automation in a limited scope to automate some manual tasks and reduce errors, which due to limited functionality would not mitigate risks and gaps in the current tool; (3) develop a custom solution to perform some of the work, which would be expected to be more costly and include higher maintenance and support costs; and (4) continue to apply program patches and fees from SAP to the current tool, which may not address or fulfill all requirements, and requires support from an external vendor, which increases support costs. The option to replace the existing tool with a new solution: (1) meets Company requirements; (2) avoids hiring two additional resources; (3) is more cost effective in the long term; (4) adds more process automation; and (5) increases SOX compliance.	29-Oct-20	(0.62)	109,809	0	0	0	69,280	0	45,529	250,800	165,000	0	35,640	37,620	12,340
449	2021	ServiceNow Customer Service Management	BP Functionality	Software (Intangible)	This project will implement the CSM module of ServiceNow to enable the recently defined Supply Chain ("SC") service delivery model to support SC optimization. This project will provide value to the Company and its customers through: (1) waste elimination; (2) cost savings; (3) maturing processes; and (4) movement to a sustainable service delivery model to continually improve the customer experience. This project specifically will enable the newly designed processes and service delivery model as it will: (1) enhance the delivery of SC services to both internal customers and suppliers; (2) include an integrated knowledge base and case management system, improving the customer experience by having one place to go for information and help; and (3) provide standard workflow and funnel the previously disparate intake channels for SC support into one solution. This technology solution is the backbone of an SC Support Center and will enable: (1) better management of work; (2) more efficient and accurate response to questions; and (3) improved satisfaction of internal customers and external suppliers. See Jeffrey D. Tolonen's testimony for full project synopsis.	28-Feb-20	(1.00)	0	0	0	0	0	0	62,905	47,065	0	15,840	0	0	
450	2021	Summer Peak Use Rate (SPUR) Release 2	BP Functionality	Software (Intangible)	The Summer Peak Use Rate (SPUR) Release 2 project will continue the implementation of Time of Use (TOU) rates for Net Metered and PayGo customers that were not included in SPUR R1, allowing customers to lower their bills during peak summer usage. This project will add value by: (1) providing Net Metered, PayGo, and electric vehicle rate customers with a TOU rate that allows them to manage their consumption during peak summer hours; (2) converting all non-TOU customers to new TOU rates by June 2020; (3) implementing a rate structure that can lower a monthly bill for a customer, which improves customer experience and Net Promoter Scores; and (4) shifting off-usage to off-peak and reducing on-peak demand during summer months. The scope of this project includes: (1) configuring new TOU rates for Net Meter, PayGo, and electric vehicle rate customers in SAP; and (2) modifying the conversion program and transitioning all new TOU customers to new TOU rates by June 2020. This project will also include modifications to the SAP Customer Relationship Management (CRM) application to ensure these customers are able to concurrently enroll in various customer programs (Budget Billing, Appliance Service Protection, Shut Off Protection, etc.). The alternatives considered include: (1) manually converting approximately 10,000 (Net Metered and PayGo) customers to the TOU rate; (2) restricting the scope of the project to exclude updating the CRM program, thereby preventing these customers from participating in customer programs; (3) requesting an exemption from the Michigan Public Service Commission. None of these alternatives were selected due to the increased risk of errors associated with manual conversion, limiting customer participation in critical assistance or elective programs, and risk of non-compliance of MFC billing rules.	01-Nov-20	(1.00)	0	0	0	0	0	116,000	0	0	0	116,000	0	0	
451	2021	Voxal Survey Tool	BP Functionality	Software (Intangible)	The Voxal Survey Tool project will implement an after-the-call survey platform for every customer contact for both Interactive Voice Response (IVR) and live agent customers. This platform would follow similar methodology as the current surveys but utilizes a much larger sample population and includes an integration with existing call routing and call analytics tools providing better feedback from customers to improve Customer Service Representative (CSR) capabilities. This project will provide value for the Company and its customers through: (1) increasing the number of surveys completed by the customers resulting in additional insight from the customer feedback; (2) reducing costs by eliminating the dependency on a third-party vendor to send surveys on behalf of the Company; and (3) increasing customer satisfaction due to feedback acquired from surveys. The project scope includes: (1) implementing the application and analytics components for IVR and agent surveys; (2) defining best practices to design the survey and analytics solution; (3) implementing the application and analytics components for email; (4) deploying and validating servers to support new survey tool; (5) improving routing strategy, IVR transfer key value pair (KVPI), and ability to support Testing Center Of Excellence (TCOE) testing objectives; (6) replacing the existing IVR administration tool; (7) implementing survey software that allows the business area the ability to create and edit surveys; and (8) implementing a repository for this data to reside until it is ready to be collected by the data analytics team. Two alternatives were considered and not chosen for this project: (1) Continue to use the SQM solution utilized today. This was not chosen as it requires continued maintenance of existing interfaces with SQM and costs \$60K/month to have SQM perform surveys on the Company's behalf. (2) Build a home-grown survey solution. This alternative was not chosen because Consumers Energy has a current skill gap to support this type of application. The option to implement the Voxal tool was selected because of the integration it provides to the Genesys framework. It synchronizes through Genesys software for reporting and Cyara Pulse for dashboard capabilities. Additionally, Voxal has intimate knowledge of the Company's Genesys framework configurations.	08-May-21	(1.00)	166,272	51,960	0	17,320	90,064	6,928	7,920	0	0	1,320	6,600	0	
452	2021	Website Redesign	BP Functionality	Software (Intangible)	The Website Redesign project will implement a new mobile experience through a progressive web application for key self-service experiences for customers. The new features will help to achieve a five to seven point increase in Customer Experience Index (CXI) goals through an improved mobile site. The project will address customer issues with the existing mobile experience, leading to improved overall customer satisfaction. The Company and its customers will gain value from this project through: (1) additional self-service capabilities; and (2) meeting customer needs and improving ease of use and enjoyment of the mobile experience. Both will help improve customer satisfaction and CXI scores. The project scope encompasses: (1) improving Sitecore and .Net platforms to support an upgraded architecture for progressive web applications; (2) implementing new designs and digital platform by partnering with a third-party firm; and (3) continuing integration with existing systems to maintain functionality that supports customer self-service. Alternatives considered include: (1) Leave the site in its current format. This alternative was not selected because it will not address customer pain points that result in inbound calls into the contact center. (2) Pursue native mobile application development. This alternative was not selected because requires considerable funding and resources to develop and maintain, increasing annual costs to support multiple versions of the application on multiple mobile operating systems. (3) Defer the work to a future year. This alternative was not selected because it will continue to negatively impact CXI Digital and Mobile scores until the work is completed. (4) Implement a progressive web application. This alternative was selected because it addresses key customer pain points and can be delivered in a timely fashion.	15-Nov-21	(1.00)	3,184,330	0	0	1,039,200	1,870,560	274,570	434,445	0	0	198,000	217,635	18,810	

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	SPEND YEAR	PROJECT NAME	IT PROGRAM	FERC CATEGORY	PROJECT SYNOPSIS	IMPLEMENTATION DATE	COST/BENEFIT RATIO	ELECTRIC PORTION CAPITAL SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC	ELECTRIC PORTION O&M SPEND FOR APPLICABLE YEAR	SOFTWARE COSTS-ELECTRIC	MATERIAL COSTS-ELECTRIC	LABOR COSTS-ELECTRIC	CONTRACTOR COSTS-ELECTRIC	OVERHEAD & OTHER COSTS-ELECTRIC
453	2021	Work Management Scheduling Analytics and Reporting	BP Functionality	Software (Intangible)	The Work Management Scheduling Analytics and Reporting project will implement a solution capable of scheduling long cycle, maintenance, and emergent work. This will combine data from the several excel spreadsheets that are used today to allow single views of all the information needed to effectively produce the various schedules and provide reporting. The project scope includes: (1) implementing a streamlined scheduling process across Operations, and (2) implementing associated analytics and reporting. The project will add value by providing: (1) accurate schedules tied to productivity, reducing waste, and shrinking the work backlog across work types (emergent, customer requested, project and compliance); (2) enhanced quality and integrity of scheduling process through reduction in manual scheduling steps and hours spent developing route sheets; (3) time saved from manual entry to be reallocated to better schedule analysis and alignment across disciplines, decreasing risk of missing code work and being in non-compliance; (4) improved transparency into whether the weekly schedule is meeting business objectives such as: financial scenarios, compliance requirements, and first time completion; and (5) increased Employee Engagement by having a quality product to schedule, while being able to focus on priority and strategic execution of work versus workarounds to meet daily scheduling needs. Two alternatives were considered for this project: (1) Purchase disconnected software products. This option was not chosen due to a risk of increased costs as well as it contains solutions that provide overlapping functions with existing solutions or will not meet base requirements. (2) Automating manual data movement across systems through Robotic Process Automation. This option was not chosen because it will not meet base requirements and does not provide desired insights. The option to purchase the SAP Multi Resource Scheduling module was selected as it will minimize ongoing support costs, meets base requirements, and will provide increased transparency into scheduling.	30-Jun-21	(1.00)	966,348	289,157	31,486	232,001	256,136	157,367	149,176	0	612	81,180	58,575	8,808
454	2021	Workforce Connect - Talent Roadmap Enablement	BP Functionality	Software (Intangible)	The Workforce Connect - Talent Enablement project is part of an overarching talent enablement plan which will enable the company to move forward on its goal of creating an employee experience to better serve customers. Improving the employee experience is focused on two outcomes (1) enabling employees to be more engaged and productive and (2) attracting and retaining new skill sets and talent, both of which provide for an improved customer experience and higher customer satisfaction. This particular project will provide the following value: (1) increased employee retention for continuity of knowledge and long-term customer relationships, as well as reduction in costs associated with recruiting and onboarding new employees; (2) positive direct customer interactions facilitated through strong employee engagement; (3) enhanced employee development to meet customer demands during a time where retirement eligibility is high, and risk of knowledge loss has the potential to negatively impact customer satisfaction; (4) enable the Company to operationalize a new career framework and competency model which will not only support training and development of employees to deliver strong customer service but will also enhance the ability to hire candidates capable of delivering on the Company's initiatives and goals; (5) advanced workforce analytics that provide data-driven insights to proactively optimize talent management strategies and programs in support of maintaining and enhancing the workforce to continue to deliver strong customer service and results; (6) reduce waste and defects leading to increased quality and simplifying manual processes; (7) improved insight and consistency in reporting (i.e. Ethics, Safety, Office of Federal Contract Compliance Programs); (8) provide transparency into key talent areas to identify retention risk within critical areas and develop succession strategies; and (9) ensure core systems are stable and operational interruptions are minimized. See Karen M. Gaston's testimony for full project synopsis.	15-Dec-22	(0.98)	325,946	0	0	207,840	0	122,106	1,412,714	402,914	0	396,000	528,000	85,800
455	2021 2021 SubTotal BP Functionality	BP Functionality						36,366,232	4,089,597	849,390	12,297,647	12,270,276	6,859,821	9,935,549	1,382,059	42,144	4,517,805	3,485,404	507,186
456	2021 2021 Total							73,737,900	5,825,811	25,164,310	18,266,913	14,506,972	9,993,694	36,939,686	3,505,189	1,101,506	19,459,283	9,379,057	3,460,650

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Historical 13-Month Average of IT Cloud Computing Prepaid Balance
For the years 2018 - 13 months balance ending September 30, 2019
Electric Allocation

Case No.: U-20697
Exhibit No.: A-107 (JDT-5)
Page: 1 of 2
Witness: JDTolonen
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
Line No.	Description	Bal. Ending 9/30/2018	Bal. Ending 10/31/2018	Bal. Ending 11/30/2018	Bal. Ending 12/31/2018	Bal. Ending 1/31/2019	Bal. Ending 2/28/2019	Bal. Ending 3/31/2019	Bal. Ending 4/30/2019	Bal. Ending 5/31/2019	Bal. Ending 6/30/2019	Bal. Ending 7/31/2019	Bal. Ending 8/31/2019	Bal. Ending 9/30/2019	13 mos. Average
1	IT Cloud Computing Prepaid	\$ 773,497	\$ 676,896	\$ 667,325	\$ 562,994	\$ 754,561	\$ 2,310,967	\$ 2,422,833	\$ 2,293,541	\$ 2,181,647	\$ 2,447,467	\$ 2,318,174	\$ 2,206,888	\$ 2,078,176	\$ 1,668,844
2	Total	\$ 773,497	\$ 676,896	\$ 667,325	\$ 562,994	\$ 754,561	\$ 2,310,967	\$ 2,422,833	\$ 2,293,541	\$ 2,181,647	\$ 2,447,467	\$ 2,318,174	\$ 2,206,888	\$ 2,078,176	\$ 1,668,844

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Projected 13-Month Average of IT Cloud Computing Prepaid Balance
For the years 2020 - 13 months balance ending December 31, 2021
Electric Allocation

Case No.: U-20697
Exhibit No.: A-107 (JDT-5)
Page: 2 of 2
Witness: JDTolonen
Date: February 2020

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
Line No.	Description	Bal. Ending 12/31/2020	Bal. Ending 1/31/2021	Bal. Ending 2/28/2021	Bal. Ending 3/31/2021	Bal. Ending 4/30/2021	Bal. Ending 5/31/2021	Bal. Ending 6/30/2021	Bal. Ending 7/31/2021	Bal. Ending 8/31/2021	Bal. Ending 9/30/2021	Bal. Ending 10/31/2021	Bal. Ending 11/30/2021	Bal. Ending 12/31/2021	13 mos. Average
1	IT Cloud Computing Prepaid	\$ 1,617,704	\$ 3,240,889	\$ 2,968,192	\$ 3,430,739	\$ 3,158,043	\$ 3,094,131	\$ 2,821,434	\$ 2,618,623	\$ 2,562,941	\$ 2,421,594	\$ 2,142,544	\$ 1,931,659	\$ 1,560,899	\$ 2,582,261
2	Total	\$ 1,617,704	\$ 3,240,889	\$ 2,968,192	\$ 3,430,739	\$ 3,158,043	\$ 3,094,131	\$ 2,821,434	\$ 2,618,623	\$ 2,562,941	\$ 2,421,594	\$ 2,142,544	\$ 1,931,659	\$ 1,560,899	\$ 2,582,261

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio

For the Projected Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20087

Exhibit No.: A-108 (JDT-6)

Page: 1 of 8

Witness: JDT/Idem

Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description		Alternatives Considered	Project Scope/Functionality	Project Benefit		Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio
1	800 MHz Modernization Project	Upgrades and Applications Currency	8,926,000	-	-	-	-	This project will replace antiquated head-end (the main audio routing switch for lower sites and dispatch consoles), lower site, dispatch, and radio infrastructure, as well as user sub-networked equipment with infrastructure that meets current Project 25 ("P25") standards. P25 standards are for digital mobile radio communications that are used by North American public safety and dispatch organizations. The current equipment was implemented in 1994 and is no longer manufactured or sold, so the Company is unable to find replacement parts.		The alternatives considered were to: (1) remain with the current system, and (2) fulfill the migration to other manufacturers and architecture. Option 1 was not selected because the equipment is not supported, and the organization is unable to find replacement parts. Option 2 was not chosen because there would be a larger learning curve, it would be more disruptive to business, and it would require a complete replacement of the existing system. The Company chose to go with the upgrade because it has been incrementally adopting the P25 standards through the wireless ARP, and this prudently utilizes current and ongoing investments. The upgrade is the easiest and least disruptive path to migrate to the P25 standards.	The scope of the project includes: (1) the design, (2) configuration, and (3) implementation of P25 systems for headsets, lower sites, dispatch consoles, and subscriber equipment.	This project creates value for the Company by: (1) moving the Company to current production equipment that can be replaced and is more stable, (2) enabling the organization to migrate from an unsupported to a supported platform, and (3) allowing for quicker response to electric outages, and (4) increasing customer and employee safety.		Jan-20	Mar-20	Jun-20	Jun-23	Sep-23	(1.00)
2	Bill Design and Delivery Transformation	Customer Experience and Operations	7,519,560	(1,250,000)	-	-	-	The Bill Design and Delivery Transformation project will execute a bill redesign for all rates and program contributions, including necessary software replacements and a flexible print and delivery outsourcing initiative. Following this initiative all in-house services for bill print will be outsourced, reducing internal costs for the Company while providing more efficient means to update outsourced printed materials.		Three alternatives considered but not chosen for this project: (1) Keeping the bill delivery process the same. This option was not chosen because it would continue current state issues resulting in \$1MM O&M annual expenses. Additionally, extensive time is required to modify bills which will generate 1MM calls annually into call centers. (2) Modifying the bills for Summer Time of Use (TOU) only. This option was not selected because while modifying the bill to support the recent TOU rate is needed, this still would not address any issues with customer processes, increased calls to call centers, or difficulty in modifying the bill. (3) Deferring the project until a later date. This option was not chosen because the need to spend nearly \$2M to redesign the bill for TOU would still exist, along with continuing significant cost and issues with the current process. The option to transition bill print to a third party provider was selected as it provides O&M savings and reduces the complexity of the existing process. The Company will be able to more efficiently update the customer's bill to improve customer satisfaction and resolve billing issues.	The project scope includes: (1) redesigned customer billing documents with online availability, (2) an outsourced solution for bill printing and mailing, (3) an improved eBill product, (4) integrating customer billing information with third party printing source for bill print capability, and (5) an integrated digital and interactive voice response (IVR) channel for consistent customer experience.	The Company and its customers will receive value from this project through: (1) reduced costs for bill print and delivery, (2) consistent and accurate delivery of bills and other printed materials, and (3) reduced calls to contact centers with improved bill format and content.		Jan-21	Feb-21	May-21	Nov-21	Jul-22	(0.56)
3	ARP — Workstation Asset Management (WAM)	Asset Refresh Program	7,372,696	-	-	-	-	This project will replace and install new desktops, laptops, and tablets.		The alternatives considered were to: (1) extend the replacement cycle from four years to five years for all desktops and laptops, (2) extend the replacement cycle on desktops from four years to five years, and (3) use outdated equipment. The Company did not select these options because: (1) there would be an increased risk of hardware failure and equipment outages that could impact the capacity of business partners to complete job tasks, (2) it could cause applications to run poorly or stop functioning, (3) it would increase the ARP by \$4 million in future years, and (4) it could cause an inability to apply security patches. The Company selected a four-year refresh cycle to alleviate these concerns.	The project scope consists of: (1) replacing workstation assets, and (2) installing new units for new resources.	This project creates value for the Company by: (1) improving stability and availability of business critical applications by proactively replacing workstations subject to the chance of hardware failures increasing, and (2) allowing business partners to complete their job tasks.		-	-	-	-	-	(1.00)
4	Financial Planning Transformation - Intake and Monthly Plan Management	Corporate	4,918,000	-	-	-	-	The Financial Planning Transformation - Intake and Monthly Plan Management project will improve portfolio management capabilities in financial planning processes (also referred to as integrated business planning).		The three alternatives considered for this project were: (1) leverage and make changes to the Company's home-grown Business Planning System, which was not chosen due to technology limitations in the existing system, and use of a custom-developed program that does not offer the improved capabilities; (2) consider other third-party applications to provide the functionality; and (3) use the existing SAP system to provide the functionality, which provides the advantage of including existing integrations to forecasting and budgeting in SAP, directly connecting the planning system to the financial management system, using existing investments in SAP, and realizing benefits of updating processes and eliminating manual processes. The preferred options are (2) and (3) as they would integrate more seamlessly with the Company's existing technology and provide a comprehensive planning solution. A final decision will be made after an RFP is issued and a vendor is selected.	The project scope includes: (1) managing Work Intake (the submission of programs and projects within the financial planning process); (2) developing Intake Scenarios Prioritization and Workflow (the process through which all projects are reviewed then prioritized, creating different scenarios for management review and decision making, and implementing the workflow needed to take an idea from creation to inclusion in the financial plan); (3) correcting planning, prioritization, and decision making to Company objectives and goals; (4) storing plans and scenarios for future reference of historical data; (5) bringing actuals and plan/forecast together for monthly review and forecast updates; (6) providing the ability to track and store risks and opportunities; and (7) bringing both financial and work volume (units) together for forecasting and planning.	Benefits from this project include: (1) labor savings across the Company, (2) improved data quality for decision making, (3) improved data analysis and reporting, and (4) automation of manual processes and reconciliation.		Sep-20	Oct-20	Mar-21	Nov-21	Aug-22	0.56
5	Website Redesign	Customer Experience and Operations	4,596,320	-	-	-	-	The Website Redesign project will implement a new mobile service experience for customers. The new features will help to achieve a five to seven point increase in Customer Experience Index (CXI) (goal through an improved mobile site. The project will address customer issues with the existing mobile experience, leading to improved overall customer satisfaction.		Alternatives considered include: (1) Leave the site in its current format. This alternative was not selected because it will not address customer pain points that result in inbound calls into the contact center. (2) Pursue native mobile application development. This alternative was not selected because requires considerable time and resources to develop and maintain, increasing annual costs to support multiple versions of the application on multiple mobile operating systems. (3) Defer the work to a future year. This alternative was not selected because it will continue to negatively impact CXI (Digital and Mobile scores) until the work is completed. (4) Implement a progressive web application. This alternative was selected because it addresses key customer pain points and can be delivered in a timely fashion.	The project scope encompasses: (1) improving Sitecore and Net platforms to support an upgraded architecture for progressive web applications, (2) implementing new designs and digital platform by partnering with a third party firm, and (3) continuing integration with existing systems to maintain functionality that supports customer self-service.	The Company and its customers will gain value from this project through: (1) additional self-service capabilities, and (2) meeting customer needs and improving ease of use and enjoyment of the mobile experience. Both will help improve customer satisfaction and CXI scores.		Jan-21	Mar-21	Jun-21	Jul-21	Nov-21	(1.00)
6	ARP — Server and Storage	Asset Refresh Program	3,709,339	-	-	-	-	This project will replace or augment server and storage infrastructure for the Company.		The alternative considered was to purchase extended maintenance. This solution was not selected because full support would not be offered after seven years and maintenance costs would increase. The Company continues to refresh these critical technologies based on a five to seven-year refresh cycle to mitigate the risk of failure.	The scope of this project encompasses: (1) replacement of server and storage hardware assets, and (2) installation of additional new computers and storage capacity to account for organic growth requirements.	This project creates value for the Company through: (1) improved stability and availability of business critical applications by proactively replacing server and storage hardware assets prior to the chance of hardware failures increasing, and (2) ensuring that adequate resources are available to support application demands after five to seven years of actual use.		-	-	-	-	-	(1.00)
7	Dashboard Redesign	Customer Experience and Operations	3,648,000	-	-	-	-	The Dashboard Redesign project will complete a refresh of the customer-facing website on mobile devices, including improved navigation capabilities for mobile users and progressive (2) and (3) as they would integrate more seamlessly with the Company's existing technology and provide a comprehensive planning solution. A final decision will be made after an RFP is issued and a vendor is selected.		Three alternatives were considered and not chosen for this project: (1) Leave the customer dashboard in its current state. This option was not selected because it does not address the negative Customer Experience Index (CXI) impacts resulting from customers struggling with payment confirmation, shut off indicator, past due indicator, adding payment methods, finding energy usage information, and overall navigation issues. (2) Adding WAP widgets into the bottom of the page to support the dashboard. This option was not selected because many customer issues would not be addressed or resolved, which negatively impacts CXI, and the return on investment would be minimal. (3) Deferring the effort to a future year. This alternative was not selected because it defers timely resolution to customer issues, it does not reduce inbound calls to the call center, and continues to negatively impact CXI. Redesigning the customer dashboard including support of mobile users was the selected alternative because it provides the necessary customer capabilities and results in a favorable return on investment.	The project scope includes: (1) developing a new user interface design for the ConsumersEnergy.com website; (2) integrating widgets for usage information and energy-saving tips; and (3) updating integrations for billing status and history, payment plans and options, account accessibility, and outage information.	This project will provide several benefits to the Company and its customers: (1) A new design of the customer dashboard on the mobile users to improve the Company's Customer Experience Index by two to four points.		Jan-21	Feb-21	Apr-21	Apr-21	Oct-21	(1.00)
8	ARP — Wireless Network	Asset Refresh Program	3,572,532	-	-	-	-	This project will replace portions of the company's aging wireless systems.		The alternatives considered for system replacement were: (1) running parallel systems while the new system is deployed and the old system is dismantled, and (2) leasing a system from a vendor. The first alternative was not selected because it would be highly disruptive due to systems having to run independently of one another, and would have a high cost associated with the need to acquire additional radio frequency spectrum. The second alternative was not chosen since the Company would be dependent on the response times offered by a shared vendor system that offers lower system reliability. The Company chose to replace existing assets based on its refresh cycle to avoid extended support cost, and to provide a seamless transition that allows both the old and new systems to interact with little to no disruption to the end user.	The project scope consists of: (1) replacing wireless assets annually, and (2) installing additional new wireless assets to account for organic growth requirements.	This project creates value for the Company by: (1) ensuring real-time communications between Company crews and dispatch locations, (2) ensuring efficient gas leak and electric outage response times for customers, and (3) maintaining critical infrastructure and regulatory compliance.		-	-	-	-	-	(1.00)

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio
For the Projected Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20087
Exhibit No.: A-108 (JDT-6)
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Witness: JDT/Adam
Date: February 2020

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description		Alternatives Considered	Project Scope/Functionality	Project Benefit		Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio
9	Landlord Small Business Portal	Customer Experience and Operations	3,023,450	-	-	-	-	The Landlord Small Business Portal project will modernize the existing Landlord Portal to reflect current usability standards, style sheets, and added features of the Company's latest website portal release. This will allow for increased adoption and utilization of digital services by landlords, multi-family dwelling owners, and management companies, resulting in a reduction of annual call volume to contact call centers.		The alternatives considered included: (1) Submit ad-hoc requests to incremental updates. Landlord Small Business Portal. This would require additional overhead for each request and extend the customer pain points or reduce inbound call volume to call centers, which will continue to adversely impact customer satisfaction scores and Digital CX scores. (2) Merge the Landlord Portal features into the core customer-facing portal. Providing these additional features in the existing portal would greatly increase the complexity of the solution while potentially impacting customer experiences. (4) Update the Landlord Portal to meet current standards and enable customer self-service features. This alternative was selected because it results in inbound call volume reduction and anticipated CX improvements.	The scope of this project includes: (1) updating Landlord Portal style sheets and content templates; (2) implementing single-sign on capabilities; (3) updating password security policy to reflect current Company standards and incorporating two-factor authentication; (4) implementing mobile-friendly web pages; (5) adding record logs in SAP to prevent unwanted overviews; (6) enabling multi-pane view within the Landlord Portal; (7) enabling self-service features for customers including ad-hoc bill payment; (8) enabling email and SMS notifications and verification; (9) enhancing exporting and reporting features; (10) implementing an attractive view for sub users; (11) adding heads-up display Section 9 export and formatting; and (12) integrating with smart meters to obtain interval consumption data.	The project creates value to both the Company and its customers through: (1) improved customer satisfaction utilizing the Landlord Portal; (2) reduced deflection rates based on self-service capabilities; (3) increased adoption rates in Landlord Portal; and (4) reduced annual call volume to contact call centers.		Jan-21	May-21	Jan-21	May-22	Jun-22	(1.00)
10	Field Contractor Work Management Technology Enablement	Electric Grid Integration	2,796,933	-	-	-	-	This project will provide the ability to electronically manage contractor work, increasing accuracy and timeliness of information processing for field work deliverables. This project will create new opportunities to measure and optimize field work processes that support customer on-time delivery goals.		The alternatives considered included: i) continuing with the current paper-based process; ii) using the current Company mobile application; iii) using off-platform options such as Service Bench; and iv) providing Company-funded field devices to get contractors on a common technology platform. These alternatives were not chosen because, respectively: i) this approach does not allow for the timely, data-driven work management metrics required to improve service to customers; ii) this solution is not expected to receive long term investment by the vendor and the mobile application would require more upfront investment than the proposed option; iii) to ensure contractors leverage the benefits and integrations with the existing Service Suite platform, the chosen option is preferred; and iv) the investment in hardware, management of on-boarding and off-boarding of devices to contractors, and training and change management is cost-prohibitive and introduces risk of loss of control of information security and corporate assets. The ABB Service Suite hybrid solution was chosen because it uses existing well-developed Service Suite functionality while leveraging cloud-based, BYOD capability to move short-term and long-term contractors from paper processes to the established, standard work management system.	The project scope includes: i) meeting with stakeholders to identify requirements for a Bring Your Own Device ("BYOD") field contractor work management technology solution and process; ii) developing, configuring, and testing interfaces, hardware, and software for the solution; iii) implementing the solution and process for the following work groups: Electric/HV/D, Electric LVD, Mutual Assistance, Forestry, Gas Distribution, Gas Code Compliance, and Substation Operations Construction/Metro; iv) updating the following vendor contract types to support BYOD field contractor work management: zone, specific bid, ancillary, electric storm, and mutual assistance; and v) training field contractors on new technology and processes.	The project will add value by: i) improving on-time delivery of field customer work by providing electronic work order information to contractors; ii) improving customer satisfaction through efficiency in scheduling work and reporting on the progress electronically; iii) increasing safety by tracking work and contractor status; iv) improving material management; v) making it easier to move emergent work to contractors, which will better meet customer commitments and balance workload; and vi) enabling near time updates to work order information, increasing data accuracy and reducing invoice reconciliation time.		Jan-21	Jan-21	Sep-21	Jul-22	Sep-22	(0.61)
11	ARP --- Field Device Asset Management (FDAM)	Asset Refresh Program	2,274,076	-	-	-	-	This project will replace field devices.		The alternatives considered were to: 1) extend refresh cycles from four years to five years, and 2) to let the assets to failure. The selection of a four-year cycle was deemed to be the best solution because replacement in less than four years would result in additional unnecessary expense for replacement of assets that are still in peak operating condition and replacement cycles that exceed four years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, which negatively impact employee productivity.	The project scope consists of replacing field device assets.	This project creates value for the Company by: (1) mitigating potential costs for hardware repairs, and (2) allowing field workers to complete their job tasks.		-	-	-	-	-	(1.00)
12	Move In Move Out Version 3.0	Customer Experience and Operations	2,111,473	-	-	-	-	The Move In Move Out (MIMO) Version 3.0 project will update and redesign the customer experience to remove barriers and increase ease of use for customers who desire a self-service MIMO experience. This will improve customer satisfaction and completion rates for MIMO and reduce inbound call center volume.		Three alternatives were considered but not chosen for this project: (1) Increase resources in Call Centers. This option was not chosen because it would increase costs in the call center and avoid cost-reduction goals. (2) Defers to later years. This option was not chosen because it did not reduce call volume or positively impact customer satisfaction, and it does not achieve construction goals. (3) Implement a third party tool to handle MIMO. The alternative was not chosen due to large cost associated with the implementation, including high complexity of integration with SAP. Additionally, any customer issues that could be experienced with a third party tool would still result in calls to the call centers. The option to enhance the current implementation was chosen because it continues to expand customer self-service capabilities while reducing Company cost reduction goals.	The scope of the project includes: (1) implementing fixes currently identified within the MIMO backlog; (2) reducing complexity of customer facing sites including CS.com where needed; (3) enhancing SAP back-end to MIMO functionality where needed; and (4) review and execute tasks identified in the MIMO backlog.	The Company and its customers will gain value from this project through: (1) eliminating barriers and increasing the ease of use for customers who opt for self-service in the MIMO experience; and (2) utilizing the MIMO backlog and Pareto charts to identify the highest value updates for customer self-service.		Jan-21	Feb-21	Apr-21	Jun-21	Oct-21	(1.00)
13	Physical Security Asset Refresh	Security	2,100,000	-	-	-	-	This project will enhance or replace physical security assets to provide improved visibility into potential security concerns and resolve incidents. The Company has several thousand physical security asset devices currently in use, including security cameras, motion detectors, intrusion detection systems, and card access systems.		As part of the review process the alternative considered was not to do this work, but this would assume the risk that the Company will not meet FERC requirements.	Included in the project is the enhancement or replacement of assets including: (1) advanced door systems at Company buildings; (2) security cameras for monitoring capabilities; and (3) gate and lock systems, which includes security cameras, motion detectors, intrusion detection systems, and card access systems.	The value provided by completing the project is to: (1) maintain compliance; (2) reduce redundancies and gaps in functionality; and (3) optimize overall performance. An integrated solution is efficient and allows for centralized management, situational awareness, real-time monitoring, compliance with regulations and guidelines; and faster, more effective and consistent responses to emergencies and incidents.		Jan-19	Feb-19	Apr-19	Oct-19	Nov-24	(1.00)
14	Service Suite Upgrade	Upgrades and Applications Currency	2,087,280	(360,000)	-	-	-	This project will: (1) implement a new version of Service Suite Work Management that allows for easier distribution to the field; and (2) maintain a current version on a vendor supported platform for this critical enterprise application used across Operations and Engineering. The product enhancements in Service Suite and TC Technology's Mobile Information Management System ("MIMS") will be implemented to provide additional business value and benefits.		The alternatives considered and determined not to be viable included: (1) remaining on the current Service Suite version, which would cause continuation of additional manual steps in the emergency response and work assignment; or (2) creating a custom-developed solution, which would increase waste and inefficiency without the same level of Service Suite integration. Upgrading Service Suite was chosen because it allows the Company to stay on the current and supported version of the application already being used, and also adds functionality that will provide additional business value and benefits.	The scope of the project includes: (1) implementing the new version of Service Suite Work Management; and (2) implementing MIMS for mobile mapping on the field device to improve safety, response, usability, and supportability of the critical application.	The project will add value by: (1) implementing a new version that provides the highest level of vendor support for this 24x7 critical system that serves over 100,000 work orders weekly; (2) implementing Service Suite Fieldworker Mobile for a touch-based interface and migration to cloud-based configurations; (3) providing live traffic display on Dispatch Application; and (4) improving usability and usability of Dispatch Application schedule view.		Mar-21	Mar-21	Apr-21	Nov-21	Dec-21	(0.63)
15	Field Mapping and Graphics	Electric Grid Integration	2,062,239	(63,702)	(148,637)	-	-	The Field Mapping and Graphics project will implement a replacement system for the mobile field mapping and data collection software ("ArcPad") that can search and view facility map data, view work order designs, and create work order as-built construction drawings in the field. This will make it possible to consolidate field graphics functionality into an efficient process while implementing a current and supported application and retiring the unsupported ArcPad solution.		Three alternatives were explored and determined non-viable for the project: (1) Continue to maintain the current ArcPad application. This option was not selected because the Company no longer able to make changes to the ArcPad application to mitigate issues if the application has critical defects. A total failure of the application could revert field crews back to paper-based facility maps, risking safety through the use of static, outdated data, or require the creation of as-built construction drawings on paper documents. (2) Use the existing Mobile Information Management System ("MIMS") mapping solution for facility maps portion and use other existing applications that have rudimentary drawing capabilities, like Adobe or Smart for creating as-built construction drawings. This option was not selected because it would introduce significant cost because of the complexity and customization to integrate the applications. Also, this alternative would forfeit the already-established user experience and application integration achieved with the proposed MIMS solution. (3) Rebuild ArcPad from the ground up. This option is not advisable as existing industry solutions are available at a much lower cost with much less risk. Implementing the new field mapping and graphics software will consolidate field graphics functionality into an efficient process and provide a current, supported solution.	The project scope includes: (1) installing a new mobile field mapping and graphics application; (2) creating the ability to search and view the facility map data in GIS format; and the ability to search by address; (3) producing the ability to view pre construction work order designs in a new GIS format; and (4) enabling the creation of as-built construction drawings in GIS format for assigned work orders.	The project will add value by: (1) providing more accurate geospatial data, including facility map data, pre-construction designs, and as-built construction drawings; (2) consolidating daily tasks into a more simplified process; (3) eliminating the process waste from duplicating asset data in two systems resulting from current system limitations; (4) enabling the adoption of the GIS standard; and (5) allowing for growth of functionality and capabilities to make more mapping and graphics data available on field devices.		Jan-21	Apr-21	Aug-21	Nov-21	Apr-22	(0.91)

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio
For the Projected Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20087
Exhibit No.: A-108 (JDT-6)
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Witness: JDT/abcm
Date: February 2020

No.	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency- Capital	Contingency- O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost	Benefit Ratio
16	On-Bill Financing	Customer Experience and Operations	1,309,140	-	-	-	-	The On-Bill Financing project will enable product purchases (both margin and Energy Efficiency, Demand Response) by customers utilizing on-bill financing. This process will allow the Company to increase sales and support customer investment in product offerings that drive better management of energy demand.	Three alternatives were considered and not chosen for this project: (1) Generating revenue by month to month billing. This was not chosen as it creates more internal work to keep contracts updated, payments tracked, and complicates billing. (2) Adding the full amount of the product to next bill. This was not chosen as it requires extensive financial planning for customers to ensure they can pay the full amount and could deter customers from acquiring the Company's products and services which support energy waste reduction. (3) Partnering with a third party to provide products and financing to customers on behalf of the Company. This option was not chosen as it reduced revenue and margin on these transactions limiting future financial opportunities for the Company. The alternative to implement an On-Bill financing solution was chosen as it supports our future retail strategy without a major up-front investment which would impact ROI.	The scope of the project encompasses: (1) identification and partnering with a third-party financing vendor; (2) updating the bill presentation, customer facing website, and interactive voice response systems with information related to available on-bill financing; (3) creating marketing campaigns for on-bill financing and related products and; (4) connecting to internal systems to track customer information and products.	This project provides value to the company and customer through: (1) enabling the purchase Energy Efficiency, Demand Response, and other products via a monthly payment system; (2) utilizing third-party financing organization to support revenue recovery and setup of the financing practices; and (3) integrating with Customer Information and Customer Relationship systems to track customer, financial terms, and impacts to customer bills.	Jan-21	Feb-21	Mar-21	Jun-21	Nov-21	(1.00)	
17	Data Governance	Security	1,600,000	-	-	-	-	This project will develop an enterprise-wide data governance solution as an aspect of an Enterprise Data Privacy and Management Plan.	Alternatives considered include: (1) remain at the current state and not implement a data governance solution; (2) evaluate, select, and implement a technology solution to fulfill the Company's data governance needs; or (3) determine if any existing solutions the Company has built or purchased can meet the needs. The alternative selected is to move forward with implementing a data governance solution. A part of the project will be to evaluate both existing and new technology solutions.	The scope of this project encompasses: (1) achieving consistency in collecting and reporting data across various organizational users and source systems; (2) achieving high quality in the collection, maintenance, analysis, and reporting of data; (3) responding to data issues; (4) promoting and ensuring consistent enterprise-wide data definitions, increasing controls regarding data creation, modification, and access; (5) proposing and implementing system controls for maintaining data quality (e.g., field-level validation, reporting, batch job cleansing); (6) facilitating the development of a data quality assurance process, including policy, process, and system (including system access) reviews against data; and (7) creating a process to include a holistic review of data management methodology for the effective collection, management, and access for data quality control.	Completion of this project will provide value to the Company through programmatic data governance resulting in increased data quality which will: (1) decrease operational costs; (2) increase employee satisfaction and productivity; (3) enhance system performance efficiencies; and (4) support better informed decision making.	Jan-21	Mar-21	Jun-21	Oct-21	Dec-21	(1.00)	
18	Move In/Move Out Digital Redesign	Customer Experience and Operations	1,596,150	-	-	-	-	The Move In-Move Out project will redesign the Move In and Move Out web pages to enable mobile device support, which improves the customer's ability to navigate on mobile devices and successfully complete move in and move out requests.	Three alternatives were considered but not chosen for this project: (1) Develop a mobile application instead of mobile pages. This option was not chosen because the cost is three to four times the current costs for a mobile redesign. (2) Defer to future years. This option was not selected because mobile users would continue to experience issues that result in inbound calls to the call center for issue resolution, which reduces customer satisfaction. (3) Generate a submission form for mobile users. This option was not chosen because the customer would have two different workflows for the same process, which increases waste and support costs and adversely impacts customer satisfaction scores. The option to complete a Move In-Move Out Digital Redesign was chosen because it establishes a standard mobile presence to increase customer satisfaction and is the most cost-efficient solution.	The scope of the project includes: (1) redesigning the Move In-Move Out solution to utilize mobile-friendly architecture; (2) updating the design to be consistent with the current customer-facing web pages, and (3) testing the new pages for performance and security gaps.	The Company and its customers will gain value from this project through: (1) implementing mobile friendly pages for Move In-Move Out transactions; (2) replacing the mobile format with a more appropriate technical framework to improve customer navigation capabilities; (3) increasing the Digital score for Move In-Move out to meet improved targets; and (4) enabling tagging to track failure points within digital interactions with customers.	Feb-21	Mar-21	Mar-21	May-21	Oct-21	(1.00)	
19	Digital - Data and Analytics in the Cloud	Digital Foundations	1,513,280	-	-	-	-	This project will extend the Company's current data and analytics environment into a cloud environment, which will enable data and analytics at scale and enable the delivery of outcomes for the Natural Gas Delivery Plan, Electric Grid Integration, customer programs, and other business needs.	The alternative considered to a cloud solution is to continue to expand the on-premise infrastructure and purchase multiple tools to solve individual capability gaps. The Company did not choose this alternative because it is more costly than cloud due to higher infrastructure costs and more manpower required for implementation.	The project scope includes: (1) the execution of data analytics at scale without scalability constraints; (2) flexible transitioning with technology platforms as they evolve; (3) the use of out-of-the-box ML and AI tools provided by cloud vendors; (4) new services and innovations on the cloud platform; (5) capacity pay per use; and (6) the foundation for future cloud migration and maintenance.	The project will add value by: (1) providing the ability to perform scale analytics at-scale; (2) allowing the ability to leverage the leading machine learning ("ML") and artificial intelligence ("AI") tools to enable predictive and prescriptive analytics at-scale; (3) providing the ability to provision infrastructure at-scale rapidly; (4) enabling pay for use; (5) empowering faster prototyping, testing, and deployment of analytics solutions; (6) reducing total cost of ownership ("TCO"); and (7) providing operational tools for monitoring, incident management, and resolution.	Jan-20	Mar-20	Jun-20	Sep-20	Dec-20	(1.00)	
20	Radar Intrusion Detection	Electric Grid Integration	1,500,000	-	-	-	-	This project will integrate and deploy new radar detection technology to enable the detection and tracking of individuals at Company sites. The radar would alert in real time and allow for the dispatch of Security or law enforcement to investigate.	Alternatives considered include: (1) keeping the current fence intrusion detection solution; or (2) evaluating, selecting, and implementing a new intrusion detection solution. The preferred option is to implement a new solution, given limitations with the current solution that result in multiple false alarms that must be investigated.	The scope of this project encompasses: (1) assessing radar intrusion detection solutions to obtain a solution that fits the needs of the Company; (2) establishing a priority list of sites where the solution will be rolled out; and (3) installing updated radar intrusion at physical sites in the Company's service territory with a focus on critical infrastructure sites, including but not limited to gas compressor stations, electric substations, and hydro-electric sites.	Completion of this project will provide value to the Company through evaluating and selecting a more reliable intrusion detection solution that reduces the number of false alarms and prevent resources from being sent to a site for unnecessary investigations.	Jan-20	Feb-20	Aug-20	Jul-21	Nov-22	(1.00)	
21	Work Management Scheduling Analytics and Reporting	Electric Grid Integration	1,394,844	-	-	-	-	The Work Management Scheduling Analytics and Reporting project will implement a solution capable of scheduling long cycle, maintenance, and emergent work. This will combine data from the several excel spreadsheets that are used today to allow single view of all the information needed to effectively produce the various schedules and provide reporting.	Two alternatives were considered for this project: (1) Purchase disconnected software products. This option was not chosen due to a risk of increased costs as well as it contains solutions that provide overlapping functions with existing solutions or will not meet base requirements. (2) Automating manual data movement across systems through Robotic Process Automation. This option was not chosen because it will not meet base requirements and does not provide desired insights. The option to purchase the SAP Multi Resource Scheduling module was selected as it will minimize ongoing support costs, meets base requirements, and will provide increased transparency into scheduling.	The project scope includes: (1) implementing a streamlined scheduling process across Operations; and (2) implementing associated analytics and reporting.	The project will add value by providing: (1) accurate schedules across work types (emergent, customer requested, project, and compliance); (2) enhanced quality and integrity of scheduling process through reduction in manual scheduling steps and hours spent developing route sheets; (3) time saved from manual entry to be reallocated to better schedule analysis and alignment across disciplines, decreasing risk of missing code work and being in non-compliance; (4) improved transparency into whether the weekly schedule is meeting business objectives such as financial scenarios, compliance requirements, and first time completion; and (5) increased employee engagement by having a quality product to schedule, while being able to focus on priority and execution of work versus workarounds to meet daily scheduling needs.	Jul-21	Jul-21	Sep-21	Nov-21	Jul-22	(1.00)	
22	Centralized Demand Response Management	Electric Grid Integration	1,293,000	-	-	-	-	The Centralized Demand Response Management project is intended to centralize ingesting and reporting of all Demand Responses (DR) events activities into one technical solution and will be informed by the outcomes of the Centralized Demand Response Management Assessment effort.	Alternatives considered during the assessment effort include: (1) continue to maintain the current hardware and software at the risk of continued inability to consolidate information for all DR programs; (2) enhance the existing solution to address centralization needs; (3) transfer the application to the vendor's hosted environment and configure to address requirements for centralized DR programs; and (4) replace the current solution with a new cloud solution; or (5) replace the current solution with a new internally hosted solution. The selected alternative will be made as an outcome of the Centralized Demand Response Management Assessment effort.	The scope consists of: (1) centralization of the DR event management programs including residential DR, bring your own device, electric vehicle charging and large customer DR; (2) upgrade or replacement of the Demand Response Management System (DRMS) with a potential move to a cloud-based solution; (3) provision of new DRMS system functionality to support ongoing DR programs; (4) addition of DRMS functionality to account for the new TOU rates.	The project will add value by providing functionality to ensure Demand Response (DR) events can be collectively managed from a central solution that includes all programs and potential load shed. The upgraded Demand Response Management System (DRMS) application will be able to support the various DR Programs including new Electric Charging Vehicle programs and Time of Use (TOU) rates implemented in 2019. Finally, the upgraded application will ensure accuracy of kilowatt hour (kWh) savings and peak load reduction when a DR event is called.	Jun-21	Aug-21	Oct-21	Apr-22	Jul-22	(1.00)	
23	Digital - Data Governance	Digital Foundations	1,098,430	-	-	-	-	This project will be used to establish data governance roles and responsibilities, processes, and the purchase of a tool to support best practices across the enterprise.	Alternatives considered include: (1) not implementing a data governance solution, potentially limiting the Company's productivity due to data quality and accessibility; (2) developing an internal tool to help manage the Company's data footprint; or (3) purchasing a third-party solution. Purchasing a third-party solution was selected because the skill sets required to internally develop such tools are not available and it would take a larger investment to upskill or hire individuals with this experience.	The project scope includes: (1) initiating key data domains and ownership across the Company through the creation of an overarching data governance process; and establishing processes and cadence for reintroducing new data elements into their domain; (2) enabling people to become functional and technical data stewards through education; and (3) implementing technology through the selection of a data cleansing, data quality, data extract, and transformation tool, including enterprise-wide semantic definition.	The project will add value by: (1) increasing productivity of data analysts across the Company by reducing time spent cleaning data; (2) improving business planning; (3) maximizing the use of data to make decisions; and (4) discovering where data lives and the definition of data elements.	Feb-20	Feb-20	Apr-20	May-20	Dec-20	(1.00)	
24	Electric Interconnection Billing and Payment	Electric Grid Integration	1,095,000	-	-	-	-	Entities with a desire to interconnect to the Company's electric distribution network are required to submit applications and associated fees for processing. The Electric Interconnection Billing and Payment project will research, evaluate, and implement a solution that will provide for the electronic processing of applicant fees.	The alternatives considered include: (1) implement an electronic billing and payment solution that the Company's current interconnection solution provider integrated into their solution for another client; (2) implement electronic billing and payment functionality that is currently used to process other types of payments to the Company; and (3) delay implementation of a solution until accepting electronic payments becomes a regulatory requirement. The third alternative was not selected because risks associated with handling physical checks will continue, while providing less time for solution evaluation and implementation once processing of electronic payments becomes a regulatory requirement. Alternatives one and two, both cloud-based solutions, will be evaluated once requirements are finalized.	The scope of this project includes: (1) researching industry solutions and defining requirements; (2) creating a request for proposal (RFP); (3) evaluating options and selecting a solution and; (4) designing and implementing the solution.	Providing the capability for interconnection applicants to pay application fees electronically adds value through: (1) reduced manual internal processing of checks; (2) minimized risks associated with lost payments, and (3) increased likelihood in applicant satisfaction.	Jan-21	Apr-21	Jul-21	Oct-21	Nov-21	(1.00)	

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25.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio	
25.	Application Currency and Enhancements - Corporate Services - Capital	Upgrades and Applications - Capital	1,035,000	-	-	-	-	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade applications that support the Corporate Services Portfolio.	Prior to implementing the applications, a review is completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocations for enhancements are considered. This project makes ongoing upgrades and support for the applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas such as Finance, HR, Learning & Development, Legal, Governmental, Regulatory and Public Affairs, Corporate Security, and Strategy and IT Governance.	The value of regular upgrades to applications in the Corporate Services Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Corporate Services.	-	-	-	-	-	-	(1.00)
26	ARP - Printer Asset Management (PAM)	Asset Refresh Program	1,032,558	-	-	-	-	This project will replace printers, plotters, and multi-function printing devices.	The alternatives considered for the project included looking at refresh cycles from three to seven years and running the assets to failure. The selection of a five-year cycle was deemed to be the best solution since anything less than five years would result in additional unneeded expense for replacement of assets that were still in peak operating condition, and anything greater than five years, including running the asset to failure, would result in additional expenses in maintenance of the equipment and downtime, negatively impacting employee productivity.	The project scope consists of the annual replacement of printer assets under this program.	This project creates value for the Company by: (1) improving the dependability of these printer devices for employees, (2) averting increased costs due to hardware repairs, and (3) ensuring compatibility with enterprise print applications.	-	-	-	-	-	(1.00)	
27	Fusion Center Technologies	Security	1,000,000	-	-	-	-	This project will further integrate the physical and cyber security teams through the implementation of new technologies. This project will evaluate, replace, and implement new capabilities for monitoring physical sales and technology assets, significantly improving the Company's quality and timeliness of detection and response to both physical and cyber security incidents.	The alternative considered was to remain on current toolsets and not implement additional physical security countermeasures, impacting the quality and timeliness of detection and response to both physical and cyber security incidents. The Company is choosing to replace and implement new capabilities to provide further efficiencies and capabilities.	The scope of this project encompasses: (1) all endpoints and servers, and all facilities that house sensitive data; (2) defining system scalability requirements; (3) designing new platform architecture; and (4) purchasing and implementing hardware and software once alternatives are reviewed.	Completion of this project will provide value to the Company through: (1) implementing a new toolset to move MITRE Adversarial Tactics, Techniques, & Common Knowledge (ATT&CK) coverage from 30% to 80%; (MITRE ATT&CK Framework is an industry-leading framework to detect malicious activity); (2) replacing the vulnerability management platform, Qualys, which has several capability gaps estimated to be remediated at a significant cost; and (3) evaluating and implementing a biometric toolset, such as facial recognition, to eliminate an unauthorized individual to follow an authorized individual into a facility and to proactively alert if unwanted or unknown personnel at Company facilities.	Apr-20	May-20	Aug-20	Jun-21	Aug-21	(1.00)	
28	Lock and Key Management System	Security	1,000,000	-	-	-	-	This project will identify and implement a physical smart lock and key management system throughout the Company's service territories. Current estimates show there are approximately 10,000-14,000 locks throughout the state, and the Company has no system to properly manage ownership of specific physical keys and control who uses them to access sites.	Alternatives considered include: (1) remain at the current state and forgo implementing new lock and key management capabilities; or (2) implement a smart lock and key system. The preferred option is to mitigate known and observed risks with the lock of lock and key management capabilities.	The scope of this project includes: (1) assessing the type of locks and smart lock, and at what sites they will be needed to properly plan this project; (2) determining the levels and function of different smart lock systems available; and (3) purchasing and implementing a lock and key management system based on assessment findings.	Completion of this project will provide value to the Company by: (1) assessing and taking inventory of current locks and keys that are used throughout the state; (2) determining which core functionalities needed to ensure proper lock and key management state-wide; and (3) reviewing and implementing a solution to give the physical security team a lock and key management capability.	Feb-21	Mar-21	May-21	Jul-21	Dec-21	(1.00)	
29	Electric Distribution Asset Management	Electric Grid Integration	992,000	-	-	-	-	The Electric Distribution Asset Management project will develop a Distribution Asset Management strategy with corresponding process and technology implementation plans and objectives.	One alternative considered is to continue to fund and implement electric distribution asset-based technology solutions as separate projects until (a) upgrading the Geographic Information System (GIS) with the next major data model version and integrating affected applications; (2) integrating GIS with the work management system (SAP); (3) enhancing mobile data; and (4) designing asset data processes. This alternative was not selected as it does not consider technology, process, people and data inter-dependencies. The recommended alternative is that which is outlined in this business case because it creates the opportunity to identify, prioritize and address key gaps in current asset data collection processes, addresses the technology and process inter-dependencies and provides for distribution asset management planning and investment optimization.	The project scope includes: (1) developing a Distribution Asset Management strategy with a corresponding process and technology implementation plans and objectives; (2) designing and implementing optimized processes and supporting technology solutions; (3) implementing an Asset Integrity Management program leveraging industry standards; (4) evaluating, designing and implementing tools to optimize investment planning and asset health management; (5) implementing a distribution asset system of record repository; and (6) designing and implementing the organizational change management necessary to ensure success.	The Company will gain value from this project through: (1) creating a standardized process for designers and field workers to collect and manage asset data, reducing waste and rework associated with current processes; (2) optimizing planning and management of distribution assets, resulting in investments of highest value; (3) establishing a channel for the Company to develop and manage asset health conditions on the distribution system, optimizing the operation and maintenance investments of distribution assets; and (4) improving the quality of Company electric distribution asset data, optimizing asset performance and providing opportunities to leverage quality data to support other engineering and operations efforts.	Jan-20	Apr-20	Aug-20	Oct-20	Dec-24	(1.00)	
30	ARP - Operational Technology Support	Asset Refresh Program	880,791	-	-	-	-	This project will replace dated and obsolete servers and workstations.	Extending maintenance is not a viable alternative as current systems do not provide sufficient capacity for new electric system support capabilities.	The project scope consists of: (1) the annual replacement of compute hardware under the program; and (2) installing additional new compute capacity to account for organic growth requirements.	This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support the operation of the Company's critical electric infrastructure.	-	-	-	-	-	(1.00)	
31	Digital - Application Programming Interface Foundations	Digital Foundations	870,000	-	-	-	-	This project provides foundational capabilities around APIs, which are a set of technologies used to integrate applications within the Company and, more importantly, with external third-party applications. In short, APIs allow two applications to talk to each other. This is currently a gap and is a key technical capability needed to enable multiple IT projects related to the Company's strategy for clean energy like: Electric Interconnection Billing and Payments, MISO Market User Interface Changes, Utility Analytics Miso-Phasor/Power Quality/Outage, Field Contractor Work Management Technology Enablement, Work Management Scheduling, Analytics and Reporting, DERMS, and Google X. Similarly, this API foundational capability will enable customer-focused projects like: Business Customer Internal Web Portal ("IWP"), Bring Your Own Thermostat ("BYOT") Pilot, Customer Relationship Management ("CRM") Product Suite, and Large Customer Rate Tool.	Two alternatives were explored and determined to be non-viable: (1) remain on current Trello API product that is part of their ESB platform, which does not support the future needs of the Company including efforts to modernize the grid; and (2) scale back on the implementation of the API fabric and complete over a longer period, which would negatively impact other projects. These alternatives were not selected because the API fabric is foundational and needs to be completed to support other projects. The current scope and direction is the best fit to support current Company initiatives.	The project scope includes: (1) evaluating multiple API management tool vendors; (2) executing API onboarding for a number of external partners without scalability constraints; and (3) configuring and deploying API services with out-of-the-box operations looking to achieve faster speed to market.	The project will add value by: (1) implementing functionality to perform API services; (2) allowing partners to integrate with the Company's internal applications; (3) providing the ability to reuse integrations; (4) enabling monitoring of API traffic; (5) implementing functionality to perform API throttling (i.e., traffic management); (6) visualizing API traffic and analytics through key performance indicators ("KPIs"); (7) enabling faster prototyping, testing, and deployment of integrations; and (8) providing operational tools for monitoring, incident management, and resolution.	Jan-20	Mar-20	May-20	Nov-20	Dec-21	(1.00)	
32	One Call Ticket Risk Analysis Model for Damage Prevention	Electric Grid Integration	837,500	(380,160)	-	-	-	The One Call Ticket Risk Assessment Model project implements a risk analysis and data analytics program that identifies the related excavation tickets, utilizing current one-call ticket data, damage history, and incorporating asset information from the GIS to focus damage prevention resources and activities on locate requests with the highest risk.	Alternatives considered include: (1) augmenting existing damage prevention staff to manually perform daily risk analysis on one call tickets; (2) developing an in-house, custom solution with significant consulting fees; and (3) purchasing a commercial implementation to a future year. The first alternative was not selected because it would require a significant increase in resources to perform the work. It is estimated that each ticket analysis would take 15 minutes, resulting in an additional 50 full time resources (15 minutes/ticket*445,000 tickets/year = 111,250 hours/year ~ 50 full time resources ~ approx. \$5M in labor and overheads). The second alternative was not selected because the estimated capital investment would exceed a commercial vendor solution. The first alternative was not selected because it continues to defer value realization and does not provide a timely response to mitigate safety risk. The option of implementing a cloud-based solution was chosen because it implements a solution that has been tested in the industry with success, provides an ongoing support model through a subscription-based application, and provides a path to deliver faster business value through damage reductions.	The project scope includes: (1) vendor application hosting, maintenance, and support of one or more instances of the Cloud-based platform; (2) a solution to provide gas and electric asset information from the GIS databases for use within the vendor platform as either extracts (FTP or emailed files) on a regular interval (i.e., quarterly, monthly) through integration with the platform; (3) ongoing and historical ticket locate data to vendor analysis would take 15 minutes, resulting in an additional 50 full time resources; (4) copies of ongoing and historical excavation damage records that occurred on any of the locate requests provided above by either a spreadsheet or web service; and (5) a damage risk statistical model that calculates risk scores based on multiple factors to produce output of a daily summary of higher risk tickets, reports, and web access to the customer portal for detailed risk assessment data.	Completion of this project will provide value to both the Company and its customers by: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations.	Feb-21	May-21	Jul-21	Sep-21	Dec-21	0.11	
33	Application Currency and Enhancements - Operations - Capital	Upgrades and Applications - Capital	832,800	-	-	-	-	The Application Currency and Enhancements - Operations O&M initiative will apply O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. The project will also upgrade applications that support the Operations Portfolio.	Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocations for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure, (2) an increased number of incidents, (3) paying increased support costs, and (4) preventing employees from performing their daily tasks.	The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the Operations Portfolio.	The value of regular upgrades to applications in the Operations Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for Operations.	-	-	-	-	-	-	(1.00)

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio
For the Projected Test Year 12 Months Ending December 31, 2021
(\$000)

Case No.: U-20087
Exhibit No.: A-108 (JDT-6)
Page: 5 of 8
Witness: JDT/Adam
Date: February 2020

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio
34	Renewables Supervisory Control and Data Acquisition Overlay	Electric Grid Integration	813,157	-	-	-	-	The Renewables Supervisory Control and Data Acquisition (SCADA) Overlay project is the design, procurement, installation, and commissioning of a SCADA overlay software application for the Merchant Operations Center and Renewable Generation Operations to view and control Lake Winds, Cross Winds, Grand Valley Solar, Western Michigan Solar, and Circuit West Solar on a consolidated application pane. It will also provide the platform for monitoring and control of future wind and solar assets added to the generating fleet.	Alternatives considered include: (1) an extensive manual build of screens into the existing SCADA view; (2) building an overlay into the current historian system, though it would have limited capabilities; and (3) implementing vendor software that is more related to distributed energy resources than utility scale generation assets and accept limited functionality and usefulness. These alternatives were not selected due to limitations in standard functionality that decreases usefulness and increases the cost of customizing the existing SCADA solution. The selected off-the-shelf renewables SCADA overlay software is the preferred option due to preliminary fit with requirements and cost-effective implementation.	The project scope includes: (1) merging multiple different Renewables SCADA views (i.e. control and control functions, alarms, warnings, and key performance indicators for wind turbines and solar inverters) into a single operational and dispatch view through off-the-shelf renewables SCADA overlay software; (2) installing on-premise servers and databases for daily operation and disaster recovery functions; and (3) integrating with the individual site native SCADA systems.	The project will add value by: (1) bringing the renewables generating fleet operations and dispatch up to industry standards for its projected size by managing all the assets from a single screen; (2) allowing better monitoring of the renewables generating fleet; (3) automating the NERC required reporting and associated changes; and (4) improving quality while reducing time spent on reporting and data management for analysis.	Jan-21	Feb-21	May-21	Sep-21	Oct-21	(1.00)
35	Enhancements - Cloud Automation	Upgrades and Applications Currency	783,443	-	-	-	-	The Company's existing cloud automation platform consists of a number of scripts that automate the build of infrastructure like servers, operating systems, databases, and application installations in minutes or hours. These scripts include security hardening and application of the latest patches. This project will maintain and improve these scripts to ensure they meet the latest security policy and requirements, and any changes to the way the operating systems, databases, or applications are configured.	The alternative would be to continue as-is with existing scripts; however, this introduces security risks and additional manual steps to build infrastructure, which will lead to increased costs and risk of manual errors.	The scope includes updating and improving the cloud automation platform to align with the Company's current security, database, operating system, network, and other application standards.	The value of this project is inherent in the time savings and quality improvement, as it used to take weeks or even months to manually build infrastructure servers, which is prone to human error.	-	-	-	-	-	(1.00)
36	Rate Design and Evaluation Tool	Corporate and Enterprise	752,475	-	-	-	-	The Rate Design and Evaluation Tool (RDET) project will provide the company the ability to design, analyze, and evaluate proposed electric rates with automated tools. This will permit the Rates Team to quickly adjust to electric rate changes proposed by the State of Michigan and Michigan Public Service Commission and identify which customers would be positively impacted by a rate change.	Alternatives considered include: (1) Implement an alternative vendor solution to support the rate analysis and design activities. This alternative was not selected as the tool has higher project costs than chosen solution and the Company is already engaging with the selected vendor for a separate effort. This alternative would also increase operational costs by introducing additional ongoing maintenance expenses. (2) Utilize the Company's Internal Data Lake and Tableau. This alternative was not chosen as it is not cost-effective to replicate the capabilities of the selected solution internally. (3) Implement the selected vendor platform. The Company selected this option because of the minimal impact to operational expenses by leveraging a vendor platform already implemented at the Company. The selected solution will create synergies by aligning several internal organizations on a single platform for rate analysis and comparison. In addition, the selected solution is considered industry best in class.	The scope for the project includes: (1) implementing RDET; (2) deploying integrations to send relevant rate data to RDET; and (3) training staff on utilizing RDET.	The Company and its customers will receive value through this project by: (1) streamlining processes for rate analysis and design; (2) automating data extraction and analysis; and (3) improving billing accuracy when new rates are implemented.	Feb-21	Mar-21	Mar-21	Apr-21	Jul-21	(1.00)
37	Electric Geographic Information System (GIS) Platform Upgrade	Upgrades and Applications Currency	768,000	-	-	-	-	This project will upgrade the GIS platform applications, servers, application hosting servers, and databases to the next major versions to remediate the outdated platform technology.	The alternatives considered for the project were to: (1) delay implementation to support a permit for extended Windows support; and incur cyber security and business continuity risk for the application hosting platform and the GIS platform; (2) upgrade the platform to the Utility Network as a "big bang" approach without the incremental GIS version upgrade; and (3) perform an iterative platform upgrade to the next GIS version with application platform hosting upgrade, server replacements, and database upgrade, and shift the Utility Network upgrade to a future year. Option 1 was not selected since further project delays will prolong the security and business continuity risk, as well as increase the probability of risk impact due to the level of effort required to complete the project. Option 2 was not selected because the Utility Network upgrade effort is a transformational technology project, requiring extensive data migration in addition to the scope outlined in this project. Furthermore, this alternative will also incur security and business continuity risk given the complexity of a prolonged planning and design effort. Option 3 was selected to implement an iterative upgrade approach as it decreases the overall project complexity while mitigating aging technology platforms, which enables the future upgrade to the Utility Network in addition to addressing overall platform stability and sustainability in a timely manner.	The project scope includes: (1) building GIS platform servers, software, and data; (2) expanding GIS platform capabilities; (3) re-architecting current single-point-of-failure areas; (4) designing and developing new architecture; (5) upgrading database technology; (6) re-platforming the application hosting platform (Citrix); and (7) upgrading and reconfiguring applications that utilize the GIS platform.	The project adds value by: (1) remediating GIS, Windows, and Citrix technology obsolescence, creating a stable and sustainable foundational platform for critical business functions; (2) improving platform resilience by remediating current single-point-of-failure areas; (3) sharing technology resources (servers, databases, etc.) across gas and electric platforms to optimize them, yet still ensuring the ability to meet diverse business area functional needs; and (4) adding new capabilities for spatial analytics, real-time visualization of data, and three-dimensional infrastructure network traceability. In addition, the GIS platform remediation work establishes the foundation necessary for the full implementation of the Electric Distribution Asset Management Program.	Jan-20	Jun-20	Dec-20	Dec-21	Dec-21	(1.00)
38	ARP — Collaboration	Asset Refresh Program	740,362	-	-	-	-	This project will replace the Company's collaborative tools and equipment.	The alternatives considered were to: (1) refresh visual assets and a portion of the audio assets; (2) refresh a portion of the audio assets only; and (3) refresh visual assets only. These alternatives were not chosen due to the risk inherent with a partial replacement of assets which includes: (1) a reduced supply of equivalent replacement Avaya parts that are no longer being produced; and (2) an erosion of the knowledge technicians possess on discounted systems in favor of education on the newest available technology.	The project scope consists of: (1) annually replacing collaboration assets; and (2) installing new collaboration assets to account for organic growth requirements.	This project creates value by: (1) ensuring that the Company's audio, visual, telephone, and other communications systems are stable and reliable; and (2) beginning the foundational retirement of the legacy Avaya PEX systems that have reached end-of-mainstream manufacturer support.	-	-	-	-	-	(1.00)
39	Electric Infrastructure Attachments	Electric Grid Integration	727,000	-	-	-	-	The Electric Infrastructure Attachment (IA) project will research, create a request for proposal (RFP), evaluate solution alternatives, design, develop and implement software to replace the existing functionality of the IA System Design and Standards (ISDS) Microsoft Access database. Unlike the current solution, the new solution will be IT supported.	Alternatives considered include: (1) select and implement a fully integrated, IT-supported solution that replaces the existing functionality of the IA ISDS database; (2) Develop an in-house replacement for the ISDS database through updates to the existing database or SharePoint. Alternative two was not chosen because this option would not integrate with Corporate systems, exacerbating ongoing issues with data discrepancies and client dissatisfaction making it an inferior solution. (3) Continue to use the current ISDS database. Alternative three is unacceptable due to the stated business risks. The first alternative was selected because it will provide for an integrated, IT-supported solution.	The scope of this project includes: (1) researching industry solutions; (2) creating a request for proposal (RFP); (3) evaluating options and selecting a solution; and (4) design and implement a solution.	Purchasing and implementing an IT-supported software platform that provides current functionality of the ISDS Database, as well as new functionality will benefit both the Company and its customers by: (1) integrating data across several platforms including SAP, Geographic Information System (GIS), the National Joint Utility Notification System (NJUNS), and location-specific field design related materials as documentation of the assets inspected; (2) providing a more efficient approach to fulfill information requests; (3) affording a history of pole attachments at a specific location; and (4) reducing waste through improved processes.	Jan-21	Apr-21	Aug-21	Nov-21	Dec-21	(1.00)
40	Enterprise Service Bus Application Upgrade	Upgrades and Applications Currency	718,505	-	-	-	-	The Enterprise Service Bus (ESB) Application Upgrade project will upgrade and migrate the Business Works developer application to the next version.	The alternative considered was to absorb the annual \$40,000 maintenance support for any of the applications involved. Any sustained product deficiency would impact many areas of the company such as billing, revenue collection, and remote meters. This alternative was not chosen due to these reasons, and the additional expense.	The project scope includes: (1) implementing new versions of all applications that are part of the ESB in two security zones; and (2) a server refresh. The new products will be implemented on the latest version of the Suse Enterprise Edition operating system.	This value this project provides the Company includes: (1) accelerated productivity; (2) continuous delivery and integration; (3) an open ecosystem for improved operational visibility; (4) integration with web, mobile apps, and application programming interfaces in real-time; and (5) improved administrative and operational efficiencies. The upgrade will also provide for easy integration with other cloud-based solutions such as Amazon Web Services and provides for easy migration to cloud-based containers in the future.	Sep-20	Sep-20	Nov-20	Apr-21	Dec-21	(1.00)
41	Data Lake Entry	Customer Experience and Operations	580,125	-	-	-	-	The Data Lake Entry project will enable the Customer Operations Data Analytics team to finalize the migration of all Customer Operations data from the Netrus SQL Server to the Microsoft Data Lake.	Three alternatives were considered but not chosen for this project: (1) Continue with segregated data which negatively impacts process efficiency for data gathering and analysis. This option was not chosen as it requires extensive manual processing to consolidate routine tasks; (2) Migrate a portion of the data. This option was not chosen because it provides minimal value compared to overall costs and processes would be more complicated with only part of the data migrated; (3) Defer this work to a future year. This option was not selected as it perpetuates an existing problem with no resolution. The option to migrate data into the existing Microsoft Data Lake was selected as the team already utilizes the tool for analytics with other data sets and adding this data provides the Company with access to key customer data without manual action.	The scope of the project includes: (1) extract, transformation, and loading of all Customer Operations data into the data lake automatically and in near real-time; and (2) implementing enhancements from existing Customer Operations systems or tools, like Interactive Voice Response (IVR), into the data lake for further analysis.	The Company will gain value from this project by having all Customer Operations data in the data lake, reducing manual efforts to collect, consolidate, and analyze the data.	Jan-21	Feb-21	Apr-21	May-21	Sep-21	(1.00)
42	Iron Enterprise Edition Upgrade	Upgrades and Applications Currency	550,000	-	-	-	-	This project will upgrade IEE, which collects the reads from meters to ensure accurate and non-estimated bills are provided to our customers.	As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and cyber security of the system.	The scope of this project includes: (1) an operating system refresh; (2) migrating from Windows 2012 to next Windows version; and (3) migrating the database from SQL Server version 2012 to the next SQL Server version.	This project creates value for the Company by: (1) ensuring the new features and functionality added to meet business requirements are available to business partners and IT; (2) meeting Information Security's requirement to keep applications patched to control cyber debt; and (3) allowing for validation, estimation, and testing functions for all data collected.	Jan-21	Jan-21	Feb-21	Apr-21	Nov-21	(1.00)

	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	OMB Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio	
43	SAP Data Archiving	Upgrades and Applications Currency	\$48,500	-	-	-	-	This project will move outdated data from an online database to off-site storage.	Three alternatives were explored and determined non-viable for the project: (1) allow the database to grow in size, which would put system performance at risk and result in prohibitive storage costs; (2) decrease the overall scope and archive less project, which would result in minimal positive impact to system growth and significant storage-related costs; and (3) increase the scope and archive more objects in a shorter timeframe, which would result in significant cost increase over shorter time period. After considering each of these options, it was determined that the current scope of the project was the best strategy to address the problem while balancing annual spending.	The project scope includes: (1) archiving data based on the fastest growing and largest objects in SAP; (2) building and archiving solutions that allow the business to retrieve archived data in the required form; and (3) setting up the solution so that the business areas meet compliance standards.	This project will create value for the Company by: (1) increasing system stability by reducing data growth; and (2) controlling maintenance costs associated with data storage.	-	-	-	-	-	-	1.00
44	Application Currency and Enhancements - Upgrades and CE&O - Capital	Upgrades and Applications Currency	\$40,000	-	-	-	-	The Application Currency and Enhancements Customer Experience and Operations ("CE&O") initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade the applications that support the CE&O portfolio.	Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increased number of incidents; (3) paying increased support costs; and (4) preventing employees from performing their daily tasks.	The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the CE&O Portfolio.	The value of regular upgrades to applications in the CE&O Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for CE&O.	-	-	-	-	-	-	1.00
45	Fusion Center	Security	\$33,915	-	-	-	-	This project will procure a location to combine the Security Command Center and the Cyber Security Incident Response Team ("CSIRT"). This project will focus on constructing a physical office space to facilitate these two teams working in a collaborative fashion.	As part of the review process, the alternative considered was to continue to run a separate Security Command Center for physical security incident response and a separate Cyber Security Incident Response Center. The two centers would continue to run separate, non-integrated systems and tools. This option was not chosen as it does not provide the integrated space and tools needed for physical and cyber security teams to collaborate to resolve critical incidents that pose both a physical and cyber threat.	The project scope encompasses building an integrated Security Command Center across physical and cyber security domains that takes into consideration physical space, technology, process integration, analytics, and intelligence capabilities without changing existing organizational structures.	This project will add value by: (1) co-locating physical and cyber team members for process efficiencies and better integration of tools in the event of a security incident that requires both teams' collaboration; (2) having a dedicated emergency operations area for use in the event of a security incident (cyber or physical or both) providing the tools needed to keep stakeholders informed; (3) evaluating and potentially implementing a shared intelligence capability across both areas; and (4) further integrating existing processes and workflows.	Jan-19	Jan-19	May-19	Oct-20	Jan-21	1.00	
46	Replace and Re-badge	Security	\$00,181	-	-	-	-	This project will: (1) replace and re-badge existing card readers with human interface device multistack readers; and (2) re-badge employees and contractors with more secure badges, mitigating during vulnerabilities with the current legacy technology being used.	As part of the review process, the alternative considered was to continue with current card readers, which have a potential to be cloned. This alternative was not chosen due to the potential to high likelihood of security vulnerabilities and risks.	The scope of the implementation includes: (1) replacing existing card readers; (2) re-badging to i-class badges, and (3) encrypting the new badges.	The value of completing the project is avoidance of the security risk for one of the Company's key security controls.	Jan-20	Apr-20	Jul-20	Aug-20	Jan-21	1.00	
47	AccessNOW	Security	\$00,000	-	-	-	-	This project will implement configurable identity access management functionality and best practices, and will enforce compliance.	As part of the review process the alternative considered was using manual processes. This option was not chosen because it was deemed too costly and inefficient.	The scope of the project includes: (1) the design; (2) configuration; and (3) implementation of P2S systems for headsets, lower sites, dispatch consoles, and subscriber equipment.	This project will add value by: (1) reducing waste and failure points through automation; (2) improving and standardizing the business partner experience; (3) centralizing access management; (4) ensuring regulatory compliance; and (5) ensuring system stability and continuous improvement.	Mar-19	Apr-19	May-19	Jul-19	Sep-19	0.96	
48	ARP - Operational Technology Storage Area Network	Asset Refresh Program	\$88,000	-	-	-	-	This project will refresh aging SAN with Unity SANs.	The alternative considered was to purchase extended maintenance. This alternative was not chosen due to the risk of increased downtime of critical infrastructure and maintenance costs. The cost of bringing personnel on site to make system corrections in the event of a service interruption is higher than the cost of buying new.	The project scope consists of: (1) annually replacing SAN hardware under the program; and (2) including additional new capacity to account for organic growth requirements.	This project creates value by maintaining the currency of the Company's IT infrastructure and core enterprise software that are utilized to support and enhance customer interactions, as well as ensure the stability of technology for business operations that are in service of the Company's customers.	-	-	-	-	-	1.00	
49	Workforce Connect - Talent Enablement	Corporate and Enterprise	\$76,250	-	-	-	-	The Workforce Connect - Talent Enablement project is part of an overarching talent enablement plan which will enable the Company to move forward in creating an employee experience to better serve customers. Improving the employee experience is focused on two outcomes: (1) enabling employees to be more engaged and productive; and (2) attracting and retaining new skills and talent, both of which result in providing better service to customers, an improved customer experience, and higher customer satisfaction.	Four alternatives were considered for this project: (1) Maintain employee master data in the SAP HCM system, knowing that vendor support for HCM will be retiring at the end of 2025. This approach was chosen as it poses a risk to system stability due to declining vendor support (such as maintenance, enhancements, and defect resolution) which could have Company-wide implications given the extent to which SAP HCM integrates with other systems throughout the Company and the number of functions and processes reliant on employee data; (2) Explore alternate systems instead of SAP HCM for core Human Resources data and transactions. Consumers Energy uses 15 SAP modules, which are reliant on employee data from SAP HCM. Additionally, more than 40 other systems and applications are integrated with and dependent on the SAP HCM data. Migrating to a non-SAP Human Resources system would result in customization of technical integrations back to SAP and other systems currently integrated with SAP HCM. This approach was not selected, as migrating to a non-SAP core Human Resources system would incur significant re-work costs and would have Company-wide implications. For example, Company employees would need to be retrained and new technology support contracts would need to be negotiated at a higher cost; (3) Replace the SuccessFactors solution with another talent management system. This option was not chosen since it would be costly to start over with a brand new talent management system, would result in significant organizational change management to support user adoption and loss of significant time and training invested in training up employees to use and maintain SuccessFactors, and would require retraining of the Human Resources Technology team as well as the majority of Human Resources employees who regularly use the current system in their day-to-day work; (4) Expand the current Workforce Connect (SuccessFactors) solution with SAP's	The Workforce Connect - Talent Enablement project will implement the following SAP SuccessFactors modules: Employee Central, Succession Planning, Career Development, and Compensation. This project will ensure support of the following processes after the SAP HCM module retires: Core Human Resources Management, Hiring, Onboarding, Performance Management, Succession Planning, Compensation and Benefits, Leader and Employee Development, Technical Training, Talent Programs, Workforce Analytics, and Workforce Planning. The project scope includes: (1) implementing SAP SuccessFactors Employee Central, Succession Planning, Career Development, and Compensation modules; (2) retrofitting current SuccessFactors modules to support the integrated talent management suite; (3) providing mobile capabilities; (4) retiring the SAP HCM module; and (5) providing integrations as needed from SuccessFactors to SAP.	This project will provide the following value: (1) increased employee retention for continuity of knowledge and long-term customer relationships, as well as reduction in costs associated with recruiting and onboarding new employees; (2) positive direct customer interactions facilitated through strong employee engagement; (3) enhanced employee development to meet customer needs during a time where retirement eligibility is high and risk of knowledge loss has the potential to negatively impact customer service and satisfaction; (4) enable the Company to operationalize a new career framework and competency model which will support training and development of employees to deliver strong customer service and will enhance the ability to hire candidates capable of delivering on the Company's initiatives and goals; (5) advanced data analytics that provide data-driven insights in support of maintaining and enhancing the workforce to continue to deliver strong customer service and results; (6) reduce waste and defects leading to increased quality and simplified manual processes; (7) improved insight and consistency in reporting (i.e. Ethics, Safety, Office of Federal Contract Compliance Programs); (8) provide transparency into key talent areas to identify retention risk within critical areas and develop succession strategies; and (9) ensure core systems are stable and operational interruptions are minimized. The Company currently uses the SAP Human Capital Management ("HCM") module as a master source of employee data which feeds such information to other systems throughout the Company, including Human Resources and non-Human Resources systems. However, vendor support for the SAP HCM module will be retiring at the end of 2025.	Oct-20	Feb-21	Jul-21	Mar-22	Mar-23	0.98	
50	SQL Server Database Upgrade	Upgrades and Applications Currency	\$57,680	-	-	-	-	This project, which supports critical applications such as the meter read collection systems and the customer contact center applications, will upgrade all SQL Server 2008-2014 instances to the latest version.	The alternative considered was to migrate to Azure Cloud. As part of this option, the organization would obtain three years of extended support through Microsoft on SQL Server versions 2008-2008. This option was not selected because extended support would not be provided 2012 and 2014 versions, and the organization would not reap the benefits of new features offered through the upgrade. The Company decided in favor of the upgrade to avoid these issues and ensure system stability.	The project scope includes: (1) upgrades to all SQL Server 2008-2014 instances currently in use and not identified as part of another portfolio upgrade project, test, build, or pending system retirement (approximately 400 instances); (2) installation support would be provided 2012 and 2014 versions, and the organization would not reap the benefits of new features offered through the upgrade. The Company decided in favor of the upgrade to avoid these issues and ensure system stability.	This project will create value for the Company and its customers by: (1) reducing the risk of system failure and the resulting impact to business partners and customers; and (2) ensuring that systems are secure, supported, and have the latest features and functionality.	Jan-20	Apr-20	Oct-20	Apr-22	May-22	1.00	
51	Customer Operations Commercial Theft	Customer Experience and Operations	\$50,118	-	-	-	-	The Customer Operations Commercial Theft project will develop tools and complex reports to enable the Theft and Fraud teams to proactively identify incidents of commercial theft. This effort will provide high-confidence theft detection.	An alternative considered was to identify a third-party technical vendor that specializes in theft analytics to provide a solution rather than developing the necessary functionality using Company resources; however, the cost of a vendor solution outweighed the implementation benefits.	The project scope includes: (1) defining high-value theft algorithms specific to the identification of commercial theft; (2) identifying and testing algorithms; and (3) creating and generating theft reports.	The project will add value by: (1) assuring a high-probability of actual theft prior to on-site investigations; (2) increasing the safety of our field teams, and (3) eliminating unnecessary customer visits and the associated expense of dispatching a field crew. The benefits will result in reduced uncollectible expenses for our customers and the Company.	Mar-21	Apr-21	May-21	Sep-21	Nov-22	1.00	
52	Asset Refresh Program - Cyber Security	Security	\$40,000	-	-	-	-	This project will replace cyber security infrastructure to support increasing system and application demands, and to prevent system failures and service interruptions.	As part of the review process, the alternative considered was not to upgrade or replace assets as required, which is likely to introduce security risks, system vulnerabilities, and out-of-warranty repair costs.	This project will support continued systems stability. The scope encompasses: (1) evaluation, validation, and replacement of cyber security firewalls and servers; and (2) asset and application upgrades.	This program includes projects that bring value to the Company by maintaining the currency of the security infrastructure and core enterprise software. These are used to ensure the stability of technology for business operations.	-	-	-	-	-	1.00	
53	Application Currency and Enhancements - Upgrades and TEOS - Capital	Upgrades and Applications Currency	\$40,000	-	-	-	-	The Application Currency and Enhancements Transmission, Engineering and Operations ("TEOS") Support initiative will utilize both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the TEOS Portfolio.	Prior to implementing the listed applications, a review was completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between keeping the application current and risking failure; (2) an increased number of incidents; (3) paying increased support costs; and (4) preventing employees from performing their daily tasks.	The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for the TEOS portfolio areas.	The value of regular upgrades to applications in the TEOS Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the TEOS Portfolio.	-	-	-	-	-	-	1.00

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio

For the Projected Test Year 12 Months Ending December 31, 2021

(\$000)

Case No.: U-20087

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Witness: JDT/abcm

Date: February 2020

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio
54	MISO Market System Replacement	Electric Grid Integration	440,000	-	-	-	-	The Midcontinent Independent System Operator (MISO) Market System Replacement project will implement software changes in response to the announcement that MISO has determined that its current market system will be obsolete in five to seven years given evolving cyber security standards and increasing market complexity. To address these concerns, MISO is proposing a \$134M, seven-year incremental (modular) replacement to transition to a new market platform.	Alternatives considered in order to continue to interact with MISO include: (1) replacing PCI with a new product, or (2) updating the PCI solution. The alternative to replace PCI would require significant market research, product evaluation, people resources, and financial investment. Due to the schedule for updates provided by MISO, and the significant footprint of the PCI solution, the alternative to upgrade was selected.	Project scope includes significant change requiring updates to the Power Costs Inc. (PCI) software used for market interaction and settlements with MISO. These updates will be applied incrementally as MISO and PCI releases the updates.	This project adds value by implementing software changes in response to MISO's system obsolescence to ensure the company can participate in the MISO market.	Aug-21	Dec-21	Jan-22	Oct-25	Dec-25	(1.00)
55	Digital - Foundation Enhancements	Digital Foundations	392,000	-	-	-	-	The Company's digital foundation consists of platforms that provide capabilities for analytics, cloud, integration, content management, and network communications. These capabilities are the underlying foundation for digital systems needed to help achieve the company's goals for clean energy, gas system reliability and safety, and the products, services, and experience our customers expect.	As part of the review process, the alternative considered was that providing the funding for foundational enhancements. However, this limits the Company's ability to meet customer expectations, reduce security risks, and maintain system performance.	This project ensures that we keep these foundational platforms updated with the latest software releases to keep the platforms secure while adding features like new advanced analytics capabilities or new bundled and improved cloud service offerings. In the near future, it is expected that more smart devices will exist on the grid and pipelines, and in customer's homes. The ability to communicate securely and in a timely manner in order to manage the electric and gas systems effectively will require upgrading underlying platforms to conform with the latest security and communication requirements.	The project will add value by: (1) enabling advanced and new functionality; (2) increasing the reliability and resiliency of Company applications; and (3) ensuring flexible, configurable platforms to scale and adapt with an evolving business landscape and customer expectations.	-	-	-	-	-	(1.00)
56	Electric Underground Conflation	Electric Grid Integration	392,000	-	-	-	-	The Electric Underground Conflation project will establish a single Geographic Information System (GIS) repository for electric distribution assets through conversion of Computer Aided Design (CAD) asset record files to assets in the electric GIS database and align the facility map records to aerial imagery and the GIS street and right-of-way maps ("land base"). This conversion and alignment process is referred to as conflation.	Alternatives considered include: (1) delaying implementation and (2) providing conversion services by skilling up internal resources. Delaying implementation was not chosen because this work is foundational to other GIS-based electric technology projects, including the conversion of electric service records to GIS. Delaying the conflation project will then delay the electric service conversion, which prolongs the maintenance of the legacy system, and does not mitigate the safety risk inherent in maintaining service records in two systems. Completing the conversion with internal resources was not chosen because of the significant cost needed to skill up internal resources to complete "one time" nature of work. The alternative to outsource the conversion and conflation was selected because it is the most cost-effective and efficient solution.	The project scope includes: (1) conversion of electric underground distribution assets currently stored in CAD file format to the electric GIS database; (2) conflation of electric underground data to spatially align to the GIS land base; (3) enhancing the electric GIS data model to support an enhanced system model; (4) establishing data quality assessment and quality control of electric underground distribution data; and (5) configuring the Primary Distribution Map (PDM) from a PDF solution.	Implementation will benefit the Company through: (1) increased accuracy of system planning and analysis models; (2) enhanced safety by providing a single, spatially accurate view for overhead and underground electric assets rather than multiple applications; (3) providing the framework for the conversion of underground electric services from Service Information Management System (SIMS); (3) improved GIS integrated design quality for electric underground assets through a consolidated view of electric assets that eliminates additional manual work-arounds to manipulate designs to align overhead and underground electric digital records; (4) improved productivity for electric designers, dispatchers, schedulers, and all office electric GIS users by eliminating additional manual work-arounds necessary to match the electric underground distribution records with the previously conflated land base; (5) improved safety for field operators by providing a spatially accurate representation of electric distribution assets over the previously conflated land base files, eliminating the misalignment and risk of misinterpretation of asset location between the conflated land base and un-conflated electric distribution files; (6) improved damage prevention through better analysis for MISS DGI staking by providing a more larger view of the electric underground system and; (7) improved safety and reliability during storm restoration activities through enhancement of the outage management system mapping application via inclusion of a more detailed underground electric asset location view that provides dispatchers with more accurate data when locating faults and improves switching operations.	Jan-21	Feb-21	Apr-21	Jul-21	Sep-21	(1.00)
57	Enterprise Content Management - Managing Business Records	Corporate and Enterprise	363,000	-	-	-	-	Using the Company's Enterprise Content Management ("ECM") system, the ECM - Managing Business Records project will manage business records for high-focus areas contained in SharePoint and Shared Drives. The records will be classified, categorized, and placed under formal retention rules, via metadata assignment.	Three alternatives were considered for this project: (1) Continue managing business records in SharePoint and Shared Drives without ECM integration. This option was not chosen as it would provide minimal visibility to the records from within the ECM system, retention rules will not be applied to the records, taxonomy or metadata will not be applied making it more difficult to find and manage records, the problem will continue to get worse over time as the amount of data grows, and the Company could face substantial fines if it is unable to find records. (2) Integrate all SharePoint and shared drive content in the ECM. This option was not chosen as it will be costly and not all content is considered a business record. (3) Start by focusing on integrating business critical records with ECM for the high-focus areas. Alternative (3) was selected as it mitigates risk with the high-focus areas and requires less funding than managing all SharePoint and shared drive content in ECM.	For high-focus areas, the scope includes: (1) integrating content from SharePoint and Shared Drives into ECM; (2) assigning taxonomy and records classification values through metadata on content and records; (3) assigning standard retention rules; (4) mapping, building metadata, and migrating existing content to new taxonomy and retention schedules; (5) ensuring records can be located; and (6) allowing legal holds and litigation audits to be conducted on content.	This project will add value through: (1) consistent practices for records management; (2) definable process for validating completeness and accuracy of records produced and records deleted; (3) deletion of excess, irrelevant, or inappropriate information; (4) easy generation of electronic records in the event of litigation; (5) achieving Generally Accepted Recordkeeping Principle maturity of 3.0 or greater through functional capabilities of ECM coupled with business practices to align to ARMA (formerly the Association of Records Managers and Administrators) principles; and (6) mitigating the possibility of fines as a result of being unable to produce records.	Feb-19	Jun-20	Aug-20	Nov-21	Dec-22	(1.00)
58	Electric Agricultural Services Data Base	Electric Grid Integration	292,000	-	-	-	-	The Agricultural (Ag) Services Database project will update and stabilize the system with which data for the Agricultural Services Neutral Separation Program is managed. The Neutral Separation Program tracks and manages mandatory inspections for stray voltage at agricultural locations.	Two alternatives were considered for execution of this project: (1) build a new database that uses current technology supported by the Company and includes functionality for keeping records up to date, or (2) restore the inoperative functionality and update current access database. Alternative two was not chosen because the current solution uses outdated and unsupported technology, and it would not be cost-effective to maintain it. Alternative one is the recommended solution because it will leverage current, supported database technology.	The scope of this project will: (1) replace the existing database with one that is supported by IT; (2) integrate with SAP to provide access to customer information; and (3) integrate with GIS to capture customer geographic information, including details related to connectivity, voltage, service type, management headquarters, and critical location.	Upgrading the Ag Services Database will provide value for both the Company and its customers by: (1) supporting consistent tracking of the agricultural services inspection program; (2) eliminating the need for processing paper forms which reduces rework and paper waste; and (3) maintaining up to date information on customers in the program, which ensures currency of participant data.	Jan-21	Mar-21	Jul-21	Oct-21	Nov-21	(1.00)
59	Voxal Survey Tool	Customer Experience and Operations	240,000	-	-	-	-	The Voxal Survey Tool project will implement an after-the-call survey platform for every customer contact for both Interactive Voice Response (IVR) and live agent customers. This platform would follow similar methodology as the current surveys but includes a much larger sample population and includes an integration with existing call routing and call analytics tools providing better feedback from customers to improve Customer Service Representative (CSR) capabilities.	Two alternatives were considered and not chosen for this project: (1) Continue to use the ECM solution utilized today. This was not chosen as it requires continued maintenance of existing interfaces with ECM and costs \$500,000 to have ECM perform surveys on the Company's behalf. (2) Build a home-grown survey solution. This alternative was not chosen because Consumers Energy has a current call gap to support this type of application. The option to implement the Voxal tool was selected because of the integration it provides to the Genesys framework. It synchronizes through Genesys software for reporting and Cyber Pulse for dashboard capabilities. Additionally, Voxal has intimate knowledge of the Company's Genesys framework configurations.	The project scope includes: (1) implementing the application and analytics components for IVR and agent surveys; (2) defining best practices to design the survey and analytics solution; (3) implementing the application and analytics components for email; (4) deploying and validating servers to support new survey tool; (5) improving routing strategy, IVR transfer key value pair (KVP), and ability to support Testing Center Of Excellence (TCOE) testing objectives; (6) replacing the existing IVR administration tool; (7) implementing survey software that allows the business area the ability to create and edit surveys; and (8) implementing a repository for the data to reside until it is ready to be collected by the data analytics team.	This project will provide value for the Company and its customers through: (1) increasing the number of surveys completed by the customers resulting in additional insight from the customer feedback; (2) reducing costs by eliminating the dependency on a third-party vendor to send surveys on behalf of the Company; and (3) increasing customer satisfaction due to feedback acquired from surveys.	Jan-21	Feb-21	Mar-21	Apr-21	Jun-21	(1.00)
60	Digital - Work Automation	Digital Foundations	219,000	-	-	-	-	This project will implement and enhance Robotic Process Automation ("RPA"), ML, and AI tools.	The alternative considered was to continue with existing manual processes, which prevents the business areas from reducing the risk of manual errors or improving productivity through automation. Providing a foundational automation platform allows the Company to maintain the benefits of automation in support of customers and employees.	The scope of the project will be to leverage the existing platforms to provide the foundation of RPA, ML, and AI to support use cases across all business units.	The project adds value to the Company by providing the foundation that will allow business areas to automate key processes based on individual use cases. Each use case will deliver benefits, reduce errors, and improve overall productivity to support Company customers and employees.	Jan-21	Mar-21	May-21	Nov-21	Dec-21	(1.00)
61	SAP Access Controls	Upgrades and Applications Currency	158,500	(180,000)	-	-	-	This project replaces the current software tool that manages and monitors Sarbanes-Oxley Act ("SOX") compliance for SAP transactions.	The alternatives that were considered and deemed to be non-viable are: (1) continue using the current application as-is, according to the audit risks stemming from the known issues and the risk associated with standard SAP support ending on December 31, 2023; and continue relying on current on-premise manual processes; (2) use robotic process automation in a limited scope to automate some manual tasks and reduce errors, which due to limited functionality would not mitigate risks and gaps in the current tool; (3) develop a custom solution to perform some of the work, which would be expected to be more costly and include higher maintenance and support costs; and (4) continue to apply program patches and fixes from SAP to the current tool, which may not address or fulfill all requirements, and requires support from an external vendor, which increases support costs. The option to replace the existing tool with a new solution: (1) meets Company requirements; (2) avoids hiring two additional resources; (3) is more cost effective in the long term; (4) adds more process automation; and (5) increases SOX compliance.	The project scope includes: (1) automating the SAP periodic user access reviews; (2) automating the emergency access management process; (3) displaying areas of risk and identifying how to mitigate that risk; (4) tracking mitigations; and (5) converting the current rule set into the new tool.	The project will add value by: (1) reducing risk and eliminating waste through new tools that provide higher levels of automation and process optimization; (2) decreasing maintenance and upgrade costs by implementing the out-of-the-box solution without customization; (3) increasing ability to meet requirement changes from auditors and the Public Company Accounting Oversight Board; and (4) creating complete and accurate report capabilities.	Jan-20	Apr-20	Jul-20	Oct-20	Dec-20	(0.62)

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Project Descriptions, Alternatives, Functionality, Benefits, Timelines, Cost/Benefit Ratio
For the Projected Test Year 12 Months Ending December 31, 2021
(\$000)

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Witness: JDT/bctm
Date: February 2020

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
No.	Project	Company IT Category	Capital Costs - Approved Amounts	O&M Savings	Capital Savings	Contingency - Capital	Contingency - O&M	Project Description	Alternatives Considered	Project Scope/Functionality	Project Benefit	Plan - Start Date	Define - Start Date	Execute - Start Date	Close - Start Date	Go Live - Date	Cost Benefit Ratio
62	Electric High Voltage Distribution Monitoring System Upgrade	Upgrades and Applications - Currency	142,000	-	-	-	-	The Electric High Voltage Distribution Monitoring System Upgrade project upgrades the software components of the High Voltage Distribution (HVD) Supervisory Control and Data Acquisition (SCADA) system, which is used to monitor and control generation, transmission, and high voltage distribution voltage circuits. The upgrade will provide new functionality and enable new capabilities.	The alternative considered was to defer the biennial upgrade of the SCADA application. The Company targets SCADA biennial application upgrades to stay within two versions of the current software. The alternative to delay the upgrade of the SCADA application was not chosen due to increased security risk and need for vendor support of the application.	The scope of this project is an upgrade to the HVD SCADA software which encompasses: (1) upgrading the application software, (2) assessing any new functionality for value to the Company, (3) making necessary configuration changes, and (4) updating documentation related to the integration changes.	Upgrading the HVD SCADA software adds value to the Company by ensuring that problems are promptly addressed when they occur. Additionally, the upgrades include enhancements and fixes to the core product that support reliability of electric delivery, ensuring that customers are receiving the energy they need when they need it. Also, because the reliability of the re-distribution system is dependent on availability of the HVD system, staying current with the solution provider upgrades is critical to the reliability of the distribution system.	Aug-21	Nov-21	May-22	Jul-22	Nov-22	(1.00)
63	Oracle Server Database Upgrade	Upgrades and Applications - Currency	92,136	-	-	-	-	The Oracle Server Database Upgrade project will upgrade Oracle server databases to the next version and supportability across business portfolios.	As this is an upgrade of an existing system the alternative considered was to delay the upgrade. This alternative upgrade was not chosen due to the risk to the stability and supportability of the system.	The scope of this project includes upgrading to the next version of Oracle across all business systems.	This project will create value for the Company and its customers by: (1) reducing the risk of system failure and the resulting impacts to business partners and customers; and (2) ensuring that systems are secure, supported, and have the latest features and functionality.	Jan-20	Apr-20	Jul-20	Apr-21	May-21	(1.00)
64	Electric Grid Telecom Device Management	Electric Grid Integration	91,000	-	-	-	-	The Electric Grid Telecom Device Management project will implement a solution for managing the telecommunications device component of grid modernization that includes managing device metadata (version, software) and the ability to historically manage the real-time status of telecommunications devices.	Several alternatives were considered for the execution of the project: (1) no change, which assumes the risk of failure of services for reliability and system optimization; (2) develop an in-house application consolidating device information from vendor-specific systems into a single view; (3) deploy a commercial solution (cloud based or on premise) that will consolidate and normalize vendor-specific data. Both alternatives two and three will be evaluated in the project planning phase to determine the most cost-effective solution, as they both support achievement of the investment objectives.	The scope of the project consists of: (1) implementing a communications device management solution; (2) performing necessary data transformations; (3) developing analytic queries that leverage the cross-functional data; (4) defining an end-to-end business process that will support actionable findings; and (5) managing associated organizational changes.	Completion of this project will benefit the Company by: (1) reducing hours spent combining data from several platforms; (2) providing visibility into devices on the electric distribution system, including a consolidated view of multiple device types; (3) providing highly reliable communications to field-control devices in support of advanced grid functionality; and (4) improving the ability to troubleshoot and perform root cause analysis, leading to improved reliability and a reduction in dispatching field crews for troubleshooting devices. The implementation of a communications management system that can monitor and maintain multiple wide area network/field area network (WAN/FAN) device types also adds value by consolidating data from multiple disparate systems into a central repository which can then be used to manage and analyze the disposition and features of distribution grid telecommunications devices. Additionally, a platform that enables both GIS integration of asset location and telecommunication asset inventory for future substation communication needs will enable geo-based trouble analysis and proactive replacement of similar types of faulty communication equipment.	Feb-20	May-21	Feb-22	Sep-22	Nov-22	(1.00)
65	Application Currency and Enhancements - ACDA - Capital	Upgrades and Applications - Currency	82,000	-	-	-	-	The Application Currency, and Enhancements Analytics, Cloud, DevOps, and Architecture ("ACDA") initiative will allow both Capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for Capital enhancements and upgrades. The project will also upgrade applications that support the ACDA Portfolio.	Prior to implementing the listed applications, a review was conducted to identify the best solution for the Company. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements were considered. This project makes ongoing upgrades and support for the listed application possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system issue, or regulatory or compliance violation; (2) installing the manual business continuity process until the application upgrade is possible, which would increase the previously mentioned business risks; or (3) revisiting the decision to utilize eSOMS and replace the application by customizing SAP. The option to upgrade the existing eSOMS application was chosen to minimize cost and risk to the Company and its employees.	The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the ACDA Portfolio.	The value of regular upgrades to applications in the ACDA Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk. Included in the implementation are small changes and functionality improvements to existing IT software application investments for the ACDA Portfolio.	-	-	-	-	-	(1.00)
66	MISO Market User Interface Changes	Electric Grid Integration	65,000	-	-	-	-	The Midcontinent Independent System Operator (MISO) Market User Interface (MUI) Changes project will update the Power Costs Inc. (PCI) software used for market interaction and settlements with MISO. These changes are required as a result of significant MISO redesign of the MUI to modernize technology and improve cyber security.	Alternatives considered in order to continue to interact with MISO include: (1) replacing PCI with a new product; or (2) updating the PCI solution. The alternative to replace PCI would require significant investment in research, product evaluation, people resources, and financial investment. Due to the schedule for updates provided by MISO, and the significant footprint of the PCI solution, the alternative to upgrade was selected.	This is a significant change that requires updates to the PCI software used for market interaction and settlements with MISO. These updates will be applied incrementally as MISO and PCI release the updates. Each update requires installation in the test environment, regression testing, and user acceptance testing prior to implementation in production.	The project will provide benefits to the Company by meeting required changes to the MUI initiated by MISO, modernizing technology and improving cyber security.	Jan-20	Apr-20	Dec-20	Feb-21	May-21	(1.00)
67	Electronic Shift Operations Management System Upgrade	Upgrades and Applications - Currency	50,000	-	-	-	-	This project will perform a major upgrade of the eSOMS application, including servers. This project will completely rewrite the character and narrative log, and the safety critical smaller functionality to enable compatibility with the new version.	Alternatives considered include: (1) continuing to use the application without vendor support at the risk of a critical application issue that results in an employee safety incident, extended plant production outage, prolonged plant reliability issue, or regulatory or compliance violation; (2) installing the manual business continuity process until the application upgrade is possible, which would increase the previously mentioned business risks; or (3) revisiting the decision to utilize eSOMS and replace the application by customizing SAP. The option to upgrade the existing eSOMS application was chosen to minimize cost and risk to the Company and its employees.	The scope of this project includes: (1) assessing any new functionality for value to the Company; (2) replacing the servers and upgrading the application software; (3) making necessary configuration changes; (4) testing any integrations to or from the application; (5) testing the upgrade; and (6) updating documentation related to the integration changes.	The project will add value by: (1) reducing the human struggle with manual workarounds and old applications; (2) empowering employees with proper electronic tools to meet customer expectations; (3) enabling process improvements to deliver outages more effectively; (4) reducing plant downtime; and (5) increasing reliability.	Jan-21	Feb-21	Apr-21	Nov-21	Dec-21	(1.00)
68	Contract Life Cycle Management	Operations Support	44,000	(\$15,490)	-	-	-	This project will implement a new contract management solution to manage the life cycle of contracts.	As part of the review process, alternatives considered included: (1) delay implementation of a new solution; or (2) remain on the current platform. The alternative to delay implementation was not selected because it defers a significant reduction in support costs and opportunities to reduce manual efforts. The alternative to remain on the current platform was not considered due to the intensive manual effort to route physical documents for approvals, inability to streamline source-to-pay ("S2P") workflow, and costly support model. The option to implement a new contract management life cycle management cloud-based solution was chosen after evaluation of leading vendor software applications for S2P solutions through a request for information and vendor demonstration process.	The scope of this project includes: (1) implementing the new contract life cycle management solution; (2) integrating this solution with SAP supply chain; (3) transitioning active contract information from current solution; (4) discontinuing use of current solution for contract management.	This project will provide value to the Company and its customer through: (1) an improved user experience; (2) standardization of the supply chain platform for sourcing and contracts; (3) reduction of manual steps to select approvers; (4) integration with SAP; and (5) reduced annual subscription fee for the solution.	Jun-21	Aug-21	Sep-21	Feb-22	Apr-22	1.22
69	Application Currency and Enhancements - Operational Technology - Capital	Upgrades and Applications - Currency	25,000	-	-	-	-	This project will utilize both capital and O&M funding to keep applications current for security and reliability, and make enhancements to existing software and to address requests generated by changing business requirements. O&M is included in this project to complete the preliminary planning phase for capital enhancements and upgrades. The project will also upgrade the various applications that support the OT Portfolio.	Prior to implementing the applications, a review is completed to identify the best solution. During that review, the alternatives of delaying the timing of the individual upgrades and zero budget allocation for enhancements are considered. This project makes ongoing upgrades and support for the applications possible and fortifies the Company's ability to make software changes as part of process improvements and regulatory changes, and to meet legally required system changes. Timing for an application upgrade is based on: (1) maintaining an optimal balance between OT, keeping the application current and reliable; (2) an increased number of incidents; (3) paying increased support costs; and (4) preventing employees from performing their daily tasks.	Included in the implementation are small changes and functionality improvements to existing IT software application investments for OT. The scope of upgrading these applications encompasses: (1) upgrading the application software; (2) assessing any new functionality for value to the Company; (3) making necessary configuration changes; and (4) updating documentation related to the integration changes. Additionally, enhancement requests are fulfilled to provide functionality for areas supported by the OT Portfolio.	The value of regular upgrades to applications in the OT Portfolio lies in: (1) enhancing security protections; (2) lessening the number of incidents associated with outdated software; (3) increasing application stability, leading to fewer incidents due to outdated software; and (4) allowing the Company to leverage new functionality available in the upgrades. Requests for this funding are governed by a cross-functional board comprised of representatives from each area. The board meets monthly to evaluate and prioritize the work and to assess requests for value using benefits that are categorized into hard cost savings, cost avoidance, safety, achieving corporate goals, and mitigating risk.	-	-	-	-	-	(1.00)

70 "h". The percentage of total Information Technology Budget that the top 25 capital projects represents, and total number of projects that fall outside of the Top 25 are 77% and 44 projects respectively.

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

KEITH G. TROYER

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

2021 Power Supply Cost Recovery Forecast

YEAR		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	2021
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
	ENERGY (MWH)													
1	COAL STEAM	993,882	951,168	955,765	269,633	782,879	928,374	1,004,572	948,985	933,954	746,820	825,011	1,014,190	10,355,233
2	GAS & OIL	560,789	513,694	562,277	495,340	399,358	471,184	639,436	536,630	448,404	218,608	481,244	516,648	5,843,613
3	NUCLEAR PPA	602,565	542,909	591,573	566,448	580,536	548,280	562,762	561,669	555,960	585,453	573,575	600,693	6,872,423
4	STATION POWER	8,114	6,805	8,927	17,570	11,048	7,472	7,181	8,710	7,901	11,338	9,614	7,915	112,594
5	CE OWNED RENEWABLES	293,957	252,902	284,113	278,935	242,247	179,438	153,652	150,406	169,895	236,194	258,969	287,868	2,788,578
6	PEAKERS	144,315	135,872	164,001	182,395	57,472	54,666	172,237	70,758	22,734	33,788	18,648	125,720	1,182,607
7	PUMPED STORAGE	159,949	131,198	158,847	146,291	107,224	110,099	127,082	130,709	128,495	69,264	76,347	160,201	1,505,706
8	TOTAL GENERATED	2,763,572	2,534,548	2,725,504	1,956,612	2,180,764	2,299,514	2,666,920	2,407,867	2,267,342	1,901,466	2,243,408	2,713,235	28,660,753
9	LESS : PUMPING	-207,307	-170,387	-202,488	-185,766	-143,764	-147,766	-156,383	-167,331	-170,261	-96,384	-96,947	-204,960	-1,949,745
10	TOTAL GENERATED	2,556,266	2,364,161	2,523,016	1,770,846	2,037,000	2,151,748	2,510,537	2,240,536	2,097,081	1,805,082	2,146,460	2,508,275	26,711,008
11	PURCHASED (NUGs)	840,816	747,701	846,385	807,427	720,595	658,571	784,599	701,662	643,020	729,630	617,086	681,663	8,779,155
12	NET INTERCHANGE	-397,289	-334,744	-482,402	58,079	5,187	134,727	16,843	326,979	153,674	222,795	11,217	-169,227	-454,162
13	TOTAL SYSTEM REQUIREMENTS	2,999,793	2,777,119	2,886,999	2,636,352	2,762,782	2,945,045	3,311,979	3,269,177	2,893,775	2,757,507	2,774,764	3,020,711	35,036,002
	VARIABLE EXPENSES (\$*1000)													
14	COAL STEAM	21,392	20,666	21,394	6,402	17,596	20,910	22,303	21,055	20,813	16,606	18,453	22,682	230,273
15	GAS & OIL	13,595	12,177	12,983	10,150	8,640	9,973	14,404	11,281	9,562	5,456	10,226	11,686	130,134
16	NUCLEAR PPA VARIABLE	4,679	3,706	3,955	3,778	3,862	4,479	4,981	5,010	4,478	3,824	3,772	4,141	50,665
17	STATION POWER	0	0	0	0	0	0	0	0	0	0	0	0	0
18	CE OWNED RENEWABLES	9,342	7,967	8,559	8,076	6,939	5,271	5,242	5,295	5,196	7,351	8,074	9,193	86,506
19	PEAKERS	4,292	3,982	4,649	4,713	1,489	1,462	4,571	1,895	675	1,047	646	4,126	33,547
20	PUMPED STORAGE	0	0	0	0	0	0	0	0	0	0	0	0	0
21	TOTAL GENERATED	53,301	48,499	51,540	33,117	38,527	42,094	51,502	44,536	40,724	34,285	41,171	51,829	531,125
22	LESS : PUMPING	0	0	0	0	0	0	0	0	0	0	0	0	0
23	TOTAL GENERATED	53,301	48,499	51,540	33,117	38,527	42,094	51,502	44,536	40,724	34,285	41,171	51,829	531,125
24	PURCHASED (NUGs) VARIABLE COST¹	33,136	29,697	31,501	29,572	26,698	25,616	31,150	28,227	22,607	23,371	19,288	21,458	322,321
25	NET INTERCHANGE, EXCLUDING ZRC	-16,573	-13,413	-15,641	227	-551	1,291	-1,687	6,613	2,214	4,776	418	-6,369	-38,696
26	TOTAL FUEL, VARIABLE PURCHASED AND NET INTERCHANGE	69,863	64,784	67,400	62,916	64,674	69,001	80,964	79,376	65,545	62,431	60,877	66,919	814,750
27	ZONAL RESOURCE CREDIT PURCHASE	102	92	102	99	102	0	0	0	0	0	0	0	497
28	OWNED RENEWABLE CAPACITY	5,442	4,641	4,986	4,704	4,042	3,070	3,054	3,085	3,027	4,282	4,703	5,355	50,392
29	NUCLEAR PPA CAPACITY	33,363	26,425	28,199	26,935	27,539	31,933	35,515	35,719	31,930	27,264	26,897	29,526	361,246
30	PURCHASED (NUG) CAPACITY	22,241	20,415	21,953	21,371	21,667	21,168	22,122	21,952	20,717	21,090	20,063	20,411	255,170
31	PURCHASED (NUG) FIXED ENERGY	6,119	5,585	6,135	6,202	6,444	6,242	6,414	6,402	6,209	6,408	6,232	6,433	74,825
32	INDEPENDENT ADMINISTRATOR EXPENSE	40	40	0	0	0	0	0	0	0	40	40	40	200
33	TOTAL CAPACITY AND NUG FIXED COSTS	67,308	57,199	61,375	59,311	59,794	62,413	67,106	67,157	61,883	59,084	57,935	61,766	742,330
34	TOTAL TRANSMISSION EXPENSE	36,778	35,381	35,220	33,257	37,463	47,009	49,654	47,001	45,927	35,319	36,213	38,334	477,556
35	ACTIVATED CARBON	101	101	101	101	101	101	101	101	101	101	101	101	1,207
36	MISO - SCHEDULE 2 (REACTIVE)	-375	-375	-375	-375	-375	-375	-375	-375	-375	-375	-375	-375	-4,500
37	AQUEOUS AMMONIA EXPENSE	90	90	90	90	90	90	90	90	90	90	90	90	1,075
38	UREA EXPENSE	152	152	152	152	152	152	152	152	152	152	152	152	1,819
39	LIME EXPENSE	639	639	639	639	639	639	639	639	639	639	639	639	7,669
40	TOTAL POWER SUPPLY COSTS	174,555	157,969	164,601	156,090	162,537	179,029	198,330	194,139	173,961	157,440	155,631	167,624	2,041,905
41	Long-term Industrial Load Rate (LTILR)	2,691	2,982	4,558	3,940	4,549	3,930	2,209	3,800	5,872	6,881	4,318	3,438	49,168
42	TOTAL POWER SUPPLY COSTS LESS LTILR PAYMENTS	171,864	154,987	160,043	152,150	157,988	175,099	196,121	190,339	168,089	150,559	151,312	164,186	1,992,737

¹Purchased (NUG) variable costs include costs associated with PURPA variable energy payments, non-capacity renewable energy plan transfer costs, the green generation program, energy only NUGs and certain hydro plant contract costs

2021 Power Supply Cost Recovery Forecast

YEAR		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	2021
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
	PURCHASED AND NET INTERCHANGE RECEIVED (MWH)													
43	MARKET ON PEAK	480	807	760	20,337	109,817	179,717	122,461	230,378	140,822	75,851	95,414	33,267	1,010,110
44	MARKET OFF PEAK	124,612	143,813	76,668	266,661	135,310	185,060	174,283	265,121	213,702	260,695	89,311	156,746	2,091,981
45	PURCHASED (NUGs)	840,816	747,701	846,385	807,427	720,595	658,571	784,599	701,662	643,020	729,630	617,086	681,663	8,779,155
46	TOTAL RECEIVED	965,908	892,321	923,813	1,094,425	965,722	1,023,348	1,081,343	1,197,161	997,544	1,066,176	801,811	871,676	11,881,246
	NET INTERCHANGE DELIVERED (MWH)													
47	EXTERNAL SALES	522,381	479,364	559,830	228,919	239,940	226,935	237,287	168,520	200,849	113,751	173,508	359,240	3,510,523
48	MISO RAC	0	0	0	0	0	3,115	42,614	0	0	0	0	0	45,730
49	TOTAL DELIVERED	522,381	479,364	559,830	228,919	239,940	230,050	279,901	168,520	200,849	113,751	173,508	359,240	3,556,252
50	NET (MWH)	443,527	412,957	363,983	865,506	725,782	793,298	801,442	1,028,641	796,694	952,425	628,303	512,436	8,324,994

2021 Power Supply Cost Recovery Forecast

YEAR		JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	2021
		PURCHASED	AND	INTERCHANGE	POWER	REPORT								
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)
	VARIABLE PURCHASED AND NET INTERCHANGE EXPENSE (\$*1000)													
51	MARKET ON PEAK ENERGY	855	862	859	1,389	3,659	5,252	4,213	6,686	4,293	2,747	3,302	1,763	35,880
52	MARKET OFF PEAK ENERGY	3,265	3,511	1,714	5,628	2,589	3,415	3,986	5,485	4,031	5,358	1,857	3,439	44,277
53	PURCHASED (NUGs) ENERGY	31,886	28,447	30,251	28,322	25,448	24,366	29,900	26,977	21,357	22,121	18,038	20,208	307,320
54	<u>CASE NO. U-16048 COST RECOVERY</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>1,250</u>	<u>15,000</u>
55	TOTAL EXPENSE	37,256	34,070	34,075	36,588	32,946	34,283	39,349	40,397	30,931	31,476	24,446	26,659	402,477
	NET INTERCHANGE CREDIT (\$*1000)													
56	EXTERNAL SALE ENERGY	20,693	17,786	18,215	6,790	6,799	6,971	8,224	5,557	6,111	3,330	4,740	11,570	116,786
57	EXTERNAL SALE CAPACITY	0	0	0	0	0	0	0	0	0	0	0	0	0
58	<u>MISO RAC</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>405</u>	<u>1,662</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>0</u>	<u>2,066</u>
59	TOTAL CREDIT	20,693	17,786	18,215	6,790	6,799	7,376	9,886	5,557	6,111	3,330	4,740	11,570	118,852
60	NET EXPENSE	16,562	16,284	15,860	29,799	26,147	26,908	29,463	34,840	24,821	28,147	19,706	15,089	283,625

Line	Formula	(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)		
		COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
1		Crystal Flash Renewable Energy LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	251.90	242.85	340.32	400.03	173.11	167.52	173.10	173.10	161.94	173.11	167.52	173.11	167.52	173.10
2		Crystal Flash Renewable Energy LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	51.56	51.56	51.56	51.70	50.03	51.70	50.03	51.70	50.03	51.70	50.03	51.70	50.03	51.70
3		Crystal Flash Renewable Energy LLC	Subtotal \$	0.00	0.00	0.00	0.00	0.00	303.46	294.41	391.88	451.59	224.80	217.55	224.80	224.80	210.30	224.80	217.55	224.80	217.55	224.80
4		Crystal Flash Renewable Energy LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5		Crystal Flash Renewable Energy LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6		Crystal Flash Renewable Energy LLC	Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7		Commonwealth Power Co.	Energy \$	0.00	0.00	0.00	0.00	0.00	1,338.51	1,244.05	904.74	1,111.03	699.19	904.82	944.94	944.94	883.98	944.94	914.46	944.93	914.45	673.87
8		Commonwealth Power Co.	Capacity \$	0.00	0.00	0.00	0.00	0.00	206.24	206.24	206.24	206.24	104.59	101.22	104.59	104.59	97.84	104.59	101.22	104.59	101.22	104.59
9		Commonwealth Power Co.	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,544.75	1,450.29	1,110.97	1,317.26	803.78	1,006.04	1,049.53	1,049.53	981.82	1,049.53	1,015.67	1,049.53	1,015.67	778.46
10		Grenfell, Inc.	Energy \$	548.67	449.77	565.97	489.80	664.85	639.65	412.65	328.65	489.35	355.95	416.33	492.60	518.05	483.87	543.47	610.22	594.31	518.06	387.71
11		Grenfell, Inc.	Capacity \$	154.68	154.68	154.68	154.68	103.12	103.12	103.12	103.12	51.70	50.03	51.70	50.03	51.70	48.36	51.70	50.03	51.70	50.03	51.70
12		Grenfell, Inc.	Subtotal \$	703.34	604.45	720.64	644.48	819.52	742.77	515.77	431.76	592.47	407.65	466.36	544.30	569.75	532.23	595.17	660.25	646.01	568.09	439.41
13		North American Natural Resources, Inc.	Energy \$	2,671.15	2,412.92	2,671.16	2,585.08	665.19	1,405.92	777.96	1,347.37	1,431.20	2,671.16	2,585.08	2,671.17	2,724.86	2,549.33	2,724.85	2,637.05	2,724.86	2,637.05	2,724.88
14		North American Natural Resources, Inc.	Capacity \$	419.53	378.93	419.53	406.00	419.53	419.53	419.53	419.53	419.53	406.00	419.53	419.53	419.53	392.47	419.53	406.00	419.53	406.00	419.53
15		North American Natural Resources, Inc.	Subtotal \$	3,090.69	2,791.85	3,090.69	2,991.08	1,084.73	1,811.92	1,197.49	1,766.90	1,837.20	3,090.69	2,991.08	3,090.71	3,144.39	2,941.80	3,144.39	3,043.05	3,144.39	3,043.05	3,144.41
16		STS Hydropower, LLC	Energy \$	2,230.54	2,224.10	2,415.46	2,547.24	2,418.60	2,123.74	1,568.48	1,308.11	1,267.34	1,436.74	1,847.69	2,224.12	2,340.18	2,303.54	2,415.46	2,547.24	2,418.59	2,123.75	1,568.47
17		STS Hydropower, LLC	Capacity \$	374.61	338.36	374.61	362.53	374.61	362.53	374.61	374.61	362.53	374.61	362.53	374.61	350.44	374.61	362.53	374.61	362.53	374.61	374.61
18		STS Hydropower, LLC	Subtotal \$	2,605.15	2,562.46	2,790.08	2,909.77	2,793.21	2,486.27	1,943.09	1,682.72	1,629.86	1,811.36	2,210.22	2,598.73	2,714.79	2,653.98	2,790.07	2,909.77	2,793.20	2,486.27	1,943.09
19		STS Hydropower, LLC	Energy \$	658.73	644.86	625.82	430.16	459.51	444.38	1,016.06	822.38	835.07	1,022.42	1,305.00	1,514.56	1,524.10	1,456.86	1,762.25	1,755.91	1,682.88	1,397.09	1,016.06
20		STS Hydropower, LLC	Capacity \$	225.69	203.85	225.69	218.41	225.69	218.41	225.69	225.69	218.41	225.69	218.41	225.69	211.13	225.69	211.13	225.69	218.41	225.69	218.41
21		STS Hydropower, LLC	Subtotal \$	884.42	848.71	851.51	648.57	685.21	662.79	1,241.75	1,048.07	1,053.48	1,248.11	1,523.42	1,740.26	1,749.80	1,667.99	1,987.94	1,974.32	1,908.58	1,615.50	1,241.75
22		Black River Limited Partnership	Energy \$	0.00	0.00	0.00	0.00	0.00	1,995.75	858.42	303.78	782.08	857.43	1,170.56	1,323.43	981.95	860.56	1,151.88	1,725.98	1,446.48	974.37	592.97
23		Black River Limited Partnership	Capacity \$	0.00	0.00	0.00	0.00	0.00	136.88	141.44	141.44	136.88	141.44	136.88	141.44	132.31	141.44	136.88	141.44	136.88	141.44	141.44
24		Black River Limited Partnership	Subtotal \$	0.00	0.00	0.00	0.00	0.00	2,132.62	999.86	445.22	918.96	998.87	1,307.44	1,464.87	1,123.39	992.87	1,293.32	1,862.85	1,587.92	1,111.25	734.40
25		Michiana Hydroelectric Company	Energy \$	0.00	0.00	0.00	0.00	0.00	93.73	87.68	39.05	37.81	49.17	69.64	98.33	90.12	93.36	98.32	98.32	98.32	73.75	53.25
26		Michiana Hydroelectric Company	Capacity \$	0.00	0.00	0.00	0.00	0.00	13.77	13.77	13.77	13.77	13.77	13.98	13.53	13.98	13.08	13.98	13.53	13.98	13.53	13.98
27		Michiana Hydroelectric Company	Subtotal \$	0.00	0.00	0.00	0.00	0.00	107.50	101.45	52.82	51.58	63.15	83.17	112.32	104.10	106.44	112.30	111.85	107.82	67.23	67.23
28		White's Bridge Hydro Company	Energy \$	0.00	0.00	0.00	0.00	0.00	1,809.86	893.75	778.09	1,056.56	844.02	901.34	1,266.00	1,311.09	1,234.85	1,376.64	1,569.26	1,528.25	1,282.40	905.49
29		White's Bridge Hydro Company	Capacity \$	0.00	0.00	0.00	0.00	0.00	137.69	137.69	137.69	137.69	139.78	139.78	139.78	139.78	130.76	139.78	135.27	139.78	135.27	139.78
30		White's Bridge Hydro Company	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,947.55	1,031.44	915.79	1,194.25	983.80	1,036.61	1,405.78	1,450.86	1,365.61	1,516.41	1,704.52	1,668.02	1,417.67	1,045.26
31		Elk Rapids Hydroelectric Power, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	730.67	668.62	376.16	381.80	583.61	796.74	900.76	659.72	578.17	773.91	1,159.57	971.82	654.64	398.39
32		Elk Rapids Hydroelectric Power, LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	165.23	165.23	165.23	165.23	168.09	162.66	168.09	168.09	157.24	168.09	162.66	168.09	162.66	168.09
33		Elk Rapids Hydroelectric Power, LLC	Subtotal \$	0.00	0.00	0.00	0.00	0.00	895.91	833.85	541.39	547.03	751.70	959.40	1,068.85	827.81	735.41	941.99	1,322.24	1,139.91	817.31	566.48
34		City of Beaverton	Energy \$	0.00	0.00	0.00	0.00	0.00	1,256.87	939.96	682.67	894.73	663.57	816.41	883.51	819.56	778.41	1,102.79	1,200.99	1,053.71	864.89	676.04
35		City of Beaverton	Capacity \$	0.00	0.00	0.00	0.00	0.00	68.85	68.85	68.85	68.85	70.22	67.96	70.22	67.96	65.69	70.22	67.96	70.22	67.96	70.22
36		City of Beaverton	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,325.72	1,008.81	751.52	963.58	733.79	884.36	953.73	889.79	844.10	1,173.01	1,268.95	1,123.93	932.85	746.26
37		Boyce Hydro Power, LLC '1	Energy \$	0.00	0.00	0.00	0.00	0.00	1,035.31	597.93	284.91	465.74	692.27	945.11	1,068.51	796.89	698.37	934.80	1,400.68	1,173.87	790.76	481.20
38		Boyce Hydro Power, LLC '1	Capacity \$	0.00	0.00	0.00	0.00	0.00	688.47	688.47	688.47	688.47	690.33	668.06	690.33	690.33	645.79	690.33	668.06	690.33	668.06	690.33
39		Boyce Hydro Power, LLC '1	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,723.78	1,286.41	973.38	1,154.21	1,382.60	1,613.17	1,758.84	1,487.22	1,344.16	1,625.13	2,068.74	1,864.20	1,458.82	1,171.53
40		Boyce Hydro Power, LLC '1	Energy \$	0.00	0.00	0.00	0.00	0.00	1,035.33	597.96	284.91	465.72	666.41	773.26	874.24	652.00	571.38	764.85	1,145.99	960.44	646.99	393.69
41		Boyce Hydro Power, LLC '1	Capacity \$	0.00	0.00	0.00	0.00	0.00	619.63	619.63	619.63	619.63	621.30	601.26	621.30	621.30	581.21	621.30	601.26	621.30	601.26	621.30
42		Boyce Hydro Power, LLC '1	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,654.95	1,217.58	904.53	1,085.35	1,187.70	1,374.51	1,495.54	1,273.30	1,152.60	1,386.15	1,747.25	1,581.73	1,248.25	1,014.99
43		Boyce Hydro Power, LLC '1	Energy \$	0.00	0.00	0.00	0.00	0.00	2,857.46	1,650.29	786.33	1,285.42	1,519.86	2,074.94	2,345.87	1,749.52	1,533.23	2,052.30	3,075.12	2,577.18	1,736.05	1,056.45
44		Boyce Hydro Power, LLC '1	Capacity \$	0.00	0.00	0.00	0.00	0.00	1,652.34	1,652.34	1,652.34	1,652.34	1,656.79	1,603.35	1,656.79	1,656.79	1,549.90	1,656.79	1,603.35	1,656.79	1,603.35	1,656.79
45		Boyce Hydro Power, LLC '1	Subtotal \$	0.00	0.00	0.00	0.00	0.00	4,509.80	3,302.63	2,438.67	2,937.76	3,176.65	3,678.29	4,002.66	3,406.31	3,083.13	3,709.09	4,678.47	4,233.97	3,339.40	2,713.24
46		Boyce Hydro Power, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
47		Boyce Hydro Power, LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
48		Boyce Hydro Power, LLC	Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
49		Energy Developments Byron Center, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	8,379.92	8,088.42	8,794.86	9,161.12	5,690.59	6,655.96	7,875.37	8,281.84	7,735.65	8,688.30	9,755.29	9,501.25	8,281.83	6,198.67
50		Energy Developments Byron Center, LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	1,													

Line		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
Formula																					
	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
67	Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,434.68	1,257.29	1,682.98	1,933.10	1,997.52	1,423.64	866.33
68		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,434.68	1,257.29	1,682.98	1,933.10	1,997.52	1,423.64	866.33
70	Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	717.17	628.54	841.32	1,260.58	1,056.45	711.64	433.06
71		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
72		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	717.17	628.54	841.32	1,260.58	1,056.45	711.64	433.06
73	Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
74		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
75		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76	Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
77		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
78		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79	Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
80		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
81		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82	Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
83		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
84		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85	Good Fruit Storage, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	20.60	14.64	11.11	2.65	3.92	12.03	15.24	19.45	25.54	26.56	25.88
86		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	66.81	69.03	66.81	69.03	69.03	64.58	69.03	66.81	69.03	66.81	69.03
87		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.41	83.67	77.91	71.68	72.96	76.61	84.27	86.26	94.57	93.37	94.91
88	NextSun Energy LLC (Workman Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
89		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
90		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
91	NextSun Energy LLC (Surrey Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
92		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
93		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
94	NextSun Energy LLC (Morey Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
95		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
96		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
97	NextSun Energy LLC (Lake City Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
98		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
99		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	Hazel Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
101		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
102		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
103	Hendershot Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
104		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
105		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
106	Jack Francis Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
107		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
108		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
109	May Shannon Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
110		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
111		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
112	13 Mile Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
113		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
114		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
115	Angola Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
116		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
117		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
118	Captain Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
119		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
120		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
121	Coldwater Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
122		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
123		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
124	Geddes 2 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00															

Line Formula		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
133	Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
134		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
135		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
136	Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
137		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
138		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
139	Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
140		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
141		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
142	Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
143		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
144		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
145	Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
146		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
147		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
148	Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
149		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
150		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
151	2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
152		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
153		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
154	2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
155		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
157	2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
158		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
159		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
160	STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
161		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
163	STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
164		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
165		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
166	Total Compensation \$	Energy \$	6,109.09	5,731.65	6,278.41	6,052.28	4,208.15	79,539.73	77,093.39	71,990.32	71,279.63	67,007.85	74,074.22	79,421.87	78,052.17	73,718.42	82,787.33	92,218.14	103,966.62	82,947.23	70,223.73
167		Capacity \$	1,174.51	1,075.82	1,174.51	1,141.62	1,174.51	19,495.22	19,532.68	19,532.68	19,562.03	19,606.71	18,974.24	19,606.71	19,606.71	18,341.76	19,606.71	18,974.24	19,606.71	18,974.24	19,606.71
168		Subtotal \$	7,283.61	6,807.48	7,452.92	7,193.89	5,382.66	99,034.95	96,626.07	91,523.00	90,841.66	86,614.56	93,048.46	99,028.58	97,658.88	92,060.18	102,394.04	111,192.38	123,573.33	101,921.47	89,830.44

Formula			(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak) Σ [(a)-(aj)]
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
1	Crystal Flash Renewable Energy LLC (Bay Windpower 1st)	Energy \$	173.10	167.52	173.11	167.52	173.10	173.11	156.36	173.11	167.52	173.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,635.77
2		Capacity \$	51.70	50.03	51.70	50.03	51.70	51.70	46.69	51.70	50.03	51.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.84
3		Subtotal \$	224.80	217.55	224.81	217.55	224.80	224.80	203.05	224.81	217.55	224.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,857.61
4	Crystal Flash Renewable Energy LLC (Bay Windpower 2nd)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	284.13	254.81	295.38	356.31	500.62	534.43	615.62	2,841.31
5		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	284.13	254.81	295.38	356.31	500.62	534.43	615.62
7	Commonwealth Power Co.	Energy \$	572.65	594.76	699.18	904.80	944.94	944.97	853.50	944.95	914.46	944.93	914.45	673.87	572.65	594.76	699.17	904.82	944.95	26,992.65
8		Capacity \$	104.59	101.22	104.59	101.22	104.59	104.59	94.47	104.59	101.22	104.59	101.22	104.59	104.59	101.22	104.59	101.22	104.59	3,601.70
9		Subtotal \$	677.24	695.98	803.78	1,006.02	1,049.53	1,049.56	947.97	1,049.54	1,015.68	1,049.53	1,015.67	778.46	677.24	695.98	803.76	1,006.04	1,049.54	30,594.35
10	Grenfell, Inc.	Energy \$	340.05	352.77	355.96	416.32	492.63	518.06	467.19	543.45	610.22	594.31	518.04	387.70	340.06	352.78	355.96	416.32	492.65	17,064.39
11		Capacity \$	51.70	50.03	51.70	50.03	51.70	51.70	46.69	51.70	50.03	51.70	50.03	51.70	51.70	50.03	51.70	50.03	51.70	2,558.31
12		Subtotal \$	391.75	402.80	407.65	466.35	544.33	569.75	513.88	595.15	660.25	646.01	568.07	439.40	391.75	402.81	407.65	466.35	544.35	19,622.70
13	North American Natural Resources, Inc.	Energy \$	2,724.88	2,637.05	2,724.85	2,637.05	2,724.87	2,779.18	2,510.46	2,779.17	2,689.56	2,779.14	2,689.59	2,779.14	2,779.14	2,689.59	2,779.15	2,689.58	2,779.15	88,789.78
14		Capacity \$	419.53	406.00	419.53	406.00	419.53	419.53	378.93	419.53	406.00	419.53	406.00	419.53	419.53	406.00	419.53	406.00	419.53	14,832.55
15		Subtotal \$	3,144.41	3,043.05	3,144.39	3,043.05	3,144.41	3,198.71	2,889.40	3,198.70	3,095.56	3,198.67	3,095.59	3,198.67	3,198.67	3,095.59	3,198.68	3,095.58	3,198.68	103,622.33
16	STS Hydropower, LLC (Cascade Hydro Plant)	Energy \$	1,308.10	1,267.35	1,436.75	1,847.70	2,224.14	2,340.18	2,224.10	2,415.49	2,547.28	2,418.59	2,123.75	1,568.50	1,308.10	1,267.33	1,436.76	1,847.70	2,224.12	71,135.34
17		Capacity \$	374.61	362.53	374.61	362.53	374.61	374.61	338.36	374.61	362.53	374.61	362.53	374.61	374.61	362.53	374.61	362.53	374.61	13,244.32
18		Subtotal \$	1,682.72	1,629.87	1,811.36	2,210.23	2,598.75	2,714.79	2,562.46	2,790.11	2,909.81	2,793.20	2,486.27	1,943.11	1,682.72	1,629.85	1,811.37	2,210.22	2,598.73	84,379.65
19	STS Hydropower, LLC (Fallsburg Hydro Plant)	Energy \$	822.38	835.05	1,022.42	1,304.99	1,514.57	1,524.07	1,406.62	1,762.25	1,755.91	1,682.88	1,397.09	1,016.06	822.38	835.04	1,022.41	1,305.01	1,514.58	41,917.79
20		Capacity \$	225.69	218.41	225.69	218.41	225.69	225.69	203.85	225.69	218.41	225.69	218.41	225.69	225.69	218.41	225.69	218.41	225.69	7,979.34
21		Subtotal \$	1,048.07	1,053.47	1,248.11	1,523.40	1,740.26	1,749.76	1,610.47	1,987.94	1,974.32	1,908.57	1,615.50	1,241.75	1,048.07	1,053.45	1,248.10	1,523.42	1,740.27	49,897.13
22	Black River Limited Partnership	Energy \$	536.31	615.63	868.65	1,185.91	1,340.76	994.79	841.77	1,166.94	1,748.58	1,465.36	987.12	600.73	543.33	623.67	880.00	1,201.38	1,358.27	31,984.87
23		Capacity \$	141.44	136.88	141.44	136.88	141.44	141.44	127.75	141.44	136.88	141.44	136.88	141.44	141.44	136.88	141.44	136.88	141.44	4,311.59
24		Subtotal \$	677.75	752.51	1,010.09	1,322.78	1,482.20	1,136.23	969.53	1,308.38	1,885.46	1,606.80	1,124.00	742.17	684.77	760.55	1,021.44	1,338.25	1,499.71	36,296.46
25	Michiana Hydroelectric Company	Energy \$	36.88	40.98	49.18	69.63	98.31	90.09	90.13	98.32	98.31	98.31	73.74	53.28	36.85	40.98	49.20	69.66	98.32	2,730.01
26		Capacity \$	13.98	13.53	13.98	13.53	13.98	13.98	12.63	13.98	13.53	13.98	13.53	13.98	13.98	13.53	13.98	13.53	13.98	426.28
27		Subtotal \$	50.86	54.51	63.16	83.16	112.29	104.07	102.75	112.31	111.84	112.30	87.28	67.26	50.84	54.51	63.18	83.19	112.30	2,699.28
28	White's Bridge Hydro Company	Energy \$	717.04	770.26	844.02	901.34	1,265.99	1,311.08	1,192.27	1,376.65	1,569.19	1,528.25	1,282.37	905.50	717.05	770.26	843.94	901.34	1,266.04	34,920.18
29		Capacity \$	139.78	135.27	139.78	135.27	139.78	139.78	126.25	139.78	135.27	139.78	135.27	139.78	139.78	135.27	139.78	135.27	139.78	4,261.60
30		Subtotal \$	856.82	905.53	983.80	1,036.61	1,405.76	1,450.86	1,318.52	1,516.42	1,704.45	1,668.02	1,417.64	1,045.27	856.83	905.53	983.72	1,036.61	1,405.81	39,181.78
31	Elk Rapids Hydroelectric Power, LLC	Energy \$	360.29	413.62	583.60	796.76	900.80	659.75	558.26	773.91	1,159.58	971.86	654.65	398.40	360.28	413.61	583.55	796.75	900.80	20,211.04
32		Capacity \$	168.09	162.66	168.09	162.66	168.09	168.09	151.82	168.09	162.66	168.09	162.66	168.09	168.09	162.66	168.09	162.66	168.09	5,123.36
33		Subtotal \$	528.38	576.28	751.68	959.42	1,068.89	827.83	710.08	942.00	1,322.24	1,139.95	817.32	566.49	528.37	576.27	751.64	959.41	1,068.89	26,044.39
34	City of Beaverton	Energy \$	547.65	570.28	672.26	827.12	895.11	830.22	761.43	1,117.25	1,216.73	1,067.49	876.20	684.87	554.79	577.73	681.08	837.93	906.81	26,259.05
35		Capacity \$	70.22	67.96	70.22	67.96	70.22	70.22	63.43	70.22	67.96	70.22	67.96	70.22	70.22	67.96	70.22	67.96	70.22	2,139.65
36		Subtotal \$	617.87	638.24	742.48	895.08	965.33	900.44	824.85	1,187.47	1,284.69	1,137.71	944.16	755.09	625.01	645.68	751.30	905.89	977.03	28,398.71
37	Boyce Hydro Power, LLC '1 (Secord Hydroelectric Plant)	Energy \$	435.23	499.57	704.93	962.40	1,088.07	811.20	686.39	951.59	1,425.83	1,194.97	804.96	489.85	443.05	508.55	717.61	979.68	1,107.60	25,177.79
38		Capacity \$	690.33	668.06	690.33	668.06	690.33	690.33	623.52	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	21,081.05
39		Subtotal \$	1,125.56	1,167.63	1,395.26	1,630.46	1,778.40	1,501.53	1,309.91	1,641.92	2,093.89	1,885.30	1,473.02	1,180.18	1,133.38	1,176.61	1,407.94	1,647.74	1,797.93	46,258.84
40	Boyce Hydro Power, LLC '1 (Smallwood Hydroelectric Plant)	Energy \$	356.09	408.75	576.76	787.41	890.23	663.71	561.59	778.57	1,166.57	977.70	658.60	400.76	362.48	416.08	587.13	801.56	906.21	21,033.34
41		Capacity \$	621.30	601.26	621.30	601.26	621.30	621.30	561.17	621.30	601.26	621.30	601.26	621.30	621.30	601.26	621.30	601.26	621.30	18,972.95
42		Subtotal \$	977.38	1,010.00	1,198.05	1,388.66	1,511.52	1,285.00	1,122.76	1,399.86	1,767.82	1,599.00	1,259.85	1,022.06	983.78	1,017.33	1,208.43	1,402.82	1,527.51	40,006.29
43	Boyce Hydro Power, LLC '1 (Sanford Hydroelectric Plant)	Energy \$	955.51	1,096.83	1,547.65	2,112.89	2,388.78	1,780.92	1,506.94	2,089.14	3,130.33	2,623.45	1,767.22	1,075.42	972.67	1,116.52	1,575.44	2,150.81	2,431.67	56,622.18
44		Capacity \$	1,656.79	1,603.35	1,656.79	1,603.35	1,656.79	1,656.79	1,496.46	1,656.79	1,603.35	1,656.79	1,603.35	1,656.79	1,656.79	1,603.35	1,656.79	1,603.35	1,656.79	50,594.53
45		Subtotal \$	2,612.30	2,700.18	3,204.44	3,716.24	4,045.57	3,437.71	3,003.40	3,745.93	4,733.68	4,280.24	3,370.57	2,732.22	2,629.46	2,719.86	3,232.23	3,754.16	4,088.46	107,216.72
46	Boyce Hydro Power, LLC (Edenville Hydroelectric Plant)																			

Line	Formula		(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak) Σ [(a)-(aj)]	
		COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
67		Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	783.54	899.43	1,269.12	1,732.66	1,958.87	1,453.17	1,229.56	1,704.66	1,958.00	2,023.27	1,442.01	877.49	793.65	910.98	1,285.44	1,754.98	1,984.10	34,656.46
68			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69			Subtotal \$	783.54	899.43	1,269.12	1,732.66	1,958.87	1,453.17	1,229.56	1,704.66	1,958.00	2,023.27	1,442.01	877.49	793.65	910.98	1,285.44	1,754.98	1,984.10	34,656.46
70		Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	391.69	449.63	634.42	866.10	979.20	726.48	614.74	852.09	1,276.81	1,070.10	720.80	438.64	396.73	455.43	642.65	877.31	991.80	18,033.39
71			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
72			Subtotal \$	391.69	449.63	634.42	866.10	979.20	726.48	614.74	852.09	1,276.81	1,070.10	720.80	438.64	396.73	455.43	642.65	877.31	991.80	18,033.39
73		Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
74			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
75			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76		Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
77			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
78			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79		Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
80			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
81			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82		Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
83			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
84			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85		Good Fruit Storage, LLC	Energy \$	24.15	19.82	14.09	10.69	2.55	4.06	12.07	15.82	20.22	26.56	27.59	26.90	25.08	20.60	14.64	11.11	2.65	456.21
86			Capacity \$	69.03	66.81	69.03	66.81	69.03	69.03	62.35	69.03	66.81	69.03	66.81	69.03	69.03	66.81	69.03	66.81	69.03	1,899.53
87			Subtotal \$	93.18	86.62	83.12	77.49	71.59	73.10	74.42	84.86	87.03	95.59	94.40	95.93	94.11	87.41	83.67	77.91	71.68	2,355.74
88		NextSun Energy LLC (Workman Road Solar)	Energy \$	0.00	0.00	684.49	517.95	123.93	197.29	586.86	768.06	982.46	1,288.47	1,340.81	1,305.57	1,218.99	1,000.57	711.68	538.53	128.86	11,394.52
89			Capacity \$	0.00	0.00	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	10,176.85
90			Subtotal \$	0.00	0.00	1,374.82	1,186.01	814.27	887.62	1,210.39	1,458.40	1,650.52	1,978.80	2,008.87	1,995.90	1,909.32	1,668.63	1,402.01	1,206.59	819.19	21,571.36
91		NextSun Energy LLC (Surrey Road Solar)	Energy \$	0.00	0.00	684.49	517.95	123.93	197.29	586.86	768.06	982.46	1,288.47	1,340.81	1,305.57	1,218.99	1,000.57	711.68	538.53	128.86	11,394.52
92			Capacity \$	0.00	0.00	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	10,176.85
93			Subtotal \$	0.00	0.00	1,374.82	1,186.01	814.27	887.62	1,210.39	1,458.40	1,650.52	1,978.80	2,008.87	1,995.91	1,909.32	1,668.63	1,402.01	1,206.59	819.19	21,571.36
94		NextSun Energy LLC (Morey Road Solar)	Energy \$	0.00	0.00	684.49	517.95	123.93	197.29	586.86	768.06	982.46	1,288.47	1,340.81	1,305.57	1,218.99	1,000.57	711.68	538.53	128.86	11,394.52
95			Capacity \$	0.00	0.00	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	10,176.85
96			Subtotal \$	0.00	0.00	1,374.82	1,186.01	814.27	887.62	1,210.39	1,458.40	1,650.52	1,978.80	2,008.87	1,995.91	1,909.32	1,668.63	1,402.01	1,206.59	819.19	21,571.36
97		NextSun Energy LLC (Lake City Solar)	Energy \$	0.00	0.00	684.49	517.95	123.93	197.29	586.86	768.06	982.46	1,288.47	1,340.81	1,305.57	1,218.99	1,000.57	711.68	538.53	128.86	11,394.51
98			Capacity \$	0.00	0.00	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	10,176.85
99			Subtotal \$	0.00	0.00	1,374.82	1,186.01	814.27	887.62	1,210.39	1,458.40	1,650.52	1,978.80	2,008.87	1,995.90	1,909.32	1,668.63	1,402.01	1,206.59	819.19	21,571.36
100		Hazel Solar, LLC	Energy \$	857.66	870.69	619.28	468.64	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.22	116.59	12,037.66
101			Capacity \$	345.17	668.06	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	11,190.08
102			Subtotal \$	1,202.83	1,538.75	1,309.61	1,136.71	802.44	868.82	1,154.49	1,385.22	1,556.98	1,856.12	1,881.18	1,871.57	1,793.25	1,573.33	1,334.23	1,155.29	806.92	23,227.74
103		Hendershot Solar, LLC	Energy \$	857.66	870.69	619.28	468.64	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.22	116.59	12,037.66
104			Capacity \$	345.17	668.06	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	11,190.08
105			Subtotal \$	1,202.83	1,538.75	1,309.61	1,136.71	802.44	868.82	1,154.49	1,385.22	1,556.98	1,856.12	1,881.18	1,871.57	1,793.25	1,573.33	1,334.23	1,155.29	806.92	23,227.74
106		Jack Francis Solar, LLC	Energy \$	1,060.75	870.69	619.28	468.64	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.23	116.59	12,240.75
107			Capacity \$	621.30	668.06	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	11,466.21
108			Subtotal \$	1,682.05	1,538.75	1,309.61	1,136.71	802.44	868.82	1,154.49	1,385.22	1,556.97	1,856.12	1,881.18	1,871.57	1,793.25	1,573.33	1,334.24	1,155.29	806.92	23,706.96
109		May Shannon Solar, LLC	Energy \$	1,060.75	870.69	619.28	468.64	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.22	116.59	12,240.75
110			Capacity \$	621.30	668.06	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	11,466.21
111			Subtotal \$	1,682.05	1,538.75	1,309.61	1,136.71	802.44	868.82	1,154.49	1,385.22	1,556.97	1,856.12	1,881.18	1,871.57	1,793.25	1,573.33	1,334.24	1,155.29	806.92	23,706.96
112		13 Mile Solar, LLC	Energy \$	857.66	870.69	619.28	468.64	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.23	116.59	12,037.66
113			Capacity \$	345.17	668.06	690.33	668.06	690.33	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.0		

Line			(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)	
Formula			Σ [(a)-(aj)]																		
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$	
133	Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	905.10	643.76	487.17	116.54	2,152.57	
134		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	400.84	690.33	668.06	690.33	2,449.57
135		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,305.94	1,334.10	1,155.23	806.87	4,602.14
136	Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	112.10	178.49	530.96	694.88	888.91	1,165.78	1,213.11	1,181.24	1,102.92	905.27	643.90	487.22	116.59	9,221.39	
137		Capacity \$	0.00	0.00	0.00	0.00	552.27	690.33	623.53	690.33	668.06	690.33	668.06	690.33	690.33	668.06	690.33	668.06	690.33	8,680.38	
138		Subtotal \$	0.00	0.00	0.00	0.00	664.37	868.82	1,154.49	1,385.22	1,556.98	1,856.12	1,881.18	1,871.57	1,793.26	1,573.33	1,334.23	1,155.29	806.92	17,901.77	
139	Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	485.30	1,186.54	1,890.98	5,703.72	7,483.97	9,588.82	12,595.49	13,109.78	12,763.62	11,913.02	9,766.82	6,928.27	5,227.13	1,234.90	99,878.36	
140		Capacity \$	0.00	0.00	0.00	200.42	6,903.33	6,903.33	6,235.27	6,903.33	6,680.64	6,903.33	6,680.64	6,903.33	6,903.33	6,680.64	6,903.33	6,680.64	6,903.33	88,384.92	
141		Subtotal \$	0.00	0.00	0.00	685.72	8,089.87	8,794.31	11,938.99	14,387.30	16,269.46	19,498.82	19,790.42	19,666.95	18,816.35	16,447.46	13,831.60	11,907.78	8,138.23	188,263.27	
142	Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	485.30	1,200.78	1,913.64	5,772.88	7,574.51	9,704.56	12,747.30	13,267.81	12,917.47	12,056.71	9,884.75	7,012.14	5,290.59	1,249.72	101,078.16	
143		Capacity \$	0.00	0.00	0.00	200.42	6,903.33	6,903.33	6,235.27	6,903.33	6,680.64	6,903.33	6,680.64	6,903.33	6,903.33	6,680.64	6,903.33	6,680.64	6,903.33	88,384.92	
144		Subtotal \$	0.00	0.00	0.00	685.72	8,104.11	8,816.97	12,008.15	14,477.84	16,385.20	19,650.63	19,948.45	19,820.80	18,960.05	16,565.39	13,915.47	11,971.23	8,153.06	189,463.08	
145	Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,136.93	
146		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,794.87	
147		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,931.80	
148	Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	369.17	262.57	198.70	47.54	877.97	
149		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	400.84	690.33	668.06	690.33	2,449.57	
150		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	770.01	952.90	866.76	737.87	3,327.54	
151	2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43,627.00	31,835.46	27,734.81	28,126.14	35,323.59	39,100.97	46,009.13	251,757.11	
152		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,626.53	2,714.08	2,714.08	2,626.53	2,714.08	2,626.53	2,714.08	18,735.92	
153		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46,253.53	34,549.55	30,448.89	30,752.67	38,037.67	41,727.51	48,723.22	270,493.03	
154	2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
155		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
156		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
157	2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
158		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
159		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
160	STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
161		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
162		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
163	STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	795.93	798.16	855.59	844.57	871.08	807.28	748.66	667.48	708.05	719.34	847.86	892.97	9,556.97	
164		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
165		Subtotal \$	0.00	0.00	0.00	0.00	0.00	795.93	798.16	855.59	844.57	871.08	807.28	748.66	667.48	708.05	719.34	847.86	892.97	9,556.97	
166	Total Compensation \$	Energy \$	72,969.19	71,981.81	77,338.39	84,071.83	86,516.73	85,381.58	91,232.78	108,898.45	125,400.19	147,195.73	171,523.59	145,720.12	132,944.65	129,073.23	130,959.15	136,505.37	135,748.56	3,066,161.57	
167		Capacity \$	23,817.74	24,986.82	28,581.04	28,059.91	42,939.97	43,078.04	38,909.20	43,078.04	41,688.43	43,078.04	44,264.93	45,740.43	45,740.43	45,868.28	48,501.76	46,937.18	50,296.62	962,335.20	
168		Subtotal \$	96,786.93	96,968.63	105,919.44	112,131.74	129,456.71	128,459.62	130,141.98	151,976.49	167,088.61	190,273.77	215,788.52	191,460.54	178,685.07	174,941.51	179,460.90	183,442.56	186,045.19	4,028,496.77	

¹ Boyce Power actual MWh for 06/2019 - 09/2019 were received as an aggregate amount and divided amongst the three identified facilities based on their nameplate capacities.

Formula			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
1	Crystal Flash Renewable Energy LLC (Bay Windpower 1st)	Energy \$	0.00	0.00	0.00	0.00	0.00	293.48	281.16	414.83	497.38	187.10	181.06	187.10	187.10	175.02	187.10	181.06	187.10	181.06	187.09
2		Capacity \$	0.00	0.00	0.00	0.00	0.00	60.07	59.69	62.85	64.11	55.88	54.07	55.88	55.88	52.27	55.88	54.07	55.88	54.07	55.88
3		Subtotal \$	0.00	0.00	0.00	0.00	0.00	353.55	340.85	477.68	561.48	242.97	235.13	242.97	242.97	227.30	242.97	235.13	242.97	235.13	242.97
4	Crystal Flash Renewable Energy LLC (Bay Windpower 2nd)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7	Commonwealth Power Co.	Energy \$	0.00	0.00	0.00	0.00	0.00	1,192.88	1,097.59	757.80	963.80	625.54	837.01	875.03	875.03	818.58	875.03	846.81	875.03	846.81	599.97
8		Capacity \$	0.00	0.00	0.00	0.00	0.00	183.80	181.95	172.74	178.91	93.58	93.63	96.85	96.85	90.61	96.85	93.73	96.85	93.73	93.12
9		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,376.68	1,279.54	930.54	1,142.71	719.12	930.64	971.89	971.89	909.19	971.89	940.54	971.89	940.54	693.09
10	Grenfell, Inc.	Energy \$	439.81	343.91	456.73	382.61	554.24	566.05	339.23	257.03	415.32	319.38	381.92	458.11	484.05	452.03	509.94	579.51	561.82	485.47	351.52
11		Capacity \$	123.99	118.27	124.82	120.83	128.94	91.25	84.77	80.65	87.52	46.38	45.89	48.08	48.30	45.18	48.51	47.51	48.87	46.88	46.87
12		Subtotal \$	563.80	462.18	581.55	503.43	683.19	657.30	424.00	337.68	502.84	365.76	427.81	506.19	532.35	497.21	558.44	627.02	610.70	532.35	398.39
13	North American Natural Resources, Inc.	Energy \$	2,801.96	2,531.13	2,801.96	2,711.69	495.10	1,324.05	613.42	1,247.04	1,353.20	2,801.96	2,711.69	2,801.99	2,809.48	2,628.57	2,809.48	2,718.96	2,809.48	2,718.96	2,809.51
14		Capacity \$	440.08	397.50	440.08	425.88	312.26	382.36	330.80	388.30	383.87	440.08	425.89	440.08	432.56	404.66	432.56	418.61	432.56	418.61	432.56
15		Subtotal \$	3,242.04	2,928.63	3,242.04	3,137.57	807.36	1,706.40	944.23	1,635.34	1,737.08	3,242.04	3,137.57	3,242.08	3,242.04	3,033.23	3,242.04	3,137.57	3,242.04	3,137.57	3,242.08
16	STS Hydropower, LLC (Cascade Hydro Plant)	Energy \$	1,988.20	2,009.68	2,177.00	2,321.40	2,180.21	1,888.54	1,318.07	1,058.62	1,025.90	1,186.38	1,608.02	1,981.65	2,100.08	2,081.45	2,177.00	2,321.41	2,180.21	1,888.54	1,318.07
17		Capacity \$	333.91	305.74	337.63	330.39	337.69	322.38	314.81	303.16	293.46	309.33	315.50	333.77	336.18	316.66	337.63	330.39	337.69	322.38	314.81
18		Subtotal \$	2,322.11	2,315.42	2,514.63	2,651.79	2,517.89	2,210.91	1,632.88	1,361.79	1,319.36	1,495.71	1,923.53	2,315.42	2,436.25	2,398.11	2,514.63	2,651.79	2,517.89	2,210.91	1,632.88
19	STS Hydropower, LLC (Fallasburg Hydro Plant)	Energy \$	504.62	503.94	473.07	293.44	316.94	306.44	663.69	680.82	861.42	1,149.79	1,355.74	1,365.37	1,308.73	1,606.75	1,606.75	1,526.20	1,242.66	855.10	
20		Capacity \$	172.89	159.30	170.61	148.99	155.67	150.61	189.94	182.14	178.07	190.15	192.44	202.03	202.19	189.66	205.78	199.79	204.68	194.27	189.94
21		Subtotal \$	677.51	663.24	643.68	442.43	472.61	457.05	1,045.04	845.83	858.89	1,051.57	1,342.22	1,557.76	1,567.56	1,498.39	1,812.53	1,805.99	1,730.88	1,436.93	1,045.04
22	Black River Limited Partnership	Energy \$	0.00	0.00	0.00	0.00	0.00	1,636.13	645.62	181.58	583.08	644.76	918.09	1,047.43	742.19	644.97	887.12	1,382.82	1,139.37	738.75	414.00
23		Capacity \$	0.00	0.00	0.00	0.00	0.00	112.21	106.38	84.54	102.05	106.36	107.35	111.94	106.90	99.17	108.93	109.66	111.41	103.78	98.75
24		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,748.34	751.99	266.12	685.12	751.12	1,025.45	1,159.37	849.10	744.13	996.05	1,492.48	1,250.78	842.53	512.76
25	Michiana Hydroelectric Company	Energy \$	0.00	0.00	0.00	0.00	0.00	65.14	60.42	23.01	22.09	30.51	46.49	68.62	62.20	65.28	68.62	68.90	68.62	49.67	33.63
26		Capacity \$	0.00	0.00	0.00	0.00	0.00	9.57	9.49	8.11	8.04	8.68	9.03	9.76	9.65	9.15	9.76	9.48	9.76	9.11	8.83
27		Subtotal \$	0.00	0.00	0.00	0.00	0.00	74.71	69.91	31.12	30.13	39.19	55.52	78.38	71.85	74.43	78.38	78.38	78.38	58.78	42.45
28	White's Bridge Hydro Company	Energy \$	0.00	0.00	0.00	0.00	0.00	1,340.61	617.28	526.96	745.06	577.16	624.68	908.75	944.36	890.02	996.15	1,151.55	1,116.05	924.65	625.22
29		Capacity \$	0.00	0.00	0.00	0.00	0.00	101.99	95.10	93.25	97.10	95.58	93.75	100.33	100.68	94.24	101.14	99.26	102.08	97.53	96.51
30		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,442.60	712.38	620.21	842.16	672.74	718.43	1,009.08	1,045.04	984.26	1,097.29	1,250.82	1,218.13	1,022.18	721.73
31	Elk Rapids Hydroelectric Power, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	474.99	427.32	208.32	212.41	361.16	527.39	605.07	419.08	362.30	506.80	810.57	660.38	417.95	223.33
32		Capacity \$	0.00	0.00	0.00	0.00	0.00	107.41	105.60	91.51	91.93	104.02	107.67	112.91	106.77	98.53	110.07	113.71	114.22	103.85	94.23
33		Subtotal \$	0.00	0.00	0.00	0.00	0.00	582.40	532.92	299.82	304.34	465.17	635.06	717.98	525.85	460.83	616.87	924.27	774.60	521.80	317.56
34	City of Beaverton	Energy \$	0.00	0.00	0.00	0.00	0.00	1,043.87	767.22	543.24	727.81	525.68	660.24	717.00	652.74	620.72	896.51	982.89	854.22	693.41	529.56
35		Capacity \$	0.00	0.00	0.00	0.00	0.00	57.18	56.20	54.79	56.00	55.63	54.96	56.99	55.93	52.38	57.09	55.62	56.93	54.48	55.01
36		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,101.05	823.42	598.02	783.82	581.30	715.20	773.98	708.67	673.10	953.60	1,038.51	911.14	747.89	584.57
37	Boyce Hydro Power, LLC ¹ (Secord Hydroelectric Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	915.08	409.01	122.72	276.56	510.11	814.86	955.28	617.11	524.40	777.12	1,370.60	1,068.27	619.47	285.65
38		Capacity \$	0.00	0.00	0.00	0.00	0.00	608.53	470.94	296.55	408.82	508.68	575.99	617.18	534.59	484.92	573.89	653.71	628.23	523.35	409.79
39		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,523.61	879.95	419.27	685.39	1,018.78	1,390.85	1,572.46	1,151.70	1,009.31	1,351.01	2,024.31	1,696.49	1,142.82	695.44
40	Boyce Hydro Power, LLC ¹ (Smallwood Hydroelectric Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	953.16	432.15	132.06	294.10	397.51	640.18	752.09	482.51	409.37	609.93	1,086.29	842.83	484.66	220.69
41		Capacity \$	0.00	0.00	0.00	0.00	0.00	570.45	447.81	287.21	391.29	436.04	497.78	534.49	459.79	416.41	495.46	569.93	545.22	450.39	348.27
42		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,523.61	879.95	419.27	685.39	833.55	1,137.95	1,286.58	942.30	825.78	1,105.39	1,656.23	1,388.04	935.05	568.96
43	Boyce Hydro Power, LLC ¹ (Sanford Hydroelectric Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	2,664.44	1,213.58	373.13	827.70	1,070.13	1,722.54	2,023.31	1,298.64	1,101.93	1,641.16	2,921.16	2,267.12	1,304.34	594.49
44		Capacity \$	0.00	0.00	0.00	0.00	0.00	1,540.73	1,215.09	784.07	1,063.97	1,166.55	1,331.04	1,428.98	1,229.81	1,113.92					

Line	Formula
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(q)	(r)	(s)
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Formula		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)		
	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020	
67	Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.55	1,070.51	1,432.98	1,645.94	1,700.80	1,212.15	737.64	
68		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
69		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.55	1,070.51	1,432.98	1,645.94	1,700.80	1,212.15	737.64
70	Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	610.63	535.16	716.34	1,073.32	899.52	605.93	368.74
71		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
72		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	610.63	535.16	716.34	1,073.32	899.52	605.93	368.74
73	Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
74		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
75		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76	Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
77		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
78		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79	Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
80		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
81		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82	Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
83		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
84		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85	Good Fruit Storage, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6.61	3.49	2.16	0.13	0.30	2.68	3.90	6.22	9.77	10.70	9.99	
86		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	21.44	16.46	12.96	3.49	5.25	14.37	17.68	21.35	26.41	26.92	26.65	
87		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	28.05	19.95	15.12	3.62	5.55	17.05	21.59	27.56	36.18	37.62	36.64	
88	NextSun Energy LLC (Workman Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
89		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
90		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
91	NextSun Energy LLC (Surrey Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
92		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
93		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
94	NextSun Energy LLC (Morey Road Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
95		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
96		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
97	NextSun Energy LLC (Lake City Solar)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
98		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
99		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
100	Hazel Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
101		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
102		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
103	Hendershot Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
104		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
105		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
106	Jack Francis Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
107		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
108		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
109	May Shannon Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
110		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
111		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
112	13 Mile Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
113		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
114		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
115	Angola Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
116		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
117		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
118	Captain Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
119		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
120		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
121	Coldwater Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
122		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
123		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
124	Geddes 2 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
125		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
126		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
127	Interchange Solar, LLC	Energy \$																				

Line	Formula		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
		COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
133		Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
134			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
135			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
136		Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
137			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
138			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
139		Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
140			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
141			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
142		Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
143			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
144			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
145		Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
146			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
147			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
148		Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
149			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
150			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
151		2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
152			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
153			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
154		2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
155			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
157		2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
158			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
159			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
160		STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
161			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
163		STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
164			Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
165			Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
166		Total Compensation \$	Energy \$	5,734.59	5,388.66	5,908.77	5,709.14	3,546.50	59,189.34	56,169.24	51,799.63	51,324.86	48,127.04	54,814.38	59,254.60	57,689.05	54,539.82	61,893.40	70,952.01	80,240.51	62,076.08	50,647.13
167			Capacity \$	1,070.87	980.81	1,073.13	1,026.09	934.56	14,650.91	14,115.71	13,260.65	13,679.31	13,826.96	14,002.86	14,600.25	14,128.23	13,204.25	14,408.74	14,584.89	15,157.52	13,968.13	13,508.41
168			Subtotal \$	6,805.46	6,369.47	6,981.90	6,735.22	4,481.06	73,840.26	70,284.95	65,060.28	65,004.17	61,953.99	68,817.23	73,854.85	71,817.28	67,744.07	76,302.14	85,536.90	95,398.04	76,044.21	64,155.53

Formula		(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)	
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
1	Crystal Flash Renewable Energy LLC (Bay Windpower 1st)	Energy \$	187.09	181.06	187.10	181.06	187.10	187.10	168.99	187.10	181.06	187.10	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,162.37
2		Capacity \$	55.88	54.07	55.88	54.07	55.88	55.88	50.47	55.88	54.07	55.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,344.41
3		Subtotal \$	242.97	235.13	242.97	235.13	242.97	242.97	219.46	242.97	235.13	242.97	0.00	0.00	0.00	0.00	0.00	0.00	0.00	6,506.78
4	Crystal Flash Renewable Energy LLC (Bay Windpower 2nd)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	439.93	394.54	457.36	551.70	775.16	827.50	953.23	4,399.42
5		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	439.93	394.54	457.36	551.70	775.16	827.50	953.23
7	Commonwealth Power Co.	Energy \$	498.02	522.75	625.54	836.98	875.03	875.07	790.35	875.03	846.81	875.03	846.81	599.97	498.02	522.75	625.54	837.01	875.03	24,512.64
8		Capacity \$	90.96	88.96	93.58	93.63	96.85	96.86	87.48	96.85	93.73	96.85	93.73	93.12	90.96	88.96	93.58	93.63	96.85	3,249.82
9		Subtotal \$	588.98	611.71	719.12	930.61	971.89	971.92	877.83	971.89	940.54	971.89	940.54	693.09	588.98	611.71	719.12	930.64	971.89	27,762.46
10	Grenfell, Inc.	Energy \$	303.32	317.48	319.38	381.89	458.15	484.05	436.44	509.94	579.54	561.82	485.44	351.52	303.32	317.48	319.38	381.92	458.15	15,307.90
11		Capacity \$	46.11	45.02	46.38	45.89	48.08	48.30	43.62	48.51	47.51	48.87	46.88	46.87	46.11	45.02	46.38	45.89	48.08	2,227.06
12		Subtotal \$	349.44	362.50	365.76	427.78	506.22	532.35	480.07	558.44	627.06	610.70	532.32	398.39	349.44	362.50	365.76	427.81	506.22	17,534.96
13	North American Natural Resources, Inc.	Energy \$	2,809.51	2,718.96	2,809.48	2,718.96	2,809.51	2,816.85	2,544.55	2,816.85	2,726.03	2,816.82	2,726.06	2,816.82	2,726.06	2,816.82	2,726.06	2,816.82	2,726.06	90,532.65
14		Capacity \$	432.56	418.61	432.56	418.61	432.57	425.22	384.08	425.22	411.51	425.22	411.51	425.22	425.22	411.51	425.22	411.51	425.22	14,920.89
15		Subtotal \$	3,242.08	3,137.57	3,242.04	3,137.57	3,242.08	3,242.08	2,928.63	3,242.08	3,137.54	3,242.04	3,137.57	3,242.04	3,242.04	3,137.57	3,242.04	3,137.57	3,242.04	105,453.54
16	STS Hydropower, LLC (Cascade Hydro Plant)	Energy \$	1,058.62	1,025.90	1,186.38	1,608.05	1,981.68	2,100.08	2,009.68	2,177.03	2,321.47	2,180.21	1,888.54	1,318.08	1,058.62	1,025.90	1,186.41	1,608.03	1,981.65	62,526.74
17		Capacity \$	303.16	293.46	309.33	315.51	333.77	336.18	305.74	337.63	330.39	337.69	322.38	314.80	303.16	293.46	309.34	315.50	333.77	11,528.76
18		Subtotal \$	1,361.79	1,319.36	1,495.71	1,923.56	2,315.45	2,436.25	2,315.42	2,514.66	2,651.86	2,517.89	2,210.91	1,632.88	1,361.79	1,319.36	1,495.75	1,923.53	2,315.42	74,055.50
19	STS Hydropower, LLC (Fallsburg Hydro Plant)	Energy \$	663.69	680.79	861.42	1,149.79	1,355.74	1,365.34	1,263.60	1,606.75	1,606.20	1,526.20	1,242.66	855.10	663.69	680.79	861.42	1,149.79	1,355.74	36,364.73
20		Capacity \$	182.14	178.07	190.15	192.44	202.03	202.19	183.13	205.78	199.79	204.68	194.27	189.94	182.14	178.07	190.15	192.43	202.02	6,748.58
21		Subtotal \$	845.83	858.86	1,051.57	1,342.22	1,557.76	1,567.53	1,446.73	1,812.53	1,805.99	1,730.88	1,436.93	1,045.04	845.83	858.86	1,051.57	1,342.22	1,557.76	43,113.30
22	Black River Limited Partnership	Energy \$	366.96	435.52	645.95	919.34	1,048.74	743.40	623.82	888.38	1,384.13	1,140.65	739.96	415.04	367.95	436.54	647.12	920.53	1,050.00	24,379.96
23		Capacity \$	96.78	96.83	105.18	106.11	110.63	105.70	94.67	107.68	108.35	110.10	102.60	97.72	95.79	95.81	104.01	104.88	109.34	3,221.58
24		Subtotal \$	463.74	532.35	751.12	1,025.45	1,159.37	849.10	718.50	996.05	1,492.48	1,250.75	842.56	512.76	463.74	532.35	751.12	1,025.41	1,159.34	27,601.54
25	Michiana Hydroelectric Company	Energy \$	21.31	24.55	30.51	46.46	68.59	62.17	63.02	68.62	68.87	68.62	49.67	33.63	21.28	24.55	30.54	46.49	68.62	1,530.68
26		Capacity \$	8.08	8.11	8.68	9.03	9.76	9.65	8.83	9.76	9.48	9.76	9.11	8.83	8.07	8.11	8.68	9.03	9.76	281.13
27		Subtotal \$	29.39	32.66	39.19	55.49	78.35	71.81	71.85	78.38	78.35	78.38	58.78	42.45	29.36	32.66	39.22	55.52	78.38	1,811.81
28	White's Bridge Hydro Company	Energy \$	478.30	522.25	577.16	624.68	908.75	944.36	859.34	996.15	1,151.49	1,116.05	924.62	625.22	478.30	522.25	577.10	624.68	908.78	24,828.00
29		Capacity \$	93.24	91.71	95.58	93.75	100.33	100.68	91.00	101.14	99.26	102.08	97.53	96.51	93.24	91.71	95.58	93.75	100.33	3,005.98
30		Subtotal \$	571.54	613.96	672.74	718.43	1,009.08	1,045.04	950.33	1,097.29	1,250.75	1,218.13	1,022.15	721.73	571.54	613.96	672.68	718.43	1,009.12	27,833.98
31	Elk Rapids Hydroelectric Power, LLC	Energy \$	195.83	236.62	361.15	527.41	605.10	419.11	349.82	506.80	810.57	660.42	417.95	223.34	195.81	236.62	361.12	527.39	605.10	13,457.20
32		Capacity \$	91.36	93.06	104.02	107.68	112.91	106.78	95.14	110.07	113.71	114.22	103.85	94.23	91.35	93.06	104.02	107.67	112.91	3,218.45
33		Subtotal \$	287.19	329.68	465.17	635.09	718.01	525.88	444.96	616.87	924.27	774.64	521.80	317.56	287.16	329.68	465.14	635.06	718.01	16,675.64
34	City of Beaverton	Energy \$	419.75	440.62	526.33	660.93	717.71	653.37	599.94	897.24	983.61	854.91	694.06	530.21	420.33	441.23	527.00	661.55	718.36	20,962.25
35		Capacity \$	53.82	52.51	54.98	54.30	56.30	55.26	49.97	56.39	54.94	56.24	53.83	54.36	53.20	51.90	54.34	53.65	55.63	1,700.79
36		Subtotal \$	473.57	493.13	581.30	715.23	774.02	708.64	649.92	953.63	1,038.54	911.14	747.89	584.57	473.53	493.13	581.34	715.20	773.98	22,663.05
37	Boyce Hydro Power, LLC ¹ (Secord Hydroelectric Plant)	Energy \$	243.23	308.90	514.72	820.99	962.09	622.20	510.62	782.99	1,378.45	1,075.32	624.51	288.65	245.89	312.06	519.28	826.96	968.72	20,271.80
38		Capacity \$	385.79	413.09	504.06	569.90	610.40	529.50	463.85	568.02	645.86	621.21	518.30	406.79	383.13	409.94	499.54	563.92	603.77	15,992.24
39		Subtotal \$	629.02	721.99	1,018.78	1,390.88	1,572.49	1,151.70	974.47	1,351.01	2,024.31	1,696.53	1,142.82	695.44	629.02	721.99	1,018.82	1,390.88	1,572.49	36,264.04
40	Boyce Hydro Power, LLC ¹ (Smallwood Hydroelectric Plant)	Energy \$	187.49	239.07	401.28	645.27	757.74	486.70	398.80	614.77	1,092.93	848.73	488.79	223.10	189.61	241.59	405.01	650.26	763.27	16,371.92
41		Capacity \$	327.13	351.67	432.27	492.72	528.83	455.60	398.50	490.59	563.30	539.34	446.23	345.86	325.00	349.12	428.58	487.76	523.30	13,936.33
42		Subtotal \$	514.62	590.74	833.55	1,137.98	1,286.58	942.30	797.30	1,105.36	1,656.23	1,388.08	935.02	568.96	514.62	590.71	833.58	1,138.02	1,286.58	30,308.26
43	Boyce Hydro Power, LLC ¹ (Sanford Hydroelectric Plant)	Energy \$	505.10	643.90	1,080.25	1,736.13	2,038.49	1,309.85	1,073.45	1,654.19	2,938.92	2,282.88	1,315.48	600.96	510.82	650.72	1,090.19	1,749.42	2,053.32	44,257.75
44		Capacity \$	875.82	941.26	1,156.43	1,317.45	1,413.84	1,218.56	1,065.98	1,311.86	1,505.31	1,441.71	1,193.50	925.84	870.10	934.45	1,146.49	1,304.13	1,399.01	37,338.25
45		Subtotal \$	1,380.92	1,585.16	2,236.68	3,053.58	3,452.32	2,528.41	2,139.43	2										

Formula		(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)	
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
67	Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	667.13	765.82	1,080.60	1,475.27	1,667.88	1,221.55	1,033.61	1,432.98	1,645.94	1,700.80	1,212.18	737.64	667.16	765.79	1,080.57	1,475.27	1,667.88	29,319.66
68		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69		Subtotal \$	667.13	765.82	1,080.60	1,475.27	1,667.88	1,221.55	1,033.61	1,432.98	1,645.94	1,700.80	1,212.18	737.64	667.16	765.79	1,080.57	1,475.27	1,667.88	29,319.66
70	Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	333.50	382.84	540.19	737.44	833.75	610.70	516.77	716.28	1,073.32	899.55	605.93	368.74	333.50	382.84	540.22	737.47	833.75	15,256.42
71		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
72		Subtotal \$	333.50	382.84	540.19	737.44	833.75	610.70	516.77	716.28	1,073.32	899.55	605.93	368.74	333.50	382.84	540.22	737.47	833.75	15,256.42
73	Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
74		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
75		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76	Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
77		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
78		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79	Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
80		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
81		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82	Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
83		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
84		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85	Good Fruit Storage, LLC	Energy \$	8.86	6.43	3.38	2.09	0.13	0.31	2.67	4.02	6.40	10.05	10.99	10.27	9.10	6.61	3.49	2.16	0.13	143.04
86		Capacity \$	25.33	21.66	16.57	13.04	3.50	5.24	13.79	17.53	21.16	26.13	26.60	26.37	25.06	21.44	16.46	12.96	3.49	489.34
87		Subtotal \$	34.19	28.09	19.95	15.12	3.62	5.55	16.46	21.55	27.56	36.18	37.59	36.64	34.16	28.05	19.95	15.12	3.62	632.38
88	NextSun Energy LLC (Workman Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
89		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
90		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
91	NextSun Energy LLC (Surrey Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
92		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
93		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
94	NextSun Energy LLC (Morey Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
95		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
96		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
97	NextSun Energy LLC (Lake City Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
98		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.75
99		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
100	Hazel Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
101		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,296.39
102		Subtotal \$	1,214.86	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,560.33
103	Hendershot Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
104		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.95	383.91	135.90	7,296.39
105		Subtotal \$	1,214.86	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,560.33
106	Jack Francis Solar, LLC	Energy \$	947.53	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,345.23
107		Capacity \$	554.98	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.72	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,502.75
108		Subtotal \$	1,502.51	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,847.98
109	May Shannon Solar, LLC	Energy \$	947.53	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,345.23
110		Capacity \$	554.98	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.72	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,502.75
111		Subtotal \$	1,502.51	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,847.98
112	13 Mile Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
113		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.96	383.91	135.90	7,296.39

Line	Formula		(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)
																				Σ [(a)-(aj)]
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
133	Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	854.77	423.28	279.94	22.93	1,580.92
134		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	378.55	453.90	383.89	135.85	1,352.18
135		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,233.31	877.18	663.83	158.78	2,933.10
136	Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	26.79	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	7,015.98
137		Capacity \$	0.00	0.00	0.00	0.00	131.99	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.96	383.91	135.90	5,555.18
138		Subtotal \$	0.00	0.00	0.00	0.00	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	12,571.15
139	Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	499.23	252.96	567.98	3,753.58	5,348.88	7,752.25	11,143.00	11,891.33	11,343.91	10,332.76	7,954.75	4,772.63	3,165.00	261.75	79,040.03
140		Capacity \$	0.00	0.00	0.00	206.17	1,471.72	2,073.49	4,103.39	4,933.89	5,401.09	6,107.25	6,059.73	6,135.47	5,987.61	5,441.16	4,755.46	4,045.09	1,463.25	58,184.78
141		Subtotal \$	0.00	0.00	0.00	705.40	1,724.68	2,641.47	7,856.97	10,282.78	13,153.34	17,250.26	17,951.05	17,479.38	16,320.37	13,395.92	9,528.09	7,210.09	1,725.00	137,224.81
142	Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	499.23	258.61	580.17	3,822.51	5,444.27	7,883.83	11,324.41	12,082.56	11,528.18	10,502.64	8,089.37	4,858.88	3,224.66	267.59	80,366.90
143		Capacity \$	0.00	0.00	0.00	206.17	1,486.77	2,092.94	4,128.68	4,961.85	5,427.25	6,132.76	6,083.84	6,160.87	6,013.51	5,467.23	4,783.48	4,071.91	1,478.12	58,495.39
144		Subtotal \$	0.00	0.00	0.00	705.40	1,745.38	2,673.12	7,951.19	10,406.12	13,311.07	17,457.18	18,166.40	17,689.04	16,516.15	13,556.59	9,642.36	7,296.57	1,745.71	138,862.29
145	Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	615.88	615.88
146		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	972.29	972.29
147		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,588.17	1,588.17
148	Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	241.18	98.58	62.06	4.17	406.00
149		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	261.87	259.18	208.67	60.59	790.31
150		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	503.06	357.76	270.73	64.76	1,196.31
151	2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	59,947.61	42,735.24	36,803.14	37,475.16	47,788.31	53,377.54	63,293.38	341,420.39
152		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	3,609.10	3,643.33	3,601.49	3,499.58	3,671.81	3,585.53	3,733.68	25,344.52
153		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	63,556.72	46,378.56	40,404.64	40,974.74	51,460.12	56,963.07	67,027.07	366,764.91
154	2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
155		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
157	2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
158		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
159		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
160	STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
161		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
163	STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	1,216.98	1,220.40	1,308.19	1,291.32	1,331.85	1,210.19	1,122.30	1,000.60	1,061.45	1,078.35	1,271.00	1,338.64	14,451.26
164		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
165		Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,216.98	1,220.40	1,308.19	1,291.32	1,331.85	1,210.19	1,122.30	1,000.60	1,061.45	1,078.35	1,271.00	1,338.64	14,451.26
166	Total Compensation \$	Energy \$	53,847.30	52,042.37	55,438.46	61,803.13	62,637.11	60,821.32	66,357.19	80,981.29	98,022.31	118,626.99	163,032.40	132,417.62	118,138.83	113,906.37	116,458.57	123,788.54	128,783.64	2,452,108.18
167		Capacity \$	17,165.17	17,819.05	19,909.26	19,546.82	19,479.50	21,034.91	26,513.18	30,972.93	32,737.46	35,718.21	37,938.93	37,770.30	36,918.02	36,152.38	34,999.43	32,411.06	24,539.75	683,808.66
168		Subtotal \$	71,012.47	69,861.42	75,347.72	81,349.95	82,116.62	81,856.24	92,870.36	111,954.21	130,759.77	154,345.20	200,971.33	170,187.92	155,056.86	150,058.75	151,458.00	156,199.60	153,323.38	3,135,916.84

¹ Boyce Power actual MWh for 06/2019 - 09/2019 were received as an aggregate amount and divided amongst the three identified facilities based on their nameplate capacities.

Line	Formula		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	
		COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
1		Crystal Flash Renewable Energy LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	251.90	242.85	340.32	400.03	173.11	167.52	173.10	173.10	161.94	173.11	167.52	173.11	167.52	173.10
2		Crystal Flash Renewable Energy LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	51.56	51.56	51.56	51.56	51.70	50.03	51.70	51.70	48.36	51.70	50.03	51.70	50.03	51.70
3		(Bay Windpower 1st)	Subtotal \$	0.00	0.00	0.00	0.00	0.00	303.46	294.41	391.88	451.59	224.80	217.55	224.80	224.80	210.30	224.80	217.55	224.80	217.55	224.80
4		Crystal Flash Renewable Energy LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
5		Crystal Flash Renewable Energy LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
6		(Bay Windpower 2nd)	Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
7		Commonwealth Power Co.	Energy \$	0.00	0.00	0.00	0.00	0.00	1,192.88	1,097.59	757.80	963.80	625.54	837.01	875.03	875.03	818.58	875.03	846.81	875.03	846.81	599.97
8		Commonwealth Power Co.	Capacity \$	0.00	0.00	0.00	0.00	0.00	183.80	181.95	172.74	178.91	93.58	93.63	96.85	96.85	90.61	96.85	93.73	96.85	93.73	93.12
9		Commonwealth Power Co.	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,376.68	1,279.54	930.54	1,142.71	719.12	930.64	971.89	971.89	909.19	971.89	940.54	971.89	940.54	693.09
10		Grenfell, Inc.	Energy \$	439.81	343.91	456.73	382.61	554.24	566.05	339.23	257.03	415.32	319.38	381.92	458.11	484.05	452.03	509.94	579.51	561.82	485.47	351.52
11		Grenfell, Inc.	Capacity \$	123.99	118.27	124.82	120.83	128.94	91.25	84.77	80.65	87.52	46.38	45.89	48.08	48.30	45.18	48.51	47.51	48.87	46.88	46.87
12		Grenfell, Inc.	Subtotal \$	563.80	462.18	581.55	503.43	683.19	657.30	424.00	337.68	502.84	365.76	427.81	506.19	532.35	497.21	558.44	627.02	610.70	532.35	398.39
13		North American Natural Resources, Inc.	Energy \$	2,671.15	2,412.92	2,671.16	2,585.08	665.19	1,405.92	777.96	1,347.37	1,431.20	2,671.16	2,585.08	2,671.17	2,724.86	2,549.33	2,724.85	2,637.05	2,724.86	2,637.05	2,724.88
14		North American Natural Resources, Inc.	Capacity \$	419.53	378.93	419.53	406.00	419.53	406.00	419.53	419.53	406.00	419.53	406.00	419.53	419.53	392.47	419.53	406.00	419.53	406.00	419.53
15		North American Natural Resources, Inc.	Subtotal \$	3,090.69	2,791.85	3,090.69	2,991.08	1,084.73	1,811.92	1,197.49	1,766.90	1,837.20	3,090.69	2,991.08	3,090.71	3,144.39	2,941.80	3,144.39	3,043.05	3,144.39	3,043.05	3,144.41
16		STS Hydropower, LLC	Energy \$	1,988.20	2,009.68	2,177.00	2,321.40	2,180.21	1,888.54	1,318.07	1,058.62	1,025.90	1,186.38	1,608.02	1,981.65	2,100.08	2,081.45	2,177.00	2,321.41	2,180.21	1,888.54	1,318.07
17		STS Hydropower, LLC	Capacity \$	333.91	305.74	337.63	330.39	337.69	322.38	314.81	303.16	293.46	309.33	315.50	333.77	336.18	316.66	337.63	330.39	337.69	322.38	314.81
18		(Cascade Hydro Plant)	Subtotal \$	2,322.11	2,315.42	2,514.63	2,651.79	2,517.89	2,210.91	1,632.88	1,361.79	1,319.36	1,495.71	1,923.53	2,315.42	2,436.25	2,398.11	2,514.63	2,651.79	2,517.89	2,210.91	1,632.88
19		STS Hydropower, LLC	Energy \$	504.62	503.94	473.07	293.44	316.94	306.44	855.10	663.69	680.82	861.42	1,149.79	1,355.74	1,365.37	1,308.73	1,606.75	1,606.20	1,526.20	1,242.66	855.10
20		STS Hydropower, LLC	Capacity \$	172.89	159.30	170.61	148.99	155.67	150.61	189.94	182.14	178.07	190.15	192.44	202.03	202.19	189.66	205.78	199.79	204.68	194.27	189.94
21		(Fallsburg Hydro Plant)	Subtotal \$	677.51	663.24	643.68	442.43	472.61	457.05	1,045.04	845.83	858.89	1,051.57	1,342.22	1,557.76	1,567.56	1,498.39	1,812.53	1,805.99	1,730.88	1,436.93	1,045.04
22		Black River Limited Partnership	Energy \$	0.00	0.00	0.00	0.00	0.00	1,636.13	645.62	181.58	583.08	644.76	918.09	1,047.43	742.19	644.97	887.12	1,382.82	1,139.37	738.75	414.00
23		Black River Limited Partnership	Capacity \$	0.00	0.00	0.00	0.00	0.00	112.21	106.38	84.54	102.05	106.36	107.35	111.94	106.90	99.17	108.93	109.66	111.41	103.78	98.75
24		Black River Limited Partnership	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,748.34	751.99	266.12	685.12	751.12	1,025.45	1,159.37	849.10	744.13	996.05	1,492.48	1,250.78	842.53	512.76
25		Michiana Hydroelectric Company	Energy \$	0.00	0.00	0.00	0.00	0.00	65.14	60.42	23.01	22.09	30.51	46.49	68.62	62.20	65.28	68.62	68.90	68.62	49.67	33.63
26		Michiana Hydroelectric Company	Capacity \$	0.00	0.00	0.00	0.00	0.00	9.57	9.49	8.11	8.04	8.68	9.03	9.76	9.65	9.15	9.76	9.48	9.76	9.11	8.83
27		Michiana Hydroelectric Company	Subtotal \$	0.00	0.00	0.00	0.00	0.00	74.71	69.91	31.12	30.13	39.19	55.52	78.38	71.85	74.43	78.38	78.38	78.38	58.78	42.45
28		White's Bridge Hydro Company	Energy \$	0.00	0.00	0.00	0.00	0.00	1,340.61	617.28	526.96	745.06	577.16	624.68	908.75	944.36	890.02	996.15	1,151.55	1,116.05	924.65	625.22
29		White's Bridge Hydro Company	Capacity \$	0.00	0.00	0.00	0.00	0.00	101.99	95.10	93.25	97.10	95.58	93.75	100.33	100.68	94.24	101.14	99.26	102.08	97.53	96.51
30		White's Bridge Hydro Company	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,442.60	712.38	620.12	842.16	672.74	718.43	1,009.08	1,045.04	984.26	1,097.29	1,250.82	1,218.13	1,022.18	721.73
31		Elk Rapids Hydroelectric Power, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	474.99	427.32	208.32	212.41	361.16	527.39	605.07	419.08	362.30	506.80	810.57	660.38	417.95	223.33
32		Elk Rapids Hydroelectric Power, LLC	Capacity \$	0.00	0.00	0.00	0.00	0.00	107.41	105.60	91.51	91.93	104.02	107.67	112.91	106.77	98.53	110.07	113.71	114.22	103.85	94.23
33		Elk Rapids Hydroelectric Power, LLC	Subtotal \$	0.00	0.00	0.00	0.00	0.00	582.40	532.92	299.82	304.34	465.17	635.06	717.98	525.85	460.83	616.87	924.27	774.60	521.80	317.56
34		City of Beaverton	Energy \$	0.00	0.00	0.00	0.00	0.00	1,043.87	767.22	543.24	727.81	525.68	660.24	717.00	652.74	620.72	896.51	982.89	854.22	693.41	529.56
35		City of Beaverton	Capacity \$	0.00	0.00	0.00	0.00	0.00	57.18	56.20	54.79	56.00	55.63	54.96	56.99	55.93	52.38	57.09	55.62	56.93	54.48	55.01
36		City of Beaverton	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,101.05	823.42	598.02	783.82	581.30	715.20	773.98	708.67	673.10	953.60	1,038.51	911.14	747.89	584.57
37		Boyce Hydro Power, LLC ¹	Energy \$	0.00	0.00	0.00	0.00	0.00	915.08	409.01	122.72	276.56	510.11	814.86	955.28	617.11	524.40	777.12	1,370.60	1,068.27	619.47	285.65
38		Boyce Hydro Power, LLC ¹	Capacity \$	0.00	0.00	0.00	0.00	0.00	608.53	470.94	296.55	408.82	508.68	575.99	617.18	534.59	484.92	573.89	653.71	628.23	523.35	409.79
39		(Secord Hydroelectric Plant)	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,523.61	879.95	419.27	685.39	1,018.78	1,390.85	1,572.46	1,151.70	1,009.31	1,351.01	2,024.31	1,696.49	1,142.82	695.44
40		Boyce Hydro Power, LLC ¹	Energy \$	0.00	0.00	0.00	0.00	0.00	953.16	432.15	132.06	294.10	397.51	640.18	752.09	482.51	409.37	609.93	1,086.29	842.83	484.66	220.69
41		Boyce Hydro Power, LLC ¹	Capacity \$	0.00	0.00	0.00	0.00	0.00	570.45	447.81	287.21	391.29	436.04	497.78	534.49	459.79	416.41	495.46	569.93	545.22	450.39	348.27
42		(Smallwood Hydroelectric Plant)	Subtotal \$	0.00	0.00	0.00	0.00	0.00	1,523.61	879.95	419.27	685.39	833.55	1,137.95	1,286.58	942.30	825.78	1,105.39	1,656.23	1,388.04	935.05	568.96
43		Boyce Hydro Power, LLC ¹	Energy \$	0.00	0.00	0.00	0.00	0.00	2,664.44	1,213.58	373.13	827.70	1,070.13	1,722.54	2,023.31	1,298.64	1,101.93	1,641.16	2,921.16	2,267.12	1,304.34	594.49
44		Boyce Hydro Power, LLC ¹	Capacity \$	0.00	0.00	0.00	0.00	0.00	1,540.73	1,215.09	784.07	1,063.97	1,166.55	1,331.04	1,428.98	1,229.81	1,113.92	1,324.89	1,523.07	1,457.47	1,204.64	932.32
45		(Sanford Hydroelectric Plant)	Subtotal \$	0.00	0.00	0.00	0.00	0.00	4,205.17	2,428.67												

		(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)		
Formula	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020	
67	Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.55	1,070.51	1,432.98	1,645.94	1,700.80	1,212.15	737.64	
68		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
69		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.55	1,070.51	1,432.98	1,645.94	1,700.80	1,212.15	737.64
70	Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	610.63	535.16	716.34	1,073.32	899.52	605.93	368.74
71		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
72		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	610.63	535.16	716.34	1,073.32	899.52	605.93	368.74
73	Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
74		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
75		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76	Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
77		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
78		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79	Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
80		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
81		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82	Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
83		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	
84		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85	Good Fruit Storage, LLC	Energy \$	0.00	0.00	0.00	0.00																

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Line			(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)
Formula																					
	COUNTERPARTY	MONTH	01/2019	02/2019	03/2019	04/2019	05/2019	06/2019	07/2019	08/2019	09/2019	10/2019	11/2019	12/2019	01/2020	02/2020	03/2020	04/2020	05/2020	06/2020	07/2020
133	Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
134		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
135		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
136	Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
137		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
138		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
139	Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
140		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
141		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
142	Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
143		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
144		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
145	Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
146		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
147		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
148	Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
149		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
150		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
151	2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
152		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
153		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
154	2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
155		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
157	2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
158		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
159		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
160	STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
161		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
163	STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
164		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
165		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
166	Total Compensation \$	Energy \$	5,603.78	5,270.44	5,777.96	5,582.53	3,716.59	59,229.64	56,295.47	51,825.45	51,305.51	47,982.24	54,674.23	59,109.78	57,590.44	54,447.50	61,794.79	70,856.57	80,141.90	61,980.63	50,548.50
167		Capacity \$	1,050.33	962.24	1,052.59	1,006.20	1,041.84	14,666.05	14,196.30	13,280.60	13,688.89	13,802.24	13,978.93	14,575.53	14,111.03	13,188.15	14,391.53	14,568.23	15,140.32	13,951.47	13,491.20
168		Subtotal \$	6,654.11	6,232.69	6,830.55	6,588.73	4,758.42	73,895.69	70,491.77	65,106.04	64,994.39	61,784.47	68,653.15	73,685.31	71,701.46	67,635.65	76,186.32	85,424.80	95,282.22	75,932.11	64,039.70

Formula		(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)	T [(a)-(aj)]
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
1	Crystal Flash Renewable Energy LLC (Bay Windpower 1st)	Energy \$	173.10	167.52	173.11	167.52	173.10	173.11	156.36	173.11	167.52	173.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4,635.77
Capacity \$		51.70	50.03	51.70	50.03	51.70	51.70	46.69	51.70	50.03	51.70	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,221.84
Subtotal \$		224.80	217.55	224.81	217.55	224.80	224.80	203.05	224.81	217.55	224.81	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	5,857.61
2	Crystal Flash Renewable Energy LLC (Bay Windpower 2nd)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	284.13	254.81	295.38	356.31	500.62	534.43	615.62	2,841.31
Capacity \$		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Subtotal \$		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	284.13	254.81	295.38	356.31	500.62	534.43	615.62	2,841.31
7	Commonwealth Power Co.	Energy \$	498.02	522.75	625.54	836.98	875.03	875.07	790.35	875.03	846.81	875.03	846.81	599.97	498.02	522.75	625.54	837.01	875.03	24,512.64
Capacity \$		90.96	88.96	93.58	93.63	96.85	96.86	87.48	96.85	93.73	96.85	93.73	93.12	90.96	88.96	93.58	93.63	96.85	3,249.82	
Subtotal \$		588.98	611.71	719.12	930.61	971.89	971.92	877.83	971.89	940.54	971.89	940.54	693.09	588.98	611.71	719.12	930.64	971.89	27,762.46	
10	Grenfell, Inc.	Energy \$	303.32	317.48	319.38	381.89	458.15	484.05	436.44	509.94	579.54	561.82	485.44	351.52	303.32	317.48	319.38	381.92	458.15	15,307.90
Capacity \$		46.11	45.02	46.38	45.89	48.08	48.30	43.62	48.51	47.51	48.87	46.88	46.87	46.11	45.02	46.38	45.89	48.08	2,227.06	
Subtotal \$		349.44	362.50	365.76	427.78	506.22	532.35	480.07	558.44	627.06	610.70	532.32	398.39	349.44	362.50	365.76	427.81	506.22	17,534.96	
13	North American Natural Resources, Inc.	Energy \$	2,724.88	2,637.05	2,724.85	2,637.05	2,724.87	2,779.18	2,510.46	2,779.17	2,689.56	2,779.14	2,689.59	2,779.14	2,779.14	2,689.59	2,779.15	2,689.58	2,779.15	88,789.78
Capacity \$		419.53	406.00	419.53	406.00	419.53	419.53	378.93	419.53	406.00	419.53	406.00	419.53	419.53	406.00	419.53	406.00	419.53	14,832.55	
Subtotal \$		3,144.41	3,043.05	3,144.39	3,043.05	3,144.41	3,198.71	2,889.40	3,198.70	3,095.56	3,198.67	3,095.59	3,198.67	3,198.67	3,095.59	3,198.68	3,095.58	3,198.68	103,622.33	
16	STS Hydropower, LLC (Cascade Hydro Plant)	Energy \$	1,058.62	1,025.90	1,186.38	1,608.05	1,981.58	2,100.08	2,009.68	2,177.03	2,321.47	2,180.21	1,888.54	1,318.08	1,058.62	1,025.90	1,186.41	1,608.03	1,981.63	62,526.74
Capacity \$		303.16	293.46	309.33	315.51	333.77	336.18	305.74	337.63	330.39	337.69	322.38	314.80	303.16	293.46	309.34	315.50	333.77	11,528.76	
Subtotal \$		1,361.79	1,319.36	1,495.71	1,923.56	2,315.45	2,436.25	2,315.42	2,514.66	2,651.86	2,517.89	2,210.91	1,632.88	1,361.79	1,319.36	1,495.75	1,923.53	2,315.42	74,055.50	
19	STS Hydropower, LLC (Fallsburg Hydro Plant)	Energy \$	663.69	680.79	861.42	1,149.79	1,355.74	1,365.34	1,263.60	1,606.75	1,606.20	1,526.20	1,242.66	855.10	663.69	680.79	861.42	1,149.79	1,355.74	36,364.73
Capacity \$		182.14	178.07	190.15	192.44	202.03	202.19	183.13	205.78	199.79	204.68	194.27	189.94	182.14	178.07	190.15	192.43	202.02	6,748.58	
Subtotal \$		845.83	858.86	1,051.57	1,342.22	1,557.76	1,567.53	1,446.73	1,812.53	1,805.99	1,730.88	1,436.93	1,045.04	845.83	858.86	1,051.57	1,342.22	1,557.76	43,113.30	
22	Black River Limited Partnership	Energy \$	366.96	435.52	645.95	919.34	1,048.74	743.40	888.38	1,384.13	1,140.65	739.96	415.04	367.95	436.54	647.12	920.53	1,050.00		24,379.96
Capacity \$		96.78	96.83	105.18	106.11	110.63	105.70	94.67	107.68	108.35	110.10	102.60	97.72	95.79	95.81	104.01	104.88	109.34		3,221.58
Subtotal \$		463.74	532.35	751.12	1,025.45	1,159.37	849.10	718.50	996.05	1,492.48	1,250.75	842.56	512.76	463.74	532.35	751.12	1,025.41	1,159.34		27,601.54
25	Michiana Hydroelectric Company	Energy \$	21.31	24.55	30.51	46.46	68.59	62.17	63.02	68.62	68.87	68.62	49.67	33.63	21.28	24.55	30.54	46.49	68.62	1,530.68
Capacity \$		8.08	8.11	8.68	9.03	9.76	9.65	8.83	9.76	9.48	9.76	9.11	8.83	8.07	8.11	8.68	9.03	9.76		281.13
Subtotal \$		29.39	32.66	39.19	55.49	78.35	71.81	71.85	78.38	78.35	78.38	58.78	42.45	29.36	32.66	39.22	55.52	78.38		1,811.81
28	White's Bridge Hydro Company	Energy \$	478.30	522.25	577.16	624.68	908.75	944.36	859.34	996.15	1,151.49	1,116.05	924.62	625.22	478.30	522.25	577.10	624.68	908.78	24,828.00
Capacity \$		93.24	91.71	95.58	93.75	100.33	100.68	91.00	101.14	99.26	102.08	97.53	96.51	93.24	91.71	95.58	93.75	100.33		3,005.98
Subtotal \$		571.54	613.96	672.74	718.43	1,009.08	1,045.04	950.33	1,097.29	1,250.75	1,218.13	1,022.15	721.73	571.54	613.96	672.68	718.43	1,009.12		27,833.98
31	Elk Rapids Hydroelectric Power, LLC	Energy \$	195.83	236.62	361.15	527.41	605.10	419.11	349.82	506.80	810.57	660.42	417.95	223.34	195.81	236.62	361.12	527.39		13,457.20
Capacity \$		91.36	93.06	104.02	107.68	112.91	106.78	95.14	110.07	113.71	114.22	103.85	94.23	91.35	93.06	104.02	107.67	112.91		3,218.45
Subtotal \$		287.19	329.68	465.17	635.09	718.01	525.88	444.96	616.87	924.27	774.64	521.80	317.56	287.16	329.68	465.14	635.06	718.01		16,675.64
34	City of Beaverton	Energy \$	419.75	440.62	526.33	660.93	717.71	653.37	599.94	897.24	983.61	854.91	694.06	530.21	420.33	441.23	527.00	661.55	718.36	20,962.25
Capacity \$		53.82	52.51	54.98	54.30	56.30	55.26	49.97	56.39	54.94	56.24	53.83	54.36	53.20	51.90	54.34	53.65	55.63		1,700.79
Subtotal \$		473.57	493.13	581.30	715.23	774.02	708.64	649.92	953.63	1,038.54	911.14	747.89	584.57	473.53	493.13	581.34	715.20	773.98		22,663.05
37	Boyce Hydro Power, LLC ¹ (Secord Hydroelectric Plant)	Energy \$	243.23	308.90	514.72	820.99	962.09	622.20	510.62	782.99	1,378.45	1,075.32	624.51	288.65	245.89	312.06	519.28	826.96	968.72	20,271.80
Capacity \$		385.79	413.09	504.06	569.90	610.40	529.50	463.85	568.02	645.86	621.21	518.30	406.79	383.13	409.94	499.54	563.92	603.77		15,992.24
Subtotal \$		629.02	721.99	1,018.78	1,390.88	1,572.49	1,151.70	974.47	1,351.01	2,024.31	1,696.53	1,142.82	695.44	629.02	721.99	1,018.82	1,390.88	1,572.49		36,264.04
40	Boyce Hydro Power, LLC ¹ (Smallwood Hydroelectric Plant)	Energy \$	187.49	239.07	401.28	645.27	757.74	486.70	398.80	614.77	1,092.93	848.73	488.79	223.10	189.61	241.59	405.01	650.26	763.27	16,371.92
Capacity \$		327.13	351.67	432.27	492.72	528.83	455.60	398.50	490.59	563.30	539.34	446.23	345.86	325.00	349.12	428.58	487.76	523.30		13,936.33
Subtotal \$		514.62	590.74	833.55	1,137.98	1,286.58	942.30	797.30	1,105.36	1,656.23	1,388.08	935.02	568.96	514.62	590.71	833.58	1,138.02	1,286.58		30,308.26
43	Boyce Hydro Power, LLC ¹ (Sanford Hydroelectric Plant)	Energy \$	505.10	643.90	1,080.25	1,736.13	2,038.49	1,309.85	1,073.45	1,654.19	2,938.92	2,282.88	1,315.48	600.96	510.82	650.72	1,090.19	1,749.42	2,053.32	44,257.75
Capacity \$		875.82	941.26	1,156.43	1,317.45	1,413.84	1,218.56	1,065.98	1,311.86	1,505.31	1,441.71	1,193.50	925.84	870.10	934.45	1,146.49	1,304.13	1,399.01		37,338.25
Subtotal \$		1,380.92	1,585.16	2,236.68	3,053.58	3,452.32	2,528.41	2,139.43	2,966.05	4,444.23	3,724.59	2,508.98	1,526.80	1,380.92	1,585.16	2,236.68	3,053.54	3,452.32		81,595.99
46	Boyce Hydro Power, LLC (Edenville Hydroelectric Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Capacity \$		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Subtotal \$		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
49	Energy Developments Byron Center, LLC	Energy \$	3,726.59	3,921.32	3,937.76	4,773.09	5,774.96	6,119.74	5,517.80	6,465.18	7,405.47	7,157.73	6,149.31	4,361.94	3,726.59	3,921.32	3,937.76	4,773.10	5,774.96	170,898.78
Capacity \$		994.28	976.007																	

			(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak)
																				I [(a)-(aj)]
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
67	Tower Kleber Limited Partnership (Kleber Hydro Plant)	Energy \$	667.13	765.82	1,080.60	1,475.27	1,667.88	1,221.55	1,033.61	1,432.98	1,645.94	1,700.80	1,212.18	737.64	667.16	765.79	1,080.57	1,475.27	1,667.88	29,319.66
68		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
69		Subtotal \$	667.13	765.82	1,080.60	1,475.27	1,667.88	1,221.55	1,033.61	1,432.98	1,645.94	1,700.80	1,212.18	737.64	667.16	765.79	1,080.57	1,475.27	1,667.88	29,319.66
70	Tower Kleber Limited Partnership (Tower Hydro Plant)	Energy \$	333.50	382.84	540.19	737.44	833.75	610.70	516.77	716.28	1,073.32	899.55	605.93	368.74	333.50	382.84	540.22	737.47	833.75	15,256.42
71		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
72		Subtotal \$	333.50	382.84	540.19	737.44	833.75	610.70	516.77	716.28	1,073.32	899.55	605.93	368.74	333.50	382.84	540.22	737.47	833.75	15,256.42
73	Cement City Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
74		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
75		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
76	Letts Creek Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
77		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
78		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
79	Thorn Lake Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
80		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
81		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
82	Pullman Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
83		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
84		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
85	Good Fruit Storage, LLC	Energy \$	8.86	6.43	3.38	2.09	0.13	0.31	2.67	4.02	6.40	10.05	10.99	10.27	9.10	6.61	3.49	2.16	0.13	143.04
86		Capacity \$	25.33	21.66	16.57	13.04	3.50	5.24	13.79	17.53	21.16	26.13	26.60	26.37	25.06	21.44	16.46	12.96	3.49	489.34
87		Subtotal \$	34.19	28.09	19.95	15.12	3.62	5.55	16.46	21.55	27.56	36.18	37.59	36.64	34.16	28.05	19.95	15.12	3.62	632.38
88	NextSun Energy LLC (Workman Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
89		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
90		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
91	NextSun Energy LLC (Surrey Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
92		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
93		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
94	NextSun Energy LLC (Morey Road Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
95		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.74
96		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
97	NextSun Energy LLC (Lake City Solar)	Energy \$	0.00	0.00	482.72	320.40	26.72	59.75	387.71	551.18	796.85	1,143.20	1,219.44	1,163.68	1,060.45	817.53	492.25	327.52	27.62	8,877.01
98		Capacity \$	0.00	0.00	486.84	413.26	148.82	209.08	411.94	495.40	541.85	612.50	607.59	615.31	600.55	545.85	477.49	406.30	147.98	6,720.75
99		Subtotal \$	0.00	0.00	969.57	733.65	175.53	268.84	799.65	1,046.58	1,338.70	1,755.70	1,827.02	1,778.99	1,660.99	1,363.39	969.73	733.81	175.60	15,597.75
100	Hazel Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
101		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,296.39
102		Subtotal \$	1,214.86	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,560.33
103	Hendershot Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
104		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.95	383.91	135.90	7,296.39
105		Subtotal \$	1,214.86	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,560.33
106	Jack Francis Solar, LLC	Energy \$	947.53	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,345.23
107		Capacity \$	554.98	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.72	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,502.75
108		Subtotal \$	1,502.51	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,847.98
109	May Shannon Solar, LLC	Energy \$	947.53	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,345.23
110		Capacity \$	554.98	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.55	523.78	453.95	383.91	135.90	7,502.75
111		Subtotal \$	1,502.51	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,847.98
112	13 Mile Solar, LLC	Energy \$	866.24	697.86	414.79	273.68	22.18	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	9,263.94
113		Capacity \$	348.62	535.45	462.39	390.14	136.60	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.95	383.91	135.90	7,296.39
114		Subtotal \$	1,214.86	1,233.31	877.18	663.83	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	16,560.33
11																				

Formula			(t)	(u)	(v)	(w)	(x)	(y)	(z)	(aa)	(ab)	(ac)	(ad)	(ae)	(af)	(ag)	(ah)	(ai)	(aj)	(ak) Σ [(a)-(aj)]
	COUNTERPARTY	MONTH	08/2020	09/2020	10/2020	11/2020	12/2020	01/2021	02/2021	03/2021	04/2021	05/2021	06/2021	07/2021	08/2021	09/2021	10/2021	11/2021	12/2021	Total Compensation \$
133	Geddes 1 Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	854.77	423.28	279.94	22.93	1,580.92
134		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	378.55	453.90	383.89	135.85	1,352.18
135		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,233.31	877.18	663.83	158.78	2,933.10
136	Stoneheart Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	26.79	49.96	332.74	474.99	691.52	997.70	1,065.96	1,015.87	924.32	709.76	423.42	279.99	22.95	7,015.98
137		Capacity \$	0.00	0.00	0.00	0.00	131.99	193.24	390.75	471.88	519.71	590.80	587.03	593.69	578.54	523.78	453.96	383.91	135.90	5,555.18
138		Subtotal \$	0.00	0.00	0.00	0.00	158.78	243.20	723.49	946.87	1,211.23	1,588.49	1,652.99	1,609.56	1,502.87	1,233.54	877.38	663.89	158.85	12,571.15
139	Bingham Solar, LLC	Energy \$	0.00	0.00	0.00	499.23	252.96	567.98	3,753.58	5,348.88	7,752.25	11,143.00	11,891.33	11,343.91	10,332.76	7,954.75	4,772.63	3,165.00	261.75	79,040.03
140		Capacity \$	0.00	0.00	0.00	206.17	1,471.72	2,073.49	4,103.39	4,933.89	5,401.09	6,107.25	6,059.73	6,135.47	5,987.61	5,441.16	4,755.46	4,045.09	1,463.25	58,184.78
141		Subtotal \$	0.00	0.00	0.00	705.40	1,724.68	2,641.47	7,856.97	10,282.78	13,153.34	17,250.26	17,951.05	17,479.38	16,320.37	13,395.92	9,528.09	7,210.09	1,725.00	137,224.81
142	Temperance Solar, LLC	Energy \$	0.00	0.00	0.00	499.23	258.61	580.17	3,822.51	5,444.27	7,883.83	11,324.41	12,082.56	11,528.18	10,502.64	8,089.37	4,858.88	3,224.66	267.59	80,366.90
143		Capacity \$	0.00	0.00	0.00	206.17	1,486.77	2,092.94	4,128.68	4,961.85	5,427.25	6,132.76	6,083.84	6,160.87	6,013.51	5,467.23	4,783.48	4,071.91	1,478.12	58,495.39
144		Subtotal \$	0.00	0.00	0.00	705.40	1,745.38	2,673.12	7,951.19	10,406.12	13,311.07	17,457.18	18,166.40	17,689.04	16,516.15	13,556.59	9,642.36	7,296.57	1,745.71	138,862.29
145	Macbeth Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	615.88	615.88
146		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	972.29	972.29
147		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1,588.17	1,588.17
148	Woodley Solar, LLC	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	241.18	98.58	62.06	4.17	406.00
149		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	261.87	259.18	208.67	60.59	790.31
150		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	503.06	357.76	270.73	64.76	1,196.31
151	2021 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	43,627.00	31,835.46	27,734.81	28,126.14	35,323.59	39,100.97	46,009.13	251,757.11
152		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2,626.53	2,714.08	2,714.08	2,626.53	2,714.08	2,626.53	2,714.08	18,735.92
153		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	46,253.53	34,549.55	30,448.89	30,752.67	38,037.67	41,727.51	48,723.22	270,493.03
154	2022 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
155		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
156		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
157	2023 Proxy Facility	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
158		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
159		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
160	STS Hydropower, LLC (Ada Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
161		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
162		Subtotal \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
163	STS Hydropower, LLC (Morrow Hydro Plant)	Energy \$	0.00	0.00	0.00	0.00	0.00	795.93	798.16	855.59	844.57	871.08	807.28	748.66	667.48	708.05	719.34	847.86	892.97	9,556.97
164		Capacity \$	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
165		Subtotal \$	0.00	0.00	0.00	0.00	0.00	795.93	798.16	855.59	844.57	871.08	807.28	748.66	667.48	708.05	719.34	847.86	892.97	9,556.97
166	Total Compensation \$	Energy \$	53,748.67	51,946.92	55,339.85	61,707.68	62,538.49	60,348.60	65,888.23	80,477.02	97,525.55	118,114.54	146,116.60	120,966.78	108,537.73	103,972.09	103,322.62	108,759.29	110,678.43	2,353,723.03
167		Capacity \$	17,147.96	17,802.40	19,892.05	19,530.17	19,462.29	21,025.05	26,504.26	30,963.06	32,727.91	35,708.35	36,950.86	36,835.37	36,024.92	35,273.83	34,036.02	31,446.55	23,514.46	676,989.16
168		Subtotal \$	70,896.63	69,749.32	75,231.90	81,237.85	82,000.78	81,373.65	92,392.49	111,440.08	130,253.45	153,822.89	183,067.45	157,802.16	144,562.65	139,245.92	137,358.64	140,205.84	134,192.89	3,030,712.19

¹ Boyce Power actual MWh for 06/2019 - 09/2019 were received as an aggregate amount and divided amongst the three identified facilities based on their nameplate capacities

MICHIGAN PUBLIC SERVICE COMMISSION
CONSUMERS ENERGY COMPANY

Case No.: U-20697
Exhibit No.: A-111 (KGT-3)
Page: 1 of 1
Witness: KGTroyer
Date: February 2020

Forecast of Consumers Energy Transmission Expense

	(a)	(b)	(c)	(d)
	Contract Year	Transmission costs including Hemlock Contract	Transmission costs without Hemlock Contract	Incremental transmission cost to serve Hemlock Contract
1	2021 ₁	477,555,848	472,127,560	5,428,289
2	2022	500,112,429	495,102,629	5,009,800
3	2023	511,368,530	506,532,923	4,835,607
4	2024	523,723,655	518,734,541	4,989,114
5	2025	526,338,351	521,307,709	5,030,642
6	2026	535,133,183	530,136,652	4,996,531
7	2027	543,540,881	538,572,160	4,968,721
8	2028	551,002,177	546,071,053	4,931,124
9	2029	557,576,089	552,659,104	4,916,985
10	2030 ₂	568,727,611	563,712,286	5,015,325
11	2031	580,102,163	574,986,532	5,115,631
12	2032	591,704,206	586,486,263	5,217,944
13	2033	603,538,290	598,215,988	5,322,303
14	2034	615,609,056	610,180,308	5,428,749
15	2035	627,921,237	622,383,914	5,537,324
16	2036	640,479,662	634,831,592	5,648,070
17	2037	653,289,255	647,528,224	5,761,031
18	2038	666,355,040	660,478,788	5,876,252
19	2039	679,682,141	673,688,364	5,993,777
20	2040	693,275,784	687,162,131	6,113,653
21	2041	707,141,300	700,905,374	6,235,926
22	2042	721,284,126	714,923,481	6,360,644

Sources:

- 1 Forecast expenses from 2021 through 2029 from transmission expense forecast model
- 2 Forecast expenses from 2030 through 2042 escalated at 2% annually from a 2029 base

STATE OF MICHIGAN
BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

BRIAN J. VANBLARCUM

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
Development of the Property Tax Rate for the Test Year

Case No.: U-20697
Exhibit No.: A-112 (BJV-1)
Page: 1 of 3
Witness: BJVanBlarcum
Date: February 2020

Line No.	(a) Description	(b) Amount (millions)	(c) Amount (millions)	(d) Amount (millions)	(e) Source
1	Electric Property Taxes Paid - 2020 Estimate			\$ 191.7	WP BJV-5, L-5
2	Electric Property Taxes on 2020 Plant Investment			14.4	Ex. A-112 (BJV-1), p.2, L-9
3	2021 Coal Plant Valuation Tax Reduction			(1.0)	WP BJV-8, L-4
4	Electric Property Taxes on Real Property Taxable Value Increases - Inflation			1.0	Ex. A-112 (BJV-1), p.3, L-4
5	Electric Property Taxes to be Paid - 2021			\$ 206.1	
6	2020 Fiscal Year Property Taxes expensed in 2021			96.2	(Line 1 * 50.2%) ¹
7	2021 Property Taxes expensed in 2022			(103.5)	(Line 5 * 50.2%) ¹
8	Electric Property Tax Expense - 2021			\$ 198.8	
9	2020 Estimated Year-End Plant-in-Service		\$ 16,614.6		²
10	2020 Estimated Construction Work-in-Progress	\$ 320.3			³
11	@ 50%	50.00%			
12	2020 Estimated Construction Work-in-Progress		\$ 160.2		
13	Estimated Taxable Plant			\$16,774.8	(L-9 + L-12)
14	Property Tax Rate			0.011851110	(L-8/L-13)

Footnotes

¹ The 50.2% factor is from the 2019 CE Property Tax Fiscal Year Study

² Plant in Service Balance as of 12-31-2020; Plant Model U-20697.xlsx; PIS (WP-HJM-67); Line 17

³ Construction Work in Progress Balance as of 12-31-2020; Plant Model U-20697.xlsx; CWIP (WP-HJM-68); Line 17

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Development of the 2021 Property Taxes
on 2020 Plant Investment

Case No.: U-20697

Exhibit No.: A-112 (BJV-1)

Page: 2 of 3

Witness: BJVanBlarcum

Date: February 2020

Line No.	(a) Description	(b) Amount (millions)	(c) Source
1	Assessed Value of Personal Property (millions)		
2	Year-end 2020 Estimated Taxable Closings less Retirements	\$ 608.7	¹
3	First Year STC Multiplier	<u>96.0%</u>	²
4	True Cash Value of Personal Property (millions)	\$ 584.4	
5	Statutory Factor for Assessed Value	<u>50%</u>	³
6	New Plant - Assessed Value	\$ 292.2	
7	Estimated Property Tax (millions)		
8	Estimated Composite Millage Rate	<u>49.1226</u>	⁴
9	Estimated 2021 Property Taxes Paid	<u>\$ 14.4</u>	

Footnotes¹ 2021 Electric Rate Case Property Tax Model² Michigan Dept of Treasury Form 3589 - 1st year multiplier for Electric Distribution and Equipment³ Article IX, Section 3 of Constitution of Michigan of 1963⁴ CE Composite Millage Rate.xls

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy CompanyDevelopment of the 2021 Estimated Electric Portion of
Real Property Taxable Value IncreaseCase No.: U-20697
Exhibit No.: A-112 (BJV-1)
Page: 3 of 3
Witness: BJVanBlarcum
Date: February 2020

Line No.	(a) Description	(b) Amount (millions)	(c) Source
1	2021 Estimated Electric Portion of Real Property Tax Increase (millions)		
2	2021 Real Property Taxable Value - Estimated Electric Portion	\$ 20.2 ¹	
3	2021 Estimated Composite Millage Rate	<u>49.1226</u> ²	
4	2021 Estimated Electric Portion of Real Property Taxable Value increase	<u>\$ 1.0</u>	

Footnotes¹ 2021 Electric Rate Case Property Tax Model² CE Composite Millage Rate.xls

MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Amortization of Excess Deferred Federal

Income Taxes for the Test Year

Case No.: U-20697

Exhibit No.: A-113 (BJV-2)

Page: 1 of 2

Witness: BJVanBlarcum

Date: February 2020

	(a)	(b)	(c)
Line No.	Description	Projected Regulatory ¹ Liability Amortization	Projected Annual ² Excess Deferred Tax
1	Protected Plant Balance - Subject to ARAM		
2	2021	(30,187,000)	(22,510,000)
3			
4	TCJA Amortization - ARAM		
5			
6			
7			
8	Non-Protected Plant Balance - 27 years		
9	2021	3,145,000	2,345,000
10			
11			
12	Non-Protected Other Balance - 10 years		
13	2021	(8,708,000)	(6,493,000)
14			
15	TCJA Amortization - Non ARAM		(4,148,000)
16			
17			
18			
19	Total Test Year Excess Deferred Taxes		<u>(26,658,000)</u>
20			

¹ Based on the Company's regulatory liability amortization schedule approved in Case No. U-20309.² Adjustment to pre-tax based on 1.341056 gross-up factor used in Case No. U-20309.

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company

Case No.: U-20697
Exhibit No.: A-113 (BVJ-2)
Page: 2 of 2
Witness: BJVanBlarcum
Date: February 2020

MICHIGAN PUBLIC SERVICE COMMISSION
Consumers Energy Company
CE Electric
Projected Tax Reform Regulatory Liability & Amortization
(\$000)

Case No.: U-20309
Exhibit No.: A-5 (SBM-3)
Page: 1 of 1
Witness: SBMcIntosh
Date: April 2019

	(a)	(b)	(c)	(d)	(e)
Line No.	Description	Plant Diff *	Oth Plant Diff	All Other	Total Electric
1	Amortization Period	ARAM	27	10	
2					
3	Tax Reform Regulatory Asset (Liability) After Gross Up	\$ (1,172,029)	\$ 84,923	\$ (87,076)	\$ (1,174,181)
4					
5	Amortization Schedule				
6	**2019	(8,953)	1,048	(2,903)	(10,807)
7	**2020	(28,209)	3,145	(8,708)	(33,771)
8	2021	(30,187)	3,145	(8,708)	(35,749)
9	2022	(31,436)	3,145	(8,708)	(36,998)
10	2023	(35,546)	3,145	(8,708)	(41,108)
11	2024	(36,898)	3,145	(8,708)	(42,460)
12	2025	(37,086)	3,145	(8,708)	(42,648)
13	2026	(37,134)	3,145	(8,708)	(42,697)
14	2027	(39,904)	3,145	(8,708)	(45,467)
15	2028	(42,747)	3,145	(8,708)	(48,309)
16	2029	(44,426)	3,145	(5,805)	(47,085)
17	2030	(45,859)	3,145	-	(42,713)
18	2031	(46,326)	3,145	-	(43,181)
19	2032	(47,316)	3,145	-	(44,170)
20	2033	(49,093)	3,145	-	(45,948)
21	2034	(51,089)	3,145	-	(47,944)
22	2035	(44,865)	3,145	-	(41,719)
23	2036	(42,334)	3,145	-	(39,189)
24	2037	(29,201)	3,145	-	(26,055)
25	2038	(26,019)	3,145	-	(22,874)
26	2039	(25,183)	3,145	-	(22,037)
27	2040	(24,370)	3,145	-	(21,225)
28	2041	(24,256)	3,145	-	(21,110)
29	2042	(24,259)	3,145	-	(21,114)
30	2043	(24,313)	3,145	-	(21,168)
31	2044	(24,345)	3,145	-	(21,199)
32	2045	(24,242)	3,145	-	(21,097)
33	2046	(24,063)	2,097	-	(21,966)
34	2047	(23,974)	-	-	(23,974)
35	2048 - 2056	(198,398)	-	-	(198,398)
36		\$ (1,172,029)	\$ 84,923	\$ (87,076)	\$ (1,174,181)

* Subject to the normalization provisions of the Internal Revenue Code.

** Assumes Calculation C Interim Credit beginning September 2019 through December 2020.

STATE OF MICHIGAN

BEFORE THE MICHIGAN PUBLIC SERVICE COMMISSION

In the matter of the application of)
CONSUMERS ENERGY COMPANY)
for authority to increase its rates for)
the generation and distribution of)
electricity and for other relief.)
_____)

Case No. U-20697

EXHIBITS

OF

TODD A. WEHNER

ON BEHALF OF

CONSUMERS ENERGY COMPANY

February 2020

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

Page: 1 of 12

Witness: TAWehner

Date: February 2020

Proxy Companies

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
			Regulated Generation Capacity (MW)	Net PP&E (\$)	LTM Payout Ratio ≥ 55%	S&P IG Rated Bond?	Moody's IG Rated Bond?
<u>Line</u>	<u>Company</u>	<u>Ticker</u>					
1	Alliant Energy Corporation	LNT	6,353	12,462,400	64	✓	✓
2	Ameren Corporation	AEE	11,096	22,810,000	59	✓	✓
3	American Electric Power Company, Inc.	AEP	24,053	55,099,100	63	✓	✓
4	Dominion Energy, Inc.	D	23,345	54,560,000	276	✓	✓
5	DTE Energy Company	DTE	11,870	21,279,000	65	✓	✓
6	Evergy, Inc.	EVRG	7,690	18,951,700	75	✓	✓
7	NiSource Inc.	NI	3,333	15,542,500	65	✓	✓
8	OGE Energy Corp.	OGE	6,709	8,591,300	66	✓	✓
9	Pinnacle West Capital Corporation	PNW	6,400	13,766,668	67	✓	✓
10	Portland General Electric Company	POR	3,972	6,474,000	67	✓	✓
11	WEC Energy Group, Inc.	WEC	9,611	22,000,900	68	✓	✓
12	Xcel Energy Inc.	XEL	18,716	36,944,000	64	✓	✓
13	Average		11,096	24,040,131	83	✓	✓
14	Consumers Energy	CMS	6,111	17,570,000	70	✓	✓

Proxy Group Selection Criteria:

Regulated generation capacity must be greater than 2,000 MW.

Net Property Plant and Equipment ("PP&E") must be greater than \$5 billion but less than \$60 billion.

Last Twelve Months ("LTM") dividend payout ratio must be greater than or equal to 55%.

Company must not be selling its business as a part of a corporate acquisition or be a restructuring entity.

Company must have investment grade ("IG") rated bonds.

Sources:

Columns (d) through (f):

S&P Global Market Intelligence as of December 31, 2019.

Columns (g), (h):

Bloomberg as of December 31, 2019.

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697
Exhibit No.: A-14 (TAW-1)
Schedule: D-5
Page: 2 of 12
Witness: TAWehner
Date: February 2020

Empirical Capital Asset Pricing Model Application

Equation: $K_e = R_f + \alpha + F + B \times (R_p - \alpha)$

Where:

K_e = The annual required return on equity

R_f = The risk free rate

α = The alpha of the risk-return line

F = The flotation cost adjustment, not included in the calculation

B = The beta, or covariance of the stock price to market

R_p = The expected equity risk premium

Report	Date	Test Year Average		
IHS Markit	Dec. 2019	3.04%	→	2021 3.04%
Blue Chip	Dec. 1, 2019	2.80%	⇒	2021 2.80%

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Company	Ticker	Current Beta (B)	Alpha (α)	Test Year Risk-Free Rate (R _f)	Projected Risk Premium (R _p)	Projected Risk Premium ECAPM ROE
1	Alliant Energy Corporation	LNT	0.60	1.50%	2.92%	10.33%	9.72%
2	Ameren Corporation	AEE	0.55	1.50%	2.92%	10.33%	9.28%
3	American Electric Power Company, Inc.	AEP	0.55	1.50%	2.92%	10.33%	9.28%
4	Dominion Energy, Inc.	D	0.55	1.50%	2.92%	10.33%	9.28%
5	DTE Energy Company	DTE	0.55	1.50%	2.92%	10.33%	9.28%
6	Evergy, Inc.	EVERG	0.49	1.50%	2.92%	10.33%	8.75%
7	NiSource Inc.	NI	0.55	1.50%	2.92%	10.33%	9.28%
8	OGE Energy Corp.	OGE	0.75	1.50%	2.92%	10.33%	11.04%
9	Pinnacle West Capital Corporation	PNW	0.55	1.50%	2.92%	10.33%	9.28%
10	Portland General Electric Company	POR	0.60	1.50%	2.92%	10.33%	9.72%
11	WEC Energy Group, Inc.	WEC	0.50	1.50%	2.92%	10.33%	8.84%
12	Xcel Energy Inc.	XEL	0.50	1.50%	2.92%	10.33%	8.84%
13	Average		0.56				9.38%
14	Minimum		0.49				8.75%
15	Maximum		0.75				11.04%

Sources: Column (d): Beta per the Value Line Investment Survey (PNW, POR and XEL as of October 25, 2019; D and FE as of November 15, 2019; NI as of November 29, 2019; LNT, AEE, AEP, DTE, OGE, & WEC as of December 13, 2019). EVERG sourced from Bloomberg.
Column (e): Alpha, mid-point of reasonable range of 1% to 2% cited by Roger A. Morin, "New Regulatory Finance" (2006).
Column (f): Average of IHS Markit U.S. Economic Outlook (December 2019) & Blue Chip Financial Forecasts (December 1, 2019).
Column (g): Exhibit A-14 (TAW-1), Schedule D-5, page 11.
Column (h) = Column (f) + Column (g).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

Page: 3 of 12

Witness: TAWehner

Date: February 2020

Capital Asset Pricing Model Application

Equation: $K_e = R_f + F + B_T \times (R_p)$

$B_T =$

$B_T =$

$B_T = \frac{\text{Standard Deviation}_{\text{Proxy Group}}}{\text{Standard Deviation}_{\text{Market}}}$

Where:

K_e = The annual required return on equity

R_f = The risk free rate

F = The flotation cost adjustment, not included in the calculation

B_T = The total beta

R_p = The expected equity risk premium

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)
Line	Proxy Group	Company	Proxy Group Standard Deviation	Market Standard Deviation	Total Beta (B_T)	Test Year Risk-Free Rate (R_f)	Projected Risk Premium (R_p)
1	Proxy Group		14.80%	13.43%	1.10	2.92%	10.33%
							CAPM ROE 14.30%

Sources: Column (b): See exhibit A-14 (TAW-1), Schedule D-5, page 1.
Column (c): Bloomberg data, 5-yr estimate, rebalanced daily
Column (d): Bloomberg 5-yr estimate.
Column (e) = Column (c) / Column (d).
Column (f): Exhibit A-14 (TAW-1), Schedule D-5, page 2, Column (f).
Column (g): Exhibit A-14 (TAW-1), Schedule D-5, page 2, Column (g).
Column (h) = Column (f) + Column (g) x Column (e).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

Page: 4 of 12

Witness: TAWehner

Date: February 2020

Projected Risk Premium Analysis Over Utility Bonds

(a)	(b)	(c)	(d)	(e)	(f)
<u>Line</u>	<u>Description</u>	<u>S&P Bond Rating</u>			
		<u>A</u>	<u>A-</u>	<u>BBB+</u>	<u>BBB</u>
Projected Risk Premium Analysis (Appropriate Use of Spread and Projected Long-Term Bond Rates)					
1	Current Spread of Electric Utility Common Stock Over Utility Bonds	11.36%	11.36%	11.36%	11.36%
2	Projected Long-Term Government Bond Return	2.92%	2.92%	2.92%	2.92%
3	Corporate Spread	<u>1.11%</u>	<u>1.22%</u>	<u>1.37%</u>	<u>1.87%</u>
4	Current Estimated Bond Yield (Line 2 + Line 3)	4.03%	4.14%	4.29%	4.79%
5	Cost of Equity (Line 1 + Line 4)	15.39%	15.50%	15.65%	16.15%
6	Average				15.67%
7	Minimum				15.39%
8	Maximum				16.15%

Sources: Line 1: Exhibit A-14 (TAW-1), Schedule D-5, page 9, line 92.
Line 2, 13: Exhibit A-14 (TAW-1), Schedule D-5, page 2, column (f).
Line 3: Exhibit A-25 (MRB-9), page 5, lines 109-113.

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

Page: 5 of 12

Witness: TAWehner

Date: February 2020

Discounted Cash Flow ("DCF") Model Application

$$\text{Equation: } K_e = D_1 / P_0 + g + F$$

Where:

K_e = Annual required rate of return on equity

D_1 = Expected annual dividend per share at the end of first year.

P_0 = Current price of stock

g = Growth rate

F = The flotation cost adjustment, not included in the calculation

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(o)
Line	Company	Ticker	Avg of 30-day Closing (\$)	Last Qtrly Dividend Payment	Current Annual Div (D_0)	Current Dividend Yield	Number of Analyst Estimates	Consensus Analyst Growth (%)	Expected Dividend Yield	Analyst Consensus DCF ROE	Flotation Cost Adjust. (F)
1	Alliant Energy Corporation	LNT	53.44	0.355	1.42	2.66%	4	6.2%	2.82%	9.02%	0.13%
2	Ameren Corporation	AEE	75.20	0.495	1.98	2.63%	5	4.4%	2.75%	7.15%	0.13%
3	American Electric Power Company, Inc.	AEP	92.35	0.700	2.80	3.03%	10	5.6%	3.20%	8.80%	0.15%
4	Dominion Energy, Inc.	D	81.93	0.918	3.67	4.48%	9	5.5%	4.73%	10.23%	0.22%
5	DTE Energy Company	DTE	126.10	1.013	4.05	3.21%	6	7.1%	3.44%	10.54%	0.16%
6	Evergy, Inc.	EVRG	63.63	0.505	2.02	3.17%	5	8.4%	3.44%	11.84%	0.16%
7	NiSource Inc.	NI	26.86	0.200	0.80	2.98%	7	4.6%	3.12%	7.72%	0.15%
8	OGE Energy Corp.	OGE	43.10	0.388	1.55	3.60%	3	6.2%	3.82%	10.02%	0.18%
9	Pinnacle West Capital Corporation	PNW	87.40	0.783	3.13	3.58%	8	6.2%	3.80%	10.00%	0.18%
10	Portland General Electric Company	POR	55.51	0.385	1.54	2.77%	6	6.2%	2.95%	9.15%	0.14%
11	WEC Energy Group, Inc.	WEC	89.83	0.590	2.36	2.63%	8	6.4%	2.80%	9.20%	0.13%
12	Xcel Energy Inc.	XEL	62.23	0.405	1.62	2.60%	6	5.8%	2.75%	8.55%	0.13%
13	Average								3.30%	9.35%	0.16%
14	Minimum								2.75%	7.15%	0.13%
15	Maximum								4.73%	11.84%	0.22%

Sources:

Column (d): Factset data from November 18, 2019 through December 31, 2019.

Column (e): Yahoo!Finance as of December 31, 2019.

Column (f) = 4 x Column (e).

Column (g) = Column (f) / Column (d).

Column (h): Number of I/B/E/S 3-year consensus analyst dividend per share ("DPS") growth estimate as of December 31, 2019.

Column (i): I/B/E/S 3-year consensus analyst dividend per share ("DPS") growth estimate as of December 31, 2019.

Column (j) = Column (g) x (1 + Column (i)).

Column (k) = Column (i) + Column (j).

Column (l): Flotation cost adjustment of 5% of current dividend yield, as described by Roger A. Morin, "New Regulatory Finance" (2006).

While flotation cost adjustment are estimated and demonstrated in this analysis, the adjustment is not included in any of the calculations comprising the analysis.

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

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Witness: TAWehner

Date: February 2020

Comparable Earnings Analysis

(a)	(b)	(c)	(d)	(e)	(f) 2022-2024	(g)
Line	Company	Ticker	Current Beta (B)	Earnings Per Share	Book Value Per Share	Implied ROE
1	Alliant Energy Corporation	LNT	0.60	2.80	27.55	10.16%
2	Ameren Corporation	AEE	0.55	4.25	41.50	10.24%
3	American Electric Power Company, Inc.	AEP	0.55	5.00	48.50	10.31%
4	Dominion Energy, Inc.	D	0.55	5.00	39.75	12.58%
5	DTE Energy Company	DTE	0.55	7.25	75.75	9.57%
6	Evergy, Inc.	EVRG	0.49	3.50	41.50	8.43%
7	NiSource Inc.	NI	0.55	1.80	20.00	9.00%
8	OGE Energy Corp.	OGE	0.75	2.75	24.00	11.46%
9	Pinnacle West Capital Corporation	PNW	0.55	5.75	55.75	10.31%
10	Portland General Electric Company	POR	0.60	3.00	32.50	9.23%
11	WEC Energy Group, Inc.	WEC	0.50	4.50	36.75	12.24%
12	Xcel Energy Inc.	XEL	0.50	3.25	30.25	10.74%
13	Average					10.36%
14	Minimum					8.43%
15	Maximum					12.58%

Sources: Column (d), (e) & (f): Value Line Investment Survey (PNW, POR and XEL as of October 25, 2019; D and FE as of November 15, 2019; NI as of November 29, 2019; LNT, AEE, AEP, DTE, OGE, & WEC as of December 13, 2019). EVRG sourced from Bloomberg.
Column (g) = Column (e) / Column (f).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

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Witness: TAWehner

Date: February 2020

Cumulative Annual Interest Savings

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
Line			2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	Total
1	S&P Senior Secured Debt Credit Rating		BBB-	BBB	BBB	BBB	BBB+	BBB+	BBB+	A-	A	A	A	A	A	A	
2	Debt Issued - First Mortgage Bonds (\$ millions)	A		\$ -	\$ 600	\$ 500	\$ 600	\$ -	\$ 725	\$ 750	\$ 500	\$ 250	\$ 450	\$ 535	\$1,900	\$1,001	\$ 7,811
3	Interest Spread Differential vs. BBB-	B		1.31%	1.31%	1.31%	0.93%	0.99%	1.24%	0.82%	0.82%	0.53%	0.72%	0.61%	1.71%	0.85%	
4	Annual Interest Savings (\$ millions)	A x B		\$ -	\$ 8	\$ 7	\$ 6	\$ -	\$ 9	\$ 6	\$ 4	\$ 1	\$ 3	\$ 3	\$ 32	\$ 9	
5	Cumulative Annual Interest Savings (\$ millions)			\$ -	\$ 8	\$ 14	\$ 20	\$ 20	\$ 29	\$ 35	\$ 39	\$ 41	\$ 44	\$ 47	\$ 80	\$ 88	\$ 88
<i>Annual savings repeats going forward</i>																	
6	BBB- Rating Issuance Spread	C	NA	NA	NMF	4.40%	2.17%	2.55%	2.47%	1.71%	1.69%	1.63%	1.83%	1.47%	2.64%	1.97%	
7	Current Rating Issuance Spread	D	NA	NA	NMF	3.09%	1.24%	1.56%	1.23%	0.90%	0.87%	1.11%	1.11%	0.86%	0.93%	1.12%	
8	Issuance Spread Differential vs. BBB-	C - D	-	-	-	1.31%	0.93%	0.99%	1.24%	0.82%	0.82%	0.53%	0.72%	0.61%	1.71%	0.85%	

Source: All issuance spreads per the Barclays Bank Utility Deal listing. Initial data began during calendar year 2008.

Line 3: Annual average fixed rate issuance spread versus BBB-, calculated in line 8.

Line 6: Annual average fixed rate BBB- issuance spread.

Line 7: Annual average fixed rate issuance spread for current ratings shown in Line 1.

Line 8 = Line 6 - Line 7.

Column (q): Debt issuance as of December 31, 2019; spread data as of November 15, 2019.

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

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Equity Risk Premium Analysis

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)
Line	Period	Large Company Total Returns	Long Term Gov Bonds Income Returns	Difference	Period	Large Company Total Returns	Long Term Gov Bonds Income Returns	Difference
1	1926	11.62%	3.73%	7.89%	1976	23.93%	7.89%	16.04%
2	1927	37.49%	3.41%	34.08%	1977	-7.16%	7.14%	-14.30%
3	1928	43.61%	3.22%	40.39%	1978	6.57%	7.90%	-1.33%
4	1929	-8.42%	3.47%	-11.89%	1979	18.61%	8.86%	9.75%
5	1930	-24.90%	3.32%	-28.22%	1980	32.50%	9.97%	22.53%
6	1931	-43.34%	3.33%	-46.67%	1981	-4.92%	11.55%	-16.47%
7	1932	-8.19%	3.69%	-11.88%	1982	21.55%	13.50%	8.05%
8	1933	53.99%	3.12%	50.87%	1983	22.56%	10.38%	12.18%
9	1934	-1.44%	3.18%	-4.62%	1984	6.27%	11.74%	-5.47%
10	1935	47.67%	2.81%	44.86%	1985	31.73%	11.25%	20.48%
11	1936	33.92%	2.77%	31.15%	1986	18.67%	8.98%	9.69%
12	1937	-35.03%	2.66%	-37.69%	1987	5.25%	7.92%	-2.67%
13	1938	31.12%	2.64%	28.48%	1988	16.61%	8.97%	7.64%
14	1939	-0.41%	2.40%	-2.81%	1989	31.69%	8.81%	22.88%
15	1940	-9.78%	2.23%	-12.01%	1990	-3.10%	8.19%	-11.29%
16	1941	-11.59%	1.94%	-13.53%	1991	30.47%	8.22%	22.25%
17	1942	20.34%	2.46%	17.88%	1992	7.62%	7.26%	0.36%
18	1943	25.90%	2.44%	23.46%	1993	10.08%	7.17%	2.91%
19	1944	19.75%	2.46%	17.29%	1994	1.32%	6.59%	-5.27%
20	1945	36.44%	2.34%	34.10%	1995	37.58%	7.60%	29.98%
21	1946	-8.07%	2.04%	-10.11%	1996	22.96%	6.18%	16.78%
22	1947	5.71%	2.13%	3.58%	1997	33.36%	6.64%	26.72%
23	1948	5.50%	2.40%	3.10%	1998	28.58%	5.83%	22.75%
24	1949	18.79%	2.25%	16.54%	1999	21.04%	5.57%	15.47%
25	1950	31.71%	2.12%	29.59%	2000	-9.10%	6.50%	-15.60%
26	1951	24.02%	2.38%	21.64%	2001	-11.89%	5.53%	-17.42%
27	1952	18.37%	2.66%	15.71%	2002	-22.10%	5.59%	-27.69%
28	1953	-0.99%	2.84%	-3.83%	2003	28.68%	4.80%	23.88%
29	1954	52.62%	2.79%	49.83%	2004	10.88%	5.02%	5.86%
30	1955	31.56%	2.75%	28.81%	2005	4.91%	4.69%	0.22%
31	1956	6.56%	2.99%	3.57%	2006	15.79%	4.68%	11.11%
32	1957	-10.78%	3.44%	-14.22%	2007	5.49%	4.86%	0.63%
33	1958	43.36%	3.27%	40.09%	2008	-37.00%	4.45%	-41.45%
34	1959	11.96%	4.01%	7.95%	2009	26.46%	3.47%	22.99%
35	1960	0.47%	4.26%	-3.79%	2010	15.06%	4.25%	10.81%
36	1961	26.89%	3.83%	23.06%	2011	2.11%	3.82%	-1.71%
37	1962	-8.73%	4.00%	-12.73%	2012	16.00%	2.46%	13.54%
38	1963	22.80%	3.89%	18.91%	2013	32.39%	2.88%	29.51%
39	1964	16.48%	4.15%	12.33%	2014	13.69%	3.41%	10.28%
40	1965	12.45%	4.19%	8.26%	2015	1.38%	2.47%	-1.09%
41	1966	-10.06%	4.49%	-14.55%	2016	11.96%	2.30%	9.66%
42	1967	23.98%	4.59%	19.39%	2017	21.83%	2.67%	19.16%
43	1968	11.06%	5.50%	5.56%	2018	-4.38%	2.82%	-7.20%
44	1969	-8.50%	5.95%	-14.45%				
45	1970	3.86%	6.74%	-2.88%				
46	1971	14.30%	6.32%	7.98%				
47	1972	18.99%	5.87%	13.12%				
48	1973	-14.69%	6.51%	-21.20%				
49	1974	-26.47%	7.27%	-33.74%				
50	1975	37.23%	7.99%	29.24%				
51				1926-2018 Average:		11.88%	4.97%	6.91%
52						Equity Risk Premium:		6.91%
53				1942-1951 Average		18.01%	2.30%	15.71%
54						Equity Risk Premium:		15.71%
55				2011-2018 Average:		11.87%	2.85%	9.02%
56						Equity Risk Premium:		9.02%
57				Low Interest Period, 1942-1951 & 2011-2018 Average:		15.28%	2.55%	12.73%
58						Equity Risk Premium:		12.73%

Source: Columns (b), (c), (d), (f), (g) & (h): 2018 Stocks, Bonds, Bills, and Inflation (SBBI) Yearbook, Roger Ibbotson, et al.

Column (e) = Column (c) - Column (d).

Column (i) = Column (g) - Column (h).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

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Date: February 2020

Line	(a) Year	(b) Year End Avg Pr/Sh	(c) Moody's Electric Utility Common Stocks Average Div/Share		(d) % Gain	(e) Dividend Yield	(f) Total Return	(g) Yields on "A" Rated Utility Bonds (Year End)	(h) Stock Spread Over "A" Rated Bond Yields
1	1931	43.23	3.47						
2	1932	39.42	2.63	-8.81%	6.08%	-2.73%	5.85%	-8.58%	
3	1933	28.73	1.95	-27.12%	4.95%	-22.17%	7.22%	-29.39%	
4	1934	21.06	1.60	-26.70%	5.57%	-21.13%	5.36%	-26.49%	
5	1935	36.06	1.32	71.23%	6.27%	77.49%	4.29%	73.20%	
6	1936	41.60	1.48	15.36%	4.10%	19.47%	3.83%	15.64%	
7	1937	24.24	1.74	-41.73%	4.18%	-37.55%	4.03%	-41.58%	
8	1938	27.55	1.50	13.66%	6.19%	19.84%	3.74%	16.10%	
9	1939	28.85	1.48	4.72%	5.37%	10.09%	3.38%	6.71%	
10	1940	22.22	1.54	-22.98%	5.34%	-17.64%	3.0%	-20.74%	
11	1941	13.45	1.44	-38.47%	6.48%	-32.99%	3.0%	-36.05%	
12	1942	14.29	1.26	9.25%	9.37%	15.1%	3.0%	12.55%	
13	1943	21.01	1.28	47.03%	8.96%	55.98%	2.9%	52.99%	
14	1944	21.09	1.31	0.38%	6.24%	6.62%	2.97%	3.65%	
15	1945	31.14	1.30	47.65%	6.16%	53.82%	2.75%	51.07%	
16	1946	32.71	1.43	5.04%	4.59%	8.62%	2.7%	6.87%	
17	1947	25.60	1.56	-21.74%	4.77%	-16.97%	3.0%	-20.02%	
18	1948	26.20	1.60	2.34%	6.25%	8.59%	3.0%	5.53%	
19	1949	30.57	1.66	16.68%	6.34%	23.02%	2.7%	20.24%	
20	1950	30.81	1.76	0.79%	5.76%	6.54%	2.8%	3.68%	
21	1951	33.85	1.88	9.87%	6.10%	15.97%	3.2%	12.6%	
22	1952	37.85	1.91	11.82%	5.64%	17.44%	3.22%	14.24%	
23	1953	39.61	2.01	4.65%	5.31%	9.96%	3.3%	6.5%	
24	1954	47.56	2.13	20.07%	5.38%	25.45%	3.11%	22.34%	
25	1955	49.35	2.21	3.76%	4.65%	8.41%	3.3%	5.0%	
26	1956	48.96	2.32	-0.79%	4.70%	3.91%	3.3%	0.0%	
27	1957	50.30	2.43	2.74%	4.96%	7.70%	4.36%	3.34%	
28	1958	66.37	2.50	31.95%	4.97%	36.92%	4.49%	32.4%	
29	1959	65.77	2.61	-0.90%	3.93%	3.03%	4.6%	-1.93%	
30	1960	76.82	2.68	16.80%	4.07%	20.88%	4.65%	16.23%	
31	1961	99.32	2.81	29.29%	3.66%	32.9%	4.65%	28.3%	
32	1962	96.49	2.97	-2.85%	2.99%	0.14%	4.44%	-4.30%	
33	1963	102.31	3.21	6.03%	3.33%	9.36%	4.4%	4.9%	
34	1964	115.54	3.43	12.93%	3.35%	16.28%	4.54%	11.74%	
35	1965	114.86	3.6	-0.59%	3.34%	2.75%	4.63%	-2.08%	
36	1966	105.99	4.11	-7.72%	3.58%	-4.14%	5.7%	-9.81%	
37	1967	98.19	4.34	-7.36%	4.0%	-3.2%	6.67%	-9.93%	
38	1968	104.04	4.50	5.96%	4.58%	10.54%	6.87%	3.6%	
39	1969	84.62	4.61	-18.67%	4.43%	-14.23%	8.59%	-22.82%	
40	1970	88.59	4.70	4.69%	5.55%	10.25%	8.48%	1.77%	
41	1971	85.56	4.77	-3.42%	5.38%	1.96%	7.9%	-5.94%	
42	1972	83.61	4.87	-2.28%	5.69%	3.41%	7.48%	-4.07%	
43	1973	60.87	5.01	-27.20%	5.9%	-21.21%	8.24%	-29.45%	
44	1974	41.17	4.83	-32.36%	7.93%	-24.43%	10.27%	-34.70%	
45	1975	55.66	4.97	35.20%	12.07%	47.27%	10.11%	37.16%	
46	1976	66.29	5.18	19.10%	9.31%	28.40%	8.62%	19.78%	
47	1977	68.19	5.54	2.87%	8.36%	11.22%	8.64%	2.58%	
48	1978	59.75	5.81	-12.38%	8.52%	3.86%	9.70%	-13.56%	
49	1979	56.41	6.22	-5.59%	10.41%	4.82%	11.7%	-6.97%	
50	1980	54.42	6.58	-3.53%	11.66%	8.14%	14.63%	-6.49%	
51	1981	57.20	6.99	5.11%	12.84%	17.95%	16.2%	1.6%	
52	1982	70.26	7.43	22.83%	12.99%	35.82%	14.4%	21.3%	
53	1983	72.03	7.87	2.52%	11.20%	13.72%	13.52%	0.20%	
54	1984	80.16	8.26	11.29%	11.47%	22.75%	13.11%	9.64%	
55	1985	94.98	8.61	18.49%	10.74%	29.23%	10.97%	18.26%	
56	1986	113.65	8.89	19.67%	9.36%	29.03%	9.12%	19.91%	
57	1987	94.24	9.12	-17.09%	8.02%	-9.0%	10.9%	-20.04%	
58	1988	100.94	8.87	7.11%	9.41%	16.52%	10.8%	6.46%	
59	1989	122.52	8.82	21.38%	8.74%	30.12%	9.44%	20.68%	
60	1990	117.77	8.79	-3.88%	7.17%	3.3%	9.3%	-6.43%	
61	1991	144.02	8.95	22.29%	7.60%	29.89%	8.6%	21.01%	
62	1992	141.06	9.05	-2.06%	6.28%	4.23%	8.4%	-4.20%	
63	1993	146.70	8.99	4.00%	6.37%	10.37%	7.34%	3.0%	
64	1994	115.50	8.96	-21.27%	6.11%	-15.16%	8.76%	-23.92%	
65	1995	142.90	9.02	23.72%	7.81%	31.53%	7.23%	24.30%	
66	1996	136.00	9.06	-4.83%	6.34%	1.51%	7.59%	-6.08%	
67	1997	155.73	9.06	14.51%	6.66%	21.17%	7.15%	14.01%	
68	1998	181.84	7.83	16.77%	5.03%	21.79%	6.91%	14.88%	
69	1999	137.30	8.10	-24.49%	4.45%	-20.04%	8.14%	-28.18%	
70	2000	227.09	8.27	65.40%	6.02%	71.42%	7.84%	63.58%	
71	2001	200.50	8.69	-11.71%	3.83%	-7.88%	7.83%	-15.71%	
72	2002	169.50	9.13	-15.46%	4.55%	-10.91%	6.93%	-17.84%	
73	2003					25.74%	5.78%	19.85%	
74	2004					26.34%	5.46%	20.88%	
75	2005					19.62%	5.56%	14.06%	
76	2006					19.83%	5.8%	14.00%	
77	2007					20.59%	6.0%	14.53%	
78	2008					-27.06%	5.9%	-33.04%	
79	2009					8.73%	5.8%	2.85%	
80	2010					4.83%	5.64%	-0.81%	
81	2011					19.67%	4.0%	15.58%	
82	2012					0.80%	3.95%	-3.16%	
83	2013					11.20%	4.75%	6.45%	
84	2014					29.67%	3.94%	25.73%	
85	2015					-4.48%	4.39%	-8.87%	
86	2016					16.54%	4.22%	12.32%	
87	2017					11.9%	3.75%	8.23%	
88	2018					3.30%	4.35%	-1.05%	
89					1932-2018 Average:	10.76%	6.39%	4.37%	
90					1942-1951 Average:	17.88%	2.96%	14.92%	
91					2011-2018 Average:	11.09%	4.18%	6.91%	
92					Low Interest Period, 1942-1951 & 2011-2018 Average	14.86%	3.50%	11.36%	

Sources: Columns (b) & (c): Mergent Public Utility Manual. Per Moody's & Mergent, Moody's Electric Utility Index is no longer maintained.

Column (d) = (current year Column (b) - prior year Column (b)) / prior year Column (b).

Column (e) = current year Column (c) / prior year Column (b).

Column (f) = Column (d) + Column (e). For 2003 - 2018, the total return is the average of the total returns from Bloomberg for the

S&P 500 Utilities & Electric Utilities Index and the Dow Jones Utilities Index (See Exhibit A-14 (TAW-1), Schedule D5, page 10).

Column (g): 1932 - 2002 Mergent Public Utility Manual; 2003 - 2018 Bloomberg.

Column (h) = Column (f) - Column (g).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

Page: 10 of 12

Witness: TAWehner

Date: February 2020

Utility Index Total Returns

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Line	Year	S&P 500 Electric Utilities Index			S&P 500 Utilities Index			Dow Jones Utilities Index			Average Total Return
		Dec 31, Price	Dividends	Total Return	Dec 31, Price	Dividends	Total Return	Dec 31, Price	Dividends	Total Return	
	2000	179.91			216.03			412.16			
1	2001	144.69	5.620	-16.45%	145.88	5.376	-29.98%	293.94	11.698	-25.84%	-24.09%
2	2002	117.42	5.843	-14.81%	97.76	5.036	-29.53%	215.18	11.411	-22.91%	-22.42%
3	2003	139.72	5.093	23.33%	118.39	4.254	25.45%	266.90	9.446	28.43%	25.74%
4	2004	170.17	5.721	25.89%	141.60	4.789	23.65%	334.95	10.668	29.49%	26.34%
5	2005	193.16	6.655	17.42%	159.66	5.511	16.65%	405.11	12.886	24.79%	19.62%
6	2006	229.94	6.991	22.66%	186.60	5.841	20.53%	456.77	14.331	16.29%	19.83%
7	2007	274.48	7.853	22.79%	216.11	6.175	19.12%	532.53	14.972	19.86%	20.59%
8	2008	196.27	8.570	-25.37%	147.93	6.632	-28.48%	370.76	16.281	-27.32%	-27.06%
9	2009	193.33	8.799	2.99%	157.99	6.652	11.30%	398.01	16.883	11.90%	8.73%
10	2010	190.39	9.067	3.17%	159.34	6.845	5.19%	404.99	17.402	6.13%	4.83%
11	2011	219.63	9.547	20.37%	182.98	7.301	19.42%	464.68	18.134	19.22%	19.67%
12	2012	208.67	9.774	-0.54%	177.66	7.655	1.28%	453.09	19.256	1.65%	0.80%
13	2013	215.55	9.515	7.86%	193.21	7.782	13.13%	490.57	19.675	12.61%	11.20%
14	2014	271.96	9.429	30.54%	240.14	7.976	28.42%	618.08	19.965	30.06%	29.67%
15	2015	247.30	9.802	-5.46%	220.00	8.418	-4.88%	577.82	21.085	-3.10%	-4.48%
16	2016	274.98	10.324	15.37%	246.83	8.782	16.19%	659.61	22.668	18.08%	16.54%
17	2017	293.61	10.575	10.62%	267.37	9.209	12.05%	723.37	23.767	13.27%	11.98%
18	2018	295.01	10.549	4.07%	268.61	9.345	3.96%	712.93	24.080	1.89%	3.30%

Sources: Columns (c), (d), (f), (g), (i) & (j): Bloomberg.
 Column (e) = (Column (c(t)) - Column (c(t-1)) + Column (d(t))) / Column (c(t-1)).
 Column (h) = (Column (f(t)) - Column (f(t-1)) + Column (g(t))) / Column (f(t-1)).
 Column (k) = (Column (i(t)) - Column (i(t-1)) + Column (j(t))) / Column (i(t-1)).
 Column (l) = Average of Columns (e), (h) & (k).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

Schedule: D-5

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Witness: TAWehner

Date: February 2020

Projected Equity Risk Premium S&P 500

Equation: $K_e = \text{Dividend Yield} + g$

Where:

K_e = Annual required rate of return on equity

Dividend Yield = Expected dividend yield of Market

g = Growth rate

(a)	(b)	(c)
<u>Line</u>		
1	S&P 500 Index Expected Dividend Yield	1.86%
2	S&P 500 Index Expected Long-Term EPS Growth Rate	+ <u>11.39%</u>
3	Market Expected ROE (Lines 1+2)	13.25%
4	Less Risk Free Rate	(-) <u>2.92%</u>
5	Estimated Market Risk Premium (Line 3 - Line 4)	10.33%

Sources:

Rows 1 and 2: Bloomberg as of December 31, 2019.

Row 4: Exhibit A-14 (TAW-1). Schedule D-5, page 2, column (f).

Schedule D-5

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-14 (TAW-1)

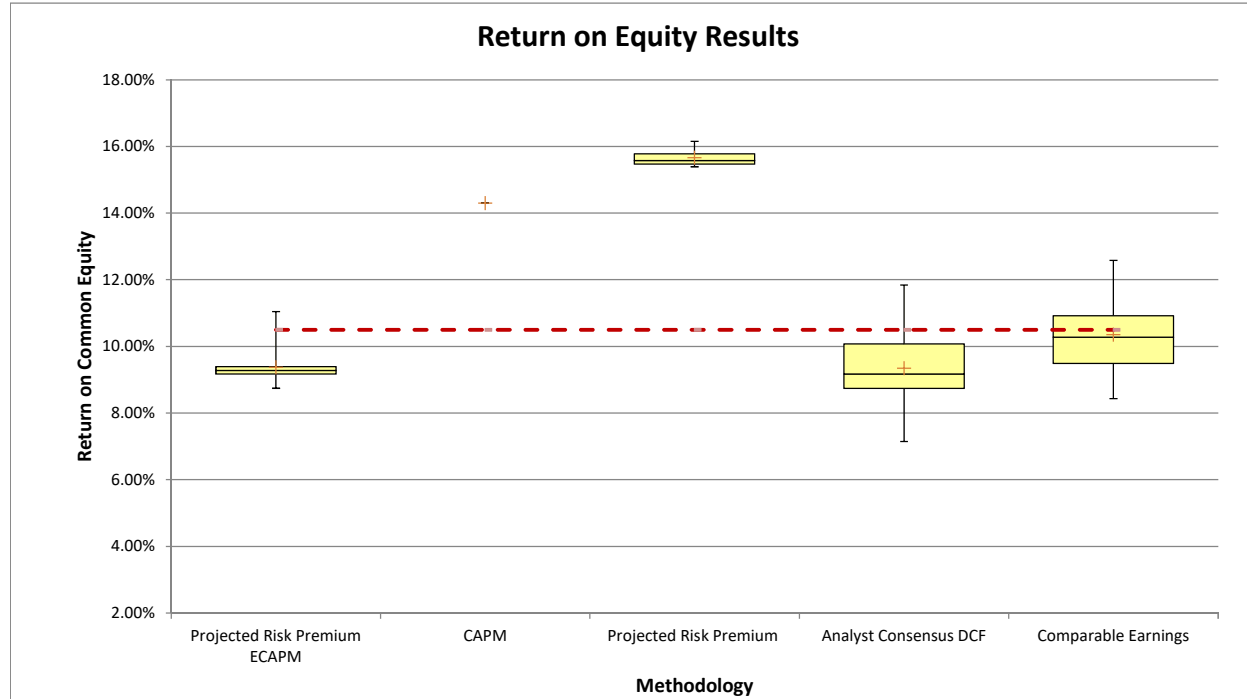
Schedule: D-5

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Witness: TAWehner

Date: February 2020

Summary of Return on Equity Results



Numerical Summary of ROE Results

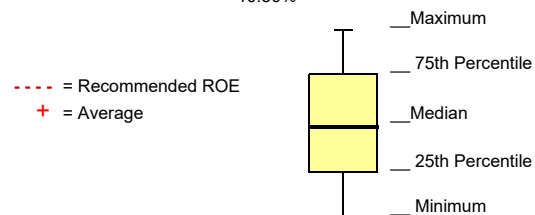
		Min	25th%	Median	75th%	Max	Avg		
1	Projected Risk Premium ECAPM	8.75%	9.17%	9.28%	9.39%	11.04%	9.38%	Exhibit A-14 (TAW-1)	Page 2 of 12
2	CAPM	14.30%	14.30%	14.30%	14.30%	14.30%	14.30%	Exhibit A-14 (TAW-1)	Page 3 of 12
3	Projected Risk Premium	15.39%	15.47%	15.58%	15.78%	16.15%	15.67%	Exhibit A-14 (TAW-1)	Page 4 of 12
4	Analyst Consensus DCF	7.15%	8.74%	9.17%	10.07%	11.84%	9.35%	Exhibit A-14 (TAW-1)	Page 5 of 12
5	Comparable Earnings	8.43%	9.49%	10.28%	10.92%	12.58%	10.36%	Exhibit A-14 (TAW-1)	Page 6 of 12

6 Recommended Cost of Equity Range for Consumers Energy:

10.0% - 11.0%

7 Recommended Ratemaking Cost of Equity for Consumers Energy:

10.50%





18 September 2019 | 10:14AM EDT

Global Markets Daily: Repo Stress Could Lead to Early Resumption of Fed Purchases

- While the spike in repo and other short-term rates can be attributed to a confluence of events that resulted in a large swing in the US Treasury's cash balances, the extent of the move suggests either that reserves may be closer to their "terminal level" than previously thought, and/or that intermediation is currently constrained. The stress could persist in the near term without Fed intervention, but is not a "systemic risk" issue in our view.
- Of the various Fed options, we believe resuming permanent OMOs (asset purchases) to offset a passive decline in reserves is the most likely solution. While the Fed may consider an IOER adjustment, without injecting reserves via open market operations, this in itself may prove insufficient to curb front-end volatility in the near term. A standing repo facility may take longer to get off the ground, and is therefore less likely than the other options.
- Our projections for reserve balances and recent price action suggest that reserve injection on the order of \$100-\$150bn may be necessary over the next few weeks to keep the policy rate within the target range. While we expect a pick-up in reserve balances in late October/early November, this pick-up is likely to prove temporary. Therefore, rather than engage in temporary OMOs over long periods, the Fed may choose to expand its balance sheet by an amount offsetting non-reserve liability growth as early as October, when it announces its next round of Treasury reinvestment purchases.

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Focus: Repo Stress Could Lead to Early Resumption of Fed Purchases

Dollar funding rates surged at the start of the week, with overnight GC repo spiking to highs of 10% intraday on Tuesday. These rate levels were higher than those observed over 2018 year-end, leading to concerns about the Fed losing control of short-term rates. Even the policy rate, fed funds, rose sharply, by 11bp on Monday to set at 2.25%, the top of its target range. On Tuesday, the Fed offered dealers up to \$75bn in repurchase operations, and followed this with another temporary Open Market Operation (OMO) this morning of similar magnitudes. Even with the Fed's intervention, fed funds effective set at 2.3% for Tuesday, above the target range. Today's operation resulted in a take-up of the full \$75bn amount, and we suspect the Fed may have to increase the size further or offer term repo.

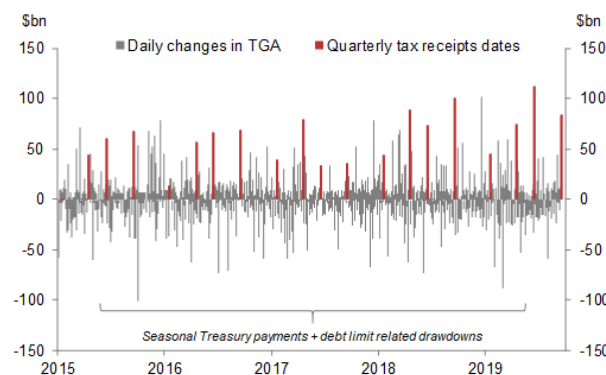
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What is driving the spike in short-term rates? We believe a confluence of events may have been to blame. The US Treasury's operating cash balances from tax payments and debt settlement jumped \$83bn on Monday alone. Since last Thursday, the Treasury General Account (TGA) has likely increased by over \$100bn, with possibly a commensurate drop in reserves. In addition to large swings in reserves and cash balances, the amount of gross coupon settlement on a single day also played a role in pressuring repo rates higher, in our view. Money fund AUM over the same period declined by roughly \$35bn according to Crane data, with outflows to a large extent mirroring the inflows into the TGA (though to a smaller magnitude). The simultaneous decline in supply of and increase in demand for funds, combined with constrained intermediation capacity, appear to have pressured funding costs higher.

Exhibit 1: Large swings in Treasury cash balances are not atypical on quarterly tax receipt dates

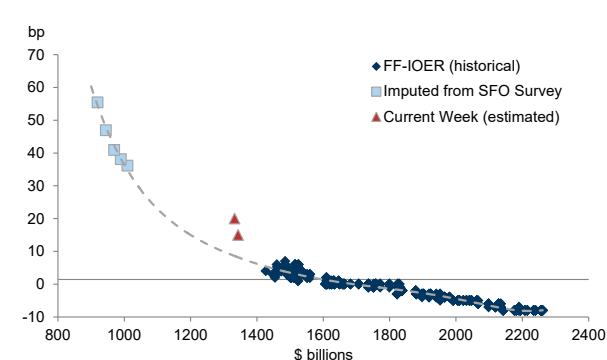
Daily changes in Treasury cash balances



Source: U.S. Treasury, Goldman Sachs Global Investment Research

Exhibit 2: The steep part of the demand curve for reserves may be closer than previously thought

Fed funds-IOER spread (bp) versus aggregate reserves (\$bn)



Source: Federal Reserve, Goldman Sachs Global Investment Research

An interesting point to note is these swings, while large, are not abnormally so, and not atypical of quarterly tax and large payment dates (Exhibit 1). Why then have markets reacted so violently? One reason could be that we are closer to the "lowest comfortable level of reserves" in aggregate than the Fed's Senior Financial Officer Survey implied. Exhibit 2 backs out an implied demand curve for reserves from the survey, and the latest fed funds (FF)-IOER spread versus our estimate of where reserve balances are likely to be today. As can be seen, until last week the relationship between FF-IOER and aggregate reserves largely seemed to track the curve imputed from the survey, which suggested a substantial buffer (i.e., the "steep" part of the curve was probably at reserve levels below \$1.2tn). However, at current levels of aggregate reserves (which we estimate had dropped to about \$1.34tn this week), the realized FF-IOER spread, at 15bp on Monday and 20bp on Tuesday, is nearly 7bp to 12bp above where we would have anticipated it to be based on the survey. This suggests one of two possibilities: 1) that the survey may have underestimated the aggregate level of reserves needed, or 2) the distribution of reserves is skewed, and banks with surplus reserves are unable to redistribute them because of other constraints (T1 leverage, SLR, G-SIB buffer). If the issue is the former, the solution is straightforward. The Fed simply needs to ensure that aggregate reserve levels remain closer to \$1.5tn. If the issue is the latter, supplying reserves through OMOs with primary dealers may only partially alleviate broader funding stresses in the system from now through year-end, given that some large banks

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that provide Dollar financing need to deleverage their 2Q19 balance sheets to stay at the same buffer level as in 4Q2018.

Looking ahead, if the Treasury meets its target cash balance of about \$350bn by the end of September, we project that aggregate reserve balances would be close to \$1.3tn, all else being equal (Exhibit 3). That level, based on recent behavior, suggests the Fed might need to intervene. What form might the Fed's intervention take? There are two issues the Fed might need to solve here. First, using our model for repo dynamics, we estimate that, under current projections for issuance and other inputs, both broad repo (SOFR) and fed funds could average between 10bp and 15bp above IOER. These estimates could be lower than that realized given the aforementioned G-SIB constraints through year-end. If this were the primary issue, the Fed would be able to address this largely via IOER adjustments—as we discuss here, this is a fairly strong tool to anchor average levels *when there are surplus reserves*. If, however, reserves are scarce, or there are intermediation constraints, lowering IOER may not be very effective in bringing fed funds back to the middle of the target range. Further, even with a modest aggregate reserve buffer, IOER adjustments wouldn't really be able to dampen volatility induced by swings in Fed liabilities (like TGA).

If our projections for reserve balances are right, Exhibit 2 suggests that the Fed will be able to get fed funds back towards the middle of the range over the next few months by injecting about \$100-\$150bn in reserves. This reserve injection could occur either on an ad hoc temporary basis, or a more permanent basis. In the short run, temporary OMOs will likely suffice, though the Fed may also make a simultaneous 5bp or 10bp IOER (and RRP rate) adjustment. If our reserve projections are correct, the Fed will likely have to engage in OMOs fairly consistently at least over the next month. Further, eventually the Fed will have to grow its balance sheet to compensate for passive erosion of reserves from the growth in non-reserve liabilities (~\$100bn annually, by our estimate). Therefore, the most effective option the Fed has, in our view, is to follow up the temporary OMOs announced yesterday (and today) with permanent OMOs sooner rather than later, perhaps when it announces its next round of Treasury reinvestment purchases. We expect that this additional amount of purchases (roughly \$10-15bn per month) will be distributed across maturity buckets in the same proportion as current Fed purchases.

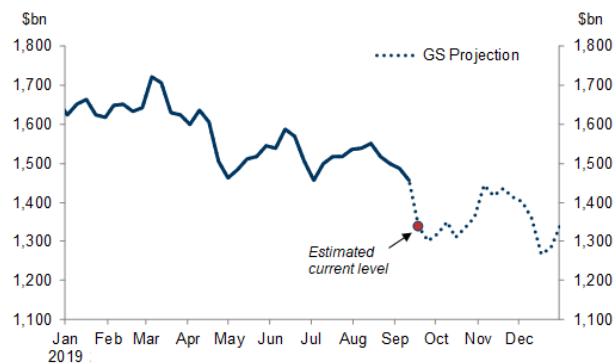
Finally, we note that, while a repo facility appears to be an effective solution in theory, as we discuss here, it will probably require the most leg work of all of the Fed's options. In addition to the Fed having to calibrate various design choices, it isn't clear that such a facility might be immediately effective in suppressing volatility. As our bank analysts note, banks would have to resubmit their resolution plans if a standing repo facility were in place, which could delay market impact until the plans were cleared (which could be several months). Exhibit 4 shows a summary of various Fed options, and our views on their efficacy/impact. A permanent steady expansion of the balance sheet tied to growth of non-reserve liabilities appears to be not only the easiest option to get off the ground, but also the most comprehensive in maintaining orderly policy rates.

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Exhibit 3: Aggregate reserve levels are likely to remain depressed over the next month, likely necessitating injections from the Fed
Total reserve balances held at the Fed



Source: Goldman Sachs Global Investment Research

Exhibit 4: Permanent OMOs appear to be the most comprehensive solution to resolving funding stresses
Summary of Fed policy options

Policy Option	Potential effect on	
	Overnight rates	Term repo rates
Cut IOER	One-off shift, spikes remain a risk and upward drift continues as reserves decline	None
Temporary OMO	Should mute volatility as and when Fed intervenes, but reserve draining continues	Some compression of premium in term rates
Permanent OMO	Balance sheet starts to grow, end of reserve draining sets structural level and diminishes risk that volatility increases further	Material compression of term premium
Standing Repo Facility	Truncates upper tail of trades on volatile days, dampening volatility	Compression of term premium (offer rate penalty-dependent)

Source: Goldman Sachs Global Investment Research

TRADE IDEAS

Best Trade Ideas Across Assets

For live pricing, charts, and a list of previous recommendations, please visit our [Trade Ideas page](#).

1. Stay long the Bloomberg-Barclays Investment Grade Wirelines index vs. the Bloomberg Barclays US Corporate IG Index, notional-neutral and rates-hedged, opened on April 8, 2019 with a target of 2.00% and a stop of -2.00%, with a PnL of 0.36%.
2. Stay short 5y5y RPI inflation swaps, opened on May 3, 2019 at 3.58%, with a target of 3.35% and a stop of 3.75%, currently trading at 3.65% with a PnL of -7bp.
3. Stay long INR 5-year bonds, pay 5-year NDOIS, opened on May 24, 2019 at 125bp, with a target of 70bp and a stop on a close above 155bp, currently trading at 137bp.
4. Stay long INR vs. short CNH, indexed at 100 on June 7, 2019, with a total return target of 106 and a stop of 97, currently trading at 100.35.
5. Stay short EUR vs. long MXN, indexed at 100 on June 21, 2019, with a total return target of 108 and a revised stop of 101, currently trading at 103.60.
6. Stay long JGB 10s30s flattener, opened on June 23, 2019 at 0.49%, with a target of 0.30% and a stop of 0.65%, currently trading at 0.51%, with a PnL of -2bp.
7. Receive 10y USD swap vs. pay 10y EUR swap, opened on August 8, 2019 at 1.70%, with a target of 1.40% and a stop of 1.95%, currently trading at 1.75%, with a PnL of -5bp.
8. Stay long SONIA 1y1y/2y1y steepeners, opened on August 9, 2019 at -0.02%, with a target of 0.10% and a stop of -0.10%, currently trading at -0.02%, with a PnL of 0bp.

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9. Stay UST 5s30s curve steepener, opened on August 16, 2019 at 0.55%, with a target of 0.80% and a stop of 0.35%, currently trading at 0.61%, with a PnL of 6bp.
10. Sell USD 6m30y straddles versus 3y30y straddles, opened on August 30, 2019 at 13abp, with a target of -5abp and a stop of 20abp, currently trading at 15abp with a PnL of -11bp.
11. Buy Apr20 TIPS hedged w/ Mar20 RBOB, opened on August 30, 2019 at 0.00%, with a target of 0.40% and a stop of -0.20%, currently trading at 0.02%.
12. Stay short USD vs. long JPY, opened on September 6, 2019 at 106.81, with a target of 103 and a stop of 109, currently trading at 108.13.
13. Stay long BRL vs. short CLP, indexed at 100 on September 13, 2019, with a total return target of 106 and a stop of 97, currently trading at 100.9.

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Disclosure Appendix

Reg AC

We, Praveen Korapaty, William Marshall and Avisha Thakkar, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

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Funds from operations (“FFO”) is a vital credit metric referenced by the credit rating agencies.

In the long-term, FFO is simply the net income of a Company plus depreciation.

$$\mathbf{FFO = Net\ Income + Depreciation \quad (1)}$$

This equation can be further developed by using the follow relationships

$$\mathbf{Net\ Income = ROE \times Equity, \text{ and } Depreciation = \frac{Assets}{Life\ of\ Assets}}$$

Substituting these equations into (1) results in the following relationship

$$\mathbf{FFO = ROE \times Equity + \frac{Assets}{Life\ of\ Assets} \quad (2)}$$

Furthermore, $\mathbf{Assets = Debt + Equity}$, and $\frac{1}{Life\ of\ Assets} = \mathbf{Depreciation\ Rate}$

Using these relationships and equation (2) results in the following

$$\mathbf{FFO = ROE \times Equity + Depreciation\ Rate \times (Debt + Equity)}$$

The equation for the FFO to Debt metric can therefore be derived as follows

$$\frac{\mathbf{FFO}}{\mathbf{Debt}} = \frac{\mathbf{ROE \times Equity + Depreciation\ Rate \times (Debt + Equity)}}{\mathbf{Debt}} \quad (3)$$

Simple mathematic distribution arrives at the following

$$\frac{\mathbf{FFO}}{\mathbf{Debt}} = \frac{\mathbf{ROE \times Equity}}{\mathbf{Debt}} + \frac{\mathbf{Depreciation\ Rate \times (Debt + Equity)}}{\mathbf{Debt}} \quad (4)$$

Further mathematical manipulation results in the following

$$\frac{\mathbf{FFO}}{\mathbf{Debt}} = \frac{\mathbf{ROE \times Equity}}{\mathbf{Debt}} + \mathbf{Depreciation\ Rate \times \left(1 + \frac{Equity}{Debt}\right)} \quad (5)$$

Using the Company's depreciation rate¹, the 52.5% equity ratio described in Company witness Mr. Bleckman's direct testimony and my recommended ROE of 10.50%, results in a 20% long-term FFO to debt as demonstrated below.

$$\frac{FFO}{Debt} = \frac{10.50\% \times 52.5\%}{(1-52.5\%)} + 3.9\% \times \left(1 + \frac{52.5\%}{(1-52.5\%)}\right) = 20\% \quad (Ex. 1)$$

Alternatively, if the Commission maintains the ROE at 10.0% and the current equity ratio of 52.5%, the resulting long-term FFO to Debt would only be 19.3%.

$$\frac{FFO}{Debt} = \frac{10.0\% \times 52.5\%}{(1-52.5\%)} + 3.9\% \times \left(1 + \frac{52.5\%}{(1-52.5\%)}\right) = 19.3\% \quad (Ex. 2)$$

If the Commission maintains the ROE at 10.0%, the Commission could approve an equity ratio of 53.67% to remain supportive of the long-term FFO to Debt at 20.0%.

$$\frac{FFO}{Debt} = \frac{10.0\% \times 53.67\%}{(1-53.67\%)} + 3.9\% \times \left(1 + \frac{53.67\%}{(1-53.67\%)}\right) = 20\% \quad (Ex. 3)$$

¹ Page 141 of Company 2018 10-K depreciation rates for the Electric and Gas Utility Property for 2018 are 3.9% and 2.9%, respectively. This example applies the electric rate of 3.9% which results in a higher FFO to Debt for a given equity ratio and ROE combination. While credit rating agencies look at credit metrics on a company-wide basis instead of segment by segment, subsequent versions of this analysis may need to be revised to reflect more accurately the lower depreciation rates and the implied lower FFO to Debt ratio.

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**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Belmont Municipal Light Department;)
Braintree Electric Light Department;)
Concord Municipal Light Plant;)
Georgetown Municipal Light Department;)
Groveland Electric Light Department;)
Hingham Municipal Lighting Plant;)
Littleton Electric Light & Water)
Department; Middleborough Gas & Electric)
Department; Middleton Electric Light)
Department; Reading Municipal Light)
Department; Rowley Municipal Lighting)
Plant; Taunton Municipal Lighting Plant;)
Wellesley Municipal Light Plant,)

Complainants,)

v.)

Central Maine Power Company; Emera)
Maine (formerly known as Bangor Hydro-)
Electric Company); Eversource Energy)
Service Company and its operating)
company affiliates: The Connecticut Light)
and Power Company, Western)
Massachusetts Electric Company, Public)
Service Company of New Hampshire, and)
NSTAR Electric Company; New England)
Power Company d/b/a National Grid; New)
Hampshire Transmission LLC d/b/a)
NextEra; The United Illuminating)
Company; Fitchburg Gas and Electric Light)
Company; and Vermont Transco, LLC,)

Respondents.)

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ANSWERING TESTIMONY AND EXHIBITS OF

JOHN D. QUACKENBUSH, CFA

**ON BEHALF OF
THE NEW ENGLAND TRANSMISSION OWNERS**

MARCH 23, 2017

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EXHIBITS TO ANSWERING TESTIMONY

<u>Exhibit No.</u>	<u>Description</u>
NET-02301	Qualifications of John D. Quackenbush
NET-02302	Federal Funds Effective Rate
NET-02303	Yields on U.S. Treasury Securities
NET-02304	Yields on Ten-Year U.S. Treasury Securities
NET-02305	Moody's Baa Utility Bond Yields
NET-02306	Yields on Global Government Securities
NET-02307	Securities Held Outright by the Federal Reserve Bank
NET-02308	RRA State-Authorized ROEs

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1 **I. INTRODUCTION**

2 **Q1. PLEASE STATE YOUR NAME AND BUSINESS ADDRESS.**

3 A1. My name is John D. Quackenbush and my business address is 46320
4 Station Road, New Buffalo, Michigan 49117. I am the President of JQ
5 Resources, LLC.

6 **Q2. HAVE YOU PREVIOUSLY APPEARED AS A WITNESS IN ANY**
7 **CASES BEFORE THIS COMMISSION?**

8 A2. No, but I have testified as a witness before state regulatory commissions
9 including the Florida Public Service Commission, the Kansas Corporation
10 Commission, the Illinois Commerce Commission, the Missouri Public
11 Service Commission, the Nevada Public Service Commission, the New
12 Jersey Board of Public Utilities, the North Carolina Utilities Commission,
13 the Oregon Public Utility Commission, the South Carolina Public Service
14 Commission, the Tennessee Public Service Commission, and the Public
15 Utility Commission of Texas. Additionally, I have served as the Chairman
16 of the Michigan Public Service Commission and the Chief Financial
17 Analyst of the Illinois Commerce Commission.

18 A listing of my qualifications is presented as Exhibit No. NET-
19 02301.

20 **Q3. WHAT IS THE PURPOSE OF YOUR TESTIMONY?**

21 A3. I am appearing on behalf of the Respondents in this proceeding, a group of
22 New England Transmission Owners (NETOs), to respond to the direct
23 testimony of Dr. Lesser and Dr. Peters filed on behalf of Eastern
24 Massachusetts Consumer-Owned Systems (EMCOS). In particular, my
25 testimony will rebut the erroneous assertions of Dr. Lesser that the

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1 existence of DCF model risk is incompatible with the Commission's
2 finding that the Efficient Market Hypothesis is valid. I will demonstrate
3 that the anomalous capital market conditions that prevailed in the time
4 period relevant to this proceeding have not changed from those that the
5 Commission found to be anomalous in Opinion No. 531 and Opinion No.
6 551.¹ I will also rebut Dr. Peter's assertion that the New England region is
7 in danger of building too much transmission infrastructure and Dr. Peters'
8 recommendation that Dr. Lesser's recommended base ROE of 8.59%
9 should be adjusted downward by 39 basis points to 8.20% due to capital
10 structure issues. I will also explain my view that Dr. Lesser and Dr. Peters
11 misjudge the relative risk of transmission and distribution investments and
12 that vertically integrated electric utility state-authorized ROEs, rather than
13 distribution-only state-authorized ROEs, are the most relevant state-
14 regulated authorized ROEs on which the Commission should focus in this
15 proceeding. In conclusion, I explain why the base returns on equity (ROEs)
16 recommended by Dr. Lesser and Dr. Peters would not provide an adequate
17 return relative to the risks of building electric transmission infrastructure
18 and do not satisfy the requirements of the U.S. Supreme Court's guidance
19 in the *Hope*² and *Bluefield*³ decisions.

¹ *Coakley v. Bangor Hydro-Elec. Co.*, Opinion No. 531, 147 FERC ¶ 61,234, *order on paper hearing*, Opinion No. 531-A, 149 FERC ¶ 61,032 (2014), *order on reh'g*, Opinion No. 531-B, 150 FERC ¶ 61,165 (2015), *appeals docketed*, *Emera Me. v. FERC*, No. 15-1118 (D.C. Cir. Apr. 30, 2015), *Braintree Elec. Light Dep't v. FERC*, No. 15-1119 (D.C. Cir. May 1, 2015), *Mass. v. FERC*, No. 15-1121 (D.C. Cir. May 1, 2015). *Ass'n of Businesses Advocating Tariff Equity v. Midcontinent Indep. Sys. Operator, Inc.*, Opinion No. 551, 156 FERC ¶ 61,234 (2016), *reh'g pending*.

² *FPC v. Hope Natural Gas Co.*, 320 U.S. 591 (1944) ("*Hope*").

³ *Bluefield Water Works & Improvement Co. v. Pub. Serv. Comm'n of W. Va.*, 262 U.S. 679 (1923) ("*Bluefield*").

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1 **Q4. WHAT IS THE BASIS FOR YOUR TESTIMONY?**

2 A4. My testimony is based on my 35 years of experience working in the field of
3 utility regulation. My career includes more than four years supporting state
4 utility regulators as a finance staff member of the Illinois Commerce
5 Commission; 14 years performing regulatory and treasury functions in the
6 telecommunications industry for Sprint Corporation partially during the
7 application of utility cost of service regulation to incumbent local exchange
8 carriers and partially during the transition from cost of service regulation to
9 price cap regulation; 11 years in the investment community covering
10 approximately 80 North American companies including regulated utilities,
11 building U.S. and Canadian domestic portfolios, and leading the global
12 utilities team in building global utility portfolios for UBS Global Asset
13 Management (UBS); more than four years regulating utilities as a state
14 utility regulatory commissioner at the Michigan Public Service
15 Commission; and most recently providing consulting services for the last
16 year to participants in regulated utility industries.

17 In preparing my testimony, I relied on my own knowledge of both
18 U.S. and global financial markets and areas of investment with which the
19 NETOs compete in the capital markets for investor funds. Also, I regularly
20 meet and interact with institutional investors and sell-side security analysts
21 that focus on the utility sector and I continue to monitor how investors
22 currently perceive and evaluate utility investment opportunities and risks.

23 **Q5. WHAT DO YOU MEAN WHEN YOU SAY YOU COVERED 80**
24 **NORTH AMERICAN COMPANIES INCLUDING REGULATED**
25 **UTILITIES AT UBS?**

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1 A5. My duties at UBS during 2001 through 2011 included building a five-year
2 forecasted income statement, balance sheet, and cash flow statement for
3 each covered company. The exact number of covered companies varied
4 over time with mergers, acquisitions, and divestitures, but I generally
5 covered approximately 40 regulated electric utilities at any time. I also met
6 regularly with the senior management, customers, suppliers, and regulators
7 of each covered company. During my time at UBS, I directed the
8 investment of significant amounts of client funds in several owners of
9 NETOs, including Northeast Utilities (a predecessor of Eversource Energy
10 and parent of The Connecticut Light and Power Company, Western
11 Massachusetts Electric Company, and Public Service Company of New
12 Hampshire) in the U.S. portfolio, Emera (parent of Emera Maine) in the
13 Canadian portfolio, NextEra Energy and its predecessor FPL Group (owner
14 of New Hampshire Transmission LLC) in the U.S. portfolio, and National
15 Grid (parent of New England Power Company) in the global portfolio.

16 **Q6. WHILE WORKING IN THE TELECOMMUNICATIONS**
17 **INDUSTRY, WHAT TREASURY DUTIES DID YOU PERFORM**
18 **THAT ARE RELEVANT TO THE ISSUES IN THIS PROCEEDING?**

19 A6. At Sprint Corporation, during the period from 1995 through 2000, I
20 prepared risk-adjusted cost of capital estimates on a quarterly basis that
21 were used for capital investment, valuation, mergers and acquisitions,
22 Economic Value Added (EVA), and product / service costing analysis
23 across divisions. These risk-adjusted cost of capital estimates varied by
24 division and were utilized as hurdle rates for capital budgeting decisions
25 across the Local, Long Distance, and Wireless Divisions.

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1 Additionally, during 1995 through 2000, I was responsible for
2 managing Sprint's relationships with four rating agencies. In providing the
3 quantitative and qualitative information required by the rating agencies to
4 rate the parent and several separately-rated subsidiaries, I became familiar
5 with how the rating agencies differentiated risk among different companies
6 and subsidiaries of the same company.

7 **II. SUMMARY OF CONCLUSIONS**

8 **Q7. PLEASE SUMMARIZE YOUR CONCLUSIONS.**

9 A7. I conclude that the anomalous capital market conditions that the
10 Commission previously found to exist in Opinion No. 531 and Opinion No.
11 551 still persist. I disagree with the conclusion of Dr. Peters that there has
12 been too much transmission investment in New England. To the contrary,
13 transmission investment in New England occurs under the direction of ISO-
14 NE and only after a rigorous needs and solutions assessment. I also
15 disagree with both Dr. Lesser and Dr. Peters on the relative risk of
16 transmission and distribution investment and concur with the Commission's
17 previous conclusion that transmission investment is more risky than
18 distribution investment. Furthermore, Dr. Peters' proposal to reduce Dr.
19 Lesser's already inadequate recommended base ROE by 39 basis points is
20 inappropriate. Finally, I demonstrate that both Dr. Lesser's and Dr. Peters'
21 base ROE recommendations are grossly inadequate for the NETOs to meet
22 the *Hope* and *Bluefield* standards.

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1 **III. ANOMALOUS CAPITAL MARKET CONDITIONS PERSIST**

2 **Q8. IN OPINION NO. 531 AND OPINION NO. 551, DID THE**
3 **COMMISSION REACH ANY FINDINGS THAT ARE RELEVANT**
4 **TO YOUR TESTIMONY IN THIS PROCEEDING?**

5 A8. Yes. These two opinions contain a number of findings that are relevant to
6 my testimony. With respect to capital market conditions, the Commission
7 stated in Paragraph 142 of Opinion No. 531:

8 [W]e conclude that a mechanical application of the DCF
9 methodology with the use of the midpoint here would result
10 in an ROE that does not satisfy the requirements of *Hope* and
11 *Bluefield*. Therefore, based on the record in this case,
12 including the unusual capital market conditions present, we
13 conclude that the just and reasonable base ROE for the
14 NETOs should be set halfway between the midpoint of the
15 zone of reasonableness and the top of the zone of
16 reasonableness.

17 The Commission continued in paragraph 145 of Opinion No. 531:

18 We are concerned that capital market conditions in the record
19 are anomalous, thereby making it difficult to determine the
20 return necessary for public utilities to attract capital. In these
21 circumstances, we have less confidence that the midpoint of
22 the zone of reasonableness established in this proceeding
23 accurately reflects the equity returns necessary to meet the
24 *Hope* and *Bluefield* capital attraction standards. We find it
25 necessary and reasonable to consider additional record
26 evidence, including evidence of alternative benchmark
27 methodologies and state commission-approved ROEs, to gain
28 insight into the potential impacts of these unusual capital
29 market conditions on the appropriateness of using the
30 resulting midpoint.

31 Turning to Opinion No. 551, the Commission found in paragraph 122:

32 Because the evidence in this proceeding indicates that capital
33 market conditions continue to reflect the type of unusual
34 conditions that the Commission identified in Opinion No.

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1 531, we remain concerned that a mechanical application of
2 the DCF methodology would result in a return inconsistent
3 with *Hope* and *Bluefield*.

4 Furthermore, the Commission concluded in paragraph 137 of Opinion No.
5 551:

6 [D]ue to the presence of unusual capital market conditions,
7 we find it appropriate to look to other record evidence to
8 inform the just and reasonable placement of the ROE within
9 the zone of reasonableness produced by the DCF
10 methodology.

11 **Q9. BASED ON YOUR STATE REGULATORY EXPERIENCE, IS WHAT**
12 **THE COMMISSION DID BY CONSIDERING MULTIPLE ROE**
13 **METHODOLOGIES AND SETTING THE BASE ROE ABOVE THE**
14 **MIDPOINT OF A ZONE AT ALL UNUSUAL?**

15 A9. No, it is not. In making ROE decisions, it is typical for regulatory
16 commissions to be confronted with a record consisting of multiple
17 methodologies from multiple witnesses. Amid the plethora of evidence
18 before it, the regulatory commission is charged with considering and
19 weighing all the evidence and determining a specific authorized base ROE.
20 The “weighing” part is challenging and can be different in each
21 commissioner’s reasoning, but the task at hand for commissioners is to
22 agree on an authorized base ROE that is within the zone defined by the
23 evidence. There are circumstances that may lead a commission to conclude
24 that the midpoint of the zone is appropriate, but at other times, the weight
25 of the evidence dictates that there is reason to select a different point in the
26 zone. It is not surprising that under certain circumstances, commissions
27 may choose to emphasize a particular methodology while downplaying that
28 same methodology in different circumstances. Similarly, it is not surprising

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1 that under certain circumstances, a commission may find that it is
2 appropriate to give more weight to the upper part or even the very top of
3 the zone. Given this perpetual challenge that faces regulatory commissions
4 in general, it is not surprising that the Commission decided to rely on
5 multiple methods and set the base ROE above the midpoint of a zone of
6 reasonableness.

7 **Q10. IN ANOMALOUS MARKET CONDITIONS, SHOULD THE**
8 **COMMISSION BE CONCERNED ABOUT MODEL RISK?**

9 A10. Yes, it should. Model risk is the risk that a model used to evaluate real-
10 world situations will fail to predict or represent the real phenomenon that is
11 being modeled. For example, the DCF model is often used to estimate the
12 cost of equity. The implementation of the DCF model requires inputs
13 including the dividend yield and the expected growth rate. If a financial
14 analyst implements the DCF model but relies on unusual or anomalous
15 dividend yields or growth rates, the model outputs are unlikely to represent
16 an accurate estimate of the cost of equity.

17 An ROE recommendation by a witness or an ROE decision by a
18 regulator requires both the application of financial models and the use of
19 informed judgment. An ROE based solely on judgment would be
20 inappropriate, as would be an ROE that relied solely on the mechanical
21 application of theoretical financial models. In my opinion, it is common for
22 regulatory commissions to acknowledge that any theoretical model, no
23 matter how conceptually appealing and well-supported, needs to be
24 supplemented with informed judgment. Commissions are on a constant
25 quest to balance the theoretical with the practical.

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1 **Q11. HOW DO INVESTORS VIEW MODEL RISK?**

2 A11. Investors use many valuation approaches, and the traditional DCF
3 methodology is among the most important. However, investors do not have
4 homogeneous expectations. Not all investors use the same tools or inputs.
5 Differing expectations are what result in different investors placing
6 different valuations on the same investment, thus creating a marketplace. I
7 will focus my comments on large institutional investors that are the primary
8 price-determining force in the marketplace, as these large institutional
9 investors tend to engage in more complex, independent analysis than retail
10 investors do. I want to point out, though, that even sophisticated investors
11 do not have homogeneous expectations.

12 While at UBS, our primary valuation approach incorporated DCF
13 and CAPM methodologies. To enhance investment comparisons across
14 industries, all analysts covering different industries used a specific type of
15 multi-stage DCF model, but I know of other large institutional investors
16 that used different and simpler single-stage or two-stage DCF models. At
17 UBS, analysts initially developed individual cash flow projections for each
18 company generally for the next five years and industry “normal” growth,
19 which by default began in year ten. Years six through ten were modeled as
20 a transition from the company-specific growth rate toward the industry
21 growth rate in year ten. The analysts had discretion to deviate from the
22 default five year initial period and year ten start of the “normal” period in
23 the DCF model if justified by specific circumstances. At UBS, our DCF
24 inputs of expected dividends and expected growth rates were driven largely
25 by the financial modeling we did to forecast income statements, balance

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1 sheets, and cash flow statements. I know that my DCF growth rate
2 estimates differed from other investors and often did not match consensus.
3 I also know analysts and investors that give more weight to non-DCF
4 valuation tools. But regardless of model differences and different inputs to
5 the model, investors apply judgment to the results in making investment
6 decisions. Thoughtful investors do not rely exclusively on mechanical
7 application of a single theoretical model. As with regulators, investors are
8 constantly balancing theory and the real world.

9 **Q12. DR. LESSER CRITICIZES COMMISSION FINDINGS ON MODEL**
10 **RISK FOUND IN OPINION NO. 531 AND OPINION NO. 551.**
11 **PLEASE COMMENT.**

12 A12. Dr. Lesser quotes from Footnote 286 of Opinion No. 531, which states:

13 As the NETOs' witness Lapson testified, "There is 'model
14 risk' associated with excessive reliance or mechanical
15 application of a model when the surrounding conditions are
16 outside the normal range. 'Model risk' is the risk that a
17 theoretical model that is used to value real-world transactions
18 fails to predict or represent the real phenomenon that is being
19 modeled."

20 Dr. Lesser also quotes from Finding 125 of Opinion No. 551, which states:

21 Consistent with Opinion No. 531, we find that the DCF
22 methodology is subject to model risk of providing unreliable
23 outputs in the presence of unusual capital market conditions.

24 Dr. Lesser takes issue with the Commission's finding that model risk exists.
25 Dr. Lesser attempts to prepare a theoretical critique of model risk and
26 concludes model risk theoretically cannot exist. However, Dr. Lesser
27 completely misses the point that the Commission does not need more

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1 theory, but rather, it is concerned, as are investors, with balancing the
2 theoretical and the practical.

3 **Q13. WHAT ARE YOUR VIEWS ON DR. LESSER'S DISCUSSION OF**
4 **THE EFFICIENT MARKET HYPOTHESIS?**

5 A13. Dr. Lesser postulates a false argument when he asserts that the efficient
6 market hypothesis is repudiated by the FERC's past findings of the
7 existence of model risk and anomalous capital markets conditions.
8 Mechanical application of any model can entail model risk depending on
9 the inputs and the model's ability to reflect reality. Dr. Lesser assumes that
10 theoretical models do not have practical limitations and thus his comments
11 on model risk are ill-informed, as judgment must always be applied to
12 assess how well the mechanical application of a theoretical model reflects
13 the real world.

14 Model risk exists in the real world, is a practical consideration for
15 both investors and commissions, and does not attack or invalidate the
16 efficient market hypothesis. Mechanically plugging data into a model, no
17 matter how theoretically robust, can result in outputs that do not reflect the
18 real world. Model risk and the lack of a perfect cost of capital model is
19 further evidenced by the continual quest of academics and practitioners to
20 discover new models.

21 Furthermore, Dr. Lesser's extensive exercise related to the lambda
22 factor fundamentally misses the point that a theoretical model, by
23 definition, is never a true reflection of all the parameters that investors
24 consider when making investment decisions. Any model abstracts from
25 reality and makes simplifying assumptions to get to a practicable result.

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1 Dr. Lesser essentially treats a simplified model as the ultimate truth. Dr.
2 Lesser's lambda exercise ignores the crux of the issue - the Commission
3 intuitively found that outputs of alternative models were deemed more
4 representative of reality than the two-stage DCF model results given
5 prevailing anomalous market conditions.

6 **Q14. DO YOU AGREE WITH THE COMMISSION'S PREVIOUS**
7 **FINDING ON THE EFFICIENT MARKET HYPOTHESIS?**

8 A14. Yes. The Commission found in paragraph 132 of Opinion No. 551 that:

9 The finding that mechanical application of the DCF
10 methodology may produce results inconsistent with *Hope* and
11 *Bluefield* in certain circumstances is not inconsistent with the
12 efficient market theory underlying the typical application of
13 the DCF methodology in normal circumstances.

14 I agree with the Commission that a finding of anomalous capital
15 market conditions is not inconsistent with the Efficient Market Hypothesis,
16 and I disagree with Dr. Lesser's assertion on this point.

17 **Q15. WHAT LEADS YOU TO CONCLUDE THAT THE ANOMALOUS**
18 **CAPITAL MARKET CONDITIONS RECOGNIZED BY THE**
19 **COMMISSION IN OPINION NO. 531 AND OPINION NO. 551 ARE**
20 **STILL IN EFFECT?**

21 A15. In response to the global financial crisis of 2008, the U.S. Federal Reserve
22 Bank and other global central banks began a massive monetary stimulus
23 program in late 2008 / early 2009 that created and have perpetuated
24 anomalous capital market conditions. According to the Federal Reserve
25 Bank of New York Staff Report No. 441 entitled "Large Scale Asset
26 Purchases by the Federal Reserve: Did They Work?" dated March 2010,
27 the Federal Reserve Bank's traditional policy instrument, the target federal

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1 funds rate, had effectively been driven to its lower bound of zero. In order
2 to further ease the stance of monetary policy as the economic outlook
3 deteriorated, the Federal Reserve Bank purchased massive quantities of
4 assets with medium and long maturities. These purchased securities are
5 reported on the Federal Reserve Bank balance sheet as “Securities Held
6 Outright.” These purchases led to significant and long-lasting reductions in
7 longer-term interest rates on a range of securities, including securities that
8 were not included in the purchase programs. Many other countries faced
9 the same policy dilemma, making the impact global.

10 Janet Yellen, Chair of the Board of Governors of the Federal
11 Reserve System, confirmed in a speech on March 3, 2017:

12 [O]nce the Committee had cut the federal funds rate to near
13 zero in late 2008, it became necessary to deploy new tools to
14 supply the considerable monetary accommodation required
15 by the extremely weak state of the job market and persistently
16 low inflation. Those tools—especially our large-scale
17 securities purchases and increasingly forward guidance
18 pertaining to the likely future path of the federal funds rate—
19 enabled the Federal Reserve to provide necessary additional
20 support to the U.S. economy by pushing down longer-term
21 interest rates and easing financial conditions more generally.⁴

22 Chair Yellen continued by mentioning that the Federal Reserve Bank
23 completed its latest round of large-scale asset purchases, sometimes
24 referred to as quantitative easing, or QE, in 2014. The Federal Open
25 Market Committee (FOMC) then issued a set of “normalization principles”
26 that indicated its intention “to maintain the overall size of the Federal

⁴ Janet L. Yellen, Chair, Bd. of Governors of Fed. Reserve Sys., From Adding Accommodation to Scaling It Back (Mar. 3, 2017), <https://www.federalreserve.gov/newsevents/speech/yellen20170303a.htm>.

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1 Reserve's balance sheet at an elevated level until sometime after the FOMC
2 had begun to raise its target for the federal funds rate." Chair Yellen's
3 March 3, 2017 speech contained footnote 9 that explained:

4 Large Federal Reserve holdings of longer-term securities
5 reduce the total amount of such securities available for
6 purchase by the public, exerting upward pressure on their
7 prices and, thus, depressing their yields and contributing to
8 lower borrowing costs for American families and businesses.

9 U.S. and global financial markets continue demonstrably to exhibit the
10 effects during the relevant Complaint IV period of a massive exercise of
11 monetary policy which has produced anomalous capital market conditions.
12 This anomaly impacts monetary aggregates, interest rates, and the valuation
13 of financial assets due to the significant and unprecedented amount of
14 monetary stimulus that has been applied by the U.S. Federal Reserve Bank
15 and other global central banks including the European Central Bank, the
16 Bank of Japan, the Bank of England, the Swiss National Bank, and the
17 Bank of Canada.

18 U.S. and global interest rates and financial markets remain subject to
19 powerful and unprecedented monetary policy actions by the Federal
20 Reserve Bank and other global central banks that continue to affect capital
21 market conditions during the refund period of this proceeding beginning on
22 April 29, 2016 and during the time periods utilized by Dr. Lesser and
23 NETOs' Witness McKenzie for their DCF analyses in this proceeding. The
24 abnormal capital market conditions include very low U.S. and global long-
25 term and short-term interest rates and monetary supply significantly in
26 excess of its normal use. The anomaly is evident in unusually low U.S.
27 Treasury bond yields and utility bond yields.

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1 Besides very low interest rates, a hugely significant component of
2 the monetary stimulus has been quantitative easing, or the ramp up and
3 maintenance of an unprecedented \$4.2 trillion level of U.S. Treasuries and
4 mortgage-backed securities purchased and held outright by the Federal
5 Reserve Bank. Dr. Lesser focuses his comments solely on low interest
6 rates and downplays the impact of the massive amount of Treasury and
7 mortgage-backed securities held outright by the Federal Reserve Bank.

8 **Q16. WHAT ARE THE IMPORTANT TIME PERIODS OF THIS**
9 **PROCEEDING?**

10 A16. The FERC has set a refund effective date of April 29, 2016 for this
11 complaint. The refund period is therefore from approximately May 2016
12 through July 2017. In addition, Dr. Lesser has used a DCF analysis period
13 of July 1, 2016 to December 31, 2016. NETOs' Witness McKenzie has
14 utilized a DCF analysis period of September 2016 to February 2017. Both
15 witnesses are using DCF study periods during the refund period.
16 Prospective rates from this proceeding will be effective upon the
17 Commission's decision in this proceeding, likely in mid-2018. Anomalous
18 market conditions have existed during the refund period to date and
19 continue. I cannot predict with certainty the market conditions that will
20 exist in mid-2018 and beyond, but there have been no indications at this
21 time that the Federal Reserve plans to sell the \$4.2 trillion of securities it
22 currently holds, even if there may be several more increases in the federal
23 funds rate by mid-2018.

24 **Q17. PLEASE DESCRIBE CURRENT U.S. INTEREST RATE LEVELS.**

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1 A17. Despite recent increases from record lows, interest rates are still
2 extraordinarily low. I draw this conclusion after comparing short-term and
3 long-term U.S. and global interest rates.

4 The federal funds rate is an important benchmark in financial
5 markets and is the primary policy tool of the Federal Reserve Bank. The
6 federal funds effective rate is the interest rate at which depository
7 institutions lend to each other overnight. The Federal Open Market
8 Committee (FOMC) establishes the federal funds target rate and then the
9 Federal Reserve Bank uses open market operations to influence the U.S.
10 money supply to ensure that the federal funds effective rate follows the
11 federal funds target rate. A time series of the federal funds effective rate
12 since 1954 is shown on Exhibit No. NET-02302. The current federal funds
13 effective rate is low at 0.66% as of February 28, 2017. By comparison, the
14 federal funds effective rate ranged from 4.24% to 5.26% during the pre-
15 crisis base period of 2006 to 2007; 0.14% to 0.16% during the Opinion No.
16 531 study period of October 1, 2012 to March 31, 2013; 0.07% to 0.12%
17 during the Opinion No. 551 study period of November 12, 2013 to
18 February 11, 2015; and 0.37% to 0.66% during the refund period to date of
19 this proceeding, April 29, 2016 to February 28, 2017. When observers say
20 that the FOMC is expected to raise interest rates three times during 2017,
21 three times during 2018, and three times during 2019, it is the target federal
22 funds rate to which they refer. On March 15, 2017, the FOMC decided to
23 raise the target federal funds rate 25 basis points to a range of 0.75% to
24 1.00%. The Committee disclosed in a press release that “the federal funds
25 rate is likely to remain, for some time, significantly below levels that are

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1 expected to prevail in the longer run.” Changes in the federal funds rate
2 have a direct influence on short-term market interest rates and a limited
3 influence on long-term market interest rates.

4 A frequently cited short-term market interest rate is the yield on one-
5 month U.S. Treasury securities. One-month U.S. Treasury yields are low at
6 0.40% as of February 28, 2017. A time series of one-month U.S. Treasury
7 yields since July 2001 is shown on Exhibit No. NET-02303, along with ten-
8 year U.S. Treasury yields. By comparison, one-month U.S. Treasury yields
9 ranged from 2.42% to 5.27% during the pre-crisis base period of 2006 to
10 2007; 0.00% to 0.17% during the Opinion No. 531 study period of October
11 1, 2012 to March 31, 2013; 0.00% to 0.13% during the Opinion No. 551
12 study period of November 12, 2013 to February 11, 2015; and 0.09% to
13 0.53% during the portion of the refund period to date of this proceeding
14 from April 29, 2016 through the end of February 2017.

15 A frequently cited long-term market interest rate is the yield on ten-
16 year U.S. Treasury securities. Ten-year U.S. Treasury yields are low at
17 2.36% as of February 28, 2017. A time series of ten-year U.S. Treasury
18 yields since January 1962 is shown on Exhibit No. NET-02304, as well as
19 since July 2001 on Exhibit No. NET-02303. By comparison, ten-year U.S.
20 Treasury yields ranged from 3.83% to 5.26% during the pre-crisis base
21 period of 2006 to 2007; 1.58% to 2.07% during the Opinion No. 531 study
22 period of October 1, 2012 to March 31, 2013; 1.68% to 3.04% during the
23 Opinion No. 551 study period of November 12, 2013 to February 11, 2015;
24 and 1.37% to 2.60% during the refund period to date of this proceeding,
25 April 29, 2016 to February 28, 2017.

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1 Moody's Baa-rated utility bond yields are low at 4.58% as of
2 February 2017. A time series of Moody's Baa-rated utility bond yields
3 since January 1968 is shown on Exhibit No. NET-02305. By comparison,
4 Moody's Baa-rated utility bond yields ranged from 6.04% to 6.61% during
5 the pre-crisis base period of 2006 to 2007; 4.51% to 4.74% during the
6 Opinion No. 531 study period of October 1, 2012 to March 31, 2013;
7 4.39% to 5.25% during the Opinion No. 551 study period of November 12,
8 2013 to February 11, 2015; and 4.16% to 5.28% during the refund period to
9 date of this proceeding, April 29, 2016 to February 28, 2017.

10 **Q18. PLEASE DESCRIBE GLOBAL INTEREST RATE LEVELS AND**
11 **HOW THEY RELATE TO U.S. INTEREST RATES.**

12 A18. Many countries have short-term and long-term interest rates significantly
13 lower than the U.S., due to intervention by global central banks that have,
14 in many ways, paralleled that of the U.S. Federal Reserve Bank. A time
15 series of global interest rates for five key countries is shown on Exhibit No.
16 NET-02306. As of February 2017, the short-term yield on government
17 securities was 0.88% in Canada, negative 0.33% in Germany, 0.34% in the
18 United Kingdom, 0.06% in Japan, and negative 0.73% in Switzerland. The
19 yields all fell precipitously in 2008 except for Japan, which experienced
20 anomalous capital market conditions earlier than the other countries, as
21 shown on page 1 of Exhibit No. NET-02306. Short-term central bank
22 interest rates hovering near zero include 0.00% at the Bank of Japan and
23 0.00% at the European Central Bank. Negative interest rates are
24 unsustainable and indicate that investors are paying for the privilege of
25 holding government debt. In the U.S., as can be seen from Exhibit NET-
26 02303, the one-month U.S. Treasury yield scraped down to 0.00% for

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1 several days at several times between December 2008 and October 2015,
2 but never fell negative.

3 Likewise, as of February 2017, the ten-year yield on government
4 securities was 1.71% in Canada, 0.26% in Germany, 1.31% in the United
5 Kingdom, 0.08% in Japan, and negative 0.21% in Switzerland, as shown on
6 page 2 of Exhibit No. NET-02306. The negative ten-year government yield
7 in Switzerland is especially notable. Negative interest rates on ten-year
8 government securities are even more anomalous and unsustainable than
9 negative interest rates on short-term government securities.

10 Global interest rates even lower than U.S. interest rates motivate
11 foreign investors to invest in U.S. securities. The capital marketplace is
12 globally competitive. These extraordinarily low global interest rates help
13 explain why global investors are attracted to U.S. debt and dividend-paying
14 equities and further show the persistence of anomalous capital market
15 conditions in the U.S. and globally.

16 **Q19. PLEASE SUMMARIZE THE INTEREST RATES THAT YOU HAVE**
17 **DISCUSSED.**

18 A19. The following table summarizes some relevant interest rate comparisons.
19 For comparison purposes, I began with a pre-crisis base period of 2006 to
20 2007. I also show the interest rate that prevailed during the Opinion No.
21 531 and Opinion No. 551 study periods. Finally, I show interest rates that
22 have existed during the relevant refund period of this proceeding to date
23 along with the most recent rates available at the time I prepared my
24 testimony.

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INTEREST RATE TABLE
(% per annum)

<u>Interest Rates</u>	<u>Pre-Crisis 2006 to 2007</u>	<u>531 Study Period 10-1-12 to 3-31-13</u>	<u>551 Study Period 11-12-13 to 2-11-15</u>	<u>Refund Period To Date 4-29-16 to 2-28-17</u>	<u>Recent 2/28/2017</u>
<u>U.S. Interest Rates</u>					
Federal Funds Effective Rate	4.24 to 5.26	0.14 to 0.16	0.07 to 0.12	0.37 to 0.66	0.66
One-Month US Treasury Yield	2.42 to 5.27	0.00 to 0.17	0.00 to 0.13	0.09 to 0.53	0.40
Ten-Year US Treasury Yield	3.83 to 5.26	1.58 to 2.07	1.68 to 3.04	1.37 to 2.60	2.36
Moody's Baa Utility Yield	6.04 to 6.61	4.51 to 4.74	4.39 to 5.25	4.16 to 5.28	4.58
<u>Global Short-Term Governments Yields</u>					
Germany	2.51 to 4.85	0.19 to 0.22	0.05 to 0.33	-0.33 to -0.25	-0.33
Japan	0.10 to 0.87	0.25 to 0.33	0.17 to 0.22	0.06 to 0.06	0.06
Canada	3.62 to 5.12	1.16 to 1.16	0.89 to 1.19	0.81 to 0.88	0.88
United Kingdom	4.52 to 6.58	0.49 to 0.54	0.50 to 0.55	0.34 to 0.57	0.34
Switzerland	1.02 to 2.90	0.01 to 0.03	-0.85 to 0.02	-0.78 to -0.73	-0.73
<u>Global Ten-Year Governments Yields</u>					
Germany	3.32 to 4.56	1.30 to 1.54	0.30 to 1.80	-0.15 to 0.26	0.26
Japan	1.50 to 1.96	0.49 to 0.78	0.28 to 0.69	-0.24 to 0.08	0.08
Canada	3.98 to 4.61	1.74 to 1.97	1.38 to 2.67	1.04 to 1.73	1.71
United Kingdom	4.08 to 5.43	1.77 to 2.18	1.59 to 2.95	0.74 to 1.57	1.31
Switzerland	2.15 to 3.19	0.53 to 0.79	-0.07 to 1.25	-0.54 to -0.07	-0.21

1 This table demonstrates that U.S. and global interest rates to date
2 continue to be extremely, unusually, and anomalously low. Refund period
3 interest rates are similar to or lower than interest rates during the Opinion
4 No. 531 and Opinion No. 551 study periods and are significantly lower than
5 interest rates observed during the pre-crisis base period of 2006 to 2007.

6 **Q20. PLEASE DESCRIBE THE OTHER SIGNIFICANT COMPONENT**
7 **OF THE U.S. FEDERAL RESERVE BANK'S EXTREME**
8 **MONETARY POLICY: THE SECURITIES PURCHASED AND**

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1 **HELD OFF OF THE OPEN MARKET BY THE FEDERAL**
2 **RESERVE BANK.**

3 A20. The balance of Federal Reserve Bank securities held outright is still
4 massive at \$4.2 trillion. A time series of securities held outright since 2002
5 is shown on Exhibit No. NET-02307. The massive Federal Reserve Bank
6 purchases of securities during 2009 to 2014 is often referred to as
7 quantitative easing (QE) and consists primarily of longer-term U.S.
8 Treasury and mortgage-backed securities. The massive balance of
9 securities held outright is a huge overhang on the Federal Reserve Bank
10 balance sheet. The \$4.2 trillion balance has been maintained since 2014 by
11 the reinvestment of interest and principal payments. The March 15, 2017
12 Federal Reserve press release stated: “The Committee is maintaining its
13 existing policy of reinvesting principal payments from its holdings of
14 agency debt and agency mortgage-backed securities in agency and
15 mortgage-backed securities and of rolling over maturing Treasury securities
16 at auction, and it anticipates doing so until normalization of the level of the
17 federal funds rate is well under way. This policy, by keeping the
18 Committee’s holdings of longer-term securities at sizable levels, should
19 help maintain accommodative financial conditions.”

20 This policy of maintaining massive amounts of securities held
21 outright on the Federal Reserve Bank balance sheet indicates that
22 anomalous capital market conditions will persist even after several
23 increases in the federal funds target rate.

24 **Q21. DO YOU HAVE ANY INDICATION OF HOW LOW U.S. INTEREST**
25 **RATES MIGHT HAVE GONE IF THE FEDERAL RESERVE BANK**

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**DID NOT PURCHASE THE MASSIVE AMOUNTS OF SECURITIES
HELD OUTRIGHT?**

A21. On February 9, 2017, Dr. Charles Evans, President of the Federal Reserve Bank of Chicago and member of the Federal Open Market Committee made a presentation to the CFA Society of Chicago entitled “Risk Management in a Low Interest Rate Environment.” During the presentation, Dr. Evans indicated that the U.S. Federal Reserve Bank desired to avoid negative interest rates in the U.S. like those that were being experienced due to central bank monetary intervention in other global markets. The Federal Reserve Bank’s massive purchases of long-term securities were an attempt to continue flooding the economy with liquidity instead of permitting interest rates to go negative, while more directly influencing long-term interest rates. Dr. Evans also indicated that, in his view, the level of negative interest rates needed to provide the Federal Reserve Bank’s desired level of economic stimulus was negative 4.0%. In other words, the Federal Reserve’s desired monetary stimulus was equivalent to negative interest rates at the level of negative 4.0%, but the Federal Reserve Bank found negative interest rates unpalatable to the U.S. economy. Instead, the Federal Reserve Bank achieved its desired level of economic stimulus by maintaining slightly positive interest rates and pursuing the unprecedented massive security purchases that grew the Federal Reserve Bank’s securities balance from \$0.5 trillion in 2008 to the \$4.2 trillion it is today. This is evidence of the extraordinary lengths gone to by the Federal Reserve and why anomalous capital market conditions resulted. This is also evidence that anomalous market conditions are unlikely to disappear immediately just because short-term interest rates

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1 such as the federal funds rate and the one-month Treasury yield, are
2 observed to be increasing. It will also be necessary to see evidence that the
3 massive balance sheet overhang is eliminated. On February 9, 2017, Dr.
4 Evans also mentioned that equilibrium interest rates are likely to be lower
5 than they have been in the past. He cites a recent study that indicates that
6 the effective real federal funds rate today is 325 basis points lower than the
7 average in the 1970s, 1980s, and 1990s. He continues by observing that
8 current forecasts of the equilibrium federal funds rate, including those of
9 the FOMC, would imply a 250 basis point increase to get to equilibrium.

10 **Q22. WHEN THE TIME COMES, HOW WILL THE FEDERAL**
11 **RESERVE BANK LIKELY BEGIN REDUCING THE \$4.2**
12 **TRILLION BALANCE OF SECURITIES HELD OUTRIGHT?**

13 A22. The Federal Reserve Bank currently maintains the high balance of
14 securities holdings by reinvesting interest payments and maturities into the
15 purchase of new long-term securities. The Fed will likely begin to
16 gradually reduce the \$4.2 trillion balance by stopping the reinvestment of
17 interest payments and maturities into new securities. When this
18 reinvestment ceases, the massive balance sheet overhang will begin a
19 gradual reduction.

20 **Q23. HAS THE FEDERAL RESERVE BANK YET BEGUN TO CEASE**
21 **THE REINVESTMENT OF INTEREST PAYMENTS AND**
22 **MATURITIES?**

23 A23. No.

24 **Q24. WHEN MIGHT THE REINVESTMENT CEASE?**

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1 A24. On February 9, 2017, Dr. Evans indicated that, in his opinion, the Federal
2 Reserve Bank may hike the federal funds rate three times per year for the
3 next three years, and that the Federal Reserve Bank would be unlikely to
4 entertain the notion of ceasing reinvestment until at least two or three more
5 interest rate hikes occur.

6 **Q25. WHAT OTHER EVIDENCE OF ANOMALOUS CAPITAL MARKET**
7 **CONDITIONS EXIST?**

8 A25. Dr. Lesser's Exhibit No. EMC-7 provides ample evidence that anomalous
9 capital market conditions persist. Exhibit No. EMC-7 is a Client Alert from
10 Duff & Phelps entitled "Duff & Phelps Increases U.S. Equity Risk
11 Premium Recommendation to 5.5%, Effective January 31, 2016." I am not
12 making a market risk premium recommendation in this proceeding and, as
13 such, I do not endorse the Duff & Phelps market risk premium
14 recommendation, but its client alert is instructive in acknowledging the
15 prevalence of anomalous capital market conditions. This Duff & Phelps
16 client alert recommends that a normalized risk-free rate of 4.0% be used in
17 a CAPM analysis rather than a spot risk-free rate of 2.4%. The difference
18 of 160 basis points is Duff & Phelps' estimated impact of anomalous capital
19 market conditions. Duff & Phelps provides an extensive explanation of
20 anomalous capital market conditions on page 9 through 33 of Exhibit No.
21 EMC-7. The key point is summarized on page 31 of the Client Alert:

22 As stated earlier, in most circumstances we would prefer to
23 use the "spot" yield on U.S. government bonds available in
24 the market as a proxy for the U.S. risk-free rate. However,
25 during times of flight to quality and/or high levels of central
26 bank intervention, those lower observed yields imply a lower
27 cost of capital (all other factors held the same) – just the
28 opposite of what one would expect in times of relative

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1 economic distress – so a “normalization” adjustment may be
2 considered appropriate. By “normalization” we mean
3 estimating a rate that more likely reflects the sustainable
4 average return of long-term risk-free rates. *If spot yield-to-*
5 *maturity were used at these times, without any other*
6 *adjustments, one would arrive at an overall discount rate that*
7 *is likely inappropriately low vis-à-vis the risks currently*
8 *facing investors.*

9 Duff & Phelps concludes that mechanistic ROE calculations
10 determined in the manner that Dr. Lesser advocates are likely to be
11 inappropriately low vis-à-vis the risk currently facing investors. Duff &
12 Phelps’ recommended normalization is remarkably conceptually similar to
13 this Commission’s findings of anomalous capital market conditions and the
14 decision to deviate from the DCF midpoint in Opinion No. 531 and
15 Opinion No. 551.

16 **Q26. PLEASE SUMMARIZE YOUR VIEW OF ANOMALOUS CAPITAL**
17 **MARKET CONDITIONS.**

18 A26. The Commission found that the conditions I describe above produced
19 anomalous capital market conditions in Opinion No. 531 and Opinion No.
20 551. For all the reasons discussed above, the Commission’s conclusions in
21 those Opinions have applied since the beginning of the refund period in this
22 Complaint IV and still apply today.

23 **Q27. WHEN DO YOU ANTICIPATE THAT ANOMALOUS CAPITAL**
24 **MARKET CONDITIONS CAUSED BY EXTREME MONETARY**
25 **POLICY WILL BE OVER?**

26 A27. I expect that there will come a day when the anomalous market conditions
27 caused by extreme monetary policy will unwind and the extreme monetary
28 policy will no longer have the distorting impact on capital market pricing

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1 that it does today. In my opinion, the evidence of this happening will
2 consist of all of the following: (1) short-term interest rates rise significantly
3 to a more normal level; (2) long-term interest rates rise significantly to a
4 more normal level; (3) the Federal Reserve Bank discontinues rolling over
5 interest payments on and maturities of its securities held outright; and (4)
6 the Federal Reserve Bank sells on the open market most, if not all, of its
7 \$4.2 trillion securities held outright. None of these four events has
8 happened yet, and there is no evidence that they will happen in the near
9 future.

10 **Q28. WHEN THE ANOMALOUS CONDITIONS CURRENTLY PRESENT**
11 **IN THE CAPITAL MARKETS ARE RESOLVED, DO YOU**
12 **BELIEVE IT WOULD BE APPROPRIATE FOR THE**
13 **COMMISSION TO RELY SOLELY ON THE RESULTS OF A**
14 **SINGLE DCF METHOD TO EVALUATE A FAIR ROE?**

15 A28. No. As I noted earlier, no single methodological approach can be
16 considered a wholly reliable indicator of investors' required return. In my
17 experience, it is common practice for regulators to consider the results of
18 alternative methods, along with their assessment of the merits of each
19 approach, in arriving at a just and reasonable ROE that meets the
20 requirements of regulatory standards.

21 **IV. THE NETOS' EXPANSION OF THE NEW ENGLAND**
22 **TRANSMISSION GRID IS CONSISTENT WITH PUBLIC POLICY**
23 **AND SUBJECT TO THOROUGH REVIEW**

24 **Q29. DO YOU AGREE WITH DR. PETERS' CONCERN THAT THE NEW**
25 **ENGLAND REGION IS IN DANGER OF BUILDING TOO MUCH**
26 **TRANSMISSION?**

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1 A29. No, not at all. Transmission projects in New England are a result of a
2 rigorous ISO New England (ISO-NE) planning process designed to ensure
3 that system upgrades necessary to meet appropriate reliability standards are
4 constructed. These reliability standards include those of North American
5 Electric Reliability Corporation (NERC) and the Northeast Power
6 Coordinating Council (NPCC). The ISO-NE-led planning process may be
7 thought of in two distinct phases with the first being the “needs” phase and
8 the second being the “solutions” phase. The initial needs assessment
9 evaluates the transmission system’s performance against mandatory
10 national and regional standards (NERC and NPCC) and if performance
11 deficiencies are found, then a solution study is initiated. The solution study
12 includes development and evaluation of a comprehensive list of mitigating
13 alternatives, of which one ultimately is recommended as “preferred” to
14 ISO-NE stakeholders.

15 It is important to note that throughout the entire process, New
16 England stakeholders are given multiple opportunities to provide input and
17 feedback both to the transmission owner and to ISO-NE staff. In addition,
18 at any time throughout the needs or solutions phase, ISO-NE can, and has,
19 declared the need to re-assess the study needs or solutions as a result of
20 material forecasted system changes; for example, generation additions and
21 retirements, and load forecast updates. This continual re-assessment of
22 needs and solutions throughout the study process ensures that only justified
23 transmission upgrades are constructed. Since May 2015, ISO-NE’s Open
24 Access Transmission Tariff provides for competitive solicitations to

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1 determine solutions. Nevertheless, the ISO-NE process remains rigorous
2 and focused on adherence to mandatory national and regional standards.

3 While serving as Chairman of the Michigan Public Service
4 Commission, I was familiar with the MISO and PJM transmission planning
5 processes, as different Michigan regions participate in MISO or PJM. I
6 observe that the ISO-NE transmission planning process shares many
7 positive attributes of the rigorous MISO and PJM planning processes.

8 Most industry observers, including regulators and investors, are
9 rightly concerned about the implications of too little energy infrastructure,
10 including electric transmission in New England. As New England pursues
11 additional renewable and low carbon energy sources, sufficient new
12 transmission infrastructure must be built to allow access to those new
13 resources.

14 **Q30. HAVE CONGRESS AND THIS COMMISSION ENCOURAGED THE**
15 **DEVELOPMENT OF TRANSMISSION?**

16 A30. Yes, they have. Congress passed the Energy Policy Act of 2005 that set
17 forth several statutory requirements intended to support transmission
18 investment. The Commission, through a 2012 Policy Statement, reaffirmed
19 its pricing reform encouraging transmission investment through incentive
20 rate treatments to assist in mitigating the risks associated with developing,
21 constructing, operating, and maintaining transmission infrastructure. The
22 Commission also enabled regional and interregional coordination processes
23 and supporting cost recovery processes through Order 1000.

24 Dr. Peters offers no compelling reason for Congress and the
25 Commission to abandon their support for enhancement and expansion of

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1 transmission infrastructure. Transmission remains the smallest percentage
2 of electricity costs when compared to distribution and generation costs.
3 Only 11% of the average U.S. price of electricity results from the costs of
4 transmission services.⁵ Recently, the American Society of Civil Engineers
5 (ASCE) released its 2017 Infrastructure Report Card. Using the simple “A
6 to F” school report card format, the ACSE assigns a “D+” to energy
7 infrastructure. Specific to transmission infrastructure, ASCE states:

8 Much of the U.S. energy system predates the turn of the 20th
9 century. Most electric transmission and distribution lines
10 were constructed in the 1950s and 1960s with a 50-year life
11 expectancy, and the more than 640,000 miles of high-voltage
12 transmission lines in the lower 48 states’ power grids are at
13 full capacity. Energy infrastructure is undergoing increased
14 investment to ensure long-term capacity and sustainability; in
15 2015, 40% of additional power generation came from natural
16 gas and renewable systems. Without greater attention to
17 aging equipment, capacity bottlenecks, and increased
18 demand, as well as increasing storm and climate impacts,
19 Americans will likely experience longer and more frequent
20 power interruptions.⁶

21 Specific to New England, the New England States Committee on
22 Electricity (NESCOE) recently studied the impact of the clean energy
23 policy goals of the New England states on transmission needs. The
24 NESCOE-sponsored report concluded that, if new transmission build is
25 limited only to reliability-related upgrades that are currently in progress,
26 “the region is forecast to be under-supplied with Renewable Energy

⁵ *Annual Energy Outlook 2017*, U.S. Energy Information Administration, January 2017, Table 8.

⁶ *2017 Infrastructure Report Card, Energy Overview*, American Society of Civil Engineers, <http://www.infrastructurereportcard.org/wp-content/uploads/2017/01/Energy-Final.pdf>.

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1 Certificates (REC) relative to Renewable Portfolio Standard (RPS) targets”
2 by 10.5% in 2025 and by 17.0% in 2030.⁷ It is clear that more transmission
3 is needed to achieve the New England states’ RPS and other clean energy
4 requirements.

5 **V. SHORTCOMINGS OF DR. LESSER’S AND DR. PETERS’**
6 **ANALYSIS**

7 **A. Dr. Lesser and Dr. Peters Misjudge the NETOs’ Risks,**
8 **Particularly in Comparison to Distribution Investment**

9 **Q31. DO YOU AGREE WITH DR. PETERS’ CONCLUSION THAT**
10 **TRANSMISSION INVESTMENT IS LESS RISKY THAN**
11 **DISTRIBUTION INVESTMENT?**

12 A31. No, Dr. Peter’s views are contrary to investors’ views and directly challenge
13 the findings of the Commission in Opinion No. 531 and Opinion No. 551.

14 On page 16 of Exhibit EMC-12, Dr. Peters lists 13 sources of risk
15 and makes the blanket conclusion that the magnitude of each risk does not
16 differ between transmission and distribution. However, at least four of his
17 enumerated risks differ markedly between transmission and distribution,
18 including permitting risk, cost overrun risk, schedule delays, and local
19 opposition to construction. Transmission projects are larger and have a
20 longer lead time to construction than distribution projects. Permitting
21 requirements for transmission projects are more significant and provide
22 more opportunities for local opposition to construction than distribution
23 projects. The larger size and longer lead time of transmission projects
24 contribute to higher risk of cost overruns and schedule delays.

⁷ *Renewable and Clean Energy Scenarios and Mechanisms 2.0 Study Base Case Results*,
New England States Committee on Electricity, November 17, 2016.

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1 **Q32. DR. PETERS OPINES THAT THERE IS A “GUARANTEE” THAT**
2 **THE NETOS WILL EARN THEIR AUTHORIZED ROE. DO YOU**
3 **AGREE?**

4 A32. No, I do not. There is no guarantee that the NETOs will earn their
5 authorized ROE on transmission investment. Under the Commission’s
6 abandoned plant precedent, there is a risk that the NETOs may spend
7 considerable sums developing projects that are never placed in service, yet
8 not fully recover their costs. This Commission’s Order 1000 contemplates
9 that the NETOs could propose transmission projects in competitive
10 solicitations and if not selected, the NETOs would not recover the
11 associated development costs.

12 **Q33. DR. LESSER APPEARS TO CONCLUDE THAT DISTRIBUTION**
13 **INVESTMENTS ARE MORE RISKY THAN TRANSMISSION**
14 **INVESTMENTS BECAUSE THE INTRODUCTION OF**
15 **DISTRIBUTED ENERGY RESOURCES MAKES THE**
16 **DISTRIBUTION GRID OBSOLETE AND THUS MORE RISKY. DO**
17 **YOU AGREE?**

18 A33. No, I do not. The distribution grid is ripe with investment opportunities
19 due to the growth of distributed energy resources. As a result, the
20 distribution grid needs to become even more robust and vibrant with
21 investments in smart meters and sensors to enable the growth of distributed
22 energy resources. Distributed energy resource owners are reliant on the
23 distribution grid for the two-way flow of electricity. Also, the distribution
24 grid relies upon the transmission grid for its own reliability and to ensure a
25 reliable supply of energy. The growth of distributed energy resources does

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1 not place the distribution grid or the transmission grid in danger of
2 becoming irrelevant.

3 **Q34. WHAT HAS THE COMMISSION PREVIOUSLY CONCLUDED**
4 **ABOUT THE RELATIVE RISK OF TRANSMISSION AND**
5 **DISTRIBUTION INVESTMENT?**

6 A34. The Commission, in Opinion No. 531, properly recognized that
7 transmission investment is riskier than distribution investment. In fact, the
8 Commission found in paragraph 149 of Opinion No. 531 that state-
9 regulated electric distribution has lower business risks than electric
10 transmission investment.

11 Some of the risks that the Commission noted for electric transmission are:

12 For example, investors providing capital for electric
13 transmission infrastructure face risk including the following:
14 long delays in transmission siting, greater project complexity,
15 environmental impact proceedings, requiring approval from
16 multiple jurisdictions overseeing permits and rights of way,
17 liquidity risk from financing projects that are large relative to
18 the size of the balance sheet, and shorter investment history.
19 We find that these factors increase the NETOs' risk relative
20 to the state-regulated distribution companies.

21 Several of these transmission risks identified by the Commission
22 have clear parallels to electric generation risks but not distribution risks,
23 including the potential for long delays, greater project complexity, the
24 burdensome impact of environmental regulations, multiple jurisdictions
25 overseeing siting, environmental compliance decisions, and financing
26 projects that are large relative to the size of the corporate balance sheet.

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1 Nothing has changed to reverse the Commission's previous
2 conclusion that transmission investment is riskier than distribution
3 investment.

4 **Q35. HOW DO INVESTORS PERCEIVE THE THREAT OF PANCAKED**
5 **ROE COMPLAINTS TO IMPACT THE RELATIVE RISK OF**
6 **TRANSMISSION?**

7 A35. Investors perceive that the prospect of never-ending pancaked ROE
8 complaints heightens both the risk of transmission investment and
9 transmission's relative risk to distribution. For example, Value Line
10 provides an example of the filing of an ROE complaint that caused an
11 immediate 6% stock price drop for a transmission provider.⁸ NETOs'
12 Witness McKenzie cites a Wolfe Research report that observes that
13 "pancaking of ROE challenges against the same transmission owners"
14 represented one of the "real risks" to investors in transmission.⁹ Moreover,
15 a recent research note from UBS, in discussing the Commission's Final
16 Order in the first MISO ROE complaint (EL14-12-002), stated "it's notable
17 a third subsequent pancaked case has not been filed, a positive in our
18 view."¹⁰ It is clear that pancaked ROE complaints increase the risk
19 associated with transmission investment.

⁸ The Value Line Investment Survey, ITC Holdings Corp., December 20, 2013.

⁹ Wolfe Research, *Don't you FERCEdabout ROE, Don't Don't Don't Don't!*, Utilities & Power (Apr. 6, 2015).

¹⁰ *FERC Affirms the MISO Win*, UBS Global Research, September 29, 2016, at 1.

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1 **B. Capital Structure Differences Should Not Be Used to Reduce the**
2 **Understated Base ROE Recommendation of Dr. Lesser**

3 **Q36. DO YOU AGREE WITH DR. PETERS' PROPOSAL TO REDUCE**
4 **THE BASE ROE BY 39 BASIS POINTS TO ACCOUNT FOR**
5 **CAPITAL STRUCTURE DIFFERENCES BETWEEN THE NETOS**
6 **AND THE PROXY GROUP?**

7 A36. No, I do not. To begin with, Dr. Peters' vague academic references about
8 optimal capital structure do not offer any empirical evidence to pinpoint
9 one. In the real world of practical corporate finance, academic theoretical
10 references are interesting and may provide a helpful guide but do not
11 provide a useful tool to fine tune a company's capital structure. Dr. Peters
12 ignores real-world practical corporate financial realities.

13 A utility must be permitted latitude in managing capital structure
14 ratios. Since there is no practical methodology to pinpoint theoretically
15 optimal capital structure ratios, targeted ratios can only be broadly
16 conceptualized. Appropriate ratios may shift over time as capital market
17 conditions or business risk characteristics change. Additionally, the timing
18 of upcoming issuances and maturities may influence the capital structure
19 ratios because both the size and frequency of issuances are affected by the
20 relative cost-effectiveness of various issuance increments. Treasury
21 professionals need an adequate degree of flexibility to perform their duties.
22 Given these practical considerations, capital structure ratios cannot be
23 deemed to be inappropriate unless the ratios significantly diverge from
24 sound industry practice and cause a lack of financial flexibility that may
25 lead to higher overall costs. As Dr. Peters shows on his Figure 4 entitled
26 "Least Cost Capital Structure" on page 22 of Exhibit No. EMC-12, the

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1 Weighted Average Cost of Capital curve is shaped like a very shallow dish
2 such that large variances in capital structure ratios lead to minimal change
3 in overall costs.

4 Moreover, the Commission's proxy group selection criteria are
5 meant to determine companies of comparable risk. The Commission has
6 chosen to exclude capital structure as an explicit factor when determining
7 the comparable risk proxy group. The Commission does include credit
8 ratings in its criteria, and the credit rating agencies evaluate capital
9 structure among other risk factors when determining credit ratings. As a
10 result, the impact of capital structure is already included in the
11 Commission's proxy group selection criteria. Dr. Peter's proposed
12 adjustment would be redundant and is clearly inappropriate.

13 In Opinion No. 551, the Commission affirmed that it has never
14 encouraged utilities to feature more debt in their capital structure and found
15 that it would be inappropriate to encourage additional debt leveraging of
16 utilities, many of which are undertaking large investments or do not have
17 high credit ratings. The Commission points out in paragraph 286:

18 [Complainants] seek a risk adjustment based upon a single
19 factor, an alleged equity-rich capital structure, without
20 consideration of any other risk factor. This is contrary to
21 Commission policy.

22 Further, the Commission realized the redundant nature of this capital
23 structure adjustment in paragraph 288:

24 In any event, Complainants' position fails to take into account
25 the fact that our criteria for selecting members of the proxy
26 group are intended to produce a proxy group make up of
27 companies of similar risk. Those criteria include screens to
28 ensure that the proxy group contains only utilities with similar
29 credit ratings to the utility at issue. . . . Consequently,

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1 additional reductions to the ROEs that are proposed by
2 Complainants essentially reduce the ROE twice for featuring
3 equity-rich capital structures.

4 Finally, the Commission concluded in paragraph 289:

5 Furthermore, as a policy matter, the Commission does not
6 directly incentivize utilities to adjust their preferred capital
7 structures. The Commission has not previously directly
8 encouraged utilities to feature more debt in their capital
9 structure. We find that it would be inappropriate to
10 encourage additional debt leveraging of utilities, many of
11 which are undertaking large investments or do not have high
12 credit ratings.

13 Dr. Peters' recommendation that the Commission depart from its
14 well-conceived policy stance on capital structure is ill-advised and should
15 be rejected.

16 **C. Dr. Lesser's and Dr. Peters' Base ROE Recommendations Are**
17 **Inadequate**

18 **Q37. WILL THE BASE ROES RECOMMENDED BY DRS. LESSER AND**
19 **PETERS PROVIDE INVESTORS WITH A RETURN**
20 **COMMENSURATE WITH THE ASSOCIATED RISK AND**
21 **ATTRACT NEW CAPITAL TO TRANSMISSION INVESTMENT?**

22 A37. No, the base ROEs of 8.59% recommended by Dr. Lesser and 8.20%
23 recommended by Dr. Peters are way too low to attract investors to provide
24 capital for electric transmission investments. Coming so soon after
25 Opinion No. 531 in which the Commission established a base ROE of
26 10.57% and potential decisions in the second and third complaints,
27 investors would react with surprise and alarm if the Commission
28 determined a base ROE in this proceeding consistent with either Dr.
29 Lesser's or Dr. Peters' recommendations. Measured against the Opinion

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1 No. 531 authorized base ROE of 10.57%, their proposed decreases are in
2 the range of 198 to 237 basis points and are even larger than the 175 basis
3 point differential between the then-current authorized base ROE and the
4 midpoint of the mechanically-applied DCF methodology that troubled the
5 Commission in paragraph 150 of Opinion No. 531. An ROE consistent
6 with either Dr. Lesser's or Dr. Peters' recommendations would discourage
7 investment in transmission projects, and would have a chilling effect on all
8 FERC-jurisdictional transmission providers, discouraging new capital
9 investments in transmission assets.

10 **Q38. HOW WOULD THE BASE ROES PROPOSED BY DRS. LESSER**
11 **AND PETERS IMPACT THE NETOS' ABILITY TO COMPETE FOR**
12 **CAPITAL IN THE GLOBAL INVESTMENT MARKETS?**

13 A38. U.S. electric transmission investments compete in the financial market with
14 other sectors and other geographies, including utilities and non-utility
15 businesses. The most directly comparable sector is the state-regulated
16 electric utility investments, and more specifically, the state-regulated
17 vertically integrated electric utilities. The recommended base ROEs of Dr.
18 Lesser and Dr. Peters are significantly below the lowest of the base ROE
19 determinations over the last two years for vertically integrated electric
20 utilities in state jurisdictions. Such a low base ROE determination would
21 put transmission infrastructure at a competitive disadvantage in the capital
22 market in comparison to investments in vertically integrated electric
23 utilities.

24 **Q39. DID YOU PERFORM AN ANALYSIS OF STATE JURISDICTIONAL**
25 **BASE ROE DETERMINATIONS FOR ELECTRIC UTILITIES?**

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1 A39. Yes, I performed an analysis using jurisdictional allowed ROEs published
2 by S&P Global Market Intelligence's Regulatory Research Associates
3 (RRA). RRA is a respected source that is relied on for accurate
4 jurisdictional authorized ROE information by both investors and expert
5 witnesses in utility regulatory matters. RRA characterizes vertically
6 integrated electric utilities as those that provide distribution, transmission,
7 and regulated generation services.

8 **Q40. TO YOUR KNOWLEDGE, DOES FERC RELY ON THE RESULTS**
9 **OF STATE ROE DETERMINATIONS AS A BASIS FOR ITS OWN**
10 **ROE DETERMINATIONS?**

11 A40. The Commission has repeatedly affirmed the use of the DCF methodology
12 as its primary model for determining the base ROE and the range or zone of
13 reasonable ROEs. In Opinion No. 531, the Commission stated that the
14 substantial difference between state ROE determinations and the midpoint
15 of the modeled DCF range calls into question the sole reliance on the DCF
16 midpoint without adjustment during a period of anomalous capital market
17 conditions. The Commission stated in paragraph 148:

18 Although we are not using state commission approved ROEs
19 to establish the NETOs' ROE in this proceeding, the
20 discrepancy between state ROEs and the 9.39 percent
21 midpoint serves as an indicator that an upward adjustment to
22 the midpoint here is necessary to satisfy *Hope* and *Bluefield*.

23 In other words, a significant difference between state-authorized
24 ROEs and the results of the mechanical application of the DCF model is in
25 itself further evidence that capital market conditions are anomalous.
26 Furthermore, as the Commission explained in Opinion No. 531, its ROE
27 determinations are guided by the Supreme Court's decisions in the *Hope*

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1 and *Bluefield* cases to allow returns on invested capital that are comparable
2 to returns available to investors in businesses of similar risk. Therefore,
3 evidence of state-authorized ROEs for companies in a related industry
4 group is an important source of information to which the Commission
5 should give weight when determining where the ROE should be placed
6 within a range or zone of reasonableness. Furthermore, investors are
7 clearly aware of the state-authorized ROEs, and transmission owners must
8 compete for capital in the marketplace generally, as well as among
9 divisions within a specific utility (some of which are global companies),
10 against other types of utility investments, as well as against the entire range
11 of investment opportunities in the capital markets. Thus, the information
12 provided by ROEs recently authorized by a wide sample of state utility
13 regulators is relevant to the Commission's decision in this proceeding.

14 **Q41. HOW DID YOU PREPARE YOUR ANALYSIS OF THE STATE-**
15 **AUTHORIZED ROES?**

16 A41. To perform this analysis, I began with all cases reported by RRA in all
17 jurisdictions. Because some cases are decided by the jurisdiction without
18 an ROE finding, I captured only those cases in which RRA identified an
19 ROE finding. Next, I reviewed orders to determine if any explicit
20 incentives or penalties were identified in the applicable order. If applicable,
21 I separated the authorized ROE into a base ROE and incentive adders or
22 penalties. I then focused only on state-authorized base ROEs during the
23 most recent 24 month period.

24 **Q42. PLEASE EXPLAIN HOW YOU TREAT THE INCENTIVE ADDERS**
25 **OR PENALTIES.**

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1 A42. In Exhibit No. NET-02308, I separate each authorized ROE reported by
2 RRA into two components: the base ROE and any explicit ROE incentives
3 or penalties. This separation allows me to focus on the base ROE for
4 comparison purposes to the Commission base ROE. It is important to
5 capture the base ROE information in the analysis and is conceptually
6 similar to separating out the base ROE from electric transmission ROEs
7 that contain incentive adders.

8 **Q43. PLEASE DESCRIBE THE ADJUSTMENTS TO THE STATE ROE**
9 **ANALYSIS ON PAGE 2 OF EXHIBIT NO. NET-02308.**

10 A43. In several instances during the 24 month period, the Virginia Corporation
11 Commission (VCC) issued multiple orders within a short period of time
12 containing similar ROE determinations that relate to individual projects, not
13 for the entire utility. These orders contain valuable information, but to
14 include each and every order separately would over-represent the VCC's
15 decisions. Therefore, as shown on page 2 of Exhibit No. NET-02308, I
16 compress the VCC ROE determinations of individual projects that
17 contained the same base ROE in close proximity into one observation. In
18 other words, I replaced five ROE determinations of 10.00% for Virginia
19 Electric and Power Company during the first half of 2015 with a single
20 observation of 10.00%. Likewise, I replaced five ROE determinations of
21 9.60% for Virginia Electric and Power during the first quarter of 2016 with
22 a single observation of 9.60%, as well as replacing two ROE
23 determinations of 9.60% during the second quarter of 2016 with a single
24 observation of 9.60%. Besides the Virginia orders, I identified one ROE
25 determination for Indianapolis Power & Light that required adjustment as

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1 shown on page 2 of Exhibit No. NET-02308. With these adjustments, the
2 number of observations is slightly reduced and the range is unchanged, but
3 the results are more representative.

4 **Q44. ARE YOUR ADJUSTMENTS TO THE STATE ROES CONSISTENT**
5 **WITH ADJUSTMENTS MADE BY THE NETOS IN DOCKET NOS.**
6 **EL13-33 AND EL14-86?**

7 A44. Yes, they are.

8 **Q45. WHY DO YOU CHOOSE A 24-MONTH PERIOD FOR YOUR**
9 **STATE-AUTHORIZED ROE ANALYSIS?**

10 A45. In each quarter of a year, there are typically only a limited number of
11 decisions in a small number of state jurisdictions for RRA to report. For
12 example, RRA reported electric utility ROE decisions per quarter between
13 2 and 12 during 2015 and between 7 and 18 during 2016. In most quarters,
14 only a few jurisdictions are represented. The sample group from quarter to
15 quarter over 24 months is comprised of a greater variety of companies in a
16 greater variety of jurisdictions. Some utilities are involved in frequent rate
17 cases, while other utilities have multiple-year rate orders or have other
18 means to avoid regular rate cases, and are rarely reported on the list of ROE
19 determinations. Thus, the reported ROE determinations from quarter to
20 quarter, or even year to year, do not represent a constant population of
21 states or companies. Extending the data to eight quarters makes the sample
22 more representative. In my opinion, using 24 months of data provides an
23 appropriate balance between choosing a representative sample and ensuring
24 the sample is meaningfully recent. A 24 month period also is consistent
25 with the Commission's finding in paragraph 148 of Opinion No. 531.

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1 **Q46. WHY DO YOU CONSIDER THE VERTICALLY INTEGRATED**
2 **ELECTRIC UTILITIES TO BE THE MOST APPROPRIATE**
3 **GROUP FOR THE COMPARISON OF STATE-AUTHORIZED**
4 **ROES?**

5 A46. RRA reports ROE decisions at several different levels of aggregation,
6 including separately for electric utilities and natural gas utilities, and further
7 splitting the electric utility cases into vertically integrated cases and
8 delivery only cases. Vertically integrated electric utilities are electric
9 utilities that own transmission, distribution, and regulated generation assets.
10 As I discuss in Section V.A. “Dr. Lesser and Dr. Peters Misjudge the
11 NETOs’ Risks, Particularly in Comparison to Distribution Investment,”
12 nothing has changed to alter the Commission’s previous conclusion that
13 transmission investment is more risky than distribution investment. The
14 natural gas and electric delivery-only utilities are generally regarded by
15 investors and state regulators as having lower business risk than vertically
16 integrated electric utilities. Thus, the vertically integrated electric utilities
17 group is the most representative sample group for this analysis because it is
18 the group most similar in risk to the NETOs. For that reason, these are the
19 utilities that I include in Exhibit No. NET-02308.

20 **Q47. IS THE STATE-AUTHORIZED ROE METHODOLOGY THAT YOU**
21 **APPLIED FOR VERTICALLY INTEGRATED UTILITIES THE**
22 **SAME AS THAT USED BY NETO WITNESS LAPSON IN DOCKET**
23 **NOS. EL11-66 AND EL13-33/EL14-86?**

24 A47. Yes, it is.

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1 **Q48. WHAT DO THE STATE ROE CASES IN EXHIBIT NO. NET-02308**
2 **SHOW ABOUT DR. LESSER’S AND DR. PETERS’ ROE**
3 **PROPOSALS IN THIS CASE?**

4 A48. The sample group of 44 state ROE cases is shown on page 1 of Exhibit No.
5 NET-02308. Dr. Lesser’s recommended ROE of 8.59% is a dramatic 64
6 basis points below the lowest of the 44 observations. Dr. Peters’
7 recommended ROE of 8.20% is an astounding 103 basis points below the
8 lowest of the 44 observations. If either of their proposals were adopted, it
9 would send a strong signal that the Commission does not want capital
10 invested in transmission.

11 **Q49. DO YOU HAVE ANY COMMENTS ABOUT DR. PETERS’**
12 **CONCLUSION ON STATE-AUTHORIZED ROES?**

13 A49. On page 37 of Exhibit No. EMC-12, Dr. Peters concludes that the state-
14 authorized ROE “most similar” to those approved by state commissions
15 generally is 9.00%. To begin with, his conclusion about state-authorized
16 ROEs is significantly flawed by his sole reliance on lower-risk electric
17 distribution ROEs. But his conclusion of 9.00% is especially puzzling
18 when considering how his own data fails to support his conclusion. Dr.
19 Peters’ Table 5 shows an average state-authorized ROE of 9.31% and his
20 Table 6 shows a state-authorized ROE range of 9.17% to 9.90%. I am
21 uncertain how Dr. Peters can conclude that the ROE “most similar” to this
22 data is 9.00%.

23 **Q50. PLEASE SUMMARIZE YOUR CONCLUSIONS REGARDING THE**
24 **VERTICALLY INTEGRATED UTILITIES’ STATE-AUTHORIZED**
25 **ROES.**

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1 A50. The state-authorized ROEs demonstrate that the ROE recommendations of
2 Dr. Lesser and Dr. Peters are much too low to attract investment to
3 transmission infrastructure. If limited to the ROE recommendations of Drs.
4 Lesser and Peters, the NETOs would be unable to achieve returns on
5 transmission investment that meet the *Hope* and *Bluefield* standards and
6 would not be able to raise capital for transmission investment.

7 **VI. SUMMARY AND CONCLUSION**

8 **Q51. PLEASE SUMMARIZE YOUR CONCLUSIONS.**

9 A51. I conclude that the anomalous capital market conditions that the
10 Commission previously found to exist still persist. I disagree with the
11 conclusion of Dr. Peters that there has been too much transmission
12 investment in New England. To the contrary, transmission investment in
13 New England occurs under the direction of ISO-NE and only after a
14 rigorous needs and solutions assessment. I also disagree with both Dr.
15 Lesser and Dr. Peters on the relative risk of transmission and distribution
16 investment and concur with the Commission's previous conclusion that
17 transmission investment is more risky than distribution investment.
18 Furthermore, Dr. Peters' proposal to reduce Dr. Lesser's already inadequate
19 recommended base ROE by 39 basis points for capital structure
20 considerations is inappropriate. Finally, I demonstrate that both Dr.
21 Lesser's and Dr. Peters' base ROE recommendations are grossly inadequate
22 for the NETOs to meet the *Hope* and *Bluefield* standards.

23 **Q52. DOES THIS CONCLUDE YOUR TESTIMONY?**

24 A52. Yes, it does.

UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Belmont Municipal Light Department,)
et al.)

v.)

Central Maine Power Company, *et al.*)

Docket No. EL16-64-002

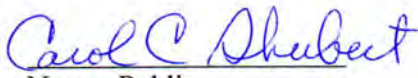
AFFIDAVIT OF JOHN D. QUACKENBUSH

John D. Quackenbush, being first duly sworn, deposes and says that he is the John D. Quackenbush referred to in the foregoing testimony, that he has read such testimony and is familiar with the contents thereof and that the answers therein are true and correct to the best of his knowledge, information, and belief.



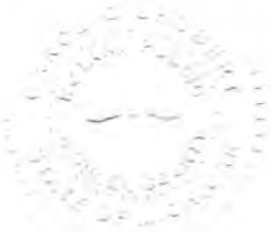
John D. Quackenbush

Subscribed and sworn to before me this 23rd day of March, 2017, by John D. Quackenbush, proved to me on the basis of satisfactory evidence to be the person who appeared before me.


Notary Public

CAROL C SHUBERT
Notary Public, State of Michigan
County of Berrien

Commission Expires on: My Commission Expires 12-23-2019
Acting in the County of Berrien



**PENNSYLVANIA
PUBLIC UTILITY COMMISSION
Harrisburg, PA 17105-3265**

Public Meeting Held December 5, 2012

Commissioners Present:

Robert F. Powelson, Chairman
John F. Coleman, Jr., Vice Chairman
Wayne E. Gardner, Dissenting in Part & Concurring in Result Only Statements
James H. Cawley, Dissenting in Part Statement
Pamela A. Witmer

Pennsylvania Public Utility Commission	R-2012-2290597
Office of Consumer Advocate	C-2012-2300266
Office of Small Business Advocate	C-2012-2301063
PP&L Industrial Customer Alliance	C-2012-2306728
William Andrews	C-2012-2300402
Tracey Andrews	C-2012-2328596
Eric Joseph Epstein	C-2012-2313283
Dave A. Kenney	C-2012-2299539
Roberta A. Kurrell	C-2012-2304870
Donald Leventry	C-2012-2304903
John G. Lucas	C-2012-2298593
Helen Schwika	C-2012-2299335

v.

PPL Electric Utilities Corporation

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OPINION AND ORDER

BY THE COMMISSION:

I. Matter Before the Commission

Before the Pennsylvania Public Utility Commission (Commission) for consideration and disposition is the Recommended Decision of Administrative Law Judge (ALJ) Susan D. Colwell, issued on October 19, 2012, relative to the above-captioned general rate increase proceeding. Also before the Commission are the Exceptions and Replies to Exceptions filed with respect thereto.

Exceptions to the Recommended Decision were filed on November 8, 2012, by the following Parties: PPL Electric Utilities Corporation (PPL or Company), the Commission's Bureau of Investigation and Enforcement (I&E), the Office of Consumer Advocate (OCA), the Office of Small Business Advocate (OSBA), the Commission on Economic Opportunity (CEO), Direct Energy Services LLC (Direct Energy), Dominion Retail, Inc. d/b/a Dominion Energy Solutions (DR), and the PP&L Industrial Customer Alliance (PPLICA). Replies to Exceptions were filed on November 19, 2012, by the following Parties: PPL, the OCA, the OSBA, DR, and PPLICA. I&E filed Replies to Exceptions on November 29, 2012.

II. History of the Proceeding¹

On March 30, 2012, PPL filed Supplement No. 118 to Tariff – Electric Pa. P.U.C. No. 201, to become effective June 1, 2012, containing proposed changes in rates, rules, and regulations calculated to produce approximately \$104.6 million in additional annual revenues. This proposed rate change represents an average increase in the Company’s distribution rates of approximately 13%, which equates to an average increase in total rates (distribution, transmission, and generation charges) of approximately 2.9%. The filing was suspended by Commission Order entered on May 24, 2012.

Formal Complaints against this proposed tariff were filed by the following: the OCA, on April 23, 2012; the OSBA, on April 25, 2012; PPLICA, on May 25, 2012; John G. Lucas, on April 9, 2012; Helen Schwika, on April 11, 2012; Dave A. Kenney, on April 16, 2012; William Andrews, on April 19, 2012; Tracey Andrews, on May 1, 2012; Roberta Kurrell, on May 3, 2012; Donald Leventry, on May 15, 2012;² and Eric Joseph Epstein, on July 5, 2012. Petitions to intervene were filed by the following: DR, on April 9, 2012; the CEO, on April 30, 2012; the International Brotherhood of Electrical Workers Local 1600, on May 1, 2012; the Sustainable Energy Fund (SEF), on May 3, 2012; Direct Energy, on May 24, 2012; and Granger Energy of Honey Brook LLC and Granger Energy of Morgantown LLC (collectively, Granger), on May 24, 2012. I&E filed a Notice of Appearance on April 10, 2012.

¹ For a full and complete history, as well as information regarding the testimony provided during the Public Input hearings, please refer to the Recommended Decision at 2-10.

² By letter received on June 19, 2012, Mr. Leventry indicated that he did not want to be involved in the litigation and asked that he be removed from the service list.

A Prehearing Conference was held on May 31, 2012. On June 1, 2012, ALJ Colwell issued a Scheduling Order which adopted the schedule agreed to by the Parties at the Prehearing Conference.

On June 11, 2012, the Company filed a Motion for a Protective Order. No Party filed a responsive pleading, and the Protective Order was granted on July 3, 2012.

On June 18, 20, and 21, Public Input Hearings were held in Scranton, Wilkes-Barre, Bethlehem, Allentown, and Harrisburg.

On July 13, 2012, Richards Energy Group, Inc. (REG) submitted a late-filed Petition to Intervene. The ALJ granted the intervention by Order issued July 26, 2012.

The evidentiary hearings were held on August 6, 7, 9, and 10, 2012. A hearing was also held on October 11, 2012, to hear the testimony of Tracey Andrews, whose Formal Complaint was filed on May 1, 2012, but was not properly associated with this rate case until October 10, 2012. The record consists of a transcript of 613 pages and numerous statements and exhibits presented by various Parties, as detailed in Appendix A of the Recommended Decision.

On August 29, 2012, the Parties filed Main Briefs and the record was thereupon closed. In addition, PPL filed a Petition to Reopen the Record in order to provide updated information regarding the long-term debt issued on August 24, 2012. As no objections were received, by Order issued September 10, 2012, the ALJ reopened the record for the purpose of accepting the updated information.

On September 14, 2012, the Parties filed Reply Briefs. The record closed upon the receipt of the Reply Briefs.

By way of Recommended Decision, issued on October 19, 2012, ALJ Colwell recommended, *inter alia*, that the company be permitted to file tariffs or tariff supplements containing rates designed to produce a \$63,830,000 increase to the Company's present revenues. I.D. at 141. As previously noted, PPL, I&E, the OCA, the OSBA, the CEO, Direct Energy, DR, and PPLICA filed Exceptions. PPL, the OCA, the OSBA, DR, and PPLICA filed Replies to Exceptions. I&E filed Replies to Exceptions on November 29, 2012, as well as a letter requesting that the Commission accept its Replies to Exceptions as timely filed.³

³ In its letter, I&E stated that on November 19, 2012, it electronically served its Replies to Exceptions on all Parties and the Office of Administrative Law Judge and served hard copies upon all internal Commission offices. I&E averred that it did not discover until November 29, 2012, that due to an administrative error, its Replies to Exceptions were inadvertently uploaded for e-filing on November 19, 2012, rather than submitted for e-filing. Under these circumstances, we find it appropriate to consider I&E's Replies to Exceptions in the interest of securing a just, speedy and inexpensive determination in this proceeding. See, 52 Pa. Code § 1.2(a). We do not believe that any of the Parties to this proceeding will be prejudiced by our consideration of I&E's Replies to Exceptions, as the Parties and this Commission were timely served with them.

III. Discussion

A. Description of the Company

PPL is a jurisdictional electrical distribution company (EDC) providing electric distribution service to approximately 1.4 million customers in all or portions of twenty-nine counties in eastern and central Pennsylvania. Under its present corporate structure, it is a wholly owned subsidiary of PPL Corporation (PPL Corp.). Another subsidiary of PPL Corporation is PPL Services Corporation, which provides various administrative and general services to the utility, including legal services, human resources, auditing, and community affairs.

B. Legal Standards

In deciding this or any other general rate increase case brought under Section 1308(d) of the Public Utility Code (Code), 66 Pa. C.S. § 1308(d), certain general principles always apply. A public utility is entitled to an opportunity to earn a fair rate of return on the value of the property dedicated to public service. *Pa. PUC v. Pennsylvania Gas and Water Co.* 341 A.2d 239, 251 (Pa. Cmwlth. 1975). In determining a fair rate of return, the Commission is guided by the criteria provided by the United States Supreme Court in the landmark cases of *Bluefield Water Works and Improvement Co. v. Public Service Comm'n of West Virginia*, 262 U.S. 679 (1923) and *Federal Power Comm'n v. Hope Natural Gas Co.*, 320 U.S. 591 (1944). In *Bluefield*, the Court stated:

A public utility is entitled to such rates as will permit it to earn a return on the value of the property which it employs for the convenience of the public equal to that generally being made at the same time and in the same general part of the country on investments in other business undertakings which are attended by corresponding risks and uncertainties; but it has no constitutional right to profits such as are realized or

anticipated in highly profitable enterprises or speculative ventures. The return should be reasonably sufficient to assure confidence in the financial soundness of the utility and should be adequate, under efficient and economical management, to maintain and support its credit and enable it to raise the money necessary for the proper discharge of its public duties. A rate of return may be too high or too low by changes affecting opportunities for investment, the money market and business conditions generally.

Bluefield, 262 U.S. at 692-693.

The burden of proof to establish the justness and reasonableness of every element of a public utility's rate increase request rests solely upon the public utility in all proceedings filed under Section 1308(d) of the Code. The standard to be met by the public utility is set forth in Section 315(a) of the Code, 66 Pa. C.S. § 315(a), as follows:

Reasonableness of rates. – In any proceeding upon the motion of the commission, involving any proposed or existing rate of any public utility, or in any proceedings upon complaint involving any proposed increase in rates, the burden of proof to show that the rate involved is just and reasonable shall be upon the public utility.

In reviewing Section 315(a) of the Code, the Pennsylvania Commonwealth Court interpreted a public utility's burden of proof in a rate proceeding as follows:

Section 315(a) of the Public Utility Code, 66 Pa. C.S. § 315(a), places the burden of proving the justness and reasonableness of a proposed rate hike squarely on the public utility. *It is well-established that the evidence adduced by a utility to meet this burden must be substantial.*

Lower Frederick Twp. Water Co. v. Pa. PUC, 409 A.2d 505, 507 (Pa. Cmwlth. 1980) (emphasis added). *See also, Brockway Glass Co. v. Pa. PUC*, 437 A.2d 1067 (Pa. Cmwlth. 1981).

In general rate increase proceedings, it is well established that the burden of proof does not shift to parties challenging a requested rate increase. Rather, the utility's burden of establishing the justness and reasonableness of every component of its rate request is an affirmative one, and that burden remains with the public utility throughout the course of the rate proceeding. There is no similar burden placed on parties to justify a proposed adjustment to the Company's filing. The Pennsylvania Supreme Court has held:

[T]he appellants did not have the burden of proving that the plant additions were improper, unnecessary or too costly; on the contrary, that burden is, by statute, on the utility to demonstrate the reasonable necessity and cost of the installations, and that is the burden which the utility patently failed to carry.

Berner v. Pa. PUC, 382 Pa. 622, 631, 116 A.2d 738, 744 (1955).

This does not mean, however, that in proving that its proposed rates are just and reasonable, a public utility must affirmatively defend every claim it has made in its filing, even those which no other party has questioned. As the Pennsylvania Commonwealth Court has held:

While it is axiomatic that a utility has the burden of proving the justness and reasonableness of its proposed rates, it cannot be called upon to account for every action absent prior notice that such action is to be challenged.

Allegheny Center Assocs. v. Pa. PUC, 570 A.2d 149, 153 (Pa. Cmwlth. 1990) (citation omitted). *See also, Pa. PUC v. Equitable Gas Co.*, 73 Pa. P.U.C. 310, 359-360 (1990).

Additionally, Section 315(a) of the Code, 66 Pa. C.S. § 315(a), cannot reasonably be read to place the burden of proof on the utility with respect to an issue the

utility did not include in its general rate case filing and which, frequently, the utility would oppose. Inasmuch as the Legislature is not presumed to intend an absurd result in interpretation of its enactments,⁴ the burden of proof must be on the party who proposes a rate increase beyond that sought by the utility. The mere rejection of evidence contrary to that adduced by the public utility is not an impermissible shifting of the evidentiary burden. *United States Steel Corp. v. Pa. PUC*, 456 A.2d 686 (Pa. Cmwlth. 1983).

In analyzing a proposed general rate increase, the Commission determines a rate of return to be applied to a rate base measured by the aggregate value of all the utility's property used and useful in the public service. The Commission determines a proper rate of return by calculating the utility's capital structure and the cost of the different types of capital during the period in issue. The Commission is granted wide discretion, because of its administrative expertise, in determining the cost of capital. *Equitable Gas Co. v. Pa. PUC*, 405 A.2d 1055, 1059 (Pa. Cmwlth. 1979) (determination of cost of capital is basically a matter of judgment which should be left to the regulatory agency and not disturbed absent an abuse of discretion).

As we proceed in our review of the various positions of the Parties in this proceeding, we are reminded that any issue or Exception that we do not specifically address shall be deemed to have been duly considered and denied without further discussion. The Commission is not required to consider expressly or at length each contention or argument raised by the parties. *Consolidated Rail Corp. v. Pa. PUC*, 625 A.2d 741 (Pa. Cmwlth. 1993); also see, generally, *University of Pennsylvania v. Pa. PUC*, 485 A.2d 1217 (Pa. Cmwlth. 1984).

⁴ 1 Pa. C.S. § 1922(1), *PA Financial Responsibility Assigned Claims Plan v. English*, 541 Pa. 424, 430-431, 64 A.2d 84, 87 (1995).

C. Rate Base

1. Depreciation Reserve

a. Positions of the Parties

In its filing, PPL claimed \$1.813 billion in its Accumulated Reserve for Depreciation based on plant in service and amortization of net salvage for the test year ending December 31, 2012. PPL Future 1-Revised, Sch. C-1. PPL reflected depreciation accruals of \$155.248 million and proposed that the Commission recognize annual depreciation expenses of \$168.92 million. PPL Exh. Future 1-Revised, Sch.D-10.

PPL explained that rate base items are not annualized but are the balances projected to be in effect at the end of the test year. PPL also explained that annualization applies only to revenue and expense items, and not to rate base items. PPL M.B. at 22. PPL averred that the OCA's approach of using a non-annualized level of plant in service with an annualized level of depreciation reserve would create a mismatch between plant in service and the accumulated reserve for depreciation, which would result in an overstatement of the accumulated depreciation reserve and an understatement of rate base. PPL further asserted that the OCA's approach is inconsistent with the fundamentals of test year ratemaking, because by including annualized depreciation expense in the calculation of the accumulated depreciation reserve, the OCA's adjustment would add depreciation expense to the reserve that has not and will not be accrued at the end of the future test year (FTY). *Id.* at 23.

The OCA recommended that the Company's proposed level of Accumulated Reserve for Depreciation be increased by \$10.417 million to better match the claimed depreciation expense, resulting in a corresponding reduction to PPL's rate base of \$10.417 million. OCA M.B. at 12; OCA St. 1-REV. at 11-12; Exh. KC-1-REV. Sched. 2 at 3. The OCA averred that, since ratepayers are being asked to pay for the full

level of depreciation expense, it is appropriate for ratepayers to have the full amount of that expense applied to accumulated depreciation. OCA M.B. at 13; OCA St. 1-SR at 4.

b. ALJ's Recommendation

The ALJ recommended adoption of PPL's position to use the accrued depreciation amount of \$155.248 million for calculating the depreciation reserve, rather than the claimed \$168.92 million in depreciation expense. R.D. at 17. The ALJ agreed with the Company's reasoning that rate base items are not annualized but are the balances which are projected to be in effect at the end of the year. *Id.* at 16, 17. The ALJ found the Company's following argument persuasive:

The reserve for depreciation is built up by recording depreciation expense, but the expense recorded is the expense per books for a particular period of time, here calendar year 2012. OCA's proposal to ignore the projected per books depreciation expense and use instead the theoretical, annualized level of expense is not correct. The annualized depreciation expense as of December 31, 2012 will not be recorded on PPL Electric's books during calendar year 2012. Therefore, it is not part of the "build-up" of the depreciation reserve by recording depreciation expense related to plant in service.

Id. at 17-18 (citing PPL R.B. at 9-10). Accordingly, the ALJ recommended that the OCA's proposed \$10.417 million adjustment be rejected. *Id.* at 18.

c. Exceptions

In its Exceptions, the OCA avers that the ALJ erred by rejecting the OCA's accumulated reserve for depreciation adjustment. The OCA states that it recommended an adjustment to PPL's accumulated reserve for depreciation to match PPL's claimed depreciation expense. OCA Exc. at 2; OCA St. 1-REV. at 11-12; Exh. KC-1-REV.

Sched. 2 at 3. The OCA explains its position that the depreciation expense included in the cost of service and the additions to the depreciation reserve, which are deducted from rate base, should be based on the level of plant the Company claims will be in service at the end of the FTY and the depreciation expense claimed for the FTY that is related to that plant. OCA Exc. at 2-3. The OCA asserts that ratepayers should receive the full benefit of the depreciation expense for which they are being charged by receiving the corresponding full benefit of accumulated depreciation reserve. *Id.* at 3.

In its Replies to Exceptions, PPL states that the ALJ properly rejected the OCA's proposal. PPL R.Exc. at 11. PPL submits that the accumulated reserve for depreciation, plant in service, and retirements as of December 31, 2012, are determined by bringing forward the book balances as of December 31, 2011, by reflecting the projected plant additions, annual depreciation expense per books, projected retirements per books, and projected net salvage per books. *Id.* at 11; PPL Exh. JJS-2 at III-6-III-7; PPL Exh. 1, Part V-A-3 at 1-3. PPL also submits that the OCA is proposing to change only one of these elements in determining net plant in service – the projected depreciation expense per books for 2012. PPL avers that the OCA's proposed adjustment is flawed, because the use of the annualized depreciation expense would be a mismatch with every other component of net plant in service, as those components are based on projected transactions per books. PPL asserts that there is not an annualized level of plant in service as of December 31, nor are there annualized retirements or annualized net salvage. PPL R.Exc. at 11. PPL further avers that its method of determining the accumulated reserve for depreciation was approved in its prior rate proceeding and has been accepted by the Commission for all major electric, gas, and water public utilities. *Id.*; PPL St. 13-R at 4.

d. Disposition

Based on our review of the record, the Parties' positions, and the Recommended Decision, we find that the ALJ properly adopted PPL's claim and rejected the OCA's proposal to use an annualized level of depreciation. PPL has met its burden of proof by showing that its method of determining the accumulated reserve for depreciation is reasonable, is consistent with the fundamentals of test year ratemaking, and is consistent with the methods used by other major public utilities. We agree with PPL that rate base items are not annualized but are balances to be in effect at the end of the test year. PPL is correct that the OCA's proposed adjustment to use a non-annualized level of plant in service with an annualized level of depreciation reserve would create a mismatch between plant in service and the accumulated reserve for depreciation, which would result in an overstatement of the accumulated depreciation reserve and an understatement of rate base. For these reasons, we shall deny the OCA's Exceptions and adopt the ALJ's decision on this issue.

2. Cash Working Capital – Lag Days for Payments to Affiliates

a. Positions of the Parties

PPL explained that its expense lag days for payments to its affiliate for support services is thirty-five days, consisting of the sum of fifteen days, which is the midpoint of the monthly service period, and twenty days, which is a standard accounting transaction for the preceding month. PPL M.B. at 24. PPL stated that it treats its payments to affiliates in the same manner that it treats its payments to non-affiliated vendors, and that it should not discriminate in favor of, or against, its affiliates. PPL also stated that a payment lag of thirty days is commercially reasonable and typical of the terms required by PPL's vendors. PPL asserted that it has consistently incorporated a thirty-five day payment lag for its affiliates in previous rate cases, and the Commission and the other parties to those proceedings have accepted the thirty-five day payment lag

for affiliated services in calculating cash working capital (CWC) requirements. *Id.* at 25; PPL St. 7-R at 3.

I&E recommended a reduction in the CWC operation and maintenance (O&M) claims based on its position that PPL unnecessarily pays its affiliate substantially in advance of the required due date under the Company's service agreement with its affiliate. I&E submitted that, under the service agreement, PPL is billed monthly and has sixty days to pay its affiliate. Therefore, I&E argued that PPL has an allowable payment lag of seventy-five days pursuant to contract. I&E M.B. at 12. I&E proposed changing the payment date to the affiliate which, when weighted with the other expense groups, would result in an overall average expense lag payment of approximately forty-eight days, compared to PPL's claimed average expense payment lag of approximately thirty-five days. *Id.*; I&E St. 2 at 56. Application of I&E's recommendation would result in a \$13,021,000 reduction to the Company's CWC claim to rate base. I&E M.B. at 11; I&E St. 2 at 56. I&E further argued that PPL did not provide any evidence that it has consistently incorporated a thirty-five day affiliate payment lag in its prior rates cases. I&E R.B. at 10. According to I&E, no prior litigated case addressed CWC generally or this O&M expense lag specifically, and there are no prior applicable Commission Orders providing the Company with Commission approval for this expense lag. *Id.* at 10-11.

b. ALJ's Recommendation

The ALJ adopted I&E's recommendation for a \$13.021 million reduction to O&M in the CWC component of the Company's claimed rate base. The ALJ found persuasive I&E's argument that PPL did not have to pay its affiliate for services within the time period that the Company claimed but had the discretion to take advantage of a longer payment period of up to sixty days under the terms of the contract with its affiliate. The ALJ did not believe that PPL met its burden of proving its claim was reasonable, because the Company was causing the ratepayers a substantial amount of

money due to a practice it could not otherwise justify except by saying that it has always been done that way. R.D. at 20.

c. Exceptions

In its Exceptions, PPL avers that the Recommended Decision's proposed adjustment to its lag days for payments to its affiliate should be rejected. PPL Exc. at 35. PPL submits that it uses a computerized system to pay all of its invoices from PPL Services and non-affiliated vendors. The Company notes that it pays its affiliates on the twentieth day of the month after services are received, which results in a thirty-five day payment lag for services it receives from its affiliates. PPL asserts that the ALJ's reliance on the payment terms in the agreement with its affiliate is not an adequate basis for the adjustment, because the agreement does not require a sixty-day payment period and clearly authorizes a twenty-day payment period. The Company explains that the agreement was entered into seventeen years ago when computers were not used to the extent they are currently and a longer time period for invoice payment was more common. *Id.* at 36.

I&E rejoins that the ALJ properly rejected PPL's calculation of its expense lag days based on the evidence presented by I&E, which demonstrated that the Company paid its affiliate well in advance of the due date, thereby resulting in a significantly shorter expense payment lag and an unnecessary annual ratepayer CWC contribution of \$1.1 million. I&E R.Exc. at 3; I&E St. 2-SR at 62. I&E believes that PPL should be required to save its ratepayers \$1.1 million annually by paying its affiliate as permitted under the agreement. I&E asserts that the manner in which PPL pays its affiliate disadvantages ratepayers and benefits its affiliate. I&E submits that, as a regulated monopoly with captive ratepayers, PPL should be held to a strict standard regarding the manner in which it handles payments to affiliates. I&E R.Exc. at 4.

d. Disposition

We agree with the ALJ's decision to adopt I&E's recommended \$13.021 million O&M reduction to the CWC component of the Company's claimed rate base. PPL did not meet its burden of proving that its expense lag days for payments to its affiliate are reasonable. Since PPL has up to sixty days to pay its affiliate under the agreement, it would have been reasonable for PPL to take advantage of the longer payment period and, by doing so, to minimize the rate impact on its customers. PPL has control over when it pays its affiliate and can alter its computerized system to change the date on which it pays its affiliate. The evidence presented by I&E demonstrated that PPL's choice to pay its affiliate forty days early resulted in an annual ratepayer CWC contribution of \$1.1 million. I&E St. 2-SR at 62. PPL's customers should not be burdened with this expense when it can be avoided. For these reasons, we shall deny PPL's Exception and adopt the ALJ's decision on this matter.

3. Cash Working Capital – Prepayment of Postage Expense

a. Positions of the Parties

PPL averred that it is proper for postage expense to be reflected in both the operation and maintenance expense component of working capital and prepayments, because each component addresses the expense during separate and distinct time periods. PPL M.B. at 26. PPL explained that the first time period related to postage expense is the prepayment, which begins when it makes prepayments to the United States Postal Service for postage to be used by the postage meter and ends when the postage meter adds postage to an envelope. According to PPL, the second time period is the payment lag, which begins when the postage is used. During the second time period, the expense appears in the working capital requirement as an O&M expense to reflect the period between when the postage meter adds postage to an envelope and when customers pay PPL. PPL's position was that there is no double recovery, because the inclusion of

postage expense as a prepayment is separate from its treatment as an O&M expense in the working capital calculation. *Id.* at 27; PPL St. 7-R at 6-7. In its Reply Brief, PPL stated that its position that there is no double recovery was consistent with controlling Commission precedent, particularly the Commission's decision in the Company's 2004 rate case. PPL R.B. at 14 (citing *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255, at 11-12 (Order entered December 22, 2004)).

I&E did not recommend a specific adjustment to the Company's treatment of postage expense, but stated that the Company should be ordered to discontinue this practice in future proceedings because it is an improper CWC calculation that overstates the Company's CWC needs. I&E M.B. at 18. I&E averred that PPL includes a full twelve-month expense dollar amount claim for postage in its total CWC O&M expense, and also includes a twelve-month average prepayment dollar amount for postage in the Prepayment CWC component. *Id.* at 17. I&E's position was that this practice overstates the actual CWC requirement for postage, because the inclusion of two different CWC components results in a funding claim that is greater than what is incurred on an annual basis. *Id.* at 17-18.

b. ALJ's Recommendation

ALJ Colwell agreed with I&E's position. The ALJ found that PPL should discontinue its practice of including the same CWC need for postage in both the O&M expense and prepayment components of the CWC calculation, because this practice improperly inflates the CWC calculation. R.D. at 22. The ALJ distinguished this case from PPL's 2004 rate case. The ALJ stated that, in the 2004 case, the Commission accepted ALJ Turner's finding that the evidence did not support a conclusion that the Company prepaid its postage, which ALJ Turner admitted would have changed her recommendation. In this case, ALJ Colwell noted that PPL admitted to prepaying for postage and using the prepaid postage in its postage meter. R.D. at 21.

c. Exceptions

In its Exceptions, PPL avers that it should be permitted to continue to calculate the postage expense component of working capital as it has been calculating it. PPL states that it has fully explained its treatment of postage expense in rate base in its briefs. PPL also states that the Commission previously approved its treatment of postage expense and that nothing has changed since the Commission's previous approval. PPL Exc. at 37 (citing *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255, at 11-12 (Order entered December 22, 2004)).

In its Replies to Exceptions, I&E states that the ALJ correctly found that the Company overstated postage and that the Company should correctly calculate its postage expense in future proceedings. I&E avers that PPL improperly included postage expense as both an O&M expense and a prepayment, which resulted in a funding claim greater than the Company incurred. I&E believes that PPL's 2004 rate case is distinguishable from this case because, in that case, the OCA did not provide evidence that the Company included a prepayment and an expense for the same item, whereas, the Company admitted that it did in this case. I&E submits that PPL's CWC claim for postage is overstated, because, whether loaded into a meter or directly expensed, postage is paid only once. I&E R.Exc. at 5.

d. Disposition

Based on our review of the record, the Parties' positions, and the ALJ's decision, we find that PPL improperly included the same postage expense in two CWC components by listing it as both an O&M expense and a prepayment, resulting in an overstatement of that expenditure. We do not find merit in PPL's reliance on our Order in the Company's 2004 rate case. We agree with the ALJ and I&E that this case is

distinguishable from the 2004 rate case. In the 2004 case, PPL included a claim for the net lag in recovery of operating expenses based upon a lead/lag study and a separate claim for average prepayments. In that case, PPL stated that the time period captured in its lead/lag study was from the date the bills were mailed to the date payment was received from customers, thus, excluding the time period from when the postage was paid to when it was expensed. *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255, at 11 (Order entered December 22, 2004). In the 2004 case, we concluded that the Company's position refuted the OCA's argument of double counting, because the time period from when the postage was paid to when it was expensed was excluded. *Id.* at 12. In the present case, PPL is expressly claiming that it properly included a prepayment and an expense for the same postage item. Accordingly, we shall adopt the ALJ's recommendation that PPL discontinue its practice of including the same CWC need for postage in both the O&M expense and prepayment components of the CWC calculation and deny PPL's Exception.

4. Cash Working Capital – Prepayment of Regulatory Assessments

a. Positions of the Parties

PPL stated that, consistent with Commission precedent, it included the Commission assessment in the prepayment component of its working capital requirement. PPL M.B. at 28 (citing *Pa. PUC v. National Fuel Gas Distribution Corp.*, 1994 Pa. PUC Lexis 134). PPL stated that, while the Commission's assessment is calculated based on a utility's jurisdictional revenue for the prior calendar year, the assessment applies for the forthcoming fiscal year as provided in the Commission's June 21, 2012 invoice. PPL quoted the language in the Commission's invoice as follows:

The Commission is submitting a request for **pre-payment** of PPL Electric's estimated Public Utility Commission assessment for the fiscal year 2012-2013. The requested pre-payment amount is an estimate based on the revenues shown

on your Company's GAO-11 submission and the Commission's **fiscal year 2012-2013 budget** request. When the assessment invoices are issued in August for the **fiscal year 2012-2013** your invoice will be adjusted to reflect the payment made in response to this letter.

PPL M.B. at 29; PPL Exh. BLJ-1 (emphasis added).

PPL averred that its position that the assessment is for the fiscal year beginning on the following July 1 is also supported by the language in Section 510 of the Code, 66 Pa.C.S. § 510. PPL explained that, under Section 510, the Commission budget is proposed to the Governor and the General Assembly the preceding November 1, and the General Assembly is expected to approve a Commission budget for the upcoming fiscal year by the preceding March 30. PPL stated that, based on the approved budget, the Commission allocates the assessment among public utilities according to each utility's jurisdictional revenues for the preceding calendar year. PPL stated that, once the Commission makes the calculations, it prepares payment requests that the utilities receive in June prior to the fiscal year for which the assessment is made. PPL M.B. at 29.

I&E recommended removing the Company's claimed Commission assessments from the prepayments component of its CWC claim, which would result in an allowable working capital prepayment of \$394,000, a reduction of \$2.78 million to the Company's working capital prepayment claim. I&E averred that the Commission assessment is not a prepayment. I&E explained that the assessment is calculated as a proportion of Commission, OCA, and OSBA services that have been provided to PPL's utility type in the prior year, and it is billed as a percentage assessed on PPL's prior calendar year jurisdictional revenue and payable to the Commission, the OCA, and the OSBA in the subsequent calendar year. I&E M.B. at 15. I&E opined that the assessment is akin to a tax and, thus, should be treated as an expense with an associated lag. I&E argued that the assessment should be matched against the revenue generation time period

on which the expense was based, namely, the prior year's jurisdictional revenue. *Id.* at 16.

b. ALJ's Recommendation

The ALJ recommended that I&E's proposal to remove PPL's Commission assessment expense as a prepayment under its CWC calculation be denied. The ALJ stated that several large utilities, including PPL, pay their assessments, or a portion of them, early in order to assure continued funding of the Commission's activities for the first quarter of the fiscal year. The ALJ found that it was clear that the assessment is based on a prior year's revenues, but the application period is the following fiscal year. R.D. at 23.

c. Exceptions

In its Exceptions, I&E avers that the ALJ erred by recommending rejection of I&E's adjustment to remove PPL's claimed regulatory assessments from the prepayments component of its CWC claim. I&E Exc. at 4. I&E states that the ALJ's finding that the regulatory assessment is a prepayment due to the time period in which the actual funds are spent is erroneous. I&E Exc. at 5. I&E contends that, under Section 510(b) of the Code, 66 Pa. C.S. § 510(b), although the assessment is paid in the subsequent fiscal year, the assessment covers the regulatory expenses incurred in the prior year. As such, I&E asserts that the assessment is not a prepayment for the next year's expenses, and it should be treated as an expense with an associated lag. *Id.* at 6.

I&E also distinguishes assessments from prepayments because prepayments are paid in advance of a service and may be refunded if the service is terminated before the end of the service period, whereas a utility's assessments are representative of the proportion of agency services rendered to the utility in the prior year

and are not subject to a refund if the utility ceases operations the following year. *Id.* I&E believes that for ratemaking purposes, the assessment, which is a billed expense, must be matched against the revenue generation time period on which the expense was based, which is the prior year's jurisdictional revenue. I&E avers that this practice is consistent with the manner in which the assessment is made and with the accrual accounting concept of matching expenses with the revenue earning period that manifested the expenses, or matching revenues with the expenses that result from the production of those revenues. *Id.* at 7; I&E St. 2-SR at 63.

I&E further submits that the Commission's June assessment letter does not support the ALJ's recommendation. I&E describes the assessment process and states that the assessment is based upon the utilities' prior calendar year revenues, which must be reported by March of the following calendar year. I&E Exc. at 7. While assessments are made in August of a fiscal year, the Commission issues letters in June, such as the one issued to PPL, asking certain larger utilities to submit an early payment of the fiscal year's assessment based on a preliminary early assessment provided by the Commission. *Id.* at 7-8; PPL Exh. BLJ-1. Thus, I&E avers that the Commission's use of the word "prepayment" in the June assessment letter is merely a request for an early payment to assure the continuous funding of regulatory agencies, and is not determinative of the status of the assessment payment for purposes of the proper calculation of PPL's CWC requirements. *Id.* at 8.

In its Replies to Exceptions, PPL avers that the ALJ properly included regulatory assessments as a prepayment in the working capital calculation. PPL states that I&E's proposed adjustment is inconsistent with the Commission's invoice for assessments, the relevant law, and the manner in which the Commission operates. According to PPL, the language in the Commission's invoice supports its position that regulatory assessments are a prepayment. PPL R.Exc. at 14. PPL states that Section

511(b) of the Code, 66 Pa. C.S. § 511(b), also supports its position. *Id.* at 14-15.⁵ PPL asserts that I&E's position suggests that regulatory assessments are paid after the fact and, if this were true, the Commission would have to borrow money to fund operations while collections of assessments were pending. PPL believes that I&E's position ignores reality and the way the Commission operates. *Id.* at 15.

d. Disposition

We find that PPL properly included the Commission assessment in the prepayment component of its working capital requirement. PPL presented evidence to show that, based on the language in the Commission's June 21, 2012 invoice, the assessment applies for the forthcoming fiscal year, July 1 through June 30. *See*, PPL St. 7-R at 3-4; PPL Exh. BLJ-1. PPL also presented evidence demonstrating that, pursuant to the assessment process set forth in Section 510 of the Code, 66 Pa. C.S. § 510, the assessment payment qualifies as a prepayment. PPL St. 7-R at 4. While it is clear under Section 510 that the assessment is calculated based on operating revenues for the preceding calendar year, the assessment that a utility pays is for the upcoming fiscal year. Moreover, PPL paid its assessment early, as requested in the Commission's invoice, and based its prepayment calculation on the manner in which it handles its assessment payments. *Id.* PPL's inclusion of the assessment as a prepayment is consistent with our prior decisions. *See, Pa. PUC v. National Fuel Gas Distribution Corp.*, 1994 Pa. PUC Lexis 134, *29-30 (permitting the public utility to include in rate base a prepayment

⁵ PPL quotes Section 511(b) of the Code, which provides the following:

All such assessments and fees, having been **advanced** by public utilities for the purpose of defraying the cost of administering this part, shall be held in trust solely for that purpose, and shall be earmarked for the use of, and annually appropriated to, the commission for disbursement solely for that purpose.

PPL R.Exc. at 15 (quoting 66 Pa. C.S. § 511(b) (emphasis added)).

balance that included the Commission's assessment). For these reasons, we shall deny I&E's Exceptions and adopt the ALJ's decision on this issue.

D. Expenses

1. Incentive Compensation

a. Positions of the Parties

PPL provides three types of compensation to its employees: base pay, benefits, and eligibility for incentive compensation. PPL makes incentive compensation payments to its own employees and reimburses PPL Services for its share of PPL Services' incentive compensation, which enables PPL Services to make incentive payments to its eligible employees. PPL St. 3-R at 15-26; PPL M.B. at 33.

The OCA recommended disallowing half of the incentive compensation expense, thereby requiring the shareholders to share equally in the cost of the compensation plans. The OCA recommendation is to adjust the expenses of \$4.468 million for the Company's incentive compensation plan and \$4.902 million related to the PPL Services' incentive compensation plan downward. OCA Exh. KC-1-SR, Sch. 4 at 4; Sch. 1 at 2.

I&E recommended an equal sharing of the claimed incentive compensation expenses between shareholders and ratepayers, resulting in a jurisdictional allowance of \$4.459 million and a reduction of the same amount from PPL's claim. I&E asserted that PPL has provided no evidence that the incurrence of this cost is necessary for the provision of safe and reliable service at just and reasonable rates. I&E M.B. at 28-29.

PPL argued that the incentive compensation payments are a part of the total compensation package that was developed and is maintained based, at least in part, on a

comparison with those of other employers for comparable positions. PPL stated that, if the incentive compensation payments to employees were eliminated, the fixed compensation would have to be raised in order to remain competitive with other employers, and “[t]here would be no savings to ratepayers.” PPL St. 3-R at 16-17; PPL M.B. at 34.

Further, PPL stated that the Commission has approved incentive compensation programs in numerous prior rate cases. PPL M.B. at 36-37 (citing *Pa. PUC v. Aqua Pennsylvania, Inc.*, Docket No. R-00072711, at 20-21 (Order entered July 31, 2008); *Pa. PUC v. PPL Gas Utilities Corporation*, Docket No. R-00061398, at 40 (Order entered February 8, 2007)).

b. ALJ’s Recommendation

The ALJ stated that, because the Parties have not challenged the reasonableness of the total compensation expense, the overall amount was not at issue; rather, only the method of recovery was at issue. While the two public advocates rely on the inherent fairness of having shareholders fund half of the incentive program, since they too receive a benefit, the ALJ found that the law does not support that concept. Rather, the ALJ found that a utility is entitled to recover in rates all expenses reasonably necessary to provide service to its customers and to earn a fair return on its investment in plant used and useful in providing service. The ALJ stated that to require a sharing of expense is to deny that portion in a rate case, which is simply not permitted under case law. R.D. at 28 (citing *Butler Township Water Co. v. Pa. PUC*, 473 A.2d 219, 221, 222 (Pa. Cmwlth. 1984); *T.W. Phillips Gas and Oil Co. v. Pa. PUC*, 474 A.2d 355 (Pa. Cmwlth. 1984)). Based upon the above rationale, the ALJ recommended that PPL be permitted full recovery of its incentive compensation plan. R.D. at 27-28.

c. Exceptions

The OCA and I&E excepted to the ALJ's recommended full recovery of PPL's incentive compensation plan. As presented in its Main and Reply Briefs and in its Exceptions, the OCA asserted that there is ample case law to support the OCA's position that shareholders should fund a portion of the incentive compensation plan. OCA Exc. at 3 (citing *Pa. PUC v. Philadelphia Gas Works*, 2007 Pa. PUC Lexis 45; *Pa. PUC v. UGI Utilities, Inc. - Electric Division*, 82 Pa. P.U.C. 488, 508 (1994); *Pa. PUC v. Roaring Creek Water Co.*, 1994 Pa. PUC Lexis 41 (1994)). The OCA believes that the ALJ erred in failing to recommend a sharing of PPL's incentive compensation plans. *Id.*

In its Exceptions, I&E contends that neither the evidence nor the case relied upon by the ALJ supports the recommendation that PPL be permitted to recover the entire incentive compensation program expense from ratepayers. I&E Exc. at 9. I&E argues that while PPL is entitled to recover all reasonably incurred expenses, necessary for the provision of safe, reliable and adequate utility service, it must first satisfy its burden of proof. *Id.* I&E contends that PPL did not meet this burden. I&E opines that, absent sufficient data to determine the relative ratepayer and shareholder values, its proposed equal sharing of the expense is fair because the Company's earnings per share performance and other financial measures directly impact shareholder value, I&E's. *Id.* at 10. I&E also contends that the ALJ erroneously concluded that *Butler Township Water Co. v. Pa. PUC*, 473 A.2d 219 (Pa. Cmwlth. 1984), prohibited, as a matter of law adoption of I&E's proposal to disallow half of PPL's incentive compensation program. *Id.* at 11.

In its Replies to Exceptions, PPL averred that this adjustment would ignore the fact that almost everything PPL does will provide a benefit to both shareholders and ratepayers. PPL R.Exc. at 12. Further, PPL argues that this adjustment is unlawful because a public utility is entitled to recover expenses reasonably necessary to provide

service to customers and to earn a fair rate of return. *Id.* A public utility is also entitled to recover operating expenses that are prudently incurred to provide service to customers. *Id.* In *PGW*, *UGI*, and *Roaring Creek*, as cited by the OCA and I&E, incentive compensation was disallowed in total because the utilities could not demonstrate that the program would provide a benefit to ratepayers. *Id.* at 13. In further support of its incentive compensation plan, PPL notes the plan's three overarching objectives: to achieve operational excellence; to optimize workforce readiness and engagement and to increase shareholder value. *Id.*

d. Disposition

We agree with the ALJ's interpretation of *Butler*. We find that, because PPL's incentive compensation plan is reasonable, prudently incurred, and is not excessive in amount, PPL is permitted full recovery of this expense. *See, Butler*, 473 A.2d at 221. PPL correctly notes that many of the cases the OCA and I&E rely on are distinguishable from this case because, in those cases there was not adequate evidence that the incentive compensation expense was reasonable or that there was a benefit to ratepayers. *See, Pa. PUC v. Philadelphia Gas Works*, 2007 Pa. PUC Lexis at *73-75; *Pa. PUC v. Roaring Creek Water Co.*, 1994 Pa. PUC Lexis at *37-38. Our decision to allow this incentive compensation expense is consistent with our prior decisions approving incentive compensation programs that are focused on improving operational effectiveness. *See, e.g., Pa. PUC v. Aqua Pennsylvania, Inc.*, 2008 Pa. PUC Lexis 50 at *24; *Pa. PUC v. Duquesne Light Co.*, 1987 Pa. PUC Lexis 342 at *99-100. Accordingly, the exceptions of the OCA and I&E on this issue are denied.

2. PPL Services

a. Environmental Management

i. Positions of the Parties

PPL's FTY claim of \$467,000 is based upon the adoption of new federal, state and local environmental regulations that require PPL to undertake greater levels of environmental management activities. More specifically, federal and state environmental rules mandate that routine inspection of storm water and erosion, and sedimentation control measures continue beyond project completion. PPL further asserted that its budgeted increase in construction carries with it an increased need for environmental management services. For these reasons, PPL asserted that the past years' variability of this expense does not support the use of an historic average because, in this instance, the past is not representative of the future. PPL St. 3-R at 2-5; PPL M.B. at 41, 42.

I&E recommended a four-year average of actual annual jurisdictional direct support fees from 2009 through 2011, and the 2012 budget amount, resulting in a ratemaking allowance of \$364,000, or a reduction of \$103,000 from PPL's FTY claim. It is I&E's position that PPL's claimed level of expense is unsubstantiated. I&E's analysis includes PPL's FTY claim, which I&E believes recognizes an increase over PPL's historic level by giving consideration to the equivalent of 1.5 new full time employees. I&E is also of the opinion that PPL failed to substantiate how new environmental regulations may impact the expenses of operating PPL's distribution system. I&E M.B. at 25-26. I&E further contended that PPL ignored the fact that costs for the implementation of a new software system will not recur, and should not be included within the FTY claim. I&E M.B. at 34.

PPL asserted that I&E's rationale for its proposed disallowance, which relies upon the variability of the expense, the nonrecurring nature of the cost of the new

computer system and that PPL does not expect its FTY level of expense to be sustained in subsequent years, was either incorrect or irrelevant, or both. PPL explained that while its expenses for the new software will not extend beyond the FTY, PPL will require additional licenses for employees using the software and additional environmental management support as more employees become authorized to use the software. PPL M.B. at 42. Further, as indicated in the data provided to I&E in response to discovery, PPL's business plan anticipates an increase in environmental management expense as follows: \$485,000 for 2013; \$494,000 for 2014; \$508,000 for 2015; and \$549,000 for 2016 and 2017. *Id.* at 43.

ii. ALJ's Recommendation

The ALJ concluded that PPL did not provide citations to the new regulations, nor any specific cost estimates for specific requirements to support its claim that there will be additional costs for environmental compliance. Further, the ALJ found that PPL did not sustain its burden of proving entitlement to the level of support fees sought. In the absence of record evidence to support its claim, the ALJ recommended adoption of I&E's proposal to reduce PPL's FTY claim by \$103,000. R.D. at 29-30.

iii. Exceptions

In its Exception, PPL argues that the ALJ's recommendation is in error. PPL claims that, due to the adoption of new regulations, it will be required to undertake greater levels of environmental management activities due to the increase in construction activity throughout its system. This increase in construction activity elevates PPL's expenses related to environmental permitting and the need for additional employees. PPL Exc. at 32.

I&E rejoins that the ALJ correctly rejected PPL's claim for payment to its affiliate for environmental management services and recommended adoption of I&E's \$103,000 reduction. I&E argues that PPL's claim contained costs that were irregular, erratic, and unsupported in the FTY. I&E R.Exc. at 9. I&E submits that despite PPL's claims that environmental compliance costs will increase substantially, PPL Corp. contended otherwise in its reports to investors, stating there will be no environmental downside for its distribution system, noting no significant exposure to currently proposed environmental regulations. *Id.* at 10.

iv. Disposition

We agree with I&E and the ALJ on this issue and shall grant the \$103,000 expense reduction proposed by I&E. We find that PPL failed to carry its burden of proof that adoption of new regulations will require PPL to undertake greater levels of environmental management activities due to the increase in construction throughout its system. PPL did not refer to any newly adopted environmental regulations to which it is, or will become subject to, in the FTY. Absent this type of support we find the position of I&E to be reasonable. Accordingly, we shall deny PPL's Exception on this issue.

b. External Affairs

i. Positions of the Parties

PPL's budget for 2012 includes \$2.602 million for direct services from the External Affairs⁶ Department of PPL Services, which is an increase of \$1.17 million, or 81% above the \$1.432 million 2011 expense. PPL St. 3-R at 6; PPL M.B. at 43. The indirect expenses from this department totaled \$1.252 million for the Historical Test Year

⁶ External Affairs provides, in part, for the coordination of government relations activities, corporate communications, such as media and public relations services, as well as community and economic development activities. PPL St. 2 at 21-22.

(HTY) and are budgeted at \$1.368 million for the FTY. I&E St. 2-SR at 17. The total charges to PPL represent 25% of the annual corporate budget for the HTY and 36% for the FTY. *Id.*

PPL explained that the reason for the increase from 25% to 36% of the annual corporate budget is two-fold. First, a review of the day-to-day activities of the regional community relations directors, who are part of the External Affairs Department, revealed that their activities center around reliability, connections and disconnections, billing and payment, street lighting and requests related to economic development. All of these activities directly benefit PPL, not other members of the PPL corporate system. Therefore, these expenses now are being directly charged to PPL instead of being allocated as indirect charges among all members of the PPL corporate system. Second, PPL stated that increases in line siting and upgrading work, tree trimming and enhanced storm damage communication protocols have also added to the responsibilities of this department. PPL St. 3-R at 6-7; PPL R.B. at 36-37.

I&E contended that the proposed percentage increase would shift an inordinate portion to the rate-regulated entity, PPL, without express consideration of the broader nature of the function of the External Affairs Department. I&E RB at 27. I&E stated that while External Affairs may become involved in billing and connection issues on occasion, PPL has other divisions specifically designed to address these functions on a daily basis. *Id.* at 27-28. In further support of its position, I&E explained that there is very little nexus, if any, between community development activities and the safe and reliable provision of utility service. *Id.* at 28. At a minimum, I&E contended that PPL's efforts with respect to community development enhance the corporate brand at least as much as they affect the provision of electric distribution service. *Id.*

I&E's original recommendation was to allow only the HTY level of directly assigned costs, or \$1.432 million representing an expense adjustment of \$1.170

million. However, upon review of PPL's explanation of the increase in this cost element from the HTY to the FTY, I&E revised its original expense adjustment. I&E R.B. at 27. I&E's revised expense allowance is based upon an average of the HTY percentage of 25% and the FTY proposed percentage of 36%, for an average of 30.5%. This average percentage, as developed in the table above, was then applied to the total FTY External Affairs Division budget of \$10.982 million, providing a recommended allowance of \$3.350 million. I&E's revised adjustment, therefore, is \$3.970 million - \$3.350 million, or \$620,000. I&E St. 2-SR at 18.

ii. ALJ's Recommendation

The ALJ found that PPL did not adequately support the proposed increase in its allocated share of the External Affairs Division's FTY budget. The ALJ also found that PPL's only reference was to a schedule attached to its rebuttal testimony. The ALJ recommended that I&E's revised adjustment of \$620,000, be adopted based upon I&E's rationale to support its calculated disallowance. R.D. at 31.

iii. Exceptions

In its Exceptions, PPL argues that the Commission should reverse the ALJ and allow the total claim of \$2.6 million. PPL Exc. at 33. PPL explains that the increased costs for external affairs is driven primarily by refinements to the process of identifying the affiliates who benefit from the services provided, rather than a dollar increase in the overall costs of those services, which was only 0.8% from 2011 to 2012. *Id.* PPL explains that starting with the FTY, more of the costs for external affairs are directly assigned rather than being allocated as an indirect cost. *Id.* at 34.

In reply, I&E states that PPL provided no evidence to connect monies spent on community and economic development (\$865,000 for 2011 and \$1.7 million for 2012)

and government relations (\$463,000 for 2011 and \$727,000 for 2012) to the provision of safe and reliable utility service. I&E R.Exc. at 10; I&E Exh. No. 2, Schedule 13, at 2. I&E also states that while logic dictated that as the allocation of direct costs rose, the allocation of indirect costs should have decreased, because overall expenses of PPL Services for this account increased by only 0.8% I&E R.Exc. at 11.

iv. Disposition

Based upon our review of the record evidence, we shall reverse the ALJ's recommendation on this issue. I&E's position is based upon its opinion that this expense lacks any nexus to PPL's provision of safe and reliable utility service and that the proposed percentage increase would shift an inordinate portion to the rate-regulated entity, PPL, without express consideration of the broader nature of the function of the External Affairs Department. I&E has also taken the position that since there was a very small increase in the total expense, the significant rise in direct expenses should have caused the indirect expense allocation to shrink. As shown in the table above, the allocated indirect costs increased from 2011 to the FTY by 9.2%, or \$116,000, while the total indirect and other expenses to be allocated increased by 75.0%, or \$21.951 million.

PPL Exhibit DAC-1, Schedule 4, page 2, indicates that the indirect and other costs to be allocated increased from \$29.241 million to \$51.192 million from 2011 to the FTY. The \$29.241 includes a Storm Insurance recovery of \$15.501 million. Without this significant insurance recovery, the increase in this account would be only 14% or \$6.45 million. I&E did not present any issue regarding the amount of indirect and other costs to be allocated until after it adopted PPL's explanation for the increase in direct assignment of costs relative to this account.

I&E's final position is to 'split the baby' by taking an average percentage of the jurisdictional expense level for 2011 and the FTY, as they are compared with the

total amount of expense as shown in the table above. We believe that this mathematical adjustment is not supported by I&E's contentions of an insufficient nexus or that the percentage increase in the direct assignment portion represents an excessive shift of expense to PPL, the regulated entity. Accordingly, we shall grant PPL's Exception and reverse the ALJ's recommendation on this issue.

c. Office of General Counsel

i. Positions of the Parties

Legal services to PPL Electric are provided by PPL Corporation's Office of General Counsel (OGC), and PPL's jurisdictional FTY claim for OGC is \$6.083 million. I&E Exh. 2, Sch. 17 p. 2. According to I&E, PPL's claim is based on its HTY expense increased by \$1.2 million in estimated costs for outside counsel fees related to this proceeding. Because of this, I&E recommended a ratemaking allowance of \$4.833 million for OGC expense, which is a \$1.2 million reduction to PPL's claim. The basis for I&E's adjustment is to eliminate the additional expense associated with outside counsel for this proceeding since the Company also includes a claim for rate case expense in its pro forma adjustments. I&E M.B. at 38.

PPL agreed with the adjustment but argued that it was more appropriate to eliminate the duplication from O&M expenses because the expense in question will be incurred by the OGC and then charged directly to PPL. PPL St. 8-R, at 41-42. PPL M.B. at 47.

I&E acknowledged PPL's acceptance of the expense reduction, but contended that it is appropriate to reflect the reduction as a part of the affiliate support allocation, and not as a rate case expense reduction. I&E M.B. at 39. I&E explained that keeping the expense as a part of PPL's affiliate support allocation will overstate the level of OGC affiliate support dedicated to the provision of electric distribution service in

years when there is no rate case. *Id.* In other words, ratepayers will be allocated an inflated portion of OGC expenses based upon rate proceeding expenses that are not provided annually or regularly by OGC. *Id.* Further, the overstated level of OGC affiliate support allocated to PPL in this proceeding will then be used in future proceedings to support similarly overstated OGC allocations. *Id.*

ii. ALJ's Recommendation

The ALJ found merit in I&E's rationale and recommended that in order to prevent the overstatement of legal expenses in non-rate case years, this reduction should be to the Affiliate Support (Direct) – Office of General Counsel expense claim. R.D. at 32.

iii. Exceptions

Exceptions were not filed by the Parties on this issue.

iv. Disposition

Finding it otherwise reasonable, we will adopt the recommendation of the ALJ. However, some accounting clarification is in order.

In PPL's Exhibit Future 1-Revised, Sch. D-6, an adjustment was made to O&M expenses to reflect its revision to rate case expense. In rebuttal testimony, PPL explained its adjustment. The original rate case expense claim of \$2.025 million was normalized over a two-year period, providing for an annual expense of \$1.013 million. Based upon opposing testimony, PPL revised this claim by removing the remaining \$674,000, representing its 2010 rate case expense, and by \$1.2 million, representing a duplicate entry. The \$1.2 million was budgeted by the OGC for this proceeding. PPL St. 8-R at 42. With these two adjustments, PPL's original O&M expense claim of \$1.687 million was revised to be a reduction to FTY O&M of \$0.861 million. Based upon these

two adjustments, which include rate case expense and a direct assignment of cost from the OGC, PPL's reduction to its collective O&M expenses for the FTY would appear to be properly reflected in Exhibit Future 1-Revised.⁷

The adjustment proposed by I&E and recommended by the ALJ to reduce the OGC allocated expense and to leave the \$1.2 million in rate case expense will not change the outcome of the revenue allowance in this proceeding. This proposed change would effectively reverse the decrease in rate case expense already included by PPL in its Future 1-Revised by \$1.2 million and reduce the OGC expense by that same amount. The impact would be an increase in rate case expense of \$1.2 million and a decrease in OGC expense of \$1.2 million.

Accordingly, we shall adopt the recommendation of the ALJ on this issue.

3. Storm Damage Expense Recovery

i. Positions of the Parties

PPL revised its total storm damage expense recovery claim due to the unavailability of insurance beyond the FTY. PPL Exc. at 20-26. PPL stated that without storm damage insurance, PPL's initial FTY expense claim as it related to insurance is moot. PPL's revised FTY storm damage expense of \$23.199 million includes the following: \$17.875 million for annual storm damage expenses and a proposal to amortize over five years the extraordinary storm expenses in excess of insurance recoveries of \$26.620 million incurred during major storms in August 2011, Hurricane Irene, and October 2011 at \$5.324 million per year for five years. PPL Exc. at 24-25; PPL Exh. GLB-9.

⁷ See, Exhibit Future 1-Revised, Schedule D-6.

PPL stated that among the details to be agreed upon before a rider may become effective are (1) provisions for interest on under and over collections; (2) timing of reconciliation; (3) reporting of storm damage expenses and revenue for their recovery; (4) methods for adjusting the annual level of the expense in rates; and (5) exact categories of storm damage expense that would be subject to the reconciliation. PPL M.B. at 71.

I&E recommended a simple five year average of total storm damage expenses, which would account for yearly fluctuations to determine an appropriate level of expense for ratemaking purposes. I&E's calculated five-year average of PPL's storm expenses from 2009 to 2011 inclusive is \$23.785 million. I&E St. 2 at 35. I&E also recommended that PPL establish either a reserve account or a rider to recover storm damage expenses. I&E St. 2-SSR at 4-5.

ii. ALJ's Recommendation

The ALJ recommended that PPL be directed to establish a storm damage reserve account, as proposed by I&E, to be submitted to the Commission for approval. R.D. at 39. If approved by the Commission, the ALJ found that the reserve account should be implemented when the insurance coverage provided by PPL's present provider expires. The ALJ also recommended that the statutory advocates be included in the development of this storm damage reserve account. R.D. at 39. The ALJ also approved PPL's original storm damage expense claim of \$26.699 million, which includes \$12.625 million for annual storm damage expenses not covered by insurance, \$8.75 million for insurance premiums and a five-year \$5.324 million amortization of PPL's 2011 extraordinary storm damage expense claim.

iii. Exceptions

In its Exceptions, PPL supports the ALJ's recommendation to establish a reserve/tracker mechanism with reconciliation for over and under collections. PPL states that it intends to propose such a mechanism in a filing to be made as soon after the Commission decision in this proceeding as practicable. PPL will request that the proposal be given expedited consideration so that it can become effective at the earliest possible date. PPL Exc. at 23. PPL also revised its expense claim because it will be unable to purchase insurance beyond 2012. PPL Exc. at 24-26. PPL's revised claim is comprised of \$12.625 million for expected storm damage not covered by current insurance; \$5.25 million for the normal ongoing level of storm damage previously covered by insurance beyond 2012; and a five-year amortization of \$5.324 million for the extraordinary loss incurred in 2011, for a total revised expense claim of \$23.199 million.

In reply, I&E encourages the Commission to require PPL to meet with the statutory advocates to develop a rider within ninety days of Order entry. I&E R.Exc. at 13.

iv. Disposition

Based upon our review of the record and the Parties' Exceptions and Replies to this issue, we agree with the ALJ's recommendation to adopt I&E's proposal for PPL to propose a Storm Damage Expense Rider for Commission review. R.D. at 39. The issues to be discussed between PPL and the public advocates shall include, but not be limited to, the following: (1) provisions for interest on under and over collections; (2) timing of reconciliation; (3) reporting of storm damage expenses and revenue for their recovery; (4) methods for adjusting the annual level of the expense in rates; and (5) exact categories of storm damage expense that would be subject to the reconciliation. Additionally, we approve I&E's recommendation, and so direct, that PPL file a rider for

storm damage expense recovery within ninety days of the date of entry of this Opinion and Order. PPL has stated its intention to file as soon as practicable after the Commission's entry of a final decision in this proceeding.

Recovery of PPL's revised FTY storm damage expenses of \$23.199 million shall be through base rates. Any recovery through a Storm Damage Rider shall be permitted only to the extent that such expense exceeds the amount included within base rates.

4. Payroll - Employee Complement

i. Positions of the Parties

PPL has proposed a budget for payroll based upon an employee complement of 2,002, which it states is necessary for the management and maintenance of the Company's transmission and distribution systems in order to meet the needs of customers. PPL St. 2-R at 8-9; PPL M.B. at 71.

The OCA has proposed reducing the payroll budget to allow for an employee complement of 1,943, which is PPL's average number of employees over the sixteen-month period prior to March 2012. OCA M.B. at 18. The OCA's proposal would reduce PPL's FTY wages, payroll taxes and benefits by \$3.740 million. OCA St. 1-REV at 17. In response, PPL asserted that it is in the process of filling 106 positions. PPL St. 2-R at 8; PPL M.B. at 72.

The OCA argued that the budgeted staff levels should be reasonably based on historic data. *See e.g., Pa. PUC v. PPL Gas Utilities Corporation*, 255 P.U.R. 4th 209, 242 (Pa. PUC 2007) (utility's complement claim was reasonable and supported by the record where at times the actual number of employees was greater than budgeted, because the number was supported by historic data). OCA M.B. at 18. The OCA also

noted that PPL's employee complement had declined from 1,974 in December 2010, to 1,943 in June 2012. OCA M.B. at 19. Thus, it is the OCA's opinion that since PPL had neither claimed nor proven that the lower complement had resulted in inadequate service, there is no evidence of record to support a need for the higher number of employees. OCA M.B. at 19.

ii. ALJ's Recommendation

The ALJ took notice that PPL's actual employee complement for the first three months of the FTY was, on average, seventy-one employees less than budgeted and that, as of June 2012, the Company's complement was 1,942, which was still one person lower than the OCA recommendation. OCA R.B. at 6. However, the ALJ found that PPL is most familiar with its own needs in terms of staffing, and that PPL's historical payroll supports a finding that the Company's claim is reasonable. R.D. at 41. Accordingly, the ALJ rejected the OCA's adjustment and recommended adoption of PPL's employee complement. *Id.*

iii. Exceptions

In its Exceptions, the OCA states that the ALJ erred in granting PPL's employee complement of 2,002 because, according to the OCA, it is not supported by the record. OCA Exc. at 9. The OCA further asserts that it is unlikely that PPL's complement will increase by three percent to achieve the budgeted 2,002 employee level by December 31, 2012. *Id.* at 11.

PPL replies that the OCA failed to recognize the appropriate level of staffing needed to maintain and manage PPL's system and instead relied upon a sixteen-month average complement ending March 2012 as the basis for its adjustment. PPL R.Exc. at 15.

iv. Disposition

We agree with the ALJ that PPL is most familiar with its needs in terms of staffing, and that PPL's historical payroll supports a finding that the Company's claim is reasonable. Further, we believe that the basis for the OCA's adjustment, while mathematically accurate, does not envision an appropriate level of staff needed to maintain and manage PPL's system. Accordingly, we shall deny the OCA's Exception on this issue.

5. Uncollectible Expenses

i. Positions of the Parties

PPL's total FTY uncollectible accounts percentage is 2.23%, representing an expense of \$42.099 million. This amount includes expected write-offs plus any change in the reserve for doubtful accounts due to increased accounts receivable, which are subject to write-off. PPL M.B. at 72.

I&E's position is that PPL's proposed reserve allowance for uncollectible accounts expense should be rejected because that methodology is subject to manipulation and does not reflect PPL's actual expense or historic percentage write-off factor. I&E St. 2 at 5-6. Further, I&E stated that the Commission has no authority to permit recovery of hypothetical expenses not actually incurred by PPL, pursuant to *Barasch v. Pa. PUC*, 493 A.2d 653, 655 (Pa. 1985):

Although the Commission is vested with broad discretion in determining what expenses incurred by a utility may be charged to the ratepayers, the Commission has no authority to permit, in the rate-making process, the inclusion of hypothetical expenses not actually incurred. When it does so,

as it did in this case, it is an error of law subject to reversal on appeal.

I&E's analysis presents PPL's actual net write-off uncollectible percentages from 2007 to 2011, which is based upon the following data supplied by PPL in response to interrogatory I&E-RE-10:

Actual Net Write-Off Uncollectible Percent				
2007	↑ 2008	↓ 2009	↓ 2010	↑ 2011
1.57%	1.72%	1.63%	1.49%	1.97%

I&E Exh. No. 2, Sch. 1 and 2; I&E MB at 22. Additionally, I&E stated that its analysis clearly showed that PPL's proposed 2.23% write-off factor is unsupported by record evidence. I&E notes that in determining the Purchase of Receivables program administrative factor percentage in PPL's 2010 base rate case, the ALJ found that use of a five-year average, as proposed by I&E here, is appropriate. *Id.* at 23 (citing *Pa. PUC v. PPL Electric Utilities Corporation*, Docket No. R-2010-2161694 (Order entered December 16, 2010)).

ii. ALJ's Recommendation

The ALJ concluded that PPL's use of a FTY permits forecasting in terms of using real data to forecast the final uncollectibles for 2012, which is sufficient to ensure that the Company's uncollectibles will be covered. Doubtful accounts, however, present an unmeasurable, and unsupported, factor which the ALJ disallowed. R.D. at 42.

I&E used five years of data in its calculation, which includes four years of recession and two years post-rate cap. The final I&E recommendation is based on the 2009-2011 three-year average, which is confirmed by I&E's five-year average, each yielding a 1.70% uncollectible rate. The ALJ stated that it is evident that the highest

historic percentage of uncollectible accounts between 2007 and 2011 is below PPL's requested 2.23% recovery rate. Further, the ALJ found that PPL's proposed increase in the uncollectible rate is unjustified. Accordingly, the ALJ found that the methodology and result proposed by I&E is reasonable and should be adopted by the Commission.

iii. Exceptions

In its Exceptions, PPL states that an historic three-year average, as proposed by I&E and recommended for approval by the ALJ, is not appropriate because it is inconsistent with the ongoing increase in write-offs over the last three years and because the three-year average is inconsistent with actual, current data. PPL Exc. at 30. PPL explains that the goal in this proceeding should be to set rates which reasonably reflect future conditions. The three-year average relied upon by the ALJ included 2009, when PPL's generation supply rates were capped. Since then, PPL's electric supply rates for provider-of-last-resort service have increased significantly, when compared to prior periods where the generation supply rate cap was in effect. Not surprisingly, PPL experienced increases in the number and dollar amounts of uncollectible accounts since the generation rate cap has ended. *Id.* In addition, PPL and its customers continue to experience the effects of the recession. *Id.* PPL, therefore, asserts that the unfavorable economic conditions adversely affect uncollectible accounts expense and the use of a three-year average where uncollectible accounts expenses are increasing will, by definition, understate current costs. Accordingly, PPL believes that there is no basis for using a three-year history to calculate PPL's FTY uncollectible accounts allowance. *Id.* Lastly, PPL excepts to the ALJ's disallowance of its proposed increase in bad debt reserve. PPL states that elimination of this adjustment would be improper because the reserve includes charges for the increase in accounts receivable that are subject to eventual write-off. *Id.* at 31.

In reply, I&E contends that PPL ignores the facts, cited by the ALJ, that the five-year average, commencing in 2007 and extending through 2011, includes not only two years of data following removal of the generation rate cap (2010 and 2011), but also four years of data from the continuing recession (2008, 2009, 2010, and 2011). I&E R.Exc. at 6-7. I&E believes that while citing an increase in the number of accounts and uncollectible dollars from 2009 through 2011, PPL has misconstrued those facts to claim there is an ongoing increase over the last three years. I&E R.Exc. at 7.

I&E also contends that the facts do not support PPL's claimed 2.23% uncollectible accounts expense rate unless the Commission looks at only a snapshot of six months of experience in the first part of the FTY and then extrapolates that to an assumed level. I&E R.Exc. at 7. However, I&E notes that this Commission has never calculated an allowed uncollectibles expense rate on this basis. *Id.* Further, I&E claims that its calculation comports with the Commission's Regulations, the Company's own calculation of other claims, and PPL's calculation of its uncollectibles expense in both its 1985 and 2010 rate cases. *Id.* I&E submits that PPL's claims that the ALJ's allowance understates PPL's experience and that a three-year average fails to reflect ongoing increases is inaccurate. *Id.*

iv. Disposition

Based upon our review of the record evidence, the ALJ's recommendation and the Exceptions and Replies filed thereto, we shall adopt PPL's position on this issue as it reflects the level of uncollectible accounts on a going forward basis. In this proceeding, a FTY is the basis for ongoing utility expenses. We believe that I&E's historic analysis, although used by the Commission in prior decisions, is not warranted in this instance as it will not reasonably reflect future conditions. Accordingly, we shall deny I&E's Exception on this issue.

6. Revised Rate Case Expense and Normalization Period

i. Positions of the Parties

PPL's original rate case expense of \$1.687 million for the FTY was comprised of \$2.025 million for the instant proceeding and \$674,000 as an amortization recovery of its 2010 base rate case expense. PPL proposed to recover the \$2.025 million over a two-year period, or \$1.013 million per year. This two-year normalized amount of \$1.013 million plus the amortization portion of \$674,000, totaled \$1.687 million. Subsequently, PPL revised its rate case expense claim to remove its proposed amortization expense of \$674,000 and \$1.2 million, which PPL inadvertently included in both rate case expense and PPL Services-Office of General Counsel. These two adjustments have been reflected in PPL Exhibit Future 1-Revised, Schedule D-6, and result in a reduction to O&M expense of \$861,000.

PPL proposed a two-year normalization period to recover the rate case expense associated with the instant proceeding and argued that a two-year recovery period was appropriate given the pressure that its capital spending program will place on earnings. PPL's planned rate base capital expenditures of approximately \$1.7 billion over the next two calendar years represent an increase in PPL's total net measure of value as of December 31, 2012, exceeding fifty percent. PPL MB at 76. PPL asserted that with such a significant capital investment over the next two years, it seems more likely than not that its next base rate case could be filed during or before 2014. Further, PPL stated that even though it may request a distribution system improvement charge (DSIC), that mechanism is capped at five percent of revenue, which would do little to offset the incremental revenue requirement associated with the significant investment in rate base projected over the next two years. For these reasons, PPL believes that a two-year normalization of rate case expense is appropriate. *Id.*

The OCA advocates using a three-year period because PPL's last three rate cases filed in 2004, 2007, and 2010, were held exactly three years apart. The OCA's position is that it is the historical filings, not the actual intentions of the utility, which will guide the determination of the normalization period. OCA M.B. at 26. (citing *Pa. PUC v. City of Lancaster*, Docket No. R-2010-2179103 (Order entered July 14, 2011); *Pa. PUC v. Metropolitan Edison Company*, Docket No. R-00061366 (Order entered January 4, 2007)).

I&E agreed that the normalization period should be determined by the historical filings and, accordingly, recommended a thirty-two month normalization period based upon PPL's last four base rate filings.

Thus, I&E's recommended allowance for expenses associated with the instant rate proceeding is \$759,375. This is calculated by dividing PPL's \$2.025 million rate case expense claim by thirty-two months and then multiplying the result by twelve months to arrive at a normalized level of expense. [$\$2,025,000 / 32 \text{ months} = \$63,281.25$; $\$63,281.25 \times 12 \text{ months} = \$759,375$] This reduces PPL's FTY claim by \$253,625. ($\$1,013,000 - \$759,375 = \$253,625$).

ii. ALJ's Recommendation

PPL has agreed to two adjustments regarding its claimed rate case expense. First, PPL has removed the prior base rate expense claim of \$647,000 from its FTY total. Second, PPL has removed from total FTY expenses the \$1.2 million double count of rate case expense as described above.

As discussed above,⁸ the ALJ recommended the double count of \$1.2 million of legal fees included by PPL in both its rate case expense claim and its PPL Services-OGC, be removed from the PPL Services expense and not the rate case expense as requested by PPL.

Regarding the normalized recovery period for allowable rate case expense, the ALJ found that the OCA and I&E used the appropriate historic analysis methodology. The ALJ found I&E's analysis to be more accurate because it used the filing date of each of the last four base rate cases to develop a normalized period reflective of PPL's actual base rate filing frequency. Therefore, the ALJ recommended adoption of I&E's thirty-two month recovery period.

iii. Exceptions

In its Exceptions, PPL notes that in late 2008, it conducted a comprehensive study to assess the age, condition and performance of plant in order to develop a strategy for capital replacements in order to avoid the cost and reliability of service effects of aging infrastructure. Based on this study, PPL embarked on a ten-year capital plan to replace, maintain and improve plant and anticipates adding \$1.6 billion in plant from 2012 through 2016. Rate case history prior to 2010 does not reflect this construction program. PPL believes that plant expenditures of this magnitude will necessitate a base rate case within two years, if not sooner. Based upon PPL's capital improvement plan, PPL also believes it is unreasonable to rely on an historic pattern of rate cases that extends back eight years to 2004 to determine the appropriate period for normalization of rate case expenses. PPL Exc. at 35.

⁸ See discussion in the Office of General Counsel section above.

In reply, I&E states that citing no error by the ALJ, PPL repeats the same argument rejected by the ALJ, namely, that because of infrastructure plans, rate case history prior to 2010 is not an accurate reflection of the Company's future rate case plans. I&E R.Exc. at 8.

I&E explains that the law is well-settled that, absent exceptional circumstances, rate case expense is normalized based upon a party's filing history and not its presently stated intentions, no matter how unequivocally declared. *Id.* (citing *Popowsky v. Pa. PUC*, 674 A.2d 1149, 1154 (Pa. Cmwlth. 1996); *Pa. P.U.C. v. Borough of Media Water Works*, 72 Pa. P.U.C. 144 (1990)). I&E believes that there are no exceptional circumstances here. Conversely, I&E asserts that there are mitigating circumstances in the form of the effect of the DSIC. *Id.*

I&E contends that PPL has been finely attuned to its infrastructure needs since 2004 when it began regularly filing rate cases and, contrary to PPL's characterization, the current infrastructure improvement plan is not a sudden development that renders its recent rate case history irrelevant. *Id.* I&E notes that, recently, the Commission rejected a similar argument in which the Borough of Quakertown disputed a seven-year normalization based on filing history because anticipated intensive capital construction was under contract and had broken ground with an estimated 2013 completion date. *Id.* In affirming the ALJ, the Commission found that if the Borough filed sooner it "may be appropriate to consider a shorter normalization period going forward." *Id.* (citing *Pa. PUC v. Borough of Quakertown*, Docket No. R-2011-2251181 (Order entered September 13, 2012)).

iv. Disposition

Based upon our review of the record established in this proceeding, the ALJ's recommendation, the Exceptions and the Replies filed thereto, we shall reverse the

ALJ and grant the Exception of PPL on this issue. As previously discussed, this proceeding is premised upon a FTY and, based upon that criterion, certain expenses may now be based upon future expectations. We believe that the normalization period for rate case expense is one of those expenses. We fully support PPL's capital expenditure program and expect that it will proceed into the future as explained by PPL. Further, we can reasonably expect that PPL will file its next base rate case much closer to a twenty-four month interval than to a thirty-two month interval as proposed by I&E and the OCA. Accordingly, we shall grant the exceptions of PPL on this issue.

7. CEO's Proposed Increase in LIURP Funding

i. Positions of the Parties

PPL has proposed no changes in its universal service programs (USPs) nor to the funding for them, as these are subject to separate proceedings. *PPL Electric Utilities Corporation Universal Service and Energy Conservation Plan for 2011-2013*, Docket No. M-2010-2179796 (Order entered May 5, 2011). This was a litigated proceeding, with the participation of interested parties.

PPL's USPs include OnTrack (PPL's customer assistance program), WRAP (PPL's free weatherization program or Low Income Usage Reduction Program), Operation Help (PPL's hardship fund for customers with incomes at or below 200 percent of the federal poverty level, and CARES (PPL's Customer Assistance and Referral Evaluation Services, which connects customers with local community based organizations offering short-term help to customers at or below 200 percent of the federal poverty level). PPL St. 9 at 3-4.

PPL's currently effective USPs were approved by Commission Order entered May 5, 2011, at Docket No. M-2010-2179796, and run through December 2013. In June 2013, PPL will submit to the Commission for review and approval its USP plan

for years 2014 through 2016, and will include therein proposals for any necessary or appropriate changes to the current programs and services available to low-income customers. PPL M.B. at 77.

CEO argued that PPL's last increase of \$250,000 in the 2011-2013 USP case was inadequate to serve the needs of the low-income customer base and suggests that funding increase from \$8.0 million to \$9.5 million for PPL's WRAP Program. CEO disagreed with PPL's position that a base rate case is not the proper place for this argument, citing former rate cases that have evaluated the low-income plan budgets.

CEO pointed out that the funding for WRAP increased only 3% in the USP case, which translates into an additional 106 customers per year at the average cost of \$2,349, an increase not consistent with the increased number of low income customers in PPL Electric's territory, which CEO argues is 44% based on the 2008 census. CEO M.B. at 5; CEO St. 1 at 7. CEO continues that the usefulness of a well-funded LIURP program has long been recognized by the Commission as a tool for lowering heating bills, thus creating a heating bill that the customer is more likely to pay. CEO M.B. at 5-6. In addition, CEO states that the higher prices resulting from this proceeding will be effective January 1, 2013, a full year prior to the end of the effective period from the current USP case. CEO R.B. at 2. It is CEO's opinion that refraining from addressing this issue now will deprive low-income customers of timely relief from a rate increase. CEO R.B. at 3.

PPL countered that the increase in low-income customers in its service territory should not be viewed in isolation. Rather, consideration needs to be given to the cost impact on other residential customers, the ability of the community based organizations (CBOs), which administer the programs, to deliver additional services, and the availability of funding from other sources. PPL advocated for the consideration of all of these issues within the triennial filings for approval of the plans themselves, where all entities involved may participate. PPL St. 9-R at 6; PPL M.B. at 79.

I&E opposed CEO's proposal because it fails to consider the total increase in the funding of universal service benefits in recent years. Since 2004, over three base rate cases, the funding for the OnTrack program increased from \$9.5 million to \$41.2 million, and from 2000 to 2008, weatherization funding grew from \$5.7 million to \$8 million. I&E M.B. at 66-67. I&E stated the following:

Through 2012 PPL ratepayers will be compelled to contribute \$75.35 million annually to the funding of PPL's USP benefits. That mandatory ratepayer funding is projected to increase to \$78 million by 2014. The trajectory of mandatory ratepayer funding of PPL's universal service benefits has skyrocketed upward, increasing 122% from 2008 to 2011 and projected to increase by 145% through 2014. I&E submits that PPL's ratepayers are contributing sufficiently towards relief for their low-income neighbors. PPL's LIURP funding should remain at its current \$8 million.

I&E M.B. at 68.

ii. ALJ's Recommendation

The ALJ found that base rate cases are the traditional forum for budgets of low-income plans, but in recent years, the Commission has required companies to file separate cases to address the USP budgets. R.D. at 44-45. PPL has a Commission-approved plan in place, including a budget. R.D. at 45.

The ALJ continued by observing that the USPs for EDCs, including PPL, are filed every three years and concentrate on the programs included in the customer assistance portfolio. After noting that, in a base rate case, any part of the Company's tariff may be brought into question, the ALJ stated that as an issue raised by another party, the burden of proving that the universal service issues deserve additional funding belongs to the party raising it – here, CEO. *Id.*

The ALJ concluded that the Commission's institution of separate proceedings for these plans is indicative of a preference to address the issues within those proceedings. Therefore, the ALJ recommended that CEO's proposed increase in funding be denied. However, the ALJ encouraged CEO to participate in the triennial plan reviews. *Id.* at 46.

iii. Exceptions

In its Exceptions, CEO submits that the Commission has a statutory duty to ensure that a company's USPs are appropriately funded and available. Further, CEO contends that a proceeding that results in a rate increase to low-income customers would require the Commission to determine the effect of the rate increase on whether those USPs are, or remain, appropriately funded and available. CEO Exc. at 6. CEO alleges that to postpone consideration of universal service funding to a time after a rate increase takes effect, and to a non-adversarial proceeding, is contrary to the Commission's past practice and its statutory duty. *Id.*

PPL responds that the ALJ properly rejected CEO's proposal because the USP costs are no longer recovered through base rates. PPL R.Exc. at 22-23. I&E also supports the ALJ's recommendation on this issue. I&E R.Exc. at 14-15.

iv. Disposition

We agree with the ALJ, PPL and I&E on this issue. Recent Commission practice is to address all aspects of USPs through the triennial filing process and to collect all revenues through a rider to base rates. We believe this process has provided, and will continue to provide, the customers who rely upon USPs with appropriate funding levels on a timely basis. Accordingly, we deny the Exceptions of CEO on this issue.

8. Consumer Education Expenses

i. Positions of the Parties

PPL's consumer education program was mandated and authorized by the Commission's Final Order in *PPL Electric Utilities Corporation Consumer Education Plan for 2008-2012*, Docket No. M-2008-2032279 (Order entered July 18, 2008), which was designed to communicate Energy Education Standards to customers. The goal was to educate consumers in each EDC's service territory regarding (1) the expiration of rate caps; (2) ways to reduce energy consumption and, thereby, lower bills; and (3) the availability of retail competition.

PPL's FTY consumer education expense claim of \$7.976 million is comprised of \$5.482 million associated with the final year of PPL's Commission-approved Consumer Education Plan (CEP), plus \$2.494 million for three Retail Markets Investigation (RMI) mailings and customer protections regarding the Eligible Customer List (ECL), which PPL proposed to collect through a CER. PPL St. 5-R at 28-29.

I&E and the OCA opposed portions of PPL's proposal. I&E pointed out that PPL's proposed CER is designed to recover costs of the RMI initiatives, and that any costs related to education regarding those initiatives should be recovered through that rider and not included in base rates. While I&E does not object to recovery of the Commission mandated RMI costs and costs related to the ECL mailings, it notes that these should be recovered under the CER, if it is approved, and removed from base rates. I&E points out that the Commission and its EDCs are moving into the next phase of retail competition and that shopping and energy efficiency are more effectively addressed by the Act 129 Energy Efficiency and Conservation (EE&C) Plan and the RMI mandates. These are funded through the Act 129 Rider and the proposed CER. I&E St. 2 at 44; I&E M.B. at 62-63.

The OCA recommends that the Company's consumer education funding be set at \$5,400,000, annually, based on the budget amount approved in the 2008-2012 Consumer Education Plan. OCA MB at 29.

ii. ALJ's Recommendation

The ALJ found that the Commission's mandates must be funded, and the issue here is the best method of funding. While PPL must be reimbursed fully for its prudent expense, there must be a limit to the amount that should be spent. The ALJ concluded that the I&E proposal to recover the costs through a CER is the best choice, as it fully funds the Commission's mandates but does not waste ratepayer money on duplication.

Accordingly, the ALJ recommended that funding for PPL's CEP lapse at the end of the FTY and that the education costs of \$2.494 million incurred in carrying out the RMI mandates be recovered using the CER and, thus, removed from the allowed increase in base rates associated with this proceeding. R.D. at 49.

iii. Exceptions

In its Exception, PPL explains that the ALJ would disallow complete recovery of costs associated with PPL's Commission-approved Consumer Education Plan, which promotes and encourages the competitive retail market for electric generation in PPL's service territory and encourages conservation, beyond 2012. PPL Exc. at 26. The issue presented here, as viewed by PPL, is whether the Commission recognizes the need for the Energy Education Standards established in the Commission's Final Order on *Policies to Mitigate Potential Electricity Price Increases*, at Docket No. M-00061957, and wants the Consumer Education Plan to continue. According to PPL, if the Plan is to

continue, the Commission should approve PPL's claim of \$5.482 million for that Plan, in addition to other consumer education expenses. If not, PPL states that the ALJ's recommendation should be adopted on this issue, and PPL will discontinue the program. *Id.*

I&E rejoins that despite PPL's assertion otherwise, the Act 129 Plan provides both financial incentives as well as education about energy efficiency, rendering the CEP duplicative. See I&E St. 2-SR at 47-48, citing PPL's *Final Report for Year 2 of PPL Electric Utilities Corporation's Act 129 Plan*, at Docket No. M-2009-2093216. I&E R.Exc. at 14. In addition, I&E states that while the specific activities and programs may differ, the goals under all of these programs are the same: (1) to educate customers about shopping and efficiency; and (2) to provide financial incentives to modify behavior. Accordingly, I&E continues to urge that PPL's five-year plan and its \$5.4 million annual cost should be allowed to lapse naturally at the end of year 2012. *Id.*

iv. Disposition

As discussed above, we agree that Commission mandates must be funded. With regard to the recovery of Act 129 costs, we believe that it is proper to recover these costs through a rider to base rates. It is unknown whether the Act 129 expenses discussed in this section will be in place for many years or for only a few years, which supports recovery through a rider to base rates. Accordingly, we shall approve the education costs incurred in carrying out RMI mandates as expenses to be recovered through the CER Rider.

Regarding continued recovery of PPL's CEP costs of \$5.482 million, we find that the record supports allowing these pre-Act 129 expenses to lapse at the end of the FTY. Accordingly, we shall deny the Exceptions of PPL on this issue and reduce PPL's O&M expenses by \$5.482 million.

9. CAP (Customer Assistance Program) Outreach

i. Positions of the Parties

In its Exceptions, the OCA states that the ALJ did not address its recommendations regarding CAP outreach initiatives. OCA Exc. at 12-13. The OCA proposed three specific outreach initiatives: (1) that PPL engage in a direct-contact outreach program aimed at a population of customers that are both confirmed low-income and 120 days or more in arrears; (2) that all shut-off notices to confirmed low-income customers be modified so that they also contain a notice of CAP availability and the means of accessing CAP; and (3) that PPL engage in a direct-contact outreach program focused on customers 120 days or more in arrears whether or not those customers are confirmed low-income customers. OCA St. 4 at 33-34; OCB M.B. at 115.

PPL noted that it is not opposed to modifying its termination notice to include information about CAP so long as it does not add another page to the termination notice because that would increase the cost. PPL St. 9-R at 22. Further, PPL would not consider a requirement to have two separate termination notices, one for confirmed low-income customers and one all other residential customers. *Id.* at 23. PPL further stated that it is willing to propose the content and format of the new information on the termination notice and review it with Commission staff and interested Parties. *Id.*

Regarding the OCA's first and third recommendations, PPL states that these should not be adopted. PPL asserts that its current outreach programs are sufficient and that the OCA has not provided evidence that more outreach is needed to contact confirmed low-income customers who are 120 days or more in arrears. PPL St. 9-R at 22. Further, most residential customers with overdue balances or terminated accounts call PPL to address their concerns. *Id.* at 23. Depending on a customers' status in the

collection process, PPL has concerns about sending them a mixed messages regarding the requirements stated in the collection notices versus the content of the targeted outreach.

Id. at 23-24.

ii. ALJ's Recommendation

As noted by the OCA, the ALJ's Recommended Decision did not address this issue.

iii. Disposition

Based upon the testimony of the Parties, we shall grant the OCA's Exception, in part, with regard to its second recommendation that shut off notices to confirmed low income customers include information about CAP. However, we expressly acknowledge and accept PPL's willingness to propose the content and format of the new information on the termination notice and review it with Commission Staff and interested Parties. We encourage PPL to proceed in a timely manner, in this regard. Further, PPL should submit its proposed content and format of the new notice to the OCA and the Commission's Bureau of Consumer Services for review. Lastly, we agree with PPL that their current outreach programs, as discussed in testimony, are well designed and that the OCA has not provided sufficient evidence to support its first and third recommendations. Accordingly, we shall grant the OCA's exception in part, as discussed above.

E. Rate of Return

1. Introduction

The overall rate of return position of the Parties in this proceeding is summarized in the following tables:

PPL

Capital Type	Percent of Total %	Cost Rate %	Weighted Cost %
Debt	48.98	5.58	2.73
Common Equity	51.02	11.25	5.74
Total	100		8.47

PPL St. 11, Exh. PRM-1, Sch. 1.

PPL modified its overall return to reflect the actual issuance of \$250 million of long-term debt on August 24, 2012, at an interest rate of 2.61%. This update resulted in the following revised rate of return position of PPL:

PPL Revised

Capital Type	Percent of Total %	Cost Rate %	Weighted Cost %
Debt	49.22	5.50	2.71
Common Equity	50.78	11.25	5.71
Total	100		8.42

PPL M.B. at 91.

OCA

Capital Type	Percent of Total %	Cost Rate %	Weighted Cost %
Debt	52.84	5.58	2.95
Common Equity	47.16	9.00	4.24
Total	100		7.19

OCA St. 2, Exh. SGH-1, Sch. 11 at 1.

I&E

Capital Type	Percent of Total %	Cost Rate %	Weighted Cost %
Debt	54.89	5.58	3.07
Common Equity	45.11	8.38	3.77
Total	100		6.84

I&E St. 1 at 12.

The Company argued that the public advocates' recommendations relied on historically low interest rates instituted during the recent recession in an attempt to justify returns on common equity that are far below any allowed by this Commission in decades. Even in these difficult financial times, allowed ROEs have ranged between 9.75% and 10.99%. PPL M.B. at 87-88; PPL St. 12-R at 3-5. The Company averred that if either of these is adopted, Pennsylvania utilities will be placed at a disadvantage compared to other utilities in the country in terms of raising capital during what it terms to be a critical infrastructure replacement phase, PPL St. 12-R at 3-5, as well as at risk for another downgrade in its credit rating.⁹ Of course, accompanying this would be higher debt costs and potential limits to access to capital in difficult markets. PPL M.B. at 87-88.

2. Capital Structure

Capital structure involves a determination of the appropriate proportions of debt and equity used to finance the rate base. This is crucial to developing the weighted cost of capital, which, in turn, determines the overall rate of return in the revenue requirement equation.

⁹ Note that in presenting its 2004 rate case, PPL had an A minus rating, which it sought to retain at that time. See, Docket No. R-00049255, Recommended Decision of Administrative Law Judge Allison K. Turner, at 94.

a. Positions of the Parties

The Capital Structure recommendations of the Parties in this proceeding are summarized in the following table:

Capital Type	PPL (1)	I&E (2)	OCA (3)
	(%)	(%)	(%)
Debt	49.22	54.89	52.84
Common Equity	50.78	45.11	47.16
Total	100.00	100.00	100.00

- (1) PPL M.B. at 91 fn. 16
- (2) I&E St. 1 at 12
- (3) OCA St. 2 at 25

As noted above, PPL proposed the use of its actual capital structure of 49.22% long-term debt and 50.78% common equity. According to the Company, the legal standard in Pennsylvania for deciding whether to use a hypothetical capital structure in setting rates is that if a utility's actual capital structure is within the range of a similarly situated barometer group of companies, rates are set based on the utility's actual capital structure. PPL stated that only if the capital structure is atypical, outside of the range of the barometer group, should a hypothetical capital structure be used to set rates for a utility. PPL R.B. at 41 (citing *Pa. PUC v. City of Lancaster – Water*, 1999 Pa. PUC Lexis 37 at *17; *Pa. PUC v. City of Bethlehem*, 84 Pa. P.U.C. 275, 304 (1995); *Carnegie Natural Gas Co. v. Pa. PUC*, 433 A.2d 938, 940 (Pa. Cmwlth. 1981) (where a utility's actual capital structure is too heavily weighted on either the debt or equity side, the Commission must make adjustments)).

Both I&E and the OCA sought to utilize a hypothetical capital structure in this proceeding. I&E stated that a capital structure should be representative of the

industry norm and be an efficient use of capital. According to I&E, the use of a capital structure that is significantly outside the range of the industry's capital structure may result in an overstated overall rate of return. I&E advocated for the use of a hypothetical capital structure based upon an industry average for ratemaking purposes if the use of the utility's actual capital structure has the potential to overstate the overall cost of capital. I&E M.B. at 82.

The OCA submitted that the Commission should adopt a hypothetical capital structure for PPL as the Company's proposed capital structure is unnecessary to attract capital and would create an unreasonable cost burden for ratepayers. The OCA averred that its proposed capital structure of 47.16% equity and 52.84% debt is reasonable, consistent with how PPL has been capitalized over the last few years prior to this current rate filing, and similar to the manner in which the electric industry is capitalized. The OCA noted that of particular concern in this case is the percentage of common equity in the capital structure, since common equity commands a higher return than debt financing. OCA M.B. at 32-42

The I&E and the OCA recommendations to utilize a hypothetical capital structure are based upon use of a barometer group of companies with characteristics similar to PPL. The three Parties' barometer groups all contain comparison companies which are higher and lower than PPL's capital structure in this case. Error! Bookmark not defined. A barometer (or proxy) group is a group of companies that act as a benchmark for determining the utility's rate of return. I&E M.B. at 79. I&E noted that a barometer group is necessary because PPL is a private wholly owned subsidiary of PPL Corp. and is not publicly traded. According to I&E, using data from a group of companies is more reliable than data from a single company in that it smooths short-term anomalies and the use of a barometer group satisfies the long-established principle of utility regulation that seeks to provide the utility the opportunity to earn a return equal to that of similar companies with corresponding risks. I&E M.B. at 79 – 80.

PPL selected two barometer groups, an Electric Distribution Group (EDG) and an Integrated Electric Group (IEG). PPL's EDG group was based upon the following criteria:

1. Their stock is traded on the New York Stock Exchange;
2. They are listed in the Electric Utility (East) section of *The Value Line Investment Survey*;
3. They are not currently the target of a publicly-announced merger or acquisition; and
4. They do not have a significant amount of electric generation.

PPL's criteria for its IEG are identical except for criterion four, which requires that at least 75% of the companies' identifiable assets are subject to public regulation. PPL St. 11 at 4-5.

I&E used a barometer group comprised of Consolidated Edison, Dominion Resources, Nextera Energy, TECO Energy, PEPCO Holdings, and UIL Holdings. I&E St. 1 at 9-11. These were chosen by I&E based on the following criteria:

1. 50% or more of the company's revenue were generated from the electric distribution industry;
2. The company's stock was publicly traded;
3. Investment information for the company was available from more than one source;
4. The company was not currently involved/targeted in an announced merger or acquisition; and

5. The company had six consecutive years of historic earnings data.

I&E M.B. at 80.

The equity ratios for I&E's barometer group for 2011 range from 39.34% equity to 52.47% equity. I&E Exh. 1, Sch. 1 at 2. I&E then averaged the companies in its barometer group and developed a hypothetical capital structure based upon the average of 54.89% long-term debt and 45.11% equity for the FTY, or 55% debt/45% equity. I&E M.B. at 82.

The OCA used sixteen companies that had at least 70% of revenues from electric operations, did not have a pending merger, did not have a recent dividend cut, had stable book values and a senior bond rating between "A" and "BBB-". The OCA used "wires" companies as well as those with generation, and all were listed in *Value Line*. OCA St. 2 at 29-30. OCA M.B. at 52.

I&E argued that PPL's selected EDG and IEG barometer groups are flawed. According to I&E, Northeast Utilities must be excluded from PPL's EDG and Duke must be excluded from its IEG because their inclusion violates the Company's own presumably objective criteria number three in that Northeast is the subject of an announced merger with NSTAR and Duke is the subject of an announced merger with Progress Energy. I&E M.B. at 81. Also, I&E maintained that TECO Energy and Dominion Resources should be excluded from the Company's IEG and, instead, included in its EDG, because they derive more than 50% of their revenues from their regulated electric distribution sector. I&E further contended that the Company's IEG group should be disregarded in its entirety, because the group is too dissimilar in terms of business lines to be comparable to PPL in this proceeding. Specifically, I&E stated that PPL does not have regulated generation or gas distribution, properties common to SCANA Corp.

and Southern Co. included in the IEG, and neither company's revenues are derived more than 50% from electric distribution only. I&E St. 1 at 11-12. I&E M.B. at 80-82.

I&E asserted that PPL's claimed capital structure, if left unadjusted, overstated its capital needs by \$15 million. I&E M.B. at 83. According to the OCA, the Company's equity-rich common equity ratio would cost its ratepayers an additional \$10.6 million annually compared to the more economically efficient capital structure it has employed in recent years. OCA M.B. at 41.

b. ALJ's Recommendation

The ALJ concluded that the appropriate capital structure is the Company's actual capital structure of 49.22% long-term debt and 50.78% common equity. R.D. at 60.

c. Exceptions

In its Exceptions, the OCA states that PPL's proposed capital structure is unnecessarily burdensome to ratepayers, contains more common equity capital than the electric industry on average and is inconsistent with how PPL has been capitalized over the last several years prior to this rate case being filed. The OCA avers that its proposed capital structure of 47.16% equity/52.4% debt is reasonable, consistent with how PPL has been capitalized over the last few years and similar to the manner in which the electric utility industry is capitalized. The OCA notes that PPL's proposed capital structure is not really an "actual" capital structure, but rather a projection based on 2012 year-end data. OCA Exc. at 12-13.

Next, the OCA avers that the ALJ erred by finding that PPL's capital structure is not atypical, as the Company's proposed capital structure contains

significantly more equity than comparable utilities. According to the OCA, the average common equity ratio for publicly-traded electric and combination gas and electric utilities is 45.9% as reported by AUS Utility Reports in its May 2012 publication. Also, the OCA submits that the average common equity ratio of PPL's IEG sample group, and the S&P Public Utilities was 44.4% and 45% in 2010, respectively. The OCA opines that these ratios are far below the 50.78% common equity ratio requested by PPL. According to the OCA, the Company's own barometer group shows that a 45% common equity ratio is common in the industry for publicly traded companies. OCA Exc. at 13-14.

The OCA submits that Pennsylvania courts have upheld the use of a hypothetical capital structure where the utility's management adopts an actual capital structure that imposes an unfair cost burden on ratepayers. The OCA refers to *T.W. Phillips Gas and Oil Co. v. Pa. PUC*, 474 A.2d 355, 362 (Pa. Cmwlth.1984) and *Carnegie Natural Gas Co. v. Pa. PUC*, 433 A.2d 938 (Pa. Cmwlth. 1981) as support for its assertion. OCA Exc. at 14.

Next, the OCA reiterates that PPL's average common equity ratio from 2006 through 2010 was 43.7% of permanent capital per PPL's Exhibit PRM 1, Schedule 2. According to the OCA, PPL's requested ratemaking capital structure contains considerably more common equity than that with which it has been successfully capitalized historically. The OCA states that PPL plans to reduce its reliance on preferred stock and increase its reliance on more expensive common equity by means of a \$150 million capital contribution to PPL by its parent company, which is a management decision at PPL Corporation that changes the regulated capital structure of PPL. The OCA avers that this new test year capitalization will cost the Company's ratepayers approximately \$10.6 million more every year than the capital structure the Company has relied on for many years. The OCA submits that ratepayers should not bear this unnecessary and unfair burden and that the ALJ's recommendation should be rejected. OCA Exc. at 14-15.

I&E also excepts to the ALJ's recommendation on capital structure, stating that the ALJ erred in not applying a more cost-efficient capital structure for PPL, using I&E's calculated industry average, particularly because PPL's more expensive equity ratio is assigned by its affiliate. I&E avers that a hypothetical capital structure based upon an industry average should be used for ratemaking purposes if use of the utility's actual capital structure has the potential to overstate the overall cost of capital. I&E recommends a hypothetical capital structure based upon its industry average of 54.89% long-term debt and 45.11% equity for the FTY. According to I&E, PPL's proposed capital structure is neither representative of the industry norm nor an efficient use of capital. I&E Exc. at 15-16.

I&E submits that while the differences between PPL's and I&E's proposed capital structures are nuanced, PPL's actual capital structure includes sufficiently more expensive equity than less expensive debt, such that I&E's proposed adjustment is appropriate. According to I&E, imposing the industry average capital structure upon PPL saves ratepayers an annual \$15 million while still providing the Company competitive and effective means to finance its capital needs. This is particularly true, alleges I&E, given today's economic environment where debt rates have been and remain at all-time lows, and where PPL's capitalization is controlled by its affiliate, which is financially accountable to PPL's corporate parent and not PPL's ratepayers. I&E offers that if the corporate family is unwilling to take advantage of historically low interest rates to benefit its affiliated rate-regulated entity's ratepayers, then it is incumbent upon this Commission to do so. I&E Exc. at 17.

Next, I&E avers that contrary to PPL's characterization, the legal standard for employment of a hypothetical capital structure is not that the actual capital structure is "atypical." Rather, I&E maintains that use of a capital structure that is representative of the industry average presents a better option for PPL's efficient capitalization than the

capital structure assigned to PPL by its corporate family. According to I&E, use of a barometer group average is more reliable than comparing data from individual companies as individual company data may be subject to short-term anomalies that distort its return on equity. I&E notes that its industry average, as well as the common equity ratio averages from PPL's own barometer groups (44.8% for EDG, 45.1% for IEG and 45.3% for the S&P Public Utilities) more closely support I&E's recommended capital structure of 45% equity and 55% debt. I&E Exc. at 18.

In conclusion, I&E submits that while it agrees that PPL's actual capital structure does not deviate substantially from the industry range, the applicable legal standard is not that the capital structure must be "atypical" before a hypothetical structure should be considered. I&E notes that Commission decisions have specifically avoided setting numeric standards to define efficient capital structures, instead using standards such as "in proper proportions," "on balance," and not "too heavily weighted" one way or another. I&E opines that a \$15 million ratepayer expense based solely upon a capital structure chosen by the same PPL affiliates that benefit from the profitability of the rate regulated entity is unfair and unreasonable to ratepayers because it can be moderated without financial harm to PPL through a minor adjustment to the rate-regulated entity's capital structure. Therefore, I&E requests that its capital structure be adopted to impartially achieve a fair balance of ratepayer and shareholder interests. I&E Exc. at 21-22.

In reply, PPL states that the ALJ's recommendation should be accepted as its actual capital structure is not atypical and, pursuant to precedent, provides no basis to employ a hypothetical capital structure. Also, PPL states that it requires an equity ratio near the high end of the historic range employed by the barometer group companies to support its expanded infrastructure replacement program and its credit rating. According to PPL, the OCA and I&E misstate the circumstances that authorize the use of a hypothetical capital structure. PPL avers that both Parties rely on statements in cases

where the utility's equity ratio was outside the range of the equity ratios of barometer group companies to contend that a hypothetical capital structure should be employed in this proceeding where the actual equity ratio is clearly within the historic range of equity ratios employed by barometer group companies. PPL opines that while the cases cited identify the Commission's power to employ a hypothetical capital structure where the actual capital structure is extreme and atypical, they do not address how to determine when the actual capital structure is atypical. PPL R. Exc. at 3-4.

Next, PPL avers that its equity ratio is not atypical and provides no basis for use of a hypothetical capital structure and that its equity ratio is necessary to support its ability to attract capital and maintain its credit rating. PPL maintains that these are important considerations as it continues to ramp up its infrastructure replacement program. PPL avers that the OCA and I&E are ignoring the fact that PPL's unsecured bond was downgraded from Baa1 to Baa2 by Moody's Investors Service in April of 2010 due to Moody's opinion that PPL's cash flow metrics will decline from their recent levels due, in part, to the increased expenditures for capital investments needed to maintain PPL's aging delivery systems. According to PPL, the modest increase in the equity ratio was designed to avoid any further downgrade of PPL's rating to Baa3 and is consistent with projections of increasing equity ratios for other electric utilities as they expand their infrastructure replacement programs. PPL R. Exc. at 4-5.

Finally, PPL avers that the OCA's and I&E's claimed savings calculations are illusory because they incorrectly assume that PPL can undertake a dramatically expanded infrastructure program without strengthening its equity ratio. PPL states that it should not be placed at a disadvantage in raising capital and be placed at risk of a further downgrade by adopting a hypothetical equity ratio. Also, PPL avers that the OCA's and I&E's calculations are erroneous because they ignore the fact that a substantial part of the increase in PPL's equity ratio results from refinancing preference stock, which does not receive a tax deduction on dividends, with 50% equity and 50% tax deductible debt at a

small net savings to ratepayers. As a result, PPL explains that the Parties alleged savings from a lower equity ratio are significantly overstated because they incorrectly assume that the increased equity to refinance preference stock increases costs to ratepayers. PPL R. Exc. at 6.

d. Disposition

Upon our consideration of the evidence of record, the Recommended Decision and the Exceptions and Reply Exceptions filed by the Parties, we are persuaded by the position of PPL to adopt the Company's actual capital structure and affirm the recommendation of the ALJ. It is important to note that the actual capital structure represents the Company's decision, in which it has full discretion, on how to capitalize its rate base. This actual capitalization forms the basis upon which PPL attracts capital. PPL's debt cost rate of 5.50%, which all Parties have accepted for ratemaking purposes, fully reflects the capitalization determined by the Company to be appropriate. Absent a finding by the Commission that a utility's actual capital structure is atypical or too heavily weighted on either the debt or equity side, we would not normally exercise our discretion with regard to implementing a hypothetical capital structure. *See, Pa. PUC v. City of Lancaster –Water*, 1999 Pa. PUC Lexis 37 at *17; *Carnegie Natural Gas Co. v. Pa. PUC*, 433 A.2d 938, 940 (Pa. Cmwlth. 1981). With regard to these factors, we are persuaded by the arguments of PPL that its actual capital structure is not atypical, is within a range of reasonableness, and, pursuant to precedent, provides no basis to employ a hypothetical capital structure. Also, we are further swayed by PPL's assertion that it requires an equity ratio near the high end of the historic range employed by the barometer group companies to support its expanded infrastructure replacement program and its credit rating.

Accordingly, based upon the foregoing discussion, we shall deny the Exceptions of I&E and the OCA, and adopt the recommendation of the ALJ to utilize PPL's actual capital structure of 49.22% long-term debt and 50.78% common equity.

3. Cost of Debt

PPL proposed to use its expected cost of long-term debt and amortization of loss on reacquired debt for the FTY of 5.50%. PPL M.B. at 91. Both I&E and the OCA agree with PPL that 5.50% is the appropriate cost of long-term debt for purposes of this proceeding.¹⁰ I&E M.B. at 83; OCA M.B. at 46. This cost of debt was unopposed by any Party. R.D. at 60. No Exceptions were filed on this issue. Finding the PPL proposed cost of debt to be reasonable and appropriate, we adopt it without further comment.

4. Cost of Equity

a. Overview

Although there are various models used to estimate the cost of equity, the Discounted Cash Flow (DCF) method applied to a barometer group of similar utilities, has historically been the primary determinant utilized by the Commission. *Pa. PUC v. City of Lancaster – Bureau of Water*, Docket No. R-2010-2179103, at 56 (Order entered July 14, 2011); *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255, at 59 (Order entered December 22, 2004). The DCF model assumes that the market price of a stock is the present value of the future benefits of holding that stock. These benefits are the future cash flows of holding the stock, *i.e.*, the dividends paid and the proceeds from the ultimate sale of the stock. Because dollars received in the future are worth less than

¹⁰ As noted above, PPL adjusted its long-term debt cost to reflect the results of the Company's actual issuance of \$250 million of long-term debt, which reduced its weighted average long-term debt cost to 5.50%. PPL M.B. at 91.

dollars received today, the cash flow must be “discounted” back to the present value at the investor’s rate of return.

b. Summary

In the instant proceeding, only PPL, I&E and the OCA presented a position on a reasonable rate of return on equity (ROE). The Parties’ positions were generally developed through comparison groups’ market data, costing models, reflection or rejection of risk and leverage adjustments and a management performance adjustment, as will be further addressed, *infra*. The following table summarizes the cost of common equity claims made and the methodologies used by the Parties in this proceeding:

	DCF (%)	RP (%)	CAPM (%)	CE (%)	Risk (%)	Leverage (%)	MEA (%)	ROE (%)
PPL-EDG	9.67	10.75	10.58	11.60	1.20	0.7	0.12	11.13
PPL-IEG	9.69	10.75	11.28	11.60	1.20	1.18	0.12	11.43
OCA	8.97	7.3	-----	-----	0	0	0	9.00
I&E	8.38	-----	8.68	-----	0	0	0	8.38
ALJ	9.68	-----			0	0	0.06	9.74

PPL proposed a common equity cost rate of 11.25% based on the results of the DCF, Risk Premium (RP), Capital Asset Pricing Model (CAPM) and Comparable Earnings (CE) methodologies. PPL included a risk adjustment of 120 basis points, a leverage adjustment of 70 basis points, and a management performance adjustment of 12 basis points to arrive at its total request. PPL stated that the use of more than one method

provides a superior foundation to arrive at the cost of equity. According to PPL, at any point in time, reliance on a single method can provide an incomplete measure of the cost of equity. PPL St. 11 at 5-6.

Both the OCA and I&E argued that an 11.25% return on equity is excessive. The OCA stated that it would result in a shareholder windfall at the expense of ratepayers and would result in rates that are unjust and unreasonable. The OCA stated that “the current and near-term future economic outlook is one that includes a low cost of capital.” OCA St. 2 at 11-19. The OCA proposed a common equity cost rate of 9.00%, based primarily on the results of the DCF analysis without consideration of any of the additional adjustments proposed by the Company. The OCA utilized a CAPM, a Modified Earnings-Price Ratio (MEPR) and a Market-to-Book Ratio analyses as a check on the reasonableness of the DCF results. The OCA also cited numerous other jurisdictions which have awarded less than 10% returns on equity. OCA M.B. at 47-52 (citing *e.g.*, *In re PEPCO*, Order No. 85028 (MD PSC, July 20, 2012) (authorizing a 9.31% ROE)).

I&E recommended a cost of common equity of 8.38% based on the DCF methodology, with consideration of CAPM as a check, with no additional adjustments. I&E’s analysis used a spot dividend yield and a 52-week dividend yield, and a combination of earnings growth forecasts and a log-linear regression analysis growth rate. Using the standard DCF model formula,¹¹ I&E recommended a dividend yield of 4.89% and a recommended growth rate of 3.49%. I&E M.B. at 84-86.

¹¹ I&E St. 1 at 24.

c. Cost Rate Models

i. Positions of the Parties

PPL performed a RP analysis to determine the cost of equity, based upon the basic financial tenet that an equity investor in a company has greater risk than a bond holder in a company. PPL explained this is because all interest on bonds is paid before any return is received by the equity investor, and, upon bankruptcy or dissolving a company, the bond holder receives his capital before any capital is provided to the equity investors. PPL M.B. at 109-110; PPL St. 11 at 44; Appendix G at G-2.

PPL claimed that the RP method has common sense appeal to investors, who would expect to earn equity returns in excess of bond returns, as has been the case for any extended period in the capital markets. Accordingly, the Company explained the RP method as determining the cost of equity by summing the expected public utility bond yield and the return of equities over bond returns (the “equity premium”) over a historic period, as adjusted to reflect lower risk of utilities compared to the common equity of all corporations. PPL St. 11 at 49-50; PPL M.B. at 110.

The Company determined the RP cost of equity to be 10.75% as follows:

Interest Rate		Risk Premium		Cost Rate
5.25%	+	5.50%	=	10.75%

Id.

PPL also performed a CAPM analysis to estimate the cost of equity for the EDG and IEG and determined the risk free rate to be 3.75% based on current and near term project yields on long term treasury bonds. PPL St. 11 at 53-54. According to PPL,

the CAPM analysis determines a “risk-free” interest rate based on U.S. Treasury obligations and an equity risk premium that is proportional to the systematic (*i.e.*, beta) risk of a stock, which are combined to produce cost rate of equity. PPL St. 11 at 50-52.

PPL determined the market or equity premium to be 8.76% based upon an average of historic and projected market premiums. PPL St. 11 at 54; Appendix H at H-4 - H-6. PPL stated that betas are applied to the market premiums to adjust for electric company risks relative to the total market, and the betas are adjusted for the same reasons as the leverage adjustment to the DCF. PPL St. 11 at 52-53. Finally, the Company added a size adjustment to reflect greater risk for smaller firms relative to the market. PPL St. 11 at 54-55. The result of the PPL CAPM analysis was 11.78% for the EDG and 12.48% for the IEG. According to PPL, the results of the CAPM analysis indicate the upper range of the cost of equity analysis using the theoretical models typically employed in utility rate cases. PPL M.B. at 113.

PPL further performed a CE analysis. PPL noted that because regulation is a substitute for competitively determined prices, returns realized by non-regulated firms with similar risks can be used as a guide to determine a fair rate of return. PPL St. 11 at 56. Based on the PPL analysis, the comparable earnings group yielded an historical return of 10.9% and a forecasted return of 12.3%, which resulted in an average return of 11.6%. PPL M.B. at 113-114.

I&E stated that while it was not opposed to using the CAPM results as a comparison to the results of the DCF calculation, it is inappropriate to give the CAPM, RP and CE models comparable weight. I&E St. 1 at 38. I&E recommended against using the RP method and averred that it cannot be used because it relies on historic risk premiums achieved over bond yields which may not be applicable to the future. I&E St. 1 at 19.

Both the OCA and I&E used a CAPM as a check of reasonableness for their DCF calculations. However, both also believe there are shortcomings to this model, express concerns regarding its use and note their preference for using the DCF model to determine the cost of equity capital. I&E M.B. at 85; OCA St. 2 at 39.

I&E also performed an analysis of a return on equity using the CAPM methodology but gave no specific weight to its CAPM results because of its concerns that unlike the DCF, which measures the cost of equity directly by measuring the discounted present value of future cash flows, the CAPM measures the cost of equity indirectly and can be manipulated by the time period used. However, having presented two analyses – historic and forecasted – both of which are comprehensive in the time periods covered, I&E submitted that for purposes of providing another point of comparison, the 8.68% simple average of those two analyses confirmed the reasonableness of the I&E 8.38% return under its DCF calculation. I&E St. 1 at 31-36; I&E M.B. at 88-89.

In its CAPM analysis, OCA chose a risk free rate based on the long-term trend for Treasury Bonds, which it determined to be 4% for a forward looking CAPM analysis. Based on historical Morningstar data which shows an 11.8% return on stocks and a 5.8% return on long-term Treasury bonds since 1926, the OCA determined a risk premium of 6%; yielding an overall expected stock market return of 10% (4% + 6%). The OCA determined a beta of 0.69 based on *Value Line* beta coefficients for its electric group. Based on this analysis, the OCA's CAPM analysis yielded a cost of equity of 8.14% ($4\% + (0.69 \times 6\%)$). OCA St. 2 at 41-44.

I&E did not perform a CE analysis. I&E stated that the CE methodology is subjective in terms of the selection of comparable companies, has generally been rejected

by the Commission and, in PPL's particular analysis, compares projected returns of companies of dissimilar business and financial risks.¹² I&E M.B. at 92.

ii. ALJ's Recommendation

Based on the I&E position, the ALJ recommended that reliance on the RP method be denied. The ALJ concluded that the Commission's preferred method of determining a utility's ROE is the DCF model. The ALJ recommended utilization of the I&E DCF analysis. R.D. at 78, 93.

d. Dividend yields

i. Positions of the Parties

PPL derived the dividend yield by calculating the six month average dividend yields for each group and adjusting those yields for expected growth in the following year to produce the 4.67% for the Electric Delivery Group and 4.69% for the Integrated Electric Group. PPL St. 11 at 26; PPL M.B. at 104.

I&E stated that a representative yield must be calculated over a time frame sufficient to avoid short-term anomalies and stale data. The I&E's dividend yield calculation placed equal emphasis on the most recent spot (4.78%) and 52-week average (5%) dividend yields resulting in an average dividend yield of 4.89%. I&E St. 1 at 40-41; I&E M.B. at 86.

The OCA employed a 4.44% DCF adjusted yield, based upon the average dividend yield of its proxy group of similar companies. OCA St. 2 at 38; OCA M.B. at 55.

¹² I&E St. 1 at 19-23, 38-39.

ii. ALJ's Recommendation

For the reasons set forth by I&E, the ALJ recommended the adoption of the I&E proxy group and methodology for determining a 4.89% dividend yield. R.D. at 66.

e. Growth Rates

i. Positions of the Parties

PPL reviewed various methods of calculating investor expected growth rates and concluded that analysts' projections of growth rates are the best indicator of expected growth. PPL St. 11 at 34. PPL arrived at a range of growth rates from 4.50% to 5.08% for the EDG and from 4.59% to 6.00% for the IEG. PPL chose a growth rate of 5.00% based upon an average EDG growth rate of 4.87% and an average IEG growth rate of 5.14%. PPL M.B. at 105.

I&E used both earnings growth forecasts and a log-linear regression analysis data to calculate its expected growth rate. The I&E earnings forecasts were developed from projected growth rates using five-year estimates from established forecasting entities for the selected barometer group of companies, yielding an average five-year growth forecast of 4.79%. I&E St. 1 at 25-26.

I&E averred that investor forecasts may be biased and/or distorted by misestimates and, therefore, used a log-linear regression analysis to determine a more appropriate long term growth rate. I&E's log-linear regression analysis used historic earnings per share (EPS) from *Value Line* for the years 2006-2011, and the financial analysts forecasted growth rate to project EPS values for the FTY (2012) through 2016. The result of this log-linear regression analysis provided an average growth rate of 3.49%. I&E St. 1 at 25-30; I&E M.B. at 85-86.

ii. ALJ's Recommendation

The ALJ recommended using the 4.79% growth rate of I&E without the log-linear analysis. R.D. at 68.

Based upon the ALJ's recommendation with regard to her dividend yield recommendation of 4.89% and her 4.79% recommendation for PPL's growth rate, the ALJ recommended utilization of a DCF based 9.68% cost of equity, prior to the adoption of any of PPL's proposed adjustments.

iii. Exceptions

PPL excepts to the ALJ's conclusion with regard to its cost rate for common equity, stating that the ALJ's recommendation is far too low and should be increased to at least 10.5%. PPL avers that the principal error in the ALJ's analysis contained in the Recommended Decision is its sole reliance on an unadjusted DCF cost rate without any check on its validity. PPL submits that the ALJ simply rejects the results of other cost rate models based on alleged flaws in the models without recognition of the flaws of the DCF model. PPL Exc. at 6-7.

With regard to the ALJ's rejection of the RP method, PPL states that the RP method has particular applicability in this case because it reflects the prospective A-rated public utility bond yield under current market conditions. Therefore, PPL alleges, it reflects interest rates to be experienced by public utilities during the period rates will be in effect. According to PPL, using an A-rated bond yield produces an equity cost rate below PPL's cost rate because PPL is rated Baa2, indicating a higher cost of debt and equity. PPL notes that the OCA witness admitted that risk premiums tend to increase during periods of lower interest rates. PPL Exc. at 8; Tr. at 329-330. Accordingly, PPL

submits that it is likely that the lower interest rates currently being experienced indicate that the average historic premium understates the premium expected by investors for the future. This, PPL asserts, makes the RP analysis in this case conservatively low under current market conditions. PPL opines that the 10.75% RP provides a clear demonstration of the inadequacy of the unadjusted DCF analysis. PPL Exc. at 7-8.

With regard to the ALJ's rejection of the CAPM analysis as a check on the ROE recommendation, PPL submits that the ALJ simply accepted the OCA's and I&E's contention that there are "shortcomings" in the model. PPL avers that its CAPM analysis resulted in a cost rate of 10.58%, after removal of the 120 basis point size adjustment which the ALJ's rejects. PPL maintains that the ALJ did not provide any basis for rejecting the revised CAPM analysis excluding the size adjustment. PPL notes that the ALJ herself noted that the Commission has concluded that it is necessary to use other methods as a check on the results of the DCF, citing the Commission decision in PPL's 2004 rate case. PPL proffers that based on that decision, the ALJ's sole reliance on a DCF analysis with no leverage adjustment should not be adopted. According to PPL, the ALJ failed to follow the Commission precedent by either adding the leverage adjustment to the unadjusted DCF result or relying on other methods, such as the RP. PPL Exc. at 9-10.

PPL further excepts to the ALJ's apparent reliance on the Maryland *In re PEPCO* decision, *supra*, to justify an ROE less than 10%. PPL avers that neither the ALJ nor the OCA cites a further quote from the *In re PEPCO* decision provided in the Company's Reply Brief, which explained that the ROE that was approved for PEPCO reflected poor service quality and the effects of a revenue decoupling mechanism employed by PEPCO. PPL maintains that neither of those circumstances apply to PPL and, as such, the 9.31% ROE does not demonstrate the reasonableness of the ALJ's recommended allowance for PPL in this proceeding. PPL notes that the ROE should reflect prospective conditions, as relying too much on the past can risk under-estimating

the cost of equity capital that PPL will face as it seeks to raise capital to fund its expanded infrastructure improvement program during the period that rates set in this proceeding will be in effect. PPL Exc. at 16-19.

In reply, the OCA avers that the ALJ was correct in primarily relying on the DCF results to arrive at a reasonable ROE for PPL. The OCA states that the ALJ spent considerable effort in her Recommended Decision reviewing and discussing the results of the Parties' various ROE estimating studies, other than the DCF, and that PPL's criticism of the ALJ for relying on an unadjusted DCF result without any check on its validity is unwarranted. According to the OCA, the ALJ correctly concluded that the Commission primarily relies on the DCF method to establish a reasonable ROE. The OCA points out that the ALJ provided an extensive discussion and review of the results of the RP analysis, the CAPM analysis, and PPL's CE study, which led the ALJ to conclude that they should not be relied upon in this proceeding. The OCA submits that the ALJ's conclusion to rely primarily on the DCF method to arrive at an ROE recommendation is consistent with well-established Commission precedent and should be accepted. OCA R.Exc. at 4-8.

In response to PPL's Exception with regard to the *PEPCO* decision referenced by the ALJ, the OCA states that PPL's attempt to differentiate the *PEPCO* decision from its situation is without merit. The OCA submits that the quoted portions of the *PEPCO* Order only serve to reinforce the fact that the ALJ's recommendation of a 9.68% ROE is adequate and reasonable. According to the OCA, PPL is similar to PEPCO as it owns no generation, has no competition for distribution service and serves a heavily residential customer base, so PPL's attempts to distance itself from PEPCO is without merit and should be rejected. OCA R.Exc. at 10-14.

In its Reply Exceptions, I&E asserts that the ALJ's 9.68% calculated ROE is supported by the record and should be adopted. I&E asserts that as this Commission

recently confirmed, although it may review other results as a check, the Commission relies primarily on the DCF methodology. I&E R.Exc. at 18 (citing *Pa. PUC v. City of Lancaster – Bureau of Water*, Docket No. R-2010-2179103, at 56 (Order entered July 14, 2011)). Therefore, I&E avers that PPL's assertions are erroneous as the DCF has always been the primary standard. Notwithstanding this position, I&E posits that the reasonableness of the ALJ's recommendation was confirmed by I&E's two CAPM analyses, the historic and forecasted. According to I&E, its 8.68% simple average of its two CAPM studies, employing the same simple averaging PPL undertook of its four methodologies, confirmed the reasonableness of I&E's DCF return of 8.38%. I&E points out that since the ALJ rejected its log linear regression analysis, the ALJ's recommended 9.68% recommended ROE is substantially higher than the 8.68% check provided by its CAPM analysis. I&E R.Exc. at 17-19.

iv. Disposition

Upon our consideration of the record evidence, we agree with the finding of the ALJ that the Company's cost of equity in this proceeding should primarily be based upon the use of the DCF methodology. We also are persuaded by the arguments of PPL that it is important to temper the results of the unadjusted DCF results in comparison to the results from the other cost of equity methodologies as presented by the Parties in the context of this proceeding. Sole reliance on one methodology without checking the validity of the results of that methodology with other cost of equity analyses does not always lend itself to responsible ratemaking. We conclude that methodologies other than the DCF can be used as a check upon the reasonableness of the DCF derived equity return calculation. *See, Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255, at 67 (Order entered December 22, 2004). It is important to recognize that each of the Parties presenting a cost of equity position in this proceeding have done so. We also note that we historically have primarily relied upon the DCF methodology in arriving at previous determinations of the proper cost of equity and utilized the results of

methods, such as the CAPM and RP methods, as a check upon the reasonableness of the DCF derived equity return amount, tempered by informed judgment. As such, where evidence based on the CAPM and RP methods suggests that the DCF-only results may understate the utility's current cost of equity capital, we will give consideration to those other methods, to some degree, in determining the appropriate range of reasonableness for our equity return determination. Therefore, we are not in agreement with the ALJ that the proper ROE in this proceeding should be determined based strictly on the reliance of the unadjusted DCF calculations presented by the Parties.

In *Lower Paxton Township v. Pa. PUC*, 317 A.2d 917, 920-921 (Pa. Cmwlth. 1974), the Commonwealth Court recognized that the Commission may consider its judgment as well as other factors which affect the cost of capital, including the utility's financial structure, credit standing, dividends, risks, regulatory lag and any peculiar features of the utility involved. The Court stated that "the cost of capital is basically a matter of judgment governed by the evidence presented and the regulatory agency's expertise." *Id.* at 921. Here, we are guided by the legal analysis in *Lower Paxton*. In this case, we will rely upon the DCF methodology and informed judgment in arriving at our determination of the proper cost of common equity. In particular, we note that the evidence presented in this case based on the CAPM and RP methods produced a range of results that was consistently higher than the results produced by a DCF-only approach. This suggests that, while properly computed in the abstract, the DCF-only results understate the current cost of equity for PPL and that consideration should be given to the CAPM and RP evidence in determining the appropriate range of reasonableness. Furthermore, we note that the setting of the proper return on equity is even more critical in this proceeding as our Pennsylvania jurisdictional utilities implement plans to accelerate the greatly needed replacement of aging infrastructure. Attracting capital to Pennsylvania at reasonable rates to accomplish this infrastructure replacement has never been more important to PPL, its customers and the Commonwealth of Pennsylvania.

Based upon our analysis and review of the record evidence, we find that a range of reasonableness for the cost of equity in this proceeding is from 9.0% to 11.25%. We conclude that within that range, considering PPL's need to fund \$1.6 billion of planned distribution improvements between 2012 and 2016, a cost of common equity of 10.28% is reasonable and appropriate to incorporate into our return determinations under the circumstances of this proceeding. We note that this return on equity is exclusive of any of the PPL-requested adjustments to be discussed, *infra*. We note, further, that (1) the DCF-derived cost of equity ranged from 8.38% (I&E) to 9.69% (PPL); (2) the range determined from the RP methodology was 7.3% (OCA) to 10.75% (PPL); and (3) the range of the CAPM calculations was 8.14% (OCA) to 11.28% (PPL). Based upon our consideration and analysis of this evidence, as explained herein, we are of the opinion that an equity return of 10.28% is reasonable and appropriate for PPL.

Accordingly, the Exceptions of PPL are granted, in part, to the extent consistent with the foregoing discussion.

f. Leverage Adjustment

i. Positions of the Parties

PPL promoted a leverage adjustment in this proceeding, which it explained was designed to adjust the DCF cost rate for the different percentage level of debt in the capital structure when capital structure is calculated at the market prices of equity and debt securities as opposed to book value. PPL M.B. at 105.

PPL proposed a 70 basis point leverage adjustment to its EDG and a 118 basis point leverage adjustment to its IEG. PPL theorized that if regulators use the results of the DCF to compute the weighted average cost of capital based on a book value capital structure used for ratemaking purposes, the utility will not, by definition, recover its risk-

adjusted capital cost. PPL believed this is because market valuations of equity are based on market value capital structures, which in general have more equity, less debt and, therefore, less risk than the capitalization measured at its book value. PPL St. 11 at 35.

The Company pointed out that the Commission has accepted the leverage adjustment in a number of cases, including PPL's last fully litigated rate case in 2004. PPL M.B. at 107 (citing *Pa. PUC v. Pa. American Water Co.*, Docket No. R-0001639 (Order entered January 10, 2012) (60 basis point adjustment); *Pa. PUC v. Philadelphia Suburban Water Company*, Docket No. R-00016750 (Order entered August 1, 2002) (80 basis points); *Pa. PUC v. Pa. American Water Co.*, Docket No. R-00038304 (Order entered November 8, 2004) (60 basis points affirmed); *Popowsky v. Pa. PUC*, 868 A.2d 606 (Pa. Cmwlth. 2004); *Pa. PUC v. Aqua Pa. Inc.*, Docket No. R-00038805 (Order entered August 5, 2004) (60 basis point adjustment); *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-00049255 (Order entered December 22, 2004) (45 basis point adjustment); *Pa. PUC v. PPL Gas Utilities Corp.*, Docket No. R-00061398 (Order entered February 8, 2007) (70 basis points)).

According to PPL, use of the DCF alone, and without consideration of the leverage adjustment, significantly understates the cost of equity. PPL opined that when investors' expectations of future earnings are pessimistic due to factors including future regulatory allowances, there is the potential for the DCF to be circular and not market based. PPL St. 11 at 24; PPL M.B. at 108.

I&E argued that rating agencies assess financial risk based upon the Company's booked debt obligations and the ability of its cash flow to cover the interest payments on those obligations by using financial statements, particularly income statements, for their analyses, not market capitalization.

I&E pointed out that, while the Commission has granted this adjustment on occasion, it has also rejected it:

In a Blue Mountain Water Company case on remand from Commonwealth Court to clarify findings concerning fair rate of return, the Commission identified seven principles that were applied to analyze the company's required and lawful rate of return. The Commission's third identified principle stated that "[m]arket price-book value ratios are not a goal of regulation but a result of regulation, general economic factors and individual company's characteristics of management, operations and perceived future. *In general, we view a market-book ratio in the area of one-to-one as appropriate for regulated industry.*"¹³

In a 2008 case involving Aqua Pennsylvania, Inc., the Commission rejected the ALJ's recommendation for a leverage adjustment stating, "the fact that we have granted leverage adjustments in the past does not mean that such adjustments are indicated in all cases."¹⁴ In a 2007 Metropolitan Edison Company case, the Commission rejected the Company's financial risk increment related to the leverage difference between market capital structures and book value capital structures.¹⁵ Most recently in a City of Lancaster case, the Commission agreed with Ms. Sears' recommendation to reject the leverage adjustment, stating "any adjustment to the results of the market based DCF as we have previously adopted are unnecessary and will harm ratepayers. Consistent with our determination in *Aqua 2008* there is no need to add a leverage adjustment."¹⁶

I&E M.B. at 73-74.

¹³ *Pa. PUC v. Blue Mountain Consolidated Water Co.*, 1982 WL 213115, at 1 (emphasis added).

¹⁴ *Pa. PUC v. Aqua Pennsylvania, Inc.*, Docket No. R-00072711, at 38 (Order entered July 31, 2008).

¹⁵ *Pa. PUC v. Metropolitan Edison Co.*, Docket No. R-00061366, at 34 (Order entered January 11, 2007).

¹⁶ *Pa. PUC v. City of Lancaster – Bureau of Water*, Docket No. R-2010-2179103, at 79 (Order entered July 14, 2011).

I&E determined that there are six cases in which the Commission accepted the leverage adjustment, most recently in 2007. According to I&E, the adjustment has been proposed in sixty-eight cases over a twenty-three year period, yielding six successful results. Finally, I&E charged that PPL's formulae for the adjustment are flawed as it used formulae which do not appear in the research cited to support it. I&E M.B. at 100.

The OCA recommended against the Company's leverage adjustment because there was no evidence to support a risk difference between a market-based capital structure and a book value capital structure. Rather, according to the OCA, the claim that the DCF results should be increased by 70-118 basis points due to PPL's leverage adjustment is "not sound ratemaking." OCA M.B. at 60. The OCA submitted that no ROE-enhancing adder is needed or appropriate for PPL based on the facts of this matter. As the OCA witness testified:

While there are certainly many aspects of rate of return analysis that are subject to judgment and, thus, debate regarding the proper application of a particular technique, Mr. Moul's use of an imaginary risk difference between a market-based capital structure and a book value capital structure is not one of them. There is no evidence available in the literature of financial economics to support any risk difference between market-value and book-value capital structures. Miller and Modigliani (supposedly the source of Mr. Moul's "leverage" adjustment) do *not* compare market-value and book value capital structures.

OCA St. 2-SR at 4. (emphasis in original).

According to the OCA, PPL testified that when utility market prices exceed book values, a risk difference exists between market-value capital structures and book-value capital structures, and market-based cost of equity estimates should, therefore, be

adjusted upwards to account for that risk difference. The OCA noted that this is the basis for PPL's "leverage adjustment." OCA St. 2 at 55-56. The OCA witness testified as to the flawed nature of this theory, in relevant part:

There simply is no difference in financial risk when the market-value capital structure of a firm is different from the book-value capital structure. Financial risk is a function of the interest payments on the debt issued by the firm. That is, a firm's debt payments create financial risk and when the amount of debt used to finance plant investment increases relative to common equity the financial risk increases. Whether the capital structure is measured with market values or book values, the debt interest payments do not change and, therefore, financial risk does not change. As a result, market-value capital structures are useful as indicators of financial risk only when they are compared with other market-value capital structures (as Miller and Modigliani do in their treatise), and Mr. Moul's mixed-metaphor comparison of market-value and book-value capital structures has no economic meaning.

OCA St. 2 at 56.

The Company is making an improper comparison between market value capital structures and book value capital structures in order to claim that a financial risk difference exists. When utility common equity market prices are above book value, the capital structure measured with market values will have a higher equity percentage and lower debt percentages than the capital structure measured with book value. That does not mean, as the Company claims, that those different capital structure measures signify any difference whatsoever in financial risk.

OCA St. 2 at 61.

The OCA acknowledged that, in some cases, the Commission made an adjustment to a DCF based cost of equity such as that proposed by PPL.

However, the OCA claimed that, more recently, the Commission has not adopted PPL's leverage adjustment, as Mr. Hill testified:

[I]t is important to note that this Commission has rejected "financial risk adders" in Docket No. R-00061366 (Metropolitan Edison (Met Ed), Pennsylvania Electric, Opinion and Order, January 11, 2007, p. 136). The "financial risk adders" in the Met Ed case were based on the leverage/risk difference between market-value capital structures and book value capital structures, just as Mr. Moul's are. In addition, in Docket No. R-00072711, Aqua Pennsylvania, Inc., July 17, 2008, at pages 35 through 39, this Commission specifically rejected Mr. Moul's leverage/risk analysis—the same leverage/financial risk adjustment Mr. Moul uses in his testimony in this proceeding.

OCA St. 2 at 57. The OCA argued that other state commissions have uniformly recognized this type of adjustment as unwarranted in their decisions. OCA M.B. at 62 (citing *West Virginia Public Service Comm'n v. West Virginia-American Water Works*, 2004 W. Va. PUC Lexis 6 at *18 (2004)). In addition to the West Virginia Public Service Commission, other Commissions have rejected similar market-to-book adjustments to the DCF model. The District of Columbia Public Service Commission rejected a company's arguments that an adjustment to the DCF was appropriate to meet investors' requirements. OCA M.B. at 62 (citing *In the Matter of the Application of Washington Gas Light Company, District of Columbia Division, for Authority to Increase Existing Rates and Charges for Gas Service*, 2003 D.C. PUC Lexis 220 at *72 (2003)).

In its surrebuttal testimony in this proceeding, the OCA summarized the reasons this Commission should reject PPL's "fictional leverage" adjustment:

- The comparison of market value capital structures and book value capital structure to measure financial risk differences, is not supported in the literature of finance;

- There is no financial risk difference between market value and book value capital structures because interest expense (the actual source of financial risk) doesn't change, regardless of the capital structure measurement perspective;
- One company cannot have two levels of financial risk (i.e., one based on book value and one based on market value);
- The DCF model does not "mis-specify" the cost of equity when market prices are different from book value, and utilities are able to attract capital on reasonable terms absent any so-called "leverage" adjustment;
- Moul's "leverage" adjustment is, fundamentally, a market-to-book ratio adjustment, and this Commission has rejected market-to-book ratio adjustments in the past;
- The "leverage" adjustment is based on the "fair value" of the capital employed in financing the utility operation, as such it is a surrogate for "fair value" rate base, which results in a revenue requirement higher than that required by law in a regulatory jurisdiction in which rates are to be based on original cost (depreciated book value);
- A utility market price significantly above book value indicates that investors expect that firm to earn a return above its cost of equity, but according to Mr. Moul's "leverage" adjustment the higher the market price, the greater the upward adjustment necessary, which would exacerbate the over-recovery;
- The "leverage" adjustment recommended by Mr. Moul has been presented in dozens of regulatory jurisdictions. It has been rejected by all of those jurisdictions (including, recently, Pennsylvania).

OCA St. 2-SR at 11. The OCA submitted that for the reasons just discussed, and taking the record as a whole, such an adjustment should not be considered in this matter.

OCA M.B. at 60-64.

ii. ALJ's Recommendation

For the reasons developed by the OCA and I&E, the ALJ recommended that the Company's leverage adjustment be denied. R.D. at 76.

iii. Exceptions

PPL excepts to the ALJ's rejection of its proposed leverage adjustment, noting that the Commission has accepted a leverage adjustment in a number of cases, including PPL's last fully litigated rate case in 2004, where the Commission adopted a forty-five basis point adjustment. PPL avers that the ALJ appears to conclude that the OCA's and I&E's criticisms of the leverage adjustment are a basis to reject the leverage adjustment, despite the fact that it has been accepted on numerous occasions in the past and each of these criticisms have been offered in the past. PPL points out that the principal criticism offered by the OCA and I&E is that there is no risk difference between a capital structure where equity is valued at market as compared to book prices, because the amount of interest that must be paid on debt remains the same. PPL opines that the error of this argument is that the interest amounts are greater as a percentage of book equity capitalization than they are as a percentage of market equity capitalization. Therefore, asserts PPL, the risk of debt payments is less as a percentage of market equity capitalization than it is at book equity capitalization. PPL states that because the DCF sets the equity cost rate at market capitalization, it understates the investor cost rate when applied to the rate base. According to PPL, the ALJ erred in declining to include a leverage adjustment when relying solely on the DCF analysis to arrive at the recommended cost of equity. PPL Exc. at 11-16.

In reply, the OCA states that the ALJ was correct to reject the leverage adjustment, as she accepted the fact that artificially increasing the ROE based on a technique that finds no support in the financial literature, does not represent sound ratemaking. Additionally, the OCA avers that PPL's leverage adjustment has been thoroughly reviewed and rejected in virtually every regulatory jurisdiction where it has been proposed, noting that since 2007, PPL's witness has testified in twenty-four regulatory jurisdictions, and none has specifically accepted and utilized the "leverage/risk" adjustment. According to the OCA, there is no need for a leverage adjustment within the confines of standard regulatory practice or a need for such a mechanism in Pennsylvania. OCA R. Exc. at 8-10.

In its Reply Exceptions, I&E asserts that the leverage adjustment is wholly discretionary and, in this case, fundamentally unnecessary, not only for the reasons directly noted by the ALJ, but also because PPL's inputs into its 9.68% DCF calculation are already overstated. Further, I&E opines that today's investment market does not support PPL's ROE. According to I&E, both PPL's calculated growth and dividend rates within its DCF analysis already provide the equity boost that PPL seeks through its leverage adjustment. I&E explains that the PPL 5% growth rate was based on its average barometer group growth rates, which were flawed in I&E's opinion because they did not satisfy even its own criteria. I&E submits that though accepting its unadjusted growth rate of 4.79%, the ALJ nonetheless arrived at a calculated return on equity of 9.6%, the same DCF return calculated by PPL using inflated growth rates. I&E avers that because PPL's DCF calculation already has inflated inputs, a further upward boost from the leverage adjustment is unnecessary. I&E R. Exc. at 19-20.

iv. Disposition

Based upon our analysis of the evidence of record, we are persuaded by the arguments of the OCA and I&E that PPL's requested leverage adjustment is not reasonable and should be denied. The fact that we have granted leverage adjustments in a few select cases in the past as noted by PPL does not mean that such adjustments are warranted in all cases. The award of such an adjustment is not precedential but discretionary with the Commission. In fact, the Commission has rejected leverage/financial risk adjustments that are similar to the one proposed by PPL in this proceeding. *See, e.g., Pa. PUC v. Aqua Pennsylvania, Inc.*, Docket No. R-00072711, at 38-39 (Order entered July 31, 2008). Moreover, in the context of our determination, *supra*, of a reasonable return on equity for PPL of 10.28%, we conclude that there is no need to have an artificial upwards adjustment to compensate for any perceived risk related to PPL's market-to-book ratio. Accordingly, we shall deny the Exceptions of PPL and adopt the ALJ's recommendation to reject PPL's requested leverage adjustment.

g. Risk Adjustment

i. Positions of the Parties

PPL proposed a 120 basis point upward adjustment because the Company believes that as the size of a firm decreases, its risk and, hence, its required return, increases. Further, PPL used the SBBI Yearbook to argue that the returns for stocks in lower deciles had returns in excess of those shown by the simple CAPM. PPL St. 11 at 54-55.

Alternatively, I&E charged that PPL's rate of return recommendations are also grossly overstated by its assignment of several faulty assumptions of risk to PPL. I&E noted:

While some technical market literature supports adjustments relating to a company's size, in a critical point of distinction, this literature is *not* specific to the utility industry. On the other hand, utility-specific academic literature specifically argues against a size adjustment for utilities. A specific study of utility stocks and the size effect concluded as follows:

The objective of this study is to examine if the size effect exists in the utility industry. After controlling for equity values, there is some weak evidence that firm size is a missing factor from the CAPM for the industrial but not for utility stocks. This implies that although the size phenomenon has been strongly documented for the industrials, the findings suggest that *there is no need to adjust for the firm size in utility rate regulation*.¹⁷

As to unpredictability, I&E stated that "one cannot expect risky companies to always outperform less risky companies; otherwise they would not be risky." I&E M.B. at 101-103.

ii. ALJ's Recommendation

The ALJ recommended that PPL's proposed size adjustment be denied. R.D. at 82.

iii. Exceptions

No Party filed Exceptions on this issue with regard to the ALJ's recommendation. Finding the ALJ's recommendation to be reasonable, we adopt it without further comment. Accordingly, PPL's proposed size adjustment is denied.

¹⁷ I&E M.B. n. 220; I&E St. 1 at 55, citing Dr. Annie Wong, "Utility Stocks and the Size Effect: An Empirical Analysis," *Journal of Midwest Finance Association*, 1993, at 95-101 (emphasis added), reproduced in I&E Exh. I, Sch. 15.

h. Management Effectiveness Adjustment

i. Positions of the Parties

PPL included a twelve basis points management effectiveness adjustment to its return on equity claim. Both I&E and the OCA oppose any allowance for management effectiveness.

The Company summarized its evidence in support of this adjustment as follows:

PPL Electric's management is effectively controlling costs, while at the same time, providing customers with high quality service and expanded service options. As detailed in the Statement of Reasons, the Company has taken substantial efforts to improve productivity and manage costs, including, but not limited to: (1) new technology to improve productivity and including advanced meters; (2) a smart grid distribution automation system, which will provide direct reliability benefits to over 60,000 customers in the project area and lead to increased reliability benefits to all customers by providing system operators advanced and timely situational awareness and control capabilities through a wider deployment throughout PPL Electric's service territory; (3) a work and asset management system, which is a new large scale software solution that will improve associated work management business processes in order to more effectively and efficiently manage the portfolio of work; (4) several initiatives to improve storm processes including call handling time and volume; (5) increased investment to address aging infrastructure, which will have a positive, long-term benefit in controlling reactive operating costs; and (6) capital investment in information systems to support customer choice and to provide expanded self-service options for customers, which improves service to customers while controlling operating costs. In addition, the Company is testing and evaluating a variety of applications and features that will expand the capabilities of the current system and equipment over the next five years.

PPL M.B. at 116-117.

I&E argued that the twelve basis points sought by PPL translates into an additional \$3 million in rate revenues. Tr. at 335; I&E M.B. at 116. I&E argued further that there is considerable room for improvement in several areas, including preventable major outages, customer service calls answered within thirty seconds, the number and percentage of bills not rendered to residential customers and small businesses, and the number of disputes with no response within thirty days. I&E M.B. at 119-120. As I&E saw it, PPL's requested twelve basis point upward adjustment to the cost of equity is neither warranted nor supported. I&E opined that it should be rejected. I&E M.B. at 123.

The OCA agreed with I&E. The OCA referred to the \$832,000 that PPL has either agreed to pay or was ordered to pay in fines and penalties. OCA M.B. at 65.

ii. ALJ's Recommendation

The ALJ stated that PPL presented substantial evidence of management effectiveness in a number of areas, including advanced metering infrastructure, operating initiatives, customer contact center, customer education, energy efficiency programs, and customer assistance programs. According to the ALJ, the provision of safe, reliable, adequate and reasonable service is the minimum required by the Code, and simply meeting that standard does not warrant excessive rewards. However, the ALJ concluded that the actions taken by PPL in its response to Commission initiatives, and in providing excellent, albeit imperfect, service, in meeting the needs of its ratepayers and customers, merited a management effectiveness increase of six basis points. R.D. at 89

iii. Exceptions

In its Exceptions, PPL notes that the ALJ correctly summarized PPL's evidence presented to support its management performance adjustment. However, PPL criticizes the ALJ for recommending a six basis point adjustment in lieu of its twelve basis point request, as she relied on certain criticisms of PPL where the Company agreed to negotiate payments to resolve certain alleged violations of the Code or Commission Regulations. PPL avers that these limited circumstances do not provide a basis for denying PPL's requested twelve basis point adjustment to the cost of equity. PPL Exc. at 19.

The OCA excepted to the ALJ's recommendation, stating that the ALJ erred by awarding any management performance bonus as the evidence of record does not support such a conclusion. The OCA maintains that all regulated utilities in Pennsylvania are required to provide safe, adequate, reasonable and efficient service as a matter of law. 66 Pa. C.S. § 1501. The OCA avers that a utility must be doing more than providing efficient and reasonable service in order to receive more than the indicated rate of return. The OCA references its Cross Exhibit 1, which listed five separate dockets where the Commission's Prosecutory Staff had investigated PPL for potential violations of the Code and avers that such actions do not support the award recommended by the ALJ. OCA Exc. at 15-18.

I&E also excepted to the ALJ's recommendation, alleging that it is not supported by the evidence. I&E avers that PPL selectively presented evidence of "high quality" service and alleges that PPL essentially sought an investor reward for implementing statutorily-mandated programs that were purely ratepayer funded through Commission-mandated rates that guaranteed PPL recovery with interest through separate surcharges and riders. I&E opines that while the Commission has the discretion to reward management, because such action essentially sanctions approval of a ratepayer premium, the Commission should exercise that discretion circumspectly. According to I&E, circumstances warranting investor rewards should be the exception not the norm.

I&E opines that PPL's service is not exceptional, finding instead that it was at times above average, at other times below average, and sometimes just average. According to I&E, PPL presented no clear evidence of any particular shareholder commitment that justifies gratuitous ratepayer funding. I&E Exc. at 23-26.

In reply, PPL states that clearly a public utility has a statutory duty to provide adequate, efficient, safe and reasonable service at just and reasonable rates. However, PPL posits that it is the efforts and manner in which the utility meets the statutory requirements that the Commission considers when determining if a management performance adder is appropriate. For example, PPL provides that the Commission awarded a twenty-five basis point adder to compensate a utility where it "promoted and accomplished cost efficiencies in several operational aspects". PPL R.Exc. at 9 (citing *Pa. PUC v. West Penn Power Co.*, 1994 Pa. PUC Lexis 144 at *147). Similarly, PPL notes that the Commission awarded a twenty-two basis point adder where a utility's "managerial performance related to its water quality, customer service and low income program continues to be laudable." PPL R.Exc. at 8-9 (citing *Pa. PUC v. Aqua Pennsylvania, Inc.*, Docket No. R-00072711, at 50 (Order entered July 31, 2008)).

PPL avers that in this proceeding, I&E and the OCA ignore the record evidence of the exceptional manner in which PPL has exceeded its statutory obligation to provide adequate, efficient, safe and reasonable service and facilities at just and reasonable rates. According to PPL, the record evidence demonstrates that PPL's management is effectively controlling costs, while at the same time, providing customers with high quality service and expanded service options. In response to the Parties' allegations with regard to the five instances over the last four years where PPL paid a civil penalty, PPL responds that these parties overlook that PPL has 1.4 million customers and has millions of interactions with these customers annually. PPL points out that in only four instances has any penalty been applied, and in three of those cases the Company settled the matter without any finding of any violation. PPL submits that in

only one instance in the past four years has it been found to have violated the Code, and on that occasion, it was assessed a civil penalty of \$100. Given the Company's efforts, PPL opines that the requested twelve basis point adder clearly is modest and within the range previously awarded by the Commission. PPL R.Exc. at 9-10.

In its Replies to Exceptions, the OCA notes its continuing opposition to the management performance bonus in its entirety and requests that the Commission modify the ALJ's recommendation and remove the six basis point ROE adder. OCA R.Exc. at 14.

In its Replies to Exceptions, I&E similarly notes that PPL's evidence does not support any management bonus. I&E R.Exc. at 17.

iv. Disposition

Pursuant to the Code, the Commission may reward utilities through rates for their performance. In pertinent part, Section 523 of the Code, 66 Pa. C.S. § 523 provides:

§ 523. Performance factor consideration.

(a) **Considerations.** – The Commission shall consider, in addition to all other relevant evidence of record, the efficiency, effectiveness and adequacy of service of each utility when determining just and reasonable rates under this title. On the basis of the commission's consideration of such evidence, it shall give effect to this section by making such adjustments to specific components of the utility's claimed cost of service as it may determine to be proper and appropriate. Any adjustment made under this section shall be made on the basis of specific findings upon evidence of record, which findings shall be set forth explicitly, together with their underlying rationale, in the final order of the commission.

(b) **Fixed utilities.** – As part of its duties pursuant to subsection (a), the commission shall set forth criteria by which it will evaluate future fixed utility performance and in assessing the performance of a fixed utility pursuant to subsection (a), the commission shall consider specifically the following:

(1) Management effectiveness and operating efficiency as measured by an audit pursuant to Section 516 (relating to audits of certain utilities) to the extent that the audit or portions of the audit have been properly introduced by a party into the record of the proceeding in accordance with applicable rules of evidence and procedure.

* * *

(4) Action or failure to act to encourage development of cost-effective energy supply alternatives such as conservation or load management, cogeneration or small power production for electric and gas utilities.

* * *

(7) Any other relevant and material evidence of efficiency, effectiveness and adequacy of service.

Based upon our analysis of the evidence of record, we are persuaded by the arguments of the Company that its management performance related to its advanced metering infrastructure, operating initiatives, customer contact center, electric competition, customer education, energy efficiency programs, and customer assistance programs is laudable and warrants consideration as a factor in our final cost of equity allowance. Accordingly, we shall grant PPL's Exception and adopt its twelve basis point management effectiveness adjustment to our prior return on equity recommendation in recognition of its exemplary managerial performance. In the context of the evidentiary record developed in this proceeding, we conclude that this adjustment is reasonable, appropriate and conservative based on Section 523 of the Code and the similar allowances in the prior Commission decisions cited by the Company. The ALJ's

recommendation of a six basis point allowance shall be modified, consistent with the foregoing and the Exceptions of I&E and the OCA are denied.

i. Summary on Common Equity

i. Positions of the Parties

As noted above, there are four methods of determining the cost of equity: DCF, RP, CAPM, and CE. PPL relied on each of these methodologies in presenting its recommended return on equity of 11.25%.

I&E argued that equal weight should not be given to the four different methodologies as PPL did in its evaluation.

Both the OCA and I&E took issue with the Company's analysis in arriving at the proposed cost of equity and capital structure. The OCA pointed out that the Commission has indicated a preference for using the DCF method to establish reasonable common equity costs.

While calculating average returns on equity for its respective groups of 11.13% and 11.43%, PPL's indicated cost of common equity reflects an upward adjustment of seventy basis points for its EDG and 118 basis points for its IEG to account for the leverage claim. It further reflects an upward adjustment of 120 basis points for both EDG and IEG to reflect its claim that PPL has higher business risk due to its small size relative to its proxy group. Finally, the indicated cost of common equity reflects PPL's upward adjustment of another twelve basis points to reflect PPL's requested award for claimed management efficiency.

I&E opposed PPL's calculated return on equity for several reasons. First, I&E averred that PPL's selected barometer group was flawed in that several of its

selections failed to meet PPL's own purportedly objective selection criteria. Second, I&E maintained that PPL gave undue weight to the RP and CE methods. Third, I&E claimed that PPL employed an inflated DCF growth rate and a dividend yield adjustment that was unnecessary. Fourth, according to I&E, PPL employed inflated CAPM betas. Finally, I&E rejected PPL's extra-method adjustments for leverage, size (business risk), and management efficiency as they are unsupported and inappropriate.

ii. ALJ's Recommendation

The ALJ concluded that the Commission's preferred method of determining a utility's ROE is the DCF model. Consequently, the ALJ recommended adoption of I&E's DCF analysis, consisting of a dividend yield of 4.89% and a growth rate, prior to I&E's log-linear adjustment, of 4.79%. Additionally, the ALJ recommended adoption of a six basis point adjustment to PPL's ROE for management effectiveness. The result of the ALJ's recommendations equates to an overall ROE of 9.74%. R.D. at 93.

iii. Disposition

The ALJ recommended that the Company's position of an actual capital structure consisting of 49.22% long-term debt and 50.78% common equity along with a long-term debt cost rate of 5.50% be adopted. Additionally, the ALJ recommended adoption of the I&E position on PPL's cost of equity capital of 9.74%. According to the ALJ, the evidence overwhelmingly demonstrated that PPL's claim for a return on equity of 11.25% and an overall rate of return of 8.47% overstated what reasonable investors should expect from a regulated public utility and is not necessary for PPL to safely and reliably provide electric distribution service to its captive ratepayers. Based on these recommendations, the resulting overall rate of return per the ALJ is 7.65%.

Capital Type	Ratio (%)	Cost Rate (%)	Weighted Cost (%)
Debt	49.22	5.50	2.71
Equity	50.78	9.74	4.95
Total	100.00		7.65

R.D. at 93-94.

Based upon the foregoing, we conclude that PPL's capital structure should be based upon the Company's actual capital structure of 49.22% debt and 50.78% equity. PPL's cost of equity capital is properly determined by the DCF analysis performed by the Parties, with other methods utilized as a check on the reasonableness of the DCF results. Accordingly, we adopt a cost of equity rate of 10.4%. In addition, the 10.4% approved ROE is inclusive of the twelve basis point management efficiency adjustment as requested by the Company. Each of the remaining PPL requested ROE adjustments are rejected as unreasonable.

The following table summarizes our final determinations concerning PPL's capital structure, cost of debt and cost of common equity, as well as the resulting weighted costs and overall rate of return of 7.99%:

Capital Type	Ratio (%)	Cost Rate (%)	Weighted Cost (%)
Debt	49.22	5.50	2.71
Equity	50.78	10.4	5.28
Total	100.00		7.99

F. Taxes – Gross Receipts Tax

1. Positions of the Parties

PPL's total FTY gross receipts tax (GRT) expense claim is \$50.102 million, which consists of two components. The first component is a pro forma calculation of gross receipts tax for the FTY at present rates of \$43.930 million. PPL Exh. Future 1, Sch. D-11 at 3. The second component is \$6.172 million, resulting from the proposed rate increase. PPL M.B. at 133; PPL Exh. Future 1, Sch. D-12 at 6.

I&E recommended a total GRT allowance of \$49.168 million, which is a \$934,000 reduction to the Company's total claim. The recommendation consists of a pro forma allowance of \$43.1 million and a rate increase allowance of \$6.068 million, assuming a full rate increase. The recommended GRT adjustments are reductions of \$830,000 to the pro forma claim and \$104,000 to the rate increase claim. I&E's recommendation is based on the fact that PPL's tax liability for the GRT is limited to the actual revenues PPL receives. As such, I&E recommended that the GRT tax allowance in rates should be calculated using the net revenues collected by PPL. I&E M.B. at 69; I&E St. 2 at 46-48.

2. ALJ's Recommendation

The ALJ found that I&E's recommendation to calculate the GRT allowance using net revenues was reasonable and should be approved, because it is a better match of the claimed actual receipts of revenue that will produce the Company's actual GRT tax liability. R.D. at 95, 96; I&E St. 2 at 46-48. The ALJ stated that the Pennsylvania Department of Revenue (DOR) Corporation Tax Bulletin 2011-02, issued July 20, 2011 (Bulletin), confirmed that the Company's net uncollected revenues would not reduce its GRT tax liability. The ALJ also stated that the Company did not provide any evidence to support that the cost of documentation would exceed the overvaluation of GRT, and the

Company's witness confirmed that the Company maintains records of customers' bad debts. R.D. at 97.

3. Exceptions

In its Exceptions, PPL avers that its GRT should be recovered in full. PPL Exc. at 37. PPL states that the ALJ's recommendation should be rejected because it disregards changes in the calculations of GRT imposed by the DOR in the Bulletin. *Id.* at 37-38. PPL opines that the Bulletin makes use of the deduction from gross receipts for uncollectible accounts almost impossible. PPL explains that, under the Bulletin, its liability for GRT is no longer limited to actual revenues received, but, instead, PPL must file GRT using the accrual method of accounting. As such, a reduction against taxable gross income for an uncollectible account requires PPL to match each write-off to the tax period when the receipts are reported as taxable to Pennsylvania. PPL indicates that it does not have the capability to perform this tracking for the write-offs of amounts for its 1.4 million customers. *Id.* at 38; PPL St. 8-RJ, Part 1, at 36-37. PPL submits, while it is correct that it does maintain records of its customers' bad debts, this does not enable PPL to meet the onerous reporting and accounting requirements that the Bulletin requires. PPL Exc. at 38.

In its Replies to Exceptions, I&E argues that the ALJ correctly determined that the Bulletin confirmed I&E's adjustment to PPL's GRT claim on the basis that PPL's tax liability is the net of uncollectibles. I&E notes that, using the accrual methodology, PPL will deduct from its accrued billed revenues accounts that are written off. I&E asserts that PPL did not present any evidence to prove there are obstacles to it following the requirements in the Bulletin and distinguishing between billed and collected revenues. I&E R.Exc. at 24. I&E avers that, absent evidence that PPL pays taxes on uncollected revenues and that the cost of avoidance exceeds the benefit, the ALJ's decision should be adopted. *Id.* at 25.

4. Disposition

We agree with the ALJ's determination that I&E's recommendation to calculate the GRT allowance using net revenues is reasonable and should be adopted, as it is a better match of the claimed actual receipts of revenue that will produce the Company's actual GRT tax liability. The Bulletin supports I&E's adjustment on the basis that PPL's tax liability is billed revenues net of write-offs and recoveries. PPL will use the accrual method of accounting to deduct from its accrued billed revenues accounts that are written off. The Bulletin states the following, in pertinent part:

If a taxpayer uses the accrual method of accounting to report its gross receipts, then the taxable gross receipts shall be calculated as follows:

Billed revenues on an accrual basis (no reserves for bad debts)
Less: Accounts actually written off for previously taxed
Pennsylvania bad debts
Plus: Collections of previously written off
Pennsylvania taxable bad debts

Taxable Gross Receipts

I&E Exh. 2-SR, Schedule 1, at 1.

Additionally, as PPL has explained, the Bulletin requires taxpayers claiming a deduction for bad debts to provide the DOR, upon request, with the following documentation: (1) the type and amount of receipts being written off; (2) the customer's location; and (3) the tax period during which the receipts were reported as taxable to Pennsylvania. *Id.* at 2. PPL submits that the DOR's reporting requirements are onerous and would require significant and costly changes to its billing and payment system. PPL M.B. at 134. Nevertheless, as I&E asserts, PPL has not presented any concrete evidence to show that it could not comply with the DOR's reporting requirements, such as cost analyses, evidence of system testing, or evidence of actual complexities. *See*, I&E Exc.

at 24; I&E M.B. at 70-71. PPL has also indicated that it does maintain records of customers' bad debts. Based on the evidence, we find that I&E's adjustment is reasonable. Accordingly, we shall deny PPL's Exception and adopt the ALJ's recommendation on this issue.

G. Rate Structure and Rate Design

This section of the Opinion and Order addresses cost of service, rate design and rate structure allocation issues. When a utility files for a rate increase, it must file a cost-of-service study (COSS) assigning to each customer class a rate based upon operating costs that it incurred in providing that service. 52 Pa. Code § 53.53; *Lloyd v. Pa. PUC*, 904 A.2d 1010, 1015 (Pa. Cmwlth. 2006). Public utility rates should enable the utility to recover its cost of service and should allocate this cost among its customers. These rates are required by statute to be just, reasonable and non-discriminatory. 66 Pa. C.S. §§ 1301, 2804(10).

1. Cost of Service Methodology

a. Positions of the Parties

PPL stated that the fundamental purpose of a cost allocation study is to aid in revenue allocation and the design of rates to be charged by identifying all of the capital and operating costs incurred by a utility to provide service to all of its customers, and then assigning or allocating those costs to individual rate classes on the basis of how those rate classes cause the cost to be incurred. PPL maintained that as a result of the *Lloyd* decision, *supra*, cost of service studies have assumed a greater degree of importance in utility ratemaking, but it still should be recognized that cost allocation is not an exact science, that there is no single correct cost allocation methodology and that the Court did not hold that all other considerations are to be disregarded. PPL M.B. at 136-137.

PPL presented a fully-allocated COSS, showing the allocation of its distribution costs among the various rate classes at both present and proposed rates for the historic (PPL Exh. JMK-1) and future (PPL Exh. JMK-12) test years. According to PPL, the filed COSS in this proceeding is virtually identical to the methodology adopted by the Commission in its 2010 base rate proceeding using the class maximum non-coincident peak (NCP) demand method, which is based on the highest demand imposed by each rate class on its distribution system, to allocate its demand-related distribution costs. PPL St. 8 at 19.

As in 2010, PPL's COSS utilized a "heightened" level of data analysis, using allocators to classify primary voltage level distribution facilities into their demand-related and minimum or no-load customer-related cost components. PPL stated that this method more accurately reflects cost causation than the method used in preceding rate cases, which allocated primary voltage level distribution facilities solely on the basis of demand. PPL St. 8-R at 9. PPL stated that prior to its 2010 case, the Company's cost allocation studies were criticized because not all of the primary voltage level distribution facilities used in its minimum size system studies had been classified into their applicable customer related and demand related components. PPL claimed that this modification is consistent with the National Association of Regulatory Commissioners (NARUC) Electric Utility Cost Allocation Manual (Manual) recommendations "that primary voltage level overhead and underground conductors be classified into their demand-related and customer-related cost components." *Id.*; PPL M.B. at 137-138.

Only the OCA opposed the Company's COSS, and on substantially the same grounds as it opposed the Company's COSS in the last base rate case. The OCA argued that primary plant should be classified on a 100% demand basis, with only secondary plant allocated to both demand and customer components. OCA St. 3 at 18. The OCA presented density studies which it claimed do not support allocation of

distribution plant based on customer count. As a “compromise” position, OCA recommended that the Commission allocate 100% of primary plant on a demand basis and apply the OCA’s minimum size study to allocate secondary plant on a customer and demand basis. OCA M.B. at 77-82.

The OCA further argued that the Parties have misinterpreted the Commission’s 2010 Order that NARUC has updated its cost of service principles since issuing the 1992 NARUC Manual, and argued that its recommendation reflects a compromise. OCA R.B. at 33-38.

The OSBA, PPLICA and REG supported PPL’s position on COSS allocation and believe that it is consistent with the NARUC Manual and reflects a more realistic operation of PPL’s system than the OCA counterproposal.

The OSBA stated that its primary focus in this case has been to determine whether the COSS presented by the Company conformed to the COSS approved by the Commission in the 2010 base rate case. The OSBA concluded that it did, and therefore, there was no need to re-litigate it in this proceeding. OSBA M.B. at 7.

REG agreed with the Company’s classification of distribution plant as partially customer-related and partially demand-related, and the Company’s allocation of the plant. This, REG argued, is consistent with the Commission’s disposition of the Company’s last rate case as well. REG M.B. at 4-5.

PPLICA argued in favor of the Company’s COSS, which it believed properly allocates primary distribution facilities costs in both a customer and demand component and is consistent with NARUC policies. PPLICA characterized the OCA approach as “a results-driven density analysis with no meaningful relation to the cost of service principles historically applied by the Commission and supported by NARUC.”

PPLICA M.B. at 7. PPLICA averred that PPL's COSS provides a reasonable basis for assessing distribution-related rates of return for each rate schedule, consistent with Commission precedent and NARUC recommendations. *Id.* at 8.

According to PPLICA, there are two recognized methodologies to estimate the customer component of distribution costs: (1) the minimum intercept method; and (2) the minimum size method, which is the method used by PPL. Each is designed to estimate the component of distribution plant cost incurred by a utility to connect a customer to the system. The minimum size method is designed to reflect costs associated with changes in both the number of distribution customers and the loads of these customers. It reflects a classification of the distribution facilities that would be required to simply interconnect a customer to the system, regardless of the kW load of that customer. PPLICA St. 1-R at 4-5; PPLICA M.B. at 9.

b. ALJ's Recommendation

The ALJ concluded that the other Parties rejected the OCA's arguments most persuasively. For the reasons set forth above by the Parties, the ALJ recommended that the Company's COSS be approved, and the OCA alternative be denied. R.D. at 108.

c. Exceptions

In its Exceptions, the OCA opines that the ALJ erred in recommending the use of PPL's COSS to allocate the revenue increase. The OCA avers that the PPL COSS is flawed because it does not accurately reflect cost causation, is inconsistent with the 1992 NARUC Manual and the updated NARUC Report, and is inconsistent with the historical method that PPL used prior to 2010. The OCA submits that, prior to 2010, PPL classified primary distribution plant as 100% demand related and further classified secondary distribution plant as partially demand and partially customer related.

According to the OCA, this method was approved by the Commission in the 2004 and 2007 PPL rate cases, and is the same COSS method that the OCA has proposed in the present case. The major change, starting with the 2010 case, is that PPL now classified primary distribution plant as 63% customer related and 37% demand related. This change, avers the OCA, has caused over one billion dollars of such costs to be shifted from a demand basis to a customer count basis. OCA Exc. at 18-19.

Next, the OCA notes that the ALJ relied on the arguments of the Company and the other Parties in adopting PPL's COSS, whereby the central point made was that the Commission had already ruled against the OCA in PPL's 2010 rate case and should do the same here. The OCA avers that PPL's COSS method does not follow the 1992 NARUC Manual in many respects, and is inconsistent with the more recent 2000 NARUC Report. The OCA states that in the 2010 rate case, PPL's recommended allocation of the entire increase to the residential class was adopted by the Commission, partially because the Commission found the OCA's approach did not accurately reflect the costs incurred to serve the residential class. *Id.* at 20 (citing *Pa. PUC v. PPL Electric Utilities Corporation*, Docket No. R-2010-2161694, at 46 (Order entered December 21, 2010) (*PPL 2010*)). However, the OCA avers that in the 2010 case its approach was identical to PPL's own COSS method used in 2004 and 2007. According to the OCA, PPL's proposed COSS in the instant proceeding contains bias to the residential class that negates any possibility of that class reaching "cost of service" anytime in the foreseeable future. OCA Exc. at 19-20.

Additionally, the OCA maintains that both primary and secondary distribution plant should be classified as 100% demand related, consistent with how regulatory bodies in over thirty states classify such plant. The OCA avers that it has recommended a reasonable and appropriate compromise COSS that maintains a customer/demand split for the secondary distribution plant but allocates primary plant on demand only, which is exactly what PPL did prior to 2010. Further, the OCA then

classified secondary distribution plant as partially demand and partially customer related, just like PPL's current and prior COSSs, but with a more appropriate customer component than PPL based on its revisions to PPL's minimum size study and consistent with how such a study is to be performed as per the 1992 NARUC Manual. OCA Exc. at 20-21.

The OCA submits that the ALJ's and other Parties' reliance on *PPL 2010* is misplaced as the Commission has substantial evidence in this record that it did not have in 2010, specifically: (1) PPL's proposed COSS is an outlier in its classification of primary distribution plant as having a customer component; (2) to the extent that a customer component should be a part of distribution plant cost assignment, PPL's minimum size study fails to follow the 1992 NARUC Manual's specific instructions for performing such a study; and (3) the fact that adhering to PPL's proposed COSS will always result in the residential class being allocated a substantial portion of future rate increases with little to no hope of ever achieving cost of service. OCA Exc. at 21.

The OCA maintains that using the method that has been accepted in over thirty states, a 100% demand allocation, the indexed rate of return for the RS class, at present rates, would be 124%. Using the method that PPL proposed in this proceeding, the indexed rate of return for the RS class would be only 63%, per the OCA. The OCA avers that its compromise position, 100% demand allocation only for primary plant, shows the Residence Service (RS) class at an indexed rate of return of 112% at present rates. Therefore, according to the OCA, at current rates under an accurate and reasonable COSS, the RS class is paying more than its cost to serve. As a result, the OCA avers that the Commission's holding in *PPL 2010* should not be controlling here. OCA Exc. at 23-24.

In reply, PPL states that its COSS is virtually identical to the methodology adopted by the Commission in the 2010 base rate proceeding, which was fully litigated

on this issue. PPL avers that the Commission fully considered and rejected the OCA's proposal in the 2010 base rate proceeding and that the OCA has offered no change in law or fact that would warrant a departure from that decision. PPL maintains that the ALJ properly approved its COSS. PPL R.Exc. at 16.

In its Replies to Exceptions, the OSBA first notes that, contrary to the OCA's argument, PPL has actually proposed to reduce the customer component of distribution plant costs in the instant proceeding relative to the method that was explicitly approved by the Commission in the Company's 2010 case, to the benefit of residential customers. The OSBA avers that for the OCA to prevail on the issue of cost allocation, it must demonstrate both that the Commission erred in its decision in the 2010 case to allow for the classification of primary system distribution plant costs into both demand and customer components and that the Commission has consistently erred over the past decades in approving PPL's cost classification methodology for secondary distribution system costs. The OSBA notes that the Commission considered virtually all of the evidence presented by the OCA in this proceeding in the 2010 case and rejected the OCA's conclusion. Moreover, the OSBA notes that, in objecting to PPL's method for classifying secondary system plant costs, the OCA is challenging an approach PPL has used for years if not decades. According to the OSBA, in the OCA's view, Commission precedent prior to 2010 is relevant only if it favors residential customers, which is both wrong and inconsistent. OSBA R.Exc. at 4-6.

In response to the OCA's assertion that regulatory bodies in thirty states do not include any customer component in classifying either primary system or secondary system distribution costs, the OSBA states that cost allocation is often hotly debated among the parties to a regulatory proceeding. The OSBA explains that the economic issue of the classification of distribution plant costs is essentially an issue involving residential and small to medium business customers, as large industrial customers are generally served at transmission voltage and have no stake in this issue. According to the

OSBA, the smaller business customers are generally unrepresented in utility regulatory proceedings, so it is unclear whether the regulatory pattern alleged by the OCA results from hard cost analysis, or simply a lack of representation. The OSBA maintains that in either event, the thirty jurisdictions are ignoring the basic principle that this Commission has accepted. As this principle has long been followed in Pennsylvania, the OSBA submits that the alleged practices of other jurisdictions are irrelevant. OSBA R.Exc. at 6-7.

Next, the OSBA submits that the OCA characterization of the “updated NARUC report” as an update to the 1992 NARUC Cost Allocation Manual is deceptive at best. The OSBA explains that the 1992 NARUC Manual was published as a NARUC Report. The report to which the OCA refers to as an update is nothing of the kind, but, in fact, a report prepared by the Regulatory Assistance Project entitled “Charging for Distribution Utility Services: Issues in Rate Design.” The OSBA avers that this document contains little in the way of specifics for distribution cost classification and allocation and does not necessarily reflect the positions of NARUC. As a result, the OSBA recommends that the Commission give no weight to this consultant’s report. OSBA R.Exc. at 7-8.

In its Replies to Exceptions, PPLICA states that PPL’s proposed COSS is firmly supported by NARUC principles, designed to achieve cost of service rates and should be approved. PPLICA points out that while the OCA refers to the updated report as a “NARUC” report, the document is not an official NARUC publication. Also, PPLICA states that the OCA’s claim that this document establishes PPL’s minimum size system COSS as an outlier is specious. According to PPLICA, this report’s statement that allocating primary distribution plant on a 100% demand basis “is used in more than thirty states” dates back to 2000, almost thirteen years ago. PPLICA avers just as PPL classified primary distribution plant on a 100% demand basis before updating its classification methods in 2010, many of the states referenced in the report may have

modified their methodologies. Therefore, PPLICA asserts that the Commission should not accord significant weight to stale data. PPLICA R.Exc. at 4-5.

PPLICA further replies that PPL's minimum size study is completely consistent with the NARUC Manual, and that it is worth noting that the same study employed by PPL in this proceeding was fully litigated in the Company's 2010 case and adopted by the Commission. Additionally, PPLICA notes that PPL's minimum size study reflects the Company's actual installations rather than the theoretical adjustments applied by the OCA. Lastly, PPLICA argues that PPL's proposed COSS contains no inherent bias towards any rate class as alleged by the OCA. PPLICA R.Exc. at 5-7.

d. Disposition

Based upon our review of the record evidence, we are in agreement with the ALJ that PPL's proposed COSS should be approved. It is important to note that the PPL COSS methodology is supported by all the Parties which offered a position on this issue, with the exception of the OCA. We have reviewed the OCA's position and Exceptions on this issue and are not persuaded by the arguments it presented in support of its recommended COSS methodology. The position presented by the OCA was considered and rejected by the Commission in the litigation of PPL's 2010 base rate proceeding. *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-2010-2161694, at 46 (Order entered December 21, 2010). We conclude that the OCA has not presented convincing arguments in this proceeding that would cause us to re-evaluate our determination in PPL's prior proceeding. PPL's proposed COSS in the instant proceeding is virtually identical to the COSS approved in 2010, is consistent with the NARUC Manual and more accurately reflects cost-causation principles than the COSS methodology the Company utilized prior to the 2010 base rate case. We conclude that PPL has carried its burden of proof on this issue, and we shall adopt the ALJ's recommendation. Accordingly, we shall deny the Exceptions of the OCA.

2. Revenue Allocation

a. Positions of the Parties

PPL explained that its proposed allocation of revenue requirement among the various rate classes in this proceeding is driven largely by the Commonwealth Court's decision in *Lloyd, supra*. PPL stated that this case is the fourth in a series that have purportedly attempted to move PPL's distribution rates to true cost of service. PPL St. 5 at 8. The Company sought to establish cost of service, and then to apply those costs to the appropriate rate schedules. Because that approach produced a distribution rate increase to customers served under Rate Schedule Residential Thermal Storage (RTS) of about 165 percent, which PPL considered to be unjust and unreasonable, it developed an alternative allocation, which limited the increase to Rate RTS from 165% to approximately 78%. According to PPL, the goal was to bring all rate classes closer to the system average rate of return, while still considering the principle of gradualism. *Id.* at 10; PPL M.B. at 152-153.

The Company's proposal is as follows:

Rate Classes	<u>Relative Rate of Return</u>	
	<u>Present Rates</u>	<u>Proposed Rates</u>
RS	63.03%	83.81%
RTS	-65.31%	23.05%
GS-1	133.55%	99.05%
GS-3	285.18%	196.34%
LP-4	163.36%	118.44%
LP-5	-90.72%	98.94%
LPEP	353.09%	256.26%

GH-2	86.64%	103.55%
SL/AL	100.49%	99.65%
Total	100%	100%

PPL Exh. JMK-2 at 8-11; PPL M.B. at 154.

The OCA stated that for the second time in two years, PPL has proposed to allocate nearly the entire revenue increase to the RS and RTS rate classes. The OCA noted that of PPL's \$104.6 million increase requested, PPL proposed to allocate over \$99 million to the residential class with over \$3.5 million of that amount allocated to RTS customers. The OCA averred that these increases amount to an annual increase to distribution rates of 20.9% and 77.6%, respectively. OCA M.B. at 66.

The OCA recommended an alternative revenue allocation that it claims reflects the results of a properly conducted, reasonable and equitable cost of service study. The OCA submitted that while cost of service should guide the Commission when setting rates in this proceeding, other ratemaking principles such as gradualism, avoidance of rate shock and basic fairness must not be abandoned. As such, the OCA recommended that no rate class receive a revenue decrease and that no class sustain an increase greater than 150% of the system-wide percentage increase, or no more than 21.45%. *Id.* at 95.

The OCA's proposal results in the following indexed rate of returns by class:

<u>Relative Rate of Return</u>		
<u>Rate Classes</u>	<u>Present Rates</u>	<u>Proposed Rates</u>
RS	112%	111%

RTS	-93%	-53%
GS-1	180%	131%
GS-3	104%	109%
LP-4	-13%	11%
LP-5	-88%	-4%
LPEP	399%	289%
GH-2	30%	50%
SL/AL	90%	111%
Total	100%	100%

OCA St. 3 at 37, 41; OCA M.B. at 96-97. According to the OCA, this proposed revenue allocation results in a reasonable movement of all classes to cost of service at PPL's proposed revenue increase, while also recognizing the need for gradualism. OCA M.B. at 97.

REG and PPLICA supported PPL's proposed revenue allocation as consistent with the COSS. According to these Parties, the Company's proposed revenue allocation moves all rate classes closer to cost of service in accordance with the Company's COSS and consistent with *Lloyd*. REG M.B. at 5; PPLICA M.B. at 13-17.

PPLICA pointed out that Rate Schedule LP-5 customers will experience a 59.1% increase, and although Rate Schedule LP-4 customers do not experience an increase, their current rates remain above cost of service. PPLICA recognized that the movement towards actual cost of service rates as set forth is reasonable, and did not oppose this allocation. PPLICA M.B. at 16.

PPLICA argued that the Commission should not give any credence to OCA's COSS, and as the OCA's proposed allocation is based on its flawed COSS,

neither should the Commission give any credence to the OCA's recommendation.
PPLICA St. 1-R at 8; PPLICA M.B. at 15.

b. ALJ's Recommendation

The ALJ concluded that as the OCA alternative was based on its COSS, and not on the Company's, which she recommended be adopted, the OCA alternative should be denied. The ALJ recommended adoption of the Company's revenue allocation, with the actual numbers to be based on the proportionate adoption of the actual revenue requirement approved. R.D. at 110.

c. Exceptions

In its Exceptions, the OCA submits that its COSS should be adopted as a guide to set rates in this proceeding and for purposes of establishing a fair and reasonable allocation of the revenue increase. The OCA avers that PPL's COSS is unduly discriminatory against residential customers and PPL's proposed revenue allocation is based on that study. The OCA maintains that its proposed allocation is based upon a more reasonable COSS and recognizes gradualism and fairness and caps increases to any one rate class at no greater than 150% of the system-wide percentage increase, or 21.45%. The OCA opines that its revenue allocation method results in a reasonable movement of all classes to cost of service at PPL's proposed revenue increase, while also recognizing the need for gradualism. OCA Exc. at 31-34.

In reply, PPL states that its proposed revenue allocation follows the Company's COSS and substantially moves all rate schedules toward the system average rate of return. PPL avers that since the OCA's revenue allocation is premised on its flawed COSS, its resulting revenue allocation was properly rejected by the ALJ. PPL R. Exc. at 16-17.

In its Replies to Exceptions, the OSBA points out that, as the OCA readily admits, the adoption of the OCA's revenue allocation proposal requires the Commission to agree to the OCA's version of the COSS. According to the OSBA, because the ALJ correctly rejected the OCA cost allocation methodology, the OCA's revenue allocation methodology should similarly be rejected. OSBA R.Exc. at 13-14.

In its reply, PPLICA states that the OCA's proposed COSS is contrary to Commission precedent and unsupported by the NARUC Manual and, as such, any revenue allocation based on the OCA's proposed COSS must be summarily rejected. In response to the OCA's argument about gradualism, PPLICA acknowledges that gradualism is a legitimate ratemaking construct designed to mitigate unreasonable rate increases. However, according to PPLICA, because PPL's COSS shows that residential customers are paying rates significantly below cost-of-service, PPL's revenue allocation limits gradualism adjustments to ensure that customers paying above-cost rates move reasonably closer to cost-of-service. PPLICA posits that as the ALJ's recommendation to approve PPL's revenue allocation incorporates gradualism, it should be approved by the Commission without modification. PPLICA R.Exc. at 9-10.

d. Disposition

Based upon our prior determination and discussion, *supra*, with respect to the rejection of the OCA COSS, we are in agreement with the ALJ that PPL's proposed revenue allocation should be approved. As the OCA's revenue allocation recommendation is based upon its COSS, which we have rejected, we conclude that its allocation proposal should similarly be denied. Additionally, we find that PPL's revenue allocation proposal is consistent with *Lloyd*, moves all rate classes closer to cost of service in a reasonable manner and considers the principle of gradualism. Accordingly,

we shall adopt the recommendation of the ALJ and deny the OCA Exceptions on this issue.

3. Revenue Scaleback

a. Positions of the Parties

As the Commission is approving a lesser revenue requirement than sought by PPL, an important consideration is the determination of how the proposed revenue allocation will be affected by the scaleback in rates.

In this proceeding, PPL and the OCA support a proportional scaleback, with no decrease in revenues for classes that do not receive a rate increase. PPL St. 5-R at 4; OCA St. 3 at 42.

I&E proposed applying the first \$1,784,000 to lower the revenue requirement for Rate Schedule RTS customers, with any further reductions applied to Rate Schedules RS, GH-2, SL/AL, and on a conditional basis, LP-5. I&E St. 3 at 16-17.

The OSBA recommended a revenue-based scaleback which would allocate any overall rate increase approved by the Commission to each rate class in proportion to the Company's proposed revenues from each class. OSBA St. 1 at 13.

PPLICA supported the scaleback recommendation proposed by the OSBA in the event that the Commission approves an overall revenue increase lower than the Company's requested \$104.6 million increase. PPLICA argued that application of a proportional scaleback in this proceeding would hinder progress to cost of service rates by reducing rate increases for customers paying below cost of service rates pursuant to PPL's COSS, but not allowing correlating adjustments for customers whose present rates are above cost of service. PPLICA M.B. at 19, PPLICA R.B. at 9.

PPLICA further asked that should the Commission not adopt the OSBA recommendation, then the scaleback should be applied to all rate classes receiving an increase as proposed by the Company and the OCA, with no further exclusions, as would apply under I&E's proposal. PPLICA opposed the restrictions on the scaleback for Rate Schedule LP-5 that I&E recommended, since that rate schedule is already targeted for a substantial increase. PPLICA M.B. at 18-19.

b. ALJ's Recommendation

The ALJ stated that in the *Lloyd* decision, the Commonwealth Court disapproved the setting of rates according to a flat across-the-board percentage, because there was no dispute that the cost of serving each rate class varied and that rates for certain classes were subsidizing rates for others in the interest of keeping the increase in the total bills of each class to 10% or less. Accordingly, the ALJ found that any scaleback should be utilized to bring the rates of each rate schedule closer to the cost of service. R.D. at 111.

However, the ALJ concluded that this concept, applied blindly, would result in reductions to customers who were not expecting an increase, or greater reductions to some customers than were originally proposed, to the detriment of those whose rates will rise more than necessary. Therefore, the ALJ recommended that PPL's proposal to apply any scaleback on a proportional basis to only those rate schedules that receive increases should be adopted by the Commission. R.D. at 112.

c. Exceptions

In its Exceptions, I&E states that it agrees with the ALJ, but believes the Commission should moderate the increases proposed for the Rate RTS usage rate and the LP-5 customer charge before the proportionate scale-back is applied. I&E Exc. at 29-30.

In its Exceptions, the OCA stated that, as a general principle, it has no disagreement with PPL's proportional scaleback approach. However, the OCA disagrees with using PPL's revenue allocation as a starting point for the proportional scaleback. The OCA submits that its revenue allocation be used as a starting point for a proportional scaleback in this proceeding. OCA Exc. at 34.

The OSBA also excepted to the ALJ's recommendation, stating that the ALJ erred in recommending a proportional scaleback of the rate increase for only those customer classes that were assigned rate increases by PPL. The OSBA avers that Rate Schedule GS-3 is significantly overpaying its cost of service at current rates, and only received mild relief under PPL's original proposed revenue allocation.¹⁸ The OSBA avers that the problem with the proportional scaleback is the progress toward cost-based rates that was part of the original intent of the Company's revenue allocation will not be retained. Under the method adopted by the ALJ, certain customer classes will not benefit from the reduction in PPL's proposed rate increase. The OSBA alleges that the I&E scaleback proposal results in the same unacceptable result. The OSBA recommends that any reduction in the overall increase be shared among the rate classes in proportion to the Company's originally proposed revenues. According to the OSBA, its recommended

¹⁸ The OSBA included Tables showing that the GS-3 class rate of return at present rates is 11.4 percentage points above system average, and, even with the proposed rate decrease, remains 8.2 percentage points above system average at PPL's proposed rates. OSBA Exc. at 8.

scaleback methodology maintains the progress towards cost-based rates that was present in PPL's original revenue allocation proposal. OSBA Exc. at 5-12.

Exceptions to the ALJ's recommendation were also filed by PPLICA, wherein it states that the ALJ erred in rejecting the OSBA recommendation of a revenue-based scaleback. PPLICA observes that PPL has now filed four base rate cases since *Lloyd*, without achieving cost-based rates for certain rate schedules. PPLICA avers that it is imperative that any scaleback applied to the lower revenue requirement also reflect continued progress towards cost-based rates. PPLICA opines that despite the ALJ explicitly acknowledging the directives and principles from *Lloyd*, the ALJ inexplicably declined to adopt the revenue-based scaleback proposed by the OSBA. PPLICA echoes the comments of the OSBA that approval of a proportional scaleback would reverse progress towards cost-based rates by reducing rates for customers receiving an increase, but still paying below cost rates. At the same time, rate schedules currently paying above-cost rates, but not receiving an increase, would be excluded from a scaleback, explains PPLICA. According to PPLICA, no reasonable basis exists for approving a scaleback that reverses progress towards cost-based rates. PPLICA Exc. at 3-6.

In reply to the arguments of the OSBA and PPLICA, PPL states that the scaleback method recommended by the ALJ is fair and should be approved. PPL maintains that the ALJ's recommended scaleback is the same method the Company proposed in its 2010 case, which was litigated and adopted by the Commission. PPL avers that both the scaleback recommended by the ALJ and the method proposed by the OSBA would move rate classes towards the system average return. However, PPL opines that as a matter of fairness, any scaleback of revenues should be applied to those customer classes that would have received a rate increase under the Company's original proposal. PPL R.Exc. at 17.

In its Replies to Exceptions, the OCA states that the ALJ was correct in recommending the use of a proportional scaleback. The OCA notes that the OSBA recommendation was directly addressed in PPL's 2010 case and rejected by the ALJ and the Commission, which stated that asking one class to pay more of an increase than the final total increase in revenue would be unreasonable. According to the OCA, the OSBA's proposed scaleback methodology would impose additional costs on certain rate classes, over and above the total revenue increase authorized, in order to provide additional rate decreases to other rate classes. The OCA avers that neither the OSBA nor PPLICA provide evidence to support the idea of what constituted unreasonable rates in 2010 is now acceptable. OCA R.Exc. at 15-17.

In its Replies to Exceptions, PPLICA first states that since the OCA's proposed revenue allocation is *per se* unreasonable, any scaleback based upon it should be disregarded by the Commission. Additionally, PPLICA avers that an increase-based scaleback will significantly hinder progress towards cost-based rates. PPLICA requests that the Commission deny any proposal to apply a proportional increase-based scaleback and adopt the revenue-based scaleback proposed by the OSBA. PPLICA R.Exc. at 10-11.

d. Disposition

Based upon our review of the record evidence, we are in agreement with the recommendation of the ALJ that PPL's proposed proportional scaleback only to those classes that were proposed to receive rate increases, of the requested revenue increase, are fair, reasonable and should be approved. We find that the OCA's Exceptions with regard to the proper starting point are without merit, as we have herein previously rejected the OCA recommended allocation proposals. We further conclude that the I&E's Exceptions with regard to first providing relief to certain designated rate classes before the proportional scaleback is applied are also without merit.

The OSBA, as well as PPLICA, filed Exceptions opposed to the adoption of a proportional scaleback. These Parties are of the opinion that a revenue based scaleback should be adopted and applied to all customer classes, whether they were to originally receive no increase or a rate decrease. On this point, we are persuaded by the comments of PPL that the ALJ's recommended scaleback is the same method the Company proposed in its 2010 case, which was litigated and adopted by the Commission. Neither the OSBA nor PPLICA have presented sufficient evidence to warrant our reconsideration of this issue in this proceeding. We find that, as a matter of fairness, those customer classes that have not been allotted any rate increase via the Company's original revenue allocation should not receive rate decreases as argued by the OSBA and PPLICA. We conclude that PPL's proposed scaleback methodology maintains the gradual movement to cost based rates and is appropriate under the unique circumstances in this proceeding. Accordingly, the Exceptions of I&E, the OCA, the OSBA and PPLICA are denied, and the ALJ's recommendation is adopted.

4. Residential Customer Charge

a. Positions of the Parties

PPL's current residential distribution schedules are RS, RTS, and Residential Time-of-Day (RTD). In PPL's presently effective residential Rate Schedule RS, a large portion of the distribution revenue is being collected through usage or kWh charges. PPL's minimum size system study indicated that residential customers should be paying a much greater monthly customer charge than the current monthly charge of \$8.75. In this proceeding, PPL has proposed raising the Rate Schedule RS customer charge from its present \$8.75 per month to \$16.00 per month and decreasing the kWh charges from \$0.03364 to \$0.03340. PPL St. 5 at 11-14. The Company pointed out that its COSS supports a charge of \$36.70, and this increase moves the rate schedule closer to the cost of serving it. PPL M.B. at 162-163.

The OCA, the CEO and I&E opposed PPL's proposal to increase the Rate Schedule RS customer charge.

The OCA opposed the increase to residential customers because it is based on the Company's COSS, which it also opposed. The OCA objected further that the Company's proposal will disproportionately impact low-income, low-usage customers and would result in a "significant disincentive" for customers to conserve. OCA M.B. at 106.

The CEO opposed the increase in the fixed monthly customer charge because it takes away a customer's motive and ability to conserve. The CEO stated that one of the only defenses that a family has against sharp increases in energy costs is conservation, CEO St. 1 at 5, and this proposal eliminates the ability to reduce that cost through conservation efforts. CEO M.B. at 7.

The Company pointed out that there is an energy charge component that is being reduced by 0.7%, and that the distribution charge is small in the context of the energy portion of the bill, which comprises 86% of the charges on the average customer's bill. According to PPL, this still provides an adequate opportunity for savings due to conservation. PPL St. 5-R at 6; Exh. DAK4; PPL M.B. at 164.

I&E developed its own offering based upon a direct customer analysis. In preparing its direct customer cost analysis, I&E stated that it was guided by long-standing Commission precedent that identifies the appropriate items to be included in a customer charge. According to I&E, those items that change with the addition or loss of a customer are the direct customer costs that were identified in the Company's cost of service study and are as follows: meter expenses, expenses for services and customer installations, expenses for meter reading and customer records & collection, other customer accounting expenses, depreciation expense and net salvage amortized for

meters and services, and the rate base related return and income taxes on customer-based rate base. I&E maintained that the Commission has long held these costs to be those most appropriately included in a customer cost study. I&E M.B. at 131 (citing *Pa. PUC v. West Penn Power Company*, 59 Pa. P.U.C. 552 (1985)). I&E noted that recently the Commission accepted a direct customer cost analysis identical to the analysis it presented in this proceeding¹⁹ in the Columbia Gas of Pennsylvania base rate case at Docket No. R-2010-2251623 (Order entered October 14, 2011). I&E recommended that the RS customer charge remain unchanged at \$8.75 per month. I&E M.B. at 130-133.

The Company countered that the OCA and I&E alternative customer cost analyses include only meters and services and exclude all other customer costs, which should be included in a customer charge. PPL M.B. at 170. Further, PPL pointed out that “conservation cannot and does not trump cost of service.” PPL M.B. at 164.

However, in response to the positions of the other Parties, PPL proposed an alternative plan that includes a residential customer charge of \$14.09 per month, consistent with the recent Commission decision in *Pa. PUC v. Aqua Pennsylvania, Inc.*, Docket No. R-00038805 (Order entered August 5, 2004) (*Aqua*). PPL stated that the costs included in its alternative Rate Schedule RS customer charge of \$14.09 per month properly reflect meters and services net plant and related O&M expenses; meter reading and billing and collection expenses, and the Company’s Meter Data Management System; as well as related employee benefits, administrative and general expenses and other O&M expenses related to the above items. These revenue requirement cost components represent the same type of direct and indirect cost components as those approved in *Aqua*. PPL M.B. at 172. The only difference is that PPL also included \$12,678,000 for customer call center-related expense. PPL averred that this expense was not specifically addressed in *Aqua*, but it is consistent with the expenses included in the

¹⁹ Tr. at 541-42.

customer charge in *Aqua*, because it is a directly assignable customer service-related expense, and it varies with the number of customer calls and the number of customers. *Id.* at 173; PPL St. 8-RJ (Part 2) at 8.

While PPL opined that the customer component of each rate schedule should include all customer-related costs determined by the cost of service study, if the Commission wished to consider an alternative compromise customer charge, PPL posits that its alternative proposal of \$14.09 would be acceptable as it would recover the same type of direct and indirect cost components as those approved in *Aqua*, and would provide some improvement in the level of fixed cost recovery in the customer charge. PPL St. 5-R at 15; PPL M.B. at 172-173.

I&E responded that it is improper to offer a compromise outside the context of a settlement, and that without an actual settlement the Company's position still needs to rely on substantial evidence to support it, which it did not provide. I&E R.B. at 107.

b. ALJ's Recommendation

The ALJ stated that while it would be improper to propose a compromise position for the first time in a brief or exception, it is not improper to propose an alternative during the litigation, when the supporting data already appears in the record, as PPL did in this proceeding. The ALJ recommended approval of the PPL alternative as it is based on an approved cost of service study, which clearly illustrates that customer-related costs for the residential class include elements that I&E ignored in its own analysis and determination of a proper residential customer charge. The ALJ found that it is reasonable to include some of these additional elements in calculating the residential customer charge, as the Commission allowed in the *Aqua* case. R.D. at 120.

According to the ALJ, the Company will be ensured recovery of more of its fixed costs, which are clearly more customer-related than usage-related, while still allowing some revenue to be recovered through usage-based charges. Thus, customers will be provided with more accurate price signals, while still being afforded some opportunity to control their monthly distribution bills through conservation. For these reasons, the ALJ concluded it is appropriate and reasonable to accept PPL's compromise position regarding the residential customer charge. R.D. at 120.

c. Exceptions

In its Exceptions, I&E states that the ALJ's recommendation to adopt the Company's compromise lacks legal support. I&E avers that as originally proposed, PPL's entire residential increase was to be recovered from an 82% increase to its RS customer charge without providing any direct customer cost analysis. I&E notes that PPL provided only a COSS, which is an entirely different cost analysis. According to I&E, PPL found very few, if any, distribution system-related costs that were a function of usage and proposed to recover essentially all fixed costs in the customer charge. I&E states that PPL included all fixed costs that it classified as customer related, as opposed to demand related, in the customer charge and made no distinction between direct and indirect costs. I&E avers that fixed costs and customer costs are not synonymous and opines that fixed costs assigned to the customer charge should be limited to those fixed costs for which there is a direct impact from an individual customer, such as metering and billing. I&E Exc. at 30-31.

Next, I&E notes that although PPL moderated its Rate RS proposal in rebuttal, it still failed to conduct an appropriate customer cost analysis. Rather, I&E asserts that PPL presented a "study" that included both direct and indirect costs that it claimed authorized a \$36.70 RS customer charge, but under which PPL only claimed a

“compromise” RS customer charge of \$14.09.²⁰ I&E avers that the ALJ’s reliance on one aberrant Commission order from 2004, *Aqua*, to support her recommendation to adopt the PPL compromise position lacks adequate legal support. I&E opines that the *Aqua* case is not controlling as the holding of that case, with respect to the inclusion of indirect costs in the calculation of a customer charge, has not been reaffirmed or reapplied since 2004. I&E maintains that since 1985 and most recently in 2011, with the one exception being *Aqua*, the Commission affirmed the basic customer cost analysis it originally articulated in 1985. I&E Exc. at 31-33.

Lastly, I&E states that as *Aqua* formed the sole basis presented by the ALJ for adoption of the Rate RS customer charge, the ALJ’s recommendation should be rejected. I&E maintains that PPL’s “compromise” RS customer charge fails to meet the parameters of a properly constructed customer cost analysis. Additionally, I&E asserts that the ALJ’s recommendation is not supported by the overwhelming Commission precedent and, unless prepared to enunciate a new standard, the Commission should reject it. Further, I&E notes that customers will lose control over a substantial part of their bill, very likely deterring conservation efforts despite the millions of dollars customers have invested in energy conservation efforts. I&E Exc. at 35-36.

The OCA also excepts to the ALJ’s recommendation, arguing that PPL’s proposed customer charge is based on its flawed COSS results, does not represent the results of a direct customer cost analysis, would disproportionately impact low-income, low-usage customers and would result in a significant disincentive for customers to engage in conservation activities. The OCA recommends that the Rate RS customer charge continue to be set at its correct level of \$8.75. The OCA avers that the ALJ erred by accepting PPL’s alternative RS customer charge without a direct cost study as support

²⁰ I&E asserted that PPL produced no such “study” and made no such “compromise” offer with respect to its originally proposed non-residential customer charges. I&E Exc. at 32.

for such a charge. According to the OCA, the Commission has repeatedly expressed its preference for a direct cost study, which includes only direct costs and not indirect costs, as PPL has done in its alternative proposal as a basis to set customer charges. The OCA notes that, in contrast to the decades of Commission precedent on this issue, PPL supports its alternative customer charge with the lone case of *Aqua*. The OCA submits that the *Aqua* decision was fact specific and provides no support for PPL's current proposal. OCA Exc. at 34-36.

Next, the OCA explains that it performed a direct customer cost analysis, consistent with Commission precedent, and found that the direct residential customer costs ranged from \$7.70 per month to \$8.24 per month. Therefore, the OCA is of the opinion that the current RS customer charge of \$8.75 is reasonable and should not be increased. The OCA avers that PPL's proposed customer charge will disproportionately impose adverse impacts on the customers least able to afford those bill increases and should not be accepted. OCA Exc. at 36-37.

In its Replies to Exceptions, PPL states that unlike the case relied upon by I&E, nothing in *Aqua* limits the Commission's holding only to that case. PPL avers that the Commission clearly stated that requests to include allocated indirect costs, such as employee benefits, local and payroll taxes, and other general and administrative costs, should be reviewed on a case-by-case basis, citing *Aqua*, at 70-72. PPL further submits that there is no order from either the Commission or the appellate courts overturning or otherwise limiting the Commission's conclusion in *Aqua*. PPL maintains that it followed the Commission's conclusion in *Aqua* and proposed the inclusion of the same type of direct and indirect cost components approved by the Commission in *Aqua*. PPL continues that I&E and the OCA failed to offer any criticisms or reasons to exclude from the customer cost study and customer charge the indirect costs that PPL allocated for employee benefits, local and payroll taxes, and other general and administrative costs.

For these reasons, PPL opines that the ALJ properly rejected the positions of I&E and the OCA. PPL R.Exc. at 17-18.

With regard to the Parties' comments on the impact on low income/low usage customers, PPL agrees that increasing the monthly charge while essentially maintaining the usage charge at its current level will result in a greater than average percentage increase to low use customers, regardless of their income level. However, PPL avers that as a utility with an obligation to serve, it must provide infrastructure to serve the needs of those customers. PPL states that utility rates should be designed based upon cost of service, not customers' income levels. According to PPL, ability to pay issues should be addressed through USPs, not by setting rates that disregard the cost of service. PPL R.Exc. at 19.

d. Disposition

Upon our consideration of the evidence of record herein, we shall adopt the ALJ's Recommendation on this issue that PPL's compromise proposal is reasonable and should be approved. In this regard, we conclude that PPL's original proposal is excessive, disregards the principle of gradualism and is not reasonable. Additionally, we conclude that the recommendations of I&E and the OCA that the residential customer charge not be increased at all in this proceeding are equally unreasonable as they are not based on a proper cost analysis. We further conclude that the ALJ correctly recommended that, consistent with *Aqua*, other customer-related costs are properly includable in a customer charge cost analysis. We find that the I&E proposed limitation of costs to only services and meters excludes all other customer costs that should be included in a customer charge and is unreasonably narrow.

With regard to the concerns expressed by the opposing Parties that PPL's compromise proposal discourages conservation, we note our agreement with the

Company's observation that the distribution charge is relatively small in the context of the energy portion of a customer's bill, which comprises approximately 86% of the charges on the average customer's bill. Therefore, we find that this will provide a more than adequate opportunity for customer savings due to energy conservation.

Therefore, we find that PPL has met its statutory burden of establishing the reasonableness of its compromise proposal. Accordingly, we adopt the recommendation of the ALJ and deny the Exceptions of I&E and the OCA on this issue.

5. Non-Residential Customer Charges

a. Positions of the Parties

PPL proposed increases to the customer charges in the Small General Service -- Rate Schedule GS-1 (GS-1), Large General Service -- Rate Schedule GS-3 (GS-3), Large Power Firm Service at 12 kV -- Rate Schedule LP-4 (LP-4), and Large Power Service at 69 kV -- Rate Schedules LP-5, LP-6, and IS-T (LP-5, LP-6 and IS-T).

PPL proposed to increase the customer charge for Rate GS-1 from \$14.00 to \$16.00 per month and decrease the demand charge from \$4.530 to \$4.258 per kW. PPL stated it has installed demand meters on all GS-1 customer premises, except for small unmetered constant load accounts. PPL St. 5 at 15; PPL Exhs. DAK 1, DAK 2; PPL Exh. 1, Exhibits Regs.

PPL proposed to increase the customer charge for Rate GS-3 from \$30.00 to \$40.00 per month and decrease the demand charge from \$4.510 to \$4.192 per kW. PPL St. 5 at 15; PPL Exhs. DAK 1, DAK 2; PPL Exh. 1, Exhibits Regs.

PPL proposed to increase the customer charge for Rate LP-4 from \$160.19 to \$170.00 per month and decrease the demand charge from \$2.136 to \$2.127 per kW. PPL St. 5 at 16; PPL Exhs. DAK 1, DAK 2; PPL Ex. 1, Exhibits Regs.

PPL proposed to increase the customer charge for Rate Schedule LP-5 from \$709.00 to \$1,125.00 per month. PPL stated that presently, there are only two customers on Rate Schedule LP-6. As there is no difference between Rate Schedules LP-6 and LP-5, PPL proposed to eliminate LP-6 and move the two remaining customers to Rate Schedule LP-5. Finally, PPL proposed to eliminate Rate Schedule IS-T because there are no customers on this interruptible service program. According to PPL, all of its interruptible service programs have been superseded by PJM Interconnection LLC's (PJM) programs. PPL St. 5 at 17; PPL Exhs. DAK 1, DAK 2; PPL Exh. 1, Exhibits Regs. PPL M.B. at 157-162.

According to PPL, its proposals to increase the customer charges and reduce the demand charge for these rate schedules are consistent with *Lloyd*, which held that rate structures should be adjusted to reflect the cost of service to each rate class and to eliminate cross-subsidization. *Id.*

I&E argued that the customer charges for these rate schedules should not be increased. I&E used its own direct customer cost analysis which, the Company argued, excludes certain items that the Company evaluation includes. I&E St. 3 at 12-14.

The Company averred that its minimum size system study is the appropriate basis for determining the fixed customer costs that are incurred to serve customers, and that those fixed costs should be recovered through a fixed customer charge. PPL argued that I&E's approach to setting the fixed monthly customer charges ignores the customer costs of the fixed and permanent infrastructure that the electric

distribution company is obligated to provide and which exists between a customer's service and the transmission substation from which the customer's load is served. PPL M.B. at 174.

b. ALJ's Recommendation

The ALJ stated that as she accepted the Company's cost of service-based evaluation for residential customers, it was consistent to accept it for the commercial and industrial customers as well. Therefore, the ALJ recommended that PPL's proposals be approved. R.D. at 121.

c. Exceptions

In its Exceptions, I&E states that the ALJ's recommendation to adopt the Company's non-residential customer charges to be consistent with the recommendation regarding the residential customer charge lacks factual support. I&E avers that its customer cost analysis did not distinguish between residential and non-residential classes, but was guided solely by the results of the properly constructed direct customer cost analysis. I&E points out that, while PPL proposed an alternative customer charge for the residential class, the Company produced no study or compromise offer with respect to its originally proposed non-residential customer charges. I&E asserts that the ALJ's recommendation to adopt PPL's non-residential customer charges to be consistent with the residential class is actually inconsistent since PPL did not present a compromise analysis applicable to the non-residential customer charges. Therefore, I&E opines that on the basis of that error, the ALJ's non-residential recommendation should not be adopted. I&E Exc. at 30-36.

In reply, PPL acknowledges that it has the burden of proof to establish that its proposed non-residential customer charges are just and reasonable; however, it is not

required to develop and present alternatives that it does not support. PPL avers that the evidence demonstrated that I&E's non-residential customer charges are based on its own direct customer cost analysis, which is based on a flawed process. PPL R.Exc. at 19-20.

d. Disposition

Upon our consideration of the record evidence, we conclude that PPL's proposed non-residential customer charges are reasonable and should be approved.

While the ALJ's comment concerning consistency may not be entirely accurate²¹, we find that her recommendation to approve PPL's non-residential customer charges is correct. It is important to note that none of the other Parties directly affected by these increased customer charges were opposed to the increase. Only I&E filed Exceptions to the ALJ's recommendation based on its own customer charge cost analysis that we have previously rejected. Accordingly, finding the ALJ's recommendation to be otherwise reasonable and duly supported by the evidence of record herein, it is adopted. The Exceptions of I&E on this issue are denied.

6. Net Metering for Renewable Customer-Generators Rider

a. Positions of the Parties

PPL proposed two changes to its Net Metering tariff provisions for Renewable Customer-Generators. First, PPL proposed to establish a limitation on the size of generator relative to the associated customer usage that would be eligible for net metering. Second, PPL proposed to clarify that, for eligible customer-generators served under PPL's Time of Use default service rate option, a weighted average of the on-peak

²¹ The ALJ stated that as she accepted the Company's cost of service-based evaluation for residential customers, it was consistent to accept it for the commercial and industrial customers as well. However, we note that PPL did not present a compromise analysis for the non-residential customer charge as it did for the residential customer charge.

and off-peak hour prices would be used to derive the Price to Compare for the purpose of compensating customer-generators for excess generation. PPL St. 5 at 25; PPL Exh. DAK 2; PPL M.B. at 180-181.

Both SEF and Granger opposed PPL's proposal to limit the eligibility for net metering based on the size of the generator relative to the associated customer usage. Granger opposed the as-filed proposal, as the Company proposed to limit the generation in all new net-metering applications to 110% of the customer-generator's connected load. SEF pointed out that the Company had provided no evidence to support an allegation that net metering customers cause PPL to incur costs that support an increase in the customer charge and asked that this allegation be rejected. Granger M.B. at 9, SEF R.B. at 1-2.

In response to this opposition, PPL withdrew this proposal and instead proposed a tariff revision to comply with the wording from the policy adopted by the Commission in the Commission's Final Order entered March 29, 2012, at Docket No. M-2011-2249441.²² This revision limits the 110% restriction to the business model where a third-party developer builds, owns, operates and maintains an alternative energy generation system on or near a customer's property and sells power and/or alternative energy credits to that customer. PPL St. 5-R.

Granger stated that the Company's revised proposal incorporated language from that Commission Order, and, consequently, it did not oppose the proposal. Granger M.B. at 9.

No party opposed PPL's second proposal, which was to revise the tariff to use the weighted average of the on-peak and off-peak hour TOU prices to derive the

²² *Net Metering – Use of Third Party Operators*, Docket No. M-2011-2249441 (Order entered March 29, 2012).

Price to Compare for customers served under PPL's Time of Use default service rate option. PPL explained that the stated purpose of this proposal was to ensure that compensation for excess generation by TOU customer-generators more closely reflects their actual on-peak and off-peak usage and generation. PPL M.B. at 180-182.

b. ALJ's Recommendation

The ALJ recommended that the revised net metering proposals be approved. R.D. at 126.

c. Disposition

No Party filed exceptions to the ALJ's recommendation. Finding it to be reasonable, we adopt it without further comment.

7. Competitive Enhancement Rider (CER)

a. Positions of the Parties

The Company proposed a new rider, the CER, to recover the costs of all customer education programs. PPL will estimate the total costs it expects to incur, on a calendar-year basis, to provide consumer education programs and competitive retail electricity market enhancement initiatives for all customers who receive distribution service from PPL. According to PPL, the CER will be a Section 1307(e), 66 Pa. C.S. § 1307(e), cost recovery mechanism developed to recover the Company's education and retail market enhancement (RME) related costs. PPL St. 8 at 30-32; PPL Exh. DAK 2; PPL M.B. at 180.

PPL argued that the Commission and the appellate courts have held that an automatic adjustment clause is appropriate when the expenses to be recovered are

substantial, subject to variation and beyond the control of the utility. PPL M.B. at 206 (citing *Popowsky v. Pa. PUC*, 869 A.2d 1144, 1159 (Pa. Cmwlth. 2005); *Pennsylvania Industrial Energy Coalition v. Pa. PUC*, 653 A.2d 1336 (Pa. Cmwlth. 1995); *Pa. PUC v. Newtown Artesian Water Co.*, Docket No. R-2009-2117550 (Order entered April 15, 2010); *Pa. PUC v. Philadelphia Thermal Energy Corp.*, 1991 Pa. PUC Lexis 80). According to PPL, its competitive enhancement expenses meet each of these standards. PPL M.B. at 206.

The Company estimated that the costs of the mandates in the RMI and other proceedings will be more than \$6 million annually, at least at the beginning, but will depend on the Commission's direction and are not within the control of the Company. PPL M.B. at 206.

The OCA, the OSBA, and Direct Energy have raised various issues and concerns regarding the proposed CER.

The OCA cautioned that care must be taken to prevent double recovery of these costs. In addition, the OCA noted that the Commission had recently held that the competitive enhancement costs should not be collected from ratepayers but from the EGSs. OCA M.B. at 125 (citing *Petition of FirstEnergy*, Docket No. P-2011-2273650, at 136 (Order entered August 16, 2012)). The OCA recommended three safeguards: (1) that the allowed costs must conform to the standards in the Commission's May 10, 2007 Order at Docket No M-000061957; (2) that competitive enhancements costs incurred by PPL, consistent with the Commission's directive, be collected from EGSs; and (3) that there be quantifiable assurances in place to prevent double recovery of these costs, such as through the CER and within the approved revenue requirement in this case. *Id.* at 125-126.

The OCA also recommended that the costs be allocated on a per kWh basis instead of per customer, reasoning that those with higher usage will benefit more from the information. According to the OCA, costs are incurred on a per customer basis and should be allocated accordingly. OCA M.B. at 126.

REG avers that Rate CER should be applied only to those customers and customer classes that benefit from the programs, activities, and enhancements funded by Rate CER. As customers already shopping know that they can shop and that Rate CER provides an incentive to customers to shop to the extent that it is imposed on them, Rate CER is best imposed on non-shopping customers to provide them with an incentive to shop and should not be imposed upon customers who have already selected alternative suppliers. REG M.B. at 6; REG R.B. at 1.

PPLICA limited its argument to cautioning the Commission to ensure that the Company's costs are not duplicated in multiple education programs. PPLICA M.B. at 21. PPLICA noted further that the Company's proposal to recover costs of RME programs from the EGSs that benefit from them is consistent with the Commission's Final Order in *Investigation of Pennsylvania's Retail Electricity Market: Intermediate Work Plan*, Docket No. I-2011-2237952 (Order entered March 2, 2012) and, therefore, PPLICA supported this proposal. PPLICA M.B. at 21.

Regarding recovery of costs, PPLICA opined that the costs allocated to a customer class should be recovered per customer, not per kWh, as proposed by the OCA as this is contrary to cost causation principles. PPLICA did not oppose approval of the proposal to recover CER costs through a fixed monthly customer charge. PPLICA M.B. at 23.

PPL asserted that to the extent it recovers these costs from EGSs, they would not be recovered through the CER. PPL M.B. at 209.

b. ALJ's Recommendation

The ALJ recommended that the CER be approved, and that the costs incurred by the Company in implementing the RME programs, including consumer education costs not recoverable from the EGSs, be recovered using the CER. The ALJ further recommended that as all customers benefit from the robust competitive market, then all customers should bear the costs involved in its development, on a per customer basis. R.D. at 128.

c. Exceptions

In its Exceptions, the OCA states that it opposes the ALJ's recommendation regarding retail market enhancement programs, and submits that this type of cost recovery for RME programs is inconsistent with the Commission's directives in this area. The OCA cites to the Commission's recent decision wherein the Commission held that EGSs should pay for RME costs. *Petition of FirstEnergy*, Docket No. P-2011-2273650, at 136 (Order entered August 16, 2012). The OCA avers that *FirstEnergy* is consistent with the Commission's decision to require EGSs to pay for the costs of opt-in auction programs in *Investigation of Pennsylvania's Retail Electricity Market: Intermediate Work Plan*, *supra*, at 79. OCA Exc. at 37-38.

The OCA also states that consumers that use more energy clearly have greater potential to benefit from these customer education programs than consumers who use very little electricity. Therefore, the OCA opines that a per kWh based rider better equates the costs and benefits of these programs. The OCA submits that whatever consumer education costs are ultimately recovered from ratepayers should be done on a kWh basis. OCA Exc. at 38-39.

The OSBA also excepted to the ALJ's recommendation, stating that there is no need at this time for yet another PPL reconcilable charge. The OSBA avers that implementing another rider will simply lead to the need for enhanced regulatory oversight to ensure that the costs claimed under the new rider include only those costs that were specifically identified as being associated with that rider. The OSBA notes that it agrees with PPL that it should be allowed to fully recover these costs, that many of these costs should be recovered from EGSs and that other Pennsylvania EDCs have similar riders. However, the OSBA does not believe that these costs should be recovered in the context of the instant distribution rate proceeding. The OSBA opines that a rate rider designed to recover RME costs would be better addressed in the Company's pending default service proceeding at Docket No. P-2012-2302074. According to the OSBA, it is established Commission policy that RME costs should be borne by EGSs and that this issue should be resolved in default service proceedings. OSBA Exc. at 15 (citing *FirstEnergy, supra*, at 136). OSBA Exc. at 13-15.

Next, the OSBA maintains that if the Commission does decide that the CER is necessary, then PPL's rate design for recovering the costs of the CER program should be changed. Instead of recovering these costs equally across all of the Company's customers as recommended by the ALJ, the OSBA submits that these costs should be directly assigned to PPL's rate classes for which costs can clearly be attributed. Furthermore, the OSBA avers that costs not specifically associated with a rate class should be allocated using some reasonable cost-based allocation factor. Then the Company should develop a separate CER charge for each rate class or rate class group, based on the allocated costs. OSBA Exc. at 16-17.

Finally, the OSBA submits that it is much more reasonable to directly assign costs, where possible, so that the cost-causing customer class pays. The OSBA asserts that in light of the high level of shopping that already exists among PPL's non-residential customers, it is not clear that there is any benefit to be gained by developing

RME programs for these customers. Additionally, if RME programs apply only to the residential class, PPL's proposal to effectively allocate those costs among all customers is clearly at odds with both cost causation and fairness considerations. OSBA Exc. 17.

In reply, PPL states that its proposed CER is appropriate for three principal reasons. First, PPL avers that such automatic adjustment clauses are appropriate for expenses that are substantial, vary and are beyond the utility's control. According to PPL, initially the CER annual expenses will total more than \$6.0 million and, thus, are substantial. PPL opines that they are subject to variation because they will change depending on Commission mandates in the RMI and other proceedings, and they are beyond PPL's control as they are incurred under Commission directives. Second, PPL avers that a CER permits a more flexible approach because it can be adjusted annually should the need for spending levels change in the future. PPL notes that such flexibility is not available if these costs are recovered through base rates. Third, PPL avers that other EDCs are employing Commission approved rider mechanisms to recover expenses incurred in response to the RMI. PPL R.Exc. at 23.

In response to the concerns expressed regarding the double recovery of costs, PPL maintains that the use of a specific reconcilable rider for all customer education expenses would assure that all costs are recovered only once. PPL opines that the possibility of double recovery would be eliminated as these expenses would all be reviewed annually in one reconciliation proceeding, and these expenses and revenues would be trued-up annually to make sure that only actual expenses are recovered. PPL R. Exc. at 23-24.

In response to the rate design issue expressed by the Parties, PPL avers that customer education costs should be recovered as it proposes on a per customer basis. PPL submits that this is consistent with cost causation because it costs the same to send a notice to an industrial customer as to a residential customer. PPL R.Exc. at 24.

Finally, PPL notes that the OSBA's proposal that the CER be addressed in PPL's default service proceeding is impractical, as it is too late for such matters to be considered in that proceeding since the record is closed. Also, PPL submits that it is important for PPL's proposed CER to be considered in this base rate case because, if it is adopted, it will have a direct impact on the level of base rates charged to customers. If it is not adopted, PPL claims that these costs would have to be recovered through base rates. PPL R.Exc. at 24.

In its Replies to Exceptions, PPLICA states that the ALJ correctly approved recovery of the costs included within the CER on a per customer basis. PPLICA avers that the costs potentially recoverable through the CER are generally customer costs and therefore rightfully recovered on a per customer basis. According to PPLICA, potential CER costs comprise broad marketing and education programs, which are readily distinguishable from the more consumption or demand-oriented energy efficiency and conservation plans administered under Act 129 of 2008. PPLICA R.Exc. at 11.

d. Disposition

We are in agreement with the ALJ that PPL's proposed CER is appropriate and should be approved. The CER is meant to recover the costs incurred by PPL to implement the RME Programs, including consumer education costs, not recoverable from EGSs, and should be designed on a per customer basis as proposed by PPL. We are persuaded by the arguments in favor of the CER presented by the Company. We agree that the costs proposed to be recovered through the CER qualify for recovery under an automatic adjustment clause, consistent with the Commonwealth Court's reasoning in *Pennsylvania Industrial Energy Coalition v. Pa. PUC*, 653 A.2d at 1349. We also concur that the CER provides a more flexible methodology for the Company to recover these Commission mandated expenses, and the CER is consistent with Commission approved

recovery mechanisms we have adopted in other EDC proceedings. Furthermore, we agree with PPL that these costs are properly recoverable on a per customer basis, consistent with cost-causation principles. Accordingly, we shall adopt the recommendation of the ALJ and deny the Exceptions of the OCA and the OSBA on this issue.

8. Purchase of Receivables

a. Positions of the Parties

PPL purchases, at a discount, the accounts receivable of EGS customers who participate in the Purchase of Receivables (POR) program. This discount is composed of an uncollectible accounts percentage factor and a development, implementation, and administrative factor. Uncollectible expenses are those costs that result from customers not paying for service, and the amount of the non-payment is written off. Uncollectible accounts expense associated with generation supply and transmission service for default service customers is separated from the Company's distribution rates and recovered through the Merchant Function Charge (MFC) and included in its Price to Compare. The cost of uncollectible expense is recovered from default customers through the MFC and from shopping customers through the discounted rate at which PPL purchases the accounts receivable within the POR program. PPL M.B. at 184-185.

The MFC percentages for the residential and small C&I customer classes have been calculated on the Company's expected 2012 uncollectible accounts expense for those customer classes. Based thereon, PPL proposed to change the MFC for the residential class from 1.80% to 2.23% and for small C&I customers from 0.10% to 0.23%. PPL St. 8 at 29-30; PPL St. 8-R at 43-44; PPL Exh. JMK 4.

PPL stated that in the ordinary course of business, the entity rendering the service is responsible for the costs and actions associated with billing and collection of payments, and also bears the risk of non-payment or late payments. Under a POR program, the EGS sells its accounts receivable to PPL and receives immediate payment for the amount due minus a discount meant to reflect collection risk and the time value of money. A POR program, therefore, allows the seller of the receivables to receive payment sooner and avoid the costs and risks associated with collecting any delinquent amounts owed by the customer. PPL M.B. at 184.

PPL explained that the existing POR program was authorized by the Commission's Order in *Petition of PPL Utilities Corporation Requesting Approval of a Voluntary Purchase of Accounts Receivables Program and Merchant Function Charge*, Docket No. P-2009-2129502 (Order entered November 19, 2009). In that Order, the Commission approved a settlement of the following factors: (1) the discount rate for residential service was 1.37%, consisting of an uncollectible accounts expense percentage factor of 1.32% and a POR administrative factor of .05%; (2) in order to participate, an EGS would sell all of its residential customer accounts receivables to the Company; (3) participating EGSs agreed to not reject new customers based on credit-related issues and would not require a deposit; (4) budget billing would be available to customers of participating EGSs; and (5) for small commercial and industrial shopping customers, the discount rate was 0.17%, reflecting an uncollectible accounts expense percentage factor of 0.12% and a POR administrative factor of 0.05%. PPL stated that the percentages were increased in the 2010 base rate case, *Pa. PUC v. PPL Electric Utilities Corporation*, Docket No. R-2010-2161694 (Order entered December 21, 2010). *Id.* at 185.

The Company noted that, in this proceeding, it based its proposed numbers on its actual write-offs from 2011, which were approximately \$40 million. PPL M.B. at 187; PPL St. 8-R at 43. To calculate the amount sought, PPL used its proposed 2012

budget amount, which is the sum of projected write-offs and the projected change in the reserve for doubtful accounts for 2012. PPL M.B. at 187; PPL St. 8-R at 44.

Direct Energy and DR opposed PPL's expected 2012 uncollectible accounts expense. Direct Energy recommended, instead, that PPL be permitted to recover 100% of its uncollectible accounts expense by implementing a non-bypassable/non-reconcilable charge applicable to all customers. In the alternative, Direct Energy recommended modifying the Company's proposal in the following manner: (1) by reducing the discount rate to reflect the amount of late payment charges that the Company collects and which offset its net uncollectible accounts expense; and (2) by reducing the discount factor by an administrative cost credit to return to the EGSs the amounts that have been collected through the administrative cost adder but which the Company did not track. Direct Energy St. 1 at 9-11; Direct Energy M.B. at 9.

Direct Energy averred that the Company's proposal must be rejected because there is no record basis to support allocation of the proposed uncollectible accounts expense percentage to generation service customers. Direct Energy claimed that while PPL has proposed that shopping and default customers pay at the same percentage level, it has not provided evidence to support a finding that this is just and reasonable. In fact, the Company admitted that it did not track write-offs by the shopping/default categories. Direct Energy M.B. at 12; Tr. at 404. While Direct Energy pointed out that it is possible that one category may be more reliable in paying bills than the other, and that the shoppers may be unfairly charged here, it is just as likely that the default customers are effectively subsidizing shopping customers. Direct Energy M.B. at 1.

Direct Energy also stated that the Company's proposal will stall development of a fully robust competitive retail market. Direct Energy noted that "[t]he level of competition in PPL's service territory is good, but it could be much better. The current levels of shopping need not only be sustained but increased in order to meet the

Commonwealth's goal of a fully competitive retail electric market. PPL's service territory presents the best opportunity to do that, but only if the Commission continues to remain vigilant about properly allocating costs to EGSs." *Id.* at 14 (footnotes omitted).

According to Direct Energy, the levels of uncollectible discount that PPL is proposing to charge through the POR program will have a significant negative effect on the development of competition because EGSs cannot administer their own programs efficiently and inexpensively and have no real choice but to rely on the Company. *Id.* at 15.

PPL denied that this increase will have a negative effect on the competitive market. While Direct Energy and DR argued that the EGSs would have to bear the difference in cost until the expiration of existing fixed-price contracts, PPL pointed out that there should have been no reasonable expectation that the discount rate would remain static indefinitely. According to PPL, such risk was willingly undertaken by the EGSs, is a business risk, and cannot be used to shift the risk of doing business as an EGS to PPL and its customers. PPL R.B. at 105-106.

Direct Energy stated that the Company's failure to properly support its own proposal opens the door for the Commission to consider the Direct Energy alternative, which is to collect total projected uncollectible accounts expense through a non-bypassable charge for all distribution customers. According to Direct Energy, this eliminates the need for determining the actual uncollectible expense. Direct Energy opined that this approach is superior to the Company's because it is consistent across shopping lines and does not contain the possibility of shoppers subsidizing default customers. Direct Energy M.B. at 18.

PPL argued that the dual MFC/POR method appropriately unbundles the uncollectibles charge and properly assigns risk of nonpayment and that Direct Energy's

proposal to refund all amounts that PPL has received under the administrative component of the POR should be rejected as impermissible retroactive ratemaking. PPL M.B. at 189-193. PPL also argued that the Commission has no authority to direct a change in its POR program due to its voluntary nature. *Id.* at 185.

Direct Energy responded that the POR is a tariffed program, which results in the requirement that it be just and reasonable. Direct Energy R.B. at 6-7. Direct Energy and DR further claimed that PPL should be required to use late payment charges to reduce the POR and MFC percentages. Direct Energy R.B. at 2, DR R.B. at 3.

PPL responded that late payment charges are paid, and are, therefore, not uncollectible but are revenue, as reflected in its accounting for decades and repeatedly approved by the Commission. PPL M.B. at 188; PPL St. 8-RJ at 8. In addition, PPL pointed out that late payment charges are used to reduce the overall distribution of revenue requirement for customer rate classes that bear the working capital requirement associated with overdue accounts receivable. PPL averred that granting this request would result in double counting. PPL M.B. at 188. Therefore, according to PPL, should the request be granted, the late payment fees would need to be split between the POR and MFC customers, accompanied by an adjustment in base rate revenues, which would increase rates for all distribution customers. PPL St. 8-RJ; PPL M.B. at 189.

b. ALJ's Recommendation

The ALJ recommended that the Company be required to track uncollectibles by default customers and shopping customers separately, and the correct percentage can be discerned from there. The ALJ noted that the proposed percentage is supported by the past uncollectibles in total, but there is no calculation of which uncollectibles are from default customers and which are from shopping customers.

According to the ALJ, this is not consistent with the terms of the settlement from which the POR program was conceived:

25. The Company will monitor individual EGS uncollectible percentages for small C&I customers pursuant to Section 12.9.2.6 of the tariff supplement provided in Appendix A and will adjust the discount rate for an individual EGS based upon the provisions contained therein.

R.D. at 131 (quoting *Petition of PPL Electric Utilities Corporation Requesting Approval of a Voluntary Purchase of Accounts Receivables Program and Merchant Function Charge*, Docket No. P-2009-2129502 (Order entered November 19, 2009)).

The ALJ expressed concern that PPL's procedure does not require the Company to determine the actual amount of its uncollectible expenses in order to recover them. The ALJ concluded that the actual amount of the uncollectible expenses is required in order to fairly charge customers the correct amount. Therefore, the ALJ found that PPL should be directed to take the next step and determine that amount for shoppers and to determine that amount for default customers, and to collect it accordingly. The ALJ recommended that PPL's proposed increase in the POR discount rate should be delayed for ninety days until the Company provides data indicating the proportions of uncollectibles attributable to default customers and to shopping customers, to support the proper discount rate. R.D. at 133, 142, O.P. # 10.

The ALJ further recommended that if PPL does not comply with this directive then the percentage discount rates currently in effect in its POR Program should remain in effect. R.D. at 142, Ordering Paragraph No. 11.

The ALJ also stated that Direct Energy and DR had not sustained their burden of proving that their alternatives were appropriate choices for the Commission to adopt in this case. R.D. at 133.

Additionally, the ALJ concluded that late payment fees are presently added to revenues, and that is where they should remain. *Id.* at 134.

c. Exceptions

In its Exceptions, Direct Energy avers that although the ALJ correctly concluded that PPL has failed to prove its increase for the POR discount rate, the ALJ erred in directing PPL to continue the current POR/MFC discount mechanism. Instead of continuing PPL's problematic mechanism, Direct Energy recommended that PPL be required to recover the currently unbundled uncollectible accounts expense in a non-bypassable charge applicable to all customers. Direct Energy avers that PPL's POR program, which reflects total uncollectible expense in the POR discount rate, has resulted in continuing and significant increases to the POR discount rate. Direct Energy compared the January 1, 2010, POR rate of 1.32% to the proposed rate in this proceeding of 2.23%. Direct Energy further notes that if the PPL proposal is adopted, then PPL's POR program would have the highest discount rate of all the Pennsylvania EDCs. Direct Energy Exc. at 3-5.

Direct Energy avers that its proposed non-bypassable mechanism would eliminate the need to determine the specific uncollectible accounts expense for shopping customers, while allocating the uncollectible accounts expense across all customers consistent with traditional rate-making principles. According to Direct Energy, while the ALJ criticizes its proposal because it does not require a calculation of actual uncollectible accounts expense for shopping customers, the fact here is that PPL cannot make that calculation. Direct Energy opines that even the ALJ acknowledged that when the actual

uncollectible accounts expense cannot be calculated, Direct Energy's approach is better than the one used by PPL, as she stated that it is "less unfair in its inherent unfairness." Direct Energy Exc. at 8 (quoting R.D. at 133).

Next, Direct Energy states that even if the ALJ's recommendation to continue PPL's current POR discount is adopted, the ALJ erred in failing to recommend adjustments to the calculation of the POR discount rate. According to Direct Energy, the Commission must direct that the initial starting point for the uncollectible accounts expense portion of the POR discount must be the same level of uncollectible accounts expense used to determine PPL's revenue requirement. From there, Direct Energy posits that the Commission should further adjust the POR discount rate to: (1) offset the uncollectible accounts expense percentage factor by the unbundled portion of the revenue PPL receives from late payment charges related to generation rates; and (2) create an administrative credit of 0.05% to the POR discount rate to return to EGSs the money PPL has collected during the POR program through the administrative component based on PPL's admitted failure to track actual incremental administrative costs and to quantify them. Direct Energy Exc. at 10-11.

In its Exceptions, DR first asserts that the ALJ should have set the POR discount at the 1.7% uncollectibles rate she adopted for ratemaking purposes. DR states that there is no real dispute in this case that the POR discount is the same as the uncollectibles rate and that PPL currently does not track uncollectibles separately as between shopping and non-shopping customers. Tr. at 404-405. Therefore, DR posits that PPL does not possess the historical data that would allow the immediate development of an appropriate uncollectible expense level, based on actual experience, for residential or commercial customers and differentiate between shopping and non-shopping customers. DR opines that any PPL proposed differentiation would be speculative, which is not permitted. According to DR, the more certain path would be to

require PPL to implement a POR discount based upon an uncollectible expense rate of 1.7%, which the ALJ accepted as reasonable. DR Exc. at 3-4.

Next, DR excepts to the ALJ's decision not to require PPL to use late payment fee revenue to reduce the POR discount. DR asserts that PPL cannot, and does not, reasonably dispute the fact that applying late payment fee revenue from shopping customers to offset the CWC expense for default service results in a subsidy to default service. DR submits that it proposed a reasonable means of eliminating this subsidization by using the late payment fee revenue from shopping customers to offset the uncollectibles expense of shopping customers. According to DR, under the methodology used today, shopping customers subsidize non-shopping or default service customers with every dollar of late payment fee revenue. DR asserts that this revenue should instead be used in a manner that provides at least some benefit to shopping customers, not an exclusive benefit to default service customers as it does today. DR Exc. at 5-6.

In reply, PPL states that it fully explained why Direct Energy's non-bypassable proposal should be rejected, including the fact that the Commission recently considered and rejected the very same proposal in PPL's 2010 base rate case. Also, PPL states that if the ALJ recommendation is approved by the Commission, the Company can and fully intends to promptly comply with the recommendation to track and separately determine the uncollectible accounts expense for shopping customers. In response to the Parties' proposal that the POR discount rate be set at the 1.7% three-year average of uncollectible accounts expense accepted by the ALJ, PPL opines that the 1.7% rate understates PPL's projected uncollectible accounts expense. PPL R. Exc. at 20-21.

PPL next notes that Direct Energy and DR continue to argue that late payment charges from shopping customers offset or reduce uncollectible accounts expense. PPL asserts that is not the case as these charges represent an addition to a utility's revenues and offset accounts receivable. PPL explains that late payment charges

are actually paid by customers and the revenues received from late payments are, by definition, not uncollectible. According to PPL, the proposal advanced by Direct Energy and DR would result in double counting of late payment revenues by crediting these revenues to customers twice. PPL R. Exc. at 21.

Lastly, in response to Direct Energy's proposal in regard to the administrative component of the POR discount rate, PPL claims that Direct Energy ignores the record evidence that the Company has incurred incremental expenses with its POR program. PPL asserts that the POR is a Section 1308 rate and cannot be retroactively changed. PPL R. Exc. at 21.

In its Reply Exceptions, PPLICA states that the ALJ correctly rejected the proposal that PPL implement a non-bypassable charge for recovery of uncollectibles expense currently recovered through the POR discount. PPLICA asserts that the ALJ's rejection of a non-bypassable charge reflects the many flaws inherent in this proposal, including the potential for double charging customers not eligible for PPL's POR program and the rebundling of generation, transmission and distribution charges. PPLICA requests that the Commission adopt the ALJ's recommendation. Further, PPLICA avers that the ALJ's rejection of this proposal is fully consistent with Commission precedent and the Code. PPLICA explains that the Commission addressed a similar proposal from the Retail Energy Supply Association in PPL's 2010 rate case and held that "EGSs should bear the collection risk for their own customers, either by including it in the charges to those customers or by selling their receivables to PPL at a discount." PPLICA R. Exc. at 13 (quoting *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-2010-2161694, at 153 (Order entered December 21, 2010)). PPLICA further asserts that adoption of Direct Energy's proposal would violate Section 2804(3) of the Competition Act, 66 Pa. C.S. § 2804(3), which requires EDCs to unbundle generation, transmission and distribution rates. PPLICA R. Exc. at 12-13.

In its Replies to Exceptions on this issue, the OSBA states that although the Direct Energy language it quotes in its Exceptions does not say so, Direct Energy is addressing the residential class uncollectibles rate. The OSBA explains that for the majority of Direct Energy's Exceptions, the 1.7% is referred to as "the uncollectibles rate" when it is, in fact, just the rate for the residential customers. While the OSBA agrees with Direct Energy that the uncollectibles rate determined for the residential class should be used to develop both the residential MFC and the residential POR discount, the OSBA cautions that the 1.7% factor is not appropriate for the non-residential classes. According to the OSBA, the Small C&I and Large C&I MFC and POR discount rates should reflect the uncollectibles rates applicable to those classes. OSBA R. Exc. at 14-15.

d. Disposition

First, with regard to Direct Energy's recommendation for the use of a non-bypassable distribution charge applicable to all customers to collect uncollectible expenses, we find that PPL correctly explained that the use of a non-bypassable charge is improper and has previously been rejected in PPL's prior 2010 base rate proceeding. In that Order we held that the collection risk for shopping customers should remain with the EGSs. *Pa. PUC v. PPL Electric Utilities Corp.*, Docket No. R-2010-2161694, at 95. We affirm that position in the instant proceeding. Therefore, the Exceptions of Direct Energy are denied on this issue.

Next, we agree with the ALJ's recommendation to delay the implementation of the Company's proposed increase in the POR discount percentage for ninety days. We concur with the ALJ's directive that the currently effective rates remain in effect until PPL provides the required breakdown on these expenses between shopping and non-shopping customers. Once this information is developed, the Commission will have thirty additional days to finalize an appropriate course of action. We note that the

Company stated in its Replies to Exceptions that it can and fully intends to promptly comply with the ALJ's recommendation to track and separately determine the uncollectible accounts expense for shopping customers. We also agree with the ALJ that if PPL fails to provide this information, then the currently effective discount rates shall remain unchanged. Therefore, the Exceptions of Direct Energy and DR are denied on this issue.

In response to the Direct Energy and DR recommendation to offset uncollectible accounts expense with late payment fees, we are persuaded by the arguments of PPL that late payment fees do not reduce uncollectibles. We agree with PPL that late payment charges are actually paid by customers and are used to reduce the overall distribution of revenue requirement for customer rate classes that bear the working capital requirement associated with overdue accounts receivable. Accordingly, we adopt the recommendation of the ALJ on this issue and deny the Exceptions of Direct Energy and DR.

In conclusion, we address the recommendation of Direct Energy that since PPL did not track the incremental expenses under the 0.05% administrative cost component of the POR discount rate, then PPL should be directed to refund all amounts collected to date under this component until the amount PPL has collected is returned. We find it disappointing that PPL did not track these costs. The administrative component of the POR rate was designed with cost recovery of incremental costs in mind. However, the tariff did not provide for these refunds. In order to avoid a repetition of this failure, the Parties should address the issue in future proceedings so as to provide a more equitable outcome.

Going forward, we direct PPL to track and make an appropriate filing with the Commission describing all revenues and incremental costs incurred to develop, implement, and administer the POR service, including costs since inception, associated

with implementation of its POR service if it desires to seek any further administrative cost recovery in the future. If, at that time, it is determined that PPL over-recovered historical administrative costs, future cost recovery will only be allowed once the historical over-recovery is netted out. Accordingly, we shall adopt the ALJ's recommendation, as modified by this Opinion and Order, and deny the Exceptions of Direct Energy.

In summary, we hold that PPL's proposed POR program discount rates remain as currently in effect for ninety days and that PPL is directed to provide the breakdown of uncollectible expenses between shopping and non-shopping customers within ninety days. If PPL does not comply with this directive, then the percentage rates currently in effect in its POR program shall remain in effect. Furthermore, the recommendations of the intervening Parties with regard to the implementation of a non-bypassable charge, the offset of late payment fees and the refund of the administrative cost component are denied, consistent with the discussion herein.

IV. CONCLUSION

We have reviewed the record as developed in this proceeding, including the ALJ's Recommended Decision and the Exceptions and Replies to Exceptions filed thereto. Based upon our review, evaluation and analysis of the record evidence, the Exceptions filed by the various Parties hereto are granted or denied, and the ALJ's Recommended Decision is modified, consistent with the discussion in this Opinion and Order; **THEREFORE,**

V. ORDERING PARAGRAPHS

IT IS ORDERED:

1. That the Exceptions of the Office of Small Business Advocate, Direct Energy Services, PP&L Industrial Customer Alliance, the Commission on Economic Opportunity and Dominion Resources, filed on November 8, 2012, are denied, consistent with this Opinion and Order.
2. That the Exceptions of PPL Electric Utilities Corporation, the Office of Consumer Advocate, the Bureau of Investigation and Enforcement, filed on November 8, 2012, are granted in part, consistent with this Opinion and Order.
3. That the Recommended Decision of Administrative Law Judge Susan D. Colwell, issued on October 19, 2012, is adopted as modified by this Opinion and Order.
4. That PPL Electric Utilities Corporation shall not place into effect the rates, rules and regulations contained in Supplement No. 118 to Tariff – Electric Pa. P.U.C. No. 201, as filed.
5. That PPL Electric Utilities Corporation is authorized to file tariffs, tariff supplements and/or tariff revisions, on less than statutory notice, and pursuant to the provisions of 52 Pa. Code §§ 53.1, *et seq.*, and 53.101, designed to produce an annual distribution rate revenue increase of approximately \$71.065 million, to become effective for service rendered on and after January 1, 2013.

6. That PPL Electric Utilities Corporation shall file detailed calculations with its tariff filing, which shall demonstrate to the Commission's satisfaction that the filed tariff adjustments comply with the provisions of this final Opinion and Order.

7. That PPL Electric Utilities Corporation shall allocate the authorized increase in operating distribution revenue to each customer class, and rate schedule within each customer class, in the manner prescribed in this Opinion and Order.

8. That PPL Electric Utilities Corporation shall comply with all directives, conclusions, and recommendations contained in the body of this Opinion and Order, which are not the subject of an individual directive in these ordering paragraphs, as fully as if they were the subject of a specific ordering paragraph.

9. That the Formal Complaints filed by the Office of Consumer Advocate, the Office of Small Business Advocate and PP&L Industrial Customer Alliance are sustained in part, consistent with this Order.

10. That the Formal Complaints filed by William Andrews; Tracey Andrews; Eric Joseph Epstein; Dave A. Kenney; Roberta Kurrell; Donald Leventry; John G. Lucas and Helen Schwika, and any other Formal Complaint not specifically noted but filed prior to issuance of this Opinion and Order, are hereby dismissed.

11. That, upon Commission approval of the tariff, tariff supplements and/or tariff revisions, submitted in compliance with this Opinion and Order, the investigation at Docket Number R-2012-2290597 shall be marked closed.

BY THE COMMISSION,

A handwritten signature in black ink, appearing to read "Rosemary Chiavetta", written in a cursive style.

Rosemary Chiavetta
Secretary

(SEAL)

ORDER ADOPTED: December 5, 2012

ORDER ENTERED: December 28, 2012

Pennsylvania Public Utility Commission

v.

PPL Electric Utilities Corporation
Docket No. R-2012-2290597

Commission Tables Calculating Allowed Revenue Increase

Table I	Income Summary
Table II	Rate of Return
Table III	Revenue Factor
Table IV	Adjustments
Table V	Interest Synchronization
Table VI	Cash Working Capital: O&M Component
Table VII	Cash Working Capital: Accrued Taxes
Table VIII	Cash Working Capital: Interest Payments
Table IX	Cash Working Capital: Summary
Table X	Capital Stock Tax
Table XI	Gross Receipts Tax
Table XII	Reconciliation

TABLE I
PPL Electric Utilities Corporation
INCOME SUMMARY
R-2012-2290597
(\$000s)

<u>Commission Final Allowance</u>					
	Pro Forma Present Rates (Revised) ⁽¹⁾ (1)	Commission Adjustments ⁽²⁾ (2)	Commission Pro Forma Present Rates (3) = (1) + (2)	Commission Revenue Increase (4)	Total Allowable Revenues (5) = (3) + (4)
1. Operating Revenue	780,425	0	780,425	71,030	851,455
2. Expenses:					
3. O & M Expense	417,869	(11,966)	405,903	1,759	407,662
4. Depreciation	139,719	0	139,719	0	139,719
5. Taxes, Other	53,516	(734)	52,782	4,149	56,931
6. Income Taxes:					
7. State	1,571	1,346	2,917	6,506	9,423
8. Federal	(7,321)	4,245	(3,076)	20,516	17,440
9. Deferred Inc.	28,861	0	28,861	0	28,861
10. ITC	(915)	0	(915)	0	(915)
11. Total Expenses	633,300	(7,109)	626,191	32,930	659,120
12. Income Available for Return	147,125	7,109	154,234	38,100	192,334
13. Rate Base	2,420,963	(13,237)	2,407,726	38,100	2,407,726
14. Rate of Return	6.08%		6.41%	(0)	7.98822%

⁽¹⁾ PPL Exh. Future-1 Revised 7-16-12; Schedule D-1, Column 6.

⁽²⁾ From Table IV - Adjustments

TABLE II
PPL Electric Utilities Corporation
RATE OF RETURN
R-2012-2290597

Commission Final Allowance

	Structure (1)	Cost (2)	After-Tax Weighted Cost (3)=(1)X(2)	Effective Tax Rate Complement (4)	Pre-Tax Weighted Cost Rate (5)=(3)X(4)
1. Total Cost of Debt			2.70710%		2.70710%
2. Long-term Debt	49.22%	5.50%	2.70710%		2.71%
3. Short-term Debt	0.00%	0.00%	0.00000%		0.00%
4. Preferred Stock	0.00%	0.00%	0.00000%	0.585065	0.00%
5. Common Equity	50.78%	10.40%	5.28112%	0.585065	9.03%
6. Totals	100.00%		7.98822%		11.74%
7. Pre-Tax Interest Coverage (11.74% / 2.70710%)	4.336%				
8. After-Tax Interest Coverage (7.872% / 2.70710%)	2.951%				
9. Tax Rate Complement (1-(35%+(9.99% X (1-35%)))	58.50650%				

TABLE III
PPL Electric Utilities Corporation
REVENUE FACTOR
R-2012-2290597

Commission Final Allowance

1.	100%	100.0000%
2.	Uncollectible Accounts Factor	-2.23000%
3.	(Line 1-Line 2)	97.7700%
4.	PUC, OCA, OSBA Assessment Factors	0.2460%
5.	Gross Receipts Tax (GRT) (Modified per Commission Order)	5.7684%
6.	Other Tax Factors (PA CST)	0.0746%
7.	(Sum of Lines 4, 5 and 6)	6.0890%
8.	Effective Assmt/GRT/CST (Line 7)	6.0890%
9.	Factor after Assmt, GRT and CST (Line 3 - Line 8)	91.681%
10.	State Corporate Net Income Tax Rate	9.990%
11.	Effective State Income Tax Rate (Line 9 x Line 10)	9.1589%
12.	Factor After Local and State Taxes (Line 9 - Line 11)	82.5220%
13.	Federal Corporate Income Tax Rate	35.00%
14.	Effective Federal Income Tax Rate (Line 13 x Line 12)	28.883%
15.	Revenue Factor (100% - Effective Tax Rates); (Line 1 - (Lines 2, 3, 8, 11 and 14))	53.6393%

TABLE IV
PPL Electric Utilities Corporation
SUMMARY OF COMMISSION ADJUSTMENTS
R-2012-2290597
(\$000)

Commission Final Adjustments

Adjustments	Rate Base	Revenues	Expenses	Depreciation	Taxes-Other	State Income Tax	Federal Income Tax
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
1. RATE BASE:							
2. CWC:							
3. Int. & Div. (Table VI)	(63)						
4. Taxes (Table V)	174						
5. O & M (Table VI)	(13,346)						
6. TAXES OTHER:							
7. Gross Receipts Tax (Table XI)					(259)	26	82
8. Capital Stock Tax (Table X)					(475)	47	150
9. REVENUES:		0				0	0
10. EXPENSES:							
11. Rate Case Normalization - PPL Revision			1,200			(120)	(378)
12. Environmental Management			(103)			10	33
13. Storm Damage Claim - PPL Revision			(3,500)			350	1,103
14. Office of General Counsel			(1,200)			120	378
15. Office of the Chairman			(387)			39	122
16. Consumer Education: Act 129 Move to CER Rider			(2,494)			249	786
17. Consumer Education: Pre Act 129			(5,482)			548	1,727
18. DEPRECIATION:						0	0
19. TAXES:						0	0
20. Interest Synchronization (Table V)						0	0
21. TOTALS	(13,237)	0	(11,956)	0	(734)	1,346	4,245

Notes:

Rate Case Expense and Office of General Counsel: PPL Exhibit Future 1-Revised presents PPL's revised rate case expense which removes \$1.2 million representing the double counting of OGC expense. See PPL Future Future 1-Revised Schedule D-6. This \$1.2 needs to be added back to Rate Case Expense to reflect adoption of the ALJ's recommended treatment of this expense.

Storm Damage Claim: PPL reduced its claim due to the unavailability of this type of insurance.

Office of General Counsel: This duplicated expense was removed from the Rate Case Expense claim in PPL Future 1-Revised Schedule D-6. The ALJ's R.D. reflected this as a reduction to the OGC expenses.

Consumer Education Pre Act 129: The ALJ's R.D. allowed this expense only to the end of the FTY but the adjustment was inadvertently excluded from the Tables attached to the R.D.

TABLE V
PPL Electric Utilities Corporation
INTEREST SYNCHRONIZATION
R-2012-2290597

<u>Commission Final Adjustments</u>		Amount (000)
1.	Company Rate Base Claim (PPL Exh. Furure 1-Revised Sch. C-1)	\$2,420,963
2.	Commission Rate Base Adjustments (From Table IV)	<u>(\$13,237)</u>
3.	Commission Rate Bas (Line 1 - Line 2)	\$2,407,726
4.	Weighted Cost of Debt (From Table II)	<u>2.7071%</u>
5.	Commission Interest Expense (Line 4 x Line 3)	\$65,180
6.	Company Claim (PPL Exh. JMK 2 p. 28, Line 1)	<u>\$65,947</u>
7.	Commission Adjustment (Line 6 - Line 5)	\$767
8.	Company Adjustment	<u>\$0</u>
	Net Commission Interest Adjustment (Line 7 - Line 8)	\$767
8.		\$767
10.	State Corporate Net Income Tax Rate	<u>9.99%</u>
	State Corporate Net Income Tax Adjustment (Line 10 x Line 9) (Flow to Table IV)	<u>\$77</u>
11.		
	Net Commission Adjustment for Federal	
12.	Taxable Income (Line 9 - Line 11)	\$690
13.	Federal Income Tax Rate	<u>35.00%</u>
14.	Federal Income Tax Adjustment (Line 13 x Line 12) (Flow to Table IV)	<u><u>\$242</u></u>

TABLE VI
PPL Electric Utilities Corporation
CASH WORKING CAPITAL: O & M COMPONENT
R-2012-2290597

Commission Allowance

(000)

1.	Total O&M Expense per PPL (PPL Exh Future 1-Revised Sch. C-4, p. 2)	\$465,055
2.	Jurisdictional Factor (See Line 15 Below)	<u>85.973%</u>
3.	Jurisdictional O&M Expense (Line 1 X Line 2)	\$399,823
4.	Commission O&M Adjustments (From Table IV)	<u>(\$11,966)</u>
5.	Net O&M Expense (Line 3 - Line 4)	\$387,857
6.	O&M Expense per Day (Line 5 / 365 days)	\$1,063
7.	Average Lag Days (ALJ R.D. at 19-20; Order at Section II C.)	9.60
8.	Commission Allowed O&M CWC Requirement (Line 6 X Line 7)	\$10,201
9.	Company Claim (PPL Future 1-Revised Sch. C-4, p. 2)	\$27,391
10.	Jurisdictional Portion of Company Claim (Line 2 X Line 15)	\$23,549
11.	Commission Adjustment to Rate Base (Line 8 - Line 10); (Flow to Table IV)	<u><u>(\$13,348)</u></u>
12.	<u>O&M Expense Per Company Filing:</u>	
13.	Total O&M (PPL Future 1-Revised Sch. D-1, Col. 5)	\$486,045
14.	Jurisdictional O&M (PPL Future 1-Revised. D-1, Col. 6)	<u>\$417,869</u>
15.	Jurisdictional Factor (To Line 2 above)	<u><u>85.973%</u></u>

TABLE VII
PPL Electric Utilities Corporation
CASH WORKING CAPITAL: ACCRUED TAXES
R-2012-2290597

Commission Allowance

	Pro Forma Taxes (000) (1)	Twelve-Month Accrued Factor per Company (2)	Accrued Taxes (000) (3) = (1) X (2)
1. Federal Income Tax	(\$1,312)	(5.95000%)	\$78
2. PA Corporate Net Income Tax	\$11,864	(3.86000%)	(\$458)
3. PA Gross Receipts Tax (See Below)	\$47,861	33.64000%	\$16,100
4. PA Capital Stock Tax	\$2,017	(3.86000%)	(\$78)
5. PA PURTA Tax	\$2,832	21.14000%	\$599
6. Total Accrued Taxes (Sum of Lines 1 - 5)			\$16,241
7. Accrued Taxes per ppl (PPL Exh. Future 1-Revised, Sch. C-4, p. 4, Line 6)			\$16,068
8. Adjustment to Accrued Taxes (Line 6 - Line 7) (Flow to Table IX)			<u>\$173</u>

PA Gross Receipts Tax

9. Per Company (Future Revised Schedule D-11 p 13)	\$43,670
Adjustment Due To Allowed Revenue	
10. Increase (Allowed Increase X 0.059)	\$4,191
	<u>\$47,861</u>

PA Capital Stock Tax

11. Per Company (Future Revised Schedule D-11 p. 2)	\$1,954
Adjustment Due To Allowed Revenue	
12. Increase (Allowed Increase X 0.0089)	\$63
	<u>\$2,017</u>

TABLE VIII
PPL Electric Utilities Corporation
CASH WORKING CAPITAL: INTEREST PAYMENTS
R-2012-2290597

Commission Allowance

	(000)
1. Rate Base (Table I)	\$2,407,726
2. Weighted Average Cost of Debt (Table II)	<u>2.70710%</u>
3. Interest Expense (Line 1 x Line 2)	\$65,180
4. Daily Amount of Interest Expense (Line 3 / 365)	\$178.57
5. Interest Payment Lag Days	<u>32.90</u>
6. Commission CWC Interest (Line 4 x Line 5)	\$5,875
7. Company Claimed CWC Interest	<u>\$5,938</u>
8. Commission Adjustment (Flow to Table IX)	<u><u>(\$63)</u></u>

TABLE IX
PPL Electric Utilities Corporation
CASH WORKING CAPITAL
R-2012-2290597

	(000)
1. O&M Expense (PPL Future 1-Revised, Sch. C-4 p. 2, Line 4)	\$27,391
2. Average Prepayments (PPL Future 1-Revised Sch. C-4, p. 3, Line 15)	\$3,174
3. Accrued Taxes (PPL Future 1-Revised, Sch C-4 p. 4, Line 6)	\$16,068
4. Interest Payments (PPL Future 1-Revised Sch. C-4, p. 5, Line 9)	<u>(\$8,061)</u>
5. Total CWC per Company (Sum of Lines 1 through 4)	\$38,572
6. Jurisdictional CWC	<u>\$31,593</u>
<u>Commission Adjustments</u>	
7. Calc O&M Difference (From Table VI Row 11)	(\$13,348)
8. Calc Accrued Tax Difference (From Table VII)	\$173
9. Calc Interest Payment Difference (From Table VIII)	<u>(\$63)</u>
10. Commission CWC Adjustments (Lines 7 + 8 + 9)	(\$13,237)
11. Total CWC (Line 6 + Line 10)	<u>\$18,356</u>

TABLE X
PPL Electric Utilities Corporation
CAPITAL STOCK TAX (CST)
R-2012-2290597

	(000)	(000)
<u>Net Income</u>	Present Rate Adjustment	Proposed Rate Adjustment
1. 2008	\$87,403	\$87,403
2. 2009	\$103,885	\$103,885
3. 2010	\$80,572	\$80,572
4. 2011	\$129,591	\$129,591
5. 2012	\$97,491	\$140,630
6.	<u>\$498,942</u>	<u>\$542,081</u>
7. Average (Line 6 / 5)	<u>\$99,788</u>	\$108,416
8. Net Worth at December 31, 2012 (Note 2)	<u>\$1,790,672</u>	<u>\$1,833,811</u>
9. Pa Capital Stock value (Per Statutory Formula)	\$1,196,704	\$1,258,269
10. Statutory Exemption	<u>(\$160)</u>	<u>(\$160)</u>
11. Value of Capital Stock less Statutory Exemption	\$1,196,544	\$1,258,109
12. Apportionment Percentage	<u>95.4053%</u>	<u>95.4053%</u>
13. Pa CST Value (Line 11 x Line 12)	<u>\$1,141,566</u>	<u>\$1,200,302</u>
14. PA CST at 0.89 mills (Line 13 x 0.89 mills)	\$1,016	\$1,068
15. Less: PA Education tax credit	<u>(\$217)</u>	<u>(\$217)</u>
16. PA CST at Proposed Rates (Line 14 + Line 15)	\$799	\$851
17. Less: PA CST at Present Rates (PPL Sch D-11 p. 2)	<u>\$1,941</u>	<u>\$799</u>
18. Jurisdictional Allocation (Note 3 Below)	41.6070%	41.6070%
19. Adjustment to PA CST (Forward to Table IV)	<u>(\$475)</u>	<u>\$52</u>
Note 1		
20. Net Income at Present Rates	\$97,491	
21. Net Income from Proposed Rate Increase (Table XI Line 9)	<u>\$ 38,100</u>	
22. Total 2012 Net Income (Line 19 + Line 20)	<u>\$135,591</u>	
Note 2		
23. Net Worth at Present Rates	\$1,790,672	
24. Net Worth from Proposed Rate Increase (Table XI Line 9)	<u>\$ 38,100</u>	
25. Total 2012 Net Worth (Line 22 + 23)	<u>\$1,828,772</u>	
Note 3		
26. Total PPL Electric (PPL Exhibit JMK-2 p. 22)	\$1,954	
27. PA Jurisdictional (PPL Exhibit JMK-2 p. 22)	<u>\$813</u>	
28. Allocation for PA Jurisdictional (Line 26 / Line 25)	<u>41.6070%</u>	

TABLE XI
PPL Electric Utilities Corporation
GROSS RECEIPTS TAX (GRT)
R-2012-2290597

<u>Commission Allowance</u>	(\$000)
1. Base for GRT (PPI Exh JMK2 p. 26)	\$744,568
2. Less Uncollectible Accounts Expense (PPL Exh JMK 2 p. 51)	<u>(\$14,055)</u>
3. Net Gross Receipts (Line 1 - Line 2)	\$730,513
4. GTR Rate	<u>5.90%</u>
5. GRT on Net Gross Receipts (Line 4 X Line 5)	\$43,100
6. PA Jurisdictional Base for GRT (Line 1 X (1 - 0.0059))	\$740,175
7. Less Uncollectible Accounts Expense (PPL Exh JMK 2 p. 51)	<u>(\$14,055)</u>
8.	\$726,120
9. GTR Rate	<u>5.90%</u>
10. GRT on Jurisdictional Net Gross Receipts (Line 9 X Line 8)	\$42,841
11. Reduction to GRT by Excluding Uncollectible Accounts Expense (Line 5 - Line 10) (Flow to Sch. IV)	<u><u>(\$259)</u></u>

Table XII
PPL Electric Utilities Corporation
R-2012-2290597
RECONCILIATION OF
Operating Revenue and Applicable Tax
Commission Allowed Rate Increase
(000)

Commission Allowance

<u>Line No.</u>	<u>Description</u>	<u>Amount</u>
1.	Commission Allowed Increase (Table I)	\$71,030
	Provision for Uncollectibles (Line 1 x 2.23%)	\$1,584
2.	PUC, OCA, OSBA Assessment	\$175
	PA Gross Receipts Tax (Line 1 - Line 3) * 59 mills	\$4,097
3.	PA Capital Stock Tax (Table X Line 16)	\$52
4.	Sub Total	\$5,908
5.	Taxable Income for PA CNI Tax (Line 1 - Line 4)	\$65,122
6.	PA CNI Tax (Line 5 * 9.99%)	\$6,506
7.	Federal Taxable Income (Line 5 - Line 6)	\$58,616
8.	Federal Income Tax (Line 7 * 35%)	\$20,516
9.	Operating Income (Line 7 - Line 8) (See Table I, Col. 4, Line 12)	\$38,100

RRA Regulatory Focus

Major Rate Case Decisions — January-September 2019

Average authorized electric and gas ROEs edged up slightly overall in the first three quarters of 2019 versus calendar 2018.

The average ROE authorized electric utilities was 9.63% in rate cases decided in the first nine months of 2019, somewhat above the 9.60% average for cases decided in calendar 2018. There were 31 electric ROE determinations in the first nine months of 2019, versus 48 in the full year 2018. Even so, the average is still hovering around the lowest levels witnessed in the industry, and with the recent rate cuts by the U.S. Federal Reserve and the potential for further cuts, lower authorized returns may be on the horizon.

This data includes several limited-issue rider cases. Excluding these cases, the average authorized ROE was 9.57% in electric rate cases decided in the first nine months of 2019, largely in line with the 9.56% average observed in 2018. The difference between the ROE averages including rider cases and those excluding the rider cases is driven by ROE premiums allowed in certain states for riders that address recovery of specific generation projects.

The average ROE authorized gas utilities was 9.68% in cases decided during the first nine months of 2019, just above the 9.59% in full-year 2018. There were only 10 gas cases that included an ROE determination in the first three quarters of 2019, versus 40 in 2018.

In the first nine months of 2019, the median authorized ROE for all electric utilities was 9.60%, versus 9.58% in full-year 2018; for gas utilities, the metric was 9.72%, versus 9.60% in 2018.

From a longer-term perspective, interest rates, as measured by the 30-year U.S. Treasury bond yield, fell almost steadily from the early 1980s until 2015 or so, placing downward pressure on authorized ROEs. Even though the decline has been less dramatic in the period since 1990, average authorized ROEs fell below 10% for gas utilities in 2011 and for electric utilities in 2014.

After reaching an almost 30-year high in 2018, when almost 140 cases were decided, rate case activity has moderated somewhat in 2019, with only about 75 electric and gas cases resolved in the first nine months. Since 2010, rate case activity has been robust, with 100 or more cases adjudicated in seven of the last nine calendar years. This count includes electric and gas cases where no ROEs have been specified; however, withdrawn cases are not included. With about 110 cases currently pending, it appears that the tally for 2019 will again top 100 cases decided.

For Detailed Data

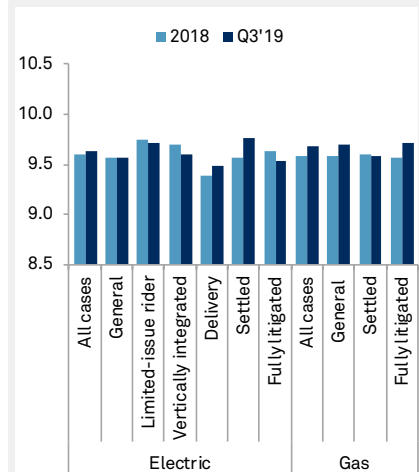
Click [here](#) to see supporting data tables.

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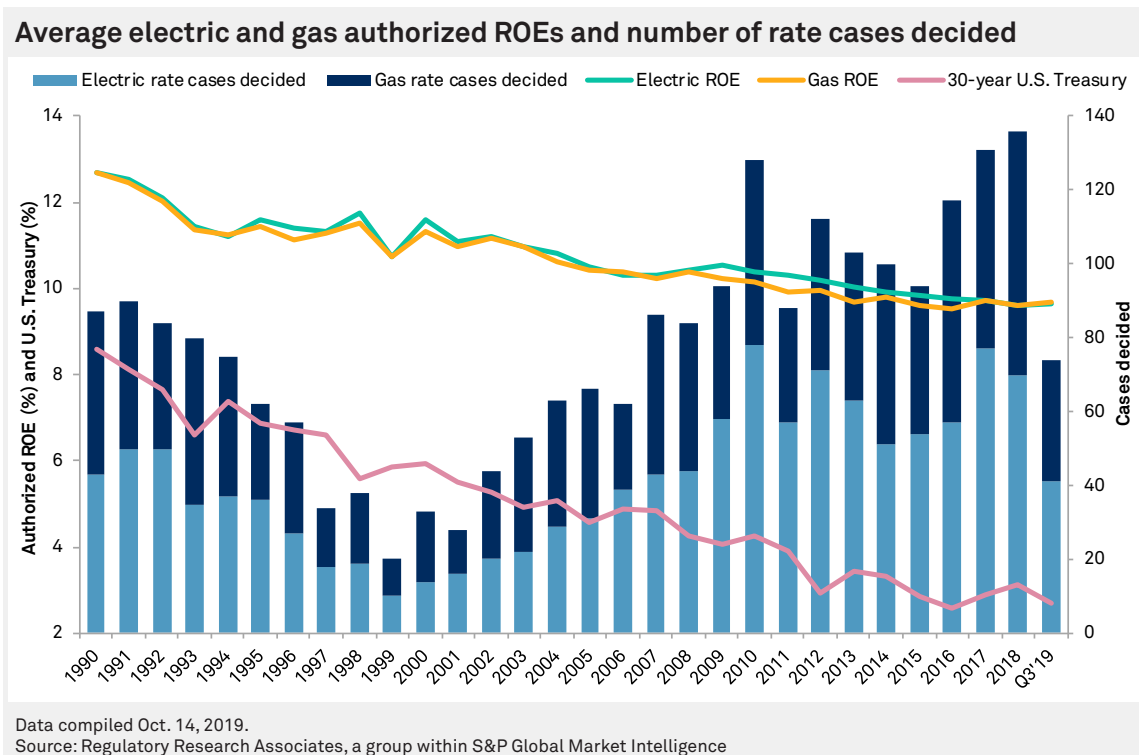
Enquiries
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Average authorized return on equity (%) Dashboard



Electric average		
2018	Q3'19	
All cases	9.60	9.63 ▲
General rate cases	9.56	9.57 ▲
Limited-issue rider cases	9.74	9.72 ▼
Vertically integrated cases	9.70	9.60 ▼
Delivery cases	9.38	9.49 ▲
Settled cases	9.57	9.77 ▲
Fully litigated cases	9.63	9.53 ▼
Gas average		
2018	Q3'19	
All cases	9.59	9.68 ▲
General rate cases	9.59	9.70 ▲
Settled cases	9.60	9.66 ▲
Fully litigated cases	9.57	9.71 ▲
U.S. Treasury		
2018	H1'19	
30-year bond yield	3.11	2.69 ▼

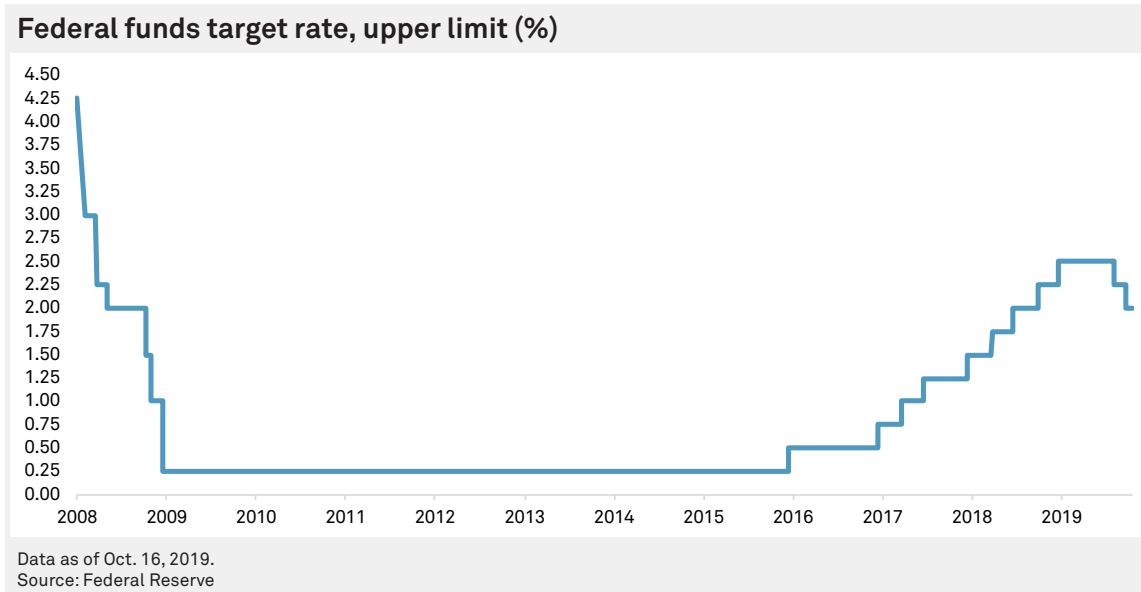
Data compiled Oct. 14, 2019.
Source: Regulatory Research Associates, a group within S&P Global Market Intelligence



Increased costs associated with environmental compliance, generation and delivery infrastructure upgrades and expansion, renewable generation mandates, storm and disaster recovery, cybersecurity and employee benefits have contributed to an active rate case agenda over the last decade. In the last couple of years, the need to address the impacts of the 2017 federal tax reform has caused rate case agendas to be more active than previously expected. Regulatory Research Associates, a group within S&P Global Market Intelligence, expects this trend to continue for the foreseeable future in light of the utilities' robust capital spending plans.

Rising interest rates over the past several years also likely contributed to the increased rate case activity. In 2015, the Federal Reserve began to gradually raise the federal funds rate. The Fed increased rates by 25 basis points nine times, with the last hike occurring in December 2018. However, with concerns of slowing growth, fears of a global recession and the impact of U.S.-China trade tensions negatively weighing on the U.S. economy, the Fed, after more than a decade without a cut, has lowered rates twice by a quarter point in July and again in September; the new target range is now 1.75% to 2%. Fed watchers expect a third cut of similar magnitude later in October.

While increases in the federal funds rate do not move in lockstep with longer-term treasuries and authorized ROEs do not move in lockstep with interest rates, the expectation is that as interest rates change, authorized ROEs would also begin to change in similar fashion. However, several factors impact the timing and magnitude of this anticipated shift. Normal regulatory lag, i.e., the amount of time it takes for a utility to put together a rate case filing and tender it to the commission and then for the commission to process the case, would without any other influences delay a change in average authorized ROEs relative to interest rates.



It is also worth noting that while both interest rates and authorized ROEs have generally been declining since 1990, the gap between authorized ROEs and interest rates widened somewhat over this period, largely as a result of an often-unstated understanding by regulators that the drop in interest rates caused by Federal Reserve intervention was unusual.

Capital structure trends

To offset the negative cash flow impact of federal tax reform, many utilities sought higher common equity ratios, and the average authorized equity ratios adopted by utility commissions in the first nine months of 2019 were modestly higher than the levels observed in 2018 and 2017. The average authorized equity ratio for electric utility cases nationwide was 50.48% in the first nine months of 2019, 49.02% in full-year 2018 and 48.90% in 2017. The average allowed equity ratio for gas utilities nationwide was 52.52% in the first nine months of 2019, 50.09% in 2018 and 49.88% in 2017.

The aforementioned averages include allowed equity ratios adopted by utility commissions in Arkansas, Florida, Indiana and Michigan — jurisdictions that authorize capital structures that include cost-free items or tax credit balances. Excluding these jurisdictions, the average authorized equity ratio for electric utilities nationwide was 51.65% for the first nine months of 2019, 50.53% in cases decided during 2018 and 50.02% in 2017. For gas utilities, excluding these jurisdictions, the average authorized equity ratio was 53.72% in the first three quarters of 2019, 51.47% in cases decided in 2018 and 51.13% in 2017.

Taking a longer-term view, equity ratios have generally increased over the last 15 years — the average equity ratio approved in electric rate cases decided during 2004 was 46.96%, while the average for gas utilities was 45.81%. Many commissions began approving more equity-rich capital structures in the wake of the 2008 financial crisis.

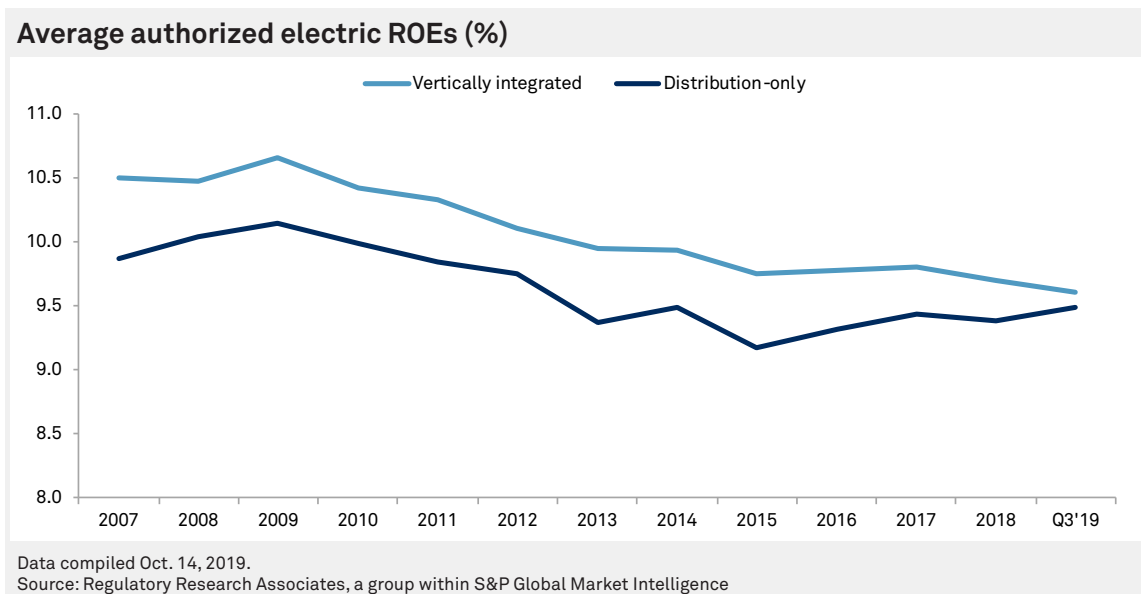
A more granular look at ROE trends

The discussion thus far has looked broadly at trends in authorized ROEs; the sections that follow provide a more granular view based upon the types of proceedings/decisions in which these ROEs were established.

RRA has observed that there can be significant differences between the average ROEs from one subcategory of cases to another.

As a result of electric industry restructuring, certain states unbundled electric rates and implemented retail competition for generation. Commissions in those states now have jurisdiction only over the revenue requirement and return parameters for delivery operations.

Comparing electric vertically integrated cases versus delivery-only proceedings over the past 12 years, RRA finds that the annual average authorized ROEs in vertically integrated cases typically are about 30 to 65 basis points higher than in delivery-only cases, arguably reflecting the increased risk associated with ownership and operation of generation assets.

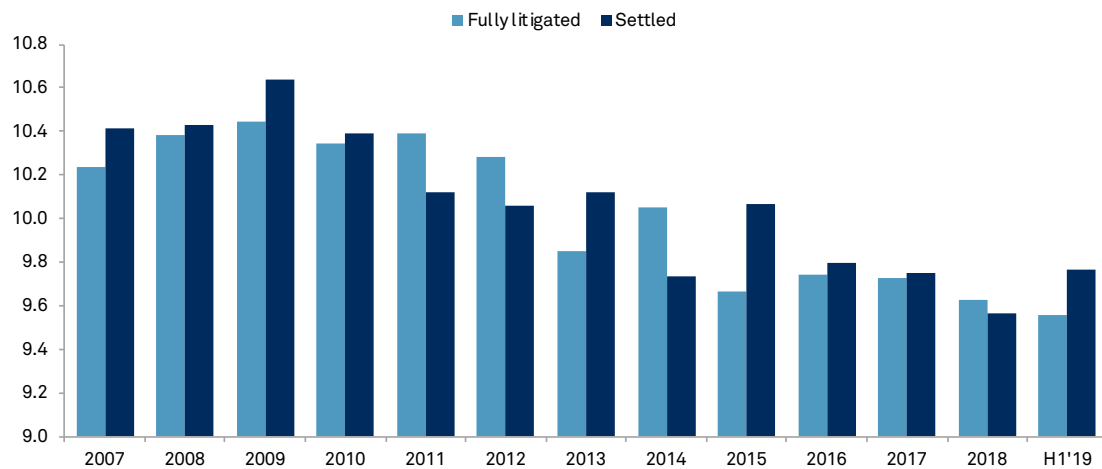


The industry average ROE for vertically integrated electric utilities was 9.60%, based on rate cases concluded in the first nine months of 2019, versus 9.70% in full-year 2018. For electric distribution-only utilities, the industry average ROE authorized in the first nine months of 2019 was 9.49%, versus 9.38% in 2018.

Settlements have frequently been used to resolve rate cases over the last several years, and in many cases, these settlements are “black box” in nature and do not specify the ROE and other typical rate case parameters underlying the stipulated rate change. However, some states preclude this type of treatment, and so, settlements must specify these values, if not the specific adjustments from which these values were derived.

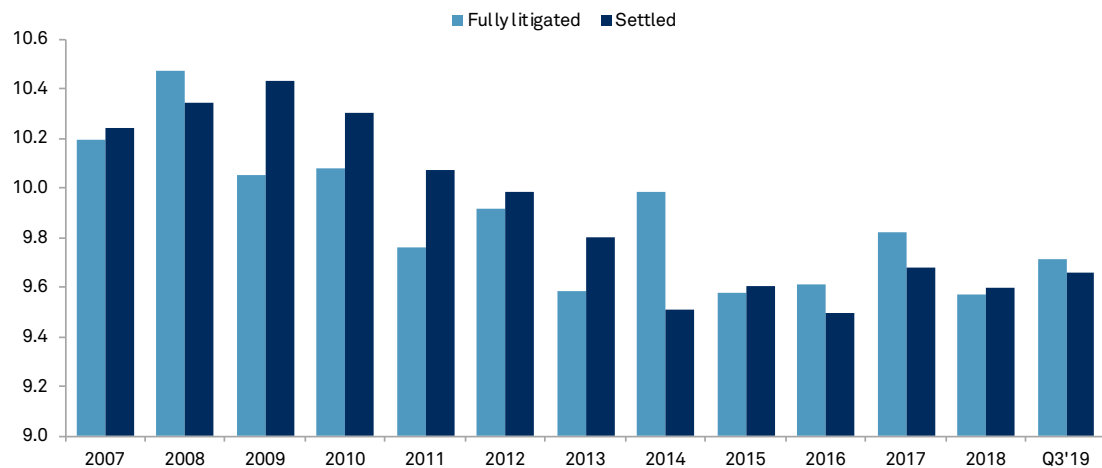
For both electric and gas cases, RRA has found no discernible pattern in the average authorized ROEs in cases that were settled versus those that were fully litigated. In some years, the average authorized ROE was higher for fully litigated cases, in others, it was higher for settled cases, and in a handful of years, the authorized ROE was similar for both fully litigated and settled cases.

Average authorized electric ROEs, settled versus fully litigated cases (%)



Data compiled Oct. 14, 2019.
Source: Regulatory Research Associates, a group within S&P Global Market Intelligence

Average authorized gas ROEs, settled versus fully litigated cases (%)



Data compiled Oct. 14, 2019.
Source: Regulatory Research Associates, a group within S&P Global Market Intelligence

Over the last several years, the annual average authorized ROEs in electric cases that involve limited-issue riders were typically meaningfully higher than those approved in general rate cases, driven by the ROE premiums authorized in Virginia. Limited-issue rider cases in which a separate ROE is determined have had limited use in the gas industry, as most of the gas riders rely on ROEs approved in a previous base rate case.

The following discussion focuses on the corresponding tables available [here](#).

Table 1 shows the average ROE authorized in major electric and gas rate decisions annually since 1990 and by quarter since 2015, followed by the number of observations in each period. **Table 2** indicates the composite electric and gas industry data for all major cases, summarized annually since 2004 and by quarter for the past six quarters.

Included in **Tables 3** and **4** are comparisons since 2007 of average authorized ROEs for settled versus fully litigated cases, general rate cases versus limited-issue rider proceedings and vertically integrated cases versus delivery-only cases for electric and gas utilities, respectively.

The individual electric and gas cases decided in 2019 are listed in **Table 5**, with the decision date shown first, followed by the company name, the abbreviation for the state issuing the decision, the authorized rate of return, the ROE and the percentage of common equity in the adopted capital structure. Next, RRA indicates the month and year in which the adopted test year ended, whether the commission utilized an average or a year-end rate base and the amount of the permanent rate change authorized. The dollar amounts represent the permanent rate change ordered at the time decisions were rendered. Fuel adjustment clause rate changes are not reflected in this study.

The simple mean is utilized for the return averages. In addition, the average equity returns indicated in this report reflect the ROEs approved in cases that were decided during the specified time periods and are not necessarily representative of either the average currently authorized ROEs for utilities industrywide or the returns actually earned by the utilities.

Please note: In an effort to align data presented in this report with data available in S&P Global Market Intelligence's online database, earlier historical data provided in previous reports may not match historical data in this report due to certain differences in presentation, including the treatment of cases that were withdrawn or dismissed.

Subsequent to the publication of our year-end 2018 report, the New Mexico Public Regulation Commission issued a revised decision in an electric rate case decision, the outcome of which impacted several of our electric ROE statistics for 2018. The instant report reflects the revised numbers.

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Global Research

29 November 2018

North America Power & Utilities Roll On (Erratum)

Equities

North America
Electric Utilities

Moving to 2021 for Valuation and Updating our Outlook

We are rolling forward our methodology to 2021 price to earnings as a basis for valuation. We refreshed our EPS forecasts and updated the UBS Regulatory Rankings for 2018 data. Regulated Utilities stand within a standard deviation of fair value on key relative yield and P/E metrics. Therefore we expect investors to capture a total return of approximately 9%. This consists of an average dividend yield of 3.5% and a return for growth of 5.3% (the average of 5.4% 5 year EPS CAGR and 5.2% dividend growth CAGR). *This version updates Figure 1 to show the current price target for EXC as \$51 and for H as \$18.*

Raising D and EMA to Buy from Neutral; Lowering AEE to Neutral

We believe D has taken the appropriate actions to overcome the surprise MLP tax ruling by FERC earlier this year and the likelihood of closing the merger with SCG has increased. Our sum of the parts price target goes to \$84 from \$75. Other ratings changes driven by valuation include AEE to Neutral from Buy while maintaining our \$71 price target and CUP to Neutral from Sell with a \$1 increase in the price target to \$13. We are also upgrading EMA to Buy from Neutral and increasing our PT to C\$51 from C\$42 as we believe the asset sale process that started with the New England Generation sale on Monday will address the balance sheet and valuation overhang.

Recommendations and Focus Stocks

Our recommendations fall into 4 categories: higher quality total return compounds; higher growth multi-utilities; special situations; and integrated power. Our top choices in each category are ETR, D, FE, and PEG.

Sell Recommendations: POR, PNM, HE and H

We lowered our price target on POR by \$1 to \$44. POR is a 4th quartile growing utility with -5% total return to our target. We remain cautious on HE at the current valuation while the commission establishes the framework for Performance-Based Ratemaking. The process will be ongoing through 2019. HE and PNM operate in states that fall in the lowest tier of our Regulatory Rankings. We are lowering our H target by C\$1 to C\$18 and believe that the AVA merger will close by 1Q'19 despite being dilutive. Even post potential AVA merger close, H will predominately operate in Ontario which we rank at the bottom of Tier 4 as a regulatory jurisdiction.

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Figure 1: Regulated Utility Price Target Changes

Price Target Old				Price Target New													
UBS 2020E EPS	2020 P/E Ratio	Old UBS Price Target	UBS 2021E EPS	2021 P/E Ratio	% Discount	New UBS Price Target	Current Dividend Yield	Total Return inc. Div. Yld	5 Yr EPS Growth	5 Yr DPS Growth	Regulatory Quartile	Old Quartile					
Rating	Ticker	Company															
Buy	ACO	ATCO Ltd. (CS)	\$3.38	11.7x	\$47	\$3.34	11.8x	(28%)	\$49	3.8%	28%	4th	4th				
Buy	EMA	Emera Inc (CS)	\$3.05	14.2x	\$42	\$3.26	13.3x	(19%)	\$51	5.4%	23%	2nd	2nd				
Buy	NEE	NextEra Energy	\$9.08	19.6x	\$195	\$9.84	18.1x	10%	\$209	2.5%	20%	1st	1st				
Buy	PEG	Public Service Ent Group	\$3.66	14.8x	\$60	\$3.77	14.4x	(12%)	\$63	3.3%	19%	2nd	2nd				
Buy	FE	FirstEnergy Corp	\$2.46	15.4x	\$43	\$2.60	14.6x	(11%)	\$43	4.0%	19%	2nd	2nd				
Buy	FTS	Fortis Inc (CS)	\$2.84	16.2x	\$49	\$3.01	15.3x	(7%)	\$53	3.9%	19%	3rd	2nd				
Buy	ETR	Entergy Corp	\$5.40	16.0x	\$94	\$5.73	15.1x	(8%)	\$99	4.2%	18%	2nd	3rd				
Buy	D	Dominion Energy	\$4.35	17.0x	\$75	\$4.71	15.7x	(4%)	\$84	4.5%	18%	1st	1st				
Buy	SRE	Sempra Energy	\$6.99	16.1x	\$130	\$7.55	14.9x	(9%)	\$129	3.2%	18%	3rd	3rd				
Buy	DUK	Duke Energy	\$5.29	16.6x	\$92	\$5.59	15.7x	(4%)	\$99	4.2%	17%	1st	1st				
Buy	PPL	PPL Corporation	\$2.56	12.1x	\$34	\$2.59	12.0x	(27%)	\$34	5.3%	15%	1st	2nd				
Buy	EXC	Exelon	\$3.16	14.4x	\$50	\$3.25	14.0x	(14%)	\$51	3.0%	15%	3rd	2nd				
Buy	AEP	American Electric Power	\$4.46	17.2x	\$83	\$4.77	16.0x	(2%)	\$85	3.5%	15%	2nd	2nd				
Buy	ES	Eversource Energy	\$3.72	18.2x	\$69	\$3.95	17.1x	4%	\$75	3.0%	14%	3rd	3rd				
Neutral	CUP	Caribbean Utilities Corp Ltd	\$0.83	14.8x	\$12	\$0.89	13.9x	(15%)	\$13	5.7%	13%	2nd	2nd				
Neutral	CU	Canadian Utilities Ltd (CS)	\$2.35	13.2x	\$33	\$2.31	13.5x	(18%)	\$33	5.1%	12%	4th	4th				
Neutral	ED	Consolidated Edison	\$4.60	17.2x	\$80	\$4.93	16.0x	(2%)	\$85	3.6%	11%	4th	4th				
Neutral	SO	Southern Company	\$3.17	14.5x	\$49	\$3.30	13.9x	(15%)	\$49	5.2%	11%	1st	1st				
Neutral	CMS	CMS Energy	\$2.73	18.7x	\$54	\$2.92	17.5x	7%	\$55	2.8%	10%	1st	1st				
Neutral	LNT	Alliant Energy	\$2.45	18.3x	\$46	\$2.54	17.6x	8%	\$48	3.0%	10%	1st	2nd				
Neutral	SCG	SCANA Corp	\$2.90	16.2x	\$41	\$2.66	17.7x	8%	\$50	1.1%	7%	3rd	1st				
Neutral	DTE	DTE Energy	\$6.62	17.8x	\$118	\$7.01	16.8x	2%	\$121	3.0%	6%	1st	1st				
Neutral	PNW	Pinnacle West Capital Corp	\$5.28	17.0x	\$90	\$5.52	16.3x	(1%)	\$92	3.3%	6%	4th	3rd				
Neutral	WEC	WEC Energy Group	\$3.74	19.0x	\$70	\$3.97	17.9x	9%	\$73	3.1%	6%	1st	1st				
Neutral	AEE	Ameren Corp	\$3.50	19.6x	\$71	\$3.85	17.8x	9%	\$71	2.8%	6%	2nd	2nd				
Neutral	AES	AES Corp.	\$1.45	10.5x	\$16	\$1.61	9.5x	(42%)	\$16	3.4%	5%	2nd	2nd				
Neutral	OGE	OGE Energy Corp	\$2.30	17.1x	\$37	\$2.39	16.5x	0%	\$40	3.7%	5%	4th	4th				
Neutral	XEL	Xcel Energy	\$2.77	18.7x	\$50	\$2.95	17.5x	7%	\$53	2.9%	5%	2nd	2nd				
Neutral	EVERG	Evergy	\$3.38	17.7x	\$58	\$3.44	17.4x	6%	\$61	3.1%	5%	3rd	3rd				
Neutral (CBE)	EIX	Edison International	\$4.87	11.4x	\$55	\$5.19	10.7x	(35%)	\$55	4.4%	4%	3rd	3rd				
Sell	H	Hydro One Ltd (CS)	\$1.34	14.5x	\$19	\$1.41	13.8x	(16%)	\$18	4.7%	(3%)	4th	4th				
Neutral (CBE)	PCG	PG&E Corp	\$4.24	6.4x	\$26	\$4.41	6.1x	(63%)	\$26	0.0%	(4%)	3rd	3rd				
Sell	POR	Portland General	\$2.64	18.2x	\$45	\$2.65	18.1x	11%	\$44	3.0%	(5%)	2nd	3rd				
Sell	PNM	PNM Resources	\$2.19	19.5x	\$37	\$2.29	18.6x	14%	\$39	2.5%	(7%)	4th	4th				
Sell	HE	Hawaiian Electric Industries	\$2.20	17.3x	\$33	\$2.19	17.3x	6%	\$33	3.3%	(9%)	4th	4th				
Overall Average (unnormalized, inc. Gas Midstream)			16.5x				15.6x										
Electric Utility Normalized (including Gas Midstream) (a)			17.4x				16.4x										
(a) Normalized multiple excludes AES, EIX, EXC, PCG, PPL, SCG and SO.																	

Source: Company reports, UBS estimates, FactSet

Rolling On to 2021 for Valuation

We updated our price targets and the underlying drivers for valuation incorporating 2021 as a basis for P/E. To determine the valuation multiple we assign to stocks we consider: the sector valuation, EPS growth, weighted average regulatory ranking, and unregulated business contributions. For sector valuation we consider relative yield to the Baa corporate bond in a series back to 1981 and for relative P/E we look at valuation back to 2008. Currently, the group is within a standard deviation of fair value using both measures. Using relative yield Regulated Utilities are +6% beyond the mean within a 7% standard deviation and +7% using relative P/E within a 10% standard deviation. Therefore, we expect one year total return for the group to approximate 9% including a 3.5% dividend yield and a 5.3% growth return (the average of a 5.4% 5 year EPS CAGR and a 5.2% 5 year dividend growth CAGR). However, our forecast for earnings leaves the group offering 5% return over the year.

With regard to EPS growth and quality of regulation we rank the companies in quartiles. Based on historical stock performance we assign +5% for top quartile, +2% for second quartile, -2% for third quartile and -5% for fourth quartile which are summarized below. For companies with significant unregulated businesses we value them on a sum of the parts basis. We also allow for special situation company specific adjustments.

Figure 2: Regulated Valuation Methodology Matrix

Investment Opinion	Ticker	Overall Reg Group Premium Discount	Regulatory Quartile Premium Discount	Regulated EPS Growth Premium Discount	ESG Premium	Specific Adjustments	Net Prem Disc. Reg. Valuation	Business Value
Neutral	AES	5%	2%	5%	0%	(5%)	7%	\$9
Buy	ACO	10%	(5%)	(5%)	0%	(5%)	(5%)	\$5
Neutral	LNT	5%	5%	5%	0%	0%	15%	\$0
Neutral	AEE	5%	2%	5%	0%	0%	12%	\$0
Buy	AEP	5%	2%	2%	0%	0%	9%	\$0
Neutral	CU	10%	(5%)	(5%)	0%	(10%)	(10%)	\$0
Neutral	CUP	0%	2%	2%	0%	(10%)	(6%)	\$0
Neutral	CMS	5%	5%	5%	0%	0%	15%	\$0
Neutral	ED	5%	(5%)	(2%)	8%	0%	6%	\$0
Buy	D	5%	5%	2%	0%	0%	12%	\$40
Neutral	DTE	5%	5%	2%	0%	0%	12%	\$30
Buy	DUK	5%	5%	(2%)	0%	0%	8%	\$0
Neutral (CBE)	EIX	5%	(2%)	(2%)	0%	(30%)	(29%)	\$0
Buy	EMA	10%	2%	(2%)	0%	(5%)	5%	\$1
Buy	ETR	5%	2%	(2%)	0%	0%	5%	\$0
Buy	ES	5%	(2%)	2%	10%	0%	15%	\$0
Neutral	EVRG	5%	(2%)	5%	0%	0%	8%	\$0
Buy	EXC	5%	(2%)	5%	0%	0%	8%	\$11
Buy	FE	5%	2%	(5%)	0%	0%	2%	\$0
Buy	FTS	10%	(2%)	2%	0%	0%	10%	\$0
Sell	HE	5%	(5%)	2%	0%	0%	2%	\$8
Sell	H	10%	(5%)	(5%)	0%	(15%)	(15%)	\$0
Buy	NEE	5%	5%	5%	0%	0%	15%	\$95
Neutral	OGE	5%	(5%)	(2%)	0%	0%	(2%)	\$10
Neutral (CBE)	PCG	5%	(2%)	(2%)	0%	(63%)	(62%)	\$0
Neutral	PNW	5%	(5%)	2%	0%	0%	2%	\$0
Sell	PNM	5%	(5%)	(5%)	0%	8%	3%	\$0
Sell	POR	5%	2%	(5%)	0%	0%	2%	\$0
Buy	PPL	5%	5%	(5%)	0%	0%	5%	\$13
Buy	PEG	5%	2%	5%	0%	0%	12%	\$11
Neutral	SCG	5%	(2%)	(5%)	0%	0%	(2%)	\$0
Buy	SRE	5%	(2%)	(2%)	0%	0%	1%	\$38
Neutral	SO	5%	5%	(5%)	0%	(15%)	(10%)	\$0
Neutral	WEC	5%	5%	2%	0%	0%	12%	\$0
Neutral	XEL	5%	2%	2%	0%	0%	9%	\$0

Note: CUP is domiciled in Grand Cayman and should not move with US or CDN interest rates

Source: FactSet, UBS Equity Research

Our Recommendations

Our focus on stock selection as we head toward the end of 2018 and into the new year is for Regulated Utilities at a value as the top quality has outperformed in 2018. As shown below, companies that delivered top quartile earnings growth outperformed the rest of the group year-to-date, offering 14% upside. Overall quality which is a ranking we are basing on the average of EPS growth and our UBS weighted average regulatory ranking also performed this year. Special events like California wildfires, major project construction and international drove the bottom quality.

Our recommendations by style are listed below. Our focus stocks by category are ETR for higher quality, total return compounders; D for higher growth multi utilities; FE for special situations; and PEG for integrated power.

Figure 3: Recommended Stocks

Rating	Ticker	Current Price	UBS Price Target	Total Return inc. Div. Yld	UBS 2018E EPS	UBS 2019E EPS	UBS 2020E EPS	UBS 2021E EPS	2020 P/E Ratio	2020 Prem/Disc	Current Dividend Yield	5 Yr EPS Growth	5 Yr DPS Growth
Higher Quality Total Return Compounders													
Buy	EMA	\$43.41	\$51	23%	\$2.85	\$2.80	\$3.10	\$3.26	14.0x	(17%)	5.4%	6.2%	5.9%
Buy	FTS	\$46.04	\$53	19%	\$2.52	\$2.64	\$2.83	\$3.01	16.2x	(4%)	3.9%	4.8%	6.4%
Buy	ETR	\$86.29	\$99	18%	\$4.70	\$5.09	\$5.40	\$5.73	16.0x	(8%)	4.2%	5.5%	2.0%
Buy	DUK	\$87.83	\$99	17%	\$4.74	\$5.03	\$5.33	\$5.59	16.5x	(5%)	4.2%	5.2%	4.0%
Buy	AEP	\$76.58	\$85	15%	\$3.93	\$4.18	\$4.46	\$4.77	17.2x	(2%)	3.5%	6.0%	6.6%
Buy	ES	\$67.45	\$75	14%	\$3.28	\$3.48	\$3.72	\$3.95	18.2x	4%	3.0%	6.1%	6.0%
Buy	AWK	\$92.11	\$101	12%	\$3.31	\$3.57	\$3.85	\$4.17	23.9x	(3%)	2.0%	8.3%	10.0%
Higher Growth Multi-Utilities													
Buy	NEE	\$178.00	\$209	20%	\$7.77	\$8.43	\$9.08	\$9.84	19.6x	13%	2.5%	9.5%	12.0%
Buy	D	\$74.11	\$84	18%	\$4.10	\$4.25	\$4.46	\$4.71	16.6x	(5%)	4.5%	6.7%	6.0%
Buy	SRE	\$112.69	\$129	18%	\$5.46	\$5.89	\$6.99	\$7.55	16.1x	(7%)	3.2%	8.0%	9.0%
Special Situation													
Buy	ACO	\$39.53	\$49	28%	\$2.95	\$3.13	\$3.20	\$3.34	12.4x	(27%)	3.8%	3.6%	5.2%
Buy	FE	\$37.89	\$43	19%	\$2.55	\$2.57	\$2.46	\$2.60	15.4x	(12%)	4.0%	3.0%	3.5%
Buy	PPL	\$31.12	\$34	15%	\$2.34	\$2.40	\$2.56	\$2.59	12.2x	(30%)	5.3%	3.3%	3.3%
Integrated Power													
Buy	PEG	\$54.29	\$63	19%	\$3.08	\$3.35	\$3.69	\$3.77	14.7x	(16%)	3.3%	6.6%	4.9%
Buy	EXC	\$45.53	\$51	15%	\$3.14	\$3.18	\$3.14	\$3.25	14.5x	(17%)	3.0%	5.9%	5.3%
Sells													
Sell	AWR	\$67.82	\$48	(28%)	\$1.75	\$1.93	\$2.04	\$2.15	33.2x	37%	1.6%	5.3%	6.0%
Sell	CWT	\$46.30	\$39	(14%)	\$1.27	\$1.45	\$1.65	\$1.76	28.0x	15%	1.6%	5.8%	5.0%
Sell	HE	\$37.92	\$33	(9%)	\$1.86	\$1.99	\$2.12	\$2.19	17.9x	3%	3.3%	6.7%	0.0%
Sell	PNM	\$42.60	\$39	(7%)	\$1.97	\$2.14	\$2.19	\$2.29	19.5x	12%	2.5%	4.3%	6.0%
Sell	POR	\$47.99	\$44	(5%)	\$2.37	\$2.48	\$2.57	\$2.65	18.7x	7%	3.0%	3.6%	6.4%
Sell	CTWS	\$69.75	\$66	(4%)	\$2.17	\$2.27	\$2.48	\$2.69	28.1x	16%	1.8%	5.8%	5.0%
Sell	H	\$19.47	\$18	(3%)	\$1.27	\$1.27	\$1.35	\$1.41	14.4x	(17%)	4.7%	4.8%	7.1%

Source: FactSet, UBS Estimates, Prices as of 11/27/18

Figure 4: 2018 Stock Performance by Measure

EPS Growth	3 Month	6 Month	YTD
1st	4.8%	11.8%	14.2%
2nd	4.5%	10.3%	-0.2%
3rd	-3.7%	1.1%	-4.2%
4th	4.7%	10.4%	2.0%
Regulation			
1st	4.4%	11.6%	3.0%
2nd	4.1%	10.5%	8.5%
3rd	-2.8%	3.8%	-0.4%
4th	3.7%	6.4%	-2.3%
Quality			
1st	3.3%	10.6%	7.1%
2nd	6.7%	12.6%	6.0%
3rd	3.0%	9.9%	5.2%
4th	-2.4%	1.3%	-6.4%

Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Earnings Growth

We updated our EPS estimates with the roll-forward in methodology and summarize our changes by company below. Our utility valuation methodology utilizes 5 year utility, parent and other EPS growth as an input. Top quartile growth begins at 7.1% while median growth is 5.8%. Investors are currently awarding a double digit premium for stocks that offer the highest growth.

Figure 5: Earnings Growth Quality Quartiles

Quality Quartile	Low Growth	High Growth	Current Growth	Current P/E Ratio	Current Prem/Disc
1st	7.10%		8.2%	18.2x	11.4%
2nd	5.80%	6.60%	6.3%	18.2x	10.9%
3rd	4.40%	5.60%	5.0%	15.8x	-3.5%
4th	0.00%	4.30%	1.5%	16.2x	-1.0%

Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Figure 6: Regulated Utility EPS Estimate Changes

Rating	Ticker	Company	UBS 2018E EPS	UBS 2019E EPS	UBS 2020E EPS	UBS 2021E EPS	UBS 2018E EPS Old	UBS 2019E EPS Old	UBS 2020E EPS Old	UBS 2021E EPS Old
Buy	ACO	ATCO Ltd. (C\$)	\$2.95	\$3.13	\$3.20	\$3.34	\$2.96	\$3.29	\$3.38	\$3.54
Neutral	AEE	Ameren Corp	\$3.40	\$3.18	\$3.48	\$3.85	\$3.40	\$3.18	\$3.50	\$3.87
Neutral	CMS	CMS Energy	\$2.33	\$2.50	\$2.70	\$2.92	\$2.33	\$2.53	\$2.73	\$2.95
Neutral	CU	Canadian Utilities Ltd (C\$)	\$2.06	\$2.15	\$2.21	\$2.31	\$2.07	\$2.27	\$2.35	\$2.46
Neutral	CUP	Caribbean Utilities Corp Ltd	\$0.71	\$0.77	\$0.80	\$0.89	\$0.71	\$0.78	\$0.83	\$0.94
Buy	D	Dominion Energy	\$4.10	\$4.25	\$4.46	\$4.71	\$4.10	\$4.16	\$4.35	\$4.58
Buy	DUK	Duke Energy	\$4.74	\$5.03	\$5.33	\$5.59	\$4.74	\$5.03	\$5.29	\$5.56
Buy	EMA	Emera Inc (C\$)	\$2.85	\$2.80	\$3.10	\$3.26	\$2.68	\$2.91	\$3.05	\$3.21
Buy	ES	Eversource Energy	\$3.28	\$3.48	\$3.72	\$3.95	\$3.28	\$3.48	\$3.72	\$3.95
Buy	EXC	Exelon	\$3.14	\$3.18	\$3.14	\$3.25	\$3.14	\$3.19	\$3.16	\$3.29
Buy	FTS	Fortis Inc (C\$)	\$2.52	\$2.64	\$2.83	\$3.01	\$2.52	\$2.65	\$2.84	\$3.01
Sell	HE	Hawaiian Electric Industries	\$1.86	\$1.99	\$2.12	\$2.19	\$1.89	\$2.05	\$2.20	\$2.29
Sell	H	Hydro One Ltd (C\$)	\$1.27	\$1.27	\$1.35	\$1.41	\$1.23	\$1.32	\$1.34	\$1.51
Buy	NEE	NextEra Energy	\$7.77	\$8.43	\$9.08	\$9.84	\$7.88	\$8.41	\$9.06	\$9.74
Neutral	OGE	OGE Energy Corp	\$2.07	\$2.12	\$2.31	\$2.39	\$2.08	\$2.14	\$2.30	\$2.38
Sell	POR	Portland General	\$2.37	\$2.48	\$2.57	\$2.65	\$2.37	\$2.53	\$2.64	\$2.78
Buy	PPL	PPL Corporation	\$2.34	\$2.40	\$2.56	\$2.59	\$2.33	\$2.42	\$2.56	\$2.59
Buy	PEG	Public Service Ent Group	\$3.08	\$3.35	\$3.69	\$3.77	\$3.08	\$3.47	\$3.66	\$3.76
Neutral	SCG	SCANA Corp	\$2.67	\$2.30	\$2.44	\$2.66	\$3.13	\$2.76	\$2.90	\$3.12
Neutral	SO	Southern Company	\$3.06	\$3.04	\$3.17	\$3.30	\$3.06	\$3.04	\$3.17	\$3.30
Buy	SRE	Sempra Energy	\$5.46	\$5.89	\$6.99	\$7.55	\$5.31	\$5.89	\$6.99	\$7.55

Source: Company reports, UBS estimates, FactSet

Regulatory Ranking Refresh

We did a refresh of the UBS regulatory rankings to include 2018 rate cases, data updates on the average customer bill and rates, and to note other changes in regulation this year. The states with material moves (beyond 6 slots plus or minus) in the United States were Louisiana, Kansas, Massachusetts, and Oregon on the plus side and Nebraska, New Hampshire, New York, and South Carolina on the negative side. As a reminder our regulatory rankings consider 6 factors in a simple

average: 1) Appointed or elected commissions; 2) Allowed return spread history, 3) Mechanisms that reduce regulatory lag; 4) Rates and customer levels compared to region; 5) Tendency to settle versus litigate rate cases; and 6) A subjective investor friendliness factor.

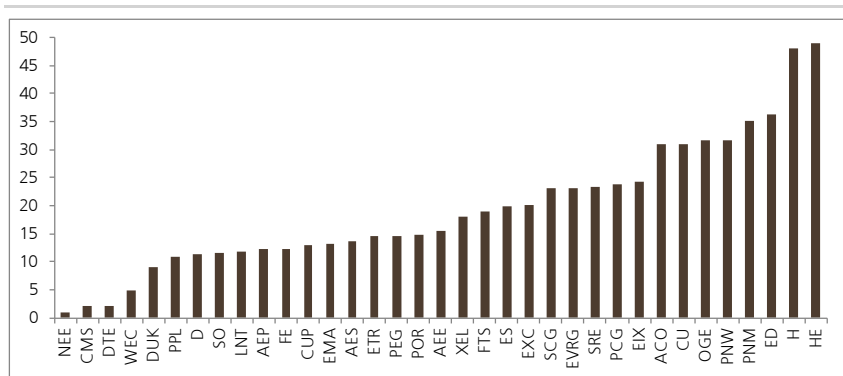
Figure 7: UBS Regulatory Rankings

<u>TIER 1</u>	<u>TIER 2</u>	<u>TIER 3</u>	<u>TIER 4</u>	<u>TIER 5</u>
		Nova Scotia		
		North Dakota		
FERC		Iowa		
		Kentucky		
		Washington		
		Tennessee		
		Texas		
		Missouri		
		Massachusetts		
		South Carolina		
	Pennsylvania	Wyoming	Prince Edward Island	
	Illinois	Kansas	Nevada	
	Arkansas	Rhode Island	New Hampshire	
	Ohio	California	New York	
Florida	Louisiana	Alberta	Oklahoma	New Mexico
Michigan	Georgia	Newfoundland & Labrador	Alaska	Maine
Utah	Idaho	Delaware	West Virginia	Maryland
Wisconsin	British Columbia	Minnesota	South Dakota	Montana
Alabama	Indiana	Connecticut	Nebraska	Hawaii
Colorado	Virginia	New Jersey	Mississippi	Vermont
North Carolina	Oregon	Arizona	Ontario	District of Columbia
JD Power Average Customer Service Scores				
727	722	707	624	695

Source: Canadian Provincial Regulatory Websites, S&P Global Market Intelligence, FactSet, JD Power, UBS Equity Research

Below we provide a weighted average regulatory ranking by company. We believe that constructive regulation is good for the customer and the shareholder. Our ranking of states is positively correlated to JD Power customer service scores.

Figure 8: Weighted Average Regulatory Ranking by Company



Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Companies with better weighted average regulatory rankings also have higher investment in the system and experience less regulatory lag which contributes to a premium valuation. Companies that had a positive move of 6 slots or more were PPL, AEE and ETR. SCG had a notably negative move.

Figure 9: Regulated Utility Quartiles

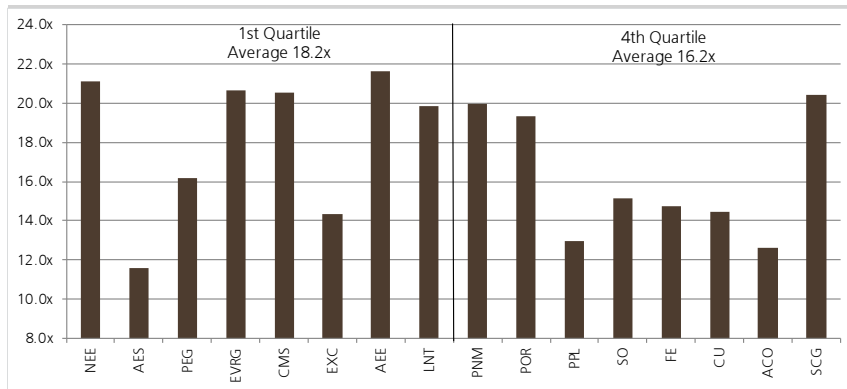
Metric	1st Quartile	2nd Quartile	3rd Quartile	4th Quartile
Ratebase Growth '17-'22	7.6%	6.4%	4.8%	2.7%
Price/Book	2.45x	2.40x	1.56x	1.72x
Earned ROE	11.5%	10.0%	9.8%	9.6%

Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Back Testing the Methodology

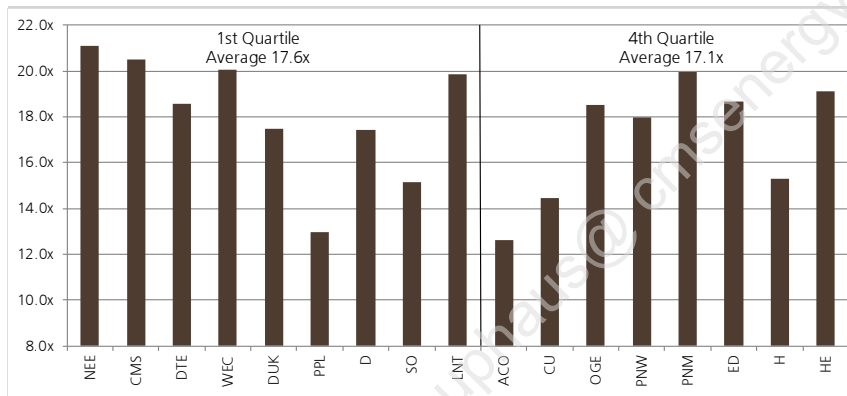
Below we show the top and bottom quartile names by earnings growth, regulatory jurisdiction and overall quality. The spread on forward year P/E premium from top to bottom quartile is 16% which is higher than the spread for growth (+12%) or regulation alone (+3%).

Figure 10: Utility, Parent and Other EPS Growth 1st/4th Quartiles



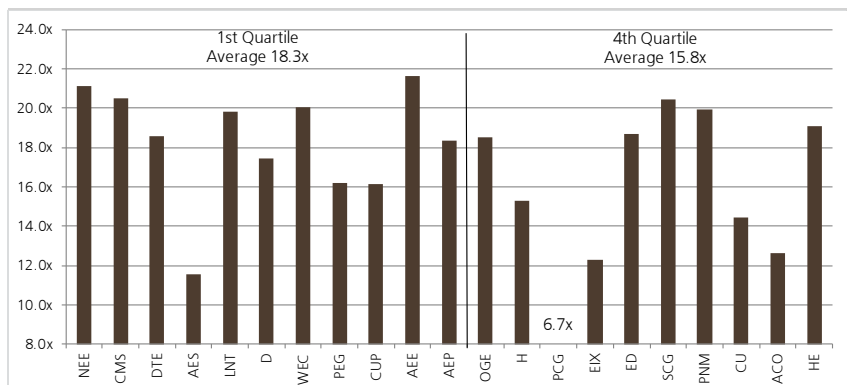
Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Figure 11: Regulatory Ranking 1st/4th Quartiles



Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Figure 12: Overall Quality Ranking 1st/4th Quartiles



Source: FactSet, S&P Global Market Intelligence, UBS Equity Research

Company Price Target and EPS Estimate Changes

AES Corp.

We are updating our AES \$16 price target methodology. AES expects to achieve investment grade metrics in 2019 and 8-10% free cash flow growth to 2020 or \$800M. We forecast 5 year EPS growth of 9% of which ½ comes from contracted renewable investments at double digit returns and could include some asset sales to achieve the growth. The company is de-risking AES Gener in Chile with the 270 MW Candelaria solar and wind project which reduces by 40% Gener's expiring hedged position in 2022.

Our current target of \$16 is a sum of parts which reflects \$13 for North America utilities, \$4 each for South America, MCAC, and EurAsia regions less \$8 for debt and 660M shares. This includes a 7% P/E premium of the 2021 Regulated Utility average or 17.7x \$0.34 for the U.S. utilities, publicly traded market values for Gener in Chile and Tiete in Brazil, 7.6x 2021 EBITDA for EurAsia and 6.5x for the Mexico/Central America/Caribbean segment. The 7% premium reflects 5% for group undervaluation, 5% for top quartile growth, 2% for regulation, and -5% for sum of the parts discount.

Figure 13: AES Sum of the Parts - Current - Dollars in Millions

2021 Forecasts	EBITDA	Adjusted PTC	EPS	Ownership Debt	UBS Comps (f)	Multiple (g)	Value
U.S. Utilities	\$609	\$197	\$0.34	\$3,225	U.S. Regulated	17.5x	\$6.04
U.S. Generation	\$593	\$389	\$199	\$199	U.S. Multi Utility	7.2x	\$6.19
El Salvador	\$76	\$52		\$237	Regulated Utility	6.2x	\$0.36
United States & Utilities	\$1,279	\$594		\$3,660			\$12.59
South America	\$959	\$574		\$3,487			\$4.22
Gener (b)	\$665	\$378		\$2,883	Market Value	6.6x	\$2.27
Argentina	\$216	\$165		\$302	Lat Am	6.0x	\$1.48
Tiete	\$41	\$17		\$119	Market Value		\$0.36
Other	\$38	\$14		\$184		6.6x	\$0.10
MCAC	\$596	\$349		\$1,504		6.5x	\$3.59
EurAsia	\$394	\$217		\$822	60% Europe/40% Asia	7.6x	\$3.26
Corporate	-\$27	-\$287		\$3,664		6.0x	-\$8.14
Total	\$3,201	\$1,447		\$11,883		6.9x	\$15.52
Shares							660
Unlisted Subs	\$1,886			\$6,911		6.1x	\$6.85
Listed/DPL&IPL	\$1,315			\$4,973		8.1x	\$8.67

(b) Publicly traded market values 11/27/18

(d) Price paid.

(f) Reflects 11/20 values from UBS Global Electric Utility Valuation

(g) Multiples are 2021 EV/EBITDA except U.S. Utilities which is P/E.

Source: Company reports, UBS equity research estimates, Factset

Figure 14: AES Sum of the Parts – Prior – Dollars in Millions

2020 Forecasts	EBITDA	Adjusted PTC	EPS	Ownership Debt	UBS Comps (f)	Multiple	Value
U.S. Utilities	\$601	\$197	\$0.31	\$3,347	U.S. Regulated	17.8x	\$5.47
U.S. Generation	\$566	\$345		\$277	U.S. Multi Utility	7.2x	\$5.75
El Salvador	\$76	\$52		\$237	Regulated Utility	6.2x	\$0.36
United States & Utilities	\$1,243	\$594		\$3,861			\$11.58
South America	\$929	\$574		\$3,393			\$4.50
Gener (b)	\$632	\$378		\$2,749	Market Value	6.9x	\$2.46
Argentina	\$216	\$165		\$310	Lat Am	6.2x	\$1.56
Tiete	\$43	\$17		\$151	Market Value		\$0.40
Other	\$38	\$14		\$184		6.2x	\$0.08
MCAC	\$596	\$349		\$1,503		6.5x	\$3.59
EurAsia	\$407	\$217		\$822	60% Europe/40% Asia	8.7x	\$4.12
Corporate	-\$40	-\$287		\$3,664		6.2x	-\$8.24
Total	\$3,136	\$1,447		\$11,883		7.1x	\$15.55
Shares							660
Unlisted Subs	\$1,860			\$6,997		6.3x	\$7.21
Listed/DPL&PL	\$1,276			\$4,887		8.1x	\$8.34

(b) Publicly traded market value 11/6/18

(d) Price paid.

(f) Reflects 10/31 values from UBS Global Electric Utility Valuation

Source: Company reports, UBS equity research estimates, Factset

Our prior \$16 target reflected \$12 for North America utilities, \$4 each for South America, MCAC, and EurAsia regions less \$8 for debt and 660M shares. This includes a 12% P/E premium of the 2020 Regulated Utility 15.9x average for the U.S. utilities, publicly traded market values for Gener in Chile and Tiete in Brazil, 8.7x 2020 EBITDA for EurAsia and 6.5x for the Mexico/Central America/Caribbean segment. The 12% premium included 10% for group undervaluation.

Alliant Energy

We are increasing our price target on LNT to \$48 from \$46. Our EPS estimates remain unchanged at \$2.15 in 2018, \$2.26 in 2019, \$2.45 in 2020 and \$2.54 in 2021. LNT is earning its allowed return in Wisconsin and under-earning slightly (20 bp) in Iowa for management compensation non-recovery which is consistent with our forecast. LNT expects to file 2 rate cases in Iowa in 2019 including one with a forward test year. LNT will likely request an increase in the equity ratio potentially to 51% from 49%.

Our current \$48 price target is premised upon a 15% premium to the 2021E Regulated Utility P/E multiple or 18.9x 2021E EPS of \$2.54. The premium includes 5% for the group undervaluation, 5% for EPS growth and 5% for regulation.

Previously our \$46 target reflected a 17% premium and was 18.7x \$2.45 in 2020. The prior multiple reflected a premium of 10% for group undervaluation, 5% for EPS growth and 2% for regulation.

Ameren Corp.

We are lowering our rating on Ameren to Neutral from Buy following good performance this year on the passage of Missouri legislation and delivering on guidance. See our separate note ([LINK](#)). Since passage of SB 564 in Missouri the stock is up 21% (+10% versus the XLU and +22% versus the S&P 500). We are fine-tuning our EPS estimates to \$3.48 in 2020 and \$3.85 in 2021 versus \$3.50 and \$3.87 previously due to the timing of the Missouri investments. The reduction

in earnings relates to the timing lag for the deferral of return on wind investment under SB 564 versus the renewable tracker.

Our \$71 price target is a 12% premium to the Regulated Utility average or 18.4x 2021E EPS of \$3.85. It includes top quartile growth (+5%), above average regulation (+2%) and the impact of the Regulated Utility group discount (+5%).

The prior methodology for our \$71 price target reflected a 17% premium applied to the average Regulated Utility P/E or 18.3x 2021E EPS of \$3.87.

American Electric Power

We are increasing our price target on AEP to \$85 from \$83. Up next for AEP are integrated resource plan filings at PSO by 12/21/18 and SWEPCO (Arkansas) by 12/1/18. We also believe there is the opportunity to increase transmission spending at PSO and SWEPCO and in batteries in Texas. We maintain our EPS estimates of \$3.93 for 2018, \$4.18 for 2020, \$4.46 in 2021 and \$4.77 in 2022.

Our current price target of \$85 is at a 9% premium to the Regulated Utility group and is 17.9x our \$4.77 2021 EPS estimate. The valuation includes a 5% premium for the group's undervaluation, 2% for second quartile regulation and 2% for second quartile growth.

Our prior price target of \$83 was 18.6x our \$4.46 2020 EPS estimate based on a 14% group premium multiple. The premium previously included 10% for the group's undervaluation.

ATCO Ltd.

We reiterate our Buy rating. Upcoming catalysts include potential for Structures & Logistics to be sanctioned work for LNG Canada, sale of the unregulated power plants in Alberta, and potential for a PBR reopener at the Alberta Utilities Commission. We continue to believe that the Canadian Utilities business remains solid and see upside to shares from the unregulated Structures & Logistics and Naltume ports segments as well as any further infrastructure investments from capital proceeds from asset sales and incremental leverage at the ATCO Ltd. level.

We are updating our eps estimates to C\$2.95/C\$3.13/C\$3.20/C\$3.34 from C\$2.96/C\$3.29/C\$3.38/C\$3.54 for '18-'21E respectively. We are increasing our price target to C\$49 from C\$47 premised upon multiple expansion and a roll to the 2021 valuation year.

Our prior price target of C\$47 was premised upon a SOTP analysis on our UPO eps in 2020E of C\$2.96, the group multiple of 15.2x at a 5% discount. A comparable group multiple of 16.8x our Structures & Logistics eps estimate of C\$0.32 in 2020. And finally a 7.5x ports 2020E EV/EBITDA multiple times our Naltume Ports 2020E EBITDA of C\$56mln on allocated debt of C\$340mln.

Our current price target of C\$49 is premised upon a 5% discount to the 15.9x 2021E normalized non-gas midstream multiple and our UPO (utility & parent only) 2021E eps of C\$2.91 which yields C\$44/share. To this we add 15.1x our 2020E S&L eps of C\$0.30 which yields C\$4.60/share and 7.6x our Naltume Ports EBITDA of C\$56mln with C\$340mln of allocated debt which yields C\$0.70/share.

Figure 15: ATCO Sum of the Parts

UPO EPS	\$ 2.91	S&L EPS	\$ 0.30
Multiple	15.9x	Multiple	15.1x
Cdn Value vs. Bonds	10%	S&L Value	\$ 4.60
Regulatory Ranking	-5%		
Growth	-5%	Neltume '20E EBITDA	56
Company Specific	-5%	EV/EBITDA	7.6x
Total Net Prem/Disc	-5%	Allocated Debt	\$ 340
Utility Value	\$ 43.95	Neltume Value	\$ 0.70
SOTP Value	\$ 49		

Source: Company reports, UBS equity research

Canadian Utilities Ltd.

We reiterate our Neutral rating. While we see potential upside from continued investment in Alberta Power Line and continued growth in Alberta we see risks to the potential for a PBR reopener in Alberta and the less constructive Alberta regulatory environment.

We are updating our eps estimates to C\$2.06/C\$2.15/C\$2.21/C\$2.31 from C\$2.07/C\$2.27/C\$2.35/C\$2.46 for '18-'21E respectively. We are maintaining our price target at C\$33 premised upon multiple expansion and a roll to the 2021 valuation year, offset by an increased discount for reopener regulatory risk in Alberta.

Our prior target of C\$33 was premised upon a 5% discount to the 15.2x group multiple on 2020 estimated earnings per share of C\$2.35.

Our current target of C\$33 is premised upon a 10% net discount to the 15.9x 2021E normalized non-gas midstream multiple and our 2021E eps of C\$2.31.

Caribbean Utilities

We are upgrading Caribbean Utilities to Neutral from Sell on valuation. See our separate note ([LINK](#)). We are updating our eps estimates to \$0.71/\$0.77/\$0.80/\$0.89 from \$0.71/\$0.78/\$0.83/\$0.94 for '18-'21E respectively. We are increasing our price target to \$13 from \$12 due to multiple expansion and a roll to the 2021 valuation year.

Our prior target of \$12 was premised upon a group multiple of 15x our 2020E eps of \$0.83.

Our current target of \$13 is premised upon a 6% net discount to the 15.9x 2021E normalized non-gas midstream multiple and our 2021E eps of \$0.89.

CMS Energy

We are raising our price target to \$55 from \$54 and fine-tuning our EPS estimates. Our new estimates reflect 8% EPS growth off the high-end of the 2019 \$2.46-\$2.50 EPS guidance range. Our revised EPS estimates are \$2.50 for 2019, \$2.70 for 2020 and \$2.92 for 2021 versus \$2.53/\$2.73/\$2.95 previously.

CMS expects to receive a proposal for decision in the electric rate case in December and an order in March 2019 and an order in the integrated resource

plan proceeding in April 2019 although the company has a track record for achieving partial settlements. We also look for more use of renewable development with the green tariffs in Michigan.

Our \$55 price target for CMS is a 15% premium to the Regulated Utility group and is 18.9x our \$2.92 2021 EPS estimate. The premium includes a 5% premium for the group's undervaluation, 5% for first quartile Michigan regulation and 5% for first quartile growth.

Our prior \$54 target was 19.6x 2020E EPS of \$2.73 and included 10% for group undervaluation, 5% for first quartile Michigan regulation and 5% for first quartile growth.

Consolidated Edison

We are increasing our price target on ED to \$85 from \$80. We expect ED to close the Semptra Solar transaction this year and to file the Consolidated Edison of New York general rate case in early 2019. Despite the fourth quartile ranking for regulation the company has done 8-10 Reforming the Energy Vision projects which have been treated with reasonable regulation. This includes a regulatory mechanism where the company retains 30% of the savings achieved on investments. REV projects also have a 10 year amortization.

Our current price target is at a 5.5% premium to the Regulated Utility average multiple or 17.2x our 2021 EPS estimate of \$4.93. The premium includes 5% for the group's undervaluation, -5% for fourth quartile New York regulation, -2% for third quartile growth and a 7.5% ESG (Environmental, Social and Governance) premium for not owning generation.

Our prior \$80 price target was 17.5x our 2020 EPS estimate of \$4.60. The methodology included 10% for group undervaluation, 7.5% for ESG, -5% for regulation and -2% for growth.

Dominion Energy

We are upgrading our rating on D to Buy from Neutral on valuation and resolution of the financing uncertainty experienced during 2018. See our separate note ([LINK](#)). Our estimates exclude SCG; however, the probability of deal close seems increasingly likely with the BLRA lawsuit settlement, providing upside to our estimates. Our price target increases to \$84 from \$75, based on a sum of the parts methodology that includes \$44 for VEPCO, \$37 for Gas Infrastructure (including Cove Point) and \$4 for Merchant Generation. The outcome of the CT Zero Carbon RFP could impact our estimates. We assume Millstone sells all production at an average energy price of ~\$47/MWh in 2021 versus the forward energy price of \$43/MWh. Every \$1/MWh change in energy price is approximately \$0.01 to EPS.

Our estimates are revised up to \$4.10 in 2018, \$4.25 in 2019, \$4.46 in 2020 and \$4.71 in 2021, from \$4.10, \$4.16, \$4.35 and \$4.58, respectively.

Our current price target of \$84 is premised upon VEPCO valued at a 12% net premium to the 16.4x 2021e Regulated Utility multiple, Gas Infrastructure (including Cove Point) valued at the average Gas Midstream 2021e P/E of 16.5x, and Merchant Generation is assigned a 7.2x mid-cycle EV/EBITDA multiple.

Our prior price target of \$75 was premised upon VEPCO valued at a 17% net premium to the 15.9x 2020e Regulated Utility multiple, Gas Infrastructure (including Cove Point) valued at a 10% discount to the average Gas Midstream

2020e P/E of 15.9x, and Merchant Generation is assigned a 7.2x mid-cycle EV/EBITDA multiple.

DTE Energy

We are increasing our price target on DTE to \$121 from \$118. We believe DTE is in a good position having replaced expiring Power and Industrial tax credits in 2020 and 2022. At the margin we look for DTE to continue working to add contracts and to fill the NEXUS pipeline; to continue working on renewable investments through green tariffs and to add investments in tax advantaged renewable natural gas which is animal gas and biodegradable vegetation.

Our current \$121 price target is a sum of the parts valuation which includes \$91 for utility, parent and other at a 12% premium 2021E P/E multiple of 18.4x \$4.98, plus \$24 for unregulated EPS of \$1.45 at a 16.5x gas conglomerate multiple, \$4 for co-gen and renewables and \$2 for the NPV of renewable energy fuel tax credits.

Our prior price target of \$118 assumed \$87 for the utility, parent and other which included a 10% benefit for the group's undervaluation. The UP&O was a 17% P/E premium 19.2x applied to 2020E EPS of \$4.53, \$27 for unregulated EPS of \$1.44 at a 19x gas conglomerate multiple, \$3 for co-gen and renewables and \$1.50 for the NPV of REF tax credits.

Figure 16: DTE Sum of the Parts

DTE	EPS 2021E	Premium/ Multiple	Value
UP&O	\$4.98	12.0%	\$91
Unregulated	\$1.45	16.5x	\$24
Co-Gen	\$0.27	16.4x	\$4
Renewable Energy Fuel Tax Credits			\$2
Total			\$121

Source: Company reports, UBS equity research

Duke Energy

We are increasing our price target on DUK to \$99 from \$92. Our new price target is premised upon a 17.7x multiple to our 2021e EPS of \$5.59, or a net 8% premium to the Regulated Utility P/E multiple. Previously our \$92 price target reflected a 13% P/E premium to the Regulated Utility P/E multiple on 2020e EPS of \$5.29.

Our estimates are revised up to \$4.74 in 2018, \$5.03 in 2019, \$5.33 in 2020 and \$5.59 in 2021, from \$4.74, \$5.03, \$5.29 and \$5.56, respectively. The revisions reflect increased confidence in DUK's ability to meet capital investment and growth targets.

Emera Inc.

We are upgrading Emera Inc. to Buy from Neutral on valuation and the likelihood of accretive asset sales filling in the C\$1.4Bln block equity need through 2021. See our separate note ([LINK](#)). We are updating our eps estimates to C\$2.68/C\$2.80/C\$3.10/C\$3.26 from C\$2.68/C\$2.91/C\$3.05/C\$3.21 for '18-'21E respectively. We are increasing our price target to C\$51 from C\$42 on multiple

expansion, a roll to the 2021 valuation year, and including generation asset sale proceeds in lieu of block equity.

Our prior price target of C\$42 was premised upon SOTP valuation which is a 10% premium to the group multiple of 14.5x on utility and parent only (UPO) EPS of \$2.55 in 2020E yielding \$40 to which we added \$2/share for the Energy segment at 7.2x 2020E EBITDA.

Our current price target of C\$51 is premised upon a net 5% premium to the 15.9x 2021E normalized non-gas midstream multiple and our '21E UPO eps of C\$3.01 which yields C\$50/share. To this we add 6.7x our '21E Energy EBITDA of C\$51mln with no allocated net debt which yields C\$1/share.

Figure 17: EMA Sum of Parts

UPO EPS	\$	3.01	Energy EBITDA	51
Multiple		15.9x	Multiple	6.7x
Overall Reg. Group		10%	Enterprise Value	336
Regulatory Prem/Disc		2%	Net Debt	-
Earnings Growth Prem/Disc		-2%	Equity Value	336
Net Premium		5%	Shares Out	277.1
Utility Valuation	\$	50	Energy Valuation	\$ 1
Consolidated Valuation	\$	51		

Source: Company reports, UBS equity research

Entergy Corp

We are increasing our price target on ETR to \$99 from \$94. Our new price target is premised upon a 17.2x multiple to our 2021e EPS of \$5.73, or a net 5% premium to the Regulated Utility P/E multiple.

Our previous \$94 price target reflected a 10% premium to the Regulated Utility P/E multiple on 2020e EPS of \$5.40. Our EPS estimates remain unchanged at \$4.70 in 2018, \$5.09 in 2019, \$5.40 in 2020 and \$5.73 in 2021.

Evergy

We are increasing our price target on EVRG to \$61 from \$58. EVRG is in the process of executing on the 60M (22% of shares) stock buyback and we expect an update on the implications of Missouri's SB 564 on the fourth quarter conference call. We also expect a cap-ex update including with regard to grid modernization in Missouri.

Our current \$61 price target is at an 8% premium to the Regulated Utility average or 17.7x 2021E EPS of \$3.44 plus \$0.42/share for the NPV of corporate owned life insurance. The 8% premium includes +5% for the group's undervaluation, +5% for top quartile EPS growth of 8% and -2% for below average regulation.

Our prior \$58 price target reflected a +10% for valuation and was 17.8x \$3.25 in 2020E plus \$0.42/share for COLI.

Eversource Energy

We are increasing our price target on ES to \$75 from \$69. Our new price target is premised upon an 18.9x multiple to our 2021e EPS of \$3.95, or a net 15%

premium to the Regulated Utility P/E multiple. Previously our \$69 price target reflected an 18% P/E premium to the Regulated Utility P/E multiple on 2020e EPS of \$3.72. Our EPS estimates remain unchanged at \$3.28 in 2018, \$3.48 in 2019, \$3.72 in 2020 and \$3.95 in 2021.

Exelon Corp.

We reiterate our Buy rating. We continue to believe that Exelon has solid utility growth driven by capital spending and closing the gap between earned and allowed ROEs particularly at the Pepco Holdings utility subsidiaries. We are updating our eps forecast to \$3.14/\$3.18/\$3.14/\$3.25 from \$3.14/\$3.19/\$3.16/\$3.29 for '18-21E respectively. We are increasing our price target to \$51 from \$50 premised upon multiple expansion and a roll to the 2021 valuation year for the utility and the 2020 valuation year for ExGen.

Our prior price target of \$50 was premised upon a net 17% premium to the group multiple of 16.1x our 2020E UPO eps of \$2.05, and 6.7x '19E open EBITDA of \$2,585mln for ExGen.

Our current price target of \$51 is premised upon a net 8% premium to the 16.4x normalized multiple and our 2021E UPO eps of \$2.21 which yields \$40/share. To this we add 7.2x our 2020E ExGen open EBITDA of \$2,257mln, NPV of hedges of \$196mln, net debt of \$6,126mln, and 949mln shares outstanding which yields \$11/share.

Figure 18: EXC Sum of Parts – Dollars in Millions

SOTP Valuation Methodology		ExGen Valuation	
UPO Valuation		EXGEN EBITDA	2,383
20 UPO EPS	\$ 2.21	GM Value of Hedges	126
Regulated Utility Group Multiple	16.4x	EXGEN Open EBITDA	2,257
Overall Regulated Group	5%	Net Multiple for ExGen	7.2x
Regulatory Ranking	-2%	Total Enterprise Value	16,250
Earnings Growth	5%	NPV of Hedges	196
Net Premium/Discount	8%	Net Debt at ExGen	6,126
UPO Value per Share	\$ 40	Equity Value	10,320
		Shares	949
Total Value per Share	\$ 51	ExGen Value per Share	\$ 11

Source: Company reports, UBS equity research

FirstEnergy Corp.

Our price target on FE remains unchanged at \$43. Management has stated they plan to provide additional cap-ex guidance on the fourth quarter conference call which could include an additional year (2022). Before increasing cap-ex FE wants to see outcomes on the New Jersey \$0.4B infrastructure plan proposal (potentially in Q1'19) and the Ohio SEET (significantly excessive earnings test).

Our price target is updated to reflect a 2% premium to the Regulated Utility P/E multiple or 16.7x 2021E EPS of \$2.60. The 2% premium includes 5% for the group's undervaluation, 2% for regulation and -5% for fourth quartile EPS growth.

Previously our \$43 reflected a 7% P/E premium to the Regulated Utility group or 17.2x \$2.46 in 2020E plus \$0.28 for Ohio rider NPV. The premium included 10% for the group's undervaluation.

Fortis Inc.

We are reiterating our Buy rating and continue to believe that Fortis Inc remains undervalued to achievable 6% long term dividend per share growth with lower relative risk resulting from no need block equity, and no significantly large scale capital projects. Further, regulatory matters are limited over the next 12 month with FERC transmission ROEs now settled for the ITC subsidiary.

We are updating our eps estimates to C\$2.52/C\$2.64/C\$2.83/C\$3.01 from C\$2.52/C\$2.65/C\$2.84/C\$3.01 for '18-'21E respectively. We are updating our price target to C\$53 from C\$49 on multiple expansion and a roll to the 2021 valuation year.

Our prior price target of C\$49 was premised upon a net 14% premium to the group multiple of 15.1x our 2020E eps of C\$2.84.

Our current price target of C\$53 is premised upon a net 10% premium to the 15.7x 2021E normalized non-gas midstream multiple and our 2021E eps of C\$3.01.

Hawaiian Electric Industries

Our price target for HE remains unchanged at \$33. Our price target is updated to reflect a 16.7x multiple to our 2021e EPS of \$2.19, or a net 2% premium to the Regulated Utility P/E multiple. Previously our \$33 target reflected no premium to the Regulated Utility P/E multiple on 2020e EPS of \$2.29.

Our EPS estimates are revised up to \$1.86 in 2018, \$1.99 in 2019, \$2.12 in 2020 and \$2.19 in 2021, from \$1.89, \$2.05, \$2.20 and \$2.29, respectively. The revisions reflect increased confidence in HE's ability to meet capital investment and growth targets.

Hydro One Ltd.

We reiterate our Sell rating. We now believe that, despite being dilutive that the merger with Avista Inc. will close after likely regulatory approvals are received on December 14. As a result we have incorporated the AVA merger dilution into our valuation methodology.

Our eps forecast remains Hydro One Ltd. on a stand-alone basis as we are updating it to C\$1.27/C\$1.27/C\$1.35/C\$1.41 from C\$1.23/C\$1.32/C\$1.34/C\$1.51 for '18-'21E respectively. We are updating our price target to C\$18 from C\$19 as a result of the inclusion of the AVA merger dilution in our valuation and lower stand-alone Hydro One eps estimates more than offsetting multiple expansion and the roll to the 2021 valuation year.

Our prior price target of C\$19 was premised upon a 14.8x multiple on our 2020E eps of C\$1.34.

Our current price target of C\$18 is premised upon a net 15% discount to the 15.9x 2021E normalized non-gas midstream multiple and our 2021E eps of C\$1.37 inclusive of (\$0.04)/share of dilution from the AVA merger.

NextEra Energy

We are increasing our price target on NEE to \$209 from \$195. Our new price target is premised upon an 18.9x multiple to our 2021e EPS of \$9.84, or a net 15% premium to the Regulated Utility P/E multiple. Previously our \$195 price

target reflected a 20% P/E premium to the Regulated Utility P/E multiple on 2020e EPS of \$9.06.

Our EPS estimates are revised up to \$7.77 in 2018, \$8.43 in 2019, \$9.08 in 2020 and \$9.84 in 2021, from \$7.88, \$8.41, \$9.06 and \$9.74, respectively. The revisions incorporate the Gulf Power acquisition which is expected to close 1Q19, offset somewhat by a more conservative growth trajectory than we had previously modelled at NEER.

OGE Energy

We are raising our price target to \$40 from \$37. OGE expects to have 1 or more rate cases in 2019 to address investments in the Sooner scrubbers and potentially in grid modernization. We fine-tuned our EPS estimates for cap-ex guidance to \$2.07 for 2018, \$2.12 for 2019, \$2.31 for 2020 and \$2.39 in 2021 versus \$2.08/\$2.14/\$2.30/\$2.38.

Our current \$40 price target is a sum of the parts which includes \$30 for the utility, parent and other at a 2% discount to the Regulated Utility average or 16.1x \$1.86 in 2021 plus \$10 for the company's 25.6% ownership in ENBL. The ENBL valuation reflects UBS MLP and Gas Pipeline analyst Shneur Gershuni's \$18 price target. "['19 Guidance Flexes Op Leverage & SCOOP Crude](#)" (11/7/18). The 2% discount multiple includes 5% for the group undervaluation, -2% for third quartile utility, parent and other EPS growth and -4% for fourth quartile regulation.

Our prior \$37 price target reflected \$30 for UP&O which at a 3% premium to the average Regulated Utility 2020 P/E or 16.7x \$1.75 at UP&O and \$7/share for ENBL using a mark-to-market for the stock.

Figure 19: OGE Sum of the Parts

OGE	EPS 2021E \$/Share	Premium/ Multiple	Value
UP&O	\$1.86	-2.0%	\$30
ENBL	\$18.00		\$10
Total			\$40

Source: Company reports, UBS equity research

Pinnacle West Capital

We are increasing our price target on PNW to \$92 from \$90. Our current \$92 target is a 2% premium to the Regulated Utility group of 16.7x our 2021 EPS estimate of \$5.52. The premium is 5% for the group's undervaluation, 2% for second quartile EPS growth, and -5% for fourth quartile regulation.

Our prior \$90 price target was a 5% premium to the Regulated Utility group average P/E of 17.0x \$5.28 in 2020E. The valuation multiple had included a -2% discount for third quartile regulation.

PNM Resources

We are increasing our price target on PNM to \$39 from \$37. Our new price target is premised upon a 16.9x multiple to our 2021e EPS of \$2.29, or a net 3% premium to the Regulated Utility P/E multiple.

Previously our \$37 price target reflected an 8% premium to the Regulated Utility P/E multiple on 2020e EPS of \$2.19. Our EPS estimates remain unchanged at \$1.97 in 2018, \$2.14 in 2019, \$2.19 in 2020 and \$2.29 in 2021.

Portland General

We are lowering our price target on POR to \$44 from \$45. Consistent with guidance we are revising our EPS estimates to reflect 80 bp of regulatory lag versus 50 bp previously (8.7% ROE versus 9.0% earned ROE). Our revised EPS estimates are \$2.48 for 2019, \$2.57 for 2020, \$2.65 for 2021 and \$2.73 for 2022 versus \$2.53 for 2019, \$2.64 for 2020, \$2.78 for 2021 and \$2.87 for 2022.

Our \$44 price target is a 2% premium to the Regulated Utility average or 16.7x \$2.65 in 2021E. The valuation includes 5% for the group's undervaluation, 2% for second quartile Oregon regulation, and -5% for 4th quartile EPS growth.

Our prior \$45 price target reflected a 6% Regulated Utility premium or 17.1x \$2.64 in 2020E and included a 10% benefit the group's undervaluation and 2% discounts for 3rd quartile growth and regulation.

PPL Corporation

We maintain our Buy rating and our \$34 price target. We are fine-tuning our EPS estimates for exposure to the British pound and for the gradual expiration of pension revenues in the U.K. at WPD. Our new EPS estimates are \$2.34 for 2018, \$2.40 for 2019, and \$2.56 for 2020 versus \$2.33/\$2.42/\$2.56. Our estimates include 7% 5 year growth (2017-2022) in Pennsylvania, 4% in Kentucky and 1% at WPD in the U.K.

F/X. Our EPS forecast assumes a 1.30x US\$/British pound exchange in 2020 and 1.37x for 2021-2022 which is the UBS Global Macro Strategy year-end 2020 forecast "[2019 Markets Outlook: Something wicked this way comes?](#)" (pp. 34-35, 11/12/18). This is a -\$0.04/share to -\$0.06/share impact from 2020 to 2023 versus our prior forecast.

Pension. PPL receives \$0.20/share in revenue to fund the pension deficit, but the plan has been outperforming. Pending a review at regulator OFGEM we expect the pension revenue will begin a 4 year calendar year expiration in April 2021. This is a -\$0.05 to -\$0.07/share annual impact.

EPS growth-U.K. and Pennsylvania. We continue to forecast 5 year 1% EPS growth in the U.K. as we believe we have been understating the impact of 6% rate base growth and incentive opportunities. We also believe PPL's PPL Electric in Pennsylvania can grow EPS 6-7% versus 5-6% previously due to transmission investments.

Our current \$34 price target reflects the sum of \$21 for a 5% premium to the U.S. utility average or 17.2x \$1.23 in 2021E, \$15 for a European average utility multiple of 12.5x \$1.22, less \$4 for the NPV of a -\$0.24 exposure in the 2023 RII0 2 case at a \$13.6x P/E. The 5% U.S. premium includes 5% for the group's undervaluation, 2% for second quartile regulation, and -2% for third quartile EPS growth.

Figure 21: PPL Sum of the Parts

PPL	EPS 2021E	Premium/ Multiple	Value
U.S. Utilities	\$1.23	5.0%	\$21
WPD	\$1.36	12.5x	\$17
NPV of RIIO 2 Rate Exposure			(\$4)
Total			\$34

Source: Company reports, UBS equity research

Public Service Enterprise Group

We are raising our price target on PEG to \$63 from \$60. We are fine-tuning our EPS estimates for changes in the forward power and gas curves. Forward around the clock power prices rose in PJM East \$5/MWhr around-the-clock in 2019 and \$1/MWhr in 2020 from early October to mid-November according to S&P Global Market Intelligence quotes. Gas prices spiked at Henry hub and Leidy hub 40 cents. These impacts tapered off in 2021 and 2022.

We are lowering our EPS estimates to reflect lower profitability on combined cycle gas plants and an offsetting benefit for higher power prices. Our revised estimates are \$3.35 in 2019, \$3.69 in 2020, \$3.77 in 2021 and \$4.02 in 2022 versus \$3.47/\$3.66/\$3.76/\$4.04. PEG's baseload generation is 100% hedged through 2019 and 75-80% in 2020 and the intermediate, combined cycle and peaking output is 35-40% hedged in 2019.

Our current \$63 sum of the parts based price target includes: \$51 for UP&O or 18.4x our 2021 EPS estimate of \$2.79 plus \$11 for PSEG Power at 6.8x 2021 EBITDA of \$1.05B. The premium includes: first quartile EPS growth for utility net of parent 12% (+5%), above-average New Jersey regulation (+2%) and a premium for the Regulated Utility group's undervaluation (+5%).

Previously our \$60 price target included \$47 for PSE&G using a 12% Regulated Utility premium 2020E P/E multiple of 16.1x applied to \$2.59 of UP&O EPS and \$13 for PSEG Power at 7.2x \$1.1B less \$1.7B of debt and 508M shares.

Figure 22: PEG Sum of the Parts – Dollars in Millions

PEG	EPS 2021E	Premium/ Multiple	Value
UP&O	\$2.79 \$/Share	12.0%	\$51
PSEG Power	\$1,046	6.8x	\$11
Total			\$63

Source: Company reports, UBS equity research

SCANA Corp.

We are updating our price target to \$49 from \$41. Given the pending Dominion merger we are not rolling our valuation to the 2021 year rather we are reflecting the value of the Dominion merger which is Dominion's closing price of

\$74.11/share at the 0.669 exchange ratio. See our separate note ([LINK](#)). Our 2020E SCANA Corp. stand-alone valuation is \$41 premised upon 2020E eps under the ORS securitization scenario of \$2.35 in 2020 and our updated normalized utility comp group multiple of 17.3x.

We are reiterating our Neutral rating. We are updating our eps estimates to be reflective of the earnings power from the latest Dominion Energy rate proposal before the South Carolina Public Service commission commensurate with no upfront refund but an approximate 15% on-going rate cut as a result of the abandoned V.C. Summer Unit 2 and Unit 3 new nuclear construction project. Our eps forecast is updated to \$2.67/\$2.30/\$2.44/\$2.66 from \$3.13/\$2.76/\$2.90/3.12 for '18-'21E respectively.

Our prior price target of \$41 was premised upon a 50/50 view of acquisition break/close which results in our \$41 price target premised upon the \$46/share full value takeout price and our \$35/share value on break and the ORS terms with securitization.

Sempra Energy

Our current price target of \$129 is premised upon a sum of the parts. We value SDG&E, SoCalGas, Oncor and Parent at a 1% net premium to the 16.4x 2021e Regulated Utility multiple (\$75), Cameron at the average Gas Midstream 2021e P/E of 16.5x (\$22), other LNG & Midstream at a 5% discount to the average Gas Midstream P/E (-\$2), and approximate the mark-to-market for SRE Mexico (\$11) and SA Utilities (\$6). This derives a value of \$112 for the shares. We calculate an upside value of ~\$146 under execution of an Elliott-inspired strategy. Our price target is based on an average of these outcomes.

Our prior target of \$130 was derived with the same methodology, using 2020 estimates in the base case to arrive at values of \$68 for the Utilities and Parent, \$24 for Cameron, (\$4) for other LNG & Midstream, \$15 for SRE Mexico, and \$8 for SA Utilities, for total base case value of \$111. Our Elliott Plan upside value was \$149. Our price target reflected the average of those two outcomes.

Our 2018 EPS estimate is revised up to \$5.46 from \$5.31. Our other estimates remain unchanged at \$5.89 in 2019, \$6.99 in 2020 and \$7.55 in 2021.

Southern Company

We are reiterating our Neutral rating. Our eps estimates remain unchanged at \$33.06/\$3.04/\$3.17/\$3.30 for '18-'21E respectively. We are reiterating our price target of \$49 as inclusion of an additional 5% discount for the Georgia triennial rate case year in 2019 balances the impact of multiple expansion and a roll to the 2021 valuation year.

Our prior price target of \$49 was premised upon a net 5% discount to the group multiple of 15.7x our 2020E eps of \$3.17.

Our current price target of \$49 is premised upon a net 10% discount to the 16.4x normalized multiple and our 2021E eps of \$3.30.

WEC Energy Group

We are increasing our price target on WEC to \$73 from \$70. Our new price target is premised upon an 18.4x multiple to our 2021e EPS of \$3.97, or a net 12% premium to the Regulated Utility P/E multiple. Previously our \$70 price target

reflected a 17% premium to the Regulated Utility P/E multiple on 2020e EPS of \$3.74. Our EPS estimates remain unchanged at \$3.34 in 2018, \$3.54 in 2019, \$3.74 in 2020 and \$3.97 in 2021.

Xcel Energy

We are increasing our price target on XEL to \$53 from \$50. Our new price target is premised upon a 17.9x multiple to our 2021e EPS of \$2.95, or a net 9% premium to the Regulated Utility P/E multiple. Previously our \$50 price target reflected a 14% premium to the Regulated Utility P/E multiple on 2020e EPS of \$2.77. Our EPS estimates remain unchanged at \$2.47 in 2018, \$2.60 in 2019, \$2.77 in 2020 and \$2.95 in 2021.

Valuation Method and Risk Statement

Our valuation methodology for the group is price to earnings based. The adjustments applied fall into 5 categories. These are as follows: 1) Group Valuation Bias: Flowing from our valuation work comparing Baa corporate yields to group dividend yields and RU price to earnings ratios to those for the S&P 500, we incorporate a positive or negative adjustment to our group multiple representing the gap we calculate to the nearest 5%; 2) Growth Adjustment: We adjust our valuations based on the growth quartile each utility occupies. First quartile receives a 5% premium, second quartile a 2% premium, third quartile a 2% discount and fourth quartile a 5% discount; 3) Regulatory Adjustment: Our valuation adjustments for regulation are based on our proprietary Regulatory Rankings. First quartile jurisdictions receive 5%, second quartile 2%, third quartile -2% and fourth quartile -5%; 4) Multi Utility Diversified Valuation: For multi utilities (those with more than 15% diversified or foreign earnings), we perform a sum-of-parts analysis applying business/region appropriate valuations to those diversified businesses; 5) One-off Adjustments: In special situations, we value risk on an issue specific basis. Common areas where we apply such an adjustment include: ESG advantage, large project construction risk, legal risk, and announced M&A completion risk.

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12-Month Rating	Definition	Coverage ¹	IB Services ²
Buy	FSR is > 6% above the MRA.	48%	24%
Neutral	FSR is between -6% and 6% of the MRA.	37%	21%
Sell	FSR is > 6% below the MRA.	15%	12%
Short-Term Rating	Definition	Coverage ³	IB Services ⁴
Buy	Stock price expected to rise within three months from the time the rating was assigned because of a specific catalyst or event.	<1%	<1%
Sell	Stock price expected to fall within three months from the time the rating was assigned because of a specific catalyst or event.	<1%	<1%

Source: UBS. Rating allocations are as of 30 September 2018.

1: Percentage of companies under coverage globally within the 12-month rating category.

2: Percentage of companies within the 12-month rating category for which investment banking (IB) services were provided within the past 12 months.

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UBS Securities LLC: Daniel Ford, CFA; Ross Fowler, CFA; Gregg Orrill; Rose-Lynn Armstrong.

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Company Name	Reuters	12-month rating	Short-term rating	Price	Price date
AES Corp ¹⁶	AES.N	Neutral	N/A	US\$15.51	28 Nov 2018
Alliant Energy Corp ¹⁶	LNT.N	Neutral	N/A	US\$44.77	28 Nov 2018
Ameren Corp ¹⁶	AEE.N	Neutral	N/A	US\$68.11	28 Nov 2018
American Electric Power Inc ^{2, 4, 6a, 7, 16}	AEP.N	Buy	N/A	US\$76.22	28 Nov 2018
ATCO Ltd	ACOX.TO	Buy	N/A	C\$40.11	28 Nov 2018
Canadian Utilities Ltd	CU.TO	Neutral	N/A	C\$31.28	28 Nov 2018
Caribbean Utilities Corp	CUPU.TO	Neutral	N/A	US\$12.50	28 Nov 2018
CMS Energy Corp ¹⁶	CMS.N	Neutral	N/A	US\$51.23	28 Nov 2018
Consolidated Edison Inc ¹⁶	ED.N	Neutral	N/A	US\$78.84	28 Nov 2018
Dominion Energy Inc ^{4, 6a, 6c, 7, 16}	D.N	Buy	N/A	US\$73.32	28 Nov 2018
DTE Energy Co ^{4, 6a, 7, 16}	DTE.N	Neutral	N/A	US\$117.40	28 Nov 2018
Duke Energy Corp ^{2, 4, 6a, 7, 16}	DUK.N	Buy	N/A	US\$87.60	28 Nov 2018
Emera Inc	EMA.TO	Buy	N/A	C\$44.11	28 Nov 2018
Enable Midstream Partners LP ¹⁶	ENBL.N	Buy	N/A	US\$13.44	28 Nov 2018
Entergy Corp ^{7, 16}	ETR.N	Buy	N/A	US\$86.08	28 Nov 2018
Eversource Energy ^{7, 16}	EVRG.N	Neutral	N/A	US\$59.13	28 Nov 2018
Exelon Corp ^{7, 16}	EXC.N	Buy	N/A	US\$45.82	28 Nov 2018
FirstEnergy Corp ¹⁶	FE.N	Buy	N/A	US\$37.60	28 Nov 2018
Fortis Inc ^{7, 16}	FTS.TO	Buy	N/A	C\$45.70	28 Nov 2018
Hawaiian Electric Industries Inc ¹⁶	HE.N	Sell	N/A	US\$37.94	28 Nov 2018
Hydro One	H.TO	Sell	N/A	C\$19.53	28 Nov 2018
NextEra Energy Inc ^{4, 6a, 7, 16, 26}	NEE.N	Buy	N/A	US\$178.05	28 Nov 2018
OGE Energy Corp ¹⁶	OGE.N	Neutral	N/A	US\$39.11	28 Nov 2018
PNM Resources Inc ^{7, 16}	PNM.N	Sell	N/A	US\$42.76	28 Nov 2018
Portland General Electric Co ¹⁶	POR.N	Sell	N/A	US\$47.96	28 Nov 2018
PPL Corp ^{2, 4, 6a, 6b, 6c, 7, 16}	PPL.N	Buy	N/A	US\$30.86	28 Nov 2018
Public Service Enterprise Group ^{7, 16}	PEG.N	Buy	N/A	US\$54.71	28 Nov 2018

Source: UBS. All prices as of local market close.

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The Cross-Section of Expected Stock Returns

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ABSTRACT

Two easily measured variables, size and book-to-market equity, combine to capture the cross-sectional variation in average stock returns associated with market β , size, leverage, book-to-market equity, and earnings-price ratios. Moreover, when the tests allow for variation in β that is unrelated to size, the relation between market β and average return is flat, even when β is the only explanatory variable.

THE ASSET-PRICING MODEL OF Sharpe (1964), Lintner (1965), and Black (1972) has long shaped the way academics and practitioners think about average returns and risk. The central prediction of the model is that the market portfolio of invested wealth is mean-variance efficient in the sense of Markowitz (1959). The efficiency of the market portfolio implies that (a) expected returns on securities are a positive linear function of their market β s (the slope in the regression of a security's return on the market's return), and (b) market β s suffice to describe the cross-section of expected returns.

There are several empirical contradictions of the Sharpe-Lintner-Black (SLB) model. The most prominent is the size effect of Banz (1981). He finds that market equity, ME (a stock's price times shares outstanding), adds to the explanation of the cross-section of average returns provided by market β s. Average returns on small (low ME) stocks are too high given their β estimates, and average returns on large stocks are too low.

Another contradiction of the SLB model is the positive relation between leverage and average return documented by Bhandari (1988). It is plausible that leverage is associated with risk and expected return, but in the SLB model, leverage risk should be captured by market β . Bhandari finds, however, that leverage helps explain the cross-section of average stock returns in tests that include size (ME) as well as β .

Stattman (1980) and Rosenberg, Reid, and Lanstein (1985) find that average returns on U.S. stocks are positively related to the ratio of a firm's book value of common equity, BE, to its market value, ME. Chan, Hamao, and Lakonishok (1991) find that book-to-market equity, BE/ME, also has a strong role in explaining the cross-section of average returns on Japanese stocks.

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Finally, Basu (1983) shows that earnings-price ratios (E/P) help explain the cross-section of average returns on U.S. stocks in tests that also include size and market β . Ball (1978) argues that E/P is a catch-all proxy for unnamed factors in expected returns; E/P is likely to be higher (prices are lower relative to earnings) for stocks with higher risks and expected returns, whatever the unnamed sources of risk.

Ball's proxy argument for E/P might also apply to size (ME), leverage, and book-to-market equity. All these variables can be regarded as different ways to scale stock prices, to extract the information in prices about risk and expected returns (Keim (1988)). Moreover, since E/P, ME, leverage, and BE/ME are all scaled versions of price, it is reasonable to expect that some of them are redundant for describing average returns. Our goal is to evaluate the joint roles of market β , size, E/P, leverage, and book-to-market equity in the cross-section of average returns on NYSE, AMEX, and NASDAQ stocks.

Black, Jensen, and Scholes (1972) and Fama and MacBeth (1973) find that, as predicted by the SLB model, there is a positive simple relation between average stock returns and β during the pre-1969 period. Like Reinganum (1981) and Lakonishok and Shapiro (1986), we find that the relation between β and average return disappears during the more recent 1963–1990 period, even when β is used alone to explain average returns. The appendix shows that the simple relation between β and average return is also weak in the 50-year 1941–1990 period. In short, our tests do not support the most basic prediction of the SLB model, that average stock returns are positively related to market β s.

Unlike the simple relation between β and average return, the univariate relations between average return and size, leverage, E/P, and book-to-market equity are strong. In multivariate tests, the negative relation between size and average return is robust to the inclusion of other variables. The positive relation between book-to-market equity and average return also persists in competition with other variables. Moreover, although the size effect has attracted more attention, book-to-market equity has a consistently stronger role in average returns. Our bottom-line results are: (a) β does not seem to help explain the cross-section of average stock returns, and (b) the combination of size and book-to-market equity seems to absorb the roles of leverage and E/P in average stock returns, at least during our 1963–1990 sample period.

If assets are priced rationally, our results suggest that stock risks are multidimensional. One dimension of risk is proxied by size, ME. Another dimension of risk is proxied by BE/ME, the ratio of the book value of common equity to its market value.

It is possible that the risk captured by BE/ME is the relative distress factor of Chan and Chen (1991). They postulate that the earning prospects of firms are associated with a risk factor in returns. Firms that the market judges to have poor prospects, signaled here by low stock prices and high ratios of book-to-market equity, have higher expected stock returns (they are penalized with higher costs of capital) than firms with strong prospects. It is

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also possible, however, that BE/ME just captures the unraveling (regression toward the mean) of irrational market whims about the prospects of firms.

Whatever the underlying economic causes, our main result is straightforward. Two easily measured variables, size (ME) and book-to-market equity (BE/ME), provide a simple and powerful characterization of the cross-section of average stock returns for the 1963–1990 period.

In the next section we discuss the data and our approach to estimating β . Section II examines the relations between average return and β and between average return and size. Section III examines the roles of E/P, leverage, and book-to-market equity in average returns. In sections IV and V, we summarize, interpret, and discuss applications of the results.

I. Preliminaries

A. Data

We use all nonfinancial firms in the intersection of (a) the NYSE, AMEX, and NASDAQ return files from the Center for Research in Security Prices (CRSP) and (b) the merged COMPUSTAT annual industrial files of income-statement and balance-sheet data, also maintained by CRSP. We exclude financial firms because the high leverage that is normal for these firms probably does not have the same meaning as for nonfinancial firms, where high leverage more likely indicates distress. The CRSP returns cover NYSE and AMEX stocks until 1973 when NASDAQ returns also come on line. The COMPUSTAT data are for 1962–1989. The 1962 start date reflects the fact that book value of common equity (COMPUSTAT item 60), is not generally available prior to 1962. More important, COMPUSTAT data for earlier years have a serious selection bias; the pre-1962 data are tilted toward big historically successful firms.

To ensure that the accounting variables are known before the returns they are used to explain, we match the accounting data for all fiscal yearends in calendar year $t - 1$ (1962–1989) with the returns for July of year t to June of $t + 1$. The 6-month (minimum) gap between fiscal yearend and the return tests is conservative. Earlier work (e.g., Basu (1983)) often assumes that accounting data are available within three months of fiscal yearends. Firms are indeed required to file their 10-K reports with the SEC within 90 days of their fiscal yearends, but on average 19.8% do not comply. In addition, more than 40% of the December fiscal yearend firms that do comply with the 90-day rule file on March 31, and their reports are not made public until April. (See Alford, Jones, and Zmijewski (1992).)

We use a firm's market equity at the end of December of year $t - 1$ to compute its book-to-market, leverage, and earnings-price ratios for $t - 1$, and we use its market equity for June of year t to measure its size. Thus, to be included in the return tests for July of year t , a firm must have a CRSP stock price for December of year $t - 1$ and June of year t . It must also have monthly returns for at least 24 of the 60 months preceding July of year t (for

“pre-ranking” β estimates, discussed below). And the firm must have COMPUSTAT data on total book assets (A), book equity (BE), and earnings (E), for its fiscal year ending in (any month of) calendar year $t - 1$.

Our use of December market equity in the E/P, BE/ME, and leverage ratios is objectionable for firms that do not have December fiscal yearends because the accounting variable in the numerator of a ratio is not aligned with the market value in the denominator. Using ME at fiscal yearends is also problematic; then part of the cross-sectional variation of a ratio for a given year is due to market-wide variation in the ratio during the year. For example, if there is a general fall in stock prices during the year, ratios measured early in the year will tend to be lower than ratios measured later. We can report, however, that the use of fiscal-year-end MEs, rather than December MEs, in the accounting ratios has little impact on our return tests.

Finally, the tests mix firms with different fiscal yearends. Since we match accounting data for all fiscal yearends in calendar year $t - 1$ with returns for July of t to June of $t + 1$, the gap between the accounting data and the matching returns varies across firms. We have done the tests using the smaller sample of firms with December fiscal yearends with similar results.

B. Estimating Market β s

Our asset-pricing tests use the cross-sectional regression approach of Fama and MacBeth (1973). Each month the cross-section of returns on stocks is regressed on variables hypothesized to explain expected returns. The time-series means of the monthly regression slopes then provide standard tests of whether different explanatory variables are on average priced.

Since size, E/P, leverage, and BE/ME are measured precisely for individual stocks, there is no reason to smear the information in these variables by using portfolios in the Fama-MacBeth (FM) regressions. Most previous tests use portfolios because estimates of market β s are more precise for portfolios. Our approach is to estimate β s for portfolios and then assign a portfolio's β to each stock in the portfolio. This allows us to use individual stocks in the FM asset-pricing tests.

B.1. β Estimation: Details

In June of each year, all NYSE stocks on CRSP are sorted by size (ME) to determine the NYSE decile breakpoints for ME. NYSE, AMEX, and NASDAQ stocks that have the required CRSP-COMPUSTAT data are then allocated to 10 size portfolios based on the NYSE breakpoints. (If we used stocks from all three exchanges to determine the ME breakpoints, most portfolios would include only small stocks after 1973, when NASDAQ stocks are added to the sample.)

We form portfolios on size because of the evidence of Chan and Chen (1988) and others that size produces a wide spread of average returns and β s. Chan and Chen use only size portfolios. The problem this creates is that size and the β s of size portfolios are highly correlated (-0.988 in their data), so

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asset-pricing tests lack power to separate size from β effects in average returns.

To allow for variation in β that is unrelated to size, we subdivide each size decile into 10 portfolios on the basis of pre-ranking β s for individual stocks. The pre-ranking β s are estimated on 24 to 60 monthly returns (as available) in the 5 years before July of year t . We set the β breakpoints for each size decile using only NYSE stocks that satisfy our COMPUSTAT-CRSP data requirements for year $t - 1$. Using NYSE stocks ensures that the β breakpoints are not dominated after 1973 by the many small stocks on NASDAQ. Setting β breakpoints with stocks that satisfy our COMPUSTAT-CRSP data requirements guarantees that there are firms in each of the 100 size- β portfolios.

After assigning firms to the size- β portfolios in June, we calculate the equal-weighted monthly returns on the portfolios for the next 12 months, from July to June. In the end, we have post-ranking monthly returns for July 1963 to December 1990 on 100 portfolios formed on size and pre-ranking β s. We then estimate β s using the full sample (330 months) of post-ranking returns on each of the 100 portfolios, with the CRSP value-weighted portfolio of NYSE, AMEX, and (after 1972) NASDAQ stocks used as the proxy for the market. We have also estimated β s using the value-weighted or the equal-weighted portfolio of NYSE stocks as the proxy for the market. These β s produce inferences on the role of β in average returns like those reported below.

We estimate β as the sum of the slopes in the regression of the return on a portfolio on the current and prior month's market return. (An additional lead and lag of the market have little effect on these sum β s.) The sum β s are meant to adjust for nonsynchronous trading (Dimson (1979)). Fowler and Rorke (1983) show that sum β s are biased when the market return is autocorrelated. The 1st- and 2nd-order autocorrelations of the monthly market returns for July 1963 to December 1990 are 0.06 and -0.05 , both about 1 standard error from 0. If the Fowler-Rorke corrections are used, they lead to trivial changes in the β s. We stick with the simpler sum β s. Appendix Table AI shows that using sum β s produces large increases in the β s of the smallest ME portfolios and small declines in the β s of the largest ME portfolios.

Chan and Chen (1988) show that full-period β estimates for portfolios can work well in tests of the SLB model, even if the true β s of the portfolios vary through time, if the variation in the β s is proportional,

$$\beta_{jt} - \beta_j = k_t(\beta_j - \beta), \quad (1)$$

where β_{jt} is the true β for portfolio j at time t , β_j is the mean of β_{jt} across t , and β is the mean of the β_j . The Appendix argues that (1) is a good approximation for the variation through time in the true β s of portfolios (j) formed on size and β . For diehard β fans, sure to be skeptical of our results on the weak role of β in average stock returns, we can also report that the results stand up to robustness checks that use 5-year pre-ranking β s, or 5-year post-ranking β s, instead of the full-period post-ranking β s.

We allocate the full-period post-ranking β of a size- β portfolio to each stock in the portfolio. These are the β s that will be used in the Fama-MacBeth cross-sectional regressions for individual stocks. We judge that the precision of the full-period post-ranking portfolio β s, relative to the imprecise β estimates that would be obtained for individual stocks, more than makes up for the fact that true β s are not the same for all stocks in a portfolio. And note that assigning full-period portfolio β s to stocks does not mean that a stock's β is constant. A stock can move across portfolios with year-to-year changes in the stock's size (ME) and in the estimates of its β for the preceding 5 years.

B.2. β Estimates

Table I shows that forming portfolios on size and pre-ranking β s, rather than on size alone, magnifies the range of full-period post-ranking β s. Sorted on size alone, the post-ranking β s range from 1.44 for the smallest ME portfolio to 0.92 for the largest. This spread of β s across the 10 size deciles is smaller than the spread of post-ranking β s produced by the β sort of *any* size decile. For example, the post-ranking β s for the 10 portfolios in the smallest size decile range from 1.05 to 1.79. Across all 100 size- β portfolios, the post-ranking β s range from 0.53 to 1.79, a spread 2.4 times the spread, 0.52, obtained with size portfolios alone.

Two other facts about the β s are important. First, in each size decile the post-ranking β s closely reproduce the ordering of the pre-ranking β s. We take this to be evidence that the pre-ranking β sort captures the ordering of true post-ranking β s. (The appendix gives more evidence on this important issue.) Second, the β sort is not a refined size sort. In any size decile, the average values of $\ln(\text{ME})$ are similar across the β -sorted portfolios. Thus the pre-ranking β sort achieves its goal. It produces strong variation in post-ranking β s that is unrelated to size. This is important in allowing our tests to distinguish between β and size effects in average returns.

II. β and Size

The Sharpe-Lintner-Black (SLB) model plays an important role in the way academics and practitioners think about risk and the relation between risk and expected return. We show next that when common stock portfolios are formed on size alone, there seems to be evidence for the model's central prediction: average return is positively related to β . The β s of size portfolios are, however, almost perfectly correlated with size, so tests on size portfolios are unable to disentangle β and size effects in average returns. Allowing for variation in β that is unrelated to size breaks the logjam, but at the expense of β . Thus, when we subdivide size portfolios on the basis of pre-ranking β s, we find a strong relation between average return and size, but no relation between average return and β .

A. Informal Tests

Table II shows post-ranking average returns for July 1963 to December 1990 for portfolios formed from one-dimensional sorts of stocks on size or β . The portfolios are formed at the end of June each year and their equal-weighted returns are calculated for the next 12 months. We use returns for July to June to match the returns in later tests that use the accounting data. When we sort on just size or 5-year pre-ranking β s, we form 12 portfolios. The middle 8 cover deciles of size or β . The 4 extreme portfolios (1A, 1B, 10A, and 10B) split the bottom and top deciles in half.

Table II shows that when portfolios are formed on size alone, we observe the familiar strong negative relation between size and average return (Banz (1981)), and a strong positive relation between average return and β . Average returns fall from 1.64% per month for the smallest ME portfolio to 0.90% for the largest. Post-ranking β s also decline across the 12 size portfolios, from 1.44 for portfolio 1A to 0.90 for portfolio 10B. Thus, a simple size sort seems to support the SLB prediction of a positive relation between β and average return. But the evidence is muddled by the tight relation between size and the β s of size portfolios.

The portfolios formed on the basis of the ranked market β s of stocks in Table II produce a wider range of β s (from 0.81 for portfolio 1A to 1.73 for 10B) than the portfolios formed on size. Unlike the size portfolios, the β -sorted portfolios do not support the SLB model. There is little spread in average returns across the β portfolios, and there is no obvious relation between β and average returns. For example, although the two extreme portfolios, 1A and 10B, have much different β s, they have nearly identical average returns (1.20% and 1.18% per month). These results for 1963–1990 confirm Reinganum's (1981) evidence that for β -sorted portfolios, there is no relation between average return and β during the 1964–1979 period.

The 100 portfolios formed on size and then pre-ranking β in Table I clarify the contradictory evidence on the relation between β and average return produced by portfolios formed on size or β alone. Specifically, the two-pass sort gives a clearer picture of the separate roles of size and β in average returns. Contrary to the central prediction of the SLB model, the second-pass β sort produces little variation in average returns. Although the post-ranking β s in Table I increase strongly in each size decile, average returns are flat or show a slight tendency to decline. In contrast, within the columns of the average return and β matrices of Table I, average returns and β s decrease with increasing size.

The two-pass sort on size and β in Table I says that variation in β that is tied to size is positively related to average return, but variation in β unrelated to size is not compensated in the average returns of 1963–1990. The proper inference seems to be that there is a relation between size and average return, but controlling for size, there is no relation between β and average return. The regressions that follow confirm this conclusion, and they produce another that is stronger. The regressions show that when one allows

Table I
Average Returns, Post-Ranking β s and Average Size For Portfolios Formed on
Size and then β : Stocks Sorted on ME (Down) then Pre-Ranking β (Across):
July 1963 to December 1990

Portfolios are formed yearly. The breakpoints for the size (ME, price times shares outstanding) deciles are determined in June of year t ($t = 1963-1990$) using all NYSE stocks on CRSP. All NYSE, AMEX, and NASDAQ stocks that meet the CRSP-COMPUSTAT data requirements are allocated to the 10 size portfolios using the NYSE breakpoints. Each size decile is subdivided into 10 β portfolios using pre-ranking β s of individual stocks, estimated with 2 to 5 years of monthly returns (as available) ending in June of year t . We use only NYSE stocks that meet the CRSP-COMPUSTAT data requirements to establish the β breakpoints. The equal-weighted monthly returns on the resulting 100 portfolios are then calculated for July of year t to June of year $t + 1$.

The post-ranking β s use the full (July 1963 to December 1990) sample of post-ranking returns for each portfolio. The pre- and post-ranking β s (here and in all other tables) are the sum of the slopes from a regression of monthly returns on the current and prior month's returns on the value-weighted portfolio of NYSE, AMEX, and (after 1972) NASDAQ stocks. The average return is the time-series average of the monthly equal-weighted portfolio returns, in percent. The average size of a portfolio is the time-series average of monthly averages of $\ln(\text{ME})$ for stocks in the portfolio at the end of June of each year, with ME denominated in millions of dollars.

The average number of stocks per month for the size- β portfolios in the smallest size decile varies from 70 to 177. The average number of stocks for the size- β portfolios in size deciles 2 and 3 is between 15 and 41, and the average number for the largest 7 size deciles is between 11 and 22.

The All column shows statistics for equal-weighted size-decile (ME) portfolios. The All row shows statistics for equal-weighted portfolios of the stocks in each β group.

	Panel A: Average Monthly Returns (in Percent)									
	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	High- β
All	1.25	1.34	1.29	1.36	1.31	1.33	1.28	1.24	1.21	1.14
Small-ME	1.52	1.71	1.57	1.79	1.61	1.50	1.50	1.37	1.63	1.42
ME-2	1.29	1.25	1.42	1.36	1.39	1.65	1.61	1.37	1.31	1.34
ME-3	1.24	1.12	1.31	1.17	1.70	1.29	1.10	1.31	1.36	1.26
ME-4	1.25	1.27	1.13	1.54	1.06	1.34	1.06	1.41	1.17	1.35
ME-5	1.29	1.34	1.42	1.39	1.48	1.42	1.18	1.13	1.27	1.18
ME-6	1.17	1.08	1.53	1.27	1.15	1.20	1.21	1.18	1.04	1.07
ME-7	1.07	0.95	1.21	1.26	1.09	1.18	1.11	1.24	0.62	1.32
ME-8	1.10	1.09	1.05	1.37	1.20	1.27	0.98	1.18	1.02	1.01
ME-9	0.95	0.98	0.88	1.02	1.14	1.07	1.23	0.94	0.82	0.59
Large-ME	0.89	1.01	0.93	1.10	0.94	0.93	0.89	1.03	0.71	0.74

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Table I – Continued

	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	β -9	High- β
	Panel B: Post-Ranking β s										
All		0.87	0.99	1.09	1.16	1.26	1.29	1.35	1.45	1.52	1.72
Small-ME	1.44	1.05	1.18	1.28	1.32	1.40	1.40	1.49	1.61	1.64	1.79
ME-2	1.39	0.91	1.15	1.17	1.24	1.36	1.41	1.43	1.50	1.66	1.76
ME-3	1.35	0.97	1.13	1.13	1.21	1.26	1.28	1.39	1.50	1.51	1.75
ME-4	1.34	0.78	1.03	1.17	1.16	1.29	1.37	1.46	1.51	1.64	1.71
ME-5	1.25	0.66	0.85	1.12	1.15	1.16	1.26	1.30	1.43	1.59	1.68
ME-6	1.23	0.61	0.78	1.05	1.16	1.22	1.28	1.36	1.46	1.49	1.70
ME-7	1.17	0.57	0.92	1.01	1.11	1.14	1.26	1.24	1.39	1.34	1.60
ME-8	1.09	0.53	0.74	0.94	1.02	1.13	1.12	1.18	1.26	1.35	1.52
ME-9	1.03	0.58	0.74	0.80	0.95	1.06	1.15	1.14	1.21	1.22	1.42
Large-ME	0.92	0.57	0.71	0.78	0.89	0.95	0.92	1.02	1.01	1.11	1.32
	Panel C: Average Size (ln(ME))										
All	4.11	3.86	4.26	4.33	4.41	4.27	4.32	4.26	4.19	4.03	3.77
Small-ME	2.24	2.12	2.27	2.30	2.30	2.28	2.29	2.30	2.32	2.25	2.15
ME-2	3.63	3.65	3.68	3.70	3.72	3.69	3.70	3.69	3.69	3.70	3.68
ME-3	4.10	4.14	4.18	4.12	4.15	4.16	4.16	4.18	4.14	4.15	4.15
ME-4	4.50	4.53	4.53	4.57	4.54	4.56	4.52	4.52	4.58	4.52	4.56
ME-5	4.89	4.91	4.91	4.93	4.95	4.93	4.92	4.93	4.92	4.92	4.95
ME-6	5.30	5.30	5.33	5.34	5.34	5.33	5.33	5.33	5.33	5.34	5.36
ME-7	5.73	5.73	5.75	5.77	5.76	5.73	5.77	5.77	5.76	5.72	5.76
ME-8	6.24	6.26	6.27	6.26	6.24	6.24	6.27	6.24	6.24	6.24	6.26
ME-9	6.82	6.82	6.84	6.82	6.82	6.81	6.81	6.81	6.81	6.80	6.83
Large-ME	7.93	7.94	8.04	8.10	8.04	8.02	8.02	7.94	7.80	7.75	7.62

Table II
Properties of Portfolios Formed on Size or Pre-Ranking β :
July 1963 to December 1990

At the end of June of each year t , 12 portfolios are formed on the basis of ranked values of size (ME) or pre-ranking β . The pre-ranking β s use 2 to 5 years (as available) of monthly returns ending in June of t . Portfolios 2–9 cover deciles of the ranking variables. The bottom and top 2 portfolios (1A, 1B, 10A, and 10B) split the bottom and top deciles in half. The breakpoints for the ME portfolios are based on ranked values of ME for all NYSE stocks on CRSP. NYSE breakpoints for pre-ranking β s are also used to form the β portfolios. NYSE, AMEX, and NASDAQ stocks are then allocated to the size or β portfolios using the NYSE breakpoints. We calculate each portfolio's monthly equal-weighted return for July of year t to June of year $t + 1$, and then reform the portfolios in June of $t + 1$.

BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t , with ME denominated in millions of dollars.

The average return is the time-series average of the monthly equal-weighted portfolio returns, in percent. $\ln(\text{ME})$, $\ln(\text{BE}/\text{ME})$, $\ln(\text{A}/\text{BE})$, E/P, and E/P dummy are the time-series averages of the monthly average values of these variables in each portfolio. Since the E/P dummy is 0 when earnings are positive, and 1 when earnings are negative, E/P dummy gives the average proportion of stocks with negative earnings in each portfolio.

β is the time-series average of the monthly portfolio β s. Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). These individual firm β s are averaged to compute the monthly β s for each portfolio for July of year t to June of year $t + 1$.

Firms is the average number of stocks in the portfolio each month.

	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel A: Portfolios Formed on Size												
Return	1.64	1.16	1.29	1.24	1.25	1.29	1.17	1.07	1.10	0.95	0.88	0.90
β	1.44	1.44	1.39	1.34	1.33	1.24	1.22	1.16	1.08	1.02	0.95	0.90
$\ln(\text{ME})$	1.98	3.18	3.63	4.10	4.50	4.89	5.30	5.73	6.24	6.82	7.39	8.44
$\ln(\text{BE}/\text{ME})$	-0.01	-0.21	-0.23	-0.26	-0.32	-0.36	-0.36	-0.44	-0.40	-0.42	-0.51	-0.65
$\ln(\text{A}/\text{BE})$	0.73	0.50	0.46	0.43	0.37	0.32	0.32	0.24	0.29	0.27	0.17	-0.03
$\ln(\text{A}/\text{BE})$	0.75	0.71	0.69	0.69	0.68	0.67	0.68	0.67	0.69	0.70	0.68	0.62
E/P dummy	0.26	0.14	0.11	0.09	0.06	0.04	0.04	0.03	0.03	0.02	0.02	0.01
E(+)/P	0.09	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.09	0.09
Firms	772	189	236	170	144	140	128	125	119	114	60	64

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Table II—Continued

	1A	1B	2	3	4	5	6	7	8	9	10A	10B
	Panel B: Portfolios Formed on Pre-Ranking β											
Return	1.20	1.20	1.32	1.26	1.31	1.30	1.30	1.23	1.23	1.33	1.34	1.18
β	0.81	0.79	0.92	1.04	1.13	1.19	1.26	1.32	1.41	1.52	1.63	1.73
ln(ME)	4.21	4.86	4.75	4.68	4.59	4.48	4.36	4.25	3.97	3.78	3.52	3.15
ln(BE/ME)	-0.18	-0.13	-0.22	-0.21	-0.23	-0.22	-0.22	-0.25	-0.23	-0.27	-0.31	-0.50
ln(A/ME)	0.60	0.66	0.49	0.45	0.42	0.42	0.45	0.42	0.47	0.46	0.46	0.31
ln(A/BE)	0.78	0.79	0.71	0.66	0.64	0.65	0.67	0.67	0.70	0.73	0.77	0.81
E/P dummy	0.12	0.06	0.09	0.09	0.08	0.09	0.10	0.12	0.12	0.14	0.17	0.23
E(+)/P	0.11	0.12	0.10	0.10	0.10	0.10	0.10	0.09	0.10	0.09	0.09	0.08
Firms	116	80	185	181	179	182	185	205	227	267	165	291

for variation in β that is unrelated to size, the relation between β and average return is flat, even when β is the only explanatory variable.

B. Fama-MacBeth Regressions

Table III shows time-series averages of the slopes from the month-by-month Fama-MacBeth (FM) regressions of the cross-section of stock returns on size, β , and the other variables (leverage, E/P, and book-to-market equity) used to explain average returns. The average slopes provide standard FM tests for determining which explanatory variables on average have non-zero expected premiums during the July 1963 to December 1990 period.

Like the average returns in Tables I and II, the regressions in Table III say that size, $\ln(\text{ME})$, helps explain the cross-section of average stock returns. The average slope from the monthly regressions of returns on size alone is -0.15% , with a t -statistic of -2.58 . This reliable negative relation persists no matter which other explanatory variables are in the regressions; the average slopes on $\ln(\text{ME})$ are always close to or more than 2 standard errors from 0. The size effect (smaller stocks have higher average returns) is thus robust in the 1963–1990 returns on NYSE, AMEX, and NASDAQ stocks.

In contrast to the consistent explanatory power of size, the FM regressions show that market β does not help explain average stock returns for 1963–1990. In a shot straight at the heart of the SLB model, the average slope from the regressions of returns on β alone in Table III is 0.15% per month and only 0.46 standard errors from 0. In the regressions of returns on size and β , size has explanatory power (an average slope -3.41 standard errors from 0), but the average slope for β is negative and only 1.21 standard errors from 0. Lakonishok and Shapiro (1986) get similar results for NYSE stocks for 1962–1981. We can also report that β shows no power to explain average returns (the average slopes are typically less than 1 standard error from 0) in FM regressions that use various combinations of β with size, book-to-market equity, leverage, and E/P.

C. Can β Be Saved?

What explains the poor results for β ? One possibility is that other explanatory variables are correlated with true β s, and this obscures the relation between average returns and measured β s. But this line of attack cannot explain why β has no power when used alone to explain average returns. Moreover, leverage, book-to-market equity, and E/P do not seem to be good proxies for β . The averages of the monthly cross-sectional correlations between β and the values of these variables for individual stocks are all within 0.15 of 0.

Another hypothesis is that, as predicted by the SLB model, there is a positive relation between β and average return, but the relation is obscured by noise in the β estimates. However, our full-period post-ranking β s do not seem to be imprecise. Most of the standard errors of the β s (not shown) are

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Table III
Average Slopes (*t*-Statistics) from Month-by-Month Regressions of
Stock Returns on β , Size, Book-to-Market Equity, Leverage, and E/P:
July 1963 to December 1990

Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t . In the regressions, these values of the explanatory variables for individual stocks are matched with CRSP returns for the months from July of year t to June of year $t + 1$. The gap between the accounting data and the returns ensures that the accounting data are available prior to the returns. If earnings are positive, $E(+)/P$ is the ratio of total earnings to market equity and E/P dummy is 0. If earnings are negative, $E(+)/P$ is 0 and E/P dummy is 1.

The average slope is the time-series average of the monthly regression slopes for July 1963 to December 1990, and the t -statistic is the average slope divided by its time-series standard error.

On average, there are 2267 stocks in the monthly regressions. To avoid giving extreme observations heavy weight in the regressions, the smallest and largest 0.5% of the observations on $E(+)/P$, BE/ME, A/ME, and A/BE are set equal to the next largest or smallest values of the ratios (the 0.005 and 0.995 fractiles). This has no effect on inferences.

β	$\ln(\text{ME})$	$\ln(\text{BE}/\text{ME})$	$\ln(\text{A}/\text{ME})$	$\ln(\text{A}/\text{BE})$	E/P Dummy	$E(+)/P$
0.15 (0.46)						
	-0.15 (-2.58)					
-0.37 (-1.21)	-0.17 (-3.41)					
		0.50 (5.71)				
			0.50 (5.69)	-0.57 (-5.34)		
					0.57 (2.28)	4.72 (4.57)
	-0.11 (-1.99)	0.35 (4.44)				
	-0.11 (-2.06)		0.35 (4.32)	-0.50 (-4.56)		
	-0.16 (-3.06)				0.06 (0.38)	2.99 (3.04)
	-0.13 (-2.47)	0.33 (4.46)			-0.14 (-0.90)	0.87 (1.23)
	-0.13 (-2.47)		0.32 (4.28)	-0.46 (-4.45)	-0.08 (-0.56)	1.15 (1.57)

0.05 or less, only 1 is greater than 0.1, and the standard errors are small relative to the range of the β s (0.53 to 1.79).

The β -sorted portfolios in Tables I and II also provide strong evidence against the β -measurement-error story. When portfolios are formed on pre-ranking β s alone (Table II), the post-ranking β s for the portfolios almost perfectly reproduce the ordering of the pre-ranking β s. Only the β for portfolio 1B is out of line, and only by 0.02. Similarly, when portfolios are formed on size and then pre-ranking β s (Table I), the post-ranking β s in each size decile closely reproduce the ordering of the pre-ranking β s.

The correspondence between the ordering of the pre-ranking and post-ranking β s for the β -sorted portfolios in Tables I and II is evidence that the post-ranking β s are informative about the ordering of the true β s. The problem for the SLB model is that there is no similar ordering in the average returns on the β -sorted portfolios. Whether one looks at portfolios sorted on β alone (Table II) or on size and then β (Table I), average returns are flat (Table II) or decline slightly (Table I) as the post-ranking β s increase.

Our evidence on the robustness of the size effect and the absence of a relation between β and average return is so contrary to the SLB model that it behooves us to examine whether the results are special to 1963–1990. The appendix shows that NYSE returns for 1941–1990 behave like the NYSE, AMEX, and NASDAQ returns for 1963–1990; there is a reliable size effect over the full 50-year period, but little relation between β and average return. Interestingly, there is a reliable simple relation between β and average return during the 1941–1965 period. These 25 years are a major part of the samples in the early studies of the SLB model of Black, Jensen, and Scholes (1972) and Fama and MacBeth (1973). Even for the 1941–1965 period, however, the relation between β and average return disappears when we control for size.

III. Book-to-Market Equity, E/P, and Leverage

Tables I to III say that there is a strong relation between the average returns on stocks and size, but there is no reliable relation between average returns and β . In this section we show that there is also a strong cross-sectional relation between average returns and book-to-market equity. If anything, this book-to-market effect is more powerful than the size effect. We also find that the combination of size and book-to-market equity absorbs the apparent roles of leverage and E/P in average stock returns.

A. Average Returns

Table IV shows average returns for July 1963 to December 1990 for portfolios formed on ranked values of book-to-market equity (BE/ME) or earnings-price ratio (E/P). The BE/ME and E/P portfolios in Table IV are formed in the same general way (one-dimensional yearly sorts) as the size and β portfolios in Table II. (See the tables for details.)

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The relation between average return and E/P has a familiar U-shape (e.g., Jaffe, Keim, and Westerfield (1989) for U.S. data, and Chan, Hamao, and Lakonishok (1991) for Japan). Average returns decline from 1.46% per month for the negative E/P portfolio to 0.93% for the firms in portfolio 1B that have low but positive E/P. Average returns then increase monotonically, reaching 1.72% per month for the highest E/P portfolio.

The more striking evidence in Table IV is the strong positive relation between average return and book-to-market equity. Average returns rise from 0.30% for the lowest BE/ME portfolio to 1.83% for the highest, a difference of 1.53% per month. This spread is twice as large as the difference of 0.74% between the average monthly returns on the smallest and largest size portfolios in Table II. Note also that the strong relation between book-to-market equity and average return is unlikely to be a β effect in disguise; Table IV shows that post-ranking market β s vary little across portfolios formed on ranked values of BE/ME.

On average, only about 50 (out of 2317) firms per year have negative book equity, BE. The negative BE firms are mostly concentrated in the last 14 years of the sample, 1976–1989, and we do not include them in the tests. We can report, however, that average returns for negative BE firms are high, like the average returns of high BE/ME firms. Negative BE (which results from persistently negative earnings) and high BE/ME (which typically means that stock prices have fallen) are both signals of poor earning prospects. The similar average returns of negative and high BE/ME firms are thus consistent with the hypothesis that book-to-market equity captures cross-sectional variation in average returns that is related to relative distress.

B. Fama-MacBeth Regressions

B.1. BE/ME

The FM regressions in Table III confirm the importance of book-to-market equity in explaining the cross-section of average stock returns. The average slope from the monthly regressions of returns on $\ln(\text{BE/ME})$ alone is 0.50%, with a t -statistic of 5.71. This book-to-market relation is stronger than the size effect, which produces a t -statistic of -2.58 in the regressions of returns on $\ln(\text{ME})$ alone. But book-to-market equity does not replace size in explaining average returns. When both $\ln(\text{ME})$ and $\ln(\text{BE/ME})$ are included in the regressions, the average size slope is still -1.99 standard errors from 0; the book-to-market slope is an impressive 4.44 standard errors from 0.

B.2. Leverage

The FM regressions that explain returns with leverage variables provide interesting insight into the relation between book-to-market equity and average return. We use two leverage variables, the ratio of book assets to market equity, A/ME, and the ratio of book assets to book equity, A/BE. We interpret A/ME as a measure of market leverage, while A/BE is a measure

Table IV
**Properties of Portfolios Formed on Book-to-Market Equity (BE/ME) and Earnings-Price Ratio (E/P):
July 1963 to December 1990**

At the end of each year $t - 1$, 12 portfolios are formed on the basis of ranked values of BE/ME or E/P. Portfolios 2-9 cover deciles of the ranking variables. The bottom and top 2 portfolios (1A, 1B, 10A, and 10B) split the bottom and top deciles in half. For E/P, there are 13 portfolios; portfolio 0 is stocks with negative E/P. Since BE/ME and E/P are not strongly related to exchange listing, their portfolio breakpoints are determined on the basis of the ranked values of the variables for all stocks that satisfy the CRSP-COMPUSTAT data requirements. BE is the book value of common equity plus balance-sheet deferred taxes, A is total book assets, and E is earnings (income before extraordinary items, plus income-statement deferred taxes, minus preferred dividends). BE, A, and E are for each firm's latest fiscal year ending in calendar year $t - 1$. The accounting ratios are measured using market equity ME in December of year $t - 1$. Firm size $\ln(\text{ME})$ is measured in June of year t , with ME denominated in millions of dollars. We calculate each portfolio's monthly equal-weighted return for July of year t to June of year $t + 1$, and then reform the portfolios at the end of year t .

Return is the time-series average of the monthly equal-weighted portfolio returns (in percent). $\ln(\text{ME})$, $\ln(\text{BE}/\text{ME})$, $\ln(\text{A}/\text{ME})$, $\ln(\text{A}/\text{BE})$, $\text{E}(+)/\text{P}$, and E/P dummy are the time-series averages of the monthly average values of these variables in each portfolio. Since the E/P dummy is 0 when earnings are positive, and 1 when earnings are negative, E/P dummy gives the average proportion of stocks with negative earnings in each portfolio.

β is the time-series average of the monthly portfolio β s. Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of June of year t (Table I). These individual-firm β s are averaged to compute the monthly β s for each portfolio for July of year t to June of year $t + 1$. Firms is the average number of stocks in the portfolio each month.

Portfolio	0	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel A: Stocks Sorted on Book-to-Market Equity (BE/ME)													
Return	0.30	0.67	0.87	0.87	0.97	1.04	1.17	1.30	1.44	1.50	1.59	1.92	1.83
β	1.36	1.34	1.32	1.30	1.30	1.28	1.27	1.27	1.27	1.27	1.29	1.33	1.35
$\ln(\text{ME})$	4.53	4.67	4.69	4.56	4.47	4.47	4.38	4.23	4.06	3.85	3.51	3.06	2.65
$\ln(\text{BE}/\text{ME})$	-2.22	-1.51	-1.09	-0.75	-0.51	-0.51	-0.32	-0.14	0.03	0.21	0.42	0.66	1.02
$\ln(\text{A}/\text{ME})$	-1.24	-0.79	-0.40	-0.05	0.20	0.20	0.40	0.56	0.71	0.91	1.12	1.35	1.75
$\ln(\text{A}/\text{BE})$	0.94	0.71	0.68	0.70	0.71	0.71	0.71	0.70	0.68	0.70	0.70	0.70	0.73
E/P dummy	0.29	0.15	0.10	0.08	0.08	0.08	0.08	0.09	0.09	0.11	0.15	0.22	0.36
$\text{E}(+)/\text{P}$	0.03	0.04	0.06	0.08	0.08	0.09	0.10	0.11	0.11	0.12	0.12	0.11	0.10
Firms	89	98	209	222	226	226	230	235	237	239	239	120	117

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Table IV—Continued

Portfolio	0	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Panel B: Stocks Sorted on Earnings-Price Ratio (E/P)													
Return	1.46	1.04	0.93	0.94	1.03	1.18	1.22	1.33	1.42	1.46	1.57	1.74	1.72
β	1.47	1.40	1.35	1.31	1.28	1.26	1.25	1.26	1.24	1.23	1.24	1.28	1.31
$\ln(\text{ME})$	2.48	3.64	4.33	4.61	4.64	4.63	4.58	4.49	4.37	4.28	4.07	3.82	3.52
$\ln(\text{BE}/\text{ME})$	-0.10	-0.76	-0.91	-0.79	-0.61	-0.47	-0.33	-0.21	-0.08	0.02	0.15	0.26	0.40
$\ln(\text{A}/\text{ME})$	0.90	-0.05	-0.27	-0.16	0.03	0.18	0.31	0.44	0.58	0.70	0.85	1.01	1.25
$\ln(\text{A}/\text{BE})$	0.99	0.70	0.63	0.63	0.64	0.65	0.64	0.65	0.66	0.68	0.71	0.75	0.86
E/P dummy	1.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
E(+)/P	0.00	0.01	0.03	0.05	0.06	0.08	0.09	0.11	0.12	0.14	0.16	0.20	0.28
Firms	355	88	90	182	190	193	196	194	197	195	195	95	91

of book leverage. The regressions use the natural logs of the leverage ratios, $\ln(A/ME)$ and $\ln(A/BE)$, because preliminary tests indicated that logs are a good functional form for capturing leverage effects in average returns. Using logs also leads to a simple interpretation of the relation between the roles of leverage and book-to-market equity in average returns.

The FM regressions of returns on the leverage variables (Table III) pose a bit of a puzzle. The two leverage variables are related to average returns, but with opposite signs. As in Bhandari (1988), higher market leverage is associated with higher average returns; the average slopes for $\ln(A/ME)$ are always positive and more than 4 standard errors from 0. But higher book leverage is associated with lower average returns; the average slopes for $\ln(A/BE)$ are always negative and more than 4 standard errors from 0.

The puzzle of the opposite slopes on $\ln(A/ME)$ and $\ln(A/BE)$ has a simple solution. The average slopes for the two leverage variables are opposite in sign but close in absolute value, e.g., 0.50 and -0.57 . Thus it is the difference between market and book leverage that helps explain average returns. But the difference between market and book leverage is book-to-market equity, $\ln(BE/ME) = \ln(A/ME) - \ln(A/BE)$. Table III shows that the average book-to-market slopes in the FM regressions are indeed close in absolute value to the slopes for the two leverage variables.

The close links between the leverage and book-to-market results suggest that there are two equivalent ways to interpret the book-to-market effect in average returns. A high ratio of book equity to market equity (a low stock price relative to book value) says that the market judges the prospects of a firm to be poor relative to firms with low BE/ME . Thus BE/ME may capture the relative-distress effect postulated by Chan and Chen (1991). A high book-to-market ratio also says that a firm's market leverage is high relative to its book leverage; the firm has a large amount of market-imposed leverage because the market judges that its prospects are poor and discounts its stock price relative to book value. In short, our tests suggest that the relative-distress effect, captured by BE/ME , can also be interpreted as an involuntary leverage effect, which is captured by the difference between A/ME and A/BE .

B.3. E/P

Ball (1978) posits that the earnings-price ratio is a catch-all for omitted risk factors in expected returns. If current earnings proxy for expected future earnings, high-risk stocks with high expected returns will have low prices relative to their earnings. Thus, E/P should be related to expected returns, whatever the omitted sources of risk. This argument only makes sense, however, for firms with positive earnings. When current earnings are negative, they are not a proxy for the earnings forecasts embedded in the stock price, and E/P is not a proxy for expected returns. Thus, the slope for E/P in the FM regressions is based on positive values; we use a dummy variable for E/P when earnings are negative.

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The U-shaped relation between average return and E/P observed in Table IV is also apparent when the E/P variables are used alone in the FM regressions in Table III. The average slope on the E/P dummy variable (0.57% per month, 2.28 standard errors from 0) confirms that firms with negative earnings have higher average returns. The average slope for stocks with positive E/P (4.72% per month, 4.57 standard errors from 0) shows that average returns increase with E/P when it is positive.

Adding size to the regressions kills the explanatory power of the E/P dummy. Thus the high average returns of negative E/P stocks are better captured by their size, which Table IV says is on average small. Adding both size and book-to-market equity to the E/P regressions kills the E/P dummy and lowers the average slope on E/P from 4.72 to 0.87 ($t = 1.23$). In contrast, the average slopes for $\ln(\text{ME})$ and $\ln(\text{BE}/\text{ME})$ in the regressions that include E/P are similar to those in the regressions that explain average returns with only size and book-to-market equity. The results suggest that most of the relation between (positive) E/P and average return is due to the positive correlation between E/P and $\ln(\text{BE}/\text{ME})$, illustrated in Table IV; firms with high E/P tend to have high book-to-market equity ratios.

IV. A Parsimonious Model for Average Returns

The results to here are easily summarized:

- (1) When we allow for variation in β that is unrelated to size, there is no reliable relation between β and average return.
- (2) The opposite roles of market leverage and book leverage in average returns are captured well by book-to-market equity.
- (3) The relation between E/P and average return seems to be absorbed by the combination of size and book-to-market equity.

In a nutshell, market β seems to have no role in explaining the average returns on NYSE, AMEX, and NASDAQ stocks for 1963–1990, while size and book-to-market equity capture the cross-sectional variation in average stock returns that is related to leverage and E/P.

A. Average Returns, Size and Book-to-Market Equity

The average return matrix in Table V gives a simple picture of the two-dimensional variation in average returns that results when the 10 size deciles are each subdivided into 10 portfolios based on ranked values of BE/ME for individual stocks. Within a size decile (across a row of the average return matrix), returns typically increase strongly with BE/ME: on average, the returns on the lowest and highest BE/ME portfolios in a size decile differ by 0.99% (1.63% – 0.64%) per month. Similarly, looking down the columns of the average return matrix shows that there is a negative relation between average return and size: on average, the spread of returns across the size portfolios in a BE/ME group is 0.58% per month. The average return matrix gives life to the conclusion from the regressions that,

Table V
Average Monthly Returns on Portfolios Formed on Size and
Book-to-Market Equity; Stocks Sorted by ME (Down) and then
BE/ME (Across); July 1963 to December 1990

In June of each year t , the NYSE, AMEX, and NASDAQ stocks that meet the CRSP-COMPUSTAT data requirements are allocated to 10 size portfolios using the NYSE size (ME) breakpoints. The NYSE, AMEX, and NASDAQ stocks in each size decile are then sorted into 10 BE/ME portfolios using the book-to-market ratios for year $t - 1$. BE/ME is the book value of common equity plus balance-sheet deferred taxes for fiscal year $t - 1$, over market equity for December of year $t - 1$. The equal-weighted monthly portfolio returns are then calculated for July of year t to June of year $t + 1$.

Average monthly return is the time-series average of the monthly equal-weighted portfolio returns (in percent).

The All column shows average returns for equal-weighted size decile portfolios. The All row shows average returns for equal-weighted portfolios of the stocks in each BE/ME group.

	Book-to-Market Portfolios										
	All	Low	2	3	4	5	6	7	8	9	High
All	1.23	0.64	0.98	1.06	1.17	1.24	1.26	1.39	1.40	1.50	1.63
Small-ME	1.47	0.70	1.14	1.20	1.43	1.56	1.51	1.70	1.71	1.82	1.92
ME-2	1.22	0.43	1.05	0.96	1.19	1.33	1.19	1.58	1.28	1.43	1.79
ME-3	1.22	0.56	0.88	1.23	0.95	1.36	1.30	1.30	1.40	1.54	1.60
ME-4	1.19	0.39	0.72	1.06	1.36	1.13	1.21	1.34	1.59	1.51	1.47
ME-5	1.24	0.88	0.65	1.08	1.47	1.13	1.43	1.44	1.26	1.52	1.49
ME-6	1.15	0.70	0.98	1.14	1.23	0.94	1.27	1.19	1.19	1.24	1.50
ME-7	1.07	0.95	1.00	0.99	0.83	0.99	1.13	0.99	1.16	1.10	1.47
ME-8	1.08	0.66	1.13	0.91	0.95	0.99	1.01	1.15	1.05	1.29	1.55
ME-9	0.95	0.44	0.89	0.92	1.00	1.05	0.93	0.82	1.11	1.04	1.22
Large-ME	0.89	0.93	0.88	0.84	0.71	0.79	0.83	0.81	0.96	0.97	1.18

controlling for size, book-to-market equity captures strong variation in average returns, and controlling for book-to-market equity leaves a size effect in average returns.

B. The Interaction between Size and Book-to-Market Equity

The average of the monthly correlations between the cross-sections of $\ln(\text{ME})$ and $\ln(\text{BE/ME})$ for individual stocks is -0.26 . The negative correlation is also apparent in the average values of $\ln(\text{ME})$ and $\ln(\text{BE/ME})$ for the portfolios sorted on ME or BE/ME in Tables II and IV. Thus, firms with low market equity are more likely to have poor prospects, resulting in low stock prices and high book-to-market equity. Conversely, large stocks are more likely to be firms with stronger prospects, higher stock prices, lower book-to-market equity, and lower average stock returns.

The correlation between size and book-to-market equity affects the regressions in Table III. Including $\ln(\text{BE/ME})$ moves the average slope on $\ln(\text{ME})$ from -0.15 ($t = -2.58$) in the univariate regressions to -0.11 ($t = -1.99$) in the bivariate regressions. Similarly, including $\ln(\text{ME})$ in the regressions

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lowers the average slope on $\ln(\text{BE}/\text{ME})$ from 0.50 to 0.35 (still a healthy 4.44 standard errors from 0). Thus, part of the size effect in the simple regressions is due to the fact that small ME stocks are more likely to have high book-to-market ratios, and part of the simple book-to-market effect is due to the fact that high BE/ME stocks tend to be small (they have low ME).

We should not, however, exaggerate the links between size and book-to-market equity. The correlation (-0.26) between $\ln(\text{ME})$ and $\ln(\text{BE}/\text{ME})$ is not extreme, and the average slopes in the bivariate regressions in Table III show that $\ln(\text{ME})$ and $\ln(\text{BE}/\text{ME})$ are both needed to explain the cross-section of average returns. Finally, the 10×10 average return matrix in Table V provides concrete evidence that, (a) controlling for size, book-to-market equity captures substantial variation in the cross-section of average returns, and (b) within BE/ME groups average returns are related to size.

C. Subperiod Averages of the FM Slopes

The message from the average FM slopes for 1963–1990 (Table III) is that size on average has a negative premium in the cross-section of stock returns, book-to-market equity has a positive premium, and the average premium for market β is essentially 0. Table VI shows the average FM slopes for two roughly equal subperiods (July 1963–December 1976 and January 1977–December 1990) from two regressions: (a) the cross-section of stock returns on size, $\ln(\text{ME})$, and book-to-market equity, $\ln(\text{BE}/\text{ME})$, and (b) returns on β , $\ln(\text{ME})$, and $\ln(\text{BE}/\text{ME})$. For perspective, average returns on the value-weighted and equal-weighted (VW and EW) portfolios of NYSE stocks are also shown.

In FM regressions, the intercept is the return on a standard portfolio (the weights on stocks sum to 1) in which the weighted averages of the explanatory variables are 0 (Fama (1976), chapter 9). In our tests, the intercept is weighted toward small stocks (ME is in millions of dollars so $\ln(\text{ME}) = 0$ implies $\text{ME} = \$1$ million) and toward stocks with relatively high book-to-market ratios (Table IV says that $\ln(\text{BE}/\text{ME})$ is negative for the typical firm, so $\ln(\text{BE}/\text{ME}) = 0$ is toward the high end of the sample ratios). Thus it is not surprising that the average intercepts are always large relative to their standard errors and relative to the returns on the NYSE VW and EW portfolios.

Like the overall period, the subperiods do not offer much hope that the average premium for β is economically important. The average FM slope for β is only slightly positive for 1963–1976 (0.10% per month, $t = 0.25$), and it is negative for 1977–1990 (-0.44% per month, $t = -1.17$). There is a hint that the size effect is weaker in the 1977–1990 period, but inferences about the average size slopes for the subperiods lack power.

Unlike the size effect, the relation between book-to-market equity and average return is so strong that it shows up reliably in both the 1963–1976 and the 1977–1990 subperiods. The average slopes for $\ln(\text{BE}/\text{ME})$ are all more than 2.95 standard errors from 0, and the average slopes for the

Table VI
Subperiod Average Monthly Returns on the NYSE
Equal-Weighted and Value-Weighted Portfolios and Subperiod
Means of the Intercepts and Slopes from the Monthly FM
Cross-Sectional Regressions of Returns on (a) Size ($\ln(\text{ME})$) and
Book-to-Market Equity ($\ln(\text{BE}/\text{ME})$), and (b) β , $\ln(\text{ME})$, and
 $\ln(\text{BE}/\text{ME})$

Mean is the time-series mean of a monthly return, Std is its time-series standard deviation, and $t(\text{Mn})$ is Mean divided by its time-series standard error.

Variable	7/63-12/90 (330 Mos.)			7/63-12/76 (162 Mos.)			1/77-12/90 (168 Mos.)		
	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$
NYSE Value-Weighted (VW) and Equal-Weighted (EW) Portfolio Returns									
VW	0.81	4.47	3.27	0.56	4.26	1.67	1.04	4.66	2.89
EW	0.97	5.49	3.19	0.77	5.70	1.72	1.15	5.28	2.82
$R_{it} = a + b_{2t}\ln(\text{ME}_{it}) + b_{3t}\ln(\text{BE}/\text{ME}_{it}) + e_{it}$									
a	1.77	8.51	3.77	1.86	10.10	2.33	1.69	6.67	3.27
b_2	-0.11	1.02	-1.99	-0.16	1.25	-1.62	-0.07	0.73	-1.16
b_3	0.35	1.45	4.43	0.36	1.53	2.96	0.35	1.37	3.30
$R_{it} = a + b_{1t}\beta_{it} + b_{2t}\ln(\text{ME}_{it}) + b_{3t}\ln(\text{BE}/\text{ME}_{it}) + e_{it}$									
a	2.07	5.75	6.55	1.73	6.22	3.54	2.40	5.25	5.92
b_1	-0.17	5.12	-0.62	0.10	5.33	0.25	-0.44	4.91	-1.17
b_2	-0.12	0.89	-2.52	-0.15	1.03	-1.91	-0.09	0.74	-1.64
b_3	0.33	1.24	4.80	0.34	1.36	3.17	0.31	1.10	3.67

subperiods (0.36 and 0.35) are close to the average slope (0.35) for the overall period. The subperiod results thus support the conclusion that, among the variables considered here, book-to-market equity is consistently the most powerful for explaining the cross-section of average stock returns.

Finally, Roll (1983) and Keim (1983) show that the size effect is stronger in January. We have examined the monthly slopes from the FM regressions in Table VI for evidence of a January seasonal in the relation between book-to-market equity and average return. The average January slopes for $\ln(\text{BE}/\text{ME})$ are about twice those for February to December. Unlike the size effect, however, the strong relation between book-to-market equity and average return is not special to January. The average monthly February-to-December slopes for $\ln(\text{BE}/\text{ME})$ are about 4 standard errors from 0, and they are close to (within 0.05 of) the average slopes for the whole year. Thus, there is a January seasonal in the book-to-market equity effect, but the positive relation between BE/ME and average return is strong throughout the year.

D. β and the Market Factor: Caveats

Some caveats about the negative evidence on the role of β in average returns are in order. The average premiums for β , size, and book-to-market

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equity depend on the definitions of the variables used in the regressions. For example, suppose we replace book-to-market equity ($\ln(\text{BE}/\text{ME})$) with book equity ($\ln(\text{BE})$). As long as size ($\ln(\text{ME})$) is also in the regression, this change will not affect the intercept, the fitted values or the R^2 . But the change, in variables increases the average slope (and the t -statistic) on $\ln(\text{ME})$. In other words, it increases the risk premium associated with size. Other redefinitions of the β , size, and book-to-market variables will produce different regression slopes and perhaps different inferences about average premiums, including possible resuscitation of a role for β . And, of course, at the moment, we have no theoretical basis for choosing among different versions of the variables.

Moreover, the tests here are restricted to stocks. It is possible that including other assets will change the inferences about the average premiums for β , size, and book-to-market equity. For example, the large average intercepts for the FM regressions in Table VI suggest that the regressions will not do a good job on Treasury bills, which have low average returns and are likely to have small loadings on the underlying market, size, and book-to-market factors in returns. Extending the tests to bills and other bonds may well change our inferences about average risk premiums, including the revival of a role for market β .

We emphasize, however, that different approaches to the tests are not likely to revive the Sharpe-Lintner-Black model. Resuscitation of the SLB model requires that a better proxy for the market portfolio (a) overturns our evidence that the simple relation between β and average stock returns is flat and (b) leaves β as the only variable relevant for explaining average returns. Such results seem unlikely, given Stambaugh's (1982) evidence that tests of the SLB model do not seem to be sensitive to the choice of a market proxy. Thus, if there is a role for β in average returns, it is likely to be found in a multi-factor model that transforms the flat simple relation between average return and β into a positively sloped conditional relation.

V. Conclusions and Implications

The Sharpe-Lintner-Black model has long shaped the way academics and practitioners think about average return and risk. Black, Jensen, and Scholes (1972) and Fama and MacBeth (1973) find that, as predicted by the model, there is a positive simple relation between average return and market β during the early years (1926–1968) of the CRSP NYSE returns file. Like Reinganum (1981) and Lakonishok and Shapiro (1986), we find that this simple relation between β and average return disappears during the more recent 1963–1990 period. The appendix that follows shows that the relation between β and average return is also weak in the last half century (1941–1990) of returns on NYSE stocks. In short, our tests do not support the central prediction of the SLB model, that average stock returns are positively related to market β .

Banz (1981) documents a strong negative relation between average return and firm size. Bhandari (1988) finds that average return is positively related to leverage, and Basu (1983) finds a positive relation between average return

and E/P. Stattman (1980) and Rosenberg, Reid, and Lanstein (1985) document a positive relation between average return and book-to-market equity for U.S. stocks, and Chan, Hamao, and Lakonishok (1992) find that BE/ME is also a powerful variable for explaining average returns on Japanese stocks.

Variables like size, E/P, leverage, and book-to-market equity are all scaled versions of a firm's stock price. They can be regarded as different ways of extracting information from stock prices about the cross-section of expected stock returns (Ball (1978), Keim (1988)). Since all these variables are scaled versions of price, it is reasonable to expect that some of them are redundant for explaining average returns. Our main result is that for the 1963-1990 period, size and book-to-market equity capture the cross-sectional variation in average stock returns associated with size, E/P, book-to-market equity, and leverage.

A. Rational Asset-Pricing Stories

Are our results consistent with asset-pricing theory? Since the FM intercept is constrained to be the same for all stocks, FM regressions always impose a linear factor structure on returns and expected returns that is consistent with the multifactor asset-pricing models of Merton (1973) and Ross (1976). Thus our tests impose a rational asset-pricing framework on the relation between average return and size and book-to-market equity.

Even if our results are consistent with asset-pricing theory, they are not economically satisfying. What is the economic explanation for the roles of size and book-to-market equity in average returns? We suggest several paths of inquiry.

- (a) The intercepts and slopes in the monthly FM regressions of returns on $\ln(\text{ME})$ and $\ln(\text{BE}/\text{ME})$ are returns on portfolios that mimic the underlying common risk factors in returns proxied by size and book-to-market equity (Fama (1976), chapter 9). Examining the relations between the returns on these portfolios and economic variables that measure variation in business conditions might help expose the nature of the economic risks captured by size and book-to-market equity.
- (b) Chan, Chen, and Hsieh (1985) argue that the relation between size and average return proxies for a more fundamental relation between expected returns and economic risk factors. Their most powerful factor in explaining the size effect is the difference between the monthly returns on low- and high-grade corporate bonds, which in principle captures a kind of default risk in returns that is priced. It would be interesting to test whether loadings on this or other economic factors, such as those of Chen, Roll, and Ross (1986), can explain the roles of size and book-to-market equity in our tests.
- (c) In a similar vein, Chan and Chen (1991) argue that the relation between size and average return is a relative-prospects effect. The earning prospects of distressed firms are more sensitive to economic

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conditions. This results in a distress factor in returns that is priced in expected returns. Chan and Chen construct two mimicking portfolios for the distress factor, based on dividend changes and leverage. It would be interesting to check whether loadings on their distress factors absorb the size and book-to-market equity effects in average returns that are documented here.

- (d) In fact, if stock prices are rational, BE/ME, the ratio of the book value of a stock to the market's assessment of its value, should be a direct indicator of the relative prospects of firms. For example, we expect that high BE/ME firms have low earnings on assets relative to low BE/ME firms. Our work (in progress) suggests that there is indeed a clean separation between high and low BE/ME firms on various measures of economic fundamentals. Low BE/ME firms are persistently strong performers, while the economic performance of high BE/ME firms is persistently weak.

B. Irrational Asset-Pricing Stories

The discussion above assumes that the asset-pricing effects captured by size and book-to-market equity are rational. For BE/ME, our most powerful expected-return variable, there is an obvious alternative. The cross-section of book-to-market ratios might result from market overreaction to the relative prospects of firms. If overreaction tends to be corrected, BE/ME will predict the cross-section of stock returns.

Simple tests do not confirm that the size and book-to-market effects in average returns are due to market overreaction, at least of the type posited by DeBondt and Thaler (1985). One overreaction measure used by DeBondt and Thaler is a stock's most recent 3-year return. Their overreaction story predicts that 3-year losers have strong post-ranking returns relative to 3-year winners. In FM regressions (not shown) for individual stocks, the 3-year lagged return shows no power even when used alone to explain average returns. The univariate average slope for the lagged return is negative, -6 basis points per month, but less than 0.5 standard errors from 0.

C. Applications

Our main result is that two easily measured variables, size and book-to-market equity, seem to describe the cross-section of average stock returns. Prescriptions for using this evidence depend on (a) whether it will persist, and (b) whether it results from rational or irrational asset-pricing.

It is possible that, by chance, size and book-to-market equity happen to describe the cross-section of average returns in our sample, but they were and are unrelated to expected returns. We put little weight on this possibility, especially for book-to-market equity. First, although BE/ME has long been touted as a measure of the return prospects of stocks, there is no evidence that its explanatory power deteriorates through time. The 1963-1990 relation between BE/ME and average return is strong, and remarkably similar

for the 1963–1976 and 1977–1990 subperiods. Second, our preliminary work on economic fundamentals suggests that high-BE/ME firms tend to be persistently poor earners relative to low-BE/ME firms. Similarly, small firms have a long period of poor earnings during the 1980s not shared with big firms. The systematic patterns in fundamentals give us some hope that size and book-to-market equity proxy for risk factors in returns, related to relative earning prospects, that are rationally priced in expected returns.

If our results are more than chance, they have practical implications for portfolio formation and performance evaluation by investors whose primary concern is long-term average returns. If asset-pricing is rational, size and BE/ME must proxy for risk. Our results then imply that the performance of managed portfolios (e.g., pension funds and mutual funds) can be evaluated by comparing their average returns with the average returns of benchmark portfolios with similar size and BE/ME characteristics. Likewise, the expected returns for different portfolio strategies can be estimated from the historical average returns of portfolios with matching size and BE/ME properties.

If asset-pricing is irrational and size and BE/ME do not proxy for risk, our results might still be used to evaluate portfolio performance and measure the expected returns from alternative investment strategies. If stock prices are irrational, however, the likely persistence of the results is more suspect.

Appendix **Size Versus β : 1941–1990**

Our results on the absence of a relation between β and average stock returns for 1963–1990 are so contrary to the tests of the Sharpe-Lintner-Black model by Black, Jensen, and Scholes (1972), Fama and MacBeth (1973), and (more recently) Chan and Chen (1988), that further tests are appropriate. We examine the roles of size and β in the average returns on NYSE stocks for the half-century 1941–1990, the longest available period that avoids the high volatility of returns in the Great Depression. We do not include the accounting variables in the tests because of the strong selection bias (toward successful firms) in the COMPUSTAT data prior to 1962.

We first replicate the results of Chan and Chen (1988). Like them, we find that when portfolios are formed on size alone, there are strong relations between average return and either size or β ; average return increases with β and decreases with size. For size portfolios, however, size ($\ln(\text{ME})$) and β are almost perfectly correlated (-0.98), so it is difficult to distinguish between the roles of size and β in average returns.

One way to generate strong variation in β that is unrelated to size is to form portfolios on size and then on β . As in Tables I to III, we find that the resulting independent variation in β just about washes out the positive simple relation between average return and β observed when portfolios are formed on size alone. The results for NYSE stocks for 1941–1990 are thus much like those for NYSE, AMEX, and NASDAQ stocks for 1963–1990.

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This appendix also has methodological goals. For example, the FM regressions in Table III use returns on individual stocks as the dependent variable. Since we allocate portfolio β s to individual stocks but use firm-specific values of other variables like size, β may be at a disadvantage in the regressions for individual stocks. This appendix shows, however, that regressions for portfolios, which put β and size on equal footing, produce results comparable to those for individual stocks.

A. Size Portfolios

Table AI shows average monthly returns and market β s for 12 portfolios of NYSE stocks formed on the basis of size (ME) at the end of each year from 1940 to 1989. For these size portfolios, there is a strong positive relation between average return and β . Average returns fall from 1.96% per month for the smallest ME portfolio (1A) to 0.93% for the largest (10B) and β falls from 1.60 to 0.95. (Note also that, as claimed earlier, estimating β as the sum of the slopes in the regression of a portfolio's return on the current and prior month's NYSE value-weighted return produces much larger β s for the smallest ME portfolios and slightly smaller β s for the largest ME portfolios.)

The FM regressions in Table AI confirm the positive simple relation between average return and β for size portfolios. In the regressions of the size-portfolio returns on β alone, the average premium for a unit of β is 1.45% per month. In the regressions of individual stock returns on β (where stocks are assigned the β of their size portfolio), the premium for a unit of β is 1.39%. Both estimates are about 3 standard errors from 0. Moreover, the β s of size portfolios do not leave a residual size effect; the average residuals from the simple regressions of returns on β in Table AI show no relation to size. These positive SLB results for 1941–1990 are like those obtained by Chan and Chen (1988) in tests on size portfolios for 1954–1983.

There is, however, evidence in Table AI that all is not well with the β s of the size portfolios. They do a fine job on the relation between size and average return, but they do a lousy job on their main task, the relation between β and average return. When the residuals from the regressions of returns on β are grouped using the pre-ranking β s of individual stocks, the average residuals are strongly positive for low- β stocks (0.51% per month for group 1A) and negative for high- β stocks (–1.05% for 10B). Thus the market lines estimated with size-portfolio β s exaggerate the tradeoff of average return for β ; they underestimate average returns on low- β stocks and overestimate average returns on high- β stocks. This pattern in the β -sorted average residuals for individual stocks suggests that (a) there is variation in β across stocks that is lost in the size portfolios, and (b) this variation in β is not rewarded as well as the variation in β that is related to size.

B. Two-Pass Size- β Portfolios

Like Table I, Table AII shows that subdividing size deciles using the (pre-ranking) β s of individual stocks results in strong variation in β that is

Table AI
Average Returns, Post-Ranking β s and Fama-MacBeth Regression Slopes for
Size Portfolios of NYSE Stocks: 1941–1990

At the end of each year $t - 1$, stocks are assigned to 12 portfolios using ranked values of ME. Included are all NYSE stocks that have a CRSP price and shares for December of year $t - 1$ and returns for at least 24 of the 60 months ending in December of year $t - 1$ (for pre-ranking β estimates). The middle 8 portfolios cover size deciles 2 to 9. The 4 extreme portfolios (1A, 1B, 10A, and 10B) split the smallest and largest deciles in half. We compute equal-weighted returns on the portfolios for the 12 months of year t using all surviving stocks. Average Return is the time-series average of the monthly portfolio returns for 1941–1990, in percent. Average firms is the average number of stocks in the portfolios each month. The simple β s are estimated by regressing the 1941–1990 sample of post-ranking monthly returns for a size portfolio on the current month's value-weighted NYSE portfolio return. The sum β s are the sum of the slopes from a regression of the post-ranking monthly returns on the current and prior month's VW NYSE returns.

The independent variables in the Fama-MacBeth regressions are defined for each firm at the end of December of each year $t - 1$. Stocks are assigned the post-ranking (sum) β of the size portfolio they are in at the end of year $t - 1$. ME is price times shares outstanding at the end of year $t - 1$. In the individual-stock regressions, these values of the explanatory variables are matched with CRSP returns for each of the 12 months of year t . The portfolio regressions match the equal-weighted portfolio returns with the equal-weighted averages of β and $\ln(\text{ME})$ for the surviving stocks in each month of year t . Slope is the average of the (600) monthly FM regression slopes and SE is the standard error of the average slope. The residuals from the monthly regressions for year t are grouped into 12 portfolios on the basis of size (ME) or pre-ranking β (estimated with 24 to 60 months of data, as available) at the end of year $t - 1$. The average residuals are the time-series averages of the monthly equal-weighted portfolio residuals, in percent. The average residuals for regressions (1) and (2) (not shown) are quite similar to those for regressions (4) and (5) (shown).

	Portfolios Formed on Size											
	1A	1B	2	3	4	5	6	7	8	9	10A	10B
Ave. return	1.96	1.59	1.44	1.36	1.28	1.24	1.23	1.17	1.15	1.13	0.97	0.93
Ave. firms	57	56	110	107	107	108	111	113	115	118	59	59
Simple β	1.29	1.24	1.21	1.19	1.16	1.13	1.13	1.12	1.09	1.05	1.00	0.98
Standard error	0.07	0.05	0.04	0.03	0.02	0.02	0.02	0.02	0.01	0.01	0.01	0.01
Sum β	1.60	1.44	1.37	1.32	1.26	1.23	1.19	1.17	1.12	1.06	0.99	0.95
Standard error	0.10	0.06	0.05	0.04	0.03	0.03	0.03	0.02	0.02	0.01	0.01	0.01

Table AI—Continued

	Portfolio Regressions			Individual Stock Regressions					
	(1) β	(2) $\ln(\text{ME})$	(3) β and $\ln(\text{ME})$		(4) β	(5) $\ln(\text{ME})$	(6) β and $\ln(\text{ME})$		
Slope	1.45	-0.137	3.05	0.149	1.39	-0.133	0.71	-0.060	
SE	0.47	0.044	1.51	0.115	0.46	0.043	0.81	0.062	
Average Residuals for Stocks Grouped on Size									
	1A	1B	2	3	4	5	6	7	8
Regression (4)	0.17	0.00	-0.04	-0.06	-0.05	-0.04	0.00	-0.03	0.03
Standard error	0.11	0.06	0.04	0.04	0.04	0.04	0.03	0.03	0.03
Regression (5)	0.30	0.02	-0.05	-0.06	-0.08	-0.07	-0.03	-0.04	0.02
Standard error	0.14	0.07	0.04	0.04	0.04	0.04	0.04	0.03	0.03
Regression (6)	0.20	0.02	-0.05	-0.07	-0.08	-0.06	-0.01	-0.02	0.04
Standard error	0.10	0.06	0.04	0.04	0.04	0.04	0.03	0.03	0.03
Average Residuals for Stocks Grouped on Pre-Ranking β									
	1A	1B	2	3	4	5	6	7	8
Regression (4)	0.51	0.61	0.38	0.32	0.16	0.12	0.03	-0.10	-0.27
Standard error	0.21	0.19	0.13	0.08	0.04	0.03	0.04	0.05	0.09
Regression (5)	-0.10	0.00	0.02	0.09	0.05	0.07	0.05	0.00	-0.03
Standard error	0.11	0.10	0.07	0.05	0.04	0.03	0.03	0.04	0.05
Regression (6)	0.09	0.25	0.13	0.19	0.11	0.14	0.09	0.01	-0.11
Standard error	0.41	0.37	0.24	0.14	0.07	0.04	0.04	0.09	0.16

Table AII
Properties of Portfolios Formed on Size and Pre-Ranking β : NYSE Stocks
Sorted by ME (Down) then Pre-Ranking β (Across): 1941 - 1990

At the end of year $t - 1$, the NYSE stocks on CRSP are assigned to 10 size (ME) portfolios. Each size decile is subdivided into 10 β portfolios using pre-ranking β s of individual stocks, estimated with 24 to 60 monthly returns (as available) ending in December of year $t - 1$. The equal-weighted monthly returns on the resulting 100 portfolios are then calculated for year t . The average returns are the time-series averages of the monthly returns, in percent. The post-ranking β s use the full 1941-1990 sample of post-ranking returns for each portfolio. The pre- and post-ranking β s are the sum of the slopes from a regression of monthly returns on the current and prior month's NYSE value-weighted market return. The average size for a portfolio is the time-series average of each month's average value of $\ln(\text{ME})$ for stocks in the portfolio. ME is denominated in millions of dollars. There are, on average, about 10 stocks in each size- β portfolio each month. The All column shows parameter values for equal-weighted size-decile (ME) portfolios. The All rows show parameter values for equal-weighted portfolios of the stocks in each β group.

	All	Low- β	β -2	β -3	β -4	β -5	β -6	β -7	β -8	β -9	High- β
Panel A: Average Monthly Return (in Percent)											
All		1.22	1.30	1.32	1.35	1.36	1.34	1.29	1.34	1.14	1.10
Small-ME	1.78	1.74	1.76	2.08	1.91	1.92	1.72	1.77	1.91	1.56	1.46
ME-2	1.44	1.41	1.35	1.33	1.61	1.72	1.59	1.40	1.62	1.24	1.11
ME-3	1.36	1.21	1.40	1.22	1.47	1.34	1.51	1.33	1.57	1.33	1.21
ME-4	1.28	1.26	1.29	1.19	1.27	1.51	1.30	1.19	1.56	1.18	1.00
ME-5	1.24	1.22	1.30	1.28	1.33	1.21	1.37	1.41	1.31	0.92	1.06
ME-6	1.23	1.21	1.32	1.37	1.09	1.34	1.10	1.40	1.21	1.22	1.08
ME-7	1.17	1.08	1.23	1.37	1.27	1.19	1.34	1.10	1.11	0.87	1.17
ME-8	1.15	1.06	1.18	1.26	1.25	1.26	1.17	1.16	1.05	1.08	1.04
ME-9	1.13	0.99	1.13	1.00	1.24	1.28	1.31	1.15	1.11	1.09	1.05
Large-ME	0.95	0.99	1.01	1.12	1.01	0.89	0.95	0.95	1.00	0.90	0.68

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Table AII—Continued

	All	Low- β	β_2	β_3	β_4	β_5	β_6	β_7	β_8	β_9	High- β
				Panel B: Post-Ranking β							
All		0.76	0.95	1.05	1.14	1.22	1.26	1.34	1.38	1.49	1.69
Small-ME	1.52	1.17	1.40	1.31	1.50	1.46	1.50	1.69	1.60	1.75	1.92
ME-2	1.37	0.86	1.09	1.12	1.24	1.39	1.42	1.48	1.60	1.69	1.91
ME-3	1.32	0.88	0.96	1.18	1.19	1.33	1.40	1.43	1.56	1.64	1.74
ME-4	1.26	0.69	0.95	1.06	1.15	1.24	1.29	1.46	1.43	1.64	1.83
ME-5	1.23	0.70	0.95	1.04	1.10	1.22	1.32	1.34	1.41	1.56	1.72
ME-6	1.19	0.68	0.86	1.04	1.13	1.20	1.20	1.35	1.36	1.48	1.70
ME-7	1.17	0.67	0.88	0.95	1.14	1.18	1.26	1.27	1.32	1.44	1.68
ME-8	1.12	0.64	0.83	0.99	1.06	1.14	1.14	1.21	1.26	1.39	1.58
ME-9	1.06	0.68	0.81	0.94	0.96	1.06	1.11	1.18	1.22	1.25	1.46
Large-ME	0.97	0.65	0.73	0.90	0.91	0.97	1.01	1.01	1.07	1.12	1.38
				Panel C: Average Size (ln(ME))							
All		4.39	4.39	4.40	4.40	4.39	4.40	4.38	4.37	4.37	4.34
Small-ME	1.93	2.04	1.99	2.00	1.96	1.92	1.92	1.91	1.90	1.87	1.80
ME-2	2.80	2.81	2.79	2.81	2.83	2.80	2.79	2.80	2.80	2.79	2.79
ME-3	3.27	3.28	3.27	3.28	3.27	3.27	3.28	3.29	3.27	3.27	3.26
ME-4	3.67	3.67	3.67	3.67	3.68	3.68	3.67	3.68	3.66	3.67	3.67
ME-5	4.06	4.07	4.06	4.05	4.06	4.07	4.06	4.05	4.05	4.06	4.06
ME-6	4.45	4.45	4.44	4.46	4.45	4.45	4.45	4.45	4.44	4.45	4.45
ME-7	4.87	4.86	4.87	4.86	4.87	4.87	4.88	4.87	4.87	4.85	4.87
ME-8	5.36	5.38	5.38	5.38	5.35	5.36	5.37	5.37	5.36	5.35	5.34
ME-9	5.98	5.96	5.98	5.99	6.00	5.98	5.98	5.97	5.95	5.96	5.96
Large-ME	7.12	7.10	7.12	7.16	7.17	7.20	7.29	7.14	7.09	7.04	6.83

independent of size. The β sort of a size decile always produces portfolios with similar average $\ln(\text{ME})$ but much different (post-ranking) β s. Table AII also shows, however, that investors are not compensated for the variation in β that is independent of size. Despite the wide range of β s in each size decile, average returns show no tendency to increase with β . AII

The FM regressions in Table AIII formalize the roles of size and β in NYSE average returns for 1941–1990. The regressions of returns on β alone show that using the β s of the portfolios formed on size and β , rather than size alone, causes the average slope on β to fall from about 1.4% per month (Table AI) to about 0.23% (about 1 standard error from 0). Thus, allowing for variation in β that is unrelated to size flattens the relation between average return and β , to the point where it is indistinguishable from no relation at all.

The flatter market lines in Table AIII succeed, however, in erasing the negative relation between β and average residuals observed in the regressions of returns on β alone in Table AI. Thus, forming portfolios on size and β (Table AIII) produces a better description of the simple relation between average return and β than forming portfolios on size alone (Table AI). This improved description of the relation between average return and β is evidence that the β estimates for the two-pass size- β portfolios capture variation in true β s that is missed when portfolios are formed on size alone.

Unfortunately, the flatter market lines in Table AIII have a cost, the emergence of a residual size effect. Grouped on the basis of ME for individual stocks, the average residuals from the univariate regressions of returns on the β s of the 100 size- β portfolios are strongly positive for small stocks and negative for large stocks (0.60% per month for the smallest ME group, 1A, and -0.27% for the largest, 10B). Thus, when we allow for variation in β that is independent of size, the resulting β s leave a large size effect in average returns. This residual size effect is much like that observed by Banz (1981) with the β s of portfolios formed on size and β .

The correlation between size and β is -0.98 for portfolios formed on size alone. The independent variation in β obtained with the second-pass sort on β lowers the correlation to -0.50 . The lower correlation means that bivariate regressions of returns on β and $\ln(\text{ME})$ are more likely to distinguish true size effects from true β effects in average returns.

The bivariate regressions (Table AIII) that use the β s of the size- β portfolios are more bad news for β . The average slopes for $\ln(\text{ME})$ are close to the values in the univariate size regressions, and almost 4 standard errors from 0, but the average slopes for β are negative and less than 1 standard error from 0. The message from the bivariate regressions is that there is a strong relation between size and average return. But like the regressions in Table AIII that explain average returns with β alone, the bivariate regressions say that there is no reliable relation between β and average returns when the tests use β s that are not close substitutes for size. These uncomfortable SLB results for NYSE stocks for 1941–1990 are much like those for NYSE, AMEX, and NASDAQ stocks for 1963–1990 in Table III.

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C. Subperiod Diagnostics

Our results for 1941–1990 seem to contradict the evidence in Black, Jensen, and Scholes (BJS) (1972) and Fama and MacBeth (FM) (1973) that there is a reliable positive relation between average return and β . The β s in BJS and FM are from portfolios formed on β alone, and the market proxy is the NYSE equal-weighted portfolio. We use the β s of portfolios formed on size and β , and our market is the value-weighted NYSE portfolio. We can report, however, that our inference that there isn't much relation between β and average return is unchanged when (a) the market proxy is the NYSE EW portfolio, (b) portfolios are formed on just (pre-ranking) β s, or (c) the order of forming the size- β portfolios is changed from size then β to β then size.

A more important difference between our results and the earlier studies is the sample periods. The tests in BJS and FM end in the 1960s. Table AIV shows that when we split the 50-year 1941–1990 period in half, the univariate FM regressions of returns on β produce an average slope for 1941–1965 (0.50% per month, $t = 1.82$) more like that of the earlier studies. In contrast, the average slope on β for 1966–1990 is close to 0 (-0.02 , $t = 0.06$).

But Table AIV also shows that drawing a distinction between the results for 1941–1965 and 1966–1990 is misleading. The stronger tradeoff of average return for β in the simple regressions for 1941–1965 is due to the first 10 years, 1941–1950. This is the only period in Table AIV that produces an average premium for β (1.26% per month) that is both positive and more than 2 standard errors from 0. Conversely, the weak relation between β and average return for 1966–1990 is largely due to 1981–1990. The strong negative average slope in the univariate regressions of returns on β for 1981–1990 (-1.01 , $t = -2.10$) offsets a positive slope for 1971–1980 (0.82, $t = 1.27$).

The subperiod variation in the average slopes from the FM regressions of returns on β alone seems moot, however, given the evidence in Table AIV that adding size always kills any positive tradeoff of average return for β in the subperiods. Adding size to the regressions for 1941–1965 causes the average slope for β to drop from 0.50 ($t = 1.82$) to 0.07 ($t = 0.28$). In contrast, the average slope on size in the bivariate regressions (-0.16 , $t = -2.97$) is close to its value (-0.17 , $t = -2.88$) in the regressions of returns on $\ln(\text{ME})$ alone. Similar comments hold for 1941–1950. In short, any evidence of a positive average premium for β in the subperiods seems to be a size effect in disguise.

D. Can the SLB Model Be Saved?

Before concluding that β has no explanatory power, it is appropriate to consider other explanations for our results. One possibility is that the variation in β produced by the β sorts of size deciles is just sampling error. If so, it is not surprising that the variation in β within a size decile is unrelated to average return, or that size dominates β in bivariate tests. The standard errors of the β s suggest, however, that this explanation cannot save the SLB

Table AIII
Average Slopes, Their Standard Errors (SE), and Average Residuals from
Monthly FM Regressions for Individual NYSE Stocks and for Portfolios Formed
on Size and Pre-Ranking β : 1941–1990

Stocks are assigned the post-ranking β of the size- β portfolio they are in at the end of year $t - 1$ (Table AII). $\ln(\text{ME})$ is the natural log of price times shares outstanding at the end of year $t - 1$. In the individual-stock regressions, these values of the explanatory variables are matched with CRSP returns for each of the 12 months in year t . The portfolio regressions match the equal-weighted portfolio returns for the size- β portfolios (Table AII) with the equal-weighted averages of β and $\ln(\text{ME})$ for the surviving stocks in each month of year t . Slope is the time-series average of the monthly regression slopes from 1941–1990 (600 months); SE is the time-series standard error of the average slope.

The residuals from the monthly regressions in year t are grouped into 12 portfolios on the basis of size or pre-ranking β (estimated with 24 to 60 months of returns, as available) as of the end of year $t - 1$. The average residuals are the time-series averages of the monthly equal-weighted averages of the residuals in percent. The average residuals (not shown) from the FM regressions (1) to (3) that use the returns on the 100 size- β portfolios as the dependent variable are always within 0.01 of those from the regressions for individual stock returns. This is not surprising given that the correlation between the time-series of 1941–1990 monthly FM slopes on β or $\ln(\text{ME})$ for the comparable portfolio and individual stock regressions is always greater than 0.99.

	Portfolio Regressions			Individual Stock Regressions									
	(1) β	(2) $\ln(\text{ME})$	(3) β and $\ln(\text{ME})$	(4) β	(5) $\ln(\text{ME})$	(6) β and $\ln(\text{ME})$							
Slope	0.22	-0.128	-0.13	-0.143	0.24	-0.133	-0.14	0.21	-0.147				
SE	0.24	0.043	0.21	0.039	0.23	0.043	0.21	0.039	0.039				
Average Residuals for Stocks Grouped on Size													
	1A	1B	2	3	4	5	6	7	8	9	10A	10B	
Regression (4)	0.60	0.26	0.13	0.06	-0.01	-0.03	-0.03	-0.09	-0.10	-0.11	-0.25	-0.27	
Standard error	0.21	0.10	0.06	0.04	0.04	0.04	0.04	0.04	0.04	0.05	0.06	0.08	
Regression (5)	0.30	0.02	-0.05	-0.06	-0.08	-0.07	-0.03	-0.04	0.02	0.08	0.01	0.13	
Standard error	0.14	0.07	0.04	0.04	0.04	0.04	0.04	0.03	0.03	0.03	0.04	0.07	
Regression (6)	0.31	0.02	-0.05	-0.06	-0.09	-0.07	-0.03	-0.04	0.02	0.08	0.01	0.13	
Standard error	0.14	0.07	0.04	0.04	0.04	0.04	0.04	0.03	0.03	0.03	0.04	0.07	

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Table AIII – Continued

	Portfolio Regressions			Individual Stock Regressions									
	(1) β	(2) ln(ME)	(3) β and ln(ME)	(4) β	(5) ln(ME)	(6) β and ln(ME)							
	Average Residuals for Stocks Grouped on Pre-Ranking β												
	1A	1B	2	3	4	5	6	7	8	9	10A	10B	
Regression (4)	-0.08	0.03	-0.01	0.08	0.04	0.08	0.04	0.02	-0.03	0.02	-0.11	-0.32	
Standard error	0.07	0.05	0.03	0.03	0.03	0.03	0.03	0.04	0.04	0.04	0.06	0.07	
Regression (5)	-0.10	0.00	0.02	0.09	0.05	0.07	0.05	0.00	-0.03	-0.01	-0.11	-0.33	
Standard error	0.11	0.10	0.07	0.05	0.04	0.03	0.03	0.04	0.05	0.07	0.10	0.13	
Regression (6)	-0.17	-0.07	-0.02	0.07	0.04	0.06	0.05	0.03	0.00	0.04	-0.04	-0.23	
Standard error	0.05	0.04	0.03	0.03	0.03	0.03	0.03	0.03	0.04	0.04	0.06	0.07	

Table A1V
Subperiod Average Returns on the NYSE Value-Weighted and
Equal-Weighted Portfolios and Average Values of the
Intercepts and Slopes for the FM Cross-Sectional Regressions
of Individual Stock Returns on β and Size ($\ln(\text{ME})$)

Mean is the average VW or EW return or an average slope from the monthly cross-sectional regressions of individual stock returns on β and/or $\ln(\text{ME})$. Std is the standard deviation of the time-series of returns or slopes, and $t(\text{Mn})$ is Mean over its time-series standard error. The average slopes (not shown) from the FM regressions that use the returns on the 100 size- β portfolios of Table A1I as the dependent variable are quite close to those for individual stock returns. (The correlation between the 1941-1990 month-by-month slopes on β or $\ln(\text{ME})$ for the comparable portfolio and individual stock regressions is always greater than 0.99.)

Variable	1941-1990 (600 Mos.)			1941-1965 (300 Mos.)			1966-1990 (300 Mos.)		
	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$	Mean	Std	$t(\text{Mn})$
NYSE Value-Weighted (VW) and Equal-Weighted (EW) Portfolio Returns									
VW	0.93	4.15	5.49	1.10	3.58	5.30	0.76	4.64	2.85
EW	1.12	5.10	5.37	1.33	4.42	5.18	0.91	5.70	2.77
a	0.98	3.93	6.11	$R_{it} = a + b_{1i}\beta_{it} + e_{it}$			1.13	4.57	4.26
b ₁	0.24	5.52	1.07	0.84	3.18	4.56	-0.02	6.19	-0.06
$R_{it} = a + b_{2i}\ln(\text{ME}_{it}) + e_{it}$									
a	1.70	8.24	5.04	0.50	4.75	1.82	1.51	9.72	2.69
b ₂	-0.13	1.06	-3.07	1.88	6.43	5.06	-0.10	1.11	-1.54
$R_{it} = a + b_{1i}\beta_{it} + b_{2i}\ln(\text{ME}_{it}) + e_{it}$									
a	1.97	6.16	7.84	1.80	4.77	6.52	2.14	7.29	5.09
b ₁	-0.14	5.05	-0.66	0.07	4.15	0.28	-0.34	5.80	-1.01
b ₂	-0.15	0.96	-3.75	-0.16	0.94	-2.97	-0.13	0.99	-2.34

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Table A1V – Continued

Panel B:										
Return	1941-1950		1951-1960		1961-1970		1971-1980		1981-1990	
	Mean	<i>t</i> (Mn)	Mean	<i>t</i> (Mn)	Mean	<i>t</i> (Mn)	Mean	<i>t</i> (Mn)	Mean	<i>t</i> (Mn)
VW	1.05	2.88	1.18	3.95	0.66	1.84	0.72	1.67	1.04	2.40
EW	1.59	3.16	1.13	3.76	0.88	1.96	1.04	1.82	0.95	2.01
a	0.24	0.66	1.41	6.36	$R_{it} = a + b_1\beta_{it} + e_{it}$		0.27	0.62	2.35	5.99
b ₁	1.26	2.20	-0.19	-0.63	0.64	1.94	0.82	1.27	-1.01	-2.10
a	2.63	3.47	1.08	2.73	$R_{it} = a + b_2\ln(ME_{it}) + e_{it}$		2.18	2.03	0.82	1.20
b ₂	-0.37	-2.90	0.03	0.53	0.32	0.72	-0.20	-1.57	0.04	0.57
a	2.14	3.93	1.38	4.03	$R_{it} = a + b_1\beta_{it} + b_2\ln(ME_{it}) + e_{it}$		1.50	2.12	2.84	4.25
b ₁	0.34	0.75	-0.17	-0.53	2.01	4.16	0.41	0.75	-1.14	-2.16
b ₂	-0.34	-2.92	0.01	0.20	-0.11	-0.27	-0.16	-1.50	-0.07	-0.84

model. The standard errors for portfolios formed on size and β are only slightly larger (0.02 to 0.11) than those for portfolios formed on size alone (0.01 to 0.10, Table AI). And the range of the post-ranking β s within a size decile is always large relative to the standard errors of the β s.

Another possibility is that the proportionality condition (1) for the variation through time in true β s, that justifies the use of full-period post-ranking β s in the FM tests, does not work well for portfolios formed on size and β . If this is a problem, post-ranking β s for the size- β portfolios should not be highly correlated across subperiods. The correlation between the half-period (1941–1965 and 1966–1990) β s of the size- β portfolios is 0.91, which we take to be good evidence that the full-period β estimates for these portfolios are informative about true β s. We can also report that using 5-year β s (pre- or post-ranking) in the FM regressions does not change our negative conclusions about the role of β in average returns, as long as portfolios are formed on β as well as size, or on β alone.

Any attempt to salvage the simple positive relation between β and average return predicted by the SLB model runs into three damaging facts, clear in Table AII. (a) Forming portfolios on size and pre-ranking β s produces a wide range of post-ranking β s in every size decile. (b) The post-ranking β s closely reproduce (in deciles 2 to 10 they exactly reproduce) the ordering of the pre-ranking β s used to form the β -sorted portfolios. It seems safe to conclude that the increasing pattern of the post-ranking β s in every size decile captures the ordering of the true β s. (c) Contrary to the SLB model, the β sorts do not produce a similar ordering of average returns. Within the rows (size deciles) of the average return matrix in Table AII, the high- β portfolios have average returns that are close to or less than the low- β portfolios.

But the most damaging evidence against the SLB model comes from the univariate regressions of returns on β in Table AIII. They say that when the tests allow for variation in β that is unrelated to size, the relation between β and average return for 1941–1990 is weak, perhaps nonexistent, even when β is the only explanatory variable. We are forced to conclude that the SLB model does not describe the last 50 years of average stock returns.

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The CAPM is Wanted, Dead or Alive

EUGENE F. FAMA and KENNETH R. FRENCH*

ABSTRACT

Kothari, Shanken, and Sloan (1995) claim that β s from annual returns produce a stronger positive relation between β and average return than β s from monthly returns. They also contend that the relation between average return and book-to-market equity (BE/ME) is seriously exaggerated by survivor bias. We argue that survivor bias does not explain the relation between BE/ME and average return. We also show that annual and monthly β s produce the same inferences about the β premium. Our main point on the β premium is, however, more basic. It cannot save the Capital asset pricing model (CAPM), given the evidence that β alone cannot explain expected return.

FAMA AND FRENCH (FF 1992) PRODUCE two negative conclusions about the empirical adequacy of the capital asset pricing model (CAPM) of Sharpe (1964) and Lintner (1965): (i) when one allows for variation in CAPM market β s that is unrelated to size, the univariate relation between β and average return for 1941–1990 is weak; (ii) β does not suffice to explain average return. Size (market capitalization) captures differences in average stock returns for 1941–1990 that are missed by β . For the post-1962 period where we have book equity data, BE/ME (the ratio of book to market equity) and other variables also help explain average return.

Kothari, Shanken, and Sloan (KSS 1995) have two main quarrels with these conclusions. First, they claim that using β s estimated from annual rather than monthly returns produces a stronger positive relation between average return and β . Second, KSS contend that the relation between average return and BE/ME observed by FF and others is seriously exaggerated by survivor bias in the COMPUSTAT sample.

We argue (Section II) that survivor bias does not explain the relation between BE/ME and average return. We also show (Section III) that annual and monthly β s produce the same inferences about the presence of a β premium in expected returns. But our main point on the β premium (Section I) is more basic: It cannot save the CAPM, given the evidence that β alone cannot explain expected return.

I. The Logic of Tests of the CAPM

As emphasized by Fama (1976), Roll (1977), and others, the main implication of the CAPM is that in a market equilibrium, the value-weight market port-

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folio, M , is mean-variance-efficient. The mean-variance-efficiency of M in turn says that:

- (i) β , the slope in the regression of a security's return on the market return, is the only risk needed to explain expected return;
- (ii) There is positive expected premium for β risk.

Our main point is that evidence of (ii), a positive relation between β and expected return, is support for the CAPM only if (i) also holds, that is, only if β suffices to explain expected return. Confirming Banz (1981), however, and like FF (1992), KSS find that size adds to the explanation of average return provided by β . Moreover, size is no longer the prime embarrassment of the CAPM. Variables that (unlike size) do not seem to be correlated with β (such as earnings/price, cashflow/price, BE/ME, and past sales growth) add even more significantly to the explanation of average return provided by β (Basu (1983), Chan, Hamao, and Lakonishok (1991), FF (1992, 1993, 1996), and Lakonishok, Shleifer, and Vishny (1994)).

The average-return anomalies of the CAPM suggest that, if asset pricing is rational, a multifactor version of Merton's (1973) intertemporal CAPM (ICAPM) or Ross' (1976) arbitrage pricing theory (APT) can provide a better description of average returns. The excess market return of the CAPM is a relevant risk in many multifactor alternatives, like the ICAPM and Connor's (1984) equilibrium version of the APT. Thus, evidence of a positive relation between β and expected return does not favor the CAPM over these alternatives.

The three-factor model in Fama and French (1993, 1994, 1995, 1996) illustrates our point. The model provides a better description of average returns than the CAPM, and it captures most of the average-return anomalies missed by the CAPM. Because of its strong theoretical standing, the excess market return is one of the three risk-factors in the model, and our tests confirm that it is important. It captures strong common times-series variation in returns, and the market premium is needed to explain the large differences between the average returns on stocks and bills. Moreover, as in the CAPM, the market premium in our multifactor model is just the average return on M in excess of the risk-free rate. Tests on long sample periods say that this premium is reliably positive. In short, our tests of the CAPM against a multifactor alternative illustrate that a positive β premium does not in itself resuscitate the CAPM, or justify using it in applications.

KSS are not misled on this basic point. But their focus on the univariate β premium may confuse some of their readers. Indeed, because the CAPM is such a simple and attractive tool, we think that many of our colleagues *want* to be confused on this point. Otherwise, we can't explain the strong interest in the KSS β tests, given that, like many others (including Amihud, Christensen, and Mendelson (1992) and Jagannathan and Wang (1996)), KSS consistently find that β does not suffice to explain expected return.

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II. Survivor Bias and BE/ME

KSS argue that survivor bias in COMPUSTAT data is important in the strong positive relation between average return and book-to-market-equity (BE/ME) observed by FF (1992) and others. COMPUSTAT is more likely to add distressed (high-BE/ME) firms that ultimately survive and to miss distressed firms that die. The survivors are likely to have unexpectedly high returns in the turnaround years immediately preceding their inclusion on COMPUSTAT. Since COMPUSTAT typically includes some historical data when it adds firms, there can be positive survivor bias in the returns of high-BE/ME firms on COMPUSTAT.

There are counter arguments. In the most detailed study of the issue, Chan, Jegadeesh, and Lakonishok (1995) conclude that survivor bias cannot explain the strong relations between average return and BE/ME observed by Lakonishok, Shleifer, and Vishny (1994) and FF (1992) in tests on the post-1968 and post-1976 periods. After 1968, and certainly after 1976, almost all the traded securities on Center for Research in Security Prices (CRSP) that are not on COMPUSTAT are missing for reasons that have nothing to do with survivor bias. Many of the missing firms are closed-end investment companies, REITs, ADRs, primes, and scores that produce no accounting information or produce information that is not comparable to that of other firms. Many financial companies are also missing because, judging that their accounting data are different from that of other firms, COMPUSTAT limited its coverage of financials for many years. These omissions, which are the result of COMPUSTAT's ex ante policy decisions, are not a source of survivor bias. Finally, some of the securities that seem to be on CRSP but not COMPUSTAT in fact appear on both, but with different identifiers.

There is other evidence that survivor bias cannot explain the relation between average return and BE/ME. Lakonishok, Shleifer, and Vishny (1994) find a strong positive relation between average return and BE/ME for the largest 20 percent of NYSE-AMEX stocks on COMPUSTAT, where survivor bias is not an issue. FF (1993) find that the relation between BE/ME and average return is strong for value-weight portfolios of COMPUSTAT stocks formed on BE/ME. Since value-weight portfolios give most weight to larger stocks, any survivor bias in these portfolios is probably trivial. In three different sets of comparisons (Table VII), KSS themselves find that the relation between average return and BE/ME is strong and strikingly similar for value-weight and equal-weight portfolios of COMPUSTAT stocks formed on BE/ME. KSS concede that survivor bias cannot explain the results for value-weight portfolios.

To support their survivor-bias story, KSS make much of the fact that stocks on CRSP but not COMPUSTAT have lower average returns than stocks on COMPUSTAT. When they risk-adjust returns using a three-factor model like that in FF (1993), however, only the smallest two size deciles of the NYSE-AMEX stocks missing from COMPUSTAT have strong negative abnormal

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returns (Table IV). This suggests that survivor bias is limited to tiny stocks; the average market cap of the stocks in the second decile is \$13 million, while the average for the first decile is between \$3 million and \$7 million. The remaining 80 percent of the stocks missing on COMPUSTAT, which account for almost all the combined value of the missing stocks, have three-factor abnormal returns that are close to zero and random in sign. In other words, these missing stocks behave like stocks that are on COMPUSTAT. Similarly, Chan, Jegadeesh, and Lakonishok (1995) fill in missing COMPUSTAT book equity (BE) data for the largest 20 percent of the NYSE-AMEX firms on CRSP. The survivor-bias story says that the relation between BE/ME and average return should be weak for the firms missing on COMPUSTAT. They find that it is as strong for the missing firms as for the included firms.

KSS also speculate that the positive relation between book-to-market-equity and average return is the result of data dredging and so is special to the post-1962 COMPUSTAT period. Using a hand-collected sample of large firms that is not subject to survivor bias, however, Davis (1994) finds a strong relation between BE/ME and average return in the 1941–1962 period.

In the end, the KSS survivor-bias story rests on their evidence that there is little relation between average return and BE/ME for the rather limited industry portfolios in the *S&P Analyst's Handbook*. Their results for the S&P industries are strange since FF (1994) document a strong positive relation between average return and BE/ME for value-weight industry portfolios that include *all* NYSE, AMEX, and Nasdaq stocks on CRSP. (We use COMPUSTAT firms only to estimate industry BE/ME.)

KSS do not say that the relation between average return and BE/ME is entirely the result of survivor bias. They push so hard on the survivor-bias story, however, that serious readers are led to strong conclusions. For example, in the lead article to volume 38 of the *Journal of Financial Economics*, MacKinlay (1995, p. 5) concludes,

"Their analysis suggests that deviations from the CAPM such as those documented by Fama and French (1993) can be explained by sample selection biases."

III. Minor Points

KSS claim that using β s estimated from annual rather than monthly returns explains why they measure somewhat stronger relations between β and average return than FF (1992). They also claim that although the explanatory power of size is statistically reliable, for practical purposes, size adds little to the explanation of average return provided by β . The tests that follow explore these issues.

A. Portfolios Formed on β

Table I summarizes returns for 1928–1993 on β deciles of NYSE stocks. Like KSS, we weight the stocks equally. We form the portfolios in June of each year,

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Table I

**Summary Statistics and Cross-Section Regressions for
Postformation Equal-Weight Returns on NYSE β Deciles: 1928–1993**

Starting in 1927, ten portfolios of NYSE stocks on CRSP are formed in June of each year based on VW-S β s, the sum of the slopes from regressions of monthly returns on the current and one lag of the value-weight NYSE market return. The formation period β s use 24 to 60 months of past returns (as available), except for 1927, where 18 months are used. Equal-weight monthly postformation returns on the β deciles are calculated from July to June of the following year, yielding time-series of postformation returns for July 1927 to December 1993. The equal-weight monthly decile returns are compounded to get annual returns. The β s shown in Panel A are estimated using all postformation monthly (VW, VW-S, EW, EW-S) or annual (VWA, EWA) returns for 1928–1993 and the value-weight (VW, VW-S, VWA) or equal-weight (EW, EW-S, EWA) NYSE market portfolio. VW-S and EW-S β s are the sums of the slopes from the regressions of the monthly postformation decile returns on the current and one lag of the market return. VW, VWA, EW, and EWA β s use only the contemporaneous market returns. Average Ln (Size) is the average across postformation months of the average monthly value of the natural log of size (price times shares) of the stocks in a β decile. Panel B shows the average slopes (Means) and the t -statistics for the average slopes from univariate cross-section regressions of postformation monthly or annual returns on the ten β portfolio on each of the six different estimates of their postformation β s.

Panel A: Summary Statistics												
	Low β	2	3	4	5	6	7	8	9	High β		
Monthly Postformation Returns: 792 Months												
Mean	1.10	1.16	1.23	1.34	1.39	1.32	1.33	1.44	1.45	1.39		
Std	5.12	5.76	6.30	6.99	7.66	7.98	8.31	9.50	9.84	11.10		
Annual Postformation Returns: 66 Years												
Mean	14.91	15.53	16.43	17.73	17.98	17.03	17.19	18.44	18.44	17.40		
Std	25.69	26.39	28.25	29.85	29.52	30.76	32.04	35.96	36.03	40.55		
Postformation β Estimates												
VW	0.80	0.95	1.05	1.16	1.27	1.32	1.35	1.52	1.57	1.73		
VW-S	0.91	1.03	1.14	1.24	1.36	1.41	1.44	1.63	1.70	1.87		
VWA	1.06	1.13	1.23	1.33	1.33	1.37	1.43	1.56	1.60	1.71		
EW	0.61	0.72	0.80	0.90	0.99	1.03	1.07	1.23	1.26	1.41		
EW-S	0.65	0.73	0.81	0.89	0.98	1.02	1.06	1.21	1.26	1.41		
EWA	0.78	0.82	0.89	0.96	0.95	0.99	1.02	1.16	1.14	1.27		
Average Ln (Size)												
Mean	12.70	12.76	12.74	12.50	12.36	12.16	12.03	11.73	11.36	11.00		
Panel B: Cross-Section Regressions, $R = a + b\beta + e$												
Monthly Dependent Returns, R						Annual Dependent Returns, R						
	VW	VW-S	VWA	EW	EW-S	EWA	VW	VW-S	VWA	EW	EW-S	EWA
Means						Means						
a	0.86	0.85	0.63	0.90	0.90	0.67	12.94	12.89	10.85	13.37	13.32	11.24
b	0.36	0.34	0.50	0.41	0.42	0.65	3.28	3.07	4.56	3.72	3.78	5.88
t -Statistics for Means						t -Statistics for Means						
a	4.78	4.53	1.94	5.57	5.28	2.14	3.81	3.72	2.27	4.13	4.04	2.45
b	1.31	1.29	1.29	1.29	1.27	1.25	1.03	1.00	1.01	1.01	0.98	0.98

using β s on the NYSE value-weight market portfolio estimated with two to five years of past monthly returns (as available). Panel A of the table shows that average monthly and annual postformation returns initially increase with post-formation β s, but the relation between average return and β is rather flat from the fourth to the tenth β decile. This pattern in average returns on β portfolios is similar to KSS' Table I. The spread in average returns for their β portfolios is larger than ours, but including AMEX stocks also makes their β sort more like a sort on size than ours.

Panel A of Table I confirms KSS' evidence that β s estimated from monthly and annual returns are different. For the purposes of inferences about the average slopes from cross-section Fama-MacBeth (FM 1973) regressions of return on β , however, the important fact is that postformation β s estimated on annual or monthly returns, and using either an equal-weight or value-weight NYSE market, are near perfect linear transforms of one another. Rounded to two decimals, the correlations between the different β s range from 0.98 to 1.00.

Panel B of Table I shows that in univariate FM regressions of return on β , different β s produce different average slopes. In particular, the average slopes for the regressions that use annual β s to explain returns are about 50 percent larger than the averages for the regressions that use monthly β s. Why? The spread in the monthly β s is about 50 percent larger than the spread in the annual β s. Since the regressions are asked to explain the same dependent returns, and since the different β s are almost perfectly correlated, the smaller spreads in the annual β s lead to larger average slopes. However, although the average slopes in the annual- β regressions are larger, the near-perfect correlation among the β s leads to t -statistics for the average slopes that are nearly identical. (Jegadeesh (1992) reports similar results. See also Chopra, Lakonishok, and Ritter (1992), and Chan and Lakonishok (1993)).

In their cross-section regressions, KSS explain monthly returns with β s from regressions of annual returns on an equal-weight market return. One can argue that this combination is far from the spirit of the CAPM. In light of the recent articles of Roll and Ross (1994) and Kandel and Stambaugh (1995), however, we doubt that there is much future in debates about which approach produces bigger β premiums in cross-section regressions. Kandel and Stambaugh (1995) show that if the market proxy is not exactly mean-variance-efficient *with respect to parameters computed from the sample data*, it is possible to form portfolios that produce essentially any univariate β premium in ordinary-least-squares cross-section regressions. This conclusion about β premiums also applies to cross-section regressions that use β and other variables to explain average return. Moreover, generalized-least-squares (GLS) regressions, like those in Amihud, Christensen, and Mendelson (1992), are no cure. Roll and Ross (1994) note that a positive β premium in univariate GLS cross-section regressions simply says that the market proxy has a higher expected return than the global-minimum-variance portfolio.

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B. Portfolios Formed on Size and β

Kandel and Stambaugh (1995) and Roll and Ross (1994) teach us to be wary of inferences about β premiums from FM cross-section regressions. Cross-section regressions can, however, be useful in judging whether β suffices to explain expected return. Like KSS, we address this issue by testing whether size adds to the explanation of average return provided by β . Summary statistics and FM cross-section regressions for 100 portfolios formed yearly first on size (deciles) and then on β are in Tables II and III.

The KSS cross-section regressions (their Table II) consistently show that, on a statistical basis, size improves on β 's explanation of average return. This evidence is inconsistent with the CAPM, but KSS argue that the errors of β 's predictions of average return are not large. The message from our Tables II and III, however, is that size does make a large incremental contribution to the explanation of average return.

Table II shows that the size sort produces a strong spread in both postformation β and average return. The average return for 1928–1993 on the smallest decile of NYSE stocks exceeds that for the largest decile by 1.31 percent per month; the spread in β s is a healthy 0.93. Table II also shows that the second pass sort on preformation β s produces large spreads in postformation β s that are independent of size. As in FF (1992), however, the β sorts do not produce much spread in average return.

It seems safe to predict that a single average premium for β cannot capture both the strong positive relation between β and average return produced by the size sort in Table II and the weak relation between β and average return in the β sorts. Table III confirms this prediction. Like KSS, our univariate regressions of return on β produce positive average premiums that are more than 2.0 standard errors from 0.0. But the univariate β regressions leave an unexplained size effect. The spread in the average residuals from the smallest to the largest size decile is 0.66 percent in the regressions for monthly returns and 7.02 percent in annual returns. There is an even more systematic pattern in the average residuals for the β sorts, which are large and positive for low- β portfolios and strongly negative for high- β portfolios. The spread in the average residuals from the lowest to the highest β deciles of NYSE stocks is 0.56 percent in the regressions for monthly returns and 7.25 percent in annual returns.

In short, for portfolios formed on size and β , the average β premiums from univariate regressions of return on β underestimate the positive relation between β and average return produced by the size sort and overestimate the relation between β and average return produced by the β sort. For the extreme size and β deciles, these CAPM pricing errors are large. These results suggest that β alone cannot explain average return. As in KSS, the bivariate regressions of return on β and size in Table III confirm that size always adds substantially to β 's description of average stock returns.

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Table II

**Summary Statistics for 100 Equal-Weight NYSE Portfolios Formed
on Size and Then β : 1928–1993, $N = 792$**

Starting in 1927, ten portfolios of the NYSE stocks on CRSP are formed in June of each year based on size (market capitalization, price times shares outstanding). Each size decile is then subdivided into β deciles using β s for individual stocks that are the sum of the slopes from regressions of monthly returns on the current and one lag of the value-weight NYSE market return. The formation period β s use 24 to 60 months of past returns (as available), except for 1927, where 18 months of returns are used. Equal-weight monthly returns on the 100 portfolios are calculated from July to June of the following year, yielding time-series of returns for July 1927 to December 1993. The VW-S β s shown are the sums of the slopes from regressions of monthly postformation returns for the February 1928 to December 1993 period on the current and one lag of the value-weight NYSE market return. Average Ln (Size) is the average across postformation months of the average monthly value of the natural log of size for the stocks in each portfolio. The Ave column of each block is the average across β deciles of the parameter values for a size decile. The Ave row of each block is the average across size deciles of the parameter values for a β decile.

	Low β	2	3	4	5	6	7	8	9	High β	Ave
Panel A: Average Monthly Postformation Returns											
Small	2.15	2.30	2.20	2.66	2.16	1.89	2.27	2.12	2.23	1.82	2.18
2	1.43	1.65	1.55	1.54	1.65	1.91	1.64	1.63	1.41	1.31	1.57
3	1.42	1.53	1.30	1.56	1.56	1.39	1.51	1.48	1.53	1.26	1.45
4	1.15	1.23	1.40	1.31	1.46	1.41	1.10	1.42	1.26	1.24	1.30
5	1.14	1.37	1.35	1.45	1.22	1.46	1.33	1.39	1.13	1.21	1.31
6	1.24	1.20	1.05	1.15	1.34	1.31	1.29	1.10	1.32	1.27	1.23
7	0.93	1.20	1.06	1.14	1.38	1.08	1.08	1.01	1.28	1.15	1.13
8	1.11	1.12	1.08	1.15	1.11	1.21	1.21	0.79	1.25	1.16	1.12
9	0.84	0.94	1.07	1.12	1.17	1.22	1.01	1.14	1.05	0.95	1.05
Big	0.80	0.82	0.85	0.85	0.91	0.97	0.92	0.84	0.86	0.84	0.87
Ave	1.22	1.34	1.29	1.39	1.40	1.38	1.33	1.29	1.33	1.22	
Panel B: Postformation VW-S β s											
Small	1.47	1.90	1.56	2.03	2.06	1.69	1.97	1.99	2.21	2.15	1.90
2	1.45	1.43	1.71	1.59	1.52	1.74	1.89	1.84	1.76	1.89	1.68
3	1.29	1.33	1.21	1.49	1.58	1.52	1.68	1.70	1.81	1.91	1.55
4	1.16	1.20	1.16	1.32	1.49	1.41	1.49	1.51	1.76	1.92	1.44
5	0.97	1.13	1.22	1.34	1.36	1.50	1.38	1.70	1.47	1.75	1.38
6	0.74	1.01	1.09	1.30	1.22	1.36	1.37	1.60	1.59	1.77	1.31
7	0.72	0.93	1.03	1.08	1.32	1.25	1.31	1.39	1.47	1.83	1.23
8	0.64	0.76	0.93	1.10	1.17	1.25	1.24	1.26	1.52	1.58	1.15
9	0.63	0.78	0.92	1.11	1.10	1.08	1.14	1.29	1.30	1.53	1.09
Big	0.61	0.81	0.76	0.88	0.93	0.98	1.05	1.11	1.15	1.42	0.97
Ave	0.97	1.13	1.16	1.32	1.38	1.38	1.45	1.54	1.60	1.78	
Panel C: Average Ln (Size)											
Small	8.59	8.61	8.62	8.59	8.57	8.55	8.54	8.54	8.55	8.42	8.56
2	9.40	9.42	9.41	9.42	9.40	9.41	9.39	9.40	9.38	9.36	9.40
3	9.94	9.93	9.93	9.92	9.90	9.91	9.92	9.91	9.89	9.88	9.91
4	10.33	10.34	10.34	10.34	10.34	10.35	10.33	10.34	10.31	10.32	10.34
5	10.73	10.75	10.75	10.76	10.74	10.75	10.74	10.73	10.72	10.73	10.74
6	11.16	11.14	11.13	11.14	11.14	11.15	11.14	11.12	11.13	11.12	11.14
7	11.57	11.57	11.57	11.58	11.58	11.56	11.55	11.54	11.56	11.55	11.56
8	12.09	12.09	12.08	12.08	12.08	12.08	12.09	12.05	12.05	12.06	12.08
9	12.71	12.73	12.73	12.71	12.74	12.72	12.71	12.71	12.70	12.69	12.71
Big	14.17	14.18	14.15	14.14	14.31	14.27	14.24	14.10	14.07	13.76	14.14
Ave	11.07	11.08	11.07	11.07	11.08	11.08	11.07	11.04	11.04	10.99	

V. Conclusions

Confirming Banz (1981), sorts on size and β like those in KSS or our Tables II and III consistently reject the central CAPM hypothesis that β suffices to explain expected return. Moreover, in recent years the size effect has been displaced as the prime embarrassment of the CAPM. There is much evidence

Table III
Cross-Section Regressions for 100 Equal-Weight Size- β Portfolios:
1928–1993, $R = a + b\beta + s\text{Ln}(\text{Size}) + e$

In June of each year beginning in 1927, NYSE stocks on CRSP are sorted into size deciles, which are then subdivided into β deciles (see Table II). Equal-weight monthly returns on the 100 portfolios are calculated from July to June of the following year. Annual returns are obtained by compounding the equal-weight monthly returns. Panel A shows the average slopes (Means) and their t -statistics from univariate cross-section regressions of monthly and annual postformation simple returns on the 100 size- β portfolios on six estimates of their 1928–1993 postformation β s, and bivariate regressions of the 100 size- β portfolio returns on their β s and $\text{Ln}(\text{Size})$, the natural log of the average size of the stocks in each of the 100 portfolios at the end of the previous month or year. The β s are estimated using all postformation monthly (VW, VW-S, EW, EW-S) or annual (VWA, EWA) returns for 1928–1993 and the value-weight (VW, VW-S, VWA) or equal-weight (EW, EW-S, EWA) NYSE market portfolio. VW-S and EW-S β s are the sums of the slopes from the regressions of the monthly postformation size- β portfolio returns on the current and one lag of the market return. VW, VWA, EW, and EWA β s use only the contemporaneous market returns. The average slopes in the cross-section regressions of returns on $\text{Ln}(\text{Size})$ alone (not shown) are -0.19 ($t = -3.60$) for monthly returns and -2.67 ($t = -3.23$) for annual returns. Panel B shows the average residuals from the univariate regressions of monthly (annual) returns on the VW-S (VWA) β s. The Ave column of each block in part B is the average across β deciles of the average residuals for a size decile. The Ave row of each block is the average across size deciles of the average residuals for a β decile.

Panel A: Regression Coefficients												
Monthly Dependent Returns: 792 Months							Annual Dependent Returns: 66 Years					
	VW	VW-S	VWA	EW	EW-S	VWA	VW	VW-S	VWA	EW	EW-S	EWA
Means							Means					
<i>a</i>	0.50	0.36	0.17	0.43	0.43	0.38	7.26	4.68	0.58	5.61	5.33	3.47
<i>b</i>	0.65	0.70	0.84	0.89	0.88	0.94	7.83	9.13	12.08	11.56	11.84	13.73
<i>t</i> -Statistics for Means							<i>t</i> -Statistics for Means					
<i>a</i>	2.55	1.72	0.68	2.26	2.28	1.84	2.69	1.71	0.16	2.23	2.16	1.24
<i>b</i>	2.18	2.64	2.98	2.67	2.86	3.27	2.13	2.58	2.76	2.62	2.77	2.89
Means							Means					
<i>a</i>	3.01	2.86	2.76	2.83	2.68	2.30	45.50	43.46	37.27	43.41	40.97	29.30
<i>b</i>	0.10	0.15	0.18	0.20	0.26	0.37	-0.75	0.29	2.71	0.37	1.48	6.29
<i>s</i>	-0.18	-0.17	-0.16	-0.17	-0.15	-0.13	-2.75	-2.65	-2.28	-2.64	-2.48	-1.69
<i>t</i> -Statistics for Means							<i>t</i> -Statistics for Means					
<i>a</i>	5.92	5.73	5.33	5.66	5.40	4.98	4.61	4.58	4.82	4.50	4.43	4.33
<i>b</i>	0.40	0.62	0.72	0.65	0.83	1.46	-0.27	0.11	0.79	0.11	0.43	1.43
<i>s</i>	-4.16	-4.12	-3.99	-4.02	-3.78	-3.47	-3.63	-3.67	-3.88	-3.61	-3.56	-3.63

Table III
Continued

Panel B: Average Residuals from Univariate β Regressions											
	Low β	2	3	4	5	6	7	8	9	High β	Ave
Monthly Returns: VW-S β s											
Small	0.76	0.61	0.75	0.87	0.36	0.34	0.52	0.37	0.31	-0.05	0.49
2	0.06	0.29	-0.01	0.06	0.22	0.33	-0.04	-0.03	-0.18	-0.38	0.03
3	0.15	0.24	0.09	0.16	0.10	-0.04	-0.03	-0.07	-0.09	-0.44	0.00
4	-0.02	0.03	0.23	0.03	0.05	0.05	-0.31	-0.00	-0.33	-0.47	-0.07
5	0.10	0.22	0.14	0.16	-0.09	0.06	0.00	-0.16	-0.27	-0.38	-0.02
6	0.36	0.13	-0.08	-0.12	0.12	-0.00	-0.03	-0.37	-0.16	-0.32	-0.05
7	0.07	0.19	-0.01	0.03	0.09	-0.16	-0.20	-0.32	-0.11	-0.49	-0.09
8	0.30	0.23	0.07	0.02	-0.08	-0.03	-0.02	-0.45	-0.18	-0.31	-0.04
9	0.04	0.04	0.06	-0.02	0.04	0.10	-0.15	-0.12	-0.22	-0.48	-0.07
Big	0.01	-0.11	-0.03	-0.13	-0.10	-0.08	-0.18	-0.30	-0.31	-0.51	-0.17
Ave	0.18	0.19	0.12	0.11	0.07	0.06	-0.04	-0.14	-0.15	-0.38	
Annual Returns: VWA β s											
Small	5.88	7.95	9.85	11.66	6.12	2.37	7.68	4.97	4.21	-2.90	5.78
2	1.51	1.74	-0.31	-0.00	2.33	3.56	-1.53	-3.07	-6.10	-5.37	-0.72
3	2.24	2.42	2.25	2.28	0.60	-1.21	-1.00	-1.20	-1.97	-5.93	-0.15
4	0.20	-0.38	3.56	-0.31	0.52	0.40	-3.33	-0.58	-5.47	-5.75	-1.11
5	0.37	3.16	1.47	2.48	0.44	0.79	-0.28	-1.20	-3.66	-4.55	-0.10
6	4.91	1.94	-0.72	-1.06	0.58	0.00	-0.89	-5.07	-1.80	-3.97	-0.61
7	1.40	4.06	0.41	-0.34	0.28	-1.66	-3.16	-3.83	-1.33	-5.68	-0.99
8	4.97	2.77	1.19	0.45	-0.93	0.29	-0.20	-5.92	-3.10	-3.00	-0.35
9	1.31	1.07	0.75	0.65	1.08	1.59	-1.80	-1.31	-2.59	-5.85	-0.51
Big	0.96	-0.33	0.87	-0.61	-0.38	-0.73	-0.67	-2.62	-3.11	-5.77	-1.24
Ave	2.37	2.44	1.93	1.52	1.06	0.54	-0.52	-1.98	-2.49	-4.88	

that other variables (like earnings/price, cashflow/price, BE/ME, and past sales growth), add even more significantly to the explanation of average return (Basu (1983), Chan, Hamao, and Lakonishok (1991), FF (1992, 1993, 1996), and Lakonishok, Shleifer, and Vishny (1994)). Unlike size, these other variables (especially BE/ME) show little relation to estimates of market β s (FF (1992, 1993, 1994), Lakonishok, Shleifer, and Vishny (1994)), so a CAPM β -proxy story for their role in average returns is unlikely.

It is, of course, possible that the apparent empirical failures of the CAPM are due to bad proxies for the market portfolio. In other words, the true market is mean-variance-efficient, but the proxies used in empirical tests are not. In this view, revival of the CAPM awaits the coming of M . The true market portfolio will cast aside the average return anomalies of existing tests and reveal that β suffices to explain expected return.

This bad-market-proxy argument, however, does not justify the way the CAPM is currently applied, for example, to estimate the cost of capital or to evaluate portfolio managers. The bad market proxies used in tests of the CAPM are similar to those used in applications of the model. If the common

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market proxies are inefficient, then applications that use them rely on the same flawed estimates of expected return that undermine empirical tests of the CAPM. Like the redemptive empirical tests, valid applications of the CAPM await the coming of M .

In our view, the evidence that β does not suffice to explain expected return is compelling. And the average-return anomalies of the CAPM are serious enough to infer that the model is not a useful approximation. Our bet, made concrete in FF (1993, 1994, 1995, 1996), is that the payoffs in empirical asset pricing are in showing that the failures of the CAPM can be explained by multifactor ICAPM or APT alternatives—or that they are consistent with specific irrational-asset-pricing stories.

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Personal Finance

Tony Tassell: The time has come for the CAPM to RIP

FEBRUARY 9 2007

Few theories are more influential or important in driving financial markets as the inelegantly-named capital asset pricing model. Too bad it does not appear to work very well.

The CAPM, as it is widely known, is a cornerstone of modern financial market analysis, studied like a rosary by analysts and executives at business school. Most financial directors use it to assess everything from the viability of a new project to their cost of capital. Most stock market analysts consider it an essential tool.

But it has faced increasing criticism in recent years as unworkable in the real world, even from luminary market academics such as Harry Markowitz who laid the groundwork for the CAPM with research in 1950s on efficient portfolios.

CAPM is basically a model for valuing stocks or securities by relating risk and expected return. Developed separately by William Sharpe, John Lintner and Jack Treynor, it is based on the idea that investors demand additional expected return to take on additional risk.

It then assumes markets are efficiently priced to reflect greater returns for greater risk. The risk is assessed on a stock or security's so-called beta, a measure of a company's volatility and correlation with the market as a whole. A company with a share price that tends to rise and fall more than the market will have a high beta and vice versa.

It is a seductively simple, catch-all theory to quantify risk and forecast returns. It has spurred the development of quantitative investing.

But there is a problem. James Montier, analyst at Dresdner Kleinwort, says CAPM has become the financial theory equivalent of Monty Python's famous dead parrot sketch. He says the model is empirically bogus – it does not work in any way, shape or form. But like the shopkeeper who insists to a customer with a dead parrot in the sketch that the bird is merely resting, financial markets are in denial.

“The CAPM is, in actual fact, Completely Redundant Asset Pricing (CRAP),” he says.

Some of the most damning evidence came from an exhaustive 2004 study by Eugene Fama and Kenneth French, the academics who helped develop the efficient markets theory in the early 1970s, that argued stocks

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are always correctly priced as everything that is publicly known about the stock is reflected in its market price.

The study looked at all stocks on the New York Stock Exchange, the American Stock Exchange and Nasdaq from 1923 to 2003. As Montier states, the study shows CAPM woefully underpredicts the returns to low beta stocks and massively overstates the returns to high beta stocks. “Over the long run there has been essentially no relationship between beta and return,” he says.

Fama and French themselves concluded that while CAPM was a theoretical tour de force, its empirical track record was so poor that its use in “applications” was probably invalid. In others words, CAPM is a fine theory but useless in the real world.

A similar study of the 600 largest US stocks by Jeremy Grantham, the value investor, last year yielded similar results. It showed from 1969 to the end of 2005, the lowest decile of beta stocks – notionally the lowest risk – outperformed by an average 1.5 per cent a year. The highest beta stocks, or the riskiest, actually underperformed by 2.7 per cent a year.

The problems in the CAPM lie in its assumptions, particularly those used to derive the efficient portfolio that is used as a benchmark for the model in theory. The most commonly-cited criticism is an implicit assumption that that all investors can borrow or lend funds on equal terms.

Other assumptions that have been criticised include: that there are no transaction costs, that all investors have a “homogeneity” of expectations and risk appetites and that investors can take any market exposure without affecting prices. It also assumes no taxes so investors are indifferent between dividends and capital gains.

Markowitz himself noted that the CAPM is like studying “the motions of objects on Earth under the assumption that the Earth has no air”.

“The calculations and results are much simpler if this assumption is made. But at some point, the obvious fact that on Earth, cannonballs and feathers do not fall at the same rate should be noted,” he says.

Current market conditions might be exacerbating problems. Vineer Bhansali, head of portfolio management analytics at Pimco, adds that the increasing availability of leverage for some investors may actually drive all risky security prices higher.

Grantham says the flaws in the CAPM are probably inconvenient enough for the academic financial establishment to want to ignore it. But there ought to be more debate, particularly in using beta as a risk benchmark.

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The concept of pursuing absolute returns rather than relative performance is now widely debated. There needs to be a similar evolution in market thinking on how risk is defined, measured and dealt with.

Montier cites a quote from legendary investor Ben Graham: “What bothers me is that authorities now equate the beta with the concept of risk. Price variability, yes; risk, no. Real investment risk is measured not by the per cent a stock may decline in price in relation to the general market in a given period but by the danger of a loss of quality and earning power through economic changes or deterioration in management.”

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THE CASE AGAINST THE USE OF THE CAPITAL ASSET
PRICING MODEL IN PUBLIC UTILITY RATEMAKING

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and Walter A. Smith, Jr. ‡*

INTRODUCTION

In the mid-1960's, the academic community introduced a mathematical formula called the Capital Asset Pricing Model ("CAPM") which, it was contended, could be reliably employed to estimate any given asset's expected rate of return. In the succeeding years, the formula has received substantial attention and been lent considerable credence, not only in academia, but in the business and investment worlds as well. In addition, and most importantly for purpose of this comment, CAPM has been used extensively by regulatory commissions to establish rates of return for utilities.

The most recent evidence, however, strongly suggests that there are serious shortcomings associated with CAPM, and that, in the utility rate-making context at least, it can produce totally unreliable results. And yet, at the very time when the case against CAPM has become so formidable, the ratemaking bodies' reliance on it appears to have reached its zenith. The purpose of this comment, therefore, is to marshal the evidence concerning the inappropriateness of CAPM's use as an estimator of a utility's required rate of return, and to question the ratemakers' continued reliance upon it.

The comment is organized as follows. First, a description is offered of the CAPM formula, its theoretical underpinnings, and its key components. This is followed by a survey of CAPM's use by the regulatory commissions in establishing utility rates of return. Finally, the case against that use is made.

Before beginning, one other point should be made. The analysis offered below is based not so much upon our own independent expertise, as upon the work of financial economists, investment analysts, statisticians, and utility managers who are expert in, and have a thorough command of, the complexities of the CAPM theory. The case made here against CAPM, however, is a lawyer's case. It is intended to summarize, in laymen's terms, the evidence which is available to a regulatory body considering the use of CAPM. In our opinion, the evidence shows that CAPM is a highly suspect ratemaking tool, and that until its reliability is more clearly established, it should either be rejected, or at the very least modified, in future rate proceedings.

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I. *The CAPM Theory*

Developed and refined over the past two decades,¹ the theory of CAPM, generally stated, is that the expected rate of return on any given asset is a function of the risk which investors perceive to be associated with that particular asset, as compared with the risk of all other assets in which they might invest; that the only risk for which investors expect to be compensated is that risk which cannot be "diversified away" (cancelled out) by holding the asset as part of a "well-diversified portfolio"; that this "nondiversifiable risk" is composed solely of "market risk," *i.e.*, risk which affects the market as a whole, and which therefore affects every asset in the market; that all risk except this non-diversifiable market risk is in fact diversified away when an asset is held as part of a well-diversified portfolio; that investors therefore regard as irrelevant any risks peculiar to a particular asset or to the business or industry of which that asset is a part, and expect compensation only for the way in which the particular asset is affected by market risk.

It is further said, according to the CAPM theory, that the expected compensation on the particular asset has two components — first, the current return due on a riskless investment (such as a U.S. treasury bill) and, second, a "risk premium" above the riskless rate; that this "risk premium" component (or "excess return" component) of a given asset's expected return should be measured in terms of how volatile these excess returns have been on the particular asset as compared with the volatility of the excess returns on the average - risk asset in the market; that this volatility-comparison measurement may be taken by correlating the past excess returns on the given asset with the past excess returns on the average asset, the latter being represented by some broad market index of equity securities such as the Standard & Poor's 500; that this correlation of past excess returns results in a coefficient called "beta," which quantifies the volatility (and therefore the riskiness) of the asset in question relative to the volatility of the average asset; that every asset's risk premium is directly proportional to how much more or less volatile than the market's excess returns have been the asset's excess returns; that beta reflects this proportionality and fully captures the asset's risk premium; and therefore, that once an asset's beta has been determined, nothing more about the asset need be known in order to reliably estimate its expected rate of return.

Based on the foregoing CAPM assumptions, the expected rate of return (or cost of equity) for a given asset is computed according to the following formula:

$$\begin{aligned} \text{expected rate of return} &= \text{risk-free rate} + \text{beta} \times \\ &(\text{market rate} - \text{risk-free rate}) + \text{alpha} \end{aligned}$$

The *risk-free rate* in the formula is usually determined by examining the current yields offered on treasury bills, which constitute essentially risk-free investments.

¹The original development of CAPM is usually attributed to W.F. Sharpe ("Capital Asset Prices: A Theory of Market Equilibrium Under Conditions of Risk," 19 *Journal of Finance* 425 (Sept. 1964)) and J. Lintner ("The Valuation of Risk Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets," 47 *Review of Economics and Statistics* 13 (Feb. 1965)).

The quantity *market rate — risk-free rate* (the excess return) represents the risk premium on the average asset. It is normally assumed to be equal to the average long-term return on the market index in excess of the average long-term risk-free rate.

Beta, as described above, purports to assess the riskiness of the given asset in relation to the riskiness of the average asset, and thereby indicates the amount of the asset's risk premium. A beta of 1.0 is said to indicate that the asset is equally risky as the market index because its excess returns fluctuate in line with and by the same percentage amount as the market's excess return; hence, the asset will have the same risk premium as the market itself. By the same token, a beta of 1.5 indicates that the asset is 50% riskier than the market index (because its excess returns increase (or decrease) at 1.5 times the magnitude of increases (or decreases) in the market's excess return) and therefore has an expected risk premium 50% higher than that of the market. It follows that an asset with a beta of zero has no market risk, and is therefore expected to return the riskless rate.

Finally, *alpha* represents that part of a security's expected return which cannot be attributed to, or explained by, the security's response to fluctuations in the market index. CAPM assumes that alpha is zero.

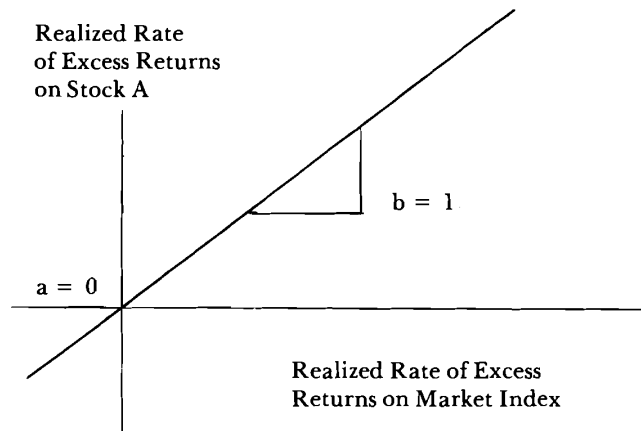
In other words, all that need be done to determine a particular asset's expected return, says CAPM, is, first, determine the asset's beta, then look up the current risk-free return, determine the current market (average asset) return, compute the resulting market "risk premium" by subtracting the risk-free return from the market return, next determine the given asset's risk premium by multiplying the market risk premium by the asset's beta, and, finally, add the current riskless return to the asset's risk premium. The sum, according to the theory, equals the asset's expected rate of return (or, in the case of a utility, its cost of equity).

Plainly, CAPM's centerpiece, and the key to its reliability, is beta. It is important, therefore, to understand how beta is computed, and how, in practice, it is used to estimate an expected rate of return or cost of equity for a particular asset.

Both the beta and the alpha of a particular asset are calculated through the use of a bivariate regression model. Such a model can be used to measure the relationship between any pair of variables. In connection with CAPM, the model is typically used to plot the relationship between the excess returns (*i.e.*, returns in excess of the riskless rate) on the asset in question and the contemporaneous excess returns on a general market index. Beta is determined by a line drawn which best "fits" the pattern of the plotted returns, and is the *slope coefficient* of the regression model; alpha, on the other hand, is determined by the point at which the "beta line" crosses the asset's excess rate-of-return line, and is the *intercept coefficient* of the regression model. Thus, every computation of beta also produces a computation of alpha.

Table 1 illustrates an asset (Stock A) that has a beta of 1.0 and an alpha of 0.

Table 1



Based on its beta of 1.0, Stock A is assumed to have the same nondiversifiable risk (sometimes referred to as “systematic” risk) as the market index, and, according to CAPM, any rise or fall in the market’s excess return should lead to a like rise or fall in the excess return on the stock. Therefore, the stock is expected to have a risk premium identical to that for the market. Furthermore, because its alpha is 0, it will be assumed that Stock A’s beta captures all of the return (above the riskless rate) which is expected by investors.

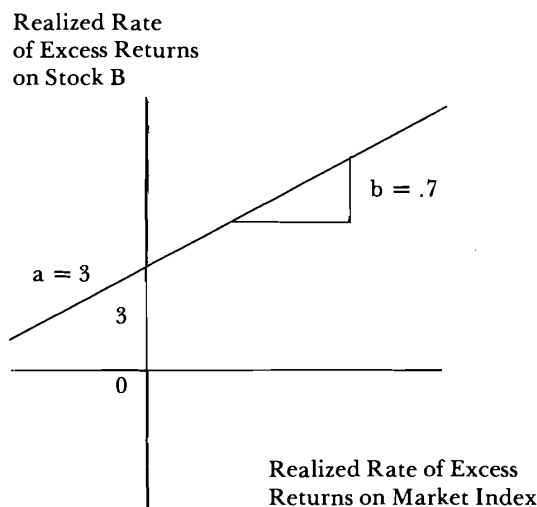
Using the previously-stated CAPM formula, and assuming that the appropriate risk-free rate and market premium can be determined, the cost of equity of Stock A can be easily computed. That equity cost (or required rate of return) is equal to the risk-free rate, plus beta times the risk premium (the return in excess of the risk-free rate) achieved by the market index, plus alpha. Assuming a risk-free rate of 10 percent and a market premium (excess return) of 5 percent, the formula produces the following result:

$$\begin{aligned} \text{return (equity cost)} &= 10\% + \\ 1 (\text{beta}) \times 5\% + 0 (\text{alpha}) &= 15\% \end{aligned}$$

Hence the return required on Stock A, and the issuing company’s cost of equity, is 15 percent.

Table 2 depicts an asset (Stock B) with a different degree of risk, and one that, as discussed below, is reflective of many utility stocks.

Table 2



Stock B is considered to have less systematic risk than the market index since its excess returns increase or decrease only 70 percent of the extent to which excess returns on the market index increase or decrease. Hence, using CAPM, Stock B will be expected to return a risk premium of only 70 percent of the market risk premium. However, Stock B has an alpha of + 3, indicating that it has earned a 3 percent return premium which is not captured, or measured, by beta. As will be discussed in Part III, experts have advanced several possible explanations for this positive alpha: for example, it could be a measurement of the extent to which beta misestimates the total nondiversifiable risk of the stock, owing to the fact that the particular market index contains less than all the available capital assets in which one might invest; it could demonstrate that the "true" market portfolio of all capital assets is not an "efficient" portfolio; it could indicate that although an asset's premium for nondiversifiable market risk is related to the asset's relative volatility, the premium is not necessarily directly proportional to that volatility; it could capture certain firm-specific (unsystematic) risks of the stock for which investors expect to be compensated; or, stated another way, it could represent the degree to which the well-diversified portfolio which investors could (or do) hold fails to "diversify away" all firm-specific risks; or it could reflect the fact that one of CAPM's necessary assumptions — that investors can freely borrow and lend at the risk-free rate — is not true. Whatever alpha's explanation as to a given asset, if it is not zero it means that beta does not tell the whole story of the asset's risk premium.

Using the previously-stated formula, and assuming the same 10 percent risk-free return and 5 percent excess return as assumed in the Stock-A

example, the return required on Stock B (and the cost of equity to the issuer) is computed as follows:

$$\begin{aligned} \text{expected return (cost of equity)} &= 10\% \\ + (.70 \times 5\%) + 3\% &= 16.5\% \end{aligned}$$

But CAPM ignores alpha, declaring that beta captures all nondiversifiable risk, and that such risk is the only thing for which investors expect to be compensated (above the riskless rate). Hence, CAPM assumes that alpha is (or in the long run will be) zero, and therefore dictates that Stock B's expected return is not 16.5%, but 13.5%. Clearly, the decision to rely on beta, but to disregard alpha, can be critical in the case of industries which have member-companies with persistent, positive alphas in their past returns. This is the case for the utility industry.

An examination of the data reported by Merrill Lynch in its January 1981 Quantitative Analysis indicates that utilities as a group, on average, had a beta of .6 and an alpha of .2 percent based on market prices as of December 31, 1980. Merrill Lynch Pierce Fenner & Smith Inc., *Quantitative Analysis* 22 (Jan. 1981). More specifically, communication utilities showed, on average, a beta of .5 and an alpha of 1 percent; electric utilities a beta of .6 and an alpha of .4 percent; gas distribution utilities a beta of .8 and an alpha of -1.2 percent; and gas pipeline utilities a beta of 1 and an alpha of -3.3 percent. *Id.* at 7. While these data are of course not definitive — before conclusions could be drawn with respect to the representative beta and alpha of a utility group or an individual utility, a long-term analysis of past betas and alphas would have to be made — nonetheless, the beta and alpha information provided in the Merrill Lynch report is indicative of the importance of alpha's treatment in conjunction with a CAPM analysis.

This importance is highlighted where the alphas of particular securities are examined. Table 3 provides a sample of utilities with high reported alphas.

Table 3

Issuer	Alpha (%)	Beta
Communications		
American Tel. & Tel.	1.1	.5
Cincinnati Bell.	2.3	.5
Gen'l Tel. & Elec.	1.5	.7
Western Union.	3.7	1.1
Electric		
Carolina Pwr. & Lt.	1.1	.7
Central Ill. Lgt.	2.5	.6
Commonwealth Ed.	3.5	.6
Consumers Power.	2.3	.6
Fitchburg G&E	2.0	.7

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Maine Pub. Svc.	2.5	.6
Middle South Utl.	1.1	.5
Pennsylvania P&L	2.9	.6
Rochester G&E	3.0	.6
Savannah Elec.	3.1	.6
United Illuminat	2.3	.6

Gas Distribution

Atlanta Gas Lt.	2.0	.8
Brooklyn Union	1.6	.5

Gas Pipeline

Pacific Lighting	1.1	.6
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Source: Merrill Lynch Pierce Fenner & Smith Inc., *Quantitative Analysis* (Jan. 1981).*

*It should be noted that the alphas and betas reported by Merrill Lynch are estimated alphas and betas, and are therefore subject to estimation errors.

If the positive alphas in Table 3 were representative of the long-term alphas of the indicated companies, and were CAPM employed as a method for establishing those companies' costs of equity, then plainly any decision to ignore the companies' alphas would have a significant downward impact on their resulting rates of return. As will be next discussed (Part II), that, in fact, describes the kind of decisions utility ratemakers have been rendering: in none of the rate cases where CAPM has been advanced or actually employed as a method for establishing a utility's cost of equity has alpha been taken into account. Indeed, in only one case was alpha even mentioned, and there it was ignored. Equally important, the decided rate cases failed to examine the mounting evidence (often because the evidence was not yet available) that the CAPM formula — with or without alpha — can yield highly arbitrary, suspect results.

II. CAPM's Application in Rate Proceedings

In the early 1970s, when the issue of CAPM's use in ratemaking was first debated, one of its proponents reported that the CAPM formula "ha[d] not yet been used in a regulatory proceeding."² However, over the past decade, CAPM has received increasing attention in rate proceedings before state and

²Stewart C. Myers, "The Application of Finance Theory to Public Utility Rate Cases," 3 *The Bell Journal of Economics and Management Science* 58, 69 (Spring 1972). Dr. Myers is Professor of Finance at the Sloan School of Management, Massachusetts Institute of Technology.

federal regulatory agencies. Moreover, it has often been relied upon by those agencies as a justification for lower equity costs than other methodological approaches would have indicated.

For example, as early as 1972, the FCC relied heavily on a CAPM analysis advanced by a trial staff witness in establishing AT&T's cost of equity. *American Telephone & Telegraph Co.*, 38 F.C.C. 2d 213 (1972). In that proceeding, CAPM indicated that AT&T (with a beta of approximately .7) was less risky than the average equity security in the marketplace, and hence that the cost of equity assigned AT&T should be lower than the returns provided on such a security. *Id.* at 237-38. The CAPM analysis endorsed by the Commission supported an equity cost finding substantially lower than that sought by AT&T.³

In recent years, state utility commissions have also begun to credit CAPM evidence in assigning equity costs to public utilities. For example, in *Portland General Electric Co.*, 23 PUR 4th 209 (Ore. Pub. Util. Comm'r 1977), Portland General Electric ("PGE") argued for a cost of equity in the range of 13.5 to 16 percent, whereas a witness for the Commissioner's staff, using CAPM, advocated a cost of equity of 11.1 to 12.25 percent. *Id.* at 219-21. After discussing the CAPM approach in detail and the staff's finding that the beta on the utility's stock was between .695 and .781, the Commissioner embraced the staff's CAPM evidence and premised his equity cost finding of 12.5 percent thereon.

The Commissioner's comments reveal how compelling he found the CAPM approach to be:

Unlike the several common equity returns estimators employed and criticized by company witnesses, the CAP model [*i.e.*, CAPM] has been subjected to the most intensive examination and testing in this proceeding and has been shown to be a reliable and useful measure of the only relevant risk component — systematic (nondiversifiable) risk.

* * * * *

The commissioner finds the CAP method and associated concepts advanced by staff to have general application to utilities whose common equity shares are traded and reported publicly. [*Id.* at 222.]⁴

³Interestingly, three years later when CAPM was advanced by Comsat witness Stewart Myers (the same witness who had offered CAPM testimony for the Commission's trial staff in the AT&T proceeding) to support the proposition that Comsat was significantly more risky than the average equity investment, the FCC was totally unresponsive and assigned Comsat a cost of equity substantially lower than that supported by the CAPM analysis. *Communications Satellite Corp.*, 56 F.C.C. 2d 1101 (1975), *remanded on other grounds sub nom.* *Communications Satellite Corp. v. FCC*, 611 F.2d 883 (D.C. Cir. 1977). The Commission ruled that the CAPM analysis was unreliable on a number of grounds, including the fact that the beta calculated for Comsat varied materially depending upon the data used in the calculation and the Commission's doubt that the basic premise of CAPM that risk should be assessed in the context of a diversified portfolio of investments was relevant to the Commission's task of determining Comsat's cost of equity. 56 F.C.C. 2d at 1169-70.

⁴In two earlier proceedings, the Commissioner had cautiously premised his cost-of-equity findings upon CAPM analyses advanced by the staff, but in each instance had called upon the staff to provide a more detailed evidentiary development of the CAPM approach in the future, a development ultimately provided in the *PGE* proceeding discussed above. *See California-Pacific Utils. Co.*, 20 PUR 4th 479, 487-90 (Ore. Pub. Util. Comm'r 1977); *Cascade Natural Gas Corp.*, 19 PUR 4th 170, 182-85 (Ore. Pub. Util. Comm'r 1977).

Similarly, in *Arkansas Western Gas Co.*, 16 PUR 4th 103 (Ark. Pub. Serv. Comm'n 1976), the Commission explicitly adopted the CAPM evidence sponsored by a witness for the Commission staff to establish the cost of equity for the utility in question. Noting that the "capital asset pricing model is a widely accepted model identifying the relationship between risk and return," the Commission found it "to be the appropriate method for establishing the cost of equity."⁵ *Id.* at 108-09.

In *Southern Bell Telephone & Telegraph Co.*, 18 PUR 4th 623 (S.C. Pub. Serv. Comm'n 1977), CAPM received further regulatory endorsement. Southern Bell had petitioned for an increase in its rates and offered evidence that it was entitled to a return on equity ranging from 14 to 15.2 percent. *Id.* at 628. Examining various indicators of AT&T and Southern Bell financial performance, the Commission concluded that a reasonable return on Southern Bell's common equity was less than that sought. The Commission explained that its conclusion was "confirmed" by the CAPM testimony offered by the staff's witness, which indicated that 10.45 percent was the upper limit to Southern Bell's appropriate equity cost (the range being 8.04 - 10.45 percent). *Id.* at 630, 634. The Commission was unimpressed by Southern Bell's claim that CAPM "is an academician's theory, rather than a 'decision maker's' tool" (*id.* at 630), noting that "[w]hile it may be that CAPM requires further development before it is completely accepted in rate making, we believe the theory underlying it is sound and the results produced more in touch with reality than those furnished to us by Southern Bell witnesses." *Id.* at 631. Because CAPM indicated that Southern Bell had already been authorized rate increases which would permit it to earn a return in excess of the CAPM-derived estimate of its cost of equity, the Commission denied outright Southern Bell's petition for a rate increase.⁶ *Id.* at 634.

In a substantial number of other rate proceedings, commission staffs have advanced, and regulatory commissions have relied, at least in part, upon CAPM evidence to support a lower cost of equity than that being sought by the regulated utilities. For example, cost of equity findings were importantly influenced by staff-submitted CAPM analyses in *New York State Electric & Gas Corp.*, 38 PUR 4th 220, 245-47 (N.Y. Pub. Serv. Comm'n 1980); *New York Telephone Co.*, 32 PUR 4th 353, 360-62 (N.Y. Pub. Serv. Comm'n 1979); *National Fuel Gas Distribution Corp.*, 28 PUR 4th 42, 62-63 (N.Y. Pub. Serv. Comm'n 1978); *Otter Tail Power Co.*, 30 PUR 4th 26, 47-51 (S.D. Pub. Utils. Comm'n 1979); *Northwestern Bell Telephone Co.*, 20 PUR 4th 462, 464-70

⁵*But cf.* *Southwestern Bell Tel. Co.*, 22 PUR 4th 209, 217-18 (Ark. Pub. Serv. Comm'n 1977), where the Commission rejected the CAPM evidence sponsored by a witness for Southwestern Bell on grounds that the beta he had calculated had arbitrarily been skewed upward to produce a higher risk assessment.

⁶CAPM has been advanced by the Commission staff in three other proceedings before the South Carolina Public Service Commission, and in each case the Commission has relied significantly, although not exclusively, upon the CAPM evidence in establishing the cost of equity of the utility under consideration. See *Southern Bell Tel. & Tel. Co.*, 35 PUR 4th 1, 23-31 (S.C. Pub. Serv. Comm'n 1980); *South Carolina Elec. & Gas Co.*, 34 PUR 4th 458, 477-85 (S.C. Pub. Serv. Comm'n 1979); *Southern Bell Tel. & Tel. Co.*, 30 PUR 4th 263, 277-85 (S.C. Pub. Serv. Comm'n 1979).

(Iowa Commerce Comm'n 1977);⁷ and *Intermountain Gas Co.*, 18 PUR 4th 79, 87-89 (Idaho Pub. Utils. Comm'n 1976).⁸

In none of the proceedings identified above was any mention made of the alpha of the utilities whose equity costs were analyzed in those proceedings. Indeed, only one case was found where the use of alpha was even addressed by a ratemaking body.

In the recent decision of *New York Telephone Company — Telephone Rates*, Case Nos. 27651 and 27710 (N.Y. Pub. Serv. Comm'n., Jan. 19, 1981), New York Telephone ("NYT") based its equity cost case in part upon a CAPM analysis that took into account a positive alpha calculation.⁹ NYT's evidence showed that beta alone supported a 15.1% equity cost, and that where an alpha of .9% was taken into account, that equity cost became 16%. Administrative Law Judges' Recommended Decision, Case Nos. 27651 and 27710 (Oct. 31, 1980) at 65. The alpha adjustment argued for by NYT was not strictly a reflection of NYT's (or AT&T's) own specific alpha, but rather was an adjustment based on the premium above the riskless rate which has consistently been earned (over the 1926-1978 period) by stocks with a beta of zero. NYT argued that the persistence of this alpha premium was clear evidence that stocks with a zero beta provide a return higher than the riskless rate, thereby demonstrating that beta alone does not adequately reflect the required return on investment. *Id.* at 64-65.

In their recommended decision, the Administrative Law Judges (the "Judges") evaluated the described NYT evidence and a CAPM analysis submitted in rebuttal by the Commission staff and then noted, first, that "[i]f anything is clear from the mass of material presented to us with respect to the CAPM method, it is that in its present stage of development we are far from

⁷Although basically adopting the staff's recommended cost of equity, the Iowa Commerce Commission was somewhat cautious in endorsing the CAPM methodology:

Although we find the theory to be consistent with cost of capital and fair rate of return determinations, much of it is still theoretical and subject to great debate.

* * * * *

[T]here is the question of the validity of beta as a measure of the cost of capital for an individual stock, not a part of a portfolio. While on each of these points we would have to agree that the record tends to support the staff position, we cannot accept the exact figure generated through this methodology as being the one and only correct cost of equity supported by the record. [20 PUR 4th at 469.]

⁸Commission staffs also submitted CAPM evidence in the following proceedings, although the role that that evidence played in the cost-of-equity determination ultimately made by the utility commission is not entirely clear: *Pennsylvania Pub. Util. Comm'n v. Philadelphia Elec. Co.*, 33 PUR 4th 319, 339-44 (Pa. Pub. Util. Comm'n 1980); *Southwestern Bell Tel. Co.*, 34 PUR 4th 224, 237-262 (Tex. Pub. Util. Comm'n 1979); *Granite State Elec. Co.*, 28 PUR 4th 240, 244 (N.H. Pub. Util. Comm'n 1978); *Idaho Power Co.*, 23 PUR 4th 299, 310-13 (Idaho Pub. Utils. Comm'n 1977); *Southern Bell Tel. & Tel. Co.*, 21 PUR 4th 451, 480-85 (Fla. Pub. Serv. Comm'n 1977).

In a survey as of February 1979, one commentator found that CAPM had been used in rate cases in 13 different states (Georgia, Iowa, Montana, New Hampshire, New Jersey, New Mexico, Nebraska, Ohio, Oregon, Rhode Island, South Carolina, Texas, and Washington). Harrington, "The Changing Use of the Capital Asset Pricing Model in Utility Regulation," 105 *Public Utilities Fortnightly* 28, 29 (Feb. 14, 1980). That survey revealed that in Oregon use of the CAPM model was required, that in South Carolina the commission staff would use CAPM in all future cases, and that only in Texas was there any predisposition against use of the model. *Id.* The survey further indicated that, in total, "38 states were considering or had seen the CAPM used." *Id.*

⁹AT&T offered less than a whole-hearted endorsement of the CAPM method, stating in its opening brief that "[w]hether CAPM can be used accurately in rate cases is still debatable. . . ." and taking the position that, if CAPM is to be used at all, it must take into account alpha. Administrative Law Judges' Recommended Decision, Case Nos. 27651 and 27710 (Oct. 31, 1980) at 64.

any consensus as to its proper use for regulatory purposes.” *Id.* at 68. With this preface, the Judges endorsed the theoretical validity of the alpha adjustment argued for by NYT, but rejected the precise computation of alpha on evidentiary grounds. *Id.* at 64-65, 69. Nevertheless, in order to take account of their view that to ignore alpha would cause the “CAPM-indicated cost of equity” to be understated “in some undetermined amount” (*id.* at 69), in assigning NYT a cost of equity the Judges based their decision on a weighted composite of the cost indicated by the DCF method¹⁰ and that indicated by the CAPM method (without the alpha adjustment), giving two-thirds weight to the former and only one-third weight to the latter. *Id.* at 73.

On appeal to the New York State Public Service Commission, NYT challenged the Judges’ failure to take specifically into account the alpha adjustment supported by its evidence when the Judges had conceded the theoretical validity of the adjustment. On grounds that NYT had failed to prove the validity of the specific alpha adjustment for which it contended, the Commission affirmed the Judges’ decision without addressing the correctness of that adjustment as a matter of theory. Slip Op. No. 81-3 (Jan. 19, 1981) at 31.¹¹ The Public Service Commission repeated this view on NYT’s Petition for Rehearing, stating that “[t]he possible need for such an adjustment [based on alpha] was a factor that we considered, but the record did not support a particular adjustment.” Order Responding to Petitions for Rehearing (May 14, 1981) at 5.

Thus, a review of the regulatory decisions reveals that the CAPM has achieved widespread acceptance among ratemaking bodies and is now routinely used by those bodies to establish rates of return for utilities. For the reasons that follow, we submit that that acceptance and use should be reconsidered.

III. *The Case Against CAPM*

In our estimation, those contending that regulatory commissions should establish utility rates of return based on CAPM should be asked first to show at least the following:

- (A) that the CAPM theory is sound, *i.e.*, that its underlying assumptions have been tested and are valid;
- (B) that the theory can be appropriately applied in the utility rate-making context; and
- (C) that, in practice, the theory produces reliable, objective estimates of a utility’s cost of equity.

We make this suggestion on the assumption that in any given rate proceeding the proponents of a particular rate-producing methodology can fairly be required to establish that methodology’s credentials. The required showing

¹⁰The DCF (discounted cash flow) method, simply stated, assumes that the cost of equity for a particular stock is equal to its dividend yield (current dividend divided by current price) plus the annual expected growth rate in dividends.

¹¹However, using updated return data, the Commission assigned NYT a higher cost of equity than that recommended by the Judges, and, although basing its finding upon both the DCF and CAPM methods, did not adopt the weighting approach employed by the Judges.

would of course vary with the proven reliability of the method at issue — we do not suggest that every rate proceeding should be burdened by the reinvention of the wheel as to every methodology employed in the proceeding. But our experience with and review of the regulatory case law is that the CAPM wheel appears never to have been invented at all. Rather, CAPM's basic credentials as a reliable ratemaking tool have simply been assumed.

This, we submit, is a mistake. Some minimal showing — such as the three elements suggested above — should be fairly well established before rates are built on a CAPM foundation.¹² Our review convinces us that the minimal showing has not been made; that, in fact, the weight of the available evidence is heavily against CAPM as to all three elements of that showing. That is the evidence we summarize below.

A. *The Validity of the CAPM Theory*

When we assert that the validity of CAPM's underlying assumptions has not been shown, we intend nothing sweeping. Rather, for purposes of the limited (but, we think, important) point we are attempting to make here, we do not presume to challenge the important principle that CAPM borrows from modern portfolio theory — that, generally speaking, the expected return on a particular asset may properly be assessed as if that asset were part of a portfolio, and that by holding the asset in a portfolio an investor can “diversify away” certain of that asset's risk. Neither do we question what we understand to be the basic proposition of CAPM itself — that investors, being risk-averse, tend to price assets so that the riskier ones have higher expected rates of return. Finally, we do not dispute the basic idea of beta — that assets' returns can generally be expected to move up and down in a pattern related to the movements of “the market” as a whole, and that the way in which a particular asset “responds” to market movements is an important indicator of that asset's risk.

But regulatory bodies should require some degree of *proof* before ascribing to these admittedly sound *general* principles a mathematical precision they have never been shown to have. One can accept that portfolios tend, through diversification, to cancel risks, that assets are generally priced to reflect nondiversifiable risks, that one such risk is market risk, and that beta is designed to capture how market risk affects a particular asset, without having to agree that once one has estimated an asset's beta, all that remains to be done to reliably *quantify* that asset's entire expected return is (in effect) to multiply the asset's beta times the market's expected return.

Rather than catalogue all the material assumptions that underlie the proposition that beta is sufficient to produce such a quantification, we discuss here only two that appear to us to be obvious and particularly important: the first necessary assumption is that there not only be a *relationship* between the market's excess return and every asset's excess return, but that every asset's

¹²The absence of any such showing has persuaded some utility commissions to reject CAPM. See, e.g., Connecticut Nat. Gas Corp., 37 PUR 4th 287, 328-29 (Conn. Div. Pub. Util. Control 1980) (characterizing CAPM as an elegant “black box” that “has not survived extensive regulatory scrutiny”).

expected excess return is *directly and exactly proportional* to the market's expected excess return, *i.e.*, that the trade off between market risk and return is a straight line; and the second necessary assumption is that once this measurement of expected return owing to market risk has been made (via beta), all expected return has been captured, *i.e.*, there is, by definition, no other risk but beta-risk for which investors expect to be compensated.

It seems to us plain that these two assumptions would have to be well-founded before the regulatory bodies' reliance on beta can be thought sound; and yet, we can find no specific evidence validating either assumption. Indeed, what evidence there is indicates that both assumptions are invalid.

1. *Beta Does Not Capture an Asset's Market-Risk*

Among the most recent and telling criticisms directed at beta have been those articulated by Richard W. Roll, a Professor of Finance at the UCLA Graduate School of Management and a leading advocate for the large (and growing) cadre of professionals who believe that CAPM provides an unreliable measure of asset risks and returns.¹³ Through Professor Roll's work, it has been demonstrated¹⁴ that CAPM depends upon a critical assumption which may or may not be true and which, unfortunately, cannot be (or, at least, to date has not been) tested.

The underlying premise of CAPM's reliance on beta is that every stock's expected excess return is directly proportional to the market's expected excess return, and that the degree of proportionality is exactly measured by beta. But, as a matter of mathematics, beta can be relied upon as a measurement of this assumed direct proportionality if, and only if, the market portfolio itself is what is called "mean-variance efficient." This means that, for beta to be sound, the market portfolio of all available assets must in fact provide the highest possible average (mean) return, given the variability (volatility) associated with that portfolio's return. And therein lies the problem.

As noted above, the market index conventionally employed in connection with a CAPM analysis is a broad index of stocks like the Standard & Poor's 500. However, use of such an index would be correct only if (i) it were a valid proxy (in the sense that it would be expected to yield the same beta) for a portfolio of all invested assets, including such difficult-to-measure assets as human capital and other non-traded assets, and (ii) if that portfolio of all invested assets were "mean-variance efficient."

Thus, there are really two problems, and neither one of them has been solved: first, it must be known whether the particular index being used is

¹³See generally Wallace, "Is Beta Dead?," *Institutional Investor* 23 (July 1980); Blustein, "Money Managers' Bedrock Theory of Investing Comes Under Attack," *Wall St. J.*, Sept. 8, 1980, at 13; Baron, "Assault on Beta Theory Jolting Money Managers," *L.A. Times*, Oct. 6, 1980, § B, at 1; and 7 *The Journal of Portfolio Management* (Winter 1981), an issue devoted to the "Is Beta Dead?" controversy.

¹⁴See Roll, "Performance Evaluation and Benchmark Errors (I)," 6 *The Journal of Portfolio Management* 5 (Summer 1980); Roll, "Performance Evaluation and Benchmark Error (II)," 7 *The Journal of Portfolio Management* 17 (Winter 1981); Roll, "Testing a Portfolio for Ex Ante Mean/Variance Efficiency," in E. Elton and M. Graber, eds., 11 *TIMS Studies in the Management Sciences* 135 (1979); Roll, "A Reply to Mayers and Rice," 7 *Journal of Financial Economics* 391 (1979); Roll, "Ambiguity When Performance Is Measured by the Securities Market Line," 33 *The Journal of Finance* 1051 (Sept. 1978); Roll, "A Critique of the Asset Pricing Theory's Tests," 4 *Journal of Financial Economics* 129 (Mar. 1977).

indeed a valid (or the best available) proxy for the true market portfolio; and second, it must be known that that true market portfolio is in fact efficient. The reason neither problem has been solved is that a portfolio of all assets (or an acceptable proxy therefor) has never been reliably formed and tested.

On the other hand, it *can* be tested whether the particular index being relied upon to estimate an asset's beta is itself efficient. Professor Roll has demonstrated, for example, that the S&P 500 — ordinarily relied upon in rate proceedings — is not efficient and therefore cannot produce sound betas. Moreover, each inefficient index (such as the S&P 500) will produce a different beta, and a different expected return — none of them reliable.

Since no valid, efficient proxies for the market as a whole have been developed, the beta of a stock as measured against a broad market index is not a reliable indicator and is, in one sense, wholly arbitrary in that it is solely a function of the index chosen. "For every asset, an index can be found to produce a beta of any desired magnitude, however large or small." Roll, "Ambiguity When Performance Is Measured by the Securities Market Line," 33 *The Journal of Finance* 1051, 1056 (Sept. 1978). Thus, since the beta of a stock varies depending upon the market index employed, since there is no dependable index which is truly reflective of the market as a whole, and since there is no way of knowing that that "market as a whole" is efficient, the most recent evidence strongly indicates that no reliable conclusions regarding the risk of an asset or the return required on it can be derived through the beta produced by any given market index. Hence, we submit, conclusions previously reached by the ratemakers through their application of CAPM (relying solely on beta) may have been totally arbitrary.

Moreover, the lack of an efficient, measurable market portfolio is not the only problem with the regulatory reliance on beta. There is considerable evidence that beta in any event does not necessarily capture all of an asset's expected return.

2. *Beta Does Not Fully Capture an Asset's Expected Return*

CAPM "says that [beta] is a complete and sufficient risk measure, that the expected risk premium demanded by investors is zero when beta is zero. . . ."¹⁵ But recent evidence shows that this critical CAPM assumption is simply not correct. In fact, the author of the quoted statement, who has offered evidence in several rate proceedings based on beta measurements, and who has generally been a proponent of such a use of beta,¹⁶ recently testified that CAPM "is probably not a complete description of equilibrium trade-off between risks and returns that actually prevails in capital markets."¹⁷

¹⁵Myers "On the Use of Modern Portfolio Theory in Public Utility Rate Cases: Comment," *Financial Management* 66, 67 (Autumn 1978).

¹⁶*See id.*; Myers, "On the Use of Beta in Regulatory Proceedings: A Comment," 3 *The Bell Journal of Economics and Management Science* 622 (Winter 1972).

¹⁷Testimony of Stewart C. Myers In the Matter of the Valuation Proceedings Under §§ 303(c) and 306 of the Regional Rail Reorganization Act, Tr. at 609 (June 6, 1979).

Some of the recent evidence shows that if an asset's beta is to be relied on to estimate its expected return, the asset's alpha also must be taken into account; other evidence shows that some assets consistently earn higher returns than that predicted by beta, and that those higher returns can be predicted by non-CAPM methods. Both kinds of evidence are wholly contrary to the assumption that beta is a complete measurement of an asset's expected return, that an asset with a zero beta has a zero risk premium, and that an asset with a beta of one has a risk premium identical to the market risk premium.

We will first describe the alpha-related evidence, and thereafter explain the evidence concerning beta's inability to predict returns.

The use of a market index to compute a given stock's beta and alpha requires, by definition, that the weighted (by market value) average beta of all the stocks in the index will be exactly 1.0, and the weighted average alpha will be zero. But just as a given stock's beta need not be exactly 1.0, neither need its alpha be exactly zero. Those who take into account a stock's estimated beta in estimating its expected return, but ignore its estimated alpha, do so on the theory that over the long term the particular stock's alpha will average to zero and that, therefore, it should not be treated as statistically significant. It is this theory which has been shown to be highly questionable, if not completely unsound.

In the current litigation designed to value the railroad properties taken by the United States for Conrail, Dr. Richard Meyer, Professor of Managerial Economics at the Harvard University Graduate School of Business Administration, presented a comprehensive survey demonstrating the relative predictive powers of beta and alpha in estimating assets' future returns. Beginning with the five-year period 1960-1965 and going up through the five-year period 1968-1973, Professor Meyer calculated the alpha and beta for every United States company for which dividend-adjusted stock return data were available. This ranged from a low of 666 firms for the 1960-1965 period to a high of 909 firms for the 1968-1973 period. He then compared the actual returns on the stocks for various multi-year periods following each of the five-year estimation periods in order to assess the predictive power of alpha and beta. *See* Testimony of Richard F. Meyer, pp. 58-61 (January 30, 1980). *In the Matter of the Valuation Proceedings under Sections 303(c) and 306 of the Regional Rail Reorganization Act of 1973*, Special Court Misc. No. 76-1 ("The Railroad Valuation Proceedings").

Using standard statistical measurements, the comparison showed that in some periods beta was more significant, and that in others alpha was more significant, but "that alpha and beta have approximately equal significance considered overall." *Id.* at 60. Crucial to these results is that "there is no evidence that alpha is systematically insignificant," even though CAPM assumes that it is. *Id.* Moreover, the study showed that "alpha tends to gain significance with the length of the forecasting period," *id.*, a fact of considerable importance in utility ratemaking where future costs of equity are often estimated for several years or more. Hence, Dr. Meyer concluded, notwithstanding CAPM's assumption that beta fully captures expected returns, "alpha may not arbitrarily be ignored. . . ." *Id.*

There are several theories for why particular assets evidence persistent alphas. One view, stated in Dr. Meyer's testimony, is that beta fails to capture all market-related risk associated with a particular asset since the market index used to compute beta does not contain all available assets. Alpha may measure the effect of these "missing" assets on investors' risk perceptions.¹⁸

A second explanation, presented by Dr. Robert H. Litzenger in *New York Telephone Company — Telephone Rates, supra*,¹⁹ is that alphas consistently appear for two reasons.²⁰ First, according to Dr. Litzenger, CAPM assumes incorrectly that investors are able to borrow and lend in unrestricted amounts at the risk-free rate, and the fact that they are not means that the relationship between risk and return is not directly proportional to the betas of individual securities. Second, the value-weighted portfolio of all New York Stock Exchange stocks often used to calculate the beta of individual stocks is not a representative surrogate for the market as a whole, but instead is relatively more risky than that market on average.²¹ As a consequence, contrary to the CAPM theory, stocks with betas of zero have, over the past 50 years, consistently earned a risk premium above the riskless rate. This additional risk premium is captured by alpha.²²

A third explanation for the appearance of alphas, as described by Dr. Richard Roll, is that alpha is simply a measurement of the inefficiency of the market index being used to compute beta; that is to say, if the market index were, as earlier discussed, mean-variance efficient, beta would be a reliable measurement of expected return and there would be no alphas. But since the index is not efficient, alpha will not necessarily be zero, and will capture that part of an asset's expected return not captured by beta.²³

Yet another view of alpha, expressed by a strong proponent of CAPM — Dr. Barr Rosenberg²⁴ — is that beta is a reliable measurement of market risk, and that such risk is an important factor affecting expected return; however,

¹⁸Testimony of Richard F. Meyer in The Railroad Valuation Proceedings, Tr. at 636 (Aug. 25, 1980).

¹⁹Dr. Litzenger is the C.O.G. Miller Distinguished Professor of Finance at the Graduate School of Business, Stanford University.

²⁰See Testimony of Robert H. Litzenger, New York Telephone Co. — Telephone Rates, Case Nos. 27651 and 27710 (N.Y. Pub. Serv. Comm'n), Tr. at 2122-28 (May 15, 1980). See also Litzenger, Ramaswamy, & Sosin, "On the CAPM Approach to the Estimation of a Public Utility's Cost of Equity Capital," 35 *The Journal of Finance* 369 (May 1980).

²¹As previously mentioned, beta is usually measured against a broad index of stocks, but theoretically should be assessed in light of all potential investments.

²²The actual alpha adjustment developed by Dr. Litzenger to account for this additional risk premium called for a .9% upward adjustment in the 15.1% NYT equity cost indicated by beta alone (or a total equity cost of 16.0%). The Public Service Commission's analysis of Dr. Litzenger's evidence suggests that had he demonstrated the value of NYT's particular alpha, rather than simply demonstrating the general inability of beta to capture an asset's total expected return, it might have adopted Dr. Litzenger's cost of equity figure. See Order Responding to Petitions for Rehearing (May 14, 1981) at 5 ("The possible need for such an [alpha] adjustment was a factor that we considered, but the record did not support a particular adjustment.").

Recently, Dr. Litzenger filed testimony in support of New York Telephone's pending application for a rate increase. His testimony included evidence showing that the alpha adjustment necessary for zero-beta stocks may be appropriately applied to stocks with betas in the more common ranges (from .5 to 1.5). See Testimony of Robert H. Litzenger, New York Telephone Co. General Rate Case, Case No. 27795 (N.Y. Pub. Serv. Com'n), Vol. V at 33-36 and Exhibit at Section 11 (May 1981). A decision by the Public Service Commission is not expected in that case until May 1982.

²³Roll, "Performance Evaluation and Benchmark Errors (II)," *supra*, at pp. 19-20; Testimony of Richard Roll in The Railroad Valuation Proceedings, pp. 72-73 (January 30, 1980), Tr. 102 (July 21, 1980), Tr. 240-44 (July 22, 1980), and Tr. 1277-78, 1290, 1376 (August 19, 1980).

²⁴Dr. Rosenberg is Professor and Director of the Berkeley Program in Finance at the Berkeley Business School, University of California.

since it may be that factors *other* than market risk affect investors' perceptions regarding expected returns, by definition there may be an expected return in excess of that measured by beta. It is this excess which, according to Dr. Rosenberg, alpha captures: "There is nothing mysterious about this alpha; it is simply an expression of judgment on the security's expected return." Rosenberg, "The Capital Asset Pricing Model and the Market Model," 7 *Journal of Portfolio Management* 5, 10 (Winter 1981).

Thus, there is substantial evidence indicating that alpha captures expected return missed by beta. There is, in addition, substantial evidence that CAPM simply misestimates expected return, whether or not the misestimation stems from the failure to consider alpha.

For example, in the previously mentioned railroad valuation proceedings, Dr. Richard Roll presented a study in which numerous portfolios of railroad stocks were specifically constructed so that they had betas exactly equal to 1.0. In all, 40 portfolios were observed over two successive five-year periods, and their returns measured relative to the market index (S&P 500) returns. Contrary to CAPM's assumption, all 40 portfolios consistently outperformed the market index, by an average of 5-8 percentage points²⁵ — meaning that if CAPM had been relied on (*i.e.*, the beta of 1.0 had been used to compute expected returns), the railroads' cost of equity would have been understated by those 5-8 points. Moreover, through the use of certain non-CAPM methods for estimating the railroads' cost of equity, Dr. Roll confirmed that that cost was in fact 5-8 points higher than the figure indicated by CAPM.²⁶

Furthermore, in those same railroad valuation proceedings, Dr. Stewart Myers, who *did* rely on beta in estimating the railroads' cost of equity, and who also rejected the need to take alpha into account, nevertheless testified on cross-examination that "the research suggests that the capital asset pricing model formula misses a measure of risk, and that measure of risk may be the standard deviation."²⁷ "Standard deviation" is a measurement of *total* volatility of the particular asset's returns, as contrasted with beta which is a measurement only of the asset's volatility *relative* to the market's volatility. Thus, to say that standard deviation may capture expected return not captured by beta is to agree that not *all* asset-specific risk is necessarily cancelled out (diversified away) by holding assets in portfolios. This necessarily means that beta-risk (nondiversifiable market risk) is not the only risk affecting investors' expectations — an idea plainly at odds with a key underpinning of the CAPM theory.

²⁵Testimony of Richard W. Roll in The Railroad Valuation Proceedings, pp. 74-80 (Jan. 30, 1980).

²⁶The two non-CAPM methods relied upon by Dr. Roll for estimating costs of equity were, first, measurements over two five-year periods of the actual differences in return between the market index and specially-constructed railroad portfolios ("minimum variance portfolios"), and, second, measurement of the actual difference in returns between the market index and general railroad portfolios over the past 50 years. *Id.* at pp. 42-67. Significantly, by relying on both beta and alpha Dr. Richard F. Meyer estimated virtually the same railroad cost of equity as did Dr. Roll. Testimony of Richard F. Meyer, *supra*, at p. 101. It should be noted that the Court in the Railroad Valuation Proceedings ultimately gave no weight to either Dr. Roll's or Dr. Meyer's testimony. The Court offered no explanation as to why it was unpersuaded by Dr. Meyer's testimony and, as to Dr. Roll's testimony, stated without analysis that his findings were not "statistically significant." Special Court for Regional Rail Reorganization, Opinion with respect to Valuation for Rail Use at 61-62. (Nov. 24, 1981).

²⁷Testimony of Stewart C. Myers in The Railroad Valuation Proceedings, Tr. at 565 (July 5, 1979).

In addition, there is recent evidence to suggest that market risk is not the only kind of systematic risk that affects assets' expected returns. Rather, as Professors Roll and Stephen Ross²⁸ have shown, through application of what they call the arbitrage pricing theory, that there are at least three, and possibly four, systematic factors — not just beta — which should be taken into account in estimating expected returns.²⁹ See Roll and Ross, "An Empirical Investigation of the Arbitrage Pricing Theory," 35 *The Journal of Finance* 1073 (Dec. 1980); Ross, "The Arbitrage Theory of Capital Asset Pricing," 13 *Journal of Economic Theory* 341 (Dec. 1976). As will be discussed below, one such additional systematic factor — interest-rate risk — has been clearly demonstrated to have a risk impact on expected utility returns which is not captured by beta.

Finally, the most recent empirical research demonstrates that portfolios of stocks with higher price/earnings ratios consistently earn higher returns than portfolios of stocks with low price/earnings ratios, even though all the portfolios have identical betas. And, similarly, portfolios of stocks in small firms consistently earn higher returns than portfolios of stocks in larger firms, again notwithstanding that both kinds of portfolios had identical betas. See Reinganum, "Misspecification of Capital Asset Pricing: Empirical Anomalies Based on Earnings Yields and Market Values," 9 *Journal of Financial Economics* 19 (Mar. 1981) 19-46; Banz, "The Relationship Between Return and Market Value of Common Stocks," 9 *Journal of Financial Economics* 3 (Mar. 1981).

This research does not necessarily demonstrate that a high P/E or small size is equated by investors with greater risk — it may be that the higher returns are equated with some other factor that happens fortuitously to be associated with high P/E and/or smaller firms. But what is necessarily demonstrated by this research is that some factor other than beta-measured risk affects investors' expected returns. *Id.* at 16-17; Reinganum, *supra*, at 44-45. That is enough to make reliance on CAPM questionable.

B. CAPM's Applicability in the Ratemaking Context

In addition to the foregoing general problems with the CAPM theory, there are additional problems with the theory when it is specifically considered in the ratemaking context. To date, these problems have not been resolved, nor even addressed by the ratemakers. We raise three of them here.

The first is that while CAPM assumes that an asset's "response" to market movements is all that need be assessed in estimating the asset's expected rate of return, in the case of utilities there is in fact another important risk to consider — the actions of the ratemakers themselves. The second problem is that utilities are peculiarly affected by a nondiversifiable risk other than market risk — interest-rate risk. Finally, the third problem is that CAPM's assump-

²⁸Stephen A. Ross is Professor of Finance at the School of Organization and Management, Yale University.

²⁹Moreover, the arbitrage pricing theory (APT) has two distinct advantages over CAPM. First, unlike CAPM, APT does not require that the universe of all available assets be included in the measured market portfolio; rather, APT yields statements of relative pricing based on "subsets" of all available assets. Second, unlike CAPM, APT does not require that the market portfolio be mean variance efficient in order that accurate systematic-factor estimates be made.

tion that everything but market (beta) risk will be cancelled out in a well-diversified portfolio depends on the further assumption that the potential up-side and down-side risk of the various assets in the portfolio is largely symmetrical — an assumption that may not be true of utilities, whose up-side potential is subject to regulatory ceilings.

1. *"Regulatory Risk" Is Not Captured by CAPM*

CAPM asserts that the way in which an asset's returns previously moved up or down in relation to the market's movements — a relationship captured by beta — is a fair estimate of the future volatility of the asset's returns. This future volatility, in turn, is said to be a fair indication of the asset's risk, and therefore of its expected return (relative to the market's risk and expected return). Whatever may be the accuracy of these propositions in the case of an unregulated company, there is serious doubt that they can be correct in the case of regulated utilities.

There are at least two reasons for this special doubt in the case of regulated assets. The first is that, as Messrs. Breen and Lerner³⁰ have pointed out, "[i]t is reasonable to believe . . . that regulatory decisions themselves directly affect the value of beta, for they influence the corporation's growth rate, stability, size, and payout."³¹ If this is so, the regulatory body must determine how its own decision may *change* the way in which the company's returns will in the future respond to the market, *i.e.*, how beta may change as a result of the rate-of-return decision.

Though Dr. Stewart Myers criticized some of the Breen and Lerner description of beta's use in regulatory proceedings, he expressly agreed that: "Breen and Lerner are correct in pointing out that regulatory decisions can affect utilities' risks. Further work is needed to identify situations in which the effect is empirically significant and to devise ways of taking the effect into account in such situations."³² Yet, to date the regulatory bodies have taken no account of the effect of regulatory risk on utility betas, and no study has been done showing how great might be the distortion in the utility betas which the regulatory bodies have relied on.

Second, the distortion in those utility betas may be severe if, as Professor Carleton³³ has argued, "regulation itself has been the main source of [utility] investor risk in recent years."³⁴ This investor risk stems from, among other things, uncertainty regarding what decision the regulators will make, when it

³⁰Dr. William J. Breen and Dr. Eugene M. Lerner are both Professors of Finance at the Graduate School of Management, Northwestern University.

³¹Breen and Lerner, "On the Use of Beta in Regulatory Proceedings," 3 *The Bell Journal of Economics and Management Science* 612, 621 (Winter 1972).

³²Myers, "On the Use of Beta in Regulatory Proceedings: A Comment," 3 *The Bell Journal of Economics and Management Science* 622, 626 (Winter 1972).

³³Dr. William T. Carleton is William R. Kenan, Jr., Professor of Business Administration at the University of North Carolina.

³⁴Carleton, "A Highly Personal Comment on the Use of the CAPM in Public Utility Rate Cases," *Financial Management* 57, 59 (Autumn 1978).

will be made, and the built-in lag between investor-expected and company-realized rates of return. This implies both a changing ability of utilities to react to market-wide risks, a changing character in those reactions, and, certainly, "non-constant expected rates of return to the firm and to investors over future periods."³⁵

All of this is at odds with CAPM's assumption that a given asset's returns react in a relatively constant manner to market-wide risks. In large part, in the case of a regulated utility, clearly this is not so; rather, the "regulatory risk" skews the reaction to market-wide risks. And there is no reason to suppose, nor evidence to demonstrate, that past betas capture the impact of that regulatory risk.

2. *Market Risk Is Not the Only Systematic Risk Affecting Utilities*

As suggested by the previously-mentioned work of Professors Ross and Roll, there may be several systematic (nondiversifiable) factors which affect a given asset's expected return. While market risk is plainly one such factor, it need not be the only one. For example, some assets may be more sensitive to interest-rate risks than is the market (or market index) as a whole, and, therefore, this greater sensitivity would not necessarily be captured by such assets' market (beta) risk. Recent empirical evidence has shown, as would be expected, that utilities are among those assets that are in fact more interest-rate sensitive than is the market as a whole. Chance, "Progress in Modeling Utility Stock Holding Period Returns," 107 *Public Utilities Fortnightly* 34 (May 7, 1981).

In the cited article, Professor Chance³⁶ shows that, if CAPM is to be used, an additional systematic-risk factor should be added to the CAPM formula in order to capture the interest-rate risk that beta misses. The value of this factor is determined through use of a bond index and represents the given utility's "response" to movements in that index, in much the same way that beta represents the utility's "response" to movements in a stock market index.³⁷ This additional nonstock market factor is generally referred to as "extra-market covariance."

What other "extra-market" factors would have to be added to the CAPM formula (in addition to the interest-rate factor) for it to be a full and fair measure of expected utility returns is not yet clear. What does seem clear, however, is that the CAPM formula which has been relied on in the past is not such a measure.

3. *Non-Market Utility Risk Is Not "Diversified Away"*

The third major problem peculiar to CAPM's application to regulated companies turns on a pivotal CAPM assumption — that all risk other than

³⁵*Id.* at 59; see Christy & Christy, "Who Says Utilities Are Less Risky?" 105 *Public Utilities Fortnightly* 11, 17-18 (May 8, 1980).

³⁶Don M. Chance is Assistant Professor of Finance at Virginia Polytechnic Institute and State University.

³⁷Addition of the interest-rate factor is based on the work of Professor Bernell Stone in "Systematic Interest Rate Risk in a Two-Index Model of Returns," 9 *Journal of Financial and Quantitative Analysis* 709-21 (November 1974).

market-related (beta) risk is cancelled out when the particular asset is held in a portfolio. The underpinning of this assumption, as Messrs. Brigham and Crum³⁸ have explained, is the further assumption that all assets' returns

are randomly distributed, and that the distribution of returns for each security is reasonably symmetrical. Symmetry means that the random losses on one security can be offset by random gains on another. This makes it possible for investors to diversify away unsystematic, or non-market, risk, leaving systematic risk, which is measured by beta, as the only relevant risk. However, if the distribution of returns on a group of securities is skewed to the left (some probability of large losses but no probability of large gains), then the CAPM breaks down. Diversification can no longer eliminate all unsystematic risk, so market risk as measured by beta is not a complete risk measure.³⁹

But as Brigham and Crum further explain, this necessary assumption is simply not true in the case of utility returns. Owing primarily to the fact that utilities' returns are regulated and therefore have a fixed ceiling — but no fixed floor — “investors have reason to view utilities as having more downside risk than upside potential, which translates into a probability distribution of future returns skewed toward negative returns.”⁴⁰ The result is that utilities — unlike unregulated companies — cannot be assumed to have all their “down-side” non-market risk cancelled out, and, therefore, their “CAPM cost of capital estimates will be downward biased.”⁴¹

Based on the foregoing, there are substantial reasons to conclude that the CAPM theory, both as a general proposition and when specifically considered as a tool for ratemakers, is unsound. Moreover, when actually applied, the theory is no more persuasive, since it has been shown to produce arbitrary, unreliable results.

C. The CAPM In Practice

To this point we have tried to show that there currently exists such considerable evidence questioning the validity of the CAPM theory, both generally and in the utility ratemaking context particularly, that regulatory bodies should seriously consider whether to accord it further use. We now wish to make a different point — that even if one were convinced that the CAPM, as theory, had been shown sound enough to warrant regulatory use, there are such significant problems associated with the theory's practical application that the results it produces do not merit regulatory reliance.

Two types of problems will be mentioned here. The first is that the CAPM formula, elegant and simple on its face, is really not simple at all; rather, it requires that choices be made about such matters as holding periods, measurement intervals, the proper risk-free rate, the proper market rate, the proper market index, etc. — all debatable, and some bordering on arbitrary — the resolution of which can produce dramatically differing re-

³⁸Dr. Eugene F. Brigham is Professor of Finance and Director of the Public Utility Research Center at the University of Florida. Dr. Roy C. Crum is Assistant Professor of Finance at the University of Florida.

³⁹Brigham and Crum, “Reply to Comments on ‘Use of the CAPM in Public Utility Rate Cases,’ ” 7 *Financial Management* 72, 73-74 (Autumn 1978).

⁴⁰*Id.* at 75.

⁴¹*Id.*

sults. The second problem that must be confronted — perhaps the most important of all — is that significant evidence exists showing that CAPM produces utility costs of equity out of keeping with common sense.

1. *Application of CAPM Is Prone to Arbitrariness*

Since CAPM states that an asset's expected return is equal to the risk-free rate plus beta times the difference between the market rate and the risk-free rate, one must know only three numbers in order to compute a given asset's expected rate of return: the expected risk-free rate; the expected market rate; and the asset's expected beta. But what each of those numbers should be for any given asset is anything but clear.

Taking the easiest of the three first — the risk-free rate — it is not difficult to determine the expected rate of return on risk-free instruments for any given date. For this purpose, *The Wall Street Journal* may be consulted. But therein lies a two-fold problem for utility regulation: the expected rates of return vary considerably from instrument to instrument, and, at least in the current volatile market, a given instrument's return can vary considerably from day to day (or, in any event, from week to week).

Assuming a current Treasury bill or note is to be relied on, what maturity should be used? In the case of utility ratemaking, it is arguable that, since a cost of equity is being established for a long period — say, five years — the riskless rate on a five-year instrument should be used.⁴² On the other hand, CAPM theory provides that whatever measurement interval was used to record periodic returns on the market index and the asset in question must also be used as the maturity period of the riskless asset.⁴³ This would dictate use of no greater than a 30-day or 90-day bill since measurement intervals less than monthly or quarterly are rarely used for CAPM.

But if only a 30-day or 90-day bill rate is employed, then arguably only a 30-day or 90-day utility cost of equity is being computed.⁴⁴ Since regulatory bodies are obviously not going to recompute allowable rates of return every few months, perhaps, alternatively, they should study the differences among prevailing riskless rates of varying maturities, and estimate a different riskless rate (and a different total cost of equity) for each of several successive periods.⁴⁵

However the regulators elect to resolve this risk-free-rate dilemma — particularly if interest rates remain volatile and the spreads among varying maturities remain large — it will potentially have a significant impact on the cost of equity. For example, the rates on all riskless instruments are currently several points higher than they were a few short months ago, and the spread among current riskless instruments of varying maturities is also several points.

⁴²See, e.g., Hyman & Egan, "The Utility Stock Market: Regulation, Risk, and Beta," 105 *Public Utilities Fortnightly* 21, 25 (Feb. 14, 1980).

⁴³See Brigham and Crum, *supra*, at 75; Carleton, *supra*, at 57-58.

⁴⁴*Id.* at 58.

⁴⁵Messrs. Litzenger, Ramaswamy, and Sosin have suggested that the risk-free rate should be computed "as a simple average of monthly forward Treasury Bill rates for the period the pending rate order is expected to be in effect." "On the CAPM Approach to the Estimation of a Public Utility's Cost of Equity Capital," 35 *The Journal of Finance* 369, 377 (May 1980).

Hence, using CAPM, the particular risk-free instrument selected and date as of which it is selected can make a substantial difference. And, unfortunately, the end result may produce a cost of equity that bears no resemblance to the utility's *actual* equity costs during the lengthy period for which the rate of return has been set.

The judgmental problems associated with determination of the expected market rate of return are even more formidable. A case can be made that one should determine the expected market return on the basis of long-term historical returns on a broad market index. But, if so, what index, and what historical period should be used? Unfortunately, arguments can be made for various periods and various indices, all resulting in radically different past market returns.⁴⁶ And yet, primarily because they *are* past returns, all these numbers now bear little or no resemblance to current returns. Indeed, Ibbotson and Sinquefeld, one of the most widely-used sources of determining average market returns, estimate only a 12.5 percent average annual return on the market for the period 1977-2000.⁴⁷ If that figure were currently used in the CAPM formula — taking into account that current *risk-free* rates are higher than that 12.5 figure — expected utility rates of return would be computed at *below* Treasury bill rates. Plainly, this is not a sound result.

For several reasons, we do not dwell further here on the problems associated with the expected risk-free rate and the expected market rate: first, because these problems have already been explored in the authorities previously cited in the margin; second, because some of the problems with those two rates are not altogether peculiar to CAPM, but would also affect other methodologies used in ratemaking; and third, because those problems are dwarfed by the difficulties associated with estimating the number that is peculiar to CAPM — beta.

As has already been stated, if one elects to compute a cost of equity based on beta, one can get any number that seems desirable simply by selecting a different market index. But the arbitrariness and ambiguity that surround beta are even more acute than that. Recent evidence demonstrates, unequivocally, that betas — particularly betas for an individual asset (such as, for example, a utility stock) — can change dramatically depending on: the length of the holding period over which the asset's beta is measured; the particular beginning date of the holding period; the ending date of the holding period; and the periodic interval (daily, weekly, monthly, quarterly) at which the asset's and the market's returns are assessed.⁴⁸ Because there is no clear-cut reason for necessarily preferring one particular index, holding period, or

⁴⁶See, e.g., Testimonies of Stewart C. Myers, pp. 28-45 (Dec. 1, 1978), and Richard W. Roll, p. 81 (Jan. 30, 1980) in *The Railroad Valuation Proceedings*; Hyman and Egan, *supra*, at 25; Glassman, "Discounted Cash Flow Versus the Capital Asset Pricing Model (Is g better than b?)," 102 *Public Utilities Fortnightly* 30, 33-34 (Sept. 14, 1978); Vandell & Malernee, "The Capital Asset Pricing Model and Utility Equity Returns," 102 *Public Utilities Fortnightly* 22, 28 (July 6, 1978).

⁴⁷R. Ibbotson and R. Sinquefeld, *Stocks, Bonds, Bills, and Inflation: The Past (1926-1976) and the Future (1977-2000)* 58 (1977).

⁴⁸See, e.g., H. Levy, "The CAPM and the Investment Horizon," 7 *The Journal of Portfolio Management* 32 (1981); B. Fielitz and M. Greene, "Shortcomings in Performance Evaluation via MPT [Modern Portfolio Theory]," 6 *The Journal of Portfolio Management* 13 (Summer 1980); Testimony of Richard F. Meyer, p. 51 (Jan. 30, 1980) in *The Railroad Valuation Proceedings* (showing that the average beta for the railroad industry for the five-year period July 1969 - June 1974 rose by 40% (from .907 to 1.253) solely by changing from daily to monthly observation of returns).

measurement interval over another, and because the selection among these variables produces altogether different betas, which in turn produce altogether different costs of equity, the process takes on an air of arbitrariness.

Moreover, even if one thought that CAPM and its reliance on beta were sound, and thought further that a reliable beta concerning a particular utility could be estimated from past data (notwithstanding the differing betas produced by different indices, holding periods, and measurement intervals),⁴⁹ there would still be no assurance that the *past* utility beta was a reasonable estimate of the *future* utility beta. In other words, no evidence has been adduced showing that utility betas are stable over time.

On the other hand, there has been evidence showing that these betas are *not* stable. For example, a recent empirical study demonstrates that, over any given five-year period, only a beta estimate for a portfolio of at least 100 assets or more can be expected to stay within 90% of the beta estimate at the beginning of the period. Thus, in the case of a single utility equity, or even in the case of a group (under 100) of reasonably comparable utility equities, there can be no confidence at all that an historical beta is a reliable guide to the utility's expected return for the next five years. See Tole,⁵⁰ "How to Maximize Stationarity of Beta," 7 *The Journal of Portfolio Management* 45 (Winter 1981).

Vivid illustration of the instability of utility betas appears in an analysis performed by Messrs. Hyman and Egan.⁵¹ They computed betas for utilities over the period 1958-1978 by observing returns for three utility indices (Moody's electric utilities, Moody's natural gas industry, and S&P's telephones). By using such portfolios, greater stability in betas can be obtained than is possible through observation of returns on individual utilities. Even so, the result was that, over the 20-year period, wide annual variations in beta appeared in all three groups, ranging from .139 in 1959 to 1.190 in 1965, with a 20-year average of .582, .596, and .664 for the electric, telephone, and natural gas groups, respectively.⁵² Whether the 20-year average, the immediately previous 5-year average, an average of the 20-year high and low, the previous year alone, or any other of the 20 single-year betas would constitute a reasonable estimate of the future betas — all of which were different, sometimes very different, numbers⁵³ — would thus be no more than a guess.⁵⁴

⁴⁹The problem of reliability is further compounded by the fact that the betas produced through this process are typically very uncertain as a statistical matter, *i.e.*, their standard errors (an indication of how "far off" the estimate could be) are high, and their R^2 's (an expression of how much of the asset's movements are in fact "explained" by the market's movements) are low. See Hyman and Egan, *supra*, at 24-27; Vandell and Malernee, *supra*, at 27; Myers, "On the Use of Modern Portfolio Theory in Public Utility Rate Cases: Comment," *supra*, at 68.

⁵⁰Dr. Tole is an Associate Professor at Auburn University and an experienced management development specialist and stock broker.

⁵¹Leonard Hyman is Vice President and head of the Utility Research Group at Merrill Lynch. Joseph Egan is a senior utility analyst at Merrill Lynch.

⁵²Hyman and Egan, *supra*, at 24.

⁵³For example, the electric utilities' annual betas ranged from .139 to .952 in the 20-year period; their previous 5-year average was .655, and its 20-year average was .582. Telephone betas ranged from .251 to 1.190, with a previous 5-year average of .528 and a 20-year average of .596. Similarly, the natural gas betas were .419 at the low end, .927 at the high end, with a previous 5-year average of .571 and a 20-year average of .664. *Id.* at 24.

⁵⁴The testimony of Stewart Myers in The Railroad Valuation Proceedings shows similar instability in the betas for the railroad industry. Thus, for purposes of determining an expected cost of equity for that industry as of 1974, Professor Myers had to reckon with betas for the S&P Railroad Index as follows: 1969, 1.02; 1970, .98; 1971, 1.37; 1972, .65; and 1973, .82. Myers Testimony, *supra*, at p. 51.

And that, we submit, is no basis upon which to determine utility rates of return.

2. *The CAPM Produces Results Unsound on Their Face*

As the foregoing indicates, there is substantial evidence that CAPM is both theoretically and empirically unsound. In light of this evidence, we submit that regulators should give serious thought to abandoning the model's use in utility ratemaking unless and until its credibility is affirmatively rehabilitated. There is one final argument that supports this view — CAPM is not only unreliable in theory, and unreliable in practice, but, regardless of one's views about the merits or demerits of CAPM itself, the results it produces are too often simply implausible. Several recent studies have elaborated upon this issue, addressing both the general beta-based proposition that utilities are less risky than the market itself (the average asset), as well as the specific indications of risk changes implied by particular utility beta measurements.

First, if the betas of utilities are to be believed, and if CAPM's assumption that beta-risk is the only risk that matters to investors is to be accepted, then, since utility betas are, by and large, consistently less than 1.0, it must follow that utilities are a less risky proposition than, say, unregulated companies on average or the Dow Jones Industrials in particular. To our knowledge, no rate proceeding has included evidence confirming (independent of CAPM) the reliability of this proposition. However, two recent analyses of this proposition declare that it simply is not so.

In "Who Says Utilities Are Less Risky?"⁵⁵ and "Competition for the Funds of Investors and the Cost of Capital for Utilities,"⁵⁶ the Messrs. Christy⁵⁷ and Dr. Lerner, respectively, studied the question whether, notwithstanding what beta may suggest, utilities are more risky than the average company. For several reasons, the authors concluded that utilities are more risky.

First, the Christys showed that utilities are riskier in light of two obvious objective measurements of comparative risk — the consistently greater volatility in price changes displayed by utilities when compared to the Dow Jones and the consistently greater rates of return to investors afforded by utilities as compared to the Dow Jones.⁵⁸ Second, in comparing utilities with unregulated companies generally, the authors again found utilities by and large to be riskier, on two broad subjective grounds: first, their higher capital intensity and their corresponding greater sensitivity to inflation and interest-rate uncertainties; and second, the greater risk presented by regulation itself, which not only prevents utilities from reacting to market changes on a daily

⁵⁵105 *Public Utilities Fortnightly* 11 (May 8, 1980).

⁵⁶105 *Public Utilities Fortnightly* 15 (Feb. 28, 1980).

⁵⁷Dr. George A. Christy is Professor of Finance at North Texas State University and served for nine years in telephone company commercial and public relations departments. J. Gordon Christy is Assistant Professor of Law at the University of Oklahoma.

⁵⁸For purposes of analyzing volatility, percent price movements were studied over the 1965-1979 period, and for purposes of analyzing returns received, quarterly earnings/price ratios were studied over the 1973-1979 period. 105 *Public Utilities Fortnightly* at 12-13 (May 8, 1980).

basis, but also raises uncertainties concerning whether, and, if so, when, their increased costs can be recovered.⁵⁹

Dr. Lerner, in a separate study, confirmed these views. First, he showed that if past stability (degree of volatility) is to be taken as an indicator of risk — which CAPM assumes it to be — then regardless “whether stability is measured by the deviations from trend of earnings, sales, dividends, or equity, industrials have been either more stable or comparable to utilities over the past five years.”⁶⁰ Similarly, he showed that the percentage price changes of utilities have been greater than those of the S&P Industrial Index.⁶¹ Hence, he concluded, “[t]he argument that an industrial commands a higher return than a utility because it has more risk is simply not true.”⁶²

Finally, in their article “The Utility Stock Market: Regulation, Risk and Beta” (previously referred to), Messrs. Hyman and Egan studied the implications of the various beta changes of utilities (electric, telephone, and natural gas) over the 1958-1978 period. Their judgment was that these implications were simply not credible. Two examples are worth mentioning.

The telephone company betas showed those companies to have “greater risk in the middle 1960’s than at present.” As to this, said the authors, “[c]onsidering that the financial situation of the industry has not improved and that competition has become a threat to the telephone monopoly, we reject that interpretation altogether.”⁶³ Regarding electric utility betas, at least if CAPM is to be credited, those utilities “involved more risk in the mid-1960’s than they do today.” Yet, as the authors point out, “[c]onsidering that industry conditions deteriorated substantially between the 1960’s and the present time, we reject that interpretation and consider it absurd.”⁶⁴

Thus, even putting to one side the considerable theoretical problems associated with CAPM — problems that are intensified in the ratemaking context — and even overlooking that betas tend to be unstable, statistically suspect, and given to arbitrary measurement, it would still be true that the results it produces are too often inconsistent with real-world perceptions about utility risk.

CONCLUSION

For the foregoing reasons, we submit that unless and until those who favor the continued use of CAPM in the utility-ratemaking setting affirmatively establish the model’s reliability in that setting, the results produced by the model should not be used further as a basis for determining fair rates of return. In our view, the case which can be made against CAPM’s continued

⁵⁹*Id.* at 14-18.

⁶⁰105 *Public Utilities Fortnightly* at 15-16 (Feb. 28, 1980).

⁶¹*Id.* at 16-18.

⁶²*Id.* at 16.

⁶³Hyman & Egan, *supra*, at 24.

⁶⁴*Id.*

use for such ratemaking is now of such proportions that the proponents' burden cannot be met.

This is not to say that we propose abandonment of CAPM altogether. Rather, the case we have tried to make here is that CAPM in its present form — that is, a formula which relies on beta as the only important systematic factor and which assumes that alpha is zero — is so fraught with error, and has led to such improbable results, that it does not merit further credence. However, it may well be that a modified CAPM — one that recognizes the value of alpha and/or takes into account other systematic factors (such as interest-rate risk) known to affect utilities — can be shown to be a valid rate-making tool. Such a showing, it seems to us, should be of high priority in future proceedings.

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Cost of Equity for Energy Utilities: Beyond the CAPM[☆]

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Abstract

The Capital Asset Pricing Model (CAPM) is applied in regulatory cases to estimate the required rate of return, or cost of equity, for low-beta, value-style energy utilities, despite the model's well documented mispricing of investments with similar characteristics. This paper examines CAPM-based estimates for a sample of American and Canadian energy utilities to assess the risk premium error. We find that the CAPM significantly underestimates the risk premium for energy utilities compared to its historical value by an annualized average of more than 4%. Two CAPM extensions, the Fama-French model and an adjusted CAPM, provide econometric estimates of the risk premium that do not present a significant misevaluation.

JEL Classifications: G12, L51, L95, Q49

Keywords: Cost of Capital, Rate of Returns, Energy Utilities

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Cost of Equity for Energy Utilities: Beyond the CAPM

Abstract

The Capital Asset Pricing Model (CAPM) is applied in regulatory cases to estimate the required rate of return, or cost of equity, for low-beta, value-style energy utilities, despite the model's well documented mispricing of investments with similar characteristics. This paper examines CAPM-based estimates for a sample of American and Canadian energy utilities to assess the risk premium error. We find that the CAPM significantly underestimates the risk premium for energy utilities compared to its historical value by an annualized average of more than 4%. Two CAPM extensions, the Fama-French model and an adjusted CAPM, provide econometric estimates of the risk premium that do not present a significant misevaluation.

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1. Introduction

An important aspect of the regulatory process for energy utilities is the determination of their equity rate of return. This return, also known as the cost of equity capital, represents the expected remuneration of the shareholders of the utilities. It is a crucial component of their total cost of capital, which is central to their investment policy and serves as a basis for setting up the rates to their customers. The purpose of this paper is to highlight the problems of the most commonly used model to determine the equity rate of return for energy utilities and to propose two alternative models that empirically improve on the estimation. By providing new direct and focused evidence for energy utilities, our analysis contributes to the knowledge of energy, regulatory and financial economists, as well as regulators, who are concerned with rate determination.

Regulatory bodies, like the National Energy Board in Canada or the Federal Energy Regulatory Commission in the United States, have the mandate to set the equity rate of return so that it is fair and reasonable. Specifically, according to Bonbright, Danielsen and Kamerschen (1988, Chap. 10), the return should provide the ability to attract and retain capital (the capital-attraction criterion), encourage efficient managerial practice (the management-efficiency criterion), promote consumer rationing (the consumer-rationing criterion), give a reasonably stable and predictable rate level to ratepayers (the rate-level stability and predictability criterion) and ensure fairness to investors (the fairness to investors criterion). While the first four criteria are designed primarily in the interest of the consuming public, the last criterion acts as an equally-important protection for private owners against confiscatory regulation. Its requirement involves determining the return available from the application of the capital to other enterprises of like risk, which demands an understanding of the risk-return relationship in the equity market.

Traditionally, the regulated return has been set through hearings, where arguments on the issue of fairness could be debated. But since the 1990s, numerous boards have adopted an annual mechanism known as a “rate of return formula” or a “rate adjustment formula”. This mechanism determines automatically the allowed rate of return through a calculation that explicitly accounts for

the risk-return relationship in the equity market. The use of rate adjustment formulas is particularly prevalent in Canada since the landmark March 1995 decision by the National Energy Board (Decision RH-2-94), which sets the stage for the widespread adoption of closely related formulas by provincial regulators.

Most rate adjustment formulas use a method known as the Equity Risk Premium method.¹ This method can be summarized as calculating a utility's equity rate of return as the risk-free rate of return plus a premium that reflects its risk. The risk-free rate is usually related to the yield on a long-term government bond. The risk premium is obtained from the Capital Asset Pricing Model (CAPM) of Sharpe (1964) and Lintner (1965), a classic model of capital market equilibrium. It is equal to the utility's beta, a measure of its systematic risk, multiplied by the market portfolio risk premium. The Equity Risk Premium method has a number of advantages. First, it is supported by a solid theoretical foundation in the academic literature, thus providing a sound basis for understanding the risk-return relationship. Second, it can be estimated based on stock returns, thereby making it more objective than other methods, and relating it to current market conditions. Third, it is relatively simple to apply and requires data that can be obtained easily.

The Equity Risk Premium method is not, however, without shortcomings. Arguably its most criticized feature is the use of the CAPM as the basis to determine the risk premium. While the CAPM is one of the most important developments in finance, research over the last forty years has produced a large body of work critical of the model. On the theoretical side, Cochrane (1999) summarizes the current most prevalent academic view: "In retrospect, it is surprising that the CAPM worked so well for so long. The assumptions on which it is built are very stylized and simplified."² For example, at least since Merton (1973), it is recognized that factors, state variables or sources of priced risk beyond the movements in the market portfolio (the only risk factor in the CAPM) might be needed to explain why some risk premiums are higher than others. On the

¹ There exist other methods for estimating the rate of return, most notably the Comparable Earnings method and the Discounted Cash Flows method. See Morin (2006) for a description. These methods are generally not directly incorporated in the rate adjustment formulas.

² Cochrane (1999), p. 39.

empirical side, the finance literature abounds with CAPM deficiencies (so-called “anomalies”). Fama and French (2004) review this literature to highlight that the CAPM is problematic in the estimation of the risk premium of low-beta firms, small-capitalisation firms and value (or low-growth) firms. While these problems have been well documented in the finance literature, their effects have not yet been fully explored for energy utilities, which may be part of the reasons why the CAPM is still widely used in rate adjustment formulas. In particular, as the CAPM does not empirically provide a valid risk-return relationship for the equity market, it might fall short of the requirement associated with the fairness to investors criterion.

Considering the importance of the CAPM in determining the regulated equity rate of return, the objectives of this paper are two-folds. First, we re-examine the use of the model in the context of energy utilities to determine if it is problematic. As utilities are typically low-beta, value-oriented investments, the finance literature suggests that the model will have difficulties in estimating their risk premiums. We analyze the issue empirically by estimating the model and its resulting risk premiums for a sample of Canadian and American energy utilities mostly related to the gas distribution sector, and by testing for the presence of significant differences between the model’s risk premium estimates and the historical ones.

Second, we implement two alternative models that are designed to circumvent some of the empirical problems of the CAPM. The first alternative is a three-factor model proposed by Fama and French (1993) (the Fama-French model hereafter). This model has been used to estimate the cost of equity by Fama and French (1997) for general industrial sectors and by Schink and Bower (1994) for the utilities sector in particular. The second alternative is a modified CAPM that includes the adjustments proposed by Blume (1975) and Litzenberger, Ramaswamy and Sosin (1980) (the Adjusted CAPM hereafter). The Fama-French model and the Adjusted CAPM provide useful comparisons with the CAPM on the estimation of the risk premiums of energy utilities.

Our empirical results can be summarized as follows. First, the CAPM significantly underestimates the risk premiums of energy utilities compared to their historical values. The

underestimations are economically important, with annualized averages of respectively 4.5% and 6.2% for the Canadian and American gas utilities we consider, and are consistent with the finance literature on the mispricing of low-beta, value-oriented stocks. Second, the Fama-French model and the Adjusted CAPM are both able to provide costs of equity that are not significantly different from the historical ones. Our results show that the value premium, in the case of the Fama-French model, and a bias correction, in the case of the Adjusted CAPM, are important in eliminating the CAPM underestimations. Both models suggest average risk premiums between 4% and 8% for gas utilities portfolios, and are relevant at the individual utility level as well as at the utilities sector level.

Overall, we conclude that the CAPM is problematic in estimating econometrically the cost of equity of energy utilities. The Fama-French model and the Adjusted CAPM are well specified for this purpose as they reduce considerably the estimation errors. These models could thus be considered as alternatives to the CAPM in the Equity Risk Premium method employed by regulatory bodies to obtain the risk-return relationship for the fairness to investors criterion.

The CAPM dates back to the mid-1960s. While the model is tremendously important, there has been a lot of progress over the last 45 years in the understanding of the cross-section of equity returns. It should be clear that the goals of this paper are not to implement full tests of asset pricing models or examine comprehensively the numerous models in the equity literature. Focusing on energy utilities, this paper is an application of the CAPM and two reasonable and relevant alternatives to the problem of cost of equity estimation, using a standard methodology. Our findings show that it is potentially important to go beyond the CAPM for energy utilities. They represent an invitation to further use the advances in the literature on the cross-section of returns to better understand their equity rate of return.

The rest of the paper is divided as follows. The next section presents our sample of energy utilities and reference portfolios. The third, fourth and fifth sections examine the risk premium estimates with the CAPM, the Fama-French model and the Adjusted CAPM, respectively. Each

section provides an overview of the model, presents its empirical estimation and results, and discusses the implications of our findings. The last section concludes.

2. Sample Selection and Descriptive Statistics

This section examines the sample of firms and portfolios for our estimation of the cost of equity of energy utilities. We focus on the gas distribution sector to present complete sector-level and firm-level results, but we also consider utilities indexes to ensure the robustness to other utilities. We provide Canadian and American results for comparison, as both energy markets are relatively integrated and investors might expect similar returns. We first discuss sample selection issues and then present descriptive statistics.

2.1. Sample Selection

Two important choices guide our sample selection process. First, we use monthly historical data in order to have sufficient data for estimating the parameters and test statistics, while avoiding the microstructure problems of the stock markets (low liquidity for numerous securities, non-synchronization of transactions, etc.) in higher frequency data.³ We then annualized our results for convenience. Second, we emphasize reference portfolios (such as sector indexes) over individual firms. Reference portfolios reduce the potentially large noise (or diversifiable risk) in the stock market returns of individual firms. They allow for an increased statistical accuracy of the estimates, an advantage recognized since (at least) Fama and MacBeth (1973), and alleviate the problem that we do not observe the returns on utilities directly and must rely on utility holding companies.

To represent the gas distribution sector in Canada and the U.S., we use a published index and a constructed portfolio for each market. The independently-calculated published indexes are widely available and consider the entire history of firms having belonged to the gas distribution sector. The constructed portfolios use the most relevant firms at present in the gas distribution or

³ See Fowler, Rorke and Jog (1979, 1980) for an analysis of these problems in the Canadian stock markets.

energy utility sector. The data collection also allows an examination of the robustness of our results at the firm level. The resulting four gas distribution reference portfolios are described below.

- *DJ_GasDi*: A Canadian gas distribution index published by Dow Jones, i.e. the “Dow Jones Canada Gas Distribution Index.” The firms in the index are weighted by their market value. Monthly returns (180) are available from January 1992 to December 2006;
- *CAindex*: An equally-weighted constructed portfolio formed of 13 Canadian energy utilities, most with activities that are related to the gas distribution sector, i.e. ATCO Ltd., Algonquin Power Income Fund, Canadian Utilities Limited, EPCOR Power, Emera Incorporated, Enbridge Inc., Fort Chicago Energy Partners, Fortis Inc., Gaz Métro Limited Partnership, Northland Power Income Fund, Pacific Northern Gas, TransAlta Corporation and TransCanada Pipelines.⁴ Monthly returns (263) are available from February 1985 to December 2006;
- *DJ_GasUS*: A U.S. gas distribution index published by Dow Jones, i.e. the “Dow Jones US Gas Distribution Index.” The firms in the index are weighted by their market value. Monthly returns (180) are available from January 1992 to December 2006;
- *USindex*: An equally-weighted constructed portfolio formed of nine U.S. firms whose activities are heavily concentrated in local gas distribution, i.e. AGL Resources Inc., Atmos Energy Corp., Laclede Group, New Jersey Resources Corp., Northwest Natural Gas Co., Piedmont Natural Gas Co., South Jersey Industries, Southwest Gas Corp. and WGL Holdings Inc. Monthly returns (407) are available from February 1973 to December 2006.

To confirm the validity of our analysis to other energy utilities, we also consider four utilities reference portfolios, which consist of the utilities sector indexes described below.

- *DJ_Util*: A Canadian utilities index published by Dow Jones, i.e. the “Dow Jones Canada Utilities Index.” The firms in the index are weighted by their market value. Monthly returns (180) are available from January 1992 to December 2006;
- *TSX_Util*: A Canadian utilities index published by S&P/TSX, i.e. the “S&P/TSX Utilities Index.” The firms in the index are weighted by their market value. Monthly returns (228) are available from January 1988 to December 2006;
- *DJ_UtiUS*: A U.S. utilities index published by Dow Jones, i.e. the “Dow Jones US Utilities Index.” The firms in the index are weighted by their market value. Monthly returns (180) are available from January 1992 to December 2006;
- *FF_Util*: A U.S. utilities index formed by Profs. Fama and French, or the University of Chicago and Dartmouth College, respectively. The firms in the index are weighted by their market value. Monthly returns (407) are available from February 1973 to December 2006.

Depending on their availability, the reference portfolio series have different starting dates.

In our econometric estimation, we keep the maximum number of observations for each series. Fama

⁴ We also considered AltaGas Utility Group, Enbridge Income Fund, Westcoast Energy, Nova Scotia Power and Energy Savings Income Fund. We did not retain the first four because they had a returns history of less than 60 months. We eliminated the last one because it is a gas broker and its average monthly return of more than 3% was a statistical outlier. Our results are robust to variations in the formation of the CAindex portfolio, like the inclusion of these five firms or the exclusion of income funds and limited partnerships.

and French (1997) find that such a choice results in costs of equity more precisely estimated and with more predictive ability than costs of equity obtained from rolling five-year estimation windows, a common choice in practice. The data are collected from the Canadian Financial Markets Research Center (CFMRC), Datastream and the web sites of Prof. French⁵ and Dow Jones Indexes⁶.

2.2. Descriptive Statistics

Descriptive statistics for the monthly returns are presented in Table 1. Panel A shows the results for the 13 Canadian energy utilities and their equally-weighted portfolio (CAindex). Panel B shows the results for nine U.S. gas distribution utilities and their equally-weighted portfolio (USindex). Panel C shows the statistics for Canadian and U.S. indexes for the utilities sector (DJ_Util, DJ_UtilUS, TSX_Util and FF_Util) and the gas distribution sub-sector (DJ_GasDi and DJ_GasUS).⁷

< INSERT TABLE 1 HERE >

For the Canadian energy utilities, the monthly average return of all 13 firms is 1.0% with a standard deviation of 3.1%. The Dow Jones Canada Gas Distribution Index, the Dow Jones Canada Utilities Index and the S&P/TSX Utilities Index have mean returns of 1.2%, 0.7% and 1.0%, respectively. The monthly average return of the nine U.S. gas distribution utilities is 1.2% with a standard deviation of 4.1%. The Dow Jones US Gas Distribution Index, the Dow Jones US Utilities Index and the Fama-French U.S. Utilities Index show mean returns of 1.2%, 0.9% and 1.0%, respectively. Correlations between the four gas distribution reference portfolios (not tabulated) are between 0.29 and 0.80. These correlations indicate that the portfolios show some commonality, but are not perfect substitutes. We next start our analysis of the equity risk premium models.

⁵ http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

⁶ <http://www.djindexes.com/mdsidx/index.cfm?event=showtotalMarketIndexData&perf=Historical%20Values>.

⁷ The returns from August to November 2001 of the Dow Jones U.S. indexes are strongly influenced by the Enron debacle, which started with the resignation of its CEO, Jeffrey Skilling, on August 14, 2001 and ended with the bankruptcy of the company on December 2, 2001. During those four months, the DJ_GasUS and DJ_UtiUS indices lost 68.9% and 16.2% of their value, respectively. By comparison, the equally-weighted portfolio of U.S. gas distributors (USindex) gained 1.2% and the Fama-French utilities index (FF_Util) lost 6.2 %. In order to soften the impact of that statistical aberration (caused by an unprecedented fraud) on the estimation of the risk premium, the returns from August to November 2001 of DJ_GasUS and DJ_UtilUS are replaced by those of USindex and FF_Util, respectively.

3. Equity Risk Premium with the CAPM

This section examines the use of the Capital Asset Pricing Model (CAPM) for estimating the rate of return for energy utilities. The CAPM is the model the most often associated with the Equity Risk Premium method that is the basis of the rate adjustment formulas of regulatory bodies. We first present the model and its relevant literature. Then we estimate the model for our sample of energy utilities. Finally, we discuss the implications of our findings.

3.1. Model and Literature

The CAPM is a model proposed by Sharpe (1964) and Lintner (1965) in which the expected equity return or cost of equity for a gas utility is given by

$$E(R_{GAS}) = R_f + \beta \times \lambda_m,$$

where R_f is the risk-free rate, β is the firm's beta or sensitivity to the market returns and λ_m is the market risk premium. In this model, a higher beta results in a higher risk premium.

The CAPM is the best known model of expected return. In spite of its undeniable importance in the field of finance, it has long been rejected by numerous empirical tests in the academic literature. The empirical rejections start with the first tests (Black, Jensen and Scholes, 1972, Fama and MacBeth, 1973, and Blume and Friend, 1973) that find that the relation between beta and average return is flatter than predicted by the model. They continue with the discovery of numerous "anomalies" (like the price-to-earnings effect of Basu, 1977, the size effect of Banz, 1981, etc.). Finally, in the 1990s, based on high-impact articles, including Fama and French (1992, 1993, 1996a and 1996b), Jegadeesh and Titman (1993) and Jagannathan and Wang (1996), the academic profession reaches a relative consensus that the CAPM is not valid empirically. In Canada, like elsewhere in the world, the literature reaches similar conclusions (see Morin, 1980, Bartholdy, 1993, Bourgeois and Lussier, 1994, Elfakhani, Lockwood and Zaher, 1998, L'Her, Masmoudi and Suret, 2002, 2004.).

A complete review of the literature on the problems of the CAPM is beyond the scope of this paper. It is nevertheless important to point out the two characteristics of energy utilities that suggest the CAPM might be problematic in estimating their equity return. First, energy utilities have typically low betas, significantly below one. Second, they are known as value investments, in the sense that they have high earnings-to-price, book-to-market, cash flows-to-price or dividend-to-price ratios. In a summary article requested for a symposium on the 40th anniversary of the CAPM, Fama and French (2004) highlight the result of using the model to estimate the cost of equity capital for firms with these two characteristics:

“As a result, CAPM estimates of the cost of equity for high beta stocks are too high (relative to historical average returns) and estimates for low beta stocks are too low (Friend and Blume, 1970). Similarly, if the high average returns on value stocks (with high book-to-market ratios) imply high expected returns, CAPM cost of equity estimates for such stocks are too low.”⁸

As Fama and French (2004) indicate, the low-beta and value characteristics of energy utilities will probably lead the CAPM to estimate a rate of return that is too low. We next examine whether this undervaluation in fact exists in our sample of reference portfolios and utilities.

3.2. Risk Premium Estimates

This section empirically estimates the risk premium with the CAPM using the previously described Canadian and U.S. monthly data.⁹ More specifically, we estimate the model using the time-series regression approach pioneered by Black, Jensen and Scholes (1972) with the following equation:

$$R_{GAS,t} - R_{f,t} = \alpha_{GAS} + \beta \times \lambda_{m,t} + \varepsilon_{GAS,t},$$

where $\lambda_{m,t} = R_{m,t} - R_{f,t}$ is the return on the market portfolio in excess of the risk-free return

and $\varepsilon_{GAS,t}$ is the mean-zero regression error, at time t . In this equation, the CAPM predicts that the

alpha (or intercept) is zero ($\alpha_{GAS} = 0$) and the risk premium is $E(R_{GAS,t} - R_{f,t}) = \beta \times E(\lambda_{m,t})$.

⁸ Fama and French (2004), p. 43-44.

⁹ Our focus is on the estimation of the equity risk premium for energy utilities. To obtain their full cost of equity, we would need to add an appropriate risk-free rate, which could depend on the circumstances. For example, one common choice advocates adding to their equity risk premium the yield on a long-term government bond. But other choices for an appropriate risk-free rate are possible.

An alpha different from zero can be interpreted as the risk premium error of the CAPM (see Pastor and Stambaugh, 1999). A positive alpha indicates the CAPM does not prescribe a large enough risk premium compared to its historical value (an underestimation), whereas a negative alpha indicates the CAPM prescribes a risk premium that is too large (an overestimation). It is therefore possible to determine the CAPM risk premium error for energy utilities based on the estimates of the alpha.¹⁰

We use Hansen's (1982) Generalized Method of Moments technique in order to estimate jointly the parameters α_{GAS} and β of the model and the market risk premium $E(\lambda_{m,t})$. As Cochrane (2001, Section 12.1) shows, this method has the necessary flexibility to correct the results for possible econometric problems in the data.¹¹ We take the monthly returns on portfolios of all listed securities weighted by their market value for the market portfolio returns and on the Treasury bills for the risk-free returns.¹² The annualized mean market risk premiums are 5.2% for Canada from February 1985 to December 2006 and 6.0% for the U.S. from February 1973 to December 2006.

Table 2 shows the results of the regressions using each of the four gas distribution reference portfolios. The estimates of the annualized risk premium error (or annualized α_{GAS}), the beta β and the risk premium $\beta \times E(\lambda_{m,t})$ are presented in Panels A, B and C, respectively. For each estimate, the table also shows its standard error, t-statistic and associated p-value.

< INSERT TABLE 2 HERE >

The estimates in Panel A of Table 2 indicate that the risk premium errors are positive. Hence, the CAPM underestimates the risk premium for the gas distribution reference portfolios. The underestimation is not small – a minimum of 4.52% (for CAindex) and a maximum of 8.43%

¹⁰ The time series regression approach is commonly used when the model factors are returns. Cochrane (2001, Chapter 12) emphasizes that the approach implicitly imposes the restriction that the factors (chosen to fully represent the cross section of returns in the modeling) should be priced correctly in the estimation. While there are other ways to estimate a model like the CAPM, one advantage of the times series regression approach is that it can be easily applied to a restricted set of assets (like energy utilities) as the cross-sectional variations in asset returns are already captured by the correct pricing of the traded factors. Cochrane (2001, Chapter 12) also shows that the approach is identical to a Generalized Least Square cross-sectional regression approach.

¹¹ All standard errors and statistical tests have been estimated using the Newey and West (1987) method, which takes account of the potential heteroscedasticity and autocorrelation in the errors of the statistical models.

¹² The data sources are CFMRC (until 2004) and Datastream (thereafter) for the Canadian returns and the web site of Prof. French for U.S. returns.

(for DJ_GasDi) – and is statistically greater than zero for all portfolios. Also, as expected, the underestimation comes with low beta estimates, with values between 0.21 and 0.46 in Panel B. For example, for CAindex, the beta is 0.34 and the annualized risk premium predicted by the CAPM is 1.76%, an underestimation of the historical risk premium $\alpha_{GAS} = 4.52\%$.

To verify the underestimation is not an artifact of the utilization of the reference portfolios and is robust to other energy utilities, Figure 1 shows the risk premium errors for the utilities that make up the CAindex portfolio (Figure 1a), the gas distributors in the USindex portfolios (Figure 1b) and the four utilities reference portfolios (Figure 1c). Once again, the alphas are always positive, with values between 2.1% and 8.9% for the Canadian utilities, between 3.5% and 8.4% for the U.S. gas distributors, and between 2.1% and 5.0% for the utilities reference portfolios. The constantly positive and often significant errors support the notion that the CAPM might not be appropriate for determining the risk premium in the utilities sector.

< INSERT FIGURE 1 HERE >

3.3. Discussion

Our results show that the CAPM underestimates the risk premium for the gas distribution sub-sector in particular and for the utilities sector in general. This finding is consistent with the empirical literature that finds that the CAPM tends to underestimate the risk premium of securities or sectors associated with low-beta, value and small-cap investments. In the terminology of asset pricing, the returns on energy utilities are “anomalous” with respect to the CAPM. As the application of the model would not be sensible in evaluating the performance of value-type mutual funds, given the related anomaly, it could be unwarranted in evaluating the cost of equity for energy utilities.

While the magnitude of the underestimation for the utilities is large, it is not unexpected. Fama and French (2004) review the evidence on the large CAPM literature for the *full cross-section* of equity returns. Their figures 2 and 3, in particular, illustrate well the findings for portfolios of stocks formed on their beta and their book-to-market ratio value indicator, respectively. In the

cross-section of all stock returns, their figure 2 show visually that the CAPM underestimation is about 3% for the lowest beta portfolio (a beta of about 0.6), while its overestimation is about 3% for the highest beta portfolio (a beta of about 1.8). Their figure 3 indicates that the CAPM underestimation is about 5% for the highest book-to-market ratio portfolio, while its overestimation is about 2% for the lowest book-to-market ratio portfolio. As energy utilities are low-beta and value-oriented stocks, our estimates of the CAPM underestimation for this segment are consistent with the evidence from the full cross-section of equity returns.

Our results are related to numerous studies documenting that the CAPM alphas are different from zero. As a consequence of these rejections, finance researchers have considered various models that generalized the CAPM as well as various empirical improvements to the estimates of the CAPM. Based on this literature, we explore two alternative ways of estimating the risk premium of energy utilities in the next two sections.

4. Equity Risk Premium with the Fama-French model

The CAPM claims that a single factor, the market portfolio return, can explain expected returns. The most natural extension is to take multiple factors into account. Clearly, if factors other than the market return have positive risk premiums that contribute to explaining expected returns, then the inclusion of those factors should provide a better estimate of the risk premium and potentially eliminate the CAPM errors (see Merton, 1973, and Ross, 1976, for formal theoretical justifications). This section considers one of the most common generalization of the CAPM, a multifactor model by Fama and French (1993). We first describe the model and then use it to estimate the risk premium of energy utilities. We finally discuss the interpretation of our findings.

4.1. Model and Literature

The Fama-French model is a three-factor model developed to capture the anomalous returns associated with small-cap, value and growth portfolios by including risk premiums for size and value. For a gas utility, the expected equity return is given by

$$E(R_{GAS}) = R_f + \beta \times \lambda_m + \beta_{SIZE} \times \lambda_{SIZE} + \beta_{VALUE} \times \lambda_{VALUE},$$

where R_f is the risk-free rate, β , β_{SIZE} and β_{VALUE} are respectively the firm's market, size and value betas, and λ_m , λ_{SIZE} and λ_{VALUE} are respectively the market, size and value risk premiums. The three betas represent sensitivities to the three sources of risk, and the higher are their values, the higher is a firm's risk premium. In cases when the size and value risk factors are not relevant, then the Fama-French model reduces to the CAPM. Theoretical justifications for the size and value premiums are provided by Berk, Green and Naik (1999), Gomez, Kogan and Zhang (2003), and Carlson, Fisher and Giammarino (2004). Fama and French (1993, 1996a) are the two of the most influential empirical tests of the model.

Like the CAPM, the Fama-French model has been used in applications ranging from performance measurement to abnormal return estimation and asset valuation. For the calculation of the cost of equity capital, the model is studied by, among others, Schink and Bower (1994), Fama and French (1997), and Pastor and Stambaugh (1999). It has also proven to be relevant for explaining stock market returns in most countries where it has been examined. For example, in Canada, the model is validated by Elfakhani, Lockwood and Zaher (1998) and L'Her, Masmoudi and Suret (2002). Given that energy utilities are associated with value investments, the Fama-French model has the potential to improve the estimation of their rates of returns. We next assess this possibility for our sample of reference portfolios and utilities.

4.2. Risk Premium Estimates

The risk premium with the Fama-French model is estimated with a methodology that is similar to the one followed for the CAPM using the following equation:

$$R_{GAS,t} - R_{f,t} = \alpha_{GAS}^{FF} + \beta \times \lambda_{m,t} + \beta_{SIZE} \times \lambda_{SIZE,t} + \beta_{VALUE} \times \lambda_{VALUE,t} + v_{GAS,t},$$

where $\lambda_{m,t} = R_{m,t} - R_{f,t}$ is the return on the market portfolio in excess of the risk-free return,

$\lambda_{SIZE,t} = R_{SMALL,t} - R_{LARGE,t}$ is the return on a small-cap portfolio in excess of the return on a large-

cap portfolio, $\lambda_{VALUE,t} = R_{VALUE,t} - R_{GROWTH,t}$ is the return on a value portfolio in excess of the return on a growth portfolio and $\nu_{GAS,t}$ is the mean-zero regression error, at time t . The alpha α_{GAS}^{FF} is still interpreted as the risk premium error. The three beta parameters give the sensitivities to the market, size and value factors. Finally, $\beta \times E(\lambda_{m,t}) + \beta_{SIZE} \times E(\lambda_{SIZE,t}) + \beta_{VALUE} \times E(\lambda_{VALUE,t})$ represents the risk premium from the Fama-French model.

The data for the market portfolio returns and the risk-free returns are the same used in the CAPM estimation. For the Canadian regressions, the small-cap portfolio returns are from a portfolio of all listed securities weighted equally whereas the large-cap portfolio returns are from a portfolio of all listed securities weighted by their market value.¹³ The value and growth portfolios are determined from the earnings-to-price ratio. Specifically, the value (growth) portfolio contains firms having an earnings/price ratio in the highest (lowest) 30%.¹⁴ For U.S. regressions, the size and value premiums are the Fama and French (1993, 1996a) SMB and HML variables, which are computed from market capitalization (size) and book-to-market ratio (value).¹⁵ The annualized mean size and value risk premiums are respectively 8.9% and 6.4% for Canada from February 1985 to December 2006 and 2.7% and 6.0% for the U.S. from February 1973 to December 2006.

Table 3 presents the results of the estimates of the coefficients and the risk premium with the Fama-French model for the four gas distribution reference portfolios previously described. Panel A shows that the annualized risk premium errors are still positive for the four portfolios, ranging from 0.31% (for USIndex) to 4.45% (for DJ_GasDi), but the underestimation is now

¹³ These indexes are taken from CFMRC for returns up to 2004 and then completed by the returns of the S&P/TSX Composite Index and the MSCI Barra Smallcap Index, respectively.

¹⁴ Data come from the web site of Prof. French, who also provides specific instructions on the composition of the portfolios. The site gives returns for value and growth portfolios based on four indicators – earnings-to-price, book-to-market, cash flows-to-price and dividend-to-price. Fama and French (1996a) show that these indicators contain the same information about expected returns. Fama and French (1998) confirm the relevance of these indicators in explaining the returns in 12 major international financial markets and emerging financial markets. We chose the earnings-to-price indicator because it is more effective in capturing the premium of value securities compared to growth securities in Canada (see Bartholdy, 1993, and Bourgeois and Lussier, 1994). The indicator book-to-market is less effective in Canada because the value effect is mainly concentrated in more extreme portfolios (highest and lowest 10%) than in those available on the site (see L'Her, Masmoudi and Suret, 2002).

¹⁵ Data again come from the web site of Prof. French. Detailed instructions on the composition of the SMB and HML variables are also provided.

statistically negligible. Panel D confirms that the inclusion of the value risk premium is instrumental in the reduction of the errors. The value betas are highly significant, with values between 0.30 and 0.71. The size betas (Panel C) are low and often not statistically different from zero, whereas the market betas (Panel B) are 0.54 on average. The estimated risk premiums vary between 4.23% and 8.83%.

< INSERT TABLE 3 HERE >

Figure 2 compares the Fama-French and CAPM results. Figure 2a illustrates the risk premium errors of the two models, while Figure 2b shows their explanatory power given by the adjusted R^2 . The errors have substantially fallen with the Fama-French model for all reference portfolios. Furthermore, the Fama-French model explains a much larger proportion of the variation in the reference portfolio returns.

< INSERT FIGURE 2 HERE >

Figures 3 and 4 present the risk premium errors and the value betas, respectively, for the utilities that make up the CAindex portfolios (Figures 3a and 4a), the gas distributors in the USindex portfolios (Figures 3b and 4b) and the four utilities reference portfolios (Figures 3c and 4c). A comparison of Figure 3 with Figure 1 shows that the risk premium errors have decreased in all cases. None of the errors are now significantly different from zero. Figure 4 confirms that the reductions in the risk premium errors are caused by the inclusion of the value risk premium. All value betas are greater than 0.23 and statistically significant. For example, the TSX_Util portfolio has a value beta of 0.41 that contributes to reduce its risk premium error from 5.0% with the CAPM to 0.7% with the Fama-French model.

< INSERT FIGURE 3 HERE >

< INSERT FIGURE 4 HERE >

4.3. Discussion

Our results support the notion that the Fama-French model is well suited to estimate the risk premium for energy utilities, consistent with the findings of Schink and Bower (1994). We obtain

lower risk premium errors with the Fama-French model than with the CAPM and significant value betas, similar to the results reported by Schink and Bower (1994), Fama and French (1997) and Pastor and Stambaugh (1999).

While the model is being increasingly considered in practice, an often mentioned limitation is that the economic interpretation of the size and value premiums is still under debate. On one side, starting with Fama and French (1993), the size and value factors are presented as part of a rational asset pricing model, where they reflect either state variables that predict investment opportunities following the theory of Merton (1973), or statistically useful variables to explain the returns following the theory of Ross (1976). On the other side, as first advocated by Lakonishok, Shleifer and Vishny (1994), the size and value factors are thought to be related to investors' irrationality in the sense that large-cap and growth stocks tend to be glamorized whereas small-cap and value stocks tend to be neglected. There is a vast literature on both sides of this debate.¹⁶

While the debate is important to improve our understanding of capital markets, Stein (1996) demonstrates that the theoretical interpretation of the model is not relevant to its application to determine the cost of capital. On one side, if the Fama-French model is rational, then the size and value factors capture true risks and should be accounted for in the risk premiums of energy utilities. On the other side, if the size and value factors are irrational, then the significant value betas of energy utilities indicate that they are neglected or undervalued firms. In this case, Stein (1996) shows that rational firms should not undertake a project that provides an expected return lower than the return estimated by the potentially irrational Fama-French model. They are better off in rejecting the project and simply buying back their own shares for which they expect an inflated future return because of the undervaluation. Thus, the potentially irrational Fama-French estimates serve as the appropriate hurdle rate for project investments. Hence, for both interpretations, the equity cost of

¹⁶ A third interpretation, following Lo and MacKinlay (1990) and Kothari, Shanken and Sloan (1995), is that the results of the Fama-French model are spurious, due to biases like data snooping or survivorship. However, the fact that similar size and value premiums have been found in countries outside the U.S. has rendered this explanation less appealing.

capital of energy utilities generated by the Fama-French model is a useful guideline of a fair rate of return for regulators.

Arguably, the Fama-French model is one of the most widely used models of expected returns in the academic finance literature (Davis, 2006). Nevertheless, the literature on the cross-section of equity returns has identified numerous other factors that could be relevant in the multifactor approach. For examples, other influential factors include the labor income factor of Jagannathan and Wang (1996), the momentum factor of Jegadeesh and Titman (1993) and Carhart (1997), the liquidity factor of Pastor and Stambaugh (2003) and the idiosyncratic volatility factor of Ang *et al.* (2006, 2009). These advances in the literature on the cross-section of returns could eventually lead to a better understanding of the equity risk premium for energy utilities.¹⁷ The next section looks at a second approach that goes beyond the CAPM to estimate the equity risk premium.

5. Equity Risk Premium with the Adjusted CAPM

This section considers two empirical adjustments to the CAPM estimates proposed in the academic literature to account for their deficiencies. We call the CAPM with the addition of the two modifications the “Adjusted CAPM”. Unlike the CAPM and the Fama-French model, the Adjusted CAPM is not an equilibrium model of expected returns. It contains adjustments to the CAPM that are empirically justified in a context where the known difficulties of a theoretical model need to be lessened for improved estimation. We first introduce the Adjusted CAPM. Then we implement it to estimate the risk premium of energy utilities. We finally offer a brief discussion of our findings.

5.1. Model and Literature

The Adjusted CAPM is based on the CAPM but provides more realistic estimates of the rate of return by considering the empirical problems of the CAPM. More specifically, the Adjusted CAPM is a model in which the expected equity return of a gas utility is arrived at by

¹⁷ Some of the documented effects, like momentum, are short-lived. Hence, their related factor might be irrelevant for estimates of the cost of equity capital.

$$E(R_{GAS}) = R_f + \alpha_{GAS} \times (1 - \beta^{Adj}) + \beta^{Adj} \times \lambda_m.$$

Compared to the CAPM, this equation incorporates a modification to take into account that estimated betas can be adjusted for better predictive power and a modification to take account of the fact the alpha (risk premium error) is high for low-beta value-oriented firms in the CAPM.

The first modification originates from the works of Blume (1971, 1975). Blume (1971) examines historical portfolio betas over two consecutive periods and finds that the historical betas, from one period to another, regress towards one, the average of the market. He also shows that the historical betas adjusted towards one predict future betas better than unadjusted betas. Blume (1975) builds a historical beta adjustment model to capture the tendency to regress towards one. He discovers that the best adjustment is to use a beta equal to $0.343 + 0.677 \times \beta^{His}$, a finding that led to the concept of “adjusted beta”. Merrill Lynch, which popularized the use of adjusted betas based on Blume (1975)’s results, advocates the adjustment $\beta^{Adj} = 0.333 + 0.667 \times \beta^{His}$. Merrill Lynch’s adjusted beta, now widely used in practice, represents a weighted-average between the beta of the market and the historical beta, with a two-thirds weighting on the historical beta.

The second adjustment is initially proposed by Litzenberger, Ramaswamy and Sosin (1980), who consider solutions to the problem that the CAPM gives a cost of equity capital with a downward bias for low beta firms, as discussed in section 3.1. They note that one way of remedying the problem is to add a bias correction to the CAPM risk premium. To be effective, the correction must take account of the importance of the risk premium error and the level of the firm’s beta because these two elements influence the magnitude of the problem. To do this for low beta securities, Litzenberger, Ramaswamy and Sosin (1980) propose the bias correction $\alpha_{GAS} \times (1 - \beta)$. As desired, the correction increases with the risk premium error of the CAPM, and decreases with the beta. The correction is nil for a firm for which the CAPM already works well (when $\alpha_{GAS} = 0$) or for a firm having a beta of one, two cases where the CAPM produces a fair rate of return on

average. Morin (2006, Section 6.3) presents an application of this adjustment in regulatory finance through a model he calls the empirical CAPM.

In summary, the two modifications incorporated in the Adjusted CAPM involve first using the adjusted beta instead of the historical beta and second including the bias correction in the risk premium calculation. Considering the documented usefulness of the two adjustments, the Adjusted CAPM has the potential to estimate a reasonable risk premium for the energy utilities.

5.2. Risk Premium Estimates

To compute the Adjusted CAPM estimates for our utilities, the starting point is the estimates of the CAPM of Section 3.2, given in Table 2. The beta estimates are now understood as the unadjusted historical betas β^{His} . The gas utility risk premium with the Adjusted CAPM can then be expressed as $\alpha_{GAS} \times (1 - \beta^{Adj}) + \beta^{Adj} \times E(\lambda_{m,t})$, where $\beta^{Adj} = 0.333 + 0.667 \times \beta^{His}$. The Adjusted CAPM risk premium error is arrived at by $\alpha_{GAS}^{Adj} = E(R_{GAS,t} - R_{f,t}) - [\alpha_{GAS} \times (1 - \beta^{Adj}) + \beta^{Adj} \times E(\lambda_{m,t})]$.

Table 4 shows the Adjusted CAPM estimates using the four gas distribution reference portfolios. The estimates of the risk premium error α_{GAS}^{Adj} , the adjusted beta β^{Adj} , the bias correction $\alpha_{GAS} \times (1 - \beta^{Adj})$ and the risk premium are shown in Panels A, B, C and D, respectively. The risk premium errors are still positive for the four portfolios, with values ranging from 1.39% (for CAindex) to 2.89% (for USindex), but the underestimation is only significant for USindex. The reduction in errors comes from the use of adjusted betas, which are 0.56 on average, and the bias corrections, which are 2.96% on average. Lastly, the risk premiums vary between 4.88% and 8.27%, findings comparable to the estimates obtained with the Fama-French model.

< INSERT TABLE 4 HERE >

Figure 5 shows the risk premium errors for the utilities that make up the CAindex portfolios (Figure 5a), the gas distributors in the USindex portfolios (Figure 5b) and the four utilities reference portfolios (Figure 5c). The errors are generally insignificant and a comparison with Figure 1

indicates that they have decreased considerably for all portfolios. For example, for the TSX_Util portfolio, the error is down from 5.0% with the CAPM to 0.9% with the Adjusted CAPM.

< INSERT FIGURE 5 HERE >

5.3. Discussion

Our results support the validity of the Adjusted CAPM for determining the rate of return on energy utilities. While its risk premium estimates are in the same range as the Fama-French estimates, it arrives at its results from a different perspective. The Fama-French model advocates the use of additional risk factors to reduce the CAPM risk premium errors. The Adjusted CAPM, through its bias correction, effectively estimates the risk premium as a weighted-average of the CAPM risk premium and the realized historical risk premium, with a weighting of beta on the former.

The Adjusted CAPM thus recognizes that the CAPM is an imperfect model that can be improved with the information contained in the historical returns. Pastor and Stambaugh (1999) propose a similar strategy by demonstrating how to estimate the cost of equity by using Bayesian econometrics to incorporate the CAPM risk premium error (or alpha) in an optimal manner based on the priors of the evaluator. Consistent with our results, they also show evidence of higher costs of equity for energy utilities using their technique than using the CAPM alone.¹⁸ As the Adjusted CAPM does not require additional risk factors like size and value, the model might be easier to interpret for regulators already familiar with the standard CAPM in their decisions.

6. Conclusion

It is difficult to overstate the importance of the evaluation of the expected rate of return in finance. For a firm's management group, the expected rate of return on equity (or the equity cost of capital) is central to its overall cost of capital, i.e. the rate used to determine which projects will be undertaken. For portfolio managers, the expected rate of return on equity is an essential ingredient

¹⁸ Pastor and Stambaugh (1999) obtain risk premiums that vary between the CAPM estimates, when they assume that there is zero prior uncertainty on the CAPM, and the historical estimates, when they assume that there is infinite prior uncertainty on the CAPM. Our bias correction corresponds approximately to a prior uncertainty on the CAPM between 3% and 6% in their setup.

in portfolio decisions. For regulatory bodies, the expected return on equity is the basis for determining the fair and reasonable rate of return of a regulated enterprise. This paper is interested in evaluating the rate of return in the context of regulated energy utilities.

The academic literature contains numerous theories for determining the expected rate of return on equity. As those theories are based on simplified assumptions of the complex world in which we live, they cannot be perfect. Even if the theoretical merit of the different models can be debated, the determination of the most valid approach to explain the financial markets really becomes an empirical question – it is necessary to answer the question “which theory best explains the information about actual returns?” This paper empirically examines the validity of the model the most often used in the rate adjustment formula of regulatory bodies, the CAPM, one of the most prominent academic alternatives, the Fama-French model, and a version of the CAPM modified to account for some of its empirical deficiencies, the Adjusted CAPM.

Our empirical results show that the risk premiums for energy utilities estimated with the CAPM are rejected as too low compared to the historical risk premiums. The rejections are related to the well-documented CAPM underestimation of the average returns of low-beta firms and value firms. The Fama-French model and the Adjusted CAPM appear statistically better specified, as we cannot reject the hypothesis that their risk premium errors are equal to zero. They suggest equity risk premiums for gas distribution utilities between 4% and 8%. Overall, our findings demonstrate that models that go beyond the CAPM have the potential to improve the estimation of the cost of equity capital of energy utilities. They are thus interesting avenues for regulators looking to set fair and reasonable equity rates of return.

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TABLE 1
Descriptive Statistics of Monthly Returns

Variable	N	Mean	St Dev	Min	Max	Brief Description
Panel A: Canadian Energy Utilities						
ATCO	263	0.013	0.067	-0.301	0.279	ATCO Ltd.
Algonqui	108	0.009	0.054	-0.163	0.166	Algonquin Power Income Fund
CanUtili	263	0.012	0.043	-0.107	0.159	Canadian Utilities Limited
EPCOR	114	0.008	0.046	-0.201	0.108	EPCOR Power
Emera	143	0.009	0.043	-0.137	0.115	Emera Incorporated
Enbridge	263	0.011	0.054	-0.365	0.205	Enbridge Inc.
FortChic	107	0.009	0.054	-0.119	0.210	Fort Chicago Energy Partners
Fortis	228	0.013	0.041	-0.134	0.146	Fortis Inc.
GazMetro	166	0.010	0.037	-0.134	0.084	Gaz Métro Limited Partnerships
NorthPow	104	0.011	0.063	-0.202	0.205	Northland Power Income Fund
PacNorth	263	0.010	0.070	-0.400	0.507	Pacific Northern Gas
TransAlt	263	0.009	0.048	-0.217	0.188	TransAlta Corporation
TransCan	258	0.008	0.054	-0.214	0.254	TransCanada Pipelines
CAindex	263	0.010	0.031	-0.130	0.087	Equally-weighted portfolio
Panel B: U.S. Gas Distribution Utilities						
AGL_Res	407	0.013	0.052	-0.138	0.253	AGL Resources Inc.
Atmos	277	0.013	0.063	-0.302	0.269	Atmos Energy Corp.
Laclede	407	0.012	0.056	-0.148	0.374	Laclede Group
NJ_Res	407	0.013	0.063	-0.171	0.577	New Jersey Resources Corp.
Northwes	407	0.012	0.060	-0.236	0.274	Northwest Natural Gas Co.
Piedmont	407	0.013	0.059	-0.188	0.315	Piedmont Natural Gas Co.
SouthJer	407	0.012	0.058	-0.194	0.486	South Jersey Industries
Southwes	407	0.011	0.070	-0.304	0.234	Southwest Gas Corp.
WGL_Hold	407	0.012	0.071	-0.232	0.807	WGL Holdings Inc.
USindex	407	0.012	0.041	-0.121	0.338	Equally-weighted portfolio
Panel C: Sector Indexes						
TSX_Util	228	0.010	0.037	-0.101	0.114	S&P/TSX Utilities Index
DJ_GasDi	180	0.012	0.043	-0.139	0.137	Dow Jones Canada Gas Distribution Index
DJ_Util	180	0.007	0.036	-0.139	0.101	Dow Jones Canada Utilities Index
DJ_GasUS	180	0.012	0.039	-0.120	0.143	Dow Jones US Gas Distribution Index
DJ_UtiUS	180	0.009	0.042	-0.127	0.136	Dow Jones US Utilities Index
FF_Util	407	0.010	0.041	-0.123	0.188	Fama-French US Utilities Index

NOTES: This table presents descriptive statistics on the monthly returns of 13 Canadian utilities and their equally-weighted portfolio (CAindex) in Panel A, of nine U.S. gas distribution utilities and their equally-weighted portfolio (USindex) in Panel B, and on selected utilities sector indexes in Panel C. The columns labelled N, Mean, St Dev, Min and Max correspond respectively to the number of observations, the mean, the standard deviation, the minimum value and the maximum value. The column labelled Brief Description gives the full name of the utility holding companies or the utilities sector indexes.

TABLE 2
CAPM Risk Premium Estimates for the Gas Distribution Reference Portfolios

Portfolio	Estimate	SE	t-stat	Prob > t
Panel A: Risk Premium Error (Alpha)				
DJ_GasDi	8.43	3.79	2.22	0.028
CAindex	4.52	2.33	1.94	0.053
DJ_GasUS	7.39	3.34	2.21	0.028
USindex	6.23	1.95	3.19	0.002
Panel B: Beta				
DJ_GasDi	0.21	0.11	1.95	0.053
CAindex	0.34	0.07	4.60	<.0001
DJ_GasUS	0.37	0.09	4.16	<.0001
USindex	0.46	0.06	7.37	<.0001
Panel C: Risk Premium				
DJ_GasDi	1.66	1.28	1.30	0.195
CAindex	1.76	1.11	1.58	0.116
DJ_GasUS	2.74	1.46	1.87	0.063
USindex	2.72	1.33	2.04	0.042

NOTES: This table reports the results of the estimation of the CAPM for the gas distribution reference portfolios. Panels A to C look at the annualized risk premium error or alpha (in percent), the market beta and the annualized risk premium (in percent), respectively. The columns labelled Estimate, SE, t-stat and Prob > |t| give respectively the estimates, their standard errors, their t-statistics and their p-values. The four gas distribution reference portfolios and their sample are described in section 2 and table 1. The annualized mean market risk premiums for their corresponding sample period are 8.1% for DJ_GasDi, 5.2% for CAindex, 7.5% for DJ_GasUS and 6.0% for USindex.

TABLE 3
Fama-French Risk Premium Estimates for the Gas Distribution Reference Portfolios

Portfolio	Estimate	SE	t-stat	Prob > t
Panel A: Risk Premium Error (Alpha)				
DJ_GasDi	4.45	3.11	1.43	0.155
CAindex	2.04	1.85	1.11	0.270
DJ_GasUS	1.31	3.01	0.43	0.665
USindex	0.31	1.80	0.17	0.863
Panel B: Beta				
DJ_GasDi	0.41	0.08	5.06	<.0001
CAindex	0.48	0.05	10.38	<.0001
DJ_GasUS	0.63	0.07	9.64	<.0001
USindex	0.64	0.06	11.18	<.0001
Panel C: Size Beta				
DJ_GasDi	-0.01	0.08	-0.11	0.912
CAindex	-0.02	0.05	-0.51	0.613
DJ_GasUS	0.00	0.09	0.04	0.971
USindex	0.20	0.07	2.9	0.004
Panel D: Value Beta				
DJ_GasDi	0.33	0.06	5.12	<.0001
CAindex	0.30	0.04	7.64	<.0001
DJ_GasUS	0.59	0.13	4.41	<.0001
USindex	0.71	0.10	7.21	<.0001
Panel E: Risk Premium				
DJ_GasDi	5.64	1.78	3.17	0.002
CAindex	4.23	1.52	2.78	0.006
DJ_GasUS	8.83	2.32	3.81	0.000
USindex	8.64	2.16	4	<.0001

NOTES: This table reports the results of the estimation of the Fama-French model for the gas distribution reference portfolios. Panels A to E look at the annualized risk premium error or alpha (in percent), the market beta, the size beta, the value beta and the annualized risk premium (in percent), respectively. The columns labelled Estimate, SE, t-stat and Prob > |t| give respectively the estimates, their standard errors, their t-statistics and their p-values. The four gas distribution reference portfolios and their sample are described in section 2 and table 1. The annualized mean market risk premiums for their corresponding sample period are 8.1% for DJ_GasDi, 5.2% for CAindex, 7.5% for DJ_GasUS and 6.0% for USindex. The annualized mean size risk premiums for their corresponding sample period are 12.4% for DJ_GasDi, 8.9% for CAindex, 2.7% for DJ_GasUS and 2.7% for USindex. The annualized mean value risk premiums for their corresponding sample period are 7.4% for DJ_GasDi, 6.4% for CAindex, 6.9% for DJ_GasUS and 6.0% for USindex.

TABLE 4
Adjusted CAPM Risk Premium Estimates for the Gas Distribution Reference Portfolios

Portfolio	Estimate	SE	t-stat	Prob > t
Panel A: Risk Premium Error (Alpha)				
DJ_GasDi	1.82	2.00	0.91	0.365
CAindex	1.39	1.54	0.9	0.366
DJ_GasUS	2.68	1.97	1.36	0.176
USindex	2.89	1.37	2.11	0.035
Panel B: Adjusted Beta				
DJ_GasDi	0.47	0.07	6.69	<.0001
CAindex	0.56	0.05	11.38	<.0001
DJ_GasUS	0.58	0.06	9.84	<.0001
USindex	0.64	0.04	15.44	<.0001
Panel C: Bias Correction				
DJ_GasDi	4.46	2.28	1.96	0.052
CAindex	1.99	1.10	1.81	0.071
DJ_GasUS	3.12	1.61	1.94	0.054
USindex	2.26	0.77	2.94	0.004
Panel D: Risk Premium				
DJ_GasDi	8.27	2.71	3.05	0.003
CAindex	4.88	2.11	2.31	0.021
DJ_GasUS	7.45	2.52	2.96	0.004
USindex	6.05	1.89	3.21	0.002

NOTES: This table reports the results of the estimation of the Adjusted CAPM for the gas distribution reference portfolios. Panels A to D look at the annualized risk premium error or alpha (in percent), the adjusted market beta, the bias correction and the annualized risk premium (in percent), respectively. The columns labelled Estimate, SE, t-stat and Prob > |t| give respectively the estimates, their standard errors, their t-statistics and their p-values. The four gas distribution reference portfolios and their sample are described in section 2 and table 1. The annualized mean market risk premiums for their corresponding sample period are 8.1% for DJ_GasDi, 5.2% for CAindex, 7.5% for DJ_GasUS and 6.0% for USindex.

FIGURE 1 **Risk Premium Errors with the CAPM for Various Utilities**

Figure 1a: Firms in the CAindex Portfolio

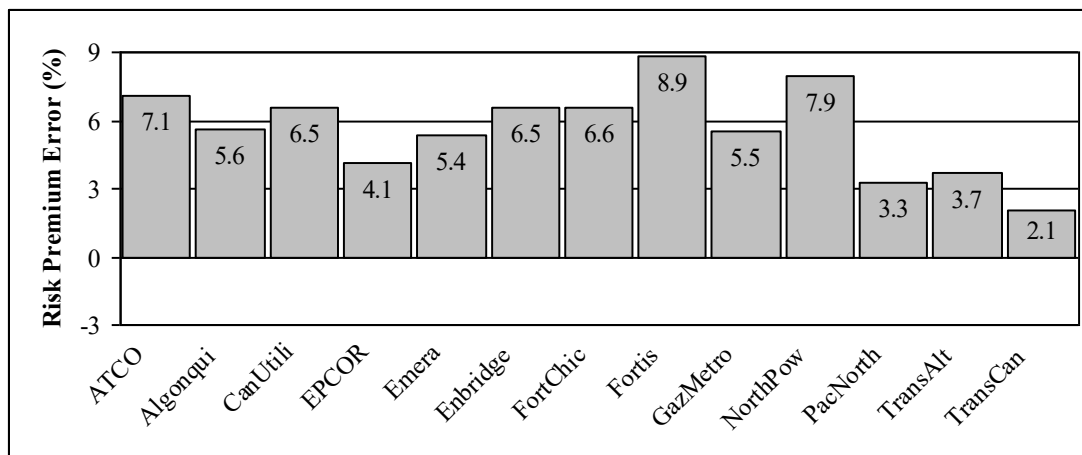


Figure 1b: Firms in the USindex Portfolio

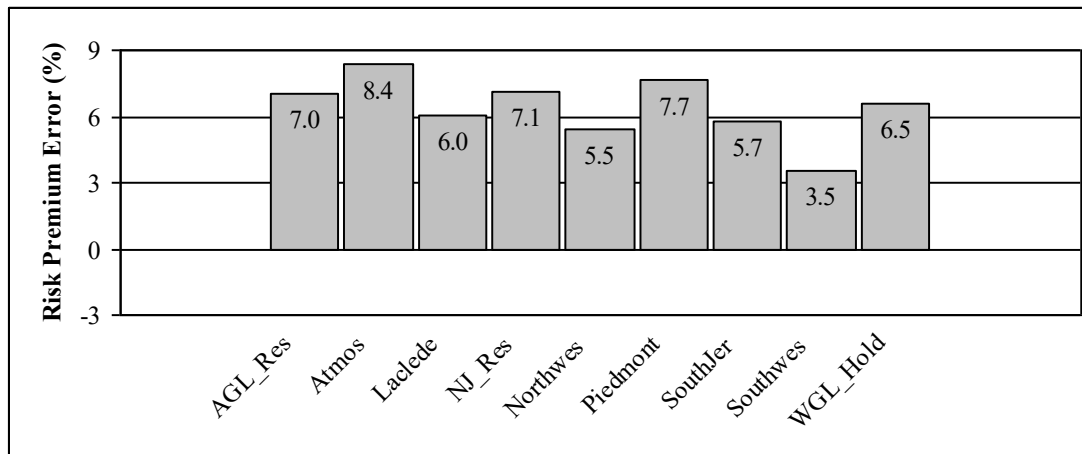
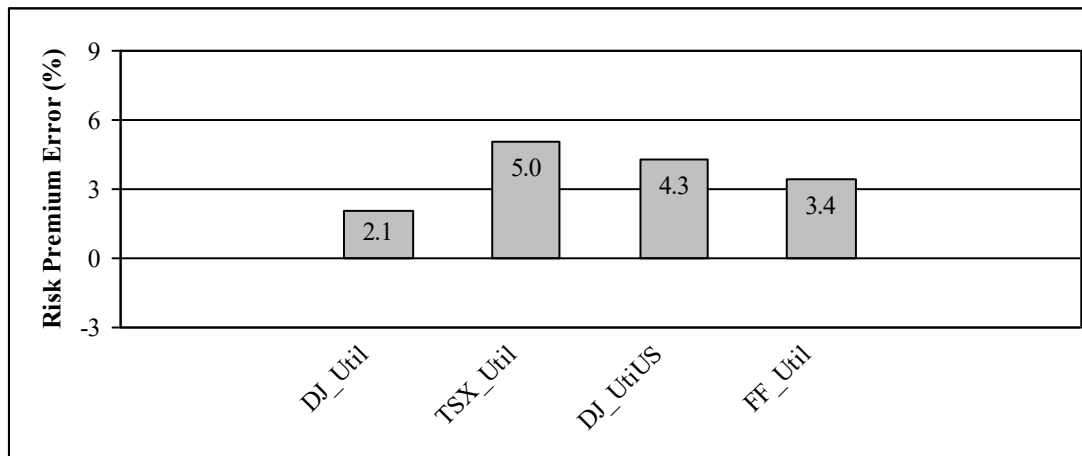


Figure 1c: Utilities Reference Portfolios



NOTES: This figure shows the annualized risk premium errors (or alphas) with the CAPM for the Canadian utilities in the CAindex portfolio (Figure 1a), the U.S. gas distributors in the USindex portfolio (Figure 1b) and the utilities reference portfolios (Figure 1c).

FIGURE 2 Comparison of the Fama-French and CAPM Results

Figure 2a: Risk Premium Errors

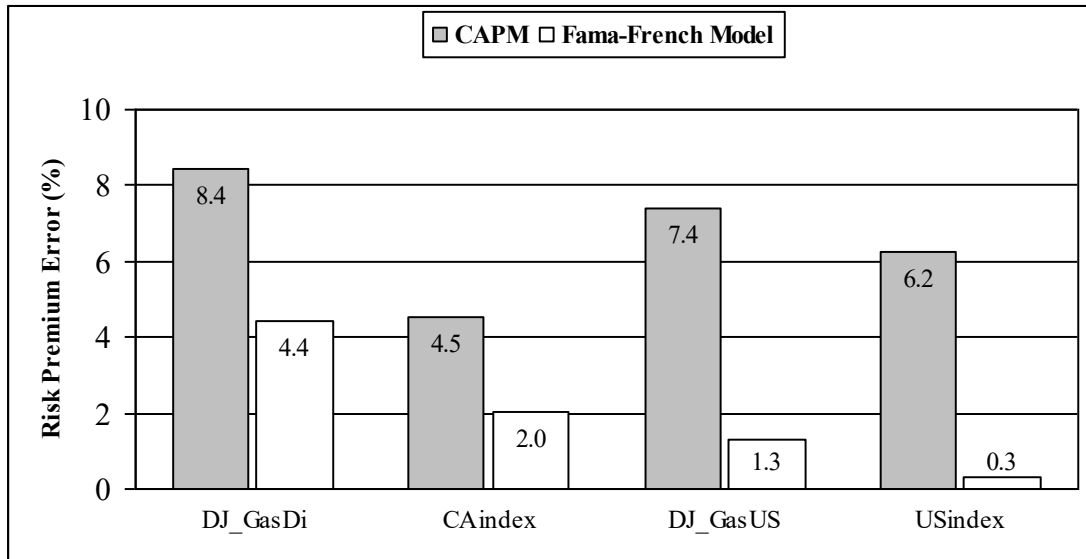
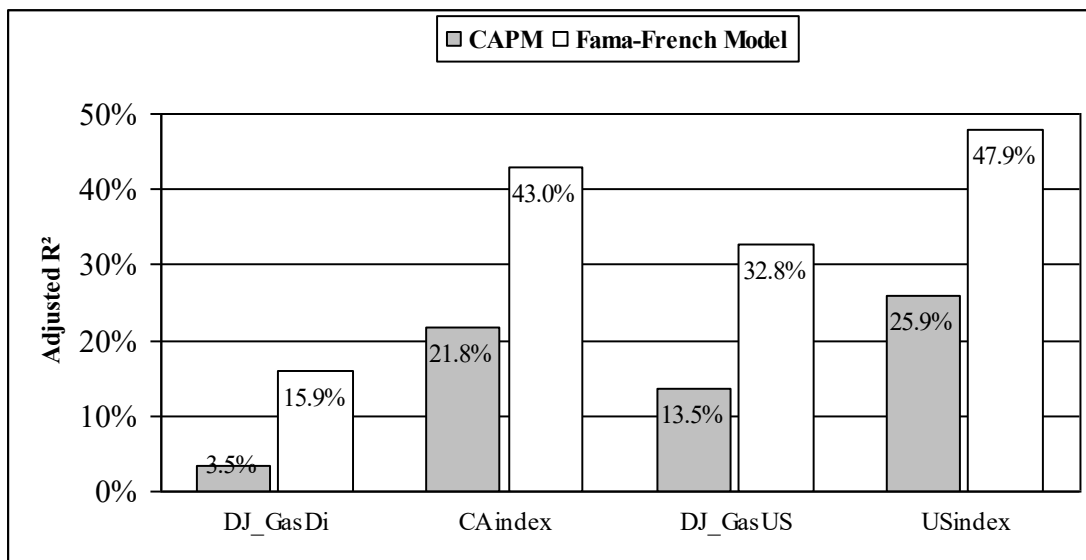


Figure 2b: Adjusted R²s



NOTES: This figure compares the results of the CAPM (gray bars) and the Fama-French model (white bars) in terms of annualized risk premium errors (or alphas) (Figure 2a) and adjusted R² (Figure 2b) for the gas distribution reference portfolios.

FIGURE 3 Risk Premium Errors with the Fama-French Model for Various Utilities

Figure 3a: Firms in the CAindex Portfolio

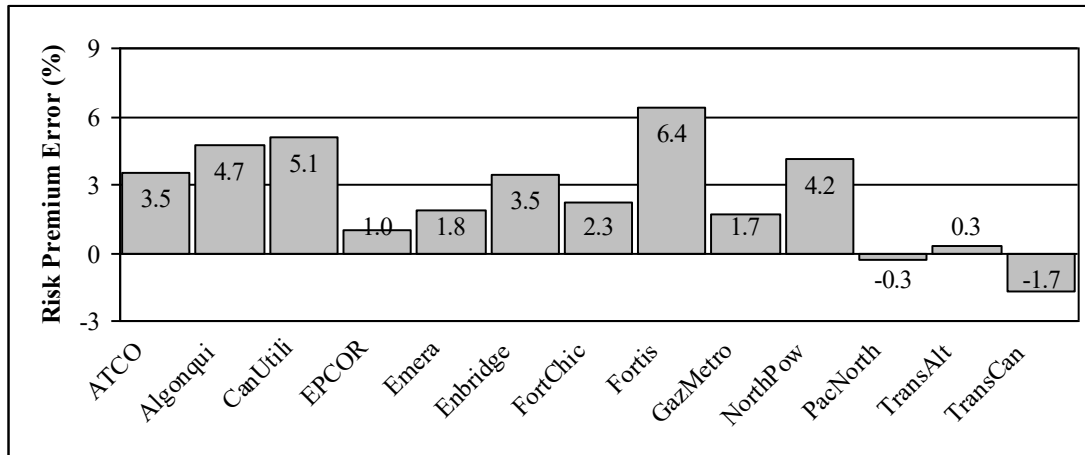


Figure 3b: Firms in the USindex Portfolio

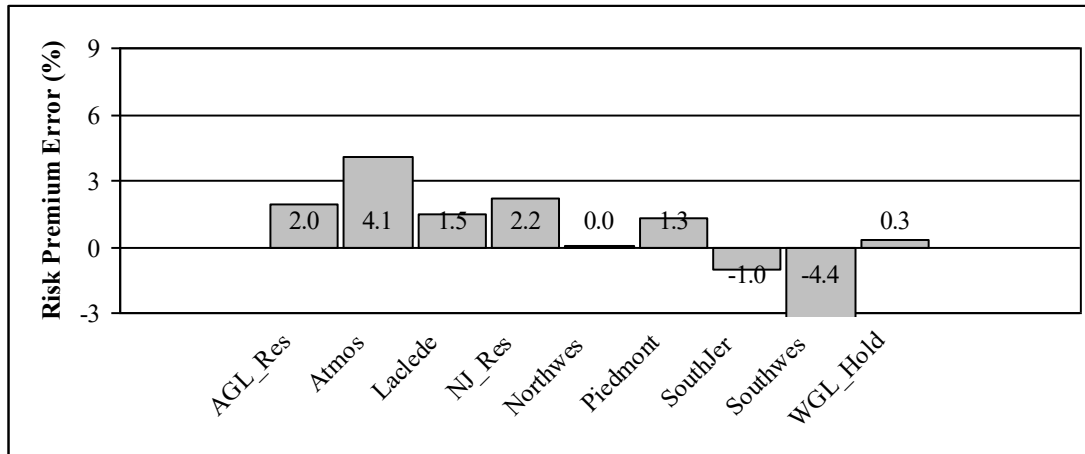
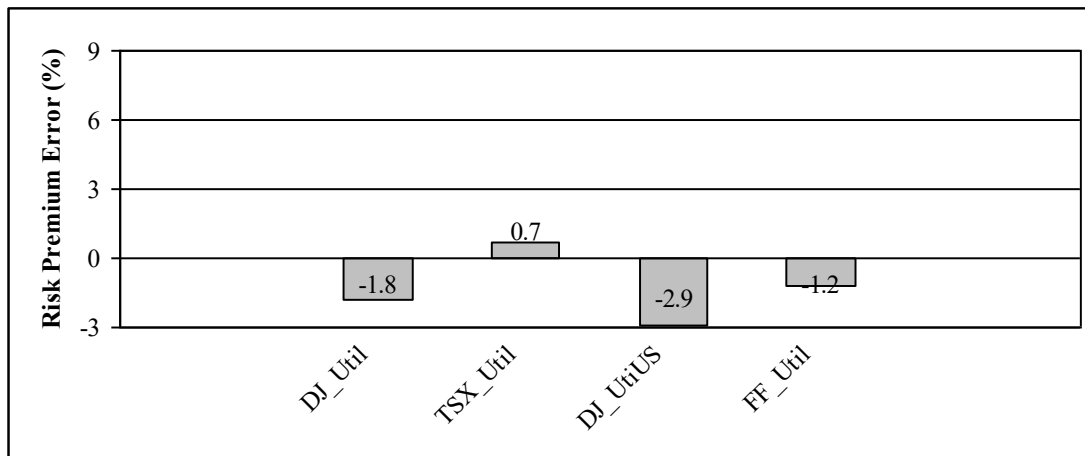


Figure 3c: Utilities Reference Portfolios



NOTES: This figure shows the annualized risk premium errors (or alphas) with the Fama-French model for the Canadian utilities in the CAindex portfolio (Figure 3a), the U.S. gas distributors in the USindex portfolio (Figure 3b) and the utilities reference portfolios (Figure 3c).

FIGURE 4 Value Betas for Various Utilities

Figure 4a: Firms in the CAindex Portfolio

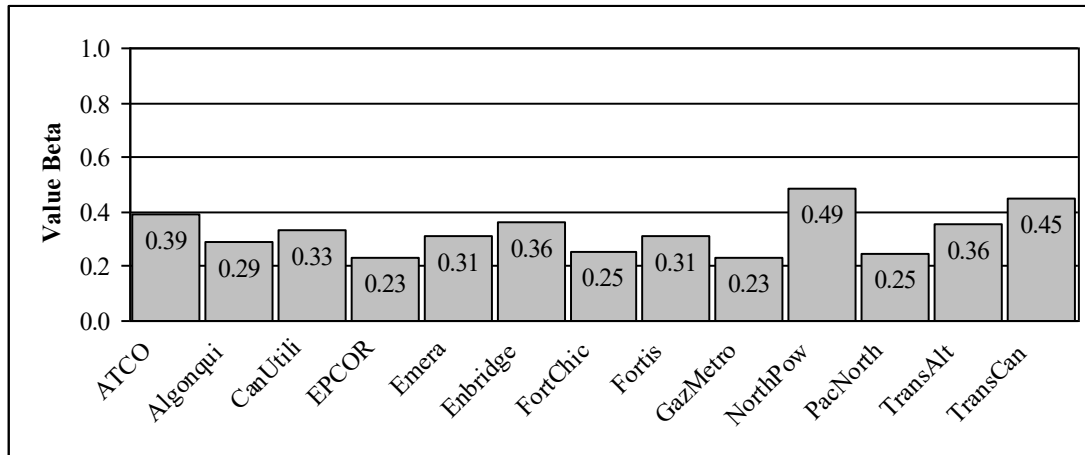


Figure 4b: Firms in the USindex Portfolio

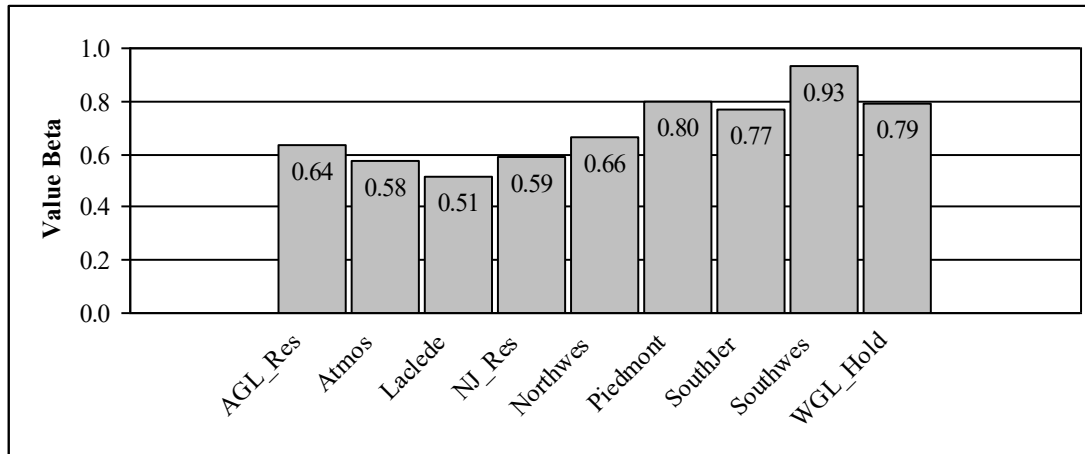
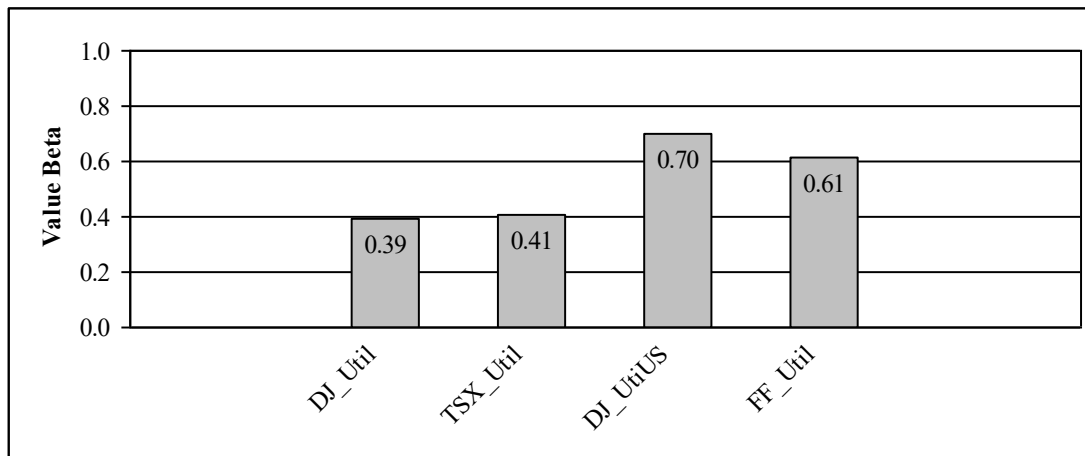


Figure 4c: Utilities Reference Portfolios



NOTES: This figure shows the value betas in the Fama-French model for the Canadian utilities in the CAindex portfolio (Figure 4a), the U.S. gas distributors in the USindex portfolio (Figure 4b) and the utilities reference portfolios (Figure 4c).

FIGURE 5
Risk Premium Errors with the Adjusted CAPM for Various Utilities

Figure 5a: Firms in the CAindex Portfolio

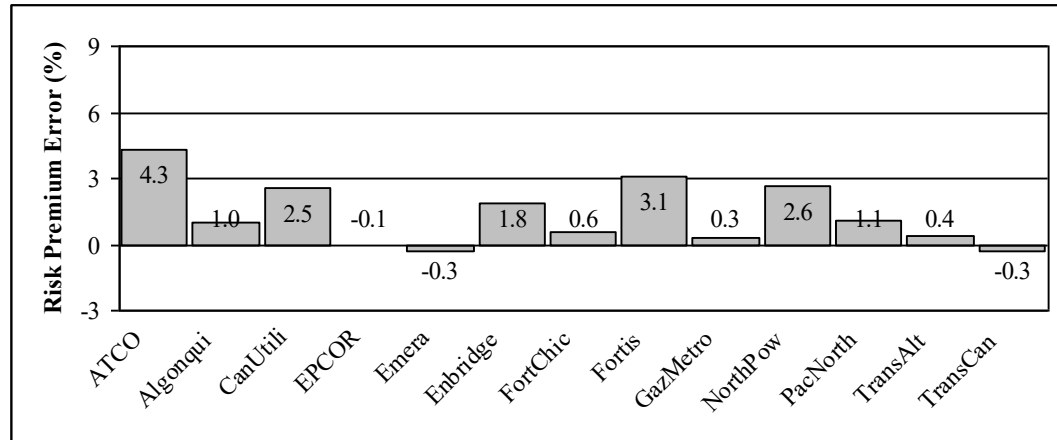


Figure 5b: Firms in the USindex Portfolio

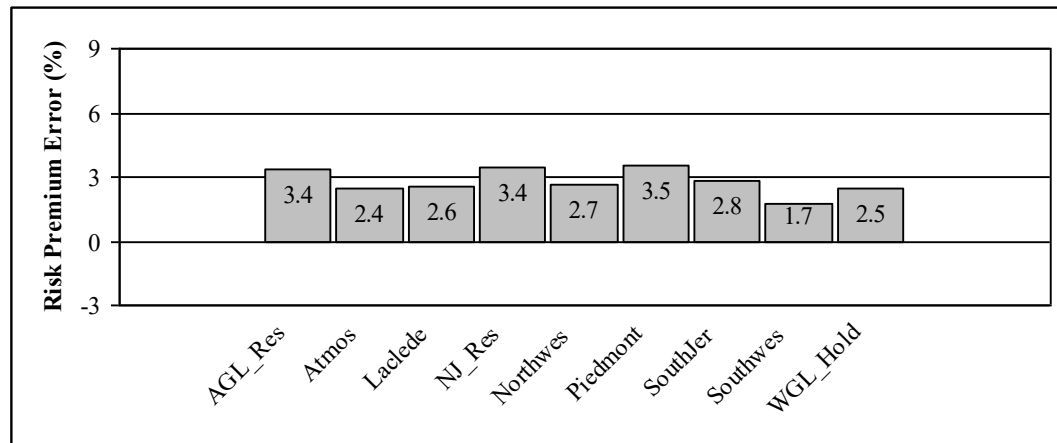
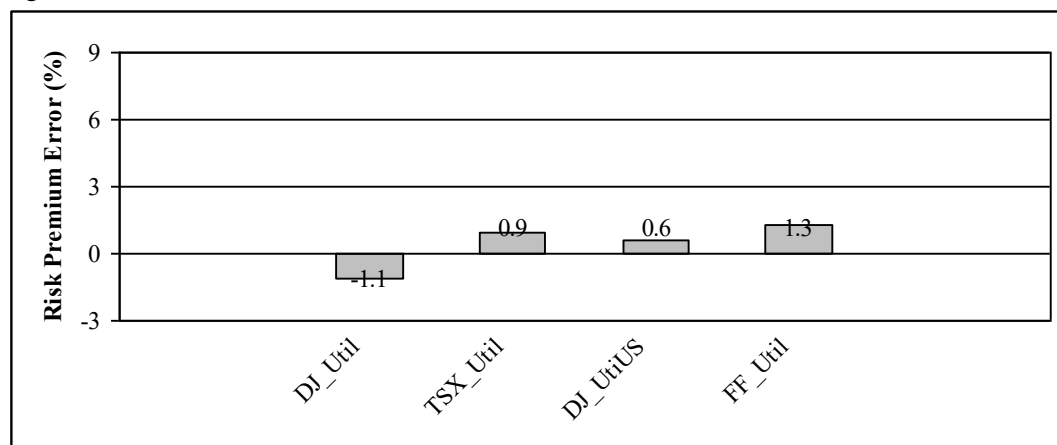


Figure 5c: Utilities Reference Portfolios



NOTES: This figure shows the annualized risk premium errors (or alphas) with the Adjusted CAPM for the Canadian utilities in the CAindex portfolio (Figure 5a), the U.S. gas distributors in the USindex portfolio (Figure 5b) and the utilities reference portfolios (Figure 5c).

150 FERC ¶ 61,165
UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Before Commissioners: Cheryl A. LaFleur, Chairman;
Philip D. Moeller, Tony Clark,
Norman C. Bay, and Colette D. Honorable.

Martha Coakley, Massachusetts Attorney General;
Connecticut Public Utilities Regulatory Authority;
Massachusetts Department of Public Utilities;
New Hampshire Public Utilities Commission;
Connecticut Office of Consumer Counsel; Maine Office
of the Public Advocate; George Jepsen, Connecticut
Attorney General; New Hampshire Office of Consumer
Advocate; Rhode Island Division of Public Utilities and
Carriers; Vermont Department of Public Service;
Massachusetts Municipal Wholesale Electric Company;
Associated Industries of Massachusetts; The Energy
Consortium; Power Options, Inc.; and the Industrial
Energy Consumer Group

Docket Nos. EL11-66-002

v.

Bangor Hydro-Electric Co.; Central Maine Power Co.;
New England Power Co.; New Hampshire Transmission
LLC; NSTAR Electric and Gas Corp.; Northeast
Utilities Service Co.; The United Illuminating Co.;
Unitil Energy Systems, Inc. and Fitchburg Gas and
Electric Light Co.; Vermont Transco, LLC

Martha Coakley, Massachusetts Attorney General;
Connecticut Public Utilities Regulatory Authority;
Massachusetts Department of Public Utilities; New
Hampshire Public Utilities Commission; Connecticut
Office of Consumer Counsel; Maine Office of the Public
Advocate; George Jepsen, Connecticut Attorney
General; New Hampshire Office of Consumer Advocate;
Rhode Island Division of Public Utilities and Carriers;
Vermont Department of Public Service; Massachusetts
Municipal Wholesale Electric Company; Associated
Industries of Massachusetts; The Energy Consortium;

EL11-66-003

Docket Nos. EL11-66-002 and EL11-66-003

- 2 -

Power Options, Inc.; and the Industrial Energy
Consumer Group

v.

Bangor Hydro-Electric Co.; Central Maine Power Co.;
New England Power Co.; New Hampshire Transmission
LLC; NSTAR Electric and Gas Corp.; Northeast
Utilities Service Co.; The United Illuminating Co.;
Unitil Energy Systems, Inc. and Fitchburg Gas and
Electric Light Co.; Vermont Transco, LLC

OPINION NO. 531-B

ORDER ON REHEARING

(Issued March 3, 2015)

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1. On July 21, 2014, the New England Transmission Owners (NETOs),¹ a group of complainants (Complainants) and intervenors (collectively, Petitioners),² and the Eastern

¹ The NETOs include Bangor Hydro-Elec. Co.; Central Maine Power Co.; New England Power Co.; New Hampshire Transmission LLC; NSTAR Electric & Gas Corp.; Northeast Utilities Service Co.; United Illuminating Co.; Unitil Energy Systems, Inc. and Fitchburg Gas & Electric Light Co.; and Vermont Transco, LLC.

² Complainants include Martha Coakley, Massachusetts Attorney General; Connecticut Public Utilities Regulatory Authority; Massachusetts Department of Public Utilities; New Hampshire Public Utilities Commission; Connecticut Office of Consumer Counsel; Maine Office of the Public Advocate; George Jepsen, Connecticut Attorney General; New Hampshire Office of Consumer Advocate; Rhode Island Division of Public Utilities and Carriers; Vermont Department of Public Service; Massachusetts Municipal Wholesale Electric Co.; Associated Industries of Massachusetts; the Energy Consortium; Power Options, Inc.; and the Industrial Energy Consumer Group. Intervenors
(continued...)

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Massachusetts Consumer-Owned Systems (EMCOS), filed requests for rehearing of the Commission's June 19, 2014 order on initial decision³ concerning a complaint, filed pursuant to section 206 of the Federal Power Act (FPA),⁴ challenging the NETOs' base return on equity (ROE) reflected in ISO New England Inc.'s (ISO-NE) open access transmission tariff (OATT).⁵ In this order, we deny rehearing.

I. Background

2. The NETOs recover their transmission revenue requirements through formula rates included in ISO-NE's OATT. The revenue requirements for Regional Network Service⁶ and Local Network Service⁷ that the NETOs provide are calculated using the same single base ROE. On October 31, 2006, the Commission, in Opinion No. 489, established the base ROE at 11.14 percent, which consisted of an initial base ROE of 10.4 percent plus an upward adjustment of 74 basis points to account for changes in capital market conditions that took place between the issuance of the Administrative Law Judge's initial decision in that proceeding and the issuance of Opinion No. 489,⁸ as reflected in changes in U.S. Treasury bond yields during that time period.

New Hampshire Electric Cooperative, Inc. and Maine Public Utilities Commission requested rehearing jointly with the Complainants.

³ *Martha Coakley, Mass. Attorney Gen. v. Bangor Hydro-Elec. Co.*, Opinion No. 531, 147 FERC ¶ 61,234 (2014) (Opinion No. 531), *order on paper hearing*, Opinion No. 531-A, 149 FERC ¶ 61,032 (2014) (Opinion No. 531-A).

⁴ 16 U.S.C. § 824e (2012).

⁵ ISO-NE's OATT is section II of ISO-NE's Transmission, Markets, and Services Tariff (Tariff). *See* ISO-NE, Tariff, § II.

⁶ Regional Network Service is the transmission service over the pool transmission facilities described in Part II.B of the OATT. ISO-NE, Tariff, § I.2 (50.0.0); *see also* ISO-NE, Tariff, § II.B Regional Network Service (0.0.0), *et seq.*

⁷ Local Network Service is the network service provided under Schedule 21 and the Local Service Schedules of ISO-NE's OATT. ISO-NE, Tariff, § I.2 (50.0.0); *see also* ISO-NE, Tariff, Schedule 21 Local Service (1.0.0), *et seq.*

⁸ *Bangor Hydro-Elec. Co.*, Opinion No. 489, 117 FERC ¶ 61,129 (2006) (Opinion No. 489), *order on reh'g*, 122 FERC ¶ 61,265 (2008), *order granting clarification*, 124 FERC ¶ 61,136 (2008), *aff'd sub nom. Conn. Dep't of Pub. Util. Control v. FERC*, 593 F.3d 30 (2010).

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3. On September 30, 2011, the Complainants filed a complaint alleging that the NETOs' 11.14 percent base ROE was unjust and unreasonable because capital market conditions had significantly changed since that base ROE was established in 2006. The Complainants argued that the bubble in the U.S. housing market, the subsequent financial crisis and economic recession, and the fiscal and monetary policies of the U.S. government had caused a "flight to quality"⁹ in the capital markets. The Complainants contended that these market conditions had lowered bond yields and, as a result, capital costs for utilities.¹⁰ The Complainants argued that, as a result, the NETOs' 11.14 percent base ROE now exceeded the level necessary to satisfy the Supreme Court's standards in *Bluefield*¹¹ and *Hope*.¹² The Complainants asserted that, based on a discounted cash flow (DCF) analysis conducted by their expert witness, the just and reasonable base ROE for the NETOs should not exceed 9.2 percent.

4. On May 3, 2012, the Commission issued an order on the complaint, establishing hearing and settlement judge procedures.¹³ The Hearing Order also set a refund effective date of October 1, 2011. The hearing commenced on May 6, 2012 and was completed on May 10, 2013.¹⁴ In accordance with the hearing's procedural schedule, the participants each first submitted an ROE analysis,¹⁵ based on data from a 6-month study period in

⁹ The "flight to quality" refers to investors seeking low-risk investment vehicles.

¹⁰ Complaint, Ex. C-1 at 5-12.

¹¹ *Bluefield Water Works & Improvement Co. v. Pub. Serv. Comm'n*, 262 U.S. 679 (1923) (*Bluefield*).

¹² *FPC v. Hope Natural Gas Co.*, 320 U.S. 591 (1944) (*Hope*).

¹³ *Martha Coakley, Mass. Attorney Gen.. v. Bangor Hydro-Elec. Co.*, 139 FERC ¶ 61,090 (2012) (Hearing Order).

¹⁴ The parties conducted settlement negotiations but reached an impasse, leading to termination of the settlement procedures in August 2012. *Martha Coakley, Mass. Attorney Gen. v. Bangor Hydro-Elec. Co.*, 144 FERC ¶ 63,012, at P 28 (2013) (Initial Decision).

¹⁵ The following expert witnesses submitted ROE analyses: Dr. William E. Avera, for the NETOs; Ms. Sabina U. Joe, for Trial Staff; Dr. John Wilson, for the EMCOS; and Dr. Randall Woolridge, for the Complainants.

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2012,¹⁶ and then filed an updated ROE analysis, using the same DCF methodology that each participant used in its initial analysis but with data based on the 6-month study period from October 2012 through March 2013.

5. On August 6, 2013, the Presiding Judge issued the initial decision, finding the NETOs' current 11.14 percent base ROE to be unjust and unreasonable.¹⁷ The Presiding Judge adopted the DCF methodology used by the NETOs and found that it is appropriate to establish two different base ROEs in this proceeding—one for the 15-month refund period from October 1, 2011 (i.e., the refund effective date) to December 31, 2012, and one for the prospective period commencing when the Commission issues its order setting the going-forward base ROE. Thus, the Presiding Judge considered two separate DCF analyses relying on overlapping data from each period, the first using data from May 2012 through October 2012 and the second using data from October 2012 through March 2013. The Presiding Judge found the just and reasonable base ROE for the refund period to be 10.6 percent and the just and reasonable base ROE for the prospective period to be 9.7 percent.¹⁸

6. On June 19, 2014, the Commission issued Opinion No. 531, affirming in part and reversing in part the initial decision.¹⁹ In Opinion No. 531, the Commission changed its approach on the DCF methodology to be applied in public utility rate cases, by adopting the two-step DCF methodology in place of the one-step DCF methodology the Commission had historically used. The Commission explained that the two-step DCF formula is $k = D/P (1 + .5g) + g$, where "D/P," the dividend yield, is calculated using a single, average dividend yield based on the indicated dividend and the average monthly high and low stock prices over a six-month period; and "g," the constant dividend growth rate, is calculated by averaging short-term and long-term growth estimates, with the short-term estimate receiving two-thirds weight and the long-term estimate receiving one-third weight.²⁰

¹⁶ Due to the different due dates for the parties' initial briefs, which ranged from October 2012 to January 2013, each party's initial ROE analysis was based on a different 6-month period in 2012.

¹⁷ Initial Decision, 144 FERC ¶ 63,012 at P 544.

¹⁸ *Id.*

¹⁹ See generally Opinion No. 531, 147 FERC ¶ 61,234.

²⁰ *Id.* PP 15, 17, 39.

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7. The Commission, after finding that there should be only one base ROE applicable to both the refund period and the prospective period in this proceeding, then applied the two-step DCF methodology to the facts of this proceeding, using a national proxy group of companies the Commission found were of comparable risk to the NETOs, to determine the NETOs' base ROE; however, because the parties had not litigated one input to the two-step DCF methodology—i.e., the appropriate long-term growth projection—the Commission instituted a paper hearing on that narrow issue. The Commission also found that, due to the anomalous capital market conditions reflected in the record, mechanically applying the DCF methodology and placing the NETOs' base ROE at the midpoint of the zone of reasonableness produced by that methodology would not satisfy the requirements of *Hope* and *Bluefield*.²¹ Therefore, the Commission found it appropriate, based on the record evidence in the proceeding, to place the NETOs' base ROE halfway between the midpoint of the zone of reasonableness and the top of that zone.²² However, the Commission explained that its finding on the specific numerical just and reasonable ROE for the NETOs was subject to the outcome of the paper hearing on the appropriate long-term growth projection to be used in the two-step DCF methodology.²³ The Commission also explained that, according to Commission precedent, “when a public utility’s ROE is changed, either under section 205 or section 206 of the FPA, that utility’s total ROE, inclusive of transmission incentive ROE adders, should not exceed the top of the zone of reasonableness produced by the two-step DCF methodology.”²⁴

8. On October 16, 2014, the Commission issued Opinion No. 531-A, the order on the paper hearing instituted by Opinion No. 531, finding that long-term projected growth in gross domestic product (GDP) is the appropriate long-term growth projection to use in the two-step DCF methodology.²⁵ Accordingly, the Commission found that a just and reasonable ROE for the NETOs is 10.57 percent, and that the NETOs' total or maximum ROE, including transmission incentive ROE adders, cannot exceed 11.74 percent, i.e., the top of the zone of reasonableness in this proceeding.²⁶ The Commission also ordered the

²¹ *Id.* P 142.

²² *Id.*

²³ *Id.*

²⁴ *Id.* P 165.

²⁵ Opinion No. 531-A, 149 FERC ¶ 61,032 at P 10.

²⁶ *Id.* PP 10-11.

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NETOs to issue refunds for the 15-month refund period from October 1, 2011 through December 31, 2012.²⁷

II. Discussion

A. Procedural Matters

9. On July 21, 2014, the NETOs, Petitioners, and EMCOS filed requests for rehearing of Opinion No. 531. On November 17, 2014, the NETOs requested rehearing of Opinion No. 531-A, in Docket No. EL11-66-003, by submitting the same pleading that they filed on July 21, 2014 as a request for rehearing of Opinion No. 531.²⁸ Because the NETOs submitted the same pleading as a request for rehearing of both Opinion Nos. 531 and 531-A and, therefore, presented identical arguments in those two proceedings, our merits determinations in the instant order apply to the NETOs' requests for rehearing in both Docket Nos. EL11-66-002 and EL11-66-003. Thus, we also deny the NETOs' request for rehearing of Opinion No. 531-A.

1. Answers to Rehearing Requests, and Related Answers to Answers

10. On August 5, 2014, the Petitioners filed an answer to the NETOs' request for rehearing (Petitioners' August 5 Answer), and the NETOs filed an answer to the Petitioners' request for rehearing (NETOs' August 5 Answer). On August 20, 2014, the NETOs filed an answer to the Petitioners' August 5 Answer (NETOs' August 20 Answer).²⁹ On August 22, 2014, the Petitioners filed an answer to the NETOs' answer to the Petitioners' request for rehearing (Petitioners' August 22 Answer). On September 4, 2014, the Petitioners filed an answer to the NETOs' August 20 Answer (Petitioners' September 4 Answer).

²⁷ *Id.* PP 12, Ordering Paragraph (C).

²⁸ See NETOs, Request for Rehearing, Docket No. EL11-66-003, at 3 (filed Nov. 7, 2014) ("the NETOs seek rehearing of Opinion No. 531-A with respect to the same issues and on the same grounds upon which they sought rehearing of Opinion No. 531. These issues and grounds are set forth in the NETOs' 'Request for Rehearing and Motion for Clarification of the New England Transmission Owners,' which the NETOs filed with the Commission on July 21, 2014, and which is attached hereto and incorporated herein (see Attachment A)").

²⁹ While the NETOs' August 20 Answer was styled as a motion to clarify the record, the filing was, in substance, an answer to the Complainants' August 5 Answer.

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11. Rule 713(d) of the Commission's Rules of Practice and Procedure, 18 C.F.R. § 385.713(d) (2014), prohibits answers to a request for rehearing. Therefore, we reject the Petitioners' August 5 Answer and the NETOs' August 5 Answer. Accordingly, we also reject the answers to those answers—specifically, the NETOs' August 20 Answer, the Petitioners' August 22 Answer, and the Petitioners' September 4 Answer.

2. Motion to Strike

12. On August 5, 2014, the NETOs filed a motion to strike certain extra-record evidence from the Petitioners' request for rehearing. On August 20, 2014, the Petitioners filed an answer opposing the NETOs' motion to strike. We grant in part and deny in part the NETOs' motion to strike. The Commission has consistently held that the submission of additional factual information in a request for rehearing is inappropriate.³⁰ Therefore, we grant the NETOs' motion with respect to the extra-record evidence in Petitioners' request for rehearing. However, we deny the NETOs' motion with respect to the evidence that was already in the record and that Petitioners have merely reframed through graphical representation or basic arithmetic.³¹

3. Motions to Intervene Out-of-Time

13. On July 21, 2014, American Municipal Power, Inc. (AMP) filed a motion to intervene out-of-time and a request for rehearing,³² and the American Public Power Association (APPA) and the National Rural Electric Cooperative Association (NRECA) jointly filed a motion to intervene out-of-time. APPA and NRECA also joined in the Petitioners' request for rehearing. On August 5, 2014, the NETOs filed an answer

³⁰ *E.g., Transcontinental Gas Pipe Line Corp.*, 94 FERC ¶ 61,066, at 61,278 (2001).

³¹ Specifically, we deny the NETOs' motion with respect to (1) the altered version of the NETOs' risk premium analysis, at page 38 and Attachment A of Petitioners' request for rehearing; (2) the altered version of Opinion No. 531's Appendix showing an alternate source of growth rate projections, at pages 43 and 51, and at Attachment B, of Petitioners' request for rehearing; (3) the altered version of Opinion No. 531's Appendix reflecting an alternate low-end outlier adjustment, at pages 14, 62, and 63, and at Attachment C, of Petitioners' request for rehearing; (4) the altered version of Exhibit SC-524, at pages 26 and 27 of Petitioners' request for rehearing; (5) the histogram at pages 2-3 of Petitioners' request for rehearing; and (6) the histogram on pages 24-25 of Petitioners' request for rehearing.

³² While AMP styled its filing as a motion for clarification, it is in substance a request for rehearing.

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opposing AMP's, APPA's, and NRECA's motions to intervene out-of-time, and AMP's request for rehearing. On August 12, 2014, APPA and NRECA filed an answer to the NETOs' answer to the motions to intervene out of time and AMP's request for rehearing. On December 5, 2014, the Maine Public Advocate Office filed a motion to intervene out-of-time.

14. In ruling on a late-filed motion to intervene, the Commission applies the criteria set forth in Rule 214(d) of the Commission's Rules of Practice and Procedure, 18 C.F.R. § 385.214(d) (2014), and considers, among other things, whether the movant had good cause for failing to file the motion within the time prescribed, whether any disruption to the proceeding might result from permitting the intervention, and whether any prejudice to or additional burdens upon the existing parties might result from permitting the intervention. A petitioner for late intervention bears a higher burden to show good cause for late intervention after the Commission has issued a final order in a proceeding, and it is the Commission's policy to deny late intervention at the rehearing stage, even when the movant claims that the decision establishes a broad policy of general application.³³

15. We find that AMP, APPA, NRECA, and the Maine Public Advocate Office have not met their burden of justifying late intervention. The Complainants filed the complaint in this proceeding on September 30, 2011, alleging that the capital market conditions following the collapse of the housing bubble and the resulting economic recession were such that the NETOs' existing ROE was no longer just and reasonable; the Commission then set the complaint for hearing on May 30, 2012 and issued a dispositive order on June 19, 2014, nearly three years after the complaint was filed. Thus, AMP, APPA, NRECA, and the Maine Public Advocate Office had ample notice that this proceeding involved the Commission's approach to determining public utilities' ROE, that the effect of recent capital market conditions on that approach was an issue central to the complaint, and that a Commission order in this proceeding would have precedential effect on similar proceedings before the Commission. AMP, APPA, NRECA, and the Maine Public Advocate Office have not shown good cause for failing to file their motions to intervene during the statutory comment period, or subsequent to that period but prior to the Commission's issuance of Opinion No. 531. AMP's, APPA's, and NRECA's statements that they did not anticipate the specific outcome in this proceeding, without more, do not suffice to make that showing.³⁴ We therefore deny their late-filed motions to intervene.

³³ See, e.g., *Williston Basin Interstate Pipeline Co.*, 112 FERC ¶ 61,038, at P 12 (2005).

³⁴ APPA and NRECA cite *Duke Energy Carolinas, LLC, et al.*, 147 FERC ¶ 61,241 (2014) (*Duke*), as an example of an instance where the Commission has allowed a national organization's late intervention due to an order's far-reaching impacts. However, we find *Duke* to be distinguishable from the instant case. In *Duke*, the National
(continued...)

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Correspondingly, we also deny AMP's, APPA's, and NRECA's requests for rehearing, because under Rule 713(b) the Commission's Rules of Practice and Procedure only a party to a proceeding may seek rehearing.³⁵

B. Substantive Matters

16. The arguments raised on rehearing involve issues concerning the burden of proof, placement of the NETOs' base ROE within the zone of reasonableness, the impact of the change in DCF methodology on the NETOs' existing transmission incentive ROE adders, and the timing of the Commission's establishment of the just and reasonable rate in this proceeding. As discussed below, we deny rehearing on these issues.

1. Burden of Proof

a. Opinion No. 531

17. The Commission in Opinion No. 531 affirmed the Presiding Judge's determination on the burden of proof,³⁶ explaining that under FPA section 206 the burden to show that a rate is unjust and unreasonable "shall be on the Commission or the complainant,"³⁷ and, in the context of an ROE proceeding, the burden entails finding that the existing ROE is not "commensurate with returns on investments in other enterprises having corresponding risks . . . [and] sufficient to assure confidence in the financial integrity of the enterprise, so as to maintain its credit and to attract capital."³⁸ The Commission explained that, to estimate the return necessary to attract equity investors, the Commission uses the DCF

Association of Regulatory Utility Commissioners (NARUC) failed to intervene in an Order No. 1000 proceeding. NARUC explained that it had intervened in multiple Order No. 1000 proceedings, but that its failure to intervene in *Duke* could only have been avoided if NARUC had intervened in every Order No. 1000 proceeding. Unlike *Duke*, the instant proceeding was the first case of its kind to challenge utilities' base ROEs during the economic recession of 2007-2009, and AMP, APPA, and NRECA should have known that the proceeding could have precedential effect on other proceedings.

³⁵ 16 U.S.C. § 825(l) (2012); 18 C.F.R. § 385.713(b) (2014); *see, e.g., Southern Company Servs., Inc.*, 92 FERC ¶ 61,167 (2000).

³⁶ Opinion No. 531, 147 FERC ¶ 61,234 at P 49.

³⁷ *Id.* P 50 (quoting 16 U.S.C. § 824e (2012)).

³⁸ *Id.* P 50 (quoting *Hope*, 320 U.S. at 603).

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model, which identifies a zone of reasonable returns.³⁹ The Commission rejected the NETOs' argument that the Commission "does not have the authority under FPA section 206 to change the existing base ROE unless the evidence shows that it is entirely outside the zone of reasonableness."⁴⁰ The Commission explained that not every ROE within the zone of reasonableness is just and reasonable, and that the zone of reasonableness identified by the DCF model "is simply the first step in the determination of a just and reasonable ROE for a utility or group of utilities."⁴¹

b. Request for Rehearing

18. The NETOs argue that, because the NETOs' existing ROE of 11.14 percent falls within the zone of reasonableness, the Commission erred in finding that the Complainants and Trial Staff have carried their burden of establishing that the existing ROE is unjust and unreasonable.⁴² According to the NETOs, court and Commission precedent support a finding that an ROE within the zone of reasonableness remains just and reasonable.⁴³ The NETOs state that the Commission misunderstood their contention with respect to FPA sections 205 and 206. They assert that FPA section 206 carries a two-prong burden, the first of which is to show that the existing rate is unjust and unreasonable. The NETOs assert that interpreting FPA section 206 otherwise would eliminate the difference between the burdens of proof under FPA sections 205 and 206 by requiring a complainant to show only that its proposed rate is more just and reasonable than the existing rate. The NETOs concede that not all rates within the zone of reasonableness are equally just and reasonable, but also argue that it is not enough to show that there is a more just and reasonable rate than the existing rate; rather the complainant must demonstrate through substantial evidence that the existing rate does not fall within the zone of just and reasonable rates.⁴⁴ The NETOs contend that no party satisfied the first prong of FPA section 206.

³⁹ *Id.*

⁴⁰ *Id.* P 51.

⁴¹ *Id.*

⁴² NETOs Request for Rehearing at 26-27.

⁴³ *Id.* at 27-30 (citing *Me. Pub. Utils. Comm'n v. FERC*, 520 F.3d 464, 470-71 (D.C. Cir. 2008) (*Maine PUC*), *rev'd in part on other grounds sub nom. NRG Power Mktg., LLC v. Me. Pub. Utils. Comm'n*, 558 U.S. 165 (2010); *Calpine Corp. v. Cal. Indep. Sys. Operator Corp.*, 128 FERC ¶ 61,271, at P 41 (2009)).

⁴⁴ *Id.* at 36.

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19. The NETOs assert that, in accordance with Commission and federal court precedent, any ROE within the zone of reasonableness cannot be found to be unjust and unreasonable. The NETOs further assert that the Commission erred in finding that the DCF zone of reasonableness is different from the zone of reasonableness under FPA section 206, and that the Commission has never before drawn a distinction between the DCF zone of reasonableness and the zone of reasonableness referred to when applying FPA section 206. The NETOs argue that determining the zone of reasonableness is not merely an intermediate step in a Commission-created DCF analysis whose final step is identification of a “pinpoint” just and reasonable ROE that the Commission believes is optimal in the context of that specific proceeding, but rather is identical to the zone of reasonableness used in FPA section 206 analyses. The NETOs state that in *Northeast Utilities Service Co.*, 124 FERC ¶ 61,044 (2008) (*Northeast Utilities*), *Central Maine Power Co.*, 125 FERC ¶ 61,079 (2008) (*Central Maine*), and *Desert Southwest Power, LLC*, 135 FERC ¶ 61,143 (2011) (*Desert Southwest*) the Commission explicitly identified the DCF zone of reasonableness with the more general zone of reasonableness used in the FPA section 206 context and treated the two as one and the same.⁴⁵

20. The NETOs further argue that the Commission’s reliance on *Bangor Hydro* to distinguish the DCF zone of reasonableness from the range of reasonableness under FPA section 206 is inappropriate because *Bangor Hydro* involved application of the last clean rate doctrine after the rate under consideration had been found to be unjust and unreasonable.⁴⁶ The NETOs argue that, if *Bangor Hydro* does mean that the DCF zone of reasonableness is not really a zone of reasonableness, then that case was wrongly decided because it would contradict Commission and court precedent, particularly the D.C. Circuit’s decision in *City of Winnfield*.⁴⁷ The NETOs argue that, although Opinion No. 531 refers to the guidance on this issue in *City of Winnfield* as dicta, the Commission has relied on that guidance in previous decisions.⁴⁸ The NETOs argue that FPA section 206 carries a stricter burden of proof than FPA section 205, that the dual burden of proof

⁴⁵ *Id.* at 39.

⁴⁶ *Id.* at 41-42 (citing *Bangor Hydro-Electric Co.*, 122 FERC ¶ 61,038 (2008) (*Bangor Hydro*)).

⁴⁷ *Id.* at 42-43 (citing *City of Winnfield v. FERC*, 744 F.2d 871, 875-76 (D.C. Cir. 1984) (*City of Winnfield*)).

⁴⁸ *Id.* at 42 (citing *Texas Eastern Transmission Corp.*, 32 FERC ¶ 61,056, at 61,150 (1985); *Promoting Transmission Investment through Pricing Reform*, Order No. 679-A, FERC Stats. & Regs. ¶ 31,236, at P 98 (2006) (cross-referenced at 117 FERC ¶ 61,345, at P 98 (2006)) (Order No. 679-A); *New Dominion Energy Coop.*, 118 FERC ¶ 63,024, at n.154 (2007)).

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under section 206 provides statutory protection to utility companies, and therefore that Congress intended to create asymmetry between FPA sections 205 and 206.⁴⁹ Lastly, the NETOs argue that Opinion No. 531 reduces the clarity and predictability of the zone of reasonableness determination by instituting a method that is no longer limited by an objective formula. The NETOs argue that the resultant lack of predictability increases the perceived risk which is counter to *Hope* and *Bluefield*.⁵⁰

c. Commission Determination

21. We deny rehearing on the issue of the burden of proof. The NETOs once again assert that an existing base ROE cannot be found unjust and unreasonable as long as it is within the zone of reasonableness produced by a DCF analysis, and that the Commission's rejection of this argument in Opinion No. 531 is contrary to court and Commission precedent. We disagree.

22. The NETOs cite precedent setting forth a general ratemaking principle that "there is not a single 'just and reasonable rate' but rather a zone of rates that are just and reasonable; a just and reasonable rate is one that falls within that zone."⁵¹ The NETOs equate references to a "zone of rates that are just and reasonable" or a "zone of reasonableness" in those cases to the "zone of reasonableness" produced by the DCF analysis we use to determine the ROE to include in a public utility's cost of service. On that basis, the NETOs contend that the Commission must show that the NETOs' existing ROE is outside the DCF zone of reasonableness in order to satisfy its FPA section 206 burden to show that their ROE is unjust and unreasonable.

23. In *City of Winnfield* and *Maine PUC*, which did not involve the determination of ROE, the term "zone of reasonableness" was used to express the general principle that under the FPA there can be more than one just and reasonable rate for a service. For example, in the portion of *City of Winnfield* cited by the NETOs, the court addressed the issue of whether the rate for a power sale should be based on an incremental fuel cost or a system average fuel cost, and the court explained that if either methodology was just and

⁴⁹ *Id.* at 44 (citing *City of Winnfield*, 744 F.2d at 875).

⁵⁰ *Id.* at 45-46.

⁵¹ *See, e.g., Maine PUC*, 520 F.3d at 470-71 (upholding Commission determination that transition payments agreed to in a settlement redesigning New England's capacity market fell within a reasonable range of capacity prices); *City of Winnfield*, 744 F.2d at 875-76.

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reasonable, the Commission could not force the utility to shift from one to the other in a section 206 proceeding.⁵²

24. In determining the ROE component of a public utility's cost of service pursuant to a DCF analysis, however, the term "zone of reasonableness" has a particular, more technical meaning that differs from its meaning when used in general descriptions of what constitutes a just and reasonable rate charged by a public utility for jurisdictional service, such as in *City of Winnfield* and *Maine PUC*. The Commission uses a three-step process to determine the just and reasonable ROE component of the cost of service of a public utility or a group of public utilities. First, the Commission establishes a proxy group of companies of comparable risk. Second, the Commission performs a DCF analysis of each member of the proxy group in order to determine a "zone of reasonableness," within which to set a just and reasonable ROE. That DCF zone of reasonableness is the range from the lowest proxy member ROE to the highest proxy member ROE. Finally, the Commission establishes a just and reasonable ROE at a single point within the DCF zone of reasonableness.

25. Thus, in the context of determining an ROE, the establishment of the DCF zone of reasonableness is simply one step in the process of determining a just and reasonable ROE for inclusion in the cost of service of the subject public utility or utilities. Typically, the DCF zone of reasonableness is relatively broad. For example, in *Bangor Hydro*⁵³ setting the NETOs' existing ROE, the DCF zone of reasonableness was from 7.3 percent to 13.1 percent, or almost 600 basis points. In this case, the zone of reasonableness is from 7.03 percent to 11.74 percent, or nearly 500 basis points. Not every ROE within that relatively broad DCF "zone of reasonableness" is a just and reasonable ROE for the particular public utility or utilities at issue. As the Commission held in *Bangor Hydro*, "[c]ertain rates, though within the zone, may not be just and reasonable given the circumstances of the case."⁵⁴

⁵² See *City of Winnfield*, 744 F.2d at 875 ("in that circumstance the agency is effectively using § 205, which is intended for the benefit of the utility—i.e., as a means of enabling it to increase its rates within what has been called the 'zone of reasonableness'—for the quite different purpose of *depriving the utility* of the statutory protection contained in § 206, that its existing rates be found to be entirely outside the zone of reasonableness before the agency can dictate their level or form.") (emphasis in original) (citation omitted).

⁵³ 122 FERC ¶ 61,038 at PP 10-15.

⁵⁴ *Id.* P 11 (quoting *Montana-Dakota Utils. Co. v. Nw. Pub. Serv. Co.*, 341 U.S. 246, 251 (1951) (*Montana-Dakota*)).

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26. The decision of the United States Court of Appeals for the District of Columbia Circuit in *S. Cal. Edison Co. v. FERC*,⁵⁵ recognized that, in the context of determining ROE, not every ROE within the DCF zone of reasonableness is just and reasonable. In that case, the utility filed to modify its rates under FPA section 205. The court stated that section 205 required the Commission to approve the utility's rate proposal "as long as the new rates are just and reasonable."⁵⁶ Nevertheless, the court also held that the Commission had authority to require the utility's ROE to be set at the median of the zone of reasonableness, even though the midpoint of the zone, proposed by the utility, was also within the DCF zone of reasonableness. In short, the court recognized that the Commission need not treat every ROE within the zone of reasonableness as a just and reasonable ROE. If the Commission were required to find any and every ROE within the zone of reasonableness to be just and reasonable, the requirement that the Commission approve any section 205 rate proposal "as long as the new rates are just and reasonable"⁵⁷ would require the Commission to accept any ROE proposed by a utility in a section 205 rate case, as long as that ROE did not exceed the top of the range of reasonableness. However, the FPA has never been understood to require such a result, which would be contrary to the consumer protection purpose of the FPA.⁵⁸

27. In Opinion No. 531, the Commission stated that the NETOs were erroneously seeking to apply a different just and reasonable standard in FPA section 206 cases than in section 205 cases. The Commission stated, "Despite the fact FPA section 205 does not require that every ROE within the zone of reasonableness be considered just and reasonable for purposes of a utility rate filing under FPA section 205, the NETOs would

⁵⁵ *S. Cal. Edison Co. v. FERC*, 717 F.3d 177, 181-82 (D.C. Cir. 2013) (finding that the Commission had authority to set a utility's ROE at the median of the zone of reasonableness even though the utility proposed using the midpoint, which was also within the zone of reasonableness); *accord Montana-Dakota*, 341 U.S. at 251 (explaining that while statutory reasonableness is an abstract concept represented by an area rather than a pinpoint the Commission must translate that concept into a concrete rate, and it is the rate—not the abstract concept—that governs the rights of the buyer and seller).

⁵⁶ *S. Cal. Edison Co. v. FERC*, 717 F.3d at 181.

⁵⁷ *Wis. Pub. Power, Inc. v. FERC*, 493 F.3d 239, 254 (D.C. Cir. 2007).

⁵⁸ Given that the FPA was intended to be a consumer-protection statute, *see, e.g., Pub. Sys. v. FERC*, 606 F.2d 973, 979 n.27 (D.C. Cir. 1979), it is hard to find persuasive an argument that would allow, under FPA section 205, a utility to propose an increase in its ROE to anywhere in the zone, but would effectively bar, under FPA section 206, a customer from seeking to decrease the ROE being challenged merely because the ROE falls somewhere within the zone.

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require us to treat every existing ROE within the zone of reasonableness as just and reasonable in a section 206 case. Nothing in the FPA, however, supports such a different understanding of the phrase “just and reasonable” as between those two sections of the FPA when establishing a utility’s ROE.”

28. On rehearing, the NETOs do not challenge Opinion No. 531’s interpretation of FPA section 205 as not requiring the Commission to treat any ROE proposed by the utility within the DCF zone of reasonableness as a just and reasonable ROE which the Commission must accept. However, the NETOs contend that Opinion No. 531 fails to recognize that the Commission’s burden of proof under FPA section 206 contains two prongs: first, the burden to show that an existing rate is unjust and unreasonable; second, the burden to show that the replacement rate is just and reasonable. The NETOs agree that the showing the Commission must make under the second prong of section 206 in order to establish a replacement ROE “is identical to the required section 205 showing, as Opinion No. 531 states.”⁵⁹ However, they assert that the showing of unjustness and unreasonableness which the Commission must make under the first prong of its section 206 burden “is very different from and more difficult to satisfy” than the showing of justness and reasonableness that must be made under either the second prong of section 206 or under section 205. As a result they assert that any ROE within the zone of reasonableness cannot be found unjust and unreasonable.

29. In making these arguments, the NETOs are confusing differences in who bears the burden of persuasion as between FPA sections 205 and 206 with the substantive “just and reasonable” standard contained in both those sections. The two sections of course differ as to who bears the burden of persuasion, because under FPA section 206 the Commission or complainant must show that the utility’s existing rate is unjust and unreasonable and the Commission must show that its replacement rate is just and reasonable, whereas under FPA section 205 the utility need only show that its proposed rate is just and reasonable. However, as the Supreme Court has stated, sections 205 and 206 are “parts of a single statutory scheme under which . . . all rates are subject to being modified by the Commission upon a finding that they are unlawful.”⁶⁰ While the party bearing the burden of persuasion is different under FPA section 205 and FPA

⁵⁹ NETOs Request for Rehearing at 35.

⁶⁰ *United Gas Pipe Line Co. v. Mobile Gas Serv. Co.*, 350 U.S. 332, 341 (1956). While this case involved the Natural Gas Act, the Supreme Court held in a companion case that the provisions of the FPA relevant to this question are substantially identical to the equivalent sections under the Natural Gas Act. *FPC v. Sierra Pacific Power Co.*, 350 U.S. 348, 353 (1956).

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section 206, “the scope and purpose of the Commission’s review remains the same – to determine whether the rate fixed by the [utility] is lawful.”⁶¹

30. Because sections 205 and 206 are part of a single statutory scheme, it follows that a rate that is lawful under one section must also be lawful under the other and a rate that is unlawful under one section must also be unlawful under the other. For this to be true, the substantive standard to determine lawfulness under each section – the just and reasonable standard – must be applied in the same manner under each section. Therefore, if every ROE within the DCF zone of reasonableness must be treated as a lawful just and reasonable ROE which cannot be modified under the first prong of the Commission’s FPA section 206 burden, as the NETOs contend, then every ROE within that zone must also be treated as a lawful just and reasonable ROE for all other purposes under the FPA, including a section 205 filing. This would require the Commission to find just and reasonable any ROE proposed by a utility in a section 205 proceeding that was within the DCF zone of reasonableness. However, as already discussed, the D.C. Circuit rejected that proposition in *SoCal Edison*.

31. The NETOs next contend that failing to treat all ROEs within the DCF zone of reasonableness as just and reasonable for purposes of the first prong of the Commission’s 206 burden would erase the difference between the burden of proof under FPA sections 205 and 206, because the ROE determination in a section 206 proceeding would be the same as in a section 205 proceeding. We disagree. We recognize that in situations where the Commission has found that more than one methodology may be used to design a just and reasonable rate for a service, such as the incremental rate situation in *City of Winnfield* discussed above, the utility may choose one of the just and reasonable ratemaking methodologies in a section 205 proceeding, and the Commission then cannot require the utility to shift to a different just and reasonable methodology in a subsequent

⁶¹ *United Gas Pipe Line Co. v. Mobile Gas Serv. Co.*, 350 U.S. at 341. The effect of the NETOs’ argument, if that argument were to be accepted, would turn the statute on its head. Section 206 would no longer be a tool to challenge an ROE that was no longer reasonable, but rather would serve to insulate that ROE from challenge as long as it fell somewhere—anywhere—within the zone of reasonableness produced by a DCF analysis. A statute that was intended to protect ratepayers from exploitation, *see, e.g., Pub. Sys. v. FERC*, 606 F.2d at 979 n.27, would protect and preserve just such exploitation. But, as the Commission has recognized, as recently as last year the D.C. Circuit has already rejected just such an approach. *See* Opinion No. 531, 147 FERC ¶ 61,234 at P 52 (citing *S. Cal. Edison Co. v. FERC*, 717 F.3d 177).

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section 206 proceeding.⁶² However, the statute does not require that we approve multiple just and reasonable methodologies to resolve every ratemaking issue. In fact, the D.C. Circuit held in *S. Cal. Edison Co.* that the Commission may require the use of a particular methodology to determine the just and reasonable ROE to be included in a utility's cost of service, despite the existence of other possible methodologies for determining ROE.⁶³

32. The Commission has long required the use of a DCF methodology (here the two-step DCF methodology adopted in Opinion No. 531) to determine a zone of reasonableness, with the lawful just and reasonable ROE set at a single numerical point within that range based on the circumstances and record of that case.⁶⁴ Therefore, when the Commission finds a utility's base ROE to be just and reasonable in a particular case, it finds *only* that single point to be just and reasonable given the facts and circumstances of that case.⁶⁵ It does *not* find any other base ROE within the DCF zone of reasonableness, either above or below the approved ROE, to be a just and reasonable base ROE for that utility or group of utilities. Thus, the DCF zone of reasonableness does not establish a continuum of just and reasonable base ROEs, any one of which the utility would equally be free to charge to ratepayers; rather, only the single point approved by the Commission within the DCF zone of reasonableness is the just and reasonable base

⁶² See *Consolidated Edison of New York, Inc. v. FERC*, 165 F.3d 992, 216-17 (D.C. Cir. 1999). (“While incremental treatment may be required at one end of the rate-setting continuum, and rolled-in pricing required at the other, in between the two extremes lie a series of intermediate points in which both cost-recovery methods would satisfy section 4’s just and reasonable test. At each of these places along the continuum, the pricing mechanism will essentially lie in the hands of the initiating pipeline. It is only when the proposed rate crosses the boundary separating the just from the unjust that FERC can act under its section 5 authority to order a rate of its own formulation.”)

⁶³ *S. Cal. Edison Co. v. FERC*, 717 F.3d at 182 (“In order to discharge its statutory duty of ensuring that ‘[a]ll rates . . . [are] just and reasonable’ the Commission may require the use of a particular ratemaking methodology so long as its embrace of that methodology is not arbitrary and capricious.”).

⁶⁴ See, e.g., *Williston Basin Interstate Pipeline Co. v. FERC*, 165 F.3d 54, 57 (D.C. Cir. 1999).

⁶⁵ Cf. *Montana-Dakota*, 341 U.S. at 251 (explaining that while statutory reasonableness is an abstract concept represented by an area rather than a pinpoint the Commission must translate that concept into a concrete rate, and it is the rate—not the abstract concept—that governs the rights of the buyer and seller).

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ROE.⁶⁶ It follows that showing the existing base ROE established in the prior case is unjust and unreasonable merely requires showing that the Commission's ROE methodology now produces a numerical value below the existing numerical value. Contrary to the NETOs' assertion, the fact that both of the burdens of proof under FPA section 206 can be satisfied using a single ROE analysis—one that generates an ROE that both is below the existing ROE (thus demonstrating that the existing ROE is excessive) and that also is a just and reasonable ROE (thus demonstrating what the new ROE should be)—does not alter those two burdens.⁶⁷

33. In short, the statute does not require that we treat all ROEs within the DCF zone of reasonableness as just and reasonable. Rather, the statute requires that, under section 206, before we may change an ROE we must find it unjust and unreasonable. And, in Opinion No. 531, that we did. Our ROE analysis showing that the NETOs' base ROE is 10.57 percent demonstrates both that their existing 11.14 percent ROE is unjust and unreasonable and that 10.57 percent is the NETOs' just and reasonable replacement base ROE.⁶⁸ Thus, we met both burdens under section 206.

34. The NETOs cite precedent that, while correctly stating the general principle of the FPA section 206 burden, is distinguishable from the facts of this case because that precedent did not discuss the FPA section 206 burden in the context of determining a utility's base ROE.⁶⁹ Whether a particular rate is just and reasonable, and what the range

⁶⁶ As discussed below in P 35, the addition of an incentive adder for a project can justify a higher overall just and reasonable ROE (i.e., the base ROE plus the incentive adder) for that project.

⁶⁷ Further, we reject the NETOs' contention that the Commission's determination on the burden of proof in this proceeding broadens the Commission's discretion and will lead to increased uncertainty and litigation. *See* NETOs Request for Rehearing at 45-46. We are following our long-standing practice with regard to the zone of reasonableness identified by a DCF analysis.

⁶⁸ A utility's ROE is simply one component of the cost-of-service reflected in its overall rates for the services it provides. Typically, each component of the cost of service is a single number, based on the utility's actual costs during the relevant test period. For example, if a utility's existing cost of service includes a cost of labor of \$10 million, a showing that its actual test period cost of labor is \$9 million satisfies both the burden to show that the existing \$10 million labor cost is unjustly and unreasonably high and the new just and reasonable labor cost is \$9 million. Our treatment of ROE is no different.

⁶⁹ *See, e.g., Maine PUC*, 520 F.3d 464, *rev'd in part on other grounds sub nom. NRG Power Mktg., LLC v. Me. Pub. Utils. Comm'n*, 558 U.S. 165 (2010) (upholding Commission determination that transition payments agreed to in a settlement redesigning

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of reasonableness is for that rate, largely depends on the nature of the rate at issue. While a utility's base ROE is a single, specific numerical value that is determined by using a well-known methodology, a tariff provision setting forth an energy market rule might produce a numerical result only in conjunction with many other associated market rules. A determination of what is an appropriate range of reasonableness, and what is just and reasonable, in these two disparate contexts requires different analyses and the balancing of different interests. As a result, the Commission uses different approaches to determining the just and reasonable resolution in different circumstances. In determining a utility's base ROE, the Commission has long used a methodology that produces a single, specific numerical value, not a range of reasonable values, and the Commission has therefore interpreted FPA section 206 to protect that specific numerical value, rather than a zone around that value.

35. The NETOs are correct that, in the context of incentive ROE adders authorized for projects, the Commission has capped the overall ROE for a particular project (i.e., the sum of the utility's base ROE and the incentive ROE adder for that project) at the top of the DCF zone of reasonableness.⁷⁰ However, it does not follow from this fact that all ROEs within the DCF zone of reasonableness must be treated as just and reasonable for purposes of the first prong of FPA section 206. The Commission awards an incentive adder based on a separate, independent showing that a particular project is of a type that qualifies for such an adder, and—as directed by Congress—the Commission allows the adder to be added to the base ROE and charged to ratepayers so long as the sum of the

New England's capacity market fell within a reasonable range of capacity prices); *Calpine Corp. v. Cal. Indep. Sys. Operator Corp.*, 128 FERC ¶ 61,271 (finding tariff provisions setting forth a method of socializing the costs of a market participant's financial default to be unjust and unreasonable); *Cal. Mun. Utils. Ass'n v. Cal. Indep. Sys. Operator Corp.*, 126 FERC ¶ 61,315 (2009) (finding that complainants failed to show tariff unjust and unreasonable due to a lack of sufficient safeguards to protect against the risk of anomalous settlements); *Cal. Indep. Sys. Operator Corp.*, 140 FERC ¶ 61,168 (2012) (finding tariff provisions concerning the repayment of an interconnection customers' network upgrade costs to be just and reasonable under FPA section 205).

⁷⁰ See, e.g., *Northeast Utils. Serv. Co.*, 124 FERC ¶ 61,044, at P 71 (2008); *Central Maine Power Co.*, 125 FERC ¶ 61,079, at P 74 (2008); *Desert Southwest Power, LLC*, 135 FERC ¶ 61,143, at P 96 (2011). The Commission uses the DCF zone of reasonableness in the same manner to ensure that the sum of a utility's base ROE plus an incentive adder for joining an RTO is just and reasonable.

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add and base ROE for that project is just and reasonable under FPA section 205.⁷¹ The Commission makes that determination by looking at whether the utility's base ROE plus the incentive ROE adder for that project remain within the zone of reasonableness. That is, the Commission looks to whether the sum of the base ROE and the adder for that project falls within the DCF-determined zone of reasonableness, or does that sum instead fall outside the zone of reasonableness, for that project. Absent both a showing that the particular project qualifies for such an adder, and a Commission finding that the resulting overall ROE satisfies the just and reasonable standard laid out in the FPA, the increased overall ROE for the project produced by summing the adder and the base ROE would not be just and reasonable.⁷² This use of the DCF-determined zone of reasonableness to place an outer limit on the overall ROE that a utility may earn on a particular project does not in any way suggest that any base ROE up to the top of the DCF-determined zone of reasonableness must be treated as just and reasonable for purposes of FPA section 206. To the contrary, it is only the separate, independent finding that the project qualifies for an incentive adder that justifies increasing the overall ROE for that project to a point within the DCF-determined zone of reasonableness above the point at which the utility's base ROE is set.

2. Placement of the Base ROE within the Zone of Reasonableness

a. Placement of the Base ROE above the Midpoint

i. Opinion No. 531

36. The Commission in Opinion No. 531 found that, although it typically sets the base ROE for a group of utilities at the midpoint of the zone of reasonableness identified by the DCF methodology, “a mechanical application of the DCF methodology with the use of the midpoint here would result in an ROE that does not satisfy the requirements of *Hope and Bluefield*.”⁷³ Therefore, the Commission explained that, “based on the record in this case, including the unusual capital market conditions present, . . . the just and reasonable base ROE for the NETOs should be set halfway between the midpoint of the zone of reasonableness and the top of the zone of reasonableness,” i.e., 10.57 percent.⁷⁴

⁷¹ See 16 U.S.C. § 824s(d) (2012) (“All rates approved under the rules adopted pursuant to [FPA section 219] . . . are subject to the requirements of sections [205 and 206] of this title that all rates . . . be just and reasonable.”).

⁷² See generally, e.g., *NSTAR Elec. Co.*, 125 FERC ¶ 61,313 (2008); *Northeast Utils. Serv. Co.*, 124 FERC ¶ 61,044 (2008).

⁷³ Opinion No. 531, 147 FERC ¶ 61,234 at P 142.

⁷⁴ *Id.*

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The Commission explained that, as “[p]arties on both sides of the instant ROE issue argue that the unique capital market conditions have impacted the level of equity return the NETOs’ require to meet the capital attraction standards of *Hope* and *Bluefield*,” the Commission was “concerned that capital market conditions in the record are anomalous, thereby making it more difficult to determine the return necessary for public utilities to attract capital.”⁷⁵ The Commission explained that “[i]n these circumstances, we have less confidence that the midpoint of the zone of reasonableness established in this proceeding accurately reflects the equity returns necessary to meet the *Hope* and *Bluefield* capital attraction standards.”⁷⁶

37. As a result of the anomalous capital market conditions reflected in the record, and their potential impact on the DCF model, the Commission found it “necessary and reasonable to consider additional record evidence, including evidence of alternative benchmark methodologies and state commission-approved ROEs, to gain insight into the potential impacts of these unusual capital market conditions on the appropriateness of using the [midpoint of the zone of reasonableness identified by the DCF methodology].”⁷⁷ The Commission found the additional record evidence—specifically the NETOs’ risk premium analysis, Capital Asset Pricing Model (CAPM) analysis, expected earnings analysis, and evidence of state commission-authorized ROEs—supported a finding that an upward adjustment from the midpoint was warranted.⁷⁸

38. After determining that the just and reasonable base ROE for the NETOs was above the midpoint, the Commission found that, because it “has traditionally looked to the central tendency to identify the appropriate return within the zone of reasonableness,” it is appropriate to “look to the central tendency for the top half of the zone of reasonableness.”⁷⁹ The Commission explained that “[w]hen placing a base ROE above the central tendency of the zone of reasonableness, the Commission has in the past placed the base ROE at the midpoint of the upper half of the zone.”⁸⁰ The Commission therefore

⁷⁵ *Id.* P 145.

⁷⁶ *Id.*

⁷⁷ *Id.*

⁷⁸ *Id.* PP 146-150.

⁷⁹ *Id.* P 151.

⁸⁰ *Id.* P 152 (citing *S. Cal. Edison Co.*, Opinion No. 445, 92 FERC ¶ 61,070, at 61,266 (2000); *Consumers Energy Co.*, Opinion No. 429, 85 FERC ¶ 61,100, at 61,363-64 (1998)).

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found that “a base ROE halfway between the midpoint of the zone of reasonableness and the top of that zone represents a just and reasonable ROE for the NETOs.”⁸¹

ii. Requests for Rehearing

39. Petitioners and EMCOS argue that the Commission’s placement of the NETOs’ base ROE three-quarters of the way up the zone of reasonableness is contrary to record evidence and Commission precedent, and is therefore arbitrary and capricious. Petitioners assert that the only basis for establishing a base ROE above the central tendency of the zone of reasonableness is that the utility or utilities whose base ROE is at issue are riskier than the proxy group. Petitioners argue that the Commission’s 38-member national proxy group is far more risky than the NETOs because the average corporate credit rating of the proxy group was between BBB and BBB+, whereas 80 percent of the NETOs are rated between BBB and A.⁸² Petitioners further state that, using the appropriate weighting to reflect the relative size of each of the NETOs, the fair average of the NETOs’ credit ratings is “A-/BBB+.” Petitioners therefore argue that the Commission should place the NETOs’ base ROE in the lower half of the zone of reasonableness.⁸³

40. EMCOS assert that the Commission has previously and consistently concluded that the midpoint of the zone of reasonableness produces a just and reasonable ROE for a diverse group of utilities because it fairly and accurately evaluates risk. EMCOS further state that Opinion No. 531 acknowledges that the midpoint of the zone of reasonableness yields an appropriate ROE for a diverse group of utilities, but then rejects the use of the 9.39 percent midpoint in favor of the higher 10.57 percent figure.⁸⁴ EMCOS state that Opinion No. 531 cites only two cases in which the Commission adopted an ROE at the midpoint of the upper half of the zone of reasonableness, and in each of those cases the utility at issue had a higher risk profile than the proxy group.⁸⁵ Petitioners and EMCOS

⁸¹ *Id.*

⁸² Petitioners Request for Rehearing 16-18.

⁸³ *Id.* at 19 (citing Ex. SC-207). Petitioners also cite several other sources claiming that the NETOs have a high level of rate certainty.

⁸⁴ EMCOS Request for Rehearing at 10 (citing Opinion No. 531, 147 FERC ¶ 61,234 at P 142).

⁸⁵ *Id.* at 13-14 (citing *Consumers Energy Co.*, 85 FERC ¶ 61,100; *S. Cal. Edison Co.*, 88 FERC ¶ 61,254 (1999)).

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argue that those two cases resulted in upward adjustments of 18 and 58 basis points, compared to the 118 basis point increase in this proceeding.⁸⁶

41. EMCOS state that Opinion No. 531 rejects the use of the midpoint of the zone of reasonableness asserting that capital market conditions here are “unique” and “anomalous.” EMCOS state that the ROE awarded must reflect the capital market conditions under which the NETOs operate and that Commission precedent recognizes the importance of basing an ROE on current market data.⁸⁷ Petitioners and EMCOS state that Opinion No. 531 asserts it must adopt an ROE higher than the midpoint because *Hope* and *Bluefield* require the Commission to identify an ROE that will attract sufficient capital; however, this position fails to recognize that market conditions must be reflected in an ROE in order for it to be just and reasonable. EMCOS explain that they made this argument in their Initial Brief, and that Opinion No. 531 acknowledged it, but did not provide any explanation of why it does not apply here.⁸⁸ EMCOS argue that this case covers “the Great Recession” which had an effect on all companies and consumers, but Opinion No. 531’s decision to upwardly adjust the base ROE in this proceeding uniquely shields the NETOs from the economic realities of that time period at the expense of New England consumers.⁸⁹

42. Petitioners state that Opinion No. 531’s reliance on a single issuance from UBS Financial Services (UBS) included in the testimony of the NETOs’ witness, Ms. Lapson, is neither well-founded nor consistent with the record. Petitioners also state that the reports in Ms. Lapson’s testimony were not selected by her, but were hand-picked by the NETOs’ counsel and that the testimony includes almost nothing addressing the views of specific investment analysts as to the potential impact of an ROE reduction in this proceeding on future transmission investment. Petitioners further argue that, a few months after the UBS report, UBS changed its mind and stated that the outcome of this proceeding “impacts only the generic New England rates.” Petitioners explain that there were many different views taken by other analysts which were unrebutted, which they state explains why there is no well-founded basis for a concern that a base ROE reduction

⁸⁶ *Id.* at 14-15.

⁸⁷ *Id.* at 16-17 (citing *Portland Natural Gas Transmission Sys.*, Opinion No. 510-A, 142 FERC ¶ 61,198, at P 233 (2013); *Consumer Advocate Div. of the Pub. Serv. Comm’n of West Virginia v. Allegheny Generating Co.*, 68 FERC ¶ 61,207, at 61,998 (1994) (*West Virginia Consumer Advocate*)).

⁸⁸ *Id.* at 17-18 (citing *Bluefield*, 262 U.S. at 692; *Hope*, 320 U.S. at 614).

⁸⁹ *Id.* at 19.

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to the central result of the national proxy group could undermine the NETOs' ability to attract capital.⁹⁰

43. Petitioners and EMCOS also assert that the Commission erred in relying on certain record evidence—i.e., the evidence of state commission-authorized ROEs and the NETOs' alternative methodologies for estimating the cost of equity—to corroborate the placement of the base ROE within the zone of reasonableness. Petitioners and EMCOS argue that, in relying on these alternative methodologies, Opinion No. 531 departed from Commission precedent without providing an explanation for doing so. Petitioners contend that the Commission has repeatedly found that non-DCF approaches to determining transmission ROEs are “unlikely to produce a just and reasonable result.”⁹¹ For example, Petitioners contend that, in the case that recently concluded with the D.C. Circuit affirming the Commission's sole reliance on the electric utility DCF median, Southern California Edison Company had sought to bolster its case for a high ROE by relying on the CAPM analysis.⁹² Petitioners note that the Commission refrained from according the non-DCF analyses even the little weight sought by Southern California Edison Company. Petitioners argue that the use of the NETOs' alternative methodologies should have been subject to the well-established test for an above-center ROE: no upward movement should be undertaken unless those methodologies make “a very persuasive case” that the central result of a conventional DCF study fails to identify the subject utility's true equity cost.⁹³ Petitioners contend that the Commission failed to state a reasoned basis for not applying the “very persuasive case” standard.

44. Petitioners and EMCOS further argue that the Commission's reliance on the NETOs' alternative benchmark methodologies without scrutinizing their flaws is inconsistent with reasoned decision-making and constitutes judicially-reversible error.⁹⁴ Petitioners and EMCOS also argue that the Commission's DCF analysis contains certain

⁹⁰ Petitioners Request for Rehearing 53-57.

⁹¹ *Id.* at 30 (citing *Xcel Energy Servs., Inc.*, 122 FERC ¶ 61,098 at P 73, *clarified*, 125 FERC ¶ 61,092 (2008) (*Xcel*)).

⁹² *Id.* at 30-31 (citing *S. Cal. Edison Co.*, 131 FERC ¶ 61,020, at P 114 (2010) (*SoCal Edison*), *reh'g denied*, 137 FERC ¶ 61,016 (2011), *petition for review granted in part and denied in part*, *S. Cal. Edison Co. v. FERC*, 717 F.3d 177).

⁹³ *Id.* at 32.

⁹⁴ *Id.*; EMCOS Request for Rehearing at 20 (citing *Ill. Pub. Telecomm. Ass'n v. FCC*, 117 F.3d 555, 564 (D.C. Cir. 1997)).

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flaws that undermine the Commission's decision to place the base ROE above the midpoint of the zone of reasonableness.

45. Petitioners' and EMCOS's arguments as to the specific, alleged flaws in both the Commission's DCF analysis and the record evidence on which the Commission relied to corroborate the placement of the base ROE above the midpoint are described below.

iii. Commission Determination

46. We deny rehearing on the issue of where to place the NETOs' base ROE within the zone of reasonableness produced by the Commission's DCF analysis.

47. As an initial matter, we disagree with Petitioners' and EMCOS's arguments concerning the circumstances under which the Commission may set a base ROE at a point other than the central tendency of the zone of reasonableness.⁹⁵ Petitioners assert that the Commission may only do so by comparing the NETOs' risks to the risks of the proxy group produced by the DCF methodology—i.e., by conducting a comparison that the Commission has historically referred to as the “relative risk analysis.” We disagree. In this case, the Commission found the proxy group to be comparable in risk to the NETOs,⁹⁶ but determined that it was necessary to adjust the NETOs' base ROE above the midpoint based on considerations other than the relative risk analysis.⁹⁷ While the Commission has indeed adjusted a company's base ROE above or below the central

⁹⁵ We also disagree with Petitioners' argument that the two precedents the Commission cited in support of using the midpoint of the upper half of the DCF-produced zone of reasonableness are distinguishable from the instant case because the upward adjustments in those two cases—*S. Cal. Edison Co.*, Opinion No. 445, 92 FERC ¶ 61,070, and *Consumers Energy Co.*, Opinion No. 429, 85 FERC ¶ 61,100—were of 58 and 18 basis points, respectively, compared to the 118 basis adjustment in Opinion No. 531. Nothing in those cases indicates that the Commission made those adjustments because they were for 58 or 18 basis points. Instead, the Commission in Opinion Nos. 445 and 429 placed the ROE at the midpoint of the upper half of the zone after finding that an upward adjustment was warranted, which is what the Commission did in Opinion No. 531.

⁹⁶ See Opinion No. 531, 147 FERC ¶ 61,234 at P 96.

⁹⁷ *Id.* PP 144-145.

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tendency of the zone or reasonableness based on the relative risk analysis,⁹⁸ the Commission is not limited to making adjustments based only on the relative risk analysis. Petitioners' argument to the contrary is inconsistent with both court and Commission precedent showing that the Commission has the discretion to make,⁹⁹ and has in fact made, adjustments to a rate based on the particular circumstances of a case, including whether unique circumstances render the results of the Commission's DCF analysis less reliable than usual.¹⁰⁰

48. We disagree with Petitioners' argument that the NETOs' are less risky than the proxy group. While Petitioners assert that 80 percent of the NETOs' have credit ratings between BBB to A, whereas the average credit rating of the proxy group company is between BBB and BBB+, this alone does not show that the NETOs are less risky than the proxy group. As explained in Opinion No. 531, the Commission uses the credit rating band because it "include[s] in the proxy group only those companies whose credit ratings *approximate* those of the utilities whose rates are at issue."¹⁰¹ We thus reiterate that Commission's finding that the credit rating band of the proxy group is comparable to the NETOs' credit ratings.¹⁰² Further, Petitioners' argument is based on a flawed comparison of the two groups' credit ratings. Assuming *arguendo* that it is helpful to compare the distribution of the NETOs' credit ratings to the average credit rating of proxy group companies, that analysis should be accompanied by a comparison of how the distribution of the proxy group companies' credit ratings compare to the average credit rating of the proxy group. In other words, the distribution of the NETOs' credit ratings should be compared to the distribution of the proxy companies' credit ratings. Petitioners' comparison is misleading because it fails to do this. In this case 34 of the 38

⁹⁸ See, e.g., *Williston Basin Interstate Pipeline Co. v. FERC*, 165 F.3d at 57 ("Once the Commission has defined a zone of reasonableness [using the DCF model], it then assigns the pipeline a rate within that range to reflect specific investment risks associated with that pipeline as compared to the proxy group companies.").

⁹⁹ See, e.g., *Fed. Power Comm'n v. Natural Gas Pipeline Co.*, 315 U.S. 575, 586 (1942) ("The Constitution does not bind rate-making bodies to the service of any single formula or combination of formulas. Agencies to whom this legislative power has been delegated are free, within the ambit of their statutory authority, to make the pragmatic adjustments which may be called for by particular circumstances.")

¹⁰⁰ See, e.g., *Town of Norwood, Mass. v. FERC*, 80 F.3d 526, 534-535 (D.C. Cir. 1996).

¹⁰¹ Opinion No. 531, 147 FERC ¶ 61,234 at P 106 (emphasis added).

¹⁰² See *id.* P 108.

companies in the proxy group—i.e., 89 percent of the proxy companies—have credit ratings between BBB and A, compared to the 80 percent of the NETOs within that band.¹⁰³ This indicates that the credit ratings of the proxy group companies and the NETOs are similarly distributed, and supports a finding that the two groups have comparable risk profiles.

49. Petitioners and EMCOS argue that the Commission erred in basing its decision to set the NETOs' base ROE above the central tendency of the zone of reasonableness produced by the DCF analysis on the presence of anomalous capital market conditions. Petitioners specifically argue that the slow economic growth reflected in the record is not anomalous, but is instead a "new normal" and should, therefore, not justify adjusting the base ROE above the midpoint. We are not persuaded by Petitioners' argument. In Opinion No. 531, the Commission acknowledged that parties on both sides of the issue had cited to unique capital market conditions.¹⁰⁴ The Commission also referenced U.S. Treasury bond yields, not economic growth, as an indicator of current capital market conditions. Given the undisputed presence of such anomalous capital market conditions, the Commission stated that it had "less confidence that the midpoint of the zone of reasonableness established in this proceeding accurately reflects the equity returns necessary to meet the *Hope* and *Bluefield* capital attraction standards."¹⁰⁵ However, we did not stop there in our analysis of whether it was appropriate to establish a base ROE above the midpoint. Rather, the record evidence of unusual capital market conditions served as an impetus for the Commission's consideration of additional record evidence. This consideration was necessary to evaluate, in this proceeding, whether setting the NETOs' ROE at the midpoint of the zone of reasonableness satisfied the requirements of *Hope* and *Bluefield*. Therefore, the Commission conducted a further analysis by analyzing the additional record evidence, including evidence of alternative benchmark methodologies and state commission-approved ROEs, to gain insight into the potential impacts of the unusual capital market conditions on the appropriateness of using the resulting midpoint. We then used this additional record evidence to corroborate our determination that placement at a point above the midpoint was warranted.¹⁰⁶

50. We also reject EMCOS's argument that, even if the capital market conditions reflected in the record are anomalous, adjusting the NETOs' ROE based on an economic anomaly ignores the *Hope* and *Bluefield* requirement that a utility's ROE must reflect

¹⁰³ See Ex. NET-701.

¹⁰⁴ Opinion No. 531, 147 FERC ¶ 61,234 at P 145.

¹⁰⁵ *Id.*

¹⁰⁶ *Id.* PP 146-149.

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current market conditions. EMCOS specifically argue that whether capital market conditions in the record are anomalous from a historical perspective is irrelevant to the determination of a just and reasonable base ROE, because the base ROE must reflect the capital market conditions under which the NETOs operate, even if those conditions are historically anomalous. We disagree. The EMCOS's argument assumes that DCF analyses are immune to ever being skewed by economic anomalies. This assumption is unrealistic, as all methods of estimating the cost of equity are susceptible to error when the assumptions underlying them are anomalous.¹⁰⁷ The Commission, in fact, acknowledged this limitation in Opinion No. 531,¹⁰⁸ and was concerned that a mechanical application of the two-step DCF methodology with the use of the midpoint in such circumstances would produce a return that would not satisfy the requirements of *Hope* and *Bluefield*.¹⁰⁹ Therefore, based on the presence of anomalous capital market conditions, the Commission considered additional record evidence that supported an upward adjustment. Contrary to EMCOS's assertions, the Commission is not constrained to a mechanical application of the DCF methodology where the Commission determines that such an approach will not produce a just and reasonable result.¹¹⁰ We further reject

¹⁰⁷ Roger A. Morin, *New Regulatory Finance* 28 (Public Utilities Reports, Inc. 2006) ("For instance, by relying solely on the DCF model at a time when the fundamental assumptions underlying the DCF model are tenuous, a regulatory body greatly limits its flexibility and increases the risk of authorizing unreasonable rates of return. The same is true for any one specific model."). We note that participants on both sides of the instant ROE issue in this proceeding have relied upon Dr. Morin's *New Regulatory Finance*. See, e.g., Ex. S-1 at 59-60 (Trial Staff exhibit quoting *New Regulatory Finance*); Ex. NET-300 at 67 (NETOs exhibit quoting *New Regulatory Finance*); Tr. 580-581 (Complainants' cross-examination relying on *New Regulatory Finance*).

¹⁰⁸ Opinion No. 531, 147 FERC ¶ 61,234 at PP 41, 145.

¹⁰⁹ *Id.* PP 150-152.

¹¹⁰ See *Fed. Power Comm'n v. Natural Gas Pipeline Co.*, 315 U.S. at 586 ("The Constitution does not bind rate-making bodies to the service of any single formula or combination of formulas. Agencies to whom this legislative power has been delegated are free, within the ambit of their statutory authority, to make the pragmatic adjustments which may be called for by particular circumstances.").

We note that neither of the Commission precedents to which Complainants cite in support of their argument—*Portland Nat. Gas Transmission Sys.*, Opinion No. 510-A, 142 FERC ¶ 61,198 (2013) (Opinion No. 510-A) and *West Virginia Consumer Advocate*, 68 FERC ¶ 61,207—constrain the Commission to mechanically apply a particular ratemaking approach without regard to economic anomalies. *West Virginia Consumer*
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EMCOS's argument that this analysis should be affected by the fact that the NETOs can subsequently request a rate increase under FPA section 205. The NETOs' ability to subsequently request a rate increase if economic conditions change does not excuse the Commission from establishing an ROE under FPA section 206 that meets the requirements of *Hope* and *Bluefield*.

51. Petitioners argue that the Commission erred in finding that a base ROE of 9.39 percent could undermine the NETOs' ability to attract capital for new investment, because the finding was based on only one analyst's report, from UBS, which is contradicted by record evidence of other analysts' reports. We disagree. Petitioners specifically cite analysts' reports from Credit Suisse; Goldman Sachs; Brean Murray, Carret & Co. (Brean Murray); Deutsche Bank; and a subsequent report from UBS. But none of the reports Petitioners cite contradicts the finding that a base ROE of 9.39 percent—i.e., a reduction of 175 basis points from the existing base ROE—could undermine the NETOs' ability to attract capital.¹¹¹

52. The Deutsche Bank report and the subsequent report from UBS provide no analysis of how a reduced base ROE would impact the NETOs and, therefore, do not contradict the UBS report the Commission relied upon in Opinion No. 531. The Deutsche Bank report merely states the possibility that the Commission could reduce the

Advocate did not involve any unusual capital market conditions. See generally *West Virginia Consumer Advocate*, 68 FERC ¶ 61,207. While Opinion No. 510-A did involve allegations of economic anomalies, the Commission in that case, in fact, weighed the evidence of anomalous conditions in determining whether to apply its policy of using the most recent record data or to use an alternative data set. See Opinion No. 510-A, 142 FERC ¶ 61,198 at P 233. Thus, Opinion No. 510-A demonstrates that the Commission may indeed consider, as it has here in Opinion No. 531, whether to apply or adjust an established policy based on anomalous economic conditions.

¹¹¹ We also reject Petitioners' argument that the NETOs' expert witness was not qualified to present testimony on this issue. The NETOs' expert witness has 43 years of experience as a financial professional, including 38 years focused on financial analysis and securities evaluation within the utilities sector, and was formerly the Managing Director of the utilities, power, and gas analytical team at Fitch Ratings, where she "supervised and wrote the credit rating criteria applied in the electric, gas, and water sector." Ex. NET-400 at 1-3.

The Presiding Judge, furthermore, admitted this witness's testimony into the record and found it "to have moderate probative value." See Initial Decision, 144 FERC ¶ 63,012 at P 576; *Entergy Servs., Inc.*, 109 FERC ¶ 61,108, at P 7 (2004) (citing 18 C.F.R. § 385.209 (2004)).

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NETOs' base ROE as a result of the low interest rate environment, while the later UBS report describes the scope of the proceeding and predicts a general trend of lower ROEs for regulated utilities, without discussing the magnitude of the potential ROE reductions or their impact on utilities' ability to attract capital.

53. The reports from Credit Suisse, Goldman Sachs, and Brean Murray provide limited analysis of two holding companies that are parent companies to certain NETOs, and none of that analysis undermines the UBS report the Commission cited in Opinion No. 531. The Credit Suisse report states that a 50 to 100 basis point reduction in Northeast Utilities' ROE in this proceeding would be a "positive" for the company.¹¹² This statement, which we interpret to mean simply that a reduction of 50 to 100 basis points would be better for Northeast Utilities than would an even greater reduction, is silent on the impacts that a reduced ROE would have on Northeast Utilities' ability to attract capital. The Goldman Sachs report, which also only addresses Northeast Utilities, states that a 100 basis point reduction to Northeast Utilities' ROE would have a minimal impact on Northeast Utilities' earnings per share and that the impact could be overcome by adding \$200-\$300 million in transmission projects to Northeast Utilities' rate base. This evidence is solely focused on the impact that an ROE reduction would have on Northeast Utilities' earnings per share and, therefore, provides insufficient evidence to determine how such a reduction would impact Northeast Utilities' ability to attract capital.¹¹³ Because the Credit Suisse and the Goldman Sachs reports only address the impact of ROE reductions of up to 100 basis points, neither is probative on the issue of how a significantly greater 175 basis point ROE reduction to 9.39 percent would affect the NETOs' ability to attract capital.

54. The Brean Murray report, which states that "[a] negative impact to [UIL Holdings] from an adverse decision would be minimal, in our view," is the least probative of these three reports. What would constitute an "adverse decision," for example, is unclear. Whether and to what magnitude an adverse ruling in this proceeding would impact the NETOs' ability to attract capital, moreover, cannot be determined with any certainty based on the magnitude of the impact the ruling might have on the much larger and more diversified parent company of one of the NETOs.

¹¹² Petitioners Request for Rehearing at 57; Ex. SC-518 at 5. We further note that the 10.57 percent base ROE established in this proceeding reduced the NETOs' base ROE by 57 basis points, which is within the 50 to 100 basis point range that Credit Suisse reported would be a positive outcome for Northeast Utilities.

¹¹³ While a company's earnings are undeniably relevant to its ability to attract capital, it is merely one of multiple factors investors rely on in determining whether to invest in the company. For example, looking at earnings in isolation provides no information about the company's dividend yield.

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55. We are also unpersuaded by Petitioners' arguments that, if the Commission concludes that the NETOs' base ROE should be set above the midpoint of the zone of reasonableness, the base ROE should be placed at the true 75th percentile of the zone of reasonableness, i.e., 9.84 percent, rather than at the 10.57 percent midpoint of the upper half of the zone. As the Commission explained in Opinion No. 531, the Commission has traditionally used measures of central tendency to determine an appropriate return in ROE cases and, in cases involving the placement of the base ROE above the central tendency of the zone of reasonableness, the Commission has used the central tendency of the top half of the zone. Our decision to utilize the midpoint of the upper half of the zone is based on the record evidence in this proceeding and is consistent with the Commission's established policy of using the midpoint of the ROEs in a proxy group when establishing a central tendency for a region-wide group of utilities.¹¹⁴ Further, we reject Petitioners' assertion that *Northwest Pipeline Corp.*, 99 FERC ¶ 61,305 (2002), requires the Commission to consider the distribution of results within the proxy group when determining where in the upper half of the zone to place the NETOs' base ROE. *Northwest Pipeline Corp.* does not bear on the Commission's decision in this proceeding to place the NETOs' base ROE above the midpoint of the zone of reasonableness, as that case involved the issue of which particular measure of central tendency should be used in setting a single pipeline's ROE *at the middle* of the zone of reasonableness.¹¹⁵

56. Lastly, we disagree with Petitioners that the Commission erred in relying on the NETOs' alternative methodologies to support its decision that an upward adjustment from the midpoint was warranted in this case. While Petitioners cite *Xcel*, 122 FERC ¶ 61,098, *Pacific Gas & Electric Company*, 141 FERC ¶ 61,168 (2012) (*PG&E*), *SoCal Edison*, 131 FERC ¶ 61,020, and *ITC Holdings Corp.*, 121 FERC ¶ 61,229 (2007)

¹¹⁴ *SoCal Edison*, 131 FERC ¶ 61,020 at P 92, *aff'd in relevant part*, *S. Cal. Edison Co. v. FERC*, 717 F.3d at 185-87.

¹¹⁵ *Northwest Pipeline Corp.*, 99 FERC ¶ 61,305 at 62,276. The Commission typically looks to the central tendency as the "most just and reasonable" and "most appropriate" return that best considers that range, and typically uses the median as the measure of central tendency in cases involving a single utility's ROE and uses the midpoint as the measure of central tendency in cases involving the ROE for a group of utilities. See *Midwest Indep. Transmission Sys. Operator, Inc.*, 106 FERC ¶ 61,302, at PP 9-10 (2004), *aff'd in relevant part sub nom. Pub. Serv. Comm'n of Ky. v. FERC*, 397 F.3d 1004, 1010-11 (D.C. Cir. 2005) (*PSC of Kentucky*); *SoCal Edison*, 131 FERC ¶ 61,020 at P 92, *aff'd in relevant part*, *S. Cal. Edison Co. v. FERC*, 717 F.3d at 185-87. *Northwest Pipeline Corp.*, in contrast, merely explains the rationale for selecting the median as the appropriate measure of central tendency in a case involving a single utility's ROE.

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(*ITC Holdings*), as precedent in which the Commission has declined to rely on alternative methodologies, we find the precedent to be distinguishable from the instant case because in none of those four cases did the record contain evidence of unique capital market conditions that called into question the rote application of the midpoint of the zone of reasonableness resulting from the Commission's DCF methodology. Additionally, in *PG&E*, the Commission set the ROE issue for hearing without any reference to the reliability of the alternative methodologies the utility submitted in support of its filing.¹¹⁶ Further, Petitioners are mistaken that the Commission in *SoCal Edison* did not give weight to the alternative methodologies. As the Commission in that case explained, the three alternative methodologies submitted in that case "were not used by the Commission in setting a base ROE for SoCal Edison," but "were used to corroborate the results of its DCF analysis."¹¹⁷ With regard to *ITC Holdings*, as discussed below, the CAPM analysis presented in that case contained methodological shortcomings that distinguish it from the NETOs' CAPM analysis in this case.¹¹⁸

57. Petitioners and EMCOS also allege that the Commission's DCF analysis and the evidence the Commission relied upon to corroborate it contain various flaws. Those arguments are addressed in turn below.

b. Discounted Cash Flow Analysis

i. Opinion No. 531

58. In Opinion No. 531, the Commission conducted a DCF analysis using a national proxy group of companies listed as Electric Utilities by *Value Line* and that had credit ratings within one notch above or below the NETOs' credit ratings (referred to as the "credit rating screen"), had paid 6-months of dividend yields without making or announcing a dividend cut, were not involved in merger and acquisition activity significant enough to distort the DCF results, and were not low-end or high-end outliers.

59. In using the national proxy group, rather than a regional proxy group, the Commission explained that "widening the geographic range of the proxy group allows for the application of more stringent screening criteria, to refine the proxy group to a level of risk more comparable, while maintaining a group of proxy companies that is sufficiently

¹¹⁶ *PG&E*, 141 FERC ¶ 61,168 at P 23.

¹¹⁷ *SoCal Edison*, 131 FERC ¶ 61,020 at P 116. And here they were similarly used to "gain insight" and "inform" our thinking on whether an upward adjustment was reasonable. Opinion No. 531, 147 FERC ¶ 61,234 at PP 145-149.

¹¹⁸ See *infra* P 115.

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large and diverse to reliably capture the range of reasonable returns.”¹¹⁹ In applying the credit rating screen, the Commission explained that “the purpose of the credit rating band screen is to include in the proxy group only those companies whose credit ratings approximate those of the utilities whose rates are at issue.”¹²⁰ The Commission found that, because investors rely on credit ratings from both Standard & Poor’s (S&P) and Moody’s, “basing the credit rating screen on data only from S&P does not necessarily provide an accurate estimate of the NETOs’ risk.”¹²¹ Therefore, the Commission found that “in applying the credit rating proxy group screen to exclude companies more than one notch above or below the NETOs’ credit ratings, it is appropriate to use both the S&P corporate credit ratings and the Moody’s issuer ratings *when both are available*.”¹²² Because the NETOs’ S&P credit ratings ranged from A- to BBB and Moody’s credit ratings ranged from A2 to Baa2, the Commission excluded companies from the proxy group that were more than one notch above or below either of those credit rating bands.¹²³

60. In screening the proxy groups for outliers, the Commission affirmed the Presiding Judge’s application of the Commission’s low-end outlier test in this proceeding, explaining that the “purpose of the low-end outlier test is to exclude from the proxy group those companies whose ROE estimates are below the average bond yield or are above the average bond yield but are sufficiently low that an investor would consider the stock to yield essentially the same return as debt.”¹²⁴ The Commission explained that “[i]n public utility ROE cases, the Commission has used 100 basis points above the cost of debt as an approximation of this threshold, but has also considered the distribution of the proxy group companies to inform its decision on which companies are outliers.”¹²⁵ The Commission explained that the cost of debt for the relevant study period was 4.61 percent and, therefore, the Commission eliminated three companies whose DCF results failed the low-end outlier test—Edison International (3.11 percent), Ameren Corp.

¹¹⁹ Opinion No. 531, 147 FERC ¶ 61,234 at P 96.

¹²⁰ *Id.* P 106.

¹²¹ *Id.* P 107.

¹²² *Id.*

¹²³ *Id.* P 108.

¹²⁴ *Id.* P 121.

¹²⁵ *Id.*

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(5.26 percent), and Public Service Enterprise Group Inc. (PSEG) (5.62 percent).¹²⁶ The Commission explained that PSEG's DCF result was only one basis point above the 100 basis point threshold, and that the Commission's decision to eliminate PSEG was informed by the fact that there was a 141 basis point break between PSEG's DCF result and that of the next lowest proxy group company.¹²⁷

61. With regard to the high-end outlier test, the Commission found that "the high-end outlier issue in this proceeding is moot,"¹²⁸ explaining that "[u]nder the two-step DCF methodology, it is unnecessary to screen the proxy group for unsustainable growth rates because the methodology assumes the long-term growth rate for each company is equal to GDP." The Commission explained that, as a result, "no company in the proxy group we are adopting here has a composite growth rate under the two-step DCF methodology in excess of the 7.66 percent growth rate of PNM Resources, Inc., or an ROE in excess of the 11.74 percent ROE of UIL Holdings," which are "well within any high-end outlier test we have previously applied in utility rate cases."¹²⁹

ii. Requests for Rehearing

62. Petitioners assert that the Commission's DCF analysis in Opinion No. 531 contained flaws that undermine the Commission's decision to place the base ROE above the midpoint of the zone of reasonableness.

63. Petitioners argue that the Commission erred in relying on a short-term growth estimate for UIL Holdings, Inc. (UIL Holdings) of 8.07 percent, which Petitioners allege was based on only one analyst estimate.¹³⁰ According to Petitioners, Commission precedent indicates that, when calculating the dividend growth rate, the Commission's analysis should be based upon as much independently calculated data as possible, and that IBES growth estimates are reliable only insofar as they represent the consensus of

¹²⁶ *Id.* P 123.

¹²⁷ *Id.*

¹²⁸ *Id.* P 118.

¹²⁹ *Id.*

¹³⁰ Petitioners Request for Rehearing at 48 (citing Exs. SC-313 and SC-314 (showing that 8.07 percent long-term growth projection for UIL Holdings represents the forecast of one analyst)).

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multiple analysts.¹³¹ In addition, Petitioners state that the Commission has made clear that its approval of the Yahoo! reported growth estimates that represent a consensus is not exclusive of other credible sources¹³² and that comparable growth projections from other sources could be considered along with Value Line projections and what was then IBES.¹³³

64. Petitioners state that it is critical in this case, and in future cases, that the Commission follow its precedent by requiring that the short-term growth rate for each proxy company be based on multiple projections. Petitioners argue that UIL Holdings's New England transmission business is smaller than its natural gas distribution business,¹³⁴ and it is therefore a less-than-ideal proxy for setting an electric transmission ROE.¹³⁵ Petitioners also assert that, during the relevant period, the Moody's credit rating for UIL Holdings was Baa3, lower than the Baa2 rating of its transmission subsidiary, United Illuminating Company, and the lowest rank among all retained proxy

¹³¹ *Id.* at 45 (citing *Yankee Atomic Elec. Co.*, Opinion No. 285, 40 FERC ¶ 61,372, at 62,210 (1987) (*Yankee Atomic*), *reh'g denied*, Opinion No. 285- A, 43 FERC ¶ 61,232 (1988) (rejecting sole reliance on Zacks' predictions of earnings growth in favor of multiple data sources for projecting earnings); *Northwest Pipeline Corp.*, 87 FERC ¶ 61,266, at 62,059 (1999) (*Northwest Pipeline*) (“[t]he IBES data is a compilation of projected growth rates from various knowledgeable financial advisors with the industry.”)).

¹³² *Id.* at 46 (citing *Composition of Proxy Groups for Determining Gas and Oil Pipeline Return on Equity*, 123 FERC ¶ 61,048, at PP 83-84 (2008), *reh'g dismissed*, 123 FERC ¶ 61,259 (2008) (conditionally allowing, but not requiring, reference to growth forecasts published by Yahoo!)).

¹³³ *Id.* (citing *ISO New England, Inc.*, 109 FERC ¶ 61,147, at P 205 (2004), *petition for review denied sub nom. Me. Pub. Utils. Comm'n v. FERC*, 454 F.3d 278 (D.C. Cir. 2006); *ISO New England, Inc.*, 110 FERC ¶ 61,111, at P 23, *reh'g denied*, 111 FERC ¶ 61,344 (2005)).

¹³⁴ *Id.* at 49.

¹³⁵ *Id.* (citing *Consumers Energy Co.*, Opinion No. 456, 98 FERC ¶ 61,333 (2002)).

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companies.¹³⁶ Petitioners assert that these considerations provide reasons to avoid undue reliance on the forecasts of one analyst.¹³⁷

65. Petitioners state that because the IBES projection for UIL Holdings was the opinion of a single analyst, Opinion No. 531 erred in failing to apply any of the other growth estimates available in the record to check whether the IBES projection for UIL Holdings produced reasonable results. Petitioners contend that neither Opinion No. 531 nor any participant identified a prior case in which the Commission placed the base ROE three-quarters of the way up the zone of reasonableness based on a high-end proxy result that was driven by the forecast of just one analyst. Petitioners state that using Value Line or Reuters data for UIL Holdings's short-term growth rate in the two-step DCF methodology provides a more appropriate benchmark than the NETOs' alternative cost of equity studies, and shows that a base ROE of 9.39 percent is sufficient for the NETOs.¹³⁸

66. Petitioners also argue that UIL Holdings's DCF result reflects a "circularity problem" that counsels against placing the base ROE at the upper quarter of the zone of reasonableness, and instead supports placing the base ROE no higher than the true 75th percentile of the proxy group companies' DCF results. Petitioners state that the "circularity problem" is that much of UIL Holdings's dividends, earnings, and earnings growth are a result of ROE incentive adders, and UIL Holdings's DCF result reflects investors' expected revenues from those ROE incentive adders. Petitioners assert that the NETOs' base ROE should be determined exclusive of the transmission incentive revenues of the proxy group companies.

67. Petitioners also state that this circularity problem should have been mitigated by placing the base ROE closer to the true "75th percentile" of the proxy group DCF results, i.e., based on 75 percent of the 38 proxy company results (interpolated between the 28th-highest and 29th-highest results), rather than at the upper quarter of the zone of reasonableness. Petitioners state that the key difference between the actual 75th percentile and the top-quarter approach that Opinion No. 531 labels as the "75th percentile" is that the actual percentile reflects the distribution of proxy group results, whereas the Commission's top-quarter approach discards all of that information and relies on the 3:1 weighted average of the two most extreme results. Petitioners assert that discarding information on the distribution of proxy results and considering only their

¹³⁶ *Id.* (citing Ex. NET-600 at 9).

¹³⁷ *Id.*

¹³⁸ *Id.* at 45.

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extremes is statistically indefensible and inconsistent with precedent applying Opinion No. 531's two-step DCF methodology.¹³⁹

68. Lastly, Petitioners argue that the Commission erred in eliminating PSEG's DCF result of 5.61 percent as a low-end outlier, thereby raising the bottom of the zone of reasonableness produced by the Commission's DCF analysis. Petitioners state that this error reinforces the arguments against raising the base ROE within the zone of reasonableness. Petitioners state that Opinion No. 531 discarded PSEG's DCF result on the grounds that, although it was above the average bond yield by more than 100 basis points, it fell below a "natural break" in the proxy group's DCF results. Petitioners argue that, while Opinion No. 531 states that this rationale "buttressed" the decision to exclude PSEG, the natural break was actually the sole basis for the Commission's decision.¹⁴⁰

69. Petitioners argue that the "natural break" standard must be applied evenhandedly to low-end and high-end outliers alike, but in Opinion No. 531 the Commission ignored the fact that there was a comparable "natural break" at the high end of the range of DCF results. Specifically, Petitioners assert that the 5.62 percent result for PSEG should not have been discarded unless the 11.74 percent result for UIL Holdings was also discarded.¹⁴¹

iii. Commission Determination

70. We deny rehearing on the various issues that Petitioners and EMCOS raise concerning the Commission's DCF analysis and their related objections to setting the base ROE above the midpoint of the zone of reasonableness.

71. Petitioners argue that the Commission erred in using UIL Holdings's DCF result to set the top of the zone of reasonableness in the Commission's DCF analysis, because UIL Holdings's DCF result was based on an IBES short-term growth projection that reflected only one analyst's growth rate projection. We reject this argument as it is contrary to years of established Commission precedent approving the use of IBES short-term growth projections in the two-step DCF methodology. For example, in *Transcontinental Gas Pipe Line Corp.*¹⁴² the Commission rejected contentions that IBES

¹³⁹ *Id.* at 58-59 (citing *Northwest Pipeline Corp.*, 99 FERC ¶ 61,305).

¹⁴⁰ *Id.* at 60.

¹⁴¹ *Id.* at 61.

¹⁴² *Transcontinental Gas Pipe Line Corp.*, Opinion No. 414-B, 85 FERC ¶ 61,323, at 62,268-9 (1998) (Opinion No. 414-B).

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growth projections should not be used in the two-step DCF methodology, because the analysts making those projections allegedly are overly optimistic in their projections. The Commission pointed to substantial evidence in the record of that case that investors rely on IBES growth projections in making investment decisions. The Commission also noted that the appropriate dividend growth rate to include in a DCF analysis is the growth rate expected by the market. While the market may be wrong in its expectations as reflected in the IBES growth projections, the cost of common equity to a regulated enterprise depends upon what the market expects, not upon precisely what is actually going to happen.

72. We recognize that the Commission has supported its use of IBES growth projections based on the fact that the IBES data is a compilation of projected growth rates from various knowledgeable financial advisors.¹⁴³ However, the Commission has not required that the IBES growth projection for each member of the proxy group reflect a minimum number of analyst growth estimates.¹⁴⁴ IBES, which the Commission has long relied on as the source of the growth rate projections to be used in the Commission's DCF analyses, does not publish the number of analyst estimates on which a company's growth rate estimate is based.¹⁴⁵ As a result, there seems little reason to conclude that investors' reliance on IBES growth projections necessarily varies depending upon the exact number of analysts contributing to any particular IBES growth projection. On balance, we find it preferable to use a consistent source of dividend growth projections for all members of the proxy group as provided by IBES, rather than to use different sources of growth projections depending upon the number of analysts contributing to each IBES growth projection, which, as discussed below, could produce skewed results. Accordingly, if a proxy company has a growth rate estimate from IBES, as does UIL Holdings, that growth rate is acceptable for purposes of the Commission's DCF analysis, regardless of the number of analysts on which it was based.

73. Contrary to Petitioners' assertion, *Yankee Atomic* and *Northwest Pipeline* do not require a different result. *Yankee Atomic* involved a much different analysis than in the instant case, because the Commission found that the small proxy group in *Yankee Atomic* was "not a valid indicator of the Yankee companies' cost of capital because the five companies are different from the Yankees in too many significant respects."¹⁴⁶ Because

¹⁴³ *Northwest Pipeline*, 87 FERC ¶ 61,266 at 62,059.

¹⁴⁴ *E.g.*, *SoCal Edison*, 131 FERC ¶ 61,020 at P 36.

¹⁴⁵ We also note that the Value Line data—which the Commission has similarly long relied upon as the source of earnings estimates in ROE proceedings—for any company consists of an earnings estimate from only one analyst.

¹⁴⁶ *Yankee Atomic*, 40 FERC ¶ 61,372 at 62,211.

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the record did not contain a valid proxy group, the Commission had to project the Yankee Companies' dividend growth based solely on projections of those companies' own dividend growth. Therefore, the Commission determined that it should base the Yankee Companies' dividend growth projection on as many independent growth projections as possible. In contrast, this case involves a robust proxy group of companies that are comparable to the NETOs, for which dividend growth projections are available to enable the Commission to conduct a full DCF analysis. This provides the Commission a significant amount of information concerning the NETOs' cost of equity. As to *Northwest Pipeline*, in that case the Commission actually rejected the very argument on which Petitioners rely, as the Commission found that it would be inappropriate to use multiple sources of growth rate data, rather than IBES alone, in determining the short-term growth projection in the two-step DCF methodology.¹⁴⁷

74. Petitioners argue that the Commission erred in placing the base ROE halfway between the midpoint of the zone of reasonableness and the top of the zone because UIL Holdings's high-end result is affected by a "circularity problem," i.e., that UIL Holdings's dividends, earnings, and earnings growth are impacted by its incentive ROE adders. The Commission has rejected this argument in the past, and we do so here for the same reasons. In Order No. 679-A, the Commission rejected the argument "that incentive ROEs will 'destabilize' the DCF methodology," explaining that

First, . . . all ROEs approved pursuant to section 219 will be within the range of reasonableness, as determined consistent with our precedents. Second, any incentive ROEs granted under section 219 should have minimal effect, if any, on the overall range of reasonableness derived from the appropriate proxy group. The DCF methodology uses proxy groups of entire companies, not individual transmission projects. In other words, the "cash flows" being measured in the DCF method are the cash flows of entire companies. These cash flows should not be significantly affected by an incentive return for any particular transmission project for one company within the proxy group. Moreover, to the extent there is any small effect on the overall range of reasonableness, it will appropriately reflect the substantial risks associated with constructing new transmission[.]¹⁴⁸

75. Further, even assuming *arguendo* that this circularity problem exists, it exists for any proxy group company that receives incentive adders and Petitioners have presented no methodology for determining whether or how much a company's incentive adders

¹⁴⁷ See *Northwest Pipeline*, 87 FERC ¶ 61,266 at 62,058-59.

¹⁴⁸ Order No. 679-A, FERC Stats. & Regs. ¶ 31,236 at P 62 (cross-referenced at 117 FERC ¶ 61,345 at P 62).

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might impact investors' expectations for a particular company, particularly where the proxy company at issue is involved in diverse business activities, as is UIL Holdings. Thus, absent more evidence, we are not persuaded that this potential "circularity problem" warrants an adjustment to the NETOs' base ROE. Further, even if Petitioners had shown this alleged circularity to be a legitimate problem warranting an adjustment to the base ROE, Petitioners have not shown that placing the base ROE at their proposed true 75th percentile of the proxy group results would be an appropriate solution.

76. We also reject Petitioners' argument that the Commission should have compared UIL Holdings's IBES growth rate against the Reuters data Trial Staff provided and the "br+sv"¹⁴⁹ data in the record. We relied only on IBES data for the DCF analysis in this proceeding, because that is the only short-term growth data available in the record for all the proxy companies. As the Commission explained in Opinion No. 531, "[u]sing different sources of growth rate data for different companies in a proxy group could produce skewed results, because those sources may take different approaches to calculating growth rates."¹⁵⁰ A comparison between UIL Holdings's IBES data and other non-IBES data in the record would be susceptible to this same skewing effect, and therefore would not provide a reliable comparison. Further, as the Commission explained in Opinion No. 531, while "the purpose of the 'br+sv' growth estimate is to act as a check on the reasonableness of the IBES forecasts," in practice the two sources often produce "widely divergent growth rates that do not engender much confidence in the reliability of the estimates."¹⁵¹ We are, therefore, not persuaded that it is necessary to compare the IBES growth rate data to the "br+sv" data. In addition, we disagree with Petitioners that declining to mix growth rate sources is inconsistent with Opinion No. 531's allowance of credit ratings from both Moody's and S&P. The purpose of using data from both Moody's and S&P is to identify a group of comparable risk companies. In contrast, the purpose of not using multiple sources of growth rate data is to ensure that the cost of equity for each company in the proxy group is estimated using the same protocols.

77. We also reject Petitioners' argument that the Commission should have used the "br+sv" growth rate as the short-term growth rate in the two-step DCF methodology.

¹⁴⁹ The term "br+sv" represents the sustainable growth formula, in the one-step DCF methodology that the Commission used for public utilities prior to Opinion No. 531, where "b" is the percentage of earnings expected to be retained (after the payment of dividends), "r" is the expected rate of return on book equity, "s" is the percent of common equity expected to be issued annually as new common stock, and "v" is the equity accretion rate.

¹⁵⁰ Opinion No. 531, 147 FERC ¶ 61,234 at P 90.

¹⁵¹ *Id.* P 37.

While the “br+sv” growth formula relies on short-term Value Line projections of five years or less for the various inputs to the formula, it seeks to estimate a company’s “sustainable growth rate.” For that reason, although the Commission has stated that the formula “only produces a projection of short-term growth, similar to the IBES projections,”¹⁵² the Commission finds the formula unreasonable for use as the short-term growth projection in the two-step DCF methodology. By seeking to estimate a “sustainable growth rate,” the “br+sv” growth formula also contains some elements of a long-term growth projection, in addition to a short-term growth projection, and thus is inappropriate for use as a purely short-term growth projection in a two-step DCF methodology. The Commission adopted the two-step DCF methodology because, among other reasons, its incorporation of a long-term growth projection in the cost of equity calculation would have the effect of ascribing sustainable long-term growth to all members of a proxy group.¹⁵³ Thus, the Commission’s adoption of the two-step DCF methodology accomplishes what the use of the “br+sv” formula was intended to accomplish.¹⁵⁴

78. We reject Petitioners’ arguments that the Commission erred in its application of the low-end and high-end outlier tests. We reiterate that it is appropriate—and consistent with Commission precedent—to eliminate PSEG as a low-end outlier in this case because PSEG’s DCF result is a mere 101 basis points above the applicable bond yield and there is a 141 basis point break between PSEG’s DCF result and the next lowest result. Further, we reject as inconsistent with Commission precedent Petitioners’ argument that the Commission should have adopted the NETOs’ proposed adjustment to the low-end outlier test instead of placing the base ROE above the midpoint of zone of reasonableness. Petitioners have identified no precedent in which the Commission has adopted such an adjustment to the low-end outlier test, and we are not persuaded to do so in this case.

¹⁵² *Id.* P 34.

¹⁵³ *Id.* PP 38, 40.

¹⁵⁴ We also note that the Commission’s rationale for adopting the two-step DCF methodology in Opinion No. 531 was, in part, to use a methodology that is more consistent with the methodology the Commission has applied in natural gas and oil pipeline cases. *See id.* P 36. However, using “br+sv” in place of IBES growth rates, as Complainants request, would produce a DCF methodology that is less closely aligned with the methodology the Commission uses in natural gas and oil pipeline cases, where the Commission has rejected the use of the “br+sv” formula. *Composition of Proxy Groups for Determining Gas and Oil Pipeline Return on Equity*, 123 FERC ¶ 61,048 at P 100.

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79. Petitioners next argue that, if the Commission eliminates PSEG as a low-end outlier, it must also eliminate UIL Holdings as a high-end outlier because UIL Holdings's DCF result is 112 basis points above the next highest DCF result, and the Commission must apply the same "natural break" analysis in both the low-end and high-end outlier tests. We disagree. The low-end outlier test and the high-end outlier test serve very different purposes: the low-end outlier test is intended to screen out companies whose ROE estimates are low enough that an investor would consider the stock to yield essentially the same return as debt,¹⁵⁵ whereas the high-end outlier test is intended to screen out companies whose growth rates are unsustainably high and therefore fail a threshold test of economic logic.¹⁵⁶ As the Commission explained in Opinion No. 531, the high-end outlier issue in this proceeding is moot because the two-step DCF methodology assumes that the long-term growth rate of all proxy companies is equal to GDP, and is therefore sustainable.

c. State Commission-Authorized ROEs

i. Opinion No. 531

80. The Commission in Opinion No. 531 found that the record evidence of state commission-approved ROEs supported the Commission's determination that a base ROE at the midpoint of the zone of reasonableness would not satisfy *Hope* and *Bluefield*. The Commission explained that, while it has "repeatedly held that it does not establish utilities' ROE based on state commission ROEs for state-regulated electric distribution assets,"¹⁵⁷ this proceeding presents "circumstances under which the midpoint of the zone of reasonableness established in this proceeding has fallen below state commission-approved ROEs, even though transmission entails unique risks that state-regulated electric distribution does not."¹⁵⁸ More specifically, the Commission explained that "while the midpoint in this case is 9.39 percent, the record indicates that, over the 24-month period from October 1, 2010 through September 30, 2012, approximately 85 percent to 91 percent of state commission authorized ROEs were between 9.8 percent and 10.74 percent."¹⁵⁹ Accordingly, the Commission found that "[a]lthough we are not using the state commission-approved ROEs to establish the NETOs' ROE in this

¹⁵⁵ See *S. Cal. Edison Co.*, 92 FERC ¶ 61,070 at 61,266.

¹⁵⁶ See, e.g., *ISO New England, Inc.*, 109 FERC ¶ 61,147 at P 205.

¹⁵⁷ Opinion No. 531, 147 FERC ¶ 61,234 at P 148.

¹⁵⁸ *Id.*

¹⁵⁹ *Id.*

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proceeding, the discrepancy between state ROEs and the 9.39 percent midpoint serves as an indicator that an upward adjustment to the midpoint here is warranted to satisfy *Hope* and *Bluefield*.”¹⁶⁰

ii. **Requests for Rehearing**

81. Petitioners and EMCOS argue that the Commission erred in relying on state commission-authorized ROEs in Opinion No. 531, because comparisons to state-authorized ROEs are not relevant to this proceeding and do not support raising the NETOs’ base ROE from the 9.39 percent midpoint to the 10.57 percent upper quartile figure.¹⁶¹ Petitioners argue that Opinion No. 531 erroneously relies on the spin that the NETOs placed on Ex. NET-403’s data, repeating their argument that “approximately 85 percent to 91 percent of state commission authorized ROEs were between 9.8 percent and 10.74 percent.”¹⁶² EMCOS argue that the fact that some state commission-approved ROEs are higher than the midpoint in this proceeding is insufficient evidence to support Opinion No. 531’s decision to ignore the Commission’s strong preference for the use of the midpoint.¹⁶³ Petitioners contend that reference points presented in the exhibit show that 89 percent of the past-period state commission ROE outcomes collected by the NETOs fall below 10.57 percent.¹⁶⁴ Petitioners further contend that the central tendency values of the state commission-authorized ROEs presented by the NETOs are a mode of 10 percent, median of 10.13 percent, a mean of 10.14 percent, and a midpoint of 10.25 percent. Petitioners argue that Opinion No. 531 does not explain how these data justify a 10.57 percent base ROE.¹⁶⁵

82. Petitioners and EMCOS contend that the state commission-authorized ROEs upon which the Commission relied were tainted by substantial lag, and that relying on them is therefore inconsistent with Opinion No. 531’s emphasis on using the most recent

¹⁶⁰ *Id.*

¹⁶¹ Petitioners Request for Rehearing at 23; EMCOS Request for Rehearing at 25-26.

¹⁶² Petitioners Request for Rehearing at 23.

¹⁶³ EMCOS Request for Rehearing at 11, 25-26 (citing *Fla. Gas Transmission Co. v. FERC*, 604 F.3d 636, 639 (D.C. Cir. 2010)).

¹⁶⁴ Petitioners Request for Rehearing at 23.

¹⁶⁵ *Id.* at 25.

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information available in the record.¹⁶⁶ Petitioners argue that the record shows that more recent state-authorized base ROEs have averaged below 10 percent. For example, Petitioners state that the Regulatory Research Associates data for the first quarter of 2013 show that the average authorized state electric ROE “approximated 9.75 [percent], 25 [basis points] below the analogous adjusted average ROE for calendar-2012 (which approximated 10 percent).”¹⁶⁷ Petitioners state that Exhibit SC-423 shows that, on March 15, 2013, the New York State Public Service Commission approved an ROE of 9.3 percent for Niagara Mohawk Power Corporation, finding the rate to be “consistent with investor expectations while being slightly below other recently authorized rate plans.”¹⁶⁸ In addition, Petitioners state that Exhibit SC-505 shows that, at around the same time, Northeast Utilities’ retail ROE was set at 9.38 percent.¹⁶⁹

83. Petitioners contend that the Commission should have made its own independent finding of the current cost of equity, based on financial market data, rather than being constrained by stale decisions reached elsewhere. Petitioners note that the Commission has previously rejected efforts to use state commission-authorized ROEs as a benchmark for setting regional transmission ROEs.¹⁷⁰ Petitioners argue that if state commission-authorized ROEs are irrelevant when they are lower than the result of the Commission’s DCF analysis, then they are also irrelevant when they are higher than the result of the Commission’s DCF study. Petitioners argue that the Commission’s failure to recognize this symmetry in Opinion No. 531 or to offer any justification for ignoring it renders the decision arbitrary and capricious.¹⁷¹ Similarly, EMCOS contend that Opinion No. 531 is inconsistent with *Missouri Public Service Commission v. FERC*, 337 F.3d 1066, 1077 (D.C. Cir. 2003) (*Missouri*), which explained that “[w]hen FERC relies upon a state agency’s prior approval to support the conclusion that rates are in the public interest, the Commission must at least say something about the prior regulator’s rationale for approving those rates.”

¹⁶⁶ *Id.* at 25-26 (citing Opinion No. 531, 147 FERC ¶ 61,234 at PP 55, 88); EMCOS Request for Rehearing at 11, 26 (citing *NorAm Gas Transmission Co. v. FERC*, 148 F.3d 1158, 1165 (1998)).

¹⁶⁷ Petitioners Request for Rehearing at 26.

¹⁶⁸ *Id.* at 28 (citing Ex. SC-423 at 18).

¹⁶⁹ *Id.*

¹⁷⁰ *Id.* at 29.

¹⁷¹ *Id.* at 29-30.

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iii. Commission Determination

84. We disagree with Petitioners' and EMCOS's arguments that the record evidence concerning state commission-authorized ROEs does not support placing the NETOs' base ROE above the midpoint of the zone of reasonableness. The Commission did not use the evidence of state commission-authorized ROEs to determine the level at which the NETOs' base ROE should be set. As explained below, the Commission merely relied on the state commission-authorized ROEs—in conjunction with evidence that interstate transmission is riskier than state-level distribution—as evidence that the 9.39 percent midpoint of the DCF-produced zone of reasonableness was insufficient to satisfy the requirements of *Hope* and *Bluefield* and, therefore, that an adjustment above 9.39 percent was warranted.¹⁷²

85. Contrary to Petitioners' and EMCOS's arguments, applying other measures of central tendency to the NETOs' data on state commission-authorized ROEs does not undermine the Commission's conclusion that an upward adjustment was warranted. Petitioners point to various measures of central tendency for the state commission-authorized ROEs: mode of 10 percent, median of 10.13 percent, a mean of 10.14 percent, and a midpoint of 10.25 percent. But all of these figures are above the 9.39 percent midpoint of the zone of reasonableness; in light of the record evidence showing that interstate transmission is riskier than state-level distribution,¹⁷³ all of these figures support adjusting the NETOs' base ROE above that level. Further, while Petitioners focus on the fact that 89 percent of the state commission-authorized ROEs in the NETOs' study are below 10.57 percent, that fact is irrelevant to how the midpoint of the DCF-produced zone of reasonableness compares to the state commission-authorized ROEs. The more relevant fact is that almost 93 percent of the state commission-authorized ROEs are above the 9.39 percent midpoint produced by the Commission's two-step DCF methodology in this case.¹⁷⁴

86. We reject Petitioners' and EMCOS's arguments that the Commission's reliance on the state ROE figures despite their time-lag is inconsistent with the Commission's preference for the most recent data in the record. The evidence of state commission-

¹⁷² See Opinion No. 531, 147 FERC ¶ 61,234 at PP 148-149.

¹⁷³ See *id.* P 149. We note that Petitioners have not refuted the record evidence that interstate transmission is riskier than state-level distribution. Petitioners' request for rehearing discusses the Commission's finding on the relative risks of transmission and distribution only in the context of whether the NETOs are more or less risky than the companies in the DCF proxy group. See Petitioners Request for Rehearing at 19-22.

¹⁷⁴ See Ex. NET-403.

authorized ROEs that the Commission relied upon is, in fact, the most recent complete study in the record. While the record does contain some more recent evidence of state commission-authorized ROEs, that evidence does not represent a data set comparable to the NETOs' 24-month study,¹⁷⁵ but is rather data for only one quarter in 2013 from Regulatory Research Associates concerning the recent trend in average authorized ROEs. According to Petitioners, the report from Regulatory Research Associates indicates that the average state commission-authorized ROE in the first quarter of 2013 "approximated 9.75 [percent], 25 [basis points] below the analogous adjusted average ROE for calendar-2012 (which approximated 10 percent)."¹⁷⁶ This evidence does not undermine, but supports, the Commission's conclusion that the 9.39 percent midpoint, determined by using the DCF methodology, is below most of the state ROEs.

87. We also reject Petitioners' argument that, in using state commission-authorized ROEs to corroborate the outcome of the DCF analysis, the Commission failed to make its own finding on the cost of equity. To the contrary, the Commission conducted its own DCF analysis and did make its own finding, based on the financial market data in the record. That the Commission looked to the state commission-authorized ROEs and alternative methodologies to corroborate the accuracy of its finding, does not undermine the Commission's finding on the cost of equity. Rather, the Commission's analysis of not only the DCF results but also additional record evidence demonstrates that the Commission fully reviewed the record to ensure a just and reasonable ROE sufficient to meet the capital attraction standards required by *Hope* and *Bluefield*.

88. We disagree that the Commission's use of state commission-authorized ROEs in Opinion No. 531 is inconsistent with Commission precedent. As the Commission explained in Opinion No. 531, while the Commission has rejected the use of state ROEs

¹⁷⁵ The NETOs' study of state commission-allowed ROEs covered the time period from October 1, 2010 through September 30, 2012. See Ex. NET-400; Ex. NET-403.

¹⁷⁶ Petitioners Request for Rehearing at 26 (citing Ex. SC-524). We note that the Regulatory Research Associates' report states that the average state commission-allowed ROE for the first quarter of 2013 is 10.24 percent. The 9.75 percent figure to which Petitioners refer was calculated by excluding from the ROE decisions issued in that quarter those from one particular state commission and, as noted, would be 10.24 percent without that exclusion. Further, we note that the record evidence also shows that the average state commission-allowed ROE for the fourth quarter of 2012, i.e., the quarter immediately following the time period of the NETOs' state ROE study, was 10.10 percent. Thus, the data concerning state commission allowed-ROEs for the fourth quarter of 2012 (10.10 percent) and the first quarter of 2013 (10.24 percent) are consistent with the data in the NETOs' study of state commission-allowed ROEs, and do not indicate a downturn in state ROEs as Petitioners allege.

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in the past, it has done so on the grounds that the state ROEs alone provide an insufficient basis for determining Commission-jurisdictional rates. Those cases are distinguishable from the instant proceeding, where the Commission instead compared the evidence provided by a significant number of state commission-authorized ROEs to the midpoint produced by the application of the Commission's traditional methodology and concluded that their levels, relative to each other, were illogical in light of the record evidence concerning the comparative risks of state-level electric distribution and interstate electric transmission. We also reject Petitioners' argument that, if state commission-approved ROEs are irrelevant when they are below Commission ROEs, then they are also irrelevant when they are above Commission ROEs. The Commission has not found state commission-approved ROEs to be irrelevant when they are lower than Commission-approved ROEs. As the Commission explained in Opinion No. 531, the relevance of the state commission-approved ROEs was determined in conjunction with the record evidence on the elevated risks of interstate transmission, compared to state-regulated distribution.

89. Lastly, we disagree with EMCOS's assertion that the Commission ignored *Missouri*, 337 F.3d 1066. *Missouri* is inapposite to the facts of this case as it involved the Commission's *adoption* of a specific rate, for a gas pipeline's sales under the Commission's jurisdiction, that had been "approved by [a state commission] under the regulatory regime that governed the pipeline prior to FERC's assertion of jurisdiction."¹⁷⁷ By comparison, in Opinion No. 531, the Commission did not adopt any rate approved by a state commission.

d. Risk Premium Analysis

i. Opinion No. 531

90. In Opinion No. 531, the Commission explained that the risk premium methodology is "based on the simple idea that since investors in stocks take greater risk than investors in bonds, the former expect to earn a return on a stock investment that reflects a 'premium' over and above the return they expect to earn on a bond investment."¹⁷⁸ The Commission further explained that "investors' required risk premiums expand with low interest rates and shrink at higher interest rates," and found that this link "provides a helpful indicator of how investors' required returns on equity

¹⁷⁷ *Missouri*, 337 F.3d at 1076.

¹⁷⁸ Opinion No. 531, 147 FERC ¶ 61,234 at P 147 (quoting Roger A. Morin, New Regulatory Finance 108 (Public Utilities Reports, Inc. 2006)) (internal quotations omitted).

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have been impacted by the interest rate environment.”¹⁷⁹ The Commission explained that it has in the past rejected the use of risk premium analyses to estimate investor-required returns on equity, but “those cases are distinguishable from the instant proceeding because they involved proposals to establish a constant risk premium based on the average difference between state commission ROEs and bond rates over multi-year periods.”¹⁸⁰

91. The Commission found the NETOs’ risk premium analysis “informative,”¹⁸¹ as it indicated that the NETOs’ cost of equity “is between 10.7 percent and 10.8 percent, which is higher than the 9.39 percent midpoint produced by our DCF analysis.”¹⁸² The Commission explained that, in relying on the NETOs’ risk premium analysis, “we do not depart from our use of the DCF methodology; rather, we use the record evidence to inform the just and reasonable placement of the ROE within the zone of reasonableness established in the record by the DCF methodology.”¹⁸³

ii. Requests for Rehearing

92. EMCOS argue that Opinion No. 531 erred by adopting the NETOs’ risk premium analysis despite the fact that the Commission has repeatedly rejected the use of risk premium analysis for determining a just and reasonable ROE for a public utility.¹⁸⁴ EMCOS assert that the Commission in Opinion No. 531 attempted to distinguish those precedents from this proceeding on the basis that the risk premium analyses in those cases relied on “the average state commission ROEs and bond rates over multi-year periods.”¹⁸⁵ However, EMCOS contend that the Commission’s rationale is flawed because the Commission’s rejection of risk premium analyses in the past was not due to the involvement of state commission ROEs, but rather was due to concerns regarding the

¹⁷⁹ *Id.*

¹⁸⁰ *Id.* n.290.

¹⁸¹ *Id.* P 146.

¹⁸² *Id.* P 147.

¹⁸³ *Id.* P 146.

¹⁸⁴ EMCOS Request for Rehearing at 20-21 (citing *Consumers Energy Co.*, 64 FERC ¶ 63,029 (1993), *aff’d*, 85 FERC ¶ 61,100 at 61,361 (1998); *New England Power Co.*, 31 FERC ¶ 61,378, at 61,841 (1985)).

¹⁸⁵ *Id.* at 21 (citing Opinion No. 531, 147 FERC ¶ 61,234 at P 147 n.290).

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reliability of the methodology to produce reliable results in fluctuating market conditions.¹⁸⁶ Additionally, EMCOS argue that Opinion No. 531 fails to respond to criticism that parties presented about the NETOs' risk premium analysis. EMCOS argue that Opinion No. 531's failure to respond to—or even acknowledge—the substantive arguments against the NETOs' specific risk premium analysis renders the decision arbitrary and capricious.¹⁸⁷

93. Petitioners argue that the NETOs' version of a risk premium analysis contains multiple flaws. Petitioners argue that the NETOs' risk premium analysis detaches the ROEs from the regulatory contexts in which they were approved, and this disconnect should have rendered the NETOs' risk premium study irrelevant as a matter of law.¹⁸⁸ In addition, Petitioners assert that, even if it were acceptable to detach the allowed ROEs from their regulatory contexts, the NETOs' risk premium study's attempt to discern regulatory outcomes and assign dates to those outcomes contains numerous errors. Specifically, Petitioners contend that the risk premium study was performed by a person who did not appear at trial, lacked professional expertise in reading Commission decisions, and used examples supplied by the NETOs' counsel rather than a random or representative sample. Petitioners also argue that the NETOs' risk premium study is flawed because it assumes that the outcomes of Commission proceedings represent equity costs on the day the Commission issued its order approving the ROE, thereby ignoring both regulatory lag and the reality that many Commission decisions that identify an ROE do not involve finding a new, currently cost-based ROE.¹⁸⁹

94. Additionally, Petitioners argue that the NETOs' risk premium study is flawed because the study makes no attempt to screen its inputs for comparable risk.¹⁹⁰ As an

¹⁸⁶ *Id.* (citing *Consumers Energy Co.*, 64 FERC ¶ 63,029, *aff'd*, 85 FERC ¶ 61,100 at 61,361; *New England Power Co.*, 31 FERC ¶ 61,378 at 61,841).

¹⁸⁷ *Id.* at 22 (citing *Ill. Pub. Telecomm. Ass'n v. FCC*, 117 F.3d at 564).

¹⁸⁸ *Id.* at 33-34 (citing *Pepco Holdings, Inc.*, 124 FERC ¶ 61,176, at P 127 (2008), *reh'g denied*, 139 FERC ¶ 61,144 (2012)).

¹⁸⁹ Similarly, EMCOS note that Trial Staff and the Complainants argued that the NETOs' risk premium analysis is based on Commission-allowed returns, which are not the same as the market indicated ROEs that this methodology claims to address. Moreover, EMCOS explain that the NETOs' analysis includes ROEs that are the result of settlements, which further skew the results. In addition, EMCOS explain that the NETOs' analysis is rife with errors regarding the applicable dates of the Commission approved ROEs upon which they rely.

¹⁹⁰ Petitioners Request for Rehearing at 35.

example illustrating this flaw, Petitioners state that the NETOs' risk premium study treated as representative of June 2012 risk premiums—without making any finding as to the current equity cost—a Commission order that merely extended to a new MISO participant the 12.38 percent ROE that was established for the MISO region more than a decade earlier.¹⁹¹ Petitioners further state that the Commission, in Opinion No. 489, rejected the NETOs' reliance on MISO's 12.38 percent ROE as a benchmark for New England.¹⁹² Petitioners also argue that the NETOs' risk premium study treats orders and data from 2008-2009 as comparable to the NETOs' ROE, which was established in 2006 based on data from 2004. The Petitioners further assert that the NETOs' study failed to include orders after June 2012, and that these omissions skewed the NETOs' results by missing the trend towards lower ROEs.

95. Petitioners argue that, although the NETOs' failed to present an informative risk premium study, they did provide a basis to construct a more useful one that accords with Opinion No. 531's discussion of the theory underlying the risk premium methodology. Specifically, Petitioners note Opinion No. 531's explanation that "investors' required risk premiums expand with low interest rates and shrink at high interest rates,"¹⁹³ and assert that the NETOs' risk premium study used an incorrect ratio in determining the rate at which risk premiums change in response to changes in interest rates. Petitioners argue that the NETOs' risk premium study relied on an inferred rate at which risk premiums expand when interest rates drop is about 93:100—i.e., a 100 basis points decline in interest rates is deemed to be offset by a risk premium increase of about 93 basis points—which leaves a net decline in the cost of equity of only 7 basis points for every 100 basis point change interest rates. However, Petitioners contend that the NETOs' witness disavowed that ratio at trial, by clarifying that "generally, one half of the move in equity returns [is] related to the move in bond returns," so "if bond returns go up 100 basis points, your best guess of equity costs is 50 or 60 basis points."¹⁹⁴ Therefore, Petitioners state that it is more appropriate to use 45:100¹⁹⁵ as the rate at which risk premiums expand when interest rates drop—i.e., a 100 basis points decline in interest rates is deemed to be offset by a risk premium increase of about 45 basis points—which leaves a net decline in the cost of equity of 55 basis points for every 100 basis point change in

¹⁹¹ *Id.*

¹⁹² *Id.* at 36.

¹⁹³ *Id.* at 37 (citing Opinion No. 531, 147 FERC ¶ 61,234 at P 147).

¹⁹⁴ *Id.* at 38.

¹⁹⁵ Petitioners calculate this ratio by taking the average of the 50-60 basis point range indicated by the NETOs' witness at trial.

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interest rates. Petitioners argue that substituting that relationship in Ex. NET-704 for the implausible 93:100 ratio, indicates an ROE of 9.67 percent to 9.91 percent.¹⁹⁶

96. Petitioners contend that Opinion No. 531's reliance on a stale and poorly designed study of past Commission orders was inconsistent with its finding that ROEs should reflect the most recent information available at the time of trial.

iii. Commission Determination

97. We deny rehearing on the issue of whether the NETOs' risk premium analysis is flawed. As the Commission explained in Opinion No. 531, the theory behind the risk premium methodology is that "since investors in stocks take greater risk than investors in bonds, the former expect to earn a return on a stock investment that reflects a 'premium' over and above the return they expect to earn on a bond investment."¹⁹⁷ There are multiple approaches that have been advanced to determine this equity risk premium for a utility.¹⁹⁸ For example, a risk premium can be developed directly, by conducting a risk premium analysis for the company at issue, or indirectly by conducting a risk premium analysis for the market as a whole and then adjusting that result to reflect the risk of the company at issue.¹⁹⁹ Another approach that investors might choose to look to in the utility context is to "examin[e] the risk premiums implied in the returns on equity allowed by regulatory commissions for utilities over some past period relative to the contemporaneous level of the long-term Treasury bond yield."²⁰⁰ In the instant case, the NETOs followed the latter approach, developing their risk premium study by analyzing the ROEs allowed by this Commission since April 2006,²⁰¹ relative to the

¹⁹⁶ Petitioners Request for Rehearing at 38.

¹⁹⁷ Roger A. Morin, New Regulatory Finance 108 (Public Utilities Reports, Inc. 2006).

¹⁹⁸ See generally *id.* at 107-130.

¹⁹⁹ *Id.* at 110.

²⁰⁰ *Id.* at 123.

²⁰¹ See Ex. NET-704 at 3-4. We note that, although Petitioners assert that the NETOs failed to include any Commission orders issued after June 2012, Petitioners have not cited any final Commission orders establishing a utility's ROE between June 2012 and the date the Presiding Judge set as the deadline for the parties to update their exhibits prior to the hearing. While Petitioners correctly note that the Commission issued such an ROE order on May 6, 2013, that decision was issued after the final updating of exhibits.

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contemporaneous level of the long-term Treasury bond yield,²⁰² to determine the risk premium implied by those regulatory decisions.²⁰³

98. Petitioners allege that the NETOs' risk premium analysis is flawed because it assigned arbitrary dates to the regulatory decisions on which it was based, ignored the fact that some of the decisions involved rates agreed to by settlement, ignored regulatory lag, and ignored the reality that some of the decisions did not involve the calculation of a current cost of equity. Given the varying duration of regulatory proceedings, it is difficult, if not impossible, to ensure precise contemporaneity between long-term Treasury bond yields and the cost of equity allowed by a regulator. Assigning approximate dates to the cost of equity determinations made in those regulatory proceedings, as the NETOs have done, is often unavoidable, and this fact alone does not undermine the relevance of risk premium analyses.²⁰⁴ Similarly, whether the regulatory decision involved a settlement agreement or the application of a cost of equity that was calculated in the past, e.g., the 12.38 percent ROE established for the MISO region, does not affect the reliability of a risk premium analysis.²⁰⁵ Risk premiums allowed by

²⁰² NETOs also analyzed the ROEs allowed by regulatory decisions relative to long-term utility bond yields.

²⁰³ See Ex. NET-704 at 1-2.

²⁰⁴ We disagree with Petitioners that the Commission's reliance on the NETOs' risk premium analysis, despite the regulatory lag reflected therein, is inconsistent with Opinion No. 531's finding that "ROEs should reflect the most recent information available at the time of trial." Petitioners Request for Rehearing at 37. The NETOs' risk premium study upon which the Commission relied is indeed the most recent such study in the record.

²⁰⁵ Further, contrary to Petitioners' assertion, the fact that the Commission, in Opinion No. 489, declined to use the 12.38 percent ROE from the MISO region as a benchmark in establishing the NETOs base ROE has no bearing on this proceeding. Using the ROE from the MISO region as a benchmark in establishing the just and reasonable ROE for the NETOs' is much different than using the ROE from the MISO region as one data point, among many, in a risk premium analysis that is then used to corroborate the results of the Commission's analysis. Additionally, assuming *arguendo* that (1) the 12.38 percent ROE for the MISO region was "stale" in June 2012, see Petitioners' Request for Rehearing at 35, (2) the 11.14 percent base ROE for the NETOs' was "stale" in August, November, and December of 2008, see *id.* at 36, and (3) it is therefore appropriate to exclude those data points from the NETOs' risk premium study, Petitioners have not shown that excluding those data points would materially affect the results of the NETOs' risk premium study or undermine its usefulness in
(continued...)

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regulators “are presumably based on the results of market-based methodologies presented to regulators in rate hearings and on the actions of objective unbiased investors in a competitive marketplace.”²⁰⁶ This is no less true in the case of settlement agreements, as settling parties rely upon the same market-based methodologies in determining the rates they are willing to accept. In short, while the approach the NETOs used in their risk premium analysis, like any methodology for estimating the cost of equity, is not without inherent weaknesses, it is nonetheless an approach that investors routinely rely upon.²⁰⁷ We similarly find the NETOs’ risk premium analysis sufficiently reliable—not to set the ROE itself—but rather to corroborate our decision to place the NETOs’ base ROE above the midpoint of the zone of reasonableness produced by the DCF analysis.

99. We also reject Petitioners’ argument that the NETOs’ risk premium study does not support placing the NETOs’ base ROE above the midpoint of the zone of reasonableness because the NETOs’ assumption regarding the inferred rate at which risk premiums expand when interest rates drop—i.e., the assumption that risk premiums expand by 93 basis points for every 100 basis point drop in interest rates—is unsupported. Petitioners assert that, if the NETOs’ study is adjusted to reflect a more appropriate ratio than 93:100, the NETOs’ risk premium study produces a result between 9.67 and 9.91 percent. While the rate at which risk premiums change as interest rates change is indeed important in a risk premium analysis, we find the alleged flaw to be immaterial in this context in

corroborating the results of the Commission’s DCF analysis. The NETOs’ risk premium analysis compared the ROEs established in 66 cases from April 2006 through June 2012 to the contemporaneous 10-year U.S. Treasury bond yields to determine 66 risk premiums, which averaged 7.33 percent. Excluding the alleged stale ROEs would eliminate five of the 66 risk premiums from the NETO’s analysis. The remaining 61 risk premiums average 7.28 percent, only marginally less than the average of the 66 risk premiums used in the NETOs’ analysis. This indicates that exclusion of the allegedly stale ROEs would not materially reduce the 10.7 percent to 10.8 percent cost of equity produced by the NETOs’ risk premium analysis.

²⁰⁶ Roger A. Morin, *New Regulatory Finance* 125 (Public Utilities Reports, Inc. 2006).

²⁰⁷ *Id.* at 123-125. We reject Petitioners’ assertion that the NETOs’ risk premium study was conducted by an unqualified analyst who did not appear at trial. The analyst to whom Petitioners refer did not conduct the NETOs’ risk premium analysis, but rather assisted the NETOs’ expert witness in conducting the analysis. *See* Tr. 647:9-648:10. Further, the analyst at issue is a chartered financial analyst, with a Masters Degree in Business Administration, who has assisted the NETOs’ expert witness in preparing testimony in over 100 Commission proceedings. *See id.* at 648:14-22.

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this case. As an initial matter, the alternative inferred rate—a ratio of 45:100—that Complainants put forth based on the NETOs’ witness’s testimony at hearing was based on state commission-allowed ROEs, not interstate transmission ROEs allowed by this Commission.²⁰⁸ In light of the record evidence on the risk differential between state-regulated distribution and Commission-regulated interstate transmission, we are not persuaded that it is appropriate to apply to the NETOs, for the time period at issue in this proceeding, the inferred rate relationship between risk premiums and interest rates that was observed in state commission-allowed ROEs over a time period dating back a quarter century, to 1987. Further, the NETOs’ determined the inferred rate relationship between risk premiums and interest rates in their risk premium study by conducting empirical observations and regression analysis of bond yields and Commission-allowed ROEs.²⁰⁹ In sum, we are not persuaded that the NETOs’ empirical results are invalid simply because they differ from the inferred rate relationship reflected in historical state commission-approved ROEs, particularly where anomalous capital market conditions exist that may impact the inferred relationship between risk premiums and interest rates.

100. EMCOS argue that the Commission erred in relying on the NETOs’ risk premium analysis because doing so is inconsistent with precedent in which the Commission has rejected the use of risk premium analyses.²¹⁰ EMCOS assert that the Commission in Opinion No. 531 attempted to distinguish those precedents on the grounds that the risk premium analyses therein involved state commission-allowed ROEs. EMCOS contend that the Commission’s interpretation of those precedents is incorrect, because the Commission in fact rejected the use of risk premium analyses in those past cases due to concerns that risk premium analyses are unreliable under fluctuating market conditions.

101. In Opinion No. 531, the Commission explained that the Commission’s rejection of the risk premium analysis in a number of past cases, including *New England Power Co.*, is distinguishable from the instant case because those cases involved “*proposals to establish a constant risk premium based on the average difference between state commission ROEs and bond rates over multi-year periods.*”²¹¹ EMCOS mischaracterize the Commission’s interpretation of *New England Power Co.* and other similar precedents

²⁰⁸ See Tr. 606:5-7 (“this is based on state returns, and state returns have marched to a slightly different drummer than FERC returns over the years.”).

²⁰⁹ See generally Ex. NET-704.

²¹⁰ EMCOS Request for Rehearing at 20-21 (citing *Consumers Energy Co.*, 64 FERC ¶ 63,029, *aff’d*, 85 FERC ¶ 61,100 at 61,361; *New England Power Co.*, 31 FERC ¶ 61,378 at 61,841).

²¹¹ Opinion No. 531, 147 FERC ¶ 61,234 at n.290 (emphasis added).

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by focusing on Opinion No. 531's reference to the fact that the risk premium analyses in the past cases relied upon state commission ROEs. As the italicized language in the above quote makes clear, however, the Commission's rationale for rejecting the proposal in *New England Power Co.* was not merely reliance on state commission-set ROEs, but was, as EMCOS correctly acknowledge, based on the Commission's finding that "[t]here is no direct relationship between historical risk premiums and a current cost of equity under constantly changing financial conditions."²¹² In *New England Power Co.*, the utility proposed to calculate a risk premium based on the difference between the most recent 20-year average yield for certain money market indicators and the most recent 20-year average annual yield for Moody's Electric Utility common stocks plus the 10-year growth in dividends for those stocks. Thus, the utility assumed a constant risk premium for a 20-year period. In the instant case, the NETOs' risk premium analysis does not assume a constant risk premium over any length of time. Rather, the NETOs calculated a varying risk premium based on variations in the difference between allowed ROEs and bond yields during the time period from April 2006 through June 2012. Those cases in which the Commission rejected risk premium analyses in the past are thus distinguishable from the instant case, because unlike the proposals in those cases the NETOs have not proposed their risk premium analysis to establish a constant risk premium.²¹³

e. **CAPM Analysis**

i. **Opinion No. 531**

102. In Opinion No. 531, the Commission explained that "[s]imilar to the risk premium analysis, the NETOs' CAPM uses interest rates as the input for the risk-free rate, which makes it useful in determining how the interest rate environment has impacted investors' required returns on equity."²¹⁴ The Commission also explained that "CAPM is utilized by investors as a measure of the cost of equity relative to its risk."²¹⁵ The Commission

²¹² *New England Power Co.*, 31 FERC at 61,841.

²¹³ Moreover, unlike other cases, the Commission here is *not* setting investor-required ROEs based on this risk premium, but is instead looking to it merely as "a helpful indicator" of the impact of the "interest rate environment" on "investors' required returns on equity." And from this analysis (and others discussed elsewhere in Opinion No. 531 and here) the Commission concludes only that the ROE should indeed be set above the midpoint. See Opinion No. 531, 147 FERC ¶ 61,234 at P 147 & n.290.

²¹⁴ Opinion No. 531, 147 FERC ¶ 61,234 at P 147.

²¹⁵ *Id.*

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explained that it has in the past rejected the use of CAPM analyses, but “those cases are distinguishable from the instant proceeding because they involved CAPM analyses that were based on historic market risk premiums,” whereas the NETOs’ CAPM analysis “is based on forward-looking investor expectations for the market risk premium.”²¹⁶

103. The Commission found the NETOs’ CAPM analysis “informative,”²¹⁷ as it produced a midpoint of 10.4 percent and a median of 10.9 percent, both of which are above the 9.39 percent midpoint produced by the Commission’s DCF analysis.²¹⁸ The Commission explained that, in relying on the NETOs’ CAPM analysis, “we do not depart from our use of the DCF methodology; rather, we use the record evidence to inform the just and reasonable placement of the ROE within the zone of reasonableness established in the record by the DCF methodology.”²¹⁹

ii. Requests for Rehearing

104. Petitioners assert that the NETOs’ CAPM study is flawed because its assumption that the market as a whole (i.e., most of the S&P 500 companies) will grow at an annual rate of 10.3 percent is overly optimistic, unsustainable, double the historical norms and projections, and inconsistent with the GDP estimate the Commission relied upon in Opinion No. 531 for other purposes.²²⁰ Petitioners argue that the NETOs calculated the unsustainable 10.3 percent growth rate by screening out almost a quarter of the market and placing excessive weight on the projections of non-utility companies’ medium-term earnings per share growth while ignoring the fact that those estimates reflect unsustainable short-term stock repurchase programs and are not long-term projections.

105. Petitioners contend that the NETOs’ CAPM study is also flawed because it relies on stock betas, which Petitioners assert are unreliable and do not meaningfully measure the risk differential between the proxy group and the dividend paying portion of the S&P 500 companies.²²¹ Petitioners state that the Commission in *ITC Holdings* found betas to

²¹⁶ *Id.* n.292.

²¹⁷ *Id.* P 146.

²¹⁸ *Id.* P 147.

²¹⁹ *Id.* P 146.

²²⁰ Petitioners Request for Rehearing at 39.

²²¹ *Id.* at 40 (citing *ITC Holdings Corp.*, 121 FERC ¶ 61,229, at P 43 (2007)); EMCOS Request for Rehearing at 23.

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be an unreliable predictor of risk and, as a result, found the CAPM methodology to be inappropriate for determining a company's ROE.²²² Petitioners assert that, while the Commission in Opinion No. 531 attempted to distinguish *ITC Holdings* on the basis that it involved historical risk premiums, Opinion No. 531 did not attempt to address *ITC Holdings*'s finding that betas are unreliable. Similarly, EMCOS assert that, because the NETOs' CAPM analysis relied on betas, that analysis failed to incorporate forward-looking expectations, which undermines the Commission's claim that the NETOs' CAPM analysis is based on "forward-looking investor expectations" and is, therefore, distinguishable from CAPM analyses the Commission has rejected in the past.²²³ Petitioners assert that their witness and Trial Staff's witness both presented more credible, forward-looking CAPM studies indicating a cost of equity of 7.5 percent and 8.2 percent, respectively, but that the Commission ignored both of these CAPM studies in Opinion No. 531.

106. In addition, Petitioners contend that the NETOs' CAPM study is flawed because it includes a "size adjustment" based on the theory that smaller companies are riskier and should, therefore, have higher growth and higher returns than the average company in the sample set. Petitioners argue that the NETOs' rationale is undermined by the Petitioners' calculation showing that the smaller firms in the NETOs' sample set have lower-than-average growth—an unweighted average of 9.8 percent, compared to the NETOs' weighted average of 10.3 percent.²²⁴ Petitioners also argue that academic studies have shown that it is improper to apply this type of "size adjustment" to utilities.²²⁵ Petitioners state that, without the size adjustment, the median and midpoint of the NETOs' CAPM analysis is 9.7 percent.²²⁶

107. EMCOS argue that the NETOs' CAPM analysis is flawed because it used a risky 30-year bond interest rate for the risk-free component of the calculation and inappropriately used a DCF result for the risk premium element of the analysis.

²²² Petitioners Request for Rehearing at 31 (citing *ITC Holdings*, 121 FERC ¶ 61,229 at P 43; *Orange & Rockland Utils., Inc.*, Opinion No. 314, 44 FERC ¶ 61,253 (*Orange & Rockland*), *order on reh'g*, Opinion No. 314-A, 45 FERC ¶ 61,252 (1988), *reh'g denied*, 46 FERC ¶ 61,036 (1989)).

²²³ EMCOS Request for Rehearing at 23 (citing Opinion No. 531, 147 FERC ¶ 61,234 at P 147 n.292).

²²⁴ Petitioners Request for Rehearing at 41 (citing Ex. SC-514).

²²⁵ *Id.* at 42 (citing SC-200 at 35-36).

²²⁶ *Id.*

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iii. Commission Determination

108. We deny rehearing on the issue of whether the NETOs' CAPM analysis is flawed. The CAPM methodology has three inputs: the risk-free rate, betas, and the market risk premium.²²⁷ The risk-free rate and betas used in a CAPM study are generally not controversial. The risk-free rate is represented by a proxy, typically the yield on 30-year Treasury bonds.²²⁸ Betas, which measure a stock's risk relative to the market, are published by several commercial sources. The market risk premium, which is where most CAPM studies diverge, can be estimated either using a backward-looking approach, a forward-looking approach, or a survey of academics and investment professionals.²²⁹ A CAPM analysis is backward-looking if its market risk premium component is determined based on historical, realized returns.²³⁰ A CAPM analysis is forward-looking if its market risk premium component is based on a DCF study of a large segment of the market.²³¹ In a forward-looking CAPM analysis, the market risk premium is calculated by subtracting the risk-free rate from the result produced by the DCF study.²³²

109. In this proceeding, the NETOs submitted a forward-looking CAPM study, using 30-year Treasury bonds for the risk-free rate, betas published by Value Line, and a market risk premium based on a DCF study of all S&P 500 companies that were paying dividends. The NETOs' CAPM approach is a generally accepted methodology routinely relied upon by investors and, therefore, one appropriately used to corroborate our own analysis. As discussed below, we reject the arguments that the NETOs' CAPM analysis contains flaws that undermine its usefulness as corroborative evidence, in determining whether the midpoint of the zone of reasonableness produced by the Commission's DCF analysis provides the NETOs a return that satisfies the requirements of *Hope* and *Bluefield*.

110. As an initial matter, we reject EMCOS's argument that the NETOs' CAPM analysis is flawed because it used a DCF study to determine the market risk premium. As

²²⁷ Roger A. Morin, *New Regulatory Finance* 150 (Public Utilities Reports, Inc. 2006).

²²⁸ *Id.* at 151.

²²⁹ *Id.* at 155-162.

²³⁰ *Id.* at 155-156.

²³¹ *Id.* at 159-160.

²³² *See id.* at 150, 155.

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explained above, using a DCF study is the standard method of calculating the market risk premium in a forward-looking CAPM analysis.²³³ We are, therefore, unpersuaded that the use of a DCF study renders the NETOs' CAPM analysis deficient. We also disagree with Petitioners' argument that the NETOs' CAPM analysis relied on an overly optimistic growth rate input in determining the market risk premium. The growth rate in the NETOs' CAPM analysis is based on IBES data, which the Commission has long relied upon as a reliable source of growth rate data.²³⁴

111. While Petitioners' assert that the growth rate input is inflated because the NETOs calculated it based on only those S&P 500 companies that were paying dividends, we are not persuaded that the exclusion of those companies not paying dividends skewed the growth rate input. As the NETOs' witness correctly explained during the hearing, a DCF analysis can only be conducted for companies that pay dividends.²³⁵ Accordingly, the proxy group in our DCF analysis consists of companies that pay dividends. Basing a CAPM study on only dividend-paying companies is therefore appropriate in this context, where the Commission is looking to the CAPM study to corroborate the results of a DCF analysis, because doing so produces a growth rate input that is more representative of the DCF proxy group than a CAPM study based on non-dividend-paying companies would be. Further, we are not persuaded by Petitioners' argument that non-dividend-paying companies have lower growth rate estimates than dividend-paying companies, because in many situations the opposite is true due to non-dividend-paying companies decision to retain and reinvest more of their earnings, rather than pay dividends.

112. We are also unpersuaded that the growth rate projection in the NETOs' CAPM study was skewed by the NETOs' reliance on analysts' projections of non-utility companies' medium-term earnings growth, or that the study failed to consider that those analysts' estimates reflect unsustainable short-term stock repurchase programs and are not long-term projections. As explained above, the NETOs based their growth rate input on data from IBES, which the Commission has found to be a reliable source of such data. Thus, the time periods used for the growth rate projections in the NETOs' CAPM study are the time periods over which IBES forecasts earnings growth. Petitioners' arguments against the time period on which the NETOs' CAPM analysis is based are, in effect, arguments that IBES data are insufficient in a CAPM study. We disagree. We acknowledge that CAPM analyses may be based on different time periods; however, without more evidence, i.e., a CAPM analysis based on a longer time period, we are not persuaded that the time period on which the NETOs' based their CAPM analysis

²³³ See *supra* P 108.

²³⁴ See *supra* PP 71-72.

²³⁵ See Tr. 740: 3-4.

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undermines the relevance of that analysis in corroborating the results of the Commission's DCF analysis.

113. Further, the fact that the Commission's two-step DCF methodology incorporates a long-term growth rate does not necessitate the incorporation of a long-term growth rate in the DCF study the NETOs used to develop the market risk premium for their CAPM analysis. The Commission's rationale for incorporating a long-term growth rate estimate in DCF analyses for public utilities was that it is often unrealistic and unsustainable for high short-term growth rates to continue in perpetuity.²³⁶ Under the CAPM model, the market risk premium is based on the difference between the "required return on the overall market" and the risk-free rate.²³⁷ The required return on the overall market is determined by conducting a DCF study of "a representative market index, such as the Standard & Poor's 500 Index."²³⁸ As noted above, the NETOs developed the market risk premium in their CAPM analysis in exactly this way, by conducting a DCF analysis of the dividend-paying companies in the S&P 500 to determine the required return on the overall market. The rationale for incorporating a long-term growth rate estimate in conducting a two-step DCF analysis of a specific group of utilities does not necessarily apply when conducting a DCF study of the companies in the S&P 500. That is because the S&P 500 is regularly updated to include only companies with high market capitalization. While an individual company cannot be expected to sustain high short-term growth rates in perpetuity, the same cannot be said for a stock index like the S&P 500 that is regularly updated to contain only companies with high market capitalization, and the record in this proceeding does not indicate that the growth rate of the S&P 500 stock index is unsustainable.

114. We also reject EMCOS's argument that the NETOs' CAPM analysis was flawed because it relied on a "risky 30-year bond interest" to calculate the risk-free rate. As noted above, 30-year U.S. Treasury bond yields are a generally accepted proxy for the risk-free rate in a CAPM analysis, and are also considered superior to short- and intermediate-term bonds for this purpose.²³⁹ Therefore, absent record evidence to the

²³⁶ Opinion No. 531, 147 FERC ¶ 61,234 at P 36 n.63 (citing Roger A. Morin, New Regulatory Finance 308 (Public Utilities Reports, Inc. 2006)).

²³⁷ Roger A. Morin, New Regulatory Finance 146 (Public Utilities Reports, Inc. 2006).

²³⁸ *Id.* at 159.

²³⁹ See Roger A. Morin, New Regulatory Finance 151-152 (Public Utilities Reports, Inc. 2006) ("the yield on very long-term government bonds, namely, the yield on 30-year Treasury bonds, is the best measure of the risk-free rate for use in the CAPM and Risk Premium methods.").

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contrary, we find 30-year Treasury bond yields to be an appropriate basis for the risk-free rate in the NETOs' CAPM analysis.

115. We also disagree with Petitioners' argument that the NETOs' CAPM study does not support placing the NETOs' base ROE above the midpoint because the study relies on betas. Petitioners' assertion is based on a misinterpretation of Commission precedent. While Petitioners correctly state that the Commission in *ITC Holdings* and *Consumers Energy Co.* found that "betas, *in isolation*, [are] unreliable predictors of risk,"²⁴⁰ Petitioners ignore the qualifier "in isolation," which highlights an important distinction between the CAPM analyses at issue in those cases and the NETOs' CAPM analysis. In both *ITC Holdings* and *Consumers Energy Co.*, the parties submitted CAPM studies that analyzed only the utility whose rates were at issue. As the Commission explained in *Consumers Energy Co.*, "CAPM is more appropriately used for determining the composition of a portfolio of stocks."²⁴¹ In the instant proceeding, the NETOs' CAPM study analyzed, as a portfolio, a proxy group of electric utilities. Thus, the NETOs' CAPM study and associated use of betas do not raise the same concerns as did the studies in *ITC Holdings* and *Consumers Energy Co.*

116. We further disagree with EMCOS's argument that the NETOs' CAPM analysis is not forward-looking because it relies on betas. As explained above, whether a CAPM analysis is forward-looking or backward-looking depends on how the market risk premium—not the betas—are calculated.²⁴² Although it is true that betas are based on historical data, reliance on betas does not render a CAPM analysis backward-looking, as that term is commonly used in the CAPM context. As explained above, a CAPM study is backward-looking if its market risk premium component is determined based on historical, realized returns,²⁴³ and a CAPM study is forward-looking if its market risk premium component is based on a DCF study of a large segment of the market.²⁴⁴ Unlike the market risk premium component of the CAPM methodology, betas are necessarily

²⁴⁰ *ITC Holdings Corp.*, 121 FERC ¶ 61,229 at P 43 (emphasis added); *Consumers Energy Co.*, 85 FERC ¶ 61,100 at 61,362 (emphasis added).

²⁴¹ *Consumers Energy Co.*, 85 FERC ¶ 61,100 at 61,362 n.26 (noting Trial Staff's testimony that, according to Value Line, beta should not be used to determine the ROE for a single company).

²⁴² See *supra* P 108.

²⁴³ Roger A. Morin, *New Regulatory Finance* 155-156 (Public Utilities Reports, Inc. 2006).

²⁴⁴ *Id.* at 159-160.

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based on historical data, because “[t]he true beta of a security can never be observed.”²⁴⁵ Therefore, we disagree with EMCOS’s assertion that the use of betas renders a CAPM analysis backward-looking. We reiterate that a CAPM study is forward-looking, notwithstanding its use of betas, if its market risk premium component is based on an appropriate DCF study.

117. We disagree with Petitioners’ argument that the NETOs’ CAPM analysis is flawed due to the fact that the NETOs applied a size adjustment to account for the difference in size between the NETOs and the dividend-paying companies in the S&P 500. This type of size adjustment is a generally accepted approach to CAPM analyses,²⁴⁶ and we are not persuaded that it was inappropriate to use a size adjustment in this case. The purpose of the NETOs’ size adjustment is to render the CAPM analysis useful in estimating the cost of equity for companies that are smaller than the companies that were used to determine the market risk premium in the CAPM analysis. While Petitioners assert that the record shows that smaller firms have lower growth,²⁴⁷ Petitioners’ assertion rests on a comparison of companies *within* the S&P 500—all of which have large market capitalization—rather than a comparison of the S&P 500 companies to companies smaller than the S&P 500 companies. While it may be true that larger dividend-paying members of the S&P 500 are growing faster than the smaller dividend-paying members of the S&P 500, this does not indicate how the growth rates of the dividend-paying members of the S&P 500 compare to the NETOs or to other groups of companies with smaller market capitalization (e.g., the companies in either the S&P 400, which consists of companies with mid-capitalization, or the S&P 600, which consists of companies with small capitalization). Further, Petitioners’ assertion is contradicted by other record evidence indicating, and supporting the generally accepted principle,²⁴⁸ that smaller firms are riskier than larger firms, and therefore experience faster growth.²⁴⁹

118. Petitioners also argue that the Commission erred in ignoring Complainants’ CAPM study, which indicated a 7.5 percent cost of equity, and Trial Staff’s CAPM study, which indicated an 8.2 percent cost of equity. However, we find both Complainants’ and

²⁴⁵ *Id.* at 79.

²⁴⁶ *Id.* at 187.

²⁴⁷ Petitioners Request for Rehearing at 41 (citing Ex. SC-514).

²⁴⁸ Roger A. Morin, *New Regulatory Finance* 187 (Public Utilities Reports, Inc. 2006).

²⁴⁹ See Ex. NET-300 at 68 (citing *Morningstar*, “Ibbotson SBBI 2012 Valuation Yearbook,” at 85).

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Trial Staff's CAPM studies to be flawed. Complainants did not determine the market risk premium by using a DCF study to determine the required return on the overall market and then subtracting the risk-free rate from the DCF result, but instead estimated the market risk premium directly, using market risk premium studies. This approach is acceptable, in theory, as it is a valid method of determining market risk premium; however, it is not clear that Complainants executed the approach as a forward-looking analysis. While Complainants' approach is purportedly forward-looking, it is not clear from the record that their estimated market risk premium is, in fact, based on prospective data. Complainants used a market risk premium of 5.00 percent,²⁵⁰ which appears to be determined using market risk premium data based on a mix of historical, prospective, and survey approaches.²⁵¹ While the record is not clear about how Complainants used these three categories of market risk premium studies to determine the market risk premium, if Complainants' market risk premium is based on a compilation of the three categories we do not consider the resulting market risk premium to be forward-looking. Further, even assuming *arguendo* that Complainants relied only on the prospective market risk premium studies, we are not persuaded that their CAPM study is sufficiently representative of the capital market conditions during this proceeding, as—importantly—all but one of the prospective studies listed in Complainants' exhibit pre-date the Great Recession.²⁵²

119. We find Trial Staff's CAPM analysis also to be flawed. Similar to Complainants' CAPM analysis, Trial Staff did not calculate the market risk premium by conducting a DCF analysis and subtracting the risk-free rate from the result, but by estimating the market risk premium directly. However, Trial Staff did not provide a study to support its estimated market risk premium,²⁵³ and Trial Staff based its CAPM analysis on only 20 companies. Further, those 20 companies are members of the NETOs' proxy group. Because the purpose of the CAPM methodology is to calculate the cost of equity using a risk-return relationship based entirely on market risk,²⁵⁴ the index of companies used in determining the market risk premium must be large enough to capture the market risk.²⁵⁵

²⁵⁰ Ex. SC-112 at 1.

²⁵¹ *Id.* at 4-6.

²⁵² *Id.* at 5-6.

²⁵³ See Ex. S-1 at 98.

²⁵⁴ Roger A. Morin, New Regulatory Finance 145-146 (Public Utilities Reports, Inc. 2006).

²⁵⁵ *Id.* at 159-160.

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We do not consider a group of 20 companies, all of comparable risk, sufficiently large or diverse to accurately reflect the risks of the market as a whole, and we are therefore not persuaded that such a group accurately reflects the market risk premium to be used in a CAPM study. In addition, we note that, unlike the NETOs, neither Complainants nor Trial Staff updated their CAPM studies during the hearing; as a result, the CAPM evidence provided by the NETOs represents the most recent CAPM evidence in the record. In sum, for the above reasons, we find Complainants' and Trial Staff's CAPM analyses to be unreliable as corroborative evidence in this proceeding.

f. Expected Earnings Analysis

i. Opinion No. 531

120. In Opinion No. 531, the Commission explained that the NETOs' expected earnings analysis "can be useful in validating" the ROE determination," given the expected earnings analysis's "close relationship to the comparable earnings standard that originated in *Hope*, and the fact that it is used by investors to estimate the ROE that a utility will earn in the future."²⁵⁶ The Commission found the NETOs' expected earnings analysis "informative,"²⁵⁷ as it produced a midpoint of 12.1 percent and a median of 10.2 percent, both of which are above the 9.39 percent midpoint produced by the Commission's DCF analysis.²⁵⁸ The Commission explained that, in relying on the NETOs' expected earnings analysis, "we do not depart from our use of the DCF methodology; rather, we use the record evidence to inform the just and reasonable placement of the ROE within the zone of reasonableness established in the record by the DCF methodology."²⁵⁹

ii. Requests for Rehearing

121. Petitioners argue that the NETOs' version of an expected earnings analysis is flawed because it "attempts to forecast returns on *book* equity, rather than investor-required returns on equity purchased at above-book study-period stock prices."²⁶⁰ Petitioners state that the NETOs' analysis forecasts returns on book equity because the

²⁵⁶ Opinion No. 531, 147 FERC ¶ 61,234 at P 147.

²⁵⁷ *Id.* P 146.

²⁵⁸ *Id.* P 147.

²⁵⁹ *Id.* P 146.

²⁶⁰ Petitioners Request for Rehearing at 43.

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analysis turns on the “expected earnings/book equity ratio (“r”) in Value Line’s five-year forecast.”²⁶¹ Petitioners contend that the Commission has long rejected setting ROEs “at the rate of return investors expect [the subject utility] to earn on [book] common equity (r), rather than the market cost of common equity (k).”²⁶² Petitioners assert that the Commission in Opinion No. 531 failed to address this inconsistency between the NETOs’ expected earnings analysis and Commission precedent.

122. Petitioners further assert that the Commission’s reliance on the NETOs’ expected earnings analysis was especially unreasonable in this case because, in adopting the two-step DCF methodology, the Commission discarded the “br+sv” element of the one-step DCF methodology, which placed the “r” input in proper context by factoring it with other components of utility firm growth. Petitioners contend that, although the record contained the necessary data for the NETOs to place their “r” input in the appropriate context, the NETOs’ expected earnings analysis ignored that data and instead emphasized “more speculative and optimistic” inputs.²⁶³ Petitioners argue that the Commission in *Kern River Gas Transmission Co.*, Opinion No. 486, 117 FERC ¶ 61,077 (2006) (Opinion No. 486) held that dividends and payout ratios “should be considered in order to account for going-concern utilities’ need to reinvest earnings instead of paying them all to shareholders;” however, Petitioners assert that the NETOs’ have failed to do so.

123. Petitioners argue that, by relying on forecasted returns on book equity, rather than forecasted returns on the market cost of equity, the NETOs’ expected earnings analysis ignores the market/book ratios of the proxy companies, which range from about 1.0 to 2.3.²⁶⁴ Petitioners assert that, as a result, the NETOs’ approach “simply reflects the perpetuation of a high market/book ratio, as was rejected in *Orange & Rockland*.”²⁶⁵ Petitioners also contend that the midpoint of the NETOs’ expected earnings analysis is particularly unreliable because it was skewed upwards by Dominion’s “unusually high earnings/book equity projection . . . which in turn reflected Dominion’s exceptionally high market/book ratio.”²⁶⁶ Petitioners argue that the Commission’s precedent on the use of midpoints in a cost of equity study is confined to DCF studies, and should not be used

²⁶¹ *Id.* at 42-43.

²⁶² *Id.* at 43 (citing *Orange & Rockland*, 44 FERC ¶ 61,253 at 61,952).

²⁶³ *Id.* at 44.

²⁶⁴ *Id.*

²⁶⁵ *Id.*

²⁶⁶ *Id.* at 44-45.

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in the context of the NETOs' expected earnings analysis because relying on a midpoint value that is distorted by a high market-to-book ratio would not help reveal the market cost of equity.²⁶⁷ Petitioners contend that, if the Commission does give weight to the NETOs' expected earnings analysis, the NETOs' analysis "points no higher than its median result, which was 10.2 percent."²⁶⁸

124. EMCOS argue that the NETOs' expected earnings analysis fails to recognize the critical link that "when actual or forecasted earnings are considered as a guide to an appropriate ROE allowance, they must be evaluated in conjunction with actual or forecasted stock prices."²⁶⁹ EMCOS further argue that Opinion No. 531 adopted and relied upon the NETOs' expected earnings analysis without addressing any of the concerns raised by Trial Staff or Complainants. For example, EMCOS note that Trial Staff argued that the NETOs' analysis "inappropriately relies on accounting return results, which are not reflective of the market's required return as indicated by actual equity stock investors."²⁷⁰ In addition, EMCOS note that the Complainants raised concerns that the NETOs' analysis included several flaws that rendered it unreliable and "overly optimistic."²⁷¹ EMCOS argue that failure to address these arguments is the definition of arbitrary and capricious decision-making.²⁷²

iii. Commission Determination

125. A comparable earnings analysis is a method of calculating the earnings an investor expects to receive on the book value of a particular stock. A comparable earnings analysis can be based either on the stock's historical earnings on book value, as reflected on the company's accounting statements, or on forward-looking estimates of earnings on book value, as reflected in analysts' earnings forecasts for the company. The latter approach is often referred to as an "expected earnings analysis" and is the approach the NETOs used in conducting their comparable earnings analysis in this proceeding. Petitioners' and EMCOS's argue that the NETOs' expected earnings analysis is flawed and does not support the Commission's decision to place the NETOs' base ROE above

²⁶⁷ *Id.* at 45.

²⁶⁸ *Id.* at 42.

²⁶⁹ EMCOS Request for Rehearing at 24 (citing Ex. No. EMC-3 at 8:15-18).

²⁷⁰ *Id.* at 24-25 (citing Trial Staff Initial Brief at 60).

²⁷¹ *Id.* (citing Complainants Initial Brief at 62).

²⁷² *Id.* at 25 (citing *Ill. Pub. Telecomm. Ass'n v. FCC*, 117 F.3d at 564).

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the midpoint of the zone of reasonableness produced by the Commission's DCF analysis. We disagree.

126. The NETOs conducted their expected earnings analysis by using the return on book equity that *Value Line* forecasted for the national group of companies that *Value Line* lists as Electric Utilities. The NETOs then multiplied each of those forecasted returns by an adjustment factor to determine each utility's average return, rather than its year-end return, explaining that using the year-end return would understate actual returns because of growth in common equity over the year.²⁷³ We consider the NETOs' expected earnings analysis to be sound, as it is forward-looking, based on a reliable source of earnings data, and appropriately converts the proxy companies' earnings to reflect average returns.²⁷⁴

127. While Petitioners correctly state that the Commission in *Orange & Rockland* rejected a proposal that "would, in effect, set the allowed rate of return on common equity at the rate of return investors expect [the utility] to earn on common equity (r), rather than the market cost of common equity (k)," that precedent is inapposite to this case for two reasons. First, *Orange & Rockland* did not involve a comparable earnings analysis; it involved a proposal to alter the DCF model by adjusting the dividend yield to reflect the expected earnings of the company whose rates were at issue, i.e., Orange & Rockland. Specifically, Orange & Rockland proposed to calculate the dividend yield in its DCF study by dividing dividend payments by book value, instead of by a current stock price. By comparison, the NETOs have not proposed to alter the DCF model to reflect expected earnings, but rather submitted an expected earnings study based on a national proxy group of utilities whose risk profiles are comparable to the NETOs.

128. Second, *Orange & Rockland* is inapposite because the Commission in that case rejected a proposal that would have had the effect of *setting* Orange & Rockland's base ROE at Orange & Rockland's own expected return on book equity. In the instant case, the Commission did not *set* the NETOs' base ROE at their own expected return on book equity or endorse an ROE analysis that would have that effect. Rather, the Commission in Opinion No. 531 used the DCF methodology to determine the NETOs' market cost of equity, and found that the NETOs' expected earnings analysis of a national proxy group was used to determine—and only relevant to—whether the midpoint of the DCF-determined zone of reasonableness provided a market cost of equity sufficient to meet the

²⁷³ See Ex. NET-300 at 73, 32.

²⁷⁴ See, e.g., *S. Cal. Edison Co.*, 92 FERC ¶ 61,070 at 61,263 (finding it necessary to adjust Value Line's forecasted returns on book equity to reflect average returns rather than year-end returns); see also Roger A. Morin, *New Regulatory Finance* 305-306 (Public Utilities Reports, Inc. 2006).

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requirements of *Hope* and *Bluefield*. The returns on book equity that investors expect to receive from a group of companies with risks comparable to those of a particular utility are relevant to determining that utility's market cost of equity, because those returns on book equity help investors determine the opportunity cost of investing in that particular utility instead of other companies of comparable risk. Such a calculation is consistent with the requirement in *Hope* that "the return to the equity owner should be commensurate with returns on investments in other enterprises having corresponding risks."²⁷⁵ As the NETOs' expert witness explained at trial, investors compare each investment alternative with the next best opportunity. If the utility is unable to offer a return similar to that available from other opportunities of comparable risk, investors will become unwilling to supply the capital on reasonable terms.²⁷⁶

129. Investors rely on both the market cost of equity and the book return on equity in determining whether to invest in a utility, because investors are concerned with both the return the regulator will allow the utility to earn *and* the company's ability to actually earn that return.²⁷⁷ If, all else being equal, the regulator sets a utility's ROE so that the utility does not have the opportunity to earn a return on its book value comparable to the amount that investors expect that other utilities of comparable risk will earn on their book equity, the utility will not be able to provide investors the return they require to invest in that utility.²⁷⁸ Thus, all else being equal, an investor is more likely to invest in a utility that it expects will have the opportunity to earn a comparable amount on its book equity as other enterprises of comparable risk are expected to earn. Because investors rely on expected earnings analyses to help estimate the opportunity cost of investing in a particular utility, we find this type of analysis useful in corroborating whether the results produced by the DCF model may have been skewed by the anomalous capital market conditions reflected in the record.

130. We also reject Petitioners' argument that it was unreasonable for the Commission to rely on the NETOs' expected earnings analysis without also considering the "br+sv" formula in the Commission's DCF analysis. Whether "r" is directly used in the Commission's calculation of the short-term growth rate in the DCF methodology does not bear on the validity of the NETOs' expected earnings analysis or on its relevance in

²⁷⁵ *Hope*, 320 U.S. at 603; see also *Petal Gas Storage, L.L.C.*, 496 F.3d 695 (D.C. Cir. 2007).

²⁷⁶ Ex. No. NET-300 at 71.

²⁷⁷ See Tr. 637:6-12.

²⁷⁸ As the NETOs' witness testified, returns on book value are analogous to the allowed return on a utility's rate base. *Id.*

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corroborating the results of the Commission's DCF analysis.²⁷⁹ As explained below, the expected earnings analysis and DCF analysis are used to estimate two different types of returns, each valid in its own right, that investors rely upon in determining whether to invest in a particular company.²⁸⁰

131. As to the argument that the midpoint of the NETOs' expected earnings analysis is skewed upwards by the results of one company, i.e., Dominion, Petitioners conclusory statements that Dominion's expected earnings are "unusually high" and that Dominion's market-to-book ratio is "exceptionally high" are insufficient to show that Dominion's results skewed the NETOs' analysis. Petitioners state that Dominion has a market-to-book ratio of 2.255, that this value skewed the result of the NETOs' expected earnings analysis, and that there is no evidence that the NETOs have market-to-book ratios comparable to Dominion's.²⁸¹ However, Petitioners have provided no evidence demonstrating that Dominion's 2.255 market-to-book ratio is "exceptionally high," and there is no evidence that the NETOs' market-to-book ratios are not comparable to those of the proxy group companies. Lastly, even assuming *arguendo* that it would be more appropriate to eliminate Dominion or to use the median, rather than the midpoint, of the NETOs' expected earnings analysis, the result would be 11.2 percent or 10.2 percent, respectively. Both of these results are above the 9.39 percent midpoint of the DCF-produced zone of reasonableness and, therefore, corroborate the Commission's decision to place the NETOs' base ROE above the 9.39 percent midpoint.

132. While Petitioners and EMCOS²⁸² assert that the NETOs' expected earnings study ignores the proxy companies' market-to-book ratios, considering market-to-book ratios in an expected earnings study is inconsistent with the purpose of the comparable earnings model. The comparable earnings model is intended to estimate the return on book equity

²⁷⁹ We also reject Petitioners' assertion that Opinion No. 486 is relevant to the validity of the NETOs' expected earnings analysis. The language from Opinion No. 486 to which Complainants cite does not involve an expected earnings analysis; rather it concerns whether it is appropriate to base the dividend yield in a DCF analysis of a master limited partnership on its earnings, rather than on dividend payments in excess of earnings. See Opinion No. 486, 117 FERC ¶ 61,077 at P 153.

²⁸⁰ See *infra* P 132.

²⁸¹ Petitioners Request for Rehearing at 44-45.

²⁸² EMCOS argue that the NETOs' expected earnings analysis is flawed because it does not evaluate forecasted earnings in conjunction with forecasted stock prices. This is merely another way of saying that the NETOs' expected earnings analysis failed to consider market-to-book ratios.

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that investors expect the *utility* will earn; the market cost of equity, by comparison, is the estimated return to *investors* that an investor requires to invest in the utility. Petitioners and EMCOS seek to adjust the estimated return on book equity produced by the NETOs' expected earnings analysis into the market cost of equity, by applying a market-to-book adjustment. However, as noted above, the return on book equity is relied upon by investors to determine the opportunity cost of investing in a particular company, and investors rely upon expected earnings analysis for this purpose without attempting to convert that opportunity cost into the market cost of equity. We, therefore, find the NETOs' expected earnings analysis reliable as corroborative evidence in this proceeding, notwithstanding the lack of a market-to-book adjustment in that analysis. Further, even assuming *arguendo* that a market-to-book adjustment was appropriate, we are not persuaded that Petitioners' approach of simply dividing a utility's book return on equity by its market-to-book ratio would accurately estimate the utility's market cost of equity. We also disagree with EMCOS's argument that the NETOs' expected earnings analysis relies on accounting return results, and is therefore not corroborative of the market cost of equity. As noted above, the NETOs' expected earnings analysis is based on forecasted earnings, not historical returns reflected on accounting statements.

3. Impact of the DCF Methodology Change on Existing ROE Transmission Incentive Adders

a. Opinion No. 531

133. Opinion No. 531 explained that, “[b]ased on the Commission’s policy that the total ROE including any incentive ROE is limited to the zone of reasonableness, the Commission has found in the past that an incentive ROE may not be implemented in full by the utility if the total ROE exceeds the zone of reasonableness.”²⁸³ The Commission found that “[n]othing in [Opinion No. 531] changes this Commission policy,”²⁸⁴ and, therefore, “when a public utility’s ROE is changed, either under section 205 or section 206 of the FPA, that utility’s total ROE, inclusive of transmission incentive ROE adders, should not exceed the top of the zone of reasonableness produced by the two-step DCF methodology.”²⁸⁵

²⁸³ Opinion No. 531, 147 FERC ¶ 61,234 at P 164.

²⁸⁴ *Id.*

²⁸⁵ *Id.* P 165.

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b. Request for Rehearing

134. The NETOs request that the Commission clarify that adjustments to the NETOs' ROE incentive adders are outside the scope of this proceeding.²⁸⁶ The NETOs state that the base ROE was the sole matter set for hearing, no party submitted evidence relating to incentive adders, and the issue was not discussed in the Initial Decision.²⁸⁷ The NETOs assert that Opinion No. 531 does not state with specificity that the NETOs' total ROE must not exceed the top of the zone of reasonableness, and the NETOs interpret the Commission's language concerning capping the total ROE as a statement of a general ratemaking principle.²⁸⁸

135. If the Commission did intend to require the NETOs to reduce the total ROE to the top of the zone of reasonableness, the NETOs request rehearing of that decision. The NETOs state that the "ROE incentive adders were approved based upon a detailed record of the benefits and risks of the relevant projects and the nexus between the incentive adders and the projects, which included consideration of the ability of the incentive to facilitate construction of the project."²⁸⁹ The NETOs state that, when the adders were approved, they were below the top end of the then-current zone of reasonableness. The NETOs argue that the Commission placed no conditions on the adders' continued effectiveness, and that the adders do not automatically change when the Commission determines a new zone of reasonableness.²⁹⁰

136. The NETOs state that the base ROE was the only matter at issue in this case, and that incentive adders were explicitly excluded by the complaint.²⁹¹ The NETOs argue that modifying the incentive adders in this proceeding would violate the Constitution's Due Process Clause and the Administrative Procedure Act.²⁹² The NETOs state that, in a similar case, the Commission granted an ROE adder without notice to the parties that the issue would be decided during the hearing and the D.C. Circuit found that the

²⁸⁶ NETOs Request for Rehearing at 6-7.

²⁸⁷ *Id.* at 7-8.

²⁸⁸ *Id.* at 8-9.

²⁸⁹ *Id.* at 11-12.

²⁹⁰ *Id.* at 13.

²⁹¹ *Id.* at 14-15.

²⁹² *Id.* at 15-16.

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Commission had violated the parties' due process rights.²⁹³ The NETOs state that, in each of the cases that the Commission cited in Opinion No. 531, the ROE incentive adders were implicated prior to the hearing, thus providing the parties with notice and the opportunity to submit evidence on the incentive adders. The NETOs assert that, assuming the Hearing Order had addressed incentive adders, the Commission erred in ruling that the adders must be reduced without accepting evidence on the issue.²⁹⁴

137. The NETOs request that the Commission clarify that the term "total ROE" refers to the total transmission assets of a utility rather than project-specific ROEs.²⁹⁵ The NETOs argue that as long as the ultimate rate charged to consumers is just and reasonable, FPA section 219 is satisfied and the Commission has no basis to look at project-specific ROEs to determine whether they are below the top of the zone of reasonableness.²⁹⁶

138. The NETOs argue that, if the term "total ROE" includes incentive ROEs, Opinion No. 531 should be reversed as inconsistent with statutory requirements and Commission precedent. The NETOs state that in Order No. 679 the Commission stated that the test for reviewing a rate is whether the end result is reasonable.²⁹⁷ The NETOs argue that such an evaluation necessarily involves review of the overall rate inclusive of all components, not merely a review of one component such as an individual project's incentive ROE.²⁹⁸

c. Commission Determination

139. We deny rehearing on this issue. As an initial matter, it is worth noting that Opinion No. 531 does not change the incentive ROE adders that the Commission previously granted to the NETOs. Rather, Opinion No. 531 follows Commission policy that a utility's ROE, even if it includes an incentive ROE adder, would be capped at the

²⁹³ *Id.* at 17-18 (citing *PSC of Kentucky*, 397 F.3d at 1011-12).

²⁹⁴ *Id.* at 18-20.

²⁹⁵ *Id.* at 20-22.

²⁹⁶ *Id.* at 22-23.

²⁹⁷ *Id.* at 23-24.

²⁹⁸ *Id.* at 24-26 (citing *Northeast Utils. Serv. Co.*, 52 FERC ¶ 61,097 (1990), *reh'g denied*, 52 FERC ¶ 61,336; *Florida Power & Light Co.*, 24 FERC ¶ 61,171, at 61,408 (1983); *Florida Power & Light Co.*, 32 FERC ¶ 61,059 at 61,162 (1985)).

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upper end of the transmission owner's DCF-determined zone of reasonableness. For example, in Order No. 679, the Commission made clear that the total ROE including any incentive ROE adder sought by an applicant must be within the utility's DCF-determined zone of reasonableness.²⁹⁹ In the orders in which the Commission granted the NETOs' incentive ROE adders, the Commission also made clear that the total ROE including such adders would be capped at the high end of the NETOs' zone of reasonableness.³⁰⁰ The fact that a transmission owner may not be able to implement in full its awarded incentive ROE adder because the resulting total ROE would exceed the high end of the transmission owner's zone of reasonableness is nothing new.³⁰¹ In addition, the

²⁹⁹ *Promoting Transmission Investment through Pricing Reform*, Order No. 679, FERC Stats. & Regs. ¶ 31,222, at PP 2, 93 (2006), *order on reh'g*, Order No. 679-A, FERC Stats. & Regs. ¶ 31,236, *order on reh'g*, 119 FERC ¶ 61,062 (2007); *Pac. Gas & Elec. Co.*, 141 FERC ¶ 61,168, at P 26 (2012); *see also Town of Norwood, Mass. v. FERC*, 80 F.3d at 534-535 (supporting the principle that ROE should be cabined within the bounds of the zone of reasonableness, by reversing a Commission decision to set ROE at the bottom of the zone of reasonableness that was established in the utility's prior rate case and explaining that the Commission cannot rely on a zone of reasonableness established in a prior rate case if the utility's circumstances have since changed); 16 U.S.C. § 824s(d) (2012) ("All rates approved under the rules adopted pursuant to [FPA section 219] . . . are subject to the requirements of sections [205 and 206] of this title that all rates . . . be just and reasonable."); Order No. 679-A, FERC Stats. & Regs. ¶ 61,236 at P 15 (cross-referenced at 117 FERC ¶ 61,345 at P 15) (indicating that the Commission will keep any incentive ROE adder within the zone of reasonableness as a means to ensure the Commission comply with its regulatory responsibilities under the FPA). The courts have also recognized that utilities cannot charge rates that exceed the DCF-determined zone of reasonableness. *See, e.g., Union Elec. Co. v. FERC*, 890 F.2d 1193, 1204 (D.C. Cir. 1989).

³⁰⁰ *See, e.g., Ne. Utils. Serv. Co.*, 124 FERC ¶ 61,044 at P 83.

³⁰¹ *See, e.g., NSTAR Elec. Co.*, 125 FERC ¶ 61,313 at PP 81-87 (granting a New England transmission owner an incentive ROE adder, to be bound by the upper end of the zone of reasonableness previously established for the New England transmission owners; and determining, based on an updated DCF analysis, that the overall ROE including the incentive ROE adders remained within the zone of reasonableness); *accord Me. Pub. Utils. Comm'n v. FERC*, 454 F.3d at 288-89 (affirming the Commission's decision to grant transmission owners that join ISO New England a 50 basis point incentive ROE adder for RTO participation, and the Commission's decision to cap the overall ROE at the top of the zone of reasonableness); *Proposed Pricing Policy for Efficient Operation and Expansion of Transmission Grid*, 102 FERC ¶ 61,032, at P 37 (2003) (noting that, in implementing ROE-based incentives, including the RTO

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Commission has summarily applied this policy in rate cases initiated after an ROE adder was approved. For example, in establishing a hearing on a section 205 rate filing by Pacific Gas and Electric Co. (PG&E), the Commission held that a 200 basis point adder originally granted to PG&E ten years earlier³⁰² and a 50 basis point ROE adder for RTO participation granted two years earlier³⁰³ would be limited to within the new zone of reasonableness determined at the hearing.³⁰⁴ Thus, whether the merits of a utility's incentive ROE adders are implicated by a proceeding is a much different issue than whether the utility can fully implement its incentive ROE adders due to changes in the zone of reasonableness for that utility. This proceeding involves only the latter of these two issues; it does not involve the merits of the NETOs' existing incentive ROE adders.

140. Contrary to the NETOs' assertion, the Commission's ruling in Opinion No. 531 on this issue was not merely a general statement of ratemaking principle, it was a continuation of a Commission policy that the NETOs' total ROE cannot exceed the zone of reasonableness calculated in this proceeding.³⁰⁵

141. The NETOs argue that the precedent cited by Opinion No. 531 concerning ROE incentive adders, such as *PG&E*, is distinguishable from the instant proceeding because in the incentives cases the incentives were implicated before the hearing, and the parties therefore had notice and opportunity to submit evidence on the issue. We disagree. In the cases cited by Opinion No. 531, the Commission did not set for hearing the issue of whether an existing incentive adder should be reduced to no higher than the top of the

participation adder, those incentives would be subject to a cap on the overall ROE equal to the top of the range of reasonable ROEs for a proxy group).

³⁰² See *Western Area Power Admin.*, 100 FERC ¶ 61,331, at PP 12-13 (2002).

³⁰³ *Pac. Gas & Elec. Co.*, 132 FERC ¶ 61,272, at P 23 (2010).

³⁰⁴ *PG&E*, 141 FERC ¶ 61,168 at P 26 (continuing to grant a 200 basis point ROE adder for the PATH 15 upgrade project, granted prior to Order No. 679, and a 50 basis point adder for RTO participation, granted subsequent to Order No. 679, and in doing so "remind[ing] PG&E that any ROE adder is limited to within the range of reasonableness of the ROE.").

³⁰⁵ This is reaffirmed by the Commission's determination in Opinion No. 531-A, the order on the paper hearing that the Commission established in Opinion No. 531, in which the Commission found that the zone of reasonableness produced by the DCF methodology in this proceeding is 7.03 percent to 11.74 percent and, therefore, that "the NETOs' total or maximum ROE, including transmission incentive ROE adders, cannot exceed 11.74 percent." Opinion No. 531-A, 149 FERC ¶ 61,032 at P 11.

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new zone of reasonableness. Rather, the Commission summarily ruled on that issue before the hearing. Because the Commission has an established policy that incentive adders must be within the zone of reasonableness in order to comply with section 219(a) of the FPA, the issue of whether to reduce an incentive adder that would otherwise exceed the top of the zone of reasonableness does not present any issue of material fact that would be appropriate for consideration in a hearing.

142. In any event, the NETOs' did in fact have notice and opportunity to present argument on the issue of their total ROE. Because it is well established both that a proceeding to determine a utility's base ROE involves a determination of the utility's zone of reasonableness under the DCF methodology, and that a transmission owner's awarded incentive ROE adder could not exceed the high end of the zone of reasonableness, the NETOs had notice to present evidence regarding the zone and thus the ultimate just and reasonable total ROE.

143. We disagree with the NETOs' argument that *PSC of Kentucky*, 397 F.3d 1004, is relevant to this issue. That case involved the Commission's post-hearing decision to grant an incentive ROE that the Commission, in setting the case for hearing, explicitly declined to grant and stated would not be at issue in the proceeding. Those facts are distinguishable from the facts here.

144. In *PSC of Kentucky*, the court found that the Commission violated the parties' due process rights because the Commission, having initially determined that it would not grant an incentive ROE adder, at the end of the proceeding granted the incentive ROE adder, and thus failed to place the parties on notice at the outset that, post-hearing, its order might grant the incentive ROE adder.³⁰⁶ The court explained that, while the Commission considered the petitioners' arguments regarding the incentive ROE adder on rehearing, the Commission did not allow them to present evidence at hearing on the relevant factual issue, i.e., the need for, or appropriate size of, the incentive ROE adder.³⁰⁷ In contrast, here the parties had both opportunities to make their case. The NETOs had notice of the Commission's already-well-established policy that a utility's total ROE must remain within the zone of reasonableness identified by the DCF analysis, and the NETOs had the opportunity to submit—and did, in fact, submit—evidence at hearing on the relevant factual issue, i.e., the zone of reasonableness identified by the

³⁰⁶ *PSC of Kentucky*, 397 F.3d at 1012.

³⁰⁷ *Id.*

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DCF analysis. Further, they also have had the opportunity to raise their arguments concerning this issue on rehearing.³⁰⁸

145. The NETOs assert that the Commission's use of the term "total ROE" in Opinion No. 531 may be read to refer only to "the overall ROE of the utility (inclusive of all transmission assets), rather than project-specific ROEs," because the Commission did not "address the meaning of 'total ROE' in the context of a multiple-asset utility."³⁰⁹ Contrary to the NETOs' assertion, Opinion No. 531 did address the meaning of the term "total ROE" both in the context of ROEs that apply to specific projects³¹⁰ and in the context of ROEs that apply to multiple utility assets.³¹¹ To be clear, the term "total ROE" applies to, and has identical meaning in, both contexts. Requests for incentive ROE adders are typically presented to the Commission in one of three ways: (1) a request for incentive ROE adders that apply to all of a utility's transmission assets;³¹² (2) a request for incentive ROE adders that apply only to specific transmission projects;³¹³ or (3) a request for a combination of incentive ROE adders, some of which apply to all of the utility's transmission assets and some of which apply only to specific transmission projects.³¹⁴ In each type of incentive ROE case, the Commission has explained that the total ROE, i.e., the base ROE plus any incentive adders, for the transmission assets to which the adder applies is capped at the top of the zone of reasonableness.³¹⁵ In other

³⁰⁸ See, e.g., *State of Cal. ex rel. Lockyer v. FERC*, 329 F.3d 700, 711 (2003) ("the Commission provided all the procedural protections required by the Fifth Amendment and FPA when it carefully considered all the evidence and arguments that the petitioners offered in their petitions for rehearing and motions to intervene."); see also *ANR Pipeline Co. and TC Offshore LLC*, 143 FERC ¶ 61,225, at PP 57, 60 (2013).

³⁰⁹ NETOs Request for Rehearing at 20-21.

³¹⁰ Opinion No. 531, 147 FERC ¶ 61,234 at P 164 (citing *Trans Bay Cable LLC*, 145 FERC ¶ 61,151 (2013), and *Atlantic Path 15, LLC*, 135 FERC ¶ 61,037 (2011)).

³¹¹ *Id.* (citing *PG&E*, 141 FERC ¶ 61,168).

³¹² See, e.g., *Oklahoma Gas & Elec. Co.*, 122 FERC ¶ 61,071 (2008).

³¹³ See, e.g., *NSTAR Elec. Co.*, 125 FERC ¶ 61,313; see also *RITELine Illinois, LLC & RITELine Indiana, LLC*, 137 FERC ¶ 61,039 (2011).

³¹⁴ See, e.g., *PG&E*, 141 FERC ¶ 61,168.

³¹⁵ See, e.g., *Oklahoma Gas & Elec. Co.*, 122 FERC ¶ 61,071 at P 36 n.26; *NSTAR Elec. Co.*, 125 FERC ¶ 61,313 at P 81; *PG&E*, 141 FERC ¶ 61,168 at P 26.

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words, incentive ROE adders are capped by the top of the DCF-determined zone of reasonableness, regardless of the particular incentive ROE adder authorized or the transmission assets to which it applies. This is appropriate because all incentives ultimately must be evaluated according to the same methodology, i.e., they must be evaluated against a zone of reasonableness above which the record does not support the total ROE including any incentive ROE adders as just and reasonable.

146. We also reject the NETOs' argument that FPA section 219 is satisfied, and the Commission has no basis to change a project-specific ROE, as long as the utility's ultimate rate is just and reasonable. This argument is inconsistent with the Commission's precedent on project-specific ROE incentives, in which the Commission has held that the utility's total ROE for the project cannot exceed the zone of reasonableness.³¹⁶ In addition, the practical effect of the NETOs' argument—"even if an incentive ROE for a particular project exceeds a utility's zone of reasonableness, so long as the entire utility's ROE (inclusive of all transmission assets) falls within the utility's zone of reasonableness, no change would be needed to the project-specific incentive ROE"—appears to result in incentive ROE adders applying to facilities to which the Commission has not granted the adders. An incentive ROE adder may not serve to increase the ROE for a transmission asset that has not been granted an incentive. Lastly, we disagree with the NETOs that *Northeast Utilities Service Co.*, 52 FERC ¶ 61,097 (1990), *Florida Power & Light Co.*, 24 FERC ¶ 61,171 (1983), and *Florida Power & Light Co.*, 32 FERC ¶ 61,059 (1985), support allowing project-specific ROEs above the zone of reasonableness. Those cases did not involve an analysis of the utilities' ROE relative to the zone of reasonableness produced by a DCF methodology; rather, those cases involved analyses of the equity returns at issue relative to either the utilities' costs³¹⁷ or to other rate designs that the utility could have used.³¹⁸

4. Establishment of a Just and Reasonable Rate

a. Opinion No. 531

147. The Commission in Opinion No. 531 did not establish the NETOs' just and reasonable ROE. As the Commission explained, the "finding concerning the specific numerical just and reasonable ROE for the NETOs is subject to the outcome of the paper

³¹⁶ See, e.g., *Pepco Holdings, Inc.*, 125 FERC ¶ 61,130, at PP 75-79, 91-94 (2008).

³¹⁷ See *Florida Power & Light*, 24 FERC ¶ 61,171 at 61,408; *Florida Power & Light Co.*, 32 FERC ¶ 61,059 at 61,162.

³¹⁸ See *Northeast Utils. Serv. Co.*, 52 FERC ¶ 61,097 at 61,485-486.

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hearing on the appropriate long-term growth projection to be used in the two-step DCF methodology.”³¹⁹

b. Requests for Rehearing

148. EMCOS requests that the Commission clarify that it intended for Opinion No. 531 to establish 10.57 percent as the prospective base ROE in effect from the date of issuance of Opinion No. 531, pending the outcome of the paper hearing on the long-term growth rate for use in the two-step DCF methodology. Similarly, Petitioners argue that the Commission erred by not directing the NETOs to prospectively reduce their rates as of June 19, 2014, based on the tentative findings in Opinion No. 531.³²⁰ Petitioners also argue that it was arbitrary and inconsistent with the section 206 “bond of protection” for the Commission to rely on the 4.39 percent long-term growth rate for purposes of excluding PSEG, while not relying on a 4.39 percent second-step growth rate for purposes of setting an interim or final ROE to be observed.³²¹

149. Petitioners assert that the paper hearing is unlikely to materially alter the conclusions reached in Opinion No. 531 and that any refinement of the NETOs’ ROE could be implemented as a refund or surcharge against the 10.57 percent base ROE. Petitioners argue that FPA section 206 requires the Commission to fix the rate to be observed as of the date of Opinion No. 531.³²² Petitioners further argue that courts have found that the Commission has “fixed” a rate when parties are in a position to supply their own inputs to a formula and thereby know the numerical rates. Petitioners contend that Opinion No. 531 provides such a formula by supplying a 10.57 percent base ROE and an 11.74 percent maximum ROE.³²³

150. Petitioners argue that implementing interim rates is required by the Commission’s obligation to “act as speedily as possible” on FPA section 206 complaints.³²⁴ Petitioners

³¹⁹ Opinion No. 531, 147 FERC ¶ 61,234 at P 152.

³²⁰ Petitioners Request for Rehearing at 66 (citing *New England Power Generators Association v. ISO New England Inc.*, 146 FERC ¶ 61,038, at P 26 (2014); *Georgia Power Co.*, 57 FERC ¶ 61,353 (1991)).

³²¹ *Id.* at 61-62.

³²² *Id.* at 69-70.

³²³ *Id.* at 70-71.

³²⁴ *Id.* at 71 (quoting 16 U.S.C. § 824e(b)).

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state that, as an alternative to making the NETOs' new ROE prospectively effective as of June 19, 2014, the Commission could direct the NETOs to use the final ROE in its true-up calculation for the 2014 rate year.³²⁵ Petitioners note that if the Commission uses this alternative method, the Commission must issue its order on the paper hearing before July 31, 2015 to ensure that the true-up filing is implemented with the correct ROE.³²⁶

c. Commission Determination

151. We deny Petitioners' and EMCOS's requests to prospectively establish the NETOs' replacement rate as of June 19, 2014.³²⁷ FPA section 206 requires that "[w]henever the Commission, after a hearing held upon its own motion or upon complaint, shall find that any rate . . . is unjust, unreasonable, unduly discriminatory or preferential, the Commission shall determine the just and reasonable rate . . . to be thereafter observed and in force, and shall fix the same by order."³²⁸ As the Commission explained in Opinion No. 531, however, its findings regarding the justness and reasonableness of the NETOs' rates were "tentative because [they were] subject to the submission of the record evidence at the paper hearing . . . as to the appropriate long-term growth rate."³²⁹ While the appropriate long-term growth rate itself was a narrow issue, that input had the potential to materially affect the NETOs' ROE by altering the DCF results of the companies in the proxy group.³³⁰ As a result, the Commission could not satisfy the requirement of FPA section 206 that it "fix" the just and reasonable rate to be in effect prospectively until after the paper hearing established by Opinion No. 531. Only

³²⁵ *Id.* at 72 (citing *South Carolina Elec. & Gas Co.*, 132 FERC ¶ 61,043 (2010)).

³²⁶ *Id.* at 74.

³²⁷ The Commission established the just and reasonable ROE for the NETOs on October 16, 2014, in Opinion No. 531-A. See Opinion No. 531-A, 149 FERC ¶ 61,032 at PP 10-12.

³²⁸ 16 U.S.C. § 824e (2012).

³²⁹ Opinion No. 531, 147 FERC ¶ 61,234 at P 142.

³³⁰ We reject Petitioners' assertion that it was inconsistent for the Commission to rely on the 4.39 percent GDP growth rate in eliminating PSEG from the proxy group as a low-end outlier and not rely on that GDP growth rate to establish a just and reasonable rate in Opinion No. 531. If the paper hearing had modified the 4.39 percent GDP growth rate, the Commission could have been required to reconsider its low-end outlier ruling based on the revised DCF results. However, the paper hearing did not change the 4.39 percent GDP growth rate and, therefore, no such reconsideration was required.

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with the issuance of Opinion No. 531-A, on October 16, 2014, did the Commission establish the prospective just and reasonable rate.³³¹

152. We similarly disagree with Petitioners that the Commission fixed the just and reasonable rate in Opinion No. 531 by providing a formula by which the parties could supply their own inputs and know the numerical rate. The Commission in Opinion No. 531 provided no such formula. Further, even assuming *arguendo* that the Commission's analysis could be characterized as a formula, a key input—the long-term growth rate—was unsettled pending the outcome of the paper hearing. Lastly, we reject Petitioners' request that we direct the NETOs to include the ROE established in this proceeding in their true-up calculation for the 2014 rate year. When the NETOs make the annual Regional Network Service true-up filing in 2015 to update the formula rates to reflect calendar year 2014 actual data, consistent with the requirements of the Regional Network Service formula, the filing should reflect the relevant ROEs in effect for any month within the 2014 time period. As mentioned above, the prospective effective date for the ROE determined in this proceeding is October 16, 2014, the issuance date of Opinion No. 531-A. Accordingly, Petitioners' alternative request to direct the NETOs to include the ROE determined in this proceeding for the entire 2014 calendar year is inconsistent with the effective date established in Opinion 531-A. We note that there are other complaints involving the NETOs' ROEs pending before the Commission in Docket Nos. EL13-33 and EL14-86 that may affect the ROE ultimately charged under the Regional Network Service formula for other months in 2014; however, any changes to the formula as a result of those complaints will not be effective until the Commission issues final orders in those proceedings.

The Commission orders:

(A) Petitioners', EMCOS's, and the NETOs' requests for rehearing of Opinion No. 531 are hereby denied, as discussed in the body of this order.

³³¹ See Opinion No. 531-A, 149 FERC ¶ 61,032. The Commission in Opinion No. 531-A also directed the NETOs to issue refunds for the 15-month refund period in this proceeding, i.e., from October 1, 2011 through December 31, 2012.

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(B) The NETOs' request for rehearing of Opinion No. 531-A is hereby denied, as discussed in the body of this order.

By the Commission. Commissioner Honorable is concurring with a separate statement attached.

(S E A L)

Kimberly D. Bose,
Secretary.

UNITED STATES OF AMERICA
FEDERAL ENERGY REGULATORY COMMISSION

Martha Coakley, Massachusetts Attorney General;
Connecticut Public Utilities Regulatory Authority;
Massachusetts Department of Public Utilities; New
Hampshire Public Utilities Commission; Connecticut
Office of Consumer Counsel; Maine Office of the Public
Advocate; George Jepsen, Connecticut Attorney
General; New Hampshire Office of Consumer Advocate;
Rhode Island Division of Public Utilities and Carriers;
Vermont Department of Public Service; Massachusetts
Municipal Wholesale Electric Company; Associated
Industries of Massachusetts; The Energy Consortium;
Power Options, Inc.; and the Industrial Energy
Consumer Group

Docket Nos. EL11-66-002
EL11-66-003

v.

Bangor Hydro-Electric Company; Central Maine Power
Company; New England Power Company d/b/a National
Grid; New Hampshire Transmission LLC d/b/a NextEra;
NSTAR Electric and Gas Corporation; Northeast
Utilities Service Company; The United Illuminating
Company; Until Energy Systems, Inc. and Fitchburg
Gas and Electric Light Company; Vermont Transco,
LLC

(Issued March 3, 2015)

HONORABLE, Commissioner, *concurring*:

In denying the requests for rehearing, the Commission sets forth a cogent defense of Opinion No. 531 and duly considers and adequately addresses the arguments of the petitioners in the numerous requests for rehearing. Additionally, it is within the Commission's discretion to alter the DCF methodology for determining the just and reasonable rates of return for the NETOs.

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I write separately to emphasize two important points to ensure that they are not lost in the shift in the DCF methodology and the placement of the base ROE above the central tendency of the zone of reasonableness. These points relate to: (1) the determination of the just and reasonable rate; and (2) the anomalous market conditions that prompted the consideration of alternative methodologies which ultimately led to the placement of the base ROE halfway between the midpoint and the top of the zone of reasonableness.

The just and reasonable rate of return for a public utility necessarily must consider *both* the protection of the consumer and the capital attraction standards set forth in *Hope* and *Bluefield*. The Commission appropriately relies upon the landmark *Hope* and *Bluefield* decisions to make the point that the allowed return should be adequate to enable it to secure the funding necessary for the proper discharge of its public duties. The duty to ensure the NETOs' ability to attract capital prompted consideration of additional record evidence and led to the use of alternative methodologies as benchmarks against which the DCF results were measured. However, while the Commission in Opinion No. 531 tacitly recognizes that a just and reasonable rate protects consumers, it does not emphasize consumer protection as forcefully as it could have. The primary purpose of the authority granted to the Commission to ensure a just and reasonable rate is to protect the consumer.¹ Indeed, the *Hope* decision, relied upon by this Commission to articulate the just and reasonable standard, explicitly provides that the Commission must balance both "investor and consumer interests."² In finding that balance, the Commission dedicates significant effort to ensuring that the NETOs are able to attract sufficient capital. While capital attraction is essential, Opinion No. 531 should not be interpreted as tipping the scale in favor of investor interests. As intended by Congress and confirmed by the Courts, consumer protection is in the DNA of FERC's ratemaking authority. Opinion No. 531 does not, and cannot, change that fact.

Keeping in mind the delicate balance that the Commission must strike when weighing investor and consumer interests, it is important to note that the finding of "anomalous market conditions" in Opinion No. 531 did not create a bright line test nor did it create a presumption that market conditions will be found to be anomalous going forward. The anomalous, or unusual, market conditions that were found in the original order to justify the placement of the base ROE above the central tendency of the zone of

¹ See, e.g., *Morgan Stanley Capital Grp. Inc. v. Pub. Util. Dist. No. 1 of Snohomish Cnty., Wash.*, 554 U.S. 527, 564 (2008) ("Congress enacted the FPA precisely because it concluded that regulation was necessary to protect consumers from deficient markets.").

² [*FPC v. Hope Natural Gas Co.*, 320 U.S. 591, 603 \(1944\).](#)

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reasonableness were, by definition, atypical. Any public utility that seeks to rely upon anomalous market conditions to justify placement of its base ROE in the upper end of the zone of reasonableness will be tasked with demonstrating, in each case, that market conditions are indeed anomalous and that the adequacy of a base ROE set at the midpoint of the zone of reasonableness should be scrutinized. The utility should expect a rigorous analysis of the record when it attempts to make such a demonstration.

The decision in Opinion No. 531 is within the Commission's broad discretion to determine the just and reasonable rate. I concur with this denial of the requests for rehearing to emphasize the points discussed above.

Colette D. Honorable
Commissioner

Federal Reserve Bank of New York
Staff Reports

The Equity Risk Premium: A Review of Models

Fernando Duarte
Carlo Rosa

Staff Report No. 714
February 2015



This paper presents preliminary findings and is being distributed to economists and other interested readers solely to stimulate discussion and elicit comments. The views expressed in this paper are those of the authors and do not necessarily reflect the position of the Federal Reserve Bank of New York or the Federal Reserve System. Any errors or omissions are the responsibility of the authors.

The Equity Risk Premium: A Review of Models

Fernando Duarte and Carlo Rosa

Federal Reserve Bank of New York Staff Reports, no. 714

February 2015

JEL classification: C58, G00, G12, G17

Abstract

We estimate the equity risk premium (ERP) by combining information from twenty models. The ERP in 2012 and 2013 reached heightened levels—of around 12 percent—not seen since the 1970s. We conclude that the high ERP was caused by unusually low Treasury yields.

Key words: equity premium, stock returns

Duarte, Rosa: Federal Reserve Bank of New York (e-mail: fernando.duarte@ny.frb.org, carlo.rosa@ny.frb.org). The authors thank Tobias Adrian and James Egelhof for helpful comments on earlier drafts. This article is an update of, and a more comprehensive and rigorous treatment than, our blog post in Liberty Street Economics (<http://libertystreeteconomics.newyorkfed.org/2013/05/are-stocks-cheap-a-review-of-the-evidence.html>). The views expressed in this paper are those of the authors and do not necessarily reflect the position of the Federal Reserve Bank of New York or the Federal Reserve System.

1. Introduction

The equity risk premium —the expected return on stocks in excess of the risk-free rate— is a fundamental quantity in all of asset pricing, both for theoretical and practical reasons. It is a key measure of aggregate risk-aversion and an important determinant of the cost of capital for corporations, savings decisions of individuals and budgeting plans for governments. Recently, the equity risk premium (ERP) has also returned to the forefront as a leading indicator of the evolution of the economy, a potential explanation for jobless recoveries and a gauge of financial stability³.

In this article, we estimate the ERP by combining information from twenty prominent models used by practitioners and featured in the academic literature. Our main finding is that the ERP has reached heightened levels. The first principal component of all models—a linear combination that explains as much of the variance of the underlying data as possible— places the one-year-ahead ERP in June 2012 at 12.2 percent, above the 10.5 percent that was reached during the financial crisis in 2009 and at levels similar to those in the mid and late 1970s. Since June 2012 and until the end of our sample in June 2013, the ERP has remained little changed, despite substantial positive realized returns. It is worth keeping in mind, however, that there is considerable uncertainty around these estimates. In fact, the issue of whether stock returns are predictable is still an active area of research.⁴ Nevertheless, we find that the dispersion in estimates across models, while quite large, has been shrinking, potentially signaling increased agreement

³ As an indicator of future activity, a high ERP at short horizons tends to be followed by higher GDP growth, higher inflation and lower unemployment. See, for example, Piazzesi and Schneider (2007), Stock and Watson (2003), and Damodaran (2012). Bloom (2009) and Duarte, Kogan and Livdan (2013) study connections between the ERP and real aggregate investment. As a potential explanation of the jobless recovery, Hall (2014) and Kuehn, Petrosky-Nadeau and Zhang (2012) propose that increased risk-aversion has prevented firms from hiring as much as would be expected in the post-crisis macroeconomic environment. Among many others, Adrian, Covitz and Liang (2013) analyze the role of equity and other asset prices in monitoring financial stability.

⁴ A few important references among a vast literature are Ang and Bekaert (2007), Goyal and Welch (2008), Campbell and Thompson (2008), Kelly and Pruitt (2013), Chen, Da and Zhao (2013), Neely, Rapach, Tu and Zhou (2014).

even when the models are substantially different from each other and use more than one hundred different economic variables.

In addition to estimating the level of the ERP, we investigate the reasons behind its recent behavior.

Because the ERP is the difference between expected stock returns and the risk-free rate, a high estimate can be due to expected stock returns being high or risk-free rates being low. We conclude the ERP is high because Treasury yields are unusually low. Current and expected future dividend and earnings growth play a smaller role. In fact, expected stock returns are close to their long-run mean. One implication of a bond-yield-driven ERP is that traditional indicators of the ERP like the price-dividend or price-earnings ratios, which do not use data from the term structure of risk-free rates, may not be as good a guide to future excess returns as they have been in the past.

As a second contribution, we present a concise and coherent taxonomy of ERP models. We categorize the twenty models into five groups: predictors that use historical mean returns only, dividend-discount models, cross-sectional regressions, time-series regressions and surveys. We explain the methodological and practical differences among these classes of models, including the assumptions and data sources that each require.

2. The Equity Risk Premium: Definition

Conceptually, the ERP is the compensation investors require to make them indifferent at the margin between holding the risky market portfolio and a risk-free bond. Because this compensation depends on the future performance of stocks, the ERP incorporates expectations of future stock market returns, which are not directly observable. At the end of the day, any model of the ERP is a model of investor expectations. One challenge in estimating the ERP is that it is not clear what truly constitutes the market return and the risk-free rate in the real world. In practice, the most common measures of total market returns are based on broad stock market indices, such as the S&P 500 or the Dow Jones Industrial

Average, but those indices do not include the whole universe of traded stocks and miss several other components of wealth such as housing, private equity and non-tradable human capital. Even if we restricted ourselves to all traded stocks, we still have several choices to make, such as whether to use value or equal-weighted indices, and whether to exclude penny or infrequently traded stocks. A similar problem arises with the risk-free rate. While we almost always use Treasury yields as measures of risk-free rates, they are not completely riskless since nominal Treasuries are exposed to inflation⁵ and liquidity risks even if we were to assume there is no prospect of outright default. In this paper, we want to focus on how expectations are estimated in different models, and not on measurement issues regarding market returns and the risk-free rate. Thus, we follow common practice and always use the S&P 500 as a measure of stock market prices and either nominal or real Treasury yields as risk-free rates so that our models are comparable with each other and with most of the literature.

While implementing the concept of the ERP in practice has its challenges, we can precisely define the ERP mathematically. First, we decompose stock returns⁶ into an expected component and a random component:

$$R_{t+k} = E_t[R_{t+k}] + error_{t+k}. \quad (1)$$

In equation (1), R_{t+k} are *realized* returns between t and $t+k$, and $E_t[R_{t+k}]$ are the returns that were expected from t to $t+k$ using information available at time t . The variable $error_{t+k}$ is a random variable that is unknown at time t and realized at $t+k$. Under rational expectations, $error_{t+k}$ has a mean of zero and is orthogonal to $E_t[R_{t+k}]$. We keep the discussion as general as possible and do not assume rational

⁵ Note that inflation risk in an otherwise risk-free nominal asset does not invalidate its usefulness to compute the ERP. If stock returns and the risk-free rate are expressed in nominal terms, their difference has little or no inflation risk. This follows from the following formula, which holds exactly in continuous time and to a first order approximation in discrete time: real stock returns – real risk-free rate = (nominal stock returns – expected inflation) – (nominal risk-free rate – expected inflation) = nominal stock returns – nominal risk-free rate. Hence, there is no distinction between a nominal and a real ERP.

⁶ Throughout this article, all returns are *net* returns. For example, a five percent return corresponds to a net return of 0.05 as opposed to a *gross* return of 1.05.

expectations at this stage, although it will be a feature of many of the models we consider. The ERP at time t for horizon k is defined as

$$ERP_t(k) = E_t[R_{t+k}] - R_{t+k}^f, \quad (2)$$

where R_{t+k}^f is the risk-free rate for investing from t to $t + k$ (which, being risk-free, is known at time t).

This definition shows three important aspects of the ERP. First, future expected returns and the future ERP are stochastic, since expectations depend on the arrival of new information that has a random component not known in advance⁷. Second, the ERP has an investment horizon k embedded in it, since we can consider expected excess returns over, say, one month, one year or five years from today. If we fix t , and let k vary, we trace the *term structure* of the equity risk premium. Third, if expectations are rational, because the unexpected component $error_{t+k}$ is stochastic and orthogonal to expected returns, the ERP is always less volatile than realized excess returns. In this case, we expect ERP estimates to be smoother than realized excess returns.

3. Models of the Equity Risk Premium

We describe twenty models of the equity risk premium, comparing their advantages, disadvantages and ease of implementation. Of course, there are many more models of the ERP than the ones we consider. We selected the models in our study based on the recent academic literature, their widespread use by practitioners and data availability. Table I describes the data we use and their sources, all of which are either readily available or standard in the literature⁸. With a few exceptions, all data is monthly from January 1960 to June 2013. Appendix A provides more details.

[Insert Table I here]

⁷ More precisely, $E_t[R_{t+k}]$ and $ERP_t(k)$ are known at time t but random from the perspective of all earlier periods.

⁸ In fact, except for data from I/B/E/S and Compustat, all sources are public.

We classify the twenty models into five categories based on their underlying assumptions; models in the same category tend to give similar estimates for the ERP. The five categories are: models based on the historical mean of realized returns, dividend discount models, cross-sectional regressions, time-series regressions and surveys.

All but one of the estimates of the ERP are constructed in real time, so that an investor who lived through the sample would have been able to construct the measures at each point in time using available information only⁹. This helps minimize look-ahead bias and makes any out-of-sample evaluation of the models more meaningful. Clearly, most of the models themselves were designed only recently and were not available to investors in real time, potentially introducing another source of forward-looking and selection biases that are much more difficult to quantify and eliminate.

3.1 Historical mean of realized returns

The easiest approach to estimating the ERP is to use the historical mean of realized market returns in excess of the contemporaneous risk-free rate. This model is very simple and, as shown in Goyal and Welch (2008), quite difficult to improve upon when considering out-of-sample predictability performance measures. The main drawbacks are that it is purely backward looking and assumes that the future will behave like the past, i.e. it assumes the mean of excess returns is either constant or very slow moving over time, giving very little time-variation in the ERP. The main choice is how far back into the past we should go when computing the historical mean. Table II shows the two versions of historical mean models that we use.

[Insert Table II here]

⁹ The one exception is Adrian, Crump and Moench's (2014) cross-sectional model, which is constructed using full-sample regression estimates.

3.2 Dividend discount models (DDM)

All DDM start with the basic intuition that the value of a stock is determined by no more and no less than the cash flows it produces for its shareholders, as in Gordon (1962). Today's stock price should then be the sum of all expected future cash flows, discounted at an appropriate rate to take into account their riskiness and the time value of money. The formula that reflects this intuition is

$$P_t = \frac{D_t}{\rho_t} + \frac{E_t[D_{t+1}]}{\rho_{t+1}} + \frac{E_t[D_{t+2}]}{\rho_{t+2}} + \frac{E_t[D_{t+3}]}{\rho_{t+3}} + \dots, \quad (3)$$

where P_t is the current price of the stock, D_t are current cash flows, $E_t[D_{t+k}]$ are the cash flows k periods from now expected as of time t , and ρ_{t+k} is the discount rate for time $t+k$ from the perspective of time t . Cash flows to stockholders certainly include dividends, but can also arise from spin-offs, buy-outs, mergers, buy-backs, etc. In general, the literature focuses on dividend distributions because they are readily available data-wise and account for the vast majority of cash flows. The discount rate can be decomposed into

$$\rho_{t+k} = 1 + R_{t+k}^f + ERP_t(k). \quad (4)$$

In this framework, the risk-free rate captures the discounting associated with the time value of money and the ERP captures the discounting associated with the riskiness of dividends. When using a DDM, we refer to $ERP_t(k)$ as the *implied* ERP. The reason is that we plug in prices, risk-free rates and estimated expected future dividends into equation (3), and then derive what value of $ERP_t(k)$ makes the right-hand side equal to the left-hand side in the equation, i.e. what ERP value is *implied* by equation (3).

DDM are forward looking and are consistent with no arbitrage. In fact, equation (3) must hold in any economy with no arbitrage¹⁰. Another advantage of DDM is that they are easy to implement. A drawback of DDM is that the results are sensitive to how we compute expectations of future dividends. Table III displays the DDM we consider and a brief description of their different assumptions.

[Insert Table III here]

3.3 Cross-sectional regressions

This method exploits the variation in returns and exposures to the S&P 500 of different assets to infer the ERP¹¹. Intuitively, cross-sectional regressions find the ERP by answering the following question: what is the level of the ERP that makes expected returns on a variety of stocks consistent with their exposure to the S&P 500? Because we need to explain the relationship between returns and exposures for multiple stocks with a single value for the ERP (and perhaps a small number of other variables), this model imposes tight restrictions on estimates of the ERP.

The first step is to find the exposures of assets to the S&P 500 by estimating an equation of the following form:

$$R_{t+k}^i - R_{t+k}^f = \alpha^i \times \text{state variables}_{t+k} + \beta^i \times \text{risk factors}_{t+k} + \text{idiosyncratic risk}_{t+k}^i. \quad (5)$$

In equation (5), R_{t+k}^i is the realized return on a stock or portfolio i from time t to $t + k$.

State variables _{$t+k$} are any economic indicators that help identify the state of the economy and its likely future path. *Risk factors* _{$t+k$} are any measures of systematic contemporaneous co-variation in returns across all stocks or portfolios. Of course, some economic indicators can be both state variables and risk

¹⁰ Note that when performing the infinite summation in equation (3) we have not assumed the n^{th} term goes to zero as n tends to infinity, which allows for rational bubbles. In this sense, DDM do allow for a specific kind of bubble.

¹¹ See Polk, Thompson and Vuolteenaho (2006) and Adrian, Crump and Moench (2014) for a detailed description of this method.

factors at the same time. Finally, *idiosyncratic risk* $_{t+k}^i$ is the component of returns that is particular to each individual stock or portfolio that is not explained by *state variables* $_{t+k}$ or *risk factors* $_{t+k}$ (both of which, importantly, are common to all stocks and hence not indexed by i). Examples of state variables are inflation, unemployment, the yield spread between Aaa and Baa bonds, the yield spread between short and long term Treasuries, and the S&P 500's dividend-to-price ratio. The most important risk factor is the excess return on the S&P 500, which we must include if we want to infer the ERP consistent with the cross-section of stock returns. Other risk-factors usually used are the Fama-French (1992) factors and the momentum factor of Carhart (1997). The values in the vector α^i give the strength of asset-specific return predictability and the values in the vector β^i give the asset-specific exposures to risk factors¹². For the cross-section of assets indexed by i , we can use the whole universe of traded stocks, a subset of them, or portfolios of stocks grouped, for example, by industry, size, book-to-market, or recent performance. It is important to point out that equation (5) is not a predictive regression; the left and right-hand side variables are both associated with time $t + k$.

The second step is to find the ERP associated with the S&P 500 by estimating the cross-sectional equations

$$R_{t+k}^i - R_{t+k}^f = \lambda_t(k) \times \hat{\beta}^i, \quad (6)$$

where $\hat{\beta}^i$ are the values found when estimating equation (5). Equation (6) attempts to find, at each point in time, the vector of numbers $\lambda_t(k)$ that makes exposures β^i as consistent as possible with realized excess returns of all stocks or portfolios considered. The element in the vector $\hat{\lambda}_t(k)$ that is multiplied by

¹² The vectors α^i and β^i could also be time-varying, reflecting a more dynamic relation between returns and their explanatory variables. In this case, the estimation of equation (5) is more complicated and requires making further assumptions. The model by Adrian, Crump and Moench (2014) is the only cross-sectional model we examine that uses time-varying α^i and β^i .

the element in the $\hat{\beta}^i$ vector corresponding to the S&P 500 is $ERP_t(k)$, the equity risk premium we are seeking.

One advantage of cross-sectional regressions is that they use information from more asset prices than other models. Cross-sectional regressions also have sound theoretical foundations, since they provide one way to implement Merton's (1973) Intertemporal Capital Asset Pricing Model. Finally, this method nests many of the other models considered. The two main drawbacks of this method are that results are dependent on what portfolios, state variables and risk factors are used (Harvey, Liu and Zhu (2014)), and that it is not as easy to implement as most of the other options. Table IV displays the cross-sectional models in our study, together with the state variables and risk factors they use.

[Insert Table IV here]

3.4 Time-series regressions

Time-series regressions use the relationship between economic variables and stock returns to estimate the ERP. The idea is to run a predictive linear regression of realized excess returns on lagged “fundamentals”:

$$R_{t+k} - R_{t+k}^f = a + b \times Fundamental_t + error_t. \quad (7)$$

Once estimates $\hat{a}$ and $\hat{b}$ for a and b are obtained, the ERP is obtained by ignoring the error term:

$$ERP_t(k) = \hat{a} + \hat{b} \times Fundamental_t. \quad (8)$$

In other words, we estimate only the forecastable or expected component of excess returns. This method attempts to implement equations (1) and (2) as directly as possible in equations (7) and (8), with the assumption that “fundamentals” are the right sources of information to look at when computing expected returns, and that a linear equation is the correct functional specification.

The use of time-series regressions requires minimal assumptions; there is no concept of equilibrium and no absence of arbitrage necessary for the method to be valid¹³. In addition, implementation is quite simple, since it only involves running ordinary least-square regressions. The challenge is to select what variables to include on the right-hand side of equation (7), since results can change substantially depending on what variables are used to take the role of “fundamentals”. In addition, including more than one predictor gives poor out-of-sample predictions even if economic theory may suggest a role for many variables to be used simultaneously (Goyal and Welch (2008)). Finally, time-series regressions ignore information in the cross-section of stock returns. Table V shows the time-series regression models that we study.

[Insert Table V here]

3.5 Surveys

The survey approach consists of asking economic agents about the current level of the ERP. Surveys incorporate the views of many people, some of which are very sophisticated and/or make real investment decisions based on the level of the ERP. Surveys should also be good predictors of excess returns because in principle stock prices are determined by supply and demand of investors such as the ones taking the surveys. On the other hand, Greenwood and Shleifer (2014) document that investor expectations of future stock market returns are positively correlated with past stock returns and with the current level of the stock market, but strongly *negatively* correlated with model-based expected returns and future realized stock market returns. Other studies such as Easton and Sommers (2007) also argue that survey measures of the ERP can be systematically biased. In this paper, we use the survey of CFOs by Graham and Harvey (2012), which to our knowledge is the only large-scale ERP survey that has more than just a few years of data (see Table VI).

[Insert Table VI here]

¹³ However, the Arbitrage Pricing Theory of Ross (1976) provides a strong theoretical underpinning for time-series regressions by using no-arbitrage conditions.

4. Estimation of the Equity Risk Premium

We now study the behavior of the twenty models we consider by conducting principal component analysis. Since forecast accuracy can be substantially improved through the combination of multiple forecasts¹⁴, the optimal strategy to forecast excess stock returns may consist of combining together all these models. The first principal component of the twenty models that we use is the linear combination of ERP estimates that captures as much of the variation in the data as possible. The second, third, and successive principal components are the linear combinations of the twenty models that explain as much of the variation of the data as possible and are also uncorrelated to all the preceding principal components. If the first few principal components —say one or two— account for most of the variation of the data, then we can use them as a good summary for the variation in all the measures over time, reducing the dimensionality from twenty to one or two. In addition, in the presence of classical measurement error, the first few principal components can achieve a higher signal-to-noise ratio than other summary measures like the cross-sectional mean of all models (Geiger and Kubin (2013)).

To compute the first principal component, we proceed in three steps. We first de-mean all ERP estimates and find their variance-covariance matrix. In the second step, we find the linear combination that explains as much of the variance of the de-meaned models as possible. The weights in the linear combination are the elements of the eigenvector associated with the largest eigenvalue of the variance-covariance matrix found in the first step. In the third step, we add to the linear combination just obtained, which has mean zero, the average of ERP estimates across all models and all time periods. Under the assumption that each of the models is an unbiased and consistent estimator of the ERP, the average across all models and all time periods is an unbiased and consistent estimator of the unconditional mean of the ERP. The time

¹⁴ See, *inter alia*, Clemen (1989), Diebold and Lopez (1996) and Timmermann (2006).

variation in the first principal component then provides an estimate of the conditional ERP¹⁵. The share of the variance of the underlying models explained by this principal component is 76 percent, suggesting that there is not too much to gain from examining principal components beyond the first¹⁶.

We now focus on the one-year-ahead ERP estimates and study other horizons in the next section.

The first two columns in Table VII show the mean and standard deviation of each model's estimates. The unconditional mean of the ERP across all models is 5.7 percent, with an average standard deviation of 3.2 percent. DDM give the lowest mean ERP estimates and have moderate standard deviations. In contrast, cross-sectional models tend to have mean ERP estimates on the high end of the distribution and very smooth time-series. Mean ERP estimates for time-series regressions are mixed, with high and low values depending on the predictors used, but uniformly large variances. The survey of CFOs has a mean and standard deviation that are both about half as large as in the overall population of models. The picture that emerges from Table VII is that there is considerable heterogeneity across model types, and even sometimes within model types, thereby underscoring the difficulty inherent in finding precise estimates of the ERP.

¹⁵ As is customary in the literature, we perform the analysis using ERP estimates in levels, even though they are quite persistent. Results in first-differences do not give economically reasonable estimates since they feature a pro-cyclical ERP and unreasonable magnitudes.

One challenge that arises in computing the principal component is when we have missing observations, either because some models can only be obtained at frequencies lower than monthly or because the necessary data is not available for all time periods (Appendix A contains a detailed description of when this happens). To overcome this challenge, we use an iterative linear projection method, which conceptually preserves the idea behind principal components. Let X be the matrix that has observations for different models in its columns and for different time periods in its rows. On the first iteration, we make a guess for the principal component and regress the non-missing elements of each row of X on the guess and a constant. We then find the first principal component of the variance-covariance matrix of the fitted values of these regressions, and use it as the guess for the next iteration. The process ends when the norm of the difference between consecutive estimates is small enough. We thank Richard Crump for suggesting this method and providing the code for its implementation.

¹⁶ The second and third principal components account for 13 and 8 percent of the variance, respectively.

[Insert Table VII here]

Figure 1 shows the time-series for all one-year-ahead ERP model estimates, with each class of models in a different panel. The green lines are the ERP estimates from the twenty underlying models. The black line, reproduced in each of the panels, is the principal component of all twenty models. The shaded areas are NBER recessions. The figure gives a sense of how the time-series move together, and how much they co-vary with the first principal component. Table VIII shows the correlations among models. Figure 1 and Table VIII give the same message: despite some outliers, there is a fairly strong correlation within each of the five classes of models. Across classes, however, correlations are small and even negative.

Interestingly, the correlation between some DDM and cross-sectional models is as low as -91 percent. This negative correlation, however, disappears if we look at lower frequencies. When aggregated to quarterly frequency, the smallest correlation between DDM and cross-sectional models is -22 percent, while at the annual frequency it is 12 percent.

[Insert Figure 1 here]

[Insert Table VIII here]

Figure 1 also shows that the first principal component co-varies negatively with historical mean models, but positively with DDM and cross-sectional regression models. Time-series regression models are also positively correlated with the first principal component, although this is not so clearly seen in Panel 4 of Figure 1 because of the high volatility of time-series ERP estimates. The last panel shows that the survey of CFOs does track the first principal component quite well at low frequencies (e.g. annual), although any conclusions about survey estimates should be interpreted with caution given the short length of the sample.

As explained earlier, the first principal component is a linear combination of the twenty underlying ERP models:

$$PC_t^{(1)} = \sum_{m=1}^{20} w^{(m)} ERP_t^{(m)}. \quad (9)$$

In the above equation, m indexes the different models, $PC_t^{(1)}$ is the first principal component, $ERP_t^{(m)}$ is the estimate from model m and $w^{(m)}$ is the weight that the principal component places on model m . The third column in Table VII, labeled “PC coefficients”, shows the weights $w^{(m)}$ normalized to sum up to one to facilitate comparison, i.e. the table reports the weights $\hat{w}^{(m)}$ where

$$\hat{w}^{(m)} = \frac{w^{(m)}}{\sum_{m=1}^{20} w^{(m)}}. \quad (10)$$

The first principal component puts positive weight on models based on the historical mean, cross-sectional regressions and the survey of CFOs. It weights DDM and time-series regressions mostly negatively. The absolute values of the weights are very similar for many of the models, and there is no single model or class of models that dominates. This means that the first principal component uses information from many of the models.

The last column in Table VII, labeled “Exposure to PC”, shows the extent to which models *load* on the first principal component. By construction, each of the twenty ERP models can be written as a linear combination of twenty principal components:

$$ERP_t^{(m)} = \sum_{i=1}^{20} load_i^{(m)} PC_t^{(i)}, \quad (11)$$

where m indexes the model and i indexes the principal components. The values in the last column of Table VII are the loadings on the first principal component ($i = 1$) for each model ($m = 1, 2, \dots, 20$), again normalized to one for ease of comparability:

$$\widehat{load}_1^{(m)} = \frac{load_1^{(m)}}{\sum_{m=1}^{20} load_i^{(m)}}. \quad (12)$$

Most models have a positive loading on the first principal component; whenever the loading is negative, it tends to be relatively small. This means the first principal component, as expected, is a good explanatory variable for most models. Looking at the third and fourth columns of Table VII together, we can obtain additional information. For example, a model with a very high loading (fourth column) accompanied by a very small PC coefficient (third column) is likely to mean that the model is almost redundant, in the sense that it is close to being a linear combination of all other models and does not provide much independent information to the principal component. On the other hand, if the PC coefficient and loading are both high, the corresponding model is likely providing information not contained in other measures.

Figure 2 shows the first principal component of all twenty models in black, with recessions indicated by shaded bars (the black line is the same principal component shown in black in each of the panels of Figure 1). As expected, the principal component tends to peak during financial turmoil, recessions and periods of low real GDP growth or high inflation. It tends to bottom out after periods of sustained bullish stock markets and high real GDP growth. Evaluated by the first principal component, the one-year-ahead ERP reaches a local peak in June of 2012 at 12.2 percent. The surrounding months have ERP estimates of similar magnitude, with the most recent estimate in June 2013 at 11.2 percent. This behavior is not so clearly seen by simply looking at the collection of individual models in Figure 1, highlighting the usefulness of principal components analysis. Similarly high levels were seen in the mid and late 1970s, during a period of stagflation, while the recent financial crisis had slightly lower ERP estimates closer to 10 percent.

[Insert Figure 2 here]

Figure 2 also displays the 10th, 25th, 75th and 90th percentiles of the cross-sectional distribution of models. These bands can be interpreted as confidence intervals, since they give the range of the distribution of ERP estimates at each point in time. However, they do not incorporate other relevant sources of uncertainty, such as the errors that occur during the estimation of each individual model, the degree of doubt in the correctness of each model, and the correlation structure between these and all other kinds of errors. Standard error bands that capture all sources of uncertainty are therefore likely to be wider.

The difference in high and low percentiles can also be interpreted as measures of agreement across models. The interquartile range—the difference between the 25th and 75th percentiles—has compressed, mostly because the models in the bottom of the distribution have had higher ERP estimates since 2010. It is also interesting to note that the 75th percentile has remained fairly constant over the last 10 years at a level somewhat below its long-run mean. The cross-sectional standard deviation in ERP estimates (not shown in the graph) also decreased from 10.2% in January of 2000 to 4.3% in June of 2013, confirming that the disagreement among models has decreased.

Another *a priori* reasonable summary statistic for the ERP is the cross-sectional mean of estimates across models. In Figure 3, we can see that by this measure the ERP has also been increasing since the crisis. However, unlike the principal component, it has not reached elevated levels compared to past values. The cross-sectional mean can be useful, but it has a few undesirable features as an overall measure of the ERP compared to the first principal component. First, it is procyclical, which contradicts the economic intuition that expected returns are highest in recessions, when risk aversion is high and future prospects look brighter than current ones. Second, it overloads on DDM simply because there is a higher number of DDM models in our sample. Lastly, it has a smaller correlation with the realized returns it is supposed to predict.

[Insert Figure 3 here]

5. The Term Structure of Equity Risk Premia

In Section 2, we described the term structure of the ERP – what expected excess returns are over different investment horizons. In practical terms, we estimate the ERP at different horizons by using the inputs for all the models at the corresponding horizons¹⁷. For example, if we want to take the historical mean of returns as our estimate, we can take the mean of returns over one month, six months, or a one-year period. In cross-sectional and time-series regressions, we can predict monthly, quarterly or annual returns using monthly, quarterly or annual right-hand side variables. DDM, on the other hand, have little variation across horizons. In fact, all the DDM we consider have a constant term structure of expected stock returns, and the only term structure variation in ERP estimates comes from risk-free rates¹⁸.

Figure 4 plots the first principal components of the ERP as a function of investment horizon for some selected dates. We picked the dates because they are typical dates for when the ERP was unusually high or unusually low at the one-month horizon. As was the case for one-year-ahead ERP estimates, we can capture the majority of the variance of the underlying models at all horizons by a single principal component. The shares of the variance explained by the first principal components at horizons of one month to three years range between 68 and 94 percent. The grey line in Figure 4 shows the average of the term structure across all periods. It is slightly upward sloping, with a short-term ERP at just over 6 percent and a three-year ERP at almost 7 percent.

[Insert Figure 4 here]

¹⁷ For other ways to estimate the term structure of the ERP using equilibrium models or derivatives, see Ait-Sahalia, Karaman and Mancini (2014), Ang and Ulrich (2012), van Binsbergen, Hueskes, Koijen and Vrugt (2014), Boguth, Carlson, Fisher and Simutin (2012), Durham (2013), Croce, Lettau and Ludvigson (2014), Lemke and Werner (2009), Lettau and Wachter (2011), Muir (2013), among others.

¹⁸ In equation (3), ρ_{t+k} is assumed to be the same for all k , while risk-free rates are allowed to vary over the investment horizon k in equation (4). Of course, with additional assumptions, it is possible to have DDM with a non-constant term structure of expected excess returns.

The first observation is that the term structure of the ERP has significant time variation and can be flat, upward or downward sloping. Figure 4 also shows some examples that hint at lower future expected excess returns when the one-month-ahead ERP is elevated and the term structure is downward sloping, and higher future expected excess returns when the one-month-ahead ERP is low and the term structure is upward sloping. In fact, this is generally true: There is a strong negative correlation between the level and the slope of the ERP term structure of -71 percent. Figure 5 plots monthly observations of the one-month-ahead ERP against the slope of the ERP term structure (the three-year-ahead minus the one-month-ahead ERP) together with the corresponding ordinary least squares regression line in black. Of course, this is only a statistical pattern and should not be interpreted as a causal relation.

[Insert Figure 5 here]

6. Why is the Equity Risk Premium High?

There are two reasons why the ERP can be high: low discount rates and high current or expected future cash flows.

Figure 6 shows that earnings are unlikely to be the reason why the ERP is high. The green line shows the year-on-year change in the mean expectation of one-year-ahead earnings per share for the S&P 500.

These expectations are obtained from surveys conducted by the Institutional Brokers' Estimate System (I/B/E/S) and available from Thomson Reuters. Expected earnings per share have been declining from 2010 to 2013, making earnings growth an unlikely reason for why the ERP was high in the corresponding period. The black line shows the realized monthly growth rates of real earnings for the S&P 500 expressed in annualized percentage points. Since 2010, earnings growth has been declining, hovering around zero for the last few months of the sample. It currently stands at 2.5 percent, which is near its long-run average.

[Insert Figure 6 here]

Another way to examine whether a high ERP is due to discount rates or cash flows is shown in Figure 7. The black line is the same one-year-ahead ERP estimate shown in Figure 2. The green line simply adds the realized one-year Treasury yield to obtain expected stock returns. The figure shows expected stock returns have increased since 2000, similarly to the ERP. However, unlike the ERP, expected stock returns are close to their long-run mean, and nowhere near their highest levels, achieved in 1980. The discrepancies between the two lines are due to exceptionally low bond yields since the end of the financial crisis.

[Insert Figure 7 here]

Figure 8 displays the term structure of the ERP under a simple counterfactual scenario, in addition to the mean and current term structures already displayed in Figure 4. In this scenario, we leave expected stock returns unmodified but change the risk-free rates in June 2012 from their actual values to the average nominal bond yields over 1960-2013. In other words, we replace R_{t+k}^f in equation (2) by the mean of R_{t+k}^f over t . The result of this counterfactual is shown in Figure 8 in green. Using average levels of bond yields brings the whole term structure of the ERP much closer to its mean level (the grey line), especially at intermediate horizons. This shows that a “normalization” of bond yields, everything else being equal, would bring the ERP close to its historical norm. This exercise shows that the current environment of low bond yields is capable, quantitatively speaking, of significantly contributing to an ERP as high as was observed in 2012-2013.

[Insert Figure 8 here]

7. Conclusion

We have analyzed twenty different models of the ERP by considering the assumptions and data required to implement them, and how they relate to each other. When it comes to the ERP, we find that there is substantial heterogeneity in estimation methodology and final estimates. We then extract the first

principal component of the twenty models, which signals that the ERP in 2012 and 2013 is at heightened levels compared to previous periods. Our analysis provides evidence that the current level of the ERP is consistent with a bond-driven ERP: expected excess stock returns are elevated not because stocks are expected to have high returns, but because bond yields are exceptionally low. The models we consider suggest that expected stock returns, on their own, are close to average levels.

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Appendix A: Data Variables

Fama and French (1992)	http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html Monthly frequency; 1/1/1960 to 6/30/2013. We use 25 portfolios sorted on size and book to market, 10 portfolios sorted on momentum, realized excess market returns, HML, SMB, and the momentum factor.
Shiller (2005)	http://www.econ.yale.edu/~shiller/data.htm Monthly frequency; 1/1/1960 to 6/30/2013. We use the nominal and real price, nominal and real dividends and nominal and real earnings for the S&P 500, CPI, and 10 year nominal treasury yield.
Baker and Wurgler (2007)	http://people.stern.nyu.edu/jwurgler/data/Investor_Sentiment_Data_v23_POST.xlsx Monthly frequency; 7/1/1965 to 12/1/2010. We use the “sentiment measure”.
Graham and Harvey (2012)	http://www.cfosurvey.org/index.htm Quarterly frequency; 6/6/2000 to 6/5/2013. We use the answer to the question “Over the next 10 years, I expect the average annual S&P 500 return will be: Expected return:” and the analogous one that asks about the next year.
Damodaran (2012)	http://www.stern.nyu.edu/~adamodar/pc/datasets/histimpl.xls Annual frequency; 1/1/1960 to 12/1/2012. We use the ERP estimates from his dividend discount models (one uses free-cash flow, the other one doesn’t).
Gurkaynak, Sack and Wright (2007)	http://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html Daily frequency; starting on 6/14/61 for one- to seven-year yields; 8/16/71 for nine- and ten-year yields; 11/15/71 for eleven- to fifteen-year yields; 7/2/81 for sixteen- to twenty-year yields; 11/25/85 for twenty-one- to thirty-year yields. We use all series until 6/30/2013.
Gurkaynak, Refet, Sack and Wright (2010)	http://www.federalreserve.gov/econresdata/researchdata.htm Monthly frequency; 1/1/2003 to 7/1/2013. We use yields on TIPS of all maturities available.
Compustat	Variable BKVLPS Annual frequency; 12/31/1977 to 12/31/2012.
Thomson Reuters I/B/E/S	Variables EPS 1 2 3 4 5 Monthly frequency; 1/14/1982 to 4/18/2013 for current and next year forecasts; 9/20/84 to 4/18/2013 for two-year-ahead forecasts; 9/19/85 to 3/15/2012 for three-year-ahead forecasts; 2/18/88 to 3/15/07 for four-year-ahead forecasts.
FRED (St. Louis Federal Reserve)	http://research.stlouisfed.org/fred2/graph/?g=D9J and http://research.stlouisfed.org/fred2/graph/?g=KKk Monthly frequency. 1/1/1960 to 7/1/2013 for Baa minus Aaa bond yield spread and recession indicator.

Tables and Figures

Table I: Data sources	
Fama and French (1992)	Fama-French factors, momentum factor, twenty-five portfolios sorted on size and book-to-market
Shiller (2005)	Inflation and ten-year nominal treasury yield. Nominal price, real price, earnings, dividends and cyclically adjusted price-earnings ratio for the S&P 500
Baker and Wurgler (2007)	Debt issuance, equity issuance, sentiment measure
Graham and Harvey (2012)	ERP estimates from the Duke CFO survey
Damodaran (2012)	ERP estimates
Gurkaynak, Sack and Wright (2007)	Zero coupon nominal bond yields for all maturities ¹⁹
Gurkaynak, Refet, Sack and Wright (2010)	Zero coupon TIPS yields for all maturities
Compustat	Book value per share for the S&P 500
Thomson Reuters I/B/E/S	Mean analyst forecast of expected earnings per share
FRED (St. Louis Federal Reserve)	Corporate bond Baa-Aaa spread and the NBER recession indicator

Note: All variables start in January 1960 (or later, if unavailable for early periods) and end in June 2013 (or until no longer available). CFO surveys are quarterly; book value per share and ERP estimates by Damodaran (2012) are annual; all other variables are monthly. Appendix A provides more details.

¹⁹ Except for the 10-year yield, which is from Shiller (2005). We use the 10-year yield from Shiller (2005) for ease of comparability with the existing literature. Results are virtually unchanged if we use all yields, including the 10-year yield, from Gurkaynak, Sack and Wright (2007).

Table II: Models based on the historical mean of realized returns

Long-run mean	Average of realized S&P 500 returns minus the risk-free rate using all available historical data
Mean of the previous five years	Average of realized S&P 500 returns minus the risk-free rate using only data for the previous five years

Table III: Dividend Discount Models

Gordon (1962) with nominal yields	S&P 500 dividend-to-price ratio minus the ten-year nominal Treasury yield
Shiller (2005)	Cyclically adjusted price-earnings ratio (CAPE) minus the ten-year nominal Treasury yield
Gordon (1962) with real yields	S&P 500 dividend-to-price ratio minus the ten year real Treasury yield (computed as the ten-year nominal Treasury rate minus the ten year breakeven inflation implied by TIPS)
Gordon (1962) with earnings forecasts	S&P 500 expected earnings-to-price ratio minus the ten-year nominal Treasury yield
Gordon (1962) with real yields and earnings forecasts	S&P 500 expected earnings-to-price ratio minus the ten-year real Treasury yield (computed as the ten-year nominal Treasury rate minus the ten-year breakeven inflation implied by TIPS)
Panigirtzoglou and Loeys (2005)	Two-stage DDM. The growth rate of earnings over the first five years is estimated by using the fitted values in a regression of average realized earnings growth over the last five years on its lag and lagged earnings-price ratio. The growth rate of earnings from years six and onwards is 2.2 percent
Damodaran (2012)	A six-stage DDM. Dividend growth the first five stages are estimated from analyst's earnings forecasts. Dividend growth in the sixth stage is the ten-year nominal Treasury yield
Damodaran (2012) free cash flow	Same as Damodaran (2012), but uses free-cash-flow-to-equity as a proxy for dividends plus stock buybacks

Table IV: Models with cross-sectional regressions

Fama and French (1992)	Uses the excess returns on the market portfolio, a size portfolio and a book-to-market portfolio as risk factors
Carhart (1997)	Identical to Fama and French (1992) but adds the momentum measure of Carhart (1997) as an additional risk factor
Duarte (2013)	Identical to Carhart (1997) but adds an inflation risk factor
Adrian, Crump and Moench (2014)	Uses the excess returns on the market portfolio as the single risk factor. The state variables are the dividend yield, the default spread, and the risk free rate

Table V: Models with time-series regressions	
Fama and French (1988)	Only predictor is the dividend-price ratio of the S&P 500
Goyal and Welch (2008)	Uses, at each point in time, the best out-of-sample predictor out of twelve predictive variables proposed by Goyal and Welch (2008)
Campbell and Thompson (2008)	Same as Goyal and Welch (2008), but imposes two restrictions on the estimation. First, the coefficient b in equation (9) is replaced by zero if it has the “wrong” theoretical sign. Second, we replace the estimate of the ERP by zero if the estimation otherwise finds a negative ERP
Fama and French (2002)	Uses, at each point in time, the best out-of-sample predictor out of three variables: the price-dividend ratio adjusted by the growth rate of earnings, dividends or stock prices
Baker and Wurgler (2007)	The predictor is Baker and Wurgler’s (2007) sentiment measure. The measure is constructed by finding the most predictive linear combination of five variables: the closed-end fund discount, NYSE share turnover, the number and average first-day returns on IPOs, the equity share in new issues, and the dividend premium

Table VI: Surveys	
Graham and Harvey (2012)	Chief financial officers (CFOs) are asked since 1996 about the one and ten-year-ahead ERP. We take the mean of all responses

Table VII: ERP models

		Mean	Std. dev.	PC coefficients $\hat{w}^{(m)}$	Exposure to PC $\widehat{load}_1^{(m)}$
Based on historical mean	Long-run mean	9.3	1.3	0.78	-0.065
	Mean of previous five years	5.7	5.8	0.42	-0.160
DDM	Gordon (1926): E/P minus nominal 10yr yield	-0.1	2.1	-0.01	0.001
	Shiller (2005): 1/CAPE minus nominal 10yr yield	-0.4	1.8	-0.10	0.011
	Gordon (1962): E/P minus real 10yr yield	3.5	2.1	0.69	-0.077
	Gordon (1962): Expected E/P minus real 10yr yield	5.3	1.7	-0.78	0.208
	Gordon (1962): Expected E/P minus nominal 10yr yield	0.4	2.3	-0.79	0.077
	Panigirtzoglou and Loeys (2005): Two-stage DDM	-1.0	2.3	0.07	-0.011
	Damodaran (2012): Six-stage DDM	3.4	1.3	-0.26	0.032
	Damodaran (2012): Six-stage free cash flow DDM	4.0	1.1	-0.62	0.053
Cross-sectional regressions	Fama and French (1992)	12.6	0.7	0.80	-0.040
	Carhart (1997): Fama-French and momentum	13.1	0.8	0.81	-0.042
	Duarte (2013): Fama-French, momentum and inflation	13.1	0.8	0.82	-0.044
	Adrian, Crump and Moench (2014)	6.5	6.9	-0.05	0.114
Time-series regressions	Fama and French (1988): D/P	2.4	4.0	-0.27	0.069
	Best predictor in Goyal and Welch (2008)	14.5	5.2	-0.07	0.023
	Best predictor in Campbell and Thompson (2008)	3.1	9.8	-0.12	0.081
	Best predictor in Fama French (2002)	11.9	6.8	-0.72	0.321
	Baker and Wurgler (2007) sentiment measure	3.0	4.7	-0.32	0.184
Surveys	Graham and Harvey (2012) survey of CFOs	3.6	1.8	0.72	0.264
	All models	5.7	3.2	0.78	-0.065

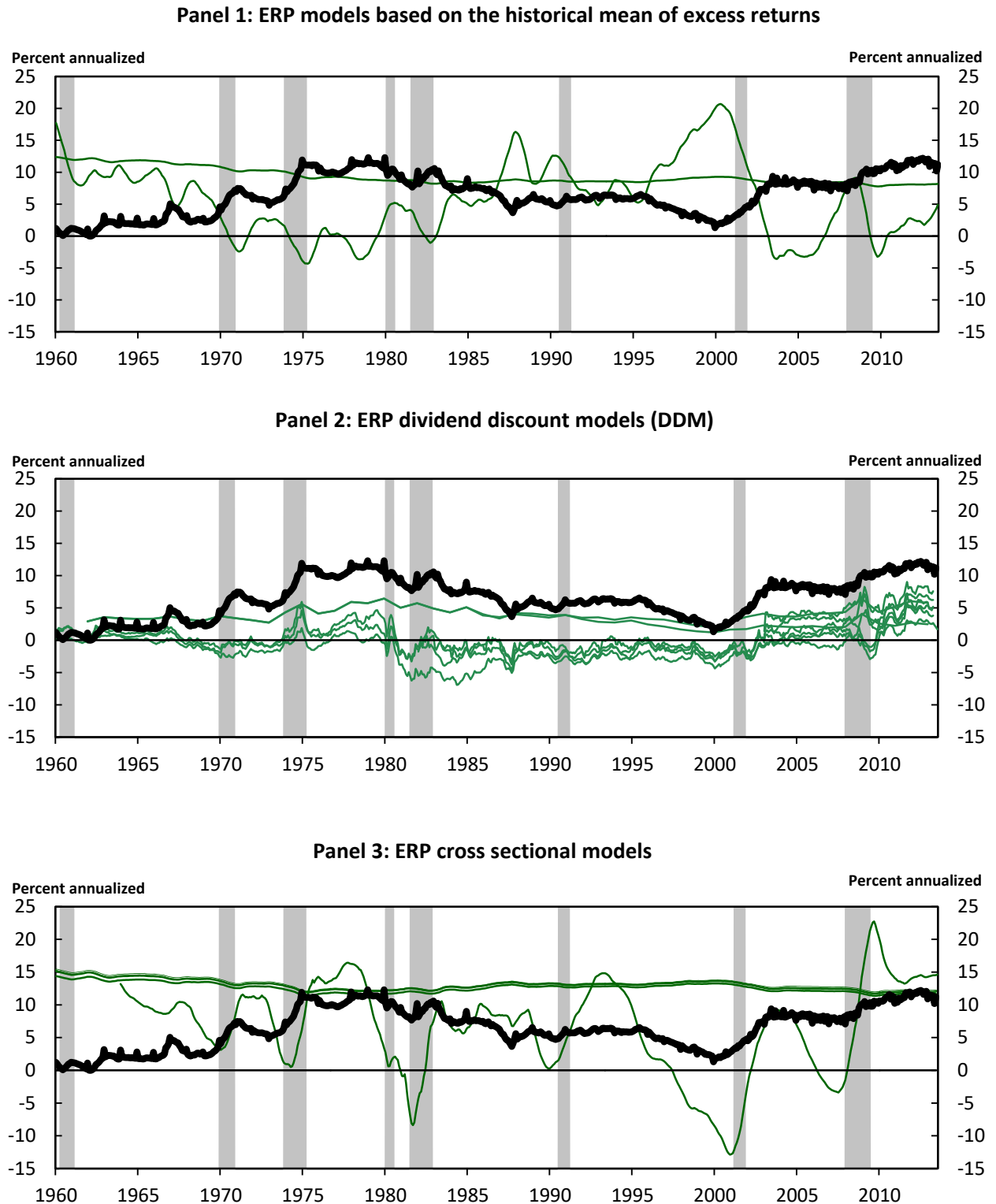
For each of the twenty models of the equity risk premium, we show four statistics. The first two are the time-series means and standard deviations for monthly observations from January 1960 to June 2013 (except for surveys, which are quarterly). The units are annualized percentage points. The third statistic, “PC coefficients $\hat{w}^{(m)}$ ”, is the weight that the first principal component places on each model (normalized to sum to one). The fourth is the “Exposure to PC $\widehat{load}_1^{(m)}$ ”, the weight on the first principal component when each model is written as a weighted sum of all principal components (also normalized to sum to one).

Table VIII: Correlation of ERP models

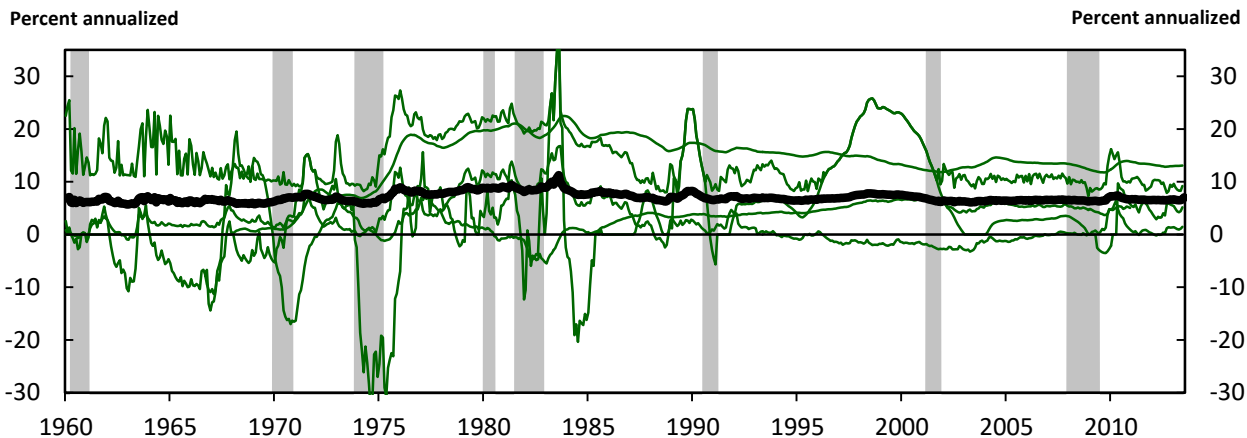
	LR mean	Mean past 5yr	E/P - 10yr	1/CAPE-10yr	E/P-real 10yr	Exp E/P-real 10yr	Exp E/P- 10yr	Two-stage DDM	Six-stage DDM	Free cash flow	FF	Carhart	Duarte	ACM	D/P	G and W	C and T	FF	Sentiment	CFO Survey
LR mean	100																			
Mean past 5yr	32	100																		
E/P - 10yr	8	15	100																	
1/CAPE-10yr	-9	0	78	100																
E/P-real 10yr	-11	25	98	23	100															
Exp E/P-real 10yr	-58	42	70	84	60	100														
Exp E/P- 10yr	-83	-61	84	95	46	98	100													
Two-stage DDM	17	27	88	54	89	66	79	100												
Six-stage DDM	3	-38	26	39	-30	32	52	-31	100											
Free cash flow	-43	-55	59	70	35	80	94	27	62	100										
FF	69	29	-8	-36	-21	-69	-91	9	-29	-77	100									
Carhart	71	30	-5	-31	-24	-71	-91	10	-25	-75	99	100								
Duarte	71	30	-3	-29	-22	-70	-91	11	-28	-74	99	100	100							
ACM	-1	-52	36	62	6	54	63	27	23	33	-28	-28	-25	100						
D/P	49	12	27	12	27	42	54	24	74	42	44	54	55	21	100					
G and W	25	12	25	21	-7	-36	-60	20	29	-9	7	13	14	-24	61	100				
C and T	27	31	14	-7	81	49	-60	28	-51	-40	60	57	58	-33	54	50	100			
FF	1	-30	-24	-29	37	-27	-37	-18	22	38	36	38	37	-9	40	23	43	100		
Sentiment	-10	33	-4	-20	68	-23	-29	27	-38	-20	18	17	18	-12	-38	-8	21	6	100	
CFO survey	-43	-33	12	30	1	1	13	16	5	-3	-36	-37	-39	60	14	-21	-32	-3	-36	100

This table shows the correlation matrix of the twenty equity risk premium models we consider. Numbers are rounded to the nearest integer. Thick lines group models by their type (see Tables II to VI). Except for the CFO survey, the observations used to compute correlations are monthly for January 1960 to June 2013. For the CFO survey, correlations are computed by taking the last observation in the quarter for monthly series and then computing quarterly correlations.

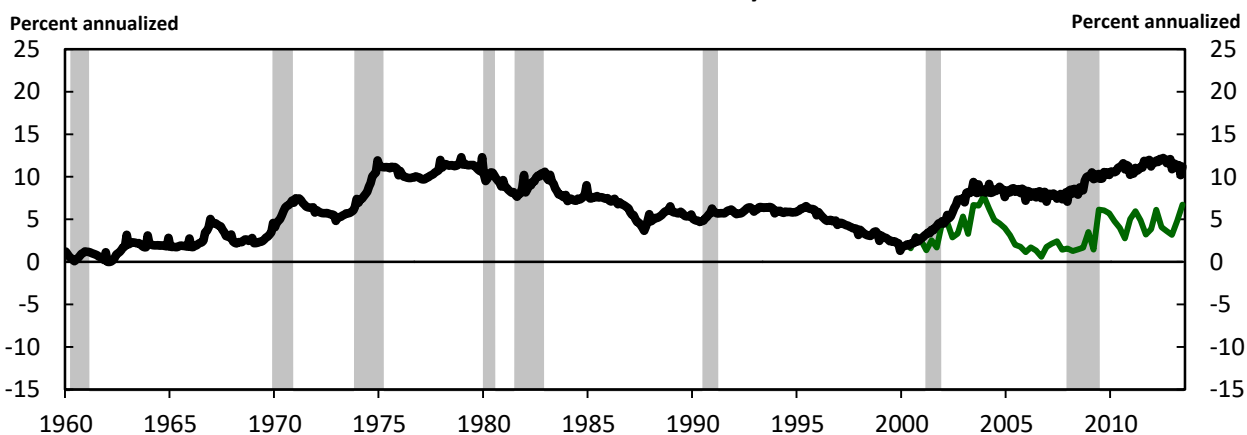
Figure 1: ERP estimates for all models



Panel 4: ERP time series models



Panel 5: ERP surveys

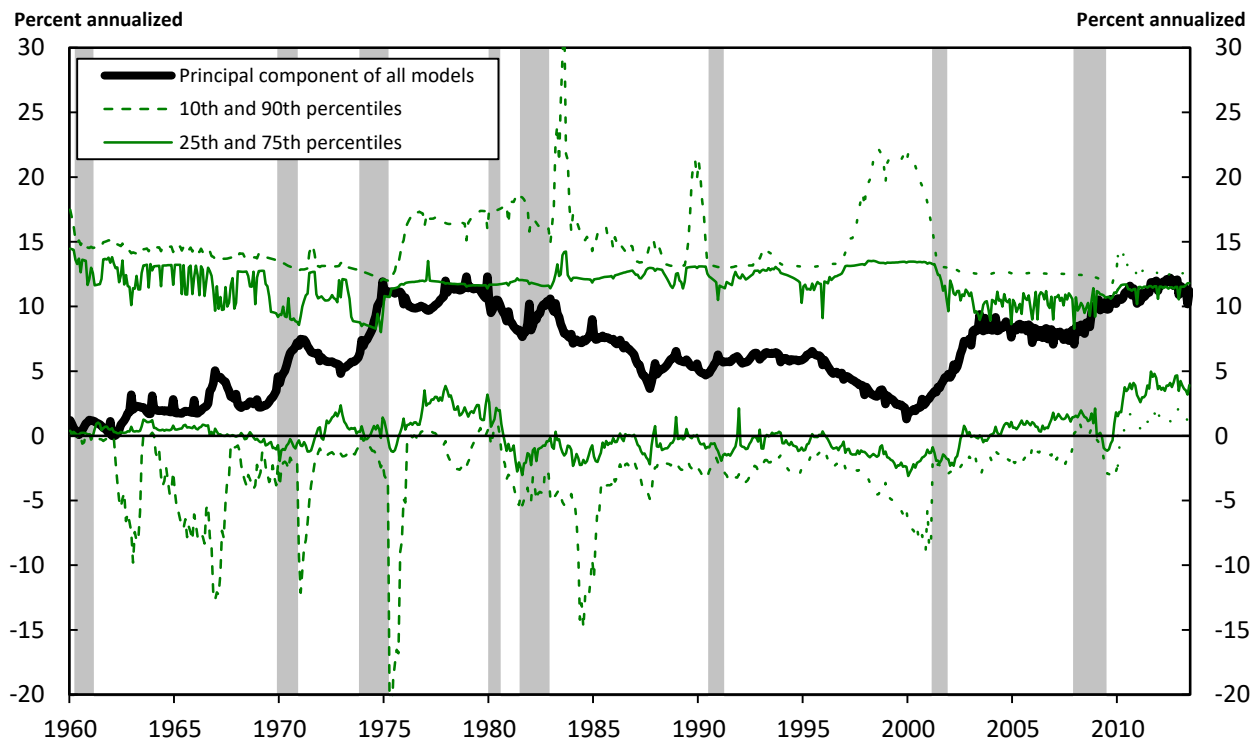


Each green line gives the one-year-ahead equity risk premium from each of the models listed in Tables II to VI. All numbers are in annualized percentage points.

Panel 1 shows the estimates for models based on the historical mean of excess returns, which are listed in Table II. Panel 2 shows estimates computed by the dividend discount models in Table III. Panel 3 uses the cross-sectional regression models from Table IV. Panel 4 shows the equity risk premium computed by the time-series regression models in Table V. Panel 5 gives the estimate obtained from the survey cited in Table VI.

In all panels, the black line is the first principal component of all twenty models (it can look different across panels due to different scales in the y-axis).

Figure 2: One-year-ahead ERP

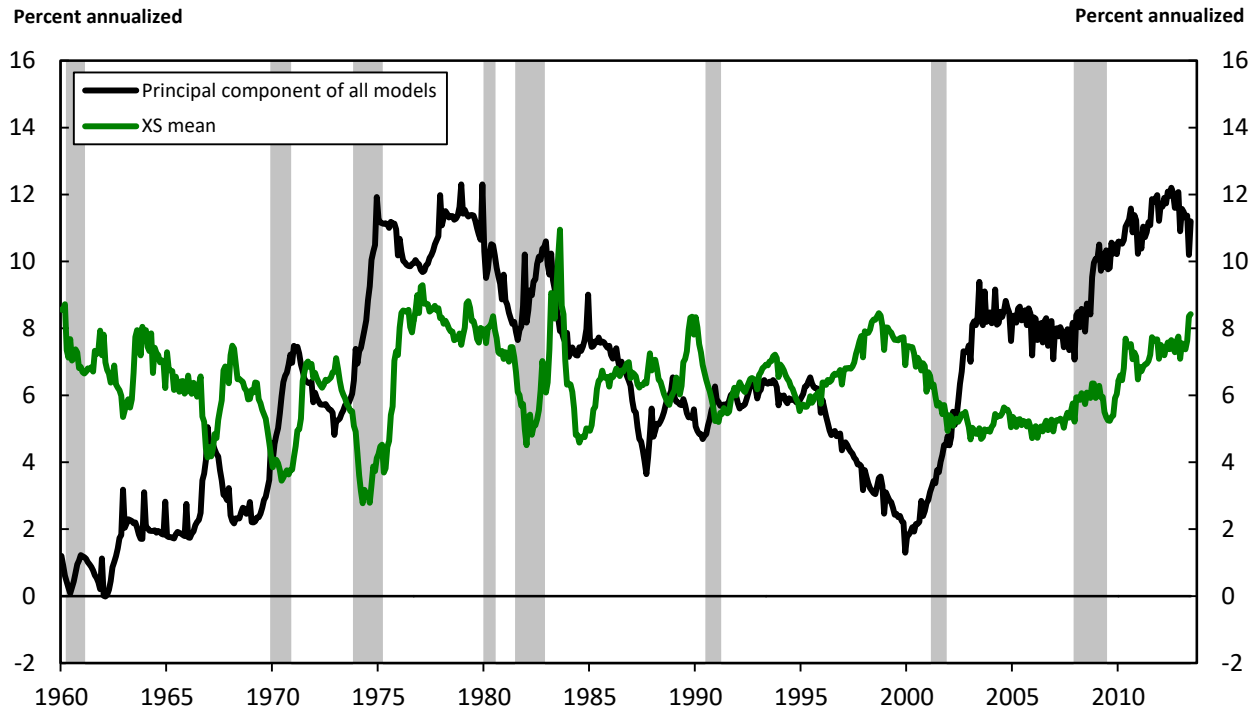


The black line is the first principal component of twenty models of the one-year-ahead equity risk premium (this is the same principal component shown in black in all panels of Figure 1). The models are listed in Tables II to VI.

The 25th and 75th percentiles (solid green lines) give the corresponding quartile of the 20 estimates for each time period, and similarly for the 10th and 90th percentiles (dashed green line).

Shaded bars indicate NBER recessions.

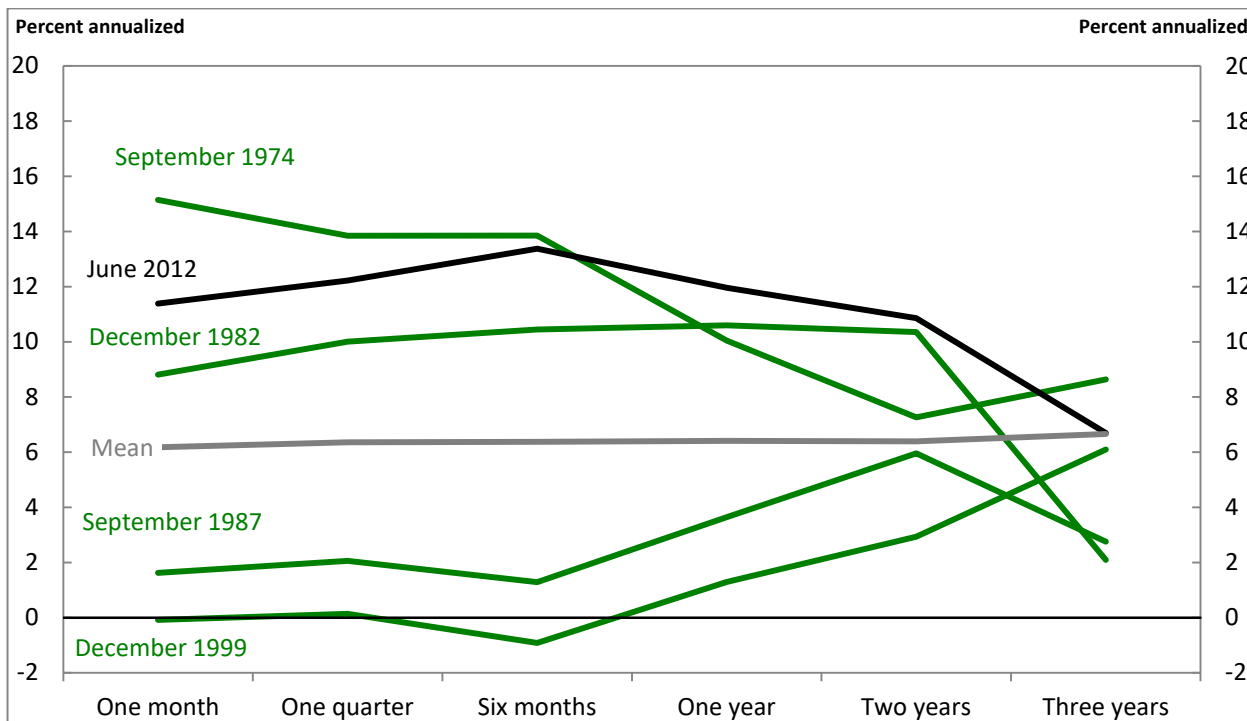
Figure 3: One-year-ahead ERP and cross-sectional mean of models



The black line is the first principal component of twenty models of the one-year-ahead equity risk premium (also shown in Figures 1 and 2). The green line is the cross-sectional average of models for each time period.

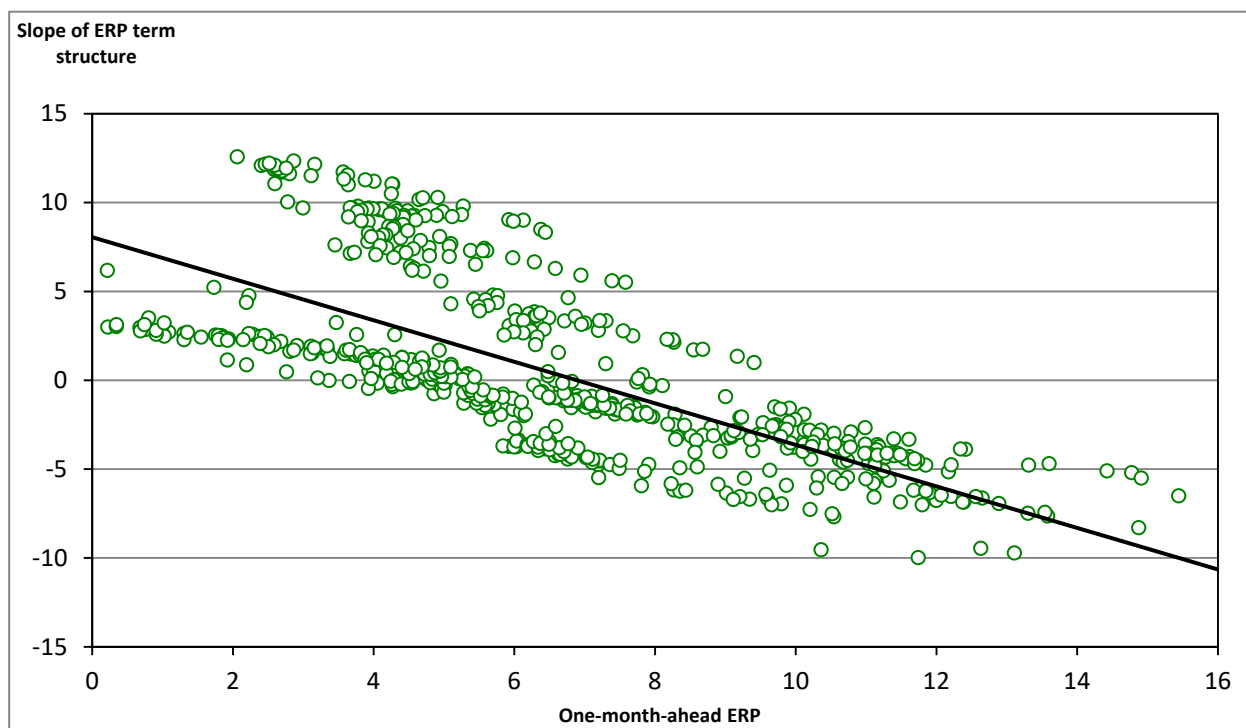
Shaded bars are NBER recessions.

Figure 4: Term structure of the ERP



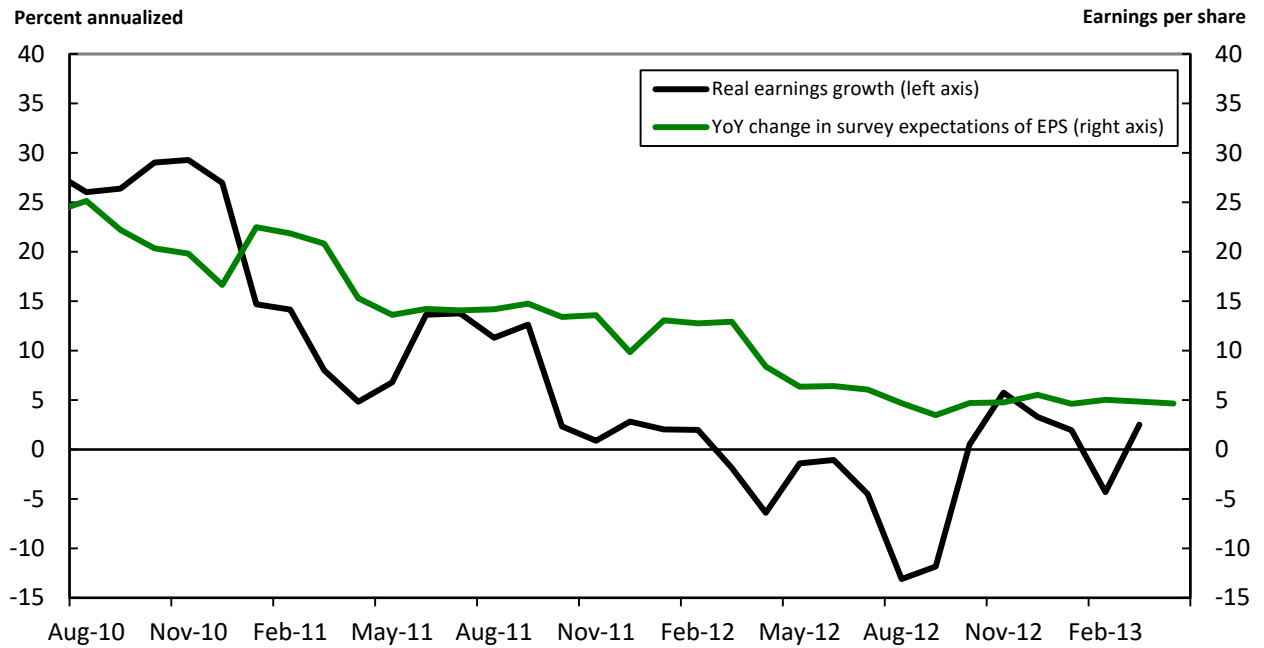
Each line, except for the grey one, shows equity risk premia as a function of investment horizon for some specific months in our sample. We consider horizons of one month, one quarter, six months, one year, two years and three years. The grey line (labeled “Mean”) shows the average risk premium at different horizons over the whole sample January 1960 to June 2013. September 1987 and December 1999 were low points in one-month-ahead equity premia. In contrast, September 1974, December 1982 and June 2012 were peaks in the one-month-ahead equity premium.

Figure 5: Regression of the slope of the ERP term structure on one-month-ahead ERP



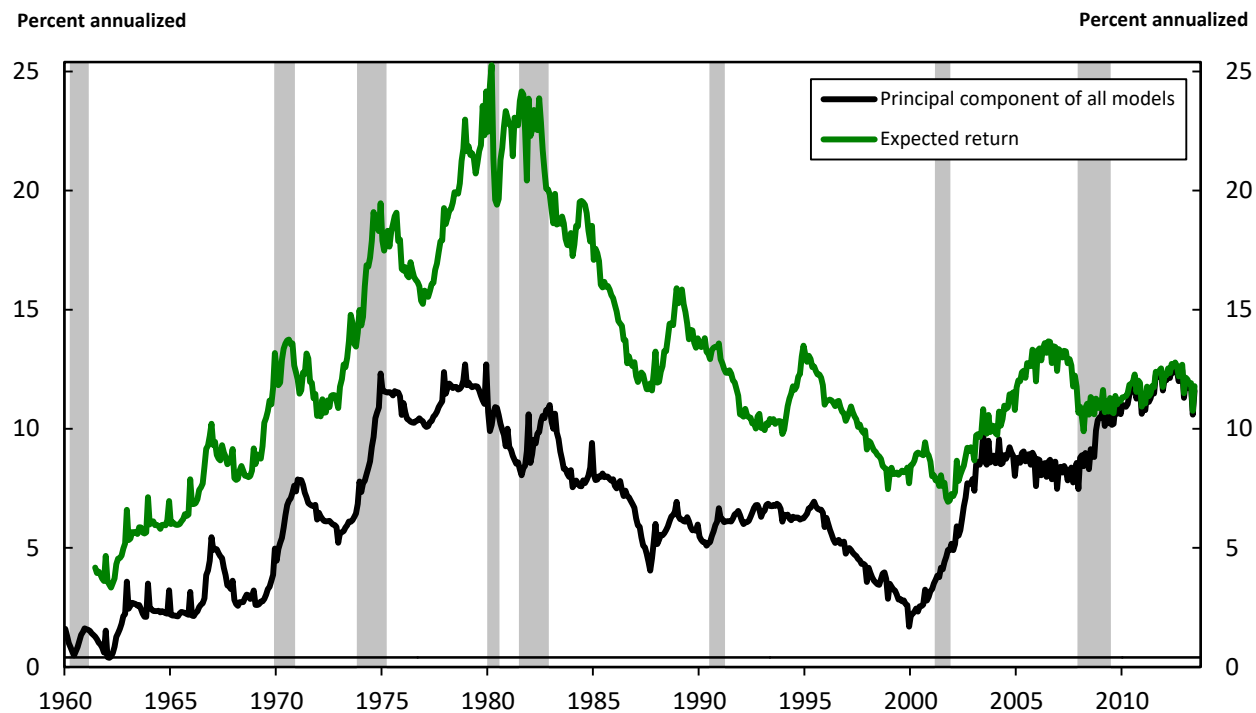
The figure shows monthly observations and the corresponding OLS regression for of the one-month-ahead ERP plotted against the slope of the ERP term structure for the period January 1960 to June 2013. The slope of the ERP term structure is the difference between the three-year-ahead ERP and the one-month-ahead ERP. All units are in annualized percentage points. The one-month-ahead and three-year-ahead ERP estimates used are the first principal components of twenty one-month-ahead or three-year-ahead ERP estimates from models described in Tables II-VI. The OLS regression slope is -1.17 (significant at the 99 percent level) and the R^2 is 50.1 percent.

Figure 6: Earnings behavior



The black line shows the monthly growth rate of real S&P 500 earnings, annualized and in percentage points. The green line shows the year-on-year change in the mean expectation of one-year-ahead earnings per share for the S&P 500 from a survey of analysts provided by Thomson Reuters I/B/E/S.

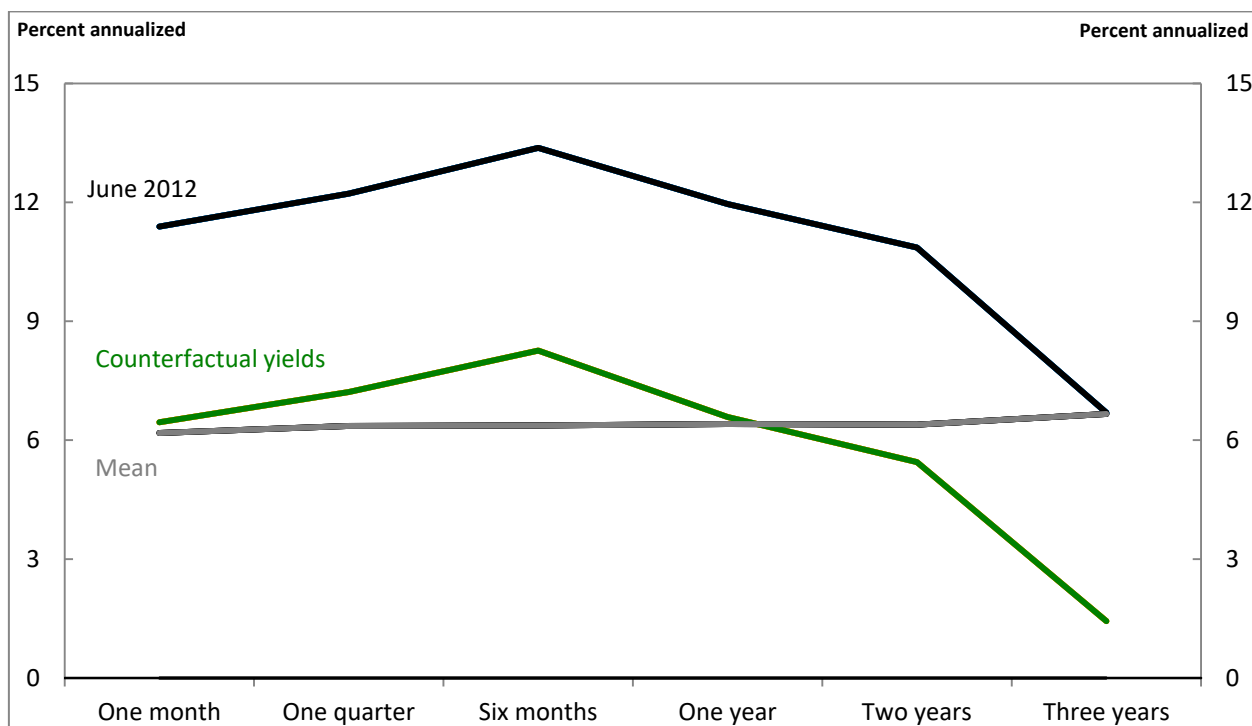
Figure 7: One-year-ahead ERP and expected returns



The black line is the first principal component of twenty models of the one-year-ahead equity risk premium (also shown in Figures 1, 2 and 3). The green line is the one-year-ahead expected return on the S&P 500, obtained by adding the realized one-year maturity Treasury yield from the principal component (the black line).

Shaded bars are NBER recessions.

Figure 8: Term structure of ERP using counterfactual bond yields



The grey line, labeled “Mean”, shows the mean term structure of the equity risk premium over the sample January 1960 to June 2013. The black line, labeled “June 2012”, shows the term structure for the most recent peak in the one-month-ahead ERP. These two lines are the same as in Figure 4. The green line, labeled “Counterfactual yields”, shows what the term structure of equity risk premia would be in June 2012 if instead of subtracting June 2012’s yield curve from expected returns we subtracted the average yield curve for January 1960 to June 2013.

The Brattle Group

Estimating the Cost of Equity for Regulated Companies

Date : 17 February 2013

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Prepared for

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EXECUTIVE SUMMARY

In this report, we discuss the models available for estimating the cost of equity for the purpose of the Natural Gas Rules in Australia. Given that the new Rule 87 requires relevant estimation methods, financial models and market data to be considered, as well as the “prevailing conditions in the market for equity funds”, this report focuses on the characteristics of the various models, how they perform under various market conditions, and therefore how to assign weight to a method, model or other data based on prevailing market or industry conditions. Further, the report finds that practitioners, regulators, and textbooks commonly look to several models or data sources before reaching a conclusion on the cost of equity.

All models have relative strengths and weaknesses, with the result that there is no one model that is the most suitable for estimating the cost of equity at any given time or for any given company. As our colleague and MIT professor Stewart Myers has put it eloquently “Use more than one model when you can. Because estimating the opportunity cost of capital is difficult, only a fool throws away useful information.” This report provides a set of guidelines that can be used in deciding which models should have more weight than others under different market, industry, or company-specific circumstances.

The focus of the report is on the key characteristics of the various cost of equity estimation methods available for a decision maker and circumstances under which each method may be more or less suitable. It is imperative that the choice of model(s) and their implementation take into account the prevailing economic conditions, industry specifics as well as characteristics of the firm for which the cost of equity is being determined, because, according to the circumstances, each model can show bias. We therefore emphasize that there is no single or formulaic approach to estimating the cost of equity. Evidence from academics, practitioners and regulators alike agree that a mechanistic reliance on a single model, without regard to changing market or industry conditions, may deliver spurious results.

The different models should be applied to a set of comparable firms, rather than the single firm for which the cost of equity is to be determined, because all methods for estimating the cost of

equity introduce significant noise or uncertainty. Applying the models to a set of comparator firms generates a range of cost of equity estimates for each model. Consideration of prevailing economic conditions, industry specifics, and characteristics of the firm for which the cost of equity is to be determined should go to the weight that is put on each model in deriving an overall reasonable range for the cost of equity.

For example, a dividend growth model might have more weight and the Sharpe–Lintner CAPM less weight when (as currently) interest rates on government bonds are unusually low. Conversely, a dividend growth model might have less weight, and the CAPM more weight, in a sector where growth forecasts are considered to be less reliable. In addition, empirical results from the Sharpe–Lintner CAPM suggest that results may be biased for firms with beta significantly different from one. In addition to the traditional Sharpe–Lintner CAPM and dividend growth models, the report also discusses other models such as the Black CAPM, the Fama–French model, the Consumption CAPM, and the Arbitrage Pricing Theory. We also touch upon new developments in implementing the dividend discount model and on other data and evidence that is sometimes used in combination with the models mentioned above.

Once a reasonable range for the cost of equity has been identified, selecting a point within that range is a matter of judgment, but that judgment can be guided by considering the riskiness of the firm at hand relative to the riskiness of the comparable firms used to generate the cost of equity estimates. Only non-diversifiable risks should be included—for example, variation in demand, which might be more highly correlated with general economic growth for a utility with significant industrial load than for a utility serving mostly residential customers.

I. INTRODUCTION

The Australian Energy Market Commission recently changed the rules that guide the regulation of pipelines (and other regulated entities) in Australia. The Australian Pipeline Industry Association (APIA) has therefore asked *The Brattle Group (Brattle)* to review the methods that are currently used or could be used to estimate the cost of equity capital for the purposes of the National Gas Rules in Australia. As part of this exercise, the APIA has asked us to review how academics, practitioners and regulators worldwide think models should be used, and how they have been used in determining the cost of equity for regulated entities. Thus, in this report, we discuss examples of regulatory approaches in the U.S., Canada and the U.K. where regulators have considered a number of methods for estimating the cost of equity capital, and have determined the optimal use of these multiple evidence sources in order to provide greater confidence in their results. The report also includes a discussion of the recommendations of academics and practitioners with regards to the use of several cost of equity estimation models.

The report focuses on the new Rule 87 and the new *allowed rate of return objective*, which, in order to be achieved, requires that “regard must be had to relevant estimation methods, financial models, market data and other evidence”¹ in determining the overall rate of return, and that “regard must be had to the prevailing conditions in the market for equity funds”² in determining the cost of equity component of the overall rate of return. We therefore focus on introducing a broad set of methods for cost of equity estimation, the risk positioning of a company relative to the industry or other companies, and methods relied upon by regulators and practitioners around the globe.

Section II provides some background for cost of equity estimation. *Section III* focuses on the evolution, theoretical underpinnings, and characteristics of various cost of equity estimation methods including (a) the Sharpe-Lintner Capital Asset Pricing Model (CAPM), (b) variations of the CAPM such as the Empirical CAPM (ECAPM) and the Consumption-Based CAPM, (c) the Fama-French Three-Factor Model, (d) the Arbitrage Pricing Theory, (e) Dividend Discount

¹ Rule 87, s.5a.

² Rule 87, s.7.

Models including both Single-Stage and Multi-Stage models, and (f) Other Models including the so-called Risk Premium method, Residual Income Valuation model, Ibbotson's Build-up method, the Comparable Earnings model, Market-to-Book and Earnings Multiples approaches. We note that the above is not intended to be an exhaustive list of models that regulators or practitioners could feasibly rely upon in determining the cost of equity. We also note that as finance evolves, new estimation methods, financial models, market data and other evidence may become available that could be informative for the purpose of estimating the cost of equity. *Section IV* discusses implementation issues, summarizes the characteristics of the various cost of equity estimation methods, and discusses how to use the models under different market conditions. Additionally, this section includes a description of how to position the target entity relative to a sample based on its relative risk.

II. METHODS, FINANCIAL MODELS, MARKET DATA AND OTHER EVIDENCE USED TO ESTIMATE THE COST OF EQUITY CAPITAL

A. INTRODUCTION

To determine the cost of capital, one must evaluate the cost of equity, the cost of debt (possibly both long-term and short-term) and the capital structure of the company subject to regulation. This report focuses on the estimation of the cost of equity component of a regulated entity's cost of capital.

To determine the cost of equity for a specific utility, decision makers typically look at a range of evidence presented to them. In the case of regulators, they commonly review expert evidence, models and other information presented by experts, the utility and other stakeholders, and also evidence that the regulator itself generates. Ultimately, a degree of judgment is used to arrive at a final determination having considered this evidence. The evidence considered might include different financial models which are used to extract estimates of the cost of equity for similar utilities from market data (stock prices). It might also include estimates from models that take equity analyst forecasts as inputs. For example, three regulators, the Alberta Utilities Commission (AUC), the Ontario Energy Board (OEB), and the U.S. Surface Transportation Board (STB), recently reviewed their cost of equity estimation approach. These three regulators noted that each methodology has its own strengths and weaknesses and subsequently decided to

rely on more than one model or approach to determine the cost of equity.³ We further note here that in discussing the characteristics of each model or practice, we are pointing to advantages or disadvantages of the models assuming they will inform the ultimate decision, but we do not expect any one model to be the only piece of evidence considered and used by either regulators or practitioners in determining the cost of equity.

This report describes a number of models that can be used to inform the regulator's judgment in determining the cost of equity. It also discusses the views of academics and practitioners with regards to the determination of the cost of equity from multiple estimation models.

Below, we describe methodologies that regulators and practitioners use in Australia, Canada, Europe, the U.K., and the U.S., as well as some more recent methods that have been proposed, albeit it is not clear from the record the extent to which regulators have used these methods. It is important to realize that in many jurisdictions the regulator does not look to a single model, but considers all the evidence in front of it and then makes a decision. In North America, where the consideration of more than one model and possibly other evidence is common, the ultimate decision is often not explicit about the weight assigned to each model or other pieces of evidence.⁴

B. THE USE OF MODELS FOR COST OF CAPITAL ESTIMATION

1. Context

The National Gas Rules set the framework for how the AER (and the ERAWA) determine access arrangements for covered gas pipelines, including the rate of return on capital which is a component of the charges paid by pipeline customers. We understand that the regulators are

³ Alberta Utilities Commission, Decision 2011-474, p. 27-28, Ontario Energy Board, EB-2009-084, p. 38, Surface Transportation Board, Ex Parte 664 (Sub-No. 1), pp. 3-5.

⁴ There are exceptions to this rule such as the Federal Energy Regulatory Commission and the Surface Transportation Board in the U.S., and the Canadian Transportation Agency. However, most U.S. state and Canadian federal and provincial regulators do not have a specified cost of equity estimation method. Instead, they commonly hear evidence from a number of different parties on cost of equity (often including regulatory staff). Based on this information the regulator then makes its decision.

currently developing guidelines as to how the rate of return provisions of the NGR will be applied in future determinations.

The NGR state that —. the rate of return for a service provider is to be commensurate with the efficient financing costs of a benchmark efficient entity with a similar degree of risk... ”.⁵ In addition, the NGR require that “[i]n determining the allowed rate of return, regard must be had to: (a) relevant estimation methods, financial models, market data and other evidence;...”⁶ and that “[i]n estimating the return on equity under subrule (6), regard must be had to the prevailing conditions in the market for equity funds.”⁷

In this report, we describe the estimation methods, financial models, market data and other evidence that may be relevant for setting the cost of equity in future access arrangement determinations in Australia.

a) The cost of capital

The cost of capital is a key parameter in regulatory settings, because it contributes to determining the return to the company’s investors. Defined as *the expected rate of return in capital markets on alternative investments of equivalent risk*, it is the expected rate of return investors require based on the risk-return alternatives available in competitive capital markets. Stated differently, the cost of capital is a type of opportunity cost: it represents the rate of return that investors could expect to earn elsewhere without bearing more risk.^{8,9}

While the details of energy network regulation are different in different jurisdictions, regulators are in many jurisdictions required to set a cost of capital which provides investors in rate-regulated entities a reasonable opportunity to earn a return on their investment equal to the opportunity cost of capital.

⁵ Rule 87(3).

⁶ Rule 87(5).

⁷ Rule 87(7).

⁸ —Expected” is used in the statistical sense: the mean of the distribution of possible outcomes. The terms —expect” and —expected” in this Report, as in the definition of the cost of capital itself, refer to the probability-weighted average over all possible outcomes.

⁹ The cost of capital is a characteristic of the investment itself, not the investor.

In the U.K., the Gas Act 1986 requires the regulator to have regard to ~~the~~ need to secure that licence holders are able to finance the[ir] activities....”¹⁰ Ofgem has also said:

In setting price controls, we are required to have regard to the ability of efficient network companies to secure financing in a timely way and at a reasonable cost in order to facilitate the delivery of their regulatory obligations.¹¹

In Canada, the National Energy Board has explained the ~~fair~~ return standard” as follows:

The Board is of the view that the fair return standard can be articulated by having reference to three particular requirements. Specifically, a fair or reasonable return on capital should:

- be comparable to the return available from the application of the invested capital to other enterprises of like risk (the comparable investment standard);
- enable the financial integrity of the regulated enterprise to be maintained (the financial integrity standard); and
- permit incremental capital to be attracted to the enterprise on reasonable terms and conditions (the capital attraction standard).¹²

Finally, in the U.S., the starting point for the Federal Energy Regulatory Commission’s approach to determining the cost of equity is Supreme Court precedent, which states that:

the return to the equity owner should be commensurate with the return on investments in other enterprises having corresponding risks. That return, moreover, should be sufficient to assure confidence in the financial integrity of the enterprise, so as to maintain its credit and to attract capital.¹³

While these legal standards are differently worded, a common thread is that regulated entities are allowed to earn a return that is comparable to that of other enterprises of similar risks and which enables the regulated entity to finance its operations. The legal standards in North America and Europe are not specific about how to accomplish the goal(s).

¹⁰ Gas Act 1986, s. 4AA(2)(b).

¹¹ *RIIO-T1: Final Proposals for National Grid Electricity Transmission and National Grid Gas*, Ofgem (December 2012), paragraph 4.6.

¹² RH-2-2004, p. 17. See also the Supreme Court of Canada’s decision in *Northwestern Utilities Limited v. City of Edmonton* [1929] S.C.R. 186.

¹³ *FPC v. Hope Natural Gas Co.*, 320 U.S. 591 (1944). *Bluefield Water Works & Improvement Co. v. Public Service Comm’n*, 262 U.S. 679 (1923), cited in FERC policy statement on the *Composition of Proxy Groups for Determining Gas and Oil Pipeline Return on Equity*, April 17 2008, p. 2.

b) What should we expect from models?

It is useful to recognize explicitly at the outset that models are imperfect. All are simplifications of reality, and this is especially true of financial models. Simplification, however, is also what makes them useful. By filtering out various complexities, a model can illuminate the underlying relationships and structures that are otherwise obscured. After all, while a perfect scale model representation of the city might be highly accurate, it would make a poor road map. It is therefore imperative that regulators and other users of the models use sound judgment when implementing and using the models — there is no one model or set of models that are perfect.

The gap between financial models and reality can sometimes be quite significant (as was painfully demonstrated by the recent financial crisis). There is no single, widely accepted, best pricing model to estimate the cost of capital — just as there is still no consensus on some fundamental issues, such as the degree to which markets are efficient. Analysts have a host of potential models at their disposal, and it must be acknowledged that cost of capital estimation continues to require the exercise of judgment. Practitioners, regulators, as well as textbooks therefore often recommend that the “best practice” for ensuring robustness is to look at a totality of information.¹⁴ These practitioners, regulators and texts therefore use or present a variety of methodologies that may be applicable for the determination of the cost of equity in a specific circumstance.

While no model is perfect, there are certain features that make models more useful from a regulatory perspective. For example, it is desirable to have models and methods that i) are consistent with the goal being pursued, ii) are transparent, iii) produce consistent results, iv) are robust to small deviations or sampling error, v) are as simple as possible (while maintaining reliability), vi) can be replicated by others (*e.g.*, data is widely available), and vii) recognize the regulatory context and legislative requirements in which the regulatory body operates. Clearly different models will satisfy these criteria to differing degrees, and different models may be better suited to different regulatory jurisdictions.

¹⁴ See, for example, the Ontario Energy Board’s EB-2009-084 decision, December 2009, the U.S. Surface Transportation Board’s Ex. Parte 664 (Sub-No. 1) decision, January 2009, Morningstar *Ibbotson Cost of Capital 2012 Yearbook*, and Roger A. Morin, *New Regulatory Finance*, Public Utilities Reports Inc., 2006, Chapter 15.

For example, the CAPM and the Dividend Discount Model (DDM) both are transparent and developed from economic theory. Their results can be replicated easily, since the data required are widely available from many public sources. However, the implementation of the CAPM and DDM requires a number of subjective decisions – decisions which can be hotly contested and can lead to significantly different results. The CAPM, for instance, relies on a risk-free rate that is currently driven unusually low by the recent flight to quality and the easing of monetary policy. The model also requires an estimate of the market risk premium, which may pose difficulties in times of high market volatility.

The single-stage DDM is especially sensitive to the growth rate estimates used, which can vary widely among analysts and over time, contradicting the underlying assumption of growth stability inherent in this model. The variability in growth rates and stock prices may increase when industries are in transition, making the reliability of the DDM more questionable in such periods. In addition, it has become more common to distribute cash to shareholders in a form other than dividends. For example, regulated entities in both the U.S. and the U.K. have had share buyback programs that substantially affected the number of shares, and these are not captured in the standard DDM.¹⁵ Some of the growth rate problems in the DDM are alleviated by the reliance on a multi-stage version of the model as done by, for example, *The Brattle Group*, Morningstar *Ibbotson Cost of Capital Yearbook*, and the U.S. Surface Transportation Board (STB).¹⁶

Similar problems arise in other models that inherently rely on data for a sample of companies and data for economic phenomena that may be changing quickly; the latter is especially true for models such as the Fama-French, where the reliance on three risk factors can lead to highly variable results across time. As a result, no single model is ideal and the implementation of any model necessarily requires choices that involve subjective judgments. Therefore, it is important to look to the totality of relevant information available from methods, models, market data and

¹⁵ See, for example, National Grid Share Buyback Programme and Spectra Energy Corp's 2008 form 10-K.

¹⁶ *The Brattle Group* is a consulting firm, Morningstar is a commercial provider of data and the STB is a U.S. federal regulator.

other evidence. The relative strengths and weaknesses of the various cost of equity estimation models are outlined in further detail in *Section III* of this report.

c) Model stability and robustness

For an estimation model used to determine the cost of equity, stability and robustness over time are desirable unless economic conditions have truly changed. Stability means that cost of capital estimates done in similar economic environments should be similar, not only period-to-period but also company-to-company within a comparable sample. Robustness is meant here as the ability of a model to estimate the cost of capital across different economic conditions.

In general, all of the models discussed here have characteristics that make them more or less suited to one economic environment versus another. As such, all individual models can be, and often are, subject to some instability over time. For example, the currently very low government bond yields lead to very low cost of equity estimates using the CAPM — sometimes less than the costs of debt of investment-grade companies! During the early 2000s, the DDM was subject to substantial criticism due to allegations of analysts' optimism bias. Similarly, the risk premium model¹⁷ has produced very different results in times of high and low inflation that did not necessarily reflect the true cost of capital. Thus, estimates at any given point of time may seem too high or too low, and it is important to understand whether the estimated figures are driven by actual changes in the systematic risk of the regulated entities, or by something else (e.g., data irregularities). It is for these reasons that regulators in the U.S. and Canada often rely on and analysts recommend relying on the results from at least two estimation models.¹⁸

A notable example of a regulator that has acknowledged the difficulty in relying on only one model or method is the U.S. Surface Transportation Board. The STB in 1982 started to rely on a single-stage DDM to determine the cost of equity for U.S. railroads. However, in 2006, the shippers on the railroads complained that the estimated cost of equity was out of line with reality,

¹⁷ The risk premium used in the risk premium model is different from the market risk premium used in the CAPM. The model is frequently used in U.S. regulatory proceedings.

¹⁸ See, for example, U.S. Surface Transportation Board, Ex Parte 664 (Sub-No. 1), served January 28, 2009; Mississippi Power, Performance Evaluation Plan, Rate Schedule 'PEP-5', November 9, 2009 (<http://www.mississippipower.com/pricing/pdf/pep-5.pdf>); Ontario Energy Board, EB-2009-0084, Report of the Board on the Cost of Capital for Ontario's Regulated Utilities, Issued December 11, 2009.

because forecasted growth rates for railroad companies were substantially higher than the economy-wide forecasted growth. The shippers argued successfully that such high growth rates could not be sustained forever as assumed by the single-stage DDM, and the STB thus initiated a rulemaking proceeding to review and eventually determine how to set the allowed cost of equity going forward. Following several years of expert submissions and proceedings, the STB decided to rely on an equally-weighted average of the Sharpe-Lintner Capital Asset Pricing Model and a specific version of the multi-stage DDM. In doing so, the STB concluded:

if our exploration of this issue has revealed nothing else, it has shown that there is no single simple or correct way to estimate the cost of equity for the railroad industry, and countless reasonable options are available. Both the CAPM and the multi-stage DCF [DDM] models we propose to use have their own strengths and weaknesses, and both take different paths to estimate the same illusory figure. By using an average of the results produced by both models, we harness the strengths of both models while minimizing their respective weaknesses. The result should be a stable yet precise estimate of the cost of equity that we can use in future regulatory proceedings and to gauge the financial health of the railroad industry.¹⁹

2. Risk-Return Tradeoff

At its most basic level, an asset (security) is a claim to a stream of future (risky) cash flows and sometimes with potential rights to exert some control over those flows. Financial markets allow investors to exchange these claims, and therefore risks. Through trade, investors are able to create different packages of risks and returns than could be achieved by holding individual securities (or fixed packages of securities), and investors can change their risk exposure over time. Because investors are assumed to be risk-averse, they evaluate the universe of risky investments on the basis of a risk-return trade-off. Investors can only be induced to hold a riskier investment if they expect to earn a higher rate of return on that investment. The essential tradeoff between risk and the cost of capital is depicted in Figure 1 below.

¹⁹ *U.S. Surface Transportation Board, Ex Parte 664 (Sub-No. 1)*, served January 28, 2009, p. 15.

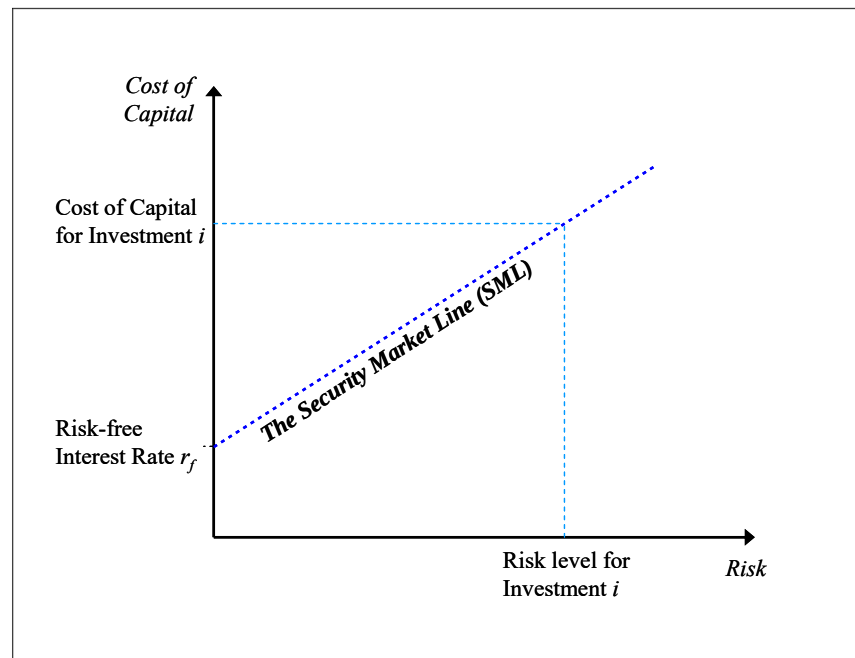


Figure 1: Security Market Line

III. COST OF EQUITY ESTIMATION MODELS

A. SHARPE-LINTNER CAPITAL ASSET PRICING MODEL

One of the most common pricing models used in business valuation and regulatory jurisdictions is the Sharpe-Lintner CAPM, which in its simplest form is depicted in Figure 2 below.

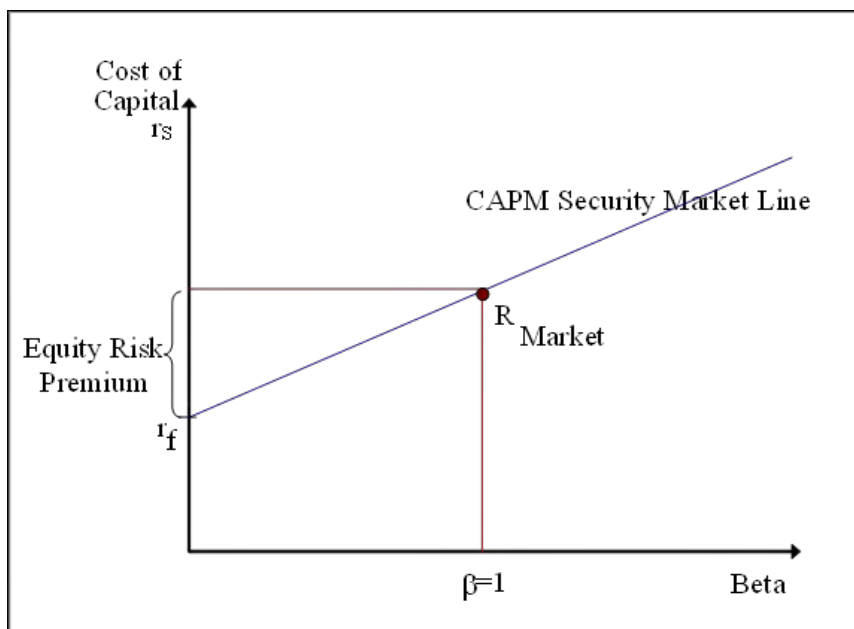


Figure 2: Capital Asset Pricing Model

Thus, in the world in which the CAPM holds, the expected cost of (equity) capital for an investment is a function of the risk-free rate, a measure of systematic risk (beta), and an expected market risk premium (MRP).²⁰

$$E(r_S - r_f) = \beta_S \times E(r_M - r_f) \quad (1)$$

where r_S is the cost of capital for investment S ; r_M is the return on the market portfolio, r_f is the risk-free rate, and β_S is the measure of systematic risk for the investment S . The $(r_M - r_f)$ term is known as the market risk premium (MRP),²¹ and β_S measures the response of the stock S to systematic risk. Re-arranging this equation produces the CAPM's formula for the cost of (equity) capital of a traded asset:

$$r_S - r_f = \beta_S \times MRP \quad (2)$$

²⁰ While the CAPM model frequently is applied to equity capital, it applies to all assets.

²¹ We note that some European regulators use the term Equity Risk Premium (ERP) instead of MRP.

To implement the CAPM, it is necessary to determine the risk-free rate, r_f , and to estimate the MRP and beta, β_S .

1. Evolution of the CAPM

The CAPM was developed as a theoretical equilibrium model and fits with the intuition of a risk-return tradeoff. The development of the CAPM signaled the first time that economists were able to quantify risk and the reward for bearing it. Under the CAPM, the expected return of an asset must be linearly related to the covariance of its return with the return of the market portfolio.²²

Markowitz (1959)²³ first laid the groundwork for the CAPM. In his seminal research, he expressed the investor's portfolio selection problem in terms of expected return and variance of return. He argued that investors would optimally hold a mean-variance efficient portfolio, that is, a portfolio with the highest expected return for a given level of variance. Sharpe (1964)²⁴ and Lintner (1965)²⁵ built on Markowitz's work to develop economy-wide implications. They showed that if investors have homogeneous expectations and optimally hold mean-variance efficient portfolios, then, in the absence of market frictions, the portfolio of all invested wealth, or the market portfolio, will itself be a mean-variance efficient portfolio. This is the heart of the Sharpe-Lintner CAPM. The standard CAPM equation (as expressed in Equation (2)) is a direct implication of this statement.

The Sharpe-Lintner CAPM assumes unrestricted lending and borrowing at a risk-free rate of interest. In the absence of a risk-free asset, Black (1972)²⁶ derived a more general version of the CAPM which did not rely on this potentially problematic assumption. In this version, known as the Black CAPM, the expected return of an asset in excess of the "zero-beta" return is linearly

²² For a basic introduction to risk-return models, see R.A. Brealey, S.C. Myers, and F. Allen, *Principles of Corporate Finance*, 10ed, 2011 (Brealey, Myers & Allen (2011)), pp. 192-203.

²³ H. Markowitz, "Portfolio Selection: Efficient Diversification of Investments," 1959, John Wiley, New York.

²⁴ W. Sharpe, "Capital Asset Prices: A Theory of Market Equilibrium under Conditions of Risk," *Journal of Finance* 19, 1964, pp. 425-442.

²⁵ J. Lintner, "The Valuation of Risky Assets and the Selection of Risky Investments in Stock Portfolios and Capital Budgets," *Review of Economics and Statistics* 47, 1965, pp. 13-37.

²⁶ F. Black, "Capital Market Equilibrium with Restricted Borrowing," *Journal of Business* 45, 1972, pp. 444-455.

related to its market beta. In essence, the return on the risk-free asset in Equation (2) above is substituted with a return on a zero-beta portfolio associated with the market portfolio. This zero-beta portfolio is defined to be the portfolio that has the minimum variance of all portfolios uncorrelated with the market portfolio. The empirical implementation of the Black CAPM is often referred to as the Empirical CAPM or ECAPM.

Empirical tests of the Sharpe-Lintner CAPM have focused on three implications of equation (2): (i) The intercept is zero; (ii) The market beta completely captures the cross-sectional variation of expected excess returns; and (iii) The market risk premium is positive.

There is substantial literature on empirical tests of the CAPM since its development in the 1960s, with mixed results. Black, Jensen and Scholes (1972)²⁷, Fama and Macbeth (1973),²⁸ and Blume and Friend (1973)²⁹ found empirical evidence to be consistent with the mean-variance efficiency of the market portfolio. However, Black, Jensen and Scholes (1972) and Fama and MacBeth (1973) identified a fundamental challenge to the CAPM; namely that low-beta stocks have higher average returns than predicted by the CAPM, and high-beta stocks lower average returns. In other words, the empirical estimates are consistent with pivoting the Security Market Line (SML) around $\beta = 1$ compared to the Sharpe-Lintner CAPM. This suggests that the cost of capital for regulated companies, which often have a beta less than one, will be underestimated by the traditional CAPM.³⁰

Several subsequent studies confirmed the robustness of this result and proposed explanations revolving around market frictions, such as different borrowing and lending rates, and the role of

²⁷ F. Black, M.C. Jensen, and M. Scholes, —The Capital Asset Pricing Model: Some Empirical Tests,” *Studies in the Theory of Capital Markets*, Praeger Publishers, 1972, pp. 79-121.

²⁸ E. Fama and J. Macbeth, —Risk, Return, and Equilibrium: Empirical Tests,” *Journal of Political Economy* 81, 1973, pp. 607-636.

²⁹ M. Blume and I. Friend, —A New Look at the Capital Asset Pricing Model,” *Journal of Finance* 28, 1973, pp. 19-33.

³⁰ Implementing a long-run version of the CAPM which uses (annualized) long-horizon returns (e.g., with long bond rates as risk-free rate) generally produces a flatter SML than obtained by using short-rates, due to the general presence of an upward sloping yield curve. While this partially compensates for the empirically observed flattening, it is not sufficient to explain all of the observed flattening of the SML. That is, even implementations that utilize a long-run risk-free interest rate require a further, albeit smaller, adjustment to match the empirical SML.

taxes. Nevertheless, the empirical evidence suggested significant movement in the SML, often flattening, to the point that Fama and French (1992) found a zero slope in the empirical SML.³¹ Fama and French (1992, 1993³²) in turn suggested that factors other than the risk relative to the market, such as size and book-to-market value ratios (among others) were significant in explaining the observed SML. Fama and French found that firms with high book-to-market ratios and small size have higher average returns than is predicted by the standard CAPM, and vice versa. Their work culminated in the model now known as the Fama-French three-factor model.

The Fama-French papers cited above continued in the vein of the so-called “anomalies” literature that had arisen in the late 1970s. These anomalies can be thought of as firm characteristics that provide incremental explanatory power for the sample’s mean returns beyond the market. Earlier anomalies included the price-earnings ratio effect (first reported by Basu (1977)³³) and the detection of the size effect (Banz (1981)³⁴). For example, Basu found that firms with low price-earnings ratios have higher sample returns than those predicted by the standard CAPM. The price-earnings ratio and size anomalies are at least partially related, as low price-earnings-ratio firms tend to be small.

The Empirical CAPM (ECAPM), described further in the section below on variations of the standard CAPM, is an alternative method of correcting for the empirical flattening of the SML. The ECAPM can be viewed from the positive school of thought as a practical adjustment that can be made to measure the cost of capital. It can be applied without knowing the “cause” of the increased intercept and decreased slope of the SML relative to the Sharpe-Lintner CAPM.

To sum up, there has been a wealth of statistical evidence contradicting the Sharpe-Lintner CAPM over the past 40 years or so and controversy remains about how the evidence should be

³¹ E.F. Fama and K.R. French, —The Cross-Section of Stock Expected Returns,” *Journal of Finance* 47, 1992, pp. 427-465.

³² E.F. Fama and K.R. French, —Common risk factors in the returns on stocks and bonds,” *Journal of Financial Economics* 33, 1993, pp. 3-56.

³³ S. Basu, —The Investment Performance of Common Stocks in Relation to Their Price to Earnings Ratios: A Test of the Efficient Market Hypothesis,” *Journal of Finance* 32, 1977, pp. 663-682.

³⁴ R. Banz, —The Relationship Between Return and Market Value of Common Stocks,” *Journal of Financial Economics* 9, 1981, pp. 3-18.

interpreted. Some argue that the standard CAPM should be replaced by multifactor models with several sources of risk, such as the Fama-French model. Others argue that evidence against the CAPM is overstated due to potential mis-measurement of the market portfolio, data mining or sample selection biases. One further key deficiency in the CAPM is that it is a static model which ignores consumption decisions, and treats asset prices as being determined by the portfolio choices of investors who have preferences defined over wealth one period in the future. Implicitly, these models assume that investors consume all their wealth after one period or at least that wealth uniquely determines consumption. This assumption does not match with reality. Therefore, to make the model more realistic, intertemporal equilibrium asset pricing models have been developed that model consumption and portfolio choices simultaneously. An example of such a model is the consumption-based CAPM, which is described further in *Section III.B.2* below.

2. CAPM Implementation Issues

Fundamentally, an analyst using the CAPM must determine three parameters to implement the model: the risk-free rate (r_f), the MRP, and the asset's beta (β_S) as shown in Equation (2) above. Through the determination (or estimation) of the parameters on the right-hand side of Equation (2), the analyst obtains an estimate of the cost of equity, r_S .

It is common to choose (i) a forecasted yield on government bonds (as is often done in Canada), (ii) a current measure of local government bond yields (a common practice in the U.S.), or (iii) a regional or global measure of the current yield on government bonds (e.g., the Netherlands).

Like the risk-free rate, the choice of market proxy is local, regional, or global. The choice of risk-free rate and market index should be consistent, so the cost of equity is estimated as either a local, regional, or global figure.

For many years it was common to estimate the MRP from an arithmetic average of historical realized MRPs, measured as the long-term excess of market returns over the risk-free rate in the country or region of interest. European decision makers have in recent years often looked to the study of Dimson, Marsh, and Staunton to determine the MRP, while many in the U.S. commonly

look to evidence from Morningstar (formerly Ibbotson).³⁵ Some decision makers and analysts also look to either forecasted MRPs or survey results.³⁶ The estimation of the MRP remains controversial and the resulting cost of equity estimates generated by the standard CAPM are sensitive to the choice of MRP.

3. Characteristics of the CAPM

While the strengths and weaknesses of the CAPM inherently depend on its exact implementation, the following are some generic strengths:

- The model is transparent, well-documented and relies on economic theory.
- Data needed for the model are readily available if applied to companies with a reasonable trading history in well-developed markets. It is therefore also auditable.
- The model is sensitive to economic conditions through risk-free rates and market performance, as well as to changes in companies' systematic risk.

Among the weaknesses of the CAPM are the following:

- The model is very sensitive to developments in the risk-free rate that may reflect monetary policy rather than economic conditions.
- The model is sensitive to different estimation procedures for the MRP.
- Because beta estimates rely on historical data, there may be a delay in incorporating changes in systematic risk. MRP estimates based on historical data are also backward-looking.
- The model may downward bias cost of equity estimates for low-beta stocks and vice versa (see section on ECAPM below).

³⁵ Texts such as Morningstar, *Ibbotson SBBI 2012 Yearbook*, p. 55-56 recommends to use the income return rather than total return or yield as the risk-free rate. The income return consists of the coupon payment divided by the bond price rather than the total return as this is the true risk-free component of the bond return. Capital gains or losses carry risk.

³⁶ For examples, see Bank of England, "Financial Stability Report," June 2012, Chart 1.11 and P. Fernandez, J. Aguirreamolla and L. Corres (2013), "Market Risk Premium used in 82 countries in 2012: a survey with 7,192 answers," IESE Business School, University of Navarra, SSRN 2084213.

- The model incorporates only one source of risk (the market), and therefore does not reflect the effects of, for e.g., consumption or economic growth, technological or regulatory risks.
- The CAPM is a static model and therefore ignores the dynamics of investment behavior and hedging.
- The model is based on the assumption that all investors optimally hold well-diversified portfolios and therefore only care about systematic risks. This assumption does not necessarily hold, however, when investor expectations about returns and investment opportunities are heterogeneous.

Because the model was developed as a generic approach to determining the cost of capital for companies, it does not specifically take industry factors or the context in which it is being used into account. However, the CAPM is a well-founded and commonly used model that relies primarily on readily available information. It may be less stable than ideal because changes in interest rates affect the risk-free rate and market volatility affects the beta estimates. Furthermore, determination of which sample companies to rely upon and the MRP remains controversial.

The CAPM has been widely used for a long period of time for a variety of reasons. The primary reason for the model's widespread use is its solid economic foundation, making it taught in every introductory finance class. The model is also relatively simple to implement. Most market-based models that have been developed since the CAPM take the CAPM as their point of departure to generalize the model. Also, academic researchers have not found any one alternative to the model that is easily applied in practice.

B. VARIATIONS ON THE CAPM

1. The Empirical CAPM

As described above, the ECAPM is one way of correcting for the empirical flattening of the Security Market Line (SML). Specifically, the ECAPM directly adjusts the CAPM SML by a parameter, alpha, that can be controlled for sensitivities, *etc.* Formally, the ECAPM relation is given by Equation (3) below:

$$r_s = r_f + \alpha + \beta_s \times (MRP - \alpha) \quad (3)$$

where α is the “alpha” adjustment of the risk-return line, a constant, and the other symbols are as defined above. The alpha adjustment has the effect of increasing the intercept but reducing the slope of the SML, which results in a security market line that more closely matches the results of empirical tests.

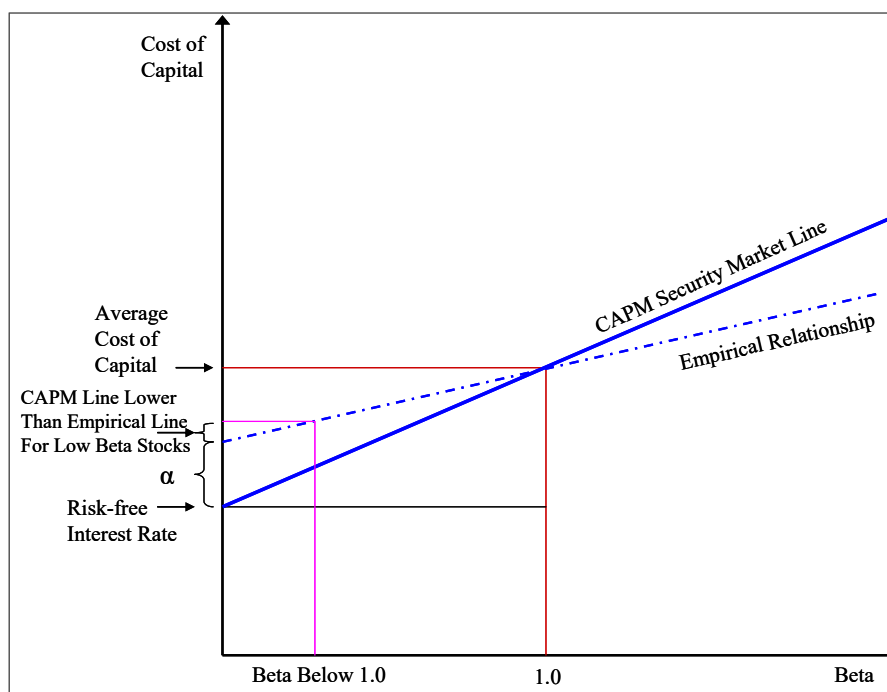


Figure 3: The Empirical Security Market Line

The academic literature has estimated a fairly wide range of alpha parameters, using primarily U.S. data, of approximately 1 to 7 percent.³⁷ While this is a rather large range, much of the variation between studies arises from differences in methodology and time periods so that the alpha estimates are not strictly comparable. The ECAPM is included among the models relied upon by some decision makers and experts including U.S. state and Canadian provincial regulators.³⁸

³⁷ See Appendix A for details.

³⁸ The Mississippi Public Service Commission in the U.S. and the Alberta Utilities Commission in Canada have included the ECAPM as one of the models used to determine the cost of equity.

2. The Consumption-Based CAPM

The Consumption CAPM is an example of an intertemporal equilibrium model. This model aggregates investors into a single representative agent and considers a changing investment opportunity set over time, unlike the static standard CAPM. The representative agent is assumed to derive utility from the aggregate consumption of the economy. In this model, the stochastic discount factor, (defined such that the expected product of any asset return with the stochastic discount factor is equal to one), is equal to the intertemporal marginal rate of substitution for the representative agent.³⁹ Through mathematical equations, (the so-called Euler equations), asset returns and consumption can be linked. Using this setup, the model explains the risk premia on assets using the covariance between their returns and the intertemporal aggregate consumption marginal rate of substitution. As a result, the consumption-based pricing model can help explain the observed phenomenon of predictable variations in asset risk premia over time, and expands the risk-return relation to allow for a time-varying relationship between a stock's risk and return.

An important feature of the consumption model is that the expected conditional risk premium on an asset is related to its predicted conditional volatility. In particular, the relationship between a stock's risk premium and its conditional volatility could be positive or negative, depending on the extent to which the stock is an intertemporal hedge against shocks to the marginal utility of consumption. Furthermore, hedging assets have volatility patterns that could lead to expected rates of return lower than the risk-free rate. Note that this would generally not be the case for public utility stocks, since they are not viewed as defensive stocks.

Several versions of the consumption-based CAPM have been developed. In one of the more applicable versions, the addition of assumptions about the preferences of investors allows the model to explain the risk premia on assets through their covariance with consumption growth, so that the model, to a degree, can explain variations in the excess returns of risky assets over time. Other versions of the model allow time-varying investor risk aversion to explain predictable movements in risk premia.

³⁹ This is equal to the discounted ratio of marginal utilities for the representative agent in two successive periods.

In a regulatory setting, the consumption CAPM can be used to either project the expected risk premium over the risk-free rate or verify the relied-upon market risk premium. The model has not commonly been used in a regulatory setting, but a recent implementation of Ahern, *et al.* (2012)⁴⁰ was developed explicitly to estimate the cost of equity for regulated entities. The description below therefore focuses on this version of the model.

The Ahern model is estimated using a so-called GARCH-in-mean (GARCH-M) model, which unlike the Sharpe-Lintner CAPM allows for the stock returns to depend on a volatility (variance) measure. In particular, the GARCH-M specification is such that the expected risk premium on a stock is a linear function of its conditional volatility. In this model, the parameter of interest, α , which represents the linear relationship between the risk premium on the stock and the conditional volatility in the GARCH-M model, can be translated into the following implication of the theoretical asset pricing model described above:

$\alpha = -\frac{vol_t[M_{t+1}]}{E_t[M_{t+1}]}corr_t[M_{t+1}, R_{t+1}]$	(4)
---	-----

where R_{t+1} is the expected total return on the public utility stock index or individual utility stock, and M_{t+1} is the stochastic discount factor (SDF), *i.e.*, the (aggregate) consumption intertemporal marginal rate of substitution. The equation above implies that the coefficient on volatility will be positive (*i.e.*, returns and conditional volatility will be positively correlated) if the conditional correlation between the SDF and the asset return is negative, *i.e.*, if the stock is not a hedging asset.

Ahern, *et al.* (2012) estimate the conditional risk-return model using monthly total returns from January 1928 to December 2007 on the S&P Public Utilities stock index, and the monthly Moody's public utility Aa, A, and Baa yields for the cost of debt. The authors then compare the model's performance with the performance of, for example, the Sharpe-Lintner CAPM. The estimates of the cost of common equity from the model are similar to the CAPM values and

⁴⁰ P.A. Ahern, F.J. Hanley, R.A. Michelfelder, —New Approach to Estimating the Cost of Common Equity Capital for Public Utilities,” *Journal of Regulatory Economics*, 2012 (Ahern, *et al.* 2012)

appear to be stable and consistent over time. Thus, the empirical implementation of the theoretical model resulted in cost of equity estimates that appeared to be within a range of reasonableness. The model has been presented in some U.S. regulatory jurisdictions but regulatory decisions based on the model are either still pending or it is not clear how the regulator used the information. Ahern, *et al.* conclude that the consumption-based asset pricing model “should be used in combination with other cost of common equity pricing models as additional information in the development of a cost of common equity capital recommendation”.⁴¹

3. Characteristics of CAPM Variations

As for the CAPM, the strengths and weaknesses of the variations discussed above depend on the implementation of the models. However, some strengths of the models are:

- Both the ECAPM and the Consumption CAPM allow for empirically observed phenomena to be modeled:
 - The ECAPM recognizes the flatter-than-predicted-by-CAPM Security Market Line.
 - The Consumption-CAPM allows for the expected risk premium to vary with asset and investor characteristics, such as conditional volatility and risk aversion.
- Data needed for the models are usually available if applied to companies with a reasonable trading history in well-developed markets. The models are therefore also auditable.
- The models are sensitive to economic conditions. The Consumption-CAPM considers more factors than does the CAPM.

Among the weaknesses of the models are the following:

⁴¹ Ahern, *et al.* (2012), p. 17.

- The ECAPM has not been tested extensively outside the U.S. or in recent market conditions.
- The Consumption CAPM relies on the use of more data than does the CAPM and requires a refined estimation process, which makes it less accessible to a broader audience.

C. THE FAMA-FRENCH THREE-FACTOR MODEL

The Fama-French model holds that the expected return of a security is described by an augmented CAPM relationship:

$$E(r_S - r_f) = \beta_S \cdot E(r_M - r_f) + s_S \cdot E(SMB) + h_S \cdot E(HML) \quad (5)$$

where $E(r_M - r_f)$ is the market risk premium (MRP) as used in the CAPM, SMB is the difference in returns between small companies and big companies (*Small Minus Big*), and HML is the difference in returns between securities of firms with a high book-to-market equity ratio and a low one (*High Minus Low*). The factor loadings s_S and h_S represent security S 's ~~holding~~ of each of these risk factors, which is to say they are the regression coefficients of r_S on each of the factors.

Evolution of the Fama-French Three-Factor Model

Fama and French (1992) was the last influential paper in a series of academic research into the placement of the empirical SML relative to the theoretical CAPM. Controlling for firm size, the authors found no relationship between the market and expected return (zero beta). Stated differently, any explanatory power that the market beta in the CAPM might have is absorbed by using size to explain the cross-sectional variation in returns. Fama and French interpreted this to mean that market beta (and by extension the CAPM) had zero explanatory power for expected returns. Moreover, they found that all of the variation in returns that were (in other research) associated with size, earnings/price ratios, book-to-market equity ratios, and leverage, could be captured by size and the book-to-market equity ratio alone. Fama and French (1993) ultimately settled on a three-factor model that brought the market return back into the model (size, book-to-market ratio, and market return). Their 1993 paper found that this model explained 90 percent of

the variations in the cross-section of returns, and it has since become known as the Fama-French three-factor model.

From an empirical perspective, the Fama-French model is an alternative to the ECAPM – one should not employ a Fama-French model with an alpha adjustment (Equation (3)). However, the interpretation of the findings of Fama and French has been critiqued by many academics as the size and book-to-market factors may proxy for other phenomena.⁴²

Standard Implementation:

The SMB factor and HML factor are typically created following Fama & French's (1993) approach. Specifically, at each point in time one allocates each firm into the small or big category, according to whether its market cap is in the top or bottom half of all firms considered. The firms in each half are then value-weighted to form two portfolios: small firms and big firms. The difference in realized returns between each of these portfolios is then taken as the SMB realization in that period. Creation of the HML series is similar, but firms are allocated to the "high" category if their book-to-market ratio is in the top 30th percentile and to the "low" category if their book-to-market ratio is in the bottom 30th percentile. These two time series can then be used to estimate the average SMB and HML, as well as the factor loadings for a given security; *i.e.*, the factors in the regression version of Equation (5), β_S , s_S , and h_S are estimated.

As a practical matter, the SMB and HML factors can be obtained free of charge from Professor Kenneth French's website,⁴³ where he maintains a database of the factors for regional areas such as Asia-Pacific, Europe, and North America.

⁴² For a discussion of this critique, see, for example, Black, F., "Beta and return," *Journal of Portfolio Management* 20, 1993, pp. 8-18; A.C. MacKinlay, "Multifactor Models Do Not Explain Deviations from the CAPM," *Journal of Financial Economics* 38, 1995, pp. 3-28; A. Lo and A.C. MacKinlay, "Data-Snooping Biases in Tests of Financial Asset Pricing Models," *Review of Financial Studies* 3, 1990, pp. 431-467; Fama, E. and K.R. French, "Size and Book-to-Market Factors in Earnings and Returns," *Journal of Finance* 50, 1995, pp. 131-155; and Fama, E., and K.R. French, "Industry costs of equity," *Journal of Financial Economics* 43(2), 1997, pp. 153-193.

⁴³ The website is located at http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Regulatory Use

The Fama-French model has been submitted in Australia, North America, and the U.K.⁴⁴ While U.S. decisions are only rarely explicit about how evidence was weighted, we are not aware of a U.S. decision that primarily relied on the Fama-French model. However, the U.K. Competition Commission used the model to determine whether a small company premium should be included in the cost of capital.⁴⁵ The Régie de l'énergie in Québec considered the Fama-French approach and found that the model had not been sufficiently examined to date to be used as a basis for setting the rate of return for a gas distributor.⁴⁶

Characteristics of the Fama-French Three-Factor Model

Many of the Fama-French model characteristics are similar to those of the CAPM. It relies on a risk-free rate and an estimate of the market risk premium, so like the CAPM it is sensitive to developments in risk-free rates. Like the ECAPM, the Fama-French model captures the empirical observation that the Security Market Line predicted by the CAPM is too steep. The Fama-French model has two additional factors, which vary over time and therefore add to the variations in the cost of equity estimates over time.

D. ARBITRAGE PRICING THEORY

The Arbitrage Pricing Theory (APT) was developed by Ross (1976a, 1976b)⁴⁷ as a multifactor alternative to the CAPM. The model is a theoretical approach to explaining the cross-section of returns with additional factors beyond the standard market portfolio in the Sharpe-Lintner CAPM. It is a one-period model in which all investors believe the stochastic properties of capital assets' returns are consistent with a factor structure. Assuming equilibrium prices offer no arbitrage opportunities, the expected returns on these capital assets are approximately linearly

⁴⁴ See, for example, Jemena Gas Networks (NSW) Ltd - Initial response to the draft decision - Appendix 5.2 - NERA: Cost of Equity – Fama-French Model; California Public Utilities Commission, "Decision 07-12-049," December 20, 2007; and U.K. Competition Commission, "Market Investigation into Supply of Bulk Liquefied Petroleum Gas for Domestic Use: Provisional Findings Report," August 2005, Appendix K.

⁴⁵ See, for example, U.K. Competition Commission, "Market Investigation into Supply of Bulk Liquefied Petroleum Gas for Domestic Use: Provisional Findings Report," August 2005, Appendix K.

⁴⁶ Régie de l'énergie, Décision D-2007-116, Gaz Métropolitain, pp. 23-24.

⁴⁷ S.A. Ross, "Options and Efficiency," *Quarterly Journal of Economics* 90, 1976, pp. 75-89 and S.A. Ross, "The Arbitrage Theory of Capital Asset Pricing," *Journal of Economic Theory* 13, 1976, pp. 341-360.

related to the factor loadings. The factor loadings are proportional to the returns' covariances with the factors - much like in the CAPM.⁴⁸

The empirical specification of the model is

$$E(r_s) = \beta_1 \cdot E(\text{Factor}1) + \beta_2 \cdot E(\text{Factor}2) + \dots + \beta_N \cdot E(\text{Factor}N) \quad (6)$$

The APT is a generalization of the standard CAPM in that it allows for multiple risk factors and does not require the identification of the market portfolio. However, the theoretical APT only provides an approximate relation between expected asset returns and a combination of factors. Therefore, testability of the model depends on imposing several additional assumptions on the conditional distribution of returns. For example, exact factor pricing holds in an equilibrium intertemporal asset pricing framework. In this general model specification, the market portfolio is one pricing factor as in the standard CAPM, and additional factors arise from investors' need to hedge uncertainty about future investment opportunities. These factors can be specified as traded portfolios of assets, or macroeconomic variables that reflect the systematic risks of the economy, such as industrial production growth, changes in bond yield spreads or unanticipated inflation.

The key difference between factor specification in the APT versus the Fama-French model described above, is that the factors in the APT are theoretically motivated as hedging variables that capture economy-wide non-diversifiable risks, whereas the factors in the Fama-French model are empirically motivated, and are instead selected based on observing the firm characteristics that best explain the cross-section of returns over a specific sample period.

E. DIVIDEND DISCOUNT MODEL

Although there are several versions of the Dividend Discount Model (DDM), all versions determine today's stock price as a sum of discounted cash flows that are expected to accrue to shareholders. Assuming that dividends are the only type of cash payment to shareholders, the pricing formula becomes:

⁴⁸ For a brief introduction, see Gur Huberman, —Arbitrage Pricing Theory,” in *The New Palgrave: Finance*, eds. J. Eatwell, M. Milgate, and P. Newman, 1989, pp. 72-80.

$$P_t = \frac{E_t(D_1)}{(1+r_s)} + \frac{E_t(D_2)}{(1+r_s)^2} + \frac{E_t(D_3)}{(1+r_s)^3} + \dots \quad (7)$$

where P_t is the market price of the stock; D_i is the dividend cash flow at the end of period i ; r_s is the cost of capital of asset/security S (as before); and the sum is into the infinite future.⁴⁹ The formula above says that the current stock price is equal to the sum of the expected future dividends (or cash flows), each discounted for the time and risk between now and the time the dividend is expected to be received – with the cost of capital r_s as the appropriate discount rate. The notion that the current stock price equals the present value of expected future dividends” was first developed in 1938 by Williams and was then rediscovered by Gordon and Shapiro in 1956.⁵⁰

1. Single-Stage DDM

If the dividend growth rate is constant, then we obtain the standard Gordon Growth model,⁵¹ which can be shown to determine the cost of capital on security S as:

$$r_s = \frac{D_o \times (1+g)}{P} + g \quad (8)$$

where g is the constant, periodical growth rate.

This equation says that the cost of capital equals the expected dividend yield (dividend divided by price) plus the (perpetual) expected future growth rate of dividends. As is readily seen from Equation (8) above, an implementation of the constant growth DDM requires a determination of the current stock price, current dividends, and the applicable growth rate.

⁴⁹ With the convention that D_i is zero for periods beyond the expected life of the asset.

⁵⁰ See Brealey, Myers, and Allen (2011), p. 82.

⁵¹ Named after Myron J. Gordon, who published an early version of the model in “Dividends, Earnings and Stock Prices,” *Review of Economics and Statistics*, Vol. 41, 1959, pp. 99-105.

2. Multi-Stage DDM

If the assumption of constant growth is not considered reasonable for several years before settling down to a constant rate, variations of the general present value formula can be used instead. For example, if there is reason to believe that investors do *not* expect a steady growth rate forever, but rather have different growth rate forecasts in the near term (*e.g.*, over the next five or ten years) converging to a constant terminal growth, these forecasts can be used to specify the early dividends in Equation (7). Once the near-term dividends are specified, Equation (8) can be used to specify the share price value at the end of the near term (*e.g.*, at the end of five or ten years), and the resulting cost of capital can be determined using a numerical solver. A standard “multi-stage” DDM approach solves the following equation for r_s :

$$P = \frac{D_1}{(1+r_s)} + \frac{D_2}{(1+r_s)^2} + \dots + \frac{D_T + P_{TERM}}{(1+r_s)^T} \quad (9)$$

The terminal price, P_{TERM} , is just the discounted value of all of the future dividends after constant growth is reached and T is the last of the periods in which a near-term dividend forecast is made. The implementation of the multi-stage growth model requires, in addition to a current price and current dividend, the selection of growth rates for each stage of the model and a determination of the length of each period.

More recent DDM implementations have focused on variations of the multi-stage model described above. For example, the U.S. Surface Transportation Board relies on a version of the multi-stage DDM that uses cash flow rather than dividends and specifies three growth rates – a near-term company-specific growth rate, an intermediate industry-specific growth rate and a long-term economy-wide growth rate.⁵² The STB version is identical to the model developed by Morningstar / Ibbotson, Ibbotson’s “three-stage” DDM, which is one of five models calculated for all U.S. SIC codes annually. In Ibbotson’s version, dividends are replaced by cash flow (excluding extraordinary items) and the figure is normalized over a three-year period. The model then uses company-specific growth rates from analysts over the first five years, industry growth rates over the next five year and the GDP growth rate after year 10.

⁵² See Surface Transportation Board, STB Ex Parte No. 664 (Sub-No. 1), “Use of a Multi-Stage Discounted Cash Flow Model in Determining the Railroad Industry’s Cost of Capital,” January 28, 2009. The Alberta Utilities Commission, Decision 2009-216 (¶271) also specifies a preference for the multi-stage model.

Another example of more recent multi-stage DDMs used is the version frequently estimated by *Brattle*, where company-specific growth rates are used for the first five years while the long-term GDP growth rate is used from year 10 onwards. In the in-between years (6-10), the model assumes that the growth rates converge linearly from the company-specific rates to the GDP growth rate. Similarly, Professor Myers' report suggests that in many industries it is important to look at the total cash flow that accrues to shareholders rather than on a per share basis, because stock buyback programs make the per share figures less reliable. In this model, the fundamental variable being determined is the market value (total price) of a company rather than the price per share, and instead of looking to dividends per share the model uses total cash flow to shareholders.⁵³

3. DDM Implementation Issues

To implement the DDM it is necessary to specify one or more growth rates and to determine whether (i) dividends accurately reflect cash flow to shareholders, (ii) the horizon over which to apply each growth rate if using a multi-stage model, and (iii) the exact determination of the initial stock price. In most applications, the choice of growth rate is the most controversial part of the DDM implementation and the determination of the stock price is the least controversial.

4. Characteristics of the DDM

As for the other models, many of the strengths and weaknesses of the DDM depend on its implementation. However, assuming a reliable implementation, some strengths of the DDM are:

- Both the single-stage and the multi-stage DDM rely on forward-looking information and hence estimate a forward-looking cost of equity.
- The models are usually easily replicated and are therefore easy to audit.

Among the weaknesses of the DDM are the following:

- The DDM relies on growth forecasts, which frequently are available only for 2-5 years.

⁵³ This revised method is explained in R. A. Brealey, S. C. Myers and F. Allen (2013), *Principles of Corporate Finance*, 11th Ed., McGraw-Hill Irwin, Ch. 16 (forthcoming).

- Because stock prices (and to a degree forecasted growth rates) change frequently, the model results often vary substantially over time.

Among the other issues to consider is the prevalence of stock buybacks, which means that dividends do not reflect all cash payments to shareholders. As mentioned above, some regulated entities have share buyback programs. In the pipeline industry, Spectra Energy, a U.S. based pipeline company, recently authorized share buybacks of \$600 million for a little over 6% of its equity capital.⁵⁴

Therefore, it is necessary to modify the model to take into account these cash transfers. In addition, for many companies, growth rates are only available on an infrequent basis, making the cost of equity estimates less forward-looking than ideal.

Both the single-stage and multi-stage DDM are frequently used in U.S. rate regulation to estimate the cost of equity. However, it is important to recognize that few U.S. regulators have a pre-specified methodology, but instead hear and review evidence from a variety of parties prior to issuing a decision on the cost of equity. Therefore, estimates from DDMs are only one of several pieces of evidence considered by most U.S. regulators. In addition, U.S. regulation was in place prior to the development of more market-based models such as the CAPM, and there is therefore a tradition to rely on the DDM.

5. Residual Income Model

One model that can be viewed as an extension of the multi-stage DDM is the residual income model, which relies on earnings or abnormal earnings instead of dividends. Broadly speaking, the model defines price as the sum of the book value of equity and the discounted present value of ~~abnormal~~ or ~~residual~~ earnings.⁵⁵ The model is a forward-looking methodology in that it generally uses analysts' forecasts to determine growth rates, although it uses historical earnings information to derive the current ~~residual income~~. The model is based on the so-called Ohlson-Juettner method, which like the multi-stage DDM allows growth rates to vary over time.

⁵⁴ See Spectra Energy, Form 10-K, 2008 p. 31.

⁵⁵ For an early exposition, see J. Ohlson, ~~Earnings~~, book values, and dividends in equity valuation," *Contemporary Accounting Research* 11, pp. 661-687.

Abnormal earnings are typically forecast using earnings estimates for one or two years ahead. Assuming that abnormal earnings in the long run grow at the assumed long-run rate, the model allows for a high short-term earnings growth rate that gradually declines to the long-term level. Technically, the model is appealing because it provides a closed form solution to the cost of equity based on few inputs, so that it is simple to implement.⁵⁶

The Residual Income Valuation (RIV) method has been debated substantially in the accounting literature in recent years. Variations on this model have been cited in recent Australian cases – for example, the “residual income model” proposed by the DBNGP in its most recent access arrangement.⁵⁷ The model was also proposed to the STB, albeit the STB instead adopted Ibbotson’s three-stage DDM model based on cash flows rather than dividends.

In a recent paper by Nekrasov & Shroff (2009)⁵⁸ the authors propose a valuation methodology that applies risk measures based on economic fundamentals directly into the valuation model, aiming to assess the differences in valuation derived from the use of fundamentals-based risk adjustments instead of the commonly used asset pricing models (estimated using historical returns). Note that this paper does not specifically address valuation and cost of equity for the regulated entities.⁵⁹

The authors use the RIV model to derive an accounting-based risk adjustment, which is equal to the covariance between a firm’s ROE and economic factors. Accounting risk factors are identified and used to construct a measure of risk adjustment, then applied to calculate firm value. Two components of value are estimated separately: the risk-free present value (RFPV) and

⁵⁶ The model was also submitted for consideration to the U.S. STB; P.S. Mohanram, *Determining an Appropriate Cost of Capital for Railroads*, submission to the Surface Transportation Board, September 2007.

⁵⁷ See *Draft Decision on Proposed Revisions to the Access Arrangement for the Dampier to Bunbury Natural Gas Pipeline*, paragraphs 458-467. Tristan Fitzgerald, Stephen Gray, Jason Hall and Ravi Jeyaraj, 2010 —“Unconstrained estimates of the equity risk premium,” Working paper, The University of Queensland, <http://ssrn.com/abstract=1551748> (—Fitzgerald et al.”).

⁵⁸ A. Nekrasov & P. Shroff, —“Fundamentals-Based Risk Measurement in Valuation,” *The Accounting Review* 84, 2009, pp. 1983-2011.

⁵⁹ See example of models submitted in regulatory settings; see Fitzgerald et al. and Partha Mohanram, —*Determining an Appropriate Cost of Capital for Railroads*,” Submission to the U.S. Surface Transportation Board, September 2007.

the covariance risk adjustment. The RFPV is calculated using a forecast of earnings, book value of equity and the risk-free rate as inputs to the model, while the covariance risk adjustment is estimated by calculating betas on the different risk factors and corresponding factor risk premia.

The authors acknowledge that this methodology ~~may~~ be more complex to implement than the returns-based cost of equity.”⁶⁰ However, the authors conclude that the strong empirical performance of the one-factor accounting-beta model, combined with the need of few additional inputs for the estimation, justify its use in valuation applications.

6. Characteristics of the Residual Income Model

The pros and cons of the Residual Income Model are generally similar to those of the DDM model, but we note that the model considers earnings instead of dividends, so that if earnings and cash flows are reasonably consistent, this model better captures the totality of cash flow that accrues to shareholders.

F. OTHER MODELS, METHODS, MARKET DATA AND EVIDENCE

1. Risk Premium Approaches

Some regulators in North America use a simplified version of the CAPM, the so-called risk-premium approach, which collapses the beta and risk premium to one figure and adds this figure to an interest rate. The debt instrument is either government bonds or utility bonds. The risk premium approach calculates the cost of equity, r_S , as:

$$r_S = r_D + \text{estimated risk premium} \quad (10)$$

where r_D is the return on a selected debt instrument. There are many versions of this model depending on the choice of the debt instrument, r_D , and the estimation of the risk premium. It is important to note here that the risk premium approach, while a generalized form of the CAPM, does not have the same level of theoretical support as the standard CAPM. This is because the return on the selected debt instrument used is not necessarily equal to the risk-free rate, and the

⁶⁰ *Ibid.* p. 1986.

estimated risk premium used is not explicitly based upon the product of the market beta and the MRP.

Equation (10) is frequently implemented using either a historical estimate of the risk premium, or a forward-looking or expected risk premium. The historical risk premium is commonly determined as the historical spread between equity and debt returns, so the primary choices for the analyst become which equity returns and debt instrument to use, as well as the period over which the spread (*i.e.*, the risk premium) is to be measured. It is not uncommon to see this model implemented using long-term government bonds or utility/corporate bonds to measure the cost of debt, while the equity investments used are typically either (a) realized accounting returns of regulated entities in the same industry, (b) realized stock returns of companies in the same industry, or (c) allowed returns on equity for the industry. In choosing a debt instrument to determine r_D , it is important that it be consistent with the debt instrument used to determine the risk premium. In other words, if a 10-year government bond is used to determine the historical risk premium, then r_D must also be measured using a 10-year government bond. The realized risk premium is highly dependent on the time period over which it is estimated, so that choice is also important.

The forward-looking model requires that the analyst determine a proper measure of the expected cost of debt and estimates the expected risk premium going forward, rather than relying on historical data. Determining the expected equity return is more difficult and requires reliance on an estimation technique. It is common to rely on DDM models to determine the risk premium in the forward-looking version of the model. One result originating from these analyses of historical or forward-looking risk-premium approaches is that empirically there is a negative relationship between the risk premium and the yield-to-maturity. Historically, a 1% increase in the yield-to-maturity of government bonds results in less than a 1% increase in the estimated (or realized) return on common equity.⁶¹ The relationship between the return on equity and

⁶¹ For example, Roger A. Morin, *New Regulatory Finance*, Public Utilities Reports, Inc., 2006 pp. 128-129 summarizes several studies and found that the realized ROE changes approximately 50 basis points when government bond rates change 100 basis points. Regulatory agencies such as the Ontario Energy Board relied on this empirical finding as well as data submitted by experts in its recent hearing to update its annual change in the estimated cost of equity for Ontario utilities by less than the change in government bond rates.

(government or utility) bond yields is depicted in Figure 4 below. The figure is for illustrative purposes only and does not reflect an actual analysis of the relationship.

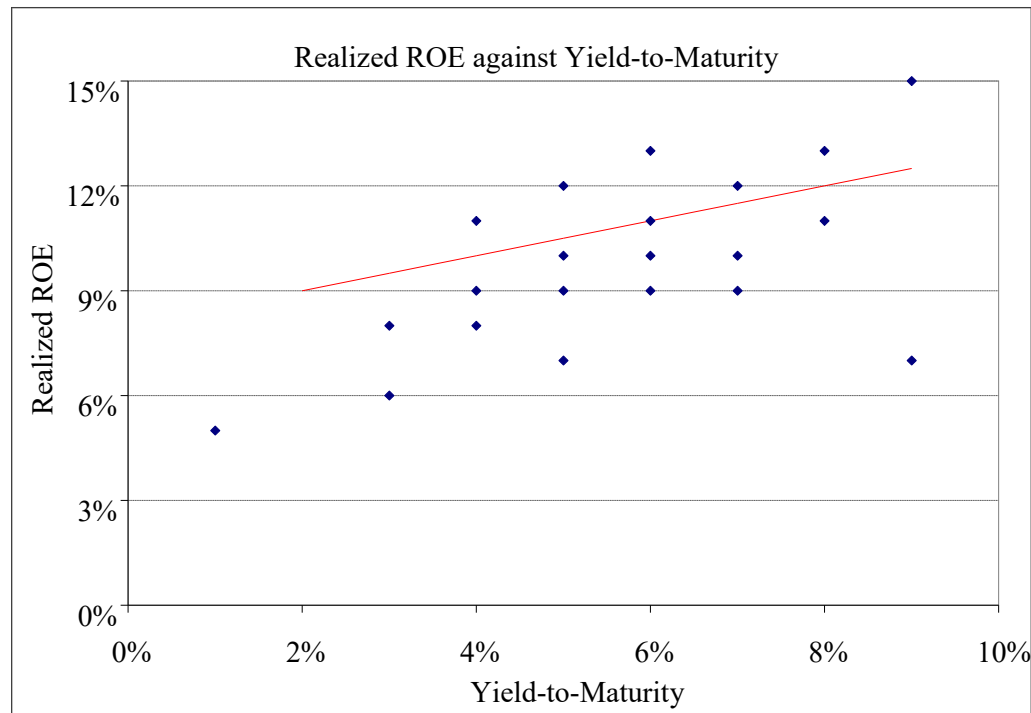


Figure 4

This is a reason why, for example, the Ontario Energy Board (OEB) took evidence from the risk premium approach into consideration when determining its baseline cost of equity in 2009.

2. Build-up Method

The build-up method estimates the return on an asset as the sum of a risk-free rate and one or more risk premia that represent the rewards an investor receives for taking on a specific risk:⁶²

$$\begin{aligned} \text{Cost of Equity} = & \text{Risk-Free Rate} + \text{Market Risk Premium} \\ & + \text{Firm Size Premium} + \text{Industry Premium} \\ & + \text{potentially other factors} \end{aligned}$$

⁶² *Morningstar Ibbotson SBBI Valuation Edition 2012 Yearbook*, p. 27.

Each of the components of the build-up method is discussed in detail below:

- The Risk-Free Rate is calculated using either Treasury bills (“T-bills”) or long-term government bonds.
- The Market Risk Premium reflects the compensation above the return on a risk-free asset that investors require for the additional market risk they bear by investing in a well-diversified market portfolio of risky assets. Ibbotson calculates this as the difference between the total expected return on the market portfolio and the risk-free rate.
- The Firm Size Premium may be included to account for the additional risk inherent in small company stocks. A firm size premium can either be adjusted or unadjusted for the effect that a small company stock’s higher beta has on its excess return. To illustrate the magnitude of the size premium, Table 1 below shows the empirically observed size premium for U.S. companies as reported by Ibbotson Associates.

	Beta- Adjusted Size Premia (%)	Non-Beta- Adjusted Small Stock Premia (%)
Mid-Cap	1.1	1.9
Low-Cap	1.9	3.4
Micro-Cap	3.9	6.3
Small Company Stocks	3.1	4.7

Table 1: Ibbotson Associates’ Size Premia on a Beta-Adjusted versus Non-Beta-Adjusted Basis, 1926-2011⁶³

- An Industry Premium can be determined based on the characteristics of the regulated entity’s industry. Research has produced no consensus on this figure and Ibbotson notes that it is important to avoid double-counting industry risk by using other beta-adjusted (hence industry dependent) risk premia (positive or negative) and at the same time adding an industry premium.

⁶³ *Morningstar Ibbotson SBBI Valuation Edition 2012 Yearbook*, p. 27.

In addition to the factors discussed above, some argue for the inclusion of minority discount premia, control premia, key person discount, *etc.* However, these additional premia (positive or negative) are very difficult to measure and we know of no regulator that has included such additional factors. The New Mexico Public Regulation Commission in the U.S. has in the past used the build-up method as one of its methods to estimate the cost of equity.

3. Comparable Earnings

The comparable earnings method requires the analyst to go through three steps. First, a group of *unregulated* companies is required because the realized accounting rate of return of a regulated company depends on its allowed return. Using regulated companies to estimate the comparable earnings cost of capital would be circular, *i.e.*, the allowed rate of return is used to determine the allowed rate of return. However, the use of unregulated companies requires careful consideration of the risk characteristics of the companies and the comparability to those of the target utility.

Second, a time period over which to estimate the return on equity must be selected. Because a company's achieved earnings fluctuate from year to year and depend substantially on both company-specific and economy-wide factors, it is necessary to include companies from several industries, averaged over several periods.

Third, because the comparable companies are unregulated entities, it is necessary to adjust for any risk differences between the sample companies and the target company. There are many ways to adjust for risk differences, so the following is a simplified description of some common approaches rather than an exhaustive review. Analysts often collect information on the comparable companies' and the target company's bond ratings, asset betas, DDM estimates of the cost of equity, and other measurable risk factors. In many instances, this information is also collected for a sample of regulated companies in the same industry as the target company. If the sample companies are found to be consistently more (less) risky than the target company and its industry peers, then an adjustment is made to the required return on equity. This can sometimes be done formally. For example, if the sample companies' DDM estimates of cost of equity are consistently 25 basis points higher (lower) than the DDM estimates for the target company (or industry peers), then a downward (upward) adjustment of 25 basis points is made. For other

measures, it is more difficult to determine the exact adjustment, so it is usually made based on the analyst's experience. For example, does a two notch difference in bond rating require a specific upward or downward adjustment? Thus, while the differences are relatively easy to measure, the adjustment for such differences requires subjective judgment.

A major issue is whether realized book returns are a good proxy for the returns that investors expect going forward. From a statistical perspective, the realized accounting return on book equity for any given period is the realization of a single outcome of a distribution, whereas the expected return represents the probability-weighted average of all possible outcomes of the distribution. These two figures can differ substantially. In addition, there are practical problems with the implementation of this model because financial reporting occurs with a lag, which during times of change can mean that the results are out of date.

4. Market-to-Book and Earnings Multiples

In some regulatory decisions on the cost of capital, regulators have sought to “cross check” a proposed cost of capital estimate by examining the market value of the firms they regulate relative to the value of the regulatory asset base (RAB). The theory behind this approach would be that the only capital on which the regulated firm is earning a return (at the regulator-determined cost of capital) is the RAB. Therefore, if the market value of the firm's returns is greater than the RAB, the belief is that it is a signal that investors are discounting future returns at a lower discount rate than the regulator's cost of capital determination — or, in other words, the regulator's cost of capital is “too high”.

This kind of cross check approach was cited by the Australian Energy Regulator in its June 2011 determination on Envestra.⁶⁴ In that decision, the AER considered two kinds of evidence: premiums paid in takeover transactions relative to the value of the RAB, and market values (based on share prices) relative to RAB.

⁶⁴ *Final Decision - Envestra Ltd Access arrangement proposal for the Qld gas network*, AER (June 2011), p. 35-37.

a) Takeover premiums

The AER reviewed premiums paid in takeover transactions, where the premium was assessed as the sale price relative to the value of the underlying RAB. Premiums were in the range of 20% to 120%. The AER considered that these premiums were too large to be explained by factors such as expected synergies, and instead considered this as evidence that the cost of capital determined by regulators has been at least as high and likely higher, than the actual cost of capital faced by the businesses.

However, there are conceptual problems with this approach so that it has no value as a cross check on a regulator's cost of capital determination. First, the reliance on the approach implicitly assumes that (i) the company to which it is applied consists entirely of regulated businesses and (ii) that the regulator's cost of capital determination is the only factor impacting the market value of the company. In reality the cost of equity is only one component of a broader determination on what the firm's regulated rates should be. Thus, even if it were possible to estimate the impact of the regulator's decision on the market value of the firm, this impact would be associated with the overall decision, not with any one specific component (like the cost of capital). The market value of a regulated firm can be thought of as the expected future cash flows (from providing services at regulated rates), discounted at the firm's actual cost of capital. However, the regulator's cost of capital determination is only one of many factors which determine expected future cash flows, particularly where price determinations are forward-looking (as in Australia):

- If investors expect the firm to "beat" regulator assumptions on any of operating costs, capital costs, or revenue growth, expected future cash flows would be larger than the RAB in net present value terms, even if the discount rate is equal to the regulator-determined cost of capital.
- Investor expectations, which are implicit within the firm's market value, encompass expected cash flows beyond the end of the current price control period.
- Expected future cash flows are also affected by firm-specific factors such as idiosyncratic volatility, which would not be captured in the discount rate.

In addition, there are likely to be other more practical difficulties: for example, many regulated firms have at least some unregulated activities. These activities are valued by investors but are not part of the RAB or the regulator's cost of capital decision.

b) Trading premiums

The AER also considered premiums measured on the basis of market value of listed firms (from share prices) relative to RAB. The AER estimated market-to-RAB trading multiples for four firms (including Envestra).⁶⁵ The trading multiples were in the range of 1.21 to 1.81.

The AER stated that these premiums were too high to be the result of factors such as expected synergies, and instead considered this as evidence that the cost of capital determined by regulators has been at least as high and likely higher, than the actual cost of capital faced by the businesses. However, the same difficulties described above for takeover premiums also apply to the consideration of trading premiums. In addition to the takeover premiums difficulties, the use of trading premiums suffers from bias in circumstances where the market is very volatile, where day-to-day changes reflect investor reactions to news such as the collapse of Lehman Brothers in September 2008, the ongoing European debt crisis, or industry factors such as cap and trade initiatives, etc. Therefore, trading premiums also have no value as a cross check on the regulator's cost of capital determination.

5. Other Evidence

Other evidence is a very broad category that does not readily lend itself to a short introduction by method. However, expert evidence can be highly valuable if of high quality, so it will be necessary to use judgment and consider how the expert arrived at his or her recommendations. Similarly, academic research may provide insights into the cost of equity, but bear in mind that most academic research focuses on finding or explaining "interesting facts" and often considers all companies and industries for which data are available. Because a result pertains to the market

⁶⁵ The four firms were SP Ausnet, Spark, Duet and Envestra (*Ibid.*, Table 5.5).

as a whole, it does not necessarily pertain to a specific industry, which may have unique characteristics.

Other types of evidence that are sometimes considered are equity analysts' reports on a specific company, an industry, or a market. When such evidence is reviewed, it is important to consider the purpose for which the evidence was produced. For example, equity analysts often produce research documents aimed at stock-buying investors and only rarely are concerned with the cost of equity over, for example, a regulatory period. Instead, equity analysts attempt to determine the current (or future) stock price as the discounted sum of future cash flows with the discount rate being the weighted average sum of the cost of debt and equity; *i.e.*, the focus is not on what the best estimate of the cost of equity is – it is merely one of many inputs to determining the stock price. In addition, because a lower cost of equity increases the estimated stock price, equity analysts have an incentive to, if anything, bias the cost of equity estimates downward.

6. Characteristics of Other Methods, Models, Market Data and Evidence

The methods, models, market data and other evidence in this section differ, so the advantages and disadvantages listed below are method-specific:

- The risk premium model is simple and data for its implementation are readily available.
- If the benchmark interest rate is a utility or corporate bond index, then the risk premium model tends to provide relatively stable results over time and is less impacted by monetary policy or country-specific risks than the CAPM.
- The build-up method recognizes size effects and potentially other risks.
- The comparable earnings method's strength is that it incorporates information from non-regulated entities.

Among the weaknesses of the methods we note the following:

- None of the methods are founded in economic or finance theory.
- The risk premium approach does not consider systematic risk specifically and does not allow for company-specific information to be considered.

- The build-up method generally does not consider systematic risks and treats size effects the same across industries.
- The comparable earnings model relies on historic accounting information, which may not be consistent with investor expectations. Also, the historic accounting information may reflect accounting choices rather than economic fundamentals and may be subject to significant variability over time.

As for other evidence such as expert reports and investment reports, the merits of the derived estimates are highly dependent upon the quality of the reports and the purpose for which the estimates were derived. We caution against placing weights on estimates where the purpose for their derivation is not known, and against placing substantial weight on estimates that were derived for purposes other than to provide an independent assessment of the cost of equity. For example, estimates derived for accounting purposes, stock recommendations, *etc.* may not be suitable for other uses.

This section has summarized the major models, methods and evidence that are currently used and considered by regulators and practitioners. The models described above are not intended to comprise an exhaustive list of all possible methods and evidence that could be relied upon in determining the cost of equity capital. Indeed, as the practice of finance continues to evolve, further relevant evidence may still be found, and certain models may become outdated or less relevant.

IV. USING THE METHODS

In this section, we first discuss implementation issues for estimating the cost of capital and summarize the key characteristics of the models described above in *Section III*. We then address the issue of how and when to use the models to determine an appropriate regulatory return on equity, or range for the regulatory return on equity for the industry or benchmark, based on the views of academic, practitioners and regulators. Finally, we discuss how to position a target entity relative to a sample of companies.

A. IMPLEMENTATION ISSUES

Regardless of the cost of equity estimation method that is used to estimate the cost of capital, there are some key elements of the cost of capital estimation process that must be addressed. This section discusses some of the important issues.

Most analysts rely on a “comparable sample” to determine the cost of equity for the target entity, so it becomes important to determine what is meant by comparable.⁶⁶ Although the selection of comparable companies is method and context-specific, it is generally viewed as ideal to have sample companies with business risk similar to that of the target company. Similar business risk generally implies selecting companies in the same line of business. Most researchers and practitioners rely on additional criteria to exclude sample companies that have the potential to bias the cost of capital estimation methodologies. For screening, it is preferable to rely on objective information from publicly available data sources; however, the determination of exactly which criteria to use is subject to the constraint that the sample be “large enough.” This, in turn, requires a determination of which criteria are the most important from the many possible criteria that could be considered. Among the criteria typically employed are combinations of the following:

- Include companies with similar business risks (e.g., companies in the same or similar industries);
- Exclude companies that face financial distress;
- Exclude companies that are or have recently been involved in substantial merger and acquisition activity;
- Exclude companies with unique circumstances that may bias the cost of capital estimation (e.g., restatements of financial statements); and
- Exclude companies with insufficient data.

⁶⁶ A comparable sample can be used to assess the cost of capital for the target entity by (i) estimating the individual companies’ cost of capital and placing the target company’s cost of capital in relation to the sample using the average, median, range, or other measure to assess the cost of capital or (ii) using a portfolio approach, where the cost of capital for the portfolio of companies (rather than individual companies) is estimated to assess the cost of capital for the target entity.

There is, however, controversy about how to implement the criteria above. Each element of the sample selection criteria requires some judgment. For example, what size sample is ~~large~~ enough”? Should the sample include both Australian and foreign companies?⁶⁷ How is financial distress measured? How is ~~substantial~~ merger and acquisition” activity to be defined? The selection criteria are interrelated, because selection of the sample based upon one criterion may immediately reduce the potential sample to a small number of companies. The sample selection process is, therefore, a balancing act between selecting a sample that is ~~more~~ comparable” and one that is ~~too~~ small.”

Second, decision makers must decide how the components of the cost of capital will be determined. For example, it is possible to estimate (a) the cost of debt, the cost of equity and the capital structure, each separately or (b) an overall cost of capital or (c) a combination of these. Another component of the cost of capital is the allowance for income taxes, which we ignore in this report. Finally, because the dollar amount that accrues to investors in a regulated entity ultimately depends on not only the allowed cost of equity and the size of the rate base but also on the relative share of equity and debt in the capital structure, it is important to consider the overall impact of these capital structure decisions on the individual components. Specifically, it is important to note that cost of equity estimation models provide estimates that reflect both the underlying business risk of the assets but also the financial risk inherent in how those assets have been financed.

B. SUMMARY CHARACTERISTICS OF THE MODELS

Before we discuss how to use the various models and other information that may be available to a decision maker, we summarize in Table 2 below the key characteristics of the discussed models in the form of their economic underpinnings, any potential empirical bias, sensitivity to economic or industry factors, and whether the models are forward or backward-looking.

⁶⁷ For example, several Canadian regulators have used beta estimates from U.S. companies. See, for example, the National Energy Board’s RH-1-2008 decision p. 67 and Ontario Energy Board’s EB-2009-0084 decision, pp. 22-23.

Table 2: Characteristics of Cost of Equity Methods

Evaluation Criteria			
Cost of Capital Methods	Economic Underpinnings	Bias	Impact of Market Conditions
Sharpe-Lintner CAPM	An equilibrium model: Under no arbitrage and in a mean-variance-optimizing world, the expected cost of equity is a function of the risk-free rate, systematic risk (beta) and the expected MRP. Transparent and sensitive to market performance and risk-free rates. Empirical support for explaining cross-sectional returns of average-beta stocks, but failure for low-beta/high-beta/small/high book-to-market firms.	Empirical evidence that CAPM under-estimates the expected return for low-beta stocks. A Portfolio approach to estimate betas provides more consistent results. MRP estimation controversial with some historical measures potentially biased.	Beta estimates are backward-looking. Historical MRP is backward-looking, but forecast MRP (e.g., DDM) are forward-looking.
ECAPM	Same as above, but captures the empirical observation that the SML predicted by CAPM is too steep. Tested in the U.S., but not extensively outside the U.S. or in recent market conditions.	Corrects for the empirical bias induced by the flatter-than-predicted-by-CAPM Security Market Line.	Same as above.

(Table 2 ctd.) Characteristics of Cost of Equity Methods

Evaluation Criteria				
Cost of Capital Methods	Economic Underpinnings	Bias	Impact of Market Conditions	Forward or Backward-Looking
Consumption-Based CAPM	Generalization of the Sharpe-Lintner CAPM that relates the risk premium on the investment to the covariance between the asset return and the intertemporal marginal rate of substitution of the decision maker. The expected conditional risk premium on an asset is related to its predicted conditional volatility.	Allows for expected risk premium to vary with asset and investor characteristics, including conditional volatility and covariance with consumption growth; (considers more factors than the Sharpe-Lintner CAPM). Potentially mitigates empirical biases.	Empirical results appear stable and consistent over time (i.e. more robust to market conditions than standard CAPM).	Models forward-looking equity risk premia based on predicted conditional volatility. More forward-looking than Sharpe-Lintner CAPM.
	Requires more data and more refined estimation techniques than the Sharpe-Lintner CAPM. Lack of empirical support for most commonly used version (where covariance factor is aggregate consumption growth), but more support for versions with market frictions time-varying risk aversion.			

(Table 2 ctd.) Characteristics of Cost of Equity Methods

Evaluation Criteria				
Cost of Capital Methods	Economic Underpinnings	Bias	Impact of Market Conditions	Forward or Backward-Looking
Fama-French Model	Corrects for empirical biases of Sharpe-Lintner CAPM by adding 2 explanatory risk factors (size and book-to-market). Empirical support for explaining cross-sectional returns of size- and book-to-market-sorted portfolios. Weak empirical support for explaining returns of other portfolios and for out-of-sample predictive power. Fama-French factors (SMB and HML) are empirically motivated.¹	Captures empirical observation that SML predicted by CAPM is too steep. Adds cross-sectional explanatory power to the standard CAPM.	Sensitive to monetary policy. Estimates of the 3 factor risk premia vary substantially over time and more so when the 3 factors interact.	Betas and factor risk premia estimates are backward-looking.
APT Model	Equilibrium multifactor model which holds under competitive markets, factor structure for asset returns, and absence of arbitrage in large economies. Corrects for empirical biases of Sharpe-Lintner CAPM by adding explanatory risk factors. APT factors are theoretically motivated. Model implemented empirically as intertemporal CAPM (see above for empirical issues).	Same as above.	Sensitive to market uncertainty and economic turmoil via market-related factor. Estimates of factor risk premia can vary substantially over time.	Forward-looking model in theory, but betas and factor risk premia are backward-looking if estimated using historical data.

¹) See Fama and French (1993), Kothari, Shanken, and Sloan (1995), Black (1993), MacKinlay (1995), and Lakonishok, Shleifer, and Vishny (1994) for additional detail.

(Table 2 ctd.) Characteristics of Cost of Equity Methods

Cost of Capital Methods	Evaluation Criteria		
	Economic Underpinnings	Bias	Impact of Market Conditions
Risk Premium Model	Simplified version of the CAPM, which collapses the beta and market risk premium to one figure and adds this figure to an interest rate. Based on empirical estimation. Does not capture systematic risk or company-specific information.	The relied upon interest rate may be biased due to, e.g., monetary policy. Does not account for changes in the risk premium.	Using a utility or corporate bond index as the benchmark, the risk premium model tends to provide relatively stable results. Reliance on government interest rates makes the model more sensitive to monetary policy. Inflation leads to bias in the risk premium model, because the historical data underlying the estimate of the risk premium may not be consistent with the current level of inflation.
Single-Stage DDM	Determines today's stock price as the sum of the discounted cash flows that are expected to accrue to shareholders going forward. Assumes that dividends (or cash flows) grow at a constant rate forever. Lack of empirical support for constant dividends/earnings growth rates in perpetuity.	Sensitive to bias in analyst forecasts of earnings growth rates which at best reflect 5 years. Less of an issue for utilities than most other industries. Sensitive to the exact implementation as dividends may not reflect all cash flow if the company engages in share repurchases or borrows to fund dividends. Does not take real options into account and will underestimate the cost of equity for companies with substantial real options.	Relies on forecasted (i.e. forward-looking) growth rates and current stock prices. Hence estimates are forward-looking. Estimates are sensitive to changes in stock prices and forecasted growth rates, which is especially an issue if the industry is in transition. Stock prices are influenced by the information available to investors. Information about financial distress or merger and acquisition activities may overwhelm fundamental information about growth.
			Mostly backward-looking due to reliance on a historic spread of realized equity returns over debt returns to measure the risk premium, and thus does not capture expected changes in the economy.

(Table 2 ctd.) Characteristics of Cost of Equity Methods

Evaluation Criteria			
Cost of Capital Methods	Economic Underpinnings	Bias	Impact of Market Conditions
Multi-Stage DDM	Extension of single-stage DDM which allows for different growth forecasts over time. Stronger empirical support than constant growth version.	Same as above, but less sensitive to any bias in analyst forecasts than the single-stage DDM, as growth rates usually converge to the GDP growth rate over time.	Requires a series of growth rates; commonly near-term, interim, and for the very long-term. Sensitivity to growth rates is moderated. Similarly to single-stage DDM, less applicable to companies in financial distress or engaged in merger or acquisition activities.
Residual Income Method	Version of the multi-stage DDM that values abnormal or unforeseen earnings instead of dividends. Abnormal earnings are forecast using earnings estimates for one or two years ahead. Allows growth rates to vary over time. Abnormal earnings are based on empirical estimates.	Considers earnings instead of dividends or cash, so if earnings reflect expectations better than dividends or cash, it reduces bias.	Same as above. Performance relative to multi-stage DDM depends on implementation of each.
			Forward or Backward-Looking Same as above.

(Table 2 ctd.) Characteristics of Cost of Equity Methods

Evaluation Criteria				
Cost of Capital Methods	Economic Underpinnings	Bias	Impact of Market Conditions	Forward or Backward-Looking
Build-up Method	Estimates the return on an asset as the sum of a risk-free rate and several risk premia that measure risks associated with size, industry, etc. Based on empirical estimation.	Recognizes size effects and other industry or company-specific risks. Exposed to same potential biases as standard CAPM and Fama-French models.	Exposed to same market uncertainties as standard CAPM and Fama-French models.	If risk factors and factor loadings are estimated from historical data, then the model is backward-looking.
Comparable Earnings	The model calculates the realized accounting rate of return on book equity of comparable (usually non-regulated) companies. Based on empirical estimation. Uses accounting returns rather than market data.	Selecting a sample of non-regulated comparable companies may lead to bias. Accounting changes can produce changes in model estimates without any change in the underlying cost of capital. The choice of estimation period may bias the accounting return on equity as accounting returns vary with economy, industry and company-specific factors.	Realized accounting returns are sensitive to economic, industry and company-specific events as well as to changes in accounting rules.	Uses backward-looking realized accounting rates of return, hence backward-looking. Can be difficult to find a time period that accurately reflects the expected horizon of the regulated entity.

C. HOW TO USE THE MODELS AND OTHER INFORMATION

In this section we discuss how academics, practitioners and regulators think models should be used and how they have been used. The section also discusses the impact of economic conditions, industry factors and company-specific issues on the choice of models. The weight assigned to each model naturally depends on the key characteristics of the cost of equity estimation models described above. Finally, the section discusses how certain regulators have decided to use the models in specific economic environments.

1. Views of Academics, Practitioners and Regulators

Academics, practitioners and regulators have all acknowledged that there is no one way to determine the cost of equity. In the academic literature, several prominent researchers have commented that the use of more than one method is important. For example, Professor Myers of the Massachusetts Institute of Technology commented:

Use more than one model when you can. Because estimating the opportunity cost of capital is difficult, only a fool throws away useful information. That means you should not use any one model or measure mechanically or exclusively. Beta is helpful as one tool in a kit, to be used in parallel with DCF models or other techniques for interpreting capital market data.⁶⁸

Professors Berk and DeMarzo of Stanford University in their corporate finance textbook comment on the use of the CAPM, DDM, and other models by practitioners, and state:

In short, there is no clear answer to the question of which technique is used to measure risk in practice — it very much depends on the organization and the sector. It is not difficult to see why there is so little consensus in practice about which technique to use. All the techniques we covered are imprecise. Financial economics has not yet reached the point where we can provide a theory of expected returns that gives a precise estimate of the cost of capital. Consider, too, that all techniques are not equally simple to implement. Because the tradeoff between simplicity and precision varies across sectors, practitioners apply the technique that best suit their particular circumstances.⁶⁹

⁶⁸ Stewart C. Myers, —“the Use of Modern Portfolio Theory in Public Utility Rate Cases: Comment,” *Financial Management*, Autumn 1978.

⁶⁹ Jonathan Berk and Peter DeMarzo, *Corporate Finance: The Core*, 2009, (Berk & DeMarzo 2009) p. 420.

Looking to practitioners' views, the widely used text, *Ibbotson Cost of Capital Yearbook*,⁷⁰ reports results on the cost of equity (and associated weighted average cost of capital) by SIC code in the U.S. and other countries. In doing so, the yearbook reports the estimated cost of equity using five estimation methods: Sharpe-Lintner CAPM, CAPM plus/minus a size premium, Fama-French 3-Factor model, Single-Stage DDM, and 3-Stage DDM. The data source does not provide specifics on how to use the data but states that:

[r]eaders can select cost of equity from five different models explored in this book. Given the size of the database being analyzed, there will clearly be instances where certain cost of equity models will fail to produce useable numbers. When NMF is displayed in a cost of equity column, it indicates that the model is producing unreasonable numbers, and greater emphasis should be placed on other models.⁷¹

Similarly, Roger A. Morin, in the context of U.S. regulation, mentions the use of the CAPM, DDM, risk premium models, and the comparable earnings method, concluding:

No one individual method provides the necessary level of precision for determining a fair return, but each method provides useful evidence to facilitate the exercise of an informed judgment. Reliance on any single method or preset formula is inappropriate when dealing with investor expectations because of possible measurement difficulties and vagaries in individual companies' market data.⁷²

Looking to regulators, the U.S. Surface Transportation Board (STB) undertook a review of its cost of equity estimation methodology in 2007-09 in two rounds, focused on the CAPM and DDM respectively. The STB's review resulted in two decisions with detailed instructions on how to estimate the cost of capital for the railway industry.⁷³

In connection with this review, the STB noted:

⁷⁰ The most recent version is Morningstar, *Ibbotson Cost of Capital 2012 Yearbook* (Ibbotson 2012).

⁷¹ Ibbotson 2012, p. 6. The text views cost of equity estimates below the risk-free rate and above 50 percent as being not meaningful.

⁷² Roger A. Morin, *New Regulatory Finance*, Public Utilities Reports, Inc., 2006, (Morin 2006) p. 428.

⁷³ Surface Transportation Board, STB Ex Parte No. 664, "Methodology to be Employed in Determining the Railroad Industry's Cost of Capital," January 17, 2008 (STB 2008) and STB Ex Parte No. 664 (Sub-No. 1), "Use of a Multi-Stage Discounted Cash Flow Model in Determining the Railroad Industry's Cost of Capital," January 28, 2009 (STB 2009).

While CAPM is a widely accepted tool for estimating the cost of equity, it has certain strengths and weaknesses, and it may be complemented by a DCF model. In theory, both approaches seek to estimate the true cost of equity for a firm, and if applied correctly should produce the same expected result. The two approaches simply take different paths towards the same objective. **Therefore, by taking an average of the results from the two approaches, we might be able to obtain a more reliable, less volatile, and ultimately superior estimate than by relying on either model standing alone** [emphasis added].⁷⁴

In arriving at this conclusion, the STB took notice of comments from the Federal Reserve that ~~multiple~~ models will improve estimation techniques when each model provides new information,”⁷⁵ and also stated that there is ~~robust~~ economic literature confirming that, in many cases, combining forecasts from different models is more accurate than relying on a single model.”⁷⁶

Similarly, the Ontario Energy Board (OEB) reviewed its cost of capital estimation methodology in 2009 following a year-long process. For context, the OEB does not focus on the cost of equity, but instead determines the premium over the risk-free rate that rate-regulated utilities are allowed. Regarding the methods used to determine the so-called Equity Risk Premium (ERP), the OEB concluded:

the use of multiple tests to directly and indirectly estimate the ERP is a superior approach to informing its judgment than reliance on a single methodology.⁷⁷

Additional examples of regulators who have relied upon multiple cost of equity estimation models and/or judgment based on a range of evidence are discussed in the section below.

To sum up, as clearly illustrated above, many academics, practitioners and regulators find that it is preferable to use more than one estimation method to determine the cost of equity. We agree that it is important to use more than one estimation method and stress that in determining how to

⁷⁴ STB 2008, p. 2.

⁷⁵ STB 2009, p. 15.

⁷⁶ STB 2009, p. 15.

⁷⁷ Ontario Energy Board, ~~EB-2009-0084~~, Report of the Board on the Cost of Capital for Ontario’s Regulated Utilities,” Issued December 11, 2009, p. 36 (emphasis in the original).

weigh the estimation results, it is important to consider the degree to which the information from the methods overlaps versus providing additional information, the economic and financial environment that gave rise to the estimates, and the context in which they are being used.

2. Regulatory Practice in using Multiple Models

a) The U.S.

In the U.S., rates for rate-regulated entities are determined by several federal entities as well as regulators in each of the fifty states and the District of Columbia. Federal regulators tend to have well-specified methods to determine the cost of equity although they review all the information put to them. However, state regulators typically do not specify one single method and commonly have evidence from several estimation methods and parties in front of them before issuing a decision on the allowed cost of equity. In most cases the state regulator does not specify which weight was assigned to each method or other evidence. An exception is the determination of the cost of equity in Mississippi Power's Performance Evaluation Plan (PEP), where the Mississippi Public Service Commission annually updated the cost of equity for the company using a combination of the CAPM, ECAPM, risk positioning, and the DDM. In this specific circumstance, the weights assigned to each method are predetermined.⁷⁸ Some other examples of U.S. regulators' thought processes are provided below.

Surface Transportation Board

The STB used the constant growth model to track the cost of equity for U.S. railroads for a number of years. However, by 2005 the largest railroads were expanding rapidly and profitability was increasing. Security analysts were forecasting "long-run" earnings growth for some railroads at 15% per year. Such growth could not be sustained, so the constant growth model overstated the true cost of equity by a wide margin. The STB therefore initiated a cost of capital proceeding to consider how to change the determination of the cost of equity. After hearing evidence from academics and practitioners, the STB found that:

⁷⁸ <http://www.psc.state.ms.us/>.

if our exploration of this issue has revealed nothing else, it has shown that there is no single simple or correct way to estimate the cost of equity for the railroad industry, and countless reasonable options are available.⁷⁹

As a result of its deliberations the STB eventually settled on a blend of the CAPM and a multi-stage DDM.⁸⁰

Georgia

The following example pertaining to Georgia Power, an integrated electric utility, illustrates a common approach in U.S. state regulation.

Georgia Power is regulated by the Georgia Public Service Commission (Georgia PSC), which has no pre-set method to determine the cost of equity. In Georgia Power's 2010 rate case, an expert for Georgia Power as well as for the Georgia PSC submitted evidence on the cost of equity for the company. The company's expert estimated the cost of equity using the Sharpe-Lintner CAPM, a single-stage DDM, and a risk premium approach, and recommended a return on equity of 11.0 to 11.2%. The PSC staff expert estimated the cost of equity using the Sharpe-Lintner CAPM, a sustainable growth DDM and also a comparable earnings model for a recommendation of 9.50 to 10.75%. The Georgia PSC approved a settlement including a cost of equity of 11.15%, but did not specify how it was arrived at.⁸¹

b) Canada

Until the early 1990s, Canadian regulators, much like U.S. state regulators, heard evidence on a multitude of methods and from various experts before arriving at a decision on the allowed cost of equity. However, starting in British Columbia in 1994, the British Columbia Utilities Commission in the first generic cost of capital proceeding in Canada established a benchmark ROE and a formulaic approach to updating the allowed ROE annually.⁸² Shortly thereafter, other Canadian regulators followed suit and similarly established a benchmark ROE and an

⁷⁹ U.S. Surface Transportation Board, *Ex Parte 664 (Sub-No. 1)*, issued January 28, 2009, p. 15.

⁸⁰ *Ibid.*

⁸¹ Direct Testimony of J.H. Vande Weide in Docket No. 31958; Direct Testimony of D. Parcell in Docket No. 31958, and Settlement Agreement in Docket No. 31958.

⁸² BCUC Decision in the Matter of Return on Common Equity BC Gas Utility Ltd., Pacific Northern Gas Ltd., West Kootenay Power Ltd., June 10, 1994 (BCUC 1994 Decision), pp. 39-40.

annual updating formula. These formulae were linked to the change or forecasted change in government bond yields.

While the formula used to update the allowed ROE annually was mechanical, the methods used to estimate the benchmark ROE varied across jurisdictions, and in many jurisdictions, the regulator looked to more than one estimation method.⁸³

As the yield on government bonds declined, so did the allowed cost of equity, and as the financial crisis of 2008 impacted financial markets, regulators in Canada abandoned or modified the formula or the relied-upon benchmark. As was the case for the originally developed benchmark, the regulators heard evidence on multiple methods from several experts and implicitly or explicitly weighted these methods to arrive at a new or modified cost of equity methodology.⁸⁴ Some examples of this regulatory approach in Canada are provided below.

British Columbia

British Columbia Utilities Commission's (BCUC) views on how to determine the appropriate cost of equity capital have evolved over time. In the BCUC 1994 Decision,⁸⁵ the BCUC "placed primary reliance on the various risk premium tests presented" whereas the "comparable earnings and DCF test results have been used primarily as a check upon reasonableness."⁸⁶ However, in the BCUC 2006 Decision, the BCUC assigned weight to the DCF model and found the comparable earnings methodology useful.⁸⁷ The BCUC 2006 Decision did not state how much weight it assigned to each model it considered. The BCUC's views evolved as the various

⁸³ For example, the BCUC 1994 Decision at p. 17 indicated that while primary reliance should be placed on risk premium tests, comparable earnings and the DDM should be used as checks.

⁸⁴ For example, the National Energy Board abandoned the formulaic approach, the Alberta Utilities Board modified the benchmark, and the Ontario Energy Board modified both the benchmark and the formula. Both the Alberta Utilities Board and the Ontario Energy Board used several cost of equity estimation methods to arrive at their revised benchmark. The British Columbia Utilities Commission is in the midst of a generic cost of capital proceeding that will determine the approach going forward.

⁸⁵ BCUC Decision in the Matter of Return on Common Equity BC Gas Utility Ltd., Pacific Northern Gas Ltd., West Kootenay Power Ltd., June 10, 1994 (BCUC 1994 Decision).

⁸⁶ BCUC 1994 Decision, p. 17.

⁸⁷ BCUC In the Matter of Terasen Gas Inc. *et al.* Return on Equity and Capital Structure Decision, December 16, 2009 (BCUC 2009 Decision), pp. 44-45.

models arrived at more or less plausible results. For example, in its 2009 decision, the BCUC found:

The Commission Panel agrees that a single variable is unlikely to capture the many causes of changes in ROE and that in particular the recent flight to quality has driven down the yield on long-term Canada bonds, while the cost of risk has been priced upwards.⁸⁸

Having acknowledged the influence of the current economic environment, the BCUC in 2009 gave the most weight to the DDM, less weight to the Equity Risk Premium method and CAPM, and a low weight to the comparable earnings model. While the BCUC acknowledged giving weight to the DDM, ERP, CAPM and comparable earnings method, it did not specify the exact weights used.⁸⁹ The BCUC is currently undertaking a review of its cost of capital estimation methodology.

Ontario Energy Board

The Ontario Energy Board (“OEB”) regulates electric and gas utilities in Ontario and sets rates for electric and natural gas distribution and transmission. The OEB also regulates other aspects of the electric and natural gas sector, but it does not regulate competitive electric or gas supply. In addition to determining the allowed cost of capital, the OEB also determines a deemed (allowed) capital structure for the utilities it regulates, and the allowed cost of equity is applied to the deemed equity portion of the allowed rate base, which is based on historical cost.

The OEB reviewed its approach to determining the cost of capital for Ontario utilities and in December 2009 issued a report on its estimation procedures going forward.⁹⁰ Prior to the review, the OEB relied on a formula-based approach using a version of the risk premium approach, or Equity Risk Premium (ERP) method to determine the return on common equity. Although a number of concerns were raised with this approach, the OEB decided to continue

⁸⁸ BCUC 2009 Decision, p. 73.

⁸⁹ BCUC 2009 Decision, p. 45.

⁹⁰ Ontario Energy Board, “EB-2009-0084, Report of the Board on the Cost of Capital for Ontario’s Regulated Utilities,” Issued December 11, 2009 (OEB Report 2009).

relying on a formula-based methodology and the ERP method, but the review led to a resetting of the risk premium and an adjustment to the formula used to update the ROE.

The OEB's current approach to cost-of-capital estimation requires that the Board determine a baseline ROE and subsequently update the estimate annually using the determined formula. The baseline ROE was most recently determined in 2009 during the generic proceeding. To arrive at its initial estimate of the ERP for determining the baseline ROE, the OEB reviewed the recommendations of the submissions as part of the 2009 proceeding, and determined each submission's Low, Medium, and High ERP.⁹¹ In determining the initial ERP, the OEB found that:

the use of multiple tests to directly and indirectly estimate the ERP is a superior approach to informing its judgment than reliance on a single methodology.⁹²

As a result, the OEB considered all submissions, which included estimates based on the CAPM, DDM, risk premium model, econometric ERP analyses, realized ERP analyses, the difference between awarded ROEs and realized government bond yields, and various forecasts. The OEB averaged the experts' calculations of the risk premium over the long-term government bond and used judgment to determine that an appropriate premium over long-term government bonds was in the low-end of the range determined by the averages of the experts' ranges.

c) The U.K.

The U.K. regulator Ofgem has for many years made its cost of equity decisions within a CAPM framework, and, at least in a formal sense, has published CAPM parameters which correspond to its cost of equity determinations. However, it is also clear that Ofgem does not treat the CAPM estimates mechanistically, and, in any case, Ofgem uses a degree of judgment in determining the equity beta parameter, since there is little direct market evidence that can be relied on. While some of Ofgem's analysis and discussion of utility submissions is framed in terms of the CAPM parameters, it is clear that Ofgem focuses much more on the final cost of equity figure than on

⁹¹ OEB 2009, p. 38.

⁹² OEB 2009, p. 36 (emphasis in the original).

the mechanistic derivation of that figure, whether in a CAPM framework or otherwise. For example, Ofgem has said: “Overall, our Final Proposals retain the cost of equity assumptions in our Initial Proposals of 7.0 percent for NGET and 6.8 percent for NGGT. Table 3.5 shows our Final Proposals for the cost of equity in terms of the CAPM components. We note, however, that **it is the overall allowed return that matters.** [emphasis added]”⁹³

3. Impact of Economic, Industry or Company Factors

It makes sense that multiple cost of equity estimation methods have been developed and remain in use for a variety of reasons as articulated by Professors Berk and DeMarzo: “[a]ll the techniques ... are imprecise” and “practitioners apply the technique that best suit their particular circumstances.”⁹⁴ Because economic, industry, and firm-specific factors vary, it is important to assess the circumstances under which the models discussed in *Section III* are and should be used.

a) Economic Factors

As a pertinent example, due to the flight to quality following the financial crisis and subsequent monetary policy initiatives in many countries, the risk-free rate has been suppressed and is unusually low. Thus, in a standard implementation of the CAPM, the current risk-free rate results in a low cost of equity estimate. At the same time, investors have in recent years faced unusually high market volatility as measured by, for example, the S&P / ASX volatility index or the S&P 500 volatility index.⁹⁵ Academic literature finds that investors expect a higher risk premium during more volatile periods. For example, French, Schwert, and Stambaugh (1987) find a positive relationship between the expected market risk premium and volatility:

We find evidence that the expected market risk premium (the expected return on a stock portfolio minus the Treasury bill yield) is positively related to the predictable volatility of stock returns. There is also evidence that unexpected stock returns are negatively related to the unexpected change in the volatility of

⁹³ *RIIO-T1: Final Proposals for National Grid Electricity Transmission and National Grid Gas*, Ofgem, 17 December 2012, paragraph 3.45.

⁹⁴ Berk & DeMarzo 2009, p. 420.

⁹⁵ The S&P/ASX Volatility Index and the S&P 500 Volatility Index reflect the markets’ expected volatility in the benchmark Australian and American equity indices, respectively.

stock returns. This negative relation provides indirect evidence of a positive relation between expected risk premiums and volatility.⁹⁶

And Kim, Morley and Nelson (2004) find:

When the effects of volatility feedback are fully taken into account, the empirical evidence supports a significant positive relationship between stock market volatility and the equity premium.⁹⁷

Other academic papers have found a relationship between general economic conditions and the MRP. Constantinides (2008) studies a classical utility model where consumers are risk-averse and also summarizes some of the empirical literature. Empirical evidence shows that consumers become more risk-averse in times of economic recession or downturn, and equity investments accentuate this risk.⁹⁸ Increased risk aversion leads to a higher expected return for investors before they will invest. Specifically, equities are pro-cyclical and their performance is positively correlated with the economy's performance. Thus, unlike government bonds, equities fail to hedge against income shocks that are more likely to occur during recessions.⁹⁹ As a result, investors require an added risk premium to hold equities during economic downturns.

The very low current risk-free rates make the cost of equity estimates from a standard implementation of the Sharpe-Lintner CAPM also very low at a time when volatility measures indicate that the MRP has increased as well. Therefore, these market circumstances call for a serious consideration of economic factors or other models rather than a mechanical implementation of the Sharpe-Lintner CAPM.

Conditional models such as the Consumption CAPM attempt to incorporate the relationship between market volatility and the MRP in determining the cost of equity. As the model

⁹⁶ K. French, W. Schwert and R. Stambaugh (1987), "Expected Stock Returns and Volatility," *Journal of Financial Economics*, Vol. 19, pp. 3.

⁹⁷ C-J. Kim, J.C. Morley and C.R. Nelson (2004), "Is There a Positive Relationship Between Stock Market Volatility and the Equity Premium?," *Journal of Money, Credit and Banking*, Vol. 36, p. 357.

⁹⁸ Constantinides, G.M. (2008), "Understanding the equity risk premium puzzle," In R. Mehra, ed., *Handbook of the Equity Risk Premium*, Elsevier, Amsterdam.

⁹⁹ Constantinides, G.M., and D. Duffie (1996), "Asset Pricing with Heterogeneous Consumers," *Journal of Political Economy*, pp. 219-240. See also E.S. Mayfield (2004), "Estimating the market risk premium," *Journal of Financial Economics*, vol. 73, pp. 465-496.

estimates a relationship between the risk premium of a stock and its conditional volatility, the model allows for a time-varying relationship between risk and return; *i.e.*, the implied cost of equity varies with the degree to which (i) the underlying stock can serve as a hedge against the market and (ii) market volatility. As rate-regulated entities commonly move with the market, the cost of equity estimate usually moves in the same direction as the volatility of the market. Thus, the consumption-based model addresses the finding that volatility impacts the required risk premium. As such, it may be particularly useful to implement this model when market volatility is unusually high or low.¹⁰⁰

Given the currently very low risk-free rates and the recent market volatility, the DDM may additionally provide useful insights into the cost of equity. This is especially true for versions of the model that take into account (i) all cash that flows to shareholders through not only dividends but also share buybacks and (ii) changes in the forecasted growth rates in the near term and the longer term (*i.e.* multi-stage versions of the DDM).

Table 3 below displays the impact of two key economic factors discussed above, market volatility and risk-free rates, on the choice of cost of equity estimation model. While there is no specific formula that can be proposed to select a particular model under given market circumstances, or a method that can be used to combine the various models mechanistically, there are certain market scenarios under which it is more appropriate to use one model rather than another. For example, in times of either extremely high or low market volatility, (or extreme values of other macroeconomic indicators such as inflation), the consumption-based CAPM becomes more relevant. The DDM model and especially the multi-stage DDM is also less sensitive to variations in the risk-free rate than the standard CAPM, but it can be sensitive to market volatility. This is because in times of economic turmoil, the growth estimates for companies, including rate-regulated entities, are less likely to be stable going forward. Because the multi-stage DDM has more realistic characteristics and is less sensitive to analysts' short-term forecasts, the tables in this section use the term DDM to reflect the multi-stage DDM.

¹⁰⁰ See Ahern, *et al.* (2012) for a discussion of its use in a regulatory setting.

The effect of the risk-free rate and market volatility on model choice is reflected in Table 3 below, which should be viewed as an illustration on the directional choice rather than a prescription.

Table 3: Relationship Between Key Economic Conditions and Weights to be Given to Models

		Prevailing Risk-free Rate in Economy		
Market Volatility		High	Average	Low
	High	Consumption CAPM		
	Average	Consumption CAPM / DDM	CAPM / ECAPM	Consumption CAPM / DDM
	Low	Consumption CAPM / DDM		

b) Industry Factors

As discussed above, empirical research has consistently found that the Security Market Line determined by the Sharpe-Lintner CAPM (as depicted in Figure 2) is too steep.¹⁰¹ This result is also consistent with the findings of Fama & French (1992), which estimated a zero slope in the empirical SML.¹⁰² Thus, the ECAPM as well as the Fama-French model attempt to find a model that is a better fit with empirical data from tests of the Sharpe-Lintner CAPM, showing that the latter tends to under estimate the cost of equity for companies with beta estimates below one, and over estimate the cost of equity for companies with beta estimates above one. A better-fitting model flattens the Security Market Line as depicted in Figure 3. Because most rate-regulated entities have beta estimates below one, reliance on the Sharpe-Lintner CAPM tends to bias the

¹⁰¹ See, for example, F. Black, M.C. Jensen, and M. Scholes, —The Capital Asset Pricing Model: Some Empirical Tests,” *Studies in the Theory of Capital Markets*, Praeger Publishers, 1972, pp. 79-121 and E.F. Fama and J.D. MacBeth, —Risk, Returns and Equilibrium: Empirical Tests,” *Journal of Political Economy* 81 (3), 1972, pp. 607-636.

¹⁰² E.F. Fama and K.R. French, —The Cross-Section of Expected Returns,” *Journal of Finance* 47, 1992, pp. 427-465.

cost of equity estimates for these companies downwards. Therefore, for entities whose beta estimates are farther from one, it becomes important to look to the ECAPM to accurately reflect the cost of equity for the entity.¹⁰³ In many countries or regions, including Australia, Canada, Europe and the U.S, estimated betas for rate-regulated entities declined and become statistically insignificant in the early 2000s as the dot.com bubble burst. In such circumstances, the downward bias in the cost of equity estimates from the Sharpe-Lintner CAPM becomes more pronounced and models such as the ECAPM can improve the estimation.

For some industries the future may look like the past, but for others this is not the case. As an example, the outlook for the U.S shale gas industry today is different than it was in 2008. Similarly, the outlook for the nuclear industry in Japan changed dramatically after the 2011 tsunami. In such circumstances, forward-looking estimates of the industry's cost of capital as obtained through, for example, versions of the DDM, may be especially useful. As noted above, the DDM implementation should carefully consider not only the current economic environment but also industry and firm-specific factors, such as the sustainability of the current growth forecasts and whether dividends truly reflect all cash distribution to shareholders. For example, the multi-stage models discussed in *Section III* rely on several growth rates and therefore enable the analyst to consider near-term, intermediate, and long-term growth prospects for the individual company, industry, and economy. Therefore, a multi-stage DDM model, unlike the Sharpe-Lintner CAPM, can capture both near-term and longer-term changes in an industry. This becomes especially important when an industry's expected risk characteristics differ from its past characteristics.

Rate-Regulated Entities vs. Other Industries

According to empirical studies, the Sharpe-Lintner CAPM remains the most commonly used model across the full spectrum of companies.¹⁰⁴ However, the utility industry and rate-regulated entities have some unique characteristics that make it plausible that the methods that serve other

¹⁰³ See Table A-1 in the Appendix for details. Much of the academic literature estimating alpha dates back to the 1980s. Academic research has since turned to the Fama-French multifactor model, which attempts to explicitly capture the empirical pivot of the SML as a function of additional pricing factors.

¹⁰⁴ J.R. Graham and C.R. Harvey, —Theory and Practice of Corporate Finance: Evidence from the Field,” *Journal of Financial Economics* 60, 2001, pp. 187-243.

industries well do not serve this industry nearly as well. For example, the utility industry tends to be relatively stable, so that the DDM (and especially the multi-stage DDM) is much more likely to provide usable results for this industry than for more volatile industries. As the residual income valuation model is a variation of the multi-stage DDM, the same comments pertain to this model.

Prior to the financial crisis, models such as the single-stage DDM, *Brattle's* multi-stage DDM, the CAPM, and versions of the ECAPM resulted in fairly similar results. Figure 5 below illustrates this for the gas distribution industry in the U.S. towards the end of 2006. Specifically, the figure is based on implementing the constant growth DDM, a 3-stage DDM, the Sharpe-Lintner CAPM, the ECAPM with an alpha of 0.5% and an ECAPM with an alpha of 1.5% for seven gas distribution companies. Figure 5 then shows the range of the cost of equity estimates assuming a 50-50 gearing for the target company. The figure also indicates the average cost of equity obtained from the sample, which is at the split of each bar.

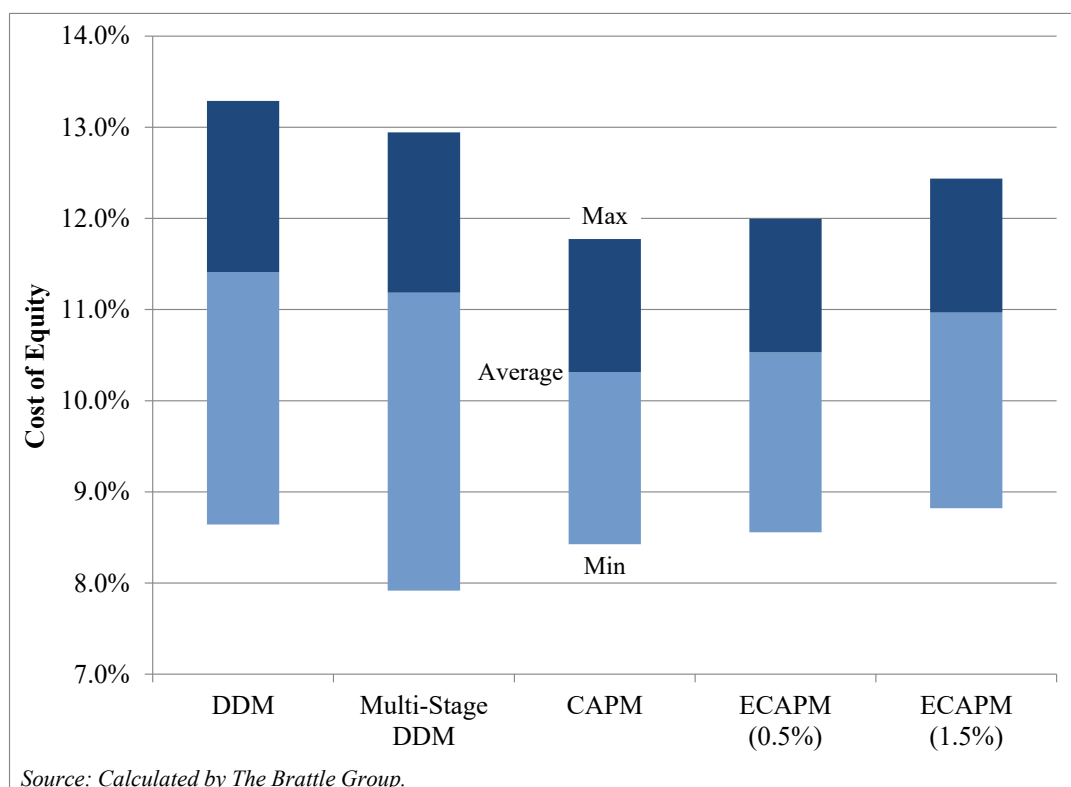


Figure 5

It is clear from Figure 5 above that there is substantial overlap in the estimates. We attribute this effect to the fact that the economy was relatively stable in 2006 and so was the gas distribution industry. At the time, these models largely confirmed the range of the cost of equity estimates.

As discussed above, rate-regulated companies also tend to be low-beta entities, so the empirical finding that the SML predicted by the Sharpe-Lintner CAPM is too steep is a serious concern for this industry; *i.e.*, it becomes important to use the ECAPM or other models to ensure that this empirical observation is accurately reflected in the cost of equity estimates.

Analogously to Table 3, Table 4, Panels A and B below summarize the directional weighing of the models depending on various industry characteristics. The two industry factors considered in Table 4, Panel A below are the stability of growth rate forecasts and the average market beta of the industry. For example, as mentioned above, rate-regulated entities tend to have relatively more stable growth forecasts over time and low betas (*i.e.*, beta estimates below one). Therefore, for this industry, the use of the ECAPM or variations of the multi-stage DDM might become valuable in determining the cost of equity capital. The effect of the stability of growth forecasts and the beta value on model choice is reflected in Table 4, Panel A below, which should be viewed as an illustration on the directional choice rather than a prescription.

Table 4: Relationship Between Key Industry Factors and Weights to be given to Models—Panel A

		Stability of Industry Growth Forecasts Over Time		
Market Beta of Industry		High	Average	Low
	High	ECAPM / DDM		ECAPM / Other
	Average	CAPM / DDM		CAPM / Other
	Low	ECAPM / DDM		ECAPM / Other

Another characteristic of the industry that should be considered is whether companies in the industry are exposed to financial distress and/or significant merger and acquisition activity, and

the prevalence of share buybacks. As discussed above, market-based estimation models are relatively more affected when a given company faces financial distress, or unique circumstances that may lead to its stock price decoupling from fundamentals. Therefore, if many companies in an industry are subject to such effects, the whole industry may be affected. Further, companies that engage in a substantial amount of share buybacks will end up distributing cash to shareholders in a form other than dividends, which makes a DDM based on a per share dividend ratio less appropriate. Panel B below illustrates these effects.

Table 4: Relationship Between Key Industry Factors and Weights to be given to Models– Panel B

		Industry Exposure to Financial Distress and/or M&A	
Prevalence of Share Buybacks		High	Low
	High	Other Models: Risk Premium, comparable earnings, maybe use other industries	CAPM, ECAPM, DDM that includes all cash that accrues to shareholders
	Low		CAPM, ECAPM, DDM

c) Company Factors

In many instances company-specific issues are better dealt with via sample selection or through risk positioning than through the determination of how to estimate the cost of equity. A company that is a potential member of the benchmark sample is often dropped if it faces unique circumstances that may bias the cost of capital estimation process. This is the case if, for example, a company is undergoing significant merger or acquisition activity, which inherently affects the information available in the market and therefore drives the stock price (and thus the results from all market-based models, including the CAPM, ECAPM, Fama-French and DDM).

After a range of cost of equity estimates has been obtained, it is necessary to consider where, from within this range, the final determination on the cost of equity will be. Provided that the range has been developed in an appropriate way that takes account of the market and industry factors described in this section, the final step is to consider the relative risk of the target company compared to the sample of companies from which the cost of equity range has been developed. The cost of equity is adjusted upward or downward depending on the target entity's risk characteristics relative to those of the sample. This issue is the topic of the next section.

D. RISK POSITIONING OF THE TARGET ENTITY

The discussion in the preceding sections covered various models that produce cost of equity estimates. Typically the cost of equity will be estimated for a sample of firms, or all firms in a particular sector, because it usually is not possible to estimate the cost of equity for a single firm with a useful degree of accuracy. To determine a single value for the cost of equity for a specific firm from a range of values for a set of comparator firms, it makes sense to consider the riskiness of the specific firm relative to the riskiness of the sample, since the cost of equity itself is compensating investors for risk.

In the regulatory context, in some cases this process is implicit in the regulator's decision, while in others it is an explicit step in the cost of equity determination process. This step can conveniently be termed "risk positioning", because the regulator considers the risk characteristics of the specific utility relative to the benchmark.

1. Why risk positioning is necessary

While the precise details and wording of the regulator's objective in setting the cost of equity vary from one jurisdiction to another, the underlying idea is that investors will expect a return equivalent to the return that they would expect from other investments of like risk. Utilities generally have low risk relative to the market as a whole, but within the utilities sector, different firms are likely to have somewhat different risk characteristics. "Risk positioning" acknowledges the possibility that different utilities can have somewhat different risks. In this context, "risk" is defined as the characteristic of an investment which determines expected returns which would usually include "systematic" exposure to the wider economy, but not "idiosyncratic" risks associated with specific projects that can be diversified away in an investment portfolio. While

the cost of equity solely captures investors' compensation for bearing systematic risk, the cost of debt reflects total risk, including idiosyncratic risks. Therefore, there are instances of regulatory mechanisms, such as decoupling, which reduce the variability of total revenues and therefore also total risk, (affecting the cost of debt), but which may not impact the cost of equity for a given utility.

One way in which a utility is exposed to systematic risk is through variations in demand. End-user demand tends to be at least somewhat correlated with wider economic activity, and is thus a source of exposure to systematic risk. One utility might have more exposure than other, for example if it has a greater proportion of price-sensitive industrial load.

In some jurisdictions, leverage is considered a source of "financial risk", which affects the risk positioning analysis. This could be so, for example, where the rate of return is generally determined on the basis of actual capital structure. A utility with more debt than the benchmark will require a higher return on equity than the benchmark, even if it otherwise has similar business risk exposure as the benchmark (just as two utilities with the same asset beta would have different equity betas if one has higher gearing than the other). Where this approach is taken, the term "business risk" is used to refer to the other sources of relevant risk differences that are taken into account in the risk positioning analysis.

Once a benchmark rate of return has been defined (whether a point estimate or a range), the risk positioning approach requires an analysis of the particular utility's risk relative to that benchmark. To the extent that the utility is found to have more (or less) risk than the benchmark, the rate of return would be set higher (or lower) than the benchmark rate of return.

2. What risk characteristics are relevant?

The characteristics relevant to risk positioning are those which expose the utility to systematic risk and which therefore have an impact on the rate of return required by investors. Some important sources of uncertainty in revenues and returns to investors may not have an impact on the required return to the extent that investors are able to diversify away exposure to those risks. For example, the weather may be an important source of variability in revenues and returns, but may not be an important source of risk to investors because it is diversifiable.

A good way to think about risk positioning is to consider the extent to which different utilities are protected from risks. A distribution utility can in principle be protected from risks to the extent that it is able to pass on risk to its customers (which depends on the detail of the regulatory framework being applied). Demand risk (which is at least partly non-diversifiable), for example, can be borne by the utility if the regulatory regime sets prices and does not “true up” revenues to account for the difference between forecast and actual demand. Alternatively, demand risk can be passed on to customers through a true-up or balancing account process, which would allow the utility to recover in one year any “missing” revenue from the prior year caused by demand forecasting errors. Protection from demand risk in this way depends on both a regulatory framework that allows for such true-ups and on the existence of franchise customers that will bear the risks passed on to them. Therefore, other things equal, a utility with true-ups for demand risk would be considered less risky than one without.

Distribution utilities typically have franchise customers that rely on the utility and have no alternative supply of energy. However, this is typically not the case for gas pipelines: in many jurisdictions, gas pipelines do not have “franchise” customers: customers may be free to switch to competing pipelines. Even if there is no prospect of competition from other pipelines, it may still be difficult for pipelines to pass on demand risk to their customers, since large end-users may be price sensitive (i.e., if the pipeline increases price in response to a fall in demand, the price increase itself could further cut demand).

Pipeline regulators in both the US and Canada apply a risk-positioning approach in determining the cost of equity.

3. FERC Approach

The Federal Energy Regulatory Commission (FERC) has a standard approach to determining the cost of equity for gas pipelines, set out in a “policy statement”,¹⁰⁵ which, together with precedent from prior decisions, guides all decisions on the cost of equity for gas pipelines. The FERC’s approach is to use a form of the dividend growth model (typically termed the “DCF” model in

¹⁰⁵ *Composition of Proxy Groups for Determining Gas and Oil Pipeline Return on Equity*, FERC (April 2008).

the US) to estimate the cost of equity for a benchmark group of publicly-traded pipeline companies. The results of the model are a cost of equity estimate for each of the companies in the benchmark (or ~~proxy~~) group.

FERC starts by assuming that the median company in the proxy group is the appropriate cost of equity, unless either the pipeline or an intervener in the case demonstrates that the instant pipeline has risk factors which mean that the cost of equity should be set above or below the median:

*after defining the zone of reasonableness through development of the appropriate proxy group for the pipeline, the Commission assigns the pipeline a rate within that range or zone, to reflect specific risks of that pipeline as compared to the proxy group companies. [f/n omitted] The Commission has historically presumed that existing pipelines fall within a broad range of average risk. A pipeline or other litigating party has to show highly unusual circumstances that indicate anomalously high or low risk as compared to other pipelines to overcome the presumption.*¹⁰⁶

And

*unless a party makes a very persuasive case in support of the need for an adjustment and the level of the adjustment proposed, the Commission will set the pipeline's [ROE] at the median of the range of reasonable returns.*¹⁰⁷

In line with this approach, most FERC decisions result in the pipeline receiving a cost of equity equal to the median of the proxy group. A recent decision for El Paso Natural Gas (EPNG),¹⁰⁸ however, illustrates how FERC assesses relative risk and may, on occasion, move away from the median. In this case, the FERC ALJ¹⁰⁹ characterized EPNG's business risk on two related dimensions: competitive risk and regulatory risk. US natural gas pipelines typically secure long-term contractual commitments from shippers to use the pipeline capacity (with relatively high fixed charges, equivalent to a take-or-pay commitment). EPNG had long-term contracts for a smaller proportion of its capacity than did the pipelines in the proxy group, and its contracts were typically shorter. Furthermore, EPNG's throughput had been declining. This is symptomatic of higher business risk, because in the absence of contractual commitments and in the absence of

¹⁰⁶ *Ibid.*, p. 4.

¹⁰⁷ *El Paso Natural Gas Company*, Initial Decision, docket no. RP10-1398 (June 18, 2012), paragraph 40, quoting prior FERC decisions.

¹⁰⁸ *El Paso Natural Gas Company*, Initial Decision, docket no. RP10-1398 (June 18, 2012).

¹⁰⁹ A FERC rate case typically results in an "initial decision" issued by an Administrative Law Judge (ALJ). The ALJ's decision is subsequently reviewed by the FERC commissioners, and may be affirmed or varied.

franchise customers, the pipeline is no longer able to pass on risks to its customers. In the limit, the pipeline may be unable to charge rates high enough to recover its authorized revenue requirement (as increasing rates drives throughput lower still).

The ALJ found that EPNG was exposed to competition in its major downstream markets from new pipeline projects, and that this competition was to an extent the result of regulatory policies that favor new pipeline projects to foster competition (possibly harming existing pipelines).

Based on this analysis (and also a finding that EPNG had above-average financial risk, as evidenced by a credit rating of BBB-, lower than all but one of the proxy group companies), the ALJ determined that EPNG's cost of equity should be set ~~well~~ above the median ROE [of the proxy group]".¹¹⁰

4. NEB approach

In Canada, the approach taken by energy regulators (both provincial and national) historically was to set the cost of equity on a formula basis and to use the same cost of equity for all pipelines. Risk positioning was then used to vary the authorized proportion of equity in the capital structure, thereby increasing the overall return on capital for those utilities judged to be riskier. However, in the most recent decision by Canada's National Energy Board (NEB), the NEB moved to an approach which focuses on the overall after-tax return directly, rather than separately determining the cost of equity, the cost of debt, and the proportion of each in the capital structure.¹¹¹ The NEB takes a systematic approach to assessing business risk under the headings ~~supply risk~~", ~~market~~ [downstream] risk", ~~regulatory risk~~", ~~competitive risk~~" and ~~operating risk~~", although the NEB said ~~The various forms of risk are in some cases inextricably linked, and the boundaries between them are subjective~~".¹¹² In the RH-1-2008 case,¹¹³ the NEB was concerned with whether the business risk of the pipeline had increased

¹¹⁰ *Ibid.*, p. 45. The ALJ did not specify an ROE. The final decision on ROE rests with the FERC commissioners.

¹¹¹ See RH-1-2008, discussed further below.

¹¹² *Reasons for Decision, Trans Quebec and Maritimes Pipelines Inc., RH-1-2008*, NEB (March 2009), p. 30.

¹¹³ Concerning the Trans Quebec and Maritimes Pipelines, which predominantly move supplies sourced from the Western Canadian Sedimentary Basin (WCSB) via the TransCanada Mainline, into Quebec and on into New Hampshire.

since the last time that a decision on the cost of capital for the pipeline had been taken. The NEB identified a number of factors as contributing to an increased overall business risk.

- **Supply risk:** the pipeline was mainly supplied from a region with declining conventional production and rising costs. While it was possible that new sources of unconventional supply (shale gas) would be developed, the result was increased uncertainty over the availability of competitively-priced supplies, and hence concerns over the possibility for reduced throughput.
- **Market and competitive risk:** because a large and increased proportion of the pipeline's throughput went to large industrial and electric power generation load, which is more variable than domestic and commercial load. In addition, competition with cheap hydro-power in the Quebec also contributed to increased market risk. Market risk was also increased as a result of the potential for competition with LNG imports in the US market.

Overall, the NEB concluded that business risk had increased as a result of these factors relative to the previous cost of capital decision for the pipeline. Whereas the FERC in the US uses a risk positioning approach to determine the cost of equity relative to a benchmark, the NEB estimated the after-tax weighted average cost of capital directly, principally on the basis of market-based estimates of the cost of capital of various comparator companies. The business risk analysis described above was part of the NEB's determination of where the pipeline's cost of capital should be relative to the sample data.¹¹⁴

5. Implementation

In the FERC and NEB examples given above, risk positioning of the target utility within the range of comparator or proxy companies is not analytically precise: the regulator considers evidence (which could be quantitative, such as the proportion of price-sensitive industrial load, or more qualitative) as to exposure to various relevant risk factors. Weighing the risk factors, and determining how the analysis of risk should be reflected in the final cost of equity determination is necessarily imprecise, and relies on judgment. For example, a regulator might determine that a

¹¹⁴ The NEB's analysis is summarized on p.79 of the decision.

particular utility, having an unusually high proportion of industrial load, was of above average risk, and that as a result the cost of equity should be 50 basis points above the mid-point of a range determined for a sample of utilities. The direction of the adjustment (upwards) is clear, but the magnitude is more a matter of judgment than something that can be derived quantitatively.

APPENDIX: ADDITIONAL TABLES AND FIGURES

Table A-1: Empirical Evidence On The Alpha Factor in ECAPM

AUTHOR	RANGE OF ALPHA	PERIOD RELIED UPON
Black (1993) ¹	1% for betas 0 to 0.80	1931-1991
Black, Jensen and Scholes (1972) ²	4.31%	1931-1965
Fama and MacBeth (1972)	5.76%	1935-1968
Fama and French (1992) ³	7.32%	1941-1990
Fama and French (2004) ⁴	N/A	
Litzenberger and Ramaswamy (1979) ⁵	5.32%	1936-1977
Litzenberger, Ramaswamy and Sosin (1980)	1.63% to 3.91%	1926-1978
Pettengill, Sundaram and Mathur (1995) ⁶	4.6%	1936-1990

* The figures reported in this table are for the longest estimation period available and, when applicable, use the authors' recommended estimation technique. Many of the articles cited also estimate alpha for sub-periods and those alphas may vary.

¹ Black estimates alpha in a one-step procedure rather than in an un-biased two-step procedure.

² Estimate a negative alpha for the sub period 1931-39 which contain the depression years 1931-33 and 1937-39.

³ Calculated using Ibbotson's data for the 30-day treasury yield.

⁴ The article does not provide a specific estimate of alpha; however, it supports the general finding that the CAPM underestimates returns for low-beta stocks and overestimates returns for high-beta stocks.

⁵ Relies on Litzenberger and Ramaswamy's before-tax estimation results. Comparable after-tax alpha estimate is 4.4%.

⁶ Pettengill, Sundaram and Mathur rely on total returns for the period 1936 through 1990 and use 90-day treasuries. The 4.6% figure is calculated using auction averages 90-day treasuries back to 1941 as no other series were found this far back.

Sources:

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PERFORMANCE EVALUATION PLAN RATE SCHEDULE "PEP-5A"

Mississippi Public Service Commission Schedule No. 28.1		
PAGE	EFFECTIVE DATE	DATE OF VERSION SUPERSEDED
24 of 31	January 1, 2015	January 9, 2009

APPENDIX C

Where:

- R_f = The Risk Free Rate is the normalized Bond Yield on 30 Year U.S. Treasury bonds as used in the Equity Risk Premium calculation.
- Beta = The Beta is the average of the betas as stated in Value Line for the same group of comparable utilities in the DCF test above. Those companies eliminated by the DCF truncation are also eliminated from this calculation
- R_m = Two historical Market Risk Premiums shall be used. The first is the total return market equity risk premium for the most current year listed and the second is the income equity risk premium for the same time period. Both are found in Ibbotson Associated Yearbook.

Two projected market risk premiums shall be used. The first is the Value Line Indicated Total Return less the normalized Yield on 30 Year U.S. Treasury bonds. The second is the S&P 500 Indicated Total Return less the normalized Yield on 30 Year U.S. Treasury bonds.

2. R_m Four calculations are performed for the Standard CAPM using each of the four market risk premiums.

Empirical CAPM

3. The following version of the Empirical CAPM model shall be used

$$K = R_f + 0.25 (R_m - R_f) + 0.75 \text{ Beta } (R_m - R_f)$$

Where:

- R_f = The Risk Free Rate is the normalized Yield on 30 Year U.S. Treasury bonds as used in the Equity Risk Premium calculation.
- Beta = The Beta is the average of the betas as stated in Value Line for the same group of comparable utilities used in the DCF test above. Those companies eliminated by the DCF truncation are also eliminated from this calculation.

193. In applying his version of the ECAPM I, Mr. Hevert used an X factor of 0.25, based on published work of Dr. Morin.²⁴² The resulting estimates were an average ROE of 8.91 per cent and 10.54 per cent for his Canadian and U.S. proxy groups, respectively, which were approximately 80 bps larger than his estimates using CAPM.²⁴³ Mr. Hevert's resulting estimates do not include any amounts for flotation costs.²⁴⁴

194. Dr. Villadsen used an alpha factor of 1.5 per cent, which was based on an average adjustment factor from academic literature.²⁴⁵ This factor was adjusted downwards to account for differences in government bond maturities and to be conservative.²⁴⁶ Dr. Villadsen's resulting ROE estimates for her Canadian and U.S. utility proxy groups are presented in Table 5 below. Consistent with her CAPM estimates, Dr. Villadsen included flotation costs and generated results under two scenarios of risk free rates and MERP.

Table 5. Dr. Villadsen's ECAPM estimates

	ROE	
	Scenario 1	Scenario 2
	(%)	
Canadian utility sample	9.0 - 9.5	10.2 - 10.9
U.S. gas utility sample	8.4	9.2
U.S. electric utility sample	8.2 - 8.3	9.0 - 9.1

Source: Exhibit 20622-X0104, evidence of Dr. Villadsen, PDF pages 54-55.

195. Dr. Booth did not use ECAPM to generate ROE estimates, but he did discuss alternatives to CAPM. Dr. Booth observed that there are a wide variety of multi-factor models, which essentially extend the one factor CAPM to include additional factors. The current 'standard' multifactor model, known as the Fama-French three factor model, includes a size premium to address the return difference between small firms and large firms and a value premium to address the return difference between value and growth stocks.²⁴⁷ Dr. Booth did not use this model or advocate for its use, as he stated this model is unlikely to generate any significant value over the use of the CAPM. He noted that he included this information in his evidence to demonstrate academic support for other risk premium based models.

Commission findings

196. The use of ECAPM is an approach recognized in the academic literature and is used to address a perceived issue with the CAPM, when the CAPM-based SML is steeper than empirical evidence suggests it should be. The ECAPM adjusts the SML by introducing an empirical adjustment factor to flatten the SML.

²⁴² Exhibit 20622-X0215, response to AML/EDTI-AUC-2016FEB18-007, PDF pages 79-80. Transcript, Volume 1, pages 139-140.

²⁴³ Exhibit 20622-X0082, evidence of Mr. Hevert, PDF page 76.

²⁴⁴ Exhibit 20622-X0082, evidence of Mr. Hevert, PDF page 124.

²⁴⁵ The academic literature references are listed in Exhibit 20622-X0105, evidence of Dr. Villadsen, Appendices, PDF page 27.

²⁴⁶ Transcript, Volume 5, PDF pages 647-648.

²⁴⁷ Exhibit 20622-X0242, evidence of Dr. Booth, PDF pages 42-43.

197. In exchanges with Commission counsel, both Mr. Hevert²⁴⁸ and Dr. Villadsen²⁴⁹ agreed that the empirical adjustment factor used in their respective ECAPMs is a function of the sample used and the time period over which the returns were examined. During the oral hearing, Commission counsel asked Mr. Hevert if there are any kinds of standards or best practices that are employed by professionals in determining what the dataset should be when estimating the empirical adjustment factor. In response, Mr. Hevert described that there have been different studies that produce a range of estimates for the empirical adjustment factor and in his view, the selection of the empirical adjustment factor will inevitably be a matter of judgement.²⁵⁰

198. Mr. Hevert's view is supported by the evidence in this proceeding with respect to the empirical adjustment factors selected by the experts who employed an ECAPM. Mr. Hevert relied on an adjustment factor based on Dr. Morin's 1989 empirical study that used data from 1926 to 1984 and Dr. Villadsen used an empirical adjustment factor based on average estimated adjustment factors from academic studies that she then adjusted downwards in order to be conservative. The studies relied upon by Dr. Villadsen used different timeframes, with none of the studies including years beyond 1991.²⁵¹

199. **In the Commission's view, the ECAPM appears to be a model that could contribute to the Commission's determination of a fair allowed ROE. Generally speaking, the Commission is supportive of models and methods that attempt to improve upon CAPM results.** The Commission agrees with Mr. Hevert that the selection of an empirical adjustment factor is a matter of judgement. Based on the evidence in this proceeding, however, the Commission has been unable to assess adequately the empirical adjustment factors employed by the experts in exercising their judgement. Consequently, the Commission will not rely heavily on the ECAPM results in this proceeding. In order for the Commission to adequately assess the judgement exercised by the experts, the Commission would require a full explanation justifying the sample and time periods adopted.

200. The Commission also notes that the empirical adjustment factors to CAPM used in the ECAPMs in this proceeding does not resolve the issues discussed in Section 6.1.4 regarding the reasonable degree of confidence in the estimated ranges for beta.

6.3 Bond yield plus risk premium model and the predictive risk premium model

201. In addition to relying on their CAPM results in estimating a fair allowed ROE, Mr. Hevert, Dr. Villadsen and Dr. Cleary presented results generated by risk premium models. All of the risk premium models presented in this proceeding are based on the fundamental assumption of modern corporate finance that risk averse investors require higher returns for bearing higher risk. In their general form, risk premium models add a premium to account for equity risk to a measure of interest rates.²⁵²

202. Mr. Hevert gave primary weight to the results of his CAPM and risk premium models in arriving at his recommended ROE range, and less weight to the results of his DCF model.²⁵³

²⁴⁸ Transcript, Volume 1, page 138, lines 10-20.

²⁴⁹ Transcript, Volume 5, page 646, lines 7-24.

²⁵⁰ Transcript, Volume 1, page 139.

²⁵¹ Exhibit 20622-X0457, rebuttal evidence of Dr. Villadsen, PDF page 27.

²⁵² Exhibit 20622-X0082, evidence of Mr. Hevert, PDF page 76. Exhibit 20622-X0164, response to AML/EDTI-UCA-2016FEB18-010, PDF page 29.

²⁵³ Exhibit 20622-X0082, evidence of Mr. Hevert, PDF page 159.

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Using Beta

Andrew J. Cueter | October 02, 2012



In finance, the Beta of a security (or portfolio) is used as an indicator of its historical volatility in regards to a benchmark, generally the New York Stock Exchange (NYSE) Composite Index or the S&P 500 Index. At Value Line, we derive the Beta coefficient from a regression analysis of the relationship between weekly percentage changes in the price of a stock and weekly percentage changes in the NYSE Composite Index over a period of five years. In the case of shorter price histories, a shorter time period is used, but two years is the minimum. Value Line then adjusts these Betas to account for their long-term tendency to converge toward 1.00. (Though the scope of this convergence is beyond our purposes here, readers can refer to M. Blume, "On the Assessment of Risk," Journal of Finance, March 1971 for further details.)

Now that we have our Beta number, what does it mean? If an equity mirrors the benchmark, then it carries a Beta of 1.00. If Stock X has a Beta of 2.00, it is expected to rise (or fall) twice as much as the movement of the benchmark. For example, if the NYSE Composite Index rises (falls) 10%, Stock X will likely rise (fall) 20%. (For a more detailed overview, see [Understanding Beta](#).) Beta can also be negative (infrequent but possible), which would mean that the equity's return tends to move in the opposite direction from the market's move. Moreover, there is no upper or lower bound to Beta, although it typically does not stray too far from 1.00. Finally, a Beta of zero does not mean the asset is risk-free, just that the correlation of that asset's return to the market's return is zero.

Now that we know what Beta is and its implications, how can we use it? If we were able to predict the movements of the overall market, we would simply buy high Beta stocks while the market rises, and low Beta stocks while the market is falling. However, no one is capable of timing the market over the long term. So, what should we do?

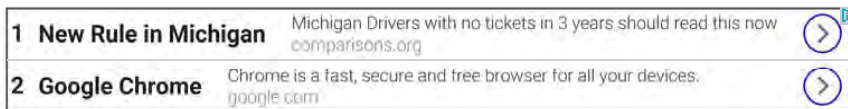
If we define a high risk asset in terms of the movement of its price, we can look towards Beta as one indicator of this riskiness. Though Beta by itself does not give a perfect indication of volatility, it does imply the direction and magnitude of movements. Using Beta as a measure of risk, we can relate this to a basic tenet of finance theory, which states that investors demand a return in exchange for assuming risk. Therefore, high-risk (or high-Beta) investments should provide a higher payout, and conversely, low-risk (or low-Beta) investments should provide a lower payout. This proposition seems reasonable and intuitive, but it may not always hold.

In a paper entitled "Re-Thinking Risk: What the Beta Puzzle Tells Us about Investing," written by David Cowan and Sam Wilderman of GMO LLC, they show just the opposite. For the paper, Beta was measured using 250-day returns of a universe of 1,000 stocks, regressed against 250-day returns of that universe. Low- and high-Beta Portfolios were then formed monthly and weighted by market capitalization, with the universe used as the benchmark. Their results present data starting in December, 1969 and show that high-Beta stocks have significantly underperformed the market (average annualized return of 7.2% vs. 10.6% for low-Beta and 9.8% for the universe), and done so with substantially higher annualized volatility (24.5% vs. 12.5% and 16.0%, respectively) and larger drawdown (-84.4% vs. -39.5% and -50.3%, respectively).

Though low-Beta may trump high-Beta over longer periods, there are some problems with solely relying on the Beta coefficient. It is a backward looking metric, and therefore may not be an accurate predictor of the future. The markets change all the time and just because a relationship held in the past does not mean it is certain to continue into the future. Also, since it is solely a statistical measure, it fails to consider underlying business fundamentals or economic developments. Consider [Altria Group \(MO\)](#). This stock has a Beta of 0.55 and the company primarily sells cigarettes. Due to the low Beta, we may say this is a low-risk stock. However, if for some reason cigarettes were deemed illegal to sell, this company would probably not stick around very long and any investment in the stock will likely become worthless. Solely looking at a stock's Beta will not uncover this risk.

So, back to our question posed earlier; what should we do? We propose Beta should be used as one factor in the equity analysis framework. Investors should also look at our Safety rank and Price Stability score when making investment decisions. Considered in conjunction with Value Line's fundamental research and valuation ratios, we believe investors can create a portfolio that may provide superior risk-adjusted returns over the long haul.

At the time of this article's writing, the author did not have positions in any of the companies mentioned.



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CAPITAL EQUIPMENT ANALYSIS: THE REQUIRED RATE OF PROFIT

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The interest in capital equipment analysis that has been evident in the business literature of the past five years is the product of numerous social, economic, and business developments of the postwar period. No conclusive listing of these developments can be attempted here. However, four should be mentioned which are of particular importance in this search for a more systematic method for discovering, evaluating, and selecting investment opportunities. These are: (1) the high level of capital outlays (in absolute terms); (2) the growth in the size of business firms; (3) the delegation of responsibility for initiating recommendations from top management to the profit center, which has been part of the general movement toward decentralization; and (4) the growing use of "scientific management" in the operations of the business firm.

These developments have motivated the current attempt to develop objective criteria whereby the executive committee in a decentralized firm can arrive at a capital budget. Since each of its profit centers submits capital proposals, the executive committee must screen these and establish an allocation and a level of capital outlays that is consistent with top management's criteria for rationing the firm's funds. Capital budgeting affords the promise that this screening process can be made amenable to some established criteria that are understandable to all the component parts of the firm. Consequently, capital budgeting appeals to top management, for, in the first place, each plant manager can see his proposal in the light of all competing proposals for the funds of the enterprise. This may not completely eliminate irritation among the various parts of the firm, but a rational capital budgeting program can go a long way toward maintaining initiative on the part of a plant manager, even though the executive committee may veto one or all of his proposals. In the second place, the use of a capital budgeting program serves to satisfy top management that each accepted proposal meets adequate predetermined standards and that the budget as a whole is part of a sound, long-run plan for the firm.

What specifically does a capital budgeting program entail? The focal points of capital budgeting are: (1) ascertaining the profit abilities of the array of capital outlay alternatives, and (2) determining the least profitability required to make an investment, i.e., a cut-off point. Capital budgeting also involves administrative procedures and organization designed to discover investment opportunities, process information, and carry out the budget; however, these latter aspects of the subject have been discussed in detail by means of case studies that have appeared in publications of the American Management Association and the

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National Industrial Conference Board and in periodicals such as the *N.A.C.A. Bulletin*.¹ Hence, we will not concern ourselves with them here.

There are at least four methods for establishing an order-preference array of the capital expenditure suggestions. They are: (1) the still popular "payoff period"; (2) the average investment formula; (3) the present value formula with the rate of interest given; and (4) the present value formula used to find the rate of profit. It is not our intention in this paper to discuss these various methods specifically, since critical analyses of these alternatives are to be found in papers by Dean, by Lorie and Savage, and by Gordon in a recent issue of the *Journal of Business*,² which is devoted exclusively to the subject of capital budgeting.

However, it is of interest to note that in each of these methods the future revenue streams generated by the proposed outlays must be amenable to measurement if the method is to be operational. However, improvements in quality, more pleasant working conditions, strategic advantages of integration, and other types of benefits from a capital outlay are still recognized only in qualitative terms, and there is a considerable hiatus in the literature of capital budgeting with respect to the solution of this problem. Hence, in the absence of satisfactory methods for quantifying these types of benefits, the evaluation of alternative proposals is still characterized by intuitive judgments on the part of management, and a general quantitative solution to the capital budgeting problem is not now feasible. It appears to us that this problem affords one of the most promising opportunities for the application of the methods of management science. In fact, we anticipate that techniques for the quantification of the more important factors now treated qualitatively will soon be found.

Given the rate of profit on each capital outlay proposal, the size of the budget and its allocation are automatically determined with the establishment of the rate of profit required for the inclusion of a proposal in the budget. In the balance of this paper, a method for determining this quantity is proposed and its use in capital budgeting is analyzed.

II

We state that the objective of a firm is the maximization of the value of the stockholders' equity. While there may be legitimate differences of opinion as to whether this is the sole motivation of management, we certainly feel that there can be no quarrel with the statement that it is a dominant variable in manage-

¹ American Management Association, *Tested Approaches to Capital Equipment Replacement*, Special Report No. 1, 1954; American Management Association, *Capital Equipment Replacement*; *AMA Special Conference*, May 3-4, 1954 (New York, 1954, American Management Association, 105 pp.); J. H. Watson, III, National Industrial Conference Board, *Controlling Capital Expenditures*, Studies in Business Policy, No. 62, April, 1953; C. I. Fellers, "Problems of Capital Expenditure Budgeting", *N.A.C.A. Bulletin*, 26 (May, 1955), 918-24; E. N. Martin, "Equipment Replacement Policy and Application", *N.A.C.A. Bulletin*, 35 (February, 1954), 715-30.

² *Journal of Business*, Vol. XXVIII, No. 3 (October, 1955).

ment's decisions. It has been shown by Lutz and Lutz in their *Theory of the Investment of the Firm*³ and by others⁴ that this objective is realized in capital budgeting when the budget is set so as to equate the marginal return on investment with the rate of return at which the corporation's stock is selling in the market. The logic and operation of this criterion will be discussed later. Now, we only wish to note the role assigned in capital budgeting to the rate of profit that is required by the market.

At the present time, the dividend yield (the current dividend divided by the price) and the earnings yield (the current income per share divided by the price) are used to measure the rate of profit at which a share is selling. However, both these yields fail to recognize that a share's payments can be expected to grow, and the earnings yield fails to recognize that the corporation's earnings per share are not the payments made to the stockholder.

The practical significance of these failures is evidenced by the qualifications with which these two rate-of-profit measures are used by investment analysts. In the comparative analysis of common stocks for the purpose of arriving at buy or sell recommendations, the conclusions indicated by the dividend and/or the earnings yield are invariably qualified by the presence or absence of the prospect of growth. If it is necessary to qualify a share's yield as a measure of the rate of profit one might expect to earn by buying the share, then it must follow that current yield, whether income or dividend, is inadequate for the purposes of capital budgeting, which is also concerned with the future. In short, it appears to us that the prospective growth in a share's revenue stream should be reflected in a measure of the rate of profit at which the share is selling. Otherwise, its usefulness as the required rate of profit in capital budgeting is questionable.

In his *Theory of Investment Value*⁵, a classic on the subject, J. B. Williams tackled this problem of growth. However, the models he developed were arbitrary and complicated so that the problem of growth remained among the phenomena dealt with qualitatively. It is our belief that the following proposal for a definition of the rate of profit that takes cognizance of prospective growth has merit.

The accepted definition of the rate of profit on an asset is the rate of discount that equates the asset's expected future payments with its price. Let P_0 = a share's price at $t = 0$, let D_t = the dividend expected at time t , and let k = the rate of profit. Then, the rate of profit on a share of stock is the value of k that satisfies

$$(1) \quad P_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+k)^t}.$$

³ Friedrich and Vera Lutz, *The Theory of Investment of the Firm* (Princeton, N. J., 1951, Princeton University Press, 253 pp.), 41-43.

⁴ Joel Dean, *Capital Budgeting: Top Management Policy on Plant, Equipment, and Product Development* (New York, 1951, Columbia University Press, 174 pp.); Roland P. Soule, "Trends in the Cost of Capital", *Harvard Business Review*, 31 (March, April, 1953), 33-47.

⁵ J. B. Williams, *The Theory of Investment Value*, (Cambridge, Massachusetts, 1938, Harvard University Press), 87-96.

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It is mathematically convenient to assume that the dividend is paid and discounted continuously at the annual rates D_t and k , in which case

$$(2) \quad P_0 = \int_0^{\infty} D_t e^{-kt} dt.$$

Since P_0 is known, estimating the rate of profit at which a share of stock is selling requires the determination of D_t , $t = 1, 2, \dots, \infty$.

At the outset it should be made clear that our objective is not to find the rate of profit that *will actually be earned* by buying a share of stock. This requires knowledge of the dividends that will be paid in the future, the price at which the share will be sold, and when it will be sold. Unfortunately, such information is not available to us. The rate of profit of interest here is a relation between the present known price and the *expected future dividends*. The latter will vary among individuals with the information they have on a host of variables and with their personality. Therefore, by expected future dividends we mean an estimate that (1) is derivable from known data in an objective manner, (2) is derived by methods that appear reasonable, i.e., not in conflict with common sense knowledge of corporation financial behavior, and (3) can be used to arrive at a manageable measure of the rate of profit implicit in the expectation.

We arrive at D_t by means of two assumptions. One, a corporation is expected to retain a fraction b of its income after taxes; and two, a corporation is expected to earn a return of r on the book value of its common equity. Let Y_t equal a corporation's income per share of common after taxes at time t . Then the expected dividend at time t is

$$(3) \quad D_t = (1 - b)Y_t$$

The income per share at time t is the income at $(t - 1)$ plus r percent of the income at $(t - 1)$ retained, or

$$(4) \quad Y_t = Y_{t-1} + rbY_{t-1}$$

Equation (4) is simply a compound interest expression so that, if Y_t grows continuously at the rate $g = br$,

$$(5) \quad Y_t = Y_0 e^{gt}.$$

From Equations (3) and (5)

$$(6) \quad D_t = D_0 e^{gt}.$$

Substituting this expression for D_t in Equation (2) and integrating, yields

$$\begin{aligned} (7) \quad P_0 &= \int_0^{\infty} D_0 e^{gt} e^{-kt} dt \\ &= D_0 \int_0^{\infty} e^{-t(k-g)} dt \\ &= \frac{D_0}{k - g}. \end{aligned}$$

The condition for a solution is $k > g$, a condition that is easily satisfied, for otherwise, P_0 would be infinite or negative.

Solving Equation (7) for k we find that

(8)

$$k = \frac{D_0}{P_0} + g.$$

Translated, this means that the rate of profit at which a share of common stock is selling is equal to the current dividend, divided by the current price (the dividend yield), plus the rate at which the dividend is expected to grow. Since there are other possible empirical definitions of the market rate of profit on a share of stock, we will refer to k as the growth rate of profit.

III

Let us now review and evaluate the rationale of the model we have just established. Estimating the rate of profit on a share of stock involves estimating the future dividend stream that it provides, and the fundamental difference between this model and the dividend yield is the assumption of growth. The latter, as can be seen, assumes that the dividend will remain constant. Since growth is generally recognized as a factor in the value of a share and since it is used to explain differences in dividend yield among shares, its explicit recognition appears desirable. Future dividends are uncertain, but the problem cannot be avoided by ignoring it. To assume a constant rate of growth and estimate it to be equal to the current rate appears to be a better alternative.

Under this model the dividend will grow at the rate br , which is the product of the fraction of income retained and the rate of return earned on net worth. It is mathematically true that the dividend will grow at this rate if the corporation retains b and earns r . While we can be most certain that the dividend will not grow uniformly and continuously at some rate, unless we believe that an alternative method for estimating the future dividend stream is superior, the restriction of the model to the assumption that it will grow uniformly at some rate is no handicap. Furthermore, the future is discounted; hence, an error in the estimated dividend for a year in the distant future results in a considerably smaller error in k than an error in estimating the dividend in a near year.

It should be noted that this measure of the rate of profit is suspect, when both income and dividend are zero, and it may also be questioned when either falls to very low (or negative) values. In such cases, the model yields a lower rate of profit than one might believe that the market requires on a corporation in such difficulties. It is evident that the dividend and the income yields are even more suspect under these conditions and, hence, are subject to the same limitations.

There are other approaches to the estimation of future dividends than the extrapolation of the current dividend on the basis of the growth rate implicit in b and r . In particular, one can arrive at g directly by taking some average of the past rate of growth in a corporation's dividend. Whether or not this or some other measure of the expected future dividends is superior to the one presented earlier will depend on their relative usefulness in such purposes as the analysis

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of variation in prices among shares and the preferences of those who want an objective measure of a share's rate of profit.

So far, we have compared the growth rate of profit with the income and dividend yields on theoretical grounds. Let us now consider how they differ in practice, using the same measurement rules for the variables in each case. The numerical difference between the growth rate of profit and the dividend yield is simply the growth rate. However, the income yield, which is the measure of the rate of profit commonly recommended for capital budgeting, differs from the growth rate of profit in a more complex manner, and to establish this difference we first note that

$$(9) \quad b = \frac{Y - D}{Y} \text{ and } r = \frac{Y}{B}$$

where B = the net worth or book value per share. The growth rate of profit, therefore, may be written as

$$(10) \quad k = \frac{D}{P} + br = \frac{D}{P} + \frac{Y - D}{B}$$

Next, the income yield can be decomposed as follows:

$$(11) \quad y = \frac{Y}{P} = \frac{D}{P} + \frac{Y - D}{P}$$

We see then that y and k will be equal when book and market values are equal. It can be argued that the income yield overstates a share's payment stream by assuming that each payment is equal to the income per share and understates the payment stream by assuming that it will not grow. Hence, in this special case where book and market values are equal, the two errors exactly compensate each other.

Commonly market and book values differ, and y will be above k when market is below book, and it will be below k when market is above book. Hence, a share of IBM, for example, that is priced far above book had had an earnings yield of two to three percent in 1955. We know that the market requires a higher rate of profit on a common stock, even on IBM, and its growth rate of profit, k , is more in accord with the value suggested by common sense. Conversely, when U. S. Steel was selling at one-half of book value in 1950, the high income yield grossly overstated the rate of profit that the market was, in fact, requiring on the stock.

Furthermore, the growth rate of profit will fluctuate in a narrower range than the earnings yield. For instance, during the last few years, income, dividends, and book value have gone up more or less together, but market price has gone up at a considerably higher rate. Consequently, the growth rate of profit, dependent in part on book value, has fallen less than the earnings yield. Conversely, in a declining market k would rise less rapidly than y .

There is a widespread feeling that many accounting figures, particularly book value per share, are insensitive to the realities of the world, and some may feel

that the comparative stability of k is merely a consequence of the limitations of accounting data. This is not true! The behavior of k is not a consequence of the supposed lack of realism in accounting data. Rather, book value appears in the model because it, and not market value, is used to measure the rate of return the corporation earns on investment, which, we have seen, is the rate of return that enters into the determination of the rate at which the dividend will grow. The comparative stability of k follows from the simple fact that, when a revenue stream is expected to grow, a change in the required rate of profit will give rise to a more than proportional change in the asset's price. Conversely, a change in the price reflects a less than proportional change in the rate of profit.

IV

Given the rate of profit expected on each item in the schedule of available investment opportunities and given the rate of profit at which the corporation's stock is selling, what should the capital budget be? As stated earlier, the accepted theory is that the budget should be set so as to equate the marginal return on investment with the rate of profit at which the stock is selling. The reasoning is, if the market requires, let us say, a 10 percent return on investment in the corporation's stock, and if the corporation can earn 15 percent on additional investment, obtaining the funds and making the investment will increase the earnings per share. As the earnings and the dividend per share increase or as the market becomes persuaded that they will increase, the price of the stock will rise. The objective, it will be recalled, is the maximization of the value of the stockholder's equity.

The conclusion drawn implicitly assumes that the corporation can sell additional shares at or above the prevailing market, or if a new issue depresses the market, the fall will be slight, and the price will soon rise above the previous level. However, some other consideration may argue against a new stock issue; for example, the management may be concerned with dilution of control, or the costs of floating a new issue may be very high, or a new issue may be expected to depress the price severely and indefinitely for reasons not recognized in the theory. Hence, it does not automatically follow that a new issue should be floated when a firm's demand for funds exceeds, according to the above criterion, those that are internally available.

In determining whether the required rate of profit is above or below r' , the marginal return on investment, one can use y , the earnings yield, or k , the growth rate of profit as the required rate of profit. If y and k differ and if the reasoning in support of k presented earlier is valid, using y to estimate the direction in which a new issue will change the price of the stock may result in a wrong conclusion.

In arriving at the optimum size of a stock issue, the objective is to equate r' and y or k , depending on which is used. Internal data may be used to estimate the marginal efficiency of capital schedule. If the required rate of profit is considered a constant, its definition, $y = Y/P$ or $k = D/P + br$, provides its value. However, the required rate of profit may vary with the size of the stock issue or with the variables that may change as a consequence of the issue. In this event,

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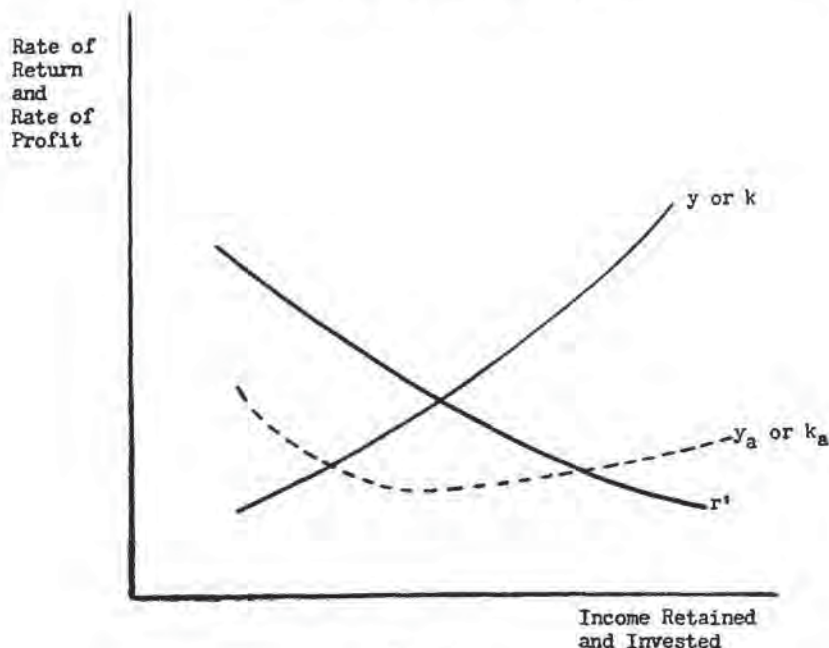


FIG. 1

finding the optimum size of a stock issue requires a model that predicts the variation in the required rate of profit with the relevant variables.

Borrowing is an alternative source of funds for investment. However, an analysis of this alternative requires the measurement of both (1) the variation in risk with debt, and (2) the difference between the rate of profit and the rate of interest needed to cover a given increase in risk. This has not been done as yet, which may explain the widespread practice of arbitrarily establishing a "satisfactory" financial structure and only borrowing to the extent allowed by it.

It has been stated by Dean⁶ and Terborgh⁷ that the long-term ceiling on a firm's capital outlays is the amount of its internally available funds. However, the share of its income a corporation retains is not beyond the control of its management; and, among the things we want from a capital budgeting model is guidance on whether the share of a corporation's income that is retained for investment should be raised or lowered.

Proceeding along traditional lines, the problem may be posed as follows. A firm estimates its earnings and depreciation allowances for the coming year and deducts the planned dividend to arrive at a preliminary figure for the capital budget. The marginal rate of return on investment in excess of this amount may be above or below the required rate of profit. We infer from theory that the two rates should be equated by (1) raising the budget and reducing the dividend

⁶ Dean, *op. cit.*, 53-55.

⁷ George Willard Terborgh, *Dynamic Equipment Policy* (New York, 1949, McGraw-Hill, 290 pp.), 228-29.

when the marginal return on investment is above the required rate of return, and (2) raising the dividend and reducing the budget when the reverse holds. The conditions under which this process yields an equilibrium are illustrated in Figure 1. The marginal return on investment, r' , should fall as the budget is increased, and the required rate of profit, y or k , should increase or it should fall at a lower rate than r' . The latter case is illustrated by the line y_a or k_a .

Changing the dividend so as to equate r' and say y should maximize the price of the stock. For instance, if r' is above y , the company can earn a higher return on investment than stockholders require, and a dollar used this way is worth more to the stockholders than the dollar distributed in dividends. In other words, the price should go up by more than the income retained.

There are, of course, a number of problems connected with the use of this model for arriving at the optimum dividend rate. First, there is the question whether y or k should be used to measure the required rate of profit. Second, there is no question that the required rate of profit varies with the dividend rate. Hence, the current rate of profit given by the definition does not tell what profit rate will be required with a different dividend rate. This requires a model which predicts the variation in y or k with the dividend rate and other variables. Third, there is a very nasty problem of the short and the long run. It is widely believed, though the evidence has limitations, that the price of a share of stock varies with the dividend rate, in which case a corporation should distribute all of its income. However, it is quite possible that a change in the dividend gives rise to the expectation that earnings and future dividends are changing in the same direction. Further, in the short run, the market is not likely to be informed on a firm's marginal efficiency of capital schedule. For these and other reasons, it is likely that the dividend rate should not be made to vary with short-run changes in the marginal efficiency of capital, and more sophisticated methods than those now in use are needed to establish the variation in price or required rate of profit with the dividend rate.

V

The major points developed in this paper may be summarized as follows. We presented a definition of the rate of profit required by the market on a share of common stock, and we noted some of its advantages. It is theoretically superior to the income and dividend yields because it recognizes that the revenue stream provided by a share can be expected to grow. Furthermore, its empirical characteristics are also superior to those of the income and dividend yields since its value is generally in closer agreement with common sense notions concerning the prevailing rate of profit on a share of stock and since its value fluctuates in a narrower range over time. We next examined some of the problems involved in using this definition of the rate of profit and the earnings yield in capital budgeting models. Finally, we saw that, before capital budgeting theory can be made a reliable guide to action, we must improve our techniques for estimating the future revenue on a capital outlay proposal, and we must learn a good deal more about how the rate of profit the market requires on a share of stock varies with the dividend, the growth rate, and other variables that may influence it.

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MICHIGAN PUBLIC SERVICE COMMISSIONConsumers Energy Company

Cost of Common Shareholders' Equity

For the Projected 12 Month Period Ending December 31, 2021

Case No.: U-20697

Exhibit No.: A-132 (TAW-20)

Page: 1 of 3

Witness: TAWehner

Date: February 2020

Capital Asset Pricing Model ApplicationEquation: $K_e = R_f + F + B \times (R_p)$

Where:

 K_e = The annual required return on equity R_f = The risk free rate F = The flotation cost adjustment, not included in the calculation B = The beta, or covariance of the stock price to market R_p = The expected equity risk premium

(a)	(b)	(c)	(d)	(e)	(f)	(g)
<u>Line</u>	<u>Company</u>	<u>Ticker</u>	<u>Current Beta (B)</u>	<u>Test Year Risk-Free Rate (R_f)</u>	<u>Projected Risk Premium (R_p)</u>	<u>Projected CAPM ROE</u>
1	Alliant Energy Corporation	LNT	0.60	2.92%	10.33%	9.12%
2	Ameren Corporation	AEE	0.55	2.92%	10.33%	8.60%
3	American Electric Power Company, Inc.	AEP	0.55	2.92%	10.33%	8.60%
4	Dominion Energy, Inc.	D	0.55	2.92%	10.33%	8.60%
5	DTE Energy Company	DTE	0.55	2.92%	10.33%	8.60%
6	Evergy, Inc.	EVRG	0.49	2.92%	10.33%	7.98%
7	NiSource Inc.	NI	0.55	2.92%	10.33%	8.60%
8	OGE Energy Corp.	OGE	0.75	2.92%	10.33%	10.67%
9	Pinnacle West Capital Corporation	PNW	0.55	2.92%	10.33%	8.60%
10	Portland General Electric Company	POR	0.60	2.92%	10.33%	9.12%
11	WEC Energy Group, Inc.	WEC	0.50	2.92%	10.33%	8.09%
12	Xcel Energy Inc.	XEL	0.50	2.92%	10.33%	8.09%
13	Average		0.56			8.72%
14	Minimum		0.49			7.98%
15	Maximum		0.75			10.67%

Sources: Column (d): Exhibit A-14 (TAW-1), Schedule D-5, page 2, Column (d).

Column (e): Exhibit A-14 (TAW-1), Schedule D-5, page 2, Column (f).

Column (f): Exhibit A-14 (TAW-1), Schedule D-5, page 2, Column (g).

Column (g) = Column (e) + Column (d) x Column (f).

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Discounted Cash Flow ("DCF") Model Application

$$\text{Equation: } K_e = D_1 / P_0 + g + F$$

Where:

K_e = Annual required rate of return on equity

D_1 = Expected annual dividend per share at the end of first year.

P_0 = Current price of stock

g = Growth rate

F = The flotation cost adjustment, not included in the calculation

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
Line	Company	Ticker	Avg of 30-day Closing \$	Last Qtrly Dividend Payment	Current Annual Div (D_0)	Current Dividend Yield	Mid-point Company Guidance (%)	Expected Dividend Yield	Company Guidance DCF ROE	Flotation Cost Adjust. (F)
1	Alliant Energy Corporation	LNT	53.44	0.355	1.42	2.66%	6.0%	2.82%	8.82%	0.13%
2	Ameren Corporation	AEE	75.20	0.495	1.98	2.63%	7.0%	2.82%	9.82%	0.13%
3	American Electric Power Company, Inc.	AEP	92.35	0.700	2.80	3.03%	6.0%	3.21%	9.21%	0.15%
4	Dominion Energy, Inc.	D	81.93	0.918	3.67	4.48%	5.0%	4.70%	9.70%	0.22%
5	DTE Energy Company	DTE	126.10	1.013	4.05	3.21%	7.0%	3.44%	10.44%	0.16%
6	Evergy, Inc.	EVRG	63.63	0.505	2.02	3.17%	6.0%	3.36%	9.36%	0.16%
7	NiSource Inc.	NI	26.86	0.200	0.80	2.98%	6.0%	3.16%	9.16%	0.15%
8	OGE Energy Corp.	OGE	43.10	0.388	1.55	3.60%	5.0%	3.78%	8.78%	0.18%
9	Pinnacle West Capital Corporation	PNW	87.40	0.783	3.13	3.58%	6.5%	3.81%	10.31%	0.18%
10	Portland General Electric Company	POR	55.51	0.385	1.54	2.77%	6.0%	2.94%	8.94%	0.14%
11	WEC Energy Group, Inc.	WEC	89.83	0.590	2.36	2.63%	6.0%	2.78%	8.78%	0.13%
12	Xcel Energy Inc.	XEL	62.23	0.405	1.62	2.60%	6.0%	2.76%	8.76%	0.13%
13	Average								9.34%	0.16%
14	Minimum								8.76%	0.13%
15	Maximum								10.44%	0.22%

Sources:

Column (d): Factset data from November 18, 2019 through December 31, 2019

Column (e): Yahoo!Finance as of December 31, 2019.

Column (f) = 4 x Column (e).

Column (g) = Column (f) / Column (d).

Column (h): LNT, AEP, DTE, EVRG, NI, OGE, POR, WEC and XEL company dividend guidance.

AEE, D, and FE company earnings guidance. PNW company rate base guidance.

Column (i) = Column (g) x (1 + Column (h)).

Column (j) = Column (h) + Column (i).

MICHIGAN PUBLIC SERVICE COMMISSION

Consumers Energy Company

Cost of Common Shareholders' Equity

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Case No.: U-20697

Exhibit No.: A-132 (TAW-20)

Page: 3 of 3

Witness: TAWehner

Date: February 2020

Discounted Cash Flow ("DCF") Model Application

$$\text{Equation: } K_e = D_1 / P_0 + g + F$$

Where:

K_e = Annual required rate of return on equity

D_1 = Expected annual dividend per share at the end of first year.

P_0 = Current price of stock

g = Growth rate

F = The flotation cost adjustment, not included in the calculation

(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
Line	Company	Ticker	Avg of 30-day Closing \$	Last Qtrly Dividend Payment	Current Annual Div (D ₀)	Current Dividend Yield	Consensus Analyst EPS Growth	Expected Dividend Yield	Consensus Earnings Growth DCF ROE	Payout Ratio (%)	Analyst EPS Yield	Earnings Based DCF ROE	Flotation Cost Adjust. (F)
1	Alliant Energy Corporation	LNT	53.44	0.355	1.42	2.66%	5.6%	2.81%	8.41%	63.7%	4.4%	10.01%	0.13%
2	Ameren Corporation	AEE	75.20	0.495	1.98	2.63%	4.2%	2.74%	6.94%	59.1%	4.6%	8.84%	0.13%
3	American Electric Power Company, I	AEP	92.35	0.700	2.80	3.03%	5.8%	3.21%	9.01%	62.9%	5.1%	10.90%	0.15%
4	Dominion Energy, Inc.	D	81.93	0.918	3.67	4.48%	4.6%	4.69%	9.29%	275.9%	1.7%	6.30%	0.22%
5	DTE Energy Company	DTE	126.10	1.013	4.05	3.21%	3.9%	3.34%	7.24%	64.6%	5.2%	9.07%	0.16%
6	Evergy, Inc.	EVERG	63.63	0.505	2.02	3.17%	9.6%	3.48%	13.08%	74.5%	4.7%	14.27%	0.16%
7	NiSource Inc.	NI	26.86	0.200	0.80	2.98%	3.9%	3.09%	6.99%	65.0%	4.8%	8.66%	0.15%
8	OGE Energy Corp.	OGE	43.10	0.388	1.55	3.60%	4.3%	3.75%	8.05%	65.6%	5.7%	10.02%	0.18%
9	Pinnacle West Capital Corporation	PNW	87.40	0.783	3.13	3.58%	4.2%	3.73%	7.93%	67.5%	5.5%	9.73%	0.18%
10	Portland General Electric Company	POR	55.51	0.385	1.54	2.77%	4.8%	2.91%	7.71%	67.1%	4.3%	9.13%	0.14%
11	WEC Energy Group, Inc.	WEC	89.83	0.590	2.36	2.63%	6.2%	2.79%	8.99%	68.2%	4.1%	10.29%	0.13%
12	Xcel Energy Inc.	XEL	62.23	0.405	1.62	2.60%	6.2%	2.76%	8.96%	63.8%	4.3%	10.53%	0.13%
13	Average								8.55%			9.81%	0.16%
14	Minimum								6.94%			6.30%	0.13%
15	Maximum								13.08%			14.27%	0.22%

Sources:

Column (d): Factset data from November 18, 2019 through December 31, 2019

Column (e): Yahoo!Finance as of December 31, 2019.

Column (f) = 4 x Column (e).

Column (g) = Column (f) / Column (d).

Column (h): I/B/E/S 3-year consensus analyst earnings per share ("EPS") growth estimate as of December 31, 2019.

Column (i) = Column (g) x (1 + Column (h)).

Column (j) = Column (h) + Column (i).

Column (k): S&P Global Market Intelligence LTM dividend payout ratio as of December 31, 2019

Column (l): Column (i) / Column (k).

Column (m): Column (h) + Column (l).

Column (n): Flotation cost adjustment of 5% of current dividend yield, as described by Roger A. Morin, "New Regulatory Finance" (2006)

While flotation cost adjustment are estimated and demonstrated in this analysis, the adjustment is not included in any of the calculations comprising the analysis: